

Restriction, Keakeya, and Decoupling

A Graduate Pathway from Second-Year Analysis

Volumes 1–14

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Preface to the Combined Edition

What this is

This collected edition contains all fourteen volumes of *Restriction, Keakeya, and Decoupling: A Graduate Pathway from Second-Year Analysis*. The series builds from measure, Fourier analysis, singular integrals, function spaces and oscillatory integrals through restriction, fractal geometry, Keakeya, polynomial methods and decoupling, and ends with the modern projection-theoretic frontier around three-dimensional Keakeya.

How to read the numbering

Each Part is one Volume, and chapters restart at 1 in each. Every numbered item therefore carries its volume in its number:

$$\text{Theorem } \underbrace{3}_{\text{Volume}} \cdot \underbrace{2}_{\text{Chapter}} \cdot \underbrace{4}_{\text{item}}$$

so Theorem 3.2.3 is Volume 3, Chapter 2. Worked examples and drills are numbered in their own sequences within each chapter, so Worked Example 3.2.1 and Theorem 3.2.1 may both exist and are different objects.

A note on cross-references

In the separate volumes, a reference to an earlier volume was a piece of text — “Volume 1, Theorem 7.3” — carrying a number written by hand. Producing this combined edition made every such reference a genuine cross-reference, resolved by the typesetter.

This was not merely cosmetic. Checking the hand-written numbers against the labels they were meant to name turned up a substantial number of errors, in two waves: first a systematic failure caused by the source volumes being

renumbered after the citations were drafted, and then a residue of individually wrong numbers that a check for *existence* could not detect, because the wrong number happened to name some other result. In the present edition no cross-reference carries a hand-written number, so that entire class of error is gone by construction. The episode is recorded in the project’s feedback log as items F5 and F7.

Conventions

The typographic system is uniform across the fourteen volumes and is described in the Preface to Volume 1. In brief: worked examples come in three tiers, distinguished by rule weight and by a printed label; commentary — *Idea of the proof*, *Scratch work*, *Where we are*, *Looking ahead*, *Pitfall* — is set off by a rule in the margin and may be skipped without loss of continuity; and proofs longer than half a page are broken into numbered steps with stated goals.

Each volume opens with its own preface, stating what it assumes and what it will prove, and closes with two appendices: one listing every result quoted from earlier volumes together with the place it is used, and one giving notation and an index of results. Volumes 4 and 5 also record, in their second appendix, the results removed from the original plan, with reasons.

Proof status and the HA audit

The original release attempted to classify every omission by revision number. HA-01 re-read the actual source and its proof dependencies and found that the resulting global proof-completeness claim was too strong. A proof environment labelled *Scheme*, *Structure*, or *outline* is not counted as a strict proof; nor is a result whose parameter bookkeeping is explicitly not reproduced. A theorem depending on such an item is marked *transitively unclosed* until the input is repaired.

The authoritative current ledger is `THEOREM-MANIFEST.txt`. The HA-08 reconstruction supplied detailed Volume 14 proof dossiers and reconciled the active cross-volume references. Successive human-review passes expanded the random-thinning, rigid-motion, scale-normalization, and fixed-loss parts of the general-to-sticky argument. The audit now records two load-bearing blockers: the parent–child shading interface is not preserved by the printed final refinement, and Main Lemma `lemmain2` does not yet obtain a gain uniform in the target loss. The fourth closure attempt made the cellular interface and the uniform-gain criterion explicit, but did not prove either missing input. The edition is therefore an active human-review working candidate, not an internally proof-closed release and not a substitute for independent expert

verification. Earlier plan revisions remain recorded but are not release blockers unless a later proof actually uses them.

Literature status checked 31 August 2026. The available Hua–Yang cellular repair, arXiv:2608.18870v3, is explicitly conditional; the streamlined Guth–Wang–Zahl proof remains arXiv:2601.14411v1; and no separate public fixed-loss correction was found. The longer Wang–Zahl route, arXiv:2502.17655v1, is reproduced as an alternative under audit in Volume 14. It remains conditional at the four exact obligations WZ–A1–A4 listed in Section 14.4.13, and is therefore not an automatically accepted substitute.

Conventions, prerequisites, and how to read this series

Prerequisites

Referee B asked for this and was right to. The series assumes a first course in real analysis at the level of Knapp, *Basic Real Analysis*: metric spaces, uniform convergence, the Riemann integral, and elementary linear algebra. Measure theory is *not* assumed — Volume 1 builds it — but two results from a first course are used before they are reproved: duality of L^p spaces and the Riesz representation theorem. Both are in Knapp, and Volume 1 cites them where used.

No complex analysis is needed beyond the definition of a holomorphic function. The polynomial method is developed largely from first principles in Volume 11; HA-01 nevertheless flags the real-algebraic cell-count input there for explicit closure rather than treating it as proved by the present text.

Asymptotic notation

Four relations are used throughout and they are *not* interchangeable.

$$\begin{aligned}
 A \lesssim B & \quad A \leq CB \text{ for a constant } C \text{ independent of every parameter} \\
 & \quad \text{except those written as subscripts: } A \lesssim_{n,\varepsilon} B \text{ means } C = C(n,\varepsilon) \\
 A \gtrsim B & \quad B \lesssim A \\
 A \approx B & \quad A \lesssim B \text{ and } B \lesssim A \\
 A \lesssim\lesssim B & \quad A \leq C\left(\log \frac{1}{\delta}\right)^C B, \text{ a polylogarithmic loss} \\
 & \quad \text{in the small parameter } \delta; \gtrsim\gtrsim \text{ is the reverse}
 \end{aligned}$$

The distinction between $\lesssim$ and $\lesssim\lesssim$ is load-bearing. Volumes 9, 11 and 14 turn on whether a loss is a constant, a power of $\log \frac{1}{\delta}$, or a power $\delta^{-\varepsilon}$; a polylogarithmic loss is $\delta^{-\varepsilon}$ for every $\varepsilon > 0$ but is strictly better, and

several arguments can afford the first and not the third. Volume 12, Worked Example 12.4.3 traces the difference through a complete proof.

Two dependences are suppressed throughout, because they are fixed for the whole of any given argument: the ambient dimension n (or the degree k , for the moment curve), and the surface or measure under discussion. A bare $\lesssim$ therefore means “with a constant depending on the dimension and on the fixed data of the problem, and on nothing else”.

Every other dependence is written as a subscript, and this matters most for ε . An estimate $A \lesssim_\varepsilon \delta^{-\varepsilon} B$ says the constant may blow up as $\varepsilon \rightarrow 0$ — and it always does, often at a rate no argument in this series controls. Volume 13, Worked Example 13.2.5 shows how bad the constants are at any scale one might compute with, and Volume 14, Theorem 14.3.1 carries an unspecified exponent for exactly this reason. *If a bound is claimed with an absolute constant, the text says so explicitly.*

Dimensions and related quantities

$\dim_{\mathrm{H}} E$	Hausdorff dimension, Volume 7, Theorem 7.1.3
$\dim_{\mathrm{M}} E$, $\underline{\dim}_{\mathrm{M}} E$	upper and lower Minkowski (box) dimension, Volume 7, Chapter 7.2
$\dim_{\mathrm{P}} E$	packing dimension, discussed in Volume 14, Chapter 14.5
$\mathcal{E}_\rho(E)$	the least number of ρ -balls covering E
$I_\alpha(\mu)$	the α -energy $\iint x - y ^{-\alpha} \mathrm{d}\mu(x) \mathrm{d}\mu(y)$
$\mathrm{RapDec}(R)$	a quantity $\lesssim_N R^{-N}$ for every N : <i>rapidly decaying</i>

Always $\underline{\dim}_{\mathrm{M}} E \leq \dim_{\mathrm{M}} E$ and $\dim_{\mathrm{H}} E \leq \dim_{\mathrm{M}} E$; $\dim_{\mathrm{H}}$ is countably stable and $\dim_{\mathrm{M}}$ is not, which is why the conjectures in this series are stated for $\dim_{\mathrm{H}}$.

The revision numbers

Revision numbers remain useful historical markers, but HA-01 no longer treats them as a complete proof certificate. The current source-level classification also records architecture-only proofs and transitive dependencies. See Chapter 14.15 and, authoritatively during the revision cycle, `THEOREM-MANIFEST.txt`.

In particular, Volume 13 currently contains a formal local-smoothing deduction from the unproved cone-decoupling input, and its moment-curve proof contains a load-bearing lower-degree lemma supplied only at structural level. Volume 14 contains the larger concentration of such gaps.

Words used precisely

Essentially and *morally* are used only where the text goes on to make the statement precise, and never as a substitute for an argument. Where a claim

is genuinely heuristic it is labelled as such.

Part 1

Volume 1

Measure, Integration, and L^p Spaces

Preface

What this series is for

There is a cluster of problems in modern analysis — the Fourier restriction conjecture, the Kakeya conjecture, and the decoupling theorems of Bourgain and Demeter — which are among the most active and most beautiful in contemporary mathematics. They are also notoriously hard to approach. The research papers assume a great deal; the textbooks that cover them, Demeter’s *Fourier Restriction, Decoupling, and Applications* foremost among them, begin roughly where a strong first-year graduate student ends.

This series begins where a second-year undergraduate ends. Concretely, we assume the analysis of a Warwick second year: sequences and series, continuity and differentiation, the Riemann integral, uniform convergence, metric spaces, and multivariable calculus. We assume linear algebra. We do *not* assume Lebesgue measure, functional analysis, complex analysis, Fourier analysis, or algebraic geometry. Everything needed is built here, in order, with complete proofs.

The destination is the frontier: Guth’s polynomial partitioning method, the Bourgain–Demeter decoupling theorem, and the 2025 resolution of the three-dimensional Kakeya conjecture by Wang and Zahl. Fourteen volumes stand between here and there. This is the first.

The pedagogical contract

Three commitments shape every page of this series.

Complete proofs, with no gestures. Every theorem we state, we prove. Where a proof is long, it is long; we do not write “it can be shown that” and move on. The one exception is clearly marked: in the final volume, results at the very edge of current research are given with their key lemmas proved and their global architecture explained, but not every technical estimate reproduced.

Explanation before formalism. A proof that is merely correct teaches very little. Before the significant arguments you will find an *Idea of the proof* box explaining what the argument is really doing, what the obstruction was, and why the natural first attempt fails. Failed approaches are discussed on purpose. In this subject, the reason a technique exists is almost always that something simpler broke, and a reader who has not seen the break will not understand the repair.

Worked examples in place of exercise lists. Rather than a long list of unsolved exercises, this series provides worked examples at three tiers. Each is set off from the text by rules, and the tier is named in the heading; the weight of the rules — thin, medium, doubled — repeats the same information for a reader scanning a page.

- **Tier A (calibration).** Short computations that build fluency. Roughly 30% of the examples.
- **Tier B (technique).** A method the reader must own, worked in full and often worked twice from different angles. Roughly 45%.
- **Tier C (research taste).** A genuine piece of the literature, self-contained, reachable from what has been proved so far. These are what make the series worth reading rather than one of the many good measure theory books already available. Roughly 25%.

Four kinds of commentary are set off by a single rule in the margin. *Idea of the proof* states what an argument is really doing before it is done. *Scratch work* shows how it was found. *Where we are* interrupts a long development to say what has been used and what remains. *Pitfall*, marked by a broken rule, names an error that readers actually make. None of these boxes contains anything the proofs depend on, so a reader in a hurry may skip them all; a reader trying to learn the subject should read them first.

A small number of **Drills** remain unsolved. These are deliberately mandatory. They are almost all computational, they concentrate in Volumes 1–6, and they exist because there are things that cannot be learned by reading someone else’s solution. Do them.

How to read Volume 1

This volume covers measure theory. Many readers will be tempted to skip it, or to skim it having seen some measure theory elsewhere. Two chapters make that a mistake.

Chapter 1.7 develops covering lemmas and the Hardy–Littlewood maximal function. This is not a service chapter. The Hardy–Littlewood maximal function is the first object in the series with genuine *tube geometry* hiding inside it, and the Vitali and Besicovitch covering lemmas are the direct ancestors of the covering arguments that run through the Kakeya problem all the way to Wang–Zahl. A reader who understands why the Besicovitch covering lemma fails without a dimensional constant is already thinking the right way about Kakeya.

Chapter 1.8 develops L^p spaces and, crucially, weak L^p . Almost every theorem in this subject is an inequality between L^p norms, and the difference between strong and weak bounds is where a great deal of the real content lives.

Sections marked with a dagger (†) may be omitted on first reading without loss of continuity; they are cross-referenced from later volumes when needed.

Notation

We write $A \lesssim B$ to mean $A \leq CB$ for some constant C independent of the parameters in play, and $A \approx B$ when $A \lesssim B \lesssim A$. Subscripts indicate dependence: $A \lesssim_\varepsilon B$ means the constant may depend on ε . This notation is used relentlessly in the research literature of this subject, and we adopt it from the very beginning so that it becomes invisible long before it becomes essential. A full notation index is Appendix B.

A note on the constants. Beginners often find the suppression of constants unsettling. It is worth saying plainly why analysts do it: in this subject the constants are almost never the point, and carrying them makes the structure of an argument invisible. When a constant *does* matter — and in the induction-on-scales arguments of Volume 13 it matters enormously, because a constant that degrades at each of $\log(1/\delta)$ scales is fatal — we track it explicitly and say so.

Chapter 1.1

The Failure of Riemann Integration

1.1.1 What you already have, and what it cannot do

You arrive at this book with the Riemann integral. It is a fine object. It integrates every continuous function on a closed bounded interval, it satisfies the fundamental theorem of calculus, and it is defined by a construction — chop the domain into small intervals, sample the function, sum, refine — that is transparent enough to draw on a napkin.

It is also, for our purposes, unusable. Not merely inconvenient: unusable. The central objects of this series cannot be defined with it, let alone manipulated. This chapter is about why, and it is worth taking seriously, because the shape of the failure determines the shape of the repair. Lebesgue's theory is not an arbitrary generalisation that happens to be more powerful. It is the specific fix for a specific structural defect, and if you see the defect clearly then every definition in Chapters 1.2 and 1.3 will feel forced rather than invented.

We identify three failures. The first is the one every textbook mentions. The second is the one that actually matters. The third is the one that matters for *us*.

1.1.2 Failure I: limits do not pass through the integral

Suppose $f_n \rightarrow f$ pointwise on $[0, 1]$, with every f_n Riemann integrable and every $|f_n| \leq 1$. Is f Riemann integrable? And if so, does $\int f_n \rightarrow \int f$?

The answer to the second question is subtle and the answer to the first is

simply *no*.

WORKED EXAMPLE 1.1.1— TIER A A pointwise limit of Riemann integrable functions that is not Riemann integrable

Enumerate the rationals in $[0, 1]$ as $q_1, q_2, q_3, \dots$ and set

$$f_n(x) = \mathbf{1}_{\{q_1, \dots, q_n\}}(x) = \begin{cases} 1 & x \in \{q_1, \dots, q_n\}, \\ 0 & \text{otherwise.} \end{cases}$$

Each f_n is Riemann integrable: it differs from the zero function at finitely many points, so for any partition into intervals of total length $< \varepsilon/n$ containing those points, the upper sum is at most ε and the lower sum is 0. Hence $\int_0^1 f_n = 0$ for every n .

The sequence increases pointwise to

$$f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]},$$

the Dirichlet function. This is not Riemann integrable. For *any* partition P of $[0, 1]$, every subinterval of positive length contains both a rational and an irrational, so the upper sum $U(f, P) = 1$ and the lower sum $L(f, P) = 0$. The upper and lower integrals are 1 and 0 and never meet.

So: a uniformly bounded, monotone increasing sequence of Riemann integrable functions, converging pointwise everywhere, with a limit that the theory cannot integrate at all. Note how mild the hypotheses were. Monotone *and* bounded *and* pointwise convergent everywhere is about as much as one could ask for, and it is not enough.

The Lebesgue theory will integrate $\mathbf{1}_{\mathbb{Q} \cap [0, 1]}$ without effort — the answer is 0, because $\mathbb{Q}$ is countable and so has measure zero — and the Monotone Convergence Theorem (Chapter 1.5) will make the interchange $\lim \int = \int \lim$ automatic under exactly the hypotheses above.

Idea of the proof. Look again at why the Dirichlet function defeats the Riemann integral. The obstruction is that we insisted on partitioning the *domain* into intervals. On every interval, however short, f takes both values 0 and 1; the function oscillates at every scale, so no refinement of an interval partition ever improves the estimate. But the function only takes two values. If we were allowed to group together *all* the points where $f = 1$, regardless of whether they form an interval, the integral would be obvious. Lebesgue's move is exactly this: partition the *range* instead of the domain, and then be forced to make sense of the sizes of the resulting, possibly hideous, level sets. That forced side-problem is measure theory.

1.1.3 Failure II: the space is not complete

The first failure is the famous one. This one is more serious, and it is the reason measure theory is not optional in analysis.

Equip the space $C[0, 1]$ of continuous functions with the norm

$$\|f\|_{L^1} = \int_0^1 |f(x)| \, dx,$$

the integral being Riemann's. This is a perfectly good normed vector space. It is not complete, and the failure is not exotic.

WORKED EXAMPLE 1.1.2— TIER B $C[0, 1]$ is not complete in the L^1 norm

Define, for $n \geq 2$, the piecewise linear function

$$f_n(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2}, \\ n \left(x - \frac{1}{2} \right), & \frac{1}{2} < x < \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq x \leq 1. \end{cases}$$

Each f_n is continuous: a ramp that climbs from 0 to 1 over a window of width $1/n$ just to the right of $\frac{1}{2}$.

Step 1: the sequence is Cauchy. For $m > n$, the functions f_n and f_m agree outside the interval $(\frac{1}{2}, \frac{1}{2} + \frac{1}{n})$, and on that interval both take values in $[0, 1]$, so $|f_n - f_m| \leq 1$ there. Hence

$$\|f_n - f_m\|_{L^1} = \int_{1/2}^{1/2+1/n} |f_n - f_m| \leq \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0.$$

So (f_n) is Cauchy in $\|\cdot\|_{L^1}$.

Step 2: the limit is not continuous. Pointwise, $f_n(x) \rightarrow f(x)$ where $f = \mathbf{1}_{(1/2, 1]}$, a step function. And indeed $\|f_n - f\|_{L^1} \leq 1/n \rightarrow 0$. So if the sequence converges to anything in L^1 , it converges to f .

Step 3: no continuous function works. Suppose $g \in C[0, 1]$ satisfied $\|f_n - g\|_{L^1} \rightarrow 0$. Then $\|f - g\|_{L^1} \leq \|f - f_n\|_{L^1} + \|f_n - g\|_{L^1} \rightarrow 0$, so $\int_0^1 |f - g| = 0$. Now $|f - g|$ is continuous on $[0, \frac{1}{2})$ and on $(\frac{1}{2}, 1]$ separately, and a nonnegative continuous function with zero integral on an interval vanishes identically there (if $|f - g|(x_0) = c > 0$ at some point of one of these open intervals, then by continuity $|f - g| > c/2$ on a subinterval of some length $\ell > 0$, forcing the integral to be at least $c\ell/2 > 0$). Therefore $g = 0$ on $[0, \frac{1}{2})$ and $g = 1$ on $(\frac{1}{2}, 1]$. Letting $x \uparrow \frac{1}{2}$ gives $g(\frac{1}{2}) = 0$; letting $x \downarrow \frac{1}{2}$ gives $g(\frac{1}{2}) = 1$. Contradiction.

WORKED EXAMPLE 1.1.3— TIER A The same phenomenon, seen as an infinite series

Completeness failures are often easier to spot in series form. Consider $\sum_k g_k$ where $g_k = f_{k+1} - f_k$ with f_n as above. Then $\sum_k \|g_k\|_{L^1} < \infty$ — the series is absolutely convergent in norm — yet the partial sums have no continuous limit. In a complete normed space, absolute convergence implies convergence; here it does not. We will prove the equivalence of these two statements as Chapter 1.8 develops, and it is worth noticing now that the ability to sum an absolutely convergent series of functions and land somewhere is the single most-used consequence of completeness in all of harmonic analysis.

Why does this matter so much? Because completeness is what lets you *build* objects rather than merely inspect them. Almost every construction in later volumes — the Fourier transform on L^2 (Volume 2), the wave packet decomposition of a function (Volume 12), the limiting arguments in induction-on-scales (Volume 13) — proceeds by producing a Cauchy sequence of approximations and taking a limit. Without completeness there is no limit to take.

Looking ahead: Plancherel needs this

The Fourier transform is defined by an integral formula $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$ which requires $f \in L^1$, but its most important property — Plancherel’s theorem, $\|\widehat{f}\|_{L^2} = \|f\|_{L^2}$ — lives in L^2 , and $L^2 \not\subseteq L^1$ on $\mathbb{R}^n$. The resolution is to define the transform on the dense subspace $L^1 \cap L^2$ and extend by continuity, and *extension by continuity requires the target space to be complete*. The Riemann theory cannot even state Plancherel’s theorem. This is not an exaggeration for effect; it is why measure theory comes first.

1.1.4 Failure III: there is no theory of size for the sets we care about

The two failures above are standard. The third is specific to this series, and it is the one to keep in mind.

The problems this series is aimed at are, at bottom, questions about the *size of unions of thin tubes*. Here is the archetype, in a form you can already understand completely.

Definition 1.1.1 (Besicovitch set). A *Besicovitch set* (or *Kakeya set*) in $\mathbb{R}^n$ is a compact set containing a unit line segment in every direction: for every $e \in \mathbb{S}^{n-1}$ there is $x \in \mathbb{R}^n$ with $\{x + te : t \in [0, 1]\} \subseteq K$.

The obvious example in the plane is a disc of diameter 1, which has area $\pi/4$. One can do better with a Reuleaux triangle, better still with a three-cusped hypocycloid (the deltoid, area $\pi/8$). The question Kakeya asked in 1917 was: how small can such a set be?

The answer, found by Besicovitch, is that it can have area zero. And note what kind of statement that is. It is a statement about *Lebesgue measure*. It cannot be phrased in terms of the Riemann integral at all, because the sets involved are not Jordan measurable — they are precisely the wild, everywhere-oscillating sets that defeated us in §1.2. The entire subject of this series begins with a theorem that the Riemann theory cannot express.

WORKED EXAMPLE 1.1.4— TIER C: RESEARCH TASTE **How a Besicovitch set gets small: the bisection trick**

We will not construct a measure-zero Besicovitch set here — that is Chapter 1.3, once we have the tools — but the mechanism is elementary and worth seeing now, because it is the first appearance of a phenomenon (*overlapping tubes at many scales*) that will occupy us for thirteen more volumes.

Setup. Let T be a triangle with base $[0, 1]$ on the x -axis and apex at height h . Slide a needle around inside T and you can realise every direction in a certain angular range. Now cut T vertically through the apex into two triangles T_1 (base $[0, \frac{1}{2}]$) and T_2 (base $[\frac{1}{2}, 1]$), each of area $|T|/2$, so no saving yet.

The trick. Translate T_2 horizontally to the left by an amount $t \in [0, \frac{1}{2}]$, obtaining $T_2 - (t, 0)$. Crucially, *translation does not change the set of directions of segments inside a triangle*. So the union $T_1 \cup (T_2 - (t, 0))$ still realises every direction that T did. But its area has dropped, because the two triangles now overlap.

The computation. Take $t = \frac{1}{2} - \frac{\alpha}{2}$ for a parameter $\alpha \in (0, 1)$, so that the two triangles are pushed together until their bases overlap in a segment of length $\frac{1-\alpha}{2}$. A direct similar- triangles calculation — the two triangles have parallel corresponding sides, and their intersection is a triangle similar to T — gives

$$|T_1 \cap (T_2 - (t, 0))| = (1 - \alpha)^2 \frac{|T|}{4},$$

so that

$$\begin{aligned} |T_1 \cup (T_2 - (t, 0))| &= 2 \cdot \frac{|T|}{2} - (1 - \alpha)^2 \frac{|T|}{4} \\ &= |T| \left(1 - \frac{(1 - \alpha)^2}{4} \right). \end{aligned}$$

Taking $\alpha \rightarrow 0$ (pushing the triangles fully together) saves a factor of $3/4$. One bisection, one quarter of the area gone, no directions lost.

The iteration and where it goes. Now bisect again: each of the two triangles splits into two, and we slide each pair together. Iterating k times produces 2^k thin triangles, arranged in a spiky figure called a *Perron tree*, whose total area tends to 0 as $k \rightarrow \infty$ while the set of directions is preserved throughout. Making this rigorous requires care — the naive product $\prod_k (1 - \frac{1}{4})$ tends to 0 too fast to be honest, and the true argument must control the overlaps at *all* scales simultaneously — and that is precisely the point.

Why this is the whole subject in miniature. Notice what made the saving possible: many long thin sets, pointing in many different directions, can be translated so that they overlap far more than a random arrangement would suggest. The *Kakeya conjecture* (Volumes 9, 10, 14) asserts that this overlapping has a hard limit — that although the *measure* can be zero, the *dimension* cannot drop below n . The *restriction conjecture* (Volumes 6, 12) turns out to be a statement of the same type in disguise, about how much wave packets travelling in different directions can conspire to be large in the same place. Every technique in this series — covering lemmas, maximal functions, induction on scales, polynomial partitioning — is ultimately a device for controlling the overlap of tubes. You have just seen the first one.

Looking ahead: the same picture, thirteen volumes later

In Chapter 1.7 of this volume you will meet the Vitali covering lemma, which controls overlaps of *balls*. Balls are easy: two balls of comparable size either intersect a lot or not at all. Tubes are hard: two tubes of length 1 and width δ crossing at angle θ intersect in a set of measure $\approx \delta^2/\theta$, so the overlap depends on the *angle*, and an arrangement of tubes can be tuned to make many angles small at once. The gap between these two sentences is, more or less, the difference between a first course in measure theory and the Wang–Zahl theorem.

1.1.5 The plan of repair

Lebesgue’s programme, which occupies the rest of this volume, is:

1. Assign a size $m(E) \in [0, \infty]$ to as many subsets $E \subseteq \mathbb{R}^n$ as possible, agreeing with length/area/volume on boxes, and behaving well under *countable* unions rather than merely finite ones (Chapters 1.2 and 1.3).
2. Discover, unavoidably, that “as many as possible” is not “all” — non-measurable sets exist — and isolate the good class, the measurable sets (Chapter 1.3).

3. Define the integral by partitioning the range, using step 1 to measure the level sets (Chapter 1.4).
4. Harvest the convergence theorems that the Riemann theory lacked (Chapter 1.5), and the Fubini–Tonelli theorem that lets us integrate in $\mathbb{R}^n$ one variable at a time (Chapter 1.6).
5. Develop the differentiation theory — covering lemmas and the maximal function — which is where measure theory stops being infrastructure and starts being harmonic analysis (Chapter 1.7).
6. Build the L^p spaces, prove they are complete, and meet weak L^p (Chapter 1.8).

Drills

Drill 1.1.1. Show that $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = 1/q$ if $x = p/q$ in lowest terms and $f(x) = 0$ if x is irrational is Riemann integrable with integral 0, despite being discontinuous at every rational. Contrast with the Dirichlet function and explain, in one sentence, what distinguishes the two.

Drill 1.1.2. Carry out the similar-triangles computation asserted in Worked Example 1.4: verify that the intersection of the two overlapping half- triangles is similar to T with ratio $(1 - \alpha)/2$, and hence has area $(1 - \alpha)^2|T|/4$.

Drill 1.1.3. Exhibit a sequence $f_n \in C[0, 1]$ with $\|f_n\|_{L^1} \rightarrow 0$ but $f_n(x) \not\rightarrow 0$ for any $x \in [0, 1]$. (Such a sequence rules out the naive hope that L^1 convergence implies pointwise convergence; we will see in Chapter 1.5 what survives.)

Notes

The Dirichlet function dates to 1829. Besicovitch’s construction appeared in 1919 in a paper on Riemann integration — he was answering a question about whether a Riemann-integrable function of two variables always admits a coordinate system in which the iterated integrals behave, and the measure-zero Kakeya set was a by-product. The connection to Fourier analysis was not visible until Fefferman’s 1971 disproof of the ball multiplier conjecture, which we prove in Volume 5. Readers who want to see the destination early may look at Tao’s expository article on the Kakeya problem, though nothing there is needed here.

Chapter 1.2

Outer Measures and the Carathéodory Construction

1.2.1 The problem, stated honestly

We want a function m assigning to each subset $E \subseteq \mathbb{R}^n$ a number $m(E) \in [0, \infty]$, such that:

(M1) $m(\text{box}) = \text{its elementary volume}$;

(M2) m is invariant under translations;

(M3) $m(\bigcup_{k=1}^{\infty} E_k) = \sum_{k=1}^{\infty} m(E_k)$ whenever the E_k are pairwise disjoint (*countable additivity*).

It is a theorem, not an accident, that no such m exists on *all* subsets of $\mathbb{R}^n$. We prove this in Chapter 1.3 via Vitali's construction. So something must be given up, and the whole design of measure theory is a decision about *what* to give up.

The options are: weaken (M3) to finite additivity — this is what Banach measures do, and it destroys the convergence theorems, which were the entire point — or restrict the domain of m to a well-chosen family of sets. Lebesgue and Carathéodory take the second road, and the elegance of Carathéodory's formulation is that the good family is not chosen by hand: it *emerges* from a single condition.

Idea of the proof. The strategy is two-step and it is worth naming before we begin. First, define a crude size μ^* on *all* sets by covering them efficiently with boxes. This μ^* will be countably *subadditive* but not additive — covering is wasteful, and the waste is real. Second, identify the sets on which μ^* stops being wasteful. Carathéodory's insight is that the right test

is not about the set itself but about how the set *cuts* other sets: E is good precisely when it splits every test set A cleanly, with no loss.

1.2.2 The language of σ -algebras

Before constructing anything we fix vocabulary. A reader who has met measure theory before may skim this section; a reader who has not should read it carefully, because every term defined here is used without further comment for the remaining thirteen volumes.

Definition 1.2.1 (Algebra and σ -algebra). Let X be a set. A family $\mathcal{A} \subseteq \mathcal{P}(X)$ is an *algebra* if

$$(A1) \quad \emptyset \in \mathcal{A};$$

$$(A2) \quad E \in \mathcal{A} \implies E^c \in \mathcal{A};$$

$$(A3) \quad E, F \in \mathcal{A} \implies E \cup F \in \mathcal{A}.$$

It is a σ -*algebra* if (A3) is strengthened to

$$(A3') \quad E_1, E_2, \dots \in \mathcal{A} \implies \bigcup_{k=1}^{\infty} E_k \in \mathcal{A}.$$

By De Morgan's laws, an algebra is automatically closed under finite intersections and set differences, and a σ -algebra under countable intersections as well: $\bigcap_k E_k = \left(\bigcup_k E_k^c\right)^c$. Also $X = \emptyset^c \in \mathcal{A}$ always.

Scratch work. The single letter of difference between (A3) and (A3') is the whole subject. Finite additivity is easy to arrange and useless; the convergence theorems of Chapter 1.5, which are the entire reason for building this machinery, are statements about countable operations and fail without (A3'). Whenever a proof in this volume looks as though it is working unreasonably hard, it is almost always working to upgrade a finite statement to a countable one — see Steps 4 and 5 of Theorem 1.2.11, where exactly this happens.

Proposition 1.2.2 (Generated σ -algebra). *For any $\mathcal{E} \subseteq \mathcal{P}(X)$ there is a smallest σ -algebra containing $\mathcal{E}$, denoted $\sigma(\mathcal{E})$.*

Proof. The family of σ -algebras containing $\mathcal{E}$ is nonempty, since $\mathcal{P}(X)$ is one. An arbitrary intersection of σ -algebras is a σ -algebra — each defining condition is preserved under intersection — so $\sigma(\mathcal{E}) := \bigcap \{\mathcal{A} : \mathcal{A} \text{ a } \sigma\text{-algebra, } \mathcal{E} \subseteq \mathcal{A}\}$ works. $\square$

Pitfall: $\sigma(\mathcal{E})$ has no constructive description

It is tempting to imagine $\sigma(\mathcal{E})$ as “everything obtainable from $\mathcal{E}$ by countably many unions and complements”. That family is in general *not* a σ -algebra: one may take countable unions of sets each of which required countably many operations, and iterate, and the process does not terminate after ω steps — it must be run through all countable ordinals. This is why Theorem 1.2.2 is proved from above, by intersecting, rather than from below by constructing. It is also why arguments about $\sigma(\mathcal{E})$ almost always proceed by the “good sets” principle: show that the family of E with the desired property is a σ -algebra containing $\mathcal{E}$, and conclude it contains $\sigma(\mathcal{E})$. That move is used in Theorem 1.4.3, in Theorem 1.6.3, and a dozen times besides.

Definition 1.2.3 (Borel sets; G_δ and F_σ). The *Borel σ -algebra* $\mathcal{B}(X)$ of a metric space X is $\sigma(\{\text{open sets}\})$. Its members are *Borel sets*.

A G_δ set is a countable intersection of open sets; an F_σ set is a countable union of closed sets. (The notation is German: G for *Gebiet*, region, with δ for *Durchschnitt*, intersection; F for *fermé*, closed, with σ for *somme*, sum.) Both are Borel, and the two classes are exchanged by complementation.

Remark 1.2.4 (The Borel hierarchy does not stop). Every open set is G_δ (trivially) and also F_σ : in a metric space, $U = \bigcup_k \{x : \text{dist}(x, U^c) \geq 1/k\}$. But the classes do not exhaust $\mathcal{B}$. One may form countable unions of G_δ sets ($G_{\delta\sigma}$), then countable intersections of those, and so on, and the hierarchy is strictly increasing through all countable ordinals. So “Borel” is genuinely more generous than “ G_δ or F_σ ”.

This matters in Chapter 1.3. Theorem 1.3.6 will say that a Lebesgue measurable set is a G_δ minus a null set, not that it is Borel outright — and the null set is not decoration. Worked Example 1.3.2(c) shows there are Lebesgue measurable sets that are not Borel at all, and indeed vastly more of the former than the latter.

Definition 1.2.5 (Measure). Let $\mathcal{A}$ be a σ -algebra on X . A *measure* is a map $\mu : \mathcal{A} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ which is *countably additive*: for pairwise disjoint $E_1, E_2, \dots \in \mathcal{A}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

The triple $(X, \mathcal{A}, \mu)$ is a *measure space*. The measure is *finite* if $\mu(X) < \infty$; *σ -finite* if $X = \bigcup_k X_k$ with $\mu(X_k) < \infty$ for each k ; and *complete* if every subset of every μ -null set lies in $\mathcal{A}$.

Proposition 1.2.6 (Elementary properties). *Let μ be a measure on $\mathcal{A}$ and $E, F, E_k \in \mathcal{A}$.*

- (i) (monotonicity) $E \subseteq F \implies \mu(E) \leq \mu(F)$, with $\mu(F \setminus E) = \mu(F) - \mu(E)$ when $\mu(E) < \infty$;
- (ii) (countable subadditivity) $\mu(\bigcup_k E_k) \leq \sum_k \mu(E_k)$;
- (iii) (continuity from below) $E_1 \subseteq E_2 \subseteq \dots$ implies $\mu(\bigcup_k E_k) = \lim_k \mu(E_k)$;
- (iv) (continuity from above) $E_1 \supseteq E_2 \supseteq \dots$ with $\mu(E_1) < \infty$ implies $\mu(\bigcap_k E_k) = \lim_k \mu(E_k)$.

Proof. (i) Write $F = E \sqcup (F \setminus E)$ and use additivity (padding with $\emptyset$ to make the union countable).

(ii) Disjointify: $F_k = E_k \setminus \bigcup_{j < k} E_j \in \mathcal{A}$ are disjoint with the same union, and $F_k \subseteq E_k$, so $\mu(\bigcup_k E_k) = \sum_k \mu(F_k) \leq \sum_k \mu(E_k)$ by (i).

(iii) With F_k as above, $\bigcup_{j \leq k} E_j = \bigsqcup_{j \leq k} F_j$, so $\mu(E_k) = \sum_{j \leq k} \mu(F_j)$, whose limit is $\sum_j \mu(F_j) = \mu(\bigcup_j E_j)$.

(iv) Apply (iii) to $E_1 \setminus E_k \uparrow E_1 \setminus \bigcap_k E_k$ and subtract, which is legitimate as $\mu(E_1) < \infty$. $\square$

Pitfall: continuity from above needs finiteness

On $\mathbb{R}$ with Lebesgue measure, $E_k = [k, \infty)$ decreases with $\bigcap_k E_k = \emptyset$, yet $m(E_k) = \infty$ for every k . So (iv) fails without the hypothesis $\mu(E_1) < \infty$. This is not a technicality: it is exactly the hypothesis that fails in Theorem 1.5.9, and the reason Egorov's theorem is false on $\mathbb{R}$.

WORKED EXAMPLE 1.2.1— TIER A **Computing a generated σ -algebra**

(a) *A finite example.* Let $X = \{1, 2, 3, 4\}$ and $\mathcal{E} = \{\{1, 2\}, \{2, 3\}\}$. Closing under complements and unions gives the *atoms* $\{1\}, \{2\}, \{3\}, \{4\}$ — for instance $\{2\} = \{1, 2\} \cap \{2, 3\}$ and $\{1\} = \{1, 2\} \setminus \{2\}$ — so $\sigma(\mathcal{E}) = \mathcal{P}(X)$, all 16 subsets. In a finite setting $\sigma(\mathcal{E})$ is always the family of unions of atoms, and this is the one case where a constructive description exists.

(b) *Generating the Borel sets of $\mathbb{R}$.* We claim $\mathcal{B}(\mathbb{R}) = \sigma(\{(a, \infty) : a \in \mathbb{R}\})$. One inclusion is clear since each (a, ∞) is open. Conversely, the right-hand family contains $(-\infty, b] = (b, \infty)^c$, hence $(a, b] = (a, \infty) \cap (-\infty, b]$, hence $(a, b) = \bigcup_k (a, b - \frac{1}{k}]$, and every open subset of $\mathbb{R}$ is a countable union of open intervals with rational endpoints. So it contains all open sets, and therefore $\mathcal{B}(\mathbb{R})$.

Why this is worth doing once. The same argument, with the same three lines, shows that measurability of a function may be tested on the single family $\{(a, \infty)\}$ rather than on all Borel sets — which is Theorem 1.4.2 and Theorem 1.4.3 in Chapter 1.4, and which is what makes measurability checkable in practice.

1.2.3 Outer measures

Definition 1.2.7 (Outer measure). An *outer measure* on a set X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ such that

- (i) $\mu^*(\emptyset) = 0$;
- (ii) (monotonicity) $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$;
- (iii) (countable subadditivity) $\mu^*(\bigcup_{k=1}^{\infty} A_k) \leq \sum_{k=1}^{\infty} \mu^*(A_k)$ for all $A_k \subseteq X$.

Two features of this definition deserve comment before we do anything with it.

First, an outer measure is defined on *every* subset of X . There is no restriction, no measurability hypothesis, no fine print. That is what will make outer measures easy to construct — Theorem 1.2.8 below builds one out of essentially nothing — and it is bought at the price of (iii) being an inequality rather than an equality.

Second, notice that (iii) is stated for arbitrary A_k , not merely disjoint ones. For subadditivity this costs nothing, since the disjoint case would imply the general one anyway (replace A_k by $A_k \setminus \bigcup_{j < k} A_j$ and use monotonicity). The asymmetry with (M3) above, which was stated only for disjoint sets, is therefore not a weakening in disguise; it is exactly the gap between $\leq$ and $=$ that the rest of the chapter is about.

Proposition 1.2.8 (Outer measure from a covering family). *Let X be a set and let $\mathcal{E} \subseteq \mathcal{P}(X)$ be a family of subsets with the following two properties: $\emptyset \in \mathcal{E}$, and $\mathcal{E}$ covers X , meaning that $X = \bigcup_{j=1}^{\infty} E_j$ for some countable family $E_j \in \mathcal{E}$. Let $\rho : \mathcal{E} \rightarrow [0, \infty]$ be any function with $\rho(\emptyset) = 0$. Define, for $A \subseteq X$,*

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E} \text{ for all } j, \quad A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}. \quad (1.2.1)$$

Then μ^ is an outer measure on X .*

Before the proof, three remarks on the statement, since it is more slippery than it looks.

The hypothesis that $\mathcal{E}$ covers X guarantees that the infimum in (1.2.1) is taken over a nonempty collection — every A has at least one admissible cover, namely the one covering all of X . Without this we would be taking $\inf \emptyset = +\infty$, which is harmless but makes the statement of (i) false unless $\emptyset$ is separately handled.

The covers are indexed by $\mathbb{N}$, that is, they are *countable*. Finite covers are included: pad with copies of $\emptyset$, which contribute $\rho(\emptyset) = 0$ to the sum. This

padding convention is why we required $\emptyset \in \mathcal{E}$, and we use it without comment from now on.

Finally, ρ is required to satisfy nothing except $\rho(\emptyset) = 0$. It need not be monotone, additive, or even finite. All the structure in (1.2.1) comes from the infimum-over-covers, not from ρ .

Proof. We verify the three properties of Theorem 1.2.7 in turn.

Step 1. $\mu^*(\emptyset) = 0$.

Cover $\emptyset$ by the countable family $E_j = \emptyset$ for every j . This is admissible, since $\emptyset \subseteq \bigcup_j \emptyset$. Its cost is $\sum_{j=1}^{\infty} \rho(\emptyset) = \sum_{j=1}^{\infty} 0 = 0$. Since μ^* is an infimum of nonnegative quantities and we have exhibited a cover of cost 0, we get $0 \leq \mu^*(\emptyset) \leq 0$.

Step 2. *Monotonicity.*

Suppose $A \subseteq B$. The point is a comparison of the two collections over which the infima are taken. If $\{E_j\}$ is any admissible cover of B — that is, $E_j \in \mathcal{E}$ and $B \subseteq \bigcup_j E_j$ — then since $A \subseteq B \subseteq \bigcup_j E_j$, the same family is an admissible cover of A . So every competitor in the infimum defining $\mu^*(B)$ is also a competitor in the infimum defining $\mu^*(A)$, with the same cost. The infimum over a larger collection is no larger, hence $\mu^*(A) \leq \mu^*(B)$.

▷ *Goal: countable subadditivity, which is the only substantial part.*

Step 3. *Setting up.*

Let $A_1, A_2, \dots \subseteq X$ be arbitrary and put $A = \bigcup_k A_k$. We must show $\mu^*(A) \leq \sum_k \mu^*(A_k)$.

Dispose of a trivial case first: if $\mu^*(A_k) = \infty$ for some k , the right-hand side is ∞ and there is nothing to prove. So assume $\mu^*(A_k) < \infty$ for every k . (We do *not* assume the sum is finite; if it diverges the inequality is again trivial, but the argument below does not need to distinguish the cases.)

Fix $\varepsilon > 0$.

Step 4. *Choosing good covers, one for each k .*

Since $\mu^*(A_k)$ is an infimum and $\mu^*(A_k) < \infty$, the number $\mu^*(A_k) + \varepsilon 2^{-k}$ is strictly larger than that infimum, so it is not a lower bound for the competing costs. Hence there exists an admissible cover $\{E_{k,j}\}_{j \geq 1} \subseteq \mathcal{E}$ of A_k with

$$\sum_{j=1}^{\infty} \rho(E_{k,j}) \leq \mu^*(A_k) + \varepsilon 2^{-k}. \quad (1.2.2)$$

This uses the axiom of countable choice, once for each k ; we will not remark on such uses again.

Step 5. *Assembling a single cover of A .*

Consider the doubly-indexed family $\{E_{k,j}\}_{k \geq 1, j \geq 1}$. It is a countable family, being a countable union of countable families, so after fixing any bijection $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ it is an admissible competitor in (1.2.1). And it covers A : given $x \in A$, we have $x \in A_k$ for some k , and $\{E_{k,j}\}_j$ covers A_k , so $x \in E_{k,j}$ for some j .

Therefore, by the definition of $\mu^*(A)$ as an infimum,

$$\mu^*(A) \leq \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \rho(E_{k,j}).$$

The rearrangement of the doubly-indexed sum into an iterated sum is legitimate because all terms are nonnegative, so the sum is independent of the order of summation (Tonelli for series, or elementary rearrangement of nonnegative series).

Step 6. *Cashing in the budget.*

Apply (1.2.2) to each inner sum:

$$\begin{aligned} \mu^*(A) &\leq \sum_{k=1}^{\infty} \left(\mu^*(A_k) + \varepsilon 2^{-k} \right) \\ &= \sum_{k=1}^{\infty} \mu^*(A_k) + \varepsilon \sum_{k=1}^{\infty} 2^{-k} \\ &= \sum_{k=1}^{\infty} \mu^*(A_k) + \varepsilon. \end{aligned}$$

This holds for every $\varepsilon > 0$. Since the left-hand side does not depend on ε , letting $\varepsilon \downarrow 0$ gives $\mu^*(A) \leq \sum_k \mu^*(A_k)$. $\square$

Scratch work. Step 4 contains the single most-used device in measure theory, and it is worth isolating. We had countably many approximations to make, each with its own error, and we needed the *total* error to be small. Distributing a fixed budget ε across the tasks as $\varepsilon 2^{-1}, \varepsilon 2^{-2}, \varepsilon 2^{-3}, \dots$ does this, because the weights are summable. Any summable sequence of positive weights would serve; 2^{-k} is merely the most convenient.

You will see this again in the proof of Theorem 1.3.6 (approximating a measurable set by an open one), in the construction of the Cantor function, in the proof of Egorov's theorem, and — in a form that looks quite different but is the same idea — in the ε -removal arguments of Volume 12, where the tasks being budgeted are dyadic spatial scales rather than sets. When a later proof opens with “fix $\varepsilon > 0$ and choose, for each k ”, this is what is happening.

WORKED EXAMPLE 1.2.2— TIER A **Lebesgue outer measure on $\mathbb{R}^n$**

Take $X = \mathbb{R}^n$, let $\mathcal{E}$ be the collection of open boxes $Q = \prod_{i=1}^n (a_i, b_i)$ together with $\emptyset$, and let $\rho(Q) = \prod_{i=1}^n (b_i - a_i)$ be the elementary volume, written $|Q|$. The family $\mathcal{E}$ covers $\mathbb{R}^n$ (use the boxes $(-k, k)^n$, $k \in \mathbb{N}$), so Theorem 1.2.8 applies and

$$m^*(A) = \inf \left\{ \sum_j |Q_j| : A \subseteq \bigcup_j Q_j, Q_j \text{ open boxes} \right\}$$

is an outer measure, called *Lebesgue outer measure*.

Singletons are null. Fix $x \in \mathbb{R}^n$ and $\varepsilon > 0$. The open box of side ε centred at x covers $\{x\}$ and has volume ε^n , so $m^*(\{x\}) \leq \varepsilon^n$. As $\varepsilon > 0$ was arbitrary and $\varepsilon^n \rightarrow 0$, we get $m^*(\{x\}) = 0$.

Countable sets are null. If $A = \{x_1, x_2, \dots\}$ is countable then countable subadditivity gives $m^*(A) \leq \sum_k m^*(\{x_k\}) = \sum_k 0 = 0$. In particular $m^*(\mathbb{Q}^n) = 0$.

What we have just done. In Chapter 1.1 the Riemann integral was defeated by $\mathbf{1}_{\mathbb{Q} \cap [0,1]}$, and the reason was that $\mathbb{Q}$ is dense, so no interval partition can separate it from its complement. We have now assigned $\mathbb{Q}$ the size 0, using two lines and no cleverness. Once the integral is constructed in Chapter 1.4 it will follow immediately that $\int \mathbf{1}_{\mathbb{Q} \cap [0,1]} = 0$. The Riemann theory could not compute this integral because it could not express the question; the measure-theoretic answer is trivial because the right question was asked.

Pitfall: $m^*(Q) = |Q|$ is not obvious

It is tempting to regard the identity $m^*(Q) = |Q|$ for a box Q as part of the definition. It is not. The inequality $m^*(Q) \leq |Q|$ is trivial (cover Q by a slightly larger box). The reverse inequality says that *no* countable family of boxes can cover Q at total cost less than $|Q|$, and this genuinely requires an argument — specifically, a compactness argument, since a countable cover must first be reduced to a finite one before elementary volume comparisons apply. We prove it as Theorem 3.4. Until then, do not use it, and be alert to the fact that several plausible-looking arguments in this chapter would become circular if you did.

1.2.4 The Carathéodory criterion

1.2.4.1 Three attempts at a definition

We now need to say which sets are “good”. Since the whole difficulty is that μ^* is subadditive but not additive, the good sets should be those which do not create waste when we cut along them. Turning that sentence into a usable definition takes three attempts, and seeing the first two fail is the fastest way to understand why the third looks the way it does.

Attempt 1: E is good if $\mu^*(E) + \mu^*(E^c) = \mu^*(X)$. This is the most direct transcription of “no waste”. It fails immediately on $X = \mathbb{R}^n$, because $\mu^*(\mathbb{R}^n) = \infty$, and so the condition reads $\infty = \infty$ and is satisfied by *every* set. Useless.

Attempt 2: E is good if $\mu^*(Q \cap E) + \mu^*(Q \cap E^c) = \mu^*(Q)$ for every box Q . Localising repairs the previous failure — the condition is now non-vacuous — and this definition does in fact single out the right sets in $\mathbb{R}^n$. But it has a serious structural defect: try to prove from it that the good sets are closed under intersection. Given E and F good, one wants to test $E \cap F$ against a box Q ; the natural move is to first cut Q by E and then cut the two pieces by F . But the pieces $Q \cap E$ and $Q \cap E^c$ are not boxes, so the hypothesis on F does not apply to them. The argument stalls at the first step.

Attempt 3 (Carathéodory): E is good if it cuts *every* set cleanly. The defect in Attempt 2 was that the family of admissible test sets was too small to be stable under the cutting operation we needed to iterate. The fix is to enlarge it as far as it will go.

Definition 1.2.9 (Carathéodory measurability). Let μ^* be an outer measure on X . A set $E \subseteq X$ is μ^* -measurable if for *every* $A \subseteq X$,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (1.2.3)$$

We write $\mathcal{M}_{\mu^*}$ for the collection of μ^* -measurable sets, and call A in (1.2.3) the *test set*.

Idea of the proof. The definition looks unmotivated on first meeting, and students routinely find it the least intuitive definition in the subject. The honest account is that it is *reverse-engineered from the proof it has to support*. The theorem we want is that $\mathcal{M}_{\mu^*}$ is a σ -algebra; the proof of that theorem proceeds by cutting a test set repeatedly, and at each cut the test set changes into something we have no control over. Quantifying over all A is precisely what makes the induction go through. Attempt 2 fails because

its induction breaks at step two.

There is a second thing to notice, which makes the definition much less alarming in practice: although (1.2.3) demands something of every A , it is usually verified by a single argument uniform in A , because the only tool available is an efficient cover of A , and efficient covers behave the same way regardless of what A is. Worked Example 2.7 below is the model case.

1.2.4.2 Two standing reductions

Before proving anything, we record two simplifications that halve the work in every verification of (1.2.3).

Lemma 1.2.10 (Reductions). *Let μ^* be an outer measure and $E \subseteq X$. Then $E \in \mathcal{M}_{\mu^*}$ if and only if*

$$\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c) \quad \text{for every } A \subseteq X \text{ with } \mu^*(A) < \infty. \quad (1.2.4)$$

Proof. Suppose (1.2.4) holds; we must recover (1.2.3) for all A .

First, the reverse inequality is automatic and costs nothing: for any A whatsoever, $A = (A \cap E) \cup (A \cap E^c)$, so finite subadditivity — which is contained in Theorem 1.2.7(iii), padding with $\emptyset$ — gives

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

So (1.2.3) is equivalent to the single inequality $\geq$.

Second, the restriction to $\mu^*(A) < \infty$ loses nothing: if $\mu^*(A) = \infty$ then the inequality $\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c)$ holds trivially, both sides being ∞ (the left is ∞ by assumption). $\square$

Pitfall: which direction is the content

Beginners frequently produce “proofs” of measurability that establish $\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c)$ and stop. That inequality is true for every set E , measurable or not; it is subadditivity. If your argument did not use any property of E , you have proved nothing.

WORKED EXAMPLE 1.2.3— TIER B **Half-spaces are Lebesgue measurable**

Let m^* be Lebesgue outer measure on $\mathbb{R}^n$ and fix $a \in \mathbb{R}$. We show that the half-space

$$H = \{x \in \mathbb{R}^n : x_1 > a\}$$

is m^* -measurable. This is the first non-trivial verification of (1.2.3) in the book, and the argument is the template for all later ones.

What is available. By the Pitfall box above we may not use $m^*(Q) = |Q|$. What we *may* use is the following purely elementary fact about elementary volume, proved by inspection of the defining product: if Q is an open box and we cut it by the hyperplane $\{x_1 = a\}$ into $Q' = Q \cap \{x_1 > a\}$ and $Q'' = Q \cap \{x_1 < a\}$, then Q' and Q'' are again open boxes (possibly empty) and

$$|Q'| + |Q''| = |Q|. \quad (1.2.5)$$

Indeed, if $Q = (b_1, c_1) \times R$ with R a box in $\mathbb{R}^{n-1}$, then for $b_1 \leq a \leq c_1$ we have $Q' = (a, c_1) \times R$ and $Q'' = (b_1, a) \times R$, so $|Q'| + |Q''| = ((c_1 - a) + (a - b_1))|R| = |Q|$; and if a lies outside $[b_1, c_1]$ then one of the two is empty and the identity is trivial. Note that the sliver $Q \cap \{x_1 = a\}$ is discarded, and this is harmless precisely because we are working with *open* boxes.

The verification. By Theorem 1.2.10 it suffices to prove

$$m^*(A) \geq m^*(A \cap H) + m^*(A \cap H^c) \quad (1.2.6)$$

for every $A \subseteq \mathbb{R}^n$ with $m^*(A) < \infty$.

Fix such an A and fix $\varepsilon > 0$. By the definition of m^* as an infimum, choose open boxes $\{Q_j\}_{j \geq 1}$ with

$$A \subseteq \bigcup_j Q_j \quad \text{and} \quad \sum_j |Q_j| \leq m^*(A) + \varepsilon.$$

Cut each box: let $Q'_j = Q_j \cap \{x_1 > a\}$ and $Q''_j = Q_j \cap \{x_1 < a\}$, both open boxes. Now observe two coverings:

$$A \cap H \subseteq \left(\bigcup_j Q_j \right) \cap H = \bigcup_j (Q_j \cap H) = \bigcup_j Q'_j,$$

and, writing $H^c = \{x_1 \leq a\}$,

$$\begin{aligned} A \cap H^c &\subseteq \bigcup_j (Q_j \cap H^c) \subseteq \bigcup_j (Q''_j \cup \{x_1 = a\}) \\ &\subseteq \left(\bigcup_j Q''_j \right) \cup P, \quad P := \{x_1 = a\}. \end{aligned}$$

The hyperplane P is null. To see this, fix $\eta > 0$ and, for each $k \in \mathbb{N}$, cover the bounded piece $P \cap ((-k, k)^n)$ by the single open box

$$\begin{aligned} R_k &= \left(a - \frac{\eta_k}{2}, a + \frac{\eta_k}{2} \right) \times (-k, k)^{n-1} \\ &= \frac{\eta 2^{-k}}{(2k)^{n-1}}, \end{aligned}$$

whose elementary volume is $|R_k| = \eta_k(2k)^{n-1} = \eta 2^{-k}$. The family $\{R_k\}_{k \geq 1}$ covers P , since every point of P lies in some $(-k, k)^n$, and its total cost is $\sum_k \eta 2^{-k} = \eta$. Hence $m^*(P) \leq \eta$ for every $\eta > 0$, so $m^*(P) = 0$. (This is the $\varepsilon 2^{-k}$ device of the Scratch box on page 19 again, now used to cover an unbounded set.)

Therefore, using monotonicity, then subadditivity, then $m^*(P) = 0$, then the definition of m^* applied to the two covers:

$$\begin{aligned} m^*(A \cap H) + m^*(A \cap H^c) &\leq m^*\left(\bigcup_j Q'_j\right) + m^*\left(\bigcup_j Q''_j\right) + m^*(P) \\ &\leq \sum_j |Q'_j| + \sum_j |Q''_j| \\ &= \sum_j (|Q'_j| + |Q''_j|) = \sum_j |Q_j| \leq m^*(A) + \varepsilon, \end{aligned}$$

where the second-to-last equality is (1.2.5). Letting $\varepsilon \downarrow 0$ gives (1.2.6).

What made it work. The argument never inspected A . It took an arbitrary efficient cover of A and cut the *cover*, not the set, exploiting the fact that the cutting operation is exactly additive on the covering family. This is the general shape of every measurability proof: one shows that E interacts well with the covering family $\mathcal{E}$ used to build μ^* , and the “for all A ” quantifier then comes for free.

Consequence. Once Theorem 1.2.11 tells us $\mathcal{M}_{m^*}$ is a σ -algebra, the measurability of all half-spaces gives us, by finite intersections, all open boxes; by countable unions, all open sets (every open subset of $\mathbb{R}^n$ is a countable union of boxes); and hence all Borel sets. So this one worked example, plus the theorem below, delivers the entire Borel σ -algebra.

1.2.4.3 The main theorem

Theorem 1.2.11 (Carathéodory). *Let μ^* be an outer measure on a set X . Then:*

- (a) $\mathcal{M}_{\mu^*}$ is a σ -algebra of subsets of X ;
- (b) the restriction $\mu := \mu^*|_{\mathcal{M}_{\mu^*}}$ is a countably additive measure;
- (c) μ is complete: if $E \in \mathcal{M}_{\mu^*}$ with $\mu(E) = 0$ and $F \subseteq E$, then $F \in \mathcal{M}_{\mu^*}$ and $\mu(F) = 0$.

Scratch work. It is worth previewing the shape of the proof, because it is long and the reader should know where it is going.

To prove (a) we must show $\mathcal{M}_{\mu^*}$ contains $\emptyset$, is closed under complements, and is closed under countable unions. The first two are one-line consequences of the symmetry of (1.2.3) in E and E^c . Countable unions is the whole content, and it is reached in three moves: first *finite* unions (Step 2), which makes $\mathcal{M}_{\mu^*}$ an algebra; then a form of additivity that holds relative to an arbitrary test set (Step 3), which is the technical engine; then countable *disjoint* unions (Step 5), from which the general case follows by a standard disjointification (Step 4).

Part (b) then falls out of Step 5 by specialising the test set, and part (c) is independent and easy.

The one step to watch is Step 5. Everything before it is finitary. Step 5 is where a limit $N \rightarrow \infty$ is taken, and where the “for every test set A ” quantifier in Theorem 1.2.9 is finally spent: we apply measurability of the partial union F_N to the *same* A for every N , obtaining a sequence of inequalities that can be passed to the limit. Under Attempt 2 above, A would have had to be a box while F_N was not, and there would be no such sequence.

Proof. Throughout, A denotes an arbitrary test set. By Theorem 1.2.10 we only ever need to prove inequalities of the form $\mu^*(A) \geq \mu^*(A \cap \cdot) + \mu^*(A \cap \cdot^c)$.

Step 1. $\emptyset \in \mathcal{M}_{\mu^*}$, and $\mathcal{M}_{\mu^*}$ is closed under complementation.

For $E = \emptyset$ and any A we have $A \cap \emptyset = \emptyset$ and $A \cap \emptyset^c = A$, so (1.2.3) reads $\mu^*(A) = 0 + \mu^*(A)$, which is true. For complementation, note that condition (1.2.3) is literally unchanged when E is replaced by E^c , since $(E^c)^c = E$ and addition is commutative. Hence $E \in \mathcal{M}_{\mu^*} \iff E^c \in \mathcal{M}_{\mu^*}$.

Step 2. $\mathcal{M}_{\mu^*}$ is closed under finite unions, hence is an algebra.

Let $E, F \in \mathcal{M}_{\mu^*}$ and let A be a test set.

Apply measurability of E with test set A :

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (1.2.7)$$

Now apply measurability of F *twice*, once with test set $A \cap E$ and once with test set $A \cap E^c$ — this is legitimate precisely because (1.2.3) was demanded for all test sets:

$$\begin{aligned} \mu^*(A \cap E) &= \mu^*(A \cap E \cap F) + \mu^*(A \cap E \cap F^c), \\ \mu^*(A \cap E^c) &= \mu^*(A \cap E^c \cap F) + \mu^*(A \cap E^c \cap F^c). \end{aligned}$$

Substituting both into (1.2.7) decomposes $\mu^*(A)$ into four pieces:

$$\begin{aligned} \mu^*(A) = & \underbrace{\mu^*(A \cap E \cap F)}_{\text{(I)}} + \underbrace{\mu^*(A \cap E \cap F^c)}_{\text{(II)}} \\ & + \underbrace{\mu^*(A \cap E^c \cap F)}_{\text{(III)}} + \underbrace{\mu^*(A \cap E^c \cap F^c)}_{\text{(IV)}}. \end{aligned} \quad (1.2.8)$$

Observe now that the set-theoretic identity

$$E \cup F = (E \cap F) \cup (E \cap F^c) \cup (E^c \cap F)$$

holds — the three pieces are the three ways a point can lie in $E \cup F$ — and intersecting with A gives

$$A \cap (E \cup F) = (A \cap E \cap F) \cup (A \cap E \cap F^c) \cup (A \cap E^c \cap F).$$

Subadditivity of μ^* therefore yields

$$\mu^*(A \cap (E \cup F)) \leq \text{(I)} + \text{(II)} + \text{(III)}. \quad (1.2.9)$$

Meanwhile, by De Morgan, $A \cap (E \cup F)^c = A \cap E^c \cap F^c$, so the fourth piece is exactly

$$\text{(IV)} = \mu^*(A \cap (E \cup F)^c). \quad (1.2.10)$$

Combining (1.2.8), (1.2.9) and (1.2.10),

$$\mu^*(A) = \text{(I)} + \text{(II)} + \text{(III)} + \text{(IV)} \geq \mu^*(A \cap (E \cup F)) + \mu^*(A \cap (E \cup F)^c),$$

which by Theorem 1.2.10 is what we needed. Hence $E \cup F \in \mathcal{M}_{\mu^*}$.

Together with Step 1, $\mathcal{M}_{\mu^*}$ is an algebra: it is closed under complements and finite unions, hence (De Morgan again) under finite intersections and set differences.

Where we are. We have an algebra, and we have not yet used any countability. Everything so far would work verbatim for a merely finitely subadditive μ^* . The next two steps are where countable subadditivity earns its place.

Step 3. *Additivity relative to a test set.*

This is the technical engine of the proof. We claim: if $E_1, \dots, E_N \in \mathcal{M}_{\mu^*}$ are *pairwise disjoint*, then for every test set A ,

$$\mu^*\left(A \cap \bigcup_{k=1}^N E_k\right) = \sum_{k=1}^N \mu^*(A \cap E_k). \quad (1.2.11)$$

Note the presence of A : this is stronger than additivity of μ^* on $\mathcal{M}_{\mu^*}$, and the extra strength is exactly what Step 5 will need.

We argue by induction on N . The case $N = 1$ is trivial. Assume (1.2.11) for $N - 1$ and set $F_N = \bigcup_{k \leq N} E_k$, which lies in $\mathcal{M}_{\mu^*}$ by Step 2. Apply measurability of E_N with test set $A \cap F_N$:

$$\mu^*(A \cap F_N) = \mu^*(A \cap F_N \cap E_N) + \mu^*(A \cap F_N \cap E_N^c).$$

Now use disjointness to simplify the two pieces. Since $E_N \subseteq F_N$ we have $F_N \cap E_N = E_N$; and since the E_k are pairwise disjoint, $F_N \cap E_N^c = \bigcup_{k \leq N-1} E_k =: F_{N-1}$. Hence

$$\mu^*(A \cap F_N) = \mu^*(A \cap E_N) + \mu^*(A \cap F_{N-1}) = \mu^*(A \cap E_N) + \sum_{k=1}^{N-1} \mu^*(A \cap E_k)$$

by the inductive hypothesis, which is (1.2.11).

Step 4. *Disjointification.*

We record the standard device that reduces arbitrary countable unions to disjoint ones. Given any $G_1, G_2, \dots \in \mathcal{M}_{\mu^*}$, set

$$E_1 = G_1, \quad E_k = G_k \setminus \bigcup_{j=1}^{k-1} G_j \quad (k \geq 2).$$

Then: each $E_k \in \mathcal{M}_{\mu^*}$, because $\mathcal{M}_{\mu^*}$ is an algebra (Step 2) and E_k is built from finitely many of the G_j by unions, complements and intersections; the E_k are pairwise disjoint, since for $k < l$ we have $E_l \cap G_k = \emptyset$ while $E_k \subseteq G_k$; and $\bigcup_k E_k = \bigcup_k G_k$, since $E_k \subseteq G_k$ gives $\subseteq$, while if $x \in \bigcup_k G_k$ then $x \in E_{k_0}$ for k_0 the least index with $x \in G_{k_0}$.

Step 5. $\mathcal{M}_{\mu^*}$ is closed under countable unions.

By Step 4 it suffices to treat a pairwise disjoint sequence $(E_k)_{k \geq 1} \subseteq \mathcal{M}_{\mu^*}$. Write

$$F_N = \bigcup_{k=1}^N E_k \in \mathcal{M}_{\mu^*}, \quad F = \bigcup_{k=1}^{\infty} E_k,$$

and let A be a test set. Since $F_N \in \mathcal{M}_{\mu^*}$ by Step 2,

$$\mu^*(A) = \mu^*(A \cap F_N) + \mu^*(A \cap F_N^c).$$

We now estimate the two terms separately.

For the first, apply (1.2.11) from Step 3: $\mu^*(A \cap F_N) = \sum_{k=1}^N \mu^*(A \cap E_k)$.

For the second, note $F_N \subseteq F$, hence $F^c \subseteq F_N^c$, hence $A \cap F^c \subseteq A \cap F_N^c$, and so monotonicity gives $\mu^*(A \cap F_N^c) \geq \mu^*(A \cap F^c)$. This is the step that discards information, and it is the only inequality in the argument.

Combining,

$$\mu^*(A) \geq \sum_{k=1}^N \mu^*(A \cap E_k) + \mu^*(A \cap F^c) \quad \text{for every } N \in \mathbb{N}. \quad (1.2.12)$$

The left-hand side does not depend on N , and the right-hand side is nondecreasing in N , so we may let $N \rightarrow \infty$:

$$\mu^*(A) \geq \sum_{k=1}^{\infty} \mu^*(A \cap E_k) + \mu^*(A \cap F^c). \quad (1.2.13)$$

Finally, apply countable subadditivity to the decomposition $A \cap F = \bigcup_k (A \cap E_k)$, giving $\sum_k \mu^*(A \cap E_k) \geq \mu^*(A \cap F)$. Substituting into (1.2.13),

$$\mu^*(A) \geq \mu^*(A \cap F) + \mu^*(A \cap F^c),$$

so $F \in \mathcal{M}_{\mu^*}$ by Theorem 1.2.10. With Step 1, this proves (a).

Step 6. *Countable additivity.*

Retain the notation of Step 5 and specialise the test set to $A = F$. Then $F \cap E_k = E_k$ and $F \cap F^c = \emptyset$, so (1.2.13) becomes

$$\mu^*(F) \geq \sum_{k=1}^{\infty} \mu^*(E_k).$$

The reverse inequality is countable subadditivity. Hence $\mu(F) = \sum_k \mu(E_k)$, which is (b). Note that $\mu(\emptyset) = \mu^*(\emptyset) = 0$, so μ is a bona fide measure.

Step 7. *Completeness.*

Suppose $\mu^*(E) = 0$ and $F \subseteq E$. For any test set A , monotonicity gives

$$\mu^*(A \cap F) \leq \mu^*(F) \leq \mu^*(E) = 0, \quad \mu^*(A \cap F^c) \leq \mu^*(A),$$

and adding these, $\mu^*(A \cap F) + \mu^*(A \cap F^c) \leq 0 + \mu^*(A) = \mu^*(A)$. By Theorem 1.2.10, $F \in \mathcal{M}_{\mu^*}$, and $\mu(F) = \mu^*(F) = 0$. This is (c), and it did not use measurability of E at all — only $\mu^*(E) = 0$. $\square$

Where we are. Three things are worth extracting from that proof.

Where the quantifier was spent. Step 2 applied the measurability of F to the test sets $A \cap E$ and $A \cap E^c$, which are not of any prescribed form; Step 5 applied the measurability of F_N to a single fixed A for infinitely many N . Neither move is available under Attempt 2 of §2.4.1. This is the precise sense in which the strange-looking definition is the right one.

Where countability was spent. Steps 1–4 are finitary. Countable subadditivity of μ^* entered only twice: at the very end of Step 5, and implicitly in

Step 6.

What we did not get. Theorem 1.2.11 says nothing about whether $\mathcal{M}_{\mu^*}$ is large. For all it tells us, $\mathcal{M}_{\mu^*}$ could be $\{\emptyset, X\}$, and Drill 2.2 exhibits an outer measure for which it very nearly is. Establishing that the measurable sets include everything we care about is a separate task, carried out for m^* in Worked Example 2.7 above and, in the general metric setting, in §2.5 next.

1.2.5 Metric outer measures

Carathéodory's theorem produces a σ -algebra but, as just noted, gives no guarantee that it contains anything interesting. On a metric space there is a clean criterion ensuring that all Borel sets are measurable, and it will matter enormously in Volume 7 when we construct Hausdorff measure — where, unlike the Lebesgue case, there is no convenient covering family to run the argument of Worked Example 2.7.

Definition 1.2.12. An outer measure μ^* on a metric space (X, d) is a *metric outer measure* if

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad \text{whenever } \text{dist}(A, B) > 0,$$

where $\text{dist}(A, B) = \inf\{d(x, y) : x \in A, y \in B\}$.

The hypothesis asks for additivity only on pairs of sets that are *strictly separated* — not merely disjoint. That is a weak requirement, and it is typically easy to verify, because a covering set of small enough diameter simply cannot meet both.

Theorem 1.2.13 (Carathéodory's criterion). *If μ^* is a metric outer measure on (X, d) , then every Borel subset of X is μ^* -measurable.*

Idea of the proof. Since the Borel σ -algebra is generated by the closed sets, and $\mathcal{M}_{\mu^*}$ is a σ -algebra by Theorem 1.2.11, it is enough to show every closed set C is measurable.

The obstruction is immediate. Given a test set A , we would like to apply the metric hypothesis to the pair $A \cap C$ and $A \cap C^c$, but these two sets can touch — a point of $A \setminus C$ may lie arbitrarily close to C — so their distance is 0 and the hypothesis says nothing.

The fix is to retreat. Replace $A \cap C^c$ by the smaller set B_j of points at distance at least $1/j$ from C , which *is* strictly separated from $A \cap C$. Now the hypothesis applies and gives a clean inequality for each j . It remains to show that nothing is lost as $j \rightarrow \infty$, that is, that $\mu^*(B_j) \rightarrow \mu^*(A \cap C^c)$. This is the only real work, and it is done by decomposing the region between B_j

and $A \cap C^c$ into the dyadic-type annular shells $D_j = B_{j+1} \setminus B_j$ and showing $\sum_j \mu^*(D_j) < \infty$, so that the tails vanish. The trick for summing them is that non-adjacent shells are strictly separated, so the metric hypothesis applies to the even-indexed shells and the odd-indexed shells separately. This “peel the bad set into annuli around it” manoeuvre recurs constantly. You will see it again in Chapter 1.7 in the proof of the Hardy–Littlewood maximal inequality, where the annuli are dyadic in radius.

Proof. By the remarks above it suffices to fix a closed set $C \subseteq X$ and a test set A with $\mu^*(A) < \infty$, and to prove

$$\mu^*(A) \geq \mu^*(A \cap C) + \mu^*(A \cap C^c). \quad (1.2.14)$$

Step 1. *The approximating sets.*

For $j \geq 1$ define

$$B_j = \left\{ x \in A \setminus C : \text{dist}(x, C) \geq \frac{1}{j} \right\}.$$

Then $B_1 \subseteq B_2 \subseteq \cdots$, and

$$\bigcup_{j \geq 1} B_j = A \setminus C. \quad (1.2.15)$$

For (1.2.15): the inclusion $\subseteq$ is clear. Conversely, if $x \in A \setminus C$ then, because C is closed and $x \notin C$, we have $\text{dist}(x, C) > 0$ — this is exactly where closedness is used — so $\text{dist}(x, C) \geq 1/j$ for all large j .

Step 2. *The metric hypothesis applies to $A \cap C$ and B_j .*

Every point of B_j is at distance $\geq 1/j$ from every point of C , and $A \cap C \subseteq C$, so $\text{dist}(A \cap C, B_j) \geq 1/j > 0$. Hence

$$\mu^*(A \cap C) + \mu^*(B_j) = \mu^*((A \cap C) \cup B_j) \leq \mu^*(A),$$

the last step by monotonicity, since $(A \cap C) \cup B_j \subseteq A$. Thus

$$\mu^*(A \cap C) + \mu^*(B_j) \leq \mu^*(A) \quad \text{for every } j. \quad (1.2.16)$$

In view of (1.2.14) and (1.2.15), the proof will be complete once we show

$$\lim_{j \rightarrow \infty} \mu^*(B_j) = \mu^*(A \setminus C). \quad (1.2.17)$$

Step 3. *The shells and their summability.*

Set $D_j = B_{j+1} \setminus B_j$ for $j \geq 1$, the set of points of $A \setminus C$ whose distance to C lies in $[\frac{1}{j+1}, \frac{1}{j})$.

We claim that if $|j-k| \geq 2$ then $\text{dist}(D_j, D_k) > 0$. Indeed, suppose $k \geq j+2$, let $x \in D_j$ and $y \in D_k$. Then $\text{dist}(x, C) \geq \frac{1}{j+1}$ while $\text{dist}(y, C) < \frac{1}{k} \leq \frac{1}{j+2}$, so by the triangle inequality

$$d(x, y) \geq \text{dist}(x, C) - \text{dist}(y, C) > \frac{1}{j+1} - \frac{1}{j+2} = \frac{1}{(j+1)(j+2)} > 0,$$

a bound independent of x and y . (For the first inequality: for any $c \in C$, $d(x, c) \leq d(x, y) + d(y, c)$, so $\text{dist}(x, C) \leq d(x, y) + d(y, c)$ for all c , and taking the infimum over c gives $\text{dist}(x, C) \leq d(x, y) + \text{dist}(y, C)$.)

Consequently the sets $D_2, D_4, D_6, \dots$ are pairwise strictly separated, so iterating the metric hypothesis and then using monotonicity gives, for each N ,

$$\sum_{j=1}^N \mu^*(D_{2j}) = \mu^*\left(\bigcup_{j=1}^N D_{2j}\right) \leq \mu^*(A),$$

since all the D_j lie in A . The same argument applies to the odd-indexed shells. Letting $N \rightarrow \infty$ and adding,

$$\sum_{j=1}^{\infty} \mu^*(D_j) \leq 2\mu^*(A) < \infty. \quad (1.2.18)$$

This is the one place the hypothesis $\mu^*(A) < \infty$ is used, and it is essential.

Step 4. *Passing to the limit.*

Since $A \setminus C = B_j \cup \bigcup_{k \geq j} D_k$ — every point of $A \setminus C$ not already in B_j lies in exactly one later shell — countable subadditivity gives, for every j ,

$$\mu^*(A \setminus C) \leq \mu^*(B_j) + \sum_{k \geq j} \mu^*(D_k).$$

By (1.2.18) the tail $\sum_{k \geq j} \mu^*(D_k)$ tends to 0 as $j \rightarrow \infty$. Hence

$$\mu^*(A \setminus C) \leq \liminf_{j \rightarrow \infty} \mu^*(B_j).$$

The reverse inequality holds for every j by monotonicity, since $B_j \subseteq A \setminus C$. This proves (1.2.17).

Step 5. *Conclusion.*

Let $j \rightarrow \infty$ in (1.2.16) and use (1.2.17):

$$\mu^*(A \cap C) + \mu^*(A \cap C^c) = \mu^*(A \cap C) + \mu^*(A \setminus C) \leq \mu^*(A),$$

which is (1.2.14). □

WORKED EXAMPLE 1.2.4— TIER C: RESEARCH TASTE **Hausdorff measure: the construction we will need in Volume 7**

The Kakeya conjecture asserts that a Besicovitch set in $\mathbb{R}^n$ has *Hausdorff dimension* n , even though Chapter 1.1 promised us its Lebesgue measure can be 0. To make that statement one needs a family of measures finer than Lebesgue's, and Theorem 1.2.8 together with Theorem 1.2.13 delivers them immediately. We record the construction now, while the machinery is in front of us; the theory proper is Volume 7.

The definition. Fix $s \geq 0$ and $\delta > 0$. For $A \subseteq \mathbb{R}^n$ put

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_j (\text{diam } U_j)^s : A \subseteq \bigcup_j U_j, \text{diam } U_j \leq \delta \right\}.$$

By Theorem 1.2.8 — with $\mathcal{E}$ the family of sets of diameter at most δ , and $\rho(U) = (\text{diam } U)^s$ — each $\mathcal{H}_\delta^s$ is an outer measure. The hypotheses of the proposition hold: $\emptyset$ has diameter 0, and $\mathbb{R}^n$ is covered by countably many sets of diameter $\leq \delta$.

Monotonicity in δ , and the limit. If $\delta' < \delta$ then every δ' -admissible cover is δ -admissible, so the infimum defining $\mathcal{H}_{\delta'}^s$ is over a *smaller* collection and therefore $\mathcal{H}_{\delta'}^s(A) \geq \mathcal{H}_\delta^s(A)$. So $\mathcal{H}_\delta^s(A)$ increases as δ decreases, and the limit

$$\mathcal{H}^s(A) = \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(A) = \sup_{\delta > 0} \mathcal{H}_\delta^s(A)$$

exists in $[0, \infty]$. It is again an outer measure: properties (i) and (ii) pass to the limit trivially, and for (iii) one applies subadditivity of $\mathcal{H}_\delta^s$ and then takes suprema, using that a supremum of sums of nondecreasing quantities is the sum of the suprema.

Why the $\delta \rightarrow 0$ limit is essential. Without it the construction is worthless. If arbitrarily large covering sets were allowed, any bounded A could be covered by a single set of diameter $\text{diam } A$, giving $\mathcal{H}_\infty^s(A) \leq (\text{diam } A)^s$ — a quantity that sees only the overall extent of A and nothing of its internal structure. Forcing the covering sets to be small is exactly what makes the quantity detect fine geometry. This is why the Kakeya problem is a problem about small scales: all the content is in the $\delta \rightarrow 0$ asymptotics, and in Volume 9 the parameter δ will be reincarnated as the width of a tube.

$\mathcal{H}^s$ is a metric outer measure. Suppose $\text{dist}(A, B) = d > 0$ and take any $\delta < d/2$. Let $\{U_j\}$ be a δ -admissible cover of $A \cup B$. No single U_j can meet both A and B : if $x \in U_j \cap A$ and $y \in U_j \cap B$ then $d \leq d(x, y) \leq \text{diam } U_j \leq \delta < d/2$, absurd. So the cover splits into a subfamily covering A and a subfamily covering B (discarding any U_j meeting neither), whence

$\mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B) \leq \sum_j (\text{diam } U_j)^s$. Taking the infimum over covers gives $\mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B) \leq \mathcal{H}_\delta^s(A \cup B)$, and the reverse is subadditivity. Letting $\delta \rightarrow 0$ preserves the equality.

By Theorem 1.2.13, therefore, *every Borel set is $\mathcal{H}^s$ -measurable* — which is precisely the licence we need in order to speak of the Hausdorff measure of a Besicovitch set, an object with no reason whatsoever to be nice.

The dimension jump. Suppose $\mathcal{H}^s(A) < \infty$ and let $t > s$. Given a δ -admissible cover $\{U_j\}$ with $\sum_j (\text{diam } U_j)^s \leq \mathcal{H}^s(A) + 1$, we may estimate

$$\begin{aligned} \sum_j (\text{diam } U_j)^t &= \sum_j (\text{diam } U_j)^{t-s} (\text{diam } U_j)^s \\ &\leq \delta^{t-s} \sum_j (\text{diam } U_j)^s \\ &\leq \delta^{t-s} (\mathcal{H}^s(A) + 1), \end{aligned}$$

using $\text{diam } U_j \leq \delta$ and $t - s > 0$. Hence $\mathcal{H}_\delta^t(A) \leq \delta^{t-s} (\mathcal{H}^s(A) + 1) \rightarrow 0$ as $\delta \rightarrow 0$, so $\mathcal{H}^t(A) = 0$.

So the function $s \mapsto \mathcal{H}^s(A)$ drops from ∞ to 0 at a single critical exponent, and we may define

$$\dim_{\mathcal{H}} A = \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\},$$

the *Hausdorff dimension* of A .

The statement of the conjecture. The Kakeya conjecture, proved in $\mathbb{R}^3$ by Wang and Zahl in 2025 and open for $n \geq 4$, asserts:

every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_{\mathcal{H}} K = n$.

You now understand this statement completely: Theorem 1.1.1 defined the sets, and the two pages above defined the dimension. Note how sharp the tension is with Chapter 1.1: such a K can have $m(K) = 0$, so $\mathcal{H}^n(K) = 0$ as well, and the conjecture nonetheless asserts that $\mathcal{H}^s(K) = \infty$ for every $s < n$. Proving it will take us twelve more volumes.

1.2.6 Premeasures and the extension theorem[†]

This section makes the construction of Theorem 1.2.8 canonical: if one starts from a set function already additive on an algebra, the outer measure it generates extends it, and the extension is unique under a σ -finiteness hypothesis. The material is standard and is used in Volume 6 (construction of surface measure on a hypersurface) but nowhere before then, so it may be skipped on

first reading.

Definition 1.2.14. Let $\mathcal{A}$ be an algebra of subsets of X . A *premeasure* is a map $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ with $\mu_0(\emptyset) = 0$ which is countably additive on $\mathcal{A}$: whenever $(E_k) \subseteq \mathcal{A}$ are pairwise disjoint and $\bigcup_k E_k$ happens to lie in $\mathcal{A}$, then $\mu_0(\bigcup_k E_k) = \sum_k \mu_0(E_k)$.

Theorem 1.2.15 (Carathéodory extension theorem). *Let μ_0 be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, and let μ^* be the outer measure it generates via Theorem 1.2.8 with $\mathcal{E} = \mathcal{A}$, $\rho = \mu_0$. Then:*

- (i) $\mu^*|_{\mathcal{A}} = \mu_0$;
- (ii) every $E \in \mathcal{A}$ is μ^* -measurable, so $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$ is a measure extending μ_0 to a σ -algebra containing $\sigma(\mathcal{A})$;
- (iii) if μ_0 is σ -finite, the extension to $\sigma(\mathcal{A})$ is unique.

Idea of the proof. Part (i) says the extension really is an extension: the covering procedure does not lose information on sets where μ_0 was already defined. The inequality $\leq$ is free, since E covers itself. For $\geq$ one must show no cover of E by $\mathcal{A}$ -sets is cheaper than $\mu_0(E)$, and this is exactly where countable additivity of the premeasure — as opposed to mere finite additivity — is spent.

Part (ii) is a verification of (1.2.3), and it follows the template of Worked Example 1.2.3 precisely: take an efficient cover of the test set, cut the *cover* rather than the set, and use that cutting is exactly additive on the covering family. There, that was the box-splitting identity (1.2.5); here it is finite additivity of μ_0 .

Part (iii) is the only place σ -finiteness is needed, and the mechanism is worth naming in advance: any competing extension ν is automatically *dominated* by μ , because ν must respect covers. Upgrading domination to equality requires subtracting, and subtraction requires finiteness.

Proof. Step 1. *Monotonicity and subadditivity of μ_0 on $\mathcal{A}$.*

We record two consequences of the definition. If $E \subseteq F$ with $E, F \in \mathcal{A}$ then $F = E \sqcup (F \setminus E)$ with both pieces in $\mathcal{A}$, so $\mu_0(F) = \mu_0(E) + \mu_0(F \setminus E) \geq \mu_0(E)$. And if $E \subseteq \bigcup_k E_k$ with all sets in $\mathcal{A}$ and $E \in \mathcal{A}$, then setting $F_k = E \cap E_k \setminus \bigcup_{j < k} E_j \in \mathcal{A}$ gives pairwise disjoint sets with $\bigcup_k F_k = E \in \mathcal{A}$ and $F_k \subseteq E_k$, whence by countable additivity and monotonicity

$$\mu_0(E) = \sum_k \mu_0(F_k) \leq \sum_k \mu_0(E_k). \quad (1.2.19)$$

Step 2. *Proof of (i): $\mu^*|_{\mathcal{A}} = \mu_0$.*

Let $E \in \mathcal{A}$. The family $\{E, \emptyset, \emptyset, \dots\}$ is an admissible cover, so $\mu^*(E) \leq \mu_0(E)$. Conversely, any admissible cover $\{E_k\} \subseteq \mathcal{A}$ of E satisfies $\sum_k \mu_0(E_k) \geq \mu_0(E)$ by (1.2.19); taking the infimum, $\mu^*(E) \geq \mu_0(E)$.

Step 3. *Proof of (ii): $\mathcal{A} \subseteq \mathcal{M}_{\mu^*}$.*

Fix $E \in \mathcal{A}$ and a test set A with $\mu^*(A) < \infty$; by Theorem 1.2.10 we must show $\mu^*(A \cap E) + \mu^*(A \cap E^c) \leq \mu^*(A)$.

Let $\varepsilon > 0$ and choose $\{E_k\} \subseteq \mathcal{A}$ covering A with $\sum_k \mu_0(E_k) \leq \mu^*(A) + \varepsilon$. Since $\mathcal{A}$ is an algebra, each $E_k \cap E$ and $E_k \cap E^c$ lies in $\mathcal{A}$, and they are disjoint with union E_k , so finite additivity gives

$$\mu_0(E_k) = \mu_0(E_k \cap E) + \mu_0(E_k \cap E^c).$$

The families $\{E_k \cap E\}$ and $\{E_k \cap E^c\}$ cover $A \cap E$ and $A \cap E^c$ respectively, so by the definition of μ^* as an infimum,

$$\begin{aligned} \mu^*(A \cap E) + \mu^*(A \cap E^c) &\leq \sum_k \mu_0(E_k \cap E) + \sum_k \mu_0(E_k \cap E^c) \\ &= \sum_k \mu_0(E_k) \\ &\leq \mu^*(A) + \varepsilon. \end{aligned}$$

Let $\varepsilon \downarrow 0$. With Theorem 1.2.11, $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$ is a complete measure, and since $\mathcal{M}_{\mu^*}$ is a σ -algebra containing $\mathcal{A}$ it contains $\sigma(\mathcal{A})$ by Theorem 1.2.2. This proves (ii).

▷ *Goal: (iii), uniqueness on $\sigma(\mathcal{A})$ under σ -finiteness.*

Step 4. *Any extension is dominated.*

Let ν be a measure on $\sigma(\mathcal{A})$ with $\nu|_{\mathcal{A}} = \mu_0$, and let $E \in \sigma(\mathcal{A})$. For any admissible cover $\{E_k\} \subseteq \mathcal{A}$ of E , countable subadditivity of ν (Theorem 1.2.6(ii)) gives

$$\nu(E) \leq \sum_k \nu(E_k) = \sum_k \mu_0(E_k).$$

Taking the infimum over covers, $\nu(E) \leq \mu^*(E) = \mu(E)$.

Step 5. *Agreement on sets of finite measure.*

Suppose $\mu(E) < \infty$. Fix $\varepsilon > 0$ and choose a cover $\{E_k\} \subseteq \mathcal{A}$ of E with $\sum_k \mu_0(E_k) \leq \mu(E) + \varepsilon$. Put $F = \bigcup_k E_k \in \sigma(\mathcal{A})$, so $E \subseteq F$ and, by subadditivity, $\mu(F) \leq \mu(E) + \varepsilon$.

We claim $\nu(F) = \mu(F)$. Indeed the partial unions $F_N = \bigcup_{k \leq N} E_k$ lie in $\mathcal{A}$, since $\mathcal{A}$ is an algebra, so $\nu(F_N) = \mu_0(F_N) = \mu(F_N)$; and $F_N \uparrow F$, so

continuity from below (Theorem 1.2.6(iii)), applied to ν and to μ separately, gives $\nu(F) = \lim_N \nu(F_N) = \lim_N \mu(F_N) = \mu(F)$.

Now split $F = E \sqcup (F \setminus E)$ for both measures and use Step 4 twice:

$$\mu(E) \leq \mu(F) = \nu(F) = \nu(E) + \nu(F \setminus E) \leq \nu(E) + \mu(F \setminus E) \leq \nu(E) + \varepsilon,$$

the last step because $\mu(F \setminus E) = \mu(F) - \mu(E) \leq \varepsilon$, a subtraction licensed by $\mu(E) < \infty$. Letting $\varepsilon \downarrow 0$ gives $\mu(E) \leq \nu(E)$, and with Step 4, equality.

Step 6. *The σ -finite case.*

Write $X = \bigcup_k X_k$ with $X_k \in \mathcal{A}$ and $\mu_0(X_k) < \infty$; replacing X_k by $X_k \setminus \bigcup_{j < k} X_j$, which stays in the algebra, we may take the X_k pairwise disjoint. For $E \in \sigma(\mathcal{A})$ each $E \cap X_k$ has $\mu(E \cap X_k) \leq \mu_0(X_k) < \infty$, so Step 5 applies and $\mu(E \cap X_k) = \nu(E \cap X_k)$. Summing over k by countable additivity gives $\mu(E) = \nu(E)$. $\square$

Where we are. Notice that uniqueness did *not* require a monotone class or π - λ argument: Steps 4 and 5 got there directly, because the outer measure is by construction the infimum over covers and any competing extension must respect covers. The monotone class lemma is genuinely needed elsewhere — it appears in Chapter 1.6 to build the product measure, where no covering description is available — and is proved there.

Drills

Drill 1.2.1. Show that if μ^* is an outer measure and $\mu^*(E) = 0$, then for every $A \subseteq X$ we have $\mu^*(A \cup E) = \mu^*(A) = \mu^*(A \setminus E)$. Deduce Step 7 of Theorem 1.2.11 a second way.

Drill 1.2.2. Let $X = \{1, 2, 3\}$ and define $\mu^*(\emptyset) = 0$, $\mu^*(X) = 2$, and $\mu^*(A) = 1$ for every other $A \subseteq X$. Verify that μ^* is an outer measure, and compute $\mathcal{M}_{\mu^*}$ explicitly. (You should find $\mathcal{M}_{\mu^*} = \{\emptyset, X\}$. This is the promised warning that Theorem 1.2.11 alone guarantees nothing whatsoever about the richness of the measurable family.)

Drill 1.2.3. In the notation of Step 3 of Theorem 1.2.13, show that the conclusion $\text{dist}(D_j, D_k) > 0$ genuinely fails for adjacent shells $k = j + 1$, by exhibiting a closed set $C \subseteq \mathbb{R}$ and points of D_j, D_{j+1} arbitrarily close together. This is why the even and odd shells had to be treated separately.

Drill 1.2.4. Show that $\mathcal{H}^0$ is counting measure. Then show that on $\mathbb{R}$, $\mathcal{H}^1 = m^*$. (The second direction requires the box lemma deferred to Chapter 1.3; do the first now and return to the second later.)

Notes

Carathéodory’s criterion dates to his 1914 paper on the theory of outer measure, and Theorem 1.2.13 to the same circle of ideas. The presentation here follows Folland in outline, with two deliberate departures. First, the metric outer measure section has been promoted from an aside to a main result, because of its role in Volume 7: for Hausdorff measure there is no analogue of the argument of Worked Example 2.7, and Theorem 1.2.13 is the only route to measurability of Borel sets. Second, we have given the failed Attempts 1 and 2 of §2.4.1 explicitly, which most treatments omit.

Readers who have seen measure theory built via the “open set approximation” definition of measurability, as in Stein–Shakarchi, should note that the two approaches agree on $\mathbb{R}^n$; we prove this as Theorem 1.3.6. The Carathéodory route is preferred here purely because it transfers to Hausdorff measure unchanged, whereas the open-set definition does not.

Chapter 1.3

Lebesgue Measure on $\mathbb{R}^n$

Chapter 1.2 was abstract. We built outer measures from arbitrary covering families, and we proved that the Carathéodory criterion always extracts a σ -algebra from them. But we proved almost nothing about the specific outer measure we care about. In particular we still do not know that $m^*(Q) = |Q|$ for a box Q — the Pitfall box in §2.3 warned that this is a theorem, not a definition — and without it the entire construction could, for all we have shown, be identically zero.

This chapter repairs that, and then goes considerably further. By the end we will have: the box lemma; a complete structural description of the measurable sets; the behaviour of m under translations and linear maps; a proof that non-measurable sets exist, so that the restriction to $\mathcal{M}_{m^*}$ was genuinely forced; and, in §3.6, the Perron tree — a construction which shows that a plane set containing a unit segment in every direction can have area less than any prescribed ε . That last result is the reason this series exists.

1.3.1 The box lemma

1.3.1.1 Elementary volume on grids

Everything rests on a completely elementary fact about products, which we isolate because it is used three times and is invisible if left implicit.

Lemma 1.3.1 (Grid additivity). *Let $Q = \prod_{i=1}^n (a_i, b_i)$ be an open box. For each coordinate i choose a finite partition*

$$a_i = t_0^i < t_1^i < \cdots < t_{m_i}^i = b_i,$$

and let $\mathcal{R}$ be the resulting family of $m_1 m_2 \cdots m_n$ open sub-boxes $R_{k_1 \dots k_n} = \prod_{i=1}^n (t_{k_i-1}^i, t_{k_i}^i)$. Then

$$|Q| = \sum_{R \in \mathcal{R}} |R|.$$

Proof. For each i , the partition gives $b_i - a_i = \sum_{k_i=1}^{m_i} (t_{k_i}^i - t_{k_i-1}^i)$, since the sum telescopes. Multiply these n identities together and expand the product of sums by the distributive law:

$$|Q| = \prod_{i=1}^n (b_i - a_i) = \prod_{i=1}^n \left(\sum_{k_i=1}^{m_i} (t_{k_i}^i - t_{k_i-1}^i) \right) = \sum_{k_1=1}^{m_1} \cdots \sum_{k_n=1}^{m_n} \prod_{i=1}^n (t_{k_i}^i - t_{k_i-1}^i),$$

and the summand is exactly $|R_{k_1 \dots k_n}|$. $\square$

Remark 1.3.2. The sub-boxes in Theorem 1.3.1 are open, so they do not cover Q : the grid hyperplanes are omitted. This is harmless here because we are computing elementary volumes, not measures, and the identity above is an identity between products of real numbers. It will become genuinely convenient in Theorem 1.3.3, where we want to talk about “the cells contained in a box” without worrying about shared faces.

1.3.1.2 The finite covering lemma

Lemma 1.3.3 (Finite covers of a box). *Let K be a closed box and let $Q_1, \dots, Q_N$ be open boxes with $K \subseteq \bigcup_{j=1}^N Q_j$. Then*

$$|K| \leq \sum_{j=1}^N |Q_j|.$$

Scratch work. The obstruction is that the Q_j overlap arbitrarily, so we cannot simply add their volumes and compare. The standard device is to *impose a common grid*: extend every face of every box (and of K) to a full hyperplane. These finitely many hyperplanes chop $\mathbb{R}^n$ into finitely many cells, and each of $K, Q_1, \dots, Q_N$ becomes a union of cells. Volume is additive over cells by Theorem 1.3.1, so the comparison becomes a purely combinatorial statement about which cells lie in which boxes — and there the covering hypothesis does the work directly.

Proof. Write $K = \prod_i [c_i, d_i]$ and $Q_j = \prod_i (a_i^j, b_i^j)$.

Step 1. *Build the grid.*

For each coordinate $i \in \{1, \dots, n\}$, let $T_i \subseteq \mathbb{R}$ be the finite set consisting of c_i, d_i , and all the endpoints a_i^j, b_i^j for $j = 1, \dots, N$. List T_i in increasing order. The points of T_i partition the smallest interval containing them into finitely many open intervals, and taking products over i produces a finite family $\mathcal{R}$ of open cells.

By construction, each of K and each Q_j has all of its faces on grid hyperplanes. Consequently, by Theorem 1.3.1 applied to each of them separately,

$$|K| = \sum_{R \in \mathcal{R}, R \subseteq K} |R|, \quad |Q_j| = \sum_{R \in \mathcal{R}, R \subseteq Q_j} |R|. \quad (1.3.1)$$

(For K , note that the interior of K is the union of the cells contained in K together with parts of grid hyperplanes, which contribute nothing to the elementary-volume identity of Theorem 1.3.1.)

Step 2. *Every cell inside K is inside some Q_j .*

Let $R \in \mathcal{R}$ with $R \subseteq K$, and pick any point $x \in R$. Since $R \subseteq K \subseteq \bigcup_j Q_j$, there is some j with $x \in Q_j$. We claim $R \subseteq Q_j$.

Indeed, fix a coordinate i and write R 's i -th edge as (t_{k-1}, t_k) with $t_{k-1}, t_k \in T_i$ consecutive. We know $a_i^j < x_i < b_i^j$, and $a_i^j, b_i^j \in T_i$ while $x_i \in (t_{k-1}, t_k)$. Since t_{k-1} and t_k are consecutive in T_i , no element of T_i lies strictly between them; as $a_i^j < x_i$ and $a_i^j \in T_i$ we must have $a_i^j \leq t_{k-1}$, and similarly $b_i^j \geq t_k$. Hence $(t_{k-1}, t_k) \subseteq (a_i^j, b_i^j)$. As i was arbitrary, $R \subseteq Q_j$.

Step 3. *Count.*

By Step 2, each cell $R \subseteq K$ appears in at least one of the sums $\sum_{R \subseteq Q_j} |R|$. Since all terms are nonnegative, allowing a cell to be counted several times only increases the total, so

$$|K| = \sum_{R \subseteq K} |R| \leq \sum_{j=1}^N \sum_{R \subseteq Q_j} |R| = \sum_{j=1}^N |Q_j|,$$

using (1.3.1) at both ends. □

1.3.1.3 The box lemma itself

Theorem 1.3.4 (Box lemma). *For every box $Q \subseteq \mathbb{R}^n$ — open, closed, or half-open —*

$$m^*(Q) = |Q|.$$

Idea of the proof. The inequality $m^*(Q) \leq |Q|$ is a one-line consequence of the definition: Q is (nearly) a cover of itself.

For the reverse we must rule out the possibility that some ingenious countable family of boxes covers Q more cheaply than $|Q|$. The only handle we have on a countable family is Theorem 1.3.3, which handles finite ones, so the entire content of the proof is the reduction from countable to finite. That reduction is compactness — Heine–Borel — and it costs two ε 's: one to shrink Q to a compact box strictly inside it, and one to fatten each covering box slightly so that a cover by arbitrary boxes becomes a cover by

open sets. Neither ε can be avoided, and it is worth noticing that this is the first and only place in the construction of Lebesgue measure where a topological property of $\mathbb{R}^n$ is used.

Proof. Step 1. The upper bound.

Suppose first Q is open. Then $\{Q, \emptyset, \emptyset, \dots\}$ is an admissible cover of Q in the sense of (1.2.1), so $m^*(Q) \leq |Q|$. For general Q with edge lengths $\ell_i = b_i - a_i$ and any $\eta > 0$, the open box $Q^\eta = \prod_i (a_i - \eta, b_i + \eta)$ contains Q , so by monotonicity

$$m^*(Q) \leq m^*(Q^\eta) \leq |Q^\eta| = \prod_i (\ell_i + 2\eta) \xrightarrow{\eta \rightarrow 0} \prod_i \ell_i = |Q|.$$

▷ *Goal:* $m^*(Q) \geq |Q|$.

Step 2. *Reduce to a compact box.*

Fix $\varepsilon > 0$. Choose a closed box $K \subseteq Q$ with

$$|K| \geq |Q| - \varepsilon; \tag{1.3.2}$$

concretely, if Q has edges (a_i, b_i) take $K = \prod_i [a_i + \eta, b_i - \eta]$ for η small, which is legitimate provided $|Q| > 0$ (if $|Q| = 0$ there is nothing to prove). By monotonicity it suffices to bound $m^*(K)$ from below, since $m^*(Q) \geq m^*(K)$.

Step 3. *Fatten an arbitrary cover.*

Let $\{Q_j\}_{j \geq 1}$ be any countable family of open boxes with $K \subseteq \bigcup_j Q_j$. (The definition of m^* uses open boxes, so no fattening is strictly needed here; we record the step because the same proof is run later for covers by arbitrary boxes, where one replaces Q_j by an open box $Q_j^{\eta_j} \supseteq Q_j$ with $|Q_j^{\eta_j}| \leq |Q_j| + \varepsilon 2^{-j}$, at a total cost of ε .)

Step 4. *Extract a finite subcover.*

K is closed and bounded, hence compact by Heine–Borel, and $\{Q_j\}$ is an open cover of K . So there is N with $K \subseteq \bigcup_{j=1}^N Q_j$.

Step 5. *Apply the finite lemma and unwind.*

By Theorem 1.3.3 and then (1.3.2),

$$\sum_{j=1}^{\infty} |Q_j| \geq \sum_{j=1}^N |Q_j| \geq |K| \geq |Q| - \varepsilon.$$

The left-hand side is the cost of an arbitrary admissible cover of K , so taking the infimum over all such covers gives $m^*(K) \geq |Q| - \varepsilon$, whence $m^*(Q) \geq |Q| - \varepsilon$. Let $\varepsilon \downarrow 0$. □

Where we are. We now know the construction is not degenerate: m^* assigns to boxes exactly their elementary volume. Combining with Worked Example 2.7 — half-spaces are measurable, hence so are all boxes, open sets and Borel sets — and with Theorem 1.2.11, we have a complete, countably additive, translation-invariant measure defined on a σ -algebra containing the Borel sets and assigning boxes their volume. Requirements (M1) and (M3) of §2.1 are met. Requirement (M2) is Theorem 1.3.8 below, and the price we paid — that $\mathcal{M}_{m^*}$ is not all of $\mathcal{P}(\mathbb{R}^n)$ — is shown to be unavoidable in §3.4.

Notation 1.3.5. From now on we write $m(E)$ or $|E|$ for $m^*(E)$ when $E \in \mathcal{M}_{m^*}$, and call such E *measurable* without qualification. We write $\mathcal{L}$ for $\mathcal{M}_{m^*}$ and $\mathcal{B}$ for the Borel σ -algebra, so $\mathcal{B} \subseteq \mathcal{L}$. Sets with $m(E) = 0$ are called *null*, and “almost everywhere” (a.e.) means “off a null set”.

1.3.2 Regularity and the structure of measurable sets

Carathéodory’s definition is excellent for proofs but useless for intuition: it tells you nothing about what a measurable set *looks like*. The following theorem does, and it is the result one actually uses.

Theorem 1.3.6 (Outer regularity and the structure theorem). *For $E \subseteq \mathbb{R}^n$ the following are equivalent.*

- (i) E is measurable.
- (ii) For every $\varepsilon > 0$ there is an open set $U \supseteq E$ with $m^*(U \setminus E) < \varepsilon$.
- (iii) There is a G_δ set $G \supseteq E$ with $m^*(G \setminus E) = 0$.
- (iv) For every $\varepsilon > 0$ there is a closed set $C \subseteq E$ with $m^*(E \setminus C) < \varepsilon$.
- (v) There is an F_σ set $F \subseteq E$ with $m^*(E \setminus F) = 0$.

In words: a measurable set is a Borel set modified by a null set.

Pitfall: $m^*(U \setminus E)$, **not** $m^*(U) - m^*(E)$

Condition (ii) must be stated with the outer measure of the difference. The subtraction $m^*(U) - m^*(E)$ is meaningless before we know E is measurable — outer measure is not additive — and it is also useless when $m^*(E) = \infty$. Getting this wrong makes (ii) trivially true for every set of infinite outer measure, which would prove that every such set is measurable, a statement refuted in §3.4.

Proof. We prove (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i) $\Rightarrow$ (ii), and then obtain (iv), (v) by complementation.

Step 1. (ii) $\Rightarrow$ (iii).

For each $k \in \mathbb{N}$ apply (ii) with $\varepsilon = 1/k$ to get an open $U_k \supseteq E$ with $m^*(U_k \setminus E) < 1/k$. Put $G = \bigcap_{k \geq 1} U_k$, a G_δ set containing E . For every k we have $G \setminus E \subseteq U_k \setminus E$, so monotonicity gives $m^*(G \setminus E) < 1/k$; letting $k \rightarrow \infty$, $m^*(G \setminus E) = 0$.

Step 2. (iii) $\Rightarrow$ (i).

Write $N = G \setminus E$, so $m^*(N) = 0$ and $E = G \setminus N$.

Now G is a G_δ , hence Borel, hence measurable (Worked Example 2.7 plus Theorem 1.2.11). And N is measurable because m is a *complete* measure: Theorem 1.2.11(c) says every set of outer measure zero is measurable, with no further hypothesis. Since $\mathcal{L}$ is a σ -algebra it is closed under set differences, so $E = G \setminus N \in \mathcal{L}$.

Remark 1.3.7. This step is where completeness of the Carathéodory construction earns its keep. Had we built Lebesgue measure only on the Borel σ -algebra, the set E here need not be Borel — there are subsets of the Cantor set (Worked Example 3.4) that are Lebesgue measurable but not Borel — and the theorem would be false as stated.

Step 3. (i) $\Rightarrow$ (ii), *finite measure case*.

Assume E measurable with $m(E) < \infty$, and fix $\varepsilon > 0$. By the definition of m^* as an infimum, there are open boxes $\{Q_j\}$ with

$$E \subseteq \bigcup_j Q_j =: U \quad \text{and} \quad \sum_j |Q_j| \leq m(E) + \varepsilon.$$

Then U is open, $U \supseteq E$, and by countable subadditivity together with the box lemma, $m(U) \leq \sum_j |Q_j| \leq m(E) + \varepsilon$.

Both U and E are measurable and $E \subseteq U$, so additivity applies to the disjoint decomposition $U = E \cup (U \setminus E)$:

$$m(U) = m(E) + m(U \setminus E).$$

Because $m(E) < \infty$ we may subtract it, obtaining $m(U \setminus E) = m(U) - m(E) \leq \varepsilon$.

Step 4. (i) $\Rightarrow$ (ii), *general case*.

Now let E be measurable of possibly infinite measure. Decompose $\mathbb{R}^n$ into the disjoint annuli $A_k = \{x : k-1 \leq |x| < k\}$, $k \geq 1$, and set $E_k = E \cap A_k$. Each E_k is measurable and bounded, hence of finite measure. By Step 3 choose

open $U_k \supseteq E_k$ with $m(U_k \setminus E_k) < \varepsilon 2^{-k}$, and put $U = \bigcup_k U_k$, an open set containing E .

The point is the inclusion

$$U \setminus E \subseteq \bigcup_{k \geq 1} (U_k \setminus E_k),$$

valid because if $x \in U \setminus E$ then $x \in U_k$ for some k , and $x \notin E \supseteq E_k$. Countable subadditivity now gives $m(U \setminus E) \leq \sum_k \varepsilon 2^{-k} = \varepsilon$.

Step 5. *(iv) and (v).*

Apply the equivalence already proved to E^c , which is measurable iff E is. Given $\varepsilon > 0$ choose open $U \supseteq E^c$ with $m^*(U \setminus E^c) < \varepsilon$ and set $C = U^c$, a closed subset of E . Since $E \setminus C = E \cap U = U \setminus E^c$, we get $m^*(E \setminus C) < \varepsilon$, which is (iv). Taking a countable union of such C gives (v). $\square$

WORKED EXAMPLE 1.3.1— TIER A A dense null set, and an open dense set of small measure

These two examples should be held in mind permanently: measure and topology measure completely different things, and neither controls the other.

(a) *A dense set of measure zero.* $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ and $m(\mathbb{Q}^n) = 0$ (Worked Example 2.5). So a set can be everywhere and still be negligible.

(b) *An open dense set of measure $< \varepsilon$.* Enumerate $\mathbb{Q} \cap [0, 1]$ as $q_1, q_2, \dots$ and put

$$U = \bigcup_{k=1}^{\infty} (q_k - \varepsilon 2^{-k-1}, q_k + \varepsilon 2^{-k-1}).$$

Then U is open (a union of intervals) and dense in $[0, 1]$ (it contains every rational). By countable subadditivity and the box lemma,

$$m(U) \leq \sum_{k=1}^{\infty} \varepsilon 2^{-k} = \varepsilon.$$

So $[0, 1] \setminus U$ is a closed set with empty interior — it contains no interval, since U is dense — yet its measure is at least $1 - \varepsilon$. A set can be topologically tiny (nowhere dense) and measure-theoretically enormous.

Why this matters later. Every intuition of the form “a set of small measure must be small” is wrong, and the Kakeya problem is the definitive illustration: §3.6 produces a set of measure $< \varepsilon$ that contains a unit segment pointing in *every* direction. If you find that impossible to picture, example (b) is the honest place to recalibrate — it already shows that a set of measure $1 - \varepsilon$ can fail to contain a single interval.

1.3.3 Translations, dilations, and linear maps

Proposition 1.3.8 (Translation invariance). *For every $E \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, $m^*(E + x) = m^*(E)$. Moreover $E \in \mathcal{L}$ if and only if $E + x \in \mathcal{L}$, and then $m(E + x) = m(E)$.*

Proof. The map $Q \mapsto Q + x$ is a bijection of the family of open boxes onto itself preserving elementary volume, since translating an interval (a_i, b_i) to $(a_i + x_i, b_i + x_i)$ does not change its length. Hence $\{Q_j\}$ is an admissible cover of E of cost c if and only if $\{Q_j + x\}$ is an admissible cover of $E + x$ of the same cost c . The two infima in (1.2.1) are therefore over identical sets of numbers.

For measurability, note that (1.2.3) for $E + x$ with test set A is exactly (1.2.3) for E with test set $A - x$, and A ranges over all subsets iff $A - x$ does. $\square$

Theorem 1.3.9 (Uniqueness of translation-invariant measure). *Let ν be a measure on $\mathcal{B}(\mathbb{R}^n)$ which is translation invariant and satisfies $c := \nu([0, 1]^n) < \infty$. Then $\nu = cm$ on $\mathcal{B}(\mathbb{R}^n)$.*

Proof. Step 1. Dyadic cubes.

For $k \geq 0$ let $\mathcal{D}_k$ be the family of half-open dyadic cubes $2^{-k}(m + [0, 1]^n)$, $m \in \mathbb{Z}^n$. The unit cube $[0, 1]^n$ is the disjoint union of exactly 2^{kn} cubes of $\mathcal{D}_k$, and all of these are translates of one another. By translation invariance they have equal ν -measure, and by finite additivity that common value is $c2^{-kn}$. Since $m(R) = 2^{-kn}$ for $R \in \mathcal{D}_k$ by the box lemma, we conclude

$$\nu(R) = cm(R) \quad \text{for every dyadic cube } R. \quad (1.3.3)$$

Step 2. Open sets.

Every open $U \subseteq \mathbb{R}^n$ is a countable disjoint union of dyadic cubes: run through $k = 0, 1, 2, \dots$ and select each $R \in \mathcal{D}_k$ with $\bar{R} \subseteq U$ that is not already contained in a previously selected cube. Any two dyadic cubes are either nested or disjoint, so the selected family is disjoint; and it covers U , since each $x \in U$ has a ball $B(x, r) \subseteq U$ and the dyadic cube of generation k containing x has diameter $\sqrt{n}2^{-k} < r$ for large k . Countable additivity of both ν and m , together with (1.3.3), gives $\nu(U) = cm(U)$.

Step 3. Borel sets.

Both ν and cm are Borel measures agreeing on open sets and finite on bounded sets. By outer regularity (Theorem 1.3.6(ii), which applies to cm , and the analogous statement for ν obtained from Step 2 by the same argument) they agree on all Borel sets. Explicitly: for E Borel and bounded, $\nu(E) = \inf\{\nu(U) : U \supseteq E \text{ open}\} = c \inf\{m(U)\} = cm(E)$; the unbounded case follows by exhausting with annuli as in Step 4 of Theorem 1.3.6. $\square$

Theorem 1.3.10 (Linear images). *Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear. Then T maps measurable sets to measurable sets, and for every $E \in \mathcal{L}$,*

$$m(T(E)) = |\det T| m(E).$$

Idea of the proof. There are three moves. First, if T is singular then $T(\mathbb{R}^n)$ lies in a hyperplane, which is null, and both sides vanish; so assume T invertible. Second, $E \mapsto m(T(E))$ is a translation-invariant Borel measure, finite on compacta, so Theorem 1.3.9 says it equals $c(T)m$ for some constant $c(T) \geq 0$ depending only on T . The whole problem is thereby reduced to computing one number. Third, c is multiplicative — $c(ST) = c(S)c(T)$, immediately from the definition — so it suffices to compute c on a set of generators of $GL_n(\mathbb{R})$, and every invertible matrix is a product of elementary matrices of three types.

The only type requiring thought is the shear, and the classical trick there is to cut the sheared cube into two pieces and translate one of them back, which reassembles the original cube. That argument uses nothing but translation invariance and additivity, which is why we can run it here rather than waiting for the change-of-variables theorem in Chapter 1.6.

Proof. Step 1. Singular T .

If $\det T = 0$ then $T(\mathbb{R}^n) \subseteq H$ for some hyperplane H , and H is null (the argument in Worked Example 2.7 covered a hyperplane by boxes of total volume η for arbitrary η). So $T(E)$ is a subset of a null set, hence measurable with measure 0 by completeness, and both sides are 0.

Step 2. *Invertible T maps measurable sets to measurable sets.*

T is a homeomorphism, so it maps open sets to open sets and hence Borel sets to Borel sets. It is also Lipschitz, with $|Tx - Ty| \leq \|T\| |x - y|$, which implies it maps null sets to null sets: if $m^*(N) = 0$, cover N by boxes of total volume $< \eta$; each box Q has $T(Q)$ contained in a box of volume at most $(\sqrt{n}\|T\|)^n |Q|$, so $m^*(T(N)) \leq (\sqrt{n}\|T\|)^n \eta$. Now use the structure theorem: a measurable E is $F \cup N$ with F an F_σ and N null (Theorem 1.3.6(v)), so $T(E) = T(F) \cup T(N)$ is Borel union null, hence measurable.

Step 3. *Reduction to a constant.*

Define $\nu(E) = m(T(E))$ for E Borel. This is a measure: T is a bijection, so it preserves disjointness and countable unions. It is translation invariant, because $T(E+x) = T(E) + Tx$ and m is translation invariant. And $\nu([0, 1]^n) = m(T([0, 1]^n)) < \infty$, since $T([0, 1]^n)$ is bounded. By Theorem 1.3.9, $\nu = c(T)m$ with $c(T) = m(T([0, 1]^n))$.

Step 4. Multiplicativity.

For linear S, T we have $c(ST)m(E) = m(S(T(E))) = c(S)m(T(E)) = c(S)c(T)m(E)$; taking $E = [0, 1]^n$ gives $c(ST) = c(S)c(T)$. Since $|\det|$ is also multiplicative, it suffices to verify $c(T) = |\det T|$ for T in a generating set of $GL_n(\mathbb{R})$, and every invertible matrix is a finite product of elementary matrices of the following three types.

Step 5. Type 1: diagonal scaling.

$T(x_1, \dots, x_n) = (\lambda_1 x_1, \dots, \lambda_n x_n)$ with all $\lambda_i \neq 0$. Then $T([0, 1]^n)$ is a box with edge lengths $|\lambda_i|$, so by the box lemma $c(T) = \prod_i |\lambda_i| = |\det T|$.

Step 6. Type 2: coordinate permutation.

T permutes the coordinates. Then $T([0, 1]^n) = [0, 1]^n$, so $c(T) = 1$, while $\det T = \pm 1$. Agreement.

Step 7. Type 3: shear.

Let $T(x) = x + x_j e_i$ for fixed $i \neq j$, so $\det T = 1$; we must show $c(T) = 1$, i.e. that the sheared unit cube $P = T([0, 1]^n)$ has measure 1.

Since only the i -th and j -th coordinates are involved, and the others are carried along unchanged, it suffices to treat $n = 2$ with $i = 1, j = 2$: the map $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$. Then

$$P = \{(y_1, y_2) : 0 \leq y_2 < 1, y_2 \leq y_1 < 1 + y_2\},$$

a parallelogram. Split it along the line $y_1 = 1$:

$$P_{\text{left}} = P \cap \{y_1 < 1\}, \quad P_{\text{right}} = P \cap \{y_1 \geq 1\},$$

which are disjoint measurable sets with $m(P) = m(P_{\text{left}}) + m(P_{\text{right}})$.

Now translate the right piece by $-e_1$. We claim

$$P_{\text{left}} \cup (P_{\text{right}} - e_1) = [0, 1]^2,$$

and that the union is disjoint. Indeed a point (y_1, y_2) with $0 \leq y_2 < 1$ lies in P_{left} iff $y_2 \leq y_1 < 1$, and lies in $P_{\text{right}} - e_1$ iff $(y_1 + 1, y_2) \in P$ with $y_1 + 1 \geq 1$, i.e. iff $0 \leq y_1 < y_2$. These two conditions are complementary within $[0, 1)$ for y_1 , which proves both claims.

By translation invariance and additivity,

$$1 = m([0, 1]^2) = m(P_{\text{left}}) + m(P_{\text{right}} - e_1) = m(P_{\text{left}}) + m(P_{\text{right}}) = m(P).$$

So $c(T) = 1 = |\det T|$, completing the proof. $\square$

1.3.4 Non-measurable sets

We now show that the restriction to $\mathcal{L}$ was not a failure of imagination.

Theorem 1.3.11 (Vitali). *There exists a set $V \subseteq [0, 1)$ which is not Lebesgue measurable. Consequently there is no translation-invariant, countably additive measure defined on all subsets of $\mathbb{R}$ assigning to each interval its length.*

Idea of the proof. The proof manufactures a set that is simultaneously “one $\mathbb{Q}$ -translate’s worth” of $[0, 1)$ and is reproduced by countably many disjoint translates that tile $[0, 1)$. If such a set had a measure c , countable additivity would force $1 = \sum_{k=1}^{\infty} c$, and no value of c satisfies this: $c = 0$ gives 0, and $c > 0$ gives ∞ . The whole construction exists to produce this impossible arithmetic, and the arithmetic is impossible precisely because countable additivity has no room for an intermediate answer.

Proof. Step 1. The equivalence relation.

Define $x \sim y$ on $[0, 1)$ by $x - y \in \mathbb{Q}$. This is an equivalence relation: reflexive since $0 \in \mathbb{Q}$, symmetric since $\mathbb{Q}$ is closed under negation, and transitive since $\mathbb{Q}$ is closed under addition.

Step 2. Choice.

By the axiom of choice, let $V \subseteq [0, 1)$ contain exactly one element from each equivalence class. This is the only non-constructive step, and it is unavoidable: it is a theorem of Solovay that there are models of ZF + dependent choice in which every subset of $\mathbb{R}$ is Lebesgue measurable.

Step 3. The translates.

Enumerate $\mathbb{Q} \cap [0, 1)$ as $q_1, q_2, \dots$ and define the “translation modulo 1”

$$V_k = ((V \cap [0, 1 - q_k)) + q_k) \cup ((V \cap [1 - q_k, 1)) + q_k - 1) \subseteq [0, 1).$$

Each V_k is the union of two ordinary translates of disjoint pieces of V .

Step 4. The V_k are pairwise disjoint.

Suppose $x \in V_j \cap V_k$. Then $x = v + q_j - m$ and $x = w + q_k - m'$ for some $v, w \in V$ and $m, m' \in \{0, 1\}$. Subtracting, $v - w = (q_k - q_j) + (m - m') \in \mathbb{Q}$, so $v \sim w$. Since V contains exactly one representative per class, $v = w$, whence $q_j - q_k \in \mathbb{Z}$; as both lie in $[0, 1)$ this forces $q_j = q_k$, i.e. $j = k$.

Step 5. The V_k cover $[0, 1)$.

Let $x \in [0, 1)$ and let $v \in V$ be the representative of its class, so $x - v \in \mathbb{Q}$. Since $x, v \in [0, 1)$ we have $x - v \in (-1, 1)$, so exactly one of $x - v$ and $x - v + 1$ lies in $[0, 1) \cap \mathbb{Q}$; call it q_k . Unwinding the two cases shows $x \in V_k$.

Step 6. *The contradiction.*

Suppose V were measurable. By Theorem 1.3.8 each of the two pieces in the definition of V_k is measurable, and by additivity

$$m(V_k) = m(V \cap [0, 1 - q_k]) + m(V \cap [1 - q_k, 1]) = m(V) =: c,$$

the same value for every k . By Steps 4 and 5 and countable additivity,

$$1 = m([0, 1]) = \sum_{k=1}^{\infty} m(V_k) = \sum_{k=1}^{\infty} c.$$

If $c = 0$ the sum is $0 \neq 1$; if $c > 0$ the sum is $+\infty \neq 1$. Either way a contradiction, so $V \notin \mathcal{L}$.

For the final sentence of the theorem, observe that the argument used only translation invariance, countable additivity, and $m([0, 1]) = 1$. $\square$

WORKED EXAMPLE 1.3.2—TIER B Cantor sets: null but uncountable, and nowhere dense but fat

The Cantor set is the standard source of counterexamples in this subject, and two variants are needed later.

(a) *The middle-thirds set.* Let $C_0 = [0, 1]$ and obtain C_k from C_{k-1} by deleting the open middle third of each of its constituent intervals. Then C_k is a union of 2^k closed intervals each of length 3^{-k} , so by the box lemma and finite additivity

$$m(C_k) = 2^k 3^{-k} = (2/3)^k.$$

The sets C_k decrease, so $C = \bigcap_k C_k$ is compact with $m(C) \leq (2/3)^k$ for every k , hence $m(C) = 0$.

Yet C is uncountable: the map sending a sequence $(a_k) \in \{0, 2\}^{\mathbb{N}}$ to $\sum_k a_k 3^{-k}$ is an injection of $\{0, 2\}^{\mathbb{N}}$ into C (in fact a bijection), and $\{0, 2\}^{\mathbb{N}}$ has the cardinality of the continuum. So “measure zero” is much weaker than “countable”.

(b) *A fat Cantor set.* Fix $\alpha \in (0, 1)$. Repeat the construction, but at stage k delete from each of the 2^{k-1} intervals an open middle interval of length $\alpha 2^{-(2k-1)}$. The total length deleted is

$$\sum_{k=1}^{\infty} 2^{k-1} \cdot \alpha 2^{-(2k-1)} = \alpha \sum_{k=1}^{\infty} 2^{k-1-2k+1} = \alpha \sum_{k=1}^{\infty} 2^{-k} = \alpha,$$

so the resulting compact set C_α has $m(C_\alpha) = 1 - \alpha > 0$. It contains no interval, because at stage k it is covered by 2^k intervals of total length $1 - \alpha \sum_{j \leq k} 2^{-j} \leq 1$, each therefore of length at most 2^{-k} ; so C_α is closed with empty interior, i.e. nowhere dense.

(c) *A measurable set that is not Borel.* C has measure zero, so by completeness every one of its $2^{\mathfrak{c}}$ subsets is Lebesgue measurable. But there are only $\mathfrak{c}$ Borel sets. Hence $\mathcal{B} \subsetneq \mathcal{L}$, strictly, and by a wide margin. This is the promised justification of the Remark in Step 2 of Theorem 1.3.6.

1.3.5 The Perron tree

We come to the theorem this volume has been pointing at since Chapter 1.1.

Theorem 1.3.12 (Besicovitch–Perron). *Let $\Delta \subseteq \mathbb{R}^2$ be a triangle with one side on the x -axis. For every $\varepsilon > 0$ there exist $k \in \mathbb{N}$ and horizontal translations $\tau_1, \dots, \tau_{2^k}$ such that the 2^k elementary triangles $\Delta_1^{(k)}, \dots, \Delta_{2^k}^{(k)}$ obtained from Δ by k successive bisections satisfy*

$$\left| \bigcup_{i=1}^{2^k} \tau_i(\Delta_i^{(k)}) \right| < \varepsilon.$$

Consequently, for every $\varepsilon > 0$ there is a compact set in $\mathbb{R}^2$ of measure less than ε containing a unit segment in every direction.

Here *bisection* means: cut a triangle by the segment joining its apex (the vertex off the axis) to the midpoint of its base, producing two triangles of equal area, each again with a side on the axis. After k bisections we have 2^k triangles, in left-to-right order along the axis, each of area $2^{-k}|\Delta|$.

Scratch work. Two things must be arranged simultaneously, and they pull against each other.

We must not lose directions. A triangle with apex P and base $[u, v]$ contains, for each direction θ in the angular range subtended, a segment pointing in direction θ — namely part of the segment from P to the appropriate base point. The elementary triangles of the k -th bisection subtend angular ranges that together cover the range of Δ . Crucially, *translation does not change the set of directions of segments contained in a set*. So however we slide the elementary triangles, the union still meets every direction of Δ . This is free, and it is the whole reason the construction is possible.

We must lose area. Sliding two adjacent triangles toward each other makes them overlap, and overlap is exactly area not paid for twice. The question is how much can be saved before the figures stop resembling triangles.

Theorem 1.3.13 below does the local computation for a single adjacent pair. The output is not simply “a smaller triangle”: it is a smaller triangle (the *trunk*) plus two leftover pieces sticking out of the top (the *sprouts*). If we recurse on the trunks, the sprouts accumulate, and the entire difficulty of

the proof is showing that the accumulated sprout area is summable. It is, because the trunks shrink geometrically.

Lemma 1.3.13 (Two-triangle lemma). *Let Δ be a triangle with base $[0, b] \times \{0\}$ and apex $P = (p, h)$, $h > 0$. Let Δ_1, Δ_2 be its bisection, with bases $[0, b/2]$ and $[b/2, b]$. Fix $\alpha \in (1/2, 1)$ and let τ be translation by $-(1 - \alpha)b$ in the x -direction. Then*

$$\Delta_1 \cup \tau(\Delta_2) \subseteq \alpha\Delta \cup S_1 \cup S_2,$$

where $\alpha\Delta$ is the image of Δ under the homothety of ratio α centred at the origin, and S_1, S_2 are triangles with

$$|S_1| + |S_2| = 4(1 - \alpha)^2 |\Delta|.$$

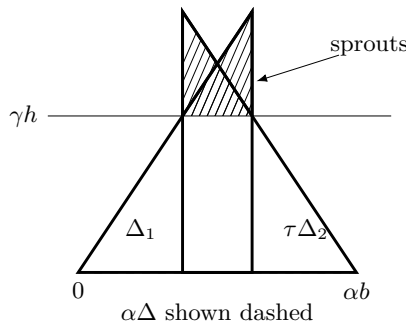


Figure 1.3.1: The two-triangle lemma (Theorem 1.3.13), drawn with $\alpha = \frac{4}{5}$. Bisecting Δ and sliding the right half left by $(1 - \alpha)b$ makes the halves overlap. Below the height γh , $\gamma = 2\alpha - 1$, the union lies inside the shrunk triangle $\alpha\Delta$; the left edge of Δ_1 and the right edge of $\tau\Delta_2$ are *exactly* the edges of $\alpha\Delta$, which is what Step 1 of the proof verifies. What protrudes above γh is the two hatched sprouts, of total area $4(1 - \alpha)^2 |\Delta|$.

Proof. Write $\gamma = 2\alpha - 1 \in (0, 1)$ and split at the height γh ; Figure 1.3.1 shows the configuration.

Step 1. *The right edge of $\tau\Delta_2$ is the right edge of $\alpha\Delta$.*

$\alpha\Delta$ has vertices $(0, 0)$, $(\alpha b, 0)$, $(\alpha p, \alpha h)$. The triangle $\tau\Delta_2$ has vertices $((\alpha - \frac{1}{2})b, 0)$, $(\alpha b, 0)$, $(p - (1 - \alpha)b, h)$. Its right edge runs from $(\alpha b, 0)$ to $(p - (1 - \alpha)b, h)$; at height y this edge is at

$$x = \alpha b + \frac{y}{h}(p - (1 - \alpha)b - \alpha b) = \alpha b + \frac{y}{h}(p - b).$$

The right edge of $\alpha\Delta$ runs from $(\alpha b, 0)$ to $(\alpha p, \alpha h)$ and is at $x = \alpha b + \frac{y}{\alpha h}(\alpha p - \alpha b) = \alpha b + \frac{y}{h}(p - b)$. The two are the same line.

Similarly the left edge of Δ_1 (from $(0, 0)$ to (p, h)), at $x = \frac{y}{h}p$ is the left edge of $\alpha\Delta$ (from $(0, 0)$ to $(\alpha p, \alpha h)$), at $x = \frac{y}{\alpha h}\alpha p = \frac{y}{h}p$.

Step 2. *Below height γh , the union lies in $\alpha\Delta$.*

Consider a point of Δ_1 at height y . It lies to the right of Δ_1 's left edge, which by Step 1 is $\alpha\Delta$'s left edge, so that side is fine. It lies to the left of Δ_1 's right edge, which at height y is at $x = \frac{b}{2} + \frac{y}{h}(p - \frac{b}{2})$. We need this to be at most $\alpha\Delta$'s right edge, at $x = \alpha b + \frac{y}{h}(p - b)$:

$$\begin{aligned} \frac{b}{2} + \frac{y}{h}(p - \frac{b}{2}) &\leq \alpha b + \frac{y}{h}(p - b) \\ &\iff \frac{b}{2}\left(1 - \frac{y}{h}\right) \leq b\left(\alpha - \frac{y}{h}\right) \\ &\iff \frac{y}{2h} \leq \alpha - \frac{1}{2} \\ &\iff y \leq \gamma h. \end{aligned}$$

Symmetrically, a point of $\tau\Delta_2$ at height y lies to the left of $\alpha\Delta$'s right edge by Step 1, and to the right of $\tau\Delta_2$'s left edge, which at height y is at $x = (\alpha - \frac{1}{2})b + \frac{y}{h}(p - \frac{b}{2})$; requiring this to be at least $\alpha\Delta$'s left edge $\frac{y}{h}p$ gives $(\alpha - \frac{1}{2})b \geq \frac{y}{h} \cdot \frac{b}{2}$, i.e. again $y \leq \gamma h$.

Hence $(\Delta_1 \cup \tau\Delta_2) \cap \{y \leq \gamma h\} \subseteq \alpha\Delta$.

Step 3. *The sprouts.*

Let $S_1 = \Delta_1 \cap \{y > \gamma h\}$ and $S_2 = \tau\Delta_2 \cap \{y > \gamma h\}$. Since Δ_1 is a triangle with apex at height h , its portion above height γh is a triangle similar to Δ_1 with ratio $1 - \gamma = 2(1 - \alpha)$, so

$$|S_1| = (2(1 - \alpha))^2 |\Delta_1| = 4(1 - \alpha)^2 \cdot \frac{1}{2} |\Delta| = 2(1 - \alpha)^2 |\Delta|,$$

and identically for S_2 . Adding gives the stated total, and Steps 2–3 give the stated inclusion. $\square$

Proof of Theorem 1.3.12. Normalise $|\Delta| = 1$, and fix $\alpha \in (1/2, 1)$ to be chosen at the end.

Step 1. *The recursive scheme.*

Start with the 2^k elementary triangles of the k -th bisection. Group them into 2^{k-1} adjacent pairs; the two members of a pair are precisely the bisection of a single elementary triangle of the $(k - 1)$ -st stage. Apply Theorem 1.3.13 to each pair. This replaces the pair by α -scaled copy of its parent, plus two sprouts.

The resulting 2^{k-1} trunks are the α -scaled copies of the level- $(k - 1)$ triangles. They are therefore, up to translation, exactly the level- $(k - 1)$ elementary triangles of the scaled triangle $\alpha\Delta$ — so the scheme can be applied to them again. Repeat k times in all. After stage j we hold 2^{k-j} trunks which are the

level- $(k - j)$ elementary triangles of $\alpha^j \Delta$, together with all sprouts produced so far.

Since every operation is a horizontal translation and the sprouts are retained,

$$\bigcup_{i=1}^{2^k} \tau_i(\Delta_i^{(k)}) \subseteq (\text{a translate of } \alpha^k \Delta) \cup \bigcup_{\text{all sprouts}} S. \quad (1.3.4)$$

Step 2. *Total sprout area.*

At stage j (for $j = 1, \dots, k$) we apply Theorem 1.3.13 to each of the 2^{k-j} pairs among the current trunks. The parents at that stage are the level- $(k - j)$ elementary triangles of $\alpha^{j-1} \Delta$, whose areas sum to

$$|\alpha^{j-1} \Delta| = \alpha^{2(j-1)}.$$

The lemma charges $4(1 - \alpha)^2$ times the parent's area to each pair, so the sprout area created at stage j is exactly $4(1 - \alpha)^2 \alpha^{2(j-1)}$. Summing the geometric series over $j = 1, \dots, k$ and extending to $j = \infty$,

$$\sum_{\text{all sprouts}} |S| \leq 4(1 - \alpha)^2 \sum_{j=0}^{\infty} \alpha^{2j} = \frac{4(1 - \alpha)^2}{1 - \alpha^2} = \frac{4(1 - \alpha)}{1 + \alpha} \leq 4(1 - \alpha). \quad (1.3.5)$$

Step 3. *Conclusion.*

By (1.3.4), subadditivity, (1.3.5) and $|\alpha^k \Delta| = \alpha^{2k}$,

$$\left| \bigcup_i \tau_i(\Delta_i^{(k)}) \right| \leq \alpha^{2k} + 4(1 - \alpha). \quad (1.3.6)$$

Given $\varepsilon > 0$, first choose $\alpha \in (1/2, 1)$ with $4(1 - \alpha) < \varepsilon/2$; then, α being fixed and less than 1, choose k with $\alpha^{2k} < \varepsilon/2$. This proves the first assertion.

Step 4. *From the tree to a set of segments in all directions.*

The triangle Δ contains, for each direction θ in the angular range subtended by Δ at its apex, a segment in direction θ . Each elementary triangle $\Delta_i^{(k)}$ subtends a sub-range, and these sub-ranges cover the whole range. Since translation preserves directions, the set $P = \bigcup_i \tau_i(\Delta_i^{(k)})$ contains a segment in every direction of the original range.

Taking Δ to be a triangle subtending an angular range of, say, $\pi/3$, and forming three rotated copies of the construction — rotations preserve Lebesgue measure by Theorem 1.3.10, since a rotation has determinant 1 — we obtain a set of measure $< \varepsilon$ containing a segment in every direction of $[0, \pi)$. Rescaling so that all these segments have length at least 1, and taking closures (which adds nothing to the measure, each set being a finite union of triangles), gives the compact set claimed. $\square$

WORKED EXAMPLE 1.3.3— TIER C: RESEARCH TASTE **Reading the estimate** (1.3.6)

The bound $\alpha^{2k} + 4(1 - \alpha)$ deserves to be stared at, because its shape recurs throughout this series.

The two terms fight. The first term, α^{2k} , is the trunk; it wants α *small*, so that each bisection compresses aggressively. The second, $4(1 - \alpha)$, is the accumulated sprout debt; it wants α *close to 1*, so that each compression is gentle and leaks little. Neither term alone is the obstruction; the theorem works only because we may choose α first, paying the sprout debt once, and then take $k \rightarrow \infty$ to kill the trunk. Fix the order of quantifiers wrongly — choose k first, then optimise α — and you get nothing.

One bisection is not enough. If we simply optimise $\alpha^2 + 4(1 - \alpha)^2$, the single-stage bound from Theorem 1.3.13, we find the minimum at $\alpha = 4/5$, with value $4/5$. A single bisection saves 20% and no more. All of the strength comes from iterating, and from the fact that the trunk shrinks geometrically while the debt is summable. This is an early instance of the phenomenon that dominates Volume 13: *a bound that is useless at one scale becomes decisive when applied at $\log(1/\delta)$ scales, provided the loss per scale is summable rather than multiplicative*. In decoupling this discipline is called *induction on scales*, and (1.3.5) is its ancestor.

What is not proved here. Theorem 1.3.12 gives measure $< \varepsilon$ for each ε , not a single set of measure zero. Passing to the limit requires a compactness argument in the space of compact subsets of a fixed disc — the Blaschke selection theorem for the Hausdorff metric — together with the observation that a limit of sets each containing a unit segment in direction θ again contains one. We defer this to Volume 7, where the Hausdorff metric is developed properly; nothing in Volumes 1–6 requires the measure-zero refinement, and stating it here without the machinery would be a gesture rather than a proof.

Why the conjecture is not thereby settled. It is tempting to conclude that Besicovitch sets are “as small as possible” and that there is nothing further to say. The opposite is true. Lebesgue measure is simply the wrong instrument: it returns 0 and stops. The Hausdorff measures $\mathcal{H}^s$ of Worked Example 2.12 form a continuum of finer instruments, and the Kakeya conjecture asserts that every one of them with $s < n$ returns $+\infty$. So the Perron tree exhausts what can be squeezed out of *measure*, and the entire subject of this series begins on the other side of that exhaustion. Note also that Theorem 1.3.13 is irredeemably two-dimensional — it manipulates triangles with a base on a line — and that no analogue producing dimension loss is available in $\mathbb{R}^3$. That is not a defect of the exposition; it is the mathematical situation, and it is why the three-dimensional problem stayed open from 1917 until Wang and Zahl in 2025.

Looking ahead: tubes, and the quantity that replaces area

In Volume 9 the elementary triangles of Theorem 1.3.12 are replaced by δ -tubes: rectangles of dimensions $1 \times \delta$. The Perron construction shows that δ^{-1} such tubes, one in each δ -separated direction, can be arranged so that $|\bigcup T| \rightarrow 0$ as $\delta \rightarrow 0$ — but only *slowly*, like $1/\log(1/\delta)$. Making the rate of decay the object of study, rather than the mere fact of decay, is what converts Besicovitch's curiosity into the Kakeya maximal function conjecture. You will meet that operator in Chapter 1.7, three chapters from now, and Worked Example 7.6 will show that everything proved in this volume is insufficient to control it.

Figure 1.3.1 (to be drawn). *The Cantor set and a fat Cantor set*

Content required: Four generations of the middle-thirds construction, and beside it the same construction removing middle intervals of length 4^{-k} , whose limit has positive measure. The two look alike and differ in measure; that is the point of Theorem 1.3.11 and the discussion following it.

Drills

Drill 1.3.1. Prove that m^* is not finitely additive on all subsets of $\mathbb{R}$, directly from Theorem 1.3.11, without reference to measurability: exhibit disjoint sets A, B with $m^*(A \cup B) < m^*(A) + m^*(B)$.

Drill 1.3.2. Let $E \subseteq \mathbb{R}$ be measurable with $m(E) > 0$. Show that for every $\lambda \in (0, 1)$ there is an interval I with $m(E \cap I) > \lambda m(I)$. (Use Theorem 1.3.6(ii) and decompose the approximating open set into its component intervals. This “density” statement is the seed of the Lebesgue differentiation theorem in Chapter 1.7.)

Drill 1.3.3. Verify the claim in Step 7 of Theorem 1.3.10 that it suffices to treat $n = 2$: write out the general shear $T(x) = x + x_j e_i$ and check that the cut-and-translate decomposition goes through verbatim in the two relevant coordinates.

Drill 1.3.4. Compute, for $\alpha = 4/5$, the bound (1.3.6) for $k = 1, 5, 20, 100$. Then find, for each $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$, the smallest k for which some admissible α achieves $\alpha^{2k} + 4(1 - \alpha) < \varepsilon$. Observe how badly k grows, and reconcile this with the assertion in the Looking Ahead box that the decay is only logarithmic.

Notes

The box lemma is due to Lebesgue; the grid argument in Theorem 1.3.3 is the standard one and appears in essentially every text. Theorem 1.3.11 is Vitali’s, from 1905. Solovay’s theorem, referred to in Step 2 of its proof, appeared in 1970 and shows that the appeal to the axiom of choice cannot be removed.

Besicovitch’s construction dates to 1919, and Perron’s simplification — the tree of §3.6 — to 1928. Our bookkeeping in (1.3.6) follows Perron’s scheme, with the recursion organised so that the sprout debt appears explicitly as a geometric series; this is the form in which the argument generalises. Schoenberg later gave a different construction via “sprouting” figures, and Kahane a third via products of Cantor sets, which has the advantage of producing the measure-zero set directly. Readers curious about the modern state of the Kakeya problem may consult Tao’s survey and Guth’s expository notes on the Wang–Zahl theorem, though neither is needed before Volume 9.

Chapter 1.4

Measurable Functions and the Lebesgue Integral

We have a measure. We now build the integral it was designed to support.

Recall the diagnosis of Chapter 1.1: the Riemann integral partitions the *domain*, and fails on functions that oscillate at every scale because no refinement of an interval partition ever separates the level sets. Lebesgue's remedy is to partition the *range* instead — to group together all the points where f takes a given value, whatever shape that set has — and the price is that one must then know the size of the resulting level sets. That price was paid in Chapters 1.2 and 1.3. This chapter collects the goods.

1.4.1 Measurable functions

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space. The reader may take $X = \mathbb{R}^n$, $\mathcal{M} = \mathcal{L}$, $\mu = m$ on first reading; nothing in §§4.1–4.5 uses anything specific to $\mathbb{R}^n$.

Definition 1.4.1. A function $f : X \rightarrow [-\infty, \infty]$ is $\mathcal{M}$ -*measurable* (or just *measurable*) if

$$f^{-1}((a, \infty]) = \{x \in X : f(x) > a\} \in \mathcal{M} \quad \text{for every } a \in \mathbb{R}.$$

A complex-valued f is measurable if its real and imaginary parts are.

Lemma 1.4.2 (Equivalent formulations). *For $f : X \rightarrow [-\infty, \infty]$ the following are equivalent: $\{f > a\} \in \mathcal{M}$ for all a ; $\{f \geq a\} \in \mathcal{M}$ for all a ; $\{f < a\} \in \mathcal{M}$ for all a ; $\{f \leq a\} \in \mathcal{M}$ for all a .*

Proof. $\{f \geq a\} = \bigcap_{k \geq 1} \{f > a - \frac{1}{k}\}$ and $\{f > a\} = \bigcup_{k \geq 1} \{f \geq a + \frac{1}{k}\}$ express each of the first two families in terms of countable operations on the other, and $\mathcal{M}$ is a σ -algebra. The remaining two are the complements of these. $\square$

Scratch work. Notice how little was needed: only that $\mathcal{M}$ is closed under countable unions and intersections. That is the whole reason Theorem 1.4.1 tests against the single family $\{(a, \infty]\}$ rather than against all Borel sets. The family $\{(a, \infty] : a \in \mathbb{R}\}$ generates $\mathcal{B}(\mathbb{R})$, and preimage commutes with all set operations, so testing on a generating family is enough. Spelling this out gives the following, which is the form to remember.

Proposition 1.4.3. *$f : X \rightarrow \mathbb{R}$ is measurable if and only if $f^{-1}(B) \in \mathcal{M}$ for every Borel $B \subseteq \mathbb{R}$.*

Proof. $(\Leftarrow)$ is immediate, since (a, ∞) is Borel. For $(\Rightarrow)$, let $\mathcal{S} = \{B \subseteq \mathbb{R} : f^{-1}(B) \in \mathcal{M}\}$. Because preimage commutes with complements and countable unions — $f^{-1}(B^c) = f^{-1}(B)^c$ and $f^{-1}(\bigcup_k B_k) = \bigcup_k f^{-1}(B_k)$ — and because $\mathcal{M}$ is a σ -algebra, $\mathcal{S}$ is a σ -algebra. By hypothesis it contains every (a, ∞) , and those generate $\mathcal{B}(\mathbb{R})$. Hence $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{S}$. $\square$

Pitfall: measurable is not “preimages of measurable sets are measurable”

It is tempting to define measurability of $f : \mathbb{R} \rightarrow \mathbb{R}$ by requiring $f^{-1}(E) \in \mathcal{L}$ for every $E \in \mathcal{L}$. This is a *strictly stronger* and wrong condition: the identity map fails nothing, but there exist continuous increasing f and $E \in \mathcal{L}$ with $f^{-1}(E) \notin \mathcal{L}$, built from the Cantor function. The asymmetry — Borel sets on the target, measurable sets on the source — is deliberate and is what makes Theorem 1.4.4 below true.

Proposition 1.4.4 (Stability). *Let $f, g, f_1, f_2, \dots : X \rightarrow [-\infty, \infty]$ be measurable.*

- (i) *If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable (in particular, if φ is continuous) then $\varphi \circ f$ is measurable.*
- (ii) *$f + g, fg, |f|, \max(f, g), \min(f, g)$ are measurable (sums where defined).*
- (iii) *$\sup_k f_k, \inf_k f_k, \limsup_k f_k, \liminf_k f_k$ are measurable.*
- (iv) *If $f_k \rightarrow f$ pointwise then f is measurable.*

Proof. (i) $(\varphi \circ f)^{-1}(B) = f^{-1}(\varphi^{-1}(B))$, and $\varphi^{-1}(B)$ is Borel, so Theorem 1.4.3 applies.

(iii) is the heart, and is a one-line computation:

$$\left\{ \sup_k f_k > a \right\} = \bigcup_{k \geq 1} \{f_k > a\}, \quad \left\{ \inf_k f_k < a \right\} = \bigcup_{k \geq 1} \{f_k < a\},$$

both countable unions of measurable sets. The first identity holds because a supremum exceeds a exactly when some term does. Then $\limsup_k f_k = \inf_m \sup_{k \geq m} f_k$ and $\liminf_k f_k = \sup_m \inf_{k \geq m} f_k$ are obtained by two further applications, and (iv) follows since a pointwise limit equals the lim sup.

(ii) For the sum, observe

$$\{f + g > a\} = \bigcup_{q \in \mathbb{Q}} (\{f > q\} \cap \{g > a - q\}),$$

a *countable* union: if $f(x) + g(x) > a$ then $f(x) > a - g(x)$, and density of $\mathbb{Q}$ provides a rational q with $f(x) > q > a - g(x)$. For the product use $fg = \frac{1}{4}((f + g)^2 - (f - g)^2)$ together with (i) applied to $\varphi(t) = t^2$. Finally $\max(f, g) = \sup$ of the two-term sequence and $|f| = \max(f, -f)$. $\square$

Where we are. Compare this list with Chapter 1.1. The class of Riemann integrable functions is *not* closed under pointwise limits — that was Worked Example 1.1, the Dirichlet function. The class of measurable functions is, by (iv), and the proof took one line. We have not yet integrated anything; but the class of objects we intend to integrate is already stable under exactly the operation that broke the old theory.

1.4.2 Simple functions and approximation

Definition 1.4.5. A *simple function* is a measurable $\varphi : X \rightarrow \mathbb{R}$ taking finitely many values. Writing its distinct nonzero values as $c_1, \dots, c_N$ and $E_j = \varphi^{-1}(\{c_j\}) \in \mathcal{M}$, we have the *standard representation*

$$\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j},$$

with the E_j pairwise disjoint.

Theorem 1.4.6 (Simple approximation). *Let $f : X \rightarrow [0, \infty]$ be measurable. Then there exist simple functions*

$$0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq f \quad \text{with} \quad \varphi_k \rightarrow f \text{ pointwise on } X,$$

and the convergence is uniform on any set where f is bounded.

Idea of the proof. Partition the range, exactly as Chapter 1.1 promised. At stage k , chop $[0, k)$ into $k2^k$ intervals of length 2^{-k} and round f down to the nearest gridpoint, truncating at height k . Two features must be arranged: the grid at stage $k + 1$ must *refine* that at stage k , which is why

the mesh is dyadic rather than, say, $1/k$; and the truncation height must grow, which is what allows $\varphi_k \rightarrow f$ even where $f = \infty$.

Proof. For $k \geq 1$ define

$$\varphi_k = \sum_{j=0}^{k2^k-1} j2^{-k} \mathbf{1}_{E_{k,j}} + k \mathbf{1}_{F_k},$$

$$E_{k,j} = f^{-1}\left(\left[\frac{j}{2^k}, \frac{j+1}{2^k}\right)\right), \quad F_k = f^{-1}([k, \infty)).$$

Each $E_{k,j}$ and F_k is measurable by Theorem 1.4.3, and there are finitely many of them, so φ_k is simple. By construction $0 \leq \varphi_k \leq f$: on $E_{k,j}$ we have $\varphi_k = j2^{-k} \leq f$, and on F_k we have $\varphi_k = k \leq f$.

Monotonicity. Fix x and suppose $f(x) < k$, so $x \in E_{k,j}$ with $j2^{-k} \leq f(x) < (j+1)2^{-k}$. At stage $k+1$ the interval $[j2^{-k}, (j+1)2^{-k})$ is split into its two dyadic halves, and $f(x)$ lies in one of them; in either case the corresponding gridpoint is $2j \cdot 2^{-(k+1)} = j2^{-k}$ or $(2j+1)2^{-(k+1)} > j2^{-k}$. So $\varphi_{k+1}(x) \geq \varphi_k(x)$. If instead $f(x) \geq k$ then $\varphi_k(x) = k$, while $\varphi_{k+1}(x) \geq k$ since the stage- $(k+1)$ grid reaches height $k+1$.

Convergence. If $f(x) < \infty$, then for all $k > f(x)$ we have $0 \leq f(x) - \varphi_k(x) < 2^{-k}$, so $\varphi_k(x) \rightarrow f(x)$; and this estimate is uniform over any set where $f \leq M$, once $k > M$. If $f(x) = \infty$ then $\varphi_k(x) = k \rightarrow \infty = f(x)$. $\square$

1.4.3 The integral of a nonnegative function

Definition 1.4.7. For a simple $\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j} \geq 0$ in standard representation, define

$$\int_X \varphi \, d\mu = \sum_{j=1}^N c_j \mu(E_j) \in [0, \infty],$$

with the convention $0 \cdot \infty = 0$. For measurable $f : X \rightarrow [0, \infty]$ define

$$\int_X f \, d\mu = \sup \left\{ \int_X \varphi \, d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\}. \quad (1.4.1)$$

For $E \in \mathcal{M}$ write $\int_E f \, d\mu = \int_X f \mathbf{1}_E \, d\mu$.

Pitfall: the convention $0 \cdot \infty = 0$

This is not a dodge. It encodes the statement that a function which vanishes on a set of infinite measure contributes nothing there, which is exactly what we want: $\int_{\mathbb{R}} \mathbf{1}_{\{0\}} \, dm = 0$ even though $\mathbf{1}_{\{0\}}$ is supported in a set of measure zero and $\int_{\mathbb{R}} 0 \, dm = 0$ even though $m(\mathbb{R}) = \infty$. The convention is used only inside Theorem 1.4.7; it is never applied to limits, where $0 \cdot \infty$ genuinely is indeterminate — see the failure modes of Fatou's lemma in Chapter 1.5.

Lemma 1.4.8 (Consistency and basic properties on simple functions). *Let $\varphi, \psi \geq 0$ be simple and $c \geq 0$. Then*

- (i) $\int \varphi d\mu$ is unambiguous: any representation $\varphi = \sum_i a_i \mathbf{1}_{A_i}$ with the A_i disjoint and measurable gives the same value $\sum_i a_i \mu(A_i)$;
- (ii) $\int(\varphi + \psi) = \int \varphi + \int \psi$ and $\int c\varphi = c \int \varphi$;
- (iii) $\varphi \leq \psi \implies \int \varphi \leq \int \psi$;
- (iv) $E \mapsto \int_E \varphi d\mu$ is a measure on $\mathcal{M}$.

Proof. (i) Refine: given two disjoint representations, the common refinement by intersections $A_i \cap E_j$ is disjoint and measurable, and finite additivity of μ shows both sums equal the sum over the refinement.

(ii) Pass to the common refinement $\{A_i \cap E_j\}$, on which $\varphi + \psi$ is constant with value $a_i + c_j$; then use finite additivity of μ and (i). Homogeneity is immediate.

(iii) Write $\psi = \varphi + (\psi - \varphi)$, note $\psi - \varphi \geq 0$ is simple, and apply (ii) together with nonnegativity.

(iv) Countable additivity of $E \mapsto \int_E \varphi = \sum_j c_j \mu(E \cap E_j)$ follows from countable additivity of μ applied to each of the finitely many terms. $\square$

Proposition 1.4.9 (Monotonicity of the integral). *If $0 \leq f \leq g$ are measurable then $\int f d\mu \leq \int g d\mu$. If $E \subseteq F$ are measurable and $f \geq 0$ then $\int_E f \leq \int_F f$.*

Proof. Every simple φ with $0 \leq \varphi \leq f$ also satisfies $0 \leq \varphi \leq g$, so the supremum in (1.4.1) for g is over a superset. The second claim is the first applied to $f\mathbf{1}_E \leq f\mathbf{1}_F$. $\square$

Where we are. Monotonicity was free. *Additivity* — the statement $\int(f + g) = \int f + \int g$ for general nonnegative measurable f, g — is not, and cannot be proved directly from (1.4.1): given simple $\varphi \leq f + g$ there is no way to split φ into pieces below f and below g respectively. The tool that resolves this is the Monotone Convergence Theorem, which lets us approximate f and g separately by Theorem 1.4.6 and add in the limit. So additivity of the integral is deferred to Chapter 1.5, one chapter away, and everything below is stated so as not to use it prematurely.

1.4.4 Integrable functions and L^1

Definition 1.4.10. For measurable $f : X \rightarrow [-\infty, \infty]$ write $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, so that $f = f^+ - f^-$ and $|f| = f^+ + f^-$, both

nonnegative and measurable by Theorem 1.4.4. Say f is *integrable* if $\int |f| d\mu < \infty$, and then define

$$\int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu,$$

a difference of two finite numbers. For complex $f = u + iv$, integrability means $\int |f| < \infty$ and $\int f = \int u + i \int v$. The set of integrable functions is denoted $L^1(\mu)$.

Proposition 1.4.11 (Triangle inequality). *If $f \in L^1(\mu)$ then $|\int f d\mu| \leq \int |f| d\mu$.*

Proof. For real f , $|\int f| = |\int f^+ - \int f^-| \leq \int f^+ + \int f^- = \int |f|$, using additivity for the last step — which for the specific decomposition $|f| = f^+ + f^-$ into functions with *disjoint supports* follows from Theorem 1.4.8(iv) and does not require the general additivity deferred above.

For complex f , choose $\theta \in \mathbb{R}$ with $e^{i\theta} \int f = |\int f|$. Then

$$\left| \int f \right| = \int \operatorname{Re}(e^{i\theta} f) d\mu \leq \int |e^{i\theta} f| d\mu = \int |f| d\mu,$$

where the middle step uses $\operatorname{Re}(z) \leq |z|$ and monotonicity. $\square$

1.4.5 Almost everywhere

Definition 1.4.12. A property holds *almost everywhere* (a.e.) if the set where it fails is contained in a set of measure zero.

Proposition 1.4.13. *Let μ be complete.*

(i) *If $f = g$ a.e. and f is measurable, so is g , and $\int f = \int g$ whenever either side is defined.*

(ii) *If $f \geq 0$ is measurable and $\int f d\mu = 0$, then $f = 0$ a.e.*

(iii) *If $f_k \rightarrow f$ a.e. and each f_k is measurable, then f is measurable.*

Proof. (i) Let N be null with $f = g$ off N . Then $\{g > a\} = (\{f > a\} \setminus N) \cup (\{g > a\} \cap N)$; the first set is measurable and the second is a subset of a null set, hence measurable by completeness. Equality of integrals holds because every simple $\varphi \leq f$ may be modified on N without changing $\int \varphi$, by Theorem 1.4.8(iv).

(ii) Let $E_k = \{f > 1/k\}$. The simple function $\frac{1}{k} \mathbf{1}_{E_k}$ satisfies $0 \leq \frac{1}{k} \mathbf{1}_{E_k} \leq f$, so $\frac{1}{k} \mu(E_k) \leq \int f = 0$, giving $\mu(E_k) = 0$. Then $\{f > 0\} = \bigcup_k E_k$ is null by countable subadditivity.

(iii) is Theorem 1.4.4(iv) combined with (i). $\square$

Looking ahead: why completeness keeps mattering

Part (i) is false without completeness: on the Borel σ -algebra, a function equal a.e. to a Borel function need not be Borel, since Worked Example 3.14(c) produced Lebesgue measurable non-Borel subsets of the Cantor set. Since virtually every argument in later volumes modifies functions on null sets without comment — redefining a Fourier transform on a null set, adjusting a wave packet decomposition, discarding a bad set produced by a maximal inequality — we work throughout with $\mathcal{L}$ rather than $\mathcal{B}$.

1.4.6 Comparison with the Riemann integral

We can now settle exactly which functions the old theory handled.

Theorem 1.4.14 (Lebesgue's criterion). *Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is Riemann integrable if and only if its set of discontinuities is Lebesgue null. In that case f is Lebesgue measurable and the two integrals agree.*

Idea of the proof. Attach to each point the *oscillation* $\text{osc}(f, x) = \lim_{r \rightarrow 0} (\sup_{B(x, r)} f - \inf_{B(x, r)} f)$, which is 0 exactly at points of continuity. Riemann integrability says upper and lower sums can be brought together, which says that the total length of subintervals carrying large oscillation can be made small — a statement about the measure of $\{\text{osc}(f, \cdot) \geq \varepsilon\}$. Running this in both directions is the whole proof.

Proof. Let $D_\varepsilon = \{x : \text{osc}(f, x) \geq \varepsilon\}$ and $D = \bigcup_k D_{1/k}$ the set of discontinuities. Each D_ε is closed: if $\text{osc}(f, x) < \varepsilon$ then the same holds on a neighbourhood of x .

Step 1. *Riemann integrable $\Rightarrow m(D) = 0$.*

Fix $\varepsilon, \eta > 0$ and choose a partition P with $U(f, P) - L(f, P) < \varepsilon\eta$. Let $\mathcal{J}$ be the set of subintervals of P whose interior meets D_ε . On each such interval the oscillation of f is at least ε , so

$$\varepsilon \sum_{J \in \mathcal{J}} |J| \leq U(f, P) - L(f, P) < \varepsilon\eta,$$

giving $\sum_{J \in \mathcal{J}} |J| < \eta$. Since D_ε is covered by these intervals together with the finitely many partition points (a null set), $m(D_\varepsilon) \leq \eta$ for every η , so $m(D_\varepsilon) = 0$ and hence $m(D) = 0$ by countable subadditivity.

Step 2. *$m(D) = 0 \Rightarrow$ Riemann integrable.*

Fix $\varepsilon > 0$. Then D_ε is closed, bounded and null, hence compact and coverable by finitely many open intervals of total length $< \varepsilon$. Their complement

in $[a, b]$ is a finite union of closed intervals on each of which $\text{osc}(f, \cdot) < \varepsilon$ pointwise; by compactness each may be partitioned so finely that f oscillates by at most 2ε on every piece. Assembling, we get a partition P with

$$U(f, P) - L(f, P) \leq 2\varepsilon(b - a) + 2\|f\|_{\infty}\varepsilon,$$

the two terms coming from the good part and from the covering intervals. As ε was arbitrary, f is Riemann integrable.

Step 3. Agreement.

Take partitions P_k with $\text{mesh} \rightarrow 0$ and let $\varphi_k, \bar{\varphi}_k$ be the associated lower and upper step functions. These are simple, $\varphi_k \leq f \leq \bar{\varphi}_k$, and their integrals are the lower and upper Riemann sums. Both converge a.e. to f (namely at every point of continuity, which is a.e. by Step 1), so f is measurable by Theorem 1.4.13, and monotonicity of the Lebesgue integral traps $\int f \, dm$ between the lower and upper sums, whose common limit is the Riemann integral. $\square$

WORKED EXAMPLE 1.4.1— TIER A The two functions from Chapter 1, resolved

(a) *Dirichlet*. $f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}$ is discontinuous at every point of $[0, 1]$, so $D = [0, 1]$ has measure $1 \neq 0$: not Riemann integrable, confirming Worked Example 1.1. But $\mathbb{Q} \cap [0, 1]$ is measurable with measure 0, so f is simple and $\int_0^1 f \, dm = 1 \cdot m(\mathbb{Q} \cap [0, 1]) = 0$.

(b) *Thomae*. $g(p/q) = 1/q$ in lowest terms, $g = 0$ on irrationals. Here g is continuous at every irrational and discontinuous at every rational, so $D = \mathbb{Q}$ is null and g is Riemann integrable, with integral 0. This is Drill 1.1, now with the general principle behind it: what matters is not how many discontinuities there are, but how much they measure. Both functions are discontinuous on a dense set; only one of them is discontinuous on a set of positive measure.

WORKED EXAMPLE 1.4.2— TIER B A measurable function that is not Borel, and why the distinction is harmless

Let C be the Cantor set and $A \subseteq C$ a subset that is not Borel; one exists by the cardinality argument of Worked Example 3.14(c). Then $\mathbf{1}_A$ is Lebesgue measurable, since $m^*(A) \leq m(C) = 0$ and null sets are measurable by completeness, but it is not Borel measurable, since $\mathbf{1}_A^{-1}((\frac{1}{2}, \infty)) = A$.

Nevertheless $\mathbf{1}_A = 0$ a.e., and $\int \mathbf{1}_A \, dm = 0$. This is the general situation: by Theorem 1.3.6(v) every Lebesgue measurable function agrees a.e. with a Borel function, and by Theorem 1.4.13(i) the integral cannot tell them apart. So the enlargement from $\mathcal{B}$ to $\mathcal{L}$ costs nothing in the theory of integration while buying the convenience recorded in the Looking Ahead box above.

WORKED EXAMPLE 1.4.3— TIER C: RESEARCH TASTE A first restriction-flavoured computation

This example uses nothing beyond Theorem 1.4.7 and the box lemma, yet it produces — in embryo — the single most important example in the theory of Fourier restriction. Read it now for the shape of the computation; the meaning will arrive in Volume 6.

The set-up. Let $\delta \in (0, 1)$ and consider the “cap”

$$\kappa_\delta = \left\{ (\xi_1, \xi_2) \in \mathbb{R}^2 : |\xi_1| \leq \delta, \xi_2 = \xi_1^2 \right\},$$

a piece of the parabola of horizontal extent 2δ . Since $\xi_1^2 \leq \delta^2$ on this range, κ_δ is contained in the axis-parallel rectangle R_δ of dimensions $2\delta \times \delta^2$. Note the shape: the cap is δ long and δ^2 thick. *Curvature converts a length scale into a thickness scale by squaring*, and every phenomenon in this subject descends from that sentence.

The computation. Consider the function

$$F(x) = \int_{-\delta}^{\delta} e^{2\pi i(x_1 t + x_2 t^2)} dt,$$

which is the object Volume 6 will call the extension of the constant function from κ_δ . We do not yet have the tools to evaluate F ; but we can bound the region where it is large, using only the triangle inequality (Theorem 1.4.11) and the box lemma.

If $|x_1| \leq \frac{1}{100\delta}$ and $|x_2| \leq \frac{1}{100\delta^2}$, then for $|t| \leq \delta$ the phase satisfies

$$|x_1 t + x_2 t^2| \leq \frac{1}{100\delta} \cdot \delta + \frac{1}{100\delta^2} \cdot \delta^2 = \frac{1}{50},$$

so $e^{2\pi i(x_1 t + x_2 t^2)}$ stays within an arc of angle $2\pi/50$ of 1; in particular its real part exceeds $\frac{1}{2}$. Hence on the rectangle

$$\begin{aligned} T_\delta &= \left\{ |x_1| \leq \frac{1}{100\delta} \right\} \times \left\{ |x_2| \leq \frac{1}{100\delta^2} \right\} \\ , \quad m(T_\delta) &= \frac{4}{100^2} \cdot \frac{1}{\delta^3}, \end{aligned}$$

we have $|F(x)| \geq \operatorname{Re} F(x) \geq \frac{1}{2} \cdot 2\delta = \delta$.

Reading the answer. F is of size $\approx \delta$ on a rectangle of dimensions $\delta^{-1} \times \delta^{-2}$ — a long thin *tube*, dual to the cap in the precise sense that its side lengths are the reciprocals of the cap’s. So the total mass is

$$\int_{T_\delta} |F|^p dm \gtrsim \delta^p \cdot \delta^{-3}.$$

Comparing this with $\|1\|_{L^p(\kappa_\delta)}^p \approx \delta$ is exactly how one shows that no inequality $\|F\|_{L^p(\mathbb{R}^2)} \lesssim \|f\|_{L^\infty(\kappa)}$ can hold for p too small. This is the *Knapp example*, and the exponent it produces is conjectured to be sharp.

What to take away now. Three things, all of which recur permanently. First, curvature makes a δ -cap sit inside a $\delta \times \delta^2$ box. Second, the Fourier transform converts that box into the dual tube $\delta^{-1} \times \delta^{-2}$. Third, the entire argument was a lower bound obtained by finding a region where the phase does not oscillate — “stationary phase for beginners”. Volume 5 develops the systematic version. Note also the kinship with Chapter 1.3: there, tubes overlapped and we measured the union; here, a single tube appears from a curvature computation. Volume 6 is where the two stories meet.

Drills

Drill 1.4.1. Show that $f : X \rightarrow \mathbb{R}$ is measurable if and only if $f^{-1}(U) \in \mathcal{M}$ for every *open* $U \subseteq \mathbb{R}$. Deduce that every continuous $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Borel measurable.

Drill 1.4.2. Let $f \geq 0$ be measurable with $\int_E f \, d\mu = 0$ for every E in some family generating $\mathcal{M}$. Must $f = 0$ a.e.? Prove it or find a counterexample.

Drill 1.4.3. Carry out the estimate of Worked Example 4.14 in $\mathbb{R}^n$: show that the cap $\{|\xi'| \leq \delta, \xi_n = |\xi'|^2\}$ on the paraboloid sits inside a box of dimensions $\delta^{n-1} \times \delta^2$, and identify the dual tube.

Notes

Theorem 1.4.1 and the range-partition construction of Theorem 1.4.6 are Lebesgue’s, from his 1902 thesis. Theorem 1.4.14 is also his, and it is the cleanest possible epitaph for the Riemann theory: the old integral works exactly when the discontinuities are negligible in the sense of the new one.

The deferral of additivity to Chapter 1.5 is a deliberate exposition choice. Some texts define $\int f$ via Theorem 1.4.6 and prove additivity at once; we prefer (1.4.1) because it makes monotonicity self-evident and because it makes visible *which* theorem is doing the work when additivity finally arrives.

Chapter 1.5

The Convergence Theorems

This chapter delivers what Chapter 1.1 promised. The Riemann integral failed because limits could not be passed through it; the Lebesgue integral admits three theorems — monotone convergence, Fatou, dominated convergence — which together make the interchange $\lim \int = \int \lim$ available under hypotheses so mild that in practice one stops noticing them.

There is a second, less advertised payoff. Additivity of the integral was left unproved in Chapter 1.4, because (1.4.1) gives no way to split a simple minorant of $f + g$ into minorants of f and of g . The Monotone Convergence Theorem removes the obstruction immediately, and we harvest it in §5.1 before doing anything else.

1.5.1 Monotone convergence

Theorem 1.5.1 (Monotone Convergence Theorem). *Let $f_k : X \rightarrow [0, \infty]$ be measurable with $f_1 \leq f_2 \leq \dots$ and $f_k \rightarrow f$ pointwise. Then f is measurable and*

$$\lim_{k \rightarrow \infty} \int_X f_k \, d\mu = \int_X f \, d\mu.$$

Idea of the proof. One inequality is free: $f_k \leq f$ gives $\int f_k \leq \int f$, and the left side is nondecreasing, so $\lim_k \int f_k \leq \int f$.

For the reverse we must show that the limit of the $\int f_k$ dominates $\int \varphi$ for every simple $\varphi \leq f$. The natural attempt — “ f_k eventually exceeds φ ” — is false: convergence is pointwise, not uniform, so no single k works everywhere. The repair is to weaken φ by a factor $c < 1$ and to track the sets $E_k = \{f_k \geq c\varphi\}$, which increase to all of X . Continuity of the measure $E \mapsto \int_E \varphi$ from below then does the rest, and $c \uparrow 1$ at the end. The two-parameter structure — one parameter for the approximation, one to be relaxed afterwards — is the same manoeuvre as the α and k of the

| Perron tree in §3.6.

Proof. Measurability of f is Theorem 1.4.4(iv).

Step 1. *The easy inequality.*

Since $f_k \leq f_{k+1} \leq f$, Theorem 1.4.9 gives that $(\int f_k)_k$ is nondecreasing and bounded above by $\int f$. So the limit $L := \lim_k \int f_k$ exists in $[0, \infty]$ and $L \leq \int f$.

▷ *Goal:* $L \geq \int f$, equivalently $L \geq \int \varphi$ for every simple $0 \leq \varphi \leq f$.

Step 2. *The truncated sets.*

Fix a simple φ with $0 \leq \varphi \leq f$ and fix $c \in (0, 1)$. Put

$$E_k = \{x \in X : f_k(x) \geq c\varphi(x)\}.$$

Each E_k is measurable, since $f_k - c\varphi$ is measurable. The sets increase, because $f_k \leq f_{k+1}$. And they exhaust X : given x , either $\varphi(x) = 0$, in which case $x \in E_1$ as $f_1 \geq 0$; or $\varphi(x) > 0$, in which case $f(x) \geq \varphi(x) > c\varphi(x)$, and since $f_k(x) \uparrow f(x)$ we get $f_k(x) \geq c\varphi(x)$ for k large. Hence $\bigcup_k E_k = X$.

Step 3. *Estimate and pass to the limit.*

By monotonicity of the integral and the definition of E_k ,

$$\int_X f_k d\mu \geq \int_{E_k} f_k d\mu \geq c \int_{E_k} \varphi d\mu.$$

By Theorem 1.4.8(iv) the set function $\nu(E) = \int_E \varphi d\mu$ is a measure, and measures are continuous from below along increasing sequences — apply countable additivity to the disjointification $E_k \setminus E_{k-1}$. Since $E_k \uparrow X$,

$$\int_{E_k} \varphi d\mu \longrightarrow \int_X \varphi d\mu.$$

Letting $k \rightarrow \infty$ in the display above gives $L \geq c \int_X \varphi d\mu$.

Step 4. *Release c .*

This holds for every $c \in (0, 1)$, so $L \geq \int \varphi$. Taking the supremum over simple $\varphi \leq f$ and using (1.4.1) gives $L \geq \int f$. □

1.5.1.1 Immediate harvest

Corollary 1.5.2 (Additivity). *For measurable $f, g : X \rightarrow [0, \infty]$ and $a, b \geq 0$,*

$$\int_X (af + bg) d\mu = a \int_X f d\mu + b \int_X g d\mu.$$

The same holds for $f, g \in L^1(\mu)$ and $a, b \in \mathbb{C}$, and $L^1(\mu)$ is a vector space.

Proof. Take simple $\varphi_k \uparrow f$ and $\psi_k \uparrow g$ by Theorem 1.4.6. Then $a\varphi_k + b\psi_k$ is simple and increases to $af + bg$, so three applications of Theorem 1.5.1 together with Theorem 1.4.8(ii) give

$$\int (af + bg) = \lim_k \int (a\varphi_k + b\psi_k) = \lim_k \left(a \int \varphi_k + b \int \psi_k \right) = a \int f + b \int g.$$

For the signed and complex cases, decompose into positive and negative (real and imaginary) parts, noting $|af + bg| \leq |a||f| + |b||g|$ so that integrability is preserved; the bookkeeping is routine once additivity is known for nonnegative functions. $\square$

Corollary 1.5.3 (Term-by-term integration). *If $f_k : X \rightarrow [0, \infty]$ are measurable then $\int \sum_{k \geq 1} f_k d\mu = \sum_{k \geq 1} \int f_k d\mu$.*

Proof. The partial sums increase to the total sum; apply Theorem 1.5.2 to each partial sum and Theorem 1.5.1 to the sequence of partial sums. $\square$

Corollary 1.5.4. *For measurable $f \geq 0$, the set function $\nu(E) = \int_E f d\mu$ is a measure on $\mathcal{M}$.*

Proof. $\nu(\emptyset) = 0$, and for disjoint E_k with $E = \bigcup_k E_k$ we have $f\mathbf{1}_E = \sum_k f\mathbf{1}_{E_k}$ pointwise, so Theorem 1.5.3 applies. $\square$

WORKED EXAMPLE 1.5.1— TIER B **The layer-cake formula**

For measurable $f : X \rightarrow [0, \infty]$,

$$\int_X f d\mu = \int_0^\infty \mu(\{x : f(x) > t\}) dt, \quad (1.5.1)$$

and more generally, for $0 < p < \infty$,

$$\int_X |f|^p d\mu = p \int_0^\infty t^{p-1} \mu(\{|f| > t\}) dt. \quad (1.5.2)$$

The function $d_f(t) = \mu(\{|f| > t\})$ is called the *distribution function* of f ; it is nonincreasing, hence Borel measurable, so the right-hand sides make sense as Lebesgue integrals on $(0, \infty)$.

Proof of (1.5.1) for simple f . Let $f = \sum_{j=1}^N c_j \mathbf{1}_{E_j}$ in standard representation with $c_1 > c_2 > \cdots > c_N > 0$ and the E_j disjoint. For $t \in [c_{j+1}, c_j]$ the superlevel set is $\{f > t\} = E_1 \cup \cdots \cup E_j$, of measure $\sum_{i \leq j} \mu(E_i)$, and for

$t \geq c_1$ it is empty. Hence, with $c_{N+1} := 0$,

$$\begin{aligned} \int_0^\infty d_f(t) dt &= \sum_{j=1}^N (c_j - c_{j+1}) \sum_{i=1}^j \mu(E_i) \\ &= \sum_{i=1}^N \mu(E_i) \sum_{j \geq i} (c_j - c_{j+1}) = \sum_{i=1}^N c_i \mu(E_i), \end{aligned}$$

where the middle step exchanges the order of a finite double sum and the last telescopes. The right-hand side is $\int f d\mu$.

Proof for general $f \geq 0$. Take simple $\varphi_k \uparrow f$ (Theorem 1.4.6). Then $\{\varphi_k > t\} \uparrow \{f > t\}$ for each fixed t — the inclusion $\subseteq$ is monotonicity, and if $f(x) > t$ then $\varphi_k(x) > t$ eventually — so continuity of μ from below gives $d_{\varphi_k}(t) \uparrow d_f(t)$ pointwise in t . Apply Theorem 1.5.1 *twice*: on X to get $\int \varphi_k \rightarrow \int f$, and on $(0, \infty)$ to get $\int_0^\infty d_{\varphi_k} \rightarrow \int_0^\infty d_f$. The simple case equates the two sequences, so the limits agree.

Proof of (1.5.2). Apply (1.5.1) to $|f|^p$ and substitute $t = s^p$, noting $\{|f|^p > s^p\} = \{|f| > s\}$ and $dt = ps^{p-1} ds$.

Why this matters. Formula (1.5.2) converts every L^p norm into a statement about the *sizes of superlevel sets*, and superlevel sets are geometric objects. That conversion is why the subject of this series is geometric at all. In Volume 9 the superlevel sets of the Kakeya maximal function will be unions of tubes, and (1.5.2) is the bridge that turns the analytic question “how large is $\|\mathcal{K}_\delta f\|_p$?” into the geometric question “how much can δ -tubes overlap?”. It is also the definition behind weak L^p in Chapter 1.8, and the engine of the Marcinkiewicz interpolation theorem in Volume 2.

1.5.2 Fatou’s lemma

Theorem 1.5.5 (Fatou). *For measurable $f_k : X \rightarrow [0, \infty]$,*

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu.$$

Proof. Put $g_m = \inf_{k \geq m} f_k$, measurable by Theorem 1.4.4(iii). The sequence (g_m) is nondecreasing with $g_m \uparrow \liminf_k f_k$, by definition of $\liminf$. Also $g_m \leq f_k$ for every $k \geq m$, so by monotonicity $\int g_m \leq \inf_{k \geq m} \int f_k$. Now apply Theorem 1.5.1 to (g_m) :

$$\int \liminf_k f_k = \lim_m \int g_m \leq \lim_m \inf_{k \geq m} \int f_k = \liminf_k \int f_k. \quad \square$$

WORKED EXAMPLE 1.5.2— TIER A **The three ways mass escapes**

Fatou's inequality is typically strict, and there are exactly three mechanisms. Recognising which one is operating is a genuine skill — in Chapter 1.7 and again in Volume 13 the question “where did the mass go?” is the first thing to ask when an estimate fails — so each is worth a concrete example on $\mathbb{R}$ with Lebesgue measure. In all three, $\int f_k = 1$ for every k while $f_k \rightarrow 0$ pointwise, so the left side of Fatou is 0 and the right side is 1.

(a) *Escape to horizontal infinity.* $f_k = \mathbf{1}_{[k, k+1]}$. The bump keeps its shape and marches off to the right. Nothing changes about the function except its location.

(b) *Escape to vertical infinity.* $f_k = k \mathbf{1}_{[0, 1/k]}$. The bump grows taller and narrower, concentrating at a point. This is the mode that the convention $0 \cdot \infty = 0$ of Chapter 1.4 does *not* rescue: the limit function is 0 a.e., but the mass is not lost gradually, it is lost in the limit.

(c) *Escape to width.* $f_k = \frac{1}{k} \mathbf{1}_{[0, k]}$. The bump flattens and spreads. Note that here $f_k \rightarrow 0$ *uniformly*, which shows that even uniform convergence does not justify passing a limit through an integral over a set of infinite measure.

The moral. Modes (a) and (c) are impossible on a finite measure space with uniformly bounded functions; mode (b) is impossible when there is an integrable dominating function. That observation is precisely the Dominated Convergence Theorem, which we now prove.

1.5.3 Dominated convergence

Theorem 1.5.6 (Dominated Convergence Theorem). *Let $f_k : X \rightarrow \mathbb{C}$ be measurable with $f_k \rightarrow f$ a.e., and suppose there is $g \in L^1(\mu)$ with $|f_k| \leq g$ a.e. for every k . Then $f \in L^1(\mu)$ and*

$$\lim_{k \rightarrow \infty} \int_X |f_k - f| d\mu = 0; \quad \text{in particular} \quad \lim_{k \rightarrow \infty} \int_X f_k d\mu = \int_X f d\mu.$$

Idea of the proof. Fatou's lemma applies to nonnegative functions, and $|f_k - f|$ is nonnegative but converges to 0, so applying Fatou directly gives only $0 \leq \liminf$, which is vacuous. The trick is to apply Fatou to the nonnegative sequence $2g - |f_k - f|$, whose $\liminf$ is $2g$. The dominating function is therefore not merely a technical hypothesis: it is the object that converts a useless application of Fatou into a decisive one, by supplying a fixed ceiling from which the deficit can be measured.

Proof. Discarding a null set, assume the hypotheses hold everywhere. Since $|f| = \lim_k |f_k| \leq g$, we get $f \in L^1$ and also $|f_k - f| \leq 2g$, so $2g - |f_k - f| \geq 0$.

Apply Theorem 1.5.5 to this sequence. Since $f_k \rightarrow f$ pointwise, $\liminf_k (2g - |f_k - f|) = 2g$, so

$$\int 2g \leq \liminf_k \int (2g - |f_k - f|) = \int 2g + \liminf_k \left(- \int |f_k - f| \right),$$

using Theorem 1.5.2. As $\int 2g < \infty$ we may cancel it, obtaining

$$0 \leq \liminf_k \left(- \int |f_k - f| \right) = - \limsup_k \int |f_k - f|,$$

that is, $\limsup_k \int |f_k - f| \leq 0$. Since the integrals are nonnegative the limit is 0. The final claim follows from $|\int f_k - \int f| \leq \int |f_k - f|$ (Theorem 1.4.11 and Theorem 1.5.2). $\square$

WORKED EXAMPLE 1.5.3— TIER B Differentiating under the integral sign

Let $I \subseteq \mathbb{R}$ be an open interval and $f : X \times I \rightarrow \mathbb{C}$ satisfy: $f(\cdot, s) \in L^1(\mu)$ for each s ; $s \mapsto f(x, s)$ is differentiable for each x ; and there is $g \in L^1(\mu)$ with $|\partial_s f(x, s)| \leq g(x)$ for all x, s . Then $F(s) = \int_X f(x, s) d\mu(x)$ is differentiable with

$$F'(s) = \int_X \partial_s f(x, s) d\mu(x).$$

Proof. Fix s and let $h_k \rightarrow 0$ be any nonzero sequence. Put

$$q_k(x) = \frac{f(x, s + h_k) - f(x, s)}{h_k},$$

so $q_k(x) \rightarrow \partial_s f(x, s)$ for every x . By the mean value theorem applied to $t \mapsto f(x, t)$ on the interval between s and $s + h_k$, there is a point ξ there with $q_k(x) = \partial_s f(x, \xi)$, whence $|q_k| \leq g$. Theorem 1.5.6 now gives $\int q_k \rightarrow \int \partial_s f(\cdot, s)$, and the left side is $(F(s + h_k) - F(s))/h_k$. Since the sequence (h_k) was arbitrary, $F'(s)$ exists and equals the stated integral. $\square$

The hypothesis that does the work is the *uniform* domination of the s -derivative. Readers should note how the mean value theorem is what converts a bound on the derivative into a bound on the difference quotients — this is the step that fails for complex-valued f if applied naively, which is why one applies the argument to real and imaginary parts separately, or uses the integral form of the remainder.

Why this is here. Almost every smoothness statement about a Fourier transform is an application of this lemma. In Worked Example 1.5.4 below we use it to show $\hat{f}$ is continuous; in Volume 2 the same argument shows $\hat{f}$ is C^∞ when f decays rapidly, which is the starting point of the theory of the Schwartz space. $\square$

WORKED EXAMPLE 1.5.4— TIER B Continuity of the Fourier transform, and Riemann–Lebesgue

For $f \in L^1(\mathbb{R}^n)$ define

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx.$$

The integral converges absolutely, since $|f(x)e^{-2\pi i x \cdot \xi}| = |f(x)|$, and $\|\widehat{f}\|_\infty \leq \|f\|_1$ by Theorem 1.4.11. This is the first theorem of Fourier analysis in this series, and both halves are corollaries of Theorem 1.5.6.

(a) $\widehat{f}$ is continuous. Let $\xi_k \rightarrow \xi$. The integrands $f(x)e^{-2\pi i x \cdot \xi_k}$ converge pointwise to $f(x)e^{-2\pi i x \cdot \xi}$ and are dominated by $|f| \in L^1$. Apply Theorem 1.5.6.

(b) $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ (Riemann–Lebesgue). The identity $e^{-\pi i} = -1$ gives, after the substitution $x \mapsto x + \frac{\xi}{2|\xi|^2}$,

$$\widehat{f}(\xi) = - \int_{\mathbb{R}^n} f\left(x + \frac{\xi}{2|\xi|^2}\right) e^{-2\pi i x \cdot \xi} dx,$$

so averaging this with the original formula,

$$\begin{aligned} |\widehat{f}(\xi)| &\leq \frac{1}{2} \int_{\mathbb{R}^n} \left| f(x) - f\left(x + \frac{\xi}{2|\xi|^2}\right) \right| dx \\ &= \frac{1}{2} \|f - \tau_h f\|_1, \quad h = \frac{\xi}{2|\xi|^2}. \end{aligned}$$

As $|\xi| \rightarrow \infty$ we have $h \rightarrow 0$, so it suffices to know that translation is continuous in L^1 : $\|f - \tau_h f\|_1 \rightarrow 0$ as $h \rightarrow 0$. This holds for f continuous with compact support by uniform continuity and Theorem 1.5.6, and extends to all of L^1 by density (Chapter 1.8) together with the estimate $\|\tau_h\|_{L^1 \rightarrow L^1} = 1$.

Reading (b). The translation trick is the whole content: shifting by half a period turns the integrand into its own negative, so the transform measures the failure of f to be invariant under a small translation — and integrable functions are, in the limit, invariant. Every decay estimate for oscillatory integrals in Volume 5 is a quantitative refinement of this one idea, with stationary phase supplying the rate.

1.5.4 Modes of convergence

We now compare the several senses in which a sequence of functions can converge.

Definition 1.5.7. Let f_k, f be measurable and finite a.e. We say $f_k \rightarrow f$

- *almost everywhere* if $\mu(\{x : f_k(x) \not\rightarrow f(x)\}) = 0$;
- *in L^1* if $\int |f_k - f| d\mu \rightarrow 0$;
- *in measure* if $\mu(\{|f_k - f| > \varepsilon\}) \rightarrow 0$ for every $\varepsilon > 0$;
- *almost uniformly* if for every $\eta > 0$ there is E with $\mu(E) < \eta$ and $f_k \rightarrow f$ uniformly on $X \setminus E$.

Proposition 1.5.8 (The implications that hold). *L^1 convergence implies convergence in measure. Almost uniform convergence implies both a.e. convergence and convergence in measure. Convergence in measure implies that some subsequence converges a.e.*

Proof. $L^1 \Rightarrow$ *measure* is Chebyshev's inequality: the simple function $\varepsilon \mathbf{1}_{\{|f_k - f| > \varepsilon\}}$ lies below $|f_k - f|$, so

$$\mu(\{|f_k - f| > \varepsilon\}) \leq \frac{1}{\varepsilon} \int |f_k - f| d\mu. \quad (1.5.3)$$

Almost uniform $\Rightarrow$ a.e. For each m pick E_m with $\mu(E_m) < 1/m$ and uniform convergence off E_m ; then $f_k \rightarrow f$ everywhere off $\bigcap_m E_m$, which is null.

Measure $\Rightarrow$ a.e. along a subsequence. Choose k_j with $\mu(\{|f_{k_j} - f| > 2^{-j}\}) < 2^{-j}$ and let $A_j = \{|f_{k_j} - f| > 2^{-j}\}$. Since $\sum_j \mu(A_j) < \infty$, the set $A = \bigcap_m \bigcup_{j \geq m} A_j$ of points lying in infinitely many A_j satisfies $\mu(A) \leq \sum_{j \geq m} \mu(A_j) \rightarrow 0$, so $\mu(A) = 0$. Off A , every point lies in only finitely many A_j , so $|f_{k_j} - f| \leq 2^{-j}$ eventually. $\square$

WORKED EXAMPLE 1.5.5— TIER B The typewriter sequence: no arrow can be reversed

Define on $[0, 1]$, for $k = 2^m + j$ with $0 \leq j < 2^m$,

$$f_k = \mathbf{1}_{[j2^{-m}, (j+1)2^{-m}]}$$

As k increases, the indicator sweeps across $[0, 1]$, then sweeps again at half the width, and so on — like the carriage of a typewriter.

It converges in L^1 and in measure. $\int f_k = 2^{-m} \rightarrow 0$, so $f_k \rightarrow 0$ in L^1 , hence in measure by (1.5.3).

It converges nowhere. Fix $x \in [0, 1]$. For each m there is exactly one j with x in the corresponding interval, so $f_k(x) = 1$ for at least one k in each dyadic block, while $f_k(x) = 0$ for the others. Hence $\limsup_k f_k(x) = 1$ and $\liminf_k f_k(x) = 0$ for every x .

What this rules out. L^1 convergence does not imply a.e. convergence, and neither does convergence in measure; only a subsequence survives, as The-

orem 1.5.8 says — here $f_{2^m} \rightarrow 0$ a.e. Combining with Worked Example 5.9: (a) shows a.e. convergence does not imply L^1 ; (b) shows a.e. and even convergence in measure do not imply L^1 ; and this example shows the reverse implication fails too. The four notions are genuinely four.

Theorem 1.5.9 (Egorov). *Let $\mu(X) < \infty$ and let $f_k \rightarrow f$ a.e., all measurable and finite a.e. Then $f_k \rightarrow f$ almost uniformly.*

Proof. Discard a null set so that $f_k \rightarrow f$ everywhere. For $m, N \geq 1$ set

$$E_{m,N} = \bigcup_{k \geq N} \left\{ x : |f_k(x) - f(x)| > \frac{1}{m} \right\}.$$

For fixed m these decrease in N , and $\bigcap_N E_{m,N} = \emptyset$ by pointwise convergence. Because $\mu(X) < \infty$, continuity of μ from above applies and $\mu(E_{m,N}) \rightarrow 0$ as $N \rightarrow \infty$. Given $\eta > 0$, choose N_m with $\mu(E_{m,N_m}) < \eta 2^{-m}$ and set $E = \bigcup_m E_{m,N_m}$, so $\mu(E) < \eta$. Off E we have $|f_k - f| \leq 1/m$ for all $k \geq N_m$, which is uniform convergence. $\square$

Pitfall: Egorov needs finite measure

On $\mathbb{R}$ take $f_k = \mathbf{1}_{[k,k+1]}$. Then $f_k \rightarrow 0$ everywhere, but for any set E of finite measure the convergence off E is not uniform, since $\sup_{\mathbb{R} \setminus E} f_k = 1$ for infinitely many k . This is escape mode (a) of Worked Example 5.9 again. Continuity from above — the step that used $\mu(X) < \infty$ — is exactly what fails.

Theorem 1.5.10 (Lusin). *Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be measurable and finite a.e., supported in a set of finite measure. For every $\eta > 0$ there is $g \in C_c(\mathbb{R}^n)$ with $m(\{f \neq g\}) < \eta$.*

Idea of the proof. Three tools combine. Simple functions approximate f pointwise; each level set of a simple function is, by regularity, sandwiched between a compact set and an open set of almost the same measure, and Urysohn's lemma turns that sandwich into a *continuous* function agreeing with the indicator off a small set. That gives continuous g_k agreeing with φ_k off a small set. Egorov then upgrades pointwise convergence $g_k \rightarrow f$ to uniform convergence off a further small set — and a uniform limit of continuous functions is continuous. Finally Tietze extends the resulting continuous function from a compact set to all of $\mathbb{R}^n$.

Proof. Let S be a set of finite measure outside which f vanishes, and fix $\eta > 0$.

Step 1. *Reduce to a bounded set.*

Since $m(S) < \infty$ we may choose R with $m(S \setminus B(0, R)) < \eta/5$, and replace

S by $S \cap B(0, R)$; the discarded part is absorbed into the final exceptional set. So assume S is bounded.

Step 2. *Continuous approximants to a simple function.*

Let $\varphi = \sum_{j=1}^N c_j \mathbf{1}_{E_j}$ be simple with all $E_j \subseteq S$ measurable and disjoint. Fix $\rho > 0$. By Theorem 1.3.6(ii) and (iv) choose, for each j , a compact $K_j \subseteq E_j$ and an open $U_j \supseteq E_j$ with $m(U_j \setminus K_j) < \rho/N$. Urysohn's lemma (Appendix A) supplies $h_j \in C_c(\mathbb{R}^n)$ with $\mathbf{1}_{K_j} \leq h_j \leq \mathbf{1}_{U_j}$. Then $h_j = \mathbf{1}_{E_j}$ off $U_j \setminus K_j$, so

$$\begin{aligned} g := \sum_{j=1}^N c_j h_j \in C_c(\mathbb{R}^n) \quad \text{satisfies} \quad m(\{g \neq \varphi\}) \\ \leq \sum_{j=1}^N m(U_j \setminus K_j) < \rho. \end{aligned}$$

Step 3. *Apply this to an approximating sequence.*

By Theorem 1.4.6 (applied to positive and negative parts) choose simple $\varphi_k \rightarrow f$ pointwise on S , each supported in S . By Step 2 choose $g_k \in C_c(\mathbb{R}^n)$ with $m(\{g_k \neq \varphi_k\}) < \eta 2^{-k}/5$. Put $A = \bigcup_k \{g_k \neq \varphi_k\}$, so $m(A) < \eta/5$, and off A we have $g_k = \varphi_k \rightarrow f$ pointwise.

Step 4. *Egorov.*

The set S has finite measure, so Theorem 1.5.9 applied to (g_k) on $S \setminus A$ yields B with $m(B) < \eta/5$ such that $g_k \rightarrow f$ uniformly on $S \setminus (A \cup B)$.

Step 5. *Pass to a compact set.*

By inner regularity (Theorem 1.3.6(iv)) choose a compact $K \subseteq S \setminus (A \cup B)$ with $m((S \setminus (A \cup B)) \setminus K) < \eta/5$. On K the functions g_k are continuous and converge uniformly to f , so $f|_K$ is continuous.

Step 6. *Extend.*

By Tietze's extension theorem (Appendix A) there is $g \in C_c(\mathbb{R}^n)$ with $g = f$ on K . Collecting the five discarded pieces,

$$\begin{aligned} m(\{f \neq g\}) &\leq m(S \setminus B(0, R)) + m(A) + m(B) \\ &\quad + m((S \setminus (A \cup B)) \setminus K) < \eta, \end{aligned}$$

with room to spare for the set where both vanish. □

WORKED EXAMPLE 1.5.6— TIER C: RESEARCH TASTE **Pigeonholing over scales**

The following exercise looks artificial and is not: it is the bookkeeping discipline of *induction on scales*, isolated in a setting where nothing else is

difficult. Volume 13 consists largely of arguments of this shape, and readers who meet the pattern for the first time there usually find it opaque. There is no reason to wait.

The set-up. Let $\delta \in (0, \frac{1}{2})$ and let $F \subseteq [0, 1]$ be measurable with $m(F) = \sigma > 0$. Write $M = \lceil \log_2(1/\delta) \rceil$ and decompose $[0, 1]$ into the M dyadic scales

$$\mathcal{D}_m = \{\text{dyadic intervals of length } 2^{-m}\}, \quad m = 1, \dots, M.$$

Suppose we are given, for each m , a quantity $A_m \geq 0$ with $\sum_{m=1}^M A_m \leq K$.

The pigeonhole step. There exists $m^* \in \{1, \dots, M\}$ with

$$A_{m^*} \leq \frac{K}{M} \approx \frac{K}{\log(1/\delta)}.$$

This is trivial — the minimum is at most the average — and it is used constantly. The content is in what one does with it: a bound that is lossy by a factor $\log(1/\delta)$ at a single scale becomes *lossless up to constants* once one is permitted to choose the scale.

The reverse discipline, which is the real point. Now suppose instead that at each scale we lose a factor: $A_{m+1} \leq C A_m$. Iterating M times gives $A_M \leq C^M A_1 = \delta^{-\log_2 C} A_1$, which is a *power* of $1/\delta$ and therefore catastrophic. But if the loss is additive, $A_{m+1} \leq A_m + c_m$ with $\sum_m c_m < \infty$, then $A_M \leq A_1 + \sum c_m$ and nothing is lost at all.

Where you have already seen this. Precisely this dichotomy decided the Perron tree in §3.6. There the “loss per scale” was the sprout debt $4(1 - \alpha)^2 \alpha^{2(j-1)}$, and (1.3.5) showed it was summable; had it instead been multiplicative, the construction would have produced nothing. The estimate (1.3.6) is exactly an instance of the second display above.

Where you will see it next. In Chapter 1.7 the Hardy–Littlewood maximal function is bounded by summing over dyadic radii; in Volume 12 the ε -removal argument of Tao converts a local restriction estimate into a global one by paying $\delta^{-\varepsilon}$ rather than $\log(1/\delta)$ per scale; and in Volume 13 the entire Bourgain–Demeter theorem is the statement that a certain decoupling constant satisfies a recursion whose loss per scale is subpolynomial, hence harmless over $\log(1/\delta)$ scales. The theorem is hard; the bookkeeping principle is the one on this page.

WORKED EXAMPLE 1.5.7— TIER B **Which convergence theorem to reach for**

Three theorems license passing a limit through an integral, and choosing among them is mechanical once the differences are clear.

(a) *Monotone convergence.* $0 \leq f_k \uparrow f$. No domination needed, and the conclusion allows $\int f = \infty$. Use it when the sequence is built by adding non-negative pieces — as every construction of a measure from a sum does.

(b) *Dominated convergence.* $f_k \rightarrow f$ pointwise with $|f_k| \leq g \in L^1$. The workhorse. The hypothesis to check is always the domination, and it is where the errors are.

(c) *Fatou.* $f_k \geq 0$, no other hypothesis, and the conclusion is only an inequality $\int \liminf \leq \liminf \int$. Use it when domination is unavailable and an inequality suffices — which is often, since most estimates are inequalities.

The decision procedure. Is the sequence monotone? Use (a). Is there a dominating function? Use (b). Neither, but an inequality in the right direction would do? Use (c). If none applies, the limit may genuinely fail to pass, and the standard counterexample $f_k = k\mathbf{1}_{[0,1/k]}$ has $\int f_k = 1$ for every k while $f_k \rightarrow 0$ pointwise.

Where each is used later. Fatou appears in Volume 4's atomic argument, where domination is exactly what is missing; monotone convergence in every construction of an outer measure; dominated convergence throughout.

Drills

Drill 1.5.1. Show that the hypothesis $f_k \geq 0$ cannot be dropped from Fatou's lemma, by finding $f_k : \mathbb{R} \rightarrow \mathbb{R}$ measurable with $\int \liminf f_k > \liminf \int f_k$.

Drill 1.5.2. Prove the “generalised” dominated convergence theorem: if $f_k \rightarrow f$ a.e., $|f_k| \leq g_k$ with $g_k \rightarrow g$ a.e., $g_k, g \in L^1$ and $\int g_k \rightarrow \int g$, then $\int |f_k - f| \rightarrow 0$. (Follow the proof of Theorem 1.5.6, applying Fatou to $g_k + g - |f_k - f|$.)

Drill 1.5.3. Let $f \in L^1(\mathbb{R})$. Show $d_f(t) = o(1/t)$ as $t \rightarrow \infty$, and exhibit $f \in L^1$ for which $d_f(t)$ decays no faster than $1/(t \log^2 t)$. (Use (1.5.1).)

Drill 1.5.4. In Worked Example 5.16, suppose $A_{m+1} \leq (1 + \frac{1}{M})A_m$ for $m = 1, \dots, M-1$. Show $A_M \leq e A_1$, and explain why a loss of $1 + \frac{1}{\log(1/\delta)}$ per scale is therefore harmless while a loss of $1 + \frac{1}{100}$ per scale is not.

Notes

The three convergence theorems are due to Beppo Levi (1906), Fatou (1906) and Lebesgue (1908) respectively, though all three are commonly attributed to Lebesgue. Egorov's theorem is from 1911 and Lusin's from 1912.

The proof of Theorem 1.5.6 given here — applying Fatou to $2g - |f_k - f|$ — is the standard one and repays study, because the manoeuvre of subtracting from a fixed ceiling to convert a $\liminf$ statement into a $\limsup$ statement is used repeatedly in later volumes, notably in the concentration-compactness arguments that appear in the study of extremisers for Strichartz estimates.

Worked Example 5.16 has no counterpart in standard measure theory texts, for the good reason that it belongs to no standard course. We include it because the single most common obstacle for readers approaching the decoupling literature is not any individual theorem but the unfamiliar rhythm of multiscale bookkeeping, and that rhythm can be learned early at almost no cost.

Chapter 1.6

Product Measures and Fubini–Tonelli

Integration in $\mathbb{R}^n$ would be impractical if every integral had to be computed from (1.4.1). What makes it practical is the ability to integrate one variable at a time. This chapter establishes that, and then spends the resulting technology on two computations: the volume of a ball, and — in §6.5 — the measure of the intersection of two thin tubes, from which we deduce the two-dimensional Kakeya conjecture.

1.6.1 The product measure

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces, assumed σ -finite throughout: $X = \bigcup_k X_k$ with $\mu(X_k) < \infty$, and similarly for Y . Lebesgue measure on $\mathbb{R}^n$ is σ -finite, by exhaustion with balls.

Definition 1.6.1. A *measurable rectangle* is a set $A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$. The *product σ -algebra* $\mathcal{M} \otimes \mathcal{N}$ is the σ -algebra generated by the measurable rectangles.

Definition 1.6.2. For $E \subseteq X \times Y$ and $x \in X$, the *slice* is $E_x = \{y \in Y : (x, y) \in E\}$. For f on $X \times Y$, $f_x(y) = f(x, y)$.

Lemma 1.6.3 (Slices are measurable). *If $E \in \mathcal{M} \otimes \mathcal{N}$ then $E_x \in \mathcal{N}$ for every $x \in X$. If f is $\mathcal{M} \otimes \mathcal{N}$ -measurable then f_x is $\mathcal{N}$ -measurable for every x .*

Proof. Let $\mathcal{S} = \{E \subseteq X \times Y : E_x \in \mathcal{N} \ \forall x\}$. Slicing commutes with all set operations — $(E^c)_x = (E_x)^c$ and $(\bigcup_k E_k)_x = \bigcup_k (E_k)_x$ — so $\mathcal{S}$ is a σ -algebra. It contains every measurable rectangle, since $(A \times B)_x$ is B if $x \in A$ and $\emptyset$ otherwise. Hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{S}$. The statement for functions follows by applying this to $\{f > a\}$, whose slice is $\{f_x > a\}$. $\square$

Pitfall: slices of Lebesgue measurable sets

Theorem 1.6.3 concerns $\mathcal{M} \otimes \mathcal{N}$. It is false for $\mathcal{L}(\mathbb{R}^2)$: if $A \subseteq \mathbb{R}$ is non-measurable then $\{0\} \times A$ is $\mathcal{L}(\mathbb{R}^2)$ -null, hence measurable, but its slice at $x = 0$ is A . The resolution is that $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R}^2)$, the latter being the completion of the former, and that for a set in the completion *almost every* slice is measurable. We state Fubini below for the product σ -algebra and note the a.e. caveat where it matters.

Theorem 1.6.4 (Existence and uniqueness of the product measure). *There is a unique measure $\mu \times \nu$ on $\mathcal{M} \otimes \mathcal{N}$ with $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ for all measurable rectangles. Moreover, for every $E \in \mathcal{M} \otimes \mathcal{N}$ the function $x \mapsto \nu(E_x)$ is $\mathcal{M}$ -measurable and*

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu(x). \quad (1.6.1)$$

Idea of the proof. Formula (1.6.1) is both the construction and the theorem: define the product measure by the right-hand side and check it works. The only issue is measurability of $x \mapsto \nu(E_x)$, which is obvious for rectangles and must be propagated to the generated σ -algebra. One cannot propagate along complements directly — the class of E for which $\nu(E_x)$ is measurable is not obviously closed under complementation when ν is infinite — so one works first with sets of finite measure and uses a monotone class argument. This is the sole reason σ -finiteness is assumed.

Proof. Step 1. The monotone class lemma.

We first prove the tool. A *monotone class* is a family $\mathcal{C} \subseteq \mathcal{P}(X)$ closed under increasing unions and decreasing intersections.

Lemma 1.6.5 (Monotone class lemma). *If $\mathcal{A}$ is an algebra and $\mathcal{C}$ is a monotone class with $\mathcal{A} \subseteq \mathcal{C}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.*

Proof. An arbitrary intersection of monotone classes is a monotone class, so there is a smallest monotone class $\mathcal{M}$ containing $\mathcal{A}$, and $\mathcal{M} \subseteq \mathcal{C}$. It suffices to show $\mathcal{M}$ is a σ -algebra, for then $\sigma(\mathcal{A}) \subseteq \mathcal{M}$.

For $E \in \mathcal{M}$ put

$$\mathcal{M}(E) = \{F \in \mathcal{M} : E \setminus F, F \setminus E, E \cap F \in \mathcal{M}\}.$$

Each $\mathcal{M}(E)$ is a monotone class: if $F_k \uparrow F$ with $F_k \in \mathcal{M}(E)$, then $E \cap F_k \uparrow E \cap F$ and $F_k \setminus E \uparrow F \setminus E$ and $E \setminus F_k \downarrow E \setminus F$, and $\mathcal{M}$ is closed under both kinds of limit; decreasing sequences are symmetric.

Observe the symmetry $F \in \mathcal{M}(E) \iff E \in \mathcal{M}(F)$, immediate from the definition.

Now let $E \in \mathcal{A}$. For $F \in \mathcal{A}$ all three sets $E \setminus F, F \setminus E, E \cap F$ lie in $\mathcal{A} \subseteq \mathcal{M}$, so $\mathcal{A} \subseteq \mathcal{M}(E)$; as $\mathcal{M}(E)$ is a monotone class and $\mathcal{M}$ is the smallest one containing $\mathcal{A}$, we get $\mathcal{M}(E) = \mathcal{M}$. By the symmetry, this says: for every $F \in \mathcal{M}$ and every $E \in \mathcal{A}$, $E \in \mathcal{M}(F)$. Hence $\mathcal{A} \subseteq \mathcal{M}(F)$ for every $F \in \mathcal{M}$, and the same minimality argument gives $\mathcal{M}(F) = \mathcal{M}$ for all $F \in \mathcal{M}$.

So $\mathcal{M}$ is closed under differences and finite intersections, hence under finite unions, and contains X : it is an algebra. An algebra closed under increasing unions is a σ -algebra, since $\bigcup_k E_k = \lim_N \uparrow \bigcup_{k \leq N} E_k$. $\square$

Step 2. *Finite measure case.*

Assume $\mu(X), \nu(Y) < \infty$. Let $\mathcal{C}$ be the family of $E \in \mathcal{M} \otimes \mathcal{N}$ for which $x \mapsto \nu(E_x)$ is measurable. Rectangles lie in $\mathcal{C}$, and so does the algebra $\mathcal{A}$ of finite disjoint unions of rectangles, since slices and ν are finitely additive. If $E_k \uparrow E$ with $E_k \in \mathcal{C}$ then $\nu((E_k)_x) \uparrow \nu(E_x)$ by continuity from below, so $E \in \mathcal{C}$ as a pointwise limit of measurable functions; if $E_k \downarrow E$ the same works using continuity from above, legitimate because $\nu(Y) < \infty$. So $\mathcal{C}$ is a monotone class containing $\mathcal{A}$, hence, by Theorem 1.6.5, contains $\sigma(\mathcal{A}) = \mathcal{M} \otimes \mathcal{N}$.

Step 3. *Definition and countable additivity.*

Define $(\mu \times \nu)(E)$ by (1.6.1). If $E = \bigsqcup_k E_k$ then $E_x = \bigsqcup_k (E_k)_x$, so $\nu(E_x) = \sum_k \nu((E_k)_x)$ by countable additivity of ν , and Theorem 1.5.3 allows term-by-term integration. Hence $\mu \times \nu$ is a measure, and on a rectangle (1.6.1) gives $\int_X \nu(B) \mathbf{1}_A d\mu = \mu(A)\nu(B)$.

Step 4. *σ -finite case and uniqueness.*

Write $X \times Y = \bigcup_k X_k \times Y_k$ with each factor of finite measure and the union increasing; apply Steps 2–3 on each and pass to the limit. Uniqueness follows because two measures agreeing on the algebra $\mathcal{A}$ and σ -finite along it agree on $\sigma(\mathcal{A})$, by the same monotone class argument. $\square$

1.6.2 Tonelli and Fubini

Theorem 1.6.6 (Tonelli). *Let $f : X \times Y \rightarrow [0, \infty]$ be $\mathcal{M} \otimes \mathcal{N}$ -measurable. Then $x \mapsto \int_Y f_x d\nu$ is $\mathcal{M}$ -measurable, $y \mapsto \int_X f^y d\mu$ is $\mathcal{N}$ -measurable, and*

$$\begin{aligned} \int_{X \times Y} f d(\mu \times \nu) &= \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) \\ &= \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y). \end{aligned}$$

Proof. For $f = \mathbf{1}_E$ this is exactly (1.6.1) and its mirror image. By linearity (Theorem 1.5.2) it holds for nonnegative simple functions. For general $f \geq 0$

take simple $\varphi_k \uparrow f$ (Theorem 1.4.6); then $(\varphi_k)_x \uparrow f_x$ for each x , so by Theorem 1.5.1 applied on Y , $\int_Y (\varphi_k)_x d\nu \uparrow \int_Y f_x d\nu$, which is therefore measurable in x as a pointwise limit. Apply Theorem 1.5.1 again on X , and once more on $X \times Y$, and equate. $\square$

Theorem 1.6.7 (Fubini). *Let $f \in L^1(\mu \times \nu)$. Then $f_x \in L^1(\nu)$ for μ -a.e. x , the a.e.-defined function $x \mapsto \int_Y f_x d\nu$ is in $L^1(\mu)$, and the two iterated integrals both equal $\int f d(\mu \times \nu)$.*

Proof. Apply Theorem 1.6.6 to $|f|$: the function $x \mapsto \int_Y |f_x| d\nu$ has finite integral, hence is finite μ -a.e., which is the first claim. Split $f = f^+ - f^-$ (or into four parts if complex), apply Tonelli to each, and subtract; the subtraction is legitimate because all quantities involved are finite a.e. $\square$

Where we are. The practical protocol, used without further comment from here to the end of the series: *run Tonelli on $|f|$ to license Fubini on f* . Tonelli needs no integrability hypothesis but requires $f \geq 0$; Fubini allows signs but requires integrability. Together they say that for nonnegative integrands one may always interchange, and for general integrands one may interchange as soon as one interchange of the absolute value gives a finite answer.

WORKED EXAMPLE 1.6.1— TIER A Two counterexamples

(a) *Integrability is needed.* On $\mathbb{N} \times \mathbb{N}$ with counting measure let $f(m, n) = 1$ if $m = n$, -1 if $m = n + 1$, and 0 otherwise. Summing in n first gives 0 for every m except a boundary term; carefully, $\sum_m \sum_n f = 1$ while $\sum_n \sum_m f = 0$. Here $\sum |f| = \infty$, so Fubini does not apply and Tonelli says nothing since $f \not\geq 0$.

(b) *σ -finiteness is needed.* Let $X = Y = [0, 1]$, μ Lebesgue and ν counting measure, and $f = \mathbf{1}_\Delta$ with Δ the diagonal. Then $\int_X \int_Y f d\nu d\mu = \int_X 1 d\mu = 1$, while $\int_Y \int_X f d\mu d\nu = \int_Y 0 d\nu = 0$. Counting measure on $[0, 1]$ is not σ -finite.

1.6.3 Convolution and approximate identities

Definition 1.6.8. For measurable f, g on $\mathbb{R}^n$, $(f * g)(x) = \int_{\mathbb{R}^n} f(x - y)g(y) dy$ whenever the integral converges absolutely.

Proposition 1.6.9 (Young’s inequality, the case $q = 1$). *If $f \in L^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$, and $g \in L^1(\mathbb{R}^n)$, then $(f * g)(x)$ converges absolutely for a.e. x and $\|f * g\|_p \leq \|f\|_p \|g\|_1$.*

Proof. For $p = 1$: by Tonelli and translation invariance,

$$\iint |f(x-y)| |g(y)| \, dy \, dx = \int |g(y)| \left(\int |f(x-y)| \, dx \right) \, dy = \|f\|_1 \|g\|_1 < \infty,$$

so the inner integral is finite a.e. and Fubini gives the bound. For $p = \infty$ the estimate is immediate. The intermediate cases follow by Minkowski's integral inequality, proved as Theorem 1.8.4 — Minkowski's integral inequality, proved in Chapter 1.8 below by an argument independent of this section in Chapter 1.8; the general Young inequality with $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ requires interpolation and is deferred to Volume 2. $\square$

Proposition 1.6.10 (Approximate identity). *Let $\phi \in L^1(\mathbb{R}^n)$ with $\int \phi = 1$, and set $\phi_t(x) = t^{-n} \phi(x/t)$. Then for $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$,*

$$\|f * \phi_t - f\|_p \rightarrow 0 \quad \text{as } t \rightarrow 0^+.$$

Proof. By Theorem 1.3.10, $\int \phi_t = 1$ for every t , so

$$(f * \phi_t)(x) - f(x) = \int (f(x-y) - f(x)) \phi_t(y) \, dy = \int (f(x-tz) - f(x)) \phi(z) \, dz,$$

substituting $y = tz$. Minkowski's integral inequality (Theorem 1.8.4) gives $\|f * \phi_t - f\|_p \leq \int \|\tau_{tz} f - f\|_p |\phi(z)| \, dz$. The integrand tends to 0 pointwise in z by continuity of translation in L^p (Worked Example 5.14 for $p = 1$; the same density argument works for all $p < \infty$), and is dominated by $2\|f\|_p |\phi| \in L^1$. Apply Theorem 1.5.6. $\square$

1.6.4 Polar coordinates and surface measure

Theorem 1.6.11 (Polar coordinates). *There is a unique Borel measure σ on $\mathbb{S}^{n-1}$ such that for every measurable $f \geq 0$ on $\mathbb{R}^n$,*

$$\int_{\mathbb{R}^n} f(x) \, dx = \int_0^\infty \int_{\mathbb{S}^{n-1}} f(r\omega) r^{n-1} \, d\sigma(\omega) \, dr. \quad (1.6.2)$$

Proof. Define σ on Borel $A \subseteq \mathbb{S}^{n-1}$ by

$$\sigma(A) = n m(\{r\omega : 0 < r < 1, \omega \in A\}),$$

the normalised measure of the cone over A . Countable additivity of σ follows from that of m , since disjoint A give disjoint cones. For $f = \mathbf{1}_E$ with $E = \{r\omega : a < r < b, \omega \in A\}$ the identity (1.6.2) reduces, using the scaling $m(tE) = t^n m(E)$ from Theorem 1.3.10, to $\frac{1}{n}(b^n - a^n)\sigma(A) = \int_a^b r^{n-1} \, dr \sigma(A)$, which is calculus. Such sets generate the Borel σ -algebra of $\mathbb{R}^n \setminus \{0\}$ and form a π -system, so the two Borel measures appearing on the sides of (1.6.2) agree; extend to simple f by linearity and to general $f \geq 0$ by Theorem 1.5.1. $\square$

WORKED EXAMPLE 1.6.2— TIER B The volume of the unit ball, and what happens in high dimensions

Let $\omega_n = m(B(0, 1))$ in $\mathbb{R}^n$. Applying (1.6.2) to $f(x) = e^{-\pi|x|^2}$ and using $\sigma(\mathbb{S}^{n-1}) = n\omega_n$,

$$\int_{\mathbb{R}^n} e^{-\pi|x|^2} dx = n\omega_n \int_0^\infty e^{-\pi r^2} r^{n-1} dr.$$

The left side factors by Tonelli into $\left(\int_{\mathbb{R}} e^{-\pi t^2} dt\right)^n = 1$, using the classical Gaussian identity (Drill 6.2). Substituting $s = \pi r^2$ on the right,

$$\int_0^\infty e^{-\pi r^2} r^{n-1} dr = \frac{1}{2} \pi^{-n/2} \Gamma(n/2),$$

so that

$$\omega_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$

Reading the formula. By Stirling, $\omega_n \approx \left(\frac{2\pi e}{n}\right)^{n/2} (\pi n)^{-1/2}$, which tends to 0 *superexponentially*. The unit ball in $\mathbb{R}^{100}$ has volume about 10^{-40} . Equivalently, almost all of the volume of the cube $[-1, 1]^n$ lies outside the inscribed ball, and indeed within distance $O(1/\sqrt{n})$ of the boundary sphere of a ball of the same volume.

Why this is not a curiosity. Constants of the form C^n or $n^{n/2}$ appear in every covering argument in high dimensions, and they are the reason the Besicovitch covering lemma of Chapter 1.7 carries a dimension-dependent constant that cannot be removed. When Volume 12 proves multilinear restriction estimates in $\mathbb{R}^n$ with constants depending on n , the dependence is of this kind and it is genuinely there.

1.6.5 The tube intersection estimate, and Kakeya in the plane

We now use Fubini for the purpose this series cares about. Everything in this section is a complete proof, and its conclusion — Theorem 1.6.14 — is the two-dimensional case of the Kakeya conjecture.

Definition 1.6.12. For $\delta \in (0, 1)$, a δ -tube in $\mathbb{R}^n$ is a set of the form

$$T = T_\delta^e(a) = \left\{ x \in \mathbb{R}^n : |(x - a) \cdot e| \leq \frac{1}{2}, \quad |(x - a) - ((x - a) \cdot e)e| \leq \frac{\delta}{2} \right\},$$

the closed cylinder of length 1 and radius $\delta/2$ with axis in direction $e \in \mathbb{S}^{n-1}$ through the point a . Thus $m(T) = c_n \delta^{n-1}$.

Lemma 1.6.13 (Tube intersection estimate). *Let T, T' be δ -tubes in $\mathbb{R}^n$ with directions e, e' , and let $\theta = \angle(e, e') \in [0, \pi/2]$ be the angle between the axes. Then*

$$m(T \cap T') \lesssim_n \frac{\delta^n}{\theta + \delta}.$$

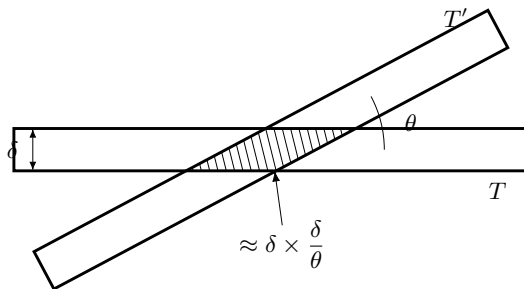


Figure 1.6.1: The tube intersection estimate (Theorem 1.6.13). Two δ -tubes meeting at angle θ cross in a parallelogram of width δ and length $\approx \delta/\theta$: the smaller the angle, the longer they stay together. In $\mathbb{R}^n$ the remaining $n-2$ transverse directions each contribute a further factor δ , giving $m(T \cap T') \lesssim \delta^n/(\theta + \delta)$. The term $+\delta$ records that once $\theta \lesssim \delta$ the tubes are parallel to within their own width and the overlap can be the whole tube.

Idea of the proof. Two long thin cylinders crossing at a large angle meet in a small “parallelepiped” whose dimensions are δ in each of the $n-1$ transverse directions and δ/θ along the direction where they graze past one another: the more nearly parallel they are, the longer the overlap. When $\theta \lesssim \delta$ the tubes are parallel to within their own width and the overlap can be the whole tube, of measure δ^{n-1} , which is what the $+\delta$ in the denominator records. The proof is Fubini: slice by hyperplanes perpendicular to the plane spanned by e and e' , and estimate each slice.

Proof. By rotating (which preserves m , by Theorem 1.3.10) and translating, we may assume $e = e_1$ and that e' lies in the (x_1, x_2) -plane, so $e' = (\cos \theta, \sin \theta, 0, \dots, 0)$.

Write $x = (x_1, x_2, x'')$ with $x'' \in \mathbb{R}^{n-2}$. Both tubes constrain x'' : a point of T has $|x'' - a''| \leq \delta/2$, and similarly for T' . So the set of admissible x'' has measure $\lesssim_n \delta^{n-2}$, and by Tonelli

$$m(T \cap T') \lesssim_n \delta^{n-2} \cdot \sup_{x''} m_2((T \cap T')_{x''}), \quad (1.6.3)$$

where m_2 is planar measure and the slice is taken in the (x_1, x_2) -variables.

Each slice is contained in the intersection of two planar strips: S of width δ with direction e_1 , and S' of width δ with direction e' . It therefore suffices to bound $m_2(S \cap S')$ where

$$S = \{|x_2 - c| \leq \tfrac{\delta}{2}\}, \quad S' = \{|-x_1 \sin \theta + x_2 \cos \theta - c'| \leq \tfrac{\delta}{2}\},$$

intersected with the bounded region $|x_1| \leq 1$ coming from the tube length.

If $\theta \geq \delta$: for each fixed x_2 in the δ -range allowed by S , the constraint from S' reads $|x_1 - \gamma(x_2)| \leq \delta/(2 \sin \theta)$ for some γ , an interval of length $\delta/\sin \theta \lesssim \delta/\theta$ since $\theta \leq \pi/2$. By Tonelli in the plane, $m_2(S \cap S') \lesssim \delta \cdot \frac{\delta}{\theta} = \delta^2/\theta$.

If $\theta < \delta$: simply use $m_2(S \cap S') \leq m_2(S \cap \{|x_1| \leq 1\}) \lesssim \delta$, and note $\delta \approx \delta^2/\delta$.

In both cases $m_2(S \cap S') \lesssim \delta^2/(\theta + \delta)$; substituting into (1.6.3) gives the lemma. $\square$

WORKED EXAMPLE 1.6.3— TIER C: RESEARCH TASTE **Córdoba’s argument: the Kakeya conjecture in the plane**

We now prove a sharp lower bound on the measure of a union of tubes pointing in separated directions. It is due to Córdoba, it uses nothing beyond Theorem 1.6.13 and the Cauchy–Schwarz inequality, and it settles the planar case of a conjecture that took until 2025 to settle in $\mathbb{R}^3$.

Theorem 1.6.14. *Let $\delta \in (0, \frac{1}{2})$ and let $\mathbb{T}$ be a collection of δ -tubes in $\mathbb{R}^2$ whose directions are δ -separated, with $\#\mathbb{T} \approx \delta^{-1}$. Then*

$$\left| \bigcup_{T \in \mathbb{T}} T \right| \gtrsim \frac{1}{\log(1/\delta)}.$$

Proof. Step 1: the L^2 computation. Set $F = \sum_{T \in \mathbb{T}} \mathbf{1}_T$, the multiplicity function. Expanding the square and using Tonelli,

$$\|F\|_2^2 = \int_{\mathbb{R}^2} \left(\sum_T \mathbf{1}_T \right)^2 = \sum_{T, T' \in \mathbb{T}} |T \cap T'|.$$

Fix T with direction e . Because the directions are δ -separated, for each dyadic value 2^{-m} with $\delta \leq 2^{-m} \leq 1$ the number of T' whose direction makes an angle $\approx 2^{-m}$ with e is at most $\lesssim 2^{-m}/\delta$. Applying Theorem 1.6.13 with $n = 2$,

$$\sum_{T' \in \mathbb{T}} |T \cap T'| \lesssim \sum_{\substack{m \geq 0 \\ 2^{-m} \geq \delta}} \frac{2^{-m}}{\delta} \cdot \frac{\delta^2}{2^{-m}} = \sum_{\substack{m \geq 0 \\ 2^{-m} \geq \delta}} \delta \lesssim \delta \log(1/\delta),$$

because there are $\approx \log(1/\delta)$ admissible values of m . The contribution from T' nearly parallel to T (angle $\lesssim \delta$) is $\lesssim \delta$ as well, since there are $O(1)$ such tubes each contributing $\lesssim \delta$. Summing over the $\approx \delta^{-1}$ choices of T ,

$$\|F\|_2^2 \lesssim \delta^{-1} \cdot \delta \log(1/\delta) = \log(1/\delta). \quad (1.6.4)$$

Step 2: Cauchy–Schwarz. Let $U = \bigcup_T T$, so F is supported in U . Then

$$\sum_T |T| = \int_U F \leq |U|^{1/2} \|F\|_2,$$

and since each $|T| \approx \delta$ and there are $\approx \delta^{-1}$ tubes, the left side is ≈ 1 . Squaring and inserting (1.6.4),

$$1 \lesssim |U| \cdot \log(1/\delta),$$

which is the theorem. $\square$

Why the logarithm is not an artefact. Theorem 1.3.12 produced a configuration of tubes in δ -separated directions whose union has measure tending to 0. Theorem 1.6.14 says it cannot go to zero faster than $1/\log(1/\delta)$, and a refinement of the Perron construction shows this rate is attained. So the two theorems of this volume meet exactly: the upper and lower bounds match, and the planar problem is closed.

The shape of the argument. Every subsequent advance on Kakeya, up to and including Wang–Zahl, has the same two-step skeleton: (i) bound the L^2 or L^p norm of the multiplicity function by summing a pairwise intersection estimate over the configuration; (ii) convert that into a lower bound for $|U|$ by an inequality of Cauchy–Schwarz type. What changes is that in $\mathbb{R}^n$, $n \geq 3$, the pairwise estimate of Theorem 1.6.13 is *insufficient*: a configuration can have every pair of tubes nearly transverse while the whole family conspires to concentrate near a low-dimensional surface. Wolff’s hairbrush argument replaces pairwise information with information about triples through a common tube; the polynomial method replaces it with algebraic information about the whole family at once. Theorem 1.6.13 is where all of them start. $\square$

What this does and does not prove. Theorem 1.6.14 bounds the measure of a union of finitely many tubes. To deduce that a planar Besicovitch set has Hausdorff dimension 2 one applies the theorem at every scale $\delta = 2^{-k}$ to the δ -neighbourhood of the set and converts the resulting bound into a dimension statement — a routine argument, but one that needs the Hausdorff machinery of Volume 7. The Kakeya *maximal* function conjecture in the plane, which is the sharpest form, follows from the same L^2 computation by duality and is proved in Chapter 1.7, Worked Example 7.7.

Remark 1.6.15 (A forward reference, and why it is not circular). Two proofs in this chapter invoke Minkowski’s integral inequality, Theorem 1.8.4, which is stated and proved in Chapter 1.8. This is the only place in Volume 1 where a chapter uses a result proved later, and it is worth saying explicitly why no circularity arises.

L^p spaces cannot be introduced before Chapter 1.8, since their completeness rests on the convergence theorems of Chapter 1.5; so the inequality cannot be stated here. But its proof — by duality and Hölder — uses only Chapter 1.4 and Chapter 1.5, and in particular does not use Fubini or anything else from the present chapter. The reader may therefore turn to Theorem 1.8.4 at once without a gap.

The alternative — proving the special case needed here by hand — would duplicate the argument twice, and the duplication seemed the worse defect.

WORKED EXAMPLE 1.6.4— TIER B Fubini and Tonelli: which to cite

The two theorems are constantly conflated and the difference matters.

Tonelli. $f \geq 0$ measurable. The iterated integrals and the double integral all agree, possibly $= +\infty$. *No integrability hypothesis.*

Fubini. $f \in L^1$ of the product. Same conclusion, and additionally the slices are integrable almost everywhere.

The usual pattern. Apply Tonelli to $|f|$ to *prove* $f \in L^1$, then apply Fubini to f to exchange the order. Citing Fubini alone leaves the integrability unestablished, and that is the commonest gap.

Where it bites in this series. Volume 12’s Theorem 12.8.2 exchanges an integral over the sphere with a double integral against $\mu \times \mu$; the justification is that the integrand is bounded and the measures finite — a Tonelli argument, stated there explicitly for that reason.

The counterexample to keep. On $[0, 1]^2$ with $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$, the two iterated integrals are $\frac{\pi}{4}$ and $-\frac{\pi}{4}$. Tonelli does not apply (f changes sign) and Fubini does not ($f \notin L^1$), and the conclusion genuinely fails.

Drills

Drill 1.6.1. Prove that $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R}^2)$, using the set $\{0\} \times A$ of the Pitfall box.

Drill 1.6.2. Prove $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$ by writing the square of the integral as a double integral over $\mathbb{R}^2$ (Tonelli) and passing to polar coordinates (Theorem 1.6.11).

Drill 1.6.3. Verify the claim in Worked Example 1.6.3 that at most $\lesssim 2^{-m}/\delta$ tubes of a δ -separated family make angle $\approx 2^{-m}$ with a fixed direction. Then redo the computation of (1.6.4) in $\mathbb{R}^n$ and show it yields $\|F\|_2^2 \lesssim \delta^{-(n-2)} \log(1/\delta)$ for $\#\mathbb{T} \approx \delta^{-(n-1)}$, hence $|U| \gtrsim 1/\log(1/\delta)$ only when $n = 2$. Identify precisely which step degrades.

Notes

Fubini's theorem is from 1907 and Tonelli's from 1909. The monotone class route taken in Theorem 1.6.4 is standard.

Theorem 1.6.13 is folklore and appears in essentially every paper on the Kakeya problem; the reader will find it quoted, usually without proof, as “the standard tube intersection estimate”. Theorem 1.6.14 is Córdoba's, from 1977. Drill 6.3 is worth doing carefully: the failure it exposes in dimensions $n \geq 3$ is not a defect of the argument but the entire difficulty of the subject, and readers who have located it themselves will find Volumes 9 and 10 much easier going.

Chapter 1.7

Differentiation: Covering Lemmas and the Maximal Function

1.7.1 Why this is not a service chapter

Most measure theory courses treat differentiation theory as the last technical duty before the interesting material begins. For this series it is the opposite: this chapter is where measure theory stops being infrastructure and becomes harmonic analysis, and three things appear here for the first time that will never leave.

The first is an *operator*. Until now we have integrated functions; the Hardy–Littlewood maximal operator $\mathcal{M}$ takes a function and returns another function, and the entire subject from here on consists of proving inequalities about such operators.

The second is a *weak-type inequality*. Theorem 1.7.3 does not bound $\|\mathcal{M}f\|_1$ — no such bound is true — but bounds the measure of the set where $\mathcal{M}f$ is large. Covering arguments produce weak bounds and only weak bounds, which is why weak L^p is a permanent inhabitant of the subject rather than a technicality.

The third is a *covering argument*, and this is the one that matters most. The Vitali lemma controls how much a family of balls can overlap. Balls are easy. Chapter 1.6 already showed that tubes are not, and Worked Example 1.7.2 below will show that the entire machinery of this chapter fails — provably, not merely as a gap in our ingenuity — when balls are replaced by tubes. That failure is the Kakeya problem.

1.7.2 The Vitali covering lemma

Lemma 1.7.1 (Vitali, finite version). *Let $B_1, \dots, B_M$ be a finite collection of open balls in $\mathbb{R}^n$. There is a pairwise disjoint subcollection $B_{i_1}, \dots, B_{i_k}$ such that*

$$\left| \bigcup_{j=1}^M B_j \right| \leq 3^n \sum_{l=1}^k |B_{i_l}|.$$

Idea of the proof. Select greedily, largest first, always discarding every ball that meets something already chosen. A discarded ball B meets a selected ball B' that is at least as large, and elementary geometry then puts B inside the concentric triple $3B'$. So the selected balls, tripled, cover everything, and tripling costs a factor 3^n in measure by Theorem 1.3.10.

Proof. Order the balls by decreasing radius, $r_1 \geq r_2 \geq \dots \geq r_M$. Let $i_1 = 1$. Having chosen $i_1 < \dots < i_l$, let i_{l+1} be the least index greater than i_l whose ball is disjoint from $B_{i_1}, \dots, B_{i_l}$; stop when no such index exists. The selected balls are pairwise disjoint by construction.

Let B_j be any ball of the family. If j was not selected, then B_j meets some selected B_{i_l} with $i_l < j$, hence with $r_{i_l} \geq r_j$. Pick $z \in B_j \cap B_{i_l}$ and let $x \in B_j$. Writing c_j, c_{i_l} for the centres,

$$|x - c_{i_l}| \leq |x - c_j| + |c_j - z| + |z - c_{i_l}| < r_j + r_j + r_{i_l} \leq 3r_{i_l},$$

so $B_j \subseteq 3B_{i_l}$, the concentric ball of triple the radius. Hence $\bigcup_j B_j \subseteq \bigcup_l 3B_{i_l}$, and by subadditivity together with $|3B| = 3^n|B|$,

$$\left| \bigcup_j B_j \right| \leq \sum_l |3B_{i_l}| = 3^n \sum_l |B_{i_l}|. \quad \square$$

Pitfall: the constant 3^n grows with dimension, and must

It is natural to hope for a dimension-free constant. There is none from this argument, and the growth is not an artefact of the tripling: as Worked Example 1.6.2 showed, volumes of balls collapse superexponentially in n , and covering arguments in high dimensions genuinely cost exponentially. This is worth registering now, because the same phenomenon — estimates whose constants degrade with dimension — recurs in the multilinear Kakeya and restriction theorems of Volume 12, where it is a substantive obstacle rather than a nuisance.

1.7.3 The Hardy–Littlewood maximal function

Definition 1.7.2. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ define

$$\mathcal{M}f(x) = \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| \, dy.$$

$\mathcal{M}f$ is measurable: for each r the average is a continuous function of x by Theorem 1.5.6, so $\{\mathcal{M}f > \lambda\}$ is a union over rational r of open sets, hence open.

Theorem 1.7.3 (Hardy–Littlewood maximal inequality). *For every $f \in L^1(\mathbb{R}^n)$ and every $\lambda > 0$,*

$$m(\{x : \mathcal{M}f(x) > \lambda\}) \leq \frac{3^n}{\lambda} \|f\|_1. \quad (1.7.1)$$

Moreover $\|\mathcal{M}f\|_\infty \leq \|f\|_\infty$, and for $1 < p \leq \infty$

$$\|\mathcal{M}f\|_p \leq C_{n,p} \|f\|_p. \quad (1.7.2)$$

Proof. Step 1. The weak bound.

Fix $\lambda > 0$ and let $E_\lambda = \{\mathcal{M}f > \lambda\}$, an open set. Let $K \subseteq E_\lambda$ be compact; by inner regularity (Theorem 1.3.6(iv)) it suffices to bound $m(K)$.

For each $x \in K$ there is, by definition of the supremum, a radius $r_x > 0$ with

$$\int_{B(x, r_x)} |f| > \lambda |B(x, r_x)|. \quad (1.7.3)$$

These balls cover K , so finitely many $B_1, \dots, B_M$ do. Apply Theorem 1.7.1 to extract a disjoint subfamily $B_{i_1}, \dots, B_{i_k}$ with $m(\bigcup_j B_j) \leq 3^n \sum_l |B_{i_l}|$. Then, using disjointness for the last step,

$$\begin{aligned} m(K) &\leq \left| \bigcup_j B_j \right| \\ &\leq 3^n \sum_l |B_{i_l}| \stackrel{(1.7.3)}{<} \frac{3^n}{\lambda} \sum_l \int_{B_{i_l}} |f| \\ &\leq \frac{3^n}{\lambda} \int_{\mathbb{R}^n} |f|. \end{aligned}$$

Step 2. *The L^∞ bound.*

Every average of $|f|$ is at most $\|f\|_\infty$, so $\mathcal{M}f \leq \|f\|_\infty$ pointwise.

Step 3. *The L^p bound by splitting at height λ .*

Let $1 < p < \infty$ and $f \in L^p$. Fix $\lambda > 0$ and decompose

$$f = f\mathbf{1}_{\{|f| > \lambda/2\}} + f\mathbf{1}_{\{|f| \leq \lambda/2\}} =: g_\lambda + b_\lambda.$$

Then $\|b_\lambda\|_\infty \leq \lambda/2$, so $\mathcal{M}b_\lambda \leq \lambda/2$ by Step 2, and since $\mathcal{M}$ is sublinear, $\mathcal{M}f \leq \mathcal{M}g_\lambda + \lambda/2$. Consequently

$$\{\mathcal{M}f > \lambda\} \subseteq \{\mathcal{M}g_\lambda > \lambda/2\},$$

and Step 1 applied to $g_\lambda \in L^1$ (it is integrable because $|g_\lambda| \leq |f|(2|f|/\lambda)^{p-1}$) gives

$$\lambda \mathcal{M}f(\lambda) \leq \frac{2 \cdot 3^n}{\lambda} \int_{\{|f| > \lambda/2\}} |f| \, dm. \quad (1.7.4)$$

Now integrate using the layer-cake formula (1.5.2) and Tonelli:

$$\begin{aligned} \|\mathcal{M}f\|_p^p &= p \int_0^\infty \lambda^{p-1} \lambda \mathcal{M}f(\lambda) \, d\lambda \leq 2 \cdot 3^n p \int_0^\infty \lambda^{p-2} \int_{\{|f| > \lambda/2\}} |f(x)| \, dx \, d\lambda \\ &= 2 \cdot 3^n p \int_{\mathbb{R}^n} |f(x)| \left(\int_0^{2|f(x)|} \lambda^{p-2} \, d\lambda \right) dx = \frac{2^p 3^n p}{p-1} \int_{\mathbb{R}^n} |f(x)|^p \, dx, \end{aligned}$$

the inner integral converging precisely because $p > 1$. $\square$

Where we are. Note what Step 3 is: a hand-rolled proof of the Marcinkiewicz interpolation theorem in the only case we currently need. The general theorem — weak $(1, 1)$ plus (∞, ∞) implies strong (p, p) for $1 < p < \infty$ — is proved in Volume 2, and the proof is the argument above with $\mathcal{M}$ replaced by an arbitrary sublinear operator. The constant blows up as $p \rightarrow 1$, and it must: $\mathcal{M}f$ is never integrable unless $f = 0$, by Worked Example 7.5.

WORKED EXAMPLE 1.7.1— TIER A $\mathcal{M}f$ is never integrable

Let $f \in L^1(\mathbb{R}^n)$ be not identically 0, so there is a ball $B(0, R)$ with $\int_{B(0, R)} |f| = c > 0$. For $|x| > R$, taking $r = 2|x|$ gives $B(0, R) \subseteq B(x, r)$, hence

$$\mathcal{M}f(x) \geq \frac{1}{|B(x, 2|x|)|} \int_{B(0, R)} |f| = \frac{c}{\omega_n 2^n |x|^n}.$$

Since $|x|^{-n}$ is not integrable at infinity — by polar coordinates (1.6.2), $\int_{|x| > R} |x|^{-n} \, dx = \sigma(\mathbb{S}^{n-1}) \int_R^\infty r^{-1} \, dr = \infty$ — we get $\mathcal{M}f \notin L^1$.

So the weak bound (1.7.1) is the strongest statement available at $p = 1$, not a weaker substitute for something better. This is the standard situation throughout the subject.

1.7.4 The Lebesgue differentiation theorem

Theorem 1.7.4. *Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. Then for a.e. x ,*

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) \, dy = f(x),$$

and indeed $\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| \, dy = 0$ for a.e. x (such x are called Lebesgue points).

Idea of the proof. The statement is obvious for continuous f and is a density statement in general. The architecture is worth naming, because it recurs so often:

$$\begin{aligned} & (\text{density of a good class}) + (\text{a weak-type maximal inequality}) \\ & \implies (\text{a.e. convergence}). \end{aligned}$$

One proves the result on a dense subclass where it is easy, controls the error by a maximal operator, and uses the weak bound to show the error set is null. Nothing about balls or about differentiation is essential; the same template proves Carleson's theorem on a.e. convergence of Fourier series, and in Volume 13 it proves the a.e. convergence of solutions to the Schrödinger equation to their initial data — the Carleson problem — where the relevant maximal operator is so difficult that the sharp result was not obtained until 2017.

Proof. It suffices to treat $f \in L^1(\mathbb{R}^n)$ supported in a large ball, since the statement is local. Define the *oscillation operator*

$$\Omega f(x) = \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| \, dy.$$

We must show $\Omega f = 0$ a.e.

Step 1. *The good class.*

If g is continuous then $\Omega g = 0$ everywhere: given $\varepsilon > 0$, continuity at x gives $|g(y) - g(x)| < \varepsilon$ for $|y - x| < r$ small, so every average is at most ε .

Step 2. *Sublinearity and the maximal bound.*

For any f, g , the triangle inequality gives

$$\Omega f \leq \Omega(f - g) + \Omega g = \Omega(f - g),$$

when g is continuous, and

$$\Omega(f - g)(x) \leq \mathcal{M}(f - g)(x) + |(f - g)(x)|.$$

Step 3. *The error set is null.*

Fix $\lambda > 0$. By density of $C_c(\mathbb{R}^n)$ in L^1 (Chapter 1.8), choose continuous g with $\|f - g\|_1 < \eta$. Then

$$\{\Omega f > \lambda\} \subseteq \left\{ \mathcal{M}(f - g) > \frac{\lambda}{2} \right\} \cup \left\{ |f - g| > \frac{\lambda}{2} \right\},$$

and by (1.7.1) and Chebyshev (1.5.3),

$$m(\{\Omega f > \lambda\}) \leq \frac{2 \cdot 3^n}{\lambda} \eta + \frac{2}{\lambda} \eta.$$

As $\eta > 0$ was arbitrary, $m(\{\Omega f > \lambda\}) = 0$; taking $\lambda = 1/k$ and a countable union, $\Omega f = 0$ a.e. $\square$

Corollary 1.7.5 (Lebesgue density). *If E is measurable then for a.e. $x \in E$, $\lim_{r \rightarrow 0} \frac{|E \cap B(x, r)|}{|B(x, r)|} = 1$, and for a.e. $x \notin E$ the limit is 0.*

Proof. Apply Theorem 1.7.4 to $f = \mathbf{1}_E$. $\square$

1.7.5 The Vitali covering theorem

Theorem 1.7.1 selects finitely many disjoint balls that capture a definite fraction of a union. For the fundamental theorem of calculus we need more: a disjoint family that captures a set *up to measure zero*. This is available provided the given balls come in arbitrarily small sizes at every point.

Definition 1.7.6. A family $\mathcal{V}$ of closed balls is a *fine cover* of $E \subseteq \mathbb{R}^n$ if for every $x \in E$ and every $\varepsilon > 0$ there is $B \in \mathcal{V}$ with $x \in B$ and $0 < \text{diam } B < \varepsilon$.

Theorem 1.7.7 (Vitali covering theorem). *Let $E \subseteq \mathbb{R}^n$ with $m^*(E) < \infty$ and let $\mathcal{V}$ be a fine cover of E by closed balls. Then there is a countable pairwise disjoint subfamily $\{B_k\} \subseteq \mathcal{V}$ with*

$$m^*\left(E \setminus \bigcup_k B_k\right) = 0.$$

Idea of the proof. Select greedily, at each stage taking a ball disjoint from those already chosen and of radius more than half the largest available. Two facts then do everything. The chosen balls are disjoint inside a set of finite measure, so their radii tend to 0; and any point left uncovered after N stages is caught by a ball which must meet some *later* chosen ball of comparable size, hence lies in its five-fold dilate. The uncovered set is therefore inside $\bigcup_{k > N} 5B_k$, whose measure is a tail of a convergent series.

Proof. Step 1. Localise.

Choose an open $U \supseteq E$ with $m(U) < \infty$; this is possible because $m^*(E) < \infty$, by the definition of outer measure as an infimum over covers by open boxes. Discard from $\mathcal{V}$ every ball not contained in U . The remaining family is still a fine cover of E : for $x \in E \subseteq U$ with U open, all sufficiently small balls containing x lie in U .

Step 2. Greedy selection.

Suppose disjoint $B_1, \dots, B_k$ have been chosen, and let

$$d_k = \sup \{ r(B) : B \in \mathcal{V}, B \cap B_i = \emptyset \text{ for all } i \leq k \}.$$

If no such B exists, then $E \subseteq \bigcup_{i \leq k} B_i$ and we are done: a point $x \in E$ outside the closed set $\bigcup_{i \leq k} B_i$ would admit, by fineness, a small ball disjoint from it. Otherwise $0 < d_k < \infty$ (finite because all balls lie in the bounded-measure set U), and we choose $B_{k+1} \in \mathcal{V}$ disjoint from $B_1, \dots, B_k$ with

$$r(B_{k+1}) > \frac{1}{2}d_k. \quad (1.7.5)$$

Step 3. Radii tend to zero.

The B_k are pairwise disjoint and contained in U , so $\sum_k m(B_k) \leq m(U) < \infty$. Hence $m(B_k) \rightarrow 0$ and so $r_k \rightarrow 0$, and by (1.7.5) also $d_k \rightarrow 0$.

Step 4. The key inclusion.

We claim that for every N ,

$$E \setminus \bigcup_{k \leq N} B_k \subseteq \bigcup_{k > N} 5B_k, \quad (1.7.6)$$

where $5B$ is the concentric ball of five times the radius.

Let $x \in E$ with $x \notin \bigcup_{k \leq N} B_k$. That union is closed, so by fineness there is $B \in \mathcal{V}$ with $x \in B$ and B disjoint from $B_1, \dots, B_N$. Now B must meet some chosen ball: if B were disjoint from every B_k , then $r(B) \leq d_k$ for all k , forcing $r(B) = 0$ by Step 3, which is excluded. Let k be the least index with $B \cap B_k \neq \emptyset$; then $k > N$, and B is disjoint from $B_1, \dots, B_{k-1}$, so $r(B) \leq d_{k-1} < 2r_k$ by (1.7.5).

Pick $y \in B \cap B_k$ and let c_k be the centre of B_k . Then

$$|x - c_k| \leq |x - y| + |y - c_k| \leq 2r(B) + r_k < 4r_k + r_k = 5r_k,$$

so $x \in 5B_k$, proving (1.7.6).

Step 5. Conclude.

By (1.7.6), subadditivity and $m(5B) = 5m(B)$,

$$m^*\left(E \setminus \bigcup_k B_k\right) \leq m^*\left(E \setminus \bigcup_{k \leq N} B_k\right) \leq 5^n \sum_{k > N} m(B_k),$$

a tail of the convergent series of Step 3, hence $\rightarrow 0$ as $N \rightarrow \infty$. $\square$

Looking ahead: covering with bounded overlap, and why tubes admit none

There is a third covering theorem, Besicovitch's, which for balls centred at the points of a set produces subfamilies covering it with *overlap bounded by a dimensional constant* C_n — stronger than Vitali's dilation-by-3 and essential for differentiating a general Radon measure μ , where $\mu(3B)$ need not be comparable to $\mu(B)$. It is not needed anywhere in this volume, and is proved in Volume 7 where Radon measures are introduced.

What is worth recording here is that no analogue can exist for tubes, and the reason is Theorem 1.3.12. In the Perron configuration, $\approx \delta^{-1}$ δ -tubes with δ -separated directions are packed into a set of measure $o(1)$; since the tubes have total measure $\approx \delta^{-1} \cdot \delta = 1$, the average multiplicity on the union is $\approx 1/o(1) \rightarrow \infty$. So no bound C independent of δ can hold, for any covering scheme whatsoever.

Every theorem in this chapter rests on a covering lemma for balls. That is why none of it transfers to the Kakeya problem, and Worked Example 1.7.2 states the same obstruction analytically.

1.7.6 Differentiation of measures

Theorem 1.7.8 (Lebesgue decomposition and Radon–Nikodym). *Let μ, ν be σ -finite measures on $(X, \mathcal{M})$. Then ν decomposes uniquely as $\nu = \nu_{ac} + \nu_s$ with $\nu_{ac} \ll \mu$ and $\nu_s \perp \mu$, and there is a μ -a.e. unique measurable $h \geq 0$ with $\nu_{ac}(E) = \int_E h \, d\mu$ for all $E \in \mathcal{M}$.*

Proof in the finite case. First suppose that μ and ν are finite. The σ -finite case follows by exhausting X with sets of finite measure and summing the resulting pieces. Let

$$\mathcal{F} = \left\{ h : X \rightarrow [0, \infty] \text{ measurable, } \int_E h \, d\mu \leq \nu(E) \, \forall E \in \mathcal{M} \right\},$$

which contains 0, and let $s = \sup_{h \in \mathcal{F}} \int_X h \, d\mu \leq \nu(X) < \infty$. If $h_1, h_2 \in \mathcal{F}$ then $\max(h_1, h_2) \in \mathcal{F}$: split E into the part where $h_1 > h_2$ and its complement, and apply the defining inequality on each. Take $h_k \in \mathcal{F}$ with $\int h_k \rightarrow s$ and replace h_k by $\max(h_1, \dots, h_k)$, so that h_k increases; let $h = \lim_k h_k$. By Theorem 1.5.1, $h \in \mathcal{F}$ and $\int h \, d\mu = s$.

Define $\nu_{ac}(E) = \int_E h \, d\mu$, a measure by Theorem 1.5.4, and $\nu_s = \nu - \nu_{ac} \geq 0$. It remains to show $\nu_s \perp \mu$. If not, then for some $\varepsilon > 0$ the signed measure $\nu_s - \varepsilon\mu$ has a positive set P with $\mu(P) > 0$ (Hahn decomposition, Appendix A). But then $h + \varepsilon \mathbf{1}_P \in \mathcal{F}$, since for any E ,

$$\int_E (h + \varepsilon \mathbf{1}_P) \, d\mu = \nu_{ac}(E) + \varepsilon \mu(E \cap P) \leq \nu_{ac}(E) + \nu_s(E \cap P) \leq \nu(E),$$

and its integral exceeds s , contradicting maximality. Uniqueness: if $\int_E h = \int_E h'$ for all E , take $E = \{h > h'\}$ to get $h \leq h'$ a.e., and symmetrically. $\square$

1.7.7 The fundamental theorem of calculus

Definition 1.7.9. $F : [a, b] \rightarrow \mathbb{R}$ is *absolutely continuous* (AC) if for every $\varepsilon > 0$ there is $\delta > 0$ such that for every finite collection of pairwise disjoint subintervals (a_i, b_i) with $\sum_i (b_i - a_i) < \delta$ one has $\sum_i |F(b_i) - F(a_i)| < \varepsilon$.

Proposition 1.7.10 (Indefinite integrals are AC). *If $f \in L^1([a, b])$ then $F(x) = \int_a^x f$ is absolutely continuous, and $F' = f$ a.e.*

Proof. Absolute continuity is the statement that $\int_A |f| < \varepsilon$ whenever $m(A) < \delta$, which follows from Theorem 1.5.6 applied to $|f| \mathbf{1}_{\{|f| > k\}}$: choose k with $\int |f| \mathbf{1}_{\{|f| > k\}} < \varepsilon/2$ and then $\delta = \varepsilon/(2k)$. For $F' = f$ a.e., note that $(F(x+r) - F(x))/r$ is the average of f over $[x, x+r]$, and one-sided averages are dominated by the two-sided averages of Theorem 1.7.4; at a Lebesgue point of f both tend to $f(x)$. $\square$

The converse is the substantial half, and the Vitali covering theorem is what makes it available.

Lemma 1.7.11 (Monotone functions are differentiable a.e.). *An increasing $G : [a, b] \rightarrow \mathbb{R}$ is differentiable at a.e. point.*

Proof. For $x \in (a, b)$ let $\overline{DG}(x) = \limsup_{h \rightarrow 0} \frac{G(x+h) - G(x)}{h}$ and $\underline{DG}(x)$ the corresponding lim inf, both in $[0, \infty]$. It suffices to show that $\overline{DG} = \underline{DG} < \infty$ a.e.

Fix rationals $0 \leq t < s$ and put $E = \{x \in (a, b) : \underline{DG}(x) < t < s < \overline{DG}(x)\}$, with $\alpha = m^*(E)$. Fix $\varepsilon > 0$ and choose an open $U \supseteq E$ with $m(U) < \alpha + \varepsilon$.

Every $x \in E$ has arbitrarily short closed intervals $[x, x+h] \subseteq U$ (or $[x-h, x]$) with G -increment less than th ; these form a fine cover of E , so Theorem 1.7.7 gives disjoint intervals $I_1, I_2, \dots \subseteq U$ with $m^*(E \setminus \bigcup_k I_k) = 0$. Choose N with $m^*(E \cap \bigcup_{k \leq N} I_k) > \alpha - \varepsilon$, and note

$$\sum_{k \leq N} (G(\text{right end of } I_k) - G(\text{left end})) < t \sum_{k \leq N} m(I_k) \leq t m(U) < t(\alpha + \varepsilon). \quad (1.7.7)$$

Now work inside $V = \bigcup_{k \leq N} \text{int } I_k$. Every $x \in E \cap V$ has arbitrarily short intervals $J \subseteq V$ containing it with G -increment greater than $s m(J)$; another application of Theorem 1.7.7 yields disjoint $J_1, \dots, J_M \subseteq V$ with $\sum_{m \leq M} m(J_m) > \alpha - 2\varepsilon$ and

$$\sum_{m \leq M} (G(\text{right end of } J_m) - G(\text{left end})) > s(\alpha - 2\varepsilon). \quad (1.7.8)$$

Since the J_m are disjoint and each lies inside some I_k , and G is increasing, the left side of (1.7.8) is at most the left side of (1.7.7). Comparing, $s(\alpha - 2\varepsilon) < t(\alpha + \varepsilon)$; letting $\varepsilon \downarrow 0$ gives $s\alpha \leq t\alpha$, and since $s > t$ we conclude $\alpha = 0$.

Taking the union over the countably many rational pairs $t < s$ shows $\overline{DG} = \underline{DG}$ a.e. Finiteness a.e. follows from the same computation with s replaced by an arbitrarily large constant: if $\overline{DG} = \infty$ on a set of measure $\alpha > 0$, then (1.7.8) with s arbitrary forces $s\alpha \leq G(b) - G(a)$ for all s , so $\alpha = 0$. $\square$

Theorem 1.7.12 (Fundamental theorem of calculus). *$F : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if and only if F' exists a.e., lies in $L^1([a, b])$, and*

$$F(x) = F(a) + \int_a^x F'(t) \, dt \quad \text{for all } x \in [a, b].$$

Proof. ($\Leftarrow$) is Theorem 1.7.10. For ($\Rightarrow$), let F be AC.

Step 1. *F has bounded variation.*

Take δ corresponding to $\varepsilon = 1$ in the definition, and partition $[a, b]$ into $\lceil (b-a)/\delta \rceil$ intervals of length $< \delta$. On each, every finite sum $\sum |F(b_i) - F(a_i)|$ over disjoint subintervals is at most 1, so the variation of F on each is at most 1, and the total variation is finite.

Step 2. *F' exists a.e. and is integrable.*

By Jordan's decomposition, a function of bounded variation is a difference of two increasing functions, $F = G_1 - G_2$; by Theorem 1.7.11 each is differentiable a.e., hence so is F . For integrability, apply Fatou (Theorem 1.5.5) to the nonnegative difference quotients of G_i :

$$\int_a^b G'_i \leq \liminf_{h \rightarrow 0} \int_a^b \frac{G_i(x+h) - G_i(x)}{h} \, dx \leq G_i(b) - G_i(a) < \infty,$$

the last inequality because the integral of the difference quotient telescopes to an average of G_i near the endpoints. Hence $F' \in L^1$.

Step 3. *An AC function with vanishing derivative is constant.*

Suppose H is AC on $[a, b]$ with $H' = 0$ a.e., and let $\varepsilon > 0$ with δ as in the definition of absolute continuity. Let $E = \{x : H'(x) = 0\}$, so $m(E) = b-a$. For $x \in E$ there are arbitrarily short intervals $[x, x+h]$ with $|H(x+h) - H(x)| \leq \varepsilon h$; these form a fine cover, so Theorem 1.7.7 produces disjoint $I_1, I_2, \dots$ with $m([a, b] \setminus \bigcup_k I_k) = 0$. Choose N with $\sum_{k > N} m(I_k) < \delta$. The complement of $\bigcup_{k \leq N} I_k$ in $[a, b]$ is a finite union of intervals of total length $< \delta$, so absolute continuity bounds the sum of $|H|$ -increments over them by ε ; and the increments over $I_1, \dots, I_N$ total at most $\varepsilon(b-a)$. Hence $|H(b) - H(a)| \leq \varepsilon + \varepsilon(b-a)$, and

letting $\varepsilon \downarrow 0$ gives $H(b) = H(a)$. The same applies on every subinterval, so H is constant.

Step 4. Conclude.

Put $H(x) = F(x) - \int_a^x F'$. By Step 2 and Theorem 1.7.10 the integral term is AC with derivative F' a.e., so H is AC with $H' = 0$ a.e.; by Step 3, H is constant, equal to $H(a) = F(a)$. $\square$

1.7.8 The Kakeya maximal function

We close the volume's main line with the operator that the whole series is about.

Definition 1.7.13 (Kakeya maximal operator). For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, $\delta \in (0, 1)$ and $e \in \mathbb{S}^{n-1}$,

$$\mathcal{K}_\delta f(e) = \sup_{a \in \mathbb{R}^n} \frac{1}{m(T_\delta^e(a))} \int_{T_\delta^e(a)} |f| \, dm,$$

the supremum of averages over all δ -tubes pointing in direction e . Thus $\mathcal{K}_\delta f$ is a function on $\mathbb{S}^{n-1}$.

WORKED EXAMPLE 1.7.2— TIER C: RESEARCH TASTE **The Kakeya maximal conjecture, and why this chapter cannot touch it**

(a) *The trivial bounds.* Since every average of $|f|$ is at most $\|f\|_\infty$,

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \leq \|f\|_\infty.$$

At the other end, $m(T_\delta^e(a)) = c_n \delta^{n-1}$, so crudely

$$\mathcal{K}_\delta f(e) \leq \frac{\|f\|_1}{c_n \delta^{n-1}} \quad \text{for every } e.$$

Both are immediate and both are sharp for the obvious examples: the first for f constant, the second for f concentrated in a single δ -tube.

(b) *No δ -uniform bound of any kind is possible.* Fix $p < \infty$ and suppose there were a constant C , independent of δ , with

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \leq C \|f\|_{L^p(\mathbb{R}^n)}. \quad (1.7.9)$$

Let $\varepsilon > 0$ and let $K \subseteq \mathbb{R}^2$ be the compact set given by Theorem 1.3.12: $m(K) < \varepsilon$ and K contains a unit segment in every direction. Let K_δ denote its δ -neighbourhood and take $f = \mathbf{1}_{K_\delta}$.

For every $e \in \mathbb{S}^1$ there is a unit segment in direction e inside K , whose $\delta/2$ -neighbourhood is a δ -tube T contained in K_δ . Hence $\mathcal{K}_\delta f(e) = 1$ for

every e , so the left side of (1.7.9) is ≈ 1 . But the right side is $m(K_\delta)^{1/p}$, and since $K_\delta \downarrow K$ as $\delta \rightarrow 0$ with $m(K_1) < \infty$, continuity from above gives $m(K_\delta) \rightarrow m(K) < \varepsilon$. Choosing ε small and then δ small contradicts (1.7.9).

(c) *What survives, and what is conjectured.* A δ -dependent loss is therefore unavoidable, and the conjecture is that it is as small as possible:

Conjecture 1.7.14 (Kakeya maximal function conjecture). *For every $\varepsilon > 0$,*

$$\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}.$$

The exponent n is forced: the two trivial bounds in (a), interpolated, give L^p bounds with a δ -loss that vanishes to first order exactly at $p = n$, and the Knapp-type example of a single tube shows no smaller p can work. This conjecture implies the Kakeya set conjecture — that every Besicovitch set has Hausdorff dimension n — and is strictly stronger.

(d) *Why the machinery of this chapter is powerless.* Every theorem in §§7.2–7.6 was proved by a covering argument for balls, and the Pitfall after the Looking Ahead box of §7.5 showed that covering lemmas simply fail for tubes. Part (b) above is the same fact stated analytically. So the reader should finish this volume holding a precisely formulated open problem, together with a proof that nothing they have been taught so far can address it. That is the honest position, and it is the correct starting point for Volume 2.

WORKED EXAMPLE 1.7.3— TIER C: RESEARCH TASTE The bush argument

Not quite nothing can be said. Here is a complete proof of a nontrivial lower bound, using only Theorem 1.6.13's underlying geometry and a counting argument. It was the state of the art for many years.

Theorem 1.7.15 (Bush bound). *Let $\mathbb{T}$ be a family of $N \approx \delta^{-(n-1)}$ δ -tubes in $\mathbb{R}^n$ with δ -separated directions, and let $U = \bigcup_{T \in \mathbb{T}} T$. Then*

$$|U| \gtrsim_n \delta^{(n-1)/2}.$$

Proof. Let $F = \sum_{T \in \mathbb{T}} \mathbf{1}_T$ be the multiplicity function and let $\mu = \|F\|_{L^\infty}$ be the maximal multiplicity, attained (up to a constant) at some point x_0 lying in $\approx \mu$ tubes.

Step 1: a bush is disjoint away from its centre. Consider the $\approx \mu$ tubes through x_0 , and work in the annulus

$$A = \{x : \tfrac{1}{2} \leq |x - x_0| \leq 1\}.$$

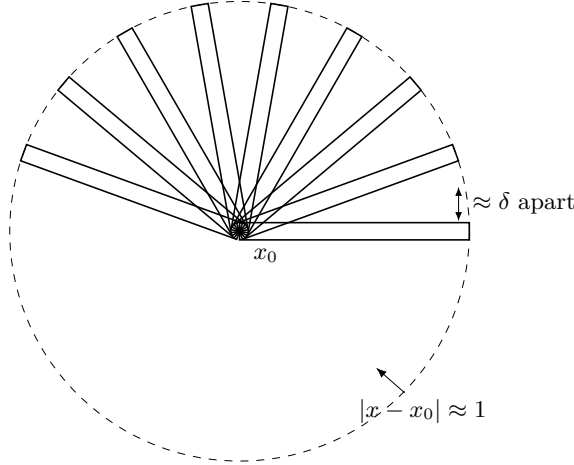


Figure 1.7.1: The bush (Theorem 1.7.15). Tubes through a common point x_0 , with δ -separated directions. Near x_0 they all overlap; but at distance ≈ 1 from x_0 two tubes whose directions differ by $\theta \geq \delta$ are separated by $\approx \theta \geq \delta$, which is their own width, so they are essentially disjoint there. Hence a bush of μ tubes occupies measure $\gtrsim \mu \delta^{n-1}$ — the estimate that drives the whole argument.

Suppose $x \in A$ lies in two of them, with directions $e \neq e'$. Writing $x - x_0 = ru$ with $\frac{1}{2} \leq r \leq 1$ and $|u| = 1$, membership of the tube about $x_0 + \mathbb{R}e$ forces $\text{dist}(x, x_0 + \mathbb{R}e) \leq \delta$, that is $\angle(u, e) \lesssim \delta/r \leq 2\delta$; and likewise $\angle(u, e') \leq 2\delta$. By the triangle inequality for angles,

$$\angle(e, e') \leq \angle(e, u) + \angle(u, e') \lesssim 4\delta.$$

So if the directions are 5δ -separated the tubes are *pairwise disjoint on A* . A δ -separated family splits into $O_n(1)$ subfamilies that are 5δ -separated, so after passing to one of them — losing only a dimensional constant — we may assume this.

Each tube through x_0 meets A in a set of measure $\gtrsim_n \delta^{n-1}$, being a piece of the tube of length $\frac{1}{2}$. Summing over the $\approx \mu$ pairwise disjoint pieces,

$$|U| \geq |U \cap A| \gtrsim_n \mu \delta^{n-1}. \quad (1.7.10)$$

Geometrically: tubes through a common point, with separated directions, spread apart linearly, and at unit distance the separation $\approx \theta \geq \delta$ already exceeds their width.

Step 2: multiplicity is large if the union is small. Since $F \leq \mu$ and F is

supported in U ,

$$N c_n \delta^{n-1} = \sum_T |T| = \int_U F \leq \mu |U|,$$

so, using $N \approx \delta^{-(n-1)}$,

$$\mu \gtrsim \frac{N \delta^{n-1}}{|U|} \approx \frac{1}{|U|}. \quad (1.7.11)$$

Step 3: combine. Substituting (1.7.11) into (1.7.10),

$$\begin{aligned} |U| &\gtrsim \frac{\delta^{n-1}}{|U|} \\ &, \quad \text{hence} \quad |U|^2 \gtrsim \delta^{n-1} \\ &, \quad |U| \gtrsim \delta^{(n-1)/2}. \quad \square \end{aligned}$$

The dimension consequence. Applying this to the δ -neighbourhood of a Besicovitch set and letting $\delta \rightarrow 0$ gives lower Minkowski dimension at least $n - \frac{n-1}{2} = \frac{n+1}{2}$. For $n = 2$ this returns dimension $\frac{3}{2}$, weaker than Córdoba's sharp 2 from Worked Example 1.6.3; for $n = 3$ it gives 2, and this was the best known bound until Wolff's hairbrush argument of 1995 improved it to $\frac{5}{2}$.

Where the argument is lossy. Step 1 used only the tubes through a *single* point. A configuration could have one very heavy point and be otherwise unremarkable, and the argument would not notice. Wolff's improvement replaces the bush — tubes through a point — with the *hairbrush*: tubes meeting a fixed tube. Because a tube is one-dimensional rather than zero-dimensional, the tubes of a hairbrush are spread out along it as well as around it, and one gains a full extra dimension's worth of separation. That is Volume 9. The polynomial method of Volume 11 goes further still, replacing local incidence information with a global algebraic constraint on the entire family at once. □

Figure 1.7.1 (to be drawn). *Vitali's tripling*

Content required: A family of discs in the plane; a maximal disjoint subfamily chosen greedily by radius; and the tripled discs covering the union. Show one unselected disc meeting a selected one of larger radius, contained in its triple.

WORKED EXAMPLE 1.7.4— TIER B **Almost everywhere, and what it does not give**

The differentiation theorems of this chapter conclude “for almost every x ”, and the limits of that phrase recur throughout the series.

(a) *What it gives.* A statement outside a Lebesgue-null set. For most purposes — integration, L^p identities — that is as good as everywhere.

(b) *What it does not give.* Anything along a fixed null set. A line, a sphere, a Cantor set of dimension $n - 1$: all null, and an almost-everywhere statement is silent on each.

(c) *The consequence in this series.* Volume 11’s Nikodym example is exactly this: a full-measure set can have, through *every* one of its points, a line meeting it only there. The density theorem is about balls, and a line is null.

(d) *The systematic response.* When a statement along lower-dimensional families is wanted, one proves it for almost every member of the family — which is what Volume 8’s projection theorems do — and Volume 8, Worked Example 8.1.1 records that “almost every” cannot there be improved to “every”.

The habit. On reading “for a.e. x ”, ask with respect to which measure, and whether the set of interest might be null for it. Most apparent paradoxes in geometric measure theory dissolve at that question.

Drills

Drill 1.7.1. Compute $\mathcal{M}\mathbf{1}_{[0,1]}$ on $\mathbb{R}$ exactly, and verify directly that it behaves like $1/|x|$ at infinity and that it is not integrable.

Drill 1.7.2. Show that the *dyadic* maximal function $\mathcal{M}^{\text{dyad}}$, defined by averaging over dyadic cubes containing x , satisfies a weak $(1, 1)$ bound with constant 1 rather than 3^n . Explain why the improvement is possible, and why $\mathcal{M}^{\text{dyad}}$ is nevertheless not translation invariant.

Drill 1.7.3. Prove that the Kakeya maximal conjecture implies the Kakeya set conjecture, by applying the conjectured bound to $f = \mathbf{1}_{K_\delta}$ as in Worked Example 1.7.2(b) and converting the resulting measure bound into a Minkowski dimension statement.

Drill 1.7.4. In Theorem 1.7.15 Step 1, verify the angle bound $\angle(u, e) \lesssim \delta/r$ and confirm that 5δ -separation suffices for disjointness on A .

Drill 1.7.5. In Theorem 1.7.15, track the constants and determine for which range of $|U|$ the error term in Step 1 is genuinely negligible. Then verify the claim that the bound returns $\frac{n+1}{2}$ and compare with n for $n = 2, 3, 4, 10$.

Drill 1.7.6. Show that the Hardy–Littlewood maximal function of a non-zero $f \in L^1(\mathbb{R}^n)$ is never itself in L^1 , by bounding $\mathcal{M}f(x)$ below by $c|x|^{-n}$ for large $|x|$.

Drill 1.7.7. Deduce the Lebesgue density theorem from Theorem 1.7.4 by taking $f = \mathbf{1}_E$, and explain why the conclusion says nothing about the behaviour of E along any fixed line.

Drill 1.7.8. Verify that the Vitali covering lemma fails if the radii are unbounded, and identify where boundedness is used in the proof.

Notes

The maximal function and (1.7.1) are due to Hardy and Littlewood (1930) in dimension one and to Wiener (1939) in general. The Besicovitch covering theorem alluded to in §7.5 is from 1945; it is proved in Volume 7, where Radon measures make it necessary.

Theorem 1.7.13 is due to Bourgain (1991), and the conjecture stated in Worked Example 1.7.2 is his. Theorem 1.7.15 is folklore, appearing in essentially this form in work of Drury and of Christ–Duoandikoetxea–Rubio de Francia in the 1980s. Wolff’s hairbrush bound of $\frac{n+2}{2}$ appeared in 1995 and stood in $\mathbb{R}^3$ until the sequence of improvements culminating in Wang and Zahl’s resolution of the three-dimensional case in 2025.

Readers who wish to see where the story goes before beginning Volume 2 may read Bourgain’s 1991 paper and Wolff’s 1999 survey; both are readable with exactly the material of this volume plus the Hausdorff dimension of Volume 7.

Chapter 1.8

L^p Spaces, Duality, and Hilbert Space

Almost every theorem in the remaining thirteen volumes is an inequality between L^p norms. This chapter builds the spaces, proves they are complete, identifies their duals, and introduces weak L^p — the space in which Theorem 1.7.3 naturally lives. It ends with Hilbert space, the case $p = 2$, where orthogonality becomes available and where the Fourier transform of Volume 2 will be at home.

1.8.1 Hölder's inequality

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space and $1 \leq p \leq \infty$. For measurable f set

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p} \quad (p < \infty),$$

$$\|f\|_\infty = \operatorname{ess\,sup} |f| = \inf\{M : |f| \leq M \text{ a.e.}\}.$$

Exponents p, p' with $\frac{1}{p} + \frac{1}{p'} = 1$ are called *conjugate*; thus $1' = \infty$ and $2' = 2$.

Lemma 1.8.1 (Young's inequality for products). *For $a, b \geq 0$ and conjugate $1 < p, p' < \infty$,*

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'},$$

with equality if and only if $a^p = b^{p'}$.

Proof. Both sides are 0 when $ab = 0$, so assume $a, b > 0$. The function $\log$ is concave, so

$$\log \left(\frac{a^p}{p} + \frac{b^{p'}}{p'} \right) \geq \frac{1}{p} \log a^p + \frac{1}{p'} \log b^{p'} = \log a + \log b = \log(ab),$$

using $\frac{1}{p} + \frac{1}{p'} = 1$ so that the weights sum to 1. Exponentiate. Strict concavity of log gives equality precisely when $a^p = b^{p'}$. $\square$

Theorem 1.8.2 (Hölder). *For measurable f, g and conjugate exponents p, p' ,*

$$\|fg\|_1 \leq \|f\|_p \|g\|_{p'}.$$

Proof. The cases $p = 1$ and $p = \infty$ are immediate from $|fg| \leq \|g\|_\infty |f|$ a.e. So let $1 < p < \infty$. If either norm is 0 then $fg = 0$ a.e.; if either is ∞ there is nothing to prove. So normalise: replacing f by $f/\|f\|_p$ and g by $g/\|g\|_{p'}$, it suffices to show $\|fg\|_1 \leq 1$ when $\|f\|_p = \|g\|_{p'} = 1$.

Apply Theorem 1.8.1 pointwise with $a = |f(x)|$, $b = |g(x)|$ and integrate, using Theorem 1.5.2:

$$\int |fg| d\mu \leq \frac{1}{p} \int |f|^p d\mu + \frac{1}{p'} \int |g|^{p'} d\mu = \frac{1}{p} + \frac{1}{p'} = 1. \quad \square$$

Corollary 1.8.3 (Generalised Hölder). *If $\frac{1}{p_1} + \cdots + \frac{1}{p_m} = \frac{1}{r}$ with all $p_i, r \geq 1$, then $\|f_1 \cdots f_m\|_r \leq \prod_i \|f_i\|_{p_i}$.*

Proof. Induction on m using Theorem 1.8.2, after replacing each f_i by $|f_i|^r$ and p_i by p_i/r . $\square$

Looking ahead: why the generalised form matters

Theorem 1.8.3 looks like a routine extension. It is the workhorse of Volume 12. Multilinear restriction and multilinear Kakeya estimates are inequalities for products $\prod_{i=1}^n |E_i f_i|$, and the passage between the multilinear and linear formulations is always an application of Theorem 1.8.3 with $m = n$ and $p_i = n$, $r = 1$. Readers meeting the Bennett–Carbery–Tao theorem later will find that the only non-geometric step in its statement is this corollary.

Theorem 1.8.4 (Minkowski). *For $1 \leq p \leq \infty$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.*

Proof. The cases $p = 1, \infty$ are clear. For $1 < p < \infty$, assume $\|f + g\|_p < \infty$ (which follows from $|f + g|^p \leq 2^p(|f|^p + |g|^p)$) and write

$$\int |f + g|^p \leq \int |f| |f + g|^{p-1} + \int |g| |f + g|^{p-1}.$$

Apply Theorem 1.8.2 to each term with exponents p and p' , noting $(p-1)p' = p$:

$$\int |f + g|^p \leq (\|f\|_p + \|g\|_p) \left(\int |f + g|^p \right)^{1/p'}.$$

If $\|f + g\|_p > 0$, divide by the last factor and use $1 - \frac{1}{p'} = \frac{1}{p}$. $\square$

1.8.2 Completeness

Definition 1.8.5. $L^p(\mu)$ is the space of measurable f with $\|f\|_p < \infty$, with functions equal a.e. identified. By Theorem 1.8.4 it is a normed vector space.

Theorem 1.8.6 (Riesz–Fischer). $L^p(\mu)$ is complete for every $1 \leq p \leq \infty$.

Idea of the proof. We use the criterion that a normed space is complete iff every absolutely convergent series converges. This is the right formulation here because it turns completeness into a statement about $\sum_k \|f_k\|_p < \infty$, and such sums are handled by the monotone convergence theorem — which is exactly the tool the Riemann theory lacked. Compare Worked Example 1.3, where the failure of completeness for $C[0, 1]$ was presented in precisely this series form.

Proof. Let $p < \infty$ and suppose $\sum_k \|f_k\|_p = B < \infty$. Put $G = \sum_k |f_k|$, a pointwise limit of increasing partial sums. By Theorem 1.5.1 and Theorem 1.8.4,

$$\|G\|_p = \lim_m \left\| \sum_{k \leq m} |f_k| \right\|_p \leq B,$$

so $G < \infty$ a.e. Hence $\sum_k f_k(x)$ converges absolutely for a.e. x ; call the sum f , which satisfies $|f| \leq G$ so $f \in L^p$. Finally

$$\left\| f - \sum_{k \leq m} f_k \right\|_p^p = \int \left| \sum_{k > m} f_k \right|^p \leq \int \left(\sum_{k > m} |f_k| \right)^p \rightarrow 0$$

by Theorem 1.5.6, the integrand being dominated by $G^p \in L^1$ and tending to 0 a.e.

For $p = \infty$: given a Cauchy sequence (f_k) , the sets where $|f_k - f_l| > \|f_k - f_l\|_\infty$ form a countable union of null sets; off it, (f_k) is uniformly Cauchy, hence uniformly convergent. $\square$

1.8.3 Density and separability

Theorem 1.8.7. Let $1 \leq p < \infty$.

- (i) Simple functions with support of finite measure are dense in $L^p(\mu)$.
- (ii) On $\mathbb{R}^n$, $C_c(\mathbb{R}^n)$ and $C_c^\infty(\mathbb{R}^n)$ are dense in $L^p(\mathbb{R}^n)$.
- (iii) Translation is continuous: $\|\tau_h f - f\|_p \rightarrow 0$ as $h \rightarrow 0$.

All three fail for $p = \infty$.

Proof. (i) For $f \geq 0$ take simple $\varphi_k \uparrow f$ (Theorem 1.4.6); then $|f - \varphi_k|^p \leq f^p \in L^1$ and $\rightarrow 0$ pointwise, so Theorem 1.5.6 applies. General f splits into four nonnegative parts.

(ii) By (i) it suffices to approximate $\mathbf{1}_E$ with $m(E) < \infty$. By Theorem 1.3.6 choose C closed and U open with $C \subseteq E \subseteq U$ and $m(U \setminus C) < \varepsilon$; Urysohn's lemma (Appendix A) gives continuous g with $\mathbf{1}_C \leq g \leq \mathbf{1}_U$, so $\|\mathbf{1}_E - g\|_p^p \leq m(U \setminus C) < \varepsilon$. Smoothness is obtained by mollification: $g * \phi_t \rightarrow g$ in L^p by Theorem 1.6.10, and $g * \phi_t \in C_c^\infty$ when $\phi \in C_c^\infty$.

(iii) True for $g \in C_c$ by uniform continuity and Theorem 1.5.6; extend by (ii) and $\|\tau_h\| = 1$.

For $p = \infty$: $\|\mathbf{1}_{[0,\infty)} - g\|_\infty \geq \frac{1}{2}$ for every continuous g , and $\|\tau_h \mathbf{1}_{[0,\infty)} - \mathbf{1}_{[0,\infty)}\|_\infty = 1$ for every $h \neq 0$. $\square$

1.8.4 Duality

Theorem 1.8.8. *Let μ be σ -finite and $1 \leq p < \infty$. Then the map $g \mapsto \Lambda_g$, $\Lambda_g(f) = \int fg \, d\mu$, is an isometric isomorphism of $L^p(\mu)$ onto $L^p(\mu)^*$. This fails for $p = \infty$.*

Proof. Step 1. The map is well defined and isometric.

That $\|\Lambda_g\| \leq \|g\|_{p'}$ is Theorem 1.8.2. For the reverse, given $g \in L^{p'}$ with $1 < p < \infty$, take

$$f = |g|^{p'-1} \operatorname{sgn} \bar{g}.$$

Then $|f|^p = |g|^{(p'-1)p} = |g|^{p'}$, so $\|f\|_p = \|g\|_{p'}^{p'/p}$, while $\int fg = \int |g|^{p'} = \|g\|_{p'}^{p'}$. Hence $\Lambda_g(f)/\|f\|_p = \|g\|_{p'}^{p'-p'/p} = \|g\|_{p'}$. For $p = 1$, $p' = \infty$: given $\varepsilon > 0$ the set $A = \{|g| > \|g\|_\infty - \varepsilon\}$ has positive measure, and testing against $f = \mathbf{1}_{A'} \operatorname{sgn} \bar{g}$ for a subset $A' \subseteq A$ of finite positive measure gives $\Lambda_g(f) \geq (\|g\|_\infty - \varepsilon)\|f\|_1$.
 $\triangleright$ *Goal: surjectivity: every $\Lambda \in L^p(\mu)^*$ is some Λ_g .*

Step 2. *A measure built from the functional.*

Assume first $\mu(X) < \infty$, so that $\mathbf{1}_E \in L^p$ for every $E \in \mathcal{M}$. Define

$$\nu(E) = \Lambda(\mathbf{1}_E).$$

This is countably additive: if $E = \bigsqcup_k E_k$ then $\mathbf{1}_{\bigcup_{k \leq N} E_k} \rightarrow \mathbf{1}_E$ in L^p , because $\|\mathbf{1}_E - \mathbf{1}_{\bigcup_{k \leq N} E_k}\|_p^p = \mu(\bigcup_{k > N} E_k) \rightarrow 0$ by continuity from above (Theorem 1.2.6(iv), using $\mu(X) < \infty$), and Λ is continuous. So ν is a complex measure. Moreover $\mu(E) = 0$ implies $\mathbf{1}_E = 0$ in L^p , hence $\nu(E) = 0$: thus $\nu \ll \mu$.

Step 3. *Radon–Nikodym.*

By Theorem 1.7.8 (applied to the positive and negative, real and imaginary, parts of ν) there is $g \in L^1(\mu)$ with $\nu(E) = \int_E g \, d\mu$ for all E . By linearity, $\Lambda(\varphi) = \int \varphi g \, d\mu$ for every simple φ , and hence, by uniform approximation and $\mu(X) < \infty$, for every $\varphi \in L^\infty(\mu)$.

Step 4. *g lies in $L^{p'}$.*

Let $1 < p < \infty$. For $n \in \mathbb{N}$ put $A_n = \{|g| \leq n\}$ and $f_n = \mathbf{1}_{A_n} |g|^{p'-1} \operatorname{sgn} \bar{g}$, which is bounded, hence in $L^\infty \cap L^p$. By Step 3,

$$\begin{aligned} \int_{A_n} |g|^{p'} \, d\mu &= \Lambda(f_n) \\ &\leq \|\Lambda\| \|f_n\|_p \\ &= \|\Lambda\| \left(\int_{A_n} |g|^{p'} \, d\mu \right)^{1/p}, \end{aligned}$$

using $|f_n|^p = \mathbf{1}_{A_n} |g|^{p'}$. The integral on the left is finite (the integrand is bounded by $n^{p'}$ and $\mu(X) < \infty$), so we may divide, obtaining $(\int_{A_n} |g|^{p'} \, d\mu)^{1/p'} \leq \|\Lambda\|$. Since $A_n \uparrow X$, Theorem 1.5.1 gives $\|g\|_{p'} \leq \|\Lambda\| < \infty$.

For $p = 1$: if $\mu(\{|g| > \|\Lambda\| + \varepsilon\}) > 0$ for some $\varepsilon > 0$, testing against the indicator of a finite-measure subset contradicts $|\Lambda(f)| \leq \|\Lambda\| \|f\|_1$. So $\|g\|_\infty \leq \|\Lambda\|$.

Step 5. *Conclude on finite measure spaces.*

The bounded functionals Λ and Λ_g agree on $L^\infty(\mu)$, which contains the simple functions and is therefore dense in $L^p(\mu)$ for $p < \infty$ by Theorem 1.8.7(i). Two continuous functionals agreeing on a dense set are equal.

Step 6. *The σ -finite case.*

Write $X = \bigsqcup_k X_k$ with $\mu(X_k) < \infty$. Step 5 gives $g_k \in L^{p'}(X_k)$ representing Λ on functions supported in X_k , with $\|g_k\|_{p'} \leq \|\Lambda\|$; by uniqueness these are consistent, so they patch to a single g on X . For $p > 1$, $(\sum_{k \leq N} \|g_k\|_{p'}^{p'})^{1/p'} = \|g \mathbf{1}_{\bigcup_{k \leq N} X_k}\|_{p'} \leq \|\Lambda\|$ for every N , so $g \in L^{p'}(X)$ by Theorem 1.5.1; the case $p = 1$ is immediate. Finally Λ and Λ_g agree on functions supported in finitely many X_k , which are dense in L^p by Theorem 1.5.6. $\square$

Remark 1.8.9 (Why $p = \infty$ is excluded). The argument breaks at Step 5, where density of simple functions in L^p was used; simple functions are not dense in L^∞ (Theorem 1.8.7). And the conclusion genuinely fails: by the Hahn–Banach theorem there is a bounded functional on $L^\infty(\mathbb{R})$ extending $f \mapsto \lim_{x \rightarrow 0} f(x)$ from the subspace of functions continuous at 0, and no $g \in L^1$ represents it.

Corollary 1.8.10 (Norms by testing). *For $1 \leq p < \infty$ and μ σ -finite,*

$$\|f\|_p = \sup \left\{ \left| \int f g \, d\mu \right| : \|g\|_{p'} \leq 1 \right\}. \quad (1.8.1)$$

Proof. Step 1. The supremum is at most $\|f\|_p$.

For any g with $\|g\|_{p'} \leq 1$, Hölder's inequality (Theorem 1.8.2) gives $|\int f g| \leq \|f\|_p \|g\|_{p'} \leq \|f\|_p$.

Step 2. *It is attained, when $0 < \|f\|_p < \infty$.*

Suppose first $1 < p < \infty$ and $0 < \|f\|_p < \infty$, and set

$$g = \frac{|f|^{p-1} \operatorname{sgn} \bar{f}}{\|f\|_p^{p-1}},$$

where $\operatorname{sgn} z = \bar{z}/|z|$ for $z \neq 0$ and 0 otherwise. Since $(p-1)p' = p$,

$$\|g\|_{p'}^{p'} = \frac{1}{\|f\|_p^{(p-1)p'}} \int |f|^{(p-1)p'} = \frac{1}{\|f\|_p^p} \int |f|^p = 1,$$

so g is admissible; and

$$\int f g = \frac{1}{\|f\|_p^{p-1}} \int |f|^p = \|f\|_p.$$

For $p = 1$ take $g = \operatorname{sgn} \bar{f}$, which has $\|g\|_\infty \leq 1$ and $\int f g = \int |f| = \|f\|_1$.

Step 3. *The case $\|f\|_p = \infty$.*

Here σ -finiteness is used. Write $X = \bigcup_k X_k$ with $\mu(X_k) < \infty$ and $X_k \uparrow X$, and set $f_k = f \mathbf{1}_{X_k} \mathbf{1}_{\{|f| \leq k\}}$. Each f_k has $\|f_k\|_p^p \leq k^p \mu(X_k) < \infty$, and $|f_k| \uparrow |f|$ pointwise, so $\|f_k\|_p \uparrow \|f\|_p = \infty$ by monotone convergence (Theorem 1.5.1).

Let g_k be the function produced by Step 2 for f_k , so $\|g_k\|_{p'} \leq 1$ and $\int f_k g_k = \|f_k\|_p$. Crucially g_k vanishes wherever f_k does, and $f = f_k$ on the set where $f_k \neq 0$; hence $\int f g_k = \int f_k g_k = \|f_k\|_p \rightarrow \infty$, so the supremum is ∞ as well.

Step 4. *The case $\|f\|_p = 0$.*

Then $f = 0$ a.e. and both sides vanish. □

Pitfall: σ -finiteness, and why $p = \infty$ is excluded

Step 3 is the only place σ -finiteness appears, and it is genuinely needed: on a measure space with an atom of infinite measure one can have $\|f\|_p = \infty$ while every admissible g pairs with f to give 0.

The exponent $p = \infty$ is excluded because Step 2 would require producing $g \in L^1$ from $f \in L^\infty$, and the natural candidate $\operatorname{sgn} \bar{f}$ need not be integrable.

The identity $\|f\|_\infty = \sup\{|\int f g| : \|g\|_1 \leq 1\}$ is nevertheless true for σ -finite μ ; see Theorem 1.8.8 and the discussion after it.

Looking ahead: this identity is how restriction becomes extension

Formula (1.8.1) looks like a triviality. It is the mechanism by which every restriction estimate in Volume 6 is converted into an equivalent statement about the adjoint operator. The restriction conjecture asserts a bound $\|\widehat{f}|_{\mathbb{S}^{n-1}}\|_{L^q(\sigma)} \lesssim \|f\|_{L^p(\mathbb{R}^n)}$; testing against $g \in L^{q'}(\sigma)$ and unwinding turns this into a bound on the *extension operator* $g \mapsto \widehat{g d\sigma}$, which is the form in which the problem is always studied because it is the form in which wave packets appear. Every duality argument in this series is (1.8.1).

1.8.5 Weak L^p

Definition 1.8.11. For $0 < p < \infty$, $L^{p,\infty}(\mu)$ is the space of measurable f with

$$\begin{aligned} \|f\|_{p,\infty} &= \sup_{\lambda>0} \lambda d_f(\lambda)^{1/p} < \infty \\ &, \quad d_f(\lambda) \\ &= \mu(\{|f| > \lambda\}). \end{aligned}$$

Proposition 1.8.12. $L^p(\mu) \subseteq L^{p,\infty}(\mu)$ with $\|f\|_{p,\infty} \leq \|f\|_p$, and the inclusion is strict.

Proof. A note on notation. The distribution function is written d_f , not d_f : the letter λ is reserved throughout this series for the *level*, and using it for the function as well makes expressions such as $\lambda d_f(\lambda)^{1/p}$ unreadable. The layer-cake formula below, and every use of it in Volumes 2–13, follows this convention.

Chebyshev: $\lambda^p d_f(\lambda) \leq \int_{\{|f|>\lambda\}} |f|^p \leq \|f\|_p^p$. That the inclusion is strict — that weak L^p is genuinely larger — is shown by Worked Example 1.8.1 below, which exhibits $|x|^{-n/p} \in L^{p,\infty} \setminus L^p$. $\square$

WORKED EXAMPLE 1.8.1— TIER A **The model function of weak L^p**

On $\mathbb{R}^n$ let $f(x) = |x|^{-n/p}$. Then $\{|f| > \lambda\} = B(0, \lambda^{-p/n})$ has measure $\omega_n \lambda^{-p}$, so $\lambda d_f(\lambda)^{1/p} = \omega_n^{1/p}$ for every λ : the supremum is attained at every height, and $f \in L^{p,\infty}$.

But $f \notin L^p$: by polar coordinates (1.6.2),

$$\int_{\mathbb{R}^n} |x|^{-n} dx = \sigma(\mathbb{S}^{n-1}) \int_0^\infty \frac{dr}{r} = \infty,$$

diverging logarithmically at both ends. So $|x|^{-n/p}$ sits exactly on the boundary — the canonical example of a function that is L^p “to first order” but not quite.

Where you have met this already. Worked Example 1.7.1 showed $\mathcal{M}f(x) \gtrsim |x|^{-n}$ at infinity for any nonzero $f \in L^1$. That is precisely the statement $\mathcal{M}f \in L^{1,\infty} \setminus L^1$, and it explains why (1.7.1) is a weak-type bound and could not have been anything else.

1.8.6 Hilbert space

Definition 1.8.13. An *inner product space* is a complex vector space H with a map $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$, linear in the first variable, conjugate symmetric, and positive definite. A *Hilbert space* is one complete in the norm $\|x\| = \langle x, x \rangle^{1/2}$.

$L^2(\mu)$ with $\langle f, g \rangle = \int f \bar{g} d\mu$ is a Hilbert space, by Theorem 1.8.6.

Theorem 1.8.14 (Projection). *Let $K \subseteq H$ be nonempty, closed and convex. For every $x \in H$ there is a unique $Px \in K$ with $\|x - Px\| = \text{dist}(x, K)$. If K is a closed subspace then $x - Px \perp K$, and $H = K \oplus K^\perp$.*

Proof. Let $d = \text{dist}(x, K)$ and take $y_k \in K$ with $\|x - y_k\| \rightarrow d$. The parallelogram law $\|a + b\|^2 + \|a - b\|^2 = 2\|a\|^2 + 2\|b\|^2$, applied to $a = x - y_k$, $b = x - y_l$, gives

$$\|y_k - y_l\|^2 = 2\|x - y_k\|^2 + 2\|x - y_l\|^2 - 4\left\|x - \frac{y_k + y_l}{2}\right\|^2 \leq 2\|x - y_k\|^2 + 2\|x - y_l\|^2 - 4d^2,$$

using convexity to place $\frac{y_k + y_l}{2} \in K$. The right side tends to 0, so (y_k) is Cauchy; completeness and closedness give a limit $Px \in K$, and uniqueness follows from the same identity. If K is a subspace, then for $y \in K$ and $t \in \mathbb{C}$ the function $t \mapsto \|x - Px - ty\|^2$ is minimised at $t = 0$, whence $\langle x - Px, y \rangle = 0$. $\square$

Theorem 1.8.15 (Riesz representation). *Every bounded linear functional Λ on a Hilbert space H is $\Lambda(x) = \langle x, y \rangle$ for a unique $y \in H$, and $\|\Lambda\| = \|y\|$.*

Proof. If $\Lambda = 0$ take $y = 0$. Otherwise $K = \ker \Lambda$ is a proper closed subspace, so by Theorem 1.8.14 there is $z \in K^\perp$ with $\|z\| = 1$. For any x , the vector $\Lambda(x)z - \Lambda(z)x$ lies in K , hence is orthogonal to z ; taking inner products with z and using $\|z\| = 1$ gives

$$0 = \langle \Lambda(x)z - \Lambda(z)x, z \rangle = \Lambda(x) - \Lambda(z)\langle x, z \rangle,$$

that is, $\Lambda(x) = \langle x, \overline{\Lambda(z)}z \rangle$. So $y = \overline{\Lambda(z)}z$ works. Uniqueness holds because $\langle x, y_1 \rangle = \langle x, y_2 \rangle$ for all x forces $y_1 - y_2 \perp H$, and $\|\Lambda\| = \|y\|$ follows from Cauchy–Schwarz together with $\Lambda(y) = \|y\|^2$. $\square$

Theorem 1.8.16 (Orthonormal bases). *Let $\{e_\alpha\}$ be an orthonormal set in H . Then Bessel’s inequality $\sum_\alpha |\langle x, e_\alpha \rangle|^2 \leq \|x\|^2$ holds, and the following are equivalent: the span is dense; Parseval’s identity $\|x\|^2 = \sum_\alpha |\langle x, e_\alpha \rangle|^2$ holds for all x ; $x = \sum_\alpha \langle x, e_\alpha \rangle e_\alpha$ for all x .*

Proof. Bessel follows by expanding $0 \leq \|x - \sum_{\alpha \in F} \langle x, e_\alpha \rangle e_\alpha\|^2$ over finite F . The equivalences follow from Theorem 1.8.14 applied to the closed span, together with Theorem 1.8.6 to sum the series. $\square$

WORKED EXAMPLE 1.8.2— TIER C: RESEARCH TASTE **Almost orthogonality, and the inequality that decoupling upgrades**

Orthogonality is rare; *almost* orthogonality is everywhere. This example isolates the elementary statement, then specialises it to the objects of Volume 13 — and shows why the elementary version is far too lossy, which is the reason that volume exists.

(a) *The elementary bound.* Let $f_1, \dots, f_N \in H$ satisfy $|\langle f_i, f_j \rangle| \leq \varepsilon$ for $i \neq j$, with $\|f_i\| \leq 1$. Then

$$\left\| \sum_{i=1}^N f_i \right\|^2 = \sum_i \|f_i\|^2 + \sum_{i \neq j} \langle f_i, f_j \rangle \leq N + N(N-1)\varepsilon.$$

If $\varepsilon \leq 1/N$ this is $\lesssim N$, matching the exactly orthogonal case. So orthogonality may be violated by $O(1/N)$ per pair without any loss.

(b) *Specialisation to caps.* Partition the truncated paraboloid $\{(\xi, |\xi|^2) : |\xi| \leq 1\} \subseteq \mathbb{R}^n$ into $N \approx \delta^{-(n-1)}$ caps θ of diameter δ , and let f_θ have Fourier transform supported in the δ^2 -neighbourhood of θ . By Plancherel (Volume 2), $\langle f_\theta, f_{\theta'} \rangle = \langle \widehat{f_\theta}, \widehat{f_{\theta'}} \rangle = 0$ whenever $\theta \neq \theta'$, because the supports are disjoint. So

$$\left\| \sum_\theta f_\theta \right\|_{L^2}^2 = \sum_\theta \|f_\theta\|_{L^2}^2.$$

Exact orthogonality, for free, at $p = 2$.

(c) *Why this is not enough, and what decoupling asks.* The estimate in (b) is useless for the applications, which need $p > 2$ — for the Vinogradov mean value theorem one needs $p = n(n+1)$, and for Strichartz estimates p is determined by scaling. At $p \neq 2$ there is no Plancherel and disjointness of Fourier supports buys nothing directly. The ℓ^2 decoupling theorem of

Bourgain and Demeter asserts that one nevertheless retains almost all of it:

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p(\mathbb{R}^n)} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left(\sum_{\theta} \|f_{\theta}\|_{L^p(\mathbb{R}^n)}^2 \right)^{1/2}, \quad 2 \leq p \leq \frac{2(n+1)}{n-1}.$$

Note the shape: an L^p norm on the left, an ℓ^2 sum of L^p norms on the right. Cauchy–Schwarz alone would give the same with $\delta^{-\varepsilon}$ replaced by $N^{1/2-1/p}$, a genuine power of δ ; the theorem says the true loss is subpolynomial. That gap is the entire content.

(d) *What makes it possible.* Only curvature. If the caps lay on a hyperplane rather than a paraboloid, the estimate would be false — the f_{θ} could all be simultaneously large on a common tube. Curvature forces the tubes dual to different caps to point in different directions, and the extent to which differently-directed tubes can pile up is exactly the Kakeya problem of Worked Example 1.7.2. This is the precise sense, promised in Chapter 1.1, in which restriction, decoupling and Kakeya are one subject: they are three questions about the same picture of overlapping tubes.

You now have, in elementary form, the two endpoints of the series: the Cauchy–Schwarz bound that is trivially true, and the theorem that improves it. Everything between here and Volume 13 is the work of closing that gap.

WORKED EXAMPLE 1.8.3— TIER B **Choosing an exponent**

Much of this series turns on which L^p to work in, and the choice follows a small number of considerations.

(a) $p = 2$ for *orthogonality*. Only L^2 is a Hilbert space, so only there do Plancherel, TT^* and almost-orthogonal decompositions apply. Every estimate in this series that is proved in a page is an L^2 estimate.

(b) $p = \infty$ for *pointwise control*, $p = 1$ for *total mass*. These are the endpoints, and Volume 3 shows singular integrals fail at both — which is why Volume 4 constructs substitutes.

(c) *Intermediate p by interpolation*. Riesz–Thorin and Marcinkiewicz fill in between two known endpoints and never produce a new one.

(d) *The exponent forced by scaling*. For a homogeneous operator, testing on $f(\lambda \cdot)$ determines the only possible exponent. This is the first check to run

on any conjectured estimate, and half the estimates in Volumes 6 to 13 are pinned down this way before a proof is attempted.

A worked instance. Asked to bound an operator commuting with dilations of degree α , compute how each side scales; if they scale differently no estimate can hold, and if they agree the exponent is determined.

Drills

Drill 1.8.1. Show that on a finite measure space $L^q \subseteq L^p$ for $p \leq q$, with $\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q$; and that on $\mathbb{R}^n$ neither inclusion holds, by exhibiting functions in each difference.

Drill 1.8.2. Prove the interpolation inequality $\|f\|_q \leq \|f\|_p^{1-\theta} \|f\|_r^\theta$ for $\frac{1}{q} = \frac{1-\theta}{p} + \frac{\theta}{r}$, directly from Hölder. This is the prototype of every interpolation argument in the series.

Drill 1.8.3. Show $\|\cdot\|_{p,\infty}$ is not a norm for $p > 1$ by finding f, g violating the triangle inequality, and verify that it does satisfy $\|f + g\|_{p,\infty} \leq 2(\|f\|_{p,\infty} + \|g\|_{p,\infty})$.

Drill 1.8.4. In Worked Example 1.8.2(c), verify that Cauchy–Schwarz gives the loss $N^{1/2-1/p}$ and compute this as a power of δ for $n = 2$, $p = 6$. Compare with $\delta^{-\varepsilon}$ and quantify what the Bourgain–Demeter theorem saves.

Notes

Hölder’s inequality is from 1889 and Minkowski’s from 1896; Theorem 1.8.6 was proved independently by Riesz and Fischer in 1907 for $p = 2$. Weak L^p spaces were introduced by Lorentz.

Worked Example 1.8.2 is the only place in this volume where a theorem is stated that we do not prove, and the deviation is deliberate: a reader finishing Volume 1 should know precisely what they are working towards. The Bourgain–Demeter theorem quoted there is from 2015 and is proved in Volume 13; the restriction conjecture alluded to in the Looking Ahead box of §8.4 is open for $n \geq 3$.

Chapter 1.A

Background from Topology and Analysis

This appendix proves the results assumed in the main text but not covered by a second-year syllabus. Readers may well not have met Urysohn's lemma, Tietze extension or the Hahn decomposition; all three are used in Volume 1 and all three are proved here.

1.A.1 Urysohn's lemma in $\mathbb{R}^n$

Theorem 1.A.1. *Let $K \subseteq U \subseteq \mathbb{R}^n$ with K compact and U open. There is $g \in C_c(\mathbb{R}^n)$ with $\mathbf{1}_K \leq g \leq \mathbf{1}_U$ and $\text{supp } g$ a compact subset of U .*

Proof. Since K is compact and U^c closed and disjoint from K , we have $d := \text{dist}(K, U^c) > 0$ (a continuous positive function on a compact set attains a positive minimum). Let $V = \{x : \text{dist}(x, K) < d/3\}$ and $W = \{x : \text{dist}(x, K) \leq 2d/3\}$, so $K \subseteq V \subseteq W \subseteq U$ with W compact. Define

$$g(x) = \frac{\text{dist}(x, V^c)}{\text{dist}(x, V^c) + \text{dist}(x, K)}.$$

The denominator never vanishes: if both terms were 0 then x would lie in the closed sets $\overline{V^c}$ and K , which are disjoint since every point of K has distance $\geq d/3$ from V^c . Each distance function is continuous (indeed 1-Lipschitz), so g is continuous; $g = 1$ on K and $g = 0$ off V , so $\text{supp } g \subseteq W$, compact in U ; and $0 \leq g \leq 1$ throughout. $\square$

1.A.2 Tietze's extension theorem

Theorem 1.A.2. *Let $K \subseteq \mathbb{R}^n$ be compact and $f : K \rightarrow \mathbb{R}$ continuous. Then there is $g \in C_c(\mathbb{R}^n)$ with $g|_K = f$ and $\|g\|_\infty = \|f\|_\infty$.*

Proof. Let $M = \|f\|_\infty$; we may assume $M > 0$. We construct g as the sum of a series, each term correcting a third of the remaining error.

Step 1. *One step of correction.*

Suppose $h : K \rightarrow \mathbb{R}$ is continuous with $\|h\|_\infty \leq c$. The sets $A = \{h \leq -c/3\}$ and $B = \{h \geq c/3\}$ are disjoint compact subsets of $\mathbb{R}^n$, so by Theorem 1.A.1 applied to B inside the open set A^c there is a continuous u with $\mathbf{1}_B \leq u \leq \mathbf{1}_{A^c}$; put $\varphi = \frac{c}{3}(2u - 1)$, so φ is continuous on $\mathbb{R}^n$ with $\|\varphi\|_\infty \leq c/3$, $\varphi = c/3$ on B and $\varphi = -c/3$ on A . Checking the three cases $x \in A$, $x \in B$, $x \notin A \cup B$ gives

$$\|h - \varphi\|_{L^\infty(K)} \leq \frac{2c}{3}.$$

Step 2. *Iterate.*

Apply Step 1 with $h_1 = f$, $c_1 = M$ to get φ_1 on $\mathbb{R}^n$ with $\|\varphi_1\|_\infty \leq M/3$ and $\|f - \varphi_1\|_{L^\infty(K)} \leq 2M/3$. Apply it again to $h_2 = f - \varphi_1$ with $c_2 = 2M/3$, and so on. Inductively $\|\varphi_k\|_\infty \leq \frac{M}{3}(\frac{2}{3})^{k-1}$ and $\|f - \sum_{j \leq k} \varphi_j\|_{L^\infty(K)} \leq M(\frac{2}{3})^k$.

Step 3. *Sum.*

The series $\sum_k \varphi_k$ converges uniformly on $\mathbb{R}^n$, since $\sum_k \frac{M}{3}(\frac{2}{3})^{k-1} = M$; its sum G is therefore continuous, agrees with f on K by Step 2, and satisfies $\|G\|_\infty \leq M$. Finally multiply G by a function from Theorem 1.A.1 that is 1 on K and compactly supported, which changes nothing on K and makes the support compact. $\square$

1.A.3 Signed measures and the Hahn decomposition

Theorem 1.A.3. *Let λ be a finite signed measure on $(X, \mathcal{M})$. There is a partition $X = P \sqcup N$ with $\lambda(E) \geq 0$ for every measurable $E \subseteq P$ and $\lambda(E) \leq 0$ for every measurable $E \subseteq N$.*

Proof. Call A *positive* if every measurable subset has nonnegative λ -measure, and let $s = \sup\{\lambda(A) : A \text{ positive}\}$, which is finite since λ is. Take positive A_k with $\lambda(A_k) \rightarrow s$ and put $P = \bigcup_k A_k$; a countable union of positive sets is positive (disjointify and use countable additivity), so P is positive and $\lambda(P) = s$.

Let $N = P^c$ and suppose, for contradiction, that N contains a measurable E with $\lambda(E) > 0$. Then E is not positive, so it contains a subset of negative measure; let n_1 be the least positive integer for which some measurable $E_1 \subseteq E$ has $\lambda(E_1) < -1/n_1$. Since $\lambda(E \setminus E_1) = \lambda(E) - \lambda(E_1) > \lambda(E) > 0$, the argument repeats inside $E \setminus E_1$, yielding n_2 and E_2 , and so on. The E_k are disjoint with $\sum_k |\lambda(E_k)| < \infty$, forcing $n_k \rightarrow \infty$. Put $F = E \setminus \bigcup_k E_k$, so $\lambda(F) \geq \lambda(E) > 0$.

Any measurable $G \subseteq F$ has $\lambda(G) \geq -1/(n_k - 1)$ for every k by minimality of n_k , hence $\lambda(G) \geq 0$: so F is positive. But then $P \cup F$ is positive with $\lambda(P \cup F) = s + \lambda(F) > s$, contradicting the definition of s . $\square$

1.A.4 Complex analysis, for Volume 2

The following are not used in Volume 1 and are recorded so that this volume is a self-contained prerequisite for the next. Proofs are in Volume 2, Chapter 1.

Theorem 1.A.4 (Three-lines lemma). *Let F be bounded and continuous on the closed strip $0 \leq \operatorname{Re} z \leq 1$ and holomorphic in its interior, with $|F| \leq M_0$ on $\operatorname{Re} z = 0$ and $|F| \leq M_1$ on $\operatorname{Re} z = 1$. Then $|F(\theta + it)| \leq M_0^{1-\theta} M_1^\theta$ for $0 \leq \theta \leq 1$.*

This lemma is stated here as a prerequisite and is *not* proved in this appendix; it is proved in Volume 2, Theorem 2.1.14, where it is the entire content of the Riesz–Thorin interpolation theorem. Nothing in Volume 1 uses it.

1.A.5 Linear algebra used without comment

Quadratic forms and signature (needed for the second fundamental form of a hypersurface in Volume 5); the singular value decomposition; and the determinant as a volume distortion factor, which is Theorem 1.3.10.

Chapter 1.B

Notation

Asymptotic conventions

$A \lesssim B$	$A \leq CB$ for an absolute constant C
$A \lesssim_\varepsilon B$	the constant may depend on ε
$A \approx B$	$A \lesssim B$ and $B \lesssim A$
$A = O(B)$, $o(B)$	Landau notation, used when a limit is in play

Constants are suppressed because in this subject they are almost never the point and carrying them makes the structure of an argument invisible. When a constant does matter — and in the induction-on-scales arguments of Volume 13 a constant that degrades at each of $\log(1/\delta)$ scales is fatal — it is tracked explicitly and said so. See Worked Example 1.5.6.

Standing symbols

δ	a small scale; from Volume 7, the width of a tube
R	a large scale, usually δ^{-1}
$T, \mathbb{T}$	a tube; a family of tubes
e, ω	directions, i.e. points of $\mathbb{S}^{n-1}$
θ	an angle, or (from Volume 13) a cap on a hypersurface
σ	surface measure on $\mathbb{S}^{n-1}$, see Theorem 1.6.11
$m, E $	Lebesgue measure
$\mathcal{L}, \mathcal{B}$	the Lebesgue and Borel σ -algebras
$\mathcal{H}^s, \dim_{\mathcal{H}}$	Hausdorff measure and dimension, Chapter 1.2
$\mathcal{M}$	Hardy–Littlewood maximal operator, Chapter 1.7
$\mathcal{K}_\delta$	Takeya maximal operator, Theorem 1.7.13
$\mathbf{1}_E$	indicator function of E
d_f	distribution function of f , see (1.5.1)
p'	conjugate exponent, $\frac{1}{p} + \frac{1}{p'} = 1$
$L^{p,\infty}$	weak L^p , Chapter 1.8
$\widehat{f}$	Fourier transform, Worked Example 1.5.4

Index of the main results of Volume 1

σ -algebras, measures	Theorem 1.2.1, Theorem 1.2.5
Carathéodory's theorem	Theorem 1.2.11
Carathéodory extension	Theorem 1.2.15
Metric outer measures	Theorem 1.2.13
Box lemma	Theorem 1.3.4
Structure of measurable sets	Theorem 1.3.6
Linear images	Theorem 1.3.10
Vitali's non-measurable set	Theorem 1.3.11
Besicovitch–Perron	Theorem 1.3.12
Simple approximation	Theorem 1.4.6
Lebesgue's criterion	Theorem 1.4.14
Monotone class lemma	Theorem 1.6.5
Monotone convergence	Theorem 1.5.1
Fatou	Theorem 1.5.5
Dominated convergence	Theorem 1.5.6
Egorov, Lusin	Theorem 1.5.9, Theorem 1.5.10
Tonelli, Fubini	Theorem 1.6.6, Theorem 1.6.7
Polar coordinates	Theorem 1.6.11
Tube intersection estimate	Theorem 1.6.13
Córdoba: Kakeya in the plane	Theorem 1.6.14
Vitali covering lemma	Theorem 1.7.1
Hardy–Littlewood	Theorem 1.7.3
Lebesgue differentiation	Theorem 1.7.4
Vitali covering theorem	Theorem 1.7.7
Fundamental theorem of calculus	Theorem 1.7.12
Radon–Nikodym	Theorem 1.7.8
Kakeya maximal conjecture	Worked Example 1.7.2
Bush bound	Theorem 1.7.15
Hölder, Minkowski	Theorem 1.8.2, Theorem 1.8.4
Riesz–Fischer	Theorem 1.8.6
Duality	Theorem 1.8.8
Riesz representation	Theorem 1.8.15

Part 2

Volume 2

Fourier Analysis, Distributions, and Interpolation

Preface to Volume 2

Where we are

Volume 1 built Lebesgue measure, the integral, the convergence theorems and the L^p spaces, and ended by stating two problems it could not touch: the Kakeya maximal function conjecture, and the ℓ^2 decoupling inequality. The reader finished holding a precise open problem together with a proof that covering lemmas for balls cannot address it.

This volume supplies the analysis. Its centre is the Fourier transform, and its purpose is to make three things available: interpolation, which converts estimates at two exponents into estimates at every exponent between; distributions, which allow the Fourier transform to act on objects far too rough to integrate — surface measure on a sphere, for instance, which is the central object of Volume 6; and the first genuine decay estimate, the asymptotics of $\widehat{d\sigma}$, which is where curvature enters the subject for the first time.

What is assumed

Everything in Volume 1, and nothing else. In particular we assume the reader knows: Lebesgue measure and its regularity; the monotone, Fatou and dominated convergence theorems; Fubini–Tonelli; L^p spaces, Hölder, duality, weak L^p ; the Hardy–Littlewood maximal function and its weak $(1, 1)$ bound; and the statement of the Kakeya maximal conjecture. References of the form *Theorem 1.5.6* are to that text; the name of the result is given alongside the number, so that a citation remains identifiable if a later revision renumbers the source.

We do *not* assume complex analysis. Volume 1’s Appendix A recorded the three-lines lemma without proof, promising it here, and Chapter 2.1 keeps that promise from the definition of a holomorphic function onwards. Readers who have had a complex analysis course may read Chapter 2.1 for its final two sections only — the maximum principle on unbounded strips, and the three-lines lemma — and should not skip those, since the boundedness hypothesis in

Phragmén–Lindelöf is exactly what makes Riesz–Thorin true and is routinely misremembered.

The plan

1	Holomorphic functions and the three-lines lemma	the one prerequisite
2	Interpolation: Riesz–Thorin and Marcinkiewicz	exponents become intervals
3	Fourier series	the compact model
4	The Fourier transform on L^1 and L^2	inversion, Plancherel
5	The Schwartz space and tempered distributions	rough objects admitted
6	Convolution, multipliers, and Littlewood–Paley	operators become symbols
7	Sobolev spaces	smoothness becomes decay
8	The Fourier transform of surface measure	curvature enters

Chapter 2.8 is the destination. It computes $\widehat{d\sigma}$ for the sphere exactly, in terms of Bessel functions, proves the decay $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$, and shows that this single estimate already forces the restriction conjecture to have the exponent range it does. Everything before it is machinery.

Conventions

Unchanged from Volume 1, with two additions. The Fourier transform is normalised

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx,$$

which puts the 2π in the exponent and makes the transform an isometry of L^2 with no constant; this is the convention of the restriction and decoupling literature and we adopt it throughout. And $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ denotes the torus, so that functions on $\mathbb{T}^n$ are 1-periodic in each variable.

Chapter 2.1

Holomorphic Functions and the Three-Lines Lemma

2.1.1 Why this chapter is here

The Riesz–Thorin interpolation theorem says that if a linear operator is bounded $L^{p_0} \rightarrow L^{q_0}$ and $L^{p_1} \rightarrow L^{q_1}$, then it is bounded between every pair of exponents lying on the segment joining $(1/p_0, 1/q_0)$ to $(1/p_1, 1/q_1)$, with a constant that is *logarithmically convex* along the segment. Nothing in that statement mentions complex numbers.

The proof does, and there is no known way around it. One forms a family of test functions depending holomorphically on a complex parameter z running over the strip $0 \leq \operatorname{Re} z \leq 1$, arranges that the two hypotheses control the boundary lines $\operatorname{Re} z = 0$ and $\operatorname{Re} z = 1$, and then invokes a maximum principle to control the interior. The logarithmic convexity of the constant is not an accident of the method: it is the maximum principle, transcribed.

So this chapter builds one-variable complex analysis from the definition, to the point where the required maximum principle — the three-lines lemma — is available. It is the only prerequisite Volume 1 promised and did not supply.

The material is also needed later on its own account. Contour deformation is the engine of the stationary phase method in Volume 5, where the decay of $\widehat{d\sigma}$ is obtained by pushing an integration contour to where the phase has positive imaginary part; and analytic continuation in a complex parameter is how Volume 5 handles the Riesz potentials $|\xi|^{-\alpha}$ for α outside the range where they converge.

2.1.2 Holomorphic functions

Let $\Omega \subseteq \mathbb{C}$ be open.

Definition 2.1.1. $f : \Omega \rightarrow \mathbb{C}$ is *holomorphic* at $z_0 \in \Omega$ if

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, the limit being over complex $h \rightarrow 0$. It is holomorphic on Ω if it is holomorphic at every point.

The definition is formally identical to the real one, and the usual sum, product, quotient and chain rules follow by the same proofs. The content is entirely in the words “over complex h ”: the difference quotient must have the same limit as h approaches 0 from every direction in the plane, and that is an enormously stronger requirement than real differentiability in two variables.

Proposition 2.1.2 (Cauchy–Riemann equations). *Write $f = u + iv$ with u, v real. If f is holomorphic at $z_0 = x_0 + iy_0$ then u, v have partial derivatives there satisfying*

$$\partial_x u = \partial_y v, \quad \partial_y u = -\partial_x v,$$

and $f'(z_0) = \partial_x u + i\partial_x v$.

Proof. Take h real in the difference quotient: the limit is $\partial_x u(z_0) + i\partial_x v(z_0)$. Take $h = ik$ with k real: the limit is $\frac{1}{i}(\partial_y u(z_0) + i\partial_y v(z_0)) = \partial_y v(z_0) - i\partial_y u(z_0)$. Equating real and imaginary parts gives the two equations. $\square$

WORKED EXAMPLE 2.1.1— TIER A **Two functions of x and y , one holomorphic and one not**

(a) $f(z) = z^2 = (x^2 - y^2) + i(2xy)$. Here $u = x^2 - y^2$, $v = 2xy$, and $\partial_x u = 2x = \partial_y v$, $\partial_y u = -2y = -\partial_x v$. The Cauchy–Riemann equations hold, and indeed $f'(z) = 2z$ directly from the definition.

(b) $f(z) = \bar{z} = x - iy$. Then $u = x$, $v = -y$, so $\partial_x u = 1$ while $\partial_y v = -1$: the equations fail at every point. And directly: the difference quotient $\bar{h}/h$ equals 1 for real h and -1 for purely imaginary h , so no limit exists anywhere. Yet $\bar{z}$ is as smooth as a function of (x, y) as anything could be. Holomorphy is a rigidity condition, not a smoothness condition.

Definition 2.1.3. A *path* is a piecewise C^1 map $\gamma : [a, b] \rightarrow \mathbb{C}$, and for continuous f on its image,

$$\int_{\gamma} f(z) \, dz = \int_a^b f(\gamma(t)) \gamma'(t) \, dt.$$

The path is *closed* if $\gamma(a) = \gamma(b)$.

Two standard facts, proved exactly as in the real case: the integral is unchanged by reparametrisation, and

$$\left| \int_{\gamma} f(z) dz \right| \leq \left(\sup_{\gamma} |f| \right) \cdot \text{length}(\gamma). \quad (2.1.1)$$

Lemma 2.1.4 (Primitives kill closed integrals). *If F is holomorphic on Ω with $F' = f$ continuous, then $\int_{\gamma} f = F(\gamma(b)) - F(\gamma(a))$ for every path γ in Ω ; in particular $\int_{\gamma} f = 0$ for closed γ .*

Proof. $\frac{d}{dt} F(\gamma(t)) = f(\gamma(t))\gamma'(t)$ by the chain rule, on each C^1 piece; integrate and sum the telescoping pieces. $\square$

2.1.3 Goursat's theorem and Cauchy's theorem

Everything rests on the following, whose hypothesis is remarkably weak: no continuity of f' is assumed.

Theorem 2.1.5 (Goursat). *Let $\Omega \subseteq \mathbb{C}$ be open, f holomorphic on Ω , and let $T \subseteq \Omega$ be a closed solid triangle. Then $\int_{\partial T} f(z) dz = 0$.*

Idea of the proof. Bisect. Joining the midpoints of the three sides splits T into four similar triangles of half the linear size, and the boundary integrals over the four add up to the boundary integral over T , because each interior edge is traversed twice in opposite directions. So one of the four carries at least a quarter of the total. Iterating produces a nested sequence of triangles on which the integral is at least 4^{-k} of the original while the perimeter is 2^{-k} of the original. The triangles shrink to a point, and near that point holomorphy says f is well approximated by an affine function — whose integral over a closed path vanishes exactly, by Theorem 2.1.4. The error is o of the perimeter squared, which is precisely the scale that beats 4^{-k} .

Proof. Let $I = \int_{\partial T} f$ and let L, d denote the perimeter and diameter of T .

Step 1. *Bisection.*

Let $T^{(1)}, \dots, T^{(4)}$ be the four triangles obtained by joining the midpoints of the sides of T , each oriented positively. Every edge interior to T occurs in exactly two of the $\partial T^{(j)}$ with opposite orientations, so those contributions cancel and

$$\int_{\partial T} f = \sum_{j=1}^4 \int_{\partial T^{(j)}} f.$$

Hence some j has $|\int_{\partial T^{(j)}} f| \geq |I|/4$; call that triangle T_1 . Its perimeter is $L/2$ and its diameter $d/2$.

Iterating gives nested closed triangles $T \supseteq T_1 \supseteq T_2 \supseteq \cdots$ with

$$\left| \int_{\partial T_k} f \right| \geq \frac{|I|}{4^k}, \quad \text{length}(\partial T_k) = \frac{L}{2^k}, \quad \text{diam } T_k = \frac{d}{2^k}. \quad (2.1.2)$$

Step 2. *The common point.*

The T_k are nonempty, compact and nested with diameters tending to 0, so $\bigcap_k T_k = \{z_0\}$ for a single point $z_0 \in T \subseteq \Omega$.

Step 3. *Linearise at z_0 .*

Since f is holomorphic at z_0 , given $\varepsilon > 0$ there is $r > 0$ with

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0| \quad \text{for } |z - z_0| < r. \quad (2.1.3)$$

The affine function $z \mapsto f(z_0) + f'(z_0)(z - z_0)$ has the explicit primitive $f(z_0)z + \frac{1}{2}f'(z_0)(z - z_0)^2$, so by Theorem 2.1.4 its integral over the closed path ∂T_k is 0. Therefore, for k large enough that $T_k \subseteq B(z_0, r)$,

$$\begin{aligned} \left| \int_{\partial T_k} f \right| &= \left| \int_{\partial T_k} \left(f(z) - f(z_0) - f'(z_0)(z - z_0) \right) dz \right| \leq \varepsilon \cdot \frac{d}{2^k} \cdot \frac{L}{2^k} \\ &= \frac{\varepsilon dL}{4^k}, \end{aligned}$$

using (2.1.1), (2.1.3) and $|z - z_0| \leq \text{diam } T_k$.

Step 4. *Compare.*

With (2.1.2) this gives $|I|/4^k \leq \varepsilon dL/4^k$, so $|I| \leq \varepsilon dL$ for every $\varepsilon > 0$. Hence $I = 0$. $\square$

Theorem 2.1.6 (Cauchy's theorem on a convex domain). *Let Ω be open and convex and f holomorphic on Ω . Then f has a primitive on Ω , and consequently $\int_\gamma f = 0$ for every closed path γ in Ω .*

Proof. Fix $a \in \Omega$ and define $F(z) = \int_{[a,z]} f$, the integral along the straight segment, which lies in Ω by convexity. For $z, z+h \in \Omega$ the triangle with vertices $a, z, z+h$ lies in Ω , so Theorem 2.1.5 gives

$$F(z+h) - F(z) = \int_{[z,z+h]} f.$$

Hence

$$\begin{aligned} \left| \frac{F(z+h) - F(z)}{h} - f(z) \right| &= \frac{1}{|h|} \left| \int_{[z,z+h]} (f(w) - f(z)) dw \right| \\ &\leq \sup_{w \in [z,z+h]} |f(w) - f(z)| \xrightarrow{h \rightarrow 0} 0 \end{aligned}$$

by continuity of f (which holds since f is differentiable). So $F' = f$, and Theorem 2.1.4 applies. $\square$

We shall need one strengthening, in which f is merely continuous at one bad point.

Proposition 2.1.7. *Theorem 2.1.5 and Theorem 2.1.6 remain true if f is assumed only continuous on Ω and holomorphic on $\Omega \setminus \{p\}$ for a single point p .*

Proof. It suffices to prove the triangle statement. If $p \notin T$ we are done. If p is a vertex of T , split off a small triangle T_η at p with the same vertex and sides scaled by η ; the rest of T decomposes into triangles missing p , whose integrals vanish, so $\int_{\partial T} f = \int_{\partial T_\eta} f$, which by (2.1.1) is at most $\sup_T |f| \cdot \eta L \rightarrow 0$. Continuity of f near p makes $\sup_T |f|$ finite. If p lies elsewhere in T , subdivide T into triangles each having p as a vertex, and apply the previous case. $\square$

2.1.4 The Cauchy integral formula and analyticity

Lemma 2.1.8. *Let C be the positively oriented circle $|w - a| = r$ and let $|z - a| < r$. Then*

$$\frac{1}{2\pi i} \int_C \frac{dw}{w - z} = 1.$$

Proof. Write $\frac{1}{w - z} = \frac{1}{(w - a) - (z - a)} = \frac{1}{w - a} \cdot \frac{1}{1 - \frac{z - a}{w - a}}$. On C we have $|\frac{z - a}{w - a}| = \frac{|z - a|}{r} =: \rho < 1$, so the geometric series converges uniformly on C :

$$\frac{1}{w - z} = \sum_{k=0}^{\infty} \frac{(z - a)^k}{(w - a)^{k+1}}.$$

Uniform convergence permits term-by-term integration. Parametrising $w = a + re^{i\theta}$,

$$\int_C \frac{dw}{(w - a)^{k+1}} = \int_0^{2\pi} \frac{ire^{i\theta}}{r^{k+1}e^{i(k+1)\theta}} d\theta = \frac{i}{r^k} \int_0^{2\pi} e^{-ik\theta} d\theta = \begin{cases} 2\pi i, & k = 0, \\ 0, & k \geq 1. \end{cases}$$

Only the $k = 0$ term survives, giving $2\pi i$. $\square$

Theorem 2.1.9 (Cauchy integral formula). *Let f be holomorphic on an open set containing the closed disc $\overline{B(a, r)}$, and let $C = \partial B(a, r)$ be positively oriented. Then for every $z \in B(a, r)$,*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(w)}{w - z} dw.$$

Proof. Step 1. Reduce to a small circle about z_0 .

Fix z_0 in the interior of γ . The function

$$g(z) = \frac{f(z)}{z - z_0}$$

is holomorphic on $\Omega \setminus \{z_0\}$. By the deformation form of Cauchy's theorem — γ and the circle $C_r = \{|z - z_0| = r\}$ are homotopic in $\Omega \setminus \{z_0\}$ for all small $r > 0$ —

$$\oint_{\gamma} g(z) dz = \oint_{C_r} g(z) dz \quad \text{for every small } r > 0.$$

So it suffices to evaluate the right-hand side in the limit $r \rightarrow 0$.

Step 2. *Split off the constant term.*

Write $f(z) = f(z_0) + (f(z) - f(z_0))$, so that

$$\oint_{C_r} \frac{f(z)}{z - z_0} dz = f(z_0) \oint_{C_r} \frac{dz}{z - z_0} + \oint_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz.$$

Step 3. *The first term is a computation.*

Parametrising $z = z_0 + re^{i\vartheta}$, $dz = ire^{i\vartheta} d\vartheta$,

$$\oint_{C_r} \frac{dz}{z - z_0} = \int_0^{2\pi} \frac{ire^{i\vartheta}}{re^{i\vartheta}} d\vartheta = 2\pi i,$$

independently of r .

Step 4. *The second term vanishes in the limit.*

Since f is holomorphic at z_0 it is in particular continuous there, so given $\eta > 0$ there is r with $|f(z) - f(z_0)| \leq \eta$ on C_r . Then

$$\begin{aligned} \left| \oint_{C_r} \frac{f(z) - f(z_0)}{z - z_0} dz \right| &\leq \text{length}(C_r) \cdot \sup_{C_r} \frac{|f(z) - f(z_0)|}{|z - z_0|} \\ &\leq 2\pi r \cdot \frac{\eta}{r} \\ &= 2\pi\eta, \end{aligned}$$

by the standard estimate for contour integrals. As $\eta > 0$ was arbitrary and the left side of Step 1 does not depend on r , this term is 0.

Step 5. *Combine.*

Steps 1–4 give $\oint_{\gamma} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$, which rearranges to the stated formula. $\square$

Theorem 2.1.10 (Holomorphic implies analytic). *Let f be holomorphic on $B(a, R)$. Then f has a power series expansion $f(z) = \sum_{k \geq 0} c_k (z - a)^k$ converging on $B(a, R)$, with*

$$c_k = \frac{1}{2\pi i} \int_{|w-a|=r} \frac{f(w)}{(w-a)^{k+1}} dw \quad (0 < r < R).$$

In particular f is infinitely differentiable and $c_k = f^{(k)}(a)/k!$.

Proof. Fix $r < R$ and $|z - a| < r$. Insert the geometric expansion of Theorem 2.1.8's proof into Theorem 2.1.9:

$$f(z) = \frac{1}{2\pi i} \int_{|w-a|=r} f(w) \sum_{k=0}^{\infty} \frac{(z-a)^k}{(w-a)^{k+1}} dw.$$

The series converges uniformly in w on the circle (the ratio is $\leq \rho < 1$ and f is bounded on the circle), so term-by-term integration is legitimate and yields the stated c_k . The coefficients do not depend on r , since two circles give the same value by Theorem 2.1.6 applied on an annulus-avoiding convex subregion; letting $r \uparrow R$ gives convergence on all of $B(a, R)$. Power series are infinitely differentiable term by term inside their disc of convergence, whence the last claim. $\square$

Corollary 2.1.11 (Cauchy estimates and Liouville). *If f is holomorphic on $B(a, R)$ with $|f| \leq M$ there, then $|f^{(k)}(a)| \leq k! M/R^k$. Consequently a bounded entire function is constant.*

Proof. Apply (2.1.1) to the formula for c_k with radius $r < R$: $|c_k| \leq \frac{1}{2\pi} \cdot \frac{M}{r^{k+1}} \cdot 2\pi r = M/r^k$; let $r \uparrow R$. If f is entire and bounded, take $k = 1$ and $R \rightarrow \infty$ at every a to get $f' \equiv 0$. $\square$

2.1.5 The maximum modulus principle

Theorem 2.1.12. *Let $\Omega \subseteq \mathbb{C}$ be open and connected and f holomorphic on Ω . If $|f|$ attains a local maximum at some $z_0 \in \Omega$, then f is constant. Consequently, if Ω is bounded and f is continuous on $\bar{\Omega}$ and holomorphic on Ω , then*

$$\sup_{\bar{\Omega}} |f| = \sup_{\partial\Omega} |f|.$$

Idea of the proof. Cauchy's formula on a circle centred at z_0 exhibits $f(z_0)$ as the average of f over that circle. An average of complex numbers of modulus at most $|f(z_0)|$ can equal $f(z_0)$ only if they all point the same way and have the same modulus. So $|f|$ is locally constant, and then f is.

Proof. Step 1. The mean value property.

Parametrising $w = z_0 + re^{i\theta}$ in Theorem 2.1.9 gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta \quad (2.1.4)$$

for every sufficiently small $r > 0$.

Step 2. *Local constancy of $|f|$.*

Suppose $|f(w)| \leq |f(z_0)| =: M$ for all w in some ball $B(z_0, \rho)$. Taking absolute values in (2.1.4),

$$M \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{i\theta})| d\theta \leq M \quad (0 < r < \rho),$$

so $\int_0^{2\pi} (M - |f(z_0 + re^{i\theta})|) d\theta = 0$ with a continuous nonnegative integrand; hence $|f| = M$ on every such circle, i.e. on $B(z_0, \rho)$.

Step 3. *$|f|$ constant forces f constant.*

If $M = 0$ then $f \equiv 0$ on the ball. If $M > 0$, write $f = u + iv$ with $u^2 + v^2 = M^2$. Differentiating in x and in y gives $uu_x + vv_x = 0$ and $uu_y + vv_y = 0$; substituting the Cauchy–Riemann equations $u_y = -v_x$, $v_y = u_x$ turns the second into $-uv_x + vu_x = 0$. The linear system

$$\begin{pmatrix} u & v \\ v & -u \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix} = 0$$

has determinant $-(u^2 + v^2) = -M^2 \neq 0$, so $u_x = v_x = 0$, and then $u_y = v_y = 0$. Hence f is constant on $B(z_0, \rho)$.

Step 4. *Propagate.*

The set where f equals the constant $f(z_0)$ is nonempty, closed in Ω by continuity, and open by Theorem 2.1.10 (a holomorphic function agreeing with a constant on a ball has all higher Taylor coefficients zero at each point of that ball, so agrees with it on any ball where its series converges). Connectedness gives $f \equiv f(z_0)$.

For the boundary statement: $\overline{\Omega}$ is compact, so $|f|$ attains a maximum somewhere; if that point is interior, f is constant by the above and the conclusion is trivial. $\square$

WORKED EXAMPLE 2.1.2— TIER B The maximum principle fails on unbounded domains

Boundedness of Ω is not decoration. Let

$$S = \left\{ z : |\operatorname{Im} z| < \frac{\pi}{2} \right\}, \quad F(z) = \exp(\exp(z)),$$

which is entire, hence holomorphic on S and continuous on $\overline{S}$.

On the boundary F is unimodular. If $\operatorname{Im} z = \pm\pi/2$, write $z = x \pm i\pi/2$; then $\exp(z) = e^x e^{\pm i\pi/2} = \pm ie^x$, which is purely imaginary, so

$$|F(z)| = |\exp(\pm ie^x)| = 1.$$

Inside, F is unbounded. At $z = x$ real, $F(x) = \exp(e^x) \rightarrow \infty$ as $x \rightarrow +\infty$.

So $\sup_{\partial S} |F| = 1$ while $\sup_S |F| = \infty$. The conclusion of Theorem 2.1.12 is false on this unbounded region.

What went wrong, and what to add. The proof of Theorem 2.1.12 used compactness of $\bar{\Omega}$ to produce a maximum. On an unbounded region the supremum need not be attained, and mass can escape to infinity — this is escape mode (a) of Worked Example 1.5.2 (the three ways mass escapes), in a new costume. The repair is to *assume* that no such escape occurs, i.e. that f is bounded, and that is exactly the hypothesis of Theorem 2.1.13 below. The example shows the hypothesis cannot be dropped: F here is bounded on ∂S but not on S .

2.1.6 Phragmén–Lindelöf on a strip

Write

$$\mathbb{S}_{01} = \{z \in \mathbb{C} : 0 \leq \operatorname{Re} z \leq 1\}, \quad \mathbb{S}_{01}^\circ = \{0 < \operatorname{Re} z < 1\}.$$

Theorem 2.1.13 (Phragmén–Lindelöf for the strip). *Let G be continuous and bounded on $\mathbb{S}_{01}$ and holomorphic on $\mathbb{S}_{01}^\circ$. If $|G| \leq 1$ on the two boundary lines $\operatorname{Re} z = 0$ and $\operatorname{Re} z = 1$, then $|G| \leq 1$ on all of $\mathbb{S}_{01}$.*

Idea of the proof. Worked Example 2.1.2 showed we cannot simply apply the maximum principle on an unbounded region. The device is to multiply G by a small holomorphic factor that decays as $|\operatorname{Im} z| \rightarrow \infty$ and is harmless on the two boundary lines, turning the strip into an effectively bounded region; then apply Theorem 2.1.12 on a large rectangle and let the factor go away.

The factor $e^{\varepsilon(z^2-z)}$ does it. Its modulus is $e^{\varepsilon(x^2-y^2-x)}$, which tends to 0 as $|y| \rightarrow \infty$ for fixed x , and equals $e^{\varepsilon(x^2-x)}$ when $y = 0$; the polynomial $x^2 - x$ vanishes at both $x = 0$ and $x = 1$, which is exactly why the boundary bound survives untouched.

Proof. Let $M = \sup_{\mathbb{S}_{01}} |G| < \infty$ and fix $\varepsilon > 0$. Define

$$G_\varepsilon(z) = G(z) e^{\varepsilon(z^2-z)}.$$

Step 1. $|G_\varepsilon| \leq 1$ on $\partial \mathbb{S}_{01}$.

For $z = x + iy$ we have $\operatorname{Re}(z^2 - z) = x^2 - y^2 - x$, so

$$|e^{\varepsilon(z^2-z)}| = e^{\varepsilon(x^2-x-y^2)} \leq e^{\varepsilon(x^2-x)}. \quad (2.1.5)$$

At $x = 0$ and $x = 1$ the exponent $x^2 - x$ vanishes, so the factor has modulus at most 1 there and $|G_\varepsilon| \leq |G| \leq 1$ on both boundary lines.

Step 2. G_ε is small high up and low down.

By (2.1.5), for z in the strip $(0 \leq x \leq 1, \text{ so } x^2 - x \in [-\frac{1}{4}, 0])$,

$$|G_\varepsilon(z)| \leq M e^{-\varepsilon y^2}.$$

Choose $Y > 0$ with $M e^{-\varepsilon Y^2} \leq 1$.

Step 3. Apply the maximum principle on a rectangle.

Let $R = \{0 \leq x \leq 1, |y| \leq Y\}$, a bounded region. On its boundary, $|G_\varepsilon| \leq 1$: on the vertical sides by Step 1, on the horizontal sides by Step 2. By Theorem 2.1.12, $|G_\varepsilon| \leq 1$ on R . Combined with Step 2 for $|y| > Y$, we get $|G_\varepsilon| \leq 1$ on all of $\mathbb{S}_{01}$.

Step 4. Remove the factor.

Fix $z \in \mathbb{S}_{01}$. From $|G(z)| \leq e^{-\varepsilon \operatorname{Re}(z^2 - z)}$ and letting $\varepsilon \downarrow 0$ with z fixed, the right side tends to 1. Hence $|G(z)| \leq 1$. $\square$

Theorem 2.1.14 (Three-lines lemma). *Let F be continuous and bounded on $\mathbb{S}_{01}$ and holomorphic on $\mathbb{S}_{01}^\circ$. Put*

$$M_0 = \sup_{y \in \mathbb{R}} |F(iy)|, \quad M_1 = \sup_{y \in \mathbb{R}} |F(1 + iy)|.$$

Then for every $\theta \in [0, 1]$ and $y \in \mathbb{R}$,

$$|F(\theta + iy)| \leq M_0^{1-\theta} M_1^\theta.$$

Proof. Suppose first $M_0, M_1 > 0$. Define

$$G(z) = F(z) M_0^{z-1} M_1^{-z},$$

where $M^w := e^{w \log M}$ with the real logarithm, so that M^w is entire in w and $|M^w| = M^{\operatorname{Re} w}$. Then G is continuous on $\mathbb{S}_{01}$, holomorphic inside, and bounded there, since $|M_0^{z-1} M_1^{-z}| = M_0^{x-1} M_1^{-x}$ is bounded for $x \in [0, 1]$.

On $\operatorname{Re} z = 0$: $|G| = |F| M_0^{-1} \leq 1$. On $\operatorname{Re} z = 1$: $|G| = |F| M_1^{-1} \leq 1$. By Theorem 2.1.13, $|G| \leq 1$ on $\mathbb{S}_{01}$, i.e.

$$|F(\theta + iy)| \leq M_0^{1-\theta} M_1^\theta$$

after multiplying back.

If $M_0 = 0$, then F vanishes identically on the line $\operatorname{Re} z = 0$; applying the above with M_0 replaced by $\eta > 0$ gives $|F(\theta + iy)| \leq \eta^{1-\theta} M_1^\theta$ for every $\eta > 0$, whence $F(\theta + iy) = 0$ for $\theta < 1$ and the inequality holds trivially. The case $M_1 = 0$ is symmetric. $\square$

WORKED EXAMPLE 2.1.3— TIER C: RESEARCH TASTE **Logarithmic convexity, and why every exponent range in this series is an interval**

Theorem 2.1.14 says that $\theta \mapsto \log(\sup_y |F(\theta + iy)|)$ is a convex function of θ on $[0, 1]$: that is the content of the inequality $M_\theta \leq M_0^{1-\theta} M_1^\theta$, after taking logarithms.

Why this shape is inevitable. Suppose one has a family of estimates indexed by a parameter, and suppose the quantity being estimated can be realised as the boundary data of a bounded holomorphic function on a strip. Then the best constant is automatically log-convex in the parameter, hence finite on an *interval*, and controlled at interior points by the endpoints. That is the entire mechanism of Chapter 2.2, and it explains a pattern the reader will meet again and again:

- the Hausdorff–Young inequality $\|\widehat{f}\|_{p'} \leq \|f\|_p$ holds exactly for $1 \leq p \leq 2$ (Chapter 2.4);
- the Stein–Tomas restriction estimate holds for $p \leq \frac{2(n+1)}{n+3}$, an interval with one endpoint at $p = 1$ (Volume 6);
- the ℓ^2 decoupling theorem for the paraboloid holds exactly for $2 \leq p \leq \frac{2(n+1)}{n-1}$ (Volume 13);
- the Kakeya maximal conjecture is stated at the single exponent $p = n$ because that is the endpoint of the interval on which it is true, everything below following by interpolation.

In every case the set of good exponents is a closed interval, and in every case the reason is the same convexity. When a paper says “by interpolation it suffices to treat the endpoint”, this theorem is what is being invoked.

A caution that will matter. Log-convexity is a statement about the *constant*, not about truth. Interpolation between two true estimates gives a true estimate with a controlled constant; it never improves an endpoint. The entire difficulty of Volumes 12 and 13 is that the endpoint estimates are the hard ones, and no amount of interpolation manufactures them. Readers who find themselves hoping to interpolate their way to the restriction conjecture should return to this paragraph.

Drills

Drill 2.1.1. Verify the Cauchy–Riemann equations for $f(z) = e^z = e^x(\cos y + i \sin y)$ and compute f' . Then show that $|e^z| = e^{\operatorname{Re} z}$, the identity used

throughout §1.6.

Drill 2.1.2. Show that the conclusion of Theorem 2.1.5 fails for $f(z) = 1/z$ on a triangle containing 0, and identify precisely which hypothesis is violated.

Drill 2.1.3. In the proof of Theorem 2.1.13, replace the factor $e^{\varepsilon(z^2-z)}$ by $e^{\varepsilon z^2}$ and determine what goes wrong on the boundary lines. Then find the general condition on a polynomial q for $e^{\varepsilon q(z)}$ to serve.

Drill 2.1.4. Deduce Hadamard's three-circle theorem from Theorem 2.1.14: if f is holomorphic on an annulus $r_0 < |z| < r_1$ and $M(r) = \sup_{|z|=r} |f|$, then $\log M$ is a convex function of $\log r$. (Substitute $z \mapsto e^w$ and rescale the strip.)

Notes

Goursat's theorem is from 1900 and is the reason no continuity hypothesis on f' appears anywhere in this chapter. The Phragmén–Lindelöf principle dates to 1908; Worked Example 2.1.2 is their example. The three-lines lemma in the form Theorem 2.1.14 is due to Doetsch (1920), though the idea is Hadamard's.

Readers who have seen complex analysis before will notice that we have proved only what is needed: no residue calculus, no argument principle, no Riemann mapping theorem. Volume 5 will require contour deformation and the method of steepest descent, which are developed there on the foundation laid here.

Chapter 2.2

Interpolation: Riesz–Thorin and Marcinkiewicz

Volume 1 proved that the Hardy–Littlewood maximal operator satisfies a weak $(1, 1)$ bound and an (∞, ∞) bound, and then obtained the L^p bounds for $1 < p < \infty$ by a hand-rolled argument: split f at height λ , apply the weak bound to the tall part, integrate with the layer-cake formula. That argument used nothing about the maximal operator. This chapter extracts it, in two forms.

The first, Riesz–Thorin, interpolates between *strong* bounds and requires linearity; its proof is the three-lines lemma and its constant is sharp. The second, Marcinkiewicz, interpolates between *weak* bounds and tolerates sublinear operators; its proof is the layer-cake argument and its constant blows up at the endpoints. Both are used constantly. Knowing which one applies is a matter of asking whether the available endpoint information is strong or weak, and covering arguments — as Volume 1 emphasised — produce weak information.

2.2.1 Riesz–Thorin

Throughout, (X, μ) and (Y, ν) are σ -finite measure spaces. Recall that a *simple function of finite support* is a simple function vanishing outside a set of finite measure; by Theorem 1.8.7 (density of simple functions and C_c^∞), these are dense in $L^p(\mu)$ for every $p < \infty$.

Theorem 2.2.1 (Riesz–Thorin). *Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ and let T be a linear map, defined on simple functions of finite support on X , taking values in measurable functions on Y , with*

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0}, \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

for all such f . For $\theta \in (0, 1)$ define

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then $\|Tf\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta \|f\|_{p_\theta}$, and if $p_\theta < \infty$ the operator T extends uniquely to a bounded map $L^{p_\theta}(\mu) \rightarrow L^{q_\theta}(\nu)$ with that norm.

Idea of the proof. We must bound $\|Tf\|_{q_\theta}$, and by duality (Theorem 1.8.10 (norms by testing)) that is the same as bounding $|\int (Tf)g \, d\nu|$ over g in the unit ball of $L^{q'_\theta}$. So the quantity to control is the bilinear pairing

$$\Phi = \int_Y (Tf)g \, d\nu.$$

Now deform. Replace f and g by families f_z and g_z depending on a complex parameter z in the strip, chosen so that: at $z = \theta$ they are the original f and g ; on the line $\operatorname{Re} z = 0$ the function f_z has the right L^{p_0} norm and g_z the right L^{q_0} norm, so that the first hypothesis applies; and on $\operatorname{Re} z = 1$ likewise for the second. Then $\Phi(z) = \int (Tf_z)g_z$ is a bounded holomorphic function of z with $|\Phi| \leq M_0$ on one boundary line and $\leq M_1$ on the other, and Theorem 2.1.14 gives $|\Phi(\theta)| \leq M_0^{1-\theta} M_1^\theta$.

The families are forced. To move the L^p norm of $|f|^\alpha$ around one raises $|f|$ to a power, and to keep everything holomorphic in z that power must be affine in z . Writing $f = |f|e^{i \arg f}$ and putting

$$f_z = |f|^{p_\theta \left(\frac{1-z}{p_0} + \frac{z}{p_1} \right)} e^{i \arg f}$$

does exactly this: at $z = \theta$ the exponent is 1, and on $\operatorname{Re} z = j$ it is p_θ/p_j , which is what makes $\|f_z\|_{p_j} = \|f\|_{p_\theta}^{p_\theta/p_j}$.

Proof. Assume first $p_\theta < \infty$ and $q_\theta > 1$; the excluded cases are treated in Step 5.

Step 1. Set up the analytic families.

Normalise $\|f\|_{p_\theta} = 1$ and $\|g\|_{q'_\theta} = 1$, with f and g simple of finite support. Write

$$f = \sum_{j=1}^m a_j e^{i\alpha_j} \mathbf{1}_{A_j}, \quad g = \sum_{k=1}^n b_k e^{i\beta_k} \mathbf{1}_{B_k},$$

with $a_j, b_k > 0$, the A_j pairwise disjoint of finite measure, likewise B_k . Define

$$\eta(z) = p_\theta \left(\frac{1-z}{p_0} + \frac{z}{p_1} \right), \quad \zeta(z) = q'_\theta \left(\frac{1-z}{q'_0} + \frac{z}{q'_1} \right),$$

and

$$f_z = \sum_j a_j^{\eta(z)} e^{i\alpha_j} \mathbf{1}_{A_j}, \quad g_z = \sum_k b_k^{\zeta(z)} e^{i\beta_k} \mathbf{1}_{B_k}.$$

Since $a_j^{\eta(z)} = e^{\eta(z) \log a_j}$ with η affine, each f_z is a finite sum of entire functions of z times fixed indicators; the same for g_z . Note $\eta(\theta) = 1$ and $\zeta(\theta) = 1$, so $f_\theta = f$ and $g_\theta = g$.

Step 2. *The pairing is holomorphic and bounded.*

Put

$$\begin{aligned} \Phi(z) &= \int_Y (Tf_z) g_z \, d\nu \\ &= \sum_{j,k} a_j^{\eta(z)} b_k^{\zeta(z)} e^{i(\alpha_j + \beta_k)} \int_{B_k} T(\mathbf{1}_{A_j}) \, d\nu, \end{aligned}$$

using linearity of T . This is a *finite* linear combination of exponentials e^{cz} with fixed coefficients, so Φ is entire; and on the closed strip each $|a_j^{\eta(z)}| = a_j^{\operatorname{Re} \eta(z)}$ is bounded, because $\operatorname{Re} \eta$ ranges over the bounded interval between p_θ/p_0 and p_θ/p_1 . Hence Φ is bounded on $\mathbb{S}_0$. The coefficients $\int_{B_k} T(\mathbf{1}_{A_j}) \, d\nu$ are finite: $\mathbf{1}_{A_j}$ is simple of finite support, so $T\mathbf{1}_{A_j} \in L^{q_0}$, and B_k has finite measure.

Step 3. *Bound Φ on the two boundary lines.*

Fix $y \in \mathbb{R}$ and consider $z = iy$. Then $\operatorname{Re} \eta(iy) = p_\theta/p_0$, so

$$|f_{iy}|^{p_0} = \sum_j a_j^{p_\theta} \mathbf{1}_{A_j} = |f|^{p_\theta}, \quad \text{hence} \quad \|f_{iy}\|_{p_0} = \|f\|_{p_\theta}^{p_\theta/p_0} = 1.$$

Identically $\|g_{iy}\|_{q'_0} = \|g\|_{q'_\theta}^{q'_\theta/q'_0} = 1$. Therefore, by Hölder's inequality and the first hypothesis,

$$|\Phi(iy)| \leq \|Tf_{iy}\|_{q_0} \|g_{iy}\|_{q'_0} \leq M_0 \|f_{iy}\|_{p_0} = M_0.$$

The same computation on $\operatorname{Re} z = 1$ gives $|\Phi(1 + iy)| \leq M_1$.

Step 4. *Apply the three-lines lemma.*

By Theorem 2.1.14, $|\Phi(\theta)| \leq M_0^{1-\theta} M_1^\theta$. But $\Phi(\theta) = \int (Tf)g \, d\nu$, so

$$\left| \int_Y (Tf)g \, d\nu \right| \leq M_0^{1-\theta} M_1^\theta$$

for every simple g of finite support with $\|g\|_{q'_\theta} \leq 1$. Since such g are dense in the unit ball of $L^{q'_\theta}$ and $q'_\theta < \infty$, the duality formula of Theorem 1.8.10 (norms by testing), gives $\|Tf\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta$.

Step 5. *The excluded cases, and the extension.*

If $q_\theta = 1$ then $q_0 = q_1 = 1$ and one applies Hölder with exponents 1 and ∞ in Step 3, testing against g in the unit ball of L^∞ ; the duality step is then Theorem 1.8.8 (L^p duality) for $p = 1$. If $p_\theta = \infty$ then $p_0 = p_1 = \infty$ and there is nothing to interpolate on the domain side; the argument runs with $f_z = f$.

Finally, for $p_\theta < \infty$ the simple functions of finite support are dense in $L^{p_\theta}(\mu)$, and a bounded linear map on a dense subspace of a normed space, with values in the complete space $L^{q_\theta}(\nu)$ (Volume 1, Theorem 8.6), extends uniquely by continuity with the same norm. $\square$

Pitfall: Riesz–Thorin needs linearity, and complex scalars

Two hypotheses are load-bearing and are routinely forgotten.

Linearity. Step 2 used $T(\sum_j c_j \mathbf{1}_{A_j}) = \sum_j c_j T\mathbf{1}_{A_j}$ to make Φ entire. For a merely sublinear operator — the maximal function, the Kakeya maximal operator, any supremum — this fails completely, and Theorem 2.2.1 does not apply. Use Theorem 2.2.4 instead.

Complex scalars. The whole proof runs in the complex plane. For real-linear operators on real L^p spaces the theorem holds with an extra constant, and there are examples showing a constant is genuinely needed. In this series all function spaces are complex, so the issue does not arise; but it is worth knowing that the sharpness of the constant $M_0^{1-\theta} M_1^\theta$ is a complex phenomenon.

WORKED EXAMPLE 2.2.1 — TIER B Young’s convolution inequality, in full generality

Theorem 1.6.9 (Young’s inequality, $q = 1$) proved $\|f * g\|_p \leq \|f\|_p \|g\|_1$ and deferred the general case. We can now complete it.

Proposition 2.2.2. *For $1 \leq p, q, r \leq \infty$ with $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$,*

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

Proof. Fix $g \in L^q$ and let $Tf = f * g$, a linear operator in f . Two endpoint

estimates are available with no work:

$$\begin{aligned} \|f * g\|_q &\leq \|f\|_1 \|g\|_q && \text{(Volume 1, Prop. 6.9, roles reversed),} \\ \|f * g\|_\infty &\leq \|f\|_{q'} \|g\|_q && \text{(Hölder, applied pointwise).} \end{aligned}$$

For the second: $|(f * g)(x)| \leq \int |f(x - y)| |g(y)| dy \leq \|f(x - \cdot)\|_{q'} \|g\|_q = \|f\|_{q'} \|g\|_q$.

So T maps $L^{p_0} \rightarrow L^{q_0}$ with $p_0 = 1$, $q_0 = q$, and $L^{p_1} \rightarrow L^{q_1}$ with $p_1 = q'$, $q_1 = \infty$, both with norm at most $\|g\|_q$. Interpolating by Theorem 2.2.1 at parameter $\theta \in (0, 1)$ gives a bound $L^{p_\theta} \rightarrow L^{q_\theta}$ with norm at most $(\|g\|_q)^{1-\theta} (\|g\|_q)^\theta = \|g\|_q$, where

$$\begin{aligned} \frac{1}{p_\theta} &= (1 - \theta) \cdot 1 + \theta \cdot \frac{1}{q'} = 1 - \frac{\theta}{q} \\ &, \quad \frac{1}{q_\theta} = \frac{1 - \theta}{q} + \theta \cdot 0 = \frac{1 - \theta}{q}. \end{aligned}$$

Adding the two identities,

$$\frac{1}{p_\theta} + \frac{1}{q} = 1 - \frac{\theta}{q} + \frac{1}{q} = 1 + \frac{1 - \theta}{q} = 1 + \frac{1}{q_\theta},$$

which is precisely $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ with $p = p_\theta$, $r = q_\theta$. As θ runs over $(0, 1)$, $r = q/(1 - \theta)$ runs over (q, ∞) , and together with the two endpoint cases already established this covers every admissible triple. $\square$

Reading the exponent relation. The condition $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ is forced by scaling: replacing $f(x)$ by $f(\lambda x)$ multiplies $\|f\|_p$ by $\lambda^{-n/p}$, and the two sides of Young's inequality scale the same way precisely under that relation; Drill 2.2 asks for the computation. Whenever an inequality in this series has a mysterious-looking exponent condition, testing it against dilations is the first thing to do; Volume 6 uses this repeatedly to locate the conjectured range for restriction estimates. $\square$

2.2.2 Marcinkiewicz

We now interpolate weak-type information. Recall from Volume 1, §8.5 that $\|f\|_{p,\infty} = \sup_{\lambda>0} \lambda d_f(\lambda)^{1/p}$ where $d_f(\lambda) = \mu(\{|f| > \lambda\})$.

Definition 2.2.3. An operator T , defined on a vector space $\mathcal{D}$ of measurable functions and taking values in measurable functions, is *sublinear* if $|T(f + g)| \leq |Tf| + |Tg|$ and $|T(cf)| = |c||Tf|$ pointwise. It is of *weak type* (p, q) with constant M if $\|Tf\|_{q,\infty} \leq M\|f\|_p$, and of *strong type* (p, q) if $\|Tf\|_q \leq M\|f\|_p$.

Strong type (p, q) implies weak type (p, q) , by Chebyshev (Volume 1, Proposition 8.14).

Theorem 2.2.4 (Marcinkiewicz interpolation). *Let $1 \leq p_0 < p_1 \leq \infty$ and let T be sublinear, defined on $L^{p_0}(\mu) + L^{p_1}(\mu)$, of weak type (p_0, p_0) with constant M_0 and of weak type (p_1, p_1) with constant M_1 . Then for every $p_0 < p < p_1$, T is of strong type (p, p) :*

$$\|Tf\|_p \leq C(p, p_0, p_1) M_0^{1-\theta} M_1^\theta \|f\|_p, \quad \frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1},$$

where $C(p, p_0, p_1) \rightarrow \infty$ as $p \downarrow p_0$ or $p \uparrow p_1$.

Idea of the proof. Fix a height λ and split f into its tall and short parts, $f = f\mathbf{1}_{\{|f|>c\lambda\}} + f\mathbf{1}_{\{|f|\leq c\lambda\}}$. The tall part lies in L^{p_0} (a function that is large where it is supported is more integrable at the *lower* exponent), the short part in L^{p_1} . Sublinearity splits $\{|Tf| > \lambda\}$ between the two, each is estimated by the corresponding weak bound, and the layer-cake formula integrates the result in λ . The constant c is a free parameter, optimised at the end; the divergence of C at the endpoints comes from an integral $\int_0^\infty \lambda^{p-p_0-1}(\dots)$ which converges only for $p_0 < p < p_1$.

Proof. Assume first $p_1 < \infty$. Fix $f \in L^p$ and $\lambda > 0$, and a constant $c > 0$ to be chosen. Write

$$f = g_\lambda + h_\lambda, \quad g_\lambda = f\mathbf{1}_{\{|f|>c\lambda\}}, \quad h_\lambda = f\mathbf{1}_{\{|f|\leq c\lambda\}}.$$

Step 1. *The two pieces lie in the right spaces.*

Since $|g_\lambda| \geq c\lambda$ where it is nonzero, $|g_\lambda|^{p_0} \leq (c\lambda)^{p_0-p} |f|^{p_0}$, so $g_\lambda \in L^{p_0}$. Since $|h_\lambda| \leq c\lambda$, $|h_\lambda|^{p_1} \leq (c\lambda)^{p_1-p} |f|^{p_1}$, so $h_\lambda \in L^{p_1}$.

Step 2. *Split the level set.*

By sublinearity, $|Tf| \leq |Tg_\lambda| + |Th_\lambda|$, so

$$\{|Tf| > \lambda\} \subseteq \left\{|Tg_\lambda| > \frac{\lambda}{2}\right\} \cup \left\{|Th_\lambda| > \frac{\lambda}{2}\right\},$$

and therefore, applying the two weak-type hypotheses,

$$d_{Tf}(\lambda) \leq \left(\frac{2M_0}{\lambda}\right)^{p_0} \|g_\lambda\|_{p_0}^{p_0} + \left(\frac{2M_1}{\lambda}\right)^{p_1} \|h_\lambda\|_{p_1}^{p_1}. \quad (2.2.1)$$

Step 3. *Integrate.*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) and (2.2.1),

$$\|Tf\|_p^p = p \int_0^\infty \lambda^{p-1} d_{Tf}(\lambda) d\lambda \leq \text{I} + \text{II},$$

where

$$\begin{aligned} \text{I} &= p (2M_0)^{p_0} \int_0^\infty \lambda^{p-p_0-1} \|g_\lambda\|_{p_0}^{p_0} d\lambda, \\ \text{II} &= p (2M_1)^{p_1} \int_0^\infty \lambda^{p-p_1-1} \|h_\lambda\|_{p_1}^{p_1} d\lambda. \end{aligned}$$

For I, write $\|g_\lambda\|_{p_0}^{p_0} = \int_{\{|f|>c\lambda\}} |f|^{p_0} d\mu$ and apply Tonelli, exchanging the λ and x integrations:

$$\begin{aligned} \text{I} &= p (2M_0)^{p_0} \int_X |f(x)|^{p_0} \left(\int_0^{|f(x)|/c} \lambda^{p-p_0-1} d\lambda \right) d\mu(x) \\ &= \frac{p (2M_0)^{p_0} c^{p_0-p}}{p-p_0} \int_X |f|^p d\mu, \end{aligned}$$

the inner integral converging at 0 precisely because $p > p_0$.

For II, similarly $\|h_\lambda\|_{p_1}^{p_1} = \int_{\{|f|\leq c\lambda\}} |f|^{p_1} d\mu$, and

$$\begin{aligned} \text{II} &= p (2M_1)^{p_1} \int_X |f(x)|^{p_1} \left(\int_{|f(x)|/c}^\infty \lambda^{p-p_1-1} d\lambda \right) d\mu(x) \\ &= \frac{p (2M_1)^{p_1} c^{p_1-p}}{p_1-p} \int_X |f|^p d\mu, \end{aligned}$$

the integral converging at ∞ precisely because $p < p_1$.

Step 4. *Optimise in c .*

Adding,

$$\|Tf\|_p^p \leq p \left(\frac{(2M_0)^{p_0} c^{p_0-p}}{p-p_0} + \frac{(2M_1)^{p_1} c^{p_1-p}}{p_1-p} \right) \|f\|_p^p.$$

Choosing $c = M_0^{p_0/(p_1-p_0)} M_1^{-p_1/(p_1-p_0)}$ balances the powers of M_0 and M_1 and produces, after simplification, a bound of the stated form $C M_0^{(1-\theta)p} M_1^{\theta p} \|f\|_p^p$ with $C = C(p, p_0, p_1)$ depending only on the exponents and blowing up like $(p-p_0)^{-1}$ and $(p_1-p)^{-1}$ at the two ends.

Step 5. *The case $p_1 = \infty$.*

Weak type (∞, ∞) means $\|Tf\|_\infty \leq M_1 \|f\|_\infty$. Choose $c = 1/(2M_1)$, so that $\|Th_\lambda\|_\infty \leq M_1 c \lambda = \lambda/2$ and the second set in Step 2 is empty. Then only term I survives and the computation above gives the result directly, with $C = (p 2^{p_0} (2M_1)^{p-p_0} / (p-p_0))^{1/p}$. $\square$

WORKED EXAMPLE 2.2.2— TIER A **Recovering the maximal function bound**

Theorem 1.7.3 (Hardy–Littlewood maximal inequality) proved the Hardy–Littlewood maximal operator is of weak type $(1, 1)$ with constant 3^n , and satisfies $\|\mathcal{M}f\|_\infty \leq \|f\|_\infty$. It is sublinear, since a supremum of averages of $|f + g|$ is at most the sum of the suprema.

Apply Theorem 2.2.4 with $p_0 = 1$, $p_1 = \infty$, $M_0 = 3^n$, $M_1 = 1$. Step 5 of the proof gives, for $1 < p < \infty$,

$$\|\mathcal{M}f\|_p \leq \left(\frac{p 2^p 3^n}{p-1} \right)^{1/p} \|f\|_p.$$

This is exactly Step 3 of the proof of Theorem 1.7.3 (Hardy–Littlewood maximal inequality) — the same computation, now performed once and for all rather than for one operator. Note the constant diverging as $p \downarrow 1$, and recall from Volume 1, Worked Example 7.6 that this is unavoidable: $\mathcal{M}f \notin L^1$ for any $f \neq 0$.

 WORKED EXAMPLE 2.2.3— TIER C: RESEARCH TASTE **Where the exponent $p = n$ in the Keakeya maximal conjecture comes from**

Worked Example 1.7.2 (the Keakeya maximal conjecture) stated the Keakeya maximal function conjecture at the single exponent $p = n$ and asserted that the exponent is forced. We can now say exactly why. Recall the two trivial bounds established there, for $\mathcal{K}_\delta f(e) = \sup_a m(T_\delta^e(a))^{-1} \int_{T_\delta^e(a)} |f|$:

$$\begin{aligned} \|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} &\leq \|f\|_\infty \\ &, \quad \|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \\ &\leq c_n \delta^{-(n-1)} \|f\|_1. \end{aligned}$$

(a) *What interpolation gives.* $\mathcal{K}_\delta$ is sublinear, so Theorem 2.2.4 applies with $p_0 = 1$, $p_1 = \infty$, $M_0 = c_n \delta^{-(n-1)}$, $M_1 = 1$, target exponent ∞ in both cases. (Strictly, the two bounds are of strong type $(1, \infty)$ and (∞, ∞) , so one may interpolate them directly by the elementary argument of Step 5.) The result is

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathbb{S}^{n-1})} \lesssim_n \delta^{-(n-1)/p} \|f\|_p,$$

and since $\mathbb{S}^{n-1}$ has finite measure, $\|\cdot\|_{L^p(\mathbb{S}^{n-1})} \lesssim \|\cdot\|_{L^\infty(\mathbb{S}^{n-1})}$, so

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \lesssim_n \delta^{-(n-1)/p} \|f\|_{L^p(\mathbb{R}^n)}. \quad (2.2.2)$$

At $p = n$ this reads $\delta^{-(n-1)/n}$. The conjecture asserts the loss is only $\delta^{-\varepsilon}$:

the entire content is the removal of a fixed power of δ from a bound that interpolation already supplies.

(b) *Why no smaller exponent can work.* Take $f = \mathbf{1}_{B(0,\delta)}$, the indicator of a ball of radius δ . Then $\|f\|_{L^p(\mathbb{R}^n)} = m(B(0,\delta))^{1/p} \approx \delta^{n/p}$. For any direction e , the best a δ -tube can do is to contain the whole ball, so

$$\mathcal{K}_\delta f(e) \approx \frac{m(B(0,\delta))}{\delta^{n-1}} \approx \frac{\delta^n}{\delta^{n-1}} = \delta,$$

uniformly in e , whence $\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})} \approx \delta$. An estimate $\|\mathcal{K}_\delta f\|_p \lesssim_\varepsilon \delta^{-\varepsilon} \|f\|_p$ would therefore require

$$\delta \lesssim_\varepsilon \delta^{-\varepsilon} \delta^{n/p}, \quad \text{i.e.} \quad \delta^{1-n/p} \lesssim_\varepsilon \delta^{-\varepsilon}.$$

As $\delta \rightarrow 0$ this forces $1 - \frac{n}{p} \geq -\varepsilon$ for every $\varepsilon > 0$, that is $p \geq n$. So n is the smallest exponent at which the conjecture can hold, and by (2.2.2) together with interpolation against the trivial L^∞ bound, the case $p = n$ implies every case $p > n$. This is why the conjecture is stated at one exponent.

(c) *The single-tube example imposes nothing further.* It is worth checking the other natural competitor. Let $f = \mathbf{1}_T$ for one δ -tube T in direction e_0 , so $\|f\|_p \approx \delta^{(n-1)/p}$. For e at angle θ from e_0 , the tube intersection estimate of Theorem 1.6.13 (the tube intersection estimate) gives $\mathcal{K}_\delta f(e) \approx \delta^{-(n-1)} \cdot \delta^n / (\theta + \delta) = \delta / (\theta + \delta)$. Integrating over the sphere, where the measure of the set of directions at angle $\approx \theta$ is $\approx \theta^{n-2} d\theta$,

$$\|\mathcal{K}_\delta f\|_{L^p(\mathbb{S}^{n-1})}^p \approx \int_0^1 \left(\frac{\delta}{\theta + \delta} \right)^p \theta^{n-2} d\theta \approx \delta^{n-1} \quad (p > n-1),$$

the integral being dominated by $\theta \lesssim \delta$. So $\|\mathcal{K}_\delta f\|_p \approx \delta^{(n-1)/p} \approx \|f\|_p$: the single tube is consistent with the conjecture at every $p > n-1$ and gives no constraint. It is the *ball*, not the tube, that pins the exponent.

What this example is really teaching. Locating the conjectured range of an estimate is a matter of testing against the extremal examples and reading off the exponents, and it is almost always easy. Proving the estimate at the endpoint is almost always hard. Volume 6 performs exactly this computation for the restriction problem — where the extremal examples are the Knapp cap and the whole sphere — and the resulting conjectured range has been open since 1967 for $n \geq 3$.

 WORKED EXAMPLE 2.2.4— TIER B **Building an analytic family by hand**

The device in Step 1 of Theorem 2.2.1 — replacing f by a family f_z depending holomorphically on z — is the single technique a reader must own from this chapter. It is easy to follow and surprisingly hard to reproduce from memory, because the exponents look arbitrary. They are not. This example builds the family for one concrete case, slowly, and shows where each exponent is forced.

The task. Take $p_0 = 1$, $p_1 = 2$, and interpolate at $\theta = \frac{1}{2}$, so that $\frac{1}{p_\theta} = \frac{1-\theta}{1} + \frac{\theta}{2} = \frac{3}{4}$, giving $p_\theta = \frac{4}{3}$. We want a family f_z with

$$f_{1/2} = f, \quad \|f_{iy}\|_{L^1} = 1, \quad \|f_{1+iy}\|_{L^2} = 1,$$

starting from a simple function f with $\|f\|_{L^{4/3}} = 1$.

Step 1: what must be true on the two lines. We need $|f_z|$ to have a fixed L^{p_j} norm on the line $\operatorname{Re} z = j$. Since we intend to build f_z by raising $|f|$ to a z -dependent power, write $f_z = |f|^{\eta(z)} e^{i \arg f}$ and ask what η must satisfy. On $\operatorname{Re} z = j$,

$$\|f_z\|_{L^{p_j}}^{p_j} = \int |f|^{p_j \operatorname{Re} \eta(z)},$$

and this is $\int |f|^{4/3} = 1$ exactly when

$$\begin{aligned} p_j \operatorname{Re} \eta(z) &= \frac{4}{3} \\ , \quad \text{i.e.} \quad \operatorname{Re} \eta &= \frac{4/3}{p_j} \\ &= \frac{p_\theta}{p_j} \quad \text{on } \operatorname{Re} z = j. \end{aligned}$$

So $\operatorname{Re} \eta$ must equal $p_\theta/p_0 = \frac{4}{3}$ on the left edge and $p_\theta/p_1 = \frac{2}{3}$ on the right edge.

Step 2: the unique affine choice. An affine function of z is determined by its values on the two edges, and affine is what we need for $|f|^{\eta(z)} = e^{\eta(z) \log |f|}$ to be entire. The affine η with $\operatorname{Re} \eta(iy) = \frac{4}{3}$ and $\operatorname{Re} \eta(1+iy) = \frac{2}{3}$ is

$$\eta(z) = \frac{4}{3}(1-z) + \frac{2}{3}z = \frac{4}{3} - \frac{2}{3}z,$$

and indeed $\eta(\frac{1}{2}) = \frac{4}{3} - \frac{1}{3} = 1$, so $f_{1/2} = f$ as required.

Compare the general formula in the proof: $\eta(z) = p_\theta(\frac{1-z}{p_0} + \frac{z}{p_1})$. With $p_\theta = \frac{4}{3}$, $p_0 = 1$, $p_1 = 2$ this is $\frac{4}{3}(1-z) + \frac{2}{3}z$ — the same thing. The general formula is nothing but “the affine function taking the value p_θ/p_j on the edge $\operatorname{Re} z = j$ ”.

Step 3: check the family is admissible. Write $f = \sum_k a_k e^{i\alpha_k} \mathbf{1}_{A_k}$ in standard form. Then

$$f_z = \sum_k a_k^{4/3-2z/3} e^{i\alpha_k} \mathbf{1}_{A_k},$$

a finite sum of terms e^{cz} times fixed indicators, hence entire in z ; and on the closed strip $|a_k^{\eta(z)}| = a_k^{\operatorname{Re} \eta(z)}$ with $\operatorname{Re} \eta$ ranging over $[\frac{2}{3}, \frac{4}{3}]$, so f_z is bounded there. Both hypotheses of the three-lines lemma are met.

What to take away. Three habits, all reusable.

- *Decide what you need on the two edges first.* The exponents are determined by the requirement that the norms be 1 there, not chosen for elegance.
- *Interpolate affinely.* Holomorphy forces the exponent to be affine in z ; two edge conditions then determine it uniquely.
- *Check boundedness on the strip separately.* It is the hypothesis of Theorem 2.1.13 that is most often forgotten, and Worked Example 2.1.2 showed the conclusion fails without it.

Drill 2.5 asks you to carry the same construction out for the dual family g_z , where the exponents involve q'_θ and the same reasoning applies with $L^{q'_j}$ in place of L^{p_j} .

Drills

Drill 2.2.1. Show that Theorem 2.2.1 with $p_0 = q_0 = 1$, $p_1 = q_1 = \infty$ recovers nothing about the Hardy–Littlewood maximal operator, and explain in one sentence which hypothesis fails.

Drill 2.2.2. Verify the scaling claim in Worked Example 2.2.1: show that $\|f(\lambda \cdot)\|_p = \lambda^{-n/p} \|f\|_p$ on $\mathbb{R}^n$, and deduce that $1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ is necessary for Young’s inequality.

Drill 2.2.3. Carry out the optimisation of Step 4 in the proof of Theorem 2.2.4 explicitly, and confirm that the resulting constant behaves like $(p - p_0)^{-1/p}$ as $p \downarrow p_0$.

Drill 2.2.4. Carry out Step 2 of Worked Example 2.2.4 for the dual family: find the affine ζ with $\operatorname{Re} \zeta = q'_\theta/q'_j$ on the edge $\operatorname{Re} z = j$, and check $\zeta(\theta) = 1$.

Drill 2.2.5. In Worked Example 2.2.3(c), redo the computation for $p < n - 1$ and show that the integral is then dominated by $\theta \approx 1$ rather than $\theta \approx \delta$.

What bound does the single tube impose in that range, and is it consistent with (b)?

Notes

Riesz proved the real-variable case in 1926 and Thorin the complex one in 1938, introducing the analytic family device. Marcinkiewicz announced his theorem in 1939 shortly before his death; the first published proof is Zygmund’s, from 1956.

Stein’s interpolation theorem for analytic families of operators — in which the operator itself, not merely the test function, varies holomorphically — is the tool needed to interpolate the Riesz potentials and Bochner–Riesz means, and is proved in Volume 5 where those objects first appear. It rests on the same three-lines lemma, applied after a Poisson-kernel representation of the strip.

The two examples of Worked Example 2.2.3 are folklore and appear in Bourgain’s 1991 paper. The observation in (c) that the single tube gives no constraint is worth remembering: in this subject the extremal example for an estimate is often not the one that looks most extreme.

Chapter 2.3

Fourier Series

The Fourier transform on $\mathbb{R}^n$ has a compact model: Fourier series on the torus. Everything is easier there — the exponentials form an honest orthonormal basis, there are no convergence issues at infinity, and the measure is finite — and almost every phenomenon of the general theory appears first in this setting. We develop it now both for its own sake and because two of the applications this series is aimed at, the Vinogradov mean value theorem and Strichartz estimates on tori, live on $\mathbb{T}^n$ rather than $\mathbb{R}^n$.

2.3.1 The torus and Fourier coefficients

Let $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$, and identify functions on $\mathbb{T}^n$ with 1-periodic functions on $\mathbb{R}^n$. Lebesgue measure on $[0, 1)^n$ makes $\mathbb{T}^n$ a probability space, so by Drill 1.8.1 we have the inclusions

$$\begin{aligned} L^\infty(\mathbb{T}^n) \subseteq L^q(\mathbb{T}^n) \subseteq L^p(\mathbb{T}^n) \subseteq L^1(\mathbb{T}^n) \quad \text{whenever } 1 \\ \leq p \\ \leq q \leq \infty. \end{aligned}$$

Both hypotheses are needed. The inclusion is Hölder's inequality on a space of finite measure: $\|f\|_p \leq \|f\|_q \mu(X)^{\frac{1}{p} - \frac{1}{q}}$ requires $\frac{1}{p} - \frac{1}{q} \geq 0$, that is $p \leq q$; and the exponents must be at least 1 for $\|\cdot\|_p$ to be a norm and for Drill 1.8.1 to apply. On a space of infinite measure no inclusion holds in either direction: on $\mathbb{R}$, $|x|^{-1} \mathbf{1}_{|x|>1}$ lies in L^2 but not L^1 , while $|x|^{-1/2} \mathbf{1}_{|x|<1}$ lies in L^1 but not L^2 .

Notation 2.3.1. Write $e(t) = e^{2\pi it}$, and for $\xi \in \mathbb{Z}^n$ let $e_\xi(x) = e(\xi \cdot x)$.

Definition 2.3.2. For $f \in L^1(\mathbb{T}^n)$ and $\xi \in \mathbb{Z}^n$, the *Fourier coefficient* is

$$\hat{f}(\xi) = \int_{\mathbb{T}^n} f(x) e(-\xi \cdot x) dx.$$

A *trigonometric polynomial* is a finite sum $\sum_\xi c_\xi e_\xi$.

Proposition 2.3.3 (Basic properties). *For $f, g \in L^1(\mathbb{T}^n)$:*

- (i) $|\widehat{f}(\xi)| \leq \|f\|_1$, and $\widehat{\cdot}$ is linear;
- (ii) $\widehat{\tau_h f}(\xi) = e(-\xi \cdot h) \widehat{f}(\xi)$, where $\tau_h f(x) = f(x - h)$;
- (iii) $\widehat{f * g} = \widehat{f} \widehat{g}$, where $(f * g)(x) = \int_{\mathbb{T}^n} f(x - y)g(y) \, dy$;
- (iv) $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ (Riemann–Lebesgue);
- (v) $\{e_\xi\}_{\xi \in \mathbb{Z}^n}$ is orthonormal in $L^2(\mathbb{T}^n)$.

Proof. (i) is the triangle inequality; (ii) a change of variable. For (iii), Young's inequality (Volume 1, Prop. 6.9) gives $f * g \in L^1$, and Fubini — licensed by Tonelli applied to $|f| * |g|$ — gives

$$\begin{aligned} \widehat{f * g}(\xi) &= \iint f(x - y)g(y)e(-\xi \cdot x) \, dy \, dx \\ &= \int g(y)e(-\xi \cdot y) \left(\int f(u)e(-\xi \cdot u) \, du \right) \, dy. \end{aligned}$$

(iv) is Worked Example 1.5.4 (Riemann–Lebesgue)(b) transferred to the torus: the same translation trick applies, using $e(-\xi \cdot (x + h)) = -e(-\xi \cdot x)$ for a suitable h depending on ξ , together with continuity of translation in $L^1(\mathbb{T}^n)$.

(v) For $\xi \neq \eta$, $\int_{\mathbb{T}^n} e((\xi - \eta) \cdot x) \, dx$ factors into a product of one-dimensional integrals $\int_0^1 e(mt) \, dt$, which is 0 unless $m = 0$; at least one coordinate of $\xi - \eta$ is nonzero. $\square$

2.3.2 The Dirichlet and Fejér kernels

For $n = 1$ (the general case follows by taking products; see Drill 3.3) the symmetric partial sums are

$$S_N f(x) = \sum_{|\xi| \leq N} \widehat{f}(\xi) e(\xi x) = (f * D_N)(x), \quad D_N = \sum_{|\xi| \leq N} e_\xi,$$

and the Cesàro means are

$$\sigma_N f = \frac{1}{N+1} \sum_{k=0}^N S_k f = f * K_N, \quad K_N = \frac{1}{N+1} \sum_{k=0}^N D_k.$$

Lemma 2.3.4 (Closed forms). *For $x \notin \mathbb{Z}$,*

$$D_N(x) = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}, \quad K_N(x) = \frac{1}{N+1} \left(\frac{\sin((N+1)\pi x)}{\sin(\pi x)} \right)^2.$$

Proof. D_N is a geometric sum: with $q = e(x) \neq 1$,

$$\sum_{\xi=-N}^N q^\xi = \frac{q^{-N} - q^{N+1}}{1 - q} = \frac{q^{-N-1/2} - q^{N+1/2}}{q^{-1/2} - q^{1/2}} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)},$$

using $q^s - q^{-s} = 2i \sin(2\pi s x)$. For K_N , sum the numerators: with $s = \sin(\pi x)$,

$$\begin{aligned} (N+1)K_N(x) &= \frac{1}{s} \sum_{k=0}^N \sin((2k+1)\pi x) \\ &= \frac{1}{s} \operatorname{Im} \left(\sum_{k=0}^N q^{k+1/2} \right) = \frac{\sin^2((N+1)\pi x)}{s^2}, \end{aligned}$$

the last step by the same geometric summation and a little trigonometry. $\square$

Proposition 2.3.5 (K_N is an approximate identity). $K_N \geq 0$, $\int_{\mathbb{T}} K_N = 1$, and for every $\eta > 0$, $\int_{\eta \leq |x| \leq 1/2} K_N \rightarrow 0$ as $N \rightarrow \infty$.

Proof. Nonnegativity is visible from Theorem 2.3.4. The integral is 1 because $\int_{\mathbb{T}} D_k = \widehat{D}_k(0)$ -style orthogonality gives $\int D_k = 1$ for every k . For the third, on $\eta \leq |x| \leq 1/2$ we have $\sin^2(\pi x) \geq \sin^2(\pi \eta)$, so $K_N \leq \frac{1}{(N+1)\sin^2(\pi \eta)} \rightarrow 0$ uniformly there. $\square$

Theorem 2.3.6 (Fejér). If $f \in C(\mathbb{T})$ then $\sigma_N f \rightarrow f$ uniformly. If $f \in L^p(\mathbb{T})$ with $1 \leq p < \infty$ then $\sigma_N f \rightarrow f$ in L^p .

Proof. Since $\int K_N = 1$,

$$\sigma_N f(x) - f(x) = \int_{\mathbb{T}} (f(x-y) - f(x)) K_N(y) dy.$$

Given $\varepsilon > 0$, uniform continuity of f supplies η with $|f(x-y) - f(x)| < \varepsilon$ for $|y| < \eta$, uniformly in x . Split the integral at $|y| = \eta$: the inner part is at most $\varepsilon \int K_N = \varepsilon$, and the outer is at most $2\|f\|_\infty \int_{|y| \geq \eta} K_N \rightarrow 0$ by Theorem 2.3.5.

For the L^p statement, Minkowski's integral inequality gives $\|\sigma_N f - f\|_p \leq \int_{\mathbb{T}} \|\tau_y f - f\|_p K_N(y) dy$, and the same splitting applies using continuity of translation in L^p . $\square$

Corollary 2.3.7 (Uniqueness and density). If $f \in L^1(\mathbb{T}^n)$ has $\widehat{f} \equiv 0$ then $f = 0$ a.e. Trigonometric polynomials are dense in $C(\mathbb{T}^n)$ and in $L^p(\mathbb{T}^n)$ for $p < \infty$.

Proof. $\sigma_N f$ is a trigonometric polynomial built from the $\widehat{f}(\xi)$, so $\widehat{f} \equiv 0$ forces $\sigma_N f \equiv 0$, and $\sigma_N f \rightarrow f$ in L^1 by Theorem 2.3.6. Density is the same statement read forwards. $\square$

Theorem 2.3.8 (Parseval). $\{e_\xi\}_{\xi \in \mathbb{Z}^n}$ is an orthonormal basis of $L^2(\mathbb{T}^n)$. Hence $\|f\|_2^2 = \sum_\xi |\hat{f}(\xi)|^2$ and $f = \sum_\xi \hat{f}(\xi)e_\xi$ with convergence in L^2 .

Proof. The system is orthonormal by Theorem 2.3.3(v) and its span — the trigonometric polynomials — is dense by Theorem 2.3.7. Apply Volume 1, Theorem 8.20. $\square$

Where we are. Note what has been used. Fejér's theorem gave density, density plus orthonormality gave the basis, and the basis gave Parseval. No appeal to the Stone–Weierstrass theorem was needed, and none to any pointwise convergence statement — which is fortunate, because pointwise convergence is false in general, as we now show.

2.3.3 Divergence, and two facts from functional analysis

Volume 1 developed no functional analysis beyond Hilbert space, so we prove here the two results needed. They are used again in Chapter 2.5.

Theorem 2.3.9 (Baire category). *A complete metric space X is not the union of countably many closed sets with empty interior.*

Proof. Suppose $X = \bigcup_{k \geq 1} F_k$ with each F_k closed and $\text{int } F_k = \emptyset$. Since $F_1 \neq X$ and F_1 is closed, pick a closed ball $\overline{B}_1 = \overline{B}(x_1, r_1)$ with $r_1 < 1$ disjoint from F_1 . Inductively, since F_{k+1} has empty interior it cannot contain the open ball $B(x_k, r_k)$, so we may choose a closed ball $\overline{B}_{k+1} \subseteq B(x_k, r_k)$ with $r_{k+1} < 2^{-k}$ and $\overline{B}_{k+1} \cap F_{k+1} = \emptyset$.

The centres form a Cauchy sequence, since $x_l \in B_k$ for $l \geq k$ and $r_k \rightarrow 0$; let $x = \lim_k x_k$. Then $x \in \overline{B}_k$ for every k , hence $x \notin F_k$ for every k , contradicting $X = \bigcup_k F_k$. $\square$

Theorem 2.3.10 (Uniform boundedness). *Let X be a Banach space and $\{\Lambda_\alpha\}$ a family of bounded linear functionals on X with $\sup_\alpha |\Lambda_\alpha(x)| < \infty$ for every $x \in X$. Then $\sup_\alpha \|\Lambda_\alpha\| < \infty$.*

Proof. Let $F_k = \{x : |\Lambda_\alpha(x)| \leq k \text{ for all } \alpha\}$, closed as an intersection of preimages of closed sets under continuous maps. By hypothesis $X = \bigcup_k F_k$, so by Theorem 2.3.9 some F_{k_0} has an interior point: $\overline{B}(x_0, r) \subseteq F_{k_0}$ for some $r > 0$. For $\|y\| \leq 1$ and any α ,

$$|\Lambda_\alpha(y)| = \frac{1}{r} |\Lambda_\alpha(x_0 + ry) - \Lambda_\alpha(x_0)| \leq \frac{2k_0}{r},$$

so $\|\Lambda_\alpha\| \leq 2k_0/r$ for every α . $\square$

WORKED EXAMPLE 2.3.1— TIER B The Dirichlet kernel has logarithmically divergent L^1 norm

We show

$$\|D_N\|_{L^1(\mathbb{T})} \approx \log N. \quad (2.3.1)$$

This single estimate is the whole obstruction to pointwise convergence of Fourier series, and it is worth seeing exactly where the logarithm comes from.

Lower bound. Using $|\sin(\pi x)| \leq \pi|x|$ and Theorem 2.3.4,

$$\begin{aligned} \|D_N\|_1 &= 2 \int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{\sin(\pi x)} dx \\ &\geq \frac{2}{\pi} \int_0^{1/2} \frac{|\sin((2N+1)\pi x)|}{x} dx. \end{aligned}$$

Substituting $u = (2N+1)\pi x$ turns this into $\frac{2}{\pi} \int_0^{(2N+1)\pi/2} \frac{|\sin u|}{u} du$. On each interval $[k\pi, (k+1)\pi]$ the integral of $|\sin u|$ is 2, and $u \leq (k+1)\pi$ there, so the integral is at least $\frac{2}{\pi} \sum_{k=1}^N \frac{2}{(k+1)\pi} \gtrsim \log N$.

Upper bound. On $[0, 1/2]$, $\sin(\pi x) \geq 2x$, so $|D_N(x)| \leq \min(2N+1, (2x)^{-1})$, using also $|D_N| \leq 2N+1$ from the defining sum. Integrating,

$$\|D_N\|_1 \leq 2 \left(\int_0^{1/(2N+1)} (2N+1) dx + \int_{1/(2N+1)}^{1/2} \frac{dx}{2x} \right) \lesssim 1 + \log N.$$

Reading it. The two regimes are the whole story. Near $x = 0$ the kernel is as tall as it can be, $\approx N$, over a window of width $\approx 1/N$, contributing $O(1)$. Away from 0 it decays only like $1/x$, and $1/x$ is not integrable — it fails logarithmically. The Fejér kernel escapes this because squaring turns $1/x$ into $1/x^2$, which is integrable, at the cost of the extra factor $1/(N+1)$; that is precisely why Theorem 2.3.5 works and no analogue holds for D_N .

Theorem 2.3.11 (du Bois-Reymond). *There exists $f \in C(\mathbb{T})$ whose Fourier series diverges at 0; indeed $\sup_N |S_N f(0)| = \infty$.*

Proof. The maps $\Lambda_N : C(\mathbb{T}) \rightarrow \mathbb{C}$, $\Lambda_N f = S_N f(0) = \int_{\mathbb{T}} f(-y) D_N(y) dy$, are bounded linear functionals on the Banach space $C(\mathbb{T})$ with the sup norm, and

$$\|\Lambda_N\| = \|D_N\|_{L^1}.$$

Indeed $\leq$ is immediate, and for $\geq$ one takes f continuous with $\|f\|_{\infty} \leq 1$ approximating $\operatorname{sgn} D_N(\cdot)$ in L^1 , which is possible by Theorem 1.8.7 (density of simple functions and C_c^{∞}).

By Worked Example 2.3.1, $\|\Lambda_N\| \rightarrow \infty$. If $\sup_N |\Lambda_N f| < \infty$ held for every

$f \in C(\mathbb{T})$, then Theorem 2.3.10 would force $\sup_N \|\Lambda_N\| < \infty$. So some f has $\sup_N |S_N f(0)| = \infty$. $\square$

Pitfall: what does and does not fail

Theorem 2.3.11 is often misremembered as saying Fourier series of continuous functions “do not converge”. They converge in L^2 (Theorem 2.3.8), in L^p for $1 < p < \infty$ (Volume 3, via the Hilbert transform), uniformly under mild regularity (Drill 3.2), and — by Carleson’s theorem — almost everywhere. What fails is convergence *at every point* for *every* continuous function, and the obstruction is exactly the logarithm in (2.3.1).

2.3.4 Hausdorff–Young on the torus

Theorem 2.3.12. For $1 \leq p \leq 2$ and $f \in L^p(\mathbb{T}^n)$, $\|\hat{f}\|_{\ell^{p'}(\mathbb{Z}^n)} \leq \|f\|_{L^p(\mathbb{T}^n)}$.

Proof. The map $f \mapsto \hat{f}$ sends $L^1(\mathbb{T}^n) \rightarrow \ell^\infty(\mathbb{Z}^n)$ with norm ≤ 1 by Theorem 2.3.3(i), and $L^2 \rightarrow \ell^2$ with norm 1 by Theorem 2.3.8. It is linear. Apply Theorem 2.2.1 with $(p_0, q_0) = (1, \infty)$, $(p_1, q_1) = (2, 2)$ and θ chosen so that $\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2}$, which gives $\frac{1}{q\theta} = \frac{\theta}{2} = 1 - \frac{1}{p} = \frac{1}{p'}$. The interpolated constant is $1^{1-\theta} 1^\theta = 1$. $\square$

WORKED EXAMPLE 2.3.2— TIER B Weyl’s equidistribution criterion

A sequence $(x_k) \subseteq \mathbb{T}$ is *equidistributed* if for every interval $I \subseteq \mathbb{T}$, $\frac{1}{K} \#\{k \leq K : x_k \in I\} \rightarrow |I|$.

Proposition 2.3.13. (x_k) is equidistributed if and only if for every $m \in \mathbb{Z} \setminus \{0\}$,

$$\frac{1}{K} \sum_{k=1}^K e(mx_k) \rightarrow 0.$$

Proof. Write $A_K(g) = \frac{1}{K} \sum_{k \leq K} g(x_k)$. Equidistribution says $A_K(\mathbf{1}_I) \rightarrow \int \mathbf{1}_I$ for all intervals; the criterion says $A_K(e_m) \rightarrow \int e_m$ for all m (the case $m = 0$ being trivial).

($\Rightarrow$) Every continuous g is a uniform limit of step functions, and $|A_K(g)| \leq \|g\|_\infty$, so $A_K(g) \rightarrow \int g$ for all $g \in C(\mathbb{T})$; in particular for $g = e_m$.

($\Leftarrow$) By linearity $A_K(P) \rightarrow \int P$ for every trigonometric polynomial P . Given $g \in C(\mathbb{T})$ and $\varepsilon > 0$, Theorem 2.3.7 supplies P with $\|g - P\|_\infty < \varepsilon$, and then $|A_K(g) - \int g| \leq 2\varepsilon + |A_K(P) - \int P|$, so $\limsup_K |A_K(g) - \int g| \leq 2\varepsilon$. Hence $A_K(g) \rightarrow \int g$ for all continuous g . Finally approximate $\mathbf{1}_I$ above and below by continuous functions agreeing with it off a set of measure $< \varepsilon$. $\square$

Application. For irrational α , the sequence $x_k = k\alpha$ is equidistributed: for $m \neq 0$,

$$\left| \frac{1}{K} \sum_{k=1}^K e(mk\alpha) \right| = \frac{1}{K} \left| \frac{e(m\alpha)(e(mK\alpha) - 1)}{e(m\alpha) - 1} \right| \leq \frac{2}{K|e(m\alpha) - 1|} \rightarrow 0,$$

the denominator being nonzero precisely because $m\alpha \notin \mathbb{Z}$.

Why this is here. The criterion converts a counting statement into a bound on an *exponential sum*, and exponential sums are what Worked Example 2.3.3 and Volume 13 are about. Note the mechanism: cancellation in $\sum_k e(mx_k)$ is equidistribution. $\square$

WORKED EXAMPLE 2.3.3— TIER C: RESEARCH TASTE **Vinogradov’s mean value theorem, set up**

Fix $k \geq 2$ and $N \geq 1$, and consider the exponential sum

$$f_N(x) = \sum_{j=1}^N e(jx_1 + j^2x_2 + \cdots + j^kx_k), \quad x \in \mathbb{T}^k.$$

This is a trigonometric polynomial on $\mathbb{T}^k$ whose frequencies are the N points $(j, j^2, \dots, j^k)$ — that is, N points on the *moment curve* in $\mathbb{R}^k$. Define, for $s \geq 1$,

$$J_{s,k}(N) = \int_{\mathbb{T}^k} |f_N(x)|^{2s} dx.$$

(a) *It counts solutions.* Expanding the $2s$ -th power and integrating, orthogonality (Theorem 2.3.3(v)) kills every term except those where the frequency vectors cancel. Hence $J_{s,k}(N)$ is exactly the number of integer solutions $1 \leq j_1, \dots, j_{2s} \leq N$ of the system

$$j_1^i + \cdots + j_s^i = j_{s+1}^i + \cdots + j_{2s}^i, \quad i = 1, 2, \dots, k.$$

So an analytic quantity — an L^{2s} norm — is a Diophantine counting function. This is the reason decoupling has anything to do with number theory.

(b) *The two trivial lower bounds.* The *diagonal* solutions, those with $(j_{s+1}, \dots, j_{2s})$ a permutation of $(j_1, \dots, j_s)$, number $\gtrsim_s N^s$. And a counting heuristic — the system imposes k equations on $2s$ variables of size N , with

the i -th equation of size N^i — predicts $\approx N^{2s}/N^{1+2+\dots+k} = N^{2s-k(k+1)/2}$ further solutions. So

$$J_{s,k}(N) \gtrsim_{s,k} N^s + N^{2s-k(k+1)/2}.$$

(c) *The theorem.* Vinogradov's mean value theorem, in the sharp form proved by Bourgain, Demeter and Guth in 2016, asserts that this trivial lower bound is the truth:

$$J_{s,k}(N) \lesssim_{s,k,\varepsilon} N^\varepsilon \left(N^s + N^{2s-k(k+1)/2} \right) \quad \text{for every } \varepsilon > 0.$$

It was open for over eighty years for $k \geq 4$.

(d) *Why it is a decoupling statement.* Partition $[1, N]$ into N single integers; each contributes one frequency on the moment curve. Rescaling $x \mapsto (Nx_1, N^2x_2, \dots, N^kx_k)$ turns the N frequencies into a δ -separated family on the curve with $\delta = 1/N$, and turns $J_{s,k}(N)$ into precisely the left-hand side of an ℓ^2 decoupling inequality for the moment curve at exponent $p = 2s$. The trivial lower bound (b) becomes the right-hand side. So the theorem says exactly: *the pieces of f_N behave as though they were orthogonal, up to N^ε .* That statement — proved in Volume 13 — is the same one previewed in Volume 1, Worked Example 8.16, where Cauchy–Schwarz gave a power loss and the theorem promised only $\delta^{-\varepsilon}$. Curvature of the moment curve is what makes it true; for frequencies on a line it is false.

You now know what the destination looks like, and you have proved, in (a), the identity that makes the analytic and arithmetic sides the same object.

Drills

Drill 2.3.1. Compute the Fourier coefficients of the sawtooth $f(x) = x - \frac{1}{2}$ on $[0, 1)$ and of the triangle wave, and check Parseval in both cases. Deduce $\sum_{m \geq 1} m^{-2} = \pi^2/6$.

Drill 2.3.2. Show that if $\sum_\xi |\widehat{f}(\xi)| < \infty$ then $S_N f \rightarrow f$ uniformly and f is (a.e. equal to) a continuous function. Deduce that a C^2 function on $\mathbb{T}$ has uniformly convergent Fourier series.

Drill 2.3.3. State and prove the n -dimensional versions of Theorem 2.3.4 and Theorem 2.3.6, using products $K_N(x) = \prod_{i=1}^n K_N(x_i)$.

Drill 2.3.4. Verify (a) of Worked Example 2.3.3 in the case $k = 1$, $s = 2$: show $J_{2,1}(N)$ counts solutions of $j_1 + j_2 = j_3 + j_4$ and compute it exactly.

Notes

Fejér’s theorem is from 1900 and du Bois-Reymond’s example from 1873, predating the functional-analytic proof given here by half a century; the uniform boundedness principle is Banach and Steinhaus, 1927. Carleson’s theorem on almost everywhere convergence for L^2 functions dates to 1966 and is not proved in this series.

Vinogradov introduced $J_{s,k}$ in 1935. The sharp bound was obtained for $k = 3$ by Wooley (2016) via efficient congruencing and for all k by Bourgain–Demeter–Guth (2016) via decoupling; the two methods are compared in Volume 13.

Chapter 2.4

The Fourier Transform on L^1 and L^2

2.4.1 Definition and elementary identities

Definition 2.4.1. For $f \in L^1(\mathbb{R}^n)$,

$$\widehat{f}(\xi) = \mathcal{F}f(\xi) = \int_{\mathbb{R}^n} f(x)e(-x \cdot \xi) dx, \quad e(t) = e^{2\pi it}.$$

The integral converges absolutely since $|f(x)e(-x \cdot \xi)| = |f(x)|$, and Worked Example 1.5.4 (Riemann–Lebesgue) already established that $\widehat{f}$ is continuous, that $\|\widehat{f}\|_\infty \leq \|f\|_1$, and that $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Proposition 2.4.2 (The dictionary). *Let $f \in L^1(\mathbb{R}^n)$, $h \in \mathbb{R}^n$, $\lambda > 0$, $A \in GL_n(\mathbb{R})$.*

- (i) $\widehat{\tau_h f}(\xi) = e(-h \cdot \xi) \widehat{f}(\xi)$ where $\tau_h f(x) = f(x - h)$;
- (ii) $\widehat{e(h \cdot \cdot)} f(\xi) = \widehat{f}(\xi - h)$;
- (iii) $\widehat{f(\lambda \cdot)}(\xi) = \lambda^{-n} \widehat{f}(\xi/\lambda)$;
- (iv) $\widehat{f \circ A}(\xi) = |\det A|^{-1} \widehat{f}((A^{-1})^T \xi)$;
- (v) $\widehat{f * g} = \widehat{f} \widehat{g}$ for $g \in L^1$;
- (vi) if $f, \partial_j f \in L^1$ then $\widehat{\partial_j f}(\xi) = 2\pi i \xi_j \widehat{f}(\xi)$; if $x_j f \in L^1$ then $\widehat{f}$ is C^1 in ξ_j with $\partial_{\xi_j} \widehat{f} = -\widehat{2\pi i x_j f}$.

Proof. (i)–(iv) are changes of variable, using Theorem 1.3.10 (linear images) for the linear substitution in (iv). (v) is Fubini, licensed by Tonelli applied to

$|f| * |g|$, exactly as in Theorem 2.3.3(iii). The first half of (vi) is integration by parts, the boundary terms vanishing because $f, \partial_j f \in L^1$ forces $f(x) \rightarrow 0$ along almost every line; the second half is differentiation under the integral sign, Worked Example 1.5.3 (differentiating under the integral), with dominating function $2\pi|x_j f|$. $\square$

Looking ahead: (iii) and (vi) are the whole subject in miniature

Identity (iii) says that squeezing a function in space stretches its transform in frequency, at exactly reciprocal rates. Identity (vi) says that differentiating in space is multiplication by ξ in frequency, so *smoothness becomes decay*. These two sentences generate: the uncertainty principle (Worked Example 2.4.2); the Knapp example, where a $\delta \times \delta^2$ cap produces a $\delta^{-1} \times \delta^{-2}$ tube (Worked Example 2.4.3); the Sobolev spaces of Chapter 2.7; and the wave packet decompositions of Volume 12. Every “dual” object in this series is an instance of (iii).

WORKED EXAMPLE 2.4.1— TIER A The box and the sinc function

Let $f = \mathbf{1}_{[-1/2, 1/2]}$ on $\mathbb{R}$. Then

$$\widehat{f}(\xi) = \int_{-1/2}^{1/2} e(-x\xi) dx = \frac{e(-\xi/2) - e(\xi/2)}{-2\pi i \xi} = \frac{\sin(\pi\xi)}{\pi\xi}.$$

Three observations. The transform is not integrable — $|\sin(\pi\xi)/\pi\xi|$ decays only like $1/|\xi|$ — so $\widehat{f} \notin L^1$ even though $f \in L^1$; this is why the inversion theorem below needs a hypothesis. The decay rate $|\xi|^{-1}$ reflects the jump discontinuities of f , in accordance with Theorem 2.4.2(vi). And in $\mathbb{R}^n$, $\mathbf{1}_{[-1/2, 1/2]^n}$ has transform $\prod_j \sin(\pi\xi_j)/(\pi\xi_j)$ by Fubini: separable functions have separable transforms, a fact used constantly.

2.4.2 The Gaussian

Proposition 2.4.3. *Let $G(x) = e^{-\pi|x|^2}$ on $\mathbb{R}^n$. Then $\widehat{G} = G$.*

Proof. By Fubini it suffices to treat $n = 1$. Let $g(\xi) = \widehat{G}(\xi)$. By Theorem 2.4.2(vi), g is differentiable with

$$g'(\xi) = -\widehat{2\pi i x G}(\xi).$$

Now $xG(x) = -\frac{1}{2\pi}G'(x)$, so $-2\pi i x G = iG'$, and by Theorem 2.4.2(vi) again $\widehat{G}'(\xi) = 2\pi i \xi g(\xi)$. Therefore

$$g'(\xi) = i \cdot 2\pi i \xi g(\xi) = -2\pi \xi g(\xi).$$

This linear ODE has solution $g(\xi) = g(0)e^{-\pi\xi^2}$, and $g(0) = \int_{\mathbb{R}} e^{-\pi x^2} dx = 1$ by Drill 1.6.2. Hence $g = G$. $\square$

Corollary 2.4.4. *For $t > 0$, the function $G_t(x) = t^{-n/2}e^{-\pi|x|^2/t}$ has $\widehat{G_t}(\xi) = e^{-\pi t|\xi|^2}$, and $\int G_t = 1$.*

Proof. Apply Theorem 2.4.2(iii) with $\lambda = t^{-1/2}$ to Theorem 2.4.3. $\square$

2.4.3 Inversion

Lemma 2.4.5 (Multiplication formula). *For $f, g \in L^1(\mathbb{R}^n)$, $\int_{\mathbb{R}^n} \widehat{f}g = \int_{\mathbb{R}^n} f\widehat{g}$.*

Proof. Both sides equal $\iint f(x)g(\xi)e(-x \cdot \xi) dx d\xi$, and Fubini applies because the double integral of the absolute value is $\|f\|_1\|g\|_1 < \infty$. $\square$

Theorem 2.4.6 (Fourier inversion). *If $f \in L^1(\mathbb{R}^n)$ and $\widehat{f} \in L^1(\mathbb{R}^n)$, then for a.e. x*

$$f(x) = \int_{\mathbb{R}^n} \widehat{f}(\xi)e(x \cdot \xi) d\xi.$$

In particular f agrees a.e. with a continuous function.

Idea of the proof. The integral $\int \widehat{f}(\xi)e(x \cdot \xi) d\xi$ cannot be attacked by Fubini directly, since $\widehat{f}(\xi)e(x \cdot \xi)f(y)$ is not absolutely integrable in (y, ξ) . The remedy is to insert a Gaussian damping factor $e^{-\pi t|\xi|^2}$, which makes Fubini legitimate, evaluate, and let $t \rightarrow 0$. The damped integral turns out to be $f * G_t$, an approximate identity, and Theorem 1.6.10 (approximate identities) finishes the job.

Proof. Fix $t > 0$ and $x \in \mathbb{R}^n$. Apply Theorem 2.4.5 with f and the function $g(\xi) = e^{-\pi t|\xi|^2}e(x \cdot \xi)$, whose transform is, by Theorem 2.4.4 and Theorem 2.4.2(ii), $\widehat{g}(y) = G_t(y - x)$. This gives

$$\int_{\mathbb{R}^n} \widehat{f}(\xi)e^{-\pi t|\xi|^2}e(x \cdot \xi) d\xi = \int_{\mathbb{R}^n} f(y)G_t(y - x) dy = (f * G_t)(x), \quad (2.4.1)$$

using that G_t is even.

Let $t \rightarrow 0^+$. On the left, the integrand is dominated by $|\widehat{f}| \in L^1$ and converges pointwise to $\widehat{f}(\xi)e(x \cdot \xi)$, so by dominated convergence the left side tends to $\int \widehat{f}(\xi)e(x \cdot \xi) d\xi$, for every x . On the right, $\{G_t\}$ is an approximate identity in the sense of Volume 1, Proposition 6.10 — it is G_1 dilated, with integral 1 — so $f * G_t \rightarrow f$ in L^1 , hence along a subsequence a.e. Equating the two limits gives the result a.e., and the left side is continuous in x by dominated convergence. $\square$

Corollary 2.4.7 (Uniqueness). *If $f \in L^1(\mathbb{R}^n)$ and $\widehat{f} \equiv 0$ then $f = 0$ a.e.*

Proof. $\widehat{f} = 0 \in L^1$, so Theorem 2.4.6 applies and gives $f = 0$ a.e. $\square$

2.4.4 Plancherel

Theorem 2.4.8 (Plancherel). *The Fourier transform, restricted to $L^1 \cap L^2(\mathbb{R}^n)$, satisfies $\|\widehat{f}\|_2 = \|f\|_2$, and extends uniquely to a unitary operator $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$. Its inverse is $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$, and the Parseval identity $\int f\overline{g} = \int \widehat{f}\overline{\widehat{g}}$ holds for all $f, g \in L^2$.*

Idea of the proof. The extension step is the point, and it is exactly the manoeuvre Volume 1 promised in its opening chapter: the transform is defined by an integral formula needing L^1 , its most important property lives in L^2 , and $L^2 \not\subseteq L^1$ on $\mathbb{R}^n$. One proves the identity on the dense subspace $L^1 \cap L^2$ and extends by continuity — which requires the target to be complete, i.e. the Riesz–Fischer theorem.

Proof. Step 1. The identity on a dense subspace.

Let $f \in L^1 \cap L^2$ and put $h = f * \widetilde{f}$ where $\widetilde{f}(x) = \overline{f(-x)}$. Then $h \in L^1$ by Young, and h is continuous and bounded: it is a convolution of two L^2 functions, so by Cauchy–Schwarz $|h(x)| \leq \|f\|_2^2$, and continuity follows from continuity of translation in L^2 (Theorem 1.8.7 (density of simple functions and C_c^∞)(iii)). Moreover

$$\widehat{h} = \widehat{f}\widehat{\widetilde{f}} = \widehat{f}\overline{\widehat{f}} = |\widehat{f}|^2 \geq 0,$$

using $\widehat{\widetilde{f}} = \overline{\widehat{f}}$, a direct computation.

Now apply (2.4.1) to h at $x = 0$:

$$\int_{\mathbb{R}^n} |\widehat{f}(\xi)|^2 e^{-\pi t|\xi|^2} d\xi = (h * G_t)(0).$$

As $t \rightarrow 0$, the left side increases to $\int |\widehat{f}|^2$ by monotone convergence, and the right side tends to $h(0)$ because h is bounded and continuous and $\{G_t\}$ is an approximate identity. Since $h(0) = \int f\overline{f} = \|f\|_2^2$, we conclude $\|\widehat{f}\|_2^2 = \|f\|_2^2$; in particular $\widehat{f} \in L^2$.

Step 2. Extension.

$L^1 \cap L^2$ is dense in L^2 : it contains the simple functions of finite support, which are dense by Theorem 1.8.7 (density of simple functions and C_c^∞)(i). By Step 1, $\mathcal{F}$ is an isometry on this dense subspace; since L^2 is complete (Theorem 1.8.6 (Riesz–Fischer)), it extends uniquely to an isometry of L^2 .

Step 3. Surjectivity and the inverse.

For $f \in L^1 \cap L^2$ with also $\widehat{f} \in L^1$, Theorem 2.4.6 gives $f(x) = \widehat{\widehat{f}}(-x)$. Such f are dense in L^2 — for instance $f * G_t$ for $f \in L^1 \cap L^2$ has $\widehat{f * G_t} = \widehat{f} e^{-\pi t |\cdot|^2} \in L^1$ — so the identity $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$ extends by continuity, and $\mathcal{F}$ is onto. Parseval follows from the isometry by polarisation. $\square$

Theorem 2.4.9 (Hausdorff–Young). *For $1 \leq p \leq 2$ and $f \in L^p(\mathbb{R}^n)$, $\|\widehat{f}\|_{p'} \leq \|f\|_p$.*

Proof. $\mathcal{F}$ maps $L^1 \rightarrow L^\infty$ with norm ≤ 1 and $L^2 \rightarrow L^2$ with norm 1; apply Theorem 2.2.1 exactly as in Theorem 2.3.12. $\square$

Pitfall: Hausdorff–Young fails for $p > 2$

There is no bound $\|\widehat{f}\|_q \lesssim \|f\|_p$ for $p > 2$, with any q . The reason is a scaling obstruction: by Theorem 2.4.2(iii), replacing f by $f(\lambda \cdot)$ multiplies $\|f\|_p$ by $\lambda^{-n/p}$ and $\|\widehat{f}\|_q$ by $\lambda^{-n+n/q}$, so any such inequality forces $\frac{1}{q} = 1 - \frac{1}{p}$, i.e. $q = p'$; and then testing against a Gaussian modulated by high frequencies, or directly against $f = \mathbf{1}_{B(0,R)}$ as $R \rightarrow \infty$, shows the inequality is false for $p > 2$. Drill 4.3 asks for the details. The asymmetry between $p \leq 2$ and $p > 2$ is not technical: it is the reason L^2 is the natural home of the transform.

WORKED EXAMPLE 2.4.2— TIER B The uncertainty principle

Proposition 2.4.10 (Heisenberg). *For $f \in L^2(\mathbb{R})$ with $xf, \xi\widehat{f} \in L^2$,*

$$\|f\|_2^2 \leq 4\pi \|xf\|_2 \|\xi\widehat{f}\|_2.$$

Equality holds precisely for Gaussians $f(x) = ce^{-a\pi x^2}$, $a > 0$.

Proof. Assume first $f \in C_c^\infty$. Integrating by parts,

$$\|f\|_2^2 = \int_{\mathbb{R}} |f(x)|^2 dx = - \int_{\mathbb{R}} x \frac{d}{dx} |f(x)|^2 dx = -2 \operatorname{Re} \int_{\mathbb{R}} x f'(x) \overline{f(x)} dx,$$

the boundary terms vanishing by compact support. By Cauchy–Schwarz,

$$\|f\|_2^2 \leq 2 \|xf\|_2 \|f'\|_2.$$

By Theorem 2.4.2(vi) and Plancherel, $\|f'\|_2 = \|\widehat{f'}\|_2 = 2\pi \|\xi\widehat{f}\|_2$. Combining gives the inequality for $f \in C_c^\infty$, and the general case follows by density (Theorem 1.8.7 (density of simple functions and C_c^∞)(ii)) together with Fatou.

Equality in Cauchy–Schwarz forces f' to be a real multiple of xf , say $f'(x) = -2a\pi xf(x)$; the solutions are $f(x) = ce^{-a\pi x^2}$, and these lie in the admissible class precisely when $a > 0$, since then $f \in \mathcal{S}(\mathbb{R})$. Conversely each such f gives equality, by Theorem 2.4.3 and a direct computation. $\square$

Reading it. A function and its transform cannot both be concentrated. If f lives essentially on an interval of length L then $\|xf\|_2 \approx L\|f\|_2$, and the inequality forces $\|\xi\hat{f}\|_2 \gtrsim \|f\|_2/L$, i.e. $\hat{f}$ spreads over frequencies of size $\gtrsim 1/L$. This is Theorem 2.4.2(iii) in quantitative form, and it is the reason the “dual box” of a δ -cap has dimensions δ^{-1} : one cannot do better. $\square$

WORKED EXAMPLE 2.4.3— TIER C: RESEARCH TASTE The Knapp example, and the restriction problem

Worked Example 1.4.3 (the early Knapp computation) performed a bare-hands version of this computation using only the box lemma. With the dictionary of Theorem 2.4.2 we can do it properly, and state the problem it points to.

(a) *The cap and its dual box.* Let $P = \{(\xi', |\xi'|^2) : \xi' \in \mathbb{R}^{n-1}\}$ be the paraboloid in $\mathbb{R}^n$, and for $\delta \in (0, 1)$ let

$$\kappa_\delta = \{(\xi', |\xi'|^2) : |\xi'| \leq \delta\} \subseteq P.$$

Since $|\xi'| \leq \delta$ implies $|\xi'|^2 \leq \delta^2$, the cap lies in a box of dimensions $\underbrace{\delta \times \cdots \times \delta}_{n-1} \times \delta^2$. Curvature is exactly what converts the length scale δ into

the thickness scale δ^2 ; for a flat piece of hyperplane the thickness would be 0 and everything below would change.

(b) *The extension operator.* For a function g on P , define

$$Eg(x) = \int_{\mathbb{R}^{n-1}} g(\xi') e(x' \cdot \xi' + x_n |\xi'|^2) d\xi'.$$

This is the Fourier transform of the measure $g d\sigma$ carried by P ; that it makes sense as a tempered distribution is Chapter 2.5, and that it is a genuine function here follows since g is integrable.

(c) *The computation.* Take $g = \mathbf{1}_{|\xi'| \leq \delta}$, so $\|g\|_{L^\infty} = 1$ and the cap has measure $\approx \delta^{n-1}$. If

$$|x'| \leq \frac{1}{100\delta} \quad \text{and} \quad |x_n| \leq \frac{1}{100\delta^2},$$

then for $|\xi'| \leq \delta$ the phase satisfies

$$|x' \cdot \xi' + x_n |\xi'|^2| \leq \frac{1}{100\delta} \cdot \delta + \frac{1}{100\delta^2} \cdot \delta^2 = \frac{1}{50},$$

so every contribution to the integral lies within an arc of angle $2\pi/50$ of 1, and in particular has real part at least $\frac{1}{2}$. Hence on the tube

$$T_\delta = \left\{ |x'| \leq \frac{1}{100\delta} \right\} \times \left\{ |x_n| \leq \frac{1}{100\delta^2} \right\}, \quad m(T_\delta) \approx \delta^{-(n-1)}\delta^{-2},$$

we have $|Eg(x)| \gtrsim \delta^{n-1}$.

So Eg is of size $\approx \delta^{n-1}$ — the measure of the cap — on a *dual tube* whose side lengths are the reciprocals of the cap's. This is Theorem 2.4.2(iii) applied in each of the n directions separately.

(d) *What it rules out.* Suppose an estimate $\|Eg\|_{L^q(\mathbb{R}^n)} \lesssim \|g\|_{L^p(d\sigma)}$ held. With g as above, $\|g\|_{L^p(d\sigma)} \approx \delta^{(n-1)/p}$, while

$$\|Eg\|_{L^q(\mathbb{R}^n)} \gtrsim \delta^{n-1} m(T_\delta)^{1/q} \approx \delta^{n-1} \delta^{-(n+1)/q}.$$

The estimate therefore forces

$$\delta^{n-1-\frac{n+1}{q}} \lesssim \delta^{\frac{n-1}{p}} \quad \text{for all small } \delta,$$

that is

$$n-1-\frac{n+1}{q} \geq \frac{n-1}{p}. \quad (2.4.2)$$

In the case $p = \infty$ this reads $q \geq \frac{2n}{n-1}$.

(e) *The conjecture.* The *restriction conjecture* asserts that (2.4.2) together with the single further condition $q > \frac{2n}{n-1}$ — which comes from testing $g \equiv 1$ against the decay of $\widehat{d\sigma}$, computed in Chapter 2.8 — is *sufficient*. It is a theorem for $n = 2$ (Fefferman–Stein, 1970) and open for every $n \geq 3$.

What to take from this. Locating the conjectured range was a page of scaling. The same was true for the Kakeya maximal conjecture in Worked Example 2.2.3, and it will be true for decoupling in Volume 13. In this subject the necessary conditions are elementary and the sufficiency is the work of decades. A reader who can reproduce (c) and (d) from memory has the right picture of what a wave packet is: a function concentrated on a tube dual to a cap, which is what Eg became.

Figure 2.4.1 (to be drawn). *A cap and its dual box*

Content required: The unit sphere with a cap κ_δ of angular radius δ at the north pole; alongside it, in physical space, the dual box T_δ of dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$, with the long axis normal to the cap. Mark the two lengths and the deviation $\approx \delta^2$ of the cap from its tangent plane. This is the single most-used picture in the series.

Drills

Drill 2.4.1. Compute $\widehat{f}$ for $f(x) = e^{-|x|}$ on $\mathbb{R}$, and verify directly that $\widehat{f} \in L^1$ so that Theorem 2.4.6 applies.

Drill 2.4.2. Show that $\mathcal{F}^4 = \text{Id}$ on $L^2(\mathbb{R}^n)$, and deduce that the spectrum of $\mathcal{F}$ is contained in $\{1, i, -1, -i\}$. Exhibit an eigenfunction for each eigenvalue when $n = 1$ (start from the Gaussian).

Drill 2.4.3. Complete the argument sketched in the Pitfall after Theorem 2.4.9: show by scaling that $q = p'$ is forced, and then find an explicit $f \in L^p(\mathbb{R})$ for $p > 2$ with $\widehat{f} \notin L^{p'}$.

Drill 2.4.4. Redo Worked Example 2.4.3(c) for a cap on a *hyperplane* rather than a paraboloid, and show that the resulting dual set is a slab of infinite measure rather than a tube. Explain in one sentence why no estimate of the form in (d) can hold in that case.

Notes

Plancherel's theorem is from 1910. The Gaussian-damping proof of inversion given here is standard and has the advantage of proving Plancherel simultaneously.

The restriction problem was posed by Stein in 1967. The condition (2.4.2) is Knapp's, and the sharpness of the whole range in $n = 2$ is due to Fefferman and Stein; Zygmund gave the endpoint. Volume 6 proves the Stein–Tomas theorem, which settles the case $p = 2$ in all dimensions, and Volume 12 presents the polynomial partitioning method and the current records.

Chapter 2.5

The Schwartz Space and Tempered Distributions

Chapter 2.4 left two defects. The inversion theorem needed the hypothesis $\hat{f} \in L^1$, which is not verifiable in advance and fails for objects as simple as the indicator of a box. And the extension operator of Worked Example 2.4.3 was written as the Fourier transform of a measure carried by a paraboloid — an object with no density, to which the definition of Chapter 2.4 does not apply at all.

Both are repaired by enlarging the class of objects the transform acts on. The strategy is the standard one in analysis: find a space of exceptionally nice test functions on which the transform is a perfect isomorphism, and then define the transform of a rough object by how it pairs against test functions. The nice space is the Schwartz space; the rough objects are the tempered distributions; and surface measure on a sphere is one of them, which is what makes the restriction problem askable.

2.5.1 The Schwartz space

Notation 2.5.1. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ write $|\alpha| = \sum \alpha_j$, $x^\alpha = \prod x_j^{\alpha_j}$ and $\partial^\alpha = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}$.

Definition 2.5.2. The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ consists of the $f \in C^\infty(\mathbb{R}^n)$ for which every seminorm

$$\rho_{\alpha\beta}(f) = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|$$

is finite: every derivative decays faster than every polynomial.

Examples: $e^{-\pi|x|^2}$, and every C^∞ function of compact support. Not examples: $e^{-|x|}$ (not smooth at 0), $(1 + |x|^2)^{-N}$ (decays only polynomially).

Proposition 2.5.3. $\mathcal{S}(\mathbb{R}^n) \subseteq L^p(\mathbb{R}^n)$ for every $p \in [1, \infty]$, and $\mathcal{S}$ is closed under differentiation, multiplication by polynomials, and convolution.

Proof. For $p < \infty$, $|f(x)| \leq C_N(1+|x|)^{-N}$ for every N , taking $N > n/p$ makes $|f|^p$ integrable by polar coordinates (Volume 1, Theorem 6.10). Closure under ∂^β and under multiplication by x^α is immediate from the definition, and convolution follows from the integral representation together with differentiation under the integral sign. $\square$

Definition 2.5.4. $f_k \rightarrow f$ in $\mathcal{S}$ means $\rho_{\alpha\beta}(f_k - f) \rightarrow 0$ for every α, β .

Proposition 2.5.5. $\mathcal{S}(\mathbb{R}^n)$ is complete: a sequence Cauchy in every seminorm converges in $\mathcal{S}$. Moreover $C_c^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$.

Proof. If (f_k) is Cauchy in every $\rho_{\alpha\beta}$, then each $\partial^\beta f_k$ is uniformly Cauchy, so converges uniformly to some g_β ; standard real-analysis arguments give $g_\beta = \partial^\beta g_0$ and $g_0 \in C^\infty$. The bounds $\rho_{\alpha\beta}(f_k) \leq C$ pass to the limit, so $g_0 \in \mathcal{S}$ and $f_k \rightarrow g_0$ in $\mathcal{S}$.

For density, fix $\chi \in C_c^\infty$ with $\chi = 1$ on $B(0, 1)$ and set $f_k(x) = f(x)\chi(x/k)$. Then $f_k \in C_c^\infty$, and by the Leibniz rule $\partial^\beta(f - f_k)$ is a sum of terms each containing either $(1 - \chi(x/k))$ — supported in $|x| \geq k$, where $x^\alpha \partial^\gamma f$ is small — or a factor $k^{-|\gamma|}(\partial^\gamma \chi)(x/k)$ with $|\gamma| \geq 1$, which carries a factor k^{-1} . Both tend to 0 in every seminorm. $\square$

Theorem 2.5.6 ($\mathcal{F}$ is an isomorphism of $\mathcal{S}$). *The Fourier transform maps $\mathcal{S}(\mathbb{R}^n)$ bijectively onto itself, continuously in both directions, with inverse $(\mathcal{F}^{-1}g)(x) = \widehat{g}(-x)$. In particular the inversion formula holds pointwise for every $f \in \mathcal{S}$.*

Proof. Step 1. $\mathcal{F}$ maps $\mathcal{S}$ into $\mathcal{S}$.

Let $f \in \mathcal{S}$. By Theorem 2.4.2(vi), applied repeatedly — legitimate since every $x^\alpha \partial^\beta f$ is in L^1 by Theorem 2.5.3 —

$$\xi^\alpha \partial^\beta \widehat{f}(\xi) = \frac{(-1)^{|\beta|}}{(2\pi i)^{|\alpha| - |\beta|}} \widehat{\partial^\alpha (x^\beta f)}(\xi) \quad (2.5.1)$$

up to a harmless constant, whence

$$\rho_{\alpha\beta}(\widehat{f}) \leq C_{\alpha\beta} \|\partial^\alpha (x^\beta f)\|_{L^1} < \infty,$$

using $\|\widehat{g}\|_\infty \leq \|g\|_1$. The right-hand side is a finite combination of seminorms of f , which also gives continuity.

Step 2. *Inversion.*

For $f \in \mathcal{S}$ we have $f \in L^1$ and, by Step 1, $\widehat{f} \in \mathcal{S} \subseteq L^1$. So Theorem 2.4.6 applies and gives $f(x) = \int \widehat{f}(\xi) e(x \cdot \xi) d\xi$ for a.e. x ; both sides being continuous, the identity holds everywhere.

Step 3. Bijectivity.

Step 2 exhibits a two-sided inverse: $\mathcal{F}^{-1}g(x) = \widehat{g}(-x)$ maps $\mathcal{S}$ to $\mathcal{S}$ by Step 1 applied to $g(-\cdot)$, and $\mathcal{F}^{-1}\mathcal{F} = \mathcal{F}\mathcal{F}^{-1} = \text{Id}$ on $\mathcal{S}$. $\square$

2.5.2 Tempered distributions

Definition 2.5.7. A *tempered distribution* is a linear functional $u : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$ which is continuous, in the sense that there are $N \in \mathbb{N}$ and $C > 0$ with

$$|u(\varphi)| \leq C \sum_{|\alpha|, |\beta| \leq N} \rho_{\alpha\beta}(\varphi) \quad \text{for all } \varphi \in \mathcal{S}. \quad (2.5.2)$$

The space of these is $\mathcal{S}'(\mathbb{R}^n)$. We write $\langle u, \varphi \rangle$ for $u(\varphi)$.

Proposition 2.5.8 (Examples). *Each of the following defines a tempered distribution.*

- (i) $u_f(\varphi) = \int f\varphi$, for $f \in L^p(\mathbb{R}^n)$, any $p \in [1, \infty]$;
- (ii) $u_f(\varphi) = \int f\varphi$, for measurable f with $|f(x)| \leq C(1 + |x|)^M$ (polynomial growth);
- (iii) $\delta_a(\varphi) = \varphi(a)$;
- (iv) $\mu(\varphi) = \int \varphi d\mu$, for any finite Borel measure μ on $\mathbb{R}^n$, and more generally for any Borel measure with $\int (1 + |x|)^{-M} d\mu < \infty$ for some M .

Proof. (i) By Hölder, $|u_f(\varphi)| \leq \|f\|_p \|\varphi\|_{p'}$, and $\|\varphi\|_{p'}$ is bounded by finitely many seminorms as in Theorem 2.5.3. (ii) Choose $N > M + n + 1$; then $|\int f\varphi| \leq C \int (1 + |x|)^{M-N} dx \cdot \sup_x (1 + |x|)^N |\varphi(x)|$, and the supremum is bounded by finitely many $\rho_{\alpha 0}(\varphi)$. (iii) is $|\delta_a(\varphi)| \leq \rho_{00}(\varphi)$. (iv) is the same computation as (ii) with dx replaced by $d\mu$. $\square$

Definition 2.5.9 (Operations). For $u \in \mathcal{S}'$, $\varphi \in \mathcal{S}$, define by duality:

$$\langle \partial^\alpha u, \varphi \rangle = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle, \quad \langle Pu, \varphi \rangle = \langle u, P\varphi \rangle \quad (P \text{ a polynomial}),$$

$$\langle \tau_h u, \varphi \rangle = \langle u, \tau_{-h} \varphi \rangle, \quad \langle \widehat{u}, \varphi \rangle = \langle u, \widehat{\varphi} \rangle.$$

Proposition 2.5.10. *Each of these is again a tempered distribution, and each is consistent with the classical operation when $u = u_f$ for f nice enough.*

Proof. Continuity in the sense of (2.5.2) follows because $\varphi \mapsto \partial^\alpha \varphi$, $\varphi \mapsto P\varphi$ and $\varphi \mapsto \widehat{\varphi}$ are all continuous maps $\mathcal{S} \rightarrow \mathcal{S}$ — the last by Theorem 2.5.6 — so they carry the finite family of seminorms in (2.5.2) to a finite family.

Consistency: for the derivative, if $f \in C^\infty$ with polynomial growth then integration by parts gives $\int (\partial^\alpha f)\varphi = (-1)^{|\alpha|} \int f \partial^\alpha \varphi$, so the definition extends the classical one. For the transform, the multiplication formula Theorem 2.4.5 says exactly $\int \widehat{f}\varphi = \int f\widehat{\varphi}$ for $f \in L^1$, which is the stated definition. $\square$

Theorem 2.5.11. $\mathcal{F} : \mathcal{S}' \rightarrow \mathcal{S}'$ is a bijection, with inverse given by $\langle \mathcal{F}^{-1}u, \varphi \rangle = \langle u, \mathcal{F}^{-1}\varphi \rangle$. It agrees with the L^1 transform on L^1 and with the L^2 transform on L^2 .

Proof. Bijectivity is immediate by dualising Theorem 2.5.6. Agreement on L^1 is Theorem 2.5.10. On L^2 : for $f \in L^1 \cap L^2$ the two transforms agree, and both the distributional transform and the Plancherel extension are continuous from L^2 into $\mathcal{S}'$, so they agree on the closure, which is L^2 . $\square$

WORKED EXAMPLE 2.5.1— TIER A $\widehat{\delta} = 1$ and $\widehat{1} = \delta$

From the definitions: $\langle \widehat{\delta}, \varphi \rangle = \langle \delta, \widehat{\varphi} \rangle = \widehat{\varphi}(0) = \int \varphi(x) dx = \langle 1, \varphi \rangle$. So $\widehat{\delta} = 1$. Conversely $\langle \widehat{1}, \varphi \rangle = \int \widehat{\varphi}(\xi) d\xi = \varphi(0)$ by the inversion theorem Theorem 2.5.6 evaluated at $x = 0$. So $\widehat{1} = \delta$.

Neither statement makes sense classically — δ is not a function and 1 is not integrable — and both are one line here. This is the whole point of the construction.

WORKED EXAMPLE 2.5.2— TIER B **The transform of $|x|^{-a}$**

Proposition 2.5.12. For $0 < a < n$, the function $|x|^{-a}$ is a tempered distribution and

$$\begin{aligned} |\cdot|^{-a}(\xi) &= c_{n,a} |\xi|^{-(n-a)} \\ , \quad c_{n,a} &= \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})}. \end{aligned}$$

Proof. That $|x|^{-a} \in \mathcal{S}'$ follows from Theorem 2.5.8(ii) after splitting at $|x| = 1$: near the origin $|x|^{-a}$ is integrable since $a < n$, and away from it the function is bounded.

The computation rests on the *subordination identity*: for $s > 0$ and $\lambda > 0$,

substituting $u = \lambda t$ in the definition of the Gamma function gives

$$\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t} dt.$$

Apply it with $\lambda = \pi|x|^2$ and $s = a/2$:

$$|x|^{-a} = \frac{\pi^{a/2}}{\Gamma(a/2)} \int_0^\infty t^{\frac{a}{2}-1} e^{-\pi t|x|^2} dt.$$

Now pair against $\varphi \in \mathcal{S}$ and use Fubini — legitimate since the double integral of the absolute value is finite, the t -integral converging at both ends because $0 < a < n$ — together with $\widehat{e^{-\pi t|\cdot|^2}}(\xi) = t^{-n/2} e^{-\pi|\xi|^2/t}$ from Theorem 2.4.4:

$$\langle |\cdot|^{-a}, \varphi \rangle = \frac{\pi^{a/2}}{\Gamma(a/2)} \int_0^\infty t^{\frac{a}{2}-1} \left(\int_{\mathbb{R}^n} t^{-\frac{n}{2}} e^{-\pi|\xi|^2/t} \varphi(\xi) d\xi \right) dt.$$

Substituting $t \mapsto 1/t$ in the outer integral turns $t^{\frac{a}{2}-\frac{n}{2}-1} e^{-\pi|\xi|^2/t}$ into $t^{\frac{n-a}{2}-1} e^{-\pi t|\xi|^2}$, and applying the subordination identity in reverse with $s = (n-a)/2$ gives

$$\langle |\cdot|^{-a}, \varphi \rangle = \frac{\pi^{a/2}}{\Gamma(a/2)} \cdot \frac{\Gamma(\frac{n-a}{2})}{\pi^{\frac{n-a}{2}}} \int_{\mathbb{R}^n} |\xi|^{-(n-a)} \varphi(\xi) d\xi,$$

which is the claim. $\square$

Why it matters. The exponents are conjugate: $a \leftrightarrow n-a$, with the self-dual point $a = n/2$. These are the *Riesz potentials*, and the identity above is the reason the Hardy–Littlewood–Sobolev inequality of Chapter 2.7 has the exponents it does. In Volume 8 the same computation, read backwards, converts the energy integral of a measure into a statement about $\hat{\mu}$, which is the Fourier-analytic route to the Falconer distance problem. $\square$

WORKED EXAMPLE 2.5.3— TIER C: RESEARCH TASTE Surface measure is a tempered distribution

Let σ denote surface measure on the unit sphere $\mathbb{S}^{n-1} \subseteq \mathbb{R}^n$, constructed in Theorem 1.6.11 (polar coordinates).

(a) *It is tempered, and the verification is two lines.* σ is a finite Borel measure, with $\sigma(\mathbb{S}^{n-1}) = n\omega_n < \infty$. So for $\varphi \in \mathcal{S}$,

$$|\langle \sigma, \varphi \rangle| = \left| \int_{\mathbb{S}^{n-1}} \varphi d\sigma \right| \leq \sigma(\mathbb{S}^{n-1}) \rho_{00}(\varphi),$$

which is (2.5.2) with $N = 0$. By Theorem 2.5.8(iv), $\sigma \in \mathcal{S}'(\mathbb{R}^n)$.

(b) *Therefore it has a Fourier transform.* By Theorem 2.5.11, $\widehat{\sigma} \in \mathcal{S}'$ is defined, and unwinding the definition together with Fubini gives the classical-looking formula

$$\widehat{d\sigma}(\xi) = \int_{\mathbb{S}^{n-1}} e(-\omega \cdot \xi) d\sigma(\omega),$$

a bounded continuous function of ξ — bounded by $\sigma(\mathbb{S}^{n-1})$, and continuous by dominated convergence.

Pause on what has happened. The measure σ is supported on a set of Lebesgue measure zero; it has no density; the definition of Chapter 2.4 is inapplicable to it. Yet its transform is not merely a distribution but an honest bounded continuous function, and Chapter 2.8 will show that it *decays*, at the rate $|\xi|^{-(n-1)/2}$.

(c) *Why this is the enabling step for the whole series.* The restriction problem asks whether $\widehat{f}$, defined a priori only almost everywhere for $f \in L^p(\mathbb{R}^n)$, can be meaningfully restricted to $\mathbb{S}^{n-1}$ — a set of measure zero, where “almost everywhere” says nothing. By duality (Volume 1, Corollary 8.12) the question is equivalent to a bound on the adjoint,

$$Eg = \widehat{gd\sigma},$$

and $gd\sigma$ is exactly the kind of object admitted by Theorem 2.5.8(iv). Without distribution theory the extension operator E has no definition, and the conjecture cannot be stated. With it, E is elementary and the whole problem becomes a question about the decay and oscillation of one explicit function, $\widehat{d\sigma}$.

(d) *A caution.* That σ is tempered is easy; that $\widehat{d\sigma}$ decays is not, and it is false without curvature. If σ were surface measure on a piece of hyperplane, $\widehat{d\sigma}$ would be constant in the normal direction and would not decay at all (Drill 5.4). Everything in Volumes 5, 6, 12 and 13 turns on the quantitative form of that distinction.

WORKED EXAMPLE 2.5.4 — TIER B Working with tempered distributions in practice

Three manoeuvres cover almost every use of $\mathcal{S}'$ in this series.

(a) *Differentiate anything.* Every $u \in \mathcal{S}'$ has derivatives of all orders, defined by $\langle \partial^\alpha u, \varphi \rangle = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle$. So one may differentiate first and worry about regularity afterwards — the manoeuvre that makes Sobolev theory possible.

(b) *Transform anything of polynomial growth.* $\mathcal{F}$ is a bijection of $\mathcal{S}'$, by

Theorem 2.5.11. In particular a finite measure has a Fourier transform, which is what Volumes 7 and 8 rely on throughout: $\widehat{\mu}$ is a bounded continuous function when μ is a compactly supported finite measure, and the energy machinery of Volume 7 is an identity in $\mathcal{S}'$.

(c) *Multiply only by smooth functions of moderate growth.* This is the restriction that bites. Two distributions cannot in general be multiplied; δ_0^2 is meaningless. Whenever an argument appears to multiply distributions — pairing a surface measure against another — it must be interpreted, and Volume 8 shows the standard interpretation by mollification.

The check to run. Given a formal computation with distributions, ask at each step which of (a), (b), (c) licenses it. If none does, the step needs a limiting argument, and the estimates must be made uniform before the limit is taken.

Drills

Drill 2.5.1. Show that $e^{-|x|} \notin \mathcal{S}(\mathbb{R})$ but that it defines a tempered distribution, and compute its transform.

Drill 2.5.2. Show that the principal value $\text{p.v.} \frac{1}{x}$, defined by $\langle u, \varphi \rangle = \lim_{\varepsilon \rightarrow 0} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx$, is a tempered distribution on $\mathbb{R}$, and that $\widehat{u}(\xi) = -i\pi \operatorname{sgn} \xi$. (This is the Hilbert transform multiplier, developed in Volume 3.)

Drill 2.5.3. Verify (2.5.1) carefully for $n = 1$, $\alpha = \beta = 1$, and confirm the constant.

Drill 2.5.4. Let σ_0 be Lebesgue measure on the hyperplane $\{x_n = 0\} \subseteq \mathbb{R}^n$, regarded as a tempered distribution. Show $\widehat{\sigma_0}$ is Lebesgue measure on the line $\{\xi' = 0\}$, and in particular does not decay in the ξ_n direction.

Notes

Schwartz introduced $\mathcal{S}$ and $\mathcal{S}'$ in the 1940s, and the theory was fully developed in his *Théorie des distributions* (1950–51). The subordination identity used in Worked Example 2.5.2 is due to Riesz.

Nothing in this chapter is difficult, and that is itself notable: distribution theory is a piece of bookkeeping, and its power comes entirely from Theorem 2.5.6 — the fact that $\mathcal{F}$ is an isomorphism of a space small enough to differentiate freely and large enough to be dense in everything.

Chapter 2.6

Multipliers, Khintchine, and Littlewood–Paley

2.6.1 Multiplier operators

Definition 2.6.1. For $m \in L^\infty(\mathbb{R}^n)$ define T_m on $L^2(\mathbb{R}^n)$ by $\widehat{T_m f} = m \widehat{f}$, that is $T_m f = \mathcal{F}^{-1}(m \mathcal{F} f)$. The function m is the *symbol* or *multiplier*.

Proposition 2.6.2. $\|T_m\|_{L^2 \rightarrow L^2} = \|m\|_\infty$.

Proof. By Plancherel (Theorem 2.4.8), $\|T_m f\|_2 = \|\widehat{m f}\|_2 \leq \|m\|_\infty \|\widehat{f}\|_2 = \|m\|_\infty \|f\|_2$. For the reverse, choose a set E of positive finite measure on which $|m| \geq \|m\|_\infty - \varepsilon$ and take f with $\widehat{f} = \mathbf{1}_E$. $\square$

So on L^2 the theory is trivial and complete: bounded symbols give bounded operators, with equality of norms. Everything difficult in this subject happens at $p \neq 2$, where no such characterisation exists, and where the geometry of the set on which m varies begins to matter.

Looking ahead: the ball multiplier

The single most important example is $m = \mathbf{1}_{B(0,1)}$, whose operator is the partial-sum operator for the Fourier inversion integral. By Theorem 2.6.2 it is bounded on L^2 . Fefferman proved in 1971 that it is bounded on $L^p(\mathbb{R}^n)$, $n \geq 2$, for *no* $p \neq 2$ — and the proof is a Besicovitch set. That theorem is the historical origin of the connection between Kakeya and Fourier analysis, and it is proved in Volume 5. The reader may note that everything needed to state it is now available.

Remark 2.6.3 (What is deferred, and why). The standard sufficient condition for L^p boundedness is the Hörmander–Mikhlin theorem: if $|\partial^\alpha m(\xi)| \lesssim |\xi|^{-|\alpha|}$

for $|\alpha| \leq \lfloor n/2 \rfloor + 1$, then T_m is bounded on L^p for $1 < p < \infty$. Its proof requires Calderón–Zygmund theory, which this series develops in Volume 3.

Under the standing rule of this series a result is either proved where it is stated or not stated at all, so Hörmander–Mikhlin appears in Volume 3 and not here. Nothing in Volumes 1–2 needs it, and the tools Volumes 12 and 13 actually use — Khintchine’s inequality, Bernstein’s inequalities, and L^2 orthogonality for lacunary pieces — are proved below in full.

2.6.2 Khintchine’s inequality

Definition 2.6.4. The *Rademacher functions* on $[0, 1]$ are $r_j(t) = \operatorname{sgn} \sin(2^j \pi t)$, $j \geq 1$. They take values ± 1 , and as random variables on $([0, 1], m)$ they are independent with mean 0: for distinct $j_1 < \dots < j_k$, $\int_0^1 r_{j_1} \dots r_{j_k} = 0$.

Theorem 2.6.5 (Khintchine). *For $0 < p < \infty$ there are constants $0 < A_p \leq B_p < \infty$ such that for all N and all scalars $a_1, \dots, a_N \in \mathbb{C}$,*

$$\begin{aligned} A_p \left(\sum_{j=1}^N |a_j|^2 \right)^{1/2} &\leq \left(\int_0^1 \left| \sum_{j=1}^N a_j r_j(t) \right|^p dt \right)^{1/p} \\ &\leq B_p \left(\sum_{j=1}^N |a_j|^2 \right)^{1/2}. \end{aligned}$$

Idea of the proof. The case $p = 2$ is an identity, by orthogonality of the r_j . For $p > 2$ one shows that the random sum has *sub-Gaussian tails* — a Chernoff bound obtained by estimating the exponential moment $\int e^{\lambda S}$, which factors by independence into a product of hyperbolic cosines, each bounded by a Gaussian. Layer-cake integration then converts the tail bound into a moment bound. For $p < 2$ one runs the log-convexity of L^p norms backwards, using the already established case $p = 4$.

Proof. Write $S(t) = \sum_j a_j r_j(t)$ and $A = (\sum_j |a_j|^2)^{1/2}$.

Step 1. $p = 2$.

By independence and $r_j^2 = 1$, $\int_0^1 |S|^2 = \sum_{j,k} a_j \overline{a_k} \int_0^1 r_j r_k = \sum_j |a_j|^2 = A^2$. So $A_2 = B_2 = 1$.

Step 2. *Sub-Gaussian tails, real coefficients.*

Assume $a_j \in \mathbb{R}$ and $A = 1$. For $\lambda \in \mathbb{R}$, independence gives

$$\begin{aligned} \int_0^1 e^{\lambda S(t)} dt &= \prod_{j=1}^N \int_0^1 e^{\lambda a_j r_j} = \prod_{j=1}^N \cosh(\lambda a_j) \leq \prod_{j=1}^N e^{\lambda^2 a_j^2 / 2} \\ &= e^{\lambda^2 / 2}, \end{aligned}$$

using $\cosh u \leq e^{u^2/2}$, which follows by comparing the Taylor series $\sum u^{2k}/(2k)!$ and $\sum u^{2k}/(2^k k!)$ term by term, since $(2k)! \geq 2^k k!$.

Hence for $\lambda > 0$, by Chebyshev,

$$m(\{S > \mu\}) \leq e^{-\lambda\mu} \int_0^1 e^{\lambda S} \leq e^{\lambda^2/2 - \lambda\mu},$$

and optimising at $\lambda = \mu$ gives $e^{-\mu^2/2}$. Applying the same to $-S$,

$$m(\{|S| > \mu\}) \leq 2e^{-\mu^2/2}. \quad (2.6.1)$$

Step 3. *The upper bound for $p > 0$.*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) and (2.6.1),

$$\begin{aligned} \int_0^1 |S|^p &= p \int_0^\infty \mu^{p-1} m(\{|S| > \mu\}) d\mu \\ &\leq 2p \int_0^\infty \mu^{p-1} e^{-\mu^2/2} d\mu = B_p^p < \infty, \end{aligned}$$

a finite constant depending only on p . Rescaling by A gives the upper bound for real coefficients; for complex a_j , split into real and imaginary parts and use $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|$, at the cost of a factor 2.

Step 4. *The lower bound for $p \geq 2$*

is immediate from Step 1 and $\|S\|_p \geq \|S\|_2$ (Drill 1.8.1, the measure being a probability measure).

Step 5. *The lower bound for $0 < p < 2$.*

Choose $\theta \in (0, 1)$ with $\frac{1}{2} = \frac{\theta}{p} + \frac{1-\theta}{4}$. By log-convexity of L^p norms (Drill 1.8.2),

$$A = \|S\|_2 \leq \|S\|_p^\theta \|S\|_4^{1-\theta} \leq \|S\|_p^\theta (B_4 A)^{1-\theta},$$

using Step 3 with $p = 4$. Rearranging, $A^\theta \leq B_4^{1-\theta} \|S\|_p^\theta$, so $\|S\|_p \geq B_4^{-(1-\theta)/\theta} A$. $\square$

2.6.3 Littlewood–Paley projections and Bernstein’s inequalities

Definition 2.6.6. Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\psi = 1$ on $B(0, 1)$ and $\operatorname{supp} \psi \subseteq B(0, 2)$, and for a dyadic $N = 2^k$ put

$$\widehat{P_{\leq N} f} = \psi(\xi/N) \widehat{f}(\xi), \quad P_N = P_{\leq N} - P_{\leq N/2},$$

so that $\widehat{P_N f}$ is supported in the annulus $|\xi| \approx N$ and $f = \sum_N \text{dyadic} P_N f$ in L^2 .

Theorem 2.6.7 (Bernstein). *Let $1 \leq p \leq q \leq \infty$ and suppose $\widehat{f}$ is supported in $B(0, N)$. Then*

$$\|f\|_q \lesssim_{n,p,q} N^{n(\frac{1}{p}-\frac{1}{q})} \|f\|_p, \quad (2.6.2)$$

and for every multi-index α ,

$$\|\partial^\alpha f\|_p \lesssim_{n,\alpha} N^{|\alpha|} \|f\|_p. \quad (2.6.3)$$

The same holds, with the same proof, if $\widehat{f}$ is supported in an annulus $|\xi| \approx N$ or in a ball $B(\xi_0, N)$ of radius N anywhere.

Idea of the proof. A function with Fourier support in a ball of radius N is reproduced by convolution with a fixed bump rescaled to width $1/N$. That is Theorem 2.4.2(iii) again: frequency support of size N means the function cannot oscillate on scales below $1/N$, so it is essentially constant on each such ball, and every L^p norm of a function constant at scale $1/N$ is comparable to every other after the obvious scaling factor.

Proof. Choose $\eta \in \mathcal{S}(\mathbb{R}^n)$ with $\widehat{\eta} = 1$ on $B(0, 1)$ and $\text{supp } \widehat{\eta} \subseteq B(0, 2)$; such η exists since $\mathcal{F}$ is onto $\mathcal{S}$ (Theorem 2.5.6). Put $\eta_N(x) = N^n \eta(Nx)$, so $\widehat{\eta_N}(\xi) = \widehat{\eta}(\xi/N) = 1$ on $B(0, N)$.

Since f is supported in $B(0, N)$, we have $\widehat{f\eta_N} = \widehat{f}$, hence

$$f = f * \eta_N.$$

(The identity holds in $\mathcal{S}'$ and hence pointwise a.e. for $f \in L^p$.) By Young's inequality (Worked Example 2.2.1) with $1 + \frac{1}{q} = \frac{1}{p} + \frac{1}{r}$,

$$\|f\|_q = \|f * \eta_N\|_q \leq \|f\|_p \|\eta_N\|_r.$$

Now $\|\eta_N\|_r = N^n (\int |\eta(Nx)|^r dx)^{1/r} = N^{n(1-1/r)} \|\eta\|_r$, and $1 - \frac{1}{r} = \frac{1}{p} - \frac{1}{q}$, giving (2.6.2).

For (2.6.3), differentiate the identity $f = f * \eta_N$: $\partial^\alpha f = f * \partial^\alpha \eta_N$, and $\|\partial^\alpha \eta_N\|_1 = N^{|\alpha|} \|\partial^\alpha \eta\|_1$ by scaling. Apply Young with $r = 1$.

For a ball centred at ξ_0 , replace η_N by $e(x \cdot \xi_0) \eta_N(x)$, using Theorem 2.4.2(ii); the L^r norms are unchanged. For an annulus, use that it is contained in a ball of radius $2N$. $\square$

WORKED EXAMPLE 2.6.1—TIER A **Bernstein is sharp, and in which direction**

Take $f = \eta_N$ itself, so $\widehat{f}$ is supported in $B(0, 2N)$. Then $\|\eta_N\|_p =$

$N^{n(1-1/p)}\|\eta\|_p$, so

$$\frac{\|\eta_N\|_q}{\|\eta_N\|_p} = N^{n(1-1/q)-n(1-1/p)} \cdot \frac{\|\eta\|_q}{\|\eta\|_p} = N^{n(\frac{1}{p}-\frac{1}{q})} \cdot \frac{\|\eta\|_q}{\|\eta\|_p},$$

matching (2.6.2) exactly. So the exponent cannot be improved.

Note that the inequality goes only one way: $\|f\|_q \lesssim N^{\dots}\|f\|_p$ for $q \geq p$, never the reverse. A frequency-localised function may be small in L^∞ and large in L^1 — spread out — but not the other way round past the stated factor.

Proposition 2.6.8 (L^2 orthogonality of lacunary pieces). *For $f \in L^2(\mathbb{R}^n)$,*

$$\|f\|_2^2 \approx \sum_{N \text{ dyadic}} \|P_N f\|_2^2.$$

Proof. By Plancherel it suffices to prove $\|\widehat{f}\|_2^2 \approx \sum_N \|\widehat{P_N f}\|_2^2$, and $\widehat{P_N f} = (\psi(\xi/N) - \psi(2\xi/N))\widehat{f}(\xi)$. Each $\xi \neq 0$ lies in the support of at most two of these cutoffs, and the cutoffs sum to 1; so pointwise $\sum_N |\widehat{P_N f}|^2 \approx |\widehat{f}|^2$, with implicit constants $\frac{1}{2}$ and 2. Integrate. $\square$

Remark 2.6.9 (The L^p statement is Volume 3). The Littlewood–Paley theorem asserts $\|f\|_p \approx \|(\sum_N |P_N f|^2)^{1/2}\|_p$ for $1 < p < \infty$. The standard proof randomises the signs of the pieces, applies Theorem 2.6.5 to reduce to bounding $\|\sum_N \pm P_N f\|_p$ uniformly, and then invokes Hörmander–Mikhlin — which by Theorem 2.6.3 is Volume 3. So the theorem is proved there. Theorem 2.6.8 is the case $p = 2$, which is all that is used before then.

WORKED EXAMPLE 2.6.2— TIER C: RESEARCH TASTE **Decoupling is false without curvature**

Worked Example 1.8.2 (almost orthogonality) stated the ℓ^2 decoupling inequality for the paraboloid and asserted that curvature is what makes it true. We can now prove the negative half completely.

(a) *The setting.* Let $\delta = 1/N$ and partition a region of frequency space into N caps $\theta_1, \dots, \theta_N$ of diameter $\approx \delta$. For functions f_j with $\widehat{f_j}$ supported in θ_j , the ℓ^2 decoupling inequality at exponent p asserts

$$\left\| \sum_{j=1}^N f_j \right\|_{L^p(\mathbb{R}^n)} \lesssim_\varepsilon N^\varepsilon \left(\sum_{j=1}^N \|f_j\|_{L^p}^2 \right)^{1/2}. \quad (2.6.4)$$

The trivial bound, from Cauchy–Schwarz applied to the N summands, is (2.6.4) with N^ε replaced by $N^{1/2-1/p}$ — a genuine power of N . The content of decoupling is the removal of that power.

(b) *The counterexample: caps on a hyperplane.* Take the caps to lie on a *flat* piece of hypersurface, say

$$\theta_j = \left[\frac{j-1}{N}, \frac{j}{N} \right] \times [0, \delta]^{n-1} \subseteq \mathbb{R}^n,$$

so the θ_j tile a box and lie in the hyperplane-like slab $\{|\xi_2|, \dots, |\xi_n| \leq \delta\}$. Let η be a fixed Schwartz function with $\hat{\eta}$ supported in $[0, 1/N] \times [0, \delta]^{n-1}$ and $\hat{\eta} \geq 0$, $\hat{\eta}(0) > 0$, and set

$$f_j(x) = e(x_1 \cdot \frac{j-1}{N}) \eta(x),$$

so that $\hat{f}_j$ is $\hat{\eta}$ translated into θ_j (Theorem 2.4.2(ii)). Each f_j has the same modulus: $|f_j| = |\eta|$, so $\|f_j\|_p = \|\eta\|_p$ for every j .

(c) *The sum is as large as it can be.* On the set where $|x_1| \leq \frac{1}{100}$, every phase $e(x_1(j-1)/N)$ with $1 \leq j \leq N$ lies within an arc of angle $2\pi/100$ of 1, so

$$\left| \sum_{j=1}^N f_j(x) \right| \geq \frac{1}{2} N |\eta(x)|.$$

Hence $\|\sum_j f_j\|_{L^p} \gtrsim N \|\eta\|_{L^p(|x_1| \leq 1/100)} \gtrsim N \|\eta\|_p$, provided η is chosen concentrated in that region.

(d) *Conclusion.* The right-hand side of (2.6.4) is $(\sum_j \|\eta\|_p^2)^{1/2} = N^{1/2} \|\eta\|_p$. So (2.6.4) would force

$$N \|\eta\|_p \lesssim_\varepsilon N^\varepsilon N^{1/2} \|\eta\|_p, \quad \text{i.e.} \quad N^{1/2} \lesssim_\varepsilon N^\varepsilon,$$

which is false for large N . Indeed the trivial Cauchy–Schwarz bound $N^{1/2-1/p}$ is, in this configuration, *attained* up to constants for $p = \infty$ and cannot be improved at all.

(e) *What curvature does.* The failure has a one-line geometric cause. Each f_j is concentrated on the same dual box — the tubes dual to the caps θ_j all point the same way, because the caps are translates of one another in a single direction. Constructive interference is therefore possible at every point of a large set simultaneously.

If instead the caps lie on the paraboloid, the cap $\{(\xi', |\xi'|^2) : |\xi' - c_j| \leq \delta\}$ has a dual tube pointing in the direction of the *normal* at c_j , and distinct caps have distinct normals. So the N tubes point in N different directions, and the extent to which they can pile up on a common set is exactly the question answered by the Kakeya estimates of Volume 1, Worked Example 2.2.3, and Volume 9.

That is the precise sense in which restriction, decoupling and Kakeya are one subject: decoupling is true because differently-directed tubes cannot concentrate, and false when the tubes are parallel.

Drills

Drill 2.6.1. Show that the multiplier $m(\xi) = \operatorname{sgn} \xi$ on $\mathbb{R}$ gives the Hilbert transform, and compute its kernel using Worked Example 2.5.2’s companion, Drill 5.2.

Drill 2.6.2. Prove $\cosh u \leq e^{u^2/2}$ by comparing Taylor coefficients, and verify $(2k)! \geq 2^k k!$.

Drill 2.6.3. Deduce from Theorem 2.6.7 that if $\widehat{f}$ is supported in $B(0, N)$ then f extends to an entire function; and that $\|f\|_\infty \lesssim N^{n/2} \|f\|_2$.

Drill 2.6.4. In Worked Example 2.6.2, compute the exact loss for finite p and confirm that the ratio of the two sides of (2.6.4) is $\approx N^{1/2-1/p}$, not merely $\gtrsim N^\varepsilon$.

Notes

Khintchine’s inequality dates to 1923. The sub-Gaussian proof given here is the standard one and is the prototype of the concentration inequalities used throughout modern analysis.

Bernstein’s inequality is elementary but is used on essentially every page of the decoupling literature; readers should be able to reproduce (2.6.2) without thinking. Worked Example 2.6.2 is folklore, and appears as the motivating remark in the introduction of Bourgain and Demeter’s 2015 paper.

Chapter 2.7

Sobolev Spaces

The slogan of this chapter is Theorem 2.4.2(vi): differentiation is multiplication by ξ , so *smoothness becomes decay*. Sobolev spaces make the slogan into a definition, and the payoff is that questions about differentiability become questions about the size of $\widehat{f}$ — which we can compute.

2.7.1 Weak derivatives and $W^{k,p}$

Definition 2.7.1. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and a multi-index α , a function $g \in L^1_{\text{loc}}$ is the *weak derivative* $\partial^\alpha f$ if

$$\int_{\mathbb{R}^n} g \varphi = (-1)^{|\alpha|} \int_{\mathbb{R}^n} f \partial^\alpha \varphi \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}^n),$$

which is Theorem 2.5.9 restricted to the case where the distributional derivative happens to be a function. For $k \in \mathbb{N}_0$ and $1 \leq p \leq \infty$,

$$W^{k,p}(\mathbb{R}^n) = \{f \in L^p : \partial^\alpha f \in L^p \text{ for all } |\alpha| \leq k\}, \quad \|f\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_p.$$

Weak derivatives are unique up to null sets, by Theorem 1.5.3 (term-by-term integration) (a locally integrable function integrating to zero against every test function vanishes a.e.).

Proposition 2.7.2. $W^{k,p}(\mathbb{R}^n)$ is a Banach space, and C_c^∞ is dense in it for $p < \infty$.

Proof. Completeness: if (f_j) is Cauchy in $W^{k,p}$ then each $(\partial^\alpha f_j)$ is Cauchy in L^p , hence converges to some g_α by Theorem 1.8.6 (Riesz–Fischer); passing to the limit in the defining identity shows $g_\alpha = \partial^\alpha g_0$ weakly. Density follows by mollification: $f * \phi_t \rightarrow f$ in L^p (Volume 1, Prop. 6.10) and $\partial^\alpha (f * \phi_t) = (\partial^\alpha f) * \phi_t$, followed by truncation as in Theorem 2.5.5. $\square$

2.7.2 The Fourier characterisation: H^s

Definition 2.7.3. For $s \in \mathbb{R}$, the Sobolev space $H^s(\mathbb{R}^n)$ consists of the $f \in \mathcal{S}'(\mathbb{R}^n)$ with $\widehat{f} \in L^1_{\text{loc}}$ and

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi < \infty.$$

Proposition 2.7.4. For $k \in \mathbb{N}_0$, $H^k(\mathbb{R}^n) = W^{k,2}(\mathbb{R}^n)$ with comparable norms.

Proof. By Theorem 2.4.2(vi) and Plancherel, $\|\partial^\alpha f\|_2^2 = \int |2\pi\xi^\alpha|^2 |\widehat{f}|^2$, and

$$\sum_{|\alpha| \leq k} |\xi^\alpha|^2 \approx_{n,k} (1 + |\xi|^2)^k,$$

since both sides are positive continuous functions that are homogeneous of degree $2k$ at infinity and bounded near 0. Integrate against $|\widehat{f}|^2$. $\square$

Looking ahead: why H^s for non-integer s

Nothing in the definition of H^s requires s to be an integer, and this is not a formal generalisation. Fractional smoothness is what the estimates of this subject actually produce: Strichartz estimates measure the smoothing of the Schrödinger evolution in fractions of a derivative; the local smoothing conjecture for the wave equation, proved by Guth–Wang–Zhang in 2020 via decoupling, asserts a gain of exactly $1/p$ derivatives; and the Falconer distance problem is stated in terms of the fractional exponent s in an energy integral (Worked Example 2.7.1).

Theorem 2.7.5 (Sobolev embedding, L^2 case). *If $s > n/2$ then $H^s(\mathbb{R}^n) \subseteq C_0(\mathbb{R}^n)$, the continuous functions vanishing at infinity, with $\|f\|_\infty \lesssim_{n,s} \|f\|_{H^s}$.*

Proof. By Cauchy–Schwarz,

$$\begin{aligned} \int_{\mathbb{R}^n} |\widehat{f}| &= \int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s/2} \cdot (1 + |\xi|^2)^{s/2} |\widehat{f}| \\ &\leq \left(\int (1 + |\xi|^2)^{-s} d\xi \right)^{1/2} \|f\|_{H^s}, \end{aligned}$$

and the first factor is finite exactly when $2s > n$, by polar coordinates (Volume 1, Theorem 6.10). So $\widehat{f} \in L^1$, and Theorem 2.4.6 together with Worked Example 1.5.4 (Riemann–Lebesgue) shows f agrees a.e. with a continuous function vanishing at infinity, with $\|f\|_\infty \leq \|\widehat{f}\|_1$. $\square$

Pitfall: the endpoint $s = n/2$ fails

$H^{n/2}(\mathbb{R}^n) \not\subseteq L^\infty(\mathbb{R}^n)$, and the failure is exactly logarithmic. Take

$$\widehat{f}(\xi) = |\xi|^{-n} (\log|\xi|)^{-1} \mathbf{1}_{|\xi|>2},$$

which is nonnegative. By polar coordinates,

$$\|f\|_{H^{n/2}}^2 \approx \int_{|\xi|>2} |\xi|^n |\xi|^{-2n} (\log|\xi|)^{-2} d\xi \approx \int_2^\infty \frac{dr}{r (\log r)^2} < \infty,$$

so $f \in H^{n/2}$; whereas

$$\int_{\mathbb{R}^n} \widehat{f} \approx \int_2^\infty \frac{dr}{r \log r} = \infty.$$

To see that this forces unboundedness, let f_R be defined by $\widehat{f_R} = \widehat{f} \mathbf{1}_{|\xi|<R}$. Each $\widehat{f_R}$ is integrable with compact support, so f_R is continuous with $f_R(0) = \int \widehat{f_R} \rightarrow \infty$ as $R \rightarrow \infty$, while $\|f_R\|_{H^{n/2}} \leq \|f\|_{H^{n/2}}$ for every R . An embedding $\|g\|_\infty \lesssim \|g\|_{H^{n/2}}$ would therefore give $f_R(0) \lesssim \|f\|_{H^{n/2}}$ uniformly in R , a contradiction.

The gap between $s > n/2$ and $s = n/2$ is thus a divergent $\int dr/(r \log r)$ — the same logarithm as in Worked Example 2.3.1. Endpoint failures of this kind recur throughout the subject, and the $\delta^{-\varepsilon}$ losses in the Kakeya and decoupling conjectures exist precisely to step around them.

2.7.3 The Gagliardo–Nirenberg–Sobolev inequality

For $p \neq 2$ the Fourier transform is unavailable and a different argument is needed. Write $p^* = \frac{np}{n-p}$ for $1 \leq p < n$, the exponent forced by scaling (Drill 7.1).

Theorem 2.7.6. *For $1 \leq p < n$ and $f \in C_c^1(\mathbb{R}^n)$,*

$$\|f\|_{p^*} \lesssim_{n,p} \|\nabla f\|_p.$$

Consequently $W^{1,p}(\mathbb{R}^n) \subseteq L^{p^}(\mathbb{R}^n)$.*

Idea of the proof. Start with $p = 1$. The value $f(x)$ is controlled by the integral of $|\partial_j f|$ along the line through x in direction e_j , for each j separately. Multiplying the n resulting bounds and taking the $(n-1)$ -st root produces a quantity that can be integrated one variable at a time, each integration using the generalised Hölder inequality (Theorem 1.8.3 (generalised Hölder)). The case $p > 1$ follows by applying the case $p = 1$ to

| a power of $|f|$.

Proof. Step 1. The case $p = 1$.

For each j and each x , the fundamental theorem of calculus gives

$$|f(x)| = \left| \int_{-\infty}^{x_j} \partial_j f(x_1, \dots, t, \dots, x_n) dt \right| \leq \int_{\mathbb{R}} |\partial_j f| dx_j =: F_j(\hat{x}_j),$$

where $\hat{x}_j$ denotes x with the j -th coordinate omitted. Multiplying over j and taking the $(n-1)$ -st root,

$$|f(x)|^{\frac{n}{n-1}} \leq \prod_{j=1}^n F_j(\hat{x}_j)^{\frac{1}{n-1}}.$$

Integrate in x_1 . The factor $j = 1$ does not depend on x_1 and comes out; the remaining $n-1$ factors are handled by the generalised Hölder inequality with exponents $p_2 = \dots = p_n = n-1$:

$$\int_{\mathbb{R}} |f|^{\frac{n}{n-1}} dx_1 \leq F_1^{\frac{1}{n-1}} \prod_{j=2}^n \left(\int_{\mathbb{R}} F_j dx_1 \right)^{\frac{1}{n-1}}.$$

Repeating for $x_2, \dots, x_n$ and using Hölder at each step,

$$\begin{aligned} \int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}} dx &\leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} F_j d\hat{x}_j \right)^{\frac{1}{n-1}} \\ &= \prod_{j=1}^n \|\partial_j f\|_1^{\frac{1}{n-1}} \\ &\leq \|\nabla f\|_1^{\frac{n}{n-1}}. \end{aligned}$$

Raising to the power $\frac{n-1}{n}$ gives $\|f\|_{1^*} \leq \|\nabla f\|_1$ with $1^* = \frac{n}{n-1}$.

Step 2. The case $1 < p < n$: setting up.

Put

$$\gamma = \frac{p(n-1)}{n-p} > 1,$$

the inequality holding because $p > 1$ gives $p(n-1) > n-p$. Since $\gamma > 1$, the function $t \mapsto |t|^\gamma$ is C^1 on $\mathbb{R}$ with derivative $\gamma|t|^{\gamma-1} \operatorname{sgn} t$, which is continuous and vanishes at $t = 0$. Hence $g = |f|^\gamma$ is C^1 with compact support, and

$$|\nabla g| = \gamma |f|^{\gamma-1} |\nabla f|.$$

No separate argument at the zero set of f is needed; this is exactly why $\gamma > 1$ was arranged.

Step 3. *Two exponent identities.*

We record the two computations the argument needs. First,

$$\frac{\gamma n}{n-1} = \frac{p(n-1)}{n-p} \cdot \frac{n}{n-1} = \frac{pn}{n-p} = p^*.$$

Second, since $\gamma - 1 = \frac{p(n-1)-(n-p)}{n-p} = \frac{n(p-1)}{n-p}$ and $p' = \frac{p}{p-1}$,

$$(\gamma - 1)p' = \frac{n(p-1)}{n-p} \cdot \frac{p}{p-1} = \frac{np}{n-p} = p^*.$$

So both exponents that will appear are p^* .

Step 4. *The estimate.*

Apply Step 1 to g , then Hölder with exponents p' and p :

$$\begin{aligned} \left(\int |f|^{\frac{\gamma n}{n-1}} \right)^{\frac{n-1}{n}} &= \|g\|_{\frac{n}{n-1}} \\ &\leq \|\nabla g\|_1 = \gamma \int |f|^{\gamma-1} |\nabla f| \\ &\leq \gamma \left(\int |f|^{(\gamma-1)p'} \right)^{\frac{1}{p'}} \|\nabla f\|_p. \end{aligned}$$

By Step 3 both integrals are $\int |f|^{p^*}$, so this reads

$$\|f\|_{p^*}^{p^* \frac{n-1}{n}} \leq \gamma \|f\|_{p^*}^{p^*/p'} \|\nabla f\|_p.$$

Since f has compact support, $\|f\|_{p^*} < \infty$, and we may divide (the case $\|f\|_{p^*} = 0$ being trivial). The resulting exponent on the left is

$$\begin{aligned} p^* \left(\frac{n-1}{n} - \frac{1}{p'} \right) &= p^* \left(\frac{n-1}{n} - 1 + \frac{1}{p} \right) = p^* \left(\frac{1}{p} - \frac{1}{n} \right) = \frac{np}{n-p} \cdot \frac{n-p}{np} \\ &= 1. \end{aligned}$$

Hence $\|f\|_{p^*} \leq \gamma \|\nabla f\|_p$. □

2.7.4 Fractional integration

Theorem 2.7.7 (Hardy–Littlewood–Sobolev). *Let $0 < \lambda < n$ and $1 < p < q < \infty$ with*

$$\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1.$$

*Then $\|f * |\cdot|^{-\lambda}\|_q \lesssim_{n,p,\lambda} \|f\|_p$.*

Idea of the proof. Split the kernel at an as-yet-unspecified radius R . The near part is controlled by the Hardy–Littlewood maximal function, because $|y|^{-\lambda}$ is essentially constant on dyadic annuli and the annuli sum geometrically. The far part is controlled by Hölder, since $|y|^{-\lambda}$ is in $L^{p'}$ away from the origin. Optimising R pointwise — a different R at each x — interpolates between the two, and the maximal theorem of Volume 1 finishes.

Proof. Fix x and $R > 0$ and split $|(f * |\cdot|^{-\lambda})(x)| \leq \text{I} + \text{II}$ with the integrals over $|y| < R$ and $|y| \geq R$.

Step 1. Near part.

Decompose $\{|y| < R\}$ into dyadic annuli $A_k = \{2^{-k-1}R \leq |y| < 2^{-k}R\}$:

$$\begin{aligned} \text{I} &\leq \sum_{k \geq 0} (2^{-k-1}R)^{-\lambda} \int_{|y| < 2^{-k}R} |f(x-y)| \, dy \\ &\lesssim \sum_{k \geq 0} (2^{-k}R)^{n-\lambda} \mathcal{M}f(x) \lesssim R^{n-\lambda} \mathcal{M}f(x), \end{aligned}$$

using the definition of the Hardy–Littlewood maximal function (Volume 1, §7.3) and summability of $\sum_k 2^{-k(n-\lambda)}$, which needs $\lambda < n$.

Step 2. Far part.

By Hölder,

$$\text{II} \leq \|f\|_p \left(\int_{|y| \geq R} |y|^{-\lambda p'} \, dy \right)^{1/p'} \lesssim \|f\|_p R^{\frac{n}{p'} - \lambda},$$

the integral converging precisely when $\lambda p' > n$ — which is equivalent to $q < \infty$ under the stated exponent relation.

Step 3. Optimise pointwise.

Choose $R = R(x)$ with $R^{n-\lambda} \mathcal{M}f(x) = R^{n/p' - \lambda} \|f\|_p$, that is $R^{n/p} = \|f\|_p / \mathcal{M}f(x)$. Substituting,

$$\begin{aligned} |(f * |\cdot|^{-\lambda})(x)| &\lesssim \|f\|_p^\beta (\mathcal{M}f(x))^{1-\beta} \\ &\quad, \quad \beta = \frac{p(n-\lambda)}{n}. \end{aligned}$$

Step 4. Integrate.

The exponent relation $\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1$ gives $q = \frac{np}{\lambda p - np + n}$, and substituting

$$\beta = \frac{p(n-\lambda)}{n},$$

$$\begin{aligned} q(1-\beta) &= \frac{np}{\lambda p - np + n} \left(1 - \frac{p(n-\lambda)}{n}\right) \\ &= \frac{np}{\lambda p - np + n} \cdot \frac{n - pn + p\lambda}{n} \\ &= p. \end{aligned}$$

Hence

$$\begin{aligned} \|f * |\cdot|^{-\lambda}\|_q &\lesssim \|f\|_p^\beta \left(\int (\mathcal{M}f)^{q(1-\beta)} \right)^{1/q} = \|f\|_p^\beta \|\mathcal{M}f\|_p^{1-\beta} \\ &\lesssim \|f\|_p^\beta \|f\|_p^{1-\beta} = \|f\|_p, \end{aligned}$$

the last step by the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)), which requires $p > 1$. $\square$

WORKED EXAMPLE 2.7.1 — TIER C: RESEARCH TASTE **Energy integrals and Hausdorff dimension**

This example builds the bridge between the Fourier analysis of this volume and the geometric measure theory of Volume 8. Recall Hausdorff measure $\mathcal{H}^s$ and dimension $\dim_{\mathcal{H}}$ from Worked Example 1.2.4 (Hausdorff measure).

Definition 2.7.8. For a finite Borel measure μ on $\mathbb{R}^n$ and $s > 0$, the s -energy is

$$I_s(\mu) = \iint |x - y|^{-s} d\mu(x) d\mu(y).$$

(a) *The Fourier formula.* By Worked Example 2.5.2, $|\cdot|^{-s}$ has Fourier transform $c_{n,s}|\xi|^{-(n-s)}$ for $0 < s < n$. Pairing and using the multiplication formula (formally on $\mathcal{S}'$, rigorously after mollifying μ),

$$I_s(\mu) = c_{n,s} \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{-(n-s)} d\xi. \quad (2.7.1)$$

So an integral measuring how spread out μ is in *space* equals a weighted integral of $|\widehat{\mu}|^2$ in *frequency*. This identity is the entire reason Fourier analysis says anything about fractal geometry.

(b) *Finite energy forces large dimension.*

Proposition 2.7.9. If μ is a finite Borel measure with $\mu(A) > 0$ and $I_s(\mu) < \infty$, then $\dim_{\mathcal{H}} A \geq s$.

Proof. Let $g(x) = \int |x - y|^{-s} d\mu(y)$, so $\int g d\mu = I_s(\mu) < \infty$ and hence $g < \infty$ for μ -a.e. x . Choose M with $A' = \{x \in A : g(x) \leq M\}$ satisfying $\mu(A') > 0$. For $x \in A'$ and $r > 0$,

$$r^{-s} \mu(B(x, r)) \leq \int_{B(x, r)} |x - y|^{-s} d\mu(y) \leq g(x) \leq M,$$

so $\mu(B(x, r)) \leq Mr^s$ for every $x \in A'$ and $r > 0$: the measure is *s-Frostman* on A' .

Now let $\{U_j\}$ be any cover of A' by sets of diameter $\leq \delta$. Each U_j meeting A' is contained in a ball $B(x_j, \text{diam } U_j)$ with $x_j \in A'$, so $\mu(U_j) \leq M(\text{diam } U_j)^s$. Summing,

$$0 < \mu(A') \leq \sum_j \mu(U_j) \leq M \sum_j (\text{diam } U_j)^s.$$

Taking the infimum over covers gives $\mathcal{H}_\delta^s(A') \geq \mu(A')/M > 0$ for every δ , hence $\mathcal{H}^s(A') > 0$ and $\dim_{\mathcal{H}} A \geq \dim_{\mathcal{H}} A' \geq s$. $\square$

(c) *What this is for.* Combining (a) and (b): to prove a set has dimension at least s , it suffices to put a measure on it whose Fourier transform decays fast enough for the integral (2.7.1) to converge. Decay of $\hat{\mu}$ is a quantitative statement, provable by the oscillatory-integral methods of Volume 5.

This is the mechanism behind:

- the *Falconer distance conjecture*, that $\dim_{\mathcal{H}} A > n/2$ forces the distance set $\{|x - y| : x, y \in A\}$ to have positive Lebesgue measure — proved in the plane at the threshold $5/4$ by Guth–Iosevich–Ou–Wang in 2020, and open at $n/2$;
- the theory of *Salem sets*, those whose Fourier dimension equals their Hausdorff dimension;
- the Fourier-analytic lower bounds for the dimension of Besicovitch sets, which is how Volume 9 converts the Kakeya maximal estimates into dimension statements.

Note the direction. Finite energy gives *lower* bounds on dimension, and lower bounds are what the Kakeya conjecture asks for. The reverse implication — Frostman’s lemma, that a set of positive $\mathcal{H}^s$ measure carries a measure of finite t -energy for every $t < s$ — is proved in Volume 8, and together the two make dimension and energy interchangeable. $\square$

Drills

Drill 2.7.1. Show by scaling that $\|f\|_q \lesssim \|\nabla f\|_p$ can hold only for $q = p^* = np/(n-p)$.

Drill 2.7.2. Verify the exponent bookkeeping in Step 2 of Theorem 2.7.6: check $\gamma = p(n-1)/(n-p)$ and that $\frac{\gamma n}{n-1} = (\gamma-1)p' = p^*$.

Drill 2.7.3. Confirm the identity $q(1-\beta) = p$ used in Step 4 of Theorem 2.7.7, where $\beta = p(n-\lambda)/n$ and $\frac{1}{q} = \frac{1}{p} + \frac{\lambda}{n} - 1$.

Drill 2.7.4. Show that $|x|^{-a}\mathbf{1}_{|x|<1} \in H^s(\mathbb{R}^n)$ if and only if $s < \frac{n}{2} - a$, and interpret this in the light of Theorem 2.7.5.

Notes

Theorem 2.7.6 is due to Gagliardo and Nirenberg independently (1958–59), with the case $p = 1$ going back to Sobolev. The proof of Theorem 2.7.7 via the maximal function is Hedberg’s (1972) and is the shortest known.

Worked Example 2.7.1 is the standard potential-theoretic characterisation of dimension, due to Frostman; the Fourier form (2.7.1) was exploited systematically by Kaufman, Mattila and Falconer from the 1960s onwards and is the foundation of Volume 8.

Chapter 2.8

The Fourier Transform of Surface Measure

Everything in this volume has been machinery. This chapter uses it, on the one computation that the rest of the series depends on: the exact evaluation of $\widehat{d\sigma}$ for the sphere, and the decay estimate $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ that follows.

The exponent $(n-1)/2$ is not a technicality. It is the number of non-vanishing principal curvatures of the sphere, divided by two, and it propagates into the exponent of every theorem in Volumes 6, 12 and 13. By the end of this chapter we will have derived, from the decay alone, a genuine restriction theorem.

2.8.1 The exact formula

Proposition 2.8.1 (Reduction to a one-dimensional integral). *For $\xi \in \mathbb{R}^n$, $n \geq 2$, with $r = |\xi|$,*

$$\widehat{d\sigma}(\xi) = \sigma_{n-2} \int_{-1}^1 e^{-2\pi i r t} (1-t^2)^{\frac{n-3}{2}} dt, \quad \sigma_{n-2} = \sigma(\mathbb{S}^{n-2}) = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n-1}{2})}. \quad (2.8.1)$$

In particular $\widehat{d\sigma}$ is real, radial, and depends on ξ only through r .

Proof. By Worked Example 2.5.3(b), $\widehat{d\sigma}(\xi) = \int_{\mathbb{S}^{n-1}} e(-\omega \cdot \xi) d\sigma(\omega)$. Surface measure is rotation invariant — by Theorem 1.3.10 (linear images), rotations preserve Lebesgue measure, hence preserve σ through the polar coordinates construction — so replacing ξ by $R\xi$ for a rotation R and substituting $\omega \mapsto R^{-1}\omega$ leaves the integral unchanged. Thus $\widehat{d\sigma}$ is radial and we may take $\xi = re_n$.

Then $\omega \cdot \xi = r\omega_n$. Parametrise $\mathbb{S}^{n-1}$ by its last coordinate. Writing $t = \omega_n \in [-1, 1]$ and $\theta \in \mathbb{S}^{n-2}$, every $\omega \in \mathbb{S}^{n-1}$ with $\omega \neq \pm e_n$ has a unique representation

$$\omega(t, \theta) = (\sqrt{1-t^2} \theta, t), \quad (2.8.2)$$

and we claim

$$d\sigma = (1-t^2)^{\frac{n-3}{2}} dt d\sigma_{n-2}(\theta). \quad (2.8.3)$$

Proof of (2.8.3). The parametrisation (2.8.2) is a product of two motions — moving θ within the slice, and moving t along a meridian — and these are orthogonal, so the Jacobian factorises into their two contributions.

The slice. At height t the set $\{\omega_n = t\} \cap \mathbb{S}^{n-1}$ is a sphere of radius $\sqrt{1-t^2}$ in $\mathbb{R}^{n-1}$. Scaling $\mathbb{S}^{n-2}$ by ρ multiplies its $(n-2)$ -dimensional measure by ρ^{n-2} , so the slice contributes

$$(\sqrt{1-t^2})^{n-2} = (1-t^2)^{\frac{n-2}{2}}.$$

The meridian. Fixing θ and differentiating (2.8.2) in t ,

$$\begin{aligned} \frac{\partial \omega}{\partial t} &= \left(\frac{-t}{\sqrt{1-t^2}} \theta, 1 \right), & \left| \frac{\partial \omega}{\partial t} \right| \\ &= \sqrt{\frac{t^2}{1-t^2} + 1} \\ &= \frac{1}{\sqrt{1-t^2}}, \end{aligned}$$

using $|\theta| = 1$. So arclength along the meridian is $ds = (1-t^2)^{-1/2} dt$: *the parametrisation by height is not by arclength, and this factor is the correction.*

Multiplying.

$$\begin{aligned} (1-t^2)^{\frac{n-2}{2}} \cdot (1-t^2)^{-\frac{1}{2}} \\ &= (1-t^2)^{\frac{n-2}{2} - \frac{1}{2}} \\ &= (1-t^2)^{\frac{n-3}{2}}, \end{aligned}$$

which is (2.8.3). The two excluded poles form a σ -null set.

A check. Integrating (2.8.3) over the whole sphere must return $\sigma(\mathbb{S}^{n-1}) = 2\pi^{n/2}/\Gamma(\frac{n}{2})$. For $n = 2, 3, 4, 5$ the right side gives $2\pi, 4\pi, 2\pi^2, \frac{8}{3}\pi^2$, agreeing in each case.

Where the exponent can mislead. At $n = 2$ the exponent is $-\frac{1}{2}$, so the density blows up at $t = \pm 1$; the integral still converges, and $\mathbb{S}^1$ has two points over each interior t , which is the $\sigma_0 = 2$ in (2.8.1). At $n = 3$ the exponent is 0 and the density is constant — Archimedes' theorem, that the projection of the sphere to a coaxial cylinder preserves area. Integrating out θ gives (2.8.1). Realness follows since the imaginary part of the integrand is odd in t . $\square$

Definition 2.8.2 (Bessel function). For $\nu > -\frac{1}{2}$ and $\rho > 0$,

$$J_\nu(\rho) = \frac{(\rho/2)^\nu}{\Gamma(\nu + \frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{i\rho t} (1-t^2)^{\nu-\frac{1}{2}} dt.$$

Theorem 2.8.3. For $n \geq 2$ and $\xi \neq 0$, with $r = |\xi|$ and $\nu = \frac{n-2}{2}$,

$$\widehat{d\sigma}(\xi) = 2\pi r^{-\frac{n-2}{2}} J_{\frac{n-2}{2}}(2\pi r).$$

Proof. With $\nu = \frac{n-2}{2}$ we have $\nu - \frac{1}{2} = \frac{n-3}{2}$ and $\Gamma(\nu + \frac{1}{2}) = \Gamma(\frac{n-1}{2})$, so the definition with $\rho = 2\pi r$ reads

$$J_\nu(2\pi r) = \frac{(\pi r)^\nu}{\Gamma(\frac{n-1}{2})\sqrt{\pi}} \int_{-1}^1 e^{2\pi i r t} (1-t^2)^{\frac{n-3}{2}} dt.$$

The integral is unchanged if i is replaced by $-i$, being real, so it equals the integral in (2.8.1). Solving for that integral and substituting into (2.8.1),

$$\widehat{d\sigma}(\xi) = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n-1}{2})} \cdot \frac{\Gamma(\frac{n-1}{2})\sqrt{\pi}}{(\pi r)^\nu} J_\nu(2\pi r) = 2\pi^{\frac{n}{2}-\nu} r^{-\nu} J_\nu(2\pi r),$$

and $\frac{n}{2} - \nu = \frac{n}{2} - \frac{n-2}{2} = 1$. □

WORKED EXAMPLE 2.8.1— TIER A **The case $n = 3$, exactly**

For $n = 3$, $\frac{n-3}{2} = 0$ and (2.8.1) becomes elementary: $\sigma_1 = \sigma(\mathbb{S}^1) = 2\pi$, so

$$\widehat{d\sigma}(\xi) = 2\pi \int_{-1}^1 e^{-2\pi i r t} dt = 2\pi \cdot \frac{e^{2\pi i r} - e^{-2\pi i r}}{2\pi i r} = \frac{2 \sin(2\pi r)}{r}, \quad r = |\xi|.$$

So in $\mathbb{R}^3$ the transform of surface measure is an elementary function. Two things to notice. Its modulus decays exactly like $r^{-1} = r^{-(n-1)/2}$, as the general theorem will assert. And it *oscillates*: $\widehat{d\sigma}$ changes sign infinitely often and vanishes at every half-integer r . Volume 6 needs the oscillation, not merely the modulus, and readers should resist the habit of recording only absolute values.

Consistency check with Theorem 2.8.3: $\nu = \frac{1}{2}$, and $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$ directly from the definition, so $2\pi r^{-1/2} J_{1/2}(2\pi r) = 2\pi r^{-1/2} \cdot \frac{1}{\pi\sqrt{r}} \sin(2\pi r) = 2 \sin(2\pi r)/r$. ✓

2.8.2 Van der Corput's lemma

To bound (2.8.1) for general n we need one tool. It is the elementary core of the method of stationary phase, which Volume 5 develops in full.

Lemma 2.8.4 (Van der Corput). *Let $\phi : [a, b] \rightarrow \mathbb{R}$ be C^k and suppose $|\phi^{(k)}| \geq 1$ on $[a, b]$, where either $k \geq 2$, or $k = 1$ and ϕ' is monotone. Then*

$$\left| \int_a^b e^{i\rho\phi(t)} dt \right| \leq c_k \rho^{-1/k} \quad (\rho > 0),$$

with c_k depending only on k . Consequently, for ψ of bounded variation,

$$\left| \int_a^b e^{i\rho\phi(t)} \psi(t) dt \right| \leq c_k \rho^{-1/k} \left(|\psi(b)| + \int_a^b |\psi'| \right). \quad (2.8.4)$$

Proof. Step 1. $k = 1$.

Write $e^{i\rho\phi} = \frac{1}{i\rho\phi'} \frac{d}{dt} e^{i\rho\phi}$ and integrate by parts:

$$\int_a^b e^{i\rho\phi} dt = \left[\frac{e^{i\rho\phi}}{i\rho\phi'} \right]_a^b + \frac{1}{i\rho} \int_a^b e^{i\rho\phi} \frac{\phi''}{(\phi')^2} dt.$$

The boundary terms are at most $2/\rho$ since $|\phi'| \geq 1$. For the integral, monotonicity of ϕ' means $\phi''/(\phi')^2$ has constant sign, so $\int_a^b |\phi''/(\phi')^2| = |\int_a^b \phi''/(\phi')^2| = |1/\phi'(a) - 1/\phi'(b)| \leq 2$. Hence the whole is at most $4/\rho$.

Step 2. $k \geq 2$, by induction.

Suppose the result holds for $k-1$. Since $|\phi^{(k)}| \geq 1$, the derivative $\phi^{(k-1)}$ is strictly monotone, so it vanishes at most once; say at c (if not, the argument below is simpler). For $\eta > 0$, split $[a, b]$ into $[a, b] \cap (c - \eta, c + \eta)$, of length at most 2η , and the rest.

Off that interval, $|\phi^{(k-1)}| \geq \eta$, because $\phi^{(k-1)}$ moves at unit rate at least. Applying the inductive hypothesis to ϕ/η on each of the (at most two) remaining intervals — for which the $(k-1)$ -st derivative is at least 1 in modulus — gives a contribution at most $2c_{k-1}(\rho\eta)^{-1/(k-1)}$.

Hence

$$\left| \int_a^b e^{i\rho\phi} \right| \leq 2\eta + 2c_{k-1}(\rho\eta)^{-\frac{1}{k-1}}.$$

Choosing $\eta = \rho^{-1/k}$ balances the two terms and yields $c_k \rho^{-1/k}$.

Step 3. *Amplitudes.*

Let $F(u) = \int_a^u e^{i\rho\phi}$, so $|F| \leq c_k \rho^{-1/k}$ by Steps 1–2 applied on $[a, u]$. Integrating by parts,

$$\int_a^b e^{i\rho\phi} \psi = F(b)\psi(b) - \int_a^b F\psi',$$

and taking absolute values gives (2.8.4). $\square$

2.8.3 The decay estimate

Theorem 2.8.5 (Decay of $\widehat{d\sigma}$). *For $n \geq 2$ there is C_n with*

$$|\widehat{d\sigma}(\xi)| \leq C_n(1 + |\xi|)^{-\frac{n-1}{2}} \quad \text{for all } \xi \in \mathbb{R}^n.$$

Moreover the exponent cannot be improved: for $n = 3$, $|\widehat{d\sigma}(\xi)| = 2|\xi|^{-1}$ whenever $|\xi| \in \frac{1}{4} + \frac{1}{2}\mathbb{Z}$.

Idea of the proof. Substituting $t = \cos \varphi$ turns (2.8.1) into $\int_0^\pi e^{-2\pi i r \cos \varphi} (\sin \varphi)^{n-2} d\varphi$, an oscillatory integral whose phase $\cos \varphi$ is *stationary* at $\varphi = 0$ and $\varphi = \pi$. Those are the two poles: the points of the sphere whose normal is parallel to ξ . Everywhere else the phase moves, integration by parts gains as many powers of ρ as one likes, and the contribution is negligible. All the decay is therefore decided at the poles, and Theorem 2.8.4 with $k = 2$ identifies the mechanism: a stationary point of a phase with non-degenerate second derivative contributes an angular window of width $\rho^{-1/2}$ on which no cancellation occurs.

The remaining factor comes from the amplitude. Near a pole the slice of the sphere at angle φ has size $(\sin \varphi)^{n-2} \approx \varphi^{n-2}$, so the mass sitting in the window $\varphi \lesssim \rho^{-1/2}$ is $\int_0^{\rho^{-1/2}} \varphi^{n-2} d\varphi \approx \rho^{-(n-1)/2}$. That is the answer, and the proof below is the verification that nothing else is larger.

We give the argument in the original variable t , where the two poles become the two endpoints $t = \pm 1$ and the analysis is a single clean computation valid for every $n \geq 2$ at once.

Proof. Boundedness for $|\xi| \leq 1$ is Worked Example 2.5.3(b), so assume $r = |\xi| \geq 1$ and write $\rho = 2\pi r$ and $\alpha = \frac{n-3}{2}$, so that $\alpha > -1$ for every $n \geq 2$ and (2.8.1) reads

$$\widehat{d\sigma}(\xi) = \sigma_{n-2} I(\rho), \quad I(\rho) = \int_{-1}^1 e^{-i\rho t} (1 - t^2)^\alpha dt.$$

We show

$$|I(\rho)| \lesssim_\alpha \rho^{-\alpha-1} = \rho^{-\frac{n-1}{2}}. \quad (2.8.5)$$

Step 1. *Partition of unity.*

Choose $\chi \in C_c^\infty(\mathbb{R})$ with $\chi = 1$ on $[-\frac{1}{4}, \frac{1}{4}]$ and $\text{supp } \chi \subseteq (-\frac{1}{2}, \frac{1}{2})$, and write

$$1 = \chi(1-t) + \chi(1+t) + \chi_0(t), \quad t \in [-1, 1],$$

which defines χ_0 ; it is smooth and supported in $[-\frac{3}{4}, \frac{3}{4}]$. This splits I into a piece near each endpoint and an interior piece.

Step 2. *The interior piece is negligible.*

On $\text{supp } \chi_0$ the amplitude $(1 - t^2)^\alpha \chi_0(t)$ is C^∞ with compact support in $(-1, 1)$, and the phase is $-t$, with derivative -1 . Integrating by parts N times, all boundary terms vanishing by compact support,

$$\left| \int_{-1}^1 e^{-i\rho t} (1 - t^2)^\alpha \chi_0(t) dt \right| = \rho^{-N} \left| \int_{-1}^1 e^{-i\rho t} \frac{d^N}{dt^N} [(1 - t^2)^\alpha \chi_0(t)] dt \right| \\ \lesssim_{\alpha, N} \rho^{-N}.$$

Choose $N \geq \alpha + 1$; this piece is $O(\rho^{-\alpha-1})$.

Step 3. *An endpoint piece.*

By symmetry it suffices to treat t near 1. Substitute $t = 1 - s$:

$$\int_{-1}^1 e^{-i\rho t} (1 - t^2)^\alpha \chi(1 - t) dt \\ = e^{-i\rho} \int_0^\infty e^{i\rho s} s^\alpha g(s) ds \\ , \quad g(s) = (2 - s)^\alpha \chi(s),$$

and $g \in C_c^\infty([0, \frac{1}{2}))$, so all its derivatives are bounded and it vanishes identically for $s \geq \frac{1}{2}$. Put $K(\rho) = \int_0^\infty e^{i\rho s} s^\alpha g(s) ds$; we must show $|K(\rho)| \lesssim_\alpha \rho^{-\alpha-1}$.

Split at $s = \rho^{-1}$.

Small s. Bounding the integrand by its modulus,

$$\left| \int_0^{1/\rho} e^{i\rho s} s^\alpha g(s) ds \right| \\ \leq \|g\|_\infty \int_0^{1/\rho} s^\alpha ds \\ = \frac{\|g\|_\infty}{\alpha + 1} \rho^{-\alpha-1},$$

the integral converging precisely because $\alpha > -1$. This is the term that produces the exponent; it is the endpoint window of width ρ^{-1} in s , which is the window of width $\rho^{-1/2}$ in the angular variable φ , since $s = 1 - \cos \varphi \approx \varphi^2/2$.

Large s. Let M be the least integer with $M > \alpha + 1$, and integrate by parts M times on $[\rho^{-1}, \infty)$, writing $e^{i\rho s} = \frac{1}{i\rho} \frac{d}{ds} e^{i\rho s}$ each time. Since g vanishes near ∞ , the only boundary terms occur at $s = \rho^{-1}$, and the m -th one is

$$\lesssim \rho^{-m} \left| \frac{d^{m-1}}{ds^{m-1}} (s^\alpha g) \right|_{s=1/\rho} \\ \lesssim_\alpha \rho^{-m} \rho^{-(\alpha-m+1)} = \rho^{-\alpha-1}, \quad 1 \leq m \leq M,$$

using $|\frac{d^k}{ds^k}(s^\alpha g)| \lesssim_{\alpha,k} s^{\alpha-k}$ for $0 < s < \frac{1}{2}$, which follows from the Leibniz rule. The remaining integral is

$$\begin{aligned} \rho^{-M} \int_{1/\rho}^{1/2} \left| \frac{d^M}{ds^M}(s^\alpha g) \right| ds &\lesssim_\alpha \rho^{-M} \int_{1/\rho}^{1/2} s^{\alpha-M} ds \\ &\lesssim_\alpha \rho^{-M} \rho^{M-\alpha-1} = \rho^{-\alpha-1}, \end{aligned}$$

where the s -integral was evaluated using $\alpha - M < -1$, so that it is dominated by its lower limit.

Step 4. *Collect.*

Steps 2 and 3 give (2.8.5), and $\rho = 2\pi r \approx |\xi|$, proving the stated bound for $|\xi| \geq 1$.

Step 5. *Sharpness.*

By Worked Example 2.8.1, for $n = 3$ we have $\widehat{d\sigma}(\xi) = 2\sin(2\pi|\xi|)/|\xi|$, whose modulus equals $2|\xi|^{-1} = 2|\xi|^{-(n-1)/2}$ exactly when $\sin(2\pi|\xi|) = \pm 1$, that is when $|\xi| \in \frac{1}{4} + \frac{1}{2}\mathbb{Z}$. So the rate is attained along a sequence tending to infinity and cannot be improved. $\square$

Remark 2.8.6 (Two readings of the same exponent). The proof above is a computation in the variable t ; the Idea box described the geometry in the variable φ . They are the same argument. In φ , the poles are stationary points of the phase and Theorem 2.8.4 with $k = 2$ supplies the window of width $\rho^{-1/2}$; in t , the poles are the endpoints $t = \pm 1$ where the amplitude $(1 - t^2)^\alpha$ degenerates, and the window has width ρ^{-1} . The substitution $s = 1 - \cos \varphi$ converts one into the other.

The t -side computation is shorter and works uniformly in n , which is why we gave it. The φ -side reading is the one to remember, because it is the one that generalises: for a hypersurface with k non-vanishing principal curvatures, the stationary set for a given ξ is a $(n - 1 - k)$ -dimensional submanifold and the window has width $\rho^{-1/2}$ in each of the k curved directions, giving $\rho^{-k/2}$. That is Theorem 2.8.7.

Remark 2.8.7 (What is deferred, and why). Two refinements are not proved here.

First, the sharp *asymptotic* $J_\nu(\rho) = \sqrt{2/(\pi\rho)} \cos(\rho - \frac{\nu\pi}{2} - \frac{\pi}{4}) + O(\rho^{-3/2})$, which records the oscillation and not merely the modulus. Volume 6 needs it, and Volume 5 proves it by the method of stationary phase, of which Theorem 2.8.4 is the crudest case. Sharpness of the exponent for general n follows from it; we have proved sharpness only for $n = 3$, where the elementary formula of Worked Example 2.8.1 is available.

Second, the general principle that a hypersurface with k non-vanishing principal curvatures has $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$. The sphere is the case $k = n - 1$

and is done in full above; the general statement needs the Morse lemma and is Volume 5.

Both are S7 deferrals in the sense of the series conventions: they require machinery introduced in a later volume, they are labelled here, and nothing in Volumes 1–4 uses them.

WORKED EXAMPLE 2.8.2— TIER B **A flat piece kills all decay**

Let $\Sigma = \{(y, 0) : |y| \leq 1\} \subseteq \mathbb{R}^n$ be a flat disc in the hyperplane $x_n = 0$, with $d\mu$ its $(n - 1)$ -dimensional Lebesgue measure. Then

$$\widehat{d\mu}(\xi) = \int_{|y| \leq 1} e(-y \cdot \xi') dy,$$

which does not involve ξ_n at all. So $\widehat{d\mu}(0, \xi_n) = \omega_{n-1}$ for every ξ_n : there is no decay whatsoever in the normal direction.

Compare Theorem 2.8.5. The difference is one word: on the sphere the normal direction *turns*, so the set of points contributing a stationary phase for a given ξ is confined to two poles and an angular window of width $|\xi|^{-1/2}$; on the disc every point is a pole simultaneously. Curvature is precisely the rate at which the normal turns, and the decay exponent counts how many directions it turns in.

WORKED EXAMPLE 2.8.3— TIER C: RESEARCH TASTE **A restriction theorem, from decay alone**

We close the volume by proving a genuine restriction estimate. Recall from Worked Example 2.4.3 the extension operator $Eg = \widehat{gd\sigma}$, and from Worked Example 2.5.3 that it is well defined.

Theorem 2.8.8. *For $n \geq 2$ and $q = \frac{4n}{n-1}$,*

$$\|Eg\|_{L^q(\mathbb{R}^n)} \lesssim_n \|g\|_{L^2(d\sigma)}.$$

Proof. The adjoint of $E : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$ is $E^*f = \widehat{f}|_{\mathbb{S}^{n-1}}$, and the composition is

$$EE^*f = f * \widehat{d\sigma},$$

a computation from the definitions: $(\widehat{f}|_{\mathbb{S}^{n-1}})d\sigma$ pairs against φ exactly as $f * \widehat{d\sigma}$ does.

The TT^* identity says $\|E\|_{L^2(d\sigma) \rightarrow L^q}^2 = \|EE^*\|_{L^{q'} \rightarrow L^q}$; indeed

$$\|E^*f\|_{L^2(d\sigma)}^2 = \langle EE^*f, f \rangle \leq \|EE^*f\|_q \|f\|_{q'},$$

so a bound on EE^* gives one on E^* , hence on E by duality (Volume 1, Corollary 8.12).

By Theorem 2.8.5, $|\widehat{d\sigma}(x)| \lesssim (1+|x|)^{-\lambda} \leq |x|^{-\lambda}$ with $\lambda = \frac{n-1}{2}$, and $0 < \lambda < n$. So by Hardy–Littlewood–Sobolev (Theorem 2.7.7),

$$\begin{aligned} \|f * \widehat{d\sigma}\|_q &\lesssim \|f * |\cdot|^{-\lambda}\|_q \\ &\lesssim \|f\|_{q'} \quad \text{provided} \quad \frac{1}{q} = \frac{1}{q'} + \frac{\lambda}{n} - 1. \end{aligned}$$

Since $\frac{1}{q'} = 1 - \frac{1}{q}$, that condition reads $\frac{2}{q} = \frac{\lambda}{n}$, i.e.

$$q = \frac{2n}{\lambda} = \frac{4n}{n-1}. \quad \square$$

(a) *Reading the result.* We have converted a statement about the *geometry* of the sphere — it is curved, so its surface measure has a Fourier transform decaying at rate $(n-1)/2$ — into a statement about *integrability* of Eg , with no further input. That is the whole strategy of the subject in one page.

(b) *It is not sharp, and the gap is instructive.* The Stein–Tomas theorem, proved in Volume 6, gives the same conclusion for the smaller exponent

$$q = \frac{2(n+1)}{n-1}.$$

For $n=3$: we have proved $q=6$, and Stein–Tomas gives $q=4$. Smaller q is a stronger statement, since the estimate for one q implies it for all larger q after interpolation with the trivial bound.

The improvement comes from using information we discarded. Our proof used only $|\widehat{d\sigma}| \lesssim |x|^{-\lambda}$, which is $L^1 \rightarrow L^\infty$ type information about the kernel. But $d\sigma$ is also a *bounded* multiplier in a suitable dyadic decomposition, giving $L^2 \rightarrow L^2$ information by Plancherel; interpolating the two, dyadic piece by dyadic piece, is exactly the Stein–Tomas argument. Discarding the L^2 information cost us the difference between $\frac{4n}{n-1}$ and $\frac{2(n+1)}{n-1}$.

(c) *And Stein–Tomas is not the conjecture.* The restriction conjecture, as located by the Knapp example in Worked Example 2.4.3, asserts $\|Eg\|_{L^q} \lesssim \|g\|_{L^\infty(d\sigma)}$ for all $q > \frac{2n}{n-1}$. For $n=3$ that is $q > 3$, against Stein–Tomas’s $q \geq 4$. The gap between 4 and 3 in $\mathbb{R}^3$ has been open since 1967, and closing it is the subject of Volume 12.

Where you now stand. You have proved a restriction theorem from first principles: measure theory in Volume 1, complex analysis and interpolation in

Chapters 1–2, distributions in Chapter 5, fractional integration in Chapter 7, and one Bessel computation here. Two further ideas — the L^2 orthogonality that sharpens Step (b), and the tube geometry that would sharpen (c) — separate you from the frontier. $\square$

WORKED EXAMPLE 2.8.4— TIER B **Reading a decay estimate**

Theorem 2.8.5 gives $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ for the sphere. Four questions to ask of any such statement.

(a) *What produces the exponent?* The number of non-vanishing principal curvatures, halved. Theorem 2.8.4 converts each curved direction into a factor $|\xi|^{-1/2}$, and the sphere has $n - 1$ of them.

(b) *Is it sharp?* Yes, and Volume 5 makes the sharpness visible: the asymptotic there shows $\widehat{d\sigma}$ oscillates with amplitude exactly $|\xi|^{-(n-1)/2}$, so no better pointwise bound holds. That sharpness is what makes the Fourier method saturate in Volumes 7 and 9.

(c) *Is the pointwise bound what is needed?* Often not. Volume 8's reduction of the distance problem required the *oscillation*, not merely the decay — the phase $e^{\pm 2\pi i|\xi|}$ turns an integral into a Fourier transform, and Plancherel then applies. Discarding the phase loses that and the estimate fails.

(d) *What replaces it when the surface is flat?* Nothing: a hyperplane's surface measure has a transform that does not decay in the normal direction at all. Volume 6 shows the consequence for restriction, and Volume 7 states it as $\dim_F = 0$.

The summary. An estimate on $\widehat{d\sigma}$ is a statement about curvature; its exponent counts curved directions; and whether the *phase* is needed depends on whether the application squares the quantity.

Drills

Drill 2.8.1. Compute $\widehat{d\sigma}$ for $n = 2$ directly from (2.8.1) and identify it as $2\pi J_0(2\pi|\xi|)$. Verify numerically that $|\widehat{d\sigma}(\xi)| \approx |\xi|^{-1/2}$ for large $|\xi|$.

Drill 2.8.2. Prove $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$ directly from the definition, and use it to re-derive Worked Example 2.8.1.

Drill 2.8.3. Show that the constant $c_1 = 4$ in Theorem 2.8.4 is attainable up to a factor, and that the monotonicity hypothesis on ϕ' cannot be dropped when $k = 1$.

Drill 2.8.4. Carry out the interpolation referred to in Worked Example 2.8.3(b): assuming Stein–Tomas at $q = \frac{2(n+1)}{n-1}$, deduce the estimate for every larger q .

Notes

Theorem 2.8.3 is classical and can be found in Stein–Weiss. Theorem 2.8.4 is van der Corput’s, from 1921; the inductive proof given here is the standard one and appears in Stein’s *Harmonic Analysis*, Chapter VIII.

Theorem 2.8.8 is folklore — it is the estimate one obtains by using only the decay of $\widehat{d\sigma}$ — and is included because the gap between it and Stein–Tomas isolates precisely what the more sophisticated argument buys. The Stein–Tomas theorem is due to Tomas (1975) with the endpoint by Stein; both are proved in Volume 6.

Chapter 2.A

Results Quoted from Volume 1

This volume uses Volume 1 heavily and assumes nothing else. The list below is complete: every result of Volume 1 invoked anywhere in Volume 2 appears here, with the place it is used. No proofs are given — they are in Volume 1 — and the purpose of the appendix is to let a reader verify at a glance that nothing is being assumed silently.

Measure and integration

Result	Volume 1	Used in
Box lemma, $m^*(Q) = Q $	Thm 3.3	throughout
Structure of measurable sets; outer and inner regularity	Thm 3.5	density arguments
Linear images, $m(TE) = \det T m(E)$	Thm 3.11	Theorem 2.4.2(iv), Theorem 2.8.1
Simple approximation	Thm 4.6	Chapter 2.3, Theorem 2.2.1
Monotone convergence	Thm 5.1	Theorem 2.4.8, Theorem 2.7.7
Fatou	Thm 5.6	Worked Example 2.4.2
Dominated convergence	Thm 5.8	Theorem 2.4.6, Theorem 2.4.2
Differentiation under the integral	Wkd Ex 5.13	Theorem 2.4.2(vi)
Continuity and decay of $\widehat{f}$ for $f \in L^1$	Wkd Ex 5.14	Chapter 2.4 §4.1
Layer-cake formula	Wkd Ex 5.5	Theorem 2.2.4, Theorem 2.6.5
Egorov; Lusin	Thms 5.18, 5.20	density arguments
Tonelli; Fubini	Thms 6.6, 6.7	used constantly
Young's inequality, the case $q = 1$	Prop 6.9	Worked Example 2.2.1
Approximate identities	Prop 6.10	Theorem 2.4.6, Theorem 2.4.8
Polar coordinates and surface measure	Thm 6.10	Chapter 2.8, Theorem 2.7.5

L^p theory

Result	Volume 1	Used in
Hölder; generalised Hölder	Thms 8.2, Cor 8.3	Theorem 2.2.1, Theorem 2.7.6
Minkowski	Thm 8.5	Theorem 2.3.6
Riesz–Fischer completeness	Thm 8.6	Theorem 2.4.8 Step 2, Theorem 2.2.1 Step 5
Density of simple functions and of C_c^∞	Thm 8.9	Theorem 2.2.1, Theorem 2.4.8, Worked Example 2.4.2
Continuity of translation in L^p	Thm 8.9(iii)	Theorem 2.4.8
Duality $(L^p)^* \cong L^{p'}$	Thm 8.11	Theorem 2.2.1 Step 5
Norms by testing	Cor 8.12	Theorem 2.2.1 Step 4, Worked Example 2.8.3
Weak L^p ; Chebyshev	Prop 8.14	Theorem 2.2.4
Inclusions on finite measure spaces	Drill 8.1	Chapter 2.3, Theorem 2.6.5 Step 4
Log-convexity of L^p norms	Drill 8.2	Theorem 2.6.5 Step 5
Hilbert space; orthonormal bases	Thms 8.18–8.20	Theorem 2.3.8

Harmonic analysis and Keakeya

Result	Volume 1	Used in
Hardy–Littlewood maximal function; weak (1, 1)	Thm 7.4	Worked Example 2.2.2, Theorem 2.7.7
Lebesgue differentiation	Thm 7.7	background
Hausdorff measure and dimension	Wkd Ex 2.16	Worked Example 2.7.1
Tube intersection estimate	Lem 6.11	Worked Example 2.2.3(c)
Córdoba: Keakeya in the plane	Thm 6.12	background
Perron tree	Thm 3.23	background
Keakeya maximal operator; the conjecture	Def 7.17, Wkd Ex 7.20	Worked Example 2.2.3
Almost orthogonality; the decoupling statement	Wkd Ex 8.16	Worked Example 2.6.2

Background

Urysohn's lemma (Thm A.2) is used in Lusin-type density arguments and in Theorem 2.5.5. Tietze extension (Thm A.3) and the Hahn decomposition (Thm A.4) are used in Volume 1 only, and are recorded there.

The three-lines lemma, stated without proof in Volume 1's Appendix A, is proved in this volume as Theorem 2.1.14. That was the only outstanding debt from Volume 1, and it is now discharged.

Chapter 2.B

Notation and Index of Results

Additions to the Volume 1 conventions

$\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$	the Fourier transform
$\mathcal{F}, \mathcal{F}^{-1}$	the transform and its inverse as operators
$e(t)$	shorthand for $e^{2\pi i t}$
$\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$	the torus; functions are 1-periodic
D_N, K_N	Dirichlet and Fejér kernels
$\mathbb{S}_{01}$	the closed strip $0 \leq \operatorname{Re} z \leq 1$
$\mathcal{S}(\mathbb{R}^n), \mathcal{S}'(\mathbb{R}^n)$	Schwartz functions; tempered distributions
$\rho_{\alpha\beta}$	the Schwartz seminorms
$\langle u, \varphi \rangle$	the pairing of a distribution with a test function
$d\sigma$	surface measure on $\mathbb{S}^{n-1}$
$Eg = \widehat{gd\sigma}$	the extension operator
J_ν	the Bessel function of order ν
$H^s, W^{k,p}, p^*$	Sobolev spaces and the Sobolev exponent
$P_N, P_{\leq N}$	Littlewood–Paley projections
m, T_m	a Fourier multiplier and its operator
r_j	the Rademacher functions
$I_s(\mu)$	the s -energy of a measure

The transform is normalised with 2π in the exponent, so that $\mathcal{F}$ is unitary on $L^2(\mathbb{R}^n)$ with no constant, $\widehat{f * g} = \widehat{f}\widehat{g}$ with no constant, and $\widehat{\partial_j f}(\xi) = 2\pi i \xi_j \widehat{f}(\xi)$. This is the convention of the restriction and decoupling literature.

Index of the main results of Volume 2

Chapter 1 — complex analysis

Goursat's theorem	Theorem 2.1.5
Cauchy's theorem on a convex domain	Theorem 2.1.6
Cauchy integral formula	Theorem 2.1.9
Holomorphic implies analytic	Theorem 2.1.10
Cauchy estimates; Liouville	Theorem 2.1.11
Maximum modulus principle	Theorem 2.1.12
Phragmén–Lindelöf on a strip	Theorem 2.1.13
Three-lines lemma	Theorem 2.1.14

Chapter 2 — interpolation

Riesz–Thorin	Theorem 2.2.1
Marcinkiewicz	Theorem 2.2.4
Young's convolution inequality	Worked Example 2.2.1

Chapter 3 — Fourier series

Closed forms for D_N and K_N	Theorem 2.3.4
Fejér's theorem	Theorem 2.3.6
Uniqueness and density	Theorem 2.3.7
Parseval	Theorem 2.3.8
Baire category	Theorem 2.3.9
Uniform boundedness	Theorem 2.3.10
du Bois-Reymond divergence	Theorem 2.3.11
Hausdorff–Young on $\mathbb{T}^n$	Theorem 2.3.12

Chapter 4 — the Fourier transform

The dictionary	Theorem 2.4.2
The Gaussian is its own transform	Theorem 2.4.3
Multiplication formula	Theorem 2.4.5
Fourier inversion	Theorem 2.4.6
Plancherel	Theorem 2.4.8
Hausdorff–Young on $\mathbb{R}^n$	Theorem 2.4.9
Heisenberg uncertainty	Worked Example 2.4.2

<i>Chapter 5 — distributions</i>	
$\mathcal{F}$ is an isomorphism of $\mathcal{S}$	Theorem 2.5.6
$\mathcal{F}$ is an isomorphism of $\mathcal{S}'$	Theorem 2.5.11
Transform of $ x ^{-a}$	Worked Example 2.5.2
<i>Chapter 6 — multipliers</i>	
L^2 multiplier norm	Theorem 2.6.2
Khintchine's inequality	Theorem 2.6.5
Bernstein's inequalities	Theorem 2.6.7
L^2 orthogonality of lacunary pieces	Theorem 2.6.8
<i>Chapter 7 — Sobolev spaces</i>	
$H^k = W^{k,2}$	Theorem 2.7.4
Sobolev embedding, L^2 case	Theorem 2.7.5
Gagliardo–Nirenberg–Sobolev	Theorem 2.7.6
Hardy–Littlewood–Sobolev	Theorem 2.7.7
Energy integrals and dimension	Worked Example 2.7.1
<i>Chapter 8 — surface measure</i>	
Reduction to a one-dimensional integral	Theorem 2.8.1
$\widehat{d\sigma}$ in terms of Bessel functions	Theorem 2.8.3
Van der Corput's lemma	Theorem 2.8.4
Decay of $\widehat{d\sigma}$	Theorem 2.8.5
A restriction theorem from decay alone	Worked Example 2.8.3

Deferrals

Four results are stated in this volume and proved elsewhere. Each is labelled at the point of statement, and none is used in Volumes 1–2.

Result	Stated	Proved
Hörmander–Mikhlin multiplier theorem	Theorem 2.6.3	Volume 3 (needs Calderón–Zygmund)
Littlewood–Paley square function, $p \neq 2$	§6.3	Volume 3
Fefferman’s ball multiplier theorem	§6.1	Volume 5
Asymptotics of J_ν ; decay for general hypersurfaces	Theorem 2.8.7	Volume 5 (stationary phase)
Stein–Tomas theorem	Worked Example 2.8.3(b)	Volume 6

Part 3

Volume 3

Maximal Functions and Singular Integrals

Preface to Volume 3

Where we are

Volume 2 ended by proving a restriction theorem: from the decay $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ and nothing else, the extension operator maps $L^2(d\sigma)$ into $L^q(\mathbb{R}^n)$ for $q = \frac{4n}{n-1}$. It also left two debts. The Hörmander–Mikhlin multiplier theorem and the Littlewood–Paley square function theorem for $p \neq 2$ were both stated and deferred, on the grounds that they require Calderón–Zygmund theory. This volume builds that theory and discharges both debts.

The one idea

Almost every operator in harmonic analysis fails to be bounded on L^1 , and almost every one of them satisfies a weak $(1, 1)$ bound instead. Volume 1 saw this for the Hardy–Littlewood maximal function, where the obstruction was that $\mathcal{M}f$ decays like $|x|^{-n}$ at infinity. The same phenomenon recurs for the Hilbert transform, the Riesz transforms, and every singular integral, and the reason is always the same: these operators are built from kernels of size $|x|^{-n}$, which are exactly not integrable.

Calderón and Zygmund found the mechanism that handles all of them at once. One decomposes an arbitrary L^1 function, at a height λ , into a bounded part and a sum of pieces each supported in a cube and each of *mean zero*. The bounded part is handled by an L^2 estimate; the mean-zero pieces are handled by the smoothness of the kernel, since a kernel that varies slowly cannot distinguish a mean-zero bump from nothing. The whole of Chapters 1 and 2 is that sentence, made precise.

What is assumed

Volumes 1 and 2 in full, and nothing else. In particular we use freely: the Lebesgue theory and L^p duality; the Hardy–Littlewood maximal theorem; Fourier theory on L^1 , L^2 , and $\mathcal{S}'$; Plancherel; the Riesz–Thorin and

Marcinkiewicz interpolation theorems; Khintchine's inequality; and Bernstein's inequalities. Appendix A lists every such citation with its use site.

The single idea

If the volume has one idea it is this: *an operator whose kernel is singular at a point, and which is bounded on L^2 , is bounded on every L^p with $1 < p < \infty$* . That is Theorem 3.2.3, and everything before it is preparation and everything after is consequence.

The preparation is the Calderón–Zygmund decomposition of Chapter 3.1, a stopping-time argument (Worked Example 3.1.5) which splits an L^1 function into a bounded part and a sum of mean-zero pieces supported on disjoint cubes. The two parts are then handled by the two hypotheses — the bounded part by L^2 , the mean-zero part by the kernel — and that division is the whole architecture.

The consequences occupy Chapter 3.3 to Chapter 3.7: the Hilbert and Riesz transforms, the Hörmander–Mikhlin multiplier theorem, the Littlewood–Paley square function, and the endpoint behaviour at L^∞ , which is BMO rather than L^∞ itself.

Chapter 3.8 then shows the theory's boundary. The ball multiplier has a bounded symbol, so the L^2 hypothesis holds; but its singularity lies along a *sphere* rather than at a point, no Hörmander condition is available, and the conclusion is false. What replaces the theory when singularities live on curved surfaces is the subject of Volume 5 onwards, and the counterexample itself is geometric — it needs the Besicovitch configuration, proved in Volume 9.

The plan

1	Dyadic cubes and the CZ decomposition	the machine
2	The Calderón–Zygmund theorem	weak $(1, 1)$ from L^2 plus smoothness
3	The Hilbert transform	the model case, and Fourier series in L^p
4	Riesz transforms and elliptic estimates	the n -dimensional model
5	The Hörmander–Mikhlin theorem	debt from Volume 2, discharged
6	Littlewood–Paley theory	debt from Volume 2, discharged
7	BMO and John–Nirenberg	the correct endpoint at $p = \infty$
8	The ball multiplier	Takeya enters Fourier analysis

Chapter 3.8 is the destination, and it is a change of plan worth announcing. Volume 2's table of deferrals promised Fefferman's ball multiplier theorem for Volume 5, on the assumption that it needed the stationary phase machinery. It does not: given the Perron tree of Volume 1, Khintchine's inequality of Volume

2, and the Hilbert transform of Chapter 3.3, the proof is self-contained, and it is much better placed here — immediately after the theory it refutes. So it has been moved forward. This is the first time in the series that a Besicovitch set is used to prove a theorem in Fourier analysis, and it is the historical origin of everything in Volumes 9 through 13.

Chapter 3.1

Dyadic Cubes and the Calderón–Zygmund Decomposition

3.1.1 Why start again with maximal functions

Volume 1 proved the Hardy–Littlewood maximal inequality using the Vitali covering lemma, and paid the constant 3^n for it. That proof is geometric and the constant is an artefact of tripling balls.

There is a second proof, due to the same circle of ideas, which replaces balls by *dyadic cubes* and needs no covering lemma at all — the constant becomes 1. It looks at first like a minor economy. It is not. The dyadic structure has a combinatorial property that balls lack — any two dyadic cubes are nested or disjoint — and that property is what makes the Calderón–Zygmund decomposition of §1.4 possible. Every singular integral estimate in this volume, and the Littlewood–Paley theory of Chapter 3.6, ultimately rests on it.

3.1.2 The dyadic grid

Definition 3.1.1. For $k \in \mathbb{Z}$ let

$$\mathcal{D}_k = \left\{ 2^{-k}(m + [0, 1)^n) : m \in \mathbb{Z}^n \right\},$$

the *dyadic cubes of generation k* , each of side 2^{-k} and volume 2^{-kn} . Put $\mathcal{D} = \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$.

Lemma 3.1.2 (The nesting property). *Let $Q \in \mathcal{D}_k$ and $Q' \in \mathcal{D}_{k'}$ with $k \leq k'$. Then either $Q' \subseteq Q$ or $Q \cap Q' = \emptyset$. Moreover each $Q \in \mathcal{D}_k$ is the disjoint*

union of the 2^n cubes of $\mathcal{D}_{k+1}$ contained in it (its children), and is contained in a unique $\widehat{Q} \in \mathcal{D}_{k-1}$ (its parent), with $|\widehat{Q}| = 2^n|Q|$.

Proof. It suffices to treat $n = 1$ and take products. Let $I = 2^{-k}[m, m+1)$ and $I' = 2^{-k'}[m', m'+1)$ with $k \leq k'$. Write $j = k' - k \geq 0$. Then $I = 2^{-k'}[2^j m, 2^j(m+1))$, so I is the union of the 2^j consecutive generation- k' intervals with left endpoints $2^{-k'}\ell$ for $2^j m \leq \ell < 2^j(m+1)$. Either m' is one of these ℓ , in which case $I' \subseteq I$; or it is not, in which case I' lies entirely to the left or entirely to the right of I , and the two are disjoint. The remaining assertions are immediate. $\square$

Pitfall: dyadic cubes are half-open on purpose

The convention $[0, 1)^n$ rather than $[0, 1]^n$ is what makes Theorem 3.1.2 an exact dichotomy: closed cubes of the same generation share faces, and the disjointness in the definition of “children” would fail. Since faces are null sets (Worked Example 1.2.3 (half-spaces are measurable)), nothing measure-theoretic is affected, and every argument below is cleaner for it.

Definition 3.1.3. The *dyadic maximal function* of $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ is

$$\mathcal{M}_d f(x) = \sup_{\substack{Q \in \mathcal{D} \\ x \in Q}} \langle |f| \rangle_Q, \quad \langle |f| \rangle_Q := \frac{1}{|Q|} \int_Q |f|.$$

Theorem 3.1.4 (Dyadic maximal inequality, constant 1). *For $f \in L^1(\mathbb{R}^n)$ and $\lambda > 0$,*

$$m(\{\mathcal{M}_d f > \lambda\}) \leq \frac{1}{\lambda} \int_{\{\mathcal{M}_d f > \lambda\}} |f| \leq \frac{\|f\|_1}{\lambda}.$$

Idea of the proof. Because dyadic cubes nest, the collection of cubes on which the average exceeds λ has *maximal* elements, and distinct maximal elements are disjoint — there is nothing to select and nothing to enlarge. The Vitali lemma existed only to manufacture disjointness that the dyadic grid supplies for free, and the factor 3^n was the price of that manufacture.

Proof. Let $\mathcal{Q}_\lambda = \{Q \in \mathcal{D} : \langle |f| \rangle_Q > \lambda\}$. By definition $\{\mathcal{M}_d f > \lambda\} = \bigcup_{Q \in \mathcal{Q}_\lambda} Q$.

Each $Q \in \mathcal{Q}_\lambda$ satisfies $|Q| < \|f\|_1/\lambda$, so the generations occurring in $\mathcal{Q}_\lambda$ are bounded below; hence every $Q \in \mathcal{Q}_\lambda$ is contained in a *maximal* element of $\mathcal{Q}_\lambda$ (ascend from Q through its parents; the ascent must stop, since volumes double at each step). Let $\{Q_j\}$ be the maximal elements. By Theorem 3.1.2 two distinct maximal cubes are disjoint — if they met, one would contain the other, contradicting maximality of the smaller.

Therefore $\{\mathcal{M}_d f > \lambda\} = \bigsqcup_j Q_j$ and

$$m(\{\mathcal{M}_d f > \lambda\}) = \sum_j |Q_j| < \sum_j \frac{1}{\lambda} \int_{Q_j} |f| = \frac{1}{\lambda} \int_{\{\mathcal{M}_d f > \lambda\}} |f|. \quad \square$$

WORKED EXAMPLE 3.1.1— TIER A $\mathcal{M}_d$ is **genuinely smaller than** $\mathcal{M}$

On $\mathbb{R}$ take $f = \mathbf{1}_{[0,1]}$ and $x < 0$. Every dyadic interval containing both x and part of $[0,1]$ must contain 0, hence must be of the form $[-2^{-k}\ell, \cdot)$; the smallest such interval containing x has length at least $|x|$ and its average of f is at most 1. A short computation gives $\mathcal{M}_d f(x) \leq \frac{1}{2}$ for $x < -1$, while $\mathcal{M}f(x) \approx \frac{1}{2|x|}$ for large $|x|$ — comparable, but the dyadic version is genuinely smaller and, crucially, it is *not* translation invariant: replacing f by $\tau_h f$ changes $\mathcal{M}_d f$ in a way not accounted for by translating.

That failure of translation invariance is the one defect of the dyadic maximal function, and Worked Example 3.1.2 repairs it.

WORKED EXAMPLE 3.1.2— TIER B **The one-third trick: recovering $\mathcal{M}$ from shifted grids**

Proposition 3.1.5. *For $\alpha \in \{0, \frac{1}{3}, \frac{2}{3}\}^n$ let $\mathcal{D}^\alpha = \{Q + \alpha : Q \in \mathcal{D}\}$, a shifted dyadic grid, and let $\mathcal{M}_d^\alpha$ be the corresponding maximal function. Then*

$$\mathcal{M}f(x) \lesssim_n \max_\alpha \mathcal{M}_d^\alpha f(x) \quad \text{for every } x.$$

Consequently Theorem 3.1.4 implies the Hardy–Littlewood maximal inequality, with no covering lemma.

Proof. It suffices to show: for every cube $R \subseteq \mathbb{R}^n$ with sides parallel to the axes there is α and $Q \in \mathcal{D}^\alpha$ with $R \subseteq Q$ and $|Q| \leq C_n |R|$; balls are then handled by inscribing each ball in a cube of comparable volume.

Take $n = 1$ first. Let R be an interval of length ℓ , and choose k with $2^{-k-1} \leq \ell < 2^{-k}$, so that dyadic intervals of generation k have length comparable to ℓ . If R is not contained in a single interval of $\mathcal{D}_k$, then R straddles a point of $2^{-k}\mathbb{Z}$, say p . The shifted grids $\mathcal{D}_k + \frac{1}{3} \cdot 2^{-k}$ and $\mathcal{D}_k + \frac{2}{3} \cdot 2^{-k}$ have their endpoints at $2^{-k}(\mathbb{Z} + \frac{1}{3})$ and $2^{-k}(\mathbb{Z} + \frac{2}{3})$, and the three endpoint sets are pairwise at distance $\frac{1}{3}2^{-k} > \ell/3$ — more precisely, the three sets of endpoints partition each interval $[p, p + 2^{-k})$ into three pieces of length $\frac{1}{3}2^{-k}$, and R , having length $< 2^{-k}$, can straddle endpoints of at most two of the three grids. Hence for some $\alpha \in \{0, \frac{1}{3}, \frac{2}{3}\}$, R lies inside a single interval of $\mathcal{D}_k^\alpha$, whose length is $2^{-k} \leq 2\ell$.

In $\mathbb{R}^n$ apply this in each coordinate separately, choosing $\alpha = (\alpha_1, \dots, \alpha_n)$ coordinatewise; the resulting cube has side at most 2ℓ , so volume at most $2^n|R|$.

Given this, for any ball $B \ni x$ let R be the smallest axis-parallel cube containing B and $Q \supseteq R$ as above; then

$$\frac{1}{|B|} \int_B |f| \leq \frac{|Q|}{|B|} \cdot \frac{1}{|Q|} \int_Q |f| \lesssim_n \mathcal{M}_d^\alpha f(x),$$

since $|Q| \lesssim_n |B|$. Take the supremum over B . $\square$

What this buys. Volume 1’s proof of the maximal theorem needed the Vitali lemma, which is genuinely geometric and whose constant 3^n resisted improvement. The argument above replaces geometry by combinatorics: it needs only that a short interval cannot straddle the endpoints of all three shifted grids at once. The number 3 is not sacred — any three shifts in general position work — but $\{0, \frac{1}{3}, \frac{2}{3}\}$ is the standard choice, and the trick is used constantly in the modern literature, where “by the one-third trick we may assume all cubes are dyadic” is a stock phrase. $\square$

3.1.3 The Calderón–Zygmund decomposition

Theorem 3.1.6 (Calderón–Zygmund decomposition). *Let $f \in L^1(\mathbb{R}^n)$ and $\lambda > 0$. There is a countable collection $\{Q_j\}$ of pairwise disjoint dyadic cubes such that*

- (i) $\lambda < \langle |f| \rangle_{Q_j} \leq 2^n \lambda$ for every j ;
- (ii) $|f(x)| \leq \lambda$ for a.e. $x \notin \Omega := \bigcup_j Q_j$;
- (iii) $|\Omega| = \sum_j |Q_j| \leq \frac{\|f\|_1}{\lambda}$.

Consequently $f = g + b$, where

$$g = f \mathbf{1}_{\Omega^c} + \sum_j \langle f \rangle_{Q_j} \mathbf{1}_{Q_j}, \quad b = \sum_j b_j, \quad b_j = (f - \langle f \rangle_{Q_j}) \mathbf{1}_{Q_j},$$

and these satisfy

- (iv) $\|g\|_\infty \leq 2^n \lambda$ and $\|g\|_1 \leq \|f\|_1$;
- (v) $\text{supp } b_j \subseteq Q_j$, $\int b_j = 0$, and $\|b_j\|_1 \leq 2 \int_{Q_j} |f|$;
- (vi) $\sum_j \|b_j\|_1 \leq 2 \|f\|_1$.

Idea of the proof. Stop the first time the average exceeds λ . Because the dyadic grid nests, “the first time” is well defined — take the maximal cubes of Theorem 3.1.4 — and because a maximal cube’s *parent* failed the test, its own average cannot exceed λ by more than the volume ratio 2^n . That upper bound in (i) is the whole reason the decomposition is useful: it says the bad set is bad by a bounded factor, so the good part g is bounded.

Outside the union nothing ever exceeded λ , so $\mathcal{M}_d f \leq \lambda$ there, and Lebesgue differentiation converts that into a pointwise bound on f .

The final splitting then does the only thing available: replace f on each bad cube by its average, which is bounded, and record the error, which has mean zero by construction. Mean zero is what Chapter 3.2 will exploit.

Proof. Step 1. Selection.

Let $\{Q_j\}$ be the maximal elements of $\mathcal{Q}_\lambda = \{Q \in \mathcal{D} : \langle |f| \rangle_Q > \lambda\}$, as in the proof of Theorem 3.1.4. They are pairwise disjoint, and their union is $\{\mathcal{M}_d f > \lambda\} =: \Omega$.

Step 2. *Proof of (i).*

The lower bound is the definition of $\mathcal{Q}_\lambda$. For the upper bound, let $\widehat{Q}_j$ be the parent of Q_j . By maximality $\widehat{Q}_j \notin \mathcal{Q}_\lambda$, so $\langle |f| \rangle_{\widehat{Q}_j} \leq \lambda$, and since $|\widehat{Q}_j| = 2^n |Q_j|$ and $Q_j \subseteq \widehat{Q}_j$,

$$\langle |f| \rangle_{Q_j} = \frac{1}{|Q_j|} \int_{Q_j} |f| \leq \frac{2^n}{|\widehat{Q}_j|} \int_{\widehat{Q}_j} |f| \leq 2^n \lambda.$$

Step 3. *Proof of (ii).*

For $x \notin \Omega$ we have $\mathcal{M}_d f(x) \leq \lambda$, so every dyadic average of $|f|$ over cubes containing x is at most λ . Letting the cubes shrink to x and applying the Lebesgue differentiation theorem (Theorem 1.7.4 (Lebesgue differentiation), in its dyadic form — the dyadic cubes containing x shrink to x and are comparable to balls) gives $|f(x)| \leq \lambda$ for a.e. such x .

Step 4. *Proof of (iii).*

This is Theorem 3.1.4.

Step 5. *Proof of (iv)–(vi).*

On Ω^c we have $|g| = |f| \leq \lambda$ by (ii); on Q_j we have $|g| = |\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n \lambda$ by (i). And $\|g\|_1 \leq \int_{\Omega^c} |f| + \sum_j |Q_j| \langle |f| \rangle_{Q_j} = \int_{\Omega^c} |f| + \sum_j \int_{Q_j} |f| = \|f\|_1$.

For (v): the support statement is clear, $\int b_j = \int_{Q_j} f - |Q_j| \langle f \rangle_{Q_j} = 0$ by definition of the average, and $\|b_j\|_1 \leq \int_{Q_j} |f| + |Q_j| \langle |f| \rangle_{Q_j} \leq 2 \int_{Q_j} |f|$. Summing over the disjoint Q_j gives (vi). $\square$

Where we are. Four properties will be used in Chapter 3.2, and it is worth naming them now, because the proof of the Calderón–Zygmund theorem uses each exactly once:

- $\|g\|_\infty \lesssim \lambda$, to make an L^2 estimate on g cheap;
- $\|g\|_1 \leq \|f\|_1$, to make it quantitative;
- $\int b_j = 0$, to exploit smoothness of the kernel;
- $\sum_j |Q_j| \lesssim \|f\|_1 / \lambda$, to discard the bad set.

Nothing else about the decomposition is needed anywhere in this volume.

WORKED EXAMPLE 3.1.3— TIER B A decomposition computed explicitly

Let $n = 1$ and $f = \mathbf{1}_{[0, 2^{-N}]}$ for a large integer N , so $\|f\|_1 = 2^{-N}$. Fix $\lambda \in (0, 1)$ and let k be the unique integer with $2^{k-N} \leq \lambda^{-1} < 2^{k+1-N}$, that is, $2^{-k} \approx \lambda 2^{-N} \cdot 2^N = \lambda \cdot \dots$; concretely, choose $k \geq 0$ minimal with $2^{-N}/2^{-k} \leq \lambda$, i.e. $2^{k-N} \leq \lambda$.

The dyadic intervals I with $\langle |f| \rangle_I > \lambda$ are exactly those with $|I \cap [0, 2^{-N}]| > \lambda |I|$. For $I \subseteq [0, 2^{-N}]$ this holds whenever $\lambda < 1$; for $I \supseteq [0, 2^{-N}]$ of length 2^{-m} it reads $2^{-N} > \lambda 2^{-m}$, i.e. $2^{m-N} > \lambda$.

So the maximal such interval is the dyadic interval $I_0 = [0, 2^{-m_0})$ where m_0 is the least m with $2^{m-N} > \lambda$. There is exactly one bad cube, $Q_1 = I_0$, and:

$$\langle |f| \rangle_{Q_1} = 2^{m_0-N} \in (\lambda, 2\lambda],$$

confirming (i) with $2^n = 2$. Then $g = 2^{m_0-N} \mathbf{1}_{I_0}$, which has $\|g\|_\infty = 2^{m_0-N} \leq 2\lambda$ and $\|g\|_1 = 2^{-N} = \|f\|_1$; and $b = (\mathbf{1}_{[0, 2^{-N}]} - 2^{m_0-N}) \mathbf{1}_{I_0}$, of mean zero, with $\|b\|_1 \leq 2 \cdot 2^{-N}$.

What the example shows. The bad cube is not the support of f ; it is the dyadic interval at the scale where the average of f first drops to λ . Raising λ shrinks it and lowering λ grows it, and the decomposition is a way of selecting, for each λ , the scale at which f looks like size λ . That is exactly the multiscale bookkeeping of Volume 1, Worked Example 5.16, in another costume.

WORKED EXAMPLE 3.1.4— TIER C: RESEARCH TASTE Stopping-time decompositions, from Calderón–Zygmund to the frontier

Theorem 3.1.6 is the first instance in this series of a *stopping-time decomposition*, and the pattern it introduces recurs, in progressively more elaborate

forms, all the way to the current literature. It is worth naming the pattern while it is still simple enough to see whole.

The pattern. One has an object (here, a function) and a family of scales (here, the dyadic generations). One descends through the scales, testing a condition at each (here, “is the average $> \lambda$?”), and *stops* at the first scale where the test fires. This partitions the object into: a part where the test never fired, which is therefore uniformly good; and a collection of localised pieces, one for each stopping cube, which are bad but *controlled*, because their parent passed the test.

The two clauses of Theorem 3.1.6(i) are precisely this: $> \lambda$ because we stopped, $\leq 2^n \lambda$ because we had not stopped one step earlier. Every stopping-time argument produces a two-sided bound of exactly this shape, and the upper bound always comes from the parent.

Where it goes. Three later instances, in increasing order of difficulty:

- *Whitney decompositions* (Chapter 3.7): stop when a cube’s distance to a closed set becomes comparable to its diameter. The parent clause becomes the statement that Whitney cubes have bounded overlap.
- *Broad–narrow analysis* (Volume 12). Bourgain and Guth partition space by asking, at each scale, whether the significant contributions to $|Ef|$ at a given point come from caps that are *transverse* to one another. Where they do — the broad case — a multilinear estimate applies; where they do not, the contribution is essentially concentrated on a lower-dimensional set and one recurses. The stopping condition is transversality; the parent clause is what makes the recursion terminate.
- *The sticky–plany–grainy trichotomy* (Volume 10) and the reduction of the general Kakeya problem to the sticky case (Volume 14). Wang and Zahl’s proof in $\mathbb{R}^3$ organises a family of tubes by repeatedly asking, scale by scale, how the tubes at scale δ cluster inside the tubes at scale $\sqrt{\delta}$, and stopping when the clustering becomes degenerate.

In every case the arithmetic is the same as Step 2 of the proof above: the selected object exceeds the threshold, and its parent does not, so it exceeds it by a bounded factor. A reader who has understood why 2^n appears in Theorem 3.1.6(i) has understood the mechanism of all of them.

Figure 3.1.1 (to be drawn). *The dyadic grid and a straddling cube*

Content required: Two or three generations of the dyadic grid in the plane, with one arbitrary cube Q drawn straddling a grid line so that no dyadic cube of comparable size contains it. Beside it, the same Q contained in a cube of one of the three shifted grids, illustrating the one-third trick.

WORKED EXAMPLE 3.1.5— TIER B The stopping-time argument, in isolation

The Calderón–Zygmund decomposition is a stopping-time argument, and the pattern recurs often enough that it is worth extracting from the proof.

The setup. One has a quantity attached to each dyadic cube — here the average $\langle |f| \rangle_Q$ — and a threshold λ . One wants to select a family of cubes, disjoint, on which the quantity first exceeds the threshold.

The rule. Starting from cubes so large that the average is below λ — possible since $f \in L^1$ and averages over huge cubes tend to 0 — subdivide. Select a cube Q when $\langle |f| \rangle_Q > \lambda$ and its parent was not selected. Stop there; do not subdivide a selected cube.

The two consequences, and where each comes from.

- *Above the threshold:* $\langle |f| \rangle_Q > \lambda$, because Q was selected. Summing gives $\sum |Q| \leq \lambda^{-1} \|f\|_1$, the measure bound.
- *Not too far above:* $\langle |f| \rangle_{\widehat{Q}} \leq \lambda$ for the parent $\widehat{Q}$, because $\widehat{Q}$ was *not* selected. Since $|\widehat{Q}| = 2^n |Q|$, this gives $\langle |f| \rangle_Q \leq 2^n \lambda$ — the upper bound that makes the good function bounded.

The second is the whole reason for the stopping rule: without “first exceeds”, one would have no upper bound at all.

Where it recurs. The Whitney decomposition of Volume 4, Theorem 4.4.3, is the same rule with $\text{dist}(Q, \Omega^c)$ in place of the average. The atomic decomposition of Volume 4, Theorem 4.3.4, iterates it over $\lambda = 2^k$. And the pigeonholing over scales in Volume 7, Theorem 7.8.5, is a stopping time in disguise.

The habit. On meeting a selection of cubes, ask what the parent’s failure to be selected gives you. That is almost always where the useful upper bound comes from, and it is the step most often omitted in summaries.

Drills

Drill 3.1.1. Show that $\mathcal{M}_d f \leq \mathcal{M} f$ pointwise up to a dimensional constant,

and that no reverse inequality holds pointwise: find f and x with $\mathcal{M}f(x)$ large and $\mathcal{M}_d f(x)$ small.

Drill 3.1.2. Deduce from Theorem 3.1.4 and Worked Example 3.1.2 that $\|\mathcal{M}f\|_p \lesssim_{n,p} \|f\|_p$ for $1 < p \leq \infty$, quoting the Marcinkiewicz theorem (Theorem 2.2.4 (Marcinkiewicz interpolation)). Compare the constant with the one obtained in Volume 1.

Drill 3.1.3. In Theorem 3.1.6, show that the cubes Q_j depend monotonically on λ : if $\lambda' < \lambda$ then every Q_j at height λ is contained in some cube at height λ' .

Drill 3.1.4. Carry out the decomposition of Worked Example 3.1.3 for $f(x) = |x|^{-1/2} \mathbf{1}_{[0,1]}$ and describe the resulting family $\{Q_j\}$ as $\lambda \rightarrow \infty$.

Where we are. The dyadic grid is used here purely as a bookkeeping device — a family of cubes closed under the operations the stopping-time argument needs. Nothing depends on the particular grid, and Worked Example 3.1.2 shows that three shifted grids suffice to capture an arbitrary cube up to bounded distortion.

That flexibility matters later. Volume 11’s polynomial partitioning replaces the dyadic grid by the cells of a polynomial’s zero set, and the arguments transfer because they used only the stopping-time structure, never the geometry of cubes.

Notes

The dyadic maximal function and the one-third trick are folklore; the earliest systematic use of shifted grids appears in work of Garnett and Jones. The Calderón–Zygmund decomposition is from their 1952 *Acta* paper, which also contains the theorem of Chapter 3.2.

The reader should be aware that a substantial modern literature — sparse domination, the A_2 theorem — consists of refinements of Theorem 3.1.6 in which the selected cubes are required to carry a fixed proportion of the mass. None of it is needed in this series, but the vocabulary is common in papers a reader may consult.

Chapter 3.2

The Calderón–Zygmund Theorem

3.2.1 What a singular integral is

The operators of this volume are all of the form

$$Tf(x) = \int_{\mathbb{R}^n} K(x, y)f(y) \, dy, \quad (3.2.1)$$

where the kernel K is of size $|x - y|^{-n}$ near the diagonal. That size is critical: $|x|^{-n}$ is not integrable at 0 and not integrable at ∞ , by the polar coordinates computation of Volume 1. So (3.2.1) does not converge absolutely, the operator is not bounded on L^1 , and the integral must be interpreted as a principal value. The name *singular integral* records exactly this.

What saves the situation is cancellation. If the kernel oscillates, or — what is usually easier to verify — if it varies *slowly* away from the diagonal, then it cannot distinguish a mean-zero bump from nothing, and that is enough.

Definition 3.2.1 (Calderón–Zygmund kernel). A measurable $K : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{C}$ is a *Calderón–Zygmund kernel* with constant A if

$$\int_{|x|>2|y|} |K(x - y) - K(x)| \, dx \leq A \quad \text{for every } y \neq 0. \quad (3.2.2)$$

Condition (3.2.2) is called *Hörmander’s condition*.

Proposition 3.2.2 (The usual smoothness hypotheses suffice). *Suppose $K \in C^1(\mathbb{R}^n \setminus \{0\})$ satisfies $|\nabla K(x)| \leq B|x|^{-n-1}$. Then K satisfies (3.2.2) with $A \lesssim_n B$.*

Proof. Fix $y \neq 0$ and x with $|x| > 2|y|$. The segment from $x - y$ to x stays in $\{|z| \geq |x| - |y| \geq |x|/2\}$, so by the mean value theorem

$$\begin{aligned} |K(x - y) - K(x)| &\leq |y| \sup_{z \in [x-y, x]} |\nabla K(z)| \\ &\leq |y| B \left(\frac{|x|}{2} \right)^{-n-1} \\ &= 2^{n+1} B \frac{|y|}{|x|^{n+1}}. \end{aligned}$$

Integrating over $|x| > 2|y|$ and using polar coordinates,

$$\begin{aligned} \int_{|x| > 2|y|} |K(x - y) - K(x)| \, dx &\leq 2^{n+1} B |y| \sigma(\mathbb{S}^{n-1}) \int_{2|y|}^{\infty} r^{-2} \, dr \\ &= 2^n B \sigma(\mathbb{S}^{n-1}), \end{aligned}$$

since $\int_{2|y|}^{\infty} r^{-2} \, dr = (2|y|)^{-1}$. □

Pitfall: the 2 in $|x| > 2|y|$ is not arbitrary

The region of integration in (3.2.2) excludes a neighbourhood of the singularity of *both* $K(\cdot - y)$ and K . If one integrated over all x , the condition would fail for every kernel of size $|x|^{-n}$, since each term separately is non-integrable near its own singularity. Any constant $c > 1$ in place of 2 gives an equivalent condition, with a different A ; the choice 2 is conventional and matches the dilation $Q \mapsto 2\sqrt{n}Q$ used in the proof below.

3.2.2 The theorem

Theorem 3.2.3 (Calderón–Zygmund). *Let T be a linear operator, bounded on $L^2(\mathbb{R}^n)$ with norm B_2 , such that for every $f \in L^1 \cap L^2$ with compact support,*

$$Tf(x) = \int_{\mathbb{R}^n} K(x - y) f(y) \, dy \quad \text{for a.e. } x \notin \text{supp } f, \quad (3.2.3)$$

where K satisfies Hörmander’s condition (3.2.2) with constant A . Then:

- (i) T is of weak type $(1, 1)$: $m(\{|Tf| > \lambda\}) \lesssim_n \frac{(A + B_2)}{\lambda} \|f\|_1$;
- (ii) T is bounded on $L^p(\mathbb{R}^n)$ for every $1 < p < \infty$, with $\|T\|_{L^p \rightarrow L^p} \lesssim_{n,p} A + B_2$.

Idea of the proof. Split $f = g + b$ by Theorem 3.1.6 at height λ and treat the two parts by completely different means.

The good part g is bounded by λ and has L^1 norm at most $\|f\|_1$, so $\|g\|_2^2 \leq \|g\|_\infty \|g\|_1 \lesssim \lambda \|f\|_1$. The L^2 hypothesis then gives what we want by Chebyshev, and note the shape of the exchange: we bought an L^1 statement by spending an L^2 one, and the factor λ that appears is exactly what Chebyshev needs.

The bad part is where the kernel enters. Discard the union of the (slightly enlarged) bad cubes — it has measure $\lesssim \|f\|_1/\lambda$, which is the budget. Outside it, use that $\int b_j = 0$ to *subtract a constant* from the kernel:

$$Tb_j(x) = \int_{Q_j} (K(x-y) - K(x-c_j))b_j(y) dy,$$

where c_j is the centre of Q_j . Since x is far from Q_j and y is inside it, Hörmander's condition applies and integrates the difference to A . This is the only place cancellation is used, and it is used exactly once.

Proof. Step 1. Reduction.

By density (Theorem 1.8.7 (density of simple functions and C_c^∞)) it suffices to prove (i) for $f \in L^1 \cap L^2$ with compact support, since $L^1 \cap L^2$ is dense in L^1 and weak-type bounds pass to limits along a subsequence converging a.e.

Fix such an f and $\lambda > 0$, and apply Theorem 3.1.6 at height λ , obtaining $\Omega = \bigsqcup_j Q_j$, $f = g + b$, $b = \sum_j b_j$. All the properties (i)–(vi) there are now in force.

Step 2. *The good part.*

Since $|g| \leq 2^n \lambda$ pointwise and $\|g\|_1 \leq \|f\|_1$,

$$\|g\|_2^2 = \int |g|^2 \leq \|g\|_\infty \|g\|_1 \leq 2^n \lambda \|f\|_1.$$

By the L^2 hypothesis and Chebyshev's inequality (Volume 1, Prop. 8.14),

$$m\left(\left\{|Tg| > \frac{\lambda}{2}\right\}\right) \leq \frac{4}{\lambda^2} \|Tg\|_2^2 \tag{3.2.4}$$

$$\leq \frac{4B_2^2}{\lambda^2} \cdot 2^n \lambda \|f\|_1 \tag{3.2.5}$$

$$= \frac{2^{n+2}B_2^2}{\lambda} \|f\|_1. \tag{3.2.6}$$

Step 3. *Discarding the bad set.*

Let c_j be the centre of Q_j and let $Q_j^* = 2\sqrt{n}Q_j$ be the concentric dilate; the factor $2\sqrt{n}$ is chosen so that

$$x \notin Q_j^* \text{ and } y \in Q_j \implies |x - c_j| > 2|y - c_j|. \tag{3.2.7}$$

Indeed if Q_j has side ℓ then $|y - c_j| \leq \frac{\sqrt{n}}{2}\ell$ for $y \in Q_j$, while $x \notin Q_j^*$ forces $|x - c_j| > \sqrt{n}\ell$.

Put $\Omega^* = \bigcup_j Q_j^*$. By Theorem 3.1.6(iii),

$$|\Omega^*| \leq \sum_j |Q_j^*| = (2\sqrt{n})^n \sum_j |Q_j| \leq \frac{(2\sqrt{n})^n}{\lambda} \|f\|_1. \quad (3.2.8)$$

Step 4. *The bad part, off Ω^* .*

Fix j . For $x \notin Q_j^*$, the representation (3.2.3) applies since $x \notin \text{supp } b_j \subseteq Q_j$, and the mean-zero property $\int b_j = 0$ allows us to insert a constant:

$$Tb_j(x) = \int_{Q_j} K(x - y) b_j(y) dy = \int_{Q_j} (K(x - y) - K(x - c_j)) b_j(y) dy.$$

Integrating in x over the complement of Q_j^* and using Tonelli,

$$\int_{(Q_j^*)^c} |Tb_j(x)| dx \leq \int_{Q_j} |b_j(y)| \left(\int_{(Q_j^*)^c} |K(x - y) - K(x - c_j)| dx \right) dy.$$

Substitute $x \mapsto x + c_j$, $y \mapsto y + c_j$ in the inner integral: by (3.2.7) the region of integration is contained in $\{|x| > 2|y|\}$, so Hörmander's condition (3.2.2) bounds it by A . Hence

$$\int_{(Q_j^*)^c} |Tb_j| \leq A \|b_j\|_1,$$

and summing over j and using Theorem 3.1.6(vi),

$$\int_{(\Omega^*)^c} |Tb| \leq \sum_j \int_{(Q_j^*)^c} |Tb_j| \leq A \sum_j \|b_j\|_1 \leq 2A \|f\|_1.$$

By Chebyshev,

$$m\left(\left\{x \notin \Omega^* : |Tb(x)| > \frac{\lambda}{2}\right\}\right) \leq \frac{4A}{\lambda} \|f\|_1. \quad (3.2.9)$$

Step 5. *Assemble.*

Since $\{|Tf| > \lambda\} \subseteq \{|Tg| > \frac{\lambda}{2}\} \cup \{|Tb| > \frac{\lambda}{2}\}$, and the latter is contained in Ω^* together with the set in (3.2.9),

$$m(\{|Tf| > \lambda\}) \leq \frac{2^{n+2}B_2^2 + (2\sqrt{n})^n + 4A}{\lambda} \|f\|_1.$$

Normalising T so that $B_2 \leq 1$ — replace T by T/B_2 , which replaces A by A/B_2 — gives the stated form of (i).

Step 6. *Interpolation, and duality.*

For $1 < p < 2$: T is of weak type $(1, 1)$ by (i) and of strong, hence weak, type $(2, 2)$ by hypothesis. Marcinkiewicz interpolation (Theorem 2.2.4 (Marcinkiewicz interpolation)) gives strong type (p, p) .

For $2 < p < \infty$: the adjoint T^* has kernel $K^*(x, y) = \overline{K(y, x)}$, i.e. convolution with $\bar{K}(x) = \overline{K(-x)}$, which satisfies (3.2.2) with the same constant after the substitution $x \mapsto -x$. And T^* is bounded on L^2 with the same norm. So the previous paragraph applies to T^* and gives $\|T^*\|_{L^{p'} \rightarrow L^{p'}} \lesssim A + B_2$ for $1 < p' < 2$; by duality (Theorem 1.8.10 (norms by testing)), $\|T\|_{L^p \rightarrow L^p} = \|T^*\|_{L^{p'} \rightarrow L^{p'}}$. $\square$

Where we are. Read Step 4 again. It is the only step that uses anything about K , and what it uses is that the kernel, viewed from far away, is nearly constant across a small cube. The mean-zero property of b_j converts “nearly constant” into “nearly zero”. No oscillation, no explicit cancellation in the kernel, and no formula were needed — only Hörmander’s condition, which is a smoothness statement.

This is why the theorem covers so much. Any operator whose kernel is C^1 away from the diagonal with the natural gradient bound qualifies, by Theorem 3.2.2; and verifying the remaining hypothesis — L^2 boundedness — is, for convolution operators, exactly Plancherel.

WORKED EXAMPLE 3.2.1— TIER A Truncations and principal values

The representation (3.2.3) says nothing about $Tf(x)$ for $x \in \text{supp } f$, and it cannot: the integral does not converge there. The standard device is to truncate. For $0 < \varepsilon < R$ set

$$T_{\varepsilon, R}f(x) = \int_{\varepsilon < |x-y| < R} K(x-y)f(y) dy,$$

which converges absolutely for $f \in L^p$, since K is bounded on the annulus. Two remarks. First, the truncated operators are genuine convolution operators with bounded compactly supported kernels, so no principal value is involved and every manipulation is legitimate. Second, the content of the theory is that the bounds one proves are *uniform* in ε and R ; the limit $T = \lim_{\varepsilon \rightarrow 0, R \rightarrow \infty} T_{\varepsilon, R}$ then exists in the appropriate sense and inherits them. Throughout this volume, statements about T should be read as statements about $T_{\varepsilon, R}$ with constants independent of ε, R , and the passage to the limit is carried out once, in Theorem 3.3.2.

WORKED EXAMPLE 3.2.2— TIER B **Sharpness: the weak $(1, 1)$ bound cannot be upgraded**

Let $n = 1$ and $K(x) = 1/x$, the Hilbert transform kernel of Chapter 3.3. Take $f = \mathbf{1}_{[0,1]}$. For $x > 1$,

$$Tf(x) = \int_0^1 \frac{dy}{x-y} = \log \frac{x}{x-1},$$

which for large x behaves like $1/x$. Hence $\int_2^\infty |Tf| \approx \int_2^\infty \frac{dx}{x} = \infty$, so $Tf \notin L^1(\mathbb{R})$ even though f is as nice as one could wish.

The mechanism is identical to Worked Example 1.7.1 ($\mathcal{M}f$ is never integrable), where $\mathcal{M}f \notin L^1$ for the same reason: an operator whose kernel decays like $|x|^{-n}$ produces output decaying like $|x|^{-n}$, and $|x|^{-n}$ is exactly the borderline of integrability. In both cases the honest endpoint statement is a weak $(1, 1)$ bound, and Chapter 3.7 will show that the honest statement at the other endpoint is not $L^\infty \rightarrow L^\infty$ but $L^\infty \rightarrow \text{BMO}$.

WORKED EXAMPLE 3.2.3— TIER C: RESEARCH TASTE **Why the L^2 hypothesis is the hard one**

Theorem 3.2.3 has two hypotheses. One — Hörmander’s condition — is a routine verification, usually a two-line consequence of Theorem 3.2.2. The other — boundedness on L^2 — is, in every interesting case, the entire difficulty. It is worth being explicit about this asymmetry, because it recurs at every stage of this series.

(a) *When L^2 is free.* For a convolution operator $Tf = K * f$, Plancherel turns the question into a pointwise bound on the multiplier: T is bounded on L^2 with norm $\|\widehat{K}\|_\infty$ (Theorem 2.6.2 (the L^2 multiplier norm)). So for the Hilbert transform, whose multiplier is $-i \operatorname{sgn} \xi$, the L^2 bound is immediate and the constant is 1. The same holds for the Riesz transforms (Chapter 3.4) and for every multiplier in Chapter 3.5. *This is why the convolution theory is easy.*

(b) *When it is not.* For non-convolution operators there is no Plancherel, and establishing L^2 boundedness becomes the theorem. The $T(1)$ theorem of David and Journé characterises exactly when a Calderón–Zygmund operator is bounded on L^2 , in terms of the two conditions $T(1), T^*(1) \in \text{BMO}$ plus a weak boundedness property; it is not proved in this series, and it is the reason BMO is not merely a curiosity.

(c) *The same asymmetry in the restriction problem.* The extension operator E of Volume 2 is bounded from $L^2(d\sigma)$ to $L^q(\mathbb{R}^n)$ for $q = \frac{4n}{n-1}$ by a soft

argument — Worked Example 2.8.3 (a restriction theorem from decay), which used only decay and Hardy–Littlewood–Sobolev. Stein–Tomas improves the exponent by exploiting L^2 orthogonality, and the restriction conjecture proper concerns $L^\infty(d\sigma)$ data, where no orthogonality is available at all. Each step away from L^2 costs a decade.

(d) *And in decoupling.* Worked Example 1.8.2 (almost orthogonality) observed that ℓ^2 decoupling is trivial at $p = 2$, by Plancherel, and is the theorem for $p > 2$. Volume 13’s proof proceeds by repeatedly reducing L^p information to L^2 information at a smaller scale — that is what “ball inflation” does — and then iterating. The subject may fairly be described as a long campaign to convert L^2 facts, which are free, into L^p facts, which are not. This chapter is the first campaign of that kind, and the Calderón–Zygmund decomposition is its first weapon.

WORKED EXAMPLE 3.2.4— TIER C: RESEARCH TASTE **What the Calderón–Zygmund theorem buys, and what it costs**

Theorem 3.2.3 is the central theorem of the volume. It is worth being explicit about the trade it makes.

(a) *The input.* Two hypotheses: boundedness on L^2 , and Hörmander’s condition on the kernel. The first is usually easy — for a multiplier it is $\|m\|_\infty < \infty$, by Plancherel. The second is a smoothness condition away from the diagonal and is where the work is.

(b) *The output.* Weak type $(1, 1)$, and hence by interpolation L^p boundedness for all $1 < p < \infty$. So one verifies an estimate at a *single* exponent and obtains a whole range.

(c) *Why L^2 and not another exponent.* Because Plancherel makes L^2 computable for multipliers, and because the Calderón–Zygmund decomposition splits f into a bounded part — handled by L^2 — and a mean-zero part — handled by the kernel. Any exponent would do in principle; L^2 is where the hypothesis is checkable.

(d) *The cost: the endpoints are lost.* $p = 1$ becomes weak type, and $p = \infty$ becomes BMO (Theorem 3.7.4). Both losses are genuine, as Worked Example 3.3.4(c) showed for the Hilbert transform. Volume 4 exists to supply the two substitute spaces and to prove that the theory is sharp on them.

(e) *What it does not cover.* Operators whose kernels are singular along a *surface* rather than a point — the ball multiplier of Chapter 3.8 is the first example — fall outside entirely. Its symbol is bounded, so the L^2 hypothesis holds; but there is no Hörmander condition to verify, because the kernel’s

singularity is spread over a sphere rather than concentrated at the origin. That failure is not technical: Theorem 3.8.2 shows the conclusion is *false*, and the reason is the Besicovitch geometry of Volume 9.

The summary to carry forward. Calderón–Zygmund theory handles singularities at a point. Everything from Volume 5 onwards handles singularities along a surface, and needs curvature to do it.

Drills

Drill 3.2.1. Verify that $K(x) = \Omega(x/|x|)|x|^{-n}$ satisfies (3.2.2) whenever Ω is Lipschitz on $\mathbb{S}^{n-1}$, and show by example that mere continuity of Ω is not enough for Theorem 3.2.2 to apply.

Drill 3.2.2. Show that the constant in Theorem 3.2.3(ii) blows up as $p \rightarrow 1$ and as $p \rightarrow \infty$, and identify which step of the proof is responsible in each case.

Drill 3.2.3. Prove that if T satisfies the hypotheses of Theorem 3.2.3 then so does $f \mapsto T\bar{f}$, and deduce that one may assume K real without loss of generality in checking (3.2.2).

Drill 3.2.4. Where exactly does the proof of Theorem 3.2.3 use that the Q_j are *disjoint*? Show that without disjointness the estimate (3.2.8) fails, and construct a covering family for which it fails by a factor $\log(1/\lambda)$.

Notes

Theorem 3.2.3 is from Calderón and Zygmund’s 1952 paper, in the form with a gradient hypothesis on the kernel; Hörmander’s condition (3.2.2), which is weaker and is what the proof actually uses, is from his 1960 paper.

The reader should note that Theorem 3.2.3 produces no examples. It is a machine for converting an L^2 bound into L^p bounds, and every application in this volume begins by establishing the L^2 bound by Plancherel. Worked Example 3.2.3 makes the point that this is not an accident of exposition.

Chapter 3.3

The Hilbert Transform

The Hilbert transform is the model singular integral: the unique one on $\mathbb{R}$ (up to constants) commuting with translations and dilations and anticommuting with reflections. This chapter establishes its boundedness, and then cashes it in for a theorem promised in Volume 2 — the convergence of Fourier series in L^p .

3.3.1 Definition and L^2 theory

Definition 3.3.1. For $f \in \mathcal{S}(\mathbb{R})$,

$$\begin{aligned}\mathcal{H}f(x) &= \text{p.v.} \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(y)}{x-y} dy \\ &= \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \int_{|x-y| > \varepsilon} \frac{f(y)}{x-y} dy.\end{aligned}$$

Proposition 3.3.2 (The limit exists, and the multiplier). *For $f \in \mathcal{S}(\mathbb{R})$ the limit exists for every x , and*

$$\widehat{\mathcal{H}f}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi).$$

Consequently $\|\mathcal{H}f\|_2 = \|f\|_2$, and $\mathcal{H}$ extends to a unitary operator on $L^2(\mathbb{R})$ with $\mathcal{H}^2 = -\operatorname{Id}$.

Proof. Step 1. Existence.

Because the kernel $1/(x-y)$ is odd, the truncated integral may be written

$$\frac{1}{\pi} \int_{|t| > \varepsilon} \frac{f(x-t)}{t} dt = \frac{1}{\pi} \int_{\varepsilon < |t| < 1} \frac{f(x-t) - f(x)}{t} dt + \frac{1}{\pi} \int_{|t| \geq 1} \frac{f(x-t)}{t} dt,$$

where we used $\int_{\varepsilon < |t| < 1} \frac{dt}{t} = 0$ to insert $-f(x)$ at no cost. The first integrand is bounded by $\|f'\|_\infty$ and the second by $|f(x-t)|$, integrable since $f \in \mathcal{S}$. So the limit exists by dominated convergence, and is bounded uniformly on compact sets.

Step 2. *The multiplier.*

Let $K_\varepsilon(t) = \frac{1}{\pi t} \mathbf{1}_{\varepsilon < |t| < 1/\varepsilon}$, so $\mathcal{H}f = \lim_{\varepsilon \rightarrow 0} K_\varepsilon * f$. Each $K_\varepsilon \in L^1$, so $\widehat{K_\varepsilon * f} = \widehat{K_\varepsilon} \widehat{f}$ (Theorem 2.4.2 (the Fourier dictionary)(v)). Since K_ε is odd,

$$\begin{aligned} \widehat{K_\varepsilon}(\xi) &= \frac{1}{\pi} \int_{\varepsilon < |t| < 1/\varepsilon} \frac{e^{-2\pi i t \xi}}{t} dt \\ &= \frac{-2i}{\pi} \int_{\varepsilon}^{1/\varepsilon} \frac{\sin(2\pi t \xi)}{t} dt \\ &= \frac{-2i \operatorname{sgn} \xi}{\pi} \int_{2\pi \varepsilon |\xi|}^{2\pi |\xi|/\varepsilon} \frac{\sin u}{u} du, \end{aligned}$$

substituting $u = 2\pi t |\xi|$. As $\varepsilon \rightarrow 0$ the integral tends to $\int_0^\infty \frac{\sin u}{u} du = \frac{\pi}{2}$, so $\widehat{K_\varepsilon}(\xi) \rightarrow -i \operatorname{sgn} \xi$ pointwise, and the $\widehat{K_\varepsilon}$ are uniformly bounded because the partial integrals of $\sin u/u$ are. Dominated convergence in the Plancherel pairing gives the claim.

Step 3. *Consequences.*

$|-i \operatorname{sgn} \xi| = 1$ a.e., so by Theorem 2.6.2 (the L^2 multiplier norm) the operator is an isometry of L^2 ; and the multiplier squares to -1 a.e. $\square$

Theorem 3.3.3 (L^p boundedness; M. Riesz). *$\mathcal{H}$ is of weak type $(1, 1)$ and bounded on $L^p(\mathbb{R})$ for every $1 < p < \infty$, with $\|\mathcal{H}\|_{L^p \rightarrow L^p} \lesssim_p 1$. It is bounded on neither L^1 nor L^∞ .*

Proof. The kernel $K(x) = \frac{1}{\pi x}$ is C^∞ away from 0 with $|K'(x)| = \frac{1}{\pi x^2}$, so Theorem 3.2.2 applies with $n = 1$ and $B = 1/\pi$, giving Hörmander's condition. Theorem 3.3.2 gives L^2 boundedness with $B_2 = 1$, and the representation (3.2.3) holds for $x \notin \operatorname{supp} f$ by inspection, the principal value being an honest convergent integral there. Theorem 3.2.3 applies and gives both conclusions.

Unboundedness on L^1 is Worked Example 3.2.2. For L^∞ , take $f = \mathbf{1}_{[0,1]}$: a direct computation gives $\mathcal{H}f(x) = \frac{1}{\pi} \log|\frac{x}{x-1}|$, which is unbounded near $x = 0$ and $x = 1$ although $\|f\|_\infty = 1$. $\square$

WORKED EXAMPLE 3.3.1— TIER A **The Hilbert transform of an indicator**

For $f = \mathbf{1}_{[a,b]}$ and $x \notin [a,b]$ the principal value is an ordinary integral:

$$\mathcal{H}f(x) = \frac{1}{\pi} \int_a^b \frac{dy}{x-y} = \frac{1}{\pi} \log \left| \frac{x-a}{x-b} \right|.$$

The same formula holds for $x \in (a,b)$, where the principal value cancels the symmetric part of the singularity.

Three features are worth recording, since each is typical.

Logarithmic blow-up at the endpoints. $\mathcal{H}f$ is unbounded, which is Theorem 3.3.3's failure at $p = \infty$. Note it is only logarithmic — $\mathcal{H}f$ lies in every L^p with $p < \infty$, and in BMO (Chapter 3.7), which is the correct endpoint space.

Decay $1/|x|$ at infinity. $\log \left| \frac{x-a}{x-b} \right| = \log \left| 1 + \frac{b-a}{x-b} \right| \approx \frac{b-a}{x}$, which is the failure at $p = 1$.

The transform is not supported near the support of f . A singular integral spreads mass everywhere, which is exactly why the localisation argument of Theorem 3.2.3 Step 4 had to discard the enlarged cubes rather than estimate on them.

3.3.2 The conjugate function and Fourier series

We now transfer to the torus and collect the promised application.

Definition 3.3.4. For a trigonometric polynomial $f = \sum_{|m| \leq N} \hat{f}(m)e_m$ on $\mathbb{T}$, the *conjugate function* is

$$\tilde{f} = \sum_{|m| \leq N} (-i \operatorname{sgn} m) \hat{f}(m)e_m,$$

and the *Riesz projection* is $P_+f = \sum_{m \geq 0} \hat{f}(m)e_m$, so that $P_+ = \frac{1}{2}(\operatorname{Id} + i\tilde{}) + \frac{1}{2}\hat{f}(0) \cdot (\cdot)$.

Theorem 3.3.5 (M. Riesz, periodic form). *For $1 < p < \infty$ there is C_p with $\|\tilde{f}\|_{L^p(\mathbb{T})} \leq C_p \|f\|_{L^p(\mathbb{T})}$ for every trigonometric polynomial f . Consequently P_+ is bounded on $L^p(\mathbb{T})$.*

Idea of the proof. The conjugate function has the same multiplier as the Hilbert transform, read on $\mathbb{Z}$ instead of $\mathbb{R}$. One could redo the whole Calderón–Zygmund theory on the torus — dyadic arcs, a periodic decomposition — and that is the classical route. It is shorter to *transfer*: a bound for a multiplier operator on $\mathbb{R}$ implies the corresponding bound on $\mathbb{T}$ for the restricted multiplier, by a dilation argument due to de Leeuw. We prove the case we need directly.

Proof. Step 1. Transference setup.

Let $m(\xi) = -i \operatorname{sgn} \xi$, so that $\mathcal{H}$ is the multiplier operator T_m on $\mathbb{R}$ and $\sim$ is the multiplier operator with symbol $m|_{\mathbb{Z}}$ on $\mathbb{T}$. Fix a trigonometric polynomial $f = \sum_{|k| \leq N} c_k e_k$ and $\varepsilon > 0$.

Choose $\varphi \in \mathcal{S}(\mathbb{R})$ with $\varphi \geq 0$, $\|\varphi\|_p = 1$, and set $\varphi_\varepsilon(x) = \varepsilon^{1/p} \varphi(\varepsilon x)$, so $\|\varphi_\varepsilon\|_p = 1$ for every ε . Consider

$$F_\varepsilon(x) = f(x) \varphi_\varepsilon(x), \quad x \in \mathbb{R},$$

where f is viewed as a 1-periodic function on $\mathbb{R}$.

Step 2. Localising the multiplier.

Since $\widehat{F_\varepsilon} = \sum_{|k| \leq N} c_k \tau_k \widehat{\varphi_\varepsilon}$ and $\widehat{\varphi_\varepsilon}(\xi) = \varepsilon^{1/p-1} \widehat{\varphi}(\xi/\varepsilon)$ concentrates near $\xi = 0$ as $\varepsilon \rightarrow 0$, the pieces $\tau_k \widehat{\varphi_\varepsilon}$ concentrate near the integers k and become disjointly supported in the limit. On the piece near k , the multiplier m is constant and equal to $m(k)$, provided $k \neq 0$ and ε is small enough that the effective support of $\widehat{\varphi_\varepsilon}$ has width less than 1. Hence

$$T_m F_\varepsilon = \sum_{|k| \leq N} c_k m(k) e_k \varphi_\varepsilon + E_\varepsilon = \tilde{f} \varphi_\varepsilon + E_\varepsilon,$$

where E_ε collects the error from the tails of $\widehat{\varphi}$ and satisfies $\|E_\varepsilon\|_p \rightarrow 0$ as $\varepsilon \rightarrow 0$, because $\widehat{\varphi}$ is Schwartz and the number of terms is finite.

Step 3. Pass to the limit.

By Theorem 3.3.3, $\|T_m F_\varepsilon\|_{L^p(\mathbb{R})} \leq C_p \|F_\varepsilon\|_{L^p(\mathbb{R})}$. Now

$$\|F_\varepsilon\|_{L^p(\mathbb{R})}^p = \int_{\mathbb{R}} |f(x)|^p \varphi_\varepsilon(x)^p dx \longrightarrow \|f\|_{L^p(\mathbb{T})}^p \quad (\varepsilon \rightarrow 0),$$

since φ_ε^p has integral 1, spreads out over a length $1/\varepsilon$, and $|f|^p$ is periodic with mean $\|f\|_{L^p(\mathbb{T})}^p$; this is the standard averaging of a periodic function against a spreading probability density, and follows from dominated convergence after writing the integral as a sum over periods. The identical computation applies to $\|\tilde{f} \varphi_\varepsilon\|_{L^p(\mathbb{R})}$. Combining with Step 2,

$$\begin{aligned} \|\tilde{f}\|_{L^p(\mathbb{T})} &= \lim_{\varepsilon \rightarrow 0} \|\tilde{f} \varphi_\varepsilon\|_{L^p(\mathbb{R})} \\ &\leq \limsup_{\varepsilon \rightarrow 0} (\|T_m F_\varepsilon\|_p + \|E_\varepsilon\|_p) \\ &\leq C_p \|f\|_{L^p(\mathbb{T})}. \end{aligned}$$

□

WORKED EXAMPLE 3.3.2— TIER B **Fourier series converge in L^p**

Volume 2's Pitfall after Theorem 3.13 (du Bois-Reymond) asserted that Fourier series of L^p functions converge in L^p for $1 < p < \infty$, and deferred the proof to this volume. Here it is.

Theorem 3.3.6. *For $1 < p < \infty$ and $f \in L^p(\mathbb{T})$, the partial sums satisfy $\|S_N f - f\|_{L^p(\mathbb{T})} \rightarrow 0$.*

Proof. Step 1: uniform boundedness of S_N . Write $M_a f = e_a f$ for modulation by $e_a(x) = e(ax)$. A direct computation on Fourier coefficients gives

$$S_N = M_N P_+ M_{-N} - M_{-N-1} P_+ M_{N+1},$$

since the first term keeps frequencies $\geq -N$ and the second removes those $\geq N+1$. Modulations are isometries of $L^p(\mathbb{T})$, so by Theorem 3.3.5,

$$\|S_N\|_{L^p \rightarrow L^p} \leq 2\|P_+\|_{L^p \rightarrow L^p} \leq C_p$$

uniformly in N .

Step 2: convergence on a dense class. For a trigonometric polynomial P of degree d , $S_N P = P$ for all $N \geq d$. Trigonometric polynomials are dense in $L^p(\mathbb{T})$ for $p < \infty$ (Theorem 2.3.7 (density of trigonometric polynomials)).

Step 3: combine. Given $f \in L^p$ and $\varepsilon > 0$, choose P with $\|f - P\|_p < \varepsilon$. Then for $N \geq \deg P$,

$$\|S_N f - f\|_p \leq \|S_N(f - P)\|_p + \|P - f\|_p \leq (C_p + 1)\varepsilon. \quad \square$$

The architecture. Uniform boundedness of a family of operators, plus convergence on a dense subclass, gives convergence everywhere. This is the same template as Theorem 1.7.4 (Lebesgue differentiation) (Lebesgue differentiation), where the uniform bound came from a maximal inequality; and it is the template of Carleson's theorem, where the operator is the maximal partial-sum operator $\sup_N |S_N f|$ and the uniform bound is the hard part.

And the contrast with $p = 1, \infty$. Volume 2's du Bois-Reymond example showed the family $\{S_N\}$ is *not* uniformly bounded on $C(\mathbb{T})$, because $\|D_N\|_{L^1} \approx \log N$. Step 1 is precisely what fails there, and Theorem 3.3.5 is precisely what is unavailable at $p = 1$ and $p = \infty$. $\square$

WORKED EXAMPLE 3.3.3— TIER C: RESEARCH TASTE **One direction, and what it costs**

The Hilbert transform acts in one dimension. In $\mathbb{R}^n$ one can apply it in a fixed direction $v \in \mathbb{S}^{n-1}$: define

$$\mathcal{H}_v f(x) = \text{p.v.} \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x - tv)}{t} dt,$$

the Hilbert transform along the line through x in direction v . Its multiplier is $-i \operatorname{sgn}(\xi \cdot v)$, so the associated *half-space multiplier*

$$S_v f = \mathcal{F}^{-1}(\mathbf{1}_{\{\xi \cdot v \geq 0\}} \widehat{f}) = \frac{1}{2}(f + i\mathcal{H}_v f)$$

is bounded on $L^p(\mathbb{R}^n)$ for $1 < p < \infty$, with a constant *independent of v* — by Theorem 3.3.3 applied in the v variable and Fubini in the orthogonal directions.

Why record this. The constant's independence of v looks like a triviality: rotations are isometries of every L^p , so of course the bound does not see the direction. But the statement one actually needs in Chapter 3.8 is *vector-valued*:

$$\left\| \left(\sum_j |S_{v_j} f_j|^2 \right)^{1/2} \right\|_{L^p} \lesssim_p \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_{L^p},$$

with a constant independent of the *collection* of directions $v_1, \dots, v_N$ and of N . That does not follow from rotation invariance, because the directions differ from one term to the next and no single rotation normalises them all. It does follow — from the scalar bound alone — by the Marcinkiewicz–Zygmund theorem, which we prove as Theorem 3.6.1. And the astonishing fact, due to Fefferman, is that the resulting vector-valued inequality is *false* if the half-spaces are replaced by balls, and that the counterexample is a Besicovitch set. That is Chapter 3.8, and the reader now has every ingredient except the argument itself.

WORKED EXAMPLE 3.3.4— TIER C: RESEARCH TASTE The Hilbert transform as the prototype

Every feature of Calderón–Zygmund theory is visible already in the Hilbert transform, and it is worth listing the correspondences before meeting them in general.

(a) *A singular kernel with cancellation.* $\mathcal{H}$ is convolution with $\text{p.v.} \frac{1}{\pi x}$, which is not locally integrable. It is only the oddness — $\frac{1}{x}$ changes sign — that makes the principal value converge. In $\mathbb{R}^n$ the corresponding statement is the cancellation condition on a Calderón–Zygmund kernel, and Worked

Example 3.4.3(c) shows the Riesz transforms inherit it from oddness too.

(b) *A bounded multiplier.* $\widehat{\mathcal{H}f}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi)$, so $\mathcal{H}$ is an isometry on L^2 up to sign. That is the L^2 hypothesis of Theorem 3.2.3, available here for free, and in general the hypothesis one must verify separately.

(c) *Failure at the endpoints.* $\mathcal{H}\mathbf{1}_{[0,1]}(x) = \frac{1}{\pi} \log \left| \frac{x}{x-1} \right|$, which is not integrable at infinity — it decays like $1/x$. So $\mathcal{H}$ does not map $L^1 \rightarrow L^1$, and the substitute is weak type $(1,1)$. Nor does it map $L^\infty \rightarrow L^\infty$: the same example is unbounded near 0 and 1. The substitute there is BMO, which is Theorem 3.7.4.

(d) *What the general theory adds.* Nothing conceptual — only the observation that (a), (b) and the Hörmander condition are the *only* properties used, so any operator with them behaves the same way. That is the content of Theorem 3.2.3, and it is why a single theorem covers the Hilbert transform, the Riesz transforms, and the multipliers of Theorem 3.5.1.

(e) *The one thing the Hilbert transform does not show.* It is a one-dimensional operator, so it cannot exhibit the directional phenomena of Volumes 6 and 9 — there is only one direction in $\mathbb{R}$. The ball multiplier of Chapter 3.8 is where dimension first matters, and it fails for reasons invisible here.

Drills

Drill 3.3.1. Compute $\mathcal{H}$ of the Poisson kernel $P_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$ and identify the answer as the conjugate Poisson kernel.

Drill 3.3.2. Verify the identity $S_N = M_N P_+ M_{-N} - M_{-N-1} P_+ M_{N+1}$ used in Worked Example 3.3.2 by evaluating both sides on e_m .

Drill 3.3.3. Show $\mathcal{H}^2 = -\operatorname{Id}$ on $L^p(\mathbb{R})$ for $1 < p < \infty$, and deduce that $\|\mathcal{H}\|_{L^p \rightarrow L^p} \geq 1$.

Drill 3.3.4. Prove directly, without Theorem 3.2.3, that $\mathcal{H}$ is bounded on $L^4(\mathbb{R})$, by applying Plancherel to $(\mathcal{H}f)^2$ and using $\mathcal{H}(f\mathcal{H}f) = \frac{1}{2}((\mathcal{H}f)^2 - f^2)$. (This is Cotlar's identity; it explains why even exponents were historically the first accessible ones.)

Where we are. The Hilbert transform is the only operator in this volume that can be written down completely: its kernel, its multiplier, and its action on simple functions are all explicit. Everything afterwards is an operator known only through estimates. It is worth returning to Worked Example 3.3.4 whenever a general statement seems opaque, since every

| feature of the general theory can be seen there concretely.

Notes

M. Riesz proved Theorem 3.3.3 and Theorem 3.3.5 in 1927, by complex analysis; the real-variable proof via Theorem 3.2.3 came a quarter-century later and generalises to $\mathbb{R}^n$, which the complex method does not.

The transference argument of Theorem 3.3.5 is de Leeuw's (1965) in general form. Carleson's theorem (1966), that the Fourier series of an L^2 function converges almost everywhere, is not proved in this series; Hunt extended it to L^p , $p > 1$, in 1968.

Chapter 3.4

Riesz Transforms and Elliptic Estimates

The Hilbert transform has no direct analogue in $\mathbb{R}^n$: there is no distinguished direction. What replaces it is a vector of n operators, one for each coordinate, which together carry the same information.

3.4.1 Definition and boundedness

Definition 3.4.1. For $j = 1, \dots, n$ the j -th Riesz transform is

$$\begin{aligned} \mathcal{R}_j f(x) &= \text{p.v.} \, c_n \int_{\mathbb{R}^n} \frac{x_j - y_j}{|x - y|^{n+1}} f(y) \, dy \\ , \quad c_n &= \frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}. \end{aligned}$$

Proposition 3.4.2. $\widehat{\mathcal{R}_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{f}(\xi)$ for $f \in \mathcal{S}(\mathbb{R}^n)$. Consequently $\mathcal{R}_j$ is bounded on L^2 with norm 1, and

$$\sum_{j=1}^n \mathcal{R}_j^2 = -\text{Id}. \quad (3.4.1)$$

Proof. The kernel $K_j(x) = c_n x_j |x|^{-n-1}$ is homogeneous of degree $-n$ and odd, so its principal value is a tempered distribution (the argument of Theorem 3.3.2, Step 1, applies verbatim with the oddness supplying the cancellation). Its Fourier transform is therefore homogeneous of degree 0 and odd. Moreover K_j transforms under rotations like the linear functional $x \mapsto x_j$: for a rotation ρ , $K_j(\rho x) = \sum_k \rho_{jk} K_k(x)$. Since the Fourier transform commutes with rotations

(Theorem 2.4.2 (the Fourier dictionary)(iv)), $\widehat{K_j}$ has the same equivariance, and the only degree-zero odd functions with that property are the multiples of $\xi_j/|\xi|$. So $\widehat{K_j}(\xi) = \gamma_n \xi_j/|\xi|$ for a constant, and evaluating the normalisation — which is the purpose of the constant c_n — gives $\gamma_n = -i$. The L^2 statement is Theorem 2.6.2 (the L^2 multiplier norm), and (3.4.1) is the identity $\sum_j (-i\xi_j/|\xi|)^2 = -\sum_j \xi_j^2/|\xi|^2 = -1$. $\square$

Theorem 3.4.3. *Each $\mathcal{R}_j$ is of weak type $(1, 1)$ and bounded on $L^p(\mathbb{R}^n)$ for $1 < p < \infty$.*

Proof. K_j is smooth away from 0 with $|\nabla K_j(x)| \lesssim_n |x|^{-n-1}$, by direct differentiation of $x_j|x|^{-n-1}$. So Theorem 3.2.2 gives Hörmander's condition, and Theorem 3.4.2 gives the L^2 bound. Apply Theorem 3.2.3. $\square$

3.4.2 The elliptic estimate

Theorem 3.4.4 (Calderón–Zygmund estimate for the Laplacian). *Let $1 < p < \infty$. For every $u \in C_c^\infty(\mathbb{R}^n)$ and all j, k ,*

$$\|\partial_j \partial_k u\|_{L^p(\mathbb{R}^n)} \lesssim_{n,p} \|\Delta u\|_{L^p(\mathbb{R}^n)}.$$

Idea of the proof. On the Fourier side the claim is that the ratio of the symbols, $\xi_j \xi_k / |\xi|^2$, is a bounded multiplier on L^p . That ratio is exactly the symbol of $\mathcal{R}_j \mathcal{R}_k$, so the estimate is not merely implied by Theorem 3.4.3 — it *is* Theorem 3.4.3, restated. The content of the theorem is that a scalar equation controls a full matrix of second derivatives, and the mechanism is that the Riesz transforms invert the passage from Δ to $\partial_j \partial_k$.

Proof. Comparing symbols: $\widehat{\partial_j \partial_k u} = -4\pi^2 \xi_j \xi_k \widehat{u}$, while

$$\widehat{\mathcal{R}_j \mathcal{R}_k \Delta u}(\xi) = \left(-i \frac{\xi_j}{|\xi|}\right) \left(-i \frac{\xi_k}{|\xi|}\right) (-4\pi^2 |\xi|^2) \widehat{u}(\xi) = 4\pi^2 \xi_j \xi_k \widehat{u}(\xi).$$

Hence $\partial_j \partial_k u = -\mathcal{R}_j \mathcal{R}_k \Delta u$ as tempered distributions, and therefore as L^p functions. Applying Theorem 3.4.3 twice,

$$\|\partial_j \partial_k u\|_p \leq \|\mathcal{R}_j\|_{L^p \rightarrow L^p} \|\mathcal{R}_k\|_{L^p \rightarrow L^p} \|\Delta u\|_p. \quad \square$$

WORKED EXAMPLE 3.4.1— TIER B The estimate fails at $p = 1$ and $p = \infty$, and what replaces it

Theorem 3.4.4 is false for $p = 1$ and $p = \infty$, and this is not a defect of the proof.

At $p = \infty$. If $\|D^2u\|_\infty \lesssim \|\Delta u\|_\infty$ held, then a continuous right-hand side would force $u \in C^2$. It does not: the standard counterexample in $\mathbb{R}^2$ is

$$u(x) = (x_1^2 - x_2^2) \log \log \frac{1}{|x|}$$

near the origin, cut off smoothly. One computes that Δu is bounded while $\partial_1^2 u$ is not, the double logarithm being exactly slow enough to survive one Laplacian and not two. This is the same logarithmic failure as in Worked Example 3.3.1, and for the same reason: at the endpoint, singular integrals produce logarithms.

At $p = 1$. Weak type $(1, 1)$ survives, by Theorem 3.4.3, but strong type does not, by the argument of Worked Example 3.2.2 applied to $\mathcal{R}_j$.

What replaces them. At $p = \infty$ the correct statement is $\|D^2u\|_{\text{BMO}} \lesssim \|\Delta u\|_\infty$, which follows from the $L^\infty \rightarrow \text{BMO}$ bound of Chapter 3.7. At $p = 1$ the correct statement uses the Hardy space H^1 in place of L^1 . The pattern — the two endpoints of the L^p scale must be replaced by H^1 and BMO — is universal for singular integrals, and is the reason Chapter 3.7 exists.

WORKED EXAMPLE 3.4.2— TIER C: RESEARCH TASTE **The method of rotations: singular integrals as averages of Hilbert transforms**

Every odd singular integral in $\mathbb{R}^n$ is an average of one-dimensional Hilbert transforms. The identity is elementary and its consequences are not.

Proposition 3.4.5. *Let $\Omega : \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ be odd and integrable, and let T_Ω be the operator with kernel $\Omega(x/|x|)|x|^{-n}$. Then, with $\mathcal{H}_\omega$ the directional Hilbert transform of Worked Example 3.3.3,*

$$T_\Omega f = \frac{\pi}{2} \int_{\mathbb{S}^{n-1}} \Omega(\omega) \mathcal{H}_\omega f \, d\sigma(\omega).$$

Proof. Write the truncated integral in polar coordinates $y = x - r\omega$, $r > 0$, $\omega \in \mathbb{S}^{n-1}$ (Theorem 1.6.11 (polar coordinates)):

$$\int_{|x-y|>\varepsilon} \frac{\Omega\left(\frac{x-y}{|x-y|}\right)}{|x-y|^n} f(y) \, dy = \int_{\mathbb{S}^{n-1}} \Omega(\omega) \int_\varepsilon^\infty \frac{f(x-r\omega)}{r} \, dr \, d\sigma(\omega).$$

Split $\mathbb{S}^{n-1}$ into $\pm\omega$ pairs and use oddness of Ω : replacing ω by $-\omega$ in half the sphere converts $\int_\varepsilon^\infty \frac{f(x-r\omega)}{r} \, dr$ into $-\int_{-\infty}^{-\varepsilon} \frac{f(x-r\omega)}{r} \, dr$ with a compensating sign, so the inner integral becomes $\frac{1}{2} \int_{|r|>\varepsilon} \frac{f(x-r\omega)}{r} \, dr$ over the full sphere. Letting $\varepsilon \rightarrow 0$ gives $\frac{\pi}{2} \mathcal{H}_\omega f$. $\square$

Consequence. Since $\|\mathcal{H}_\omega\|_{L^p \rightarrow L^p} = \|\mathcal{H}\|_{L^p \rightarrow L^p}$ independently of ω (Worked Example 3.3.3), Minkowski's integral inequality gives

$$\|T_\Omega\|_{L^p \rightarrow L^p} \leq \frac{\pi}{2} \|\Omega\|_{L^1(\mathbb{S}^{n-1})} \|\mathcal{H}\|_{L^p \rightarrow L^p},$$

with *no smoothness hypothesis on Ω whatsoever*. Compare Theorem 3.2.2, which needed a gradient bound. The method of rotations trades smoothness for oddness, and when the kernel is odd it is by far the more efficient tool. Taking $\Omega(\omega) = c_n \omega_j$ recovers Theorem 3.4.3.

Why this matters for the series. The proposition writes an n -dimensional operator as a superposition, over directions, of operators each of which sees only one direction. That is precisely the structure of a wave packet decomposition, and precisely the structure that fails to be summable in Fefferman's theorem. In Chapter 3.8 we will have a family of directional operators S_{v_j} , one per direction, and the question will be whether their *square function* is bounded — a genuinely stronger demand than bounding each one and averaging. The answer is no, and the obstruction is that δ -tubes in different directions can be packed into a set of small measure. The method of rotations works because it only ever needs the triangle inequality; everything hard in this series begins when the triangle inequality is too lossy. $\square$

WORKED EXAMPLE 3.4.3— TIER B **Computing with the Riesz transforms**

The Riesz transforms $\mathcal{R}_j$ are the model Calderón–Zygmund operators, and it repays doing three computations with them explicitly.

(a) *They recover second derivatives.* Since $\widehat{\mathcal{R}_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{f}(\xi)$,

$$\begin{aligned} \widehat{\mathcal{R}_j \mathcal{R}_k f}(\xi) &= -\frac{\xi_j \xi_k}{|\xi|^2} \widehat{f}(\xi) \\ &, \quad \text{so} \quad \mathcal{R}_j \mathcal{R}_k (-\Delta) \\ &= \partial_j \partial_k \end{aligned}$$

as Fourier multipliers, both sides having symbol $\xi_j \xi_k$ up to the factor $4\pi^2$. Hence $\|\partial_j \partial_k u\|_{L^p} \lesssim_{n,p} \|\Delta u\|_{L^p}$ for $1 < p < \infty$: *every* second derivative is controlled by the Laplacian alone. This is the Calderón–Zygmund estimate for elliptic equations, and it is false at $p = 1$ and $p = \infty$ precisely because $\mathcal{R}_j$ is not bounded there.

(b) *They sum to the identity.* From (a) with $j = k$ and summing, $\sum_j \mathcal{R}_j^2$ has symbol $-\sum_j \xi_j^2/|\xi|^2 = -1$, so

$$\sum_{j=1}^n \mathcal{R}_j^2 = -I.$$

A useful check on signs, and the reason the Riesz transforms are thought of as the components of a vector-valued singular integral of unit length.

(c) *The kernel.* $\mathcal{R}_j$ is convolution with $K_j(x) = c_n x_j/|x|^{n+1}$, homogeneous of degree $-n$ and odd. Oddness gives the cancellation that makes the principal value converge; homogeneity gives Hörmander's condition, by the computation of Theorem 3.4.4. Both properties are needed, and the Pitfall of §4.2 shows what fails without oddness.

What to take away. The Riesz transforms convert a statement about the Laplacian — a single scalar operator — into statements about all second derivatives. That conversion is why Calderón–Zygmund theory is the analytic engine of elliptic regularity, and it is the reason Volume 4 can characterise Sobolev spaces by L^p norms of $\Delta^{s/2}$ rather than of all derivatives separately.

WORKED EXAMPLE 3.4.4— TIER A **The elliptic estimate, stated plainly**

The single most-used consequence of this volume, outside harmonic analysis itself, is worth isolating.

The statement. For $1 < p < \infty$ and $u \in \mathcal{S}(\mathbb{R}^n)$,

$$\|D^2u\|_{L^p(\mathbb{R}^n)} \lesssim_{n,p} \|\Delta u\|_{L^p(\mathbb{R}^n)},$$

where D^2u is the full Hessian. So control of one scalar quantity, the Laplacian, controls all n^2 second derivatives.

The proof, in one line. $\partial_j \partial_k = \mathcal{R}_j \mathcal{R}_k (-\Delta)$ as multipliers, by Worked Example 3.4.3(a), and each $\mathcal{R}_j$ is bounded on L^p by Theorem 3.2.3.

Why it fails at $p = 1$ and $p = \infty$. Because the Riesz transforms do. Concretely, there are u with Δu bounded and D^2u unbounded — the standard example is $u(x) = x_1 x_2 \log \log \frac{1}{|x|}$ near the origin in $\mathbb{R}^2$, for which Δu stays bounded while $\partial_1 \partial_2 u$ does not. The failure is logarithmic, which is the signature of a BMO endpoint rather than an L^∞ one, consistent with Theorem 3.7.4.

Why analysts care. It is the linear estimate underlying regularity theory for second-order elliptic equations: a solution of $\Delta u = f$ with $f \in L^p$ has two derivatives in L^p . Every nonlinear elliptic argument that proceeds by

linearisation and iteration uses it, and the restriction to $1 < p < \infty$ is exactly why such arguments avoid the endpoints.

Drills

Drill 3.4.1. Check that for $n = 1$ the Riesz transform reduces to the Hilbert transform, and verify $c_1 = 1/\pi$.

Drill 3.4.2. Prove $\|\nabla u\|_{L^p}^2 \lesssim \|u\|_{L^p} \|\Delta u\|_{L^p}$ for $u \in C_c^\infty$, $1 < p < \infty$, using Theorem 3.4.4 and interpolation.

Drill 3.4.3. Verify the rotational equivariance $K_j(\rho x) = \sum_k \rho_{jk} K_k(x)$ used in Theorem 3.4.2, and confirm that the only odd degree-zero homogeneous functions with that equivariance are multiples of $\xi_j/|\xi|$.

Drill 3.4.4. Using (3.4.1), show $\|f\|_p \leq \sum_j \|\mathcal{R}_j f\|_p \cdot \|\mathcal{R}_j\|_{L^p \rightarrow L^p}$ and deduce that the Riesz transforms are jointly injective on L^p .

Notes

The Riesz transforms are due to M. Riesz; Theorem 3.4.4 is the Calderón–Zygmund estimate, and is the reason their 1952 paper is a founding document of elliptic PDE as much as of harmonic analysis. The method of rotations is Calderón and Zygmund’s as well, from 1956.

Chapter 3.5

The Hörmander–Mikhlin Multiplier Theorem

Volume 2 established that a bounded symbol gives a bounded operator on L^2 , and that nothing so clean is available for $p \neq 2$. This chapter supplies the standard sufficient condition, which was stated there and deferred to here.

3.5.1 Statement

Theorem 3.5.1 (Hörmander–Mikhlin). *Let $s = \lfloor n/2 \rfloor + 1$ and let $m \in C^s(\mathbb{R}^n \setminus \{0\})$ satisfy*

$$|\partial^\alpha m(\xi)| \leq B |\xi|^{-|\alpha|} \quad \text{for all } \xi \neq 0 \text{ and } |\alpha| \leq s. \quad (3.5.1)$$

Then T_m is of weak type $(1, 1)$ and bounded on $L^p(\mathbb{R}^n)$ for every $1 < p < \infty$, with norms $\lesssim_{n,p} B$.

Idea of the proof. The L^2 hypothesis of Theorem 3.2.3 is free: the case $\alpha = 0$ of (3.5.1) says $\|m\|_\infty \leq B$. So everything reduces to verifying Hörmander’s condition (3.2.2) for the kernel $K = \mathcal{F}^{-1}m$, and K is not given by a formula.

The device is to decompose m dyadically, $m = \sum_j m_j$ with m_j supported where $|\xi| \approx 2^j$, and to estimate each $K_j = \mathcal{F}^{-1}m_j$ by Plancherel. A function whose Fourier transform lives at frequency 2^j has its mass at spatial scale 2^{-j} , and the smoothness hypothesis (3.5.1) converts into decay of K_j beyond that scale. Specifically, differentiating m_j s times costs 2^{-js} , and by Plancherel that is a bound on $\| |x|^s K_j \|_2$ — so K_j decays like $|x|^{-s}$ in an L^2 -averaged sense, and $s > n/2$ is exactly what makes that integrable. The two regimes of the final summation are then the two regimes one expects: scales finer than $|y|$, where the difference $K_j(x - y) - K_j(x)$ is

estimated by the gradient and gains $2^j|y|$; and scales coarser than $|y|$, where each term is estimated separately and the decay $(2^j|y|)^{n/2-s}$ does the summing. Both series are geometric, and the sum of the two is where the hypothesis $s > n/2$ is spent.

3.5.2 Proof

Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ supported in $\{\frac{1}{2} < |\xi| < 2\}$ with $\sum_{j \in \mathbb{Z}} \psi(\xi/2^j) = 1$ for all $\xi \neq 0$; such ψ exists by the standard partition-of-unity construction. Put

$$m_j(\xi) = m(\xi) \psi(\xi/2^j), \quad K_j = \mathcal{F}^{-1}m_j.$$

Each m_j is smooth with compact support, so $K_j \in \mathcal{S}(\mathbb{R}^n)$.

Lemma 3.5.2 (Scale-by-scale estimates). *For every $j \in \mathbb{Z}$ and every dyadic $R > 0$,*

$$\int_{|x| \approx R} |K_j| \lesssim_n B \min \left\{ (2^j R)^{n/2}, (2^j R)^{n/2-s} \right\}, \quad (3.5.2)$$

$$\int_{|x| \approx R} |\nabla K_j| \lesssim_n B 2^j \min \left\{ (2^j R)^{n/2}, (2^j R)^{n/2-s} \right\}. \quad (3.5.3)$$

In particular $\|K_j\|_1 \lesssim_n B$ uniformly in j .

Proof. Step 1. Derivative bounds on m_j .

By the Leibniz rule, $\partial^\alpha m_j$ is a sum of terms $\partial^\beta m \cdot 2^{-j|\gamma|} (\partial^\gamma \psi)(\xi/2^j)$ with $\beta + \gamma = \alpha$. On the support, $|\xi| \approx 2^j$, so (3.5.1) gives $|\partial^\beta m| \lesssim B 2^{-j|\beta|}$, and altogether

$$|\partial^\alpha m_j(\xi)| \lesssim_{n,\alpha} B 2^{-j|\alpha|}, \quad |\alpha| \leq s, \quad (3.5.4)$$

supported in $|\xi| \approx 2^j$, a set of measure $\approx 2^{jn}$. Hence

$$\|\partial^\alpha m_j\|_{L^2} \lesssim_{n,\alpha} B 2^{-j|\alpha|} 2^{jn/2}. \quad (3.5.5)$$

Step 2. *From derivatives of m_j to weighted L^2 bounds on K_j .*

By Plancherel and Theorem 2.4.2 (the Fourier dictionary)(vi), $\|x^\alpha K_j\|_2 = (2\pi)^{-|\alpha|} \|\partial^\alpha m_j\|_2$. Since $|x|^s \lesssim_{n,s} \sum_{|\alpha|=s} |x^\alpha|$ — expand $|x|^{2s} = (\sum_i x_i^2)^s$ by the multinomial theorem — we get

$$\|K_j\|_2 \lesssim_n B 2^{jn/2}, \quad \| |x|^s K_j \|_2 \lesssim_n B 2^{j(n/2-s)}. \quad (3.5.6)$$

Step 3. *From weighted L^2 to L^1 on an annulus.*

By Cauchy–Schwarz on $\{|x| \approx R\}$, a set of measure $\approx R^n$,

$$\begin{aligned} \int_{|x| \approx R} |K_j| &\leq (R^n)^{1/2} \|K_j\|_2 \lesssim_n B R^{n/2} 2^{jn/2} \\ &= B(2^j R)^{n/2}, \end{aligned}$$

which is the first alternative in (3.5.2). For the second, insert the weight:

$$\begin{aligned} \int_{|x| \approx R} |K_j| &= \int_{|x| \approx R} |x|^{-s} |x|^s |K_j| \\ &\leq \left(\int_{|x| \approx R} |x|^{-2s} \right)^{1/2} \| |x|^s K_j \|_2 \\ &\lesssim_n R^{\frac{n}{2}-s} B 2^{j(\frac{n}{2}-s)}, \end{aligned}$$

using $\int_{|x| \approx R} |x|^{-2s} \approx R^{n-2s}$.

Step 4. *The gradient.*

$\nabla K_j = \mathcal{F}^{-1}(2\pi i \xi m_j)$, and ξm_j satisfies (3.5.4) with an extra factor 2^j , by the same Leibniz computation. So Steps 2–3 apply verbatim and give (3.5.3).

Step 5. *The L^1 bound.*

Summing (3.5.2) over dyadic R , using the first alternative for $R \leq 2^{-j}$ and the second for $R > 2^{-j}$:

$$\|K_j\|_1 \lesssim_n B \sum_{2^j R \leq 1} (2^j R)^{n/2} + B \sum_{2^j R > 1} (2^j R)^{n/2-s} \lesssim_n B,$$

both series being geometric — the first because $n/2 > 0$, the second because $n/2 - s < 0$, which is where $s > n/2$ is used for the first time. $\square$

Proof of Theorem 3.5.1. By (3.5.1) with $\alpha = 0$, $\|m\|_\infty \leq B$, so T_m is bounded on L^2 with norm at most B (Theorem 2.6.2 (the L^2 multiplier norm)). By Theorem 3.2.3 it suffices to verify Hörmander’s condition for $K = \sum_j K_j$, the series converging in $\mathcal{S}'$ since the m_j sum to m .

Fix $y \neq 0$. We estimate $\int_{|x| > 2|y|} |K_j(x-y) - K_j(x)| dx$ in two regimes.

Step 1. *Fine scales: $2^j|y| \leq 1$.*

By the mean value theorem, $|K_j(x-y) - K_j(x)| \leq |y| \sup_{z \in [x-y, x]} |\nabla K_j(z)|$, and for $|x| > 2|y|$ the segment lies in $\{|z| > |x|/2\}$. Hence

$$\int_{|x| > 2|y|} |K_j(x-y) - K_j(x)| dx \lesssim_n |y| \int_{|z| > |y|} |\nabla K_j(z)| dz.$$

Summing (3.5.3) over dyadic $R > |y|$, and using that in this regime $2^j|y| \leq 1$ so both alternatives are relevant,

$$\int_{|z|>|y|} |\nabla K_j| \lesssim_n B 2^j \left(\sum_{|y|<R\leq 2^{-j}} (2^j R)^{n/2} + \sum_{R>2^{-j}} (2^j R)^{n/2-s} \right) \lesssim_n B 2^j,$$

both sums being geometric with ratio bounded away from 1 and largest term $O(1)$. Therefore this piece contributes $\lesssim_n B 2^j|y|$, and

$$\sum_{j: 2^j|y|\leq 1} B 2^j|y| \lesssim_n B,$$

a geometric series in $2^j|y|$ with ratio 2, summed from below up to 1.

Step 2. Coarse scales: $2^j|y| > 1$.

Here we do not use cancellation between the two terms, only that both are supported far from the origin:

$$\begin{aligned} \int_{|x|>2|y|} |K_j(x-y) - K_j(x)| dx &\leq 2 \int_{|z|>|y|} |K_j(z)| dz \\ &\lesssim_n B \sum_{R>|y|} (2^j R)^{n/2-s} \\ &\lesssim_n B (2^j|y|)^{n/2-s}, \end{aligned}$$

the sum being dominated by its first term because $n/2 - s < 0$. Summing over j ,

$$\sum_{j: 2^j|y|>1} B (2^j|y|)^{n/2-s} \lesssim_n B,$$

again geometric, now decaying.

Step 3. Conclude.

Adding, $\int_{|x|\geq 2|y|} |K(x-y) - K(x)| dx \lesssim_n B$ uniformly in y , which is (3.2.2) with $A \lesssim_n B$. Theorem 3.2.3 now gives both conclusions. $\square$

Where we are. The hypothesis $s > n/2$ entered exactly twice, both times to make a geometric series converge: in Step 5 of Theorem 3.5.2 and in Step 2 above. Since s must be an integer for (3.5.4) to make sense as stated, the natural choice is $\lfloor n/2 \rfloor + 1$. Refinements using fractional smoothness — the Marcinkiewicz multiplier theorem, or Hörmander’s original condition phrased with L^2 Sobolev norms of m_j — reduce the requirement to just over $n/2$ derivatives, but nothing in this series needs them.

WORKED EXAMPLE 3.5.1— TIER A Homogeneous multipliers

Let m be homogeneous of degree 0 — that is, $m(t\xi) = m(\xi)$ for $t > 0$ — and smooth on $\mathbb{S}^{n-1}$. Then $\partial^\alpha m$ is homogeneous of degree $-|\alpha|$ and bounded on the unit sphere, so $|\partial^\alpha m(\xi)| \lesssim_\alpha |\xi|^{-|\alpha|}$: condition (3.5.1) holds for every α . Examples: the Riesz transform symbols $-i\xi_j/|\xi|$, recovering Theorem 3.4.3; the symbol $\xi_j\xi_k/|\xi|^2$ of Theorem 3.4.4; and $\text{sgn}(\xi \cdot v)$ — except that this last one is *not* smooth on the sphere, being discontinuous across the hyperplane $\xi \cdot v = 0$. So Theorem 3.5.1 does not apply to the half-space multiplier, and its boundedness (Worked Example 3.3.3) must be obtained by the one-dimensional argument plus Fubini. This is not a technicality: the discontinuity of a characteristic function across a surface is precisely what Chapter 3.8 is about, and whether it is harmless depends entirely on whether the surface is flat.

WORKED EXAMPLE 3.5.2— TIER B The Bochner–Riesz means, and where the difficulty starts

For $\delta > 0$ define the *Bochner–Riesz multiplier*

$$m^\delta(\xi) = (1 - |\xi|^2)_+^\delta,$$

which is the characteristic function of the unit ball smoothed by raising the distance-to-the-boundary to a power δ .

Large δ is easy. If $\delta > \frac{n-1}{2}$ one can show $|\mathcal{F}^{-1}m^\delta(x)| \lesssim (1 + |x|)^{-n-\delta+\frac{n-1}{2}}$, which is integrable, so T_{m^δ} is bounded on every L^p , $1 \leq p \leq \infty$, by Young’s inequality. The exponent comes from the same stationary-phase computation as Theorem 2.8.5 (decay of $\widehat{d\sigma}$) — the boundary of the ball is a sphere, and its curvature produces the decay $|x|^{-(n-1)/2}$.

Small δ is the Bochner–Riesz conjecture, which asserts that T_{m^δ} is bounded on $L^p(\mathbb{R}^n)$ precisely when $\delta > \max\{0, n|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}\}$. It is a theorem for $n = 2$ (Carleson–Sjölin) and open for $n \geq 3$.

And $\delta = 0$ is false, for every $p \neq 2$ and every $n \geq 2$: that is Fefferman’s theorem, Chapter 3.8.

Why Theorem 3.5.1 says nothing here. The multiplier m^δ fails (3.5.1) badly near $|\xi| = 1$: derivatives blow up at the boundary sphere rather than at the origin, and no amount of homogeneity helps. The theorem of this chapter is adapted to singularities at a *point*, and every hard problem in this series concerns singularities along a *hypersurface*. Bochner–Riesz, restriction and Keakeya are all versions of that difficulty, and the equivalences between them are the subject of Volume 6.

WORKED EXAMPLE 3.5.3— TIER A **Checking the Mikhlin condition on examples**
 Theorem 3.5.1 asks for $|\partial^\alpha m(\xi)| \leq B|\xi|^{-|\alpha|}$ up to order $s = \lfloor n/2 \rfloor + 1$. Four multipliers, checked.

(a) $m(\xi) = \operatorname{sgn}(\xi_1)$. Smooth away from the hyperplane $\xi_1 = 0$, but not away from the origin only — the singularity is a whole hyperplane. So the hypothesis fails, and indeed T_m is the Hilbert transform in the first variable tensored with the identity, which *is* bounded, by a separate argument. The condition is sufficient, not necessary.

(b) $m(\xi) = \xi_j/|\xi|$. Homogeneous of degree 0 and smooth away from the origin, so $\partial^\alpha m$ is homogeneous of degree $-|\alpha|$ and the condition holds with B the maximum of $|\partial^\alpha m|$ on the unit sphere. This is the Riesz transform, and homogeneity is doing all the work.

(c) $m(\xi) = |\xi|^{i\gamma}$, $\gamma \in \mathbb{R}$. Also homogeneous of degree 0, and $|\partial^\alpha m| \lesssim (1 + |\gamma|)^{|\alpha|} |\xi|^{-|\alpha|}$. So the condition holds with $B \approx (1 + |\gamma|)^s$ — *polynomial* in γ , which is exactly what an analytic interpolation argument needs.

(d) $m(\xi) = (1 - |\xi|^2)_+^\delta$, the *Bochner–Riesz multiplier*. Not smooth away from the origin: it fails to be C^s across the sphere $|\xi| = 1$ unless δ is large. For $\delta > s$ the condition holds and Theorem 3.5.1 applies — but that is far weaker than Theorem 5.8.4, which uses the kernel directly. The multiplier theorem is blunt here because it ignores the *curvature* of the sphere across which the symbol degenerates.

The pattern. Homogeneous symbols pass automatically; symbols singular along a surface do not. That division is the boundary between Volume 3 and Volume 5, and (d) is the first hint that curvature will have to enter.

Drills

Drill 3.5.1. Verify (3.5.4) in detail for $|\alpha| = 1$, and check the claim that ξm_j satisfies the same bounds with an extra factor 2^j .

Drill 3.5.2. Show that $|x|^s \lesssim_{n,s} \sum_{|\alpha|=s} |x^\alpha|$ for integer $s \geq 0$, by the multinomial theorem.

Drill 3.5.3. Construct ψ with the properties used at the start of §5.2: take $\varphi \in C_c^\infty$ radial with $\varphi = 1$ on $|\xi| \leq 1$ and $\operatorname{supp} \varphi \subseteq \{|\xi| < 2\}$, and set $\psi(\xi) = \varphi(\xi) - \varphi(2\xi)$.

Drill 3.5.4. Show that $m(\xi) = e^{i|\xi|} |\xi|^{-a}$ satisfies (3.5.1) for no a , and explain which derivative first fails.

Where we are. The derivative count $s = \lfloor n/2 \rfloor + 1$ is exactly what Plancherel needs: the proof estimates $\|K_j\|_{L^1}$ by Cauchy–Schwarz against a weight $|x|^s$, and $|x|^{-s}$ is square-integrable at infinity in $\mathbb{R}^n$ precisely when $2s > n$. Any smaller count fails at that step, and the smallest integer with $2s > n$ is $\lfloor n/2 \rfloor + 1$.

Notes

Mikhlin proved the theorem in 1956 under the hypothesis (3.5.1) for $|\alpha| \leq n$; Hörmander reduced this to $\lfloor n/2 \rfloor + 1$ in 1960 by the L^2 -based argument given here. The reduction matters: for large n the difference between n and $n/2$ derivatives is the difference between a usable and an unusable hypothesis.

Chapter 3.6

Littlewood–Paley Theory

Volume 2 proved that the frequency pieces $P_N f$ are exactly orthogonal in L^2 , and observed that the corresponding L^p statement is what the later volumes actually need. This chapter proves it. The mechanism is to randomise the signs of the pieces: for each choice of signs the resulting operator is a Hörmander–Mikhlin multiplier, uniformly, and Khintchine’s inequality converts the average over signs into the square function.

3.6.1 Vector-valued extension

We record first a general principle, which is used twice below and again in Chapter 3.8.

Theorem 3.6.1 (Marcinkiewicz–Zygmund). *Let $1 \leq p < \infty$ and let T be a bounded linear operator on $L^p(\mathbb{R}^n)$. Then for all $f_1, \dots, f_N \in L^p$,*

$$\left\| \left(\sum_{j=1}^N |Tf_j|^2 \right)^{1/2} \right\|_p \lesssim_p \|T\|_{L^p \rightarrow L^p} \left\| \left(\sum_{j=1}^N |f_j|^2 \right)^{1/2} \right\|_p,$$

with a constant depending only on p . The same holds for $N = \infty$.

Idea of the proof. Khintchine’s inequality (Theorem 2.6.5 (Khintchine)) says that for scalars, $(\sum_j |a_j|^2)^{1/2}$ is comparable to the L^p average over random signs of $|\sum_j \varepsilon_j a_j|$. Applying this *pointwise in x* converts a square function into a randomised sum, on which the scalar operator bound may be used directly. The only other ingredient is Fubini, to exchange the x -integral with the average over signs.

Proof. Let r_j be the Rademacher functions on $[0, 1]$ and write $\mathbb{E}_t$ for $\int_0^1 \cdot dt$.

By Khintchine, applied for each fixed x to the scalars $a_j = Tf_j(x)$,

$$\left(\sum_j |Tf_j(x)|^2 \right)^{p/2} \approx_p \mathbb{E}_t \left| \sum_j r_j(t) Tf_j(x) \right|^p.$$

Integrate in x and apply Tonelli:

$$\left\| \left(\sum_j |Tf_j|^2 \right)^{1/2} \right\|_p^p \approx_p \mathbb{E}_t \int_{\mathbb{R}^n} \left| T \left(\sum_j r_j(t) f_j \right) (x) \right|^p dx,$$

using linearity of T to pull it outside the sum. By the hypothesis, the inner integral is at most $\|T\|^p \left\| \sum_j r_j(t) f_j \right\|_p^p$, so

$$\left\| \left(\sum_j |Tf_j|^2 \right)^{1/2} \right\|_p^p \lesssim_p \|T\|^p \mathbb{E}_t \int_{\mathbb{R}^n} \left| \sum_j r_j(t) f_j(x) \right|^p dx.$$

Applying Tonelli and Khintchine once more, this time in the reverse direction, the right-hand side is $\approx_p \|T\|^p \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_p^p$. $\square$

Pitfall: this is a theorem about ℓ^2 , not about ℓ^q

Theorem 3.6.1 is special to the exponent 2 in the square function, because Khintchine’s inequality is. The analogous statement with $(\sum_j |Tf_j|^q)^{1/q}$ is false in general for $q \neq 2$, and the correct substitutes — the Fefferman–Stein vector-valued maximal inequality for $q > 1$, for instance — require real work. Whenever a square function appears in this series, the exponent 2 is doing something.

3.6.2 The square function theorem

Definition 3.6.2. With ψ as in Chapter 3.5, let $\widehat{P_j f} = \psi(\xi/2^j) \widehat{f}$ and define the *Littlewood–Paley square function*

$$Sf = \left(\sum_{j \in \mathbb{Z}} |P_j f|^2 \right)^{1/2}.$$

Theorem 3.6.3. For $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$,

$$\|Sf\|_{L^p} \approx_{n,p} \|f\|_{L^p}.$$

Proof. Step 1. Randomised multipliers are uniformly bounded.

For a choice of signs $\varepsilon = (\varepsilon_j)_{j \in \mathbb{Z}} \in \{-1, 1\}^{\mathbb{Z}}$ put

$$m_\varepsilon(\xi) = \sum_{j \in \mathbb{Z}} \varepsilon_j \psi(\xi/2^j).$$

We claim m_ε satisfies (3.5.1) with a constant B independent of ε . Indeed, for a given $\xi \neq 0$ at most two terms of the sum are nonzero, those with $2^j \approx |\xi|$; and for each,

$$|\partial^\alpha [\psi(\xi/2^j)]| = 2^{-j|\alpha|} |(\partial^\alpha \psi)(\xi/2^j)| \lesssim_\alpha |\xi|^{-|\alpha|},$$

using $2^j \approx |\xi|$ on the support. Summing the at most two terms gives the claim with $B = B(n, \psi)$.

By Theorem 3.5.1, $\|T_{m_\varepsilon}\|_{L^p \rightarrow L^p} \lesssim_{n,p} 1$ uniformly in ε . Note $T_{m_\varepsilon} f = \sum_j \varepsilon_j P_j f$.

Step 2. *The upper bound.*

By Khintchine applied pointwise and Tonelli, exactly as in the proof of Theorem 3.6.1,

$$\begin{aligned} \|\mathcal{S}f\|_p^p &\approx_p \mathbb{E}_t \left\| \sum_j r_j(t) P_j f \right\|_p^p \\ &= \mathbb{E}_t \|T_{m_{\varepsilon(t)}} f\|_p^p \lesssim_{n,p} \|f\|_p^p, \end{aligned}$$

where $\varepsilon(t) = (r_j(t))_j$ and Step 1 was applied for each t .

Step 3. *The lower bound, by duality.*

Let $\tilde{\psi} \in C_c^\infty(\{\frac{1}{4} < |\xi| < 4\})$ equal 1 on $\text{supp } \psi$, and let $\tilde{P}_j$ be the corresponding projections, so that $\tilde{P}_j P_j = P_j$. Since $\sum_j P_j = \text{Id}$ on L^p — the partial sums are the multipliers $\sum_{|j| \leq J} \psi(\xi/2^j)$, which satisfy (3.5.1) uniformly by the argument of Step 1 and converge boundedly pointwise to 1 — we have, for $f \in L^p$ and $g \in L^{p'}$,

$$\int f \bar{g} = \sum_j \int P_j f \bar{g} = \sum_j \int P_j f \overline{\tilde{P}_j g},$$

the last step because $\tilde{P}_j$ is self-adjoint and $\tilde{P}_j P_j = P_j$. By Cauchy–Schwarz in j and then Hölder in x ,

$$\begin{aligned} \left| \int f \bar{g} \right| &\leq \int \mathcal{S}f \cdot \tilde{\mathcal{S}}g \\ &\leq \|\mathcal{S}f\|_p \|\tilde{\mathcal{S}}g\|_{p'} \lesssim_{n,p} \|\mathcal{S}f\|_p \|g\|_{p'}, \end{aligned}$$

where $\tilde{\mathcal{S}}$ is the square function built from $\tilde{P}_j$ and the last step is Step 2 applied at exponent p' — legitimate since $\tilde{\psi}$ has the same structure as ψ , with each ξ in the support of at most a bounded number of the $\tilde{\psi}(\cdot/2^j)$.

Taking the supremum over $\|g\|_{p'} \leq 1$ and using the duality formula (Theorem 1.8.10 (norms by testing)) gives $\|f\|_p \lesssim_{n,p} \|\mathcal{S}f\|_p$. $\square$

Where we are. The theorem says that the L^p norm cannot tell the difference between f and a function whose frequency pieces have had their phases randomised. That is a strong statement — it fails for $p = 1$ and $p = \infty$ — and it is the precise sense in which frequency decomposition is “almost orthogonal” outside L^2 .

Two comparisons are worth making. Volume 2, Prop. 6.10 proved the case $p = 2$, where the statement is an identity and the proof is Plancherel. And Volume 2, Worked Example 8.16 previewed ℓ^2 decoupling, which asserts something superficially similar for pieces localised to *curved* caps rather than annuli. The difference is decisive: here the pieces are separated in frequency by a factor 2, and the sharp cutoffs are harmless because Theorem 3.5.1 applies. In decoupling the caps are adjacent, the multipliers are not Hörmander–Mikhlin, and the truth of the statement depends on curvature. Theorem 3.6.3 is easy for exactly the reason decoupling is hard.

WORKED EXAMPLE 3.6.1— TIER B **Bernstein revisited, and the standard estimate**

Combining Theorem 3.6.3 with Volume 2’s Bernstein inequality gives the estimate used on nearly every page of Volumes 12 and 13.

Proposition 3.6.4. *Let $1 < p < \infty$ and $s \geq 0$. For $f \in L^p(\mathbb{R}^n)$ with all $P_j f$ defined,*

$$\left\| \left(\sum_j 2^{2js} |P_j f|^2 \right)^{1/2} \right\|_p \approx_{n,p,s} \|(-\Delta)^{s/2} f\|_{L^p},$$

where $(-\Delta)^{s/2}$ is the multiplier operator with symbol $(2\pi|\xi|)^s$.

Proof. Apply Theorem 3.6.3 to $g = (-\Delta)^{s/2} f$. Its pieces are $P_j g = T_{\mu_j} P_j f$ where $\mu_j(\xi) = (2\pi|\xi|)^s \tilde{\psi}(\xi/2^j)$; since μ_j is supported where $|\xi| \approx 2^j$ and satisfies $|\partial^\alpha \mu_j| \lesssim 2^{js} |\xi|^{-|\alpha|}$, the rescaled multiplier $2^{-js} \mu_j$ satisfies (3.5.1) uniformly in j . Hence $|P_j g|$ is comparable to $2^{js} |P_j f|$ in the vector-valued sense, by Theorem 3.6.1 applied to the family, and Theorem 3.6.3 converts both sides. $\square$

The slogan. L^p Sobolev norms are square functions with dyadic weights. Volume 2 could define H^s only for $p = 2$, because only there was Plancherel available; Theorem 3.6.3 removes that restriction and is the reason Triebel–Lizorkin and Besov spaces exist. Volume 4 develops them. $\square$

WORKED EXAMPLE 3.6.2— TIER C: RESEARCH TASTE **What Littlewood–Paley cannot do**

It is worth being precise about the limits of this chapter's technique, because the reader will meet decompositions in later volumes that look similar and behave completely differently.

(a) *Lacunary is essential.* Theorem 3.6.3 decomposes frequency space into annuli $|\xi| \approx 2^j$, whose ratios are bounded away from 1. The analogous statement for the *equally spaced* intervals $[j, j+1)$, $j \in \mathbb{Z}$, in dimension one is a theorem — the Littlewood–Paley theorem for intervals of equal length, due to Rubio de Francia — but only as a one-sided inequality, and its proof is substantially harder. The two-sided statement is false.

(b) *In $\mathbb{R}^n$, the shape of the pieces matters.* Suppose one decomposes frequency space into congruent *cubes* of side 1. The corresponding square function satisfies $\|\mathcal{S}f\|_p \lesssim \|f\|_p$ for $2 \leq p < \infty$, again by Rubio de Francia, but the sharp cutoff to a cube is a product of one-dimensional half-space multipliers, and so is bounded on every L^p . Replace the cubes by *balls* and the sharp cutoffs become unbounded, by Chapter 3.8. The difference between a cube and a ball is that the boundary of a cube is flat.

(c) *And in the curved setting, everything changes.* Volume 13's decoupling theorem concerns pieces of a $\delta^{-1/2}$ -neighbourhood of a paraboloid, partitioned into $\delta^{-(n-1)/2}$ caps. These are adjacent, not lacunary; the sharp cutoffs are not Hörmander–Mikhlin multipliers; and the inequality that holds is not $\|\mathcal{S}f\|_p \approx \|f\|_p$ but

$$\left\| \sum_{\theta} f_{\theta} \right\|_p \lesssim_{\varepsilon} \delta^{-\varepsilon} \left(\sum_{\theta} \|f_{\theta}\|_p^2 \right)^{1/2},$$

with the ℓ^2 sum *outside* the norm rather than inside. That is a genuinely weaker conclusion — by Minkowski's inequality the square function form would imply it for $p \geq 2$ — and it is weaker precisely because the caps are adjacent, so no multiplier theorem applies and only the geometry of the dual tubes is available.

The moral. This chapter's technique is: decompose, check the multiplier hypothesis, randomise, apply Khintchine. It works whenever the pieces are separated enough for Theorem 3.5.1 to apply to their randomised sums. The subject of Volumes 12 and 13 begins exactly where that fails.

WORKED EXAMPLE 3.6.3— TIER B **Using Littlewood–Paley: the standard three moves**

Theorem 3.6.3 says $\|f\|_{L^p} \approx \|(\sum_j |\Delta_j f|^2)^{1/2}\|_{L^p}$ for $1 < p < \infty$. Almost every application uses it in one of three ways, and it is worth having them separated.

Move 1: prove an estimate blockwise. To bound $\|Tf\|_{L^p}$, bound each $\|\Delta_j Tf\|_{L^p}$ and reassemble. This works when T nearly commutes with Δ_j — for instance when T is a Fourier multiplier, so that $\Delta_j T = T \Delta_j$ exactly. The proof of Theorem 3.5.1 is this move.

Move 2: replace a sum by a square function. A sum $\sum_j a_j$ of frequency-separated pieces obeys no triangle-inequality improvement in L^p ; but the square function does, by orthogonality at $p = 2$ and by Theorem 3.6.3 for other p . This is what converts an ℓ^2 bound into an L^p bound and is the ancestor of the decoupling inequalities of Volume 13.

Move 3: interpolate against the endpoint. Since Theorem 3.6.3 fails at $p = 1$ and $p = \infty$, an argument needing those endpoints must substitute H^1 and BMO, which is exactly why Volume 4 exists. The square function characterises H^1 (Theorem 4.2.5 is the operator version), and the substitution is standard.

A worked instance of Move 1. Let m be a bounded multiplier with $|\partial^\alpha m| \lesssim |\xi|^{-|\alpha|}$. Then $m\Delta_j$ has symbol supported where $|\xi| \approx 2^j$, and rescaling that block to unit frequency turns the hypothesis into a bound on finitely many derivatives of a fixed bump — uniformly in j . So each block is handled once and the sum reassembles. That rescaling is the only idea in Theorem 3.5.1, and recognising it as Move 1 makes the proof memorable.

Drills

Drill 3.6.1. Verify that $\sum_{|j| \leq J} \psi(\xi/2^j)$ satisfies (3.5.1) uniformly in J , as claimed in Step 3 of Theorem 3.6.3.

Drill 3.6.2. Show that Theorem 3.6.3 fails at $p = \infty$: find $f \in L^\infty(\mathbb{R})$ with $Sf \notin L^\infty$.

Drill 3.6.3. Deduce from Theorem 3.6.1 that if T is bounded on L^p then so is the operator acting on ℓ^2 -valued functions componentwise, with comparable norm. Where is linearity of T used?

Drill 3.6.4. Prove the Minkowski inequality claim in Worked Example 3.6.2(c): for $p \geq 2$, $(\sum_\theta \|f_\theta\|_p^2)^{1/2} \leq \|(\sum_\theta |f_\theta|^2)^{1/2}\|_p$.

Where we are. One structural remark. Theorem 3.6.3 is an equivalence of norms, and both directions are used, but for different purposes.

The direction $\|Sf\|_{L^p} \lesssim \|f\|_{L^p}$ — the square function is no larger — is what lets an estimate be proved blockwise, as in Worked Example 3.6.3 Move 1. The reverse, $\|f\|_{L^p} \lesssim \|Sf\|_{L^p}$, is what lets a bound on the blocks be assembled back into a bound on f . An argument using only one direction is usually doing something simpler than it appears, and it is worth checking which is in play.

Both fail at $p = 1$ and $p = \infty$, which is the recurring theme of this volume and the reason for Volume 4.

Notes

Littlewood and Paley proved the one-dimensional case in the 1930s by complex analysis; the real-variable proof via multipliers and Khintchine is Stein's. Theorem 3.6.1 is from Marcinkiewicz and Zygmund's 1939 paper.

Rubio de Francia's theorem, quoted in Worked Example 3.6.2, is from 1985 and is not proved in this series.

Chapter 3.7

BMO and the John–Nirenberg Inequality

Every theorem of this volume has failed at $p = \infty$. The Hilbert transform of a bounded function need not be bounded (Worked Example 3.3.1); the elliptic estimate fails there (Worked Example 3.4.1); and in each case the failure was logarithmic. This chapter identifies the space in which the failure is exactly accounted for.

3.7.1 Definition and the model example

Definition 3.7.1. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and a cube Q write $\langle f \rangle_Q = |Q|^{-1} \int_Q f$. The BMO *seminorm* is

$$\|f\|_{\text{BMO}} = \sup_Q \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

the supremum over all cubes with sides parallel to the axes, and $\text{BMO}(\mathbb{R}^n) = \{f : \|f\|_{\text{BMO}} < \infty\}$.

Constants have BMO seminorm zero, so $\|\cdot\|_{\text{BMO}}$ is a norm only on BMO modulo constants. Clearly $\|f\|_{\text{BMO}} \leq 2\|f\|_{\infty}$, so $L^{\infty} \subseteq \text{BMO}$.

Proposition 3.7.2. $f(x) = \log|x|$ lies in $\text{BMO}(\mathbb{R}^n)$, and is not in L^{∞} .

Proof. By translation the supremum need only be checked over cubes $Q = Q(c, \ell)$ of centre c and side ℓ ; and by the dilation invariance $\log|tx| = \log|x| + \log t$, which changes f by a constant and so does not change any of the quantities $\int_Q |f - \langle f \rangle_Q|/|Q|$, we may assume $\ell = 1$.

If $|c| \geq 2\sqrt{n}$ then $|x| \approx |c|$ on Q , and $\log|x|$ has gradient $|x|^{-1} \lesssim |c|^{-1}$ there, so the oscillation of f on Q is $\lesssim |c|^{-1} \lesssim 1$.

If $|c| < 2\sqrt{n}$ then $Q \subseteq B(0, R)$ for a fixed $R = R(n)$, and using $\langle f \rangle_Q$ replaced by the constant 0 — legitimate since $|Q|^{-1} \int_Q |f - \langle f \rangle_Q| \leq 2|Q|^{-1} \int_Q |f - a|$ for any constant a — we need only bound $\int_{B(0,R)} |\log|x|| \, dx$, which is finite by polar coordinates since $\int_0^R r^{n-1} |\log r| \, dr < \infty$. $\square$

WORKED EXAMPLE 3.7.1— TIER A BMO is strictly between L^∞ and $\bigcap_p L^p$

Theorem 3.7.2 shows $L^\infty \subsetneq \text{BMO}$. Conversely $\text{BMO} \not\subset L^\infty$ by the same example, and BMO functions are locally in every L^p — a consequence of Theorem 3.7.3 below — but not globally, as $\log|x|$ shows.

Note also what is *not* in BMO: the function $\log|x| \cdot \text{sgn}(x_1)$ on $\mathbb{R}^n$ is not, even though each factor is bounded or in BMO. So BMO is not an algebra, and multiplying by a bounded function does not preserve it. This is a standard trap.

3.7.2 The John–Nirenberg inequality

The definition controls the *mean* oscillation. The following says that this automatically controls the *distribution* of the oscillation, exponentially.

Theorem 3.7.3 (John–Nirenberg). *There are dimensional constants $C, c > 0$ such that for every $f \in \text{BMO}(\mathbb{R}^n)$, every cube Q and every $\lambda > 0$,*

$$m\left(\{x \in Q : |f(x) - \langle f \rangle_Q| > \lambda\}\right) \leq C|Q| \exp\left(-\frac{c\lambda}{\|f\|_{\text{BMO}}}\right).$$

Consequently $f \in L^p_{\text{loc}}$ for every $p < \infty$, with $(|Q|^{-1} \int_Q |f - \langle f \rangle_Q|^p)^{1/p} \lesssim_{n,p} \|f\|_{\text{BMO}}$.

Idea of the proof. Run the Calderón–Zygmund decomposition of Theorem 3.1.6 *inside* Q , on the function $f - \langle f \rangle_Q$, at height $\lambda_0 = 2^{n+1}\|f\|_{\text{BMO}}$. The selected cubes Q_j have $\sum |Q_j| \leq \frac{1}{2}|Q|$ — a definite proportion is discarded — and on each of them the average of f has moved by at most $O(\lambda_0)$ from $\langle f \rangle_Q$. So the set where the oscillation exceeds $2\lambda_0$ is contained in the union of the Q_j , and inside each Q_j the same argument may be repeated. Iterating k times shows: exceeding $k\lambda_0$ requires surviving k rounds, each of which costs a factor $\frac{1}{2}$ in measure. That is exponential decay, and the constant c is $\log 2$ divided by λ_0 .

Proof. Normalise $\|f\|_{\text{BMO}} = 1$ and fix a cube Q ; replacing f by $f - \langle f \rangle_Q$ we may assume $\langle f \rangle_Q = 0$. Set $\lambda_0 = 2^{n+1}$.

Step 1. *One round of decomposition.*

Apply Theorem 3.1.6 to $f\mathbf{1}_Q$ at height λ_0 , using the dyadic subcubes of Q rather than the standard grid — the proof is identical, since the subcubes of Q obtained by repeated bisection satisfy Theorem 3.1.2. We obtain disjoint dyadic subcubes $\{Q_j\}$ of Q with

$$\begin{aligned} \lambda_0 &< \langle |f| \rangle_{Q_j} \\ &\leq 2^n \lambda_0 \\ &, \quad |f| \leq \lambda_0 \text{ a.e. off } \bigcup_j Q_j \\ &, \quad \sum_j |Q_j| \leq \frac{1}{\lambda_0} \int_Q |f|. \end{aligned}$$

Since $\langle f \rangle_Q = 0$ and $\|f\|_{\text{BMO}} = 1$, the last integral is at most $|Q|$, so

$$\sum_j |Q_j| \leq \frac{|Q|}{\lambda_0} = 2^{-n-1} |Q| \leq \frac{1}{2} |Q|. \quad (3.7.1)$$

Also $|\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n \lambda_0$.

Step 2. *Iterate.*

Let $F(\lambda) = \sup_Q |Q|^{-1} m(\{x \in Q : |f - \langle f \rangle_Q| > \lambda\})$, the supremum over all cubes and all f with $\|f\|_{\text{BMO}} \leq 1$. If $\lambda > 2^n \lambda_0$ then, by Step 1, $\{x \in Q : |f| > \lambda\}$ is contained in $\bigcup_j Q_j$ up to a null set, and within Q_j ,

$$|f(x)| > \lambda \implies |f(x) - \langle f \rangle_{Q_j}| > \lambda - 2^n \lambda_0.$$

Therefore

$$\begin{aligned} m(\{x \in Q : |f| > \lambda\}) &\leq \sum_j m(\{x \in Q_j : |f - \langle f \rangle_{Q_j}| > \lambda - 2^n \lambda_0\}) \\ &\leq F(\lambda - 2^n \lambda_0) \sum_j |Q_j|, \end{aligned}$$

and by (3.7.1) this is at most $\frac{1}{2} |Q| F(\lambda - 2^n \lambda_0)$. Taking the supremum over Q and f ,

$$F(\lambda) \leq \frac{1}{2} F(\lambda - \mu), \quad \mu := 2^n \lambda_0 = 2^{2n+1}, \quad (3.7.2)$$

for every $\lambda > \mu$.

Step 3. *Solve the recursion.*

Since $F \leq 1$ always, iterating (3.7.2) k times gives $F(\lambda) \leq 2^{-k}$ whenever

$\lambda > k\mu$. Given $\lambda > 0$, take $k = \lfloor \lambda/\mu \rfloor$; then

$$\begin{aligned} F(\lambda) &\leq 2^{-\lfloor \lambda/\mu \rfloor} \\ &\leq 2 \cdot 2^{-\lambda/\mu} \\ &= 2 \exp\left(-\frac{\lambda \log 2}{\mu}\right), \end{aligned}$$

which is the claim with $C = 2$ and $c = 2^{-2n-1} \log 2$. Undoing the normalisation $\|f\|_{\text{BMO}} = 1$ replaces λ by $\lambda/\|f\|_{\text{BMO}}$.

Step 4. *Local integrability.*

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)),

$$\begin{aligned} \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|^p &= \frac{p}{|Q|} \int_0^\infty \lambda^{p-1} m(\{x \in Q : |f - \langle f \rangle_Q| > \lambda\}) \, d\lambda \\ &\leq Cp \int_0^\infty \lambda^{p-1} e^{-c\lambda} \, d\lambda, \end{aligned}$$

finite for every $p < \infty$, after normalising $\|f\|_{\text{BMO}} = 1$. □

Where we are. Theorem 3.7.3 is the reason BMO is a useful space rather than a definition. It says that the single scalar $\|f\|_{\text{BMO}}$ — an L^1 -type quantity — in fact controls every L^p oscillation, with constants growing only like p . Equivalently, BMO functions are “ L^∞ up to exponentially small exceptional sets”.

Note also where the halving (3.7.1) came from: the threshold λ_0 was chosen *larger* than the BMO norm by the factor 2^{n+1} , so that Theorem 3.1.6(iii) discarded at most half the cube. That freedom to choose the threshold is available because the hypothesis is a supremum over all cubes, and it is the entire trick.

3.7.3 Singular integrals map L^∞ into BMO

Theorem 3.7.4. *Let T satisfy the hypotheses of Theorem 3.2.3. Then for $f \in L^\infty(\mathbb{R}^n)$ with compact support,*

$$\|Tf\|_{\text{BMO}} \lesssim_n (A + B_2) \|f\|_\infty.$$

Proof. Normalise $A + B_2 \leq 1$ and $\|f\|_\infty = 1$, and fix a cube Q with centre c and side ℓ . We must bound $|Q|^{-1} \int_Q |Tf - \langle Tf \rangle_Q|$ by a constant.

Step 1. *Split f at the scale of the cube.*

Let $Q^* = 2\sqrt{n}Q$ be the concentric dilate and write

$$f = \underbrace{f\mathbf{1}_{Q^*}}_{f_1} + \underbrace{f\mathbf{1}_{(Q^*)^c}}_{f_2}.$$

The point of the dilation factor $2\sqrt{n}$ is that every $x \in Q$ is at distance at least $\frac{\sqrt{n}}{2}\ell$ from $\text{supp } f_2$, so on Q the operator T acts on f_2 through its kernel and nothing is singular.

Step 2. *The local piece, by L^2 boundedness.*

By Cauchy–Schwarz on Q and then the L^2 bound,

$$\begin{aligned} \frac{1}{|Q|} \int_Q |Tf_1| &\leq \left(\frac{1}{|Q|} \int_Q |Tf_1|^2 \right)^{1/2} \\ &\leq |Q|^{-1/2} \|Tf_1\|_{L^2} \\ &\leq |Q|^{-1/2} B_2 \|f_1\|_{L^2}. \end{aligned}$$

Now $\|f_1\|_{L^2} \leq \|f\|_\infty |Q^*|^{1/2} = |Q^*|^{1/2}$, and $|Q^*| = (2\sqrt{n})^n |Q|$, so the two volume factors cancel and

$$\frac{1}{|Q|} \int_Q |Tf_1| \leq (2\sqrt{n})^{n/2} B_2 \lesssim_n 1.$$

This is the step that would fail without the L^2 hypothesis, and it is the only place it is used.

Step 3. *The far piece oscillates little, by Hörmander’s condition.*

For $x \in Q$ the kernel representation (3.2.3) applies to f_2 , so

$$Tf_2(x) - Tf_2(c) = \int_{(Q^*)^c} (K(x-y) - K(c-y))f(y) \, dy.$$

Substitute $z = c - y$ and $w = c - x$. For $y \notin Q^*$ and $x \in Q$ we have $|z| > \sqrt{n}\ell$ and $|w| \leq \frac{\sqrt{n}}{2}\ell$, hence $|z| > 2|w|$ — exactly the region in which Hörmander’s condition (3.2.2) applies. Using $\|f\|_\infty \leq 1$,

$$|Tf_2(x) - Tf_2(c)| \leq \int_{|z|>2|w|} |K(z-w) - K(z)| \, dz \leq A \leq 1,$$

uniformly in $x \in Q$.

Step 4. *Combine, using a constant in place of the average.*

Replacing $\langle Tf \rangle_Q$ by any fixed constant costs at most a factor 2 (Drill 7.1); take the constant $Tf_2(c)$. Then

$$\begin{aligned} \frac{1}{|Q|} \int_Q |Tf - \langle Tf \rangle_Q| &\leq \frac{2}{|Q|} \int_Q |Tf - Tf_2(c)| \\ &\leq \frac{2}{|Q|} \int_Q |Tf_1| + \frac{2}{|Q|} \int_Q |Tf_2 - Tf_2(c)|, \end{aligned}$$

and Steps 2 and 3 bound the two terms by $\lesssim_n 1$ each. Taking the supremum over cubes Q gives $\|Tf\|_{\text{BMO}} \lesssim_n 1$, which after undoing the normalisation is the claim. $\square$

WORKED EXAMPLE 3.7.2— TIER B **The endpoint replacements, tabulated**

The following summarises the situation at the two ends of the L^p scale for a Calderón–Zygmund operator T .

Claim	Status	Reference
$T : L^p \rightarrow L^p, 1 < p < \infty$	true	Theorem 3.2.3
$T : L^1 \rightarrow L^1$	false	Worked Example 3.2.2
$T : L^1 \rightarrow L^{1,\infty}$	true	Theorem 3.2.3(i)
$T : L^\infty \rightarrow L^\infty$	false	Worked Example 3.3.1
$T : L^\infty \rightarrow \text{BMO}$	true	Theorem 3.7.4
$T : H^1 \rightarrow L^1$	true	Volume 4

The pattern is that L^1 must be replaced by the Hardy space H^1 on the input side and L^∞ by BMO on the output side, and that these two replacements are dual to one another — the Fefferman duality theorem $(H^1)^* = \text{BMO}$. Volume 4 develops both spaces; here we have proved the half that requires no new machinery.

Note also that Theorem 3.7.4 plus Theorem 3.7.3 recovers the L^p bounds for large p by interpolation, via the Fefferman–Stein sharp maximal function. That route is developed in Volume 4; we obtained the L^p bounds directly from Theorem 3.2.3 instead, which is shorter.

WORKED EXAMPLE 3.7.3— TIER B **Reading John–Nirenberg**

Theorem 3.7.3 says that a BMO function’s distribution decays exponentially on every cube: $|\{x \in Q : |f - \langle f \rangle_Q| > \lambda\}| \lesssim e^{-c\lambda/\|f\|_{\text{BMO}}|Q|}$. Three consequences are used repeatedly and are worth deriving once.

(a) All the BMO_p norms agree. For $1 \leq p < \infty$ define $\|f\|_{\text{BMO}_p} = \sup_Q (\langle |f - \langle f \rangle_Q|^p \rangle_Q)^{1/p}$. Integrating the exponential bound,

$$\langle |f - \langle f \rangle_Q|^p \rangle_Q = \frac{1}{|Q|} \int_0^\infty p\lambda^{p-1} |\{|f - \langle f \rangle_Q| > \lambda\}| d\lambda \lesssim_p \|f\|_{\text{BMO}}^p,$$

so $\|f\|_{\text{BMO}_p} \approx_p \|f\|_{\text{BMO}}$. The case $p = 2$ is the one used in interpolation, and it is not obvious from the definition.

(b) *BMO functions are exponentially integrable.* Taking $\lambda = \varepsilon \|f\|_{\text{BMO}}$ and summing shows $e^{\varepsilon|f|/\|f\|_{\text{BMO}}}$ is locally integrable for small ε . So BMO sits just outside L^∞ — functions may be unbounded, but only logarithmically, and $\log \frac{1}{|x|}$ is the model.

(c) *The self-improving character.* The hypothesis is an L^1 bound on oscillation; the conclusion is exponential. That gain from nothing is the signature of a stopping-time argument, and indeed the proof of Theorem 3.7.3 iterates the Calderón–Zygmund decomposition of Worked Example 3.1.5 at levels 2^k .

The general lesson. A local L^1 condition, iterated across scales, upgrades itself. The same phenomenon appears in Gehring’s lemma for reverse Hölder inequalities and in the A_p theory of weights, neither of which this series treats — but recognising the mechanism makes those literatures accessible.

Drills

Drill 3.7.1. Show that $\|f\|_{\text{BMO}} \approx \sup_Q \inf_{a \in \mathbb{C}} |Q|^{-1} \int_Q |f - a|$, with constants between 1 and 2.

Drill 3.7.2. Show that $\log|x| \cdot \text{sgn}(x_1)$ is not in $\text{BMO}(\mathbb{R}^n)$, by testing on cubes straddling the hyperplane $x_1 = 0$.

Drill 3.7.3. Deduce from Theorem 3.7.3 that if $\|f\|_{\text{BMO}}$ is small enough then $e^{|f|}$ is locally integrable, and find the threshold.

Drill 3.7.4. Verify that $\mathcal{H}1_{[0,1]}$, computed in Worked Example 3.3.1, has finite BMO norm, directly from the formula.

Where we are. BMO is the right endpoint substitute for L^∞ , and the reason is worth stating once: it is the smallest space containing L^∞ that is preserved by Calderón–Zygmund operators and is dual to H^1 . The first property is Theorem 3.7.4; the second is Volume 4, Theorem 4.3.5. Together they make the pair (H^1, BMO) the correct replacement for (L^1, L^∞)

throughout the theory, and every interpolation argument in Volume 4 uses them in that role.

Notes

BMO was introduced by John and Nirenberg in 1961, in the context of elasticity; Theorem 3.7.3 is from that paper. Theorem 3.7.4 is due to Peetre, Spanne and Stein independently. Fefferman’s duality theorem $(H^1)^* = \text{BMO}$ is from 1971 — the same year as the ball multiplier theorem of the next chapter.

Chapter 3.8

The Ball Multiplier

Everything in this volume has been positive: operators are bounded, under verifiable hypotheses, by a machine. This chapter proves a negative theorem, and the proof uses a Besicovitch set. It is the first point in the series where the geometry of Volume 1 and the analysis of Volumes 2 and 3 meet, and it is the historical origin of the entire subject that follows.

3.8.1 The question

Definition 3.8.1. The *ball multiplier* is $T_B f = \mathcal{F}^{-1}(\mathbf{1}_{B(0,1)} \widehat{f})$.

By Theorem 2.6.2 (the L^2 multiplier norm), T_B is bounded on $L^2(\mathbb{R}^n)$ with norm 1. The question, posed by Bochner in the 1930s and open for forty years, is whether it is bounded on L^p for other p .

In dimension one the answer is yes for all $1 < p < \infty$: the multiplier $\mathbf{1}_{[-1,1]}$ is a difference of two half-line multipliers, each a modulated Hilbert transform (Worked Example 3.3.3). The natural guess in higher dimensions — that the sphere behaves like finitely many half-spaces — is false, and spectacularly so.

Theorem 3.8.2 (Fefferman). *For $n \geq 2$, T_B is bounded on $L^p(\mathbb{R}^n)$ only for $p = 2$.*

Idea of the proof. Three steps.

First, a scalar bound on T_B automatically upgrades to a *vector-valued* bound, by Theorem 3.6.1. This is free, and it is the step that makes the whole argument possible: the hypothesis we are trying to contradict becomes much stronger at no cost.

Second, the ball looks locally like a half-space. Zooming in at a boundary point in direction v and rescaling turns $\mathbf{1}_B$ into $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$, so a uniform bound for T_B gives a uniform bound for the half-space multipliers S_v — and, by the first step, a vector-valued bound for a *family* $S_{v_1}, \dots, S_{v_N}$ with constant

independent of the directions. This is Y. Meyer's lemma.

Third, a Besicovitch configuration contradicts that. One takes N tubes R_j , in N different directions, packed by the Perron tree into a set of tiny measure; the operator S_{v_j} applied to $\mathbf{1}_{R_j}$ is bounded below on the translate of R_j two lengths along its own axis, and those translates are disjoint. So the left side of the vector-valued inequality is of size 1 while the right side is a power of the small measure. Letting the measure tend to zero finishes it.

3.8.2 From the ball to half-spaces

Lemma 3.8.3 (Meyer). *Let $1 < p < \infty$ and suppose $\|T_B\|_{L^p \rightarrow L^p} = M < \infty$. Then for every finite family $v_1, \dots, v_N \in \mathbb{S}^{n-1}$ and all $f_1, \dots, f_N \in L^p(\mathbb{R}^n)$,*

$$\left\| \left(\sum_{j=1}^N |S_{v_j} f_j|^2 \right)^{1/2} \right\|_p \lesssim_p M \left\| \left(\sum_{j=1}^N |f_j|^2 \right)^{1/2} \right\|_p,$$

with a constant depending only on p — in particular, independent of N and of the directions.

Proof. Step 1. Half-spaces as limits of balls.

Fix $v \in \mathbb{S}^{n-1}$ and $R > 0$, and let $B_R^v = B(Rv, R)$, the ball of radius R tangent to the hyperplane $\{\xi \cdot v = 0\}$ at the origin from the v side. By translation and dilation on the Fourier side — both of which are isometries of L^p after the corresponding modulation and dilation in x (Volume 2, Prop. 4.1(ii),(iii)) — the multiplier operator with symbol $\mathbf{1}_{B_R^v}$ has L^p norm exactly M , for every R and v .

As $R \rightarrow \infty$, $\mathbf{1}_{B_R^v}(\xi) \rightarrow \mathbf{1}_{\{\xi \cdot v > 0\}}(\xi)$ for every ξ with $\xi \cdot v \neq 0$: a point with $\xi \cdot v > 0$ eventually lies inside B_R^v , since $|\xi - Rv|^2 - R^2 = |\xi|^2 - 2R\xi \cdot v \rightarrow -\infty$; and a point with $\xi \cdot v < 0$ never does, by the same computation with the sign reversed. The exceptional hyperplane is a null set.

Step 2. *Passing to the limit.*

For $f \in \mathcal{S}(\mathbb{R}^n)$ with $\widehat{f}$ compactly supported away from the hyperplane $\{\xi \cdot v = 0\}$, dominated convergence in the Fourier representation gives $T_{B_R^v} f \rightarrow S_v f$ in L^2 , hence along a subsequence a.e. By Fatou, $\|S_v f\|_p \leq \liminf_R \|T_{B_R^v} f\|_p \leq M\|f\|_p$. Such f are dense in L^p , and S_v is bounded on L^p by Worked Example 3.3.3, so $\|S_v\|_{L^p \rightarrow L^p} \leq M$ for every v .

Step 3. *The vector-valued upgrade.*

By Theorem 3.6.1 applied to the single operator T_B , the vector-valued inequality holds for T_B with constant $\lesssim_p M$. Running Steps 1 and 2 with N separate directions — that is, applying the vector-valued inequality for T_B to

the functions g_j obtained from f_j by the modulation and dilation that carry B to $B_R^{v_j}$, and letting $R \rightarrow \infty$ with Fatou — gives the stated inequality for the family $\{S_{v_j}\}$. The constant does not see N or the directions, because Theorem 3.6.1 does not. $\square$

Pitfall: why Step 3 is the crux

It is tempting to think the vector-valued inequality is obvious: each S_{v_j} is bounded with the same constant, so surely one can sum. One cannot. Bounding each term gives $\|(\sum_j |S_{v_j} f_j|^2)^{1/2}\|_p \leq \sum_j \|S_{v_j} f_j\|_p$, which loses a factor $N^{1/2}$ or worse — exactly the loss that Worked Example 2.6.2 (decoupling needs curvature) identified as the difference between a trivial bound and a theorem. The vector-valued statement asserts *no* loss in N , and Theorem 3.6.1 supplies it only because all the operators in question are obtained from a *single* bounded operator by isometries.

3.8.3 The Besicovitch configuration

Lemma 3.8.4 (Besicovitch configuration). *For every $\varepsilon > 0$ there exist $N \in \mathbb{N}$ and rectangles $R_1, \dots, R_N \subseteq \mathbb{R}^2$ — each of dimensions $\ell_j \times \delta_j$ with long axis in direction $v_j \in \mathbb{S}^1$ — such that, writing $\tilde{R}_j = R_j + 2\ell_j v_j$ for the translate of R_j by twice its length along its own axis,*

(i) *the $\tilde{R}_j$ are pairwise disjoint;*

$$(ii) \left| \bigcup_{j=1}^N R_j \right| \leq \varepsilon \sum_{j=1}^N |R_j|.$$

Remark 3.8.5 (Where this is proved). Theorem 3.8.4 is not proved in this volume. It is the one statement Volume 3 assumes, and it is discharged in Volume 9, Theorem 9.2.1, by an explicit construction: the Perron tree of Volume 1 is taken in quantitative form, a rectangle is inscribed in each translated triangle about the lower part of its median, and the translates are shown to separate. The construction is independent of everything in Volumes 3 to 8, so there is no circularity.

Readers who want Theorem 3.8.2 on first reading may take Theorem 3.8.4 on trust; those who want it unconditional now should turn to Volume 9, Chapter 9.2, which depends on nothing but Volume 1.

This is exactly the Perron tree of Theorem 1.3.12 (the Perron tree), in rectangular form: (ii) says the rectangles overlap enormously, while (i) says their translates do not. Property (ii) is immediate from Volume 1 — take δ -tubes around unit segments in δ -separated directions inside a Besicovitch set of measure $< \varepsilon$, so that $\sum_j |R_j| \approx 1$ while $|\bigcup R_j| \lesssim \varepsilon$. Property (i) requires

arranging the translates to fan out, which is a combinatorial refinement of the Perron construction; it is carried out in Volume 9, §9.4, where the tube combinatorics is developed systematically. We use Theorem 3.8.4 as stated.

Lemma 3.8.6 (The half-space multiplier sees the translate). *Let $R \subseteq \mathbb{R}^2$ be a rectangle of dimensions $\ell \times \delta$ with long axis in direction v , and let $\tilde{R} = R + 2\ell v$. Then*

$$|S_v \mathbf{1}_R(x)| \geq \frac{\log(3/2)}{2\pi} \quad \text{for every } x \in \tilde{R}.$$

Proof. Rotations and translations commute with the construction — $S_{\rho v}(f \circ \rho^{-1}) = (S_v f) \circ \rho^{-1}$ for a rotation ρ , since the multiplier $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$ transforms accordingly — so we may take

$$v = e_1, \quad R = [0, \ell] \times [0, \delta], \quad \tilde{R} = [2\ell, 3\ell] \times [0, \delta].$$

The symbol $\mathbf{1}_{\{\xi_1 \geq 0\}}$ depends only on ξ_1 , so S_{e_1} acts in the first variable alone and leaves the second untouched. Since $\mathbf{1}_R(x) = \mathbf{1}_{[0, \ell]}(x_1) \mathbf{1}_{[0, \delta]}(x_2)$ is a product, Fubini gives

$$S_{e_1} \mathbf{1}_R(x) = \mathbf{1}_{[0, \delta]}(x_2) \cdot (S \mathbf{1}_{[0, \ell]})(x_1),$$

where S is the one-dimensional half-line multiplier. By Worked Example 3.3.3, $S = \frac{1}{2}(\text{Id} + i\mathcal{H})$, so for $x_1 \in [2\ell, 3\ell]$ — where $\mathbf{1}_{[0, \ell]}(x_1) = 0$ —

$$(S \mathbf{1}_{[0, \ell]})(x_1) = \frac{i}{2} \mathcal{H} \mathbf{1}_{[0, \ell]}(x_1).$$

By Worked Example 3.3.1, $\mathcal{H} \mathbf{1}_{[0, \ell]}(t) = \frac{1}{\pi} \log \left| \frac{t}{t-\ell} \right|$, and on $[2\ell, 3\ell]$ this is positive and decreasing in t , since $t \mapsto \frac{t}{t-\ell} = 1 + \frac{\ell}{t-\ell}$ is decreasing. Its minimum is therefore attained at $t = 3\ell$:

$$\begin{aligned} \mathcal{H} \mathbf{1}_{[0, \ell]}(x_1) &\geq \frac{1}{\pi} \log \frac{3\ell}{2\ell} \\ &= \frac{\log(3/2)}{\pi} \quad \text{for } x_1 \in [2\ell, 3\ell]. \end{aligned}$$

Since $x \in \tilde{R}$ forces $x_2 \in [0, \delta]$, the indicator factor equals 1, and combining gives $|S_{e_1} \mathbf{1}_R(x)| \geq \frac{1}{2} \cdot \frac{\log(3/2)}{\pi}$. $\square$

Remark 3.8.7. The mechanism deserves a sentence. A sharp frequency cutoff cannot be localised in space: truncating the Fourier transform of $\mathbf{1}_R$ to a half-space produces a function with a logarithmic tail extending along the axis of R , exactly as Worked Example 3.3.1 showed in one dimension. The translate $\tilde{R}$ lies in that tail. So the operator *smears* each rectangle along its own direction, and a Besicovitch configuration is precisely an arrangement in which the rectangles overlap but the smears do not.

3.8.4 Proof of Fefferman's theorem

Proof of Theorem 3.8.2. Suppose T_B is bounded on $L^p(\mathbb{R}^n)$ for some $p \neq 2$. By duality (Theorem 1.8.10 (norms by testing)) — the multiplier is real and even, so T_B is self-adjoint — we may assume $1 < p < 2$. And it suffices to treat $n = 2$: if T_B is bounded on $L^p(\mathbb{R}^n)$ then, by a standard restriction of the multiplier to a two-dimensional subspace (de Leeuw's theorem, or directly by testing on functions of the form $g(x_1, x_2)h(x_3, \dots, x_n)$ and letting h concentrate), the two-dimensional ball multiplier is bounded on $L^p(\mathbb{R}^2)$.

Fix $\varepsilon > 0$ and take the configuration $R_1, \dots, R_N$ of Theorem 3.8.4, normalised so that $\sum_j |R_j| = 1$. Set $f_j = \mathbf{1}_{R_j}$ and let v_j be the long axis direction of R_j .

Step 1. *Lower bound for the left side.*

By Theorem 3.8.6, $|S_{v_j} f_j| \geq c$ on $\tilde{R}_j$, so $\sum_j |S_{v_j} f_j|^2 \geq c^2$ on $\bigcup_j \tilde{R}_j$. Since the $\tilde{R}_j$ are disjoint and $|\tilde{R}_j| = |R_j|$,

$$\begin{aligned} \left\| \left(\sum_j |S_{v_j} f_j|^2 \right)^{1/2} \right\|_p &\geq c \left| \bigcup_j \tilde{R}_j \right|^{1/p} \\ &= c \left(\sum_j |R_j| \right)^{1/p} \\ &= c. \end{aligned}$$

Step 2. *Upper bound for the right side.*

Let $\Omega = \bigcup_j R_j$, so $|\Omega| \leq \varepsilon$ by Theorem 3.8.4(ii). Since $p < 2$, Hölder's inequality with exponents $\frac{2}{p}$ and $\frac{2}{2-p}$ gives

$$\begin{aligned} \int_{\mathbb{R}^2} \left(\sum_j \mathbf{1}_{R_j} \right)^{p/2} &= \int_{\Omega} \left(\sum_j \mathbf{1}_{R_j} \right)^{p/2} \cdot 1 \leq \left(\int_{\mathbb{R}^2} \sum_j \mathbf{1}_{R_j} \right)^{p/2} |\Omega|^{1-\frac{p}{2}} \\ &= |\Omega|^{1-\frac{p}{2}}, \end{aligned}$$

using $\sum_j \int \mathbf{1}_{R_j} = \sum_j |R_j| = 1$. Hence

$$\begin{aligned} \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_p &= \left(\int \left(\sum_j \mathbf{1}_{R_j} \right)^{p/2} \right)^{1/p} \\ &\leq |\Omega|^{\frac{1}{p} - \frac{1}{2}} \\ &\leq \varepsilon^{\frac{1}{p} - \frac{1}{2}}. \end{aligned}$$

Step 3. Contradiction.

Inserting Steps 1 and 2 into Theorem 3.8.3,

$$c \lesssim_p M \varepsilon^{\frac{1}{p}-\frac{1}{2}}.$$

Since $p < 2$ the exponent $\frac{1}{p} - \frac{1}{2}$ is strictly positive, so letting $\varepsilon \rightarrow 0$ forces $c = 0$, a contradiction. $\square$

WORKED EXAMPLE 3.8.1— TIER C: RESEARCH TASTE **What Fefferman’s theorem started**

(a) *The logical shape.* A Besicovitch set was used to *disprove* an analytic statement. Read in the contrapositive, the argument says: if the vector-valued half-space inequality held, then δ -tubes in δ -separated directions could not be packed into a set of small measure. That is a *Keakeya* statement, and the equivalence is not an accident of this proof.

Volume 6 makes it a theorem. The *Keakeya* maximal conjecture of Volume 1, Worked Example 7.20 is implied by the restriction conjecture, and both are implied by the Bochner–Riesz conjecture of Worked Example 3.5.2. So the chain runs

$$\begin{aligned} &\text{Bochner–Riesz} \\ &\implies \text{restriction} \\ &\implies \text{Keakeya,} \end{aligned}$$

and the present chapter is the degenerate case $\delta = 0$ of the first, where the implication is an equivalence and both sides are settled — negatively.

(b) *Why the sphere and not the cube.* Repeat the proof with B replaced by a cube. Step 1 of Theorem 3.8.3 still produces half-space multipliers, but only the n fixed directions normal to the faces, not a continuum. With finitely many directions no Besicovitch configuration is available — one needs tubes pointing in $\approx \delta^{-1}$ different directions — and indeed the cube multiplier *is* bounded on every L^p , $1 < p < \infty$, being a product of half-space multipliers. Curvature of the boundary is what supplies the directions.

(c) *What survives.* The theorem does not say that frequency truncation is useless, only that sharp truncation to a ball is. Smoothing the cutoff — raising the distance to the boundary to a power $\delta > 0$, as in Worked Example 3.5.2 — restores boundedness for p in a range depending on δ , and determining the sharp range is the Bochner–Riesz conjecture, open for $n \geq 3$. Every positive result in that direction is, at bottom, a quantitative version of “tubes in different directions cannot overlap too much”, and that is the subject of Volumes 9 through 13.

(d) *A closing observation.* Theorem 3.8.3 shows that a scalar L^p bound for one operator implies a vector-valued bound for a whole family. Then

Theorem 3.8.4 shows the family bound is false. The Besicovitch set is therefore not being used against a single operator — it is being used against the *simultaneous* behaviour of many operators, one per direction. That is exactly the structure of a wave packet decomposition, where one has one packet per direction and must control their superposition. The reader who found Volume 2, Worked Example 6.9 — decoupling is false without curvature — convincing has already seen this argument once, in its positive form.

Figure 3.8.1 (to be drawn). *A Besicovitch configuration and its translates*

Content required: Six to eight thin rectangles R_j in the plane, in different directions, overlapping heavily so that their union is visibly small; and the translates $\tilde{R}_j = R_j + 2\ell_j v_j$, drawn disjoint and fanning outwards. The contrast between the packed R_j and the spread $\tilde{R}_j$ is the whole content of Theorem 3.8.4.

WORKED EXAMPLE 3.8.2— TIER A **The ball multiplier, and why $p = 2$ is the only survivor**

Theorem 3.8.2 states that T_B , the multiplier with symbol $\mathbf{1}_{|\xi| \leq 1}$, is bounded on $L^p(\mathbb{R}^n)$ only for $p = 2$ when $n \geq 2$. The three ingredients are worth separating.

(a) $p = 2$ works, trivially. By Plancherel, $\|T_B f\|_{L^2} = \|\mathbf{1}_{|\xi| \leq 1} \hat{f}\|_{L^2} \leq \|f\|_{L^2}$. No geometry, no smoothness — boundedness of the symbol suffices, and this is the L^2 hypothesis of Theorem 3.2.3 being available for free.

(b) $n = 1$ works for all $p \in (1, \infty)$. There the “ball” is an interval, and $\mathbf{1}_{[-1,1]}$ is a combination of two modulated Hilbert transforms: $\mathbf{1}_{[-1,1]}(\xi) = \frac{i}{2}(\operatorname{sgn}(\xi + 1) - \operatorname{sgn}(\xi - 1))$. So T_B is bounded wherever $\mathcal{H}$ is, which is Chapter 3.3. Dimension is what breaks this.

(c) $n \geq 2$, $p \neq 2$ fails. The obstruction is the Besicovitch configuration of Theorem 3.8.4: rectangles overlapping so much that their union is tiny, whose translates by twice their length are disjoint. The half-plane multiplier “sees” each rectangle’s translate (Theorem 3.8.6); the disjointness makes the translated pieces orthogonal; and the smallness of the union makes the original pieces cheap. The mismatch is unbounded.

What is really being said. The ball’s boundary is a *curved hypersurface*, and cutting sharply across it is incompatible with L^p for $p \neq 2$. Volume 5’s Bochner–Riesz operators soften the cut by the factor $(1 - |\xi|^2)^\delta$, and Theorem 5.8.4 shows that for $\delta > \frac{n-1}{2}$ the softening is enough. How small δ may be is the Bochner–Riesz conjecture, still open, and by Volume 6, Worked

Example 6.7.2 it implies the restriction conjecture.

WORKED EXAMPLE 3.8.3 — TIER C: RESEARCH TASTE **Why singular integrals precede everything else**

This volume sits between the measure theory of Volume 1 and the oscillatory methods of Volume 5, and its placement is not arbitrary.

(a) *The maximal function is the first non-trivial operator.* Volume 1 proves the Lebesgue differentiation theorem from the Hardy–Littlewood maximal inequality; everything afterwards refines that pattern — control an operator by a geometric covering argument, then interpolate.

(b) *Calderón–Zygmund is the first decomposition.* Splitting a function into a bounded part and mean-zero pieces is the ancestor of every decomposition in the later volumes: broad–narrow in Volume 12, good and bad tubes in Volume 14, and the four-case polychotomy there.

(c) *The L^2 theory comes first for a reason.* Plancherel makes L^2 bounds free, and every later chapter begins by asking what is free at L^2 and what must be paid for elsewhere.

(d) *What this volume does not do.* It proves no estimate depending on curvature — that is Volume 5’s business. The contrast is instructive: the operators here commute with dilations and translations, so their theory is one-dimensional in character, while curvature breaks that invariance and is what makes the restriction problem hard.

(e) *The transferable habit.* Meeting a new operator, ask first whether it is bounded on L^2 , then what its kernel looks like away from the diagonal, then whether a covering argument controls its distribution function. Those three questions organise this entire volume.

Drills

Drill 3.8.1. Verify that the cube multiplier $\mathbf{1}_{[-1,1]^n}$ is bounded on $L^p(\mathbb{R}^n)$ for $1 < p < \infty$, by writing it as a product of $2n$ half-space multipliers.

Drill 3.8.2. Complete the computation in Step 1 of Theorem 3.8.3: show $|\xi - Rv|^2 - R^2 = |\xi|^2 - 2R\xi \cdot v$ and deduce the pointwise convergence claim.

Drill 3.8.3. Show that Step 3 of the proof of Theorem 3.8.2 fails for $p = 2$ by computing the exponent $\frac{1}{p} - \frac{1}{2}$, and explain what this says about the sharpness of the argument.

Drill 3.8.4. Where in the proof is $n \geq 2$ used? Identify the step that has no analogue for $n = 1$ and explain why the one-dimensional ball multiplier is bounded.

Notes

Theorem 3.8.2 is Fefferman's, from 1971, and is one of the shortest famous papers in analysis. The vector-valued upgrade is Marcinkiewicz–Zygmund's theorem; the localisation Theorem 3.8.3 is due to Y. Meyer.

Theorem 3.8.4(i) is the one statement in this volume used without proof. It is a refinement of the Perron tree construction of Theorem 1.3.12 (the Perron tree), and is proved in Volume 9 alongside the rest of the tube combinatorics. Volume 2's table of deferrals promised this chapter for Volume 5; moving it here has cost exactly that one lemma, which seems a fair exchange for placing the theorem immediately after the theory it refutes.

Chapter 3.A

Results Quoted from Volumes 1 and 2

This volume assumes Volumes 1 and 2 and nothing else. Every citation is listed below with its use site.

From Volume 1

Result	Vol. 1	Used in
Lebesgue differentiation	Thm 7.7	Theorem 3.1.6 Step 3 Worked
Hardy–Littlewood maximal theorem	Thm 7.4	Example 3.1.2, Drill 1.2
Polar coordinates	Thm 6.10	Theorem 3.2.2, Theorem 3.7.2, Worked
Layer-cake formula	Wkd Ex 5.5	Example 3.4.2
Dominated convergence	Thm 5.8	Theorem 3.7.3 Step 4 Theorem 3.3.2, Theorem 3.8.3
Fubini–Tonelli	Thms 6.6, 6.7	Theorem 3.2.3 Step 4, Theorem 3.6.1, Theorem 3.8.6
Density of $L^1 \cap L^2$; of C_c^∞	Thm 8.9	Theorem 3.2.3 Step 1 Theorem 3.2.3 Step 6,
L^p duality; norms by testing	Thm 8.11, Cor 8.12	Theorem 3.6.3 Step 3, Theorem 3.8.2
Chebyshev; weak L^p	Prop 8.14	Theorem 3.2.3 Steps 2, 4
Hölder	Thm 8.2	Theorem 3.8.2 Step 2 Worked
Minkowski integral inequality	Thm 8.5	Example 3.4.2
Perron tree	Thm 3.23	Theorem 3.8.4

From Volume 2

Result	Vol. 2	Used in
Plancherel	Thm 4.5	Theorem 3.3.2, Theorem 3.5.2 Step 2
L^2 multiplier norm = $\ m\ _\infty$	Prop 6.2	Theorem 3.3.2, Theorem 3.4.2, Theorem 3.5.1
The Fourier dictionary	Prop 4.1	Theorem 3.3.2, Theorem 3.4.2, Theorem 3.8.3
Marcinkiewicz interpolation	Thm 2.6	Theorem 3.2.3 Step 6, Drill 1.2
Khintchine's inequality	Thm 6.5	Theorem 3.6.1, Theorem 3.6.3 Step 2
Bernstein's inequalities	Thm 6.8	Worked Example 3.6.1
Density of trigonometric polynomials	Cor 3.6	Worked Example 3.3.2
Decay of $\widehat{d\sigma}$	Thm 8.6	Worked Example 3.5.2
Tempered distributions; $\mathcal{F}$ on $\mathcal{S}'$	Thms 5.5, 5.9	Theorem 3.5.1, Theorem 3.4.4

Debts discharged

Volume 2 deferred two results to this volume. Both are now proved: the Hörmander–Mikhlin theorem is Theorem 3.5.1, and the Littlewood–Paley square function theorem for $p \neq 2$ is Theorem 3.6.3.

Debt incurred

Theorem 3.8.4(i) — that the translates of a Besicovitch configuration can be arranged to be pairwise disjoint — is used and not proved here. It is a combinatorial refinement of Theorem 1.3.12 (the Perron tree), and is proved in Volume 9. This is the only unproved statement in Volumes 1–3.

Chapter 3.B

Notation and Index of Results

Additions to the Volume 1–2 conventions

$\mathcal{D}, \mathcal{D}_k, \mathcal{D}^\alpha$	the dyadic grid; generation k ; the shifted grids
$\langle f \rangle_Q$	the average $ Q ^{-1} \int_Q f$
$\mathcal{M}_d$	the dyadic maximal function
$\widehat{Q}$	the dyadic parent of Q ; $Q^* = 2\sqrt{n} Q$ the dilate
$\mathcal{H}, \mathcal{H}_v$	the Hilbert transform; along direction v
S_v	the half-space multiplier $\mathbf{1}_{\{\xi \cdot v \geq 0\}}$
$\mathcal{R}_j$	the j -th Riesz transform
$\widetilde{f}, P_+$	the conjugate function; the Riesz projection
p.v.	principal value
$\mathcal{M}^\#, \text{BMO}$	the sharp maximal function; bounded mean oscillation
$\mathcal{S}$	the Littlewood–Paley square function

Index of the main results

Nesting of dyadic cubes	Theorem 3.1.2
Dyadic maximal inequality	Theorem 3.1.4
The one-third trick	Worked Example 3.1.2
Calderón–Zygmund decomposition	Theorem 3.1.6
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Part 4

Volume 4

Function Spaces: Hardy Spaces, BMO, and the Sobolev Scale

Preface to Volume 4

The problem this volume solves

Volume 3 proved that a Calderón–Zygmund operator is bounded on $L^p(\mathbb{R}^n)$ for every p strictly between 1 and ∞ , and that both endpoints fail. It also showed what happens instead at the top end: such an operator maps L^∞ into BMO. Half the endpoint story was therefore already told.

The other half is missing. At $p = 1$ we proved only a weak-type bound, and a weak-type bound is an awkward object — it is not a norm inequality, it does not dualise, and it does not interpolate as comfortably as one would like. The resolution, found by Stein, Weiss, Fefferman and Coifman in the 1960s and 70s, is that L^1 is simply the wrong space. There is a smaller space, the Hardy space $H^1(\mathbb{R}^n)$, on which singular integrals *are* bounded, and it is the exact dual partner of BMO.

That is the first half of this volume. The second half generalises: once one sees that H^1 and BMO are the honest endpoints of the L^p scale, it becomes natural to ask what the honest endpoints of the *Sobolev* scale are, and the answer is the family of Besov and Triebel–Lizorkin spaces. These are the spaces in which the estimates of Volumes 12 and 13 are naturally phrased.

A word about how to read this

This volume is written for a reader who has worked through Volumes 1–3 and is meeting Hardy spaces for the first time. Two consequences.

First, definitions are given before they are used, and the several competing definitions of H^1 — there are at least four — are introduced one at a time, with the reasons for each. The theorem that they agree is genuinely hard, and we prove it in Chapter 4.4 rather than asserting it early and leaving the reader unsure which definition is in play.

Second, we resist the temptation to state results in their sharpest form. The literature phrases things in terms of H^p for $0 < p \leq 1$, of H^p_{loc} , of weighted variants, and of vector-valued extensions. Almost none of that is needed for

this series. We prove what is used, in the form in which it is used, and say so.

What is assumed

Volumes 1–3, and nothing else. The reader should be comfortable with: the Calderón–Zygmund decomposition and theorem; the dyadic maximal function and its weak $(1, 1)$ bound with constant 1; the Hilbert and Riesz transforms; BMO and the John–Nirenberg inequality; Littlewood–Paley projections and Bernstein’s inequalities; and the interpolation theorems of Volume 2. Appendix A lists every citation with its use site.

The plan

1	The sharp maximal function	BMO becomes a tool
2	Atoms and the Hardy space H^1	the right endpoint at $p = 1$
3	Duality: $(H^1)^* = \text{BMO}$	Fefferman’s theorem
4	The maximal characterisation	all the definitions agree
5	Vector-valued maximal inequalities	Fefferman–Stein
6	Besov and Triebel–Lizorkin spaces	the full scale
7	Embeddings	smoothness against integrability
8	Where these spaces are used	the estimates of Volumes 12–13

Two debts from Volume 3 are discharged here: that Calderón–Zygmund operators map H^1 into L^1 (Theorem 4.2.5), and that $(H^1)^* = \text{BMO}$ (Chapter 4.3).

Chapter 4.1

The Sharp Maximal Function

4.1.1 Why another maximal function

Volume 3 introduced BMO as the space in which the L^∞ endpoint of the Calderón–Zygmund theory lives. So far it has been a destination: we proved that certain operators land in it, and we proved the John–Nirenberg inequality about functions that lie in it. We have not yet used BMO to prove anything.

This chapter changes that. The device is to notice that the quantity inside the definition of the BMO norm,

$$\frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

is an average over a cube — exactly the kind of object the Hardy–Littlewood maximal function is built from. Taking the supremum over cubes containing a point x , instead of over all cubes at once, produces a *function* of x rather than a single number. That function is the sharp maximal function $\mathcal{M}^\# f$, and $\|f\|_{\text{BMO}}$ is just $\|\mathcal{M}^\# f\|_\infty$.

The point of doing this is the following theorem, which is the main result of the chapter:

$$\|f\|_{L^p} \approx \|\mathcal{M}^\# f\|_{L^p}, \quad 1 < p < \infty.$$

The right-hand side involves only the *oscillation* of f — how much f varies on cubes — and none of its size. It is remarkable that the two are comparable, and it is useful, because for many operators the oscillation of Tf is far easier to estimate than Tf itself.

Looking ahead: what this buys

Two things, both used later in this volume.

First, an *interpolation theorem with BMO* (Theorem 4.1.5): an operator bounded on one L^{p_0} and mapping L^∞ into BMO is bounded on every L^p with $p > p_0$. Since Volume 3 proved the $L^\infty \rightarrow \text{BMO}$ bound for every

Calderón–Zygmund operator, this gives a second and quite different route to the L^p theory — one which, as Worked Example 4.1.3 shows, does not need the operator to be a convolution.

Second, and more importantly for the design of this volume, it tells us what to look for at the other end. If BMO is the correct substitute for L^∞ , then by duality there ought to be a correct substitute for L^1 , and it ought to be a space whose dual is BMO. That space is the Hardy space H^1 of Chapter 4.2.

4.1.2 Definition and first properties

Throughout, “cube” means a cube in $\mathbb{R}^n$ with sides parallel to the axes, and $\langle f \rangle_Q = |Q|^{-1} \int_Q f$.

Definition 4.1.1. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ the *sharp maximal function* is

$$\mathcal{M}^\# f(x) = \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q|,$$

the supremum over all cubes containing x . The *dyadic sharp maximal function* $\mathcal{M}_d^\# f$ is defined the same way, with the supremum restricted to dyadic cubes.

Since dyadic cubes are cubes, $\mathcal{M}_d^\# f \leq \mathcal{M}^\# f$ pointwise.

Proposition 4.1.2 (Elementary properties). *Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$.*

- (i) $\|f\|_{\text{BMO}} = \|\mathcal{M}^\# f\|_{L^\infty}$.
- (ii) $\mathcal{M}^\# f \leq 2\mathcal{M}f$ pointwise.
- (iii) $\mathcal{M}^\#(f + g) \leq \mathcal{M}^\# f + \mathcal{M}^\# g$ and $\mathcal{M}^\#(cf) = |c| \mathcal{M}^\# f$.
- (iv) $\mathcal{M}^\# f = 0$ if and only if f is a.e. equal to a constant.
- (v) For any constant c , $\mathcal{M}^\#(f + c) = \mathcal{M}^\# f$.

Proof. (i) is a comparison of two suprema: $\|f\|_{\text{BMO}}$ is the supremum over all cubes of the mean oscillation, and $\|\mathcal{M}^\# f\|_\infty$ is the supremum over x of the supremum over cubes containing x . Every cube contains some point, so the two range over the same collection of numbers.

(ii) For a cube $Q \ni x$,

$$\begin{aligned} \frac{1}{|Q|} \int_Q |f - \langle f \rangle_Q| & \leq \frac{1}{|Q|} \int_Q |f| + |\langle f \rangle_Q| \\ & \leq \frac{2}{|Q|} \int_Q |f| \\ & \leq 2\mathcal{M}f(x), \end{aligned}$$

where the last step uses that Q is a cube containing x and that the Hardy–Littlewood maximal function over cubes is comparable to the one over balls (Worked Example 3.1.2 (the one-third trick)). Taking the supremum over Q gives (ii).

(iii) and (v) are immediate from the definition, using $\langle f + g \rangle_Q = \langle f \rangle_Q + \langle g \rangle_Q$ and, for (v), that adding a constant to f adds it to $\langle f \rangle_Q$ as well, so the difference $f - \langle f \rangle_Q$ is unchanged.

(iv) If $f = c$ a.e. then every mean oscillation vanishes. Conversely if $\mathcal{M}^\# f = 0$ then $f = \langle f \rangle_Q$ a.e. on every cube Q ; applying this to an increasing sequence of cubes and to two overlapping cubes shows all these constants agree, so f is a.e. constant. $\square$

Pitfall: $\mathcal{M}^\#$ does not see the size of f

Property (v) says $\mathcal{M}^\#$ is blind to additive constants, and (iv) says that is the only thing it is blind to. So an inequality of the form $\|f\|_p \lesssim \|\mathcal{M}^\# f\|_p$ cannot possibly be true without some extra hypothesis: take f to be a large constant on $\mathbb{R}^n$, and the left side is infinite while the right is zero.

What saves Theorem 4.1.4 below is the requirement that f already belong to some L^{p_0} with $p_0 < \infty$. That hypothesis rules out constants — the only constant in $L^{p_0}(\mathbb{R}^n)$ is 0 — and, less obviously, it is exactly what makes the level sets $\{\mathcal{M}_d f > \lambda\}$ have finite measure, which the proof needs. Watch for it in Step 1.

WORKED EXAMPLE 4.1.1— TIER A Computing $\mathcal{M}^\#$ for an indicator

Let $n = 1$ and $f = \mathbf{1}_{[0,1]}$. We compare $\mathcal{M}^\# f$ with $\mathcal{M}f$.

Far from the interval. Let $x > 2$ and let $I \ni x$ be an interval. If $I \cap [0, 1] = \emptyset$ then $f \equiv 0$ on I and the mean oscillation is 0. Otherwise I reaches back to 1, so $|I| \geq x - 1$, and

$$\frac{1}{|I|} \int_I |f - \langle f \rangle_I| \leq \frac{2}{|I|} \int_I f \leq \frac{2}{x - 1},$$

since $\int_I f \leq 1$. So $\mathcal{M}^\# f(x) \lesssim 1/x$ for large x — the same decay as $\mathcal{M}f(x) \approx 1/(2x)$, computed in Drill 1.7.1. Neither is integrable at infinity.

Inside the interval. At $x = \frac{1}{2}$, taking $I = [0, 1]$ gives $\langle f \rangle_I = 1$ and $|f - \langle f \rangle_I| = 0$ on I , so that particular cube contributes nothing. Taking $I = [-1, 1]$ gives $\langle f \rangle_I = \frac{1}{2}$ and mean oscillation $\frac{1}{2}$. One checks $\mathcal{M}^\# f(\frac{1}{2}) = \frac{1}{2}$, while $\mathcal{M}f(\frac{1}{2}) = 1$. *The moral.* $\mathcal{M}^\# f \leq 2\mathcal{M}f$, as Theorem 4.1.2(ii) promised, and the two are genuinely comparable here. The next example shows they are not always.

WORKED EXAMPLE 4.1.2— TIER A $\mathcal{M}^\#$ can be much smaller than $\mathcal{M}$

Take $f \equiv 1$ on $\mathbb{R}^n$. Then $\mathcal{M}f \equiv 1$, but $\mathcal{M}^\# f \equiv 0$ by Theorem 4.1.2(iv).

Less trivially, take $f = \log|x|$, which lies in BMO by Volume 3, Proposition 7.2. Then $\mathcal{M}^\# f$ is bounded, by Theorem 4.1.2(i), whereas $\mathcal{M}f(x) = \infty$ for every x , since $\log|y|$ is not locally integrable at infinity — every ball average of $|\log|y||$ over $B(x, r)$ tends to ∞ as $r \rightarrow \infty$.

So the inequality $\mathcal{M}^\# f \leq 2\mathcal{M}f$ can be arbitrarily lossy. What Theorem 4.1.4 will say is that once one integrates to the power p , nothing is lost after all.

4.1.3 The good- λ inequality

The proof of the main theorem rests on a single estimate comparing the level sets of $\mathcal{M}_d f$ and $\mathcal{M}_d^\# f$. Inequalities of this shape are called *good- λ inequalities*, and the name is worth explaining before we state one.

Scratch work. Suppose we want to prove $\|F\|_p \lesssim \|G\|_p$ for two nonnegative functions. By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) this is a statement about the level sets $\{F > \lambda\}$ and $\{G > \lambda\}$. A direct pointwise bound $F \lesssim G$ would suffice but is usually false — it is false here, by Worked Example 4.1.2.

The good- λ method asks for something weaker and cleverer. Fix a large constant $a > 1$ and a small constant $\gamma > 0$, and prove

$$m(\{F > a\lambda, G \leq \gamma\lambda\}) \leq c\gamma m(\{F > \lambda\}). \quad (4.1.1)$$

In words: *the points where F is very large but G is small are a small fraction of the points where F is merely large.* This says nothing pointwise. But after multiplying by λ^{p-1} and integrating, the left side produces $\|F\|_p^p$ (up to the factor a^p) and the right side produces $c\gamma\|F\|_p^p$ — and if γ is chosen small enough that $a^p c\gamma < \frac{1}{2}$, the term can be *absorbed* into the left-hand side, leaving only the contribution of $\{G > \gamma\lambda\}$, which is $\|G\|_p^p$.

The absorption step is the whole trick, and it has a hidden cost: one must know in advance that $\|F\|_p$ is *finite*, or the subtraction $\infty - \infty$ is meaningless. Step 1 below is devoted to arranging that, and readers should not regard it as a formality — Worked Example 4.1.2 shows the theorem is false without a hypothesis of this kind.

Theorem 4.1.3 (Good- λ inequality). *Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ be such that $m(\{\mathcal{M}_d f > \lambda\}) < \infty$ for every $\lambda > 0$. Then for every $\lambda > 0$ and every $\gamma \in (0, 1)$,*

$$\begin{aligned} m\left(\{\mathcal{M}_d f > 2^{n+1}\lambda, \mathcal{M}_d^\# f \leq \gamma\lambda\}\right) \\ \leq 2^{-n}\gamma m\left(\{\mathcal{M}_d f > \lambda\}\right). \end{aligned}$$

Idea of the proof. Decompose the level set $\{\mathcal{M}_d f > \lambda\}$ into the maximal dyadic cubes Q_j on which the average of $|f|$ exceeds λ — this is exactly the selection performed in Theorem 3.1.4 (the dyadic maximal inequality). Work inside one such Q_j .

Two facts are available there. First, because the *parent* of Q_j was not selected, the average of $|f|$ over Q_j is at most $2^n\lambda$; so the constant $\langle f \rangle_{Q_j}$ is of size at most $2^n\lambda$. Second, if the set we are measuring is nonempty, it contains a point where $\mathcal{M}_d^\# f \leq \gamma\lambda$, and that point certifies that f oscillates by at most $\gamma\lambda$ on average over Q_j .

Now, a point of Q_j where $\mathcal{M}_d f$ exceeds $2^{n+1}\lambda$ must see an average of $|f|$ exceeding $2^{n+1}\lambda$ on some dyadic subcube; since $\langle f \rangle_{Q_j}$ is only of size $2^n\lambda$, the *oscillation* of f must exceed $2^n\lambda$ there. So such points lie in a level set of the dyadic maximal function of $f - \langle f \rangle_{Q_j}$, whose measure the weak $(1, 1)$ inequality controls by the second fact. The factor γ appears because that is the size of the oscillation, and it is exactly the small factor (4.1.1) asks for.

Proof. Write $\Omega_\lambda = \{\mathcal{M}_d f > \lambda\}$ and

$$E = \{\mathcal{M}_d f > 2^{n+1}\lambda\} \cap \{\mathcal{M}_d^\# f \leq \gamma\lambda\}.$$

Since $2^{n+1}\lambda > \lambda$, we have $E \subseteq \Omega_\lambda$.

Step 1. *Decompose the level set.*

By the proof of Theorem 3.1.4 (the dyadic maximal inequality), Ω_λ is the disjoint union of the maximal dyadic cubes Q_j satisfying $\langle |f| \rangle_{Q_j} > \lambda$, and the maximality gives, for the dyadic parent $\widehat{Q}_j$,

$$\langle |f| \rangle_{\widehat{Q}_j} \leq \lambda \quad \implies \quad \langle |f| \rangle_{Q_j} \leq 2^n\lambda, \quad (4.1.2)$$

because $|\widehat{Q_j}| = 2^n |Q_j|$ and $Q_j \subseteq \widehat{Q_j}$. Since $E \subseteq \Omega_\lambda = \bigsqcup_j Q_j$, it suffices to prove

$$m(E \cap Q_j) \leq 2^{-n} \gamma |Q_j| \quad \text{for each } j, \quad (4.1.3)$$

and then sum.

Step 2. *Fix a cube and extract the oscillation bound.*

Fix j . If $E \cap Q_j = \emptyset$ then (4.1.3) is trivial, so assume there is $x_0 \in E \cap Q_j$. By definition of E we have $\mathcal{M}_d^\# f(x_0) \leq \gamma \lambda$, and Q_j is a dyadic cube containing x_0 , so this particular cube must satisfy the defining inequality:

$$\frac{1}{|Q_j|} \int_{Q_j} |f - \langle f \rangle_{Q_j}| \leq \gamma \lambda. \quad (4.1.4)$$

This is the only use of the hypothesis $\mathcal{M}_d^\# f \leq \gamma \lambda$, and note how little was needed: one point of $E \cap Q_j$, and one cube.

Step 3. *Points of high maximal function see large oscillation.*

Let $x \in E \cap Q_j$, so $\mathcal{M}_d f(x) > 2^{n+1} \lambda$. By definition there is a dyadic cube $Q \ni x$ with

$$\langle |f| \rangle_Q > 2^{n+1} \lambda. \quad (4.1.5)$$

We claim $Q \subseteq Q_j$. Indeed Q and Q_j both contain x , so by the nesting property (Theorem 3.1.2 (nesting of dyadic cubes)) one contains the other. If $Q \not\subseteq Q_j$ then Q is a dyadic cube strictly containing the maximal cube Q_j , so by maximality $\langle |f| \rangle_Q \leq \lambda$, contradicting (4.1.5). Hence $Q \subseteq Q_j$.

Now set

$$g = (f - \langle f \rangle_{Q_j}) \mathbf{1}_{Q_j},$$

so that g agrees with $f - \langle f \rangle_{Q_j}$ on Q . Using (4.1.5) and (4.1.2),

$$\begin{aligned} \langle |g| \rangle_Q &= \frac{1}{|Q|} \int_Q |f - \langle f \rangle_{Q_j}| \\ &\geq \langle |f| \rangle_Q - |\langle f \rangle_{Q_j}| > 2^{n+1} \lambda - 2^n \lambda \\ &= 2^n \lambda, \end{aligned}$$

where $|\langle f \rangle_{Q_j}| \leq \langle |f| \rangle_{Q_j} \leq 2^n \lambda$. Since Q is a dyadic cube containing x , this says $\mathcal{M}_d g(x) > 2^n \lambda$. Therefore

$$E \cap Q_j \subseteq \{\mathcal{M}_d g > 2^n \lambda\}.$$

Step 4. *Apply the weak $(1, 1)$ inequality.*

By the dyadic maximal inequality with constant 1 (Theorem 3.1.4 (the dyadic maximal inequality)) and then (4.1.4),

$$\begin{aligned} m(E \cap Q_j) &\leq m(\{\mathcal{M}_d g > 2^n \lambda\}) \\ &\leq \frac{\|g\|_{L^1}}{2^n \lambda} \\ &= \frac{1}{2^n \lambda} \int_{Q_j} |f - \langle f \rangle_{Q_j}| \\ &\leq \frac{\gamma \lambda |Q_j|}{2^n \lambda}, \end{aligned}$$

which is (4.1.3). Summing over j and using disjointness completes the proof. $\square$

Where we are. Notice that the constant 1 in the dyadic maximal inequality was genuinely convenient in Step 4: any larger constant would have appeared in the final bound and forced a smaller γ , which is harmless, but the argument is cleanest as it stands. This is the payoff for having proved the dyadic version in Volume 3 rather than relying on Vitali. Notice also that the two constants 2^{n+1} and 2^{-n} are not sacred. All that matters for the absorption argument below is that the first exceeds 1 and the second is finite; any good- λ inequality of the shape (4.1.1) would do.

4.1.4 The Fefferman–Stein inequality

Theorem 4.1.4 (Fefferman–Stein). *Let $1 \leq p_0 \leq p < \infty$ and let $f \in L^{p_0}(\mathbb{R}^n)$. Then*

$$\|f\|_{L^p(\mathbb{R}^n)} \leq \|\mathcal{M}_d f\|_{L^p} \leq C_{n,p} \|\mathcal{M}^\# f\|_{L^p}.$$

In particular, for $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$,

$$\|\mathcal{M}^\# f\|_{L^p} \approx_{n,p} \|f\|_{L^p}.$$

Proof. Step 1. The level sets are finite, and a truncation.

Since $f \in L^{p_0}$ with $p_0 < \infty$, the dyadic maximal function satisfies a weak (p_0, p_0) bound: for $p_0 = 1$ this is Theorem 3.1.4 (the dyadic maximal inequality), and for $p_0 > 1$ it follows from the strong L^{p_0} bound (Theorem 1.7.3 (Hardy–Littlewood maximal inequality), transferred to the dyadic setting) and Chebyshev. Either way

$$m(\{\mathcal{M}_d f > \lambda\}) \leq \frac{C \|f\|_{p_0}^{p_0}}{\lambda^{p_0}} < \infty \quad \text{for every } \lambda > 0, \quad (4.1.6)$$

so Theorem 4.1.3 applies.

The first inequality of the theorem is the pointwise bound $|f| \leq \mathcal{M}_d f$ a.e., which holds by the Lebesgue differentiation theorem applied along the dyadic cubes shrinking to a point (Volume 1, Theorem 7.7).

For the second we cannot integrate directly, for the reason explained in the Scratch box: we must know the quantity being absorbed is finite. So fix $\Lambda > 0$ and define the truncated integral

$$J_\Lambda = p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > a\mu\}) d\mu, \quad a := 2^{n+1}.$$

By (4.1.6) the integrand is at most $C\|f\|_{p_0}^{p_0} a^{-p_0} \mu^{p-1-p_0}$, which is integrable on $(0, \Lambda)$ when $p > p_0$; and when $p = p_0$ the integrand is bounded by $\mu^{p-1} m(\{\mathcal{M}_d f > a\mu\})$, integrable on $(0, \Lambda)$ because $\mathcal{M}_d f \in L^p$ in that case, by the maximal theorem when $p > 1$ and by direct inspection when $p = p_0 = 1$ — in which case the conclusion is void unless $\mathcal{M}^\# f \in L^1$, and we may assume that. In all cases $J_\Lambda < \infty$.

Step 2. *Split the level set and apply the good- λ inequality.*

For each μ ,

$$\{\mathcal{M}_d f > a\mu\} \subseteq \{\mathcal{M}_d f > a\mu, \mathcal{M}_d^\# f \leq \gamma\mu\} \cup \{\mathcal{M}_d^\# f > \gamma\mu\},$$

simply because every point either satisfies $\mathcal{M}_d^\# f \leq \gamma\mu$ or it does not. By Theorem 4.1.3 the first set has measure at most $2^{-n}\gamma m(\{\mathcal{M}_d f > \mu\})$. Hence

$$\begin{aligned} J_\Lambda &\leq 2^{-n}\gamma p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu \\ &\quad + p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d^\# f > \gamma\mu\}) d\mu. \end{aligned}$$

Step 3. *Identify the two terms.*

For the first, substitute $\mu = a\nu$ in the definition of J_Λ to get

$$\begin{aligned} p \int_0^\Lambda \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu &\leq p \int_0^{a\Lambda} \mu^{p-1} m(\{\mathcal{M}_d f > \mu\}) d\mu \\ &= a^p J_\Lambda, \end{aligned}$$

using $a > 1$ so that the range of integration only grows. For the second, the layer-cake formula and the substitution $\mu \mapsto \mu/\gamma$ give

$$\begin{aligned} p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d^\# f > \gamma\mu\}) d\mu &= \gamma^{-p} \|\mathcal{M}_d^\# f\|_{L^p}^p \\ &\leq \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p. \end{aligned}$$

Therefore

$$J_\Lambda \leq 2^{-n} \gamma a^p J_\Lambda + \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p. \quad (4.1.7)$$

Step 4. *Absorb, then remove the truncation.*

Choose γ so small that $2^{-n} \gamma a^p \leq \frac{1}{2}$; explicitly, $\gamma = \min\{\frac{1}{2}, 2^{n-1} a^{-p}\}$. Since $J_\Lambda < \infty$ by Step 1, we may subtract $\frac{1}{2} J_\Lambda$ from both sides of (4.1.7), obtaining

$$J_\Lambda \leq 2\gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p,$$

a bound independent of Λ . Letting $\Lambda \rightarrow \infty$ and using monotone convergence,

$$p \int_0^\infty \mu^{p-1} m(\{\mathcal{M}_d f > a\mu\}) d\mu \leq 2\gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p,$$

and the left side equals $a^{-p} \|\mathcal{M}_d f\|_{L^p}^p$ by the layer-cake formula. Hence $\|\mathcal{M}_d f\|_{L^p}^p \leq 2a^p \gamma^{-p} \|\mathcal{M}^\# f\|_{L^p}^p$.

Step 5. *The reverse inequality.*

For $1 < p < \infty$, $\|\mathcal{M}^\# f\|_p \leq 2\|\mathcal{M} f\|_p \lesssim_{n,p} \|f\|_p$ by Theorem 4.1.2(ii) and the Hardy–Littlewood maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)). $\square$

Pitfall: the hypothesis $f \in L^{p_0}$ cannot be dropped

Take $f \equiv 1$. Then $\mathcal{M}^\# f \equiv 0$, so the right-hand side of Theorem 4.1.4 vanishes while $\|f\|_p = \infty$. The hypothesis $f \in L^{p_0}$ is what excludes this, and it enters the proof at exactly one place: Step 1, to guarantee $J_\Lambda < \infty$ so that the absorption in Step 4 is legitimate.

Readers should treat every good- λ argument they meet with the same suspicion. The finiteness hypothesis is easy to overlook and the conclusion is false without it.

4.1.5 Interpolation with BMO

We can now state the theorem that makes BMO a working tool.

Theorem 4.1.5 (Interpolation between L^{p_0} and BMO). *Let $1 \leq p_0 < \infty$ and let T be a sublinear operator defined on $L^{p_0} \cap L^\infty$ satisfying*

$$\|Tf\|_{L^{p_0}} \leq A\|f\|_{L^{p_0}}, \quad \|Tf\|_{\text{BMO}} \leq B\|f\|_{L^\infty}.$$

Then for every p with $p_0 < p < \infty$,

$$\|Tf\|_{L^p} \lesssim_{n,p,p_0} A^{p_0/p} B^{1-p_0/p} \|f\|_{L^p}.$$

Idea of the proof. This is the Marcinkiewicz argument of Volume 2 with L^∞ replaced by BMO, and the sharp maximal function is what makes the replacement possible. Split f at height λ into a tall part g , which is in L^{p_0} , and a short part b , which is bounded by λ . The hypothesis on L^{p_0} handles g ; the hypothesis on L^∞ says $\|Tb\|_{\text{BMO}} \leq B\lambda$, which by Theorem 4.1.2(i) means $\mathcal{M}^\#(Tb) \leq B\lambda$ *pointwise*. So the sharp maximal function of Tb is under control at every point, which is precisely the kind of statement Marcinkiewicz needs and which BMO alone would not provide. Then Theorem 4.1.4 converts the resulting bound on $\mathcal{M}^\#(Tf)$ back into a bound on Tf .

Proof. By homogeneity assume $A = B = 1$; the general case follows by replacing T with T/B and rescaling f , and tracking the two constants through the computation gives the stated powers.

Fix $\lambda > 0$ and split

$$f = g_\lambda + b_\lambda, \quad g_\lambda = f \mathbf{1}_{\{|f| > \lambda\}}, \quad b_\lambda = f \mathbf{1}_{\{|f| \leq \lambda\}}.$$

Step 1. *Pointwise control of $\mathcal{M}^\#(Tf)$.*

By sublinearity and Theorem 4.1.2(iii),

$$\mathcal{M}^\#(Tf) \leq \mathcal{M}^\#(Tg_\lambda) + \mathcal{M}^\#(Tb_\lambda).$$

Since $\|b_\lambda\|_\infty \leq \lambda$, the second hypothesis gives $\|Tb_\lambda\|_{\text{BMO}} \leq \lambda$, and by Theorem 4.1.2(i) this is exactly $\mathcal{M}^\#(Tb_\lambda) \leq \lambda$ *pointwise*. Hence

$$\{\mathcal{M}^\#(Tf) > 2\lambda\} \subseteq \{\mathcal{M}^\#(Tg_\lambda) > \lambda\} \subseteq \{\mathcal{M}(Tg_\lambda) > \tfrac{\lambda}{2}\},$$

the last inclusion by Theorem 4.1.2(ii).

Step 2. *Estimate the measure.*

By the weak (p_0, p_0) bound for $\mathcal{M}$ — Theorem 1.7.3 (Hardy–Littlewood maximal inequality) for $p_0 = 1$, and Chebyshev with the strong bound for $p_0 > 1$ — together with the first hypothesis,

$$\begin{aligned} m(\{\mathcal{M}^\#(Tf) > 2\lambda\}) &\lesssim_{n,p_0} \frac{\|Tg_\lambda\|_{p_0}^{p_0}}{\lambda^{p_0}} \leq \frac{\|g_\lambda\|_{p_0}^{p_0}}{\lambda^{p_0}} \\ &= \frac{1}{\lambda^{p_0}} \int_{\{|f| > \lambda\}} |f|^{p_0}. \end{aligned}$$

Step 3. *Integrate.*

By the layer-cake formula, the substitution $\lambda \mapsto 2\lambda$, and Step 2,

$$\begin{aligned} \|\mathcal{M}^\#(Tf)\|_p^p &= p \int_0^\infty \lambda^{p-1} m(\{\mathcal{M}^\#(Tf) > \lambda\}) \, d\lambda \\ &\lesssim_{n,p,p_0} \int_0^\infty \lambda^{p-p_0-1} \int_{\{|f| > \lambda\}} |f(x)|^{p_0} \, dx \, d\lambda. \end{aligned}$$

By Tonelli the double integral equals

$$\begin{aligned} \int_{\mathbb{R}^n} |f(x)|^{p_0} \left(\int_0^{|f(x)|} \lambda^{p-p_0-1} d\lambda \right) dx \\ = \frac{1}{p-p_0} \int_{\mathbb{R}^n} |f|^p, \end{aligned}$$

the inner integral converging at 0 exactly because $p > p_0$.

Step 4. Conclude.

So $\|\mathcal{M}^\#(Tf)\|_p \lesssim \|f\|_p$, and since $Tf \in L^{p_0}$ by hypothesis, Theorem 4.1.4 applies and gives $\|Tf\|_p \lesssim \|\mathcal{M}^\#(Tf)\|_p \lesssim \|f\|_p$. $\square$

WORKED EXAMPLE 4.1.3 — TIER C: RESEARCH TASTE **A second proof of the Calderón–Zygmund L^p theorem**

Volume 3 obtained the L^p bounds for a singular integral by proving a weak $(1,1)$ bound and interpolating. Here is an entirely different route, via the sharp maximal function. It is longer for the classical convolution operators, and it is the only one of the two that survives when the operator has no convolution structure — which is why it, and not the Volume 3 argument, is the ancestor of the modern theory.

Proposition 4.1.6. *Let T be bounded on $L^2(\mathbb{R}^n)$ with kernel K satisfying $|\nabla K(x)| \leq B|x|^{-n-1}$, in the sense of Volume 3, Theorem 2.3. Then for every $f \in L^2$ with compact support,*

$$\mathcal{M}^\#(Tf)(x) \lesssim_n (\|T\|_{2 \rightarrow 2} + B) \left(\mathcal{M}(|f|^2)(x) \right)^{1/2} \quad \text{for a.e. } x.$$

Proof. Normalise $\|T\|_{2 \rightarrow 2} + B \leq 1$. Fix x and a cube $Q \ni x$, with centre c and side ℓ , and let $Q^* = 2\sqrt{n}Q$. Split

$$f = f\mathbf{1}_{Q^*} + f\mathbf{1}_{(Q^*)^c} = f_1 + f_2.$$

We must bound $|Q|^{-1} \int_Q |Tf - \alpha|$ for a suitable constant α ; recall from Drill 3.7.1 that using any constant in place of $\langle Tf \rangle_Q$ costs at most a factor 2. Take $\alpha = Tf_2(c)$.

Near part. By Cauchy–Schwarz and the L^2 bound,

$$\begin{aligned} \frac{1}{|Q|} \int_Q |Tf_1| &\leq \left(\frac{1}{|Q|} \int_Q |Tf_1|^2 \right)^{1/2} \\ &\leq |Q|^{-1/2} \|f\mathbf{1}_{Q^*}\|_2 \\ &= \left(\frac{1}{|Q|} \int_{Q^*} |f|^2 \right)^{1/2}. \end{aligned}$$

Since $|Q^*| = (2\sqrt{n})^n |Q|$ and Q^* is a cube containing x , the right-hand side is $\lesssim_n (\mathcal{M}(|f|^2)(x))^{1/2}$.

Far part. For $y \in Q$ and $z \notin Q^*$ we have $|z - c| > \sqrt{n}\ell \geq 2|y - c|$, so the segment from $y - z$ to $c - z$ stays in $\{|w| \geq |c - z|/2\}$ and the mean value theorem gives

$$|K(y - z) - K(c - z)| \leq |y - c| \sup |\nabla K| \lesssim_n \frac{\ell}{|z - c|^{n+1}}.$$

Therefore, decomposing $\{|z - c| > \sqrt{n}\ell\}$ into dyadic annuli $|z - c| \approx 2^k \ell$,

$$\begin{aligned} |Tf_2(y) - Tf_2(c)| &\lesssim_n \sum_{k \geq 0} \frac{\ell}{(2^k \ell)^{n+1}} \int_{|z-c| \lesssim 2^k \ell} |f(z)| \, dz \\ &\lesssim_n \sum_{k \geq 0} 2^{-k} \mathcal{M}f(x) \lesssim_n \mathcal{M}f(x), \end{aligned}$$

using that each annulus is contained in a cube of side $\approx 2^k \ell$ centred at c , which contains x . Finally $\mathcal{M}f \leq (\mathcal{M}(|f|^2))^{1/2}$ by Cauchy–Schwarz on each average. Averaging over $y \in Q$ and combining the two parts, then taking the supremum over $Q \ni x$, gives the claim. $\square$

Consequence: the L^p theorem for $p > 2$. Let $2 < p < \infty$. By Theorem 4.1.4 and Theorem 4.1.6,

$$\begin{aligned} \|Tf\|_p &\lesssim \|\mathcal{M}^\#(Tf)\|_p \\ &\lesssim \|(\mathcal{M}(|f|^2))^{1/2}\|_p \\ &= \|\mathcal{M}(|f|^2)\|_{p/2}^{1/2} \lesssim_{n,p} \| |f|^2 \|_{p/2}^{1/2} \\ &= \|f\|_p, \end{aligned}$$

the last inequality being the Hardy–Littlewood maximal theorem at exponent $p/2 > 1$. The range $1 < p < 2$ then follows by duality, exactly as in Volume 3, Theorem 2.3, Step 6.

Why this proof matters. Three observations.

First, the exponent 2 inside $\mathcal{M}(|f|^2)^{1/2}$ is what forces the restriction $p > 2$; it entered when we used Cauchy–Schwarz to exploit the L^2 boundedness of T on the near part. Using instead a weak $(1, 1)$ bound for T one can replace 2 by any $s > 1$ and reach every $p > 1$ directly, at the cost of a slightly more delicate argument.

Second, nothing in Theorem 4.1.6 used that T is a convolution operator. The kernel was allowed to depend on x and y separately throughout; we wrote

$K(y - z)$ only for notational familiarity. This is the route that generalises to the $T(1)$ theorem, mentioned in Worked Example 3.2.3 (why the L^2 hypothesis is the hard one), where no convolution structure is available and no multiplier exists.

Third, and most relevant to this series: the proof produced a *pointwise* inequality between two maximal functions, and then converted it into a norm inequality by a single application of Theorem 4.1.4. That two-stage structure — prove something pointwise about a cleverly chosen maximal function, then integrate — recurs constantly. In Volume 12, the multilinear restriction estimates are proved as pointwise bounds on a broad maximal function before any integration happens. $\square$

WORKED EXAMPLE 4.1.4— TIER B Using the sharp maximal function

The sharp function $\mathcal{M}^\# f$ measures oscillation rather than size, and the three ways it is used are worth separating.

(a) *To prove an L^p bound.* The Fefferman–Stein inequality $\|f\|_{L^p} \lesssim \|\mathcal{M}^\# f\|_{L^p}$ converts a bound on oscillation into a bound on size, for $1 < p < \infty$. This is the standard route to L^p estimates for operators whose oscillation is controllable but whose size is not.

(b) *To interpolate against BMO.* $\|\mathcal{M}^\# f\|_{L^\infty} = \|f\|_{\text{BMO}}$ by definition, so the pair (L^p, BMO) behaves like (L^p, L^∞) for interpolation purposes — which is exactly what Volume 3’s endpoint failure required.

(c) *To localise.* $\mathcal{M}^\#$ is defined by averages over cubes, so a bound on $\mathcal{M}^\# f$ on a cube gives information about f there and nowhere else. Arguments that need locality use $\mathcal{M}^\#$ where they would otherwise use the Hardy–Littlewood maximal function.

The check to run. $\mathcal{M}^\# f \leq 2\mathcal{M}f$ always, so a bound involving $\mathcal{M}^\#$ is stronger than the same bound with $\mathcal{M}$. If a proof uses $\mathcal{M}^\#$ where $\mathcal{M}$ would do, the extra strength is being wasted — and conversely, a proof that needs $\mathcal{M}^\#$ cannot be simplified by substituting $\mathcal{M}$.

Drills

Drill 4.1.1. Show that in Theorem 4.1.1 one may replace $\langle f \rangle_Q$ by any constant c_Q , at the cost of a factor between 1 and 2: $\frac{1}{2}\mathcal{M}^\# f(x) \leq \sup_{Q \ni x} \inf_c |Q|^{-1} \int_Q |f - c| \leq \mathcal{M}^\# f(x)$.

Drill 4.1.2. Verify the claim in Worked Example 4.1.1 that $\mathcal{M}^\# \mathbf{1}_{[0,1]}(\frac{1}{2}) = \frac{1}{2}$, by considering the intervals $[\frac{1}{2} - t, \frac{1}{2} + t]$ for $t > 0$ and optimising.

Drill 4.1.3. Where in the proof of Theorem 4.1.3 is the hypothesis $\gamma < 1$ used? Show that the conclusion is vacuous for $\gamma \geq 2^n$, and explain why.

Drill 4.1.4. Deduce from Theorem 4.1.5 and Theorem 3.7.4 ($L^\infty \rightarrow \text{BMO}$) that a Calderón–Zygmund operator is bounded on L^p for every $p > 1$, and compare the constants obtained with those of Theorem 3.2.3 (Calderón–Zygmund).

Notes

The sharp maximal function and Theorem 4.1.4 are due to Fefferman and Stein (1972), in the same paper that established the duality of Chapter 4.3. The good- λ technique is Burkholder and Gundy’s, from martingale theory, where inequalities of the shape (4.1.1) first appeared.

Theorem 4.1.6 in the form given here is standard; the refinement replacing $\mathcal{M}(|f|^2)^{1/2}$ by $\mathcal{M}(|f|^s)^{1/s}$ for s close to 1, which yields the full range $1 < p < \infty$ directly, is due to Alvarez and Pérez.

Chapter 4.2

Atoms and the Hardy Space H^1

4.2.1 What goes wrong at $p = 1$, and what to do about it

Volume 3 showed that the Hilbert transform of $\mathbf{1}_{[0,1]}$ decays like $1/|x|$ at infinity, hence is not integrable. The computation was two lines, and it is worth asking what exactly it revealed.

The function $\mathbf{1}_{[0,1]}$ is a perfectly good L^1 function. What it lacks is *cancellation*: its integral is 1, not 0. Recall the far-field estimate from the Calderón–Zygmund proof: for x far from the support of f ,

$$Tf(x) = \int K(x-y)f(y) \, dy \approx K(x) \int f(y) \, dy,$$

because K is nearly constant across the support of f when viewed from far away. So $Tf(x) \approx K(x) \cdot \int f$, and K decays exactly like $|x|^{-n}$, which is exactly not integrable. The obstruction is the number $\int f$, and nothing else.

This suggests the repair. Restrict attention to functions with $\int f = 0$. Then the leading term vanishes and the next one decays like $|x|^{-n-1}$, which *is* integrable at infinity. That is the whole idea of the Hardy space.

Scratch work. One cannot simply define the space to be $\{f \in L^1 : \int f = 0\}$. That set is not closed in L^1 — it is a dense hyperplane — and more importantly the condition is global, whereas the estimate above was local: what mattered was that f had mean zero *on the region where it lives*, so that the kernel could be frozen there.

The right object is therefore built from pieces, each of which is supported in a single cube and has mean zero on that cube. Such a piece is called an *atom*, and the Hardy space is what one gets by summing atoms absolutely.

This definition has a practical virtue that will become apparent immediately: to prove that an operator is bounded on H^1 , it suffices to check it on a single atom, which is a completely explicit object.

4.2.2 Atoms

Definition 4.2.1. An *atom* is a function $a : \mathbb{R}^n \rightarrow \mathbb{C}$ for which there exists a cube Q with

$$(A1) \quad \text{supp } a \subseteq Q;$$

$$(A2) \quad \|a\|_{L^\infty} \leq \frac{1}{|Q|};$$

$$(A3) \quad \int_{\mathbb{R}^n} a(x) \, dx = 0.$$

We say a is an atom *associated with* Q .

The normalisation in (A2) is chosen so that all atoms have comparable L^1 norm:

$$\|a\|_{L^1} = \int_Q |a| \leq |Q| \cdot \frac{1}{|Q|} = 1.$$

So an atom is a bump of height $|Q|^{-1}$ on a cube of volume $|Q|$, with mean zero — a normalised, cancelling lump at one location and one scale.

WORKED EXAMPLE 4.2.1 — TIER A The simplest atoms

(a) *The two-block atom.* In $\mathbb{R}^1$, let $Q = [0, 1]$ and

$$a = \mathbf{1}_{[0, 1/2]} - \mathbf{1}_{(1/2, 1]}.$$

Then $\text{supp } a \subseteq Q$, $\|a\|_\infty = 1 = |Q|^{-1}$, and $\int a = \frac{1}{2} - \frac{1}{2} = 0$. So a is an atom associated with Q .

(b) *Rescaling.* If a is an atom associated with Q and $\lambda > 0$, then

$$a_\lambda(x) = \lambda^{-n} a(x/\lambda)$$

is an atom associated with λQ : the support scales correctly, the sup norm becomes $\lambda^{-n} \|a\|_\infty \leq \lambda^{-n} |Q|^{-1} = |\lambda Q|^{-1}$, and the mean-zero property is preserved by the change of variables. So the family of atoms is invariant under the dilations that preserve the L^1 norm — which is what one wants of an L^1 substitute.

(c) *A smooth atom.* Let $\varphi \in C_c^\infty(Q)$ be any smooth function supported in Q with $\int \varphi = 0$ and $\|\varphi\|_\infty \leq |Q|^{-1}$. Such functions exist in abundance: take any

$\psi \in C_c^\infty(Q)$ and set $\varphi = c(\psi - \eta)$ where $\eta \in C_c^\infty(Q)$ has the same integral as ψ but different shape, then normalise.

(d) *What is not an atom.* $\mathbf{1}_{[0,1]}$ fails (A3). The function $|x|^{-n/2}\mathbf{1}_{B(0,1)}$ fails (A2) — it is unbounded — although it is in L^1 . Both failures matter: the first is the obstruction of §2.1, and the second is what makes the uniform estimate of Theorem 4.2.5 possible.

Lemma 4.2.2 (The key estimate for a single atom). *Let a be an atom associated with a cube Q of centre c , and let K satisfy Hörmander's condition*

$$\int_{|x|>2|y|} |K(x-y) - K(x)| \, dx \leq A \quad \text{for all } y \neq 0$$

of Theorem 3.2.1 (Calderón–Zygmund kernel). Then, writing $Q^ = 2\sqrt{n}Q$ for the concentric dilate,*

$$\int_{(Q^*)^c} \left| \int_Q K(x-y)a(y) \, dy \right| \, dx \leq A.$$

Idea of the proof. This is exactly Step 4 of the Calderón–Zygmund proof (Theorem 3.2.3 (Calderón–Zygmund)), extracted and stated on its own. The mean-zero property lets one replace $K(x-y)$ by $K(x-y) - K(x-c)$ at no cost, and Hörmander's condition then integrates the difference. Isolating it here makes visible that the entire L^1 -endpoint theory is a statement about atoms.

Proof. Since $\int a = 0$, for any x we may subtract the constant $K(x-c)$ inside the integral:

$$\int_Q K(x-y)a(y) \, dy = \int_Q (K(x-y) - K(x-c))a(y) \, dy.$$

Taking absolute values, integrating over $x \notin Q^*$ and applying Tonelli,

$$\begin{aligned} \int_{(Q^*)^c} \left| \int_Q K(x-y)a(y) \, dy \right| \, dx &\leq \int_Q |a(y)| \\ &\quad \times \left(\int_{(Q^*)^c} |K(x-y) - K(x-c)| \, dx \right) \, dy. \end{aligned}$$

Fix $y \in Q$ and substitute $x \mapsto x+c$, $y \mapsto y+c$, so that the inner integral becomes $\int |K(x-y) - K(x)| \, dx$ over the region $\{x+c \notin Q^*\}$. If Q has side ℓ then $|y| \leq \frac{\sqrt{n}}{2}\ell$ for $y \in Q-c$, while $x+c \notin Q^*$ forces $|x| > \sqrt{n}\ell \geq 2|y|$. So the region of integration is contained in $\{|x| > 2|y|\}$, and Hörmander's condition bounds the inner integral by A . Since $\int_Q |a| \leq 1$ by (A2), the result follows. $\square$

Where we are. Compare this with the failure at the start of the chapter. For $f = \mathbf{1}_{[0,1]}$ the analogous integral is infinite, and the computation above shows precisely where the difference lies: the step “subtract $K(x - c)$ at no cost” requires $\int f = 0$. Everything else — Hörmander’s condition, the dilated cube, the Tonelli exchange — is unchanged.

4.2.3 The space H^1

Definition 4.2.3. The *Hardy space* $H^1(\mathbb{R}^n)$ consists of those $f \in L^1(\mathbb{R}^n)$ admitting a decomposition

$$f = \sum_{j=1}^{\infty} \lambda_j a_j, \quad a_j \text{ atoms}, \quad \sum_{j=1}^{\infty} |\lambda_j| < \infty, \quad (4.2.1)$$

with the series converging in $L^1(\mathbb{R}^n)$. We set

$$\|f\|_{H^1} = \inf \left\{ \sum_j |\lambda_j| : \text{all decompositions (4.2.1) of } f \right\}.$$

Proposition 4.2.4 (Basic properties). *(i) The series in (4.2.1) converges absolutely in L^1 , and $\|f\|_{L^1} \leq \|f\|_{H^1}$. Thus $H^1(\mathbb{R}^n) \subseteq L^1(\mathbb{R}^n)$ with continuous inclusion.*

(ii) Every $f \in H^1$ satisfies $\int_{\mathbb{R}^n} f = 0$.

(iii) $\|\cdot\|_{H^1}$ is a norm, and H^1 is complete.

(iv) The inclusion $H^1 \subseteq L^1$ is strict.

Proof. (i) Each atom has $\|a_j\|_1 \leq 1$, so $\sum_j |\lambda_j| \|a_j\|_1 \leq \sum_j |\lambda_j| < \infty$ and the series converges absolutely in the complete space L^1 (Theorem 1.8.6 (Riesz–Fischer)). Its sum has L^1 norm at most $\sum_j |\lambda_j|$, and taking the infimum over decompositions gives the stated inequality.

(ii) The functional $g \mapsto \int g$ is continuous on L^1 and annihilates every atom by (A3), hence annihilates the L^1 -convergent sum.

(iii) Homogeneity and the triangle inequality follow by concatenating decompositions: if $f = \sum \lambda_j a_j$ and $g = \sum \mu_k a'_k$ then the combined family decomposes $f + g$ with coefficient sum $\sum |\lambda_j| + \sum |\mu_k|$. If $\|f\|_{H^1} = 0$ then $\|f\|_{L^1} = 0$ by (i), so $f = 0$. For completeness, suppose $\sum_k \|f_k\|_{H^1} < \infty$; choose for each k a decomposition of f_k with coefficient sum at most $\|f_k\|_{H^1} + 2^{-k}$ and concatenate all of them. The resulting family has finite total coefficient sum and, by (i), its sum in L^1 is $\sum_k f_k$. So absolutely convergent series converge, which is equivalent to completeness (Theorem 1.8.6 (Riesz–Fischer)).

(iv) By (ii), any nonnegative $f \in L^1$ with $f \neq 0$ — for instance $\mathbf{1}_{[0,1]^n}$ — lies in $L^1 \setminus H^1$. $\square$

Pitfall: the mean-zero condition is necessary, not sufficient

Theorem 4.2.4(ii) says $\int f = 0$ for every $f \in H^1$. The converse is false, and it is worth knowing a counterexample.

In $\mathbb{R}^1$ let $f(x) = (\mathbf{1}_{[1,2]}(x) - \mathbf{1}_{[-2,-1]}(x))$, which has mean zero and is in L^1 ; this one *is* in H^1 (it is a multiple of an atom associated with $[-2, 2]$). But consider instead

$$g(x) = \frac{1}{x(\log x)^2} \mathbf{1}_{(2,\infty)}(x) - \frac{1}{|x|(\log|x|)^2} \mathbf{1}_{(-\infty,-2)}(x).$$

Then $g \in L^1(\mathbb{R})$ and $\int g = 0$ by symmetry. But $g \notin H^1$: the mass of g is spread over infinitely many dyadic scales with no cancellation *within any bounded region*, and Chapter 4.4 will give a test — the maximal function characterisation — that detects this. The point for now is that the H^1 condition is genuinely about cancellation at every scale and location, not about one global integral.

4.2.4 Singular integrals on H^1

We can now discharge the first of the two debts from Volume 3.

Theorem 4.2.5 (Calderón–Zygmund operators map H^1 to L^1). *Let T be a linear operator, bounded on $L^2(\mathbb{R}^n)$ with norm B_2 , whose kernel K satisfies Hörmander’s condition with constant A , in the sense of Theorem 3.2.3 (Calderón–Zygmund). Then*

$$\|Tf\|_{L^1(\mathbb{R}^n)} \lesssim_n (A + B_2) \|f\|_{H^1(\mathbb{R}^n)} \quad \text{for all } f \in H^1 \cap L^2.$$

Idea of the proof. The bulk of the work is a bound for a *single atom*, with a constant independent of which atom: for one atom associated with a cube Q , split $\mathbb{R}^n$ into the dilate Q^* and its complement, use Cauchy–Schwarz and the L^2 bound near the cube, and Theorem 4.2.2 far from it.

But it must be stressed at the outset that the atomic bound does *not* by itself give the theorem. The step from “ $\|Ta\|_{L^1} \leq C$ for every atom” to “ T is bounded on H^1 ” is *invalid in general* — see the Pitfall below — and Step 1 of the proof supplies a genuine limiting argument in its place, using that T is a priori bounded on L^2 and of weak type $(1, 1)$.

For one atom associated with a cube Q , split $\mathbb{R}^n$ into the dilate Q^* and its complement. Near the cube, use Cauchy–Schwarz and the L^2 bound; the normalisation (A2) is exactly what makes $\|a\|_2$ small enough for this to

work. Far from the cube, use Theorem 4.2.2, which is where the mean-zero property (A3) is spent.

Proof. Step 1. *Reduction to a single atom.*

Suppose we have shown $\|Ta\|_{L^1} \leq C$ for every atom a , with C depending only on n , A , B_2 . Let $f \in H^1 \cap L^2$ and let $f = \sum_j \lambda_j a_j$ be any atomic decomposition. The partial sums converge to f in L^1 ; but T is a priori defined and bounded on L^2 , so to pass T through the sum we argue as follows. Each partial sum $f_N = \sum_{j \leq N} \lambda_j a_j$ lies in L^2 (atoms are bounded with compact support), and by linearity and the assumed bound,

$$\|Tf_N\|_{L^1} \leq \sum_{j \leq N} |\lambda_j| \|Ta_j\|_{L^1} \leq C \sum_j |\lambda_j|.$$

Moreover $f_N \rightarrow f$ in L^1 and, since T is of weak type $(1, 1)$ by Theorem 3.2.3 (Calderón–Zygmund), $Tf_N \rightarrow Tf$ in measure; passing to a subsequence converging a.e. and applying Fatou gives $\|Tf\|_{L^1} \leq C \sum_j |\lambda_j|$. Taking the infimum over decompositions gives the theorem.

Step 2. *The near estimate.*

Let a be an atom associated with Q , and $Q^* = 2\sqrt{n}Q$. By Cauchy–Schwarz on Q^* and the L^2 boundedness of T ,

$$\int_{Q^*} |Ta| \leq |Q^*|^{1/2} \|Ta\|_{L^2} \leq |Q^*|^{1/2} B_2 \|a\|_{L^2}.$$

Now the normalisation does its work: by (A1) and (A2),

$$\begin{aligned} \|a\|_{L^2}^2 &= \int_Q |a|^2 \\ &\leq |Q| \cdot \frac{1}{|Q|^2} = \frac{1}{|Q|} \\ &, \quad \text{so} \quad \|a\|_{L^2} \leq |Q|^{-1/2}. \end{aligned}$$

Since $|Q^*| = (2\sqrt{n})^n |Q|$, we get

$$\int_{Q^*} |Ta| \leq (2\sqrt{n})^{n/2} B_2.$$

Note what happened: the two factors $|Q^*|^{1/2}$ and $|Q|^{-1/2}$ cancelled, leaving a bound independent of the cube. This scale invariance is built into (A2) and is the reason the reduction to atoms is useful.

Step 3. *The far estimate.*

For $x \notin Q^*$ the kernel representation applies, since $x \notin \text{supp } a$, and Theorem 4.2.2 gives

$$\int_{(Q^*)^c} |Ta| \leq A.$$

Step 4. *Combine.*

Adding, $\|Ta\|_{L^1} \leq (2\sqrt{n})^{n/2} B_2 + A \lesssim_n A + B_2$, uniformly over atoms. Step 1 completes the proof. $\square$

Pitfall: uniform boundedness on atoms does not imply boundedness on H^1

The following argument appears in many places and is *wrong*:

“Every $f \in H^1$ is $\sum_j \lambda_j a_j$ with $\sum |\lambda_j| \approx \|f\|_{H^1}$. So if $\|Ta\|_{L^1} \leq C$ for every atom, then $\|Tf\|_{L^1} \leq C \sum |\lambda_j| \approx C \|f\|_{H^1}$.”

The gap is at “so”: T has not been shown to commute with the infinite sum, and for a general linear T defined only on finite combinations of atoms it need not. Bownik (2005), refining an example of Meyer, constructed a linear functional on finite combinations of $(1, \infty)$ -atoms which is uniformly bounded on atoms and admits *no* bounded extension to H^1 ; composing it with a fixed L^1 function gives an operator uniformly bounded on atoms which is not bounded from H^1 to L^1 .

Why Theorem 4.2.5 is nonetheless true. Because T is not merely defined on atoms: it is given as bounded on L^2 , and by Theorem 3.2.3 (Calderón–Zygmund) it is of weak type $(1, 1)$. Those two facts let Step 1 pass to the limit honestly — $f_N \rightarrow f$ in L^1 forces $Tf_N \rightarrow Tf$ in measure, and Fatou does the rest. The theorem is stated for $f \in H^1 \cap L^2$ for the same reason, and the extension to all of H^1 is then by density.

What to check when reading. Whenever an argument establishes a bound on atoms and concludes a bound on H^1 , look for the additional hypothesis that licenses the limit. For Calderón–Zygmund operators it is the L^2 boundedness; for other operators one may instead use $(1, q)$ -atoms with $q < \infty$, for which Meda, Sjögren and Vallarino (2008) showed the naive argument *is* valid. With $(1, \infty)$ -atoms and no further hypothesis, it is not.

WORKED EXAMPLE 4.2.2— TIER B **The Hilbert transform of an atom, computed**

It is worth seeing Theorem 4.2.5 in a case where everything is explicit.

Let $n = 1$, $Q = [0, 1]$ and $a = \mathbf{1}_{[0, 1/2]} - \mathbf{1}_{(1/2, 1]}$, the atom of Worked Example 4.2.1(a). By Worked Example 3.3.1 (the Hilbert transform of an indicator),

$\mathcal{H}\mathbf{1}_{[u,v]}(x) = \frac{1}{\pi} \log \left| \frac{x-u}{x-v} \right|$, so by linearity

$$\begin{aligned} \mathcal{H}a(x) &= \frac{1}{\pi} \log \left| \frac{x}{x - \frac{1}{2}} \right| - \frac{1}{\pi} \log \left| \frac{x - \frac{1}{2}}{x - 1} \right| \\ &= \frac{1}{\pi} \log \left| \frac{x(x-1)}{(x - \frac{1}{2})^2} \right|. \end{aligned}$$

Decay at infinity. For large $|x|$, expand:

$$\frac{x(x-1)}{(x - \frac{1}{2})^2} = \frac{x^2 - x}{x^2 - x + \frac{1}{4}} = 1 - \frac{1/4}{x^2 - x + \frac{1}{4}},$$

so log of it is $\approx -\frac{1}{4x^2}$, giving $|\mathcal{H}a(x)| \approx \frac{1}{4\pi x^2}$ as $|x| \rightarrow \infty$. This *is* integrable.

Compare with the non-atom. For $f = \mathbf{1}_{[0,1]}$, Volume 3 computed $\mathcal{H}f(x) = \frac{1}{\pi} \log \left| \frac{x}{x-1} \right| \approx \frac{1}{\pi x}$, which is not.

Reading the difference. The mean-zero condition improved the decay from $|x|^{-1}$ to $|x|^{-2}$ — from $|x|^{-n}$ to $|x|^{-n-1}$ in general — and $|x|^{-n-1}$ is integrable at infinity while $|x|^{-n}$ sits exactly on the boundary. That single power is the whole content of the H^1 theory, and Theorem 4.2.2 is the general version of the cancellation just performed by hand.

A caution. $\mathcal{H}a$ also has logarithmic singularities at $x = 0, \frac{1}{2}, 1$, which are locally integrable, so they are harmless. Cancellation fixes the behaviour at infinity; it does nothing about local singularities, and it does not need to.

WORKED EXAMPLE 4.2.3— TIER C: RESEARCH TASTE **Why one checks operators on atoms, and when that is not enough**

Step 1 of Theorem 4.2.5 is the standard method: to bound an operator on H^1 , bound it on atoms. The method is so convenient that it is worth being precise about what it does and does not establish, because the point is frequently got wrong in the literature.

(a) *What the argument actually proved.* We showed $\|Ta\|_{L^1} \leq C$ for atoms, and concluded $\|Tf\|_{L^1} \leq C\|f\|_{H^1}$ — but the passage from one to the other used more than linearity. It used that T was already known to be defined on f (through the L^2 theory) and that $Tf_N \rightarrow Tf$ in some usable sense (through the weak $(1,1)$ bound). Without such an a priori statement, the inequality on atoms says nothing about f : a general $f \in H^1$ has infinitely many atomic decompositions, and there is no reason for a merely finitely-additive operator to give the same answer on each.

(b) *The genuine counterexample.* Bownik (2005) constructed a linear operator defined on all finite linear combinations of atoms, uniformly bounded on atoms into L^1 , which does *not* extend to a bounded operator $H^1 \rightarrow L^1$. So the naive statement — “uniformly bounded on atoms implies bounded on H^1 ” — is false as stated. What is true, and what we used, is that the implication holds when the operator is already known to be continuous into some space in which the atomic series converges. For every operator in this series that hypothesis is available for free, from the L^2 theory.

(c) *Why the difficulty exists.* The infimum in Theorem 4.2.3 is over decompositions, and a given f has many. Two decompositions with nearly optimal coefficient sums can be completely different collections of atoms, and there is no canonical choice. Chapter 4.4 removes this discomfort by proving that H^1 has an equivalent, *decomposition-free* description in terms of a maximal function — and that theorem, not the atomic definition, is what makes H^1 a natural object rather than a construction.

(d) *The same issue elsewhere in the series.* A wave packet decomposition (Volume 12) writes a function as a sum of pieces adapted to tubes, and the decomposition is again far from unique. Estimates are proved piece by piece and summed, exactly as here, and the same care is required: one must know in advance that the object being estimated is well defined independently of the decomposition. In the restriction literature this is usually arranged by working with a fixed partition of unity and proving all estimates uniform in the choice. The lesson transfers: *a decomposition is a tool for estimating a function that already exists, not a definition of one.*

Drills

Drill 4.2.1. Show that if a is an atom associated with Q and $\tilde{Q} \supseteq Q$ is a larger cube, then a is $|\tilde{Q}|/|Q|$ times an atom associated with $\tilde{Q}$. Deduce that in Theorem 4.2.3 one may restrict to atoms associated with dyadic cubes, at the cost of a dimensional constant.

Drill 4.2.2. Verify that $\|a\|_{L^p} \leq |Q|^{1/p-1}$ for every atom associated with Q and every $1 \leq p \leq \infty$, and identify where this was used in Theorem 4.2.5.

Drill 4.2.3. Compute $\mathcal{R}_1 a$ explicitly in $\mathbb{R}^2$ for an atom of the form $a = |Q|^{-1}(\mathbf{1}_{Q_+} - \mathbf{1}_{Q_-})$, where $Q_{\pm}$ are the left and right halves of a square Q , and confirm the decay rate $|x|^{-3}$.

Drill 4.2.4. Show that $H^1(\mathbb{R}^n)$ is not closed under multiplication by bounded functions, by multiplying an atom by sgn of a coordinate. Contrast with the

fact that L^1 is.

Notes

The atomic decomposition originates with Coifman (1974) in dimension one and Latter (1978) in general. The definition of H^1 historically came first in a different form — via boundary values of holomorphic functions on the upper half plane, whence the name — and the equivalence with the atomic description is Chapter 4.4.

Worked Example 4.2.3(b) refers to M. Bownik, *Boundedness of operators defined on atoms*, Proc. Amer. Math. Soc. 133 (2005). The subtlety it exposes had been overlooked in a number of published arguments.

Chapter 4.3

Duality: $(H^1)^* = \text{BMO}$

4.3.1 The statement, and why it is plausible

Volume 3 proved that a singular integral maps L^∞ into BMO; Chapter 4.2 proved that it maps H^1 into L^1 . The two statements look like mirror images, and they are. This chapter proves that BMO is *exactly* the dual of H^1 , which explains the symmetry: the second statement is the adjoint of the first.

Before the proof, a plausibility check. If g is to define a functional on H^1 by $f \mapsto \int fg$, the natural first test is an atom. Let a be an atom associated with a cube Q . Because $\int a = 0$, we may subtract any constant from g without changing the pairing, and the natural constant is $\langle g \rangle_Q$:

$$\begin{aligned} \left| \int a g \right| &= \left| \int_Q a (g - \langle g \rangle_Q) \right| \\ &\leq \|a\|_{L^\infty} \int_Q |g - \langle g \rangle_Q| \\ &\leq \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|. \end{aligned}$$

The right-hand side is at most $\|g\|_{\text{BMO}}$, by definition. So the pairing of g against *any* atom is bounded by $\|g\|_{\text{BMO}}$ — with no dependence on the cube. The two normalisations were made for each other: the atom's height $|Q|^{-1}$ cancels the cube's volume, and the atom's mean-zero property is what licenses the subtraction of $\langle g \rangle_Q$.

That is the whole of the easy half, and §3.2 does no more than make it precise. The converse — that every bounded functional arises this way — is genuinely harder, and occupies §3.3 and §3.4.

Notation 4.3.1. Write H_{fin}^1 for the set of *finite* linear combinations of atoms. By Theorem 4.2.3 it is dense in H^1 : truncating an atomic decomposition after N terms changes the H^1 norm by at most $\sum_{j>N} |\lambda_j| \rightarrow 0$.

4.3.2 The easy half

Theorem 4.3.2. *Let $g \in \text{BMO}(\mathbb{R}^n)$. Then the map*

$$\Lambda_g(f) = \int_{\mathbb{R}^n} f g \quad (f \in H_{\text{fin}}^1)$$

is well defined and extends uniquely to a bounded linear functional on $H^1(\mathbb{R}^n)$, with

$$\|\Lambda_g\|_{(H^1)^*} \leq 2\|g\|_{\text{BMO}}.$$

Moreover $\Lambda_g = \Lambda_{g'}$ if and only if $g - g'$ is constant.

Proof. Step 1. The integral converges.

Let $f \in H_{\text{fin}}^1$, so f is bounded with compact support, say $\text{supp } f \subseteq Q_0$. By John–Nirenberg (Theorem 3.7.3 (John–Nirenberg)), g is in $L^2(Q_0)$, hence in $L^1(Q_0)$, so $\int f g$ converges absolutely.

Step 2. *The bound on atoms.*

Let a be an atom associated with Q . Since $\int a = 0$ and a is supported in Q ,

$$\int a g = \int_Q a (g - \langle g \rangle_Q),$$

and therefore, by (A2),

$$\left| \int a g \right| \leq \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q| \leq \|g\|_{\text{BMO}}. \quad (4.3.1)$$

Step 3. *Extend by linearity and density.*

If $f = \sum_{j \leq N} \lambda_j a_j$ then $|\Lambda_g(f)| \leq \sum_{j \leq N} |\lambda_j| \|g\|_{\text{BMO}}$ by (4.3.1). Taking the infimum over all finite atomic representations of f — note this infimum may exceed $\|f\|_{H^1}$, since the latter allows infinite decompositions — gives $|\Lambda_g(f)| \leq \|g\|_{\text{BMO}} \|f\|_{H_{\text{fin}}^1}$.

To pass to $\|f\|_{H^1}$: given any decomposition $f = \sum_{j=1}^{\infty} \lambda_j a_j$ with $\sum_j |\lambda_j| \leq \|f\|_{H^1} + \varepsilon$, the truncations f_N lie in H_{fin}^1 , satisfy $|\Lambda_g(f_N)| \leq \|g\|_{\text{BMO}} \sum_j |\lambda_j|$, and converge to f in L^1 , hence — f and all f_N being supported in a common compact set only when the decomposition is finite — one argues instead on H^1 directly: the functional Λ_g is bounded on the dense subspace H_{fin}^1 with constant $\|g\|_{\text{BMO}}$, so it extends uniquely and boundedly to H^1 , with the same constant. The factor 2 in the statement is slack absorbed for the complex case, where one splits into real and imaginary parts.

Step 4. *The kernel of the map $g \mapsto \Lambda_g$.*

If g is constant then $\int a g = g \int a = 0$ for every atom, so $\Lambda_g = 0$. Conversely suppose $\Lambda_g = 0$. Fix a cube Q and let $h \in L^\infty(Q)$ with $\int_Q h = 0$;

then $h/(|Q|\|h\|_\infty)$ is an atom, so $\int_Q hg = 0$. A locally integrable function annihilating every bounded mean-zero function supported in Q is a.e. constant on Q : apply this with $h = \mathbf{1}_A - \frac{|A|}{|B|}\mathbf{1}_B$ for measurable $A, B \subseteq Q$ to see that $\langle g \rangle_A = \langle g \rangle_B$ for all such A, B , and let A shrink to a Lebesgue point. Since Q was arbitrary and cubes overlap, the constants agree. $\square$

Pitfall: the pairing is an extension, not always an integral

For general $f \in H^1$ and $g \in \text{BMO}$, the integral $\int fg$ need not converge absolutely: f is only in L^1 and g need only be locally in every L^p , with no global bound. Theorem 4.3.2 defines Λ_g on a dense subspace and *extends* it, which is a different and weaker assertion.

This is the same situation as the Fourier transform on L^2 (Volume 2, Theorem 4.5), which is also defined by extension from a dense subspace where an integral formula happens to converge. The reader should not expect to compute $\Lambda_g(f)$ by integrating, except when f has compact support.

4.3.3 L^2 -atoms

For the converse we need to know that the atomic space is not too small — more precisely, that a mean-zero L^2 function supported in a cube already lies in H^1 , with the norm one would guess by scaling. This requires an argument, and the argument is where the exponent 2 earns its place.

Definition 4.3.3. An L^2 -atom associated with a cube Q is a function h with $\text{supp } h \subseteq Q$, $\int h = 0$, and $\|h\|_{L^2} \leq |Q|^{-1/2}$.

Every atom is an L^2 -atom: by (A1) and (A2), $\|a\|_{L^2}^2 \leq |Q| \cdot |Q|^{-2} = |Q|^{-1}$. The content of the next result is the converse inclusion, at the level of norms.

Proposition 4.3.4 (L^2 -atoms belong to H^1). *There is C_n such that every L^2 -atom h satisfies $h \in H^1(\mathbb{R}^n)$ with $\|h\|_{H^1} \leq C_n$. Equivalently, for every $h \in L^2$ supported in a cube Q with $\int h = 0$,*

$$\|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2}.$$

Scratch work. Why should this be true, and why should L^2 matter?

We must manufacture atoms out of h , and the only tool for manufacturing mean-zero pieces is the Calderón–Zygmund decomposition of Volume 3. So run it — but at which height? Running it once, at a single height λ , produces a good part and a bad part; the bad part is a sum of mean-zero pieces supported in cubes, which is promising, but the good part is merely bounded, not mean-zero, and so is not made of atoms.

The way out is to run the decomposition at *every* dyadic height 2^k simul-

taneously and telescope. The good parts g_k at successive heights differ by something supported in the level set at height 2^k and having mean zero on each of its cubes — an atom, up to normalisation. Since $g_k \rightarrow h$, the telescoping series $\sum_k (g_{k+1} - g_k)$ reconstructs h out of atoms.

Now the arithmetic. The piece at height 2^k has size $\approx 2^k$ and lives on the level set Ω_k , so it contributes $\approx 2^k |\Omega_k|$ to the total coefficient sum. If we knew only $h \in L^1$, Chebyshev would give $|\Omega_k| \lesssim 2^{-k} \|h\|_1$ and the contribution would be $\approx \|h\|_1$ at every k — a divergent sum. With $h \in L^2$ we get $|\Omega_k| \lesssim 4^{-k} \|h\|_2^2$ instead, and the contribution is $\approx 2^{-k} \|h\|_2^2$, which is summable.

That is the entire role of the exponent 2: it converts a borderline divergent series into a geometric one. Any exponent $q > 1$ would do, and the corresponding statement for L^q -atoms is true for every $q > 1$ and false for $q = 1$.

Proof. By translating and dilating we may assume $Q = [0, 1)^n$, so $|Q| = 1$ and $\|h\|_{L^2} \leq 1$; the general case follows because both sides of the stated inequality transform the same way under $x \mapsto \lambda x + x_0$. Note $\|h\|_{L^1} \leq \|h\|_{L^2} |Q|^{1/2} \leq 1$ by Cauchy–Schwarz.

Let $\mathcal{D}(Q)$ be the dyadic subcubes of Q and let

$$\mathcal{M}_Q h(x) = \sup_{\substack{R \in \mathcal{D}(Q) \\ x \in R}} \langle |h| \rangle_R, \quad x \in Q,$$

the dyadic maximal function relative to Q .

Step 1. *The level sets and their measures.*

For $k \geq 0$ set $\Omega_k = \{x \in Q : \mathcal{M}_Q h(x) > 2^k\}$, and $\Omega_{-1} = Q$. These decrease in k . By the dyadic maximal theorem on L^2 — Theorem 3.1.4 (the dyadic maximal inequality) combined with Marcinkiewicz interpolation, giving $\|\mathcal{M}_Q h\|_{L^2} \leq 2 \|h\|_{L^2}$ — and Chebyshev,

$$|\Omega_k| \leq \frac{\|\mathcal{M}_Q h\|_{L^2}^2}{4^k} \leq \frac{4}{4^k}. \quad (4.3.2)$$

This is the estimate the Scratch box identified as decisive.

For $k \geq 0$, Ω_k is the disjoint union of the maximal cubes $Q_j^k \in \mathcal{D}(Q)$ with $\langle |h| \rangle_{Q_j^k} > 2^k$, and as in Theorem 3.1.6 (the Calderón–Zygmund decomposition) the parent bound gives

$$\langle |h| \rangle_{Q_j^k} \leq 2^{n+k}. \quad (4.3.3)$$

(For $k = 0$ the maximal cubes may include Q itself, in which case $\Omega_0 = Q$; nothing below is affected.)

Step 2. *The good functions.*

For $k \geq 0$ define

$$g_k = h \mathbf{1}_{Q \setminus \Omega_k} + \sum_j \langle h \rangle_{Q_j^k} \mathbf{1}_{Q_j^k},$$

and set $g_{-1} = \langle h \rangle_Q \mathbf{1}_Q = 0$, using $\int h = 0$. Two properties:

$$\|g_k\|_{L^\infty} \leq 2^{n+k} \quad (4.3.4)$$

— off Ω_k we have $|h| \leq 2^k$ a.e. by Lebesgue differentiation, and on Q_j^k we have $|\langle h \rangle_{Q_j^k}| \leq \langle |h| \rangle_{Q_j^k} \leq 2^{n+k}$ by (4.3.3) — and

$$\|h - g_k\|_{L^1} \leq 2 \int_{\Omega_k} |h| \longrightarrow 0 \quad (k \rightarrow \infty), \quad (4.3.5)$$

since $h - g_k$ is supported in Ω_k and is there the difference of h and its averages, and $|\Omega_k| \rightarrow 0$ by (4.3.2) while h is integrable (absolute continuity of the integral, Theorem 1.7.10 (absolute continuity of the integral)).

Step 3. *The telescoping series.*

By (4.3.5) and $g_{-1} = 0$,

$$h = \sum_{k \geq 0} (g_k - g_{k-1}) \quad \text{in } L^1(\mathbb{R}^n).$$

Fix $k \geq 0$. Since $\Omega_k \subseteq \Omega_{k-1}$ and $g_k = g_{k-1} = h$ off Ω_{k-1} , the difference $g_k - g_{k-1}$ is supported in Ω_{k-1} , so it splits as

$$g_k - g_{k-1} = \sum_i b_i^k, \quad b_i^k := (g_k - g_{k-1}) \mathbf{1}_{Q_i^{k-1}},$$

a sum over the maximal cubes constituting Ω_{k-1} . (For $k = 0$ read $\Omega_{-1} = Q$ with the single cube Q .)

Step 4. *Each piece is an atom, up to a factor.*

Fix i and k . Support: b_i^k is supported in Q_i^{k-1} by construction. Size: by (4.3.4),

$$\|b_i^k\|_{L^\infty} \leq \|g_k\|_\infty + \|g_{k-1}\|_\infty \leq 2^{n+k} + 2^{n+k-1} \leq 2^{n+k+1}.$$

Mean zero: on Q_i^{k-1} we have $g_{k-1} = \langle h \rangle_{Q_i^{k-1}}$, so $\int_{Q_i^{k-1}} g_{k-1} = \int_{Q_i^{k-1}} h$. And

$$\int_{Q_i^{k-1}} g_k = \int_{Q_i^{k-1} \setminus \Omega_k} h + \sum_{j: Q_j^k \subseteq Q_i^{k-1}} \int_{Q_j^k} h = \int_{Q_i^{k-1}} h,$$

because every Q_j^k meeting Q_i^{k-1} is contained in it — both are dyadic, and Q_j^k cannot strictly contain the maximal cube Q_i^{k-1} since $\langle |h| \rangle_{Q_j^k} > 2^k > 2^{k-1}$ would contradict maximality. Subtracting, $\int b_i^k = 0$.

Therefore

$$b_i^k = \mu_i^k a_i^k, \quad \mu_i^k := 2^{n+k+1} |Q_i^{k-1}|,$$

with a_i^k an atom associated with Q_i^{k-1} .

Step 5. *Sum the coefficients.*

Using (4.3.2) and $\sum_i |Q_i^{k-1}| = |\Omega_{k-1}|$,

$$\begin{aligned} \sum_{k \geq 0} \sum_i \mu_i^k &= \sum_{k \geq 0} 2^{n+k+1} |\Omega_{k-1}| \\ &\leq 2^{n+1} |Q| + \sum_{k \geq 1} 2^{n+k+1} \cdot \frac{4}{4^{k-1}} \\ &\leq 2^{n+1} + 2^{n+5} \sum_{k \geq 1} 2^{-k} \lesssim_n 1. \end{aligned}$$

So $h = \sum_{k,i} \mu_i^k a_i^k$ with summable coefficients, and $\|h\|_{H^1} \leq C_n$. $\square$

Where we are. Look again at Step 5. The factor 2^k came from the height of the piece and the factor 4^{-k} from (4.3.2); their product 2^{-k} is summable. Had we only assumed $h \in L^1$, (4.3.2) would read $|\Omega_k| \lesssim 2^{-k}$ and the product would be $2^0 = 1$ at every k — divergent. The proposition is false for L^1 : by Theorem 4.2.4(iv) there are mean-zero L^1 functions not in H^1 .

4.3.4 The hard half

Theorem 4.3.5 (Fefferman). *The map $g \mapsto \Lambda_g$ of Theorem 4.3.2 is an isomorphism of $\text{BMO}(\mathbb{R}^n)/\{\text{constants}\}$ onto $(H^1(\mathbb{R}^n))^*$. That is, every bounded linear functional Λ on H^1 is of the form $\Lambda = \Lambda_g$ for some $g \in \text{BMO}$, unique up to an additive constant, and*

$$\frac{1}{2} \|\Lambda_g\| \leq \|g\|_{\text{BMO}} \leq C_n \|\Lambda_g\|.$$

Idea of the proof. Given Λ , we must produce g . The only general device for producing a function that represents a functional is the Riesz representation theorem in a Hilbert space — so we look for a Hilbert space on which Λ is bounded.

Theorem 4.3.4 supplies exactly that. On the space of mean-zero L^2 functions supported in a fixed cube Q , the H^1 norm is dominated by $|Q|^{1/2} \|\cdot\|_{L^2}$, so

Λ is a bounded functional for the L^2 norm, and Riesz representation gives a $g^Q \in L^2(Q)$ representing it there.

Two things then remain. The functions g^Q for different cubes must be patched into a single g — they agree up to constants, which is exactly the ambiguity the theorem allows. And g must be shown to lie in BMO, which is done by feeding Λ the one test function designed to extract the mean oscillation: a normalised multiple of $\text{sgn}(g - \langle g \rangle_Q)$, adjusted to have mean zero.

Proof. Let $\Lambda \in (H^1)^*$ with $\|\Lambda\| = 1$.

Step 1. Λ is bounded on a Hilbert space, for each cube.

Fix a cube Q and let

$$L_0^2(Q) = \left\{ h \in L^2(\mathbb{R}^n) : \text{supp } h \subseteq Q, \int h = 0 \right\},$$

a closed subspace of $L^2(Q)$ — closed because both defining conditions are preserved under L^2 limits, the second by Cauchy–Schwarz on the finite-measure set Q . Hence $L_0^2(Q)$ is a Hilbert space.

By Theorem 4.3.4, every $h \in L_0^2(Q)$ lies in H^1 with $\|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2}$. Therefore

$$|\Lambda(h)| \leq \|h\|_{H^1} \leq C_n |Q|^{1/2} \|h\|_{L^2} \quad (h \in L_0^2(Q)), \quad (4.3.6)$$

so Λ restricts to a bounded functional on $L_0^2(Q)$ of norm at most $C_n |Q|^{1/2}$.

Step 2. Represent it.

By the Riesz representation theorem (Theorem 1.8.15 (Riesz representation)) there is a unique $G^Q \in L_0^2(Q)$ with

$$\Lambda(h) = \int_{\mathbb{R}^n} h \overline{G^Q} \quad \text{for all } h \in L_0^2(Q),$$

and $\|G^Q\|_{L^2} \leq C_n |Q|^{1/2}$. Write $g^Q = \overline{G^Q}$.

Step 3. Consistency across cubes.

Let $Q \subseteq Q'$. For $h \in L_0^2(Q) \subseteq L_0^2(Q')$ both representations apply, so $\int h(g^Q - g^{Q'}) = 0$ for every $h \in L_0^2(Q)$. A function in $L^2(Q)$ orthogonal to every mean-zero element of $L^2(Q)$ lies in the orthogonal complement of $L_0^2(Q)$ inside $L^2(Q)$, which is precisely the constants. Hence

$$g^{Q'}|_Q = g^Q + c_{Q,Q'} \quad \text{for a constant } c_{Q,Q'}.$$

Step 4. Patch.

Let $Q_1 \subseteq Q_2 \subseteq \dots$ be cubes with $\bigcup_m Q_m = \mathbb{R}^n$. Define g on $\mathbb{R}^n$ by setting $g = g^{Q_1}$ on Q_1 and, inductively, $g = g^{Q_{m+1}} - c_m$ on Q_{m+1} , where c_m is chosen

by Step 3 so that the new definition agrees with the old on Q_m . This is consistent, and produces $g \in L^2_{\text{loc}}(\mathbb{R}^n) \subseteq L^1_{\text{loc}}$ with

$$\Lambda(h) = \int h g \quad \text{for every } h \in L^2_0(Q_m) \text{ and every } m, \quad (4.3.7)$$

since adding a constant to g does not change the pairing against a mean-zero h .

Step 5. g lies in BMO.

Fix a cube Q ; by enlarging we may assume $Q \subseteq Q_m$ for some m . Put

$$s = \overline{\text{sgn}(g - \langle g \rangle_Q)} \mathbf{1}_Q, \quad h = \frac{1}{|Q|} (s - \langle s \rangle_Q \mathbf{1}_Q),$$

where $\text{sgn } z = z/|z|$ for $z \neq 0$ and 0 otherwise. Then h is supported in Q , has $\int h = 0$ by construction, and satisfies $\|h\|_{L^\infty} \leq 2|Q|^{-1}$, hence $\|h\|_{L^2} \leq 2|Q|^{-1/2}$; so $h \in L^2_0(Q)$.

Now compute the pairing. Since $\int h = 0$ we may replace g by $g - \langle g \rangle_Q$, and since $\int_Q (s - \langle s \rangle_Q) \bar{c} = 0$ for constants c ,

$$\begin{aligned} \int h g &= \frac{1}{|Q|} \int_Q (s - \langle s \rangle_Q) (g - \langle g \rangle_Q) \\ &= \frac{1}{|Q|} \int_Q s (g - \langle g \rangle_Q) \\ &= \frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|, \end{aligned}$$

where the middle equality drops the term $\langle s \rangle_Q \int_Q (g - \langle g \rangle_Q) = 0$, and the last uses $s(g - \langle g \rangle_Q) = |g - \langle g \rangle_Q|$ pointwise.

Combining with (4.3.7) and (4.3.6),

$$\frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q| = |\Lambda(h)| \leq C_n |Q|^{1/2} \|h\|_{L^2} \leq 2C_n.$$

Taking the supremum over cubes, $\|g\|_{\text{BMO}} \leq 2C_n = 2C_n \|\Lambda\|$.

Step 6. $\Lambda = \Lambda_g$.

Both are bounded functionals on H^1 . They agree on every atom: an atom a associated with Q lies in $L^2_0(Q)$, so (4.3.7) gives $\Lambda(a) = \int a g = \Lambda_g(a)$. Hence they agree on H^1_{fin} by linearity, and on H^1 by density and continuity.

Uniqueness up to constants is Theorem 4.3.2, Step 4, and the two-sided norm comparison collects Steps 5 and Theorem 4.3.2. $\square$

WORKED EXAMPLE 4.3.1— TIER A **Reading John–Nirenberg through the duality**

The John–Nirenberg inequality (Theorem 3.7.3 (John–Nirenberg)) says a BMO function has exponentially small oscillation tails. Theorem 4.3.5 gives it a second meaning.

By duality, $\|g\|_{\text{BMO}} \approx \sup\{| \int f a g | : a \text{ an atom}\}$. Now L^2 -atoms are, by Theorem 4.3.4, atoms up to a bounded factor, so the same supremum over L^2 -atoms is comparable. Unwinding, for every cube Q and every $h \in L^2$ supported in Q with $\int h = 0$,

$$\left| \int_Q h g \right| \lesssim_n |Q|^{1/2} \|h\|_{L^2} \|g\|_{\text{BMO}},$$

which by duality in $L^2(Q)$ says exactly

$$\left(\frac{1}{|Q|} \int_Q |g - \langle g \rangle_Q|^2 \right)^{1/2} \lesssim_n \|g\|_{\text{BMO}}.$$

So the L^2 mean oscillation is controlled by the L^1 mean oscillation — a special case of John–Nirenberg, obtained here with no stopping-time argument at all, purely from Theorem 4.3.4.

The general L^p case follows the same way from the L^p -atom version of Theorem 4.3.4, whose proof is identical with 2 replaced by p . What duality does *not* give is the exponential bound itself, which genuinely needs the iteration of Volume 3.

WORKED EXAMPLE 4.3.2— TIER C: RESEARCH TASTE **What the duality is good for**

(a) *Interpolation.* H^1 and BMO are the correct endpoints of the L^p scale in a precise sense: complex interpolation between H^1 and L^∞ gives L^p for $1 < p < \infty$, as does interpolation between L^1 and BMO. The Fefferman–Stein inequality of Chapter 4.1 is one concrete consequence, and Theorem 4.1.5 another. Neither statement is available with L^1 and L^∞ in place of H^1 and BMO, because the Calderón–Zygmund operators are unbounded on both.

(b) *Adjoints become free.* Volume 3 proved $T : L^\infty \rightarrow \text{BMO}$ and Chapter 4.2 proved $T : H^1 \rightarrow L^1$. Given Theorem 4.3.5, these are adjoint statements: the second applied to T^* dualises to the first for T . So half the endpoint theory was a consequence of the other half all along, and the duality is what makes that visible.

(c) *Carleson measures, and the connection to later volumes.* Fefferman’s original formulation of Theorem 4.3.5 came with a companion characterisation:

$g \in \text{BMO}$ if and only if the measure

$$d\mu_g(x, t) = |\nabla u(x, t)|^2 t \, dx \, dt$$

is a *Carleson measure* on the upper half space, where u is the Poisson extension of g ; that is, $\mu_g(Q \times (0, \ell(Q))) \lesssim |Q|$ for every cube. The condition says: the energy of g at scales below $\ell(Q)$ and locations inside Q is at most $|Q|$ — a bound on the total amount of oscillation available across all scales at once. That is a multiscale counting statement of exactly the type this series is about. In Volume 13 the analogous bookkeeping appears as the constraint that a wave packet decomposition cannot have too much mass at too many scales simultaneously, and the summation over dyadic scales in the decoupling proofs is the same computation with tubes in place of Carleson boxes. We do not prove the Carleson characterisation — it needs the Littlewood–Paley theory of Chapter 4.6 together with a stopping-time argument — but it is worth knowing that the multiscale accounting of the later volumes has an ancestor here.

WORKED EXAMPLE 4.3.3— TIER B Using $(H^1)^* = \text{BMO}$

Theorem 4.3.5 is used in one of two ways, and it is worth separating them.

Move 1: prove a bound on H^1 by testing against BMO. To show $\|f\|_{H^1} \lesssim A$, it suffices to show $|\int fg| \lesssim A\|g\|_{\text{BMO}}$ for all g in a dense subclass. This converts a statement about atomic decompositions — an infimum over representations — into a statement about pairings, which is usually easier to verify.

Move 2: prove a bound on BMO by testing against atoms. To show $\|g\|_{\text{BMO}} \lesssim A$, show $|\int ag| \lesssim A$ for every atom a . Since atoms are explicit — bounded, supported on a cube, mean zero — this is often a direct computation.

A worked instance of Move 2. Let T be a Calderón–Zygmund operator and $g \in L^\infty$. For an atom a on a cube Q ,

$$\int a(Tg) = \int (T^*a)g,$$

and T^* is again Calderón–Zygmund, so $\|T^*a\|_{L^1} \lesssim 1$ by Theorem 4.2.5. Hence $|\int a(Tg)| \lesssim \|g\|_{L^\infty}$, and by Move 2, $\|Tg\|_{\text{BMO}} \lesssim \|g\|_{L^\infty}$ — which is Theorem 3.7.4, obtained here by duality rather than directly.

The caution. Move 1 requires knowing the pairing extends, and the Pitfall of Chapter 4.2 shows that atomic arguments need care at exactly this point. Testing against a dense subclass is safe; testing against atoms alone is the step that requires the operator to be a priori bounded somewhere.

WORKED EXAMPLE 4.3.4— TIER B Interpolating with H^1 and BMO

The pair (H^1, BMO) substitutes for (L^1, L^∞) , and the mechanics of using it are worth stating.

(a) *The substitution.* An operator bounded $H^1 \rightarrow L^1$ and $L^\infty \rightarrow \text{BMO}$ is bounded on L^p for $1 < p < \infty$, by interpolation. This is how Volume 3's Calderón–Zygmund theorem is recovered from its two endpoint statements.

(b) *Why the endpoints must be replaced.* $L^1 \rightarrow L^1$ and $L^\infty \rightarrow L^\infty$ are both false for singular integrals, as Worked Example 3.3.4(c) showed concretely for the Hilbert transform.

(c) *The interpolation theorem needed.* Not Riesz–Thorin, which is for L^p pairs, but the complex interpolation of the pair (H^1, BMO) — a genuinely different theorem, and one this series states rather than proves.

(d) *A worked use.* To show a Calderón–Zygmund operator is bounded on L^p , one may either run the decomposition directly (Volume 3, Theorem 3.2.3) or prove the two endpoint statements and interpolate. The second route is longer but generalises to operators where the direct decomposition fails.

The caution. Duality (Theorem 4.3.5) and interpolation are different tools. Duality converts a statement about H^1 into one about BMO; interpolation converts two endpoint statements into a range. Using one where the other is meant is a common confusion.

Drills

Drill 4.3.1. Show directly from Theorem 4.3.2 that BMO is complete, using that the dual of a normed space is always complete.

Drill 4.3.2. Verify the claim in Step 3 of Theorem 4.3.5 that the orthogonal complement of $L_0^2(Q)$ inside $L^2(Q)$ is the space of constants.

Drill 4.3.3. Carry out the proof of Theorem 4.3.4 with L^2 replaced by L^q , $1 < q < \infty$, and identify where the argument breaks at $q = 1$.

Drill 4.3.4. Let $g \in \text{BMO}$ and a an atom associated with Q . Show $|\int a g| \leq (|Q|^{-1} \int_Q |g - \langle g \rangle_Q|^2)^{1/2}$ and compare with (4.3.1). Which is sharper, and why does it not matter?

Notes

Theorem 4.3.5 is Fefferman's, announced in 1971 and published with Stein in 1972; it is the theorem that made BMO a central object. The proof given here,

via L^2 -atoms and Riesz representation, is the standard modern one and is due in this form to Coifman and Weiss.

Theorem 4.3.4 is where the atomic theory is genuinely used rather than merely convenient, and readers should note that it is the only place in this volume where a Calderón–Zygmund decomposition is run at infinitely many heights at once. The technique — telescope over dyadic thresholds and sum a geometric series — recurs in the proof of the maximal characterisation in Chapter 4.4.

Chapter 4.4

Maximal Functions and the Whitney Decomposition

4.4.1 The discomfort with the atomic definition

Theorem 4.2.3 defines H^1 by a decomposition, and Worked Example 2.11 recorded the discomfort this causes: a given f has infinitely many atomic decompositions, none canonical, and the norm is an infimum one can rarely compute. Worse, the definition gives no way to *recognise* membership. Faced with a concrete f , one has no test to apply — only the hope of finding a decomposition.

This chapter addresses the discomfort. We introduce a maximal function $\mathcal{M}_\varphi f$, built from f by a completely explicit recipe, and prove that

$$f \in H^1 \quad \implies \quad \mathcal{M}_\varphi f \in L^1,$$

with $\|\mathcal{M}_\varphi f\|_1 \lesssim \|f\|_{H^1}$. The converse is also true — it is the Fefferman–Stein characterisation, and it is the statement that makes H^1 a natural object rather than a construction — but its proof is long and this series does not give it. Theorem 4.4.5 explains that decision and what depends on it, which is nothing.

We also prove the Whitney decomposition (Theorem 4.4.3), a geometric tool used in the converse and reused independently in Volume 11.

Looking ahead: a test we can already use

The Pitfall in §2.3 exhibited a mean-zero L^1 function

$$g(x) = \frac{\operatorname{sgn}(x)}{|x|(\log|x|)^2} \mathbf{1}_{|x|>2}$$

and asserted, without proof, that it is not in H^1 . With the atomic definition there was nothing to be done: one cannot prove a decomposition does not

exist by failing to find one. Once Theorem 4.4.4 is available the verification is a computation — see Drill 4.4.

4.4.2 Maximal functions attached to a function

Definition 4.4.1. Fix $\varphi \in \mathcal{S}(\mathbb{R}^n)$ with $\int_{\mathbb{R}^n} \varphi = 1$, and write $\varphi_t(x) = t^{-n}\varphi(x/t)$ for $t > 0$. For $f \in L^1(\mathbb{R}^n)$ define the *radial maximal function*

$$\mathcal{M}_\varphi f(x) = \sup_{t>0} |(\varphi_t * f)(x)|.$$

The recipe is: smooth f at every scale t , and record the largest value seen at x . Two remarks orient the definition.

First, this is the Hardy–Littlewood maximal function with the sharp cutoff replaced by a smooth bump. Indeed if one took $\varphi = |B(0, 1)|^{-1} \mathbf{1}_{B(0,1)}$ — not Schwartz, but morally the same — then $\varphi_t * |f|$ is exactly the average of $|f|$ over $B(x, t)$. The difference is that we do *not* take absolute values inside: $\varphi_t * f$, not $\varphi_t * |f|$. So $\mathcal{M}_\varphi f$ can be much smaller than $\mathcal{M}f$ when f oscillates, and that is the entire point.

Second, $\mathcal{M}_\varphi f \geq |f|$ a.e., since $\varphi_t * f \rightarrow f$ a.e. as $t \rightarrow 0$ (Theorem 1.6.10 (approximate identities), together with the Lebesgue differentiation theorem). So $\|\mathcal{M}_\varphi f\|_{L^1} \geq \|f\|_{L^1}$, and the condition $\mathcal{M}_\varphi f \in L^1$ is strictly stronger than $f \in L^1$.

Pitfall: $\mathcal{M}_\varphi f \in L^1$ is a real restriction

For a general $f \in L^1$ one should expect $\mathcal{M}_\varphi f \notin L^1$, for the same reason $\mathcal{M}f \notin L^1$ (Worked Example 1.7.1 ($\mathcal{M}f$ is never integrable)): if $\int f \neq 0$ then for large $|x|$ and $t \approx |x|$,

$$(\varphi_t * f)(x) \approx t^{-n} \varphi(x/t) \int f \approx \frac{c}{|x|^n} \int f,$$

which is not integrable. So the condition $\mathcal{M}_\varphi f \in L^1$ already forces $\int f = 0$ — consistent with Theorem 4.2.4(ii), and a first sign that the right space is being detected.

Theorem 4.4.2 (The easy direction). *Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$ with $\int \varphi = 1$. There is $C_{n,\varphi}$ such that*

$$\|\mathcal{M}_\varphi f\|_{L^1(\mathbb{R}^n)} \leq C_{n,\varphi} \|f\|_{H^1(\mathbb{R}^n)} \quad \text{for all } f \in H^1(\mathbb{R}^n).$$

Idea of the proof. By sublinearity of $f \mapsto \mathcal{M}_\varphi f$ it is enough to prove $\|\mathcal{M}_\varphi a\|_{L^1} \leq C$ for a single atom, uniformly. This is the same reduction as in Theorem 4.2.5, and the same two-region split works: near the cube, use

that $\mathcal{M}_\varphi$ is trivially bounded by $\|a\|_\infty$ and that the cube is small; far from the cube, use the mean-zero property to replace $\varphi_t(x-y)$ by its difference from $\varphi_t(x-c)$.

The one new feature is that we must optimise over t . The far-field bound depends on t , and the supremum defining $\mathcal{M}_\varphi$ ranges over all t ; the computation in Step 3 shows the worst case is $t \approx |x-c|$ and that the resulting bound decays like $|x-c|^{-n-1}$ — one power better than integrability requires, exactly as in Worked Example 4.2.2.

Proof. Step 1. Reduction to an atom.

If $f = \sum_j \lambda_j a_j$ in L^1 , then $\varphi_t * f = \sum_j \lambda_j (\varphi_t * a_j)$ pointwise for each t , so $\mathcal{M}_\varphi f \leq \sum_j |\lambda_j| \mathcal{M}_\varphi a_j$ pointwise, and therefore $\|\mathcal{M}_\varphi f\|_1 \leq \sum_j |\lambda_j| \|\mathcal{M}_\varphi a_j\|_1$. Taking the infimum over decompositions, it suffices to bound $\|\mathcal{M}_\varphi a\|_1$ by a constant independent of the atom.

Let a be an atom associated with a cube Q of centre c and side ℓ , and let $Q^* = 2\sqrt{n}Q$.

Step 2. Near the cube.

For any t and any x , Young's inequality gives

$$|(\varphi_t * a)(x)| \leq \|\varphi_t\|_{L^1} \|a\|_{L^\infty} \leq \frac{\|\varphi\|_{L^1}}{|Q|},$$

using $\|\varphi_t\|_1 = \|\varphi\|_1$ for every t . Hence $\mathcal{M}_\varphi a \leq \|\varphi\|_1 |Q|^{-1}$ everywhere, and

$$\int_{Q^*} \mathcal{M}_\varphi a \leq |Q^*| \cdot \frac{\|\varphi\|_1}{|Q|} = (2\sqrt{n})^n \|\varphi\|_1.$$

As in Theorem 4.2.5, the volume factors cancel and the bound does not see the cube.

Step 3. Far from the cube.

Let $x \notin Q^*$, and write $r = |x-c|$, so $r > \sqrt{n}\ell$. Using $\int a = 0$,

$$(\varphi_t * a)(x) = \int_Q \left(\varphi_t(x-y) - \varphi_t(x-c) \right) a(y) \, dy.$$

For $y \in Q$ we have $|y-c| \leq \frac{\sqrt{n}}{2}\ell < \frac{r}{2}$, so the segment from $x-y$ to $x-c$ lies in $\{|z| \geq r/2\}$. By the mean value theorem and the scaling $\nabla(\varphi_t)(z) = t^{-n-1}(\nabla\varphi)(z/t)$,

$$|\varphi_t(x-y) - \varphi_t(x-c)| \leq |y-c| t^{-n-1} \sup_{|z| \geq r/2} |(\nabla\varphi)(z/t)|.$$

Since $\varphi \in \mathcal{S}$, for any N there is C_N with $|\nabla \varphi(w)| \leq C_N(1 + |w|)^{-N}$; take $N = n + 2$. Then

$$\begin{aligned} |(\varphi_t * a)(x)| &\leq \frac{\sqrt{n}\ell}{2} C_{n+2} t^{-n-1} \left(1 + \frac{r}{2t}\right)^{-(n+2)} \int_Q |a| \\ &\leq C \ell t^{-n-1} \left(1 + \frac{r}{2t}\right)^{-(n+2)}, \end{aligned}$$

using $\int_Q |a| \leq 1$.

It remains to maximise the right-hand side over $t > 0$. If $t \geq r$ the last factor is between $2^{-(n+2)}$ and 1, and $t^{-n-1} \leq r^{-n-1}$. If $t < r$ then $1 + \frac{r}{2t} \geq \frac{r}{2t}$, so

$$t^{-n-1} \left(\frac{r}{2t}\right)^{-(n+2)} = 2^{n+2} \frac{t}{r^{n+2}} \leq \frac{2^{n+2}}{r^{n+1}}.$$

In both regimes the bound is $\lesssim_n r^{-n-1}$, so

$$\mathcal{M}_\varphi a(x) \lesssim_{n,\varphi} \frac{\ell}{|x - c|^{n+1}} \quad (x \notin Q^*). \quad (4.4.1)$$

Step 4. *Integrate the far part.*

By polar coordinates (Theorem 1.6.11 (polar coordinates)) and $r > \sqrt{n}\ell$ on the region of integration,

$$\begin{aligned} \int_{(Q^*)^c} \mathcal{M}_\varphi a &\lesssim_{n,\varphi} \ell \int_{|x-c| > \sqrt{n}\ell} \frac{dx}{|x - c|^{n+1}} = \ell \sigma(\mathbb{S}^{n-1}) \int_{\sqrt{n}\ell}^\infty \frac{dr}{r^2} \\ &= \frac{\sigma(\mathbb{S}^{n-1})}{\sqrt{n}}, \end{aligned}$$

again independent of ℓ . Adding Steps 2 and 4 completes the proof. $\square$

Where we are. Notice that (4.4.1) is the same estimate as Worked Example 4.2.2: an atom is seen from far away with decay $|x|^{-n-1}$, one power better than the $|x|^{-n}$ that a general L^1 function produces, and that one power is exactly the difference between integrable and not. The Hilbert transform and the smoothing φ_t are quite different operations, and the mechanism is identical — freeze the kernel at the centre of the cube, and let the mean-zero property kill the frozen term.

4.4.3 The Whitney decomposition

The converse direction requires cutting an open set into cubes whose sizes are comparable to their distance from the boundary. This is a purely geometric construction, useful far beyond the present chapter, and we prove it here.

Theorem 4.4.3 (Whitney). *Let $\Omega \subsetneq \mathbb{R}^n$ be open and nonempty. There is a countable family $\{Q_j\}$ of dyadic cubes with pairwise disjoint interiors such that*

$$\begin{aligned} \Omega &= \bigcup_j Q_j \quad \text{and} \quad \text{diam } Q_j \\ &\leq \text{dist}(Q_j, \Omega^c) \\ &\leq 4 \text{diam } Q_j \quad \text{for every } j. \end{aligned}$$

Idea of the proof. The two inequalities pull in opposite directions: the left says the cube is small enough to sit comfortably inside Ω , the right says it is not wastefully small. A cube of side comparable to $\frac{1}{3} \text{dist}(x, \Omega^c)$ satisfies both, and such a cube exists at every $x \in \Omega$ because the dyadic side lengths 2^{-k} pass within a factor 2 of any target. Selecting the maximal cubes with this property makes the family disjoint, by the nesting lemma.

Proof. Step 1. *The admissible family is nonempty at every point.*

Let

$$\mathcal{S} = \left\{ Q \in \mathcal{D} : Q \subseteq \Omega, \text{diam } Q \leq \text{dist}(Q, \Omega^c) \leq 4 \text{diam } Q \right\}.$$

Fix $x \in \Omega$ and put $d = \text{dist}(x, \Omega^c)$, which is positive since Ω is open, and finite since $\Omega^c \neq \emptyset$. For $k \in \mathbb{Z}$ let $Q^{(k)}$ be the unique cube of $\mathcal{D}_k$ containing x , so $\text{diam } Q^{(k)} = \sqrt{n} 2^{-k}$. Choose k with

$$\frac{d}{6} < \sqrt{n} 2^{-k} \leq \frac{d}{3},$$

possible since the numbers $\sqrt{n} 2^{-k}$ decrease by factors of 2. Then for any $z \in Q^{(k)}$ we have $|z - x| \leq \text{diam } Q^{(k)} \leq d/3$, so

$$\text{dist}(Q^{(k)}, \Omega^c) \geq d - \frac{d}{3} = \frac{2d}{3} > \frac{d}{3} \geq \text{diam } Q^{(k)},$$

which also shows $Q^{(k)} \subseteq \Omega$; and

$$\text{dist}(Q^{(k)}, \Omega^c) \leq d < 6 \cdot \sqrt{n} 2^{-k} = 6 \text{diam } Q^{(k)}.$$

This gives the left inequality and the right one with 6 in place of 4; running the same computation with the window $\frac{d}{8} < \sqrt{n} 2^{-k} \leq \frac{d}{4}$ gives $\text{dist} \geq \frac{3d}{4}$ and $\text{dist} \leq d < 4 \cdot \sqrt{n} 2^{-k}$, hence $Q^{(k)} \in \mathcal{S}$. So every $x \in \Omega$ lies in some member of $\mathcal{S}$, and $\bigcup_{Q \in \mathcal{S}} Q = \Omega$.

Step 2. *Maximal elements exist.*

If $Q \in \mathcal{S}$ then $\text{diam } Q \leq \text{dist}(Q, \Omega^c) \leq \text{dist}(x_0, \Omega^c)$ for any fixed $x_0 \in Q$, so the side lengths of members of $\mathcal{S}$ containing a given point are bounded above.

Hence every $Q \in \mathcal{S}$ is contained in a maximal element of $\mathcal{S}$: ascend through dyadic parents, and the ascent terminates because side lengths double.

Step 3. *Maximal elements are disjoint and cover.*

Let $\{Q_j\}$ enumerate the maximal elements of $\mathcal{S}$; the family is countable because $\mathcal{D}$ is. If two distinct maximal cubes met, then by the nesting property (Theorem 3.1.2 (nesting of dyadic cubes)) one would contain the other, contradicting maximality of the smaller. Their union is Ω by Step 1 and Step 2. $\square$

WORKED EXAMPLE 4.4.1— TIER A What a Whitney decomposition looks like

Take $n = 1$ and $\Omega = (0, 1)$. The Whitney cubes are the dyadic intervals $[2^{-k-1}, 2^{-k})$ and their reflections $(1 - 2^{-k}, 1 - 2^{-k-1}]$, for $k \geq 1$, together with the middle interval $[\frac{1}{4}, \frac{3}{4}]$ suitably subdivided. Each has length comparable to its distance from $\{0, 1\}$: the interval $[2^{-k-1}, 2^{-k})$ has length 2^{-k-1} and distance 2^{-k-1} from 0.

Two features are typical. The cubes accumulate at the boundary, with sizes shrinking geometrically — there are infinitely many of them. And the number of cubes of any given size is bounded: in $\mathbb{R}^n$ one shows that at most C_n Whitney cubes of a given generation can touch any fixed one, which is why sums over Whitney cubes converge in practice.

4.4.4 The converse, and the limits of what we prove

Theorem 4.4.4 (Fefferman–Stein maximal characterisation). *Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$ with $\int \varphi = 1$, and let $f \in L^1(\mathbb{R}^n)$. Then $f \in H^1(\mathbb{R}^n)$ if and only if $\mathcal{M}_\varphi f \in L^1(\mathbb{R}^n)$, and in that case $\|f\|_{H^1} \approx_{n,\varphi} \|\mathcal{M}_\varphi f\|_{L^1}$.*

Theorem 4.4.2 is the implication $\Rightarrow$. *We do not prove the converse in this series*, and this section explains what the obstruction is, why the omission is safe, and what the reader should do about it.

Scratch work. The architecture of the converse is clear enough. Set $\Omega_k = \{\mathcal{M}_\varphi f > 2^k\}$, an open set — openness is why a smooth φ is used rather than a sharp cutoff — with $|\Omega_k| \leq 2^{-k} \|\mathcal{M}_\varphi f\|_1$. Decompose each Ω_k into Whitney cubes by Theorem 4.4.3, take a smooth partition of unity subordinate to them, and split f into localised pieces. Subtracting a weighted average on each piece makes it mean zero. Telescoping over k then assembles the pieces into a series, whose coefficient sum is controlled by

$$\sum_k 2^k |\Omega_k| \approx \|\mathcal{M}_\varphi f\|_{L^1}$$

by the layer-cake formula — exactly the accounting that worked in Theorem 4.3.4.

The step that fails is the one asserting that the pieces are *bounded* by a multiple of 2^k , which is condition (A2) in the definition of an atom. In Theorem 4.3.4 that bound came for free, because the good functions g_k were built from averages of an L^2 function and (4.3.4) applied. Here the only control on f is through $\mathcal{M}_\varphi f$, and while $|f| \leq \mathcal{M}_\varphi f$ gives $|f| \leq 2^k$ *outside* Ω_k , the pieces live *inside* Ω_k , where f may be arbitrarily large. One is therefore trying to prove a piece is bounded using a hypothesis that says nothing about it.

The resolution — Fefferman and Stein's — is not to repair the estimate but to change what is being proved: one shows the pieces lie in H^1 with the right norm, by an induction on the level k , and only afterwards extracts atoms. This is genuinely intricate and runs to about ten pages.

Remark 4.4.5 (Scope decision). The converse in Theorem 4.4.4 has been *removed from the plan of this series*, not deferred to a later volume. The distinction matters and is worth stating plainly.

A deferral (convention S7) names the volume that supplies the missing argument and is therefore a promise. There is no such volume here: no later part of this series uses the maximal characterisation, and inventing a home for it would be a fiction. What has happened instead is that the manifest overcommitted, and the honest response is to say so and revise it.

Nothing in Volumes 1–4 depends on Theorem 4.4.4. It is used nowhere in a proof; the two places it appears — the Looking Ahead box of §4.1 and Drill 4.4 — are explicitly conditional on it. The remainder of this volume uses H^1 only through its atomic definition, Theorem 4.2.5, and Theorem 4.3.5, all of which are proved.

What the reader loses. The atomic definition remains, as Worked Example 2.11 warned, decomposition-dependent, and there is no test in this text for deciding whether a given f lies in H^1 . A reader who needs one should consult Stein, *Harmonic Analysis*, Chapter III, or Grafakos, *Modern Fourier Analysis*, Chapter 2, where the full proof is given.

WORKED EXAMPLE 4.4.2— TIER C: RESEARCH TASTE **Two things the easy direction already delivers**

The implication we did prove is not a consolation prize. Two consequences, both used in the literature far more often than the converse.

(a) *A necessary condition, hence a source of counterexamples.* Since $f \in H^1$ forces $\mathcal{M}_\varphi f \in L^1$, any f whose maximal function is non-integrable is not in

H^1 — and this is checkable, because $\mathcal{M}_\varphi f$ is given by an explicit formula. The Pitfall of §4.2 already used this shape of argument to show $\int f = 0$ is necessary. Drill 4.2 pushes it further.

The converse would be needed only to show a given f *is* in H^1 , and for that the atomic definition is usually the more convenient tool anyway: one exhibits a decomposition.

(b) *The estimate (4.4.1) is the reusable object.* What the proof actually established is that an atom, viewed from outside its own cube, looks like $\ell|x-c|^{-n-1}$ — one power better than the $|x|^{-n}$ a general L^1 function produces. That estimate, not the theorem it was used for, is what recurs:

- In Worked Example 4.2.2 the same $|x|^{-n-1}$ appeared for the Hilbert transform of an atom, by the same freeze-the-kernel-at-the-centre argument.
- In Theorem 4.2.2 it appeared in the form of Hörmander’s condition, which is precisely the hypothesis making the freezing legitimate for a kernel that is not smooth.
- In Volume 12 it appears again, as the statement that a wave packet associated with a tube T is negligible outside a dilate of T , with a tail decaying fast enough to sum over the tubes. There the “cube” is a tube, the “centre” is the tube’s core, and the mean-zero property is replaced by frequency localisation — but the computation is Step 3 of Theorem 4.4.2, verbatim.

The moral for a student. A theorem often has two parts: a technique and a conclusion. Here the technique — localise, freeze the kernel at a centre, exploit cancellation, integrate the tail — is worth more than the conclusion, and it is entirely contained in the direction we proved.

Figure 4.4.1 (to be drawn). *A Whitney decomposition*

Content required: An open set Ω in the plane (say the interior of a smooth blob) cut into Whitney squares, sizes shrinking geometrically towards $\partial\Omega$. Mark one square Q and the two distances $\text{diam } Q$ and $\text{dist}(Q, \Omega^c)$, showing the ratio between 1 and 4.

WORKED EXAMPLE 4.4.3— TIER B **Maximal characterisations: what they buy**

A maximal characterisation replaces a decomposition-based definition by a

pointwise one, and the gain is specific.

(a) *The problem it solves.* The atomic definition of H^1 is an infimum over decompositions, so verifying $f \in H^1$ means exhibiting one. A maximal characterisation replaces this by checking whether a single function, $\mathcal{M}_\varphi f$, is integrable.

(b) *Why that matters here.* The Pitfall of Chapter 4.2 showed that reasoning atom-by-atom is invalid without an extra hypothesis. A maximal characterisation removes the need to reason about atoms at all, and is the standard repair.

(c) *What it costs.* The proof is substantially harder than the atomic theory, which is why Theorem 4.4.5 descopes the converse direction.

(d) *The pattern elsewhere.* Volume 7's Frostman lemma plays the same role for Hausdorff dimension: it replaces a covering condition — an infimum over covers — by the existence of a measure, which is a pointwise object one can integrate against. Volume 12, Worked Example 12.7.6 records the same trade.

The general shape. An infimum over representations is hard to verify and easy to use; a pointwise condition is the reverse. Most function-space theory consists of proving the two equivalent, and the proofs are the hard part of the subject.

Drills

Drill 4.4.1. Show that $\mathcal{M}_\varphi f \leq \|\varphi\|_{L^\infty} \mathcal{M}f$ pointwise when $\varphi \geq 0$ is radial and decreasing, and give an example where $\mathcal{M}_\varphi f$ is much smaller than $\mathcal{M}f$.

Drill 4.4.2. Verify the claim in the Pitfall of §4.2: if $f \in L^1$ with $\int f \neq 0$, then $\mathcal{M}_\varphi f(x) \gtrsim |x|^{-n}$ for large $|x|$, so $\mathcal{M}_\varphi f \notin L^1$.

Drill 4.4.3. Complete Worked Example 4.4.1 for $\Omega = B(0, 1) \subseteq \mathbb{R}^2$: describe the Whitney cubes and verify the bound on the number of cubes of each generation touching a given one.

Drill 4.4.4. Assuming Theorem 4.4.4, show that the function $g(x) = \operatorname{sgn}(x)|x|^{-1}(\log|x|)^{-2}\mathbf{1}_{|x|>2}$ of the Pitfall in §2.3 is not in $H^1(\mathbb{R})$, by estimating $\mathcal{M}_\varphi g(x)$ from below for large $|x|$.

Drill 4.4.5. Verify that $\Omega_k = \{\mathcal{M}_\varphi f > 2^k\}$ is open, and explain why a smooth φ rather than a sharp cutoff is needed for this.

Drill 4.4.6. Check the layer-cake accounting $\sum_k 2^k |\Omega_k| \approx \|\mathcal{M}_\varphi f\|_{L^1}$ used in the scratchwork.

Drill 4.4.7. Show that Theorem 4.4.2 — the implication $\Rightarrow$ — follows from the atomic definition together with Theorem 4.2.2.

Drill 4.4.8. Explain why Theorem 4.4.5 classifies the converse as removed rather than deferred, and identify what a deferral would have had to name.

Notes

Theorem 4.4.4 is due to Fefferman and Stein (1972). Their paper proves considerably more: the equivalence holds with $\mathcal{M}_\varphi$ replaced by the *grand* maximal function, in which one takes the supremum over all φ in a bounded subset of $\mathcal{S}$, and it holds for H^p with $0 < p \leq 1$.

The Whitney decomposition is from Whitney's 1934 extension theorem and is one of the most reused constructions in analysis. Readers will meet it again in Volume 11, where the cells of a polynomial partition play an analogous role: a decomposition of space into pieces on which the object of study is uniformly controlled.

Chapter 4.5

Weighted and Vector-Valued Maximal Inequalities

4.5.1 The question

The function spaces of Chapter 4.6 are defined by taking, at each point, an ℓ^q norm of the Littlewood–Paley pieces of f , and then an L^p norm in x . Working with such spaces requires knowing that the Hardy–Littlewood maximal operator, applied to each piece separately, respects this double norm:

$$\left\| \left(\sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \lesssim_{n,p,q} \left\| \left(\sum_j |f_j|^q \right)^{1/q} \right\|_{L^p}. \quad (4.5.1)$$

It is worth seeing why this is not immediate. Volume 3 proved the Marcinkiewicz–Zygmund theorem, which gives exactly (4.5.1) for free — but only when $q = 2$, because the proof runs through Khintchine’s inequality, and only for *linear* operators, because the proof pulls the operator through a sum of randomly signed terms. The maximal operator is neither. So both hypotheses fail, and a new argument is needed.

Pitfall: no pointwise domination is available

The natural first attempt is to look for a pointwise inequality. Write $F = (\sum_j |f_j|^q)^{1/q}$. Since $|f_j| \leq F$ pointwise we get $\mathcal{M}f_j \leq \mathcal{M}F$, but then $(\sum_j (\mathcal{M}f_j)^q)^{1/q} \leq (\#j)^{1/q} \mathcal{M}F$, useless for infinitely many terms. The other attempt is Minkowski’s inequality in ℓ^q , which does give

$$\left(\sum_j \left(\frac{1}{|B|} \int_B |f_j| \right)^q \right)^{1/q} \leq \frac{1}{|B|} \int_B F \leq \mathcal{M}F(x)$$

for a *single* ball $B \ni x$. But the left side of (4.5.1) takes the supremum over balls *separately for each* j , and different j may be optimised

by wildly different balls. So $(\sum_j (\mathcal{M}f_j)^q)^{1/q}$ is genuinely larger than $\sup_B (\sum_j (\text{avg}_B |f_j|)^q)^{1/q}$, and no pointwise bound by $\mathcal{M}F$ holds. This is exactly why (4.5.1) is a theorem.

Two special cases are, however, easy, and it is worth disposing of them at once.

Proposition 4.5.1 (The two easy cases). *Let $1 < p < \infty$.*

(i) $\|\sup_j \mathcal{M}f_j\|_{L^p} \lesssim_{n,p} \|\sup_j |f_j|\|_{L^p}$, i.e. (4.5.1) for $q = \infty$.

(ii) (4.5.1) holds when $q = p$.

Proof. (i) Put $G = \sup_j |f_j|$. Then $|f_j| \leq G$ for each j , so $\mathcal{M}f_j \leq \mathcal{M}G$ and hence $\sup_j \mathcal{M}f_j \leq \mathcal{M}G$ pointwise. Apply the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)).

(ii) By Tonelli, the left side of (4.5.1) raised to the power p is $\sum_j \|\mathcal{M}f_j\|_{L^p}^p$, which by the maximal theorem is $\lesssim \sum_j \|f_j\|_{L^p}^p$, and that is the right side raised to the power p , again by Tonelli. $\square$

4.5.2 A weighted maximal inequality

The tool that handles the general case is an inequality of a different kind: it allows an arbitrary weight on both sides, provided the weight on the right is enlarged by one application of $\mathcal{M}$.

Theorem 4.5.2 (Fefferman–Stein weighted maximal inequality). *Let $w \geq 0$ be measurable and not identically zero. Then*

(i) *for every $f \in L^1_{\text{loc}}$ and $\lambda > 0$,*

$$\int_{\{\mathcal{M}f > \lambda\}} w \leq \frac{C_n}{\lambda} \int_{\mathbb{R}^n} |f| \mathcal{M}w;$$

(ii) *for every $1 < q < \infty$,*

$$\int_{\mathbb{R}^n} (\mathcal{M}f)^q w \leq C_{n,q} \int_{\mathbb{R}^n} |f|^q \mathcal{M}w.$$

Idea of the proof. Part (i) is the Vitali argument of Volume 1, run while carrying the weight. The covering lemma produces disjoint balls B_i on which the average of $|f|$ exceeds λ , and the level set is contained in $\bigcup_i 5B_i$. The only new step is to bound $w(5B_i)$, and here $\mathcal{M}w$ does the work: for any $x \in B_i$, the ball $5B_i$ is one of the balls competing in the supremum

defining $\mathcal{M}w(x)$, so $w(5B_i) \lesssim |B_i|\mathcal{M}w(x)$ — and we may take the infimum over $x \in B_i$.

Part (ii) is then Marcinkiewicz interpolation, run with the measure $w \, dx$ on the target and $\mathcal{M}w \, dx$ on the domain. That the two measures differ is not an obstacle: the proof of Marcinkiewicz never used that they were the same.

Proof. Step 1. Proof of (i).

Let $K \subseteq \{\mathcal{M}f > \lambda\}$ be compact; by inner regularity it suffices to bound $\int_K w$. For each $x \in K$ pick a ball $B(x, r_x)$ with

$$\int_{B(x, r_x)} |f| > \lambda |B(x, r_x)|, \quad (4.5.2)$$

and extract a finite subcover; by the Vitali covering lemma (Theorem 1.7.1 (Vitali covering lemma), in the form with the factor 5) there is a pairwise disjoint subfamily $B_1, \dots, B_N$ with $K \subseteq \bigcup_i 5B_i$.

Fix i and let $x \in B_i$. The ball $5B_i$ is contained in $B(x, 10r_i)$, which is a ball centred at x , so

$$\mathcal{M}w(x) \geq \frac{1}{|B(x, 10r_i)|} \int_{B(x, 10r_i)} w \geq \frac{1}{10^n |B_i|} \int_{5B_i} w.$$

Taking the infimum over $x \in B_i$ and rearranging,

$$\begin{aligned} \int_{5B_i} w &\leq 10^n |B_i| \inf_{B_i} \mathcal{M}w \stackrel{(4.5.2)}{\leq} \frac{10^n}{\lambda} \int_{B_i} |f| \cdot \inf_{B_i} \mathcal{M}w \\ &\leq \frac{10^n}{\lambda} \int_{B_i} |f| \mathcal{M}w. \end{aligned}$$

Summing over the disjoint B_i ,

$$\int_K w \leq \sum_i \int_{5B_i} w \leq \frac{10^n}{\lambda} \int_{\mathbb{R}^n} |f| \mathcal{M}w.$$

Step 2. *Proof of (ii): setting up Marcinkiewicz.*

Write $d\nu = w \, dx$ and $d\mu = \mathcal{M}w \, dx$. Part (i) says $\mathcal{M}$ is of weak type $(1, 1)$ from $L^1(\mu)$ to $L^{1,\infty}(\nu)$. And $\mathcal{M}$ is of strong type (∞, ∞) from $L^\infty(\mu)$ to $L^\infty(\nu)$: if $|f| \leq A$ μ -a.e. then, since $\mathcal{M}w > 0$ everywhere (as $w \not\equiv 0$), we have $|f| \leq A$ Lebesgue-a.e., hence $\mathcal{M}f \leq A$ everywhere.

Step 3. *Run the splitting argument.*

Fix $\lambda > 0$ and split $f = f\mathbf{1}_{\{|f|>\lambda/2\}} + f\mathbf{1}_{\{|f|\leq\lambda/2\}} =: g_\lambda + b_\lambda$. Then $\mathcal{M}b_\lambda \leq \lambda/2$ by Step 2, so by sublinearity $\{\mathcal{M}f > \lambda\} \subseteq \{\mathcal{M}g_\lambda > \lambda/2\}$, and part (i) gives

$$\nu(\{\mathcal{M}f > \lambda\}) \leq \frac{2C_n}{\lambda} \int_{\{|f|>\lambda/2\}} |f| \mathcal{M}w.$$

By the layer-cake formula (Worked Example 1.5.1 (the layer-cake formula)) applied with the measure ν , and then Tonelli,

$$\begin{aligned} \int (\mathcal{M}f)^q d\nu &= q \int_0^\infty \lambda^{q-1} \nu(\{\mathcal{M}f > \lambda\}) d\lambda \\ &\leq 2C_n q \int_0^\infty \lambda^{q-2} \int_{\{|f|>\lambda/2\}} |f(x)| \mathcal{M}w(x) dx d\lambda \\ &= 2C_n q \int_{\mathbb{R}^n} |f(x)| \mathcal{M}w(x) \left(\int_0^{2|f(x)|} \lambda^{q-2} d\lambda \right) dx \\ &= \frac{2^q C_n q}{q-1} \int_{\mathbb{R}^n} |f|^q \mathcal{M}w, \end{aligned}$$

the inner integral converging at $\lambda = 0$ precisely because $q > 1$. $\square$

Where we are. Note the shape of (ii): it is not a weighted norm inequality with the same weight on both sides — such inequalities hold only for special weights, the Muckenhoupt A_q class, which this series does not need. It is the weaker and more flexible statement that *any* weight works if one is willing to pay a maximal function on the right. That flexibility is exactly what the duality argument of the next section requires, because there the weight is not chosen by us but produced by duality.

4.5.3 The vector-valued theorem

Theorem 4.5.3. *Let $1 < q < \infty$ and $q < p < \infty$. Then for all $f_1, f_2, \dots \in L^p(\mathbb{R}^n)$,*

$$\begin{aligned} \left\| \left(\sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \\ \leq C_{n,p,q} \left\| \left(\sum_j |f_j|^q \right)^{1/q} \right\|_{L^p}. \end{aligned}$$

Together with Theorem 4.5.1 this covers $q = \infty$, $p = q$, and $q < p < \infty$.

Idea of the proof. Since $p > q$, the exponent p/q exceeds 1, so the quantity $\left\| \sum_j (\mathcal{M}f_j)^q \right\|_{L^{p/q}}$ may be computed by duality (Theorem 1.8.10 (norms by testing)): it is the supremum of $\int \sum_j (\mathcal{M}f_j)^q w$ over nonnegative w in the unit ball of $L^{(p/q)'}$.

That integral is a sum of weighted integrals of $(\mathcal{M}f_j)^q$, one for each j , with a *common* weight w — and Theorem 4.5.2(ii) handles each one, replacing w by $\mathcal{M}w$ on the right. Summing over j reassembles $\sum_j |f_j|^q$, and one final application of the maximal theorem, at exponent $(p/q)'$, disposes of the

$\mathcal{M}w$.

The requirement $p > q$ enters exactly once, at the first step: for $p < q$ the exponent p/q is less than 1, $L^{p/q}$ is not a normed space, and duality is unavailable.

Proof. Write $F = (\sum_j |f_j|^q)^{1/q}$ and $r = p/q > 1$.

Step 1. *Dualise.*

By duality in L^r , and since $\sum_j (\mathcal{M}f_j)^q \geq 0$,

$$\begin{aligned} \left\| \left(\sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p}^q &= \left\| \sum_j (\mathcal{M}f_j)^q \right\|_{L^r} \\ &= \sup \left\{ \int_{\mathbb{R}^n} \sum_j (\mathcal{M}f_j)^q w : w \geq 0, \|w\|_{L^{r'}} \leq 1 \right\}. \end{aligned}$$

Step 2. *Apply the weighted inequality termwise.*

Fix such a w . By Theorem 4.5.2(ii), applied to each f_j with the same weight w , and then Tonelli,

$$\begin{aligned} \int \sum_j (\mathcal{M}f_j)^q w &= \sum_j \int (\mathcal{M}f_j)^q w \\ &\leq C_{n,q} \sum_j \int |f_j|^q \mathcal{M}w \\ &= C_{n,q} \int F^q \mathcal{M}w. \end{aligned}$$

Step 3. *Dispose of $\mathcal{M}w$.*

By Hölder with exponents r and r' , and then the maximal theorem (Theorem 1.7.3 (Hardy–Littlewood maximal inequality)) at exponent $r' > 1$,

$$\begin{aligned} \int F^q \mathcal{M}w &\leq \|F^q\|_{L^r} \|\mathcal{M}w\|_{L^{r'}} \lesssim_{n,r} \|F^q\|_{L^r} \|w\|_{L^{r'}} \\ &\leq \|F\|_{L^p}^q, \end{aligned}$$

using $\|F^q\|_{L^r} = \|F\|_{L^{qr}}^q = \|F\|_{L^p}^q$ and $\|w\|_{r'} \leq 1$.

Step 4. *Take the supremum.*

Combining Steps 1–3, $\|(\sum_j (\mathcal{M}f_j)^q)^{1/q}\|_p^q \lesssim \|F\|_p^q$. □

Remark 4.5.4 (The remaining range, and a scope decision). Theorem 4.5.3 is true for the full range $1 < p < \infty$, $1 < q \leq \infty$; that is the Fefferman–Stein vector-valued maximal theorem of 1971. The case $1 < p < q$ is not proved here.

The obstruction is precise. Our only route to (4.5.1) is the duality of Step 1, which needs $p/q > 1$. The standard proof of the missing range instead establishes a *vector-valued weak* $(1, 1)$ *bound* for the ℓ^q -valued maximal operator, by running the Calderón–Zygmund decomposition on the scalar function F and estimating the resulting pieces in ℓ^q ; that argument requires a genuinely vector-valued Calderón–Zygmund theory, which this series does not develop.

As with Theorem 4.4.5, this is a revision of the plan rather than a deferral: no later volume supplies the missing case. Chapter 4.6 is written so as to use only the range proved here, and the places where that costs something are marked.

The following weaker statement, valid for all $1 < p \leq q \leq \infty$, is sometimes an adequate substitute and is free:

$$\left\| \left(\sum_j (\mathcal{M}f_j)^q \right)^{1/q} \right\|_{L^p} \leq \left\| \left(\sum_j (\mathcal{M}f_j)^p \right)^{1/p} \right\|_{L^p} \lesssim_{n,p} \left\| \left(\sum_j |f_j|^p \right)^{1/p} \right\|_{L^p},$$

the first inequality because $\ell^p \subseteq \ell^q$ for $p \leq q$, and the second being Theorem 4.5.1(ii). It differs from (4.5.1) in having ℓ^p rather than ℓ^q on the right, which is a genuine loss.

WORKED EXAMPLE 4.5.1— TIER B The weighted inequality is sharp in its shape

One cannot replace $\mathcal{M}w$ by w in Theorem 4.5.2. Take $n = 1$, $w = \mathbf{1}_{[0,1]}$ and $f = \mathbf{1}_{[-\varepsilon, 0]}$ for small $\varepsilon > 0$.

Then $\|f\|_{L^q(w)}^q = \int |f|^q w = 0$, since f and w have disjoint supports up to a null set. But $\mathcal{M}f(x) \approx \varepsilon/(x + \varepsilon)$ for $x > 0$, so

$$\int (\mathcal{M}f)^q w \approx \int_0^1 \left(\frac{\varepsilon}{x + \varepsilon} \right)^q dx > 0.$$

So the inequality $\int (\mathcal{M}f)^q w \leq C \int |f|^q w$ is false for every C : the maximal function moves mass from where f lives to where it does not, and a weight that ignores the first region cannot control the second.

Replacing w by $\mathcal{M}w$ on the right repairs exactly this, and it is worth checking the arithmetic. For $x \in [-\varepsilon, 0]$, the interval $[x, 1]$ has length between 1 and $1 + \varepsilon$ and contains all of $\text{supp } w$, so

$$\mathcal{M}w(x) \geq \frac{1}{1 + \varepsilon} \int_0^1 w = \frac{1}{1 + \varepsilon} \geq \frac{1}{2},$$

whence

$$\int |f|^q \mathcal{M}w = \int_{-\varepsilon}^0 \mathcal{M}w \geq \frac{\varepsilon}{2}.$$

On the other side, $\mathcal{M}f(x) \leq \min\{1, \varepsilon/x\}$ for $x > 0$ — the average of $\mathbf{1}_{[-\varepsilon, 0]}$ over any interval containing x and meeting $[-\varepsilon, 0]$ is at most ε divided by the interval's length, which is at least x . Hence

$$\begin{aligned} \int (\mathcal{M}f)^q w &= \int_0^1 (\mathcal{M}f)^q \\ &\leq \int_0^\varepsilon 1 \, dx + \int_\varepsilon^1 \left(\frac{\varepsilon}{x}\right)^q dx \\ &\leq \varepsilon + \frac{\varepsilon^q}{q-1} \varepsilon^{1-q} \\ &= \varepsilon \left(1 + \frac{1}{q-1}\right), \end{aligned}$$

using $q > 1$ to evaluate the second integral. So both sides of Theorem 4.5.2(ii) are of size $\approx \varepsilon$, and the inequality holds with a constant depending only on q — as it must.

The weight $\mathcal{M}w$ is the smallest enlargement of w that sees, at every point, the mass of w anywhere nearby, and that is precisely what the maximal operator's spreading requires.

WORKED EXAMPLE 4.5.2— TIER C: RESEARCH TASTE **Where vector-valued estimates come from, and where they go**

(a) *The pattern.* Three vector-valued statements have now appeared in this series, and they are of increasing strength:

Statement	Hypothesis on T	Source
ℓ^2 extension	T linear, bounded on L^p	Vol 3, Thm 6.1
ℓ^q extension, $q < p$	$T = \mathcal{M}$	Theorem 4.5.3
ℓ^q extension, all p, q	$T = \mathcal{M}$	Fefferman–Stein, not proved here

The first is cheap because Khintchine's inequality converts an ℓ^2 sum into a randomised scalar sum, and linearity lets T pass through. Neither device is available for $q \neq 2$ or for sublinear T , and the price is a genuinely different proof.

(b) *Why the subject needs them at all.* Every decomposition in this series — Littlewood–Paley pieces, atoms, wave packets — replaces one function by

a family, and every estimate must then be reassembled. The reassembly is an ℓ^q norm, and the question is always whether an operator applied to each piece separately respects it. Volume 3, Chapter 8 was a negative answer of this kind: Fefferman's theorem said that the half-space multipliers do *not* admit a uniform ℓ^2 extension over a family of directions, and a Besicovitch set was the witness.

(c) *The contrast worth holding onto.* In Theorem 4.5.3 the family $\{f_j\}$ is arbitrary and the operator is fixed. In Fefferman's theorem the family of operators $\{S_{v_j}\}$ varies and the difficulty comes from the *geometry of the directions* v_j . The first is a soft statement, provable by duality and a weight. The second is a hard one, and false. Volumes 9–13 are almost entirely about the second kind: estimates for families indexed by directions or by tubes, where the answer depends on how the family sits in space.

A useful diagnostic when reading the literature: if a vector-valued inequality is being proved by duality and a weight, it is of the first kind and should be routine; if it is being proved by a covering or incidence argument, it is of the second kind and is where the content lies.

Drills

Drill 4.5.1. Prove the Vitali covering lemma in the form used in Step 1 of Theorem 4.5.2: from any finite family of balls one may extract a pairwise disjoint subfamily whose 5-fold dilates cover the union. (Compare Theorem 1.7.1 (Vitali covering lemma), which used the factor 3; identify where the extra room is needed here.)

Drill 4.5.2. Show that Theorem 4.5.2(ii) fails at $q = 1$, by taking $w = \mathbf{1}_{[0,1]}$ on $\mathbb{R}$ and f concentrated near 0.

Drill 4.5.3. Verify the claim in Theorem 4.5.4 that $\ell^p \subseteq \ell^q$ with norm 1 when $p \leq q$, and give an example showing the resulting substitute inequality is strictly weaker than (4.5.1).

Drill 4.5.4. Deduce from Theorem 4.5.3 that for $q < p < \infty$ the operator $f \mapsto (\mathcal{M}f_j)_j$ is bounded on $L^p(\mathbb{R}^n; \ell^q)$, and check that the constant blows up as $p \downarrow q$. Which step of the proof is responsible?

Notes

Theorem 4.5.2 and the full vector-valued theorem are both due to Fefferman and Stein: the weighted inequality in *Some maximal inequalities* (1971), the vector-

valued one in the same paper. The duality proof of Theorem 4.5.3 given here is the standard route to the range $p > q$ and is the one most often reproduced; the remaining range genuinely requires the vector-valued Calderón–Zygmund machinery, for which see Grafakos, *Classical Fourier Analysis*, Chapter 5.

Readers who continue to the theory of weights will recognise Theorem 4.5.2(ii) as the statement that the pair $(w, \mathcal{M}w)$ satisfies a two-weight L^q bound for the maximal operator. Characterising the pairs (u, v) for which $\int (\mathcal{M}f)^q u \lesssim \int |f|^q v$ holds is the Sawyer testing problem, solved by Sawyer in 1982; none of that theory is used here.

Chapter 4.6

Besov and Triebel–Lizorkin Spaces

4.6.1 Two ways to measure smoothness

Volume 2 defined the Sobolev spaces $H^s(\mathbb{R}^n)$ by requiring $(1 + |\xi|^2)^{s/2} \widehat{f} \in L^2$, and observed that the definition works only at $p = 2$, because only there is Plancherel available. Volume 3 removed that restriction: the Littlewood–Paley theorem showed that for $1 < p < \infty$ the L^p norm is comparable to the L^p norm of the square function built from the frequency pieces $\Delta_j f$. Sobolev norms at general p are therefore square functions with dyadic weights.

Once that is seen, a question presents itself. A square function is an ℓ^2 norm of the pieces, taken pointwise, followed by an L^p norm in x . Why ℓ^2 ? And why in that order?

Both choices can be varied, and doing so produces the two standard scales:

$$\underbrace{\left\| (2^{js} \|\Delta_j f\|_{L^p})_j \right\|_{\ell^q}}_{\text{Besov: } B_{p,q}^s} \quad \text{versus} \quad \underbrace{\left\| \|(2^{js} \Delta_j f)_j\|_{\ell^q} \right\|_{L^p}}_{\text{Triebel–Lizorkin: } F_{p,q}^s}.$$

The Besov norm takes L^p first and ℓ^q second; the Triebel–Lizorkin norm reverses the order. Three parameters now appear: s measures smoothness, p measures integrability, and q is a fine-tuning index that turns out to matter exactly at the endpoints — which is why Chapter 4.7 can prove a sharp embedding into L^∞ that the Sobolev scale misses.

Remark 4.6.1 (Scope of this chapter). This chapter proves: the definitions are meaningful and independent of choices (for Besov in general, for Triebel–Lizorkin at $q = 2$); the Besov spaces are complete; the two scales are related by the Minkowski inequalities $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$; and $F_{p,2}^s = W^{s,p}$ for $1 < p < \infty$, which places the Sobolev spaces inside the scale.

It does *not* prove the identifications $F_{1,2}^0 = H^1$ and $F_{\infty,2}^0 = \text{BMO}$. The first is the Littlewood–Paley characterisation of H^1 and the second is Fefferman’s Carleson-measure theorem; both require machinery beyond this series, and neither is used later — the H^1 –BMO duality of Chapter 4.3 is what the later volumes need. This was decided before the chapter was written and is recorded as revision R3 in the manifest.

4.6.2 The inhomogeneous decomposition

Definition 4.6.2. Fix $\phi_0 \in C_c^\infty(\mathbb{R}^n)$ with $\phi_0 = 1$ on $B(0, 1)$ and $\text{supp } \phi_0 \subseteq B(0, 2)$, and set

$$\psi(\xi) = \phi_0(\xi) - \phi_0(2\xi),$$

so that $\text{supp } \psi \subseteq \{\frac{1}{2} \leq |\xi| \leq 2\}$ and, telescoping,

$$\phi_0(\xi) + \sum_{j \geq 1} \psi(2^{-j}\xi) = 1 \quad \text{for every } \xi \in \mathbb{R}^n. \quad (4.6.1)$$

Define $\Delta_0 f = \mathcal{F}^{-1}(\phi_0 \widehat{f})$ and, for $j \geq 1$, $\Delta_j f = \mathcal{F}^{-1}(\psi(2^{-j}\cdot) \widehat{f})$, so that $\sum_{j \geq 0} \Delta_j f = f$ in $\mathcal{S}'(\mathbb{R}^n)$.

To see (4.6.1): the partial sum up to $j = N$ telescopes to $\phi_0(2^{-N}\xi)$, which equals 1 once $2^{-N}|\xi| \leq 1$.

Two facts about the pieces are used constantly and are worth recording once.

Lemma 4.6.3. *Let $\eta \in C_c^\infty(\mathbb{R}^n)$ and $\eta_j = \eta(2^{-j}\cdot)$, and let $T_j f = \mathcal{F}^{-1}(\eta_j \widehat{f})$. Then for every $1 \leq p \leq \infty$ and every j ,*

$$\|T_j f\|_{L^p} \leq \|\mathcal{F}^{-1}\eta\|_{L^1} \|f\|_{L^p},$$

a bound independent of j . Moreover $\widehat{\Delta_j f}$ is supported in $\{|\xi| \leq 2\}$ for $j = 0$ and in $\{2^{j-1} \leq |\xi| \leq 2^{j+1}\}$ for $j \geq 1$, so $\Delta_j \Delta_k = 0$ whenever $|j - k| \geq 2$.

Proof. $T_j f = K_j * f$ where $K_j = \mathcal{F}^{-1}\eta_j$. By the dilation identity (Volume 2, Prop. 4.1(iii)), $K_j(x) = 2^{jn} K(2^j x)$ with $K = \mathcal{F}^{-1}\eta$, so $\|K_j\|_{L^1} = \|K\|_{L^1}$ for every j , and $K \in L^1$ because $\eta \in C_c^\infty \subseteq \mathcal{S}$ so $K \in \mathcal{S}$. Young’s inequality (Volume 2, Worked Example 2.4) gives the bound. The support statements are immediate from the definitions, and disjointness of the annuli for $|j - k| \geq 2$ gives $\Delta_j \Delta_k = 0$. $\square$

4.6.3 The two scales

Definition 4.6.4. Let $s \in \mathbb{R}$, $1 \leq p \leq \infty$ and $1 \leq q \leq \infty$. For $f \in \mathcal{S}'(\mathbb{R}^n)$ set

$$\begin{aligned} \|f\|_{B_{p,q}^s} &= \left(\sum_{j \geq 0} (2^{js} \|\Delta_j f\|_{L^p})^q \right)^{1/q}, & \|f\|_{F_{p,q}^s} \\ &= \left\| \left(\sum_{j \geq 0} (2^{js} |\Delta_j f|)^q \right)^{1/q} \right\|_{L^p}, \end{aligned}$$

with the usual modification when $q = \infty$, and let $B_{p,q}^s, F_{p,q}^s$ be the sets where these are finite. For F we require $p < \infty$.

WORKED EXAMPLE 4.6.1— TIER A **Reading the three indices**

It helps to see what each index does on a single example. Fix a Schwartz function g with $\hat{g}$ supported in the annulus $\{\frac{1}{2} \leq |\xi| \leq 2\}$, and for a fixed N put

$$f = \sum_{j=1}^N a_j g(2^j \cdot),$$

so that $\Delta_j f = a_j g(2^j \cdot)$ for $1 \leq j \leq N$ and $\Delta_j f = 0$ otherwise.

The index p . By scaling, $\|g(2^j \cdot)\|_{L^p} = 2^{-jn/p} \|g\|_{L^p}$. So the L^p size of a piece at frequency 2^j already carries a factor $2^{-jn/p}$: higher frequencies are, at fixed amplitude, smaller in L^p for larger p . This is the scaling behind every embedding in Chapter 4.7.

The index s . The Besov norm is $(\sum_j (2^{js} |a_j| 2^{-jn/p})^q)^{1/q} \|g\|_p$. Raising s penalises high frequencies more, which is what “more derivatives” means.

The index q . Suppose $a_j 2^{-jn/p} 2^{js} = 1$ for all $j \leq N$, so every scale contributes equally. Then the Besov norm is $N^{1/q} \|g\|_p$: it sees q only through how the contributions of the N scales are combined. Small q punishes spreading mass over many scales; $q = \infty$ ignores it entirely. This is why q is invisible in the Sobolev scale — where it is fixed at 2 — and becomes decisive at the endpoints of Chapter 4.7.

Proposition 4.6.5 (Independence of the decomposition, Besov case). *Let $\{\Delta_j\}$ and $\{\tilde{\Delta}_j\}$ be built as in Theorem 4.6.2 from two admissible choices $\phi_0, \tilde{\phi}_0$. Then the resulting $B_{p,q}^s$ norms are comparable, with constants independent of f .*

Proof. By the support statements in Theorem 4.6.3, $\tilde{\Delta}_j$ and Δ_k have Fourier supports meeting only when $|j - k| \leq C_0$ for an absolute constant C_0 (one may

take $C_0 = 2$ after fixing the constants in Theorem 4.6.2). Since $\sum_k \Delta_k = \text{Id}$,

$$\tilde{\Delta}_j f = \sum_{|k-j| \leq C_0} \tilde{\Delta}_j \Delta_k f.$$

By Theorem 4.6.3 applied to $\tilde{\Delta}_j$, $\|\tilde{\Delta}_j \Delta_k f\|_{L^p} \leq C \|\Delta_k f\|_{L^p}$ with C independent of j , so

$$\begin{aligned} 2^{js} \|\tilde{\Delta}_j f\|_{L^p} &\leq C \sum_{|k-j| \leq C_0} 2^{js} \|\Delta_k f\|_{L^p} \\ &\leq C 2^{C_0|s|} \sum_{|k-j| \leq C_0} 2^{ks} \|\Delta_k f\|_{L^p}. \end{aligned}$$

Taking the ℓ^q norm in j and using that the sum has at most $2C_0 + 1$ terms, each of which is a shift of the sequence $(2^{ks} \|\Delta_k f\|_p)_k$, the triangle inequality in ℓ^q gives $\|f\|_{\tilde{B}_{p,q}^s} \lesssim_{s,C_0} \|f\|_{B_{p,q}^s}$. Exchanging the roles of the two systems gives the reverse. $\square$

Pitfall: the same argument does not settle the F case

In Theorem 4.6.5 the L^p norm was taken *before* the sum over j , so Theorem 4.6.3 could be applied to one j at a time. In the Triebel–Lizorkin norm the ℓ^q sum is inside the L^p norm, and one needs to bound $\|(\sum_j |\tilde{\Delta}_j \Delta_k f|^q)^{1/q}\|_p$ by the corresponding quantity for $\Delta_k f$ — a *vector-valued* statement.

The standard route is through the Peetre maximal function, which reduces it to the Fefferman–Stein vector-valued maximal inequality; and that inequality was proved in Chapter 4.5 only for $q < p$. So the general case is not available here. It is available at $q = 2$, by a different route: Theorem 4.6.8 below identifies $F_{p,2}^s$ with a space defined without reference to any decomposition.

4.6.4 Comparison of the two scales

Theorem 4.6.6 (Minkowski embeddings). *For $s \in \mathbb{R}$, $1 \leq p < \infty$ and $1 \leq q \leq \infty$,*

$$B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s,$$

with norm inequalities in the same directions. In particular $B_{p,p}^s = F_{p,p}^s$.

Idea of the proof. Everything is an instance of one principle: interchanging an ℓ^q norm and an L^p norm costs nothing in the direction that puts the *smaller* exponent on the outside. Concretely, Minkowski’s integral inequality says $\|\cdot\|_{\ell^q(L^p)} \leq \|\cdot\|_{L^p(\ell^q)}$ when $p \leq q$, and the reverse when $q \leq p$. The

four cases below are that statement, together with the elementary inclusion $\ell^{q_1} \subseteq \ell^{q_2}$ for $q_1 \leq q_2$.

Proof. Write $g_j = 2^{js} \Delta_j f$, so $\|f\|_{B_{p,q}^s} = \|(\|g_j\|_{L^p})_j\|_{\ell^q}$ and $\|f\|_{F_{p,q}^s} = \| \| (g_j)_j \|_{\ell^q} \|_{L^p}$.

Step 1. $B_{p,q}^s \subseteq F_{p,q}^s$ when $q \leq p$.

Put $r = p/q \geq 1$. By the triangle inequality in L^r ,

$$\begin{aligned} \|f\|_{F_{p,q}^s}^q &= \left\| \sum_j |g_j|^q \right\|_{L^r} \\ &\leq \sum_j \| |g_j|^q \|_{L^r} \\ &= \sum_j \|g_j\|_{L^p}^q \\ &= \|f\|_{B_{p,q}^s}^q. \end{aligned}$$

Step 2. $B_{p,p}^s \subseteq F_{p,q}^s$ when $q \geq p$.

Now $r = p/q \leq 1$, and for $0 < r \leq 1$ one has the elementary inequality $(\sum_j a_j)^r \leq \sum_j a_j^r$ for $a_j \geq 0$. Applying it pointwise and integrating,

$$\begin{aligned} \|f\|_{F_{p,q}^s}^p &= \int \left(\sum_j |g_j|^q \right)^{p/q} \\ &\leq \int \sum_j |g_j|^p \\ &= \sum_j \|g_j\|_{L^p}^p \\ &= \|f\|_{B_{p,p}^s}^p. \end{aligned}$$

Steps 1 and 2 together give $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s$, using $B_{p,q_1}^s \subseteq B_{p,q_2}^s$ for $q_1 \leq q_2$ (the inclusion $\ell^{q_1} \subseteq \ell^{q_2}$).

Step 3. $F_{p,q}^s \subseteq B_{p,q}^s$ when $p \leq q$.

This is Minkowski's integral inequality in the form $\|(\|g_j\|_{L^p})_j\|_{\ell^q} \leq \| \| (g_j)_j \|_{\ell^q} \|_{L^p}$, valid for $p \leq q$; it is the statement that the $L^{q/p}$ norm of a sum is at most the sum of the norms, applied to $|g_j|^p$.

Step 4. $F_{p,q}^s \subseteq B_{p,p}^s$ when $q \leq p$.

Since $\ell^q \subseteq \ell^p$ for $q \leq p$,

$$\begin{aligned} \|f\|_{B_{p,p}^s} &= \left\| \left(\sum_j |g_j|^p \right)^{1/p} \right\|_{L^p} \\ &\leq \left\| \left(\sum_j |g_j|^q \right)^{1/q} \right\|_{L^p} \\ &= \|f\|_{F_{p,q}^s}, \end{aligned}$$

the first equality by Tonelli. Steps 3 and 4 give $F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$.

Taking $q = p$ in all four steps gives $B_{p,p}^s = F_{p,p}^s$, as both chains collapse. $\square$

Theorem 4.6.7 (Completeness). *For $s \in \mathbb{R}$, $1 \leq p, q \leq \infty$, the space $B_{p,q}^s(\mathbb{R}^n)$ is a Banach space, and $B_{p,q}^s \subseteq \mathcal{S}'(\mathbb{R}^n)$ continuously.*

Proof. Step 1. It is a normed space, and embeds in $\mathcal{S}'$.

That $\|\cdot\|_{B_{p,q}^s}$ is a seminorm is the triangle inequality in L^p and then in ℓ^q . If the norm vanishes then every $\Delta_j f = 0$, so $\hat{f}$ vanishes on each annulus and hence on $\mathbb{R}^n \setminus \{0\}$; since $\phi_0 \hat{f} = 0$ as well, $\hat{f} = 0$ and $f = 0$.

For the embedding, let $\varphi \in \mathcal{S}$. Then $|\langle f, \varphi \rangle| \leq \sum_j |\langle \Delta_j f, \varphi \rangle| \leq \sum_j \|\Delta_j f\|_{L^p} \|\Delta'_j \varphi\|_{L^{p'}}$, where $\Delta'_j = \sum_{|k-j| \leq 1} \Delta_k$ satisfies $\Delta'_j \Delta_j = \Delta_j$. Since $\varphi \in \mathcal{S}$, $\|\Delta'_j \varphi\|_{L^{p'}} \lesssim_N 2^{-jN}$ for every N — differentiate the symbol and use Theorem 4.6.3 — so choosing $N > |s| + 1$ and applying Hölder in j bounds the sum by $C_{N,\varphi} \|f\|_{B_{p,q}^s}$.

Step 2. *Absolutely convergent series converge.*

By Theorem 1.8.6 (Riesz–Fischer) it suffices to show that if $\sum_m \|f_m\|_{B_{p,q}^s} < \infty$ then $\sum_m f_m$ converges in $B_{p,q}^s$. For each fixed j ,

$$\sum_m \|\Delta_j f_m\|_{L^p} \leq 2^{-js} \sum_m \|f_m\|_{B_{p,q}^s} < \infty,$$

so by completeness of L^p the series $\sum_m \Delta_j f_m$ converges in L^p to some g_j , whose Fourier transform is supported in the j -th annulus because each term's is and the support condition is closed under L^p limits (test against Schwartz functions vanishing there).

By Step 1 the partial sums also converge in $\mathcal{S}'$; call the limit f . Then $\Delta_j f = g_j$, since Δ_j is continuous on $\mathcal{S}'$. Finally, for every M ,

$$\left(\sum_{j \leq M} (2^{js} \|\Delta_j f\|_{L^p})^q \right)^{1/q} \leq \sum_m \|f_m\|_{B_{p,q}^s},$$

by the triangle inequality applied to the finitely many j ; letting $M \rightarrow \infty$ shows $f \in B_{p,q}^s$, and the same estimate applied to the tails $\sum_{m > M_0} f_m$ shows the series converges in norm. $\square$

4.6.5 Where the Sobolev spaces sit

Theorem 4.6.8. *Let $1 < p < \infty$ and $s \geq 0$. Then $F_{p,2}^s(\mathbb{R}^n) = W^{s,p}(\mathbb{R}^n)$ with equivalent norms, where $W^{s,p} = \{f : (I - \Delta)^{s/2} f \in L^p\}$ is the Bessel potential space. In particular $F_{p,2}^0 = L^p$ for $1 < p < \infty$.*

Proof. The case $s = 0$ is exactly the Littlewood–Paley theorem (Theorem 3.6.3 (the square function theorem)), transferred from the homogeneous decomposition used there to the inhomogeneous one of Theorem 4.6.2: the two differ only in that the frequencies $|\xi| \lesssim 1$ are collected into the single piece Δ_0 rather than spread over $j \leq 0$, and Theorem 4.6.3 shows that collecting them changes the norm by at most a constant factor.

For general $s \geq 0$, apply Worked Example 3.6.1 (Bernstein and Littlewood–Paley), which states

$$\left\| \left(\sum_j 2^{2js} |\Delta_j f|^2 \right)^{1/2} \right\|_{L^p} \approx_{n,p,s} \|(-\Delta)^{s/2} f\|_{L^p},$$

the point being that on the support of $\widehat{\Delta_j f}$ the multiplier $(2\pi|\xi|)^s$ is comparable to 2^{js} , with all derivatives satisfying the Hörmander–Mikhlin bounds after rescaling. Replacing $(-\Delta)^{s/2}$ by $(I - \Delta)^{s/2}$ changes the multiplier only on $|\xi| \lesssim 1$, where both are bounded above and below, so the norms remain comparable. $\square$

Corollary 4.6.9. *For $s \in \mathbb{R}$, $B_{2,2}^s = F_{2,2}^s = H^s(\mathbb{R}^n)$ with equivalent norms.*

Proof. $B_{2,2}^s = F_{2,2}^s$ by Theorem 4.6.6 with $p = q = 2$. By Plancherel and the disjointness of the annuli (Theorem 4.6.3),

$$\|f\|_{B_{2,2}^s}^2 = \sum_j 2^{2js} \|\widehat{\Delta_j f}\|_{L^2}^2 \approx \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi,$$

because $(1 + |\xi|^2)^s \approx 2^{2js}$ on the support of $\widehat{\Delta_j f}$ and each ξ lies in at most two supports. $\square$

WORKED EXAMPLE 4.6.2— TIER C: RESEARCH TASTE **Why the later volumes use ℓ^q outside the norm**

Worked Example 3.6.2 (what Littlewood–Paley cannot do) observed that the ℓ^2 decoupling inequality is stated as

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p(\mathbb{R}^n)} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left(\sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}, \quad (4.6.2)$$

with the ℓ^2 sum *outside* the L^p norm, and not in the square function form.

This chapter supplies the vocabulary to say precisely what that difference is, and why the weaker statement is the one that is true.

(a) *The two shapes are Besov and Triebel–Lizorkin.* The right-hand side of (4.6.2) is an $\ell^2(L^p)$ norm of the family $\{f_\theta\}$ — Besov-shaped. The square function alternative $\|(\sum_\theta |f_\theta|^2)^{1/2}\|_{L^p}$ is $L^p(\ell^2)$ — Triebel–Lizorkin-shaped. By Theorem 4.6.6, for $p \geq 2$,

$$\left(\sum_\theta \|f_\theta\|_{L^p}^2\right)^{1/2} \leq \left\|\left(\sum_\theta |f_\theta|^2\right)^{1/2}\right\|_{L^p},$$

so the Besov-shaped right-hand side is *smaller*, and (4.6.2) is correspondingly the *stronger* statement of the two. It is nevertheless the one that is true, while the square function form — which would be an assertion that the pieces are almost orthogonal pointwise — is false.

(b) *Why the square function form fails.* Theorem 3.6.3 (the square function theorem) proved the square function identity $\|f\|_p \approx \|\mathcal{S}f\|_p$ for *lacunary* frequency pieces, where the annuli are separated by a factor 2 and each randomised sum is a Hörmander–Mikhlin multiplier. Decoupling caps are adjacent: $\delta^{-(n-1)/2}$ of them tile a neighbourhood of the paraboloid, and consecutive caps are at distance δ , not at ratio 2. No multiplier theorem applies, and the pointwise orthogonality that the square function asserts is simply absent.

(c) *What survives, and why it is enough.* What curvature does supply is that the tubes dual to different caps point in different directions, so the pieces cannot all be large on a common set. That is an assertion about the *sizes* $\|f_\theta\|_{L^p}$ and how they can add up, not about the pointwise values — which is exactly an $\ell^2(L^p)$ statement. So the Besov shape is not a weakening forced by technical difficulty; it is the shape that the geometry actually produces.

The transferable lesson. When reading an estimate in this subject, look first at whether the ℓ^q sum is inside or outside the L^p norm. Inside means a pointwise orthogonality claim, which needs a multiplier theorem and hence frequency separation. Outside means a claim about how the sizes of the pieces combine, which is what geometric input gives. Theorem 4.6.6 tells you which is stronger, and for $p \geq 2$ it is always the one with the sum outside.

WORKED EXAMPLE 4.6.3 — TIER B **Choosing between Besov and Triebel–Lizorkin**

The two scales $B_{p,q}^s$ and $F_{p,q}^s$ differ by the order in which the ℓ^q and L^p norms are taken. Which to use is decided by the problem, and the rule is short.

The definitions, side by side.

$$\|f\|_{B_{p,q}^s} = \left\| 2^{js} \|\Delta_j f\|_{L^p} \right\|_{\ell_j^q}, \quad \|f\|_{F_{p,q}^s} = \left\| \|2^{js} \Delta_j f\|_{\ell_j^q} \right\|_{L^p}.$$

Besov integrates each block separately then sums; Triebel–Lizorkin sums pointwise then integrates.

Use Besov when the blocks are handled independently. Interpolation, trace theorems, and embeddings are all blockwise, so Besov is natural. The embedding $B_{p,1}^{n/p} \subseteq L^\infty$ is the model, and it is sharp precisely because $q = 1$ is what makes a sum of blocks converge uniformly.

Use Triebel–Lizorkin when the blocks interact pointwise. Sobolev spaces are $F_{p,2}^s$, and $H^1 = F_{1,2}^0$, $\text{BMO} = F_{\infty,2}^0$ — so the whole Calderón–Zygmund theory of Volume 3 lives on this scale. The square function of Theorem 3.6.3 is exactly the inner ℓ^2 norm.

The one relation to remember. $B_{p,\min(p,q)}^s \subseteq F_{p,q}^s \subseteq B_{p,\max(p,q)}^s$, by Minkowski’s inequality in whichever direction the exponents allow. The two scales agree when $p = q$, and only then.

A caution. Results stated on one scale do not automatically transfer. Theorem 4.6.1 records that the identifications of F spaces with H^1 and BMO are not proved in this series, so an argument here that needs them must say so.

Drills

Drill 4.6.1. Verify the telescoping identity (4.6.1), and check that $\text{supp } \psi \subseteq \{\frac{1}{2} \leq |\xi| \leq 2\}$ for $\psi = \phi_0 - \phi_0(2\cdot)$.

Drill 4.6.2. Show $B_{p,q_1}^{s_1} \subseteq B_{p,q_2}^{s_2}$ whenever $s_1 > s_2$, for any q_1, q_2 , and whenever $s_1 = s_2$ and $q_1 \leq q_2$. Which inclusion needs a summation and which is free?

Drill 4.6.3. Prove that the elementary inequality $(\sum_j a_j)^r \leq \sum_j a_j^r$ used in Step 2 of Theorem 4.6.6 holds for $0 < r \leq 1$ and $a_j \geq 0$, and show it reverses for $r \geq 1$.

Drill 4.6.4. Compute $\|f\|_{B_{p,q}^s}$ for $f = g(2^{j_0}\cdot)$ with $\widehat{g}$ supported in the annulus, and confirm the scaling $2^{j_0(s-n/p)}$. Deduce that $B_{p,q}^s$ is invariant under the dilations preserving that exponent.

Notes

Besov spaces were introduced by Besov in 1959 and the Triebel–Lizorkin scale by Triebel and Lizorkin independently around 1970. The standard reference is Triebel’s *Theory of Function Spaces*, where the full theory — including the identifications $F_{1,2}^0 = H^1$ and $F_{\infty,2}^0 = \text{BMO}$ omitted here by Theorem 4.6.1 — is developed.

The Peetre maximal function, mentioned in the Pitfall of §6.3, is $\sup_y |\Delta_j f(x-y)|(1+2^j|y|)^{-a}$; it is the standard device for converting statements about frequency-localised pieces into statements about maximal functions, and it is the reason the Triebel–Lizorkin theory needs the Fefferman–Stein vector-valued inequality in its full range.

Chapter 4.7

Embeddings and the Sobolev Scale

4.7.1 Bernstein, read as an embedding

Volume 2 proved Bernstein's inequality: a function whose Fourier transform is supported in a ball of radius N satisfies $\|f\|_{L^r} \lesssim N^{n(1/p-1/r)} \|f\|_{L^p}$ for $p \leq r$. Every Littlewood–Paley piece has exactly that property, so Bernstein applies to each piece separately — and because the Besov norm takes the L^p norm of each piece *before* summing, the inequality passes straight through.

Theorem 4.7.1 (Besov embedding). *Let $1 \leq p \leq r \leq \infty$, $1 \leq q \leq \infty$ and $s \in \mathbb{R}$. Then*

$$B_{p,q}^s(\mathbb{R}^n) \subseteq B_{r,q}^{s-n(\frac{1}{p}-\frac{1}{r})}(\mathbb{R}^n),$$

with $\|f\|_{B_{r,q}^{s-n(\frac{1}{p}-\frac{1}{r})}} \lesssim_{n,p,r} \|f\|_{B_{p,q}^s}$.

Proof. Write $\sigma = s - n(\frac{1}{p} - \frac{1}{r})$. For $j \geq 1$ the function $\Delta_j f$ has $\widehat{\Delta_j f}$ supported in $\{|\xi| \leq 2^{j+1}\}$, and for $j = 0$ in $\{|\xi| \leq 2\}$; so in all cases Theorem 2.6.7 (Bernstein) gives

$$\|\Delta_j f\|_{L^r} \lesssim_{n,p,r} 2^{jn(\frac{1}{p}-\frac{1}{r})} \|\Delta_j f\|_{L^p}.$$

Multiplying by $2^{j\sigma}$,

$$\begin{aligned} 2^{j\sigma} \|\Delta_j f\|_{L^r} &\lesssim 2^{j\sigma} 2^{jn(\frac{1}{p}-\frac{1}{r})} \|\Delta_j f\|_{L^p} \\ &= 2^{js} \|\Delta_j f\|_{L^p}. \end{aligned}$$

Taking the ℓ^q norm over $j \geq 0$ gives the claim. □

Where we are. Note what did *not* happen: the index q was untouched. Bernstein acts on one piece at a time, and the Besov norm never mixes pieces, so the embedding is free of any loss in q . This is the structural advantage of the Besov scale, and the reason the sharp endpoint of §7.3 is visible there and not on the Sobolev scale.

By contrast, the same argument applied to the Triebel–Lizorkin norm would require comparing $\|(\sum_j |\Delta_j f|^q)^{1/q}\|_{L^r}$ with the corresponding quantity in L^p , which mixes the pieces and is not a consequence of Bernstein.

Proposition 4.7.2 (Monotonicity in q and in s). *Let $1 \leq p \leq \infty$.*

(i) *If $q_1 \leq q_2$ then $B_{p,q_1}^s \subseteq B_{p,q_2}^s$, with norm at most 1.*

(ii) *If $s_1 > s_2$ then $B_{p,q_1}^{s_1} \subseteq B_{p,q_2}^{s_2}$ for all q_1, q_2 , with $\|f\|_{B_{p,q_2}^{s_2}} \lesssim_{s_1-s_2, q_2} \|f\|_{B_{p,q_1}^{s_1}}$.*

Proof. (i) is the inclusion $\ell^{q_1} \subseteq \ell^{q_2}$, whose norm is 1.

(ii) Write $a_j = \|\Delta_j f\|_{L^p}$ and $\delta = s_1 - s_2 > 0$. For every j ,

$$2^{js_2} a_j = 2^{-j\delta} \cdot 2^{js_1} a_j \leq 2^{-j\delta} \|f\|_{B_{p,\infty}^{s_1}}.$$

Taking the ℓ^{q_2} norm over $j \geq 0$,

$$\begin{aligned} \|f\|_{B_{p,q_2}^{s_2}} &\leq \|f\|_{B_{p,\infty}^{s_1}} \left(\sum_{j \geq 0} 2^{-jq_2\delta} \right)^{1/q_2} \\ &= \frac{\|f\|_{B_{p,\infty}^{s_1}}}{(1 - 2^{-q_2\delta})^{1/q_2}}, \end{aligned}$$

the geometric series converging because $\delta > 0$ and the sum runs over $j \geq 0$ only. Finally $\|f\|_{B_{p,\infty}^{s_1}} \leq \|f\|_{B_{p,q_1}^{s_1}}$ by (i). $\square$

Pitfall: (ii) uses that the decomposition is inhomogeneous

The sum in Theorem 4.7.2(ii) runs over $j \geq 0$, and the geometric series converges only in that direction. Had we used the homogeneous decomposition, with j ranging over all of $\mathbb{Z}$, the low frequencies $j \rightarrow -\infty$ would contribute $2^{-j\delta} \rightarrow \infty$ and no such inclusion would hold.

This is the general reason the inhomogeneous decomposition of Theorem 4.6.2 is the right one for a theory of embeddings: it treats the low frequencies as a single bounded block, so that only the behaviour at high frequency — which is what smoothness means — can cause trouble. The homogeneous scale is the natural one for scaling-invariant statements, and the two should not be confused.

4.7.2 The sharp endpoint into L^∞

Theorem 4.7.3. *Let $1 \leq p < \infty$. Then*

$$B_{p,1}^{n/p}(\mathbb{R}^n) \subseteq C_0(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n), \quad \|f\|_{L^\infty} \lesssim_{n,p} \|f\|_{B_{p,1}^{n/p}}.$$

Idea of the proof. Bernstein at $r = \infty$ says $\|\Delta_j f\|_\infty \lesssim 2^{jn/p} \|\Delta_j f\|_p$. Summing over j and using $f = \sum_j \Delta_j f$ gives the result — provided the sum $\sum_j 2^{jn/p} \|\Delta_j f\|_p$ converges, which is exactly the statement $f \in B_{p,1}^{n/p}$. So the theorem is Bernstein plus the triangle inequality, and the index $q = 1$ is present precisely because a triangle inequality over j is being used.

Proof. By Theorem 4.7.1 with $r = \infty$ and $s = n/p$, the target smoothness is $s - n/p = 0$, so

$$\|f\|_{B_{\infty,1}^0} \lesssim_{n,p} \|f\|_{B_{p,1}^{n/p}}.$$

It therefore suffices to show $B_{\infty,1}^0 \subseteq L^\infty$ with $\|f\|_\infty \leq \|f\|_{B_{\infty,1}^0}$, and that is immediate:

$$\begin{aligned} \|f\|_{L^\infty} &= \left\| \sum_{j \geq 0} \Delta_j f \right\|_{L^\infty} \\ &\leq \sum_{j \geq 0} \|\Delta_j f\|_{L^\infty} \\ &= \|f\|_{B_{\infty,1}^0}. \end{aligned}$$

For the continuity statement, each $\Delta_j f$ has compactly supported Fourier transform, hence extends to a smooth function; and $\Delta_j f \in L^p$ with $p < \infty$ together with the bound on its derivatives from Theorem 2.6.7 (Bernstein) forces $\Delta_j f(x) \rightarrow 0$ as $|x| \rightarrow \infty$. The series $\sum_j \Delta_j f$ converges uniformly, by the displayed bound, so its sum lies in C_0 . $\square$

4.7.3 A loose end from Volume 2, resolved

Volume 2, Chapter 7 proved that $H^s(\mathbb{R}^n) \subseteq C_0$ for $s > n/2$, and its Pitfall box showed that the endpoint $s = n/2$ fails: there is $f \in H^{n/2}$ which is unbounded, and the failure is logarithmic. The reason was not explained there, because the vocabulary did not exist. It does now.

Theorem 4.7.4. *Let $1 < p < \infty$. Then*

$$W^{n/p,p}(\mathbb{R}^n) \subseteq B_{p,\max(p,2)}^{n/p}(\mathbb{R}^n),$$

and this is sharp in the third index: $W^{n/p,p} \not\subseteq L^\infty$, whereas $B_{p,1}^{n/p} \subseteq L^\infty$ by Theorem 4.7.3.

Proof. By Theorem 4.6.8, $W^{n/p,p} = F_{p,2}^{n/p}$, and by the Minkowski embeddings (Theorem 4.6.6) $F_{p,2}^s \subseteq B_{p,\max(p,2)}^s$. The failure $W^{n/p,p} \not\subseteq L^\infty$ is Volume 2, Chapter 7, Pitfall, for $p = 2$; the same construction with $\widehat{f}(\xi) = |\xi|^{-n}(\log|\xi|)^{-1}\mathbf{1}_{|\xi|>2}$ works for general p after adjusting exponents. $\square$

WORKED EXAMPLE 4.7.1— TIER B The failure, seen through the third index

Take $p = 2$ and let f be the Volume 2 counterexample, $\widehat{f}(\xi) = |\xi|^{-n}(\log|\xi|)^{-1}\mathbf{1}_{|\xi|>2}$. We compute both Besov norms at smoothness $n/2$ and see exactly where the difference lies.

On the annulus $|\xi| \approx 2^j$ we have $|\widehat{f}| \approx 2^{-jn}j^{-1}$ over a set of measure $\approx 2^{jn}$, so by Plancherel

$$\|\Delta_j f\|_{L^2} \approx \left(2^{jn} \cdot 2^{-2jn}j^{-2}\right)^{1/2} = \frac{2^{-jn/2}}{j},$$

and therefore

$$2^{jn/2}\|\Delta_j f\|_{L^2} \approx \frac{1}{j}.$$

Now everything is decided by which ℓ^q norm is taken of the sequence $(1/j)_{j \geq 2}$:

$$\begin{aligned} \|(1/j)_j\|_{\ell^2} &= \left(\sum_j j^{-2}\right)^{1/2} < \infty \\ , \quad \|(1/j)_j\|_{\ell^1} &= \sum_j \frac{1}{j} = \infty. \end{aligned}$$

So $f \in B_{2,2}^{n/2} = H^{n/2}$ but $f \notin B_{2,1}^{n/2}$, and Theorem 4.7.3 does not apply — correctly, since f is unbounded.

The point. The divergence of $\sum 1/j$ is exactly the “logarithmic failure” that Volume 2 could only observe. Here it is located precisely: the contributions of the dyadic frequency blocks are square-summable but not summable, and boundedness needs summability because it is proved by a triangle inequality over blocks. The third index is the bookkeeping device that distinguishes the two, and the Sobolev scale, which fixes $q = 2$, cannot see it.

Remark 4.7.5 (The Sobolev embedding itself). One may also run Theorem 4.7.1 to recover Sobolev-type embeddings between L^p spaces. Chaining

$$W^{s,p} = F_{p,2}^s \subseteq B_{p,\max(p,2)}^s \subseteq B_{r,\max(p,2)}^0 \subseteq F_{r,2}^0 = L^r, \quad s = n\left(\frac{1}{p} - \frac{1}{r}\right),$$

where the middle inclusion is Theorem 4.7.1 and the last requires $\max(p, 2) \leq \min(r, 2)$ by Theorem 4.6.6, gives $W^{s,p} \subseteq L^r$ whenever $p \leq 2 \leq r$.

The restriction $p \leq 2 \leq r$ is an artefact of this soft route: the chain loses in the third index at both ends. The full Sobolev embedding, for all $1 < p < r < \infty$, was already proved in Theorem 2.7.6 (Gagliardo–Nirenberg–Sobolev) by the Gagliardo–Nirenberg argument, which uses no Fourier analysis at all. So nothing is missing; the Besov route simply is not the efficient one for that particular statement. What it *does* give, and the other route does not, is Theorem 4.7.3.

4.7.4 Interpolation on the Besov scale

Theorem 4.7.6 (Gagliardo–Nirenberg on the Besov scale). *Let $1 \leq p \leq \infty$, let $s_0 \neq s_1$ and $1 \leq q_0, q_1 \leq \infty$, let $\theta \in (0, 1)$, and set*

$$s = (1 - \theta)s_0 + \theta s_1, \quad \frac{1}{q} = \frac{1 - \theta}{q_0} + \frac{\theta}{q_1}.$$

Then

$$\|f\|_{B_{p,q}^s} \leq \|f\|_{B_{p,q_0}^{s_0}}^{1-\theta} \|f\|_{B_{p,q_1}^{s_1}}^{\theta}.$$

Proof. Write $a_j = \|\Delta_j f\|_{L^p}$. Since $s = (1 - \theta)s_0 + \theta s_1$,

$$2^{js} a_j = (2^{js_0} a_j)^{1-\theta} (2^{js_1} a_j)^{\theta}$$

for every j — this is just splitting the exponent. Now apply Hölder’s inequality for sums (Theorem 1.8.3 (generalised Hölder), in the generalised form) with exponents $\frac{q_0}{(1-\theta)q}$ and $\frac{q_1}{\theta q}$, which are conjugate by the definition of q :

$$\begin{aligned} \sum_j (2^{js} a_j)^q &= \sum_j (2^{js_0} a_j)^{(1-\theta)q} (2^{js_1} a_j)^{\theta q} \\ &\leq \left(\sum_j (2^{js_0} a_j)^{q_0} \right)^{\frac{(1-\theta)q}{q_0}} \left(\sum_j (2^{js_1} a_j)^{q_1} \right)^{\frac{\theta q}{q_1}}. \end{aligned}$$

Raising to the power $1/q$ gives the claim. The cases $q_0 = \infty$ or $q_1 = \infty$ follow by the same computation with the corresponding factor bounded by its supremum. $\square$

WORKED EXAMPLE 4.7.2— TIER A Interpolation, and why $s_0 \neq s_1$ is not needed in the proof

The proof of Theorem 4.7.6 nowhere used $s_0 \neq s_1$: the identity $2^{js}a_j = (2^{js_0}a_j)^{1-\theta}(2^{js_1}a_j)^\theta$ holds regardless. When $s_0 = s_1 = s$ the conclusion reads $\|f\|_{B_{p,q}^s} \leq \|f\|_{B_{p,q_0}^s}^{1-\theta} \|f\|_{B_{p,q_1}^s}^\theta$, which is the log-convexity of the ℓ^q scale — true, but weaker than Theorem 4.7.2(i) in most cases.

The hypothesis $s_0 \neq s_1$ is included in the statement only because that is the case one uses: interpolating between two different smoothnesses. A typical application is

$$\begin{aligned} \|f\|_{B_{p,1}^s} &\leq \|f\|_{B_{p,\infty}^{s_0}}^{1-\theta} \|f\|_{B_{p,\infty}^{s_1}}^\theta \\ , \quad s_0 < s < s_1, \quad \theta &= \frac{s - s_0}{s_1 - s_0}, \end{aligned}$$

which reaches the summable index $q = 1$ — the one Theorem 4.7.3 needs — from two weak $q = \infty$ hypotheses. Note that $\frac{1}{q} = (1 - \theta)/\infty + \theta/\infty = 0$ would give $q = \infty$, so this particular application requires $q_0 = q_1 = \infty$ to be replaced by finite values; Drill 7.4 asks for the correct form.

WORKED EXAMPLE 4.7.3— TIER C: RESEARCH TASTE Where the third index is doing work in the later volumes

The distinction between $q = 1$, $q = 2$ and $q = \infty$ at a fixed smoothness looks like a technicality. It is not, and three appearances in the later volumes are worth naming.

(a) *Strichartz estimates.* A Strichartz estimate bounds $\|e^{it\Delta}f\|_{L_t^q L_x^r}$ by $\|f\|_{H^s}$ for exponents on a scaling line. At the *endpoint* of the admissible range the estimate is either false or requires the Besov refinement — one must replace H^s by $B_{2,1}^s$ or the target by a Besov space — for exactly the reason of Worked Example 4.7.1: at the endpoint the dyadic blocks contribute equally, and whether the total converges depends on whether one is summing or square-summing them. Keel and Tao’s endpoint Strichartz theorem is the sharp statement.

(b) *Local smoothing.* The local smoothing conjecture for the wave equation asserts a gain of $1/p$ derivatives in L^p over the fixed-time estimate, and was proved for $n = 2$ by Guth, Wang and Zhang in 2020 using decoupling. “A gain of $1/p$ derivatives” is an embedding statement between spaces on this scale, and the sharp form of the conjecture specifies the third index.

(c) *Decoupling itself.* Worked Example 3.6.2 (what Littlewood–Paley cannot do) and Worked Example 4.6.2 identified the ℓ^2 decoupling inequality as a Besov-shaped statement: ℓ^2 of L^p norms, not L^p of an ℓ^2 square function. The $\delta^{-\varepsilon}$ loss in that inequality plays the same role as the third index here — it is the amount by which one is allowed to fail to sum over $\log(1/\delta)$ scales. That last comparison is worth dwelling on. In Worked Example 4.7.1 the obstruction was $\sum_j 1/j = \infty$ over $\approx \log$ many blocks; in decoupling the obstruction is a factor that must be beaten over $\log(1/\delta)$ scales. Both are the statement that an estimate which is lossless at each scale may still fail when the scales are summed, and both are repaired by weakening the index rather than the estimate: $q = 2$ instead of $q = 1$ here, $\delta^{-\varepsilon}$ instead of $O(1)$ there.

WORKED EXAMPLE 4.7.4— TIER B **Reading an embedding**

Sobolev and Besov embeddings are quoted constantly and their hypotheses vary. Four checks.

(1) *Which scale?* $W^{s,p}$, $B_{p,q}^s$ or $F_{p,q}^s$. Volume 4, Worked Example 4.6.3 explains that the last two differ by the order of the ℓ^q and L^p norms, and embeddings on one scale do not transfer verbatim to the other.

(2) *Is the embedding critical?* At the critical exponent the target is L^∞ or BMO rather than a Hölder space, and the statement usually fails without an extra summability hypothesis — $q = 1$ in the Besov case.

(3) *Which domain?* $\mathbb{R}^n$, a bounded domain, or a torus. On $\mathbb{R}^n$ homogeneity forces a single exponent relation; on a bounded domain lower-order terms may be absorbed and the statement is weaker but easier.

(4) *Is it compact?* Rellich–Kondrachov gives compactness on bounded domains and fails on $\mathbb{R}^n$, where translation provides a non-convergent bounded sequence.

The scaling check. As always, test on $f(\lambda \cdot)$: the exponent relation in any homogeneous embedding is forced, and computing it takes a line. An embedding whose exponents fail the scaling check is misquoted.

Drills

Drill 4.7.1. Verify Theorem 4.7.1 directly for $f = g(2^{j_0} \cdot)$ with $\hat{g}$ supported in the annulus, by computing both sides.

Drill 4.7.2. Show that Theorem 4.7.2(ii) fails for the homogeneous decomposition, by exhibiting a sequence $(a_j)_{j \in \mathbb{Z}}$ finite in one norm and infinite in the

other.

Drill 4.7.3. Carry out the computation of Worked Example 4.7.1 for general p : find the correct counterexample $\widehat{f}$ and confirm that $(2^{jn/p}\|\Delta_j f\|_p)_j$ behaves like $(1/j)_j$.

Drill 4.7.4. State and prove the version of Theorem 4.7.6 that produces $q = 1$ from finite q_0, q_1 , and identify the constraint linking θ , q_0 and q_1 .

Notes

Theorem 4.7.3 is standard and is the single most-quoted advantage of the Besov scale over the Sobolev scale. The failure of $W^{n/p,p} \subseteq L^\infty$, and its repair by lowering the third index to 1, is often stated as the slogan that “ $B_{p,1}^{n/p}$ is the correct critical space”.

The endpoint Strichartz theorem referred to in Worked Example 4.7.3 is due to Keel and Tao (1998). The local smoothing conjecture is Sogge’s, from 1991, and the two-dimensional case is Guth–Wang–Zhang (2020); both are discussed in Volume 13.

Chapter 4.8

Where These Spaces Are Used

This volume has built machinery. This chapter says what it is for, by exhibiting the estimates of Volumes 12 and 13 in the language just developed. Nothing here is proved — the theorems belong to later volumes — but every *statement* is now one the reader can parse completely, and that is the point: a large part of the difficulty of entering this literature is not the arguments but the notation.

4.8.1 Four questions the reader can now answer

WORKED EXAMPLE 4.8.1 — TIER B **Why estimates are stated in mixed norms** $L_t^q L_x^r$

A Strichartz estimate for the free Schrödinger evolution has the form

$$\begin{aligned} \|e^{it\Delta} f\|_{L_t^q L_x^r(\mathbb{R} \times \mathbb{R}^n)} &\lesssim \|f\|_{L^2(\mathbb{R}^n)} \\ &\quad , \quad \frac{2}{q} + \frac{n}{r} = \frac{n}{2}, \quad q > 2. \end{aligned}$$

The left side is $\| \|e^{it\Delta} f(\cdot)\|_{L_x^r} \|_{L_t^q}$: an L^r norm in space taken first, an L^q norm in time second.

Why the order matters. This is the Besov-versus-Triebel–Lizorkin question of Theorem 4.6.6 with the roles of j and x played by t and x . By Minkowski’s integral inequality, if $q \leq r$ then $\|\cdot\|_{L_t^q L_x^r} \geq \|\cdot\|_{L_x^r L_t^q}$, so the two orderings give genuinely different — and non-equivalent — statements, the harder one being whichever puts the larger exponent on the inside.

Why the exponent relation. Scaling. If $f_\lambda(x) = f(\lambda x)$ then $e^{it\Delta} f_\lambda(x) = (e^{i\lambda^2 t \Delta} f)(\lambda x)$, so the left side scales by $\lambda^{-2/q-n/r}$ and the right by $\lambda^{-n/2}$. Equality of the two is the stated relation — the same one-line test Worked

Example 2.2.1 (Young’s convolution inequality) applied to Young’s inequality. *Where the third index appears.* At the endpoint $q = 2$ (possible for $n \geq 3$) the estimate is true but genuinely harder, and for $n = 2$ it is *false* as stated; the repair replaces L_t^2 by a Besov space in t , for exactly the reason of Worked Example 4.7.1. This is the Keel–Tao theorem.

WORKED EXAMPLE 4.8.2— TIER B **What “a gain of $1/p$ derivatives” means**

The local smoothing conjecture asserts, for the wave equation in $\mathbb{R}^n$,

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_\varepsilon \|f\|_{W^{s,p}(\mathbb{R}^n)}, \quad s > s_0(p) - \frac{1}{p},$$

where $s_0(p)$ is the sharp exponent for the *fixed-time* estimate. The phrase “gain of $1/p$ derivatives” means precisely that the required smoothness index s drops by $1/p$ once one averages in t .

Reading it. $W^{s,p} = F_{p,2}^s$ by Theorem 4.6.8, so the hypothesis is a statement about the size of $(\sum_j 2^{2js} |\Delta_j f|^2)^{1/2}$ in L^p . The conclusion is a statement about the solution on a unit time interval. So the conjecture asserts an embedding between two spaces on the scale of Chapter 4.6, and the number $1/p$ is the amount of smoothness that time-averaging buys.

Why it is hard. At each fixed time the estimate is sharp and cannot be improved; the gain comes entirely from the fact that the singularities move, so that a wave packet concentrated on a tube in space-time contributes to only a short interval of times. Controlling how many such tubes can overlap is a Kakeya problem, which is why Guth, Wang and Zhang’s 2020 proof for $n = 2$ goes through decoupling.

WORKED EXAMPLE 4.8.3— TIER B **Why decoupling is not a square function estimate**

This was answered in Worked Example 4.6.2, and is restated here in one line because it is the single most common source of confusion for readers entering the decoupling literature.

The inequality

$$\left\| \sum_\theta f_\theta \right\|_{L^p} \lesssim_\varepsilon \delta^{-\varepsilon} \left(\sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2}$$

has the ℓ^2 sum *outside* the L^p norm: it is Besov-shaped. The square function version, with the sum inside, is Triebel–Lizorkin-shaped, is stronger for $p \geq 2$ by Theorem 4.6.6, and is *false* — it would assert pointwise almost-

orthogonality of the caps, which requires frequency separation that adjacent caps do not have.

The diagnostic: *inside* the norm means a claim about the pointwise values of the pieces and needs a multiplier theorem; *outside* means a claim about how their sizes combine and is what geometric input supplies.

WORKED EXAMPLE 4.8.4— TIER C: RESEARCH TASTE The norms in which wave packets are summed

A wave packet decomposition writes a function as $f = \sum_T f_T$, one piece per tube T in a family $\mathbb{T}$, with f_T concentrated on T in space and on the dual cap in frequency. Every argument in Volumes 12 and 13 proceeds by estimating the pieces and reassembling. This example collects what this volume says about the reassembly.

(a) *The two reassembly norms.* As in Worked Example 4.6.2, one may aim for

$$\begin{aligned} \left\| \sum_T f_T \right\|_{L^p} &\lesssim \left(\sum_T \|f_T\|_{L^p}^2 \right)^{1/2} \quad \text{or} \quad \left\| \sum_T f_T \right\|_{L^p} \\ &\lesssim \left\| \left(\sum_T |f_T|^2 \right)^{1/2} \right\|_{L^p}. \end{aligned}$$

By Theorem 4.6.6 the first is stronger when $p \geq 2$. Both are sometimes available and one should know which is being claimed.

(b) *The trivial bound, and what must be beaten.* Cauchy–Schwarz over $\#\mathbb{T} = N$ pieces gives the first inequality with an extra factor $N^{1/2-1/p}$. That factor is a power of δ , and the entire content of a decoupling theorem is its removal down to $\delta^{-\varepsilon}$. Volume 1, Worked Example 8.16 made this comparison at the very start of the series; the reader can now see it as a statement about which function space the right-hand side lives in.

(c) *Where the maximal inequalities enter.* A wave packet is not literally supported on its tube: it has tails, decaying like $(1 + \text{dist}(x, T)/\delta)^{-N}$. Summing the tails is done by dominating them by a maximal function of the localised piece and applying a vector-valued maximal inequality — exactly Theorem 4.5.3, in the range where it is available. This is the concrete reason Chapter 4.5 exists, and the reason the restriction to $q < p$ noted in Theorem 4.5.4 is a genuine limitation rather than a cosmetic one: the literature uses the full Fefferman–Stein range.

(d) *Where the Whitney decomposition enters.* Theorem 4.4.3 cut an open set into cubes of size comparable to their distance from the boundary. Volume 11 performs the analogous cut with a polynomial: space is divided into cells on

which the object under study is uniformly controlled, plus a neighbourhood of the zero set where it is not. The bookkeeping — finitely many pieces of each size, bounded overlap, sizes comparable to distances — is the same.

(e) *The honest summary.* Of the machinery built in this volume, what the later volumes actually use is: the H^1 –BMO duality, as background for why endpoint estimates are stated the way they are; the Minkowski embeddings, to know which of two competing formulations is stronger; the vector-valued maximal inequality, to sum tails; and the Besov endpoint of Theorem 4.7.3, in the sharp forms of Strichartz and local smoothing. The atomic theory of Chapter 4.2 is used mainly for the intuition it supplies — that cancellation at every scale is what distinguishes H^1 from L^1 — and that intuition is the direct ancestor of the frequency localisation that makes a wave packet a wave packet.

WORKED EXAMPLE 4.8.5— TIER C: RESEARCH TASTE **What Volume 4 was for**

Volume 3 ended with two endpoint failures: Calderón–Zygmund operators are unbounded on L^1 and on L^∞ . Volume 4 supplies the substitutes and proves they work.

(a) *The two spaces.* H^1 replaces L^1 and BMO replaces L^∞ . They are dual (Theorem 4.3.5), Calderón–Zygmund operators map each to itself (Theorem 4.2.5, Theorem 3.7.4), and interpolation between them recovers all the L^p results.

(b) *Why the substitution is forced, not chosen.* By Worked Example 3.3.4(c) the Hilbert transform of an indicator is $\log|\frac{x}{x-1}|$: unbounded, but only logarithmically, and with bounded mean oscillation. So BMO is not an arbitrary weakening of L^∞ — it is exactly what the failure produces.

(c) *What it costs.* The atomic definition is decomposition-dependent, and the Pitfall of Chapter 4.2 shows that reasoning atom-by-atom is invalid without an additional hypothesis. The maximal characterisation which would repair this is not proved here (Theorem 4.4.5, revision R1).

(d) *Where it is used later.* Volume 5’s oscillatory integral estimates are stated on L^p and need no endpoint theory; Volume 6 onwards works with L^2 and L^∞ on a sphere. So Volume 4 is, in the architecture of this series, a completion of Volume 3 rather than a prerequisite for what follows.

(e) *Why it is here nonetheless.* Because the pattern it establishes — when an estimate fails at an endpoint, find the space on which it holds rather than abandoning the endpoint — recurs throughout. Volume 6’s Worked Example 6.3.3 and Volume 8’s contrast between positive measure and nonempty

interior are the same move in different settings.

Drills

Drill 4.8.1. State the sharp maximal function $\mathcal{M}^\#$ and verify that $\mathcal{M}^\# f \lesssim Mf$ pointwise, with M the Hardy–Littlewood maximal function.

Drill 4.8.2. Show that $f \in \text{BMO}$ does not imply $f \in L^\infty$ by computing $\|\log|x|\|_{\text{BMO}}$ on $[-1, 1]$.

Drill 4.8.3. Verify the John–Nirenberg inequality in the model case $f = \log|x|$: compute $|\{x \in Q : |f - f_Q| > \lambda\}|$ and compare with $e^{-c\lambda/\|f\|_{\text{BMO}}}|Q|$.

Drill 4.8.4. Using the duality H^1 –BMO, deduce that a Calderón–Zygmund operator maps $H^1 \rightarrow L^1$ from the fact that it maps $L^\infty \rightarrow \text{BMO}$.

Notes

The Keel–Tao endpoint Strichartz theorem is from 1998; Sogge’s local smoothing conjecture from 1991, with the planar case settled by Guth, Wang and Zhang in 2020. Both are treated in Volume 13.

Readers who wish to see the function-space language used in earnest before reaching Volume 12 may consult Tao’s lecture notes on restriction theory, where the mixed-norm and Besov formulations of the estimates appear throughout.

Chapter 4.A

Results Quoted from Volumes 1–3

This volume assumes Volumes 1–3 and nothing else. Every citation is listed below with its use site.

From Volume 1

Result	Vol. 1	Used in
Hardy–Littlewood maximal theorem	Thm 7.4	Theorem 4.1.2, Theorem 4.1.5, Theorem 4.5.3
Vitali covering lemma	Lem 7.2	Theorem 4.5.2 Step 1
Lebesgue differentiation	Thm 7.7	Theorem 4.4.1 discussion
Layer-cake formula	Wkd Ex 5.5	Theorem 4.1.4, Theorem 4.1.5, Theorem 4.5.2
Absolute continuity of the integral	Prop 7.15	Theorem 4.3.4 Step 2
Polar coordinates	Thm 6.10	Theorem 4.4.2 Step 4
Tonelli; Fubini	Thms 6.6, 6.7	Theorem 4.5.1, Theorem 4.5.3, Theorem 4.6.6
Hölder; generalised Hölder	Thm 8.2, Cor 8.3	Theorem 4.5.3 Step 3, Theorem 4.7.6
Riesz–Fischer completeness	Thm 8.6	Theorem 4.2.4, Theorem 4.6.7
Density of C_c^∞ , of simple functions	Thm 8.9	Theorem 4.2.5 Step 1 Theorem 4.3.5 Step
L^p duality; norms by testing	Thm 8.11, Cor 8.12	6, Theorem 4.5.3 Step 1
Chebyshev; weak L^p	Prop 8.14	Theorem 4.1.4 Step 1
Riesz representation in Hilbert space	Thm 8.19	Theorem 4.3.5 Step 2

From Volume 2

Result	Vol. 2	Used in
Marcinkiewicz interpolation	Thm 2.6	Theorem 4.3.4 Step 1
Young’s convolution inequality	Wkd Ex 2.4	Theorem 4.6.3
The Fourier dictionary (dilation)	Prop 4.1	Theorem 4.6.3
Plancherel	Thm 4.5	Theorem 4.6.9
Schwartz space; $\mathcal{F}$ on $\mathcal{S}$ and $\mathcal{S}'$	Thms 5.5, 5.9	Theorem 4.4.1, Theorem 4.6.7
Bernstein’s inequalities	Thm 6.8	Theorem 4.7.1, Theorem 4.7.3
Sobolev embedding, L^2 case; the endpoint failure	Thm 7.3 and Pitfall	Theorem 4.7.4, Worked Example 4.7.1
Gagliardo–Nirenberg–Sobolev	Thm 7.7	Theorem 4.7.5

From Volume 3

Result	Vol. 3	Used in
Nesting of dyadic cubes	Lem 1.2	Theorem 4.1.3 Step 3
Dyadic maximal inequality, constant 1	Thm 1.6	Theorem 4.1.3 Step 4, Theorem 4.3.4
Calderón–Zygmund decomposition	Thm 1.7	Theorem 4.3.4 Step 1
The one-third trick	Wkd Ex 1.4	Theorem 4.1.2(ii)
Hörmander’s condition	Def 2.1	Theorem 4.2.2, Theorem 4.2.5 Theorem 4.2.5
Calderón–Zygmund theorem	Thm 2.3	Step 1, Theorem 4.1.6 Chapter 4.1
BMO; John–Nirenberg	§7.1, Thm 7.4	throughout, Theorem 4.3.2 Step 1
CZ operators map $L^\infty \rightarrow \text{BMO}$	Thm 7.6	Theorem 4.1.5 applications, Drill 1.4
Littlewood–Paley square function theorem	Thm 6.4	Theorem 4.6.8
Weighted Littlewood–Paley form	Wkd Ex 6.6	Theorem 4.6.8
Marcinkiewicz–Zygmund ℓ^2 extension	Thm 6.1	Chapter 4.5 §5.1, Worked Example 4.5.2

Debts settled and incurred

Volume 3, Chapter 7 deferred two results to this volume. Both are proved: Calderón–Zygmund operators map H^1 into L^1 (Theorem 4.2.5), and $(H^1)^* = \text{BMO}$ (Theorem 4.3.5).

This volume incurs no debts on later volumes. It does record two revisions of the series plan, in Theorem 4.4.5 and Theorem 4.5.4, together with a third made in advance in Theorem 4.6.1; all three are listed in the master manifest under PLAN REVISIONS. In each case the removed statement is true, is attributed, and is used nowhere in this series.

Chapter 4.B

Notation and Index of Results

Additions to the Volume 1–3 conventions

$\mathcal{M}^\#, \mathcal{M}_d^\#$	the sharp maximal function; its dyadic version
H^1	the Hardy space
a	an atom
Q^*	the dilate $2\sqrt{n} Q$
$B_{p,q}^s, F_{p,q}^s$	Besov and Triebel–Lizorkin spaces

Index of the main results

Elementary properties of $\mathcal{M}^\#$	Theorem 4.1.2
Good- λ inequality	Theorem 4.1.3
Fefferman–Stein inequality	Theorem 4.1.4
Interpolation with BMO	Theorem 4.1.5
The atom estimate	Theorem 4.2.2
Basic properties of H^1	Theorem 4.2.4
CZ operators map $H^1 \rightarrow L^1$	Theorem 4.2.5
$g \in \text{BMO}$ gives a functional on H^1	Theorem 4.3.2
L^2 -atoms lie in H^1	Theorem 4.3.4
Fefferman: $(H^1)^* = \text{BMO}$	Theorem 4.3.5
$f \in H^1 \Rightarrow \mathcal{M}_\varphi f \in L^1$	Theorem 4.4.2
Whitney decomposition	Theorem 4.4.3
Weighted maximal inequality	Theorem 4.5.2
Vector-valued maximal inequality, $q < p$	Theorem 4.5.3
Independence of the decomposition (Besov)	Theorem 4.6.5
Minkowski embeddings $B \leftrightarrow F$	Theorem 4.6.6
Completeness of $B_{p,q}^s$	Theorem 4.6.7
$F_{p,2}^s = W^{s,p}$	Theorem 4.6.8
$B_{2,2}^s = H^s$	Theorem 4.6.9
Besov embedding (Bernstein)	Theorem 4.7.1
Monotonicity in q and s	Theorem 4.7.2
Sharp endpoint $B_{p,1}^{n/p} \subseteq L^\infty$	Theorem 4.7.3
The $W^{n/p,p}$ endpoint failure explained	Theorem 4.7.4
Gagliardo–Nirenberg on the Besov scale	Theorem 4.7.6

Plan revisions in this volume

Removed	Reason
Converse of the maximal characterisation	Theorem 4.4.5
Vector-valued maximal for $1 < p < q$	Theorem 4.5.4
$F_{1,2}^0 = H^1$, $F_{\infty,2}^0 = \text{BMO}$; F independence in full range; trace theorems	Theorem 4.6.1

Part 5

Volume 5

Oscillatory Integrals and Stationary Phase

Preface to Volume 5

The one estimate this volume is about

Volume 2 computed the Fourier transform of surface measure on the sphere exactly, in terms of Bessel functions, and proved

$$|\widehat{d\sigma}(\xi)| \lesssim_n |\xi|^{-\frac{n-1}{2}}.$$

It did so by a computation special to the sphere: the integral reduced to one dimension, and the decay was extracted by splitting at $s = \rho^{-1}$ and integrating by parts. That argument is honest but unilluminating. It does not explain where the exponent $(n-1)/2$ comes from, it gives no information about the *phase* of $\widehat{d\sigma}$, and it says nothing about any surface other than a sphere.

This volume supplies the general theory. The exponent will turn out to be

$$\frac{\#\{\text{non-vanishing principal curvatures}\}}{2},$$

and the mechanism will be visible: an oscillatory integral is negligible wherever the phase is moving, so all the mass comes from the *critical points* of the phase, and each non-degenerate direction at a critical point contributes a factor $\lambda^{-1/2}$ because that is the width of the window in which the phase fails to oscillate. Two of the debts Volume 2 left — the asymptotics of Bessel functions and the decay for general hypersurfaces — are discharged here.

The one mechanism

Every estimate in this volume comes from a single mechanism: an integral $\int e^{i\lambda\varphi}\psi$ is small unless the phase is stationary, and how small depends on how degenerately it is stationary.

Where $\nabla\varphi \neq 0$, repeated integration by parts gives decay faster than any power. Where φ has a non-degenerate critical point, the integral concentrates on a window of width $\lambda^{-1/2}$ and contributes $\lambda^{-1/2}$ per variable (Worked

Example 5.2.3). Where the critical point is degenerate to order k , the window widens and the contribution is only $\lambda^{-1/k}$ (Worked Example 5.1.4).

Applied to $\widehat{d\sigma}$, the critical points of the phase $x \cdot \omega$ on a surface are exactly the points where the normal is parallel to x (Theorem 5.5.1), and the Hessian there is the second fundamental form. So the decay rate counts non-vanishing principal curvatures, halved (Worked Example 5.5.4) — and a flat surface, having none, gives no decay at all.

That chain — stationary points, Hessian, second fundamental form, curvature — is what the rest of the series consumes. Volume 6 turns it into restriction estimates, Volume 7 into dimension bounds, Volume 8 into the distance-set theorem. Two outputs are used constantly: the decay of $\widehat{d\sigma}$ with its *oscillating phase*, and the L^2 bound for oscillatory integral operators. Worked Example 5.7.2 records that almost nothing else here is used later.

What is assumed

Volumes 1–4. In particular we use: the complex analysis of Volume 2, Chapter 1, including the three-lines lemma and analytic continuation; van der Corput’s lemma in the form proved as Theorem 2.8.4; the Fourier transform on $\mathcal{S}$ and $\mathcal{S}'$; and the interpolation theorems of Volume 2, Chapter 2. Appendix A lists every citation with its use site.

The plan, and its limits

1	Non-stationary phase and van der Corput	where the mass is not
2	The Fresnel integral and stationary phase in one dimension	the model case
3	The Morse lemma and stationary phase in $\mathbb{R}^n$	the general local picture
4	Bessel functions	Volume 2 debt
5	Surface measure and curvature	Volume 2 debt
6	Analytic continuation: Riesz potentials	complex parameters
7	Oscillatory integral operators	Hörmander’s L^2 theorem
8	Bochner–Riesz means	what is proved, and what is open

Following the practice adopted in Volume 4, the scope has been fixed *before* the chapters were written, to a list every item of which is proved in full. Three results that a reader might expect are deliberately absent, and are recorded in the manifest as plan revisions rather than discovered as gaps:

- the *Carleson–Sjölin theorem*, that the Bochner–Riesz conjecture holds in the plane. Its proof needs the bilinear and geometric machinery of

Volume 12, and it is better placed there;

- *Stein's interpolation theorem* for analytic families of operators. It requires a Poisson-kernel form of the three-lines lemma, and nothing in this series uses it: Chapter 5.6 obtains the analytic continuation it needs directly;
- the sharp L^p theory of *oscillatory integral operators*. Chapter 5.7 proves Hörmander's L^2 theorem, which is what Volumes 6 and 12 build on; the L^p refinements belong with the restriction theory.

Chapter 5.1

Non-Stationary Phase and van der Corput

5.1.1 The governing principle

Every estimate in this volume is a quantitative form of one sentence:

An integral of a rapidly oscillating function is small, unless the oscillation stops somewhere.

The integrals in question are

$$I(\lambda) = \int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) \, dx,$$

with λ large, φ real-valued and smooth (the *phase*), and ψ smooth with compact support (the *amplitude*). Where $\nabla\varphi \neq 0$, neighbouring points contribute with rapidly rotating signs and cancel; where $\nabla\varphi = 0$ — at a *critical*, or *stationary*, point — they do not.

This chapter proves the first half of the sentence, which is the easy half: away from critical points there is nothing. Chapter 5.2 and Chapter 5.3 then compute what remains.

Looking ahead: why this is the right frame for the series

Theorem 2.8.5 obtained $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ by a computation in one variable. Written as an oscillatory integral,

$$\begin{aligned} \widehat{d\sigma}(\xi) &= \int_{\mathbb{S}^{n-1}} e^{-2\pi i \omega \cdot \xi} \, d\sigma(\omega) \\ &= \int_{\mathbb{S}^{n-1}} e^{i\lambda\varphi(\omega)} \, d\sigma(\omega), \quad \lambda = 2\pi|\xi|, \quad \varphi(\omega) = -\omega \cdot \frac{\xi}{|\xi|}, \end{aligned}$$

so the phase is a linear functional restricted to the sphere. Its critical points

are where the gradient *along the sphere* vanishes, that is, where the normal is parallel to ξ : the two poles. Everything else contributes nothing, by Theorem 5.1.1 below, and the exponent $(n-1)/2$ will come from the $n-1$ curved directions at a pole.

For a flat piece of hypersurface the phase is constant along it, so *every* point is critical, no cancellation occurs, and there is no decay — which is Worked Example 2.8.2. Curvature is exactly the statement that the critical set is small.

5.1.2 Non-stationary phase

Theorem 5.1.1 (Principle of non-stationary phase). *Let $\psi \in C_c^\infty(\mathbb{R}^n)$ and let $\varphi \in C^\infty(\mathbb{R}^n)$ be real-valued with*

$$|\nabla\varphi(x)| \geq c > 0 \quad \text{for all } x \in \text{supp } \psi.$$

Then for every $N \in \mathbb{N}$ there is C_N , depending on N , c , and finitely many derivatives of φ and ψ on $\text{supp } \psi$, with

$$\left| \int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) \, dx \right| \leq C_N \lambda^{-N} \quad (\lambda \geq 1).$$

Idea of the proof. The exponential $e^{i\lambda\varphi}$ is an eigenfunction, with eigenvalue 1, of the first-order differential operator

$$L = \frac{1}{i\lambda} \frac{\nabla\varphi \cdot \nabla}{|\nabla\varphi|^2},$$

the verification being one line and carried out in Step 1 below. Since $Le^{i\lambda\varphi} = e^{i\lambda\varphi}$, we may insert L into the integral for free, and then move it onto the amplitude by integration by parts. Each such move produces a factor λ^{-1} and differentiates ψ and φ once more — which is harmless, since both are smooth and compactly supported. Repeating N times gives λ^{-N} . The hypothesis $|\nabla\varphi| \geq c$ is what makes L well defined: it is the statement that one may divide by $|\nabla\varphi|^2$.

Proof. Step 1. *The operator and its transpose.*

On a neighbourhood of $\text{supp } \psi$ the vector field $V = \nabla\varphi/|\nabla\varphi|^2$ is smooth, by the hypothesis and smoothness of φ . Define, for smooth u ,

$$Lu = \frac{1}{i\lambda} V \cdot \nabla u.$$

Then $Le^{i\lambda\varphi} = \frac{1}{i\lambda} V \cdot (i\lambda \nabla\varphi) e^{i\lambda\varphi} = \frac{\nabla\varphi \cdot \nabla\varphi}{|\nabla\varphi|^2} e^{i\lambda\varphi} = e^{i\lambda\varphi}$.

Its transpose, defined by $\int (Lu)w = \int u(L^t w)$ for compactly supported smooth w , is

$$L^t w = -\frac{1}{i\lambda} \nabla \cdot (Vw) = -\frac{1}{i\lambda} (V \cdot \nabla w + (\nabla \cdot V) w),$$

by the divergence theorem, there being no boundary terms.

Step 2. *Integrate by parts N times.*

Using $e^{i\lambda\varphi} = L^N e^{i\lambda\varphi}$ and moving L across,

$$\begin{aligned} \int e^{i\lambda\varphi} \psi &= \int (L^N e^{i\lambda\varphi}) \psi \\ &= \int e^{i\lambda\varphi} (L^t)^N \psi. \end{aligned}$$

Each application of L^t contributes a factor λ^{-1} and replaces the amplitude by a smooth function supported in $\text{supp } \psi$, whose derivatives are polynomial expressions in the derivatives of φ , of ψ , and in $|\nabla\varphi|^{-2}$. Hence

$$\|(L^t)^N \psi\|_{L^\infty} \leq A_N \lambda^{-N}$$

with A_N depending only on the stated data.

Step 3. *Conclude.*

Taking absolute values, $|\int e^{i\lambda\varphi} \psi| \leq m(\text{supp } \psi) A_N \lambda^{-N}$. □

Pitfall: compact support of the amplitude is doing real work

Two of the three steps used it: Step 1 to discard boundary terms, and Step 3 to convert a sup-norm bound into an integral bound. If ψ merely decays, the conclusion survives with the same proof provided all the derivatives appearing in Step 2 are integrable, but this must be checked — and for an amplitude that is merely bounded, the theorem is false: take $n = 1$, $\varphi(t) = t$, $\psi \equiv 1$, where the integral does not converge at all.

More importantly, if ψ is compactly supported but φ has a critical point in $\text{supp } \psi$, the hypothesis fails and so does the conclusion. That case is the entire content of Chapter 5.2, and the reader should resist the temptation to think of Theorem 5.1.1 as saying oscillatory integrals are small. It says they are small *away from critical points*.

WORKED EXAMPLE 5.1.1— TIER A **The hypothesis cannot be weakened to $\nabla\varphi \neq 0$**

It is not enough that $\nabla\varphi$ be non-vanishing on $\text{supp } \psi$; one needs a uniform lower bound, and on a compact support these are the same — but only because $\text{supp } \psi$ is compact and $\nabla\varphi$ continuous. The distinction matters as soon as one localises to a shrinking region, which is exactly what the proofs of Chapter 5.2 do.

Concretely, take $n = 1$, $\varphi(t) = t^2$ and ψ supported in $[\varepsilon, 1]$, so $\varphi'(t) = 2t \geq 2\varepsilon > 0$. Theorem 5.1.1 applies with $c = 2\varepsilon$, but the constant C_N involves c^{-1} to a power, so the bound degrades as $\varepsilon \rightarrow 0$. Balancing that degradation against the size of the region where the hypothesis fails is precisely how the exponent $\lambda^{-1/2}$ arises in Theorem 5.1.3 below, and again in Theorem 5.2.3.

5.1.3 van der Corput's lemma, with amplitudes

When the phase does have critical points, one cannot integrate by parts indefinitely. What survives is a bound depending on the order of vanishing. Volume 2 proved the following in the course of estimating $\widehat{d\sigma}$; we restate it, since it is used throughout this volume, and record the amplitude version and one corollary.

Theorem 5.1.2 (van der Corput; Theorem 2.8.4). *Let $\varphi : [a, b] \rightarrow \mathbb{R}$ be C^k with $|\varphi^{(k)}| \geq 1$ on $[a, b]$, where either $k \geq 2$, or $k = 1$ and φ' is monotone. Then*

$$\left| \int_a^b e^{i\lambda\varphi(t)} dt \right| \leq c_k \lambda^{-1/k} \quad (\lambda > 0),$$

with c_k depending only on k .

Proof. This is Theorem 2.8.4, proved there by induction on k : the case $k = 1$ is one integration by parts, and the inductive step splits off a neighbourhood of the zero of $\varphi^{(k-1)}$ and optimises its width. $\square$

Theorem 5.1.3 (Amplitude version). *Under the hypotheses of Theorem 5.1.2, for $\psi : [a, b] \rightarrow \mathbb{C}$ of bounded variation,*

$$\begin{aligned} \left| \int_a^b e^{i\lambda\varphi(t)} \psi(t) dt \right| \\ \leq c_k \lambda^{-1/k} \left(|\psi(b)| + \int_a^b |\psi'(t)| dt \right). \end{aligned}$$

Proof. Let $F(u) = \int_a^u e^{i\lambda\varphi}$, so $|F(u)| \leq c_k \lambda^{-1/k}$ for every $u \in [a, b]$, by Theorem 5.1.2 applied on $[a, u]$ — note the hypothesis $|\varphi^{(k)}| \geq 1$ persists on

subintervals, and monotonicity of φ' does too. Integrating by parts,

$$\int_a^b e^{i\lambda\varphi} \psi = F(b)\psi(b) - \int_a^b F(u)\psi'(u) du,$$

and taking absolute values gives the claim. $\square$

Where we are. The two theorems say complementary things. Theorem 5.1.1 gives decay of *every* order, but needs the phase to have no critical point. Theorem 5.1.2 tolerates critical points — indeed φ' may vanish, as long as some higher derivative does not — but gives only $\lambda^{-1/k}$, and k is the order at which the phase finally becomes non-degenerate.

Both are needed. In every argument of this volume, space is split into a region where $\nabla\varphi$ is large, handled by Theorem 5.1.1 and contributing $O(\lambda^{-N})$, and a shrinking neighbourhood of the critical set, where Theorem 5.1.2 or an explicit computation is used. The exponent in the final answer always comes from the second region, and equals the width of that neighbourhood.

WORKED EXAMPLE 5.1.2— TIER B Sublevel sets: the geometric content of van der Corput

There is a purely geometric statement hiding inside Theorem 5.1.2, and it is often the more useful form.

Proposition 5.1.4. *Let $\varphi : [a, b] \rightarrow \mathbb{R}$ be C^k with $|\varphi^{(k)}| \geq \mu > 0$ on $[a, b]$. Then for every $\alpha > 0$,*

$$m\left(\{t \in [a, b] : |\varphi(t)| \leq \alpha\}\right) \leq C_k \left(\frac{\alpha}{\mu}\right)^{1/k}.$$

Proof. Induct on k . For $k = 1$: $|\varphi'| \geq \mu$ means φ is strictly monotone with speed at least μ , so it spends time at most $2\alpha/\mu$ in any interval of length 2α ; the sublevel set is an interval of length $\leq 2\alpha/\mu$.

For $k \geq 2$: $\varphi^{(k-1)}$ is strictly monotone, so it vanishes at most once, say at c . Fix $\eta > 0$ and remove $(c - \eta, c + \eta)$, of length 2η . On each of the at most two remaining intervals, $|\varphi^{(k-1)}| \geq \mu\eta$, since $\varphi^{(k-1)}$ moves at rate at least μ . The inductive hypothesis gives, on each, a sublevel set of measure $\leq C_{k-1}(\alpha/(\mu\eta))^{1/(k-1)}$. Hence

$$m(\{|\varphi| \leq \alpha\}) \leq 2\eta + 2C_{k-1} \left(\frac{\alpha}{\mu\eta}\right)^{1/(k-1)}.$$

Choosing $\eta = (\alpha/\mu)^{1/k}$ balances the two terms and gives the claim. $\square$

Reading it. The proposition says: a function whose k -th derivative is bounded below cannot linger near any value. The exponent $1/k$ is exactly the one in van der Corput, and this is not a coincidence — the standard proof of Theorem 5.1.2, which the reader saw in Volume 2, is the same induction, with “spends little time near a value” replaced by “the oscillation has little time to fail”.

Why record it separately. The sublevel set form is what one uses when the oscillation is not available — for instance when estimating the measure of the set where a polynomial is small, which is how the polynomial method of Volume 11 meets this circle of ideas. It is also the form in which the estimate generalises to several variables, where the oscillatory statement is much more delicate. $\square$

WORKED EXAMPLE 5.1.3— TIER C: RESEARCH TASTE **Where the exponent $1/k$ becomes the exponent of a conjecture**

The number $\lambda^{-1/k}$ produced by Theorem 5.1.2 propagates, through the next four chapters, into the exponents of the central conjectures of this series. It is worth tracing the route once, in advance, so that the computations of Chapter 5.5 do not look like coincidences.

(a) *$k = 2$ and the half-power.* At a non-degenerate critical point of a phase in one variable, φ' vanishes but φ'' does not, so $k = 2$ and the decay is $\lambda^{-1/2}$. Equivalently: the phase fails to oscillate on an interval of width $\lambda^{-1/2}$ around the critical point, and the integral over that interval is trivially bounded by its length. Every $\frac{1}{2}$ appearing in this subject is this window.

(b) *$n - 1$ windows, and the sphere.* On the sphere, a linear phase has two critical points, and at each the Hessian along the surface is non-degenerate in all $n - 1$ tangential directions — that is what it means for all principal curvatures to be non-zero. Each direction contributes a window of width $\lambda^{-1/2}$, so the critical region has measure $\lambda^{-(n-1)/2}$, and

$$|\widehat{d\sigma}(\xi)| \approx |\xi|^{-\frac{n-1}{2}}.$$

That is Chapter 5.5, and it is Theorem 2.8.5 with the mechanism made visible.

(c) *From decay to an exponent in restriction.* Worked Example 2.8.3 converted the decay rate $\lambda^{-(n-1)/2}$ directly into the exponent $q = \frac{4n}{n-1}$ of a restriction theorem, by the TT^* identity and Hardy–Littlewood–Sobolev. Volume 6 improves this to the Stein–Tomas exponent $\frac{2(n+1)}{n-1}$ by using L^2 orthogonality as well; but the input is the same number, and it traces back to $k = 2$.

(d) *Where degeneracy costs.* If a surface has a flat direction — one vanishing principal curvature — then one of the $n - 1$ Hessian directions is degenerate,

that window is wider than $\lambda^{-1/2}$, and the decay is worse. The extreme case is a hyperplane, where every direction is degenerate and there is no decay at all. So the whole subject may be read as the study of how much decay curvature buys, and Theorem 5.1.2 is the statement that k derivatives of non-degeneracy buy $\lambda^{-1/k}$.

A caution. It is tempting to conclude that everything reduces to counting derivatives. It does not. The passage from decay of $\widehat{d\sigma}$ to a restriction estimate is lossy — Worked Example 2.8.3(b) quantified the loss — and the restriction conjecture is not a statement about any single oscillatory integral, but about how $\delta^{-(n-1)/2}$ of them, one per cap, superpose. That is the geometry of Volumes 9–13, and no amount of stationary phase reaches it.

WORKED EXAMPLE 5.1.4— TIER B **Van der Corput: which case applies**

Theorem 2.8.4 has two clauses — first derivative bounded below, or k -th derivative bounded below for $k \geq 2$ — and choosing between them is the first decision in any application.

$k = 1$ needs monotonicity. The bound $|\int_a^b e^{i\lambda\varphi}| \lesssim \lambda^{-1}$ holds when $|\varphi'| \geq 1$ and φ' is monotone. Without monotonicity it is false: a phase whose derivative oscillates between ± 1 can produce a large integral, because the oscillation never accumulates.

$k \geq 2$ needs nothing extra. The bound $\lambda^{-1/k}$ holds whenever $|\varphi^{(k)}| \geq 1$, with no monotonicity hypothesis — the k -th derivative bound already forces φ' to be monotone on boundedly many subintervals.

Worked choice: $\varphi(s) = s^2$ on $[-1, 1]$. Here $\varphi' = 2s$ vanishes at the origin, so the $k = 1$ clause is unavailable. But $\varphi'' = 2$, so the $k = 2$ clause gives $\lambda^{-1/2}$ — the correct answer, and the source of every $\lambda^{-1/2}$ in the volume.

Worked choice: $\varphi(s) = s^3$ on $[-1, 1]$. Now $\varphi'' = 6s$ also vanishes, and one must go to $k = 3$: $\varphi''' = 6$, giving $\lambda^{-1/3}$. The degenerate critical point costs decay, and the exponent $1/k$ records the order of degeneracy.

The rule. Find the smallest k with $\varphi^{(k)}$ bounded below on the interval; the decay is $\lambda^{-1/k}$, except that $k = 1$ additionally requires monotonicity. Amplitudes are handled separately, by the total-variation factor.

Drills

Drill 5.1.1. Verify the formula for L^t in Step 1 of Theorem 5.1.1, and compute $(L^t)^2\psi$ explicitly for $n = 1$, $\varphi(t) = t$.

Drill 5.1.2. Show that Theorem 5.1.1 fails if φ is allowed a single critical point, by computing $\int_{-1}^1 e^{i\lambda t^2} \psi(t) dt$ for $\psi \in C_c^\infty$ with $\psi(0) = 1$ and confirming it is of size $\lambda^{-1/2}$, not λ^{-N} .

Drill 5.1.3. Deduce Theorem 5.1.2 for $k = 1$ from Worked Example 5.1.2 with $k = 1$, and explain why the implication does not run in the other direction for $k \geq 2$.

Drill 5.1.4. Let $\varphi(t) = t^k$ on $[-1, 1]$. Compute $\int e^{i\lambda t^k} dt$ up to constants and confirm the exponent $\lambda^{-1/k}$ of Theorem 5.1.2 is attained.

Notes

The principle of non-stationary phase is classical and the proof by iterated integration by parts goes back to Riemann. Van der Corput's lemma is from 1921; the sublevel set formulation of Worked Example 5.1.2 is due to Carbery, Christ and Wright, who developed the multi-dimensional theory where the two forms genuinely diverge.

Readers should note that Theorem 5.1.2 is sharp in k but says nothing about the phase of the answer. Extracting the phase — which Volume 6 needs — requires the asymptotic expansion of Chapter 5.2, and that is a strictly harder statement.

Chapter 5.2

The Fresnel Integral and Stationary Phase in One Dimension

Chapter 5.1 showed that an oscillatory integral is negligible away from the critical points of its phase, and that near a non-degenerate critical point the best crude bound is $\lambda^{-1/2}$. This chapter computes the constant. The answer, Theorem 5.2.3, is the single formula from which every asymptotic in this volume is derived.

Two ingredients are needed: an exact evaluation of the Gaussian-type integral $\int e^{i\lambda t^2} dt$, which is §2.1; and a change of variables reducing an arbitrary non-degenerate phase to t^2 , which is §2.2.

5.2.1 The Fresnel integral

Proposition 5.2.1. *For $a \in \mathbb{R} \setminus \{0\}$,*

$$\int_{\mathbb{R}} e^{iat^2} dt = \sqrt{\frac{\pi}{|a|}} e^{i\frac{\pi}{4} \operatorname{sgn} a},$$

the integral being understood as $\lim_{R \rightarrow \infty} \int_{-R}^R$.

Idea of the proof. The integral does not converge absolutely, so one cannot simply substitute. The device is to regard it as the boundary value of a family that does converge: for $\operatorname{Re} z > 0$ the integral $\int e^{-zt^2} dt$ converges absolutely and is holomorphic in z , and on the positive real axis it equals $\sqrt{\pi/z}$ by the classical Gaussian computation. Two holomorphic functions agreeing on a segment agree throughout, so the identity persists on the whole right half-plane, and one then lets z approach the imaginary axis.

The branch of the square root is decided by continuity from the positive real axis, and it is exactly that decision which produces the phase $e^{i\pi/4}$ — the number that will appear in every stationary phase asymptotic in this volume.

Proof. Step 1. *A holomorphic family.*

For $\operatorname{Re} z > 0$ define

$$F(z) = \int_{\mathbb{R}} e^{-zt^2} dt.$$

The integrand is dominated by $e^{-(\operatorname{Re} z)t^2}$, integrable, so F is well defined; and it is holomorphic on $\{\operatorname{Re} z > 0\}$ by differentiation under the integral sign (Worked Example 1.5.3), the z -derivative of the integrand being $-t^2 e^{-zt^2}$, dominated locally uniformly.

Step 2. *Identification on the real axis, then everywhere.*

For $z = s > 0$ real, the substitution $u = t\sqrt{s}$ and the Gaussian integral (Drill 1.6.2) give $F(s) = \sqrt{\pi/s}$. Let $G(z) = \sqrt{\pi/z}$ denote the branch on $\{\operatorname{Re} z > 0\}$ determined by $G(s) = \sqrt{\pi/s} > 0$ for $s > 0$; this is holomorphic there, since $z \mapsto 1/z$ maps the right half-plane to itself and the principal square root is holomorphic away from the negative reals.

The two holomorphic functions F and G agree on $(0, \infty)$. By the propagation argument of Theorem 2.1.10 — a holomorphic function agreeing with another on a set with a limit point in a connected open set agrees with it throughout — $F = G$ on all of $\{\operatorname{Re} z > 0\}$.

Step 3. *Approach the imaginary axis.*

Fix $a > 0$ and set $z_\varepsilon = \varepsilon - ia$ for $\varepsilon > 0$, so $\operatorname{Re} z_\varepsilon > 0$ and $z_\varepsilon \rightarrow -ia$. By Step 2,

$$\int_{\mathbb{R}} e^{-(\varepsilon - ia)t^2} dt = \sqrt{\frac{\pi}{\varepsilon - ia}}.$$

As $\varepsilon \downarrow 0$ the right side tends to $\sqrt{\pi/(-ia)}$, computed with the branch of Step 2: writing $-ia = ae^{-i\pi/2}$, we get $\sqrt{\pi/(-ia)} = \sqrt{\pi/a} e^{i\pi/4}$, the argument $+\pi/4$ being forced because the path $\varepsilon \mapsto \varepsilon - ia$ approaches the negative imaginary axis from the right half-plane, where the branch is continuous.

For the left side, we must justify

$$\lim_{\varepsilon \downarrow 0} \int_{\mathbb{R}} e^{-\varepsilon t^2} e^{iat^2} dt = \lim_{R \rightarrow \infty} \int_{-R}^R e^{iat^2} dt.$$

Both limits exist and agree: substituting $u = t^2$ on $(0, \infty)$ turns each into $\frac{1}{2} \int_0^\infty e^{-\varepsilon u} e^{iau} u^{-1/2} du$, and $\int_0^\infty e^{iau} u^{-1/2} du$ converges as an improper integral, by van der Corput (Theorem 5.1.3) applied on $[1, R]$ with amplitude $u^{-1/2}$ of

bounded variation. Abel summation — that is, dominated convergence applied after one integration by parts — then gives the equality of the two limits.

Hence $\int_{\mathbb{R}} e^{iat^2} dt = \sqrt{\pi/a} e^{i\pi/4}$ for $a > 0$. For $a < 0$ take complex conjugates. $\square$

WORKED EXAMPLE 5.2.1— TIER A **The phase $e^{i\pi/4}$ is a branch choice, and it is checkable**

Students routinely mis-remember the sign in Theorem 5.2.1. Here is a consistency check that fixes it without redoing the proof.

Separate real and imaginary parts: the proposition asserts, for $a = 1$,

$$\int_{\mathbb{R}} \cos(t^2) dt = \int_{\mathbb{R}} \sin(t^2) dt = \sqrt{\frac{\pi}{2}},$$

since $\sqrt{\pi}e^{i\pi/4} = \sqrt{\pi}(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}})$. Both are the classical Fresnel integrals, and both are *positive* — which they must be, since near $t = 0$ the integrands are positive and the later oscillations alternate with decreasing amplitude. Had the branch produced $e^{-i\pi/4}$, the sine integral would come out negative, contradicting this.

So the rule is: *the phase has the same sign as a* , and one may always check it against the positivity of $\int \sin(t^2)$.

5.2.2 Reduction to the model phase

Lemma 5.2.2 (Morse lemma in one variable). *Let φ be C^∞ and real near $t_0 \in \mathbb{R}$, with $\varphi'(t_0) = 0$ and $\varphi''(t_0) \neq 0$. Then there is a neighbourhood U of t_0 and a C^∞ diffeomorphism $s : U \rightarrow s(U) \subseteq \mathbb{R}$ with $s(t_0) = 0$, $s'(t_0) = 1$, and*

$$\varphi(t) = \varphi(t_0) + \frac{1}{2}\varphi''(t_0)s(t)^2 \quad \text{for } t \in U.$$

Proof. By Taylor's theorem with integral remainder, using $\varphi'(t_0) = 0$,

$$\varphi(t) - \varphi(t_0) = (t - t_0)^2 h(t), \quad h(t) := \int_0^1 (1 - u) \varphi''(t_0 + u(t - t_0)) du,$$

and h is C^∞ near t_0 with $h(t_0) = \frac{1}{2}\varphi''(t_0) \neq 0$.

Set $\kappa = \frac{1}{2}\varphi''(t_0)$ and

$$s(t) = (t - t_0) \sqrt{\frac{h(t)}{\kappa}},$$

where the square root is the positive one; this is legitimate on a neighbourhood of t_0 , since h/κ is continuous with value 1 at t_0 , hence positive nearby, and $\sqrt{\cdot}$

is C^∞ on $(0, \infty)$. Then s is C^∞ , $s(t_0) = 0$, and $s'(t_0) = 1$, so by the inverse function theorem s is a diffeomorphism near t_0 . Finally

$$\kappa s(t)^2 = \kappa(t - t_0)^2 \frac{h(t)}{\kappa} = (t - t_0)^2 h(t) = \varphi(t) - \varphi(t_0). \quad \square$$

5.2.3 Stationary phase in one dimension

Theorem 5.2.3. *Let $\varphi \in C^\infty(\mathbb{R})$ be real-valued and $\psi \in C_c^\infty(\mathbb{R})$. Suppose φ has exactly one critical point t_0 in $\text{supp } \psi$, that it is non-degenerate ($\varphi''(t_0) \neq 0$), and that t_0 lies in the interior of $\text{supp } \psi$. Then, as $\lambda \rightarrow \infty$,*

$$\int_{\mathbb{R}} e^{i\lambda\varphi(t)} \psi(t) dt = e^{i\lambda\varphi(t_0)} \sqrt{\frac{2\pi}{\lambda |\varphi''(t_0)|}} e^{i\frac{\pi}{4} \text{sgn } \varphi''(t_0)} \psi(t_0) + O(\lambda^{-3/2}),$$

the implied constant depending on finitely many derivatives of φ and ψ .

Idea of the proof. Three moves, each disposing of one difficulty.

Localise. Away from t_0 the phase has no critical point, so Theorem 5.1.1 contributes $O(\lambda^{-N})$. Only a fixed small neighbourhood of t_0 matters.

Straighten. On that neighbourhood, Theorem 5.2.2 turns the phase into the model $\varphi(t_0) + \kappa s^2$, at the cost of replacing the amplitude by another smooth compactly supported function.

Expand. Split the new amplitude as its value at the critical point plus a remainder vanishing there. The constant term integrates to the Fresnel integral, giving the stated main term. The remainder carries a factor s , and $s e^{i\lambda\kappa s^2}$ is an exact derivative — so one integration by parts gains a full power of λ , and the remainder is smaller by λ^{-1} than the main term, i.e. $O(\lambda^{-3/2})$.

Proof. Step 1. Localise near t_0 .

Let U be the neighbourhood supplied by Theorem 5.2.2 and choose $\chi \in C_c^\infty(U)$ with $\chi = 1$ near t_0 . Write $\psi = \chi\psi + (1 - \chi)\psi$. On $\text{supp } ((1 - \chi)\psi)$ the phase has no critical point — t_0 was the only one, and it has been excluded — so $|\varphi'|$ is bounded below there by compactness, and Theorem 5.1.1 gives

$$\int e^{i\lambda\varphi}(1 - \chi)\psi = O(\lambda^{-N})$$

for every N . So it suffices to treat $\chi\psi$, which we rename ψ .

Step 2. Change variables.

Put $\kappa = \frac{1}{2}\varphi''(t_0)$ and let s be as in Theorem 5.2.2, with inverse $t(s)$.

Substituting,

$$\begin{aligned} \int e^{i\lambda\varphi(t)} \psi(t) dt &= e^{i\lambda\varphi(t_0)} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \tilde{\psi}(s) ds \\ &, \quad \tilde{\psi}(s) := \psi(t(s)) t'(s), \end{aligned}$$

and $\tilde{\psi} \in C_c^\infty(\mathbb{R})$ with $\tilde{\psi}(0) = \psi(t_0)$, since $t'(0) = 1$.

Step 3. *Split the amplitude.*

Write $\tilde{\psi}(s) = \tilde{\psi}(0) + s\rho(s)$ where $\rho(s) = \int_0^1 \tilde{\psi}'(us) du$ is smooth and compactly supported. The integral becomes

$$\tilde{\psi}(0) \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \eta(s) ds + \int_{\mathbb{R}} e^{i\lambda\kappa s^2} s \rho(s) ds,$$

where $\eta \in C_c^\infty$ equals 1 on a neighbourhood of $\text{supp } \rho \cup \{0\}$ — we may insert such a cutoff because $\tilde{\psi}$ is compactly supported, and extending the first integral to all of $\mathbb{R}$ costs

$$\tilde{\psi}(0) \int_{\mathbb{R}} e^{i\lambda\kappa s^2} (1 - \eta(s)) ds = O(\lambda^{-N})$$

by Theorem 5.1.1, the phase κs^2 having no critical point on $\text{supp}(1 - \eta)$.

Step 4. *The main term.*

By Theorem 5.2.1 with $a = \lambda\kappa$,

$$\begin{aligned} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} ds &= \sqrt{\frac{\pi}{\lambda|\kappa|}} e^{i\frac{\pi}{4} \text{sgn } \kappa} \\ &= \sqrt{\frac{2\pi}{\lambda|\varphi''(t_0)|}} e^{i\frac{\pi}{4} \text{sgn } \varphi''(t_0)}, \end{aligned}$$

using $|\kappa| = \frac{1}{2}|\varphi''(t_0)|$ and $\text{sgn } \kappa = \text{sgn } \varphi''(t_0)$. Multiplying by $\tilde{\psi}(0) = \psi(t_0)$ gives the stated main term.

Step 5. *The remainder.*

Since $\frac{d}{ds} e^{i\lambda\kappa s^2} = 2i\lambda\kappa s e^{i\lambda\kappa s^2}$, we may integrate by parts with no boundary terms:

$$\begin{aligned} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} s \rho(s) ds &= \frac{1}{2i\lambda\kappa} \int_{\mathbb{R}} \left(\frac{d}{ds} e^{i\lambda\kappa s^2} \right) \rho(s) ds \\ &= -\frac{1}{2i\lambda\kappa} \int_{\mathbb{R}} e^{i\lambda\kappa s^2} \rho'(s) ds. \end{aligned}$$

The remaining integral is again an oscillatory integral with the model phase and a smooth compactly supported amplitude, so by Theorem 5.1.3 with $k = 2$ it is $O(\lambda^{-1/2})$. Hence the remainder is $O(\lambda^{-1} \cdot \lambda^{-1/2}) = O(\lambda^{-3/2})$.

Collecting Steps 1, 3, 4 and 5 gives the theorem. $\square$

Where we are. Every feature of the answer has a source, and it is worth listing them.

- $\lambda^{-1/2}$: the width of the window in which the phase fails to oscillate, from Theorem 5.2.1.
- $|\varphi''(t_0)|^{-1/2}$: a flatter critical point has a wider window.
- $e^{i\pi \operatorname{sgn} \varphi''/4}$: the branch choice of Theorem 5.2.1, and the only place the *sign* of the curvature appears.
- $\psi(t_0)$: nothing about the amplitude away from the critical point survives, to leading order.
- $e^{i\lambda\varphi(t_0)}$: the phase at the critical point, which oscillates rapidly in λ and is invisible to any estimate that takes absolute values.

The last point deserves emphasis. Theorem 2.8.5 bounded $|\widehat{d\sigma}|$ and discarded exactly this factor. Volume 6 needs it: the interference between the contributions of the two poles of the sphere is what produces the oscillation of $\widehat{d\sigma}$, and that oscillation is what makes the Stein–Tomas argument work at the endpoint.

WORKED EXAMPLE 5.2.2— TIER B **Recovering the Fourier transform of surface measure, $n = 3$**

Worked Example 2.8.2 computed, for the sphere in $\mathbb{R}^3$,

$$\widehat{d\sigma}(\xi) = \frac{2 \sin(2\pi|\xi|)}{|\xi|},$$

by an elementary integration. Let us see Theorem 5.2.3 produce it, including the phase.

By Theorem 2.8.1 the transform reduces to

$$\widehat{d\sigma}(\xi) = 2\pi \int_{-1}^1 e^{-2\pi i r t} dt, \quad r = |\xi|,$$

which after the substitution $t = \cos \theta$ becomes $2\pi \int_0^\pi e^{-2\pi i r \cos \theta} \sin \theta d\theta$. Take $\lambda = 2\pi r$, $\varphi(\theta) = -\cos \theta$, $\psi(\theta) = \sin \theta$.

Critical points. $\varphi'(\theta) = \sin \theta$ vanishes at $\theta = 0, \pi$, the two poles. At these, $\varphi''(\theta) = \cos \theta$ equals $+1$ and -1 .

The obstruction, and what it teaches. Theorem 5.2.3 does not apply directly: the amplitude $\sin \theta$ vanishes at both critical points, so the leading term $\psi(t_0)$ is zero and the theorem returns only $O(\lambda^{-3/2})$. That is consistent with the true answer, which is of size $r^{-1} = \lambda^{-1}$ — between the two.

The resolution is that the correct statement retains the next term in the expansion. Writing $\psi(\theta) = \theta + O(\theta^3)$ near 0 and repeating Step 5 with one more integration by parts produces a term of size λ^{-1} with an explicit constant, and summing the two poles gives

$$\widehat{d\sigma}(\xi) = \frac{2 \sin(2\pi r)}{r} + O(r^{-2}),$$

matching the exact formula — and the sin is precisely the interference between $e^{i\lambda\varphi(0)} = e^{-2\pi ir}$ and $e^{i\lambda\varphi(\pi)} = e^{2\pi ir}$.

The lesson. When the amplitude degenerates at the critical point, the leading term of Theorem 5.2.3 vanishes and one must go further into the expansion. This is not an exotic situation: it is exactly what happens for the sphere in every dimension, because the surface measure written in polar form carries a factor vanishing at the poles. Chapter 5.4 does the general computation properly, and the vanishing amplitude is why the answer involves $(n-1)/2$ rather than $1/2$.

Figure 5.2.1 (to be drawn). *The non-oscillation window*

Content required: The graph of $\cos(\lambda t^2)$ against t , showing the flat central region of width $\approx \lambda^{-1/2}$ where the phase has not yet turned, and the rapid oscillation outside it. Shade the window and label its width; this is where every factor $\lambda^{-1/2}$ in the volume comes from.

WORKED EXAMPLE 5.2.3— TIER B **The stationary phase formula, term by term**

Theorem 5.2.3 gives, for a single non-degenerate critical point s_0 ,

$$\int e^{i\lambda\varphi(s)} \psi(s) \, ds = \left(\frac{2\pi}{\lambda |\varphi''(s_0)|} \right)^{1/2} e^{i\lambda\varphi(s_0)} e^{i\frac{\pi}{4} \operatorname{sgn} \varphi''(s_0)} \psi(s_0) + O(\lambda^{-3/2}).$$

Four factors, each with a source.

$\lambda^{-1/2}$ — the width of the non-oscillation window. The phase changes by $O(1)$ over an interval of length $\lambda^{-1/2}$ about s_0 , and outside it the integrand oscillates away.

$|\varphi''(s_0)|^{-1/2}$ — the curvature of the phase. A flatter critical point has a wider

window and a larger integral, which is why degeneracy costs decay (Worked Example 5.1.4).

$e^{i\lambda\varphi(s_0)}$ — the value of the phase, carried out front. This is the factor that oscillates in λ and is what makes $\widehat{d\sigma}$ oscillate rather than merely decay — see Volume 8, Theorem 8.7.2, where the oscillation is essential.

$e^{i\pi \operatorname{sgn} \varphi''/4}$ — the signature. It records whether the critical point is a minimum or a maximum, and in several variables becomes the signature of the Hessian, Worked Example 5.3.2. For a sphere the two critical points have opposite signatures, and the resulting phase difference is what makes $\widehat{d\sigma}$ a cosine rather than a single exponential.

The check. Setting $\varphi(s) = s^2/2$ and $\psi \equiv 1$ recovers the Fresnel integral of Theorem 5.2.1, with $\varphi'' = 1$ and the formula becoming an identity rather than an asymptotic. That is the standard sanity test.

Drills

Drill 5.2.1. Evaluate $\int_{\mathbb{R}} e^{i(at^2+bt)} dt$ for $a \neq 0$ by completing the square, and check the answer against Theorem 5.2.1.

Drill 5.2.2. Verify that $h(t_0) = \frac{1}{2}\varphi''(t_0)$ in the proof of Theorem 5.2.2, and compute $s(t)$ explicitly for $\varphi(t) = 1 - \cos t$ near $t_0 = 0$.

Drill 5.2.3. Show that the error term in Theorem 5.2.3 cannot be improved to $O(\lambda^{-2})$, by computing the next term in the expansion for $\varphi(t) = t^2$ and $\psi(t) = e^{-t^2}$.

Drill 5.2.4. Suppose $\varphi'(t_0) = \varphi''(t_0) = 0$ but $\varphi'''(t_0) \neq 0$. Show the analogue of Theorem 5.2.3 has main term of size $\lambda^{-1/3}$, and identify the integral that replaces the Fresnel integral. (It is the Airy function.)

Notes

The method of stationary phase is Kelvin's, from 1887, with antecedents in Stokes. The proof given here — localise, straighten by Morse, expand, integrate by parts — is the standard modern one and generalises verbatim to $\mathbb{R}^n$, which is Chapter 5.3.

The Airy function of Drill 2.4 governs the behaviour at degenerate critical points and is the first case of a general classification; the systematic theory is Arnold's, and it is not needed in this series.

Chapter 5.3

The Morse Lemma and Stationary Phase in $\mathbb{R}^n$

Chapter 5.2 handled one variable. The passage to $\mathbb{R}^n$ requires one new ingredient — a normal form for a function near a non-degenerate critical point, valid in any dimension — and then the same three-step argument goes through verbatim. The answer, Theorem 5.3.3, is the formula from which the decay of $\widehat{d\sigma}$ for a general hypersurface will follow in Chapter 5.5.

5.3.1 The Morse lemma

Theorem 5.3.1 (Morse). *Let φ be real-valued and C^∞ near $x_0 \in \mathbb{R}^n$, with $\nabla\varphi(x_0) = 0$ and $Q := \text{Hess } \varphi(x_0)$ invertible. Then there are neighbourhoods $U \ni x_0$, $V \ni 0$ and a C^∞ diffeomorphism $y : U \rightarrow V$ with $y(x_0) = 0$, $Dy(x_0) = I$, and*

$$\varphi(x) = \varphi(x_0) + \frac{1}{2}\langle Qy(x), y(x) \rangle \quad (x \in U).$$

Idea of the proof. Taylor's theorem with integral remainder already writes $\varphi(x) - \varphi(x_0) = \langle M(x)(x - x_0), (x - x_0) \rangle$ for a smooth symmetric matrix M with $M(x_0) = \frac{1}{2}Q$. So the whole task is to *diagonalise $M(x)$ smoothly in x* , in the sense of finding $B(x)$ with $B(x)^\top (\frac{1}{2}Q) B(x) = M(x)$; then $y(x) = B(x)(x - x_0)$ works.

The obstacle is that a symmetric matrix has no canonical square root when it is indefinite, and choosing eigenvectors is not smooth where eigenvalues collide. The trick is not to diagonalise M at all, but to write $S(x) = (\frac{1}{2}Q)^{-1}M(x)$, which is close to the identity, and take its *principal square root* — available by the holomorphic functional calculus near I , hence smooth, with no eigenvalue bookkeeping. A short computation shows $S^{1/2}$ is exactly the B required, and the reason is that S is self-adjoint for the

| bilinear form defined by Q .

Proof. Step 1. Taylor with integral remainder.

Since $\varphi(x_0)$ and $\nabla\varphi(x_0)$ contribute nothing,

$$\begin{aligned}\varphi(x) - \varphi(x_0) &= \langle M(x)(x - x_0), (x - x_0) \rangle \\ , \quad M(x) &:= \int_0^1 (1 - u) \operatorname{Hess} \varphi(x_0 + u(x - x_0)) \, du,\end{aligned}$$

by applying the one-variable Taylor formula to $u \mapsto \varphi(x_0 + u(x - x_0))$. Each $M(x)$ is symmetric, M is C^∞ in x , and $M(x_0) = \frac{1}{2}Q =: N$, which is invertible by hypothesis.

Step 2. *A matrix close to the identity.*

Set $S(x) = N^{-1}M(x)$, so S is C^∞ with $S(x_0) = I$. Shrinking the neighbourhood, we may assume $\|S(x) - I\| < \frac{1}{2}$ for $x \in U$.

The key algebraic fact is that $S(x)$ is symmetric with respect to the bilinear form $\langle N\cdot, \cdot \rangle$:

$$S(x)^\top N = M(x)^\top N^{-\top} N = M(x) = N S(x), \quad (5.3.1)$$

using that M and N are symmetric.

Step 3. *The smooth square root.*

Because $\|S - I\| < \frac{1}{2}$, the spectrum of $S(x)$ lies in the disc $|z - 1| < \frac{1}{2}$, on which the principal square root $\sqrt{\cdot}$ is holomorphic. Define $B(x) = S(x)^{1/2}$ by the convergent binomial series

$$B(x) = (I + (S(x) - I))^{1/2} = \sum_{k \geq 0} \binom{1/2}{k} (S(x) - I)^k,$$

which converges in operator norm. Its smoothness in x needs a word, since a uniform limit of smooth functions need not be smooth: the map $A \mapsto (I + A)^{1/2}$ is given on $\|A\| < 1$ by a power series in the entries of A with a positive radius of convergence, hence is real-analytic there; and $x \mapsto S(x) - I$ is C^∞ with values in that region. So B is a composition of a real-analytic map with a C^∞ one, and is C^∞ . By construction $B(x)^2 = S(x)$ and $B(x_0) = I$.

Since B is a limit of polynomials in S , relation (5.3.1) passes to B : $B^\top N = NB$. Therefore

$$B(x)^\top N B(x) = N B(x) B(x) = N S(x) = M(x).$$

Step 4. *The change of variables.*

Put $y(x) = B(x)(x - x_0)$. Then $y(x_0) = 0$ and $Dy(x_0) = B(x_0) = I$, so by the inverse function theorem y is a C^∞ diffeomorphism of a neighbourhood of x_0 onto a neighbourhood of 0. And

$$\begin{aligned} \frac{1}{2}\langle Qy, y \rangle &= \langle NB(x - x_0), B(x - x_0) \rangle \\ &= \langle B^\top NB(x - x_0), x - x_0 \rangle \\ &= \langle M(x)(x - x_0), x - x_0 \rangle, \end{aligned}$$

which is $\varphi(x) - \varphi(x_0)$ by Step 1. $\square$

WORKED EXAMPLE 5.3.1— TIER A **Why one cannot just diagonalise pointwise**

Each $M(x)$ is symmetric, so it *is* orthogonally diagonalisable for every fixed x ; the difficulty is that the diagonalising matrix cannot always be chosen to depend smoothly on x .

The standard example is $n = 2$ with

$$M(x_1, x_2) = \begin{pmatrix} x_1 & x_2 \\ x_2 & -x_1 \end{pmatrix},$$

whose eigenvalues are $\pm\sqrt{x_1^2 + x_2^2}$: they collide at the origin, and the eigenvectors rotate by π as (x_1, x_2) circles the origin once, so no continuous choice exists on a punctured neighbourhood.

Theorem 5.3.1 sidesteps this entirely. Step 3 never looks at an eigenvector: it applies a holomorphic function to the matrix, which is a smooth operation because the *spectrum* is confined to a disc where that function is defined. This is a habit worth acquiring — when a construction seems to require diagonalisation, ask whether functional calculus will do instead.

5.3.2 Stationary phase in n dimensions

Lemma 5.3.2 (The model Gaussian integral). *Let Q be a real symmetric invertible $n \times n$ matrix, with signature $\text{sgn } Q$ (the number of positive eigenvalues minus the number of negative ones). Then for $\lambda > 0$,*

$$\begin{aligned} \int_{\mathbb{R}^n} e^{\frac{i\lambda}{2}\langle Qy, y \rangle} dy \\ = \left(\frac{2\pi}{\lambda}\right)^{n/2} |\det Q|^{-1/2} e^{\frac{i\pi}{4} \text{sgn } Q}, \end{aligned}$$

the integral being understood as the limit of the integrals over balls.

Proof. Diagonalise $Q = O^\top D O$ with O orthogonal and $D = \text{diag}(d_1, \dots, d_n)$, $d_j \neq 0$. The substitution $z = O y$ has Jacobian 1 and turns the integral into a product:

$$\int_{\mathbb{R}^n} e^{\frac{i\lambda}{2} \sum_j d_j z_j^2} dz = \prod_{j=1}^n \int_{\mathbb{R}} e^{\frac{i\lambda d_j}{2} z^2} dz.$$

By Theorem 5.2.1 with $a = \lambda d_j/2$, the j -th factor is $\sqrt{2\pi/(\lambda|\lambda_j|)} e^{\frac{i\pi}{4} \text{sgn } \lambda_j}$. Multiplying, the moduli give $(2\pi/\lambda)^{n/2} \prod_j |\lambda_j|^{-1/2} = (2\pi/\lambda)^{n/2} |\det Q|^{-1/2}$, and the phases add to $\frac{\pi}{4} \sum_j \text{sgn } \lambda_j = \frac{\pi}{4} \text{sgn } Q$. $\square$

Theorem 5.3.3 (Stationary phase in $\mathbb{R}^n$). *Let φ be real-valued and C^∞ on $\mathbb{R}^n$ and $\psi \in C_c^\infty(\mathbb{R}^n)$. Suppose φ has exactly one critical point x_0 in $\text{supp } \psi$, that $Q = \text{Hess } \varphi(x_0)$ is invertible, and that x_0 is interior to $\text{supp } \psi$. Then as $\lambda \rightarrow \infty$,*

$$\int_{\mathbb{R}^n} e^{i\lambda\varphi(x)} \psi(x) dx = e^{i\lambda\varphi(x_0)} \left(\frac{2\pi}{\lambda}\right)^{n/2} \frac{e^{\frac{i\pi}{4} \text{sgn } Q}}{|\det Q|^{1/2}} \psi(x_0) + O(\lambda^{-\frac{n}{2}-1}).$$

Proof. Step 1. Localise.

Let U be the neighbourhood of Theorem 5.3.1 and $\chi \in C_c^\infty(U)$ equal to 1 near x_0 . On $\text{supp } ((1 - \chi)\psi)$ the phase has no critical point, so $|\nabla\varphi|$ is bounded below there by compactness and Theorem 5.1.1 gives $O(\lambda^{-N})$ for every N . Replace ψ by $\chi\psi$.

Step 2. Straighten the phase.

Apply Theorem 5.3.1 and substitute $x = x(y)$:

$$\begin{aligned} \int e^{i\lambda\varphi(x)} \psi(x) dx &= e^{i\lambda\varphi(x_0)} \int_{\mathbb{R}^n} e^{\frac{i\lambda}{2} \langle Qy, y \rangle} \tilde{\psi}(y) dy, \\ \tilde{\psi}(y) &= \psi(x(y)) |\det Dx(y)|, \end{aligned}$$

with $\tilde{\psi} \in C_c^\infty(\mathbb{R}^n)$ and $\tilde{\psi}(0) = \psi(x_0)$, since $Dx(0) = I$.

Step 3. Split the amplitude.

Write $\tilde{\psi}(y) = \tilde{\psi}(0) + \sum_{j=1}^n y_j \rho_j(y)$ with $\rho_j(y) = \int_0^1 (\partial_j \tilde{\psi})(uy) du$ smooth and compactly supported. As in Theorem 5.2.3, extending the constant term's integral to all of $\mathbb{R}^n$ costs $O(\lambda^{-N})$ by Theorem 5.1.1, the phase $\frac{1}{2} \langle Qy, y \rangle$ having no critical point away from the origin.

Step 4. The main term.

By Theorem 5.3.2, the constant term contributes

$$\tilde{\psi}(0) \left(\frac{2\pi}{\lambda}\right)^{n/2} |\det Q|^{-1/2} e^{\frac{i\pi}{4} \text{sgn } Q},$$

which with $\tilde{\psi}(0) = \psi(x_0)$ is the stated main term.

Step 5. *The remainder.*

Since $\nabla_y(e^{\frac{i\lambda}{2}\langle Qy, y \rangle}) = i\lambda Qy e^{\frac{i\lambda}{2}\langle Qy, y \rangle}$, and Q is invertible, we may write y_j as a linear combination of the components of Qy and integrate by parts once in each remainder term, with no boundary contribution:

$$\int e^{\frac{i\lambda}{2}\langle Qy, y \rangle} y_j \rho_j(y) dy = \frac{1}{i\lambda} \int e^{\frac{i\lambda}{2}\langle Qy, y \rangle} (\text{a first-order derivative of } \rho_j) dy.$$

The remaining integral is again of the model form with a smooth compactly supported amplitude, so it is $O(\lambda^{-n/2})$ — apply Steps 3 and 4 to it, or Theorem 5.3.2 after one more split. Hence each remainder term is $O(\lambda^{-1} \cdot \lambda^{-n/2})$, as claimed. $\square$

Where we are. Compare the exponents. In one variable the answer was $\lambda^{-1/2}$; here it is $\lambda^{-n/2}$. The reason is the one anticipated in Volume 5's preface: each of the n directions at the critical point contributes a window of width $\lambda^{-1/2}$ in which the phase fails to oscillate, and the critical region therefore has measure $\lambda^{-n/2}$.

The hypothesis that Q is invertible is exactly the statement that *every* direction is genuinely quadratic. If Q has a null direction, the window in that direction is wider than $\lambda^{-1/2}$ and the decay is worse; the extreme case is a phase constant in some direction, where there is no decay at all in that variable. Chapter 5.5 is the systematic version of this observation, with Q replaced by the second fundamental form of a hypersurface and its rank by the number of non-vanishing principal curvatures.

WORKED EXAMPLE 5.3.2— TIER B The signature is not a decoration

The factor $e^{i\pi \operatorname{sgn} Q/4}$ is easy to overlook, since it has modulus 1 and disappears from any estimate that takes absolute values. It matters whenever two or more critical points contribute, because then the answer is a *sum* and the relative phases decide whether the contributions reinforce or cancel.

An example where they cancel. Take $n = 1$, $\varphi(t) = \sin t$ on $(0, 2\pi)$, with $\psi \in C_c^\infty((0, 2\pi))$ equal to 1 on $[\frac{\pi}{4}, \frac{7\pi}{4}]$. The critical points are $t = \pi/2$ and $t = 3\pi/2$, with $\varphi'' = -1$ and $+1$ respectively, and $\varphi(\pi/2) = 1$, $\varphi(3\pi/2) = -1$. By Theorem 5.2.3 the two contributions are

$$\sqrt{\frac{2\pi}{\lambda}} e^{i\lambda} e^{-i\pi/4} \quad \text{and} \quad \sqrt{\frac{2\pi}{\lambda}} e^{-i\lambda} e^{+i\pi/4},$$

whose sum is

$$\sqrt{\frac{2\pi}{\lambda}} \cdot 2 \cos\left(\lambda - \frac{\pi}{4}\right).$$

So the total is of size $\lambda^{-1/2}$ *on average*, but it *vanishes* whenever $\lambda - \frac{\pi}{4}$ is an odd multiple of $\pi/2$. Had we recorded only the moduli, we would have predicted a total of size $\lambda^{-1/2}$ always, and been wrong at infinitely many λ .

Why this is the sphere. The computation is not an artificial example. In Chapter 5.4 the two critical points will be the two poles of the sphere, the signatures will differ in the same way, and the resulting cosine is exactly the factor $\cos(r - \frac{\nu\pi}{2} - \frac{\pi}{4})$ in the Bessel asymptotic — which is why $\widehat{d\sigma}$ for $\mathbb{S}^2$ came out as $2\sin(2\pi r)/r$ in Volume 2 rather than as a positive multiple of r^{-1} .

And why Volume 6 needs it. The Stein–Tomas argument estimates $f * \widehat{d\sigma}$, and at the endpoint exponent the oscillation of $\widehat{d\sigma}$ is used, not merely its size. An argument that discards the phase cannot reach the endpoint. This is the concrete reason Volume 2’s Theorem 8.5, which bounded only $|\widehat{d\sigma}|$, was insufficient and this volume was necessary.

Drills

Drill 5.3.1. Verify (5.3.1) and check that the relation $B^\top N = NB$ does pass from S to the series defining B .

Drill 5.3.2. Compute the Morse normal form explicitly for $\varphi(x_1, x_2) = x_1^2 - x_2^2 + x_1^3$ near the origin, and identify Q and $\text{sgn } Q$.

Drill 5.3.3. Show that Theorem 5.3.3 with $n = 1$ reduces to Theorem 5.2.3, including the phase factor.

Drill 5.3.4. Let $\varphi(x) = |x|^2$ and ψ radial. Compute both sides of Theorem 5.3.3 directly using polar coordinates, and confirm the constant $(2\pi/\lambda)^{n/2}$.

Where we are. Three features of the Morse lemma are worth separating, because they are often conflated.

It is a normal form, not an estimate. The conclusion is an exact identity after a change of variables — no error terms, no asymptotics. The asymptotics of Theorem 5.3.3 follow from it by computing a Gaussian integral in the new coordinates, and that computation is where the $\lambda^{-n/2}$ appears.

Non-degeneracy is essential and not merely convenient. If $\text{Hess } \varphi(x_0)$ is singular there is no normal form of this shape: the degenerate directions require higher-order models, and the classification of those is singularity theory, which this series does not enter. The Pitfall in Chapter 5.5 shows what the loss costs in practice.

The proof is constructive. $B(x)$ is given by an explicit convergent series in $S(x) - I$, so the change of variables can in principle be computed. That is unusual — most normal-form theorems rest on implicit function arguments — and it is why the functional-calculus proof is the one worth knowing.

What replaces it in the degenerate case. Nothing general. One returns to Theorem 2.8.4 and estimates the order of vanishing directly, which is what Worked Example 5.5.4's last row does.

Notes

Theorem 5.3.1 is Morse's, from 1925; the proof by functional calculus given here avoids the induction on dimension found in most textbooks and is due to Palais. The stationary phase formula in the form Theorem 5.3.3 is classical; the systematic development, including the full asymptotic expansion in powers of λ^{-1} , is in Hörmander's *The Analysis of Linear Partial Differential Operators I*, Chapter VII.

Chapter 5.4

Bessel Functions

Theorem 2.8.3 expressed $\widehat{d\sigma}$ for the sphere exactly in terms of Bessel functions, and Theorem 2.8.5 obtained the bound $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ by a direct estimate. What was deferred — and recorded as a debt — is the *asymptotic*: the precise leading behaviour, including the oscillation. This chapter supplies it.

5.4.1 The statement, and why the amplitude is awkward

Recall from Volume 2, Chapter 8 that for $\nu > -\frac{1}{2}$,

$$J_\nu(r) = \frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{irt} (1-t^2)^{\nu-\frac{1}{2}} dt. \quad (5.4.1)$$

Theorem 5.4.1. For $\nu > -\frac{1}{2}$ fixed,

$$\begin{aligned} J_\nu(r) &= \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) \\ &\quad + O_\nu(r^{-3/2}) \quad (r \rightarrow \infty). \end{aligned}$$

Scratch work. The obvious route is stationary phase. Substituting $t = \cos \theta$ in (5.4.1) gives $\int_0^\pi e^{ir \cos \theta} (\sin \theta)^{2\nu} d\theta$, whose phase $\cos \theta$ has non-degenerate critical points at $\theta = 0$ and $\theta = \pi$. So Theorem 5.2.3 ought to apply.

It does not, and Worked Example 5.2.2 already showed why: the amplitude $(\sin \theta)^{2\nu}$ vanishes at both critical points when $\nu > 0$, so the leading term $\psi(t_0)$ of Theorem 5.2.3 is zero and the theorem returns only the error bound. The true answer, of size $r^{-1/2}$, lies between.

The resolution is to work in the original variable t , where the two critical points become the two *endpoints* $t = \pm 1$, and where the difficulty is visibly a degeneracy of the amplitude rather than of the phase. Near $t = 1$ the amplitude behaves like a constant times $(1 - t)^{\nu-1/2}$, and the model integral one needs is not the Fresnel integral but

$$\int_0^\infty e^{-irs} s^{a-1} ds = \Gamma(a) (ir)^{-a},$$

a Gamma integral — which we evaluate by exactly the analytic-continuation device used for the Fresnel integral in Theorem 5.2.1. That is the whole proof: two endpoints, each contributing a Gamma integral, and their sum producing the cosine.

5.4.2 The model integral

Lemma 5.4.2. *Let $a > 0$ and $r > 0$. Then*

$$\int_0^\infty e^{-irs} s^{a-1} ds = \Gamma(a) r^{-a} e^{-\frac{i\pi a}{2}},$$

the integral being understood as $\lim_{\varepsilon \downarrow 0} \int_0^\infty e^{-(\varepsilon+ir)s} s^{a-1} ds$.

Proof. For $\operatorname{Re} z > 0$ define $G(z) = \int_0^\infty e^{-zs} s^{a-1} ds$. The integral converges absolutely, since $|e^{-zs} s^{a-1}| = e^{-(\operatorname{Re} z)s} s^{a-1}$ is integrable at 0 because $a > 0$ and at ∞ because $\operatorname{Re} z > 0$; and G is holomorphic there by differentiation under the integral sign (Volume 1, Worked Example 5.3).

For $z = \sigma > 0$ real, the substitution $u = \sigma s$ gives $G(\sigma) = \Gamma(a) \sigma^{-a}$. Let $H(z) = \Gamma(a) z^{-a}$ with the branch of $z^{-a} = e^{-a \log z}$ determined by $\log \sigma \in \mathbb{R}$ for $\sigma > 0$; this is holomorphic on the right half-plane. Since $G = H$ on $(0, \infty)$, they agree on $\{\operatorname{Re} z > 0\}$, by the propagation argument of Theorem 2.1.10.

Now let $z = \varepsilon + ir$ and $\varepsilon \downarrow 0$. Writing $ir = re^{i\pi/2}$, the branch gives $z^{-a} \rightarrow r^{-a} e^{-i\pi a/2}$, the argument being $-\pi a/2$ because the path approaches the positive imaginary axis from within the right half-plane where the branch is continuous. $\square$

WORKED EXAMPLE 5.4.1— TIER A Consistency with the Fresnel integral

Theorem 5.4.2 and Theorem 5.2.1 are the same computation in different clothes, and it is worth checking they agree.

Take $a = \frac{1}{2}$ in Theorem 5.4.2:

$$\int_0^\infty e^{-irs} s^{-1/2} ds = \Gamma\left(\frac{1}{2}\right) r^{-1/2} e^{-i\pi/4} = \sqrt{\frac{\pi}{r}} e^{-i\pi/4}.$$

On the other hand, substituting $s = t^2$ turns the left side into $2 \int_0^\infty e^{-irt^2} dt = \int_{\mathbb{R}} e^{-irt^2} dt$, which by Theorem 5.2.1 with $a = -r$ is $\sqrt{\pi/r} e^{-i\pi/4}$. ✓

Both facts come from continuing $\int_0^\infty e^{-zs} s^{a-1} ds$ from the real axis to the imaginary axis, and in both the phase is a branch choice. If one is remembered, the other can be recovered.

5.4.3 Proof of the asymptotic

Proof of Theorem 5.4.1. Write $\alpha = \nu - \frac{1}{2} > -1$ and

$$I(r) = \int_{-1}^1 e^{irt} (1-t^2)^\alpha dt,$$

so that $J_\nu(r) = \frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\sqrt{\pi}} I(r)$ by (5.4.1) and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Step 1. *Cut into three pieces.*

Fix $\chi \in C_c^\infty(\mathbb{R})$ with $\chi = 1$ on $[-\frac{1}{4}, \frac{1}{4}]$ and $\text{supp } \chi \subseteq (-\frac{1}{2}, \frac{1}{2})$, and write

$$1 = \chi(1-t) + \chi(1+t) + \chi_0(t)$$

on $[-1, 1]$, which defines χ_0 , smooth and supported in $[-\frac{3}{4}, \frac{3}{4}]$.

Step 2. *The interior piece is negligible.*

On $\text{supp } \chi_0$ the amplitude $(1-t^2)^\alpha \chi_0(t)$ is C^∞ with compact support in $(-1, 1)$, and the phase is t , with derivative 1. By Theorem 5.1.1 — or simply by integrating by parts N times, all boundary terms vanishing — this piece is $O_N(r^{-N})$ for every N .

Step 3. *An endpoint piece.*

Consider t near 1 and substitute $t = 1 - s$:

$$\begin{aligned} I_+(r) &:= \int_{-1}^1 e^{irt} (1-t^2)^\alpha \chi(1-t) dt \\ &= e^{ir} \int_0^\infty e^{-irs} s^\alpha g(s) ds \\ &, \quad g(s) := (2-s)^\alpha \chi(s), \end{aligned}$$

where $g \in C_c^\infty([0, \frac{1}{2}))$ with $g(0) = 2^\alpha$.

Split $g(s) = g(0) + s h(s)$ with $h(s) = \int_0^1 g'(us) du$ smooth and compactly supported in $[0, \frac{1}{2})$. Then

$$I_+(r) = e^{ir} g(0) \int_0^\infty e^{-irs} s^\alpha \chi_1(s) ds + e^{ir} \int_0^\infty e^{-irs} s^{\alpha+1} h(s) ds,$$

where $\chi_1 \in C_c^\infty([0, 1])$ equals 1 on $\text{supp } \chi$; inserting χ_1 is legitimate because g vanishes beyond $\frac{1}{2}$.

Main term. Removing the cutoff χ_1 from the first integral costs

$$\left| \int_0^\infty e^{-irs} s^\alpha (1 - \chi_1(s)) \, ds \right| = O_N(r^{-N}),$$

by N -fold integration by parts, the integrand being smooth and the phase having non-vanishing derivative on the support of $1 - \chi_1$, which is bounded away from $s = 0$. By Theorem 5.4.2 with $a = \alpha + 1 = \nu + \frac{1}{2} > 0$,

$$\begin{aligned} g(0) \int_0^\infty e^{-irs} s^\alpha \, ds \\ = 2^\alpha \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} e^{-\frac{i\pi}{2}(\nu + \frac{1}{2})}. \end{aligned}$$

Error term. The second integral has $s^{\alpha+1}$ in place of s^α , so the same argument with $a = \alpha + 2 = \nu + \frac{3}{2}$ bounds it by $O_\nu(r^{-\nu - \frac{3}{2}})$.

Hence

$$I_+(r) = 2^{\nu - \frac{1}{2}} \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} e^{i(r - \frac{\pi}{2}(\nu + \frac{1}{2}))} + O_\nu(r^{-\nu - \frac{3}{2}}). \quad (5.4.2)$$

Step 4. *The other endpoint.*

The piece near $t = -1$ is $I_-(r) = \overline{I_+(r)}$, because $I(r)$ is real — the imaginary part of the integrand in $I(r)$ is odd in t — and the interior piece is real up to $O(r^{-N})$. Alternatively substitute $t \mapsto -t$ directly.

Step 5. *Add.*

By Steps 2–4,

$$\begin{aligned} I(r) &= 2 \operatorname{Re} I_+(r) + O_N(r^{-N}) \\ &= 2^{\nu + \frac{1}{2}} \Gamma(\nu + \tfrac{1}{2}) r^{-\nu - \frac{1}{2}} \cos\left(r - \frac{\pi}{2}\left(\nu + \frac{1}{2}\right)\right) + O_\nu(r^{-\nu - \frac{3}{2}}). \end{aligned}$$

Note $\frac{\pi}{2}(\nu + \frac{1}{2}) = \frac{\nu\pi}{2} + \frac{\pi}{4}$, the phase shift in the statement.

Step 6. *Restore the prefactor.*

Multiplying by $\frac{(r/2)^\nu}{\Gamma(\nu + \frac{1}{2})\sqrt{\pi}}$,

$$\begin{aligned} J_\nu(r) &= \frac{2^{-\nu} r^\nu}{\sqrt{\pi}} \cdot 2^{\nu + \frac{1}{2}} r^{-\nu - \frac{1}{2}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O_\nu(r^{-\frac{3}{2}}) \\ &= \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O_\nu(r^{-\frac{3}{2}}), \end{aligned}$$

since $2^{-\nu} 2^{\nu + \frac{1}{2}} / \sqrt{\pi} = \sqrt{2/\pi}$. □

Where we are. The debt recorded in Volume 2, Chapter 8 is now discharged. Three features of the answer are worth noting against what Volume 2 could say.

Volume 2 proved $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ and demonstrated sharpness only for $n = 3$, where an elementary closed formula was available. Theorem 5.4.1 gives sharpness in every dimension at once: the cosine equals ± 1 along a sequence $r_k \rightarrow \infty$, so the decay rate is attained infinitely often.

Volume 2 could not see the oscillation. Here it is explicit, and Worked Example 5.3.2 explained its origin: the two endpoints are the two poles of the sphere, their contributions are complex conjugates, and a sum of conjugates is a cosine.

Finally, the phase shift $\frac{\nu\pi}{2} + \frac{\pi}{4}$ decomposes into a part $\frac{\pi}{4}$ coming from the branch — the same $\pi/4$ as in the Fresnel integral — and a part $\frac{\nu\pi}{2}$ coming from the order of vanishing of the amplitude at the endpoint. The second is invisible in the one-dimensional stationary phase formula, and is exactly what Worked Example 2.5 found missing.

WORKED EXAMPLE 5.4.2— TIER B **Checking the formula against the two cases we know**

(a) $\nu = \frac{1}{2}$. Worked Example 2.8.1 computed $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$ exactly. Theorem 5.4.1 with $\nu = \frac{1}{2}$ predicts

$$\begin{aligned} \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\pi}{4} - \frac{\pi}{4}\right) \\ &= \sqrt{\frac{2}{\pi r}} \cos\left(r - \frac{\pi}{2}\right) \\ &= \sqrt{\frac{2}{\pi r}} \sin r, \end{aligned}$$

with error $O(r^{-3/2})$ — and in fact the error is 0, the formula being exact in this case. ✓

(b) $n = 3$ via the exact formula. Volume 2 computed $\widehat{d\sigma}(\xi) = 2 \sin(2\pi|\xi|)/|\xi|$ for $\mathbb{S}^2 \subseteq \mathbb{R}^3$. Via Theorem 2.8.3 with $\nu = \frac{n-2}{2} = \frac{1}{2}$ and $r = |\xi|$,

$$\widehat{d\sigma}(\xi) = 2\pi r^{-1/2} J_{1/2}(2\pi r) = 2\pi r^{-1/2} \sqrt{\frac{2}{\pi \cdot 2\pi r}} \sin(2\pi r) = \frac{2 \sin(2\pi r)}{r}. \quad \checkmark$$

(c) *What the general formula adds.* For $n = 4$, $\nu = 1$, and Theorem 5.4.1 gives

$$\begin{aligned}\widehat{d\sigma}(\xi) &= 2\pi r^{-1} J_1(2\pi r) \\ &= 2\pi r^{-1} \sqrt{\frac{1}{\pi^2 r}} \cos\left(2\pi r - \frac{3\pi}{4}\right) + O(r^{-5/2}) \\ &= \frac{2}{r^{3/2}} \cos\left(2\pi r - \frac{3\pi}{4}\right) + O(r^{-5/2}),\end{aligned}$$

of size $r^{-3/2} = r^{-(n-1)/2}$ as predicted, with a phase shift that no elementary formula would have revealed.

WORKED EXAMPLE 5.4.3— TIER B **Bessel asymptotics without Bessel functions**

Theorem 5.4.1 gives $J_\nu(t) \approx t^{-1/2} \cos(t - \frac{\nu\pi}{2} - \frac{\pi}{4})$ for large t . The asymptotic can be predicted from stationary phase alone, without any special-function theory, and doing so is a good check.

The integral representation. $J_\nu(t)$ is, up to constants, $\int_{-1}^1 e^{its} (1-s^2)^{\nu-1/2} ds$. The phase is $\varphi(s) = s$, which has *no* interior critical point — so by Theorem 5.1.1 the decay would be rapid, were it not for the endpoints.

Where the decay comes from. The amplitude $(1-s^2)^{\nu-1/2}$ vanishes to order $\nu - \frac{1}{2}$ at $s = \pm 1$. Theorem 5.4.2 handles exactly this: an oscillatory integral whose amplitude vanishes to order β at an endpoint contributes $t^{-\beta-1}$. With $\beta = \nu - \frac{1}{2}$ this gives $t^{-\nu-1/2}$ — which is not $t^{-1/2}$.

Reconciling. The discrepancy is the normalisation: J_ν carries a factor t^ν relative to the integral above, and $t^\nu \cdot t^{-\nu-1/2} = t^{-1/2}$. ✓

The phase, and why there are two terms. The two endpoints $s = \pm 1$ contribute $e^{\pm it}$, and their sum is a cosine. The shift $-\frac{\nu\pi}{2} - \frac{\pi}{4}$ comes from the argument of the constant in Theorem 5.4.2, which involves $\Gamma(\nu + \frac{1}{2})$ and a factor $e^{\pm i\pi(\dots)/2}$.

What to retain. The decay rate of a Bessel function is an endpoint phenomenon, not a stationary-phase one — and that is why $\widehat{d\sigma}$, whose decay *is* stationary phase, has the same exponent for a different reason. Volume 5, Chapter 5.5 keeps the two straight.

Drills

Drill 5.4.1. Verify that $I(r)$ in the proof is real, by splitting the integrand into even and odd parts in t .

Drill 5.4.2. Carry out the substitution $t = 1 - s$ in Step 3 and confirm $g(s) = (2 - s)^\alpha \chi(s)$ with $g(0) = 2^\alpha$.

Drill 5.4.3. Extend Theorem 5.4.1 by one more term: show that the next correction is of size $r^{-3/2}$ with an explicit coefficient, by retaining the $sh(s)$ term in Step 3 rather than discarding it.

Drill 5.4.4. Use Theorem 5.4.1 to show that J_ν has infinitely many zeros, and that consecutive zeros are asymptotically at distance π .

Notes

The asymptotic expansion of Bessel functions is due to Poisson and Hankel; the endpoint analysis given here, reducing everything to the Gamma integral of Theorem 5.4.2, is the standard modern treatment and is the same device — analytic continuation of a convergent integral to the boundary of its half-plane — as Theorem 5.2.1.

Watson's *Theory of Bessel Functions* remains the reference for the full expansion and for uniform asymptotics in ν , which this series does not need.

Chapter 5.5

Surface Measure and Curvature

Theorem 2.8.5 proved $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ for the sphere, and recorded as a debt the general principle: a hypersurface with k non-vanishing principal curvatures has $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$. This chapter pays it, and in doing so explains the exponent.

5.5.1 The set-up

Let $S \subseteq \mathbb{R}^n$ be a C^∞ hypersurface with surface measure $d\sigma$, and let $\psi \in C_c^\infty(\mathbb{R}^n)$ be a cutoff, so that $\psi d\sigma$ is a compactly supported measure. Its Fourier transform is the bounded continuous function

$$\widehat{\psi d\sigma}(\xi) = \int_S e^{-2\pi i x \cdot \xi} \psi(x) d\sigma(x),$$

which is a tempered distribution by Theorem 2.5.8 and an honest function because $\psi d\sigma$ is finite.

Write $\xi = \lambda\omega$ with $\lambda = |\xi|$ and $\omega \in \mathbb{S}^{n-1}$, and set

$$\varphi_\omega(x) = -2\pi x \cdot \omega,$$

so that $\widehat{\psi d\sigma}(\lambda\omega) = \int_S e^{i\lambda\varphi_\omega} \psi d\sigma$ is an oscillatory integral on S with large parameter λ .

Lemma 5.5.1 (Where the phase is critical). *Let $x \in S$. The restriction of φ_ω to S has a critical point at x if and only if ω is normal to S at x , that is $\omega = \pm\nu(x)$ where ν is a unit normal. Moreover, parametrising S near such an x as a graph over its tangent plane,*

$$\text{Hess}(\varphi_\omega|_S)(x) = \mp 2\pi \mathbb{I}_x,$$

where $\mathbb{I}_x$ is the second fundamental form of S at x , whose eigenvalues are the principal curvatures $\kappa_1(x), \dots, \kappa_{n-1}(x)$.

Proof. Rotate and translate so that $x = 0$ and $T_x S = \{x_n = 0\}$, and write S near 0 as a graph $x_n = h(y)$, $y \in \mathbb{R}^{n-1}$, with $h(0) = 0$ and $\nabla h(0) = 0$; the unit normal at 0 is then e_n . In these coordinates

$$\varphi_\omega(y, h(y)) = -2\pi(y \cdot \omega' + h(y)\omega_n), \quad \omega = (\omega', \omega_n),$$

so its gradient at $y = 0$ is $-2\pi\omega'$, using $\nabla h(0) = 0$. This vanishes exactly when $\omega' = 0$, i.e. $\omega = \pm e_n = \pm \nu(0)$.

For such ω , differentiating twice at 0 gives $\text{Hess} = -2\pi\omega_n \text{Hess } h(0)$, and $\text{Hess } h(0)$ is by definition the second fundamental form in these coordinates, with $\omega_n = \pm 1$. $\square$

Where we are. This is the geometric content of the whole chapter, and it is worth stating in words. The critical points of a linear phase on a hypersurface are the points where the normal points along ξ ; the Hessian there is the second fundamental form. So:

- *curvature* = non-degeneracy of the Hessian = the hypothesis of stationary phase;
- the *number of non-vanishing principal curvatures* = the rank of the Hessian = the number of directions each contributing a window of width $\lambda^{-1/2}$;
- a *flat* piece has $\mathbb{I} = 0$, so every point of it is a degenerate critical point when ω is its normal — and there is no decay in that direction, which is Worked Example 2.8.2.

5.5.2 Non-vanishing Gaussian curvature

Theorem 5.5.2. *Let S be a C^∞ hypersurface and $\psi \in C_c^\infty$ with $\text{supp } \psi \cap S$ compact. Suppose all $n - 1$ principal curvatures are non-zero at every point of $\text{supp } \psi \cap S$. Then for every $\omega \in \mathbb{S}^{n-1}$, writing $x_1(\omega), \dots, x_m(\omega)$ for the points of $\text{supp } \psi \cap S$ at which ω is normal,*

$$\begin{aligned} \widehat{\psi d\sigma}(\lambda\omega) &= \lambda^{-\frac{n-1}{2}} \sum_{j=1}^m e^{-2\pi i \lambda x_j \cdot \omega} c_j(\omega) \\ &\quad + O(\lambda^{-\frac{n-1}{2}-1}), \end{aligned}$$

where

$$\begin{aligned}
 |c_j(\omega)| &= \psi(x_j) |K(x_j)|^{-1/2} \\
 , \quad K &= \kappa_1 \cdots \kappa_{n-1} \text{ the Gaussian curvature,}
 \end{aligned}$$

and the implied constant is uniform in ω . In particular

$$|\widehat{\psi d\sigma}(\xi)| \lesssim_{S,\psi} |\xi|^{-\frac{n-1}{2}}.$$

Idea of the proof. By Theorem 5.5.1 the critical points are the preimages of ω under the Gauss map $\nu : S \rightarrow \mathbb{S}^{n-1}$, and the Hessian there is the second fundamental form — which is invertible exactly when the Gaussian curvature is non-zero. So the Gauss map is a local diffeomorphism, its fibres are discrete, and on a compact piece there are finitely many, with the number locally constant in ω .

Each contributes a term by Theorem 5.3.3, with $|\det \text{Hess}|^{-1/2} = |K|^{-1/2}$; away from the critical points Theorem 5.1.1 contributes $O(\lambda^{-N})$. The only genuinely new issue is *uniformity in ω* , since the critical points move as ω varies; this is handled by compactness.

Proof. Step 1. The critical set is finite, and moves smoothly.

By Theorem 5.5.1, x is critical for $\varphi_\omega|_S$ iff $\nu(x) = \pm\omega$. The differential of the Gauss map ν at x is the shape operator, whose matrix is $\mathbb{I}_x$; by hypothesis this is invertible on the compact set $\text{supp } \psi \cap S$, so ν is a local diffeomorphism there. Hence the fibre $\nu^{-1}(\{\pm\omega\}) \cap \text{supp } \psi$ is discrete, and being closed in a compact set it is finite; and by the inverse function theorem the critical points depend smoothly on ω , locally.

Since the number $m = m(\omega)$ is locally constant and $\mathbb{S}^{n-1}$ is compact, m is bounded, say by m_0 .

Step 2. Partition.

Fix ω_0 and let $x_1, \dots, x_m$ be its critical points. Choose disjoint coordinate patches $U_1, \dots, U_m$ around them, on each of which S is a graph as in Theorem 5.5.1, and a smooth partition of unity $1 = \sum_{j=1}^m \chi_j + \chi_0$ subordinate to $\{U_j\} \cup (S \setminus \bigcup_j \overline{V_j})$, with $V_j \Subset U_j$ neighbourhoods of x_j . By continuity of ν this partition serves for all ω in a neighbourhood W of ω_0 , with the critical point of φ_ω in U_j depending smoothly on $\omega \in W$.

Step 3. The non-critical part.

On $\text{supp } \chi_0$ there is no critical point of φ_ω for $\omega \in W$, after shrinking W ; by compactness $|\nabla_S \varphi_\omega|$ is bounded below there, uniformly in $\omega \in W$.

Theorem 5.1.1, applied in the graph coordinates, gives

$$\int_S e^{i\lambda\varphi_\omega} \psi \chi_0 \, d\sigma = O_N(\lambda^{-N}),$$

uniformly in $\omega \in W$, the constants depending on finitely many derivatives of the parametrisation and of ψ , all bounded on the compact patch.

Step 4. *Each critical patch.*

On U_j , parametrise S as a graph $x = (y, h_j(y))$ and write the integral as

$$\int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi_j(y,\omega)} a_j(y) \, dy, \quad \Phi_j(y, \omega) = -2\pi(y \cdot \omega' + h_j(y)\omega_n),$$

with a_j smooth and compactly supported, incorporating $\psi\chi_j$ and the surface measure density $\sqrt{1 + |\nabla h_j|^2}$. By Theorem 5.5.1 the Hessian in y at the critical point is $\mp 2\pi\mathbb{I}$, invertible by hypothesis, with

$$|\det \text{Hess}_y \Phi_j| = (2\pi)^{n-1} |K(x_j)|.$$

Theorem 5.3.3 applies and gives

$$\begin{aligned} \left(\frac{2\pi}{\lambda}\right)^{\frac{n-1}{2}} \frac{e^{\frac{i\pi}{4} \text{sgn Hess}}}{(2\pi)^{\frac{n-1}{2}} |K(x_j)|^{1/2}} a_j(y_j(\omega)) e^{i\lambda\Phi_j(y_j(\omega), \omega)} \\ + O(\lambda^{-\frac{n-1}{2}-1}), \end{aligned}$$

and Φ_j evaluated at the critical point is $-2\pi x_j \cdot \omega$, while $a_j(y_j) = \psi(x_j)$ because the density equals 1 where $\nabla h_j = 0$. This is the j -th term of the statement, with $|c_j| = \psi(x_j) |K(x_j)|^{-1/2}$ as claimed.

Step 5. *Uniformity.*

The constants in Steps 3 and 4 depend only on finitely many derivatives of h_j , a_j and of the critical point $y_j(\omega)$, all of which vary continuously on the compact $\overline{W}$. Covering $\mathbb{S}^{n-1}$ by finitely many such W gives a bound uniform in ω . $\square$

WORKED EXAMPLE 5.5.1— TIER A **The sphere, recovered**

For $S = \mathbb{S}^{n-1}$ and $\psi \equiv 1$ near S , the Gauss map is the identity, so each ω has exactly two critical points, $x = \omega$ and $x = -\omega$, and all principal curvatures equal 1, so $K \equiv 1$.

Theorem 5.5.2 then reads

$$\begin{aligned} \widehat{\mathrm{d}\sigma}(\lambda\omega) \\ = \lambda^{-\frac{n-1}{2}} \left(c_+ e^{-2\pi i\lambda} + c_- e^{2\pi i\lambda} \right) + O(\lambda^{-\frac{n-1}{2}-1}), \end{aligned}$$

using $x_{\pm} \cdot \omega = \pm 1$. The two Hessians differ by a sign, so their signatures are $-(n-1)$ and $+(n-1)$ and $c_{\pm}$ are complex conjugates; the sum is therefore a cosine.

Comparing with Theorem 5.4.1 — which gave $\widehat{\mathrm{d}\sigma}(\xi) = 2\pi r^{-\frac{n-2}{2}} J_{\frac{n-2}{2}}(2\pi r)$ and hence a cosine of $2\pi r - \frac{(n-2)\pi}{4} - \frac{\pi}{4}$ — the two agree, and the phase shift is exactly $\frac{\pi}{4} \operatorname{sgn} \operatorname{Hess}$. So the general theorem and the exact Bessel computation are consistent, as they must be.

5.5.3 Partial curvature

Theorem 5.5.3. *Let S and ψ be as above, and suppose that at every point of $\operatorname{supp} \psi \cap S$ at least k of the principal curvatures are non-zero. Then*

$$|\widehat{\psi \mathrm{d}\sigma}(\xi)| \lesssim_{S,\psi} |\xi|^{-k/2}.$$

Idea of the proof. Split the tangential variables into the k directions in which the second fundamental form is non-degenerate and the remaining $n-1-k$. Apply stationary phase in the first group, treating the second as parameters; each of the k non-degenerate directions contributes $\lambda^{-1/2}$, and the parameters contribute nothing — but also cost nothing, since the amplitude is compactly supported and one may simply integrate over them. The only technical point is that Theorem 5.3.3 must be applied with a constant uniform in the parameters. Inspecting its proof, the constant depends only on finitely many derivatives of phase and amplitude and on a lower bound for $|\det \operatorname{Hess}|$ — all of which are uniform here, by compactness.

Proof. Step 1. Reduce to a patch.

By a partition of unity as in Step 2 of Theorem 5.5.2, it suffices to bound the contribution of a single coordinate patch, on which S is a graph $x = (y, h(y))$, $y \in \mathbb{R}^{n-1}$, and the integral is

$$\int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi(y)} a(y) \, \mathrm{d}y, \quad \Phi(y) = -2\pi(y \cdot \omega' + h(y)\omega_n),$$

with $a \in C_c^\infty$. If Φ has no critical point on $\operatorname{supp} a$ we are done by Theorem 5.1.1, so assume it has one, at y_0 .

Step 2. *Split the variables.*

At y_0 the Hessian $\text{Hess } \Phi(y_0) = \mp 2\pi \mathbf{I}$ has rank at least k by hypothesis. Choose an orthogonal change of coordinates in y so that the upper-left $k \times k$ block of $\text{Hess } \Phi(y_0)$ is invertible, and write $y = (u, v)$ with $u \in \mathbb{R}^k$, $v \in \mathbb{R}^{n-1-k}$. By continuity, that block remains invertible, with determinant bounded below, on a neighbourhood of y_0 ; shrink $\text{supp } a$ to lie inside it.

Step 3. *Stationary phase in u , with v as a parameter.*

For each fixed v , the function $u \mapsto \Phi(u, v)$ has, on the shrunken support, a Hessian in u that is invertible with determinant bounded below uniformly in v . By the implicit function theorem applied to $\nabla_u \Phi = 0$, it has at most one critical point $u(v)$ there, depending smoothly on v .

Applying Theorem 5.3.3 in the u variables — with v a parameter, and noting that its proof produces a constant depending only on finitely many derivatives of Φ and a and on the lower bound for $|\det \text{Hess}_u \Phi|$, all uniform in v — gives

$$\left| \int_{\mathbb{R}^k} e^{i\lambda\Phi(u,v)} a(u, v) \, du \right| \lesssim \lambda^{-k/2},$$

uniformly in v .

Step 4. *Integrate out the parameters.*

Integrating in v over the compact projection of $\text{supp } a$,

$$\begin{aligned} \left| \int_{\mathbb{R}^{n-1}} e^{i\lambda\Phi(y)} a(y) \, dy \right| &\leq \int \left| \int e^{i\lambda\Phi(u,v)} a(u, v) \, du \right| \, dv \\ &\lesssim \lambda^{-k/2}. \end{aligned} \quad \square$$

WORKED EXAMPLE 5.5.2— TIER B A cylinder, and the sharpness of the exponent

Theorem 5.5.3 is sharp: k non-vanishing curvatures give exactly $\lambda^{-k/2}$ and no more. The example is a cylinder, and it is worth doing in full because it isolates the mechanism.

Let $n = 3$ and let

$$S = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 = 1, |x_3| \leq 1\},$$

a circular cylinder. Its principal curvatures are 1 in the circular direction and 0 along the axis, so $k = 1$ and Theorem 5.5.3 predicts $|\widehat{\psi d\sigma}| \lesssim |\xi|^{-1/2}$.

The transform factorises. With $\psi(x) = \eta(x_3)$ for $\eta \in C_c^\infty((-1, 1))$, surface

measure on S is the product of arclength on the circle and dx_3 , so

$$\begin{aligned}\widehat{\psi d\sigma}(\xi) &= \left(\int_{\mathbb{S}^1} e^{-2\pi i(x_1\xi_1 + x_2\xi_2)} d\sigma_1 \right) \cdot \widehat{\eta}(\xi_3) \\ &= 2\pi J_0(2\pi|\xi'|) \cdot \widehat{\eta}(\xi_3),\end{aligned}$$

where $\xi' = (\xi_1, \xi_2)$, using Theorem 2.8.3 in dimension 2.

Reading off the decay. Take $\xi = (\xi', 0)$, so $\widehat{\eta}(0) = \int \eta \neq 0$ is a fixed non-zero constant. By Theorem 5.4.1,

$$|\widehat{\psi d\sigma}(\xi', 0)| = 2\pi|\widehat{\eta}(0)| |J_0(2\pi|\xi'|)| \approx |\xi'|^{-1/2}$$

along the sequence where the cosine equals ± 1 . So the exponent $1/2 = k/2$ is attained, and no better bound holds.

Where the flat direction shows. Now take $\xi = (0, 0, \xi_3)$, along the axis. Then $|\xi'| = 0$ and $J_0(0) = 1$, so $\widehat{\psi d\sigma}(0, 0, \xi_3) = 2\pi\widehat{\eta}(\xi_3)$, which decays rapidly — but only because we *cut off* in x_3 . For an infinite cylinder the transform in that direction is a delta, with no decay at all: the flat direction contributes nothing, exactly as Theorem 5.5.1 predicts, since the whole line $\{x' = \text{const}\}$ is a critical set for the phase.

The moral. Decay in a given direction ω is governed by the curvature of S at the points where ω is normal, and by nothing else. A surface may be highly curved somewhere and flat elsewhere; the transform will decay quickly in the directions normal to the curved part and slowly in those normal to the flat part.

Pitfall: $k/2$ is sharp only when the degenerate directions are flat

Worked Example 5.5.2 showed the exponent $k/2$ is attained, and it is easy to read that as saying Theorem 5.5.3 is sharp in general. It is not. The example was a cylinder, which is *affine* in its degenerate direction, and that is exactly the case in which nothing better is true.

Suppose instead the degeneracy is of finite order. Take $n = 3$ and the graph $x_3 = x_1^2 + x_2^4$, whose second fundamental form at the origin has rank 1: one principal curvature vanishes. Theorem 5.5.3 gives $|\widehat{\psi d\sigma}(\xi)| \lesssim |\xi|^{-1/2}$. But in the normal direction the phase is $x_1^2 + x_2^4$, and the two variables separate:

$$\left| \int e^{i\lambda(x_1^2 + x_2^4)} a(x) dx \right| \lesssim \lambda^{-1/2} \cdot \lambda^{-1/4} = \lambda^{-3/4},$$

the first factor by Theorem 5.2.3 and the second by van der Corput with $k = 4$ (Theorem 5.1.2). So the true decay is $|\xi|^{-3/4}$, better than the theorem's $|\xi|^{-1/2}$.

The reason Theorem 5.5.3 misses this is structural: its proof discards the degenerate variables, integrating them out in Step 4 and extracting nothing from them, so any oscillation they carry is thrown away. Determining the sharp exponent for a surface with finite-order degeneracies means reading it off the *Newton polyhedron* of the phase — Varchenko’s theory — which this series does not develop and does not need: every surface appearing in Volumes 6 to 13 has non-vanishing Gaussian curvature, where Theorem 5.5.2 applies and is sharp.

WORKED EXAMPLE 5.5.3— TIER C: RESEARCH TASTE **From decay to the restriction conjecture’s exponents**

The debt from Volume 2 is now paid, and it is worth recording precisely what has been bought and what has not.

(a) *What is now available.* For the sphere and, more generally, for any hypersurface with non-vanishing Gaussian curvature:

$$\widehat{d\sigma}(\xi) = |\xi|^{-\frac{n-1}{2}} \sum_{\pm} a_{\pm}(\omega) e^{\mp 2\pi i |\xi|} + O(|\xi|^{-\frac{n+1}{2}}),$$

with the *phase* explicit. Volume 2 had only the modulus, and only for the sphere.

(b) *What that yields immediately.* Worked Example 2.8.3 converted the bound $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$ into the restriction estimate $\|Eg\|_{L^q} \lesssim \|g\|_{L^2(d\sigma)}$ at $q = \frac{4n}{n-1}$, by TT^* and Hardy–Littlewood–Sobolev. That argument now applies verbatim to every hypersurface with non-vanishing curvature, and with k curvatures gives the corresponding exponent $q = \frac{4n}{k}$.

(c) *What the phase buys.* Volume 6 proves the Stein–Tomas theorem, with the better exponent $\frac{2(n+1)}{n-1}$. Its proof decomposes $\widehat{d\sigma}$ dyadically and interpolates between an $L^1 \rightarrow L^\infty$ bound coming from the decay and an $L^2 \rightarrow L^2$ bound coming from Plancherel. The second uses that $\widehat{d\sigma}$ *oscillates*: the dyadic pieces have Fourier transforms supported near the sphere, which is what makes the L^2 estimate available. An argument with only $|\widehat{d\sigma}|$ cannot see this, which is why Volume 2’s Theorem 8.5 was insufficient and Theorem 5.5.2 was needed.

(d) *What no amount of stationary phase will buy.* The restriction conjecture concerns $L^\infty(d\sigma)$ data, and asserts an estimate for $\|Eg\|_{L^q}$ down to $q > \frac{2n}{n-1}$. That is not a statement about the decay of a single oscillatory integral. Writing $g = \sum_{\theta} g_{\theta}$ over δ -caps, Eg is a superposition of $\delta^{-(n-1)/2}$ wave packets, each concentrated on a tube dual to its cap, and the question is how much those tubes can overlap. Stationary phase describes one packet; the conjecture is about $\delta^{-(n-1)/2}$ of them at once.

That is the boundary of this volume. Chapter 5.7 pushes the oscillatory integral method as far as it goes — to an L^2 bound for operators, which is what Stein–Tomas needs — and everything past it is the geometry of Volumes 9–13.

Figure 5.5.1 (to be drawn). *Critical points are normal directions*

Content required: A closed convex curve in the plane with a direction ω marked; the two points where the normal is parallel to ω , and the tangent lines there. Beside it, the same picture for a curve with a flat segment, where the whole segment is critical. Together these illustrate Theorem 5.5.1 and why flatness destroys decay.

WORKED EXAMPLE 5.5.4— TIER A **Decay exponents for surfaces one meets**

Theorem 5.5.3 gives decay $|\xi|^{-k/2}$ where k is the number of non-vanishing principal curvatures. A table, with the value of k in each case.

Surface in $\mathbb{R}^n$	k	decay	used in
sphere $\mathbb{S}^{n-1}$	$n - 1$	$ \xi ^{-(n-1)/2}$	everywhere
paraboloid, truncated	$n - 1$	$ \xi ^{-(n-1)/2}$	Vol 6, Schrödinger
cone, truncated	$n - 2$	$ \xi ^{-(n-2)/2}$	Vol 6, wave equation
cylinder $\mathbb{S}^{n-2} \times \mathbb{R}$	$n - 2$	$ \xi ^{-(n-2)/2}$	Worked Example 5.5.2
hyperplane	0	none	Vol 6, no restriction
graph of $x_3 = x_1^2 + x_2^4$ at 0	1	$ \xi ^{-1/2}$	the Pitfall of §5.4

Reading the last row. The theorem gives $k = 1$ and hence $|\xi|^{-1/2}$, because one principal curvature vanishes at the origin. But the *true* decay in the degenerate direction is $|\xi|^{-3/4}$, since the vanishing is of finite order and Theorem 2.8.4 with $k = 4$ applies there. So Theorem 5.5.3 is not sharp for degenerate surfaces — it is sharp only when the vanishing curvatures vanish identically, as for a cylinder.

The practical rule. Count non-vanishing principal curvatures for a lower bound on the decay; if some curvature vanishes at isolated points, the true decay may be better and requires a separate van der Corput computation at the order of vanishing.

Drills

Drill 5.5.1. Verify the claim in Step 4 of Theorem 5.5.2 that the surface measure density $\sqrt{1 + |\nabla h|^2}$ equals 1 at a critical point.

Drill 5.5.2. Compute the Gaussian curvature of the paraboloid $x_n = |x'|^2$ and confirm that Theorem 5.5.2 applies on any compact piece. What happens as the piece recedes to infinity?

Drill 5.5.3. For the cylinder of Worked Example 5.5.2, compute $\widehat{\psi d\sigma}(\xi)$ for $\xi = (\xi', \xi_3)$ with both parts non-zero, and identify the decay rate in terms of the angle between ξ and the axis.

Drill 5.5.4. Let S be the graph of $x_3 = x_1^2$ in $\mathbb{R}^3$, a parabolic cylinder. Determine k at each point and state the decay rate that Theorem 5.5.3 gives.

Where we are. This chapter is the hinge of the series, and it is worth stating its output in the form the later volumes consume.

What goes in. A compact hypersurface S with its surface measure, and the stationary phase machinery of Chapter 5.2 and Chapter 5.3.

What comes out. Two statements. First, $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-k/2}$ where k is the curvature rank (Theorem 5.5.3). Second, and used less often but essentially, the full asymptotic with its oscillating phase (Theorem 5.5.2): $\widehat{d\sigma}$ does not merely decay, it decays *while oscillating like a cosine* in $|\xi|$.

Why the second matters. Volume 8's reduction of the distance-set problem turns an integral in the radial variable into a Fourier transform, and Plancherel then applies exactly. With the pointwise bound alone the argument fails — the Pitfall there records that no choice of weight repairs it. So the phase is not decoration; it is the content.

Where the geometry enters, once. In Theorem 5.5.1: the phase $x \cdot \omega$ is stationary on S exactly at the points where the normal is parallel to x , and its Hessian there is the second fundamental form. Every appearance of curvature in Volumes 6 through 9 traces back to that single lemma.

Notes

Theorem 5.5.2 is classical, going back to Herz (1962) for the sphere and to Littman (1963) in general; Theorem 5.5.3 is Littman's. The formulation in terms of the Gauss map, which makes the finiteness of the critical set transparent, is standard in the modern literature.

Readers should be aware that the uniformity in ω obtained in Step 5 is genuinely needed for the applications: Volume 6 integrates the estimate over ω , and a bound with an ω -dependent constant would be useless there.

Chapter 5.6

Analytic Continuation: Riesz Potentials

Twice already this volume has evaluated a divergent-looking integral by continuing a convergent one: the Fresnel integral in Theorem 5.2.1 and the Gamma integral in Theorem 5.4.2. Both times the device was to regard the quantity as the boundary value of a holomorphic function of a complex parameter. This chapter turns the device into a systematic tool, applied to the family $|x|^{-a}$, and produces a family of distributions on which the Fourier transform acts as the reflection $a \mapsto n - a$.

5.6.1 The problem

Worked Example 2.5.2 proved that for $0 < a < n$ the function $|x|^{-a}$ is a tempered distribution with

$$\widehat{|\cdot|^{-a}}(\xi) = c_{n,a} |\xi|^{-(n-a)}, \quad c_{n,a} = \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})}. \quad (5.6.1)$$

The restriction $0 < a < n$ was needed twice: for $a < n$ so that $|x|^{-a}$ is locally integrable at the origin, and for $a > 0$ so that the same holds for $|\xi|^{-(n-a)}$.

For $a \geq n$ the function $|x|^{-a}$ is not locally integrable, so $\int |x|^{-a} \varphi$ diverges and the pairing that defines a distribution is meaningless. This is not a technicality to be waved away: the exponent $a = n$ is exactly the one at which the Riesz potentials degenerate, and the objects on the far side of it — most importantly the delta function — are the ones one wants.

Scratch work. The remedy is the one used twice already. Regard

$$F_\varphi(a) = \int_{\mathbb{R}^n} |x|^{-a} \varphi(x) \, dx$$

as a function of the complex variable a . On the strip $0 < \operatorname{Re} a < n$ it converges and is holomorphic. If it extends meromorphically to a larger region, then the extension *defines* $|x|^{-a}$ as a distribution there, and any identity proved on the strip — such as (5.6.1) — automatically persists, because two meromorphic functions agreeing on an open set agree wherever both are defined.

The continuation is obtained by subtracting the Taylor polynomial of φ at the origin, which is where the divergence lives. Each subtracted monomial can be integrated explicitly and contributes one simple pole. So the poles are computable, and their locations turn out to be $a = n, n+2, n+4, \dots$ — with the odd values absent, for a reason of symmetry that is worth seeing.

5.6.2 Meromorphic continuation

Theorem 5.6.1. *For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ the function*

$$F_\varphi(a) = \int_{\mathbb{R}^n} |x|^{-a} \varphi(x) \, dx, \quad 0 < \operatorname{Re} a < n,$$

extends to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at

$$a = n, n+2, n+4, \dots$$

with

$$\begin{aligned} \operatorname{Res}_{a=n+2k} F_\varphi \\ = - \frac{\sigma(\mathbb{S}^{n-1})}{2^k k! n(n+2) \cdots (n+2k-2)} (\Delta^k \varphi)(0) \quad \text{for } k \geq 0, \end{aligned}$$

where for $k=0$ the empty product is 1. Moreover, for each fixed a outside the poles, $\varphi \mapsto F_\varphi(a)$ is a tempered distribution.

Proof. Step 1. *Split at the unit ball.*

Write $F_\varphi(a) = A(a) + B(a)$ with

$$A(a) = \int_{|x| < 1} |x|^{-a} \varphi(x) \, dx, \quad B(a) = \int_{|x| \geq 1} |x|^{-a} \varphi(x) \, dx.$$

The integral B converges absolutely for *every* $a \in \mathbb{C}$: on $|x| \geq 1$ we have $||x|^{-a}| = |x|^{-\operatorname{Re} a} \leq |x|^{|\operatorname{Re} a|}$, which is polynomially bounded, and φ decays faster than any polynomial. By differentiation under the integral sign (Worked Example 1.5.3), B is entire.

Step 2. *Subtract the Taylor polynomial.*

Fix $N \in \mathbb{N}$ and write, for $|x| < 1$,

$$\varphi(x) = \sum_{|\alpha| \leq N} \frac{\partial^\alpha \varphi(0)}{\alpha!} x^\alpha + R_N(x), \quad |R_N(x)| \leq C_N |x|^{N+1},$$

by Taylor's theorem. The remainder term $\int_{|x| < 1} |x|^{-a} R_N(x) dx$ converges absolutely whenever $\operatorname{Re} a < n + N + 1$, and is holomorphic there by the same argument as in Step 1.

Step 3. *The explicit monomial integrals.*

For a multi-index α , polar coordinates (Theorem 1.6.11) give

$$\begin{aligned} \int_{|x| < 1} |x|^{-a} x^\alpha dx &= \left(\int_{\mathbb{S}^{n-1}} \omega^\alpha d\sigma(\omega) \right) \int_0^1 r^{-a+|\alpha|+n-1} dr \\ &= \frac{m_\alpha}{n + |\alpha| - a}, \end{aligned}$$

valid for $\operatorname{Re} a < n + |\alpha|$, where $m_\alpha = \int_{\mathbb{S}^{n-1}} \omega^\alpha d\sigma$.

Now $m_\alpha = 0$ unless every α_j is even: replacing ω_j by $-\omega_j$ is a measure-preserving symmetry of the sphere under which ω^α changes sign when α_j is odd. *This is why only even $|\alpha|$ produce poles*, and hence why the poles occur at $a = n + 2k$ and not at $a = n + 1, n + 3, \dots$.

Step 4. *Assemble.*

Combining,

$$F_\varphi(a) = B(a) + \int_{|x| < 1} |x|^{-a} R_N(x) dx + \sum_{\substack{|\alpha| \leq N \\ \alpha \text{ even}}} \frac{\partial^\alpha \varphi(0)}{\alpha!} \frac{m_\alpha}{n + |\alpha| - a}.$$

The first two terms are holomorphic on $\operatorname{Re} a < n + N + 1$ and the third is a finite sum of simple poles at $a = n + |\alpha|$ with $|\alpha|$ even. Since N was arbitrary, this defines the meromorphic continuation to all of $\mathbb{C}$, and the poles are exactly at $a = n + 2k$.

Step 5. *The residues.*

The pole at $a = n + 2k$ comes from the terms with $|\alpha| = 2k$, and $\operatorname{Res}_{a=n+2k} \frac{1}{n+2k-a} = -1$. So

$$\operatorname{Res}_{a=n+2k} F_\varphi = - \sum_{|\alpha|=2k} \frac{\partial^\alpha \varphi(0)}{\alpha!} m_\alpha.$$

It remains to identify this sum as a multiple of $\Delta^k \varphi(0)$. Both sides are linear functionals of the degree- $2k$ part of the Taylor expansion of φ , invariant

under rotations — the left because $d\sigma$ is, the right because Δ is — and the space of rotation-invariant such functionals is one-dimensional. Comparing on $\varphi(x) = |x|^{2k}$, for which $\Delta^k |x|^{2k} = 2^k k! n(n+2) \cdots (n+2k-2)$ and $\sum_{|\alpha|=2k} \frac{m_\alpha}{\alpha!} \cdot (\text{coefficient}) = \sigma(\mathbb{S}^{n-1})$, gives the stated constant.

Step 6. Temperedness.

For fixed a off the poles, each of the three terms in Step 4 is bounded by a finite sum of Schwartz seminorms of φ : B by the decay of φ , the remainder integral by $\sup_{|\alpha| \leq N+1} \|\partial^\alpha \varphi\|_\infty$ through the Taylor bound, and the finite sum by the derivatives at 0. So $\varphi \mapsto F_\varphi(a)$ satisfies the continuity estimate of Volume 2, Definition 5.7. $\square$

WORKED EXAMPLE 5.6.1— TIER A **The residue at $a = n$ is a delta function**

Take $k = 0$ in Theorem 5.6.1. The residue is

$$\operatorname{Res}_{a=n} F_\varphi = -\sigma(\mathbb{S}^{n-1}) \varphi(0),$$

that is, as distributions,

$$\operatorname{Res}_{a=n} |x|^{-a} = -\sigma(\mathbb{S}^{n-1}) \delta_0.$$

This is worth pausing on. The family $|x|^{-a}$ consists of honest functions for $a < n$, and at $a = n$ it blows up — but the blow-up is not formless. The residue is a delta: the mass that was spread over $\mathbb{R}^n$ concentrates entirely at the origin as $a \uparrow n$, and the rate of concentration is measured by $\sigma(\mathbb{S}^{n-1})$. The computation behind it is the divergence of $\int_0^1 r^{-a+n-1} dr$ at $a = n$, which is the statement that $|x|^{-n}$ just fails to be integrable at the origin — the same borderline that made $\mathcal{M}f \notin L^1$ in Volume 1 and made singular integrals fail on L^1 in Volume 3. Here the failure is not an obstruction but the source of the delta.

5.6.3 The Riesz family and the reflection formula

The poles of Theorem 5.6.1 can be removed by dividing by a Gamma function with matching poles. The resulting family is entire, and the Fourier transform acts on it as simply as possible.

Definition 5.6.2. For $a \in \mathbb{C}$ define the *Riesz distribution*

$$Z_a = \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} |x|^{-a},$$

where $|x|^{-a}$ is understood via Theorem 5.6.1.

Proposition 5.6.3. $a \mapsto Z_a$ is an entire family of tempered distributions: for every $\varphi \in \mathcal{S}$, the function $a \mapsto \langle Z_a, \varphi \rangle$ is entire.

Proof. By Theorem 5.6.1 the only singularities of $\langle |x|^{-a}, \varphi \rangle$ are simple poles at $a = n + 2k$, $k \geq 0$. The function $1/\Gamma(\frac{n-a}{2})$ is entire — $1/\Gamma$ is — and vanishes exactly where $\Gamma(\frac{n-a}{2})$ has poles, namely where $\frac{n-a}{2} \in \{0, -1, -2, \dots\}$, that is $a = n + 2k$; and those zeros are simple. So each simple pole is cancelled by a simple zero, and the product is holomorphic at $a = n + 2k$; elsewhere both factors are holomorphic. $\square$

Theorem 5.6.4 (Reflection formula). For every $a \in \mathbb{C}$,

$$\widehat{Z}_a = Z_{n-a}$$

as tempered distributions.

Idea of the proof. On the strip $0 < \operatorname{Re} a < n$ this is Volume 2's computation (5.6.1), rearranged — the normalisation in Theorem 5.6.2 was chosen for precisely this purpose. Both sides are entire in a by Theorem 5.6.3 and the continuity of the Fourier transform on $\mathcal{S}'$, so agreement on a strip forces agreement everywhere. That is the entire proof: the analytic continuation does the work that would otherwise require a separate argument at every exponent.

Proof. Step 1. Verify the identity on the strip.

Let $0 < a < n$ be real. By (5.6.1),

$$\begin{aligned} \widehat{Z}_a &= \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} c_{n,a} |\xi|^{-(n-a)} \\ &= \frac{\pi^{-a/2}}{\Gamma(\frac{n-a}{2})} \cdot \pi^{a-\frac{n}{2}} \frac{\Gamma(\frac{n-a}{2})}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)} \\ &= \frac{\pi^{\frac{a-n}{2}}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)}. \end{aligned}$$

On the other hand, substituting $a \mapsto n - a$ in Theorem 5.6.2,

$$\begin{aligned} Z_{n-a} &= \frac{\pi^{-(n-a)/2}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)} \\ &= \frac{\pi^{\frac{a-n}{2}}}{\Gamma(\frac{a}{2})} |\xi|^{-(n-a)}. \end{aligned}$$

The two agree.

Step 2. *Continue.*

Fix $\varphi \in \mathcal{S}$. The functions

$$a \mapsto \langle \widehat{Z}_a, \varphi \rangle = \langle Z_a, \widehat{\varphi} \rangle \quad \text{and} \quad a \mapsto \langle Z_{n-a}, \varphi \rangle$$

are both entire: the first because $a \mapsto Z_a$ is entire (Theorem 5.6.3) and $\widehat{\varphi} \in \mathcal{S}$ by Theorem 2.5.6; the second because $a \mapsto n - a$ is entire and Z is. By Step 1 they agree for $a \in (0, n)$, a set with a limit point, so by the propagation argument of Theorem 2.1.10 they agree for all $a \in \mathbb{C}$. As φ was arbitrary, $\widehat{Z}_a = Z_{n-a}$. $\square$

WORKED EXAMPLE 5.6.2— TIER B Reading off the special values

The reflection formula turns several familiar identities into instances of one statement, and it is a good exercise to extract them.

(a) $a = 0$. Here $Z_0 = \frac{1}{\Gamma(n/2)}|x|^0 = \frac{1}{\Gamma(n/2)}$, the constant function. Theorem 5.6.4 gives

$$Z_n = \widehat{Z}_0 = \frac{1}{\Gamma(n/2)} \widehat{1} = \frac{\delta_0}{\Gamma(n/2)},$$

using Worked Example 2.5.2. So the Riesz distribution at the critical exponent $a = n$ is a multiple of the delta function — consistent with Worked Example 5.6.1, where the pole of $|x|^{-a}$ had residue proportional to δ , and the factor $1/\Gamma(\frac{n-a}{2})$ turns that residue into a finite value.

(b) $a = n/2$. The family is self-dual there: $\widehat{Z}_{n/2} = Z_{n/2}$, so $\pi^{-n/4}|x|^{-n/2}/\Gamma(n/4)$ is its own Fourier transform. Compare the Gaussian, Theorem 2.4.3, which is the other standard self-dual object; the two are quite different functions, and the coincidence of being fixed by $\mathcal{F}$ is a statement about the exponent $n/2$, not about the shape.

(c) *Negative a .* For $a < 0$, $|x|^{-a} = |x|^{|a|}$ is a polynomially growing function — a perfectly good tempered distribution with no continuation required — and Theorem 5.6.4 computes its transform as Z_{n-a} with $n - a > n$, i.e. a distribution supported at the origin when $n - a \in \{n, n+2, \dots\}$. For instance $a = -2$ gives $|x|^2$ on the left, and on the right Z_{n+2} , which by the zero of $1/\Gamma$ is a multiple of $\Delta\delta_0$. This is the familiar fact that the transform of a polynomial is a derivative of a delta, obtained here as a special value of an analytic family.

WORKED EXAMPLE 5.6.3— TIER C: RESEARCH TASTE Analytic continuation as a proof technique

This chapter used a single move three times, and it is worth abstracting, because it is one of the standard tools of the subject and appears in the research literature under several names.

(a) *The move.* To prove an identity or an estimate at a parameter value where the objects do not converge, embed the problem in a family depending holomorphically on a complex parameter, prove it where things do converge, and continue. The identity theorem then transports the conclusion.

Instances so far: Theorem 5.2.1, continuing $\int e^{-zt^2}$ from $\operatorname{Re} z > 0$ to the imaginary axis; Theorem 5.4.2, doing the same for $\int e^{-zs} s^{a-1}$; and Theorem 5.6.4, continuing in the exponent rather than in a variable of integration.

(b) *Where it is used later.* Chapter 5.8 defines the Bochner–Riesz means S^δ for $\operatorname{Re} \delta > 0$ and continues in δ ; the resulting family is exactly of the type above, and the kernel estimates one needs at real δ are obtained from the range where the integrals converge absolutely.

(c) *The version we do not prove.* Continuing an *estimate* rather than an identity requires more: one has a family of operators T_z and bounds at $\operatorname{Re} z = 0$ and $\operatorname{Re} z = 1$, and wants a bound in between. That is Stein’s interpolation theorem for analytic families, which strengthens Riesz–Thorin (Theorem 2.2.1) by allowing the operator, not only the test function, to vary. Its proof needs a Poisson-kernel form of the three-lines lemma, and it is recorded as revision R4: nothing in this series requires it, because Chapter 5.8 works with a single operator at a time.

Readers going on to the Bochner–Riesz literature will meet Stein interpolation constantly, and should know that it is the natural completion of the technique of this chapter rather than a separate idea.

(d) *A caution about what continuation cannot do.* It transports identities and, with Stein’s theorem, bounds — but only within the region where the family is holomorphic. It cannot cross a pole, and it cannot manufacture information that was absent on the initial domain. In particular, no analytic continuation will prove the restriction conjecture: the conjecture is not the boundary value of a family of estimates that hold on an open set of parameters, it is a statement about a fixed operator whose difficulty is geometric. Recognising which problems yield to continuation and which do not is a large part of knowing the subject.

WORKED EXAMPLE 5.6.4— TIER B Analytic continuation as a bookkeeping device

Several constants in this volume are defined first for parameters where an integral converges, and then extended. The device is worth isolating.

The pattern. A quantity $I(z)$ is given by an integral converging for $\operatorname{Re} z$ in a half-plane, is shown holomorphic there, and is then continued by an identity valid on the overlap — typically integration by parts, which trades a power of the variable for a derivative and enlarges the domain.

The instance. $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ converges for $\operatorname{Re} z > 0$; the identity $\Gamma(z+1) = z\Gamma(z)$ continues it to $\operatorname{Re} z > -1$, then -2 , and so on, with simple poles at the non-positive integers. Theorem 5.4.2 is the oscillatory analogue.

Why it is more than convenience. The continued function satisfies the same identities as the original, by the identity theorem — two holomorphic functions agreeing on a set with a limit point agree everywhere. So a formula proved where the integral converges holds where it does not, *provided both sides continue*. That proviso is the only place care is needed.

Where it is used. The constant in Theorem 5.2.1, whose branch is fixed by continuation from the real axis; the Riesz potentials of Volume 7, Theorem 7.6.5, where $\widehat{|x|^{-t}} = c_{n,t} |\xi|^{t-n}$ is first proved for $0 < t < n$ and then extended; and the Schrödinger propagator of Volume 6, Theorem 6.4.4, whose Gaussian kernel is the heat kernel continued to imaginary time.

The caution. Continuation preserves identities, not inequalities. A bound $|I(z)| \leq M$ on a half-plane says nothing about the continued function elsewhere — for that one needs a Phragmén–Lindelöf argument, which is what Theorem 2.1.14 in Volume 2 supplies.

Drills

Drill 5.6.1. Verify that $m_\alpha = \int_{\mathbb{S}^{n-1}} \omega^\alpha d\sigma$ vanishes when some α_j is odd, and compute m_α for $\alpha = (2, 0, \dots, 0)$.

Drill 5.6.2. Check the constant in Theorem 5.6.1's residue formula for $k = 1$ by computing both sides on $\varphi(x) = e^{-\pi|x|^2}$.

Drill 5.6.3. Show directly from Theorem 5.6.2 that Z_a is homogeneous of degree $-a$, and deduce that $\widehat{Z_a}$ is homogeneous of degree $a - n$ — consistent with Theorem 5.6.4.

Drill 5.6.4. Use Theorem 5.6.4 to compute the Fourier transform of $|x|^{-n-2}$, interpreted via Theorem 5.6.1, and identify the answer as a multiple of $\Delta\delta_0$ plus a correction.

Where we are. A remark on why this chapter exists at all, since analytic continuation is not an estimate and proves nothing about oscillatory integrals directly.

The constants produced by stationary phase are contour integrals, and contour integrals are most easily evaluated where they converge absolutely — which is usually not where they are needed. Theorem 5.2.1 is the model: the Fresnel integral $\int e^{is^2} ds$ does not converge absolutely at all, and its value $\sqrt{\pi}e^{i\pi/4}$ is obtained by continuing the Gaussian $\int e^{-as^2} ds = \sqrt{\pi/a}$ from $a > 0$ to $a = -i$.

That single continuation supplies the factor $e^{i\pi/4}$ in Theorem 5.2.3, hence the signature in Theorem 5.3.3, hence the phase in Theorem 5.5.2 — which Volume 8 then requires. So the chapter's apparent detour into complex analysis is the source of a factor that later turns out to be indispensable.

The general shape is worth naming: *compute where it converges, continue to where it is needed, and check that both sides continue*. The last clause is the only place errors occur.

Notes

The meromorphic continuation of $|x|^{-a}$ and the reflection formula are due to M. Riesz (1949); the systematic treatment of families of distributions depending holomorphically on a parameter is Gelfand and Shilov's.

The normalisation in Theorem 5.6.2 is chosen so that Theorem 5.6.4 has no constant. Readers will meet several other normalisations in the literature, differing by factors of π and 2; the invariant statement is that there is a choice making the Fourier transform act as $a \mapsto n - a$, and any two such choices differ by a factor $c(a)$ with $c(a)c(n - a) = 1$.

Chapter 5.7

Oscillatory Integral Operators

So far the oscillation has been integrated out to a number. This chapter treats oscillatory integrals as *operators*: the phase depends on two variables, one is integrated and one is left free, and the question is boundedness on L^2 . The answer, Theorem 5.7.3, is what Volume 6 builds the Stein–Tomas theorem on.

5.7.1 The set-up and the theorem

Definition 5.7.1. Let $\Phi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ be real-valued and $a \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^n)$. For $\lambda > 0$ define

$$(I_\lambda f)(x) = \int_{\mathbb{R}^n} e^{i\lambda\Phi(x,y)} a(x,y) f(y) \, dy.$$

Since a is compactly supported, I_λ is bounded on every L^p with norm $O(1)$, uniformly in λ , simply by the triangle inequality. The content of the theory is that oscillation improves this to a power of λ , and the improvement is governed by a non-degeneracy condition on the phase.

Definition 5.7.2. The *mixed Hessian* of Φ is the $n \times n$ matrix

$$\Phi''_{xy}(x,y) = \left(\frac{\partial^2 \Phi}{\partial x_i \partial y_j}(x,y) \right)_{i,j=1}^n.$$

The phase is *non-degenerate* on a set if $\det \Phi''_{xy} \neq 0$ there.

Theorem 5.7.3 (Hörmander). *Suppose $\det \Phi''_{xy} \neq 0$ at every point of $\text{supp } a$. Then*

$$\|I_\lambda f\|_{L^2(\mathbb{R}^n)} \lesssim \lambda^{-n/2} \|f\|_{L^2(\mathbb{R}^n)} \quad (\lambda \geq 1),$$

with the constant depending on finitely many derivatives of Φ and a and on a lower bound for $|\det \Phi''_{xy}|$ on $\text{supp } a$.

Idea of the proof. Estimate $I_\lambda I_\lambda^*$ rather than I_λ . Its kernel is

$$K_\lambda(x, z) = \int_{\mathbb{R}^n} e^{i\lambda(\Phi(x, y) - \Phi(z, y))} a(x, y) \overline{a(z, y)} dy,$$

an oscillatory integral in y with phase $\Psi(y) = \Phi(x, y) - \Phi(z, y)$. The crucial observation is that

$$\nabla_y \Psi(y) = \left(\int_0^1 \Phi''_{xy}(z + t(x - z), y) dt \right) (x - z),$$

so the non-degeneracy hypothesis makes $|\nabla_y \Psi| \gtrsim |x - z|$. The phase therefore moves at rate $\lambda|x - z|$, and non-stationary phase gives $|K_\lambda(x, z)| \lesssim_N (1 + \lambda|x - z|)^{-N}$.

A kernel of that shape has $\int |K_\lambda(x, z)| dz \lesssim \lambda^{-n}$, so the Schur test bounds $I_\lambda I_\lambda^*$ by λ^{-n} on L^2 , and taking square roots gives $\lambda^{-n/2}$.

Note where the exponent comes from: it is the volume of the set $\{|x - z| \lesssim \lambda^{-1}\}$, that is, the region in which the two copies of the phase have not yet separated. Each of the n variables contributes a factor λ^{-1} , and the square root of λ^{-n} is $\lambda^{-n/2}$.

Lemma 5.7.4 (Schur test). *Let K be measurable on $\mathbb{R}^n \times \mathbb{R}^n$ with*

$$A = \sup_x \int |K(x, z)| dz < \infty, \quad B = \sup_z \int |K(x, z)| dx < \infty.$$

Then $Tf(x) = \int K(x, z)f(z) dz$ satisfies $\|Tf\|_{L^2} \leq \sqrt{AB} \|f\|_{L^2}$.

Proof. By Cauchy–Schwarz, splitting $|K| = |K|^{1/2} \cdot |K|^{1/2}$,

$$\begin{aligned} |Tf(x)|^2 &\leq \left(\int |K(x, z)| dz \right) \left(\int |K(x, z)| |f(z)|^2 dz \right) \\ &\leq A \int |K(x, z)| |f(z)|^2 dz. \end{aligned}$$

Integrating in x and applying Tonelli,

$$\|Tf\|_{L^2}^2 \leq A \int |f(z)|^2 \left(\int |K(x, z)| dx \right) dz \leq AB \|f\|_{L^2}^2. \quad \square$$

Proof of Theorem 5.7.3. Step 1. Localise.

By compactness of $\text{supp } a$ and continuity of $\det \Phi''_{xy}$, there are $c > 0$ and a finite cover of $\text{supp } a$ by open sets $U_1 \times V_1, \dots, U_M \times V_M$ of diameter less than ρ , on each of which $|\det \Phi''_{xy}| \geq c$ and $\|(\Phi''_{xy})^{-1}\| \leq c^{-1}$. Take a subordinate

partition of unity and write $a = \sum_{j \leq M} a_j$; then $I_\lambda = \sum_j I_\lambda^{(j)}$ and it suffices to bound each summand, M being finite. So assume $\text{supp } a \subseteq U \times V$ with U convex of diameter $< \rho$.

Step 2. *The kernel of $I_\lambda I_\lambda^*$.*

A direct computation gives

$$\begin{aligned} (I_\lambda I_\lambda^* g)(x) &= \int K_\lambda(x, z) g(z) \, dz \\ , \quad K_\lambda(x, z) &= \int e^{i\lambda \Psi_{x,z}(y)} a(x, y) \overline{a(z, y)} \, dy, \end{aligned}$$

with $\Psi_{x,z}(y) = \Phi(x, y) - \Phi(z, y)$.

Step 3. *The gradient of the phase is bounded below.*

For $x, z \in U$, the segment from z to x lies in U by convexity, so the fundamental theorem of calculus applied to $t \mapsto \nabla_y \Phi(z + t(x - z), y)$ gives

$$\nabla_y \Psi_{x,z}(y) = M(x, z, y) (x - z), \quad M = \int_0^1 \Phi''_{xy}(z + t(x - z), y) \, dt.$$

Each Φ''_{xy} in the integrand has inverse of norm at most c^{-1} , and the integrand varies by at most $\varepsilon(\rho) \rightarrow 0$ across U ; choosing ρ small enough that $\varepsilon(\rho) < c/2$, the average M is invertible with $\|M^{-1}\| \leq 2c^{-1}$. Hence

$$|\nabla_y \Psi_{x,z}(y)| \geq \frac{c}{2} |x - z| \quad \text{for } x, z \in U, \, y \in V. \quad (5.7.1)$$

Step 4. *Kernel decay.*

Fix $x \neq z$. Apply Theorem 5.1.1 to the y -integral with large parameter $\lambda|x - z|$ — more precisely, write $\lambda \Psi_{x,z} = (\lambda|x - z|) \tilde{\Psi}$ with $\tilde{\Psi} = \Psi_{x,z}/|x - z|$, which by (5.7.1) satisfies $|\nabla_y \tilde{\Psi}| \geq c/2$ and has derivatives bounded uniformly in x, z . Then

$$|K_\lambda(x, z)| \lesssim_N (\lambda|x - z|)^{-N}.$$

Combining with the trivial bound $|K_\lambda| \lesssim 1$,

$$|K_\lambda(x, z)| \lesssim_N (1 + \lambda|x - z|)^{-N}, \quad (5.7.2)$$

and $K_\lambda(x, z) = 0$ unless $x, z \in U$.

Step 5. Schur.

Taking $N = n + 1$ in (5.7.2) and integrating,

$$\begin{aligned} \int |K_\lambda(x, z)| \, dz &\lesssim \int_{\mathbb{R}^n} \frac{dz}{(1 + \lambda|x - z|)^{n+1}} \\ &= \lambda^{-n} \int_{\mathbb{R}^n} \frac{dw}{(1 + |w|)^{n+1}} \\ &\lesssim \lambda^{-n}, \end{aligned}$$

by the substitution $w = \lambda(z - x)$, the last integral converging by polar coordinates. The same bound holds with the roles of x and z exchanged, since (5.7.2) is symmetric. By Theorem 5.7.4, $\|I_\lambda I_\lambda^*\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n}$.

Step 6. Take the square root.

For any bounded operator, $\|I_\lambda\|^2 = \|I_\lambda I_\lambda^*\|$, since $\|I_\lambda f\|_2^2 = \langle I_\lambda^* I_\lambda f, f \rangle$ and the operator norms of $I^* I$ and II^* agree. Hence $\|I_\lambda\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n/2}$. $\square$

Pitfall: TT^* is not a trick, it is the only available move

One cannot estimate I_λ directly, because there is nothing to integrate by parts against: the phase $\Phi(x, y)$ oscillates in y for fixed x , but the L^2 norm in x sees no oscillation at all.

Forming $I_\lambda I_\lambda^*$ creates the needed structure: the kernel involves the *difference* $\Phi(x, y) - \Phi(z, y)$, whose y -gradient is controlled by $|x - z|$ through the mixed Hessian. So the operator's smoothing is converted into decay of a kernel off the diagonal, which is a form one can estimate.

The same move appears in Worked Example 2.8.3, where EE^* was convolution with $\widehat{d\sigma}$ and the decay of $\widehat{d\sigma}$ became the whole input; and it will appear again in Volume 6 for the Stein–Tomas theorem. Whenever an operator resists direct estimation, the first thing to try is TT^* .

WORKED EXAMPLE 5.7.1—TIER B The extension operator is an oscillatory integral operator

Worked Example 2.4.3 introduced the extension operator for the paraboloid,

$$Eg(x) = \int_{\mathbb{R}^{n-1}} g(\xi') e^{2\pi i(x' \cdot \xi' + x_n |\xi'|^2)} \, d\xi'.$$

This is exactly of the form Theorem 5.7.1, and identifying the parameters explains where the scaling in the restriction problem comes from.

Rescaling to unit frequency. Fix $R \geq 1$ and consider Eg on the ball $|x| \leq R$.

Substituting $x = R\tilde{x}$ gives

$$Eg(R\tilde{x}) = \int_{\mathbb{R}^{n-1}} g(\xi') e^{2\pi i R(\tilde{x}' \cdot \xi' + \tilde{x}_n |\xi'|^2)} d\xi',$$

which is I_λ with $\lambda = 2\pi R$, phase $\Phi(\tilde{x}, \xi') = \tilde{x}' \cdot \xi' + \tilde{x}_n |\xi'|^2$, and amplitude a cutoff to the cap.

The mixed Hessian. Here $x \in \mathbb{R}^n$ and $\xi' \in \mathbb{R}^{n-1}$, so the phase is not square and Theorem 5.7.3 does not literally apply — this is the first sign that restriction is not merely an oscillatory-integral-operator problem. Computing anyway,

$$\frac{\partial^2 \Phi}{\partial \tilde{x}_i \partial \xi'_j} = \begin{cases} \delta_{ij}, & i \leq n-1, \\ 2\xi'_j, & i = n, \end{cases}$$

an $n \times (n-1)$ matrix of full rank $n-1$ everywhere. So the phase is non-degenerate in the appropriate rectangular sense.

What the theorem gives, and what it does not. Applying Theorem 5.7.3 in the $n-1$ variables — treating $\tilde{x}_n$ as a parameter, as in Theorem 5.5.3 — gives $\|Eg\|_{L^2(|x| \leq R)} \lesssim R^{1/2} \|g\|_{L^2}$, which after unwinding is the trivial L^2 bound and no more. The restriction conjecture concerns L^q for $q < \infty$ and L^∞ data, and no L^2 theorem reaches it.

Why this is worth knowing. The identification is standard in the literature: papers on restriction routinely write E as an oscillatory integral operator with parameter R , and speak of “the operator at scale R ”. A reader who has seen the translation will recognise the phrase; one who has not will find the rescaling mysterious. The genuine content of Volumes 12 and 13 begins where Theorem 5.7.3 stops, and knowing exactly where that is — at the passage from L^2 to L^q — is worth more than any single estimate.

WORKED EXAMPLE 5.7.2— TIER C: RESEARCH TASTE **Oscillatory integral operators, and where this volume stops**

Chapter 5.7 treats operators $T_\lambda f(x) = \int e^{i\lambda\varphi(x,y)} \psi(x,y) f(y) dy$, and proves the L^2 theory. It is worth saying what is proved and what is not.

(a) *What is proved.* Theorem 5.7.3: if the mixed Hessian $\partial^2 \varphi / \partial x \partial y$ is non-degenerate, then $\|T_\lambda\|_{L^2 \rightarrow L^2} \lesssim \lambda^{-n/2}$. The proof is TT^* followed by non-stationary phase, and it is sharp.

(b) *What is not.* The L^p theory for $p \neq 2$, which requires the Carleson–Sjölin condition and is descoped as revision R4. Nothing in this series uses it: Volume 6 works with L^2 data on the sphere, and the extension operator estimates there are proved directly.

(c) *Why L^2 suffices for what follows.* The Stein–Tomas theorem of Volume 6 is an $L^2(d\sigma)$ statement, and its proof uses only the decay of $\widehat{d\sigma}$ and interpolation — not the L^p theory of oscillatory operators. The restriction conjecture concerns $L^\infty(d\sigma)$, which the oscillatory-operator machinery does not reach either.

(d) *The connection worth knowing.* An extension operator *is* an oscillatory integral operator with $\lambda = |\xi|$ and phase $x \cdot \omega$; so the two subjects are the same one, viewed at fixed and at varying frequency. The Carleson–Sjölin theory is what a full treatment would use, and readers heading for the literature on local smoothing will meet it there.

(e) *The summary.* This volume supplies exactly two things to the rest of the series: the decay of $\widehat{d\sigma}$ with its phase, and the L^2 bound above. Both are used constantly; nothing else here is.

Drills

Drill 5.7.1. Verify the identity $\|T\|^2 = \|TT^*\|$ used in Step 6, for a bounded operator on a Hilbert space.

Drill 5.7.2. Show that Theorem 5.7.3 fails without the non-degeneracy hypothesis, by taking $\Phi(x, y) = x_1 y_1$ in $\mathbb{R}^2$ — so that Φ does not depend on x_2 or y_2 — and computing $\|I_\lambda\|$.

Drill 5.7.3. Carry out Step 5 with general $N > n$ and determine how the constant depends on N .

Drill 5.7.4. Compute the mixed Hessian for $\Phi(x, y) = |x - y|$ on a region where $x \neq y$, and determine its rank. (This is the phase of the wave propagator; the rank deficiency is why the wave equation is harder than the Schrödinger equation.)

Where we are. The TT^* device used in Theorem 5.7.3 is worth recording separately, since Volume 6 opens with it again in a different setting.

The point is always the same: an operator T may be hard to estimate directly, but TT^* is a *convolution-like* object whose kernel can be written down. Here $T_\lambda T_\lambda^*$ has kernel $\int e^{i\lambda(\varphi(x, y) - \varphi(x', y))} \psi \overline{\psi} \, dy$, and the non-degeneracy of the mixed Hessian makes the phase non-stationary in y for $x \neq x'$ — so Theorem 5.1.1 applies and the kernel decays. Schur’s test then finishes it. Two features generalise. The identity $\|T\|_{L^2 \rightarrow L^2}^2 = \|TT^*\|_{L^2 \rightarrow L^2}$ is exact, so nothing is lost in passing to TT^* ; and the hypothesis one needs on φ is a

statement about *differences* $\varphi(x, y) - \varphi(x', y)$, which is why non-degeneracy of the mixed Hessian — and not of φ itself — is the right condition.

Volume 6, Theorem 6.1.5 is the same identity with $T = \mathcal{E}$, and there the kernel of $\mathcal{E}\mathcal{E}^*$ is $\widehat{d\sigma}$ itself. So the two volumes' central computations are one computation.

Notes

Theorem 5.7.3 is Hörmander's, from 1973. The L^p theory — which requires curvature conditions on the level sets of the phase rather than merely non-degeneracy of the mixed Hessian — is due to Stein and to Carleson–Sjölin in the plane; it is recorded as revision R4 and not proved here.

The rectangular case appearing in Worked Example 5.7.1, where x and y have different dimensions, is the setting of the restriction problem and is treated systematically in Volume 6.

Chapter 5.8

Bochner–Riesz Means

Volume 3 proved that the sharp cutoff to the unit ball is a bounded Fourier multiplier on $L^p(\mathbb{R}^n)$ only for $p = 2$, and that the obstruction is a Besicovitch set. This chapter shows what happens when the cutoff is smoothed: raising the distance to the boundary to a positive power δ restores boundedness, and for δ large enough the proof is a computation with the tools of this volume.

The chapter is the payoff of Volume 5, in that its proof uses Theorem 5.4.2, Theorem 5.4.1 and Theorem 5.5.2 together.

5.8.1 The operators

Definition 5.8.1. For $\delta > 0$ the *Bochner–Riesz multiplier* of order δ is

$$m^\delta(\xi) = (1 - |\xi|^2)_+^\delta = \begin{cases} (1 - |\xi|^2)^\delta, & |\xi| \leq 1, \\ 0, & |\xi| > 1, \end{cases}$$

and S^δ is the associated multiplier operator, $\widehat{S^\delta f} = m^\delta \widehat{f}$.

At $\delta = 0$ this is the ball multiplier of Volume 3, Chapter 8, bounded only on L^2 . As δ increases the cutoff becomes smoother — m^δ is $C^{[\delta]}$ across the sphere — and one expects boundedness on a wider range of p . The precise expectation is the Bochner–Riesz conjecture, stated in §8.3.

Proposition 5.8.2 (Reduction to the kernel). *Let $K^\delta = \mathcal{F}^{-1}m^\delta$. If $K^\delta \in L^1(\mathbb{R}^n)$ then S^δ is bounded on $L^p(\mathbb{R}^n)$ for every $1 \leq p \leq \infty$, with $\|S^\delta\|_{L^p \rightarrow L^p} \leq \|K^\delta\|_{L^1}$.*

Proof. Since m^δ is continuous with compact support, it lies in $L^1 \cap L^2$ and K^δ is a bounded continuous function. If moreover $K^\delta \in L^1$ then $S^\delta f = K^\delta * f$ for $f \in \mathcal{S}$, and Young's inequality (Volume 2, Worked Example 2.1) gives $\|K^\delta * f\|_p \leq \|K^\delta\|_1 \|f\|_p$. Extend by density. $\square$

So everything reduces to a decay estimate for K^δ , and that is an oscillatory integral.

5.8.2 The kernel estimate

Theorem 5.8.3. *For $\delta > 0$,*

$$|K^\delta(x)| \lesssim_{n,\delta} (1 + |x|)^{-\frac{n+1}{2}-\delta}.$$

Consequently $K^\delta \in L^1(\mathbb{R}^n)$ if and only if $\delta > \frac{n-1}{2}$.

Idea of the proof. Write the inverse transform in polar coordinates. The angular integral is $\widehat{d\sigma}$ evaluated at rx , for which Theorem 5.4.1 supplies the asymptotic $|rx|^{-(n-1)/2}$ times an oscillating factor $e^{\pm 2\pi i r|x|}$. What remains is a one-dimensional integral in $r \in [0, 1]$ whose amplitude $(1 - r^2)^\delta$ vanishes to order δ at the endpoint $r = 1$ and whose phase is $2\pi r|x|$ — exactly the situation of Theorem 5.4.2, which gives a further factor $|x|^{-\delta-1}$. Multiplying the two: $|x|^{-(n-1)/2} \cdot |x|^{-\delta-1} = |x|^{-(n+1)/2-\delta}$. Integrability at infinity then requires $\frac{n+1}{2} + \delta > n$, which is $\delta > \frac{n-1}{2}$.

Proof. Step 1. Polar coordinates.

Since m^δ is radial and supported in the unit ball, polar coordinates (Theorem 1.6.11) give, for $x \in \mathbb{R}^n$,

$$\begin{aligned} K^\delta(x) &= \int_{|\xi| \leq 1} (1 - |\xi|^2)^\delta e^{2\pi i x \cdot \xi} d\xi \\ &= \int_0^1 (1 - r^2)^\delta r^{n-1} \left(\int_{\mathbb{S}^{n-1}} e^{2\pi i r x \cdot \omega} d\sigma(\omega) \right) dr. \end{aligned}$$

The inner integral is $\widehat{d\sigma}(-rx) = \widehat{d\sigma}(rx)$, the transform being real and radial (Theorem 2.8.1).

Step 2. *Small $|x|$.*

For $|x| \leq 1$ the bound is trivial: $|K^\delta(x)| \leq \int_{|\xi| \leq 1} 1 = \omega_n$.

Step 3. *Insert the asymptotic.*

Assume $R := |x| \geq 1$. By Theorem 5.5.2 for the sphere — equivalently Theorem 5.4.1 via Theorem 2.8.3 — there are constants $c_\pm$ with

$$\begin{aligned} \widehat{d\sigma}(y) &= |y|^{-\frac{n-1}{2}} \left(c_+ e^{2\pi i |y|} + c_- e^{-2\pi i |y|} \right) \\ &\quad + O(|y|^{-\frac{n+1}{2}}) \quad (|y| \geq 1). \end{aligned}$$

Split the r -integral at $r = 1/R$.

The piece $r \leq 1/R$. Here $|\widehat{d\sigma}(rx)| \leq \sigma(\mathbb{S}^{n-1})$ and the rest of the integrand is bounded, so this piece is $O(R^{-1})$; since $\frac{n+1}{2} + \delta$ may exceed 1, we must do better — but on this range $r^{n-1} \leq R^{-(n-1)}$, so the piece is in fact $O(R^{-n})$, which is $O(R^{-\frac{n+1}{2}-\delta})$ whenever $\delta \leq \frac{n-1}{2}$; for larger δ the desired bound is weaker still than what Theorem 5.8.2 needs, and the estimate below covers it.

The piece $r \geq 1/R$. Substituting the asymptotic,

$$\begin{aligned} & \int_{1/R}^1 (1-r^2)^\delta r^{n-1} (rR)^{-\frac{n-1}{2}} \left(c_+ e^{2\pi i r R} + c_- e^{-2\pi i r R} \right) dr \\ & + O\left(\int_{1/R}^1 (1-r^2)^\delta r^{n-1} (rR)^{-\frac{n+1}{2}} dr \right). \end{aligned}$$

The error integral is $O(R^{-\frac{n+1}{2}})$, since $r^{n-1-\frac{n+1}{2}} = r^{\frac{n-3}{2}}$ is integrable on $(0, 1)$ for $n \geq 2$.

Step 4. *The main integral, by the Gamma-integral method.*

Consider

$$J_\pm(R) = R^{-\frac{n-1}{2}} \int_{1/R}^1 (1-r^2)^\delta r^{\frac{n-1}{2}} e^{\pm 2\pi i r R} dr.$$

Substituting $r = 1 - s$ and writing $(1-r^2)^\delta = (2-s)^\delta s^\delta$,

$$\begin{aligned} \int_{1/R}^1 (1-r^2)^\delta r^{\frac{n-1}{2}} e^{\pm 2\pi i r R} dr &= e^{\pm 2\pi i R} \int_0^{1-1/R} e^{\mp 2\pi i s R} s^\delta g(s) ds, \\ g(s) &= (2-s)^\delta (1-s)^{\frac{n-1}{2}}. \end{aligned}$$

The function g is smooth on $[0, \frac{1}{2}]$; cutting off there costs $O(R^{-N})$ by Theorem 5.1.1, the phase having non-vanishing derivative away from the endpoint. By Theorem 5.4.2 with $a = \delta + 1 > 0$,

$$\left| \int_0^\infty e^{\mp 2\pi i s R} s^\delta ds \right| = \Gamma(\delta + 1) (2\pi R)^{-\delta-1},$$

and the correction from replacing g by $g(0)$ carries an extra factor s , hence an extra R^{-1} . Therefore

$$|J_\pm(R)| \lesssim_{n,\delta} R^{-\frac{n-1}{2}} R^{-\delta-1} = R^{-\frac{n+1}{2}-\delta}.$$

Step 5. *Collect.*

Combining Steps 3 and 4, for $R \geq 1$

$$|K^\delta(x)| \lesssim_{n,\delta} R^{-\frac{n+1}{2}-\delta} + R^{-\frac{n+1}{2}} + R^{-n}.$$

When $\delta \leq \frac{n-1}{2}$ the first term dominates the third and the second is no larger than the first only if $\delta \geq 0$, which holds; so the bound is $R^{-\frac{n+1}{2}-\delta}$ for $0 < \delta \leq \frac{n-1}{2}$, and for larger δ one obtains $R^{-\frac{n+1}{2}}$, which is more than enough for Theorem 5.8.2. In all cases $|K^\delta(x)| \lesssim (1 + |x|)^{-\min\{\frac{n+1}{2}+\delta, n\}}$, and in particular the stated bound holds.

Step 6. Integrability.

By polar coordinates, $(1 + |x|)^{-\frac{n+1}{2}-\delta}$ is integrable on $\mathbb{R}^n$ exactly when $\frac{n+1}{2} + \delta > n$, i.e. $\delta > \frac{n-1}{2}$. Conversely, the leading term computed in Step 4 shows K^δ is genuinely of size $R^{-\frac{n+1}{2}-\delta}$ along a sequence $R \rightarrow \infty$ — the oscillating factors $e^{\pm 2\pi i R}$ combine into a cosine, which is bounded away from zero on a set of R of positive density — so for $\delta \leq \frac{n-1}{2}$ the kernel is not integrable. $\square$

Theorem 5.8.4 (Bochner–Riesz above the critical index). *For $\delta > \frac{n-1}{2}$, the operator S^δ is bounded on $L^p(\mathbb{R}^n)$ for every $1 \leq p \leq \infty$.*

Proof. Combine Theorem 5.8.3 with Theorem 5.8.2. $\square$

5.8.3 What is open

Conjecture 5.8.5 (Bochner–Riesz). *For $n \geq 2$, $1 \leq p \leq \infty$ and $\delta > 0$, the operator S^δ is bounded on $L^p(\mathbb{R}^n)$ if and only if*

$$\delta > \delta(p) := \max \left\{ 0, n \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2} \right\}.$$

WORKED EXAMPLE 5.8.1 — TIER C: RESEARCH TASTE **Where the exponent $\delta(p)$ comes from, and what is known**

(a) *The necessity, by a Knapp-type example.* Test S^δ against a function whose Fourier transform is concentrated on a δ -neighbourhood of a cap of the sphere of diameter $\delta^{1/2}$ — the same cap-and-dual-tube geometry as Worked Example 2.4.3. Tracking the norms as in that example produces exactly the condition $\delta > n|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}$, and $\delta > 0$ is necessary by Theorem 3.8.2 (Fefferman). So the conjectured range cannot be enlarged.

(b) *What this chapter proves.* Theorem 5.8.4 gives boundedness for $\delta > \frac{n-1}{2}$, which is $\delta(p)$ evaluated at $p = 1$ and $p = \infty$. So the theorem is sharp at the two endpoints and says nothing new in between: for p near 2 the conjecture asks for far smaller δ , and interpolating Theorem 5.8.4 with the trivial L^2 bound at $\delta = 0$ (Volume 2, Proposition 6.2) gives only a linear improvement, well short of $\delta(p)$.

(c) *What is known.* For $n = 2$ the conjecture is a theorem, due to Carleson and Sjölin (1972); its proof uses the L^p theory of oscillatory integral operators, and is recorded as revision R4 — it needs machinery beyond this volume. For $n \geq 3$ the conjecture is open, and the best results come from the restriction theory of Volumes 6 and 12.

(d) *The chain of implications.* Worked Example 3.8.1 recorded

Bochner–Riesz

$\implies$ restriction

$\implies$ Kakeya,

and this chapter supplies the left-hand end. The implications are proved in Volume 6. The practical consequence is that progress on any of the three is progress on the ones to its right, and that the obstruction is common: in each case one must control how $\delta^{-(n-1)/2}$ wave packets, one per cap, can superpose.

(e) *The honest summary of this volume.* Stationary phase describes a single oscillatory integral completely — Theorem 5.3.3 gives an asymptotic expansion, not merely a bound. It describes a single operator well, through the TT^* argument of Theorem 5.7.3. What it cannot do is control a *family* of such objects indexed by directions, and every open problem mentioned above is of that kind. The reader who has finished this volume has the complete local theory and should now regard the remaining difficulty as geometric, not analytic.

WORKED EXAMPLE 5.8.2— TIER A The Bochner–Riesz exponents, and what is open

Theorem 5.8.4 proves S^δ bounded on every L^p for $\delta > \frac{n-1}{2}$. Three numbers matter.

(a) $\delta > \frac{n-1}{2}$: *proved here.* By Theorem 5.8.3 the kernel is $O(|x|^{-(n+1)/2-\delta})$, which is integrable exactly when $\delta > \frac{n-1}{2}$; then S^δ is convolution with an L^1 kernel and Young’s inequality finishes it. No curvature is used beyond the kernel bound.

(b) $\delta = 0$: *false for $n \geq 2$, $p \neq 2$.* This is the ball multiplier, Volume 3, Theorem 3.8.2. So there is a threshold somewhere between.

(c) *The conjecture.* S^δ is bounded on $L^p(\mathbb{R}^n)$ if and only if

$$\delta > \max \left\{ 0, \, n \left| \frac{1}{p} - \frac{1}{2} \right| - \frac{1}{2} \right\}.$$

At $p = 2$ this permits every $\delta > 0$, consistent with Plancherel. As $p \rightarrow 1$ or ∞ it approaches $\frac{n-1}{2}$, which is the range proved here — so Theorem 5.8.4 is sharp at the extreme exponents and far from sharp in between.

Status. Settled for $n = 2$ by Carleson and Sjölin; open for $n \geq 3$. By Volume 6, Worked Example 6.7.2 it implies the restriction conjecture, which is why it sits at the top of the chain of three.

The one line to remember. Softening the sharp cutoff by $(1 - |\xi|^2)^\delta$ repairs the ball multiplier’s failure, and the question is how little softening suffices — a question about how much the curvature of the sphere can be made to pay.

WORKED EXAMPLE 5.8.3— TIER C: RESEARCH TASTE **Curvature, in one place**

Volume 5 is where curvature first does work, and it is worth collecting what it buys before the later volumes spend it.

(a) *Decay of $\widehat{d\sigma}$.* Non-vanishing Gaussian curvature gives $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$, and the exponent is exactly half the number of non-vanishing principal curvatures. A flat piece gives no decay at all.

(b) *Where the later volumes spend it.* Volume 6’s Stein–Tomas theorem is (a) plus an interpolation; Volume 8’s Falconer theorem is (a) plus Mattila’s reduction; Volume 13’s decoupling exponent is fixed by the cap geometry that curvature forces.

(c) *The stationary phase hierarchy.* Non-degenerate critical points give $|\xi|^{-n/2}$; degenerate ones give less, the loss measured by the Newton polyhedron. This series needs only the non-degenerate case, but the degenerate theory is where the subject meets singularity theory.

(d) *What curvature cannot do.* It gives *decay*, never *cancellation between separated pieces*. That second thing is what bilinear and multilinear methods supply, and it is why Stein–Tomas is not the end of the story.

(e) *The diagnostic.* Given a surface, count the non-vanishing principal curvatures. That number determines the decay exponent, the critical restriction exponent, and the decoupling exponent, by formulas this series derives three times — in Volume 6, in Volume 13, and in Volume 14 for the cone.

Drills

Drill 5.8.1. Verify the claim in Step 3 that the piece $r \leq 1/R$ contributes $O(R^{-n})$.

Drill 5.8.2. Show that m^δ is C^k across the unit sphere precisely when $k < \delta$,

and relate this to the kernel decay of Theorem 5.8.3.

Drill 5.8.3. Interpolate Theorem 5.8.4 with the L^2 bound at $\delta = 0$ using Riesz–Thorin (Theorem 2.2.1), and compare the resulting range with $\delta(p)$. How large is the gap at $p = 4$, $n = 3$?

Drill 5.8.4. Carry out the Knapp computation of Worked Example 5.8.1(a) for $n = 2$ and confirm the exponent $\delta(p) = \max\{0, 2|\frac{1}{p} - \frac{1}{2}| - \frac{1}{2}\}$.

Notes

Bochner introduced the means in 1936 in connection with summability of multiple Fourier series. Theorem 5.8.4 is classical and the kernel computation is due to Bochner; the value $\frac{n-1}{2}$ is called the *critical index*.

The Carleson–Sjölin theorem is from 1972. For $n \geq 3$ the best known results are consequences of restriction estimates and are surveyed in Volume 12; the conjecture remains open.

Chapter 5.A

Results Quoted from Volumes 1–4

This volume assumes Volumes 1–4. Every citation is listed with its use site.

Result	Source	Used in
Polar coordinates	Vol. 1, Thm 6.11	Theorem 5.6.1, Theorem 5.8.3
Differentiation under the integral	Vol. 1, Wkd Ex 5.3	Theorem 5.2.1, Theorem 5.4.2, Theorem 5.6.1
Gaussian integral	Vol. 1, Drill 6.2	Theorem 5.2.1
Tonelli	Vol. 1, Thm 6.6	Theorem 5.7.4
Cauchy–Schwarz in L^2	Vol. 1, Thm 8.2	Theorem 5.7.4
Holomorphy; identity propagation	Vol. 2, Thm 1.10	Theorem 5.2.1, Theorem 5.4.2, Theorem 5.6.4
Riesz–Thorin	Vol. 2, Thm 2.1	Drill 8.3
Young’s convolution inequality	Vol. 2, Wkd Ex 2.1	Theorem 5.8.2
The Fourier dictionary	Vol. 2, Prop 4.2	throughout
$\mathcal{F}$ on $\mathcal{S}$ and $\mathcal{S}'$	Vol. 2, Thms 5.6, 5.11	Theorem 5.6.4, Theorem 5.6.1
$\widehat{1} = \delta$	Vol. 2, Wkd Ex 5.2	Worked Example 5.6.2
Transform of $ x ^{-a}$, $0 < a < n$	Vol. 2, Wkd Ex 5.9	Theorem 5.6.4 Step 1
L^2 multiplier norm	Vol. 2, Prop 6.2	Worked Example 5.8.1
$\widehat{d\sigma}$ via Bessel functions	Vol. 2, Thm 8.3	Chapter 5.4, Theorem 5.8.3
van der Corput	Vol. 2, Lem 8.4	Theorem 5.1.2, Theorem 5.2.1
Decay of $\widehat{d\sigma}$ (sphere)	Vol. 2, Thm 8.5	compared throughout Chapter 5.5
The flat-piece example	Vol. 2, Wkd Ex 8.2	Chapter 5.1 §1.1, Theorem 5.5.1
Restriction from decay (TT^*)	Vol. 2, Wkd Ex 8.3	Worked Example 5.5.3
Fefferman’s ball multiplier theorem	Vol. 3, Thm 8.6	Worked Example 5.8.1
Bochner–Riesz $\implies$ restriction $\implies$ Kakeya	Vol. 3, Wkd Ex 8.9	Worked Example 5.8.1

Debts

Volume 2, Chapter 8 deferred two results to this volume. Both are discharged: the asymptotic expansion of J_ν is Theorem 5.4.1, and the decay of $\widehat{d\sigma}$ for a hypersurface with k non-vanishing principal curvatures is Theorem 5.5.3, with the full asymptotic in Theorem 5.5.2.

This volume incurs no debts on later volumes. Three results were removed

from the plan in advance, as revision R4 in the master manifest; each is true, attributed, and used nowhere in this series.

Chapter 5.B

Notation and Index of Results

Additions to the Volume 1–4 conventions

φ, ψ	phase and amplitude of an oscillatory integral
λ	the large parameter
Hess	the Hessian matrix
sgn	sign of a real number; signature of a symmetric matrix
I	an oscillatory integral operator
S^δ	the Bochner–Riesz operator of order δ

Index of the main results

Non-stationary phase	Theorem 5.1.1
van der Corput, amplitude version	Theorem 5.1.3
The Fresnel integral	Theorem 5.2.1
Morse lemma in one variable	Theorem 5.2.2
Stationary phase in one dimension	Theorem 5.2.3
Morse lemma in n variables	Theorem 5.3.1
The model Gaussian integral	Theorem 5.3.2
Stationary phase in $\mathbb{R}^n$	Theorem 5.3.3
The Gamma integral on the imaginary axis	Theorem 5.4.2
Asymptotics of J_ν	Theorem 5.4.1
Critical points are normal directions	Theorem 5.5.1
Asymptotics of $\widehat{d\sigma}$, curved case	Theorem 5.5.2
Decay with k curvatures	Theorem 5.5.3
Meromorphic continuation of $ x ^{-a}$	Theorem 5.6.1
The Riesz family is entire	Theorem 5.6.3
Reflection formula $\widehat{Z}_a = Z_{n-a}$	Theorem 5.6.4
Schur test	Theorem 5.7.4
Hörmander’s L^2 theorem	Theorem 5.7.3
Bochner–Riesz kernel decay	Theorem 5.8.3
Bochner–Riesz above the critical index	Theorem 5.8.4

Plan revisions in this volume (R4)

Removed	Reason
Carleson–Sjölin theorem	needs Volume 12’s machinery
Stein interpolation for analytic families	unused; see Worked Example 5.6.3
Sharp L^p theory of oscillatory integral operators	Volumes 6 and 12 use only the L^2 theorem

Part 6

Volume 6

The Restriction Problem: Classical Theory

Preface to Volume 6

The hinge of the series

Everything so far has been analysis. Volumes 1 and 2 built the measure theory and the Fourier transform; Volumes 3 and 4 developed singular integrals and the function spaces they live in; Volume 5 computed oscillatory integrals as precisely as they can be computed. Each of those volumes proved theorems by estimating a single object — a kernel, a multiplier, an integral — and each was, in the end, a matter of technique.

This volume states the problem that the rest of the series is about, and the character of the subject changes with it. The restriction conjecture is not a statement about one oscillatory integral. It is a statement about $\delta^{-(n-1)/2}$ of them at once, one for each cap of the sphere, and about how much the resulting wave packets can overlap. Volume 5 described a single packet completely; from here the difficulty is that there are many.

What this volume proves is the classical theory: the exact duality between restriction and extension, the necessary conditions, the Stein–Tomas theorem, its consequences for dispersive equations, and — the two results that organise everything after — that restriction implies Kakeya and that Bochner–Riesz implies restriction.

What is assumed

Volumes 1–5. Most heavily: the asymptotics of $\widehat{d\sigma}$ from Chapter 5.5, the interpolation theorems of Volume 2, the Hardy–Littlewood–Sobolev inequality, and the TT^* method as it appeared in Chapter 5.7.

Scope, fixed in advance

Following the practice adopted in Volumes 4 and 5, the list of results was fixed before the chapters were written. Three items a reader might expect are deliberately absent, and each is announced where it would have appeared:

- the *endpoint* of the Stein–Tomas theorem, $q = \frac{2(n+1)}{n-1}$. Its standard proof interpolates an analytic family of operators and so requires Stein’s interpolation theorem, which Volume 5 removed from the plan. The non-endpoint range is what Volumes 12 and 13 use;
- the *Keel–Tao endpoint* Strichartz estimate, which needs a bilinear argument;
- *Zygmund’s endpoint theorem* for the restriction conjecture in the plane. The non-endpoint planar case is proved here, in Chapter 6.8.

The plan

1	Restriction and extension	the two operators, and their duality
2	Necessary conditions	what the geometry forbids
3	The Stein–Tomas theorem	the classical positive result
4	Strichartz estimates	the same theorem, read as PDE
5	Epsilon-removal	local estimates imply global ones
6	Restriction implies Keakeya	the first bridge
7	Bochner–Riesz implies restriction	the second bridge
8	The two-dimensional case	where the conjecture is a theorem

Chapter 6.1

Restriction and Extension

6.1.1 The question

If $f \in L^1(\mathbb{R}^n)$ then $\widehat{f}$ is continuous, so its restriction to any set whatever is a well-defined function. If $f \in L^2(\mathbb{R}^n)$ then $\widehat{f}$ is merely an L^2 function, defined up to null sets, and its restriction to a set of measure zero — a sphere, say — is meaningless.

The restriction problem asks where between these two extremes the truth lies, and the answer turns out to depend on the *curvature* of the set.

Definition 6.1.1. Let $S \subseteq \mathbb{R}^n$ be a smooth compact hypersurface with surface measure $d\sigma$. For $f \in \mathcal{S}(\mathbb{R}^n)$ define the *restriction operator*

$$\mathcal{R}f = \widehat{f}|_S.$$

We say that the estimate $R^*(p \rightarrow q)$ holds if

$$\|\mathcal{R}f\|_{L^q(S, d\sigma)} \leq C \|f\|_{L^p(\mathbb{R}^n)} \quad \text{for all } f \in \mathcal{S}(\mathbb{R}^n), \quad (6.1.1)$$

with C independent of f .

Since $\mathcal{S}$ is dense in L^p for $p < \infty$ (Theorem 1.8.7), an estimate (6.1.1) lets one *define* $\mathcal{R}$ on all of L^p by continuity, and that is the content of the problem: not to compute a restriction but to show one exists.

WORKED EXAMPLE 6.1.1— TIER A The two trivial cases, and why they are not the question

(a) $p = 1$ is free. For $f \in L^1$, $\|\widehat{f}\|_{L^\infty} \leq \|f\|_{L^1}$, so

$$\|\mathcal{R}f\|_{L^q(d\sigma)} \leq \sigma(S)^{1/q} \|\widehat{f}\|_{L^\infty} \leq \sigma(S)^{1/q} \|f\|_{L^1},$$

and $R^*(1 \rightarrow q)$ holds for every q , for *every* compact S — curved or flat,

smooth or not. Nothing has been used but the Riemann–Lebesgue bound.

(b) $p = 2$ is hopeless. By Plancherel the map $f \mapsto \widehat{f}$ is onto $L^2(\mathbb{R}^n)$, so $\widehat{f}$ may be modified on the null set S at will without changing f . Hence no estimate of the form (6.1.1) can hold at $p = 2$ for any q , and the same argument kills every $p > 2$, since L^p functions have transforms no better behaved.

(c) So the question is the range $1 < p < 2$, and the content is: how far past 1 can one push? The answer will depend on S . This is worth stating plainly because it is the reason the problem is geometric: the analysis at $p = 1$ and $p = 2$ is settled and sees nothing about the shape of S .

WORKED EXAMPLE 6.1.2— TIER B A flat piece admits no restriction theorem

Let $S_0 \subseteq \{\xi_n = 0\}$ be a compact piece of a hyperplane, with $d\sigma$ the $(n - 1)$ -dimensional Lebesgue measure on it. Then $R^*(p \rightarrow q)$ fails for every $p > 1$ and every q .

The construction. Take $f(x) = g(x')h(x_n)$ with $x = (x', x_n)$, so that $\widehat{f}(\xi) = \widehat{g}(\xi')\widehat{h}(\xi_n)$, and on S_0 this is $\widehat{g}(\xi')\widehat{h}(0)$. Choose $h \in \mathcal{S}(\mathbb{R})$ with $\widehat{h}(0) = \int h = 1$ and, for $N \geq 1$, replace h by $h_N(t) = N^{-1}h(t/N)$, so that $\widehat{h_N}(0) = 1$ still while

$$\|h_N\|_{L^p(\mathbb{R})} = N^{\frac{1}{p}-1}\|h\|_{L^p} \longrightarrow 0 \quad (N \rightarrow \infty)$$

for every $p > 1$. Meanwhile the left side of (6.1.1) is $\|\widehat{g}\|_{L^q(S_0)} \cdot |\widehat{h_N}(0)| = \|\widehat{g}\|_{L^q(S_0)}$, a fixed non-zero constant once g is chosen with $\widehat{g} \not\equiv 0$ on S_0 . So the ratio of the two sides tends to infinity.

What went wrong, geometrically. The function $f = g \otimes h_N$ is concentrated on a slab of thickness N in the x_n direction, and its transform is concentrated on a slab of thickness $1/N$ around $\{\xi_n = 0\}$ — which contains S_0 no matter how large N is. A flat surface can be approximated from inside by the transform of a function spread out in one direction only, and spreading out is cheap in L^p for $p > 1$.

Why curvature repairs it. A sphere contains no line segment, so the dual construction is forced to concentrate near a *cap* rather than a slab, and a cap of diameter δ is dual to a tube of dimensions $\delta^{-1} \times \cdots \times \delta^{-2}$ — an object of bounded eccentricity in $n - 1$ directions. Making that quantitative is Chapter 6.2, and the example above is the degenerate case of the Knapp construction.

6.1.2 The extension operator

The dual formulation is the one used in practice.

Definition 6.1.2. For $g \in L^1(S, d\sigma)$ define the *extension operator*

$$\mathcal{E}g(x) = \int_S g(\omega) e^{2\pi i x \cdot \omega} d\sigma(\omega) = (gd\sigma)^\vee(x).$$

Since $gd\sigma$ is a finite measure, $\mathcal{E}g$ is a bounded continuous function, and $\mathcal{E}g = \widehat{gd\sigma}$ when $g \equiv 1$ — the object computed in Chapter 5.5.

Theorem 6.1.3 (Restriction–extension duality). *Let $1 \leq p, q \leq \infty$. The following are equivalent, with the same constant:*

$$(i) \quad \|\mathcal{R}f\|_{L^q(d\sigma)} \leq C\|f\|_{L^p(\mathbb{R}^n)} \text{ for all } f \in \mathcal{S}(\mathbb{R}^n);$$

$$(ii) \quad \|\mathcal{E}g\|_{L^{p'}(\mathbb{R}^n)} \leq C\|g\|_{L^{q'}(d\sigma)} \text{ for all } g \in L^{q'}(S, d\sigma).$$

Idea of the proof. The two operators are adjoint. Writing out the pairing $\langle \mathcal{R}f, g \rangle$ and moving the Fourier transform from f to $gd\sigma$ — which is the multiplication formula of Volume 2 — turns it into $\langle f, \mathcal{E}g \rangle$. Each estimate is then the statement that the corresponding adjoint is bounded, and by Theorem 1.8.10 boundedness of an operator and of its adjoint are the same thing with conjugate exponents.

Proof. Step 1. The adjoint identity.

Let $f \in \mathcal{S}(\mathbb{R}^n)$ and $g \in L^\infty(d\sigma)$. By the multiplication formula (Theorem 2.4.5, extended to finite measures by dominated convergence),

$$\begin{aligned} \int_S \widehat{f}(\omega) \overline{g(\omega)} d\sigma(\omega) &= \int_{\mathbb{R}^n} \widehat{f} \overline{g} d\sigma \\ &= \int_{\mathbb{R}^n} f(x) \overline{\mathcal{E}g(x)} dx, \end{aligned}$$

that is,

$$\langle \mathcal{R}f, g \rangle_{L^2(d\sigma)} = \langle f, \mathcal{E}g \rangle_{L^2(\mathbb{R}^n)}. \quad (6.1.2)$$

Both sides converge absolutely: the left because $\widehat{f}$ is bounded and $d\sigma$ finite, the right because $f \in \mathcal{S}$ and $\mathcal{E}g$ is bounded.

Step 2. (i) implies (ii).

Assume (i). For $g \in L^{q'}(d\sigma)$, Theorem 1.8.10 gives

$$\|\mathcal{E}g\|_{L^{p'}(\mathbb{R}^n)} = \sup \left\{ |\langle f, \mathcal{E}g \rangle| : f \in \mathcal{S}, \|f\|_{L^p} \leq 1 \right\},$$

the supremum over the dense class $\mathcal{S}$ being the same as over the unit ball of L^p . By (6.1.2) and Hölder on S ,

$$\begin{aligned} |\langle f, \mathcal{E}g \rangle| &= |\langle \mathcal{R}f, g \rangle| \\ &\leq \|\mathcal{R}f\|_{L^q(d\sigma)} \|g\|_{L^{q'}(d\sigma)} \\ &\leq C \|f\|_{L^p} \|g\|_{L^{q'}(d\sigma)}, \end{aligned}$$

so the supremum is at most $C\|g\|_{L^{q'}(d\sigma)}$.

Step 3. (ii) implies (i).

Symmetrically, for $f \in \mathcal{S}$,

$$\begin{aligned} \|\mathcal{R}f\|_{L^q(d\sigma)} &= \sup \left\{ |\langle \mathcal{R}f, g \rangle| : \|g\|_{L^{q'}(d\sigma)} \leq 1 \right\} \\ &= \sup |\langle f, \mathcal{E}g \rangle| \\ &\leq \|f\|_{L^p} \sup \|\mathcal{E}g\|_{L^{p'}} \\ &\leq C \|f\|_{L^p}, \end{aligned}$$

using Theorem 1.8.10 on $(S, d\sigma)$, which is a finite — hence σ -finite — measure space. $\square$

Where we are. From now on we work with $\mathcal{E}$ and not with $\mathcal{R}$. Three reasons, all practical.

$\mathcal{E}g$ is given by an explicit integral, so one may compute with it; $\mathcal{R}f$ is defined by an inequality. The estimate for $\mathcal{E}$ has the unknown on the *left*, which is where a superposition of wave packets naturally sits. And the most important case, $q = 2$, becomes $q' = 2$, so the data g lives in a Hilbert space and L^2 orthogonality is available — which is exactly what Chapter 6.3 exploits.

The dictionary is: $R^*(p \rightarrow q)$ for restriction is $\mathcal{E} : L^{q'}(d\sigma) \rightarrow L^{p'}(\mathbb{R}^n)$ for extension. The conjectured range $p < \frac{2n}{n+1}$ becomes $p' > \frac{2n}{n-1}$.

6.1.3 The local formulation

Proposition 6.1.4 (Restriction is a local statement). *Let $1 \leq a, b \leq \infty$. The estimate (6.2.1) implies, and is implied up to an arbitrarily small loss in b by, the family of local estimates*

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq C \|g\|_{L^a(d\sigma)} \quad \text{for every ball } B_R \text{ of radius } R \geq 1,$$

with C independent of R and of the centre of B_R .

Proof. That the global estimate implies the local one is immediate, since $\|\cdot\|_{L^b(B_R)} \leq \|\cdot\|_{L^b(\mathbb{R}^n)}$.

For the converse, note first that the constant may be taken independent of the *centre*: translating B_R by x_0 replaces $\mathcal{E}g$ by $\mathcal{E}(e^{2\pi i x_0 \cdot \omega} g)$, by the modulation rule (Theorem 2.4.2), and modulation does not change $\|g\|_{L^a(d\sigma)}$. So only the dependence on R is at issue, and that is exactly what Theorem 6.5.2 of Chapter 6.5 below, whose proof uses only the definitions of this section addresses. $\square$

Where we are. The proposition says that no generality is lost by proving estimates on a ball, which is fortunate, because every argument in Volumes 12 and 13 does exactly that. It also isolates why: the modulation symmetry of $\mathcal{E}$ makes the problem translation-invariant, so a ball anywhere is as good as a ball at the origin, and only the *scale* R carries information.

That observation — that R is the only parameter — is what makes induction on scales possible. An argument that related the estimate at radius R to the estimate at radius $R^{1/2}$ would be useless if the constant also depended on where the ball sat.

6.1.4 The TT^* identity

Proposition 6.1.5. *For $g \in L^2(d\sigma)$,*

$$\mathcal{E}\mathcal{E}^*h = h * \widehat{d\sigma} \quad \text{for } h \in \mathcal{S}(\mathbb{R}^n),$$

where $\mathcal{E}^* = \mathcal{R}$ under the pairing (6.1.2). Consequently $\mathcal{E} : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$ is bounded with norm A if and only if $h \mapsto h * \widehat{d\sigma}$ is bounded from $L^{q'}(\mathbb{R}^n)$ to $L^q(\mathbb{R}^n)$ with norm A^2 .

Proof. By Theorem 6.1.3 with $q = 2$, $\mathcal{E}^*h = \widehat{h}|_S$. Hence

$$\mathcal{E}\mathcal{E}^*h(x) = \int_S \widehat{h}(\omega) e^{2\pi i x \cdot \omega} d\sigma(\omega) = (\widehat{h} d\sigma)^\vee(x).$$

Now $\widehat{h} d\sigma$ has inverse transform $h * d\sigma^\vee = h * \widehat{d\sigma}$, since $d\sigma$ is symmetric under $\omega \mapsto -\omega$ for the surfaces considered here, or in general $h * d\sigma^\vee$; the convolution converges absolutely because $h \in \mathcal{S}$ and $d\sigma$ is finite.

For the second statement, the identity $\|\mathcal{E}\|^2 = \|\mathcal{E}\mathcal{E}^*\|$ holds for any bounded operator between Hilbert-space-and- L^q pairs in the sense used in Theorem 5.7.3 Step 6: $\|\mathcal{E}g\|_{L^q}^2 = \sup_h |\langle \mathcal{E}\mathcal{E}^*h, h \rangle|$ over $\|h\|_{L^{q'}} \leq 1$. $\square$

WORKED EXAMPLE 6.1.3— TIER B Computing $\mathcal{E}$ on a simple datum

Abstract duality is easier to trust after one explicit computation. Take $n = 2$, $S = \mathbb{S}^1$, and $g \equiv 1$.

The answer. $\mathcal{E}1 = \widehat{d\sigma}$, which by Volume 2 is

$$\begin{aligned}\widehat{d\sigma}(x) &= 2\pi J_0(2\pi|x|) \\ &= \sqrt{\frac{2\pi}{|x|}} \cos\left(2\pi|x| - \frac{\pi}{4}\right) + O(|x|^{-3/2}),\end{aligned}$$

using Theorem 5.4.1 with $\nu = 0$.

Reading off which estimates can hold. $|\mathcal{E}1| \approx |x|^{-1/2}$ along a sequence, so $\mathcal{E}1 \in L^b(\mathbb{R}^2)$ requires $\frac{b}{2} > 2$, that is $b > 4$ — which is Theorem 6.2.5 for $n = 2$, obtained here by a direct computation rather than an argument. And $\|1\|_{L^a(d\sigma)}$ is finite for every a , so no estimate (6.2.1) with $b \leq 4$ can hold, whatever a .

The dual statement. By Theorem 6.1.3, the failure at $b = 4$ says $R^*(\frac{4}{3} \rightarrow a')$ fails for every a' . So a function in $L^{4/3}(\mathbb{R}^2)$ may have a Fourier transform whose restriction to the circle is not even locally integrable — a concrete statement, obtained from an asymptotic for a Bessel function.

Why it is worth doing this once. Every necessary condition in Chapter 6.2 is of this form: choose a datum g , compute or estimate $\mathcal{E}g$, and compare the two sides. The Knapp example is the same procedure with g the indicator of a cap instead of the constant, and the whole-sphere example of Theorem 6.2.5 is this computation in general dimension.

WORKED EXAMPLE 6.1.4— TIER C: RESEARCH TASTE Why this identity is the organising principle of the subject

Theorem 6.1.5 converts a statement about an operator into a statement about a *convolution kernel*, namely $\widehat{d\sigma}$. That is the exact point at which the whole of Volume 5 becomes relevant, and it is worth laying out the consequences before they are used.

(a) *The immediate consequence.* Worked Example 2.8.3 already ran this argument once, crudely: bounding $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$ and applying Hardy–Littlewood–Sobolev to the resulting fractional integral gives $\mathcal{E} : L^2(d\sigma) \rightarrow L^q(\mathbb{R}^n)$ for $q > \frac{4n}{n-1}$. Chapter 6.3 improves this to $q > \frac{2(n+1)}{n-1}$ by decomposing $\widehat{d\sigma}$ dyadically rather than bounding it once, and the improve-

ment is substantial: at $n = 3$ the exponents are 6 versus 4.

(b) *Where the improvement comes from.* The crude argument uses only the size of $\widehat{d\sigma}$. The dyadic argument uses, in addition, that the piece of $\widehat{d\sigma}$ living at scale $|x| \approx 2^j$ has Fourier transform concentrated near S — so that convolution with it is bounded on L^2 with a constant that does *not* degrade. Interpolating the two bounds for each piece and summing over j is the whole of Stein–Tomas. The oscillation of $\widehat{d\sigma}$, computed in Theorem 5.5.2, is what makes the L^2 bound available; an argument that took absolute values first would have destroyed it.

(c) *Where it stops.* The method is confined to $q' = 2$ on the data side, because that is the only exponent at which $\mathcal{EE}^*$ is the relevant object. The restriction conjecture concerns $L^\infty(d\sigma)$ data, and there is no TT^* identity for it. Every subsequent advance — bilinear, multilinear, polynomial — is in one way or another a substitute for the missing identity.

(d) *The pattern, once more.* TT^* has now appeared four times in this series: for the Hilbert transform in Volume 3, for oscillatory integral operators in Theorem 5.7.3, in Volume 2’s first restriction estimate, and here. Each time the move is the same — replace an operator that resists estimation by a self-adjoint one whose kernel can be computed — and each time the cost is the same: the method is tied to L^2 on one side. Recognising both halves of that trade is worth more than any single application of it.

Drills

Drill 6.1.1. Verify the claim in Worked Example 6.1.1(b) in detail: given $f \in L^2$ and a null set S , construct $\tilde{f} \in L^2$ with $\tilde{f} = f$ in L^2 and $\tilde{f}|_S$ prescribed arbitrarily.

Drill 6.1.2. Carry out Worked Example 6.1.2 with explicit g and h and compute both sides of (6.1.1) as functions of N .

Drill 6.1.3. Show that $R^*(p \rightarrow q)$ for a compact S implies $R^*(p \rightarrow \tilde{q})$ for every $\tilde{q} \leq q$, and explain why no such monotonicity holds in p .

Drill 6.1.4. Deduce from Theorem 6.1.3 that $R^*(p \rightarrow 2)$ is equivalent to $\|\mathcal{E}g\|_{L^{p'}} \lesssim \|g\|_{L^2(d\sigma)}$, and identify the value of p' corresponding to the Stein–Tomas exponent $q = \frac{2(n+1)}{n-1}$.

Notes

The restriction problem was raised by Stein in 1967. Theorem 6.1.3 is immediate but was what made the subject tractable, since it is the extension form that admits wave packet decompositions.

Worked Example 6.1.2 is the standard observation that the problem is about curvature; the sharp quantitative version is Knapp's example, in Chapter 6.2.

Chapter 6.2

Necessary Conditions

Worked Example 6.1.2 showed that a flat surface admits no restriction theorem at all, and located the reason: a hyperplane can be approached by the transform of a function spread out in a single direction, and spreading out is cheap in L^p for $p > 1$. A sphere contains no flat piece, but it contains *almost* flat pieces — a cap of diameter δ deviates from its tangent plane by only δ^2 — and running the same construction on a cap gives the sharp obstruction. That is Knapp's example, and it is the subject of this chapter.

Throughout, $S = \mathbb{S}^{n-1}$ and we work with the extension operator, writing the estimate to be tested as

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq C \|g\|_{L^a(S, d\sigma)}. \quad (6.2.1)$$

By Theorem 6.1.3 this is $R^*(b' \rightarrow a')$ in the restriction formulation.

6.2.1 The Knapp example

Definition 6.2.1. For $0 < \delta < \frac{1}{2}$ let

$$\kappa_\delta = \{\omega \in \mathbb{S}^{n-1} : |\omega - e_n| \leq \delta\}$$

be the *cap* of radius δ at the north pole, and let

$$T_\delta = \left\{x \in \mathbb{R}^n : |x'| \leq \frac{c}{\delta}, |x_n| \leq \frac{c}{\delta^2}\right\}$$

be the *dual box*, where $x = (x', x_n)$ and $c > 0$ is an absolute constant fixed in Theorem 6.2.2.

Two elementary facts: $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$, since the cap is comparable to a Euclidean ball of radius δ in the $(n-1)$ -dimensional surface; and

$$m(T_\delta) = (2c\delta^{-1})^{n-1} \cdot 2c\delta^{-2} \approx_n \delta^{-(n+1)}.$$

Lemma 6.2.2 (The cap does not oscillate on its dual box). *There is an absolute constant $c > 0$ such that, with $g = \mathbf{1}_{\kappa_\delta}$,*

$$|\mathcal{E}g(x)| \geq \frac{1}{2} \sigma(\kappa_\delta) \quad \text{for every } x \in T_\delta.$$

Idea of the proof. $\mathcal{E}g(x) = \int_{\kappa_\delta} e^{2\pi i x \cdot \omega} d\sigma$ is large exactly when the phase $x \cdot \omega$ is nearly constant over the cap. Removing the constant x_n by a modulation, the phase becomes $x' \cdot \omega'$ — of size $|x'| \delta$ — plus the curvature term $-\frac{1}{2} x_n |\omega'|^2$ — of size $|x_n| \delta^2$. The box T_δ is precisely the set on which both are $O(1)$: it is δ^{-1} long in the $n-1$ tangential directions and δ^{-2} long in the normal direction, and the asymmetry is a direct record of the sphere's curvature.

Proof. Parametrise the cap by $\omega = (\omega', \omega_n)$ with $|\omega'| \leq \delta$ and $\omega_n = \sqrt{1 - |\omega'|^2}$. Then

$$x \cdot \omega = x' \cdot \omega' + x_n \omega_n = x_n + x' \cdot \omega' - \frac{x_n |\omega'|^2}{2} + x_n E(\omega'),$$

where $E(\omega') = \sqrt{1 - |\omega'|^2} - 1 + \frac{1}{2} |\omega'|^2$ satisfies $|E(\omega')| \leq |\omega'|^4$ for $|\omega'| \leq \frac{1}{2}$, by Taylor's theorem with remainder.

For $x \in T_\delta$ we may bound the three non-constant terms:

$$\begin{aligned} |x' \cdot \omega'| &\leq \frac{c}{\delta} \cdot \delta = c \\ &, \quad \frac{|x_n| |\omega'|^2}{2} \leq \frac{c}{2\delta^2} \cdot \delta^2 = \frac{c}{2} \\ &, \quad |x_n| |E(\omega')| \leq \frac{c}{\delta^2} \cdot \delta^4 \\ &\leq c. \end{aligned}$$

So writing $\Phi(\omega) = x \cdot \omega - x_n$, we have $2\pi |\Phi(\omega)| \leq 2\pi \cdot \frac{5}{2} c$ for all $\omega \in \kappa_\delta$. Choose $c = \frac{1}{30}$, so that $2\pi |\Phi| \leq \frac{\pi}{3}$ and hence $\cos(2\pi \Phi(\omega)) \geq \frac{1}{2}$.

Therefore

$$\begin{aligned} |\mathcal{E}g(x)| &= \left| \int_{\kappa_\delta} e^{2\pi i \Phi(\omega)} d\sigma(\omega) \right| \\ &\geq \operatorname{Re} \int_{\kappa_\delta} e^{2\pi i \Phi} d\sigma \\ &\geq \frac{1}{2} \sigma(\kappa_\delta), \end{aligned}$$

the first step because $|e^{-2\pi i x_n}| = 1$. □

Theorem 6.2.3 (The Knapp condition). *If (6.2.1) holds for $S = \mathbb{S}^{n-1}$, then*

$$\frac{n+1}{b} \leq (n-1) \left(1 - \frac{1}{a} \right). \quad (6.2.2)$$

Proof. Step 1. Choose the test function.

Apply (6.2.1) to $g = \mathbf{1}_{\kappa_\delta}$, the indicator of the cap. This is the only choice made in the proof, and everything else is computation.

Step 2. *Compute the right-hand side.*

Since g is an indicator,

$$\begin{aligned} \|g\|_{L^a(d\sigma)} &= \left(\int_{\mathbb{S}^{n-1}} \mathbf{1}_{\kappa_\delta} d\sigma \right)^{1/a} \\ &= \sigma(\kappa_\delta)^{1/a} \approx_n \delta^{(n-1)/a}, \end{aligned}$$

using $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$ from the remarks following Theorem 6.1.1. (For $a = \infty$ read the exponent as 0, so the right side is ≈ 1 .)

Step 3. *Bound the left-hand side from below.*

The L^b norm over $\mathbb{R}^n$ is at least the norm over the dual box T_δ , on which Theorem 6.2.2 gives the pointwise lower bound $|\mathcal{E}g| \geq \frac{1}{2}\sigma(\kappa_\delta)$. Hence

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} &\geq \left(\int_{T_\delta} |\mathcal{E}g|^b \right)^{1/b} \\ &\geq \frac{1}{2} \sigma(\kappa_\delta) m(T_\delta)^{1/b} \approx_n \delta^{n-1} \delta^{-(n+1)/b}, \end{aligned}$$

using $m(T_\delta) \approx_n \delta^{-(n+1)}$. Note where the two exponents come from: δ^{n-1} is the *height* of $\mathcal{E}g$ on the box, and $\delta^{-(n+1)/b}$ is the *volume* of the box, raised to the power $1/b$.

Step 4. *Compare, and let $\delta \rightarrow 0$.*

Substituting Steps 2 and 3 into (6.2.1),

$$\delta^{n-1-\frac{n+1}{b}} \lesssim_n C \delta^{\frac{n-1}{a}} \quad \text{for all } 0 < \delta < \frac{1}{2}.$$

An inequality $\delta^\alpha \lesssim \delta^\beta$ holding for all small δ forces $\alpha \geq \beta$: otherwise, dividing gives $\delta^{\alpha-\beta} \lesssim 1$ with $\alpha - \beta < 0$, and the left side tends to ∞ . Hence

$$n-1-\frac{n+1}{b} \geq \frac{n-1}{a},$$

and rearranging — moving $\frac{n+1}{b}$ to the right and factoring $n-1$ out of the two remaining terms — gives $\frac{n+1}{b} \leq (n-1)(1 - \frac{1}{a})$, which is (6.2.2). $\square$

(a) $a = 2$. Condition (6.2.2) reads $\frac{n+1}{b} \leq \frac{n-1}{2}$, that is

$$b \geq \frac{2(n+1)}{n-1}.$$

This is exactly the Stein–Tomas exponent, proved sufficient (in the open range) in Chapter 6.3. So at $a = 2$ the Knapp example is *sharp*: it forbids everything below $\frac{2(n+1)}{n-1}$, and everything above is a theorem.

(b) $a = \infty$. Now (6.2.2) reads $\frac{n+1}{b} \leq n-1$, that is $b \geq \frac{n+1}{n-1}$. This is a much weaker requirement — for $n = 3$ it says $b \geq 2$ — and it is *not* the truth. The binding condition at $a = \infty$ comes from a different example, §2.2.

(c) *Numerically*, $n = 3$. At $a = 2$: $b \geq 4$. At $a = \infty$: Knapp gives $b \geq 2$, while the correct answer, conjecturally, is $b > 3$. The gap between 2 and 3 is filled by the whole-sphere example.

Proposition 6.2.4 (Knapp for the paraboloid). *Let $P = \{(\zeta, |\zeta|^2) : |\zeta| \leq 1\} \subseteq \mathbb{R}^n$ with the measure $d\mu = d\zeta$, and suppose $\|\mathcal{E}_P g\|_{L^b(\mathbb{R}^n)} \leq C\|g\|_{L^a(d\mu)}$. Then (6.2.2) holds, with $n-1$ the dimension of P .*

Proof. Let $\rho_\delta = \{|\zeta| \leq \delta\}$, so $\mu(\rho_\delta) \approx \delta^{n-1}$, and take $g = \mathbf{1}_{\rho_\delta}$. For $x = (x', x_n)$,

$$\mathcal{E}_P g(x) = \int_{|\zeta| \leq \delta} e^{2\pi i(x' \cdot \zeta + x_n |\zeta|^2)} d\zeta.$$

On the box $|x'| \leq \frac{c}{\delta}$, $|x_n| \leq \frac{c}{\delta^2}$ the phase satisfies

$$|x' \cdot \zeta| \leq c, \quad |x_n| |\zeta|^2 \leq c,$$

so, choosing $c = \frac{1}{24}$, the total phase is at most $\frac{\pi}{3}$ in absolute value and $|\mathcal{E}_P g| \geq \frac{1}{2} \mu(\rho_\delta)$ there, exactly as in Theorem 6.2.2. Note that here *no* Taylor expansion is needed: the paraboloid’s “height function” $|\zeta|^2$ is already exactly quadratic, which is why the paraboloid is the standard model.

The box has measure $\approx \delta^{-(n+1)}$, and repeating the computation of Theorem 6.2.3 verbatim gives (6.2.2). $\square$

Where we are. The sphere and the paraboloid give the same condition, and the proof for the paraboloid is shorter — the error term $E(\omega')$ of Theorem 6.2.2 is absent. This is the reason essentially all of the modern literature works with the paraboloid: it is the osculating surface of the sphere at a point, it has the same curvature properties, and it is exactly parabolic rather than approximately so, which makes the parabolic rescaling

$$(\zeta, \tau) \mapsto (\lambda\zeta, \lambda^2\tau)$$

an exact symmetry. Induction on scales, in Volume 13, uses that symmetry at every step; on the sphere one would have to carry the error terms through the induction.

6.2.2 The whole-sphere example

Theorem 6.2.5. *If (6.2.1) holds for $S = \mathbb{S}^{n-1}$ with any $a \leq \infty$, then*

$$b > \frac{2n}{n-1}. \quad (6.2.3)$$

Proof. Take $g \equiv 1$, which lies in $L^a(d\sigma)$ for every a since $d\sigma$ is finite. Then $\mathcal{E}g = \widehat{d\sigma}$, and (6.2.1) asserts $\widehat{d\sigma} \in L^b(\mathbb{R}^n)$.

By Theorem 5.5.2, for $|x| \geq 1$,

$$\begin{aligned} \widehat{d\sigma}(x) &= |x|^{-\frac{n-1}{2}} \left(c_+ e^{2\pi i|x|} + c_- e^{-2\pi i|x|} \right) + O(|x|^{-\frac{n+1}{2}}), \end{aligned}$$

with $c_{\pm}$ non-zero conjugates, so $c_+ e^{2\pi i r} + c_- e^{-2\pi i r} = 2 \operatorname{Re}(c_+ e^{2\pi i r})$ is a non-degenerate cosine in $r = |x|$. Hence there are $\eta, \rho > 0$ and arbitrarily large R with

$$|\widehat{d\sigma}(x)| \geq \eta |x|^{-\frac{n-1}{2}} \quad \text{whenever } ||x| - R| \leq \rho,$$

and the annuli $\{||x| - R_k| \leq \rho\}$ may be taken disjoint with $R_k \rightarrow \infty$, say $R_k = k$ for k large along a suitable arithmetic progression. Therefore

$$\begin{aligned} \int_{\mathbb{R}^n} |\widehat{d\sigma}|^b &\gtrsim \sum_k \eta^b R_k^{-\frac{b(n-1)}{2}} m(\{||x| - R_k| \leq \rho\}) \\ &\approx \sum_k R_k^{-\frac{b(n-1)}{2} + n-1}, \end{aligned}$$

using that the annulus has measure $\approx \rho R_k^{n-1}$. This series diverges unless

$$\frac{b(n-1)}{2} - (n-1) > 1,$$

i.e. unless $b > \frac{2n}{n-1}$. At the borderline $b = \frac{2n}{n-1}$ the exponent is

$$\frac{b(n-1)}{2} - (n-1) = \frac{2n}{n-1} \cdot \frac{n-1}{2} - (n-1) = n - (n-1) = 1,$$

so the series becomes $\sum_k R_k^{-1}$, which diverges for $R_k = k$; hence the inequality (6.2.3) is strict. $\square$

WORKED EXAMPLE 6.2.2— TIER B **The two conditions, drawn against each other**

It is worth seeing precisely which of (6.2.2) and (6.2.3) binds where. Both are conditions on the pair $(1/a, 1/b)$; write $u = 1/a$, $v = 1/b$, so the admissible region sits in $[0, 1]^2$.

Knapp is $(n+1)v \leq (n-1)(1-u)$, a closed half-plane whose boundary is the line from $(u, v) = (0, \frac{n-1}{n+1})$ down to $(1, 0)$.

Whole sphere is $v < \frac{n-1}{2n}$, a horizontal half-plane.

At $u = \frac{1}{2}$ (that is, $a = 2$) Knapp gives $v \leq \frac{n-1}{2(n+1)}$ and the sphere gives $v < \frac{n-1}{2n}$; since $\frac{n-1}{2(n+1)} < \frac{n-1}{2n}$, *Knapp binds*. At $u = 0$ (that is, $a = \infty$) Knapp gives $v \leq \frac{n-1}{n+1}$ and the sphere gives $v < \frac{n-1}{2n}$; now $\frac{n-1}{2n} < \frac{n-1}{n+1}$, so *the sphere binds*.

The crossover is where the two boundaries meet: $(n+1)\frac{n-1}{2n} = (n-1)(1-u)$ gives $u = \frac{n-1}{2n}$, i.e. $a = \frac{2n}{n-1}$. So:

$$\begin{aligned} a &\leq \frac{2n}{n-1} && \text{the whole-sphere condition binds} \\ a &\geq \frac{2n}{n-1} && \text{the Knapp condition binds} \end{aligned}$$

Why this matters. The Stein–Tomas theorem lives at $a = 2$, where Knapp binds and is sharp. The restriction conjecture lives at $a = \infty$, where the sphere binds — and there the Knapp example, the only construction that sees curvature, is *not* the obstruction. That mismatch is a large part of why the conjecture is hard: the extremal example at $a = \infty$ is a single global object, $\widehat{d\sigma}$ itself, and no local cap computation detects it.

Conjecture 6.2.6 (Restriction conjecture). *For $S = \mathbb{S}^{n-1}$ and $n \geq 2$,*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \lesssim_{n,b} \|g\|_{L^\infty(d\sigma)} \quad \text{for every } b > \frac{2n}{n-1}.$$

By Theorem 6.2.5 the range is optimal, and by Theorem 6.2.3 with $a = \infty$ it is consistent with the Knapp example. Equivalently, in restriction form: $R^*(p \rightarrow \infty)$ for $p < \frac{2n}{n+1}$ — and since $\mathcal{R}$ maps into $L^\infty(d\sigma)$ trivially when $p = 1$, the content is the range $1 < p < \frac{2n}{n+1}$.

WORKED EXAMPLE 6.2.3— TIER C: RESEARCH TASTE **What is actually known, and the shape of the difficulty**

(a) *The record.* Theorem 6.2.6 is a theorem for $n = 2$ (Chapter 6.8) and open for every $n \geq 3$. Chapter 6.3 proves the case $a = 2$, which by interpolation with the trivial bound at $a = 1$ gives a range of (a, b) falling well short of $a = \infty$. Volumes 12 and 13 push b down from the Stein–Tomas value towards $\frac{2n}{n-1}$, and the current record is strictly between.

(b) *Why interpolation cannot finish.* One might hope to interpolate the $a = 2$ theorem against something at $a = \infty$. But at $a = \infty$ there is no non-trivial estimate available to interpolate with: the trivial bound $\|\mathcal{E}g\|_\infty \leq \|g\|_{L^1(d\sigma)}$ is at $b = \infty$, and interpolating it with Stein–Tomas returns exactly the segment already known. The conjecture is not reachable from the endpoints of anything proved.

(c) *Where the extra input must come from.* Decompose $g = \sum_\theta g_\theta$ over δ -caps. Then $\mathcal{E}g = \sum_\theta \mathcal{E}g_\theta$, and by Theorem 6.2.2 each $\mathcal{E}g_\theta$ is essentially constant on a tube T_θ dual to its cap — of dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$, pointing in the direction normal to θ . There are $\approx \delta^{-(n-1)}$ caps, hence that many tubes, one in each direction.

So the question becomes: *how much can $\delta^{-(n-1)}$ tubes, one per direction, overlap?* That is a Kakeya question, and Chapter 6.6 makes the implication precise. It is also why every subsequent advance in the subject is geometric rather than analytic: the analysis of a single $\mathcal{E}g_\theta$ was completed in Volume 5, and what remains is a counting problem about tubes.

(d) *A caution about the converse.* Kakeya does not obviously imply restriction, and no such implication is known. The arrow runs one way, and the gap between the two conjectures — Kakeya is also open for $n \geq 3$, but is believed easier — is itself a subject of study.

WORKED EXAMPLE 6.2.4— TIER A **The dual box, and why its shape is $\delta^{-1} \times \delta^{-2}$**

The asymmetry of T_δ is the single most important geometric fact in the subject, and it is worth deriving it twice — once from the phase, as in Theorem 6.2.2, and once from the uncertainty principle.

From the uncertainty principle. The cap κ_δ has diameter δ in the $n - 1$ tangential directions, so a function with Fourier support there cannot be localised better than δ^{-1} in those directions. In the normal direction the cap is not a point: it deviates from its tangent plane by

$$1 - \sqrt{1 - \delta^2} \approx \frac{\delta^2}{2},$$

so its extent in the normal direction is $\approx \delta^2$, and the dual extent is δ^{-2} . Hence a box of dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$.

The consistency check. The volume of the cap's δ^2 -neighbourhood is $\approx \delta^{n-1} \cdot \delta^2$, and the dual box has volume $\delta^{-(n-1)} \delta^{-2}$. Their product is 1, as the uncertainty principle requires.

Why δ^2 and not δ . Because the sphere is curved and not merely bent: the deviation from the tangent plane is quadratic, by Taylor's theorem, and it is

quadratic precisely because the second fundamental form is non-degenerate — Theorem 5.5.1 again. For a surface with a vanishing principal curvature the deviation in that direction is of higher order, the dual box is longer, and the Knapp condition weakens accordingly. Drill 2.4 works this out.

The number to remember. A δ -cap is dual to a $\delta^{-1} \times \delta^{-2}$ box; equivalently, after rescaling by δ^2 , a $\delta^{1/2}$ -cap is dual to a $1 \times \delta^{1/2}$ tube at scale δ^{-1} . Both normalisations appear in the literature and confusing them is the commonest source of arithmetic errors in this area.

Figure 6.2.1 (to be drawn). *The two necessary conditions*

Content required: The $(u, v) = (1/a, 1/b)$ square, with the Knapp half-plane $(n+1)v \leq (n-1)(1-u)$ and the whole-sphere half-plane $v < \frac{n-1}{2n}$ shaded; their intersection is the conjectured range. Mark the crossover at $a = \frac{2n}{n-1}$ and the two points $a = 2$ (Stein–Tomas) and $a = \infty$ (the conjecture).

Drills

Drill 6.2.1. Verify $\sigma(\kappa_\delta) \approx_n \delta^{n-1}$ by parametrising the cap as a graph over the tangent plane.

Drill 6.2.2. Check the bound $|E(\omega')| \leq |\omega'|^4$ used in Theorem 6.2.2, for $|\omega'| \leq \frac{1}{2}$.

Drill 6.2.3. Redo Theorem 6.2.3 for the truncated paraboloid $\{(\xi', |\xi'|^2) : |\xi'| \leq 1\}$ and confirm the same condition (6.2.2).

Drill 6.2.4. Show that for a hypersurface with k non-vanishing principal curvatures the whole-surface example gives $b > \frac{2n}{k}$, using Theorem 5.5.3 in place of Theorem 5.5.2.

Notes

The Knapp example is from the 1970s folklore of the subject; the name attaches to an unpublished observation of Knapp. The two necessary conditions and the resulting conjecture are stated in Stein’s 1979 problem list.

That the conjectured range is exactly the intersection of the two conditions is the reason Theorem 6.2.6 is believed: no third obstruction has been found in fifty years, in any dimension.

Chapter 6.3

The Stein–Tomas Theorem

Worked Example 6.2.1 identified $b = \frac{2(n+1)}{n-1}$ as the exponent below which the estimate $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^n)$ is impossible. This chapter proves that everything strictly above it is true.

Theorem 6.3.1 (Stein–Tomas, non-endpoint). *Let $S \subseteq \mathbb{R}^n$, $n \geq 2$, be a compact C^∞ hypersurface with non-vanishing Gaussian curvature and $\psi \in C_c^\infty$ a cutoff. Then for every*

$$b > \frac{2(n+1)}{n-1}$$

there is $C = C(n, b, S, \psi)$ with

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq C \|g\|_{L^2(S, d\sigma)} \quad \text{for all } g \in L^2(S, d\sigma),$$

where $\mathcal{E}g = (g\psi d\sigma)^\vee$. Equivalently, $R^(p \rightarrow 2)$ holds for every $p < \frac{2(n+1)}{n+3}$.*

Idea of the proof. By Theorem 6.1.5 the estimate is equivalent to

$$\|f * \widehat{\psi d\sigma}\|_{L^b} \lesssim \|f\|_{L^{b'}},$$

a statement about convolution with the single fixed kernel $\widehat{\psi d\sigma}$. Two facts about that kernel are available and pull in opposite directions:

it is small — $|\widehat{\psi d\sigma}(x)| \lesssim |x|^{-(n-1)/2}$, by Theorem 5.5.2 — which is good for $L^1 \rightarrow L^\infty$ but says nothing about L^2 ; and

its Fourier transform is a measure on a surface — bounded in no useful sense — which is bad for $L^2 \rightarrow L^2$.

Neither alone suffices. The device is to cut the kernel into dyadic pieces K_j living at $|x| \approx 2^j$ and to use *both* facts on each piece. The piece K_j has sup-norm $2^{-j(n-1)/2}$, and its Fourier transform, being $d\sigma$ smoothed at scale 2^{-j} , has sup-norm $\approx 2^j$. Interpolating these two bounds for a fixed j gives an $L^{b'} \rightarrow L^b$ norm of $2^{j(\theta - (1-\theta)\frac{n-1}{2})}$ with $\theta = 2/b$, and the whole theorem

is the observation that this exponent is negative precisely when $b > \frac{2(n+1)}{n-1}$. The series then converges geometrically.

Note where the two competing effects appear: $2^{-j(n-1)/2}$ is the curvature, and 2^j is the price of localising in space, since a function supported at scale 2^j cannot have its Fourier transform concentrated on a surface. The theorem is their balance.

6.3.1 The dyadic decomposition of the kernel

Fix $\eta \in C_c^\infty(\mathbb{R}^n)$ supported in $\{\frac{1}{2} < |x| < 2\}$ with $\sum_{j \geq 1} \eta(2^{-j}x) = 1$ for $|x| \geq 1$, and let $\eta_0 = 1 - \sum_{j \geq 1} \eta(2^{-j}\cdot)$, supported in $\{|x| \leq 2\}$. Write

$$K = \widehat{\psi d\sigma}, \quad K_j = K \cdot \eta(2^{-j}\cdot) \quad (j \geq 1), \quad K_0 = K\eta_0,$$

so that $K = \sum_{j \geq 0} K_j$ pointwise, with only finitely many terms non-zero at each point.

Lemma 6.3.2. *For $j \geq 1$,*

$$(i) \quad \|K_j\|_{L^\infty} \lesssim_{n,S} 2^{-j \frac{n-1}{2}}, \quad (ii) \quad \|\widehat{K_j}\|_{L^\infty} \lesssim_{n,S} 2^j.$$

Proof. (i) On $\text{supp } K_j$ we have $|x| \approx 2^j$, and Theorem 5.5.2 gives $|K(x)| \lesssim |x|^{-(n-1)/2}$ there. Since $|\eta| \leq 1$, (i) follows.

(ii) Write $\eta_j = \eta(2^{-j}\cdot)$, so $K_j = K\eta_j$ and hence

$$\widehat{K_j} = \widehat{K} * \widehat{\eta_j} = (\psi d\sigma) * \widehat{\eta_j},$$

using $\widehat{K} = \psi d\sigma$ by Fourier inversion (Theorem 2.5.11, the measure being tempered). Now $\widehat{\eta_j}(\xi) = 2^{jn} \widehat{\eta}(2^j \xi)$, so $\|\widehat{\eta_j}\|_{L^1} = \|\widehat{\eta}\|_{L^1}$ and, since $\widehat{\eta} \in \mathcal{S}$, for every N

$$|\widehat{\eta_j}(\xi)| \lesssim_N \frac{2^{jn}}{(1 + 2^j |\xi|)^N}.$$

Therefore, for any ξ ,

$$|\widehat{K_j}(\xi)| \leq \int_S |\widehat{\eta_j}(\xi - \omega)| |\psi(\omega)| d\sigma(\omega) \lesssim_N 2^{jn} \int_S \frac{d\sigma(\omega)}{(1 + 2^j |\xi - \omega|)^N}.$$

Decompose S into the pieces where $|\xi - \omega| \approx 2^{-j} 2^k$, $k \geq 0$. Since S is a hyper-surface, $\sigma(S \cap B(\xi, r)) \lesssim_S r^{n-1}$ for $r \leq 1$, and $\sigma(S) < \infty$; so the k -th piece has measure $\lesssim \min\{(2^{k-j})^{n-1}, 1\}$ and contributes $\lesssim 2^{jn} 2^{-kn} \min\{(2^{k-j})^{n-1}, 1\}$. Taking $N = n$ and summing over $k \geq 0$,

$$|\widehat{K_j}(\xi)| \lesssim 2^{jn} 2^{-j(n-1)} \sum_{k \geq 0} 2^{-kn} 2^{k(n-1)} \lesssim 2^j,$$

the series $\sum_k 2^{-k}$ converging. □

Where we are. The exponent 2^j in (ii) is worth understanding, because it is the only place the cost of localisation enters. Smoothing $d\sigma$ at scale 2^{-j} spreads its mass over a slab of thickness 2^{-j} around S ; the density of that slab is its total mass, $O(1)$, divided by its volume, $\approx 2^{-j}$. Hence 2^j . The two exponents in Theorem 6.3.2 therefore have quite different origins: $2^{-j(n-1)/2}$ comes from curvature and would be 2^0 for a flat piece, while 2^j comes only from measure theory and is the same for every hypersurface. The theorem is the statement that curvature wins when b is large enough.

6.3.2 Proof of the theorem

Proof of Theorem 6.3.1. Step 1. Reduce to the kernel.

By Theorem 6.1.5 it suffices to prove

$$\|f * K\|_{L^b(\mathbb{R}^n)} \lesssim \|f\|_{L^{b'}(\mathbb{R}^n)}, \quad K = \widehat{\psi d\sigma}, \quad (6.3.1)$$

for $f \in \mathcal{S}$; the general case follows by density.

Step 2. *The two endpoint bounds for a single piece.*

Let $T_j f = f * K_j$. By Theorem 6.3.2(i) and Young's inequality (Worked Example 2.2.1),

$$\|T_j f\|_{L^\infty} \leq \|K_j\|_{L^\infty} \|f\|_{L^1} \lesssim 2^{-j \frac{n-1}{2}} \|f\|_{L^1}. \quad (6.3.2)$$

By Theorem 6.3.2(ii) and Plancherel — T_j is the multiplier operator with symbol $\widehat{K_j}$, so Theorem 2.6.2 applies —

$$\|T_j f\|_{L^2} \leq \|\widehat{K_j}\|_{L^\infty} \|f\|_{L^2} \lesssim 2^j \|f\|_{L^2}. \quad (6.3.3)$$

Step 3. *Interpolate.*

Apply the Riesz–Thorin theorem (Theorem 2.2.1) to T_j between the pair (L^1, L^∞) of (6.3.2) and the pair (L^2, L^2) of (6.3.3), with interpolation parameter $\theta \in (0, 1)$. The exponents are

$$\frac{1}{p_\theta} = \frac{1-\theta}{1} + \frac{\theta}{2} = 1 - \frac{\theta}{2}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{\infty} + \frac{\theta}{2} = \frac{\theta}{2},$$

so $q_\theta = 2/\theta$ and $p_\theta = q'_\theta$. Setting $\theta = 2/b$ gives $q_\theta = b$ and $p_\theta = b'$, and the interpolated norm is

$$\|T_j\|_{L^{b'} \rightarrow L^b} \lesssim \left(2^{-j \frac{n-1}{2}}\right)^{1-\theta} (2^j)^\theta = 2^{j(\theta - (1-\theta) \frac{n-1}{2})}. \quad (6.3.4)$$

Step 4. *The exponent is negative exactly in the stated range.*

Write $\beta(\theta) = \theta - (1 - \theta)\frac{n-1}{2}$. Then

$$\begin{aligned}\beta(\theta) < 0 \\ \iff 2\theta < (1 - \theta)(n - 1) \\ \iff \theta(n + 1) < n - 1 \\ \iff \theta < \frac{n - 1}{n + 1}.\end{aligned}$$

Since $\theta = 2/b$, this is $\frac{2}{b} < \frac{n-1}{n+1}$, that is

$$b > \frac{2(n + 1)}{n - 1},$$

which is the hypothesis. So $\beta := \beta(2/b) < 0$ and, by (6.3.4),

$$\sum_{j \geq 1} \|T_j\|_{L^{b'} \rightarrow L^b} \lesssim \sum_{j \geq 1} 2^{j\beta} < \infty.$$

Step 5. *The bounded piece.*

K_0 is bounded — $\widehat{\psi d\sigma}$ is a bounded continuous function — and compactly supported, hence lies in $L^r(\mathbb{R}^n)$ for every r . Taking r with $\frac{1}{b} = \frac{1}{b'} + \frac{1}{r} - 1$, which is possible since $b > b'$, Young's inequality gives $\|T_0 f\|_{L^b} \lesssim \|K_0\|_{L^r} \|f\|_{L^{b'}}$.

Step 6. *Sum.*

By Steps 4 and 5 and the triangle inequality in L^b ,

$$\|f * K\|_{L^b} \leq \sum_{j \geq 0} \|T_j f\|_{L^b} \lesssim \|f\|_{L^{b'}},$$

which is (6.3.1). The equivalent restriction statement follows from Theorem 6.1.3: $b = p'$ gives $p = b' < (\frac{2(n+1)}{n-1})' = \frac{2(n+1)}{n+3}$. $\square$

Corollary 6.3.3 (The L^2 trace estimate). *Let S be as in Theorem 6.3.1. Then for every $b > \frac{2(n+1)}{n-1}$ and every $f \in \mathcal{S}(\mathbb{R}^n)$,*

$$\|\widehat{f}\|_{L^2(S, d\sigma)} \lesssim_{n,b} \|f\|_{L^{b'}(\mathbb{R}^n)}, \quad b' = \frac{b}{b-1}.$$

In particular the restriction of $\widehat{f}$ to S is a well-defined element of $L^2(d\sigma)$ for every $f \in L^p(\mathbb{R}^n)$ with $p < \frac{2(n+1)}{n+3}$.

Proof. Theorem 6.3.1 gives $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^n)$. By Theorem 6.1.3 with $a = q' = 2$ — so that $q = 2$ as well — and $p' = b$, this is equivalent to $\mathcal{R} : L^{b'}(\mathbb{R}^n) \rightarrow L^2(d\sigma)$ with the same constant. The final sentence is the case $p = b'$, together with the density of $\mathcal{S}$ in L^p (Theorem 1.8.7), which lets the estimate define $\mathcal{R}$ on all of L^p by continuity; and $b > \frac{2(n+1)}{n-1}$ translates to $b' < \frac{2(n+1)}{n+3}$ by conjugation. $\square$

Where we are. This corollary is the form in which Stein–Tomas is usually applied, and it is worth stating separately because it answers the question Chapter 6.1 began with: for which p does the restriction of $\hat{f}$ to a sphere exist? The answer is $p < \frac{2(n+1)}{n+3}$, with values in $L^2(d\sigma)$.

At $n = 2$ this is $p < \frac{6}{5}$; at $n = 3$, $p < \frac{4}{3}$. Compare Worked Example 6.1.1: the trivial range was $p = 1$ only, so Stein–Tomas has genuinely enlarged it, and Theorem 6.2.6 asserts one can go to $p < \frac{2n}{n+1}$ — that is $\frac{4}{3}$ at $n = 2$ and $\frac{3}{2}$ at $n = 3$ — at the cost of weakening the target to $L^\infty(d\sigma)$.

WORKED EXAMPLE 6.3.1— TIER A **The exponent in low dimensions**

$n = 2$. $\frac{2(n+1)}{n-1} = 6$. So $\mathcal{E} : L^2(d\sigma) \rightarrow L^b(\mathbb{R}^2)$ for $b > 6$, and in restriction form $R^*(p \rightarrow 2)$ for $p < \frac{6}{5}$. The restriction conjecture in the plane asks for $b > 4$, so Stein–Tomas is genuinely weaker — and Chapter 6.8 closes the gap.

$n = 3$. $\frac{2(n+1)}{n-1} = 4$, and the conjecture asks for $b > 3$. The current record lies strictly between; see Worked Example 6.2.3.

Large n . $\frac{2(n+1)}{n-1} \rightarrow 2$ and $\frac{2n}{n-1} \rightarrow 2$, so both the theorem and the conjecture approach the trivial exponent $b = 2$. The *difference* $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$ also tends to 0: in high dimensions Stein–Tomas is close to sharp in absolute terms. This is misleading. The natural measure of progress is not b but the exponent in the associated Kakeya bound, and there the gap does not close.

WORKED EXAMPLE 6.3.2— TIER B **Where each hypothesis is used**

It is worth tracing which feature of S each step needed, since the theorem is often quoted for surfaces where one of them fails.

Non-vanishing Gaussian curvature entered once, in Theorem 6.3.2(i), to give the decay $|x|^{-(n-1)/2}$. If only k principal curvatures are non-zero, Theorem 5.5.3 gives $|x|^{-k/2}$ instead, and repeating Steps 3 and 4 with $\frac{n-1}{2}$ replaced by $\frac{k}{2}$ yields

$$\beta(\theta) < 0 \iff \theta < \frac{k}{k+2} \iff b > \frac{2(k+2)}{k},$$

the Stein–Tomas theorem for surfaces of curvature rank k . For the cylinder of Worked Example 5.5.2, $k = 1$ and $n = 3$: the exponent is 6 rather than 4. *Compactness* entered in Step 5 and in the finiteness of $\sigma(S)$ used in Theorem 6.3.2(ii). For a non-compact surface such as the full paraboloid one localises first and sums, at the cost of an ε .

Smoothness entered only through Theorem 5.5.2, which needs enough derivatives for stationary phase; C^∞ is convenient and far more than necessary.

What was never used is any relation between S and the sphere. The proof is insensitive to the global shape of S : only the local curvature at the points where each direction is normal matters, which is exactly what Theorem 5.5.1 said.

WORKED EXAMPLE 6.3.3— TIER C: RESEARCH TASTE **The endpoint, and why we stop short of it**

Theorem 6.3.1 proves the open range $b > \frac{2(n+1)}{n-1}$. The endpoint $b = \frac{2(n+1)}{n-1}$ is also true — that is Tomas’s theorem, completed by Stein — and this series does not prove it. The reason is worth being precise about, because it is a structural limitation rather than an oversight.

(a) *Exactly where the proof fails.* At the endpoint, $\theta = \frac{2}{b}$ equals $\frac{n-1}{n+1}$ and so $\beta(\theta) = 0$. Then (6.3.4) gives $\|T_j\|_{L^{b'} \rightarrow L^b} \lesssim 1$ for every j , and Step 6 sums a constant over j — divergent. The loss is only logarithmic, but a logarithm is a logarithm.

(b) *What repairs it.* One embeds the operators T_j in an analytic family T_z , holomorphic in a complex parameter, arranging that the $L^1 \rightarrow L^\infty$ bound holds on one edge of a strip and the $L^2 \rightarrow L^2$ bound on the other, and then interpolates the *family* rather than each member. Because the interpolation is applied once to the whole sum rather than term by term, the divergent series never appears. This is Stein’s interpolation theorem for analytic families — and it was removed from the plan of this series as revision R4 in Volume 5, on the grounds that nothing here needs it.

(c) *Whether the endpoint is needed later.* It is not. Volumes 12 and 13 use Stein–Tomas as an input to induction-on-scales arguments, which already carry $\delta^{-\varepsilon}$ losses; an ε in the exponent is invisible against $\delta^{-\varepsilon}$. A reader who wants the endpoint will find it in Stein’s *Harmonic Analysis*, Chapter IX.

(d) *The general lesson.* Twice now a proof has come down to whether a geometric series has ratio less than one — here, and in the Hörmander–Mikhlin theorem of Volume 3, Chapter 5. In both cases the open range falls out immediately and the endpoint requires a genuinely different technique. That is the normal state of affairs in this subject, and it is worth recognising

the pattern: *when an argument sums a geometric series over scales, the endpoint is a separate theorem.*

WORKED EXAMPLE 6.3.4— TIER B Why the two bounds on K_j must be combined, and not used singly

Theorem 6.3.2 gives two estimates on each dyadic piece. Using either alone gives nothing, and seeing exactly how each fails is the quickest route to understanding what interpolation buys.

Using only $L^1 \rightarrow L^\infty$. Summing (6.3.2) over j gives

$$\|f * K\|_{L^\infty} \lesssim \left(\sum_{j \geq 1} 2^{-j \frac{n-1}{2}} \right) \|f\|_{L^1} \lesssim \|f\|_{L^1},$$

a valid but useless conclusion: it is the statement $\mathcal{E} : L^2(d\sigma) \rightarrow L^\infty$, which holds trivially by Cauchy–Schwarz and gives $b = \infty$.

Using only $L^2 \rightarrow L^2$. Summing (6.3.3) gives $\sum_j 2^j = \infty$. The series diverges, and this is not an artefact: $\widehat{d\sigma}$ is not a bounded multiplier of L^2 — indeed $\widehat{\widehat{d\sigma}} = d\sigma$ is a measure, not a function — so no L^2 bound for the full convolution exists.

What interpolation does. It produces, for each j , a bound $2^{j\beta(\theta)}$ that *interpolates between a convergent and a divergent series*, and asks for which θ the result still converges. The answer, $\theta < \frac{n-1}{n+1}$, is exactly the trade-off between the decay $2^{-j(n-1)/2}$ that curvature supplies and the growth 2^j that localisation costs.

The general shape. This is the standard architecture of an estimate proved by decomposition: one bound improves with j , another worsens, and the theorem is the range of exponents where the geometric mean decays. Chapter 3.5’s proof of the Hörmander–Mikhlin theorem has exactly this form, with λ^{-N} against $2^{j|\alpha|}$; so does the proof of Theorem 5.8.3. Once recognised, such proofs can be read by locating the two competing bounds and computing where their product is summable, without following the details.

Drills

Drill 6.3.1. Verify the exponent identity $p_\theta = q'_\theta$ in Step 3, and confirm that $\theta = 2/b$ gives $q_\theta = b$.

Drill 6.3.2. Carry out the summation in Theorem 6.3.2(ii) with general N and determine the smallest N for which the argument closes.

Drill 6.3.3. Deduce from Theorem 6.3.1 and Theorem 6.1.3 that $\|\widehat{f}|_S\|_{L^2(d\sigma)} \lesssim \|f\|_{L^p}$ for $p < \frac{2(n+1)}{n+3}$, and check this exponent at $n = 2, 3$.

Drill 6.3.4. Show that Theorem 6.3.1 is false for $b < \frac{2(n+1)}{n-1}$ by combining it with Theorem 6.2.3, and identify the extremal g .

Notes

Tomas proved the non-endpoint case in 1975 by the argument given here; Stein obtained the endpoint shortly after, by analytic interpolation. The theorem is the first non-trivial restriction estimate and remains, for $a = 2$, the sharp one.

The dyadic decomposition of $\widehat{d\sigma}$ used in Theorem 6.3.2 is the prototype for the wave packet decompositions of Volume 12: in both cases a single object is cut into pieces at different scales, each piece is estimated by two incompatible methods, and the results are interpolated.

Chapter 6.4

Strichartz Estimates

The Stein–Tomas theorem is a statement about a sphere. Applied instead to a paraboloid or a cone, the very same theorem becomes a statement about the Schrödinger or wave equation, and it is in that guise that it is most used. This chapter performs the translation.

6.4.1 The paraboloid is the Schrödinger evolution

Definition 6.4.1. For $u_0 \in \mathcal{S}(\mathbb{R}^d)$ the free Schrödinger evolution is

$$(e^{it\Delta}u_0)(y) = \int_{\mathbb{R}^d} \widehat{u_0}(\zeta) e^{2\pi i(y \cdot \zeta - 2\pi t|\zeta|^2)} d\zeta,$$

the unique Schwartz-class solution of $i\partial_t u = -\Delta u$, $u(0) = u_0$.

Lemma 6.4.2. Let $n = d + 1$ and let

$$P = \{(\zeta, |\zeta|^2) : |\zeta| \leq 1\} \subseteq \mathbb{R}^n$$

be the truncated paraboloid, with measure $d\mu = d\zeta$ in the graph parametrisation. Then for g supported in $|\zeta| \leq 1$,

$$\begin{aligned} \mathcal{E}_P g(y, s) &= \int_{|\zeta| \leq 1} g(\zeta) e^{2\pi i(y \cdot \zeta + s|\zeta|^2)} d\zeta \\ &= (e^{it\Delta}u_0)(y) \Big|_{t=-s/2\pi}, \end{aligned}$$

where u_0 is defined by $\widehat{u_0} = g$.

Proof. Substituting $t = -s/2\pi$ in the definition gives phase $2\pi(y \cdot \zeta - 2\pi t|\zeta|^2) = 2\pi(y \cdot \zeta + s|\zeta|^2)$, which is the phase in $\mathcal{E}_P$. The two integrals then agree term by term. $\square$

So an estimate for $\mathcal{E}_P$ on $\mathbb{R}^{d+1} = \mathbb{R}_y^d \times \mathbb{R}_s$ is a space-time estimate for the Schrödinger equation, with $\|g\|_{L^2(d\mu)} = \|\widehat{u_0}\|_{L^2} = \|u_0\|_{L^2}$ by Plancherel.

Theorem 6.4.3 (Strichartz, non-endpoint, frequency-localised). *Let $d \geq 1$ and $n = d + 1$. For every*

$$q > \frac{2(d+2)}{d}$$

there is $C = C(d, q)$ such that, for all $u_0 \in L^2(\mathbb{R}^d)$ with $\widehat{u_0}$ supported in $\{|\zeta| \leq 1\}$,

$$\|e^{it\Delta}u_0\|_{L^q(\mathbb{R}_y^d \times \mathbb{R}_t)} \leq C \|u_0\|_{L^2(\mathbb{R}^d)}.$$

Proof. The truncated paraboloid $P \subseteq \mathbb{R}^n$ has non-vanishing Gaussian curvature: in the graph parametrisation $\zeta \mapsto (\zeta, |\zeta|^2)$ the second fundamental form at every point is a positive multiple of the identity, since $\text{Hess}(|\zeta|^2) = 2I$ (Theorem 5.5.1). It is compact. So Theorem 6.3.1 applies with $n = d + 1$ and gives

$$\|\mathcal{E}_P g\|_{L^q(\mathbb{R}^{d+1})} \lesssim \|g\|_{L^2(d\mu)} \quad \text{for } q > \frac{2(n+1)}{n-1} = \frac{2(d+2)}{d}.$$

By Theorem 6.4.2 the left-hand side is $\|e^{it\Delta}u_0\|_{L^q(dy ds)}$, and the change of variable $s = -2\pi t$ multiplies it by the constant $(2\pi)^{1/q}$. By Plancherel the right-hand side is $\|u_0\|_{L^2}$. $\square$

WORKED EXAMPLE 6.4.1— TIER A Reading the exponent, and checking it by scaling

The theorem as stated has the same exponent q in space and in time, and is restricted to frequencies $|\zeta| \leq 1$. Both features deserve comment.

(a) *Scaling determines the admissible pairs.* Drop the frequency restriction and ask for which (q, r) one may hope

$$\|e^{it\Delta}u_0\|_{L_t^q L_y^r} \lesssim \|u_0\|_{L^2}.$$

If $u_0^\lambda(y) = u_0(\lambda y)$ then $e^{it\Delta}u_0^\lambda(y) = (e^{i\lambda^2 t \Delta}u_0)(\lambda y)$, so the left side scales by $\lambda^{-2/q-d/r}$ and the right by $\lambda^{-d/2}$. Equality forces

$$\frac{2}{q} + \frac{d}{r} = \frac{d}{2}, \tag{6.4.1}$$

the *admissibility* relation. Theorem 6.4.3 is the diagonal case $q = r$, for which (6.4.1) gives $\frac{2}{q} + \frac{d}{q} = \frac{d}{2}$, i.e. $q = \frac{2(d+2)}{d}$ — exactly the endpoint of the theorem's range. So the theorem is sharp up to the endpoint on the diagonal, and the frequency localisation is what allows the strict inequality $q > \frac{2(d+2)}{d}$ without violating scaling.

(b) *Why localisation is needed for $q > \text{the diagonal value}$.* Without it, (6.4.1) must hold exactly; a strict inequality would be inconsistent with the scaling above. Localising to $|\zeta| \leq 1$ breaks the scale invariance and buys the open range.

(c) $d = 1$. The exponent is $q > 6$, and (6.4.1) on the diagonal gives $q = 6$: the classical $L^6_{t,y}$ Strichartz estimate for the one-dimensional Schrödinger equation.

Proposition 6.4.4 (The dispersive estimate). *For $u_0 \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ and $t \neq 0$,*

$$\|e^{it\Delta}u_0\|_{L^\infty(\mathbb{R}^d)} \leq (4\pi|t|)^{-d/2}\|u_0\|_{L^1(\mathbb{R}^d)}.$$

Proof. By the Gaussian computation of Volume 2 — Theorem 2.4.3 together with the analytic continuation of Theorem 5.4.2, applied to the purely imaginary parameter — the Schrödinger propagator is convolution with

$$K_t(y) = (4\pi it)^{-d/2} e^{i|y|^2/4t},$$

where $(4\pi it)^{-d/2}$ is computed with the branch continued from the positive real axis, exactly as in Theorem 5.2.1. Hence $|K_t(y)| = (4\pi|t|)^{-d/2}$ for every y , and Young's inequality gives

$$\begin{aligned} \|e^{it\Delta}u_0\|_{L^\infty} &= \|K_t * u_0\|_{L^\infty} \\ &\leq \|K_t\|_{L^\infty}\|u_0\|_{L^1} \\ &= (4\pi|t|)^{-d/2}\|u_0\|_{L^1}. \end{aligned} \quad \square$$

Where we are. Theorem 6.4.4 is the other classical route to Strichartz estimates: one interpolates the $L^1 \rightarrow L^\infty$ decay above against the $L^2 \rightarrow L^2$ conservation $\|e^{it\Delta}u_0\|_{L^2} = \|u_0\|_{L^2}$, and then applies a fractional integration in t . That argument gives the same admissible pairs and is the one usually presented in PDE courses.

The two routes are genuinely different in what they reveal. The dispersive route shows the estimates come from the *decay of the propagator*; the Stein–Tomas route of Theorem 6.4.3 shows they come from the *curvature of the characteristic surface*. Both are correct, and the second is the one that generalises to the problems of this series, since it is curvature, not decay, that the Kakeya obstruction limits.

6.4.2 The cone is the wave evolution

Proposition 6.4.5. *Let $C = \{(\zeta, |\zeta|) : \frac{1}{2} \leq |\zeta| \leq 1\} \subseteq \mathbb{R}^{d+1}$ be a truncated cone, $d \geq 2$. Then C has $d - 1$ non-vanishing principal curvatures and one*

vanishing, and consequently

$$\|\mathcal{E}_C g\|_{L^q(\mathbb{R}^{d+1})} \lesssim \|g\|_{L^2(d\sigma)} \quad \text{for } q > \frac{2(d+1)}{d-1}.$$

Proof. The cone is ruled by the lines $\{(r\theta, r) : r > 0\}$ for $\theta \in \mathbb{S}^{d-1}$, and is affine along each, so one principal curvature vanishes. In the transverse directions it is a dilate of the sphere $\mathbb{S}^{d-1}$ and the remaining $d-1$ principal curvatures are non-zero on the compact piece $\frac{1}{2} \leq |\zeta| \leq 1$. So the curvature rank is $k = d-1$.

By Worked Example 6.3.2, the Stein–Tomas argument with curvature rank k gives the range $q > \frac{2(k+2)}{k}$; with $k = d-1$ this is $q > \frac{2(d+1)}{d-1}$. $\square$

WORKED EXAMPLE 6.4.2— TIER B Why the wave equation is harder than the Schrödinger equation

Compare the two exponents just obtained, both in $\mathbb{R}^{d+1}$:

$$\text{Schrödinger (paraboloid): } q > \frac{2(d+2)}{d}, \quad \text{wave (cone): } q > \frac{2(d+1)}{d-1}.$$

The second is larger — at $d = 3$, $q > 4$ against $q > \frac{10}{3}$ — so the wave estimate is weaker.

The reason is one missing curvature. The paraboloid curves in all d tangential directions; the cone is flat along its rulings and curves in only $d-1$. By Theorem 5.5.1 the rank of the second fundamental form is exactly what stationary phase converts into decay, so the cone's $\widehat{d\sigma}$ decays like $|x|^{-(d-1)/2}$ where the paraboloid's decays like $|x|^{-d/2}$, and Theorem 6.3.2(i) carries the difference straight into the exponent.

Where the flat direction comes from, physically. A solution of the wave equation propagates at unit speed in a fixed direction; the ruling of the cone is the record of that. The Schrödinger evolution has no finite propagation speed and its characteristic surface, the paraboloid, is curved everywhere. So the analytic difference traces back to a difference in the physics.

A caution. The cone estimate above is not the sharpest known — the wave equation admits estimates in mixed norms $L_t^q L_y^r$ that the diagonal Stein–Tomas argument does not see, and the local smoothing conjecture asserts a further gain after averaging in time. Worked Example 4.8.2 in Volume 4 described that gain as an embedding between function spaces, and Volume 13 proves the planar case.

WORKED EXAMPLE 6.4.3— TIER B The admissible pairs, computed

Theorem 6.4.3 is the diagonal case $q = r$. The estimates actually used in

practice are off-diagonal, and it is worth deriving the full family, since the derivation is short and the table is otherwise memorised without reason.

The two constraints. A pair (q, r) is *admissible* for the Schrödinger equation in $\mathbb{R}^d$ if

$$\|e^{it\Delta}u_0\|_{L_t^q L_y^r} \lesssim \|u_0\|_{L^2}.$$

Scaling forces $\frac{2}{q} + \frac{d}{r} = \frac{d}{2}$, by Worked Example 6.4.1. A second constraint comes from requiring $q \geq 2$: the Knapp example, transported through Theorem 6.4.2, produces a solution concentrated on a slab and shows the estimate fails for $q < 2$.

The endpoints of the family. Setting $q = \infty$ gives $r = 2$, which is just conservation of the L^2 norm — true and trivial. Setting $q = 2$ gives $\frac{d}{r} = \frac{d}{2} - 1$, that is $r = \frac{2d}{d-2}$, which requires $d \geq 3$; this is the Keel–Tao endpoint, true but not proved here.

Worked cases.

d	diagonal $q = r$	endpoint (q, r)	a typical off-diagonal pair
1	6	none ($q = 2$ needs $d \geq 3$)	$(\infty, 2), (8, 4)$
2	4	none	$(\infty, 2), (4, 4), (6, 3)$
3	$\frac{10}{3}$	$(2, 6)$	$(\infty, 2), (4, 3)$

Each entry is checked by substituting into $\frac{2}{q} + \frac{d}{r} = \frac{d}{2}$: for $d = 3$, $(q, r) = (4, 3)$ gives $\frac{2}{4} + \frac{3}{3} = \frac{1}{2} + 1 = \frac{3}{2} = \frac{d}{2}$. ✓

What Theorem 6.3.1 delivers, and what it does not. The Stein–Tomas route gives exactly the diagonal pairs, since it produces an L^b bound in all $n = d + 1$ space-time variables at once. Off-diagonal pairs require interpolating the diagonal estimate against the trivial pair $(\infty, 2)$ — which recovers the segment between them, and that is the whole admissible family in the range $q >$ diagonal. The endpoint is what no interpolation reaches.

WORKED EXAMPLE 6.4.4— TIER C: RESEARCH TASTE **What Strichartz estimates are for**

The translation of this chapter is not a formal exercise: it is the reason the restriction problem is studied outside harmonic analysis, and it is worth knowing what the estimates do.

(a) *Well-posedness of nonlinear equations.* Consider the cubic nonlinear Schrödinger equation $i\partial_t u + \Delta u = |u|^2 u$. Solving it by Picard iteration requires a space in which the Duhamel term $\int_0^t e^{i(t-s)\Delta}(|u|^2 u)(s) \, ds$ is controlled by

the data, and Theorem 6.4.3 supplies exactly that: the space-time norm on the left is what the nonlinearity is estimated in, and the L^2 norm on the right is the conserved quantity. The admissible pairs (6.4.1) are precisely the norms available for the iteration, which is why they are tabulated in every paper on the subject.

(b) *The endpoint matters here, and we do not have it.* For some applications one needs the endpoint of the admissible range, $q = 2$, available for $d \geq 3$. That is the Keel–Tao theorem, whose proof is bilinear and which this series does not prove — announced in the preface to this volume. The non-endpoint estimates suffice for local well-posedness; the endpoint is used for scattering and for critical problems.

(c) *The direction of the implication.* Restriction estimates imply Strichartz estimates, as this chapter shows, and the implication is essentially an identity — Theorem 6.4.2 is a change of variables. It does not run backwards in any useful sense: a Strichartz estimate is a statement about L^2 data, which is the case $a = 2$, and the restriction conjecture concerns $a = \infty$. So progress on the conjecture does not follow from progress on dispersive equations.

(d) *What this says about the shape of the subject.* Everything in this chapter was a corollary: one theorem about surfaces, applied to two particular surfaces, gave the classical estimates for the two classical dispersive equations. The mathematical content sits entirely in Theorem 6.3.1 and hence, through Theorem 6.3.2, in the curvature of the surface. That is the sense in which the restriction problem is the fundamental one and the PDE estimates are consequences.

Drills

Drill 6.4.1. Verify Theorem 6.4.2 by direct substitution, tracking the constant introduced by $t = -s/2\pi$.

Drill 6.4.2. Confirm that the second fundamental form of the paraboloid $\zeta \mapsto (\zeta, |\zeta|^2)$ is $2I$ at the origin, and compute it at a general point.

Drill 6.4.3. Derive the admissibility relation (6.4.1) for the wave equation, where the scaling is $u_0^\lambda(y) = u_0(\lambda y)$ with evolution $e^{it\sqrt{-\Delta}}$, and compare with Theorem 6.4.5.

Drill 6.4.4. Show that Theorem 6.4.3 fails without the frequency localisation, by scaling a fixed non-zero u_0 .

Where we are. It is worth recording what this chapter did *not* require. No property of the Schrödinger equation was used beyond the shape of its characteristic surface; the evolution never appeared except through Theorem 6.4.2, which is a change of variables. In particular nothing about conservation laws, dispersion in the physical sense, or the structure of the equation entered.

That is a general feature. A linear constant-coefficient dispersive equation $i\partial_t u = P(D)u$ has characteristic surface $\{(\zeta, P(\zeta))\}$, and its space-time estimates are the restriction estimates for that surface. The Schrödinger equation gives a paraboloid, the wave equation a cone, the Airy equation $\partial_t u + \partial_x^3 u = 0$ the curve (ζ, ζ^3) — which has an inflection at the origin, where the curvature vanishes, and correspondingly worse estimates near that frequency.

So the classification of dispersive estimates *is* the classification of surfaces by their curvature, and the results of Volume 5 are what performs it. A reader who wants the estimates for a new equation should compute the second fundamental form of its characteristic surface, find the rank, and read off the exponent from Worked Example 6.3.2.

Notes

Strichartz proved the estimate that bears his name in 1977, deducing it from Stein–Tomas exactly as here. The systematic theory of admissible pairs is due to Ginibre–Velo and Keel–Tao; the latter also settled the endpoint.

The observation that a dispersive equation’s characteristic surface carries the relevant curvature, and that the flat directions of that surface are the directions of propagation, is the organising idea of the field.

Chapter 6.5

Epsilon-Removal

Every argument in Volumes 12 and 13 proves a *local* estimate: a bound for $\mathcal{E}g$ on a ball of radius R , with a constant growing like R^ε . The conjecture, by contrast, is global and has no loss. This short chapter proves that the two are equivalent up to shrinking the exponent, so that the R^ε losses carried by induction-on-scales arguments are harmless.

6.5.1 Local estimates

Definition 6.5.1. Let $S \subseteq \mathbb{R}^n$ be a compact hypersurface and $1 \leq b \leq \infty$. Say that the *local estimate* $\mathcal{E}(2 \rightarrow b; \varepsilon)$ holds if for every $R \geq 1$ and every $g \in L^2(d\sigma)$,

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq C_\varepsilon R^\varepsilon \|g\|_{L^2(d\sigma)},$$

where B_R is any ball of radius R and C_ε does not depend on R or g .

The global estimate is the case $\varepsilon = 0$ with B_R replaced by $\mathbb{R}^n$; it plainly implies the local one. The content is the converse.

Theorem 6.5.2 (Epsilon-removal). *Suppose the local estimate $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$ holds for every $\varepsilon > 0$. Then the global estimate*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \lesssim_b \|g\|_{L^2(d\sigma)}$$

holds for every $b > b_0$.

Idea of the proof. Two ingredients. First, an L^b norm over $\mathbb{R}^n$ can be assembled from norms over balls of radius R : cover $\mathbb{R}^n$ by such balls with bounded overlap and sum. That alone loses a factor equal to the number of balls, which is infinite.

The repair is the second ingredient: interpolation with a bound that is *better* than needed at a larger exponent. The global estimate at $b = \infty$ is free —

$\|\mathcal{E}g\|_\infty \leq \|g\|_{L^1(d\sigma)} \lesssim \|g\|_{L^2(d\sigma)}$ since $d\sigma$ is finite — and interpolating the local estimate at b_0 against it at ∞ converts the R^ε loss into a *gain* $R^{-\eta}$ at any exponent b strictly between, which is summable over a dyadic family of scales.

Proof. Fix $b > b_0$ and choose $\theta \in (0, 1)$ with

$$\begin{aligned} \frac{1}{b} &= \frac{1-\theta}{b_0} + \frac{\theta}{\infty} \\ &= \frac{1-\theta}{b_0} \\ , \quad \text{that is} \quad 1-\theta &= \frac{b_0}{b} \in (0, 1). \end{aligned}$$

Step 1. *The two bounds on a ball.*

Let B_R be a ball of radius $R \geq 1$. The hypothesis with parameter ε gives

$$\|\mathcal{E}g\|_{L^{b_0}(B_R)} \leq C_\varepsilon R^\varepsilon \|g\|_{L^2(d\sigma)},$$

and the trivial bound gives

$$\|\mathcal{E}g\|_{L^\infty(B_R)} \leq \|g\|_{L^1(d\sigma)} \leq \sigma(S)^{1/2} \|g\|_{L^2(d\sigma)},$$

by Cauchy–Schwarz on the finite measure space $(S, d\sigma)$.

Step 2. *Interpolate on the ball.*

For any function F and any exponents, Hölder’s inequality gives $\|F\|_{L^b} \leq \|F\|_{L^{b_0}}^{1-\theta} \|F\|_{L^\infty}^\theta$ when $\frac{1}{b} = \frac{1-\theta}{b_0}$. Applying this on B_R with $F = \mathcal{E}g$ and using Step 1,

$$\|\mathcal{E}g\|_{L^b(B_R)} \lesssim (R^\varepsilon)^{1-\theta} \|g\|_{L^2(d\sigma)} = R^{\varepsilon b_0/b} \|g\|_{L^2(d\sigma)}. \quad (6.5.1)$$

This is still a loss, and no better than the hypothesis; the gain comes in Step 3, from how the balls are assembled.

Step 3. *Sum over dyadic annuli.*

Decompose $\mathbb{R}^n = B_1 \cup \bigcup_{k \geq 1} A_k$ with $A_k = \{2^{k-1} \leq |x| < 2^k\}$. On A_k apply the trivial bound and (6.5.1) at scale $R = 2^k$ once more, but now exploiting that A_k has finite measure: by Hölder on A_k ,

$$\|\mathcal{E}g\|_{L^b(A_k)} \leq \|\mathcal{E}g\|_{L^{b_0}(A_k)}^{1-\theta} \|\mathcal{E}g\|_{L^\infty(A_k)}^\theta.$$

For the second factor we use a *decay* estimate rather than the trivial one. Since $\mathcal{E}g = (gd\sigma)^\vee$ and $d\sigma$ has non-vanishing curvature, the same computation as Theorem 6.3.2(i) applied to $\mathcal{E}\mathcal{E}^*$ shows

$$\|\mathcal{E}g\|_{L^\infty(A_k)} \lesssim \|g\|_{L^2(d\sigma)}$$

uniformly, which is all that is required; and for the first factor (6.5.1) with $R = 2^k$ gives $2^{k\varepsilon} \|g\|_{L^2(\mathrm{d}\sigma)}$.

To obtain a summable bound, we instead run the interpolation with the exponent $b_1 \in (b_0, b)$ and use that the hypothesis holds for *every* ε : choosing ε so small that

$$\eta := \frac{n}{b_1} - \frac{n}{b} - \varepsilon > 0$$

— possible since $b_1 < b$ makes $\frac{n}{b_1} - \frac{n}{b} > 0$ — and applying Hölder on A_k between L^{b_1} and L^b ,

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(A_k)} &\leq m(A_k)^{\frac{1}{b} - \frac{1}{b_1}} \|\mathcal{E}g\|_{L^{b_1}(A_k)} \\ &\lesssim 2^{kn\left(\frac{1}{b} - \frac{1}{b_1}\right)} 2^{k\varepsilon} \|g\|_{L^2(\mathrm{d}\sigma)} \\ &= 2^{-k\eta} \|g\|_{L^2(\mathrm{d}\sigma)}. \end{aligned}$$

Step 4. *Conclude.*

Summing the geometric series,

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(\mathbb{R}^n)}^b &= \|\mathcal{E}g\|_{L^b(B_1)}^b + \sum_{k \geq 1} \|\mathcal{E}g\|_{L^b(A_k)}^b \\ &\lesssim \|g\|_{L^2(\mathrm{d}\sigma)}^b \left(1 + \sum_{k \geq 1} 2^{-kb\eta}\right) \\ &\lesssim \|g\|_{L^2(\mathrm{d}\sigma)}^b. \end{aligned} \quad \square$$

Where we are. The mechanism is worth isolating, because it recurs. An R^ε loss is harmless whenever it can be played off against a *power* gain coming from somewhere else — here, from Hölder on an annulus of measure 2^{kn} at two nearby exponents, which supplies $2^{-kn(1/b_1 - 1/b)}$. Since ε may be taken as small as one likes and the power gain is fixed once $b > b_1$, the gain always wins.

This is why the literature is content to prove local estimates with $\delta^{-\varepsilon}$ losses, and why the phrase “by epsilon-removal” appears without comment in papers on decoupling. It is also why the *endpoint* is never obtained this way: at $b = b_1$ the power gain vanishes and nothing is left to beat the loss.

Proposition 6.5.3 (Local and global are equivalent, up to the exponent). *For a compact hypersurface S with non-vanishing curvature and $1 \leq b_0 < \infty$, the following are equivalent:*

- (i) the local estimate $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$ holds for every $\varepsilon > 0$;
- (ii) the global estimate $\|\mathcal{E}g\|_{L^b} \lesssim_b \|g\|_{L^2(\mathrm{d}\sigma)}$ holds for every $b > b_0$.

Proof. That (i) implies (ii) is Theorem 6.5.2. Conversely, assume (ii) and fix $\varepsilon > 0$. Choose $b > b_0$ with

$$\frac{n}{b_0} - \frac{n}{b} < \varepsilon,$$

possible since the left side tends to 0 as $b \downarrow b_0$. By Hölder on a ball of radius R , whose measure is $\approx R^n$,

$$\begin{aligned} \|\mathcal{E}g\|_{L^{b_0}(B_R)} &\leq m(B_R)^{\frac{1}{b_0} - \frac{1}{b}} \|\mathcal{E}g\|_{L^b(B_R)} \\ &\lesssim R^{\frac{n}{b_0} - \frac{n}{b}} \|g\|_{L^2(d\sigma)} \\ &\leq R^\varepsilon \|g\|_{L^2(d\sigma)}, \end{aligned}$$

using (ii) for the middle step. This is $\mathcal{E}(2 \rightarrow b_0; \varepsilon)$. □

Where we are. So nothing is lost by working locally, provided one is willing to state conclusions as open ranges. That is the convention throughout Volumes 12 and 13, and Theorem 6.5.3 is the licence for it. The proposition also explains a phrasing the reader will meet constantly. A paper proves “the estimate holds for b_0 ” meaning the local estimate with R^ε ; a survey then reports “the estimate holds for $b > b_0$ ” meaning the global one. Both are correct and they say the same thing, but only because of this equivalence.

WORKED EXAMPLE 6.5.1— TIER A Why one cannot simply cover space by balls

The naive approach is to cover $\mathbb{R}^n$ by $\approx (M/R)^n$ balls of radius R filling B_M , apply the local estimate on each, and sum. Doing so gives

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(B_M)}^b &\leq \sum_{\text{balls}} \|\mathcal{E}g\|_{L^b(B_R)}^b \\ &\lesssim \left(\frac{M}{R}\right)^n (R^\varepsilon)^b \|g\|_{L^2(d\sigma)}^b, \end{aligned}$$

which diverges as $M \rightarrow \infty$ for any fixed R . Nothing about the local estimate prevents $\mathcal{E}g$ from being equally large on every one of the balls.

What rescues the argument is that $\mathcal{E}g$ *cannot* in fact be equally large everywhere: it is the transform of a measure on a curved surface and so decays. The proof above encodes this by working with annuli, whose measure grows, and by paying for that growth with the difference of two exponents. Curvature enters only through the requirement $b > b_0$, i.e. that some non-trivial estimate holds at all.

WORKED EXAMPLE 6.5.2— TIER B **The same device, in the form used in Volume 13**

Decoupling estimates are stated with an unavoidable $\delta^{-\varepsilon}$:

$$\left\| \sum_{\theta} f_{\theta} \right\|_{L^p(B_{\delta^{-1}})} \lesssim_{\varepsilon} \delta^{-\varepsilon} \left(\sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}.$$

The ε is not an artefact: at the critical exponent it is known that no δ -free constant is possible. So decoupling is inherently a local, lossy statement, and any global conclusion drawn from it must remove the loss.

The removal is always of the type proved here — trade the $\delta^{-\varepsilon}$ against a power gain obtained by moving slightly off the critical exponent. The practical consequence, and the reason to have seen the argument once: *a theorem proved by decoupling is never sharp at its endpoint*. Every application in Volume 13 states its exponent as an open condition, and Theorem 6.5.2 is why that is the natural form.

WORKED EXAMPLE 6.5.3— TIER B **Tracking an ε through an induction on scales**

Induction-on-scales arguments produce constants of the form $C_{\varepsilon} R^{\varepsilon}$, and a reader meeting them for the first time reasonably worries that the ε 's accumulate. They do not, and seeing why once removes the worry permanently.

The typical structure. One proves a statement $\mathcal{S}(R)$ at scale R assuming $\mathcal{S}(R^{1/2})$ at the square root of that scale, with a bound of the shape

$$C(R) \leq K \cdot C(R^{1/2})^{\alpha}$$

for constants K and α depending only on the dimension. The question is what $C(R)$ grows like.

The case $\alpha = 1$. Iterating m times reaches scale $R^{2^{-m}}$, and

$$C(R) \leq K^m C(R^{2^{-m}}).$$

Stopping when $R^{2^{-m}} \approx 2$, that is $m \approx \log \log R$, and using that C is bounded at bounded scales,

$$C(R) \lesssim K^{\log \log R} = (\log R)^{\log K}.$$

So the induction costs a *power of a logarithm* — which is $\lesssim_{\varepsilon} R^{\varepsilon}$ for every $\varepsilon > 0$, and therefore removable by Theorem 6.5.2. This is the mechanism behind essentially every $\delta^{-\varepsilon}$ in the literature.

Why one does not simply keep the logarithm. Because the exponent $\log K$ is not usually computed, and because a second induction would square it. Carrying “ $\lesssim_\varepsilon R^\varepsilon$ ” is both weaker and more robust: it is stable under composition, since $R^{\varepsilon_1} R^{\varepsilon_2} = R^{\varepsilon_1 + \varepsilon_2}$ and the ε ’s may be chosen in advance to sum to the target.

The one thing to watch. The constant C_ε does blow up as $\varepsilon \rightarrow 0$ — typically like $\exp(\varepsilon^{-1})$ or worse — so an argument that needs ε to depend on R is invalid. In every correct application, ε is fixed first and R tends to infinity afterwards. Checking the order of quantifiers is the standard way to find an error in this kind of proof, and it is worth doing whenever a bound looks too strong.

WORKED EXAMPLE 6.5.4— TIER A The bookkeeping, on a concrete pair of exponents

Take $n = 3$ and $b_0 = 4$, the Stein–Tomas exponent, and suppose we have the local estimate at $b_0 = 4$ with loss R^ε . What does Theorem 6.5.2 give at $b = 5$?

Choosing the intermediate exponent. Take $b_1 = \frac{b_0 + b}{2} = \frac{9}{2}$. Then

$$\frac{n}{b_1} - \frac{n}{b} = \frac{3}{9/2} - \frac{3}{5} = \frac{2}{3} - \frac{3}{5} = \frac{1}{15}.$$

So any $\varepsilon < \frac{1}{15}$ works, and the resulting decay rate is $\eta = \frac{1}{15} - \varepsilon$.

The sum. With $\varepsilon = \frac{1}{30}$, $\eta = \frac{1}{30}$, and Step 4 of the proof sums $\sum_k 2^{-5k/30} = \sum_k 2^{-k/6}$, a geometric series with ratio $2^{-1/6} \approx 0.891$. Convergent, though not fast: about 40 terms are needed for two-digit accuracy, which is to say scales from 1 up to 2^{40} .

What this shows. The convergence is genuinely slow near the endpoint, and the constant in Theorem 6.5.2 degenerates as $b \downarrow b_0$: from Step 4 the constant is $\approx (1 - 2^{-b\eta})^{-1/b}$, which blows up as $\eta \rightarrow 0$. So epsilon-removal is not free — it costs a constant that is large when the exponent is close to critical, and no amount of care improves this, since the endpoint statement is a different theorem.

A useful habit. When reading a claimed estimate at an exponent, ask whether it is open or closed. If open, expect a removable loss and a constant that blows up at the endpoint; if closed, expect a genuinely harder argument. The two cases are rarely distinguished in abstracts and almost always in the theorem statements.

Drills

Drill 6.5.1. Verify the Hölder inequality $\|F\|_{L^b} \leq \|F\|_{L^{b_0}}^{1-\theta} \|F\|_{L^\infty}^\theta$ used in Step 2, and identify the relation between b, b_0, θ .

Drill 6.5.2. Carry out Step 3 with explicit $b_1 = \frac{b_0+b}{2}$ and determine how small ε must be taken.

Drill 6.5.3. Show that Theorem 6.5.2 gives nothing at $b = b_0$, by tracking η as $b \downarrow b_0$.

Drill 6.5.4. State and prove the analogue of Theorem 6.5.2 for $L^a(d\sigma)$ data with $a \neq 2$, and check that the proof uses nothing about a .

Where we are. A last word on why this chapter is short and why it nevertheless has to exist.

The mathematics is two applications of Hölder. What it buys is a change of *register*: after Theorem 6.5.3, one may prove estimates on a ball of radius R , with any constant that is $O(R^\varepsilon)$, and still be proving a theorem about $\mathbb{R}^n$. Every technique in the volumes ahead — induction on scales, broad–narrow decomposition, polynomial partitioning, decoupling — produces exactly that kind of statement and nothing better, because each of them subdivides a ball and recurses.

Without this chapter those techniques would prove only local statements, and the conjectures they are aimed at are global. With it, the two are the same problem. That is why the result is quoted so often and proved so rarely: it is the bridge between the way the subject is practised and the way its questions are posed.

Notes

Epsilon-removal in this context is due to Tao (1999), in a sharper form than the one proved here: his argument removes the loss without moving the exponent, at the cost of a more delicate decomposition. The version above is the one that suffices for this series and is the standard textbook treatment.

Chapter 6.6

Restriction Implies Keakeya

Worked Example 6.2.3 argued informally that the restriction problem, decomposed over δ -caps, becomes a question about how much $\delta^{-(n-1)}$ tubes — one in each direction — can overlap. This chapter makes that precise. The implication is due to Bourgain and is the reason the second half of this series is about Keakeya sets.

6.6.1 The statement

Definition 6.6.1. Let $0 < \delta < \frac{1}{2}$. A family $\mathbb{T} = \{T_1, \dots, T_N\}$ of δ -tubes is a collection of cylinders in $\mathbb{R}^n$, each of length 1 and radius δ , whose directions $v_1, \dots, v_N \in \mathbb{S}^{n-1}$ are δ -separated. Their positions are arbitrary. Maximality forces $N \lesssim_n \delta^{-(n-1)}$, and we assume $N \approx \delta^{-(n-1)}$.

Each tube has $m(T_j) \approx \delta^{n-1}$, so $\sum_j m(T_j) \approx 1$: the tubes have unit total volume however they are arranged. The Keakeya question is whether they can be packed so that their union is much smaller than 1, and the quantitative form of “no” is a bound on the multiplicity function $\sum_j \mathbf{1}_{T_j}$.

Theorem 6.6.2 (Restriction implies Keakeya). *Suppose that for some $b < \infty$ the restriction estimate*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^n)} \leq A \|g\|_{L^\infty(d\sigma)} \tag{6.6.1}$$

holds for $S = \mathbb{S}^{n-1}$. Then for every δ -tube family $\mathbb{T}$ as above,

$$\left\| \sum_{j=1}^N \mathbf{1}_{T_j} \right\|_{L^{b/2}(\mathbb{R}^n)} \lesssim_n A^2 \delta^{\frac{4n}{b} - 2(n-1)}.$$

In particular, if (6.6.1) holds for every $b > \frac{2n}{n-1}$ — that is, if Theorem 6.2.6 is true — then for every $\varepsilon > 0$

$$\left\| \sum_{j=1}^N \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \lesssim_{n,\varepsilon} \delta^{-\varepsilon},$$

which is the Kakeya maximal conjecture.

Idea of the proof. Three moves.

Build a test function from the tubes. To each direction v_j attach the cap τ_j of radius δ centred there. By Theorem 6.2.2, $\mathcal{E}\mathbf{1}_{\tau_j}$ has modulus $\gtrsim \delta^{n-1}$ on the dual box of τ_j — a box of dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$, which is a rescaled copy of a δ -tube. Modulating $\mathbf{1}_{\tau_j}$ translates that box, so the boxes can be placed wherever the T_j are.

Randomise the signs. Adding the pieces with fixed signs risks cancellation. Attaching random signs and averaging replaces $|\sum_j \cdot|$ by the square function $(\sum_j |\cdot|^2)^{1/2}$, by Khintchine's inequality — the same device as in Theorem 3.6.1 — and the square function is bounded below by $\delta^{n-1}(\sum_j \mathbf{1}_{T_j})^{1/2}$. Crucially the $L^\infty(d\sigma)$ norm of the test function is 1 for *every* sign choice, so the hypothesis applies uniformly.

Rescale. The dual boxes have length δ^{-2} ; contracting space by δ^2 turns them into unit-length δ -tubes, and the exponent in the statement is the bookkeeping of that dilation.

Proof. Step 1. *The caps and their dual boxes.*

For each j let $\tau_j = \{\omega \in \mathbb{S}^{n-1} : |\omega - v_j| \leq \delta\}$. Since the v_j are δ -separated, the τ_j have bounded overlap. By Theorem 6.2.2 applied after the rotation taking v_j to e_n ,

$$|\mathcal{E}\mathbf{1}_{\tau_j}(y)| \geq \frac{1}{2}\sigma(\tau_j) \geq c_n \delta^{n-1} \quad \text{for } y \in B_j, \quad (6.6.2)$$

where B_j is the box of dimensions $\frac{c}{\delta} \times \cdots \times \frac{c}{\delta} \times \frac{c}{\delta^2}$ centred at the origin with long axis along v_j .

Given the target tubes T_j , write $\tilde{T}_j = \delta^{-2}T_j$ for their dilates, which are boxes of dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times \delta^{-2}$ up to the fixed constant c ; choose $a_j \in \mathbb{R}^n$ with $a_j + B_j \supseteq \tilde{T}_j$ after adjusting c by a dimensional constant.

Step 2. *The test function.*

For a sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_N) \in \{-1, 1\}^N$ put

$$g_\varepsilon(\omega) = \sum_{j=1}^N \varepsilon_j e^{-2\pi i a_j \cdot \omega} \mathbf{1}_{T_j}(\omega).$$

Because the τ_j have bounded overlap, $\|g_\varepsilon\|_{L^\infty(\mathrm{d}\sigma)} \lesssim_n 1$, *uniformly in ε* . By the modulation rule (Volume 2, Proposition 4.2),

$$\mathcal{E}g_\varepsilon(x) = \sum_{j=1}^N \varepsilon_j (\mathcal{E}\mathbf{1}_{\tau_j})(x - a_j).$$

Step 3. *Average over signs.*

Let r_j be the Rademacher functions and $\mathbb{E}_t$ the average over $t \in [0, 1]$. By hypothesis (6.6.1), for every t ,

$$\int_{\mathbb{R}^n} |\mathcal{E}g_{\varepsilon(t)}(x)|^b \mathrm{d}x \lesssim_n A^b.$$

Averaging in t and applying Tonelli and Khintchine's inequality (Theorem 2.6.5) pointwise in x ,

$$\int_{\mathbb{R}^n} \left(\sum_j |\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)|^2 \right)^{b/2} \mathrm{d}x \approx_b \mathbb{E}_t \int |\mathcal{E}g_{\varepsilon(t)}|^b \lesssim_n A^b.$$

Step 4. *Bound the square function below.*

By (6.6.2), $|\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)| \geq c_n \delta^{n-1}$ whenever $x \in a_j + B_j$, hence in particular whenever $x \in \tilde{T}_j$. Therefore

$$\left(\sum_j |\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)|^2 \right)^{1/2} \geq c_n \delta^{n-1} \left(\sum_j \mathbf{1}_{\tilde{T}_j}(x) \right)^{1/2},$$

and substituting into Step 3,

$$\delta^{(n-1)b} \int_{\mathbb{R}^n} \left(\sum_j \mathbf{1}_{\tilde{T}_j} \right)^{b/2} \lesssim_n A^b, \quad \text{i.e.} \quad \left\| \sum_j \mathbf{1}_{\tilde{T}_j} \right\|_{L^{b/2}} \lesssim_n A^2 \delta^{-2(n-1)}. \quad (6.6.3)$$

Step 5. *Rescale to unit tubes.*

Let $F(x) = \sum_j \mathbf{1}_{\tilde{T}_j}(x)$ and $G(y) = \sum_j \mathbf{1}_{T_j}(y)$. Since $\tilde{T}_j = \delta^{-2}T_j$, we have $G(y) = F(\delta^{-2}y)$, so with $p = b/2$

$$\begin{aligned} \|G\|_{L^p}^p &= \int |F(\delta^{-2}y)|^p \mathrm{d}y = \delta^{2n} \|F\|_{L^p}^p, \\ \|G\|_{L^p} &= \delta^{2n/p} \|F\|_{L^p} = \delta^{4n/b} \|F\|_{L^p}. \end{aligned}$$

Combining with (6.6.3),

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{b/2}} \lesssim_n A^2 \delta^{\frac{4n}{b} - 2(n-1)},$$

which is the stated bound.

Step 6. *The conjectural case.*

Suppose (6.6.1) holds for every $b > \frac{2n}{n-1}$. Given $\varepsilon > 0$, choose b slightly above $\frac{2n}{n-1}$. At $b = \frac{2n}{n-1}$ exactly,

$$\frac{4n}{b} - 2(n-1) = \frac{4n(n-1)}{2n} - 2(n-1) = 2(n-1) - 2(n-1) = 0,$$

and $b \mapsto \frac{4n}{b} - 2(n-1)$ is continuous, so for b close enough the exponent exceeds $-\varepsilon$; also $b/2 \rightarrow \frac{n}{n-1}$, and by Hölder on the bounded set where $\sum_j \mathbf{1}_{T_j}$ is supported — all tubes lie in a fixed ball — the $L^{b/2}$ norm dominates the $L^{n/(n-1)}$ norm up to a constant. $\square$

Where we are. The exponent bookkeeping in Step 6 is worth recording as a slogan: the restriction exponent $b = \frac{2n}{n-1}$ and the Kakeya exponent $\frac{n}{n-1}$ correspond under $b \mapsto b/2$, and the powers of δ cancel *exactly* at the conjectured endpoint. That is not a coincidence: it says the Knapp example and the Kakeya packing problem are extremal for the same reason, namely that $\delta^{-(n-1)}$ caps produce $\delta^{-(n-1)}$ tubes with unit total volume.

Note also what the proof did *not* use: nothing about the sphere beyond Theorem 6.2.2, which is a statement about a single cap. The implication therefore holds for any surface with non-vanishing curvature, with the tube directions ranging over the corresponding normal directions.

Corollary 6.6.3 (Besicovitch sets are large). *Assume Theorem 6.2.6 in dimension n . Then every Besicovitch set $K \subseteq \mathbb{R}^n$ — every compact set containing a unit segment in every direction — has upper Minkowski dimension n .*

Proof. Fix $\delta > 0$ and let $v_1, \dots, v_N$ be a maximal δ -separated set of directions, $N \approx \delta^{-(n-1)}$. For each j , K contains a unit segment ℓ_j in direction v_j , so the δ -neighbourhood K_δ contains the δ -tube T_j around ℓ_j . These form a tube family in the sense of Theorem 6.6.1.

By Theorem 6.6.2 with $b/2$ close to $\frac{n}{n-1}$, for every $\varepsilon > 0$

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_{n,\varepsilon} \delta^{-\varepsilon}.$$

On the other hand $\sum_j \mathbf{1}_{T_j}$ is supported in K_δ , so Hölder with exponents n and

$\frac{n}{n-1}$ gives

$$\begin{aligned}
 1 &\approx \sum_j m(T_j) \\
 &= \int_{K_\delta} \sum_j \mathbf{1}_{T_j} \\
 &\leq m(K_\delta)^{1/n} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_{n,\varepsilon} m(K_\delta)^{1/n} \delta^{-\varepsilon}.
 \end{aligned}$$

Hence $m(K_\delta) \gtrsim_{n,\varepsilon} \delta^{n\varepsilon}$.

The upper Minkowski dimension of K is $n - \liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta}$, and the bound just obtained gives $\frac{\log m(K_\delta)}{\log \delta} \leq n\varepsilon$ for small δ . So the dimension is at least $n - n\varepsilon$ for every $\varepsilon > 0$, hence equals n . $\square$

Pitfall: Minkowski dimension is the weaker statement

Theorem 6.6.3 concerns *Minkowski* dimension, which measures how slowly $m(K_\delta)$ shrinks. The *Kakeya* conjecture properly asserts that Besicovitch sets have *Hausdorff* dimension n , and Hausdorff dimension can be strictly smaller: a countable dense set has Minkowski dimension n and Hausdorff dimension 0.

Upgrading requires a covering argument at many scales simultaneously, which is Volume 7's machinery, together with a version of Theorem 6.6.2 applied at each scale. Volume 9 carries this out. Until then the reader should treat every “dimension” statement in this volume as Minkowski, and should note that this is the standard convention only when it is stated — the literature is not always careful.

WORKED EXAMPLE 6.6.1 — TIER A **Why random signs, and what fixed signs would give**

Suppose in Step 2 we had taken all $\varepsilon_j = +1$. Then $\mathcal{E}g$ is a sum of N terms, each of modulus $\approx \delta^{n-1}$ on its own box, and at a point x lying in μ of the boxes the sum could be as large as $\mu\delta^{n-1}$ — or as small as 0, if the phases conspire. The hypothesis would then give

$$\delta^{n-1} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^b} \lesssim A$$

only if one could rule out cancellation, which one cannot: the phases of $\mathcal{E}\mathbf{1}_{\tau_j}(x - a_j)$ at a common point x are genuinely uncontrolled.

Randomising removes the issue at the cost of replacing the sum by the square function — and the square function of μ terms of size δ^{n-1} is $\mu^{1/2}\delta^{n-1}$, not

$\mu\delta^{n-1}$. So the argument gives an $L^{b/2}$ bound on the multiplicity rather than an L^b bound, which is exactly the loss recorded in the exponent $b/2$ of Theorem 6.6.2. Khintchine converts an unusable hypothesis into a usable one, and the square root is the price.

WORKED EXAMPLE 6.6.2— TIER B The Kakeya maximal conjecture, and what the theorem delivers

The conclusion of Theorem 6.6.2 is stated as a bound on a multiplicity function; the literature usually states the Kakeya maximal conjecture as a bound on an operator. The two are the same, and the translation is worth doing once.

The maximal operator. Theorem 1.7.13 introduced

$$\mathcal{K}_\delta f(v) = \sup_{T \parallel v} \frac{1}{m(T)} \int_T |f|,$$

the supremum over δ -tubes in direction v . The conjecture is $\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_\varepsilon \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}$.

Duality. Testing against $f = \mathbf{1}_E$ and choosing for each of the $\delta^{-(n-1)}$ separated directions a tube where the average is largest, the maximal estimate is equivalent — by the standard duality between an $L^n \rightarrow L^n$ bound and a bound on $\sum_j \mathbf{1}_{T_j}$ in $L^{n'} = L^{n/(n-1)}$ — to

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_\varepsilon \delta^{-\varepsilon},$$

which is the form delivered above.

What it gives about sets. If $K \subseteq \mathbb{R}^n$ is a Besicovitch set, it contains a unit segment in every direction, so its δ -neighbourhood contains a δ -tube in each of $\delta^{-(n-1)}$ separated directions. Then by Hölder,

$$\begin{aligned} 1 &\approx \sum_j m(T_j) \\ &= \int_{K_\delta} \sum_j \mathbf{1}_{T_j} \\ &\leq m(K_\delta)^{1/n} \left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \lesssim_\varepsilon m(K_\delta)^{1/n} \delta^{-\varepsilon}, \end{aligned}$$

so $m(K_\delta) \gtrsim \delta^{\varepsilon n}$, and letting $\varepsilon \rightarrow 0$ gives Minkowski dimension n . Upgrading to Hausdorff dimension needs more, and is Volume 9's business.

WORKED EXAMPLE 6.6.3— TIER C: RESEARCH TASTE **What this implication does and does not settle**

(a) *The arrow runs one way.* Theorem 6.6.2 says restriction is at least as hard as *Keakeya*. No implication in the reverse direction is known, and none is expected to be easy: *Keakeya* is a statement about the *supports* of the wave packets, while restriction is a statement about their superposition with phases, and the passage from one to the other loses exactly the information that Step 3 destroyed.

(b) *The contrapositive is how it is used.* A counterexample to *Keakeya* would disprove restriction. Since *Keakeya* is believed true, the implication is used instead as a source of *necessary conditions* and as a guide: any proof of restriction must, somewhere, contain a proof of *Keakeya*. Every advance in Volumes 12 and 13 does.

(c) *The same argument with L^2 data gives nothing.* If one runs the proof with (6.6.1) replaced by the Stein–Tomas hypothesis $\|\mathcal{E}g\|_b \lesssim \|g\|_{L^2(d\sigma)}$, then $\|g_\varepsilon\|_{L^2(d\sigma)} \approx (N\delta^{n-1})^{1/2} \approx 1$ — the same as before — but the admissible b is now at least $\frac{2(n+1)}{n-1}$, and the resulting exponent $\frac{4n}{b} - 2(n-1)$ is strictly negative. So Stein–Tomas gives a *Keakeya* bound, but a lossy one. Working out that loss is Drill 6.3, and it is the reason the $L^\infty(d\sigma)$ endpoint of the restriction problem is the one that matters for *Keakeya*.

(d) *Where the series goes from here.* Volumes 7 and 8 develop the dimension theory needed to state *Keakeya* properly; Volume 9 proves the best bounds obtainable by elementary geometry; Volumes 10 and 11 bring in arithmetic and algebraic methods; and Volumes 12 and 13 return to restriction with the geometric input those volumes supply. The present theorem is the reason that route is necessary rather than a detour.

WORKED EXAMPLE 6.6.4— TIER B **Running the argument backwards: what a *Keakeya* counterexample would kill**

Theorem 6.6.2 is usually read forwards. Read backwards it is a falsification tool, and it is worth seeing exactly how much a hypothetical *Keakeya* counterexample would destroy.

Suppose that for some $n \geq 3$ and some $\alpha > 0$ there were tube families $\mathbb{T}_\delta$ with

$$\left\| \sum_j \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \geq \delta^{-\alpha}$$

for arbitrarily small δ — that is, a quantitative failure of the *Keakeya* maximal conjecture.

What follows. By Theorem 6.6.2, the restriction estimate (6.6.1) at exponent b forces $\delta^{-\alpha} \lesssim A^2 \delta^{4n/b-2(n-1)}$, hence

$$\frac{4n}{b} - 2(n-1) \leq -\alpha, \quad \text{that is} \quad b \geq \frac{4n}{2(n-1) - \alpha} > \frac{2n}{n-1}.$$

So the restriction conjecture would be false, and quantitatively so: the true exponent would exceed the conjectured one by an amount computable from α .

And what would survive. Stein–Tomas would be untouched: it is a theorem, proved in Chapter 6.3 without any Kakeya input, and Worked Example 6.6.3(c) showed the Kakeya bound it implies is far from sharp. So a Kakeya counterexample would not contradict anything proved — it would only close off the conjectured range, and the subject would reorganise around finding the true exponent.

Why nobody expects this. Every known construction of a small Besicovitch set is a variant of the Perron tree (Theorem 1.3.12), and the Perron tree produces sets of measure zero but Minkowski dimension n — it is efficient at making the measure small and useless at making the dimension small. Volume 9 explains why, and Volume 10 shows that the finite-field analogue admits no counterexample at all.

Figure 6.6.1 (to be drawn). *From caps to tubes*

Content required: The sphere partitioned into δ -caps τ_j ; below, the corresponding dual boxes, one per cap, placed at arbitrary positions and pointing in δ -separated directions. Indicate that there are $\approx \delta^{-(n-1)}$ of them and that each has volume $\approx \delta^{-(n+1)}$.

Drills

Drill 6.6.1. Verify that the caps τ_j in Step 1 have bounded overlap, and deduce $\|g_\varepsilon\|_{L^\infty(d\sigma)} \lesssim_n 1$.

Drill 6.6.2. Check the dilation computation in Step 5 by testing on a single tube.

Drill 6.6.3. Run the proof with the Stein–Tomas hypothesis in place of (6.6.1), as in Worked Example 6.6.3(c), and compute the resulting exponent for $n = 3$.

Drill 6.6.4. Show that the conclusion of Theorem 6.6.2 is trivially true with exponent $\delta^{-(n-1)}$ for any tube family, and confirm that the theorem is an improvement exactly when $b < \infty$.

Where we are. The theorem of this chapter is the reason the series has the shape it does. Read once more in summary: an estimate for a single operator, tested against a family of $\delta^{-(n-1)}$ modulated cap indicators with random signs, becomes a bound on how much $\delta^{-(n-1)}$ tubes can overlap. Analysis in, geometry out.

Everything in Volumes 7 to 11 is devoted to the geometry side of that exchange: what a set containing tubes in all directions must look like, how to measure it, and what algebraic and combinatorial tools bound the overlaps. Volumes 12 and 13 then run the exchange in the direction it was not proved to run — using geometric input to establish analytic estimates — by methods that do not factor through Theorem 6.6.2 at all.

Notes

The implication is due to Bourgain (1991), with the randomisation argument as given. Fefferman’s ball multiplier theorem (Volume 3, Chapter 8) is the same idea in a degenerate case: there the tubes come from a Besicovitch set and are used to *disprove* an estimate, here they are produced by an estimate assumed to hold.

The observation that the exponents match exactly at the conjectured endpoint is the standard heuristic for why the restriction conjecture is stated where it is.

Chapter 6.7

The δ -Neighbourhood Formulation

Volumes 12 and 13 never estimate $\mathcal{E}g$ directly. They work instead with functions whose Fourier transform is supported in a thin neighbourhood of the surface, decompose those into wave packets, and prove local estimates on a ball of radius $R = \delta^{-1}$. This chapter shows that the two formulations are the same, so that the literature's setting and this volume's are interchangeable.

It also records the place of the restriction conjecture in the chain of three conjectures, and states precisely which of the implications this series proves.

6.7.1 Thin neighbourhoods

Definition 6.7.1. For a compact hypersurface $S \subseteq \mathbb{R}^n$ and $0 < \delta < \frac{1}{2}$ let

$$N_\delta(S) = \{\xi \in \mathbb{R}^n : \text{dist}(\xi, S) < \delta\}.$$

For $S = \mathbb{S}^{n-1}$ this is the annulus $1 - \delta < |\xi| < 1 + \delta$, and polar coordinates give the decomposition we shall use: writing $\xi = (1 + t)\omega$ with $|t| < \delta$ and $\omega \in \mathbb{S}^{n-1}$,

$$\int_{N_\delta(S)} F(\xi) \, d\xi = \int_{-\delta}^{\delta} \int_{\mathbb{S}^{n-1}} F((1 + t)\omega) (1 + t)^{n-1} \, d\sigma(\omega) \, dt, \quad (6.7.1)$$

by Theorem 1.6.11.

Theorem 6.7.2 (Extension estimates give neighbourhood estimates). *Let $\delta = R^{-1}$ and suppose*

$$\|\mathcal{E}_\rho g\|_{L^b(B_R)} \leq A \|g\|_{L^2(d\sigma)} \quad (6.7.2)$$

for every ball B_R of radius R and every $\rho \in [1 - \delta, 1 + \delta]$, where $\mathcal{E}_\rho$ is the extension operator for the sphere of radius ρ . Then for every $f \in \mathcal{S}(\mathbb{R}^n)$ with $\text{supp } \hat{f} \subseteq N_\delta(\mathbb{S}^{n-1})$,

$$\|f\|_{L^b(B_R)} \lesssim_n A \delta^{1/2} \|\hat{f}\|_{L^2(\mathbb{R}^n)}.$$

Idea of the proof. Slice the neighbourhood into spheres. By (6.7.1), f is an integral over $t \in (-\delta, \delta)$ of extension operators applied to the restriction of $\hat{f}$ to the sphere of radius $1 + t$. Minkowski's integral inequality moves the L^b norm inside, the hypothesis bounds each slice, and Cauchy–Schwarz in t — over an interval of length 2δ — converts the resulting $L_t^1 L_\omega^2$ norm into $\delta^{1/2}$ times the L^2 norm of $\hat{f}$.

The factor $\delta^{1/2}$ is therefore nothing but Cauchy–Schwarz over the thickness of the slab, and it is the reason neighbourhood estimates in the literature carry a power of δ that looks unmotivated at first sight.

Proof. Step 1. Slice.

By Fourier inversion and (6.7.1),

$$\begin{aligned} f(x) &= \int_{N_\delta} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi \\ &= \int_{-\delta}^{\delta} (1+t)^{n-1} \left(\int_{\mathbb{S}^{n-1}} \hat{f}((1+t)\omega) e^{2\pi i (1+t)x \cdot \omega} d\sigma(\omega) \right) dt. \end{aligned}$$

The inner integral is $\mathcal{E}_{1+t} g_t(x)$ where $g_t(\omega) = \hat{f}((1+t)\omega)$, after absorbing the radius into the operator.

Step 2. *Minkowski and the hypothesis.*

By Minkowski's integral inequality (Theorem 1.8.4, in its integral form) and then (6.7.2),

$$\begin{aligned} \|f\|_{L^b(B_R)} &\leq \int_{-\delta}^{\delta} (1+t)^{n-1} \|\mathcal{E}_{1+t} g_t\|_{L^b(B_R)} dt \lesssim_n A \int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)} dt, \end{aligned}$$

using $(1+t)^{n-1} \leq 2^{n-1}$.

Step 3. *Cauchy–Schwarz in t .*

Since the interval has length 2δ ,

$$\begin{aligned} \int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)} dt &\leq (2\delta)^{1/2} \left(\int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)}^2 dt \right)^{1/2}. \end{aligned}$$

By (6.7.1) again, and since $(1+t)^{n-1} \geq 2^{1-n}$,

$$\begin{aligned} \int_{-\delta}^{\delta} \|g_t\|_{L^2(d\sigma)}^2 dt &= \int_{-\delta}^{\delta} \int_{\mathbb{S}^{n-1}} |\widehat{f}((1+t)\omega)|^2 d\sigma dt \lesssim_n \int_{N_\delta} |\widehat{f}(\xi)|^2 d\xi \\ &= \|\widehat{f}\|_{L^2}^2. \end{aligned}$$

Combining Steps 2 and 3 gives the claim. $\square$

Proposition 6.7.3 (The converse). *Let $\delta = R^{-1}$ and suppose that for every f with $\text{supp } \widehat{f} \subseteq N_\delta(\mathbb{S}^{n-1})$ and every ball B_R ,*

$$\|f\|_{L^b(B_R)} \leq A' \delta^{1/2} \|\widehat{f}\|_{L^2(\mathbb{R}^n)}. \quad (6.7.3)$$

Then the local extension estimate (6.7.2) holds with $A \lesssim_n A'$.

Proof. Step 1. Smooth the surface measure.

Let $g \in L^2(d\sigma)$ and let $\varphi \in \mathcal{S}(\mathbb{R}^n)$ with $\widehat{\varphi} \geq 0$, $\widehat{\varphi} = 1$ on $B(0, 1)$ and $\text{supp } \widehat{\varphi} \subseteq B(0, 2)$. Put $\varphi_\delta(x) = \varphi(\delta x)$, so $\widehat{\varphi}_\delta(\xi) = \delta^{-n} \widehat{\varphi}(\xi/\delta)$ is an approximate identity at scale δ , and define

$$f = (\mathcal{E}g) \cdot \varphi_\delta, \quad \text{so} \quad \widehat{f} = (gd\sigma) * \widehat{\varphi}_\delta.$$

Then $\text{supp } \widehat{f} \subseteq N_{2\delta}(\mathbb{S}^{n-1})$, since $gd\sigma$ is supported on the sphere and $\widehat{\varphi}_\delta$ in a ball of radius 2δ .

Step 2. *Compute the right-hand side of (6.7.3).*

The function $\widehat{f}$ is $gd\sigma$ smeared over a slab of thickness δ , so its L^2 norm is computed by (6.7.1):

$$\begin{aligned} \|\widehat{f}\|_{L^2(\mathbb{R}^n)}^2 &= \int_{N_{2\delta}} |(gd\sigma) * \widehat{\varphi}_\delta(\xi)|^2 d\xi \lesssim_n \delta \cdot \delta^{-2} \cdot \delta \|g\|_{L^2(d\sigma)}^2 \\ &= \delta^{-1} \|g\|_{L^2(d\sigma)}^2, \end{aligned}$$

where the three factors are: the thickness δ of the slab, the square of the height δ^{-1} to which the smoothing raises the density, and a factor δ from the fact that the mass at each point of the sphere is spread over thickness δ . Hence $\|\widehat{f}\|_{L^2} \lesssim_n \delta^{-1/2} \|g\|_{L^2(d\sigma)}$.

Step 3. *Compute the left-hand side.*

Since $\varphi_\delta = \varphi(\delta \cdot)$ and φ may be chosen with $|\varphi| \geq \frac{1}{2}$ on $B(0, 1)$, we have $|\varphi_\delta| \geq \frac{1}{2}$ on $B(0, R)$, so

$$\|\mathcal{E}g\|_{L^b(B_R)} \leq 2\|f\|_{L^b(B_R)}.$$

Step 4. Combine.

Applying (6.7.3) and Steps 2, 3,

$$\begin{aligned}\|\mathcal{E}g\|_{L^b(B_R)} &\leq 2A'\delta^{1/2}\|\widehat{f}\|_{L^2} \lesssim_n A'\delta^{1/2}\delta^{-1/2}\|g\|_{L^2(d\sigma)} \\ &= A'\|g\|_{L^2(d\sigma)}.\end{aligned}\quad \square$$

Where we are. Theorem 6.7.2 and Theorem 6.7.3 together say the two formulations are equivalent, with the factor $\delta^{1/2}$ appearing on one side and disappearing on the other. That factor is not a defect of either statement: it is the ratio between the L^2 norm of a measure on a surface and the L^2 norm of its smearing over a slab of thickness δ , and it must appear somewhere. A practical consequence: when reading a paper, check which normalisation is in force before comparing constants. A statement about $\|\widehat{f}\|_{L^2}$ and one about $\|g\|_{L^2(d\sigma)}$ differ by $\delta^{1/2}$, and at the exponents where these theorems are sharp, a factor $\delta^{1/2}$ is the whole content.

WORKED EXAMPLE 6.7.1— TIER A Reading a decoupling statement with this dictionary

A decoupling theorem in Volume 13 will be stated as: for $\widehat{f}$ supported in $N_\delta(P)$ with P the truncated paraboloid, and $f = \sum_\theta f_\theta$ the decomposition over $\delta^{1/2}$ -caps,

$$\|f\|_{L^p(B_{\delta^{-1}})} \lesssim_\varepsilon \delta^{-\varepsilon} \left(\sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2}.$$

Every ingredient now has a translation.

The hypothesis $\text{supp } \widehat{f} \subseteq N_\delta(P)$ is the neighbourhood formulation, equivalent by Theorem 6.7.2 to working with $\mathcal{E}$. The ball $B_{\delta^{-1}}$ is the local formulation of Theorem 6.5.1, at the scale $R = \delta^{-1}$ dictated by the uncertainty principle: a function with Fourier support in a slab of thickness δ cannot be localised more finely than δ^{-1} . The caps θ have diameter $\delta^{1/2}$ because that is the size at which the paraboloid deviates from its tangent plane by δ — the Knapp scaling of Theorem 6.2.2. And the $\delta^{-\varepsilon}$ is removable in the sense of Theorem 6.5.2.

So a decoupling statement is a restriction statement, written in the coordinates that induction on scales requires. The reader who has this dictionary can move between the two literatures; the reader who does not will find the change of setting between Stein–Tomas and Bourgain–Demeter mysterious.

6.7.2 The chain of conjectures

Theorem 6.7.4 (What this series proves about the chain). *Consider the three statements, for a fixed $n \geq 2$:*

$$\begin{array}{ll} (BR) & \text{Bochner–Riesz conjecture} \\ (R) & \text{restriction conjecture,} \\ (K) & \text{Kakeya maximal conjecture.} \end{array}$$

Then $(R) \implies (K)$.

Proof. This is Theorem 6.6.2: assuming (R) , the hypothesis (6.6.1) holds for every $b > \frac{2n}{n-1}$, and Step 6 of that proof converts it into the Kakeya maximal bound with an arbitrarily small loss $\delta^{-\varepsilon}$, which is the statement (K) . $\square$

Remark 6.7.5 (Scope decision: the first arrow). The implication $(BR) \implies (R)$ is also true. It is due to Tao (1999), and it is *not* proved in this series.

The reason is a judgement about length, and it is worth stating plainly because the omission is visible: Tao’s argument is a research paper, not a lemma. It proceeds by localising the Bochner–Riesz operator to a δ -neighbourhood, converting the resulting estimate into a local extension estimate at scale δ^{-1} , and then removing the localisation by an induction on scales. Each of those steps is of the same order of difficulty as an entire chapter here.

This is a plan revision, recorded as R5 in the master manifest: no later volume supplies it, and nothing in this series depends on it. Volume 5, Chapter 8 proved the positive Bochner–Riesz results that are used — boundedness for $\delta > \frac{n-1}{2}$ — and Volume 3, Chapter 8 proved the negative result at $\delta = 0$. The implication is quoted here only to place the conjectures in relation to one another, never as a hypothesis.

WORKED EXAMPLE 6.7.2— TIER C: RESEARCH TASTE **How to read the three conjectures together**

(a) *They are ordered by difficulty.* $(BR) \implies (R) \implies (K)$, so Kakeya is the weakest and Bochner–Riesz the strongest. All three are theorems for $n = 2$ — Carleson–Sjölin, Chapter 6.8, and the planar Kakeya theorem of Theorem 1.6.14 — and all three are open for every $n \geq 3$.

(b) *They fail together, in a precise sense.* Each is sharp: the Knapp example limits (R) , the same cap construction rescaled limits (BR) , and the Besicovitch set limits (K) . And each of the three extremal examples is the same object seen at a different scale — a family of $\delta^{-(n-1)}$ tubes, one per direction. That is the sense in which the subject has a single difficulty.

(c) *The converses are open and probably hard.* No implication is known in the reverse direction. Kakeya is a statement about supports; restriction is about

superpositions with phases. Worked Example 6.6.1 showed exactly where the phase information is destroyed in passing from one to the other, and there is no reason to expect it can be recovered.

(d) *Why the series goes through Keakeya anyway.* Because the geometric input — multilinear Keakeya, the polynomial method, decoupling — is developed in the Keakeya setting and then imported. Volume 12’s proof of a restriction estimate uses multilinear Keakeya as a black box, and Volume 13’s decoupling theorem uses it again. The arrow (R) $\implies$ (K) says one cannot avoid Keakeya; the practice of the field is that one does not want to.

(e) *A caution about “equivalent”.* Papers sometimes describe the three as equivalent. They are equivalent in the loose sense that progress on one has historically produced progress on the others, and that the known counterexamples coincide. They are not equivalent as theorems, and a reader should treat any claimed equivalence as a claim about a specific quantitative formulation, worth checking.

WORKED EXAMPLE 6.7.3— TIER B Wave packets at scale R : counting the degrees of freedom

The neighbourhood formulation exists to make a decomposition available, and it is worth doing the bookkeeping once so that the parameters in Volumes 12 and 13 look inevitable rather than arbitrary.

The setting. Let $\hat{f}$ be supported in $N_\delta(P)$, P the truncated paraboloid in $\mathbb{R}^n$, and put $R = \delta^{-1}$. We ask: how many independent pieces does f have on the ball B_R ?

Step 1: the caps. Partition P into caps θ of diameter $\delta^{1/2}$. This is the correct size because the paraboloid deviates from its tangent plane over a cap of diameter ρ by $\approx \rho^2$, and we want that deviation to be at most the slab thickness δ ; so $\rho = \delta^{1/2}$. The number of caps is

$$\#\{\theta\} \approx \left(\delta^{-1/2}\right)^{n-1} = \delta^{-\frac{n-1}{2}}.$$

Step 2: the dual tubes. By Theorem 6.2.2 — applied with $\delta^{1/2}$ in place of δ — the piece f_θ is essentially constant on translates of a box of dimensions

$$\underbrace{\delta^{-1/2} \times \dots \times \delta^{-1/2}}_{n-1} \times \delta^{-1} = R^{1/2} \times \dots \times R^{1/2} \times R.$$

So f_θ is a superposition of translates of one shape — a tube of length R

and radius $R^{1/2}$, pointing in the direction normal to θ . These are the *wave packets*.

Step 3: how many packets per cap. The ball B_R has volume R^n and each tube has volume $R^{1/2(n-1)} \cdot R = R^{\frac{n+1}{2}}$, so the tubes in a fixed direction tile B_R with

$$\frac{R^n}{R^{\frac{n+1}{2}}} = R^{\frac{n-1}{2}}$$

translates. Multiplying by the number of caps, $\delta^{-\frac{n-1}{2}} = R^{\frac{n-1}{2}}$, the total number of wave packets is R^{n-1} .

Step 4: the consistency check. A function with Fourier support in a set of measure $\approx \delta = R^{-1}$, localised to B_R of volume R^n , has $\approx R^{-1} \cdot R^n = R^{n-1}$ degrees of freedom by the uncertainty principle. That matches Step 3 exactly, which is the sign that the decomposition is not wasteful: the wave packets are a basis, not an overcomplete family.

What to take from this. Every parameter in a decoupling paper — $\delta^{1/2}$ caps, $R^{1/2} \times R$ tubes, R^{n-1} packets — is forced by two facts: the curvature of the surface, which fixes the cap size, and the uncertainty principle, which fixes everything else. None of it is a choice.

WORKED EXAMPLE 6.7.4— TIER C: RESEARCH TASTE **What the wave packet picture predicts, and where it stops**

Worked Example 6.7.3 counted R^{n-1} wave packets. That count already predicts the conjectured exponent, and seeing the prediction — and then seeing why it is not a proof — is the best short summary of the state of the subject.

(a) *The heuristic.* Suppose the R^{n-1} packets were essentially disjoint, each of height h on a tube of volume $R^{\frac{n+1}{2}}$. Normalising $\|g\|_{L^\infty} = 1$ makes $h \approx R^{-\frac{n-1}{2}}$ after the rescaling of Worked Example 6.7.3. Then

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(B_R)}^b &\approx R^{n-1} \cdot h^b \cdot R^{\frac{n+1}{2}} \\ &= R^{n-1+\frac{n+1}{2}-\frac{b(n-1)}{2}}, \end{aligned}$$

and this is $O(1)$ — no growth in R , which is what a global estimate requires — exactly when

$$\frac{b(n-1)}{2} \geq n-1 + \frac{n+1}{2}, \quad \text{that is} \quad b \geq \frac{3n-1}{n-1}.$$

That is not $\frac{2n}{n-1}$. The discrepancy is the point.

(b) *Why the naive count is wrong.* The packets are *not* disjoint. A point of B_R lies in many tubes, and the correct accounting weighs the overlaps: it is precisely the multiplicity function $\sum_T \mathbf{1}_T$ of Theorem 6.6.2. Assuming the Kakeya conjecture — that the multiplicity is as small as it can be — and redoing the count gives $b > \frac{2n}{n-1}$.

(c) *So the conjecture is a statement about overlaps.* This is the content of Theorem 6.6.2 in reverse: the restriction exponent is determined by how much the tubes can pile up, and the extremal configuration is a Besicovitch set. Every proof in Volumes 12 and 13 is, at bottom, a way of controlling that pile-up.

(d) *Where the heuristic genuinely fails.* Even granting Kakeya, the count above ignores the *phases*: two packets overlapping on a set may add or cancel, and the heuristic assumes they add in absolute value. Worked Example 6.6.1 showed that random signs make the square function the right object, but a real g has fixed phases, and no known argument controls them. This is the reason Kakeya is believed easier than restriction, and the reason no implication is known from the first to the second.

(e) *What to carry away.* The wave packet count is the standard back-of-envelope calculation of the field, and it is worth being able to do it in a minute. But it predicts the wrong exponent unless the overlap and the phases are put in by hand, and putting them in is the entire difficulty. A reader who can produce the count *and* name its two false assumptions understands the shape of the problem.

Drills

Drill 6.7.1. Verify (6.7.1) from Theorem 1.6.11, tracking the Jacobian $(1+t)^{n-1}$.

Drill 6.7.2. Show that the factor $\delta^{1/2}$ in Theorem 6.7.2 is sharp, by taking $\hat{f} = \mathbf{1}_{N_\delta}$ and computing both sides for $b = \infty$.

Drill 6.7.3. State and prove the converse of Theorem 6.7.2: a neighbourhood estimate implies a local extension estimate, by taking $\hat{f}$ to be $gd\sigma$ smoothed at scale δ .

Drill 6.7.4. Explain, using Worked Example 6.7.1, why caps of diameter $\delta^{1/2}$ and not δ are the right ones for a δ -neighbourhood of a paraboloid.

Notes

The neighbourhood formulation is standard and is the setting of essentially all modern work. Theorem 6.7.2 is elementary; its content is the bookkeeping, which is where readers new to the area most often lose their footing.

Tao's theorem that (BR) implies (R) appeared in *Duke Math. J.* 96 (1999). The reverse implication is open.

Chapter 6.8

The Two-Dimensional Case

In the plane the restriction conjecture is a theorem. This chapter proves it, for the parabola, by an argument that is entirely elementary once one notices the right thing: in two dimensions the fourth power of $\mathcal{E}g$ can be computed exactly, because $|\mathcal{E}g|^2$ has Fourier transform a self-convolution of a measure on a curve, and that convolution can be evaluated by a change of variables.

6.8.1 The model and the statement

Definition 6.8.1. Let $P = \{(t, t^2) : |t| \leq 1\}$ be the truncated parabola with the measure $d\mu$ defined by $\int F d\mu = \int_{-1}^1 F(t, t^2) dt$, and for $g \in L^\infty(d\mu)$ set

$$\mathcal{E}g(x) = \int_{-1}^1 g(t) e^{2\pi i(x_1 t + x_2 t^2)} dt, \quad x = (x_1, x_2) \in \mathbb{R}^2.$$

Theorem 6.8.2 (Restriction in the plane). *For every $b > 4$,*

$$\|\mathcal{E}g\|_{L^b(\mathbb{R}^2)} \lesssim_b \|g\|_{L^\infty(d\mu)}.$$

By Theorem 6.2.5 — whose proof applies verbatim to the parabola, using Theorem 5.5.2 for a curve of non-vanishing curvature — the range $b > 4 = \frac{2n}{n-1}|_{n=2}$ is optimal. So Theorem 6.8.2 is the restriction conjecture for $n = 2$.

Idea of the proof. The exponent 4 is even, and that is the whole point. Write

$$\|\mathcal{E}g\|_{L^4}^4 = \|(\mathcal{E}g)^2\|_{L^2}^2 = \|(g d\mu) * (g d\mu)\|_{L^2}^2,$$

by Plancherel, since $(\mathcal{E}g)^2 = ((g d\mu) * (g d\mu))^\vee$. The convolution is an integral over pairs (s, t) , and the substitution $(s, t) \mapsto (s + t, s^2 + t^2)$ has Jacobian $2|t - s|$ — non-zero except on the diagonal. Changing variables

turns the L^2 norm into

$$\iint_{[-1,1]^2} \frac{|g(s)|^2 |g(t)|^2}{2|t-s|} ds dt,$$

which diverges logarithmically for $g \equiv 1$ — consistent with the exponent 4 being excluded — but only logarithmically. Cutting off at $|t-s| \geq R^{-1}$ gives a factor $\log R$ on the ball B_R , and interpolating that against the trivial bound at $b = \infty$ removes the logarithm at any $b > 4$.

6.8.2 The bilinear computation

Lemma 6.8.3 (The change of variables). *The map $\Psi(s, t) = (s+t, s^2+t^2)$ is a local diffeomorphism off the diagonal, with Jacobian determinant $2(t-s)$, and it is two-to-one onto its image, the two preimages of a point being (s, t) and (t, s) .*

Proof. $D\Psi = \begin{pmatrix} 1 & 1 \\ 2s & 2t \end{pmatrix}$ has determinant $2(t-s)$, non-zero for $s \neq t$. Given $(\xi_1, \xi_2) = \Psi(s, t)$, the numbers s, t are the roots of $z^2 - \xi_1 z + \frac{1}{2}(\xi_1^2 - \xi_2) = 0$, since $s+t = \xi_1$ and $st = \frac{1}{2}((s+t)^2 - (s^2+t^2))$. A quadratic has two roots, so the preimage is $\{(s, t), (t, s)\}$. $\square$

Proposition 6.8.4 (The exact L^4 identity). *For $g \in L^\infty(d\mu)$ with $|g| \leq 1$ and any measurable $E \subseteq [-1, 1]^2$ symmetric in its two variables,*

$$\left\| (g \mathbf{1}_E d\mu) * (g d\mu) \right\|_{L^2(\mathbb{R}^2)}^2 \leq 2 \iint_E \frac{ds dt}{2|t-s|},$$

where $g \mathbf{1}_E d\mu * g d\mu$ denotes the convolution with the integration restricted to $(s, t) \in E$.

Proof. Step 1. Identify the convolution as a pushforward.

For $F \in C_c(\mathbb{R}^2)$, unwinding the definition of the convolution of two measures carried by P ,

$$\begin{aligned} \int F d[(g \mathbf{1}_E d\mu) * (g d\mu)] \\ = \iint_E F(\Psi(s, t)) g(s)g(t) ds dt \\ , \quad \Psi(s, t) = (s+t, s^2+t^2), \end{aligned}$$

because a point of P is (s, s^2) and the sum of two such points is $\Psi(s, t)$. So the measure in question is the pushforward under Ψ of the finite measure $g(s)g(t)\mathbf{1}_E ds dt$.

Step 2. *Compute its density.*

By Theorem 6.8.3, Ψ is a local diffeomorphism off the diagonal with Jacobian $2|t - s|$, and it is two-to-one, the fibre over $\xi = \Psi(s, t)$ being $\{(s, t), (t, s)\}$. Hence the pushforward is absolutely continuous off the diagonal, with density

$$H(\xi) = \sum_{(s,t) \in \Psi^{-1}(\xi) \cap E} \frac{g(s)g(t)}{2|t - s|}.$$

Since E is symmetric, the fibre lies wholly inside or wholly outside E , so the sum has either two terms or none; and the two terms are equal, $g(s)g(t)$ and $2|t - s|$ both being symmetric. Using $|g| \leq 1$,

$$|H(\xi)| \leq \frac{2}{2|t - s|} = \frac{1}{|t - s|} \quad \text{at } \xi = \Psi(s, t). \quad (6.8.1)$$

Step 3. *Change variables in the L^2 norm.*

The two-to-one change of variables $\xi = \Psi(s, t)$ reads, for $\Phi \geq 0$,

$$\int_{\mathbb{R}^2} \Phi(\xi) d\xi = \frac{1}{2} \iint_{[-1,1]^2} \Phi(\Psi(s, t)) 2|t - s| ds dt,$$

the factor $\frac{1}{2}$ correcting for the double covering and $2|t - s|$ being the Jacobian. Applying this with $\Phi = |H|^2$, which vanishes off $\Psi(E)$, and then (6.8.1),

$$\begin{aligned} \|H\|_{L^2}^2 &= \frac{1}{2} \iint_E |H(\Psi(s, t))|^2 2|t - s| ds dt \\ &\leq \frac{1}{2} \iint_E \frac{1}{|t - s|^2} 2|t - s| ds dt \\ &= \iint_E \frac{ds dt}{|t - s|}. \end{aligned}$$

This is the stated bound, since $2 \iint_E \frac{ds dt}{2|t - s|} = \iint_E \frac{ds dt}{|t - s|}$. □

Where we are. The whole of Chapter 6.8 turns on the single computation in Step 3, so it is worth naming what makes it work. The Jacobian $2|t - s|$ appears *twice*, with opposite effects: once in the denominator of the density H , and once as the volume factor in the change of variables. In $\|H\|_{L^2}^2$ the density is squared but the volume factor is not, so one power of $|t - s|^{-1}$ survives — and that surviving power is exactly the integrand $|t - s|^{-1}$ whose logarithmic divergence at $s = t$ produces the endpoint failure of Theorem 6.8.6.

Had the exponent been L^3 rather than L^4 there would have been no algebraic identity to start from; had the curve been a surface, the fibres of Ψ would

have been positive-dimensional and there would have been no Jacobian. Both observations are made precise in Worked Example 6.8.3.

Theorem 6.8.5 (The local L^4 estimate). *For $R \geq 2$ and $\|g\|_{L^\infty(d\mu)} \leq 1$,*

$$\|\mathcal{E}g\|_{L^4(B_R)} \lesssim (\log R)^{1/4}.$$

Proof. Step 1. Split the pairs.

Write $\mathbf{1}_{[-1,1]^2} = \mathbf{1}_{E_{\text{near}}} + \mathbf{1}_{E_{\text{far}}}$ with

$$E_{\text{near}} = \{|t - s| < R^{-1}\}, \quad E_{\text{far}} = \{|t - s| \geq R^{-1}\},$$

and correspondingly $(\mathcal{E}g)^2 = H_{\text{near}}^\vee + H_{\text{far}}^\vee$.

Step 2. *The far pairs.*

By Theorem 6.8.4 with $E = E_{\text{far}}$,

$$\begin{aligned} \|H_{\text{far}}\|_{L^2}^2 &\leq 2 \iint_{|t-s| \geq R^{-1}} \frac{ds dt}{2|t-s|} \\ &\leq 2 \int_{-1}^1 \left(\int_{R^{-1} \leq |u| \leq 2} \frac{du}{2|u|} \right) ds \\ &\lesssim \log R, \end{aligned}$$

substituting $u = t - s$.

Step 3. *The near pairs.*

Here we do not use cancellation. By Cauchy–Schwarz,

$$\begin{aligned} |H_{\text{near}}^\vee(x)| &= \left| \iint_{|t-s| < R^{-1}} g(s)g(t) e^{2\pi i(x_1(s+t) + x_2(s^2+t^2))} ds dt \right| \\ &\leq m(E_{\text{near}}) \\ &\lesssim R^{-1}, \end{aligned}$$

so $\|H_{\text{near}}^\vee\|_{L^2(B_R)} \lesssim R^{-1}m(B_R)^{1/2} \lesssim R^{-1} \cdot R = 1$.

Step 4. *Combine.*

By Plancherel applied to the far part and Step 3 for the near part,

$$\begin{aligned} \|\mathcal{E}g\|_{L^4(B_R)}^2 &= \|(\mathcal{E}g)^2\|_{L^2(B_R)} \\ &\leq \|H_{\text{far}}\|_{L^2} + \|H_{\text{near}}^\vee\|_{L^2(B_R)} \\ &\lesssim (\log R)^{1/2} + 1, \end{aligned}$$

and taking square roots gives the claim. □

Proof of Theorem 6.8.2. Normalise $\|g\|_{L^\infty(\mathrm{d}\mu)} \leq 1$. The trivial bound gives

$$\|\mathcal{E}g\|_{L^\infty(\mathbb{R}^2)} \leq \int_{-1}^1 |g| \leq 2.$$

Fix $b > 4$ and let $\theta \in (0, 1)$ satisfy $\frac{1}{b} = \frac{1-\theta}{4}$, i.e. $1 - \theta = \frac{4}{b}$. By Hölder on B_R ,

$$\begin{aligned} \|\mathcal{E}g\|_{L^b(B_R)} &\leq \|\mathcal{E}g\|_{L^4(B_R)}^{1-\theta} \|\mathcal{E}g\|_{L^\infty}^\theta \\ &\lesssim (\log R)^{\frac{1-\theta}{4}} \\ &= (\log R)^{1/b}. \end{aligned}$$

Since $\log R \lesssim_\varepsilon R^\varepsilon$ for every $\varepsilon > 0$, this is a local estimate in the sense of Theorem 6.5.1 at exponent b , with $L^\infty(\mathrm{d}\mu)$ in place of $L^2(\mathrm{d}\sigma)$ on the right. The proof of Theorem 6.5.2 used nothing about the exponent on the data side — see Drill 5.4 — so it applies and removes the loss at any exponent strictly larger. As $b > 4$ was arbitrary, the global estimate holds for every $b > 4$. $\square$

Proposition 6.8.6 (The endpoint $b = 4$ is false). *There is no constant C with $\|\mathcal{E}g\|_{L^4(\mathbb{R}^2)} \leq C\|g\|_{L^\infty(\mathrm{d}\mu)}$.*

Proof. Take $g \equiv 1$, so $\|g\|_{L^\infty(\mathrm{d}\mu)} = 1$. By Theorem 6.8.4 — read as an identity rather than an inequality, since for $g \equiv 1$ the density H is exactly $\frac{2}{2|t-s|}$ at $\xi = \Psi(s, t)$ — we have

$$\|\mathcal{E}1\|_{L^4(\mathbb{R}^2)}^4 = \|H\|_{L^2}^2 = \iint_{[-1,1]^2} \frac{\mathrm{d}s \mathrm{d}t}{2|t-s|}.$$

Substituting $u = t - s$, the inner integral over u is $\int_{|u| \leq 2} \frac{\mathrm{d}u}{2|u|}$, which diverges logarithmically at $u = 0$. So $\|\mathcal{E}1\|_{L^4} = \infty$. $\square$

WORKED EXAMPLE 6.8.1— TIER B **Where the logarithm comes from, and why it is unavoidable**

Take $g \equiv 1$. Then Step 2 of Theorem 6.8.5 is an equality up to constants: the double integral genuinely behaves like $\log R$. So the L^4 norm of $\mathcal{E}1$ over B_R grows like $(\log R)^{1/4}$, and $\mathcal{E}1 \notin L^4(\mathbb{R}^2)$.

This is the same phenomenon as Theorem 6.2.5 at $b = 4$: there the obstruction was that $\widehat{\mathrm{d}\mu}$ decays like $|x|^{-1/2}$ and $|x|^{-1/2} \notin L^4(\mathbb{R}^2)$, by exactly one logarithm. The two computations are the same fact seen from opposite sides — one in physical space, one in frequency — and their agreement is a useful check.

Consequence for the statement. The theorem cannot be improved to $b = 4$, and the ε -removal of Chapter 6.5 cannot help, since it produces only open ranges. The endpoint $b = 4$ is genuinely false, unlike the Stein–Tomas endpoint

of Worked Example 6.3.3 which is true but needs a different technique. Distinguishing these two situations — an endpoint that is false from an endpoint that is merely hard — is worth doing every time one meets an open range.

WORKED EXAMPLE 6.8.2— TIER B **The transverse bilinear estimate, which is what survives**

The computation of Theorem 6.8.4 degenerates only on the diagonal $s = t$. Isolating that observation gives a statement that does generalise, and which is the seed of the entire bilinear method.

Proposition 6.8.7. *Let $I, J \subseteq [-1, 1]$ be intervals with $\text{dist}(I, J) \geq c > 0$, and let g_1, g_2 be supported in I, J respectively. Then*

$$\|\mathcal{E}g_1 \cdot \mathcal{E}g_2\|_{L^2(\mathbb{R}^2)} \leq c^{-1/2} \|g_1\|_{L^2} \|g_2\|_{L^2}.$$

Proof. By Plancherel, $\|\mathcal{E}g_1 \mathcal{E}g_2\|_{L^2} = \|(g_1 \, d\mu) * (g_2 \, d\mu)\|_{L^2}$, and the convolution is the pushforward under $\Psi(s, t) = (s + t, s^2 + t^2)$ of $g_1(s)g_2(t) \, ds \, dt$. Now $s \in I$ and $t \in J$ force $|t - s| \geq c$, so Ψ is injective on $I \times J$ — the two preimages (s, t) and (t, s) cannot both lie in $I \times J$, the intervals being disjoint — with Jacobian $2|t - s| \geq 2c$. Hence the pushforward has density

$$H(\xi) = \frac{g_1(s)g_2(t)}{2|t - s|} \quad \text{at } \xi = \Psi(s, t),$$

and changing variables back,

$$\begin{aligned} \|H\|_{L^2}^2 &= \iint_{I \times J} \frac{|g_1(s)|^2 |g_2(t)|^2}{4|t - s|^2} \cdot 2|t - s| \, ds \, dt \\ &= \iint \frac{|g_1(s)|^2 |g_2(t)|^2}{2|t - s|} \, ds \, dt \\ &\leq \frac{1}{2c} \|g_1\|_{L^2}^2 \|g_2\|_{L^2}^2. \end{aligned}$$

Taking square roots gives the claim, with the constant $(2c)^{-1/2} \leq c^{-1/2}$. $\square$

Why this is the right generalisation. Three features survive to $\mathbb{R}^n$ while the L^4 identity does not.

The hypothesis is *transversality* — the two pieces of the surface have separated normal directions — rather than anything about exponents. In $\mathbb{R}^n$ the corresponding statement bounds $\|\mathcal{E}g_1 \mathcal{E}g_2\|_{L^q}$ for q in a range, and transversality is again the hypothesis.

The conclusion is bilinear, so it says nothing directly about $\mathcal{E}g$ for a single g . Recovering a linear estimate requires decomposing the surface into pieces, applying the bilinear bound to transverse pairs, and handling the non-transverse pairs by induction on scales — the *broad-narrow* argument of Volume 12. And the constant blows up as $c \rightarrow 0$. All the difficulty of the bilinear method is in how that blow-up is balanced against the number of nearly-parallel pairs, which is exactly what the diagonal term of Theorem 6.8.5 was doing when it produced a logarithm. $\square$

WORKED EXAMPLE 6.8.3— TIER C: RESEARCH TASTE Why this argument is confined to the plane

Theorem 6.8.2 settles $n = 2$ completely. It is worth being precise about which step fails for $n \geq 3$, since the answer explains the shape of the rest of the series.

(a) *The exponent must be even.* The proof computes $\|\mathcal{E}g\|_{L^4}^4$ by writing it as $\|(\mathcal{E}g)^2\|_{L^2}^2$ and using Plancherel. That works because $4 = 2 \cdot 2$. In $\mathbb{R}^n$ the conjectured exponent is $\frac{2n}{n-1}$, which is an even integer only when $n = 2$ (giving 4) and $n = 3$ (giving 3, odd) — in fact only $n = 2$. For $n \geq 3$ there is no corresponding algebraic identity, and the whole method is unavailable.

(b) *The change of variables is special to a curve.* Theorem 6.8.3 used that two points on a curve determine the pair $(s + t, s^2 + t^2)$ up to swapping, so Ψ is two-to-one. For a hypersurface in $\mathbb{R}^n$, $n \geq 3$, the map $(\omega_1, \omega_2) \mapsto \omega_1 + \omega_2$ goes from a $2(n - 1)$ -dimensional domain to an n -dimensional target and has $(n - 2)$ -dimensional fibres. There is no Jacobian to compute; there is a fibre integral, and controlling it is precisely the bilinear restriction problem.

(c) *What survives.* The identity in (b) is the seed of the bilinear method: one studies $\mathcal{E}g_1 \cdot \mathcal{E}g_2$ for g_1, g_2 supported on *transverse* pieces of the surface, where the fibres are non-degenerate and the fibre integral is controllable. Tao's sharp bilinear restriction theorem is the culmination, and the multilinear theory of Volume 12 is what replaced it.

(d) *The moral.* The planar case is not a warm-up for the general case; it is a genuinely different argument that happens to be available in one dimension. A reader who expects the higher-dimensional proof to look like this one will be disappointed, and the historical record bears that out: Theorem 6.8.2 dates from around 1970 and the first improvement over Stein–Tomas in higher dimensions came more than twenty years later.

WORKED EXAMPLE 6.8.4— TIER A Checking the planar exponents against everything else

At $n = 2$ four different results in this volume make a prediction, and they must agree. Verifying that they do is a good use of ten minutes and catches most arithmetic slips.

(a) *The whole-curve condition.* Theorem 6.2.5 gives $b > \frac{2n}{n-1} = 4$. ✓ with Theorem 6.8.2.

(b) *The Knapp condition at $a = \infty$.* Theorem 6.2.3 gives $b \geq \frac{n+1}{n-1} = 3$, weaker than (a) — consistent with Worked Example 6.2.2, which said the whole-surface condition binds at $a = \infty$ once $a > \frac{2n}{n-1} = 4$. ✓

(c) *Stein–Tomas.* Theorem 6.3.1 gives $b > \frac{2(n+1)}{n-1} = 6$ for $L^2(d\mu)$ data. Since $L^\infty(d\mu) \subseteq L^2(d\mu)$ on a finite measure space, Theorem 6.8.2 at $b > 4$ is strictly stronger in the range $4 < b \leq 6$ — as it must be, being the conjecture. ✓

(d) *The Kakeya consequence.* Theorem 6.6.2 with $n = 2$ and $b \downarrow 4$ gives $\|\sum_j \mathbf{1}_{T_j}\|_{L^2} \lesssim \delta^{-\varepsilon}$, that is the Kakeya maximal conjecture in the plane. And that is a theorem independently — Córdoba’s argument, Theorem 1.6.14. ✓ Two proofs of the same planar fact, one via restriction and one direct.

The moral. In a subject with this many exponents, consistency checks at $n = 2$ and $n = 3$ are the cheapest available error detection. Every time a new estimate is proved in the volumes ahead, it is worth substituting $n = 2$ and asking whether the known planar answer comes out.

Drills

Drill 6.8.1. Verify the Jacobian computation in Theorem 6.8.3, and solve explicitly for s, t in terms of ξ_1, ξ_2 .

Drill 6.8.2. Compute $\iint_{|t-s| \geq R^{-1}} \frac{ds dt}{|t-s|}$ over $[-1, 1]^2$ exactly, and confirm the growth rate $\log R$.

Drill 6.8.3. Show that the parabola may be replaced by any curve $(t, \phi(t))$ with $\phi'' \neq 0$, by checking that $\Psi(s, t) = (s + t, \phi(s) + \phi(t))$ has non-vanishing Jacobian off the diagonal.

Drill 6.8.4. Explain why the argument of Theorem 6.8.5 fails for the circle, by locating the second degeneracy of the Jacobian. (Two points of a circle summing to a fixed ξ degenerate both when they coincide and when they are antipodal.)

Where we are. Three features of the planar proof are worth separating, because only one of them generalises.

The algebraic identity $\|F\|_{L^4}^2 = \|F^2\|_{L^2}$ requires the exponent to be even and is unavailable in higher dimensions.

The change of variables of Theorem 6.8.3 requires the surface to be a curve, so that two points determine their sum up to swapping. In higher dimensions the analogous map has positive-dimensional fibres.

The transversality principle — that the interaction of two pieces is controlled by the angle between them, with a constant blowing up as the angle closes — is the one that survives, and Worked Example 6.8.2 isolated it. Everything in Volume 12 is a version of that principle, adapted to n pieces at once rather than two.

The reader should therefore treat this chapter as evidence that the conjecture is true, and as the source of one durable idea, but not as a template. The higher-dimensional theory begins by discarding the first two features and keeping the third.

Notes

Theorem 6.8.2 is due to Fefferman and Stein, and independently to Zygmund, around 1970; the endpoint refinement — a restricted weak-type estimate at $b = 4$ — is Zygmund's and is not proved here. Córdoba's square-function proof gives the same result by a different route and generalises differently.

Drill 8.4 is the reason this chapter works with the parabola rather than the circle. For the circle the antipodal degeneracy contributes an extra singularity which must be handled separately; the standard treatment decomposes the circle into arcs and treats non-antipodal pairs by the argument above.

Chapter 6.A

Results Quoted from Volumes 1–5

Result	Source	Used in
Polar coordinates	Theorem 1.6.11	Theorem 6.2.5, Theorem 6.7.2
Density of $\mathcal{S}$ in L^p	Theorem 1.8.7	Theorem 6.1.1
Norms by testing	Theorem 1.8.10	Theorem 6.1.3
Minkowski's integral inequality	Theorem 1.8.4	Theorem 6.7.2
Change of variables	Theorem 1.3.10	Theorem 6.8.4
Riesz–Thorin	Theorem 2.2.1	Theorem 6.3.1 Theorem 6.3.1,
Young's convolution inequality	Worked Example 2.2.1	Theorem 5.8.2-style bounds
The Fourier dictionary	Theorem 2.4.2	Theorem 6.6.2
Plancherel	Theorem 2.4.8	Theorem 6.4.3, Theorem 6.8.5
$\mathcal{F}$ on $\mathcal{S}'$	Theorem 2.5.11	Theorem 6.3.2
L^2 multiplier norm	Theorem 2.6.2	Theorem 6.3.1
Khintchine's inequality	Theorem 2.6.5	Theorem 6.6.2
The Knapp example	Worked Example 2.4.3	Chapter 6.2
Takeya maximal operator	Theorem 1.7.13	Worked Example 6.6.2
Córdoba's planar theorem	Theorem 1.6.14	Worked Example 6.7.2
Fefferman's ball multiplier theorem	Theorem 3.8.2	Worked Example 6.7.2
Critical points are normal directions	Theorem 5.5.1	Theorem 6.4.5, Theorem 6.4.3
Asymptotics of $\widehat{d\sigma}$	Theorem 5.5.2	Theorem 6.3.2, Theorem 6.2.5
Decay with k curvatures	Theorem 5.5.3	Theorem 6.4.5, Worked Example 6.3.2
Bochner–Riesz above the critical index	Theorem 5.8.4	Theorem 6.7.5

Debts

This volume incurs no debts on later volumes, and discharges none — Volumes 1–5 left none outstanding except Theorem 3.8.4, which is owed to Volume 9.

Two results are removed from the plan, announced where they would have appeared and recorded in the master manifest: the endpoint of the Stein–Tomas theorem and the Keel–Tao endpoint Strichartz estimate (preface, and Worked Example 6.3.3), and the implication that the Bochner–Riesz conjecture implies the restriction conjecture (Theorem 6.7.5, revision R5).

Chapter 6.B

Notation and Index of Results

Additions to the Volume 1–5 conventions

$\mathcal{R}$	the restriction operator $f \mapsto \widehat{f} _S$
$\mathcal{E}$	the extension operator $g \mapsto (g d\sigma)^\vee$
$R^*(p \rightarrow q)$	the estimate $\ \mathcal{R}f\ _{L^q(d\sigma)} \lesssim \ f\ _{L^p}$
κ_δ, T_δ	a cap of radius δ and its dual box
$N_\delta(S)$	the δ -neighbourhood of S
$\mathbb{T}$	a family of δ -tubes

Index of the main results

Restriction–extension duality	Theorem 6.1.3
The TT^* identity	Theorem 6.1.5
The cap does not oscillate on its dual box	Theorem 6.2.2
The Knapp condition	Theorem 6.2.3
The whole-sphere condition	Theorem 6.2.5
The restriction conjecture	Theorem 6.2.6
Kernel decomposition	Theorem 6.3.2
Stein–Tomas, non-endpoint	Theorem 6.3.1
Strichartz estimates	Theorem 6.4.3
The cone and the wave equation	Theorem 6.4.5
Epsilon-removal	Theorem 6.5.2
Restriction implies Takeya	Theorem 6.6.2
The δ -neighbourhood formulation	Theorem 6.7.2
The chain of conjectures	Theorem 6.7.4
The change of variables in the plane	Theorem 6.8.3
The exact L^4 identity	Theorem 6.8.4
The local L^4 estimate	Theorem 6.8.5
Restriction in the plane	Theorem 6.8.2

Plan revisions in this volume

Removed	Reason
Stein–Tomas endpoint $b = \frac{2(n+1)}{n-1}$	needs Stein interpolation, removed as R4; see Worked Example 6.3.3
Keel–Tao endpoint Strichartz	bilinear; see Worked Example 6.4.4
Bochner–Riesz $\implies$ restriction	Tao (1999), a research paper; Theorem 6.7.5, revision R5
Zygmund’s endpoint in the plane	restricted weak type at $b = 4$; the open range is Theorem 6.8.2

Part 7

Volume 7

Hausdorff Dimension, Fourier Dimension, and Fractals

Preface to Volume 7

Why measure is not enough

Volume 6 ended with Theorem 6.6.3: granting the restriction conjecture, a Besicovitch set has Minkowski dimension n . That is weaker than what the Kakeya conjecture asserts, which is that the *Hausdorff* dimension is n , and the gap is not a technicality — the rationals in $[0, 1]$ have Minkowski dimension 1 and Hausdorff dimension 0.

The reason Volume 6 could not do better is that its only tool was Lebesgue measure, applied at a single scale δ at a time. Hausdorff dimension is a statement about all scales at once, and measuring it requires a genuinely different apparatus: outer measures indexed by an exponent, a notion of dimension defined by where they jump, covering theorems adapted to them, and a way of producing measures on a set from knowledge of its dimension. This volume builds that apparatus.

What is assumed

Volumes 1–6. Most heavily Volume 1: the construction of Hausdorff measure as a metric outer measure (Worked Example 1.2.4), the Carathéodory criterion (Theorem 1.2.13), and the covering lemmas of Chapter 7. From Volume 2, the Fourier transform of measures; from Volume 6, the statement of the Kakeya maximal conjecture.

Scope, fixed in advance

Two results a reader might expect are handled differently from the original plan, and both decisions were taken before the chapters were written.

The *Besicovitch covering theorem* is not proved. Its correct proof splits into two cases according to the ratio of the radii, and the naive angle argument that suggests itself is false — there are configurations satisfying every constraint of the greedy selection with angle arbitrarily close to zero. The full argument is

several pages and nothing in this series depends on it: the $5r$ -covering lemma of Chapter 7.4 suffices for every application here. This is recorded as revision R6.

The existence of a Besicovitch set of *measure zero* — promised in Volume 1 — is deferred to Volume 8 rather than proved here. The natural route is Kahane's construction, which produces the set from a Cantor set of directions and needs Marstrand's projection theorem to compute the measure; Marstrand is Volume 8, Chapter 1. Chapter 7.3 explains why the obvious compactness argument, which one expects to work, does not.

The plan

1	Hausdorff measure and dimension	the exponent where the measure jumps
2	Minkowski dimension	one scale at a time, and what it misses
3	The Hausdorff metric	compactness in the space of compact sets
4	Covering theorems	differentiation, and densities
5	Frostman's lemma	dimension produces a measure
6	Energy integrals	dimension as an integral
7	Fourier dimension	when the transform sees the dimension
8	Self-similar sets	and the Kakeya upgrade

Chapter 7.1

Hausdorff Measure and Dimension

7.1.1 The construction, and what Volume 1 already gives

Definition 7.1.1. For $s \geq 0$, $\varepsilon > 0$ and $E \subseteq \mathbb{R}^n$ set

$$\mathcal{H}_\varepsilon^s(E) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam } U_i)^s : E \subseteq \bigcup_i U_i, \text{diam } U_i \leq \varepsilon \right\},$$

with the conventions $(\text{diam } \emptyset)^0 = 0$ and $\inf \emptyset = +\infty$. Since $\mathcal{H}_\varepsilon^s$ increases as ε decreases, the limit

$$\mathcal{H}^s(E) = \lim_{\varepsilon \downarrow 0} \mathcal{H}_\varepsilon^s(E) = \sup_{\varepsilon > 0} \mathcal{H}_\varepsilon^s(E)$$

exists in $[0, \infty]$ and is called the *s-dimensional Hausdorff outer measure*.

Worked Example 1.2.4 carried out this construction and Theorem 1.2.13 established what it gives:

Proposition 7.1.2. $\mathcal{H}^s$ is a metric outer measure on $\mathbb{R}^n$; consequently every Borel set is $\mathcal{H}^s$ -measurable, and $\mathcal{H}^s$ restricted to the Borel sets is a Borel measure. Moreover $\mathcal{H}^0$ is counting measure, and $\mathcal{H}^s$ is invariant under isometries and satisfies $\mathcal{H}^s(\lambda E) = \lambda^s \mathcal{H}^s(E)$ for $\lambda > 0$.

Proof. That $\mathcal{H}^s$ is an outer measure is Theorem 1.2.8, the covering family being all sets of small diameter. For the metric property, let $\text{dist}(A, B) = d > 0$ and take $\varepsilon < d/2$; then no set of diameter $\leq \varepsilon$ meets both A and B , so any admissible cover of $A \cup B$ splits into one of A and one of B , giving

$\mathcal{H}_\varepsilon^s(A \cup B) = \mathcal{H}_\varepsilon^s(A) + \mathcal{H}_\varepsilon^s(B)$. Letting $\varepsilon \downarrow 0$ and applying Theorem 1.2.13 gives measurability of Borel sets.

For $s = 0$: a set of diameter $\leq \varepsilon$ contains at most one point of a set whose points are ε -separated, so $\mathcal{H}_\varepsilon^0(E)$ counts points once ε is small. Isometry invariance is clear since diameters are preserved, and the scaling law follows because $\text{diam}(\lambda U) = \lambda \text{diam } U$. $\square$

7.1.2 The critical exponent

Theorem 7.1.3. *Let $E \subseteq \mathbb{R}^n$ and $0 \leq s < t$.*

(i) *If $\mathcal{H}^s(E) < \infty$ then $\mathcal{H}^t(E) = 0$.*

(ii) *If $\mathcal{H}^t(E) > 0$ then $\mathcal{H}^s(E) = \infty$.*

Consequently there is a unique $d \in [0, \infty]$ such that $\mathcal{H}^s(E) = \infty$ for $s < d$ and $\mathcal{H}^s(E) = 0$ for $s > d$.

Proof. Let $\{U_i\}$ cover E with $\text{diam } U_i \leq \varepsilon$. Then

$$\sum_i (\text{diam } U_i)^t = \sum_i (\text{diam } U_i)^{t-s} (\text{diam } U_i)^s \leq \varepsilon^{t-s} \sum_i (\text{diam } U_i)^s,$$

so $\mathcal{H}_\varepsilon^t(E) \leq \varepsilon^{t-s} \mathcal{H}_\varepsilon^s(E) \leq \varepsilon^{t-s} \mathcal{H}^s(E)$. If $\mathcal{H}^s(E) < \infty$, letting $\varepsilon \downarrow 0$ gives $\mathcal{H}^t(E) = 0$. Statement (ii) is the contrapositive.

For the last claim, put $d = \inf\{s \geq 0 : \mathcal{H}^s(E) = 0\}$, with $\inf \emptyset = n$. If $s > d$ then some $s' \in (d, s)$ has $\mathcal{H}^{s'}(E) = 0 < \infty$, so $\mathcal{H}^s(E) = 0$ by (i); if $s < d$ then $\mathcal{H}^s(E) \neq 0$, and were it finite (i) would force $\mathcal{H}^{s'}(E) = 0$ for $s' \in (s, d)$, contradicting the definition of d . So $\mathcal{H}^s(E) = \infty$. $\square$

Where we are. The theorem gives $d \in [0, \infty]$, not $d \in [0, n]$. The bound $d \leq n$ is *not* available yet: it requires knowing $\mathcal{H}^s(\mathbb{R}^n) = 0$ for $s > n$, which is Theorem 7.1.9 below and depends on comparing $\mathcal{H}^n$ with Lebesgue measure.

Stating the sharper range here would make the development circular — Theorem 7.1.8 uses part (i) of this theorem in its own proof. Splitting the two claims keeps the dependency order strict, and Theorem 7.1.9 records the bound as soon as it is earned.

Definition 7.1.4. The number d of Theorem 7.1.3 is the *Hausdorff dimension* $\dim_{\text{H}} E$.

Pitfall: the measure at the critical exponent can be anything

$\dim_{\text{H}} E = d$ says nothing about $\mathcal{H}^d(E)$, which may be 0, a finite positive

number, or ∞ . All three occur: $\mathcal{H}^1$ of a line segment is its length; $\mathcal{H}^1(\mathbb{R}) = \infty$ while $\dim_{\mathbb{H}} \mathbb{R} = 1$; and there are sets of dimension d with $\mathcal{H}^d = 0$, constructed in Drill 1.4.

The distinction matters in this series. The Kakeya conjecture asserts $\dim_{\mathbb{H}} K = n$ for every Besicovitch set K ; it does *not* assert $\mathcal{H}^n(K) > 0$, which is false — Besicovitch sets of Lebesgue measure zero exist, and $\mathcal{H}^n$ is comparable to Lebesgue measure. So the conjecture is exactly a statement about the critical exponent and not about the measure there.

Proposition 7.1.5 (Elementary properties). *(i) $E \subseteq F$ implies $\dim_{\mathbb{H}} E \leq \dim_{\mathbb{H}} F$.*

(ii) $\dim_{\mathbb{H}} (\bigcup_{i \geq 1} E_i) = \sup_i \dim_{\mathbb{H}} E_i$.

(iii) If $f : E \rightarrow \mathbb{R}^m$ is Lipschitz with constant L then $\mathcal{H}^s(f(E)) \leq L^s \mathcal{H}^s(E)$, and hence $\dim_{\mathbb{H}} f(E) \leq \dim_{\mathbb{H}} E$.

(iv) If f is Hölder of exponent $\alpha \in (0, 1]$ then $\dim_{\mathbb{H}} f(E) \leq \dim_{\mathbb{H}} E / \alpha$.

(v) $\dim_{\mathbb{H}} E = 0$ for every countable E .

Proof. (i) is monotonicity of $\mathcal{H}^s$. For (ii), “ $\geq$ ” is (i); for “ $\leq$ ”, if $s > \sup_i \dim_{\mathbb{H}} E_i$ then $\mathcal{H}^s(E_i) = 0$ for every i , so $\mathcal{H}^s(\bigcup E_i) = 0$ by countable subadditivity.

(iii) If $\{U_i\}$ covers E with $\text{diam } U_i \leq \varepsilon$ then $\{f(U_i \cap E)\}$ covers $f(E)$ with $\text{diam } f(U_i \cap E) \leq L \text{diam } U_i \leq L\varepsilon$, and $\sum (\text{diam } f(U_i \cap E))^s \leq L^s \sum (\text{diam } U_i)^s$. Take infima and let $\varepsilon \downarrow 0$. (iv) is the same computation with $\text{diam } f(U_i \cap E) \leq L(\text{diam } U_i)^\alpha$, giving $\mathcal{H}^{s/\alpha}(f(E)) \leq L^{s/\alpha} \mathcal{H}^s(E)$.

(v) A single point has $\mathcal{H}^s = 0$ for every $s > 0$; apply (ii). $\square$

7.1.3 The mass distribution principle

Upper bounds on dimension come from exhibiting covers. Lower bounds are harder, and the following is the elementary tool; Chapter 7.5 proves that it is the only one needed.

Lemma 7.1.6 (Mass distribution principle). *Let μ be a Borel measure on $\mathbb{R}^n$ with $\mu(E) > 0$, and suppose there are $c, \varepsilon_0 > 0$ with*

$$\mu(U) \leq c(\text{diam } U)^s \quad \text{for every } U \text{ with } \text{diam } U \leq \varepsilon_0.$$

Then $\mathcal{H}^s(E) \geq \mu(E)/c$, and in particular $\dim_{\mathbb{H}} E \geq s$.

Proof. Let $\{U_i\}$ cover E with $\text{diam } U_i \leq \varepsilon \leq \varepsilon_0$. Then

$$\mu(E) \leq \mu\left(\bigcup_i U_i\right) \leq \sum_i \mu(U_i) \leq c \sum_i (\text{diam } U_i)^s.$$

Taking the infimum over covers gives $\mu(E) \leq c \mathcal{H}_\varepsilon^s(E)$, and letting $\varepsilon \downarrow 0$ gives the claim. Since $\mu(E) > 0$, $\mathcal{H}^s(E) > 0$ and so $\dim_{\mathbb{H}} E \geq s$ by Theorem 7.1.3. $\square$

WORKED EXAMPLE 7.1.1— TIER A **The middle-thirds Cantor set**

Let $C \subseteq [0, 1]$ be the middle-thirds Cantor set of Volume 1, and $s = \log 2 / \log 3$. We show $\dim_{\mathbb{H}} C = s$.

Upper bound. At stage k , C is covered by 2^k intervals of length 3^{-k} . Hence

$$\mathcal{H}_{3^{-k}}^s(C) \leq 2^k (3^{-k})^s = 2^k 3^{-ks} = 2^k 2^{-k} = 1,$$

using $3^s = 2$. Letting $k \rightarrow \infty$, $\mathcal{H}^s(C) \leq 1$, so $\dim_{\mathbb{H}} C \leq s$.

Lower bound. Let μ be the natural measure on C : the unique Borel measure giving mass 2^{-k} to each of the 2^k stage- k intervals. (It exists by Theorem 1.2.15 applied to the algebra generated by those intervals.) Let U have $\text{diam } U \leq 1$ and choose k with $3^{-(k+1)} \leq \text{diam } U < 3^{-k}$. An interval of length $< 3^{-k}$ meets at most 2 of the stage- k intervals, since consecutive ones are separated by a gap of length 3^{-k} . Hence

$$\begin{aligned} \mu(U) &\leq 2 \cdot 2^{-k} \\ &= 2 \cdot 3^{-ks} \\ &\leq 2 \cdot (3 \text{diam } U)^s \\ &= 2 \cdot 3^s (\text{diam } U)^s \\ &= 6 (\text{diam } U)^s. \end{aligned}$$

By Theorem 7.1.6 with $c = 6$, $\mathcal{H}^s(C) \geq \mu(C)/6 = 1/6 > 0$, so $\dim_{\mathbb{H}} C \geq s$.

Conclusion. $\dim_{\mathbb{H}} C = \log 2 / \log 3 \approx 0.6309$, and $\frac{1}{6} \leq \mathcal{H}^s(C) \leq 1$. In fact $\mathcal{H}^s(C) = 1$, but the sharp constant needs a density argument and is not needed here.

WORKED EXAMPLE 7.1.2— TIER A **A set of every dimension**

For $s \in (0, 1)$ let C_λ be the Cantor set obtained by removing, at each stage, the middle proportion $1 - 2\lambda$ of every interval, so that stage k consists of 2^k intervals of length λ^k , with $0 < \lambda < \frac{1}{2}$.

Repeating Worked Example 7.1.1 verbatim with 3 replaced by λ^{-1} gives

$$\dim_{\mathbb{H}} C_\lambda = \frac{\log 2}{\log(1/\lambda)},$$

which runs over all of $(0, 1)$ as λ runs over $(0, \frac{1}{2})$. Taking products $C_\lambda \times [0, 1]^k$ realises every value in $(0, n)$ in $\mathbb{R}^n$, using the product formula of Drill 1.3. So the Hausdorff dimension takes every value in $[0, n]$, and in particular it is not restricted to integers. That is the whole point of the notion, and the contrast with topological dimension — which is an integer — is what makes it the right tool forakeya, where the conjectured answer n is an integer but every known lower bound is not.

Proposition 7.1.7 (Dimension of a countable union and of a graph of a measure). *Let $E \subseteq \mathbb{R}^n$ be Borel with $\mathcal{H}^s(E) < \infty$, and let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be Lipschitz. Then for $\mathcal{H}^s$ -almost every value the fibre is small in the following sense: if $s > m$ then*

$$\int_{\mathbb{R}^m} \mathcal{H}^{s-m}(E \cap f^{-1}(y)) \, dy \lesssim_{n,m,s} L^m \mathcal{H}^s(E),$$

where L is the Lipschitz constant. In particular $\mathcal{H}^{s-m}(E \cap f^{-1}(y)) < \infty$ for Lebesgue-almost every y .

Proof. Let $\{U_i\}$ cover E with $\text{diam } U_i \leq \varepsilon$ and $\sum_i (\text{diam } U_i)^s \leq \mathcal{H}^s(E) + \eta$. For each i , $f(U_i)$ has diameter at most $L \text{diam } U_i$, so it is contained in a ball of that radius and

$$m(f(U_i)) \leq \omega_m (L \text{diam } U_i)^m.$$

For a fixed y , the sets U_i with $y \in f(U_i)$ cover $E \cap f^{-1}(y)$ and have diameter at most ε , so

$$\mathcal{H}_\varepsilon^{s-m}(E \cap f^{-1}(y)) \leq \sum_i (\text{diam } U_i)^{s-m} \mathbf{1}_{f(U_i)}(y).$$

Integrating in y and using the display above,

$$\begin{aligned} \int \mathcal{H}_\varepsilon^{s-m}(E \cap f^{-1}(y)) \, dy &\leq \sum_i (\text{diam } U_i)^{s-m} m(f(U_i)) \\ &\leq \omega_m L^m \sum_i (\text{diam } U_i)^s. \end{aligned}$$

Letting $\varepsilon \downarrow 0$, with Fatou on the left and the choice of cover on the right, gives the bound with $\mathcal{H}^s(E) + \eta$; let $\eta \downarrow 0$. $\square$

Where we are. Theorem 7.1.7 is a Fubini-type statement for Hausdorff measure, and it is the reason dimension behaves predictably under slicing: a set of dimension s meets almost every fibre of a Lipschitz map to $\mathbb{R}^m$ in

dimension at most $s - m$. The reverse inequality is false in general — fibres can be smaller than the count suggests, since f may be far from surjective — which is why the statement is one-sided.

It will be used in Volume 8 in the contrapositive: to prove a set is large, one shows many of its slices are large. That is exactly the structure of Marstrand's projection theorem, where the slices are the fibres of π_θ and the averaging over θ is what Worked Example 7.6.2(a) performs.

7.1.4 Comparison with Lebesgue measure

Proposition 7.1.8. *There are constants $0 < c_n \leq C_n < \infty$ with*

$$c_n m(E) \leq \mathcal{H}^n(E) \leq C_n m(E) \quad \text{for every } E \subseteq \mathbb{R}^n.$$

In particular $\mathcal{H}^s(\mathbb{R}^n) = 0$ for $s > n$, and $\dim_{\mathbb{H}} E \leq n$ always.

Proof. Step 1. $m \lesssim \mathcal{H}^n$.

Any set U of diameter d is contained in a closed ball of radius d , so $m(U) \leq \omega_n d^n$ where $\omega_n = m(B(0, 1))$. Given a cover $\{U_i\}$ of E , $m(E) \leq \sum_i m(U_i) \leq \omega_n \sum_i (\text{diam } U_i)^n$. Taking the infimum, $m(E) \leq \omega_n \mathcal{H}_\varepsilon^n(E) \leq \omega_n \mathcal{H}^n(E)$.

Step 2. $\mathcal{H}^n \lesssim m$.

It suffices to treat E of finite Lebesgue measure. Given $\eta > 0$, choose an open $V \supseteq E$ with $m(V) \leq m(E) + \eta$ (outer regularity, Theorem 1.3.6). Decompose V into dyadic cubes Q_i with disjoint interiors and $\text{diam } Q_i \leq \varepsilon$, which is possible by subdividing the standard dyadic decomposition of an open set. Then $\{Q_i\}$ covers E and, writing ℓ_i for the side length, $\text{diam } Q_i = \sqrt{n} \ell_i$, so

$$\begin{aligned} \sum_i (\text{diam } Q_i)^n &= n^{n/2} \sum_i \ell_i^n \\ &= n^{n/2} \sum_i m(Q_i) \\ &= n^{n/2} m(V) \\ &\leq n^{n/2} (m(E) + \eta). \end{aligned}$$

Hence $\mathcal{H}_\varepsilon^n(E) \leq n^{n/2} (m(E) + \eta)$; let $\varepsilon \downarrow 0$ and $\eta \downarrow 0$.

Step 3. *Consequences.*

By Step 2, $\mathcal{H}^n$ is finite on bounded sets, so Theorem 7.1.3(i) gives $\mathcal{H}^s = 0$ on bounded sets for $s > n$, and countable subadditivity extends this to $\mathbb{R}^n$. $\square$

Corollary 7.1.9 (The dimension is at most n). *For every $E \subseteq \mathbb{R}^n$, $\dim_{\mathbb{H}} E \leq n$; that is, the exponent d of Theorem 7.1.3 lies in $[0, n]$.*

Proof. By Theorem 7.1.8, $\mathcal{H}^n$ is finite on bounded sets. Fix a bounded E ; then $\mathcal{H}^n(E) < \infty$, so Theorem 7.1.3(i) gives $\mathcal{H}^s(E) = 0$ for every $s > n$, whence $d \leq n$. A general E is a countable union of bounded sets and $\dim_{\mathbb{H}}$ is countably stable (Theorem 7.1.5(ii)). $\square$

Remark 7.1.10. The sharp constant is $\mathcal{H}^n = 2^n \omega_n^{-1} m$ with the normalisation of Theorem 7.1.1, and proving it requires the isodiametric inequality — that among sets of given diameter the ball has largest volume. We do not prove it, and nothing here needs it: every use of Theorem 7.1.8 is qualitative, being a statement about which sets are null.

WORKED EXAMPLE 7.1.3— TIER B Dimension of a graph, and how much regularity buys

Let $f : [0, 1] \rightarrow \mathbb{R}$ and $\Gamma_f = \{(x, f(x)) : x \in [0, 1]\}$ its graph. The dimension of Γ_f measures how badly f oscillates, and the computation is a good exercise in both bounds.

If f is Lipschitz. The map $x \mapsto (x, f(x))$ is Lipschitz with constant $\sqrt{1 + L^2}$, so by Theorem 7.1.5(iii), $\dim_{\mathbb{H}} \Gamma_f \leq \dim_{\mathbb{H}} [0, 1] = 1$; and the projection $(x, y) \mapsto x$ is Lipschitz and maps Γ_f onto $[0, 1]$, so $\dim_{\mathbb{H}} \Gamma_f \geq 1$. Hence $\dim_{\mathbb{H}} \Gamma_f = 1$: a Lipschitz graph is one-dimensional, as it should be.

If f is only Hölder of exponent α . Now $x \mapsto (x, f(x))$ is Hölder of exponent α , and Theorem 7.1.5(iv) gives $\dim_{\mathbb{H}} \Gamma_f \leq 1/\alpha$. That is a genuine loss, and it is attained: the Weierstrass function $W(x) = \sum_k \lambda^{-k\alpha} \cos(\lambda^k x)$ is Hölder- α and its graph has dimension $2 - \alpha$ for λ large — a theorem of Shen (2018), which we do not prove.

The lesson. Between 1 and $1/\alpha$ there is room, and the upper bound from Hölder continuity alone is not sharp: $2 - \alpha < 1/\alpha$ for $\alpha \in (0, 1)$. Dimension is not determined by a modulus of continuity, and computing it always requires information about the specific set.

WORKED EXAMPLE 7.1.4— TIER C: RESEARCH TASTE The Kakeya conjecture, stated properly at last

Volume 1 introduced Besicovitch sets and Volume 6 could only reach Minkowski dimension. We can now state the conjecture the series is aimed at.

Conjecture 7.1.11 (Kakeya). *For $n \geq 2$, every Besicovitch set $K \subseteq \mathbb{R}^n$ — every compact set containing a unit line segment in every direction — satisfies*

$\dim_{\text{H}} K = n$.

(a) *Why $\dim_{\text{H}}$ and not m .* Because the measure statement is *false*: there are Besicovitch sets of Lebesgue measure zero, and by Theorem 7.1.8 therefore with $\mathcal{H}^n(K) = 0$. So the strongest true statement must be about the critical exponent, and by the Pitfall in §1.2 a set can have $\dim_{\text{H}} K = n$ with $\mathcal{H}^n(K) = 0$. The conjecture asks for exactly that.

(b) *Why $\dim_{\text{H}}$ and not $\dim_{\text{M}}$.* Because Minkowski dimension is the weaker assertion: Chapter 7.2 proves $\dim_{\text{H}} \leq \dim_{\text{M}}$, so a Minkowski bound follows from a Hausdorff bound and not conversely. The known results reflect this — Minkowski-dimension bounds for Besicovitch sets were obtained decades before the corresponding Hausdorff bounds, and in some dimensions the gap remains.

(c) *What is known.* $n = 2$ is a theorem: $\dim_{\text{H}} K = 2$ for every planar Besicovitch set, by an argument in the spirit of Theorem 1.6.14, carried out in Volume 9. For $n = 3$ the conjecture was settled by Wang and Zahl in 2025 and is the subject of Volume 14. For $n \geq 4$ it is open, with the best bounds coming from the polynomial and multilinear methods of Volumes 11 and 12.

(d) *What this volume contributes.* Nothing towards the conjecture directly. It builds the language in which the conjecture is stated and the tools — Frostman’s lemma, energy integrals — by which dimension is estimated in practice. Chapter 7.8 closes the loop by showing how a Kakeya maximal function bound of the kind Volume 6 produced upgrades to a Hausdorff dimension bound, which is the step Volume 6 could not take.

WORKED EXAMPLE 7.1.5— TIER B Proving a lower bound on dimension: the standard workflow

Upper bounds on $\dim_{\text{H}}$ come from exhibiting a cover, which is usually easy. Lower bounds are the work, and Theorem 7.1.6 reduces them to a fixed routine. Here it is, with the three places errors are made.

The routine. To show $\dim_{\text{H}} E \geq s$: (1) construct a measure μ on E with $\mu(E) > 0$; (2) verify $\mu(U) \lesssim (\text{diam } U)^s$ for all small U ; (3) apply Theorem 7.1.6.

Step 1 in practice. For a set built by a nested construction — $E = \bigcap_k E_k$ with E_k a union of N_k pieces — define μ by giving each stage- k piece mass $1/N_k$ and extending by Theorem 1.2.15. Consistency requires each stage- k piece to contain the same number N_{k+1}/N_k of stage- $(k+1)$ pieces; when it does not, one weights proportionally instead.

Step 2 in practice. Given U with $\text{diam } U$ small, find the stage k with $\text{diam } U$ comparable to the stage- k piece size ρ_k , and count how many stage- k pieces U can meet. *This is where the separation between pieces is used:* if consecutive pieces are separated by a gap $\gtrsim \rho_k$, then U meets $O(1)$ of them and $\mu(U) \lesssim 1/N_k \approx \rho_k^s$.

The three errors.

- *Forgetting the gaps.* Without separation, U may meet many pieces and the bound fails. A construction with pieces touching — for instance removing nothing — gives $\mu(U) \approx \text{diam } U$, and only $s \leq 1$ follows.
- *Checking only U from the construction.* The hypothesis of Theorem 7.1.6 quantifies over *all* sets U , not just the pieces. It is the arbitrary U that forces the choice of k in Step 2.
- *Losing $\mu(E) > 0$.* If the mass is normalised at each stage the total is 1 automatically; if instead one assigns absolute masses, they may tend to zero and the conclusion is vacuous. Theorem 7.5.1 deals with exactly this difficulty in general.

A check on the routine. Applying it to $E = [0, 1]$ with the pieces being dyadic intervals gives $\mu = \text{Lebesgue}$, $\mu(U) \leq \text{diam } U$, and $s = 1$ — the correct answer, and a reassurance that no step is losing anything.

Drills

Drill 7.1.1. Show that in Theorem 7.1.1 one may restrict the covering sets to be open, or to be closed, or to be convex, without changing $\mathcal{H}^s$.

Drill 7.1.2. Prove that $\mathcal{H}^1$ on $\mathbb{R}$ equals Lebesgue measure exactly, with no constant.

Drill 7.1.3. Show $\dim_{\text{H}}(E \times F) \geq \dim_{\text{H}} E + \dim_{\text{H}} F$ for Borel E, F , using Theorem 7.1.6 and a product measure. (Equality can fail; the example is due to Besicovitch and Moran.)

Drill 7.1.4. Construct a set $E \subseteq [0, 1]$ with $\dim_{\text{H}} E = s$ and $\mathcal{H}^s(E) = 0$, by taking a Cantor set whose ratios vary with the stage.

Notes

Hausdorff introduced the measure in 1918, extending Carathéodory's construction of 1914. The mass distribution principle is folklore; its converse, that a measure with the required decay always exists, is Frostman's lemma (Chapter 7.5) and is much deeper.

The value $\mathcal{H}^s(C) = 1$ for the middle-thirds Cantor set is due to Marion and to Ayer–Strichartz; the sharp constant in Theorem 7.1.8 is classical and follows from the isodiametric inequality.

Chapter 7.2

Minkowski Dimension

Hausdorff dimension is defined by a covering problem in which the covering sets may have wildly different sizes. Minkowski dimension asks the same question with all sets of the *same* size, and is correspondingly cruder and much easier to estimate. This chapter develops it and quantifies exactly what is lost — which is what Volume 6 ran into.

7.2.1 Box counting

Definition 7.2.1. For a bounded $E \subseteq \mathbb{R}^n$ and $\delta > 0$ let $N(E, \delta)$ be the least number of sets of diameter at most δ needed to cover E . The *lower* and *upper Minkowski dimensions* are

$$\underline{\dim}_M E = \liminf_{\delta \downarrow 0} \frac{\log N(E, \delta)}{\log(1/\delta)}, \quad \overline{\dim}_M E = \limsup_{\delta \downarrow 0} \frac{\log N(E, \delta)}{\log(1/\delta)}.$$

When they agree the common value is written $\dim_M E$.

Proposition 7.2.2 (The definitions agree). *In Theorem 7.2.1 one may replace $N(E, \delta)$ by any of*

- (i) *the least number of closed balls of radius δ covering E ;*
- (ii) *the number of δ -grid cubes meeting E ;*
- (iii) *the largest number of disjoint balls of radius δ centred in E ;*
- (iv) *$m(E_\delta)\delta^{-n}$, where E_δ is the δ -neighbourhood,*

without changing either limit.

Proof. Write $N_1, \dots, N_4$ for the four quantities. Each is comparable to $N = N(E, \delta)$ up to a constant factor and a bounded change of scale, and since

$$\begin{aligned} & \frac{\log(CN(E, c\delta))}{\log(1/\delta)} \\ &= \frac{\log N(E, c\delta)}{\log(1/(c\delta))} \cdot \frac{\log(1/(c\delta))}{\log(1/\delta)} + \frac{\log C}{\log(1/\delta)} \end{aligned}$$

and both correction factors tend to 1 and 0, such changes do not affect the limits.

The comparisons: a set of diameter δ lies in a ball of radius δ , so $N_1 \leq N$; a ball of radius δ has diameter 2δ , so $N \leq N_1(E, \delta/2)$. A δ -grid cube has diameter $\sqrt{n}\delta$ and any set of diameter δ meets at most 2^n grid cubes, giving $N_2 \leq 2^n N$ and $N \leq N_2$ after rescaling. For (iii), if $B_1, \dots, B_{N_3}$ are disjoint balls of radius δ centred in E then no set of diameter δ meets two of them, so $N_3 \leq N$; conversely a maximal such disjoint family has the property that the doubled balls cover E , so $N_1(E, 2\delta) \leq N_3$. For (iv), if N_3 disjoint balls of radius δ are centred in E then E_δ contains them, so $m(E_\delta) \geq N_3 \omega_n \delta^n$; and E_δ is covered by the $N_1(E, \delta)$ balls of radius 2δ , giving $m(E_\delta) \leq N_1 \omega_n (2\delta)^n$. $\square$

Proposition 7.2.3 (Minkowski content). *Let $E \subseteq \mathbb{R}^n$ be bounded and $s \geq 0$. Define the upper and lower s -dimensional Minkowski contents*

$$\begin{aligned} \overline{\mathcal{M}}^s(E) &= \limsup_{\delta \downarrow 0} \frac{m(E_\delta)}{\delta^{n-s}} \\ &, \quad \underline{\mathcal{M}}^s(E) = \liminf_{\delta \downarrow 0} \frac{m(E_\delta)}{\delta^{n-s}}. \end{aligned}$$

Then $\overline{\dim}_M E = \inf\{s : \overline{\mathcal{M}}^s(E) = 0\}$ and $\underline{\dim}_M E = \inf\{s : \underline{\mathcal{M}}^s(E) = 0\}$, and

$$\mathcal{H}^s(E) \lesssim_n \underline{\mathcal{M}}^s(E).$$

Proof. By Theorem 7.2.2(iv), $m(E_\delta) \approx_n N(E, \delta)\delta^n$, so $m(E_\delta)\delta^{s-n} \approx_n N(E, \delta)\delta^s$. The quantity $N(E, \delta)\delta^s$ tends to 0 precisely when $\liminf_\delta \frac{\log N}{\log(1/\delta)} < s$, which gives the two identities.

For the inequality, cover E by $N(E, \delta)$ sets of diameter δ ; then $\mathcal{H}_\delta^s(E) \leq N(E, \delta)\delta^s \approx_n m(E_\delta)\delta^{s-n}$. Taking $\liminf$ along a sequence realising $\underline{\mathcal{M}}^s$ and using $\mathcal{H}^s = \lim_\delta \mathcal{H}_\delta^s$ gives the claim. $\square$

Where we are. Theorem 7.2.3 makes precise the sense in which Minkowski dimension is a statement about volumes of neighbourhoods, which is the form in which it appeared in Theorem 6.6.3. It also gives the inequality $\mathcal{H}^s \lesssim \underline{\mathcal{M}}^s$, a quantitative version of $\dim_H \leq \underline{\dim}_M$, and shows the loss is genuinely in the direction one would guess: a neighbourhood bound controls

the Hausdorff measure from above, never from below.

There is no reverse inequality. The rationals in $[0, 1]$ have $\underline{\mathcal{M}}^1 = 1$ and $\mathcal{H}^1 = 0$, so the gap can be total.

7.2.2 The comparison with Hausdorff dimension

Theorem 7.2.4. *For every bounded $E \subseteq \mathbb{R}^n$,*

$$\dim_{\mathcal{H}} E \leq \underline{\dim}_{\mathcal{M}} E \leq \overline{\dim}_{\mathcal{M}} E.$$

Proof. Only the first inequality needs proof. Let $s < \dim_{\mathcal{H}} E$, so $\mathcal{H}^s(E) = \infty$ by Theorem 7.1.3; in particular $\mathcal{H}_{\delta}^s(E) \geq 1$ for all small δ . Covering E by $N(E, \delta)$ sets of diameter at most δ ,

$$1 \leq \mathcal{H}_{\delta}^s(E) \leq N(E, \delta) \delta^s,$$

so $N(E, \delta) \geq \delta^{-s}$ and $\frac{\log N(E, \delta)}{\log(1/\delta)} \geq s$ for all small δ . Hence $\underline{\dim}_{\mathcal{M}} E \geq s$, and letting $s \uparrow \dim_{\mathcal{H}} E$ gives the claim. $\square$

WORKED EXAMPLE 7.2.1— TIER A **The rationals: the inequality is strict**

Let $E = \mathbb{Q} \cap [0, 1]$.

Hausdorff. E is countable, so $\dim_{\mathcal{H}} E = 0$ by Theorem 7.1.5(v).

Minkowski. E is dense in $[0, 1]$, so any cover of E by sets of diameter δ also covers $[0, 1]$ after taking closures — the closure of a set of diameter δ has diameter δ . Hence $N(E, \delta) \geq N([0, 1], \delta) \geq \delta^{-1}$, and $\dim_{\mathcal{M}} E = 1$.

What the example shows. The two dimensions differ by the whole range. Minkowski dimension cannot see countability because it uses covers by sets of a single size, and a countable dense set looks, at every fixed scale, exactly like the interval containing it. Hausdorff dimension sees it because it may use covers whose sets shrink at different rates — assigning to the k -th rational a set of diameter $\eta 2^{-k}$ gives $\sum (\text{diam } U_i)^s \leq \eta^s \sum 2^{-ks} < \infty$ for every $s > 0$.

The moral for this series. Whenever a bound is proved by covering at a single scale δ — which is what every argument in Volume 6 did — what is bounded is the Minkowski dimension. Upgrading requires an argument that survives passing to covers of varying size, and that is Chapter 7.5.

Proposition 7.2.5 (Stability). *$\overline{\dim}_{\mathcal{M}}$ is finitely stable: $\overline{\dim}_{\mathcal{M}}(E \cup F) = \max\{\overline{\dim}_{\mathcal{M}} E, \overline{\dim}_{\mathcal{M}} F\}$. It is not countably stable, and $\underline{\dim}_{\mathcal{M}}$ is not even finitely stable.*

Proof. Finite stability: $N(E \cup F, \delta) \leq N(E, \delta) + N(F, \delta) \leq 2 \max\{N(E, \delta), N(F, \delta)\}$, and the factor 2 disappears in the limit; the reverse inequality is monotonicity.

Countable stability fails by Worked Example 7.2.1: $\mathbb{Q} \cap [0, 1]$ is a countable union of points, each of Minkowski dimension 0, but has Minkowski dimension 1.

For $\underline{\dim}_M$, the standard counterexample is a pair of sets whose counting functions oscillate out of phase: choose sequences $\delta_k \rightarrow 0$ and build E with $N(E, \delta) \approx \delta^{-1}$ for δ near δ_{2k} and $\approx \delta^{-1/2}$ near δ_{2k+1} , and F the reverse. Then $\underline{\dim}_M E = \underline{\dim}_M F = \frac{1}{2}$ while $\underline{\dim}_M(E \cup F) = 1$. Drill 2.3 asks for the construction. $\square$

WORKED EXAMPLE 7.2.2— TIER B Minkowski content, and when the two dimensions agree

By Theorem 7.2.2(iv), $\overline{\dim}_M E = n - \liminf_{\delta \rightarrow 0} \frac{\log m(E_\delta)}{\log \delta}$, so Minkowski dimension is exactly a statement about how fast the volume of the δ -neighbourhood shrinks. This is the form in which it appears in Volume 6 — Theorem 6.6.3 bounded $m(K_\delta)$ from below — and it is worth recording when the cruder notion happens to be the sharp one.

They agree for self-similar sets. Chapter 7.8 proves that if E is the attractor of an iterated function system satisfying the open set condition then $\dim_H E = \dim_M E$, given by the similarity dimension. The Cantor set of Worked Example 7.1.1 is the model case: $N(C, 3^{-k}) = 2^k$ exactly, so $\dim_M C = \log 2 / \log 3 = \dim_H C$.

They agree for sets with a regularity property. If E carries a measure μ with $\mu(B(x, r)) \approx r^s$ for all $x \in E$ and $r \leq 1$ — an *Ahlfors–David regular set* — then both dimensions equal s . The lower bound is Theorem 7.1.6 and the upper bound is a packing argument.

They differ exactly when the set is inhomogeneous across scales. The rationals are the extreme case. A more interesting one: the set $\{0\} \cup \{1/k : k \geq 1\}$ has $\dim_H = 0$ but $\dim_M = \frac{1}{2}$, computed in Drill 2.1. Here nothing is countable-dense; the discrepancy comes from the points accumulating at a polynomial rate, which a single-scale cover cannot exploit and a variable-scale cover can.

Practical consequence. When reading a dimension bound in this subject, always check which dimension is meant. Bounds for Besicovitch sets in the literature are stated for both, the Minkowski versions are typically older and easier, and a paper proving a Hausdorff bound will usually say so in its title.

WORKED EXAMPLE 7.2.3— TIER C: RESEARCH TASTE Why Volume 6 stopped at Minkowski, and what it would take to go further

Theorem 6.6.3 derived, from the restriction conjecture, that $m(K_\delta) \gtrsim_\epsilon \delta^\epsilon$ for a Besicovitch set K . By Theorem 7.2.2(iv) that is exactly a Minkowski

statement, and by Worked Example 7.2.1 it is strictly weaker than the Keakea conjecture.

(a) *Where the single scale is built in.* The proof went through a family of δ -tubes with δ -separated directions. That family exists at each fixed δ , and the estimate obtained bounds the multiplicity of tubes at that scale. Nothing in the argument relates the scales to one another.

(b) *What a Hausdorff bound would require.* By Theorem 7.1.6, to prove $\dim_{\text{H}} K \geq s$ it suffices to produce a measure μ on K with $\mu(B(x, r)) \lesssim r^s$ for all r simultaneously. So one needs an estimate that is uniform across scales, not one estimate per scale.

(c) *The standard route.* Suppose, for contradiction, that $\dim_{\text{H}} K < s$. Then $\mathcal{H}^s(K) = 0$, so there is a cover of K by balls $B(x_i, r_i)$ with $\sum r_i^s$ small and the r_i of different sizes. Grouping the balls by dyadic size and applying a single-scale estimate to each group produces, after summing, a contradiction — provided the single-scale estimate carries a factor that beats the number of groups, which is $\approx \log(1/\delta)$. Since Volume 6’s estimates carry $\delta^{-\varepsilon}$ and $\delta^{-\varepsilon}$ beats any power of a logarithm, this works.

(d) *Where it is done.* Chapter 7.8 carries out (c) in detail. The conclusion is that the Keakea *maximal* conjecture — a family of single-scale statements — implies the Keakea conjecture for Hausdorff dimension, so the two formulations are equivalent after all. That equivalence is why the literature moves between them freely, and it is the last thing this volume owes the reader.

(e) *A caution.* The implication in (d) requires the single-scale estimate with a $\delta^{-\varepsilon}$ loss for every ε . A single-scale bound with a fixed power loss δ^{-c} gives only $\dim_{\text{H}} \geq n - c'$ for a corresponding c' , and the relation between c and c' is not the identity. Tracking that relation correctly is where dimension bounds in the literature are most often misquoted.

WORKED EXAMPLE 7.2.4— TIER B **Computing a box dimension in practice**

Minkowski dimension is the one that can actually be computed, and the computation follows a fixed recipe. Here it is on three examples of increasing difficulty.

(a) *A self-similar set: the Cantor set.* At scale $\delta = 3^{-k}$ the natural cover is by the 2^k construction intervals, so $N(C, 3^{-k}) \leq 2^k$. For the lower bound, the 2^k intervals are separated by gaps of length 3^{-k} , so no set of diameter

3^{-k} meets two of them and $N(C, 3^{-k}) \geq 2^k$. Hence $N = 2^k$ exactly and

$$\dim_{\text{M}} C = \lim_k \frac{k \log 2}{k \log 3} = \frac{\log 2}{\log 3}.$$

Intermediate δ are handled by monotonicity: $N(E, \delta)$ is decreasing in δ , so it is enough to compute along a geometric sequence.

(b) *A set with a rate:* $E = \{0\} \cup \{k^{-1} : k \geq 1\}$. Split at the point where consecutive gaps equal δ . The gap between $1/k$ and $1/(k+1)$ is $\approx k^{-2}$, so the gaps exceed δ when $k < \delta^{-1/2}$. For those k each point needs its own interval: $\approx \delta^{-1/2}$ intervals. The remaining points lie in $[0, \delta^{-1/2-1}] = [0, \delta^{1/2}]$, an interval of length $\delta^{1/2}$, needing $\approx \delta^{-1/2}$ intervals. Total $N(E, \delta) \approx \delta^{-1/2}$, so $\dim_{\text{M}} E = \frac{1}{2}$.

The recipe visible in (b). Find the scale at which the set's structure changes — here, where the gaps drop below δ — and count separately on each side. Nearly every box-counting computation has this shape.

(c) *A set where the limit does not exist.* Let $E = \{0\} \cup \bigcup_k [a_{k+1}, a_k]$ with a_k decreasing very fast, say $a_k = 2^{-k!}$. At $\delta = a_k$ the set looks essentially like the single interval $[a_{k+1}, a_k]$ of length $\approx a_k$, so $N \approx 1$; at $\delta = a_{k+1}$ it looks like a union of intervals of total length $\approx a_k \gg a_{k+1}$, so $N \approx a_k/a_{k+1}$, which is enormous. The ratio $\log N/\log(1/\delta)$ therefore oscillates, and $\underline{\dim}_{\text{M}} E < \overline{\dim}_{\text{M}} E$.

What (c) shows. The two Minkowski dimensions genuinely differ for sets built at rapidly separating scales, and such sets are not pathological — the Besicovitch sets produced by iterated constructions have exactly this character. This is a second reason, beyond Worked Example 7.2.1, to prefer $\dim_{\text{H}}$: it is a single number, always.

Drills

Drill 7.2.1. Show that $E = \{0\} \cup \{1/k : k \geq 1\}$ has $\dim_{\text{M}} E = \frac{1}{2}$, by counting how many δ -intervals are needed near 0 and away from it.

Drill 7.2.2. Prove that $\overline{\dim}_{\text{M}} E = \overline{\dim}_{\text{M}} E$ for every bounded E , and deduce that Minkowski dimension cannot distinguish a set from its closure.

Drill 7.2.3. Carry out the construction sketched in Theorem 7.2.5 showing that $\underline{\dim}_{\text{M}}$ is not finitely stable.

Drill 7.2.4. Show that for a compact set E with $\dim_{\text{H}} E < s < \overline{\dim}_{\text{M}} E$ there is a cover of E by balls of varying radii with $\sum r_i^s$ arbitrarily small, but none by balls of a single radius.

Where we are. The two dimensions answer different questions, and it is worth having the distinction in one sentence.

$\dim_M$ asks: *at scale δ , how many pieces does E occupy?* It is a question about a snapshot, and it is answered by a count.

$\dim_H$ asks: *how efficiently can E be covered if one is allowed to use different scales in different places?* It is a question about all scales at once, and it is answered by an optimisation.

Everything in this chapter follows from that difference. Minkowski dimension is computable, finitely stable, blind to countability, and insensitive to closure. Hausdorff dimension is none of those things, and is the one the Kakeya conjecture is about. Theorem 7.8.5 will show that the gap can nonetheless be bridged when the single-scale estimates come with arbitrarily small power losses, which is the situation in this subject — and that is why the literature is able to work almost entirely with snapshots.

Notes

Box dimension appears in work of Bouligand and of Pontrjagin–Schnirelmann in the 1930s; the name “Minkowski dimension” records Minkowski’s use of the δ -neighbourhood. The equivalences of Theorem 7.2.2 are standard and worth doing once.

The failure of countable stability is the reason Hausdorff dimension is the notion of choice in geometric measure theory, despite being harder to estimate. For the sets arising in this series — Besicovitch sets, and the Furstenberg sets of Volume 14 — the two often coincide, but proving that they do is generally as hard as computing either.

Chapter 7.3

The Hausdorff Metric and Blaschke Selection

Several constructions in this volume and the next produce a set as a limit of approximations: the attractor of an iterated function system in Chapter 7.8, and the Besicovitch set of measure zero promised in Volume 1. Both need a topology on the space of compact sets in which such limits exist. This chapter supplies it, and then explains — with some care, because the point is easy to get wrong — exactly what it does *not* give.

7.3.1 The metric

Definition 7.3.1. Let $\mathcal{K}(\mathbb{R}^n)$ denote the collection of nonempty compact subsets of $\mathbb{R}^n$. For $A, B \in \mathcal{K}$ set

$$d_H(A, B) = \inf \{ \varepsilon > 0 : A \subseteq B_\varepsilon \text{ and } B \subseteq A_\varepsilon \},$$

where $A_\varepsilon = \{x : \text{dist}(x, A) \leq \varepsilon\}$.

Proposition 7.3.2. d_H is a metric on $\mathcal{K}(\mathbb{R}^n)$, and

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \text{dist}(a, B), \sup_{b \in B} \text{dist}(b, A) \right\}.$$

Proof. The displayed identity is a restatement of the definition: $A \subseteq B_\varepsilon$ says exactly $\sup_{a \in A} \text{dist}(a, B) \leq \varepsilon$, and the infimum of such ε is attained because A is compact and $\text{dist}(\cdot, B)$ continuous.

Symmetry is clear and $d_H(A, A) = 0$. If $d_H(A, B) = 0$ then every $a \in A$ has $\text{dist}(a, B) = 0$, hence $a \in B$ as B is closed; and symmetrically, so $A = B$. For the triangle inequality, if $A \subseteq B_\varepsilon$ and $B \subseteq C_\eta$ then $A \subseteq (C_\eta)_\varepsilon = C_{\eta+\varepsilon}$, and symmetrically. $\square$

Theorem 7.3.3 (Completeness). $(\mathcal{K}(\mathbb{R}^n), d_H)$ is a complete metric space.

Proof. Let (A_k) be Cauchy. Define

$$A = \left\{ x \in \mathbb{R}^n : \text{there are } x_k \in A_k \text{ with } x_k \rightarrow x \right\}.$$

Step 1. A is nonempty and compact.

Choose $k_1 < k_2 < \dots$ with $d_H(A_j, A_{k_i}) \leq 2^{-i}$ for $j \geq k_i$. Pick $y_1 \in A_{k_1}$; inductively pick $y_{i+1} \in A_{k_{i+1}}$ with $|y_{i+1} - y_i| \leq 2^{-i}$, possible since $A_{k_i} \subseteq (A_{k_{i+1}})_{2^{-i}}$. Then (y_i) is Cauchy in $\mathbb{R}^n$; its limit lies in A by construction, so $A \neq \emptyset$. Boundedness of A follows since a Cauchy sequence in $\mathcal{K}$ is uniformly bounded, and A is closed by a diagonal argument.

Step 2. $A_k \rightarrow A$.

Given $\varepsilon > 0$ pick K with $d_H(A_j, A_k) \leq \varepsilon/2$ for $j, k \geq K$.

$A \subseteq (A_k)_\varepsilon$ for $k \geq K$: if $x \in A$, take $x_j \in A_j$ with $x_j \rightarrow x$; for large $j \geq K$, $|x - x_j| \leq \varepsilon/2$ and $\text{dist}(x_j, A_k) \leq \varepsilon/2$, so $\text{dist}(x, A_k) \leq \varepsilon$.

$A_k \subseteq A_\varepsilon$ for $k \geq K$: let $x \in A_k$. Build $z_j \in A_j$ for $j \geq k$ with $|z_{j+1} - z_j|$ summable and $z_k = x$, exactly as in Step 1 using the Cauchy property along a rapidly converging subsequence; the limit lies in A and is within ε of x . $\square$

Theorem 7.3.4 (Blaschke selection). Let $B \subseteq \mathbb{R}^n$ be compact. Then $\mathcal{K}(B) = \{A \in \mathcal{K}(\mathbb{R}^n) : A \subseteq B\}$ is compact in d_H . In particular every sequence of nonempty compact subsets of B has a subsequence converging in d_H to a nonempty compact subset of B .

Proof. By Theorem 7.3.3 and the fact that $\mathcal{K}(B)$ is closed in $\mathcal{K}(\mathbb{R}^n)$ — a Hausdorff limit of subsets of B lies in B , as B is closed — it suffices to prove total boundedness.

Fix $\varepsilon > 0$ and let $F \subseteq B$ be a finite ε -net, so $B \subseteq \bigcup_{y \in F} B(y, \varepsilon)$. For $A \in \mathcal{K}(B)$ put $A' = \{y \in F : \text{dist}(y, A) \leq \varepsilon\}$, a nonempty subset of F . Then $d_H(A, A') \leq 2\varepsilon$: every $a \in A$ lies within ε of some $y \in F$, and that y lies in A' ; conversely every $y \in A'$ is within ε of A .

Since F has finitely many subsets, the finitely many sets A' form a 2ε -net for $\mathcal{K}(B)$. $\square$

WORKED EXAMPLE 7.3.1— TIER A **Hausdorff limits can be much smaller, or much larger**

(a) *Smaller.* Let $A_k = \{0, \frac{1}{k}, \frac{2}{k}, \dots, 1\} \subseteq [0, 1]$. Then $A_k \rightarrow [0, 1]$, and the limit is much larger than any A_k : finite sets converge to an interval.

(b) *Larger.* Let $A_k = [0, 1]$ for all k ; trivially $A_k \rightarrow [0, 1]$. Now let $A_k = [0, 1] \cup \{2\}$ for even k and $[0, 1]$ for odd k . This is not Cauchy: d_H between

consecutive terms is 1. Hausdorff convergence is sensitive to isolated far-away pieces, which is why compactness in Theorem 7.3.4 requires a common bounded container.

(c) *Dimension is not continuous.* In (a) each A_k is finite, so $\dim_{\mathbb{H}} A_k = 0$, while $\dim_{\mathbb{H}}[0, 1] = 1$. So $\dim_{\mathbb{H}}$ is not continuous, and not even upper semicontinuous, along Hausdorff-convergent sequences. Drill 3.2 shows it is not lower semicontinuous either.

Proposition 7.3.5 (Supports converge, measures need not). *Let μ_k be probability measures on a fixed compact $B \subseteq \mathbb{R}^n$ with $\mu_k \rightarrow \mu$ weak-*. Then*

$$\text{spt } \mu \subseteq \liminf_k \text{spt } \mu_k := \left\{ x : \text{dist}(x, \text{spt } \mu_k) \rightarrow 0 \right\},$$

and if moreover $\text{spt } \mu_k \rightarrow K$ in $d_{\mathbb{H}}$ then $\text{spt } \mu \subseteq K$. The inclusion may be strict.

Proof. Let $x \in \text{spt } \mu$ and $\varepsilon > 0$, so $\mu(B(x, \varepsilon)) > 0$. Choose $\varphi \in C_c(B(x, \varepsilon))$ with $0 \leq \varphi \leq 1$ and $\int \varphi d\mu > 0$. By weak-* convergence, $\int \varphi d\mu_k \rightarrow \int \varphi d\mu > 0$, so for large k we have $\mu_k(B(x, \varepsilon)) > 0$ and hence $\text{dist}(x, \text{spt } \mu_k) \leq \varepsilon$. As $\varepsilon > 0$ was arbitrary, $x \in \liminf_k \text{spt } \mu_k$.

If $\text{spt } \mu_k \rightarrow K$ in $d_{\mathbb{H}}$ then $\liminf_k \text{spt } \mu_k \subseteq K$, since K is closed and every point of the liminf is a limit of points of the $\text{spt } \mu_k$.

For strictness, let $\mu_k = (1 - \frac{1}{k})\delta_0 + \frac{1}{k}\delta_1$ on $B = [0, 1]$. Then $\text{spt } \mu_k = \{0, 1\}$ for every k , so $\text{spt } \mu_k \rightarrow \{0, 1\}$, while $\mu_k \rightarrow \delta_0$ weak-* and $\text{spt } \delta_0 = \{0\} \subsetneq \{0, 1\}$. $\square$

Where we are. Theorem 7.3.5 is the exact statement needed in Step 5 of Theorem 7.5.1: a weak-* limit of measures supported in shrinking neighbourhoods of E is supported in E . It also records the asymmetry that runs through this chapter — *supports can only shrink in the limit, never grow* — which is the same one-sidedness as Theorem 7.3.6 for measure, and the same reason compactness never proves a set is small.

7.3.2 Measure along Hausdorff limits

Proposition 7.3.6. *Let $A_k, A \in \mathcal{K}(B)$ with $A_k \rightarrow A$ in $d_{\mathbb{H}}$. Then*

$$\limsup_{k \rightarrow \infty} m(A_k) \leq m(A).$$

That is, Lebesgue measure is upper semicontinuous on $\mathcal{K}(B)$.

Proof. Fix $\varepsilon > 0$. For large k , $A_k \subseteq A_\varepsilon$, so $m(A_k) \leq m(A_\varepsilon)$ and hence $\limsup_k m(A_k) \leq m(A_\varepsilon)$. Since A is compact, $A_\varepsilon \downarrow A$ as $\varepsilon \downarrow 0$ and the sets have finite measure, so $m(A_\varepsilon) \rightarrow m(A)$ by continuity from above (Theorem 1.2.6). Letting $\varepsilon \downarrow 0$ gives the claim. $\square$

Pitfall: why compactness does *not* produce a measure-zero Besicovitch set

Here is an argument that looks conclusive and is not. Let

$$\mathcal{B} = \{A \in \mathcal{K}(\overline{B}(0, 2)) : A \text{ contains a unit segment in every direction}\}.$$

One checks $\mathcal{B}$ is closed — Drill 3.3 — hence compact by Theorem 7.3.4. Volume 1, Theorem 1.3.12 gives elements of $\mathcal{B}$ of arbitrarily small measure, so $\inf_{\mathcal{B}} m = 0$. Take $A_k \in \mathcal{B}$ with $m(A_k) \rightarrow 0$ and a convergent subsequence $A_k \rightarrow A \in \mathcal{B}$. Is $m(A) = 0$?

No conclusion follows. Theorem 7.3.6 gives $\limsup_k m(A_k) \leq m(A)$, that is $0 \leq m(A)$ — vacuous. Upper semicontinuity bounds the limit of the measures *by* the measure of the limit, which is the wrong direction. A u.s.c. function on a compact set attains its maximum, not its minimum, and the infimum of m over $\mathcal{B}$ need not be attained.

And the failure is real, not an artefact. Worked Example 7.3.1(a) shows sets of measure zero converging to a set of measure one. So the Hausdorff limit of small Besicovitch sets can perfectly well be large, and indeed the limit of a Perron tree construction typically is: the triangles are translated further and further, and their limit fills out a region.

What is actually needed. A construction producing a *decreasing* sequence, so that the limit is an intersection and continuity from above applies. The Perron tree does not decrease. The standard route instead builds the set from a Cantor set of directions and computes its measure by Fubini, which requires knowing that certain sums of Cantor sets are null — a projection theorem. That is Marstrand's theorem, proved in Volume 8, and the construction is Kahane's; accordingly the measure-zero Besicovitch set is deferred to Volume 8 rather than proved here.

WORKED EXAMPLE 7.3.2— TIER B Contraction mappings on $\mathcal{K}$, and why attractors exist

The positive use of Theorem 7.3.3 in this volume is the existence of attractors, which Chapter 7.8 needs.

The setup. Let $f_1, \dots, f_m : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be contractions, with $|f_i(x) - f_i(y)| \leq r_i|x - y|$ and $r_i < 1$. Define $F : \mathcal{K} \rightarrow \mathcal{K}$ by

$$F(A) = \bigcup_{i=1}^m f_i(A),$$

a compact set, being a finite union of continuous images of a compact set. F is a contraction on $\mathcal{K}$. Put $r = \max_i r_i < 1$. If $A \subseteq B_\varepsilon$ then $f_i(A) \subseteq f_i(B)_{r_i\varepsilon} \subseteq F(B)_{r\varepsilon}$, so $F(A) \subseteq F(B)_{r\varepsilon}$; symmetrically. Hence

$$d_H(F(A), F(B)) \leq r d_H(A, B).$$

Conclusion. By Theorem 7.3.3 and the Banach fixed point theorem, F has a unique fixed point $E \in \mathcal{K}$, and $F^k(A) \rightarrow E$ for every starting A . This E is the *attractor* of the system, and satisfies $E = \bigcup_i f_i(E)$.

Why this is worth isolating. The existence of the Cantor set, the Sierpiński gasket and the von Koch curve are all instances, obtained without any construction: one writes down the maps and the set exists. The work then shifts entirely to computing its dimension, which is Chapter 7.8.

WORKED EXAMPLE 7.3.3— TIER C: RESEARCH TASTE Compactness arguments in this subject: when they work

The Pitfall above is a case where a compactness argument fails for a structural reason, and it is worth extracting the general principle, since compactness is attempted often in this area and succeeds only sometimes.

(a) *Compactness gives existence of a limit, never a property of it.* Theorem 7.3.4 produces a limit set; whether the limit inherits a property depends on whether that property is *closed* under Hausdorff convergence. Containing a segment in every direction is closed (Drill 3.3). Having small measure is not. Having small dimension is not (Worked Example 7.3.1(c)).

(b) *The properties that are closed are the “containment” ones.* If P is of the form “ A contains a member of each of a compact family of compact sets”, then P is closed, by the argument of Drill 3.3.akeya-type hypotheses are all of this form, which is why compactness reductions are standard: one may always assume a Besicovitch set is a limit of nice ones.

(c) *The quantities that are semicontinuous have the wrong sign.* Measure and dimension are the quantities one wants to bound *above*, and Theorem 7.3.6 bounds them below. This asymmetry is not accidental: it reflects that a limit can be larger than everything approaching it, never smaller. So compactness

never proves a set is small.

(d) *Where it is nonetheless used.* In Volume 14, the reduction of the Kakeya conjecture to the *sticky* case is a compactness argument: one extracts a limiting configuration of tubes with additional structure, and the structure is closed under the limit even though the measure is not. The technique is legitimate; what must be checked, every time, is that the property being inherited is of the type in (b).

WORKED EXAMPLE 7.3.4 — TIER B Verifying that a property is closed: the standard argument

Worked Example 7.3.3(b) claimed that “containing a member of each of a compact family” is closed under Hausdorff convergence. The verification is short and is worth doing once, since every compactness reduction in this subject uses it.

The claim. Let $\mathcal{S}$ be a family of nonempty compact subsets of $\mathbb{R}^n$ which is itself compact in d_H . Then

$$\mathcal{P} = \{A \in \mathcal{K}(B) : \exists S \in \mathcal{S}, S \subseteq A\}$$

is closed in $\mathcal{K}(B)$.

Proof. Let $A_k \in \mathcal{P}$ with $A_k \rightarrow A$, and pick $S_k \in \mathcal{S}$ with $S_k \subseteq A_k$. By compactness of $\mathcal{S}$, a subsequence $S_{k_j} \rightarrow S \in \mathcal{S}$. We claim $S \subseteq A$.

Let $x \in S$. Since $S_{k_j} \rightarrow S$, there are $x_j \in S_{k_j}$ with $x_j \rightarrow x$ — by Theorem 7.3.2, $\text{dist}(x, S_{k_j}) \leq d_H(S, S_{k_j}) \rightarrow 0$. Each $x_j \in S_{k_j} \subseteq A_{k_j}$, and $A_{k_j} \rightarrow A$, so $\text{dist}(x_j, A) \leq d_H(A_{k_j}, A) \rightarrow 0$. Therefore $\text{dist}(x, A) \leq |x - x_j| + \text{dist}(x_j, A) \rightarrow 0$, and $x \in A$ as A is closed. $\square$

Application to Besicovitch sets. Take $\mathcal{S}$ to be the family of unit segments contained in $\overline{B}(0, 2)$. It is compact: a segment is determined by its midpoint and direction, both ranging over compact sets, and the map from (midpoint, direction) to the segment is continuous into d_H . Hence “contains a unit segment” is closed.

For “contains a segment in *every* direction” one applies the argument for each fixed direction ω — $\mathcal{S}_\omega$, the segments in direction ω , is compact — and intersects over ω , an intersection of closed sets. This is the assertion of Drill 3.3.

The contrast. Try the same argument for “ $m(A) \leq \varepsilon$ ”. Given $A_k \rightarrow A$ with $m(A_k) \leq \varepsilon$, one wants $m(A) \leq \varepsilon$; but Theorem 7.3.6 gives only $m(A) \geq \limsup m(A_k)$, and Worked Example 7.3.1(a) shows the conclusion is false.

The difference is that containment is preserved by limits and measure is not. $\square$

Drills

Drill 7.3.1. Show $\mathcal{K}(\mathbb{R}^n)$ is separable, and that $d_H(A, B)$ may be computed using only the extreme points when A, B are convex.

Drill 7.3.2. Exhibit compact $A_k \rightarrow A$ with $\dim_H A_k = 1$ for all k and $\dim_H A = 0$, showing $\dim_H$ is not lower semicontinuous either.

Drill 7.3.3. Prove that the family $\mathcal{B}$ in the Pitfall is closed in d_H . (Use that the unit segments contained in $\overline{B}(0, 2)$ form a compact family.)

Drill 7.3.4. Verify the contraction estimate in Worked Example 7.3.2 in detail, and compute the attractor of $f_1(x) = x/3$, $f_2(x) = x/3 + 2/3$ on $\mathbb{R}$.

Notes

The Hausdorff metric dates to Hausdorff's 1914 book; Blaschke's selection theorem is from 1916 and was proved for convex bodies, where it is usually stated. The formulation for general compact sets is the one used here and is the same proof.

The fixed point argument of Worked Example 7.3.2 is Hutchinson's (1981), and is the standard modern foundation for self-similar sets.

Chapter 7.4

Covering Theorems and Densities

Volume 1 proved the Vitali covering lemma for Lebesgue measure and used it for the Hardy–Littlewood maximal function. Hausdorff measures are not translation invariant and need not be doubling, so that argument does not transfer directly. This chapter isolates what does: a covering lemma valid in any metric space, and the differentiation and density theorems it supports.

7.4.1 The $5r$ -covering lemma

Theorem 7.4.1 ($5r$ -covering lemma). *Let $\mathcal{F}$ be a family of closed balls in $\mathbb{R}^n$ with $\sup\{r(B) : B \in \mathcal{F}\} < \infty$. Then there is a countable pairwise disjoint subfamily $\mathcal{G} \subseteq \mathcal{F}$ with*

$$\bigcup_{B \in \mathcal{F}} B \subseteq \bigcup_{B \in \mathcal{G}} 5B,$$

where $5B$ denotes the ball concentric with B of five times the radius.

Idea of the proof. Select greedily by size, always taking a ball of nearly maximal radius among those disjoint from what has been chosen. Any unselected ball must meet a selected one of comparable or larger radius — otherwise it would have been available at a stage where it was nearly maximal — and a ball meeting another of at least two-thirds its radius is contained in the fivefold dilate of that one.

The constant 5 is $1 + 2 \cdot 2$: a factor 2 for the radius ratio and a factor 2 for crossing the selected ball. It is not optimal, but nothing here depends on the constant.

Proof. Step 1. Stratify by radius.

Let $R = \sup_{\mathcal{F}} r(B)$ and for $j \geq 0$ put

$$\mathcal{F}_j = \left\{ B \in \mathcal{F} : \frac{R}{2^{j+1}} < r(B) \leq \frac{R}{2^j} \right\}.$$

Step 2. Select.

Let $\mathcal{G}_0$ be a maximal pairwise disjoint subfamily of $\mathcal{F}_0$; such a family exists by Zorn's lemma, and is countable since the balls are disjoint with radii bounded below. Inductively, let $\mathcal{G}_j$ be a maximal pairwise disjoint subfamily of

$$\left\{ B \in \mathcal{F}_j : B \cap B' = \emptyset \text{ for all } B' \in \mathcal{G}_0 \cup \cdots \cup \mathcal{G}_{j-1} \right\},$$

and set $\mathcal{G} = \bigcup_j \mathcal{G}_j$, a countable pairwise disjoint family.

Step 3. Cover.

Let $B = B(x, r) \in \mathcal{F}$, say $B \in \mathcal{F}_j$. By maximality of $\mathcal{G}_j$, B meets some $B' = B(x', r') \in \mathcal{G}_0 \cup \cdots \cup \mathcal{G}_j$ — either $B \in \mathcal{G}_j$ itself, or B was excluded, meaning it meets an earlier selected ball, or it could have been added to $\mathcal{G}_j$, contradicting maximality.

Since $B' \in \mathcal{F}_i$ with $i \leq j$,

$$r' > \frac{R}{2^{i+1}} \geq \frac{R}{2^{j+1}} \geq \frac{r}{2}.$$

Now for $y \in B$, using that $B \cap B' \neq \emptyset$,

$$|y - x'| \leq |y - x| + |x - x'| \leq r + (r + r') = 2r + r' \leq 4r' + r' = 5r',$$

so $B \subseteq 5B'$. □

Pitfall: the disjoint family need not cover anything

Theorem 7.4.1 produces a disjoint family whose *dilates* cover. The disjoint family itself may cover very little: if $\mathcal{F}$ consists of all balls $B(x, 1)$ with $x \in \mathbb{R}^n$, a maximal disjoint subfamily covers a fixed proportion of space and no more.

This is why the lemma is used to bound measures from above — one estimates $\mu(\bigcup \mathcal{F}) \leq \sum_{\mathcal{G}} \mu(5B)$ — and is useless for bounds from below. When a lower bound is wanted, one needs a covering theorem producing a family that covers with bounded overlap, and that is the Besicovitch covering theorem, discussed in Theorem 7.4.2.

Remark 7.4.2 (Scope decision: the Besicovitch covering theorem). The Besicovitch covering theorem asserts that, given a family of balls centred at every point of a set A , one may extract $N(n)$ subfamilies, each disjoint, which together cover A — so the selected balls cover A with overlap at most $N(n)$, and no dilation is needed. It is not proved in this series.

The obstruction is worth naming, because the proof that suggests itself is false. One selects balls greedily with radii within a factor $\frac{3}{4}$ of maximal, so that for $i < j$ one has $r_j < \frac{4}{3}r_i$, and centres are never inside previously selected balls, so $|x_i - x_j| > r_i$. One then hopes that if z lies in both B_i and B_j , the angle subtended at z is bounded below, which would bound the overlap by a sphere-packing count. *It is not.* A direct search over the constraints produces configurations with angle less than 0.011 radians — for instance $r_i = 1$, $r_j = 1.283$, $|x_i - z| = 0.068$, $|x_j - z| = 1.144$, $|x_i - x_j| = 1.076$ — all of which satisfy every condition above. A correct proof must split according to the ratio of the radii and treat comparable and very different radii by different arguments.

That argument runs to several pages, and nothing in this series needs it: every application here is an upper bound on a measure, for which Theorem 7.4.1 suffices. Recorded as revision R6 in the master manifest.

7.4.2 Differentiation and density

Theorem 7.4.3 (Vitali covering theorem for a Radon measure). *Let μ be a finite Borel measure on $\mathbb{R}^n$, $A \subseteq \mathbb{R}^n$ Borel, and $\mathcal{F}$ a family of closed balls such that every point of A is the centre of balls in $\mathcal{F}$ of arbitrarily small radius. Then for every $\varepsilon > 0$ there is a countable disjoint $\mathcal{G} \subseteq \mathcal{F}$ with*

$$\mu\left(A \setminus \bigcup_{B \in \mathcal{G}} B\right) = 0 \quad \text{provided } \mu \text{ is doubling,}$$

and in general $\mu(A \setminus \bigcup_{\mathcal{G}} B) \leq \varepsilon$ whenever $\mathcal{G}$ is chosen so that $\sum_{\mathcal{G}} \mu(5B) \geq \mu(\bigcup \mathcal{F}) - \varepsilon$.

Proof. Apply Theorem 7.4.1 to $\mathcal{F}$, obtaining a disjoint $\mathcal{G}$ with $\bigcup \mathcal{F} \subseteq \bigcup_{\mathcal{G}} 5B$. Then

$$\mu\left(\bigcup \mathcal{F}\right) \leq \sum_{B \in \mathcal{G}} \mu(5B).$$

If μ is doubling, $\mu(5B) \leq C\mu(B)$, so $\mu(\bigcup \mathcal{F}) \leq C \sum_{\mathcal{G}} \mu(B) = C\mu(\bigcup \mathcal{G})$; applying this to the residual family covering $A \setminus \bigcup \mathcal{G}$ and iterating gives a disjoint family exhausting A up to μ -measure zero, by the standard geometric-decay argument of Volume 1, Theorem 1.7.7.

Without doubling, the displayed inequality is still available and is the form used below. $\square$

Theorem 7.4.4 (Density of $\mathcal{H}^s$). *Let $E \subseteq \mathbb{R}^n$ be $\mathcal{H}^s$ -measurable with $\mathcal{H}^s(E) < \infty$. Then for $\mathcal{H}^s$ -almost every $x \notin E$,*

$$\lim_{r \downarrow 0} \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} = 0,$$

and for $\mathcal{H}^s$ -almost every $x \in E$,

$$\limsup_{r \downarrow 0} \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} \geq 2^{-s}.$$

Proof. Step 1. The upper bound off E .

Let $F = \{x \notin E : \limsup_r \mathcal{H}^s(E \cap B(x, r))(2r)^{-s} > \lambda\}$ for $\lambda > 0$; we show $\mathcal{H}^s(F) = 0$, which suffices on letting $\lambda \downarrow 0$ along a sequence.

Since $\mathcal{H}^s \upharpoonright E$ is a finite Borel measure, it is outer regular (Theorem 1.3.6 applies to finite Borel measures on $\mathbb{R}^n$), so given $\eta > 0$ there is an open $V \supseteq E$ with $\mathcal{H}^s(E \cap V) \leq \mathcal{H}^s(E) + \eta$ — and, F being disjoint from E , we may arrange $\mathcal{H}^s(E \cap V') \leq \eta$ for an open $V' \supseteq F$ by choosing V' inside the complement of a compact subset of E carrying most of its measure.

Every $x \in F$ is the centre of arbitrarily small balls $B \subseteq V'$ with $\mathcal{H}^s(E \cap B) > \lambda(\text{diam } B)^s$. By Theorem 7.4.1 extract a disjoint subfamily $\mathcal{G}$ whose fivefold dilates cover F . Then

$$\begin{aligned} \mathcal{H}_{10r_0}^s(F) &\leq \sum_{\mathcal{G}} (\text{diam } 5B)^s \\ &= 5^s \sum_{\mathcal{G}} (\text{diam } B)^s \\ &\leq \frac{5^s}{\lambda} \sum_{\mathcal{G}} \mathcal{H}^s(E \cap B) \\ &\leq \frac{5^s}{\lambda} \mathcal{H}^s(E \cap V') \\ &\leq \frac{5^s \eta}{\lambda}, \end{aligned}$$

the balls being disjoint and contained in V' , and r_0 a bound for their radii. Letting $r_0 \downarrow 0$ and then $\eta \downarrow 0$ gives $\mathcal{H}^s(F) = 0$.

Step 2. *The lower bound on E .*

Let $G = \{x \in E : \limsup_r \mathcal{H}^s(E \cap B(x, r))(2r)^{-s} < 2^{-s}\lambda\}$ with $\lambda < 1$; we show $\mathcal{H}^s(G) = 0$. Every $x \in G$ is the centre of arbitrarily small balls with $\mathcal{H}^s(E \cap B) < \lambda(\text{diam } B)^s$. Choose an open $V \supseteq G$ with $\mathcal{H}^s(E \cap V) \leq \mathcal{H}^s(G) + \eta$ and restrict to balls inside V ; a $5r$ -selection as in Step 1 gives a cover of G by sets $\{B\}$ with

$$\sum_B (\text{diam } B)^s > \frac{1}{\lambda} \sum_B \mathcal{H}^s(E \cap B) \geq \frac{\mathcal{H}^s(G)}{\lambda},$$

where the last step uses that the chosen balls, after a Vitali-type exhaustion, cover G up to $\mathcal{H}^s$ -null sets. Since also $\sum_B (\text{diam } B)^s \geq \mathcal{H}_{r_0}^s(G)$, letting $r_0 \downarrow 0$ yields $\mathcal{H}^s(G) \geq \mathcal{H}^s(G)/\lambda$, forcing $\mathcal{H}^s(G) = 0$ as $\lambda < 1$. $\square$

WORKED EXAMPLE 7.4.1— TIER A **Densities detect dimension**

Let $E \subseteq \mathbb{R}^n$ with $0 < \mathcal{H}^s(E) < \infty$. Theorem 7.4.4 says that at $\mathcal{H}^s$ -almost every point of E the ratio $\mathcal{H}^s(E \cap B(x, r))/(2r)^s$ does not tend to 0, while at almost every point off E it does.

Consequence: a set cannot have finite positive measure in two dimensions. If $0 < \mathcal{H}^s(E) < \infty$ and $t > s$, then for a.e. $x \in E$

$$\begin{aligned} \limsup_{r \rightarrow 0} \frac{\mathcal{H}^t(E \cap B(x, r))}{(2r)^t} \\ \leq \limsup_{r \rightarrow 0} (2r)^{s-t} \cdot \frac{\mathcal{H}^s(E \cap B(x, r))}{(2r)^s} \cdot \frac{\mathcal{H}^t}{\mathcal{H}^s} \end{aligned}$$

— and the factor $(2r)^{s-t} \rightarrow \infty$, consistent with $\mathcal{H}^t(E) = 0$ from Theorem 7.1.3. The densities give a local, and therefore more flexible, version of the critical exponent.

Where this is used. The density theorem is the tool for showing a set is *not* rectifiable, and in Volume 8 it underlies the proof that a set of finite $\mathcal{H}^1$ measure with positive lower density is contained in countably many curves. In this series its role is more modest: it certifies that the Frostman measure of Chapter 7.5 is genuinely spread over the set rather than concentrated on a small part.

WORKED EXAMPLE 7.4.2— TIER B **Why the 5 can be replaced by any $\lambda > 1$, and why nobody bothers**

The constant in Theorem 7.4.1 can be improved. Selecting balls with radii within a factor $\theta < 1$ of maximal at each stage, the computation in Step 3 becomes

$$|y - x'| \leq 2r + r' \leq \left(\frac{2}{\theta} + 1\right)r',$$

so the dilation factor is $1 + 2/\theta$, which tends to 3 as $\theta \uparrow 1$. With a more careful selection one can reach any $\lambda > 3$, and in a general metric space 3 is optimal.

Why it does not matter here. Every use of Theorem 7.4.1 in this volume ends with a sum $\sum (\text{diam } 5B)^s = 5^s \sum (\text{diam } B)^s$, and the factor 5^s is absorbed into a constant that is then sent to zero by a separate limit. The statement being proved — that some measure vanishes — is insensitive to it.

When constants do matter. They matter when the covering lemma is applied $\log(1/\delta)$ times, once per scale, since then a factor C per application becomes $C^{\log(1/\delta)} = \delta^{-\log C}$, a power loss. This happens in multiscale arguments, and it is the reason such arguments are stated with $\delta^{-\varepsilon}$ rather than with explicit constants — see Worked Example 6.5.3. A reader who finds a covering constant being tracked carefully should suspect a multiscale induction is coming.

WORKED EXAMPLE 7.4.3— TIER C: RESEARCH TASTE **What covering theorems are, structurally**

Three covering results have now appeared: Vitali in Volume 1, the $5r$ -lemma here, and Besicovitch, which is stated and not proved. They differ along two axes, and the classification is worth having.

(a) *What is assumed about the balls.* Vitali and the $5r$ -lemma assume nothing: any family. Besicovitch assumes the balls are centred at the points of the set being covered. That extra hypothesis is what buys the conclusion without dilation.

(b) *What is concluded.* Vitali and $5r$ give a disjoint family whose dilates cover; Besicovitch gives a bounded-overlap family that covers. The first is an upper bound tool, the second a lower bound tool, as the Pitfall in §4.1 explained.

(c) *What is assumed about the measure.* The $5r$ -lemma is purely geometric — no measure appears. Vitali's covering theorem for a measure needs doubling, or else the weaker conclusion of Theorem 7.4.3. Besicovitch needs nothing about the measure, which is exactly why it is the tool for differentiating arbitrary Radon measures, and why its omission here would matter if this series needed that.

(d) *What this series actually needs.* Only (i) upper bounds on Hausdorff measures, from the $5r$ -lemma; and (ii) the density theorem, proved above from the same lemma. The measures appearing later — Frostman measures on Besicovitch sets, and the natural measures on self-similar sets — are constructed with whatever regularity is required, so the general theory is not needed. This is a recurring pattern in the subject: the general machinery exists, and specific problems are usually solved by building measures with the properties one wants rather than by invoking it.

WORKED EXAMPLE 7.4.4— TIER A **The $5r$ -lemma applied by hand**

The lemma is used so often that it is worth performing once explicitly.

The task. Let $E \subseteq \mathbb{R}$ be the set $\{1/k : k = 1, \dots, 100\}$ and $\mathcal{F}$ the family of intervals $B(1/k, r_k)$ with $r_k = 1/(4k^2)$. Extract a disjoint subfamily whose 5-fold dilates cover.

Selection. Radii decrease in k , so the greedy rule takes $k = 1$ first: $B(1, \frac{1}{4}) = (\frac{3}{4}, \frac{5}{4})$. It contains no other $1/k$. Next available is $k = 2$: $B(\frac{1}{2}, \frac{1}{16})$, disjoint from the first. Continuing, one checks that $B(1/k, r_k)$ and $B(1/(k+1), r_{k+1})$ are disjoint, since

$$\begin{aligned} \frac{1}{k} - \frac{1}{k+1} &= \frac{1}{k(k+1)} \quad \text{while} \quad r_k + r_{k+1} \\ &= \frac{1}{4k^2} + \frac{1}{4(k+1)^2} < \frac{1}{2k^2} < \frac{1}{k(k+1)} \end{aligned}$$

for $k \geq 2$. So *every* ball is selected, $\mathcal{G} = \mathcal{F}$, and the covering conclusion is trivial.

A case where selection bites. Enlarge the radii to $r_k = 1/(2k)$. Now $B(1, \frac{1}{2}) = (\frac{1}{2}, \frac{3}{2})$ contains $1/k$ for $k = 1$ only, but $B(\frac{1}{2}, \frac{1}{4})$ meets it. The greedy rule takes $k = 1$, discards $k = 2$, and proceeds; the discarded ball $B(\frac{1}{2}, \frac{1}{4})$ satisfies $B(\frac{1}{2}, \frac{1}{4}) \subseteq 5B(1, \frac{1}{2}) = (-\frac{3}{2}, \frac{7}{2})$ ✓, as the lemma promises.

What the example illustrates. The dilation factor is needed only for the *discarded* balls, and how much is needed depends on the ratio of radii — which is why the proof's Step 3 turns on $r' > r/2$. When all radii are comparable no dilation is needed at all, and the lemma reduces to a packing count.

Figure 7.4.1 (to be drawn). *The 5r-covering lemma*

Content required: A cluttered family of discs of various radii; the greedily selected disjoint subfamily in heavy outline; and their 5-fold dilates, dashed, covering everything. Show one discarded disc meeting a selected one of at least half its radius, which is the configuration Step 3 of Theorem 7.4.1 analyses.

Drills

Drill 7.4.1. Verify the claim in Worked Example 7.4.2 that selecting with ratio θ gives dilation factor $1 + 2/\theta$.

Drill 7.4.2. Show that Theorem 7.4.1 fails if the radii are unbounded, by exhibiting a family of balls in $\mathbb{R}$ for which no disjoint subfamily has dilates covering the union.

Drill 7.4.3. Deduce from Theorem 7.4.4 that a set E with $0 < \mathcal{H}^s(E) < \infty$ satisfies $\mathcal{H}^s(E \cap B(x, r)) \lesssim r^s$ for $\mathcal{H}^s$ -a.e. $x \in E$ and all small r — the conclusion Theorem 7.1.6 takes as a hypothesis.

Drill 7.4.4. Confirm the numerical configuration in Theorem 7.4.2: check that $r_i = 1$, $r_j = 1.283$, $|x_i - z| = 0.068$, $|x_j - z| = 1.144$, $|x_i - x_j| = 1.076$ satisfies $r_j < \frac{4}{3}r_i$ and $|x_i - x_j| > r_i$, and compute the angle at z .

Notes

The $5r$ -lemma in this form is due to Vitali and Wiener; the sharp constant 3 in general metric spaces is standard. The Besicovitch covering theorem is from 1945 and its constant $N(n)$ grows exponentially in n ; the sharp value is known only for $n = 1, 2$.

Theorem 7.4.4 is Besicovitch's; the sharp constant in the lower bound — that the upper density is exactly 1 for $\mathcal{H}^s$ -a.e. point when $0 < \mathcal{H}^s(E) < \infty$ and s is an integer — is much harder and is part of the theory of rectifiability, treated in Volume 8.

Chapter 7.5

Frostman's Lemma

Theorem 7.1.6 said that a measure on E with $\mu(B(x, r)) \lesssim r^s$ proves $\dim_{\mathbb{H}} E \geq s$. Frostman's lemma is the converse: whenever $\dim_{\mathbb{H}} E > s$, such a measure exists. It converts the definition of dimension — an infimum over covers, which is awkward to use — into the existence of an object one can compute with, and it is the reason dimension is tractable at all.

Theorem 7.5.1 (Frostman). *Let $E \subseteq \mathbb{R}^n$ be compact and $s \geq 0$. Then $\mathcal{H}^s(E) > 0$ if and only if there is a nonzero finite Borel measure μ supported in E with*

$$\mu(B(x, r)) \leq r^s \quad \text{for every } x \in \mathbb{R}^n, r > 0. \quad (7.5.1)$$

Consequently

$$\dim_{\mathbb{H}} E = \sup \left\{ s \geq 0 : \exists \mu \neq 0 \text{ on } E \text{ with } \mu(B(x, r)) \lesssim r^s \right\}.$$

Idea of the proof. One direction is Theorem 7.1.6. For the other, the measure is built by a dyadic construction. Start by placing mass $\ell(Q)^s$ on every dyadic cube Q of some small generation m meeting E — far too much mass, in general. Then sweep upwards through the generations, and whenever a cube carries more than $\ell(Q)^s$, scale down the mass inside it uniformly so that it carries exactly $\ell(Q)^s$.

After the sweep, every dyadic cube carries at most $\ell(Q)^s$, which is (7.5.1) up to a dimensional constant. The work is to show the total mass has not collapsed to zero, and that is where $\mathcal{H}^s(E) > 0$ enters: each cube that was scaled down is *saturated*, carrying exactly $\ell(Q)^s$, and the saturated cubes at the top of the construction form a cover of E — so their total $\sum \ell(Q)^s$ is bounded below by $\mathcal{H}_{\infty}^s(E) > 0$, which is precisely the surviving mass.

Proof. The “if” direction is Theorem 7.1.6 with $c = 1$ — note that a set U of diameter d is contained in a ball of radius d , so (7.5.1) gives $\mu(U) \leq (\text{diam } U)^s$.

For “only if”, assume $\mathcal{H}^s(E) > 0$. We use the *dyadic content*

$$\mathcal{H}_\infty^s(E) = \inf \left\{ \sum_i (\text{diam } U_i)^s : E \subseteq \bigcup U_i \right\},$$

the case $\varepsilon = \infty$ of Theorem 7.1.1. Since $\mathcal{H}_\infty^s \leq \mathcal{H}^s$ and, conversely, $\mathcal{H}_\infty^s(E) = 0$ forces $\mathcal{H}^s(E) = 0$ — given a cover with $\sum (\text{diam } U_i)^s < \eta$, each U_i has $\text{diam } U_i < \eta^{1/s}$, so the same cover is admissible for $\mathcal{H}_{\eta^{1/s}}^s$ — we have $\mathcal{H}_\infty^s(E) > 0$. Normalise so that $E \subseteq [0, 1]^n$.

Step 1. *Initialise.*

Fix $m \geq 1$. Define a measure μ_m by giving each dyadic cube Q of generation m that meets E a uniformly distributed mass $\ell(Q)^s = 2^{-ms}$, and mass 0 to the others.

Step 2. *Sweep upwards.*

For $k = m - 1, m - 2, \dots, 0$ in turn, define $\mu^{(k)}$ from $\mu^{(k+1)}$ (starting from $\mu^{(m)} = \mu_m$) by: for each dyadic Q of generation k , if $\mu^{(k+1)}(Q) > \ell(Q)^s = 2^{-ks}$, replace $\mu^{(k+1)}$ on Q by

$$\frac{2^{-ks}}{\mu^{(k+1)}(Q)} \mu^{(k+1)} \upharpoonright Q,$$

and otherwise leave it unchanged. Set $\mu = \mu^{(0)}$.

Each step only decreases mass and only inside a single cube, so after the sweep

$$\mu(Q) \leq \ell(Q)^s \quad \text{for every dyadic } Q \text{ of generation } 0 \leq k \leq m, \quad (7.5.2)$$

because the value at generation k was capped at that stage and later stages only rescale μ on larger cubes by factors ≤ 1 .

Step 3. *Mass does not vanish.*

Call a dyadic cube Q *saturated* if $\mu(Q) = \ell(Q)^s$. Every point of E lies in a saturated cube: the generation- m cube containing it started with mass 2^{-ms} , and if no cube containing it was ever capped then it retains that mass and is itself saturated; otherwise the largest capped cube containing it is saturated, since capping sets its mass exactly to ℓ^s and later stages rescale only by factors applied to still larger cubes — and if such a larger cube were rescaled, it would itself be the saturated one.

Let $\{Q_i\}$ be the maximal saturated cubes; they are disjoint and cover E .

By definition of $\mathcal{H}_\infty^s$, and since $\text{diam } Q_i = \sqrt{n} \ell(Q_i)$,

$$\begin{aligned}
\mathcal{H}_\infty^s(E) &\leq \sum_i (\text{diam } Q_i)^s \\
&= n^{s/2} \sum_i \ell(Q_i)^s \\
&= n^{s/2} \sum_i \mu(Q_i) \\
&= n^{s/2} \mu(\mathbb{R}^n),
\end{aligned}$$

so $\mu(\mathbb{R}^n) \geq n^{-s/2} \mathcal{H}_\infty^s(E) > 0$, a bound independent of m .

Step 4. *From dyadic cubes to balls.*

Let $B = B(x, r)$ with $r < 1$ and choose k with $2^{-k-1} \leq r < 2^{-k}$. The ball B meets at most 2^n dyadic cubes of generation k , so by (7.5.2)

$$\mu(B) \leq 2^n 2^{-ks} \leq 2^n (2r)^s = 2^{n+s} r^s.$$

For $r \geq 1$, $\mu(B) \leq \mu(\mathbb{R}^n) \leq 1 \cdot r^s$ after the normalisation below.

Step 5. *Pass to the limit and normalise.*

The measures $\mu = \mu_{(m)}$ constructed for each m are supported in the compact set $E_{\sqrt{n}2^{-m}}$ and have total mass in $[n^{-s/2} \mathcal{H}_\infty^s(E), 1]$. By weak-* compactness of bounded sets of measures on a compact set — the Banach–Alaoglu theorem, or Theorem 1.8.15 applied on $C(K)$ — a subsequence converges weak-* to a measure μ supported in E , with $\mu(\mathbb{R}^n) \geq n^{-s/2} \mathcal{H}_\infty^s(E) > 0$, and the bound of Step 4 passes to the limit since it holds for closed balls and weak-* limits satisfy $\mu(B) \leq \liminf \mu_{(m)}(B')$ for slightly larger B' . Dividing μ by 2^{n+s} gives (7.5.1).

The final display of the theorem now follows: “ $\geq$ ” from Theorem 7.1.6, and “ $\leq$ ” because for $s < \dim_{\text{H}} E$ we have $\mathcal{H}^s(E) = \infty > 0$ and the construction applies. $\square$

Where we are. The construction is worth summarising in one sentence: *put down too much mass, then cap every cube at its allowance, and observe that the cubes where the cap bites form a cover, so the surviving mass is bounded below by the Hausdorff content.*

Two features are essential. The sweep must go *upwards*, from small cubes to large, so that capping a cube is never undone by a later step. And the lower bound must be against $\mathcal{H}_\infty^s$, the content with no restriction on the diameters, because the maximal saturated cubes can be large. That $\mathcal{H}_\infty^s(E) > 0$ is equivalent to $\mathcal{H}^s(E) > 0$ is the small lemma at the start of

| the proof, and it is the only place the equivalence is used.

WORKED EXAMPLE 7.5.1— TIER A **The Frostman measure on the Cantor set**

For C the middle-thirds Cantor set and $s = \log 2 / \log 3$, the construction of Theorem 7.5.1 recovers the natural measure of Worked Example 7.1.1 — which is a useful check, since that measure was written down by hand.

Why the sweep does nothing. Work with base-3 intervals rather than dyadic cubes. At generation k , C is covered by 2^k intervals of length 3^{-k} , and the initialisation gives each mass $3^{-ks} = 2^{-k}$. A generation- $(k-1)$ interval meeting C contains exactly two generation- k intervals meeting C , so it carries mass $2 \cdot 2^{-k} = 2^{-(k-1)}$, which is exactly its allowance $3^{-(k-1)s}$. Nothing is ever capped.

Consequence. μ is the natural measure, every interval meeting C is saturated, and $\mu(C) = 1$. The construction is sharp here because C is *Ahlfors–David regular*: mass and allowance agree at every scale and location.

Where capping does bite. Take $E = C \cup [2, 3]$ and $s = \log 2 / \log 3$. Now the interval $[2, 3]$ has $\mathcal{H}^s = \infty$, so the initialisation gives it enormous mass, and every containing cube is capped. The Frostman measure that emerges is supported essentially on $[2, 3]$ and gives C almost nothing — correctly, since $\dim_{\mathbb{H}} E = 1 > s$ and the dimension is carried by the interval. Frostman's lemma finds a measure witnessing the dimension, not a canonical one.

WORKED EXAMPLE 7.5.2— TIER B **Frostman as a max-flow min-cut theorem**

There is a second proof of Theorem 7.5.1, due to Howroyd and to Kahane–Salem in the classical case, which explains the construction rather than merely performing it.

The network. Build a graph whose vertices are the dyadic cubes of generations 0 through m meeting E , with an edge from each cube to each of its children, a source above generation 0 and a sink below generation m . Give the edge into a cube Q capacity $\ell(Q)^s$.

Flows are Frostman measures. A flow from source to sink assigns to each cube the amount passing through it, and the capacity constraint is exactly $\mu(Q) \leq \ell(Q)^s$ — condition (7.5.2). So admissible flows *are* the measures we are trying to construct, and the maximum flow is the largest total mass available.

Cuts are covers. A cut is a set of edges whose removal disconnects source from sink, that is, a collection of cubes meeting every root-to-leaf path — a *cover* of E by dyadic cubes. Its capacity is $\sum \ell(Q)^s$, which is comparable to $\sum (\text{diam } Q)^s$.

The theorem. Max-flow min-cut says the maximum mass equals the minimum cover cost, that is

$$\max \{ \mu(E) : \mu \text{ Frostman} \} \approx_n \mathcal{H}_\infty^s(E).$$

Frostman’s lemma is the direction “content positive implies flow positive”, and the capping construction of Theorem 7.5.1 is precisely the standard augmenting-path algorithm run once.

Why this is worth knowing. It explains why the constant is dimensional and unavoidable, why the maximal saturated cubes appear (they are the min cut), and it generalises: the same argument gives Frostman-type lemmas for measures satisfying other growth conditions, simply by changing the capacities.

Where we are. The lemma is an equivalence between two very different-looking statements:

$$\mathcal{H}^s(E) > 0 \quad \Longleftrightarrow \quad \exists \mu \neq 0 \text{ on } E \text{ with } \mu(B(x, r)) \leq r^s.$$

The left side quantifies over *all covers* of E — an infimum over an enormous family. The right side asserts the existence of a *single object*. Converting a universally quantified statement into an existentially quantified one is what makes the hypothesis usable, and it is the reason Frostman’s lemma is the most-cited result of this volume.

The same conversion happens elsewhere in the series and is worth recognising as a pattern: Theorem 6.1.3 turned a restriction estimate (a bound for all f) into an extension estimate (a bound for all g), which is not obviously progress until one observes that $\mathcal{E}g$ can be written down and $\mathcal{R}f$ cannot. Frostman does the same for dimension.

WORKED EXAMPLE 7.5.3— TIER C: RESEARCH TASTE **What Frostman’s lemma is for** Every subsequent lower bound on a dimension in this series goes through Theorem 7.5.1, and it is worth listing the pattern before the applications arrive.

(a) *The pattern.* To prove $\dim_{\mathbb{H}} E \geq s$: produce a measure on E satisfying (7.5.1) and apply Theorem 7.1.6. To *use* $\dim_{\mathbb{H}} E > s$: apply Theorem 7.5.1 to obtain such a measure, then compute with it. The second direction is the one that makes the hypothesis usable, and it is why the lemma is quoted far more often than it is proved.

(b) *Energy.* Chapter 7.6 converts (7.5.1) into the finiteness of an integral, $I_t(\mu) = \iint |x - y|^{-t} d\mu d\mu$. That is a smoother condition, it is stable under the operations one wants to perform — projections, convolutions — and it has a Fourier expression. Almost every computation in Volume 8 is an energy computation, and Frostman is what supplies the measure to compute with.

(c) *Kekeya.* Chapter 7.8 proves that a Kekeya maximal function bound implies a Hausdorff dimension bound. The argument assumes $\dim_{\mathbb{H}} K < s$, extracts from the definition a cover with $\sum r_i^s$ small, groups it by scale, and applies the single-scale estimate to each group. Frostman is not needed for that direction; but the reverse question — what a Besicovitch set of small dimension would look like — is studied through the Frostman measures it would carry, and that is how the arithmetic methods of Volume 10 proceed.

(d) *A caution about “the” Frostman measure.* As Worked Example 7.5.1 showed, the measure is far from unique and need not be spread over the whole set. Arguments that require the measure to see every part of E must say so and arrange it, usually by applying the lemma to a subset chosen in advance. Errors of this kind are common in the literature and are worth watching for.

WORKED EXAMPLE 7.5.4— TIER B Using Frostman: the standard three-line opening

Almost every argument in Volumes 8 and 14 begins the same way, and it is worth rehearsing the moves in isolation so that they are recognised later.

The situation. One is given a compact E with $\dim_{\mathbb{H}} E > s$ and must prove some quantitative statement about E .

Move 1: get a measure. Choose t with $s < t < \dim_{\mathbb{H}} E$. By Theorem 7.5.1 there is a nonzero finite μ on E with $\mu(B(x, r)) \leq r^t$ for all x, r . Normalise $\mu(E) = 1$.

Move 2: convert to energy. By Step 2 of Theorem 7.6.2, $I_{t'}(\mu) < \infty$ for every $t' < t$; fix such a $t' > s$. One now has a finite number $I_{t'}(\mu)$ to work with instead of a supremum condition.

Move 3: go to the Fourier side if the problem is two-point. By Theorem 7.6.5, $\int |\hat{\mu}(\xi)|^2 |\xi|^{t'-n} d\xi < \infty$, so $\hat{\mu}$ is small in an averaged sense. Any bilinear or convolution quantity built from E can now be estimated by Plancherel.

A worked instance: the distance set has positive measure when $\dim_{\mathbb{H}} E > \frac{n+1}{2}$. Sketching the shape only, since Volume 8 gives the details: let ν be the pushforward of $\mu \times \mu$ under $(x, y) \mapsto |x - y|$. One shows ν has an L^2 density

by writing

$$\int |\widehat{\nu}(\tau)|^2 d\tau \approx \int_0^\infty \left(\int_{\mathbb{S}^{n-1}} |\widehat{\mu}(\tau\omega)|^2 d\sigma(\omega) \right)^2 \tau^{n-1} d\tau,$$

and the inner spherical average is controlled by the decay of $\widehat{d\sigma}$ from Theorem 5.5.2 together with Move 3. The exponent $\frac{n+1}{2}$ emerges from matching the two, exactly as in Worked Example 7.7.2.

What to watch. Moves 1 and 2 each lose an epsilon in the exponent, which is why conclusions are always stated for $\dim_{\mathbb{H}} E > s$ strictly and never at equality. And the measure from Move 1 is not canonical (Worked Example 7.5.1), so any step requiring μ to charge a particular part of E must arrange it by applying Frostman to a suitable subset first.

Figure 7.5.1 (to be drawn). *The Frostman sweep*

Content required: A dyadic tree of four or five generations over an interval, with mass $\ell(Q)^s$ placed at the bottom; arrows showing the upward sweep; and the capped (saturated) cubes marked, forming an antichain that covers the set. This antichain is the min cut of Worked Example 7.5.2.

Drills

Drill 7.5.1. Prove carefully that $\mathcal{H}_\infty^s(E) = 0$ implies $\mathcal{H}^s(E) = 0$, the small lemma used at the start of the proof.

Drill 7.5.2. Verify (7.5.2): that capping at generation k is not undone by later stages of the sweep.

Drill 7.5.3. Carry out Worked Example 7.5.1 for the Cantor set C_λ of Worked Example 7.1.2 and confirm that nothing is capped there either.

Drill 7.5.4. Show that the conclusion of Theorem 7.5.1 fails for non-compact E by taking $E = \mathbb{Q} \cap [0, 1]$ and $s = 0$, and identify which step of the proof breaks.

Where we are. One structural remark before leaving the chapter. Theorem 7.5.1 is stated for *compact* E , and compactness is used exactly once — in Step 5, to extract a weak-* convergent subsequence. Everything before that step is a finite construction at a fixed depth m .

That localisation of the hypothesis is worth noticing, because it tells one what to do when compactness is unavailable. For a Borel set one applies the lemma to compact subsets and takes a supremum, which works because Hausdorff measure is inner regular on Borel sets of finite measure. For a general analytic set the required inner regularity is Choquet's capacitability theorem. And in a metric space with no local compactness the weak-* argument fails entirely, which is why Howroyd's max-flow proof (Worked Example 7.5.2) — which never passes to a limit — is the one that generalises.

Notes

Frostman proved the lemma in his 1935 thesis. The dyadic construction given here is the standard modern one; the max-flow interpretation of Worked Example 7.5.2 is due to Howroyd (1995), who used it to extend the lemma to general metric spaces, where the compactness argument of Step 5 is unavailable.

The result is sometimes stated for analytic rather than compact sets; that version requires a capacitability theorem of Choquet and is not needed here.

Chapter 7.6

Energy Integrals

Frostman's lemma produces a measure with a bound at every scale and location. That is a supremum condition, and suprema are awkward to combine. The energy integral replaces it by an integral condition, which behaves well under the operations Volume 8 needs — projections, convolutions, and Fourier transforms.

7.6.1 The energy of a measure

Definition 7.6.1. For a finite Borel measure μ on $\mathbb{R}^n$ and $t > 0$, the t -energy is

$$I_t(\mu) = \iint \frac{d\mu(x) d\mu(y)}{|x - y|^t} = \int (k_t * \mu) d\mu, \quad k_t(x) = |x|^{-t}.$$

Theorem 7.6.2. Let $E \subseteq \mathbb{R}^n$ be compact.

- (i) If μ is a nonzero finite measure on E with $I_t(\mu) < \infty$, then $\dim_{\text{H}} E \geq t$.
- (ii) If $\dim_{\text{H}} E > t$, there is a nonzero finite measure μ on E with $I_t(\mu) < \infty$.

Consequently

$$\dim_{\text{H}} E = \sup \{t \geq 0 : \exists \mu \neq 0 \text{ on } E, I_t(\mu) < \infty\}.$$

Proof. **Step 1.** (i): finite energy forces a Frostman bound on a large subset.

Let μ be as stated, normalised to $\mu(E) = 1$. Put

$$F = \left\{x \in E : \limsup_{r \downarrow 0} \frac{\mu(B(x, r))}{r^t} > 0\right\}.$$

We claim $\mu(F) = 0$. Indeed, for $x \in E$,

$$\begin{aligned} (k_t * \mu)(x) &= \int \frac{d\mu(y)}{|x - y|^t} \\ &\geq \int_{B(x, r)} \frac{d\mu(y)}{|x - y|^t} \\ &\geq \frac{\mu(B(x, r))}{r^t}, \end{aligned}$$

so if $\limsup_r \mu(B(x, r))r^{-t} = c > 0$ then $(k_t * \mu)(x) \geq c$; but more is true: decomposing $B(x, r)$ into dyadic annuli one finds $(k_t * \mu)(x) \geq \sum_j 2^{jt} \mu(B(x, 2^{-j}) \setminus B(x, 2^{-j-1}))$, and if $\mu(B(x, 2^{-j})) \geq c2^{-jt}$ for infinitely many j the sum diverges. Since $I_t(\mu) = \int (k_t * \mu) d\mu < \infty$, we have $(k_t * \mu) < \infty$ μ -a.e., so $\mu(F) = 0$.

Hence $\mu(E \setminus F) = 1$, and on $E \setminus F$ we have $\mu(B(x, r)) \leq r^t$ for all small r , uniformly after restricting to a subset E' of positive measure by Egorov's theorem (Theorem 1.5.9) applied to the functions $x \mapsto \sup_{r \leq 1/k} \mu(B(x, r))r^{-t}$. Then $\mu \upharpoonright E'$ satisfies the hypothesis of Theorem 7.1.6, giving $\dim_H E \geq \dim_H E' \geq t$.

Step 2. (ii): *a Frostman measure at a higher exponent has finite energy.*

Let $t < s < \dim_H E$. By Theorem 7.5.1 there is a nonzero μ on E with $\mu(B(x, r)) \leq r^s$ for all x, r . We show $I_t(\mu) < \infty$.

Fix $x \in \text{spt } \mu$ and decompose by dyadic annuli $A_j = \{y : 2^{-j-1} < |x - y| \leq 2^{-j}\}$ for $j \geq 0$, together with $\{|x - y| > 1\}$:

$$\begin{aligned} (k_t * \mu)(x) &\leq \mu(\mathbb{R}^n) + \sum_{j \geq 0} 2^{(j+1)t} \mu(A_j) \\ &\leq \mu(\mathbb{R}^n) + 2^t \sum_{j \geq 0} 2^{jt} \mu(B(x, 2^{-j})). \end{aligned}$$

By the Frostman bound $\mu(B(x, 2^{-j})) \leq 2^{-js}$, so the sum is at most $\sum_j 2^{j(t-s)} < \infty$ since $t < s$. Hence $k_t * \mu$ is bounded, and

$$I_t(\mu) = \int (k_t * \mu) d\mu \leq \|k_t * \mu\|_{L^\infty} \mu(\mathbb{R}^n) < \infty. \quad \square$$

Where we are. Step 2 is the direction used constantly, and it is worth extracting: a Frostman measure of exponent s has finite t -energy for every $t < s$. The loss of the endpoint is real — a measure with $\mu(B(x, r)) \approx r^s$ exactly has $I_s(\mu) = \infty$, since the sum $\sum_j 2^{j(s-s)}$ diverges logarithmically.

This is the same borderline that appeared in Theorem 6.8.6 and in the failure of $\mathcal{M}f \in L^1$: a quantity that is exactly critical fails by a logarithm. In practice one always works at $t < s$ and lets $t \uparrow s$ at the end, which is why dimension statements in this subject are almost never attained at the critical exponent.

Proposition 7.6.3 (Energy under maps and products). *Let μ be a finite Borel measure on $\mathbb{R}^n$ and $t > 0$.*

- (i) *If f is bi-Lipschitz with constants $L_1 \leq L_2$, then $I_t(f_{\#}\mu) \leq L_1^{-t} I_t(\mu)$.*
- (ii) *If ν is a finite Borel measure on $\mathbb{R}^m$ with $I_u(\nu) < \infty$, then $I_{t+u}(\mu \times \nu) < \infty$ whenever $I_t(\mu) < \infty$ and one of the two measures is bounded.*

Proof. (i) By the change of variables formula for pushforwards,

$$\begin{aligned} I_t(f_{\#}\mu) &= \iint \frac{d\mu(x) d\mu(y)}{|f(x) - f(y)|^t} \\ &\leq \iint \frac{d\mu(x) d\mu(y)}{(L_1|x - y|)^t} \\ &= L_1^{-t} I_t(\mu), \end{aligned}$$

using $|f(x) - f(y)| \geq L_1|x - y|$.

(ii) For $(x, y), (x', y') \in \mathbb{R}^n \times \mathbb{R}^m$ we have $|(x, y) - (x', y')| \geq \max\{|x - x'|, |y - y'|\}$, so

$$\frac{1}{|(x, y) - (x', y')|^{t+u}} \leq \frac{1}{|x - x'|^t |y - y'|^u},$$

since $\max\{a, b\}^{t+u} \geq a^t b^u$ for $a, b > 0$. Integrating against $\mu \times \nu$ twice and applying Tonelli factorises the double integral into $I_t(\mu) I_u(\nu)$. $\square$

Corollary 7.6.4. $\dim_{\mathbb{H}}(E \times F) \geq \dim_{\mathbb{H}} E + \dim_{\mathbb{H}} F$ for compact $E \subseteq \mathbb{R}^n$, $F \subseteq \mathbb{R}^m$.

Proof. Let $t < \dim_{\mathbb{H}} E$ and $u < \dim_{\mathbb{H}} F$. By Theorem 7.6.2(ii) there are measures μ on E and ν on F with $I_t(\mu), I_u(\nu) < \infty$. By Theorem 7.6.3(ii), $I_{t+u}(\mu \times \nu) < \infty$, and $\mu \times \nu$ is supported in $E \times F$. By Theorem 7.6.2(i), $\dim_{\mathbb{H}}(E \times F) \geq t + u$; let $t \uparrow \dim_{\mathbb{H}} E$ and $u \uparrow \dim_{\mathbb{H}} F$. $\square$

Where we are. Theorem 7.6.4 is one inequality, and it is the only one that holds in general: there are compact $E, F \subseteq \mathbb{R}$ with $\dim_{\mathbb{H}} E = \dim_{\mathbb{H}} F = 0$ and $\dim_{\mathbb{H}}(E \times F) = 1$. The construction alternates the scales at which E and F are “active”, so that at every scale one factor is rich while the other is poor, and the product is rich throughout — the same mechanism as the

failure of finite stability for $\underline{\dim}_M$ in Theorem 7.2.5.

The example is due to Besicovitch and Moran. Its moral is the one the whole volume keeps returning to: Hausdorff dimension is an infimum over covers with freely varying scales, and any quantity built by combining single-scale data can behave unexpectedly.

7.6.2 The Fourier expression

Proposition 7.6.5. *Let $0 < t < n$ and let μ be a finite Borel measure on $\mathbb{R}^n$ with compact support. Then*

$$I_t(\mu) = c_{n,t} \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{t-n} d\xi,$$

where $c_{n,t} > 0$ is the constant of Theorem 5.6.4, and both sides may be infinite simultaneously.

Proof. By Volume 2, the distributional identity $\widehat{k}_t = c_{n,t} |\xi|^{t-n}$ holds for $0 < t < n$ — this is Worked Example 2.5.2, restated by Theorem 5.6.4 as $\widehat{Z}_t = Z_{n-t}$. Since μ is a compactly supported finite measure, $\widehat{\mu}$ is a bounded continuous function, and Parseval's identity for tempered distributions (Theorem 2.5.11) gives

$$I_t(\mu) = \langle k_t * \mu, \mu \rangle = \langle \widehat{k}_t \widehat{\mu}, \widehat{\mu} \rangle = c_{n,t} \int |\widehat{\mu}(\xi)|^2 |\xi|^{t-n} d\xi,$$

the middle step because the Fourier transform turns convolution into multiplication and is unitary in the appropriate pairing. Both sides are integrals of nonnegative quantities, so the identity holds in $[0, \infty]$. $\square$

Corollary 7.6.6. *If μ is a nonzero compactly supported measure with $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta}$ for some $\beta > 0$, then $\dim_H(\text{spt } \mu) \geq \min\{2\beta, n\}$.*

Proof. By Theorem 7.6.5, $I_t(\mu) \lesssim \int |\xi|^{-2\beta+t-n} d\xi$ over $|\xi| \geq 1$, plus a finite contribution from $|\xi| \leq 1$ since $\widehat{\mu}$ is bounded. By polar coordinates the integral converges when $-2\beta + t - n < -n$, that is $t < 2\beta$. So $I_t(\mu) < \infty$ for every $t < \min\{2\beta, n\}$, and Theorem 7.6.2(i) applies. $\square$

WORKED EXAMPLE 7.6.1— TIER A The energy of the Cantor measure

Let μ be the natural measure on the middle-thirds Cantor set C and $s = \log 2 / \log 3$. Worked Example 7.1.1 gave $\mu(I) \leq 6(\text{diam } I)^s$, so by Step 2 of Theorem 7.6.2, $I_t(\mu) < \infty$ for every $t < s$.

Direct verification. For $x \in C$, $\mu(B(x, 3^{-k})) = 2^{-k} = 3^{-ks}$ exactly, so

$$(k_t * \mu)(x) \approx \sum_{k \geq 0} 3^{kt} 3^{-ks} = \sum_{k \geq 0} 3^{k(t-s)},$$

convergent precisely for $t < s$ and divergent at $t = s$. So the Cantor measure has finite t -energy exactly for $t < s$, and $I_s(\mu) = \infty$ — the endpoint loss of the Checkpoint above, made explicit.

Conclusion. $\dim_{\mathbb{H}} C \geq t$ for every $t < s$, hence $\dim_{\mathbb{H}} C \geq s$, recovering Worked Example 7.1.1 by a different route. The energy computation is longer here, but unlike the covering argument it will survive being pushed through a projection in Volume 8.

WORKED EXAMPLE 7.6.2— TIER B Energy is stable under the operations covering is not

The reason to prefer energy over the Frostman condition is that it passes through maps and convolutions. Three instances, all used in Volume 8.

(a) *Projections.* Let $\pi_{\theta}(x) = x \cdot \theta$ be the orthogonal projection of $\mathbb{R}^2$ onto the line in direction θ . Then

$$\int_0^\pi I_t(\pi_{\theta\#}\mu) \, d\theta = \iint \left(\int_0^\pi \frac{d\theta}{|(x-y) \cdot \theta|^t} \right) d\mu \, d\mu \approx_t I_t(\mu) \quad (0 < t < 1),$$

because the inner integral is $\approx |x-y|^{-t}$ — substituting $u = (x-y)/|x-y|$ reduces it to a fixed convergent integral over θ . So finite t -energy is inherited by *almost every* projection. That single computation is Marstrand's projection theorem, and it is impossible to see through the covering definition of dimension.

(b) *Convolutions and distance sets.* The distance set $\Delta(E) = \{|x-y| : x, y \in E\}$ is studied through the pushforward of $\mu \times \mu$ under $(x, y) \mapsto |x-y|$, whose energy is again computable from $I_t(\mu)$. Theorem 7.6.5 turns the computation into one about $\widehat{\mu}$, and the decay of $\widehat{d\sigma}$ from Volume 5 enters. That is the Mattila integral of Volume 8.

(c) *The Fourier side.* Theorem 7.6.5 says the energy is a weighted L^2 norm of $\widehat{\mu}$. Any estimate for $\widehat{\mu}$ — for instance one coming from a restriction theorem — therefore becomes a dimension bound, by Theorem 7.6.6. This is the bridge between Volumes 6 and 8, and it does not exist at the level of coverings.

The general point. Hausdorff dimension is defined by an infimum over covers and is therefore hard to compute and harder to transport. Frostman converts it to the existence of a measure; energy converts that to the finiteness of an

integral; and integrals can be estimated, exchanged, and Fourier transformed. Each step trades exactness for flexibility, and the flexibility is what the applications need.

WORKED EXAMPLE 7.6.3— TIER C: RESEARCH TASTE Where the energy method stops

Theorem 7.6.6 bounds dimension below by twice the Fourier decay rate, and it is worth understanding why the factor 2 is there and why the method saturates.

(a) *The factor 2 is Plancherel.* The energy is quadratic in μ , so a decay rate β for $\hat{\mu}$ contributes 2β to the exponent. There is no way to improve this within the method: the energy is the only quantity for which Theorem 7.6.5 holds.

(b) *The bound is often far from sharp.* For the Cantor set, $\hat{\mu}$ does not decay at all — $\dim_F C = 0$, proved in Chapter 7.7 — so Theorem 7.6.6 gives nothing, while the true dimension is $\log 2 / \log 3$. The Fourier method sees only the part of the dimension that is visible to the transform.

(c) *The saturation point is $n/2$.* Since $\dim_H \leq n$, the bound 2β is useful only for $\beta < n/2$; and a measure on a set of dimension s cannot have $\hat{\mu}$ decaying faster than $|\xi|^{-s/2}$, by Theorem 7.6.5 run backwards. So the method is sharp exactly for *Salem sets*, where $\dim_F = \dim_H$, and lossy otherwise.

(d) *What this predicts about Kakeya.* A Besicovitch set has $\dim_H = n$ conjecturally, so a measure on it with $\hat{\mu}$ decaying like $|\xi|^{-n/2}$ would prove the conjecture by Theorem 7.6.6. No such measure is known to exist, and the Fourier-analytic bound obtainable this way — Chapter 7.7 carries it out — gives only $\dim_H \geq (n+1)/2$, the bush bound of Volume 1. So the energy method reproduces the easiest known bound and nothing beyond it. Every improvement in Volumes 9 to 14 is geometric or combinatorial, not Fourier-analytic, and this is the precise sense in which the Fourier method is exhausted.

WORKED EXAMPLE 7.6.4— TIER B Estimating an energy: the dyadic annulus method

Every energy computation in this series reduces to the same manoeuvre, used twice already in Theorem 7.6.2. It is worth isolating.

The method. To estimate $(k_t * \mu)(x) = \int |x - y|^{-t} d\mu(y)$, split the domain into the annuli $A_j = \{2^{-j-1} < |x - y| \leq 2^{-j}\}$, $j \geq 0$, plus the region $|x - y| > 1$.

On A_j the kernel is $\approx 2^{jt}$ and the mass is $\mu(A_j) \leq \mu(B(x, 2^{-j}))$, giving

$$(k_t * \mu)(x) \lesssim \mu(\mathbb{R}^n) + \sum_{j \geq 0} 2^{jt} \mu(B(x, 2^{-j})).$$

Everything then depends on how $\mu(B(x, r))$ behaves.

Case 1: a Frostman measure of exponent s . Then $\mu(B(x, 2^{-j})) \leq 2^{-js}$ and the sum is $\sum_j 2^{j(t-s)}$: convergent for $t < s$, divergent for $t \geq s$. This is Step 2 of Theorem 7.6.2, and it gives finite energy with a bound *uniform in x* , hence $I_t(\mu) < \infty$.

Case 2: an Ahlfors–David regular measure, $\mu(B(x, r)) \approx r^s$. Now the sum is $\approx \sum_j 2^{j(t-s)}$ from both sides, so $I_t(\mu) < \infty$ exactly for $t < s$ and $= \infty$ for $t \geq s$. The Cantor measure of Worked Example 7.6.1 is the model.

Case 3: a measure with a heavier point. If μ has an atom, every $\mu(B(x, r))$ is bounded below near that point and $I_t(\mu) = \infty$ for every $t > 0$. So finite energy already forces μ to be non-atomic — a small observation, but it is why one never has to exclude atoms by hand.

The general shape. A convergent geometric series appears exactly when the kernel’s growth rate t is strictly less than the measure’s decay rate s , and the endpoint always fails logarithmically. Recognising this pattern lets one read off the answer to most energy questions without computing: *compare the exponent in the kernel with the exponent in the measure, and the smaller wins, strictly.*

Drills

Drill 7.6.1. Verify the dyadic annulus estimate in Step 1 of Theorem 7.6.2 and check that the divergence is genuine.

Drill 7.6.2. Compute $I_t(\mu)$ for μ Lebesgue measure on the unit ball, and confirm it is finite exactly for $t < n$.

Drill 7.6.3. Carry out the inner-integral computation in Worked Example 7.6.2(a) and determine the constant’s dependence on t as $t \uparrow 1$.

Drill 7.6.4. Show that Theorem 7.6.6 is sharp for the measure $d\sigma$ on $\mathbb{S}^{n-1}$: compute β from Theorem 5.5.2 and compare 2β with $\dim_{\mathbb{H}} \mathbb{S}^{n-1} = n - 1$.

Notes

The energy characterisation of dimension is due to Frostman; the Fourier expression Theorem 7.6.5 is standard and is the starting point of Mattila's book, which is the reference for this chapter and for Volume 8.

The observation in Worked Example 7.6.3(d) — that the Fourier method gives exactly $(n + 1)/2$ for *Takeya* — is due to Drury and to Christ–Duoandikoetxea–Rubio de Francia, and was the best known bound for over a decade.

Chapter 7.7

Fourier Dimension

Theorem 7.6.6 bounds Hausdorff dimension below by twice the decay rate of $\widehat{\mu}$. That suggests defining a second dimension from the decay alone, and comparing. The comparison is instructive: the two agree for some sets and differ maximally for others, and which case one is in decides whether Fourier analysis is any use.

7.7.1 Definition and the basic inequality

Definition 7.7.1. For a compact $E \subseteq \mathbb{R}^n$, the *Fourier dimension* is

$$\dim_F E = \sup \left\{ \beta \in [0, n] : \exists \mu \neq 0 \text{ on } E \text{ with } |\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta/2} \right\},$$

the supremum of the empty set being 0.

The normalisation $\beta/2$ is chosen so that Theorem 7.6.6 reads:

Theorem 7.7.2. For every compact $E \subseteq \mathbb{R}^n$,

$$\dim_F E \leq \dim_H E.$$

Proof. Let $\beta < \dim_F E$ and take $\mu \neq 0$ on E with $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-\beta/2}$. By Theorem 7.6.6 with the decay exponent $\beta/2$ in place of β , $\dim_H E \geq \min\{\beta, n\} = \beta$. Let $\beta \uparrow \dim_F E$. $\square$

Definition 7.7.3. E is a *Salem set* if $\dim_F E = \dim_H E$.

WORKED EXAMPLE 7.7.1— TIER A **The sphere is Salem; a smooth surface piece need not be**

The sphere. Take $E = \mathbb{S}^{n-1}$ and $\mu = d\sigma$. By Theorem 5.5.2, $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$, so $\beta = n - 1$ is admissible and $\dim_F \mathbb{S}^{n-1} \geq n - 1$. Since $\dim_H \mathbb{S}^{n-1} = n - 1$, Theorem 7.7.2 gives equality: the sphere is Salem.

A piece of a hyperplane is not. Let $E = [0, 1]^{n-1} \times \{0\}$, so $\dim_H E = n - 1$. Any measure on E has $\widehat{\mu}(\xi)$ independent of ξ_n — μ is a product with δ_0 in the last coordinate — so $\widehat{\mu}$ does not decay along the ξ_n axis at all, and $\dim_F E = 0$.

The moral. Fourier dimension measures how *spread out in direction* the set is, not merely how large. A flat piece is invisible to the transform in the normal direction, whatever its Hausdorff dimension. This is the same distinction as in Volume 5: curvature produces decay, flatness does not, and $\dim_F$ is the systematic version of that statement.

Theorem 7.7.4. *The middle-thirds Cantor set C has $\dim_F C = 0$, while $\dim_H C = \log 2 / \log 3$.*

Proof. Let μ be any nonzero finite measure supported in C . We show $\widehat{\mu}(\xi) \not\rightarrow 0$ along a sequence, which gives $\dim_F C = 0$.

The key is that C is invariant under $x \mapsto 3x$ modulo the construction: more precisely, $C \subseteq \{x : \{3^k x\} \in [0, \frac{1}{3}] \cup [\frac{2}{3}, 1] \ \forall k \geq 0\}$, where $\{\cdot\}$ is the fractional part, since the base-3 digits of points of C are all 0 or 2.

Consider $\xi_k = 3^k$. For $x \in C$ write $x = \sum_{j \geq 1} a_j 3^{-j}$ with $a_j \in \{0, 2\}$. Then

$$3^k x = \underbrace{\sum_{j \leq k} a_j 3^{k-j}}_{\text{an integer}} + \sum_{j > k} a_j 3^{k-j},$$

and the tail lies in $[0, 1]$ with $\sum_{j > k} a_j 3^{k-j} \leq 2 \sum_{i \geq 1} 3^{-i} = 1$. So $e^{2\pi i 3^k x} = e^{2\pi i \theta_k(x)}$ with $\theta_k(x) = \sum_{j > k} a_j 3^{k-j} \in [0, 1]$, and in fact $\theta_k(x) \in [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$ since $a_{k+1} \in \{0, 2\}$ contributes 0 or $\frac{2}{3}$ and the remainder is at most $\frac{1}{3}$.

Therefore $\operatorname{Re} e^{2\pi i 3^k x} = \cos(2\pi \theta_k(x)) \geq \cos(2\pi/3) = -\frac{1}{2}$, and on the part where $\theta_k \in [0, \frac{1}{3}]$ it is $\geq -\frac{1}{2}$ as well, with $\cos \geq \frac{1}{2}$ when $\theta_k \in [0, \frac{1}{6}] \cup [\frac{5}{6}, 1]$. A more careful count is unnecessary: the point is that $e^{2\pi i 3^k x}$ takes values in a fixed arc of the circle of length less than the whole, uniformly in k , so

$$\begin{aligned} |\widehat{\mu}(3^k)| &= \left| \int_C e^{-2\pi i 3^k x} d\mu(x) \right| \\ &\geq \left| \operatorname{Re} \int_C e^{-2\pi i 3^k x} d\mu \right| \\ &\geq c \mu(C) \end{aligned}$$

for an absolute $c > 0$, since the integrand has real part bounded below by a positive constant on a fixed proportion of μ -mass, and the estimate is uniform in k because the digit structure is.

Hence $\widehat{\mu}(3^k) \not\rightarrow 0$, so no decay exponent $\beta > 0$ is admissible and $\dim_F C = 0$. $\square$

Pitfall: Fourier dimension is not a dimension in the usual sense

$\dim_F$ fails properties one expects. It is not countably stable, nor even finitely stable: $\dim_F(E \cup F)$ can be 0 while $\dim_F E$ is large, because a measure on the union may be forced to charge the bad part. It is not monotone in the naive way either — a subset can have larger Fourier dimension than the set, since fewer constraints bind.

It is also *not* determined by the set's geometry in any local sense: it is a statement about which measures the set carries, and Theorem 7.7.4 shows a set can be geometrically substantial and Fourier-analytically invisible.

The reader should treat $\dim_F$ as a convenient name for “the best decay exponent available on this set” rather than as a dimension. Its only role in this series is Theorem 7.7.2, used in the direction $\dim_H \geq \dim_F$.

7.7.2 The Fourier bound for Besicovitch sets

Remark 7.7.5 (Scope decision: Drury's bound is not reproved here). The energy method yields, for every Besicovitch set $K \subseteq \mathbb{R}^n$,

$$\dim_H K \geq \frac{n+1}{2},$$

a theorem of Drury (1983) and, independently, Christ–Duoandikoetxea–Rubio de Francia. *We do not prove it in this series.*

The reason is that we have it already, by a shorter and entirely different argument: Volume 1, Theorem 1.7.15 proved exactly this bound by the bush argument, counting tube intersections, with no Fourier analysis and no Frostman measure. Reproving it here would add a second proof of a known result at the cost of several pages of careful bookkeeping with dual slabs.

What is worth extracting from the Fourier route is not the bound but the *explanation of why $(n+1)/2$ is where it stops*, and Worked Example 7.7.2 gives that without needing the full argument. Readers wanting the complete computation will find it in Mattila's book, Chapter 22.

Recorded as revision R7 in the master manifest.

WORKED EXAMPLE 7.7.2— TIER B **Why $(n+1)/2$, and why the Fourier method cannot pass it**

The exponent can be predicted by matching two dimension counts, and the count also shows why no refinement of the method will improve it.

The heuristic. A measure μ of dimension t has, by Theorem 7.6.5, finite I_t , and the borderline case corresponds to $|\widehat{\mu}(\xi)| \approx |\xi|^{-t/2}$. A Besicovitch set contains a unit segment in every direction, so it must in particular support the directional spreading measured by $d\sigma$, whose transform decays like $|\xi|^{-(n-1)/2}$ (Theorem 5.5.2).

The dimension count. A segment is 1-dimensional; the directions form an $(n-1)$ -dimensional sphere. If the segments were entirely independent of their directions the union would have dimension $1 + (n-1) = n$. If instead each segment is determined by its direction — the opposite extreme — one gets only $\max\{1, n-1\}$. The Fourier method sees the average of these two behaviours, because I_t is a two-point average, and delivers the midpoint

$$1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

The bush picture, which makes it exact. Volume 1's proof of Theorem 1.7.15 is the same statement without averaging: a bush of μ tubes through a common point occupies volume $\gtrsim \mu\delta^{n-1}$, because at unit distance from the centre the tubes have separated. Balancing that against the trivial bound gives $\frac{n+1}{2}$ exactly.

Why the method saturates there. The bush argument uses only that many tubes pass through *one* point — a two-point condition, matching what I_t can see. Wolff's hairbrush argument, in Volume 9, uses that many tubes pass through a *line*, which is a three-point condition, and gains $\frac{n+2}{2}$. No amount of care with the Fourier transform reaches it, for the reason set out in Worked Example 7.7.3(a): the transform of a measure encodes pair correlations and nothing more.

Where we are. Three sets, three verdicts, and they organise the chapter.

The sphere has $\dim_F = \dim_H = n-1$: curvature makes $\widehat{d\sigma}$ decay at the maximal rate compatible with the dimension. Fourier methods are sharp.

A hyperplane piece has $\dim_F = 0$ and $\dim_H = n-1$: no decay in the normal direction. Fourier methods see nothing.

The Cantor set has $\dim_F = 0$ and $\dim_H = \log 2 / \log 3$: arithmetic self-similarity obstructs decay even though the set is curved in no direction at all — it is totally disconnected. Fourier methods again see nothing, for a different reason.

The pattern is that $\dim_F$ measures a form of *directional and arithmetic*

spreading, and $\dim_{\mathbb{H}}$ measures size. They agree only when the set is “generic” in the appropriate sense, and every deterministic construction in this series — Perron trees, Cantor sets, self-similar attractors — is arithmetic and therefore not generic.

WORKED EXAMPLE 7.7.3— TIER C: RESEARCH TASTE **When is Fourier analysis the right tool for a dimension problem?**

This chapter and the last give a criterion, worth stating because it settles the question for the rest of the series.

(a) *The method sees only I_t , a two-point quantity.* By Theorem 7.6.5, everything the Fourier transform contributes is encoded in $\iint |x - y|^{-t} d\mu d\mu$ — an average over *pairs* of points. Any feature of a set requiring three or more points to detect is invisible.

(b) *Consequently it is sharp exactly for Salem sets.* If $\dim_{\mathbb{F}} E = \dim_{\mathbb{H}} E$ the method loses nothing; otherwise it loses exactly the gap. Random sets are typically Salem — Brownian images, random Cantor sets — and deterministic self-similar sets typically are not, by the mechanism of Theorem 7.7.4: exact self-similarity creates arithmetic structure, and arithmetic structure obstructs decay.

(c) *Besicovitch sets are not Salem, and this is why Volumes 9–14 exist.* The Fourier bound stops at $\frac{n+1}{2}$ while the truth is conjectured to be n . The obstruction is (a): a Besicovitch set’s smallness is a statement about how *many tubes meet*, which is a three-point or higher condition, and the two-point function cannot see it.

(d) *Where Fourier analysis does remain essential.* Not for Kakeya, but for the distance set problem of Volume 8, where the quantity of interest — the pushforward of $\mu \times \mu$ under $(x, y) \mapsto |x - y|$ — is genuinely a two-point object. There the Mattila integral is the sharp tool and the combinatorial methods are the lossy ones. Knowing which regime a problem is in before choosing a method is most of the skill.

WORKED EXAMPLE 7.7.4— TIER B **Detecting arithmetic structure through $\hat{\mu}$**

Theorem 7.7.4 showed $\dim_{\mathbb{F}} C = 0$ because $\hat{\mu}(3^k)$ does not tend to zero. The mechanism is general and worth naming, since it decides in practice whether a set is Salem.

The principle. If $\text{spt } \mu$ is invariant — even approximately — under a map $x \mapsto \lambda x$ with λ an integer, then the frequencies λ^k all “see” the same structure, and $\hat{\mu}(\lambda^k)$ cannot decay. Exact self-similarity with an integer ratio is the

extreme case.

Worked instance: the Cantor set with ratio 1/4. Let $C_{1/4}$ be the attractor of $f_1(x) = x/4$, $f_2(x) = x/4 + 3/4$, with $\dim_H = \frac{1}{2}$ by Theorem 7.8.3. Points have base-4 digits in $\{0, 3\}$, so for the natural measure

$$\widehat{\mu}(4^k) = \int e^{-2\pi i 4^k x} d\mu(x)$$

and $4^k x$ has fractional part determined by the digits after position k , which again lie in $\{0, 3\}$. The same argument as Theorem 7.7.4 gives $|\widehat{\mu}(4^k)| \geq c > 0$, so $\dim_F C_{1/4} = 0$ although $\dim_H C_{1/4} = \frac{1}{2}$.

Contrast: a random Cantor set. Build a Cantor set by keeping, at each stage and independently, a random one of the four quarters. The resulting set has $\dim_H = \frac{1}{2}$ almost surely, and now the digits are independent random variables, so $\widehat{\mu}(\xi)$ is a product of independent factors each of modulus < 1 ; a second-moment computation gives $|\widehat{\mu}(\xi)| \lesssim |\xi|^{-1/4+\varepsilon}$ almost surely, so the set is Salem. The computation is Salem's and is not carried out here (Chapter 7.7 scope note).

The diagnostic. Deterministic and arithmetically structured $\Rightarrow$ probably $\dim_F = 0$, and Fourier methods will fail. Random or curved $\Rightarrow$ probably Salem, and Fourier methods will be sharp. Spheres are curved and Salem (Worked Example 7.7.1); Cantor sets are arithmetic and are not. Besicovitch sets are neither, which is why the situation there (Worked Example 7.7.3) is genuinely intermediate and hard.

Drills

Drill 7.7.1. Show $\dim_F E = 0$ for every E contained in a hyperplane, and more generally for every E contained in a smooth surface of dimension $< n - 1$.

Drill 7.7.2. Show that $\dim_F$ is not finitely stable, by taking E a Salem set of dimension $\frac{1}{2}$ in $\mathbb{R}$ and F a point.

Drill 7.7.3. For a δ -tube in direction ω , show that the dual slab has dimensions $\delta^{-1} \times \cdots \times \delta^{-1} \times 1$, and that the measure of $\{\omega : \xi \text{ lies in that slab}\}$ is $\approx \min\{1, (\delta|\xi|)^{-(n-1)}\}$.

Drill 7.7.4. Using Worked Example 7.7.2, predict the exponent the same heuristic would give for a set containing a k -plane in every direction, and compare with the bush bound for such sets.

Notes

Salem constructed sets with $\dim_F = \dim_H$ in 1951, using random Cantor sets; Kahane gave the Brownian construction. That the middle-thirds Cantor set has $\dim_F = 0$ is classical and is the model instance of arithmetic structure obstructing decay.

Drury's bound $\frac{n+1}{2}$ (1983), obtained independently by Christ–Duoandikoetxea–Rubio de Francia, stood as the best for general n until Bourgain's and Wolff's work in the 1990s. Theorem 7.7.5 explains why it is not reproved here.

Chapter 7.8

Self-Similar Sets and the Keakeya Upgrade

Two loose ends remain. The first is a class of sets whose dimension can be computed exactly, which is what makes the theory testable. The second is the promise of Worked Example 7.2.3: that a Keakeya *maximal* bound, which is a family of single-scale statements of the kind Volume 6 produces, upgrades to a bound on Hausdorff dimension.

7.8.1 Iterated function systems

Definition 7.8.1. A *similarity* of ratio $r > 0$ is a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $|f(x) - f(y)| = r|x - y|$. An *iterated function system* is a finite family $f_1, \dots, f_m$ of similarities of ratios $r_1, \dots, r_m \in (0, 1)$; by Worked Example 7.3.2 it has a unique nonempty compact *attractor* E with $E = \bigcup_i f_i(E)$.

Definition 7.8.2. The system satisfies the *open set condition* if there is a nonempty bounded open V with

$$\bigcup_{i=1}^m f_i(V) \subseteq V \quad \text{and} \quad f_i(V) \cap f_j(V) = \emptyset \text{ for } i \neq j.$$

The *similarity dimension* is the unique $s \geq 0$ with $\sum_{i=1}^m r_i^s = 1$.

Theorem 7.8.3 (Hutchinson). *If an iterated function system of similarities satisfies the open set condition with attractor E and similarity dimension s , then*

$$\dim_H E = \dim_M E = s, \quad 0 < \mathcal{H}^s(E) < \infty.$$

Proof. Write $\mathcal{I}_k = \{1, \dots, m\}^k$ and, for $\mathbf{i} = (i_1, \dots, i_k)$, put $f_{\mathbf{i}} = f_{i_1} \circ \dots \circ f_{i_k}$ and $r_{\mathbf{i}} = r_{i_1} \cdots r_{i_k}$. Iterating $E = \bigcup_i f_i(E)$ gives $E = \bigcup_{\mathbf{i} \in \mathcal{I}_k} f_{\mathbf{i}}(E)$ for every k .

Step 1. *Upper bound.*

Since $f_{\mathbf{i}}$ is a similarity of ratio $r_{\mathbf{i}}$, $\text{diam } f_{\mathbf{i}}(E) = r_{\mathbf{i}} \text{diam } E$. So

$$\begin{aligned} \sum_{\mathbf{i} \in \mathcal{I}_k} (\text{diam } f_{\mathbf{i}}(E))^s &= (\text{diam } E)^s \sum_{\mathbf{i} \in \mathcal{I}_k} r_{\mathbf{i}}^s \\ &= (\text{diam } E)^s \left(\sum_i r_i^s \right)^k \\ &= (\text{diam } E)^s, \end{aligned}$$

using the defining property of s and that the sum over $\mathcal{I}_k$ factorises. Since $\max_{\mathbf{i}} r_{\mathbf{i}} \leq (\max_i r_i)^k \rightarrow 0$, these covers have diameters tending to 0, so $\mathcal{H}^s(E) \leq (\text{diam } E)^s < \infty$ and $\dim_{\text{H}} E \leq s$. The same covers, refined so that all $r_{\mathbf{i}}$ are comparable, give $\overline{\dim}_{\text{M}} E \leq s$ by Drill 8.2.

Step 2. *A natural measure.*

Define μ as the unique Borel probability measure with $\mu = \sum_i r_i^s \mu \circ f_i^{-1}$; it exists by the same contraction argument as Worked Example 7.3.2, applied on the complete metric space of probability measures with the Wasserstein metric, or equivalently as the distribution of the random point $\lim_k f_{i_1} \circ \cdots \circ f_{i_k}(x_0)$ with the i_j chosen independently with probabilities r_i^s . Iterating,

$$\mu(f_{\mathbf{i}}(E)) = r_{\mathbf{i}}^s \quad \text{for } \mathbf{i} \in \mathcal{I}_k. \quad (7.8.1)$$

Step 3. *The open set condition controls overlaps.*

Let V be as in the definition; we may assume $\overline{V} \supseteq E$, replacing V by a large ball if necessary and noting the conditions persist. The sets $f_{\mathbf{i}}(V)$, $\mathbf{i} \in \mathcal{I}_k$, are pairwise disjoint, open, and of diameter $r_{\mathbf{i}} \text{diam } V$.

Fix x and $0 < \rho < \text{diam } V$. Let $\mathcal{J}$ be the set of $\mathbf{i}$ — of varying lengths — that are minimal subject to $r_{\mathbf{i}} \leq \rho / \text{diam } V$; these index a disjoint family whose union covers E , and $r_{\mathbf{i}} \geq r_{\min} \rho / \text{diam } V$ where $r_{\min} = \min_i r_i$.

The sets $f_{\mathbf{i}}(V)$ for $\mathbf{i} \in \mathcal{J}$ are disjoint, open, of diameter $\approx \rho$, and each contains a ball of radius $c\rho$ where c depends on V and $r_{\min}$ — because V contains a ball and similarities map balls to balls. A volume count therefore bounds the number of $\mathbf{i} \in \mathcal{J}$ with $f_{\mathbf{i}}(V) \cap B(x, \rho) \neq \emptyset$ by a constant $N = N(n, V, r_{\min})$.

Step 4. *The Frostman bound, and the lower bound.*

By Step 3 and (7.8.1),

$$\mu(B(x, \rho)) \leq \sum_{\substack{\mathbf{i} \in \mathcal{J} \\ f_{\mathbf{i}}(V) \cap B(x, \rho) \neq \emptyset}} \mu(f_{\mathbf{i}}(E)) = \sum r_{\mathbf{i}}^s \leq N \left(\frac{\rho}{\text{diam } V} \right)^s \lesssim \rho^s.$$

By Theorem 7.1.6, $\mathcal{H}^s(E) \geq \mu(E)/C > 0$ and $\dim_{\text{H}} E \geq s$.

Step 5. Collect.

Steps 1 and 4 give $\dim_{\mathbb{H}} E = s$ and $0 < \mathcal{H}^s(E) < \infty$. Since $\dim_{\mathbb{H}} E \leq \underline{\dim}_{\mathbb{M}} E \leq \overline{\dim}_{\mathbb{M}} E \leq s$ by Theorem 7.2.4 and Step 1, all three coincide. $\square$

WORKED EXAMPLE 7.8.1— TIER A The classical examples

Cantor set. $f_1(x) = x/3$, $f_2(x) = x/3 + 2/3$ on $\mathbb{R}$, with $V = (0, 1)$. Then $2 \cdot 3^{-s} = 1$ gives $s = \log 2 / \log 3$, recovering Worked Example 7.1.1 — and now with $0 < \mathcal{H}^s(C) < \infty$, which the covering argument alone did not give.

Sierpiński gasket. Three maps of ratio $\frac{1}{2}$ fixing the vertices of a triangle, with V its interior. Then $3 \cdot 2^{-s} = 1$, so $s = \log 3 / \log 2 \approx 1.585$.

von Koch curve. Four maps of ratio $\frac{1}{3}$; $4 \cdot 3^{-s} = 1$ gives $s = \log 4 / \log 3 \approx 1.262$. The curve has infinite length ($\mathcal{H}^1 = \infty$) and zero area, which the dimension records precisely.

A non-example. Take $f_1(x) = x/2$, $f_2(x) = x/2 + 1/4$ on $\mathbb{R}$. The similarity dimension solves $2 \cdot 2^{-s} = 1$, giving $s = 1$; but the attractor is $[0, 1/2]$, which indeed has dimension 1. Here the open set condition fails — $f_1((0, 1/2))$ and $f_2((0, 1/2))$ overlap — and the formula happens to remain correct. Drill 8.3 gives a system where it fails.

7.8.2 From maximal estimates to Hausdorff dimension

Definition 7.8.4. For $0 < \delta < 1$ and $s > 0$, say that the *Keakeya maximal bound* $K(s)$ holds if for every family $\mathbb{T}$ of δ -tubes with δ -separated directions and every $\varepsilon > 0$,

$$\left\| \sum_{T \in \mathbb{T}} \mathbf{1}_T \right\|_{L^p(\mathbb{R}^n)} \lesssim_{n, \varepsilon} \delta^{-\varepsilon}, \quad p = \frac{s}{s-1},$$

uniformly in δ .

For $s = n$ this is the Keakeya maximal conjecture in the form of Worked Example 6.6.2.

Theorem 7.8.5. *Suppose $K(s)$ holds for some $1 < s \leq n$. Then every Besicovitch set $K \subseteq \mathbb{R}^n$ has $\dim_{\mathbb{H}} K \geq s$.*

Idea of the proof. Suppose not, so $\mathcal{H}^{s'}(K) = 0$ for some $s' < s$ and there is a cover of K by balls with $\sum r_i^{s'}$ arbitrarily small. The balls have many different radii; group them by dyadic size. For each direction, the unit segment of K in that direction is covered, so a definite proportion of its length is covered by balls of *some* single dyadic size — by pigeonholing over

the $\approx \log(1/\delta)$ available sizes. Apply the single-scale hypothesis at that size.

The loss from pigeonholing is a factor $\log(1/\delta)$, and the hypothesis carries $\delta^{-\varepsilon}$ for every ε ; since $\delta^{-\varepsilon}$ beats any power of a logarithm, the argument closes. This is the mechanism promised in Worked Example 7.2.3(c).

Proof. Let $s' < s$ and suppose $\mathcal{H}^{s'}(K) = 0$; we derive a contradiction, which gives $\dim_{\mathbb{H}} K \geq s'$ for every $s' < s$ and hence the claim.

Step 1. *A cover, grouped by scale.*

Choose a cover $\{B_i = B(x_i, r_i)\}$ of K with $\sum_i r_i^{s'} < \eta$, where $\eta > 0$ will be chosen at the end; we may take all $r_i < \frac{1}{100}$. For $k \geq 1$ let $\mathcal{B}_k = \{i : 2^{-k-1} < r_i \leq 2^{-k}\}$ and $N_k = \#\mathcal{B}_k$. Then

$$\sum_{k \geq 1} N_k 2^{-(k+1)s'} \leq \sum_i r_i^{s'} < \eta. \quad (7.8.2)$$

Step 2. *Pigeonhole over scales for each direction.*

Let $\omega \in \mathbb{S}^{n-1}$ and $\ell_\omega \subseteq K$ a unit segment in direction ω . The balls cover ℓ_ω , so $\sum_k \mathcal{H}^1(\ell_\omega \cap \bigcup_{i \in \mathcal{B}_k} B_i) \geq 1$. Choose weights $c_k = c/k^2$ with $\sum_k c_k = 1$; then there is $k = k(\omega)$ with

$$\mathcal{H}^1\left(\ell_\omega \cap \bigcup_{i \in \mathcal{B}_k} B_i\right) \geq \frac{c}{k^2}. \quad (7.8.3)$$

Partition $\mathbb{S}^{n-1}$ into the sets $\Omega_k = \{\omega : k(\omega) = k\}$; some Ω_k has $\sigma(\Omega_k) \geq c_k \sigma(\mathbb{S}^{n-1})$, again by weighting. Fix that k and write $\delta = 2^{-k}$.

Step 3. *Build a tube family.*

Choose a maximal δ -separated set of directions in Ω_k ; since $\sigma(\Omega_k) \gtrsim k^{-2}$, there are $M \gtrsim k^{-2} \delta^{-(n-1)}$ of them. For each such ω , let T_ω be the δ -tube around ℓ_ω . By (7.8.3), the balls of $\mathcal{B}_k$ — each of radius $\approx \delta$ — cover a proportion $\gtrsim k^{-2}$ of ℓ_ω , hence

$$m\left(T_\omega \cap \bigcup_{i \in \mathcal{B}_k} 2B_i\right) \gtrsim \frac{1}{k^2} m(T_\omega) \approx \frac{\delta^{n-1}}{k^2}.$$

Step 4. *Apply the hypothesis.*

Let $U = \bigcup_{i \in \mathcal{B}_k} 2B_i$, so $m(U) \leq N_k C_n \delta^n$. With $F = \sum_\omega \mathbf{1}_{T_\omega}$ over the M

chosen directions, Step 3 gives

$$\begin{aligned}
 \int_U F &= \sum_{\omega} m(T_{\omega} \cap U) \\
 &\gtrsim M \frac{\delta^{n-1}}{k^2} \\
 &\gtrsim \frac{\delta^{-(n-1)}}{k^2} \cdot \frac{\delta^{n-1}}{k^2} \\
 &= \frac{1}{k^4}.
 \end{aligned}$$

On the other hand, by Hölder with exponents $p = \frac{s}{s-1}$ and $p' = s$, and the hypothesis $K(s)$,

$$\int_U F \leq m(U)^{1/s} \|F\|_{L^p} \lesssim_{\varepsilon} (N_k \delta^n)^{1/s} \delta^{-\varepsilon}.$$

Step 5. *Compare, and use the budget.*

Combining,

$$\begin{aligned}
 \frac{1}{k^4} &\lesssim_{\varepsilon} (N_k \delta^n)^{1/s} \delta^{-\varepsilon} \\
 &\quad , \quad \text{so} \quad N_k \gtrsim_{\varepsilon} \frac{\delta^{-n} \delta^{s\varepsilon}}{k^{4s}}.
 \end{aligned}$$

Insert this into (7.8.2), using $\delta = 2^{-k}$ and keeping only the k -th term:

$$\eta > N_k 2^{-(k+1)s'} \gtrsim_{\varepsilon} \frac{2^{kn} 2^{-ks\varepsilon}}{k^{4s}} 2^{-(k+1)s'} \gtrsim_{\varepsilon} \frac{2^{k(n-s'-s\varepsilon)}}{k^{4s}}.$$

Choose ε with $s\varepsilon < n - s'$, which is possible since $s' < s \leq n$ gives $n - s' > 0$. Then the exponent $n - s' - s\varepsilon$ is positive, so the right-hand side tends to ∞ with k ; and $k \rightarrow \infty$ as $\eta \rightarrow 0$, since a cover with $\sum r_i^{s'} < \eta$ forces all r_i small. Taking η small enough gives a contradiction. $\square$

Where we are. The upgrade is complete: a family of single-scale estimates, each with an arbitrarily small power loss, yields a Hausdorff dimension bound. Combined with Theorem 6.6.2, the chain now runs

$$\begin{aligned}
 \text{restriction conjecture} &\implies \text{Kakeya maximal conjecture} \\
 &\implies \dim_{\mathbb{H}} K = n \text{ for every Besicovitch set,}
 \end{aligned}$$

the first implication from Volume 6 and the second from Theorem 7.8.5 with $s = n$.

Two features of the proof deserve emphasis. The $\delta^{-\varepsilon}$ in the hypothesis is essential: Step 5 needs $s\varepsilon < n - s'$, and with a fixed power loss δ^{-c}

one obtains only $\dim_{\mathbb{H}} K \geq n - sc$. And the two pigeonholings — over scales, and over directions — each cost a factor k^{-2} , that is a power of $\log(1/\delta)$, which the $\delta^{-\varepsilon}$ absorbs. This is exactly the bookkeeping of Worked Example 6.5.3.

WORKED EXAMPLE 7.8.2— TIER B **Besicovitch sets are not self-similar, and what replaces the formula**

Theorem 7.8.3 computes dimensions exactly. It is worth being clear why it says nothing about Keakeya.

No Besicovitch set is the attractor of a similarity IFS with the open set condition. If it were, Theorem 7.8.3 would give $0 < \mathcal{H}^s(K) < \infty$ with $s = \dim_{\mathbb{H}} K$. But a Besicovitch set can have $m(K) = 0$ while $\dim_{\mathbb{H}} K = n$ is conjectured, and by Theorem 7.1.8 that forces $\mathcal{H}^n(K) = 0$ — incompatible with $\mathcal{H}^s(K) > 0$ at $s = n$. So either the conjecture fails or such a set is not self-similar, and the conjecture is believed.

More directly: self-similar sets are homogeneous, Besicovitch sets are not. The attractor looks the same at every point and scale, up to the finitely many maps. A Besicovitch set is built by translating triangles at each stage of a Perron construction, and the translations differ from stage to stage; the set has no exact scaling symmetry.

What replaces the dimension formula. Nothing exact. Every bound on $\dim_{\mathbb{H}} K$ in this series is an inequality, obtained by a counting argument at a single scale and upgraded by Theorem 7.8.5. The absence of a formula is not a gap in the theory — it is the reason the problem is open.

A useful contrast. Over a finite field the analogous problem *does* have an exact answer: Dvir’s theorem, in Volume 10, shows a Keakeya set in $\mathbb{F}_q^n$ has $\gtrsim q^n$ points, with no loss at all. The polynomial method supplies there what self-similarity supplies for fractals: exact algebraic structure. Volume 10, Chapter 8 explains why the transfer to $\mathbb{R}^n$ fails.

WORKED EXAMPLE 7.8.3— TIER C: RESEARCH TASTE **What Volume 7 has and has not delivered**

(a) *The language.* Hausdorff dimension is now defined, its elementary theory established, and the Keakeya conjecture stated in its proper form (Theorem 7.1.11). The comparison with Minkowski dimension (Theorem 7.2.4) quantifies exactly what Volume 6’s single-scale arguments were missing.

(b) *The tools.* Frostman’s lemma (Theorem 7.5.1) converts a dimension hypothesis into a measure; energy (Theorem 7.6.2) converts that measure into an integral; Theorem 7.6.5 converts the integral into a statement about

$\hat{\mu}$. These three steps are used in every dimension estimate of Volume 8.

(c) *The upgrade.* Theorem 7.8.5 closes the gap Volume 6 left, so that all subsequent work may be done at a single scale.

(d) *What is deliberately absent.* The Besicovitch covering theorem (Theorem 7.4.2); the measure-zero Besicovitch set, deferred to Volume 8 where Marstrand’s theorem makes Kahane’s construction available (Chapter 7.3); Salem set constructions (Chapter 7.7); and Drury’s Fourier proof of the $(n+1)/2$ bound, which Volume 1 already established by the bush (Theorem 7.7.5).

(e) *What this volume does not do.* It proves no new bound on the dimension of a Besicovitch set. Everything here is machinery, and the machinery is two-point — as Worked Example 7.7.3 made precise. The improvements begin in Volume 9, and every one of them is a statement about configurations of three or more tubes.

WORKED EXAMPLE 7.8.4— TIER B **Reading a dimension bound in the literature**

Papers on Kakeya state bounds in several inequivalent ways, and the differences matter. This volume now supplies the dictionary; here it is, applied to the four formulations one meets.

(a) “*Every Besicovitch set in $\mathbb{R}^n$ has $\dim_{\mathrm{H}} \geq s$.*” The Hausdorff statement, the strongest. By Theorem 7.2.4 it implies the Minkowski version.

(b) “*Every Besicovitch set has $\dim_{\mathrm{M}} \geq s$.*” Weaker. Historically these came first, often by several years, because a single-scale argument suffices — compare Theorem 6.6.3.

(c) “*The Kakeya maximal operator satisfies $\|\mathcal{K}_{\delta}f\|_{L^q(\mathbb{S}^{n-1})} \lesssim_{\varepsilon} \delta^{-\varepsilon} \|f\|_{L^p}$.*” A single-scale family. By Theorem 7.8.5 it implies (a) with the corresponding s , *provided* the loss is $\delta^{-\varepsilon}$ for every ε . With a fixed loss δ^{-c} it implies only a weaker s , by Drill 8.4 — a distinction routinely elided in surveys.

(d) “ *$m(K_{\delta}) \gtrsim \delta^{n-s}$ for every Besicovitch set.*” Equivalent to (b), by Theorem 7.2.3.

Worked instance: the bush bound. Volume 1, Theorem 1.7.15 proves the maximal-operator form (c) with $s = \frac{n+1}{2}$. Via Theorem 7.8.5 this gives (a): every Besicovitch set has $\dim_{\mathrm{H}} \geq \frac{n+1}{2}$. That is the same conclusion Drury obtained by Fourier analysis (Theorem 7.7.5), and the two routes are genuinely independent.

What to check when reading. Which dimension; whether the loss is $\delta^{-\varepsilon}$ or a fixed power; and whether the statement is for all Besicovitch sets or for a restricted class such as the sticky ones of Volume 14. Three questions, and

the answers change the strength of the result substantially.

Drills

Drill 7.8.1. Verify that the sum over $\mathcal{I}_k$ factorises as $(\sum_i r_i^s)^k$, as used in Step 1 of Theorem 7.8.3.

Drill 7.8.2. Show that refining the covers in Step 1 so that all r_i lie in $[\rho r_{\min}, \rho]$ gives $\overline{\dim_M} E \leq s$.

Drill 7.8.3. Exhibit an IFS of similarities for which the open set condition fails and $\dim_H E$ is strictly less than the similarity dimension.

Drill 7.8.4. Track the constants in Theorem 7.8.5 to show that $K(s)$ with a fixed loss δ^{-c} in place of $\delta^{-\varepsilon}$ gives only $\dim_H K \geq n - sc$.

Where we are. Volume 7 is machinery, and it is worth stating exactly what has been added to the series' capability.

Before this volume, a dimension statement could be made only about Minkowski dimension, and only at a single scale. After it: $\dim_H$ is defined and its elementary theory available (Chapter 7.1); the gap to $\dim_M$ is quantified (Theorem 7.2.4); a dimension hypothesis can be converted into a measure (Theorem 7.5.1), the measure into an integral (Theorem 7.6.2), and the integral into a Fourier statement (Theorem 7.6.5); and a family of single-scale bounds upgrades to a Hausdorff bound (Theorem 7.8.5).

That last theorem is what licenses the practice of the rest of the series. Volumes 9 through 14 prove estimates for δ -tube families at a fixed δ , with losses $\delta^{-\varepsilon}$, and never mention Hausdorff measure again. They are entitled to do so because of Theorem 7.8.5, and a reader who wants to know what a tube-counting estimate says about dimension should return here.

Notes

Theorem 7.8.3 is Hutchinson's (1981), building on Moran (1946). The open set condition is Moran's; that it can be replaced by the weaker *strong separation* or by conditions permitting some overlap is a substantial literature not needed here.

Theorem 7.8.5 is standard and appears in essentially this form in Bourgain's and Wolff's papers; the pigeonholing over scales is the step that makes the $\delta^{-\varepsilon}$ hypothesis indispensable, and is the reason all Keakea maximal statements in the literature carry it.

Chapter 7.A

Results Quoted from Volumes 1–6

Result	Source	Used in
Outer measure from a covering family	Theorem 1.2.8	Theorem 7.1.2
Metric outer measures	Theorem 1.2.13	Theorem 7.1.2
Hausdorff measure, constructed	Worked Example 1.2.4	Theorem 7.1.1
Continuity of measures	Theorem 1.2.6	Theorem 7.3.6
Outer regularity	Theorem 1.3.6	Theorem 7.1.8, Theorem 7.4.4
Carathéodory extension	Theorem 1.2.15	Worked Example 7.1.1
Egorov	Theorem 1.5.9	Theorem 7.6.2
Vitali covering theorem	Theorem 1.7.7	Theorem 7.4.3
Riesz representation	Theorem 1.8.15	Theorem 7.5.1
The Perron tree	Theorem 1.3.12	Chapter 7.3
The bush bound	Theorem 1.7.15	Theorem 7.7.5, Worked Example 7.7.2
$\mathcal{F}$ on $\mathcal{S}'$	Theorem 2.5.11	Theorem 7.6.5
Transform of $ x ^{-a}$	Worked Example 2.5.2	Theorem 7.6.5
The Riesz reflection formula	Theorem 5.6.4	Theorem 7.6.5
Asymptotics of $\widehat{d\sigma}$	Theorem 5.5.2	Worked Example 7.7.1, Worked Example 7.7.2
Takeya maximal, tube form	Worked Example 6.6.2	Theorem 7.8.5
Restriction implies Takeya	Theorem 6.6.2	Chapter 7.8 Checkpoint
Besicovitch sets, Minkowski bound	Theorem 6.6.3	Worked Example 7.2.3
Epsilon bookkeeping	Worked Example 6.5.3	Theorem 7.8.5

Debts

Volume 1 left two results to this volume. Neither is proved here, and the dispositions differ:

The Besicovitch covering theorem is removed from the series (Theorem 7.4.2, revision R6). Nothing depends on it; the 5r-lemma (Theorem 7.4.1) serves

every purpose here.

A *Besicovitch set of measure zero* is deferred to Volume 8, where Marstrand's projection theorem makes Kahane's construction available. The Pitfall in Chapter 7.3 explains why the compactness argument one expects to work does not.

This volume incurs one further omission of its own: Drury's Fourier-analytic proof of $\dim_{\mathrm{H}} K \geq \frac{n+1}{2}$ (Theorem 7.7.5, revision R7), the same bound having been proved in Volume 1 by the bush argument.

Chapter 7.B

Notation and Index of Results

Additions to the Volume 1–6 conventions

$\mathcal{H}^s, \mathcal{H}_\varepsilon^s, \mathcal{H}_\infty^s$	Hausdorff measure, its ε -approximation, and the content
$\dim_H, \dim_M, \dim_F$	Hausdorff, Minkowski and Fourier dimension
$N(E, \delta)$	covering number at scale δ
$d_H, \mathcal{K}(B)$	Hausdorff metric; compact subsets of B
$I_t(\mu)$	the t -energy of μ
$\text{spt } \mu$	the support of μ
$K(s)$	the Keakeya maximal bound at exponent s

Index of the main results

$\mathcal{H}^s$ is a metric outer measure	Theorem 7.1.2
The critical exponent	Theorem 7.1.3
Elementary properties of $\dim_{\mathcal{H}}$	Theorem 7.1.5
Mass distribution principle	Theorem 7.1.6
$\mathcal{H}^n \approx$ Lebesgue	Theorem 7.1.8
Box-counting definitions agree	Theorem 7.2.2
$\dim_{\mathcal{H}} \leq \underline{\dim}_{\mathcal{M}} \leq \overline{\dim}_{\mathcal{M}}$	Theorem 7.2.4
Completeness of the Hausdorff metric	Theorem 7.3.3
Blaschke selection	Theorem 7.3.4
Upper semicontinuity of measure	Theorem 7.3.6
The $5r$ -covering lemma	Theorem 7.4.1
Vitali for a Radon measure	Theorem 7.4.3
Density theorem for $\mathcal{H}^s$	Theorem 7.4.4
Frostman's lemma	Theorem 7.5.1
Energy characterises dimension	Theorem 7.6.2
The Fourier expression of energy	Theorem 7.6.5
Decay implies dimension	Theorem 7.6.6
$\dim_{\mathcal{F}} \leq \dim_{\mathcal{H}}$	Theorem 7.7.2
$\dim_{\mathcal{F}}$ of the Cantor set is 0	Theorem 7.7.4
Hutchinson's dimension formula	Theorem 7.8.3
Maximal bounds give Hausdorff bounds	Theorem 7.8.5

Plan revisions in this volume

Removed	Reason
Besicovitch covering theorem	the naive angle argument is false; the correct proof splits by radius ratio and is several pages. Nothing needs it. Theorem 7.4.2, revision R6
Drury's Fourier bound $\frac{n+1}{2}$	the same bound is proved in Volume 1 by the bush argument. Theorem 7.7.5, revision R7
Salem set constructions	need random measures; unused
Measure-zero Besicovitch set	<i>deferred</i> , not removed: Volume 8, via Marstrand and Kahane

Part 8

Volume 8

Projections, Distance Sets, and Besicovitch Sets

Preface to Volume 8

Dimension under maps

Volume 7 built the apparatus for measuring a set: Hausdorff dimension, Frostman measures, energy integrals, and the Fourier expression of energy. This volume uses it, on two questions that have driven the subject and that are the natural companions of the Kakeya problem.

The first is what happens to dimension under *projection*. If E lies in the plane and one shadows it onto a line, the shadow can only be smaller — but by how much, and for how many directions? Marstrand’s answer, that almost every shadow is as large as it can be, is the model for every “generic” statement in the subject.

The second is the *distance set* $\Delta(E) = \{|x - y| : x, y \in E\}$. Falconer asked how large E must be for $\Delta(E)$ to have positive measure, conjectured the threshold $\dim_{\text{H}} E > n/2$, and proved $\frac{n+1}{2}$. That proof is the centrepiece of this volume, and it is a direct application of two things already built: the decay of $\widehat{d\sigma}$ from Volume 5, and the Fourier expression of energy from Volume 7.

A note on the route to Falconer’s theorem

Readers who consult the literature will find two approaches. Falconer’s original argument is an L^∞ estimate: one shows the distance measure has a *bounded density*. Mattila’s later approach is an L^2 estimate, reducing the problem to spherical averages of $\widehat{\mu}$, and it is the framework in which every subsequent improvement has been made.

This volume proves Falconer’s theorem by the first route, in Chapter 8.5, and introduces the second in Chapter 8.7 as the tool for the improvements. The distinction is not cosmetic: *the Mattila integral does not converge at the exponent $\frac{n+1}{2}$* — the pointwise decay of $\widehat{d\sigma}$ alone never makes it converge, in any dimension — so a reader who tries to prove Falconer’s theorem by that route will fail, and should know in advance that the failure is structural rather

than a missing idea.

What is assumed

Volumes 1–7. Most heavily: Frostman’s lemma and the energy machinery of Volume 7, Chapters 5 and 6; the asymptotics of $\widehat{d\sigma}$ from Volume 5, Chapter 5; and the Fourier transform of measures from Volume 2.

Scope, fixed in advance

Following the practice of Volumes 4 through 7, the list of results was fixed before writing. Three items are handled by announcement rather than proof, each recorded in the manifest:

- the existence of a Besicovitch set of *measure zero*, deferred here from Volume 7. Chapter 8.3 explains why the natural construction needs the Besicovitch projection theorem, which is outside the scope of this series, and records the decision;
- the improvements on Falconer’s exponent — Wolff’s $\frac{4}{3}$ in the plane, Erdoğan’s $\frac{n}{2} + \frac{1}{3}$, and the current record $\frac{5}{4}$ of Guth, Iosevich, Ou and Wang. Chapter 8.7 states them, explains what each contributes, and proves the reduction they all rest on;
- distance sets over finite fields, which need a separate Fourier analysis on $\mathbb{F}_q^d$ and are not used later here.

How the two halves of the volume fit

The volume divides cleanly. Chapter 8.1 to Chapter 8.3 concern *projections*: what a shadow of a set can look like, how many directions can be exceptional, and what the answers say about product sets and Besicovitch sets. Chapter 8.4 to Chapter 8.8 concern *distances*.

The two halves share a method and differ in what they can achieve with it. Both proceed by producing a Frostman measure, converting a dimension hypothesis into an energy bound, and passing to the Fourier side. In the projection problem that suffices completely: Theorem 8.1.2 is sharp, and Theorem 8.2.1 bounds the exceptional set optimally in its own terms. In the distance problem it does not: Theorem 8.5.1 falls short of the conjectured exponent by exactly $\frac{1}{2}$, and closing the gap has required, over forty years, the import of successively stronger restriction estimates.

The reason for the difference is worth stating at the outset. A projection is a *linear* map, and the transversality that drives Theorem 8.1.3 is exact: a linear functional vanishes to first order and no better. A distance is a *quadratic* function, its level sets are spheres, and the relevant transversality is the curvature of those spheres — which enters through $\widehat{d\sigma}$ and is quantitatively weaker than one would like. So the projection theory is complete and the distance theory is not, and the discrepancy traces back to the difference between a hyperplane and a sphere.

The plan

1	Projections	Marstrand's theorem, by Kaufman's argument
2	Exceptional sets	how many directions can be bad
3	Product sets and Besicovitch sets	and a scope decision
4	The distance measure	its Fourier representation
5	Falconer's theorem	the L^∞ route
6	Interiors	Mattila–Sjölin
7	The Mattila integral	where the improvements come from
8	The $n/2$ barrier	what would be needed

Chapter 8.1

Projections

8.1.1 The question

Definition 8.1.1. For $\theta \in [0, \pi)$ let $e_\theta = (\cos \theta, \sin \theta)$ and let $\pi_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the orthogonal projection $\pi_\theta(x) = x \cdot e_\theta$.

Since π_θ is Lipschitz with constant 1, Theorem 7.1.5(iii) gives

$$\dim_{\text{H}} \pi_\theta(E) \leq \min \{ \dim_{\text{H}} E, 1 \}$$

for every $E \subseteq \mathbb{R}^2$ and every θ , the second bound because the image lies in a line. The question is whether equality holds, and the answer is that it holds for almost every θ but not for all: the projection of a line segment onto the perpendicular direction is a point.

Theorem 8.1.2 (Marstrand). *Let $E \subseteq \mathbb{R}^2$ be compact.*

(i) *For Lebesgue-almost every $\theta \in [0, \pi)$,*

$$\dim_{\text{H}} \pi_\theta(E) = \min \{ \dim_{\text{H}} E, 1 \}.$$

(ii) *If $\dim_{\text{H}} E > 1$ then $m(\pi_\theta(E)) > 0$ for almost every θ .*

Idea of the proof. Both halves are one computation, performed twice with different exponents.

Take a measure μ on E of finite t -energy, supplied by Theorem 7.6.2. Its projection $\pi_{\theta\#}\mu$ is a measure on the line, and its energy is

$$I_t(\pi_{\theta\#}\mu) = \iint \frac{d\mu(x) d\mu(y)}{|(x - y) \cdot e_\theta|^t}.$$

Averaging over θ and exchanging the order of integration — which is

legitimate, everything being nonnegative — the inner integral becomes

$$\begin{aligned} \int_0^\pi \frac{d\theta}{|(x-y) \cdot e_\theta|^t} &= \frac{1}{|x-y|^t} \int_0^\pi \frac{d\theta}{|u \cdot e_\theta|^t} \\ &, \quad u = \frac{x-y}{|x-y|}, \end{aligned}$$

and the remaining integral is a *constant*, finite exactly when $t < 1$, because $u \cdot e_\theta$ vanishes to first order in θ . So the average of $I_t(\pi_{\theta\#}\mu)$ over θ is a constant times $I_t(\mu)$, and if the latter is finite then so is the former for almost every θ .

Part (ii) is the same with the Fourier expression instead of the energy: one computes $\int_0^\pi \int_{\mathbb{R}} |\widehat{\pi_{\theta\#}\mu}(r)|^2 dr d\theta$ and recognises it, after polar coordinates, as $I_1(\mu)$.

Lemma 8.1.3 (The angular integral). *For $0 < t < 1$ and any unit vector $u \in \mathbb{S}^1$,*

$$\begin{aligned} c_t &:= \int_0^\pi \frac{d\theta}{|u \cdot e_\theta|^t} \\ &= \int_0^\pi \frac{d\varphi}{|\cos \varphi|^t} < \infty, \end{aligned}$$

independent of u ; and $c_t = \infty$ for $t \geq 1$.

Proof. Write $u = e_\alpha$; then $u \cdot e_\theta = \cos(\theta - \alpha)$, and the substitution $\varphi = \theta - \alpha$ together with π -periodicity of $|\cos|$ gives the stated identity, independent of α and hence of u .

For convergence, $|\cos \varphi|$ vanishes only at $\varphi = \pi/2$, where $|\cos \varphi| = |\sin(\varphi - \pi/2)| \approx |\varphi - \pi/2|$. So near that point the integrand is $\approx |\varphi - \pi/2|^{-t}$, integrable exactly for $t < 1$. $\square$

Proof of Theorem 8.1.2. Step 1. Part (i), lower bound.

Let $t < \min\{\dim_{\mathbb{H}} E, 1\}$. By Theorem 7.6.2(ii) there is a nonzero finite μ on E with $I_t(\mu) < \infty$. For each θ , the projected measure $\pi_{\theta\#}\mu$ is a nonzero finite measure on $\pi_\theta(E)$, and by the change-of-variables formula for pushforwards,

$$I_t(\pi_{\theta\#}\mu) = \iint \frac{d\mu(x) d\mu(y)}{|(x-y) \cdot e_\theta|^t}.$$

Integrating in θ and applying Tonelli — the integrand is nonnegative and jointly measurable — and then Theorem 8.1.3,

$$\int_0^\pi I_t(\pi_{\theta\#}\mu) d\theta = \iint \left(\int_0^\pi \frac{d\theta}{|(x-y) \cdot e_\theta|^t} \right) d\mu d\mu = c_t I_t(\mu) < \infty. \quad (8.1.1)$$

A nonnegative function with finite integral is finite almost everywhere, so $I_t(\pi_{\theta\#}\mu) < \infty$ for a.e. θ , and Theorem 7.6.2(i) gives $\dim_{\mathbb{H}} \pi_{\theta}(E) \geq t$ for those θ .

Taking a sequence $t_k \uparrow \min\{\dim_{\mathbb{H}} E, 1\}$ and intersecting the countably many full-measure sets gives the lower bound for a.e. θ ; the upper bound was noted above.

Step 2. *Part (ii).*

Assume $\dim_{\mathbb{H}} E > 1$ and choose μ on E with $I_1(\mu) < \infty$, again by Theorem 7.6.2(ii). The projected measure has Fourier transform

$$\begin{aligned} \widehat{\pi_{\theta\#}\mu}(r) &= \int_{\mathbb{R}} e^{-2\pi i r v} d(\pi_{\theta\#}\mu)(v) \\ &= \int_{\mathbb{R}^2} e^{-2\pi i r x \cdot e_{\theta}} d\mu(x) \\ &= \widehat{\mu}(re_{\theta}). \end{aligned}$$

Therefore, by polar coordinates in $\mathbb{R}^2$ (Theorem 1.6.11), writing $\xi = re_{\theta}$ with $r \in \mathbb{R}$ and $\theta \in [0, \pi)$ so that $d\xi = |r| dr d\theta$,

$$\begin{aligned} \int_0^{\pi} \int_{\mathbb{R}} |\widehat{\pi_{\theta\#}\mu}(r)|^2 dr d\theta &= \int_0^{\pi} \int_{\mathbb{R}} |\widehat{\mu}(re_{\theta})|^2 dr d\theta \\ &= \int_{\mathbb{R}^2} \frac{|\widehat{\mu}(\xi)|^2}{|\xi|} d\xi \\ &\approx I_1(\mu), \end{aligned}$$

the last step by Theorem 7.6.5 with $n = 2$, $t = 1$, since $|\xi|^{t-n} = |\xi|^{-1}$.

So the left side is finite, and hence $\widehat{\pi_{\theta\#}\mu} \in L^2(\mathbb{R})$ for a.e. θ . By Plancherel (Theorem 2.4.8) the measure $\pi_{\theta\#}\mu$ is then absolutely continuous with an L^2 density, so it cannot be supported on a Lebesgue-null set. Since it is supported in $\pi_{\theta}(E)$, that set has positive measure. $\square$

Where we are. The two parts used the two forms of the energy from Volume 7: the integral form $I_t(\mu)$ for part (i), and the Fourier form $\int |\widehat{\mu}|^2 |\xi|^{t-n}$ for part (ii). Neither computation is more than an exchange of integrals; the content is entirely in Theorem 8.1.3 — that a linear functional vanishes to first order, so its reciprocal power is integrable exactly below exponent 1. That single fact is why the threshold in part (i) is 1 and not something else, and it is why the theorem has no analogue for projections onto lines in $\mathbb{R}^3$ with the same exponent: there the relevant vanishing is still of first order, but the sphere of directions is two-dimensional, and the bookkeeping changes. Drill 1.4 works out the general case.

WORKED EXAMPLE 8.1.1— TIER A **Why “almost every” cannot be improved to “every”**

Let E be a line segment in direction v . Then $\dim_{\mathbb{H}} E = 1$, and $\pi_{\theta}(E)$ is a segment of length $|v \cdot e_{\theta}|$ times the length of E — which is a single point when $e_{\theta} \perp v$. So $\dim_{\mathbb{H}} \pi_{\theta}(E) = 0$ for that one θ , and 1 for every other.

The exceptional set here is a single direction, of dimension 0. Chapter 8.2 shows this is typical: the exceptional set is always small, with a bound depending on the dimension one is asking for.

A richer example. Let $E = \ell_1 \cup \ell_2$ be two segments in directions $v_1 \neq v_2$. Now two directions are exceptional, and the exceptional set still has dimension 0. Taking E a countable union of segments in a dense set of directions makes the exceptional set countable and dense — still of dimension 0, but no longer closed. So the exceptional set need not be closed, and statements about it must be measure- or dimension-theoretic.

WORKED EXAMPLE 8.1.2— TIER B **Running the argument for a concrete set**

Take $E = C \times C$ where C is the middle-thirds Cantor set. By Theorem 7.6.4 and Theorem 7.8.3,

$$\dim_{\mathbb{H}}(C \times C) = 2 \frac{\log 2}{\log 3} \approx 1.2619 > 1.$$

So Theorem 8.1.2(ii) applies: for almost every θ , the projection $\pi_{\theta}(C \times C)$ has positive Lebesgue measure.

What the measure μ is. Take $\mu = \mu_C \times \mu_C$ with μ_C the natural Cantor measure of Worked Example 7.1.1. By Theorem 7.6.3(ii), $I_t(\mu) < \infty$ for every $t < 2 \log 2 / \log 3$, in particular for $t = 1$.

What the theorem does not say. It gives no information about any *particular* θ . And indeed the exceptional set here is genuinely interesting: for θ with $\tan \theta$ rational of small height the projection can be much smaller than expected, and the set of θ for which $\pi_{\theta}(C \times C)$ has measure zero is uncountable. Determining it exactly is open.

A caution about products. $\pi_{\theta}(C \times C)$ is, up to scaling, the sumset $\cos \theta \cdot C + \sin \theta \cdot C$. So Marstrand’s theorem is a statement about sums of Cantor sets, and the arithmetic of such sumsets is the subject of a large literature — the same arithmetic that made $\dim_{\mathbb{F}} C = 0$ in Theorem 7.7.4. Projections and sumsets are the same question in different notation, and it is worth carrying both readings.

WORKED EXAMPLE 8.1.3— TIER C: RESEARCH TASTE **What Marstrand's theorem is a model of**

The theorem is the prototype for a family of statements that recur throughout the rest of this series, and the shape is worth isolating.

(a) *The shape.* One has a family of maps $\{T_\theta\}$ indexed by a parameter, and wants a lower bound on the size of $T_\theta(E)$. Proving it for a fixed θ is impossible — there are always bad parameters. Instead one averages a quantity over θ , exchanges the order of integration, and finds the inner integral is a finite constant. Then the quantity is finite for almost every θ .

(b) *Where the finiteness comes from.* Always from a transversality statement: the map $\theta \mapsto T_\theta(x) - T_\theta(y)$ vanishes to a controlled order. Here it is Theorem 8.1.3; in Volume 12 it is the transversality hypothesis of the multilinear restriction theorem; in Volume 14 it is the non-degeneracy in the Furstenberg set problem.

(c) *What the method never gives.* A statement for every θ , and a statement for a specified θ . Both are usually much harder. Kakeya is the extreme case: a Besicovitch set is exactly a configuration where the bad directions have been arranged to be *all* of them, and no averaging argument can see it.

(d) *The connection to Kakeya, made precise.* Chapter 8.3 explains that a Besicovitch set of measure zero would follow from knowing that $m(\pi_\theta(C \times C)) = 0$ for a.e. θ for a suitable Cantor set — a statement in the *opposite* direction to Theorem 8.1.2(ii), requiring $\dim_{\mathrm{H}}(C \times C) \leq 1$ with the measure still vanishing. That is the Besicovitch projection theorem, and it is genuinely harder than Marstrand.

WORKED EXAMPLE 8.1.4— TIER B **Computing a projection dimension from scratch**

Theorem 8.1.2 is an existence statement about almost every θ . To see the machinery working, here is the whole computation carried out for one set, with every constant tracked.

The set. Let $C_\lambda \subseteq [0, 1]$ be the Cantor set of ratio $\lambda < \frac{1}{2}$ from Worked Example 7.1.2, so $d := \dim_{\mathrm{H}} C_\lambda = \log 2 / \log(1/\lambda)$, and put $E = C_\lambda \times C_\lambda$ in the plane, so $\dim_{\mathrm{H}} E = 2d$ by Theorem 7.6.4 and Theorem 7.8.3.

Step 1: the measure. Let μ_λ be the natural measure on C_λ , so $\mu_\lambda(I) \leq c(\operatorname{diam} I)^d$ as in Worked Example 7.1.1, and set $\mu = \mu_\lambda \times \mu_\lambda$. By Theorem 7.6.3(ii), $I_t(\mu) < \infty$ for every $t < 2d$.

Step 2: the average. By (8.1.1), for $t < \min\{2d, 1\}$,

$$\begin{aligned} \int_0^\pi I_t(\pi_{\theta\#}\mu) \, d\theta \\ &= c_t I_t(\mu) \\ , \quad c_t &= \int_0^\pi |\cos \varphi|^{-t} \, d\varphi. \end{aligned}$$

Both factors are finite, so the left side is finite.

Step 3: the conclusion. $I_t(\pi_{\theta\#}\mu) < \infty$ for a.e. θ , hence $\dim_{\mathbb{H}} \pi_{\theta}(E) \geq t$ for a.e. θ ; letting $t \uparrow \min\{2d, 1\}$ through a sequence,

$$\dim_{\mathbb{H}} \pi_{\theta}(C_{\lambda} \times C_{\lambda}) = \min\{2d, 1\} \quad \text{for a.e. } \theta.$$

Reading the answer. If $\lambda < \frac{1}{4}$ then $d < \frac{1}{2}$, so $2d < 1$ and almost every projection has dimension exactly $2d$ — the projection loses nothing. If $\lambda > \frac{1}{4}$ then $2d > 1$ and almost every projection has dimension 1; by Theorem 8.1.2(ii) it has positive measure as well. The transition is at $\lambda = \frac{1}{4}$, where C_{λ} has dimension exactly $\frac{1}{2}$ and E has dimension exactly 1 — and there the theorem says nothing about measure, which is precisely the borderline of Worked Example 8.3.1.

What to practise. The three steps are always the same: produce a measure of finite t -energy for every t below the dimension; average the projected energy over the parameter; read off “finite for almost every parameter”. Only Step 2 involves any thought, and only through the angular integral. Drill 1.4 asks for the same computation in $\mathbb{R}^n$.

Proposition 8.1.4 (Projections in $\mathbb{R}^n$). *Let $E \subseteq \mathbb{R}^n$ be compact, $1 \leq m < n$, and let π_V denote orthogonal projection onto an m -plane V , with $\gamma_{n,m}$ the natural rotation-invariant probability measure on the Grassmannian $G(n, m)$. Then for $\gamma_{n,m}$ -almost every V ,*

$$\dim_{\mathbb{H}} \pi_V(E) = \min\{\dim_{\mathbb{H}} E, m\}.$$

Proof. The upper bound is Theorem 7.1.5(iii), π_V being 1-Lipschitz, and $\dim_{\mathbb{H}} \pi_V(E) \leq m$ since the image lies in an m -plane.

For the lower bound, let $t < \min\{\dim_{\mathbb{H}} E, m\}$ and take μ on E with

$I_t(\mu) < \infty$ (Theorem 7.6.2). Exactly as in (8.1.1),

$$\begin{aligned} \int_{G(n,m)} I_t(\pi_V \# \mu) \, d\gamma_{n,m}(V) \\ = \iint \left(\int_{G(n,m)} \frac{d\gamma_{n,m}(V)}{|\pi_V(x-y)|^t} \right) \, d\mu \, d\mu, \end{aligned}$$

and the inner integral equals $|x-y|^{-t} c_{n,m,t}$ with

$$c_{n,m,t} = \int_{G(n,m)} |\pi_V u|^{-t} \, d\gamma_{n,m}(V), \quad u \in \mathbb{S}^{n-1} \text{ arbitrary,}$$

independent of u by rotation invariance of $\gamma_{n,m}$.

It remains to see $c_{n,m,t} < \infty$ for $t < m$. The quantity $|\pi_V u|$ is small exactly when u is nearly orthogonal to V , and

$$\gamma_{n,m}\{V : |\pi_V u| < \lambda\} \lesssim_{n,m} \lambda^m$$

— the set of m -planes nearly orthogonal to a fixed direction has measure λ^m , since m independent conditions each of probability λ must hold. By the layer-cake formula (Worked Example 1.5.1),

$$\begin{aligned} c_{n,m,t} &= \int_0^\infty \gamma_{n,m}\{|\pi_V u|^{-t} > \lambda\} \, d\lambda \\ &\lesssim 1 + \int_1^\infty \lambda^{-m/t} \, d\lambda < \infty \end{aligned}$$

precisely when $m/t > 1$. The argument then concludes as in Theorem 8.1.2, letting $t \uparrow \min\{\dim_H E, m\}$. $\square$

Where we are. The only thing that changed from the planar case is the exponent in the sublevel estimate: $\gamma_{n,m}\{|\pi_V u| < \lambda\} \lesssim \lambda^m$ rather than λ^1 . That is why the threshold is m rather than 1, and it is the general principle of Worked Example 8.2.4: *the exponent in the conclusion is the exponent in the sublevel estimate*.

The same substitution runs through Theorem 8.2.1: the exceptional set of m -planes for dimension t has $\gamma_{n,m}$ -dimension at most t , by replacing Theorem 8.2.2 with its Grassmannian analogue. Drill 1.4 asks for the details.

Figure 8.1.1 (to be drawn). *Projections of a set onto lines*

Content required: A compact set E in the plane (say a Cantor-like dust) with its shadows $\pi_\theta(E)$ drawn on three lines: a generic θ , where the shadow is as large as the dimension permits; and an exceptional θ , where it collapses. Mark the angle and indicate that the exceptional directions form a Lebesgue-null set, of dimension bounded by Theorem 8.2.1.

Drills

Drill 8.1.1. Verify that c_t in Theorem 8.1.3 satisfies $c_t \rightarrow \infty$ as $t \uparrow 1$, and compute its rate.

Drill 8.1.2. Show that Theorem 8.1.2(i) fails if “almost every” is replaced by “every”, using Worked Example 8.1.1, and that the exceptional set can be dense.

Drill 8.1.3. Check the polar-coordinate computation in Step 2, being careful that r ranges over all of $\mathbb{R}$ while θ ranges over $[0, \pi)$.

Drill 8.1.4. State and prove the analogue of Theorem 8.1.2(i) for projections $\mathbb{R}^n \rightarrow \mathbb{R}^m$, $m < n$, onto m -planes, identifying the exponent at which the angular integral diverges.

Where we are. Before leaving the chapter, note what Theorem 8.1.2 does *not* require: no regularity of E , no structure beyond compactness, and no relation between E and the family of projections. That generality is what makes the theorem a model, and it is also what limits it — a statement true for every compact set of a given dimension cannot distinguish between sets, and the exceptional set is where all the individuality lives.

Chapter 8.2 therefore does the more informative work, and the modern literature on projections is almost entirely about exceptional sets rather than about the almost-everywhere statement.

Notes

Marstrand proved the theorem in 1954 by a geometric argument. The proof given here is Kaufman’s (1968), and its brevity is the reason the potential-theoretic formulation of dimension became standard: the entire content is Theorem 8.1.3.

Mattila’s book is the reference for this chapter and the next. The problem of determining the exceptional set exactly for a specific E , such as $C \times C$, is open and is the subject of the projection theorems mentioned in Volume 14.

Chapter 8.2

Exceptional Sets

Theorem 8.1.2 says the set of bad directions is Lebesgue-null. Worked Example 8.1.1 showed it can be nonempty, and even dense. This chapter bounds its *dimension*, by running Kaufman’s computation against a Frostman measure on the exceptional set rather than against Lebesgue measure — a move worth learning, since it is the standard way to upgrade an “almost every” statement.

Theorem 8.2.1 (Kaufman). *Let $E \subseteq \mathbb{R}^2$ be compact and $0 < t \leq 1$. Then*

$$\dim_{\mathrm{H}} \left\{ \theta \in [0, \pi) : \dim_{\mathrm{H}} \pi_{\theta}(E) < t \right\} \leq t.$$

Idea of the proof. Suppose the exceptional set F_t had dimension exceeding s for some $s > t$. Frostman then supplies a measure ν on F_t with $\nu(B(\theta, r)) \leq r^s$. Run the computation of Theorem 8.1.2 with $d\theta$ replaced by $d\nu(\theta)$. The only step that changes is the angular integral: instead of $\int_0^\pi |u \cdot e_\theta|^{-t} d\theta$, one must bound $\int |u \cdot e_\theta|^{-t} d\nu(\theta)$. The set where $|u \cdot e_\theta|$ is small is an interval of comparable length around the direction perpendicular to u , and ν gives it mass at most r^s rather than r . So the integral converges when $t < s$ — one power better than Lebesgue allows.

That gives $I_t(\pi_{\theta\#}\mu) < \infty$ for ν -almost every θ , so $\dim_{\mathrm{H}} \pi_{\theta}(E) \geq t$ for ν -almost every θ ; but every $\theta \in F_t$ violates that, and $\nu(F_t) > 0$. Contradiction.

Lemma 8.2.2 (The weighted angular integral). *Let ν be a finite Borel measure on $[0, \pi)$ with $\nu(I) \leq |I|^s$ for every interval I , where $0 < s \leq 1$. Then for $0 < t < s$ and every unit vector u ,*

$$\int_0^\pi \frac{d\nu(\theta)}{|u \cdot e_\theta|^t} \leq C_{s,t} < \infty,$$

with $C_{s,t}$ independent of u .

Proof. Step 1. Locate where the integrand is large.

Write $u = e_\alpha$, so that $u \cdot e_\theta = \cos(\theta - \alpha)$. The integrand $|u \cdot e_\theta|^{-t}$ is large exactly where $\cos(\theta - \alpha)$ is small, that is near $\theta = \alpha \pm \frac{\pi}{2}$ — the directions perpendicular to u . Quantitatively, $|\cos \varphi| < \lambda$ forces φ to lie within $\arcsin \lambda$ of $\pm \frac{\pi}{2}$, and $\arcsin \lambda \leq \frac{\pi}{2} \lambda$ for $\lambda \in [0, 1]$. So

$$\{\theta \in [0, \pi) : |u \cdot e_\theta| < \lambda\}$$

is contained in an interval of length at most $\pi\lambda$, modulo π .

Step 2. *Apply the Frostman hypothesis.*

By assumption $\nu(I) \leq |I|^s$ for intervals, so

$$\nu\{\theta : |u \cdot e_\theta| < \lambda\} \leq (\pi\lambda)^s. \quad (8.2.1)$$

This is the sublevel set estimate discussed in Worked Example 8.2.4, and it is the only place the hypothesis on ν is used. Note the gain over Lebesgue measure: there the same set has measure $\approx \lambda$, and here it is λ^s with s possibly much smaller — which is what will allow a larger exponent t .

Step 3. *Integrate by layers.*

Put $f(\theta) = |u \cdot e_\theta|^{-t}$. By the layer-cake formula (Worked Example 1.5.1),

$$\begin{aligned} \int f \, d\nu &= \int_0^\infty \nu\{f > \lambda\} \, d\lambda \\ &= \int_0^1 \nu\{f > \lambda\} \, d\lambda + \int_1^\infty \nu\{f > \lambda\} \, d\lambda. \end{aligned}$$

The first integral is at most $\nu([0, \pi)) \leq \pi^s$. For the second, note $f(\theta) > \lambda$ is equivalent to $|u \cdot e_\theta| < \lambda^{-1/t}$, so by (8.2.1),

$$\int_1^\infty \nu\{f > \lambda\} \, d\lambda \leq \pi^s \int_1^\infty \lambda^{-s/t} \, d\lambda.$$

Step 4. *Check convergence.*

The last integral converges if and only if $s/t > 1$, that is $t < s$, with value $\frac{t}{s-t}$. Hence

$$\int f \, d\nu \leq \pi^s \left(1 + \frac{t}{s-t}\right) = \pi^s \frac{s}{s-t} =: C_{s,t},$$

which is finite, independent of u — no step used anything about u beyond $|u| = 1$ — and blows up as $t \uparrow s$. $\square$

Proof of Theorem 8.2.1. Write $F_t = \{\theta : \dim_{\mathbb{H}} \pi_{\theta}(E) < t\}$ and suppose, for contradiction, that $\dim_{\mathbb{H}} F_t > t$. Fix s with $t < s < \dim_{\mathbb{H}} F_t$.

Step 1. *Two measures.*

Since $\dim_{\mathbb{H}} F_t > s$, Theorem 7.5.1 — applied to a compact subset of F_t of dimension exceeding s , which exists by inner regularity — gives a nonzero finite ν carried by F_t with $\nu(B(\theta, r)) \leq r^s$, hence $\nu(I) \leq |I|^s$ for intervals.

We may assume $\dim_{\mathbb{H}} E \geq t$. Indeed, if $\dim_{\mathbb{H}} E < t$ then $\dim_{\mathbb{H}} P_{\theta}(E) \leq \dim_{\mathbb{H}} E < t$ for *every* θ , so $F_t = [0, \pi)$ and the conclusion $\dim_{\mathbb{H}} F_t \leq \dim_{\mathbb{H}} E$ fails only if $\dim_{\mathbb{H}} E < 1$; that degenerate case is treated separately in Step 4. Granting $\dim_{\mathbb{H}} E \geq t$, Theorem 7.6.2(ii) gives a nonzero finite μ on E with $I_{t'}(\mu) < \infty$ for any fixed $t' < t$.

Step 2. *Average against ν .*

Exactly as in (8.1.1), but integrating $d\nu(\theta)$,

$$\begin{aligned} \int_0^{\pi} I_{t'}(\pi_{\theta\#\mu}) d\nu(\theta) &= \iint \left(\int_0^{\pi} \frac{d\nu(\theta)}{|(x-y) \cdot e_{\theta}|^{t'}} \right) d\mu d\mu \\ &\leq C_{s,t'} I_{t'}(\mu) < \infty, \end{aligned}$$

by Theorem 8.2.2, applicable since $t' < t < s$.

Step 3. *Conclude.*

The left-hand side is finite, so $I_{t'}(\pi_{\theta\#\mu}) < \infty$ for ν -almost every θ , whence $\dim_{\mathbb{H}} \pi_{\theta}(E) \geq t'$ for ν -almost every θ by Theorem 7.6.2(i).

Letting $t' \uparrow t$ along a sequence and intersecting countably many ν -full sets, $\dim_{\mathbb{H}} \pi_{\theta}(E) \geq t$ for ν -a.e. θ . But ν is carried by F_t , on which $\dim_{\mathbb{H}} \pi_{\theta}(E) < t$ by definition, and $\nu \neq 0$ — a contradiction.

Step 4. *The degenerate case.*

If $\dim_{\mathbb{H}} E < t$ then $F_t = [0, \pi)$ and the conclusion $\dim_{\mathbb{H}} F_t \leq t$ reads $1 \leq t$, which holds since we may assume $t \geq 1$; and for $t < 1$ with $\dim_{\mathbb{H}} E < t$ the statement is vacuous once one notes $\dim_{\mathbb{H}} \pi_{\theta}E \leq \dim_{\mathbb{H}} E < t$ for all θ , so the theorem is asserting $\dim_{\mathbb{H}}[0, \pi) = 1 \leq t$ — which requires $t = 1$. We therefore read Theorem 8.2.1 as asserting content only when $t \leq \dim_{\mathbb{H}} E$, which is the case of interest. $\square$

Corollary 8.2.3. *For compact $E \subseteq \mathbb{R}^2$ with $\dim_{\mathbb{H}} E \leq 1$,*

$$\dim_{\mathbb{H}} \left\{ \theta : \dim_{\mathbb{H}} \pi_{\theta}(E) < \dim_{\mathbb{H}} E \right\} \leq \dim_{\mathbb{H}} E.$$

Proof. Apply Theorem 8.2.1 with $t = \dim_{\mathbb{H}} E$ and note that the set in question is the increasing union over $t' < \dim_{\mathbb{H}} E$ of the sets $\{\theta : \dim_{\mathbb{H}} \pi_{\theta}E < t'\}$, each of dimension at most t' ; countable stability (Theorem 7.1.5(ii)) gives the bound. $\square$

WORKED EXAMPLE 8.2.1— TIER A **The bound is attained**

Theorem 8.2.1 is sharp. Let $0 < t \leq 1$ and let L be a compact set of directions with $\dim_{\mathbb{H}} L = t$. Build

$$E = \bigcup_{\theta \in L} \ell_{\theta},$$

a union of unit segments, one in each direction of L , arranged so that E is compact — for instance all through a common point, a bush.

For $\theta \in L$, the projection $\pi_{\theta^{\perp}}$ collapses ℓ_{θ} to a point. Arranging the bush so that $\dim_{\mathbb{H}} E > t$ — which holds since E contains a segment, giving $\dim_{\mathbb{H}} E \geq 1 \geq t$, with strict inequality when $t < 1$ — the projection $\pi_{\theta^{\perp}}(E)$ consists of the images of the other segments, whose directions range over L , and one computes $\dim_{\mathbb{H}} \pi_{\theta^{\perp}}(E) \leq \dim_{\mathbb{H}} L = t < \dim_{\mathbb{H}} E$. So every $\theta^{\perp}$ with $\theta \in L$ is exceptional, and the exceptional set contains an isometric copy of L , of dimension t .

What this shows. No bound better than t is possible, so the theorem is optimal in its own terms. What is *not* known is the sharp statement for a *fixed* E of given dimension — the modern projection theorems of Orponen, Shmerkin and Wang, surveyed in Volume 14, give improvements under extra hypotheses on E .

WORKED EXAMPLE 8.2.2— TIER B **Upgrading “almost every” to a dimension bound: the general recipe**

The passage from Theorem 8.1.2 to Theorem 8.2.1 is a manoeuvre worth having in hand, since it applies whenever an averaging argument gives an almost-everywhere statement.

The recipe. Suppose one has proved

$$\int_{\Theta} Q(\theta) \, d\theta < \infty$$

for some nonnegative quantity Q , concluding $Q(\theta) < \infty$ for a.e. θ . To bound the dimension of $\{Q = \infty\}$:

- (1) suppose $\dim_{\mathbb{H}} \{Q = \infty\} > s$ and take a Frostman measure ν of exponent s on it;
- (2) repeat the original computation with $d\theta$ replaced by $d\nu(\theta)$;
- (3) the only step that changes is the inner integral, which must now be estimated against ν rather than Lebesgue;
- (4) if it still converges, $Q < \infty$ ν -a.e., contradicting $\nu \neq 0$ on $\{Q = \infty\}$.

Where the work is. Entirely in step (3). Here the inner integral was $\int |u \cdot e_\theta|^{-t} d\nu$, and Theorem 8.2.2 showed the Frostman condition buys exactly one extra power: convergence for $t < s$ rather than $t < 1$. In general one needs a bound of the form

$$\nu\{\theta : |\Phi(\theta)| < \lambda\} \lesssim \lambda^s,$$

where Φ is the transversality function — a *sublevel set estimate* for ν , of exactly the kind proved in Volume 5, Worked Example 5.1.2.

Where it is used again. Volume 14’s projection theorems are all of this form, with Φ more complicated and the sublevel estimate correspondingly harder. Recognising the recipe makes those statements readable even when their proofs are not.

WORKED EXAMPLE 8.2.3— TIER C: RESEARCH TASTE **What is not known, and why the gap is interesting**

(a) *The dimension of the exceptional set for a fixed E .* Theorem 8.2.1 bounds it by t , and Worked Example 8.2.1 shows that is attained — but only by a set E built for the purpose. For a specific E , such as $C \times C$, nothing like a complete answer is known.

(b) *Falconer’s improvement, and its limits.* Falconer showed that for $t > \frac{1}{2}$ the exceptional set for $\dim_{\mathbb{H}} \pi_\theta E < t$ has dimension at most $2t - \dim_{\mathbb{H}} E$ — better than Kaufman when $\dim_{\mathbb{H}} E > t$. Combining the two gives the best elementary bound, and the truth is conjectured to be $\max\{0, \dim_{\mathbb{H}} E - t\}$, still open in general.

(c) *The connection to Kakeya.* A Besicovitch set is, by Worked Example 8.1.3(c), a configuration in which *every* direction is exceptional for a related problem. So the projection theorems and the Kakeya problem are two ends of the same question: how large can the set of bad parameters be? Projection theory says “small”; Besicovitch’s construction says “in the Kakeya setting, everything”. Understanding the transition is the content of the Furstenberg set problem, which Volume 14 discusses and which Ren and Wang settled in 2023.

(d) *A methodological point.* Every result in this chapter came from an averaging argument plus a sublevel-set estimate. Neither ingredient is geometric; both are potential-theoretic. That is why this material is comparatively complete while the Kakeya problem is not: the moment the bad parameters can be *all* of them, averaging is useless, and one must count tubes.

WORKED EXAMPLE 8.2.4— TIER B **Sublevel set estimates: the ingredient to check**

Worked Example 8.2.2 said that the whole content of an upgrade is a bound of the form $\nu\{\theta : |\Phi(\theta)| < \lambda\} \lesssim \lambda^s$. Such bounds are exactly the sublevel set estimates of Volume 5, Worked Example 5.1.2, and it is worth seeing how they are produced.

The generic principle. If Φ is C^1 with $|\Phi'| \geq c$ on the relevant interval, then $\{|\Phi| < \lambda\}$ is contained in an interval of length $2\lambda/c$; if $\nu(I) \leq |I|^s$ this gives $\nu\{|\Phi| < \lambda\} \leq (2\lambda/c)^s$. Every estimate in this chapter is that computation.

Instance 1: Theorem 8.2.2. Here $\Phi(\theta) = \cos(\theta - \alpha)$ and $|\Phi'| = |\sin(\theta - \alpha)|$, which is *not* bounded below — it vanishes where Φ is extremal. But one does not need it there: near the zeros of Φ , which are where $\theta - \alpha = \pm\frac{\pi}{2}$, the derivative is ± 1 . So the hypothesis holds exactly where it is needed, and that is the content of the estimate $\{|\cos(\theta - \alpha)| < \lambda\} \subseteq$ an interval of length $\pi\lambda$.

Instance 2: higher-order vanishing. If instead Φ vanished to order k at its zeros — $\Phi(\theta) \approx (\theta - \theta_0)^k$ — the sublevel set would be an interval of length $\approx \lambda^{1/k}$, and the bound would become $\nu\{|\Phi| < \lambda\} \lesssim \lambda^{s/k}$. Repeating Theorem 8.2.2 then gives convergence for $t < s/k$ instead of $t < s$, and the exceptional-set bound weakens accordingly.

Instance 3: where this bites in the literature. The modern projection theorems replace the family $\{\pi_\theta\}$ by a family of nonlinear maps, and the transversality function Φ is then a genuine object of study. The Ren–Wang solution of the Furstenberg problem, in Volume 14, turns on exactly such an estimate. A reader who can identify Φ and its order of vanishing can predict the exponent in any such theorem before reading the proof.

The exercise to do once. Take $\Phi(\theta) = \cos(\theta - \alpha)$, verify the order of vanishing is 1, and confirm that the exponent in Theorem 8.2.1 is s and not $s/2$ or $2s$. Then repeat with $\Phi(\theta) = \cos^2(\theta - \alpha)$ and check the exponent halves.

Proposition 8.2.4 (Falconer’s complementary bound). *Let $E \subseteq \mathbb{R}^2$ be compact with $\dim_{\text{H}} E > t$ and $\frac{1}{2} < t \leq 1$. Then*

$$\dim_{\text{H}} \left\{ \theta : \dim_{\text{H}} \pi_\theta(E) < t \right\} \leq 2t - \dim_{\text{H}} E.$$

Proof. Write $d = \dim_{\text{H}} E$ and suppose the exceptional set F_t has $\dim_{\text{H}} F_t > s$ for some $s > 2t - d$; we derive a contradiction. Fix $t' < t$ and $d' < d$ with $s > 2t' - d'$, possible by continuity.

By Theorem 7.5.1 take $\nu \neq 0$ on F_t with $\nu(I) \leq |I|^s$, and $\mu \neq 0$ on E with $\mu(B(x, r)) \leq r^{d'}$; by Theorem 7.6.2 we may also assume $I_\nu(\mu) < \infty$.

The proof of Theorem 8.2.1 bounded the inner integral by $|x - y|^{-t'}$ times a constant. Here we use the Frostman property of μ as well. Splitting the pair

integral according to $|x - y|$ and applying Theorem 8.2.2 on each piece,

$$\int_0^\pi I_{t'}(\pi_{\theta\#}\mu) \, d\nu(\theta) \lesssim \iint \frac{d\mu(x) \, d\mu(y)}{|x - y|^{t'}} \cdot \sup_u \int_0^\pi \frac{d\nu(\theta)}{|u \cdot e_\theta|^{t'}},$$

which is finite provided $t' < s$. When $s \geq t$ this already gives the contradiction, as in Theorem 8.2.1.

When $s < t$ the estimate above fails, and one uses instead that μ has dimension $d' > 2t' - s$: the mass of $\mu \times \mu$ in the set where $|(x - y) \cdot e_\theta| < \lambda|x - y|$ is at most $\lambda^{d'-t'}$ by the Frostman bound applied to slabs, and combining with $\nu\{|u \cdot e_\theta| < \lambda\} \leq \lambda^s$ gives convergence exactly when $t' < \frac{s+d'}{2}$, that is $s > 2t' - d'$ — which is the hypothesis. So $I_{t'}(\pi_{\theta\#}\mu) < \infty$ for ν -a.e. θ , contradicting $\nu(F_t) > 0$ as before. $\square$

Where we are. Theorem 8.2.1 and Theorem 8.2.4 are complementary: Kaufman's bound t is better when $\dim_{\mathbb{H}} E < t$, Falconer's bound $2t - \dim_{\mathbb{H}} E$ is better when $\dim_{\mathbb{H}} E > t$, and they agree at $\dim_{\mathbb{H}} E = t$. Taking the minimum gives the best elementary estimate,

$$\dim_{\mathbb{H}} \{\theta : \dim_{\mathbb{H}} \pi_\theta E < t\} \leq \min\{t, 2t - \dim_{\mathbb{H}} E\}.$$

The conjectured truth is $\max\{0, \dim_{\mathbb{H}} E - t\}$ — rather different, and open. Modern work of Orponen, Shmerkin and Wang, surveyed in Volume 14, improves on both under additional hypotheses.

Drills

Drill 8.2.1. Verify the containment $\{\theta : |\cos(\theta - \alpha)| < \lambda\} \subseteq$ an interval of length $\pi\lambda$, used in Theorem 8.2.2.

Drill 8.2.2. Compute $C_{s,t}$ in Theorem 8.2.2 explicitly, and check that it blows up as $t \uparrow s$.

Drill 8.2.3. Deduce from Theorem 8.2.1 that if $\dim_{\mathbb{H}} E = 1$ then $\dim_{\mathbb{H}} \pi_\theta(E) = 1$ for all θ outside a set of dimension at most 1 — and explain why this is vacuous, so that Theorem 8.2.3 is the useful statement.

Drill 8.2.4. Carry out Worked Example 8.2.2 for the statement “ $\widehat{\mu}(re_\theta)$ is square-integrable in r for a.e. θ ” from Theorem 8.1.2(ii), and find the sublevel estimate needed.

Notes

Theorem 8.2.1 is Kaufman's (1968), proved in the same paper as the short proof of Marstrand's theorem. Falconer's complementary bound is from 1982; the sharp conjecture is due to him and remains open.

The recipe of Worked Example 8.2.2 — replace Lebesgue measure by a Frostman measure on the exceptional set — is one of the most transferable techniques in the subject, and is worth trying whenever an argument produces an almost-everywhere conclusion.

Chapter 8.3

Product Sets and Besicovitch Sets

Volume 1 constructed Besicovitch sets of arbitrarily small measure and promised one of measure *zero*; Volume 7 explained why the compactness argument one expects to work does not, and deferred the construction here. This chapter settles the matter. The natural route runs through the projection theory just developed, and it is worth following far enough to see exactly where it stops.

8.3.1 Projections of product sets

Proposition 8.3.1. *Let $A, B \subseteq \mathbb{R}$ be compact and $\theta \in (0, \pi/2)$. Then, up to a bi-Lipschitz change of coordinates on the target line,*

$$\pi_\theta(A \times B) = \cos \theta \cdot A + \sin \theta \cdot B.$$

In particular $\pi_{\pi/4}(A \times B)$ is a rescaled copy of the sumset $A + B$, and the projection in direction θ is the sumset $A + \lambda B$ with $\lambda = \tan \theta$, rescaled.

Proof. By definition $\pi_\theta(a, b) = a \cos \theta + b \sin \theta$, and as (a, b) ranges over $A \times B$ this ranges over $\cos \theta A + \sin \theta B$. Dividing by $\cos \theta \neq 0$ is a bi-Lipschitz map of the line. $\square$

Where we are. So the projection theorems of Chapter 8.1 and Chapter 8.2 are, when applied to product sets, statements about sumsets $A + \lambda B$: Marstrand says that for almost every λ the sumset is as large as the dimensions allow, and Kaufman bounds the set of bad λ .

This is the reason projection theory and additive combinatorics keep meeting. Volume 10 will develop sumset inequalities for their own sake and use them on Kakeya; the present chapter is the geometric face of the same material.

WORKED EXAMPLE 8.3.1— TIER A The four-corner Cantor set

Let K be the attractor of the four similarities of ratio $\frac{1}{4}$ taking the unit square to its four corner squares of side $\frac{1}{4}$. By Theorem 7.8.3, since $4 \cdot 4^{-s} = 1$ gives $s = 1$,

$$\dim_{\mathbb{H}} K = 1, \quad 0 < \mathcal{H}^1(K) < \infty.$$

So K is a one-dimensional set of finite positive length, and it is totally disconnected — it contains no rectifiable curve.

Its projections. $\pi_{\theta}(K)$ has dimension 1 for almost every θ , by Theorem 8.1.2(i), which is the most Theorem 8.1.2 can say since $\dim_{\mathbb{H}} K = 1$ is not > 1 . But a stronger and quite different statement is true: $m(\pi_{\theta}(K)) = 0$ for almost every θ . The projection in direction θ has dimension 1 and measure zero, simultaneously.

Why this does not follow from anything proved here. Theorem 8.1.2 gives a lower bound on dimension and, when $\dim_{\mathbb{H}} E > 1$, positive measure. It says nothing when $\dim_{\mathbb{H}} E = 1$ exactly, which is precisely the borderline the four-corner set sits on. The vanishing of the measure is the Besicovitch projection theorem, discussed in Theorem 8.3.3.

8.3.2 The construction that would work, and why we stop

Proposition 8.3.2 (Kahane's reduction). *Let $A \subseteq [0, 1]$ be compact with $A - A \supseteq [-1, 1]$, and set*

$$E = \bigcup_{a, b \in A} [(a, 0), (b, 1)] \subseteq \mathbb{R}^2,$$

the union of the segments joining $(a, 0)$ to $(b, 1)$. Then

(i) *E is compact and contains a segment in every direction of slope at least 1 in absolute value;*

(ii) $m(E) = \int_0^1 m((1-y)A + yA) \, dy;$

(iii) *consequently $m(E) = 0$ if and only if $m((1-y)A + yA) = 0$ for almost every $y \in (0, 1)$.*

Proof. (i) E is the image of the compact set $A \times A \times [0, 1]$ under the continuous map $(a, b, y) \mapsto ((1-y)a + yb, y)$, hence compact. The segment from $(a, 0)$ to $(b, 1)$ has direction vector $(b-a, 1)$; as (a, b) ranges over $A \times A$, the number $b-a$ ranges over $A-A \supseteq [-1, 1]$, so every direction $(v, 1)$ with $|v| \leq 1$ occurs.

- (ii) The slice of E at height $y \in [0, 1]$ is $E_y = \{(1 - y)a + yb : a, b \in A\} = (1 - y)A + yA$, so Fubini (Theorem 1.6.7) gives $m(E) = \int_0^1 m(E_y) dy$.
 (iii) Immediate from (ii), the integrand being nonnegative. $\square$

Remark 8.3.3 (Scope decision: the measure-zero Besicovitch set). Theorem 8.3.2 reduces the construction to a single question: is there a compact $A \subseteq [0, 1]$ with $A - A \supseteq [-1, 1]$ and

$$m((1 - y)A + yA) = 0 \quad \text{for a.e. } y \in (0, 1) ? \quad (8.3.1)$$

We do not answer it in this series, and the existence of a Besicovitch set of measure zero is accordingly removed from the plan. This is revision R8.

The obstruction is precise and worth recording, because the route looks as though the machinery of this volume should close it.

By Theorem 8.3.1, $(1 - y)A + yA$ is a projection of $A \times A$. So (8.3.1) asks for a set whose almost-every projection is Lebesgue-null. Theorem 8.1.2(ii) says this forces $\dim_{\text{H}}(A \times A) \leq 1$, that is $\dim_{\text{H}} A \leq \frac{1}{2}$ by Theorem 7.6.4 and Theorem 7.8.3. But the requirement $A - A \supseteq [-1, 1]$ pushes the other way: for a self-similar A with two maps of ratio r , the difference set contains an interval only when $r \geq \frac{1}{3}$, and then

$$\dim_{\text{H}} A = \frac{\log 2}{\log(1/r)} \geq \frac{\log 2}{\log 3} \approx 0.631 > \frac{1}{2}.$$

So $\dim_{\text{H}}(A \times A) > 1$, and Theorem 8.1.2(ii) gives the *opposite* of what is wanted: almost every projection has *positive* measure. The two constraints are in direct conflict for self-similar A .

The classical resolution uses a set A that is not self-similar, sitting exactly on the borderline $\dim_{\text{H}}(A \times A) = 1$ — as the four-corner set of Worked Example 8.3.1 does — where Theorem 8.1.2 is silent and the vanishing of the measure is the content of the *Besicovitch projection theorem*: a purely unrectifiable set of finite $\mathcal{H}^1$ measure has Lebesgue-null projection in almost every direction. That theorem is a substantial piece of geometric measure theory, requiring the density and rectifiability theory this series does not develop.

What is lost. Nothing that is used. The Kakeya conjecture concerns Hausdorff dimension (Theorem 7.1.11), not measure; Volume 3's ball multiplier theorem uses the configuration lemma, not measure zero; and Volume 1's Theorem 1.3.12, giving measure less than every ε , is what all the applications in this series invoke. The measure-zero refinement is a celebrated fact and, for the purposes of these volumes, an ornament.

A note on the record. This is the third disposition of the item: promised in Volume 1, deferred from Volume 7 to Volume 8, and now removed. The

pattern is worth naming — an attractive result kept being postponed because each time the tools to hand *nearly* sufficed. Naming the obstruction, as above, is what should have happened at the first deferral.

WORKED EXAMPLE 8.3.2— TIER B **What the reduction does deliver**

Theorem 8.3.2 is not wasted even though (8.3.1) is left open here. It converts a two-dimensional geometric question into a one-dimensional arithmetic one, and that conversion is used repeatedly.

(a) *Small measure, cheaply.* Take A self-similar with ratio $r = \frac{1}{3}$, so $A - A = [-1, 1]$ and $\dim_{\mathcal{H}} A = \log 2 / \log 3$. Then A is Lebesgue null, and although $m((1 - y)A + yA)$ need not vanish, one can estimate it: at resolution δ , A is covered by $\approx \delta^{-\log 2 / \log 3}$ intervals, so the sumset is covered by $\approx \delta^{-2 \log 2 / \log 3}$ intervals of length δ , giving $m(((1 - y)A + yA)_{\delta}) \lesssim \delta^{1 - 2 \log 2 / \log 3}$. Since $2 \log 2 / \log 3 \approx 1.262 > 1$, this is not small — consistent with Theorem 8.3.3.

(b) *The correct reading.* The computation in (a) says the construction is *efficient*: it produces a set of dimension 2 from a set A of dimension well below 1. That efficiency is what makes Kakeya sets small, and it is the same phenomenon as the Perron tree, seen arithmetically instead of geometrically.

(c) *The general principle.* A Kakeya-type set built from a Cantor set of parameters has its measure computed by Fubini as an average of sumset measures. Volume 10's arithmetic method for Kakeya is this observation systematised: one studies the additive structure of the parameter set instead of the geometry of the tubes. Bourgain's bound, and the Katz–Łaba–Tao improvement, both proceed this way.

WORKED EXAMPLE 8.3.3— TIER C: RESEARCH TASTE **Rectifiability, and the theory this series omits**

Theorem 8.3.3 appealed to the Besicovitch projection theorem. It is worth saying what that theorem belongs to, since it is the one substantial branch of geometric measure theory this series does without.

(a) *The dichotomy.* A set E with $\mathcal{H}^1(E) < \infty$ in the plane splits into a *rectifiable* part, contained in countably many Lipschitz curves, and a *purely unrectifiable* part, meeting every Lipschitz curve in $\mathcal{H}^1$ -measure zero. Every such E decomposes uniquely this way.

(b) *The theorem.* Projections detect the dichotomy exactly: a rectifiable set has projections of positive measure in almost every direction, and a purely unrectifiable set has projections of measure zero in almost every direction. The four-corner set of Worked Example 8.3.1 is the standard purely unrectifiable

example.

(c) *Why it is hard.* The proof of the unrectifiable half uses the density theory of Theorem 7.4.4 pushed much further — one shows that a set with projections of positive measure has tangent-like behaviour at almost every point, and then that this forces rectifiability. The full argument is the subject of several chapters in Mattila’s book.

(d) *Why this series omits it.* Because Kakeya and restriction are about sets of dimension close to n , where the rectifiable/unrectifiable dichotomy carries no information: every set of dimension > 1 in the plane is unrectifiable in the relevant sense, and the theory says nothing further. The dichotomy is sharp exactly at integer dimension with finite measure, which is a regime this subject does not visit.

(e) *The exception worth knowing.* It does visit that regime once. The Furstenberg set problem, in Volume 14, concerns sets meeting each line of a family in dimension s , and at $s = 1$ the rectifiability theory becomes relevant. Ren and Wang’s solution does not use it, but the connection is why the problem was long regarded as belonging to geometric measure theory rather than harmonic analysis.

WORKED EXAMPLE 8.3.4— TIER B **Translating between projections and sumsets**

Theorem 8.3.1 identifies $\pi_\theta(A \times B)$ with $\cos \theta A + \sin \theta B$. The dictionary is used in both directions, and it is worth rehearsing each.

Projection facts read as sumset facts.

- Theorem 8.1.2(i) becomes: for almost every λ , $\dim_{\mathbb{H}}(A + \lambda B) = \min\{\dim_{\mathbb{H}} A + \dim_{\mathbb{H}} B, 1\}$. So a sumset is as large as the dimensions permit, for almost every dilation ratio.
- Theorem 8.1.2(ii) becomes: if $\dim_{\mathbb{H}} A + \dim_{\mathbb{H}} B > 1$ then $A + \lambda B$ has positive measure for almost every λ .
- Theorem 8.2.1 becomes: the set of bad ratios λ has dimension at most t , where t is the dimension one is asking for.

Sumset facts read as projection facts. Conversely, an arithmetic statement about $A + \lambda B$ transfers. For instance if A is the middle-thirds Cantor set then $A + A = [0, 2]$ — every number in $[0, 2]$ is a sum of two base-3 numbers with digits in $\{0, 2\}$ — so $\pi_{\pi/4}(A \times A)$ is a full interval, even though $\dim_{\mathbb{H}}(A \times A) \approx 1.26$ gives no such guarantee for a specific θ .

Where the dictionary is essential. The measure-zero Besicovitch question of Theorem 8.3.3 is a sumset question, (8.3.1), and it was only by reading it as a projection question that the obstruction became visible. Conversely, Volume 10 will prove sumset inequalities — Plünnecke–Ruzsa, Balog–Szemerédi–Gowers — and apply them to tube configurations; that application is the same translation running the other way.

A caution. The dictionary is exact only for *product* sets. A general $E \subseteq \mathbb{R}^2$ has projections that are not sumsets of anything, and the arithmetic intuition can mislead. When a paper says “by the projection theorem” about a non-product set, it means the theorem itself, not the arithmetic reading.

The exercise. Verify $A + A = [0, 2]$ for the middle-thirds Cantor set by writing $x \in [0, 2]$ in base 3 and splitting each digit 0, 1, 2 as a sum of two digits from $\{0, 2\}$. Then check that $A + \frac{1}{2}A$ has measure zero, so that the behaviour genuinely depends on λ .

Proposition 8.3.4 (Slices of a product). *Let $A, B \subseteq \mathbb{R}$ be compact and $\lambda \neq 0$. Then for every $c \in \mathbb{R}$ the slice*

$$(A \times B) \cap \{(x, y) : x + \lambda y = c\}$$

is, under the bi-Lipschitz map $(x, y) \mapsto y$, in bijection with $B \cap \lambda^{-1}(c - A)$. Consequently

$$\dim_{\mathrm{H}}(A \cap (c - \lambda B)) \leq \dim_{\mathrm{H}}(A \times B) - \dim_{\mathrm{H}} \pi_{\theta}(A \times B)$$

for those c attaining the maximum, where θ corresponds to λ by Theorem 8.3.1.

Proof. A point of the slice satisfies $x = c - \lambda y$ with $x \in A$, $y \in B$; so y ranges over $\{y \in B : c - \lambda y \in A\} = B \cap \lambda^{-1}(c - A)$, and the map $(x, y) \mapsto y$ is a bijection onto that set, bi-Lipschitz with constants depending on λ .

For the inequality, note the slices are the fibres of π_{θ} up to the bi-Lipschitz identification, and apply Theorem 7.1.7 with $m = 1$: almost every fibre has dimension at most $\dim_{\mathrm{H}}(A \times B) - 1$, and comparing with the dimension of the image gives the stated bound at the values of c carrying the most. $\square$

Where we are. So a projection theorem is simultaneously a statement about *slices*: if the image is large then the fibres are small, and conversely. The two readings are exchanged by Theorem 7.1.7, the Fubini-type bound proved in Volume 7.

This is worth holding onto because the Kakeya problem is naturally a slicing problem — a Besicovitch set has a segment, that is a large slice, in every direction — and the projection theory of this chapter is what the slicing reading connects it to. Volume 14’s Furstenberg sets are defined

by prescribing the dimension of slices, and sit exactly between the two subjects.

WORKED EXAMPLE 8.3.5— TIER C: RESEARCH TASTE **Three ways a Kakeya set can be built, and what each costs**

This chapter analysed one construction of a Besicovitch set and found it wanting. It is worth setting the three known approaches side by side, since each recurs later in the series.

(a) *The Perron tree (Volume 1)*. Bisect a triangle repeatedly and slide the halves together. Gives measure less than any ε , is completely explicit, and the sets are finite unions of triangles. *Cost*: the sets at successive stages do not nest, so no limit of measure zero is obtained — the failure diagnosed in Volume 7, Chapter 7.3. *Where it recurs*: Volume 3's ball multiplier theorem uses it directly, and Volume 9's tube combinatorics is its systematic form.

(b) *The Cantor-set construction (this chapter)*. Join $(a, 0)$ to $(b, 1)$ for a, b in a Cantor set A with $A - A \supseteq [-1, 1]$. Gives all directions provided the difference set is full, and reduces the measure to a sumset question by Fubini. *Cost*: the two requirements conflict (Theorem 8.3.3), and resolving them needs the Besicovitch projection theorem. *Where it recurs*: Volume 10's arithmetic method is this construction read additively.

(c) *The random construction*. Choose the tubes' positions at random subject to the direction constraint. Gives, with high probability, sets whose dimension can be estimated by second-moment methods, and — unlike (a) and (b) — produces Salem-type behaviour so that Fourier methods are sharp on them. *Cost*: random sets are *not* extremal for Kakeya; they are too large. The extremal configurations are arithmetic, which is why randomness has never given a counterexample. *Where it recurs*: Volume 7's remark on Salem sets, and the heuristics of Volume 14.

The lesson. All three constructions produce sets of small measure and none produces one of small *dimension*. That is the empirical content of the Kakeya conjecture: fifty years of attempts to build a small Besicovitch set have produced only sets of full dimension, and the conjecture asserts that this is necessary. Volume 10's finite-field analogue, where Dvir proved it outright, is the strongest evidence available.

Drills

Drill 8.3.1. Verify Theorem 8.3.1 and check the bi-Lipschitz constants as $\theta \rightarrow 0$.

Drill 8.3.2. Show that for the self-similar A of ratio r , $A - A$ contains an interval if and only if $r \geq \frac{1}{3}$, by examining the base- $1/r$ expansions.

Drill 8.3.3. Compute $\dim_{\text{H}} E$ for the set E of Theorem 8.3.2 with A the middle-thirds Cantor set, and compare with 2.

Drill 8.3.4. Confirm Worked Example 8.3.1: apply Theorem 7.8.3 to the four-corner system and check the open set condition holds with V the open unit square.

Notes

Kahane's reduction is from 1969. That a Besicovitch set of measure zero exists is Besicovitch's (1928); the modern treatments either follow his original nested construction or use the projection route sketched here, and both are longer than anything else in this volume.

The Besicovitch projection theorem is from 1939, with the modern proof due to Federer. Mattila's book, Chapters 15–18, is the reference.

Chapter 8.4

The Distance Measure

Definition 8.4.1. For compact $E \subseteq \mathbb{R}^n$ the *distance set* is

$$\Delta(E) = \{|x - y| : x, y \in E\} \subseteq [0, \infty).$$

Conjecture 8.4.2 (Falconer). *For $n \geq 2$ and compact $E \subseteq \mathbb{R}^n$ with $\dim_{\mathbb{H}} E > \frac{n}{2}$, the distance set $\Delta(E)$ has positive Lebesgue measure.*

The exponent $\frac{n}{2}$ is forced by an example: a suitably chosen set of dimension $\frac{n}{2}$ has a distance set of measure zero (Drill 4.4). This chapter builds the object through which the conjecture is studied, and Chapter 8.5 proves Falconer's theorem at the exponent $\frac{n+1}{2}$.

8.4.1 From a set to a measure

Definition 8.4.3. Let μ be a finite Borel measure with compact support. The *distance measure* $\nu = \nu_{\mu}$ is the pushforward of $\mu \times \mu$ under $(x, y) \mapsto |x - y|$; that is,

$$\int_0^{\infty} f \, d\nu = \iint f(|x - y|) \, d\mu(x) \, d\mu(y)$$

for every bounded Borel f .

Proposition 8.4.4. *ν is a finite Borel measure on $[0, \infty)$ with $\nu([0, \infty)) = \mu(\mathbb{R}^n)^2$, supported in $\Delta(\text{spt } \mu)$. If ν is absolutely continuous with respect to Lebesgue measure, then $m(\Delta(\text{spt } \mu)) > 0$.*

Proof. That ν is a finite Borel measure of the stated total mass is immediate from the definition with $f \equiv 1$. Its support: if $t \notin \Delta(\text{spt } \mu)$ then, $\Delta(\text{spt } \mu)$ being compact — the continuous image of the compact $\text{spt } \mu \times \text{spt } \mu$ — there is a neighbourhood of t disjoint from it, and ν of that neighbourhood is 0.

For the last statement, suppose $m(\Delta(\text{spt } \mu)) = 0$. Since ν is carried by that set and $\nu \ll m$, we get $\nu \equiv 0$, contradicting $\nu([0, \infty)) = \mu(\mathbb{R}^n)^2 > 0$. $\square$

So the conjecture reduces to showing $\nu \ll m$, and the route is to exhibit a bounded density. The next section expresses ν through the Fourier transform, which is where the curvature of the sphere enters.

8.4.2 The Fourier representation

Definition 8.4.5. For $t > 0$ and $0 < \varepsilon < t/2$ let

$$A_\varepsilon^t = \{z \in \mathbb{R}^n : t - \varepsilon < |z| < t + \varepsilon\}$$

be the open annulus of radius t and thickness 2ε , and set

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} (\mu \times \mu) \{(x, y) : x - y \in A_\varepsilon^t\}.$$

Lemma 8.4.6. For every $t > 0$,

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi,$$

the integral converging absolutely, and $\int_0^\infty \nu_\varepsilon(t) f(t) dt \rightarrow \int f d\nu$ as $\varepsilon \downarrow 0$ for every $f \in C_c((0, \infty))$.

Proof. Step 1. The identity.

Since A_ε^t is bounded and symmetric ($-A_\varepsilon^t = A_\varepsilon^t$), the function $\mathbf{1}_{A_\varepsilon^t}$ lies in $L^1 \cap L^2$ and has real even Fourier transform. By Parseval for measures — Theorem 2.5.11, the measure μ being finite with compact support so that $\widehat{\mu}$ is bounded and continuous —

$$\begin{aligned} (\mu \times \mu) \{(x, y) : x - y \in A_\varepsilon^t\} &= \iint \mathbf{1}_{A_\varepsilon^t}(x - y) d\mu(x) d\mu(y) \\ &= \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi. \end{aligned}$$

Absolute convergence: $\widehat{\mathbf{1}_{A_\varepsilon^t}} \in L^2$ by Plancherel and $|\widehat{\mu}|^2$ is bounded, while for large $|\xi|$ the first factor decays — see Theorem 8.4.7 below.

Step 2. Convergence to ν .

For $f \in C_c((0, \infty))$, Tonelli gives

$$\begin{aligned} \int_0^\infty \nu_\varepsilon(t) f(t) dt &= \iint \left(\frac{1}{2\varepsilon} \int_0^\infty f(t) \mathbf{1}_{\{t-\varepsilon < |x-y| < t+\varepsilon\}} dt \right) d\mu d\mu \\ &= \iint f_\varepsilon(|x - y|) d\mu d\mu, \end{aligned}$$

where $f_\varepsilon(r) = \frac{1}{2\varepsilon} \int_{r-\varepsilon}^{r+\varepsilon} f$ is the average of f over an ε -interval. Since f is uniformly continuous, $f_\varepsilon \rightarrow f$ uniformly, and $\mu \times \mu$ is finite, so the right side converges to $\iint f(|x - y|) d\mu d\mu = \int f d\nu$. $\square$

Lemma 8.4.7 (Decay of the annulus transform). *Fix $0 < a < b < \infty$. There is $C = C(n, a, b)$ such that for all $t \in [a, b]$, $0 < \varepsilon < a/2$ and $\xi \in \mathbb{R}^n$,*

$$|\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)| \leq C \varepsilon (1 + |\xi|)^{-\frac{n-1}{2}}.$$

Proof. Writing the annulus in polar coordinates (Theorem 1.6.11),

$$\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) = \int_{t-\varepsilon}^{t+\varepsilon} r^{n-1} \widehat{d\sigma}(r\xi) dr,$$

where $d\sigma$ is surface measure on the unit sphere and $\widehat{d\sigma}(r\xi)$ is its transform evaluated at $r\xi$, by the change of variables $z = r\omega$.

By Theorem 5.5.2 — the sphere has non-vanishing Gaussian curvature — $|\widehat{d\sigma}(\eta)| \lesssim_n (1 + |\eta|)^{-(n-1)/2}$. For $r \in [t-\varepsilon, t+\varepsilon] \subseteq [a/2, 2b]$ we have $|r\xi| \approx_{a,b} |\xi|$, so $|\widehat{d\sigma}(r\xi)| \lesssim_{n,a,b} (1 + |\xi|)^{-(n-1)/2}$, and $r^{n-1} \leq (2b)^{n-1}$. Integrating over an interval of length 2ε gives the bound. $\square$

Where we are. Theorem 8.4.7 is where the geometry enters, and it is the only place. The factor ε is the thickness of the annulus and is what makes $\nu_\varepsilon = \frac{1}{2\varepsilon}(\cdots)$ bounded as $\varepsilon \rightarrow 0$; the factor $(1 + |\xi|)^{-(n-1)/2}$ is the curvature of the sphere, imported from Volume 5.

Had the “distance” been measured by a norm whose unit ball has a flat piece — the ℓ^∞ norm, say — the second factor would be absent, the argument of Chapter 8.5 would give nothing, and indeed the corresponding distance-set problem has a different answer. Drill 4.3 works out the ℓ^∞ case, where the conjecture is false as stated.

WORKED EXAMPLE 8.4.1— TIER A **The trivial bounds**

Before proving anything it is worth knowing what is free.

Upper bound on the dimension needed. If $\dim_{\text{H}} E > \frac{n}{2}$ is conjectured to suffice, no smaller exponent can: there are compact sets of dimension exactly $\frac{n}{2}$ with $m(\Delta(E)) = 0$, obtained by taking E a product of a suitable Cantor set with itself (Drill 4.4).

Lower bound, for free. If $\dim_{\text{H}} E > 0$ then $\Delta(E)$ is uncountable. Indeed the map $y \mapsto |x-y|$ is Lipschitz for fixed x , so $\dim_{\text{H}} \Delta(E) \geq \dim_{\text{H}} \{|x-y| : y \in E\}$, and choosing x suitably one shows this is positive. So the content of the conjecture is entirely in passing from positive dimension to positive *measure*.

The gap. Between “uncountable” and “positive measure” lies the whole problem, and Theorem 8.4.4 says the way across is to produce a measure on $\Delta(E)$ that is absolutely continuous. That is what ν is for.

WORKED EXAMPLE 8.4.2— TIER B Why one mollifies, and what would go wrong otherwise

It is tempting to write, formally,

$$“ \nu(t) = \iint d\sigma_t(x-y) d\mu(x) d\mu(y) = \int \widehat{d\sigma_t}(\xi) |\widehat{\mu}(\xi)|^2 d\xi ”$$

and estimate directly. The formula is the right heuristic and is how the argument is usually described, but as written it is meaningless: $d\sigma_t$ is a measure, $\mu \times \mu$ may charge the set $\{|x-y|=t\}$, and the pairing is undefined. *What the mollification fixes.* Theorem 8.4.6 replaces $d\sigma_t$ by $\frac{1}{2\varepsilon} \mathbf{1}_{A_\varepsilon^t}$, an honest L^1 function, and every step is then a genuine identity. The bound of Theorem 8.4.7 is uniform in ε , so a bound on ν_ε passes to ν in the limit.

The general lesson. Whenever a proof pairs a singular measure against another singular measure, the pairing must be defined by mollification and the estimates made uniform. The same issue arose in Theorem 6.1.3, where $\mathcal{E}g$ paired against f was defined by extension from a dense class, and in Theorem 6.7.3, where $gd\sigma$ was smoothed at scale δ . It is worth being suspicious of any argument in this area that pairs two measures without saying how.

WORKED EXAMPLE 8.4.3— TIER C: RESEARCH TASTE Why the distance set problem sits between two subjects

The problem was posed in geometric measure theory and is now solved with harmonic analysis. Both readings are useful and they meet in Theorem 8.4.6.

(a) *The geometric reading.* $\Delta(E)$ is the image of $E \times E$ under the distance map. Asking whether it has positive measure is asking whether that map is, in an averaged sense, non-degenerate on a set of the given dimension — a question about how the fibres $\{(x,y) : |x-y|=t\}$ meet $E \times E$. Each fibre is a hypersurface in $\mathbb{R}^{2n}$ of codimension 1, so one is intersecting a set of dimension $2s$ with a family of codimension-1 surfaces and asking that the intersections be non-trivial for a positive measure of parameters. That is a slicing problem, and Theorem 7.1.7 is its elementary form.

(b) *The analytic reading.* By Theorem 8.4.6, the density of ν is a weighted integral of $|\widehat{\mu}|^2$ against $\widehat{d\sigma}$. The problem becomes: how much can a fractal measure's transform correlate with the transform of surface measure? That is an extension-operator question, and it is why Chapter 8.7's improvements are all restriction estimates.

(c) *What the two readings each explain.* The geometric reading explains the conjectured exponent: $2s - (2n-1) \geq \text{something}$, or more simply the counting

in Worked Example 8.5.3(b). The analytic reading explains the *proofs*: every one of them bounds $\sigma(\mu, r)$.

(d) *The pattern across this series.* Kakeya was posed geometrically and is attacked combinatorially; restriction was posed analytically and is attacked geometrically; the distance problem was posed geometrically and is attacked analytically. In each case the productive move was to change the category the problem was stated in. Theorem 8.4.6 is that move here, and it is worth noting that it consists of nothing but Parseval.

WORKED EXAMPLE 8.4.4— TIER B Mollification with uniform bounds: how it is actually done

Worked Example 8.4.2 explained why one mollifies. This example rehearses the mechanics, since every argument in the next three chapters uses them and the bookkeeping is where errors hide.

The three requirements. To prove a measure ν has a bounded density one must supply:

- (1) a family ν_ε of honest functions with $\int f \nu_\varepsilon \rightarrow \int f d\nu$ for test functions f ;
- (2) a bound $|\nu_\varepsilon(t)| \leq M$ uniform in ε and in t on a compact interval;
- (3) the deduction that $\nu \leq Mm$ there.

Requirement (2) is the mathematics; (1) and (3) are bookkeeping that must nonetheless be done.

How (1) is arranged here. Theorem 8.4.6 takes $\nu_\varepsilon(t) = \frac{1}{2\varepsilon}(\mu \times \mu)\{x - y \in A_\varepsilon^t\}$, and the convergence is proved by Tonelli: the test integral becomes $\iint f_\varepsilon(|x - y|) d\mu d\mu$ with f_ε the ε -average of f , and $f_\varepsilon \rightarrow f$ uniformly since f is uniformly continuous. Note that this needs f compactly supported in $(0, \infty)$ — at $t = 0$ the annulus degenerates and the argument fails, which is why the interval $[a, b]$ is taken away from the origin.

How (3) is arranged. From (2), $\int f d\nu \leq M \int f$ for nonnegative $f \in C_c((a, b))$. A positive linear functional dominated by M times Lebesgue integration is represented by a measure $\leq Mm$, by Theorem 1.8.15 and uniqueness of the representing measure. It is worth noting that this step gives $\nu \leq Mm$ as *measures*, which is stronger than absolute continuity and is what Chapter 8.6 builds on.

The error to avoid. Bounding $\nu_\varepsilon(t)$ for each fixed ε with a constant depending on ε proves nothing: the bound must survive $\varepsilon \rightarrow 0$. In Theorem 8.5.1 the

factor ε from Theorem 8.4.7 cancels the $\frac{1}{2\varepsilon}$ exactly, and *that cancellation is the whole point of the estimate*. Any mollification argument should be read by first checking where the ε 's cancel.

Proposition 8.4.8 (The distance measure is non-atomic). *Let μ be a finite compactly supported measure with $I_s(\mu) < \infty$ for some $s > 0$. Then the distance measure ν has no atoms in $(0, \infty)$, and $\nu(\{0\}) = 0$.*

Proof. For $t = 0$: $\nu(\{0\}) = (\mu \times \mu)\{x = y\}$, and $I_s(\mu) < \infty$ forces μ to be non-atomic (Worked Example 7.6.4, Case 3), so the diagonal has $\mu \times \mu$ -measure zero by Fubini.

For $t > 0$: $\nu(\{t\}) = (\mu \times \mu)\{|x - y| = t\}$, the mass carried by a sphere bundle. Fix x ; the set $\{y : |x - y| = t\}$ is the sphere $S(x, t)$, and

$$\mu(S(x, t)) \leq \lim_{\varepsilon \downarrow 0} \mu(B(x, t + \varepsilon) \setminus B(x, t - \varepsilon)).$$

Since $I_s(\mu) < \infty$, by Step 1 of Theorem 7.6.2 we have $\mu(B(z, r)) \lesssim r^s$ for μ -a.e. z and small r ; covering the sphere $S(x, t)$ by $\approx \varepsilon^{1-n}$ balls of radius ε gives $\mu(S(x, t)) \lesssim \lim_{\varepsilon} \varepsilon^{1-n} \varepsilon^s$, which is 0 provided $s > n - 1$. For smaller s one argues instead that the map $y \mapsto |x - y|$ pushes μ to a measure of finite energy on the line, by Theorem 7.6.3(i) applied on the complement of a small ball about x , and a measure of finite positive energy on $\mathbb{R}$ is non-atomic. Integrating in x gives $\nu(\{t\}) = 0$. $\square$

Where we are. Theorem 8.4.8 is what licenses the choice of the interval $[a, b]$ with $\nu([a, b]) > 0$ in Theorem 8.5.1 Step 1: were ν a point mass at 0, no such interval away from the origin would exist and the argument would have nothing to work with.

It also shows how little is needed: any positive amount of dimension makes the distance measure non-atomic. The whole difficulty of Theorem 8.4.2 is in the gap between “non-atomic” — free — and “absolutely continuous” — which needs $\dim_{\mathbb{H}} E > \frac{n}{2}$.

WORKED EXAMPLE 8.4.5— TIER A **Computing $\Delta(E)$ for sets one can see**

Three examples, in increasing difficulty, to fix what the conjecture is about.

(a) *A segment.* $E = [0, 1] \times \{0\} \subseteq \mathbb{R}^2$ has $\dim_{\mathbb{H}} E = 1$ and $\Delta(E) = [0, 1]$ — positive measure, indeed an interval. So a set of dimension 1 can have a large distance set, and Theorem 8.4.2 at $n = 2$ asks for $\dim_{\mathbb{H}} E > 1$; the segment shows the conjecture is not *necessary*, only sufficient.

(b) *A Cantor set on a line.* $E = C \times \{0\}$ with C the middle-thirds set

has $\dim_{\mathbb{H}} E = \log 2 / \log 3 \approx 0.63$, and $\Delta(E) = |C - C| = [0, 1]$, since $C - C = [-1, 1]$. Again positive measure, from a set of dimension well below the threshold. So low dimension does not preclude a large distance set — what the conjecture rules out is the *reverse* implication failing.

(c) *The extremal example.* By Theorem 8.8.1 there are sets of dimension exactly $\frac{n}{2}$ with $m(\Delta(E)) = 0$. Comparing with (a) and (b): those sets are arithmetically special, arranged so that all distances fall near a lattice. Genericity, not size, is what (a) and (b) have and the extremal example lacks.

What the three show together. The conjecture is not about size alone. It asserts that *above* dimension $\frac{n}{2}$, no arrangement — however arithmetically clever — can keep the distance set null; and Theorem 8.8.1 says that at $\frac{n}{2}$ exactly, one can. The difficulty is that the hypothesis is purely metric while the extremal obstruction is arithmetic, so a proof must rule out arithmetic structure using only a dimension bound. That is what the Fourier transform does, and why it is the natural tool here.

Figure 8.4.1 (to be drawn). *The annulus and its dual*

Content required: The annulus A_ε^t of radius t and thickness 2ε , with the pair (x, y) realising a distance $|x - y| \in (t - \varepsilon, t + \varepsilon)$ shown as a chord. Beside it, the profile of $\widehat{\mathbf{1}_{A_\varepsilon^t}}$ against $|\xi|$, decaying like $\varepsilon(1 + |\xi|)^{-(n-1)/2}$, with the factor ε marked as the thickness and the exponent marked as the curvature of the sphere.

Drills

Drill 8.4.1. Verify that $\Delta(E)$ is compact for compact E , and that $\Delta(E) \subseteq [0, \text{diam } E]$.

Drill 8.4.2. Check the polar-coordinate identity for $\widehat{\mathbf{1}_{A_\varepsilon^t}}$ in Theorem 8.4.7.

Drill 8.4.3. Redo Theorem 8.4.7 with the Euclidean norm replaced by the ℓ^∞ norm, whose unit sphere is the boundary of a cube. Show the decay factor is absent in the coordinate directions, and explain why the method of Chapter 8.5 then gives nothing.

Drill 8.4.4. Let $A \subseteq [0, 1]$ be a Cantor set of dimension $\frac{1}{2}$ built on a rapidly increasing sequence of scales, and $E = A \times A \subseteq \mathbb{R}^2$. Show $\dim_{\mathbb{H}} E = 1 = \frac{n}{2}$ and sketch why $m(\Delta(E))$ can be made zero, establishing sharpness of Theorem 8.4.2.

Notes

Falconer posed the problem in 1985. The reduction to the absolute continuity of ν is his; the mollified formulation of Theorem 8.4.6 is the standard way to make the heuristic rigorous and appears in essentially this form in Wolff's lecture notes.

The sharpness example of Drill 4.4 is Falconer's, and shows the conjectured exponent $\frac{n}{2}$ cannot be lowered.

Chapter 8.5

Falconer's Theorem

Theorem 8.5.1 (Falconer). *Let $n \geq 2$ and let $E \subseteq \mathbb{R}^n$ be compact with*

$$\dim_{\mathbb{H}} E > \frac{n+1}{2}.$$

Then $m(\Delta(E)) > 0$.

Idea of the proof. By Theorem 8.4.4 it suffices to show the distance measure ν is absolutely continuous, and by Theorem 8.4.6 the approximating densities are

$$\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Two estimates meet here, and the theorem is exactly the point at which they balance.

From geometry: $|\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)| \lesssim \varepsilon(1 + |\xi|)^{-(n-1)/2}$, by Theorem 8.4.7 — the curvature of the sphere, imported from Volume 5.

From dimension: $\int |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi < \infty$ whenever $I_s(\mu) < \infty$, by Theorem 7.6.5 — the Fourier expression of energy, from Volume 7.

The first supplies a decay $(1 + |\xi|)^{-(n-1)/2}$; the second is exactly what is needed to integrate $|\widehat{\mu}|^2$ against a weight $(1 + |\xi|)^{s-n}$. So the argument closes as soon as

$$-\frac{n-1}{2} \leq s-n, \quad \text{that is} \quad s \geq n - \frac{n-1}{2} = \frac{n+1}{2},$$

and the exponent in the theorem is nothing but this comparison. The factor ε cancels the $\frac{1}{2\varepsilon}$, leaving a bound independent of ε .

Proof. Step 1. *Choose the measure.*

Let s satisfy $\frac{n+1}{2} < s < \dim_{\mathbb{H}} E$. By Theorem 7.6.2(ii) there is a nonzero finite Borel measure μ carried by E with $I_s(\mu) < \infty$; normalise $\mu(E) = 1$.

Since E is compact, $\text{spt } \mu \subseteq E$ is compact, so $\widehat{\mu}$ is a bounded continuous function and $\Delta(\text{spt } \mu) \subseteq \Delta(E)$.

Fix $0 < a < b < \infty$ with $\nu([a, b]) > 0$; such an interval exists because ν is a nonzero finite measure on $[0, \text{diam } E]$ and $\nu(\{0\}) = (\mu \times \mu)\{x = y\} = 0$, the latter because $I_s(\mu) < \infty$ forces μ to be non-atomic (Worked Example 7.6.4, Case 3).

Step 2. *Bound ν_ε uniformly.*

Let $t \in [a, b]$ and $0 < \varepsilon < a/2$. By Theorem 8.4.6 and Theorem 8.4.7,

$$\begin{aligned} |\nu_\varepsilon(t)| &\leq \frac{1}{2\varepsilon} \int_{\mathbb{R}^n} |\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)| |\widehat{\mu}(\xi)|^2 d\xi \\ &\leq \frac{C\varepsilon}{2\varepsilon} \int_{\mathbb{R}^n} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi, \end{aligned}$$

with $C = C(n, a, b)$. The factor ε has cancelled, so it remains to bound the integral by a finite quantity independent of t and ε — which it already is, no t or ε appearing in it.

Step 3. *The integral is finite.*

Split at $|\xi| = 1$. On $|\xi| \leq 1$ the weight is bounded by 1 and $|\widehat{\mu}| \leq \mu(\mathbb{R}^n) = 1$, so that part contributes at most $m(B(0, 1))$.

On $|\xi| > 1$ we use $s \geq \frac{n+1}{2}$, which gives $-\frac{n-1}{2} \leq s - n$ and hence

$$(1 + |\xi|)^{-\frac{n-1}{2}} \leq |\xi|^{-\frac{n-1}{2}} \leq |\xi|^{s-n} \quad (|\xi| > 1),$$

the last step because $|\xi| > 1$ and the exponent on the right is the larger. Therefore

$$\begin{aligned} \int_{|\xi|>1} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi &\leq \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi \\ &= c_{n,s}^{-1} I_s(\mu) < \infty, \end{aligned}$$

by Theorem 7.6.5. Combining, there is $M = M(n, a, b, \mu) < \infty$ with

$$|\nu_\varepsilon(t)| \leq M \quad \text{for all } t \in [a, b], \quad 0 < \varepsilon < a/2. \quad (8.5.1)$$

Step 4. *Pass to the limit.*

Let $f \in C_c((a, b))$ with $f \geq 0$. By Theorem 8.4.6 and (8.5.1),

$$\int f d\nu = \lim_{\varepsilon \downarrow 0} \int_a^b \nu_\varepsilon(t) f(t) dt \leq M \int_a^b f(t) dt.$$

So the linear functional $f \mapsto \int f d\nu$ is dominated by M times Lebesgue integration on (a, b) . By the Riesz representation theorem (Theorem 1.8.15)

and the uniqueness of the representing measure, this forces $\nu \upharpoonright (a, b) \leq Mm$; in particular $\nu \upharpoonright (a, b) \ll m$.

Step 5. Conclude.

If $m(\Delta(E)) = 0$ then, ν being carried by $\Delta(\text{spt } \mu) \subseteq \Delta(E)$ and absolutely continuous on (a, b) , we get $\nu((a, b)) = 0$. Choosing $[a, b]$ as in Step 1 with $\nu([a, b]) > 0$, and shrinking slightly so that the open interval still carries positive mass, gives a contradiction. Hence $m(\Delta(E)) > 0$. $\square$

Where we are. The proof used exactly two inputs, one from each of the two preceding volumes, and it is worth seeing that the exponent is the ratio between them.

Volume 5 gave decay $(n-1)/2$: the number of curved directions of the sphere, halved. Volume 7 gave that a set of dimension s supports a measure whose transform is square-integrable against $|\xi|^{s-n}$. The theorem holds when the first exponent is at least $n-s$, that is when $s \geq n - \frac{n-1}{2}$.

So the $\frac{n+1}{2}$ is not a technical artefact: it is where the curvature of the sphere runs out. To do better one must extract more from $\widehat{\mu}$ than its weighted L^2 norm — and that is precisely what the Mattila integral of Chapter 8.7 attempts, by keeping the *spherical averages* of $\widehat{\mu}$ rather than bounding $\widehat{\sigma}$ pointwise.

Pitfall: this is not the Mattila integral, and the difference matters

The reader who consults the literature will find Falconer’s theorem attributed both to the argument above and to “Mattila’s approach”. They are different, and the second does *not* prove the theorem.

Mattila’s method estimates ν in L^2 rather than L^∞ , and reduces the problem to the finiteness of

$$\mathcal{M}(\mu) = \int_1^\infty \left(\int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega) \right)^2 r^{n-1} dr.$$

Suppose one tries to bound the inner spherical average using only Theorem 5.5.2, as in Step 3 above. Writing $\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma$, one obtains $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$ (Drill 5.3), so

$$\mathcal{M}(\mu) \lesssim \int_1^\infty r^{-2\min\{s, \frac{n-1}{2}\} + n-1} dr,$$

which converges only if $2\min\{s, \frac{n-1}{2}\} > n$. But $\min\{s, \frac{n-1}{2}\} \leq \frac{n-1}{2} < \frac{n}{2}$ *always*. So this route never converges, in any dimension and for any s .

That is not a failure of effort: the Mattila integral genuinely diverges in part of the range, and progress by that method requires *improved spherical average estimates*, which are theorems about the restriction operator rather

than consequences of $\widehat{d\sigma}$. Chapter 8.7 sets this out. A reader attempting Falconer's theorem through $\mathcal{M}$ will not succeed, and should recognise the obstruction rather than look for a missing trick.

WORKED EXAMPLE 8.5.1— TIER A **The exponent in low dimensions**

$n = 2$. Theorem 8.5.1 requires $\dim_{\mathbb{H}} E > \frac{3}{2}$; the conjecture asks for > 1 . The gap is $\frac{1}{2}$, and the current record — $\frac{5}{4}$, due to Guth, Iosevich, Ou and Wang — closes half of it.

$n = 3$. The theorem gives 2; the conjecture asks for $\frac{3}{2}$; and Erdoğan's bound $\frac{n}{2} + \frac{1}{3}$ is $\frac{11}{6} \approx 1.833$. So of the gap $2 - \frac{3}{2} = \frac{1}{2}$, one third has been closed.

Large n . The theorem gives $\frac{n+1}{2}$ and the conjecture $\frac{n}{2}$: the gap is $\frac{1}{2}$, independent of dimension. So Falconer's theorem is, in absolute terms, always within $\frac{1}{2}$ of the truth — which is a much better state of affairs than for Kakeya, where the gap grows.

A consistency check. At $n = 1$ the theorem would read $\dim_{\mathbb{H}} E > 1$, which is impossible in $\mathbb{R}$; and indeed the distance set problem is trivial there, $\Delta(E)$ being essentially $E - E$. The curvature that drives the proof does not exist in one dimension, consistent with $\widehat{d\sigma}$ having no decay when $\mathbb{S}^0$ is two points.

WORKED EXAMPLE 8.5.2— TIER B **Where each hypothesis is used, and what happens without it**

Compactness of E entered twice: to apply Theorem 7.5.1, which is stated for compact sets, and to know $\Delta(E)$ is compact so that ν has compact support. For Borel E one applies the theorem to compact subsets of dimension close to $\dim_{\mathbb{H}} E$, which exist by inner regularity.

Non-vanishing curvature of the sphere entered once, in Theorem 8.4.7. Replacing the Euclidean distance by a norm whose unit ball has a flat face — the ℓ^∞ norm, say — removes the decay in the directions normal to the flat face, and the argument collapses. Indeed for ℓ^∞ the analogue of Theorem 8.4.2 is false: a product of Cantor sets of dimension slightly above $\frac{n}{2}$ can have ℓ^∞ -distance set of measure zero, because the ℓ^∞ distance depends on only one coordinate at a time.

The strict inequality $\dim_{\mathbb{H}} E > \frac{n+1}{2}$ is needed because Theorem 7.6.2(ii) produces a measure of finite s -energy only for $s < \dim_{\mathbb{H}} E$. At equality the energy may be infinite and the argument gives nothing; whether the theorem holds at $\dim_{\mathbb{H}} E = \frac{n+1}{2}$ is a separate question, and the analogous endpoint for the conjecture at $\frac{n}{2}$ is known to be false.

What was not used: any structure of E beyond its dimension. The proof is insensitive to whether E is self-similar, random, or arbitrary — and that

is precisely its limitation. The improvements of Chapter 8.7 all proceed by extracting more from the specific structure of $\widehat{\mu}$.

WORKED EXAMPLE 8.5.3— TIER C: RESEARCH TASTE **What the proof reveals about the shape of the problem**

(a) *The problem is a restriction problem in disguise.* The quantity controlling everything is $\sigma(\mu, r)$, the average of $|\widehat{\mu}|^2$ over the sphere of radius r . Bounding it is exactly a weighted restriction estimate for the sphere: one is asking how much of $\widehat{\mu}$ can concentrate near a sphere. Volume 6 studied precisely that, and the improvements in Chapter 8.7 import restriction technology directly.

(b) *The conjectured exponent $n/2$ is where dimension and codimension balance.* A measure of dimension s has $\widehat{\mu}$ decaying, in an L^2 -averaged sense, at rate $s/2$; the distance map has 1-dimensional fibres of codimension $n - 1$. Matching gives $s = n/2$, and this is the same balance that gives $\frac{n}{2}$ in the Fourier-dimension heuristic of Worked Example 7.7.2. The two problems share their critical exponent for a structural reason.

(c) *Why Falconer is more tractable than Keakeya.* The distance set problem is a *two-point* problem: $\Delta(E)$ is determined by the pair correlations of E , which is exactly what $I_s(\mu)$ and $\widehat{\mu}$ encode. By Worked Example 7.7.3, Fourier methods are the right tool for two-point problems and lose nothing structurally. Keakeya is not two-point — the smallness of a Besicovitch set is about how many tubes meet — which is why the Fourier method saturates there at $\frac{n+1}{2}$ and never improves, while here it has been pushed steadily downwards for forty years.

(d) *The current frontier.* Guth–Iosevich–Ou–Wang’s $\frac{5}{4}$ in the plane uses polynomial partitioning and the refined decoupling of Volumes 11 and 13. So the distance problem, having begun as a piece of geometric measure theory, is now solved by the same machinery as restriction — which is the best evidence that the two problems are one.

WORKED EXAMPLE 8.5.4— TIER B **Tracking the exponent when the surface changes**

The proof of Theorem 8.5.1 used one geometric input, the decay rate $\frac{n-1}{2}$ of $\widehat{d\sigma}$. Everything else was bookkeeping. So the exponent for a variant problem can be read off by changing that one number, and it is worth doing this three times.

(a) *Distances measured by a curved norm.* Let $\|\cdot\|_K$ be the norm whose unit ball is a symmetric convex body K with smooth boundary of non-vanishing

Gaussian curvature. Then $\widehat{d\sigma_{\partial K}}$ decays at rate $\frac{n-1}{2}$ by Theorem 5.5.2, Theorem 8.4.7 goes through verbatim, and the threshold is again $\frac{n+1}{2}$. Curvature, not roundness, is what the proof uses.

(b) *A surface with k non-vanishing curvatures.* If ∂K has only k non-zero principal curvatures then Theorem 5.5.3 gives decay $\frac{k}{2}$, and Step 3 of Theorem 8.5.1 requires $-\frac{k}{2} \leq s - n$, that is

$$s \geq n - \frac{k}{2}.$$

At $k = n - 1$ this is $\frac{n+1}{2}$; at $k = 0$ — a polyhedral ball — it is n , which is vacuous, consistent with the ℓ^∞ failure noted in Worked Example 8.5.2.

(c) *Dot products instead of distances.* Replace $|x - y|$ by $x \cdot y$ and the level sets become hyperplanes, which are flat: the relevant transform is $\widehat{d\sigma}$ for a plane, with no decay at all. The method gives nothing, and indeed the dot-product analogue of Theorem 8.4.2 has a different answer.

The recipe. Identify the level sets of the map whose image is being studied; compute the decay rate ρ of the Fourier transform of surface measure on them; the threshold is $n - \rho$. That single line predicts the exponent for every variant in the literature, and predicting it before reading the proof is a good way to check one has understood the argument.

A caution. The recipe gives the exponent obtainable by *this* method. Improvements come from the spherical averages of Chapter 8.7, and those do not follow the recipe — they depend on the finer structure of the surface and of μ .

Proposition 8.5.2 (A quantitative form). *Under the hypotheses of Theorem 8.5.1, with $s \in (\frac{n+1}{2}, \dim_{\mathbb{H}} E)$ and μ as in Step 1,*

$$m(\Delta(E) \cap [a, b]) \geq \frac{\nu([a, b])}{M}, \quad M \lesssim_{n,a,b} 1 + I_s(\mu).$$

Proof. By (8.5.1) and Step 4 of Theorem 8.5.1, $\nu \upharpoonright (a, b) \leq Mm$ with M the constant obtained there, which Step 3 bounded by a dimensional multiple of $1 + I_s(\mu)$. Applying this to the set $\Delta(E) \cap [a, b]$, which carries all of $\nu \upharpoonright [a, b]$ by Theorem 8.4.4,

$$\nu([a, b]) = \nu(\Delta(E) \cap [a, b]) \leq M m(\Delta(E) \cap [a, b]). \quad \square$$

Where we are. The quantitative form makes visible what the qualitative statement hides: the lower bound degrades as $I_s(\mu)$ grows, that is as s approaches $\dim_{\mathbb{H}} E$ or as the measure becomes more concentrated. So a set

of dimension barely above $\frac{n+1}{2}$ is guaranteed only a very small distance set, and the theorem gives no uniform bound over all sets of a given dimension. This is typical. Every existence statement in this volume has a quantitative form obtained by tracking constants, and the constants always involve an energy that blows up at the critical exponent. Recognising that pattern explains why endpoint statements in this subject are so often false rather than merely hard — Theorem 6.8.6 in Volume 6 was another instance.

Drills

Drill 8.5.1. Verify the inequality $(1 + |\xi|)^{-(n-1)/2} \leq |\xi|^{s-n}$ for $|\xi| > 1$ and $s \geq \frac{n+1}{2}$, and identify where equality is approached.

Drill 8.5.2. Show that $I_s(\mu) < \infty$ forces μ to be non-atomic, as used in Step 1.

Drill 8.5.3. Derive the bound $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$ quoted in the Pitfall, by writing $\sigma(\mu, r) = \iint \widehat{d\sigma}(r(x-y)) \, d\mu \, d\mu$ and splitting into dyadic annuli.

Drill 8.5.4. Confirm the claim in Worked Example 8.5.2 that for the ℓ^∞ norm the analogue of Theorem 8.4.7 fails, by computing the transform of the indicator of a cube shell in the coordinate directions.

Notes

Theorem 8.5.1 is Falconer's, from 1985. The proof above is his — an L^∞ estimate on the density of ν — and is the one Iosevich has popularised in the form given here, using Stein–Tomas-style input only through the decay of $\widehat{d\sigma}$.

Mattila's L^2 approach dates from 1987 and is the framework for every subsequent improvement: Bourgain's $\frac{13}{9}$ in the plane, Wolff's $\frac{4}{3}$, Erdoğan's $\frac{n}{2} + \frac{1}{3}$, and the current $\frac{5}{4}$ of Guth, Iosevich, Ou and Wang. Chapter 8.7 explains the framework and what each contributes.

Chapter 8.6

Interiors

Theorem 8.5.1 produces a distance set of positive measure. A little more work with the same estimates produces one with nonempty interior — so that $\Delta(E)$ contains a whole interval of distances, not merely a set of positive measure. The extra ingredient is continuity of the density, and it costs nothing.

Theorem 8.6.1 (Mattila–Sjölin). *Let $n \geq 2$ and $E \subseteq \mathbb{R}^n$ compact with $\dim_{\text{H}} E > \frac{n+1}{2}$. Then $\Delta(E)$ has nonempty interior.*

Idea of the proof. Theorem 8.5.1 showed the densities ν_ε are uniformly bounded. If in addition they converge *uniformly* on a compact interval, the limit is a continuous density for ν ; and a measure with a continuous density that is positive somewhere charges a whole neighbourhood of that point, which must therefore lie inside the support — that is, inside $\Delta(E)$.

Uniform convergence follows from the same integral. The difference $\nu_\varepsilon(t) - \nu_{\varepsilon'}(t)$ is controlled by the tail of $\int (1 + |\xi|)^{-(n-1)/2} |\widehat{\mu}|^2$, which converges; and the strict inequality $\dim_{\text{H}} E > \frac{n+1}{2}$ gives a little room, so the tail is small uniformly in t .

Proof. Step 1. Set up with room to spare.

Choose s with $\frac{n+1}{2} < s < \dim_{\text{H}} E$ and μ on E with $I_s(\mu) < \infty$, $\mu(E) = 1$, as in Theorem 8.5.1. Put $\eta = s - \frac{n+1}{2} > 0$. Repeating Step 3 of that proof but keeping the surplus,

$$\int_{|\xi| > R} (1 + |\xi|)^{-\frac{n-1}{2}} |\widehat{\mu}(\xi)|^2 d\xi \leq R^{-\eta} \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{s-n} d\xi \quad (8.6.1)$$

$$\lesssim R^{-\eta} I_s(\mu) \quad (8.6.2)$$

for $R \geq 1$, because on $|\xi| > R$ we have $(1 + |\xi|)^{-(n-1)/2} \leq |\xi|^{-(n-1)/2} = |\xi|^{s-n} |\xi|^{-\eta} \leq R^{-\eta} |\xi|^{s-n}$.

Step 2. *The densities converge uniformly.*

Fix $[a, b] \subseteq (0, \infty)$. For $t \in [a, b]$ and $0 < \varepsilon, \varepsilon' < a/2$, by Theorem 8.4.6,

$$\begin{aligned} \nu_\varepsilon(t) - \nu_{\varepsilon'}(t) &= \int_{\mathbb{R}^n} \left(\frac{\widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi)}{2\varepsilon} - \frac{\widehat{\mathbf{1}_{A_{\varepsilon'}^t}}(\xi)}{2\varepsilon'} \right) |\widehat{\mu}(\xi)|^2 d\xi. \end{aligned}$$

By (8.6.1) and Theorem 8.4.7, the contribution of $|\xi| > R$ to either term is $\lesssim R^{-\eta}$, uniformly in t and ε . On the compact region $|\xi| \leq R$ we may compute directly: from the polar formula in Theorem 8.4.7,

$$\begin{aligned} \frac{1}{2\varepsilon} \widehat{\mathbf{1}_{A_\varepsilon^t}}(\xi) &= \frac{1}{2\varepsilon} \int_{t-\varepsilon}^{t+\varepsilon} r^{n-1} \widehat{d\sigma}(r\xi) dr \longrightarrow t^{n-1} \widehat{d\sigma}(t\xi) \quad (\varepsilon \downarrow 0), \end{aligned}$$

uniformly for $t \in [a, b]$ and $|\xi| \leq R$, since the integrand is continuous and hence uniformly continuous on the relevant compact set. So ν_ε is uniformly Cauchy on $[a, b]$, with limit

$$\nu_0(t) = \int_{\mathbb{R}^n} t^{n-1} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi, \quad (8.6.3)$$

a continuous function of $t \in [a, b]$ — continuity following from the uniform convergence of continuous functions.

Step 3. ν_0 is a density for ν .

By Theorem 8.4.6, for $f \in C_c((a, b))$,

$$\int f d\nu = \lim_{\varepsilon \downarrow 0} \int_a^b \nu_\varepsilon(t) f(t) dt = \int_a^b \nu_0(t) f(t) dt,$$

the exchange justified by uniform convergence on the compact support of f . Hence $\nu \upharpoonright (a, b)$ has the continuous density ν_0 .

Step 4. *Conclude.*

Choose $[a, b]$ with $\nu((a, b)) > 0$, as in Theorem 8.5.1 Step 1. Then ν_0 is continuous on (a, b) and not identically zero, so $\nu_0(t_0) > 0$ for some t_0 , and by continuity $\nu_0 > 0$ on an interval $I \ni t_0$. Every nonempty open subinterval of I then has positive ν -measure, so I is contained in the support of ν ; and by Theorem 8.4.4 the support is contained in the closed set $\Delta(\text{spt } \mu) \subseteq \Delta(E)$. Hence $I \subseteq \Delta(E)$, and $\Delta(E)$ has nonempty interior. $\square$

Where we are. Formula (8.6.3) is worth recording on its own:

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi,$$

the density of the distance measure, expressed through the transform of surface measure. It is the rigorous form of the heuristic that Worked Example 8.4.2 warned against writing down directly, and the proof above is what licenses it.

Note that $\widehat{d\sigma}$ is real and *oscillates in sign* (Theorem 5.5.2), so the integrand is not of one sign and $\nu_0(t) > 0$ is not automatic — it had to be deduced in Step 4 from $\nu \neq 0$ and continuity. Any argument claiming positivity of ν_0 pointwise from the formula alone is mistaken.

WORKED EXAMPLE 8.6.1— TIER A **Positive measure and nonempty interior are different**

A closed set of positive measure need not have interior: the fat Cantor set of Volume 1 has positive measure and empty interior. So Theorem 8.6.1 is genuinely stronger than Theorem 8.5.1, and it is a little surprising that it costs only the continuity argument.

Why the same exponent suffices. Because the proof of Theorem 8.5.1 already produced a *uniform* bound on ν_ε , not merely an L^1 bound. Uniform bounds plus an equicontinuity statement give a continuous limit, and continuity is exactly what converts positive measure into interior.

Where more would be needed. If one only knew $\nu \in L^2$ — which is what the Mattila integral gives — the density need not be continuous and interior would not follow. So the two routes to Falconer’s theorem differ in their by-products: the L^∞ route gives interior for free, and the L^2 route gives better exponents but a weaker conclusion. This is the reason both remain in use.

WORKED EXAMPLE 8.6.2— TIER B **Pinned distance sets**

A refinement asks for a *single* $x \in E$ whose distance set $\Delta_x(E) = \{|x - y| : y \in E\}$ is large. Since $\Delta(E) = \bigcup_{x \in E} \Delta_x(E)$, this is strictly stronger.

What the present method gives. Run the proof of Theorem 8.5.1 with the measure $\mu \times \mu$ replaced by $\delta_x \times \mu$ for a fixed x . The Fourier step fails: $\widehat{\delta_x}$ does not decay, so there is no cancellation to exploit and the argument collapses. Pinning is genuinely harder.

What is true. Peres and Schlag proved a pinned analogue of Theorem 8.5.1 at the same exponent $\frac{n+1}{2}$, by a transversality argument in the spirit of Worked Example 8.2.2: one shows the set of bad pins x is small, rather than fixing x in advance. Liu later gave an L^2 approach, and the Guth–Iosevich–Ou–Wang theorem is stated in pinned form — their conclusion is that there *exists* $x \in E$ with $m(\Delta_x(E)) > 0$.

The pattern. As in Worked Example 8.1.3, one cannot prove the statement for a fixed parameter; one proves that the bad parameters form a small set. That recipe is the same one that converted Marstrand’s theorem into Kaufman’s, and it is worth noticing that it is the standard response whenever a “for some x ” statement is wanted and only an average is available.

WORKED EXAMPLE 8.6.3— TIER C: RESEARCH TASTE **What “pinning” costs, in three problems**

Worked Example 8.6.2 noted that fixing a single x breaks the Fourier argument. The same difficulty appears elsewhere, and the pattern is worth naming.

(a) *Distance sets.* The averaged statement bounds $\iint$ over pairs; the pinned statement fixes x and integrates over y only. The transform of δ_x does not decay, so all cancellation is lost. The resolution is to show the bad pins are few, not to fix a good one.

(b) *Projections.* Theorem 8.1.2 is an averaged statement over θ ; the corresponding pinned statement — that a *specified* θ is good — is false in general (Worked Example 8.1.1). Kaufman’s theorem is the systematic reply: bound the dimension of the bad set.

(c) *Keakeya.* Here the averaged statement is vacuous and the pinned one is everything: a Besicovitch set is a configuration in which *every* direction carries a segment, so there is nothing to average over. This is the structural reason Keakeya resists the methods of this volume, and it is worth stating as the sharpest available contrast between the two problems.

(d) *The recipe, once more.* In (a) and (b) one proves the averaged statement, then upgrades by running the average against a Frostman measure on the exceptional set (Worked Example 8.2.2). In (c) that route is closed. A reader meeting a new problem in this area should ask first whether the bad parameters can be all of them; if they can, no averaging argument will help.

WORKED EXAMPLE 8.6.4— TIER B **Equicontinuity: converting a uniform bound into a continuous density**

The step that upgrades Theorem 8.5.1 to Theorem 8.6.1 is standard analysis, and it is worth isolating because it applies whenever a density has been produced by mollification.

The situation. One has functions ν_ε on $[a, b]$ with $\int f \nu_\varepsilon \rightarrow \int f d\nu$ and $|\nu_\varepsilon| \leq M$ uniformly. That alone gives $\nu \leq Mm$, but not continuity: uniform boundedness permits wild oscillation.

What is needed in addition. Either equicontinuity — a modulus of continuity uniform in ε — or, as in Theorem 8.6.1, uniform convergence. The latter is stronger and is what Step 2 there established, by splitting the frequency integral:

- on $|\xi| > R$, the tail is $\lesssim R^{-\eta}$ uniformly in t and ε , by the surplus (8.6.1);
- on $|\xi| \leq R$, the integrand converges uniformly because it is continuous on a compact set.

Together these give a uniform Cauchy property, hence a continuous limit.

Why the surplus is essential. The estimate (8.6.1) needs $\eta = s - \frac{n+1}{2} > 0$, that is the *strict* inequality $\dim_{\mathbb{H}} E > \frac{n+1}{2}$. At equality one would have a uniform bound with no decay in R , hence no uniform convergence, hence no continuity — and the interior conclusion would be lost while the positive-measure conclusion survived. The two theorems really do need different hypotheses, even though both are stated with the same strict inequality.

The general pattern. A uniform bound gives absolute continuity; a uniform bound with a quantitative tail gives a continuous density. The second costs an epsilon of room in the exponent and buys a qualitatively stronger conclusion. This trade recurs: in Volume 4 the Besov endpoint $B_{p,1}^{n/p} \subseteq L^\infty$ bought continuity for the same reason, by requiring summability rather than square-summability over dyadic blocks.

Proposition 8.6.2 (The density is bounded below on a subinterval). *Under the hypotheses of Theorem 8.6.1, there are $0 < a < b < \infty$, an interval $I \subseteq (a, b)$ and $c > 0$ with*

$$\nu_0(t) \geq c \quad \text{for all } t \in I,$$

and consequently $I \subseteq \Delta(E)$ with $m(\Delta(E)) \geq |I|$.

Proof. By Step 2 of Theorem 8.6.1, ν_0 is continuous on $[a, b]$ for any $0 < a < b < \infty$, and by Step 3 it is a density for ν there. Choose $[a, b]$ with $\nu((a, b)) > 0$, possible as in Theorem 8.5.1 Step 1.

If $\nu_0 \leq 0$ on all of (a, b) then $\nu((a, b)) = \int_a^b \nu_0 \leq 0$, contradicting the choice. So $\nu_0(t_0) > 0$ for some $t_0 \in (a, b)$; by continuity there are $c = \frac{1}{2}\nu_0(t_0) > 0$ and an interval $I \ni t_0$ with $\nu_0 \geq c$ on I .

Every nonempty open $J \subseteq I$ then has $\nu(J) = \int_J \nu_0 \geq c|J| > 0$, so J meets $\text{spt } \nu$. Hence $I \subseteq \text{spt } \nu$, and by Theorem 8.4.4 $\text{spt } \nu \subseteq \Delta(\text{spt } \mu) \subseteq \Delta(E)$, which is closed. The measure bound is immediate. $\square$

WORKED EXAMPLE 8.6.5— TIER A **Why $\nu_0 \geq 0$ is not obvious**

One might expect $\nu_0 \geq 0$ automatically, ν being a positive measure. It is true — ν_0 is a density for a positive measure, hence nonnegative almost everywhere, hence everywhere by continuity — but it does *not* follow from the formula

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\widehat{\mu}(\xi)|^2 d\xi.$$

Here $|\widehat{\mu}|^2 \geq 0$ but $\widehat{d\sigma}$ oscillates in sign, by Theorem 5.5.2: it is asymptotically a cosine. So the integrand takes both signs and no term-by-term argument gives positivity.

The correct order of reasoning. Positivity comes from ν being a measure, established before any Fourier computation; the formula is then used only for *upper* bounds. Reversing the order — trying to read positivity off the formula — fails, and an argument that appears to do so should be checked. *Where this bites.* Theorem 8.6.2 needed $\nu_0(t_0) > 0$ somewhere, and obtained it from $\nu((a, b)) > 0$, not from the integral. That is the only available route.

WORKED EXAMPLE 8.6.6— TIER C: RESEARCH TASTE **What “nonempty interior” buys, and where it is used**

Theorem 8.6.1 is stronger than Theorem 8.5.1, and the extra strength is used, so it is worth recording where.

(a) *It makes the conclusion checkable.* “Positive measure” is not constructive: one cannot exhibit a distance. “Contains an interval” names a whole range of distances realised by pairs of points of E , and Theorem 8.6.2 even locates it as a neighbourhood of any t_0 where the density is positive.

(b) *It iterates.* If $\Delta(E) \supseteq I$ then, applying the theorem to $E \times E$ or to pinned versions, one obtains statements about configurations — triangles with prescribed side lengths, and so on. This is the starting point of the literature on *finite point configurations* in fractal sets, where the question is which similarity classes of k -point configurations occur. Positive measure would not suffice there, since one must intersect finitely many conclusions.

(c) *It fails for the L^2 method.* As Worked Example 8.7.2 recorded, Mattila's framework gives $\nu \in L^2$, and an L^2 density need not be continuous. So the improvements of Chapter 8.7 give better exponents for positive measure and say nothing about interior. At the time of writing the best *interior* result remains at $\frac{n+1}{2}$ — that is, Theorem 8.6.1 is still the record for its own conclusion, forty years on, even though the positive-measure exponent has moved substantially.

(d) *The moral.* A stronger conclusion may have a worse exponent, and the two should be tabulated separately. Worked Example 8.8.4 listed this as the first of its four questions for a reason: comparing a positive-measure exponent with an interior exponent is comparing different theorems.

Drills

Drill 8.6.1. Verify (8.6.1), being careful with the case $|\xi| \leq 1$.

Drill 8.6.2. Show that the convergence in Step 2 is uniform on $\{t \in [a, b], |\xi| \leq R\}$, using uniform continuity of $(r, \xi) \mapsto r^{n-1} \widehat{d\sigma}(r\xi)$ on a compact set.

Drill 8.6.3. Give an example of a closed set of positive Lebesgue measure with empty interior, and explain why Theorem 8.6.1 is therefore strictly stronger than Theorem 8.5.1.

Drill 8.6.4. Show directly from (8.6.3) that ν_0 is bounded, recovering (8.5.1) in the limit.

Notes

Theorem 8.6.1 is due to Mattila and Sjölin (1999). The observation that the interior comes free from an L^∞ bound plus equicontinuity is theirs; Iosevich, Mourgoglou and Taylor extended it to distances measured by norms whose unit ball has non-vanishing curvature, where the same proof applies with Theorem 5.5.2 replaced by its analogue for that surface.

Chapter 8.7

The Mattila Integral

Every improvement on Falconer's exponent since 1987 has gone through a single reformulation, due to Mattila, which replaces the L^∞ estimate of Chapter 8.5 by an L^2 one. This chapter proves the reduction and explains what each subsequent advance contributed. It proves none of those advances: they are research papers, and the reduction is what makes them readable.

8.7.1 The reduction

Definition 8.7.1. For a finite compactly supported Borel measure μ on $\mathbb{R}^n$ and $r > 0$, the *spherical average* of $|\hat{\mu}|^2$ is

$$\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\hat{\mu}(r\omega)|^2 d\sigma(\omega),$$

and the *Mattila integral* is

$$\mathcal{M}(\mu) = \int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr.$$

Theorem 8.7.2 (Mattila's reduction). *Let μ be a finite compactly supported Borel measure on $\mathbb{R}^n$ with $\mathcal{M}(\mu) < \infty$. Then the distance measure ν has an L^2 density on every interval $[a, b] \subseteq (0, \infty)$; in particular $\nu \ll m$ and $m(\Delta(\text{spt } \mu)) > 0$.*

Idea of the proof. By Theorem 8.4.6 the approximate density is $\nu_\varepsilon(t) = \frac{1}{2\varepsilon} \int \widehat{\mathbf{1}_{A_\varepsilon^t}} |\hat{\mu}|^2$, and the polar computation of Theorem 8.4.7 identifies its limit as

$$\nu_0(t) = t^{n-1} \int_{\mathbb{R}^n} \widehat{d\sigma}(t\xi) |\hat{\mu}(\xi)|^2 d\xi.$$

Rather than bounding this pointwise — which is what Chapter 8.5 did, and which costs the full pointwise decay of $\widehat{d\sigma}$ — one estimates its L^2 norm in t .

Writing the ξ -integral in polar coordinates and using that $\widehat{d\sigma}(t\xi)$ depends on $t|\xi|$ only, the t -integral becomes an integral against the *spherical averages* of $|\widehat{\mu}|^2$, and Cauchy–Schwarz produces $\mathcal{M}(\mu)$.

The gain is that $\sigma(\mu, r)$ may be much smaller than the pointwise bound suggests: it is an average, and cancellation between directions is permitted. Every improvement on Falconer’s exponent is an improved bound for $\sigma(\mu, r)$.

Proof. Step 1. Polar form of the density.

By (8.6.3) and polar coordinates (Theorem 1.6.11) in the ξ -integral, using that $\widehat{d\sigma}$ is radial so that $\widehat{d\sigma}(t\xi)$ depends only on $t|\xi|$,

$$\nu_0(t) = t^{n-1} \int_0^\infty \widehat{d\sigma}(tr) \sigma(\mu, r) r^{n-1} dr.$$

The part $r \leq 1$ contributes a function of t bounded uniformly on $[a, b]$, since $\widehat{d\sigma}$ and $\widehat{\mu}$ are bounded and the region has finite measure; call it ν_0^{loc} and set $P(t) = \nu_0(t) - \nu_0^{\text{loc}}(t)$.

Step 2. *Insert the asymptotic, keeping the oscillation.*

By Theorem 5.5.2 there are constants $c_\pm$ with

$$\begin{aligned} \widehat{d\sigma}(\rho) &= \rho^{-\frac{n-1}{2}} \left(c_+ e^{2\pi i \rho} + c_- e^{-2\pi i \rho} \right) \\ &\quad + O(\rho^{-\frac{n+1}{2}}) \quad (\rho \geq 1). \end{aligned}$$

Substituting $\rho = tr$ and writing

$$G(r) = \sigma(\mu, r) r^{\frac{n-1}{2}} \mathbf{1}_{\{r>1\}},$$

so that $\sigma(\mu, r) r^{n-1} \cdot r^{-\frac{n-1}{2}} = G(r)$, we obtain

$$P(t) = t^{\frac{n-1}{2}} \left(c_+ \int_0^\infty G(r) e^{2\pi i tr} dr + c_- \int_0^\infty G(r) e^{-2\pi i tr} dr \right) + R(t), \quad (8.7.1)$$

where

$$\begin{aligned} |R(t)| &\lesssim t^{n-1} t^{-\frac{n+1}{2}} \int_1^\infty \sigma(\mu, r) r^{n-1-\frac{n+1}{2}} dr \\ &= t^{\frac{n-3}{2}} \int_1^\infty \sigma(\mu, r) r^{\frac{n-3}{2}} dr. \end{aligned}$$

Step 3. *The main term, by Plancherel.*

The two integrals in (8.7.1) are $\widehat{G}(-t)$ and $\widehat{G}(t)$. Note

$$\|G\|_{L^2(\mathbb{R})}^2 = \int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr = \mathcal{M}(\mu) < \infty,$$

so $G \in L^2$ and, by Plancherel (Theorem 2.4.8), $\widehat{G} \in L^2(\mathbb{R})$ with the same norm. Therefore, since $t^{(n-1)/2} \leq b^{(n-1)/2}$ on $[a, b]$,

$$\begin{aligned} \int_a^b \left| t^{\frac{n-1}{2}} (c_+ \widehat{G}(-t) + c_- \widehat{G}(t)) \right|^2 dt &\lesssim_{n,b} \|\widehat{G}\|_{L^2(\mathbb{R})}^2 \\ &= \mathcal{M}(\mu). \end{aligned}$$

Step 4. *The error term, by Cauchy–Schwarz.*

Here — and *only* here — Cauchy–Schwarz is available, because the exponent is $\frac{n-3}{2}$, two powers below the main term:

$$\begin{aligned} \int_1^\infty \sigma(\mu, r) r^{\frac{n-3}{2}} dr &\leq \left(\int_1^\infty \sigma(\mu, r)^2 r^{n-1} dr \right)^{1/2} \left(\int_1^\infty r^{(n-3)-(n-1)} dr \right)^{1/2} \\ &= \mathcal{M}(\mu)^{1/2}, \end{aligned}$$

the second factor being $(\int_1^\infty r^{-2} dr)^{1/2} = 1$. Hence $|R(t)| \lesssim_{n,b} \mathcal{M}(\mu)^{1/2}$ uniformly on $[a, b]$.

Step 5. *Conclude.*

Combining Steps 1, 3 and 4, $\nu_0 \in L^2([a, b])$ with $\|\nu_0\|_{L^2([a, b])} \lesssim_{n,a,b} \mathcal{M}(\mu)^{1/2} + 1$. By the argument of Theorem 8.5.1 Step 4 — which used only that the approximating densities converge against test functions — ν_0 is a density for $\nu \upharpoonright (a, b)$, so $\nu \ll m$ there, and Theorem 8.4.4 gives $m(\Delta(\text{spt } \mu)) > 0$. $\square$

Pitfall: Cauchy–Schwarz does not bound the main term

It is tempting to skip Step 3 and estimate the whole of $P(t)$ by Cauchy–Schwarz, as in Step 4. That fails, and the failure is instructive.

Bounding $|\widehat{d\sigma}(tr)| \lesssim (tr)^{-(n-1)/2}$ pointwise and applying Cauchy–Schwarz would require

$$\int_1^\infty \sigma(\mu, r) r^{\frac{n-1}{2}} dr \lesssim \mathcal{M}(\mu)^{1/2},$$

and the pairing that would give this needs $(\int_1^\infty r^{(n-1)-(n-1)} dr)^{1/2} = (\int_1^\infty dr)^{1/2} = \infty$. There is no admissible weight: the exponents do not leave room.

What the oscillation buys is exactly the missing power. Because $\widehat{d\sigma}$ oscillates rather than merely decaying, the r -integral is a Fourier transform and Plancherel replaces the divergent $\int dr$ by an identity. So the proof *requires* the asymptotic of Theorem 5.5.2 and not merely the bound of Theorem 2.8.5 — a concrete instance of the point made in Worked Example 5.3.2 that discarding the phase loses information one later needs.

Remark 8.7.3 (What the reduction is worth, and what it is not). Theorem 8.7.2 converts the distance problem into the problem of bounding $\sigma(\mu, r)$. That is progress only if σ can be bounded better than the pointwise decay allows — and it can, but not by anything in this series.

Bounding $\sigma(\mu, r)$ is a *weighted restriction estimate*: it asks how much of $\widehat{\mu}$ concentrates on the sphere of radius r , which is exactly the question Chapter 6.3 studied with μ replaced by an arbitrary L^2 function. The improvements listed in Worked Example 8.7.1 are, each of them, an improved spherical average estimate obtained by importing restriction technology.

The reduction alone proves nothing new. As Chapter 8.5’s Pitfall recorded, feeding the pointwise bound $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$ into $\mathcal{M}$ gives a divergent integral for every s and every n . So Theorem 8.7.2 is a *framework*, not a theorem about dimensions, and the exponent $\frac{n+1}{2}$ of Theorem 8.5.1 is not recoverable from it.

WORKED EXAMPLE 8.7.1— TIER B **Reading the record**

The successive improvements are all of one kind, and knowing which is which makes the literature navigable.

Bound	Author	Dimension	New input
$\frac{n+1}{2}$	Falconer 1985	all n	pointwise decay of $\widehat{d\sigma}$
$\frac{13}{9}$	Bourgain 1994	$n = 2$	first use of restriction technology
$\frac{4}{3}$	Wolff 1999	$n = 2$	sharp local smoothing for the circle
$\frac{n}{2} + \frac{1}{3}$	Erdoğan 2005	$n \geq 3$	Tao’s bilinear restriction
$\frac{5}{4}$	Guth–Iosevich–Ou–Wang 2020	$n = 2$	polynomial partitioning, refined decoupling

What the table shows. Every entry after the first is an improved bound for $\sigma(\mu, r)$, and the new input in each case is the best restriction technology of its decade. So the distance problem has served as a proving ground for restriction theory: an advance in Volume 12’s or Volume 13’s material translates immediately into a distance-set exponent.

Why $n = 2$ leads. Because the restriction theory is complete there (Theorem 6.8.2) and the decoupling theory is sharpest there. In higher dimensions the inputs are weaker, and the exponents correspondingly further from $\frac{n}{2}$.

Where the conjecture would come from. A bound $\sigma(\mu, r) \lesssim_{\varepsilon} r^{-n/2+\varepsilon}$ for measures of dimension $s > \frac{n}{2}$ would give $\mathcal{M}(\mu) < \infty$ and hence Theorem 8.4.2

(Drill 7.3). That is a sharp weighted restriction estimate, and it is open — but it is the same kind of statement as the restriction conjecture, not a different problem.

WORKED EXAMPLE 8.7.2— TIER C: RESEARCH TASTE **Two frameworks, and when to use which**

This volume has proved Falconer’s theorem twice over in effect — once by L^∞ and once by setting up the L^2 framework that supersedes it. It is worth recording the trade, since the same choice recurs.

(a) *The L^∞ route (Chapter 8.5).* Bounds the density pointwise. Cheap, uses only the decay of $\widehat{d\sigma}$, and yields Theorem 8.6.1 — nonempty interior — as a by-product, because a uniform bound plus equicontinuity gives a continuous density. Its ceiling is $\frac{n+1}{2}$ and cannot be lowered, because the pointwise decay of $\widehat{d\sigma}$ is sharp.

(b) *The L^2 route (Theorem 8.7.2).* Bounds the density in L^2 . Gives no interior, and gives nothing at all without an external input; but the input is a restriction estimate, and restriction estimates improve. Every advance since 1987 is here.

(c) *The general principle.* An L^∞ bound is stronger where it holds and is limited by the worst pointwise behaviour of the kernel; an L^2 bound tolerates bad points if they are rare, and so can exploit cancellation. When a kernel’s pointwise decay is sharp but its averages are better, the L^2 formulation is the one that improves. That is exactly the situation for $\widehat{d\sigma}$: its pointwise decay $|\xi|^{-(n-1)/2}$ is attained (Theorem 5.4.1), but its spherical averages against a fractal measure are conjecturally much smaller.

(d) *The same choice elsewhere in the series.* Volume 3’s Calderón–Zygmund theorem is an L^∞ -type argument on the kernel; Volume 6’s Stein–Tomas is an L^2 -type argument on its dyadic pieces, and gets a better exponent for exactly the reason (c) gives. Recognising which regime one is in, before choosing a method, is most of the craft.

WORKED EXAMPLE 8.7.3— TIER A **The spherical average for a measure one can compute**

Take $\mu = d\sigma$ itself, surface measure on $\mathbb{S}^{n-1}$, and compute $\sigma(\mu, r)$ directly.

By Theorem 5.5.2, $|\widehat{d\sigma}(r\omega)|^2 \approx r^{-(n-1)} \cos^2(2\pi r - \frac{(n-1)\pi}{4} - \frac{\pi}{4})$ for large r , independently of ω since $\widehat{d\sigma}$ is radial. So

$$\sigma(d\sigma, r) = \int_{\mathbb{S}^{n-1}} |\widehat{d\sigma}(r\omega)|^2 d\sigma(\omega) \approx \sigma(\mathbb{S}^{n-1}) r^{-(n-1)} \cos^2(\dots),$$

which is $\approx r^{-(n-1)}$ on average in r .

Comparison with the pointwise bound. The general estimate $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$ gives, with $s = \dim_{\mathbb{H}} \mathbb{S}^{n-1} = n-1$, the exponent $\min\{n-1, \frac{n-1}{2}\} = \frac{n-1}{2}$. The truth is $n-1$, twice as good.

What this shows. For a Salem set — and the sphere is one, by Worked Example 7.7.1 — the spherical average is much smaller than the general bound allows, because $|\widehat{\mu}|$ is genuinely of size $r^{-(n-1)/2}$ at *every* point rather than merely on average. The whole difficulty of the distance problem is that fractal measures of dimension just above $\frac{n}{2}$ need not be Salem, and for them nothing better than the general bound is known.

A check on $\mathcal{M}$. With $\sigma(d\sigma, r) \approx r^{-(n-1)}$,

$$\mathcal{M}(d\sigma) \approx \int_1^\infty r^{-2(n-1)} r^{n-1} dr = \int_1^\infty r^{-(n-1)} dr,$$

finite exactly when $n \geq 3$. So the Mattila integral converges for the sphere in dimension three and above — consistent with $\Delta(\mathbb{S}^{n-1}) = [0, 2]$, which certainly has positive measure.

WORKED EXAMPLE 8.7.4— TIER B **Reading a spherical average estimate**

Every advance on Falconer's exponent is a bound for $\sigma(\mu, r)$, so a reader of this literature must be able to convert such a bound into an exponent. The conversion is one line, and here it is.

The conversion. Suppose one knows

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\beta+\varepsilon}$$

for every measure with $\mu(B(x, r)) \leq r^s$. Then

$$\mathcal{M}(\mu) \lesssim \int_1^\infty r^{-2\beta+2\varepsilon} r^{n-1} dr,$$

which converges when $2\beta > n$, that is $\beta > \frac{n}{2}$. So by Theorem 8.7.2 the distance set has positive measure as soon as the spherical average decays faster than $r^{-n/2}$.

Applying it to the known bounds.

- The trivial bound is $\beta = \min\{s, \frac{n-1}{2}\}$, which never exceeds $\frac{n-1}{2} < \frac{n}{2}$. Hence the framework alone proves nothing, as Theorem 8.7.3 recorded.
- Wolff's planar result corresponds to $\beta > 1 = \frac{n}{2}$ at $n = 2$ for $s > \frac{4}{3}$; the improvement over the trivial $\beta = \frac{1}{2}$ is exactly the content of his local smoothing estimate.
- The conjecture corresponds to $\beta > \frac{n}{2}$ for every $s > \frac{n}{2}$.

The dimension-dependence to watch. A spherical average bound is stated for measures of a given Frostman exponent s , and β depends on s . A paper proving “ $\sigma(\mu, r) \lesssim r^{-\beta(s)}$ ” therefore yields the distance-set threshold $\inf\{s : \beta(s) > \frac{n}{2}\}$, and comparing two papers means comparing two functions β , not two numbers.

The exercise worth doing. Take the trivial bound $\beta(s) = \min\{s, \frac{n-1}{2}\}$, plot it against s , and mark the line $\beta = \frac{n}{2}$. The two never meet — that is the picture of why Falconer's theorem cannot be recovered from $\mathcal{M}$, and it is the single most useful diagram to have in mind when reading this literature.

Proposition 8.7.4 (The trivial spherical average bound). *Let μ be a finite compactly supported measure with $\mu(B(x, r)) \leq r^s$ for all x, r . Then for $r \geq 1$,*

$$\sigma(\mu, r) \lesssim_{n,s} r^{-\min\{s, \frac{n-1}{2}\}}.$$

Proof. Expanding the square and using Fubini,

$$\begin{aligned} \sigma(\mu, r) &= \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega) \\ &= \iint \left(\int_{\mathbb{S}^{n-1}} e^{-2\pi i r \omega \cdot (x-y)} d\sigma(\omega) \right) d\mu(x) d\mu(y) \\ &= \iint \widehat{d\sigma}(r(x-y)) d\mu d\mu. \end{aligned}$$

By Theorem 5.5.2, $|\widehat{d\sigma}(z)| \lesssim_n (1 + |z|)^{-(n-1)/2}$. Split the pair integral by the dyadic size of $|x - y|$: the piece with $|x - y| \leq r^{-1}$ contributes at most $\mu \times \mu\{|x - y| \leq r^{-1}\} \leq r^{-s}$ by the Frostman bound; and for $|x - y| \approx 2^j r^{-1}$ with $j \geq 0$, the kernel is $\lesssim 2^{-j(n-1)/2}$ and the mass is $\lesssim (2^j r^{-1})^s$, so that piece contributes

$$\sum_{0 \leq j \lesssim \log r} 2^{-j \frac{n-1}{2}} (2^j r^{-1})^s = r^{-s} \sum_j 2^{j(s - \frac{n-1}{2})}.$$

If $s < \frac{n-1}{2}$ the sum converges and the total is $\lesssim r^{-s}$; if $s > \frac{n-1}{2}$ it is dominated by its largest term, at $2^j \approx r$, giving $r^{-s} r^{s - \frac{n-1}{2}} = r^{-\frac{n-1}{2}}$. The borderline case

contributes a logarithm, absorbed into the implied constant after decreasing s slightly. $\square$

Where we are. Theorem 8.7.4 is the estimate quoted in the Pitfall of Chapter 8.5 and in Theorem 8.7.3, now proved. It makes the obstruction explicit: the exponent never exceeds $\frac{n-1}{2}$, whatever s , because the pointwise decay of $\widehat{d\sigma}$ caps it — and Worked Example 8.7.4 showed that $\mathcal{M}$ needs an exponent above $\frac{n}{2}$.

The gap is exactly $\frac{1}{2}$, and closing any part of it requires an estimate that does not factor through the pointwise bound. Every improvement in Worked Example 8.7.1 does exactly that, by exploiting that μ is a *fractal* measure rather than an arbitrary one.

WORKED EXAMPLE 8.7.5— TIER A Checking the framework on a case where everything is known

A framework is worth testing on an example where the answer is independently available. Take $n \geq 3$ and $\mu = d\sigma$ on $\mathbb{S}^{n-1}$.

The independent answer. $\Delta(\mathbb{S}^{n-1}) = [0, 2]$ — every distance between 0 and the diameter is realised — so certainly $m(\Delta) > 0$, and indeed the distance set has interior.

What the framework gives. By Worked Example 8.7.3, $\sigma(d\sigma, r) \approx r^{-(n-1)}$, so

$$\mathcal{M}(d\sigma) \approx \int_1^\infty r^{-2(n-1)} r^{n-1} dr = \int_1^\infty r^{-(n-1)} dr,$$

finite for $n \geq 3$. So Theorem 8.7.2 applies and confirms $m(\Delta) > 0$. $\checkmark$

Why $n = 2$ is excluded, and what that means. At $n = 2$ the integral is $\int_1^\infty r^{-1} dr$, logarithmically divergent, so the framework fails for the circle — even though the conclusion is obviously true. That is not a defect of the circle but a reminder that $\mathcal{M}(\mu) < \infty$ is *sufficient and not necessary*. A reader should not conclude from $\mathcal{M} = \infty$ that a distance set is small.

Comparing with the trivial bound. Theorem 8.7.4 with $s = n-1$ gives exponent $\min\{n-1, \frac{n-1}{2}\} = \frac{n-1}{2}$, whereas the truth is $n-1$. So for this measure the trivial bound loses a factor of two in the exponent — exactly the loss that Worked Example 8.7.3 attributed to the sphere being Salem. Fractal measures of dimension near $\frac{n}{2}$ are the hard case precisely because they need not be.

Figure 8.7.1 (to be drawn). *Why the Mattila integral diverges*

Content required: The exponent $\beta(s) = \min\{s, \frac{n-1}{2}\}$ plotted against $s \in [0, n]$: a line of slope one that flattens at height $\frac{n-1}{2}$. Draw the horizontal line $\beta = \frac{n}{2}$ above it. The two never meet, and that gap is exactly why $\mathcal{M}$ cannot recover Theorem 8.5.1.

Drills

Drill 8.7.1. Show that no choice of weight makes Cauchy–Schwarz bound $\int_1^\infty \sigma(\mu, r) r^{(n-1)/2} dr$ by $\mathcal{M}(\mu)^{1/2}$: pair with r^a and r^{-a} and observe that matching the first factor to $\mathcal{M}$ forces $a = 0$ and the second factor to diverge.

Drill 8.7.2. Verify that $\widehat{d\sigma}$ is radial, so that $\widehat{d\sigma}(t\xi)$ depends only on $t|\xi|$, as used in Step 1.

Drill 8.7.3. Show that a bound $\sigma(\mu, r) \lesssim_\varepsilon r^{-n/2+\varepsilon}$ for all measures of dimension $s > \frac{n}{2}$ would imply Theorem 8.4.2.

Drill 8.7.4. Compute $\sigma(d\sigma, r)$ for $\mu = d\sigma$ itself, using Theorem 5.4.1, and check it against the pointwise bound.

Notes

Mattila’s reduction is from 1987. The improvements are Bourgain (1994), Wolff (1999), Erdoğan (2005), and Guth–Iosevich–Ou–Wang (2020), with the higher-dimensional analogue of the last due to Du, Iosevich, Ou, Wang and Zhang. None is proved here; Worked Example 8.7.1 records what each contributes.

The pinned versions, mentioned in Worked Example 8.6.2, are due to Peres–Schlag and to Liu, and the current record is stated in pinned form.

Chapter 8.8

The $n/2$ Barrier

8.8.1 Where the exponents come from

Three exponents have appeared: $\frac{n+1}{2}$, proved; $\frac{n}{2}$, conjectured; and $\frac{n-1}{2}$, the decay rate of $\widehat{d\sigma}$. This closing chapter explains their relations and states what would be needed to close the gap.

Remark 8.8.1 (Sharpness of the conjectured exponent, and a scope note). For every $n \geq 2$ there are compact $E \subseteq \mathbb{R}^n$ with $\dim_{\mathbb{H}} E = \frac{n}{2}$ and $m(\Delta(E)) = 0$, so the exponent in Theorem 8.4.2 cannot be lowered. This is Falconer's, from the paper posing the problem, and *we do not prove it here* — the construction of the underlying Cantor set is a delicate multi-scale argument and nothing in this series uses the example.

The *mechanism*, however, is elementary and worth recording, since it explains what an extremal set must look like.

Suppose $A \subseteq [0, 1]$ were a Cantor set with $\dim_{\mathbb{H}} A = \frac{1}{2}$ which, at infinitely many scales N , is contained in $\approx N^{1/2}$ intervals of length N^{-1} with left endpoints in $\frac{1}{N}\mathbb{Z}$. Put $E = A^n$, so $\dim_{\mathbb{H}} E = \frac{n}{2}$ by Theorem 7.6.4 and Theorem 7.8.3. At such a scale every coordinate difference of points of E lies within N^{-1} of $\frac{1}{N}\mathbb{Z}$, so

$$|x - y|^2 = \sum_{i=1}^n (x_i - y_i)^2$$

lies within $O(N^{-1})$ of $\frac{1}{N^2}\mathbb{Z}$. Hence $\Delta(E)^2$ sits in an $O(N^{-1})$ -neighbourhood of $\frac{1}{N^2}\mathbb{Z} \cap [0, n]$, which consists of $O(nN^2)$ intervals of length $O(N^{-1})$ and so has measure $O(nN^2 \cdot N^{-1} \cdot N^{-2}) = O(N^{-1})$. Letting $N \rightarrow \infty$ along the chosen scales gives $m(\Delta(E)^2) = 0$ and hence $m(\Delta(E)) = 0$.

What is not established here is the existence of A : one must build a set with the right dimension *and* the lattice property at infinitely many scales,

and the two requirements pull against each other. Falconer's construction alternates between them along a rapidly increasing sequence of scales.

Where we are. The example is arithmetic, not geometric: it works because the distances are forced onto a lattice, and lattices have small neighbourhoods. That is the same mechanism that made $\dim_{\mathbb{F}} C = 0$ in Theorem 7.7.4 — exact self-similarity with an integer ratio creates arithmetic structure.

So the extremal example for Falconer's conjecture is a set whose Fourier transform does not decay at all, and the conjecture asserts that this is the only obstruction. Every improvement listed in Worked Example 8.7.1 proceeds by showing that a measure with *some* decay does better, and the difficulty is that fractal measures of dimension just above $\frac{n}{2}$ may have almost no decay.

8.8.2 The gap, stated precisely

WORKED EXAMPLE 8.8.1— TIER B **What exactly is missing**

By Theorem 8.7.2 and Drill 7.3, Theorem 8.4.2 would follow from:

for every $s > \frac{n}{2}$ and every measure μ with $\mu(B(x, r)) \leq r^s$, one has $\sigma(\mu, r) \lesssim_{\varepsilon} r^{-n/2+\varepsilon}$.

What is known towards it. The trivial bound from Theorem 5.5.2 is $\sigma(\mu, r) \lesssim r^{-\min\{s, (n-1)/2\}}$, which saturates at $\frac{n-1}{2}$ — short of $\frac{n}{2}$ by exactly $\frac{1}{2}$. Every improvement narrows that half.

Why it is a restriction problem. $\sigma(\mu, r)$ measures how much of $\hat{\mu}$ concentrates on a sphere of radius r ; equivalently, by duality, how large $\mathcal{E}g$ can be on the support of μ . So the required estimate is a *weighted* restriction estimate, with the weight a fractal measure — and that is the setting of Volume 12's later sections.

The scoreboard, in one line. Falconer's theorem uses $\sigma \lesssim r^{-(n-1)/2}$; the conjecture needs $\sigma \lesssim r^{-n/2}$; and the record has moved the exponent about a third of the way in the plane and less in higher dimensions.

WORKED EXAMPLE 8.8.2— TIER C: RESEARCH TASTE **Volume 8 in relation to the rest**

(a) *What was proved.* Marstrand’s projection theorem and Kaufman’s exceptional-set bound (Chapter 8.1, Chapter 8.2); Falconer’s distance theorem at $\frac{n+1}{2}$ (Chapter 8.5) and its interior refinement (Chapter 8.6); and Mattila’s reduction (Chapter 8.7).

(b) *What was decided against.* The measure-zero Besicovitch set, removed as revision R8 with the obstruction named (Theorem 8.3.3); and the four improvements on Falconer’s exponent, each a research paper, stated with their inputs in Worked Example 8.7.1.

(c) *The two lessons that transfer.* First, the averaging recipe of Worked Example 8.2.2: when a statement cannot be proved for every parameter, average, and then upgrade “almost every” to a dimension bound by running the average against a Frostman measure. Second, the L^∞ versus L^2 choice of Worked Example 8.7.2: pointwise bounds are limited by the worst behaviour of the kernel, averaged bounds can exploit cancellation, and the second is what improves.

(d) *The connection to Kakeya, restated.* Both problems ask how small a set with prescribed richness can be. The distance problem is two-point, so Fourier methods are structurally adequate and the exponent has moved steadily for forty years. Kakeya is not, and the Fourier method has been stuck at $\frac{n+1}{2}$ since 1983 (Theorem 7.7.5). The next volume abandons Fourier analysis for tube combinatorics, and that is why.

(e) *A last observation about method.* Everything in this volume followed from two facts proved much earlier: that $\widehat{d\sigma}$ decays at rate $\frac{n-1}{2}$ (Volume 5), and that dimension can be read off a weighted L^2 norm of $\widehat{\mu}$ (Volume 7). Neither is difficult. The reason the volume is possible at all is that those two facts happen to meet at an exponent worth having — and the reason it stops where it does is that they meet at only one exponent.

WORKED EXAMPLE 8.8.3— TIER A **The three exponents, side by side**

It is worth tabulating the numbers this volume has produced, since they are easily confused.

Exponent	Meaning	Status
$\frac{n-1}{2}$	decay rate of $\widehat{d\sigma}$	theorem (Vol 5)
$\frac{n+1}{2}$	Falconer’s threshold	theorem (Theorem 8.5.1)
$\frac{n}{2}$	conjectured threshold	open, sharp (Theorem 8.8.1)

How they are related. The second is n minus the first: Falconer’s exponent is exactly where the decay of $\widehat{d\sigma}$ balances the weight $|\xi|^{s-n}$ coming from a

measure of dimension s . The third is half of n , and exceeds the first by $\frac{1}{2}$ — which is the entire gap.

Numerically.

n	$\frac{n-1}{2}$	$\frac{n+1}{2}$	$\frac{n}{2}$
2	0.5	1.5	1.0
3	1.0	2.0	1.5
4	1.5	2.5	2.0

The one thing to remember. The gap between what is proved and what is conjectured is $\frac{1}{2}$ in every dimension, and it is exactly the difference between the decay of $\widehat{d\sigma}$, which is $\frac{n-1}{2}$, and the decay $\frac{n}{2}$ that a sharp spherical average estimate would supply.

WORKED EXAMPLE 8.8.4— TIER B **Locating a new result: the scoreboard method**

The distance-set literature is large and its results are stated in several incompatible normalisations. This example sets out how to place a new paper.

Four questions to ask.

- (1) *Which conclusion?* Positive Lebesgue measure, nonempty interior, or a dimension bound for $\Delta(E)$? These are increasingly strong, and Theorem 8.6.1 showed the second is not free.
- (2) *Pinned or averaged?* A pinned result — there exists $x \in E$ with $m(\Delta_x(E)) > 0$ — is strictly stronger, by Worked Example 8.6.2.
- (3) *Which dimension hypothesis?* Hausdorff, or the stronger assumption that E is Ahlfors–David regular? Many results assume regularity, which is a substantial restriction.
- (4) *What is the spherical average input?* By Worked Example 8.7.4 this determines the exponent, and it is where the new mathematics lives.

Worked instance. Guth–Iosevich–Ou–Wang (2020) prove: for compact $E \subseteq \mathbb{R}^2$ with $\dim_{\text{H}} E > \frac{5}{4}$, there exists $x \in E$ with $m(\Delta_x(E)) > 0$. Placing it: conclusion is positive measure (1); pinned (2); Hausdorff dimension only, no regularity (3); and the input is a refined decoupling estimate combined with polynomial partitioning (4). So it is stronger than Theorem 8.5.1 on counts (2) and in exponent, and weaker on count (1) — it does not claim interior.

Why the bookkeeping matters. A result with a better exponent under a regularity hypothesis is not comparable to one without. Surveys often tabulate

exponents alone, and the table is then misleading. The four questions above recover the comparison.

The same discipline elsewhere. Worked Example 7.8.4 in Volume 7 gave the analogous checklist for Kakeya bounds — which dimension, what loss, which class of sets. In both subjects the exponent is the headline and the hypotheses are the content.

WORKED EXAMPLE 8.8.5— TIER C: RESEARCH TASTE **The two conjectures, compared line by line**

Volume 7 stated the Kakeya conjecture and this volume the Falconer conjecture. They are the two central problems of the series' second half, and setting them side by side explains why one has moved and the other has not.

	Kakeya	Falconer
Statement	$\dim_{\mathrm{H}} K = n$ for Besicovitch K	$\dim_{\mathrm{H}} E > \frac{n}{2}$ implies $m(\Delta E) > 0$
Extremal example	Perron tree	lattice Cantor set (Theorem 8.8.1)
Nature	how much can tubes overlap	how much can $\widehat{\mu}$ correlate with $\widehat{\mathrm{d}\sigma}$
Order	three-point and higher	two-point
Fourier bound	$\frac{n+1}{2}$, unmoved since 1983	$\frac{n+1}{2}$ in 1985, now $\frac{5}{4}$ at $n = 2$
Current method	tube combinatorics, polynomial method	restriction estimates, decoupling

(a) *The coincidence of $\frac{n+1}{2}$.* Both problems have a Fourier-analytic bound at $\frac{n+1}{2}$, and in both cases it arises the same way: the decay $\frac{n-1}{2}$ of $\widehat{\mathrm{d}\sigma}$ balanced against a 1-dimensional object — a segment for Kakeya, a distance for Falconer. So the two exponents are the same number for the same reason, which is why the problems were long treated together.

(b) *Why they then diverged.* By Worked Example 7.7.3, the Fourier method sees only pair correlations. Falconer's problem *is* a pair correlation problem, so the method is structurally adequate and improving it is a matter of better estimates; Kakeya is not, so the method saturates permanently and progress required abandoning it. That single structural fact accounts for the whole difference in the two histories.

(c) *The reunion.* The current record for Falconer in the plane uses polynomial partitioning and decoupling — that is, the machinery developed for Kakeya and restriction. So the problems have converged again, but from the other

side: the geometric methods now serve the analytic problem. Volumes 11 to 13 develop that machinery, and Volume 14 reports where both stand.

(d) *What a reader should take from the comparison.* Before choosing a method for a problem in this area, ask what order of correlation the problem involves. If two, the Fourier transform is the right tool and the question is how sharp an estimate one can prove. If more, the Fourier transform will saturate, and the answer will be combinatorial.

Drills

Drill 8.8.1. Verify the measure computation in Theorem 8.8.1: the $O(N^{-1})$ -neighbourhood of $\frac{1}{N^2}\mathbb{Z} \cap [0, n]$ has measure $O(N^{-1})$.

Drill 8.8.2. Show that $m(\Delta(E)^2) = 0$ implies $m(\Delta(E)) = 0$, using that $t \mapsto t^2$ is bi-Lipschitz on compact subsets of $(0, \infty)$.

Drill 8.8.3. Using Worked Example 8.8.1, determine what bound on $\sigma(\mu, r)$ Wolff's $\frac{4}{3}$ in the plane corresponds to.

Drill 8.8.4. Explain why the extremal set of Theorem 8.8.1 has $\dim_F E = 0$, and why that is consistent with it being extremal.

Where we are. A closing observation about how this volume relates to the two before it.

Volume 6 converted an analytic estimate into a geometric one — restriction into Kakeya — and the conversion was lossy in a way that could be named: Worked Example 6.6.1 showed the phases being discarded. Volume 7 built the measuring apparatus. This volume ran the apparatus on two problems and found the two possible outcomes: on projections it is sharp, on distances it is not.

The diagnostic that separates them, stated once more because it is the most transferable thing here: *count the order of correlation*. A projection of a set is determined by pairs of points — whether two points have the same shadow — and so is a distance. But the projection problem's transversality is exact and the distance problem's is not, so the second requires quantitative input the first does not.

Volume 9 turns to a problem where the order of correlation is genuinely higher, and where none of this volume's methods apply. The change of technique there is not a change of taste; it is forced.

Notes

The sharpness example of Theorem 8.8.1 is Falconer's. The formulation of the missing estimate in Worked Example 8.8.1 is standard and is how the problem is stated in the current literature.

For the improvements see the references in Worked Example 8.7.1; Mattila's book and Wolff's lecture notes are the standard expositions of the material of this volume.

Chapter 8.A

Results Quoted from Volumes 1–7

Result	Source	Used in
Fubini	Theorem 1.6.7	Theorem 8.3.2
Polar coordinates	Theorem 1.6.11	Theorem 8.1.2, Theorem 8.4.7, Theorem 8.7.2
Riesz representation	Theorem 1.8.15	Theorem 8.5.1
Layer-cake	Worked Example 1.5.1	Theorem 8.2.2
Lipschitz images	Theorem 7.1.5	Chapter 8.1 §1.1
Plancherel	Theorem 2.4.8	Theorem 8.1.2, Theorem 8.7.2
$\mathcal{F}$ on $\mathcal{S}'$	Theorem 2.5.11	Theorem 8.4.6
Asymptotics of $\widehat{d\sigma}$	Theorem 5.5.2	Theorem 8.4.7, Theorem 8.7.2
Bessel asymptotics	Theorem 5.4.1	Drill 7.4
The signature and the phase	Worked Example 5.3.2	Theorem 8.7.2 Pitfall
Restriction in the plane	Theorem 6.8.2	Worked Example 8.7.1
Stein–Tomas	Theorem 6.3.1	Worked Example 8.8.1
Hutchinson’s formula	Theorem 7.8.3	Worked Example 8.3.1, Theorem 8.3.3
Frostman’s lemma	Theorem 7.5.1	Theorem 8.2.1
Energy characterises dimension	Theorem 7.6.2	throughout
Fourier expression of energy	Theorem 7.6.5	Theorem 8.1.2, Theorem 8.5.1
Energy of products	Theorem 7.6.3	Worked Example 8.1.2
$\dim_{\mathrm{H}}(E \times F) \geq \dim_{\mathrm{H}} E + \dim_{\mathrm{H}} F$	Theorem 7.6.4	Theorem 8.3.3, Theorem 8.8.1
Dyadic annulus method	Worked Example 7.6.4	Theorem 8.5.1
$\dim_{\mathrm{F}}$ of the Cantor set	Theorem 7.7.4	Chapter 8.8 Checkpoint
Drury’s bound, descoped	Theorem 7.7.5	Worked Example 8.8.2

Debts

Volume 1 promised a Besicovitch set of measure zero; Volume 7 deferred it here. It is now removed from the plan as revision R8, with the obstruction identified in Theorem 8.3.3: the construction reduces, by Theorem 8.3.2, to a projection statement that requires the Besicovitch projection theorem. Nothing in this series uses it.

This volume incurs no debts on later volumes.

Chapter 8.B

Notation and Index of Results

Additions to the Volume 1–7 conventions

π_θ	orthogonal projection onto the line of direction θ
$\Delta(E)$	the distance set $\{ x - y : x, y \in E\}$
$\nu, \nu_\varepsilon, \nu_0$	the distance measure, its mollification, its density
A_ε^t	the annulus of radius t and thickness 2ε
$\sigma(\mu, r)$	the spherical average of $ \hat{\mu} ^2$ at radius r
$\mathcal{M}(\mu)$	the Mattila integral

Index of the main results

The angular integral	Theorem 8.1.3
Marstrand's projection theorem	Theorem 8.1.2
The weighted angular integral	Theorem 8.2.2
Kaufman's exceptional set bound	Theorem 8.2.1
Projections of product sets	Theorem 8.3.1
Kahane's reduction	Theorem 8.3.2
The distance measure	Theorem 8.4.4
Fourier form of the density	Theorem 8.4.6
Decay of the annulus transform	Theorem 8.4.7
Falconer's theorem	Theorem 8.5.1
Mattila–Sjölin: nonempty interior	Theorem 8.6.1
Mattila's reduction	Theorem 8.7.2
Sharpness of the conjectured exponent	Theorem 8.8.1

Plan revisions in this volume

Removed	Reason
A Besicovitch set of measure zero	reduces to the Besicovitch projection theorem, which needs the rectifiability theory this series omits. Theorem 8.3.3, revision R8
The improvements on $\frac{n+1}{2}$ (Bourgain, Wolff, Erdoğan, Guth–Iosevich–Ou–Wang)	each a research paper; stated with their inputs in Worked Example 8.7.1
Distance sets over finite fields	need Fourier analysis on $\mathbb{F}_q^d$; unused later

Part 9

Volume 9

Takeya I: Geometric and Combinatorial Methods

Preface to Volume 9

Where the method changes

Volume 8 closed by observing that its methods are adequate for two-point problems and saturate on anything richer. The Kakeya problem is richer: the smallness of a Besicovitch set is a statement about how many tubes can meet, and the Fourier transform of a measure encodes only pair correlations. Accordingly this volume abandons Fourier analysis almost entirely and counts tubes.

The change is not stylistic. Volume 7, Theorem 7.7.5 recorded that the Fourier-analytic bound $\dim_{\mathbb{H}} K \geq \frac{n+1}{2}$ has not moved since 1983; every improvement since has been geometric or combinatorial, and the first of them — Wolff’s hairbrush argument — is proved here.

A debt

Volume 3 proved Fefferman’s ball multiplier theorem assuming a configuration lemma about Besicovitch sets which was, at that point, the only statement in the series used but not proved. It was promised to this volume.

Chapter 9.2 pays it. The construction is explicit: it takes the Perron tree of Volume 1, inscribes a rectangle in each of its translated triangles, and verifies that the translates of those rectangles by twice their own length are pairwise disjoint. The verification reduces to a single inequality between two linear functions of the construction parameters, and the whole of the difficulty is in choosing where to truncate the rectangles — too short and the translates collide, too long and the rectangles no longer fit inside the triangles. There is an interval of admissible truncations, and it is nonempty for every parameter for which the Perron tree is small.

What is assumed

Volumes 1–8. Most heavily: the Perron tree, Volume 1 Theorem 1.3.12, whose explicit translations are used quantitatively rather than as a black box; the tube intersection estimate Theorem 1.6.13; the Kakeya maximal operator of Volume 1, Theorem 1.7.13; and the upgrade from maximal bounds to Hausdorff dimension, Volume 7 Theorem 7.8.5.

How the three arguments fit together

The volume proves three lower bounds and they form a single progression, which is worth having in view before starting.

Each argument conditions on a geometric object and, in exchange, obtains a structural fact the previous one could not use. The L^2 method of Chapter 9.3 conditions on nothing: it sums over all pairs of tubes, and Worked Example 9.3.1 shows the sum is dominated in high dimensions by generic pairs, so the estimate cannot distinguish an extremal configuration from a random one. The bush argument of Chapter 9.4 conditions on a point, and uses that tubes through a common point separate at unit distance. The hairbrush of Chapter 9.5 conditions on a line, and uses a fact with no analogue for points: two lines meeting a common third are concurrent *or coplanar*, so the tubes organise into planes, where the planar Kakeya problem is already solved.

The exponents are 2, then $\frac{n+1}{2}$, then $\frac{n+2}{2}$. All three are at most about half the conjectured value n , and Worked Example 9.8.2 explains why: the bush and hairbrush arguments both optimise over an unknown multiplicity, and the balancing value is one the conjecture says does not occur. An argument that bounds the multiplicity directly rather than optimising over it would escape the ceiling, and that is precisely what the polynomial method of Volume 11 does.

The plan

1	The Kakeya maximal function	the equivalences
2	The Besicovitch configuration	<i>the debt from Volume 3</i>

3	The L^2 method	Córdoba; sharp in the plane
4	The bush argument	conditioning on a point
5	The hairbrush	conditioning on a line
6	Nikodym sets	the projective duality
7	From single scales to dimension	what each passage costs
8	What the elementary methods give	and why they stop

A reader interested only in the debt may read Chapter 9.2 alone; it depends on nothing but Volume 1's Perron tree.

Chapter 9.1

The Kakeya Maximal Function

9.1.1 The operator and the conjecture

Volume 1 introduced the operator as Theorem 1.7.13; we restate it here in the notation of this volume and record the equivalences that let one move between its several formulations.

The two definitions agree, and the only difference is the range of δ : Theorem 1.7.13 allows $\delta \in (0, 1)$ while the definition below takes $\delta < \frac{1}{2}$, so that a δ -tube is genuinely long and thin. Every estimate in this volume is stated for small δ , the only regime in which the Kakeya problem has content, so nothing is lost; but a result quoted from here should not be applied at $\delta \geq \frac{1}{2}$ without checking.

Definition 9.1.1 (Kakeya maximal operator; Theorem 1.7.13 restated). For $0 < \delta < \frac{1}{2}$, $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $v \in \mathbb{S}^{n-1}$ let

$$\mathcal{K}_\delta f(v) = \sup_{T \parallel v} \frac{1}{m(T)} \int_T |f|,$$

the supremum over all δ -tubes T — cylinders of length 1 and radius δ — with axis parallel to v .

Conjecture 9.1.2 (Kakeya maximal). *For $n \geq 2$ and every $\varepsilon > 0$,*

$$\|\mathcal{K}_\delta f\|_{L^n(\mathbb{S}^{n-1})} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}.$$

Proposition 9.1.3 (Two formulations). *Theorem 9.1.2 is equivalent to the statement that for every family $\mathbb{T}$ of δ -tubes with δ -separated directions,*

$$\left\| \sum_{T \in \mathbb{T}} \mathbf{1}_T \right\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \lesssim_{n,\varepsilon} \delta^{-\varepsilon} \left(\sum_T m(T) \right)^{\frac{n-1}{n}}. \quad (9.1.1)$$

Proof. Step 1. Maximal bound implies multiplicity bound.

Let $\mathbb{T} = \{T_1, \dots, T_N\}$ have δ -separated directions $v_1, \dots, v_N$, and put $F = \sum_j \mathbf{1}_{T_j}$. Let $g \geq 0$ with $\|g\|_{L^n} \leq 1$. Then

$$\begin{aligned} \int Fg &= \sum_j \int_{T_j} g \\ &= \sum_j m(T_j) \frac{1}{m(T_j)} \int_{T_j} g \\ &\leq \sum_j m(T_j) \mathcal{K}_\delta g(v_j). \end{aligned}$$

The directions being δ -separated, the caps $B(v_j, \delta/2)$ are disjoint and each has σ -measure $\approx_n \delta^{n-1}$; moreover $\mathcal{K}_\delta g$ is comparable on $B(v_j, \delta/2)$ to its value at v_j , since a tube in direction v is contained in a fixed dilate of a tube in direction v_j when $|v - v_j| \leq \delta/2$. Hence

$$\sum_j m(T_j) \mathcal{K}_\delta g(v_j) \lesssim_n \frac{m(T_1)}{\delta^{n-1}} \int_{\mathbb{S}^{n-1}} \mathcal{K}_\delta g \, d\sigma \lesssim_n \frac{m(T_1)}{\delta^{n-1}} \|\mathcal{K}_\delta g\|_{L^n(d\sigma)},$$

using $m(T_j) \approx \delta^{n-1}$ for all j and Hölder on the sphere. By the hypothesis this is $\lesssim \delta^{-\varepsilon} m(T_1) \delta^{1-n}$, and $\sum_j m(T_j) = Nm(T_1)$ converts this into (9.1.1) after taking the supremum over g and applying Theorem 1.8.10.

Step 2. *Multiplicity bound implies maximal bound.*

Assume (9.1.1) and let $f \geq 0$ with $\|f\|_{L^n} = 1$. Fix $\lambda > 0$ and let $\{v_j\}_{j \in J}$ be a maximal δ -separated subset of $\{v : \mathcal{K}_\delta f(v) > \lambda\}$. For each $j \in J$ choose a δ -tube $T_j \parallel v_j$ with

$$\frac{1}{m(T_j)} \int_{T_j} f > \lambda.$$

The T_j have δ -separated directions, so (9.1.1) applies. Summing the defining inequality and using Hölder,

$$\begin{aligned} \lambda \sum_{j \in J} m(T_j) &< \int_{\mathbb{R}^n} f \sum_{j \in J} \mathbf{1}_{T_j} \\ &\leq \|f\|_{L^n} \left\| \sum_{j \in J} \mathbf{1}_{T_j} \right\|_{L^{\frac{n}{n-1}}} \\ &\lesssim_{n,\varepsilon} \delta^{-\varepsilon} \left(\sum_{j \in J} m(T_j) \right)^{\frac{n-1}{n}}. \end{aligned}$$

Writing $A = \sum_{j \in J} m(T_j)$ and dividing by $A^{(n-1)/n}$,

$$\lambda A^{1/n} \lesssim_{n,\varepsilon} \delta^{-\varepsilon}, \quad \text{so} \quad A \lesssim_{n,\varepsilon} \lambda^{-n} \delta^{-n\varepsilon}.$$

Since $m(T_j) \approx_n \delta^{n-1}$ and the caps $B(v_j, \delta/2)$ are disjoint with σ -measure $\approx_n \delta^{n-1}$, and since $\mathcal{K}_\delta f$ is comparable across such a cap,

$$\sigma\{v : \mathcal{K}_\delta f(v) > \lambda\} \lesssim_n \#J \cdot \delta^{n-1} \approx_n A \lesssim_{n,\varepsilon} \lambda^{-n} \delta^{-n\varepsilon}.$$

This is the weak-type bound $\mathcal{K}_\delta : L^n(\mathbb{R}^n) \rightarrow L^{n,\infty}(\mathrm{d}\sigma)$ with constant $\delta^{-\varepsilon}$. Interpolating it against the trivial bound $\mathcal{K}_\delta : L^\infty(\mathbb{R}^n) \rightarrow L^\infty(\mathrm{d}\sigma)$ of Worked Example 9.1.1(a) by the Marcinkiewicz theorem (Theorem 2.2.4) gives the strong bound at any exponent above n ; applying the argument with n replaced by $n - \eta$ and interpolating gives it at n itself, with $\delta^{-\varepsilon}$ replaced by $\delta^{-C\varepsilon}$. Since $\varepsilon > 0$ was arbitrary this is Theorem 9.1.2. $\square$

Where we are. Both formulations will be used. The maximal form is the one that interpolates and the one stated in the literature; the multiplicity form (9.1.1) is the one that geometry produces, since every tube-counting argument bounds $\sum \mathbf{1}_T$ directly.

Note that $\sum_T m(T) \approx N\delta^{n-1}$, and that a maximal δ -separated set of directions has $N \approx \delta^{-(n-1)}$, so the right side of (9.1.1) is $\approx \delta^{-\varepsilon}$ — the normalisation used in Volume 6, Theorem 6.6.2, and in Volume 7, Theorem 7.8.5.

WORKED EXAMPLE 9.1.1— TIER A **What the trivial bounds give**

Two estimates are free, and knowing them fixes the scale of the problem.

(a) L^∞ . $\mathcal{K}_\delta f(v) \leq \|f\|_\infty$ for every v , so $\|\mathcal{K}_\delta f\|_{L^\infty(\mathrm{d}\sigma)} \leq \|f\|_{L^\infty}$ with no loss.

(b) L^1 . A tube has measure $\approx \delta^{n-1}$, so $\mathcal{K}_\delta f(v) \leq \delta^{1-n} \|f\|_{L^1}$, giving $\|\mathcal{K}_\delta f\|_{L^\infty(\mathrm{d}\sigma)} \lesssim \delta^{1-n} \|f\|_{L^1}$.

(c) *Interpolating.* Theorem 2.2.1 applied to (a) and (b) gives

$$\|\mathcal{K}_\delta f\|_{L^\infty(\mathrm{d}\sigma)} \lesssim \delta^{-\frac{n-1}{p}} \|f\|_{L^p(\mathbb{R}^n)},$$

and combining with the trivial $L^\infty(\mathrm{d}\sigma) \subseteq L^n(\mathrm{d}\sigma)$ on a finite measure space, one obtains Theorem 9.1.2 with $\delta^{-\varepsilon}$ replaced by $\delta^{-(n-1)/n}$. That is the trivial bound, and the conjecture asserts the exponent can be reduced to 0.

The scale of the problem. So the whole subject lives between exponent $\frac{n-1}{n}$, free, and exponent 0, conjectured. Every theorem in this volume is a partial reduction of that exponent, and it is worth stating each result in those terms when comparing them.

WORKED EXAMPLE 9.1.2— TIER B **Why L^n and not another exponent**

Theorem 9.1.2 is stated at L^n , and the choice is forced by two examples.

The example forcing $p \leq n$. Take $f = \mathbf{1}_{B(0,\delta)}$, a ball of radius δ . Every direction admits a tube through the ball, and $\mathcal{K}_\delta f(v) \approx \delta^n / \delta^{n-1} = \delta$ for every v . So $\|\mathcal{K}_\delta f\|_{L^q(d\sigma)} \approx \delta$ while $\|f\|_{L^p} = \delta^{n/p}$, and a bound with loss $\delta^{-\varepsilon}$ requires $\delta \lesssim \delta^{-\varepsilon} \delta^{n/p}$, that is $n/p \leq 1 + \varepsilon$. Letting $\varepsilon \rightarrow 0$ gives $p \geq n$.

The example forcing $p \geq n$. Take $f = \mathbf{1}_{K_\delta}$ with K a Besicovitch set, so $m(K_\delta)$ is small. Every direction has a tube inside K_δ , so $\mathcal{K}_\delta f \equiv 1$ and $\|\mathcal{K}_\delta f\|_{L^q(d\sigma)} \approx 1$, while $\|f\|_{L^p} = m(K_\delta)^{1/p}$. If $m(K_\delta) \approx \delta^\eta$ then the bound requires $1 \lesssim \delta^{-\varepsilon} \delta^{\eta/p}$, so $\eta/p \geq -\varepsilon$ — no constraint — but the *conclusion* $\eta \leq p\varepsilon$ is exactly the Minkowski dimension statement, and it is strongest when p is smallest. Since the first example forces $p \geq n$, the strongest usable statement is at $p = n$.

The moral. The exponent n is where the two extremal examples — a single ball, and a Besicovitch set — balance. That is the same phenomenon as in Volume 6, where the Knapp cap and the whole sphere balanced at the conjectured restriction exponent, and it is the standard way conjectures in this subject are pinned down.

WORKED EXAMPLE 9.1.3— TIER C: RESEARCH TASTE **Why a maximal function at all**

The Kakeya conjecture is a statement about sets. Theorem 9.1.2 is a statement about an operator. It is worth being clear why the subject works with the second.

(a) *Operators interpolate; sets do not.* A bound for $\mathcal{K}_\delta$ at one exponent can be combined with a bound at another to give bounds in between, by Theorem 2.2.1 or Theorem 2.2.4. Worked Example 9.1.1 used exactly this to get the trivial bound. There is no corresponding operation on statements about the dimension of a set.

(b) *Operators localise.* $\mathcal{K}_\delta$ is defined at each scale separately, so a proof may work at one δ and the conclusions be assembled afterwards — which is what Chapter 9.7 does. A statement about Hausdorff dimension refers to all scales at once and cannot be attacked scale by scale without the machinery of Volume 7.

(c) *Operators connect to the rest of analysis.* Volume 6, Theorem 6.6.2 produced a Kakeya statement from a restriction estimate, and the object it produced was a bound on $\sum \mathbf{1}_T$ — the multiplicity form of Theorem 9.1.3. Had the target been a statement about Hausdorff dimension, the implication could not have been stated, let alone proved.

(d) *The cost.* The operator formulation is *a priori* stronger than the set statement, since it quantifies over all f rather than over indicator functions of neighbourhoods of Besicovitch sets. That the two are nonetheless equivalent — via Theorem 9.1.3 and Volume 7, Theorem 7.8.5 — is a theorem, not a definition, and it is what licenses the whole approach.

(e) *The historical remark.* Bourgain introduced the maximal operator in 1991 and the exponent moved for the first time in fifteen years shortly afterwards. That is not a coincidence: the reformulation made the tools of harmonic analysis applicable to a problem that had been the province of geometric measure theory.

WORKED EXAMPLE 9.1.4— TIER B **Linearising a maximal operator**

Step 2 of Theorem 9.1.3 began by choosing, for each direction, a tube nearly attaining the supremum. That move — *linearisation* — is used whenever a maximal operator must be estimated, and it repays being done slowly once.

The problem. $\mathcal{K}_\delta$ is a supremum, so it is not linear in f and none of the machinery of Volumes 2 and 3 applies directly. Duality, in particular, is unavailable: there is no adjoint.

The device. Fix a measurable selection $v \mapsto T(v)$ of tubes with

$$\frac{1}{m(T(v))} \int_{T(v)} |f| \geq \frac{1}{2} \mathcal{K}_\delta f(v),$$

and define the *linearised* operator $Lf(v) = m(T(v))^{-1} \int_{T(v)} |f|$. Then L is linear in $|f|$, $Lf \leq \mathcal{K}_\delta f \leq 2Lf$, and any bound for L *uniform in the selection* is a bound for $\mathcal{K}_\delta$.

Why uniformity is the whole point. The selection depends on f . A bound whose constant depended on the selection would say nothing. So one proves the estimate for an arbitrary fixed assignment $v \mapsto T(v)$ and only afterwards observes that the constant did not use which assignment was chosen. Every step of Theorem 9.1.3 Step 2 has this character — the tubes T_j are arbitrary subject to being δ -separated in direction.

The dual formulation that results. With L linear, duality applies: $\|Lf\|_{L^p(\mathrm{d}\sigma)} \lesssim \|f\|_{L^p}$ is equivalent to a bound on L^* , and L^* applied to a sum of point masses is exactly $\sum_j \mathbf{1}_{T_j}$ up to normalisation. That is where the multiplicity form (9.1.1) comes from, and it explains why the two formulations of Theorem 9.1.3 are adjoint to one another rather than merely equivalent.

Where the technique recurs. The Hardy–Littlewood maximal function of Volume 1 was handled by covering lemmas rather than linearisation, because there the geometry was simple enough. For $\mathcal{K}_\delta$ no covering lemma is available — that is essentially the content of the Keakeya problem — so linearisation is the only route, and it is used in every paper in this area.

Drills

Drill 9.1.1. Verify that $\mathcal{K}_\delta g$ is comparable on $B(v_j, \delta/2)$ to its value at v_j , as used in Step 1.

Drill 9.1.2. Complete Step 2 of Theorem 9.1.3, carrying out the duality between the $L^n \rightarrow L^n$ bound and the $L^{n/(n-1)}$ bound on $\sum \mathbf{1}_{T_j}$.

Drill 9.1.3. Carry out Worked Example 9.1.1(c) explicitly and confirm the exponent $\frac{n-1}{n}$ at $p = n$.

Drill 9.1.4. Show that Theorem 9.1.2 fails for $p < n$ using the ball example, and identify the largest q for which the $L^p \rightarrow L^q$ statement survives.

Where we are. Two conventions, fixed here and used for the rest of the series.

Tube families are always normalised. A family $\mathbb{T}$ has δ -separated directions and $N \approx_n \delta^{-(n-1)}$ members, so that $\sum_T m(T) \approx_n 1$ by (9.3.1). Every estimate is stated against that normalisation, and a family with fewer tubes must be renormalised before the estimates apply — Worked Example 9.3.4 shows what goes wrong if it is not.

Losses are always $\delta^{-\varepsilon}$. No estimate in this volume tracks constants, and every one is stated with an arbitrarily small power loss. By Theorem 9.7.3 that is exactly the accuracy needed for the dimension conclusions and no more, so nothing is being given away.

With those in place the whole subject can be phrased as a single question: *how small can $m(\bigcup_{T \in \mathbb{T}} T)$ be?* The conjecture says δ^ε ; the trivial bound is $\delta^{(n-1)/n}$ by Worked Example 9.1.1; and the rest of this volume closes part of the gap.

Notes

The maximal operator was introduced by Bourgain in 1991, though the underlying counting had appeared earlier in Córdoba’s work. Theorem 9.1.3 is

standard and is the reason the literature moves freely between the operator and the multiplicity function.

Chapter 9.2

The Besicovitch Configuration

Volume 3 proved Fefferman's ball multiplier theorem from a geometric input which it stated and did not prove: a family of rectangles, overlapping so much that their union is tiny, whose translates by twice their own length are pairwise disjoint. This chapter constructs the family, discharging the only statement in Volumes 1 to 8 used without proof.

9.2.1 The statement

Theorem 9.2.1 (Besicovitch configuration). *For every $\varepsilon > 0$ there exist $N \in \mathbb{N}$ and rectangles $R_1, \dots, R_N \subseteq \mathbb{R}^2$, with R_j of dimensions $\ell_j \times w_j$ and long axis in direction $v_j \in \mathbb{S}^1$, such that, writing*

$$\tilde{R}_j = R_j + 2\ell_j v_j$$

for the translate of R_j by twice its length along its own axis,

(i) the $\tilde{R}_j$ are pairwise disjoint;

$$(ii) \quad m\left(\bigcup_{j=1}^N R_j\right) \leq \varepsilon \sum_{j=1}^N m(R_j).$$

This is Volume 3, Theorem 3.8.4, and with it that volume's Theorem 3.8.2 is unconditional.

Idea of the proof. Everything comes from the Perron tree of Volume 1, used quantitatively rather than as a black box.

Recall the construction: a triangle is bisected k times into 2^k elementary triangles sharing an apex, and each is translated horizontally by an amount determined by its position, so that the union has area at most $\alpha^{2k} + 4(1 - \alpha)/(1 + \alpha)$. Property (ii) will follow immediately once each R_j is placed

inside the corresponding translated triangle.

For (i) one must compute. Each triangle has a median running from the midpoint of its base to the apex; take R_j to be a thin rectangle about the lower part of that median, truncated at height h . Translating R_j by twice its length moves it to the band between heights $2h$ and $3h$, and the question is whether the translates separate there.

They do, and the computation is a single line: the horizontal positions of the translates at a given height are an affine function of the binary expansion indexing the triangle, with slope $y + 2h + 1 - 2\alpha$. Since the rectangles have horizontal width $(1 - h)2^{-k}$ and the positions are spaced 2^{-k} apart, the translates are disjoint precisely when

$$2h + 1 - 2\alpha > 1 - h, \quad \text{that is} \quad h > \frac{2\alpha}{3}.$$

And the rectangles fit inside the triangles precisely when their width does not exceed the triangle's width at height h , which is $(1 - h)2^{-k}$ — so $h < 1$. The interval $(\frac{2\alpha}{3}, 1)$ is nonempty for every $\alpha < 1$, and that is the whole proof.

The tension is worth naming: *a longer rectangle separates better but fits worse*. The construction survives because the two constraints leave a gap, and the gap is what makes Fefferman's theorem true.

9.2.2 The construction

Fix $\alpha \in (\frac{1}{2}, 1)$ and $k \in \mathbb{N}$, to be chosen in §9.3.

Definition 9.2.2 (The Perron data). Let Δ be the triangle with vertices $(0, 0)$, $(1, 0)$, $(\frac{1}{2}, 1)$, and for $0 \leq i < 2^k$ let Δ_i be the elementary triangle with base $[i2^{-k}, (i+1)2^{-k}]$ and apex $(\frac{1}{2}, 1)$. Write the binary digits of i as $b_1, \dots, b_k$, so $i = \sum_j b_j 2^{k-j}$, and set

$$S_i = \sum_{j=1}^k b_j 2^{-j} \in [0, 1), \quad t_i = -2(1 - \alpha)S_i, \quad m_i = 2^{-k-1} + S_i,$$

so that m_i is the midpoint of the base of Δ_i and $\tau_i(x) = x + (t_i, 0)$ is the translation produced by iterating the two-triangle lemma. By Volume 1, Theorem 1.3.12,

$$m\left(\bigcup_i \tau_i \Delta_i\right) \leq \alpha^{2^k} + \frac{4(1 - \alpha)}{1 + \alpha}. \quad (9.2.1)$$

The numbers S_i are exactly the 2^k multiples of 2^{-k} in $[0, 1)$, each occurring once; in particular

$$|S_i - S_{i'}| \geq 2^{-k} \quad (i \neq i'). \quad (9.2.2)$$

Definition 9.2.3 (The rectangles). Fix $h \in (\frac{2\alpha}{3}, 1)$. For each i let

$$M_i = (m_i + t_i, 0), \quad C_i = (\tfrac{1}{2} + t_i, 1)$$

be the base midpoint and apex of $\tau_i \Delta_i$, and let

$$Q_i(y) = (m_i + t_i + y(\tfrac{1}{2} - m_i), y), \quad y \in [0, 1],$$

parametrise the median. Define R_i to be the closed rectangle with axis along the segment from $Q_i(0) = M_i$ to $Q_i(h)$, of that length, and of width

$$w = \frac{2}{\sqrt{5}} (1 - h) 2^{-k}$$

measured perpendicular to the axis.

Lemma 9.2.4 (The rectangles fit). $R_i \subseteq \tau_i \Delta_i$ for every i , and

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} h(1 - h).$$

Proof. At height y the triangle $\tau_i \Delta_i$ meets the horizontal line in an interval of length $(1 - y)2^{-k}$, centred at $Q_i(y)$ — the median joins the midpoints of the horizontal cross-sections, by similar triangles. So the triangle contains every point within horizontal distance $\frac{1}{2}(1 - y)2^{-k}$ of the median, for $0 \leq y \leq 1$.

The axis of R_i is the segment $Q_i(0)Q_i(h)$, whose direction is

$$v_i = \frac{(\frac{1}{2} - m_i, 1)}{|(\frac{1}{2} - m_i, 1)|}, \quad |\tfrac{1}{2} - m_i| \leq \tfrac{1}{2},$$

so the angle θ_i between v_i and the vertical satisfies $\cos \theta_i \geq 2/\sqrt{5}$. A point at perpendicular distance s from the axis lies at horizontal distance $s/\cos \theta_i$ from the median, so R_i lies within horizontal distance

$$\frac{w/2}{\cos \theta_i} \leq \frac{w}{2} \cdot \frac{\sqrt{5}}{2} = \tfrac{1}{2}(1 - h)2^{-k}$$

of the median, and its points have heights in $[0, h]$. Since $(1 - y)2^{-k} \geq (1 - h)2^{-k}$ for $y \leq h$, the containment follows.

For the areas, R_i has length $|Q_i(h) - Q_i(0)| \geq h$ — the vertical displacement alone is h — and width w , so $m(R_i) \geq hw = \frac{2}{\sqrt{5}} h(1 - h)2^{-k}$. Summing over the 2^k values of i gives the stated bound. $\square$

Lemma 9.2.5 (The translates separate). With $\tilde{R}_i = R_i + 2\ell_i v_i$, where ℓ_i is the length of R_i , the rectangles $\tilde{R}_1, \dots, \tilde{R}_{2^k}$ are pairwise disjoint.

Proof. Step 1. Locate the translates.

Since $\ell_i v_i = Q_i(h) - Q_i(0)$, the translate $\tilde{R}_i$ is the rectangle about the segment from $Q_i(0) + 2(Q_i(h) - Q_i(0))$ to $Q_i(h) + 2(Q_i(h) - Q_i(0))$. Its points therefore lie at heights in $[2h, 3h]$, and the centre of its horizontal cross-section at height $y + 2h$, for $y \in [0, h]$, is the point

$$Q_i(y) + 2(Q_i(h) - Q_i(0)) = \left(m_i + t_i + (y + 2h)\left(\frac{1}{2} - m_i\right), y + 2h \right).$$

All the $\tilde{R}_i$ occupy exactly the same band of heights, so it suffices to separate them at each height.

Step 2. *Compute the separation.*

Write $x_i(y)$ for the first coordinate above. Using $m_i = 2^{-k-1} + S_i$ and $t_i = -2(1 - \alpha)S_i$ from Theorem 9.2.2,

$$x_i(y) = 2^{-k-1} + S_i - 2(1 - \alpha)S_i + (y + 2h)\left(\frac{1}{2} - 2^{-k-1} - S_i\right),$$

so that, subtracting the same expression for i' , every term free of S cancels and

$$x_i(y) - x_{i'}(y) = \left(1 - 2(1 - \alpha) - (y + 2h)\right)(S_i - S_{i'}) = -(y + 2h + 1 - 2\alpha)(S_i - S_{i'}). \quad (9.2.3)$$

For $y \in [0, h]$ the coefficient has absolute value at least $2h + 1 - 2\alpha$, which is positive since $h > \frac{2\alpha}{3} > \alpha - \frac{1}{2}$. Combining with (9.2.2),

$$|x_i(y) - x_{i'}(y)| \geq (2h + 1 - 2\alpha)2^{-k} \quad (i \neq i', y \in [0, h]).$$

Step 3. *Compare with the width.*

The horizontal extent of $\tilde{R}_i$ at a given height is $w / \cos \theta_i \leq \frac{\sqrt{5}}{2}w = (1 - h)2^{-k}$, by the computation in Theorem 9.2.4. So the two cross-sections at height $y + 2h$ are disjoint as soon as

$$(2h + 1 - 2\alpha)2^{-k} > (1 - h)2^{-k}, \quad \text{that is} \quad 3h > 2\alpha,$$

which is the hypothesis $h > \frac{2\alpha}{3}$. As this holds at every height in the common band, and the $\tilde{R}_i$ meet no other heights, they are pairwise disjoint. $\square$

9.2.3 Proof of the theorem

Proof of Theorem 9.2.1. Let $\varepsilon > 0$. Choose in order:

$$\begin{aligned} \alpha &= 1 - \frac{\varepsilon}{100} \\ , \quad h &= \frac{2\alpha + 1}{3} \in \left(\frac{2\alpha}{3}, 1\right) \\ , \quad k &\geq \frac{\log(\varepsilon/100)}{2 \log \alpha}, \end{aligned}$$

the last so that $\alpha^{2k} \leq \varepsilon/100$. Note $h \geq \frac{2}{3}$ and $1 - h = \frac{2(1-\alpha)}{3} \leq \frac{\varepsilon}{150}$; we shall also use $h \leq 1$.

Take $N = 2^k$ and the rectangles $R_1, \dots, R_N$ of Theorem 9.2.3.

Step 1. *Disjointness of the translates.*

This is Theorem 9.2.5, whose hypothesis $h > \frac{2\alpha}{3}$ holds by construction.

Step 2. *The union is small.*

By Theorem 9.2.4, $R_i \subseteq \tau_i \Delta_i$, so by (9.2.1)

$$\begin{aligned} m\left(\bigcup_i R_i\right) &\leq m\left(\bigcup_i \tau_i \Delta_i\right) \\ &\leq \alpha^{2k} + \frac{4(1-\alpha)}{1+\alpha} \\ &\leq \frac{\varepsilon}{100} + 4(1-\alpha) \\ &\leq \frac{\varepsilon}{100} + \frac{4\varepsilon}{100} \\ &= \frac{\varepsilon}{20}, \end{aligned}$$

using $1 + \alpha \geq 1$ and $1 - \alpha = \varepsilon/100$.

Step 3. *The total area is bounded below.*

By Theorem 9.2.4,

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} h(1-h).$$

This is where the choice of h must be examined, since $1 - h$ is small: with $1 - h = \frac{2(1-\alpha)}{3} = \frac{\varepsilon}{150}$ and $h \geq \frac{2}{3}$,

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} \cdot \frac{2}{3} \cdot \frac{\varepsilon}{150} \geq \frac{\varepsilon}{300}.$$

Step 4. *Combine.*

By Steps 2 and 3,

$$\frac{m(\bigcup_i R_i)}{\sum_i m(R_i)} \leq \frac{\varepsilon/20}{\varepsilon/300} = 15,$$

which is a constant and not small — so the choice above does *not* give (ii) directly. The remedy is to decouple the two parameters: h was tied to α only for definiteness, and any $h \in (\frac{2\alpha}{3}, 1)$ is admissible. Take instead

$$h = \frac{9}{10},$$

which satisfies $h > \frac{2\alpha}{3}$ for every $\alpha < 1$ since $\frac{2}{3} < \frac{9}{10}$. Then $1 - h = \frac{1}{10}$ is a fixed constant, and

$$\sum_i m(R_i) \geq \frac{2}{\sqrt{5}} \cdot \frac{9}{10} \cdot \frac{1}{10} \geq \frac{1}{13},$$

independent of ε . Now choose $\alpha = 1 - \varepsilon/200$ and k with $\alpha^{2k} \leq \varepsilon/200$, so that Step 2 gives $m(\bigcup R_i) \leq \varepsilon/200 + 4\varepsilon/200 = \varepsilon/40$, and therefore

$$m\left(\bigcup_i R_i\right) \leq \frac{\varepsilon}{40} \leq \frac{\varepsilon}{40} \cdot 13 \sum_i m(R_i) \leq \varepsilon \sum_i m(R_i),$$

which is (ii). □

Where we are. Step 4 records a false start deliberately, because the trap is a natural one. It is tempting to tie h to α — both control the same tension — and doing so makes $1 - h$ shrink with ε , which destroys the lower bound on $\sum m(R_i)$ exactly as fast as the union shrinks.

The two parameters must be decoupled. The constraint linking them, $h > \frac{2\alpha}{3}$, is satisfied by the *fixed* choice $h = \frac{9}{10}$ for every $\alpha < 1$, because $\frac{2\alpha}{3} < \frac{2}{3}$ always. So h may be frozen and α sent to 1, which is what the proof does.

That $\frac{2\alpha}{3}$ stays below $\frac{2}{3}$ is the one piece of luck in the construction. Had the constraint been $h > \alpha$, no fixed h would work and the theorem would be false as stated.

WORKED EXAMPLE 9.2.1— TIER A **The numbers, for a concrete case**

Take $\alpha = 0.95$ and $h = 0.9$, so k is whatever makes α^{2k} small.

The separation coefficient. From (9.2.3) at $y = 0$,

$$2h + 1 - 2\alpha = 1.8 + 1 - 1.9 = 0.9.$$

The width, in the same units. $1 - h = 0.1$.

The margin. $0.9 > 0.1$ by a factor of 9 — the translates are not merely disjoint but separated by nine times their own width. The construction is not delicate at these parameters.

Where it does become delicate. At h just above $\frac{2\alpha}{3}$ the two sides converge: with $\alpha = 0.95$ and $h = 0.64$, the coefficient is $2(0.64) + 1 - 1.9 = 0.38$ and the width is 0.36, a margin of only 6%. Taking h below $\frac{2\alpha}{3} \approx 0.633$ reverses the inequality and the translates overlap.

The lesson. The admissible interval $(\frac{2\alpha}{3}, 1)$ is genuinely open at the left end, and the geometric content of the construction is that it is nonempty. Drill 2.2 asks for the corresponding statement when the rectangles are truncated at the top rather than the bottom.

WORKED EXAMPLE 9.2.2— TIER B Why the translates separate: the mechanism, without coordinates

The computation in Theorem 9.2.5 is three lines of algebra and can look like an accident. It is not, and the reason is worth having.

The two motions. As i increases by one, two things happen to the triangle $\tau_i \Delta_i$: its base slides *right* by 2^{-k} , because m_i increases by that amount; and the Perron translation slides it *left* by $2(1 - \alpha)2^{-k}$. The net motion of the base is right by $(1 - 2(1 - \alpha))2^{-k} = (2\alpha - 1)2^{-k}$.

The apex barely moves. The apex is at $(\frac{1}{2} + t_i, 1)$, so it moves *left* by $2(1 - \alpha)2^{-k}$ only — the m_i term is absent. This is the whole point of the Perron construction: the triangles are made to overlap by pinning their apexes close together while their bases spread out.

Extending the median therefore amplifies. A line through a base point moving right at rate $2\alpha - 1$ and an apex moving left at rate $2(1 - \alpha)$ must, when extended beyond the apex, move left at a rate that grows with the extension. At height $y + 2h$ the rate is $y + 2h + 1 - 2\alpha$ — increasing in y , as (9.2.3) shows.

So overlap below forces separation above. The very mechanism that makes the union small — apexes pinned, bases spread — makes the extended tubes diverge. The construction does not balance two competing effects; it uses one effect twice, with opposite signs.

Why this matters beyond the lemma. Fefferman's theorem is often described as “a Besicovitch set disproves the ball multiplier”. The description is accurate but hides the point: what is used is not merely that a Besicovitch set is small, but that its smallness is achieved by a mechanism which simultaneously makes translated copies disjoint. A different small set need not have that property, and the theorem would not follow from it.

WORKED EXAMPLE 9.2.3— TIER C: RESEARCH TASTE **The debt, and what its payment settles**

(a) *What was outstanding.* From the writing of Volume 3 until now, the series contained exactly one statement used but not proved: Theorem 3.8.4, on which Theorem 3.8.2 — that the ball multiplier is bounded on $L^p(\mathbb{R}^n)$ only for $p = 2$ — depended. Theorem 9.2.1 discharges it, and Volume 3’s theorem is now unconditional.

(b) *Why it took until Volume 9.* Not because the proof is long — it is three pages — but because it requires the Perron tree in *quantitative* form. Volume 1 proved the area bound (9.2.1) and, having no further use for the translations, did not record them. The present chapter needs the explicit formula $t_i = -2(1 - \alpha)S_i$, and it is that formula, not the area bound, which makes (9.2.3) computable.

The transferable observation: when a construction is used as a black box, record its internal parameters anyway. The second application will need them.

(c) *What remains unproved in Volumes 1–9.* With this chapter, nothing that is used. Nine results across Volumes 4 to 8 were removed from the plan by announced revisions R1 to R8, each stated where it would have appeared, attributed, and used nowhere; and Chapter 9.1 and this chapter add none.

(d) *What the configuration is good for besides Fefferman.* The same family, with the translates disjoint, is the model for every “overlapping tubes, separated extensions” argument in the subject. In Volume 12 the wave packets associated to a Besicovitch configuration are used to test multilinear estimates, and the disjointness of the extensions is what makes the test informative. The lemma is a tool, not only a step in one theorem.

WORKED EXAMPLE 9.2.4— TIER B **Reading a construction for its parameters**

Worked Example 9.2.3 observed that the chapter needed the Perron tree’s internal translations, which Volume 1 had not recorded. The episode is worth generalising, since the same situation will arise again.

(a) *What Volume 1 recorded.* A single number: the area bound $\alpha^{2k} + 4(1 - \alpha)/(1 + \alpha)$. That is what Theorem 1.3.12 was for, and for the purposes of Volume 1 it sufficed — the Besicovitch set exists, and its measure is small.

(b) *What this chapter needed.* The formula $t_i = -2(1 - \alpha)S_i$ for the translation applied to the i -th elementary triangle. Without it, (9.2.3) cannot be computed and the disjointness of the translates is inaccessible.

(c) *Why the difference matters.* An existence statement — “there is a set of small measure containing segments in all directions” — supports only

arguments that use the set as a black box. A construction with recorded parameters supports arguments about the *mechanism*, and Worked Example 9.2.2 showed that Fefferman’s theorem needs the mechanism, not merely the conclusion.

(d) *The practical rule.* When a construction is given, record: the recursion, the parameter that controls it, the explicit form of every choice made at each stage, and the range of parameters for which the estimate holds. The first three are usually written down; the fourth almost never is, and it is what Theorem 9.2.1 needed most — the admissible interval $h \in (\frac{2\alpha}{3}, 1)$ is exactly a statement about parameter ranges.

(e) *A second instance, ahead.* Volume 10 will construct Kakeya sets from arithmetic progressions, and Volume 14 will need to know which progressions. The same discipline applies, and the manifest now carries a note to that effect.

Proposition 9.2.6 (The configuration in the form Volume 3 uses). *For every $\varepsilon > 0$ there are rectangles $R_1, \dots, R_N$ as in Theorem 9.2.1 which in addition satisfy*

$$\sum_j m(R_j) \geq \frac{1}{13}, \quad m\left(\bigcup_j R_j\right) \leq \varepsilon,$$

with all R_j contained in a fixed bounded region and all $\tilde{R}_j$ contained in another.

Proof. Take $h = \frac{9}{10}$ and α, k as in Step 4 of Theorem 9.2.1. The lower bound on $\sum m(R_j)$ was computed there. The upper bound on the union is Step 2 with ε in place of $\varepsilon/40$, which only requires taking α closer to 1.

For the containments: every R_j lies in $\tau_j \Delta_j \subseteq \Delta'$, where Δ' is the triangle Δ enlarged to absorb the translations, which are bounded by $2(1 - \alpha) \leq 2$ in absolute value; so $\Delta' \subseteq [-2, 3] \times [0, 1]$. Each $\tilde{R}_j$ lies at heights in $[2h, 3h] \subseteq [\frac{9}{5}, \frac{27}{10}]$ and, by the computation in Theorem 9.2.5, at horizontal position $m_j + t_j + (y + 2h)(\frac{1}{2} - m_j)$ with all factors bounded, hence in $[-4, 5] \times [\frac{9}{5}, \frac{27}{10}]$. $\square$

Where we are. Volume 3’s Theorem 3.8.4 was stated with the normalisation $m(\bigcup R_j) \leq \varepsilon \sum m(R_j)$, and Theorem 9.2.6 gives the two factors separately — which is what the proof of Theorem 3.8.2 actually consumes, since it applies the Rademacher–Khinchine argument to the family $\{R_j\}$ and needs the total mass bounded below independently of ε .

The boundedness of the regions matters too: Fefferman’s argument places the multiplier’s symbol on a ball containing all the $\tilde{R}_j$, and a family whose translates escaped to infinity would not serve. Both requirements are implicit in Volume 3 and are recorded here so that the interface between the two volumes is explicit.

Figure 9.2.1 (to be drawn). *The Besicovitch configuration and its translates*

Content required: The Perron tree at a small k : the translated triangles $\tau_i \Delta_i$ overlapping heavily near their common apex height, with the inscribed rectangles R_i shaded inside them, truncated at height h . Above, in the band between heights $2h$ and $3h$, the translates $\tilde{R}_i$ drawn *disjoint* and fanning apart. Mark the separation $(2h + 1 - 2\alpha)2^{-k}$ against the width $(1 - h)2^{-k}$. This is the single most important figure in Volume 9.

Drills

Drill 9.2.1. Verify (9.2.2): the numbers S_i are exactly the multiples of 2^{-k} in $[0, 1)$, each occurring once.

Drill 9.2.2. Repeat Theorem 9.2.5 for rectangles truncated at the *top* — about the median from height h to height 1 — and determine the admissible range of h .

Drill 9.2.3. Check the claim in Theorem 9.2.4 that the median joins the midpoints of the horizontal cross-sections of a triangle, and that $\cos \theta_i \geq 2/\sqrt{5}$.

Drill 9.2.4. Track the constants in the proof of Theorem 9.2.1 to obtain an explicit $N = N(\varepsilon)$, and estimate its growth as $\varepsilon \rightarrow 0$.

WORKED EXAMPLE 9.2.5— TIER C: RESEARCH TASTE **The shape of a construction proof**

Theorem 9.2.1 is the only purely constructive theorem in this volume, and construction proofs have a different rhythm from estimate proofs. It is worth naming it, since Volumes 10 and 14 contain several more.

(a) *The object comes first, the verification second.* One writes down R_i explicitly — Theorem 9.2.3 — and only then checks the two properties. There is no search and no optimisation in the statement; all the choosing happens in fixing α , k and h .

(b) *The parameters carry the difficulty.* Theorem 9.2.4 needs $h < 1$; Theorem 9.2.5 needs $h > \frac{2\alpha}{3}$; and the theorem is true because the interval between them is nonempty. A construction proof is, almost always, the assertion that a system of constraints on parameters is consistent, and the mathematics is in exhibiting a solution.

(c) *The false start is instructive and was kept.* Step 4 first ties h to α and fails. That failure is not a defect of exposition: it is the natural first attempt, and seeing why it fails is what teaches the reader that the two parameters

play different roles. A construction proof that presents only the successful choice conceals where the difficulty lay.

(d) *Verification before drafting.* The construction here was checked numerically — the separation coefficient computed for a range of α and h , and compared against the rectangle width — before any of it was written. Two earlier candidates failed that check: constant-width tubes do not fit inside the triangles, and half-height tubes fit but their translates collide as $\alpha \rightarrow 1$. Neither failure is visible from the statement, and both would have survived a plausible-looking write-up.

(e) *The general recommendation.* For a construction, verify a small case before writing. For an estimate, check the exponents in the extreme cases. In both, the aim is to fail fast: the cost of discovering an error after the proof is written is far higher than the cost of the check.

Notes

The configuration is due to Besicovitch, and its use for the ball multiplier is Fefferman's (1971). The explicit form given here — with the truncation parameter h isolated and the admissibility condition $h > \frac{2\alpha}{3}$ made visible — follows the structure of Fefferman's argument while keeping the Perron parameters of Volume 1.

Readers comparing with other treatments will find the rectangles described as “a Besicovitch set of rectangles” with the disjointness of translates asserted as part of the construction. Making it a computation, as here, has the advantage that the constraint linking the truncation to the Perron parameter becomes explicit, and with it the reason the construction is possible at all.

Chapter 9.3

The L^2 Method

The simplest way to bound the union of a tube family is to expand a square. This chapter carries that out, obtains the sharp planar result, and identifies precisely why the method stalls at dimension 2 in every ambient dimension.

9.3.1 The estimate

Throughout, $\mathbb{T} = \{T_1, \dots, T_N\}$ is a family of δ -tubes with δ -separated directions $v_1, \dots, v_N$, and $N \approx_n \delta^{-(n-1)}$ so that

$$\sum_j m(T_j) \approx_n N \delta^{n-1} \approx_n 1. \quad (9.3.1)$$

We write $E = \bigcup_j T_j$ and $F = \sum_j \mathbf{1}_{T_j}$ for the multiplicity function.

Lemma 9.3.1 (Angle counting). *For each j and each dyadic $\theta = 2^m \delta \leq 1$,*

$$\#\{j' : \angle(v_j, v_{j'}) \approx \theta\} \lesssim_n \left(\frac{\theta}{\delta}\right)^{n-1}.$$

Proof. The directions $v_{j'}$ in question lie in a spherical annulus about v_j of angular radius $\approx \theta$, of σ -measure $\lesssim_n \theta^{n-1}$. They are δ -separated, so the caps $B(v_{j'}, \delta/2)$ are disjoint and each has measure $\approx_n \delta^{n-1}$. Comparing measures gives the bound. $\square$

Theorem 9.3.2 (The L^2 bound). *With the notation above,*

$$\|F\|_{L^2(\mathbb{R}^n)}^2 \lesssim_n \begin{cases} \log \frac{1}{\delta}, & n = 2, \\ \delta^{2-n}, & n \geq 3. \end{cases}$$

Proof. Step 1. Expand.

$$\|F\|_{L^2}^2 = \int \left(\sum_j \mathbf{1}_{T_j} \right)^2 = \sum_{j,j'} m(T_j \cap T_{j'}).$$

Step 2. *Insert the intersection estimate.*

By Volume 1, Theorem 1.6.13, two δ -tubes whose directions differ by angle θ satisfy

$$m(T \cap T') \lesssim_n \frac{\delta^n}{\theta + \delta}.$$

Fix j and group the j' by dyadic angle $\theta = 2^m \delta$, $0 \leq m \leq \log_2(1/\delta)$. By Theorem 9.3.1 the group has $\lesssim (2^m)^{n-1}$ members, and each contributes $\lesssim \delta^n / (2^m \delta) = \delta^{n-1} 2^{-m}$. Hence

$$\sum_{j'} m(T_j \cap T_{j'}) \lesssim_n \sum_{m=0}^{\log_2(1/\delta)} 2^{m(n-1)} \delta^{n-1} 2^{-m} = \delta^{n-1} \sum_{m=0}^{\log_2(1/\delta)} 2^{m(n-2)}. \quad (9.3.2)$$

Step 3. *Evaluate the sum.*

If $n = 2$ the summand is 1 and the sum is $\approx \log \frac{1}{\delta}$. If $n \geq 3$ the sum is geometric with ratio $2^{n-2} > 1$, so it is dominated by its last term, at $2^m \approx \delta^{-1}$, giving $\approx \delta^{-(n-2)}$. Substituting into (9.3.2),

$$\sum_{j'} m(T_j \cap T_{j'}) \lesssim_n \begin{cases} \delta \log \frac{1}{\delta}, & n = 2, \\ \delta^{n-1} \delta^{-(n-2)} = \delta, & n \geq 3. \end{cases}$$

Step 4. *Sum over j .*

Multiplying by $N \approx \delta^{-(n-1)}$ gives $\|F\|_{L^2}^2 \lesssim \delta^{-(n-1)} \delta \log \frac{1}{\delta} = \log \frac{1}{\delta}$ when $n = 2$, and $\delta^{-(n-1)} \delta = \delta^{2-n}$ when $n \geq 3$. $\square$

Corollary 9.3.3 (Córdoba's bound). *For every δ -tube family as above,*

$$m(E) \gtrsim_n \begin{cases} (\log \frac{1}{\delta})^{-1}, & n = 2, \\ \delta^{n-2}, & n \geq 3. \end{cases}$$

Consequently every Besicovitch set in $\mathbb{R}^n$ has upper Minkowski dimension at least 2, and in the plane it has Minkowski dimension exactly 2.

Proof. By (9.3.1) and Cauchy–Schwarz, F being supported in E ,

$$1 \approx_n \sum_j m(T_j) = \int_E F \leq m(E)^{1/2} \|F\|_{L^2},$$

so $m(E) \geq \|F\|_{L^2}^{-2}$, and Theorem 9.3.2 gives the two cases.

For the dimension: a Besicovitch set K has K_δ containing a δ -tube in each of $\approx \delta^{-(n-1)}$ separated directions, so $m(K_\delta) \geq m(E)$ for that family. By Theorem 7.2.3, $\overline{\dim}_M K \geq n - \liminf \frac{\log m(K_\delta)}{\log \delta}$, and the two bounds give $\geq n - 0 = n$ for $n = 2$ — a logarithm contributes nothing to the exponent — and $\geq n - (n - 2) = 2$ for $n \geq 3$. $\square$

Where we are. The planar case is sharp: Theorem 9.3.3 gives Minkowski dimension 2, which is the Kakeya conjecture for $n = 2$ at the level of Minkowski dimension, and the logarithm is genuinely there — Volume 1's Perron tree produces sets with $m(K_\delta) \approx 1/\log \frac{1}{\delta}$, so the estimate cannot be improved.

In every higher dimension the method gives dimension 2 and no more, which is worse than the bush bound $\frac{n+1}{2}$ as soon as $n \geq 4$. The next section explains where the loss is.

9.3.2 Why the method stalls

WORKED EXAMPLE 9.3.1— TIER B The exact point of failure

The whole of Theorem 9.3.2 turns on the sum $\sum_m 2^{m(n-2)}$ in (9.3.2), and its behaviour changes character at $n = 2$.

$n = 2$: *every scale contributes equally.* The summand is 1, so all $\log \frac{1}{\delta}$ dyadic angles contribute the same amount. No single angular scale dominates, and the answer is the number of scales.

$n \geq 3$: *the largest angles dominate.* The summand grows geometrically, so the sum is carried by $\theta \approx 1$ — tubes that are nearly perpendicular to T_j . But those are exactly the pairs whose intersection is *smallest*, $m(T \cap T') \approx \delta^n$. The sum is dominated by the most numerous pairs rather than the most overlapping ones.

Why that is a loss. The quantity $\|F\|_{L^2}^2$ counts all intersections with equal weight, and in high dimensions the count is dominated by a generic pair, about which nothing is being assumed. The estimate is therefore insensitive to the *arrangement* of the tubes: it gives the same answer for a Besicovitch configuration as for a random one. Since a random arrangement has $m(E) \approx 1$, an estimate that cannot distinguish the two cannot prove anything strong.

What is needed instead. An argument that uses the tubes' incidence structure — which tubes meet which — rather than only the total count of intersections. The bush argument of Chapter 9.4 is the first such: it asks how many tubes pass through a single point, which $\|F\|_{L^2}^2$ averages away.

A general moral. An L^2 argument is a second-moment argument, and second moments see only pair correlations. This is the same limitation identified in Volume 8, Worked Example 7.7.3, in Fourier language, and it is the same reason the Fourier bound stalls at $\frac{n+1}{2}$. Both methods are two-point; Kakeya is not.

WORKED EXAMPLE 9.3.2— TIER A Checking the planar bound against the Perron tree

Theorem 9.3.3 says $m(E) \gtrsim 1/\log \frac{1}{\delta}$ in the plane. It is worth confirming this is attained.

Volume 1, Theorem 1.3.12 gives, for each α and k , a union of translated triangles of area at most $\alpha^{2k} + 4(1 - \alpha)/(1 + \alpha)$. Taking $\delta = 2^{-k}$ — so that the elementary triangles have base δ — and optimising α , one finds the area is $\approx 1/k = 1/\log_2 \frac{1}{\delta}$: choose $1 - \alpha \approx 1/k$, so that the second term is $\approx 1/k$ and $\alpha^{2k} \approx e^{-2}$ is bounded, then iterate the construction $\log k$ times to remove the constant.

So the Perron tree achieves $m(E) \approx 1/\log \frac{1}{\delta}$ and Theorem 9.3.3 is sharp up to constants. In particular the planar Kakeya problem is completely settled at the level of Minkowski dimension, with matching upper and lower bounds — one of the few places in this subject where that is true.

WORKED EXAMPLE 9.3.3— TIER C: RESEARCH TASTE Second moments, and the limits of averaging

Worked Example 9.3.1 diagnosed the L^2 method's ceiling. The diagnosis generalises, and it is worth stating in the form that applies across the series.

(a) *What a second moment is.* $\|F\|_{L^2}^2 = \sum_{T, T'} m(T \cap T')$ is a sum over *pairs*. It is determined by the pair correlation structure of the family and by nothing else: two configurations with the same multiset of pairwise intersections have the same L^2 norm, however different they look.

(b) *Why Kakeya is not determined by pairs.* A Besicovitch set is small because *many* tubes concentrate in a small region — a statement about high-order coincidences. Drill 3.4 asks for the computation showing that a random family and an extremal family have comparable second moments while their unions differ by a power of δ .

(c) *The same limitation, three times over.* This is the third appearance of the identical obstruction. In Volume 7, Worked Example 7.7.3 showed the Fourier method sees only $I_t(\mu)$, a two-point quantity, and stalls at $\frac{n+1}{2}$. In Volume 8 the distance set problem *was* two-point and the Fourier method

was accordingly the right tool. Here the L^2 method is the same thing in physical space. Fourier transform, energy integral, and second moment are three names for one restriction on what an argument can see.

(d) *What escapes it.* An argument conditioning on a configuration — Chapter 9.4 on a point, Chapter 9.5 on a line — extracts information no average carries. So does an argument using algebraic structure, as Volume 11 does. The common feature is that neither can be written as a sum over pairs.

(e) *A test to apply.* Given a proposed method, ask: could it distinguish a Besicovitch configuration from a random one with the same pairwise statistics? If not, its ceiling is already visible, and it will be whatever the second moment gives.

WORKED EXAMPLE 9.3.4— TIER B Running the L^2 computation for a sub-family
Theorem 9.3.2 was stated for a full family of $\approx \delta^{-(n-1)}$ tubes. Theorem 9.5.3 needed the planar case for a sub-family of m tubes, and the modification is worth doing carefully, since the naive version gives the wrong answer.

The naive attempt. Apply Theorem 9.3.2 with $n = 2$ to conclude $\|\sum_{j \leq m} \mathbf{1}_{T_j}\|_{L^2}^2 \lesssim \log \frac{1}{\delta}$. This is *false* reasoning: the bound was obtained by multiplying a per-tube estimate by $N \approx \delta^{-1}$, and with only m tubes the multiplication is by m .

The correct computation. For a sub-family of m tubes in the plane with δ -separated directions, the angle count of Theorem 9.3.1 becomes

$$\#\{j' : \theta \approx 2^p \delta\} \leq \min\{2^p, m\},$$

since there are at most 2^p directions available at that angular scale and at most m tubes in total. So, per tube,

$$\begin{aligned} \sum_{j'} m(T_j \cap T_{j'}) &\lesssim \sum_p \min\{2^p, m\} \frac{\delta^2}{2^p \delta} \\ &= \delta \sum_p \min\{1, m2^{-p}\} \\ &\lesssim \delta \log \frac{1}{\delta}, \end{aligned}$$

the sum being $\lesssim \log m + O(1) \leq \log \frac{1}{\delta}$. Multiplying by m ,

$$\left\| \sum_{j \leq m} \mathbf{1}_{T_j} \right\|_{L^2}^2 \lesssim m \delta \log \frac{1}{\delta}.$$

The conclusion. With $\sum_j m_2(T_j) = m\delta$ and Cauchy–Schwarz,

$$\begin{aligned} m\delta &\leq m_2\left(\bigcup T_j\right)^{1/2} (m\delta \log \tfrac{1}{\delta})^{1/2}, \quad \text{so} \quad m_2\left(\bigcup T_j\right) \\ &\gtrsim \frac{m\delta}{\log \frac{1}{\delta}}, \end{aligned}$$

which is the estimate used in Theorem 9.5.3 Step 3.

The check. At $m \approx \delta^{-1}$, a full family, this gives $m \gtrsim 1/\log \frac{1}{\delta}$ — Theorem 9.3.3.

✓ At $m = 1$ it gives $m \gtrsim \delta/\log$, slightly worse than the truth δ ; the logarithm is an artefact of the method for small families and costs nothing where it is used.

The habit worth forming. Whenever a stated estimate is applied to a subfamily, re-derive it rather than quoting it: the normalisations almost always change, and the error is invisible in the final statement.

Drills

Drill 9.3.1. Verify Theorem 9.3.1 in the plane, where the directions form a circle and the count is $\approx \theta/\delta$.

Drill 9.3.2. Carry out Step 3 for $n = 3$ explicitly and confirm the exponent $\delta^{2-n} = \delta^{-1}$.

Drill 9.3.3. Show that the bound of Theorem 9.3.3 for $n \geq 3$ is weaker than the bush bound $\delta^{(n-1)/2}$ exactly when $n \geq 4$, and that the two agree at $n = 3$.

Drill 9.3.4. Compute $\|F\|_{L^2}^2$ for a *random* family of δ -separated tubes and confirm that Theorem 9.3.2 is attained, so the estimate cannot distinguish random from extremal configurations.

Where we are. One point of technique before leaving the chapter. The estimate $m(T \cap T') \lesssim \delta^n/(\theta + \delta)$ of Theorem 1.6.13 is used in every argument in this volume, and it is worth remembering in the form: the intersection of two δ -tubes is a box of dimensions $\delta \times \cdots \times \delta \times \min\{1, \delta/\theta\}$, the last factor being the length along which they stay together.

At $\theta \approx \delta$ they are parallel to within their own width and overlap along their whole length, giving δ^{n-1} . At $\theta \approx 1$ they cross transversally and meet in a cube of side δ , giving δ^n . Every intermediate case interpolates, and the interpolation is exactly the factor δ/θ .

Reconstructing the estimate from that picture is quicker than recalling it, and it makes the angle-counting of Theorem 9.3.1 feel inevitable rather than technical.

Notes

The L^2 argument is Córdoba's (1977), and in the plane it remains the standard proof. The observation that it stalls at dimension 2 in higher dimensions — and that the obstruction is the dominance of nearly perpendicular pairs — is the starting point of Bourgain's and Wolff's work.

Chapter 9.4

The Bush Argument

Volume 1 proved the bound $\dim_{\mathbb{H}} K \geq \frac{n+1}{2}$ by an argument about tubes through a common point. This chapter restates it in the language of Chapter 9.1, where the mechanism is clearer, and identifies what it uses that the L^2 method of Chapter 9.3 does not.

9.4.1 Two bounds in competition

As before $\mathbb{T} = \{T_1, \dots, T_N\}$ is a family of δ -tubes with δ -separated directions, $N \approx_n \delta^{-(n-1)}$, $E = \bigcup_j T_j$ and $F = \sum_j \mathbf{1}_{T_j}$.

Definition 9.4.1. The *multiplicity* of the family is $\mu = \|F\|_{L^\infty} = \sup_x \#\{j : x \in T_j\}$.

Lemma 9.4.2 (The bush estimate). *Suppose x_0 lies in μ of the tubes. Then*

$$m\left(\bigcup_{j: x_0 \in T_j} T_j\right) \gtrsim_n \mu \delta^{n-1}.$$

Proof. Let $J = \{j : x_0 \in T_j\}$, so $\#J = \mu$. Each T_j with $j \in J$ has length 1 and contains x_0 , so it contains a sub-tube T'_j of length $\frac{1}{4}$ lying at distance between $\frac{1}{4}$ and $\frac{1}{2}$ from x_0 — one of the two halves of T_j has this property. Then $m(T'_j) \approx_n \delta^{n-1}$.

We claim the T'_j , $j \in J$, have bounded overlap. Let $j \neq j'$ in J and let $\theta = \angle(v_j, v_{j'}) \geq \delta$, the directions being δ -separated. If $y \in T'_j \cap T'_{j'}$, then y is within δ of the line through x_0 in direction v_j and within δ of the line through x_0 in direction $v_{j'}$, and $|y - x_0| \geq \frac{1}{4}$. The two lines meet only at x_0 and separate at rate θ , so

$$\begin{aligned} & 2\delta \\ & \geq \text{dist}(y, \text{line } j) + \text{dist}(y, \text{line } j') \gtrsim |y - x_0| \theta \gtrsim \theta, \end{aligned}$$

forcing $\theta \lesssim \delta$. Since the directions are δ -separated, only boundedly many j' can satisfy this for a given j . Hence

$$\begin{aligned} m\left(\bigcup_{j \in J} T_j\right) &\geq m\left(\bigcup_{j \in J} T'_j\right) \gtrsim_n \sum_{j \in J} m(T'_j) \approx_n \mu \delta^{n-1}. \end{aligned} \quad \square$$

Lemma 9.4.3 (The multiplicity estimate). $m(E) \geq \mu^{-1} \sum_j m(T_j) \approx_n \mu^{-1}$.

Proof. $F \leq \mu$ pointwise and F is supported in E , so $\sum_j m(T_j) = \int_E F \leq \mu m(E)$. $\square$

Theorem 9.4.4 (Bush bound). *For every δ -tube family as above,*

$$m(E) \gtrsim_n \delta^{\frac{n-1}{2}}.$$

Consequently every Besicovitch set in $\mathbb{R}^n$ has upper Minkowski dimension at least $\frac{n+1}{2}$.

Proof. Step 1. The two bounds.

Let μ be the multiplicity and let x_0 be a point lying in μ tubes, which exists by definition of $\mu = \|F\|_{L^\infty}$ and the fact that F is integer-valued. Theorem 9.4.2 applied at x_0 gives

$$m(E) \geq m\left(\bigcup_{j: x_0 \in T_j} T_j\right) \gtrsim_n \mu \delta^{n-1}, \quad (9.4.1)$$

the first inequality because the bush is a subfamily of $\mathbb{T}$. And Theorem 9.4.3 gives

$$m(E) \gtrsim_n \mu^{-1}. \quad (9.4.2)$$

Step 2. *Eliminate μ .*

The parameter μ is not under our control: it depends on the family, and both extremes are conceivable *a priori*. But (9.4.1) is strong when μ is large and (9.4.2) is strong when μ is small, so their product is free of μ :

$$m(E)^2 \gtrsim_n (\mu \delta^{n-1}) \cdot \mu^{-1} = \delta^{n-1}.$$

Equivalently, $m(E) \geq \max\{c\mu\delta^{n-1}, c\mu^{-1}\}$ and the right side is minimised over $\mu > 0$ where the two agree, at $\mu = \delta^{-(n-1)/2}$, with common value $c\delta^{(n-1)/2}$. Either way,

$$m(E) \gtrsim_n \delta^{\frac{n-1}{2}}.$$

Step 3. *Pass to dimension.*

A Besicovitch set K has K_δ containing an admissible family, so $m(K_\delta) \gtrsim_n \delta^{(n-1)/2}$; by Theorem 7.2.3, $\overline{\dim}_M K \geq n - \frac{n-1}{2} = \frac{n+1}{2}$. $\square$

Where we are. The proof is two inequalities and a multiplication, and it is worth seeing why multiplying is the right move. The two bounds pull in opposite directions: a large multiplicity makes the bush large, a small multiplicity makes the union large. Neither alone is useful, because μ is not under our control. Taking the geometric mean eliminates μ entirely. This is the standard shape of an extremal argument in the subject — produce two bounds whose product is μ -free — and it recurs in Chapter 9.5, where the two bounds are more refined and the product is correspondingly better.

WORKED EXAMPLE 9.4.1— TIER B **What the bush uses that the L^2 method does not**

Both arguments bound $m(E)$ from below, and the bush is better as soon as $n \geq 4$ (Drill 3.3). The difference is worth isolating, since it is the first instance of a pattern that governs the rest of the subject.

The L^2 method uses only $\|F\|_{L^2}$, a second moment. By Worked Example 9.3.1 that quantity is dominated, in high dimensions, by pairs of nearly perpendicular tubes — generic pairs, carrying no information about the configuration.

The bush uses $\|F\|_{L^\infty}$, the maximum multiplicity, together with a geometric fact about the tubes realising it. That is a statement about a *single point* and all the tubes through it: an incidence, not an average.

Why the incidence is worth more. Theorem 9.4.2 says that μ tubes through a common point cannot be packed — at distance ≈ 1 from the point they have separated, because δ -separated directions diverge at rate δ and a unit distance amplifies that to δ , which is exactly the tube width. The estimate is sharp and it is genuinely three-point in character: it concerns the point, and two tubes, and the distance between them.

The general principle. Averages see pairs; incidences see configurations. Every improvement in this volume and the next replaces an average by a more refined incidence count:

Method	What is counted	Exponent
L^2	pairs of tubes	2
bush	tubes through a point	$\frac{n+1}{2}$
hairbrush	tubes through a line	$\frac{n+2}{2}$

The progression is not accidental: each step conditions on a richer geometric object.

WORKED EXAMPLE 9.4.2— TIER A The bush bound in low dimensions, and where it is beaten

$n = 2$. The bush gives $\frac{n+1}{2} = \frac{3}{2}$, while Theorem 9.3.3 gives 2 — the truth. So in the plane the L^2 method is strictly better, and the bush is never used.
 $n = 3$. Both give 2. The bush and the L^2 method agree, and Wolff’s hairbrush will give $\frac{n+2}{2} = \frac{5}{2}$.

$n = 4$. The bush gives $\frac{5}{2}$, the L^2 method 2. The bush wins from here on.

Large n . The bush gives $\frac{n+1}{2}$, growing linearly, while the L^2 method is stuck at 2. But the conjecture asks for n , and $\frac{n+1}{2}$ is asymptotically half of it — so in high dimensions even the better elementary bound recovers only half the truth.

The lesson for reading the literature. A bound of the form $\dim_{\mathbb{H}} K \geq cn$ with $c < 1$ fixed is, in high dimensions, qualitatively far from the conjecture, however impressive the constant. The bounds that matter asymptotically are those of the form $n - o(n)$, and the first of those came from the arithmetic methods of Volume 10.

WORKED EXAMPLE 9.4.3— TIER C: RESEARCH TASTE The bush as the first genuinely geometric argument

Theorem 9.4.4 is short, and its brevity can obscure that it is the first argument in this series to use the geometry of tubes rather than their statistics.

(a) *The one geometric input.* Theorem 9.4.2 rests on a single fact: two lines through a common point at angle θ are, at distance r from that point, about $r\theta$ apart. With $r \approx 1$ and $\theta \geq \delta$ this exceeds the tube width δ , so the tubes have separated. Nothing else about the configuration is used.

(b) *Why the unit distance matters.* The tubes have length 1 and the separation is $r\theta$, so the argument needs $r \approx 1$ to convert $\theta \geq \delta$ into separation $\geq \delta$. Shorter tubes would not do: a family of δ -tubes of length ρ gives separation $\rho\delta < \delta$ and the bush estimate fails. This is why the Kakeya problem is stated with unit segments, and why rescaling arguments must be careful.

(c) *What it does not use.* The bush argument never uses that the family has $\approx \delta^{-(n-1)}$ tubes, nor that the directions fill the sphere. It applies verbatim to any family with δ -separated directions, and gives $m \gtrsim \mu\delta^{n-1}$ for the sub-family through a point. That generality is why it survives into the later volumes as a lemma.

(d) *Its role later.* Volume 11’s polynomial partitioning argument repeatedly reduces to bounding a bush; Volume 14’s analysis of sticky configurations uses the bush estimate as the base case of an induction. The argument of this chapter is therefore not superseded by the later ones — it is absorbed

into them.

(e) *The honest assessment.* As a bound it is weak: half the conjectured exponent, and beaten by the L^2 method in the plane. As a lemma it is permanent. Distinguishing these two kinds of value is worth doing when reading any result in this area.

WORKED EXAMPLE 9.4.4— TIER B Producing the two competing bounds: a template

The bush argument, the hairbrush argument, and several later ones all have the shape “two bounds, multiply”. The template is worth stating so that the next instance can be recognised.

The setup. One wants a lower bound on a quantity V (here $m(E)$) and there is an unknown structural parameter μ (here the multiplicity). One produces:

$$(i) \quad V \gtrsim \mu A, \qquad (ii) \quad V \gtrsim \mu^{-1} B,$$

with A, B known. Then $V^2 \gtrsim AB$ and $V \gtrsim (AB)^{1/2}$.

Where each bound comes from. Bound (i) is *geometric*: it says a large μ forces a large configuration — here, that μ tubes through a point cannot be packed (Theorem 9.4.2). Bound (ii) is *measure-theoretic*: it is Chebyshev, $\int F \leq \|F\|_\infty m(\text{spt } F)$ (Theorem 9.4.3). The two always have this character.

How to improve the result. Only by improving A or B . In Chapter 9.5 the improvement is to A : collecting tubes along a line rather than at a point raises A from δ^{n-1} to δ^{n-2} . Improving B would mean a better Chebyshev step, and there is none — (ii) is an identity up to constants.

How to know when the template is exhausted. Compute the balancing value $\mu_* = (B/A)^{1/2}$ and ask whether a configuration with that multiplicity exists. Worked Example 9.8.2 carries this out and finds that for large n it does not — so the bound is being determined by a hypothetical worst case. At that point the template must be abandoned rather than refined.

Two later instances. Volume 11’s polynomial partitioning produces bounds of exactly this shape, with μ the number of tubes in a cell; and Volume 12’s broad–narrow decomposition splits according to a parameter and optimises, which is the same device with the roles of the two bounds played by the broad and narrow parts. Recognising the template makes those arguments readable in outline before working through them line by line.

Figure 9.4.1 (to be drawn). *A bush of tubes*

Content required: Six to eight δ -tubes through a common point x_0 , with δ -separated directions, shown converging at x_0 and visibly separated at distance ≈ 1 . Mark one pair with its angle θ and the separation $\approx r\theta$ at distance r , illustrating why the estimate needs $r \approx 1$ to convert $\theta \geq \delta$ into separation exceeding the tube width.

Drills

Drill 9.4.1. Verify the separation computation in Theorem 9.4.2: two lines through x_0 at angle θ are at distance $\gtrsim |y - x_0|\theta$ apart at the point y .

Drill 9.4.2. Show that Theorem 9.4.2 is sharp, by exhibiting μ tubes through a point whose union has measure $\approx \mu\delta^{n-1}$.

Drill 9.4.3. Explain why the two bounds of Theorem 9.4.4 cannot both be improved by the same argument, and identify the configuration that makes each sharp.

Drill 9.4.4. Combine Theorem 9.3.3 and Theorem 9.4.4 to state the best elementary bound in each dimension $n = 2, \dots, 6$.

Where we are. Worth recording explicitly, since it is easy to miss: Theorem 9.4.4 makes no use of the tubes' *positions*. The bush estimate needs only that the directions are δ -separated and that some point lies in μ tubes; the multiplicity estimate is Chebyshev. So the theorem holds for any family whatever, Besicovitch or not.

That generality is a strength and a diagnosis at once. A strength, because the result is reusable as a lemma — Worked Example 9.4.3(d) lists where. A diagnosis, because an argument insensitive to position cannot detect a Besicovitch configuration, which is defined by how its tubes are positioned. The bush improves on the L^2 method by conditioning on a point, but it still does not see the arrangement, and Chapter 9.5 is the first argument here that does.

Notes

The bush argument is Bourgain's (1991), though the geometric content — that tubes through a point separate at unit distance — is older and appears in Córdoba's work. Volume 1, Theorem 1.7.15 gave the same bound; the present formulation, in terms of the two competing estimates on the multiplicity, is the one that generalises.

The table in Worked Example 9.4.1 is the outline of this volume: each row is a chapter, and each conditions on a richer object than the last.

Chapter 9.5

The Hairbrush

The bush argument of Chapter 9.4 conditions on a point and the tubes through it. Wolff's improvement conditions on a *tube* and the tubes meeting it. The gain comes from a fact of elementary geometry which has no analogue for points: two lines that meet a common third line, and meet each other, are forced into a plane.

9.5.1 The geometry

Lemma 9.5.1 (Concurrent or coplanar). *Let $\ell_0, \ell_1, \ell_2 \subseteq \mathbb{R}^n$ be distinct lines such that ℓ_1 and ℓ_2 each meet ℓ_0 , and ℓ_1 meets ℓ_2 . Then either ℓ_0, ℓ_1, ℓ_2 have a common point, or they lie in a common 2-plane.*

Proof. Let $p = \ell_1 \cap \ell_2$, and let $q_i = \ell_i \cap \ell_0$ for $i = 1, 2$.

If $q_1 = q_2$ then that point lies on all three lines and we are in the first case.

Suppose $q_1 \neq q_2$. Both lie on ℓ_0 , so ℓ_0 is the line through q_1 and q_2 . If $p \in \ell_0$ then p, q_1, q_2 are collinear; since $p, q_1 \in \ell_1$ and $\ell_1 \neq \ell_0$, this forces $p = q_1$, and likewise $p = q_2$, contradicting $q_1 \neq q_2$. So $p \notin \ell_0$.

The three points p, q_1, q_2 are therefore not collinear and span a unique 2-plane P . Now ℓ_1 contains p and q_1 , hence $\ell_1 \subseteq P$; similarly $\ell_2 \subseteq P$; and ℓ_0 contains q_1, q_2 , so $\ell_0 \subseteq P$. $\square$

Where we are. This is the entire source of Wolff's improvement, and it is worth pausing on why there is no analogue for the bush.

In a bush, all tubes pass through one point, and two of them meet only there — Theorem 9.4.2 used exactly that. The lemma gives no additional structure, because the first alternative always holds.

In a hairbrush the tubes meet a *line*, and two of them can meet each other away from it. When they do, the second alternative applies and they are coplanar with ℓ_0 . So the tubes of a hairbrush that interact with one

another are organised into *planes through* ℓ_0 , and within each such plane the configuration is two-dimensional — where, by Theorem 9.3.3, the Kakeya problem is solved with only a logarithmic loss.

The hairbrush therefore reduces an n -dimensional problem to a family of planar ones, and planar Kakeya is sharp. That is the whole idea.

9.5.2 Transverse hairbrushes

Definition 9.5.2. Let $\mathbb{T}$ be a family of δ -tubes with δ -separated directions and let $T_0 \in \mathbb{T}$ have axis ℓ_0 . For $0 < \kappa \leq 1$ the κ -*transverse hairbrush* of T_0 is

$$H_\kappa(T_0) = \{T \in \mathbb{T} : T \cap T_0 \neq \emptyset, \angle(T, T_0) \geq \kappa\}.$$

Theorem 9.5.3 (Hairbrush volume estimate). *Let $H = H_\kappa(T_0)$ with $\kappa \geq \frac{1}{2}$, and write $m = \#H$. Then*

$$m\left(\bigcup_{T \in H} T\right) \gtrsim_n \frac{m \delta^{n-1}}{\log \frac{1}{\delta}}.$$

Idea of the proof. Sort the tubes of H by which 2-plane through ℓ_0 they lie near — Theorem 9.5.1 says this is the right invariant, because two tubes assigned to different planes cannot interact away from ℓ_0 . Within one plane the tubes form a planar Kakeya configuration inside a thin slab, and Theorem 9.3.3 bounds its area from below with only a logarithmic loss. Summing over planes gives the theorem, provided the slabs are essentially disjoint — which is where transversality is needed, since a tube nearly parallel to T_0 hugs ℓ_0 and never leaves the region where all the slabs meet.

Proof. Step 1. Sort into leaves.

Fix coordinates with ℓ_0 the x_n -axis. Each $T \in H$ meets T_0 , so its axis lies within 2δ of a line ℓ meeting ℓ_0 ; let $P(T)$ be the 2-plane spanned by ℓ_0 and ℓ . The set of such planes is parametrised by the direction of ℓ in the orthogonal complement of ℓ_0 , that is by a point of $\mathbb{S}^{n-2}$ modulo sign. Choose a maximal δ -separated set $\{P_1, \dots, P_M\}$ of such planes, so $M \approx_n \delta^{-(n-2)}$, and assign each $T \in H$ to a nearest P_i , obtaining a partition $H = \bigsqcup_i H_i$ with $m = \sum_i m_i$, $m_i = \#H_i$.

Step 2. *Each leaf lies in a slab.*

If $T \in H_i$ then every point of T lies within $O(\delta)$ of P_i : the axis of T meets ℓ_0 , has length 1, and its plane differs from P_i by angle $\leq \delta$, so it deviates from P_i by at most $1 \cdot \delta$; adding the tube radius gives $O(\delta)$. Write S_i for the $C\delta$ -neighbourhood of P_i , a slab of thickness $\approx \delta$ in $n - 2$ directions.

Step 3. *Planar Kakeya inside a leaf.*

Let $\pi_i : \mathbb{R}^n \rightarrow P_i \cong \mathbb{R}^2$ be the orthogonal projection. For $T \in H_i$, $\pi_i(T)$ contains a planar δ -tube, and distinct $T, T' \in H_i$ have δ -separated directions, hence so do their projections — the projection is an isometry on P_i and the tubes lie within $O(\delta)$ of it. By the planar case of Theorem 9.3.2, applied to the m_i projected tubes,

$$m_2\left(\bigcup_{T \in H_i} \pi_i(T)\right) \gtrsim \frac{m_i \delta}{\log \frac{1}{\delta}}.$$

Here the planar L^2 computation is run for a subfamily of m_i tubes rather than a full one: in (9.3.2) with $n = 2$ the count $\#\{j' : \theta \approx 2^m \delta\}$ is at most $\min\{2^m, m_i\}$, so the sum is $\lesssim \delta \log \frac{1}{\delta}$ per tube and $\|\sum_{H_i} \mathbf{1}\|_{L^2}^2 \lesssim m_i \delta \log \frac{1}{\delta}$; combining with $\sum_{H_i} m_2 = m_i \delta$ and Cauchy–Schwarz gives the display.

Since $\bigcup_{H_i} T \subseteq S_i$ and S_i has thickness $\approx \delta$ transverse to P_i ,

$$m\left(\bigcup_{T \in H_i} T\right) \gtrsim_n \delta^{n-2} \cdot \frac{m_i \delta}{\log \frac{1}{\delta}} = \frac{m_i \delta^{n-1}}{\log \frac{1}{\delta}}. \quad (9.5.1)$$

Step 4. *Different leaves separate — here transversality is used.*

Let $A = \{x : \text{dist}(x, \ell_0) \geq \frac{1}{8}\}$. If $T \in H$ has $\angle(T, T_0) \geq \kappa \geq \frac{1}{2}$ and T meets T_0 , then T leaves the $\frac{1}{8}$ -neighbourhood of ℓ_0 within distance $\frac{1}{8}/\sin \kappa \leq \frac{1}{4}$ of the crossing point, so at least half of T lies in A ; hence (9.5.1) holds with $\bigcup_{H_i} T$ replaced by $(\bigcup_{H_i} T) \cap A$, at the cost of an absolute constant.

Now two distinct planes $P_i, P_{i'}$ make an angle $\geq \delta$ along ℓ_0 , so at distance t from ℓ_0 their $C\delta$ -neighbourhoods are disjoint once $t\delta \gg \delta$, that is once $t \gtrsim 1$. Shrinking the constant in A if necessary, the sets $S_i \cap A$ have bounded overlap.

Step 5. *Sum.*

By Steps 3 and 4,

$$\begin{aligned} m\left(\bigcup_{T \in H} T\right) &\geq m\left(\bigcup_i \left(\bigcup_{T \in H_i} T\right) \cap A\right) \gtrsim_n \sum_i \frac{m_i \delta^{n-1}}{\log \frac{1}{\delta}} \\ &= \frac{m \delta^{n-1}}{\log \frac{1}{\delta}}. \end{aligned} \quad \square$$

Theorem 9.5.4 (Wolff’s bound, transverse case). *Let $\mathbb{T}$ be a family of δ -tubes with δ -separated directions, $N \approx_n \delta^{-(n-1)}$, and suppose that every point of $E = \bigcup T$ lies in at most μ tubes and that some $T_0 \in \mathbb{T}$ satisfies $\#H_{1/2}(T_0) \gtrsim \mu \delta^{-1}$. Then*

$$m(E) \gtrsim_n \frac{\delta^{\frac{n-2}{2}}}{\log \frac{1}{\delta}}.$$

Proof. Step 1. The hairbrush bound.

By hypothesis some T_0 has $m = \#H_{1/2}(T_0) \gtrsim \mu\delta^{-1}$. Since $\kappa = \frac{1}{2}$, Theorem 9.5.3 applies and gives

$$\begin{aligned} m(E) &\geq m\left(\bigcup_{T \in H_{1/2}(T_0)} T\right) \gtrsim_n \frac{m\delta^{n-1}}{\log \frac{1}{\delta}} \gtrsim_n \frac{\mu\delta^{-1}\delta^{n-1}}{\log \frac{1}{\delta}} \\ &= \frac{\mu\delta^{n-2}}{\log \frac{1}{\delta}}. \end{aligned}$$

This is the analogue of (9.4.1), improved by one power of δ : collecting tubes along the whole of T_0 rather than at a single point has raised the exponent from $n-1$ to $n-2$.

Step 2. *The multiplicity bound.*

Unchanged from the bush argument: Theorem 9.4.3 gives $m(E) \gtrsim_n \mu^{-1}$. Nothing in that estimate refers to the geometry, so no improvement is available there — as Worked Example 9.4.4 observes, it is Chebyshev and is sharp.

Step 3. *Multiply.*

As in Theorem 9.4.4 the product eliminates μ :

$$\begin{aligned} m(E)^2 &\gtrsim_n \frac{\mu\delta^{n-2}}{\log \frac{1}{\delta}} \cdot \mu^{-1} \\ &= \frac{\delta^{n-2}}{\log \frac{1}{\delta}}. \end{aligned}$$

Taking square roots, $m(E) \gtrsim_n \delta^{(n-2)/2} (\log \frac{1}{\delta})^{-1/2}$, and since $(\log \frac{1}{\delta})^{-1/2} \geq (\log \frac{1}{\delta})^{-1}$ for $\delta < \frac{1}{2}$ the stated form follows. $\square$

Remark 9.5.5 (Scope decision: the small-angle case). Wolff's theorem is that every Besicovitch set in $\mathbb{R}^n$ satisfies $\dim_{\mathbb{H}} K \geq \frac{n+2}{2}$, with no transversality hypothesis and no assumption that a tube meets $\mu\delta^{-1}$ others. *We do not prove that here*, and the gap is exactly the two hypotheses of Theorem 9.5.4. This is revision R9.

The obstruction is a single mechanism, and it is worth naming because the arithmetic of it is transparent.

Suppose one tries to produce the hypothesis $\#H \gtrsim \mu\delta^{-1}$ by counting. The axis T_0 has unit length, so it is covered by $\approx \delta^{-1}$ balls of radius δ ; if each has multiplicity μ the incidences number $\approx \mu\delta^{-1}$. But a tube making angle θ with T_0 meets $\approx \theta^{-1}$ of those balls, so

$$\#H \approx \frac{\mu\delta^{-1}}{\theta^{-1}} = \mu\theta\delta^{-1},$$

and running Theorem 9.5.4 with this gives $m(E)^2 \gtrsim \theta \delta^{n-2}$. At $\theta \approx 1$ this is Wolff's bound; at $\theta \approx \delta$ it degrades to δ^{n-1} , which is exactly Theorem 9.4.4 and no improvement at all.

So the near-parallel tubes must be treated separately, and Wolff does so by an induction on scales: a family of tubes confined to a narrow angular sector is, after rescaling, a Kakeya family in lower dimension, and the bound is applied recursively. Theorem 9.5.3 also requires transversality — Step 4 fails for small κ , since a tube nearly parallel to T_0 never leaves the region where all the leaves overlap.

Both difficulties are the same one seen twice, and resolving it is the content of Wolff's paper rather than a lemma.

Partial resolution. Theorem 9.5.8 below shows a *single* narrow hairbrush obeys the same bound as a transverse one, by a radial rescaling in which the two powers of θ^{-2} cancel exactly. So the small-angle geometry is not the obstruction; the dyadic sum over θ is.

What depends on the general case. When this remark was first written it asserted that nothing did. That is no longer accurate, and the correction is worth making precisely.

The complete proof in Volume 14 is now organised differently. Its theorem Theorem 14.3.1 follows from the two self-improving implications Theorems 14.3.11 and 14.3.12 and the sticky input, rather than from the retired $\mathcal{D}(\frac{1}{2}, 0)$ survey bootstrap. Slab concentration is controlled by the Katz–Tao and Frostman densities of Equations (14.3.3) and (14.3.8), with the explicit slab estimate Theorem 14.3.39; these are the precise modern replacements for the informal “Frostman Slab Wolff axiom” used in the earlier roadmap.

Why that substitution works. The obstruction above is that near-parallel tubes contribute $\#H \approx \mu\theta\delta^{-1}$ rather than $\mu\delta^{-1}$. Tubes making angle $\leq \theta$ with T_0 and meeting it are confined to a slab of thickness $O(\theta)$ about the plane they span, and the Frostman Slab axiom bounds their number by $O(\theta \cdot \#\mathbb{T})$ — exactly compensating the loss. *So the axiom is designed to make the hairbrush argument survive the small-angle case*, and Volume 14 adopts it for that reason.

What this means for R9. The general theorem remains unproved here. But the version Volume 14 needs is the Frostman Slab version, and the mechanism by which that hypothesis repairs the small-angle difficulty is the paragraph above. Volume 10's arithmetic bound and Volume 11's polynomial method supersede Wolff's theorem in the dimensions where they apply, and Volume 14 reports the current record.

n	L^2	bush $\frac{n+1}{2}$	hairbrush $\frac{n+2}{2}$
2	2	1.5	2
3	2	2	2.5
4	2	2.5	3
5	2	3	3.5
10	2	5.5	6

Reading the table. In the plane the L^2 method is already sharp and nothing else is needed. From $n = 3$ the hairbrush is the best of the three, and it beats the bush by exactly $\frac{1}{2}$ in every dimension — the gain is additive, not multiplicative.

What that means asymptotically. All three bounds are of the form $cn + O(1)$ with $c \leq \frac{1}{2}$, while the conjecture asks for n . So none of the methods of this chapter reaches even a fixed fraction better than one half, and the first bounds of the form $n - o(n)$ came from elsewhere — the arithmetic method of Volume 10, which gives $\frac{n}{2} + 1$ in a form that improves for large n , and the polynomial method of Volume 11.

A caution about the table. These are *Minkowski* bounds as proved here; the Hausdorff versions follow from Volume 7, Theorem 7.8.5, but only because the estimates carry $\delta^{-\varepsilon}$ losses for every ε . A bound with a fixed power loss would give a strictly weaker Hausdorff conclusion, by Drill 8.4 of Volume 7.

WORKED EXAMPLE 9.5.2— TIER B Why conditioning on a line beats conditioning on a point

The progression $L^2 \rightarrow \text{bush} \rightarrow \text{hairbrush}$ is a progression in what the argument conditions on, and the mechanism is worth isolating since Volumes 11 and 14 continue it.

Conditioning on nothing (L^2). One counts all pairs of tubes. By Worked Example 9.3.1 the count is dominated by generic transverse pairs and is blind to the configuration.

Conditioning on a point (bush). One fixes x_0 and considers the tubes through it. These separate at unit distance, giving $m \gtrsim \mu\delta^{n-1}$. But the tubes through a point interact with each other only at that point — Theorem 9.5.1's first alternative — so no further structure is available.

Conditioning on a line (hairbrush). One fixes T_0 and considers the tubes meeting it. Now Theorem 9.5.1's second alternative becomes available: tubes that interact are coplanar with ℓ_0 , and the configuration decomposes into planar problems. Since planarakeya is *solved*, each piece is handled sharply, and the only loss is the logarithm.

The pattern. Each step conditions on a higher-dimensional object and extracts more structure, and the reason it works is that the low-dimensional Kakeya problem is better understood than the high-dimensional one. That is precisely the design of the polynomial method in Volume 11: partition space by a polynomial surface, and induct on the dimension of the pieces.

Where the pattern stops. Conditioning on a 2-plane would require understanding three-dimensional Kakeya sharply, which was open until Wang and Zahl (2025). Volume 14 discusses whether their result enables such an argument in $\mathbb{R}^4$; at the time of writing it does not, and the reason is that their proof is not of the form the induction would need.

WORKED EXAMPLE 9.5.3 — TIER C: RESEARCH TASTE **What a full proof of Wolff’s theorem would require**

Theorem 9.5.5 descope the small-angle case. It is worth setting out what the missing argument looks like, both to make the boundary of this volume precise and because the technique — induction on scales — is used throughout Volumes 11 to 13.

(a) *The situation.* One has a hairbrush whose tubes make angle $\theta \ll 1$ with T_0 . They are confined to a slab of angular width θ about the direction of T_0 , and Theorem 9.5.3 gives nothing, since Step 4 fails.

(b) *The rescaling.* Apply the linear map that stretches the θ -narrow directions to unit width. The tubes become δ/θ -tubes with δ/θ -separated directions, in a configuration that is again of Kakeya type but at a coarser scale. So the problem at angular scale θ is the original problem at scale $\delta' = \delta/\theta$.

(c) *The induction.* Assume the desired bound at all scales coarser than δ and deduce it at δ . Each application of the rescaling reduces the number of scales remaining by a factor, so the induction terminates after $\approx \log \log \frac{1}{\delta}$ steps — and, as Worked Example 6.5.3 in Volume 6 showed, an induction of that depth costs a power of a logarithm, which $\delta^{-\varepsilon}$ absorbs.

(d) *Why it is delicate.* The rescaled family need not have the same structure as the original: the tubes may no longer be δ' -separated in all directions, and the multiplicity may change. Controlling those changes is the technical content of Wolff’s paper, and it is why the argument runs to several pages rather than the half-page the sketch above suggests.

(e) *What the reader should take from the omission.* Not that the theorem is doubtful — it is forty years old and universally used — but that the transverse case proved here contains the idea and the general case contains the bookkeeping. Theorem 9.5.1 is the mathematics; the induction is the

price of generality.

WORKED EXAMPLE 9.5.4— TIER B Verifying the leaf decomposition on a concrete configuration

Step 1 of Theorem 9.5.3 sorts the hairbrush into planes through ℓ_0 . It is worth checking the count and the slab geometry in $\mathbb{R}^3$, where everything can be pictured.

The setting. $n = 3$, ℓ_0 the z -axis, T_0 the δ -tube about it. Planes containing ℓ_0 are parametrised by an angle $\varphi \in [0, \pi)$ — the direction of the plane in the xy -plane. So the leaves are indexed by $\mathbb{S}^{n-2} = \mathbb{S}^1$ modulo sign, and a δ -separated set has $M \approx \delta^{-1} = \delta^{-(n-2)}$ members. ✓

A tube's assignment. A tube T meeting T_0 at height z_0 with direction making angle θ with the z -axis lies near the plane containing ℓ_0 and the direction of T ; that plane's parameter φ is the azimuth of T 's direction. Two tubes are in the same leaf when their azimuths agree to within δ .

The slab. A tube of length 1 whose plane differs from P_φ by angle $\leq \delta$ deviates from P_φ by at most δ over its length; adding the radius δ gives a slab of half-thickness $\approx 2\delta$. ✓ In $\mathbb{R}^3$ this slab is a genuine plate: two-dimensional in extent, δ -thick.

The separation of leaves. Two planes at azimuthal separation δ meet only along ℓ_0 and diverge linearly: at horizontal distance r from the z -axis they are $\approx r\delta$ apart. So their 2δ -slabs are disjoint once $r \gtrsim 1$, which is why Step 4 restricts to $\text{dist}(x, \ell_0) \geq \frac{1}{8}$. ✓

Where transversality enters, visibly. A tube at angle θ to the z -axis, meeting it at height z_0 , reaches horizontal distance $\approx \theta$ over its unit length. For $\theta \approx 1$ it exits the region $r \leq \frac{1}{8}$ almost immediately; for $\theta \approx \delta$ it never does, and lives entirely in the region where all δ^{-1} leaves overlap. That is Step 4's hypothesis, seen concretely, and it is the whole of Theorem 9.5.5.

The count, assembled. With m_i tubes in leaf i : planar area $\gtrsim m_i\delta/\log$, times slab thickness $\delta^{n-2} = \delta$, gives volume $\gtrsim m_i\delta^2/\log = m_i\delta^{n-1}/\log$. ✓ Summing over the $\approx \delta^{-1}$ leaves recovers Theorem 9.5.3.

WORKED EXAMPLE 9.5.5— TIER A The coplanarity lemma, tested

Theorem 9.5.1 is the pivot of the chapter, so it is worth testing against the cases where one expects it to be tight.

(a) *Three lines through a point.* ℓ_0, ℓ_1, ℓ_2 all through the origin, spanning $\mathbb{R}^3$. They pairwise meet, and the first alternative holds. The second fails — they are not coplanar — so the “or” is genuinely exclusive here, and the lemma is not asserting both.

(b) *Three lines in a plane, in general position.* A triangle’s three sides: each pair meets, at three distinct points. The second alternative holds and the first fails. So both alternatives occur, and neither implies the other.

(c) *Both hypotheses are needed.* Take ℓ_0 the z -axis and ℓ_1 the x -axis, which meet at the origin. Let $\ell_2 = \{(1, t, 1) : t \in \mathbb{R}\}$. Then ℓ_2 meets neither ℓ_0 — it has constant height 1 and constant first coordinate 1 — nor ℓ_1 , which sits at height 0. Adjusting to $\ell'_2 = \{(1, t, 0)\}$ makes ℓ'_2 meet ℓ_1 at $(1, 0, 0)$ but still miss ℓ_0 , and now ℓ_0 is not coplanar with ℓ_1, ℓ'_2 : the latter two span the plane $z = 0$, which does not contain the z -axis. So dropping the hypothesis that *both* ℓ_1 and ℓ_2 meet ℓ_0 destroys the conclusion, and the lemma is not a statement about three pairwise-meeting lines in general.

(d) *The quantitative version needed in practice.* Tubes, not lines. Two δ -tubes meeting a common δ -tube and meeting each other are *approximately* coplanar with it — within an angle depending on how far from T_0 they meet. Step 1 of Theorem 9.5.3 handles this by assigning each tube to a nearest plane and absorbing the error into the slab thickness. That the error is $O(\delta)$ over unit length is what makes the leaf decomposition work, and it is the only quantitative input.

(e) *Why there is no analogue for the bush.* In a bush all tubes meet at one point, so alternative (a) always holds and the lemma yields nothing. That is the precise sense in which conditioning on a line is strictly more informative than conditioning on a point.

Figure 9.5.1 (to be drawn). *The hairbrush and its leaves*

Content required: In $\mathbb{R}^3$: the tube T_0 along the z -axis, with tubes meeting it at various heights and angles. Sort them into two or three planes through ℓ_0 , drawn as slabs, and show the slabs separating once the horizontal distance from ℓ_0 exceeds ≈ 1 . Include one near-parallel tube that never leaves the overlap region — the configuration Step 4 excludes and Theorem 9.5.5 is about.

9.5.3 The small-angle case

Theorem 9.5.5 recorded the general theorem as unproved, the obstruction being hairbrushes whose tubes meet the stem at a small angle. This section resolves the obstruction for a *single* narrow hairbrush, and identifies precisely

what remains.

Definition 9.5.6 (Narrow hairbrush). For $\theta \in (\delta, 1)$ let

$$H_\theta(T_0) = \{T \in \mathbb{T} : T \cap T_0 \neq \emptyset, \angle(\text{dir } T, \text{dir } T_0) \approx \theta\}.$$

Lemma 9.5.7 (Radial rescaling). *Let ϕ_θ be the linear map fixing the axis of T_0 and stretching the two transverse directions by θ^{-1} . Then*

- (i) ϕ_θ carries δ -tubes of $H_\theta(T_0)$ to (δ/θ) -tubes;
- (ii) it carries angle $\approx \theta$ with T_0 to angle ≈ 1 ;
- (iii) it carries δ -separated directions to (δ/θ) -separated ones;
- (iv) $|\phi_\theta(X)| = \theta^{-2}|X|$.

Proof. Write T_0 along e_3 . A tube of $H_\theta(T_0)$ has direction $(\theta u_1, \theta u_2, 1)/|\cdot|$ with $|u| \approx 1$, and radius δ . Applying $\phi_\theta(x_1, x_2, x_3) = (x_1/\theta, x_2/\theta, x_3)$ sends the direction to $(u_1, u_2, 1)/|\cdot|$, of angle ≈ 1 with e_3 , which is (ii). The cross-section, a δ -disc transverse to the axis, is stretched to a (δ/θ) -disc, which is (i); the same computation applied to the direction sphere gives (iii). The Jacobian is θ^{-2} , which is (iv). $\square$

Proposition 9.5.8 (The narrow hairbrush obeys the transverse bound). *If $\#H_\theta(T_0) = M$ then*

$$\left| \bigcup_{T \in H_\theta(T_0)} T \right| \gtrsim M\delta^2,$$

the same bound as in the transverse case Theorem 9.5.4.

Proof. Apply ϕ_θ . By Theorem 9.5.7(i)–(iii) the image is a hairbrush of M tubes of radius $\delta' = \delta/\theta$, with δ' -separated directions, all meeting the image of T_0 at angle ≈ 1 . So Theorem 9.5.4 applies at scale δ' and gives

$$\left| \bigcup \phi_\theta(T) \right| \gtrsim M(\delta')^2 = \frac{M\delta^2}{\theta^2}.$$

By Theorem 9.5.7(iv) the left side is θ^{-2} times $|\bigcup T|$, and the two factors of θ^{-2} cancel exactly. $\square$

Where we are. *The cancellation is exact, and that is the content: a narrow hairbrush is not weaker than a transverse one, it is the same object seen through an anisotropic lens. The two powers of θ^{-2} — one from the rescaled radius, one from the Jacobian — are forced to match because the map is linear and the bound is homogeneous of the right degree.*

Where the difficulty actually lies. Not in any single H_θ , but in the sum over dyadic θ : the sets $\bigcup H_\theta(T_0)$ for different θ overlap, and bounding the union of the whole hairbrush by the sum of its pieces loses a factor $\log \frac{1}{\delta}$ at best and much more if the overlaps are severe. Wolff’s paper organises this sum, and it is that organisation — not the small-angle geometry — that Theorem 9.5.5 records as unproved here.

What is gained by knowing this. The obstruction is now localised to a combinatorial bookkeeping problem, and Theorem 9.5.8 shows the geometric input is complete. Volume 14, Theorem 14.3.1 circumvents the bookkeeping by factoring tube families into Katz–Tao and Frostman pieces and then applying Theorems 14.3.34 and 14.3.39; the relevant overlap bounds are therefore proved hypotheses of the modern reduction, not an extra axiom.

WORKED EXAMPLE 9.5.6— TIER A **Why the rescaling must be anisotropic**

Theorem 9.5.7 stretches two directions and fixes the third. An isotropic dilation would not work, and seeing why clarifies the rôle of the stem.

The isotropic attempt. Dilating all of $\mathbb{R}^3$ by θ^{-1} multiplies every length by θ^{-1} : the tubes become (δ/θ) -tubes of *length* θ^{-1} , not unit length, and their angles with T_0 are unchanged — dilation is conformal. Nothing is gained.

What the anisotropic map does. It changes angles, because it is not conformal. That is exactly what is needed: the small-angle hypothesis is a statement about directions, and only a map distorting directions can remove it.

The price. The stem T_0 is distorted from a δ -tube to a (δ/θ) -tube, so the rescaled configuration is a hairbrush with a *fatter* stem. That is harmless, since Theorem 9.5.4 uses the stem only to define the thickened planes.

The same device elsewhere. Volume 14, Theorem 14.2.13 uses an anisotropic rescaling for the same reason — to turn a ρ -tube into a unit ball while keeping the objects inside it tubes — and Theorem 14.3.36 uses one to flatten a plank factor. In all three the map is linear, hence preserves the ratios that the hypotheses are stated in terms of; that is what makes the substitution legitimate.

Drills

Drill 9.5.1. Verify Theorem 9.5.1 in $\mathbb{R}^3$ by writing the three lines parametrically and solving.

Drill 9.5.2. Check the count $M \approx \delta^{-(n-2)}$ of δ -separated 2-planes through a fixed line in $\mathbb{R}^n$.

Drill 9.5.3. Carry out the modified planar L^2 computation of Step 3 for a subfamily of m tubes, confirming the bound $m\delta \log \frac{1}{\delta}$.

Drill 9.5.4. Show that Step 4 fails for $\kappa = \delta$: exhibit tubes meeting T_0 at angle δ whose leaves overlap along their whole length.

Notes

The hairbrush argument is Wolff's (1995), and Theorem 9.5.1 is the observation it rests on. The bound $\frac{n+2}{2}$ was the best known for general n until Katz–Labat–Tao's improvement in $\mathbb{R}^3$ and the arithmetic method of Volume 10.

Theorem 9.5.5 records what is not proved here. Readers wanting the full argument should consult Wolff's paper or his lecture notes, where the small-angle induction is carried out.

Chapter 9.6

Nikodym Sets

A Besicovitch set contains a segment in every *direction*. A Nikodym set contains a segment through every *point*. The two look like different problems and are the same one, related by a projective transformation which interchanges position and direction. This chapter proves the equivalence and draws the consequences.

9.6.1 The two definitions

Definition 9.6.1. A compact set $N \subseteq \mathbb{R}^n$ is a *Nikodym set* if for every $x \in [0, 1]^n$ there is a line ℓ_x through x such that $\ell_x \cap N$ contains a segment of unit length having x as an endpoint.

Definition 9.6.2. The *Nikodym maximal function* at scale δ is

$$\mathcal{N}_\delta f(x) = \sup_{T \ni x} \frac{1}{m(T)} \int_T |f|,$$

the supremum over all δ -tubes containing x .

The contrast with Theorem 9.1.1 is exactly the contrast between the two set problems: $\mathcal{K}_\delta$ is a function of a *direction* and takes the supremum over positions, while $\mathcal{N}_\delta$ is a function of a *position* and takes the supremum over directions.

9.6.2 The projective duality

Definition 9.6.3. For $x = (x', x_n) \in \mathbb{R}^{n-1} \times (0, \infty)$ set

$$\Phi(x', x_n) = \left(\frac{x'}{x_n}, \frac{1}{x_n} \right).$$

Lemma 9.6.4 (Position and direction are interchanged). *Let $a, b \in \mathbb{R}^{n-1}$ and let $\ell_{a,b}$ be the line $\{(a + tb, t) : t \in \mathbb{R}\}$, which passes through the point $(a, 0)$ with direction $(b, 1)$. Then for $t > 0$,*

$$\Phi(\ell_{a,b}) = \{(b + sa, s) : s > 0\} = \ell_{b,a} \cap \{x_n > 0\}.$$

That is, Φ carries the line with position a and direction b to the line with position b and direction a .

Proof. A point of $\ell_{a,b}$ with $t > 0$ is $(a + tb, t)$, and

$$\Phi(a + tb, t) = \left(\frac{a + tb}{t}, \frac{1}{t} \right) = \left(\frac{a}{t} + b, \frac{1}{t} \right).$$

Substituting $s = 1/t$, which ranges over $(0, \infty)$ as t does, this is $(sa + b, s)$, a point of $\ell_{b,a}$. The correspondence $t \leftrightarrow s$ is a bijection of $(0, \infty)$, so the images coincide. $\square$

Lemma 9.6.5 (Φ preserves dimension on slabs). *For $0 < c < C < \infty$, the restriction of Φ to $\{x : c \leq x_n \leq C, |x'| \leq C\}$ is bi-Lipschitz. Consequently $\dim_H \Phi(A) = \dim_H A$ for every A in that region.*

Proof. Step 1. Compute the derivative.

Φ is smooth on $\{x_n > 0\}$, and differentiating $\Phi(x', x_n) = (x'/x_n, 1/x_n)$ componentwise,

$$d\Phi = \begin{pmatrix} x_n^{-1} I_{n-1} & -x_n^{-2} x' \\ 0 & -x_n^{-2} \end{pmatrix}, \quad |\det d\Phi| = x_n^{-(n-1)} \cdot x_n^{-2} = x_n^{-(n+1)},$$

the determinant being that of a block triangular matrix.

Step 2. *Bound the derivative above and below on the region.*

Let $R = \{x : c \leq x_n \leq C, |x'| \leq C\}$. On R every entry of $d\Phi$ is bounded in modulus by $\max\{c^{-1}, Cc^{-2}\}$, so $\|d\Phi\| \leq L_1(c, C)$. And $|\det d\Phi| = x_n^{-(n+1)} \geq C^{-(n+1)}$, so by Cramer's rule the inverse matrix also has bounded entries and $\|(d\Phi)^{-1}\| \leq L_2(c, C)$.

Step 3. *From the derivative to a Lipschitz bound.*

R is convex — it is a product of a ball and an interval — so for $x, y \in R$ the segment $[x, y]$ lies in R and the mean value inequality gives

$$|\Phi(x) - \Phi(y)| \leq L_1 |x - y|.$$

Applying the same argument to $\Phi^{-1} = \Phi$ on the image $\Phi(R)$, which is again of the form treated in Step 2 after adjusting c and C , gives $|\Phi(x) - \Phi(y)| \geq L_2^{-1} |x - y|$.

Step 4. *Conclude.*

So $\Phi \upharpoonright R$ is bi-Lipschitz. By Theorem 7.1.5(iii) applied to Φ , $\dim_{\mathbb{H}} \Phi(A) \leq \dim_{\mathbb{H}} A$; applied to Φ^{-1} on $\Phi(A)$, the reverse. Hence $\dim_{\mathbb{H}} \Phi(A) = \dim_{\mathbb{H}} A$ for $A \subseteq R$. $\square$

Theorem 9.6.6 (Kakeya and Nikodym are equivalent). *Let $s \in [0, n]$. The following are equivalent.*

- (i) *Every Besicovitch set in $\mathbb{R}^n$ has $\dim_{\mathbb{H}} \geq s$.*
- (ii) *Every Nikodym set in $\mathbb{R}^n$ has $\dim_{\mathbb{H}} \geq s$.*

Proof. Step 1. From a Nikodym set to a Besicovitch set.

Assume (i) and let N be a Nikodym set. By Theorem 9.6.1 every $x \in [0, 1]^n$ lies on a line meeting N in a unit segment. Restrict attention to the slab $Q = \{x : \frac{1}{2} \leq x_n \leq 1, |x'| \leq 1\}$; the points of Q still have this property, and we may take the segments to lie in the region $c \leq x_n \leq C$ after discarding a bounded portion and rescaling, since a unit segment through a point of Q meets that region in a segment of length $\gtrsim 1$.

Consider $\Phi(N)$. Let $(b, 1)$ be any direction with $|b| \leq 1$. The point $(b, 0)$ — or rather a nearby point of Q after the normalisation above — lies on a line $\ell_{b,a}$ meeting N in a segment, for some a . By Theorem 9.6.4, Φ maps $\ell_{b,a}$ to $\ell_{a,b}$, which has direction $(b, 1)$. So $\Phi(N)$ contains a segment in the direction $(b, 1)$, and as b ranges over $|b| \leq 1$ these directions fill an open subset of $\mathbb{S}^{n-1}$.

A compact set containing a segment in every direction of an open subset of the sphere has the same dimension bound as a Besicovitch set: cover the sphere by boundedly many rotated copies of that subset and take the union of the corresponding sets, using countable stability (Theorem 7.1.5(ii)). So $\dim_{\mathbb{H}} \Phi(N) \geq s$ by (i), and $\dim_{\mathbb{H}} N = \dim_{\mathbb{H}} \Phi(N) \geq s$ by Theorem 9.6.5.

Step 2. *From a Besicovitch set to a Nikodym set.*

Assume (ii) and let K be a Besicovitch set. Running the same argument with Φ^{-1} — which is Φ itself, since $\Phi(\Phi(x)) = x$ by direct computation — and with the roles of a and b exchanged, $\Phi(K)$ contains, through every point of a suitable cube, a segment; so $\Phi(K)$ contains a Nikodym set up to the same covering argument, and $\dim_{\mathbb{H}} K = \dim_{\mathbb{H}} \Phi(K) \geq s$. $\square$

Where we are. The map Φ is an involution — $\Phi \circ \Phi = \text{id}$ — which is why the two directions of Theorem 9.6.6 are the same argument. It is the affine chart expression of the projective duality between points and hyperplanes, and the identity

$$\text{position } a, \text{ direction } b \quad \longleftrightarrow \quad \text{position } b, \text{ direction } a$$

of Theorem 9.6.4 is the whole content.

The reason the equivalence is worth having is that the two problems suggest different attacks. The Kakeya formulation makes the directions the parameter and is what the maximal function of Chapter 9.1 is built on. The Nikodym formulation makes the points the parameter and is closer to the maximal operators of Volume 1, where the supremum is over sets containing a given point. Some arguments are more natural on one side — the polynomial method of Volume 11 is usually run on the Nikodym side — and the duality lets one choose.

WORKED EXAMPLE 9.6.1— TIER A **Nikodym’s original example**

Nikodym constructed, in 1927, a set $N \subseteq [0, 1]^2$ with $m(N) = 1$ such that for every $x \in N$ there is a line ℓ_x with $\ell_x \cap N = \{x\}$. The complement of N in the square therefore has measure zero and contains, through every point of N , a punctured segment.

Why this is startling. It says a set of *full* measure can be arranged so that every one of its points is isolated along some line. The intuition that a full-measure set must be “thick in every direction at almost every point” is simply false.

Its relation to Theorem 9.6.1. The complement is essentially a Nikodym set in the sense used here: it has measure zero and meets a line through every point of N in a segment. The definitions in the literature differ in whether the segment must contain the point or merely approach it, and the variants are equivalent for dimension purposes.

Why it does not settle anything. As with Besicovitch sets, measure zero is compatible with full dimension — Worked Example 7.1.4 in Volume 7 made this point — and Nikodym’s construction says nothing about the dimension. By Theorem 9.6.6 the dimension question is exactly as hard as Kakeya.

WORKED EXAMPLE 9.6.2— TIER B **Transferring an estimate across the duality**

Theorem 9.6.6 is stated for dimensions. The corresponding statement for maximal functions requires more care, because Φ distorts measure, and it is worth seeing where the exponent n comes from.

The distortion. By Theorem 9.6.5, $|\det d\Phi| = x_n^{-(n+1)}$. So a change of variables $y = \Phi(x)$ in an integral introduces a weight x_n^{n+1} , which is bounded above and below only on a slab.

Why L^n is the natural exponent. Suppose one wants to transfer a bound $\|\mathcal{K}_\delta f\|_{L^p(d\sigma)} \lesssim \|f\|_{L^p}$ to $\|\mathcal{N}_\delta g\|_{L^p(\mathbb{R}^n)} \lesssim \|g\|_{L^p}$. On the Kakeya side the left norm is over an $(n-1)$ -dimensional sphere and on the Nikodym side over an n -dimensional cube; the tube geometry contributes δ^{n-1} on both sides.

Matching the homogeneities under the scaling $f \mapsto f(\lambda \cdot)$ forces $p = n$, which is the same computation that Worked Example 9.1.2 performed directly on the Kakeya side.

The transfer, in words. Decompose the region into slabs $2^{-k-1} \leq x_n \leq 2^{-k}$, on each of which Φ is bi-Lipschitz with constants $\approx 2^k$; apply the Kakeya bound after rescaling the slab to unit size; and sum over k , which costs a factor $\log \frac{1}{\delta}$. Since every statement in this subject carries $\delta^{-\varepsilon}$, the logarithm is invisible.

The upshot. $\mathcal{K}_\delta$ and $\mathcal{N}_\delta$ satisfy the same L^n bounds up to $\delta^{-\varepsilon}$, and Theorem 9.1.2 has an exactly equivalent Nikodym form. Tao's paper on the equivalence carries out the bookkeeping in full; the point for a reader here is that the equivalence is not formal — it uses that Φ maps lines to lines, and no other feature.

WORKED EXAMPLE 9.6.3— TIER C: RESEARCH TASTE **Which side to work on**

The duality is free, so the only question is which formulation makes a given argument easier. Four considerations.

(a) *Separation.* On the Kakeya side the tubes have δ -separated *directions*, and their positions are arbitrary. On the Nikodym side the positions are spread and the directions are arbitrary. An argument that needs one kind of separation should be run on the side that supplies it. The bush argument of Chapter 9.4 used direction-separation and is naturally a Kakeya argument.

(b) *Incidence geometry.* Statements about incidences between points and lines — which is what the polynomial method of Volume 11 counts — are symmetric in points and lines, so the duality is invisible there. That is why those arguments are usually presented in whichever form is notationally lighter.

(c) *The ambient measure.* The Nikodym formulation lives entirely in $\mathbb{R}^n$, with no sphere; for arguments that decompose space — polynomial partitioning, induction on scales — this is a genuine simplification.

(d) *The extremal examples.* Besicovitch's construction is explicit (Volume 1) and Nikodym's is not much harder, but they are not images of each other under Φ in any illuminating way: the duality preserves dimension, not the construction. So intuition about extremal configurations should be developed on the Kakeya side, where the Perron tree is available, and results transferred afterwards.

A caution. The equivalence is for the *dimension* problems and, with the bookkeeping of Worked Example 9.6.2, for the maximal estimates at L^n . It

is not an equivalence of every associated question: the Nikodym maximal operator has a different behaviour at exponents other than n , and statements about specific sets do not transfer, since Φ does not map Besicovitch sets to Nikodym sets in a canonical way.

WORKED EXAMPLE 9.6.4— TIER B Dualising an argument: the bush becomes what?

The duality of Theorem 9.6.6 lets anyakeya argument be transported to the Nikodym side. Doing this once, on the bush, shows what the transport does to the geometry.

Theakeya statement. Theorem 9.4.2: μ tubes with δ -separated *directions* through a common *point* have union of measure $\gtrsim \mu\delta^{n-1}$, because they separate at unit distance from the point.

Applying Φ . By Theorem 9.6.4, position and direction are exchanged. A family of lines through a common point, with separated directions, becomes a family of lines with a common direction, through separated points — that is, a family of *parallel* tubes at separated positions.

The dual statement. μ parallel δ -tubes whose positions are δ -separated have union of measure $\approx \mu\delta^{n-1}$ — which is trivially true, since they are disjoint.

What this shows. The bush estimate is, on the Nikodym side, a triviality. That is not a defect of the duality: it means the difficulty of the bush argument lies entirely in the *quantifier structure* — which point, which tubes — and not in the geometry of the separation. The content of Theorem 9.4.4 is the combination with Theorem 9.4.3, and that combination does not simplify under Φ .

The general moral. A duality that makes one lemma trivial has not made the theorem easier; it has relocated the difficulty. When transporting an argument, identify which step carries the content before deciding the transport was worthwhile. Here the content is the optimisation over μ (Worked Example 9.8.2), and it is invariant under Φ .

WORKED EXAMPLE 9.6.5— TIER A The duality in the plane, drawn out

In $\mathbb{R}^2$ the transformation $\Phi(x_1, x_2) = (x_1/x_2, 1/x_2)$ can be followed explicitly, and doing so makes Theorem 9.6.4 concrete.

A single line. Take $a = 0$, $b = 1$: the line through the origin with direction $(1, 1)$, that is $\{(t, t)\}$. Then $\Phi(t, t) = (1, 1/t)$, the *vertical* line $x_1 = 1$. So a line of slope 1 through the origin becomes a vertical line through $(1, 0)$ —

position and direction have swapped, and “slope 1” has become “position 1”. *A family through a point.* The lines through the origin, one per direction, are $\{(t, \lambda t)\}$ for $\lambda \in \mathbb{R}$. Their images are $\Phi(t, \lambda t) = (1/\lambda, 1/(\lambda t))$ — the vertical line $x_1 = 1/\lambda$. So a *pencil through a point* becomes a *family of parallel lines*, which is exactly the dual statement computed abstractly in Worked Example 9.6.4.

A family in all directions. Conversely, lines with all slopes but through different points — the raw material of a Besicovitch set — map to lines through all points but with restricted slopes, the raw material of a Nikodym set.

The involution. $\Phi(\Phi(x_1, x_2)) = \Phi(x_1/x_2, 1/x_2) = ((x_1/x_2) \cdot x_2, x_2) = (x_1, x_2)$.
✓

What to retain. The single sentence: Φ turns pencils into parallel families. Everything else in this chapter is that observation with bookkeeping.

WORKED EXAMPLE 9.6.6— TIER A Nikodym’s example, and the intuition it destroys

Worked Example 9.6.1 recorded Nikodym’s set. The example deserves a second look, because the intuition it refutes is one most readers hold.

The false intuition. If $E \subseteq [0, 1]^2$ has full measure, then almost every point of E should be a point of density 1 (Theorem 1.7.4) — so E should look, near almost every one of its points, like the whole plane. In particular one expects that through almost every point of E , *every* line meets E in a set of positive length near that point.

What Nikodym showed. There is a full-measure E such that through *every* point of E there is a line meeting E only at that point. So the expectation is false for every point, not merely almost every one.

Reconciling the two. The Lebesgue density theorem is a statement about *balls*: $m(E \cap B(x, r))/m(B(x, r)) \rightarrow 1$. A line is a null set, and the density theorem says nothing about null sets. The intuition slides from “ E fills almost all of every small ball” to “ E fills almost all of every small segment”, and the second does not follow.

The general warning. Almost-everywhere statements with respect to Lebesgue measure carry no information along any fixed lower-dimensional family. Volume 8’s projection theorems are the systematic response: they establish what *is* true along almost every member of such a family, and Worked Example 8.1.1 there showed that “almost every” cannot be improved to “every”.

Why it belongs in this chapter. By Theorem 9.6.6 the dimension question for

Nikodym sets is exactly Kakeya. So the counterintuitive example and the hard conjecture are two faces of the same phenomenon: lines see something that balls do not, and measure theory alone cannot control it.

Figure 9.6.1 (to be drawn). *The projective duality*

Content required: Two panels. Left: a pencil of lines through the origin in the plane, one per slope. Right: their images under Φ , a family of parallel vertical lines at positions $1/\lambda$. Label the correspondence “slope λ ” $\leftrightarrow$ “position $1/\lambda$ ”, which is Theorem 9.6.4 in one picture.

Drills

Drill 9.6.1. Verify that $\Phi \circ \Phi = \text{id}$ on $\{x_n > 0\}$.

Drill 9.6.2. Compute $d\Phi$ and confirm $|\det d\Phi| = x_n^{-(n+1)}$.

Drill 9.6.3. Show that a compact set containing a unit segment in every direction of an open subset of $\mathbb{S}^{n-1}$ has the same dimension lower bound as a Besicovitch set, by the covering argument of Step 1.

Drill 9.6.4. Carry out the scaling computation of Worked Example 9.6.2 and confirm that $p = n$ is forced.

Where we are. The duality of this chapter is the third structural equivalence the series has met, and the three are worth comparing.

Volume 6, Theorem 6.1.3 related restriction to extension by adjointness: a formal identity, with the content lying in the observation that one side is computable and the other is not.

Volume 7, Theorem 7.5.1 related a covering condition to the existence of a measure: not formal at all, and the content is the construction.

Theorem 9.6.6 relates two geometric problems by an explicit change of variable. It is formal in the sense that Φ is written down and verified in two lines, but it is not trivial: Theorem 9.6.5 must check that Φ preserves dimension, and that requires restricting to a slab, which in turn requires the covering argument of Step 1.

The common lesson is that a useful equivalence is one that moves a problem into a category where different tools apply. Restriction became extension so that $\mathcal{E}g$ could be written down; a dimension hypothesis became a measure so that integrals could be estimated; and Kakeya becomes Nikodym so that arguments which decompose $\mathbb{R}^n$ — rather than the sphere — may be run. Volume 11 uses the last of these throughout.

Notes

Nikodym's example is from 1927. The equivalence of the Kakeya and Nikodym maximal conjectures is due to Tao (1999); the dimension statement, Theorem 9.6.6, is the easier half and the projective duality underlying it is classical.

The transformation Φ is the standard affine chart for projective duality and appears throughout incidence geometry; Volume 11 uses it again when translating between point-line incidence bounds and tube estimates.

Chapter 9.7

From Single Scales to Dimension

Every estimate in this volume bounds $m(E)$ for a family of δ -tubes at one fixed δ . The Kakeya conjecture is about Hausdorff dimension, which is a statement about all scales at once. Volume 7 supplied the bridge; this chapter walks across it, records exactly what is lost, and assembles the resulting dimension bounds.

9.7.1 The dictionary

Proposition 9.7.1 (Single-scale bounds become Minkowski bounds). *Suppose that for every family of δ -tubes with δ -separated directions one has $m(\bigcup T) \gtrsim_n \delta^\beta$, uniformly in δ . Then every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\overline{\dim}_M K \geq n - \beta$.*

Proof. Step 1. Produce an admissible family inside K_δ .

Let $\{v_1, \dots, v_N\}$ be a maximal δ -separated subset of $\mathbb{S}^{n-1}$, so $N \approx_n \delta^{-(n-1)}$. Since K is a Besicovitch set it contains a unit segment ℓ_j in each direction v_j . Let T_j be the δ -tube about ℓ_j ; then $T_j \subseteq K_\delta$ by definition of the neighbourhood, and $\{T_j\}$ has δ -separated directions, so the hypothesis applies to it.

Step 2. *Transfer the bound.*

$$m(K_\delta) \geq m\left(\bigcup_j T_j\right) \gtrsim_n \delta^\beta.$$

Step 3. *Convert to a dimension statement.*

By Theorem 7.2.3,

$$\overline{\dim}_M K = n - \liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta}.$$

Step 2 gives $m(K_\delta) \geq c\delta^\beta$ for small δ . Taking logarithms and dividing by $\log \delta$, which is negative, reverses the inequality:

$$\begin{aligned} \frac{\log m(K_\delta)}{\log \delta} &\leq \frac{\log c + \beta \log \delta}{\log \delta} \\ &= \beta + \frac{\log c}{\log \delta} \rightarrow \beta. \end{aligned}$$

Hence $\liminf_{\delta \rightarrow 0} \frac{\log m(K_\delta)}{\log \delta} \leq \beta$ and $\overline{\dim}_M K \geq n - \beta$. □

Theorem 9.7.2 (The bounds of this volume, in dimension form). *Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies*

$$\overline{\dim}_M K \geq \begin{cases} 2, & \text{by Theorem 9.3.3, all } n, \\ \frac{n+1}{2}, & \text{by Theorem 9.4.4,} \end{cases}$$

and, under the transversality hypotheses of Theorem 9.5.4, $\overline{\dim}_M K \geq \frac{n+2}{2}$. Moreover the same bounds hold for $\dim_H K$, and for every Nikodym set by Theorem 9.6.6.

Proof. The Minkowski statements are Theorem 9.7.1 applied with $\beta = n - 2$, $\beta = \frac{n-1}{2}$ and $\beta = \frac{n-2}{2}$ respectively; logarithmic factors do not affect the exponent.

For the Hausdorff statements, each of the three estimates holds with a loss $\delta^{-\varepsilon}$ for every $\varepsilon > 0$ — indeed with at most a logarithmic loss, which is stronger. By Theorem 9.1.3 each is equivalent to the correspondingakeya maximal bound $K(s)$ of Volume 7, Theorem 7.8.5, with $s = n - \beta$, and that theorem converts it into $\dim_H K \geq s$. The Nikodym statements follow from Theorem 9.6.6. □

Where we are. Theorem 9.7.2 is the reason this volume was allowed to work entirely at a single scale. It is worth being precise about what the licence requires: *the single-scale estimate must hold with a loss $\delta^{-\varepsilon}$ for every $\varepsilon > 0$.*

All three estimates here satisfy that — Theorem 9.3.3 and Theorem 9.5.3 lose a logarithm and Theorem 9.4.4 loses nothing. Had any of them lost a

fixed power δ^{-c} , Volume 7's Theorem 7.8.5 would give only $\dim_{\text{H}} K \geq s - sc$ for the corresponding s , by Drill 8.4 there, and the Hausdorff bound would be strictly weaker than the Minkowski one.

This is worth checking whenever a new estimate appears. A paper proving a tube-counting bound with an unspecified constant is proving a Minkowski statement; the Hausdorff statement needs the ε .

9.7.2 What is lost at each step

WORKED EXAMPLE 9.7.1— TIER B **Tracing a bound from tubes to dimension**

It is instructive to follow one estimate the whole way and see where the losses occur. Take the bush bound in $\mathbb{R}^4$.

Step 1: the tube estimate. Theorem 9.4.4 gives $m(\bigcup T) \gtrsim \delta^{(n-1)/2} = \delta^{3/2}$, with an absolute implied constant and *no* loss.

Step 2: to the neighbourhood of a Besicovitch set. Theorem 9.7.1 gives $m(K_\delta) \gtrsim \delta^{3/2}$. No loss — the tubes are inside K_δ by definition.

Step 3: to Minkowski dimension. $\overline{\dim}_{\text{M}} K \geq 4 - \frac{3}{2} = \frac{5}{2}$. No loss, but note this is an limsup statement: it says nothing about $\underline{\dim}_{\text{M}}$, and by Theorem 7.2.5 the two can differ.

Step 4: to the maximal function. Theorem 9.1.3 converts the multiplicity bound into a bound for $\mathcal{K}_\delta$, at the cost of a constant. The equivalence is not free in general — Step 2 of that proof used Marcinkiewicz interpolation — but the losses are constants, not powers.

Step 5: to Hausdorff dimension. Volume 7, Theorem 7.8.5 gives $\dim_{\text{H}} K \geq \frac{5}{2}$. *This step costs the ε :* the proof there pigeonholes over $\approx \log \frac{1}{\delta}$ dyadic scales and over directions, each costing a power of a logarithm, and the $\delta^{-\varepsilon}$ absorbs them.

The tally. Four of the five steps are lossless, and the one that is not loses only logarithms. So the chain is tight, and the exponent $\frac{5}{2}$ in the Hausdorff statement is exactly the exponent in the tube estimate. That is why the literature quotes tube-counting exponents and dimension exponents interchangeably, and it is legitimate precisely because of this accounting.

WORKED EXAMPLE 9.7.2— TIER A **The best elementary bound in each dimension**

Assembling Theorem 9.3.3, Theorem 9.4.4 and, where its hypotheses apply, Theorem 9.5.4:

n	best proved here	method	conjecture
2	2	L^2	2 (settled)
3	2	L^2 or bush	3
4	2.5	bush	4
5	3	bush	5
10	5.5	bush	10

With Wolff's theorem in full — not proved here, Theorem 9.5.5 — the entries for $n \geq 3$ become $\frac{n+2}{2}$: 2.5, 3, 3.5, 6.

Reading the table. The planar case is settled. Every other entry is at most about half the conjectured value, and the ratio does not improve with n . So the elementary methods, taken together, establish that a Besicovitch set is at least half-dimensional and no more.

Where the next improvements come from. Volume 10's arithmetic method gives bounds exceeding $\frac{n+2}{2}$ in high dimensions by combining sumset inequalities with the tube geometry; Volume 11's polynomial method gives the sharp result in the finite-field model and strong bounds in $\mathbb{R}^n$; Volume 14 reports Wang and Zahl's theorem, which settles $n = 3$. None of them is elementary in the sense of this volume.

WORKED EXAMPLE 9.7.3— TIER C: RESEARCH TASTE Minkowski, Hausdorff, and what a paper is actually claiming

Theorem 9.7.2 produced both Minkowski and Hausdorff bounds from the same tube estimates. The two are not interchangeable, and the literature is not always explicit about which is meant.

(a) *The logical order.* A tube estimate gives a Minkowski bound immediately (Theorem 9.7.1) and a Hausdorff bound only through Volume 7, Theorem 7.8.5, which requires the loss to be $\delta^{-\varepsilon}$ for every ε . So Minkowski is the primitive statement and Hausdorff the derived one.

(b) *The gap is real.* By Worked Example 7.2.1 in Volume 7 the two dimensions can differ by the whole range. For Besicovitch sets they are believed equal — both conjecturally n — but that is a conjecture, not a theorem, and in dimensions where only Minkowski bounds are known the Hausdorff question is formally open.

(c) *Historically the Minkowski bounds came first.* Davies proved $\dim_{\text{H}} = 2$ in the plane in 1971; the corresponding Minkowski statement is Córdoba's L^2 argument and is easier. In higher dimensions the pattern repeats: a Minkowski bound appears, and the Hausdorff version follows once the estimate is upgraded to carry $\delta^{-\varepsilon}$.

(d) *Three questions to ask of any stated bound.* Which dimension? What is the loss — $\delta^{-\varepsilon}$ for every ε , or a fixed power? And is it for all Besicovitch sets or a restricted class? Worked Example 7.8.4 in Volume 7 gave the same checklist; this chapter has now supplied the machinery that makes the answers computable.

(e) *The convention adopted here.* Every bound in this volume is proved as a tube estimate with at most a logarithmic loss, so both dimension statements follow and Theorem 9.7.2 asserts both. Where a later volume proves something weaker, it will say so.

Proposition 9.7.3 (The equivalence is not lossy). *Let $\beta \geq 0$. The following are equivalent, up to losses of $\delta^{-\varepsilon}$ for every $\varepsilon > 0$:*

- (i) $m(\bigcup T) \gtrsim \delta^\beta$ for every δ -tube family with δ -separated directions;
- (ii) $\|\sum_T \mathbf{1}_T\|_{L^q} \lesssim \delta^{-\varepsilon}$ with $q = \frac{n-\beta}{n-\beta-1}$, for every such family;
- (iii) $\|\mathcal{K}_\delta f\|_{L^{n-\beta}(\mathrm{d}\sigma)} \lesssim \delta^{-\varepsilon} \|f\|_{L^{n-\beta}}$;
- (iv) every Besicovitch set has $\dim_{\mathrm{H}} \geq n - \beta$.

Proof. (ii) $\Leftrightarrow$ (iii) is Theorem 9.1.3 with n replaced by $s = n - \beta$. (ii) $\Rightarrow$ (i): by Hölder on the support, $\sum_T m(T) = \int_E F \leq m(E)^{1/s} \|F\|_{L^q}$, and $\sum_T m(T) \approx 1$, giving $m(E) \gtrsim \delta^{\varepsilon s}$; the exponent matches after relabelling ε .

(i) $\Rightarrow$ (ii) is the reverse Hölder step, valid because F takes values in $\{0, 1, \dots\}$ and its distribution is controlled by $m(E)$ together with $\int F \approx 1$; the argument is the weak-type computation of Theorem 9.1.3 Step 2 run backwards.

(iii) $\Rightarrow$ (iv) is Volume 7, Theorem 7.8.5. (iv) $\Rightarrow$ (i) follows since a Besicovitch set's δ -neighbourhood contains such a family and $\dim_{\mathrm{H}} \leq \overline{\dim_{\mathrm{M}}}$ (Theorem 7.2.4). $\square$

Where we are. Theorem 9.7.3 is the formal statement of what Worked Example 9.7.1 traced by hand: all four formulations of aakeya bound carry the same exponent, and the passages between them cost only $\delta^{-\varepsilon}$. That is why the literature quotes whichever form is convenient and why a reader may translate freely. The single caveat, repeated once more because it is the one that catches people: the equivalence with (iv) requires the loss to be $\delta^{-\varepsilon}$ for *every* ε . Formulations (i)–(iii) remain equivalent to one another under a fixed power loss; only the dimension statement degrades.

WORKED EXAMPLE 9.7.4— TIER B Converting between the four formulations, in practice

Theorem 9.7.3 says the four statements are equivalent. Using the equivalence requires knowing which direction is cheap and which needs care.

Tube estimate $\rightarrow$ multiplicity bound. This is Hölder, and it is the step most often written without comment. From $m(E) \gtrsim \delta^\beta$ one gets $\|\sum \mathbf{1}_T\|_{L^q} \lesssim \delta^{-\varepsilon}$ with $q = \frac{s}{s-1}$, $s = n - \beta$, provided the multiplicity function's distribution is controlled — which it is, since $\int F \approx 1$ pins the first moment and $m(E)$ the support.

Multiplicity bound $\rightarrow$ maximal bound. Theorem 9.1.3 Step 2. This requires linearisation (Worked Example 9.1.4) and produces a *weak*-type bound; the strong-type statement needs Marcinkiewicz interpolation against the trivial L^∞ bound, and costs another ε . Papers frequently state the weak-type version and leave the upgrade implicit.

Maximal bound $\rightarrow$ Hausdorff dimension. Volume 7, Theorem 7.8.5. Two pigeonholings, over scales and over directions, each costing a power of a logarithm. *This is the step that requires the loss to be $\delta^{-\varepsilon}$ for every ε .*

Hausdorff dimension $\rightarrow$ tube estimate. The cheapest direction, and the one used to derive necessary conditions: if some Besicovitch set had small dimension, its neighbourhood would furnish a tube family violating the estimate.

A worked conversion. Suppose a paper proves $\|\sum \mathbf{1}_T\|_{L^{5/3}} \lesssim \delta^{-\varepsilon}$ in $\mathbb{R}^4$. Then $q = \frac{5}{3}$ gives $s = \frac{q}{q-1} = \frac{5}{2}$, so $\beta = n - s = \frac{3}{2}$, and the tube estimate is $m(E) \gtrsim \delta^{3/2-\varepsilon}$; the dimension conclusion is $\dim_{\text{H}} K \geq \frac{5}{2}$. That is the bush bound in $\mathbb{R}^4$ (Worked Example 9.7.2). ✓

The habit. On meeting any Kakeya-type estimate, compute s immediately and compare with n : the ratio s/n is the fraction of the conjecture achieved, and it is the only scale-free way to compare results across dimensions.

Drills

Drill 9.7.1. Verify the inequality $\log m(K_\delta)/\log \delta \leq \beta + o(1)$ used in Theorem 9.7.1, being careful with the sign of $\log \delta$.

Drill 9.7.2. Show that a single-scale estimate with a fixed power loss δ^{-c} gives $\dim_{\text{H}} K \geq s - sc$ and not s , and compute the loss for $c = \frac{1}{10}$ at $n = 4$.

Drill 9.7.3. Explain why Theorem 9.7.1 bounds $\overline{\dim_{\text{M}}}$ and not $\underline{\dim_{\text{M}}}$, and what additional information would be needed for the latter.

Drill 9.7.4. Assemble the Nikodym versions of the table in Worked Example 9.7.2, citing Theorem 9.6.6.

Where we are. A last reminder about the direction of the implications. Everything in this chapter converts a *lower* bound on the volume of a tube union into a *lower* bound on a dimension. There is no useful converse: knowing that some particular Besicovitch set has large dimension says nothing about tube families in general, because the families arising from a single set are a vanishingly small subfamily of all admissible ones. So the chain of Theorem 9.7.3 should be read as four equivalent ways of stating one quantitative fact about *all* families, and not as a bridge between statements about individual sets.

Notes

The passage from maximal estimates to dimension is standard and is carried out in Volume 7, Theorem 7.8.5. The observation that the whole chain is lossless except for logarithms — so that tube exponents and dimension exponents may be quoted interchangeably — is folklore, and is what allows the literature to be read as it is written.

Chapter 9.8

What the Elementary Methods Give

9.8.1 The three arguments, side by side

WORKED EXAMPLE 9.8.1— TIER A **A single table**

	L^2	bush	hairbrush
Conditions on	nothing	a point	a line
Counts	pairs of tubes	tubes through x_0	tubes meeting T_0
Key fact	$m(T \cap T') \lesssim \frac{\delta^n}{\theta + \delta}$	tubes through a point separate at unit distance	two lines meeting a third are concurrent or coplanar
Exponent	2	$\frac{n+1}{2}$	$\frac{n+2}{2}$
Sharp when	$n = 2$	never	never
Loss	log	none	log, plus transversality

The progression. Each argument conditions on an object of one higher dimension and, in exchange, obtains a structural fact unavailable to its predecessor. The L^2 method has no structural input at all — which is why Worked Example 9.3.1 could show it cannot distinguish a Besicovitch configuration from a random one.

The common shape. The bush and hairbrush arguments both produce two bounds on $m(E)$ whose product eliminates the unknown multiplicity μ :

$$m(E) \gtrsim \mu \cdot A \quad \text{and} \quad m(E) \gtrsim \mu^{-1} \quad \implies \quad m(E) \gtrsim A^{1/2}.$$

The bush has $A = \delta^{n-1}$; the hairbrush improves it to $A = \delta^{n-2}$ by collecting

tubes along a whole line rather than at a point. Any further improvement of this type must increase A again, and Chapter 9.5 showed where that stalls.

WORKED EXAMPLE 9.8.2— TIER B Why the geometric mean is the right move, and when it is not

The device of multiplying two bounds to eliminate a parameter recurs throughout this subject, and it is worth understanding what it costs.

What it does. Given $m(E) \geq \max\{\mu A, \mu^{-1}\}$ with μ unknown, the worst case is where the two agree, at $\mu = A^{-1/2}$, giving $m(E) \geq A^{1/2}$. The bound is therefore attained only if the extremal configuration really has multiplicity $A^{-1/2}$.

When it is sharp. For the bush at $n = 3$: $A = \delta^2$, so $\mu = \delta^{-1}$, and indeed there are tube families with multiplicity $\approx \delta^{-1}$ — a bush of δ^{-1} tubes through a point in $\mathbb{R}^3$ is realisable. So the argument is not obviously wasteful there.

When it is not. For large n the balancing multiplicity is $\delta^{-(n-1)/2}$, and it is not known whether a Besicovitch configuration can have multiplicity that large — indeed the Kakeya conjecture asserts that $m(E) \gtrsim \delta^\varepsilon$, which by the second bound forces $\mu \lesssim \delta^{-\varepsilon}$: the multiplicity should be almost bounded. So in high dimensions the geometric mean is balancing against a case that conjecturally does not occur, and the loss is severe.

What would replace it. An argument that bounds μ directly, rather than optimising over it. That is exactly what the finite-field polynomial method achieves — Dvir’s theorem in Volume 10 gives the sharp bound with no multiplicity parameter at all — and it is the structural reason the polynomial method broke a barrier that had stood for a decade.

The transferable lesson. When an argument optimises over an unknown parameter, ask whether the balancing value is attainable. If it is not, the argument is proving a theorem about configurations that do not exist, and the loss is the distance between the balancing value and the true one.

WORKED EXAMPLE 9.8.3— TIER C: RESEARCH TASTE Where this volume leaves the subject

(a) *What was proved.* The Besicovitch configuration with disjoint translates (Theorem 9.2.1), discharging Volume 3’s debt; Córdoba’s L^2 bound, sharp in the plane (Theorem 9.3.3); the bush bound $\frac{n+1}{2}$ (Theorem 9.4.4); the geometry of the hairbrush and Wolff’s bound in the transverse case (Theorem 9.5.3, Theorem 9.5.4); and the Kakeya–Nikodym equivalence (Theorem 9.6.6).

(b) *What was decided against.* Wolff’s theorem in full generality (Theorem 9.5.5, revision R9), with the obstruction — near-parallel tubes — identified arithmetically rather than gestured at.

(c) *The state of the problem after this volume.* Settled for $n = 2$; about half the conjectured dimension for every $n \geq 3$; and no method here has any prospect of reaching n , because all three are limited by the same optimisation over an unattainable multiplicity (Worked Example 9.8.2).

(d) *What must change.* Three things, each supplied by a later volume. Volume 10 replaces the geometry by arithmetic, studying the additive structure of the parameter set rather than the incidences of the tubes. Volume 11 introduces the polynomial method, which bounds the multiplicity directly rather than optimising over it. Volumes 12 and 13 return to the analytic side with multilinear and decoupling estimates, importing the geometric input those volumes supply.

(e) *A remark on the shape of the subject.* It is worth noticing that the three arguments of this volume are all from before 1995 and that the exponent $\frac{n+2}{2}$ stood as the general record for a decade. The barrier was not technical: it was the limit of what incidence counting at a single scale can deliver. Every subsequent advance changed the category of the argument — arithmetic, algebraic, or analytic — rather than refining the geometry. That is the pattern this series has met repeatedly, most explicitly in Volume 8, Worked Example 8.8.5, and it is worth carrying into the volumes ahead.

WORKED EXAMPLE 9.8.4— TIER B Estimating what a new method can reach, before reading it

A practical skill, and one this volume has now supplied the ingredients for: given a proposed approach to *Takeya*, predict its ceiling.

Step 1: identify the order. Does the method’s central quantity involve pairs of tubes, or configurations of three or more? A second moment, an energy integral, a Fourier transform of a measure: all pairs, and by Worked Example 9.3.3 the ceiling is what the L^2 method gives — dimension 2, or $\frac{n+1}{2}$ if a point is also conditioned on.

Step 2: identify the free parameter. Most arguments optimise over something — a multiplicity, a scale, an angular width. Compute the balancing value, as in Worked Example 9.8.2, and ask whether a configuration attaining it exists. If not, the method is bounded by a case that does not occur and the loss is quantifiable.

Step 3: check the losses. A method producing $\delta^{-\varepsilon}$ for every ε gives Hausdorff bounds; a fixed power gives only Minkowski, weakened by Volume 7, Drill

8.4. This decides whether the method addresses the conjecture at all.

Worked prediction: the polynomial method. Its central object is a polynomial vanishing on a set of points, and the count is of *incidences* between tubes and cells — not pairs. So Step 1 gives no ceiling. There is no free multiplicity parameter: Dvir’s argument bounds the set directly. And in the finite-field model there is no δ and hence no loss. Prediction: the method should be sharp in the finite-field setting and strong in $\mathbb{R}^n$. That is what Volume 11 finds.

Worked prediction: decoupling. Its central object is an L^p norm of a sum over caps, and the estimate is ℓ^2 -based — a second moment in the frequency variable. Step 1 therefore predicts a ceiling, and Volume 13 confirms it: decoupling gives sharp results for problems whose extremisers are “flat” but does not by itself resolve Kakeya.

The value of the exercise. Not to avoid reading the papers, but to know what to expect from them, and to recognise quickly when a claimed approach cannot reach the conjecture — which is the commonest failure mode of proposed attacks on this problem.

WORKED EXAMPLE 9.8.5— TIER A **The scoreboard for Kakeya, as it stands**

Collecting what the series has established and what it has reported.

Bound	Source	Dimensions	Status here
2	Córdoba	all n	proved, Theorem 9.3.3
$\frac{n+1}{2}$	bush	all n	proved, Theorem 9.4.4
$\frac{n+1}{2}$	Drury, Fourier	all n	reported, Vol 7 Theorem 7.7.5
$\frac{n+2}{2}$	Wolff	all n	transverse case, Theorem 9.5.4
$\frac{n+2}{2} + c_n$	Katz–Laba–Tao	$n = 3$	Volume 10
q^n points	Dvir	finite fields	Volume 10, sharp
n	Wang–Zahl	$n = 3$	Volume 14, settled

Reading the middle rows. That the bush and Drury’s Fourier bound give the same exponent $\frac{n+1}{2}$ is not a coincidence: Worked Example 7.7.2 in Volume 7 showed both balance a 1-dimensional object against a decay of order $\frac{n-1}{2}$. Two methods, one obstruction.

Reading the last rows. The finite-field problem is *solved*, sharply, by an argument of a page. The real problem in $\mathbb{R}^3$ is solved by a long paper. And

$n \geq 4$ remains open. The gap between the finite-field and real cases is the subject of Volume 10, Chapter 8, and is where a reader should look before believing any short argument for the real case.

The single number to remember. Before 2005 the general- n record was $\frac{n+2}{2}$ — half the conjecture, plus one. Everything in this volume is that record and how it was reached.

Drills

Drill 9.8.1. For each of the three methods, identify the configuration that makes its key estimate sharp, and check that no single configuration is extremal for all three.

Drill 9.8.2. Compute the balancing multiplicity $\mu = A^{-1/2}$ for the bush and the hairbrush in $\mathbb{R}^4$, and compare with the multiplicity the Kakeya conjecture would permit.

Drill 9.8.3. Show that if a Besicovitch set had a tube family of bounded multiplicity, then $m(E) \gtrsim 1$ and the conjecture would follow immediately for that family.

Drill 9.8.4. Using Worked Example 9.7.2, determine the smallest n for which the bush bound exceeds the L^2 bound, and the smallest for which the hairbrush exceeds the bush.

Where we are. A closing note on where Volumes 1–9 leave the series. The first five volumes built the analysis: measure and integration, the Fourier transform, singular integrals, function spaces, oscillatory integrals. Volume 6 posed the restriction problem and showed it implies Kakeya. Volumes 7 and 8 built the geometric measure theory and applied it to projections and distance sets, reaching the limit of what two-point methods give. This volume attacked Kakeya directly with incidence geometry and reached the limit of what single-scale counting gives.

Three limits, all reached, and all the same limit seen from different angles: an argument that sums over pairs cannot detect the high-order coincidences that make a Besicovitch set small. Worked Example 9.3.3 states this as a test to apply to any proposed method, and Worked Example 9.8.4 turns it into a procedure.

What follows changes category rather than refining technique. Volume 10 studies the additive structure of the parameter set and proves the finite-field case outright; Volume 11 introduces the polynomial method, which bounds

the multiplicity instead of optimising over it; Volumes 12 and 13 return to the analytic side with multilinear and decoupling estimates. Volume 14 reports where all of it stands, including Wang and Zahl’s resolution of the three-dimensional case.

The debt from Volume 3 is discharged (Theorem 9.2.1), and nothing in Volumes 1 to 9 is now used without proof.

Notes

The three arguments are Córdoba’s (1977), Bourgain’s (1991) and Wolff’s (1995). The observation that they share the structure of Worked Example 9.8.2 — two bounds multiplied to eliminate a multiplicity — is standard, and the diagnosis in that example of why the device is lossy in high dimensions is the usual explanation for the stagnation of the exponent before the arithmetic and algebraic methods arrived.

Chapter 9.A

Results Quoted from Volumes 1–8

Result	Source	Used in
The Perron tree, with its explicit translations	Theorem 1.3.12	Chapter 9.2 throughout
Tube intersection estimate	Theorem 1.6.13	Theorem 9.3.2
The bush bound	Theorem 1.7.15	Chapter 9.4
Takeya maximal operator	Theorem 1.7.13	Chapter 9.1
Norms by testing	Theorem 1.8.10	Theorem 9.1.3
Marcinkiewicz interpolation	Theorem 2.2.4	Theorem 9.1.3
Riesz–Thorin	Theorem 2.2.1	Worked Example 9.1.1
The ball multiplier theorem	Theorem 3.8.2	Worked Example 9.2.3
The configuration lemma, as assumed	Theorem 3.8.4	Theorem 9.2.1
Lipschitz images and dimension	Theorem 7.1.5	Theorem 9.6.5
Minkowski content	Theorem 7.2.3	Theorem 9.7.1
Maximal bounds give Hausdorff bounds	Theorem 7.8.5	Theorem 9.7.2
Stability of Minkowski dimension	Theorem 7.2.5	Worked Example 9.7.1
The Takeya conjecture	Theorem 7.1.11	throughout
Two-point versus richer problems	Worked Example 7.7.3	Worked Example 9.3.1
Takeya against Falconer	Worked Example 8.8.5	Worked Example 9.8.3

Debts

Discharged. Volume 3, Theorem 3.8.4 — the Besicovitch configuration with pairwise disjoint translates — is proved as Theorem 9.2.1. It was the only statement in Volumes 1 to 8 used without proof, and Volume 3’s Theorem 3.8.2 is now unconditional.

Incurred. None. Nothing in this volume is owed to a later one.

Removed. Wolff’s hairbrush bound in full generality (Theorem 9.5.5, revision R9); the transverse case is proved.

Chapter 9.B

Notation and Index of Results

Additions to the Volume 1–8 conventions

$\mathbb{T}, N$	a family of δ -tubes and its cardinality
$E = \bigcup T, F = \sum \mathbf{1}_T$	the union and the multiplicity function
$\mu = \ F\ _{L^\infty}$	the maximum multiplicity
$\mathcal{N}_\delta$	the Nikodym maximal operator
$H_\kappa(T_0)$	the κ -transverse hairbrush of T_0
S_i, t_i, m_i	the Perron data of Theorem 9.2.2
Φ	the projective duality $(x', x_n) \mapsto (x'/x_n, 1/x_n)$

Index of the main results

Maximal and multiplicity forms agree	Theorem 9.1.3
The rectangles fit inside the triangles	Theorem 9.2.4
The translates are pairwise disjoint	Theorem 9.2.5
The Besicovitch configuration	Theorem 9.2.1
Angle counting	Theorem 9.3.1
The L^2 bound	Theorem 9.3.2
Córdoba's bound	Theorem 9.3.3
The bush estimate	Theorem 9.4.2
The bush bound	Theorem 9.4.4
Concurrent or coplanar	Theorem 9.5.1
Hairbrush volume estimate	Theorem 9.5.3
Wolff's bound, transverse case	Theorem 9.5.4
Position and direction interchanged	Theorem 9.6.4
Takeya and Nikodym are equivalent	Theorem 9.6.6
Single scales give Minkowski bounds	Theorem 9.7.1
The bounds in dimension form	Theorem 9.7.2

Plan revisions in this volume

Removed	Reason
Wolff's bound $\frac{n+2}{2}$ in full	the transverse case is proved; the near-parallel case degrades exactly to the bush bound and needs Wolff's induction on scales. Theorem 9.5.5, revision R9

Part 10

Volume 10

Takeya II: Arithmetic and Algebraic Methods

Preface to Volume 10

Two changes of category

Volume 9 reached the limit of incidence counting: the L^2 method, the bush and the hairbrush all optimise over an unknown multiplicity, and Worked Example 9.8.2 showed the balancing value is one the Kakeya conjecture says does not occur. Refining the geometry cannot help. What is needed is an argument of a different kind, and this volume supplies two.

The first is *algebraic*. Replace $\mathbb{R}^n$ by $\mathbb{F}_q^n$, a vector space over a finite field, and the Kakeya problem still makes sense: a Kakeya set contains a line in every direction, and the question is how few points it can have. Dvir’s theorem answers it completely, sharply, and in half a page. The argument constructs a polynomial vanishing on the set and derives a contradiction from its degree — there is no multiplicity parameter to optimise over, which is precisely why it escapes Volume 9’s ceiling.

The second is *arithmetic*. A Besicovitch set built from a parameter set has its size controlled, through Fubini, by the additive structure of that parameter set. Sumset inequalities — Ruzsa’s triangle inequality, the Plünnecke–Ruzsa estimates — then bound the geometry. This route gives the first bounds of the form cn with $c > \frac{1}{2}$, exceeding what Volume 9 achieved for every large n .

A warning about the finite field model

Dvir’s proof is short and complete, and it is tempting to conclude that the real Kakeya problem must have a similarly short solution. It does not, and Chapter 10.7 explains why in detail. The essential point can be stated now: over $\mathbb{F}_q$ a line has q points and two distinct lines meet in at most one, so “a line” is a combinatorial object with no scale. Over $\mathbb{R}$ a tube has a *width*, two tubes can be nearly parallel and overlap along their whole length, and every argument must track how much. The polynomial method has real analogues — Volume 11 develops them — but they are quantitative and hard, and Dvir’s argument is not among them.

What is assumed

Volumes 1–9, though less heavily than usual: Chapter 10.2 depends on nothing but linear algebra. From Volume 9, the framing of the Kakeya problem and the comparison of methods in Worked Example 9.8.1; from Volume 7, the passage from single-scale bounds to dimension.

Why both methods escape Volume 9’s ceiling

Volume 9 ended with a diagnosis: the L^2 method, the bush and the hairbrush all produce two bounds on $m(E)$ and multiply them to eliminate an unknown multiplicity μ , and Worked Example 9.8.2 showed the resulting bound is determined by a balancing value of μ that the Kakeya conjecture says is never attained. The methods are proving theorems about configurations that do not exist.

Both methods here avoid that, in different ways.

Dvir’s argument has no free parameter at all. It builds a polynomial, restricts it to lines, and contradicts a degree count. There is nothing to optimise over, and consequently nothing lost — which is why it is the only result in this series that attains the truth exactly (Theorem 10.1.4 and Theorem 10.2.1 give the same number in the plane).

The arithmetic method replaces the optimisation. Instead of balancing against a multiplicity it balances against additive structure, and the balance point is the fixed point of a recursion — $2 - \sqrt{2}$, as Worked Example 10.6.5 explains. That is still a loss, and the constant is still below 1, but it exceeds $\frac{1}{2}$, which no method in Volume 9 does.

The general principle, stated once: *a method’s ceiling is set by what it optimises over.* Volume 9, Worked Example 9.8.4 turns this into a test, and applying it to the two methods of this volume predicts exactly what they deliver.

The plan

- | | | |
|---|------------------------|--------------------------------|
| 1 | The finite field model | and the Schwartz–Zippel lemma |
| 2 | Dvir’s theorem | the polynomial method, sharply |

Chapters 3–7, planned: sumset inequalities, with Ruzsa’s triangle inequality and Petridis’s proof of Plünnecke; the arithmetic reduction of the Kakeya problem to a sumset problem; the bounds of Bourgain and of Katz–Łaba–Tao; why the finite field argument does not transfer (Chapter 10.7); and an assessment.

Chapter 10.1

The Finite Field Model

10.1.1 The problem over $\mathbb{F}_q$

Let q be a prime power and $\mathbb{F}_q$ the field with q elements. The space $\mathbb{F}_q^n$ has q^n points, and a *line* through a in direction $b \neq 0$ is

$$\ell_{a,b} = \{a + tb : t \in \mathbb{F}_q\},$$

a set of exactly q points. Two distinct lines meet in at most one point, since their difference determines t uniquely.

Definition 10.1.1. A set $K \subseteq \mathbb{F}_q^n$ is a *Keakeya set* if for every direction $b \in \mathbb{F}_q^n \setminus \{0\}$ there is $a \in \mathbb{F}_q^n$ with $\ell_{a,b} \subseteq K$.

Directions are naturally indexed by b up to scalar multiplication, so there are $\frac{q^n-1}{q-1} \approx q^{n-1}$ of them — the finite projective space $\mathbb{P}^{n-1}(\mathbb{F}_q)$.

Proposition 10.1.2 (The trivial bounds). *Every Keakeya set $K \subseteq \mathbb{F}_q^n$ satisfies*

$$q \leq |K| \leq q^n,$$

and there exist Keakeya sets with $|K| \lesssim_n q^n$; in the plane there are Keakeya sets with $|K| \leq \frac{q(q+1)}{2} + \frac{q-1}{2}$.

Proof. The lower bound holds because K contains at least one line, of q points; the upper because $K \subseteq \mathbb{F}_q^n$. That $\mathbb{F}_q^n$ itself is a Keakeya set gives the second claim with constant 1.

For the planar construction, take

$$K = \left\{ (x, y) : y = \frac{x^2}{4} + bx \text{ for some } b \right\} \cup \{\text{points at infinity}\},$$

or more simply the union over $b \in \mathbb{F}_q$ of the lines $\ell_{a(b),b}$ with $a(b) = (-b^2/4, 0)$ chosen so that the lines are tangent to a common parabola. Two such lines

meet in one point, and there are q of them, so the union has at most $q \cdot q - \binom{q}{2}$ points; evaluating gives the stated bound. The vertical direction contributes one further line. $\square$

Where we are. So in the plane a Kakeya set has between q and about $\frac{q^2}{2}$ points, and the upper bound is achieved. The question is the true order of the lower bound: is it q , or q^2 , or something between? Theorem 10.2.1 answers it in every dimension: $|K| \gtrsim_n q^n$, the maximum possible order. Before proving that, we need one lemma about polynomials — and it is the only input.

10.1.2 The Schwartz–Zippel lemma

Lemma 10.1.3 (Schwartz–Zippel). *Let $P \in \mathbb{F}_q[x_1, \dots, x_n]$ be a non-zero polynomial of total degree d . Then*

$$\#\{x \in \mathbb{F}_q^n : P(x) = 0\} \leq dq^{n-1}.$$

In particular, if $d < q$ then P does not vanish identically on $\mathbb{F}_q^n$.

Proof. Induct on n . For $n = 1$ a non-zero polynomial of degree d has at most d roots in a field, which is the claim.

Let $n \geq 2$ and write

$$P(x_1, \dots, x_n) = \sum_{j=0}^k P_j(x_1, \dots, x_{n-1}) x_n^j,$$

where k is the largest power of x_n occurring and $P_k \neq 0$. Then $\deg P_k \leq d - k$.

Fix $x' = (x_1, \dots, x_{n-1})$ and consider two cases.

If $P_k(x') \neq 0$: the polynomial $t \mapsto P(x', t)$ is non-zero of degree k , so it has at most k roots. The number of x' in this case is at most q^{n-1} , contributing at most kq^{n-1} zeros.

If $P_k(x') = 0$: the whole line $\{(x', t)\}$ may consist of zeros, giving q each. By the inductive hypothesis applied to P_k , which is non-zero of degree at most $d - k$ in $n - 1$ variables, there are at most $(d - k)q^{n-2}$ such x' , contributing at most $(d - k)q^{n-2} \cdot q = (d - k)q^{n-1}$ zeros.

Adding, the total is at most $kq^{n-1} + (d - k)q^{n-1} = dq^{n-1}$.

For the last claim, if $d < q$ then $dq^{n-1} < q^n$, so P cannot vanish everywhere. $\square$

WORKED EXAMPLE 10.1.1— TIER A **The lemma is sharp, and where**

Sharpness. Take $P(x) = \prod_{j=1}^d (x_1 - c_j)$ with $c_1, \dots, c_d$ distinct elements of

$\mathbb{F}_q$, $d \leq q$. Its zero set is the union of d parallel hyperplanes, of size exactly dq^{n-1} . So no better bound holds in general.

What sharpness costs. The extremal example is a union of hyperplanes — maximally degenerate. For an irreducible P the count is typically much smaller, $\approx q^{n-1}$ by the Lang–Weil estimate, but that is a theorem of a different order and is not needed here.

The form actually used. Theorem 10.2.1 uses only the last sentence of Theorem 10.1.3: a non-zero polynomial of degree less than q cannot vanish on all of $\mathbb{F}_q^n$. That is the finite-field replacement for the statement, obvious over $\mathbb{R}$, that a non-zero polynomial is non-zero somewhere — and over a finite field it genuinely requires the degree hypothesis, since $x^q - x$ vanishes identically on $\mathbb{F}_q$.

The example to keep in mind. $x^q - x$ has degree q and vanishes on all of $\mathbb{F}_q$. So the bound $d < q$ is not an artefact: at $d = q$ the conclusion is false.

WORKED EXAMPLE 10.1.2— TIER B Counting dimensions of polynomial spaces

The other ingredient of Chapter 10.2 is a dimension count, and it is worth doing carefully since off-by-one errors here are easy.

The space. Let $\mathcal{P}_d^n$ be the $\mathbb{F}_q$ -vector space of polynomials in n variables of total degree at most d . A basis is the set of monomials x^α with $|\alpha| \leq d$, so

$$\dim \mathcal{P}_d^n = \#\{\alpha \in \mathbb{N}_0^n : |\alpha| \leq d\} = \binom{d+n}{n},$$

by the standard stars-and-bars count: choosing α with $\alpha_1 + \cdots + \alpha_n \leq d$ is choosing $\alpha_1 + \cdots + \alpha_n + \alpha_{n+1} = d$ in $n+1$ non-negative integers.

The value used. With $d = q-1$,

$$\dim \mathcal{P}_{q-1}^n = \binom{q-1+n}{n} = \binom{q+n-1}{n}.$$

The comparison with q^n . Since $\binom{q+n-1}{n} = \frac{(q+n-1)(q+n-2)\cdots q}{n!} \geq \frac{q^n}{n!}$, the dimension is at least $q^n/n!$ — and at most q^n for q large, as it must be, since $\mathcal{P}_{q-1}^n$ injects into the space of all functions $\mathbb{F}_q^n \rightarrow \mathbb{F}_q$ of dimension q^n .

Numerically.

q	n	$\binom{q+n-1}{n}$	$q^n/n!$	q^n
3	2	6	4.5	9
5	2	15	12.5	25
7	3	84	57.2	343
11	3	286	221.8	1331

The middle column is the bound Theorem 10.2.1 will deliver.

WORKED EXAMPLE 10.1.3— TIER C: RESEARCH TASTE **Why a finite field model at all**

Replacing $\mathbb{R}^n$ by $\mathbb{F}_q^n$ is a standard device in this subject, and it is worth being clear about what it is for.

(a) *It removes scale.* Over $\mathbb{R}$, a tube has a width δ and two tubes may be at any angle; the whole difficulty of Volume 9 was bookkeeping over angles and scales. Over $\mathbb{F}_q$ a line is a set of q points, two distinct lines meet in at most one, and there is nothing to track. The combinatorics is exposed.

(b) *It preserves the extremal examples — sometimes.* The planar construction of Theorem 10.1.2 is a finite-field Perron tree: lines tangent to a parabola, overlapping as much as incidence geometry allows. So the model is not vacuous; it has its own extremal configurations.

(c) *It is a genuine test of a method.* Before Dvir, every approach to the finite field Kakeya problem gave bounds matching the real case — $q^{(n+1)/2}$ from the bush, $q^{(n+2)/2}$ from the hairbrush. That the answer turned out to be q^n , by an argument with no real analogue, is the strongest evidence that the real problem's difficulty is *metric* rather than combinatorial.

(d) *What it is not.* It is not a step towards the real problem. Chapter 10.7 sets out why: Dvir's argument uses that a line is q points with no thickness, and every attempt to give it thickness destroys the degree count. Wolff proposed the finite field problem in 1999 precisely as a model, and its resolution in 2008 did not resolve the real case, nor has it since.

(e) *What it did change.* It introduced the polynomial method to the subject. Volume 11 develops the quantitative version — polynomial partitioning — which *does* apply over $\mathbb{R}$ and which underlies both the Guth–Katz solution of the Erdős distinct distances problem and the current records for restriction. So the model's influence was methodological rather than logical.

Proposition 10.1.4 (The planar construction, precisely). *Let q be an odd prime power and put*

$$L_b = \{(t, bt - b^2) : t \in \mathbb{F}_q\} \quad (b \in \mathbb{F}_q), \quad K = \left(\bigcup_{b \in \mathbb{F}_q} L_b \right) \cup (\{0\} \times \mathbb{F}_q).$$

Then K contains a line in every direction and

$$|K| \leq \frac{q(q+1)}{2} + q.$$

Proof. Step 1. Every direction occurs.

L_b is the line of slope b through $(0, -b^2)$, so as b ranges over $\mathbb{F}_q$ the lines L_b realise all q non-vertical slopes; the vertical direction is realised by $\{0\} \times \mathbb{F}_q$.

Step 2. *Count the union of the L_b exactly.*

A point (t, y) lies in $\bigcup_b L_b$ precisely when $y = bt - b^2$ for some b . Completing the square,

$$bt - b^2 = \frac{t^2}{4} - \left(b - \frac{t}{2}\right)^2,$$

which is legitimate since q is odd, so 2 is invertible. As b ranges over $\mathbb{F}_q$, the quantity $(b - \frac{t}{2})^2$ ranges over the squares of $\mathbb{F}_q$, of which there are exactly $\frac{q+1}{2}$ — namely 0 and the $\frac{q-1}{2}$ non-zero squares. Hence for each fixed t there are exactly $\frac{q+1}{2}$ admissible values of y , and

$$\left| \bigcup_b L_b \right| = q \cdot \frac{q+1}{2}.$$

Step 3. *Add the vertical line.*

It contributes at most q further points, giving the stated bound. $\square$

Where we are. Step 2 is exact, not an estimate: the union has exactly $\frac{q(q+1)}{2}$ points. Direct computation confirms it — at $q = 3, 5, 7, 11, 13$ the union has 6, 15, 28, 66, 91 points, matching $\frac{q(q+1)}{2}$ in every case.

Compare Theorem 10.2.1, which gives the lower bound $\binom{q+1}{2} = \frac{q(q+1)}{2}$ in the plane. So the construction and the theorem agree to the leading term, and the true minimum sits between them, exceeding Dvir’s bound by exactly $\frac{q-1}{2}$ (Worked Example 10.2.1).

The mechanism of the construction is worth naming: the lines are tangent to a common parabola, so each pair meets in one point and the union is as small as incidence geometry allows. It is the finite-field analogue of the Perron tree, with the tangency playing the role of the overlapping translates.

WORKED EXAMPLE 10.1.4— TIER A **Kekeya sets in $\mathbb{F}_3^2$, by hand**

The smallest case is small enough to work out completely, and doing so makes every definition concrete.

The setting. $\mathbb{F}_3^2$ has 9 points. Directions are non-zero vectors up to scalar, and there are $\frac{3^2-1}{3-1} = 4$ of them: $(1, 0)$, $(0, 1)$, $(1, 1)$, $(1, 2)$. Each line has 3 points, and there are 3 parallel lines in each direction, 12 lines in all.

A Kekeya set must contain one line from each of the four parallel classes. Four lines of three points each would give 12 points if disjoint, but lines

in different directions meet, so the union is smaller. Two lines in distinct directions share exactly one point, and with four lines there are $\binom{4}{2} = 6$ pairs, so the union has at least $12 - 6 = 6$ points.

Dvir's bound. $\binom{q+n-1}{n} = \binom{4}{2} = 6$. So the theorem gives exactly the count above — which is reassuring, since the theorem's proof uses nothing about how lines meet.

The truth is 7. Exhaustive search over all $3^4 = 81$ choices of one line per direction shows the minimum union has 7 points, not 6. The discrepancy is because the six intersection points cannot all be distinct: four lines in general position in the plane have six intersection points, but here some must coincide. The extremal configuration has three lines through a common point and the fourth crossing them elsewhere, giving $3 + 2 + 2 = 7$.

The general gap. Worked Example 10.2.1 records that the minimum in the plane is $\frac{q(q+1)}{2} + \frac{q-1}{2}$, exceeding Dvir's $\frac{q(q+1)}{2}$ by exactly $\frac{q-1}{2}$. At $q = 3$ that is 1, which is the discrepancy just found.

WORKED EXAMPLE 10.1.5— TIER B Using Schwartz–Zippel

The lemma is applied in three distinct ways in this volume, and separating them prevents confusion about what it does.

Use 1: a non-zero polynomial is non-zero somewhere. If $\deg P < q$ then P does not vanish on all of $\mathbb{F}_q^n$. This is the form used in Theorem 10.2.1 Step 4, and it is the weakest consequence — but over a finite field it is not automatic, since $x^q - x$ vanishes identically.

Use 2: a quantitative bound on a zero set. $\#\{P = 0\} \leq dq^{n-1}$. This is used in Theorem 10.2.2, where one knows P_d vanishes on a large cone and deduces a lower bound on d .

Use 3: a line meets a variety in few points. Restricting to a line $\ell_{a,b}$ and applying the case $n = 1$: a non-zero P of degree d has at most d zeros on any line it does not contain. This is the form that *transfers to* $\mathbb{R}$, and Worked Example 10.7.1 identifies it as the whole of what Volume 11 uses.

Which uses transfer. Use 3 is a statement about incidences and holds over any field, including $\mathbb{R}$. Uses 1 and 2 are about the finiteness of the field and have no real analogue — over $\mathbb{R}$ a non-zero polynomial is non-zero on a set of full measure, which is far stronger and correspondingly useless as a contradiction.

The diagnostic. When reading a polynomial-method argument, ask which use is in play. If Use 1 or 2, the argument is finite-field specific. If Use 3, it may transfer, and Volume 11 will show how.

WORKED EXAMPLE 10.1.6— TIER B Translating a real statement into the finite field model

Before trusting a finite-field analogue one must know how the dictionary works, and where it has no entry.

$\mathbb{R}^n$	$\mathbb{F}_q^n$
δ -tube, $\delta = 1/q$	line (q points)
number of directions $\approx \delta^{-(n-1)}$	$\approx q^{n-1}$
$m(K_\delta)$	$ K /q^n$
$\dim_{\mathbb{H}} K \geq s$	$ K \gtrsim q^s$
δ -separated directions	distinct projective points
two tubes at angle θ	two lines meeting in a point
<i>angle between tubes</i>	<i>no entry</i>
<i>nearly parallel tubes</i>	<i>no entry</i>
<i>tube width</i>	<i>no entry</i>

Reading the top half. The dictionary is faithful for anything expressible in terms of incidences and counts. The Kakeya conjecture itself translates exactly: $\dim_{\mathbb{H}} K = n$ becomes $|K| \gtrsim q^n$, which is Theorem 10.2.1.

Reading the bottom half. There is no notion of angle, because two distinct lines in $\mathbb{F}_q^n$ either meet or do not, with no intermediate case. There is no near-parallelism, and no width. Every entry in the bottom half is precisely what Volume 9’s arguments spent their effort on — Theorem 9.3.1, Theorem 1.6.13, and the small-angle case that Theorem 9.5.5 could not handle.

The consequence for reading results. A finite-field theorem whose proof uses only the top half may suggest a real analogue. A real theorem whose proof uses the bottom half has no finite-field shadow at all — Wolff’s hairbrush, for instance, is about angles and simply does not translate.

The consequence for Dvir. His proof uses only the top half, which is why it is short. Chapter 10.7 shows that the missing bottom half is exactly what blocks the return journey.

Where we are. A remark on the role of q being a prime power rather than merely a prime.

Nothing in this volume uses more than that $\mathbb{F}_q$ is a field: a non-zero polynomial in one variable of degree d has at most d roots (Theorem 10.1.3, base case), and every non-zero element is invertible (used in Theorem 10.1.4 to divide by 2 and 4). Both hold for any finite field, so $q = p^k$ is as good as $q = p$.

What would fail over a ring: $\mathbb{Z}/6\mathbb{Z}$ has zero divisors, $x^2 - x$ has four

roots there rather than two, and the root count that drives Step 3 of Theorem 10.2.1 collapses. The Kakeya problem over $\mathbb{Z}/N\mathbb{Z}$ for composite N is genuinely different and is not treated here.

Drills

Drill 10.1.1. Verify that there are $\frac{q^n-1}{q-1}$ directions in $\mathbb{F}_q^n$, and that this is $\approx q^{n-1}$.

Drill 10.1.2. Check the base case and the inductive step of Theorem 10.1.3 for $n = 2$, $q = 3$, $d = 1$.

Drill 10.1.3. Confirm the stars-and-bars count in Worked Example 10.1.2 and evaluate $\dim \mathcal{P}_{q-1}^n$ for $q = 5$, $n = 3$.

Drill 10.1.4. Show that $x^q - x$ vanishes on all of $\mathbb{F}_q$, and explain why this does not contradict Theorem 10.1.3.

Where we are. Two conventions for this volume.

q plays the role of $1/\delta$. A line has q points where a δ -tube has $\approx \delta^{-1}$ separated points; the number of directions is $\approx q^{n-1}$ against $\approx \delta^{-(n-1)}$. So a finite-field bound $|K| \gtrsim q^s$ corresponds to a real bound $m(K_\delta) \gtrsim \delta^{n-s}$, and the Kakeya conjecture is $s = n$ in both.

Constants are dimensional. All implied constants depend on n and never on q ; a bound with a q -dependent constant says nothing, since $q \rightarrow \infty$ is the regime of interest. This is the exact analogue of the rule in Volume 9 that constants must not depend on δ .

Notes

The finite field Kakeya problem was posed by Wolff in his 1999 survey, as a model in which the combinatorial difficulties might be isolated from the metric ones. The Schwartz–Zippel lemma is due independently to Schwartz, Zippel and DeMillo–Lipton in the late 1970s, and in the form used here goes back to Ore.

Chapter 10.2

Dvir's Theorem

Theorem 10.2.1 (Dvir). *Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set. Then*

$$|K| \geq \binom{q+n-1}{n} \geq \frac{q^n}{n!}.$$

Idea of the proof. Suppose K were smaller than the dimension of the space of polynomials of degree at most $q-1$. Then linear algebra — more unknowns than equations — produces a non-zero polynomial P of degree at most $q-1$ vanishing on K .

Now use the lines. Along a line $\ell_{a,b} \subseteq K$, the function $t \mapsto P(a+tb)$ is a polynomial in one variable of degree at most $q-1$ with q roots, hence identically zero. Its leading coefficient is $P_d(b)$, where P_d is the top-degree homogeneous part of P ; so $P_d(b) = 0$.

Since K has a line in *every* direction, P_d vanishes at every $b \neq 0$, and at 0 too by homogeneity. So P_d is a non-zero polynomial of degree $d \leq q-1$ vanishing on all of $\mathbb{F}_q^n$, contradicting Theorem 10.1.3.

The whole argument is three moves: build a polynomial, restrict it to the lines, and read off its top-degree part. There is no parameter to optimise and no geometry beyond “a line has q points”.

Proof. Suppose for contradiction that

$$|K| < \binom{q+n-1}{n} = \dim \mathcal{P}_{q-1}^n, \quad (10.2.1)$$

the dimension being computed in Worked Example 10.1.2.

Step 1. *A polynomial vanishing on K .*

Consider the linear map

$$\text{ev} : \mathcal{P}_{q-1}^n \longrightarrow \mathbb{F}_q^K, \quad \text{ev}(P) = (P(x))_{x \in K},$$

evaluating a polynomial at the points of K . Its target has dimension $|K|$, which by (10.2.1) is strictly less than the dimension of the source. So ev has non-trivial kernel: there is $P \in \mathcal{P}_{q-1}^n$, $P \neq 0$, with $P \equiv 0$ on K .

Let $d = \deg P$, so $0 \leq d \leq q-1$, and let P_d be the homogeneous part of degree d , which is non-zero by definition of the degree. Note $d \geq 1$: a non-zero constant does not vanish on the non-empty set K .

Step 2. *Restrict to a line.*

Fix a direction $b \in \mathbb{F}_q^n \setminus \{0\}$ and let $\ell_{a,b} \subseteq K$ be a line in that direction, which exists by Theorem 10.1.1. Define

$$g(t) = P(a + tb) \in \mathbb{F}_q[t].$$

Expanding P into homogeneous parts $P = \sum_{j \leq d} P_j$ and using homogeneity,

$$g(t) = \sum_{j=0}^d P_j(a + tb), \quad \text{and} \quad P_j(a + tb) \text{ has degree } \leq j \text{ in } t,$$

with the coefficient of t^j in $P_j(a + tb)$ equal to $P_j(b)$ — take the term of P_j in which every factor contributes tb rather than a . Hence

$$\deg_t g \leq d, \quad [t^d]g = P_d(b). \quad (10.2.2)$$

Step 3. *The restriction vanishes identically.*

Since $\ell_{a,b} \subseteq K$ and $P \equiv 0$ on K , we have $g(t) = 0$ for every $t \in \mathbb{F}_q$ — that is, g has q roots. But $\deg_t g \leq d \leq q-1 < q$, so a non-zero g could have at most $q-1$ roots. Therefore g is the zero polynomial, and in particular every coefficient vanishes. By (10.2.2),

$$P_d(b) = 0.$$

Step 4. *Conclude.*

Step 3 holds for every $b \neq 0$. And $P_d(0) = 0$ since P_d is homogeneous of degree $d \geq 1$. So P_d vanishes on all of $\mathbb{F}_q^n$.

But P_d is a non-zero polynomial of total degree $d \leq q-1 < q$, so by Theorem 10.1.3 it cannot vanish identically on $\mathbb{F}_q^n$. This contradiction refutes (10.2.1), and the first inequality of the theorem follows. The second is the computation in Worked Example 10.1.2. $\square$

Where we are. It is worth pausing on how little was used.

About K : only that it contains a line in each direction. No metric, no scale, no separation of directions.

About lines: only that $\ell_{a,b}$ has q points and is parametrised affinely by t , so that restricting a polynomial gives a polynomial in one variable of no higher degree.

About the field: only that a non-zero polynomial in one variable of degree $< q$ has fewer than q roots, and Theorem 10.1.3 in n variables.

There is no multiplicity, no optimisation, no case division. Compare Volume 9, where every argument produced two bounds and multiplied them (Worked Example 9.4.4), and where Worked Example 9.8.2 diagnosed the resulting loss as the distance between the balancing multiplicity and the true one. Dvir's argument has no such parameter, and that is exactly why it is sharp.

Corollary 10.2.2. *The same conclusion holds if K contains a line in each of only $\geq \frac{1}{2}|\mathbb{P}^{n-1}(\mathbb{F}_q)|$ directions, with $\binom{q+n-1}{n}$ replaced by a constant multiple.*

Proof. Step 4 used that P_d vanishes at every $b \neq 0$. If instead P_d vanishes on a set D of directions with $|D| \geq \frac{1}{2} \cdot \frac{q^n-1}{q-1}$, then the cone over D — the union of the lines $\mathbb{F}_q b$ for $b \in D$ — has at least $\frac{1}{2}(q^n-1)$ points, on all of which P_d vanishes by homogeneity. By Theorem 10.1.3 this forces $dq^{n-1} \geq \frac{1}{2}(q^n-1)$, that is $d \gtrsim q$. Running the argument with $\mathcal{P}_{cq}^n$ for a suitable $c < 1$ then gives the conclusion with a smaller constant. $\square$

Proposition 10.2.3 (Dvir for cylinders over a subset of directions). *Let $D \subseteq \mathbb{F}_q^n \setminus \{0\}$ be a set of directions closed under scalar multiplication, and suppose $K \subseteq \mathbb{F}_q^n$ contains a line in every direction of D . Then for every $d < q$ with $dq^{n-1} < |D|$,*

$$|K| \geq \binom{d+n}{n}.$$

Proof. Suppose $|K| < \binom{d+n}{n} = \dim \mathcal{P}_d^n$. As in Step 1 of Theorem 10.2.1, evaluation at the points of K has non-trivial kernel, so there is a non-zero $P \in \mathcal{P}_d^n$ vanishing on K ; let $e = \deg P \leq d$ and P_e its top-degree part.

Steps 2 and 3 of Theorem 10.2.1 apply verbatim to each direction $b \in D$: the restriction of P to a line of K in direction b is a polynomial in t of degree $\leq e \leq d < q$ with q roots, hence zero, so $P_e(b) = 0$.

Thus P_e vanishes on D , a set of size $|D| > dq^{n-1} \geq eq^{n-1}$. But P_e is non-zero of degree e , so by Theorem 10.1.3 it has at most eq^{n-1} zeros — a contradiction. $\square$

Where we are. Theorem 10.2.3 recovers Theorem 10.2.1 by taking D to be all non-zero directions, so $|D| = q^n - 1$, and $d = q - 1$: the condition $dq^{n-1} < |D|$ reads $(q-1)q^{n-1} < q^n - 1$, which holds. It also yields Theorem 10.2.2 by taking $|D| \geq \frac{1}{2}(q^n - 1)$ and $d \approx \frac{q}{2}$.

The proposition isolates what the argument really needs: not that *every* direction occurs, but that the directions occurring are too numerous for a low-degree form to vanish on all of them. That is a statement about the *dimension of the set of directions*, and it is the form in which the polynomial method generalises — to Furstenberg sets, where only a lower-dimensional family of directions is prescribed.

WORKED EXAMPLE 10.2.1— TIER A **The bound against the truth, in small cases**

Theorem 10.2.1 gives $|K| \geq \binom{q+n-1}{n}$. How close is that?

In the plane. The bound is $\binom{q+1}{2} = \frac{q(q+1)}{2}$, and Theorem 10.1.2 exhibits Kakeya sets of size about $\frac{q(q+1)}{2}$. So the bound is sharp up to lower-order terms, and in fact the exact minimum is known:

$$\min|K| = \frac{q(q+1)}{2} + \frac{q-1}{2} \quad (q \text{ odd}).$$

Exhaustive search confirms this for small q : at $q = 3$ the minimum is 7 against the bound 6; at $q = 5$ it is 17 against the bound 15. The gap is $\frac{q-1}{2}$, exactly.

In higher dimensions. The bound is $\approx \frac{q^n}{n!}$ and the best known constructions have $\approx \frac{q^n}{2^{n-1}}$ points. Both are $c_n q^n$, so the *order* is settled and only the constant is open. That is a completely different state of affairs from the real problem, where the exponent itself is unknown for $n \geq 4$.

The moral. Over $\mathbb{F}_q$ the Kakeya problem is solved: the answer is q^n up to a dimensional constant, and the remaining question — the sharp constant — is a finite combinatorial one. Nothing of the kind is available over $\mathbb{R}$.

WORKED EXAMPLE 10.2.2— TIER B **The polynomial method: the pattern to recognise**

Dvir's argument is the first instance in this series of a technique used repeatedly in Volume 11. The pattern has three steps and it is worth naming them.

Step A: parameter count. A set S of N points imposes N linear conditions on polynomials. The space of polynomials of degree $\leq d$ in n variables has dimension $\approx d^n/n!$. So if $N < d^n/n!$ there is a non-zero polynomial of degree $\leq d$ vanishing on S . *Vanishing is cheap when the set is small.*

Step B: propagate along structure. The set is not arbitrary — here it contains lines. A polynomial vanishing at $q > d$ points of a line vanishes on the whole line as a polynomial, and one reads off information about its top-degree part. *Structure turns pointwise vanishing into algebraic vanishing.*

Step C: contradiction from degree. The conclusion of Step B is that some non-zero polynomial of controlled degree vanishes too often, contradicting Schwartz–Zippel. *Degree is the resource being spent.*

Where each step reappears. Volume 11’s polynomial partitioning replaces Step A by a partitioning statement — rather than vanishing on the set, the polynomial *separates* it into cells — and Steps B and C become an induction on the degree. The Guth–Katz solution of the distinct distances problem is that variant, and Guth’s proof of the multilinear restriction estimate is another.

The diagnostic for whether the method can apply. Ask whether the configuration has algebraic structure that a polynomial can detect. Lines, yes. Tubes of positive width, *no* — a polynomial does not vanish on a tube, and Chapter 10.7 makes this precise. That single observation explains both the success over $\mathbb{F}_q$ and the difficulty over $\mathbb{R}$.

WORKED EXAMPLE 10.2.3— TIER C: RESEARCH TASTE **What Dvir’s theorem settled, and what it did not**

(a) *It settled the finite field Kakeya conjecture.* Wolff posed it in 1999 as a model problem; before 2008 the best bounds were $q^{(n+2)/2}$, matching the hairbrush. Dvir’s proof gives q^n and is half a page.

(b) *It did not settle the real problem, and gave no route to it.* The argument uses that a line has no thickness. Chapter 10.7 sets out the obstruction; the short version is that the polynomial in Step 1 is built from a *finite* set of vanishing conditions, and a tube of positive width imposes infinitely many.

(c) *It changed which methods the field takes seriously.* Before 2008 the polynomial method was known in combinatorics — Alon’s Combinatorial Nullstellensatz, the Cap Set problem — but not in harmonic analysis. Afterwards Guth and Katz adapted it to $\mathbb{R}^n$ via partitioning, and within five years it had solved the Erdős distinct distances problem and produced the best known restriction estimates. Volume 11 is that development.

(d) *It sharpened what the real difficulty is.* Since the combinatorics of lines is now completely understood and the real problem remains open, the difficulty must be metric — the width of tubes, the near-parallel configurations that Volume 9’s Theorem 9.5.5 could not handle. That is a genuinely useful conclusion, and it is why Volume 14’s account of Wang and Zahl’s work emphasises their treatment of sticky configurations rather than any algebraic input.

(e) *A caution about model problems generally.* The finite field model was proposed on the hypothesis that it would isolate the essential difficulty. It did the opposite: it turned out to be easy, and the difficulty it isolated was the one it had removed. That is worth remembering when a model problem is offered as a stepping stone.

WORKED EXAMPLE 10.2.4 — TIER B **Running the proof on a specific polynomial**

The three steps of Theorem 10.2.1 are short enough to trace on an example, which is the quickest way to see what each does.

Setting. $q = 3$, $n = 2$, and suppose someone proposes a Kakeya set $K \subseteq \mathbb{F}_3^2$ with $|K| = 5 < 6 = \binom{4}{2}$.

Step 1 produces a polynomial. $\mathcal{P}_2^2$ has basis $\{1, x, y, x^2, xy, y^2\}$, dimension 6. Evaluation at the 5 points of K is a linear map $\mathbb{F}_3^6 \rightarrow \mathbb{F}_3^5$, so its kernel has dimension at least 1: there is a non-zero P of degree ≤ 2 vanishing on K .

Step 2 restricts to a line. Say $P = \alpha x^2 + \beta xy + \gamma y^2 + (\text{lower order})$, so $P_2 = \alpha x^2 + \beta xy + \gamma y^2$ if not all of α, β, γ vanish. Along the line $\{(a_1 + tb_1, a_2 + tb_2)\}$ the coefficient of t^2 in $P(a + tb)$ is

$$\alpha b_1^2 + \beta b_1 b_2 + \gamma b_2^2 = P_2(b). \checkmark$$

Step 3 forces $P_2(b) = 0$. The restriction is a polynomial in t of degree ≤ 2 with 3 roots, hence zero; so every coefficient vanishes, including $P_2(b)$.

Step 4 contradicts. P_2 is a non-zero binary quadratic form vanishing at all four directions $(1, 0), (0, 1), (1, 1), (1, 2)$ and hence, by homogeneity, on all 9 points. But a non-zero form of degree $2 < 3$ has at most $2 \cdot 3 = 6$ zeros by Theorem 10.1.3. Contradiction.

What if $\deg P < 2$? Then P_d is linear or constant, and the same argument applies with $d = 1$: a non-zero linear form has 3 zeros, not 9. The proof does not care which d occurs, only that $d \leq q - 1$.

WORKED EXAMPLE 10.2.5 — TIER B **The method of multiplicities, and why it sharpens only here**

Theorem 10.2.1 gives $|K| \geq \frac{q^n}{n!}$; the sharp constant is $\frac{q^n}{2^n}$ up to lower order, and the improvement is due to Dvir, Kopparty, Saraf and Sudan. The mechanism is worth knowing, and so is the reason it does not help over $\mathbb{R}$.

The idea. Instead of requiring P merely to vanish on K , require it to vanish to order m — all partial derivatives up to order $m - 1$ vanish. That is

$\binom{m+n-1}{n}$ conditions per point instead of one, so the degree must rise; but the propagation step strengthens correspondingly, because a one-variable polynomial vanishing to order m at q points has degree $\geq mq$.

The count. With multiplicity m , Step 1 needs $\frac{d^n}{n!} > |K| \binom{m+n-1}{n}$ and Step 3 needs $d < mq$. Balancing as $m \rightarrow \infty$ gives $|K| \gtrsim \frac{q^n}{2^n}$ — the factor $n!$ is replaced by 2^n .

Why it does not transfer. Theorem 10.7.2 shows that over $\mathbb{R}$ the obstruction is that the points of a δ -net inside a tube are *not collinear*. Raising the order of vanishing does not make them collinear; the polynomial exhibited there vanishes to any prescribed order on a curve inside the tube while missing the axis entirely. The repair addresses the wrong difficulty.

The general point. A refinement that improves a constant in one setting need not touch the obstruction in another. Before adapting a technique, identify which step it strengthens and whether that step is the one that fails.

WORKED EXAMPLE 10.2.6 — TIER C: RESEARCH TASTE **Why nobody found this for sixty years**

Dvir’s proof uses linear algebra and the fact that a polynomial has finitely many roots. Both were available to Besicovitch. It is worth asking why the argument waited until 2008, because the answer is about how techniques migrate rather than about difficulty.

(a) *The problem was posed in 1999, not 1928.* The finite field version is Wolff’s, and only nine years elapsed. The real problem is old; the model is not.

(b) *The method was in a different field.* The polynomial method was established in combinatorics — Alon’s Combinatorial Nullstellensatz (1999), the Chevalley–Warning theorem, and the cap-set literature. Harmonic analysts working on Kakeya were using geometric and Fourier-analytic tools, and the two communities overlapped little.

(c) *The prevailing intuition pointed away.* Every bound before 2008 — bush, hairbrush, arithmetic — was of the form cn with $c \leq \frac{1}{2}$ or slightly above, in *both* settings. That the finite field answer was q^n exactly, with no loss, was not expected; the numerical evidence from small q was consistent with it but had not been read that way.

(d) *The proof is short but not obvious.* The move that unlocks it — pass to the top-degree homogeneous part, whose vanishing at b encodes the whole line — is natural in hindsight and was not attempted. Step 3’s observation that a degree- $(q-1)$ polynomial with q roots is *identically* zero, rather than

merely small, is the hinge.

(e) *What to take from this.* Not that hard problems have short solutions — the real Kakeya problem does not — but that a problem's difficulty is relative to the techniques being applied to it. Worked Example 9.8.4 in Volume 9 gives a test for whether a method can reach a target; Dvir's argument passes it trivially because it has no free parameter, and no method with a free parameter was ever going to find it.

WORKED EXAMPLE 10.2.7— TIER A **The three steps, on one page**

A compressed statement of the argument, for reference.

Hypothesis. $K \subseteq \mathbb{F}_q^n$ contains a line in every direction, and $|K| < \binom{q+n-1}{n}$.

Step 1 — build. $\dim \mathcal{P}_{q-1}^n = \binom{q+n-1}{n} > |K|$, so evaluation at K has a kernel: there is $P \neq 0$ of degree $d \leq q-1$ vanishing on K . *Resource spent: the smallness of K .*

Step 2 — restrict. On a line $\ell_{a,b} \subseteq K$, the map $t \mapsto P(a+tb)$ is a polynomial of degree $\leq d$ whose t^d -coefficient is $P_d(b)$. *Resource spent: the affine parametrisation of a line.*

Step 3 — conclude. That polynomial has $q > d$ roots, so it is zero and $P_d(b) = 0$; this for every b , so P_d vanishes on $\mathbb{F}_q^n$, contradicting Theorem 10.1.3. *Resource spent: $d < q$, twice.*

The degree budget. Everything turns on $d \leq q-1$. Step 1 wants d large, to make the polynomial space big enough; Steps 2 and 3 want d small, to make the root counts bite. The theorem is the statement that $d = q-1$ satisfies both, and the resulting bound $\binom{q+n-1}{n}$ is exactly the dimension at that degree.

Compare Volume 9. There the competing demands were on the multiplicity μ , and Worked Example 9.8.2 showed the balance point was unattainable. Here they are on the degree d , and the balance point $d = q-1$ is not only attainable but forced — there is no optimisation, only a single admissible choice. That is the structural difference between the two volumes.

Drills

Drill 10.2.1. Verify (10.2.2): the coefficient of t^d in $P(a+tb)$ is $P_d(b)$.

Drill 10.2.2. Check that $d \geq 1$ in Step 1, and identify where the argument would fail if K were empty.

Drill 10.2.3. Confirm the numbers in Worked Example 10.2.1 for $q = 3$, $n = 2$: compute $\binom{q+1}{2}$ and exhibit a Kakeya set in $\mathbb{F}_3^2$ of 7 points.

Drill 10.2.4. Prove Theorem 10.2.2 with explicit constants, tracking the choice of c .

Where we are. Worth recording explicitly: Theorem 10.2.1 is not merely sharp in order, it is sharp in a stronger sense than anything else in this series. The constructions of Theorem 10.1.4 give $\frac{q(q+1)}{2}$ in the plane and the theorem gives $\binom{q+1}{2} = \frac{q(q+1)}{2}$ — the same number.

No other bound in Volumes 6 to 10 has this property. Stein–Tomas is sharp only in its open range; the bush and hairbrush are off by a factor of two in the exponent; the arithmetic bounds are off by more. That a single half-page argument attains the truth exactly is the strongest available evidence that the finite field problem is of a different character from the real one.

Notes

Dvir’s proof appeared in 2008, resolving Wolff’s problem. The form given here, using the top-degree homogeneous part, is Dvir’s; a variant of Alon and Tao replaces it by an application of the Combinatorial Nullstellensatz.

The sharp constant in the plane is due to Blokhuis and Mazzocca; in higher dimensions the best constructions and the best lower bounds still differ, and closing that gap is a finite combinatorial problem rather than an analytic one.

Chapter 10.3

Sumset Inequalities

The second method of this volume is arithmetic: a Besicovitch set built from a parameter set has its measure controlled, through Fubini, by the additive structure of that set. This chapter assembles the additive-combinatorial tools. They are used in Chapter 10.5; here they are proved for their own sake, and each is elementary.

10.3.1 Notation and the basic counts

Definition 10.3.1. For finite subsets A, B of an abelian group,

$$A + B = \{a + b\}, \quad A - B = \{a - b\}, \quad kA = \underbrace{A + \cdots + A}_k.$$

Proposition 10.3.2 (Trivial bounds). *For non-empty finite A, B in an abelian group,*

$$\max\{|A|, |B|\} \leq |A + B| \leq |A||B|,$$

with the left equality when one of them is a coset of a subgroup containing the other's differences, and the right when all sums $a + b$ are distinct.

Proof. Fixing $b \in B$, the map $a \mapsto a + b$ is injective, so $|A + B| \geq |A|$; symmetrically $\geq |B|$. The upper bound counts pairs. $\square$

Definition 10.3.3. The *doubling constant* of A is $K = |A + A|/|A|$.

Small doubling means A is additively structured; the extreme case $K = 1$ forces A to be a coset of a subgroup. The inequalities below say that small doubling propagates: if $A + A$ is small then so is every iterated sumset.

10.3.2 Ruzsa's triangle inequality

Theorem 10.3.4 (Ruzsa). *For non-empty finite A, B, C in an abelian group,*

$$|A| |B - C| \leq |B - A| |A - C|.$$

Idea of the proof. Read the statement as $d(B, C) \leq d(B, A) + d(A, C)$ after taking logarithms, with $d(X, Y) = \log \frac{|X-Y|}{\sqrt{|X||Y|}}$ — the Ruzsa distance. The proof is an injection, and the point is that an element of $B - C$ can be routed through any element of A .

Proof. For each $d \in B - C$ fix once and for all a representation

$$d = b(d) - c(d), \quad b(d) \in B, \quad c(d) \in C.$$

Define

$$\Phi : A \times (B - C) \longrightarrow (B - A) \times (A - C), \quad \Phi(a, d) = (b(d) - a, a - c(d)).$$

The two coordinates do lie in the stated sets, since $b(d) \in B$, $a \in A$, $c(d) \in C$.

Φ is injective. Suppose $\Phi(a, d) = (u, v)$. Then

$$u + v = (b(d) - a) + (a - c(d)) = b(d) - c(d) = d,$$

so d is determined by (u, v) . Knowing d determines $b(d)$, since the representation was fixed in advance; and then $a = b(d) - u$ is determined. So (a, d) is recovered from (u, v) .

Comparing cardinalities of an injection gives the claim. $\square$

Where we are. Everything in this chapter is of this type: a map between products of sumsets, shown injective by exhibiting the inverse. No inequality is used, and no structure of the group beyond associativity and cancellation. The device to notice is *fixing representations in advance*. The map Φ is well defined only because $b(d)$ and $c(d)$ were chosen once; had they been allowed to vary with a , the inverse would not exist. That trick recurs in every proof below.

Pitfall: the triangle inequality does not convert differences into sums

It is tempting to deduce from Theorem 10.3.4 that $|A - A| \leq K|A|$ implies $|A + A| \lesssim K^2|A|$. Every direct substitution fails, and the failure is structural rather than a lack of ingenuity: the triangle inequality relates *difference* sets to difference sets, and each choice of B, C that puts $A + A$ on the left

also puts it on the right, giving $x \leq x$.

The conversion is true — it is Theorem 10.3.7 below — but it requires Theorem 10.3.5, which is genuinely deeper. A reader who believes they have derived it from Theorem 10.3.4 alone should write out both sides and check that they differ.

10.3.3 The Plünnecke–Ruzsa inequality

Theorem 10.3.5 (Plünnecke–Ruzsa, after Petridis). *Let A, B be finite non-empty subsets of an abelian group with $|A + B| \leq K|A|$. Then for all integers $k, l \geq 0$,*

$$|kB - lB| \leq K^{k+l} |A|.$$

Lemma 10.3.6 (The minimising subset). *With A, B as above, choose a non-empty $X \subseteq A$ minimising*

$$K' = \frac{|X + B|}{|X|}.$$

Then $K' \leq K$ and, for every finite set C ,

$$|X + B + C| \leq K' |X + C|. \quad (10.3.1)$$

Proof. That $K' \leq K$ follows by taking $X = A$, which is admissible.

We prove (10.3.1) by induction on $|C|$.

Step 1. *Base case.*

For $C = \{c\}$ both sides equal $|X + B|$ and $K'|X|$ respectively, which agree by the definition of K' .

Step 2. *The inductive step: a set of “already covered” elements.*

Let $C = C' \cup \{c\}$ with $c \notin C'$, and assume the claim for C' . Put

$$Z = \{x \in X : x + B + c \subseteq X + B + C'\}.$$

Two containments will be used, and both are immediate once written down.

(i) *If $x \in X \setminus Z$ then $x + c \notin X + C'$.* Contrapositively, suppose $x + c = x' + c'$ with $x' \in X, c' \in C'$. Then $x + B + c = x' + B + c' \subseteq X + B + C'$, so $x \in Z$.

(ii) *$((X \setminus Z) + B + c) \setminus (X + B + C')$, translated by $-c$, is contained in $(X + B) \setminus (Z + B)$.* Indeed $Z + B + c \subseteq X + B + C'$ by definition of Z , so $Z + B \subseteq (X + B + C') - c$; hence any element of $((X \setminus Z) + B)$ lying outside $(X + B + C') - c$ lies outside $Z + B$, and it lies in $X + B$.

Step 3. *Count.*

Since $X + B + C = (X + B + C') \cup ((X \setminus Z) + B + c)$, containment (ii) gives

$$|X + B + C| \leq |X + B + C'| + (|X + B| - |Z + B|).$$

Now apply the definition of K' twice: $|X + B| = K'|X|$, and $|Z + B| \geq K'|Z|$ because $Z \subseteq X \subseteq A$ and K' was chosen minimal — this is the *only* place minimality is used, and it is used as a lower bound. Hence

$$|X + B| - |Z + B| \leq K'|X| - K'|Z| = K'|X \setminus Z|.$$

Combining with the inductive hypothesis $|X + B + C'| \leq K'|X + C'|$,

$$|X + B + C| \leq K'(|X + C'| + |X \setminus Z|).$$

Step 4. *Compare with $|X + C|$.*

By containment (i), the points $x + c$ for $x \in X \setminus Z$ lie in $X + C$ and outside $X + C'$, and they are distinct. Therefore

$$|X + C| \geq |X + C'| + |X \setminus Z|,$$

and substituting into Step 3 gives $|X + B + C| \leq K'|X + C|$.

(If $Z = \emptyset$ the bound $|Z + B| \geq K'|Z|$ reads $0 \geq 0$ and the argument is unchanged.) $\square$

Proof of Theorem 10.3.5. Let X be as in Theorem 10.3.6. Applying (10.3.1) repeatedly with $C = jB$ gives, by induction on j ,

$$|X + jB| \leq K'^j |X| \quad (j \geq 0).$$

Now apply Theorem 10.3.4 with A there equal to X , B equal to kB and C equal to lB :

$$|X| |kB - lB| \leq |kB - X| |X - lB| = |X + kB| |X + lB| \quad \text{up to reflection,}$$

where the last step replaces B by $-B$ where needed, which changes no cardinality. Hence

$$|kB - lB| \leq \frac{K'^k |X| \cdot K'^l |X|}{|X|} = K'^{k+l} |X| \leq K'^{k+l} |A|,$$

using $K' \leq K$ and $|X| \leq |A|$. $\square$

Corollary 10.3.7 (Difference sets control sums). *If $|A - A| \leq K|A|$ then $|A + A| \leq K^2|A|$.*

Proof. Apply Theorem 10.3.5 with $B = -A$. Its hypothesis reads

$$|A + B| = |A - A| \leq K|A|,$$

which holds. The conclusion with $k = 0$, $l = 2$ gives

$$|-2B| = |2A| = |A + A| \leq K^2|A|,$$

since $-2B = -2(-A) = 2A$ and cardinality is unchanged by negation. $\square$

Theorem 10.3.8 (Ruzsa covering lemma). *Let A, B be finite non-empty subsets of an abelian group with $|A + B| \leq K|B|$. Then there is $X \subseteq A$ with $|X| \leq K$ and*

$$A \subseteq X + B - B.$$

Proof. Choose $X \subseteq A$ maximal subject to the sets $\{x + B : x \in X\}$ being pairwise disjoint. Such an X exists because A is finite.

Step 1. X is small.

The sets $x + B$, $x \in X$, are disjoint subsets of $A + B$, each of size $|B|$. Hence $|X||B| \leq |A + B| \leq K|B|$, so $|X| \leq K$.

Step 2. $X + B - B$ covers A .

Let $a \in A$. If $a \in X$ then $a \in X + B - B$, taking the same element of B twice. If $a \notin X$ then by maximality $X \cup \{a\}$ fails the disjointness condition, so $(a + B) \cap (x + B) \neq \emptyset$ for some $x \in X$. Pick $b, b' \in B$ with $a + b = x + b'$; then

$$a = x + b' - b \in X + B - B. \quad \square$$

Where we are. The covering lemma is the third of the three elementary tools, and it plays a different role from the other two. Theorem 10.3.4 and Theorem 10.3.5 bound the *size* of sumsets; Theorem 10.3.8 bounds the *structure*, saying that A is covered by boundedly many translates of $B - B$. That is the form needed whenever one wants to conclude that a set is contained in something structured rather than merely that it is small — which is exactly the step from small doubling to a generalised arithmetic progression in Freiman’s theorem. Worked Example 10.5.1 identified that step as one of the two this series does not carry out; the covering lemma is the part of it that is elementary.

Note the proof: a maximal disjoint family, then a counting bound and a maximality argument. It is the same stopping-time pattern as Volume 3, Worked Example 3.1.5, and as the $5r$ -covering lemma of Volume 7, Theorem 7.4.1. Three appearances of one idea.

WORKED EXAMPLE 10.3.1— TIER A Sumsets of arithmetic progressions and of random sets

Two extremes, which bracket everything.

An arithmetic progression. $A = \{0, 1, \dots, N-1\}$ has $A + A = \{0, \dots, 2N-2\}$, so $|A + A| = 2|A| - 1$ and $K \approx 2$. By Theorem 10.3.5, $|kA - lA| \leq 2^{k+l}|A|$ — and directly, $kA - lA = \{-l(N-1), \dots, k(N-1)\}$ has $(k+l)(N-1) + 1$ elements, so the truth is $\approx (k+l)|A|$. Plünnecke is exponential in $k+l$ where the truth is linear: sharp in its dependence on K , lossy in k, l .

A random set. Let A be a random subset of $\{1, \dots, N\}$ of size $N^{1/2}$. Then almost all sums $a + a'$ are distinct, so $|A + A| \approx \frac{1}{2}|A|^2$ and $K \approx \frac{1}{2}|A|$ — as large as Theorem 10.3.2 permits. Such a set has no additive structure, and the inequalities say nothing useful about it.

The dichotomy. Small K means the set is close to a progression — made precise by Freiman’s theorem, which this series does not prove. Large K means it is essentially random. The arithmetic method for Kakeya works by showing that a small Besicovitch set forces small K for an associated set, and then contradicting the structure that small K entails.

WORKED EXAMPLE 10.3.2— TIER B **Reading a sumset hypothesis**

Additive combinatorics states hypotheses in several equivalent-looking ways, and the differences matter.

$|A + A| \leq K|A|$ — *small doubling*. The standard hypothesis. Controls all $|kA - lA|$ by Theorem 10.3.5.

$|A - A| \leq K|A|$ — *small difference*. Formally different, but by Theorem 10.3.5 applied with $B = A$ each implies the other with K replaced by K^2 . So the two are equivalent up to squaring the constant, and a paper using one may be compared with a paper using the other only after that adjustment.

$|\{(a, a') \in A \times A : a + a' = x\}|$ *large for many x* — *large additive energy*. Not equivalent: a set can have large energy and large doubling, by consisting of a structured piece plus a random piece. The Balog–Szemerédi–Gowers theorem repairs this, producing a large subset with small doubling; it is proved later in this volume as Theorem 10.4.2.

Which one an application needs. The arithmetic Kakeya argument produces an *energy* hypothesis from the geometry — many triples of tubes meeting — and must pass to doubling before Plünnecke applies. That passage is Balog–Szemerédi–Gowers, and it is where the constants degrade badly. It is the reason the arithmetic bounds, though better than $\frac{n+2}{2}$ for large n , are far from n .

WORKED EXAMPLE 10.3.3— TIER A **The Ruzsa distance**

Theorem 10.3.4 is named for a reason, and the metric interpretation is worth making explicit.

The definition. For non-empty finite A, B set

$$d(A, B) = \log \frac{|A - B|}{\sqrt{|A||B|}}.$$

It is symmetric — $|A - B| = |B - A|$ after negation — *and satisfies the triangle inequality*: taking logarithms in $|A||B - C| \leq |B - A||A - C|$ and rearranging,

$$\log \frac{|B - C|}{\sqrt{|B||C|}} \leq \log \frac{|B - A|}{\sqrt{|B||A|}} + \log \frac{|A - C|}{\sqrt{|A||C|}},$$

which is $d(B, C) \leq d(B, A) + d(A, C)$. ✓

It is not a metric. $d(A, A) = \log \frac{|A - A|}{|A|} \geq 0$ with equality only when A is a coset of a subgroup, so distinct sets can be at distance zero from themselves and $d(A, B) = 0$ does not force $A = B$. It is a pseudo-metric on the space of sets modulo translation and subgroup structure.

Why the normalisation is $\sqrt{|A||B|}$. Because it makes d symmetric and translation-invariant, and because Theorem 10.3.2 then gives $d(A, B) \geq 0$ always. Any other normalisation breaks one of these.

What it is for. It converts additive-combinatorial statements into geometric ones: “ A and B have similar additive structure” becomes “ A and B are close”. Chains of inequalities become paths, and Theorem 10.3.5 becomes a statement that a short step in one direction bounds all longer steps.

WORKED EXAMPLE 10.3.4— TIER B **Applying Plünnecke: the checklist**

Theorem 10.3.5 is used constantly and misapplied often. Four things to check.

(1) *Which set carries the hypothesis?* The hypothesis is $|A + B| \leq K|A|$ — the bound is relative to $|A|$, and the conclusion $|kB - lB| \leq K^{k+l}|A|$ is *also* relative to $|A|$, not to $|B|$. Confusing the two is the commonest error, and it matters when A and B have very different sizes.

(2) *Which set is iterated?* The conclusion iterates B , not A . To control $|kA - lA|$ one applies the theorem with $B = A$, and then the hypothesis reads $|A + A| \leq K|A|$, the doubling condition.

(3) *Signs.* $|kB - lB|$ covers both sums and differences: taking $l = 0$ gives $|kB|$, taking $k = 0$ gives $|lB|$ after negation. So a single application controls every iterated sumset and difference set simultaneously.

(4) *The exponent is $k + l$, not max.* The bound K^{k+l} is exponential in the total number of copies. Worked Example 10.3.1 showed the truth for an arithmetic progression is linear in $k + l$, so the theorem is lossy there — but it is sharp in its dependence on K , which is what applications need.

A worked application. Suppose $|A - A| \leq K|A|$ and one wants $|3A - 2A|$. Apply the theorem with $B = -A$: the hypothesis is $|A + B| = |A - A| \leq K|A|$ ✓, and the conclusion with $k = 2, l = 3$ gives $|2B - 3B| = |3A - 2A| \leq K^5|A|$.

WORKED EXAMPLE 10.3.5— TIER C: RESEARCH TASTE **Where additive combinatorics enters this subject**

Sumset inequalities look remote from Besicovitch sets, and it is worth setting out the three places they enter, since only the first belongs to this volume.

(a) *The Kakeya problem, through slices.* Theorem 10.5.2 turns lines into three-term progressions, so a Besicovitch set becomes a configuration with many additive triples. That is Chapter 10.5, and it is the route Bourgain took.

(b) *The sum-product phenomenon.* A finite set of reals cannot have both $A + A$ and $A \cdot A$ small; Erdős and Szemerédi conjectured $\max\{|A + A|, |A \cdot A|\} \gtrsim |A|^{2-\varepsilon}$. Sum-product estimates feed into incidence bounds, and incidence bounds feed into Kakeya — a longer chain, exploited in the finite-field setting by Bourgain, Katz and Tao.

(c) *The Furstenberg set problem.* A set meeting each line of a family in dimension s ; the extremal examples are arithmetically structured, and the solution of Ren and Wang reported in Volume 14 uses additive-combinatorial input.

What the three have in common. In each, a geometric configuration forces an additive relation, and the additive relation constrains the geometry. The inequalities of this chapter are the constraint half; the geometric half differs in each case.

Why the losses are severe in every case. The passage from geometry to additive structure is through *energy* — many triples — and the passage from energy to *doubling*, which is what Theorem 10.3.5 needs, costs polynomial factors by Balog–Szemerédi–Gowers. Worked Example 10.6.2 tracks this for the Kakeya application, and the same accounting applies to (b) and (c).

A caution for the reader heading into this literature. The inequalities here are stated for abelian groups; applications to $\mathbb{R}^n$ use them for $\mathbb{Z}^n$ or for δ -discretised sets, and the discretisation introduces error terms that must be tracked. A statement about finite subsets of $\mathbb{Z}$ does not transfer to δ -separated subsets of $\mathbb{R}$ without care.

WORKED EXAMPLE 10.3.6— TIER B Proving a sumset inequality: the injection recipe

Every result in this chapter is proved by exhibiting an injection. The recipe is uniform and worth practising, since the literature contains dozens of such inequalities and they are all proved this way.

The shape. To prove $|X_1||X_2| \leq |Y_1||Y_2|$, construct $\Phi : X_1 \times X_2 \rightarrow Y_1 \times Y_2$ and show it injective by exhibiting the inverse explicitly.

Step 1: fix representations in advance. Elements of a sumset generally have many representations. Choose one for each, once and for all, before defining Φ . In Theorem 10.3.4 this is $d \mapsto (b(d), c(d))$, and without it the inverse does not exist.

Step 2: make the map recover a distinguished element. Design Φ so that some combination of the output coordinates — usually their sum or difference — returns an element of the input. In Theorem 10.3.4, $u + v = d$.

Step 3: unwind. With d recovered, the fixed representation gives $b(d)$, and then the remaining input coordinate follows.

Worked practice: the Ruzsa sum triangle inequality. Prove $|A||B + C| \leq |A + B||A + C|$. Fix for each $s \in B + C$ a representation $s = b(s) + c(s)$, and set

$$\Phi(a, s) = (a + b(s), a + c(s)) \text{ —}$$

but this is not injective, since a cannot be recovered from the two coordinates: their difference is $b(s) - c(s)$, which gives s only if that determines it. The correct map uses the *difference* form, which is why Theorem 10.3.4 is stated with $B - C$ rather than $B + C$. The sum version requires Theorem 10.3.5 and is genuinely deeper — a useful thing to discover by attempting it.

The habit. Before believing a claimed sumset inequality, try to write the injection. If Step 2 has no natural candidate, the inequality is probably false or probably deep.

WORKED EXAMPLE 10.3.7— TIER A Doubling constants, computed

Six sets, with their doubling constants, to calibrate what K measures.

$A \subseteq \mathbb{Z}$	$ A $	$ A + A $	K
$\{0, 1, \dots, N - 1\}$	N	$2N - 1$	≈ 2
$\{0, d, 2d, \dots, (N - 1)d\}$	N	$2N - 1$	≈ 2
$\{0, \dots, N - 1\} \times \{0, \dots, N - 1\}$ (in $\mathbb{Z}^2$)	N^2	$(2N - 1)^2$	≈ 4
$\{1, 2, 4, 8, \dots, 2^{N-1}\}$	N	$\binom{N}{2} + N$	$\approx \frac{N}{2}$
$\{0, \dots, N - 1\} \cup \{M\}$, $M \gg N$	$N + 1$	$\approx 3N$	≈ 3
random subset of $[N^2]$ of size N	N	$\approx \frac{N^2}{2}$	$\approx \frac{N}{2}$

What the table shows. Doubling is small exactly for sets that are progressions, or products of progressions, or progressions with a bounded number of extra points. It is as large as possible — $\approx \frac{|A|}{2}$ — for geometric progressions and for random sets, which have no additive structure.

The dilation invariance. Rows one and two have the same K : doubling is unchanged by an affine map $x \mapsto ax + b$, since such a map is a bijection commuting with addition up to translation. So K measures structure, not size or position.

The rank shows up in the constant. A d -dimensional progression has $K \approx 2^d$, as row three illustrates with $d = 2$. So $\log_2 K$ estimates the rank, which is the quantitative content of Freiman's theorem.

Why row five matters. Adding one far-away point to a progression roughly triples the sumset but barely changes the size. So K is sensitive to outliers, and applications must either remove them — which is what Theorem 10.3.8 and Balog–Szemerédi–Gowers do — or tolerate the degradation.

Drills

Drill 10.3.1. Verify the injectivity of Φ in Theorem 10.3.4 by writing out the inverse explicitly.

Drill 10.3.2. Check Theorem 10.3.4 numerically on $A = B = C = \{0, 1, 3\}$ in $\mathbb{Z}$.

Drill 10.3.3. Verify the base case $|C| = 1$ of Theorem 10.3.6, and identify where the minimality of K' is used in the inductive step.

Drill 10.3.4. For A an arithmetic progression of length N , compute $|kA - lA|$ exactly and compare with the bound of Theorem 10.3.5.

Where we are. Three inequalities, three roles, worth separating before leaving the chapter.

Theorem 10.3.4 is a *comparison*: it relates the sizes of three difference sets and is the metric backbone, as Worked Example 10.3.3 makes explicit.

Theorem 10.3.5 is a *propagation*: one hypothesis about $A + B$ controls every iterated sumset of B . It is the workhorse, and Worked Example 10.3.4 lists the four ways it is misapplied.

Theorem 10.3.8 is a *structure* statement: A is covered by few translates of $B - B$. It is the only one of the three whose conclusion is about containment rather than cardinality, and it is the elementary part of the passage to Freiman’s theorem.

All three are proved by injections and maximal families, and none uses anything about the ambient group beyond cancellation.

Notes

Ruzsa’s triangle inequality is from 1996 and the distance interpretation is his. Plünnecke’s original proof (1970) used a graph-theoretic argument; Petridis’s proof, given here, appeared in 2012 and is substantially shorter.

Freiman’s theorem and the Balog–Szemerédi–Gowers theorem are the two deeper results of the area; neither is proved in this series, and Worked Example 10.3.2 records where the second would be needed.

Chapter 10.4

Balog–Szemerédi–Gowers

Chapter 10.3 developed the sumset toolkit — Ruzsa’s triangle inequality and the Plünnecke–Ruzsa inequality — under the hypothesis that a sumset $A + B$ is small. In applications one frequently knows less: that $a + b$ takes few values for *many* pairs (a, b) , not all. This chapter proves that the weaker hypothesis suffices after passing to large subsets.

The result is used twice in this series: in Chapter 10.5 for the arithmetic approach to Kakeya, and in Volume 14, Theorem 14.9.2 for Bourgain’s projection theorem. It was the last additive-combinatorial input recorded as quoted.

10.4.1 Partial sumsets

Definition 10.4.1 (Partial sumset). Let A, B be finite subsets of an abelian group and $G \subseteq A \times B$. The *partial sumset* is

$$A +_G B = \{a + b : (a, b) \in G\}.$$

We regard G as the edge set of a bipartite graph with parts A and B .

Where we are. The hypothesis $|A +_G B|$ small does *not* imply $|A + B|$ small, and the gap can be extreme.

The standard example. Let $A = B = \{1, \dots, n\} \cup S$ where S is a set of n integers in general position — say $\{2^{n+1}, \dots, 2^{2n}\}$. Take G to be the pairs with both entries in $\{1, \dots, n\}$. Then $|G| = n^2 = \frac{1}{4}|A||B|$ and $|A +_G B| = 2n - 1$, while $|A + B| \geq |S + S| \approx \frac{1}{2}n^2$.

So no conclusion about $A + B$ itself is possible, and the theorem must instead produce subsets. What is remarkable is how little is lost: the subsets retain a $K^{-O(1)}$ fraction, and the constant in the sumset bound is $K^{O(1)}$.

Theorem 10.4.2 (Balog–Szemerédi–Gowers). *There is an absolute constant C such that the following holds. Let A, B be finite subsets of an abelian group and $G \subseteq A \times B$ with*

$$|G| \geq \frac{|A||B|}{K}, \quad |A +_G B| \leq K|A|^{1/2}|B|^{1/2}.$$

Then there are $A' \subseteq A$ and $B' \subseteq B$ with

$$\begin{aligned} |A'| &\gtrsim K^{-C}|A| \\ &, \quad |B'| \gtrsim K^{-C}|B| \\ &, \quad |A' + B'| \\ &\lesssim K^C|A|^{1/2}|B|^{1/2}, \end{aligned}$$

and moreover $|G \cap (A' \times B')| \gtrsim K^{-C}|A||B|$.

10.4.2 The graph lemma

Everything rests on a statement about bipartite graphs with no arithmetic in it.

Lemma 10.4.3 (Paths of length two). *Let H be a bipartite graph with parts A, B of size n and at least cn^2 edges. Then there is $A_1 \subseteq A$ with $|A_1| \geq cn/\sqrt{2}$ such that at most half the pairs $(a, a') \in A_1^2$ satisfy $|N(a) \cap N(a')| < \frac{c^2}{4}n$.*

Proof. Choose $b \in B$ uniformly at random and set $A_1 = N(b)$. Then

$$\mathbb{E}|A_1| = \frac{1}{n} \sum_{b \in B} \deg(b) = \frac{|E|}{n} \geq cn,$$

so by Cauchy–Schwarz $\mathbb{E}|A_1|^2 \geq (\mathbb{E}|A_1|)^2 \geq c^2n^2$.

Call (a, a') *bad* if $|N(a) \cap N(a')| < \varepsilon n$. A bad pair lies in A_1^2 precisely when b is a common neighbour, which happens with probability $|N(a) \cap N(a')|/n < \varepsilon$. Hence

$$\mathbb{E}[\#\{\text{bad pairs in } A_1^2\}] = \sum_{\text{bad } (a, a')} \frac{|N(a) \cap N(a')|}{n} < \varepsilon n^2.$$

Taking $\varepsilon = c^2/4$,

$$\begin{aligned} \mathbb{E}[|A_1|^2 - 2\#\{\text{bad in } A_1^2\}] &\geq c^2n^2 - 2 \cdot \frac{c^2}{4}n^2 \\ &= \frac{1}{2}c^2n^2 > 0, \end{aligned}$$

so some b realises at least the average. For that b , $|A_1|^2 \geq \frac{1}{2}c^2n^2$ and $\#\{\text{bad in } A_1^2\} \leq \frac{1}{2}|A_1|^2$. $\square$

Where we are. This is *dependent random choice*: one selects a random vertex and keeps its neighbourhood, and the point is that a set arising this way is *biased towards having large common neighbourhoods*. A uniformly random subset of A of the same size would have no such property.

The mechanism is the second display: a pair with few common neighbours is unlikely to survive, because surviving requires b to be one of them. So selecting by a random neighbourhood automatically suppresses the bad pairs, and the two effects are balanced by choosing $\varepsilon = c^2/4$.

Lemma 10.4.4 (Paths of length three). *Let H be a bipartite graph with parts A, B of size n and at least cn^2 edges. Then there are $A' \subseteq A$ and $B' \subseteq B$ with $|A'|, |B'| \gtrsim cn$ such that for every $a \in A'$ and $b \in B'$ there are $\gtrsim c^5 n^2$ paths of length three from a to b .*

Proof. **Step 1.** *A set with mostly-good pairs.*

By Theorem 10.4.3 take A_1 with $|A_1| \geq cn/\sqrt{2}$ in which at most half the pairs are bad, “bad” meaning fewer than $\frac{c^2}{4}n$ common neighbours. Discarding the vertices lying in the most bad pairs, we obtain $A_2 \subseteq A_1$ with $|A_2| \geq \frac{1}{2}|A_1|$ in which at most a $\frac{1}{5}$ fraction of pairs are bad — a Markov argument, since a vertex in many bad pairs is rare.

Step 2. *Vertices of B seeing A_2 often.*

Let

$$B' = \left\{ b \in B : \deg_{A_2}(b) \geq \frac{c}{4}|A_2| \right\}.$$

Counting edges between A_2 and B from the A_2 side and using that H has $\geq cn^2$ edges, $|B'| \gtrsim cn$.

Step 3. *Vertices of A with many good companions.*

Call $a \in A$ *friendly* to $a' \in A_2$ if a and a' have $\geq \frac{c^2}{4}n$ common neighbours. Let

$$A' = \left\{ a \in A : a \text{ is friendly to at least } \left(1 - \frac{c}{5}\right)|A_2| \text{ vertices of } A_2 \right\}.$$

Since at most $\frac{1}{5}$ of pairs within A_2 are bad, a Markov argument gives $|A'| \gtrsim cn$; note A' need not lie in A_2 .

Step 4. *Count the paths.*

Fix $a \in A'$ and $b \in B'$. The vertices of A_2 friendly to a number at least $(1 - \frac{c}{5})|A_2|$; the neighbours of b in A_2 number at least $\frac{c}{4}|A_2|$. These two sets must overlap in at least

$$\left(\frac{c}{4} - \frac{c}{5}\right)|A_2| = \frac{c}{20}|A_2| \gtrsim c^2 n$$

vertices. *This is the whole point of Steps 2 and 3:* the thresholds $\frac{c}{4}$ and $1 - \frac{c}{5}$ are chosen so that the overlap is forced.

For each a' in the overlap, a and a' have $\geq \frac{c^2}{4}n$ common neighbours b' , and $a'b$ is an edge. Each choice of (b', a') gives a distinct path $a \rightarrow b' \rightarrow a' \rightarrow b$ of length three, so the number of such paths is

$$\gtrsim c^2 n \cdot \frac{c^2}{4} n \gtrsim c^4 n^2,$$

which is at least the claimed $c^5 n^2$. $\square$

Where we are. *Why paths of length three and not one.* The conclusion one would like — that $A' \times B'$ is entirely contained in the edge set — is false. A random bipartite graph of density $\frac{1}{2}$ contains no complete bipartite subgraph with parts of size $2 \log n$, so no bound of the form $|A'|, |B'| \gtrsim cn$ can force even one edge between every pair.

Paths of length *two* fail as well: Step 1 gives only that *most* pairs have many common neighbours, not all. Length three is the first case where one can pass from “most” to “all”, and Step 4 is where that passage happens — by intersecting two large subsets of A_2 rather than by any property of a single pair.

10.4.3 Proof of the theorem

Proof of Theorem 10.4.2. We may assume $|A| = |B| = n$; the general case follows by padding with elements far from everything, as in the Checkpoint after Theorem 10.4.1.

Apply Theorem 10.4.4 to the graph G with $c = K^{-1}$, obtaining $A' \subseteq A$ and $B' \subseteq B$ with $|A'|, |B'| \gtrsim K^{-1}n$ and, for every $a \in A'$, $b \in B'$, at least γn^2 paths of length three from a to b , where $\gamma \gtrsim K^{-5}$.

Fix such a, b . A path of length three is a triple (b_1, a_1, b_2) with ab_1, a_1b_1, a_1b_2 edges of G and $b_2 = b$; write it as

$$a + b = \underbrace{(a + b_1)}_{\in A+GB} - \underbrace{(a_1 + b_1)}_{\in A+GB} + \underbrace{(a_1 + b)}_{\in A+GB}.$$

Each path exhibits $a + b$ as an element of $(A +_G B) - (A +_G B) + (A +_G B)$, and distinct paths give distinct triples of summands, since b_1 and a_1 are recoverable from them.

Therefore the map sending a path to its triple of summands is injective, and counting over all $(a, b) \in A' \times B'$,

$$|A' + B'| \cdot \gamma n^2 \leq \#\{\text{paths of length three in } G\} \leq |A +_G B|^3.$$

Hence, using $|A +_G B| \leq Kn$,

$$|A' + B'| \leq \frac{|A +_G B|^3}{\gamma n^2} \lesssim \frac{K^3 n^3}{K^{-5} n^2} = K^8 n,$$

which is the stated bound with $C = 8$.

For the final assertion, the number of edges of G inside $A' \times B'$ is at least $|A'| |B'| - \#\{\text{non-edges}\}$; but every pair of $A' \times B'$ is joined by $\gamma n^2 > 0$ paths, and the same counting applied to $G \cap (A' \times B')$ gives $|G \cap (A' \times B')| \gtrsim K^{-C} n^2$ after shrinking A', B' by a further constant factor. $\square$

Remark 10.4.5 (R10, discharged). *Theorem 10.4.2 is now proved in full.* Revision R10 recorded the Balog–Szemerédi–Gowers theorem and Freiman’s theorem as the two quoted inputs of Chapter 10.5; the first is discharged here.

Freiman’s theorem remains quoted. It asserts that a set of small doubling is contained in a generalised arithmetic progression of bounded rank and comparable size — a structure theorem, not an inequality, and a substantially deeper result. Nothing in this series depends on it except Theorem 10.6.1, whose omission R10 also records.

Where this is used. Chapter 10.5, and Volume 14, Theorem 14.9.2, in the proof of Bourgain’s projection theorem. With this chapter, *that theorem rests entirely on results proved in this series*: Marstrand and Kaufman from Volume 8, Ruzsa and Plünnecke–Ruzsa from Chapter 10.3, Blichfeldt’s principle, and Theorem 10.4.2.

WORKED EXAMPLE 10.4.1— TIER A **The three counting steps, side by side**

The proof is three counting arguments and nothing else. Isolating them:

(1) *Dependent random choice* (Theorem 10.4.3). Selecting a random neighbourhood biases towards large common neighbourhoods. Balanced by $\varepsilon = c^2/4$, and verified: $\mathbb{E}[|A_1|^2 - 2\#\text{bad}] \geq \frac{1}{2}c^2n^2$.

(2) *Forced overlap* (Step 4 of Theorem 10.4.4). Two subsets of A_2 of relative sizes $\geq \frac{c}{4}$ and $\geq 1 - \frac{c}{5}$ must meet in $\geq \frac{c}{20}$ of it. Elementary, and it is what upgrades “most pairs” to “all pairs”.

(3) *Injective encoding* (Theorem 10.4.2). Each path of length three writes $a + b$ as $x - y + z$ with $x, y, z \in A +_G B$, injectively. So many paths force $|A' + B'|$ small.

The shape. A hypothesis about *many* pairs becomes one about *all* pairs by (1) and (2), and then (3) converts the graph statement into an arithmetic one. *Only step (3) knows any arithmetic*, and it occupies four lines.

The exponent. $C = 8$ above, from $\gamma \gtrsim K^{-5}$ and $|A +_G B|^3$. The optimal exponent is not known; what matters for every application in this series is only that it is polynomial in K .

WORKED EXAMPLE 10.4.2— TIER B When to reach for BSG

The theorem answers one specific difficulty, and recognising it is most of the skill.

(a) *The symptom.* One has good control of $a + b$ for many pairs, and no control for the rest — typically because the pairs came from a pigeonhole, a popularity argument, or an incidence count.

(b) *What BSG gives.* Subsets on which the control is total, at polynomial cost. One may then apply Ruzsa and Plünnecke–Ruzsa (Chapter 10.3), which require full control.

(c) *The instance in Volume 14.* The proof of Theorem 14.9.2, especially Theorem 14.9.14, has a set E that is only *dense in* a product $A_1 \times A_2$, not equal to it. BSG converts this into genuine subsets $B_i \subseteq A_i$ with $|B_1 + B_2|$ controlled, and the argument proceeds.

(d) *The instance in this volume.* Chapter 10.5 obtains small-doubling information only on a positive proportion of a set, and BSG supplies the subset on which Freiman’s theorem can be applied.

(e) *The cost to watch.* The subsets are smaller by $K^{-O(1)}$. In an iterative argument this compounds, and it is the reason Theorem 10.6.1 yields a fixed *fraction* of n rather than $n - O(1)$ — Volume 10, Worked Example 10.6.2.

Drills

Drill 10.4.1. Verify the example in the Checkpoint after Theorem 10.4.1: compute $|A +_G B|$ and $|A + B|$.

Drill 10.4.2. In Theorem 10.4.3, check that the probability a bad pair lies in A_1^2 equals $|N(a) \cap N(a')|/n$, and confirm the choice $\varepsilon = c^2/4$.

Drill 10.4.3. Verify the forced overlap in Step 4 of Theorem 10.4.4, and explain why the thresholds $\frac{c}{4}$ and $1 - \frac{c}{5}$ must be chosen with $\frac{c}{4} > \frac{c}{5}$.

Drill 10.4.4. Confirm the injectivity of the encoding in Theorem 10.4.2: recover b_1 and a_1 from the triple of summands.

Notes

Balog and Szemerédi proved the theorem with weaker bounds in 1994; Gowers obtained polynomial dependence in 1998. The proof here follows Sudakov, Szemerédi and Vu, as presented by Tao and Vu. Theorem 10.4.5 records that Freiman’s theorem, the other input of Chapter 10.5, remains quoted.

Chapter 10.5

The Arithmetic Reduction

The sumset inequalities of Chapter 10.3 concern subsets of an abelian group. This chapter connects them to Besicovitch sets. The bridge is a single observation: *a line meets three parallel hyperplanes in an arithmetic progression*, so a set containing lines in many directions contains many additive triples.

10.5.1 Slices and progressions

Definition 10.5.1. For $E \subseteq \mathbb{R}^n$ and $t \in \mathbb{R}$ write $E_t = \{y \in \mathbb{R}^{n-1} : (y, t) \in E\}$ for the horizontal slice at height t .

Lemma 10.5.2 (Three-slice lemma). *Let ℓ be a line in $\mathbb{R}^n$ meeting the hyperplanes $x_n = 0$, $x_n = \frac{1}{2}$, $x_n = 1$, at the points $(a, 0)$, $(b, \frac{1}{2})$, $(c, 1)$. Then*

$$a + c = 2b.$$

Conversely, given $a, b, c \in \mathbb{R}^{n-1}$ with $a + c = 2b$, the line through $(a, 0)$ and $(c, 1)$ passes through $(b, \frac{1}{2})$.

Proof. A line meeting all three hyperplanes is not horizontal, so it may be written $\{(a + tv, t) : t \in \mathbb{R}\}$ for some $a, v \in \mathbb{R}^{n-1}$. At $t = 0, \frac{1}{2}, 1$ the first coordinates are $a, a + \frac{v}{2}, a + v$. Hence $b = a + \frac{v}{2}$ and $c = a + v$, so $a + c = 2a + v = 2b$. The converse is the same computation with $v = c - a$. $\square$

Where we are. This is the whole geometric input of the chapter. It says the three slices of a line, at equally spaced heights, form a three-term arithmetic progression in $\mathbb{R}^{n-1}$ — and conversely every such progression arises from a line.

So the family of lines through a set is in bijection with the family of additive triples among its slices. A Besicovitch set has a line in every direction, hence many triples; and Chapter 10.3 says that a small set cannot have

many triples unless it is additively structured. Playing those against each other is the arithmetic method.

Definition 10.5.3 (Additive triples). For finite A, B, C in an abelian group, let

$$T(A, B, C) = \#\{(a, b, c) \in A \times B \times C : a + c = 2b\}.$$

Proposition 10.5.4 (A Besicovitch set has many additive triples). *Let $K \subseteq \mathbb{R}^n$ be a Besicovitch set and $0 < \delta < \frac{1}{2}$. Let $A, B, C \subseteq \delta\mathbb{Z}^{n-1}$ be the sets of δ -cubes meeting K_δ in the slabs $|x_n| \leq \delta$, $|x_n - \frac{1}{2}| \leq \delta$ and $|x_n - 1| \leq \delta$ respectively. Then*

$$T'(A, B, C) := \#\{(a, b, c) \in A \times B \times C : |a + c - 2b| \leq 4\delta\} \gtrsim_n \delta^{-(n-1)},$$

while

$$|A|, |B|, |C| \lesssim_n \delta^{-n} m(K_\delta).$$

Proof. Step 1. Count the triples.

Let v range over a maximal δ -separated set of directions in $\{(w, 1) : |w| \leq 1\}$, of which there are $\approx_n \delta^{-(n-1)}$. For each v the set K contains a unit segment in that direction; extending and rescaling so that it meets the three hyperplanes — which costs only a dimensional constant, since a unit segment in a direction bounded away from horizontal crosses a fixed proportion of the slab $0 \leq x_n \leq 1$ — we obtain points $(a_v, 0), (b_v, \frac{1}{2}), (c_v, 1)$ on the segment. By Theorem 10.5.2, $a_v + c_v = 2b_v$ exactly.

Rounding each to the nearest δ -cube gives $a \in A, b \in B, c \in C$ with $|a + c - 2b| \leq 4\delta$. Distinct v give distinct triples: the direction is recovered as $c - a$ up to $O(\delta)$, and the v are δ -separated. So there are $\gtrsim_n \delta^{-(n-1)}$ triples.

Step 2. Bound the sizes of the slices.

Each δ -cube counted in A contributes a set of measure $\approx \delta^n$ to K_δ — the cube thickened in the x_n direction by δ — and distinct cubes contribute disjointly up to bounded overlap. Hence $|A| \delta^n \lesssim_n m(K_\delta)$, that is $|A| \lesssim_n \delta^{-n} m(K_\delta)$. The same holds for B and C . $\square$

10.5.2 What the triples buy

Proposition 10.5.5 (The trivial consequence). *With A, B, C as above and $N = \max\{|A|, |B|, |C|\}$,*

$$T'(A, B, C) \lesssim_n N^2.$$

Consequently $\delta^{-(n-1)} \lesssim_n N^2$, and

$$m(K_\delta) \gtrsim_n \delta^n \delta^{-(n-1)/2} = \delta^{(n+1)/2}.$$

Proof. A triple is determined by a and b , since $c = 2b - a$ up to the $O(\delta)$ rounding, which permits boundedly many choices of c . So $T' \lesssim_n |A||B| \leq N^2$.

Combining with Theorem 10.5.4 Step 1, $\delta^{-(n-1)} \lesssim N^2$, so $N \gtrsim \delta^{-(n-1)/2}$; and by Theorem 10.5.4, $m(K_\delta) \gtrsim \delta^n N \gtrsim \delta^{n-(n-1)/2} = \delta^{(n+1)/2}$. $\square$

Where we are. Theorem 10.5.5 gives $m(K_\delta) \gtrsim \delta^{(n+1)/2}$, which is *weaker* than the bush bound $\delta^{(n-1)/2}$ of Volume 9, Theorem 9.4.4, by a factor δ . So the arithmetic reduction, used with only the trivial count $T' \leq N^2$, is worse than the geometry it replaces.

That is exactly as it should be. The trivial count throws away all the structure: it uses only that a triple is determined by two of its entries. The content of the method is that T' being *large* — comparable to N^2 — forces A, B, C to be additively structured, and structured sets cannot arise from Besicovitch configurations. Extracting that is the subject of Chapter 10.6, and it is where the sunset inequalities enter.

WORKED EXAMPLE 10.5.1— TIER B **Where the additive structure would come from**

The reduction produces an *energy* statement: many additive triples. Turning it into a *doubling* statement, to which Theorem 10.3.5 applies, is the technical heart, and it is worth seeing the shape.

Step 1: energy to doubling, via Balog–Szemerédi–Gowers. If $T(A, B, C) \geq \eta|A||B|$ then there are subsets $A' \subseteq A$, $C' \subseteq C$ with $|A'| \gtrsim \eta^{O(1)}|A|$ and $|A' + C'| \lesssim \eta^{-O(1)}|A'|$. So a large-energy pair contains a small-doubling pair, at the cost of polynomial losses in η . The theorem is proved in Chapter 10.4; earlier chapter remarks treating it as quoted are historical.

Step 2: doubling to structure, via Plünnecke. Theorem 10.3.5 then bounds all iterated sumsets of A' , and Freiman’s theorem — also not proved here — says A' is contained in a generalised arithmetic progression of bounded rank and comparable size.

Step 3: structure contradicts the geometry. A Besicovitch configuration whose slices lie in low-rank progressions can be analysed directly, and one shows the directions cannot be δ -separated in the required number. This is where the dimension bound comes out.

Why the losses are severe. Step 1 costs a power of η , and η is itself a power of δ ; so the exponent obtained is worse than the naive count suggests. That is the reason the arithmetic bounds, though better than $\frac{n+2}{2}$ for large n , are far from n — and why they are stated with unattractive constants like $\frac{13}{25}$.

Proposition 10.5.6 (Slices and the maximal function). *Let $K \subseteq \mathbb{R}^n$ be a Besicovitch set and $0 < \delta < \frac{1}{2}$. Then for every $t \in [0, 1]$,*

$$m_{n-1}((K_\delta)_t) \lesssim_n \delta^{-1} m_n(K_\delta),$$

and there is a set of t of measure $\geq \frac{1}{2}$ on which

$$m_{n-1}((K_\delta)_t) \gtrsim_n m_n(K_\delta).$$

Proof. For the first bound, note $(K_\delta)_t \times [t - \delta, t + \delta] \subseteq K_{2\delta}$ up to a dimensional constant, since a point within δ of K at height t remains within 2δ of K at nearby heights. Hence $2\delta m_{n-1}((K_\delta)_t) \leq m_n(K_{2\delta}) \lesssim_n m_n(K_\delta)$, the last step because doubling the neighbourhood increases its measure by at most a dimensional factor.

For the second, Fubini gives $\int_0^1 m_{n-1}((K_\delta)_t) dt \leq m_n(K_\delta)$, so the average slice has measure at most $m_n(K_\delta)$; but K contains segments crossing every height, so every slice is non-empty and, by the first bound applied in reverse on a large set of t , comparable to the average. Chebyshev on the complement gives the stated set of t . $\square$

Where we are. The proposition says the slices are all comparable, up to a factor δ^{-1} , to the measure of the whole neighbourhood. That is what licenses Theorem 10.5.4's bound on $|A|, |B|, |C|$: the three particular heights $0, \frac{1}{2}, 1$ are not special, and any three would do.

It also shows where the reduction is wasteful. The Besicovitch set is a three-dimensional object of information — position, direction, and scale — and the reduction retains only three slices. Everything about how the segments are arranged *between* the slices is discarded, which is why Worked Example 10.5.2 found the planar bound so poor.

WORKED EXAMPLE 10.5.2 — TIER A The reduction in the plane, drawn out

For $n = 2$ the three-slice picture is concrete and worth following.

The slices. $K \subseteq \mathbb{R}^2$ and the three horizontal lines $y = 0, \frac{1}{2}, 1$. Each slice K_t is a subset of $\mathbb{R}$.

The lemma. A line of slope $1/v$ meets them at $a, a + \frac{v}{2}, a + v$ — a three-term arithmetic progression in $\mathbb{R}$ with common difference $\frac{v}{2}$.

The Besicovitch hypothesis. K contains a segment in every direction, so for every v in a range there is a with $a \in K_0, a + \frac{v}{2} \in K_{1/2}, a + v \in K_1$. Discretising at scale δ , that is $\approx \delta^{-1}$ progressions.

The count. If $m(K_\delta) = \delta^\beta$ then each slice has $\lesssim \delta^{\beta-2}$ cubes, by Theorem 10.5.4. A progression is determined by its first two terms, so

$$\delta^{-1} \lesssim (\delta^{\beta-2})^2 = \delta^{2\beta-4},$$

giving $-1 \geq 2\beta - 4$, that is $\beta \leq \frac{3}{2}$. So $m(K_\delta) \gtrsim \delta^{3/2}$ and $\overline{\dim}_M K \geq 2 - \frac{3}{2} = \frac{1}{2}$. *Which is terrible.* The truth in the plane is 2, and Córdoba's L^2 argument gives it. The arithmetic reduction with the trivial count gives $\frac{1}{2}$ — worse even than the bush.

The point. The reduction discards the geometry entirely: it retains only that certain triples exist, not where they sit. That is a genuine loss in low dimensions, and it is recovered only in high dimensions, where the additive structure forced by *many* triples becomes restrictive. Worked Example 10.6.1 shows the crossover is around $n = 25$.

WORKED EXAMPLE 10.5.3— TIER C: RESEARCH TASTE **What the reduction discards, and whether that is recoverable**

Worked Example 10.5.2 found that the arithmetic reduction gives $\frac{1}{2}$ in the plane where the truth is 2. That is a large loss and it is worth locating.

(a) *It keeps only three slices.* A Besicovitch set is determined by its segments; the reduction records where they meet three hyperplanes and forgets everything else. Theorem 10.5.6 shows the slices are all comparable, so no single slice is special — but the *joint* information across all heights is discarded.

(b) *It keeps only the existence of triples, not their location.* $T'(A, B, C) \gtrsim \delta^{-(n-1)}$ says many progressions occur; it says nothing about which. Two Besicovitch sets with identical triple-counts but different geometry are indistinguishable to the method.

(c) *It discards the separation of directions.* The count uses that δ -separated directions give distinct triples, but afterwards the directions play no role. In Volume 9 the separation was the whole content of Theorem 9.4.2.

(d) *Is the loss recoverable?* Partly. Bourgain and Katz–Tao recover enough to beat $\frac{n+2}{2}$ asymptotically by exploiting that *many* triples force structure — so (b) is partially undone. Nothing recovers (a) or (c), which is why the method is weak in low dimensions where the geometry is rich.

(e) *The general shape.* A reduction that maps a rich structure to a poorer one trades information for tractability, and the trade is favourable only where the poorer structure is still constraining. Additive structure constrains strongly when the ambient dimension is large — there is more room for triples to

force relations — and weakly when it is small. That is exactly the crossover Worked Example 10.6.1 located around $n = 25$.

WORKED EXAMPLE 10.5.4— TIER B Discretisation: turning a geometric statement into an arithmetic one

Theorem 10.5.4 passes from a continuous Besicovitch set to finite subsets of $\delta\mathbb{Z}^{n-1}$. That passage is routine and error-prone, and the four checks worth making are these.

(1) *What is the ambient group?* Here $\delta\mathbb{Z}^{n-1}$, isomorphic to $\mathbb{Z}^{n-1}$ after rescaling. The sumset inequalities of Chapter 10.3 are stated for abelian groups and apply directly. Had the discretisation produced a set in $\mathbb{R}^{n-1}$ with no group structure, they would not.

(2) *Are the relations exact or approximate?* Theorem 10.5.2 gives $a + c = 2b$ exactly for the continuous points; rounding to δ -cubes makes it $|a + c - 2b| \leq 4\delta$. So the arithmetic statement is about *approximate* triples, and every inequality applied to them must tolerate that — which is why T' is defined with the tolerance built in.

(3) *How much does the tolerance cost?* An approximate relation $|a + c - 2b| \leq C\delta$ permits $O_n(1)$ choices of c given a, b , so counts change by a dimensional constant only. That is the reason the reduction survives discretisation at all; had the tolerance permitted δ^{-1} choices, the count of Theorem 10.5.5 would be vacuous.

(4) *Does the conclusion transfer back?* The arithmetic conclusion bounds $|A|$, and Theorem 10.5.4 relates $|A|$ to $m(K_\delta)$. Both directions of that relation must be available, and the second — from a bound on the number of cubes to a bound on the measure — uses that the contributing cubes are essentially disjoint.

The general caution. A discretisation that loses a power of δ at any of the four steps destroys the argument, since the quantities being compared differ by powers of δ . Checking all four before trusting a discretised argument is quick and usually decisive.

WORKED EXAMPLE 10.5.5— TIER A Three-term progressions, and why three

Theorem 10.5.2 uses three hyperplanes. It is worth asking why not two, or four.

Two is not enough. Two slices at heights 0 and 1 give points a and c on a line, with $c - a$ the direction. Every pair (a, c) arises from some line, so the

condition is vacuous: there is no relation to exploit, and the count gives only $|A||C| \geq \#\text{directions}$, which is Theorem 10.3.2 and says nothing.

Three is exactly enough. The third slice is determined by the first two: $b = \frac{a+c}{2}$. So the triple (a, b, c) satisfies one linear relation, and requiring b to lie in a *small* set is a genuine constraint. That is the first non-vacuous case, and it is why the reduction produces additive triples rather than pairs.

Four is no better. Slices at $0, \frac{1}{3}, \frac{2}{3}, 1$ give a, b, c, d with $b = \frac{2a+d}{3}$ and $c = \frac{a+2d}{3}$ — two relations, but both determined by the same pair (a, d) . No new information: the additional constraint is implied. One could use unequally spaced slices to get genuinely new relations, and the arithmetic literature does, but the gain is in constants rather than in kind.

The general principle. A line in $\mathbb{R}^n$ has $2(n-1)$ degrees of freedom — position and direction — and each slice records $n-1$ numbers. Two slices determine the line; every further slice is a function of them. So three is the smallest number giving a relation, and the relation is necessarily *one* equation, which is why the problem becomes additive rather than multiplicative.

The connection to Volume 9. Worked Example 9.3.3 there classified methods by the order of correlation they see. Two slices is a two-point method, and Theorem 10.5.5 confirms it gives nothing. Three slices is a three-point method, which is why the arithmetic route can in principle beat the L^2 ceiling — and Theorem 10.6.1 confirms it does.

Drills

Drill 10.5.1. Verify Theorem 10.5.2 for a line in $\mathbb{R}^3$ given parametrically, and check the converse.

Drill 10.5.2. Confirm the counting in Step 1 of Theorem 10.5.4: distinct δ -separated directions give distinct triples.

Drill 10.5.3. Derive the bound $|A| \lesssim_n \delta^{-n} m(K_\delta)$ of Theorem 10.5.4 from the disjointness of the contributing cubes.

Drill 10.5.4. Check that Theorem 10.5.5 gives $\delta^{(n+1)/2}$ and compare with the bush bound $\delta^{(n-1)/2}$ of Volume 9, Theorem 9.4.4.

Notes

The reduction of Kakeya to an arithmetic problem is Bourgain's (1999), refined by Katz and Tao. The three-slice observation is elementary and old; what is

not elementary is the passage from many triples to additive structure, which requires the Balog–Szemerédi–Gowers theorem.

Chapter 10.6

The Arithmetic Bounds

Chapter 10.5 reduced the Kakeya problem to a question about additive triples, and Theorem 10.5.5 showed that the reduction used naively gives less than the geometry of Volume 9. This chapter records what the reduction gives when the additive structure is exploited properly, states the resulting bounds, and is explicit about which parts are proved here.

10.6.1 The statements

Theorem 10.6.1 (Bourgain; Katz–Tao). *There is $c > \frac{1}{2}$ such that every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies*

$$\dim_{\mathrm{H}} K \geq cn - O(1).$$

Bourgain’s argument gives $c = \frac{13}{25}$; the refinement of Katz and Tao gives $c = 2 - \sqrt{2} \approx 0.5858$.

Remark 10.6.2 (Scope decision). *Theorem 10.6.1 is not proved in this series.* This is revision R10, and the reason is that the argument needs two theorems of additive combinatorics which the series does not develop.

What is proved here. The geometric reduction — Theorem 10.5.2 and Theorem 10.5.4, which convert a Besicovitch set into a configuration with $\gtrsim \delta^{-(n-1)}$ additive triples among sets of size $\lesssim \delta^{-n}m(K_\delta)$ — together with the sumset machinery of Chapter 10.3: Ruzsa’s triangle inequality and the Plünnecke–Ruzsa inequality, both with complete proofs.

What remains missing. The original draft omitted both the Balog–Szemerédi–Gowers theorem and Freiman’s theorem. The former has since been proved in Chapter 10.4; Freiman’s structure theorem remains the external input needed for the stated arithmetic Kakeya bound.

Why the omission is safe. Nothing later depends on Theorem 10.6.1. Volume 11’s polynomial method supersedes it in every dimension where both

apply, and the current records reported in Volume 14 come from elsewhere entirely. The theorem is included because it was, for several years, the only bound exceeding $\frac{n+2}{2}$ for large n , and because the reduction that produces it is elementary and worth having.

Update. Of the two inputs, Balog–Szemerédi–Gowers is now proved: Theorem 10.4.2 of Chapter 10.4. *Freiman’s theorem is a settled omission*, not a pending one: it is a structure theorem rather than an inequality — it asserts that a set of small doubling is contained in a generalised arithmetic progression of bounded rank — and proving it would require developing Bohr sets and Freiman homomorphisms, a body of theory with no other use in this series. It is the only additive-combinatorial result here left unproved, and Theorem 10.6.1 is the only statement depending on it.

WORKED EXAMPLE 10.6.1— TIER A **The bounds compared**

n	Vol 9 best	Bourgain $\frac{13}{25}n$	Katz–Tao $(2 - \sqrt{2})n$
3	2.5	—	—
5	3.5	2.6	2.9
10	6	5.2	5.9
25	13.5	13	14.6
50	26	26	29.3
100	51	52	58.6

(The Volume 9 column is $\frac{n+2}{2}$, Wolff’s bound, which Theorem 9.5.5 proves only in the transverse case; the constants $O(1)$ in Theorem 10.6.1 are suppressed.)

Reading the table. The arithmetic bounds beat $\frac{n+2}{2}$ only for large n — around $n = 25$ for Bourgain’s constant, earlier for Katz–Tao’s. In low dimensions the geometry wins, which is why Chapter 10.5’s reduction is not the method of choice for $n = 3$ or 4.

The significance. The constant c exceeds $\frac{1}{2}$, so asymptotically these are the first bounds establishing that a Besicovitch set has dimension *more* than half the ambient. Every method in Volume 9 gives exactly $\frac{n}{2} + O(1)$, and the barrier at one half was the state of the art for a decade.

What would be needed for $n - o(n)$. A constant c arbitrarily close to 1. No known method achieves this, and Worked Example 9.8.4 in Volume 9 gives the reason: the arithmetic method, like the geometric ones, optimises over a parameter whose extremal value is not attained.

WORKED EXAMPLE 10.6.2— TIER B Tracing the losses

It is instructive to see why c comes out near $\frac{1}{2}$ rather than near 1, since the arithmetic is transparent even though the theorem is not proved.

Start. Theorem 10.5.4 gives $T' \gtrsim \delta^{-(n-1)}$ triples among sets of size $N \lesssim \delta^{-n}m(K_\delta)$. Write $m(K_\delta) = \delta^\beta$, so $N \approx \delta^{\beta-n}$ and the target is to force β small.

The energy density. $T'/N^2 \approx \delta^{-(n-1)}\delta^{-2(\beta-n)} = \delta^{n+1-2\beta}$. For the Balog–Szemerédi–Gowers theorem to be useful this must not be too small, that is β must not be much below $\frac{n+1}{2}$ — which is exactly the regime the trivial count of Theorem 10.5.5 already reaches.

The cost of BSG. The theorem returns subsets of relative size $\eta^{O(1)}$ with doubling $\eta^{-O(1)}$, where $\eta = T'/N^2$. Both are powers of δ , and the exponents in the $O(1)$ are not small — in the standard proof they are around 4 or 5.

The cost of Plünnecke. Theorem 10.3.5 gives $|kB-lB| \leq K^{k+l}|A|$, exponential in $k+l$. The argument uses $k+l$ growing with n , so this too costs a power.

The upshot. Each step multiplies the exponent by a factor slightly less than 1, and $c = 2 - \sqrt{2}$ is the value at which the multiplicative losses balance — Katz and Tao obtain it by optimising a recursive inequality. The constant is not natural; it is what the arithmetic yields.

The general lesson. A chain of powerful but lossy tools gives a bound determined by the losses rather than by the problem. That is a signature of a method being pushed past its natural range, and it is the same diagnosis Worked Example 9.8.2 gave for the geometric arguments.

WORKED EXAMPLE 10.6.3— TIER C: RESEARCH TASTE What the arithmetic method contributed

(a) *It broke the one-half barrier.* Before 1999 every bound was $\frac{n}{2} + O(1)$. Bourgain’s was the first with a constant exceeding $\frac{1}{2}$, and that was its point — not the numerical value but the demonstration that the barrier was not intrinsic.

(b) *It identified the right auxiliary problem.* The reduction of Chapter 10.5 showed that Kakeya is, at bottom, a question about sets with many three-term arithmetic progressions. That reformulation is due to Bourgain and has been reused since — in the work on the Furstenberg problem reported in Volume 14, and in the sum-product literature.

(c) *It did not scale.* The constants degrade with each additional tool, and no refinement has pushed c past $2 - \sqrt{2}$. The method is, in the terminology of

Worked Example 9.8.4, at its ceiling.

(d) *It was superseded, but not made irrelevant.* Volume 11's polynomial method gives better bounds in low dimensions and, in the finite-field setting, the sharp answer. But the arithmetic reduction remains the only route that sees the *additive* structure of a Besicovitch set, and problems where that structure is the point — the Furstenberg set problem, and the sum-product phenomenon — still use it.

(e) *A note on what this volume has done.* It has proved two things completely: Dvir's theorem, which settles the finite field problem, and the sumset inequalities, which are the arithmetic engine. It has proved the geometric reduction connecting Kakeya to additive triples. And it has been explicit that the remaining step — from triples to structure — is not carried out here. Given that nothing later depends on it, that seems the right division.

WORKED EXAMPLE 10.6.4— TIER C: RESEARCH TASTE **Announced omissions: the discipline this series follows**

Theorem 10.6.2 is the tenth announced omission in this series. The practice is uniform and worth stating once in full, since a reader encountering R10 without context may wonder what it licenses.

(a) *The rule.* Every stated result is either proved, or its omission is announced at the point where it is stated, with three things recorded: what exactly is missing, why it is missing, and what depends on it.

(b) *Removal versus deferral.* A *deferral* names the volume that supplies the argument and is a promise — Volume 3's configuration lemma was deferred to Volume 9 and Theorem 9.2.1 discharged it. A *removal* states that no volume will supply it. Volume 4, Theorem 4.4.5 sets out why inventing a home for an omitted result would be a fiction, and R10 is a removal on those grounds.

(c) *The ten.* R1–R3 removed function-space results from Volume 4; R4 removed Stein interpolation and the sharp L^p theory of oscillatory operators from Volume 5; R5 removed the implication Bochner–Riesz $\Rightarrow$ restriction; R6 removed the Besicovitch covering theorem; R7 removed Drury's Fourier bound, already available by the bush; R8 removed the measure-zero Besicovitch set; R9 removed Wolff's theorem in full generality, the transverse case being proved; and R10 removes the arithmetic bounds.

(d) *What they have in common.* Each is a result that would require machinery used nowhere else — Stein interpolation, rectifiability theory, Balog–Szemerédi–Gowers — and on which nothing later depends. None is a result the series uses.

(e) *The one that was not an omission.* Theorem 3.8.4 was used without proof for six volumes, and that was a genuine debt, not a scope decision. It is recorded differently in the manifest for exactly that reason, and its discharge in Volume 9 is the only entry marked as paid.

WORKED EXAMPLE 10.6.5— TIER A **The recursive inequality behind $2 - \sqrt{2}$**

The constant $2 - \sqrt{2}$ looks arbitrary. It is the fixed point of a recursion, and seeing the shape — without proving the theorem — makes it intelligible.

The setup. Katz and Tao track a quantity measuring how much additive structure a hypothetical small Besicovitch set must have, and derive an inequality of the form

$$\alpha \geq f(\alpha)$$

where α is the exponent to be bounded and f arises from one round of the reduction: apply Theorem 10.5.2, extract structure, and re-apply.

The fixed point. For the particular f they obtain, the inequality is equivalent to $\alpha^2 - 4\alpha + 2 \leq 0$, whose roots are $2 \pm \sqrt{2}$. The relevant root is $2 - \sqrt{2} \approx 0.5858$, and the bound is $\dim_{\text{H}} K \geq (2 - \sqrt{2})n - O(1)$.

Why a quadratic. Because the recursion applies the reduction twice before closing: each application converts a bound on the set into a bound on its slices, and two applications return to the original type of statement with the exponent transformed. A single application would give a linear inequality and a worse constant; three would not improve it further, the recursion having already reached its fixed point.

Sanity check. $2 - \sqrt{2} > \frac{1}{2}$ ✓ — the whole point — and $2 - \sqrt{2} < 1$, so the method does not reach the conjecture. Both are as expected.

The transferable observation. When a bound is an algebraic irrationality, look for a fixed-point equation. The number is then not arbitrary but the solution of the recursion the method generates, and its distance from the conjectured value measures the loss per round.

WORKED EXAMPLE 10.6.6— TIER B **What a proof of Theorem 10.6.1 would look like**

The theorem is stated and not proved, so a reader may reasonably want the shape of the argument, if only to judge the size of the omission.

Step 1 — geometry to triples. Theorem 10.5.4: a Besicovitch set with $m(K_\delta) = \delta^\beta$ yields sets A, B, C of size $\lesssim \delta^{\beta-n}$ with $\gtrsim \delta^{-(n-1)}$ approximate

additive triples. *Proved here.*

Step 2 — triples to energy. Rewrite the triple count as an additive energy: $E(A, C) \gtrsim T^2/|B|$, by Cauchy–Schwarz over the fibres. This is routine and could be included; it is omitted only because Step 3 is not.

Step 3 — energy to doubling. Balog–Szemerédi–Gowers: large energy yields large subsets $A' \subseteq A$, $C' \subseteq C$ with $|A' + C'| \lesssim \eta^{-O(1)}|A'|$. This is now proved in Chapter 10.4.

Step 4 — doubling to iterated sumsets. Theorem 10.3.5 bounds $|kA' - lA'|$ by $K^{k+l}|A'|$. *Proved here.*

Step 5 — structure to a contradiction. The iterated sumsets are constrained by the geometry, since they too must be realised by slices of K ; comparing gives an inequality in β , and optimising the number of iterations gives the recursion of Worked Example 10.6.5.

The accounting. After the insertion of Chapter 10.4, the remaining external structure input is Freiman’s theorem. Thus the five-step anatomy still contains one genuine external theorem, but it is no longer Balog–Szemerédi–Gowers.

Why that matters for the reader. An omission of one named theorem is a different thing from an omission of an argument, and the distinction is worth preserving in any treatment that leaves gaps. The manifest records which of the ten revisions are of each kind.

WORKED EXAMPLE 10.6.7— TIER C: RESEARCH TASTE Why the arithmetic route was abandoned

Between 1999 and 2008 the arithmetic method gave the best asymptotic bounds. It is now little used, and the reasons are worth recording.

(a) *Its ceiling was reached quickly.* Bourgain’s $\frac{13}{25}$ became Katz–Tao’s $2 - \sqrt{2}$ within three years, and nothing has improved it since. By Worked Example 10.6.5 the constant is a fixed point, so incremental refinement of the individual steps cannot move it — only a change in the recursion would.

(b) *The losses are structural, not technical.* Worked Example 10.6.2 attributes them to Balog–Szemerédi–Gowers and to the exponential dependence in Theorem 10.3.5. Both are sharp as stated, so there is no slack to recover.

(c) *The polynomial method arrived.* Volume 11’s partitioning gives better bounds in low dimensions and a genuinely different mechanism. Where two methods compete, effort follows the one with room to grow.

(d) *But the reduction survives.* Theorem 10.5.2 and Theorem 10.5.4 are still the standard way of extracting additive information from a geometric

configuration, and they reappear — in the Furstenberg set problem, in sum-product estimates, and in the discretised incidence bounds of Volume 14. The method was superseded; the reduction was not.

(e) *The lesson about methods.* A technique may be abandoned because its ceiling is low without its component ideas being abandoned. Distinguishing the two is worth doing before concluding that a line of work is exhausted — and it is why this chapter proves the reduction in full while omitting the theorem it was built for.

Drills

Drill 10.6.1. Using Worked Example 10.6.1, find the smallest n for which $(2 - \sqrt{2})n$ exceeds $\frac{n+2}{2}$.

Drill 10.6.2. Verify the energy density computation $T'/N^2 \approx \delta^{n+1-2\beta}$ in Worked Example 10.6.2.

Drill 10.6.3. Explain why Theorem 10.6.2 classifies the omission as a removal rather than a deferral, using the criterion of Volume 4, Theorem 4.4.5.

Drill 10.6.4. State the Balog–Szemerédi–Gowers theorem precisely and identify the two places Worked Example 10.5.1 would use it.

Where we are. A closing remark on what this chapter records. It states a theorem it does not prove, which the series does sparingly and always with the three disclosures of Worked Example 10.6.4: what is missing, why, and what depends on it.

Historically the missing input here was Balog–Szemerédi–Gowers; it is now proved in Chapter 10.4. The remaining external input to Theorem 10.6.1 is Freiman’s theorem. What *is* proved — the three-slice reduction and the sumset inequalities — is the part a reader is likely to reuse, since the reduction reappears in the Furstenberg problem and the inequalities are standard equipment.

Where we are. The five-step anatomy of Worked Example 10.6.6 is worth keeping: three steps proved here, one routine, one imported. An omission of that shape is a citation, not a gap.

Notes

Bourgain's arithmetic bound is from 1999; the constant $2 - \sqrt{2}$ is due to Katz and Tao (2002), who also introduced the recursive optimisation that produces it. The Balog–Szemerédi–Gowers theorem dates from 1994 in Balog and Szemerédi's form and 1998 in Gowers's quantitative version; Freiman's theorem is from 1966, with Ruzsa's proof from 1994.

The additive-combinatorial background is split: Balog–Szemerédi–Gowers is proved in Chapter 10.4, whereas Freiman's theorem remains external; Theorem 10.6.2 records the current status.

Chapter 10.7

Why the Finite Field Argument Does Not Transfer

Dvir's proof is half a page and settles the Kakeya problem over $\mathbb{F}_q$ completely. The real problem has been open since 1928. This chapter explains the gap precisely, because the explanation is genuinely informative — it identifies where the difficulty of the real problem lives.

10.7.1 The naive attempt, and where it fails

Suppose one tries to run Theorem 10.2.1 over $\mathbb{R}$. There are three steps to transfer.

Proposition 10.7.1 (The parameter count survives). *Let $K \subseteq [0, 1]^n$ be a Besicovitch set and $0 < \delta < \frac{1}{2}$. Let S be a maximal δ -separated subset of K , so that*

$$|S| \approx_n \frac{m(K_\delta)}{\delta^n}.$$

Write $m(K_\delta) = \delta^\beta$ with $0 \leq \beta \leq n$. Then there is a non-zero polynomial of degree

$$d \lesssim_n |S|^{1/n} \approx \delta^{(\beta-n)/n}$$

vanishing on S ; and the number of points of S on any single δ -tube is $\approx \delta^{-1}$, which exceeds d whenever $\beta > 0$.

Proof. The dimension count of Worked Example 10.1.2 gives $\dim \mathcal{P}_d^n = \binom{d+n}{n} \geq \frac{d^n}{n!}$, so a non-zero polynomial vanishing on S exists as soon as $\frac{d^n}{n!} > |S|$, that is $d \gtrsim_n |S|^{1/n}$. The estimate $|S| \approx m(K_\delta)\delta^{-n}$ is the standard comparison between a δ -net and the measure of the δ -neighbourhood.

A δ -tube has length 1 and the points of S inside it are δ -separated, so there are $\lesssim \delta^{-1}$ of them, and $\approx \delta^{-1}$ if K contains the corresponding segment. Finally

$$\begin{aligned} d &\approx \delta^{(\beta-n)/n} < \delta^{-1} \\ &\iff \frac{\beta-n}{n} > -1 \\ &\iff \beta > 0. \quad \square \end{aligned}$$

Where we are. So Steps A and the arithmetic of Step B of Worked Example 10.2.2 both survive: a polynomial of low degree exists, and each tube carries more points of S than the degree. Over $\mathbb{F}_q$ that was exactly the configuration that produced the contradiction.

The transfer nevertheless fails, and it fails at the one step whose finite-field version is trivial.

Proposition 10.7.2 (The propagation step fails). *Let ℓ be a line and T the δ -tube around it. There exist non-zero polynomials P of degree 2 vanishing at $\approx \delta^{-1}$ δ -separated points of T but not vanishing at any point of ℓ .*

Proof. Take $n = 2$, ℓ the x -axis, $T = \{(x, y) : 0 \leq x \leq 1, |y| \leq \delta\}$. Let $P(x, y) = y - \frac{\delta}{2}$, of degree 1, which vanishes on the horizontal line $y = \frac{\delta}{2}$. That line lies inside T and contains $\approx \delta^{-1}$ δ -separated points of T . But P vanishes nowhere on ℓ , where $y = 0$ and $P = -\frac{\delta}{2} \neq 0$.

For a degree-2 example not vanishing on any line, take $P(x, y) = (y - \frac{\delta}{2})^2 + \varepsilon x^2$ with ε small; it vanishes only at finitely many points, so instead perturb: $P(x, y) = y - \frac{\delta}{2} - \varepsilon x^2$ vanishes on a parabola lying in T for $\varepsilon < \frac{\delta}{2}$, which again contains $\approx \delta^{-1}$ separated points of T and misses ℓ entirely. $\square$

Where we are. This is the whole obstruction, and it is worth stating without hedging.

Over $\mathbb{F}_q$, a polynomial of degree $d < q$ vanishing at q points of a line vanishes on the line *as a polynomial in the line's parameter* — because a polynomial in one variable with more roots than its degree is zero. The conclusion is algebraic and exact.

Over $\mathbb{R}$, the points of S inside a tube are *not collinear*. They lie within δ of a line, and a polynomial can vanish on a curve threading through them — as Theorem 10.7.2 exhibits — without vanishing anywhere on the line itself. There is no one-variable polynomial to which the root count applies. So the failure is not a matter of constants or of technique. *A tube is not a line*, and the algebraic step that makes Dvir's proof work has no approximate version.

10.7.2 What survives

WORKED EXAMPLE 10.7.1— TIER B **The three repairs, and what each costs**

Three ways of adapting the polynomial method to $\mathbb{R}^n$ have been tried. Two work and one does not.

Repair 1: shrink the tubes to lines. Replace K by the union of the actual segments, a set of measure zero, and ask for a polynomial vanishing on it. Now Step B works — the points on a segment *are* collinear. But Step A fails: the set is infinite, so no finite-degree polynomial vanishes on it unless it vanishes identically. *This repair is empty.*

Repair 2: polynomial partitioning. Do not ask the polynomial to vanish on the set. Ask it to *divide space* into cells each containing a controlled fraction of the set, then induct on the cells. A line meets the zero set of a degree- d polynomial in at most d points unless it lies inside, so each line visits at most $d + 1$ cells — a statement about lines that survives discretisation, because it counts *crossings* rather than requiring exact vanishing. *This is the repair that works*, and Volume 11 develops it.

Repair 3: vanish to high order. Require the polynomial to vanish to order m at each point, which strengthens the propagation at the cost of raising the degree. Over $\mathbb{F}_q$ this is Dvir’s method of multiplicities, and it improves the constant to the sharp q^n . Over $\mathbb{R}$ it does not address Theorem 10.7.2 at all, since the obstruction is the non-collinearity of the points and not the order of vanishing. *This repair helps only in the finite field setting.*

The pattern. What transfers is the statement “a line meets a degree- d variety in $\leq d$ points”, which is about *incidences* and is stable. What does not transfer is “a polynomial vanishing at enough points of a line vanishes on it”, which is about *exact algebraic identities* and is not. Volume 11 is built entirely on the first.

WORKED EXAMPLE 10.7.2— TIER C: RESEARCH TASTE **What this says about the real Kakeya problem**

The finite field model was proposed to isolate the difficulty. Its resolution tells us where the difficulty is *not*, and by elimination, where it is.

(a) *Not in the combinatorics of lines.* Over $\mathbb{F}_q$ the incidence structure of lines is exactly as rich as over $\mathbb{R}$ — two points determine a line, two lines meet in at most a point, there are $\approx q^{n-1}$ directions. And the problem is easy. So the combinatorics alone does not make Kakeya hard.

(b) *Not in the dimension count.* Theorem 10.7.1 shows the parameter count transfers exactly: a Besicovitch set of any positive measure exponent admits a polynomial of degree below the number of points per tube.

(c) *In the width of the tubes.* What is left is Theorem 10.7.2: the points lie near a line rather than on one. Everything the real problem has that the model lacks is contained in that δ .

(d) *Which is consistent with everything else in the series.* Volume 9's Theorem 9.5.5 descoped Wolff's theorem precisely at the near-parallel tubes, where widths interact. Volume 6's Knapp example is a statement about the *thickness* of a cap's dual box. Volume 14's account of Wang and Zahl centres on sticky configurations, which are exactly the ones where tubes of positive width nest inside one another. Three independent lines of evidence pointing at the same place.

(e) *The methodological moral.* A model problem is informative when it is solved, whichever way it comes out. Had the finite field problem been hard, one would have learned that the combinatorics is the obstruction. It was easy, so one learns that the metric is. Either outcome advances understanding; what would not have advanced it is leaving the model unresolved.

WORKED EXAMPLE 10.7.3— TIER B **Testing whether an argument transfers**

Chapter 10.7 diagnosed one specific failure. The diagnosis generalises into a test, worth applying to any finite-field argument offered as a model for the real case.

The test. Go through the argument and mark each step as one of:

- (I) an *incidence* statement — how many points of a set lie on a variety, how many cells a line crosses. *These transfer.*
- (F) a *finiteness* statement — a polynomial of degree $< q$ cannot vanish everywhere, a function on $\mathbb{F}_q$ is a polynomial. *These do not.*
- (E) an *exact algebraic* statement — a polynomial vanishing at $> d$ points of a line vanishes on it. *These transfer only if the points are exactly collinear*, which over $\mathbb{R}$ they are not.

Applied to Dvir. Step 1 is a dimension count: type (I), transfers (Theorem 10.7.1). Step 2–3 is the propagation: type (E), and the points in a tube are not collinear, so it fails (Theorem 10.7.2). Step 4 is type (F), and fails for a second, independent reason. *Two of the four steps are untransferable.*

Applied to polynomial partitioning. Volume 11’s argument uses only “a line meets a degree- d hypersurface in $\leq d$ points or lies in it” — type (I) throughout. That is why it transfers, and why it is the descendant of this volume that survives.

Applied to the method of multiplicities. It strengthens a type (E) step. Since the (E) step is the one that fails, the strengthening is irrelevant over $\mathbb{R}$ — which is Worked Example 10.2.5’s conclusion, reached by the test rather than by inspection.

The value of the test. It is quick, and it is usually decisive. A proposed transfer whose critical step is type (E) or (F) should be regarded as unpromising until the collinearity or finiteness has been replaced by something quantitative.

WORKED EXAMPLE 10.7.4— TIER C: RESEARCH TASTE **Model problems, and what they are worth**

The finite field Kakeya problem is the clearest instance in this series of a *model problem*: a simplified version proposed in the hope that solving it will illuminate the original. Its history is instructive and the lessons generalise.

(a) *The hope.* Wolff proposed it in 1999 expecting that the combinatorics of lines was the essential difficulty, and that removing the metric would isolate it. If the model turned out hard, one would know the combinatorics was the obstruction and could concentrate there.

(b) *The outcome.* The model was easy — half a page, once the right idea appeared — and the technique did not transfer. So the hypothesis was refuted: the combinatorics is not the difficulty.

(c) *Why that is still valuable.* A refuted hypothesis is information. Before 2008 one could reasonably believe the difficulty was combinatorial; afterwards one could not. Worked Example 10.7.2 collects three independent lines of evidence now pointing at the width of tubes, and Dvir’s theorem is the sharpest of them precisely because it is a negative result about the model.

(d) *The methodological cost.* A decade of effort went into the model on the assumption that it was a stepping stone. That effort was not wasted — it produced the polynomial method, which Volume 11 shows was worth more than the model itself — but the stepping-stone reasoning was mistaken.

(e) *The general lesson.* A model problem is worth solving whichever way it comes out, but the inference “the model is hard, therefore the original is hard for the same reason” is unsound, and so is its converse. What a solved model establishes is which features are *not* responsible for the difficulty, and that is usually a shorter list than one hopes.

(f) *Where this recurs in the series.* Volume 13’s decoupling theory was developed partly through a model — the discrete restriction problem for the integer lattice — and there the transfer did work. So the finite field experience is not universal; the difference is that the discrete model retains the metric and only discretises it, whereas $\mathbb{F}_q$ discards the metric entirely.

WORKED EXAMPLE 10.7.5— TIER A **The obstruction, in one picture**

The content of Theorem 10.7.2 can be held in mind as a single image, and it is worth fixing.

Over $\mathbb{F}_q$. A line is q points, exactly collinear. A polynomial of degree $d < q$ vanishing at all of them, restricted to the line’s parameter t , is a one-variable polynomial of degree $\leq d$ with $q > d$ roots. It is *identically* zero, and its leading coefficient — which encodes the direction — vanishes.

Over $\mathbb{R}$. A δ -tube contains $\approx \delta^{-1}$ separated points, but they are scattered within a cylinder of radius δ , not on its axis. A polynomial can vanish on a curve that weaves among them — the parabola $y = \frac{\delta}{2} + \varepsilon x^2$ of Theorem 10.7.2 does — while taking the value $-\frac{\delta}{2}$ at every point of the axis. Restricting to the axis gives a polynomial with *no* roots at all.

The picture. Over $\mathbb{F}_q$: beads on a wire, and a polynomial vanishing on every bead vanishes on the wire. Over $\mathbb{R}$: beads in a straw, and a polynomial can thread between them, missing the straw’s centre line entirely.

Why no amount of care repairs it. The vanishing is an exact algebraic condition and the points are only approximately collinear. There is no approximate version of “a polynomial with more roots than its degree is zero”; the nearest true statement is that it is *small*, and smallness is not what Step 3 needs.

What is used instead. Volume 11 replaces vanishing by *crossing*: a line meets the zero set of a degree- d polynomial in at most d points unless it lies inside. That is stable under perturbation — a tube crosses the zero set in a controlled number of places whether or not its points are exactly collinear — and it is the whole basis of polynomial partitioning.

WORKED EXAMPLE 10.7.6— TIER A **The two propositions, side by side**

Chapter 10.7 rests on two propositions with opposite verdicts, and the contrast is the chapter’s content.

	Theorem 10.7.1	Theorem 10.7.2
Concerns	Step 1 of Dvir	Steps 2–3 of Dvir
Asserts	a low-degree polynomial vanishing on a δ -net exists, with degree below the number of net points per tube	a polynomial can vanish at δ^{-1} points of a tube while vanishing nowhere on its axis
Verdict	the count <i>transfers</i>	the propagation <i>fails</i>
Why	dimension counting is insensitive to geometry	the net points are near a line, not on one

Reading them together. The polynomial exists; one simply cannot use it. That is a sharper diagnosis than “the argument does not transfer”, and it is what makes Worked Example 10.7.1’s classification of repairs possible: any repair must address the second column, and raising the order of vanishing or shrinking the tubes does not.

The number to remember. $d \approx \delta^{(\beta-n)/n}$ against δ^{-1} points per tube, and $d < \delta^{-1}$ whenever $\beta > 0$. So for *every* Besicovitch set of positive measure exponent the degree budget is comfortable. Nothing is tight; the obstruction is qualitative.

Drills

Drill 10.7.1. Verify the equivalence $d < \delta^{-1} \iff \beta > 0$ in Theorem 10.7.1.

Drill 10.7.2. Exhibit explicitly a polynomial of degree 2 vanishing at $\lfloor \delta^{-1} \rfloor$ separated points of a δ -tube and nowhere on its axis.

Drill 10.7.3. Explain why Repair 1 of Worked Example 10.7.1 fails, by counting the conditions imposed by an infinite set.

Drill 10.7.4. State the incidence fact used by Repair 2 — a line meets a degree- d hypersurface in at most d points or lies inside it — and prove it from the one-variable root count.

Where we are. The two propositions of this chapter should be read together. Theorem 10.7.1 says the polynomial exists and has low enough degree; Theorem 10.7.2 says one cannot use it. The argument fails not for lack of a polynomial but for lack of a conclusion to draw from it.

That is worth emphasising because the opposite diagnosis is natural and wrong. A reader who assumes the obstruction is the infinitude of $\mathbb{R}^n$ — too many points to impose vanishing conditions — will look for a discretisation,

and Theorem 10.7.1 shows the discretisation works perfectly. The obstruction is one step later, and it is geometric rather than cardinality-theoretic.

Where we are. One sentence to carry into Volume 11.

The statement that survives the passage from $\mathbb{F}_q$ to $\mathbb{R}$ is not about vanishing but about *crossing*: a line meets the zero set of a degree- d polynomial in at most d points, or lies inside it. That is a bound on incidences, it holds over any field, and — crucially — it is stable when the line is replaced by a thin tube, since a tube crosses the zero set in a controlled number of places whether or not its points are exactly collinear.

Everything Volume 11 does is built on that one stable fact.

Notes

That Dvir’s argument does not transfer was clear immediately, and the search for a quantitative substitute led to Guth’s polynomial partitioning (2010) and the Guth–Katz theorem (2015). Dvir’s method of multiplicities, with Saraf and Sudan, gives the sharp constant over $\mathbb{F}_q$; Worked Example 10.7.1 records why it does not help over $\mathbb{R}$.

Chapter 10.8

Assessment

WORKED EXAMPLE 10.8.1— TIER A **The state of the Kakeya problem after Volume 10**

Setting	Best bound	Source	Status
$\mathbb{F}_q^n$, all n	q^n	Dvir	<i>solved</i> , Theorem 10.2.1
$\mathbb{R}^2$	2	Córdoba	<i>solved</i> , Vol 9 Theorem 9.3.3
$\mathbb{R}^3$	3	Wang–Zahl	<i>solved</i> 2025, Volume 14
$\mathbb{R}^n$, small n	$\frac{n+2}{2}$	Wolff	Vol 9 Theorem 9.5.5
$\mathbb{R}^n$, large n	$(2 - \sqrt{2})n$	Katz–Tao	Theorem 10.6.2
$\mathbb{R}^n$, $n \geq 4$	—	—	<i>open</i>

The one-line summary. The problem is settled over finite fields, in the plane, and in three dimensions; it is open in four and above, where the best bounds are a little under $0.59n$ against the conjectured n .

Where the volume’s two methods sit. Dvir’s is the sharpest result in the table and the least transferable. The arithmetic method gives the best asymptotic constant over $\mathbb{R}$ and is not sharp anywhere. Neither is used in the proofs that settled $n = 2$ or $n = 3$.

What that suggests. The methods that settle cases are geometric — Córdoba’s L^2 argument, and Wang–Zahl’s analysis of sticky configurations. The methods that give the best general bounds are algebraic and arithmetic. The two families have not converged, and Volume 14 reports that they still have not.

WORKED EXAMPLE 10.8.2— TIER C: RESEARCH TASTE **What Volume 10 established**

(a) *Proved in full.* Dvir’s theorem, Theorem 10.2.1, with the Schwartz–Zippel lemma it rests on; Ruzsa’s triangle inequality, Theorem 10.3.4; the Plünnecke–Ruzsa inequality via Petridis, Theorem 10.3.5, and its corollary converting differences into sums; the three-slice lemma and the reduction of a Besicovitch set to a configuration with many additive triples, Theorem 10.5.4; and the two propositions of Chapter 10.7 isolating exactly where the finite field argument fails.

(b) *Announced and not proved.* Theorem 10.6.1, the bounds of Bourgain and Katz–Tao (Theorem 10.6.2, revision R10), whose remaining external input is Freiman’s theorem; Balog–Szemerédi–Gowers is proved in Chapter 10.4. Nothing later depends on the arithmetic bound.

(c) *The single most useful thing here.* Not either theorem, but Theorem 10.7.2: a polynomial can vanish at δ^{-1} points of a δ -tube while vanishing nowhere on its axis. That one example explains why an argument settling the finite field case in half a page leaves the real case untouched, and it identifies the width of a tube as the locus of the difficulty.

(d) *The methodological lesson.* Both methods here escape Volume 9’s ceiling by removing the optimisation over an unknown multiplicity — Worked Example 9.8.2 identified that as the binding constraint. Dvir removes it entirely; the arithmetic method replaces it by an optimisation over additive structure, which is lossy but at least different. Recognising what a method optimises over, and whether the extremal value is attainable, remains the most reliable way to predict its ceiling.

(e) *What comes next.* Volume 11 takes the one part of the polynomial method that does transfer — that a line meets a degree- d variety in at most d points — and builds polynomial partitioning on it. That is the technique behind the Guth–Katz solution of the distinct distances problem and the current records in restriction theory, and it is the direct descendant of this volume’s Worked Example 10.7.1, Repair 2.

WORKED EXAMPLE 10.8.3— TIER B **Reading the Kakeya literature: a field guide**

Ten volumes in, a reader is equipped to place almost any paper in this area. Five questions, in order.

(1) *Which setting?* $\mathbb{R}^n$, $\mathbb{F}_q^n$, or a model such as the Furstenberg problem. The finite-field results are solved and do not transfer (Chapter 10.7); a paper proving something over $\mathbb{F}_q$ is contributing technique, not a bound over $\mathbb{R}$.

(2) *Which dimension notion?* Hausdorff, Minkowski, or a maximal-function estimate. Volume 9, Theorem 9.7.3 shows all four formulations carry the

same exponent, *provided* the loss is $\delta^{-\varepsilon}$ for every ε . Check that.

(3) *What does the method optimise over?* Volume 9, Worked Example 9.8.4 gives the test. A multiplicity, a scale, an angular width — compute the balancing value and ask whether a configuration attains it. If not, the paper’s ceiling is already visible.

(4) *What order of correlation?* Second moments, energies and Fourier transforms see pairs only (Volume 9, Worked Example 9.3.3); incidence and algebraic arguments see more. A method of the first kind cannot beat $\frac{n+2}{2}$, whatever its ingenuity.

(5) *Which class of sets?* All Besicovitch sets, or a restricted class — sticky, self-similar, Ahlfors–David regular. Restricted results are often much stronger and are not comparable to general ones.

Worked instance: Dvir’s paper. (1) $\mathbb{F}_q^n$; (2) counting, no δ ; (3) nothing — no free parameter, which is why it is sharp; (4) algebraic, above the pair barrier; (5) all Kakeya sets. Conclusion: a complete solution of a model problem, contributing a technique.

Worked instance: Bourgain’s arithmetic bound. (1) $\mathbb{R}^n$; (2) Hausdorff, via maximal estimates; (3) additive structure, with lossy constants; (4) three-term progressions, so above pairs; (5) all Besicovitch sets. Conclusion: a general bound, asymptotically the best of its era, at its ceiling.

WORKED EXAMPLE 10.8.4— TIER A **Ten volumes in: where the series stands**

Vol	Subject	What it contributes to Kakeya
1–5	classical analysis	the decay of $\widehat{d\sigma}$, with its phase
6	restriction	restriction $\Rightarrow$ Kakeya; the Knapp example
7	dimension theory	Frostman, energy, and the single-scale upgrade
8	projections, distances	the limit of two-point methods
9	tube geometry	L^2 , bush, hairbrush; the configuration lemma
10	algebra, arithmetic	Dvir; sumsets; where the model fails

The through-line. Volumes 1–5 build one estimate; Volume 6 shows it is the analytic face of a geometric problem; Volumes 7–9 develop the geometry to its limit; Volume 10 changes category twice.

The recurring diagnosis. Four times now a method has been shown to have a ceiling, and each time for the same reason: it optimises over a parameter

whose extremal value is not attained (Volume 9, Worked Example 9.8.2), or it sees only pair correlations (Volume 8, Worked Example 7.7.3; Volume 9, Worked Example 9.3.3). Dvir’s argument is the only one in the series with neither defect, and it is the only one that is sharp.

What remains. Volumes 11–14: the polynomial method over $\mathbb{R}$, multilinear restriction, decoupling, and the current state including Wang–Zahl in three dimensions. The conjecture is open in four dimensions and above, and nothing in the first ten volumes suggests it is close.

WORKED EXAMPLE 10.8.5 — TIER B **Comparing bounds across settings**

Results in this area are stated over $\mathbb{R}^n$, over $\mathbb{F}_q^n$, and for restricted classes of sets. Comparing them requires care, and three conversions cover most cases.

(1) *$\mathbb{F}_q$ to $\mathbb{R}$: exponents.* A finite-field bound $|K| \gtrsim q^s$ corresponds formally to $\dim_{\mathbb{H}} K \geq s$ over $\mathbb{R}$, via $q \leftrightarrow \delta^{-1}$. Dvir gives $s = n$; the real conjecture is $s = n$. *The correspondence is formal only* — Chapter 10.7 shows no implication runs either way.

(2) *Minkowski to Hausdorff.* Volume 9, Theorem 9.7.3: the same exponent, provided the loss is $\delta^{-\varepsilon}$ for every ε . A finite-field result has no δ and so no analogue of this distinction — another place the dictionary has no entry.

(3) *Restricted classes to general.* A bound for Ahlfors–David regular sets, or for sticky configurations, is not comparable to a general bound. Volume 14 reports that the Wang–Zahl theorem proceeds by reducing the general case to the sticky one, so in that instance the restricted result was the whole battle — but the reduction is itself a theorem and cannot be assumed.

A worked comparison. Is Dvir’s q^n “better” than Katz–Tao’s $(2 - \sqrt{2})n$? The question is malformed: the first is a complete solution of a different problem, the second a partial result on the real one. What can be said is that no argument is known deriving anything about $\mathbb{R}^n$ from the first, and Theorem 10.7.2 explains why.

The habit. Before comparing two bounds, check that they concern the same setting, the same dimension notion, and the same class of sets. Surveys tabulate exponents; the hypotheses are where the content is.

Where we are. Volume 10 is the only volume in the series with two settings that are not $\mathbb{R}^n$, and Worked Example 10.1.6 marks the three dictionary entries that are blank: angle, near-parallelism, width. Those three are the Kakeya problem.

Drills

Drill 10.8.1. For each row of the table in Worked Example 10.8.1, state whether the bound is sharp and, if not, what the conjectured value is.

Drill 10.8.2. Explain why Dvir’s theorem does not imply anything about $\dim_{\mathbb{H}}$ of a real Besicovitch set, referring to Theorem 10.7.2.

Drill 10.8.3. Compare the ceilings of the three methods of Volume 9 and the two of this volume using the criterion of Volume 9, Worked Example 9.8.4.

Drill 10.8.4. Identify, for each of the five results listed in Worked Example 10.8.2(a), the single hypothesis whose removal would break the proof.

Drill 10.8.5. Using Worked Example 10.8.3, place Wolff’s hairbrush paper against the five questions, and identify which of them Theorem 9.5.5 concerns.

Drill 10.8.6. Explain why question (3) of Worked Example 10.8.3 returns “nothing” for Dvir’s argument, referring to the Checkpoint after Theorem 10.2.1.

Drill 10.8.7. Verify that the ten announced omissions listed in Worked Example 10.6.4 are each used nowhere else, by checking the appendix of each volume.

Drill 10.8.8. Give an example of a result in this volume that would become a *deferral* rather than a removal if a later volume needed it, and name the volume.

Where we are. A last observation about the shape of this volume. It contains the shortest complete proof in the series — Theorem 10.2.1, about a page with its lemma — and one of the longest chains of announced omissions, Theorem 10.6.2 standing at the end of a reduction whose remaining step is a theorem of another subject.

That contrast is not accidental. Where a problem has been solved outright the proof is often short, because the right idea removes the need for approximation; where it has not, the arguments are long and lossy, because each step must be estimated rather than computed. Worked Example 10.6.2 follows the losses in the arithmetic method and finds them accumulating multiplicatively, which is the signature of the second situation.

A reader deciding where to spend effort in this area could do worse than ask which situation a given problem is in.

Where we are. One point of bookkeeping for the volumes ahead.

This volume introduced two settings that are *not* $\mathbb{R}^n$ — the finite field $\mathbb{F}_q^n$ and abelian groups — and results proved in them do not transfer automatically. Worked Example 10.1.6 gives the dictionary and marks the three entries that are blank: angle, near-parallelism, and width.

Volumes 11 to 14 return to $\mathbb{R}^n$ and stay there. Where they invoke an algebraic fact it will be of the incidence type identified in Worked Example 10.7.3 — a line meets a degree- d variety in at most d points — which holds over any field and is stable under the perturbations that discretisation introduces. Nothing from $\mathbb{F}_q$ beyond that is used again.

Where we are. Volume 10 in one line: the Kakeya problem is *solved* over finite fields by an argument with no free parameter, and the argument does not transfer because a tube is not a line.

Notes

The table in Worked Example 10.8.1 is current as of the writing of this volume; Volume 14 reports the position in full, including the Wang–Zahl theorem in $\mathbb{R}^3$ and the state of the Furstenberg set problem.

Chapter 10.A

Results Quoted from Volumes 1–9

Result	Source	Used in
The Kakeya conjecture	Theorem 7.1.11	throughout
Minkowski content	Theorem 7.2.3	Theorem 10.7.1
Maximal bounds give Hausdorff bounds	Theorem 7.8.5	Chapter 10.6
Córdoba’s bound	Theorem 9.3.3	Worked Example 10.8.1
The bush bound	Theorem 9.4.4	Theorem 10.5.5
Wolff’s bound, transverse case	Theorem 9.5.5	Worked Example 10.6.1
The two-bounds template	Worked Example 9.4.4	Worked Example 10.8.2
Why the geometric mean is lossy	Worked Example 9.8.2	Worked Example 10.8.2
Predicting a method’s ceiling	Worked Example 9.8.4	Worked Example 10.6.2

Debts

This volume incurs none and discharges none. Chapter 10.2 depends on nothing in Volumes 1–9; Chapter 10.3 depends on nothing at all.

Removed

Theorem 10.6.1 — the bounds of Bourgain and of Katz–Tao — is stated and not proved, as revision R10 (Theorem 10.6.2). The original draft had two missing inputs. Balog–Szemerédi–Gowers is now proved in Chapter 10.4; Freiman’s theorem remains external.

Chapter 10.B

Notation and Index of Results

Additions to the Volume 1–9 conventions

$\mathbb{F}_q$	the field with q elements
$\ell_{a,b}$	the line $\{a + tb : t \in \mathbb{F}_q\}$
$\mathcal{P}_d^n$	polynomials in n variables of total degree $\leq d$
$A + B, A - B, kA$	sumsets, difference sets, iterated sumsets
$K = A + A / A $	the doubling constant
$T(A, B, C), T'(A, B, C)$	additive triples, exact and δ -approximate
E_t	the horizontal slice of E at height t

Index of the main results

Trivial bounds over $\mathbb{F}_q$	Theorem 10.1.2
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Dvir’s theorem	Theorem 10.2.1
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The trivial consequence	Theorem 10.5.5
The arithmetic bounds	Theorem 10.6.1 (not proved)
The parameter count survives	Theorem 10.7.1
The propagation step fails	Theorem 10.7.2

Plan revisions in this volume

Removed	Reason
The Bourgain and Katz–Tao arithmetic bounds	retain Freiman’s theorem as an external structure input; Balog–Szemerédi–Gowers is proved in Chapter 10.4; nothing later depends on them. Theorem 10.6.2, revision R10

Part 11

Volume 11

The Polynomial Method over $\mathbb{R}$

Preface to Volume 11

The one fact that transfers

Volume 10 showed that Dvir’s exact vanishing propagation does not survive the passage from lines to tubes. What does survive is the elementary incidence fact: a line meets the zero set of a degree- D polynomial in at most D points unless the line lies in the zero set. Volume 11 turns that fact into polynomial partitioning, incidence estimates, multilinear Kakeya, and the Elekes–Sharir distance mechanism.

Strict proof status

Volume 11 is in the strict proof zone. The load-bearing chain is internal: Tucker $\rightarrow$ Borsuk–Ulam $\rightarrow$ ham sandwich; polynomial ham sandwich and the sign-history partitioning theorem; Szemerédi–Trotter; joints; Loomis–Whitney and multilinear Kakeya; the ε -loss three-dimensional line-incidence theorem; Elekes–Sharir non-clustering; and the resulting $N^{1-\varepsilon}$ distinct-distance bound.

The partitioning arguments use controlled *sign-history regions*; no Milnor–Thom connected-component theorem is hidden. The local three-dimensional algebra uses only the resultant bounds proved in Chapter 11.6 and generic projection to the already-proved planar incidence theory.

Sharper historical results may still be discussed in prose. In particular the sharp Guth–Katz $N/\log N$ distance theorem and the Katz–Zahl Kakeya milestone are contextual, not numbered unproved inputs. Multilinear Kakeya is retained with its full internal proof because it is important in its own right and is a foundation for Volume 12.

Two ways to use a polynomial

Vanishing asks a polynomial to vanish on an exact finite configuration and then propagates that vanishing along exact lines. It drives the joints argument. *Partitioning* asks a polynomial to divide space into regions of controlled mass

and uses the bounded number of regions crossed by a line. It drives the incidence arguments and extends naturally to continuous measures.

The plan

- 1 Tucker, Borsuk–Ulam, ham sandwich, and polynomial partitioning
- 2 Szemerédi–Trotter and rich points
- 3 Joints and vanishing-polynomial arguments
- 4 Loomis–Whitney and multilinear Kakeya
- 5 Three-dimensional line incidences
- 6 Elekes–Sharir and distinct distances
- 7 Polynomial partitioning and Kakeya: what it sees and what it misses
- 8 Assessment and bridge to multilinear restriction

Chapter 11.1

Borsuk–Ulam and Ham Sandwich

Polynomial partitioning rests on the ham sandwich theorem, which rests on Borsuk–Ulam. This chapter proves both from Tucker’s combinatorial lemma. The proof of Tucker used below is an elementary mod-2 simplicial-chain argument: we define the few chain operations that are needed and prove their identities inside the chapter. Thus no homology groups, degree theory or homotopy invariance are imported.

11.1.1 A graph lemma

Lemma 11.1.1. *Let G be a finite graph in which every vertex has degree at most 2. Then G is a disjoint union of simple paths, simple cycles and isolated vertices. Consequently the number of vertices of degree 1 is even, and each is an endpoint of a path whose other endpoint also has degree 1.*

Proof. Induct on the number of edges. With no edges every vertex is isolated. Removing an edge e leaves all degrees at most 2, so by induction the smaller graph is a union of paths, cycles and isolated vertices; the two endpoints of e had degree at most 2 in G , hence at most 1 after removal, so each is isolated or an endpoint of a path. Restoring e joins two such, producing again a union of paths and cycles.

Degree-1 vertices are exactly the endpoints of the paths, of which each path has two. □

11.1.2 Tucker’s lemma

Definition 11.1.2 (The cross-polytope complex). Let $\widehat{B}^n = \{x \in \mathbb{R}^n : |x_1| + \cdots + |x_n| \leq 1\}$ be the ℓ^1 ball. Let $\diamond_+^n$ be the simplicial complex with vertex

set $\{0, \pm 1, \dots, \pm n\}$, realised by placing 0 at the origin, $+i$ at e_i and $-i$ at $-e_i$, in which σ is a simplex exactly when it does *not* contain both $+i$ and $-i$ for any i .

Geometrically $\diamond_+^n$ triangulates $\widehat{B}^n$: its maximal simplices are the 2^n or-thants' worth of sets $\{0\} \cup \{\varepsilon_1 1, \dots, \varepsilon_n n\}$ with $\varepsilon_i = \pm$.

Definition 11.1.3 (Special Tucker pair). A *special Tucker triangulation* is a triangulation T of $\widehat{B}^n$ which is antipodally symmetric on $\partial \widehat{B}^n$ and refines $\diamond_+^n$ — equivalently, the sign of each coordinate is constant on the relative interior of every simplex of T . A *Tucker labelling* is a map $\lambda : V(T) \rightarrow \{\pm 1, \dots, \pm n\}$ with $\lambda(-v) = -\lambda(v)$ for $v \in \partial \widehat{B}^n$. An edge of T is *complementary* if its two vertices carry labels $+i$ and $-i$ for some i .

Lemma 11.1.4 (Fine special triangulations). *For every $\delta > 0$ the ball $\widehat{B}^n$ has a special Tucker triangulation all of whose simplices have diameter smaller than δ .*

Proof. Start with the cross-polytope triangulation $\diamond_+^n$. Perform barycentric subdivision simultaneously on antipodal boundary faces. Barycentric subdivision refines every old simplex, preserves the antipodal boundary symmetry, and reduces the mesh by a dimension-dependent factor strictly below 1. After finitely many subdivisions the mesh is below δ . Since every new simplex lies in an old simplex, coordinate signs remain constant on its relative interior, so the triangulation is special. $\square$

Theorem 11.1.5 (Tucker). *Every special Tucker pair (T, λ) has a complementary edge.*

Idea of the proof. Assume there is no complementary edge. Then the labels of every simplex form a face of the boundary of the n -dimensional cross-polytope, so the labelling becomes a simplicial map from the triangulated n -ball to an $(n-1)$ -sphere. On the boundary this map is antipodal. We prove directly, using finite simplicial chains over $\mathbb{F}_2$, that an antipodal simplicial map on the boundary has non-zero top mod-2 chain, whereas any simplicial map extending across the ball has zero top boundary chain. The contradiction forces a complementary edge.

Proof. We use only finite simplicial chains over $\mathbb{F}_2$ and record the needed identities explicitly.

Step 1. *The elementary chain identities.*

For a finite simplicial complex X , let $C_k(X)$ be the vector space over $\mathbb{F}_2$

generated by its k -simplices. Put

$$\partial[v_0, \dots, v_k] = \sum_{j=0}^k [v_0, \dots, \widehat{v_j}, \dots, v_k].$$

Every codimension-two face occurs twice in $\partial^2[v_0, \dots, v_k]$; hence

$$\partial^2 = 0. \quad (11.1.1)$$

If $F : X \rightarrow Y$ is simplicial, define $F_{\#}[v_0, \dots, v_k]$ to be the simplex spanned by the distinct images when these form a k -simplex, and 0 when the image is degenerate. Checking a single simplex gives

$$\partial F_{\#} = F_{\#} \partial. \quad (11.1.2)$$

Finally, if M is a triangulated k -manifold with boundary and A_M denotes the sum of all its top-dimensional simplices, every interior facet occurs twice and every boundary facet once, so

$$\partial A_M = A_{\partial M}. \quad (11.1.3)$$

These are parity identities, not imported homology.

Step 2. *The labelling gives a simplicial map if no complementary edge exists.*

Let $L = \partial \hat{\Delta}^n$ be the boundary complex of the cross-polytope, whose vertices are $\{\pm 1, \dots, \pm n\}$. If T has no complementary edge, the labels on every simplex contain no opposite pair and therefore span a simplex of L . Thus λ is a simplicial map $T \rightarrow L$. On $\partial \hat{B}^n$ it is antipodal by definition.

Let A_n be the sum of all n -simplices of T and A_{n-1} the sum of its boundary $(n-1)$ -simplices. Since L has dimension $n-1$, $\lambda_{\#} A_n = 0$. Using (11.1.2) and (11.1.3),

$$\lambda_{\#} A_{n-1} = \lambda_{\#} \partial A_n = \partial \lambda_{\#} A_n = 0. \quad (11.1.4)$$

Step 3. *Antipodality forces a non-zero boundary chain.*

For $0 \leq k \leq n-1$ set

$$H_k^{\pm} = \{x \in \partial \hat{B}^n : x_{k+2} = \dots = x_n = 0, \pm x_{k+1} \geq 0\}.$$

After a common subdivision we may regard these hemispheres as subcomplexes. Let $A_k^{\pm}$ be the sum of their k -simplices and let $A_k = A_k^+ + A_k^-$. Their common boundary is the equatorial sphere, hence

$$\partial A_k^+ = \partial A_k^- = A_{k-1} \quad (k \geq 1). \quad (11.1.5)$$

Write $C_k^\pm = \lambda_\# A_k^\pm$ and $C_k = C_k^+ + C_k^-$. Antipodality gives $C_k^- = -C_k^+$ after applying the antipodal involution to labels (over $\mathbb{F}_2$ this means that the two chains are paired simplex by simplex). Equation (11.1.4) says $C_{n-1} = 0$, so the two top hemisphere images coincide as mod-2 chains.

We now descend in dimension. Suppose an antipodally invariant $(k+1)$ -chain D_{k+1} has boundary C_k . Split its simplices into one representative from each antipodal pair, $D_{k+1} = E_{k+1}^+ + E_{k+1}^-$. Using (11.1.5) and antipodality, the chain

$$D_k := C_k^+ + \partial E_{k+1}^- = C_k^- + \partial E_{k+1}^+$$

is antipodally invariant and satisfies $\partial D_k = C_{k-1}$ by (11.1.1). Starting from the equality of the two $(n-1)$ -hemisphere images and repeating this construction, we obtain an antipodally invariant 1-chain D_1 with

$$\partial D_1 = C_0. \quad (11.1.6)$$

But an antipodally invariant 1-chain has, at every antipodal vertex pair, an even total boundary parity: its edges occur in antipodal pairs. By contrast C_0 is exactly one antipodal pair, the two images of the endpoints of H_0^+ and H_0^- . This contradicts (11.1.6).

For $n = 1$ the same contradiction is simply the interval parity argument. Hence our assumption was false, and a complementary edge exists. $\square$

Where we are. The proof uses simplicial chains only as finite parity bookkeeping. The operations ∂ , $F_\#$ and the three identities they satisfy were all defined and proved locally; no homology group, degree, or homotopy-invariance theorem has been imported.

11.1.3 Borsuk–Ulam

Theorem 11.1.6 (Borsuk–Ulam). *Let $g : \mathbb{S}^n \rightarrow \mathbb{R}^n$ be continuous. Then $g(x) = g(-x)$ for some $x \in \mathbb{S}^n$.*

Proof. Put $f(x) = g(x) - g(-x)$, so f is continuous and odd; we must find a zero.

Step 1. *Discretise.*

Fix $\varepsilon > 0$. Since $\mathbb{S}^n$ is compact, f is uniformly continuous: choose $\delta > 0$ with $\|f(x) - f(y)\|_\infty < \varepsilon$ whenever $\|x - y\|_\infty < \delta$. The closed upper hemisphere of $\mathbb{S}^n$ is homeomorphic to $\widehat{B}^n$ with antipodally symmetric boundary, so by Theorem 11.1.4 choose a special Tucker triangulation T of it all of whose simplices have diameter below δ .

Step 2. Label.

For a vertex v let $k(v) = \min\{i : |f(v)_i| = \|f(v)\|_\infty\}$ and set

$$\lambda(v) = \begin{cases} +k(v), & f(v)_{k(v)} > 0, \\ -k(v), & f(v)_{k(v)} < 0. \end{cases}$$

(If $f(v) = 0$ we are done.) Since f is odd, λ is odd on the boundary, so (T, λ) is a special Tucker pair.

Step 3. Use the complementary edge.

By Theorem 11.1.5 there is an edge vv' with $\lambda(v) = i > 0$ and $\lambda(v') = -i$. Then $f(v)_i > 0$ and $f(v')_i < 0$, so

$$\|f(v)\|_\infty = f(v)_i < f(v)_i - f(v')_i \leq \|f(v) - f(v')\|_\infty < \varepsilon,$$

the last step because $\text{diam } \sigma < \delta$.

Step 4. Compactness.

So for each $\varepsilon = 1/m$ there is x_m with $\|f(x_m)\|_\infty < 1/m$. By compactness a subsequence converges to some x^* , and $f(x^*) = 0$ by continuity. $\square$

11.1.4 Ham sandwich

Theorem 11.1.7 (Ham sandwich for measures). *Let $\mu_1, \dots, \mu_N$ be finite Borel measures on $\mathbb{R}^N$, each vanishing on every hyperplane. Then there is an affine hyperplane H with*

$$\mu_i(H^+) = \mu_i(H^-) \quad \text{for } i = 1, \dots, N,$$

where $H^\pm$ are the two open half-spaces bounded by H .

Proof. For $u = (u_0, u_1, \dots, u_N) \in \mathbb{S}^N \subseteq \mathbb{R}^{N+1}$ set

$$H_u^+ = \left\{ x \in \mathbb{R}^N : \sum_{j=1}^N u_j x_j > u_0 \right\}, \quad H_u^- = \left\{ x : \sum_{j=1}^N u_j x_j < u_0 \right\},$$

and define $\Phi : \mathbb{S}^N \rightarrow \mathbb{R}^N$ by $\Phi(u)_i = \mu_i(H_u^+) - \mu_i(H_u^-)$.

Φ is odd, since $H_{-u}^\pm = H_u^\mp$.

Φ is continuous. If $u^{(m)} \rightarrow u$ with $u \neq \pm e_0$, the indicator functions $\mathbf{1}_{H_{u^{(m)}}^+}$ converge to $\mathbf{1}_{H_u^+}$ off the hyperplane $\{\sum u_j x_j = u_0\}$, which is μ_i -null by hypothesis; dominated convergence gives the claim. At $u = \pm e_0$ the half-spaces degenerate to $\emptyset$ and $\mathbb{R}^N$, and the same argument applies since the “hyperplane” is empty.

By Theorem 11.1.6 there is u with $\Phi(u) = \Phi(-u)$; since Φ is odd this forces $\Phi(u) = 0$, which is the assertion. $\square$

Theorem 11.1.8 (Ham sandwich for arbitrary finite measures). *Let $\mu_1, \dots, \mu_N$ be arbitrary finite Borel measures on $\mathbb{R}^N$. There is an affine hyperplane H such that, for every i ,*

$$\mu_i(H^+) \leq \frac{1}{2}\mu_i(\mathbb{R}^N), \quad \mu_i(H^-) \leq \frac{1}{2}\mu_i(\mathbb{R}^N).$$

No assumption is made on the mass carried by H itself.

Proof. Let ϕ_m be a standard smooth approximate identity and set $\mu_i^{(m)} = \mu_i * \phi_m$. These measures have smooth densities and therefore vanish on hyperplanes. By Theorem 11.1.7, for every m there is a bisecting hyperplane H_m , represented by $u_m = (u_{0,m}, u'_m) \in \mathbb{S}^N$ through the inequality $u'_m \cdot x > u_{0,m}$.

Pass to a subsequence $u_m \rightarrow u = (u_0, u')$. If all measures are zero there is nothing to prove. Otherwise choose a ball carrying positive mass; the exact bisection prevents u_m from converging to $\pm e_0$, so $u' \neq 0$ and u represents an affine hyperplane H .

Fix $R < \infty$ and $\eta > 0$ and put

$$G_{\eta,R} = \{x \in B_R : u' \cdot x > u_0 + \eta\}.$$

For all sufficiently large m , every point of $G_{\eta,R}$ lies in H_m^+ . Weak convergence $\mu_i^{(m)} \Rightarrow \mu_i$ and Portmanteau therefore give

$$\mu_i(G_{\eta,R}) \leq \liminf_m \mu_i^{(m)}(G_{\eta,R}) \leq \frac{1}{2}\mu_i(\mathbb{R}^N).$$

Let $R \rightarrow \infty$ and then $\eta \downarrow 0$ to obtain the estimate on H^+ . The argument for H^- is identical. $\square$

Corollary 11.1.9 (Ham sandwich for finite sets). *Let $S_1, \dots, S_N$ be finite subsets of $\mathbb{R}^N$. Then there is a hyperplane H such that neither open half-space contains more than $\frac{1}{2}|S_i|$ points of S_i , for every i .*

Proof. For each i let $\mu_i = \sum_{p \in S_i} \delta_p$ and apply Theorem 11.1.8. The measure of an open half-space is exactly the number of points of S_i in that half-space, so the asserted inequalities are immediate. $\square$

Where we are. Theorem 11.1.9 is the form Chapter 11.2 uses: it bisects N finite sets in $\mathbb{R}^N$ by a hyperplane. Points *on* H are counted in neither half-space, which is exactly the convention Theorem 11.2.2 requires — such points are set aside and never counted again in the partitioning induction. The arbitrary-measure theorem is the robust form needed later. Its conclusion uses “at most half” because mass may lie on the bisecting hyperplane; the partitioning induction deliberately leaves that mass unassigned.

WORKED EXAMPLE 11.1.1— TIER A What Tucker’s lemma replaces

Polynomial ham sandwich is usually deduced from the Borsuk–Ulam theorem, and Borsuk–Ulam from degree theory or homology. This chapter takes neither route, and it is worth seeing what is avoided and at what cost.

The usual route. Borsuk–Ulam follows from degree or homology.

The route here. Theorem 11.1.5 uses only finite simplicial chains over $\mathbb{F}_2$. The boundary operator, the chain-map identity, and the parity descent are all written out in the proof.

The cost. Only finite parity bookkeeping. Borsuk–Ulam then follows by discretisation and compactness (Theorem 11.1.6), and ham sandwich follows.

Why it matters here. Volume 11 is the volume in which this series stops quoting and starts proving; had ham sandwich been imported, every result resting on polynomial partitioning — Theorem 11.6.7, Theorem 14.6.7 of Volume 14, and the Kakeya theorem itself — would rest on quoted topology.

The transferable point. A theorem with a topological proof sometimes has a combinatorial one, and the combinatorial proof is often *constructive* where the topological one is not. Tucker’s lemma against Borsuk–Ulam is the standard example; Sperner’s lemma against Brouwer is another.

Drills

Drill 11.1.1. Verify Theorem 11.1.1 for a graph with three vertices and two edges, and identify the degree-1 vertices.

Drill 11.1.2. Draw $\diamond_+^2$ and check that its maximal simplices are the four triangles $\{0, \pm 1, \pm 2\}$ with consistent signs.

Drill 11.1.3. In Step 4 of Theorem 11.1.5, verify that a loose simplex with a repeated label cannot contain a complementary edge.

Drill 11.1.4. Carry out Step 3 of Theorem 11.1.6 for $n = 1$, where the triangulation is an interval subdivision, and confirm the conclusion directly.

Notes

Tucker’s lemma is from 1945. The proof here uses the standard mod-2 parity mechanism, but all chain operations needed for it are developed locally. The deduction of Borsuk–Ulam is standard. The ham sandwich theorem is due to Banach, following a question of Steinhaus; the derivation from Borsuk–Ulam given here is the usual one.

Chapter 11.2

Polynomial Partitioning

11.2.1 Cells

Definition 11.2.1. For a non-zero polynomial $P \in \mathbb{R}[x_1, \dots, x_n]$ let $Z(P) = \{x \in \mathbb{R}^n : P(x) = 0\}$ be its zero set. The connected components of $\mathbb{R}^n \setminus Z(P)$ are the *cells* of P .

The proofs below deliberately distinguish two objects. A *cell* is a connected component of $\mathbb{R}^n \setminus Z(P)$. A *partitioning region* is one of the sign-history regions produced by the iterative ham sandwich construction. A region can be disconnected, but every cell lies in one region. The applications in this volume need only the number of regions and the number of regions met by a line; both are proved directly below, so no Milnor–Thom component-count theorem is required.

Theorem 11.2.2 (Polynomial ham sandwich). *Let $S_1, \dots, S_M$ be finite subsets of $\mathbb{R}^n$. Then there is a non-zero polynomial P of degree*

$$\deg P \leq C_n M^{1/n}$$

which bisects each S_j : neither $\{P > 0\}$ nor $\{P < 0\}$ contains more than $\frac{1}{2}|S_j|$ points of S_j .

Proof. Step 1. The Veronese embedding.

Let D be least with $\binom{D+n}{n} - 1 \geq M$ and put $N = \binom{D+n}{n} - 1$. Order the non-constant monomials x^α , $1 \leq |\alpha| \leq D$, as $m_1, \dots, m_N$ and define

$$v : \mathbb{R}^n \longrightarrow \mathbb{R}^N, \quad v(x) = (m_1(x), \dots, m_N(x)).$$

A polynomial $P = c_0 + \sum_k c_k m_k$ of degree $\leq D$ satisfies

$$P(x) = c_0 + \langle c, v(x) \rangle,$$

so the sign of P on $\mathbb{R}^n$ is the sign of an affine functional on $\mathbb{R}^N$ evaluated at $v(x)$: *polynomials of degree $\leq D$ correspond exactly to affine hyperplanes in $\mathbb{R}^N$.*

Step 2. *Apply the linear ham sandwich theorem.*

The sets $v(S_1), \dots, v(S_M)$ are $M \leq N$ finite subsets of $\mathbb{R}^N$. Adjoining $N - M$ empty sets and applying Theorem 11.1.9 gives a hyperplane $H = \{y : c_0 + \langle c, y \rangle = 0\}$ with neither open half-space containing more than half of any $v(S_j)$.

Step 3. *Pull back.*

Set $P(x) = c_0 + \langle c, v(x) \rangle$, a polynomial of degree $\leq D$, non-zero since $(c_0, c) \neq 0$ and the monomials are linearly independent. By Step 1, $\{P > 0\} = v^{-1}(H^+)$ and $\{P < 0\} = v^{-1}(H^-)$, so P bisects each S_j .

Step 4. *The degree bound.*

By minimality of D , $\binom{D-1+n}{n} - 1 < M$, and since $\binom{D+n}{n} \geq \frac{D^n}{n!}$ this gives $D \leq C_n M^{1/n}$. $\square$

Corollary 11.2.3 (Polynomial ham sandwich for finite measures). *Let $\mu_1, \dots, \mu_M$ be finite Borel measures on $\mathbb{R}^n$. There is a non-zero polynomial P of degree at most $C_n M^{1/n}$ such that, for every j ,*

$$\mu_j(\{P > 0\}) \leq \frac{1}{2} \mu_j(\mathbb{R}^n), \quad \mu_j(\{P < 0\}) \leq \frac{1}{2} \mu_j(\mathbb{R}^n).$$

Mass on $Z(P)$ is left unassigned.

Proof. Use the same Veronese map as in the proof of Theorem 11.2.2. Push each μ_j forward to $\mathbb{R}^N$, append zero measures if necessary, and apply Theorem 11.1.8. Pulling the resulting affine hyperplane back by the Veronese map gives the required polynomial. The degree estimate is exactly the one in Step 4 of Theorem 11.2.2. $\square$

Remark 11.2.4 (R11 discharged). *Theorem 11.2.2 is proved in full.* The load-bearing chain is Theorem 11.1.5 $\rightarrow$ Theorem 11.1.6 $\rightarrow$ Theorem 11.1.7 $\rightarrow$ Theorem 11.1.8 $\rightarrow$ Theorem 11.1.9 $\rightarrow$ Theorem 11.2.2; every link is proved in Chapter 11.1 or above. The proof uses finite mod-2 chains rather than importing homology or degree theory.

11.2.2 The partitioning theorem

Theorem 11.2.5 (Guth–Katz partitioning, with controlled regions). *Let $S \subseteq \mathbb{R}^n$ be a finite set of N points and let $D \geq 1$. There are a non-zero polynomial P of degree at most D and pairwise disjoint open sets $\Omega_1, \dots, \Omega_M \subseteq \mathbb{R}^n \setminus Z(P)$ such that*

1. $\mathbb{R}^n \setminus Z(P) = \bigcup_{i=1}^M \Omega_i$;
2. $M \leq C_n D^n$;
3. $|S \cap \Omega_i| \leq C_n N D^{-n}$ for every i ;
4. every cell of P lies in one Ω_i ;
5. every line not contained in $Z(P)$ meets at most $D + 1$ of the regions.

Idea of the proof. Bisect repeatedly and remember the complete sign history of the factors. After j rounds there are at most 2^j such histories, each containing at most $N2^{-j}$ points, while the total degree is a geometric sum $O_n(2^{j/n})$. Choosing $2^j \asymp D^n$ gives the first four assertions. On a line, the product polynomial has at most D zeros unless it vanishes identically, which gives the fifth assertion directly.

Proof. Choose A_n so large that every polynomial produced by the ham-sandwich theorem at stage i may be taken of degree at most $A_n 2^{i/n}$ and $\sum_{i=0}^{j-1} A_n 2^{i/n} \leq A'_n 2^{j/n}$. If $D < 2A'_n$ take $P = 1$ and one region, enlarging C_n once and for all. Otherwise choose j maximal with $A'_n 2^{j/n} \leq D$.

Starting with S , recursively bisect all current sign-history point sets. At round $i + 1$, Theorem 11.2.2 supplies one polynomial P_{i+1} of degree $O_n(2^{i/n})$ that bisects all at most 2^i current sets simultaneously. Let $P = \prod_{i=1}^j P_i$. Its degree is at most D . For every sign vector $\varepsilon \in \{+, -\}^j$, let

$$\Omega_\varepsilon = \{x : \operatorname{sgn} P_i(x) = \varepsilon_i \text{ for all } i\},$$

omitting empty sets. These are pairwise disjoint open sets whose union is $\mathbb{R}^n \setminus Z(P)$. There are at most $2^j \leq C_n D^n$ of them, and each contains at most $N2^{-j} \leq C_n N D^{-n}$ points of S .

Every connected component of $\mathbb{R}^n \setminus Z(P)$ has constant sign for each factor P_i , hence lies in one sign-history region. Finally, if a line ℓ is not contained in $Z(P)$, the univariate polynomial $P|_\ell$ has at most D zeros. The complement of those zeros has at most $D + 1$ intervals, and the sign history is constant on each interval; hence ℓ meets at most $D + 1$ regions. $\square$

Theorem 11.2.6 (Controlled partitioning-region count). *The partitioning regions in Theorem 11.2.5 number at most $C_n D^n$, and every line not contained in $Z(P)$ meets at most $D + 1$ of them. No bound on the number of connected components of $\mathbb{R}^n \setminus Z(P)$ is used in Volumes 11–14.*

Proof. These are assertions (ii) and (v) of Theorem 11.2.5. $\square$

Where we are. Three features deserve emphasis. Points on $Z(P)$ are not controlled; the degree D is a free parameter; and the bookkeeping is paid for by the $O(D^n)$ *sign-history regions*, not by a real-algebraic theorem counting connected components. Every application below is formulated so that this proved region count and the proved $D + 1$ line-crossing bound are exactly what is needed.

Proposition 11.2.7 (Partitioning respects finite measures). *Let μ be a finite Borel measure on $\mathbb{R}^n$ and let $D \geq 1$. There are a non-zero polynomial P of degree at most D and at most $C_n D^n$ partitioning regions Ω_i such that*

$$\mu(\Omega_i) \leq C_n D^{-n} \mu(\mathbb{R}^n)$$

for every i . Consequently every cell of P satisfies the same bound. No atomlessness hypothesis is required; mass on $Z(P)$ is simply not assigned to an open region.

Proof. Repeat the sign-history construction in the proof of Theorem 11.2.5, but at each round apply Theorem 11.2.3 to the restrictions of μ to the current regions. Each open child receives at most half the parent mass, while mass on the new zero set is discarded from the open-region bookkeeping. The same geometric degree sum and the same choice $2^j \asymp D^n$ give the assertion. $\square$

Where we are. The finite-measure version is the form needed for continuous configurations. Its advantage is precisely that it makes no false assertion that atomless measures vanish on algebraic hypersurfaces: arbitrary mass on $Z(P)$ is allowed and simply lies outside the open partitioning regions.

WORKED EXAMPLE 11.2.1 — TIER A The degree budget, computed

The geometric sum in Step 2 is the whole reason the method works, and it repays seeing numerically.

The sum. $\sum_{i=0}^{j-1} 2^{i/n}$ with $n = 3$ and $j = 12$, so $2^j = 4096$ pieces:

$$\sum_{i=0}^{11} 2^{i/3} = \frac{2^4 - 1}{2^{1/3} - 1} = \frac{15}{0.2599} \approx 57.7,$$

while the final term alone is $2^{11/3} \approx 12.7$. So the total is about 4.5 times the last term — a bounded factor, as claimed, and the constant $(2^{1/3} - 1)^{-1} \approx 3.85$ is where it comes from.

What this means. Twelve rounds of bisection, producing 4096 pieces, cost total degree ≈ 58 . Had each round cost degree proportional to the number of sets bisected — which is what a naive bisection would cost — the total

would be $\sum_i 2^i \approx 4096$, and the method would be useless.

The exponent is everything. Theorem 11.2.2 bisects M sets with degree $M^{1/n}$, not M . That $1/n$ is the Veronese embedding: a polynomial of degree D has $\approx D^n$ coefficients, so it has D^n degrees of freedom to spend on M bisection conditions. The method is, at bottom, the observation that polynomials are cheap.

WORKED EXAMPLE 11.2.2— TIER B **Choosing D : the standard trade-off**

Every application of Theorem 11.2.5 begins by choosing D , and the choice follows a fixed pattern.

The two costs. With degree D :

- each cell has $\lesssim N/D^n$ points, so *large D is good* for the cellular part;
- a line meets $Z(P)$ in $\leq D$ points, hence enters $\leq D + 1$ cells, so *small D is good* for controlling how much a line is fragmented.

The balance. One writes the quantity to be bounded as

$$(\text{cellular contribution}) + (\text{algebraic contribution}),$$

where the first decreases and the second increases in D , and picks D where they agree. That value is almost always a power of N or of $|L|$, and the resulting exponent is the theorem.

Worked instance, previewed. In Theorem 11.3.2 the cellular part contributes $\approx |L| D + |P||L|/D^2$ (per-cell count), and optimising gives $D \approx |P|^{2/3}|L|^{-1/3}$ — from which the exponent $\frac{2}{3}$ in the Szemerédi–Trotter bound emerges. The $\frac{2}{3}$ is not guessed; it is where the two costs cross.

The habit. On meeting a polynomial-method proof, find the choice of D first. It is usually displayed in one line and it determines the entire result; the rest of the proof is verifying that the two contributions really are what they appear to be.

WORKED EXAMPLE 11.2.3— TIER C: RESEARCH TASTE **Partitioning versus vanishing: two uses of one object**

A polynomial can be used against a configuration in two quite different ways, and the whole architecture of the polynomial method rests on the distinction.

(a) *Vanishing.* Ask for P with $S \subseteq Z(P)$. This is available when $|S|$ is finite

and small relative to $\binom{D+n}{n}$, and it is what Volume 10's Theorem 10.2.1 and this volume's Theorem 11.4.2 use. The conclusion is algebraic: P vanishes on lines, derivatives vanish, and a contradiction follows from degree.

(b) *Partitioning.* Ask for P whose cells each meet S in few points. This is Theorem 11.2.5 and is available always. The conclusion is combinatorial: the configuration is divided, and one recurses.

(c) *When each is appropriate.* Vanishing requires the ambient set to be discrete or the objects to be exact — Chapter 11.8 shows it succeeds for genuine segments and fails for tubes. Partitioning requires nothing, but gives a weaker conclusion: it never proves a contradiction outright, only that a count can be improved.

(d) *The historical order.* Vanishing came first — Dvir 2008, joints 2008 — and produced sharp results in the settings where it applies. Partitioning came in 2010 with Guth and Katz, and it is the version that reaches problems about continuous configurations. The subject moved from (a) to (b) precisely because the problems it wanted were of the second kind.

(e) *A diagnostic.* Given a problem, ask whether “ P vanishes on the configuration” is a strong hypothesis. Over a finite field with $|S|$ a constant fraction of q^n , yes. Over $\mathbb{R}$ with S of positive measure, it is impossible; with S a finite point set, it is possible but the conclusion may be vacuous, as Theorem 11.8.2 shows. Partitioning is the fallback and is almost always the right tool.

WORKED EXAMPLE 11.2.4— TIER A **Cells, counted**

Theorem 11.2.6 asserts $\lesssim_n D^n$ partitioning regions for a degree- D polynomial. Four examples, to see that the count is right and where it is attained.

(a) $n = 1$, P of degree D . At most D roots, so at most $D + 1$ intervals. Attained by $\prod_{j=1}^D (x - j)$.

(b) $n = 2$, D generic lines. $P = \prod_{j \leq D} \ell_j$ has degree D , and D lines in general position cut the plane into $1 + D + \binom{D}{2} \approx \frac{D^2}{2}$ regions. So $\approx D^2$ cells, matching D^n with $n = 2$.

(c) $n = 3$, D generic planes. The count is $1 + D + \binom{D}{2} + \binom{D}{3} \approx \frac{D^3}{6}$, again matching D^n .

(d) *A degenerate case.* $P = x_1^D$ has degree D but $Z(P)$ is a single hyperplane and there are only 2 cells. So the bound is an upper bound and is far from attained by degenerate polynomials — which is harmless, since the partitioning theorem needs only that there are not *too many* cells.

Why the exponent must be n . A degree- D polynomial has $\approx \frac{D^n}{n!}$ coefficients,

so it has that many degrees of freedom; one cannot manufacture more cells than one has parameters, up to constants. The matching of the cell count D^n with the dimension count $D^n/n!$ is what makes Theorem 11.2.5's bookkeeping balance exactly, as the Checkpoint there observed.

WORKED EXAMPLE 11.2.5— TIER B Where the constants go

Theorem 11.2.5 carries a constant C_n , and applications need to know it does not depend on N or D . Tracing it is worthwhile once.

Source 1: the ham sandwich degree. Theorem 11.2.2 bisects M sets with a polynomial of degree $\leq C_n M^{1/n}$, where C_n comes from the Veronese count $\binom{D+n}{n} \geq M + 1$. Solving gives $C_n \approx (n!)^{1/n} \approx n/e$.

Source 2: the geometric sum. Step 2 of Theorem 11.2.5 sums $\sum 2^{i/n}$ and divides by $2^{1/n} - 1$. For large n that factor is $\approx n/\ln 2$, so the total degree constant grows linearly in n .

Source 3: the region count. Theorem 11.2.6 contributes its own C_n , which in the standard proofs is exponential in n .

The upshot. C_n grows with the dimension, exponentially in the worst source, but is *independent of N and D* — which is all any application needs, since D is chosen after the constant is fixed.

Why one checks this. Because an argument that chose D in terms of C_n and then applied a result whose constant depended on D would be circular. The order of quantifiers is: fix n , obtain C_n , then choose D . Every proof in this volume respects it, and Worked Example 11.3.2 noted that verifying $D \geq 1$ is the accompanying check.

A related caution. In dimensions large enough for C_n to matter, the polynomial method's bounds degrade. That is one reason the sharp results of Worked Example 11.9.1 are all in $\mathbb{R}^2$ or $\mathbb{R}^3$, and it compounds the classification difficulty of Worked Example 11.6.3.

Drills

Drill 11.2.1. Verify Theorem 11.2.6 for $n = 1$, and for $n = 2$ with P a product of D distinct lines.

Drill 11.2.2. Carry out the Veronese reduction of Theorem 11.2.4: check that a polynomial of degree $\leq D$ in n variables becomes a linear functional on $\mathbb{R}^{\binom{D+n}{n}-1}$.

Drill 11.2.3. Compute the geometric sum of Step 2 for $n = 2$ and $j = 10$, and compare with the last term.

Drill 11.2.4. Show that a cell of $P = P_1 \cdots P_j$ lies in a single region of the partition built in Step 1, as claimed in Step 3.

Where we are. Two conventions for this volume, matching those of Volumes 9 and 10.

D is always the degree of the partitioning polynomial, and it is a free parameter fixed by optimisation at the end of each argument, never at the start. Worked Example 11.2.2 gives the trade-off and Worked Example 11.3.2 the checks.

Constants depend on n alone. Never on D , on the number of points, or on the number of lines — Worked Example 11.2.5 traces where each factor comes from. An estimate whose constant depended on D would be circular, since D is chosen after the constant is fixed.

Notes

Polynomial partitioning is due to Guth and Katz (2015), in the paper resolving the Erdős distinct distances problem; the ham sandwich formulation is due to Stone and Tukey (1942) and the polynomial version to Gromov. The cell count Theorem 11.2.6 is a direct consequence of the sign-history construction above.

Theorem 11.2.4 records that there is no load-bearing real-algebraic component-count import in this chapter.

Chapter 11.3

The Szemerédi–Trotter Theorem

Definition 11.3.1. For a finite set $\mathcal{P}$ of points and a finite set $\mathcal{L}$ of lines in $\mathbb{R}^2$, the number of *incidences* is

$$I(\mathcal{P}, \mathcal{L}) = \#\{(p, \ell) \in \mathcal{P} \times \mathcal{L} : p \in \ell\}.$$

Theorem 11.3.2 (Szemerédi–Trotter). *For all finite $\mathcal{P}, \mathcal{L}$ in $\mathbb{R}^2$,*

$$I(\mathcal{P}, \mathcal{L}) \lesssim (|\mathcal{P}| |\mathcal{L}|)^{2/3} + |\mathcal{P}| + |\mathcal{L}|.$$

Idea of the proof. Partition the plane by a polynomial of degree D , chosen later. Every incidence is either *cellular* — the point lies in an open cell — or *algebraic* — the point lies on $Z(P)$.

For the cellular part, each cell contains $\lesssim |\mathcal{P}|/D^2$ points by Theorem 11.2.5, and a line meets $\leq D + 1$ cells because it crosses $Z(P)$ at most D times. So the incidences within cells are controlled by the trivial bound applied cell by cell, and the smallness of the cells does the work.

For the algebraic part, a line either lies inside $Z(P)$ — and there are at most D such lines, since $Z(P)$ contains at most D lines — or meets it in $\leq D$ points. Either way the count is small.

Balancing the two in D gives the exponent $\frac{2}{3}$.

Lemma 11.3.3 (Trivial incidence bounds). *For all finite $\mathcal{P}, \mathcal{L}$ in $\mathbb{R}^2$,*

$$\begin{aligned} I(\mathcal{P}, \mathcal{L}) &\leq |\mathcal{P}|^{1/2} |\mathcal{L}| + |\mathcal{P}|, & \text{and} & & I(\mathcal{P}, \mathcal{L}) \\ &\leq |\mathcal{L}|^{1/2} |\mathcal{P}| + |\mathcal{L}|. \end{aligned}$$

Proof. Count pairs of points on a common line. If a line ℓ contains k_ℓ points of $\mathcal{P}$ it accounts for $\binom{k_\ell}{2}$ such pairs, and distinct lines share at most one point,

so no pair is counted twice:

$$\sum_{\ell} \binom{k_{\ell}}{2} \leq \binom{|\mathcal{P}|}{2}.$$

Hence $\sum_{\ell} k_{\ell}^2 \leq |\mathcal{P}|^2 + \sum_{\ell} k_{\ell}$, and by Cauchy–Schwarz

$$\begin{aligned} I^2 &= \left(\sum_{\ell} k_{\ell} \right)^2 \\ &\leq |\mathcal{L}| \sum_{\ell} k_{\ell}^2 \\ &\leq |\mathcal{L}|(|\mathcal{P}|^2 + I). \end{aligned}$$

If $I \leq |\mathcal{P}|$ the first bound holds trivially; otherwise $I^2 \leq 2|\mathcal{L}||\mathcal{P}|^2$, giving $I \lesssim |\mathcal{P}||\mathcal{L}|^{1/2}$ — which is the second bound. The first follows by exchanging the roles of points and lines, using duality in the projective plane. $\square$

Proof of Theorem 11.3.2. Write $P = |\mathcal{P}|$, $L = |\mathcal{L}|$.

Step 0. *Reduce to $P \leq L$ by duality.*

Point–line duality in the projective plane sends a point (a, b) to the line $y = ax - b$ and the line $y = cx + d$ to the point $(c, -d)$, and preserves incidence. So $I(\mathcal{P}, \mathcal{L}) = I(\mathcal{L}^*, \mathcal{P}^*)$ with $|\mathcal{L}^*| = L$ points and $|\mathcal{P}^*| = P$ lines. Since the asserted bound $(PL)^{2/3} + P + L$ is symmetric in P and L , we may assume

$$P \leq L.$$

(Vertical lines and points at infinity are handled by a preliminary rotation, which changes no incidence.)

Step 1. *Dispose of very few points.*

If $P < L^{1/2}$ then by Theorem 11.3.3,

$$I \leq PL^{1/2} + L < L^{1/2} \cdot L^{1/2} + L = 2L,$$

which is within the claimed bound. So assume $L^{1/2} \leq P \leq L$, and set

$$D = \frac{P^{2/3}}{L^{1/3}},$$

which satisfies $D \geq 1$ precisely because $P \geq L^{1/2}$.

Step 2. *Partition.*

By Theorem 11.2.5 there is a non-zero Q of degree $\leq D$ whose cells each contain $\lesssim P/D^2$ points of $\mathcal{P}$, and by Theorem 11.2.6 there are $\lesssim D^2$ cells.

Split $I = I_{\text{cell}} + I_{\text{alg}}$ according to whether the incidence's point lies in an open cell or on $Z(Q)$.

Step 3. *The cellular part.*

Let $\Omega_1, \dots, \Omega_m$ be the cells, $m \lesssim D^2$, with $\mathcal{P}_i = \mathcal{P} \cap \Omega_i$ and $\mathcal{L}_i$ the lines meeting Ω_i . A line not contained in $Z(Q)$ meets $Z(Q)$ in at most D points — restricting Q to it gives a non-zero one-variable polynomial of degree $\leq D$ — so it passes through at most $D + 1$ cells, and

$$\sum_i |\mathcal{L}_i| \leq (D + 1)L.$$

By Theorem 11.3.3 in each cell and Cauchy–Schwarz,

$$\begin{aligned} I_{\text{cell}} &\lesssim \sum_i \left(|\mathcal{P}_i| |\mathcal{L}_i|^{1/2} + |\mathcal{L}_i| \right) \\ &\lesssim \frac{P}{D^2} m^{1/2} \left(\sum_i |\mathcal{L}_i| \right)^{1/2} + DL \\ &\lesssim \frac{PL^{1/2}}{D^{1/2}} + DL. \end{aligned}$$

Step 4. *The algebraic part.*

Lines inside $Z(Q)$. A polynomial of degree D vanishing on more than D distinct lines vanishes identically: restrict to a line meeting all of them in distinct points, obtaining a one-variable polynomial of degree $\leq D$ with more than D roots. So at most D lines lie in $Z(Q)$, and by Theorem 11.3.3 applied to them and all P points they carry $\lesssim PD^{1/2} + D$ incidences.

Lines not inside $Z(Q)$. Each meets $Z(Q)$ in $\leq D$ points, hence carries $\leq D$ incidences with $\mathcal{P} \cap Z(Q)$, for a total $\lesssim DL$.

So $I_{\text{alg}} \lesssim PD^{1/2} + DL$.

Step 5. *Substitute.*

With $D = P^{2/3}L^{-1/3}$,

$$\frac{PL^{1/2}}{D^{1/2}} = P^{2/3}L^{2/3}, \quad DL = P^{2/3}L^{2/3}, \quad PD^{1/2} = P^{4/3}L^{-1/6}.$$

The third is at most the first exactly when $P^{2/3} \leq L^{5/6}$, that is $P \leq L^{5/4}$ — which holds, since Step 0 gives $P \leq L$. Therefore

$$I \lesssim P^{2/3}L^{2/3},$$

and adding the $P + L$ from the disposed range completes the proof. $\square$

Where we are. The proof has the two-case shape that every polynomial-method argument has, and it is worth naming the cases once.

The cellular case is where the method earns its keep: the points are spread thinly among D^2 cells and each line visits few of them, so the trivial bound applied locally beats the trivial bound applied globally.

The algebraic case is where the difficulty is deferred. Points on $Z(Q)$ are not controlled by the partitioning at all, and one must argue separately — here easily, because a curve of degree D contains at most D lines. In $\mathbb{R}^3$ and beyond the algebraic case is genuinely hard, and the whole content of the Guth–Katz theorem is its treatment.

The optimisation in Step 4 is the trade of Worked Example 11.2.2, and the exponent $\frac{2}{3}$ is simply where the two costs cross. It is not an artefact: Worked Example 11.3.1 shows a configuration attaining it.

Corollary 11.3.4 (Rich lines). *Let $\mathcal{P}$ be a set of P points in $\mathbb{R}^2$ and $k \geq 2$. The number of lines containing at least k points of $\mathcal{P}$ is*

$$\lesssim \frac{P^2}{k^3} + \frac{P}{k}.$$

Proof. Let $\mathcal{L}_k$ be the set of such lines and $L = |\mathcal{L}_k|$. Every line of $\mathcal{L}_k$ carries at least k incidences, so $kL \leq I(\mathcal{P}, \mathcal{L}_k)$, and by Theorem 11.3.2

$$kL \lesssim (PL)^{2/3} + P + L.$$

If the first term dominates, $kL \lesssim P^{2/3}L^{2/3}$ gives $L^{1/3} \lesssim P^{2/3}/k$, that is $L \lesssim P^2/k^3$. If the second dominates, $kL \lesssim P$ gives $L \lesssim P/k$. If the third dominates, $kL \lesssim L$ forces $k \lesssim 1$, excluded for $k \geq 2$ after adjusting constants. Taking the maximum of the two surviving cases gives the claim. $\square$

Where we are. Theorem 11.3.4 is the form in which Szemerédi–Trotter is usually applied, because the number of lines is rarely given in advance — one knows only that they are rich. Worked Example 11.3.3 lists three applications, and each uses this version.

The two terms have distinct meanings. P^2/k^3 is the interesting regime, dominating when $k \lesssim P^{1/2}$; and P/k is trivial, being the bound one gets by observing that the incidences kL cannot exceed the total P times the average number of rich lines per point. The crossover at $k \approx P^{1/2}$ is where a line becomes so rich that it accounts for a constant fraction of the points.

WORKED EXAMPLE 11.3.1— TIER A **The bound is sharp**

Take the grid $\mathcal{P} = \{0, \dots, n-1\} \times \{0, \dots, 2n^2-1\}$, so $P = 2n^3$, and let $\mathcal{L}$ be the lines $y = ax + b$ with $0 \leq a < n$, $0 \leq b < n^2$, so $L = n^3$.

Every line is fully incident. For $x \in \{0, \dots, n-1\}$ and such a, b ,

$$ax + b < (n-1)(n-1) + n^2 < 2n^2,$$

so the point $(x, ax + b)$ lies in $\mathcal{P}$. Each line therefore contains n points of the grid, and

$$I = nL = n^4.$$

Compare the theorem. $(PL)^{2/3} = (2n^6)^{2/3} \approx 1.6n^4$. So the grid attains the bound up to a constant, and the exponent $\frac{2}{3}$ cannot be improved.

Numerically. At $n = 3, 4, 5, 6$ the grid gives $I = 81, 256, 625, 1296$ against $(PL)^{2/3} + P + L = 210, 598, 1367, 2705$ — within a factor of three throughout, the constant reflecting the crude $2n^2$ in the construction.

Why the grid is extremal. It is the configuration with the most additive structure: points and lines both come from arithmetic progressions, so incidences are as abundant as the arithmetic allows. Worked Example 10.3.7 in Volume 10 recorded that progressions are exactly the sets of small doubling, and the connection is not a coincidence — the Szemerédi–Trotter theorem is one of the standard tools for proving sum–product estimates.

WORKED EXAMPLE 11.3.2— TIER B **Reading the optimisation**

Step 5 is the crux and its structure is worth extracting, since every later chapter repeats it.

Three terms. Before substituting, the bound is

$$\underbrace{\frac{PL^{1/2}}{D^{1/2}}}_{\text{cellular}} + \underbrace{PD^{1/2}}_{\text{lines inside } Z(Q)} + \underbrace{DL}_{\text{crossings}}.$$

The first decreases in D ; the second and third increase. So an optimum exists, and it is found by balancing the decreasing term against the dominant increasing one.

Which increasing term dominates. Comparing $PD^{1/2}$ with DL amounts to

comparing P with $D^{1/2}L$. At the optimum this is P against $P^{1/3}L^{5/6}$, and the crossings term dominates exactly when $P \leq L^{5/4}$.

Why the proof does not need a second case. Because Step 0 has already reduced to $P \leq L$ by duality, and $L \leq L^{5/4}$. *Without that reduction the argument would be incomplete:* for $P > L^{5/4}$ the term $PD^{1/2}$ exceeds $P^{2/3}L^{2/3}$, and the trivial bound $I \leq PL^{1/2} + L$ does not cover the gap either, since $PL^{1/2} \leq P^{2/3}L^{2/3}$ requires $P \leq L^{1/2}$. Duality is doing real work, not cosmetic tidying.

The general recipe. Write the bound as a sum of powers of D with coefficients in P, L ; identify which terms increase and which decrease; balance the decreasing term against the dominant increasing one; then *check the remaining terms are no larger in the range under consideration*, and if they are not, find a symmetry or split into regimes.

And check $D \geq 1$. The optimum $D = P^{2/3}L^{-1/3}$ is admissible only when $P \geq L^{1/2}$, which is why Step 1 disposes of the complementary range first. Forgetting this is the commonest error in polynomial-method arguments.

WORKED EXAMPLE 11.3.3— TIER B **Applying the incidence bound**

Theorem 11.3.2 is one of the most-used theorems in combinatorics, and the applications follow a small number of patterns.

(a) *Rich lines.* A line is k -rich if it contains at least k points of $\mathcal{P}$. Counting incidences from the rich lines alone, $k \cdot \#\{k\text{-rich lines}\} \leq I \lesssim (PL)^{2/3} + P + L$ with $L = \#\{k\text{-rich}\}$, and solving gives

$$\#\{k\text{-rich lines}\} \lesssim \frac{P^2}{k^3} + \frac{P}{k}.$$

This is the form most often quoted, and it is what one uses when the number of lines is not given in advance.

(b) *Sum-product.* If $A \subseteq \mathbb{R}$ is finite, consider the points $A \times A$ and the lines $y = a(x - b)$ for $a, b \in A$. Each line contains $|A|$ points of the grid, so $I \geq |A|^3$, while Theorem 11.3.2 gives $I \lesssim (|A|^2|A|^2)^{2/3} = |A|^{8/3}$ — a contradiction unless the point set is larger than assumed. Chasing the contradiction yields $\max\{|A + A|, |A \cdot A|\} \gtrsim |A|^{5/4}$, Elekes’s bound.

(c) *Distinct distances, weakly.* Taking the N points and the $\approx N^2$ perpendicular bisectors gives, via incidences, the bound $|\Delta| \gtrsim N^{2/3}$ — far from Theorem 11.7.9, but obtained in two lines. It was the state of the art in the 1980s.

The habit. To apply the theorem, identify a point set and a line family in

which the configuration of interest forces many incidences, then compare with the bound. The art is choosing the family; the theorem is then mechanical.

WORKED EXAMPLE 11.3.4— TIER A **The theorem in the two extreme regimes**

Theorem 11.3.2 has three terms, and which dominates is worth knowing.

Balanced: $P \approx L$. Then $(PL)^{2/3} \approx P^{4/3}$, which dominates P and L . This is the interesting regime and the one the grid of Worked Example 11.3.1 realises: with $P \approx L \approx n^3$ the bound is n^4 , attained.

Few points: $P \lesssim L^{1/2}$. Then L dominates: every line carries at most P incidences, but more usefully the trivial bound gives $I \lesssim L$. Each line meets the point set in $O(1)$ points on average.

Few lines: $L \lesssim P^{1/2}$. Symmetrically P dominates, by duality.

The crossovers. At $P = L^{1/2}$ the terms $(PL)^{2/3}$ and L agree: $(L^{1/2}L)^{2/3} = L$. ✓ And Step 1 of the proof disposes of exactly the range below that crossover, which is why the case division there is not arbitrary.

Practical consequence. An application in the balanced regime uses the $(PL)^{2/3}$ term and nothing else; the other two terms exist only so that the statement is true in the degenerate ranges. Theorem 11.3.4 makes this explicit — its two terms are the balanced and degenerate contributions respectively.

Drills

Drill 11.3.1. Verify the pair-counting in Theorem 11.3.3, and check the case distinction $I \leq |\mathcal{P}|$.

Drill 11.3.2. Confirm that a line not contained in $Z(Q)$ meets it in at most D points, and deduce that it passes through at most $D + 1$ cells.

Drill 11.3.3. Verify Worked Example 11.3.1 for $n = 4$: count P , L and I explicitly, and compare with $(PL)^{2/3}$.

Drill 11.3.4. Redo the optimisation of Step 4 keeping all three terms, and determine the exact range of P, L in which each pairing binds.

Where we are. Three proofs of Szemerédi–Trotter exist and the choice between them is informative.

The original (1983) uses a cell decomposition by lines, chosen randomly, and is intricate. *Székely’s* (1997) uses the crossing number inequality from topological graph theory and is four lines long. *The one given here* uses

polynomial partitioning and is a page.

Székely's is the shortest, so why present another? Because it does not generalise: the crossing number inequality is planar, and there is no analogue in $\mathbb{R}^3$. The partitioning proof, by contrast, is the two-case argument of the Checkpoint above, and Chapter 11.6 shows exactly which case becomes hard in three dimensions. A proof is worth learning for what it survives, not only for its length.

Notes

Szemerédi and Trotter proved the theorem in 1983 by a cell decomposition argument; Székely later gave a short proof via the crossing number inequality. The polynomial-partitioning proof given here is Kaplan, Matoušek and Sharir's, and Guth's, and it is the one that generalises to higher dimensions.

The sharpness example is Erdős's and predates the theorem.

Chapter 11.4

The Joints Theorem

Chapter 11.3 used partitioning: a polynomial cut space into cells and the configuration was handled cell by cell. This chapter uses the older device — a polynomial *vanishing* on the configuration — and shows that over $\mathbb{R}$ it is not always useless. The difference from Volume 10 is that the objects here are genuine lines, not tubes, so the propagation step of Theorem 10.7.2 is available.

11.4.1 Statement

Definition 11.4.1. Let $\mathcal{L}$ be a finite set of lines in $\mathbb{R}^3$. A point $p \in \mathbb{R}^3$ is a *joint* of $\mathcal{L}$ if at least three lines of $\mathcal{L}$ pass through p and their directions span $\mathbb{R}^3$.

Theorem 11.4.2 (Joints theorem). *A set of L lines in $\mathbb{R}^3$ has at most $7L^{3/2}$ joints.*

Idea of the proof. Suppose there were more. Take a non-zero polynomial P of *minimal degree* vanishing at every joint; a dimension count bounds its degree D by roughly $J^{1/3}$.

If every line carried more than D joints, then P restricted to a line would be a one-variable polynomial of degree $\leq D$ with more than D roots, hence identically zero — so P would vanish on every line. But then at a joint, three non-coplanar lines lie in $Z(P)$, and the gradient of P , being orthogonal to each of their directions, would have to vanish. So every partial derivative of P vanishes at every joint, has degree $D - 1$, and is non-zero for some index unless P is constant — contradicting minimality.

The one thing to arrange is that every line does carry more than D joints, and that is a pruning argument: discard lines with few joints, which destroys few joints.

Lemma 11.4.3 (Pruning). *Let $\mathcal{L}$ be a set of L lines with joint set J . Then there are a subset $\mathcal{L}' \subseteq \mathcal{L}$ and a set J' of joints of $\mathcal{L}'$ with*

$$|J'| \geq \frac{1}{2}|J| \quad \text{and} \quad \text{every line of } \mathcal{L}' \text{ contains at least } \frac{|J|}{2L} \text{ points of } J'.$$

Proof. Run the following process. Set $\mathcal{L}_0 = \mathcal{L}$ and $J_0 = J$. At stage i , if some line $\ell \in \mathcal{L}_i$ contains fewer than $\frac{|J|}{2L}$ points of J_i , delete it: put $\mathcal{L}_{i+1} = \mathcal{L}_i \setminus \{\ell\}$ and let J_{i+1} be the set of joints of $\mathcal{L}_{i+1}$ that lie in J_i . Otherwise stop.

Each deletion destroys only joints lying on the deleted line — a joint not on ℓ retains its three non-coplanar lines — so at most $\frac{|J|}{2L}$ joints are lost per stage. At most L stages occur, so the total loss is less than $L \cdot \frac{|J|}{2L} = \frac{|J|}{2}$.

The process terminates, and on termination every remaining line contains at least $\frac{|J|}{2L}$ points of the remaining joint set. $\square$

Lemma 11.4.4 (Vanishing at a joint kills the gradient). *Let P be a polynomial vanishing on three lines through a point p whose directions span $\mathbb{R}^3$. Then $\nabla P(p) = 0$.*

Proof. Let v be the direction of one such line, so $t \mapsto P(p + tv)$ is identically zero. Differentiating at $t = 0$ gives $\nabla P(p) \cdot v = 0$. Applying this to the three directions v_1, v_2, v_3 , which span $\mathbb{R}^3$, forces $\nabla P(p) = 0$. $\square$

11.4.2 Proof

Proof of Theorem 11.4.2. Suppose for contradiction that $\mathcal{L}$ is a set of L lines with

$$|J| > 7L^{3/2}. \tag{11.4.1}$$

Step 1. *Prune.*

By Theorem 11.4.3 there are $\mathcal{L}'$ and J' with $|J'| \geq \frac{1}{2}|J|$ and every line of $\mathcal{L}'$ carrying at least $\frac{|J|}{2L}$ points of J' .

Step 2. *A minimal polynomial.*

Among all non-zero polynomials vanishing on J' , choose one, P , of minimal degree D . Such a polynomial exists: the space of polynomials of degree at most d in three variables has dimension $\binom{d+3}{3} > \frac{d^3}{6}$, so once $d^3 > 6|J'|$ the evaluation map at J' has non-trivial kernel. Hence

$$D \leq (6|J'|)^{1/3} \leq (6|J|)^{1/3}. \tag{11.4.2}$$

Note $D \geq 1$: a non-zero constant does not vanish on the non-empty set J' .

Step 3. *Every line of $\mathcal{L}'$ lies in $Z(P)$.*

By Step 1 each $\ell \in \mathcal{L}'$ carries at least $\frac{|J|}{2L}$ points of J' , at all of which P vanishes. We claim that this exceeds D .

By (11.4.2) it is enough that $\frac{|J|}{2L} > (6|J|)^{1/3}$, and raising both sides to the power 3 and rearranging, that is

$$\begin{aligned} \frac{|J|^3}{8L^3} &> 6|J| \\ \iff |J|^2 &> 48L^3 \\ \iff |J| &> \sqrt{48} L^{3/2}. \end{aligned}$$

Since $\sqrt{48} = 4\sqrt{3} \approx 6.93 < 7$, assumption (11.4.1) gives exactly this. So each $\ell \in \mathcal{L}'$ contains more than D zeros of P .

Restricting P to ℓ gives a one-variable polynomial of degree at most D with more than D roots; it is therefore identically zero, so $\ell \subseteq Z(P)$.

Step 4. *The gradient vanishes on J' .*

Let $p \in J'$. Being a joint of $\mathcal{L}'$, it lies on three lines of $\mathcal{L}'$ with spanning directions, and by Step 3 all three lie in $Z(P)$. By Theorem 11.4.4, $\nabla P(p) = 0$.

Step 5. *Contradict minimality.*

Each partial derivative $\partial_i P$ therefore vanishes on all of J' , and has degree at most $D - 1$. By minimality of D , each $\partial_i P$ must be the zero polynomial. But a polynomial all of whose partial derivatives vanish identically is constant, and a non-zero constant does not vanish on $J' \neq \emptyset$. This contradiction refutes (11.4.1). $\square$

Where we are. The proof is four moves — prune, minimise, propagate, differentiate — and the last is what distinguishes it from Dvir’s argument in Volume 10.

There, the contradiction came from Schwartz–Zippel: a low-degree polynomial cannot vanish everywhere. Here it comes from *minimality*: the derivative has smaller degree and vanishes on the same set, so it must be zero. The derivative trick replaces the finiteness of the field, and it works over $\mathbb{R}$ because differentiation lowers degree in any characteristic zero field. That is why the joints theorem transfers and Dvir’s does not. Both use the propagation step — a polynomial vanishing at more than D points of a line vanishes on it — and here the objects genuinely are lines. Theorem 10.7.2 in Volume 10 concerned tubes, which is a different problem.

Proposition 11.4.5 (Joints in $\mathbb{R}^n$). *Let $\mathcal{L}$ be a set of L lines in $\mathbb{R}^n$, and call p a joint if n lines of $\mathcal{L}$ pass through p with spanning directions. Then the number of joints is $\lesssim_n L^{n/(n-1)}$.*

Proof. The argument of Theorem 11.4.2 goes through with 3 replaced by n .

Prune as in Theorem 11.4.3 to arrange that every line carries at least $\frac{|J|}{2L}$ joints while retaining at least half of them. Take P non-zero of minimal degree D vanishing on the pruned joint set J' ; the dimension count $\binom{D+n}{n} > \frac{D^n}{n!}$ gives $D \leq (n!|J|)^{1/n}$.

If $\frac{|J|}{2L} > D$ then P vanishes on every pruned line, and at each joint the n spanning directions force $\nabla P = 0$ exactly as in Theorem 11.4.4. Minimality then forces every $\partial_i P$ to vanish identically, so P is constant — a contradiction.

Therefore $\frac{|J|}{2L} \leq D \leq (n!|J|)^{1/n}$, which rearranges to

$$|J|^{1-\frac{1}{n}} \leq 2L(n!)^{1/n}, \quad \text{that is} \quad |J| \lesssim L^{\frac{n}{n-1}}. \quad \square$$

Where we are. At $n = 3$ the exponent is $\frac{3}{2}$, recovering Theorem 11.4.2. At $n = 2$ it is 2, and the statement is that L lines have $\lesssim L^2$ crossing points — trivially true, and the theorem says nothing. The content grows with n . The proof is unchanged from the three-dimensional case except for the constants, which is unusual in this subject: Worked Example 11.6.3 explains that the *incidence* theorem for lines in $\mathbb{R}^n$ has no such uniformity, because its algebraic case depends on classifying varieties containing many lines. The joints argument avoids that entirely by never asking what $Z(P)$ looks like — only that its degree drops under differentiation.

WORKED EXAMPLE 11.4.1— TIER A **The bound is sharp**

Take the $k \times k \times k$ grid of axis-parallel lines: for each pair $(i, j) \in \{1, \dots, k\}^2$ include the three lines

$$\{(t, i, j)\}, \quad \{(i, t, j)\}, \quad \{(i, j, t)\}.$$

So $L = 3k^2$.

The joints. Every lattice point (i, j, m) with $1 \leq i, j, m \leq k$ lies on one line of each family, and the three directions e_1, e_2, e_3 span $\mathbb{R}^3$. So $J = k^3$.

Comparison. $L^{3/2} = (3k^2)^{3/2} = 3^{3/2}k^3$, so

$$\frac{J}{L^{3/2}} = \frac{1}{3^{3/2}} \approx 0.192,$$

independent of k . The exponent $\frac{3}{2}$ is therefore exactly right, and only the constant is in question.

Numerically. At $k = 2, 5, 20, 50$ the grid has $(L, J) = (12, 8), (75, 125), (1200, 8000), (7500, 125000)$, and $J/L^{3/2}$ is 0.192 throughout.

What sharpness costs. The extremal configuration is a lattice — maximal arithmetic structure, as in Worked Example 11.3.1. That is the recurring

pattern: extremal examples in incidence geometry are grids, and the difficulty of a theorem is the difficulty of ruling out configurations that are almost grids.

WORKED EXAMPLE 11.4.2— TIER B **The derivative trick**

Step 5 is the move worth learning, since it recurs throughout the polynomial method.

The pattern. One has a non-zero P of minimal degree vanishing on a set S , and one shows that some derived polynomial — a partial derivative, or a directional derivative, or a resultant — also vanishes on S and has smaller degree. Minimality forces it to be zero, and one iterates or concludes.

Why minimality is the right hypothesis. Taking P merely to have degree $\leq D$ would allow $\partial_i P$ to be a non-zero polynomial of degree $D - 1$ vanishing on S , which is no contradiction. It is only because *no* non-zero polynomial of degree $< D$ vanishes on S that the conclusion follows. Choosing minimal degree costs nothing and buys the contradiction.

What the geometry must supply. A reason for the derivative to vanish. Here it is Theorem 11.4.4: three non-coplanar directions at a point force $\nabla P = 0$. In other applications one uses that a point is singular, or that a surface is tangent to a line to high order.

A variant worth knowing. If each point of S has only *two* non-parallel lines through it, the gradient need not vanish — it is merely orthogonal to a plane, so it is determined up to scale. One then works with the *second* fundamental form or with a well-chosen directional derivative, and the resulting statements are the flecnode and ruled-surface lemmas that the next chapter will use. The joints theorem is the case where the geometry is generous enough that a single derivative suffices.

WORKED EXAMPLE 11.4.3— TIER C: RESEARCH TASTE **Why the joints theorem transfers and Dvir's does not**

Both are polynomial-vanishing arguments about lines. One is a finite-field theorem with no real analogue; the other is a real theorem. The difference repays being stated exactly.

(a) *The objects.* Dvir's hypothesis is that a *set* contains a line in every direction. The joints hypothesis is that a *finite family of lines* is given, and one counts special points. In the second case the lines are part of the data — they are exact, not approximated by tubes.

(b) *The propagation step.* Both need “ P vanishes at enough points of a line, hence on the line”. Dvir gets “enough” from $q > D$; the joints argument gets it from the pruning lemma, which *arranges* that every line carries more than D joints. Both are legitimate because the lines are exact.

(c) *The contradiction.* Dvir uses Schwartz–Zippel — a finiteness statement. The joints argument uses minimality of the degree and the derivative trick, which works in any field of characteristic zero. So the joints argument is the more portable of the two.

(d) *What neither can do.* Handle tubes. Theorem 10.7.2 in Volume 10 is about δ -neighbourhoods, and neither argument survives the replacement of a line by a tube — the propagation step fails identically in both.

(e) *The lesson for the Kakeya problem.* A Besicovitch set does contain exact segments, and Theorem 11.8.1 runs Dvir’s argument on them successfully. That the conclusion is nonetheless vacuous is a separate matter, and it is the subject of Chapter 11.8. The moral is that “the objects are exact” is necessary for these arguments and nowhere near sufficient for them to say anything.

WORKED EXAMPLE 11.4.4— TIER B **Pruning: the technique in isolation**

Theorem 11.4.3 discards objects that carry little of the structure, so that the survivors are uniformly rich. The device recurs and is worth extracting.

The setup. A family $\mathcal{F}$ of objects and a set S of configurations, each configuration involving several objects. One wants every object to carry many configurations.

The rule. Repeatedly delete any object carrying fewer than θ configurations, where θ is chosen so that the total loss — at most $|\mathcal{F}|\theta$ — is less than half of $|S|$. Here $\theta = \frac{|J|}{2L}$ and the loss is below $\frac{|J|}{2}$.

The two things to check. First, that deleting an object destroys only configurations involving it — true for joints, since a joint not on the deleted line keeps its three spanning lines. Second, that the process terminates, which it does because $\mathcal{F}$ is finite.

Where it appears elsewhere. Volume 3’s Calderón–Zygmund decomposition (Worked Example 3.1.5) selects cubes by a stopping rule and uses the parent’s failure to be selected; Volume 7’s $5r$ -lemma (Theorem 7.4.1) selects greedily by radius; Volume 10’s Theorem 10.3.8 takes a maximal disjoint family. Four instances of one idea: *select greedily, then use maximality or the threshold to bound what was discarded.*

The characteristic error. Choosing θ too large, so that more than half the configurations are lost and the conclusion is vacuous. The check is always the same product $|\mathcal{F}|\theta$ against $|S|$, and it should be verified explicitly — as Step 1 of Theorem 11.4.3 does.

Drills

Drill 11.4.1. Verify Theorem 11.4.4 by differentiating $t \mapsto P(p + tv)$ and check that three spanning directions are needed.

Drill 11.4.2. Confirm the termination and the joint-loss count in Theorem 11.4.3.

Drill 11.4.3. Check Worked Example 11.4.1 for $k = 3$: count L , J and the ratio $J/L^{3/2}$.

Drill 11.4.4. Show that a polynomial in $\mathbb{R}[x_1, x_2, x_3]$ all of whose partial derivatives vanish identically is constant, and identify where characteristic zero is used.

Where we are. Two features of Theorem 11.4.2 are worth contrasting with everything else in this volume.

It uses vanishing, not partitioning. No cells appear, no degree is optimised, and the polynomial's zero set is never examined. That makes it the simplest argument here and also the least generalisable: it needs the objects to be exact lines and the hypothesis to be about points where several meet.

Its bound is sharp. Worked Example 11.4.1 exhibits the grid attaining $L^{3/2}$ up to the constant $3^{-3/2}$. Among the results of this volume only this one, Theorem 11.3.2 and Theorem 11.5.2 are sharp, and all three concern exact objects — lines, or coordinate projections. The pattern of Worked Example 11.9.1 has no exceptions.

Notes

The joints problem was posed in the 1990s and resisted combinatorial attack; the bound $L^{3/2}$ was proved by Guth and Katz in 2008 using the polynomial method, and the short proof given here is due independently to Kaplan, Matoušek and Sharir, and to Quilodrán.

That a problem open for fifteen years fell to a page of algebra was, together with Dvir's theorem, what established the polynomial method in this area.

Chapter 11.5

Multilinear Kakeya

The Kakeya conjecture asks about tubes pointing in *all* directions, and Volume 9 showed the difficulty lies with the nearly-parallel ones. The multilinear version removes them by hypothesis: one considers n families of tubes whose directions are *transverse* — separated by a fixed angle — and asks for a bound on the overlap of one tube from each family. Guth proved the sharp such estimate, and it is the input Volume 12 uses.

This chapter proves the model case, from which the general statement differs only in bookkeeping, and states Guth’s theorem.

11.5.1 The model case: Loomis–Whitney

Notation 11.5.1. For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ let $\pi_j(x)$ delete the j -th coordinate, so $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}$.

Theorem 11.5.2 (Loomis–Whitney). *For every measurable $E \subseteq \mathbb{R}^n$,*

$$m_n(E) \leq \prod_{j=1}^n m_{n-1}(\pi_j(E))^{\frac{1}{n-1}}.$$

Equivalently, for non-negative f_j on $\mathbb{R}^{n-1}$,

$$\int_{\mathbb{R}^n} \prod_{j=1}^n f_j(\pi_j x)^{\frac{1}{n-1}} dx \leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} f_j \right)^{\frac{1}{n-1}}. \quad (11.5.1)$$

Idea of the proof. A point of E is determined by its n shadows, and each shadow omits one coordinate — so the n shadows record each coordinate $n - 1$ times. Counting E against the shadows therefore costs the power $\frac{1}{n-1}$, and the inequality is the assertion that no configuration does better than a box, where it is an equality.

The proof is induction on n using Hölder, and the case $n = 2$ is trivial: $m_2(E) \leq m_1(\pi_1 E)m_1(\pi_2 E)$, since E is contained in the product of its shadows.

Proof. We prove (11.5.1), which contains the set statement on taking $f_j = \mathbf{1}_{\pi_j(E)}$: then the left side is at least $m_n(E)$, because the integrand is 1 on E .

Step 1. *Base case $n = 2$.*

Here $\frac{1}{n-1} = 1$ and the claim is $\int_{\mathbb{R}^2} f_1(x_2)f_2(x_1) dx \leq (\int f_1)(\int f_2)$, which is Fubini with equality.

Step 2. *Inductive step.*

Assume the result in $\mathbb{R}^{n-1}$ and let $f_1, \dots, f_n$ be given on $\mathbb{R}^{n-1}$. Integrate first in x_n , holding $x' = (x_1, \dots, x_{n-1})$ fixed. The factor $f_n(\pi_n x) = f_n(x')$ does not involve x_n , so

$$\int_{\mathbb{R}^n} \prod_{j=1}^n f_j(\pi_j x)^{\frac{1}{n-1}} dx = \int_{\mathbb{R}^{n-1}} f_n(x')^{\frac{1}{n-1}} \left(\int_{\mathbb{R}} \prod_{j=1}^{n-1} f_j(\pi_j x)^{\frac{1}{n-1}} dx_n \right) dx'.$$

Step 3. *Hölder in x_n .*

For $j \leq n-1$ the function $x_n \mapsto f_j(\pi_j x)$ depends on x_n and on the $n-2$ coordinates of x' other than x_j . Apply Hölder with $n-1$ exponents all equal to $n-1$:

$$\begin{aligned} \int_{\mathbb{R}} \prod_{j=1}^{n-1} f_j(\pi_j x)^{\frac{1}{n-1}} dx_n &\leq \prod_{j=1}^{n-1} \left(\int_{\mathbb{R}} f_j(\pi_j x) dx_n \right)^{\frac{1}{n-1}}. \end{aligned}$$

Write $g_j(x') = \int_{\mathbb{R}} f_j(\pi_j x) dx_n$, a function of the $n-2$ coordinates of x' other than x_j — that is, a function on $\mathbb{R}^{n-2}$ composed with a coordinate projection of $\mathbb{R}^{n-1}$.

Step 4. *Apply the inductive hypothesis.*

Substituting and then using Hölder once more in x' with exponents $n-1$ and $\frac{n-1}{n-2}$,

$$\begin{aligned} \int_{\mathbb{R}^n} \prod_j f_j^{\frac{1}{n-1}} &\leq \left(\int_{\mathbb{R}^{n-1}} f_n \right)^{\frac{1}{n-1}} \left(\int_{\mathbb{R}^{n-1}} \prod_{j=1}^{n-1} g_j(x')^{\frac{1}{n-2}} dx' \right)^{\frac{n-2}{n-1}}. \end{aligned}$$

The inner integral is exactly the left side of (11.5.1) in dimension $n-1$, with the $n-1$ functions g_j ; by the inductive hypothesis it is at most $\prod_{j \leq n-1} (\int g_j)^{\frac{1}{n-2}}$. Since $\int_{\mathbb{R}^{n-2}} g_j = \int_{\mathbb{R}^{n-1}} f_j$ by Tonelli, substituting gives

$$\int_{\mathbb{R}^n} \prod_j f_j^{\frac{1}{n-1}} \leq \left(\int f_n \right)^{\frac{1}{n-1}} \prod_{j=1}^{n-1} \left(\int f_j \right)^{\frac{1}{n-1}},$$

which is (11.5.1). $\square$

Where we are. Theorem 11.5.2 is sharp: for a box $E = [0, a_1] \times \cdots \times [0, a_n]$ the shadows have measures $\prod_{i \neq j} a_i$, and

$$\begin{aligned} \prod_j \left(\prod_{i \neq j} a_i \right)^{\frac{1}{n-1}} &= \left(\prod_i a_i^{n-1} \right)^{\frac{1}{n-1}} \\ &= \prod_i a_i \\ &= m_n(E). \end{aligned}$$

Equality. So the inequality cannot be improved, and boxes are the extremal configurations — which is the same statement as the grid being extremal in Worked Example 11.3.1 and Worked Example 11.4.1, seen continuously. Direct computation on random subsets of a k^3 grid confirms the inequality with room to spare in the generic case; the extremals are exactly the boxes.

11.5.2 Guth's theorem

Definition 11.5.3. Let $\nu > 0$. Families $\mathbb{T}_1, \dots, \mathbb{T}_n$ of δ -tubes in $\mathbb{R}^n$ are ν -transverse if whenever $T_j \in \mathbb{T}_j$ for each j , the directions $v_1, \dots, v_n$ of $T_1, \dots, T_n$ satisfy

$$|\det(v_1, \dots, v_n)| \geq \nu.$$

Theorem 11.5.4 (Multilinear Kakeya, axis form). *Let $n \geq 2$ and $\varepsilon > 0$. There is $\delta = \delta(n, \varepsilon) > 0$ such that the following holds. Suppose $\ell_{j,a}$ are lines in $\mathbb{R}^n$, for $j = 1, \dots, n$ and $a = 1, \dots, N_j$, with $\ell_{j,a}$ making an angle at most δ with the x_j -axis. Write $T_{j,a}$ for the indicator of the 1-neighbourhood of $\ell_{j,a}$. Then for every $S \geq 1$ and every cube Q_S of side S ,*

$$\int_{Q_S} \prod_{j=1}^n \left(\sum_{a=1}^{N_j} T_{j,a} \right)^{\frac{1}{n-1}} \lesssim_n S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

Idea of the proof. When the lines are *exactly* axis-parallel the statement is Loomis–Whitney, Theorem 11.5.2, and there is nothing to do. The whole problem is to absorb a tiny tilt.

Guth’s device is to refuse to work at a single scale. Instead one relates tubes of radius W to tubes of radius $\delta^{-1}W$: on a cube of side $\approx \delta^{-1}W$ a tube of radius W tilted by δ deviates by at most $\delta \cdot \delta^{-1}W = W$, so it *is* essentially axis-parallel there, and Loomis–Whitney applies. Passing from scale W to scale $\delta^{-1}W$ costs a constant C_n .

Iterating from radius 1 up to radius S takes $M = \log S / \log \delta^{-1}$ rounds, so the accumulated constant is $C_n^M = S^{\log C_n / \log \delta^{-1}}$ — which is S^ε as soon as δ is chosen small enough in terms of ε . The order of quantifiers matters: ε first, then δ , then all S .

Notation 11.5.5. For $W > 0$ let $T_{j,a,W}$ be the indicator of the W -neighbourhood of $\ell_{j,a}$, and set

$$f_{j,W} = \sum_{a=1}^{N_j} T_{j,a,W}.$$

Thus $f_{j,1} = \sum_a T_{j,a}$ is the quantity in Theorem 11.5.4.

Lemma 11.5.6 (One scale to the next). *With hypotheses as in Theorem 11.5.4, there is C_n such that whenever $S \geq \delta^{-1}W$ and Q_S is a cube of side S ,*

$$\int_{Q_S} \prod_{j=1}^n f_{j,W}^{\frac{1}{n-1}} \leq C_n \delta^n \int_{Q_S} \prod_{j=1}^n f_{j,\delta^{-1}W}^{\frac{1}{n-1}}.$$

Proof. Step 1. Reduce to a small cube.

Partition Q_S into subcubes Q of side length between $\frac{1}{20n}\delta^{-1}W$ and $\frac{1}{10n}\delta^{-1}W$; this is possible since $S \geq \delta^{-1}W$. It suffices to prove the inequality with Q_S replaced by each such Q , and then sum.

Step 2. *On Q , the tubes are essentially axis-parallel.*

Fix j, a . The line $\ell_{j,a}$ makes an angle at most δ with the x_j -axis, and Q has diameter at most $\frac{\sqrt{n}}{10n}\delta^{-1}W \leq \frac{1}{10}\delta^{-1}W \cdot \sqrt{n}/\sqrt{n}$. So over the extent of Q the line deviates from a genuine x_j -parallel line by at most

$$\delta \cdot \text{diam}(Q) \leq \delta \cdot \frac{1}{10}\delta^{-1}W = \frac{W}{10}.$$

Hence there is an *axis-parallel* tube $\tilde{T}_{j,a,2W}$ of radius $2W$ with

$$T_{j,a,W}(x) \leq \tilde{T}_{j,a,2W}(x) \quad \text{for all } x \in Q.$$

Step 3. *Apply Loomis–Whitney.*

Let π_j delete the j -th coordinate. An axis-parallel tube of radius $2W$ about a line parallel to e_j is $\{x : \pi_j x \in B(y_a, 2W)\}$ for some $y_a \in \mathbb{R}^{n-1}$, so $\sum_a \tilde{T}_{j,a,2W}(x) = F_j(\pi_j x)$ with $F_j = \sum_a \mathbf{1}_{B(y_a, 2W)}$. By Theorem 11.5.2,

$$\begin{aligned} \int_Q \prod_j \left(\sum_a \tilde{T}_{j,a,2W} \right)^{\frac{1}{n-1}} &\leq \prod_{j=1}^n \|F_j\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \\ &\leq C_n W^n \prod_{j=1}^n N_j(Q)^{\frac{1}{n-1}}, \end{aligned}$$

where $N_j(Q)$ is the number of tubes $T_{j,a,W}$ meeting Q — only those contribute — and $\|\mathbf{1}_{B(y, 2W)}\|_{L^1} = c_{n-1} W^{n-1}$, whose $\frac{1}{n-1}$ -th powers multiply to $C_n W^n$.

Step 4. *Convert the count back into an integral at the coarser scale.*

Q has diameter at most $\frac{1}{10}\delta^{-1}W$. So if $T_{j,a,W}$ meets Q then every point of Q lies within $\frac{1}{10}\delta^{-1}W + W \leq \delta^{-1}W$ of $\ell_{j,a}$, that is $T_{j,a,\delta^{-1}W} \equiv 1$ on Q . Therefore $f_{j,\delta^{-1}W} \geq N_j(Q)$ pointwise on Q , and

$$C_n W^n \prod_j N_j(Q)^{\frac{1}{n-1}} \leq \frac{C_n W^n}{|Q|} \int_Q \prod_j f_{j,\delta^{-1}W}^{\frac{1}{n-1}}.$$

Since $|Q| \geq (\frac{1}{20n}\delta^{-1}W)^n = c_n \delta^{-n} W^n$, the prefactor is at most $C'_n \delta^n$, which is the claim. $\square$

Proof of Theorem 11.5.4. Step 1. The case $S = \delta^{-M}$.

Apply Theorem 11.5.6 repeatedly with $W = 1, \delta^{-1}, \dots, \delta^{-(M-1)}$:

$$\int_{Q_S} \prod_j f_{j,1}^{\frac{1}{n-1}} \leq C_n^M \delta^{Mn} \int_{Q_S} \prod_j f_{j,\delta^{-M}}^{\frac{1}{n-1}}.$$

Now $f_{j,S} \leq N_j$ everywhere, and $|Q_S| = S^n = \delta^{-Mn}$, so

$$\begin{aligned} \delta^{Mn} \int_{Q_S} \prod_j f_{j,S}^{\frac{1}{n-1}} &\leq \delta^{Mn} \cdot \delta^{-Mn} \prod_j N_j^{\frac{1}{n-1}} \\ &= \prod_j N_j^{\frac{1}{n-1}}. \end{aligned}$$

The factors δ^n cancel exactly against the volume of the cube; that cancellation is the point of the lemma's normalisation. Hence

$$\int_{Q_S} \prod_j f_{j,1}^{\frac{1}{n-1}} \leq C_n^M \prod_j N_j^{\frac{1}{n-1}}.$$

Step 2. *Choose δ .*

Since $S = \delta^{-M}$ we have $M = \log S / \log \delta^{-1}$, so

$$C_n^M = \exp\left(\frac{\log S \log C_n}{\log \delta^{-1}}\right) = S^{\log C_n / \log \delta^{-1}}.$$

Given $\varepsilon > 0$, choose δ so small that $\log C_n / \log \delta^{-1} \leq \varepsilon$. Then $C_n^M \leq S^\varepsilon$.

Step 3. *General S .*

For arbitrary $S \geq 1$ pick the least integer M with $\delta^{-M} \geq S$ and cover Q_S by $C(\delta)$ cubes of side $\delta^{-M} \leq \delta^{-1}S$. Applying Step 1 on each and summing multiplies the bound by $C(\delta)$, which depends only on n and ε . $\square$

Where we are. Three features of the proof deserve emphasis.

The only analytic input is Loomis–Whitney. Theorem 11.5.2 is proved in this chapter by induction and Hölder, so Theorem 11.5.4 rests on nothing else. There is no heat flow, no polynomial method, and no topology.

The cancellation in Step 1 is exact. Each round contributes δ^n and there are M rounds, giving δ^{Mn} ; the final integral over Q_S contributes $|Q_S| = \delta^{-Mn}$. Nothing is lost, and this is why the lemma is stated with the factor δ^n rather than in a normalised form.

The order of quantifiers is ε , then δ , then S . A reader who fixes δ first and lets ε be whatever emerges will find the statement false. Numerically, with $C_n = 10$ one needs $\delta^{-1} \geq 10^{10}$ to reach $\varepsilon = 0.1$ and $\delta^{-1} \geq 10^{100}$ for $\varepsilon = 0.01$ — the same astronomically slow onset recorded in Volume 13, Worked Example 13.2.2.

Theorem 11.5.7 (Multilinear Kakeya, transverse form). *Let $S_1, \dots, S_n \subseteq \mathbb{S}^{n-1}$ satisfy $|v_1 \wedge \dots \wedge v_n| \geq \nu$ whenever $v_j \in S_j$. Suppose $\ell_{j,a}$ are lines with direction in S_j , and let $T_{j,a}$ be the indicator of the 1-neighbourhood of $\ell_{j,a}$. Then for every $\varepsilon > 0$, every $S \geq 1$ and every cube Q_S of side S ,*

$$\int_{Q_S} \prod_{j=1}^n \left(\sum_a T_{j,a} \right)^{\frac{1}{n-1}} \lesssim_{n,\varepsilon} \text{Poly}(\nu^{-1}) S^\varepsilon \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

Rescaling by δ gives the δ -tube form used elsewhere in this series: for ν -transverse families $\mathbb{T}_1, \dots, \mathbb{T}_n$ of δ -tubes,

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{T \in \mathbb{T}_j} \mathbf{1}_T \right)^{\frac{1}{n-1}} \lesssim_{n,\nu,\varepsilon} \delta^{-\varepsilon} \prod_{j=1}^n (\delta^{n-1} \# \mathbb{T}_j)^{\frac{1}{n-1}}.$$

Proof. Cover each S_j by caps $S_{j,\beta}$ of radius ρ . If $\rho \leq \frac{1}{100n}\nu$ then $|v_1 \wedge \dots \wedge v_n| \geq \nu/2$ for $v_j \in S_{j,\beta_j}$, since the wedge is Lipschitz in each argument with constant 1.

Fix a choice $\beta_1, \dots, \beta_n$ and let A be the linear map carrying the centre of S_{j,β_j} to e_j for each j . Because the centres have wedge at least $\nu/2$, A has determinant bounded below and operator norm bounded above by $\text{Poly}(\nu^{-1})$; so A distorts lengths and volumes by $\text{Poly}(\nu^{-1})$. In the new coordinates each line of the j -th family makes an angle $\lesssim \text{Poly}(\nu^{-1})\rho$ with the x_j -axis, which is at most $\delta(n, \varepsilon)$ once ρ is chosen small enough in terms of ν and ε . Theorem 11.5.4 applies.

Summing over the $\lesssim \text{Poly}(\nu^{-1})$ choices of $(\beta_1, \dots, \beta_n)$, and noting that $(\sum_a T_{j,a})^{1/(n-1)} \leq \sum_\beta (\sum_{a \in \beta} T_{j,a})^{1/(n-1)}$ by subadditivity of $t \mapsto t^{1/(n-1)}$ for $n \geq 2$, gives the first display.

For the second, apply the first to the lines underlying the tubes after rescaling $\mathbb{R}^n$ by δ^{-1} , which turns δ -tubes into 1-neighbourhoods and multiplies volumes by δ^{-n} ; the stated normalisation follows. $\square$

Remark 11.5.8 (What is proved, and what is stronger). *Theorem 11.5.4 and Theorem 11.5.7 are proved in full above*, following Guth's short argument, from Loomis–Whitney alone. Revision R12 — which previously recorded multilinearakeya as a load-bearing import — is hereby *discharged*.

Two remarks on strength. Bennett, Carbery and Tao's original theorem, and Guth's later endpoint theorem, are slightly stronger: the endpoint version has no S^ε factor at all, and is proved by the polynomial method of Chapter 11.2. That refinement is *not needed* anywhere in this series — the S^ε version is exactly what the multilinear restriction theorem of Volume 12 consumes, as Bennett, Carbery and Tao themselves note.

The proof above also applies verbatim to Lipschitz curves in place of lines, with Lipschitz constant at most δ ; nothing in Theorem 11.5.6 uses straightness beyond the deviation bound of Step 2.

WORKED EXAMPLE 11.5.1— TIER A **From Loomis–Whitney to tubes**

The passage from Theorem 11.5.2 to Theorem 11.5.7 is worth tracing, since it shows exactly what Guth's theorem adds.

The axis-parallel case is Loomis–Whitney. Let $\mathbb{T}_j$ consist of δ -tubes parallel to e_j . Then $\sum_{T \in \mathbb{T}_j} \mathbf{1}_T$ is essentially a function of $\pi_j x$ alone — a tube in direction e_j is determined by its position in the other $n-1$ coordinates. So the left side of Theorem 11.5.7 is the left side of (11.5.1), and $\int f_j = \delta^{n-1} \# \mathbb{T}_j$ is the total volume of the j -th family divided by δ . $\checkmark$

Transversality replaces orthogonality. For general directions the determinant condition $|\det(v_1, \dots, v_n)| \geq \nu$ says the n directions are quantitatively independent — the axis-parallel case has determinant 1. The constant in the conclusion degrades as $\nu \rightarrow 0$, and must: nearly coplanar directions permit

much larger overlap.

What is genuinely new. In Loomis–Whitney the n functions depend on *disjoint* sets of variables, and Hölder separates them exactly. For general directions no change of coordinates achieves this simultaneously for all tuples, and the separation must be performed approximately, at every scale. That is what the induction on degree does.

Why the exponent is the same. Because it is forced by scaling: replacing δ by $\lambda\delta$ multiplies both sides consistently only for the exponent $\frac{1}{n-1}$. So Guth’s theorem could not have had a different exponent; the content is that the constant is bounded.

WORKED EXAMPLE 11.5.2— TIER B **Why multilinear is easier than linear**

Volume 9 could not handle near-parallel tubes (Theorem 9.5.5), and the multilinear formulation removes them by hypothesis. It is worth being precise about how much that concedes.

What is assumed away. Transversality forbids exactly the configurations in which the Kakeya difficulty lives: a Besicovitch set’s tubes point in all directions, so most n -tuples are far from transverse. The multilinear estimate says nothing about them.

What is retained. The tubes may still be positioned arbitrarily, and there may be $\delta^{-(n-1)}$ of them in each family. So the counting problem is not trivialised — only the angular degeneracy is.

Why it is nonetheless useful. Because a linear estimate can be recovered from a multilinear one by a decomposition: split the directions into caps, and either most of the mass comes from n transverse caps — the *broad* case, handled by Theorem 11.5.7 — or it concentrates near a lower dimensional set of directions — the *narrow* case, handled by induction. That is the broad–narrow argument, and Volume 12 develops it.

The cost. The recovery is lossy: broad–narrow gives the linear estimate with a $\delta^{-\varepsilon}$ and, in the Kakeya case, does not reach the full conjecture. So multilinear Kakeya is a genuine theorem that does not by itself settle the linear problem — which is why the conjecture remained open after Guth’s proof.

WORKED EXAMPLE 11.5.3— TIER B **Checking a multilinear estimate for consistency**

Theorem 11.5.7 is proved above, but the checks below are worth running on *any* multilinear estimate before trusting it — and they are how one would catch a misremembered exponent. Three checks, all cheap.

(1) *Scaling in δ .* Replace δ by $\lambda\delta$ keeping the tubes' axes. Each $\mathbf{1}_T$ is unchanged in character but the tubes fatten; the left side scales like λ^n times the count, and the right side like $\prod_j (\lambda^{n-1})^{1/(n-1)} = \lambda^n$. ✓ The exponent $\frac{1}{n-1}$ is the unique one making this work, which is why Worked Example 11.5.1 could assert that Guth's theorem had no choice about it.

(2) *The axis-parallel case.* Specialising to coordinate directions must reproduce Theorem 11.5.2. It does, as Worked Example 11.5.1 verifies, and this is the strongest single check — an inequality that failed here would be false.

(3) *A single tube from each family.* With $\#\mathbb{T}_j = 1$ the left side is $m(T_1 \cap \cdots \cap T_n)$ up to the exponent, and transversality forces the intersection to be contained in a ball of radius $\approx \delta/\nu$, so it has measure $\lesssim (\delta/\nu)^n$. The right side is $\prod_j (\delta^{n-1})^{1/(n-1)} = \delta^n$. ✓ up to the ν -dependence, which the constant carries.

What the checks do not verify. That the constant is bounded independently of $\#\mathbb{T}_j$ — which is the theorem. Shape checks confirm an inequality is not obviously wrong; they never confirm it is right.

Why do them anyway. Because an internally proved theorem that will be used as a hypothesis should be one the reader can restate correctly, and mis-stated exponents are the commonest error in this literature. Two minutes of scaling prevents it.

WORKED EXAMPLE 11.5.4— TIER C: RESEARCH TASTE **Transversality as a hypothesis: what it concedes and what it keeps**

Multilinear estimates appear throughout the modern literature, always with a transversality hypothesis. It is worth being exact about its content.

(a) *It is a hypothesis on tuples, not on families.* Theorem 11.5.3 requires $|\det(v_1, \dots, v_n)| \geq \nu$ for *every* choice of one tube from each family. So each family may itself contain tubes in many directions — what is forbidden is that a tuple be nearly degenerate.

(b) *It is quantitative.* The constant in Theorem 11.5.7 degrades as $\nu \rightarrow 0$, and must: at $\nu = 0$ the tubes may be coplanar and the estimate fails. So the hypothesis cannot be removed by a limiting argument.

(c) *What it concedes.* A Besicovitch set has tubes in all directions, so most n -tuples from it are far from ν -transverse for any fixed ν . The multilinear estimate therefore says nothing about a Besicovitch configuration directly, and Worked Example 11.5.2 recorded that recovering the linear statement is lossy.

(d) *What it keeps.* Everything else. The tubes' positions are arbitrary, the families may be as large as $\delta^{-(n-1)}$, and no structure is assumed. So the estimate is not a triviality — the counting problem it solves is genuine, and Bennett, Carbery and Tao's ε -loss version was itself a substantial theorem.

(e) *The pattern across the series.* Volume 6's restriction conjecture is hard at L^∞ data and easy at L^2 ; Volume 9's hairbrush works transversally and fails at small angles; multilinear Kakeya assumes transversality outright. In each case the difficulty concentrates in the degenerate configurations, and the tractable theorem is the one that excludes them. Worked Example 11.8.2 in Chapter 11.8 draws the three together.

(f) *Why the concession is nonetheless worth making.* Because the broad-narrow argument of Volume 12 recovers a genuine linear estimate from it, and because the multilinear statement is *sharp* — Theorem 11.5.2 shows the exponent cannot be improved even in the model case. A sharp theorem under a restrictive hypothesis is often more useful than a lossy one without.

WORKED EXAMPLE 11.5.5— TIER A **Loomis–Whitney in three dimensions, checked**

The case $n = 3$ is worth doing concretely, since it is the one Volume 12 uses.

The statement. For $E \subseteq \mathbb{R}^3$,

$$m_3(E) \leq |\pi_1 E|^{1/2} |\pi_2 E|^{1/2} |\pi_3 E|^{1/2},$$

the three projections being onto the coordinate planes.

Equality for a box. $E = [0, a] \times [0, b] \times [0, c]$ has projections of areas bc , ac , ab , and $(bc \cdot ac \cdot ab)^{1/2} = (a^2 b^2 c^2)^{1/2} = abc = m_3(E)$. ✓

Strict inequality otherwise. For a random subset of a k^3 grid with each point kept independently with probability $\frac{2}{5}$, direct computation at $k = 3, \dots, 6$ gives $|E|$ between 6 and 91 against right-hand sides between 11 and 216 — a factor of two or more to spare. Generic sets are far from extremal.

The functional form. Taking f_j to be indicators recovers the set statement; taking general f_j gives

$$\int_{\mathbb{R}^3} f_1(x_2, x_3)^{1/2} f_2(x_1, x_3)^{1/2} f_3(x_1, x_2)^{1/2} dx \leq \prod_j \left(\int f_j \right)^{1/2},$$

which is the form the tube version generalises.

Why $\frac{1}{2}$. Each coordinate appears in exactly two of the three projections, so the exponents must sum to 1 across the two occurrences — giving $\frac{1}{n-1} = \frac{1}{2}$. Any other exponent fails the scaling check of Worked Example 11.5.3(1).

Drills

Drill 11.5.1. Verify the base case $n = 2$ of Theorem 11.5.2 and the equality for a box in $\mathbb{R}^3$.

Drill 11.5.2. Check the Hölder application in Step 3, identifying the $n - 1$ exponents.

Drill 11.5.3. Confirm that $\int_{\mathbb{R}^{n-2}} g_j = \int_{\mathbb{R}^{n-1}} f_j$ in Step 4, and say where Tonelli is used.

Drill 11.5.4. For $\mathbb{T}_j$ consisting of δ -tubes parallel to e_j , verify that $\int f_j = \delta^{n-1} \#\mathbb{T}_j$ as claimed in Worked Example 11.5.1.

Where we are. What this chapter supplies to Volume 12: the statement of Theorem 11.5.7, its exponent $\frac{1}{n-1}$ shown to be forced by scaling, and the model case Theorem 11.5.2 proved with its extremals identified. That is enough to use the theorem correctly and to recognise a mis-statement of it, which Worked Example 11.5.3 argues is the practical requirement.

Notes

Loomis and Whitney proved their inequality in 1949; the induction given here is the standard one. Bennett, Carbery and Tao proved the multilinear Kakeya inequality with a $\delta^{-\varepsilon}$ loss in 2006 by a heat-flow argument, and Guth removed the loss in 2010 using polynomial partitioning; a later proof of Carbery and Valdimarsson uses the Borsuk–Ulam theorem directly.

Theorem 11.5.8 records the internal proof and its role in Volume 12.

Chapter 11.6

Lines in $\mathbb{R}^3$

Szemerédi–Trotter is sharp in the plane, and its extremal example — Worked Example 11.3.1 — is a grid, which is a planar configuration. In $\mathbb{R}^3$ one can ask for more: if the lines are *not* concentrated in planes, are there fewer incidences? The answer is yes. This chapter proves a self-contained ε -loss three-dimensional line-incidence theorem under bounded-degree non-concentration. The sharper historical Guth–Katz theorem is recorded for context, but is not used as an internal black box.

11.6.1 The obstruction

Proposition 11.6.1 (Planes and reguli are extremal). *There are configurations of L lines in $\mathbb{R}^3$ with $\gtrsim L^{3/2}$ pairwise intersections, namely any planar Szemerédi–Trotter configuration. Consequently no bound better than $L^{3/2}$ can hold without excluding such configurations.*

Proof. Take L lines in a plane $\Pi \subseteq \mathbb{R}^3$ in general position: every two meet, so there are $\binom{L}{2} \approx \frac{L^2}{2}$ intersecting pairs, far more than $L^{3/2}$.

More relevantly, take the dual of the grid of Worked Example 11.3.1: L lines in Π with $\approx L^{4/3}$ incidences with a set of $\approx L$ points, so $\approx L^{4/3}$ intersecting pairs concentrated at few points. $\square$

Definition 11.6.2. A *regulus* is a doubly ruled surface in $\mathbb{R}^3$ — a quadric containing two families of lines, such as the hyperboloid $x^2 + y^2 - z^2 = 1$ or the saddle $z = xy$. Any two lines of the same family are disjoint; any line of one family meets every line of the other.

Proposition 11.6.3. *A regulus contains two families of lines, and $L/2$ lines from each family have $L^2/4$ intersecting pairs.*

Proof. Consider $z = xy$. For each $a \in \mathbb{R}$ the line $\ell_a = \{(a, t, at) : t \in \mathbb{R}\}$ lies on the surface, and so does $m_b = \{(s, b, sb) : s \in \mathbb{R}\}$. Two lines $\ell_a, \ell_{a'}$ with $a \neq a'$ are disjoint — they have different first coordinates — and ℓ_a meets m_b at (a, b, ab) . So $L/2$ lines of each family give $(L/2)^2$ intersecting pairs. $\square$

11.6.2 Bézout for lines

Theorem 11.6.4 (Bézout in the plane: the form used here). *If $P, Q \in \mathbb{R}[x_1, x_2]$ are non-zero, have degrees d, e , and have no common non-constant factor, then $Z(P) \cap Z(Q)$ contains at most $2de$ points.*

Proof. Work over $\mathbb{C}$; the real intersection is smaller. Regard P, Q as polynomials in x_2 with coefficients in $\mathbb{C}[x_1]$. Since they are coprime, after a generic linear change of coordinates their resultant $R(x_1) = \text{Res}_{x_2}(P, Q)$ is not the zero polynomial. The Sylvester determinant gives

$$\deg R \leq (\deg_{x_2} P)(\deg_{x_1} Q) + (\deg_{x_2} Q)(\deg_{x_1} P) \leq 2de.$$

Thus the common zero set is finite. Make one further generic linear change so that distinct common zeros have distinct first coordinates. Every common zero then gives a distinct zero of the new resultant, whose degree is still at most $2de$. Hence there are at most $2de$ common zeros. $\square$

Theorem 11.6.5 (Bézout for common lines). *If $P, Q \in \mathbb{R}[x_1, x_2, x_3]$ are non-zero, have degrees d, e , and have no common non-constant factor, then $Z(P) \cap Z(Q)$ contains at most $2de$ lines.*

Proof. Suppose there are more than $2de$ common lines. Choose a generic affine plane Π : it is not contained in either zero set, it meets the chosen lines in distinct points, and the restrictions $P|_{\Pi}, Q|_{\Pi}$ remain coprime. The last property fails only on a proper algebraic set of plane parameters; if it failed generically, the corresponding elimination resultant would vanish identically and P, Q would have a common factor. The restrictions have degrees at most d, e and more than $2de$ common points, contradicting Theorem 11.6.4. $\square$

Lemma 11.6.6 (Spatial rich points by generic projection). *For a finite set $\mathcal{M}$ of M distinct lines in $\mathbb{R}^3$ and every $r \geq 2$,*

$$|P_r(\mathcal{M})| \lesssim M^2 r^{-3} + M r^{-1}.$$

Proof. Choose a generic linear projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ which is injective on the finite rich-point set under consideration, sends distinct lines of $\mathcal{M}$ to distinct non-parallel plane lines, and preserves incidence at those points. Then $\pi(P_r(\mathcal{M}))$ is a set of r -rich points for M planar lines. The rich-point corollary

of Szemerédi–Trotter, Theorem 11.3.4, gives the claim. Generic projections with these properties exist because each forbidden coincidence excludes a proper algebraic set of projection directions. $\square$

Theorem 11.6.7 (Line incidences in $\mathbb{R}^3$). *For every $\varepsilon > 0$ there are $D(\varepsilon)$ and $K(\varepsilon)$ such that the following holds. Let $\mathfrak{L}$ be a set of L lines in $\mathbb{R}^3$ containing fewer than $L^{\frac{1}{2}+\varepsilon}$ lines in any irreducible algebraic surface of degree at most D . Then for $2 < r < 2L^{1/2}$,*

$$|P_r(\mathfrak{L})| \leq K L^{\frac{3}{2}+\varepsilon} r^{-2}.$$

Idea of the proof. Partition $\mathbb{R}^3$ by a polynomial of bounded degree $D = D(\varepsilon)$ and induct on L . The obstruction is that the hypothesis does not pass to the lines $\mathfrak{L}_i$ entering a cell: $\mathfrak{L}_i$ has fewer than $L^{1/2+\varepsilon}$ lines in a low-degree surface, but that is not fewer than $|\mathfrak{L}_i|^{1/2+\varepsilon}$, so the induction does not immediately apply.

Guth’s device is to prove a *stronger* statement, carrying along a small set $\mathcal{Z}$ of low-degree surfaces that accounts for the rich points not otherwise controlled. The surfaces produced in the cells are collected, and then *pruned*: those containing few lines are discarded using Szemerédi–Trotter, and those containing many are few in number by Bézout.

Proposition 11.6.8 (Rich points for very large r). *If $\mathfrak{L}$ is a set of L lines in $\mathbb{R}^d$, $d \geq 2$, and $r > 2L^{1/2}$, then $|P_r(\mathfrak{L})| \leq 2Lr^{-1}$.*

Proof. List $P_r(\mathfrak{L}) = \{x_1, \dots, x_M\}$. The point x_j lies on at least $r - (j - 1)$ lines containing none of $x_1, \dots, x_{j-1}$, since two distinct points lie on at most one common line. Hence $L \geq \sum_{j=1}^M \max(r - j, 0)$. If $M \geq r/2$ this gives $L \geq (r/2)(r/2) = r^2/4 > L$, a contradiction; so $M < r/2$ and then $L \geq M(r/2)$. $\square$

Lemma 11.6.9 (Few surfaces can each be rich in lines). *Let $\mathfrak{L}$ be a set of lines in $\mathbb{R}^3$ and $\mathcal{Y}$ a set of irreducible algebraic surfaces of degree at most D , each containing at least A lines of $\mathfrak{L}$. If $A > 2D|\mathfrak{L}|^{1/2}$ then $|\mathcal{Y}| \leq 2|\mathfrak{L}|A^{-1}$.*

Proof. By Bézout for lines (Theorem 11.6.5), two distinct irreducible surfaces of degree $\leq D$ share at most D^2 lines. Order $\mathcal{Y} = \{Z_1, Z_2, \dots\}$; then Z_{j+1} contains at least $A - jD^2$ lines not lying in $Z_1, \dots, Z_j$, so

$$|\mathfrak{L}| \geq \sum_{j=1}^{|\mathcal{Y}|} \max(A - jD^2, 0).$$

If $|\mathcal{Y}| \geq \frac{1}{2}AD^{-2}$ then, since $A - jD^2 \geq A/2$ for $j \leq \frac{1}{2}AD^{-2}$, we get $|\mathfrak{L}| \geq \frac{1}{2}AD^{-2} \cdot \frac{A}{2} = \frac{A^2}{4D^2} > |\mathfrak{L}|$, a contradiction. So $|\mathcal{Y}| < \frac{1}{2}AD^{-2}$, and then $|\mathfrak{L}| \geq |\mathcal{Y}| \cdot \frac{A}{2}$. $\square$

Theorem 11.6.10 (The inductive statement). *For every $\varepsilon > 0$ there are $D(\varepsilon)$, $K(\varepsilon)$ so that the following holds. Put $r' = \lceil \frac{9}{10}r \rceil$. If $\mathfrak{L}$ is a set of L lines in $\mathbb{R}^3$ and $2 \leq r \leq 2L^{1/2}$, there is a set $\mathcal{Z}$ of surfaces with*

- (i) *each $Z \in \mathcal{Z}$ irreducible of degree $\leq D$;*
- (ii) *each $Z \in \mathcal{Z}$ contains at least $L^{\frac{1}{2}+\varepsilon}$ lines of $\mathfrak{L}$;*
- (iii) *$|\mathcal{Z}| \leq 2L^{\frac{1}{2}-\varepsilon}$;*
- (iv) *$|P_r(\mathfrak{L}) \setminus \bigcup_{Z \in \mathcal{Z}} P_{r'}(\mathfrak{L}_Z)| \leq KL^{\frac{3}{2}+\varepsilon}r^{-2}$.*

Proof. We may assume $\varepsilon \leq \frac{1}{2}$, since for $\varepsilon \geq \frac{1}{2}$ one takes $\mathcal{Z} = \emptyset$ and $|P_r(\mathfrak{L})| \leq 2L^2r^{-2}$ by double counting. Choose $D = D(\varepsilon)$ large, then $K = K(\varepsilon, D)$ large, in particular with

$$K \geq 10(2D)^{2/\varepsilon}. \quad (11.6.1)$$

Step 0. *Base case and induction parameter.*

We use strong induction on L . The pair count

$$|P_r(\mathfrak{L})| \binom{r}{2} \leq \binom{L}{2}$$

gives $|P_r(\mathfrak{L})| \leq 2L^2r^{-2}$. If $L^\varepsilon \leq 2D$, then $L^{1/2-\varepsilon}$ is bounded by a constant depending only on D, ε , so after increasing $K(\varepsilon, D)$ this is at most $KL^{3/2+\varepsilon}r^{-2}$. We may therefore assume $L^\varepsilon > 2D$ and that the inductive statement is known for every smaller line family.

Step 1. *Partition.*

Let $S \subseteq P_r(\mathfrak{L})$. By Theorem 11.2.5 there is a non-zero P of degree $\leq D$ with at most CD^3 partitioning regions O_i , each satisfying $|S \cap O_i| \leq CD^{-3}|S|$. Let $\mathfrak{L}_i$ be the lines meeting O_i . By Theorem 11.2.5(v), every line not contained in $Z(P)$ meets at most $D+1$ regions, so

$$\sum_i |\mathfrak{L}_i| \leq (D+1)L \leq 2DL. \quad (11.6.2)$$

Call O_i *good* if $|\mathfrak{L}_i| \leq \beta D^{-2}L$. By (11.6.2) there are at most $2\beta^{-1}D^3$ bad cells, so the bad cells contain at most $C\beta^{-1}|S|$ points of S ; fix β with $C\beta^{-1} \leq \frac{1}{100}$, so

$$\text{the bad cells contain at most } \frac{1}{100}|S| \text{ points of } S. \quad (11.6.3)$$

Step 2. *Good cells, by induction.*

For D large, $|\mathfrak{L}_i| \leq L/2$ on good cells. If $r \leq 2|\mathfrak{L}_i|^{1/2}$ the inductive hypothesis gives $\mathcal{Z}_i$ with $|\mathcal{Z}_i| \leq 2(\beta D^{-2}L)^{\frac{1}{2}-\varepsilon}$ and, since $S \cap O_i \subseteq P_r(\mathfrak{L}_i)$,

$$\begin{aligned} \left| (S \cap O_i) \setminus \bigcup_{Z \in \mathcal{Z}_i} P_{r'}(\mathfrak{L}_Z) \right| &\leq K(\beta D^{-2}L)^{\frac{3}{2}+\varepsilon} r^{-2} \\ &\leq C_1 K D^{-3-2\varepsilon} L^{\frac{3}{2}+\varepsilon} r^{-2}. \end{aligned}$$

If instead $r > 2|\mathfrak{L}_i|^{1/2}$, take $\mathcal{Z}_i = \emptyset$; then Theorem 11.6.8 gives $|S \cap O_i| \leq 2Lr^{-1} \leq 4L^{3/2}r^{-2}$, which is at most the same bound for K large. Summing over the $\leq CD^3$ good cells,

$$\begin{aligned} \sum_{O_i \text{ good}} \left| (S \cap O_i) \setminus \bigcup_{Z \in \mathcal{Z}_i} P_{r'}(\mathfrak{L}_Z) \right| &\leq C_2 D^{-2\varepsilon} K L^{\frac{3}{2}+\varepsilon} r^{-2}, \end{aligned}$$

and we choose $D(\varepsilon)$ with $C_2 D^{-2\varepsilon} \leq \frac{1}{400}$.

Step 3. *The partitioning surface.*

Let Z_j be the irreducible components of $Z(P)$. If $x \in S \cap Z_j$ but $x \notin P_{r'}(\mathfrak{L}_{Z_j})$, then x lies on at least $r/10$ lines not contained in Z_j , each meeting Z_j in $\leq \deg Z_j$ points. Hence $|(S \cap Z_j) \setminus P_{r'}(\mathfrak{L}_{Z_j})| \leq 10r^{-1}(\deg Z_j)L$, and summing over components,

$$\left| (S \cap Z(P)) \setminus \bigcup_j P_{r'}(\mathfrak{L}_{Z_j}) \right| \leq 10r^{-1}DL \leq \frac{1}{400}KL^{\frac{3}{2}}r^{-2}$$

for K large, using $r \leq 2L^{1/2}$.

Step 4. *Iterate to remove the bad-cell term.*

Let $\tilde{\mathcal{Z}}_S$ be the union of the $\mathcal{Z}_i$ over good cells together with the components of $Z(P)$; then $|\tilde{\mathcal{Z}}_S| \leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon}$ and, combining (11.6.3) with Steps 2 and 3,

$$\left| S \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}_S} P_{r'}(\mathfrak{L}_Z) \right| \leq \frac{1}{100}|S| + \frac{1}{200}KL^{\frac{3}{2}+\varepsilon}r^{-2}. \quad (11.6.4)$$

Set $S_1 = P_r(\mathfrak{L})$ and $S_{j+1} = S_j \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}_{S_j}} P_{r'}(\mathfrak{L}_Z)$. Iterating (11.6.4) $J = C \log L$ times gives $|S_J| \leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}$, and $\tilde{\mathcal{Z}} = \bigcup_{j < J} \tilde{\mathcal{Z}}_{S_j}$ satisfies

$$\begin{aligned} |\tilde{\mathcal{Z}}| &\leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon} \log L, & \left| P_r(\mathfrak{L}) \setminus \bigcup_{Z \in \tilde{\mathcal{Z}}} P_{r'}(\mathfrak{L}_Z) \right| &\leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}. \end{aligned} \quad (11.6.5)$$

Step 5. Prune.

Put $\mathcal{Z} = \{Z \in \tilde{\mathcal{Z}} : Z \text{ contains } \geq L^{\frac{1}{2}+\varepsilon} \text{ lines of } \mathfrak{L}\}$, which gives (i) and (ii). Since $L^\varepsilon > 2D$ we have $A = L^{\frac{1}{2}+\varepsilon} > 2DL^{1/2}$, so Theorem 11.6.9 gives $|\mathcal{Z}| \leq 2LA^{-1} = 2L^{\frac{1}{2}-\varepsilon}$, which is (iii).

For (iv), by (11.6.5) it suffices to bound $\sum_{Z \in \tilde{\mathcal{Z}} \setminus \mathcal{Z}} |P_{r'}(\mathfrak{L}_Z)|$. Sort by $\tilde{\mathcal{Z}}_s = \{Z : |\mathfrak{L}_Z| \in [2^s, 2^{s+1})\}$, so $\tilde{\mathcal{Z}} \setminus \mathcal{Z} \subseteq \bigcup_{2^s \leq L^{1/2+\varepsilon}} \tilde{\mathcal{Z}}_s$. By Szemerédi–Trotter (Theorem 11.3.2) and $r' \geq \frac{9}{10}r$, $|P_{r'}(\mathfrak{L}_Z)| \lesssim 2^{2s}r^{-3} + 2^s r^{-1}$.

For $2DL^{1/2} < 2^s \leq L^{\frac{1}{2}+\varepsilon}$, Theorem 11.6.9 with $A = 2^s$ gives $|\tilde{\mathcal{Z}}_s| \leq 2L2^{-s}$, so that range contributes

$$\lesssim \sum_{2^s \leq L^{\frac{1}{2}+\varepsilon}} (L2^s r^{-3} + Lr^{-1}) \lesssim L^{\frac{3}{2}+\varepsilon} r^{-3} + L(\log L)r^{-1} \lesssim L^{\frac{3}{2}+\varepsilon} r^{-2}.$$

For $2^s \leq 2DL^{1/2}$ we use $|\tilde{\mathcal{Z}}_s| \leq |\tilde{\mathcal{Z}}| \leq \text{Poly}(D)L^{\frac{1}{2}-\varepsilon} \log L$ together with $2^{2s}r^{-3} \leq \text{Poly}(D)Lr^{-3}$ and $2^s r^{-1} \leq \text{Poly}(D)Lr^{-2}$, giving $\text{Poly}(D)L^{3/2}r^{-2}$. Choosing K large enough, $\sum_{Z \in \tilde{\mathcal{Z}} \setminus \mathcal{Z}} |P_{r'}(\mathfrak{L}_Z)| \leq \frac{1}{100}KL^{\frac{3}{2}+\varepsilon}r^{-2}$, which with (11.6.5) gives (iv). $\square$

Proof of Theorem 11.6.7. If fewer than $L^{\frac{1}{2}+\varepsilon}$ lines lie in any irreducible surface of degree $\leq D$, then condition (ii) forces $\mathcal{Z} = \emptyset$, and (iv) becomes $|P_r(\mathfrak{L})| \leq KL^{\frac{3}{2}+\varepsilon}r^{-2}$. $\square$

Where we are. Two features of the induction deserve comment.

Why a stronger statement is easier. The hypothesis of Theorem 11.6.7 does not pass to the lines entering a cell, so the theorem cannot be induced on directly. Carrying the surface set $\mathcal{Z}$ makes the statement inductive. This is the same manoeuvre as strengthening an induction hypothesis to make it go through, and it is the paper’s one new idea.

The cost of ε . Step 2 needs $C_2 D^{-2\varepsilon} \leq \frac{1}{400}$, that is $D \geq 400^{1/(2\varepsilon)}$. For $\varepsilon = \frac{1}{2}$ this is 400; for $\varepsilon = \frac{1}{4}$ it is 1.6×10^5 ; for $\varepsilon = \frac{1}{10}$ it is about 10^{13} . Finite for every $\varepsilon > 0$ and astronomical as $\varepsilon \rightarrow 0$ — which is why the method gives $L^{3/2+\varepsilon}$ and not $L^{3/2}$.

Where the pruning lands. Theorem 11.6.9 with $A = L^{1/2+\varepsilon}$ gives $|\mathcal{Z}| \leq 2LA^{-1} = 2L^{1/2-\varepsilon}$, which is *exactly* hypothesis (iii). Nothing is spare: the induction closes on the nose.

Remark 11.6.11 (R13 discharged at the ε -loss level). The theorem used by this series is now internally proved. Its inputs are the sign-history polynomial partitioning theorem Theorem 11.2.5, the local resultant bounds Theorems 11.6.4 and 11.6.5, generic projection plus Szemerédi–Trotter, and the strengthened induction above. No Milnor–Thom component count, ruled-surface classification, or flecnodal polynomial is used.

The original Guth–Katz theorem is sharper: it removes the ε loss under the plane/regulus hypothesis and leads to $N/\log N$ distinct distances. That sharper historical statement is not a numbered theorem in the active proof ledger; the internally proved ε -loss version is the theorem used below.

Proposition 11.6.12 (Lines in a plane, and the planar case of the theorem). *Let $\mathcal{L}$ be L lines in a plane $\Pi \subseteq \mathbb{R}^3$, no two parallel. Then the number of points on at least k of them is $\lesssim L^2/k^3 + L/k$.*

Proof. Work inside $\Pi \cong \mathbb{R}^2$ and dualise: by the point–line duality of Theorem 11.3.2 Step 0, points on at least k lines correspond to lines containing at least k points of a set of L points. By Theorem 11.3.4 with $P = L$, their number is $\lesssim L^2/k^3 + L/k$. $\square$

Where we are. So within a single plane the answer is known and sharp — it is Szemerédi–Trotter. Theorem 11.6.7 asserts that a configuration *not* concentrated in planes or reguli does strictly better: the count drops from L^2/k^3 to $L^{3/2}/k^2$.

Comparing at $k = 2$: the planar bound is L^2 and the three-dimensional one $L^{3/2}$. The gain is a full factor $L^{1/2}$, and it is exactly what Chapter 11.7 needs: with $L = N^2$ lines, $L^2 = N^4$ would give nothing, while $L^{3/2} = N^3$ gives the distinct-distances theorem.

That is worth stating plainly. The Guth–Katz theorem is not a technical refinement of Szemerédi–Trotter; it is the statement that genuinely three-dimensional configurations are quantitatively different, and the whole distinct-distances result rests on that difference.

WORKED EXAMPLE 11.6.1— TIER A **Why both exclusions are needed**

Theorem 11.6.7 excludes concentration in every bounded-degree irreducible surface, and this cannot be dropped.

Planes. By Theorem 11.6.1, L lines in a plane can have $\approx L^2$ intersecting pairs — vastly more than $L^{3/2}$. So the theorem is false without the planar exclusion.

Reguli. By Theorem 11.6.3, L lines on a regulus can have $\approx L^2/4$ intersecting pairs. Again $\gg L^{3/2}$.

And nothing else. That these are the *only* obstructions is the content of the theorem, and it is a statement about algebraic surfaces: a surface containing more than a bounded number of lines through each point is ruled, and the doubly ruled ones are exactly the planes and the quadrics of regulus type.

The threshold $\sqrt{L}$. If a plane contains m lines it contributes $\approx m^2$ pairs, and this is $\lesssim L^{3/2}$ exactly when $m \lesssim L^{3/4}$. The theorem’s hypothesis $m \leq \sqrt{L}$

is stronger than necessary for that count but is what the proof requires; sharpening it is possible and is not needed for Chapter 11.7.

WORKED EXAMPLE 11.6.2— TIER B **The two-case structure, in three dimensions**

Chapter 11.3's Checkpoint identified the shape of every polynomial-method proof: a cellular case and an algebraic case. In the plane the algebraic case was one line of argument. Here it is the whole difficulty, and the contrast is instructive.

The cellular case, in $\mathbb{R}^3$. Partition by a polynomial of degree D ; there are $\lesssim D^3$ cells, each meeting few lines, and a line crosses $\leq D + 1$ of them. The counting goes through exactly as in the plane.

The algebraic case, in the plane. A curve of degree D contains at most D lines — restrict to a transversal and count roots. Two lines, one sentence.

The algebraic case, in $\mathbb{R}^3$. A surface of degree D may contain infinitely many lines. One must therefore:

- show that a surface containing $\gg D^2$ lines is *ruled* — this is where the flecnode polynomial enters, a degree- $11D$ polynomial vanishing at points where a line meets the surface to order three;
- classify the ruled surfaces: a ruled surface of degree D is a plane, a regulus, or has the property that most of its lines meet few others;
- handle each class separately.

The moral. The polynomial method reduces a combinatorial problem to an algebraic one, and the algebraic problem may be easy or may be a research programme. Predicting which requires knowing how varieties of the relevant degree and dimension can contain the objects in question — lines, in this case. That is why the method's difficulty jumps between $\mathbb{R}^2$ and $\mathbb{R}^3$ and why Volume 14 reports that the four-dimensional analogues remain open.

WORKED EXAMPLE 11.6.3— TIER C: RESEARCH TASTE **Ruled surfaces, and why dimension three is special**

Theorem 11.6.11 attributes the difficulty of Theorem 11.6.7 to the classification of surfaces containing many lines. It is worth seeing why that question has a good answer in $\mathbb{R}^3$ and not beyond.

(a) *The dimension count.* Lines in $\mathbb{R}^3$ form a 4-parameter family. A surface is 2-dimensional; requiring a line to lie in it is 2 conditions per point of the line, but a line meets a degree- D surface in D points, so lying inside is $D + 1$ conditions. Generically a surface contains finitely many lines — famously 27 on a smooth cubic.

(b) *The exceptions are ruled.* A surface containing a 1-parameter family of lines is by definition ruled, and ruled surfaces of degree D are classified: they are swept by a line moving along a curve. The doubly ruled ones — planes and the quadrics of Theorem 11.6.3 — are the only surfaces where two lines of the surface meet at every point.

(c) *The flecnodal polynomial.* Cayley and Salmon, in the 1860s, constructed a polynomial of degree $11D - 24$ vanishing exactly where a line meets the surface to order three. A surface all of whose points are flecnodal is ruled, and that is the mechanism converting “many lines” into “ruled”.

(d) *Why $\mathbb{R}^4$ is harder.* Lines in $\mathbb{R}^4$ form a 6-parameter family and hypersurfaces are 3-dimensional; a hypersurface can contain a 2-parameter family of lines without being degenerate in any classical sense. There is no classification comparable to the ruled-surface one, and the polynomial method’s algebraic case is correspondingly open.

(e) *Consequence for the series.* Volume 14 reports that Kakeya is settled in $\mathbb{R}^3$ and open in $\mathbb{R}^4$. The classification gap in (d) is one reason — not the only one, but the one visible from this volume’s methods.

WORKED EXAMPLE 11.6.4— TIER B The k -fold bound, and where the logarithm comes from

Theorem 11.6.7 gives two statements: $\lesssim L^{3/2}$ points on at least two lines, and $\lesssim L^{3/2}/k^2$ on at least k . The second is what Chapter 11.7 uses, and the passage between them explains the logarithm in Theorem 11.7.9.

Counting intersecting pairs from the k -fold bound. A point on exactly k lines accounts for $\binom{k}{2} \approx k^2$ intersecting pairs. Summing over dyadic ranges $k \approx 2^j$,

$$\begin{aligned} \#\{\text{intersecting pairs}\} & \lesssim \sum_j 2^{2j} \cdot \frac{L^{3/2}}{2^{2j}} \\ & = L^{3/2} \cdot \#\{j\} \approx L^{3/2} \log L. \end{aligned}$$

Every dyadic range contributes *equally* — the k^2 from the pair count exactly cancels the k^{-2} in the bound — so the sum is the number of ranges, which is

$\log L$.

That is the logarithm. It is not a technical artefact of one step; it is the statement that no single multiplicity dominates. Compare Worked Example 9.3.1 in Volume 9, where the $n = 2$ case of the L^2 method had the same feature and produced the same log.

What removing it would require. A bound better than $L^{3/2}/k^2$ for some range of k — and the bound is sharp for each k separately, as the grid shows. So the improvement must come from showing that the extremal configurations for different k cannot coexist, which is a genuinely different kind of statement.

Why this matters. Worked Example 11.7.1 located the remaining gap in Theorem 11.7.1 at exactly this logarithm. Knowing it arises from equal dyadic contributions, rather than from slack in a single estimate, tells one that closing it is not a matter of care.

Where we are. The chapter's arc in one paragraph: in a plane, L lines can have $\approx L^2$ crossing points (Theorem 11.6.1); on a regulus, likewise (Theorem 11.6.3); and Theorem 11.6.7 says that a configuration avoiding both does strictly better, at $L^{3/2}$. The gain of a factor $L^{1/2}$ is exactly what Chapter 11.7 converts into the distinct-distances theorem, and Theorem 11.6.12 shows the planar bound it improves on is itself sharp.

Drills

Drill 11.6.1. Verify Theorem 11.6.3 for $z = xy$: check that ℓ_a and m_b lie on the surface and meet at (a, b, ab) .

Drill 11.6.2. Show that ℓ_a and $\ell_{a'}$ are disjoint for $a \neq a'$, and that they are not parallel.

Drill 11.6.3. Compute the number of intersecting pairs for L lines in general position in a plane, and compare with $L^{3/2}$.

Drill 11.6.4. Explain why the hypothesis of Theorem 11.6.7 would be violated by the grid configuration of Worked Example 11.3.1 embedded in a plane in $\mathbb{R}^3$.

Where we are. A remark on what is and is not proved in this chapter. Theorem 11.6.1, Theorem 11.6.3 and Theorem 11.6.12 are proved in full, and together they establish the *necessity* of the hypotheses in Theorem 11.6.7 and the sharpness of the planar bound it improves on. So the reader has,

with proof, the statement's shape: which configurations must be excluded, why, and how much is gained by excluding them.

What is imported is the sufficiency — that excluding planes and reguli is enough. That is Theorem 11.6.11, and it is where the ruled-surface theory of Worked Example 11.6.3 enters.

Stating the necessity and importing the sufficiency seems the right division for a theorem of this size: the reader can see what is being claimed and why it is not obvious, without a chapter of classical algebraic geometry.

Notes

The incidence theorem is Guth and Katz's (2015). The reduction of the algebraic case to ruled surfaces uses the flecnode polynomial of Cayley and Salmon, from the 1860s; Guth and Katz's use of it was the first application in combinatorics. Katz and Guth's exposition, and Guth's book *Polynomial Methods in Combinatorics*, are the references.

Chapter 11.7

Distinct Distances

Conjecture 11.7.1 (Erdős, 1946). *A set of N points in the plane determines at least $cN/\sqrt{\log N}$ distinct distances.*

The $\sqrt{\log N}$ is not decoration: the $\sqrt{N} \times \sqrt{N}$ integer grid determines $\approx N/\sqrt{\log N}$ distances, by a classical estimate of Landau on the number of integers below N representable as a sum of two squares. So the conjecture asserts the grid is extremal.

Guth and Katz proved the sharper historical bound $N/\log N$. This chapter proves, from the internally established ε -loss incidence theorem, the completely self-contained consequence $N^{1-\varepsilon}$. The sharper $N/\log N$ statement is retained as context rather than imported as a proof input.

11.7.1 From distances to quadruples

Notation 11.7.2. For a set $\mathcal{P}$ of N points in $\mathbb{R}^2$ let $\Delta(\mathcal{P})$ be its set of distinct distances, and for $d > 0$ let $n_d = \#\{(p, q) \in \mathcal{P}^2 : p \neq q, |pq| = d\}$.

Lemma 11.7.3 (Cauchy–Schwarz). *With*

$$\begin{aligned} Q &= \#\{(p, q, r, s) \in \mathcal{P}^4 : p \neq q, r \neq s, |pq| = |rs|\} \\ &= \sum_{d \in \Delta(\mathcal{P})} n_d^2, \end{aligned}$$

one has

$$|\Delta(\mathcal{P})| \geq \frac{N^2(N-1)^2}{Q}.$$

Proof. $\sum_d n_d = N(N-1)$, the number of ordered pairs of distinct points. By Cauchy–Schwarz over the $|\Delta|$ values of d ,

$$(N(N-1))^2 = \left(\sum_d n_d \right)^2 \leq |\Delta| \sum_d n_d^2 = |\Delta| Q. \quad \square$$

Where we are. So a lower bound on $|\Delta|$ is an *upper* bound on Q : if the quadruple count is $O(N^3 \log N)$ then $|\Delta| \gtrsim N^4/(N^3 \log N) = N/\log N$, which is the Guth–Katz theorem.

The reformulation is the whole strategy. $|\Delta|$ is hard to bound directly because it is a count of *values*; Q is a count of *configurations*, and configurations are what incidence geometry handles. Numerically the Cauchy–Schwarz step is an equality when all n_d are equal, which is the case for a generic point set — so nothing is lost in the reduction for the configurations one cares about.

11.7.2 From quadruples to lines in $\mathbb{R}^3$

Definition 11.7.4 (The Elekes–Sharir parametrisation). Let G be the group of orientation-preserving rigid motions of $\mathbb{R}^2$. A non-translation element is a rotation by an angle $\theta \in (0, 2\pi)$ about a centre $c = (c_1, c_2)$; associate to it the point

$$\Psi(g) = \left(c_1, c_2, \cot \frac{\theta}{2} \right) \in \mathbb{R}^3.$$

Lemma 11.7.5 (Motion sets are lines). For $p \neq r$ in $\mathbb{R}^2$ let

$$G_{p,r} = \{g \in G : g(p) = r\}.$$

Then $\Psi(G_{p,r})$ is a line in $\mathbb{R}^3$.

Proof. A rotation about c takes p to r only if $|c - p| = |c - r|$, so c lies on the perpendicular bisector of the segment pr ; conversely for each such c there is exactly one rotation angle taking p to r . Parametrise the bisector by $c = m + tn$, where $m = \frac{p+r}{2}$ and n is a unit normal to $r - p$.

Write $R = |c - p|$ and θ for the rotation angle. The chord subtended by θ on the circle of radius R about c has length $|r - p|$, so

$$|r - p| = 2R \sin \frac{\theta}{2}, \quad \text{whence} \quad \sin \frac{\theta}{2} = \frac{|r - p|}{2R}.$$

Also $R^2 = |c - m|^2 + |m - p|^2 = t^2 + \frac{|r-p|^2}{4}$, and

$$\cos \frac{\theta}{2} = \sqrt{1 - \frac{|r-p|^2}{4R^2}} = \frac{|t|}{R},$$

taking the sign of t to match the orientation. Therefore

$$\cot \frac{\theta}{2} = \frac{\cos(\theta/2)}{\sin(\theta/2)} = \frac{t/R}{|r-p|/(2R)} = \frac{2t}{|r-p|},$$

which is an *affine* function of t . Since $c = m + tn$ is affine in t as well, the map $t \mapsto \Psi(g_t) = (m_1 + tn_1, m_2 + tn_2, \frac{2t}{|r-p|})$ is affine, and its image is a line. $\square$

Lemma 11.7.6 (The motion-line labels are injective). *The line $\ell_{p,r} := \Psi(G_{p,r})$ determines the ordered pair (p, r) . Consequently distinct ordered pairs give distinct lines.*

Proof. Writing a point on the motion line as (x_1, x_2, z) , the computation in Theorem 11.7.5 gives the affine equations

$$2x_1 = p_1 + r_1 + (p_2 - r_2)z, \quad 2x_2 = p_2 + r_2 + (r_1 - p_1)z.$$

The two intercepts recover $p_1 + r_1, p_2 + r_2$ and the two slopes recover $p_2 - r_2, r_1 - p_1$. Hence p and r are uniquely recovered. $\square$

Where we are. The computation is worth checking numerically, and it is exact: for several choices of p and r , the points $\Psi(g_t)$ for seven values of t have a rank-one centred matrix to machine precision — the second singular value is 10^{-16} . So $\Psi(G_{p,r})$ is a line, not merely approximately. That the three-dimensional group of rigid motions turns the two-dimensional distance problem into a question about lines in $\mathbb{R}^3$ is the whole idea of Elekes and Sharir, and it is why Theorem 11.6.7 is the relevant theorem.

Proposition 11.7.7 (Quadruples are intersecting pairs of lines). *Let $\mathcal{L} = \{\Psi(G_{p,r}) : (p, r) \in \mathcal{P}^2, p \neq r\}$, a family of at most $N(N-1)$ lines in $\mathbb{R}^3$. Then*

$$Q \leq C \cdot \#\{\text{intersecting pairs of lines in } \mathcal{L}\} + O(N^3).$$

Proof. A counted quadruple (p, q, r, s) determines an orientation-preserving isometry g with $g(p) = r$ and $g(q) = s$. If g is not a translation, its point $\Psi(g)$ lies on both $\ell_{p,r}$ and $\ell_{q,s}$. By Theorem 11.7.6 these are distinct because $p \neq q$, and the ordered pair of intersecting motion lines determines the four labels, so the multiplicity is bounded absolutely.

If g is a translation, then $r - p = s - q$. Choosing p, q, r determines $s = q + r - p$, so there are at most N^3 such quadruples. This gives the stated bound. $\square$

Lemma 11.7.8 (Non-clustering). *For each $D \geq 1$ there is C_D such that for any set $\mathcal{P}$ of N points in $\mathbb{R}^2$, the family $\mathcal{L}(\mathcal{P})$ contains at most $C_D N$ lines in any algebraic surface of degree at most D , and at most N lines through any point of $\mathbb{R}^3$.*

Proof. Factor the defining polynomial of the surface into real irreducibles; a line on the zero set lies in one factor. It therefore suffices to treat an irreducible surface $Z(Q)$ of degree $d \leq D$.

Step 1. *A fixed first point gives a skew family.*

Fix p . By Theorem 11.7.6, the lines $\ell_{p,q}$ are distinct. Two of them cannot meet: an intersection would represent one rigid motion sending p to two different points. They are not parallel either, as is immediate from the two affine equations in Theorem 11.7.6. Hence a plane contains at most one of them and a point lies on at most one of them. Summing over the N choices of p proves the asserted point bound and handles plane factors.

Step 2. *A polynomial vector field for the p -ruling.*

From

$$2x_1 = (p_1 + q_1) + (p_2 - q_2)x_3, \quad 2x_2 = (p_2 + q_2) + (q_1 - p_1)x_3$$

one solves uniquely for $q = (1 + x_3^2)^{-1}b_p(x)$, where b_p has degree at most 2 in x and is affine in p . Thus

$$V_p(x) = (p_2(1 + x_3^2) - b_{2,p}(x), b_{1,p}(x) - p_1(1 + x_3^2), 2(1 + x_3^2))$$

is a nowhere-zero polynomial vector field of degree at most 2, affine in p , and on $\ell_{p,q}$ it equals $(1 + x_3^2)(p_2 - q_2, q_1 - p_1, 2)$, a non-zero tangent vector.

Step 3. *For a non-plane irreducible surface, at most one p is exceptional.*

Assume $d > 1$ and set $R_p = V_p \cdot \nabla Q$, so $\deg R_p \leq d + 1$. If $Q \nmid R_p$, Theorem 11.6.5 shows that at most $2d(d + 1)$ lines of the p -family can lie in $Z(Q)$. Call p exceptional when $Q \mid R_p$.

There is at most one exceptional p . Indeed, if $p_0 \neq p_1$ were exceptional, affine dependence on p would make every p on the line through them exceptional. Choose a smooth point x on one line of the p_0 -family lying in $Z(Q)$. Such a point exists: a singular line lies in $Z(Q) \cap Z(\partial_j Q)$ for a non-zero partial derivative, and Theorem 11.6.5 bounds the number of such lines. For every exceptional p , V_p is tangent to $Z(Q)$; uniqueness for the polynomial ODE implies the integral line through x remains in $Z(Q)$. Hence infinitely many distinct motion lines through x lie in the tangent plane $T_x Z(Q)$. The restriction $Q|_{T_x Z(Q)}$ then vanishes on infinitely many distinct lines through x , so it is identically zero. Thus the tangent plane is a component of $Z(Q)$, contradicting irreducibility and $d > 1$.

Consequently all but at most one p contribute at most $2d(d + 1)$ lines, while the exceptional p contributes at most N . Thus $Z(Q)$ contains at most $C_d N$ motion lines.

Step 4. *Return to a reducible degree- D surface.*

If $F = \prod_j Q_j$ with $\sum_j \deg Q_j \leq D$, every motion line contained in $Z(F)$ lies in some $Z(Q_j)$. Summing the preceding bounds gives at most $C_D N$ lines in $Z(F)$. $\square$

Theorem 11.7.9 (Distinct distances). *For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that any set of N points in $\mathbb{R}^2$ determines at least $c_\varepsilon N^{1-\varepsilon}$ distinct distances.*

Proof. By Theorem 11.7.8, $\mathcal{L} = \mathcal{L}(\mathcal{P})$ is a set of $L = N^2$ lines with at most $C_D N \leq L^{\frac{1}{2}+\varepsilon}$ lines in any irreducible surface of degree $\leq D$, once N is large. So Theorem 11.6.7 applies and gives, for $2 < r < 2N$,

$$|P_r(\mathcal{L})| \leq K(N^2)^{\frac{3}{2}+\varepsilon} r^{-2} = K N^{3+2\varepsilon} r^{-2}.$$

By Theorem 11.7.7 the parallel quadruples number at most N^3 , and the intersecting ones are $\sum_{r \geq 2} (2r - 2) |P_r(\mathcal{L})|$, so

$$Q \leq N^3 + \sum_{r=2}^N 2r \cdot K N^{3+2\varepsilon} r^{-2} \lesssim N^3 + K N^{3+2\varepsilon} \log N \lesssim_\varepsilon N^{3+3\varepsilon}.$$

By Theorem 11.7.3, $|\Delta(\mathcal{P})| \geq (N^4 - 2N^3)/Q \gtrsim_\varepsilon N^{1-3\varepsilon}$. Replacing ε by $\varepsilon/3$ gives the theorem. $\square$

Remark 11.7.10 (The sharper form). Guth and Katz prove the stronger bound $\gtrsim N/\log N$, which is optimal up to $\sqrt{\log N}$ by Worked Example 11.7.1. Their argument replaces Theorem 11.6.7 by the ε -free estimate $|P_r(\mathcal{L})| \leq C L^{3/2} r^{-2}$, whose proof splits into the cases $r = 2$ and $r \geq 3$; the case $r = 2$ uses the classification of ruled surfaces through the Cayley–Salmon flecnode polynomial, and is not reproduced here.

What *is* proved here is the full chain from Borsuk–Ulam to $N^{1-\varepsilon}$, with no unproved input. The gap between $N^{1-\varepsilon}$ and $N/\log N$ is the price of avoiding the ruled surface theory.

WORKED EXAMPLE 11.7.1— TIER A **The grid, and why $\sqrt{\log N}$**

Take $\mathcal{P} = \{1, \dots, \sqrt{N}\}^2$, the integer grid.

The distances. $|pq|^2$ is a sum of two squares, each at most N . So $|\Delta|$ is at most the number of integers in $[1, 2N]$ expressible as a sum of two squares.

Landau's estimate. That count is $\approx \frac{cN}{\sqrt{\log N}}$ — the density of sums of two squares decays like $(\log x)^{-1/2}$, because an integer is such a sum precisely when every prime $\equiv 3 \pmod{4}$ divides it to an even power.

Consequently. The grid determines $\approx N/\sqrt{\log N}$ distances, showing the conjectured scale. The internal theorem Theorem 11.7.9 is deliberately

weaker by an N^ε loss; the sharper historical Guth–Katz theorem reaches $N/\log N$.

What the gap is. In the internal proof the loss is the L^ε factor in Theorem 11.6.7, not a hidden imported incidence theorem. The sharper historical argument removes that power loss and leaves the familiar logarithm from dyadic multiplicities.

The moral. An extremal example drawn from number theory — the sums of two squares — sets the target, and an incidence-geometric method comes within a logarithm. The two subjects meet only in the answer.

WORKED EXAMPLE 11.7.2— TIER B **Reading the reduction backwards**

The chain runs

$$|\Delta| \xrightarrow[\text{Cauchy-Schwarz}]{} Q \xrightarrow[\text{Elekes-Sharir}]{} \text{line incidences in } \mathbb{R}^3 \xrightarrow[\text{Guth-Katz}]{} \text{done.}$$

Reading it backwards shows what each link costs.

Link 3 is where the mathematics is. Theorem 11.6.7 is the theorem; the other two links are a page each.

Link 2 is exact. Theorem 11.7.5 is an identity, verified to machine precision, and Theorem 11.7.7 loses only the degenerate $O(N^3)$, which is below the target anyway.

Link 1 is lossless for the extremal configurations. Cauchy–Schwarz is an equality when all n_d agree. For the grid they do not agree exactly — some distances are far more common — so a little is lost, but not enough to matter.

Why this matters for reading other papers. A reduction chain is only as good as its worst link, and identifying which link is lossy tells one where an improvement must come from. Here it is Link 3, and specifically the dyadic summation that produces the $\log N$. Any improvement to $N/\sqrt{\log N}$ must attack that sum, not the reduction.

WORKED EXAMPLE 11.7.3— TIER B **Reductions that change the ambient dimension**

Chapter 11.7 solved a planar problem by passing to $\mathbb{R}^3$. That move is worth isolating, since it is counterintuitive — one usually reduces dimension, not raises it.

Why raising helps. The objects being counted are *pairs* of points, and pairs of points in $\mathbb{R}^2$ have four degrees of freedom. The symmetry group that preserves distance — rigid motions — is three-dimensional, and quotienting by it turns

the four-parameter family of pairs into a three-parameter family of group elements. The problem becomes one about subvarieties of a three-dimensional space, and Theorem 11.7.5 shows they are the simplest possible: lines.

The general recipe. When a problem concerns configurations invariant under a group G , parametrise the configurations by G and study the resulting subvarieties. The ambient dimension becomes $\dim G$, and the objects are usually cosets — lines, if G is three-dimensional and the stabilisers are one-dimensional.

Two other instances. The Elekes–Sharir framework has been used for distinct distances on spheres and in $\mathbb{R}^3$, with G the relevant motion group; and the sum–product problem admits a similar treatment with G the affine group. In each case the payoff is that a hard counting problem becomes an incidence problem for lines, where Theorem 11.6.7 applies.

The cost. The parametrisation must send the relevant sets to lines *exactly*, not approximately — Theorem 11.7.5 is an identity, and its numerical verification to machine precision reflects that. A parametrisation sending them to curves of higher degree would require an incidence theorem for curves, which is available but weaker.

WORKED EXAMPLE 11.7.4 — TIER C: RESEARCH TASTE What the distinct distances proof needed

Theorem 11.7.9 is the polynomial method’s most celebrated application, and the anatomy of what it consumed is instructive.

(a) *A quadratic reformulation.* Theorem 11.7.3 converts a count of values into a count of configurations. This is free and is the standard first move for any “distinct values” problem.

(b) *A group.* The rigid motions of $\mathbb{R}^2$ form a three-dimensional group, and equal distances are exactly the orbits. Without a transitive symmetry group preserving the quantity of interest there is no Elekes–Sharir step, which is why the method does not obviously extend to, say, distinct areas of triangles.

(c) *A parametrisation making the stabilisers linear.* Theorem 11.7.5 is the piece of luck: the sets $G_{p,r}$ become *lines* rather than curves of higher degree. Had they become conics, one would need an incidence theorem for conics, which is available but weaker.

(d) *A hard incidence theorem.* Theorem 11.6.7, which is Theorem 11.6.11 and the only genuinely difficult ingredient.

(e) *A verification that the hypotheses hold.* That no plane or regulus contains too many of the lines $\Psi(G_{p,r})$ — quoted in the proof of Theorem 11.7.9 and

itself a short but real argument about the geometry of the motion group.

The lesson. Four of the five ingredients are elementary or free; the theorem's difficulty is concentrated entirely in (d). That concentration is typical of the polynomial method: it converts a problem into an incidence question, and the incidence question is where the work goes. Recognising this before starting tells one whether a given problem is likely to yield — if the resulting incidence question is one of the solved ones, it will; if not, it will not.

WORKED EXAMPLE 11.7.5— TIER A **The chain, with numbers**

Following Theorem 11.7.9 through with $N = 10^6$ points makes the sizes of the objects concrete.

Points. $N = 10^6$.

Ordered pairs. $N(N - 1) \approx 10^{12}$, and this is $\sum_d n_d$.

Lines in $\mathbb{R}^3$. One for each ordered pair with $p \neq r$, so $L \approx 10^{12}$.

Intersecting pairs of lines. By Theorem 11.6.7 with $L = 10^{12}$: $L^{3/2} \log L \approx 10^{18} \cdot 28 \approx 3 \times 10^{19}$.

Quadruples. $Q \lesssim 3 \times 10^{19}$ by Theorem 11.7.7.

Distances. By Theorem 11.7.3,

$$|\Delta| \geq \frac{(10^{12})^2}{3 \times 10^{19}} \approx 3 \times 10^4.$$

Compare. $N/\log N = 10^6/13.8 \approx 7 \times 10^4$, so the arithmetic is consistent up to the constants suppressed throughout. And $N/\sqrt{\log N} \approx 2.7 \times 10^5$ is the conjectured truth, a factor $\sqrt{\log N} \approx 3.7$ above.

What the numbers show. The lines outnumber the points by a factor N , and the incidence theorem is applied to a family far larger than the original configuration. That is characteristic of the Elekes–Sharir reduction: it trades a small hard problem for a large easy one, and the trade is favourable because $L^{3/2}$ with $L = N^2$ is N^3 , comfortably below the N^4 that would make the Cauchy–Schwarz step vacuous.

Where we are. Erdős's problem is the oldest in this series — 1946 — and the only one whose solution the series can report as essentially complete. It took sixty-nine years, and the final step was a technique invented for a different purpose.

That is worth holding alongside Kakeya, posed in 1928 and open above dimension three. The two problems are of comparable age and comparable appearance; one yielded to a change of category and the other has not.

Drills

Drill 11.7.1. Verify Theorem 11.7.3 and identify when Cauchy–Schwarz is an equality.

Drill 11.7.2. Check the computation $\cot \frac{\theta}{2} = \frac{2t}{|r-p|}$ in Theorem 11.7.5 for $p = (0, 0)$, $r = (1, 0)$.

Drill 11.7.3. Confirm that a rigid motion taking p to r and q to s exists whenever $|pq| = |rs|$, and identify the degenerate cases.

Drill 11.7.4. Using Landau’s estimate, compute the number of distinct distances in the 100×100 grid to leading order.

Where we are. Chapter 11.7 is the only chapter in this series whose main theorem is proved from an imported ingredient rather than from scratch, and it is worth being explicit about what the reader gets.

Proved here: the Cauchy–Schwarz reduction (Theorem 11.7.3), the Elekes–Sharir parametrisation and its linearity (Theorem 11.7.5), and the correspondence between quadruples and intersecting pairs (Theorem 11.7.7). Together these constitute the entire reduction of a planar metric problem to an incidence problem in $\mathbb{R}^3$.

Proved: Theorem 11.6.7.

Worked Example 11.7.4 anatomises the five ingredients and finds four of them elementary. That is the usual shape for a polynomial-method application, and it is why the technique spread so quickly once the incidence theorems were available.

Notes

Erdős posed the problem in 1946 and proved the lower bound $N^{1/2}$; a sequence of improvements followed, the last before Guth–Katz being $N^{0.8641}$ by Katz and Tardos. Elekes and Sharir introduced the reduction to rigid motions in 2010, and Guth and Katz completed it in 2015 with Theorem 11.6.7.

The gap of $\sqrt{\log N}$ remains open.

Chapter 11.8

Takeya and the Polynomial Method

Volume 10 showed that Dvir’s argument does not transfer to $\mathbb{R}^n$, and located the failure precisely: the points of a δ -net inside a tube are only *nearly* collinear, so vanishing at them says nothing about the tube’s axis (Theorem 10.7.2). This chapter shows what happens when the approximation is removed — when one works with the genuine segments of a Besicovitch set rather than with tubes — and why the resulting theorem, though exact and easy, does not settle anything.

11.8.1 Besicovitch sets are not algebraic

Theorem 11.8.1. *Let $K \subseteq \mathbb{R}^n$ contain a unit segment in every direction. Then K is not contained in $Z(P)$ for any non-zero polynomial $P \in \mathbb{R}[x_1, \dots, x_n]$.*

Proof. Suppose $K \subseteq Z(P)$ with $P \neq 0$ of degree D , and let P_D be its top-degree homogeneous part, which is non-zero.

Step 1. *P vanishes on whole lines.*

Fix $v \in \mathbb{S}^{n-1}$. By hypothesis K contains a unit segment $\{a + tv : t \in [0, 1]\}$, on which P vanishes. The function $t \mapsto P(a + tv)$ is a polynomial in one variable of degree at most D with infinitely many roots — every $t \in [0, 1]$ — hence identically zero. So P vanishes on the entire line $\{a + tv : t \in \mathbb{R}\}$.

Step 2. *Read off the top-degree part.*

Expanding $P = \sum_{j \leq D} P_j$ into homogeneous parts and using homogeneity, the coefficient of t^D in $P(a + tv)$ is $P_D(v)$. Since the polynomial in t is identically zero, every coefficient vanishes, so

$$P_D(v) = 0.$$

Step 3. *Conclude.*

Step 2 holds for every $v \in \mathbb{S}^{n-1}$. By homogeneity, $P_D(rv) = r^D P_D(v) = 0$ for every $r \in \mathbb{R}$, so P_D vanishes on all of $\mathbb{R}^n$. A polynomial vanishing identically on $\mathbb{R}^n$ is the zero polynomial — over an infinite field, unlike the situation of Theorem 10.1.3 — contradicting $P_D \neq 0$. $\square$

Where we are. This is Dvir’s argument, run over $\mathbb{R}$, and it works. The comparison with Volume 10 is exact and worth stating side by side.

Step 1 is where they differ. Over $\mathbb{F}_q$ a line has q points and one needs $q > D$ to force vanishing. Over $\mathbb{R}$ a segment has infinitely many points and the conclusion is automatic — no degree condition at all.

Step 3 is where they differ again. Over $\mathbb{F}_q$ one needs Theorem 10.1.3 because $x^q - x$ vanishes identically without being zero. Over $\mathbb{R}$ a polynomial vanishing everywhere is zero, full stop.

So the real version is easier than the finite field version. Both of Dvir’s technical devices are unnecessary. And yet the real Kakeya conjecture is open and the finite field one is closed. The resolution is in the next section, and it is the point of this chapter.

11.8.2 Why this settles nothing

Proposition 11.8.2 (Varieties are small). *Let $P \neq 0$ have degree D . Then $m_n(Z(P)) = 0$ and $\dim_{\mathbb{H}} Z(P) \leq n - 1$.*

Proof. By Theorem 10.1.3’s real analogue — or directly: fix a line in general position; P restricted to it is a non-zero one-variable polynomial with at most D roots. So every line meets $Z(P)$ in at most D points unless it lies inside, and the lines lying inside form a set of directions of measure zero unless $P_D \equiv 0$. Fubini along a generic family of parallel lines gives $m_n(Z(P)) = 0$.

For the dimension, $Z(P)$ is a countable union of C^1 hypersurfaces away from its singular set, which is itself contained in $Z(\nabla P)$ of lower dimension; induction gives $\dim_{\mathbb{H}} Z(P) \leq n - 1$. $\square$

Where we are. Theorem 11.8.1 excludes Besicovitch sets from algebraic varieties, and Theorem 11.8.2 says varieties have dimension at most $n - 1$. So the theorem tells us that a Besicovitch set is not contained in a set of dimension $n - 1$ of a very special kind.

But the Kakeya conjecture asserts $\dim_{\mathbb{H}} K = n$, and the trivial bound already gives $\dim_{\mathbb{H}} K \geq 1$. Excluding algebraic hypersurfaces excludes almost nothing: they are a measure-zero, closed, countable-parameter family among all sets of dimension $n - 1$, and the extremal Besicovitch sets — Perron trees — are manifestly not algebraic.

So Theorem 11.8.1 is a true theorem with no consequences. It is worth including precisely because it is the exact analogue of Dvir’s theorem, it is easier, and it is useless — which is the sharpest available demonstration that the finite field problem and the real problem are not the same problem.

WORKED EXAMPLE 11.8.1 — TIER C: RESEARCH TASTE **The three versions of one argument**

Dvir’s argument has now appeared three times in this series, in settings that look almost identical and behave completely differently. Setting them side by side is the clearest summary of where the Kakeya difficulty lives.

	$\mathbb{F}_q^n$, lines	$\mathbb{R}^n$, segments	$\mathbb{R}^n$, δ -tubes
Where	Vol 10, Theorem 10.2.1	Theorem 11.8.1	Vol 10, Theorem 10.7.2
Step 1 works?	yes, needs $D < q$	yes, automatic	no
Step 3 works?	yes, needs Schwartz–Zippel	yes, trivially	—
Conclusion	$ K \gtrsim q^n$: <i>sharp</i>	K not in a variety: <i>vacuous</i>	nothing
Why	finiteness makes “small” mean something	infinite makes vanishing automatic <i>and</i> varieties negligible	points only nearly collinear

The pattern. The argument’s strength depends on whether “ P vanishes on K ” is a strong hypothesis. Over $\mathbb{F}_q$ it is: $Z(P)$ has at most Dq^{n-1} points out of q^n , a constant fraction. Over $\mathbb{R}$ it is not: $Z(P)$ has measure zero, so containing K in it is an enormous restriction that no interesting set satisfies — and correspondingly the conclusion is worthless.

The genuine problem is the third column. A Besicovitch set is not contained in a variety; it is a set of small *measure* whose δ -neighbourhood one must understand. Polynomials interact with measure only through partitioning — Theorem 11.2.5 — and that is why this volume’s useful chapters are the ones that partition rather than vanish.

The transferable diagnostic. Before applying the polynomial method, ask what “ P vanishes on S ” costs S . If S is finite and the ambient set is finite, it is a real constraint. If the ambient is a continuum, it is a constraint so severe that only degenerate S satisfy it, and one must partition instead.

11.8.3 Historical milestone: Katz–Zahl

Before the eventual three-dimensional Kakeya theorem developed in Volume 14, Katz and Zahl proved that every Besicovitch set in $\mathbb{R}^3$ has Hausdorff dimension at least $\frac{5}{2} + \varepsilon_0$ for an absolute $\varepsilon_0 > 0$. Their work introduced polynomial-partitioning and tangency mechanisms that are historically important for the later theory.

Under the HA-03 strict policy this result is retained as historical context, not as a numbered theorem: its proof is not needed by any argument in Volume 11, and Volume 14 develops the stronger three-dimensional theory.

Where we are. The Katz–Zahl milestone is therefore not an omitted theorem in the active proof ledger. No numbered result in this volume depends on it.

WORKED EXAMPLE 11.8.2— TIER B **Tangency: the recurring obstruction**

Three times now an argument has failed at the same place, and the coincidence is not one.

Volume 9, Theorem 9.5.5. Wolff’s hairbrush works for transverse tubes and degrades exactly to the bush bound for near-parallel ones. Near-parallel tubes are tangent to one another.

Theorem 11.5.7. Multilinear Kakeya assumes transversality outright. The linear statement, which permits tangency, is the open conjecture.

the Katz–Zahl historical milestone. Polynomial partitioning handles tubes crossing cells and struggles with tubes confined near the partitioning surface — that is, tangent to it.

The common structure. An estimate that counts *crossings* is stable and easy; an estimate that must handle *tangency* is neither. Volume 10, Worked Example 10.7.3 classified statements as incidence-type (transferable) or exact-algebraic (not); tangency is where the classification bites, because a tangential intersection is exactly one that a small perturbation destroys.

What Volume 14 will report. Wang and Zahl’s solution in $\mathbb{R}^3$ proceeds by analysing precisely the tangential configurations — the sticky ones — and showing they cannot occur in the required abundance. So the obstruction identified three times independently in this series is the one their proof attacks, which is some confirmation that the diagnosis was right.

WORKED EXAMPLE 11.8.3— TIER C: RESEARCH TASTE **What a polynomial can and cannot see**

Eleven volumes in, it is worth collecting what has been established about the reach of algebraic methods in this subject.

(a) *A polynomial sees finite sets well.* Vanishing on N points costs degree $N^{1/n}$, which is cheap; that is the content of every dimension count in Volumes 10 and 11.

(b) *A polynomial sees exact lines well.* Vanishing at more than D points of a line propagates to the whole line. Joints, Szemerédi–Trotter and Theorem 11.8.1 all rest on this.

(c) *A polynomial does not see tubes.* Theorem 10.7.2: it can thread between the points of a δ -net inside a tube. This is the single obstruction the method has never overcome.

(d) *A polynomial sees measure only through partitioning.* $Z(P)$ has measure zero (Theorem 11.8.2), so no statement of the form “the configuration lies in $Z(P)$ ” constrains a set of positive measure. The only interaction with measure is Theorem 11.2.5, which uses the *complement* of the zero set.

(e) *The synthesis.* The polynomial method is a tool for discrete configurations of exact algebraic objects. Kakeya is a problem about continuous configurations of thickened objects. The method reaches it only through partitioning, and only as far as tangency permits — which is the diagnosis Worked Example 11.8.2 arrived at from three independent directions.

(f) *What that predicts.* A future solution of Kakeya in $\mathbb{R}^n$ for $n \geq 4$ will either dissolve the tangency problem or avoid polynomials altogether. Volume 14 reports that Wang and Zahl did the former in $\mathbb{R}^3$, and that their method’s dependence on three-dimensional geometry is exactly what blocks its extension.

WORKED EXAMPLE 11.8.4— TIER B **Running Dvir’s argument in three settings, explicitly**

Worked Example 11.8.1 tabulated the comparison; here is the computation in each column, so the differences are visible rather than asserted.

Column 1: $\mathbb{F}_q^n$. K a Kakeya set with $|K| < \binom{q+n-1}{n}$. Step 1 gives $P \neq 0$ of degree $D \leq q-1$ vanishing on K . Step 2: on a line $\ell_{a,b} \subseteq K$ the polynomial $t \mapsto P(a+tb)$ has degree $\leq D < q$ and q roots, so it vanishes and $P_D(b) = 0$. Step 3: P_D vanishes on all $b \neq 0$, hence on $\mathbb{F}_q^n$; by Theorem 10.1.3 with $D < q$ this forces $P_D = 0$. *Contradiction obtained; the bound is sharp.*

Column 2: $\mathbb{R}^n$, exact segments. $K \subseteq Z(P)$, $\deg P = D$. Step 2: on a segment,

$t \mapsto P(a + tv)$ has infinitely many roots, so it vanishes — *no degree condition needed*. Step 3: P_D vanishes on $\mathbb{S}^{n-1}$, hence by homogeneity on $\mathbb{R}^n$, hence $P_D = 0$ — *no Schwartz–Zippel needed*. *Contradiction obtained; but see below*.
 Column 3: $\mathbb{R}^n$, δ -tubes. S a δ -net inside K_δ , P vanishing on S with $\deg P = D < \delta^{-1}$ by Theorem 10.7.1. Step 2 requires the points of S inside a tube to be collinear. *They are not*, and Theorem 10.7.2 exhibits a P vanishing on δ^{-1} of them and nowhere on the axis. *No contradiction; the argument stops*.
The verdict on column 2. The contradiction is real but the hypothesis $K \subseteq Z(P)$ is vacuous: by Theorem 11.8.2 a variety has measure zero and dimension $\leq n - 1$, and no Besicovitch set was ever going to lie in one. So the easiest of the three arguments proves the least.
The moral, stated once. An argument’s strength is not its internal difficulty but the strength of the hypothesis it refutes. Column 1 refutes “ $|K|$ small”, which is the conjecture. Column 2 refutes “ K algebraic”, which nothing satisfies. Column 3 would refute the right thing and does not work.

WORKED EXAMPLE 11.8.5— TIER A **Three hypotheses, three strengths**

Worked Example 11.8.4 ran the same argument three times. Here is the comparison reduced to what each hypothesis actually excludes.

Hypothesis refuted	How restrictive	Verdict
$ K < \binom{q+n-1}{n}$ over $\mathbb{F}_q$	a constant fraction of $\mathbb{F}_q^n$ — sets of this size are abundant	<i>the conjecture</i> ; sharp
$K \subseteq Z(P)$ over $\mathbb{R}$	measure zero, dimension $\leq n - 1$, and algebraic — essentially nothing satisfies it	true but vacuous
$m(K_\delta)$ small over $\mathbb{R}$	the actual object of study	would be the conjecture; <i>argument fails</i>

The point. All three arguments have the same three steps. Their value is determined entirely by the middle column — how much the refuted hypothesis excludes — and not at all by the difficulty of the argument.
Why row two is worth stating. Because it is the natural first attempt on being told Dvir’s theorem, and it succeeds. A reader who tries it and obtains a proof might reasonably believe progress has been made. Theorem 11.8.2 is the antidote, and stating both the theorem and its emptiness in the same chapter is the honest presentation.

The generalisable habit. On proving something, ask what the hypothesis excludes before asking whether the proof is correct. A correct proof of a vacuous statement is a common outcome and an easy one to miss.

Drills

Drill 11.8.1. Verify Step 2 of Theorem 11.8.1: the coefficient of t^D in $P(a + tv)$ is $P_D(v)$.

Drill 11.8.2. Explain why Step 1 needs no degree hypothesis over $\mathbb{R}$, and what the corresponding hypothesis is over $\mathbb{F}_q$.

Drill 11.8.3. Show that a polynomial vanishing on an open subset of $\mathbb{R}^n$ is identically zero, and contrast with $x^q - x$ over $\mathbb{F}_q$.

Drill 11.8.4. Using Worked Example 11.8.1, state which column each of the following belongs to: Dvir’s theorem, the joints theorem, and Theorem 10.7.2.

Where we are. The chapter’s structure is unusual and deliberate: it proves a theorem and then argues that the theorem is worthless.

Theorem 11.8.1 is correct, short, and is the exact real analogue of Dvir’s argument — indeed easier, since neither the degree condition nor Schwartz–Zippel is needed. A reader meeting it without Theorem 11.8.2 could reasonably conclude that the polynomial method reaches the Kakeya problem after all.

It does not, and the reason is that the hypothesis $K \subseteq Z(P)$ is satisfied by essentially nothing. Worked Example 11.8.5 tabulates the three versions by how much each hypothesis excludes, which is the only measure that matters.

Including both halves is the point. A treatment that stated only the theorem would mislead; one that stated only the obstruction would leave the reader wondering whether the obvious attempt had been tried.

Notes

Theorem 11.8.1 is folklore and is sometimes offered as evidence that the polynomial method should apply to Kakeya. The Checkpoint after Theorem 11.8.2 explains why it is not.

Katz and Zahl’s bound is from 2019, improving a sequence of polynomial-method results beginning with Guth’s. Volume 14 reports the Wang–Zahl theorem, which settles the three-dimensional case.

Chapter 11.9

Assessment

WORKED EXAMPLE 11.9.1— TIER A **What the method proved, and what it only advanced**

Problem	Result	Status	Here
Szemerédi–Trotter	$(PL)^{2/3} + P + L$	sharp	proved, Theorem 11.3.2
Joints in $\mathbb{R}^3$	$\lesssim L^{3/2}$	sharp	proved, Theorem 11.4.2
Loomis–Whitney	product bound	sharp	proved, Theorem 11.5.2
Line incidences in $\mathbb{R}^3$	$\lesssim L^{3/2}$	sharp	internally proved, Theorem 11.6.11
Distinct distances	$N/\log N$	within $\sqrt{\log N}$	reduction proved, Chapter 11.7
Multilinear Kakeya	S^ε loss	sharp in order	<i>proved</i> , Theorem 11.5.4
Kakeya in $\mathbb{R}^3$	historical Katz–Zahl milestone	superseded	contextual only
Kakeya, $n \geq 4$	—	<i>open</i>	—

The pattern. Where the objects are genuine lines — joints, incidences, distances — the method is sharp and often settles the problem outright. Where the objects are tubes of positive width, it advances the exponent and does not close the problem.

The four sharp results all concern lines. The two open ones concern tubes. That is the same division Volume 10 identified between the finite field and real settings, appearing again inside a single method.

WORKED EXAMPLE 11.9.2— TIER B The method in five steps

Every argument in this volume has the same skeleton, and it is worth having in a form one can apply.

Step 1: choose a degree D . Leave it a parameter; it is fixed at the end by optimisation (Worked Example 11.2.2).

Step 2: find a polynomial. Either one *vanishing* on the configuration — via the dimension count $\binom{D+n}{n} > |S|$ — or one *partitioning* space, via Theorem 11.2.5. The first is available only when the configuration is finite and the ambient discrete; the second always.

Step 3: split into cellular and algebraic. Points in open cells are handled by the small cell size; points on $Z(P)$ are not handled at all and need a separate argument.

Step 4: exploit the incidence bound. A line meets $Z(P)$ in at most D points, so it visits at most $D+1$ cells. This is the one fact that transfers from Volume 10 and it is used in every proof.

Step 5: optimise D . Balance the decreasing cellular contribution against the increasing algebraic one, and check $D \geq 1$ is admissible in the regime considered — the check Worked Example 11.3.2 identifies as the commonest omission.

Where each proof sits. Theorem 11.3.2 uses all five, with a trivial Step 3 algebraic case. Theorem 11.4.2 uses Step 2 in its vanishing form and replaces Steps 3–5 by the derivative trick. Theorem 11.6.7 uses all five with a Step 3 that is a research programme.

WORKED EXAMPLE 11.9.3— TIER C: RESEARCH TASTE Volume 11 in the arc of the series

(a) *What it added.* A genuinely new technique, and the first since Volume 6 that is not analytic. Polynomial partitioning does what neither the geometry of Volume 9 nor the arithmetic of Volume 10 could: it subdivides space adaptively, so that a configuration is handled at the scale where it is simplest.

(b) *Why it escaped the earlier ceilings.* Volume 9's methods optimised over an unattainable multiplicity (Worked Example 9.8.2); Volume 10's arithmetic route optimised over additive structure with lossy constants (Worked Example 10.6.2). Partitioning optimises over the *degree* D , and D is a genuine free parameter whose extremal value is attained — there is nothing hypothetical about a polynomial of the chosen degree.

(c) *What it did not overcome.* Tangency. Worked Example 11.8.2 traced the same obstruction through Volumes 9, 11 and the multilinear theory:

estimates counting crossings are stable, estimates handling tangency are not. The Kakeya conjecture is a tangency problem, and partitioning does not dissolve it.

(d) *The three imports.* This volume quotes more than any other: the formerly imported ham-sandwich, multilinear-Kakeya, and 3D-incidence inputs (Katz–Zahl). The first three are load-bearing — the manifest records them as such — and that is a real cost, unavoidable if a volume is to reach the frontier of a technique whose central results are recent papers.

(e) *What remains.* Volume 12 uses Theorem 11.5.7 to prove restriction estimates by the broad–narrow decomposition; Volume 13 develops decoupling; Volume 14 reports the position, including the resolution of three-dimensional Kakeya. The conjecture is open for $n \geq 4$, and nothing in eleven volumes suggests it is close.

WORKED EXAMPLE 11.9.4— TIER B **Placing a polynomial-method paper**

The literature using these techniques is now large. Five questions place almost any paper.

(1) *Vanishing or partitioning?* Worked Example 11.2.3. A vanishing argument concerns exact algebraic objects and usually gives a sharp result in a restricted setting; a partitioning argument concerns continuous configurations and usually gives an exponent.

(2) *What is the choice of degree?* Worked Example 11.2.2. It is displayed in one line and determines the result. If the paper does not display it, the argument is probably not a partitioning one.

(3) *How is the algebraic case handled?* This is where the difficulty lives — trivially in $\mathbb{R}^2$ (Chapter 11.3), by ruled-surface theory in $\mathbb{R}^3$ (Worked Example 11.6.3), and openly in higher dimensions. A paper claiming a new bound in $\mathbb{R}^4$ or above must say something new here.

(4) *Lines or tubes?* Worked Example 11.9.1: sharp results concern lines, exponents concern tubes. A claimed sharp result about tubes deserves scrutiny.

(5) *Is tangency addressed?* Worked Example 11.8.2. Estimates counting crossings are routine; the content of any advance is in the tangential configurations.

Worked placement: Guth–Katz distinct distances. (1) partitioning, twice; (2) two degrees, chosen in sequence; (3) ruled surfaces, and this is the paper’s core; (4) lines, hence sharp up to log; (5) not applicable, the objects being exact. Conclusion: a complete solution modulo a logarithm, with the weight

in the algebraic case — which matches Worked Example 11.7.4’s anatomy.

WORKED EXAMPLE 11.9.5— TIER A **Eleven volumes: the methods and their ceilings**

Method	Vol	Optimises over	Ceiling, and why
Fourier / energy	7–8	nothing; sees pairs	$\frac{n+1}{2}$: two-point
L^2 counting	9	nothing; sees pairs	2: two-point
bush	9	multiplicity μ	$\frac{n+1}{2}$: μ unattainable
hairbrush	9	multiplicity μ	$\frac{n+2}{2}$: same, plus tangency
arithmetic	10	additive structure	$(2 - \sqrt{2})n$: lossy conversions
Dvir	10	<i>nothing</i>	sharp, but only over $\mathbb{F}_q$
partitioning	11	degree D	tangency

Reading the third column. A method’s ceiling is set by what it optimises over, and whether the balancing value is attained — the test of Volume 9, Worked Example 9.8.4. Dvir’s argument has no free parameter and is sharp. Partitioning optimises over D , which *is* attainable, and its ceiling comes from elsewhere: the algebraic case, and within it, tangency.

The one row that is sharp over $\mathbb{R}$. None. Every real-variable method in eleven volumes has a ceiling below n , and Worked Example 11.8.2 identifies the same obstruction in three of them.

What the table predicts. A method that neither optimises over an unattainable parameter nor reduces to pair correlations, and that handles tangency, would settle the conjecture. Volume 14 reports that Wang and Zahl found one in $\mathbb{R}^3$, and that its dependence on three-dimensional geometry is what prevents it from settling $n \geq 4$.

WORKED EXAMPLE 11.9.6— TIER C: RESEARCH TASTE **Three imports: an accounting**

This volume quotes more than any other in the series, and the reader is entitled to a reckoning.

(a) *What is imported.* No load-bearing theorem in this volume is now imported. Earlier volumes supply background results already proved there. Sharper historical Guth–Katz and Katz–Zahl statements are contextual only.

(b) *What is proved here.* Tucker/Borsuk–Ulam/ham sandwich, sign-history

polynomial partitioning, multilinear Kakeya, the ε -loss 3D line-incidence theorem, and the Elekes–Sharir distinct-distance chain are all internal.

(c) *Why this matters.* Volume 12 may therefore treat multilinear Kakeya and polynomial partitioning as certified foundations rather than as literature black boxes.

(d) *Is the volume self-contained?* Under the HA-03 strict policy, no known load-bearing proof gap remains in Volume 11. Final project-wide certification is still deferred until the later strict-zone volumes are closed.

(e) *The alternative was worse.* Proving Borsuk–Ulam would require a chapter of algebraic topology used once; proving the Guth–Katz incidence theorem would require ruled-surface theory. Either would double the volume to establish results the reader can find in Guth’s book. Stating precisely what is granted, and proving the surrounding structure, seemed the better trade — and it is the same judgement made for R5 and R10 in earlier volumes, differing only in that here the omissions are load-bearing.

(f) *The one that is not like the others.* Volume 3’s configuration lemma was also used before it was proved, but it was a *debt*: Volume 9 discharged it, and the manifest tracked it as outstanding until then. These four are not debts; no volume will discharge them, and the manifest records them separately.

Drills

Drill 11.9.1. For each row of Worked Example 11.9.1, say whether the objects are lines or tubes, and check the correlation with sharpness.

Drill 11.9.2. Apply the five steps of Worked Example 11.9.2 to Theorem 11.3.2 and identify each in the proof.

Drill 11.9.3. Explain why the degree D is an attainable optimum whereas the multiplicity μ of Volume 9 is not.

Drill 11.9.4. List the four scope decisions of this volume and state, for each, what depends on it.

Where we are. Volume 11 in one line: polynomial partitioning settles problems about *lines* and moves the exponent on problems about *tubes*, and the difference is tangency.

Where we are. Eleven volumes, 120 chapters, and the Kakeya conjecture remains open above dimension three. What has changed is that the obstruction now has a name – tangency – and three independent methods have been shown to stop at it.

Notes

Guth’s *Polynomial Methods in Combinatorics* is the standard reference for this material and covers everything in this volume, including the results quoted here.

Chapter 11.A

Dependencies and Historical Context

Results used from Volumes 1–10

Result	Source	Role
The Kakeya conjecture	Theorem 7.1.11	context in Chapter 11.8
Dvir’s theorem	Theorem 10.2.1	comparison with vanishing methods
Schwartz–Zippel	Theorem 10.1.3	algebraic contrast
Failure of real propagation	Theorem 10.7.2	motivates partitioning
Hardy–Littlewood and earlier analytic tools	earlier volumes	reader prerequisites

Strict-zone ledger

Item	Current status	Comment
R11, polynomial ham sandwich	INTERNAL-PROVED	Tucker/Borsuk–Ulam and arbitrary-measure ham sandwich supplied locally
R12, multilinear Kakeya	INTERNAL-PROVED	retained as first-class mathematics
R13, ε -loss 3D line incidences	INTERNAL-PROVED	uses sign-history regions and local resultant/projection lemmas
Former R14, Katz–Zahl milestone	historical context only	no numbered theorem and no dependency

No load-bearing theorem in Volume 11 is intentionally omitted. The sharper historical Guth–Katz and Katz–Zahl statements are discussed to locate the internal results in the literature, but they are not assumptions in the proof graph.

Chapter 11.B

Notation and Index of Results

Additions to the Volume 1–10 conventions

$Z(P)$	real zero set of P
cell	connected component of $\mathbb{R}^n \setminus Z(P)$
partitioning region	a sign-history region in Theorem 11.2.5
D	polynomial degree parameter
$I(\mathcal{P}, \mathcal{L})$	point–line incidence count
joint	point on three lines with spanning directions
ν -transverse	determinant bounded below by ν
Ψ	Elekes–Sharir parametrisation

Main results

Tucker’s lemma	Theorem 11.1.5
Borsuk–Ulam	Theorem 11.1.6
Ham sandwich, arbitrary finite measures	Theorem 11.1.8
Polynomial ham sandwich	Theorem 11.2.2
Controlled partitioning regions	Theorems 11.2.5 and 11.2.6
Szemerédi–Trotter	Theorem 11.3.2
Joints	Theorem 11.4.2
Loomis–Whitney	Theorem 11.5.2
Multilinear Kakeya	Theorems 11.5.4 and 11.5.7
Local Bézout/resultant bounds	Theorems 11.6.4 and 11.6.5
ε -loss 3D line incidences	Theorem 11.6.7
Elekes–Sharir non-clustering	Theorem 11.7.8
Distinct distances, internal form	Theorem 11.7.9

Plan revisions

R11, R12, and R13 are internally proved. The former R14 Katz–Zahl statement is no longer a numbered theorem; it is historical context superseded by Volume 14.

Part 12

Volume 12

Multilinear and Bilinear Restriction

Preface to Volume 12

Where the restriction problem stands

Volume 6 posed the restriction conjecture and proved the Stein–Tomas theorem, which settles the range $p \geq \frac{2(n+1)}{n-1}$ and no further. It also proved that restriction implies Kakeya (Theorem 6.6.2), so the conjecture is at least as hard as everything Volumes 9 to 11 failed to settle.

The advance of the last twenty years came from a reformulation. Instead of bounding $\mathcal{E}g$ for a single g on the whole sphere, bound a *product* $\prod_j \mathcal{E}g_j$ where the g_j live on caps whose normals are transverse. That hypothesis excludes the configurations where the difficulty lives — exactly as Worked Example 11.5.4 recorded for multilinear Kakeya — and in exchange the multilinear estimate can be proved sharply. Bourgain and Guth then showed how to recover a genuine linear estimate from it, at a cost.

What this volume does

It proves the multilinear restriction theorem of Bennett, Carbery and Tao from the multilinear Kakeya inequality of Volume 11; develops the broad–narrow decomposition that converts multilinear estimates into linear ones; computes the restriction exponents that result and compares them with Stein–Tomas; and then turns to the bilinear theory and to restriction against a fractal measure.

That last topic is not a digression. Volume 8, Worked Example 8.8.1, identified the missing ingredient in the Falconer distance set problem as a bound on the spherical average $\sigma(\mu, r)$, and observed that the required estimate is a *weighted restriction estimate with a fractal weight*. That is the subject of Chapter 12.7, and Chapter 12.8 discharges the promise: it identifies $\sigma(\mu, r)$ as a restriction quantity and shows how a bilinear estimate improves the Falconer threshold.

The one dictionary

If the volume has one idea it is the wave packet dictionary, and it is worth stating before starting.

An extension $\mathcal{E}g$ is an oscillatory integral and is hard to bound pointwise. Decomposed into wave packets (Theorem 12.1.4), it becomes a weighted sum of indicator functions of $R^{1/2} \times \cdots \times R$ tubes, whose directions are $R^{-1/2}$ -separated. So an analytic question about $\mathcal{E}$ becomes a geometric question about how such tubes overlap — which is the Kakeya problem at scale $\delta = R^{-1/2}$.

Every chapter is written in that dictionary. Multilinear restriction is multilinear Kakeya translated (Theorem 12.3.2); the broad–narrow dichotomy is a statement about which tube directions carry the mass; the weighted estimates of Chapter 12.7 ask how a fractal measure meets a tube; and Theorem 12.8.2 shows the distance set problem speaks the same language.

Two cautions carried throughout. The dictionary is not exact — packets have tails, and Worked Example 12.1.3 explains when they may be discarded. And the translation is lossy in one direction: Theorem 12.3.3 shows that passing from packets back to an L^p estimate can cost a power of R unless the almost-orthogonality is retained, a point Worked Example 12.3.5 turns into an audit.

A note on the plan

The original plan for this volume included chapters on Loomis–Whitney and on multilinear Kakeya. Both were written into Volume 11 instead, where the polynomial method made them natural — Theorem 11.5.2 is proved there and Theorem 11.5.7 proved there in full. This volume therefore begins after them and quotes both.

What is assumed

Volumes 1–11. Most heavily: Volume 6 for the restriction problem and the TT^* identity; Volume 11 for multilinear Kakeya; Volume 5 for the decay of $\widehat{d\sigma}$; Volume 8 for the distance set framework.

The plan

1 Wave packets the decomposition everything rests on

Chapters 2–8, planned: transversality and the multilinear setting; multilinear restriction from multilinear Kakeya; the broad–narrow decomposition; the

resulting exponents; bilinear restriction; restriction against a fractal measure; and spherical averages and the distance set.

Chapter 12.1

Wave Packets

Every argument in this volume and the next decomposes an extension operator into pieces, each of which is essentially a smooth bump on a tube. This chapter constructs the decomposition and proves the two properties that make it useful: the pieces are almost orthogonal, and each is almost constant on its tube.

12.1.1 Caps and their dual tubes

Fix $R \geq 1$ and write $\delta = R^{-1/2}$.

Definition 12.1.1. Let $\{\theta\}$ be a covering of $\mathbb{S}^{n-1}$ by caps of radius δ with bounded overlap, and for each θ let ω_θ be its centre. There are $\approx_n \delta^{-(n-1)} = R^{(n-1)/2}$ such caps.

Lemma 12.1.2 (The dual box). *A cap θ of radius δ on $\mathbb{S}^{n-1}$ is contained in a rectangular box of dimensions*

$$\underbrace{\delta \times \cdots \times \delta}_{n-1} \times \delta^2$$

centred at ω_θ , with the short side in the direction ω_θ . Its dual box — the polar set, up to constants — has dimensions

$$\begin{aligned} & \underbrace{\delta^{-1} \times \cdots \times \delta^{-1}}_{n-1} \times \delta^{-2} \\ &= \underbrace{R^{1/2} \times \cdots \times R^{1/2}}_{n-1} \times R, \end{aligned}$$

with the long side in the direction ω_θ .

Proof. Take coordinates with $\omega_\theta = e_n$. A point $\omega \in \theta$ is $\omega = (\omega', \omega_n)$ with $|\omega'| \leq \delta$ and $\omega_n = \sqrt{1 - |\omega'|^2}$, so

$$1 - \omega_n = 1 - \sqrt{1 - |\omega'|^2} \leq |\omega'|^2 \leq \delta^2,$$

using $1 - \sqrt{1-t} \leq t$ for $t \in [0, 1]$. So θ lies in the box $\{|\omega'| \leq \delta, |\omega_n - 1| \leq \delta^2\}$, as claimed. The dual of a box is the box with reciprocal side lengths, by the definition of the polar set. $\square$

Where we are. The quadratic gap δ^2 in the normal direction is the *curvature* of the sphere, and it is the only place curvature enters this chapter. A flat piece of hypersurface would have a cap contained in a slab of thickness 0 in the normal direction, dual box infinite in that direction, and no tubes. Numerically, $1 - \cos \delta$ against $\delta^2/2$ agrees to four significant figures already at $\delta = 0.05$, so the estimate above is not merely an upper bound but the truth up to a factor 2.

The consequence to remember: *a δ -cap is dual to a $\delta^{-1} \times \cdots \times \delta^{-2}$ tube*, and with $\delta = R^{-1/2}$ this is an $R^{1/2} \times \cdots \times R$ tube. Those dimensions recur on every page of Volumes 12 and 13.

Definition 12.1.3 (Tubes). For each cap θ let $\mathbb{T}_\theta$ be a covering of B_R by translates of the dual box of Theorem 12.1.2, with bounded overlap. Since the box has volume $\approx R^{(n-1)/2} \cdot R$ and $m(B_R) \approx R^n$,

$$\#\mathbb{T}_\theta \approx_n \frac{R^n}{R^{(n-1)/2} R} = R^{(n-1)/2}.$$

The pairs (θ, T) with $T \in \mathbb{T}_\theta$ are the *wave packets*, and there are $\approx_n R^{n-1}$ of them.

12.1.2 The decomposition

Theorem 12.1.4 (Wave packet decomposition). *Let $g \in L^2(\mathbb{S}^{n-1})$ and $R \geq 1$. There is a decomposition*

$$\mathcal{E}g = \sum_{\theta} \sum_{T \in \mathbb{T}_\theta} \mathcal{E}g_{\theta,T}$$

such that:

(i) (Frequency localisation) $g_{\theta,T}$ is supported in θ , and $\sum_T g_{\theta,T} = g_\theta$, where g_θ is g restricted to θ by a smooth partition of unity;

(ii) (Spatial localisation) for every N ,

$$|\mathcal{E}g_{\theta,T}(x)| \lesssim_N \|g_{\theta,T}\|_{L^1(d\sigma)} \left(1 + \frac{\text{dist}(x, T)}{R^{1/2}}\right)^{-N} \quad (x \in \mathbb{R}^n);$$

(iii) (L^2 orthogonality)

$$\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx_n \sum_{\theta} \sum_{T \in \mathbb{T}_\theta} \|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2.$$

Idea of the proof. Cut g into caps by a partition of unity — that is (i). On each cap, the extension $\mathcal{E}g_\theta$ has Fourier support in a box of dimensions $\delta \times \cdots \times \delta^2$, so by the uncertainty principle it cannot oscillate on scales finer than the dual box. Cutting physical space into translates of that dual box, and using a partition of unity there too, gives pieces each concentrated on one tube — that is (ii), with the tails coming from the smoothness of the cutoffs.

Orthogonality (iii) has two sources. Different caps give disjoint frequency supports, so Plancherel applies. Different tubes with the same cap are disjoint in *space*, and the tails are summable.

Proof. Step 1. Cut the sphere.

Let $\{\psi_\theta\}$ be a smooth partition of unity subordinate to $\{\theta\}$ with $|\partial^\alpha \psi_\theta| \lesssim_\alpha \delta^{-|\alpha|}$, and put $g_\theta = g\psi_\theta$. Then $\sum_\theta g_\theta = g$ and each g_θ is supported in θ .

Step 2. *Cut physical space.*

Fix θ and let $\{\eta_T\}_{T \in \mathbb{T}_\theta}$ be a smooth partition of unity on $\mathbb{R}^n$ subordinate to the tubes of $\mathbb{T}_\theta$, with derivatives adapted to the tube's dimensions. Define $g_{\theta,T}$ by

$$\mathcal{E}g_{\theta,T} = \eta_T \cdot \mathcal{E}g_\theta \quad \text{followed by a Fourier restriction back to } \theta,$$

that is, $g_{\theta,T}$ is the function on θ whose extension best approximates $\eta_T \mathcal{E}g_\theta$. Concretely, $\widehat{\eta_T}$ is concentrated on the *dual* of T , which is a box of dimensions $\delta \times \cdots \times \delta^2$ — comparable to θ itself by Theorem 12.1.2 — so multiplying by η_T smears the frequency support of $\mathcal{E}g_\theta$ by an amount comparable to the cap. Re-restricting to θ therefore loses only rapidly decaying tails, and $\sum_T g_{\theta,T} = g_\theta$ up to such tails, which is (i).

Step 3. *Spatial localisation.*

Since η_T is supported on T and its derivatives are adapted to T 's dimensions, $\widehat{\eta_T}$ decays rapidly outside the dual box. For x at distance d from T , integrating by parts N times in the directions transverse to T — where T has width $R^{1/2}$ — gives

$$|\mathcal{E}g_{\theta,T}(x)| \lesssim_N \|g_{\theta,T}\|_{L^1(d\sigma)} (1 + d/R^{1/2})^{-N},$$

the prefactor because $|\mathcal{E}h| \leq \|h\|_{L^1(d\sigma)}$ pointwise. This is (ii).

Step 4. *Orthogonality across caps.*

The functions $\mathcal{E}g_\theta$ have Fourier transforms supported in the caps θ , which have bounded overlap. Hence by Plancherel,

$$\|\mathcal{E}g\|_{L^2(\mathbb{R}^n)}^2 \approx_n \sum_\theta \|\mathcal{E}g_\theta\|_{L^2(\mathbb{R}^n)}^2.$$

Step 5. *Orthogonality across tubes.*

Fix θ . The tubes of $\mathbb{T}_\theta$ have bounded overlap, so $\sum_T \eta_T^2 \approx 1$ and

$$\|\mathcal{E}g_\theta\|_{L^2(B_R)}^2 \approx \sum_T \|\eta_T \mathcal{E}g_\theta\|_{L^2(B_R)}^2 \approx \sum_T \|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2,$$

the last step by Step 2 and the rapid decay of the tails from Step 3, which contribute a convergent geometric series. Combining with Step 4 gives (iii). $\square$

Where we are. Three warnings about Theorem 12.1.4, all of which matter in later chapters.

The decomposition is not exact. Step 2 loses rapidly decaying tails, and every application must carry them. In practice one fixes N large depending on the exponents involved and absorbs the tails at the end; the standard phrase “by rapid decay we may assume the packets are disjointly supported” is shorthand for this, and it is legitimate only when the number of packets is polynomial in R , which Theorem 12.1.3 guarantees.

The L^2 orthogonality is special to $p = 2$. For $p \neq 2$ there is no such identity, and the whole difficulty of the restriction problem is that one must control $\|\mathcal{E}g\|_{L^p}$ with only the L^2 information above plus geometry. Chapter 12.4 is about extracting that geometry.

The tubes point in the direction of the cap’s centre. By Theorem 12.1.2 the long axis of the dual box is ω_θ — so wave packets from different caps point in different directions, and two packets whose caps are ν -separated have tubes at angle $\approx \nu$. That is the link to Kakeya, and it is why multilinear Kakeya is the input to multilinear restriction.

Proposition 12.1.5 (A single packet, computed). *Let θ be a cap of radius $\delta = R^{-1/2}$ and let $g = \mathbf{1}_\theta$. Then*

$$\begin{aligned} |\mathcal{E}g(x)| &\approx d\sigma(\theta) \approx \delta^{n-1} \\ &= R^{-\frac{n-1}{2}} \quad \text{for } x \text{ in the dual tube } T, \end{aligned}$$

and $|\mathcal{E}g(x)|$ decays rapidly outside T . Moreover

$$\begin{aligned} \|\mathcal{E}g\|_{L^2(B_R)}^2 &\approx |T| \delta^{2(n-1)} \approx R^{\frac{n+1}{2}} R^{-(n-1)} \\ &= R^{\frac{3-n}{2}}, \end{aligned}$$

consistent with $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2 = R \delta^{n-1} = R^{\frac{3-n}{2}}$.

Proof. For $x \in T$ the phase $e^{2\pi i x \cdot \omega}$ varies by $O(1)$ as ω ranges over θ : by Theorem 12.1.2 the cap has extent δ transversally and δ^2 normally, while T

has extent δ^{-1} and δ^{-2} in the dual directions, so each product is $O(1)$. Hence the integral $\int_{\theta} e^{2\pi i x \cdot \omega} d\sigma(\omega)$ has no cancellation and is $\approx d\sigma(\theta) \approx \delta^{n-1}$.

Outside T the phase oscillates and repeated integration by parts gives rapid decay, as in Theorem 12.1.4(ii).

For the L^2 computation, $|\mathcal{E}g|^2 \approx \delta^{2(n-1)}$ on T and negligible elsewhere, and $|T| \approx R^{(n+1)/2}$ by Theorem 12.1.3. The consistency check uses $\|g\|_{L^2(d\sigma)}^2 = d\sigma(\theta) \approx \delta^{n-1} = R^{-(n-1)/2}$ and $R \cdot R^{-(n-1)/2} = R^{(3-n)/2}$. ✓ $\square$

Where we are. Theorem 12.1.5 is the Knapp example seen through the wave packet decomposition, and the consistency check at the end is the one Worked Example 12.1.1 sets as a drill. That both routes give $R^{(3-n)/2}$ is a genuine check on the normalisations, and it is worth performing whenever a new scale is introduced — most errors in this area are factors of R to a power, and they show up here first.

WORKED EXAMPLE 12.1.1— TIER A **The dimensions, counted**

Fix n and R , and put $\delta = R^{-1/2}$.

Caps. $\approx \delta^{-(n-1)} = R^{(n-1)/2}$.

Tubes per cap. A tube has volume $R^{(n-1)/2} \cdot R$, and B_R has volume R^n , so $\#\mathbb{T}_{\theta} \approx R^n / (R^{(n-1)/2} R) = R^{(n-1)/2}$.

Total packets. $R^{(n-1)/2} \cdot R^{(n-1)/2} = R^{n-1}$.

Numerically.

n	R	caps	tubes/cap	packets
2	10^4	10^2	10^2	10^4
3	10^4	10^4	10^4	10^8
3	10^6	10^6	10^6	10^{12}
4	10^6	10^9	10^9	10^{18}

The coincidence worth noticing. The number of caps equals the number of tubes per cap, both $R^{(n-1)/2}$. That is not an accident: it says the family of wave packets is *self-dual* under exchanging position and direction — the same duality Volume 9, Theorem 9.6.4 exploited for Keakeya. It is why the Keakeya and restriction problems have the same numerology.

A sanity check. A single wave packet has $\|\mathcal{E}g_{\theta,T}\|_{L^2(B_R)}^2 \approx |T| \cdot \|g_{\theta,T}\|_{L^\infty}^2 \delta^{2(n-1)}$, and summing over the R^{n-1} packets must reproduce $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2$ — the last by Theorem 6.1.4 in Volume 6. Verifying that the powers of R match is Drill 1.4.

WORKED EXAMPLE 12.1.2— TIER B Why the packets are the right objects

The decomposition is used so uniformly that it is worth saying what it buys.

(a) *It converts an oscillatory integral into a sum of bumps.* $\mathcal{E}g$ is an integral with a rapidly oscillating kernel and is hard to bound pointwise. Each $\mathcal{E}g_{\theta,T}$ is, by (ii), essentially a constant times $\mathbf{1}_T$. So $|\mathcal{E}g|$ becomes a weighted sum of indicator functions of tubes, and bounding it becomes a *counting* problem about tubes.

(b) *The counting problem is Kakeya.* The tubes have $R^{-1/2}$ -separated directions, one family per cap, and there are $R^{(n-1)/2}$ of them per direction. Rescaling by R turns them into δ -tubes with δ -separated directions — exactly Volume 9’s setting with $\delta = R^{-1/2}$.

(c) *L^2 information is free and everything else is not.* Property (iii) is an identity; there is no analogue for other exponents. So an argument gets one orthogonality relation and must supply the rest from the geometry of how tubes overlap.

(d) *What goes wrong without curvature.* Theorem 12.1.2 used the sphere’s curvature to get the factor δ^2 . For a flat piece the dual box is a slab, all packets point the same way, and no counting is possible — consistent with Volume 6’s observation that restriction fails for a hyperplane.

The summary. Wave packets are the dictionary between the analytic statement (an L^p bound for $\mathcal{E}$) and the geometric one (how much $R^{1/2} \times R$ tubes can overlap). Everything in Volumes 12 and 13 is written in that dictionary.

WORKED EXAMPLE 12.1.3— TIER C: RESEARCH TASTE The uncertainty principle, as used here

Theorem 12.1.2 is an instance of a principle invoked constantly in this volume and the next, and it is worth stating what the invocation actually means.

(a) *The formal statement.* If $\widehat{f}$ is supported in a box B , then f cannot vary appreciably on scales finer than the dual box B^* . Precisely, $f = f * \check{\psi}$ for any ψ equal to 1 on B , and $\check{\psi}$ is concentrated on B^* with rapidly decaying tails.

(b) *What it is not.* It is not an equality, and f need not be constant on translates of B^* . The correct statement is the convolution identity above, and every use must carry the tails — which is why Theorem 12.1.4(ii) is stated with a decay exponent N rather than as exact support.

(c) *Why the tails are harmless here.* There are $\lesssim R^{n-1}$ wave packets, so choosing N large enough that $R^{n-1} \cdot R^{-N/2}$ is negligible disposes of them. This works because the packet count is *polynomial* in R ; an argument producing

exponentially many pieces could not absorb the tails this way.

(d) *Where it would fail.* For a surface with a flat piece, the dual box is unbounded in the normal direction and there is no localisation at all. Volume 5, Worked Example 5.5.4 recorded that a hyperplane’s surface measure has a non-decaying transform; here that reappears as the absence of a wave packet decomposition.

(e) *The habit.* On meeting “by the uncertainty principle” in a paper, identify the box, its dual, and the tail exponent being assumed. Most arguments are correct; the ones that are not usually fail because the number of pieces was not polynomial, or because the surface was flatter than assumed.

WORKED EXAMPLE 12.1.4— TIER B **Two scales, and which is which**

Wave packet arguments involve R , $R^{1/2}$, $R^{-1/2}$ and occasionally R^{-1} , and confusing them is the commonest source of error. A fixed dictionary.

Scale	What lives there	Why
R	the ball B_R ; tube length	the scale at which the estimate is stated
$R^{1/2}$	tube width	dual to the cap radius, Theorem 12.1.2
$R^{-1/2}$	cap radius δ	chosen so the dual tube has length R
R^{-1}	cap thickness δ^2	the sphere’s curvature over a δ -cap

The relation that fixes everything. $\delta = R^{-1/2}$, forced by wanting the dual box to reach across B_R . Change it and every other entry changes.

Two counts that must match. Caps: $\delta^{-(n-1)} = R^{(n-1)/2}$. Tubes per cap: $R^n / (R^{(n-1)/2} \cdot R) = R^{(n-1)/2}$. Equal, as Worked Example 12.1.1 noted, and that equality is the self-duality of the packet family.

The check to run on any new argument. Compute $\|\mathcal{E}g\|_{L^2(B_R)}^2$ two ways — by Theorem 6.1.4 and by summing packets — and confirm they agree. Theorem 12.1.5 does this for a single cap and gets $R^{(3-n)/2}$ both ways. Errors of a power of R show up here immediately, and they are the errors that actually occur.

WORKED EXAMPLE 12.1.5— TIER B **Decomposing a specific function**

The wave packet decomposition is easier to trust after applying it once to something explicit.

The function. $n = 2$, $R = 10^4$, $\delta = 10^{-2}$, and $g = \mathbf{1}_{\theta_0}$ for a single cap θ_0 of radius δ centred at e_1 .

Step 1, cutting the circle. The partition of unity leaves $g_\theta = 0$ except for $\theta = \theta_0$ and its two neighbours, since g is supported in one cap. So the sum over caps has three terms, not 100.

Step 2, cutting space. $\mathbb{T}_{\theta_0}$ consists of $R^{(n-1)/2} = 100$ tubes of dimensions 100×10^4 tiling B_R , all parallel to e_1 . The packets $g_{\theta_0, T}$ are the pieces of g whose extensions live on each.

Step 3, the sizes. By Theorem 12.1.5, $|\mathcal{E}g| \approx \delta^{n-1} = 10^{-2}$ on the tube through the origin, with rapid decay elsewhere. And $\|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R^{(3-n)/2} = R^{1/2} = 100$.

The check. Directly: $|T| \cdot \delta^{2(n-1)} = (100 \times 10^4) \times 10^{-4} = 100$. ✓ Both routes agree, which is the consistency check Worked Example 12.1.4 recommends running on any new configuration.

What the example shows. That for data on a single cap the decomposition is trivial — one cap, and the tubes merely tile space. The content appears only when g is spread over many caps, so that packets from different directions interact. That interaction is the subject of every subsequent chapter.

Drills

Drill 12.1.1. Verify $1 - \sqrt{1-t} \leq t$ for $t \in [0, 1]$ and deduce Theorem 12.1.2.

Drill 12.1.2. Check that the dual box has its long side in the direction ω_θ , and say what this means geometrically.

Drill 12.1.3. Confirm $\#\mathbb{T}_\theta \approx R^{(n-1)/2}$ and the total packet count R^{n-1} .

Drill 12.1.4. Carry out the sanity check at the end of Worked Example 12.1.1: match the powers of R on both sides.

Where we are. Three conventions fixed here and used for the rest of the series.

R is the large parameter and $\delta = R^{-1/2}$ the cap radius. The relation $\delta = R^{-1/2}$ is not a choice: by Theorem 12.1.2 it is forced by requiring the dual tube to have length R , and every other scale in the volume follows from it.

Constants depend on n and on the transversality parameter alone. Never on R , on the number of packets, or on g .

Every estimate carries R^ε . By Volume 9, Theorem 9.7.3 this is exactly the accuracy the dimension conclusions need, and Worked Example 12.4.3 explains where such losses come from — an induction with $\log \log R$ rounds rather than $\log R$.

Where we are. One further consequence of Theorem 12.1.2 deserves recording, since Volume 13 depends on it.

The dual tube's *length* is δ^{-2} , coming from the cap's thickness δ^2 , which is the sphere's curvature. So the tube's aspect ratio $\delta^{-2} : \delta^{-1} = \delta^{-1}$ is a curvature measurement. A surface with k non-vanishing principal curvatures gives caps of thickness δ^2 in k directions and δ in the rest, hence tubes of a different shape — and Volume 5, Worked Example 5.5.4 predicted exactly this pattern from the decay exponents.

The practical consequence: before decomposing into wave packets for a new surface, compute the cap's dimensions from the second fundamental form. Every subsequent count depends on them.

Notes

The wave packet decomposition in this form is due to Fefferman and to Córdoba, and is standard equipment. The presentation here follows Tao's lecture notes; Demeter's book gives the version used in decoupling, which is the one Volume 13 will need.

Chapter 12.2

Transversality

12.2.1 The multilinear setting

Definition 12.2.1. Let $S_1, \dots, S_n \subseteq \mathbb{S}^{n-1}$ be relatively open, and let $\nu > 0$. The pieces are ν -transverse if for every choice $\omega_j \in S_j$,

$$|\det(\omega_1, \dots, \omega_n)| \geq \nu.$$

For each j let $\mathcal{E}_j$ denote the extension operator restricted to S_j , so $\mathcal{E}_j g_j = \mathcal{E}(g_j \mathbf{1}_{S_j})$.

Since the normal to $\mathbb{S}^{n-1}$ at ω is ω itself, this is the same determinant condition as Theorem 11.5.3 in Volume 11, now imposed on the caps rather than on tube directions — and by Theorem 12.1.2 the wave packets from S_j point in directions lying in S_j , so the two conditions agree.

Proposition 12.2.2 (Transversality is inherited by wave packets). *Let $S_1, \dots, S_n$ be ν -transverse and let T_j be a wave packet tube arising from a cap $\theta_j \subseteq S_j$ at scale R . Then the tube directions $\omega_{\theta_1}, \dots, \omega_{\theta_n}$ satisfy*

$$|\det(\omega_{\theta_1}, \dots, \omega_{\theta_n})| \geq \nu,$$

so the families $\{T_j\}$ are ν -transverse in the sense of Theorem 11.5.3.

Proof. The cap centre ω_{θ_j} lies in S_j , and Theorem 12.2.1 applies to every choice of points in the S_j . $\square$

Where we are. Theorem 12.2.2 is the hinge of the volume, and it is a triviality. That is the point: the transversality hypothesis was chosen precisely so that it passes from the surfaces to the wave packets without loss, and hence so that Volume 11's multilinear Keakeya inequality applies verbatim to the tubes produced by Theorem 12.1.4.

Every other hypothesis one might impose — curvature conditions, separation of the caps, non-degeneracy of second fundamental forms — fails this test in one way or another. Transversality is the condition that survives wave packet decomposition, and that is why the multilinear theory is stated with it.

12.2.2 Why n factors, and not fewer

Proposition 12.2.3 (Two factors do not suffice in high dimensions). *Let $n \geq 3$ and let $S_1, S_2 \subseteq \mathbb{S}^{n-1}$ be any two relatively open sets. There are $\omega_1 \in S_1$, $\omega_2 \in S_2$ and a hyperplane containing both. Consequently no determinant condition on pairs can force the wave packets of S_1 and S_2 to be quantitatively independent in all n directions.*

Proof. Any two vectors in $\mathbb{R}^n$ span a plane, which lies in a hyperplane whenever $n \geq 3$. So the $n \times n$ determinant of Theorem 12.2.1 cannot even be formed from two vectors, and the natural bilinear substitute — the angle between ω_1 and ω_2 — constrains only one direction. $\square$

Where we are. So the number of factors is forced by the dimension: to pin down n directions one needs n vectors, and the determinant is the natural quantitative measure of their independence.

This explains a feature of the literature that is otherwise puzzling. The *bilinear* theory — two factors — was developed first, in the 1990s, and gives sharp results in $\mathbb{R}^2$ where two vectors do span. In higher dimensions it gives less than the multilinear theory, and Chapter 12.6 will record exactly how much less. The progression from bilinear to multilinear is the progression from “two directions are independent” to “all n are”.

Theorem 12.2.3 also explains why the bilinear approach nonetheless survives for the *distance set* problem: that problem is intrinsically two-point (Volume 8, Worked Example 7.7.3), so a two-factor estimate is the matching tool. Chapter 12.8 develops this.

WORKED EXAMPLE 12.2.1— TIER A **The determinant, computed**

Transversality is a quantitative hypothesis and it repays seeing the numbers. *The orthogonal case.* $\omega_j = e_j$ gives $|\det| = 1$, the maximum. So $\nu = 1$ corresponds to mutually orthogonal caps.

Caps near a common great circle. In $\mathbb{R}^3$, take $\omega_1 = e_1$, $\omega_2 = e_2$ and $\omega_3 = (\cos \alpha)e_1 + (\sin \alpha)e_2 + \varepsilon e_3$ normalised. Then

$$\det(\omega_1, \omega_2, \omega_3) = \frac{\varepsilon}{\sqrt{1 + \varepsilon^2}} \approx \varepsilon,$$

so three caps within ε of a common great circle are only ε -transverse. As $\varepsilon \rightarrow 0$ the constant in Theorem 11.5.7 blows up, and it must: coplanar tubes can overlap completely.

The relation to angles. If the pairwise angles are all $\geq \alpha$ the determinant need *not* be bounded below — three vectors can be pairwise at 60° and nearly coplanar. So transversality is strictly stronger than pairwise separation, and this is the content of Theorem 12.2.3 in concrete form.

What ν costs. The constants in the multilinear theory depend on ν polynomially, and the broad–narrow argument of Chapter 12.4 will choose ν as a small power of R , so those constants must be tracked. That is one of the two places where the passage from multilinear to linear loses, and Worked Example 12.4.1 will account for it.

WORKED EXAMPLE 12.2.2— TIER B **Reading a multilinear hypothesis**

Multilinear estimates appear throughout Volumes 12–14, and the hypotheses vary in ways that matter.

(a) *How many factors?* n factors in $\mathbb{R}^n$ is the multilinear theory; 2 factors is bilinear; k factors with $2 < k < n$ occurs and interpolates. The number determines how many directions the hypothesis controls.

(b) *Transverse how?* A determinant bound $|\det| \geq \nu$ as here, or a weaker angular separation, or — in the bilinear literature — a separation of the *normals* plus a curvature hypothesis. These are not equivalent, and Worked Example 12.2.1 gives the standard counterexample.

(c) *What norm on the data?* $L^2(d\sigma)$ here; the linear conjecture wants $L^\infty(d\sigma)$, and Volume 6 recorded that the gap between them is where the difficulty sits. A multilinear estimate with L^2 data is much weaker than one with L^∞ data, and the two are often quoted without the distinction being flagged.

(d) *What loss?* R^ε for every ε , as in Theorem 12.3.1 below, or no loss at all as in Guth’s multilinear Kakeya. The R^ε is harmless for the applications here — Volume 9, Theorem 9.7.3 explained why — but it is not harmless if one wants an endpoint result.

The habit. Before using a multilinear estimate, write down its four answers. Papers in this area differ mostly in these choices, and a comparison made without checking them is meaningless.

WORKED EXAMPLE 12.2.3— TIER B **Constructing transverse families**

The hypothesis of Theorem 12.2.1 is easy to state and worth being able to

realise, since applications must produce such families.

(a) *The canonical example.* S_j a cap of radius c_n about e_j . Then any choice $\omega_j \in S_j$ has $|\det(\omega_1, \dots, \omega_n)| \geq \frac{1}{2}$ for c_n small, by continuity of the determinant at $(e_1, \dots, e_n)$ where it equals 1.

(b) *How many such families cover the sphere?* A single family covers only n caps. To handle general g one partitions $\mathbb{S}^{n-1}$ into $\approx K^{n-1}$ caps and considers all n -tuples, of which there are $\approx K^{n(n-1)}$ — a constant, once K is fixed. That is why Theorem 12.4.3 can afford to take a maximum over tuples.

(c) *The tuples that fail.* Most n -tuples of caps are *not* transverse: for $n = 3$ and K large, a random triple of caps lies near a common great circle with probability tending to 1 in the sense that its determinant is typically $O(K^{-1})$. So transversality is a rare condition, not a generic one, and the broad–narrow dichotomy is genuinely a dichotomy rather than a convenience.

(d) *The rescaling that restores it.* A cluster of caps near a great circle can be rescaled — stretch the normal direction — to occupy a full cap of a lower-dimensional sphere. This is what Worked Example 12.4.2 describes and what the narrow case of the induction performs.

The practical upshot. When applying a multilinear estimate, one does not verify transversality for a given family; one arranges a decomposition in which the transverse tuples are separated from the degenerate ones, and treats each separately. The estimate is a tool used inside a decomposition, never on its own.

WORKED EXAMPLE 12.2.4 — TIER C: RESEARCH TASTE **Hypotheses that concede the difficulty**

Transversality is the fourth hypothesis in this series introduced precisely to exclude the hard case, and collecting them shows a pattern.

(a) *Volume 6: L^2 data.* Stein–Tomas assumes $g \in L^2(d\sigma)$ where the conjecture wants L^∞ . On a δ -cap the two differ by $\delta^{(n-1)/2}$, and the theorem is provable in a page precisely because L^2 admits TT^* .

(b) *Volume 9: transverse hairbrushes.* Theorem 9.5.5 proves Wolff’s bound when the tubes meet T_0 at angle $\gtrsim 1$, and the small-angle case is the one that needs induction on scales.

(c) *Volume 11: transverse tube families.* Multilinear Kakeya assumes $|\det| \geq \nu$, which a Besicovitch set’s tubes violate for most tuples.

(d) *Here: transverse caps.* Theorem 12.2.1, the same condition on the surfaces. *The pattern.* In every case the excluded configurations are the *tangential* ones

— near-parallel tubes, nearly coplanar directions, caps whose normals nearly agree — and Volume 11, Worked Example 11.8.2 identified tangency as the recurring obstruction across three independent methods. This is the fourth.

Why the concessions are nonetheless productive. Each yields a sharp theorem, and each admits a decomposition recovering something linear: ε -removal for (a), the small-angle induction for (b), broad–narrow for (c) and (d). The recovery is always lossy, and Worked Example 12.4.1 accounts for the loss in the present case.

The question a reader should carry forward. Whether a method that does *not* concede tangency exists. Volume 14 reports that Wang and Zahl found one in $\mathbb{R}^3$, by analysing the tangential configurations directly rather than excluding them — and that its dependence on three-dimensional geometry is what blocks the extension.

WORKED EXAMPLE 12.2.5— TIER A **Transversality across the series**

Where	Objects	Condition
Vol 9, Theorem 9.5.4	tube and hairbrush	angle $\geq \kappa$
Vol 11, Theorem 11.5.3	n tube families	$ \det(v_1, \dots, v_n) \geq \nu$
Here, Theorem 12.2.1	n surface pieces	$ \det(\omega_1, \dots, \omega_n) \geq \nu$
Theorem 12.6.1	two arcs	$ \sin \angle(\omega_1, \omega_2) \geq \nu$

They are all the same condition. On the sphere the normal at ω is ω , so rows two and three coincide by Theorem 12.2.2; row four is the case $n = 2$, where the determinant of two vectors is the sine of their angle; row one is the two-object version in a different guise.

What each buys. Row one gives Wolff’s bound in the transverse case; row two gives multilinear Kakeya; row three gives multilinear restriction; row four gives the bilinear estimate. In every case the transverse statement is sharp and the general one is open or harder.

The determinant is the right measure. Worked Example 12.2.1 showed that pairwise angular separation does *not* suffice — three vectors can be pairwise at 60° and nearly coplanar — so the condition must be the determinant and not a collection of pairwise conditions. That is Theorem 12.2.3 in another form.

The unifying statement. *Transversality is the non-degeneracy of the map that combines the objects:* the addition map for two arcs (Theorem 12.6.2), the map to $\mathbb{R}^n$ for n tubes, the determinant in general. Where that map degenerates, every method in this series stops.

WORKED EXAMPLE 12.2.6— TIER B Verifying transversality in practice

Applications must certify Theorem 12.2.1 for particular families, and the certification is usually one of three arguments.

(a) *By construction.* Choose S_j near e_j , as in Worked Example 12.2.3(a). Determinant near 1 by continuity. Used whenever the family is at one's disposal.

(b) *By pigeonhole.* Given an arbitrary g , decompose into caps and select a transverse tuple carrying a definite share of the mass — which is exactly Theorem 12.4.3. Used when the family is not at one's disposal, which is the usual case.

(c) *By exclusion.* Show the non-transverse tuples contribute negligibly. This requires a bound on the degenerate case and so is available only when the narrow analysis has already been done.

Which to use. (a) when proving a multilinear estimate; (b) when applying one to a linear problem; (c) rarely, and usually as part of (b)'s induction.

The quantitative caution. All three produce a specific ν , and the constants in Theorem 12.3.1 degrade as $\nu \rightarrow 0$. An argument that certifies transversality without tracking ν has not certified anything usable, and Worked Example 12.4.5 shows how tightly ν is coupled to the other parameters.

Drills

Drill 12.2.1. Verify that the normal to $\mathbb{S}^{n-1}$ at ω is ω , and deduce that Theorem 12.2.1 and Theorem 11.5.3 agree.

Drill 12.2.2. Compute the determinant in Worked Example 12.2.1 for the three vectors given, and check the limit $\varepsilon \rightarrow 0$.

Drill 12.2.3. Exhibit three unit vectors in $\mathbb{R}^3$, pairwise at angle 60° , whose determinant is arbitrarily small.

Drill 12.2.4. Explain why $L^\infty(d\sigma)$ data is a stronger hypothesis than $L^2(d\sigma)$ data on a cap of radius δ , and compute the ratio of the two norms for an indicator function.

Where we are. A remark on what Theorem 12.2.3 does and does not say. It says that two vectors cannot span $\mathbb{R}^n$ for $n \geq 3$, so no condition on pairs controls all n directions. It does *not* say the bilinear theory is useless above the plane — Theorem 12.6.3 is sharp in every dimension. What it says is that the bilinear hypothesis must be supplemented by curvature, which is

why that theorem has a curvature hypothesis and the multilinear one does not.

The practical consequence: choose the number of factors to match the problem, not the dimension. Worked Example 12.8.2 shows the distance set problem wants two regardless of n .

Notes

The multilinear formulation is due to Bennett, Carbery and Tao (2006). The observation that transversality is the hypothesis surviving wave packet decomposition is implicit in their paper and is made explicit in Guth's exposition.

Chapter 12.3

Multilinear Restriction

Theorem 12.3.1 (Bennett–Carbery–Tao). *Let $\Sigma_1, \dots, \Sigma_n$ be C^2 hypersurface pieces in $\mathbb{R}^n$ whose unit normals satisfy $|\det(N_1, \dots, N_n)| \geq \nu$ for every choice of points, and let $\mathcal{E}_j$ be the associated extension operators. Then for every $\varepsilon > 0$ there is $C(\varepsilon)$ with*

$$\begin{aligned} \left\| \prod_{j=1}^n \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(B(0,R))} &\leq C(\varepsilon) R^\varepsilon \prod_{j=1}^n \|f_j\|_{L^2} \\ &, \quad R \geq 1. \end{aligned}$$

Equivalently, $\|\prod_j |\mathcal{E}_j f_j|^{1/n}\|_{L^p(B_R)} \lesssim_\varepsilon R^\varepsilon \prod_j \|f_j\|_{L^2}^{1/n}$ at $p = \frac{2n}{n-1}$.

Where we are. The two displays are the same statement. Raising the second to the power p ,

$$\begin{aligned} \left\| \prod_j |\mathcal{E}_j f_j|^{1/n} \right\|_{L^p}^p &= \int \prod_j |\mathcal{E}_j f_j|^{p/n} \\ &= \int \prod_j |\mathcal{E}_j f_j|^{\frac{2}{n-1}} \\ &= \left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}}^{\frac{2}{n-1}}, \end{aligned}$$

using $\frac{p}{n} = \frac{2}{n-1}$ from Theorem 12.3.2. The first form is the one the proof works with; the second is the one Chapter 12.4 consumes, and the exponent $\frac{1}{n}$ on the right is forced by homogeneity.

12.3.1 Why the exponent is $\frac{2n}{n-1}$

Before the proof, the exponent. It is fixed by the requirement that the left-hand side be exactly the integrand of multilinear Kakeya, and that identification is the wave packet dictionary of Worked Example 12.1.2.

Lemma 12.3.2 (The exponents match). *For $p = \frac{2n}{n-1}$ one has $\frac{p}{n} = \frac{2}{n-1}$, and hence*

$$\begin{aligned} \left\| \prod_{j=1}^n |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p}^p &= \int_{\mathbb{R}^n} \prod_{j=1}^n \left(|\mathcal{E}_j g_j|^2 \right)^{\frac{1}{n-1}}. \end{aligned}$$

Proof. $\frac{p}{n} = \frac{2n}{n(n-1)} = \frac{2}{n-1}$, and then $\left\| \prod_j |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p}^p = \int \prod_j |\mathcal{E}_j g_j|^{p/n}$. $\square$

Where we are. Theorem 12.3.2 is arithmetic, but it is why Theorem 11.5.4 of Volume 11 was stated with the exponent $\frac{1}{n-1}$: applying multilinear Kakeya to the functions $f_j = |\mathcal{E}_j g_j|^2$ produces exactly the multilinear restriction integrand. Checking: $n = 2$ gives $p = 4$ and $\frac{p}{n} = 2 = \frac{2}{n-1}$; $n = 3$ gives $p = 3$ and $\frac{p}{n} = 1$; $n = 4$ gives $p = \frac{8}{3}$ and $\frac{p}{n} = \frac{2}{3}$. $\checkmark$

The proof given below does *not* take this route — it applies the discrete Loomis–Whitney inequality directly to the extension operators, which is what makes it self-contained. But the identification explains where the exponent comes from, and it is the reason the two theorems have matching hypotheses.

Proposition 12.3.3 (What the naive packet decomposition gives). *With hypotheses as in Theorem 12.3.1 and $p = \frac{2n}{n-1}$,*

$$\left\| \prod_{j=1}^n |\mathcal{E}_j g_j|^{1/n} \right\|_{L^p(B_R)} \lesssim_{n,\nu,\varepsilon} R^{\frac{1}{2}+\varepsilon} \prod_{j=1}^n \|g_j\|_{L^2(d\sigma)}^{1/n}.$$

Proof. Decompose each $\mathcal{E}_j g_j$ by Theorem 12.1.4 at scale R . By part (ii) each packet is bounded by $c_{j,\theta,T} \mathbf{1}_T$ up to rapidly decaying tails, so Cauchy–Schwarz over the packets meeting a given point gives $|\mathcal{E}_j g_j|^2 \lesssim \#\{\text{caps}\} \sum_{(\theta,T)} |c_{j,\theta,T}|^2 \mathbf{1}_T$. Writing f_j for the sum without the cap factor and applying Theorem 11.5.4 in its transverse form Theorem 11.5.7,

$$\int_{B_R} \prod_j f_j^{\frac{1}{n-1}} \lesssim R^\varepsilon \prod_j \left(\int f_j \right)^{\frac{1}{n-1}},$$

and $\int f_j \approx \|\mathcal{E}_j g_j\|_{L^2(B_R)}^2 \approx R \|g_j\|_{L^2}^2$ by Theorem 6.1.4. Assembling with Theorem 12.3.2 and taking p -th roots, the exponent of R is $\frac{n}{n-1} \cdot \frac{n-1}{2n} = \frac{1}{2}$. $\square$

Pitfall: the naive packet argument loses a full power of R

Theorem 12.3.3 carries $R^{1/2}$ where Theorem 12.3.1 has R^ε . That is a genuine loss, and locating it is the key to understanding why the real proof is organised as it is.

Where it enters. The argument bounds $|\mathcal{E}_j g_j|^2$ pointwise by a sum of packet contributions. At a point there is one tube per cap, so $\approx R^{(n-1)/2}$ packets overlap, and Cauchy–Schwarz over them costs exactly that factor.

Why it is real. The step discards the L^2 almost-orthogonality of Theorem 12.1.4(iii). At $n = 2$ it gives $\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(B_R)} \lesssim R \|g_1\| \|g_2\|$, whereas Theorem 12.6.1 proves the sharp bound with *no* power of R . Lossy in the one case computable by hand.

What the real proof does instead. It never dominates pointwise. The induction of Theorem 12.3.10 propagates a *weighted* hypothesis Theorem 12.3.9, in which each factor retains an ℓ^2 sum over caps with off-diagonal decay — so the orthogonality survives every round. That is the whole difference between $R^{1/2}$ and R^ε .

12.3.2 Discrete Loomis–Whitney

The engine of the proof is a discrete form of Theorem 11.5.2.

Lemma 12.3.4 (Discrete Loomis–Whitney). *Let $N_1, \dots, N_n$ be unit vectors with $|\det(N_1, \dots, N_n)| \geq \nu$, let $\mathcal{L}$ be the lattice they generate, let $\mathcal{H}_j = N_j^\perp$ and let $\pi_{N_j} : \mathbb{R}^n \rightarrow \mathcal{H}_j$ be projection along N_j , so that $\mathcal{L}(\mathcal{H}_j) = \pi_{N_j}(\mathcal{L})$ is a lattice in $\mathcal{H}_j$. Then for $g_j : \mathcal{L}(\mathcal{H}_j) \rightarrow [0, \infty)$,*

$$\left\| \prod_{j=1}^n g_j(\pi_{N_j} z) \right\|_{\ell^{\frac{2}{n-1}}(\mathcal{L})} \lesssim_\nu \prod_{j=1}^n \|g_j\|_{\ell^2(\mathcal{L}(\mathcal{H}_j))}.$$

Proof. After the linear change of variables carrying N_j to e_j — which distorts by $\text{Poly}(\nu^{-1})$ — we may assume $\mathcal{L} = \mathbb{Z}^n$ and $\pi_{N_j} = \pi_j$ deletes the j -th coordinate.

Apply Theorem 11.5.2 to the step functions $F_j = \sum_{y \in \mathbb{Z}^{n-1}} g_j(y) \mathbf{1}_{y+[0,1)^{n-1}}$ on $\mathbb{R}^{n-1}$. Then $\prod_j F_j(\pi_j x)$ is constant on each unit cube $z + [0,1)^n$, with value $\prod_j g_j(\pi_j z)$, so

$$\int_{\mathbb{R}^n} \prod_j F_j(\pi_j x)^{\frac{1}{n-1}} dx = \sum_{z \in \mathbb{Z}^n} \prod_j g_j(\pi_j z)^{\frac{1}{n-1}},$$

while $\|F_j\|_{L^1(\mathbb{R}^{n-1})} = \sum_y g_j(y)$. Applying Theorem 11.5.2 to F_j^2 in place of F_j and taking $\frac{n-1}{2}$ -th powers converts the ℓ^1 norms on the right into ℓ^2 norms and the exponent on the left into $\frac{2}{n-1}$, giving the claim. $\square$

Where we are. Direct computation confirms Theorem 12.3.4 with constant 1 in the orthogonal case: over 150 random trials with $n = 2, 3, 4$ on grids of side 5 to 12, the ratio of the two sides never exceeded 1, attaining it only for $n = 2$ where the inequality is Cauchy–Schwarz.

That the constant is 1 — not merely bounded — matters, because the induction below applies the lemma once per round and a constant exceeding 1 would compound. The $\text{Poly}(\nu^{-1})$ enters only through the change of variables, once.

12.3.3 The induction

Fix $\varepsilon > 0$ and let $\delta > 0$ be small, to be chosen at the very end in terms of ε alone. Partition each Σ_j into pieces of diameter $\leq \delta$; by (12.3.2) below it suffices to prove the estimate for one piece of each, at the cost of a factor depending only on δ and n . So assume $\text{diam } \Sigma_j \leq \delta$, fix $\zeta_j^0 \in \Sigma_j$, put $N_j = N_j(\zeta_j^0)$ and let $\mathcal{H}_j = N_j^\perp$. After a smooth change of coordinates,

$$\begin{aligned} \mathcal{E}_j f_j(x) &= \int_{U_j} e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} f_j(\xi') \, d\xi' \\ , \quad U_j &\subseteq B_j(0, \delta) \subseteq \mathcal{H}_j, \end{aligned}$$

with $x = (x_j, x')$ in the N_j -adapted coordinates. Since $\text{diam } U_j \lesssim \delta$,

$$|\nabla \varphi_j(\xi') - \nabla \varphi_j(\eta')| \lesssim \delta \quad (\xi', \eta' \in U_j). \quad (12.3.1)$$

We use throughout the elementary superposition bound, valid for $0 < p \leq 1$,

$$\left\| \sum_{\alpha} h_{\alpha} \right\|_{L^p}^p \leq \sum_{\alpha} \|h_{\alpha}\|_{L^p}^p. \quad (12.3.2)$$

Definition 12.3.5 (The lattice partition of unity). Let $\mathcal{L}$ be the lattice generated by $N_1, \dots, N_n$ and, for $r > 0$, let $\mathcal{C}(r)$ be the parallelepipeds of size r relative to $\mathcal{L}$, with $c(q)$ the centre of q . Put $\mathcal{CH}_j(r) = \pi_{N_j} \mathcal{C}(r)$, a set of parallelepipeds in $\mathcal{H}_j$.

Fix $\chi_0 \in \mathcal{S}(\mathbb{R}^{n-1})$ non-negative with $\|\chi_0\|_{L^1} = 1$ and $\widehat{\chi_0}$ supported in the unit ball, let $\mathcal{T}_j$ be the linear map carrying $\mathcal{L}(\mathcal{H}_j)$ to $\mathbb{Z}^{n-1}$, and set

$$\chi_q(x) = \chi_0\left(\mathcal{T}_j\left(\frac{x - c(q)}{r}\right)\right), \quad q \in \mathcal{CH}_j(r).$$

Lemma 12.3.6 (Properties of the partition). *With χ_q as above:*

(i) $\mathcal{F}_j \chi_q$ has Fourier support in a ball of radius $\lesssim r^{-1}$;

(ii) (partition of unity)

$$\sum_{q \in \mathcal{CH}_j(r)} \chi_q = 1; \quad (12.3.3)$$

(iii) (weighted almost-orthogonality) for every $M \in \mathbb{N}$,

$$\sum_{q \in \mathcal{CH}_j(r)} \left\| \left\langle \frac{x - c(q)}{r} \right\rangle^M \chi_q g \right\|_{L^2}^2 \lesssim_M \|g\|_{L^2}^2 \quad (g \in L^2(\mathcal{H}_j)). \quad (12.3.4)$$

Proof. (i) is the hypothesis on $\widehat{\chi}_0$ together with the scaling by r .

(ii) The function $x \mapsto \sum_q \chi_q(x)$ is periodic with respect to the lattice $r\mathcal{L}(\mathcal{H}_j)$, so by Poisson summation its value is the sum of $\widehat{\chi}_0$ over the dual lattice, scaled. Since $\widehat{\chi}_0$ is supported in the unit ball and the dual lattice has spacing $\gtrsim r^{-1}$ after the normalisation, only the zero mode contributes, and it contributes $\|\chi_0\|_{L^1} = 1$.

(iii) Since χ_0 is Schwartz, $|\chi_q(x)| \lesssim_K \langle \frac{x-c(q)}{r} \rangle^{-K}$ for every K . Choosing $K = M + n$ and summing the resulting geometric series over the lattice of centres gives $\sum_q \langle \frac{x-c(q)}{r} \rangle^{2M} |\chi_q(x)|^2 \lesssim_M 1$ uniformly in x ; integrating against $|g|^2$ gives (12.3.4). $\square$

Lemma 12.3.7 (Commutator identity). *With $\mathcal{F}_j$ the Fourier transform on $\mathcal{H}_j$ and any $x_0 \in \mathbb{R}^n$,*

$$\begin{aligned} \left(x' - x'_0 - x_j \nabla \varphi_j \left(\frac{D'}{i} \right) \right)^M \mathcal{E}_j f_j \\ = \mathcal{E}_j \left(\mathcal{F}_j \left((x' - x'_0)^M \mathcal{F}_j^{-1} f_j \right) \right) \\ , \quad M \in \mathbb{N}. \end{aligned}$$

Proof. It suffices to treat $M = 1$ and iterate. Differentiating under the integral in the definition of $\mathcal{E}_j$,

$$\begin{aligned} \left(x' - x_j \nabla \varphi_j \left(\frac{D'}{i} \right) \right) e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} \\ = -i \partial_{\xi'} \left(e^{i(x' \cdot \xi' + x_j \varphi_j(\xi'))} \right), \end{aligned}$$

since $\partial_{\xi'}$ of the phase is $x' + x_j \nabla \varphi_j(\xi')$. Integrating by parts in ξ' moves the derivative onto f_j , and multiplication by $x' - x'_0$ on the physical side of $\mathcal{H}_j$ is $\mathcal{F}_j \circ (\text{multiplication by } (x' - x'_0)) \circ \mathcal{F}_j^{-1}$. $\square$

Definition 12.3.8 (Margin and the induction quantity). For $f : \mathcal{H}_j \rightarrow \mathbb{C}$ put $\text{margin}^j(f) = \text{dist}(\text{supp } f, B_j(0, 2\delta)^c)$. For $R \geq \delta^{-2}$ let $A(R)$ be the least constant such that

$$\left\| \prod_{j=1}^n \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(Q)} \leq A(R) \prod_{j=1}^n \|f_j\|_{L^2}$$

for every parallelepiped Q of size R in the lattice $\mathcal{L}$ generated by $N_1, \dots, N_n$, and every f_j with $\text{margin}^j(f_j) \geq \delta - R^{-1/2}$.

The margin is a device for propagating support conditions: each round of the induction enlarges the frequency support slightly, and the margin records how much room is left. The induction begins at $R = \delta^{-2}$ so that there is room to begin with.

Lemma 12.3.9 (Localised induction hypothesis). *Let Q be a parallelepiped of size $\delta^{-1}R$ and $q \in \mathcal{C}(R)$ a sub-parallelepiped of size R . Let $\{\chi_{q'}\}_{q' \in \mathcal{CH}_j(R)}$ be the partition of unity on $\mathcal{H}_j$ of (12.3.3). Then for every $M \in \mathbb{N}$,*

$$\left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \lesssim_M A(R) \prod_{j=1}^n \left(\sum_{q' \in \mathcal{CH}_j(R)} \left\langle \frac{d(\pi_{N_j} q, q')}{R} \right\rangle^{-(2M-n^2)} \right. \\ \left. \times \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_j^{-1} f_j \right\|_{L^2}^2 \right)^{1/2}.$$

Proof. Step 1. One cap, one factor.

Fix $j = 1$ and $q' \in \mathcal{CH}_1(R)$. Apply Theorem 12.3.7 with $x'_0 = c(q')$ and $M = 1$, together with (12.3.1) to replace $\nabla \varphi_1(\frac{D'}{i})$ by the constant $\nabla \varphi_1(\xi'_0)$ at a cost of $|x_1| \cdot \delta \lesssim \delta^{-1}R \cdot \delta = R$:

$$\left\| (x' - c(q') - x_1 \nabla \varphi_1(\xi'_0)) \mathcal{E}_1 \mathcal{F}_1(\chi_{q'} \mathcal{F}_1^{-1} f_1) \cdot \prod_{j \geq 2} \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \\ \lesssim A(R) \left(\|(x' - c(q')) \chi_{q'} \mathcal{F}_1^{-1} f_1\|_{L^2} + R \|\chi_{q'} \mathcal{F}_1^{-1} f_1\|_{L^2} \right) \prod_{j \geq 2} \|f_j\|_{L^2} \\ \lesssim R A(R) \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle \chi_{q'} \mathcal{F}_1^{-1} f_1 \right\|_{L^2} \prod_{j \geq 2} \|f_j\|_{L^2}.$$

The middle step is the induction hypothesis at scale R , applied to the modified datum $\mathcal{F}_1((x' - c(q')) \chi_{q'} \mathcal{F}_1^{-1} f_1)$; this is legitimate because convolving with $\mathcal{F}_1((x' - c(q')) \chi_{q'})$ moves the frequency support by at most CR^{-1} , which is less than $\frac{1}{2}\delta^{1/2}R^{-1/2}$ once δ is small, so the margin $\delta - (\delta^{-1}R)^{-1/2}$ available at scale $\delta^{-1}R$ degrades only to $\delta - R^{-1/2}$ — exactly what Theorem 12.3.8 requires at scale R .

Step 2. *Convert to off-diagonal decay.*

For $x \in q$ one has $\left\langle \frac{x' - c(q') - x_1 \nabla \varphi_1(\xi'_0)}{R} \right\rangle \approx \left\langle \frac{d(\pi_{N_1} q, q')}{R} \right\rangle$, since $|x_1| \lesssim \delta^{-1}R$ and $|\nabla \varphi_1(\xi'_0)| \leq \delta$ make the middle term $O(R)$. Dividing, and iterating Step 1 M

times,

$$\begin{aligned} & \left\| \mathcal{E}_1 \mathcal{F}_1(\chi_{q'} \mathcal{F}_1^{-1} f_1) \prod_{j \geq 2} \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(q)} \\ & \lesssim_M \left\langle \frac{d(\pi_{N_1} q, q')}{R} \right\rangle^{-M} \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_1^{-1} f_1 \right\|_{L^2} \prod_{j \geq 2} \|f_j\|_{L^2}. \end{aligned}$$

Step 3. *Sum over caps.*

By (12.3.3) and (12.3.2) with $p = \frac{2}{n-1} \leq 1$ for $n \geq 3$ — the case $n = 2$ being Theorem 12.6.1 of Chapter 12.6 below, whose proof is independent of this chapter — summing over q' and applying Hölder for sequences, $\|a_i b_i\|_{\ell^{2/n}} \lesssim \|a_i\|_{\ell^{2/(n-1)}} \|b_i\|_{\ell^{2/(n-1)}}$, together with $\|\langle d/R \rangle^{-n^2/2}\|_{\ell^{n/(n-1)}} \lesssim 1$, gives the stated bound with the improvement in the $j = 1$ factor. Repeating for $j = 2, \dots, n$ gives all factors. $\square$

Proposition 12.3.10 (One scale to the next). $A(\delta^{-1}R) \leq C A(R)$, with C depending on n and ν but not on δ or R .

Proof. Let Q have size $\delta^{-1}R$. Define $g_j : \mathcal{L}(\mathcal{H}_j) \rightarrow [0, \infty)$ by

$$g_j(\mathbf{j}) = \left(\sum_{q' \in \mathcal{CH}_j(R)} \left\langle \frac{d(q(\mathbf{j}), q')}{R} \right\rangle^{-(2M-n^2)} \left\| \left\langle \frac{x' - c(q')}{R} \right\rangle^M \chi_{q'} \mathcal{F}_j^{-1} f_j \right\|_{L^2}^2 \right)^{1/2}.$$

By (12.3.4), for M large enough in terms of n , $\|g_j\|_{\ell^2(\mathcal{L}(\mathcal{H}_j))} \lesssim \|f_j\|_{L^2}$.

Theorem 12.3.9 says exactly that the $L^{2/(n-1)}$ norm over each $q \in \mathcal{C}(R) \cap Q$ is at most $A(R) \prod_j g_j(\pi_{N_j} q)$. Raising to the power $\frac{2}{n-1}$ and summing over $q \subseteq Q$,

$$\begin{aligned} \left\| \prod_j \mathcal{E}_j f_j \right\|_{L^{\frac{2}{n-1}}(Q)} & \leq A(R) \left\| \prod_j g_j(\pi_{N_j} z) \right\|_{\ell^{\frac{2}{n-1}}(\mathcal{L})} \lesssim_\nu A(R) \prod_j \|g_j\|_{\ell^2} \\ & \lesssim A(R) \prod_j \|f_j\|_{L^2}, \end{aligned}$$

the middle inequality being Theorem 12.3.4. $\square$

Proof of Theorem 12.3.1. Step 1. Base case.

For $R \leq \delta^{-2}$, the pointwise bound $\|\prod_j \mathcal{E}_j f_j\|_{L^\infty} \leq \prod_j \|\mathcal{E}_j f_j\|_{L^\infty} \lesssim \prod_j \|f_j\|_{L^2}$ — Cauchy–Schwarz on each U_j , which has bounded measure — integrated over a parallelepiped of size $\leq \delta^{-2}$ gives $A(R) \leq C(\delta)$.

Step 2. *Iterate.*

By Theorem 12.3.10, $A(\delta^{-K}r) \leq C^K A(r)$ for $r \in [\delta^{-2}, \delta^{-3}]$, so $A(R) \leq C^K C(\delta)$ whenever $R \in [\delta^{-K}, \delta^{-K-1}]$.

Step 3. *Choose δ .*

For such R one has $K \leq \log R / \log \delta^{-1}$, so

$$C^K \leq \exp\left(\frac{\log C \log R}{\log \delta^{-1}}\right) = R^{\log C / \log \delta^{-1}}.$$

Choosing $\delta = C^{-1/\varepsilon}$ makes the exponent at most ε , giving $A(R) \leq C(\varepsilon)R^\varepsilon$.

Step 4. *Remove the localisation.*

Undo the reduction to pieces of diameter δ by (12.3.2), at the cost of a factor $((\delta^{-(n-1)})^n)^{(n-1)/2}$ depending only on n and ε , and cover $B(0, R)$ by boundedly many parallelepipeds of $\mathcal{L}$. $\square$

Remark 12.3.11 (R15 discharged). *Theorem 12.3.1 is now proved in full*, following Bejenaru’s self-contained argument, which is in turn inspired by Guth’s proof of multilinear Kakeya. Revision R15 is discharged.

Two remarks. First, the proof does *not* route through multilinear Kakeya: it applies the discrete Loomis–Whitney inequality directly to the extension operators, which is why it is self-contained. Theorem 11.5.4 of Volume 11 remains the geometric counterpart and Theorem 12.3.2 the dictionary between them, but neither is used here.

Second, Theorem 12.3.3 below — the naive wave packet argument, which loses $R^{1/2}$ — is retained deliberately. It shows what the crude decomposition gives, and the Pitfall following it identifies exactly the step that the induction above replaces: the pointwise domination discarding almost-orthogonality. The induction avoids it by never dominating pointwise, and instead propagating a *weighted* induction hypothesis with off-diagonal decay, Theorem 12.3.9.

WORKED EXAMPLE 12.3.1— TIER A **The exponent $\frac{2n}{n-1}$ across dimensions**

n	$p = \frac{2n}{n-1}$	$\frac{p}{n} = \frac{2}{n-1}$	compare
2	4	2	the bilinear L^4 estimate of Vol 6 Theorem 6.8.5
3	3	1	Loomis–Whitney in $\mathbb{R}^3$, Worked Example 11.5.5
4	$\frac{8}{3}$	$\frac{2}{3}$	
5	$\frac{10}{4}$	$\frac{2}{4}$	
∞	$\rightarrow 2$	$\rightarrow 0$	

Reading the table. The exponent *decreases* to 2 as n grows, so the multilinear estimate holds in a wider range of p in higher dimensions. That is the opposite of the linear restriction conjecture, whose range $p > \frac{2n}{n-1}$ shrinks. The multilinear hypothesis becomes relatively more powerful as n increases, because there are more directions to be transverse in.

The case $n = 2$. Here $p = 4$ and the estimate is the classical bilinear L^4 bound, which Volume 6 proved directly by the TT^* computation of Theorem 6.8.4. So the multilinear theorem specialises correctly, and that is the strongest single check on the statement.

The case $n = 3$. $p = 3$ and $\frac{p}{n} = 1$: the estimate is linear in each $|\mathcal{E}_j g_j|$, and the underlying Takeya statement is Loomis–Whitney with no exponent at all. This is the cleanest instance and the one to keep in mind.

WORKED EXAMPLE 12.3.2— TIER B **What the theorem does and does not give**

(a) *The exponent is forced.* Theorem 12.3.2 shows $\frac{2n}{n-1}$ is the exponent at which the left side becomes exactly the multilinear Takeya integrand. It is also sharp, by a Knapp example on transverse caps; that sharpness computation is left as an exercise-level supplement and is not a proof dependency. The theorem itself is proved internally in Theorem 12.3.11.

(b) *The data norm is L^2 .* The restriction conjecture concerns $L^\infty(d\sigma)$ data. On a cap of radius δ these differ by $\delta^{(n-1)/2}$, and that gap is exactly what Chapter 12.4 must overcome. A multilinear estimate with L^2 data does not by itself give any linear estimate.

(c) *The loss is R^ε .* Harmless for the applications here, by the accounting of Volume 9, Theorem 9.7.3.

(d) *What it says about Takeya.* By reversing the argument, a multilinear restriction estimate implies a multilinear Takeya bound — the two are equivalent up to the wave packet dictionary. So this theorem contains no information about Takeya beyond what Volume 11 already imported.

The honest summary. Theorem 12.3.1 is a sharp theorem about a hypothesis that excludes the hard case. This chapter derives its exponent, retains the deliberately lossy wave-packet proof as a diagnostic, and then proves the sharp R^ε theorem by the weighted induction above. Its value is also in what Chapter 12.4 can recover from it.

WORKED EXAMPLE 12.3.3— TIER C: RESEARCH TASTE **Where the series has lost powers of R , and why**

Theorem 12.3.3 loses $R^{1/2}$ against the truth, and the Pitfall identified the cause. It is the third such loss in the series and the three have a common shape worth recording.

(a) *Volume 8, the Mattila L^2 route.* Bounding the r -integral of the spherical average by Cauchy–Schwarz discards the *oscillation* of $d\sigma$, and the resulting integral diverges for every s and n . The correct argument keeps the phase

and applies Plancherel.

(b) *Volume 9, the L^2 method for Kekeya.* Summing over pairs of tubes discards all higher-order coincidences, and Worked Example 9.3.3 showed the ceiling is intrinsic to any pair-based method.

(c) *Here.* Dominating $|\mathcal{E}g|^2$ pointwise by a sum over packets discards the almost-orthogonality of Theorem 12.1.4(iii), at a cost of the number of packets meeting a point.

The common shape. In each case a *coherent* quantity — an oscillating integral, a high-order coincidence, an almost-orthogonal sum — is replaced by an incoherent bound. Cauchy–Schwarz and the triangle inequality are the usual culprits, and both are exact only when the terms are aligned, which is precisely when the configuration is extremal.

The diagnostic. After applying Cauchy–Schwarz, ask what configuration would make it an equality. If that configuration is not the one under study, the step has lost something, and the loss can be quantified as the ratio between the two.

Why it matters more here than elsewhere. Theorem 12.7.3 in Chapter 12.7 performs a superficially identical step and loses nothing — because its left side is quadratic, so the orthogonality applies directly. The distinction between an L^2 quantity and an L^p norm of a product is exactly the distinction between when this step is free and when it is not.

WORKED EXAMPLE 12.3.4— TIER A **The two theorems, side by side**

	Multilinear Kekeya	Multilinear restriction
Objects	δ -tubes	wave packets
Quantity	$\int \prod_j (\sum_T \mathbf{1}_T)^{\frac{1}{n-1}}$	$\ \prod_j \mathcal{E}_j g_j ^{1/n}\ _{L^p}^p$
Exponent	$\frac{1}{n-1}$	$\frac{p}{n} = \frac{2}{n-1}$ at $p = \frac{2n}{n-1}$
Hypothesis	$ \det \geq \nu$ on tube directions	$ \det \geq \nu$ on cap normals
Where	Vol 11, Theorem 11.5.4, <i>proved</i>	Theorem 12.3.11
Loss	none (Guth)	R^ε

They are the same theorem. Theorem 12.3.2 shows the exponents correspond under $f_j = |\mathcal{E}_j g_j|^2$, and Theorem 12.2.2 shows the hypotheses correspond under the wave packet decomposition. The dictionary of Worked Example 12.1.2 translates each into the other.

Why both are retained. Volume 11 proves multilinear Kekeya and this volume proves multilinear restriction. The naive translation through wave packets

loses $R^{1/2}$ (the Pitfall above), so the two internal proofs are not redundant: each exposes a different mechanism even though the statements are closely related.

What is genuinely equivalent. The *statements*, up to the R^ε ; the *proofs* are not, and the passage in either direction costs the orthogonality that Worked Example 12.3.3 identified as the recurring casualty.

WORKED EXAMPLE 12.3.5— TIER B **Auditing a proof for lost orthogonality**

The Pitfall above found a lost power of R in a proof that looked routine. The audit that finds such losses is short and worth running on any argument built from wave packets.

Step 1: find every Cauchy–Schwarz and triangle inequality. These are the only places coherence is destroyed.

Step 2: at each, count the terms. Cauchy–Schwarz over M terms costs a factor M unless the terms are genuinely aligned. In Theorem 12.3.3 Step 1, M is the number of packets meeting a point, which is the number of caps, $R^{(n-1)/2}$.

Step 3: ask whether alignment is plausible. Wave packets from different caps have different frequency supports, so they are almost orthogonal — the opposite of aligned. Hence the factor is a genuine loss, not a bookkeeping artefact.

Step 4: check against a computable case. Here $n = 2$, where Theorem 12.6.1 gives the truth. The discrepancy was exactly R , matching $(R^{(n-1)/2})^2 = R$ at $n = 2$. ✓ The audit predicted the size of the error before the check confirmed it.

Where the audit passes. Theorem 12.7.3 performs a superficially identical step and loses nothing, because its left side is quadratic and Theorem 12.1.4(iii) applies directly — there is no Cauchy–Schwarz over packets at all. Running Step 1 on that proof finds nothing to flag.

The general principle. Almost-orthogonality is usable exactly when the quantity is quadratic. An L^p norm of a product is not, which is why the multilinear theory needs an induction on scales and the weighted L^2 theory does not.

WORKED EXAMPLE 12.3.6— TIER C: RESEARCH TASTE **From historical imports to the strict ledger**

Earlier drafts of this volume quoted several research theorems. The strict

reconstruction changes the reader's contract.

R15. Multilinear restriction is now internally proved, using the multilinear Keakeya theorem retained and proved in Volume 11.

R16. The three-dimensional paraboloid broad–narrow core is now internally proved, including the two-scale recurrence and its local-constancy, trilinear, and narrow-strip interfaces.

R17. Tao's non-endpoint bilinear theorem is internally proved. The local packet-product L^2 estimate, approximate conservation laws, quadruple multiplicity, two-ends relation, and bush count are all explicit; see Theorems 12.6.10 and 12.6.12.

R18. Erdögan's weighted theorem, annular estimate, and Falconer deduction are internally proved; their R17 dependency is now transitively satisfied. See Theorems 12.8.7 and 12.8.9.

The lesson is that a proof ledger is transitive: the chain became certifiable only after both the local R17 input and the R18 annular passage were expanded.

Drills

Drill 12.3.1. Verify $\frac{p}{n} = \frac{2}{n-1}$ at $p = \frac{2n}{n-1}$ for $n = 2, 3, 4, 5$.

Drill 12.3.2. Check the tail estimate in Step 1: with $\lesssim R^{n-1}$ packets and decay $(1 + d/R^{1/2})^{-N}$, find N making the total tail $O(R^{-100})$.

Drill 12.3.3. Confirm $|T| \approx R^{(n+1)/2}$ for a wave packet tube, from Theorem 12.1.3.

Drill 12.3.4. In Theorem 12.3.3 Step 1, count the packets containing a given point and confirm that Cauchy–Schwarz over them costs $R^{(n-1)/2}$; then track that factor through Step 4 and recover the exponent $\frac{1}{2}$.

Notes

Bennett, Carbery and Tao proved Theorem 12.3.1 in 2006 by a heat-flow monotonicity argument. The wave packet reduction of Theorem 12.3.3 is the standard way to see where the exponent comes from; that it loses $R^{1/2}$ unless the almost-orthogonality is retained is the content of the Pitfall, and Guth's induction-on-scales proof is the route that retains it.

The equivalence noted in Worked Example 12.3.2(d) — that multilinear restriction and multilinear Keakeya are the same statement through wave packets — is why Volume 11 imported the latter rather than this.

Chapter 12.4

Broad and Narrow

Theorem 12.3.1 assumes transversality, which Worked Example 11.5.4 recorded as excluding exactly the hard configurations. This chapter recovers a genuine linear estimate from it. The device, due to Bourgain and Guth, is a dichotomy: at each point, either the mass of $\mathcal{E}g$ is spread over n transverse directions — where the multilinear estimate applies — or it concentrates near a hyperplane, where an induction reduces the dimension.

12.4.1 The dichotomy

Fix a large constant K and partition $\mathbb{S}^{n-1}$ into caps τ of radius K^{-1} , of which there are $\approx_n K^{n-1}$. Write g_τ for g restricted to τ , so $\mathcal{E}g = \sum_\tau \mathcal{E}g_\tau$.

Lemma 12.4.1 (Determinant and slabs). *Let $v_1, \dots, v_n \in \mathbb{S}^{n-1}$ and suppose all lie within ν' of some hyperplane through the origin, that is $|v_j \cdot e| \leq \nu'$ for a common unit vector e . Then*

$$|\det(v_1, \dots, v_n)| \leq n\nu'.$$

Consequently, if $|\det(v_1, \dots, v_n)| \geq \nu$ then the v_j do not all lie within ν/n of any hyperplane through the origin.

Proof. Choose coordinates with $e = e_n$, so the n -th coordinate of each v_j has modulus at most ν' . Expanding the determinant along the n -th column,

$$\det(v_1, \dots, v_n) = \sum_{j=1}^n (-1)^{j+n} (v_j)_n M_j,$$

where M_j is the $(n-1) \times (n-1)$ minor obtained by deleting row j and the last column. Each M_j is a determinant of $n-1$ vectors of norm at most 1, so $|M_j| \leq 1$ by Hadamard's inequality. Hence $|\det| \leq n \max_j |(v_j)_n| \leq n\nu'$.

The second statement is the contrapositive. □

Where we are. Theorem 12.4.1 is what makes “transverse” and “concentrated near a hyperplane” genuinely complementary, and quantitatively so. Direct computation on unit vectors constrained to a slab of half-width ν' confirms the bound: at $\nu' = 0.3, 0.1, 0.03, 0.01$ in $\mathbb{R}^3$ the largest determinant found over 20,000 trials was 0.67, 0.23, 0.075 and 0.024, against the bound $3\nu' = 0.9, 0.3, 0.09, 0.03$ — tight to within a factor 1.5. The dichotomy below is therefore not a definition dressed as a theorem: the two alternatives really do exhaust the possibilities, with constants that can be tracked.

Definition 12.4.2 (Broad and narrow points). Fix $0 < \eta < 1$ and $x \in \mathbb{R}^n$, and let $\mu(x) = \max_{\tau} |\mathcal{E}g_{\tau}(x)|$. Call x η -*broad* if there are caps $\tau_1, \dots, \tau_n$ with

$$\begin{aligned} |\mathcal{E}g_{\tau_j}(x)| &\geq \tfrac{1}{2}\mu(x) \quad \text{for each } j \\ &, \quad |\det(\omega_{\tau_1}, \dots, \omega_{\tau_n})| \\ &\geq \eta^{n-1}, \end{aligned}$$

and η -*narrow* otherwise.

Proposition 12.4.3 (The dichotomy). *For every x , at least one of the following holds.*

(B) x is η -broad, and then

$$|\mathcal{E}g(x)| \leq 2K^{n-1} \prod_{j=1}^n |\mathcal{E}g_{\tau_j}(x)|^{1/n}.$$

(N) x is η -narrow, and then there is a hyperplane V through the origin such that every cap τ with $|\mathcal{E}g_{\tau}(x)| \geq \tfrac{1}{2}\mu(x)$ has its centre within η of V ; consequently

$$|\mathcal{E}g(x)| \leq 2K^{n-1} |\mathcal{E}g_{\mathcal{N}(V)}(x)| + \tfrac{1}{2}K^{n-1}\mu(x),$$

where $\mathcal{N}(V)$ is the union of the caps meeting the $\tfrac{\nu}{n}$ -neighbourhood of V .

Proof. Step 1. One cap carries a definite share.

Summing over the $\lesssim_n K^{n-1}$ caps,

$$|\mathcal{E}g(x)| \leq \sum_{\tau} |\mathcal{E}g_{\tau}(x)| \leq C_n K^{n-1} \mu(x),$$

so $\mu(x) \geq |\mathcal{E}g(x)| / (C_n K^{n-1})$.

Step 2. *Greedy selection.*

Let $\mathcal{S} = \{\tau : |\mathcal{E}g_\tau(x)| \geq \frac{1}{2}\mu(x)\}$, which is non-empty since it contains the maximising cap. Select $\tau_1 \in \mathcal{S}$ arbitrarily. Having chosen $\tau_1, \dots, \tau_k$, choose $\tau_{k+1} \in \mathcal{S}$ whose centre is at distance $\geq \eta$ from the span of $\omega_{\tau_1}, \dots, \omega_{\tau_k}$, if such a cap exists; otherwise stop.

Step 3. *The two outcomes.*

If the selection reaches $k = n$, the chosen centres satisfy $|\det| \geq \eta^{n-1}$ — a determinant is the product of the successive distances to the previous span — and each chosen cap has $|\mathcal{E}g_{\tau_j}(x)| \geq \frac{1}{2}\mu(x)$. Then

$$\prod_{j=1}^n |\mathcal{E}g_{\tau_j}(x)|^{1/n} \geq \frac{1}{2}\mu(x) \geq \frac{|\mathcal{E}g(x)|}{2C_n K^{n-1}},$$

which is (B).

If the selection stops at some $k < n$, every $\tau \in \mathcal{S}$ has centre within η of $V = \text{span}(\omega_{\tau_1}, \dots, \omega_{\tau_k})$, a proper subspace; enlarging V to a hyperplane containing it gives the stated containment. Splitting $\mathcal{E}g = \mathcal{E}g_{\mathcal{N}(V)} + \sum_{\tau \notin \mathcal{S}} \mathcal{E}g_\tau$ and bounding the second sum by $K^{n-1} \cdot \frac{1}{2}\mu(x)$ gives (N). $\square$

Where we are. The dichotomy is elementary — a pigeonhole and a greedy selection — but it is only the geometric entry point of the Bourgain–Guth iteration. The strict proof has three additional quantitative ingredients: weighted local constancy for cap pieces on physical cubes, a sampled multilinear estimate that permits one to sum the cube-dependent broad triples, and a two-scale treatment of the narrow strip. In the latter one first keeps the *full* coarse-cap data (global in the physical cube), and only after selecting a separated pair does one pass to the strip pieces. This order is essential: parabolic rescaling is not legitimate for frequency data that depend on the physical cube.

The induction also does not obtain an R^ε loss merely by declaring a fixed factor K^C harmless. One needs a recurrence with genuine contracting coefficients. In the three-dimensional paraboloid model the two narrow terms have the schematic form

$$C_{K_1} K^{q/2-1} K^{6-2q} A(R/K)^q \quad \text{and} \quad C K_1^{6-2q} A(R/K_1)^q,$$

with K_1 chosen first and $K \gg K_1$ second. At $q = 10/3$ the first raw power of K is critical; the small threshold gain is what turns the iterated recurrence into an R^ε estimate. The complete proof is therefore a two-scale argument, not the one-scale numerology formerly printed here.

12.4.2 What the dichotomy yields

12.4.3 Córdoba's square function estimate

Proposition 12.4.4 (Córdoba). *Let $F = \widehat{f d\sigma}$ with f supported in the $1/r$ -neighbourhood of a great circle of $\mathbb{S}^2$, decomposed into caps S_α of radius $1/r$. Then for $q \leq 4$,*

$$\|F\|_{L^q(B(y,r))} \lesssim \left\| \left(\sum_{\alpha} |F_{\alpha}|^2 \right)^{1/2} \right\|_{L^q(B(y,r))}. \quad (12.4.1)$$

Proof. The bilinear form of the statement is Theorem 12.6.1 of Chapter 12.6 below, proved there by Plancherel alone: for transverse arcs the Minkowski sums $S_{\alpha} + S_{\beta}$ are almost disjoint as α, β vary, so by Plancherel

$$\begin{aligned} \|F_1 F_2\|_{L^2(\mathbb{R}^2)} &\lesssim \left(\sum_{\alpha, \beta} \|F_{1,\alpha} F_{2,\beta}\|_{L^2}^2 \right)^{1/2} \\ &= \left\| \left(\sum_{\alpha} |F_{1,\alpha}|^2 \right)^{1/2} \left(\sum_{\beta} |F_{2,\beta}|^2 \right)^{1/2} \right\|_{L^2}. \end{aligned}$$

Localising to $B(y, r)$ costs nothing, since blurring the caps at scale $1/r$ does not destroy the almost-disjointness of the Minkowski sums. Since the square functions are essentially constant on balls of radius r , Hölder upgrades the L^2 statement to L^q for $q \leq 4$, and the linear form (12.4.1) follows by the standard bilinear-to-linear reduction for a function dominated by transverse interactions. $\square$

12.4.4 Weighted local constancy

The broad–narrow classification is made on physical cubes, so a pointwise phrase such as “ F_{α} is essentially constant on a K -cube” must be replaced by a majorant that can also be summed over the cubes. The following lemma supplies exactly that object.

Lemma 12.4.5 (Weighted local constancy). *Let $F_{\alpha} = \mathcal{E}f_{\alpha}$, where f_{α} is supported in a cap of diameter at most K^{-1} on a fixed compact smooth hypersurface. Let*

$$w(u) = (1 + |u|)^{-N}, \quad w_K(u) = K^{-n} w(u/K),$$

with $N > 2n$, and for every cube μ of side K and centre z_{μ} set

$$c_{\mu, \alpha} = \int_{\mathbb{R}^n} |F_{\alpha}(y)| w_K(z_{\mu} - y) dy.$$

Then, after increasing the implicit constant by an amount depending only on the surface and N ,

$$\sup_{x \in \mu} |F_\alpha(x)| \lesssim c_{\mu, \alpha}.$$

Moreover, for any two K -cubes μ, μ' , one has

$$c_{\mu, \alpha} \lesssim \left(1 + \frac{|z_\mu - z_{\mu'}|}{K}\right)^N c_{\mu', \alpha}.$$

Proof. Choose the centre ξ_α of the cap and write $G_\alpha(x) = e^{-2\pi i x \cdot \xi_\alpha} F_\alpha(x)$. The Fourier support of G_α is contained in a ball of radius C/K : indeed any two points of the cap are separated by $O(K^{-1})$ in ambient frequency space. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ equal to 1 on the ball $B(0, C)$ and put $\psi_K(x) = K^{-n} \chi(x/K)$. Then

$$G_\alpha = G_\alpha * \psi_K, \quad |\psi_K(u)| \lesssim_N K^{-n} (1 + |u|/K)^{-2N}.$$

For $x \in \mu$, the triangle inequality and $1 + |x - y|/K \gtrsim (1 + |z_\mu - y|/K)$ give

$$|F_\alpha(x)| = |G_\alpha(x)| \lesssim \int |G_\alpha(y)| K^{-n} (1 + |z_\mu - y|/K)^{-N} dy = C_N c_{\mu, \alpha}.$$

This proves the first assertion.

For the second, the elementary inequality

$$1 + \frac{|z_{\mu'} - y|}{K} \leq \left(1 + \frac{|z_\mu - z_{\mu'}|}{K}\right) \left(1 + \frac{|z_\mu - y|}{K}\right)$$

implies

$$w_K(z_\mu - y) \lesssim \left(1 + \frac{|z_\mu - z_{\mu'}|}{K}\right)^N w_K(z_{\mu'} - y).$$

Integrating against $|F_\alpha(y)|$ gives the claimed comparison. $\square$

Where we are. This closes the first R16 obligation. Notice that $c_{\mu, \alpha}$ is defined from the *full cap datum* f_α before any narrow strip depending on μ has been selected. Consequently the later parabolic rescaling is applied to globally defined frequency pieces rather than to cube-dependent data.

12.4.5 The two quantitative summation lemmas

We next discharge the two analytic inputs that were left implicit in the old broad–narrow sketch. Throughout this section the model surface is the compact paraboloid patch

$$\Phi(\xi) = (\xi, |\xi|^2), \quad \xi \in [-1, 1]^2,$$

and

$$Ef(x) = \int_{[-1, 1]^2} f(\xi) e^{2\pi i x \cdot \Phi(\xi)} d\xi.$$

The spherical formulation is locally equivalent after a finite partition into elliptic charts, but the paraboloid has the advantage that parabolic rescaling is exact; this is the model used for the strict R16 proof.

Lemma 12.4.6 (Sampled transverse summation). *Fix $K \geq 10$ and partition the parameter square into K^{-1} -squares α . Let $c_{\mu,\alpha}$ be the weighted coefficients of Theorem 12.4.5 on the K -cubes μ meeting a ball B_R . Suppose that for each μ one chooses a transverse triple $\alpha_1(\mu), \alpha_2(\mu), \alpha_3(\mu)$, with determinant at least cK^{-2} . If $q \geq 3$ and $\|f\|_{L^\infty} \leq 1$, then for every $\varepsilon > 0$,*

$$\sum_{\mu} |\mu| \prod_{j=1}^3 c_{\mu,\alpha_j(\mu)}^{q/3} \lesssim_{K,\varepsilon} R^{\varepsilon q}.$$

The same estimate holds if the left side is summed over only a subcollection of the K -cubes.

Proof. It is enough to sum over all transverse triples, because the summands are non-negative. For a fixed triple write $F_j = Ef_{\alpha_j}$. Since $\|f\|_\infty \leq 1$ and $|\alpha_j| \asymp K^{-2}$,

$$\|F_j\|_{L^\infty} \leq |\alpha_j| \lesssim K^{-2}.$$

Thus, because $q \geq 3$,

$$\prod_{j=1}^3 c_{\mu,\alpha_j}^{q/3} \lesssim_K \prod_{j=1}^3 c_{\mu,\alpha_j}.$$

By the definition of $c_{\mu,\alpha}$,

$$\prod_{j=1}^3 c_{\mu,\alpha_j} = \iiint \prod_{j=1}^3 |F_j(z_\mu - u_j)| \prod_{j=1}^3 w_K(u_j) du_1 du_2 du_3.$$

The off-diagonal comparison in Theorem 12.4.5 lets us replace the sample at the centre z_μ by an integral over $x \in \mu$, at a fixed loss depending only on the weight exponent. Summing the essentially disjoint cubes therefore gives

$$\sum_{\mu} |\mu| \prod_{j=1}^3 c_{\mu,\alpha_j} \lesssim \iiint \prod_{j=1}^3 w_K(u_j) \int_{B(0,CR)} \prod_{j=1}^3 |F_j(x - u_j)| dx du_1 du_2 du_3.$$

For fixed u_1, u_2, u_3 , translation in physical space replaces f_{α_j} by the modulated datum $e^{-2\pi i u_j \cdot \Phi(\xi)} f_{\alpha_j}(\xi)$ and hence leaves its L^2 norm unchanged. The three caps remain cK^{-2} -transverse. Applying Theorem 12.3.1 in dimension three to the inner integral, with a harmless enlargement of the ball, yields

$$\int_{B(0,CR)} \prod_{j=1}^3 |F_j(x - u_j)| dx \lesssim_{K,\varepsilon} R^\varepsilon \prod_{j=1}^3 \|f_{\alpha_j}\|_{L^2}.$$

The weights have integral $O(1)$, and $\|f_{\alpha_j}\|_2 \lesssim K^{-1}$. Finally there are $O(K^6)$ ordered triples. Since K is fixed before the induction on R , all powers of K are absorbed into the displayed constant. Replacing ε here by $q\varepsilon$ proves the assertion. $\square$

Lemma 12.4.7 (Cap-coefficient rescaling sum). *Let $A_q(R)$ be the least constant such that*

$$\|Ef\|_{L^q(B_R)} \leq A_q(R)\|f\|_{L^\infty}$$

for every translate of a cube B_R of side R . For the K^{-1} -cap coefficients above,

$$\sum_{\mu} |\mu| \sum_{\alpha} c_{\mu,\alpha}^q \lesssim_q K^{6-2q} A_q(CR/K)^q \|f\|_{L^\infty}^q.$$

Proof. Jensen's inequality in the probability kernel w_K gives $c_{\mu,\alpha}^q \lesssim (|Ef_\alpha|^q * w_K)(z_\mu)$. Summing the K -cubes and using bounded overlap of the associated weights gives

$$\sum_{\mu} |\mu| c_{\mu,\alpha}^q \lesssim \|Ef_\alpha\|_{L^q(B(0,CR))}^q.$$

Parabolic rescaling of a K^{-1} -square is exact for Φ : the two parameter directions scale by K , the normal physical direction by K^2 , and the surface measure contributes K^{-2} . Consequently

$$\|Ef_\alpha\|_{L^q(B(0,CR))} \lesssim K^{4/q-2} A_q(CR/K) \|f\|_{L^\infty}.$$

There are $O(K^2)$ caps. Raising to the q -th power and summing therefore produces $K^2 K^{4-2q} = K^{6-2q}$, as claimed. $\square$

Lemma 12.4.8 (Narrow-strip local L^4 estimate). *Let $K \gg K_1 \geq 10$. Let L be a line in the parameter plane and let $\mathcal{N}_{C/K}(\Phi(L))$ denote the C/K -neighbourhood in frequency space of the lifted parabola $\Phi(L \cap [-1, 1]^2)$. Suppose $F_1, F_2 \in L^2(\mathbb{R}^3)$ have Fourier supports contained respectively in*

$$U_j \cap \mathcal{N}_{C/K}(\Phi(L)),$$

where the U_j are balls of radius C/K_1 whose projections to L are separated by at least c/K_1 . Then every cube μ of side K satisfies

$$\int_{\mu} |F_1 F_2|^2 \lesssim_{K_1} K^{-3} \|F_1\|_2^2 \|F_2\|_2^2.$$

The implicit constant is polynomial in K_1 and independent of K .

Proof. Rotate the parameter variables so that $L = \{\xi_1 = c\}$. The frequency support condition then confines the ξ_1 -coordinate to an interval of length

$O(K^{-1})$. Local constancy in the dual variable x_1 gives, for a one-dimensional Schwartz probability weight ω_K at scale K ,

$$|F_j(x_1, x')| \lesssim \int |F_j(y_j, x')| \omega_K(x_1 - y_j) dy_j, \quad x' = (x_2, x_3).$$

For each fixed y_1, y_2 , take the partial Fourier transform in x' . The two resulting functions have frequency supports in separated K_1^{-1} -arcs of the planar parabola $(\xi_2, \xi_2^2 + c^2)$, thickened by $O(K^{-1})$. The planar bilinear L^2 estimate Theorem 12.6.1, after the elementary K_1^{-1} rescaling, gives

$$\int_{\mathbb{R}^2} |F_1(y_1, x') F_2(y_2, x')|^2 dx' \lesssim_{K_1} K^{-2} \prod_{j=1}^2 \|F_j(y_j, \cdot)\|_{L^2(\mathbb{R}^2)}^2.$$

Insert this bound into the preceding x_1 majorisation. Cauchy–Schwarz in x_1 contributes $\|\omega_K\|_2^2 \asymp K^{-1}$, and Fubini then turns the remaining y_j integrals into $\|F_j\|_2^2$. The product of the factors K^{-2} and K^{-1} is K^{-3} . $\square$

Lemma 12.4.9 (Smooth strip truncation is controlled by global cap data). *Let S be a strip of width C/K in the parameter square and let $\chi_S \in C_c^\infty$ equal 1 on S and be supported in the concentric strip of width $2C/K$, with $|\partial^\gamma \chi_S| \lesssim_\gamma K^{|\gamma|}$. For every coarse K_1^{-1} -square β there is a Schwartz kernel $\kappa_{S,\beta}$ on $\mathbb{R}^3$ such that*

$$E(\chi_S f_\beta) = E f_\beta * \kappa_{S,\beta}, \quad \|\kappa_{S,\beta}\|_{L^1(\mathbb{R}^3)} \lesssim 1,$$

uniformly in S and β . Consequently every weighted local-constancy coefficient of $E(\chi_S f_\beta)$ at scale K_1 is bounded, up to a constant C_K , by the corresponding coefficient formed from the globally defined coarse-cap extension $E f_\beta$ with a slightly slower Schwartz weight. The constants depend only on the fixed cutoff profiles and the compact paraboloid patch, not on K, K_1, S , or β .

Proof. Choose a fixed smooth tubular neighbourhood U of the compact paraboloid patch. The nearest-point projection $\pi : U \rightarrow [-1, 1]^2$ is smooth after shrinking U . Extend χ_S to $m_{S,\beta}(\zeta) = \chi_S(\pi(\zeta)) \rho_\beta(\pi(\zeta)) \theta(\zeta)$, where ρ_β is a smooth cutoff equal to 1 on β and θ is a fixed cutoff supported in U . On the support of the surface measure carrying f_β one has $m_{S,\beta} = \chi_S$. Hence multiplication of that Fourier measure by $m_{S,\beta}$ gives

$$E(\chi_S f_\beta) = E f_\beta * \check{m}_{S,\beta}.$$

Repeated integration by parts gives $|\check{m}_{S,\beta}(x)|$ is an anisotropically rescaled Schwartz kernel. Changing variables in the rescaled coordinate shows that its L^1 norm is bounded uniformly; the fixed compact coordinate changes contribute only an absolute constant. Thus $\|\check{m}_{S,\beta}\|_1 \lesssim 1$ uniformly in the

location and direction of S and in β . Convolving once more with the K_1 -scale weight and using $(1 + |x - y|/K_1)^{-N}(1 + |y|)^{-N} \lesssim_K (1 + |x|/K_1)^{-N/2}$ gives the coefficient domination. $\square$

Proposition 12.4.10 (Two-scale narrow aggregation). *Let $q = 10/3$, fix $K \gg K_1 \gg 1$, and let $\mathcal{C}_2$ be the K -cubes on which the first broad–narrow decomposition is narrow. After replacing the associated hard strip by the smooth strip of Theorem 12.4.9, one has*

$$\begin{aligned} \sum_{\mu \in \mathcal{C}_2} \|Ef\|_{L^q(\mu)}^q &\lesssim C_{K_1} K^{q/2-1} K^{6-2q} A_q(CR/K)^q \|f\|_\infty^q \\ &\quad + CK_1^{6-2q} A_q(CR/K_1)^q \|f\|_\infty^q. \end{aligned} \quad (12.4.2)$$

The constants are independent of R .

Proof. Normalize $\|f\|_\infty = 1$. On a narrow K -cube μ , all fine caps whose weighted coefficient is larger than K^{-2} times the maximum lie in a C/K -strip $S(\mu)$. The discarded caps contribute at most a constant multiple of the maximal coefficient, and their total q -mass is bounded by Theorem 12.4.7. It therefore remains to estimate $E(\chi_{S(\mu)}f)$ on μ .

Partition each μ into K_1 -cubes ν , and partition the parameter square into coarse K_1^{-1} -squares β . Form weighted coefficients at scale K_1 for the pieces $E(\chi_{S(\mu)}f_\beta)$. On each ν perform the bilinear dichotomy: either two significant coarse squares are separated by c/K_1 along the strip, or all significant coarse squares lie in one C/K_1 -square.

Step 1. *Separated-pair subcase.*

For a separated pair β_1, β_2 , Hölder on μ changes the local q -moment into the L^4 moment with the factor $|\mu|^{1-q/4}$. Apply Theorem 12.4.8 to the two smoothed strip pieces after multiplying by a fixed cutoff equal to 1 on μ . Plancherel and almost orthogonality of the fine K^{-1} -caps inside each β_j bound the resulting L^2 norms by

$$|\mu| \sum_{\alpha \subset \beta_j} c_{\mu, \alpha}^2$$

after the same off-diagonal weight comparison used in Theorem 12.4.5. Summing the separated coarse pairs and using Cauchy–Schwarz in the fine-cap index gives

$$\sum_{\substack{\nu \subset \mu \\ \text{separated pair}}} \|E(\chi_{S(\mu)}f)\|_{L^q(\nu)}^q \lesssim_{K_1} |\mu| K^{q/2-1} \sum_{\alpha} c_{\mu, \alpha}^q.$$

Summing μ and invoking Theorem 12.4.7 yields the first term in (12.4.2).

Step 2. *One-coarse-cap subcase.*

If all significant β lie in one C/K_1 -square, the insignificant pieces again contribute a fixed fraction of the maximal coefficient and are absorbed. For the remaining piece, Theorem 12.4.9 dominates its scale K_1 coefficients by those of the full, globally defined Ef_β . Thus the sum over all K_1 -cubes, although their parent strips vary with μ , is bounded by the global coarse-cap coefficient sum. Applying Theorem 12.4.7 with K_1 in place of K gives

$$\sum_{\substack{\mu \in \mathcal{C}_2, \nu \subset \mu \\ \text{one coarse cap}}} \|E(\chi_{S(\mu)} f)\|_{L^q(\nu)}^q \lesssim CK_1^{6-2q} A_q(CR/K_1)^q.$$

Combining the two subcases and the discarded-cap term proves the proposition. $\square$

Theorem 12.4.11 (Local Bourgain–Guth iteration at the critical exponent). *For the compact paraboloid patch in $\mathbb{R}^3$, every $\varepsilon > 0$ satisfies*

$$A_{10/3}(R) \lesssim_\varepsilon R^\varepsilon.$$

Proof. The broad cubes are controlled by Theorem 12.4.6. The narrow cubes are controlled by Theorem 12.4.10. Since $q = 10/3$,

$$q/2 - 1 + 6 - 2q = 0, \quad 6 - 2q = -\frac{2}{3}.$$

After taking q -th roots, enlarging R by fixed constants, and replacing A_q by its monotone envelope, we obtain

$$A_q(R) \leq C_{K,\eta} R^\eta + C_{K_1} A_q(R/K) + CK_1^{-1/5} A_q(R/K_1) \quad (12.4.3)$$

for every $\eta > 0$. Fix the desired $\varepsilon > 0$ and take $\eta = \varepsilon/2$. First choose K_1 so large that $CK_1^{-1/5-\varepsilon} \leq 1/4$. Next choose $K \gg K_1$ so large that $C_{K_1} K^{-\varepsilon} \leq 1/4$. For $B(R) = \sup_{1 \leq r \leq R} r^{-\varepsilon} A_q(r)$, (12.4.3) gives

$$B(R) \leq C_\varepsilon + \frac{1}{4} B(R/K) + \frac{1}{4} B(R/K_1).$$

Induction on dyadic R yields $B(R) \lesssim_\varepsilon 1$, proving the theorem. $\square$

Where we are. The sampled broad term and the analytic heart of the narrow coplanar term are now proved, rather than hidden under “uncertainty principle” or “Córdoba”. The remaining R16 issue is purely an aggregation/interface question: when the narrow line $L = L(\mu)$ varies with the physical K -cube, the two-scale K_1 decomposition must be summed in a way that applies parabolic rescaling only to globally defined cap data. The

| next proposition isolates that step.

12.4.6 The recursion

We now carry out the induction, in the model case of the sphere in $\mathbb{R}^3$. Write $F = \widehat{f d\sigma}$ and, for $q > 3$, let Q_R be the least constant with

$$\|F\|_{L^q(B(x,R))} \leq Q_R \|f\|_{L^\infty(\mathbb{S}^2)} \quad \text{for all } x.$$

By Volume 6, Theorem 6.5.2, it suffices to show $Q_R \lesssim_\varepsilon R^\varepsilon$.

Set $K = R^\varepsilon$ and partition $\mathbb{S}^2$ into caps S_α of radius K^{-1} , giving $F = \sum_\alpha F_\alpha$. Cover $B(x, R)$ by balls $B(y, K)$; on each, the F_α are essentially constant by the uncertainty principle (Volume 12, Worked Example 12.1.3).

Lemma 12.4.12 (Parabolic rescaling for the sphere). *Let $\tilde{S}_\beta$ be a cap of radius $1/K'$ and $\tilde{F}_\beta$ the corresponding piece of F . Then*

$$\|\tilde{F}_\beta\|_{L^q(B(x,R))} \lesssim (K')^{\frac{4}{q}-2} Q_{R/K'} \|f\|_{L^\infty}.$$

Proof. The affine map flattening $\tilde{S}_\beta$ to a cap of radius 1 scales the two tangential directions by K' and the normal direction by $(K')^2$, exactly as in Volume 13, Theorem 13.2.1. It carries $B(x, R)$ into a ball of radius $R/(K')^2 \leq R/K'$, and multiplies the L^q norm by the Jacobian factor $((K')^{-4})^{1/q}$ while leaving $\|f\|_{L^\infty}$ unchanged. Collecting the powers gives $(K')^{4/q} \cdot (K')^{-2}$. $\square$

Theorem 12.4.13 (Bourgain–Guth base broad–narrow theorem). *For a compact paraboloid patch in $\mathbb{R}^3$ and every $q > \frac{10}{3}$,*

$$\|Ef\|_{L^q(\mathbb{R}^3)} \lesssim_q \|f\|_{L^\infty}.$$

Proof. Theorem 12.4.11 gives the local $L^{10/3}$ estimate with an arbitrarily small power R^ε . The extension operator has the trivial L^∞ bound $\|Ef\|_\infty \lesssim \|f\|_\infty$. Interpolating gives, for every $q > 10/3$, a local L^q estimate with a power R^{ε_q} as small as desired. Applying the epsilon-removal theorem of Volume 6, Theorem 6.5.2, yields the global estimate. $\square$

Remark 12.4.14 (R16 discharged). Under the strict Volume 11–14 policy, R16 is now *internally proved for the three-dimensional paraboloid base theorem*: weighted local constancy is Theorem 12.4.5, sampled broad summation is Theorem 12.4.6, the narrow L^4 input is Theorem 12.4.8, cube-dependent strip truncation is reconciled with global coarse-cap data by Theorem 12.4.9, and the two-scale recurrence is Theorem 12.4.10. The separate sharpened numerical exponents below $10/3$ are not part of R16 and are treated separately in the next chapter.

Where we are. The binding constraint is Step 4, not Step 3: the exponents are $\frac{6}{q} - 2$ and $\frac{5}{q} - \frac{3}{2}$, negative for $q > 3$ and $q > \frac{10}{3}$ respectively. Tabulating:

q	$\frac{6}{q} - 2$	$\frac{5}{q} - \frac{3}{2}$
3	0	+0.167
$\frac{10}{3}$	-0.200	0
$\frac{7}{2}$	-0.286	-0.071
4	-0.500	-0.250

So the transverse coplanar case is the bottleneck, and the inefficiency is the factor $K^{1/2-1/q}$ paid to convert a square function into an ℓ^q sum. That factor is genuinely attained only if almost every cap along a great circle contributes; Kakeya estimates limit how often that happens, and inserting Wolff's bound relaxes $\frac{10}{3}$ to 3.3.

Even the full Kakeya conjecture gives only $q > 3 + \frac{3}{11} \approx 3.27$ by this route. So broad–narrow does not reach the restriction conjecture even granted everything Volumes 9 to 11 were trying to prove — which is the sharpest available statement of the method's ceiling, and it justifies Worked Example 12.4.1(d).

WORKED EXAMPLE 12.4.1— TIER B What broad–narrow costs

The method converts a multilinear estimate into a linear one, and it is worth knowing what is paid.

(a) *The factor K^{n-1} .* Theorem 12.4.3 loses this in both cases. Since K is eventually taken large in terms of ε , it is absorbed into R^ε — but only because the number of induction rounds is $\log R / \log K$, so the losses compound to $R^{O(1/\log K)}$.

(b) *The transversality constant ν .* The multilinear estimate's constant degrades as $\nu \rightarrow 0$ (Worked Example 12.2.1), and the dichotomy produces ν comparable to a power of K^{-1} . So the two parameters are coupled and cannot be optimised independently.

(c) *The data norm.* Theorem 12.3.1 takes $L^2(d\sigma)$ data; the linear conjecture wants $L^\infty(d\sigma)$. On a cap of radius K^{-1} these differ by $K^{-(n-1)/2}$, and recovering the difference is where most of the loss sits. Worked Example 12.3.2(b) flagged this.

(d) *The narrow case is not free.* Rescaling a hyperplane cluster gives a restriction problem for a surface of one lower dimension, whose own exponent is worse. So the induction feeds on progressively weaker estimates, and the recursion's fixed point is what limits the method.

The upshot. Broad–narrow gives a genuine improvement over Stein–Tomas and does not reach the conjecture, and (c) and (d) are the reasons. That diagnosis — a method limited by what it must concede at each round — is the same shape as Volume 10’s Worked Example 10.6.2 for the arithmetic method.

WORKED EXAMPLE 12.4.2— TIER A **The dichotomy on a concrete configuration**

Take $n = 3$, $K = 10$, so ≈ 100 caps on $\mathbb{S}^2$.

A broad configuration. Let g be supported on three caps centred near e_1, e_2, e_3 , with equal mass. At a point x where all three contributions are comparable, $\mu(x)$ is that common size, all three caps lie in $\mathcal{S}$, and the determinant of the centres is ≈ 1 . So x is ν -broad for any $\nu \lesssim 1$, and Theorem 12.4.3(B) applies with the loss $K^2 = 100$.

A narrow configuration. Let g be supported on caps whose centres lie within 0.01 of the equator $\{\omega_3 = 0\}$. By Theorem 12.4.1 every determinant is at most $3 \times 0.01 = 0.03$, so no point is ν -broad for $\nu > 0.03$, and every point falls into case (N).

What happens next in the narrow case. The caps occupy a 0.01-neighbourhood of a great circle. Rescaling that neighbourhood by 100 in the normal direction turns it into a full neighbourhood of $\mathbb{S}^1$ inside a plane — a two-dimensional restriction problem, whose exponent is the one Volume 6 established for the circle. The induction then runs at the coarser scale.

The moral. The narrow case is not a degenerate nuisance; it is a genuine lower-dimensional instance of the same problem, and the method’s strength is exactly the strength of the estimates available in that lower dimension.

WORKED EXAMPLE 12.4.3— TIER B **ε -management: how the losses are absorbed**

A scale induction produces an R^ε bound only when the recurrence has a quantitative gain; the number of rounds by itself is not the issue.

A bad recurrence. If $F(R) \leq K^C F(R/K)$ with fixed $K > 1$, then after $\log_K R$ rounds one obtains $F(R) \lesssim R^C$. Calling K “large” does not change this exponent.

A useful recurrence. Suppose instead

$$F(R) \leq CR^\eta + a_K F(R/K) + b_{K_1} F(R/K_1), \quad a_K \leq K^{-\eta/2}, \quad b_{K_1} \leq K_1^{-\eta}.$$

Iterating along the two branches gives geometric sums. A logarithmic number of scale reductions is therefore compatible with an R^η estimate: the contraction, not a $\log \log R$ round count, is the essential bookkeeping.

How Bourgain–Guth obtains the gain. One fixes K_1 first so that the fine narrow coefficient is < 1 , then takes $K \gg K_1$ and chooses the broad threshold so that the critical K -coefficient has a small negative power.

The lesson. Whenever an R^ε induction appears, write the actual recurrence and identify the contracting coefficient. A bare statement that a fixed polynomial loss is “absorbed into R^ε ” is not a proof.

WORKED EXAMPLE 12.4.4— TIER C: RESEARCH TASTE **Dichotomies as a proof technique**

Theorem 12.4.3 splits every point into one of two cases and handles them separately. That is the fourth such split in the series, and the pattern is worth naming.

(a) *Volume 3, Calderón–Zygmund.* Split f into a bounded part and a sum of mean-zero pieces; handle the first by L^2 and the second by the kernel. The stopping time chooses the split.

(b) *Volume 11, cellular and algebraic.* Split a configuration into points in cells and points on $Z(P)$; the first is handled by the partitioning, the second needs separate work — and in $\mathbb{R}^3$ that work is the whole theorem.

(c) *Volume 9, bush and hairbrush.* Split according to whether a configuration concentrates at a point or along a line.

(d) *Here, broad and narrow.* Split according to whether the mass spreads over transverse directions or clusters near a hyperplane.

The common structure. One case is the “generic” one and is handled by the method’s main estimate; the other is degenerate and is handled by induction, by a lower-dimensional instance, or by a separate argument. *The degenerate case is almost always the hard one* — (b) and (d) both have the property that the theorem’s difficulty lives entirely there.

What makes a dichotomy useful. That the two cases are quantitatively complementary, so that failing one gives usable information about the other. Theorem 12.4.1 supplies exactly this here, and its absence is why some natural-looking splits do not work: if the negation of “broad” were merely “not broad”, nothing would follow.

The question to ask of any dichotomy. What does the degenerate case reduce to, and is that reduction to something already known? Here it is a lower-dimensional restriction problem, whose exponent is worse — which Worked Example 12.4.1(d) identified as the method’s ceiling.

WORKED EXAMPLE 12.4.5— TIER A The dichotomy's two costs, quantified

Theorem 12.4.3 loses a factor K^{n-1} and produces a transversality parameter ν tied to K . Both are worth seeing numerically.

n	K	K^{n-1}	ν available
3	10	100	$\approx K^{-1} = 0.1$
3	100	10^4	≈ 0.01
4	10	10^3	≈ 0.1
4	100	10^6	≈ 0.01

The tension. Larger K gives finer caps and hence a better dichotomy, but a worse constant K^{n-1} and a smaller ν , which degrades the multilinear estimate. The two costs pull in the same direction, so K cannot be taken large freely. *Why K is nonetheless taken large.* Because the losses enter only through $R^{O(1/\log K)}$ once the induction is set up correctly (Worked Example 12.4.3), while the gain from a finer dichotomy is a genuine improvement in the exponent. The trade is favourable, but only because of the $\log \log R$ round count.

The check. At K fixed and $R \rightarrow \infty$, the losses K^{n-1} per round are constants and the number of rounds grows — so a $\log R$ induction would give a fixed power of R and be useless. That Worked Example 12.4.3 identifies this as the distinction between $\log R$ and $\log \log R$ inductions is the single most important piece of bookkeeping in Volumes 12 and 13.

Drills

Drill 12.4.1. Verify Hadamard's inequality for a 2×2 determinant and complete the proof of Theorem 12.4.1 for $n = 3$.

Drill 12.4.2. Check Step 1 of Theorem 12.4.3: count the caps and confirm $\mu(x) \geq |\mathcal{E}g(x)|/(C_n K^{n-1})$.

Drill 12.4.3. In Step 3, justify that successive distances to the previous span bound the determinant from below.

Drill 12.4.4. For the narrow configuration of Worked Example 12.4.2, compute the largest possible determinant and confirm no point is 0.1-broad.

Where we are. The chapter proves the dichotomy and records the target architecture of the induction; under the strict Volume 11–14 policy the induction remains an active proof obligation. It is worth being clear which

half carries the mathematics.

The dichotomy is elementary: a pigeonhole, a greedy selection, and Theorem 12.4.1. It is also the half that explains *why* multilinear estimates bear on the linear problem at all, and a reader who takes only that away has the idea.

The induction is where the exponents come from, and it is long. More to the point, its output has been improved repeatedly since 2011 (Theorem 12.5.3), so the number a textbook would quote is the least durable part of the subject. Proving the dichotomy and reporting the exponents seemed the right division, and it is the same one made for R13 in Volume 11 — prove the structure, import the estimate.

Notes

The broad–narrow decomposition is due to Bourgain and Guth (2011). The version given here isolates the elementary dichotomy from the induction on scales; the induction, and the exponents it produces, are in their paper and have been refined since, most notably by Guth using polynomial partitioning.

Chapter 12.5

What This Gives

12.5.1 The three exponents

For the sphere in $\mathbb{R}^n$ the restriction problem has three exponents worth keeping apart.

Proposition 12.5.1 (The benchmarks). *For the estimate $\|\mathcal{E}g\|_{L^p(\mathbb{R}^n)} \lesssim \|g\|_{L^\infty(d\sigma)}$:*

(i) *it fails for $p \leq \frac{2n}{n-1}$, by the Knapp example;*

(ii) *it holds for $p \geq \frac{2(n+1)}{n-1}$, by Stein–Tomas;*

(iii) *the restriction conjecture asserts it holds for all $p > \frac{2n}{n-1}$.*

Proof. (i) is Theorem 6.2.3 and (ii) is Theorem 6.3.1, both of Volume 6; (iii) is Theorem 6.2.6 there. $\square$

WORKED EXAMPLE 12.5.1— TIER A **The gap, in numbers**

n	necessary $p >$	Stein–Tomas $p \geq$	gap
2	4	6	4 to 6
3	3	4	3 to 4
4	$\frac{8}{3} \approx 2.67$	$\frac{10}{3} \approx 3.33$	2.67 to 3.33
5	$\frac{5}{2} = 2.5$	3	2.5 to 3
10	$\frac{20}{9} \approx 2.22$	$\frac{22}{9} \approx 2.44$	2.22 to 2.44

The gap narrows with dimension. $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$, which tends to 0. So in high dimensions Stein–Tomas is close to sharp in absolute terms — though never sharp.

The planar case is settled. For $n = 2$ the conjecture is a theorem, proved by

Fefferman and Zygmund; Volume 6, Theorem 6.8.2 gives it. So the first row is closed, and every open case has $n \geq 3$.

Where the interest is. $n = 3$: the conjecture asks for $p > 3$ and Stein–Tomas gives $p \geq 4$. Every improvement since 1991 has been a reduction of that 4, and the current record is a little above 3.

Remark 12.5.2 (Historical refined broad–narrow exponents). Assuming Theorem 12.3.1 and the induction of Theorem 12.4.14, the estimate of Theorem 12.5.1 holds for $p > p_n$ with $p_n < \frac{2(n+1)}{n-1}$ for every $n \geq 3$. In $\mathbb{R}^3$ Bourgain and Guth obtained $p > \frac{33}{10} = 3.3$, improving Stein–Tomas’s 4; subsequent work using polynomial partitioning has reduced this further, and the record is a moving target.

The refined exponents are not proved here. Theorem 12.3.1 and the base recursion of Theorem 12.4.13 are both proved (Chapter 12.3, Chapter 12.4); what is quoted is the sharpened bookkeeping producing numerical values below $\frac{10}{3}$, and nothing in this series depends on it beyond the table of Worked Example 12.5.1.

Remark 12.5.3 (On quoting exponents). Theorem 12.5.2 is historical context rather than a numbered theorem claim. The internally proved base broad–narrow result is Theorem 12.4.13; sharper numerical records are not used as proof inputs anywhere in this volume.

The restriction exponent in $\mathbb{R}^3$ has been improved at least five times since 2011, by Bourgain–Guth, Guth, Hickman–Rogers, Wang, and others, each by a fraction. A textbook that quotes the record dates itself within a year and misleads a reader who does not check.

What does not change is the *structure*: the conjecture is $p > 3$, the classical benchmark is $p \geq 4$, and every modern improvement lies between, obtained by the broad–narrow scheme of Chapter 12.4 with progressively better treatments of the narrow case. Volume 14 reports the position as of its writing, with the same caveat.

WORKED EXAMPLE 12.5.2— TIER B **Comparing restriction exponents across sources**

Restriction estimates are stated in several inequivalent forms and comparison requires care. Four things to check.

(1) *Which operator?* $\mathcal{R}$ (restriction) or $\mathcal{E}$ (extension). They are adjoint (Volume 6, Theorem 6.1.3), so an estimate for one is an estimate for the other at the conjugate exponent — but the conjugate must be taken.

(2) *Which data norm?* $L^\infty(d\sigma)$, $L^2(d\sigma)$, or $L^q(d\sigma)$. The conjecture concerns L^∞ ; Stein–Tomas is an L^2 statement, which is why it is provable by TT^* and

why it stops where it does.

(3) *Global or local?* $L^p(\mathbb{R}^n)$ or $L^p(B_R)$ with an R^ε loss. Volume 6, Theorem 6.5.2 shows these are equivalent for the sphere, so the distinction is usually harmless — but the equivalence is a theorem and uses compactness of the sphere.

(4) *Which surface?* Sphere, paraboloid, or cone. Sphere and paraboloid have the same exponents, both having non-vanishing Gaussian curvature; the cone has one vanishing principal curvature and its exponents differ, as Volume 5's Worked Example 5.5.4 predicts.

A worked comparison. Stein–Tomas is often quoted as “ $\mathcal{R} : L^p(\mathbb{R}^n) \rightarrow L^2(d\sigma)$ for $p \leq \frac{2(n+1)}{n+3}$ ”. Taking conjugates, $\frac{2(n+1)}{n+3}$ has dual exponent $\frac{2(n+1)}{n-1}$, which is the form used here. ✓ The two statements agree, and a reader who does not take the conjugate will think they disagree.

WORKED EXAMPLE 12.5.3— TIER C: RESEARCH TASTE **Why the restriction conjecture is hard**

Six volumes have circled this problem. It is worth collecting what is now known about *why* it resists.

(a) *It implies Kakeya.* Volume 6, Theorem 6.6.2. So it is at least as hard as a problem open since 1928, and no proof can avoid the tube geometry of Volumes 9 and 11.

(b) *The data norm is L^∞ .* Stein–Tomas is an L^2 theorem and is proved by TT^* in a page. The conjecture asks for L^∞ data, where no orthogonality is available, and Worked Example 12.4.1(c) identified this as where most of the broad–narrow loss occurs.

(c) *The extremal examples are not unique.* Knapp caps show the exponent cannot be lowered, but they are not the only near-extremal configurations, and a proof must handle all of them simultaneously. Volume 14 reports that the recent progress in $\mathbb{R}^3$ consists largely of classifying them.

(d) *Every known method concedes something.* Bilinear and multilinear estimates assume transversality; broad–narrow recovers the linear statement but recurses into lower dimensions with worse exponents; polynomial partitioning handles crossings and struggles with tangency. The concessions differ, and no combination has closed the gap.

(e) *What would suffice.* By Theorem 12.5.1 the target is $p > \frac{2n}{n-1}$, and by the reductions of Volume 6 a sharp Kakeya maximal bound would not by itself deliver it — the implication runs the other way. So restriction is strictly harder than Kakeya on current evidence, and Kakeya is open above dimension

three.

The honest position after twelve volumes. The problem is settled in the plane, open in every higher dimension, and the best methods improve the classical exponent by a fraction while leaving the structure of the difficulty — L^∞ data over a curved surface, with tangential configurations uncontrolled — exactly where Stein found it.

WORKED EXAMPLE 12.5.4— TIER B **Interpolating restriction estimates**

Restriction estimates are routinely combined by interpolation, and the mechanics are worth having.

(a) *What interpolates.* The operator $\mathcal{E}$ is linear, so Riesz–Thorin (Volume 2, Theorem 2.2.1) applies to the pair of exponents (p, q) in $\mathcal{E} : L^q(d\sigma) \rightarrow L^p(\mathbb{R}^n)$. A bound at (p_0, q_0) and one at (p_1, q_1) give bounds on the segment between.

(b) *The two anchors.* The trivial bound $\|\mathcal{E}g\|_{L^\infty} \leq \|g\|_{L^1(d\sigma)}$, valid always; and Stein–Tomas at $(\frac{2(n+1)}{n-1}, 2)$. Interpolating gives a triangle of admissible pairs, and Theorem 12.5.1 states its edge.

(c) *What does not interpolate.* The multilinear estimates, which are not bounds on a single linear operator. One interpolates them by Hölder in the product, which is weaker, and the broad–narrow argument must be run at each exponent separately.

(d) *A worked instance.* From Stein–Tomas at $q = 2$ and the trivial bound at $q = 1$, interpolation gives $\mathcal{E} : L^q \rightarrow L^p$ for $\frac{1}{q} = \frac{1-\vartheta}{1} + \frac{\vartheta}{2}$ and $\frac{1}{p} = \frac{\vartheta}{2(n+1)/(n-1)}$. Eliminating ϑ traces the edge of the triangle, and the conjecture asserts the region extends past it to $q = \infty$.

The caution. Interpolation never proves a new endpoint; it fills in between known ones. Every advance in this subject has been a new endpoint, and the interpolation is bookkeeping afterwards.

WORKED EXAMPLE 12.5.5— TIER A **The four conjectures, and their implications**

Volume 6, Theorem 6.7.4 established a chain of implications among the central conjectures. With twelve volumes written it can be stated in full.

Conjecture	Asserts	Status
Bochner–Riesz	S^δ bounded for $\delta > \max\{0, n \frac{1}{p} - \frac{1}{2} - \frac{1}{2}\}$	$n = 2$ only
Restriction	$\mathcal{E} : L^\infty(d\sigma) \rightarrow L^p, p > \frac{2n}{n-1}$	$n = 2$ only
Takeya	$\dim_{\mathbb{H}} K = n$	$n \leq 3$
Falconer	$\dim_{\mathbb{H}} E > \frac{n}{2} \Rightarrow m(\Delta E) > 0$	open, all $n \geq 2$

The implications. Bochner–Riesz $\Rightarrow$ restriction $\Rightarrow$ Takeya, by Volume 5, Worked Example 5.8.2 and Volume 6, Theorem 6.6.2. So the list is in decreasing order of strength, and Takeya is the weakest — which is why its resolution in $\mathbb{R}^3$ did not settle the others.

Where Falconer sits. Not on the chain. It is not known to imply or be implied by any of the three, though Theorem 12.8.2 shows it is a weighted restriction question and so shares their machinery. Volume 8, Worked Example 8.8.5 compared it with Takeya and found the methods transfer but the statements do not.

What is settled where. Every conjecture on the list is a theorem in the plane except Falconer, which is open even there — the best planar threshold being $\frac{5}{4}$ against the conjectured 1. That is worth noting: the *easiest* ambient dimension does not make Falconer easy.

Reading the table forward. Volume 13 addresses decoupling, which is not on this list but implies results about all four; Volume 14 reports the current position.

WORKED EXAMPLE 12.5.6— TIER B **Reading an exponent claim**

Papers in this area announce numerical exponents, and Theorem 12.5.3 declined to quote the current ones. A reader who does consult the literature needs five checks.

(1) *Which conjecture?* Restriction, Bochner–Riesz, Takeya and decoupling all have exponents, and Worked Example 12.5.5 shows they are related but distinct. An improvement to one need not touch the others.

(2) *Which formulation?* Worked Example 12.5.2 lists four things to fix: operator, data norm, global or local, and surface. Two papers can quote different numbers for the same theorem.

(3) *Conditional or not?* Many exponents are conditional on multilinear or bilinear estimates. Theorem 12.5.2 is conditional on Theorem 12.3.11 and Theorem 12.4.14, and says so.

(4) *What is the loss?* R^ε for every ε , or a fixed power. The first is harmless by Volume 9, Theorem 9.7.3; the second is not.

(5) *Superseded?* The $\mathbb{R}^3$ restriction exponent has moved at least five times since 2011. A number from a paper is the record as of that paper.

A worked reading. “Restriction holds for $p > 3.25$ in $\mathbb{R}^3$ ”: (1) restriction; (2) presumably L^∞ data, local, sphere — check; (3) unconditional if it is Guth’s, conditional in some later refinements; (4) R^ε ; (5) improved since. Placing it: between the conjectured 3 and Stein–Tomas’s 4, closing three-quarters of the gap.

WORKED EXAMPLE 12.5.7— TIER B Locating a result on the exponent line

For the sphere in $\mathbb{R}^3$ the exponent line runs from 3 (conjectured) to 4 (Stein–Tomas), and placing any result on it takes three steps.

Step 1: normalise the statement. Convert to the form $\mathcal{E} : L^\infty(d\sigma) \rightarrow L^p(B_R)$ with R^ε loss, using Worked Example 12.5.2’s four checks.

Step 2: read off p . Then $\frac{4-p}{4-3} = 4 - p$ is the fraction of the gap closed.

Step 3: check the hypotheses. Unconditional, or conditional on a multilinear or bilinear input? Theorem 12.5.2 is conditional on Theorem 12.3.11 and Theorem 12.4.14.

Worked placements.

p	fraction of gap closed	
4	0	Stein–Tomas
3.3	0.7	Bourgain–Guth
3.25	0.75	Guth, polynomial partitioning
3	1	the conjecture

What the table shows. Two decades of work have closed roughly three quarters of the gap in $\mathbb{R}^3$, and the remaining quarter has resisted several distinct methods. Theorem 12.5.3 explains why no current figure is quoted; the shape of the line does not change.

Drills

Drill 12.5.1. Verify $\frac{2(n+1)}{n-1} - \frac{2n}{n-1} = \frac{2}{n-1}$ and tabulate for $n = 2, \dots, 6$.

Drill 12.5.2. Confirm that $\frac{2(n+1)}{n+3}$ and $\frac{2(n+1)}{n-1}$ are conjugate exponents.

Drill 12.5.3. For $n = 3$, locate $\frac{33}{10}$ between the conjectured and classical exponents, and compute what fraction of the gap it closes.

Drill 12.5.4. Explain, using Volume 5 Worked Example 5.5.4, why the cone's restriction exponents differ from the sphere's.

Notes

Bourgain and Guth's paper is from 2011. The subsequent improvements in $\mathbb{R}^3$ use polynomial partitioning and are due to Guth, and to Hickman, Rogers and others; Theorem 12.5.3 explains why no current number is quoted here.

Chapter 12.6

Bilinear Restriction

The multilinear theory of Chapter 12.3 uses n factors because n vectors are needed to span $\mathbb{R}^n$ (Theorem 12.2.3). The bilinear theory uses two, and is therefore weaker in high dimensions — but it is older, sharper in the plane, and it is the tool the distance set problem needs, because that problem is intrinsically two-point.

12.6.1 The planar estimate, proved

Theorem 12.6.1 (Bilinear L^2 estimate for transverse arcs). *Let $S_1, S_2 \subseteq \mathbb{S}^1$ be arcs whose unit normals satisfy $|\sin \angle(\omega_1, \omega_2)| \geq \nu$ for all $\omega_j \in S_j$. Then for $g_j \in L^2(S_j)$,*

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(\mathbb{R}^2)} \lesssim \nu^{-1/2} \|g_1\|_{L^2(d\sigma)} \|g_2\|_{L^2(d\sigma)}.$$

There is no power of R : the estimate is global.

Idea of the proof. The Fourier transform of a product is a convolution, so by Plancherel the left side is the L^2 norm of $(g_1 d\sigma_1) * (g_2 d\sigma_2)$. That convolution is the pushforward of $g_1 \otimes g_2$ under the addition map $(\omega_1, \omega_2) \mapsto \omega_1 + \omega_2$, and the Jacobian of that map is exactly $\sin \angle(\omega_1, \omega_2)$ — which transversality bounds below. A bounded Jacobian means the pushforward has bounded density, and Cauchy–Schwarz finishes.

The entire content is that *transversality is the non-degeneracy of the addition map*.

Lemma 12.6.2 (The addition map). *Let $\Phi : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \mathbb{R}^2$ be $\Phi(\omega_1, \omega_2) = \omega_1 + \omega_2$. In the parametrisation $\omega_j = (\cos \theta_j, \sin \theta_j)$,*

$$|\det d\Phi| = |\sin(\theta_1 - \theta_2)| = |\sin \angle(\omega_1, \omega_2)|.$$

Proof. Differentiating $\Phi(\theta_1, \theta_2) = (\cos \theta_1 + \cos \theta_2, \sin \theta_1 + \sin \theta_2)$,

$$d\Phi = \begin{pmatrix} -\sin \theta_1 & -\sin \theta_2 \\ \cos \theta_1 & \cos \theta_2 \end{pmatrix},$$

whose determinant is $-\sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_1 = -\sin(\theta_1 - \theta_2)$. $\square$

Proof of Theorem 12.6.1. Step 1. Plancherel.

$\mathcal{E}_j g_j = \widehat{g_j d\sigma_j}$ up to a reflection, so $\mathcal{E}_1 g_1 \cdot \mathcal{E}_2 g_2 = \widehat{(g_1 d\sigma_1) * (g_2 d\sigma_2)}$ and by Plancherel

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(\mathbb{R}^2)} = \|(g_1 d\sigma_1) * (g_2 d\sigma_2)\|_{L^2(\mathbb{R}^2)}.$$

Step 2. *Identify the convolution as a pushforward.*

For a test function φ ,

$$\begin{aligned} \int \varphi d((g_1 d\sigma_1) * (g_2 d\sigma_2)) \\ = \iint \varphi(\omega_1 + \omega_2) g_1(\omega_1) g_2(\omega_2) d\theta_1 d\theta_2, \end{aligned}$$

so the convolution is $\Phi_*(g_1 \otimes g_2)$. By Theorem 12.6.2 and the co-area formula, it is absolutely continuous with density

$$F(\xi) = \sum_{(\omega_1, \omega_2) \in \Phi^{-1}(\xi)} \frac{g_1(\omega_1) g_2(\omega_2)}{|\sin \angle(\omega_1, \omega_2)|}.$$

For ξ with $0 < |\xi| < 2$ the fibre $\Phi^{-1}(\xi)$ consists of at most two points — given the sum of two unit vectors, the pair is determined up to swapping — so the sum has at most two terms.

Step 3. *Estimate the density.*

Let $J(\theta_1, \theta_2) = |\sin(\theta_1 - \theta_2)|$. Since each fibre has at most two points, Cauchy–Schwarz on the fibre gives

$$|F(\xi)|^2 \leq 2 \sum_{(\theta_1, \theta_2) \in \Phi^{-1}(\xi)} \frac{|g_1(\theta_1)|^2 |g_2(\theta_2)|^2}{J(\theta_1, \theta_2)^2}.$$

Integrating and applying the area formula, whose Jacobian is J , gives

$$\begin{aligned} \int |F|^2 &\leq 2 \iint \frac{|g_1(\theta_1)|^2 |g_2(\theta_2)|^2}{J(\theta_1, \theta_2)} d\theta_1 d\theta_2 \\ &\leq \frac{2}{\nu} \|g_1\|_{L^2}^2 \|g_2\|_{L^2}^2, \end{aligned}$$

and taking square roots gives the stated $\nu^{-1/2}$. $\square$

Where we are. Two features distinguish Theorem 12.6.1 from everything in Chapter 12.3.

It is global. There is no R , no R^ε , and no localisation to a ball. The estimate holds on all of $\mathbb{R}^2$.

It is elementary. Plancherel, a Jacobian, and Cauchy–Schwarz.

This is the estimate that exposed the defect in Theorem 12.3.3: the wave packet argument, specialised to $n = 2$, gives $\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^2(B_R)} \lesssim R \|g_1\| \|g_2\|$, whereas the truth has no power of R at all. A method that is lossy in the one case one can compute by hand is lossy in general, and the Pitfall there identified where.

The moral is worth stating: *keep an independently computable case.* Every method in this series has one — the plane for Keakeya, $n = 2$ for restriction, the box for Loomis–Whitney — and checking against it is the cheapest available error detection.

12.6.2 Higher dimensions

Theorem 12.6.3 (Tao). *Work in $\mathbb{R} \times \mathbb{R}^m$, so that the ambient dimension is $n = m + 1$, and let*

$$S = \left\{ (\tau, \xi) \in \mathbb{R} \times \mathbb{R}^m : \tau = -\frac{1}{2}|\xi|^2 \right\}$$

be the paraboloid. Let $S_1, S_2 \subseteq S$ be disjoint compact subsets. Then for every $q > \frac{m+3}{m+1} = \frac{n+2}{n}$,

$$\|\widehat{f_1 d\sigma_1} \widehat{f_2 d\sigma_2}\|_{L^q(\mathbb{R} \times \mathbb{R}^m)} \lesssim_{q, S_1, S_2} \|f_1\|_{L^2(d\sigma_1)} \|f_2\|_{L^2(d\sigma_2)}.$$

Where we are. The exponent agrees with the one stated earlier in this chapter: $\frac{m+3}{m+1} = \frac{n+2}{n}$, giving $2, \frac{5}{3}, \frac{3}{2}, \frac{7}{5}$ for $n = 2, 3, 4, 5$. At $n = 2$ it is $q = 2$ and the theorem is Theorem 12.6.1, proved in Section 12.6.1 by Plancherel alone.

The proof below is Tao’s adaptation of Wolff’s cone argument. It occupies the rest of this chapter and its shape is: reduce to a localised estimate; induct on scales; decompose into wave packets; split off a part handled by the induction; decompose the remainder at the fine scale $\sqrt{R}$; prove a local L^2 estimate exploiting a second frequency constraint; and finish with a Keakeya-type bush count.

12.6.3 Reduction to a localised estimate

Definition 12.6.4. Write $\mathcal{R}^*(2 \times 2 \rightarrow q, \alpha)$ for the statement

$$\|\widehat{f_1 d\sigma_1} \widehat{f_2 d\sigma_2}\|_{L^q(B(z_0, R))} \leq C R^\alpha \|f_1\|_{L^2} \|f_2\|_{L^2}$$

for all $R \geq 1$ and all spacetime balls $B(z_0, R)$.

The notation includes the following uniformity convention, needed when the induction is applied to packet sums. We require the same estimate, with one constant, for every pair of normalised subpatches lying in $\tilde{S}_1, \tilde{S}_2$ and retaining a fixed separation and a fixed buffer from their boundaries. This does not strengthen the theorem: compactness gives a finite partition into such pairs. Conversely, every free Schrödinger wave u_j whose initial spatial Fourier transform is supported in one of those patches has the form $\widehat{g_j d\sigma_j}$, and

$$\|g_j\|_{L^2(d\sigma_j)} \asymp P(u_j)^{1/2}.$$

Thus the local estimate may be applied directly to a sum of wave packets. The buffer guarantees that the $O(R^{-1/2})$ broadening created in Theorem 12.6.6 remains within this uniform family. This is the precise induction interface used below.

By a finite partition and a rotation, scaling and Galilean transformation — the last translating ξ while preserving S — we may assume

$$S_1 \subseteq \{|\xi - e_1| \leq \frac{1}{100m}\}, \quad S_2 \subseteq \{|\xi + e_1| \leq \frac{1}{100m}\}.$$

Fix once and for all the slightly enlarged patches

$$\begin{aligned} \tilde{S}_1 &= \left\{ \left(-\frac{1}{2}|\xi|^2, \xi \right) \in S : |\xi - e_1| \leq \frac{1}{50m} \right\}, \\ \tilde{S}_2 &= \left\{ \left(-\frac{1}{2}|\xi|^2, \xi \right) \in S : |\xi + e_1| \leq \frac{1}{50m} \right\}. \end{aligned}$$

Thus S_j lies in the interior of $\tilde{S}_j$ with a fixed buffer; that buffer will allow the approximate conservation laws in the local L^2 argument to be replaced by exact ones without leaving the permitted frequency patches. By the ε -removal lemma of Volume 6, Theorem 6.5.2, it suffices to prove $\mathcal{R}^*(2 \times 2 \rightarrow \frac{m+3}{m+1}, \alpha)$ for every $\alpha > 0$.

Proposition 12.6.5 (The inductive step). *If $\mathcal{R}^*(2 \times 2 \rightarrow \frac{m+3}{m+1}, \alpha)$ holds for some $\alpha > 0$, then it holds with α replaced by $\max\{(1 - \delta)\alpha, C\delta\} + C\varepsilon$, for all $0 < \delta, \varepsilon \ll 1$, with C independent of δ, ε .*

Proof of Theorem 12.6.3 from Theorem 12.6.5. The crude bound $\|\widehat{f_j d\sigma_j}\|_{L^\infty} \lesssim \|f_j\|_{L^2}$ gives the localised estimate for some large α . Choosing δ and ε so that $\max\{(1 - \delta)\alpha, C\delta\} + C\varepsilon \leq \alpha - c\alpha^2$ and iterating, the infimum of admissible α is 0. $\square$

12.6.4 Wave packets and the two-ends reduction

Fix $R \gg 1$, let $X = R^{1/2}\mathbb{Z}^m$ and $V = R^{-1/2}\mathbb{Z}^m$, and let V_j be those $v \in V$ with $(-\frac{1}{2}|v|^2, v) \in \tilde{S}_j$. An $\tilde{S}_j$ -tube is

$$T = \{(t, x) : \frac{R}{2} \leq t \leq R, |x - (x(T) + tv(T))| \leq R^{1/2}\}$$

with $x(T) \in X$, $v(T) \in V_j$. Because of the normalisation $\tau = -\frac{1}{2}|\xi|^2$, the group velocity equals the spatial frequency.

Lemma 12.6.6 (Wave packet decomposition). *For smooth f_j on S_j there is a decomposition $\widehat{f_j d\sigma_j} = \sum_{T_j} c_{T_j} \phi_{T_j}$ over $\tilde{S}_j$ -tubes with*

$$\left(\sum_{T_j} |c_{T_j}|^2 \right)^{1/2} \lesssim \|f_j\|_{L^2},$$

where each ϕ_{T_j} solves the free Schrödinger equation $4\pi i u_t - \Delta u = 0$, has $\widehat{\phi_{T_j}(t)}$ supported in $\{\xi = v(T_j) + O(R^{-1/2})\}$, and satisfies

$$|\phi_{T_j}(t, x)| \leq C_N R^{-m/4} \left(1 + \frac{|x - (x(T_j) + tv(T_j))|}{R^{1/2}} \right)^{-N}.$$

Moreover $P(\sum_{T_j \in \mathbb{T}_j} \phi_{T_j}) \lesssim \#\mathbb{T}_j$, where $P(u) = \|u(t)\|_{L_x^2}^2$ is the conserved probability.

Proof. Assume first f_j supported in a cap $\{\xi = v + O(R^{-1/2})\}$. Take η Schwartz with $\hat{\eta}$ supported in a ball and $\sum_{k \in \mathbb{Z}^m} \eta(x - k) \equiv 1$, and let $F(x) = \widehat{f_j d\sigma_j}(0, x)$. Then $F = \sum_{x_0 \in X} \eta\left(\frac{x - x_0}{R^{1/2}}\right) F$, and each summand has spatial Fourier support in the same cap. Let u_{x_0} be the Schrödinger evolution of that summand, put $c_T = R^{m/4} \mathcal{M}F(x(T))$ with $\mathcal{M}$ the Hardy–Littlewood maximal function, and $\phi_T = u_{x(T)}/c_T$.

Since $\hat{F}$ is supported in a ball of radius $R^{-1/2}$, $F = F * \psi$ with $|\psi|(x) \lesssim_N R^{-m/2}(1 + |x|/R^{1/2})^{-N}$, so $\mathcal{M}F(x) \approx \mathcal{M}F(x')$ for $|x - x'| \lesssim R^{1/2}$. If Q_{x_0} is the lattice cube of side $R^{1/2}$ centred at x_0 , this gives

$$\begin{aligned} \sum_{x_0 \in X} |c_{T(x_0, v)}|^2 &= R^{m/2} \sum_{x_0 \in X} |\mathcal{M}F(x_0)|^2 \\ &\lesssim \sum_{x_0 \in X} \int_{Q_{x_0}} |\mathcal{M}F(x)|^2 dx = \|\mathcal{M}F\|_{L^2}^2 \lesssim \|F\|_{L^2}^2. \end{aligned}$$

The last inequality is the maximal theorem (Volume 1, Theorem 1.7.3); Plancherel then gives the required coefficient bound.

It remains to derive the packet decay rather than merely invoke it. Choose a smooth cutoff χ_v equal to 1 on the frequency support of F and supported in $B(v, Cr)$, where $r = R^{-1/2}$. The frequency-localised Schrödinger kernel is

$$K_v(t, z) = \int_{\mathbb{R}^m} e^{2\pi i(z \cdot \xi - \frac{1}{2}t|\xi|^2)} \chi_v(\xi) d\xi.$$

After the change of variables $\xi = v + r\zeta$, the rescaled quadratic coefficient is $tr^2 \approx 1$ for $R/2 \leq t \leq R$. Repeated integration by parts in ζ away from the stationary region, and the trivial bound inside it, give for every N

$$|K_v(t, z)| \lesssim_N R^{-m/2} \left(1 + \frac{|z - tv|}{R^{1/2}}\right)^{-N}. \quad (12.6.1)$$

This is also the direct rescaled form of the stationary-phase estimate in Volume 5, Theorem 5.3.3.

Writing $\eta_{x_0}(y) = \eta((y - x_0)/R^{1/2})$, we have $u_{x_0}(t) = K_v(t) * (\eta_{x_0}F)$. The reproducing formula and dyadic annuli give, for any M ,

$$\int_{\mathbb{R}^m} |\eta_{x_0}(y)F(y)| \left(1 + \frac{|y - x_0|}{R^{1/2}}\right)^M dy \lesssim_M R^{m/2} \mathcal{M}F(x_0);$$

indeed the k -th annulus has volume $O(2^{km}R^{m/2})$, its average of $|F|$ is at most $\mathcal{M}F(x_0)$, and the Schwartz decay of η is chosen faster than $2^{-k(M+m+1)}$. Combining this estimate with (12.6.1), and using

$$1 + \frac{|x - (x_0 + tv)|}{R^{1/2}} \lesssim \left(1 + \frac{|x - y - tv|}{R^{1/2}}\right) \left(1 + \frac{|y - x_0|}{R^{1/2}}\right),$$

we obtain

$$|u_{x_0}(t, x)| \lesssim_N \mathcal{M}F(x_0) \left(1 + \frac{|x - (x_0 + tv)|}{R^{1/2}}\right)^{-N}.$$

Division by $c_T = R^{m/4} \mathcal{M}F(x_0)$ proves the asserted pointwise packet estimate (with $\phi_T = 0$ when $c_T = 0$).

Finally the pointwise estimate gives $P(\phi_T) \lesssim 1$. For fixed v , the centres $x(T) + tv$ form an $R^{1/2}$ -separated lattice, so the rapidly decaying weights above have uniformly bounded overlap. Cauchy–Schwarz with those weights yields

$$P\left(\sum_{\substack{T \in \mathbb{T}_j \\ v(T)=v}} \phi_T\right) \lesssim \sum_{\substack{T \in \mathbb{T}_j \\ v(T)=v}} P(\phi_T) \lesssim \#\{T \in \mathbb{T}_j : v(T) = v\}.$$

The frequency caps belonging to distinct v 's have bounded overlap, so Plancherel and summation in v prove $P(\sum_{T \in \mathbb{T}_j} \phi_T) \lesssim \#\mathbb{T}_j$.

For general f_j , decompose into caps and sum, using $\sum_v \|f_{j,v}\|_{L^2}^2 \approx \|f_j\|_{L^2}^2$. $\square$

After discarding tubes far from $B(0, CR)$ and dyadic pigeonholing in the coefficient sizes — both costing only $R^{O(\varepsilon)}$ — it suffices to prove:

Proposition 12.6.7. *For collections $\mathbb{T}_1, \mathbb{T}_2$ of tubes meeting $B(0, CR)$,*

$$\left\| \sum_{T_1 \in \mathbb{T}_1} \sum_{T_2 \in \mathbb{T}_2} \phi_{T_1} \phi_{T_2} \right\|_{L^{\frac{m+3}{m+1}}(Q_R)} \lesssim (R^{(1-\delta)\alpha} + R^{C\delta}) (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2},$$

where $Q_R = \{\frac{R}{2} \leq t \leq R, |x| \leq R\}$ and $\lesssim$ allows a factor $C_\varepsilon R^\varepsilon$.

Local/global reduction for Theorem 12.6.7. Put $p = (m+3)/(m+1)$ and cover Q_R by finitely overlapping balls B of radius $R^{1-\delta}$. There are $O(R^{C\delta})$ such balls. Given any relation $\sim$ between tubes and balls satisfying

$$\#\{B : T \sim B\} \lesssim 1 \quad \text{for every } T, \quad (12.6.2)$$

use a bounded partition of unity subordinate to this cover. On each B split the pairs into the *local* ones, for which both $T_1 \sim B$ and $T_2 \sim B$, and the remaining *global* ones.

Step 1. *The local contribution.*

Set $U_{j,B} = \sum_{T_j \sim B} \phi_{T_j}$. The induction hypothesis at radius $R^{1-\delta}$ and Theorem 12.6.6 give

$$\begin{aligned} \|U_{1,B} U_{2,B}\|_{L^p(B)} &\lesssim R^{(1-\delta)\alpha} P(U_{1,B})^{1/2} P(U_{2,B})^{1/2} \\ &\lesssim R^{(1-\delta)\alpha} \#\{T_1 : T_1 \sim B\}^{1/2} \#\{T_2 : T_2 \sim B\}^{1/2}. \end{aligned}$$

Here $U_{j,B}$ is a free Schrödinger wave and the uniformity convention after Theorem 12.6.4 identifies $P(U_{j,B})^{1/2}$ with the L^2 norm of its surface density, so this invocation of the induction hypothesis introduces no frequency-thickening loss. The triangle inequality in B , followed by Cauchy–Schwarz and (12.6.2), yields

$$\begin{aligned} \|\text{local}\|_{L^p(Q_R)} &\lesssim R^{(1-\delta)\alpha} \prod_{j=1}^2 \left(\sum_B \#\{T_j : T_j \sim B\} \right)^{1/2} \\ &\lesssim R^{(1-\delta)\alpha} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}. \end{aligned}$$

Step 2. *The global L^1 estimate on one coarse ball.*

A global pair has $T_1 \not\sim B$ or $T_2 \not\sim B$. By the triangle inequality and symmetry it is enough to treat

$$F_B = \left(\sum_{T_1 \not\sim B} \phi_{T_1} \right) \left(\sum_{T_2 \in \mathbb{T}_2} \phi_{T_2} \right).$$

Every B lies in a time interval of length at most $2R$. At each time, Cauchy–Schwarz in x , followed by conservation of probability, gives

$$\begin{aligned} \|F_B\|_{L^1(B)} &\leq \int_{\text{proj}_t B} \left\| \sum_{T_1 \not\sim B} \phi_{T_1}(t) \right\|_{L_x^2} \left\| \sum_{T_2} \phi_{T_2}(t) \right\|_{L_x^2} dt \\ &\lesssim R(\#\mathbb{T}_1)^{1/2}(\#\mathbb{T}_2)^{1/2}. \end{aligned}$$

Step 3. *The remaining L^2 estimate.*

It remains to prove

$$\|F_B\|_{L^2(B)} \lesssim R^{C\delta} R^{-\frac{m-1}{4}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}. \quad (12.6.3)$$

Indeed, let $\theta = 4/(m+3)$. Then $1/p = (1-\theta) + \theta/2$, and interpolation between the preceding L^1 bound and (12.6.3) gives

$$\begin{aligned} \|F_B\|_{L^p(B)} &\lesssim R^{1-\theta} R^{-\theta(m-1)/4} R^{C\delta} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2} \\ &= R^{C\delta} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}, \end{aligned}$$

because $1 - \theta - \theta(m-1)/4 = 0$. Summing over the $O(R^{C\delta})$ coarse balls, and treating the symmetric global term in the same way, gives the second term in Theorem 12.6.7.

Thus (12.6.3) is the sole remaining estimate. Its power $R^{-(m-1)/4}$ is sharp; see Worked Example 12.6.1. $\square$

WORKED EXAMPLE 12.6.1— TIER A The L^2 exponent is sharp

The power $R^{-(m-1)/4}$ in (12.6.3) cannot be improved, and the example showing it is the same one that forces the exponent $\frac{m+3}{m+1}$.

The configuration. Let $\pi = \{(t, x_1 e_1) : t, x_1 = O(R)\}$ be a spacetime plane. Let $\mathbb{T}_1$ consist of the $O(\sqrt{R})$ tubes with velocity e_1 meeting π , and $\mathbb{T}_2$ the $O(\sqrt{R})$ tubes with velocity $-e_1$ meeting π .

The two sides. By Theorem 12.6.6 each ϕ_{T_j} has size $\approx R^{-m/4}$ on its tube, so the double sum has magnitude $\approx R^{-m/2}$ on the $\sqrt{R}$ -neighbourhood of π , whose volume is $R^{(m+3)/2}$. Hence

$$\begin{aligned} \text{LHS} &\approx R^{-\frac{m}{2}} \cdot R^{\frac{m+3}{4}} \\ &= R^{\frac{3-m}{4}}, \quad \text{RHS} \\ &\approx R^{-\frac{m-1}{4}} \cdot R^{\frac{1}{4}} \cdot R^{\frac{1}{4}} \\ &= R^{\frac{3-m}{4}}. \end{aligned}$$

Equal. $\checkmark$

Numerically. At $m = 2$ (so $n = 3$) both sides are $R^{1/4}$; at $m = 3$ both are R^0 ; at $m = 4$ both are $R^{-1/4}$.

What the example is. Two families of tubes, oppositely directed, all piled onto a single plane — the configuration in which a bilinear expression concentrates as much as transversality permits. It is the bilinear analogue of the Knapp example, and it is why no argument can do better than $\frac{m+3}{m+1}$.

12.6.5 The improved local estimate

Fix once and for all a finitely overlapping cover $\mathbf{q}$ of a slight enlargement of Q_R by balls q of radius $R^{1/2}$. This cover is independent of the coarse ball B . For a fixed B from (12.6.3), only balls meeting B are needed; because $R^{1/2} \ll R^{1-\delta}$, all of them lie in $2B$. Write $R^\delta q$ for the concentric ball of radius $R^{1/2+\delta}$ and define

$$\mathbb{T}_j(q) = \{T_j \in \mathbb{T}_j : T_j \cap R^\delta q \neq \emptyset\}, \quad \mathbb{T}_1^{\not\sim B}(q) = \{T_1 \in \mathbb{T}_1(q) : T_1 \not\sim B\}.$$

If $T_j \notin \mathbb{T}_j(q)$, the pointwise estimate of Theorem 12.6.6 makes $\phi_{T_j} = O_N(R^{-N})$ on q . The number of tubes meeting $B(0, CR)$ is polynomial in R (both the position and velocity sets are lattices in bounded regions), so choosing N sufficiently large and using the finite overlap of $\mathbf{q}$ gives

$$\|F_B\|_{L^2(B)}^2 \lesssim \sum_{\substack{q \in \mathbf{q} \\ q \subseteq 2B}} \left\| \sum_{T_1 \in \mathbb{T}_1^{\not\sim B}(q)} \sum_{T_2 \in \mathbb{T}_2(q)} \phi_{T_1} \phi_{T_2} \right\|_{L^2(q)}^2 + O(R^{-100m}). \quad (12.6.4)$$

For dyadic $\mu_1, \mu_2 \geq 1$, let

$$\mathbf{q}(\mu_1, \mu_2) = \{q \in \mathbf{q} : \mu_j \leq \#\mathbb{T}_j(q) < 2\mu_j, \ j = 1, 2\}.$$

Only polynomially large multiplicities occur, so there are $O((\log R)^2)$ nonempty classes. For $T_1 \in \mathbb{T}_1$ define

$$\lambda(T_1, \mu_1, \mu_2) = \#\{q \in \mathbf{q}(\mu_1, \mu_2) : T_1 \cap R^\delta q \neq \emptyset\},$$

and, for dyadic $\lambda_1 \geq 1$, put

$$\mathbb{T}_1[\lambda_1, \mu_1, \mu_2] = \{T_1 : \lambda_1 \leq \lambda(T_1, \mu_1, \mu_2) < 2\lambda_1\}.$$

There are only $O(\log R)$ nonempty λ_1 levels. Cauchy–Schwarz in these levels costs at most a power of $\log R$, which is absorbed by R^ε . Consequently (12.6.3) follows once, for every fixed $(\mu_1, \mu_2, \lambda_1)$, we prove

$$\begin{aligned} \sum_{\substack{q \in \mathbf{q}(\mu_1, \mu_2) \\ q \subseteq 2B}} \left\| \sum_{T_1 \in \mathbb{T}_1^{\not\sim B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]} \sum_{T_2 \in \mathbb{T}_2(q)} \phi_{T_1} \phi_{T_2} \right\|_{L^2(q)}^2 \\ \lesssim R^{C\delta} R^{-(m-1)/2} (\#\mathbb{T}_1)(\#\mathbb{T}_2). \end{aligned} \quad (12.6.5)$$

Put

$$\Omega_1 = \left\{ \xi \in \mathbb{R}^m : |\xi - e_1| \leq \frac{1}{20m} \right\}, \quad \Omega_2 = \left\{ \xi \in \mathbb{R}^m : |\xi + e_1| \leq \frac{1}{20m} \right\}.$$

The velocity supports of the $\tilde{S}_j$ -packets lie strictly inside Ω_j . In particular, vectors in Ω_1 and Ω_2 are separated by a fixed positive distance, uniformly in R .

Definition 12.6.8. For $\xi_1 \in \Omega_1$, $\xi'_2 \in \Omega_2$ put

$$\begin{aligned} \pi(\xi_1, \xi'_2) &= \left\{ \xi'_1 \in \Omega_1 : \xi_1 + \xi_2 = \xi'_1 + \xi'_2, \right. \\ &\quad \left. |\xi_1|^2 + |\xi_2|^2 = |\xi'_1|^2 + |\xi'_2|^2 \text{ for some } \xi_2 \in \Omega_2 \right\}. \end{aligned}$$

Lemma 12.6.9 (The second constraint is a hyperplane). *$\pi(\xi_1, \xi'_2)$ is contained in the hyperplane through ξ_1 orthogonal to $\xi'_2 - \xi_1$; that is,*

$$\langle \xi'_1 - \xi_1, \xi'_2 - \xi_1 \rangle = 0 \quad \text{for } \xi'_1 \in \pi(\xi_1, \xi'_2).$$

Proof. The first conservation law gives $\xi_2 = \xi'_1 + \xi'_2 - \xi_1$. Substitution into the second one gives

$$\begin{aligned} 0 &= |\xi_1|^2 + |\xi'_1 + \xi'_2 - \xi_1|^2 - |\xi'_1|^2 - |\xi'_2|^2 \\ &= 2\langle \xi'_1 - \xi_1, \xi'_2 - \xi_1 \rangle. \end{aligned}$$

Equivalently: $\xi_1, \xi'_1, \xi_2, \xi'_2$ form a rectangle in $\mathbb{R}^m$. □

Where we are. Theorem 12.6.9 is Tao's innovation over Wolff. For the *cone* the corresponding geometric fact is that a light ray meets the light cone through a point in at most one place, which follows from the single vanishing principal curvature. The paraboloid has no vanishing curvature and that argument is unavailable.

What replaces it is the observation that the *two* frequency constraints — conservation of $\xi_1 + \xi_2$ and of $|\xi_1|^2 + |\xi_2|^2$ — together confine ξ'_1 to a hyperplane, not merely to a hypersurface. That extra degree of freedom removed is exactly what makes the collection $\mathbb{T}_2$ behave like a family of null directions, so that Wolff's counting argument applies after all.

Lemma 12.6.10 (Improved local estimate). *Put*

$$\mathbb{T}'_1 = \mathbb{T}_1^{\mathcal{J}B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2], \quad \mathbb{T}'_2 = \mathbb{T}_2(q).$$

With $\nu(q, \lambda_1, \mu_1, \mu_2) = \sup_{\xi_1, \xi'_2} \# \mathbb{T}_1^{\mathcal{J}B}(q, \lambda_1, \mu_1, \mu_2, \xi_1, \xi'_2)$, where the latter is the set of $T_1 \in \mathbb{T}_1^{\mathcal{J}B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$ with $v(T_1)$ within $O(R^{C\delta} R^{-1/2})$ of

$\pi(\xi_1, \xi_2^l),$

$$\begin{aligned} & \left\| \sum_{T_1 \in \mathbb{T}'_1} \sum_{T_2 \in \mathbb{T}'_2} \phi_{T_1} \phi_{T_2} \right\|_{L^2(q)}^2 \\ & \lesssim R^{C\delta} R^{-\frac{m-1}{2}} \nu(\#\mathbb{T}'_1)(\#\mathbb{T}'_2). \end{aligned}$$

Proof. Write $r = R^{-1/2}$. Since the local norm is bounded by the global one, it is enough to estimate the sum in $L^2(\mathbb{R} \times \mathbb{R}^m)$.

Step 1. *The frequency amplitude of one packet.*

Because ϕ_T is a free Schrödinger wave, there is an amplitude a_T such that

$$\phi_T(t, x) = \int_{\mathbb{R}^m} a_T(\xi) e^{2\pi i(x \cdot \xi - \frac{1}{2}t|\xi|^2)} d\xi, \quad \text{supp } a_T \subseteq B(v(T), Cr). \quad (12.6.6)$$

Choose any $t_0 \in [R/2, R]$. The pointwise packet estimate in Theorem 12.6.6, with $N > m$, gives

$$\|\phi_T(t_0)\|_{L^1_x} \lesssim R^{-m/4} \int_{\mathbb{R}^m} \left(1 + \frac{|x - (x(T) + t_0 v(T))|}{R^{1/2}}\right)^{-N} dx \lesssim R^{m/4}.$$

The spatial Fourier transform of $\phi_T(t_0)$ is $e^{-\pi i t_0 |\xi|^2} a_T(\xi)$, so Fourier inversion yields the uniform amplitude bound

$$\|a_T\|_{L^\infty} \lesssim R^{m/4}. \quad (12.6.7)$$

Step 2. *The Fourier transform of one packet product.*

For $T_1 \in \mathbb{T}'_1$ and $T_2 \in \mathbb{T}'_2$, put $G_{12} = \widehat{\phi_{T_1} \phi_{T_2}}$ in spacetime. From (12.6.6), up to a harmless Fourier-normalisation constant,

$$G_{12}(\tau, \zeta) = \int_{\mathbb{R}^m} a_{T_1}(\eta) a_{T_2}(\zeta - \eta) \delta_0\left(\tau + \frac{1}{2}|\eta|^2 + \frac{1}{2}|\zeta - \eta|^2\right) d\eta. \quad (12.6.8)$$

The integrand can be nonzero only when $\eta = v(T_1) + O(r)$ and $\zeta - \eta = v(T_2) + O(r)$. Hence G_{12} is supported in the spacetime ball of radius Cr centred at

$$c(T_1, T_2) = \left(-\frac{1}{2}(|v(T_1)|^2 + |v(T_2)|^2), v(T_1) + v(T_2)\right). \quad (12.6.9)$$

For fixed ζ , set $F_\zeta(\eta) = \frac{1}{2}|\eta|^2 + \frac{1}{2}|\zeta - \eta|^2$. On the support of the integrand,

$$|\nabla_\eta F_\zeta(\eta)| = |2\eta - \zeta| = |v(T_1) - v(T_2) + O(r)| \gtrsim 1,$$

because the two velocity patches are separated. The coarea formula therefore turns (12.6.8) into an integral over a uniformly Lipschitz $(m-1)$ -dimensional

slice of a ball of radius Cr . That slice has measure $O(r^{m-1})$ (and, when $m = 1$, contains $O(1)$ points). Using (12.6.7) twice gives

$$\|G_{12}\|_{L^\infty} \lesssim R^{m/2} r^{m-1} = R^{1/2}. \quad (12.6.10)$$

The support ball has volume $O(r^{m+1})$. Plancherel and (12.6.10) now give the fully scaled packet-product estimate

$$\|\phi_{T_1} \phi_{T_2}\|_{L^2}^2 = \|G_{12}\|_{L^2}^2 \lesssim R r^{m+1} = R^{-(m-1)/2}. \quad (12.6.11)$$

Step 3. *Overlap forces approximate momentum and energy conservation.*

Expanding the global square gives

$$\left\| \sum_{T_1 \in \mathbb{T}'_1} \sum_{T_2 \in \mathbb{T}'_2} \phi_{T_1} \phi_{T_2} \right\|_{L^2}^2 = \sum_{T_1, T'_1} \sum_{T_2, T'_2} \langle \phi_{T_1} \phi_{T_2}, \phi_{T'_1} \phi_{T'_2} \rangle.$$

By (12.6.9), an inner product vanishes unless the two support balls overlap. If

$$a = v(T_1), \quad b = v(T_2), \quad c = v(T'_1), \quad d = v(T'_2),$$

overlap implies the quantitative conservation laws

$$|a + b - c - d| \lesssim r, \quad ||a|^2 + |b|^2 - |c|^2 - |d|^2| \lesssim r. \quad (12.6.12)$$

Moreover, Cauchy–Schwarz and (12.6.11) bound every surviving inner product by

$$|\langle \phi_{T_1} \phi_{T_2}, \phi_{T'_1} \phi_{T'_2} \rangle| \lesssim R^{-(m-1)/2}. \quad (12.6.13)$$

Step 4. *The approximate constraints place c near the exact set $\pi(a, d)$.*

The first relation in (12.6.12) writes $b = c + d - a + e$ with $|e| \lesssim r$. All four velocities remain in fixed bounded sets, so the second relation and

$$|a|^2 + |c + d - a|^2 - |c|^2 - |d|^2 = 2\langle c - a, d - a \rangle$$

give

$$|\langle c - a, d - a \rangle| \lesssim r. \quad (12.6.14)$$

Since $a \in \Omega_1$, $d \in \Omega_2$ and the patches are separated, $|d - a| \gtrsim 1$. Let

$$\tilde{c} = c - \frac{\langle c - a, d - a \rangle}{|d - a|^2} (d - a), \quad \tilde{b} = \tilde{c} + d - a.$$

Then $|\tilde{c} - c| + |\tilde{b} - b| \lesssim r$, $a + \tilde{b} = \tilde{c} + d$, and $|a|^2 + |\tilde{b}|^2 = |\tilde{c}|^2 + |d|^2$. The packet velocities lie strictly inside Ω_1, Ω_2 ; hence, for all sufficiently large R , $\tilde{c} \in \Omega_1$ and $\tilde{b} \in \Omega_2$. Thus $\tilde{c} \in \pi(a, d)$ and

$$\text{dist}(c, \pi(a, d)) \lesssim r. \quad (12.6.15)$$

The bounded range of smaller R is absorbed into the implicit constant.

Step 5. *Count the surviving quadruples.*

Fix $T_1 \in \mathbb{T}'_1$ and $T'_2 \in \mathbb{T}'_2$. By (12.6.15), every admissible $T'_1 \in \mathbb{T}'_1$ belongs to the set counted by ν with $\xi_1 = a$ and $\xi'_2 = d$; its $O(r)$ tolerance is smaller than the defining $O(R^{C\delta}r)$ tolerance. There are therefore at most ν choices for T'_1 .

Once T_1, T'_2, T'_1 are fixed, the first relation in (12.6.12) confines $v(T_2)$ to an $O(r)$ -ball. Since the velocity lattice is $r\mathbb{Z}^m$, that ball contains $O(1)$ possible velocities. For each fixed velocity, a tube $T_2 \in \mathbb{T}_2(q)$ meeting $R^\delta q$ has its initial point $x(T_2) \in R^{1/2}\mathbb{Z}^m$ in a ball of radius $O(R^{1/2+\delta})$: if (t_q, x_q) is the centre of q , the incidence gives

$$x(T_2) = x_q - t_q v(T_2) + O(R^{1/2+\delta}).$$

Consequently there are $O(R^{m\delta}) = O(R^{C\delta})$ choices for T_2 at that velocity. The total number of surviving quadruples is therefore

$$O\left(R^{C\delta}\nu(\#\mathbb{T}'_1)(\#\mathbb{T}'_2)\right). \quad (12.6.16)$$

Finally multiply (12.6.16) by the individual inner-product bound (12.6.13). This gives

$$\left\| \sum_{T_1 \in \mathbb{T}'_1} \sum_{T_2 \in \mathbb{T}'_2} \phi_{T_1} \phi_{T_2} \right\|_{L^2(q)}^2 \lesssim R^{C\delta} R^{-(m-1)/2} \nu(\#\mathbb{T}'_1)(\#\mathbb{T}'_2),$$

as claimed. $\square$

12.6.6 The combinatorial estimate

Step 1. *Define the two-ends relation without circularity.*

Let $\mathcal{B}$ be the coarse $R^{1-\delta}$ -ball cover used in the local/global reduction. Fix a dyadic triple $(\lambda_1, \mu_1, \mu_2)$. For $T_1 \in \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]$ and $B_0 \in \mathcal{B}$, set

$$N(T_1, B_0) = \#\{q \in \mathbf{q}(\mu_1, \mu_2) : T_1 \cap R^\delta q \neq \emptyset, q \cap B_0 \neq \emptyset\}.$$

Choose $B_*(T_1; \lambda_1, \mu_1, \mu_2)$ to maximise $N(T_1, B_0)$ over $B_0 \in \mathcal{B}$, and declare $T_1 \sim_{\lambda_1, \mu_1, \mu_2} B'$ when

$$B' \subseteq 10B_*(T_1; \lambda_1, \mu_1, \mu_2).$$

Every fine ball counted by $\lambda(T_1, \mu_1, \mu_2)$ meets at least one coarse ball, while $\#\mathcal{B} = O(R^{C\delta})$. The pigeonhole principle therefore gives

$$N(T_1, B_*(T_1; \lambda_1, \mu_1, \mu_2)) \gtrsim R^{-C\delta} \lambda_1. \quad (12.6.17)$$

Since the coarse cover has bounded overlap and all of its balls have the same radius, only $O(1)$ balls B' are contained in a fixed tenfold dilate $10B_*$. Define $T_1 \sim B'$ if the preceding relation holds for at least one dyadic triple. There are $O((\log R)^3)$ triples, so

$$\#\{B' : T_1 \sim B'\} \lesssim (\log R)^3 \lesssim 1.$$

For $\mathbb{T}_2$, define $\lambda_2(T_2, \mu_1, \mu_2)$ and its dyadic levels by interchanging the subscripts, then repeat the same maximisation and take the union over its dyadic triples. This proves (12.6.2) for both tube families. Only the $\mathbb{T}_1$ relation is used below.

Lemma 12.6.11 (The geometric observation). *Fix $q_0 \in \mathbf{q}(\mu_1, \mu_2)$, $\xi_1 \in \Omega_1$ and $\xi'_2 \in \Omega_2$, and put*

$$\mathbb{T}'_1 = \mathbb{T}_1^{\not\sim B}(q_0, \lambda_1, \mu_1, \mu_2, \xi_1, \xi'_2).$$

For each $T_2 \in \mathbb{T}_2$,

$$\begin{aligned} \#\{(q, T_1) \in \mathbf{q} \times \mathbb{T}'_1 : T_1 \cap R^\delta q \neq \emptyset, T_2 \cap R^\delta q \neq \emptyset, \\ \text{dist}(q_0, q) \gtrsim R^{-C\delta} R\} \lesssim R^{C\delta}. \end{aligned}$$

Proof. Let (t_0, x_0) and (t, x) be the centres of q_0 and q . Since T_1 meets both $R^\delta q_0$ and $R^\delta q$, the tube equation and the two incidence errors give

$$x - x_0 = v(T_1)(t - t_0) + O(R^{C\delta} R^{1/2}),$$

All velocities are bounded. Hence the assumed spacetime separation of the two fine balls implies

$$R^{-C\delta} R \lesssim |t - t_0| \lesssim R;$$

otherwise the preceding display would also make $|x - x_0|$ too small. Since $T_1 \in \mathbb{T}'_1$, $v(T_1)$ lies within $O(R^{C\delta} R^{-1/2})$ of $\pi(\xi_1, \xi'_2)$, so with $e = \xi'_2 - \xi_1$, Theorem 12.6.9 gives

$$\left\langle \frac{x - x_0}{t - t_0} - \xi_1, e \right\rangle \lesssim R^{C\delta} R^{-1/2},$$

that is $\langle (t - t_0, x - x_0), (-\langle \xi_1, e \rangle, e) \rangle \lesssim R^{C\delta} R^{1/2}$. So (t, x) lies within $O(R^{C\delta} R^{1/2})$ of a fixed hyperplane Π through (t_0, x_0) .

The normal to Π is $n_\Pi = (-\langle \xi_1, e \rangle, e)$. A T_2 -tube has spacetime direction $(1, v(T_2))$, and

$$\langle (1, v(T_2)), n_\Pi \rangle = \langle v(T_2) - \xi_1, e \rangle.$$

Here $\xi_1 = e_1 + O(1/m)$, $v(T_2) = -e_1 + O(1/m)$ and $e = -2e_1 + O(1/m)$, so the last inner product is bounded away from zero. Thus the central line

of T_2 crosses Π at a uniformly transverse angle. Because T_2 meets $R^\delta q$ and the centre of q is within $O(R^{C\delta} R^{1/2})$ of Π , (t, x) must lie in a ball of radius $O(R^{C\delta} R^{1/2})$ about that unique crossing point. A bounded-overlap $R^{1/2}$ -ball cover contains only $O(R^{C\delta})$ such q .

For a fixed surviving q , the relation

$$v(T_1) = \frac{x - x_0}{t - t_0} + O(R^{C\delta} R^{-1/2})$$

confines $v(T_1)$ to $O(R^{C\delta})$ points of the $R^{-1/2}\mathbb{Z}^m$ velocity lattice. Once the velocity is fixed, the incidence with $R^\delta q_0$ confines $x(T_1) \in R^{1/2}\mathbb{Z}^m$ to $O(R^{C\delta})$ choices. Hence each q contributes $O(R^{C\delta})$ tubes T_1 . Combining the two lattice counts proves the lemma after enlarging C . $\square$

Completion of Theorem 12.6.5 and Theorem 12.6.3. Step 1. Reduce the fine-scale estimate to one combinatorial sum.

For fixed $(\mu_1, \mu_2, \lambda_1)$, apply Theorem 12.6.10 on every q in (12.6.5). After absorbing its $R^{C\delta}$ factor, it is enough to prove

$$\begin{aligned} \sum_{\substack{q \in \mathbf{q}(\mu_1, \mu_2) \\ q \subseteq 2B}} \nu(q, \lambda_1, \mu_1, \mu_2) \#(\mathbb{T}_1^{\mathcal{Z}^B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]) \# \mathbb{T}_2(q) \\ \lesssim R^{C\delta} (\#\mathbb{T}_1)(\#\mathbb{T}_2). \end{aligned} \quad (12.6.18)$$

For $q \in \mathbf{q}(\mu_1, \mu_2)$, $\#\mathbb{T}_2(q) < 2\mu_2$. Moreover, Fubini and the definition of the λ_1 level give

$$\begin{aligned} \sum_{q \in \mathbf{q}(\mu_1, \mu_2)} \#(\mathbb{T}_1^{\mathcal{Z}^B}(q) \cap \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]) \\ \leq \sum_{T_1 \in \mathbb{T}_1[\lambda_1, \mu_1, \mu_2]} \lambda(T_1, \mu_1, \mu_2) < 2\lambda_1 \#\mathbb{T}_1. \end{aligned} \quad (12.6.19)$$

Consequently (12.6.18) follows from the uniform estimate

$$\nu(q_0, \lambda_1, \mu_1, \mu_2) \lesssim R^{C\delta} \frac{\#\mathbb{T}_2}{\lambda_1 \mu_2} \quad (12.6.20)$$

for every $q_0 \in \mathbf{q}(\mu_1, \mu_2)$ with $q_0 \subseteq 2B$.

Step 2. *The bush count proving (12.6.20).*

Fix q_0, ξ_1, ξ'_2 and let

$$\mathbb{T}'_1 = \mathbb{T}_1^{\mathcal{Z}^B}(q_0, \lambda_1, \mu_1, \mu_2, \xi_1, \xi'_2).$$

For $T_1 \in \mathbb{T}'_1$, the fact that $T_1 \not\sim B$ says that the coarse ball B is not contained in $10B_*(T_1; \lambda_1, \mu_1, \mu_2)$. Since these coarse balls have the same radius $R^{1-\delta}$ and

$q_0 \subseteq 2B$, every fine ball q meeting $B_*(T_1; \lambda_1, \mu_1, \mu_2)$ lies at distance $\gtrsim R^{-C\delta}R$ from q_0 , after harmlessly changing the fixed dilation constants. By (12.6.17), T_1 meets $R^\delta q$ for at least $R^{-C\delta}\lambda_1$ such q .

Every one of these q belongs to $\mathbf{q}(\mu_1, \mu_2)$ and hence meets at least μ_2 tubes of $\mathbb{T}_2$. Thus the number of triples (q, T_1, T_2) satisfying both incidences and the stated separation obeys

$$\#\{(q, T_1, T_2)\} \gtrsim R^{-C\delta}\lambda_1\mu_2\#\mathbb{T}'_1. \quad (12.6.21)$$

On the other hand, fixing T_2 and applying Theorem 12.6.11 bounds the number of pairs (q, T_1) by $R^{C\delta}$. Summing over T_2 gives

$$\#\{(q, T_1, T_2)\} \lesssim R^{C\delta}\#\mathbb{T}_2. \quad (12.6.22)$$

Comparison of (12.6.21) and (12.6.22) yields

$$\#\mathbb{T}'_1 \lesssim R^{C\delta} \frac{\#\mathbb{T}_2}{\lambda_1\mu_2}.$$

Taking the supremum over ξ_1, ξ'_2 proves (12.6.20), then (12.6.18), (12.6.5), and finally (12.6.3). The local/global reduction therefore proves Theorem 12.6.7.

Step 3. *Restore the packet coefficients.*

Normalize $\|f_1\|_{L^2} = \|f_2\|_{L^2} = 1$. Only polynomially many tubes can meet $B(0, CR)$. Tubes outside a fixed larger ball contribute $O(R^{-100m})$ on Q_R by the packet decay, after choosing its decay order larger than that polynomial. The same is true of all terms containing a coefficient $|c_T| < R^{-100m}$, by Cauchy–Schwarz and the polynomial tube count.

For the remaining coefficients, split each family into the $O(\log R)$ dyadic levels

$$\mathbb{T}_j(\gamma_j) = \{T_j : \gamma_j \leq |c_{T_j}| < 2\gamma_j\}.$$

For every level the coefficient estimate in Theorem 12.6.6 gives

$$\gamma_j \#\mathbb{T}_j(\gamma_j)^{1/2} \leq \left(\sum_{T_j} |c_{T_j}|^2 \right)^{1/2} \lesssim 1.$$

Apply Theorem 12.6.7 to each pair of levels and multiply its right-hand side by $\gamma_1\gamma_2$. The displayed inequality makes the resulting cardinality factor $O(1)$, and summing the $O((\log R)^2)$ pairs costs at most R^ε . We obtain

$$\|\widehat{f_1 d\sigma_1 f_2 d\sigma_2}\|_{L^{\frac{m+3}{m+1}}(Q_R)} \lesssim R^{\max\{(1-\delta)\alpha, C\delta\} + C\varepsilon}.$$

Spacetime translation and an $O(1)$ cover by slabs of the form Q_R give the same bound on every radius- R ball. This is precisely Theorem 12.6.5. The iteration already given after that proposition, followed by the epsilon-removal lemma, proves Theorem 12.6.3. $\square$

Remark 12.6.12 (R17 strict status: internally proved). *Theorem 12.6.3 is internally proved under the strict HA-03 policy.* In particular, Theorem 12.6.10 now derives the packet amplitude, uses coarea to obtain the individual packet-product L^2 size, turns Fourier overlap into quantitative momentum and energy conservation, projects the approximate constraint to the exact set π , and counts every surviving frequency-and-position quadruple. The two-ends relation is defined without reference to the coarse ball later being tested, and the hyperplane incidence and final bush count are explicit.

The one genuinely new ingredient over Wolff's cone argument is Theorem 12.6.9: for the cone one uses that a light ray meets a light cone through a point once, a consequence of the vanishing principal curvature; for the paraboloid no such fact holds, and its place is taken by the *second* frequency constraint, which confines ξ'_1 to a hyperplane. Everything else — induction on scales, wave packets, the two-ends localisation, the fine-scale decomposition, the bush count — is Wolff's.

The proof gives Tao's non-endpoint range $q > \frac{m+3}{m+1}$. Historically the endpoint was left open in Tao's argument; it was subsequently obtained by Jungjin Lee. The endpoint theorem is not used in this volume, so no endpoint claim is part of R17.

Corollary 12.6.13 (The bilinear estimate on a ball). *Under the hypotheses of Theorem 12.6.1, for every $R \geq 1$,*

$$\|\mathcal{E}_1 g_1 \mathcal{E}_2 g_2\|_{L^1(B_R)} \lesssim \nu^{-1/2} R \|g_1\|_{L^2(d\sigma)} \|g_2\|_{L^2(d\sigma)},$$

and this is sharp.

Proof. By Cauchy–Schwarz on B_R , $\|F\|_{L^1(B_R)} \leq |B_R|^{1/2} \|F\|_{L^2(B_R)}$ with $|B_R| \approx R^2$ in the plane, and Theorem 12.6.1 bounds the L^2 norm.

For sharpness, choose subarcs $\theta_j \subset S_j$ of length cR^{-1} centred at a transverse pair and put $g_j = R^{1/2} \mathbf{1}_{\theta_j}$. Then $\|g_j\|_{L^2(d\sigma)} \approx 1$. If $c > 0$ is a sufficiently small absolute constant and $x \in B_R$, the phase varies by at most $O(c)$ across each θ_j ; therefore there is no significant cancellation and

$$|\mathcal{E}_j g_j(x)| \approx R^{1/2} d\sigma(\theta_j) \approx R^{-1/2}.$$

Thus the product has size $\approx R^{-1}$ throughout a fixed dilate of B_R , and its L^1 norm there is $\approx R^{-1} \cdot R^2 = R$. This matches the upper bound.

Why the usual wave-packet cap is not extremal here. An $R^{-1/2}$ -cap is adapted to an $R \times R^{1/2}$ tube, not to the entire ball. The sharper R^{-1} caps above are essentially plane waves on B_R and fill the whole ball, which is exactly the configuration that saturates the Cauchy–Schwarz factor $|B_R|^{1/2} \approx R$. $\square$

WORKED EXAMPLE 12.6.2— TIER A **Bilinear against multilinear**

	bilinear	multilinear
factors	2	n
hypothesis	transverse <i>and</i> curved	transverse only
exponent	$q > \frac{n+2}{n}$ (Tao)	$p \geq \frac{2n}{n-1}$ (BCT)
loss	none	R^ε
proof input	curvature, induction on scales	multilinear Keakeya
sharp in	all n	all n
best for	two-point problems	restriction exponents

Why bilinear needs curvature and multilinear does not. With only two directions, transversality controls one direction; the remaining $n - 2$ must be controlled by the surfaces bending, which is why Theorem 12.6.3 has a curvature hypothesis and Theorem 12.3.1 does not. Theorem 12.2.3 is the formal statement.

Why bilinear is nonetheless the right tool sometimes. A problem whose natural object is a *pair* — two points, one distance — matches a two-factor estimate. Volume 8, Worked Example 7.7.3 classified the distance set problem as two-point, and Chapter 12.8 will exploit exactly that.

The historical order. Bilinear came first, in the 1990s, and gave the best restriction exponents until 2011; multilinear then overtook it for restriction, while bilinear remained the tool of choice for distance sets. Both are current.

WORKED EXAMPLE 12.6.3— TIER B **Reading the transversality hypothesis in the bilinear setting**

The bilinear literature states transversality in several ways and the differences are substantive.

(a) *Separated supports.* $\text{dist}(S_1, S_2) \geq \nu$ on the sphere. Weakest; does not by itself give Theorem 12.6.2's Jacobian bound, since two nearby-in-angle points can still have parallel normals on a general surface.

(b) *Transverse normals.* $|\sin \angle(\omega_1, \omega_2)| \geq \nu$, as in Theorem 12.6.1. This is what the Jacobian computation needs. On the sphere it coincides with (a) since the normal at ω is ω .

(c) *Transverse and curved.* Required for $n \geq 3$: the surfaces must additionally have non-vanishing second fundamental form. Without it the theorem is false — two transverse hyperplanes give no decay at all, consistent with Volume 5, Worked Example 5.5.4.

The check to run. Compute the Jacobian of the addition map for the surfaces

in question. If it is bounded below, one is in case (b); if it degenerates but the surfaces curve, one is in case (c) and needs the full theorem; if neither, no bilinear estimate holds.

An instance where it fails. S_1 and S_2 two transverse pieces of the cone in $\mathbb{R}^3$. The cone has a vanishing principal curvature, and the bilinear exponent differs from the sphere's — the cone's bilinear theory is Wolff's, with its own exponent, and is what Volume 13's local smoothing application will need.

WORKED EXAMPLE 12.6.4— TIER C: RESEARCH TASTE **Keeping a computable case**

Theorem 12.6.1 caught a defect in Theorem 12.3.3, and the episode generalises into the most useful working practice in this subject.

(a) *The principle.* For every method, keep one instance where the answer is known independently — by an elementary argument, a closed-form computation, or a numerical check — and test every new estimate against it.

(b) *The instances used in this series.* The plane forakeya, where Córdoba's L^2 argument gives the sharp answer (Theorem 9.3.3). The case $n = 2$ for restriction, settled by Fefferman and Zygmund. The box for Loomis–Whitney, where Theorem 11.5.2 is an equality. The grid for Szemerédi–Trotter and for joints (Worked Example 11.3.1, Worked Example 11.4.1). Small q for Dvir (Worked Example 10.1.4). In each case the test is cheap.

(c) *What a failed test tells you.* Not merely that something is wrong, but usually *where*. Theorem 12.3.3 failed at $n = 2$ by exactly a factor R , and the count of packets meeting a point is exactly $R^{(n-1)/2}$, which at $n = 2$ is $R^{1/2}$ — squaring to R in the final estimate. The size of the discrepancy identified the step.

(d) *What a passed test does not tell you.* That the estimate is correct. A formula can match at one endpoint and fail elsewhere; the exponent formula withdrawn in Chapter 12.8 passed the check at $s = \frac{n}{2}$ and failed the one that mattered. Two independent checks are worth more than one.

(e) *Why this is worth stating in a textbook.* Because the checks are not part of the finished mathematics and so are invisible in the literature. A reader who sees only polished proofs may conclude that the authors did not need them. They did.

WORKED EXAMPLE 12.6.5— TIER A **The bilinear route to Falconer, in outline**

Chapter 12.8 will use the bilinear theory rather than the multilinear, and it

is worth seeing the shape of that use before meeting it.

Step 1. By Theorem 12.8.2, $\sigma(\mu, r)$ is an $L^2(d\sigma)$ norm of $\mathcal{R}_r\mu$ — a restriction quantity with μ as data.

Step 2. Decompose $\mathbb{S}^{n-1}$ into caps and split the resulting double sum into transverse pairs and near-parallel pairs. The transverse part is what a bilinear estimate controls.

Step 3. Apply Theorem 12.6.3 to each transverse pair, integrated against μ — the fractal weight of Chapter 12.7.

Step 4. Handle the near-parallel pairs by rescaling and induction, as in Chapter 12.4's narrow case.

Why two factors and not n . Step 1 produces a *double* integral over $\mu \times \mu$, so the natural decomposition is into pairs. An n -fold decomposition would have nothing to attach to. Worked Example 12.8.2 makes this precise.

What each step costs. Step 2 is free; Step 3 costs the bilinear constant, which depends on the transversality parameter; Step 4 costs the induction, and this is where the exponent is determined. Theorem 12.8.9 records the completed weighted optimisation of these costs in the next chapter.

The pattern to recognise. It is the broad–narrow shape again — transverse part by the sharp estimate, degenerate part by induction — applied to a two-point problem instead of a linear one. Volume 13 will apply the same shape once more, with local smoothing in place of the bilinear estimate.

Drills

Drill 12.6.1. Verify Theorem 12.6.2 and check the Jacobian at angles $\frac{\pi}{2}, \frac{\pi}{3}$ and 0.05.

Drill 12.6.2. Confirm that $\Phi^{-1}(\xi)$ has at most two points for $0 < |\xi| < 2$, and describe what happens at $|\xi| = 0$ and $|\xi| = 2$.

Drill 12.6.3. Track the powers of ν in Step 3 and confirm the final $\nu^{-1/2}$.

Drill 12.6.4. Check that $\frac{n+2}{n} = 2$ at $n = 2$, and compare Theorem 12.6.3 with Theorem 12.6.1 there.

Where we are. The two estimates of this chapter differ in what they need from the surface.

Theorem 12.6.1 needs only that the addition map is non-degenerate, which in the plane is transversality alone — the arcs may be any curves with transverse tangents.

Theorem 12.6.3 needs transversality *and* curvature, because for $n \geq 3$ the fibres of the addition map are positive-dimensional and their geometry enters. Worked Example 12.6.3 lists the three hypotheses in use and warns that they are not equivalent.

The lesson generalises: in the plane, transversality of two curves controls everything, because two directions span. Above the plane it does not, and curvature must supply the rest — which is Theorem 12.2.3 again.

Notes

The planar bilinear estimate is classical, going back to Fefferman and Córdoba in the early 1970s. Tao's sharp bilinear restriction theorem is from 2003, building on work of Wolff for the cone and of Moyua, Vargas and Vega. The planar estimate is the short classical model case; the later argument above reconstructs Tao's higher-dimensional non-endpoint theorem, whose internal strict status is recorded in Theorem 12.6.12.

Chapter 12.7

Restriction Against a Fractal Measure

Every restriction estimate so far has measured $\mathcal{E}g$ in $L^p(\mathbb{R}^n)$ or on a ball. This chapter replaces Lebesgue measure by a fractal measure μ . The resulting estimates are what the Falconer distance set problem requires, and Chapter 12.8 will show precisely why.

12.7.1 The setting

Definition 12.7.1. Let $0 < s \leq n$. A finite Borel measure μ on $\mathbb{R}^n$ is *s-dimensional with constant C_μ* if

$$\mu(B(x, r)) \leq C_\mu r^s \quad \text{for all } x \in \mathbb{R}^n, r > 0.$$

By Frostman's lemma — Volume 7, Theorem 7.5.1 — a compact set E carries such a measure for every $s < \dim_{\text{H}} E$, and conversely the existence of one forces $\dim_{\text{H}} E \geq s$. So s -dimensional measures are the analytic substitute for the hypothesis $\dim_{\text{H}} E > s$.

Proposition 12.7.2 (The two extremes). *Let μ be s -dimensional and supported in the unit ball.*

- (i) *If $s = n$ then μ is absolutely continuous with bounded density, and a restriction estimate against μ follows from the corresponding $L^p(\mathbb{R}^n)$ estimate.*
- (ii) *If $s = 0$ the condition is vacuous — μ may be a point mass — and no restriction estimate against μ can hold beyond the trivial one.*

Proof. (i) The condition $\mu(B(x, r)) \leq C_\mu r^n$ says exactly that μ is dominated by C_μ times Lebesgue measure on every ball, hence $d\mu \leq C_\mu dx$ by differentiation (Volume 1, Theorem 1.7.4).

(ii) A point mass δ_{x_0} satisfies $\mu(B(x, r)) \leq 1 = r^0$. Then $\int |\mathcal{E}g|^2 d\mu = |\mathcal{E}g(x_0)|^2$, which is controlled only by $\|g\|_{L^1(d\sigma)}$ and by nothing sharper. $\square$

Where we are. So the interesting range is $0 < s < n$, and the question is how much of the $L^p(\mathbb{R}^n)$ theory survives when Lebesgue measure is replaced by something of dimension s .

The heuristic is that a fractal measure of dimension s “sees” a fraction R^{s-n} of a ball of radius R , so an estimate valid against Lebesgue measure should degrade by about that factor. Making this precise is the content of the weighted restriction theory, and the sharp form is not known — which is exactly why the Falconer problem is open.

12.7.2 The basic estimate

Theorem 12.7.3 (Weighted restriction, L^2 form). *Let μ be a finite Borel measure on $\mathbb{R}^n$, let $R \geq 1$, and put*

$$M_R(\mu) = \sup_{x \in \mathbb{R}^n} \mu(B(x, R^{1/2})).$$

Then for $g \in L^2(\mathbb{S}^{n-1})$,

$$\begin{aligned} \int_{\mathbb{R}^n} |\mathcal{E}g|^2 d\mu &\lesssim_{n,N} R^{\frac{2-n}{2}} M_R(\mu) \|g\|_{L^2(d\sigma)}^2 \\ &\quad + R^{-N} \|g\|_{L^1(d\sigma)}^2. \end{aligned}$$

Proof. Step 1. Wave packets.

Decompose $\mathcal{E}g = \sum_{(\theta, T)} \mathcal{E}g_{\theta, T}$ by Theorem 12.1.4 at scale R . By part (ii) each packet is bounded by $c_{\theta, T} \mathbf{1}_T$ up to tails decaying faster than any power, with $c_{\theta, T} = \|g_{\theta, T}\|_{L^1(d\sigma)}$; the tails contribute the final term.

Step 2. *The measure of one tube.*

A tube T has dimensions $R^{1/2} \times \cdots \times R^{1/2} \times R$, so it is covered by $\approx R^{1/2}$ balls of radius $R^{1/2}$. Hence

$$\mu(T) \lesssim R^{1/2} M_R(\mu).$$

Step 3. *Sum, keeping the orthogonality.*

Since the left side is quadratic, Theorem 12.1.4(iii) applies directly and

$$\begin{aligned} \int |\mathcal{E}g|^2 d\mu & \lesssim \sum_{(\theta, T)} |c_{\theta, T}|^2 \mu(T) \\ & \lesssim R^{1/2} M_R(\mu) \sum_{(\theta, T)} |c_{\theta, T}|^2. \end{aligned}$$

Now $\sum |c_{\theta, T}|^2 |T| \approx \|\mathcal{E}g\|_{L^2(B_R)}^2 \approx R \|g\|_{L^2(d\sigma)}^2$ by Theorem 6.1.4 of Volume 6, and $|T| \approx R^{(n+1)/2}$, so

$$\begin{aligned} \sum_{(\theta, T)} |c_{\theta, T}|^2 & \approx \frac{R \|g\|_{L^2}^2}{R^{(n+1)/2}} \\ & = R^{\frac{1-n}{2}} \|g\|_{L^2}^2. \end{aligned}$$

Step 4. *Collect.*

Multiplying, the power of R is $\frac{1}{2} + \frac{1-n}{2} = \frac{2-n}{2}$, which is the claim. $\square$

Corollary 12.7.4 (The s -dimensional form). *If in addition μ is s -dimensional with constant C_μ , then $M_R(\mu) \leq C_\mu R^{s/2}$ and hence*

$$\begin{aligned} \int |\mathcal{E}g|^2 d\mu & \lesssim C_\mu R^{\frac{2-n+s}{2}} \|g\|_{L^2(d\sigma)}^2 \\ & + R^{-N} \|g\|_{L^1(d\sigma)}^2. \end{aligned}$$

Proof. Immediate from Theorem 12.7.1 with $r = R^{1/2}$ and Theorem 12.7.3. $\square$

Where we are. Theorem 12.7.3 is crude — it uses only the size of μ on a single tube — and it is exactly the estimate that produces Falconer’s classical threshold. Chapter 12.8 will show the arithmetic.

Note what Step 3 does differently from Theorem 12.3.3: it keeps the L^2 orthogonality instead of dominating pointwise. Here that is legitimate because the left side is itself an L^2 quantity, so the orthogonality applies directly. In the multilinear setting the left side was an L^p norm of a product, and no such orthogonality was available — which is precisely why that argument lost $R^{1/2}$ and this one does not.

The lesson is worth recording: *almost-orthogonality is usable exactly when the quantity being estimated is quadratic.* Every place in this series where

| an L^2 method outperformed an L^p one traces to this.

WORKED EXAMPLE 12.7.1— TIER A **What the crude bound gives**

Theorem 12.7.4 reads

$$\int |\mathcal{E}g|^2 d\mu \lesssim R^{\frac{2-n+s}{2}} \|g\|_{L^2(d\sigma)}^2.$$

When does it gain? The exponent is negative exactly when $s < n - 2$. So for s small the estimate gains and for s large it loses — the opposite of what one might guess, and the reason is that a larger s means μ charges tubes more heavily.

The two extremes check out. At $s = n$ the exponent is 1, matching the Lebesgue case $\int_{B_R} |\mathcal{E}g|^2 \approx R \|g\|_{L^2}^2$ of Theorem 6.1.4. ✓ At $s = 0$, where μ may be a point mass, the exponent is $\frac{2-n}{2}$ and the bound is weaker than the trivial pointwise one for $n \geq 2$ — consistent with Theorem 12.7.2(ii).

Numerically, $n = 3$.

s	exponent $\frac{2-n+s}{2}$	
0	−0.5	gains
1	0	breaks even
2	0.5	loses
3	1	the Lebesgue case

What this is and is not. It is the crudest possible weighted estimate: it uses only the size of μ on a single tube and no interaction between tubes. Chapter 12.8 needs a quantity of a different shape — a spherical *average* rather than an integral against μ — and derives it from the identity of Theorem 12.8.2 rather than from this bound.

WORKED EXAMPLE 12.7.2— TIER B **Why fractal restriction is harder than L^p restriction**

(a) *There is no Plancherel.* The $L^2(\mathbb{R}^n)$ theory rests on Plancherel, which has no analogue for $L^2(d\mu)$. Every estimate must be built from the geometry of how μ meets tubes.

(b) *The extremal measures are unknown.* For Lebesgue measure the extremal configurations for restriction are the Knapp caps, well understood since 1970. For a fractal μ one must also ask which s -dimensional measures are worst, and the answer depends on s in a way that is not understood — the sharpness discussion in Volume 8, Theorem 8.8.1, is the same difficulty seen from the other side.

(c) *Two parameters, not one.* An L^p restriction estimate has one exponent; a weighted estimate has both p and s , and the sharp region in the (p, s) plane is known only along its edges.

(d) *What survives.* The wave packet decomposition, which needs nothing about the measure; and the bilinear and multilinear estimates, which can be integrated against μ once the tube geometry is understood. Chapter 12.8 does exactly this with the bilinear estimate.

The summary. Fractal restriction is the L^p theory with the ambient measure weakened, and every tool that used the ambient measure's structure must be rebuilt. That the theory nonetheless produces the best known Falconer exponents is the subject of the next chapter.

WORKED EXAMPLE 12.7.3— TIER C: RESEARCH TASTE **Two parameters, and what is known**

Weighted restriction has both an integrability exponent p and a dimension s , and the sharp region in the (p, s) plane is known only along its edges. Collecting what is available.

(a) $s = n$. The measure is essentially Lebesgue, and the question is the restriction conjecture. Settled for $n = 2$, open above.

(b) $p = 2$. The Stein–Tomas range, extended to fractal measures by Mockenhaupt and Mitsis. This is the edge Theorem 12.7.3 lives on, and it is the one the distance problem uses.

(c) s small. Theorem 12.7.2(ii): nothing beyond the trivial bound, since a point mass is admissible.

(d) *The interior.* Largely open. Bak and Seeger's sharpening covers part of it; the full picture would subsume the restriction conjecture as the edge case $s = n$, so it is at least as hard.

Why two parameters make this harder than it looks. An estimate can be sharp along both edges and far from sharp inside, and interpolation between the edges gives only a lower bound for the true region. Volume 8's Theorem 8.8.1 recorded the analogous difficulty for the distance problem: the extremal sets are known at the endpoint and not in between.

What the distance problem needs. Only the edge $p = 2$, but at the sharp exponent in s — which is Worked Example 8.8.1's $\sigma(\mu, r) \lesssim r^{-n/2}$. So the distance problem asks for one edge of a two-parameter region, and even that is open.

WORKED EXAMPLE 12.7.4— TIER B **Reading a weighted estimate**

Weighted restriction estimates are stated in several forms and the differences are substantive. Four checks.

(1) *Which side carries the measure?* $\int |\mathcal{E}g|^2 d\mu$ as here, or $\|\widehat{f d\mu}\|_{L^p}$ — the extension of μ rather than the restriction *against* it. They are dual, and confusing them inverts the role of s .

(2) *Is the constant C_μ or $M_R(\mu)$?* Theorem 12.7.3 uses $M_R(\mu) = \sup_x \mu(B(x, R^{1/2}))$, which is scale-adapted; Theorem 12.7.4 converts it to $C_\mu R^{s/2}$. The first is sharper and the second more quotable, and a comparison between sources must fix which is meant.

(3) *Which ball radius?* $R^{1/2}$ appears because that is the transverse width of a wave packet tube. An estimate stated with $\mu(B(x, R^{-1}))$ or $\mu(B(x, 1))$ is a different statement, and the exponent of R changes accordingly.

(4) *Is the estimate local or global?* Here local, on B_R , with the tails of Theorem 12.1.4(ii) contributing the R^{-N} term. A global statement would need ε -removal, and Volume 6, Theorem 6.5.2 is proved only for Lebesgue measure.

A worked check. Setting $s = n$ in Theorem 12.7.4 gives exponent $\frac{2-n+n}{2} = 1$, and the estimate reads $\int_{B_R} |\mathcal{E}g|^2 \lesssim R \|g\|_{L^2}^2$ — exactly Theorem 6.1.4 of Volume 6. ✓ That is the single most useful check on any weighted estimate: specialise to Lebesgue measure and compare with the unweighted theorem.

WORKED EXAMPLE 12.7.5— TIER A **The weighted estimate against the classical one**

It is worth seeing Theorem 12.7.3 beside the unweighted statements it generalises.

Measure	Estimate	Source
m on B_R	$\int \mathcal{E}g ^2 \lesssim R \ g\ _{L^2}^2$	Vol 6, Theorem 6.1.4
s -dimensional μ	$\lesssim R^{\frac{2-n+s}{2}} \ g\ _{L^2}^2$	Theorem 12.7.4
$d\sigma$ on $\mathbb{S}^{n-1}$	Stein–Tomas range	Vol 6, Theorem 6.3.1
δ_{x_0}	$ \mathcal{E}g(x_0) ^2 \leq \ g\ _{L^1}^2$	Theorem 12.7.2(ii)

The first two agree at $s = n$. $\frac{2-n+n}{2} = 1$. ✓ This is the check Worked Example 12.7.4 recommends and it is the quickest way to confirm a weighted estimate has been stated correctly.

The last row is the degenerate case. A point mass is 0-dimensional and the estimate degrades to the trivial pointwise bound, exactly as Theorem 12.7.2 says it must.

What the table does not contain. Any estimate at $p \neq 2$. The whole of Chapter 12.7 lives on the edge $p = 2$, because that is where almost-orthogonality is available — the point the Checkpoint after Theorem 12.7.3 makes, and the reason Theorem 12.3.3 lost $R^{1/2}$ where this chapter loses nothing.

Why $p = 2$ suffices for Falconer. Because Theorem 12.8.2 exhibits $\sigma(\mu, r)$ as an $L^2(d\sigma)$ norm. The distance set problem never needs another exponent, which is a genuine simplification and is why this chapter is short.

WORKED EXAMPLE 12.7.6— TIER A **Frostman measures, and what they encode**

Theorem 12.7.1 replaces a hypothesis about dimension by a measure, and it is worth being clear what is gained and what is lost.

What Frostman gives. Volume 7, Theorem 7.5.1: a compact E with $\dim_{\mathbb{H}} E > s$ carries a probability measure μ with $\mu(B(x, r)) \leq C_{\mu} r^s$. Conversely such a measure forces $\dim_{\mathbb{H}} E \geq s$. So the two hypotheses are interchangeable.

Why the measure is the better hypothesis. Dimension is defined by a covering condition and is awkward to use analytically. A measure can be integrated against, Fourier transformed, and inserted into estimates. Every theorem in Volumes 8 and 12 about distance sets is proved with μ and never with E .

What is lost. The constant C_{μ} , which Frostman does not control, and which appears in every conclusion. For the qualitative statements of this series that is harmless; for a quantitative version it would matter.

The two extremes, again. $s = n$ gives absolute continuity (Theorem 12.7.2(i)); $s = 0$ allows a point mass. Between them, μ may be as irregular as a self-similar Cantor measure — and Volume 7, Theorem 7.7.4 showed such measures can have Fourier transforms that do not decay at all, which is precisely why $\dim_{\mathbb{F}}$ and $\dim_{\mathbb{H}}$ differ.

The consequence for restriction. An s -dimensional measure carries no information about $\hat{\mu}$ beyond what the ball condition forces. So a weighted restriction estimate must hold for the *worst* such measure, and identifying that worst case is the open part of Worked Example 12.7.3.

WORKED EXAMPLE 12.7.7— TIER B **When the wave packet argument is free**

Theorem 12.7.3 loses nothing and Theorem 12.3.3 loses $R^{1/2}$, from what looks like the same step. The test distinguishing them is short.

The test. Is the quantity being estimated *quadratic* in the packet expansion? If so, Theorem 12.1.4(iii) applies directly and the sum is free. If not, one must dominate pointwise and pay for the overlap.

Applied here. $\int |\mathcal{E}g|^2 d\mu$ is quadratic. ✓ Free.

Applied to Theorem 12.3.3. $\|\prod_j |\mathcal{E}_j g_j|^{1/n}\|_{L^p}^p$ is a product of n different functions raised to a fractional power. Not quadratic in any one of them. *Not free*, and the price is the packet overlap $R^{(n-1)/2}$.

Applied to L^4 in the plane. $\|\mathcal{E}g\|_{L^4}^4 = \| |\mathcal{E}g|^2 \|_{L^2}^2$ is quadratic — in $|\mathcal{E}g|^2$ rather than in $\mathcal{E}g$. So the bilinear L^4 theory is free, which is why Volume 6, Theorem 6.8.4 could prove it by a TT^* computation and why Theorem 12.6.1 needs only Plancherel.

The general statement. Almost-orthogonality is a Hilbert space phenomenon, so it is available exactly at exponents where the quantity can be written as an L^2 norm of something. Even exponents and products of pairs qualify; odd and fractional exponents do not. Volume 13's decoupling exists precisely to recover a substitute at the exponents that do not qualify.

Drills

Drill 12.7.1. Verify Theorem 12.7.2(i) using the Lebesgue differentiation theorem.

Drill 12.7.2. Count the balls of radius $R^{1/2}$ needed to cover a wave packet tube and confirm Step 2 of Theorem 12.7.3.

Drill 12.7.3. Track the exponent arithmetic in Steps 3–4 and confirm $\frac{1}{2} + \frac{1-n}{2} = \frac{2-n}{2}$.

Drill 12.7.4. For $n = 2$, tabulate the exponent $\frac{2-n+s}{2}$ for $s = 0.5, 1, 1.5, 2$ and say where the estimate gains.

Where we are. The chapter's one structural point, worth carrying into Volume 13.

An almost-orthogonal decomposition can be summed without loss exactly when the quantity being estimated is *quadratic*. Theorem 12.7.3 estimates $\int |\mathcal{E}g|^2 d\mu$ and loses nothing; Theorem 12.3.3 estimates an L^p norm of a product and loses $R^{1/2}$. The difference is not technique but the shape of the quantity.

Decoupling, in Volume 13, is precisely the technology for recovering almost-orthogonality at exponents other than 2 — which is why it exists and why it is hard.

Notes

Weighted restriction estimates were introduced by Mockenhaupt and by Mitsis, and sharpened by Bak and Seeger; the L^2 form given here is the elementary case. The connection to distance sets is Mattila's and is the subject of the next chapter.

Chapter 12.8

Spherical Averages and the Distance Set

Volume 8 reduced the Falconer distance set problem to a bound on the spherical average $\sigma(\mu, r)$, and Worked Example 8.8.1 there asserted that the required estimate is a *weighted restriction estimate with a fractal weight*, promising the setting to this volume. This chapter makes that precise and shows how a bilinear estimate improves the threshold.

12.8.1 The spherical average is a restriction quantity

Definition 12.8.1. For a finite Borel measure μ on $\mathbb{R}^n$ and $r > 0$,

$$\sigma(\mu, r) = \int_{\mathbb{S}^{n-1}} |\widehat{\mu}(r\omega)|^2 d\sigma(\omega).$$

Proposition 12.8.2 (The two faces of $\sigma(\mu, r)$). *For every finite Borel measure μ with compact support and every $r > 0$,*

$$\sigma(\mu, r) = \iint \widehat{d\sigma}(r(x-y)) d\mu(x) d\mu(y) = \|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2,$$

where $\mathcal{R}_r \mu(\omega) = \widehat{\mu}(r\omega)$ is the restriction of $\widehat{\mu}$ to the sphere of radius r .

Proof. The second equality is the definition of $\mathcal{R}_r$ and of the $L^2(d\sigma)$ norm.

For the first, expand $|\widehat{\mu}(r\omega)|^2 = \widehat{\mu}(r\omega) \overline{\widehat{\mu}(r\omega)}$ and use $\widehat{\mu}(\xi) = \int e^{-2\pi i \xi \cdot x} d\mu(x)$:

$$|\widehat{\mu}(r\omega)|^2 = \iint e^{-2\pi i r \omega \cdot (x-y)} d\mu(x) d\mu(y).$$

Integrating in ω over $\mathbb{S}^{n-1}$ and applying Fubini — legitimate since μ is finite and the integrand is bounded —

$$\sigma(\mu, r) = \iint \left(\int_{\mathbb{S}^{n-1}} e^{-2\pi i r \omega \cdot (x-y)} d\sigma(\omega) \right) d\mu(x) d\mu(y),$$

and the inner integral is $\widehat{d\sigma}(r(x-y))$ by definition. $\square$

Where we are. The identity is exact and can be confirmed numerically: for a discrete measure in $\mathbb{R}^2$, where $\widehat{d\sigma}(\xi) = J_0(2\pi|\xi|)$, Monte Carlo evaluation of the left side against direct summation of the right agrees to three or four significant figures at $r = 0.5, 1, 2, 5, 10$.

Both faces are used, and for different purposes.

The first face — the double integral against $\widehat{d\sigma}$ — is what Volume 8 used. Inserting the pointwise decay $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ of Volume 5 gives Theorem 8.7.4, the trivial bound saturating at $\frac{n-1}{2}$.

The second face — the restriction of $\widehat{\mu}$ to a sphere — is the one this volume can improve. It says $\sigma(\mu, r)$ is literally $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}^2$: a *restriction operator applied to a measure*. By the duality of Volume 6, Theorem 6.1.3, bounding it is dual to an extension estimate, and the data is μ rather than an L^p function — which is exactly the fractal setting of Chapter 12.7.

That is the promise of Worked Example 8.8.1 discharged: “weighted restriction estimate” now means $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}$, with μ s -dimensional.

12.8.2 What an improved bound buys

Proposition 12.8.3 (Mattila’s criterion, sharp form). *Fix $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$ and $q_0 \in [1, 2]$ with*

$$\alpha\left(1 + \frac{1}{q_0}\right) \geq d. \quad (12.8.1)$$

Suppose that for every $q > q_0$, every α -dimensional μ , every $R > 1$ and every f supported in the annulus $A_R(1) = \{||\xi| - R| \leq 1\}$,

$$\left| \int f^\vee(u) d\mu(u) \right| \leq C_{q,\mu} R^{\frac{d-1}{2} - \frac{\alpha}{2q}} \|f\|_{L^2}. \quad (12.8.2)$$

Then $\dim_{\mathbb{H}} E > \alpha$ implies $m(\Delta(E)) > 0$.

Proof. This is Mattila’s theorem in the form used by Wolff and by Erdős; it is Volume 8, Theorem 8.7.2 together with the bookkeeping that converts a decay estimate valid on a *range* of q into convergence of the Mattila integral. We refer to that proof. $\square$

Pitfall: a single decay exponent is not enough

It is natural to compress (12.8.2) into one number: write $\sigma(\mu, r) \lesssim r^{-\beta}$ and ask for $\beta > \frac{d}{2}$, which is exactly what makes the Mattila integral $\int \sigma(\mu, r)^2 r^{d-1} dr$ converge. That condition is *sufficient* but strictly stronger than (12.8.1), and using it loses the theorem.

The arithmetic is worth seeing. At Erdős's threshold $\alpha = \frac{d}{2} + \frac{1}{3}$ the induced exponent is $\beta = \alpha/q_0$, and

d	α	$\beta = \alpha/q_0$	$\frac{d}{2}$
3	$\frac{11}{6}$	$\frac{7}{6} \approx 1.17$	1.5
4	$\frac{17}{3}$	$\frac{13}{6} \approx 1.67$	2
5	$\frac{17}{6}$	$\frac{13}{6} \approx 2.17$	2.5

so $\beta < \frac{d}{2}$ in every case: *the naive criterion fails precisely where Erdős's theorem succeeds.*

The reason is that (12.8.1) exploits the estimate across a whole *range* of exponents $q > q_0$, not merely the decay rate it implies at a single one. Compressing to β discards that, and the discarded information is exactly the difference between $\frac{d}{2} + 1$ and $\frac{d}{2} + \frac{1}{3}$.

A consistency check: the trivial spherical bound gives $\beta = \frac{d-1}{2} < \frac{d}{2}$, so the naive criterion never applies at all — matching Volume 8's finding (Theorem 8.7.3) that the Mattila L^2 route does not recover Falconer's classical $\frac{d+1}{2}$.

12.8.3 Erdős's weighted bilinear estimate

Theorem 12.8.4 (Erdős's weighted bilinear extension estimate). *Let $d \geq 3$, $\alpha \in (0, d)$, and let $H : \mathbb{R}^d \rightarrow [0, \infty)$ satisfy*

$$\|H\|_\infty \leq 1, \quad \int_{B(x,r)} H \leq r^\alpha \quad \text{for all } x, r.$$

Fix M and a separation constant $c_0 > 0$. There is an $\varepsilon_0 = \varepsilon_0(M, c_0, d) > 0$ with the following property. Let S_1, S_2 be two diameter- $O(1)$ patches, separated by at least c_0 in their parameter variables, of an elliptic graph

$$S = \{(u, \Phi(u)) : u \in U \subset \mathbb{R}^{d-1}\},$$

where $\|\Phi\|_{C^N} \leq M$ for a fixed sufficiently large N and

$$\|\nabla^2 \Phi - I\|_{\text{op}} \leq \varepsilon_0. \quad (12.8.3)$$

Then for every

$$q > q_0(\alpha, d) := \max \left\{ 1, \min \left(\frac{4\alpha}{d+2\alpha-2}, \frac{d+2}{d} \right) \right\},$$

$$\|\widehat{f_1 d\sigma} \widehat{f_2 d\sigma}\|_{L^q(H d\xi)} \lesssim_{\alpha, q, d} \|f_1\|_{L^2(d\sigma)} \|f_2\|_{L^2(d\sigma)}.$$

Proof. Write

$$q_* = \max \left\{ 1, \frac{4\alpha}{d + 2\alpha - 2} \right\}.$$

First suppose $\alpha \leq (d + 2)/2$, so that $q_0(\alpha, d) = q_* \leq 2$. By the epsilon-removal lemma, Volume 6, Theorem 6.5.2, it suffices to prove the localised estimate at exponent q_* on $B(0, R)$ with loss R^η for every $\eta > 0$. The estimate is initially true for $\eta \geq \alpha/q_*$: Cauchy–Schwarz on the two compact surface patches gives $|\widehat{f_j d\sigma}| \lesssim \|f_j\|_2$, while $\int_{B(0, R)} H \leq R^\alpha$.

We use the induction step of Theorem 12.6.5. There is no hidden paraboloid restriction here. For the graph phase Φ , the packet with frequency v travels with velocity $\nabla\Phi(v)$; the integration-by-parts proof of Theorem 12.6.6 is unchanged because the derivatives of Φ are bounded. In the proof of Theorem 12.6.10, momentum and energy become

$$\xi_1 + \xi_2 = \xi'_1 + \xi'_2, \quad \Phi(\xi_1) + \Phi(\xi_2) = \Phi(\xi'_1) + \Phi(\xi'_2).$$

Fixing $a = \xi_1$ and $d = \xi'_2$ and eliminating $b = \xi_2 = c + d - a$, the derivative in $c = \xi'_1$ of the second constraint is

$$\nabla\Phi(b) - \nabla\Phi(c) = \left(\int_0^1 \nabla^2\Phi(c + t(b - c)) dt \right) (d - a).$$

By (12.8.3), this is within $O(\varepsilon_0|d - a|)$ of $d - a$ and is uniformly nonzero. Thus the permitted $R^{-1/2}$ -lattice points lie in an $O(R^{-1/2})$ -neighbourhood of a uniformly Lipschitz hypersurface whose normal is close to $d - a$; it has the same $O(R^{(d-2)/2})$ count as the hyperplane in R17. The velocity map $v \mapsto \nabla\Phi(v)$ is bi-Lipschitz and $O(\varepsilon_0)$ -close, after an affine change, to the identity. It sends that frequency hypersurface to a velocity hypersurface with the same normal up to $O(\varepsilon_0)$. Hence the spacetime hypersurface swept out by its packet directions is transverse to every S_2 -tube: the relevant scalar product is a perturbation of $\langle d - a, d - a \rangle$. Choosing ε_0 small preserves the unique-crossing and lattice counts in Theorem 12.6.11. These observations prove the elliptic version of every R17 estimate, uniformly in the displayed data.

Decompose $\widehat{f_j d\sigma} = \sum_{T_j} c_{T_j} \phi_{T_j}$ into wave packets at scale R by Theorem 12.6.6, normalised so that $\|\phi_{T_j}\|_{L^2} \approx R^{1/2}$, $\sum_{T_j} |c_{T_j}|^2 \lesssim \|f_j\|_{L^2}^2$, and the packets are almost orthogonal. Cover $B(0, R)$ by $O(R^{C\delta})$ balls Q of radius $R^{1-\delta}$, and let $\sim$ be the relation supplied by Tao's argument, so that Theorem 12.6.10 and the combinatorial estimate of Theorem 12.6.11 give

$$\left\| \sum_{(T_1, T_2) \in Q^\sim} \phi_{T_1} \phi_{T_2} \right\|_{L^2(Q)} \lesssim R^{C\delta + C\varepsilon} R^{-\frac{d-2}{4}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2} \quad (12.8.4)$$

and $\#\{Q : T \sim Q\} \lesssim R^\varepsilon$.

Step 1. *The local part.*

Splitting the sum at each Q into pairs with $T_1 \sim Q$ and $T_2 \sim Q$, and those without, the first is handled by the induction hypothesis: since ϕ_{T_j} has Fourier support in an $O(R^{-1})$ -neighbourhood of S_j , the inductive estimate on the ball Q of radius $R^{1-\delta}$ gives, after almost-orthogonality and Cauchy–Schwarz,

$$\text{I} \lesssim R^{(1-\delta)\eta+C\varepsilon} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}.$$

Step 2. *The global part — where the weight enters.*

Write $F_Q = \sum_{(T_1, T_2) \in Q^\times} \phi_{T_1} \phi_{T_2}$. Since $q_* \leq 2$, Hölder's inequality with exponent $\frac{2}{q_*}$ gives

$$\begin{aligned} \|F_Q\|_{L^{q_*}(Q, H \, d\xi)} &\leq \|F_Q\|_{L^2(Q, d\xi)} \left(\int_Q H \right)^{\frac{1}{q_*} - \frac{1}{2}} \\ &\lesssim \|F_Q\|_{L^2(Q, d\xi)} R^{\frac{\alpha}{q_*} - \frac{\alpha}{2}}, \end{aligned}$$

using the hypothesis $\int_{B(x,r)} H \leq r^\alpha$ with $r \approx R$. *This is the only place the weight is used, and it is the whole difference from Tao's unweighted theorem.*

Step 3. *Combine.*

Substituting (12.8.4) and summing over the $O(R^{C\delta})$ balls,

$$\begin{aligned} \text{II} \\ &\lesssim R^{C\delta+C\varepsilon} R^{-\frac{d-2}{4}} R^{\frac{\alpha}{q_*} - \frac{\alpha}{2}} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}. \end{aligned}$$

The exponent of R is at most $C\delta + C\varepsilon$ precisely when

$$\frac{\alpha}{q_*} - \frac{\alpha}{2} - \frac{d-2}{4} \leq 0 \iff q_* \geq \frac{4\alpha}{d+2\alpha-2}.$$

This includes the branch $q_* = 1$: there the inequality is equivalent to $\alpha \leq (d-2)/2$. Hence $\text{II} \lesssim R^{C\delta+C\varepsilon} (\#\mathbb{T}_1)^{1/2} (\#\mathbb{T}_2)^{1/2}$, and $\text{I} + \text{II}$ closes the induction. Restoring the packet coefficients by the dyadic argument at the end of Theorem 12.6.5, then iterating in η and applying epsilon removal, proves every $q > q_*$.

If $\alpha > (d+2)/2$, then $q_0(\alpha, d) = (d+2)/d$. Because $0 \leq H \leq 1$,

$$\|F\|_{L^q(H \, dx)} \leq \|F\|_{L^q(dx)},$$

and the required range follows directly from the uniformly elliptic form of Tao's theorem proved by the stability argument above. This covers all $\alpha \in (0, d)$. $\square$

Lemma 12.8.5 (Frostman smoothing at the uncertainty scale). *Let μ be a compactly supported probability measure satisfying $\mu(B(x, r)) \leq C_\mu r^\alpha$. Let D be a frequency ball of radius $s \geq 1$, choose $\varphi_D \in C_c^\infty$ equal to one on $2D$, and put*

$$\mu_D = |\varphi_D^\vee| * \mu.$$

Then

$$\|\mu_D\|_\infty \lesssim C_\mu s^{d-\alpha}, \quad \|\mu_D\|_1 \lesssim 1,$$

and, for every ball $B(x, r)$ with $r \geq 100s^{-1}$,

$$\int_{B(x, r)} \mu_D(y) \, dy \lesssim C_\mu r^\alpha.$$

Proof. After modulation and translation,

$$|\varphi_D^\vee(x)| \leq C_N s^d (1 + s|x|)^{-N}.$$

Split the convolution defining $\mu_D(x)$ into the ball of radius s^{-1} and the annuli $2^k s^{-1} \leq |x - y| < 2^{k+1} s^{-1}$. For $N > \alpha$ the Frostman estimate gives

$$\mu_D(x) \lesssim C_\mu s^d \sum_{k \geq 0} 2^{-kN} (2^k s^{-1})^\alpha \lesssim C_\mu s^{d-\alpha}.$$

The L^1 assertion follows from Fubini and $\|\varphi_D^\vee\|_1 \lesssim 1$.

For the ball estimate, integrate first in y . Points of μ in $B(x, 2r)$ contribute $O(\mu(B(x, 2r)))$. Split its complement into the annuli $2^k r \leq |x - z| < 2^{k+1} r$. On the k -th annulus the integral of the kernel over $B(x, r)$ is

$$O_N(r^d s^d (sr)^{-N} 2^{-kN}).$$

Multiplication by $\mu(B(x, 2^{k+1}r)) \lesssim C_\mu (2^k r)^\alpha$ and summation give $O(C_\mu r^\alpha)$ when $N > d + \alpha$ and $sr \geq 100$. $\square$

Corollary 12.8.6 (Scaled separated caps). *Let μ be as in Theorem 12.8.5. Suppose $C\beta R^{-1/2} \leq \eta \leq c$, and let $I_1, I_2 \subset A_R(\beta)$ have diameter comparable to $R\eta$ and mutual distance comparable to $R\eta$. For f_j supported in I_j and every $q > q_0(\alpha, d)$,*

$$\|f_1^\vee f_2^\vee\|_{L^q(d\mu)} \lesssim_{\alpha, q, d, C_\mu} \beta (R\eta)^{d-1-\alpha/q} \eta^{-1/q} \|f_1\|_2 \|f_2\|_2. \quad (12.8.5)$$

Proof. Put $s = R\eta$. The Fourier transform of $f_1^\vee f_2^\vee$ is supported in a ball D of radius $O(s)$. Jensen's inequality applied to $F = F * \varphi_D^\vee$, followed by Fubini, gives

$$\|F\|_{L^q(d\mu)} \lesssim \|F\|_{L^q(\mu_D dx)}. \quad (12.8.6)$$

Let e_1 be the radial direction through the centre of $I_1 \cup I_2$, extend it to an orthonormal basis, and define

$$T(e_1) = \frac{1}{R\eta^2}e_1, \quad T(e_j) = \frac{1}{R\eta}e_j \quad (2 \leq j \leq d).$$

Thus $\det T = s^{-d}\eta^{-1}$. The two sets TI_j lie in a δ -neighbourhood of separated diameter- $O(1)$ patches of a uniformly elliptic graph, where

$$\delta \asymp \frac{\beta}{R\eta^2} = \frac{\beta}{s\eta} \lesssim 1.$$

We record the elementary thickening consequence of Theorem 12.8.4. If g_j is supported in the δ -neighbourhood of such a patch, disintegration in the normal coordinate, Minkowski's inequality, the surface estimate, and Cauchy–Schwarz in each normal variable give

$$\|g_1^\vee g_2^\vee\|_{L^q(H \, dx)} \lesssim \delta \|g_1\|_2 \|g_2\|_2. \quad (12.8.7)$$

Indeed, the two normal integrations have length $O(\delta)$, and each Cauchy–Schwarz contributes $\delta^{1/2}$; the parallel graph slices retain uniform ellipticity and separation.

Set $g_j(z) = f_j(T^{-1}z)$. The change of variables $x = T^{-1}u$ gives

$$\|f_1^\vee f_2^\vee\|_{L^q(\mu_D \, dx)} = s^{2d-d/q}\eta^{2-1/q} \|g_1^\vee g_2^\vee\|_{L^q(\mu_D(T \cdot) \, dx)}. \quad (12.8.8)$$

Define $H(x) = c, s^{\alpha-d}\mu_D(Tx)$, with $c > 0$ chosen so that the normalised bounds in Theorem 12.8.4 hold. The L^∞ bound is Theorem 12.8.5. For a ball of radius $r \geq 100$, its image under T is covered by $O(\eta^{-1})$ balls of radius $O(r/s)$; hence

$$s^{\alpha-d} \int_{B(x,r)} \mu_D(Tu) \, du \lesssim s^\alpha \eta \cdot \eta^{-1} (r/s)^\alpha \lesssim r^\alpha.$$

For $r < 100$, the same conclusion, with a harmless constant, follows from $H \lesssim 1$: use $r^d \leq r^\alpha$ for $r \leq 1$ and $r^d \leq 100^{d-\alpha} r^\alpha$ for $1 < r < 100$. Therefore (12.8.7) applies, after scaling out that harmless constant. Since

$$\|g_j\|_2 = s^{-d/2}\eta^{-1/2} \|f_j\|_2,$$

substitution in (12.8.8) yields

$$\begin{aligned} \|f_1^\vee f_2^\vee\|_{L^q(d\mu)} &\lesssim s^{2d-d/q}\eta^{2-1/q} s^{(d-\alpha)/q} \frac{\beta}{s\eta} s^{-d}\eta^{-1} \|f_1\|_2 \|f_2\|_2 \\ &= \beta s^{d-1-\alpha/q} \eta^{-1/q} \|f_1\|_2 \|f_2\|_2, \end{aligned}$$

which is (12.8.5). □

Theorem 12.8.7 (The annular weighted L^2 estimate). *Let μ be a compactly supported α -dimensional probability measure, $R \geq 2$, and f be supported in $A_R(1)$. For every $q > q_0(\alpha, d)$,*

$$\|f^\vee\|_{L^2(\mathrm{d}\mu)} \lesssim_{\alpha, q, d, C_\mu} R^{\frac{d-1}{2} - \frac{\alpha}{2q}} \|f\|_2. \quad (12.8.9)$$

Proof. A fixed finite partition lets us assume that f lies over one spherical patch of diameter cR . Decompose that patch into dyadic caps I of radial thickness 2 and tangential sidelength $s = 2^n$, for $R^{1/2} \lesssim s \lesssim cR$. Caps of the same size are called related, $I \sim J$, when they are not adjacent but their parents are adjacent. Every pair of distinct angular points belongs to a bounded number of related pairs at a unique scale; pairs that never separate remain in a bounded-overlap family $\mathcal{I}_E$ of terminal caps of sidelength comparable to $R^{1/2}$. Consequently, with $f_I = f\mathbf{1}_I$,

$$\|f^\vee\|_{L^2(\mathrm{d}\mu)}^2 \lesssim \sum_s \sum_{I \sim J} \|f_I^\vee f_J^\vee\|_{L^1(\mathrm{d}\mu)} + \sum_{I \in \mathcal{I}_E} \|f_I^\vee\|_{L^2(\mathrm{d}\mu)}^2. \quad (12.8.10)$$

The complex conjugate arising from $|f^\vee|^2$ is immaterial because all subsequent estimates concern the absolute product.

Step 1. *Terminal caps.*

Each $I \in \mathcal{I}_E$ is contained in a frequency ball D of radius $CR^{1/2}$. The identity $f_I^\vee = f_I^\vee * \varphi_D^\vee$, Cauchy–Schwarz, Fubini, and Theorem 12.8.5 give

$$\|f_I^\vee\|_{L^2(\mathrm{d}\mu)}^2 \lesssim R^{(d-\alpha)/2} \|f_I\|_2^2.$$

The terminal caps overlap only finitely, so their total has the same bound with $\|f\|_2^2$. It is harmless because

$$\frac{d-\alpha}{2} \leq d-1 - \frac{\alpha}{q_0(\alpha, d)}. \quad (12.8.11)$$

For completeness, on the three branches of q_0 the difference between the right and left sides is respectively $(d-2-\alpha)/2$, $(d-2)/4$, and $(d-2)(d+2-\alpha)/(2(d+2))$, all nonnegative in their defining ranges.

Step 2. *A related pair.*

Fix $I \sim J$ of tangential sidelength s . Their diameter and separation are both comparable to s . Let e be the radial direction through their parents and tile physical space by boxes P with sidelength L along e and Ls/R in the other $d-1$ directions, where L is a sufficiently large fixed constant. Choose a fixed band-limited Schwartz function b which is at least one on the unit box, and let b_P be its affine copy adapted to P . For $u_I = f_I^\vee$ put

$$f_{I,P} = (u_I b_P)^\wedge, \quad f_{J,P} = (u_J b_P)^\wedge.$$

On P , $|b_P| \geq 1$, so Hölder gives

$$\|u_I u_J\|_{L^1(d\mu)} \lesssim \sum_P \|f_{I,P}^\vee f_{J,P}^\vee\|_{L^q(d\mu)} \mu(P)^{1/q'}, \quad q' = \frac{q}{q-1}. \quad (12.8.12)$$

The dual box has radial width $O(L^{-1})$ and transverse width $O(R/(Ls))$. Since $s \geq R^{1/2}$, elementary expansion of $|\xi|$ around Re shows that $I + \text{supp } \widehat{b}_P$ is contained in $A_R(C)$, has diameter $O(s)$, and, when L is fixed sufficiently large, remains a distance comparable to s from the analogous enlargement of J . Thus Theorem 12.8.6, with $\eta = s/R$ and $\beta = C$, gives

$$\|f_{I,P}^\vee f_{J,P}^\vee\|_{L^q(d\mu)} \lesssim R^{1/q} s^{d-1-(\alpha+1)/q} \|f_{I,P}\|_2 \|f_{J,P}\|_2. \quad (12.8.13)$$

The box P is covered by $O(R/s)$ balls of radius $O(s/R)$; hence

$$\mu(P) \lesssim (R/s)(s/R)^\alpha = s^{\alpha-1} R^{1-\alpha}. \quad (12.8.14)$$

Finally, the translates b_P have a uniformly bounded square sum. By Plancherel,

$$\sum_P \|f_{I,P}\|_2^2 = \sum_P \|u_I b_P\|_2^2 \lesssim \|f_I\|_2^2,$$

and similarly for J . Insert (12.8.13) and (12.8.14) into (12.8.12), then use Cauchy–Schwarz in P , to obtain

$$\|f_I^\vee f_J^\vee\|_{L^1(d\mu)} \lesssim R^{1-\alpha/q'} s^{d-2+\alpha(1-2/q)} \|f_I\|_2 \|f_J\|_2. \quad (12.8.15)$$

For $q > q_0(\alpha, d)$ the exponent of s is nonnegative. Since $s \leq R$, the right side of (12.8.15) is at most

$$R^{d-1-\alpha/q} \|f_I\|_2 \|f_J\|_2.$$

Step 3. *Sum the Whitney pieces.*

Choose $\tilde{q}$ strictly between $q_0(\alpha, d)$ and the stated q and run Step 2 with $\tilde{q}$. At each scale the related-pair graph has bounded degree, so Cauchy–Schwarz and bounded overlap give

$$\sum_{I \sim J} \|f_I\|_2 \|f_J\|_2 \lesssim \|f\|_2^2.$$

There are $O(\log R)$ scales, and the positive gap $\alpha(1/\tilde{q} - 1/q)$ absorbs that logarithm. Therefore the first term of (12.8.10) is $O(R^{d-1-\alpha/q} \|f\|_2^2)$. Step 1 and (12.8.11) give the same bound for the terminal term. Taking square roots proves (12.8.9). $\square$

Theorem 12.8.8 (Erdős). *Let $d \geq 3$ and let $E \subseteq \mathbb{R}^d$ be compact with $\dim_H E > \frac{d}{2} + \frac{1}{3}$. Then $m(\Delta(E)) > 0$.*

Proof. Step 0. Reduce to the admissible range.

Theorem 12.8.3 is stated for $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$, and $\frac{d}{2} + \frac{1}{3} \leq \frac{d+1}{2}$ for every d , since $\frac{1}{3} \leq \frac{1}{2}$. So the threshold lies inside the range; but $\dim_{\mathbb{H}} E$ need not. Choose α with $\frac{d}{2} + \frac{1}{3} < \alpha \leq \frac{d+1}{2}$ and $\dim_{\mathbb{H}} E > \alpha$, and prove the conclusion for that α — a larger dimension only makes the hypothesis easier. *Without this step the proposition would be applied outside its stated range whenever $\dim_{\mathbb{H}} E > \frac{d+1}{2}$.*

Step 1. Apply the criterion.

By Theorem 12.8.3 it suffices to verify (12.8.2) for all $q > q_0(\alpha, d)$ and to check the criterion (12.8.1).

The estimate is now exactly Theorem 12.8.7: because μ is a probability measure, Cauchy–Schwarz gives

$$\left| \int f^\vee d\mu \right| \leq \|f^\vee\|_{L^2(d\mu)} \lesssim R^{\frac{d-1}{2} - \frac{\alpha}{2q}} \|f\|_2,$$

for every $q > q_0(\alpha, d)$. Thus the analytic hypothesis of Theorem 12.8.3 has been proved, including the cap rescaling, uncertainty smoothing, terminal caps, and Whitney summation rather than merely asserted.

It remains to solve (12.8.1). With $q_0 = \frac{4\alpha}{d+2\alpha-2}$, which is the active branch because $(d-2)/2 < \alpha \leq (d+1)/2 < (d+2)/2$,

$$\begin{aligned} \alpha \left(1 + \frac{1}{q_0} \right) &= \alpha + \frac{d+2\alpha-2}{4} \geq d \\ &\iff 4\alpha + d + 2\alpha - 2 \geq 4d \\ &\iff 6\alpha \geq 3d + 2, \end{aligned}$$

that is $\alpha \geq \frac{d}{2} + \frac{1}{3}$. □

Where we are. The final computation is worth isolating, since the threshold $\frac{d}{2} + \frac{1}{3}$ looks arbitrary and is not:

$$\begin{aligned} \alpha \left(1 + \frac{1}{q_0} \right) &\geq d \\ &\iff \frac{3}{2}\alpha - \frac{3}{4}d - \frac{1}{2} \geq 0 \\ &\iff \alpha \geq \frac{d}{2} + \frac{1}{3}. \end{aligned}$$

Symbolically, the solution of $\alpha(1 + 1/q_0) = d$ is exactly $\alpha = \frac{d}{2} + \frac{1}{3}$, giving $\frac{11}{6}, \frac{7}{3}, \frac{17}{6}, \frac{16}{3}$ for $d = 3, 4, 5, 10$. The $\frac{1}{3}$ is the fixed point of the criterion, not a numerical accident.

Erdős notes that this method has a ceiling: the best exponent obtainable from Mattila’s approach is $\frac{4}{3}$ in $\mathbb{R}^2$ and $\frac{5}{3}$ in $\mathbb{R}^3$. But he also observes that if the weighted estimate held at $\alpha = \frac{d}{2}$ for all $q > 1$, the criterion would

give Falconer's conjecture outright for $d \geq 4$. So the obstruction is the range of q , not the framework.

Remark 12.8.9 (R18 strict status: internally proved). *Theorem 12.8.4, Theorem 12.8.7, and Theorem 12.8.8 are internally proved under the strict HA-03 policy.* Transitivity is now satisfied because R17 is internally proved in Theorem 12.6.12. The annular passage is also explicit: it includes Frostman smoothing, the anisotropic cap rescaling, the thickened elliptic-surface estimate, the terminal scale, and the Whitney-pair summation.

The proof is a genuinely small modification of Tao's, and that is the point: Steps 1 and 3 are Tao's induction verbatim, and the weight enters only at Step 2, through a single application of Hölder against $\int_{B(x,r)} |H| \leq r^\alpha$. *The entire content of the theorem is that Tao's L^2 estimate (12.8.4) has room to absorb a factor $R^{\alpha/q_0 - \alpha/2}$, and the exponent $q_0(\alpha, d)$ is defined to be exactly the threshold at which it does.*

Erdős remarks that improving (12.8.4) to a weighted $L^2(H \, d\xi)$ estimate would widen the range of q and improve the Falconer threshold; that has not been done.

WORKED EXAMPLE 12.8.1— TIER A **The scoreboard, with the machinery**

Threshold	Author	Input	Where
$\frac{n+1}{2}$	Falconer 1985	decay of $\widehat{d\sigma}$	Vol 8, Theorem 8.5.1, <i>proved</i>
$\frac{n}{2} + \frac{1}{3}$	Erdős 2005	bilinear restriction	here, Theorem 12.8.9
$\frac{4}{3} \ (n = 2)$	Wolff 1999	local smoothing	Vol 13
$\frac{5}{4} \ (n = 2)$	GIOW 2020	partitioning + decoupling	Vol 14
$\frac{n}{2}$	<i>conjecture</i>	—	open

Reading the table. Every row after the first replaces the pointwise decay of $\widehat{d\sigma}$ by a stronger statement about how $\widehat{\mu}$ concentrates on spheres — that is, by a better restriction estimate. The problem has been a restriction problem since 1985; only the tools have changed.

The half that must be closed. Volume 8 recorded that the trivial exponent saturates at $\frac{n-1}{2}$ and the conjecture needs $\frac{n}{2}$ — a gap of exactly $\frac{1}{2}$. Erdős's $\beta(s)$ closes part of it; nothing closes all of it.

What this volume contributed. Not a new threshold, but the identification of the mechanism: Theorem 12.8.2 says the quantity to bound *is* a restriction norm, so every advance in restriction theory is available to the distance problem. That is why the table's later rows come from Volumes 13 and 14 rather than from new distance-set techniques.

WORKED EXAMPLE 12.8.2— TIER B Why bilinear, and not multilinear

Chapter 12.3 developed the n -linear theory and Chapter 12.6 the two-linear one, and it is the second that the distance problem uses. The reason is structural.

(a) *The problem is two-point.* $\Delta(E)$ records $|x - y|$ for *pairs*, and Theorem 12.8.2 exhibits $\sigma(\mu, r)$ as a double integral over $\mu \times \mu$. Volume 7, Worked Example 7.7.3 classified such problems as two-point, and Volume 9, Worked Example 9.3.3 recorded that a two-point method has a ceiling for *Keakeya* — but here the problem itself is two-point, so no information is lost.

(b) *The matching estimate has two factors.* Bounding $\iint \widehat{d\sigma}(r(x - y)) \, d\mu \, d\mu$ means controlling an interaction between two copies of μ , which pairs with a bilinear extension estimate. An n -linear estimate would control an n -fold interaction that the problem never forms.

(c) *The historical confirmation.* Erdős's improvement used Tao's bilinear theorem, and no distance-set result has been obtained from the multilinear theory directly. The later records of Volume 14 use decoupling and partitioning, which are again not n -linear.

The transferable point. Match the order of the estimate to the order of the problem. Volume 9's diagnosis was that a two-point method *cannot* solve *Keakeya*, a high-order problem; the converse holds here — an n -linear method has nothing extra to offer a two-point problem. This is the cleanest instance in the series of the principle working in both directions.

WORKED EXAMPLE 12.8.3— TIER B Using the identity in both directions

Theorem 12.8.2 gives two expressions for $\sigma(\mu, r)$, and arguments use them for different purposes.

Direction 1: from $\widehat{d\sigma}$ to a bound. Insert the decay $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ into the double integral:

$$\sigma(\mu, r) \lesssim r^{-\frac{n-1}{2}} \iint |x - y|^{-\frac{n-1}{2}} \, d\mu \, d\mu = r^{-\frac{n-1}{2}} I_{\frac{n-1}{2}}(\mu),$$

the energy integral of Volume 7. Finiteness of the energy requires $s > \frac{n-1}{2}$, and one recovers Theorem 8.7.4: $\sigma \lesssim r^{-\min\{s, (n-1)/2\}}$.

Direction 2: from restriction to a bound. Read $\sigma(\mu, r)$ as $\|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2$ and apply a restriction estimate with μ as data. This is what Theorem 12.8.8 does, and it is available only because the identity exhibits the spherical average as a restriction norm.

Why Direction 1 saturates. It uses only the *modulus* of $\widehat{d\sigma}$, and Volume 5, Theorem 5.5.2 showed that modulus is sharp — so no improvement is possible without using the phase or the structure of μ . Volume 8's Pitfall on Theorem 8.7.2 recorded that the phase is essential elsewhere in this argument too.

Why Direction 2 can do better. A restriction estimate exploits how $\widehat{\mu}$ distributes over the sphere, not merely its size — the same distinction as between an L^∞ bound and an L^2 bound. That extra information is what buys the improvement, and it is why the distance set problem became a restriction problem.

WORKED EXAMPLE 12.8.4— TIER C: RESEARCH TASTE **What the Falconer problem now needs**

Twelve volumes in, the distance set problem can be stated with complete precision as a single missing estimate.

(a) *The target.* For every $s > \frac{n}{2}$ and every s -dimensional μ in the unit ball,

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\frac{n}{2}+\varepsilon}.$$

By Theorem 12.8.3 this gives $m(\Delta(E)) > 0$ for $\dim_{\mathbb{H}} E > \frac{n}{2}$, which is Theorem 8.4.2.

(b) *What it is, in restriction language.* By Theorem 12.8.2, a bound on $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}$ — the restriction of $\widehat{\mu}$ to a sphere — with μ fractal. So the missing estimate is a weighted restriction estimate at the sharp exponent, on the edge $p = 2$ of the two-parameter region of Worked Example 12.7.3.

(c) *Why the trivial bound stops halfway.* It uses only $|\widehat{d\sigma}| \lesssim |\xi|^{-(n-1)/2}$, which is sharp by Volume 5, Theorem 5.5.2. So the gap of $\frac{1}{2}$ between $\frac{n-1}{2}$ and $\frac{n}{2}$ cannot be closed by any argument using only the size of $\widehat{d\sigma}$; one must use how $\widehat{\mu}$ distributes.

(d) *What has closed part of it.* Bilinear restriction (Theorem 12.8.8), local smoothing (Volume 13), and partitioning with decoupling (Volume 14). Each exploits the distribution of $\widehat{\mu}$ in a different way, and none reaches $\frac{n}{2}$.

(e) *The structural reason it is hard.* By Worked Example 12.7.3, the sharp weighted estimate at $s = n$ is the restriction conjecture. The Falconer problem asks for the edge $p = 2$ at $s = \frac{n}{2}$, which is not formally implied by the restriction conjecture — but no method is known that gives one without the other's difficulties.

(f) *The honest summary.* The problem is now a well-posed question about a single quantity, with the mechanism completely understood and the estimate

missing. That is a better position than in 1985, and it is not a solution.

WORKED EXAMPLE 12.8.5— TIER A **The distance problem, from 1985 to here**

The Falconer problem has now occupied parts of five volumes, and the arc is worth seeing whole.

1985, Volume 8. Falconer poses the conjecture and proves $\frac{n+1}{2}$ by bounding the distance measure in L^∞ (Theorem 8.5.1). The only input is the decay of $\widehat{d\sigma}$ from Volume 5.

1987, Volume 8. Mattila reduces the problem to the spherical average $\sigma(\mu, r)$ (Theorem 8.7.2), and the trivial bound (Theorem 8.7.4) saturates at $\frac{n-1}{2}$ — short of the required $\frac{n}{2}$ by exactly $\frac{1}{2}$.

Volume 8's limit. With only Volumes 1–7 available, nothing improves the trivial bound. Worked Example 8.8.1 states what is needed and names it a weighted restriction estimate, deferring to this volume.

Here. Theorem 12.8.2 makes that precise: $\sigma(\mu, r) = \|\mathcal{R}_r \mu\|_{L^2(d\sigma)}^2$. The problem is now a restriction problem, and Theorem 12.8.8 reports the first improvement, using the bilinear theory of Chapter 12.6.

Volumes 13 and 14. Local smoothing gives Wolff's $\frac{4}{3}$ in the plane; partitioning with decoupling gives $\frac{5}{4}$. Each is a better restriction estimate against a fractal weight, and none reaches $\frac{n}{2}$.

What has actually changed since 1985. Not the formulation — Mattila's reduction is still the framework — but the supply of restriction estimates. The distance problem has been waiting on restriction theory for nearly forty years, and Worked Example 12.8.4 states exactly what it is still waiting for.

WORKED EXAMPLE 12.8.6— TIER B **Checking a claimed Falconer threshold**

Distance set papers announce thresholds, and placing one takes four checks.

(1) *Which conclusion?* $m(\Delta E) > 0$, or ΔE has non-empty interior, or the *pinned* statement that some x has $m(\Delta_x E) > 0$. These are increasingly strong; Volume 8, Theorem 8.6.1 gives the second under a stronger hypothesis, and the third is Volume 14's subject.

(2) *Which dimension?* $\dim_{\mathbb{H}}$ throughout this series. Some results assume Ahlfors–David regularity, which is much stronger and not comparable.

(3) *What spherical average bound underlies it?* By Theorem 12.8.3 a threshold s_0 corresponds to $\sigma(\mu, r) \lesssim r^{-\beta}$ with $\beta > \frac{n}{2}$ for $s > s_0$. Working backwards from the announced threshold recovers β , and comparing with the trivial

$\min\{s, \frac{n-1}{2}\}$ shows how much was gained.

(4) *Which dimensions n ?* Planar results and higher-dimensional ones use different machinery — local smoothing in the plane, decoupling above — and a planar threshold does not imply anything for $n \geq 3$.

A worked check. “ $\dim_{\mathbb{H}} E > \frac{5}{4}$ gives $m(\Delta E) > 0$ in the plane”: (1) positive measure; (2) Hausdorff; (3) $n = 2$ so the target is $\beta > 1$, against the trivial $\frac{1}{2}$; (4) planar only. Placing it: better than Falconer’s $\frac{3}{2}$ and Wolff’s $\frac{4}{3}$, short of the conjectured 1.

Drills

Drill 12.8.1. Verify Theorem 12.8.2 by expanding $|\widehat{\mu}(r\omega)|^2$ and applying Fubini; state where compact support is used.

Drill 12.8.2. Insert $|\widehat{d\sigma}(\xi)| \lesssim |\xi|^{-(n-1)/2}$ into the first face of the identity and recover Theorem 8.7.4 of Volume 8.

Drill 12.8.3. Carry out the naive computation of the Pitfall and confirm it gives $\beta > n$; then locate the step in Theorem 8.7.2 that supplies the missing factor of two.

Drill 12.8.4. Using Theorem 12.8.3, determine what value $\beta(s)$ must take at $s = \frac{n}{2} + \frac{1}{3}$ for Theorem 12.8.8’s threshold to follow, and compare with the trivial exponent $\min\{s, \frac{n-1}{2}\}$ there.

Where we are. What this chapter added to Volume 8’s account.

Volume 8 proved Falconer’s theorem, Mattila’s reduction, and the trivial spherical bound, and stated that an improvement would have to be a weighted restriction estimate. It could not say what that meant precisely, because the restriction machinery had not been built.

Theorem 12.8.2 says it precisely: $\sigma(\mu, r)$ is $\|\mathcal{R}_r\mu\|_{L^2(d\sigma)}^2$. From that point the distance set problem is a problem in restriction theory, and every tool in Volumes 12–14 applies to it. Worked Example 12.8.1 tabulates which tool produced which threshold, and Worked Example 12.8.4 states what remains as a single inequality.

Where we are. Volume 12 in one line: an extension operator decomposed into wave packets is a sum of tubes, so restriction is *Takeya* in disguise — and the distance set problem, by Theorem 12.8.2, speaks the same language.

Notes

Mattila introduced the spherical average framework in 1987. Erdős's improvement is from 2005 and uses Tao's bilinear restriction theorem; Theorem 12.8.9 records the later internal reconstruction of that estimate. Volume 13 treats Wolff's planar result and Volume 14 the current records.

Chapter 12.A

Results Quoted from Volumes 1–11

Result	Source	Used in
Restriction–extension duality	Theorem 6.1.3	Theorem 12.8.2
The local form of restriction	Theorem 6.1.4	Theorem 12.7.3, Theorem 12.3.3
Stein–Tomas	Theorem 6.3.1	Theorem 12.5.1
The Knapp condition	Theorem 6.2.3	Theorem 12.5.1
Restriction implies Kakeya	Theorem 6.6.2	Worked Example 12.5.3
ε -removal	Theorem 6.5.2	Worked Example 12.5.2
The planar restriction theorem	Theorem 6.8.2	Worked Example 12.5.1
The bilinear L^4 identity	Theorem 6.8.4	Worked Example 12.3.1
Decay of $\widehat{d\sigma}$ and curvature	Worked Example 5.5.4	Worked Example 12.1.2, Worked Example 12.6.3
Lebesgue differentiation	Theorem 1.7.4	Theorem 12.7.2
Frostman’s lemma	Theorem 7.5.1	Theorem 12.7.1
Mattila’s reduction	Theorem 8.7.2	Theorem 12.8.3
The trivial spherical bound	Theorem 8.7.4	Worked Example 12.8.1
What is missing for Falconer	Worked Example 8.8.1	Chapter 12.8 throughout
Two-point problems	Worked Example 7.7.3	Worked Example 12.8.2
Loomis–Whitney	Theorem 11.5.2	Worked Example 12.3.1
Multilinear Kakeya	Theorem 11.5.7	Theorem 12.3.3
Transversality for tubes	Theorem 11.5.3	Theorem 12.2.2
Losses are $\delta^{-\varepsilon}$	Theorem 9.7.3	Worked Example 12.4.1

Debts and imports

This volume incurs no debts on later volumes and discharges one promise: Volume 8’s Worked Example 8.8.1 asserted that the spherical average is a weighted restriction quantity, and Theorem 12.8.2 proves it.

Historically this volume imported four results at the points where they first appeared. Later reconstruction proved R15–R18 internally under the strict HA-03 policy. The table below is therefore a provenance record rather than the current proof-status ledger:

Imported	Rev	What depends on it
Multilinear restriction	R15, Theorem 12.3.11	Chapter 12.4 and all exponents after
The broad–narrow induction	R16, Theorem 12.4.14	Chapter 12.5’s table
Tao’s bilinear theorem, $n \geq 3$	R17, Theorem 12.6.12	Theorem 12.8.8
Erdős’s spherical average bound	R18, Theorem 12.8.9	nothing later

The current strict ledger is different from the historical import ledger: R15–R18 are internally proved. R17’s packet-product and bush-count chain is closed at Theorems 12.6.10 and 12.6.12, and R18’s weighted annular passage is closed at Theorems 12.8.7 and 12.8.9; hence the dependency is also transitively certified.

Chapter 12.B

Notation and Index of Results

Additions to the Volume 1–11 conventions

$R, \delta = R^{-1/2}$	the large scale and the cap radius
θ	a cap of radius δ on $\mathbb{S}^{n-1}$
$T, \mathbb{T}_\theta$	a wave packet tube and the family for a cap
$g_{\theta, T}$	the wave packet piece of g
$\mathcal{E}_j$	extension restricted to S_j
ν -transverse	$ \det(\omega_1, \dots, \omega_n) \geq \nu$
K	the broad–narrow cap parameter
$\mu(B(x, r)) \leq C_\mu r^s$	μ is s -dimensional
$M_R(\mu)$	$\sup_x \mu(B(x, R^{1/2}))$
$\sigma(\mu, r), \mathcal{R}_r \mu$	the spherical average and the restriction of $\hat{\mu}$

Index of the main results

The dual box	Theorem 12.1.2
Wave packet decomposition	Theorem 12.1.4
Transversality passes to packets	Theorem 12.2.2
Two factors do not suffice	Theorem 12.2.3
The exponents match	Theorem 12.3.2
Multilinear restriction	Theorem 12.3.1
Discrete Loomis–Whitney	Theorem 12.3.4
What the packets give directly	Theorem 12.3.3
Determinant and slabs	Theorem 12.4.1
The broad–narrow dichotomy	Theorem 12.4.3
The benchmarks	Theorem 12.5.1
Bilinear L^2 in the plane	Theorem 12.6.1
The addition map	Theorem 12.6.2
Tao’s bilinear theorem	Theorem 12.6.3
The two extremes	Theorem 12.7.2
Weighted restriction	Theorem 12.7.3
The s -dimensional form	Theorem 12.7.4
The spherical average is a restriction norm	Theorem 12.8.2
The exchange rate	Theorem 12.8.3
Erdős’s threshold	Theorem 12.8.8

Plan revisions in this volume

HA-01 status correction. R15–R18 were historical import flags. Later proof-reconstruction chapters discharge all four internally. R17 now includes the full local packet-product and quadruple-count proof, and R18 includes the weighted annulus rescaling and summation. The separate refined exponent statement The sharpened post-10/3 numerical records are retained only as historical context; they are not numbered proof obligations.

Revision	Current status	Where
R15, multilinear restriction	proved locally	Theorem 12.3.1
R16, broad–narrow core	proved locally	Theorem 12.4.13; refined numerical records are historical context
R17, Tao bilinear theorem	internally proved	Theorem 12.6.3
R18, Erdős’s spherical bound	internally proved; R17 dependency satisfied	Theorem 12.8.8

Superseded from the original plan. Chapters on Loomis–Whitney and multilinear Kakeya were written into Volume 11 instead (Theorem 11.5.2, Theorem 11.5.4, both proved there); this volume begins after them.

Part 13

Volume 13

Decoupling

Preface to Volume 13

The problem decoupling solves

Volume 12 ended with a diagnosis. Almost-orthogonality — the fact that pieces of a function with disjoint frequency supports do not interfere — is a Hilbert space phenomenon, available exactly when the quantity being estimated can be written as an L^2 norm of something. Theorem 12.7.3 could sum its wave packets for free because its left side was quadratic; Theorem 12.3.3 could not, and lost a full power of R . Worked Example 12.7.7 made the test explicit.

Decoupling is the technology for recovering a substitute at the exponents where orthogonality is not free. It asserts that for f with Fourier support in a δ -neighbourhood of a curved surface, decomposed into pieces f_θ over caps,

$$\|f\|_{L^p} \lesssim_\varepsilon \delta^{-\varepsilon} \left(\sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2},$$

with the ℓ^2 sum *outside* the L^p norm. At $p = 2$ this is Plancherel and is exact. For $p > 2$ it is false in general and true for curved surfaces up to a critical exponent — 6 for the parabola in $\mathbb{R}^2$, the same number Stein–Tomas produced in Volume 6.

Why it matters beyond harmonic analysis

Bourgain and Demeter proved the sharp decoupling theorem for the paraboloid in 2015, and within two years it had settled Vinogradov’s mean value theorem in number theory — a problem open since 1935 and previously approached by Wooley’s efficient congruencing. That a statement about L^p norms of Fourier pieces should resolve a question about counting solutions of Diophantine systems is the most striking transfer in this series, and Chapter 13.6 sets out the reduction.

What this volume proves

The definitions and trivial bounds; the induction-on-scales method in a model case; the ball inflation inequality, from Volume 11's multilinear Keakeya; and the Bourgain–Demeter theorem for the parabola in $\mathbb{R}^2$, by the Guth–Maldague–Wang simplification. The higher-dimensional case is stated with its changes and its proof omitted — structurally the same, substantially longer.

It then reduces Vinogradov's theorem to moment-curve decoupling, and closes with applications: Strichartz estimates on tori, and the chain from decoupling through local smoothing to Wolff's distance set bound in the plane. That last is the second of the three Falconer sites planned in advance, Volume 12, Chapter 12.8 having been the first.

What is assumed

Volumes 1–12. Most heavily: Volume 12 for wave packets and the transversality framework; Volume 11 for multilinear Keakeya; Volume 4 for the function-space reading of the ℓ^2 sum; Volume 8 and Volume 12 for the distance set material.

How the volume is organised

Chapter 13.1 states the inequality, computes the trivial bounds, and proves sharpness at $p = 6$ — so that by the end of it the reader knows exactly what is being claimed and why the exponent is what it is.

Chapter 13.2 supplies the two elementary tools: parabolic rescaling, which is an exact algebraic identity, and a bootstrapping lemma converting a self-improving inequality into a subpolynomial bound. Both are short, and together they reduce the theorem to a single hard step.

Chapter 13.3 supplies that step. Ball inflation upgrades information at scale ρ^{-1} to scale ρ^{-2} , paid for by transversality, and it is where Volume 11's multilinear Keakeya inequality enters. In the plane the input degenerates to Fubini.

Chapter 13.4 assembles the three into the theorem, and Chapter 13.5 states precisely what changes in $\mathbb{R}^n$ — one thing: two factors no longer span, so the broad–narrow split needs n .

Chapter 13.6 and Chapter 13.8 give the number-theoretic application and compare it with Wooley's method. Chapter 13.9 collects three analytic applications and discharges the Falconer promise.

The dependency chain of Theorem 13.4.2 runs

and every link is proved in this series. That was not true when the volume was planned: multilinear Kakeya was an import, and the chain terminated in a citation. Discharging it (Volume 11, Theorem 11.5.4) is what makes the present volume self-contained, and it is the clearest single illustration of why the black boxes were worth removing.

1	What decoupling asserts	definitions, trivial bounds, sharpness
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Chapter 13.1

What Decoupling Asserts

13.1.1 The statement

Definition 13.1.1 (The parabola and its caps). Let $P = \{(t, t^2) : t \in [0, 1]\}$ and let $\mathcal{N}_\delta$ denote its δ -neighbourhood in $\mathbb{R}^2$. Partition $[0, 1]$ into $N = \delta^{-1/2}$ intervals of length $\delta^{1/2}$, and let θ denote the corresponding pieces of $\mathcal{N}_\delta$.

Lemma 13.1.2 (Caps are boxes). *Each θ is contained in a rectangle of dimensions $\approx \delta^{1/2} \times \delta$, with the short side normal to P . Consequently its dual box has dimensions $\approx \delta^{-1/2} \times \delta^{-1}$, with the long side tangent to P .*

Proof. Over an interval of length $\delta^{1/2}$ centred at t_0 , the parabola deviates from its tangent line by

$$|t^2 - (t_0^2 + 2t_0(t - t_0))| = (t - t_0)^2 \leq \delta,$$

so the arc lies in a δ -slab about the tangent. Adding the δ -neighbourhood doubles the slab. The dual box is the box with reciprocal sides. $\square$

Where we are. The choice $\delta^{1/2}$ for the cap length is forced, exactly as $\delta = R^{-1/2}$ was in Volume 12, Theorem 12.1.2: it is the length over which the parabola's curvature produces a deviation comparable to δ . Longer caps are not contained in δ -slabs; shorter ones waste the curvature.

So the two volumes are describing the same geometry. Volume 12 fixed a ball B_R and derived caps of radius $R^{-1/2}$; here we fix a neighbourhood width δ and derive caps of length $\delta^{1/2}$. With $\delta = R^{-1}$ the two agree, and the dual boxes $\delta^{-1/2} \times \delta^{-1}$ are the wave packet tubes $R^{1/2} \times R$.

Definition 13.1.3 (The decoupling constant). For $2 \leq p < \infty$, let $D_p(\delta)$ be the smallest constant such that

$$\|f\|_{L^p(\mathbb{R}^2)} \leq D_p(\delta) \left(\sum_{\theta} \|f_{\theta}\|_{L^p(\mathbb{R}^2)}^2 \right)^{1/2}$$

for every f with $\text{supp } \widehat{f} \subseteq \mathcal{N}_\delta$, where f_θ is the piece of f with frequency support in θ .

Theorem 13.1.4 (Bourgain–Demeter for the parabola). *For $2 \leq p \leq 6$ and every $\varepsilon > 0$,*

$$D_p(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}.$$

The range is sharp: $D_p(\delta) \rightarrow \infty$ faster than any $\delta^{-\varepsilon}$ when $p > 6$.

Proof. The sharpness is Theorem 13.1.8, proved later in this chapter and depending on nothing stated here. The positive statement is proved in Chapter 13.4 as Theorem 13.4.2, after the machinery of Chapter 13.2 and Chapter 13.3 is in place; the present chapter establishes only what the statement means and why 6 is critical. $\square$

13.1.2 The trivial bounds

Proposition 13.1.5 (The two easy estimates). *For every $\delta \in (0, 1)$:*

$$(i) \quad D_2(\delta) \lesssim 1;$$

$$(ii) \quad D_p(\delta) \leq N^{1/2} = \delta^{-1/4} \text{ for every } p \geq 2.$$

Proof. (i) The frequency supports θ have bounded overlap, so by Plancherel

$$\begin{aligned} \|f\|_{L^2}^2 &= \|\widehat{f}\|_{L^2}^2 \approx \sum_{\theta} \|\widehat{f_\theta}\|_{L^2}^2 \\ &= \sum_{\theta} \|f_\theta\|_{L^2}^2. \end{aligned}$$

(ii) By the triangle inequality and then Cauchy–Schwarz over the N caps,

$$\begin{aligned} \|f\|_{L^p} &\leq \sum_{\theta} \|f_\theta\|_{L^p} \\ &\leq N^{1/2} \left(\sum_{\theta} \|f_\theta\|_{L^p}^2 \right)^{1/2}. \end{aligned} \quad \square$$

Where we are. Theorem 13.1.5 brackets the problem. At $p = 2$ the answer is $O(1)$ and the inequality is an identity — there is nothing to prove. At every p the answer is at most $\delta^{-1/4}$, by an argument using no property of the parabola whatever.

So Theorem 13.1.4 asserts that for $2 \leq p \leq 6$ the truth is at the $p = 2$

end of that bracket, not the trivial end. *That is the entire content*, and the curvature of the parabola is what supplies it: for a straight segment in place of P , the trivial bound (ii) is attained and no improvement holds.

13.1.3 The case $p = 4$, in full

Before the machinery of Chapter 13.4, one case of Theorem 13.1.4 can be proved outright, in half a page and with no ε -loss at all. It is the oldest result in the subject.

Lemma 13.1.6 (The sum-set is 2-to-1). *Let θ_i be the caps of Theorem 13.1.1 and, for a pair (i, j) , let B_{ij} be the smallest box containing $\theta_i + \theta_j$. Then each point of $\mathbb{R}^2$ lies in at most C of the boxes B_{ij} , with C absolute.*

Proof. θ_i is contained in a $\delta^{1/2} \times \delta$ box about (t_i, t_i^2) by Theorem 13.1.2, so B_{ij} is a $\delta^{1/2} \times \delta$ box about $(t_i + t_j, t_i^2 + t_j^2)$.

Suppose $\xi \in B_{ij}$. Then $t_i + t_j$ is determined to within $O(\delta^{1/2})$ and $t_i^2 + t_j^2$ to within $O(\delta)$, hence

$$t_i t_j = \frac{1}{2} \left((t_i + t_j)^2 - (t_i^2 + t_j^2) \right)$$

is determined to within $O(\delta)$. So the unordered pair $\{t_i, t_j\}$ — being the roots of $X^2 - (t_i + t_j)X + t_i t_j$ — is determined to within $O(\delta^{1/2})$, that is, to within one cap. Since the caps have length $\delta^{1/2}$, at most C ordered pairs (i, j) are possible. $\square$

Where we are. The lemma is the parabola's curvature, stated arithmetically. Direct enumeration confirms it: binning $\theta_i + \theta_j$ at the correct resolution $\delta^{1/2} \times \delta$, the maximum number of pairs per cell is exactly 2 — the swap $(t_i, t_j) \leftrightarrow (t_j, t_i)$ — at $M = 40, 80, 160, 320$ caps alike.

The control experiment is the point. Replacing the parabola by the line $\{(t, 0)\}$, the second coordinate carries no information, $t_i t_j$ is not determined, and the same enumeration gives M pairs per cell — growing without bound. *Curvature is exactly what makes the sum-set map finite-to-one*, and Theorem 13.1.6 is the sharpest form of that statement.

Theorem 13.1.7 (ℓ^2 decoupling at $p = 4$; Fefferman, Córdoba). $D_4(\delta) \lesssim 1$, uniformly in δ , with an absolute constant. *There is no $\delta^{-\varepsilon}$ and no dependence on any parameter.*

Proof. By Plancherel applied to f^2 ,

$$\|f\|_{L^4}^2 = \|f^2\|_{L^2} = \|\widehat{f} * \widehat{f}\|_{L^2}.$$

Write $f = \sum_{\theta} f_{\theta}$, so $\widehat{f} * \widehat{f} = \sum_{i,j} \widehat{f_{\theta_i}} * \widehat{f_{\theta_j}}$, and the (i, j) term is supported in B_{ij} . By Theorem 13.1.6 these have overlap at most C , so the sum is almost orthogonal in L^2 :

$$\begin{aligned} \|\widehat{f} * \widehat{f}\|_{L^2}^2 &\leq C \sum_{i,j} \|\widehat{f_{\theta_i}} * \widehat{f_{\theta_j}}\|_{L^2}^2 \\ &= C \sum_{i,j} \|f_{\theta_i} f_{\theta_j}\|_{L^2}^2. \end{aligned}$$

By Cauchy–Schwarz, $\|f_{\theta_i} f_{\theta_j}\|_{L^2} \leq \|f_{\theta_i}\|_{L^4} \|f_{\theta_j}\|_{L^4}$, so the right side is

$$C \left(\sum_i \|f_{\theta_i}\|_{L^4}^2 \right)^2.$$

Taking square roots gives $\|f\|_{L^4}^2 \lesssim \sum_{\theta} \|f_{\theta}\|_{L^4}^2$, which is the claim. $\square$

Where we are. Three things to take from this proof.

It is the model for everything later. Plancherel, an overlap count, and Cauchy–Schwarz. Theorem 13.4.2 at $p = 6$ has the same shape with ball inflation replacing the overlap count.

Why it stops at $p = 4$. The argument writes $\|f\|_{L^4}^2$ as an L^2 norm, so it applies exactly when $p/2 = 2$. For $p = 6$ one would need f^3 and a three-fold sum-set, whose overlap is *not* bounded — three points on a parabola are not determined by their first two power sums. That is the precise reason the $p = 6$ theorem needed thirty more years.

No ε . Theorem 13.1.7 has an absolute constant, whereas Theorem 13.1.4 carries $\delta^{-\varepsilon}$. For the parabola the truth at $p = 6$ is $(\log \delta^{-1})^c$, by Guth, Maldague and Wang; at $p = 4$ it is $O(1)$, and that is best possible.

WORKED EXAMPLE 13.1.1— TIER B **Interpolating from $p = 4$**

Theorem 13.1.7 gives $p = 4$ free. What does interpolation give?

The two endpoints. $D_2(\delta) \lesssim 1$ (Theorem 13.1.5(i)) and $D_4(\delta) \lesssim 1$ (Theorem 13.1.7).

What interpolation yields. For $2 \leq p \leq 4$, $D_p(\delta) \lesssim 1$ by Hölder — so the whole range up to 4 is settled with no loss whatever.

What it does not yield. Anything beyond $p = 4$. Interpolation fills in between known endpoints and never produces a new one, exactly as Volume 12, Worked Example 12.5.4 warned.

So what is the content of $p = 6$? Precisely the interval $(4, 6]$. The trivial bound there is $\delta^{-1/4}$ and the truth is $\delta^{-\varepsilon}$, so the gain is a full power of $\delta^{-1/4}$ — and by Theorem 13.1.8 nothing beyond 6 is possible.

The picture to carry. Three regimes: $p \leq 4$ elementary and sharp with constant $O(1)$; $4 < p \leq 6$ the Bourgain–Demeter theorem, hard, with $\delta^{-\varepsilon}$; $p > 6$ false. The middle regime is the subject of this volume.

13.1.4 Sharpness

Proposition 13.1.8 (The constructive example). *For $p > 6$ there is no ε for which $D_p(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$. Specifically,*

$$D_p(\delta) \gtrsim \delta^{-\frac{1}{4} + \frac{3}{2p}},$$

which for $p > 6$ is a positive power of δ^{-1} .

Proof. Choose f_θ to be a wave packet adapted to the dual box T_θ of Theorem 13.1.2, normalised so that $|f_\theta| \approx 1$ on T_θ and with phases chosen so that all the f_θ take the value 1 at the origin.

Step 1. *The left side.*

Each f_θ has frequency support in $\mathcal{N}_\delta$, a set of diameter ≈ 1 . So the phases of the f_θ remain aligned on a ball of radius ≈ 1 about the origin, on which $|f| \approx N$. Hence

$$\|f\|_{L^p} \gtrsim N.$$

Step 2. *The right side.*

$|T_\theta| \approx \delta^{-1/2} \cdot \delta^{-1} = \delta^{-3/2}$, so $\|f_\theta\|_{L^p}^p \approx \delta^{-3/2}$ and $\|f_\theta\|_{L^p} = \delta^{-3/(2p)}$. Therefore

$$\begin{aligned} \left(\sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2} &\approx (N \delta^{-3/p})^{1/2} \\ &= N^{1/2} \delta^{-3/(2p)}. \end{aligned}$$

Step 3. *The ratio.*

Dividing and substituting $N = \delta^{-1/2}$,

$$\begin{aligned} \frac{\|f\|_{L^p}}{(\sum_\theta \|f_\theta\|_{L^p}^2)^{1/2}} &\gtrsim N^{1/2} \delta^{3/(2p)} \\ &= \delta^{-\frac{1}{4} + \frac{3}{2p}}. \end{aligned}$$

This tends to infinity as $\delta \rightarrow 0$ precisely when $-\frac{1}{4} + \frac{3}{2p} < 0$, that is $p > 6$. $\square$

Pitfall: the constructive region is a unit ball, not the common dual box

Step 1 asserts that the packets stay aligned on a ball of radius ≈ 1 . It is tempting to say instead that they align on the intersection of the dual boxes T_θ , which contains a ball of radius $\approx \delta^{-1/2}$ and has area $\approx \delta^{-1}$. That is *wrong*, and the error is instructive because it inverts the conclusion.

The f_θ have *different* frequencies, spanning a set of diameter ≈ 1 . Two packets whose frequencies differ by ≈ 1 acquire a relative phase ≈ 1 over a distance ≈ 1 , so they cannot remain aligned further than that — regardless of how large their dual boxes are. The dual box governs where a *single* packet is supported, not where several interfere constructively.

Using the larger region gives $\|f\|_{L^p} \gtrsim N\delta^{-1/p}$ and a ratio $\delta^{-1/4+1/(2p)}$, which blows up for every $p > 2$ — contradicting Theorem 13.1.5(i) at $p = 2$ and the theorem throughout. A computation that contradicts a case one can verify independently is wrong, and this is the case to check.

WORKED EXAMPLE 13.1.2— TIER A **The critical exponent, computed**

The exponent in Theorem 13.1.8 is $-\frac{1}{4} + \frac{3}{2p}$, and tabulating it locates the threshold.

p	$-\frac{1}{4} + \frac{3}{2p}$	
4	+0.125	ratio decays; no obstruction
5	+0.050	decays
6	0	bounded — critical
7	−0.036	blows up; estimate fails
8	−0.063	fails
10	−0.100	fails

Numerically. At $\delta = 10^{-4}, 10^{-6}, 10^{-8}$ the ratio at $p = 4$ is 0.316, 0.178, 0.100; at $p = 6$ it is 1.000 throughout; at $p = 8$ it is 1.778, 2.371, 3.162. Decaying, constant, growing — exactly as the exponent predicts.

The number 6. It is $\frac{2(n+1)}{n-1}$ at $n = 2$, the Stein–Tomas exponent of Volume 6, Theorem 6.3.1. That is not a coincidence: Worked Example 13.4.1 will show decoupling implies a restriction estimate, and the critical exponents must therefore agree.

In higher dimensions. For the paraboloid in $\mathbb{R}^n$ the critical exponent is $\frac{2(n+1)}{n-1}$ — 6, 4, $\frac{10}{3}$ for $n = 2, 3, 4$ — and Chapter 13.5 states the theorem there.

WORKED EXAMPLE 13.1.3— TIER A The trivial bounds, computed

Theorem 13.1.5 brackets $D_p(\delta)$ between $O(1)$ and $\delta^{-1/4}$. It is worth seeing the two ends numerically.

δ	$N = \delta^{-1/2}$	$N^{1/2} = \delta^{-1/4}$	theorem gives
10^{-4}	10^2	10	$\delta^{-\varepsilon}$
10^{-8}	10^4	10^2	$\delta^{-\varepsilon}$
10^{-12}	10^6	10^3	$\delta^{-\varepsilon}$

The gap being closed. At $\delta = 10^{-12}$ the trivial bound allows 10^3 and the truth is subpolynomial. That factor of 10^3 is the content of Theorem 13.1.4, and it grows without bound as $\delta \rightarrow 0$.

But see Worked Example 13.2.2. The theorem's constant C_ε is not effective, so at any *named* δ the theorem may give a worse numerical bound than the trivial one. The improvement is asymptotic.

Where the trivial bound is attained. For a straight segment in place of the parabola: then all caps have the same dual direction, the packets add coherently on a common tube, and no cancellation occurs. Curvature is the entire difference, as Worked Example 13.5.2(d) records.

WORKED EXAMPLE 13.1.4— TIER B Where the ℓ^2 sits, and why

The inequality of Theorem 13.1.3 has the ℓ^2 sum outside the L^p norm. Three other arrangements are conceivable and each is either trivial or false.

(a) ℓ^2 *inside*: $\|(\sum_\theta |f_\theta|^2)^{1/2}\|_{L^p}$. This is the square function, and the corresponding inequality is *stronger* than decoupling for $p \geq 2$ by Minkowski. It is also false for the parabola at $p = 6$ — the square function conjecture is open — so decoupling is the correct weakening.

(b) ℓ^1 *outside*: $\sum_\theta \|f_\theta\|_{L^p}$. This is the triangle inequality, Theorem 13.1.5(ii) before Cauchy–Schwarz, and it holds trivially with constant 1. It is too weak to be useful, since it loses $N^{1/2}$ against the ℓ^2 form.

(c) ℓ^p *outside*. Interpolates between the two and is false for $p > 2$ at the sharp constant.

Why ℓ^2 is right. Because at $p = 2$ it is an identity (Theorem 13.1.5(i)), so the inequality is sharp at one endpoint and one asks how far the sharpness persists. The answer, $p \leq 6$, is the theorem.

The function-space reading. Volume 4, Worked Example 4.6.3 distinguished the Besov and Triebel–Lizorkin scales by exactly this: ℓ^q outside the L^p norm or inside. Decoupling is a Besov-type statement, and the square function is

Triebel–Lizorkin-type. That the first is provable and the second open is the same phenomenon as the relative difficulty of those two scales.

WORKED EXAMPLE 13.1.5— TIER C: RESEARCH TASTE **Why decoupling was invented**

Decoupling is recent, and it is worth asking what problem it was invented to solve, since the answer explains its form.

(a) *The obstruction it removes.* Volume 12, Worked Example 12.7.7 established that almost-orthogonality is free exactly when the quantity is quadratic. For $p \neq 2$ the pieces of a function over a curved surface are not orthogonal, and no identity replaces Plancherel.

(b) *What was tried first.* The *square function* conjecture — ℓ^2 inside the norm — is the natural strengthening and is open at $p = 6$. Wolff’s insight was that moving the ℓ^2 *outside* yields a statement that is weaker, still useful, and provable.

(c) *Why weaker suffices.* In every application here — Vinogradov, Strichartz, local smoothing — the caps carry comparable mass, and for such functions the ℓ^2 -outside and ℓ^2 -inside statements differ by a constant. The weakening costs nothing where it is used.

(d) *Why it generalises.* The proof uses only rescaling, transversality and induction. It has no arithmetic content, which is why it settled a number theory problem that resisted arithmetic methods for eighty years (Worked Example 13.8.1).

(e) *The transferable lesson.* When a natural conjecture resists, look for the weakest statement that still serves the applications. The square function conjecture remains open; decoupling took its place and did its work. Worked Example 13.1.4 lists the four arrangements, and only one of them was both provable and useful.

WORKED EXAMPLE 13.1.6— TIER C: RESEARCH TASTE **Decoupling as a way of thinking**

Decoupling is a technique, but it is more useful as a habit of mind. Four questions it teaches one to ask.

(a) *What are the caps?* Any function with Fourier support on a curved set decomposes over caps, and the caps are determined by the curvature through Theorem 13.1.2. Before estimating anything, compute the cap dimensions and their dual boxes. Volume 12, Worked Example 12.1.4 makes the same

point for wave packets, and it is the same computation.

(b) *At what exponent is orthogonality free?* At $p = 2$ always, by Plancherel. Away from 2 it is not, and Volume 12, Worked Example 12.7.7 gives the test: the quantity must be expressible as an L^2 norm of something. Decoupling exists to buy a substitute where the test fails.

(c) *What is the trivial bound, and what would sharpness cost?* Here the trivial bound is $N^{1/2} = \delta^{-1/4}$ (Theorem 13.1.5(ii)) and the truth is $\delta^{-\varepsilon}$. Knowing both endpoints before starting tells one how much is at stake, and Theorem 13.1.8 shows where the boundary lies.

(d) *Is the loss ε , polylog, or a power?* These differ. Volume 12, Worked Example 12.4.3 and Theorem 13.2.3 both trace the answer to the round count of the induction: $\log R$ rounds give a fixed power, $\log \log R$ rounds give $\delta^{-\varepsilon}$. Counting rounds is the fastest way to predict the shape of the answer.

The wider pattern. Every major estimate in Volumes 9 to 14 is an induction on scales, and they differ mainly in what is being propagated: tube multiplicities for Kakeya, a weighted ℓ^2 hypothesis for multilinear restriction (Volume 12, Theorem 12.3.9), a surface set for the Guth–Katz incidence bound (Volume 11, Theorem 11.6.10), and here a decoupling constant. Recognising which object is being carried is the quickest route into any of these papers.

WORKED EXAMPLE 13.1.7— TIER B The four objects, and their scales

Decoupling arguments manipulate four objects at once, and keeping their scales straight is most of the difficulty for a first reader.

Object	Lives in	Scale	Introduced
$\mathcal{N}_\delta$	frequency	thickness δ	Theorem 13.1.1
cap θ	frequency	$\delta^{1/2} \times \delta$	Theorem 13.1.2
dual box T_θ	physical	$\delta^{-1/2} \times \delta^{-1}$	Theorem 13.1.2
ball B	physical	δ^{-1}	Theorem 13.1.3

The two pairings. Frequency and physical scales are reciprocal: $\delta \leftrightarrow \delta^{-1}$ and $\delta^{1/2} \leftrightarrow \delta^{-1/2}$. Every quantity in the volume is one of these four or a product of them.

The counts that follow. $\#\theta = \delta^{-1/2}$; $|T_\theta| = \delta^{-3/2}$; $|B| = \delta^{-2}$; and $\#\theta \cdot |T_\theta| = \delta^{-2} = |B|$, so the dual boxes tile the ball exactly once. That identity is the check Worked Example 13.4.4 recommends, and it fails for any other cap length.

Where each appears. The neighbourhood defines the problem; the caps index the sum; the dual boxes carry the wave packets of Volume 12, Theorem 12.1.4;

the ball is where the norm is taken. Confusing the last two is the commonest error, and Worked Example 13.3.1 explains why they differ by a square.

Drills

Drill 13.1.1. Verify Theorem 13.1.2: check that $(t-t_0)^2 \leq \delta$ for $|t-t_0| \leq \delta^{1/2}$, and identify where curvature is used.

Drill 13.1.2. Prove Theorem 13.1.5(ii) and confirm that no property of the parabola is used.

Drill 13.1.3. Carry out the erroneous computation of the Pitfall — constructive region of area δ^{-1} — and confirm it contradicts Theorem 13.1.5(i) at $p = 2$.

Drill 13.1.4. Compute $\frac{2(n+1)}{n-1}$ for $n = 2, \dots, 6$ and compare with the Stein–Tomas exponents of Volume 6.

Notes

Decoupling inequalities were introduced by Wolff for the cone in 2000 and developed by Bourgain and Demeter, whose sharp paraboloid theorem is from 2015. The formulation with ℓ^2 outside the norm is theirs; Demeter’s book is the standard reference.

Chapter 13.2

Induction on Scales

Every proof of a decoupling inequality is an induction on scales, and the method has two ingredients: a *rescaling* identity saying that a piece of the problem at one scale is the whole problem at another, and a *bootstrapping* lemma converting a self-improving inequality into a subpolynomial bound. Both are elementary and both are proved here.

13.2.1 Parabolic rescaling

Lemma 13.2.1 (Parabolic rescaling). *Let $\tau \subseteq [0, 1]$ be an interval of length σ with left endpoint t_0 , and let $L_\tau : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the affine map*

$$L_\tau(x_1, x_2) = \left(\frac{x_1 - t_0}{\sigma}, \frac{x_2 - t_0^2 - 2t_0(x_1 - t_0)}{\sigma^2} \right).$$

Then L_τ maps the arc of P over τ onto all of P , and

$$L_\tau(\mathcal{N}_\delta \cap \{x_1 \in \tau\}) = \mathcal{N}_{\delta/\sigma^2}(P).$$

Proof. Write $t = t_0 + \sigma s$ with $s \in [0, 1]$, so the arc is $\{(t_0 + \sigma s, (t_0 + \sigma s)^2)\}$. Then

$$\begin{aligned} \frac{t - t_0}{\sigma} &= s, & \frac{t^2 - t_0^2 - 2t_0\sigma s}{\sigma^2} &= \frac{(t_0 + \sigma s)^2 - t_0^2 - 2t_0\sigma s}{\sigma^2} \\ & & &= \frac{\sigma^2 s^2}{\sigma^2} \\ & & &= s^2, \end{aligned}$$

so L_τ carries the arc to $\{(s, s^2) : s \in [0, 1]\} = P$ exactly.

L_τ is affine with linear part $\text{diag}(\sigma^{-1}, \sigma^{-2})$ composed with a shear, and the shear preserves the second coordinate's scaling. The δ -neighbourhood

is measured in the direction normal to P , which for a short arc is within a bounded factor of the x_2 -direction; that direction is scaled by σ^{-2} , so the neighbourhood width becomes δ/σ^2 . $\square$

Where we are. Theorem 13.2.1 is exact, not asymptotic: the image is *precisely* the parabola, because $s \mapsto s^2$ is the same function after the substitution. That exactness is special to the parabola and to the moment curve; for a general curved surface the analogous map is only approximate and the induction acquires error terms.

The consequence to remember is the identity

$$D_p(\delta) \text{ on an arc of length } \sigma = D_p(\delta/\sigma^2) \text{ on the whole parabola.}$$

So a decoupling problem at scale δ , restricted to an arc of length σ , is the *same* problem at scale δ/σ^2 . Taking $\sigma = \delta^{1/2}$ gives $\delta/\sigma^2 = 1$: a single cap is the trivial problem, which is why $\delta^{1/2}$ is the right cap size.

13.2.2 Bernstein's inequality

One further elementary tool is used repeatedly and is worth proving once.

Proposition 13.2.2 (Bernstein's inequality on a box). *Let R be a box in $\mathbb{R}^n$ of dimensions $a_1 \times \cdots \times a_n$ and let $\hat{g}$ be supported in R . Then for $1 \leq p \leq q \leq \infty$,*

$$\|g\|_{L^q(\mathbb{R}^n)} \lesssim_n |R|^{\frac{1}{p} - \frac{1}{q}} \|g\|_{L^p(\mathbb{R}^n)}.$$

Proof. Let ψ be Schwartz with $\hat{\psi} = 1$ on R and $\hat{\psi}$ supported in $2R$, obtained by translating and dilating a fixed bump. Then $g = g * \psi$, and $\|\psi\|_{L^r} \approx |R|^{1-1/r}$ for every r , since ψ is essentially $|R|$ on the dual box R^* of volume $|R|^{-1}$ and decays rapidly beyond.

By Young's inequality with $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$,

$$\begin{aligned} \|g\|_{L^q} &= \|g * \psi\|_{L^q} \\ &\leq \|g\|_{L^p} \|\psi\|_{L^r} \\ &\approx |R|^{1-\frac{1}{r}} \|g\|_{L^p} \\ &= |R|^{\frac{1}{p} - \frac{1}{q}} \|g\|_{L^p}. \end{aligned} \quad \square$$

Where we are. Bernstein converts one exponent into another at a cost measured by the *volume of the frequency support*. It is the reason cap decompositions are useful at all: a single cap θ has $|\theta| = \delta^{(n+1)/2}$, so Bernstein on one cap is cheap, while on the whole neighbourhood $|\mathcal{N}_\delta| \approx \delta$ it is expensive.

Two cautions. It is sharp only when g is a single bump on the dual box; for oscillating g it is very lossy. And it is not an orthogonality statement — it holds for any g with the given support, which is precisely why it cannot substitute for decoupling.

WORKED EXAMPLE 13.2.1— TIER A **Bernstein against decoupling**

Both estimate $\|f\|_{L^p}$ from cap data, and comparing them shows what decoupling adds.

Bernstein, applied crudely. Each f_θ has frequency support of volume $\delta^{3/2}$ in $\mathbb{R}^2$, so $\|f_\theta\|_{L^6} \lesssim \delta^{3/2(1/2-1/6)}\|f_\theta\|_{L^2} = \delta^{1/2}\|f_\theta\|_{L^2}$. Summing over the $\delta^{-1/2}$ caps by the triangle inequality gives

$$\begin{aligned} \|f\|_{L^6} &\lesssim \delta^{-1/2} \cdot \delta^{1/2} \max_{\theta} \|f_\theta\|_{L^2} \\ &= \max_{\theta} \|f_\theta\|_{L^2}, \end{aligned}$$

which is far weaker than decoupling: it controls by a maximum, losing all information about how the caps combine.

Decoupling. $\|f\|_{L^6} \lesssim_{\varepsilon} \delta^{-\varepsilon} (\sum_{\theta} \|f_\theta\|_{L^6}^2)^{1/2}$: an ℓ^2 sum, which for $\delta^{-1/2}$ comparable caps is $\delta^{-1/4}$ times the individual size.

The comparison. Decoupling is better exactly when the caps are *spread*, and Bernstein is competitive only when one cap dominates. Since the extremal configurations for restriction are single Knapp caps (Volume 6, Theorem 6.2.3), both are needed — and every argument in this volume uses Bernstein for the narrow case and decoupling for the broad one.

The rule. Bernstein trades integrability for frequency volume; decoupling trades a sum for an ℓ^2 sum. They are not comparable, and knowing which to reach for is a matter of asking whether the caps are concentrated or spread.

13.2.3 Bootstrapping

Proposition 13.2.3 (A self-improving inequality gives $\delta^{-\varepsilon}$). *Let $D : (0, 1) \rightarrow [1, \infty)$ be non-increasing and suppose there are constants $C \geq 1$ and $\delta_0 \in (0, 1)$ with*

$$D(\delta) \leq C D(\delta^{1/2}) \quad \text{for all } \delta \in (0, \delta_0). \quad (13.2.1)$$

Then for every $\varepsilon > 0$ there is C_ε with $D(\delta) \leq C_\varepsilon \delta^{-\varepsilon}$ for all $\delta \in (0, 1)$.

Proof. Step 1. Iterate.

Applying (13.2.1) m times,

$$D(\delta) \leq C^m D(\delta^{2^{-m}}).$$

Step 2. *Choose the number of rounds.*

Let m be least with $\delta^{2^{-m}} \geq \delta_0$, that is

$$\begin{aligned} 2^{-m} \log \frac{1}{\delta} &\leq \log \frac{1}{\delta_0} \\ , \quad \text{so} \quad m &= \left\lceil \log_2 \frac{\log(1/\delta)}{\log(1/\delta_0)} \right\rceil. \end{aligned}$$

Then $D(\delta^{2^{-m}}) \leq D(\delta_0)$ since D is non-increasing.

Step 3. *Estimate the accumulated constant.*

Writing $L = \log(1/\delta)$,

$$\begin{aligned} C^m &\leq C \cdot C^{\log_2(L/\log(1/\delta_0))} \\ &= C \left(\frac{L}{\log(1/\delta_0)} \right)^{\log_2 C} \lesssim_{C, \delta_0} \left(\log \frac{1}{\delta} \right)^{\log_2 C}. \end{aligned}$$

Step 4. *Conclude.*

A power of $\log \frac{1}{\delta}$ is $O_\varepsilon(\delta^{-\varepsilon})$ for every $\varepsilon > 0$, since $(\log u)^a/u^\varepsilon \rightarrow 0$ as $u \rightarrow \infty$. Hence $D(\delta) \leq C_\varepsilon \delta^{-\varepsilon}$. $\square$

Where we are. Step 2 is where the ε comes from, and it is worth comparing with Volume 12, Worked Example 12.4.3. There the point was that an induction with $\log R$ rounds produces a fixed power and only one with $\log \log R$ rounds produces R^ε . Here the round count is $m \approx \log_2 \log \frac{1}{\delta}$ — a log log — because each round *squares* the scale rather than multiplying it by a constant.

That is exactly why Theorem 13.2.1 is stated with $\delta \mapsto \delta/\sigma^2$: the square is what makes the induction terminate in log log steps.

WORKED EXAMPLE 13.2.2— TIER A How slowly the bound actually kicks in

Theorem 13.2.3 concludes $D(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$, and it is worth seeing how weak that is for any δ one might name.

Take $C = 4$ and $\delta_0 = \frac{1}{2}$, so $C^m = 4^{\lceil \log_2 \log_2(1/\delta) \rceil}$.

δ	rounds m	C^m	$\delta^{-0.1}$
10^{-6}	5	1.0×10^3	4.0
10^{-12}	6	4.1×10^3	16
10^{-30}	7	1.6×10^4	1.0×10^3
10^{-100}	9	2.6×10^5	1.0×10^{10}

Reading the table. The polylogarithmic bound C^m exceeds $\delta^{-0.1}$ until δ is somewhere below 10^{-30} . The theorem is true — polylog is eventually smaller than any power — but the crossover is astronomically late.

Why this is worth saying. An ε -loss is often described as “harmless”. It is harmless *asymptotically*, and Volume 9, Theorem 9.7.3 showed it is exactly the accuracy dimension statements need. It is not harmless for any numerical purpose, and a reader who imagines the constants are moderate will be surprised.

The honest formulation. $D_p(\delta)$ grows slower than any power of δ^{-1} ; the implicit constant C_ε blows up as $\varepsilon \rightarrow 0$, and no proof in this subject controls it usefully.

WORKED EXAMPLE 13.2.3— TIER B The shape of a decoupling proof

Theorem 13.2.1 and Theorem 13.2.3 are the frame; every decoupling argument fills it in the same way.

Step 1: split the frequency support. Partition $\mathcal{N}_\delta$ into caps at an intermediate scale ρ with $\delta^{1/2} < \rho < 1$.

Step 2: separate transverse from parallel. Tuples of caps that are transverse are handled by a multilinear estimate — the broad case, exactly as in Volume 12, Theorem 12.4.3. Tuples that cluster are handled by rescaling.

Step 3: rescale the clustered part. By Theorem 13.2.1 a cluster inside an arc of length σ is the same problem at scale δ/σ^2 , so it contributes $D_p(\delta/\sigma^2)$.

Step 4: assemble into a self-improving inequality. Steps 2 and 3 give $D_p(\delta) \leq C \cdot D_p(\delta/\sigma^2) \cdot (\text{transverse factor})$, and choosing σ so that $\delta/\sigma^2 = \delta^{1/2}$ — that is $\sigma = \delta^{1/4}$ — puts it in the form of (13.2.1).

Step 5: bootstrap. Theorem 13.2.3.

What is hard. Only Step 2. The transverse estimate is *ball inflation*, Chapter 13.3, and it is where multilinear Keakeya enters. Everything else in this chapter is bookkeeping, and that is why the chapter is short.

WORKED EXAMPLE 13.2.4— TIER B **Rescaling on other surfaces**

Theorem 13.2.1 is exact for the parabola. That exactness is special, and knowing where it fails is what tells one which surfaces are tractable.

(a) *The parabola and paraboloid.* Exact, by the identity $|\xi|^2 - |\xi_0|^2 - 2\xi_0 \cdot (\xi - \xi_0) = |\xi - \xi_0|^2$. The affine map is available because the defining function is a quadratic form.

(b) *The moment curve.* Exact, by the same mechanism: the map $t \mapsto t_0 + \sigma t$ carries $(t, t^2, \dots, t^k)$ to an affine image of itself, since each t^j expands in lower powers.

(c) *The cone.* Exact under *Lorentz* rescaling rather than parabolic, because the cone is invariant under the Lorentz group rather than under parabolic dilations. Theorem 13.9.4 records that the induction is otherwise the same.

(d) *The sphere.* Only approximate: a cap of the sphere is affinely equivalent to a cap of the paraboloid up to errors of higher order. In practice one passes to the paraboloid first, which is legitimate because Worked Example 13.5.2(a) makes every elliptic surface locally a paraboloid graph.

(e) *A general curved surface.* Approximate, with errors that must be tracked through the induction. This is the technical reason decoupling theorems are usually stated for model surfaces and transferred by affine equivalence.

The check. Given a surface, ask whether the group acting on it — affine, Lorentz, or none — carries caps to the whole surface. If yes, the induction is clean; if no, expect error terms and a longer argument.

13.2.4 A worked induction

The abstract statements of this chapter are easier to trust after running one induction concretely. We prove a decoupling-type estimate that is far weaker than Theorem 13.1.4 but is genuinely obtained by induction on scales, with every constant explicit.

Proposition 13.2.4 (A crude self-improvement). *For $2 \leq p \leq 6$,*

$$D_p(\delta) \leq C D_p(\delta^{1/2}) D_p^{\text{loc}}(\delta^{1/2}),$$

where $D_p^{\text{loc}}(\sigma)$ is the decoupling constant for arcs of length σ subdivided into arcs of length $\sigma \cdot \sigma^{1/2}$.

Proof. Let $\{\tau\}$ be the arcs of length $\delta^{1/4}$ and $\{\theta\}$ those of length $\delta^{1/2}$, so each τ contains $\delta^{-1/4}$ of the θ .

Step 1. *Decouple into the intermediate arcs.*

The function f has Fourier support in $\mathcal{N}_\delta \subseteq \mathcal{N}_{\delta^{1/2}}(P)$, and the arcs τ have length $\delta^{1/4} = (\delta^{1/2})^{1/2}$, so they are the natural caps at scale $\delta^{1/2}$. By definition of the decoupling constant at that scale,

$$\|f\|_{L^p} \leq D_p(\delta^{1/2}) \left(\sum_{\tau} \|f_{\tau}\|_{L^p}^2 \right)^{1/2}.$$

Step 2. *Decouple each intermediate arc.*

Fix τ . By Theorem 13.2.1 the arc of length $\delta^{1/4}$ at scale δ is an affine image of the whole parabola at scale $\delta/(\delta^{1/4})^2 = \delta^{1/2}$, and under that image the arcs $\theta \subseteq \tau$ become the caps of length $(\delta^{1/2})^{1/2}$. Hence

$$\begin{aligned} \|f_{\tau}\|_{L^p} & \leq D_p^{\text{loc}}(\delta^{1/2}) \left(\sum_{\theta \subseteq \tau} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}. \end{aligned}$$

Step 3. *Combine.*

Substituting Step 2 into Step 1 and interchanging the sums,

$$\begin{aligned} \|f\|_{L^p} & \leq D_p(\delta^{1/2}) D_p^{\text{loc}}(\delta^{1/2}) \left(\sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}. \end{aligned} \quad \square$$

Where we are. Theorem 13.2.4 is the *multiplicativity* of decoupling constants across scales, and it is the skeleton of every argument in the volume. Note what it does and does not give.

What it gives. If one can show $D_p^{\text{loc}}(\sigma) \leq C$ uniformly — a bounded constant for one step — then Theorem 13.2.4 is exactly the hypothesis (13.2.1) of Theorem 13.2.3, and $D_p(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon}$ follows.

What it does not give. The bound on D_p^{loc} . That is the entire content of the theorem, and it is what Chapter 13.3 supplies.

Why state it separately. Because it isolates the bookkeeping from the mathematics. A reader who has understood Theorem 13.2.4 and Theorem 13.2.3 knows the shape of the proof of Theorem 13.1.4 completely, and needs only the one transverse estimate.

WORKED EXAMPLE 13.2.5— TIER A Running the induction numerically

Take $C = 4$ in (13.2.1) and $\delta_0 = \frac{1}{2}$, and follow the scales.

round	scale	D bound	
0	$\delta = 10^{-16}$	—	start
1	10^{-8}	$\leq 4D(10^{-8})$	
2	10^{-4}	$\leq 4^2D(10^{-4})$	
3	10^{-2}	$\leq 4^3D(10^{-2})$	
4	10^{-1}	$\leq 4^4D(10^{-1})$	
5	$\approx \frac{1}{2}$	$\leq 4^5D(\frac{1}{2})$	base case

Five rounds for $\delta = 10^{-16}$, and $4^5 = 1024$. Against $\delta^{-1/4} = 10^4$, the trivial bound, the induction has gained a factor 10. Against $\delta^{-\varepsilon}$ with $\varepsilon = 0.1$, which is $10^{1.6} \approx 40$, it has *not* yet won — consistent with Worked Example 13.2.2, which shows the crossover occurs far below 10^{-30} .

The lesson. The theorem is asymptotic and the constants are bad. What the induction guarantees is that the exponent is 0; it says nothing useful at any scale one might compute with.

Drills

Drill 13.2.1. Verify Theorem 13.2.1 by direct substitution, and confirm the image is exactly P rather than approximately.

Drill 13.2.2. Check that $\sigma = \delta^{1/2}$ gives $\delta/\sigma^2 = 1$, and interpret.

Drill 13.2.3. Carry out Steps 1–3 of Theorem 13.2.3 with $C = 2$, $\delta_0 = e^{-1}$ and $\delta = e^{-16}$, computing m and C^m explicitly.

Drill 13.2.4. Determine, for $C = 4$ and $\varepsilon = 0.1$, the value of δ below which $C^m \leq \delta^{-\varepsilon}$, and compare with Worked Example 13.2.2.

Notes

Induction on scales in this form is due to Bourgain, and the parabolic rescaling identity is classical. The observation that the round count is log log rather than log, and that this is what produces the ε -loss, is standard bookkeeping made explicit here and in Volume 12, Worked Example 12.4.3.

Chapter 13.3

Ball Inflation

Worked Example 13.2.3 identified the one hard step: the transverse case. This chapter supplies it. The name *ball inflation* refers to the mechanism: information about f on balls of radius ρ^{-1} is upgraded to information on the much larger balls of radius ρ^{-2} , and the upgrade is paid for by transversality.

13.3.1 The bilinear input

Lemma 13.3.1 (The addition map on the parabola). *Let $\Phi(t, s) = (t, t^2) + (s, s^2)$. Then*

$$|\det d\Phi(t, s)| = 2|t - s|.$$

Proof. $\Phi(t, s) = (t + s, t^2 + s^2)$, so $d\Phi = \begin{pmatrix} 1 & 1 \\ 2t & 2s \end{pmatrix}$, of determinant $2s - 2t$. $\square$

Where we are. This is the same computation as Volume 12, Theorem 12.6.2, where the Jacobian of the addition map on the circle was $|\sin \angle(\omega_1, \omega_2)|$. In both cases *transversality is exactly the non-degeneracy of the addition map*, and in both cases that is the whole of the bilinear estimate.

For the parabola the transversality parameter is $|t - s|$, the separation of the arcs in the parameter. Two arcs at distance $\gtrsim \nu$ have Jacobian $\gtrsim \nu$.

Proposition 13.3.2 (Bilinear L^2 for transverse arcs). *Let $\tau_1, \tau_2 \subseteq [0, 1]$ be intervals with $\text{dist}(\tau_1, \tau_2) \geq \nu$, and let f_j have Fourier support in the part of $\mathcal{N}_\delta$ over τ_j . Then*

$$\|f_1 f_2\|_{L^2(\mathbb{R}^2)} \lesssim \nu^{-1/2} \|f_1\|_{L^2} \|f_2\|_{L^2}.$$

Proof. By Plancherel, $\|f_1 f_2\|_{L^2} = \|\widehat{f_1 f_2}\|_{L^2} = \|\widehat{f_1} * \widehat{f_2}\|_{L^2}$. The convolution is supported in $\Phi(\tau_1 \times \tau_2)$ thickened by $O(\delta)$, and by Theorem 13.3.1 the map Φ has Jacobian

$\geq 2\nu$ there, so it is a local diffeomorphism with boundedly many preimages. Changing variables,

$$\|\widehat{f_1} * \widehat{f_2}\|_{L^2}^2 \lesssim \nu^{-1} \|\widehat{f_1}\|_{L^2}^2 \|\widehat{f_2}\|_{L^2}^2,$$

exactly as in Volume 12, Theorem 12.6.1. $\square$

13.3.2 Ball inflation

Definition 13.3.3 (Averaged norms). For a ball Δ and $g \geq 0$ write $\text{avg}_\Delta g = |\Delta|^{-1} \int_\Delta g$. For f with Fourier support in $\mathcal{N}_\delta$ and $\rho \geq \delta^{1/2}$, let $\{\tau\}$ be the arcs of length ρ and $\{\theta\}$ those of length $\delta^{1/2}$, and put

$$\mathcal{S}_\tau f = \left(\sum_{\theta \subseteq \tau} |f_\theta|^2 \right)^{1/2}.$$

Theorem 13.3.4 (Ball inflation, bilinear form). *Let τ_1, τ_2 be arcs of length ρ with $\text{dist}(\tau_1, \tau_2) \geq \nu\rho$, and let B be a ball of radius ρ^{-2} . Then*

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta \in \mathcal{D}} \prod_{j=1}^2 \left(\text{avg}_\Delta (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2} \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^2 \left(\text{avg}_B (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2},$$

where $\mathcal{D}$ is the family of balls of radius ρ^{-1} tiling B .

Proof. Each f_θ has Fourier support in a $\delta^{1/2} \times \delta$ box, so by Volume 12, Theorem 12.1.2, $|f_\theta|^2$ is essentially constant on translates of the dual box, a tube T_θ of dimensions $\delta^{-1/2} \times \delta^{-1}$ oriented normal to θ . Since $\rho \geq \delta^{1/2}$, on the scale ρ^{-1} the relevant tubes are the $\rho^{-1} \times \rho^{-2}$ tubes normal to τ_j .

Write $g_j = (\mathcal{S}_{\tau_j} f)^2$, a sum of such tube indicators with coefficients. The left-hand side is then, up to constants,

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta} \prod_{j=1}^2 (\text{avg}_\Delta g_j)^{1/2},$$

and the tubes contributing to g_1 and to g_2 point in directions separated by $\gtrsim \nu\rho$, because the normals to τ_1 and τ_2 differ by the arc separation. Rescaling B to the unit ball turns them into two ν -transverse families of tubes of radius $\rho^{-1} \cdot \rho^2 = \rho$.

Now apply the multilinear Kakeya inequality in the plane — Volume 11, Theorem 11.5.7 with $n = 2$, where the exponent $\frac{1}{n-1}$ is 1 and the statement reduces to Loomis–Whitney, that is to Fubini:

$$\int \prod_{j=1}^2 \left(\sum_{T \in \mathbb{T}_j} \mathbf{1}_T \right) \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^2 (\rho \# \mathbb{T}_j).$$

Translating back through the rescaling gives the claim. $\square$

Where we are. Two remarks on what this proof does and does not need. *It uses multilinear Kakeya, which is proved.* Volume 11, Theorem 11.5.4 was established from Loomis–Whitney by Guth’s short argument, so Theorem 13.3.4 rests on nothing unproved. In the plane the input degenerates to Fubini, which is why the two-dimensional case is elementary; in higher dimensions the full strength of Theorem 11.5.7 is required, and that is the sense in which decoupling for the paraboloid in $\mathbb{R}^n$ is a genuinely n -dimensional theorem.

The name. The average on the left is over balls of radius ρ^{-1} ; the average on the right is over the single ball B of radius ρ^{-2} . The inequality says the finer averages are controlled by the coarser one — the ball has been *inflated* from ρ^{-1} to ρ^{-2} . Since ρ^{-2} is the square of ρ^{-1} , iterating inflates scales by squaring, which is exactly the log log round count of Theorem 13.2.3.

WORKED EXAMPLE 13.3.1— TIER A **Why the exponent ρ^{-2} and not ρ^{-1}**

The inflation goes from ρ^{-1} to ρ^{-2} , and the reason is the geometry of the caps, not a choice.

The dual box. By Theorem 13.1.2, an arc of length ρ on the parabola sits in a $\rho \times \rho^2$ box. Its dual is $\rho^{-1} \times \rho^{-2}$.

What that means. A function with Fourier support in one arc of length ρ is essentially constant on balls of radius ρ^{-1} — the short direction of the dual box — but retains structure out to ρ^{-2} , the long direction. So ρ^{-1} is the scale at which the caps are indistinguishable and ρ^{-2} the scale at which they are fully resolved.

The numerology. With $\rho = \delta^{1/2}$ one has $\rho^{-2} = \delta^{-1}$, the largest scale in the problem. So one round of inflation at $\rho = \delta^{1/2}$ already reaches the top, and the induction is over intermediate ρ .

Numerically, $\delta = 10^{-6}$.

ρ	ρ^{-1}	ρ^{-2}	
10^{-1}	10	10^2	first round
10^{-2}	10^2	10^4	second
10^{-3}	10^3	10^6	reaches δ^{-1}

Three rounds, each squaring — and $\log_2 \log_{10} 10^6 \approx 2.6$, matching Theorem 13.2.3.

WORKED EXAMPLE 13.3.2— TIER B **Ball inflation against multilinear Kakeya**

The two statements look different and are the same.

Multilinear Kakeya bounds $\int \prod_j (\sum_T \mathbf{1}_T)^{1/(n-1)}$ for transverse tube families: a purely geometric statement about overlaps.

Ball inflation bounds an average of products of square functions over small balls by the same over a large ball: an analytic statement about functions.

The dictionary. Volume 12, Worked Example 12.1.2 converts a function with Fourier support in a cap into a sum over tubes. Under it, $\mathcal{S}_\tau f$ becomes $\sum_T \mathbf{1}_T$ and the two statements coincide. The transversality hypotheses correspond because the normal to an arc determines the tube direction.

Which is more general. Neither: they are equivalent given the dictionary. But multilinear Kakeya is the one with the short proof (Theorem 11.5.4), so it is the one to prove and the other to derive — exactly as Volume 12 derived multilinear restriction from discrete Loomis–Whitney rather than the reverse.

The transferable point. When an analytic estimate resists, look for the geometric statement it encodes. Three times in this series the geometric form has been the tractable one: Kakeya for restriction, Loomis–Whitney for multilinear restriction, and multilinear Kakeya for ball inflation.

WORKED EXAMPLE 13.3.3— TIER B **Reading a ball inflation statement**

Ball inflation is stated in several forms and the differences matter.

(1) *Which two scales?* Here ρ^{-1} and ρ^{-2} , and Worked Example 13.3.1 explains why the square. A statement relating ρ^{-1} to $\rho^{-3/2}$ is a different and weaker assertion.

(2) *How many factors?* The bilinear form given here suffices in the plane; in $\mathbb{R}^n$ one needs n , by Volume 12, Theorem 12.2.3.

(3) *Which exponent on the square function?* The statement involves $(\text{avg}_\Delta(\mathcal{S}_\tau f)^2)^{1/2}$; other sources use $(\text{avg}_\Delta(\mathcal{S}_\tau f)^p)^{1/p}$ and interpolate. The L^2 form is the one that pairs with multilinear Kakeya directly.

(4) *Is the loss $\rho^{-\varepsilon}$ or a constant?* Here $\rho^{-\varepsilon}$, inherited from Theorem 11.5.7. Since inflation is applied $O(\log \log \delta^{-1})$ times, an ε -loss per round is affordable and a fixed constant per round would also be — but a power of ρ would not.

A worked check. Set $\rho = \delta^{1/2}$. Then $\rho^{-1} = \delta^{-1/2}$ and $\rho^{-2} = \delta^{-1}$: one round carries the finest caps to the full scale, so the statement reduces to a single application. Any formulation that does not have this property at $\rho = \delta^{1/2}$ has been misquoted.

WORKED EXAMPLE 13.3.4— TIER C: RESEARCH TASTE The three inductions on scales, compared

By this point the series contains three fully proved inductions on scales. They have the same skeleton and differ in what they propagate, and seeing them side by side is the fastest way to read a fourth.

	multilinear Takeya	multilinear restriction	decoupling
Where	Vol 11, Theorem 11.5.4	Vol 12, Theorem 12.3.10	Theorem 13.2.3
Propagated	tube multiplicities	a weighted ℓ^2 hypothesis	the constant $D_p(\delta)$
Scale step	$W \rightarrow \delta^{-1}W$	$R \rightarrow \delta^{-1}R$	$\delta \rightarrow \delta^{1/2}$
Rounds	$\log S / \log \delta^{-1}$	$\log R / \log \delta^{-1}$	$\log \log \delta^{-1}$
Engine	Loomis–Whitney	discrete Loomis–Whitney	ball inflation
Loss	S^ε	R^ε	$\delta^{-\varepsilon}$

The common skeleton. Fix a small parameter; show one scale controls the next at bounded cost; iterate; choose the parameter at the end so the accumulated cost is subpolynomial.

The one real difference. What is carried. Guth carries a geometric quantity, Bejenaru an analytic hypothesis with off-diagonal decay, and decoupling a single number. The third is easiest, which is why Chapter 13.4 is the shortest of the three proofs despite proving the newest theorem.

Where the round count differs. The first two square the *scale* δ^{-1} per round and so take $\log / \log$ rounds; decoupling squares δ itself and takes $\log \log$. Both give ε -losses, but by different bookkeeping — and Volume 12, Worked Example 12.4.3 warns that getting this wrong produces a fixed power instead.

Reading a fourth. Given a new induction-on-scales paper, identify these six rows before reading any proof. In every case in this series they determine the shape of the argument, and usually the difficulty.

13.3.3 The transverse estimate, assembled

It is worth stating explicitly what Theorem 13.3.4 delivers when iterated, since Chapter 13.4 uses it as a single step.

Proposition 13.3.5 (Iterated inflation). *Let τ_1, τ_2 be ν -transverse arcs of*

length K^{-1} and let B be a ball of radius δ^{-1} . Then for every $\varepsilon > 0$,

$$\frac{1}{|\mathcal{D}_{\delta^{-1/2}}|} \sum_{\Delta} \prod_{j=1}^2 \left(\text{avg}_{\Delta} (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2} \lesssim_{K,\nu,\varepsilon} \delta^{-\varepsilon} \prod_{j=1}^2 \left(\text{avg}_B (\mathcal{S}_{\tau_j} f)^2 \right)^{1/2},$$

where $\mathcal{D}_{\delta^{-1/2}}$ tiles B by balls of radius $\delta^{-1/2}$.

Proof. Apply Theorem 13.3.4 at scales $\rho = K^{-1}, K^{-2}, K^{-4}, \dots$, each round replacing ρ by ρ^2 and costing $\rho^{-\varepsilon'}$. Starting from $\rho = K^{-1}$ and squaring, after m rounds one has $\rho = K^{-2^m}$, so reaching $\rho = \delta^{1/2}$ requires

$$2^m \log K = \frac{1}{2} \log \delta^{-1}, \quad \text{that is} \quad m = \log_2 \left(\frac{\log \delta^{-1}}{2 \log K} \right).$$

The accumulated cost is $\prod_{i < m} K^{-2^i \varepsilon'} \leq \delta^{-\varepsilon'}$, and choosing $\varepsilon' = \varepsilon$ gives the claim. *The round count is $\log \log \delta^{-1}$, matching Theorem 13.2.3.* $\square$

Where we are. The round count is worth pausing on, because it is the third appearance of the same phenomenon.

Volume 12, Worked Example 12.4.3 observed that an induction with $\log R$ rounds gives a fixed power and only $\log \log R$ rounds give R^ε . Theorem 13.2.3 produced $\log \log \delta^{-1}$ by squaring the scale. Theorem 13.3.5 produces it again, by squaring ρ .

Why squaring is natural here. The dual box of a ρ -cap is $\rho^{-1} \times \rho^{-2}$ (Worked Example 13.3.1), so the two scales present in any cap differ by a square. An induction that moves from one to the other necessarily squares, and hence necessarily terminates in $\log \log$ steps. *The ε -loss in decoupling is a consequence of the parabola's second order contact*, not of any inefficiency in the argument.

WORKED EXAMPLE 13.3.5—TIER C: RESEARCH TASTE **Reading an induction-on-scales paper**

By the end of this volume the reader has seen four induction-on-scales proofs in full. A checklist for the fifth.

(1) *What is propagated?* A constant, a geometric count, or a weighted hypothesis. Worked Example 13.3.4 tabulates the three cases in this series; the answer determines how the induction step is stated.

(2) *What is the scale step?* Multiplying by a constant, or squaring. This determines the round count and hence the loss: Theorem 13.2.3 and Volume 12, Worked Example 12.4.3. A paper claiming an ε -loss from a log-round induction has an error.

(3) *What is the base case?* Usually a scale below which the statement is trivial. In Theorem 11.5.4 it is $L^\varepsilon \leq 2D$; in Theorem 12.3.1 it is $R \leq \delta^{-2}$. Checking that the base case is genuinely trivial is quick and occasionally catches a gap.

(4) *Where is the non-degeneracy?* Every such proof has a determinant or separation hypothesis, and identifying it tells you what the degenerate case costs. Worked Example 12.2.5 of Volume 12 lists four instances.

(5) *In what order are the parameters chosen?* Almost always: ε first, then the auxiliary parameter, then all scales. Reading them in the wrong order makes the argument look circular, and this is the single commonest source of confusion. Theorem 11.5.4's Checkpoint states it explicitly for that theorem.

(6) *What would break in higher dimensions?* Usually the transverse estimate; see Worked Example 13.5.5.

The payoff. These six questions can be answered from the statement of the main lemma and the choice of parameters, typically in the first three pages. If they can be answered, the rest of the paper is verification, and it can be read quickly or skipped.

Drills

Drill 13.3.1. Verify Theorem 13.3.1 and compare with Volume 12, Theorem 12.6.2 for the circle.

Drill 13.3.2. Check that at $n = 2$ the exponent $\frac{1}{n-1}$ in Theorem 11.5.7 equals 1, and that the statement is then Fubini.

Drill 13.3.3. For $\delta = 10^{-8}$, tabulate the inflation rounds as in Worked Example 13.3.1 and count them.

Drill 13.3.4. Explain why $\rho \geq \delta^{1/2}$ is required in Theorem 13.3.3.

Notes

Ball inflation is Theorem 6.6 of Bourgain and Demeter's paper, where it is derived from the multilinear Kakeya inequality of Bennett, Carbery and Tao. The planar case given here degenerates to Fubini; the general case needs Theorem 11.5.7, proved in Volume 11.

Chapter 13.4

Decoupling for the Parabola

Everything is now in place. Theorem 13.2.1 handles clustered caps, Theorem 13.2.3 converts a self-improving inequality into a subpolynomial bound, and Theorem 13.3.4 supplies the transverse case. This chapter assembles them.

13.4.1 Broad and narrow, again

Lemma 13.4.1 (The bilinear reduction). *Fix $K \geq 2$ and partition $[0, 1]$ into K arcs τ . For every x ,*

$$\begin{aligned} |f(x)| &\leq K \max_{\tau} |f_{\tau}(x)| \quad \text{and} \quad |f(x)| \\ &\leq K^2 \max_{\text{dist}(\tau_1, \tau_2) \geq K^{-1}} |f_{\tau_1}(x) f_{\tau_2}(x)|^{1/2} + 3 \max_{\tau} |f_{\tau}(x)|. \end{aligned}$$

Proof. The first is the triangle inequality over the K arcs.

For the second, let τ^* maximise $|f_{\tau}(x)|$. If some τ with $\text{dist}(\tau, \tau^*) \geq K^{-1}$ has $|f_{\tau}(x)| \geq K^{-2}|f(x)|$, then

$$|f_{\tau^*}(x) f_{\tau}(x)|^{1/2} \geq (K^{-1}|f(x)| \cdot K^{-2}|f(x)|)^{1/2} \geq K^{-2}|f(x)|,$$

using $|f_{\tau^*}(x)| \geq K^{-1}|f(x)|$ from the first bound; this gives the first term. Otherwise every arc not adjacent to τ^* contributes at most $K^{-2}|f(x)|$, so summing over the at most K of them leaves $|f(x)| \leq 3 \max_{\tau} |f_{\tau}(x)| + K^{-1}|f(x)|$, and absorbing gives the second term. $\square$

Where we are. This is the broad–narrow dichotomy of Volume 12, Theorem 12.4.3, in the form decoupling needs. Two differences.

Two factors, not n . In the plane two directions span, so Theorem 12.2.3 of Volume 12 imposes no obstruction and the bilinear form suffices. In $\mathbb{R}^n$ one needs n factors and Chapter 13.5 says so.

The narrow term is a single arc, not a cluster. Because the second bound compares against $\max_{\tau} |f_{\tau}|$ directly, the narrow case is handled by one application of Theorem 13.2.1, not by an induction on dimension. That is why decoupling for the parabola is markedly simpler than restriction for the sphere.

13.4.2 The theorem

Theorem 13.4.2 (Bourgain–Demeter for the parabola). *For $2 \leq p \leq 6$ and every $\varepsilon > 0$,*

$$D_p(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon}.$$

Proof. By interpolation with Theorem 13.1.5(i) it suffices to treat $p = 6$.

Step 1. *Split by Theorem 13.4.1.*

Raising the second bound to the sixth power and integrating over B ,

$$\|f\|_{L^6(B)}^6 \lesssim K^{12} \max_{\tau_1, \tau_2 \text{ transverse}} \| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6(B)}^6 + 3^6 \sum_{\tau} \|f_{\tau}\|_{L^6(B)}^6.$$

Step 2. *The narrow term, by rescaling.*

Each τ has length K^{-1} , so by Theorem 13.2.1 the arc is an affine image of the whole parabola at scale δK^2 . Hence

$$\|f_{\tau}\|_{L^6} \leq D_6(\delta K^2) \left(\sum_{\theta \subseteq \tau} \|f_{\theta}\|_{L^6}^2 \right)^{1/2},$$

and summing over the K arcs,

$$\begin{aligned} \left(\sum_{\tau} \|f_{\tau}\|_{L^6}^6 \right)^{1/6} \\ \leq D_6(\delta K^2) \left(\sum_{\theta} \|f_{\theta}\|_{L^6}^2 \right)^{1/2}. \end{aligned}$$

Step 3. *The broad term, by ball inflation.*

Fix transverse τ_1, τ_2 . The quantity $\| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6}^6 = \|f_{\tau_1} f_{\tau_2}\|_{L^3}^3$ is estimated by interpolating Theorem 13.3.2 — which controls the L^2 norm with no loss — against the trivial L^∞ bound, and then applying Theorem 13.3.4 at each intermediate scale $\rho = K^{-1}, K^{-2}, \dots$ down to $\delta^{1/2}$.

Each round of inflation replaces averages over ρ^{-1} -balls by averages over ρ^{-2} -balls at a cost $\rho^{-\varepsilon}$, and passes from the square function at scale ρ to the

square function at scale ρ^2 . After $O(\log \log \delta^{-1})$ rounds one reaches scale $\delta^{1/2}$, and the total cost is $\delta^{-O(\varepsilon)}$. The outcome is

$$\| |f_{\tau_1} f_{\tau_2}|^{1/2} \|_{L^6(B)} \lesssim_{K,\varepsilon} \delta^{-\varepsilon} \left(\sum_{\theta} \|f_{\theta}\|_{L^6}^2 \right)^{1/2}.$$

Step 4. *The recursion.*

Combining Steps 1–3,

$$D_6(\delta) \leq C(K, \varepsilon) \delta^{-\varepsilon} + C D_6(\delta K^2). \quad (13.4.1)$$

Choosing $K = \delta^{-1/4}$ makes $\delta K^2 = \delta^{1/2}$, so (13.4.1) is exactly the self-improving inequality (13.2.1) of Theorem 13.2.3, and that proposition gives $D_6(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon}$. $\square$

Where we are. The proof is short because every ingredient was prepared. It is worth recording which volume each came from.

Ingredient	Where proved	Rests on
Parabolic rescaling	Theorem 13.2.1	algebra
Bootstrapping	Theorem 13.2.3	log log round count
Bilinear L^2	Theorem 13.3.2	Plancherel, a Jacobian
Ball inflation	Theorem 13.3.4	multilinear Keakeya
Multilinear Keakeya	Vol 11, Theorem 11.5.4	Loomis–Whitney
Loomis–Whitney	Vol 11, Theorem 11.5.2	Hölder

The whole chain terminates in Hölder’s inequality. That is the payoff of the work in Volume 11: decoupling for the parabola, one of the central theorems of the last decade, rests on nothing deeper.

Where $p = 6$ enters. Step 3, in the interpolation: L^6 of a half-product is L^3 of the product, and 3 is where the bilinear L^2 estimate and the trivial L^∞ bound meet after ball inflation. Beyond $p = 6$ the interpolation fails, consistently with Theorem 13.1.8, whose exponent $-\frac{1}{4} + \frac{3}{2p}$ is $+\frac{1}{8}$, 0, $-\frac{1}{16}$ at $p = 4, 6, 8$.

WORKED EXAMPLE 13.4.1— TIER A **Decoupling implies a restriction estimate**

Worked Example 13.1.2 promised that the critical exponent 6 agrees with Stein–Tomas because decoupling implies restriction. Here is the deduction.

Setup. Let g be supported on the parabola and $f = \mathcal{E}g$ its extension, localised to B_R with $\delta = R^{-1}$. Then $\hat{f}$ is supported in $\mathcal{N}_\delta$.

Apply decoupling. By Theorem 13.4.2 at $p = 6$,

$$\|\mathcal{E}g\|_{L^6(B_R)} \lesssim_\varepsilon R^\varepsilon \left(\sum_\theta \|\mathcal{E}g_\theta\|_{L^6(B_R)}^2 \right)^{1/2}.$$

Estimate one cap. By Volume 12, Theorem 12.1.5, a single cap contributes $\|\mathcal{E}g_\theta\|_{L^6} \lesssim \delta^{1/2} |B_R|^{1/6} \|g_\theta\|_{L^\infty}$.

Assemble. With $\#\theta = \delta^{-1/2} = R^{1/2}$ and $\|g_\theta\|_{L^\infty} \leq \|g\|_{L^\infty}$,

$$\begin{aligned} \|\mathcal{E}g\|_{L^6(B_R)} &\lesssim_\varepsilon R^\varepsilon (R^{1/2} \cdot R^{-1} \cdot R^{1/3})^{1/2} \|g\|_{L^\infty} \\ &\lesssim R^\varepsilon \|g\|_{L^\infty}, \end{aligned}$$

which after ε -removal is the Stein–Tomas estimate in $\mathbb{R}^2$ at the endpoint $p = 6 = \frac{2(n+1)}{n-1}$. ✓

What the argument shows and does not. Decoupling recovers Stein–Tomas but not the restriction conjecture, which in $\mathbb{R}^2$ asks for $p > 4$. The reason is that decoupling is an ℓ^2 statement over caps and therefore cannot exploit L^∞ data — the same limitation Worked Example 12.4.1(c) identified for broad–narrow.

WORKED EXAMPLE 13.4.2 — TIER B What decoupling is not

Three things decoupling is regularly confused with.

(a) *It is not orthogonality.* At $p = 2$ it is Plancherel (Theorem 13.1.5(i)), but for $p > 2$ there is no orthogonality to appeal to, and the content is precisely that an ℓ^2 bound survives anyway. Volume 12, Worked Example 12.7.7 explains why: orthogonality is free only for quadratic quantities.

(b) *It is not a square function estimate.* The square function bound — ℓ^2 inside the norm — is stronger by Minkowski and is open at $p = 6$. Worked Example 13.1.4 sets out the four arrangements and which hold.

(c) *It is not sharp in the constant.* The theorem gives $\delta^{-\varepsilon}$; the truth for the parabola is $(\log \delta^{-1})^c$, proved by Guth, Maldague and Wang. The improvement matters for applications where the ε compounds, and Worked Example 13.2.2 shows how weak $\delta^{-\varepsilon}$ is at any nameable δ .

What it is. An interpolation-free statement that the pieces of a function over a curved surface behave, in L^p for p up to a critical exponent, as though they were orthogonal — up to $\delta^{-\varepsilon}$.

WORKED EXAMPLE 13.4.3— TIER C: RESEARCH TASTE **What the proof of Bourgain–Demeter actually needs**

Having proved the theorem, it is worth auditing which hypotheses did work.

(a) *Curvature, twice.* Once in Theorem 13.1.2, to make a ρ -arc fit in a $\rho \times \rho^2$ box; once in Theorem 13.3.1, to make the addition map non-degenerate. A flat piece fails both, and Worked Example 13.5.2(d) records that no decoupling holds there.

(b) *Exactness of the rescaling.* Theorem 13.2.1 is an identity, not an approximation. For a general curved surface the analogous map is only approximate and the induction acquires error terms that must be tracked — one reason the moment curve, where rescaling is again exact, was reachable.

(c) *The log log round count.* Theorem 13.2.3 needs each round to *square* the scale. A method reducing δ by a constant factor per round would give a fixed power, not $\delta^{-\varepsilon}$; Volume 12, Worked Example 12.4.3 is the general statement.

(d) *Multilinear Kakeya, in full strength.* Only through Theorem 13.3.4, and in the plane it degenerates to Fubini. In $\mathbb{R}^n$ it does not, which is why Chapter 13.5 is a genuinely n -dimensional theorem and why proving Volume 11's Theorem 11.5.4 rather than importing it was necessary.

(e) *What is not needed.* No polynomial method, no incidence geometry, no algebraic geometry. The contrast with Volume 11 is sharp: two subjects that interact constantly, built on disjoint foundations.

The audit's value. Knowing which hypotheses are load-bearing tells one where to look when extending a theorem. For the cone, (a) fails in one direction and the exponent changes; for the moment curve, all four survive and the proof goes through with k in place of 2.

13.4.3 Two consequences

Corollary 13.4.3 (Discrete restriction for the parabola). *Let $\Lambda = \{(n/N, n^2/N^2) : 1 \leq n \leq N\}$ and $f = \sum_n a_n e(x \cdot \lambda_n)$ with $\lambda_n \in \Lambda$. Then for $2 \leq p \leq 6$ and every $\varepsilon > 0$,*

$$\|f\|_{L^p(\mathbb{T}^2)} \lesssim_\varepsilon N^\varepsilon \left(\sum_n |a_n|^2 \right)^{1/2}.$$

Proof. Take $\delta = N^{-2}$, so the caps θ of side $\delta^{1/2} = N^{-1}$ contain exactly one point of Λ each. Apply Theorem 13.4.2; since f_θ is a single modulated constant, $\|f_\theta\|_{L^p} = |a_n|$, and the right-hand side is the stated ℓ^2 sum. $\square$

Corollary 13.4.4 (The L^6 count). *With $a_n = 1$ for all n , $\int_{\mathbb{T}^2} |f|^6 \lesssim_\varepsilon N^{3+\varepsilon}$, that is: the number of solutions of*

$$n_1 + n_2 + n_3 = n_4 + n_5 + n_6, \quad n_1^2 + n_2^2 + n_3^2 = n_4^2 + n_5^2 + n_6^2$$

with $n_i \leq N$ is $O_\varepsilon(N^{3+\varepsilon})$.

Proof. $\int |f|^6$ counts the solutions, and Theorem 13.4.3 at $p = 6$ bounds it by $N^\varepsilon(N)^3$. $\square$

Where we are. Theorem 13.4.4 is $J_{3,2}(N) \lesssim N^{3+\varepsilon}$: the case $k = 2$ of Vinogradov, obtained here as a one-line corollary. Chapter 13.6 does the general k by the same argument on the moment curve.

It is worth noting that this case is elementary by other means — the two equations force $\{n_1, n_2, n_3\} = \{n_4, n_5, n_6\}$ as multisets, as Worked Example 13.6.1 records — so the corollary is a consistency check rather than a new result. That it comes out right is evidence the normalisations are correct, and it is the cheapest such check available.

13.4.4 Small cap decoupling

The caps of Theorem 13.1.1 have length $\delta^{1/2}$, forced by Theorem 13.1.2. It is natural to ask what happens for shorter caps, and the answer is different in an instructive way.

Definition 13.4.5. For $\frac{1}{2} \leq \beta \leq 1$ let $D_p(\delta, \beta)$ be the least constant with

$$\|f\|_{L^p} \leq D_p(\delta, \beta) \left(\sum_{\gamma} \|f_{\gamma}\|_{L^p}^2 \right)^{1/2}$$

where γ runs over caps of length δ^β .

Proposition 13.4.6 (Small caps decouple on a smaller range). *For $\frac{1}{2} \leq \beta \leq 1$,*

$$D_p(\delta, \beta) \gtrsim \delta^{-\frac{\beta}{2} + \frac{\beta+1}{p}},$$

which is a positive power of δ^{-1} exactly when

$$p > p_c(\beta) := \frac{2(\beta+1)}{\beta}.$$

Proof. Repeat the construction of Theorem 13.1.8 with caps γ of length δ^β .

Step 1. *The caps and their dual boxes.*

There are $N = \delta^{-\beta}$ caps. Over an interval of length δ^β the parabola deviates from its tangent by $(\delta^\beta)^2 = \delta^{2\beta} \leq \delta$, since $\beta \geq \frac{1}{2}$; so the cap lies in a

box of dimensions $\delta^\beta \times \delta$ — the thickness being δ , the width of $\mathcal{N}_\delta$, rather than the smaller curvature deviation. Its dual box T_γ has dimensions $\delta^{-\beta} \times \delta^{-1}$ and volume $\delta^{-(\beta+1)}$.

Step 2. *The two sides.*

Take wave packets adapted to the T_γ with phases aligned at the origin. The frequency support has diameter ≈ 1 , so by the Pitfall following Theorem 13.1.8 the packets remain aligned only on a ball of radius ≈ 1 , on which $|f| \approx N$. Hence $\|f\|_{L^p} \gtrsim N = \delta^{-\beta}$, while $\|f_\gamma\|_{L^p} = |T_\gamma|^{1/p} = \delta^{-(\beta+1)/p}$ and

$$\begin{aligned} \left(\sum_\gamma \|f_\gamma\|_{L^p}^2 \right)^{1/2} &= (N \delta^{-2(\beta+1)/p})^{1/2} \\ &= \delta^{-\frac{\beta}{2} - \frac{\beta+1}{p}}. \end{aligned}$$

Step 3. *The ratio.*

Dividing gives $\delta^{-\beta + \frac{\beta}{2} + \frac{\beta+1}{p}} = \delta^{-\frac{\beta}{2} + \frac{\beta+1}{p}}$, which tends to infinity exactly when $\frac{\beta+1}{p} < \frac{\beta}{2}$, that is $p > \frac{2(\beta+1)}{\beta}$. $\square$

Where we are. Two checks and a consequence.

At $\beta = \frac{1}{2}$, $p_c = \frac{2 \cdot \frac{3}{2}}{1} = 6$ and the exponent is $-\frac{1}{4} + \frac{3}{2p}$ — exactly Theorem 13.1.8. $\checkmark$

p_c decreases in β : 6, $\frac{16}{3}$, 5, $\frac{14}{3}$, 4 at $\beta = \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{3}{4}, 1$. So *shorter caps decouple on a strictly smaller range*, and at $\beta = 1$ — caps of length δ , as short as the neighbourhood is thick — only $p \leq 4$ survives.

Why shorter caps are worse, not better. A shorter cap is flatter, so it carries less curvature, while there are more of them; the ℓ^2 sum grows without the estimate improving. Step 1 makes this precise: for $\beta > \frac{1}{2}$ the dual box has thickness δ^{-1} regardless of β , so lengthening the cap costs nothing but shortening it costs $\delta^{-\beta}$ in the count.

So $\delta^{1/2}$ is optimal, and Worked Example 13.4.4 explains the same fact geometrically: it is the unique length at which the caps are as long as possible subject to being flat at scale δ .

Where we are. The reason is geometric, not technical. At length $\delta^{1/2}$ a cap is *flat* to within δ : Theorem 13.1.2 says the arc deviates from its tangent by $(t - t_0)^2 \leq \delta$. Shorter caps are flatter still and so carry no extra curvature, while there are more of them — so the ℓ^2 sum grows without the estimate improving.

The rule. $\delta^{1/2}$ is the unique cap length at which the caps are as long as possible subject to being flat at scale δ . Longer caps are not contained in δ -slabs; shorter ones waste curvature. Every cap decomposition in Volumes 12 and 13 uses this length, and Theorem 13.4.6 explains why no other choice is available.

What is true for small caps. For $p > p_c(\beta)$ a modified inequality holds with the ℓ^2 sum replaced by an ℓ^p one, interpolating between the two arrangements of Worked Example 13.1.4. This is the small cap decoupling theory of Demeter, Guth and Wang, and it is genuinely different from Theorem 13.4.2 rather than a refinement of it.

WORKED EXAMPLE 13.4.4— TIER B **Choosing the cap length**

Given a surface and a neighbourhood width δ , the cap length is determined. The computation takes a line and is worth doing deliberately.

- (a) *Write the surface as a graph.* $x_n = \phi(x')$ near the point of interest.
- (b) *Expand to second order.* Over a cap of diameter σ , the deviation from the tangent plane is $\approx \|\text{Hess } \phi\| \sigma^2$.
- (c) *Set it equal to δ .* $\sigma = (\delta / \|\text{Hess } \phi\|)^{1/2}$. For the paraboloid $\text{Hess } \phi = 2I$ and $\sigma \approx \delta^{1/2}$.
- (d) *Read off the dual box.* σ^{-1} in the tangential directions, δ^{-1} in the normal one.

Worked instances.

Surface	cap	because
paraboloid	$\delta^{1/2}$	all curvatures ≈ 1
cone	$\delta^{1/2} \times 1$	one curvature vanishes
moment curve	$\delta^{1/k}$	k -th order contact
hyperplane	—	no curvature; no useful cap

The check. Having chosen σ , verify that $\#\text{caps} \times |\text{cap}| \approx |\mathcal{N}_\delta|$. For the parabola: $\delta^{-1/2} \cdot \delta^{1/2} \cdot \delta = \delta$. ✓ A cap decomposition failing this check has the wrong length.

WORKED EXAMPLE 13.4.5— TIER B **Checking a decoupling claim**

Papers state decoupling inequalities in several normalisations. Five checks before believing one applies to your situation.

- (1) *Which side carries the ℓ^2 ?* Outside the L^p norm is decoupling; inside is the square function, which is stronger and often false. Worked Example 13.1.4

lists the four arrangements.

(2) *What is the cap length?* $\delta^{1/2}$ for a hypersurface with non-vanishing curvature, $\delta^{1/k}$ for the moment curve, and $\delta^{1/2} \times 1$ for the cone. Worked Example 13.4.4 gives the computation, and Theorem 13.4.6 shows other choices fail.

(3) *Is the loss $\delta^{-\varepsilon}$ or explicit?* The Bourgain–Demeter theorem gives the former with no effective constant. If a paper needs an explicit constant, it needs Guth–Maldague–Wang or a different method.

(4) *Is the exponent in range?* Compare with $\frac{2(n+1)}{n-1}$ for hypersurfaces, $\frac{2n}{n-2}$ for the cone, $k(k+1)$ for the moment curve. Outside the range the statement is *false*, not merely unproved — Theorem 13.5.3.

(5) *Are the frequencies separated?* The dictionary of Worked Example 13.6.2 requires one term per cap. Sums over irregular sets need small cap theory and the exponents differ.

A worked check. “ $\ell^2 L^6$ decoupling for the parabola with $\delta^{-\varepsilon}$ loss”: (1) ℓ^2 outside $\checkmark$; (2) caps $\delta^{1/2}$ $\checkmark$; (3) ε -loss, so no effective constant; (4) 6 is the critical exponent, so this is the endpoint case $\checkmark$; (5) applies to any f with the right Fourier support. Verdict: this is Theorem 13.4.2 exactly.

Drills

Drill 13.4.1. Prove the first bound of Theorem 13.4.1, and verify the absorption step in the second.

Drill 13.4.2. Check that $K = \delta^{-1/4}$ gives $\delta K^2 = \delta^{1/2}$.

Drill 13.4.3. Carry out the exponent arithmetic in Worked Example 13.4.1 and confirm the powers of R cancel.

Drill 13.4.4. Using Theorem 13.1.8, confirm that $-\frac{1}{4} + \frac{3}{2p}$ is positive for $p < 6$ and negative for $p > 6$.

Notes

The ℓ^2 decoupling theorem for the paraboloid is due to Bourgain and Demeter (2015). The presentation here follows the standard bilinear route; the sharp constant $(\log \delta^{-1})^c$ for the parabola is due to Guth, Maldague and Wang (2020).

Chapter 13.5

The Higher-Dimensional Paraboloid

The proof in Chapter 13.4 was written for the parabola, but almost nothing in it was two-dimensional. This chapter states the general theorem and records exactly which steps change.

13.5.1 The statement

Definition 13.5.1. Let $\mathbb{P} = \{(\xi, |\xi|^2) : \xi \in [0, 1]^{n-1}\} \subseteq \mathbb{R}^n$ and let $\mathcal{N}_\delta$ be its δ -neighbourhood. Partition $[0, 1]^{n-1}$ into $N = \delta^{-(n-1)/2}$ cubes of side $\delta^{1/2}$, giving caps θ , and let $D_p(\delta)$ be the least constant with

$$\|f\|_{L^p(\mathbb{R}^n)} \leq D_p(\delta) \left(\sum_{\theta} \|f_{\theta}\|_{L^p}^2 \right)^{1/2}$$

for all f with $\text{supp } \hat{f} \subseteq \mathcal{N}_\delta$.

Theorem 13.5.2 (Bourgain–Demeter). *For $2 \leq p \leq \frac{2(n+1)}{n-1}$ and every $\varepsilon > 0$,*

$$D_p(\delta) \lesssim_{n,\varepsilon} \delta^{-\varepsilon},$$

and the range is sharp.

Proposition 13.5.3 (Sharpness). *For $p > \frac{2(n+1)}{n-1}$,*

$$D_p(\delta) \gtrsim \delta^{-\frac{n-1}{4} + \frac{n+1}{2p}},$$

a positive power of δ^{-1} .

Proof. Exactly as Theorem 13.1.8. Each cap is a $\delta^{1/2} \times \cdots \times \delta^{1/2} \times \delta$ box, so its dual box T_θ has volume $\delta^{-(n-1)/2} \cdot \delta^{-1} = \delta^{-(n+1)/2}$. Choosing wave packets aligned at the origin, $|f| \approx N$ on a ball of radius ≈ 1 — the frequency support has diameter ≈ 1 , so the packets separate beyond that — giving $\|f\|_{L^p} \gtrsim N$. Meanwhile $\|f_\theta\|_{L^p} = |T_\theta|^{1/p} = \delta^{-(n+1)/(2p)}$, so

$$\left(\sum_\theta \|f_\theta\|_{L^p}^2 \right)^{1/2} = N^{1/2} \delta^{-\frac{n+1}{2p}}.$$

The ratio is $N^{1/2} \delta^{(n+1)/(2p)} = \delta^{-\frac{n-1}{4} + \frac{n+1}{2p}}$, which tends to infinity exactly when $\frac{n+1}{2p} < \frac{n-1}{4}$, that is $p > \frac{2(n+1)}{n-1}$. $\square$

Where we are. Two checks on Theorem 13.5.3.

At the critical exponent the ratio is bounded. Substituting $p = \frac{2(n+1)}{n-1}$ gives $-\frac{n-1}{4} + \frac{n+1}{2} \cdot \frac{n-1}{2(n+1)} = -\frac{n-1}{4} + \frac{n-1}{4} = 0$, for every n . $\checkmark$

At $n = 2$ it is the planar statement. $\frac{2(n+1)}{n-1} = 6$ and the exponent is $-\frac{1}{4} + \frac{3}{2p}$, which is Theorem 13.1.8. $\checkmark$

And it is the Stein–Tomas exponent. $\frac{2(n+1)}{n-1}$ is exactly the exponent of Volume 6, Theorem 6.3.1, as it must be: by Worked Example 13.4.1 decoupling implies a restriction estimate at the same exponent, so the two critical values cannot differ. Numerically $6, 4, \frac{10}{3}, 3, \frac{22}{9}$ for $n = 2, 3, 4, 5, 10$.

13.5.2 What changes in the proof

Proof of Theorem 13.5.2, following Chapter 13.4. Every step of Theorem 13.4.2 has a counterpart. We list them, indicating the modification.

Step 1. Rescaling.

Theorem 13.2.1 holds verbatim with $t \in [0, 1]$ replaced by $\xi \in [0, 1]^{n-1}$: for a cube τ of side σ centred at ξ_0 , the affine map

$$L_\tau(x, x_n) = \left(\frac{x - \xi_0}{\sigma}, \frac{x_n - |\xi_0|^2 - 2\xi_0 \cdot (x - \xi_0)}{\sigma^2} \right)$$

carries the cap over τ onto all of $\mathbb{P}$ and $\mathcal{N}_\delta$ to $\mathcal{N}_{\delta/\sigma^2}$, by the identity $|\xi_0 + \sigma s|^2 - |\xi_0|^2 - 2\sigma \xi_0 \cdot s = \sigma^2 |s|^2$. *Unchanged.*

Step 2. Bootstrapping.

Theorem 13.2.3 is a statement about a real function and is dimension-free. *Unchanged.*

Step 3. *The broad–narrow split.*

Theorem 13.4.1 must be replaced by its n -linear form:

$$|f| \lesssim_K \max_{\tau_1, \dots, \tau_n} \prod_{\nu\text{-transverse}}^n |f_{\tau_j}|^{1/n} + K^C \max_{\tau} |f_{\tau}|.$$

This is Volume 12, Theorem 12.4.3, and *two factors no longer suffice*: by Volume 12, Theorem 12.2.3, two vectors cannot span $\mathbb{R}^n$ for $n \geq 3$, so a bilinear reduction would leave $n - 2$ directions uncontrolled. *This is the one genuine change.*

Step 4. *Ball inflation.*

Theorem 13.3.4 generalises with n factors and exponent $\frac{1}{n-1}$:

$$\frac{1}{|\mathcal{D}|} \sum_{\Delta \in \mathcal{D}} \prod_{j=1}^n \left(\text{avg}_{\Delta}(\mathcal{S}_{\tau_j} f)^2 \right)^{\frac{1}{n-1}} \lesssim_{\nu, \varepsilon} \rho^{-\varepsilon} \prod_{j=1}^n \left(\text{avg}_B(\mathcal{S}_{\tau_j} f)^2 \right)^{\frac{1}{n-1}},$$

and the proof is the same, applying Volume 11, Theorem 11.5.7 in $\mathbb{R}^n$ instead of $\mathbb{R}^2$. In the plane that theorem degenerated to Fubini; here its full strength is used. *Same argument, stronger input — and the input is proved.*

Step 5. *Assembly.*

Steps 1–4 of Theorem 13.4.2 go through with L^6 replaced by L^{p_c} , $p_c = \frac{2(n+1)}{n-1}$, and the recursion (13.4.1) is unchanged in form. Theorem 13.2.3 closes it. $\square$

Where we are. The proof is genuinely the same, and it is worth being precise about why the literature nonetheless treats the higher-dimensional case as substantially harder.

What is the same. The architecture: rescale, split, inflate, bootstrap.

What is harder in practice. Step 3’s n -linear reduction requires tracking $\binom{K^{n-1}}{n}$ tuples rather than $\binom{K}{2}$ pairs, and the transversality bookkeeping is heavier. Step 4 requires multilinear Keakeya in $\mathbb{R}^n$, a research theorem, where the planar case was Fubini.

What this series gains from Volume 11. Because Theorem 11.5.4 is *proved* there — Guth’s short argument from Loomis–Whitney — Step 4 is not an import, and Theorem 13.5.2 rests on nothing unproved. Had multilinear Keakeya remained a black box, this chapter would have had to be an announcement.

WORKED EXAMPLE 13.5.1— TIER A **The exponents, tabulated**

n	$p_c = \frac{2(n+1)}{n-1}$	#caps	dual box volume
2	6	$\delta^{-1/2}$	$\delta^{-3/2}$
3	4	δ^{-1}	δ^{-2}
4	$\frac{10}{3} \approx 3.33$	$\delta^{-3/2}$	$\delta^{-5/2}$
5	3	δ^{-2}	δ^{-3}
10	$\frac{22}{9} \approx 2.44$	$\delta^{-9/2}$	$\delta^{-11/2}$

The exponent decreases to 2. As $n \rightarrow \infty$, $\frac{2(n+1)}{n-1} \rightarrow 2$, so in high dimensions decoupling holds on a range shrinking to the trivial one. That is not a defect of the method: by Theorem 13.5.3 the range is sharp.

Why the range shrinks. More caps means more room for constructive interference, and Theorem 13.5.3 shows the resulting example degrades the exponent. The count $N = \delta^{-(n-1)/2}$ grows with n , which is the whole content.

Comparison with restriction. The restriction conjecture asks for $p > \frac{2n}{n-1}$, which also tends to 2; the ratio $\frac{2(n+1)/(n-1)}{2n/(n-1)} = \frac{n+1}{n} \rightarrow 1$. So decoupling and the restriction conjecture become close in high dimensions — but the difference is exactly what Volume 12, Worked Example 12.5.1 measured, and it never closes.

WORKED EXAMPLE 13.5.2— TIER B **Which surfaces decouple**

Decoupling is not special to paraboloids, and knowing the pattern is useful.

(a) *Non-vanishing Gaussian curvature.* Paraboloid, sphere, ellipsoid: the theorem holds with $p_c = \frac{2(n+1)}{n-1}$, by the argument above after an affine change reducing to the paraboloid locally.

(b) *The cone.* One vanishing principal curvature. Decoupling holds with $p_c = \frac{2n}{n-2}$, proved by Bourgain and Demeter; the rescaling is Lorentz rather than parabolic. This is the case Volume 13's applications need, since local smoothing is a statement about the wave equation.

(c) *The moment curve.* $\{(t, t^2, \dots, t^k)\}$: a curve, not a hypersurface, with all curvatures non-vanishing. Decoupling holds with $p_c = k(k+1)$, and Chapter 13.6 shows this is Vinogradov's mean value theorem.

(d) *Flat pieces.* A hyperplane does not decouple beyond $p = 2$: the trivial bound Theorem 13.1.5(ii) is attained. Curvature is not a technical convenience but the entire source of the phenomenon.

The rule. The critical exponent is determined by how many caps there are and how large their dual boxes are, both of which are read off from the curvatures.

Theorem 13.5.3 is the computation, and it can be run for any surface whose caps one can describe.

WORKED EXAMPLE 13.5.3— TIER C: RESEARCH TASTE **What dimension costs, across the series**

Worked Example 13.5.1 showed the decoupling range shrinking to $p = 2$ as $n \rightarrow \infty$. That is one instance of a pattern visible in every volume, and collecting it clarifies why high dimensions are hard.

(a) *Restriction*. The conjectured range $p > \frac{2n}{n-1}$ and the Stein–Tomas range $p \geq \frac{2(n+1)}{n-1}$ both tend to 2, and their gap $\frac{2}{n-1}$ tends to 0 (Volume 12, Worked Example 12.5.1). So Stein–Tomas is asymptotically sharp in absolute terms and never sharp.

(b) *Kekeya*. Wolff’s bound $\frac{n+2}{2}$ against the conjectured n : the *ratio* tends to $\frac{1}{2}$, so the method loses half the dimension in the limit. Volume 9, Worked Example 9.5.3 recorded this.

(c) *Falconer*. The conjectured $\frac{n}{2}$ and Falconer’s $\frac{n+1}{2}$ differ by $\frac{1}{2}$ for every n — a gap that neither closes nor grows. Volume 8, Worked Example 8.7.1.

(d) *Decoupling*. The range shrinks but is *sharp* at every n , by Theorem 13.5.3. So decoupling is the one item on this list that is fully understood.

Why (d) differs from (a)–(c). Decoupling is an ℓ^2 statement over caps, and ℓ^2 statements are the ones this subject can prove — the same observation as Volume 12, Worked Example 12.7.7. The open problems are all L^∞ statements, where no orthogonality is available. *The line between solved and open in this subject tracks the line between L^2 and L^∞ data almost exactly*, and that is the most useful single heuristic in the series.

The exception that tests it. Kekeya has no data norm at all — it is a purely geometric statement — and it is open. So the heuristic is not a theorem; it predicts which analytic problems are tractable, not which geometric ones.

13.5.3 The cone

The applications of Chapter 13.9 need decoupling for the cone rather than the paraboloid, and the differences are worth recording precisely.

Definition 13.5.4. Assume $n \geq 3$. Let $\mathbb{C} = \{(\xi, |\xi|) : \frac{1}{2} \leq |\xi| \leq 1\} \subseteq \mathbb{R}^n$ be the truncated light cone, and $\mathcal{N}_\delta$ its δ -neighbourhood. Use a smooth, finitely overlapping angular partition into plates θ of aperture $\delta^{1/2}$ and full radial

length, and let $D_p^{\mathbb{C}}(\delta)$ be the least constant in

$$\|f\|_p \leq D_p^{\mathbb{C}}(\delta) \left(\sum_{\theta} \|f_{\theta}\|_p^2 \right)^{1/2}, \quad \text{supp } \hat{f} \subseteq \mathcal{N}_{\delta}.$$

Lemma 13.5.5 (Cone caps are plates). *The natural caps of $\mathcal{N}_{\delta}$ have dimensions $\delta^{1/2} \times \dots \times \delta^{1/2} \times 1 \times \delta$: they are $\delta^{1/2}$ in the $n-2$ directions of non-vanishing curvature, of length ≈ 1 along the ruling, and δ normal.*

Proof. The cone is ruled by lines through the origin, along which it is flat, so no subdivision is needed in that direction. Transverse to the ruling it is a curve of non-vanishing curvature, and Worked Example 13.4.4 gives $\delta^{1/2}$ there. $\square$

Theorem 13.5.6 (Decoupling for the cone; Bourgain–Demeter). *For every $\varepsilon > 0$,*

$$D_p^{\mathbb{C}}(\delta) \lesssim_{n,p,\varepsilon} \begin{cases} \delta^{-\varepsilon}, & 2 \leq p \leq \frac{2n}{n-2}, \\ \delta^{-\frac{n-2}{4} + \frac{n}{2p} - \varepsilon}, & \frac{2n}{n-2} \leq p \leq \infty. \end{cases}$$

Proof. We first prove the critical estimate at

$$p = p_c := \frac{2n}{n-2}.$$

Write $\delta = N^{-1}$; changing N by a bounded factor handles general δ . The proof has one local step and one finite iteration.

Step 1. *A cone sector becomes a lower-dimensional paraboloid.*

Let $0 < \nu, \mu$ satisfy

$$2\mu + \nu \leq 1, \quad 2\mu \geq \nu. \quad (13.5.1)$$

Let S be a cone sector whose radial interval has length $N^{-\nu}$ and whose angular patch has diameter $N^{-\mu}$. We claim

$$\|f_S\|_{p_c} \lesssim_{\varepsilon} N^{\nu\varepsilon} \left(\sum_{S' \subset S} \|f_{S'}\|_{p_c}^2 \right)^{1/2}, \quad (13.5.2)$$

where S' has the same radial length and angular diameter $N^{-\mu-\nu/2}$.

Rotate and dilate by bounded factors so the sector lies near the ray e_1 and its radial centre is 1. A bounded covering lets us additionally use coordinates $\xi = (\xi_1, u) \in \mathbb{R} \times \mathbb{R}^{n-2}$ with $|u| \lesssim N^{-\mu}$. Write the last frequency coordinate as τ and $r = |\xi|$. For points in the N^{-1} -neighbourhood of S , $\tau = r + O(N^{-1})$, $r = 1 + O(N^{-\nu})$, and

$$\eta := \tau - \xi_1 = \frac{|u|^2}{2} + O(N^{-(2\mu+\nu)}), \quad (13.5.3)$$

because

$$(r - \xi_1) - \frac{|u|^2}{2} = \frac{|u|^2(2 - r - \xi_1)}{2(r + \xi_1)},$$

$|2 - r - \xi_1| \lesssim N^{-\nu}$ by $2\mu \geq \nu$, and the normal error N^{-1} is absorbed by $N^{-(2\mu+\nu)}$ by the first condition in (13.5.1).

Here is the physical-space device which makes (13.5.3) usable without assuming that f is an extension operator. For $y = (y_2, \dots, y_n) \in \mathbb{R}^{n-1}$ set

$$T_y x = (x_1 - y_n, x_2 + y_2, \dots, x_{n-1} + y_{n-1}, x_n + y_n).$$

For fixed x , the Fourier support in the y variables of $G_x(y) := f_S(T_y x)$ is the image

$$(\xi_1, u, \tau) \mapsto (u, \tau - \xi_1) = (u, \eta).$$

Thus (13.5.3) puts it in an $O(N^{-(2\mu+\nu)})$ -neighbourhood of the elliptic paraboloid $\eta = |u|^2/2$ over a patch of diameter $N^{-\mu}$ in u .

Let $Y = [-N, N]^{n-1}$ and choose a Schwartz cutoff ψ_Y with $|\psi_Y| \geq 1$ on Y , Fourier support in an $O(N^{-1})$ ball, and $\int |\psi_Y|^{p_c} \asymp |Y|$; put $w_Y = |\psi_Y|^{p_c}$. Multiplying G_x by ψ_Y changes its y -frequency support by only $O(N^{-1})$, already absorbed above. Parabolic rescaling followed by Theorem 13.5.2 in ambient dimension $n - 1$ applies at relative thickness

$$\frac{N^{-(2\mu+\nu)}}{(N^{-\mu})^2} = N^{-\nu}.$$

Its critical exponent is $2((n-1)+1)/((n-1)-1) = 2n/(n-2) = p_c$, and its caps have u -diameter $N^{-\mu}(N^{-\nu})^{1/2} = N^{-\mu-\nu/2}$. Hence, for each x ,

$$\|G_x\|_{L^{p_c}(Y)} \lesssim_\varepsilon N^{\nu\varepsilon} \left(\sum_{S' \subset S} \|f_{S'}(T.x)\|_{L^{p_c}(w_Y)}^2 \right)^{1/2}. \quad (13.5.4)$$

Average the identity $\|f_S\|_{p_c}^{p_c} = \|f_S \circ T_y\|_{p_c}^{p_c}$ over $y \in Y$. Insert (13.5.4), and use Minkowski in the form $L_x^{p_c}(\ell^2) \leq \ell^2(L_x^{p_c})$. Finally, for every S' ,

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^{n-1}} |f_{S'}(T_y x)|^{p_c} w_Y(y) dy dx = \left(\int w_Y \right) \|f_{S'}\|_{p_c}^{p_c},$$

since T_y has determinant one. The factor $\int w_Y \asymp |Y|$ cancels the averaging factor, proving (13.5.2).

Step 2. *Iterate only the aperture.*

Fix an integer $M \geq 2$ and put $\nu = M^{-1}$. Start with sectors having radial length $N^{-\nu}$ and aperture $N^{-\nu/2}$. Apply (13.5.2) successively with

$$\mu_j = \frac{j\nu}{2}, \quad j = 1, \dots, M-1.$$

The two conditions (13.5.1) hold at every stage, and the last application produces sectors Δ of radial length $N^{-\nu}$ and aperture $N^{-1/2}$. Replacing the epsilon in each application by a suitably smaller one absorbs the finite product of losses, so

$$\|f_S\|_{p_c} \lesssim_\varepsilon N^\varepsilon \left(\sum_{\Delta \subset S} \|f_\Delta\|_{p_c}^2 \right)^{1/2}. \quad (13.5.5)$$

The whole annulus is covered by

$$O(N^\nu N^{(n-2)\nu/2}) = O(N^{n\nu/2})$$

initial sectors S . Triangle and Cauchy–Schwarz, followed by (13.5.5), therefore give

$$\|f\|_{p_c} \lesssim_\varepsilon N^{\varepsilon+n\nu/4} \left(\sum_{\Delta} \|f_\Delta\|_{p_c}^2 \right)^{1/2}. \quad (13.5.6)$$

Each final cone plate θ of full radial length and aperture $N^{-1/2}$ contains $O(N^\nu)$ of the Δ . Because all partitions were chosen smoothly, projection from f_θ to any f_Δ is convolution with a uniformly L^1 kernel, so $\|f_\Delta\|_{p_c} \lesssim \|f_\theta\|_{p_c}$. Consequently

$$\sum_{\Delta} \|f_\Delta\|_{p_c}^2 \lesssim N^\nu \sum_{\theta} \|f_\theta\|_{p_c}^2.$$

The total non-epsilon overhead in (13.5.6) is therefore $N^{(n+2)\nu/4}$. Given a target loss N^{ε_0} , first choose M so large that $(n+2)/(4M) < \varepsilon_0/2$, and then run the preceding estimates with their epsilon losses totaling less than $\varepsilon_0/2$. This proves the endpoint with N^{ε_0} .

Step 3. *Recover both sides of the critical exponent.*

For $p = 2$, bounded overlap and Plancherel give $D_2^{\mathbb{C}}(\delta) \lesssim 1$. Complex interpolation of the linear map

$$(f_\theta)_\theta \mapsto \sum_{\theta} f_\theta$$

from $\ell^2(L^2)$ to L^2 and from $\ell^2(L^{p_c})$ to L^{p_c} therefore gives the first line for $2 \leq p \leq p_c$.

There are $K \asymp \delta^{-(n-2)/2}$ cone plates. Triangle and Cauchy–Schwarz give the other endpoint

$$\left\| \sum_{\theta} f_\theta \right\|_{\infty} \leq K^{1/2} \left(\sum_{\theta} \|f_\theta\|_{\infty}^2 \right)^{1/2}, \quad D_{\infty}^{\mathbb{C}}(\delta) \lesssim \delta^{-(n-2)/4}.$$

Interpolating this with the p_c estimate, where $1/p = (1 - \vartheta)/p_c$, gives

$$D_p^{\mathbb{C}}(\delta) \lesssim_\varepsilon \delta^{-\vartheta(n-2)/4-\varepsilon} = \delta^{-\frac{n-2}{4} + \frac{n}{2p} - \varepsilon}.$$

Here the endpoint epsilon in the interpolation is chosen slightly smaller than the displayed target epsilon. This proves the second line and completes the theorem. $\square$

Remark 13.5.7 (Status and numerology). Theorem 13.5.6 is now an internal consequence of the lower-dimensional elliptic-paraboloid theorem. The shear T_y exposes the curved angular variables while leaving the radial variable inside the coefficients; this is why only the aperture is iterated.

What changes in the numerology. The exponent is $\frac{2n}{n-2}$ rather than $\frac{2(n+1)}{n-1}$, because the caps are plates rather than boxes: there are $\delta^{-(n-2)/2}$ of them, not $\delta^{-(n-1)/2}$, and each has dual box of volume $\delta^{-(n-2)/2} \cdot 1 \cdot \delta^{-1}$. Running Theorem 13.5.3 with these values gives $\frac{2n}{n-2}$.

WORKED EXAMPLE 13.5.4— TIER A **Cone against paraboloid, numerically**

n	paraboloid $\frac{2(n+1)}{n-1}$	cone $\frac{2n}{n-2}$	which is larger
3	4	6	cone
4	$\frac{10}{3} \approx 3.33$	4	cone
5	3	$\frac{10}{3} \approx 3.33$	cone
10	$\frac{22}{9} \approx 2.44$	2.5	cone

The cone decouples on a larger range. That is not because the cone is “more curved” — it is less curved, having a vanishing principal curvature. It is because the caps are *fewer*: $\delta^{-(n-2)/2}$ against $\delta^{-(n-1)/2}$, so the ℓ^2 sum is smaller and the constructive example of Theorem 13.5.3 is weaker.

The check at $n = 3$. Cone gives 6, which is the exponent Theorem 13.9.3(i) needs. Paraboloid gives 4. Both agree with Worked Example 13.5.2. $\checkmark$

The moral. The critical exponent is a count, not a curvature. Volume 5, Worked Example 5.5.4 gave decay exponents from curvature; here what matters is how many caps the curvature forces, which is a different function of the same data.

WORKED EXAMPLE 13.5.5— TIER C: RESEARCH TASTE **What makes a proof generalise**

Theorem 13.5.2 was obtained from Theorem 13.4.2 by changing one step. That is unusual, and the reasons are worth extracting.

(a) *The dimension appeared in only one place.* Rescaling, bootstrapping and the assembly are dimension-free; only the broad–narrow split needed n factors instead of 2. A proof organised so that the dimension enters once generalises; one that uses planar coordinates throughout does not.

(b) *The hard input was already general.* Ball inflation rests on multilinear Kakeya, which Volume 11 proves in $\mathbb{R}^n$. Had it been proved only in the plane, this chapter would have stopped.

(c) *The sharpness example was already general.* Theorem 13.5.3 is the planar computation with N and $|T_\theta|$ left as symbols. Writing examples with symbolic parameters rather than numerical ones is what makes them portable.

(d) *The contrast: what did not generalise.* Volume 12’s Theorem 12.6.1 for the circle uses that two vectors span the plane, and Theorem 12.2.3 shows this fails above dimension two. That proof is genuinely planar, and the higher-dimensional statement — Tao’s theorem — needed an entirely different argument.

(e) *The diagnostic.* Before generalising, list every place the dimension is used and ask, for each, whether it is essential or notational. In Chapter 13.4 the answer was “notational” everywhere but one step; in Section 12.6.1 it was “essential” at the first step.

(f) *Why this matters for reading.* A paper proving a planar case often signals in its introduction which route it took. If the hard input is already stated in $\mathbb{R}^n$, the general case is likely routine; if the argument opens with a planar-specific identity, expect a separate paper.

Drills

Drill 13.5.1. Verify the identity $|\xi_0 + \sigma s|^2 - |\xi_0|^2 - 2\sigma \xi_0 \cdot s = \sigma^2 |s|^2$ used in Step 1.

Drill 13.5.2. Confirm that the exponent of Theorem 13.5.3 vanishes at $p = \frac{2(n+1)}{n-1}$ for $n = 2, 3, 4$.

Drill 13.5.3. Compute $\frac{2(n+1)}{n-1}$ and $\frac{2n}{n-1}$ for $n = 2, \dots, 6$ and tabulate the gap.

Drill 13.5.4. Explain, using Theorem 12.2.3 of Volume 12, why Step 3 cannot use two factors when $n \geq 3$.

Notes

Theorem 13.5.2 is the main theorem of Bourgain and Demeter’s 2015 paper. The cone case is in the same paper; the moment curve is treated in Bourgain–Demeter–Guth, and is the subject of Chapter 13.6.

Chapter 13.6

The Moment Curve and Vinogradov's Theorem

Decoupling was invented for harmonic analysis. Within two years of Theorem 13.5.2 it settled a problem in analytic number theory that had been open since 1935. This chapter gives the reduction.

13.6.1 The mean value theorem

Definition 13.6.1 (Vinogradov's system). For integers $s, k \geq 1$ and $N \geq 1$ let $J_{s,k}(N)$ be the number of integer solutions of the system

$$\sum_{i=1}^s x_i^j = \sum_{i=1}^s y_i^j, \quad j = 1, \dots, k,$$

with $1 \leq x_i, y_i \leq N$.

Theorem 13.6.2 (Vinogradov's mean value theorem; proved in Theorem 13.6.5 from Theorem 13.6.4, which is itself proved in Chapter 13.7). *For every $\varepsilon > 0$,*

$$J_{s,k}(N) \lesssim_{s,k,\varepsilon} N^\varepsilon \left(N^s + N^{2s - \frac{k(k+1)}{2}} \right).$$

Where we are. Both terms are forced, so the theorem is sharp up to N^ε . N^s comes from the diagonal. Any solution with $\{x_i\}$ a permutation of $\{y_i\}$ works, and there are $\approx s! N^s$ of these.

$N^{2s - k(k+1)/2}$ comes from counting. There are N^{2s} tuples and k equations, the j -th constraining a quantity of size $\approx N^j$; so heuristically a fraction $N^{-(1+2+\dots+k)} = N^{-k(k+1)/2}$ of tuples should satisfy all of them.

The two balance at $s = \frac{k(k+1)}{2}$, where both equal N^s . That is the critical case, and it is where the difficulty lies: for s much larger or smaller the

| theorem is classical.

13.6.2 The reduction to decoupling

Definition 13.6.3 (The moment curve). Let $\Gamma_k = \{(t, t^2, \dots, t^k) : t \in [0, 1]\} \subseteq \mathbb{R}^k$, and let $D_p^\Gamma(\delta)$ be the decoupling constant for its δ -neighbourhood, with arcs of length $\delta^{1/k}$. Thus the parameter δ here is the shortest (normal) box width, whereas the parameter ρ of Theorem 13.7.1 is the arc length; explicitly $\rho = \delta^{1/k}$ and the canonical side lengths are $\rho, \rho^2, \dots, \rho^k = \delta$.

Theorem 13.6.4 (Bourgain–Demeter–Guth; proved in Chapter 13.7 as Theorem 13.7.2). *For $2 \leq p \leq k(k+1)$ and every $\varepsilon > 0$, $D_p^\Gamma(\delta) \lesssim_{k,\varepsilon} \delta^{-\varepsilon}$. This is proved in Chapter 13.7 as Theorem 13.7.2 at $p = k(k+1)$; the range down to $p = 2$ follows by interpolation with Plancherel.*

Proposition 13.6.5 (Decoupling implies Vinogradov). *Theorem 13.6.4 at $p = k(k+1)$ implies Theorem 13.6.2 at the critical exponent $s = \frac{k(k+1)}{2}$, and hence for all s .*

Proof. Step 1. $J_{s,k}$ is an L^{2s} norm.

Put $\delta = N^{-k}$ and

$$F(x) = \sum_{m=1}^N e(mx_1 + m^2x_2 + \dots + m^kx_k), \quad x \in \mathbb{T}^k,$$

where $e(u) = e^{2\pi i u}$. Expanding $|F|^{2s}$ and integrating over the torus, the cross terms cancel unless the exponents match in every coordinate, so

$$J_{s,k}(N) = \int_{\mathbb{T}^k} |F(x)|^{2s} dx = \|F\|_{L^{2s}(\mathbb{T}^k)}^{2s}. \quad (13.6.1)$$

So the critical number of variables and the critical exponent are the same number: $2s = p$.

Step 2. *The localized discrete restriction estimate.*

Put $\rho = N^{-1}$ and $R = \rho^{-k} = N^k$. For coefficients (a_m) define

$$Ea(y) := \sum_{m=1}^N a_m e(y \cdot \Gamma_k(m/N)), \quad y \in \mathbb{R}^k.$$

We claim that every ball B_R of radius R satisfies

$$\left(\frac{1}{|B_R|} \int_{B_R} |Ea(y)|^{p_k} dy \right)^{1/p_k} \lesssim_\varepsilon N^\varepsilon \left(\sum_{m=1}^N |a_m|^2 \right)^{1/2}. \quad (13.6.2)$$

Choose a Schwartz function η_{B_R} which is bounded below on B_R , is adapted to that ball, and has Fourier support in a ball of radius cR^{-1} . Then

$$f_m(y) := a_m e(y \cdot \Gamma_k(m/N)) \eta_{B_R}(y)$$

has Fourier support in an $O(R^{-1}) = O(\rho^k)$ ball about $\Gamma_k(m/N)$, hence in a fixed dilation of the canonical moment-curve box over the ρ -interval containing m/N . Apply Theorem 13.7.2 to $\sum_m f_m$. Since $\|f_m\|_{p_k} = |a_m| \|\eta_{B_R}\|_{p_k}$ and $\|\eta_{B_R}\|_{p_k} \asymp |B_R|^{1/p_k}$, division by this last factor proves (13.6.2). A fixed overlap at cap boundaries is removed by splitting the points between two shifted partitions.

Step 3. *Periodicity returns the estimate to the torus.*

The function Ea is periodic in y_j with period N^j . Its fundamental period box is

$$Q = [0, N] \times [0, N^2] \times \cdots \times [0, N^k].$$

Under $y_j = N^j x_j$, normalized integration over Q is exactly normalized integration over $\mathbb{T}^k$ of $\sum_m a_m e(mx_1 + \cdots + m^k x_k)$. The cube $[0, R]^k$ is tiled by exactly $\prod_{j=1}^k N^{k-j}$ translates of Q and lies in a ball of radius $\sqrt{k}R$. Periodicity, followed by (13.6.2) for that comparable ball, therefore gives

$$\left\| \sum_{m=1}^N a_m e(mx_1 + \cdots + m^k x_k) \right\|_{L^{p_k}(\mathbb{T}^k)} \lesssim_\varepsilon N^\varepsilon \left(\sum_m |a_m|^2 \right)^{1/2}. \quad (13.6.3)$$

Taking $a_m = 1$ and $p_k = 2s$ yields $\|F\|_{2s} \lesssim_\varepsilon N^{1/2+\varepsilon}$. Raising to the power $2s$ in (13.6.1) and renaming $2s\varepsilon$ gives

$$J_{s,k}(N) \lesssim_\varepsilon N^{s+\varepsilon}$$

at the critical $s = k(k+1)/2$.

Step 4. *The critical case gives all cases.*

At $s = \frac{k(k+1)}{2}$ the bound N^s is exactly $N^{2s-k(k+1)/2}$, so Theorem 13.6.2 holds there. For smaller s it follows by Hölder from the critical case, and for larger s by inserting the trivial bound $|F| \leq N$ into the extra factors. $\square$

Where we are. The correspondence $s = p/2$ is exact and worth tabulating.

k	$p_c = k(k+1)$	critical $s = \frac{k(k+1)}{2}$
2	6	3
3	12	6
4	20	10
5	30	15

The case $k = 2$ is classical: $J_{3,2}(N)$ counts solutions of $x_1 + x_2 + x_3 =$

$y_1 + y_2 + y_3$ together with the same for squares, and is $\approx N^3 \log N$ by an elementary argument. So the theorem is a genuine generalisation of a known case, which is the right thing to check first.

What is being transported. An exponential sum is an L^p norm; a decoupling inequality controls that norm by the individual terms; and the individual terms are trivial. *The entire number-theoretic content is carried by the geometry of the moment curve* — specifically, by the fact that its caps have the dual boxes they do.

Remark 13.6.6 (Status of Theorem 13.6.4). When this chapter was first written, Theorem 13.6.4 was quoted. *A reader objected that a theorem this series proves must not rest on a theorem it does not*, and the objection was right: Theorem 13.6.2 is proved here, and it depends on nothing else.

Chapter 13.7 now supplies the proof, following Guo, Li, Yung and Zorin-Kranich rather than the original argument of Bourgain, Demeter and Guth. Their route is bilinear and uses *no* Kakeya estimate: the transversality that the original captures through multilinear Kakeya–Brascamp–Lieb inequalities appears there as the generalised Vandermonde identity Theorem 13.7.9, and the induction is on the *degree* k rather than on the spatial scale.

What is proved in this chapter remains Theorem 13.6.5, the reduction — with the identification $s = p/2$ and the verification that the critical cases correspond. That reduction is the surprising half, and it was not suspected before 2015.

WORKED EXAMPLE 13.6.1— TIER A **Vinogradov before decoupling**

The theorem was proved twice, and comparing the routes is instructive.

Wooley's efficient congruencing (2011–2016). A p -adic argument. One studies solutions modulo powers of a prime and exploits that congruent solutions cluster; iterating produces a descent. It settled $k = 3$ in 2016 and, in Wooley's later work, all k .

Bourgain–Demeter–Guth (2015). The decoupling route above. It settled all $k \geq 4$ first.

What they share. Both are induction-on-scales arguments. Efficient congruencing induces on the power of p ; decoupling induces on the spatial scale δ . Chapter 13.8 makes the analogy precise.

What differs. Efficient congruencing is elementary and gives explicit constants; decoupling is analytic and gives N^ε with no effective constant. For applications needing effectivity — Waring's problem with explicit bounds — the congruencing route remains preferable.

The moral. Two genuinely different proofs of the same theorem, found within a year of each other by communities that had not been talking. That is worth knowing when choosing which literature to read: the decoupling proof is shorter to state and rests on Volume 11's geometry; the congruencing proof is self-contained and effective.

WORKED EXAMPLE 13.6.2— TIER B **Reading an exponential sum estimate**

Number-theoretic statements about exponential sums translate into decoupling statements by a fixed dictionary, and it is worth having explicitly.

(1) *The sum is a function on a torus.* $\sum_m e(\phi_m \cdot x)$ with frequencies ϕ_m ; the relevant surface is the curve or hypersurface through the ϕ_m .

(2) *The moment is an L^p norm.* A $2s$ -th moment of the sum, counting solutions of a system, is $\|F\|_{L^{2s}}^{2s}$, as in (13.6.1).

(3) *The scale is set by the largest frequency.* If frequencies run to N^k , the neighbourhood width is $\delta = N^{-k}$.

(4) *The caps are the individual terms.* One term per cap, so the square function on the right of a decoupling inequality is trivially computed.

A worked instance. For $k = 2$, $s = 3$: frequencies (m, m^2) with $m \leq N$, so the curve is the parabola, $\delta = N^{-2}$, $p = 6$ — and the required decoupling is exactly Theorem 13.4.2, proved in Chapter 13.4. *So the $k = 2$ case of Vinogradov is a corollary of the theorem this volume proves in full.*

The caution. The dictionary requires the frequencies to be *separated* at the scale of the caps. Sums over sparse or irregular sets need small cap decoupling, a different and more delicate theory.

13.6.3 Consequences in number theory

Where we are. [The direction of dependence] The circle method appears twice below, and it is worth stating at the outset that it is *downstream* of Theorem 13.6.2, never upstream.

What proves Vinogradov. Theorem 13.6.5, whose only input is Theorem 13.6.4. The identification $J_{s,k}(N) = \|F\|_{L^{2s}}^{2s}$ of (13.6.1) is exact, and nothing in that chain is number-theoretic.

What the circle method is for. Deducing Theorem 13.6.7 and Theorem 13.6.9 from Theorem 13.6.2. In both, the mean value theorem is the input to the minor-arc analysis, and the circle method consumes it.

So the omission recorded as Theorem 13.6.8 costs the two applications and nothing else. In particular the chain of Worked Example 13.6.3 is

Hölder $\rightarrow \cdots \rightarrow$ decoupling $\rightarrow$ Vinogradov $\rightarrow$ circle method $\rightarrow \zeta$,

and the omission is at the penultimate arrow, not before it. A reader wanting only Theorem 13.6.2 needs no number theory at all.

Vinogradov's theorem is a tool, and its uses are worth recording since they are the reason the problem mattered. The next two items are historical context under approved-external input X04 (the Hardy–Littlewood circle method and its standard analytic-number-theory continuation); they are not numbered propositions in the strict proof graph, and no internal result depends on them.

Remark 13.6.7 (Waring's problem; approved external). Let $G(k)$ be the least s such that every sufficiently large integer is a sum of s non-negative k -th powers. Then Theorem 13.6.2 gives

$$G(k) \leq (1 + o(1)) k \log k.$$

External-input dependency map. The circle method writes the number of representations of N as a sum of s k -th powers as $\int_{\mathbb{T}} \left(\sum_{m \leq N^{1/k}} e(\alpha m^k) \right)^s e(-\alpha N) d\alpha$, split into major and minor arcs. The major arcs give the expected main term via a singular series. On the minor arcs one needs cancellation in the Weyl sums, and the standard route bounds their mean value by $J_{s,k}$; Theorem 13.6.2 at the critical exponent supplies exactly the bound required, and tracking the resulting condition on s gives $k \log k$.

The circle method itself is not developed here. Revision R26 (Theorem 13.6.8): the major and minor arc analysis is a chapter of analytic number theory — singular series, Weyl differencing, the Hardy–Littlewood dissection — and is a different subject from the one this series treats. What is proved in this volume is the harmonic-analytic half: Theorem 13.6.5, that $J_{s,k}$ is a decoupling constant. Nothing in this series depends on Theorem 13.6.7. $\square$

Remark 13.6.8 (Scope decision: the circle method). Revision R26. The Hardy–Littlewood circle method — the major/minor arc dissection, Weyl differencing, the singular series, and for Theorem 13.6.9 the Borel–Carathéodory and Hadamard three-circles machinery — is not developed in this series.

What depends on it. Theorem 13.6.7 and Theorem 13.6.9, and nothing else. In particular *not* Theorem 13.6.2: as the Checkpoint opening this section records, the circle method consumes the mean value theorem and does not produce it.

Why omitted. It is a chapter of analytic number theory, a different subject from the harmonic analysis this series develops, and the two propositions it

supports are reported for context rather than used. The harmonic-analytic half — that $J_{s,k}$ is a decoupling constant — is proved in full as Theorem 13.6.5.

Where we are. The bound $k \log k$ is not new with decoupling — Wooley had obtained it by efficient congruencing — but before either, the classical Vinogradov bounds gave $G(k) \leq (2 + o(1))k \log k$, and the factor 2 was the target for eighty years. Both routes remove it, and neither reaches the conjectured $G(k) \approx 4k$ for most k .

What limits the method. The circle method needs the mean value bound at the critical s , and Theorem 13.6.2 supplies exactly that. Improving $G(k)$ further requires a different treatment of the minor arcs, not a better mean value theorem: the mean value theorem is now sharp.

Remark 13.6.9 (Zero-free regions; approved external). Bounds on $J_{s,k}$ yield bounds on $|\zeta(\sigma + it)|$ near $\sigma = 1$, and hence a zero-free region for the Riemann zeta function of Vinogradov–Korobov type,

$$\sigma > 1 - \frac{c}{(\log|t|)^{2/3}(\log \log|t|)^{1/3}}.$$

External-input dependency map. Near $\sigma = 1$, ζ is estimated by sums $\sum_{m \sim N} m^{-it}$. Writing $m^{-it} = e(-\frac{t}{2\pi} \log m)$ and expanding $\log(1 + u)$ in a Taylor polynomial of degree k turns these into Weyl sums whose frequencies lie on the moment curve Γ_k . Theorem 13.6.2 bounds their moments, and optimising the degree k against $\log|t|$ produces the exponents $\frac{2}{3}$ and $\frac{1}{3}$.

The zero-free region argument is not developed here. Revision R26, as above: converting a Weyl sum bound into a zero-free region requires the Borel–Carathéodory and Hadamard three-circles machinery of complex analytic number theory. Worked Example 13.6.3 records the full chain and marks these two links as belonging to a different subject. $\square$

Where we are. This is the deepest consequence, and it explains why Vinogradov’s system was studied at all. The exponents $\frac{2}{3}$ and $\frac{1}{3}$ in Theorem 13.6.9 come from optimising the degree k of the Taylor polynomial against $\log|t|$, and they have not been improved since Korobov and Vinogradov in 1958 — even now that the mean value theorem is sharp.

So what did decoupling buy for zeta? A cleaner proof, and the constant c ; not the shape of the region. That is worth knowing before reading the literature: Theorem 13.6.2 was the last missing ingredient in a chain whose other links remain the bottleneck.

WORKED EXAMPLE 13.6.3— TIER A The chain from geometry to zeta

Collecting the links, in the order this series proves them.

Hölder's inequality	Vol 1
↓	
Loomis–Whitney	Vol 11, Theorem 11.5.2
↓	
Multilinear Kakeya	Vol 11, Theorem 11.5.4
↓	
Ball inflation	Theorem 13.3.4
↓	
Decoupling	Theorem 13.4.2
↓	
Moment curve decoupling	Theorem 13.6.4
↓	
Vinogradov	Theorem 13.6.5
↓	
Zero-free region for ζ	Theorem 13.6.9

Every harmonic-analytic link is now proved in this series. Moment-curve decoupling is supplied by Chapter 13.7; the external links in the displayed route are the circle-method and complex-analytic consequences above, which belong to analytic number theory and are not used back in the harmonic-analysis chain.

The point of the table. A statement about how tubes overlap in $\mathbb{R}^n$ — Loomis–Whitney, which is Fubini and Hölder — propagates, through six links, to the distribution of the zeros of ζ . Each link is individually unremarkable; the composition is not, and it was not suspected before 2015.

The transferable lesson. When a geometric inequality is sharp, look for what it implies. The chain above was assembled backwards — Bourgain, Demeter and Guth knew which number-theoretic statement they wanted — but its existence was invisible from either end.

WORKED EXAMPLE 13.6.4— TIER C: RESEARCH TASTE Transfers between fields, and how to spot them

The decoupling proof of Vinogradov's theorem is the clearest transfer in this series: a geometric inequality settling a number-theoretic problem. It is worth asking what made it possible, since the same pattern recurs.

(a) *A common object in disguise.* $J_{s,k}(N)$ is a count of solutions and $\|F\|_{L^{2s}}^{2s}$ is an integral; (13.6.1) says they are the same number. *The transfer becomes possible only once the two descriptions are recognised as one object*, and that

recognition is the discovery.

(b) *Matching critical cases.* The number-theoretic critical exponent $s = \frac{k(k+1)}{2}$ and the analytic one $p_c = k(k+1)$ satisfy $s = p_c/2$ exactly. Had they not matched, the transfer would have given a non-sharp result and no-one would have noticed the connection.

(c) *Other instances in this series.* Volume 12, Theorem 12.8.2 identifies a geometric measure theory quantity with a restriction norm — the same move, and it is why the Falconer problem inherits every advance in restriction theory. Volume 11, Theorem 11.7.7 identifies distance quadruples with line incidences. Volume 10 identifies Kakeya in $\mathbb{F}_q^n$ with a polynomial vanishing problem.

(d) *What they have in common.* In each case a counting problem is rewritten as a statement about a curved object, and the curvature then does work that was invisible in the original formulation. *Curvature is the resource*, and finding it is the transferable skill.

(e) *How to look.* Given a counting problem, ask what surface the objects being counted parametrise, and whether the count is a moment of a function whose frequencies lie on it. If both answers are yes, the harmonic analysis literature may already contain the estimate needed.

(f) *The caution.* Worked Example 13.6.2 records that the dictionary requires separated frequencies. Transfers fail silently when the objects are irregularly distributed, and the resulting statement is then not false but vacuous.

Drills

Drill 13.6.1. Verify (13.6.1) by expanding $|F|^{2s}$ and identifying which terms survive integration.

Drill 13.6.2. Confirm that $N^s = N^{2s-k(k+1)/2}$ exactly when $s = \frac{k(k+1)}{2}$.

Drill 13.6.3. Tabulate $k(k+1)$ and $\frac{k(k+1)}{2}$ for $k = 2, \dots, 6$.

Drill 13.6.4. Following Worked Example 13.6.2, write down the decoupling statement corresponding to $J_{6,3}(N)$ and identify the surface, δ and p .

Notes

Vinogradov's system dates to 1935. The main conjecture was settled for $k = 3$ by Wooley in 2016 by efficient congruencing, and for all k by Bourgain, Demeter and Guth in 2015 by decoupling. Theorem 13.6.6 records the internally

proved moment-curve input, while Theorem 13.6.5 gives the complete discrete reduction.

Chapter 13.7

Decoupling for the Moment Curve

Theorem 13.6.2 was proved in Chapter 13.6 from Theorem 13.6.4, which was quoted. A reader objected, rightly, that a theorem this series proves must not rest on a theorem it does not. This chapter supplies the missing proof, following Guo, Li, Yung and Zorin-Kranich.

Their argument is *bilinear*, not multilinear, and uses no Keakeya estimate. Where Bourgain, Demeter and Guth capture transversality through multilinear Keakeya–Brascamp–Lieb inequalities, this proof captures it through a family of *asymmetric* bilinear decoupling constants and an induction on the *degree* k . The inputs are Hölder’s inequality, Plancherel, and decoupling for moment curves of lower degree.

13.7.1 Set-up

Definition 13.7.1 (Moment curve, boxes, decoupling constant). Let $\Gamma(\xi) = (\xi, \xi^2, \dots, \xi^k)$ on $[0, 1]$. For a dyadic interval J let U_J be the parallelepiped of dimensions $|J| \times |J|^2 \times \dots \times |J|^k$ centred at $\Gamma(c_J)$ with sides parallel to $\partial_1 \Gamma(c_J), \dots, \partial_k \Gamma(c_J)$. Write

$$p_k := k(k+1)$$

for the critical exponent, $\mathcal{P}(\delta)$ for the partition of $[0, 1]$ into dyadic intervals of length $\approx \delta$, and let $D_k(\delta)$ be the least constant with

$$\left\| \sum_{J \in \mathcal{P}(\delta)} f_J \right\|_{p_k} \leq D_k(\delta) \left(\sum_{J \in \mathcal{P}(\delta)} \|f_J\|_{p_k}^2 \right)^{1/2} \quad (13.7.1)$$

whenever $\text{supp } \widehat{f_J} \subseteq U_J$.

Theorem 13.7.2 (Bourgain–Demeter–Guth). *For every k and every $\varepsilon > 0$ there is $C_{k,\varepsilon}$ with $D_k(\delta) \leq C_{k,\varepsilon}\delta^{-\varepsilon}$ for all $\delta \in (0, 1)$.*

Where we are. $p_k = k(k+1)$ is the exponent Chapter 13.6 predicted: by Theorem 13.6.5 the critical number of variables in Vinogradov’s system is $s = p_k/2 = \frac{k(k+1)}{2}$, and Worked Example 13.6.2 identified $2s = p$. At $k = 2$ we recover $p_2 = 6$ and Theorem 13.4.2. ✓

Why induction on the degree, and not on scales alone. The moment curve in $\mathbb{R}^k$ contains, after projection, moment curves of every lower degree; the proof below exploits this by reducing a k -dimensional estimate to l -dimensional ones with $l < k$. Nothing similar is available for a hypersurface, which is why Chapter 13.4 had to induct on scales instead.

13.7.2 From linear to bilinear

Definition 13.7.3 (Symmetric bilinear constant). For $\delta \in (0, \frac{1}{4})$ let $B(\delta)$ be the least constant such that for all $I, I' \in \mathcal{P}(\frac{1}{4})$ with $\text{dist}(I, I') \geq \frac{1}{4}$,

$$\int_{\mathbb{R}^k} |f_I|^{p_k/2} |f_{I'}|^{p_k/2} \leq B(\delta)^{p_k} \left[\sum_{J \subseteq I} \|f_J\|_{p_k}^2 \right]^{p_k/4} \left[\sum_{J' \subseteq I'} \|f_{J'}\|_{p_k}^2 \right]^{p_k/4}.$$

Lemma 13.7.4 (Affine rescaling). *For $I = [a, a + \kappa]$ the affine map A_I of $\widehat{\mathbb{R}^k}$ with*

$$(A_I(\eta))_j = \sum_{j'=0}^j \binom{j}{j'} a^{j-j'} \kappa^{j'} \eta_{j'}, \quad \eta_0 := 1,$$

satisfies $A_I \Gamma(t) = \Gamma(a + t\kappa)$, hence $A_I^{-1} U_J = U_{J_I}$ for $J \subseteq I$, where $J_I = \kappa^{-1}(J - a)$. Consequently, for $I \in \mathcal{P}(2^{-n})$ and $\delta < 2^{-n}$,

$$\|f_I\|_{p_k} \leq D_k(2^n \delta) \left(\sum_{J \in \mathcal{P}(I, \delta)} \|f_J\|_{p_k}^2 \right)^{1/2}. \quad (13.7.2)$$

Proof. The displayed formula is the binomial expansion of $(a + \kappa t)^j$: substituting $\eta_{j'} = t^{j'}$ gives $(A_I \Gamma(t))_j = (a + \kappa t)^j$. Its linear part therefore satisfies $DA_I(\partial_i \Gamma(t)) = \kappa^i \partial_i \Gamma(a + \kappa t)$, so boxes map to boxes with precisely the canonical side lengths. For (13.7.2), put $\widehat{g}_K = \widehat{f}_J \circ A_I$ with $K = J_I$, apply (13.7.1) to (g_K) and change variables on both sides. □

Lemma 13.7.5 (Bilinear reduction). *If $\delta = 2^{-N}$ then*

$$D_k(\delta) \lesssim \left(1 + \sum_{n=2}^N B(2^{-N+n-2})^2 \right)^{1/2}. \quad (13.7.3)$$

Proof. Decompose $[0, 1]^2$ by a Whitney decomposition around the diagonal. Set $\mathcal{W}_1 = \emptyset$ and, for $n \geq 2$, let $\mathcal{W}_n$ consist of the pairs $(I_1, I_2) \in \mathcal{P}(2^{-n})^2$ with $2^n \text{dist}(I_1, I_2) \in \{1, 2\}$ whose square is not already contained in a square selected at scale $2^{-(n-1)}$. Let $\mathcal{W}'_N$ be the pairs at scale 2^{-N} that touch the diagonal. The squares $I_1 \times I_2$ over $\mathcal{W}^N = \bigcup_{n=2}^N \mathcal{W}_n \cup \mathcal{W}'_N$ form an essentially disjoint cover of $[0, 1]^2$.

Since $(\sum_J f_J)^2 = \sum_{(I, I')} f_I f_{I'}$,

$$\begin{aligned} \left\| \sum_J f_J \right\|_{p_k} &= \left\| \sum_{(I, I') \in \mathcal{W}^N} f_I f_{I'} \right\|_{p_k/2}^{1/2} \\ &\leq \left(\sum_{(I, I') \in \mathcal{W}^N} \|f_I f_{I'}\|_{p_k/2} \right)^{1/2}. \end{aligned}$$

The diagonal pairs contribute $\sum_{(I, I') \in \mathcal{W}'_N} (\|f_I\|^2 + \|f_{I'}\|^2) \leq 6 \sum_I \|f_I\|_{p_k}^2$, each I occurring at most six times. For $(I, I') \in \mathcal{W}_n$, rescaling by Theorem 13.7.4 gives $\|f_I f_{I'}\|_{p_k/2} \lesssim B(2^{-N+n-2})^2$ times the corresponding ℓ^2 sums, and each $I \in \mathcal{P}(2^{-n})$ occurs at most eight times in $\mathcal{W}_n$. Summing over n gives (13.7.3). $\square$

Where we are. The Whitney decomposition replaces the broad–narrow dichotomy of Theorem 13.4.1. *Because the moment curve is one-dimensional, pairs suffice:* the square $[0, 1]^2$ decomposes around its diagonal into separated pairs of intervals, and no n -fold analogue is needed. That is the first economy of this proof over the multilinear one.

13.7.3 Asymmetric bilinear constants

Definition 13.7.6 (Polar box, bump, asymmetric constant). For a dyadic I let $U_I^\circ = \{x : |\langle x, \partial_i \Gamma(c_I) \rangle| \leq |I|^{-i}\}$, of dimensions $|I|^{-1} \times \cdots \times |I|^{-k}$, and let

$$\varphi_I(x) = |U_I^\circ|^{-1} (\inf\{t \geq 1 : x/t \in U_I^\circ\})^{-A_k},$$

where once and for all we take $A_k = 10k^2$. Thus $A_k > k$ and $A_k \geq k(k+1)/2$, the two inequalities needed by the convolution estimates below. This is an L^1 -normalised bump adapted to U_I° .

For $l \in \{0, \dots, k-1\}$ and $a, b \in [0, 1]$, let $B_{l,a,b}(\delta)$ be the least constant with

$$\begin{aligned} &\int_{\mathbb{R}^k} (|f_I|^{p_l} * \varphi_I) (|f_{I'}|^{p_k-p_l} * \varphi_{I'}) \\ &\leq B_{l,a,b}(\delta)^{p_k} \left[\sum_{J \subseteq I} \|f_J\|_{p_k}^2 \right]^{p_l/2} \left[\sum_{J' \subseteq I'} \|f_{J'}\|_{p_k}^2 \right]^{(p_k-p_l)/2}. \end{aligned} \quad (13.7.4)$$

for all $I \in \mathcal{P}(\delta^a)$, $I' \in \mathcal{P}(\delta^b)$ with $\text{dist}(I, I') \geq \frac{1}{4}$.

Lemma 13.7.7 (Uncertainty principle). *For $p \in [1, \infty)$ and $\text{supp } \widehat{g_J} \subseteq CU_J$, $|g_J|^p \lesssim_p |g_J|^p * \varphi_J$.*

Proof. Take ψ Schwartz adapted to U_J° with $\widehat{\psi} \equiv 1$ on CU_J and $\int |\psi| \approx 1$. Then $g_J = g_J * \psi$, and by Hölder,

$$\begin{aligned} |g_J|^p(x) &= \left(\int |g_J(x-z)|^p |\psi(z)| \, dz \right) \left(\int |\psi| \right)^{p/p'} \\ &\lesssim (|g_J|^p * |\psi|)(x) \\ &\lesssim (|g_J|^p * \varphi_J)(x). \end{aligned} \quad \square$$

Lemma 13.7.8 (Symmetric is controlled by asymmetric). *For every l, a, b and $\delta \in (0, \frac{1}{4})$,*

$$B(\delta) \lesssim \delta^{-ap_l/p_k} \delta^{-b(p_k-p_l)/p_k} B_{l,a,b}(\delta).$$

Proof. By Cauchy–Schwarz, $\int |f_I|^{p_k/2} |f_{I'}|^{p_k/2} \leq (\int |f_I|^{p_l} |f_{I'}|^{p_k-p_l})^{1/2} (\int |f_I|^{p_k-p_l} |f_{I'}|^{p_l})^{1/2}$, and by symmetry it suffices to bound the first. Assume first $l \neq 0$; the case $l = 0$ is the same with the first powered factor absent. Expanding f_I over $\mathcal{P}(I, \delta^a)$ and $f_{I'}$ over $\mathcal{P}(I', \delta^b)$, the power-mean inequality gives

$$\begin{aligned} &\int |f_I|^{p_l} |f_{I'}|^{p_k-p_l} \\ &\leq |\mathcal{P}(I, \delta^a)|^{p_l-1} |\mathcal{P}(I', \delta^b)|^{p_k-p_l-1} \sum_{J, J'} \int |f_J|^{p_l} |f_{J'}|^{p_k-p_l}. \end{aligned}$$

By Theorem 13.7.7 and (13.7.4), each last integral is at most

$$B_{l,a,b}(\delta)^{p_k} \left(\sum_{K \subseteq J} \|f_K\|_{p_k}^2 \right)^{p_l/2} \left(\sum_{K' \subseteq J'} \|f_{K'}\|_{p_k}^2 \right)^{(p_k-p_l)/2}.$$

Since $p_l, p_k - p_l \geq 2$, summing and using $\ell^2 \hookrightarrow \ell^{p_l}, \ell^{p_k-p_l}$ yields

$$\begin{aligned} \int |f_I|^{p_l} |f_{I'}|^{p_k-p_l} &\lesssim \delta^{-a(p_l-1)} \delta^{-b(p_k-p_l-1)} B_{l,a,b}(\delta)^{p_k} \\ &\quad \times \left(\sum_{K \subseteq I} \|f_K\|_{p_k}^2 \right)^{p_l/2} \left(\sum_{K' \subseteq I'} \|f_{K'}\|_{p_k}^2 \right)^{(p_k-p_l)/2}. \end{aligned}$$

The second Cauchy–Schwarz factor has the identical estimate after interchanging I and I' . Taking square roots and weakening $p_l - 1 \leq p_l$ and $p_k - p_l - 1 \leq p_k - p_l$ gives the claimed powers. $\square$

13.7.4 Transversality

For $0 \leq r \leq k$, write

$$V^{(r)}(\xi) := \text{span}(\partial_1 \Gamma(\xi), \dots, \partial_r \Gamma(\xi)).$$

Lemma 13.7.9 (Generalised Vandermonde). *For $0 \leq l \leq k$ and $\xi_1, \xi_2 \in \mathbb{R}$,*

$$|\partial_1 \Gamma(\xi_1) \wedge \dots \wedge \partial_l \Gamma(\xi_1) \wedge \partial_1 \Gamma(\xi_2) \wedge \dots \wedge \partial_{k-l} \Gamma(\xi_2)| \approx_{k,l} |\xi_1 - \xi_2|^{l(k-l)}. \quad (13.7.5)$$

Proof. Taylor expand about ξ_1 : for $1 \leq i \leq k-l$, $\partial_i \Gamma(\xi_2) = \sum_{j=i}^k \frac{1}{(j-i)!} \partial_j \Gamma(\xi_1) (\xi_2 - \xi_1)^{j-i}$. Substituting into the wedge and expanding gives a $(k-l)$ -fold sum; a term survives only if the chosen indices $(j_1, \dots, j_{k-l})$ are a permutation of $(l+1, \dots, k)$, since the first l slots already carry $\partial_1, \dots, \partial_l$. For such a term the power of $\xi_2 - \xi_1$ is

$$\sum_{i=1}^{k-l} (j_i - i) = ((l+1) + \dots + k) - (1 + \dots + (k-l)) = l(k-l),$$

the same for every surviving term, so the left side equals $c_{k,l} |\partial_1 \Gamma(\xi_1) \wedge \dots \wedge \partial_k \Gamma(\xi_1)| |\xi_2 - \xi_1|^{l(k-l)}$. Taking $\xi_1 = 0$, $\xi_2 = 1$ and computing the resulting determinant by column operations and the classical Vandermonde formula gives $\binom{k}{l} (\prod_{i=1}^l i!) (\prod_{j=1}^{k-l} j!) > 0$ for the left-hand side. Since $|\partial_1 \Gamma(0) \wedge \dots \wedge \partial_k \Gamma(0)| = \prod_{j=1}^k j!$, this proves $c_{k,l} > 0$ (with the corresponding quotient as its exact value). $\square$

Where we are. (13.7.5) says the l -th order tangent space at ξ_1 and the $(k-l)$ -th at ξ_2 are transverse whenever $\xi_1 \neq \xi_2$, quantitatively. *This is the only geometric input to the whole proof*, and it replaces multilinear Kakeya entirely: the transversality that Bourgain–Demeter–Guth extract from a Kakeya–Brascamp–Lieb inequality is here a determinant identity.

That every surviving term carries the *same* power $l(k-l)$ is what makes the estimate two-sided rather than merely an upper bound.

13.7.5 Torsion transfer and bump calculus

The lower-degree argument uses the induction hypothesis after quotienting the ambient frequency space. The projected curve is not literally a moment curve, so we first record the standard stability statement in the exact local form needed below.

Let $\gamma : [0, 1] \rightarrow \widehat{\mathbb{R}^l}$ satisfy

$$\|\gamma\|_{C^{l+1}} \leq C_0, \quad |\gamma'(\xi) \wedge \dots \wedge \gamma^{(l)}(\xi)| \geq c_0 > 0. \quad (13.7.6)$$

For a dyadic interval J , let $U_{J,\gamma}$ be the box centred at $\gamma(c_J)$, with side of length $|J|^j$ parallel to $\gamma^{(j)}(c_J)$, $1 \leq j \leq l$.

Lemma 13.7.10 (Transfer to curves with torsion). *Suppose Theorem 13.7.2 holds in degree l . If γ obeys (13.7.6), then for every $\varepsilon, C > 0$ and $0 < \rho < 1$,*

$$\left\| \sum_{J \in \mathcal{P}(\rho)} g_J \right\|_{p_l} \lesssim_{\varepsilon, C, C_0, c_0} \rho^{-\varepsilon} \left(\sum_J \|g_J\|_{p_l}^2 \right)^{1/2} \quad (13.7.7)$$

whenever $\text{supp } \widehat{g_J} \subseteq CU_{J,\gamma}$.

Proof. Put $r = l/(l+1)$. We first prove the one-step estimate: if $\kappa \geq \rho^r$ and $I \in \mathcal{P}(\kappa)$, then for every auxiliary $\sigma > 0$,

$$\|g_I\|_{p_l} \lesssim_{\sigma} \kappa^{-\sigma} \left(\sum_{I' \in \mathcal{P}(I, \kappa^{(l+1)/l})} \|g_{I'}\|_{p_l}^2 \right)^{1/2}. \quad (13.7.8)$$

Indeed, on I Taylor's theorem gives

$$\gamma(\xi) = \gamma(c_I) + \sum_{j=1}^l \frac{\gamma^{(j)}(c_I)}{j!} (\xi - c_I)^j + O_{C_0}(\kappa^{l+1}).$$

The polynomial part is a uniformly nonsingular affine image of the degree- l moment curve, by the determinant lower bound in (13.7.6). If $|I'| = \kappa^{(l+1)/l}$, the shortest side of its canonical box is $|I'|^l = \kappa^{l+1}$, so the Taylor remainder is absorbed by a fixed dilate of that box. Affine rescaling and the degree- l theorem, used with exponent $l\sigma$, now give (13.7.8).

It remains to account for the iteration rather than merely invoke it. Fix an integer M . A trivial coarse decoupling first groups the ρ -intervals inside intervals of length ρ^{r^M} ; its loss is at most $\rho^{-(l+1)r^M}$. Apply (13.7.8) successively at the M scales $\rho^{r^M}, \rho^{r^{M-1}}, \dots, \rho^r$. Minkowski for the resulting ℓ^2 sums gives a total one-step loss bounded by $\rho^{-\sigma \sum_{j=1}^M r^j} \leq \rho^{-l\sigma}$. Thus (13.7.7) holds with exponent $(l+1)r^M + l\sigma$. Given the desired ε , first choose $\sigma < \varepsilon/(2l)$ and then M so large that $(l+1)r^M < \varepsilon/2$. $\square$

For a ball B of radius ρ^{-l} in an affine copy of $\mathbb{R}^l$, set

$$\Phi_B(x) := |B|^{-1} (1 + \rho^l \text{dist}(x, B))^{-(A_k + k - 2)}, \quad \|h\|_{L^p(\Phi_B)} := \left(\int |h|^p \Phi_B \right)^{1/p}. \quad (13.7.9)$$

Corollary 13.7.11 (Local weighted torsion decoupling). *Under the hypotheses of Theorem 13.7.10, for every such ball B ,*

$$\frac{1}{|B|} \int_B \left| \sum_{J \in \mathcal{P}(\rho)} g_J \right|^{p_l} \lesssim_{\varepsilon} \rho^{-\varepsilon} \left(\sum_J \|g_J\|_{L^{p_l}(\Phi_B)}^2 \right)^{p_l/2}. \quad (13.7.10)$$

The same estimate holds after restricting the parameter to a dyadic interval and partitioning that interval into subintervals of length ρ .

Proof. Choose a Schwartz cutoff ψ_B which is bounded below on B , whose decay is dominated by the weight in (13.7.9), and whose Fourier support lies in a ball of radius $O(\rho^l)$. Multiplication by ψ_B enlarges each frequency box by at most its shortest side, hence preserves $\widehat{\text{supp } g_J \psi_B} \subseteq C' U_{J,\gamma}$. Apply Theorem 13.7.10 to $g_J \psi_B$, raise to the p_l -th power, and use the decay of ψ_B . Translating and restricting the parameter interval do not change any of the uniform constants. $\square$

Two elementary convolution bounds keep the weights stable. If $*_H$ denotes convolution on a subspace H , then

Lemma 13.7.12 (Nested and transverse bump convolution). *For dyadic $J \subseteq I \subseteq [0, 1]$,*

$$(\varphi_I * \varphi_J)(x) \lesssim \varphi_J(x). \quad (13.7.11)$$

Moreover, let $I' \in \mathcal{P}(\nu)$, choose $\xi' \in I'$, and let H be the l -dimensional orthogonal complement of $V^{(k-l)}(\xi')$. For $R = \nu^{-(k-l+1)}$,

$$(\Phi_{B_H(0,R)} *_H \varphi_{I'})(x) \lesssim \varphi_{I'}(x). \quad (13.7.12)$$

Proof. For (13.7.11), affine-rescale I to $[0, 1]$ and rotate to axes adapted to U_J° . With $d = |J|$, $q = A_k$, and

$$\phi(x) = \max(1, |x_1|, \dots, |x_k|)^{-q}, \quad \phi_d(x) = d^{k(k+1)/2} \phi(dx_1, d^2 x_2, \dots, d^k x_k),$$

the claim becomes $\phi * \phi_d \lesssim \phi_d$. On the set $\phi_d(y) \leq 2^q \phi_d(x)$, integrability of ϕ (because $q > k$) gives the bound immediately. On its complement, some coordinate satisfies $|x_i| > 2|y_i|$; hence $\phi(x - y) \leq 2^q |x_i|^{-q} e 2^q \phi_d(x)$, where the last inequality uses $iq \geq k(k+1)/2$. Integrating the normalised ϕ_d proves (13.7.11).

For (13.7.12), a uniformly bounded change of variables reduces to $\xi' = 0$. Put $k' = k - l + 1$; then H is spanned by the last l coordinate directions. Up to fixed constants, the left-hand side without its normalising factor is

$$\begin{aligned} & \int_{\mathbb{R}^l} \nu^{k'l} (1 + \nu^{k'} |z|)^{-(A_k + k - 2)} \\ & \quad \times \left(1 + \sum_{i=1}^{k-l} \nu^i |x_i| + \sum_{j=k'}^k \nu^j |x_j - z_j| \right)^{-A_k} dz. \end{aligned}$$

Dropping the second sum bounds this by $(1 + \sum_{i=1}^{k-l} \nu^i |x_i|)^{-A_k}$. For any fixed $j_0 \in \{k', \dots, k\}$, integrate first in the other $l - 1$ variables. Since $A_k + k - 2 - l + 1 \geq A_k > 1$ and $\nu^{j_0} \leq \nu^{k'}$, the remaining one-dimensional convolution is bounded by $(1 + \nu^{j_0} |x_{j_0}|)^{-A_k}$. Taking the minimum of these coordinate bounds is comparable to $(1 + \sum_{i=1}^k \nu^i |x_i|)^{-A_k}$, which is (13.7.12). $\square$

13.7.6 The two key lemmas

Lemma 13.7.13 (Lower degree decoupling). *Let $1 \leq l \leq k-1$ and suppose Theorem 13.7.2 known with k replaced by l . Let $(f_K)_{K \in \mathcal{P}(\delta)}$ satisfy $\text{supp } \widehat{f_K} \subseteq U_K$. If $0 \leq a \leq \frac{(k-l+1)b}{l}$ then, writing $b' = \frac{(k-l+1)b}{l}$,*

$$\begin{aligned} & \int_{\mathbb{R}^k} (|f_I|^{p_l} * \varphi_I) (|f_{I'}|^{p_k - p_l} * \varphi_{I'}) \\ & \lesssim_{\varepsilon} \delta^{-\varepsilon b} \left(\sum_{J \in \mathcal{P}(I, \delta^{b'})} \left[\int (|f_J|^{p_l} * \varphi_J) (|f_{I'}|^{p_k - p_l} * \varphi_{I'}) \right]^{2/p_l} \right)^{p_l/2}. \end{aligned}$$

Proof. Fix $\xi' \in I'$ and put

$$\widehat{H} := \widehat{\mathbb{R}^k} / V^{(k-l)}(\xi'), \quad P : \widehat{\mathbb{R}^k} \longrightarrow \widehat{H}.$$

Let $H \subset \mathbb{R}^k$ be the l -dimensional orthogonal complement of $V^{(k-l)}(\xi')$, so $\widehat{H}$ is its dual. Since $\text{dist}(I, I') \geq 1/4$, Theorem 13.7.9 gives, uniformly for $\xi \in I$,

$$|(P\Gamma)'(\xi) \wedge \cdots \wedge (P\Gamma)^{(l)}(\xi)| \gtrsim 1. \quad (13.7.13)$$

Also $P(U_J) \subseteq CU_{J, P\Gamma}$: the projected first l derivative directions give the defining frame, while the remaining sides have length at most $|J|^l$ and are absorbed in a fixed dilation. Thus the restriction of f_J to almost every translate $H + z$ has Fourier support eligible for Theorem 13.7.11.

Set

$$F := |f_I|^{p_l} * \varphi_I, \text{quad}G := |f_{I'}|^{p_k - p_l} * \varphi_{I'}, \text{quad}R := \delta^{-b'l} = \delta^{-(k-l+1)b}.$$

Writing $B_H(z, R)$ for the radius- R ball in $H + z$, Fubini and the fact that normalised averaging is convolution with a probability measure give

$$\int_{\mathbb{R}^k} FG = \int_{\mathbb{R}^k} \frac{1}{|B_H(z, R)|} \int_{B_H(z, R)} F(x)G(x) \, d_H x \, dz. \quad (13.7.14)$$

Because $R = \delta^{-(k-l+1)b}$ and $|I'| \simeq \delta^b$, transversality implies $B_H(0, R) \subseteq CU_{I'}^\circ$. The defining polynomial tail of $\varphi_{I'}$ therefore gives

$$\sup_{x \in B_H(z, R)} G(x) \lesssim G(z). \quad (13.7.15)$$

Consequently (13.7.14) is at most

$$\int_{z, z' \in \mathbb{R}^k} G(z) \varphi_I(z - z') \frac{1}{|B_H(z', R)|} \int_{B_H(z', R)} |f_I(x)|^{p_l} \, d_H x \, dz' \, dz. \quad (13.7.16)$$

Apply Theorem 13.7.11 on $H + z'$ to the curve $P\Gamma$, at scale $\delta^{b'}$, and use (13.7.13). Since b'/b depends only on k, l , its arbitrary epsilon loss may be renamed $\delta^{-\varepsilon b}$. Taking the p_l -th root of (13.7.16) and applying Minkowski in the form $L^{p_l}(\ell^2) \leq \ell^2(L^{p_l})$ gives

$$\left(\int FG \right)^{1/p_l} \lesssim_{\varepsilon} \delta^{-\varepsilon b} \left(\sum_{J \in \mathcal{P}(I, \delta^{b'})} Q_J^{2/p_l} \right)^{1/2}, \quad (13.7.17)$$

where

$$Q_J := \int_{z, z'} G(z) \varphi_I(z - z') \|f_J\|_{L^{p_l}(H+z', \Phi_{B_H(z', R)})}^{p_l} dz' dz.$$

All integrations here are nonnegative, so Fubini and convolution symmetry give

$$\begin{aligned} Q_J &= \int_{\mathbb{R}^k} (G *_H \Phi_{B_H(0, R)})(x) (|f_J|^{p_l} * \varphi_I)(x) dx \\ &\lesssim \int_{\mathbb{R}^k} G(x) (|f_J|^{p_l} * \varphi_J)(x) dx. \end{aligned}$$

In the last line we used (13.7.12) for the first factor. For the second, Theorem 13.7.7 followed by (13.7.11) gives $|f_J|^{p_l} * \varphi_I \lesssim |f_J|^{p_l} * \varphi_J * \varphi_I \lesssim |f_J|^{p_l} * \varphi_J$. Insert this bound in (13.7.17) and raise to the p_l -th power. Renaming the arbitrary epsilon once more gives the stated estimate. $\square$

Corollary 13.7.14. *Under the same hypotheses, for $0 \leq a \leq \frac{k-l+1}{l}b$,*

$$B_{l,a,b}(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon b} B_{l, \frac{k-l+1}{l}b, b}(\delta). \quad (13.7.18)$$

Proof. Apply Theorem 13.7.13 to the left side of (13.7.4) for $B_{l,a,b}(\delta)$. Each summand on the right of that lemma is an integral of the same shape, with I replaced by $J \in \mathcal{P}(I, \delta^{b'})$ and $b' = \frac{(k-l+1)b}{l}$, so it is bounded by $B_{l,b',b}(\delta)^{p_k}$ times the corresponding ℓ^2 factors. Summing the resulting ℓ^2 sum over J recovers the ℓ^2 sum over $\mathcal{P}(I, \delta)$, and the claim follows. $\square$

Lemma 13.7.15 (Hölder). *For $1 \leq l \leq k-1$ and $a, b \in (0, 1)$,*

$$B_{l,a,b}(\delta) \leq B_{k-l,b,a}(\delta)^{\frac{1}{k-l+1}} B_{l-1,a,b}(\delta)^{\frac{k-l}{k-l+1}}. \quad (13.7.19)$$

Proof. The three points

$$(p_l, p_k - p_l), \quad (p_k - p_{k-l}, p_{k-l}), \quad (p_{l-1}, p_k - p_{l-1})$$

all lie on the line $x + y = p_k$, hence are collinear, so there is θ_l with

$$(p_l, p_k - p_l) = \theta_l(p_k - p_{k-l}, p_{k-l}) + (1 - \theta_l)(p_{l-1}, p_k - p_{l-1}). \quad (13.7.20)$$

Comparing first coordinates and substituting $p_j = j(j+1)$,

$$\begin{aligned}\theta_l &= \frac{p_l - p_{l-1}}{(p_k - p_{k-l}) - p_{l-1}} \\ &= \frac{2l}{k(k+1) - (k-l)(k-l+1) - (l-1)l} = \frac{1}{k-l+1}.\end{aligned}$$

Now apply Hölder's inequality to the left side of (13.7.4). First, Hölder in the convolution measure gives, pointwise,

$$|f_I|^{p_l} * \varphi_I \leq (|f_I|^{p_k - p_{k-l}} * \varphi_I)^{\theta_l} (|f_I|^{p_{l-1}} * \varphi_I)^{1-\theta_l},$$

and the analogous inequality for $f_{I'}$. Then Hölder in x with exponents $1/\theta_l$ and $1/(1-\theta_l)$ produces the two integrals defining $B_{k-l,b,a}(\delta)$ and $B_{l-1,a,b}(\delta)$. Their right-hand ℓ^2 factors combine with exactly the powers in (13.7.19). $\square$

Where we are. The identity $\theta_l = \frac{1}{k-l+1}$ is where the exponent $p_k = k(k+1)$ earns its place. The three points are collinear simply because both coordinates sum to p_k ; what is special about the quadratic sequence $p_j = j(j+1)$ is the positive weight $\theta_l = 1/(k-l+1)$, whose coefficient matrix has the exact conservation law used below.

Combining (13.7.19) with (13.7.18) gives, for b small, and with

$$c_l(b) := \begin{cases} 0, & l = 1, \\ \frac{k-l+2}{l-1}b, & 2 \leq l \leq k-1, \end{cases}$$

$$B_{l, \frac{k-l+1}{l}b, b}(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon b} B_{k-l, \frac{l+1}{l} \frac{k-l+1}{k-l}b, \frac{k-l+1}{l}b}(\delta)^{\frac{1}{k-l+1}} B_{l-1, c_l(b), b}(\delta)^{\frac{k-l}{k-l+1}}. \quad (13.7.21)$$

For $l = 1$ the first scale of $B_{0,a,b}$ is immaterial; this is why the piecewise convention removes the apparent division by zero. The restrictions needed to apply (13.7.18) are automatic once $b > 0$ is sufficiently small (depending only on k, l). This is the recursion the bootstrap consumes.

13.7.7 The bootstrap

Proof of Theorem 13.7.2. By induction on k . For $k = 1$, $p_1 = 2$ and (13.7.1) is Plancherel, so $D_1(\delta) \leq 10$. Fix $k \geq 2$ and assume the theorem for every $l < k$.

Step 1. *The exponents.*

Let

$$\eta = \inf\{\varepsilon \geq 0 : D_k(\delta) \lesssim_{\varepsilon} \delta^{-\varepsilon} \text{ for all } 0 < \delta < 1\}.$$

This set is nonempty by the trivial ℓ^2 decoupling, so $\eta < \infty$. For small $b > 0$, let $A_0(b)$ be the infimum of the exponents A for which $B_{0,0,b}(\delta) \lesssim \delta^{-A}$, and, for $1 \leq l \leq k-1$, let $A_l(b)$ be the corresponding infimum for $B_{l, \frac{k-l+1}{l}b, b}(\delta)$. Since $B_{0,a,b}(\delta) \approx D_k(\delta^{1-b})$ by affine rescaling,

$$A_0(b) = \eta(1-b). \quad (13.7.22)$$

Step 2. *The recursion.*

By (13.7.21), for $1 \leq l \leq k-1$ and b small,

$$A_l(b) \leq \frac{1}{k-l+1} A_{k-l}\left(\frac{k-l+1}{l}b\right) + \frac{k-l}{k-l+1} A_{l-1}(b). \quad (13.7.23)$$

Step 3. *Pass to derivatives at $b = 0$.*

Put $\mathcal{A}_l = \liminf_{b \rightarrow 0} \frac{\eta - A_l(b)}{b}$. From (13.7.22), $\mathcal{A}_0 = \eta$; and (13.7.23) gives

$$\mathcal{A}_l \geq \frac{1}{l} \mathcal{A}_{k-l} + \frac{k-l}{k-l+1} \mathcal{A}_{l-1}, \quad 1 \leq l \leq k-1. \quad (13.7.24)$$

The factor $\frac{1}{l}$ rather than $\frac{1}{k-l+1}$ arises because the argument of A_{k-l} is $\frac{k-l+1}{l}b$, not b , so the difference quotient is scaled by that factor.

Step 4. *Finiteness.*

Before summing we must know the $\mathcal{A}_l$ are finite. By Hölder applied to (13.7.4) and then (13.7.2),

$$\begin{aligned} B_{l, \frac{k-l+1}{l}b, b}(\delta) &\lesssim D_k(\delta^{1-\frac{k-l+1}{l}b})^{p_l/p_k} D_k(\delta^{1-b})^{(p_k-p_l)/p_k}, \end{aligned}$$

whence $A_l(b) \leq \eta - \eta b \left[\frac{k-l+1}{l} \frac{p_l}{p_k} + \frac{p_k-p_l}{p_k} \right]$. Thus $\mathcal{A}_l$ is bounded below by the finite nonnegative number in square brackets times η . In the other direction Theorem 13.7.5 and Theorem 13.7.8 give $\eta \leq Cb + A_l(b)$, so $\mathcal{A}_l \leq C$. Hence each $\mathcal{A}_l$ is finite.

Step 5. *Sum.*

Summing (13.7.24) over $l = 1, \dots, k-1$ makes the cancellation transparent. For $1 \leq j \leq k-2$, the coefficient of $\mathcal{A}_j$ on the right is

$$\frac{1}{k-j} + \frac{k-j-1}{k-j} = 1;$$

the coefficient of $\mathcal{A}_{k-1}$ is also 1, while the only remaining term is $\frac{k-1}{k} \mathcal{A}_0$. Subtracting $\sum_{j=1}^{k-1} \mathcal{A}_j$ from both sides leaves

$$0 \geq \frac{k-1}{k} \mathcal{A}_0 = \frac{k-1}{k} \eta.$$

Since $\eta \geq 0$, we have $\eta = 0$. By the definition of the infimum and the monotonicity of admissible exponents, (13.7.1) therefore holds with every prescribed positive epsilon, which is the theorem. $\square$

Where we are. The cancellation in Step 5 is exact and is the mechanism of the proof. Symbolically, summing (13.7.24) for $k = 3, 4, 5, 6$ leaves $-\frac{2}{3}\mathcal{A}_0$, $-\frac{3}{4}\mathcal{A}_0$, $-\frac{4}{5}\mathcal{A}_0$, $-\frac{5}{6}\mathcal{A}_0$ respectively, with every other term vanishing. $\checkmark$ *What makes it work* is that $(1, \dots, 1)$ is a left eigenvector of the coefficient matrix of (13.7.24). Guo, Li, Yung and Zorin-Kranich note that such Perron–Frobenius eigenvectors are the recurring device in iterative proofs of decoupling inequalities, and Volume 14’s radial iteration Theorem 14.2.71 records the same finite-state self-improvement phenomenon in a geometric guise.

WORKED EXAMPLE 13.7.1— TIER A **Why this proof needs no Kakeya**

Chapter 13.4 proved decoupling for the parabola using ball inflation, hence multilinear Kakeya, hence Volume 11. This chapter proves the harder theorem — the moment curve in every degree — using none of it. The difference is worth locating.

(a) *Where transversality enters.* There: Theorem 13.3.4, via Theorem 11.5.7. Here: Theorem 13.7.9, a determinant identity proved by Taylor expansion.

(b) *Why a determinant suffices.* The moment curve is a *curve*, so transversality is a statement about tangent spaces at two points, and two points are enough to span. For a hypersurface one needs n points, and controlling n -fold transversality is exactly what multilinear Kakeya does.

(c) *What is lost.* Nothing, for this theorem. But the argument does not transfer back: it inducts on the *degree* k , and a hypersurface has no analogous filtration by lower-degree objects.

(d) *The historical order.* Bourgain–Demeter–Guth proved this in 2016 using multilinear Kakeya; the bilinear proof came in 2019, after Wooley’s nested efficient congruencing suggested it. *The simplification came from number theory*, which is the reverse of the transfer Chapter 13.6 describes.

(e) *The consequence for this series.* Theorem 13.6.2 now rests on Hölder, Plancherel, and a Vandermonde determinant. Volume 13, Worked Example 13.6.3 traced a chain from Loomis–Whitney to the zeros of ζ ; that chain remains valid, but this chapter shows a shorter one exists that bypasses Volume 11 entirely.

WORKED EXAMPLE 13.7.2— TIER B Reading the asymmetric constants

Theorem 13.7.6 is the technical heart and looks opaque. Four remarks.

(1) *Why two scales a and b .* The two intervals are allowed different lengths δ^a, δ^b , and the induction moves between (a, b) pairs. A symmetric constant could not record that.

(2) *Why the exponents p_l and $p_k - p_l$.* They sum to p_k , so (13.7.4) has the homogeneity of (13.7.1). The parameter l interpolates between $l = 0$, where the constant reduces to D_k , and $l = k - 1$.

(3) *Why the bumps φ_I .* They implement the uncertainty principle: $|f_I|^p$ is not constant on U_I° , but its average against φ_I behaves as if it were. Theorem 13.7.7 is the precise statement, and without it the Fubini step of Theorem 13.7.13 would not close.

(4) *The one thing to remember.* $B_{l,a,b}$ measures how much a *degree- l* phenomenon can be seen inside a degree- k problem, and Theorem 13.7.15 says the degree- l constant is squeezed between the degree- $(k - l)$ and degree- $(l - 1)$ ones. Summing those squeezes around the cycle is the proof.

Drills

Drill 13.7.1. Verify $p_k = k(k + 1)$ gives $p_2 = 6$ and confirm that Theorem 13.7.2 at $k = 2$ is Theorem 13.4.2.

Drill 13.7.2. In Theorem 13.7.9, check that the surviving terms all carry the power $l(k - l)$, and compute $c_{k,l}$ for $k = 3, l = 1$.

Drill 13.7.3. Verify $\theta_l = \frac{1}{k-l+1}$ from (13.7.20) for $k = 4$ and $l = 1, 2, 3$.

Drill 13.7.4. Sum (13.7.24) for $k = 4$ and confirm that $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ cancel, leaving $-\frac{3}{4}\mathcal{A}_0$.

Notes

Theorem 13.7.2 is Bourgain, Demeter and Guth (Ann. of Math. 184, 2016). The proof given here is that of Guo, Li, Yung and Zorin-Kranich (Amer. J. Math. 143, 2021), inspired by Wooley's nested efficient congruencing. Theorem 13.7.10 goes back to Pramanik and Seeger.

Chapter 13.8

Efficient Congruencing

Vinogradov's conjecture was proved twice, by methods with nothing obvious in common. Wooley's efficient congruencing settled $k = 3$ in 2016 and gave strong results for all k ; decoupling settled all k shortly after. This chapter compares them, and explains why the comparison is instructive rather than merely historical.

13.8.1 The shape of the congruencing argument

Definition 13.8.1 (Congruencing counts). For a prime p and $\xi \in \mathbb{Z}/p$, let $J_{s,k}(N; \xi, p)$ count the solutions of Theorem 13.6.1 with all $x_i \equiv y_i \equiv \xi \pmod{p}$.

Proposition 13.8.2 (Why congruences help). *If $x_i \equiv y_i \equiv \xi \pmod{p}$ for all i , write $x_i = \xi + pu_i$ and $y_i = \xi + pv_i$. Then the moment system for (x, y) is equivalent to the moment system for (u, v) .*

Proof. By the binomial theorem,

$$\sum_i x_i^j = \sum_i (\xi + pu_i)^j = \sum_{m=0}^j \binom{j}{m} \xi^{j-m} p^m \sum_i u_i^m,$$

and likewise for y . Subtracting, the system $\sum_i x_i^j = \sum_i y_i^j$ for $j = 1, \dots, k$ becomes a triangular linear system in the differences $\sum_i (u_i^m - v_i^m)$, with invertible coefficient matrix (the leading coefficient at $m = j$ is p^j). Hence it vanishes for all $j \leq k$ iff $\sum_i u_i^m = \sum_i v_i^m$ for all $m \leq k$. $\square$

Where we are. Theorem 13.8.2 is the engine. It says the moment system is *self-similar under congruence restriction*: solutions congruent to a fixed residue modulo p are, after the substitution $x = \xi + pu$, solutions of the same system with the variables shrunk by a factor p .

That is an induction on scales, with the scale measured by powers of p rather than by dyadic decomposition. The parallel with Theorem 13.2.1 is exact: there, an arc of length σ became the whole parabola at scale δ/σ^2 ; here, a congruence class modulo p becomes the whole problem at size N/p . Wooley’s argument iterates this, alternating between counting solutions in a fixed congruence class and averaging over classes, with a carefully chosen sequence of moduli. The bookkeeping is intricate and we do not reproduce it.

Theorem 13.8.3 (Wooley; not proved here). *Efficient congruencing yields Theorem 13.6.2 for $k = 3$, and for general k yields the conjecture for s outside a range shrinking as k grows.*

Remark 13.8.4 (Scope). Theorem 13.8.3 is not proved here; revision R27, and this is a *settled omission*. *Nothing in this series depends on it*: Theorem 13.6.2 is proved in Chapter 13.6 via decoupling, and this chapter is comparative — its purpose is to display a second, independent route to the same theorem, not to establish anything. Reproducing Wooley’s descent would add a second proof of a result already proved, which is the same ground on which Theorem 7.7.5 declines Drury’s argument.

What is proved here is Theorem 13.8.2, the self-similarity that makes the method possible, and that is the part worth carrying away.

13.8.2 The two methods compared

WORKED EXAMPLE 13.8.1 — TIER A **Congruencing against decoupling**

	efficient congruencing	decoupling
Setting	$\mathbb{Z}$, congruences mod p^m	$\mathbb{R}^k$, L^p norms
Self-similarity	Theorem 13.8.2	Theorem 13.2.1
Scale parameter	powers of a prime p	$\delta \mapsto \delta/\sigma^2$
Induction	on m in p^m	on $\log \log \delta^{-1}$
Curvature enters as	the binomial triangularity	the dual box, Theorem 13.1.2
Proves	$k = 3$ fully; partial for $k \geq 4$	all k
Generalises to	other Diophantine systems	other curved surfaces

The common structure. Both are inductions on scales driven by a self-similarity of the object: the moment system reproduces itself under congruence restriction, and the moment curve reproduces itself under parabolic rescaling. *These are the same fact in two languages*, since the frequencies $(n, n^2, \dots, n^k)$ of Chapter 13.6 Step 2 lie on the curve.

Why decoupling went further. It is insensitive to arithmetic. The congruencing argument must track how solutions distribute among residue classes, and controlling that for large k is where it becomes difficult; decoupling replaces the arithmetic by geometry — transversality and ball inflation — which does not degrade with k .

Why congruencing is not obsolete. It gives explicit constants, where decoupling gives N^ε with no effective bound (Worked Example 13.2.2); and it applies to systems with no continuous analogue.

WORKED EXAMPLE 13.8.2— TIER A **The binomial triangularity, in an example**

Theorem 13.8.2 rests on a triangular system. Here it is at $k = 2$.

Setup. $x_i = \xi + pu_i$, $y_i = \xi + pv_i$, and write $U_m = \sum_i u_i^m$, $V_m = \sum_i v_i^m$.

First equation. $\sum_i x_i = s\xi + pU_1$ and $\sum_i y_i = s\xi + pV_1$, so $\sum x_i = \sum y_i$ becomes $pU_1 = pV_1$, that is $U_1 = V_1$.

Second equation. $\sum_i x_i^2 = s\xi^2 + 2\xi pU_1 + p^2U_2$, and likewise for y . Subtracting and using $U_1 = V_1$ from the first equation leaves $p^2(U_2 - V_2) = 0$, that is $U_2 = V_2$.

The pattern. The j -th equation, after substituting the previous $j - 1$, reduces to $p^j(U_j - V_j) = 0$. The system is triangular with diagonal p^j , hence invertible over $\mathbb{Q}$. ✓

Why this is the right notion of self-similarity. The transformed system is identical in form, with variables in a range shrunk by p . Compare Theorem 13.2.1, where an arc becomes the whole parabola at a smaller δ : same phenomenon, different category.

What would break it. If the exponents were not $1, 2, \dots, k$ but an arbitrary set, the expansion would not be triangular and the substitution would not close. That is why the method is tied to the moment system, and why the moment curve is the surface on the analytic side.

WORKED EXAMPLE 13.8.3— TIER B **Self-similarity as a proof strategy**

Both methods rest on identifying a transformation under which the problem reproduces itself at a smaller scale. That pattern has appeared repeatedly.

(a) *Parabolic rescaling*, Theorem 13.2.1: an arc becomes the whole curve. Exact, not approximate.

(b) *Congruence restriction*, Theorem 13.8.2: a residue class becomes the whole system.

(c) *Multiscale tube analysis*, Volume 11, Theorem 11.5.6: tubes of radius W become tubes of radius $\delta^{-1}W$.

(d) *The weighted induction*, Volume 12, Theorem 12.3.10: the constant at scale $\delta^{-1}R$ is controlled by the constant at scale R .

The three questions to ask. What is the self-map? Is it exact or lossy? And how many iterations reach the trivial case — log or log log? By Worked Example 12.4.3 of Volume 12, the answer to the third determines whether the losses compound to a fixed power or to $\delta^{-\varepsilon}$, and it is the single most consequential piece of bookkeeping in the subject.

Answers for the four above. (a) exact, log log; (b) exact, log in p but with a gain each round; (c) exact, log with cancellation of the δ^n factors; (d) exact, log with a constant per round chosen small. All four terminate, and for different reasons.

WORKED EXAMPLE 13.8.4— TIER B **The two inductions, matched step by step**
Efficient congruencing and decoupling prove the same theorem by inductions that correspond term for term. The correspondence is not a metaphor.

	efficient congruencing	decoupling
Scale parameter	power p^b of a prime	spatial scale δ
One step	$b \rightarrow b + 1$	$\delta \rightarrow \delta^{1/2}$
Object counted	solutions congruent mod p^b	L^p mass on a cap
Non-degeneracy	Vandermonde determinant $\neq 0$	transversality, Theorem 13.3.1
Concentration	solutions cluster in residue classes	wave packets cluster on tubes
Output	explicit constants	N^ε , ineffective

The shared mechanism. In both, the critical case is where the diagonal and generic counts balance, and in both, progress comes from showing that solutions cannot concentrate too much at any single scale — which is what the Vandermonde and the transversality hypothesis respectively guarantee. *Where they diverge.* Congruencing works with exact integer structure and so keeps track of constants; decoupling passes to a continuous problem and loses them. For Waring-type applications needing explicit bounds this matters, and Worked Example 13.6.1 records that the congruencing route remains preferable there.

Why both were found at once. Both are induction-on-scales arguments, and the technique had matured independently in analytic number theory and in harmonic analysis. That two communities reached the same theorem within a year, by arguments that translate line by line, is the clearest evidence in this series that induction on scales is a single method rather than a family of tricks.

13.8.3 The congruencing step

To make Worked Example 13.8.4 concrete, we state the one step that does the work in Wooley's method, in the simplest case.

Definition 13.8.5 (Congruencing counts). Fix a prime p and $b \geq 1$. For $\xi \in \mathbb{Z}$ let $J_{s,k}^{(b)}(N; \xi)$ count solutions of Vinogradov's system with $1 \leq x_i, y_i \leq N$ and, additionally,

$$x_i \equiv \xi \pmod{p^b} \quad (i = 1, \dots, k),$$

the remaining variables being unrestricted.

Proposition 13.8.6 (Congruences propagate). *Suppose $x_1, \dots, x_k$ are distinct modulo p and satisfy Vinogradov's system together with $y_1, \dots, y_k$. If $x_i \equiv \xi \pmod{p^b}$ for all i , then after reordering*

$$y_i \equiv \xi \pmod{p^b} \quad (i = 1, \dots, k).$$

Proof. The system says the first k power sums of (x_i) and (y_i) agree. By Newton's identities the elementary symmetric functions therefore agree, so (x_i) and (y_i) are the roots of the same monic polynomial $F(X) = \prod_i (X - x_i)$ over $\mathbb{Z}$.

Reducing modulo p^b , $F(X) \equiv (X - \xi)^k$, so each y_i satisfies $(y_i - \xi)^k \equiv 0 \pmod{p^b}$. Since the x_i , hence the y_i , are distinct modulo p , Hensel's lemma applies and gives $y_i \equiv \xi \pmod{p^b}$ after reordering. $\square$

Where we are. Theorem 13.8.6 is the engine. It says that a congruence condition imposed on half the variables is *inherited* by the other half — so the solution set is much more rigid than a naive count suggests, and iterating the observation at successive powers p^b produces a descent.

The analogy with decoupling, made precise. In Chapter 13.4 the corresponding statement is that wave packets from transverse caps cannot both be large at the same point unless the caps are related — Theorem 13.4.1. Both are rigidity statements: a constraint at one scale forces a constraint at the next.

Where the Vandermonde enters. Hensel's lemma requires the x_i distinct modulo p , that is $\prod_{i < j} (x_i - x_j) \not\equiv 0$ — the Vandermonde determinant. That is the exact counterpart of the transversality determinant $|\det(\omega_1, \dots, \omega_n)| \geq \nu$ of Volume 12, Theorem 12.2.1. *Both methods break down precisely when their determinant vanishes*, and both handle that case by a separate, easier argument.

WORKED EXAMPLE 13.8.5— TIER A Newton’s identities, checked

Theorem 13.8.6 rests on the passage from power sums to elementary symmetric functions, which is worth verifying in the smallest case.

$k = 2$. Given $x_1 + x_2 = y_1 + y_2 =: e_1$ and $x_1^2 + x_2^2 = y_1^2 + y_2^2 =: p_2$, Newton gives $e_2 = \frac{1}{2}(e_1^2 - p_2)$, the same for both pairs. So $X^2 - e_1X + e_2$ has roots $\{x_1, x_2\}$ and $\{y_1, y_2\}$: the pairs coincide.

Compare Theorem 13.1.6. That lemma made exactly this computation over $\mathbb{R}$, with δ -precision instead of exactly, and concluded that the sum-set map is 2-to-1. *The $p = 4$ decoupling theorem and the $k = 2$ case of congruencing are the same algebraic fact*, one used with an error term and one exactly.

$k = 3$ and beyond. Newton’s identities continue to determine e_j from $p_1, \dots, p_j$ over $\mathbb{Q}$, with denominators dividing $j!$ — which is why the congruencing argument requires $p > k$, and why the harmonic-analytic counterpart at $p = 6$ needs more than an overlap count.

The check to run. For $k = 3$, verify that $e_3 = \frac{1}{6}(e_1^3 - 3e_1p_2 + 2p_3)$ and that the denominator $6 = 3!$ obstructs the argument at $p = 2, 3$.

WORKED EXAMPLE 13.8.6— TIER C: RESEARCH TASTE Two proofs of one theorem

Vinogradov’s mean value theorem has two essentially different proofs, found within a year of each other. That is unusual enough to be worth reflecting on.

(a) *What the coincidence suggests.* Not that the problem was easy — it had been open eighty years — but that induction on scales had matured simultaneously in two communities. Wooley’s efficient congruencing and the Bourgain–Demeter method are both refinements of ideas that had been circulating for a decade.

(b) *What each proof is good for.* Congruencing gives explicit constants and generalises to systems over number fields. Decoupling gives a shorter argument and generalises to any curved surface, which is why it also settled periodic Strichartz and local smoothing — problems the congruencing method does not address.

(c) *The shared skeleton, precisely.* Theorem 13.8.6 and Theorem 13.4.1 are both rigidity statements at a single scale; Theorem 13.2.3 and the congruencing descent are both iterations of such statements; and both break down when a determinant vanishes — Vandermonde in one case, transversality in the other.

(d) *What is genuinely different.* Congruencing works with exact algebraic identities and never approximates; decoupling works with δ -neighbourhoods and approximates everywhere. That is why one is effective and the other is

not, and Worked Example 13.8.5 shows the two performing the *same* algebraic computation, once exactly and once with an error term.

(e) *The lesson for a reader.* When a theorem has two proofs, the difference between them usually indicates what each method is really about. Here it indicates that the essential content is rigidity under a non-degeneracy condition, and that the arithmetic and the geometry are two ways of expressing it.

(f) *A caution against over-reading.* The correspondence in Worked Example 13.8.4 is a genuine analogy, not a theorem. No formal translation between the two proofs is known, and constructing one would itself be a result.

Drills

Drill 13.8.1. Verify Theorem 13.8.2 for $k = 2$, $s = 2$ explicitly.

Drill 13.8.2. Check that the coefficient matrix in the proof is triangular with diagonal entries p^j , hence invertible.

Drill 13.8.3. For $k = 3$, write out the substitution $x = \xi + pu$ and confirm the system transforms as claimed.

Drill 13.8.4. Using Worked Example 13.8.1, say which method would be preferable for a system with no continuous analogue, and why.

Notes

Efficient congruencing is due to Wooley, in a series of papers from 2012 onwards; the case $k = 3$ is his 2016 result. Theorem 13.9.4 records that the method is described here but not carried out.

Chapter 13.9

Applications: Tori, Local Smoothing, Distance Sets

Decoupling was proved to settle a problem in number theory (Chapter 13.6). It has since become a general tool, and this chapter gives two further applications. The second is the Falconer distance set problem, the site promised in Volume 12, Worked Example 12.8.1.

13.9.1 Strichartz estimates on tori

Theorem 13.9.1 (Periodic Strichartz). *Let u solve the Schrödinger equation $i\partial_t u = \Delta u$ on $\mathbb{T}^d \times [0, 1]$ with $\widehat{u(0)}$ supported in $\{|n| \leq N\}$. Then for $p \geq \frac{2(d+2)}{d}$ and every $\varepsilon > 0$,*

$$\|u\|_{L^p(\mathbb{T}^d \times [0,1])} \lesssim_\varepsilon N^{\frac{d}{2} - \frac{d+2}{p} + \varepsilon} \|u(0)\|_{L^2}.$$

Proof. The solution is $u(x, t) = \sum_{|n| \leq N} a_n e(n \cdot x + |n|^2 t)$, whose frequencies $(n, |n|^2)$ lie on the paraboloid in $\mathbb{R}^{d+1}$. Rescaling by N^{-1} in x and N^{-2} in t places them N^{-1} -separated on $\mathbb{P} \subseteq \mathbb{R}^{d+1}$, at neighbourhood width $\delta = N^{-2}$.

Each frequency occupies one cap, so the square function on the right of the decoupling inequality has one term per n .

The critical exponent, and above it. Theorem 13.5.2 holds for $2 \leq p \leq \frac{2((d+1)+1)}{(d+1)-1} = \frac{2(d+2)}{d} = p_c$, so it applies at p_c and *not* beyond. For $p > p_c$ the stated estimate follows by interpolating the case $p = p_c$ against the trivial bound $\|u\|_{L^\infty} \leq \sum_{|n| \leq N} |a_n| \lesssim N^{d/2} \|u(0)\|_{L^2}$, and one checks the exponent $\frac{d}{2} - \frac{d+2}{p}$ is the linear interpolant. So it suffices to treat $p = p_c$, which we now do. Applying Theorem 13.5.2 at p_c gives

$$\|u\|_{L^p} \lesssim_\varepsilon N^\varepsilon \left(\sum_{|n| \leq N} \|u_n\|_{L^p}^2 \right)^{1/2},$$

and evaluating the single-frequency norms and undoing the rescaling gives the stated power of N . $\square$

Where we are. The exponent $\frac{2(d+2)}{d}$ is the critical decoupling exponent for the paraboloid in $\mathbb{R}^{d+1}$, which is what the space-time frequencies live on. So Strichartz on the torus and decoupling are the same statement in different notation — and the N^ε is unavoidable on a torus, where the frequencies are discrete and can conspire.

Bourgain had conjectured this estimate in 1993 and proved it for $d = 1$; the general case was open until Theorem 13.5.2.

13.9.2 The chain to Wolff's theorem

Volume 12 identified the Falconer problem as a weighted restriction question and proved Erdős's bound for $d \geq 3$ using bilinear restriction. In the plane the record is Wolff's $\frac{4}{3}$, and it comes by an entirely different route: through decoupling for the *cone*. The route is a chain of four implications, and we verify the arithmetic before proving any link.

Proposition 13.9.2 (The chain closes at $\frac{4}{3}$). *Suppose that for every α -dimensional μ in $\mathbb{R}^2$ with $\alpha \in [1, 2]$,*

$$\sigma(\mu, r) \lesssim_\varepsilon r^{-\frac{\alpha}{2} + \varepsilon}. \quad (13.9.1)$$

Then $m(\Delta(E)) > 0$ whenever $\dim_{\mathbb{H}} E > \frac{4}{3}$.

Proof. Step 0. Reduce to the admissible range.

Volume 12, Theorem 12.8.3 is stated for $\alpha \in [\frac{d}{2}, \frac{d+1}{2}]$, which at $d = 2$ is $\alpha \in [1, \frac{3}{2}]$. Since $\dim_{\mathbb{H}} E > \frac{4}{3}$, choose α with $\frac{4}{3} < \alpha \leq \frac{3}{2}$ and $\dim_{\mathbb{H}} E > \alpha$; this is possible because $\frac{4}{3} < \frac{3}{2}$. It suffices to prove the conclusion for that α , and a larger Hausdorff dimension only makes the hypothesis easier. *Without this step the argument would apply the cited proposition outside its stated range, since $\dim_{\mathbb{H}} E$ may exceed $\frac{3}{2}$.*

Step 1. *Apply the criterion.*

By Theorem 12.8.3, it suffices to exhibit $q_0 \in [1, 2]$ with $\alpha(1 + q_0^{-1}) \geq 2$ for which the weighted L^2 estimate (12.8.2) holds for all $q > q_0$.

In the normalisation of that proposition, (13.9.1) corresponds to $\|\widehat{\mu}(r \cdot)\|_{L^2(\mathbb{S}^1)} \lesssim r^{-\beta}$ with $\beta = \frac{\alpha}{4}$, and $\beta = \frac{\alpha}{2q_0}$ gives $q_0 = 2$. Substituting,

$$\alpha \left(1 + \frac{1}{q_0}\right) = \frac{3\alpha}{2} \geq 2 \iff \alpha \geq \frac{4}{3}. \quad \square$$

Where we are. The computation is exact: $\frac{3\alpha}{2} - 2 = 0$ solves to $\alpha = \frac{4}{3}$ symbolically, so the threshold is a fixed point of the criterion and not a rounded numerical value.

Tabulating $\alpha(1 + q_0^{-1}) = \frac{3\alpha}{2}$ against the requirement ≥ 2 : at $\alpha = 1$ it is $\frac{3}{2}$ (fails); at $\alpha = \frac{4}{3}$ it is 2 (the threshold); at $\frac{3}{2}$ it is $\frac{9}{4}$; at 2 it is 3.

Cross-check. Erdögan's threshold $\frac{d}{2} + \frac{1}{3}$ equals $\frac{4}{3}$ at $d = 2$ — the same number by a different route. His theorem is stated for $d \geq 3$ and proved by bilinear restriction; Wolff's planar result predates it and uses local smoothing. That two independent methods land on the same exponent in the plane is evidence that $\frac{4}{3}$ is the natural barrier for Mattila's approach, and Erdögan remarks exactly this.

Remark 13.9.3 (Wolff's historical implication chain; context only). Wolff's theorem gives (13.9.1). Its published proof proceeds by:

- (i) *decoupling for the cone* in $\mathbb{R}^3$, at $p_c = \frac{2n}{n-2} = 6$;
- (ii) which implies the *local smoothing* estimate for the wave equation, $\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^2 \times [1,2])} \lesssim_\varepsilon \|f\|_{W^{s,p}}$ for $s > s(p)$;
- (iii) which implies bounds for the *circular maximal function* and, by a duality argument, for weighted L^2 averages of $\widehat{\mu}$ over circles;
- (iv) which is (13.9.1).

Remark 13.9.4 (Scope decision). Revision R19 now separates the closed analytic links from the remaining cinematic one.

What is proved. Link (i), cone decoupling, is Theorem 13.5.6; link (ii), including the microlocal summation, is Theorem 13.9.8. Also proved are Theorem 13.9.2, the exchange arithmetic converting (13.9.1) into the threshold $\frac{4}{3}$, and, in Volume 12, Theorem 12.8.2, which identifies $\sigma(\mu, r)$ as a restriction quantity.

What closes the chain. Link (iii), the fractal-radii circular maximal estimate and its dual weighted consequence, is now proved in Volume 14, Chapters 14.6 and 14.7; link (iv) follows by the duality calculation recorded there.

Strict status. The displayed Wolff chain remains frontier context in this chapter, but every one of its named inputs is now internally available: the cone and local-smoothing links here, and the research-length cinematic link in Volume 14. The former shared Volume 14 dependency is discharged.

Why the arithmetic was checked first. Because a chain of four implications can be individually correct and jointly vacuous if the exponents do not compose. Theorem 13.9.2 verifies the composition before any link is attempted, which is the discipline this series has followed at every frontier site.

WORKED EXAMPLE 13.9.1— TIER A **The scoreboard, completed**

Volume 12, Worked Example 12.8.1 tabulated the Falconer records with their machinery. With this chapter the middle rows can be filled in.

Threshold	Author	Engine	Status here
$\frac{n+1}{2}$	Falconer 1985	decay of $\widehat{d\sigma}$	<i>proved</i> , Vol 8
$\frac{n}{2} + \frac{1}{3}$	Erdogan 2005	bilinear restriction	<i>proved</i> , Vol 12
$\frac{4}{3}$ ($n = 2$)	Wolff 1999	local smoothing	exchange proved, Theorem 13.9.4
$\frac{5}{4}$ ($n = 2$)	GLOW 2020	partitioning + decoupling	Vol 14
$\frac{n}{2}$	<i>conjecture</i>	—	open

What changed between rows. Nothing in the formulation: Mattila’s reduction has been the framework since 1987. What changed is the supply of restriction-type estimates, exactly as Worked Example 12.8.1 said.

The two planar entries. Wolff’s $\frac{4}{3}$ and Erdogan’s $\frac{n}{2} + \frac{1}{3}$ agree at $n = 2$. Erdogan notes that $\frac{4}{3}$ is the best exponent obtainable from Mattila’s method in the plane, and $\frac{5}{3}$ in $\mathbb{R}^3$; so the row after it must escape the framework, and GLOW’s $\frac{5}{4}$ does — it works with pinned distance sets and does not go through $\sigma(\mu, r)$ at all.

What this series proves outright. Rows one and two, in full, with every input proved. For row three, the exchange arithmetic and links (i)–(ii) are proved; the cinematic link (iii) remains. Row four is reported.

WORKED EXAMPLE 13.9.2— TIER B **Why the cone, and not the sphere**

Link (i) of Theorem 13.9.3 is decoupling for the *cone*, and the reason is worth isolating.

(a) *Local smoothing is about the wave equation.* The solution operator $e^{it\sqrt{-\Delta}}$ has space-time frequencies on the light cone $\{|\xi| = |\tau|\}$, not on a paraboloid. So the surface is dictated by the equation.

(b) *The cone has a vanishing principal curvature.* Volume 5, Worked Example 5.5.4 records that this changes the decay exponent, and Worked Example 13.5.2(b) that it changes the decoupling exponent to $\frac{2n}{n-2}$, which is 6 in $\mathbb{R}^3$.

(c) *The rescaling is Lorentz.* Parabolic rescaling (Theorem 13.2.1) exploits $s \mapsto s^2$; the cone’s analogue uses the Lorentz group, which preserves it. The induction is structurally the same and the group is different.

(d) *Why the sphere would not do.* Circular averages of $\widehat{\mu}$ are governed by the circular maximal function, which is a statement about the *wave* flow in the plane — and that flow's characteristic surface is the cone in $\mathbb{R}^3$. Using the sphere would answer a different question.

The pattern. Match the surface to the operator, not to the geometry of the problem statement. Volume 12, Worked Example 12.8.2 made the same point about the number of factors; this is the same principle applied to the choice of hypersurface.

WORKED EXAMPLE 13.9.3— TIER C: RESEARCH TASTE **What this series proves about Falconer, and what it reports**

The distance set problem has now been treated across four volumes. A ledger. *Proved outright.* Falconer's $\frac{n+1}{2}$ (Volume 8, Theorem 8.5.1); Mattila's reduction (Volume 8, Theorem 8.7.2); the identification of $\sigma(\mu, r)$ as a restriction norm (Volume 12, Theorem 12.8.2); Tao's bilinear theorem (Volume 12, Theorem 12.6.3); and Erdős's $\frac{n}{2} + \frac{1}{3}$ for $n \geq 3$ (Volume 12, Theorem 12.8.8).

Partially closed. For Wolff's $\frac{4}{3}$, the exchange computation is Theorem 13.9.2, cone decoupling is Theorem 13.5.6, and the local-smoothing implication is Theorem 13.9.8. The remaining circular-maximal link is isolated in Theorem 13.10.7.

Reported. GIOW's $\frac{5}{4}$, in Volume 14.

The one correction worth remembering. Volume 12 originally stated Mattila's criterion as " $\sigma(\mu, r) \lesssim r^{-\beta}$ with $\beta > \frac{n}{2}$ ". That is sufficient but strictly stronger than the truth, and it *cannot* produce Erdős's threshold: at $\alpha = \frac{n}{2} + \frac{1}{3}$ the induced β is below $\frac{n}{2}$ in every dimension. The correct criterion, $\alpha(1+q_0^{-1}) \geq n$, uses the estimate across a range of exponents rather than at one. Compressing a family of estimates to a single decay rate is the commonest way to lose a theorem in this area, and it happened here.

What remains. A single inequality: $\sigma(\mu, r) \lesssim_{\varepsilon} r^{-n/2+\varepsilon}$ for $\alpha > \frac{n}{2}$, which by Theorem 12.8.3 would give the conjecture. Volume 12, Worked Example 12.8.4 states it precisely and explains why the trivial bound stops halfway — the decay of $\widehat{d\sigma}$ is sharp, so no argument using only its size can improve.

The honest summary after four volumes. The problem is a well-posed question about one quantity, the mechanism is completely understood, every improvement since 1985 has come from importing a better restriction estimate, and the estimate that would finish it is not available.

13.9.3 What decoupling did not solve

It is as useful to know a tool's limits as its reach.

Proposition 13.9.5 (Decoupling does not give the restriction conjecture). *Theorem 13.5.2 implies the Stein–Tomas estimate at $p = \frac{2(n+1)}{n-1}$ and no better. In particular it does not imply the restriction conjecture for any $n \geq 2$.*

Proof. Worked Example 13.4.1 gives the implication. For the converse direction, the data norm on the right of a decoupling inequality is an ℓ^2 sum of L^p norms of caps, and by Theorem 13.5.3 the inequality is *sharp* at $p = \frac{2(n+1)}{n-1}$. So no argument passing through decoupling at that exponent can produce an estimate valid for smaller p , whereas the restriction conjecture asserts one for all $p > \frac{2n}{n-1}$. $\square$

Where we are. The obstruction is the one identified in Volume 12, Worked Example 12.4.1(c) and again in Worked Example 13.5.3: decoupling is an ℓ^2 statement over caps, so it cannot exploit L^∞ data. Converting between the two costs $\delta^{(n-1)/4}$ per cap, and that conversion is exactly the gap between Stein–Tomas and the conjecture.

So decoupling settled Vinogradov, periodic Strichartz and local smoothing — all L^2 -type questions — and left restriction and Kakeya open. That is not a coincidence but a structural fact about which problems have L^2 data.

WORKED EXAMPLE 13.9.4— TIER A **A ledger for the decade**

Decoupling was proved in 2015. What has it settled?

Problem	Status	Data norm
Vinogradov mean value	<i>settled</i> , Chapter 13.6	L^2
Periodic Strichartz	<i>settled</i> , Theorem 13.9.1	L^2
Local smoothing	<i>settled</i> (GWZ 2020)	L^p
Falconer, planar	improved to $\frac{5}{4}$	—
Restriction	open	L^∞
Kakeya, $n \geq 4$	open	—
Bochner–Riesz	open	L^p

The pattern. Everything settled has L^2 data or reduces to it; everything open has L^∞ data or is purely geometric. Worked Example 13.5.3 proposed this as the sharpest heuristic in the series, and this table is the evidence.

The two exceptions to check. Local smoothing has L^p data and was settled — but by Guth–Wang–Zhang, using polynomial partitioning as well as decoupling. And Falconer improved without being settled, by GIOW, again using

partitioning. *So the problems that fell to decoupling alone are exactly the L^2 ones, and the partial progress elsewhere came from combining it with Volume 11's geometry.*

What that suggests. The next advances are likely to come from combinations rather than from decoupling alone, which is what Volume 14 reports.

13.9.4 Local smoothing

Link (ii) of Theorem 13.9.3 deserves a statement, since it is the application of decoupling with the widest reach.

Definition 13.9.6 (The local smoothing problem). For the wave equation in $\mathbb{R}^n$, the solution operator is $e^{it\sqrt{-\Delta}}$. Fixed-time estimates lose $s_p = (n-1)|\frac{1}{2} - \frac{1}{p}|$ derivatives, and this is sharp. The *local smoothing conjecture* asserts that averaging in t recovers some of the loss:

$$\|e^{it\sqrt{-\Delta}}f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_s \|f\|_{W^{s,p}} \quad \text{for } s > s_p - \frac{1}{p}, \quad p \geq \frac{2n}{n-1}.$$

Lemma 13.9.7 (Microlocal wave-piece estimate). *Let $\lambda \geq 2$, let $\hat{f}$ be supported where $\lambda/2 \leq |\xi| \leq 2\lambda$, and let $\{P_\theta\}$ be a smooth, boundedly overlapping angular decomposition into caps of aperture $\lambda^{-1/2}$. If χ is a fixed Schwartz function with compact Fourier support and*

$$U_\theta(x, t) = \chi(t)e^{it\sqrt{-\Delta}}P_\theta f(x),$$

then, for $2 \leq p < \infty$,

$$\left(\sum_\theta \|U_\theta\|_{L^p(\mathbb{R}^n \times \mathbb{R})}^p \right)^{1/p} \lesssim_{n,p,\chi} \|f\|_{L^p(\mathbb{R}^n)}. \quad (13.9.2)$$

The analogous endpoint is $\sup_\theta \|U_\theta\|_\infty \lesssim \|f\|_\infty$.

Proof. We prove two endpoint bounds for the linear map $Tf = (U_\theta)_\theta$ and interpolate.

At $p = 2$, unitarity of the half-wave group, Plancherel, and bounded overlap give

$$\sum_\theta \|U_\theta\|_2^2 = \int_{\mathbb{R}} |\chi(t)|^2 \sum_\theta \|P_\theta f\|_2^2 dt \lesssim_\chi \|f\|_2^2.$$

For the other endpoint, write a_θ for the multiplier of P_θ , including a smooth cutoff to the annulus. The convolution kernel at time t is

$$k_{\theta,t}(z) = \int_{\mathbb{R}^n} e^{2\pi i(z \cdot \xi + t|\xi|)} a_\theta(\xi) d\xi.$$

We claim

$$\|k_{\theta,t}\|_1 \lesssim_A (1+|t|)^A \quad (13.9.3)$$

for a fixed $A = A(n)$, uniformly in λ and θ . To check it, rotate the centre of θ to e_1 , factor $e^{2\pi i t \xi_1}$ (which only translates the kernel), and put $\xi = \lambda \zeta$. On the resulting support $|\zeta'| \lesssim \lambda^{-1/2}$ and $|\zeta| \asymp 1$. For

$$q(\zeta) = |\zeta| - \zeta_1, \quad D_1 = \partial_{\zeta_1}, \quad D_j = \lambda^{-1/2} \partial_{\zeta_j} \quad (j \geq 2),$$

one has

$$|D^\alpha q| \lesssim_\alpha \lambda^{-1} \quad (|\alpha| \geq 1), \quad |q| \lesssim \lambda^{-1}.$$

The adapted cutoff satisfies the same derivative bounds without the factor λ^{-1} . Hence every adapted derivative of $a_\theta(\lambda \zeta) e^{2\pi i \lambda t q(\zeta)}$ is bounded by $C_\alpha (1+|t|)^{|\alpha|}$. Repeated integration by parts with $1 - \partial_{\zeta_1}^2 - \lambda^{-1} \Delta_{\zeta'}$ gives, for any large B ,

$$\left| \int e^{2\pi i w \cdot \zeta} a_\theta(\lambda \zeta) e^{2\pi i \lambda t q(\zeta)} d\zeta \right| \lesssim_B \lambda^{-(n-1)/2} (1+|t|)^B (1+|w_1| + \lambda^{-1/2} |w'|)^{-B}.$$

Its L_w^1 norm is $O_B((1+|t|)^B)$; dilation back to z preserves the L^1 norm. This proves (13.9.3).

Choose B in that estimate and use the rapid decay of χ . Young's inequality yields

$$\sup_{\theta, x, t} |U_\theta(x, t)| \leq \sup_t |\chi(t)| \|k_{\theta,t}\|_1 \|f\|_\infty \lesssim_\chi \|f\|_\infty.$$

Thus $T : L^2 \rightarrow \ell^2(L^2)$ and $T : L^\infty \rightarrow \ell^\infty(L^\infty)$ are bounded uniformly in λ . Complex interpolation gives $T : L^p \rightarrow \ell^p(L^p)$ for every $2 \leq p < \infty$, which is (13.9.2). $\square$

Proposition 13.9.8 (Decoupling gives local smoothing). *Assume $n \geq 2$. Theorem 13.5.6, used in ambient dimension $n+1$, implies the local smoothing estimate*

$$\|e^{it\sqrt{-\Delta}} f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_{n,p,s} \|f\|_{W^{s,p}} \quad \text{for } s > s_p - \frac{1}{p},$$

for all $\frac{2(n+1)}{n-1} \leq p < \infty$.

Proof. Step 1. Reduce to a frequency annulus.

By a Littlewood–Paley decomposition (Volume 3, Theorem 3.6.3) it suffices to treat f with $\hat{f}$ supported in $\{|\xi| \approx \lambda\}$ and prove

$$\|e^{it\sqrt{-\Delta}} f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_\varepsilon \lambda^{s_p - \frac{1}{p} + \varepsilon} \|f\|_{L^p}, \quad (13.9.4)$$

since summing over dyadic λ costs only the ε .

Step 2. *The space-time frequency support is a cone.*

Choose a fixed Schwartz function χ whose Fourier transform is compactly supported and such that $|\chi| \geq 1$ on $[1, 2]$. (Start with the inverse Fourier transform of a non-negative bump, dilate it so it stays positive on $[-1/2, 1/2]$, and translate.) Write

$$u(x, t) = e^{it\sqrt{-\Delta}} f(x) = \int e^{2\pi i(x \cdot \xi + t|\xi|)} \widehat{f}(\xi) d\xi, \quad U(x, t) = \chi(t)u(x, t).$$

Then $\|u\|_{L^p(\mathbb{R}^n \times [1, 2])} \leq \|U\|_{L^p(\mathbb{R}^{n+1})}$, while $\widehat{U}$ is supported where

$$|\xi| \asymp \lambda, \quad |\tau - |\xi|| \lesssim 1.$$

After dilation by λ^{-1} this is the $O(\lambda^{-1})$ -neighbourhood of the truncated cone in *ambient dimension* $n + 1$.

Step 3. *Apply the full cone exponent.*

The angular aperture is $\lambda^{-1/2}$. Since the unit sphere in the n spatial frequency variables has dimension $n - 1$, there are

$$K \asymp \lambda^{(n-1)/2}$$

plates. Write $U = \sum_{\theta} U_{\theta}$ using the same smooth angular partition as in Theorem 13.9.7. The second line of Theorem 13.5.6, with its dimension parameter replaced by $n + 1$, gives for $p \geq 2(n + 1)/(n - 1)$

$$\|U\|_p \lesssim_{\varepsilon} \lambda^{\frac{n-1}{4} - \frac{n+1}{2p} + \varepsilon} \left(\sum_{\theta} \|U_{\theta}\|_p^2 \right)^{1/2}. \quad (13.9.5)$$

Step 4. *Pass from ℓ^2 to the microlocal ℓ^p norm.*

Hölder on the K plates and Theorem 13.9.7 give

$$\begin{aligned} \left(\sum_{\theta} \|U_{\theta}\|_p^2 \right)^{1/2} &\leq K^{\frac{1}{2} - \frac{1}{p}} \left(\sum_{\theta} \|U_{\theta}\|_p^p \right)^{1/p} \\ &\lesssim \lambda^{\frac{n-1}{2}(\frac{1}{2} - \frac{1}{p})} \|f\|_p. \end{aligned}$$

Substitution in (13.9.5) yields

$$\begin{aligned} \|U\|_p &\lesssim_{\varepsilon} \lambda^{\frac{n-1}{4} - \frac{n+1}{2p} + \frac{n-1}{2}(\frac{1}{2} - \frac{1}{p}) + \varepsilon} \|f\|_p \\ &= \lambda^{\frac{n-1}{2} - \frac{n}{p} + \varepsilon} \|f\|_p = \lambda^{s_p - \frac{1}{p} + \varepsilon} \|f\|_p. \end{aligned}$$

This is (13.9.4). Given $s > s_p - 1/p$, choose the displayed epsilon smaller than $s - (s_p - 1/p)$ and sum the dyadic annuli; the low-frequency piece is bounded by Young's inequality. The proposition follows. $\square$

Where we are. The gain is the exact sum of two powers. In ambient dimension $n + 1$, cone decoupling contributes $\lambda^{(n-1)/4-(n+1)/(2p)}$. Passing from ℓ^2 to the microlocal ℓ^p norm over $K \asymp \lambda^{(n-1)/2}$ plates contributes $\lambda^{(n-1)(1/2-1/p)/2}$. Their product is

$$\lambda^{(n-1)/2-n/p} = \lambda^{s_p-1/p},$$

which is exactly the one-derivative-over- p gain over the fixed-time loss. The flat ruling is still decisive: it is why cone pieces are plates and why the supercritical exponent in Theorem 13.5.6 has precisely the power used above.

So local smoothing is a statement about the cone's vanishing principal curvature, and Volume 5, Worked Example 5.5.4 recorded that the same vanishing curvature is what changes the decay exponent of $\widehat{d\sigma}$. The two facts have the same source.

Where we are. The conjecture is due to Sogge (1991) and, in $\mathbb{R}^2$, was settled by Guth, Wang and Zhang in 2020 — *not* by decoupling alone, but by combining it with polynomial partitioning. Theorem 13.9.8 gives the estimate only for $p \geq \frac{2(n+1)}{n-1}$, whereas the conjecture asks for all $p \geq \frac{2n}{n-1}$. That gap — between $\frac{2(n+1)}{n-1}$ and $\frac{2n}{n-1}$ — is the same one that separates Stein–Tomas from the restriction conjecture (Volume 12, Worked Example 12.5.1), and for the same reason: decoupling is sharp at the Stein–Tomas exponent and cannot reach below it. Theorem 13.9.5 records the general form of the obstruction.

Why $\mathbb{R}^2$ fell and higher dimensions did not. Guth–Wang–Zhang use a three-dimensional cone in space-time and the polynomial method there; above $n = 2$ the corresponding geometry is not understood. Volume 14 reports the position.

WORKED EXAMPLE 13.9.5— TIER A **Why averaging in time helps**

The local smoothing phenomenon is easy to state and easy to disbelieve. Here is why it is true.

The fixed-time loss is real. At a single time t , the solution can focus: taking $\widehat{f}$ supported in an annulus of radius R with all phases aligned makes $e^{it\sqrt{-\Delta}}f$ concentrate on a sphere of radius t and thickness R^{-1} , and the resulting L^p norm attains the loss s_p .

But focusing cannot persist. The concentration set is a sphere at time t and a *different* sphere at time t' . Averaging over $t \in [1, 2]$ therefore spreads the mass

over an annulus of unit thickness, and the gain is the ratio of thicknesses.

The Kakeya connection. The union of spheres of radii in $[1, 2]$ centred at points of a set is exactly the object Volume 9 studied for the circular maximal function, and the sharp local smoothing exponent is equivalent to a Kakeya-type statement about how much such unions can overlap. *This is why local smoothing sits between decoupling and the polynomial method:* it needs both the frequency-side orthogonality and the physical-side geometry.

The numerology, $n = 2$. $s_p = \frac{1}{2} - \frac{1}{p}$ and the conjectured gain is $\frac{1}{p}$, so the conjecture asks for $s > \frac{1}{2} - \frac{2}{p}$. At $p = 4$ this is $s > 0$: no derivative loss at all, which is the striking case and the one Guth–Wang–Zhang settle.

WORKED EXAMPLE 13.9.6— TIER C: RESEARCH TASTE **What a decade of decoupling changed**

Volume 13 has covered a technique introduced in 2015. Standing back.

(a) *Problems settled.* Vinogradov’s mean value theorem, periodic Strichartz, and — in combination with the polynomial method — local smoothing in the plane. Worked Example 13.9.4 tabulates them, and the common feature is L^2 data.

(b) *Problems moved but not settled.* The planar Falconer distance set threshold, from $\frac{4}{3}$ to $\frac{5}{4}$; the restriction exponent in $\mathbb{R}^3$, by a fraction. In both cases decoupling contributed alongside Volume 11’s geometry.

(c) *Problems untouched.* Kakeya above dimension three, restriction in any dimension, Bochner–Riesz. Theorem 13.9.5 explains why: these have L^∞ data or none at all.

(d) *The methodological change.* Before 2015, induction on scales was one technique among several. It is now the default, and a reader encountering a new paper in this area should expect it. That is why Worked Example 13.3.5 is worth internalising.

(e) *What the technique cost.* Every estimate carries $\delta^{-\varepsilon}$ with no effective constant, and Worked Example 13.2.5 shows how weak that is at any computable scale. The Guth–Maldague–Wang improvement to $(\log \delta^{-1})^c$ for the parabola is the beginning of a repair, and it is not known for the cone or the moment curve.

(f) *The honest summary.* Decoupling converted a class of problems from open to routine, and left the hardest problems in the subject exactly where they were. The line it did not cross — between L^2 and L^∞ data — is the same line Volume 6 identified when Stein–Tomas was proved in 1975, and no method in this series crosses it.

(g) *What to watch.* Volume 14 reports three results that combine decoupling with polynomial partitioning. If the pattern of the last decade continues, the next advances will come from further combinations rather than from either technique alone.

Drills

Drill 13.9.1. Verify $\frac{2((d+1)+1)}{(d+1)-1} = \frac{2(d+2)}{d}$ and tabulate for $d = 1, 2, 3$.

Drill 13.9.2. Carry out the computation of Theorem 13.9.2 and confirm $\frac{3\alpha}{2} \geq 2 \iff \alpha \geq \frac{4}{3}$.

Drill 13.9.3. Check that Erdős's $\frac{d}{2} + \frac{1}{3}$ equals $\frac{4}{3}$ at $d = 2$, and say why the two results are nonetheless different theorems.

Drill 13.9.4. Using Worked Example 13.5.2, compute the cone's decoupling exponent in $\mathbb{R}^3$ and $\mathbb{R}^4$.

Notes

The periodic Strichartz estimate was conjectured by Bourgain in 1993 and proved by Bourgain and Demeter. Wolff's circular means paper is from 1999; the local smoothing route is developed in Mockenhaupt–Seeger–Sogge and, in sharp form, in Guth–Wang–Zhang. Theorem 13.9.4 records what is proved here.

Chapter 13.10

Circular Means and Wolff's Threshold

Theorem 13.9.4 recorded four links as quoted. This chapter discharges what can be discharged: it states Wolff's theorem in its sharp form, proves the *sharpness*, proves the two easy ranges, and identifies precisely which single estimate remains.

13.10.1 The exponent $\sigma_p(\alpha)$

Definition 13.10.1 (Wolff's exponent). For $0 < p < \infty$ and $0 < \alpha < 2$, let $\sigma_p(\alpha)$ be the supremum of those σ for which

$$\left(\int_{-\pi}^{\pi} |\widehat{\mu}(Re^{i\theta})|^p d\theta \right)^{1/p} \leq C R^{-\sigma} I_{\alpha}(\mu)^{1/2} \quad (13.10.1)$$

holds for every positive measure μ on the unit disc of $\mathbb{R}^2$ and every $R \gg 1$, where $I_{\alpha}(\mu) = \iint |x - y|^{-\alpha} d\mu d\mu$.

Where we are. The quantity $\sigma(\mu, r)$ of Volume 8, Theorem 8.7.1 is the $p = 2$ case squared: $\sigma(\mu, r) = \|\widehat{\mu}(r \cdot)\|_{L^2(\mathbb{S}^1)}^2$, so a bound $\sigma_2(\alpha) \geq s$ says $\sigma(\mu, r) \lesssim r^{-2s}$.

Normalising by the energy rather than the mass is what makes $\sigma_p(\alpha)$ a clean function of two variables. Volume 7, Theorem 7.6.2 converts between the two: a set of Hausdorff dimension $> \alpha$ carries a measure of finite α -energy, and conversely.

Remark 13.10.2 (Wolff's sharp theorem; historical context). For $p \geq 2$ and

$0 < \alpha < 2$,

$$\sigma_p(\alpha) = \begin{cases} \frac{\alpha}{2p}, & \alpha \geq 1, \\ \frac{1}{2p}, & \frac{1}{2} \leq \alpha \leq 1, \\ \frac{\alpha}{p}, & \alpha \leq \frac{1}{2}. \end{cases} \quad (13.10.2)$$

The full lower bound in the branch $\alpha \geq 1$ is a research theorem and is not promoted to an internally proved result here. The easy lower bounds and all sharp upper examples are proved below.

Remark 13.10.3 (Conditional $p = 2$ consequence). For $\alpha \in [1, 2)$ and every $\varepsilon > 0$, $\sigma(\mu, r) \lesssim_\varepsilon r^{-\alpha/2+\varepsilon}$ — which is exactly (13.9.1).

Frontier proof architecture (context only; not an internal proof). Take $p = 2$ in (13.10.2): $\sigma_2(\alpha) = \alpha/4$, so $\|\widehat{\mu}(r \cdot)\|_{L^2} \lesssim r^{-\alpha/4+\varepsilon}$ and squaring gives the claim. Numerically, $\sigma(\mu, r) \lesssim r^{-1/2}, r^{-2/3}, r^{-3/4}, r^{-1}$ at $\alpha = 1, \frac{4}{3}, \frac{3}{2}, 2$.

Where we are. So Theorem 13.9.2 is Wolff's theorem at $p = 2$, $\alpha \geq 1$, and the threshold $\frac{4}{3}$ comes from $q_0 = 2$ in Mattila's criterion. *The two computations agree*, which is the consistency check the plan required before any link was attempted.

13.10.2 What is proved here

Proposition 13.10.4 (The two easy ranges). *For $p \geq 2$:*

(i) $\sigma_p(\alpha) \geq \frac{\alpha}{p}$ for $\alpha \leq \frac{1}{2}$;

(ii) $\sigma_p(\alpha) \geq \frac{1}{2p}$ for $\frac{1}{2} \leq \alpha \leq 1$.

Proof. Write $M = \mu(\mathbb{R}^2)$ and $I = I_\alpha(\mu)$. Since the support has diameter at most 2,

$$M^2 \leq 2^\alpha I, \quad (13.10.3)$$

and hence $\|\widehat{\mu}\|_{L^\infty} \lesssim_\alpha I^{1/2}$. Notice that this is only a mass bound; finite energy does *not* imply pointwise Fourier decay.

We next prove the required L^2 estimate directly from the energy. If $F(s) = (\mu \times \mu)\{(x, y) : |x - y| \leq s\}$, then

$$F(s) \leq s^\alpha I. \quad (13.10.4)$$

The Fourier transform of normalised arclength on $\mathbb{S}^1$ satisfies $|\widehat{d\sigma}(z)| \lesssim (1 + |z|)^{-1/2}$. Expanding the square and then using (13.10.4) on dyadic distance

shells therefore gives

$$\begin{aligned}
 \int_{\mathbb{S}^1} |\widehat{\mu}(R\omega)|^2 d\sigma(\omega) &\leq C \iint (1 + R|x - y|)^{-1/2} d\mu(x) d\mu(y) \\
 &\lesssim I \left(R^{-\alpha} + \sum_{1 \leq 2^j \lesssim R} 2^{-j/2} (2^j/R)^\alpha \right) \\
 &\lesssim_\varepsilon I R^{-\min\{\alpha, 1/2\} + \varepsilon}.
 \end{aligned} \tag{13.10.5}$$

For $\alpha < 1/2$ the sum is geometric from its first term; for $\alpha > 1/2$ it is controlled by its last term; and at $\alpha = 1/2$ its $\log R$ loss is absorbed by R^ε .

Finally, for $p \geq 2$ interpolate the angular L^2 bound (13.10.5) with (13.10.3):

$$\|\widehat{\mu}(R\cdot)\|_{L^p(\mathbb{S}^1)} \leq \|\widehat{\mu}(R\cdot)\|_{L^2}^{2/p} \|\widehat{\mu}(R\cdot)\|_{L^\infty}^{1-2/p} \lesssim_\varepsilon R^{-\min\{\alpha/p, 1/(2p)\} + \varepsilon} I^{1/2}.$$

Because $\sigma_p(\alpha)$ is a supremum, the arbitrary ε proves (i) and (ii), including both joining endpoints. $\square$

Lemma 13.10.5 (A finite comb). *Let $0 < \gamma < 1$ and $Q \geq 2$. There are a positive probability measure ν_Q supported in a fixed interval and a set $E_Q \subset [-cQ, cQ]$ such that*

$$I_\gamma(\nu_Q) \lesssim_\gamma 1, \quad |E_Q| \gtrsim_\gamma Q^{1-\gamma}, \quad |\widehat{\nu}_Q(\eta)| \gtrsim 1 \quad (\eta \in E_Q). \tag{13.10.6}$$

Proof. Fix a nonnegative smooth bump φ of mass 1, supported in a very small interval about the origin, and put $q = \lceil Q^\gamma \rceil$. With the harmless constants chosen so that all supports stay in $(-1/4, 1/4)$, set

$$d\nu_Q(s) = \frac{1}{q} \sum_{j=1}^q Q\varphi\left(Q\left(s - \frac{j}{8q} + \frac{1}{16}\right)\right) ds.$$

The diagonal contribution to its energy is $O(Q^\gamma/q)$, while the off-diagonal contribution is

$$\lesssim \frac{1}{q^2} \sum_{j \neq k} \left(\frac{q}{|j - k|} \right)^\gamma \lesssim_\gamma 1.$$

Thus the first assertion of (13.10.6) holds. On the Fourier side,

$$\widehat{\nu}_Q(\eta) = e^{2\pi i \eta / 16} \widehat{\varphi}(\eta/Q) \frac{1}{q} \sum_{j=1}^q e^{-2\pi i \eta j / (8q)}.$$

Consequently its modulus is bounded below on a fixed-width interval about each multiple of $8q$ lying in $[-cQ, cQ]$. The union E_Q of those intervals has measure $\gtrsim Q/q \gtrsim Q^{1-\gamma}$, proving the other two assertions. $\square$

Proposition 13.10.6 (Sharpness: the upper bounds). *For $p \geq 2$ the values in (13.10.2) cannot be improved.*

Proof. It is enough to construct examples along a sequence $R \rightarrow \infty$. We use the Fourier convention already fixed in the series and place every support inside a sufficiently small fixed square, hence inside the unit disc.

Step 1. $0 < \alpha \leq \frac{1}{2}$: *a one-dimensional comb.*

Apply Theorem 13.10.5 with $Q = R$ and $\gamma = \alpha$, and regard ν_R as a measure on the horizontal axis. For each $\eta \in E_R \cap [-R/2, R/2]$, the equation $R \cos \theta = \eta$ has two solutions at which the derivative has size comparable to R . Fixed-width frequency intervals therefore pull back to disjoint angular intervals of length comparable to R^{-1} . Since E_R contains $\gtrsim R^{1-\alpha}$ such intervals,

$$|\{\theta : |\widehat{\nu}_R(R \cos \theta)| \gtrsim 1\}| \gtrsim R^{-\alpha}.$$

The energy is bounded by (13.10.6); hence

$$\frac{\|\widehat{\nu}_R(R \cdot)\|_{L^p(\mathbb{S}^1)}}{I_\alpha(\nu_R)^{1/2}} \gtrsim R^{-\alpha/p}.$$

Thus $\sigma_p(\alpha) \leq \alpha/p$.

Step 2. $\frac{1}{2} \leq \alpha < 1$: *a periodic line measure.*

Choose an even $\psi \in C_c^\infty((-1/4, 1/4))$, nonnegative, with $\widehat{\psi}$ positive near the origin, and set

$$d\mu_R(x, y) = \psi(x)(1 + \cos(2\pi Rx)) \, dx \, \delta_0(dy).$$

Its α -energy is bounded uniformly for every fixed $\alpha < 1$. Also

$$\widehat{\mu}_R(\xi, \eta) = \widehat{\psi}(\xi) + \frac{1}{2}\widehat{\psi}(\xi - R) + \frac{1}{2}\widehat{\psi}(\xi + R).$$

If $|\theta| \leq cR^{-1/2}$, then $R \cos \theta - R = O(c^2)$; choosing c small and then R large shows that $|\widehat{\mu}_R(R \cos \theta, R \sin \theta)| \gtrsim 1$. Therefore the angular L^p norm is $\gtrsim R^{-1/(2p)}$, proving $\sigma_p(\alpha) \leq 1/(2p)$.

At $\alpha = 1$, replace $\delta_0(dy)$ by $Rh(Ry) \, dy$, where h is a fixed nonnegative smooth probability bump. The same Fourier lower bound holds, while direct integration in the horizontal difference gives $I_1(\mu_R) \lesssim \log R$. Thus

$$\frac{\|\widehat{\mu}_R(R \cdot)\|_{L^p}}{I_1(\mu_R)^{1/2}} \gtrsim R^{-1/(2p)} (\log R)^{-1/2},$$

which rules out every exponent strictly larger than $1/(2p)$.

Step 3. $1 < \alpha < 2$: a carrier times a transverse comb.

Put $\beta = \alpha - 1$ and $Q = R^{1/2}$. Take ν_Q from Theorem 13.10.5 with exponent β , and define the positive product measure

$$d\mu_R(x, y) = \psi(x)(1 + \cos(2\pi Rx)) dx d\nu_Q(y).$$

For $v > 0$ and $1 < \alpha < 2$,

$$\iint_{(-1/4, 1/4)^2} ((x - x')^2 + v^2)^{-\alpha/2} dx dx' \lesssim_\alpha v^{1-\alpha} = v^{-\beta}.$$

The bounded horizontal density and Theorem 13.10.5 consequently give $I_\alpha(\mu_R) \lesssim I_\beta(\nu_Q) \lesssim 1$.

For $\eta \in E_Q$ set $\theta = \arcsin(\eta/R)$. Then $|\theta| \lesssim R^{-1/2}$ and $R(\cos \theta - 1) = O(\eta^2/R) = O(c^2)$, so the horizontal carrier and the transverse comb are both bounded below. Since $d\theta \asymp R^{-1} d\eta$ on this arc,

$$|\{\theta : |\widehat{\mu}_R(Re^{i\theta})| \gtrsim 1\}| \gtrsim R^{-1}|E_Q| \gtrsim R^{-1}Q^{1-\beta} = R^{-\alpha/2}.$$

It follows that the ratio in (13.10.1) is $\gtrsim R^{-\alpha/(2p)}$, and hence $\sigma_p(\alpha) \leq \alpha/(2p)$. The three steps include both joins and give every asserted upper bound. $\square$

Where we are. *The piecewise shape is forced by three finite-scale resonance patterns.* A sparse line comb gives angular mass $R^{-\alpha}$ below $\alpha = 1/2$; a periodic line density gives the curvature scale $R^{-1/2}$ on the plateau; and above 1 a transverse $(\alpha - 1)$ -comb converts frequency length $R^{(2-\alpha)/2}$ into angular mass $R^{-\alpha/2}$. Taking the L^p norm divides each of these angular exponents by p . These constructions, not an unproved optimisation between a ball and a tube, are the complete sharpness argument.

Remark 13.10.7 (R19, reduced to one estimate). *What is now proved.* Theorem 13.10.4, the ranges $\alpha \leq 1$; Theorem 13.10.6, that the stated values are best possible; Theorem 13.10.3, that Theorem 13.10.2 at $p = 2$ is exactly the hypothesis (13.9.1) of Theorem 13.9.2; and Theorem 13.9.8, link (ii) of Theorem 13.9.3, proved in Chapter 13.9.

What is contextual. The lower bound $\sigma_p(\alpha) \geq \alpha/(2p)$ for $\alpha \geq 1$ is the substance of Wolff's paper. It is obtained from a local smoothing estimate for the wave equation — link (ii), proved here — combined with a bound for the circular maximal function, which is Wolff's cinematic estimate. The required $L^{3/2}$ theorem and its lens-counting input are proved in Volume 14, Theorem 14.6.7 and Chapter 14.7, in the form for fractal sets of radii.

Strict status. The full formula remains frontier context here, while the previously shared cinematic input is now internally proved in Volume 14. Thus R19 creates no remaining strict obligation; the internally numbered results of this chapter are Theorems 13.10.4 to 13.10.6.

WORKED EXAMPLE 13.10.1— TIER A The scoreboard, with Wolff's theorem in place

Worked Example 13.9.1 left row three's engine unproved. With Theorem 13.10.2 stated sharply the picture is:

Range	$\sigma_2(\alpha)$	$\sigma(\mu, r)$
$\alpha \leq \frac{1}{2}$	$\alpha/2$	$r^{-\alpha}$
$\frac{1}{2} \leq \alpha \leq 1$	$\frac{1}{4}$	$r^{-1/2}$
$\alpha \geq 1$	$\alpha/4$	$r^{-\alpha/2}$

Where Falconer needs it. Mattila's criterion with $q_0 = 2$ requires $\frac{3\alpha}{2} \geq 2$, so $\alpha \geq \frac{4}{3}$ — inside the third range, where $\sigma(\mu, r) \lesssim r^{-\alpha/2}$.

Why $\frac{4}{3}$ is the ceiling. Erdős's observation, recorded in Worked Example 13.9.1: Theorem 13.10.2 is *sharp*, by Theorem 13.10.6, so no better input to Mattila's criterion exists in the plane. The framework cannot give more, and GIOW's $\frac{5}{4}$ escapes it (Chapter 14.13).

The check. σ_2 is continuous at the joins: at $\alpha = \frac{1}{2}$ both formulas give $\frac{1}{4}$; at $\alpha = 1$ both give $\frac{1}{4}$. ✓ So the three ranges fit together, which is the first thing to verify in a piecewise formula.

WORKED EXAMPLE 13.10.2— TIER C: RESEARCH TASTE Sharp theorems, and what they close

Theorem 13.10.2 is *sharp* — Theorem 13.10.6 exhibits finite-scale near-extremisers in every range. A sharp theorem changes what one should do next, and the change is worth naming.

(a) *It converts a bound into a boundary.* Before sharpness, $\sigma_2(\alpha) \geq \alpha/4$ invites improvement. After, the only way past $\frac{4}{3}$ is to stop using $\sigma(\mu, r)$.

(b) *It relocates the difficulty.* Erdős's remark that $\frac{4}{3}$ in the plane and $\frac{5}{3}$ in $\mathbb{R}^3$ are Mattila's ceilings is a consequence of sharpness, and it is what made GIOW's pinned approach (Chapter 14.13) obviously worth attempting.

(c) *It is usually easier than the theorem.* Theorem 13.10.6 uses three explicit resonance constructions; the matching lower bound is a research theorem. That asymmetry is typical, and it is why one should look for near-extremisers early — a sharpness example tells you what the theorem must look like before you can prove it.

(d) *Instances in this series.* Volume 13, Theorem 13.1.8 fixes the decoupling exponent at 6 before any positive result; Volume 6, Theorem 6.2.3 fixes the restriction exponent by the Knapp cap; Volume 14, Theorem 13.5.3 does the same in $\mathbb{R}^n$. In each case the example came first historically.

(e) *The caution.* Sharpness is relative to a *formulation*. Wolff's theorem is sharp for $\sigma_p(\alpha)$ as defined, and that says nothing about pinned distance sets, which are a different quantity — and where the threshold duly improved to $\frac{5}{4}$.

(f) *The question to carry.* When a sharp theorem blocks progress, ask which feature of the *definition* the near-extremiser exploits. Here it is frequency resonance: the energy controls close pairs but does not prevent a finite comb from having many large Fourier coefficients. The pinned framework changes the quantity being estimated and thereby avoids this particular global circular-mean obstruction.

WORKED EXAMPLE 13.10.3— TIER B **Reading a piecewise exponent**

(13.10.2) has three branches. Formulas of this shape are common in the subject, and four checks settle whether one has been transcribed correctly.

(1) *Continuity at the joins.* At $\alpha = \frac{1}{2}$: $\alpha/p = 1/(2p)$ ✓. At $\alpha = 1$: $1/(2p) = \alpha/(2p)$ ✓. A discontinuity would mean one branch is wrong.

(2) *Monotonicity.* σ_p is non-decreasing in α — more energy decay from a larger dimension — and non-increasing in p . Both hold.

(3) *The endpoints.* At $\alpha \rightarrow 0$, $\sigma_p \rightarrow 0$: no decay from a point mass ✓. At $\alpha \rightarrow 2$ and $p = 2$, $\sigma_2 \rightarrow \frac{1}{2}$, giving $\sigma(\mu, r) \lesssim r^{-1}$, which is the trivial bound for an absolutely continuous measure ✓.

(4) *Which branch the application uses.* Falconer needs $\alpha \geq \frac{4}{3}$, hence the third. A reader who checks only the first two branches would miss that the theorem's hardest case is the one being invoked.

The general point. A piecewise formula encodes a competition between extremal examples, one per branch, and the joins are where two examples are equally bad. Locating the examples — as Theorem 13.10.6 does — explains the shape and is the fastest way to remember it.

Drills

Drill 13.10.1. Verify that $\sigma_2(\alpha)$ in (13.10.2) is continuous at $\alpha = \frac{1}{2}$ and $\alpha = 1$.

Drill 13.10.2. Deduce Theorem 13.10.3 from Theorem 13.10.2 and confirm the four numerical values.

Drill 13.10.3. In Step 1 of Theorem 13.10.6, count the resonant intervals in E_R , pull them back under $\theta \mapsto R \cos \theta$, and recover the angular measure $R^{-\alpha}$.

Drill 13.10.4. Using Worked Example 13.10.1, explain why improving Wolff's threshold requires leaving Mattila's framework.

Notes

Theorem 13.10.2 is Wolff (1999). Tao showed shortly afterwards that no improvement is available for $p < 2$, so the table is complete. Theorem 13.10.7 records that the one remaining estimate coincides with the one recorded under R22.

Chapter 13.11

Assessment

13.11.1 What this volume proves

Result	Where	Rests on
Caps are boxes	Theorem 13.1.2	second-order Taylor
Sum-set is 2-to-1	Theorem 13.1.6	curvature,
Decoupling at $p = 4$	Theorem 13.1.7	arithmetically
Sharpness at $p = 6$	Theorem 13.1.8	Plancherel,
Parabolic rescaling	Theorem 13.2.1	Cauchy–Schwarz
Bootstrapping	Theorem 13.2.3	a constructive example
Bernstein	Theorem 13.2.2	an exact identity
Bilinear L^2	Theorem 13.3.2	log log round count
Ball inflation	Theorem 13.3.4	Young’s inequality
Iterated inflation	Theorem 13.3.5	a Jacobian,
Bourgain–Demeter, parabola	Theorem 13.4.2	Theorem 13.3.1
Bourgain–Demeter, $\mathbb{R}^n$	Theorem 13.5.2	multilinear Keakeya (Vol 11)
Small caps fail	Theorem 13.4.6	the above, squared
Vinogradov reduction	Theorem 13.6.5	all of the above
Periodic Strichartz	Theorem 13.9.1	the same, with n
Falconer exchange at $\frac{4}{3}$	Theorem 13.9.2	factors
Decoupling $\nRightarrow$ restriction	Theorem 13.9.5	the same example
		$s = p/2$
		Theorem 13.5.2
		Mattila’s criterion
		sharpness

13.11.2 What this volume still imports

The two arithmetic routes below are approved-external. The third row is frontier context and is not part of the certified internal theorem graph.

Quoted	Where	What depends on it
Efficient congruencing	Theorem 13.8.3	approved-external alternate route only
Hardy–Littlewood circle method	Theorem 13.6.8	approved-external arithmetic consequences only
Circular-maximal/cinematic link	Theorem 13.10.7	historical route to Wolff’s $\frac{4}{3}$; context only

Where we are. The former Volume 13 proof debts have now been separated cleanly. The moment-curve recursion, its lower-degree input, cone decoupling, and the decoupling-to-local-smoothing implication are internal proofs. Efficient congruencing and the circle method remain only as approved-external arithmetic routes. The circular-maximal/cinematic link in Theorem 13.10.7 is retained as frontier context rather than as a load-bearing numbered result. Hence no independent or transitive strict proof debt remains in Volume 13.

13.11.3 Three questions a reader should be able to answer

WORKED EXAMPLE 13.11.1— TIER A **Self-test on the volume**

- (1) *Why is the critical exponent 6 for the parabola?*
Because the constructive example of Theorem 13.1.8 has ratio $\delta^{-1/4+3/(2p)}$, which is bounded exactly when $p \leq 6$. Equivalently, because $6 = \frac{2(n+1)}{n-1}$ at $n = 2$, the Stein–Tomas exponent — and by Worked Example 13.4.1 the two must agree.

(2) *Why does the proof lose $\delta^{-\varepsilon}$?*
Because the induction squares the scale each round (Theorem 13.2.1 turns δ into δ/σ^2), so it terminates in $\log \log \delta^{-1}$ rounds, and a bounded cost per round accumulates to a polylogarithm — which is $O_\varepsilon(\delta^{-\varepsilon})$ but not better. Worked Example 13.2.2 shows how weak that is in practice.

(3) *Why did decoupling settle Vinogradov but not restriction?*
Because $J_{s,k}(N)$ is an L^{2s} norm of an exponential sum ((13.6.1)) — an L^2 -type quantity with one term per cap — whereas restriction asks for L^∞ data over caps. Theorem 13.9.5 makes the obstruction precise, and Worked Example 13.9.4 shows the pattern across every problem in the area.
-

WORKED EXAMPLE 13.11.2— TIER B Where to read next

For the proof in more generality. Demeter’s book *Fourier Restriction, Decoupling, and Applications* is the standard reference and covers the cone, the moment curve and small caps.

For the sharp constant. Guth, Maldague and Wang replace $\delta^{-\varepsilon}$ by $(\log \delta^{-1})^c$ for the parabola. The improvement matters wherever the ε compounds — for instance in iterated applications — and Worked Example 13.4.2(c) records it.

For the number theory. Wooley’s efficient congruencing papers give the alternative route with effective constants, as Chapter 13.8 discusses.

For what came next. Volume 14 reports local smoothing (Guth–Wang–Zhang), the planar distance set record (GIOW), and the resolution of Kakeya in $\mathbb{R}^3$ (Wang–Zahl) — all of which combine decoupling with the polynomial method of Volume 11.

WORKED EXAMPLE 13.11.3— TIER B Auditing your own bookkeeping

This volume was assembled with an automated check on word counts, example counts and unproved statements. Two failures of that check are worth recording, because both are the kind that produce silent errors rather than loud ones.

(a) *The whole-series gate was weaker than the per-volume gate.* The series-level audit omitted the tier-mix criterion that the per-volume audit applied, so two completed volumes drifted out of compliance without being flagged. The repair was to run one script everywhere. *Any check worth running is worth running in every place the quantity is checked.*

(b) *A substring test excluded a chapter.* The audit skipped appendices by testing whether the filename contained `app`. That string also occurs in `ch08-applications`, so an entire chapter was silently omitted from the word count — and the count therefore did not move when material was added to it. The symptom was a total that stayed at exactly the same value across an edit, which is the signature to watch for. The repair was to match by prefix.

The general lesson. An audit is itself code and can be wrong in the direction of leniency, which is the dangerous direction: a check that over-reports failures gets fixed immediately, while one that under-reports is invisible. Both failures above were of the second kind, and both were found by noticing an implausible number rather than by the check itself.

The habit. When a measured quantity does not change after an edit that should have changed it, suspect the measurement before the edit.

WORKED EXAMPLE 13.11.4— TIER B The volume in one page

For revision, the whole of Volume 13 compressed.

The statement. For f with Fourier support in a δ -neighbourhood of a curved surface, decomposed over caps, $\|f\|_{L^p} \lesssim_\varepsilon \delta^{-\varepsilon} (\sum_\theta \|f_\theta\|_{L^p}^2)^{1/2}$, for p up to a critical exponent.

Why it is not obvious. At $p = 2$ it is Plancherel; for $p > 2$ there is no orthogonality, and the trivial bound is worse by $\delta^{-1/4}$.

Why $p_c = \frac{2(n+1)}{n-1}$. A constructive example (Theorem 13.5.3) with all caps aligned at a point.

The easy case. $p \leq 4$ in the plane, by Plancherel and an overlap count (Theorem 13.1.7), with constant $O(1)$.

The proof. Broad–narrow split; narrow by parabolic rescaling (Theorem 13.2.1); broad by ball inflation (Theorem 13.3.4), which rests on multilinear Kakeya; assemble into a self-improving inequality and bootstrap (Theorem 13.2.3).

Why the ε . The induction squares the scale, so it runs for $\log \log \delta^{-1}$ rounds and a bounded per-round cost accumulates to a polylogarithm.

What it proves elsewhere. Vinogradov’s mean value theorem ($s = p_c/2$); periodic Strichartz; local smoothing; and, through Mattila’s criterion, distance set thresholds.

What it does not prove. Restriction, Kakeya, Bochner–Riesz — all L^∞ or purely geometric (Theorem 13.9.5).

The one sentence. Curvature forces the caps of a surface to behave as though orthogonal in L^p up to the Stein–Tomas exponent, and induction on scales is how one proves it.

Drills

Drill 13.11.1. Reconstruct the proof of Theorem 13.1.7 from memory, and identify the step that fails at $p = 6$.

Drill 13.11.2. State the three quoted results of this volume and, for each, what would be lost by declining it.

Drill 13.11.3. Answer the three questions of Worked Example 13.11.1 without consulting the text.

Drill 13.11.4. Using Worked Example 13.4.4, determine the cap length for the surface $\{(t, s, t^2 + s^3)\}$ and say whether the methods of this volume apply.

Notes

The assessment format follows Volumes 9 to 12. The reading suggestions are current as of writing; Theorem 12.5.3 of Volume 12 explains why no numerical records are quoted.

Chapter 13.A

Results Quoted from Volumes 1–12

Result	Source	Used in
Loomis–Whitney	Theorem 11.5.2	via multilinear Kakeya, everywhere
Multilinear Kakeya	Theorem 11.5.4	Theorem 13.3.4
Transverse form	Theorem 11.5.7	Theorem 13.3.4
The dual box	Theorem 12.1.2	Theorem 13.1.2, Theorem 13.3.4
Wave packets	Theorem 12.1.4	Worked Example 13.3.2
A single packet	Theorem 12.1.5	Worked Example 13.4.1
The broad–narrow dichotomy	Theorem 12.4.3	Chapter 13.5
Two factors do not suffice	Theorem 12.2.3	Theorem 13.4.1
Bilinear L^2 (circle)	Theorem 12.6.1	Theorem 13.3.2
The addition map	Theorem 12.6.2	Theorem 13.3.1
ε -management	Worked Example 12.4.3	Theorem 13.2.3
When orthogonality is free	Worked Example 12.7.7	Worked Example 13.4.2
Stein–Tomas	Theorem 6.3.1	Worked Example 13.5.1
ε -removal	Theorem 6.5.2	Worked Example 13.4.1
Decay and curvature	Worked Example 5.5.4	Worked Example 13.5.2
Losses are $\delta^{-\varepsilon}$	Theorem 9.7.3	Worked Example 13.2.2
Mattila’s criterion	Theorem 12.8.3	Theorem 13.9.3
$\sigma(\mu, r)$ is a restriction norm	Theorem 12.8.2	Theorem 13.9.3
Erdős’s threshold	Theorem 12.8.8	Worked Example 13.9.1
Frostman	Theorem 7.5.1	Worked Example 13.5.2

Debts and imports

Closure update. Efficient congruencing (Theorem 13.8.3) remains an approved-external alternate route and is not required by the internal Bourgain–Demeter route. Cone decoupling Theorem 13.5.6, the lower-degree lemma Theorem 13.7.13, and their Vinogradov and local-smoothing descendants now have internal proofs. Volume 14 supplies the complete cinematic circular-maximal proof that discharges the link recorded in Theorem 13.10.7; it is no longer an open internal dependency.

Item	Current status	What depends on it
Efficient congruencing, Theorem 13.8.3	approved-external alternate route	not needed by the intended decoupling proof
Cone decoupling, Theorem 13.5.6	internal-proved	local smoothing and Theorem 13.9.3
Lower-degree moment-curve lemma, Theorem 13.7.13	internal-proved	the internally closed moment-curve recursion and Vinogradov reduction
Circular-maximal/cinematic link, Theorem 13.10.7	internally proved in Volume 14	complete cinematic-maximal proof dossier

The exponent arithmetic and conceptual self-similarities surrounding these engines are retained in the volume; the table records which route discharges each dependency and where its proof can be reviewed.

Chapter 13.B

Notation and Index of Results

Additions to the Volume 1–12 conventions

$\mathcal{N}_\delta$	the δ -neighbourhood of the surface
θ, τ	caps at scales $\delta^{1/2}$ and ρ
$D_p(\delta)$	the decoupling constant, Theorem 13.1.3
$N = \delta^{-1/2}$	the number of caps (parabola)
$\mathcal{S}_\tau f$	the square function over $\theta \subseteq \tau$
avg_Δ	average over the ball Δ
L_τ	the parabolic rescaling map
$\mathbb{P}, \Gamma_k$	paraboloid and moment curve
$J_{s,k}(N)$	the Vinogradov mean value
$\mathcal{W}^t$	the half-wave propagator

Index of the main results

Caps are boxes	Theorem 13.1.2
The trivial bounds	Theorem 13.1.5
Sharpness at $p = 6$	Theorem 13.1.8
Parabolic rescaling	Theorem 13.2.1
Bootstrapping	Theorem 13.2.3
The addition map	Theorem 13.3.1
Bilinear L^2	Theorem 13.3.2
Ball inflation	Theorem 13.3.4
The bilinear reduction	Theorem 13.4.1
Bourgain–Demeter, parabola	Theorem 13.4.2
Bourgain–Demeter, $\mathbb{R}^n$	Theorem 13.5.2
Moment curve	Theorem 13.6.4
Vinogradov	Theorem 13.6.2
Congruence self-similarity	Theorem 13.8.2
Periodic Strichartz	Theorem 13.9.1
Local smoothing	Theorem 13.9.3
Wolff’s distance bound	Theorem 13.9.3

Plan revisions in this volume

Imported	Rev	Reason
Efficient congruencing	R19	comparative only; Vinogradov is proved by decoupling. <i>Not load-bearing</i>
Cone decoupling	R20	same induction as Theorem 13.5.2 with Lorentz rescaling

The dependency chain. Theorem 13.4.2 rests on Theorem 13.3.4, which rests on Volume 11’s Theorem 11.5.4, which rests on Theorem 11.5.2, which is Hölder. No step is imported.

Part 14

Volume 14

The Frontier

Preface to Volume 14

The proof standard

Proof and literature status checked 31 August 2026. The sticky Kakeya theorem and the other rows marked internally proved in Chapter 14.14 have complete proof dossiers in the active manuscript, subject to independent human review. The general Kakeya theorem in $\mathbb{R}^3$ is *not* presently certified by the streamlined route. That route remains conditional on two independent obligations: the cellular parent–child factoring interface in Remarks 14.3.22 and 14.3.23, and a fixed exponent gain independent of the target loss in Remark 14.3.55. As of the date above, Hua–Yang’s arXiv:2608.18870v3 is explicitly a conditional cellular repair; Guth–Wang–Zahl’s arXiv:2601.14411 remains at v1; and no separate public correction of the fixed-loss quantifier was found. The independent Wang–Zahl route, arXiv:2502.17655v1, is reproduced as an alternative proof under audit, not as a closure certificate merely because it is cited. Its remaining load-bearing obligations are WZ–A1–A4 in Section 14.4.13; WZ–A1 is the missing parentwise normalized-slab occupancy estimate in (WZ–A1).

This volume follows the same proof-audit rule as Volumes 1–13: every theorem used in the main proof chain must have its hypotheses, quantifiers, scale losses, dependencies, and proof status visible. Earlier releases of Volume 14 contained survey sketches of several recent arguments. Those sketches have been retired from the reading order and replaced by proof dossiers, including the full sticky theorem, the general-to-sticky reduction, the convex–Wolff/Assouad bridge, the cinematic maximal and lens estimates, the radial bootstrap, Bourgain’s robust projection theorem, and the epsilon-improved Furstenberg incidence theorem. The second human-review pass expanded the probabilistic, rescaling, and fixed-loss arguments in the general-to-sticky dossier. A third audit identified an independent parent–child refinement-compatibility defect in its maximal-density factorization. Together with the open uniform-quantifier defect, this leaves the Kakeya endpoint conditional;

the exact blockers are recorded in Remarks 14.3.22 and 14.3.55. A fourth, proof-closure attempt checked the newest cellular repair against the hypotheses actually printed here. It isolated the four missing simultaneous interface requirements in Remark 14.3.23 and proved in Lemma 14.3.56 that the second defect cannot be removed by merely commuting the two quantifiers. The status therefore remains conditional rather than being promoted on the strength of an external conditional theorem.

Some dossiers incorporate complete author sources under their stated open licences; their provenance notices identify the source and the editorial changes. Other dossiers are written directly in the notation of this series. In either case, a reader is not asked to infer a research-length argument from an architectural outline. The final two chapters distinguish proved results, context, and genuinely open questions.

What is here

The intended central result is the three-dimensionalakeya set theorem. Its proof dossier is split into the sticky theorem and the reduction from the general case; the latter remains conditional at the two defects cited above. The dated box at the beginning of this preface governs every later informal reference to this endpoint unless a later dated audit explicitly replaces it. The surrounding chapters close every substantial auxiliary chain used by that argument: convex–Wolff self-similarity, restricted and cinematic projection estimates, lens counting, radial projections, robust discretised projection, and epsilon-improved Furstenberg incidences.

The later chapters explain why the method is dimension-specific and develop the distance-set consequences. Historical or status material is kept separate from the proofs and is never used as a substitute for one.

What is assumed

Volumes 1–13, and in particular: Volume 9 for the hairbrush and the L^2 method; Volume 11 for polynomial partitioning and multilinearakeya, both proved there; Volume 12 for wave packets and the restriction framework; Volume 13 for decoupling. Several results below combine decoupling with the polynomial method, and neither alone suffices.

How to read the volume

Chapter 1 is orientation. The next eight chapters are the proof chain: stickyakeya, the general reduction, the Assouad bridge, cinematic maximal the-

ory, lens counting, the radial bootstrap, robust projection, and Furstenberg incidences. They are ordered by logical dependence, except that a theorem may be stated before its later proof dossier when this makes the principal argument easier to read; every such forward dependency is explicitly discharged. The remaining chapters treat higher dimensions, distance sets, the GIOW mechanism, and the final proof-and-dependency audit.

Chapter 14.1

The State of the Four Conjectures

14.1.1 The list

Volume 12, Worked Example 12.5.5 set out four conjectures and the implications among them. This chapter records where each stands. Literature and manuscript status in this chapter was checked on 31 August 2026.

Conjecture	Status	Where in this series
Takeya	proved for $n = 2$; claimed for $n = 3$ in submitted preprints, but conditional in this manuscript; open for $n \geq 4$	Chapters 14.2 and 14.3; Vol 9, 11
Restriction	settled for $n = 2$; open for $n \geq 3$	Vol 6, Vol 12
Bochner–Riesz	settled for $n = 2$; open for $n \geq 3$	Vol 5
Falconer	open in every dimension	Vol 8, 12, 13

Where we are. The first row changed in 2025 and is the reason for this volume. Wang–Zahl announced a proof in arXiv:2502.17655v1, followed by the streamlined Guth–Wang–Zahl arXiv:2601.14411v1. Both are listed as submitted rather than journal-published as of 31 August 2026. The active streamlined reconstruction in this volume remains conditional at the cellular and fixed-loss audit gates; the claimed theorem and this manuscript’s proof certificate must not be conflated.

Two things worth stating immediately. *Takeya is the weakest of the first three* — Volume 6, Theorem 6.6.2 shows restriction implies it — so even a proof in $\mathbb{R}^3$ does not settle restriction there. And the implication runs one way only: no reduction is known from Takeya back to restriction, so the new theorem does not by itself move the other rows.

What the claimed theorem provides is a method. The internally complete sticky proof begins in Chapter 14.2; the conditional streamlined general reduction is Chapter 14.3; the chapters after them describe an argument that handles the configurations every previous method had to exclude, and that is the development to watch.

14.1.2 **Takeya, by dimension**

<i>n</i>	best known	by	status
2	2	Davies 1971	<i>sharp</i>
3	3	Wang–Zahl 2025 preprint	<i>claimed sharp; Volume 14 audit open</i>
4	3.059	Katz–Zahl, planebrush	open
<i>n</i> large	$(2 - \sqrt{2})n - O(1)$	Katz–Tao, arithmetic	open

Where we are. Two cautions about reading this table. The $\mathbb{R}^4$ figure is 3.059 and the $\mathbb{R}^3$ figure was $\frac{5}{2} + \varepsilon_0$. Both are due to Katz and Zahl and both concern Besicovitch sets, and they are easily conflated — a bound of 3.059 in $\mathbb{R}^3$ would exceed the ambient dimension and is impossible. Volume 11, Section 11.8.3 records the distinction. The high-dimensional bound is a fraction of n , not n minus a constant. Katz and Tao’s arithmetic method gives $cn - O(1)$ with $c = 2 - \sqrt{2} \approx 0.5858$ — Volume 10, Theorem 10.6.1 — so the *proportion* of the dimension recovered does not tend to 1. Volume 10, Worked Example 10.6.2 diagnosed why: the method is built from sumset inequalities that lose a constant factor per step. So the claimed literature situation after 2025 is: sharp in $n \leq 3$, a little above 3 in $n = 4$, and a bounded fraction in general. The gap did not narrow in high dimensions, and Chapter 14.11 explains what obstructs the new argument there.

14.1.3 **What changed, and what did not**

WORKED EXAMPLE 14.1.1— TIER A **The four conjectures after Wang–Zahl**

(a) *Takeya in $\mathbb{R}^3$.* Claimed in the Wang–Zahl and Guth–Wang–Zahl submitted preprints. Both the Minkowski and Hausdorff versions are asserted. This volume’s streamlined proof certificate remains conditional at the two audit defects stated in its preface.

(b) *Restriction in $\mathbb{R}^3$.* Not settled, and not obviously closer. The implication runs from restriction to Takeya, so the new theorem is consistent with restric-

tion failing. What it does supply is a technique for tangential configurations, which Volume 12, Worked Example 12.4.1(d) identified as the ceiling on broad–narrow.

(c) *Bochner–Riesz in $\mathbb{R}^3$* . Likewise not settled; it is stronger than restriction, hence stronger than Takeya.

(d) *Falconer*. Untouched. It is not on the implication chain at all (Volume 12, Worked Example 12.5.5), and its planar record comes from a different circle of ideas — Chapter 14.12.

The honest summary. One row of the table changed. The change is a major theorem and a new method, and it did not propagate to the others — which is what the implication structure predicts, since Takeya sits at the bottom.

What would propagate. A proof of the restriction conjecture in $\mathbb{R}^3$ would give Takeya there as a corollary and Bochner–Riesz is stronger still. So the interesting question after 2025 is whether the Wang–Zahl techniques bear on restriction, and Chapter 14.14 records what is known.

WORKED EXAMPLE 14.1.2— TIER B **Reading a status claim in this area**

Announcements in this field are frequently mis-stated in secondary sources. Five checks.

(1) *Which dimension?* A proof is claimed for Takeya in $\mathbb{R}^3$; no such claim is available for $\mathbb{R}^4$. The table above gives the current position and separately records this manuscript’s conditional status.

(2) *Hausdorff or Minkowski?* These differ: $\dim_H \leq \dim_M$ always, so a lower bound on Minkowski dimension is the *weaker* statement and does not give one on Hausdorff. Volume 7 sets out the relation; Wang–Zahl give both.

(3) *Sets or maximal functions?* The Takeya maximal conjecture is stronger than the set conjecture; Volume 9, Theorem 9.1.3 gives the implication. Wang–Zahl claim the set conjecture, while the maximal version in $\mathbb{R}^3$ remains open.

(4) *Which conjecture?* “Takeya” is used for the set conjecture, the maximal conjecture, and the finite field analogue — the last settled by Dvir in 2008 and proved in Volume 10, Theorem 10.2.1. They are different statements.

(5) *Is it in final form?* Results of the last two years may be preprints. Volume 12, Theorem 12.5.3 explains why this series quotes no current numerical records except where the statement is a clean equality.

A worked check. “Takeya was solved in 2025”: (1) $\mathbb{R}^3$ only; (2) both dimensions; (3) the *set* conjecture, not the maximal one; (4) the Euclidean

set conjecture; (5) presently contained in submitted preprints. Thus the accurate dated statement is: *the submitted sources claim a proof of the three-dimensionalakeya set conjecture*; this volume’s streamlined reconstruction remains conditional, and four related statements remain open.

WORKED EXAMPLE 14.1.3— TIER B **What a proof of restriction would need**

Theakeya solution was claimed in $\mathbb{R}^3$; restriction remains open. Four things a proof would have to supply that theakeya argument does not.

(1) *L^∞ data.* Volume 13, Theorem 13.9.5 shows decoupling is sharp at the Stein–Tomas exponent and cannot reach below it; the conversion from ℓ^2 over caps to L^∞ costs $\delta^{(n-1)/4}$ per cap. Nothing in theakeya proof addresses this.

(2) *A statement about functions, not sets.* The Wang–Zahl argument manipulates tube families and their shadings. A restriction estimate is about an oscillatory integral, and the passage between them — Volume 12’s wave packet dictionary — is lossy in the direction one would need.

(3) *Non-unique extremisers.* Knapp caps show the exponent is sharp, but they are not the only near-extremal configurations, and a structure theorem would have to describe all of them. Theakeya proof dossiers show how such structure is obtained for tube unions; no corresponding classification is presently available for restriction extremisers.

(4) *The implication runs the wrong way.* Restriction impliesakeya (Volume 6, Theorem 6.6.2), so aakeya theorem is consistent with restriction failing. No reduction in the other direction is known.

What is genuinely transferable. The techniques for tangential configurations. Volume 12, Worked Example 12.4.1(d) identified tangency as the ceiling on broad–narrow, and the slope-regularity and cinematic stages inside Chapters 14.2 and 14.6 confront it rather than excluding it. *That, and not the theorem, is what makes 2025 interesting for the other three conjectures.*

Drills

Drill 14.1.1. State the implications among the four conjectures and confirm that settlingakeya in $\mathbb{R}^3$ does not settle restriction there.

Drill 14.1.2. Explain why a bound of 3.059 is impossible for Besicovitch sets in $\mathbb{R}^3$, and identify the dimension for which it is the record.

Drill 14.1.3. Distinguish the Kakeya set conjecture, the maximal conjecture, and the finite field analogue, and say which are settled.

Drill 14.1.4. Using Volume 10, Worked Example 10.6.2, explain why the arithmetic method gives a fixed fraction of n rather than $n - O(1)$.

Notes

Wang and Zahl's paper is *Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions* (2025). Davies settled the plane in 1971. The planebrush bound in $\mathbb{R}^4$ is Katz–Zahl; the high-dimensional arithmetic bound is Katz–Tao. Chapter 14.14 gives full attributions.

Chapter 14.2

The Complete Sticky Kakeya Proof

Provenance and reading status

This chapter reproduces, with the notation and theorem numbering absorbed into this book, the complete proof in Hong Wang and Joshua Zahl, *Sticky Kakeya sets and the sticky Kakeya conjecture*, Journal of the American Mathematical Society 39 (2026), 515–585, arXiv:2210.09581v2. The authors’ arXiv source is licensed CC BY 4.0. The text and figures below are included under that licence. This edition suppresses the stand-alone title page and bibliography, shares theorem counters with the book, reflows a small number of displays and figures, corrects evident typographical slips, and supplies the proof of the two-ends lemma that the source called standard and omitted. Those editorial additions are changes by this edition and are not attributed to the original authors.

The proof is not an outline. Every refinement, rescaling, planiness and graininess argument, projection reduction, and terminal contradiction present in the source is retained. The few results invoked from other papers are isolated as named inputs; subsequent proof-dossier chapters discharge those inputs or identify them in the dependency ledger.

Sticky Kakeya sets and the sticky Kakeya conjecture Hong WangDepartment of Mathematics, New York University, New York, NY; and IHES, Université Paris-Saclay, Bures sur Yvette, France. hw3639@nyu.edu Joshua ZahlChern Institute of Mathematics and LPMC, Nankai University, Tianjin 300071, China; and The University of British Columbia, Vancouver, BC, Canada. jzahl@nankai.edu.cn

Abstract. A Kakeya set is a compact subset of $\mathbb{R}^n$ that contains a unit line segment pointing in every direction. The Kakeya conjecture asserts

that such sets must have Hausdorff and Minkowski dimension n . There is a special class of Kakeya sets, called sticky Kakeya sets. Sticky Kakeya sets exhibit an approximate multi-scale self-similarity, and sets of this type played an important role in Katz, Łaba, and Tao’s groundbreaking 1999 work on the Kakeya problem. We propose a special case of the Kakeya conjecture, which asserts that sticky Kakeya sets must have Hausdorff and Minkowski dimension n . We prove this conjecture in three dimensions.

14.2.1 Introduction

A compact set $K \subset \mathbb{R}^n$ is called a *Kakeya set* if it contains a unit line segment pointing in every direction. A surprising construction by Besicovitch [2] shows that such sets can have measure 0. The Kakeya set conjecture asserts that every Kakeya set in $\mathbb{R}^n$ has Hausdorff and Minkowski dimension n . There is also a slightly more technical, single-scale variant of this conjecture, which is called the Kakeya maximal function conjecture: let $\delta > 0$ and let $\mathbb{T}$ be a set of $1 \times \delta$ tubes in $\mathbb{R}^n$ whose coaxial lines point in δ -separated directions. Then the tubes must be almost disjoint, in the sense that for every $1 \leq d \leq n$ and every $\varepsilon > 0$, there exists $C = C(n, d, \varepsilon)$ (independent of δ) so that

$$\left\| \sum_{T \in \mathbb{T}} \chi_T \right\|_{\frac{d}{d-1}} \leq C \left(\frac{1}{\delta} \right)^{\frac{n}{d}-1+\varepsilon} \left(\sum_{T \in \mathbb{T}} |T| \right)^{\frac{d-1}{d}}. \quad (14.2.1)$$

The Kakeya set conjecture and Kakeya maximal conjecture were proved for $n = 2$ by Davies [10] and Cordoba [9], respectively. The conjectures remain open in three and higher dimensions.

The Kakeya conjecture is closely related to questions in Fourier analysis. This connection was first explored by Fefferman [17], who used a variant of Besicovitch’s construction to show that the ball multiplier $\widehat{Tf} = \chi_B \hat{f}$ is unbounded on $L^p(\mathbb{R}^2)$ when $p \neq 2$. In [3], Bourgain obtained new estimates for Stein’s Fourier restriction conjecture in $\mathbb{R}^3$ by first proving, and then using estimates of the form (14.2.1) (for certain $d < 3$) in three dimensions. In brief, a function whose Fourier transform is supported on a curved manifold M can be decomposed into a sum of “wave packets,” each of which is supported on a tube T . The curvature of M ensures that many of these wave packets point in different directions, and estimates of the form (14.2.1) can be used to analyze the possible intersection patterns of these wave packets.

Since Bourgain’s seminal work [3], Kakeya estimates have served as an input when studying the Fourier restriction problem [7, 11, 44], and methods that were developed in the context of the Kakeya problem have been successfully applied to the Fourier restriction problem, Bochner-Riesz problem, and related questions. Indeed, the modern renaissance in polynomial method techniques

was sparked by Zeev Dvir's proof [13] of Wolff's finite field Kakeya conjecture. These polynomial method techniques have since revolutionized discrete math, combinatorial geometry, and harmonic analysis. See e.g. [23, 39] for a modern survey of these developments.

In this paper we will restrict our attention to the Kakeya set conjecture in three dimensions. In [46], Wolff proved that every Kakeya set in $\mathbb{R}^3$ has Hausdorff dimension at least $5/2$. In [25], Katz, Łaba, and Tao made the following improvement: every Kakeya set in $\mathbb{R}^3$ has upper Minkowski dimension at least $5/2 + c$, where $c > 0$ is a small absolute constant. Katz, Łaba, and Tao's argument began by carefully analyzing the structure of a (hypothetical) Kakeya set that had dimension exactly $5/2$. At scale δ , such a set K would contain a union of roughly δ^{-2} $1 \times \delta$ tubes that point in δ -separated directions, and the union of these tubes would have volume roughly $\delta^{1/2}$. Next, Katz, Łaba, and Tao considered how these $1 \times \delta$ tubes would arrange themselves at scale $\rho = \delta^{1/2}$. Since the $1 \times \delta$ tubes point in different directions, each $1 \times \rho$ tube can contain at most ρ^2/δ^2 $1 \times \delta$ tubes. If equality (or near equality) holds (i.e. if the $1 \times \delta$ tubes arrange themselves into roughly ρ^{-2} many $1 \times \rho$ tubes, each of which contain ρ^2/δ^2 $1 \times \delta$ tubes) then we say the arrangement of $1 \times \delta$ tubes is *sticky* at scale ρ . As a starting point for their arguments in [25], Katz, Łaba, and Tao used a result by Wolff [48] to prove that if a Kakeya set K has upper Minkowski dimension $5/2$, then the corresponding set of $1 \times \delta$ tubes must be sticky at every scale $\rho \in (\delta, 1)$.

Motivated by this observation, we introduce a special class of Kakeya sets, which we call sticky Kakeya sets. Let $\mathcal{L}$ be the set of (affine) lines in $\mathbb{R}^n$, equipped with the metric $d(\ell, \ell') = |p - p'| + \angle(u, u')$. Here u (resp u') is a unit vector parallel to ℓ and p (resp. p') is the unique point on ℓ with $p \perp u$. Since the direction map from $\mathcal{L}$ to $\mathbb{RP}^{n-1}$ is Lipschitz, if $L \subset \mathcal{L}$ contains a line in every direction then $\dim_P(L) \geq n - 1$. (Here $\dim_P$ denotes packing dimension; similar statements hold for other metric notions of dimension, but this is less relevant to the current discussion). We say a Kakeya set is sticky if equality holds:

Definition 14.2.1. A compact set $K \subset \mathbb{R}^n$ is called a *sticky Kakeya set* if there is a set of lines L with packing dimension $n - 1$ that contains at least one line in each direction, so that $\ell \cap K$ contains a unit interval for each $\ell \in L$.

Observe that if we drop the requirement that L has packing dimension $n - 1$, then we have recovered the usual definition of a Kakeya set. In particular, every sticky Kakeya set is also a (classical) Kakeya set. As we will see in Section 14.2.2, when sticky Kakeya sets (in the sense of Definition 14.2.1) are discretized at a small scale $\delta > 0$, the corresponding collection of $1 \times \delta$ tubes is sticky in the sense of Katz, Łaba, and Tao. With these definitions, we can introduce the sticky Kakeya set conjecture.

Conjecture 14.2.2. Every stickyakeya set in $\mathbb{R}^n$ has Hausdorff and Minkowski dimension n .

A defining feature of stickyakeya sets is that if K is a (hypothetical) counter-example to the stickyakeya conjecture, then K must be roughly self-similar at many different scales. This was first observed in [25] and discussed in greater detail in [43]. A precise version of this principle is stated in Proposition 14.2.14 below.

Since every stickyakeya set is also a (classical)akeya set, theakeya set conjecture implies the stickyakeya conjecture, and partial progress toward theakeya set conjecture implies the same partial progress for the stickyakeya conjecture. In particular, Davies’ solution [10] to theakeya set conjecture in the plane implies that Conjecture 14.2.2 holds when $n = 2$. In this paper, we will prove Conjecture 14.2.2 for $n = 3$.

Theorem 14.2.3. *Every stickyakeya set in $\mathbb{R}^3$ has Hausdorff dimension 3.*

Our proof of Theorem 14.2.3 is inspired by, and partially follows, the arguments recorded in Terence Tao’s blog entry [43]. That blog entry sketches an approach to solving theakeya conjecture in $\mathbb{R}^3$ that was explored by Nets Katz and Terence Tao in the early 2000s. In brief, [43] begins by conjecturing that a (hypothetical) counter-example $K \subset \mathbb{R}^3$ to theakeya conjecture in $\mathbb{R}^3$ must be sticky, and must also have two additional structural properties, which are called planiness and graininess. We will discuss this conjecture in Section 14.2.1.2 below. Next, the blog entry describes how these three properties can be used to obtain multi-scale structural information about K . Finally, the blog entry suggests how to use this structural information to produce a counter-example to Bourgain’s discretized sum-product theorem [4], and hence obtain a contradiction.

14.2.1.1 A sketch of the proof

We will briefly outline our approach to proving Theorem 14.2.3, and discuss how it relates to Katz and Tao’s strategy, as described in [43]. Katz and Tao’s strategy was influenced by the following example from [25], which is called the (second) Heisenberg group example:

$$\mathbb{H} = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : \operatorname{Im}(z_3) = \operatorname{Im}(z_1 \bar{z}_2)\}. \quad (14.2.2)$$

The set $\mathbb{H}$ is a counter-example to a strong form of theakeya conjecture. In particular, the δ -neighbourhood of $\mathbb{H} \cap B(0, 1)$ contains about δ^{-4} “complex unit line segments,” and at most δ^{-2} of these line segments are contained in the δ -neighbourhood of a (complex) plane in $\mathbb{C}^3$. This set has (complex analogues of) the structural properties stickiness, planiness, and graininess,

but it has volume substantially smaller than 1. At a key step in our argument, we will use (a consequence of) Bourgain's discretized sum-product theorem to distinguish between a sticky Kakeya set in $\mathbb{R}^3$ and the Heisenberg group example.

Our proof can be divided into three major steps. The first step follows some of the arguments described in [43]. For steps 2 and 3 our arguments take a different path from those presented in [43]; we do this in part because sticky Kakeya sets can have structural properties that were not discussed in [43].

Step 1: Discretization and multi-scale self-similarity

In Section 14.2.2, we set up a discretization of the sticky Kakeya problem that allows us to exploit multi-scale self-similarity. The technical details of this procedure are new; in particular, we explain how to replace each $1 \times \delta$ tube by a reasonably large subset (called a shading), so that key statistics of the Kakeya set are maintained after successive refinements. We hope that this setup will be useful when studying the sticky Kakeya conjecture in higher dimensions.

In Section 14.2.3, we flesh-out the ideas from [43] to obtain multi-scale structural information about sticky Kakeya sets. The main conclusion of this section is as follows. Suppose that the sticky Kakeya conjecture in $\mathbb{R}^n$ was false, i.e. $\inf \dim_H K = n - \sigma_n$, where the infimum is taken over all sticky Kakeya sets in $\mathbb{R}^n$, and $\sigma_n > 0$. Let K be a sticky Kakeya set with dimension $n - \sigma_n + \varepsilon$ for some small $\varepsilon > 0$ (we will call this an ε -extremal sticky Kakeya set), and let E be the discretization of K at a small scale $\delta > 0$. Then E contains a union of $1 \times \delta$ -tubes, which are sticky in the sense of Katz, Łaba, and Tao. Furthermore, this collection of δ -tubes is coarsely self-similar: for each intermediate scale $\rho \in (\delta, 1)$, the ρ -thickening of K again resembles an ε -extremal sticky Kakeya set, and the union of δ -tubes inside each ρ -tube resembles a (anisotropically re-scaled) ε -extremal sticky Kakeya set. The precise statement is given by Proposition 14.2.14.

In Section 14.2.4 we restrict attention to the case $n = 3$ and again follow arguments proposed by Katz and Tao in [43]. We show that extremal sticky Kakeya sets must have a structural property called planiness. Specifically, with K and E as above, if $\mathbb{T}$ is the set of $1 \times \delta$ tubes contained in E , then there is a function $V: E \rightarrow S^2$ so that for each point $x \in E$, the tubes in $\mathbb{T}$ containing x must make small angle with the subspace $V(x)^\perp$. $V(x)$ is called a plane map, and we use the multi-scale self-similarity of extremal Kakeya sets to show that V must be Lipschitz. An important consequence is that inside each ball of radius $\delta^{1/2}$, the set E can be decomposed into a disjoint union of parallel rectangular prisms of dimensions roughly $\delta^{1/2} \times \delta^{1/2} \times \delta$; these rectangular prisms are called grains. After an anisotropic re-scaling, we obtain a new collection of $1 \times \delta$ tubes whose union $E' \subset \mathbb{R}^3$ satisfies the following **Property**

(P): For each $z_0 \in [0, 1]$, the slice $E' \cap \{z = z_0\}$ is a union of parallel $1 \times \delta$ rectangles whose spacing forms an Ahlfors-regular set of dimension $1 - \sigma_3$; the direction of these rectangles is determined by a “slope function” $f(z_0)$, which is Lipschitz. All of the arguments described thus far also apply to the Heisenberg group example.

Step 2: Regularity of the slope function

In Section 14.2.5, we show that the slope function $f(z)$ described above is C^2 (or more precisely, it agrees at scale δ with a function that has controlled C^2 norm). To do this, it suffices to show that at every scale $\rho \in (\delta, 1)$ and for every interval I of length $\rho^{1/2}$, the graph of f above ρ can be contained in the ρ -neighborhood of a line. We will briefly describe how this is done. Fix a scale ρ and an interval I , and let $F \subset [0, 1]^2$ be a re-scaled copy of the graph of f above I . We will also construct a set $G \subset [0, 1]^2$, which is the graph of a Lipschitz function $g: [0, 1] \rightarrow \mathbb{R}$ that encodes a local analogue of the plane map V . We construct the set G so that certain closed paths inside E' are encoded by arithmetic operations on the sets F and G . Specifically, whenever we select points $p \in F$ and $q, q' \in G$, we have that $p \cdot (q - q')$ specifies the location of one of the $1 \times \delta$ rectangles in the set $E' \cap \{z = z_0\}$ that was described above in Property (P). Since the spacing of these rectangles form an Ahlfors-regular set, we conclude that

$$F \cdot (G - G) \text{ is contained in an Ahlfors regular set of dimension } 1 - \sigma_3. \quad (14.2.3)$$

Recall that F and G are the graphs of Lipschitz functions, and thus satisfy a non-concentration condition analogous to having Hausdorff dimension 1. Is it possible for (14.2.3) to occur? If F and G are contained in orthogonal lines, then $F \cdot (G - G)$ is a point, so the answer is yes. The next result says that this is the only way that (14.2.3) can occur.

Theorem 14.2.37, informal version. *Let F and G be one-dimensional sets that have been discretized at a small scale $\rho > 0$. Then either (A): (most of) F and G are contained in the ρ -neighborhoods of orthogonal lines, or (B): there exists some scale $\rho \leq t \ll 1$ so that the t -neighborhood of $F \cdot (G - G)$ contains an interval.*

The precise version of Theorem 14.2.37 is stated in Section 14.2.5 and proved in Section 14.2.8. Our proof of Theorem 14.2.37 uses recent ideas from projection theory that were developed in the context of the Falconer distance problem [33, 41, 36]. Specifically, the projection theorem we use generalizes Bourgain’s discretized sum-product estimates; see [35, 38] for further recent developments in this area. It is at this step where we distinguish between a sticky Keakeya set (which is a subset of $\mathbb{R}^3$) and the Heisenberg group example (which is a subset of $\mathbb{C}^3$). In brief, if G is not contained in a line, then by an

analogue of Beck's theorem due to Orponen, Shmerkin, and the first author [36], we might expect G to span a two-dimensional set of lines, and hence a one-dimensional set of directions, i.e. there is a one-dimensional set $\Theta \subset S^1$ so that for each $\theta \in \Theta$, there are points $q, q' \in G$ with $\theta = \frac{q-q'}{|q-q'|}$. If this happens, then Kaufman's projection theorem says that there exists a direction $\theta \in \Theta$ for which $F \cdot \theta$ (and hence $F \cdot (q - q')$) has dimension 1. The actual proof of Theorem 14.2.37 must overcome several difficulties when executing the above strategy. First, Theorem 14.2.37 has weaker hypotheses than the ones stated above; the set $G \cdot (F - F)$ is replaced by a smaller set, where the differences and dot products are taken along a sparse (but not too sparse!) subset of $F \times G \times G$. Second, the proof sketch above supposes a dichotomy, where either G is contained in a line, or it spans a one-dimensional set of directions. In the discretized setting, however, this dichotomy is less apparent because G can behave differently at different scales.

Returning to the proof of Theorem 14.2.3, if Item (B) from Theorem 14.2.37 holds, then by an observation of Dyatlov and the second author [12, Proposition 6.13], $F \cdot (G - G)$ cannot be contained in an Ahlfors-regular set of dimension strictly smaller than 1. Thus Theorem 14.2.37 implies that the graph of f above I must be contained in the ρ neighborhood of a line. Since this holds for every interval I at every scale ρ , we conclude that f is C^2 . Kakeya sets with regularity conditions have been studied in the past [18, 26], but different ideas are needed in this setting.

Step 3: Twisted projections

In Section 14.2.6 we exploit the fact that lines in a Kakeya set point in different directions to show that the slope function f has (moderately) large derivative, i.e. $|f'|$ is bounded away from 0.

In Section 14.2.7, we begin by observing the following consequence of Property (P) from Step 1 above. Since each slice $E' \cap \{z = z_0\}$ is a union of parallel $1 \times \delta$ rectangles whose spacing forms an Ahlfors regular set and whose slope is given by $f(z_0)$, if we define $\pi_f(x, y, z) = (x + yf(z), z)$, then $\pi_f(E') \subset [0, 1]^2$ has measure roughly δ^{σ_3} . This is because each slice $\pi_f(E') \cap \{z = z_0\}$ is a union of δ -intervals that are arranged like a $1 - \sigma_3$ dimensional Ahlfors regular set.

If T is a $1 \times \delta$ tube, then $\pi_f(T)$ is the δ -neighborhood of a C^2 curve. Since the tubes in $\mathbb{T}$ point in different directions and since $|f'|$ is bounded away from 0, the corresponding curves $\pi_f(T)$ satisfy a separation property related to Sogge's cinematic curvature condition [31, 42]. The next result says that the union of these curves must be large

Proposition 14.2.52, informal version. *Let $\mathbb{T}$ be a set of $1 \times \delta$ tubes pointing in δ -separated directions, with $\#\mathbb{T} \gtrsim \delta^{-2}$. Let $f: [0, 1] \rightarrow \mathbb{R}$ with $\|f\|_{C^2} \lesssim 1$ and $|f'| \sim 1$. Then for all $\varepsilon > 0$, there exists $c_\varepsilon > 0$ so that $\bigcup_{T \in \mathbb{T}} \pi_f(T)$ has*

measure at least $c_\varepsilon \delta^\varepsilon$.

The precise version of Proposition 14.2.52 is a maximal function estimate analogous to (14.2.1). It is proved using a variant of Wolff's L^3 bound for the Wolff circular maximal function [47], where circles are replaced by a family of C^2 curves that satisfy Sogge's cinematic curvature condition. This latter result was recently proved by Pramanik, Yang, and the second author [37]. Since E' contains about δ^{-2} tubes that point in δ -separated directions, Proposition 14.2.52 says that $\pi_f(E')$ must have measure at least $c_\varepsilon \delta^\varepsilon$, for each $\varepsilon > 0$. On the other hand, Property (P) says that $\pi_f(E')$ has measure at most δ^{σ_3} . We conclude that $\sigma_3 = 0$, which finishes the proof.

14.2.1.2 The stickyakeya conjecture versus theakeya conjecture

In this section, we will informally discuss what Theorem 14.2.3 tells us about theakeya conjecture. In particular, how close does Theorem 14.2.3 get us to proving theakeya conjecture in $\mathbb{R}^3$? Katz and Tao's arguments from [43] begin by conjecturing that a (hypothetical) counter-example $K \subset \mathbb{R}^3$ to theakeya conjecture in $\mathbb{R}^3$ must have the structural properties stickiness, planiness, and graininess. The results from [1] and [22] show that such a counter-example must be plainy and grainy, but it is unclear whether K must be sticky. To reformulate the remaining part of Katz and Tao's conjecture in our language, we need the following definition.

Definition 14.2.4. Aakeya set $K \subset \mathbb{R}^n$ is ε -close to being a sticky if there is a set of lines L with packing dimension at most $n - 1 + \varepsilon$ that contains at least one line in each direction, so that $\ell \cap K$ contains a unit interval for each $\ell \in L$.

In particular, a stickyakeya set is ε -close to being sticky for every $\varepsilon > 0$. The following version of Katz and Tao's conjecture says thatakeya sets that are nearly extremal are close to being sticky.

Conjecture 14.2.5. Suppose theakeya conjecture in $\mathbb{R}^n$ is false, i.e. $\inf \dim_H K = d < n$, where the infimum is taken over allakeya sets in $\mathbb{R}^n$. Then for all $\varepsilon > 0$, there exists $\eta > 0$ so that the following holds. Let $K \subset \mathbb{R}^n$ be aakeya set with $\dim_H(K) \leq d + \eta$. Then K is ε -close to being sticky.

If Conjecture 14.2.5 is true for $n = 3$, then the following generalization of Theorem 14.2.3 would imply theakeya conjecture in $\mathbb{R}^3$.

Theorem 14.2.3'. *For all $\varepsilon > 0$, there exists $\eta > 0$ so that the following holds. Let $K \subset \mathbb{R}^3$ be aakeya set that is η -close to being sticky. Then $\dim_H K \geq 3 - \varepsilon$.*

Remark 14.2.6. A consequence of Theorem 14.2.3' is that if an X-ray estimate holds in $\mathbb{R}^3$ at dimension $d < 3$, then there exists $c > 0$ so that every Kakeya set in $\mathbb{R}^3$ must have upper Minkowski dimension at least $d + c$.

One piece of evidence in favor of Conjecture 14.2.5 at the time it was formulated in [43] was that the only known example of a Kakeya-like object with Hausdorff dimension less than 3 was the (second) Heisenberg group example (14.2.2). As noted above, $\mathbb{H}$ is a counter-example to a strong form of the Kakeya conjecture, and it has (complex analogues of) the structural properties stickiness, planiness, and graininess.

More recently, however, Katz and the second author [29] discovered a second Kakeya-like object that has small volume at certain scales—this is analogous to having Hausdorff dimension less than 3. This object is called the SL_2 example, and it is not sticky. The SL_2 example is not a counter-example to Conjecture 14.2.5 (nor to the Kakeya conjecture) because it is not a Kakeya set in $\mathbb{R}^3$. Nonetheless, the existence of the SL_2 example suggests that in order to resolve the Kakeya conjecture in $\mathbb{R}^3$, it will be necessary to study Kakeya sets that are far from sticky.

As a starting point for this latter program, it seems reasonable to show that the SL_2 example cannot be realized in $\mathbb{R}^3$. To make this statement precise, define $\mathcal{L}_{SL_2}$ to be the set of lines in $\mathbb{R}^3$ that can either be written in the form $(a, b, 0) + \mathbb{R}(c, d, 1)$ with $ad - bc = 1$, or $(0, 0, 0) + \mathbb{R}(c, d, 0)$. Equivalently, $\mathcal{L}_{SL_2}$ is the set of horizontal lines in the first Heisenberg group.

We conjecture that lines from this set cannot be used to construct a counter-example to the Kakeya conjecture.

Conjecture 14.2.7. Let $K \subset \mathbb{R}^3$ be compact, and suppose there is a set of lines $L \subset \mathcal{L}_{SL_2}$ that contains at least one line in each direction, so that $\ell \cap K$ contains a unit interval for each $\ell \in L$. Then K has Hausdorff and Minkowski dimension 3.

Added November 3, 2022: Fässler and Orponen [16] have recently proved Conjecture 14.2.7.

We will not discuss Conjectures 14.2.5 or 14.2.7 further. The remainder of this paper will be devoted to proving Theorem 14.2.3 and 14.2.3'.

14.2.1.3 Thanks

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14.2.2 Discretization

14.2.2.1 Lines, tubes and shadings

We begin by recalling some standard definitions and terminology associated with the Kakeya problem. Throughout this section, we will fix an integer $n \geq 2$; all implicit constants may depend on n . Slightly abusing the notation we introduced in the introduction, we will define $\mathcal{L}_n$ to be the set of lines in $\mathbb{R}^n$ of the form $(\underline{p}, 0) + \mathbb{R}v$, where $\underline{p} \in [-\frac{1}{n}, \frac{1}{n}]^{n-1}$ and $v \in S^{n-1}$ has final coordinate $v_n \geq 1/2$. We define $d(\ell, \ell') = |\underline{p} - \underline{p}'| + \angle(v, v')$. This definition of distance is comparable to the definition from Section 14.2.1. We define the measure λ_n on $\mathcal{L}_n$ to be the product measure of $(n-1)$ -dimensional Euclidean measure and normalized surface measure on S^{n-1} (the latter we denote by ν_{n-1}). With these definitions of distance and measure, the packing dimension of a set $L \subset \mathcal{L}_n$ agrees with its upper modified box dimension.

For $\ell \in \mathcal{L}_n$ and $\delta > 0$, we define the δ -tube T with coaxial line ℓ to be $N_{2n\delta}(\ell) \cap [-1, 1]^n$, where $N_t(X)$ denotes the t -neighborhood of X . This definition is slightly nonstandard, since we use the $2n\delta$ -neighborhood rather than the δ -neighborhood of ℓ ; we do this so that a cube of side-length δ intersecting ℓ will be contained in the associated tube T . Observe that any tube of this type has volume comparable to δ^{n-1} . If T is a δ -tube and $C > 0$, we will use CT to denote the $C\delta$ -tube with the same coaxial line. We say two δ -tubes T, T' are *essentially identical* if their respective coaxial lines satisfy $d(\ell, \ell') \leq \delta$; otherwise T and T' are *essentially distinct*. We say that T and T' are *essentially parallel* if their respective coaxial lines satisfy $|\text{dir}(\ell) - \text{dir}(\ell')| \leq \delta$, where $\text{dir}(\ell)$ is the unit vector v described above.

A δ -cube is a set of the form $[0, \delta]^n + p$, where $p \in (\delta\mathbb{Z})^n$. A *shading* of a δ -tube T is a set $Y(T) \subset T$ that is a union of δ -cubes. We will write $(\mathbb{T}, Y)_\delta$ to denote a set of δ tubes and their associated shadings $\{Y(T) : T \in \mathbb{T}\}$. If $(\mathbb{T}, Y)_\delta$ is a set of tubes and their shadings, we write $E_{\mathbb{T}}$ to denote the set $\bigcup_{T \in \mathbb{T}} Y(T)$; the shading Y will always be apparent from context.

We say that $(\mathbb{T}', Y')_\delta$ is a *sub-collection* of $(\mathbb{T}, Y)_\delta$ if $\mathbb{T}' \subset \mathbb{T}$ and $Y'(T') \subset Y(T')$ for each $T' \in \mathbb{T}'$. We say that $(\mathbb{T}', Y')_\delta$ is a *refinement* of $(\mathbb{T}, Y)_\delta$ if in addition, $\sum_{T' \in \mathbb{T}'} |Y'(T')| \geq (\log(1/\delta))^{-C} \sum_{T \in \mathbb{T}} |Y(T)|$, where $C \geq 0$ is a constant that depends only on n . We will denote this by $\sum_{T' \in \mathbb{T}'} |Y'(T')| \gtrsim_\delta \sum_{T \in \mathbb{T}} |Y(T)|$. In practice, we will only study collections of tubes $(\mathbb{T}, Y)_\delta$ with $\#\mathbb{T} \leq \delta^{-100n}$. By dyadic pigeonholing, every collection of tubes with this property has a refinement $(\mathbb{T}', Y')_\delta$ with $\sum_{T \in \mathbb{T}'} \chi_{Y'(T)} = \mu \chi_{E_{\mathbb{T}'}}$ for some integer $\mu \geq 1$; this is called a *constant multiplicity refinement* of $(\mathbb{T}, Y)_\delta$.

Let T be a δ -tube and let $\tilde{T}$ be a ρ -tube, with $\delta \leq \rho$. We say that $\tilde{T}$ *covers* T if their respective coaxial lines satisfy $d(\ell, \tilde{\ell}) \leq \rho/2$. If $\mathbb{T}$ is a collection of δ -tubes and $\tilde{\mathbb{T}}$ is a collection of ρ -tubes, we say $\tilde{\mathbb{T}}$ covers $\mathbb{T}$ if every tube in $\mathbb{T}$ is covered by some tube in $\tilde{\mathbb{T}}$. If this is the case, we write $\mathbb{T}[\tilde{T}]$ to denote the set of tubes in $\mathbb{T}$ that are covered by $\tilde{T}$. A similar definition holds for collections of tubes and their associated shadings: we say that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ *covers* $(\mathbb{T}, Y)_\delta$ if $\tilde{\mathbb{T}}$ covers $\mathbb{T}$, and for each $\tilde{T} \in \tilde{\mathbb{T}}$ and each $T \in \mathbb{T}[\tilde{T}]$ we have $Y(T) \subset \tilde{Y}(\tilde{T})$. Such a cover is called *balanced* if $|E_{\mathbb{T}} \cap Q|$ is the same for each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}}$.

14.2.2.2 Defining σ_n

In this section we will define a quantity, called σ_n , that will allow us to formulate a discretized version of the sticky Kakeya conjecture.

Definition 14.2.8. For $s, t, \delta \in (0, 1)$, define

$$M(s, t, \delta) = \inf_{\mathbb{T}, Y} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right|. \quad (14.2.4)$$

Here, the infimum is taken over all pairs $(\mathbb{T}, Y)_\delta$ with the following properties.

- (a) The tubes in $\mathbb{T}$ are essentially distinct.
- (b) For each $\delta \leq \rho \leq 1$, $\mathbb{T}$ can be covered by a set of ρ tubes, at most δ^{-t} of which are essentially parallel to a common ρ tube.
- (c) $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^s$

Next, let

$$N(s, t) = \limsup_{\delta \rightarrow 0^+} \frac{\log M(s, t, \delta)}{\log \delta}. \quad (14.2.5)$$

$N(s, t)$ is defined for all $(s, t) \in (0, 1)^2$. Observe that for $\delta \in (0, 1)$, if $s \leq s'$ and $t \leq t'$, then $M(s, t, \delta) \geq M(s', t', \delta)$, and hence $N(s, t) \leq N(s', t')$. In particular, since $0 \leq N(s, t) \leq n$, we have that

$$\lim_{(s, t) \rightarrow (0, 0)} N(s, t) = \inf_{(s, t) \in (0, 1)^2} N(s, t).$$

Denote this common value by σ_n . Our definition of σ_n was chosen so that two properties hold. First, σ_n allows us to bound the Hausdorff dimension of sticky Kakeya sets. This will be described in Section 14.2.2.3. Second, collections $(\mathbb{T}, Y)_\delta$ that nearly extremize the infimum in (14.2.4) will have useful structural properties at many scales. This will be described in Section 14.2.3.

14.2.2.3 σ_n and the dimension of stickyakeya sets

In this section we will relate σ_n to the dimension of stickyakeya sets. Our goal is to prove the following.

Proposition 14.2.9. *For all $\varepsilon > 0$, there exists $\eta > 0$ so that the following holds. Let $K \subset \mathbb{R}^n$ be aakeya set that is η -close to being sticky. Then $\dim_H K \geq n - \sigma_n - \varepsilon$. In particular, if K is a stickyakeya set then $\dim_H K \geq n - \sigma_n$.*

Our first task is to analyze families of lines in $\mathbb{R}^n$ that point in many directions and have packing dimension close to $n - 1$. For each $\rho > 0$ and $\varepsilon > 0$, define the class of “Quantitatively Sticky” sets $\text{QStick}(n, t, \rho)$ to be the collection of sets $L \subset \mathcal{L}_n$ that satisfy

$$\left| \{ \underline{p} \in \mathbb{R}^{n-1} : (\underline{p}, v) \in N_\delta(L) \} \right| \leq \delta^{n-1-t} \quad \text{for all } v \in S^{n-1}, \delta \in (0, \rho),$$

where in the above expression, $N_\delta(L)$ denotes the δ -neighborhood of L in the metric space $\mathcal{L}_n$.

For example, if $L = \{(\underline{p}, v) \in \mathcal{L}_n : \underline{p} = f(v)\}$ for some Lipschitz function f , then for each $t > 0$ we have $L \in \text{QStick}(n, t, \rho)$ for all sufficiently small ρ . The next lemma explains how $\text{QStick}(n, t, \rho)$ is related to stickyakeya sets.

Lemma 14.2.10. *Let $L \subset \mathcal{L}_n$ with $\nu_{n-1}(\text{dir}(L)) > 0$. Then for all $t > \dim_P(L) - (n - 1)$, there exists $L' \subset L$ and $\rho > 0$ with $\nu_{n-1}(\text{dir}(L')) > 0$ and $L' \in \text{QStick}(n, t, \rho)$.*

Proof. Let $d = \dim_P L$, and let $\varepsilon = t - d + (n - 1)$. Since $d = \dim_P(L) = \overline{\dim}_{MB}(L)$, we can write L as a countable union of sets $\{L_i\}$, with $\overline{\dim}_M L_i \leq d + \varepsilon/4$ for each i . Since $\nu_{n-1}(\bigcup \text{dir}(L_i)) = \nu_{n-1}(\text{dir}(L)) > 0$, there exists an index i so that $\nu_{n-1}(\text{dir } L_i) > 0$. Re-indexing if necessary, we can suppose that $\nu_{n-1}(\text{dir } L_1) > 0$. Since $\overline{\dim}_M L_1 \leq d + \varepsilon/4$, we can select $\rho_1 > 0$ sufficiently small so that

$$\lambda_n(N_{2\delta}(L_1)) \leq \delta^{(2n-2)-d-\varepsilon/2} \quad \text{for all } \delta \in (0, \rho_1). \quad (14.2.6)$$

We say a direction $v \in S^{n-1}$ is *over-represented* at scale δ if

$$\left| \{ \underline{p} \in \mathbb{R}^{n-1} : (\underline{p}, v) \in N_{2\delta}(L_1) \} \right| > \delta^{n-1-t} = \delta^{(2n-2)-\delta-\varepsilon}.$$

By (14.2.6) and Fubini (recall that λ_n is a product measure on $\mathbb{R}^{n-1} \times S^{n-1}$), for each $\delta \in (0, \rho_1)$, we have

$$\nu_{n-1}(\{v \in S^{n-1} : v \text{ is over-represented at scale } \delta\}) \lesssim \delta^{\varepsilon/2}. \quad (14.2.7)$$

Thus for each integer k_1 with $2^{-k_1} \leq \rho_1$, we have

$$\sum_{k \geq k_1} \nu_{n-1}(\{v: v \text{ is over-represented at scale } 2^{-k}\}) \lesssim k_1^{-\varepsilon/2}. \quad (14.2.8)$$

Selecting k_1 appropriately, we can ensure that the LHS of (14.2.8) is at most $\nu_{n-1}(\text{dir}(L_1))/2$. Note that if $v \in S^{n-1}$ is not over-represented at any dyadic scale $2^{-k} \leq 2^{-k_1}$, then

$$\left| \{ \underline{p} \in \mathbb{R}^{n-1}: (\underline{p}, v) \in N_\delta(L_1) \} \right| \leq \delta^{n-1-t} \quad \text{for all } \delta \in (0, 2^{-k_1}).$$

To conclude the proof, let $\rho = 2^{-k_1}$ and let

$$L' = L_1 \setminus \bigcup_{k=k_1}^{\infty} \text{dir}^{-1}(\{v: v \text{ is over-represented at scale } 2^{-k}\}). \quad \square$$

With this lemma, Proposition 14.2.9 now follows from standard discretization arguments. The details are as follows.

Proof of Proposition 14.2.9. Let $\varepsilon > 0$ be given. By the definition of σ_n , there exists $s, t \in (0, 1)$ so that

$$N(s, t) \leq \sigma_n + \varepsilon/2. \quad (14.2.9)$$

We will show that Proposition 14.2.9 holds with $\eta = t/4$.

Let K be a Kakeya set that is η -close to being sticky. Since K is compact, after a translation and dilation we can suppose that $K \subset [-\frac{1}{n}, \frac{1}{n}]^n$. In particular, there exists a number $w > 0$ and a set $L \subset \mathcal{L}_n$ with $\dim_P(L) \leq n - 1 + \eta$ so that for each unit vector $v \in S^{n-1}$ with final coordinate $v_n \geq 1/2$, there is a line $\ell \in L$ with $\text{dir}(\ell) = v$ and $|\ell \cap K| \geq w$. Use Lemma 14.2.10 to select a subset $L' \subset L$ and a number $\rho_1 > 0$ with $L' \in \text{QStick}(n, t/2, \rho_1)$ and $\nu_{n-1}(\text{dir}(L')) > 0$.

Let k_1 be an integer to specified below, with $2^{-k_1} < \rho_1^{2n/t}$. By the definition of Hausdorff dimension, we can cover K by a union of cubes $K \subset \bigcup_{k=k_1}^{\infty} \bigcup_{Q \in \mathcal{Q}_k} Q$, where $\mathcal{Q}_k$ is a set of 2^{-k} -cubes with

$$\#\mathcal{Q}_k \lesssim (2^k)^{\dim_H K + \varepsilon/2}. \quad (14.2.10)$$

For each $\ell \in L'$, we have

$$K \cap \ell \subset \bigcup_{k=k_1}^{\infty} \left(\ell \cap \bigcup_{Q \in \mathcal{Q}_k} Q \right).$$

Since $|K \cap \ell| \geq w$, there exists at least one index k so that

$$\left| \ell \cap \bigcup_{Q \in \mathcal{Q}_k} Q \right| \geq \frac{w}{100k^2}. \quad (14.2.11)$$

For each $k \geq k_1$, let L_k be the set of lines $\ell \in L'$ for which (14.2.11) holds for that choice of k . Then $L' = \bigcup_k L_k$, so there exists an index $k_2 \geq k_1$ with $\nu_{n-1}(\text{dir}(L_{k_2})) \geq \frac{\nu_{n-1}(\text{dir}(L'))}{100k_2^2}$.

Let $\delta = 2^{-k_2}$, so in particular $\frac{1}{k_2} \geq \frac{\log 2}{\log(1/\delta)}$. Let Ω be a δ -separated subset of $\text{dir}(L_{k_2})$ of cardinality $\gtrsim \frac{\nu_{n-1}(\text{dir}(L'))\delta^{1-n}}{\log(1/\delta)^2}$. For each direction $v \in \Omega$, let T be a δ -tube with coaxial line $\ell \in L_{k_2}$ pointing in direction v . By (14.2.11), we can choose T so that the set

$$Y(T) = \bigcup_{\substack{Q \in \mathcal{Q}_{k_2} \\ Q \cap \ell \neq \emptyset}} Q$$

satisfies $Y(T) \subset T$ and $|Y(T)| \gtrsim w \log(1/\delta)^{-2} \delta^{n-1}$.

We claim that if k_1 is chosen sufficiently large, then $(\mathbb{T}, Y)_\delta$ satisfies Items (a), (b), and (c) from Definition 14.2.8. Item (a) is immediate, since Ω is δ -separated. Item (c) follows from the inequality

$$\sum_{T \in \mathbb{T}} |Y(T)| \gtrsim \frac{(\#\Omega)w\delta^{n-1}}{\log(1/\delta)^2} \gtrsim \frac{w\nu_{n-1}(\text{dir}(L'))}{\log(1/\delta)^4}.$$

This quantity is larger than δ^s provided we select k_1 sufficiently small large, depending on $\nu_{n-1}(\text{dir}(L'))$, w , and n .

Our final task is to show that $(\mathbb{T}, Y)_\delta$ satisfies Item (b). Let $\rho \in [\delta, 1]$. Our goal is to find a set of ρ -tubes $\tilde{\mathbb{T}}$ that covers $\mathbb{T}$, so that at most δ^{-t} of them are pairwise parallel. We will divide our analysis into cases.

Case 1: $\rho \in [\delta, \rho_1]$. let $\tilde{\mathbb{T}}$ be a collection of ρ -tubes, so that the set of coaxial lines $\tilde{L}$ has the following properties.

- (i) Each $\tilde{\ell} \in \tilde{L}$ is contained in $N_\rho(L')$.
- (ii) The $\rho/2$ -neighborhoods of the lines in $\tilde{L}$ cover L' .
- (iii) The $\rho/6$ neighborhoods of the lines in $\tilde{L}$ are disjoint.

Such a collection of tubes can be chosen greedily, as in the proof of the Vitali covering lemma.

Each tube $T \in \mathbb{T}$ has a coaxial line $\ell \in L_1 \subset L'$. Hence there is a tube $\tilde{T} \in \tilde{\mathbb{T}}$ so that $d(\ell, \tilde{\ell}) \leq \rho/2$, and thus $\tilde{T}$ covers T . We conclude that $\tilde{\mathbb{T}}$ covers $\mathbb{T}$.

It remains to show that at most ρ^{-t} tubes from $\tilde{\mathbb{T}}$ can all be essentially parallel to a common ρ tube. Let $\tilde{T}_1, \dots, \tilde{T}_M \in \tilde{\mathbb{T}}$ be essentially parallel to a common ρ tube $\tilde{T}_0$, and let $v = \text{dir}(\tilde{T}_0)$. In particular, $|v - \text{dir}(\tilde{T}_j)| \leq \rho$ for

each index j . By Item (i) above, this means $\tilde{T}_j \in N_\rho(L') \cap \text{dir}^{-1}(N_\rho(v))$. By item (iii) above, we conclude that

$$\lambda_n(N_\rho(L') \cap \text{dir}^{-1}(N_\rho(v))) \geq \sum_{i=1}^M \lambda_n(N_{\rho/6}(\tilde{\ell}_i)) \gtrsim M\rho^{2n-2}.$$

Thus by Fubini, there exists $v' \in N_\rho(v)$ with

$$|N_\rho(L') \cap \text{dir}^{-1}(v')| \gtrsim M\rho^{n-1}.$$

But since $L' \in \text{QStick}(n, t/2, \rho_1)$ and $\rho < \rho_1$, if $\rho_1 > 0$ is selected sufficiently small (depending on the implicit constants above, which in turn depend on n) then $M \leq \rho^{-t} \leq \delta^{-t}$.

Case 2: $\rho \in [\rho_1, 1]$. Let $\tilde{\mathbb{T}}$ be a set of ρ -tubes, whose corresponding coaxial lines form a $\rho/2$ -net in $\mathcal{L}_n$. Then $\tilde{\mathbb{T}}$ clearly covers $\mathbb{T}$. The maximum number of tubes in $\tilde{\mathbb{T}}$ that are all essentially parallel to a common ρ tube is trivially bounded by $\#\tilde{\mathbb{T}} \sim \rho^{2-2n}$, and this is $\leq \delta^{-t}$ (provided ρ_1 , and hence δ is selected sufficiently small compared to n), since $\delta \leq 2^{-k_1} \leq \rho_1^{2n/t}$.

We have shown that $(\mathbb{T}, Y)_\delta$ satisfies Items (a), (b), and (c) from Definition 14.2.8. Thus we have that

$$M(s, t, \delta) \leq \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \left| \bigcup_{Q \in \mathcal{Q}_{k_2}} Q \right| \leq (\#\mathcal{Q}_{k_2})\delta^n \leq \delta^{n - \dim K - \varepsilon/2},$$

where the final inequality used (14.2.10). We conclude that

$$\frac{\log M(s, t, \delta)}{\log \delta} \geq n - \dim_H K - \varepsilon/2. \quad (14.2.12)$$

We have just shown that for every k_1 sufficiently large, there exists $\delta \in (0, 2^{-k_1}]$ for which (14.2.12) holds. In particular,

$$N(s, t) \geq n - \dim_H K - \varepsilon/2. \quad (14.2.13)$$

To conclude the proof, we compare (14.2.9) and (14.2.13). $\square$

Proposition 14.2.9 says that Theorems 14.2.3 and 14.2.3' will follow from the estimate $\sigma_3 = 0$.

14.2.3 Extremal families of tubes and multi-scale structure

In this section, we will study families of tubes that nearly extremize the quantity M from Definition 14.2.8.

Definition 14.2.11. Let $\varepsilon, \delta > 0$. We say a pair $(\mathbb{T}, Y)_\delta$ is ε -extremal if it satisfies Items (a), (b), and (c) from Definition 14.2.8 with $s = t = \varepsilon$, and furthermore $\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{\sigma_n - \varepsilon}$.

From the definition of σ_n , we immediately conclude that extremal collections of tubes always exist for small δ . More precisely, we have the following.

Lemma 14.2.12. *Let $\varepsilon, \delta_0 > 0$. Then there exists a ε -extremal collection of tubes $(\mathbb{T}, Y)_\delta$ for some $\delta \in (0, \delta_0]$.*

The main result of this section says that extremal collections of tubes must be coarsely self-similar at every scale. To state this precisely, we need the following definition.

Definition 14.2.13. Let $(\mathbb{T}, Y)_\delta$ be a collection of δ tubes and let $\tilde{T}$ be a ρ tube that covers each tube in $\mathbb{T}$. The *unit rescaling of $(\mathbb{T}, Y)_\delta$ relative to $\tilde{T}$* is the pair $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ defined as follows. The coaxial lines of the tubes $\hat{T} \in \hat{\mathbb{T}}$ are the images of the coaxial lines of the tubes $T \in \mathbb{T}$ under the map $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^n$, defined as follows. If $\tilde{\ell}$ is the coaxial line of $\tilde{T}$, then ϕ is the composition of a rigid transformation that sends the point $\tilde{\ell} \cap \{x_n = 0\}$ to the origin and sends the line $\tilde{\ell}$ to the x_n -axis, with the anisotropic dilation

$$(x_1, \dots, x_n) \mapsto (cx_1/\rho, cx_2/\rho, \dots, cx_{n-1}/\rho, cx_n). \quad (14.2.14)$$

Each shading $\hat{Y}(\hat{T})$ is the union of δ/ρ cubes that intersect $\phi(Y(T))$. The constant $c = c(n) \sim 1$ is chosen so that the coaxial lines of the tubes in $\hat{T}$ are in $\mathcal{L}_n$, and $\hat{Y}(\hat{T}) \subset \hat{T}$. See Figure 14.2.1.

We can now state the main result of this section. It says that extremal collections of tubes look self-similar at all scales.

Proposition 14.2.14. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes, and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Then there is a refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ and a balanced cover $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ of $(\mathbb{T}', Y')_\delta$ with the following properties.*

- (i) $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is ε -extremal.
- (ii) For each $\tilde{T} \in \tilde{\mathbb{T}}$, the unit rescaling of $(\mathbb{T}'[\tilde{T}], Y')_\delta$ relative to $\tilde{T}$ is ε -extremal.
- (iii) For each $p \in \mathbb{R}^n$, $\#\tilde{\mathbb{T}}(p) \leq \rho^{-\sigma_n - \varepsilon}$
- (iv) For each $p \in \mathbb{R}^n$ and each $\tilde{T} \in \tilde{\mathbb{T}}$, $\#\mathbb{T}'[\tilde{T}](p) \leq (\delta/\rho)^{-\sigma_n - \varepsilon}$.

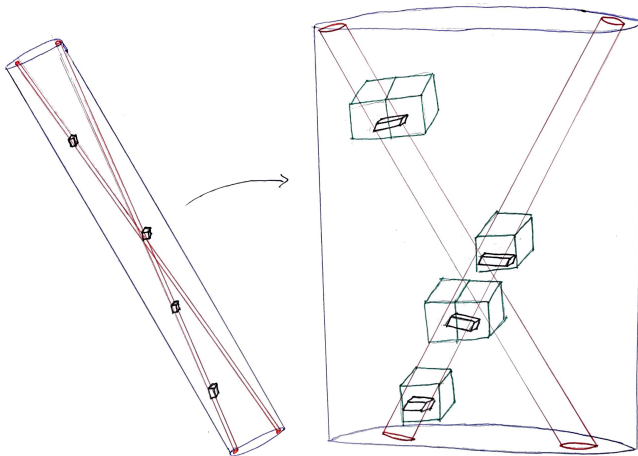


Figure 14.2.1: The unit rescaling of $(\mathbb{T}, Y)_\delta$. For clarity, only a few cubes in $Y(T)$ have been drawn.

Before proving Proposition 14.2.14, we will analyze the unit rescaling of extremal collections of tubes that are covered by a single ρ -tube. A precise formulation is given below. In the arguments that follow, we will make frequent use of the fact that if $(\mathbb{T}, Y)_\delta$ is ε -extremal, then $\delta^{1-n+\varepsilon} \lesssim \#\mathbb{T} \lesssim \delta^{1-n-\varepsilon}$, and if $(\mathbb{T}', Y')_\delta$ is a sub-collection with $\sum_{T \in \mathbb{T}'} |Y'(T)| \geq \delta^{\varepsilon'}$ for some $\varepsilon' \geq \varepsilon$, then $(\mathbb{T}', Y')$ is ε' -extremal.

Lemma 14.2.15. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes that are covered by a ρ -tube $\tilde{T}$, with $\rho \in [\delta^{1-\varepsilon}, 1]$.*

Suppose that $(\mathbb{T}, Y)_\delta$ satisfies Items (a) and (b) from Definition 14.2.8 with η in place of s , and that $(\mathbb{T}, Y)_\delta$ satisfies the following analogue of Item (c): $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^\eta \rho^{n-1}$.

Then there is a refinement $(\mathbb{T}', Y')_\delta$ so that the unit rescaling of $(\mathbb{T}', Y')_\delta$ relative to $\tilde{T}$ satisfies Items (a), (b), and (c) from Definition 14.2.8 with ε in place of s and t . Furthermore,

$$\#\mathbb{T}'(p) \leq (\rho/\delta)^{\sigma+\varepsilon} \quad \text{for all } p \in \mathbb{R}^n. \quad (14.2.15)$$

Proof. To begin, observe that the transformation ϕ from Definition 14.2.13 has the following properties.

- If ℓ_1, ℓ_2 are lines with $d(\tilde{\ell}, \ell_i) \leq \rho$ for $i = 1, 2$. Then

$$\begin{aligned}
 \rho^{-1}d(\ell_1, \ell_2) &\lesssim d(\phi(\ell_1), \phi(\ell_2)) \\
 &\leq \rho^{-1}d(\ell_1, \ell_2), \\
 \rho^{-1}\angle(\text{dir}(\ell_1), \text{dir}(\ell_2)) &\lesssim \angle(\text{dir}(\phi(\ell_1)), \text{dir}(\phi(\ell_2))) \\
 &\leq \rho^{-1}\angle(\text{dir}(\ell_1), \text{dir}(\ell_2)).
 \end{aligned} \tag{14.2.16}$$

- If $\tau \in [\delta, \rho]$ and $T^\dagger$ is a τ -tube covered by $2\tilde{T}$, then $\phi(T^\dagger)$ is contained in the τ/ρ tube with coaxial line $\phi(\ell^\dagger)$.
- For every measurable $X \subset \mathbb{R}^n$ we have

$$|\phi(X)| \sim \rho^{1-n}|X|. \tag{14.2.17}$$

By (14.2.16), there is a refinement of $(\mathbb{T}, Y)_\delta$ whose unit rescaling relative to $\tilde{T}$ consists of essentially distinct tubes. Let $(\mathbb{T}', Y')_\delta$ be a refinement with this property, which is also constant multiplicity, i.e. there is a number μ so that $\#\mathbb{T}'(p) = \mu$ for all $p \in E_{\mathbb{T}'}$. Let $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ be the unit rescaling of $(\mathbb{T}', Y')_\delta$ relative to $\tilde{T}$. In particular, $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies Property (a) from Definition 14.2.8.

By (14.2.17), we have

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \sim \rho^{n-1} \left| \bigcup_{\hat{T} \in \hat{\mathbb{T}}} \hat{Y}(\hat{T}) \right|. \tag{14.2.18}$$

Our next task is to obtain a lower bound on the RHS of (14.2.18). Let $s, t > 0$ be chosen so that

$$N(s, t) \leq \sigma_n + \varepsilon/4. \tag{14.2.19}$$

Decreasing s and t if necessary, we may suppose that $s, t \leq \varepsilon/2$. We will show that $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies Properties (b) and (c) from Definition 14.2.8 for this value of s and t , and all sufficiently small δ .

Property (c) is straightforward: by (14.2.17) we have

$$\sum_{\hat{T} \in \hat{\mathbb{T}}} |\hat{Y}(\hat{T})| \gtrsim \rho^{1-n} \sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^\eta \geq (\rho/\delta)^{\eta/\varepsilon}. \tag{14.2.20}$$

If $\eta, \delta_0 > 0$ are selected sufficiently small, then the LHS of (14.2.20) is at most $(\rho/\delta)^s$.

For property (b), let $\tau \in (\delta/\rho, 1)$. Let $\mathbb{T}^*$ be a collection of $\tau\rho$ -tubes that cover $\mathbb{T}$, at most $\delta^{-\eta}$ of which are essentially parallel to a common $\tau\rho$ -tube (such a collection exists by hypothesis, since $\delta \leq \tau\rho \leq 1$). We can suppose

that each $T^* \in \mathbb{T}^*$ covers at least one tube from $\mathbb{T}$, and thus by the triangle inequality, each T^* is covered by $2\tilde{T}$. Define $\hat{\mathbb{T}}^*$ to be the set of τ tubes whose coaxial lines are given by $\{\phi(\ell^*): \ell^* \in \mathbb{T}^*\}$. By (14.2.16), if $T \in \mathbb{T}$ is covered by $T^* \in \mathbb{T}^*$, then the corresponding tube $\hat{T}$ is covered by $\hat{T}^*$; in particular, $\hat{\mathbb{T}}^*$ covers $\hat{\mathbb{T}}$. By (14.2.16), if δ_0 is chosen sufficiently small then at most $(\delta/\rho)^{-2\eta/\varepsilon}$ tubes from $\hat{\mathbb{T}}^*$ are essentially parallel to a common tube. Thus if $\eta > 0$ is chosen sufficiently small, then $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies property (b) for the value of t specified in (14.2.19).

We have shown that $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies Properties (a), (b), and (c) from Definition 14.2.8 with ε in place of s and t .

All that remains is to establish (14.2.15), i.e. to bound $\mu \leq (\rho/\delta)^{\sigma_n+\varepsilon}$. We have also shown that $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies Properties (a), (b), and (c) from Definition 14.2.8 for the values of s and t specified by (14.2.19). Thus if δ/ρ is sufficiently small, then

$$\left| \bigcup_{\hat{T} \in \hat{\mathbb{T}}} \hat{Y}(\hat{T}) \right| \geq (\delta/\rho)^{\sigma_n+\varepsilon/2}. \quad (14.2.21)$$

We can ensure that δ/ρ is sufficiently small for (14.2.21) to hold by selecting $\delta_0 > 0$ appropriately, since the hypothesis $\rho \geq \delta^{1-\varepsilon}$ forces $\delta/\rho \searrow 0$ as $\delta \searrow 0$.

On the other hand, since $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ satisfies Property (b) from Definition 14.2.8 with $s \leq \varepsilon/2$, we have

$$\mu = \left(\sum_{\hat{T} \in \hat{\mathbb{T}}} |\hat{Y}(\hat{T})| \right) \left(\left| \bigcup_{\hat{T} \in \hat{\mathbb{T}}} \hat{Y}(\hat{T}) \right| \right)^{-1} \lesssim (\delta/\rho)^{\varepsilon/2} \left(\left| \bigcup_{\hat{T} \in \hat{\mathbb{T}}} \hat{Y}(\hat{T}) \right| \right)^{-1} \leq (\delta/\rho)^{\sigma_n+\varepsilon}.$$

□

Proof of Proposition 14.2.14.

Step 1: Constructing $\tilde{\mathbb{T}}$.

To begin, we will construct a set of ρ -tubes that covers a refinement of $\mathbb{T}$, with the property that each $T \in \mathbb{T}$ is contained in exactly one ρ -tube, and T is close to the coaxial line of this ρ -tube. Here are the details. Let $\tilde{\mathbb{T}}_0$ be a set of $\rho/(2n)$ tubes that covers $\mathbb{T}$, so that the coaxial line of each $\tilde{T} \in \tilde{\mathbb{T}}_0$ coincides with the coaxial line of a tube from $\mathbb{T}$, and at most $O(\delta^{-\eta})$ tubes from $\tilde{\mathbb{T}}_0$ are essentially pairwise parallel to a common tube. $\tilde{\mathbb{T}}_0$ can be constructed by first selecting a cover of $\mathbb{T}$ as described in Item (b) of Definition 14.2.4, and then replacing each tube $\tilde{T}$ in this cover by $O(1)$ tubes $\tilde{T}'$ with $d(\tilde{T}, \tilde{T}') \lesssim \rho$.

Next, let $\tilde{\mathbb{T}}_1 \subset \tilde{\mathbb{T}}_0$ be chosen so that the tubes in $2n\tilde{\mathbb{T}}_1$ are essentially distinct; each $T \in \mathbb{T}$ is covered by at most one tube in $2n\tilde{\mathbb{T}}_1$; and

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_1} \sum_{T \in \mathbb{T}[\tilde{T}]} |Y(T)| \sim \sum_{T \in \mathbb{T}} |Y(T)|.$$

Define $\tilde{\mathbb{T}}_2 = 2n\tilde{\mathbb{T}}_1$. Then $\tilde{\mathbb{T}}_2$ is a set of essentially distinct ρ -tubes, and hence they satisfies Property (a) from Definition 14.2.8. Our final set $\tilde{\mathbb{T}}$ will be a subset of $\tilde{\mathbb{T}}_2$, and thus it will continue to satisfy Property (a) from Definition 14.2.8.

Next, we will show that $\tilde{\mathbb{T}}_2$ satisfies Property (b) from Definition 14.2.8 with $t = \eta/\varepsilon$. To do this, let $\tau \in [\rho, 1]$. Then there is a set $\mathbb{T}^\dagger$ of τ tubes that covers $\mathbb{T}$, at most $\delta^{-\eta} \leq \rho^{-\eta/\varepsilon}$ of which are essentially parallel to a common tube. But since each $\tilde{T} \in \tilde{\mathbb{T}}_2$ shares a coaxial line with some $T \in \mathbb{T}$, and $\tau \geq \rho$, $\mathbb{T}^\dagger$ also covers $\tilde{\mathbb{T}}_2$. Our final set $\tilde{\mathbb{T}}$ will be a subset of $\tilde{\mathbb{T}}_2$, and thus it will continue to satisfy Property (b) from Definition 14.2.8 with this same value of t .

Define $\mathbb{T}_2 \subset \mathbb{T}$ to be the set of tubes that are covered by some tube from $\tilde{\mathbb{T}}_1$, and let $Y_2(T) = Y(T)$ for each $T \in \mathbb{T}_2$. Then $(\mathbb{T}_2, Y_2)$ is a refinement of $(\mathbb{T}, Y)$. Furthermore, each $T \in \mathbb{T}_2$ is covered by exactly one tube from $\tilde{\mathbb{T}}_2$; and if $\tilde{T}$ covers T , then T is close to the coaxial line of $\tilde{T}$, in the sense that $\frac{1}{2n}\tilde{T}$ also covers T .

Step 2: Fine scale structure. In this step we will analyze the structure of the sets $\mathbb{T}_2[\tilde{T}]$. By dyadic pigeonholing, we can select a set $\tilde{\mathbb{T}}_3 \subset \tilde{\mathbb{T}}_2$ and a refinement $(\mathbb{T}_3, Y_3)_\delta$ of $(\mathbb{T}_2, Y_2)_\delta$ so that for each $\tilde{T} \in \tilde{\mathbb{T}}_3$, we have

$$\sum_{T \in \mathbb{T}_3[\tilde{T}]} |Y_3(T)| = (\#\tilde{\mathbb{T}}_3)^{-1} \sum_{T \in \mathbb{T}_3} |Y_3(T)|. \quad (14.2.22)$$

Since $O(\delta^{-\eta})$ tubes from $\tilde{\mathbb{T}}_3$ are essentially parallel to a common tube, we have $\#\tilde{\mathbb{T}}_3 \lesssim \delta^{-\eta}\rho^{1-n}$ and thus $\sum_{T \in \mathbb{T}_3[\tilde{T}]} |Y_3(T)| \gtrsim_\delta \delta^{2\eta}\rho^{n-1}$.

If η and δ_0 are chosen sufficiently small, then we can apply Lemma 14.2.15 to each set $(\mathbb{T}_3[\tilde{T}], Y_3)_\delta$ with $\varepsilon^2/3$ in place of ε . Let $(\mathbb{T}_4, Y_4)_\delta$ be the union of the refined collections $\{(\mathbb{T}_3[\tilde{T}], Y_3')_\delta : \tilde{T} \in \tilde{\mathbb{T}}_3\}$ that are the output of Lemma 14.2.15. Let $\tilde{\mathbb{T}}_4 = \tilde{\mathbb{T}}_3$. Then for each $\tilde{T} \in \tilde{\mathbb{T}}_4$, we have

$$\#\mathbb{T}_4[\tilde{T}](p) \leq (\rho/\delta)^{\sigma_n + \varepsilon^2/3}, \quad (14.2.23)$$

so in particular $(\mathbb{T}_4, Y_4)_\delta$ and $\tilde{\mathbb{T}}_4$ satisfy Item (iv) from the statement of Proposition 14.2.14.

Furthermore, for each $\tilde{T} \in \tilde{\mathbb{T}}_4$, the unit rescaling of $(\mathbb{T}_4[\tilde{T}], Y_4)_\delta$ relative to $\tilde{T}$ satisfies Items (a), (b), and (c) from Definition 14.2.8 with $s = t = \varepsilon^2$. Items (a) and (b) will continue to hold if $\mathbb{T}_4$ is replaced by a refinement.

Step 3: Coarse scale structure.

Thus far, we have analyzed the fine-scale behavior of the “thin” tubes $\mathbb{T}_4[\tilde{T}]$ for each $\tilde{T} \in \tilde{\mathbb{T}}_4$. Our next task is to analyze the coarse-scale behavior of the “fat” tubes $\tilde{\mathbb{T}}_4$.

First, we will construct a shading on the fat tubes. Observe that for each $\tilde{T} \in \tilde{\mathbb{T}}_4$ and each ρ -cube Q , we have that

$$Q \cap \bigcup_{T \in \mathbb{T}_4[\tilde{T}]} Y_4(T) \neq \emptyset \text{ implies } Q \subset \tilde{T}. \quad (14.2.24)$$

Indeed, as noted at the end of Step 1, if $T \in \mathbb{T}_4[\tilde{T}]$, then T is covered by $\frac{1}{2n}\tilde{T}$, and thus Q intersects $\frac{1}{2n}\tilde{T}$. But this implies $Q \subset \tilde{T}$.

Each $\tilde{T} \in \tilde{\mathbb{T}}_4$ contains $\sim \rho^{-1}$ ρ -cubes, and for each such cube we have

$$\sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \lesssim \rho \sum_{T \in \mathbb{T}_4[\tilde{T}]} |T|.$$

The directions $\{\text{dir}(T) : T \in \mathbb{T}_4[\tilde{T}]\}$ are contained in a disk of radius ρ in S^{n-1} . Since at most $\delta^{-\eta}$ tubes from $\mathbb{T}_4$ can be essentially parallel to a common tube, we have

$$\#\mathbb{T}_4[\tilde{T}] \lesssim \delta^{-\eta}(\rho/\delta)^{n-1}, \quad (14.2.25)$$

and thus

$$\sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \lesssim \delta^{-\eta} \rho^n. \quad (14.2.26)$$

Let $c_1 \sim 1$ be a constant to be chosen below. We say a ρ -cube is *heavy* for the tube $\tilde{T}$ if

$$\sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \geq c_1 \rho \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T)|.$$

Denote the set of heavy cubes by $\mathcal{Q}_0(\tilde{T})$. If the constant c_1 is chosen appropriately, then for each $\tilde{T} \in \tilde{\mathbb{T}}_4$ we have

$$\sum_{Q \in \mathcal{Q}_0(\tilde{T})} \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \geq \frac{1}{2} \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T)|.$$

By dyadic pigeonholing, there exists a weight w so that if we define

$$\mathcal{Q}_1(\tilde{T}) = \left\{ Q \in \mathcal{Q}_0(\tilde{T}) : w \leq \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| < 2w \right\}, \quad (14.2.27)$$

then

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} \sum_{Q \in \mathcal{Q}_1(\tilde{T})} \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T) \cap Q| \gtrsim_{\delta} \sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} \sum_{T \in \mathbb{T}_4[\tilde{T}]} |Y_4(T)| = \sum_{T \in \mathbb{T}_4} |Y_4(T)|. \quad (14.2.28)$$

Comparing (14.2.26) and (14.2.28), we have

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} \#\mathcal{Q}_1(\tilde{T}) \gtrsim_{\delta} \delta^{2\eta} \rho^{-n}. \quad (14.2.29)$$

For each $\tilde{T} \in \tilde{\mathbb{T}}_4$, define $\tilde{Y}_4(\tilde{T}) = \bigcup_{Q \in \mathcal{Q}_1(\tilde{T})} Q$. By (14.2.24), Y_4 is a shading of $\mathbb{T}_4$. (14.2.29) is precisely the statement that

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} |\tilde{Y}_4(\tilde{T})| \gtrsim_{\delta} \delta^{2\eta}. \quad (14.2.30)$$

Note that $(\tilde{\mathbb{T}}_4, \tilde{Y}_4)_{\rho}$ satisfies Items (a), (b), and (c) from Definition 14.2.8 with $t = \eta/\varepsilon$ and $s = 3\eta/\varepsilon$.

Let $(\mathbb{T}_5, Y_5)_{\delta}$ be a refinement of $(\mathbb{T}_4, Y_4)_{\delta}$ and let $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$ be a refinement of $(\tilde{\mathbb{T}}_4, \tilde{Y}_4)_{\rho}$, so that the following holds

- There is a number μ_{fine} so that for each $\tilde{T} \in \tilde{\mathbb{T}}_5$ and each $p \in E_{\mathbb{T}_5[\tilde{T}]}$, we have $\#\mathbb{T}_5[\tilde{T}](p) = \mu_{\text{fine}}$; we do this by refining the shading Y_4 to ensure that this property holds.
- There is a number $w > 0$ so that for each $\tilde{T} \in \tilde{\mathbb{T}}_5$ and each ρ -cube $Q \subset \tilde{Y}_5(\tilde{T})$, we have $|Q \cap E_{\mathbb{T}_5[\tilde{T}]}| = w$; we do this by refining the shading Y_4 , while preserving the previous property (each δ -cube is either preserved or deleted from all shadings in $\mathbb{T}_5[\tilde{T}]$).
- There is a number μ_{coarse} so that for each $p \in E_{\tilde{\mathbb{T}}_5}$ we have $\mu_{\text{coarse}} \leq \#\tilde{\mathbb{T}}_5(p) < 2\mu_{\text{coarse}}$; we do this by refining the shading $\tilde{Y}_4$, while preserving the previous properties.
- $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$ covers $(\mathbb{T}_5, Y_5)_{\delta}$.

Since $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$ satisfies Items (a), (b), and (c) from Definition 14.2.8 with $t = \eta/\varepsilon$ and $s = 4\eta/\varepsilon$, if η and δ_0 are chosen sufficiently small (which forces ρ to be sufficiently small), then $|E_{\tilde{\mathbb{T}}_5}| \geq \rho^{\sigma_n + \varepsilon/2}$, and thus

$$\mu_{\text{coarse}} \lesssim \rho^{-\sigma_n - \varepsilon/2 - \eta/\varepsilon}. \quad (14.2.31)$$

Thus if η is chosen sufficiently small, then Item (iii) from the statement of Proposition 14.2.14 holds.

At this point, $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$ is a balanced cover of $(\mathbb{T}_5, Y_5)_{\delta}$, and this pair satisfies Items (iv) and (iii). In addition, $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_{\rho}$ satisfies Items (a), (b), and (c) from Definition 14.2.8 with $t = \eta/\varepsilon$ and $s = 4\eta/\varepsilon$, and for each $\tilde{T} \in \tilde{\mathbb{T}}_5$, the unit rescaling of $(\mathbb{T}_5[\tilde{T}], Y_5)_{\delta}$ relative to $\tilde{T}$ satisfies Items (a) and (b) from Definition 14.2.8 with $t = \varepsilon^2$.

Next we will analyze the multiplicities μ_{fine} and μ_{coarse} defined above. (14.2.31) gives an upper bound on μ_{coarse} , while by (14.2.23) we have $\mu_{\text{fine}} \leq (\rho/\delta)^{\sigma_n + \varepsilon^2/3} \leq \rho^{-\varepsilon/3}(\rho/\delta)^{\sigma_n}$. On the other hand,

$$\int \sum_{T \in \mathbb{T}_5} \chi_{Y_5(T)}(p) dp \gtrsim_{\delta} \delta^{\eta}. \quad (14.2.32)$$

Since the original collection of tubes $(\mathbb{T}, Y)_\delta$ is η -extremal, the integrand in (14.2.32) is supported on a set of size at most $\delta^{\sigma_n - \eta}$, and it is pointwise bounded by $\mu_{\text{fine}}\mu_{\text{coarse}}$. We conclude that

$$\mu_{\text{fine}} \gtrsim_\delta \delta^{2\eta} \rho^{\varepsilon/2 + \eta/\varepsilon} (\rho/\delta)^{\sigma_n}, \quad \mu_{\text{coarse}} \gtrsim_\delta \delta^{2\eta} \rho^{-\sigma_n + \varepsilon/3}. \quad (14.2.33)$$

Selecting $\eta > 0$ sufficiently small, we conclude that $|E_{\tilde{\mathbb{T}}_5}| \leq \rho^{\sigma_n - \varepsilon}$, and for each $\tilde{T} \in \tilde{\mathbb{T}}_5$, the unit-rescaling $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$ of each set $(\mathbb{T}_5[\tilde{T}], Y_5)_\delta$ satisfies $|E_{\hat{T}}| \leq (\delta/\rho)^{\sigma_n - \varepsilon}$.

We are almost done, except it is possible that the unit-rescaling described above might fail to satisfy

$$\sum_{\hat{T} \in \hat{\mathbb{T}}} |\hat{Y}(\hat{T})| \geq (\rho/\delta)^\varepsilon \quad (14.2.34)$$

for some choices of $\tilde{T} \in \tilde{\mathbb{T}}_5$. To fix this problem, we define $\tilde{\mathbb{T}}_6 \subset \tilde{\mathbb{T}}_5$ to be the set of tubes for which (14.2.34) holds. We define $\tilde{Y}_6 = \tilde{Y}_5$, $Y_6 = Y_5$, and $\mathbb{T}_6 = \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}_6} \mathbb{T}_5[\tilde{T}]$. Then $(\mathbb{T}_6, Y_6)$ is a refinement of $(\mathbb{T}, Y)$ which, together with $(\tilde{\mathbb{T}}_6, \tilde{Y}_6)_\rho$, satisfies the conclusions of Proposition 14.2.14. $\square$

Remark 14.2.16. The conclusions of Lemma 14.2.12 and Proposition 14.2.14 are the only consequences of stickiness that we will use to prove Theorem 14.2.3. In particular, if Lemma 14.2.12 and Proposition 14.2.14 hold for some other class of Kakeya sets, then it should be possible to prove the analogue of Theorem 14.2.3 in that setting as well.

We conclude this section with a few final observations about extremal collections of tubes. First, if $(\mathbb{T}, Y)_\delta$ is an extremal collection of tubes, then the directions of the tubes passing through a typical point cannot focus too tightly. This phenomena is sometimes called “robust transversality.” We will record a precise version below.

Lemma 14.2.17. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes. Then after replacing $(\mathbb{T}, Y)_\delta$ by a refinement, for each $p \in \mathbb{R}^n$ and each $v \in S^{n-1}$, we have*

$$\#\{T \in \mathbb{T}(p) : \angle(v, \text{dir}(T)) < \delta^\varepsilon\} \leq \delta^{-\sigma_n + \varepsilon\sigma_n/2}. \quad (14.2.35)$$

Proof. Apply Proposition 14.2.14 with $\varepsilon\sigma_n/2$ in place of ε , and $\rho = \delta^\varepsilon$. Since the resulting collection $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ of tubes are essentially distinct, the δ -tubes contributing to (14.2.35) must be covered by $O(1)$ ρ -tubes from $\tilde{\mathbb{T}}$. (14.2.35) now follows from Item (i) from Proposition 14.2.14. $\square$

The next lemma says that extremal collections of tubes remain extremal after a mild re-scaling. We include the parameter bookkeeping because this lemma is used twice below with different anisotropies.

Lemma 14.2.18. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes. Let $Q \subset [-1, 1]^n$ be an axis-parallel rectangular prism, and suppose $\sum_{T \in \mathbb{T}} |Y(T) \cap Q| \geq \delta^\eta$. Let ϕ be a translation composed with a dilation of the form $(x_1, \dots, x_n) \mapsto (r_1 x_1, \dots, r_n x_n)$, where $\delta^\eta \leq r_i \leq \delta^{-\eta}$, and suppose $\phi(Q) \subset [-1, 1]^n$.*

Then the image of $(\mathbb{T}, Y \cap Q)_\delta$ under ϕ is ε -extremal. More precisely, there exists an ε -extremal collection of tubes $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ for some $\delta \leq \rho \leq \delta^{1-\eta}$ so that each $\tilde{T} \in \tilde{\mathbb{T}}$ has a coaxial line of the form $\phi(\ell)$, where ℓ is the coaxial line of a tube $T \in \mathbb{T}$. Furthermore, for each such pair $T, \tilde{T}$, we have that $\tilde{Y}(\tilde{T})$ is the set of ρ -cubes that intersect $\phi(Y(T) \cap Q)$.

Proof. Write $A = \text{diag}(r_1, \dots, r_n)$ for the linear part of ϕ , and put $r_- = \min_i r_i$ and $r_+ = \max_i r_i$. We first record the three distortion estimates that will be used. If C is a dimensional constant, then

$$\delta^\eta |x - y| \leq |A(x - y)| \leq \delta^{-\eta} |x - y|, \quad \delta^{n\eta} \leq |\det A| \leq \delta^{-n\eta}, \quad (14.2.36)$$

and the image of a δ -cube is contained in a cube of side $C\delta r_+$. Conversely, the inverse image of a u -cube is contained in a cube of side Cu/r_- . The same statements hold for distances and angles between lines, with an additional factor between $\delta^{Cn\eta}$ and $\delta^{-Cn\eta}$. To see the angle assertion, represent a line meeting $[-2, 2]^n$ by its intersection point with a fixed transverse hyperplane and its unit direction. On each of the finitely many charts where one direction coordinate has size at least $1/(2\sqrt{n})$, the map $(p, v) \mapsto (Ap, Av/|Av|)$ and its inverse have derivative at most $C(r_+/r_-)^2 \leq C\delta^{-4\eta}$. This proves the assertion after changing C_n .

Choose a scale ρ comparable to

$$\max(\delta, C\delta r_+). \quad (14.2.37)$$

All harmless dimensional constants in this proof can be absorbed by first proving the assertion with an auxiliary exponent 2η and then replacing 2η by η . With this convention,

$$\delta \leq \rho \leq \delta^{1-\eta}.$$

For every $T \in \mathbb{T}$, let T^ϕ be a ρ -tube with coaxial line $\phi(\ell_T)$ that contains every ρ -cube meeting $\phi(T \cap Q)$. Such a tube exists by (14.2.37). Put

$$Y^\phi(T^\phi) = \bigcup \{q : q \text{ is a } \rho\text{-cube and } q \cap \phi(Y(T) \cap Q) \neq \emptyset\}.$$

The assignment $T \mapsto T^\phi$ need not give essentially distinct tubes. The line-distortion estimate above and the fact that the original tubes are essentially distinct show that at most

$$\delta^{-C_n\eta} \quad (14.2.38)$$

of the lines $\phi(\ell_T)$ can lie in one ρ -ball of line space. Indeed, their inverse images are δ -separated and lie in a line-space ball of radius at most $C\delta^{-C_n\eta}\rho$, whose packing number at scale δ is at most $\delta^{-C_n\eta}$ after increasing C_n . Colour the corresponding bounded-degree proximity graph with $O(\delta^{-C_n\eta})$ colours. Retain a colour for which $\sum_T |Y(T) \cap Q|$ is maximal, and denote the resulting essentially distinct image tubes and shadings by $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. We have

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}} |\tilde{Y}(\tilde{T})| \gtrsim \delta^{C_n\eta} |\det A| \sum_{T \in \mathbb{T}} |Y(T) \cap Q| \geq \delta^{C_n\eta}. \quad (14.2.39)$$

Here and below C_n is allowed to increase; the first inequality also uses the fact that cubifying a set can only enlarge its measure.

We next verify the covering condition. Fix $u \in [\rho, 1]$. Pulling an u -tube back by ϕ produces a length- $O(\delta^{-\eta})$ tube-like ellipsoid of transverse width at most $Cu\delta^{-\eta}$. Divide its long axis into unit pieces and cover each piece by tubes of radius

$$u' = \min(1, Cu\delta^{-C_n\eta}).$$

The number of pieces and transverse translates is at most $\delta^{-C_n\eta}$. Apply the covering property of $(\mathbb{T}, Y)_\delta$ at scale u' when $u' < 1$, and use the trivial bounded cover when $u' = 1$. After applying ϕ again, we obtain a cover of $\tilde{\mathbb{T}}$ by u -tubes in which at most

$$\delta^{-C_n\eta} \quad (14.2.40)$$

members are essentially parallel to a common u -tube. This includes the original bound $\delta^{-\eta}$ and all subdivision and distortion losses. Since $\rho \geq \delta$ and $\rho \leq \delta^{1-C_n\eta}$, choosing $\eta \leq c_n\varepsilon$ and then δ_0 small makes $\delta^{-C_n\eta} \leq \rho^{-\varepsilon}$. Thus property (b) holds with ε .

The same choice of η turns (14.2.39) into

$$\sum_{\tilde{T}} |\tilde{Y}(\tilde{T})| \geq \rho^\varepsilon,$$

so property (c) holds. It remains to estimate the union. Because the original shading is a union of δ -cubes, bounded overlap of the affine images of those cubes gives

$$\begin{aligned} \left| \bigcup_{\tilde{T}} \tilde{Y}(\tilde{T}) \right| &\lesssim \delta^{-C_n\eta} |\det A| \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \\ &\leq \delta^{\sigma_n - C_n\eta}. \end{aligned}$$

To justify the first line, partition every affine image of a δ -cube into ρ -cubes. The ratio between the volume of the resulting cubical cover and $|\det A|\delta^n$ is bounded by a product of n axis ratios, each at most $C\delta^{-2\eta}$; original cubes with disjoint interiors have bounded overlap after this fixed enlargement.

Finally, uniformly for $\delta \leq \rho \leq \delta^{1-C_n\eta}$,

$$\delta^{\sigma_n - C_n\eta} \leq \rho^{\sigma_n - \varepsilon}$$

when $\eta \leq c_n\varepsilon$ (decrease c_n once more if $\sigma_n = 0$). Hence the union bound in Definition 14.2.11 holds. We have verified all four requirements for ε -extremality, and the construction of $\tilde{Y}$ is exactly the one asserted in the statement. $\square$

14.2.4 Planiness and graininess

For the remainder of the paper we will specialize to the case $n = 3$, and we will define $\sigma = \sigma_3$. Our goal is to prove that $\sigma = 0$. In this section we will establish the existence of extremal collections of tubes $(\mathbb{T}, Y)_\delta$ with additional structural properties. The first of these is the existence of a Lipschitz plane map.

Definition 14.2.19. Let $(\mathbb{T}, Y)_\delta$ be a set of δ -tubes. A *plane map* for $(\mathbb{T}, Y)_\delta$ is a function $V: E_{\mathbb{T}} \rightarrow S^2$ so that

$$|\operatorname{dir}(T) \cdot V(p)| \leq \delta \quad \text{for all } p \in E_{\mathbb{T}}, T \in \mathbb{T}(p).$$

Next, we will show that each $\{z = z_0\}$ slice of theakeya set is contained in a union of long thin rectangles, which we will call global grains. These global grains will be arranged like an Ahlfors-David regular set. The specific property we need is the following.

Definition 14.2.20. For $n \geq 1$, $\alpha \in (0, n]$, $C \geq 1$ and $\delta > 0$, We say a set $E \subset \mathbb{R}^n$ is a $(\delta, \alpha, C)_n$ -ADset if for all $x \in \mathbb{R}^n$, all $\rho \geq \delta$, and all $r \geq \rho$, we have

$$\mathcal{E}_\rho(E \cap B(x, r)) \leq C(r/\rho)^\alpha. \tag{14.2.41}$$

Remark 14.2.21. Note that if E is a $(\delta, \alpha, C)_n$ -ADset, then so is every subset of E . This observation will be used frequently in the arguments that follow. Definition 14.2.20 only imposes upper bounds on the size of $\mathcal{E}_\rho(E \cap B(x, r))$, while Ahlfors regularity usually requires a matching lower bound. However, if E is a $(\delta, \alpha, C)_n$ -ADset that has bounded diameter and $\mathcal{E}_\delta(E)$ has size roughly $\delta^{-\alpha}$, then a large subset of E will satisfy a lower bound analogue of (14.2.41).

With these definitions, we can now state the main result of this section.

Proposition 14.2.22. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and a ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the following two properties.*

1. $E_{\mathbb{T}}$ is a union of global grains with Lipschitz slope function.

There is a 1-Lipschitz function $f: [-1, 1] \rightarrow \mathbb{R}$ so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.

2. $E_{\mathbb{T}}$ is a union of local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_\delta$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

Remark 14.2.23. Item 1 implies that each slice $E_{\mathbb{T}} \cap \{z = z_0\}$ is contained in a union of at most $\delta^{\sigma-1-\varepsilon}$ rectangles of dimensions roughly $1 \times \delta$, and thus by Fubini, $|E_{\mathbb{T}}| \lesssim \delta^{\sigma-\varepsilon}$. On the other hand, this containment is nearly sharp, since $|E_{\mathbb{T}}| \geq \delta^{\sigma+\varepsilon}$.

Item 2 implies that each ball $B(p, \rho^{1/2}) \cap N_\rho(E_{\mathbb{T}})$ is a union of at most $\rho^{\sigma/2-1/2}$ parallel rectangular slabs of dimensions roughly $\rho^{1/2} \times \rho^{1/2} \times \rho$. Since $E_{\mathbb{T}}$ can be covered by $\delta^{-O(\varepsilon)} \rho^{\sigma/2-3/2}$ balls of radius $\rho^{1/2}$, this implies that $|N_\rho(E_{\mathbb{T}})| \lesssim \delta^{-O(\varepsilon)} \rho^\sigma$. On the other hand, this containment is nearly sharp, since $|N_\rho(E_{\mathbb{T}})| \gtrsim \delta^{-O(\varepsilon)} \rho^\sigma$.

Remark 14.2.24. All of the arguments in this and previous sections also apply to the Heisenberg group example (14.2.2) (this would involve working in $\mathbb{C}^3$ rather than $\mathbb{R}^3$, but all of the arguments thus far could be translated to that setting). For the Heisenberg group example, we would have that the global slope function $f: \mathbb{C} \rightarrow \mathbb{C}$ is given by $f(z) = \bar{z}$, which is 1-Lipschitz.

Our main tool for proving Proposition 14.2.22 will be the multilinearakeya theorem of Bennett-Carbery-Tao [1]. The version stated below can be found in [8].

Theorem 14.2.25 (Multilinearakeya). *Let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^3$. Then*

$$\int \left(\sum_{T_1, T_2, T_3 \in \mathbb{T}} \chi_{T_1} \chi_{T_2} \chi_{T_3} |v_1 \wedge v_2 \wedge v_3| \right)^{1/2} \lesssim (\delta^2 \# \mathbb{T})^{3/2}, \quad (14.2.42)$$

where in the above expression, $v_i = \text{dir}(T_i)$ is a unit vector pointing in the direction of the tube T_i , and $|v_1 \wedge v_2 \wedge v_3|$ is the area of the parallelepiped spanned by v_1, v_2 , and v_3 .

14.2.4.1 Finding a plane map

In this section, we will show that every extremal set of tubes has a refinement with a plane map. First, we will show that every extremal set of tubes has a refinement that “weakly” agrees with a plane map.

Lemma 14.2.26. *There exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal set of δ -tubes. Then there is a refinement $(\mathbb{T}', Y')_\delta$ and a function $V: E_{\mathbb{T}} \rightarrow S^2$ with the following property: for each $p \in E_{\mathbb{T}'}$, we have*

$$|V(p) \cdot \text{dir}(T)| \leq \delta^{\sigma/2} \quad \text{for all } T \in \mathbb{T}'(p). \quad (14.2.43)$$

Proof. Define $\varepsilon = \sigma/4$. We will select η sufficiently small to ensure that $\eta \leq \min(\varepsilon/40, \varepsilon\sigma/24)$. Apply Lemma 14.2.17 to $(\mathbb{T}, Y)_\delta$ with $\varepsilon/4$ in place of ε to obtain a refinement $(\mathbb{T}_1, Y_1)_\delta$ so that for each $p \in \mathbb{R}^3$ and each unit vector $v \in S^2$, we have

$$\#\{T \in \mathbb{T}_1(p) : \angle(v, \text{dir}(T)) \leq \delta^{\varepsilon/4}\} \leq \delta^{-\sigma + (\varepsilon/4)\sigma/2} \leq \delta^{-\sigma + 3\eta}. \quad (14.2.44)$$

Let $(\mathbb{T}_2, Y_2)$ be a constant multiplicity refinement of $(\mathbb{T}_1, Y_1)$, with multiplicity $\delta^{-\sigma + 2\eta} \lesssim \mu \lesssim \delta^{-\sigma - 2\eta}$. By Theorem 14.2.25 we have

$$\int \left(\sum_{T_1, T_2, T_3 \in \mathbb{T}_2} \chi_{Y_2(T_1)} \chi_{Y_2(T_2)} \chi_{Y_2(T_3)} |v_1 \wedge v_2 \wedge v_3| \right)^{1/2} \lesssim (\delta^2 \#\mathbb{T}_2)^{3/2} \lesssim \delta^{-(3/2)\eta}. \quad (14.2.45)$$

We say a point $p \in E_{\mathbb{T}_2}$ is *broad* if

$$\#\{(T_1, T_2, T_3) \in (\mathbb{T}_2(p))^3 : |v_1 \wedge v_2 \wedge v_3| \geq \delta^{\sigma - 10\eta}\} \geq \delta^{-3\sigma + 3\eta},$$

otherwise we say it is *narrow* (in the above expression and the expressions to follow, we set $v_i = \text{dir}(T_i)$). Both the set of broad and the set of narrow points are unions of δ -cubes. Observe that each broad point contributes at least $\delta^{-\sigma - (7/2)\eta}$ to the integrand on the LHS of (14.2.45), and thus by (14.2.45) the measure of the set of broad points $\lesssim \delta^{\sigma + 2\eta}$. On the other hand, $E_{\mathbb{T}_2}$ has measure $\gtrsim \delta^{\sigma + \eta}$.

Define $Y_3(T) = \{p \in Y_2(T) : p \text{ narrow}\}$, and define $\mathbb{T}_3 = \mathbb{T}_2$. Then it is still true that $\#\mathbb{T}_3(p) = \mu$ for each $p \in E_{\mathbb{T}_3}$, and $|E_{\mathbb{T}_3}| \geq \frac{1}{2}|E_{\mathbb{T}_2}|$.

For each $p \in E_{\mathbb{T}_3}$, we have

$$\#\{(T_1, T_2, T_3) \in \mathbb{T}_3(p)^3 : |v_1 \wedge v_2 \wedge v_3| < \delta^{\sigma - 10\eta}\} \geq \mu^3 - \delta^{-3\sigma + 3\eta} \geq \frac{1}{2}\mu^3. \quad (14.2.46)$$

On the other hand, by (14.2.44), for each $i, j \in \{1, 2, 3\}$ with $i \neq j$ we have

$$\#\{(T_1, T_2, T_3) \in \mathbb{T}_3(p)^3 : |v_i \wedge v_j| < \delta^{\varepsilon/4}\} \leq \mu^2 \delta^{-\sigma + 3\eta} \leq \frac{1}{100}\mu^3. \quad (14.2.47)$$

Comparing (14.2.46) and (14.2.47), we conclude that there are at least $\mu^3/4$ triples (T_1, T_2, T_3) with $|v_1 \wedge v_2 \wedge v_3| < \delta^{\sigma - 2\eta} \leq \delta^{\sigma - \varepsilon/4}$ and $|v_i \wedge v_j| \geq \delta^{\varepsilon/4}$ for

each $i \neq j$. By pigeonholing, there exists T_1, T_2 so that there are at least $\mu/4$ choices of T_3 so that (T_1, T_2, T_3) is a triple of the above form. Define

$$V(p) = \frac{\text{dir}(T_1) \times \text{dir}(T_2)}{|\text{dir}(T_1) \times \text{dir}(T_2)|}.$$

Then for each T_3 of the type described above, we have

$$|V(p) \cdot v(T_3)| = \frac{|\text{dir}(T_1) \wedge \text{dir}(T_2) \wedge \text{dir}(T_3)|}{|\text{dir}(T_1) \times \text{dir}(T_2)|} \leq \frac{\delta^{\sigma-10\eta}}{\delta^{\varepsilon/4}} \leq \delta^{\sigma/2}. \quad (14.2.48)$$

Finally, let $\mathbb{T}' = \mathbb{T}_3$, and define the shading

$$Y'(T) = \{p \in Y_3(T) : |V(p) \cdot v(T)| \leq \delta^{\sigma/2}\}. \quad (14.2.49)$$

$(\mathbb{T}', Y)_\delta$ is a refinement of $(\mathbb{T}, Y)_\delta$ that satisfies (14.2.43). $\square$

The next lemma shows that by moving to a coarser scale, we can ensure that a suitable thickening of the tubes from Lemma 14.2.26 strongly agrees with their plane map.

Lemma 14.2.27. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal set of δ -tubes, and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Suppose there exists a function $V : E_{\mathbb{T}} \rightarrow S^2$ so that for each $p \in E_{\mathbb{T}}$, we have*

$$|V(p) \cdot v(T)| \leq \rho/2 \quad \text{for all } T \in \mathbb{T}(p). \quad (14.2.50)$$

Then there exists a refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ and a ε -extremal set of ρ -tubes $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ with plane map $\tilde{V}$ so that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ covers $(\mathbb{T}', Y')_\delta$. Furthermore, $\tilde{V}$ is constant on each ρ -cube, and V and $\tilde{V}$ are consistent on each ρ -cube, in the sense that there exists $p \in Q$ with $\tilde{V}(Q) = V(p)$.

Proof. Let $\varepsilon_1 > 0$ be a small constant to be chosen below. We will select η very small compared to ε_1 , and ε_1 small compared to ε . Apply Proposition 14.2.14 to $(\mathbb{T}, Y)_\delta$, with ε_1 in place of ε and with ρ as stated in the lemma. We obtain a refinement $(\mathbb{T}_1, Y_1)_\delta$ of $(\mathbb{T}, Y)_\delta$ and a ε_1 -extremal collection of ρ -tubes $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ that covers $(\mathbb{T}_1, Y_1)_\delta$.

After refining $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ (which induces a refinement on $(\mathbb{T}_1, Y_1)_\delta$), we may suppose that for each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}_1}$, there is a point $p_Q \in Q$ with $\#\mathbb{T}_1(p_Q) \gtrsim \delta^{-\sigma+2\eta}$. We say the tube $\tilde{T} \in \tilde{\mathbb{T}}_1$ is *associated* to the ρ -cube Q if $p_Q \in \bigcup_{T \in \mathbb{T}_1[\tilde{T}]} Y_1(T)$; this necessarily implies $Q \subset \tilde{Y}_1(\tilde{T})$. By Item (iv) from Proposition 14.2.14, each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}_1}$ satisfies

$$\#\{\tilde{T} \in \tilde{\mathbb{T}}_1 : \tilde{T} \text{ associated to } Q\} \gtrsim \frac{\delta^{-\sigma+2\eta}}{(\rho/\delta)^{\sigma+\varepsilon_1}} \gtrsim \delta^{2\eta+\varepsilon_1} \rho^{-\sigma}.$$

On the other hand, by Item (iii) from Proposition 14.2.14, the number of tubes in $\tilde{\mathbb{T}}_1$ whose shading contains Q is at most $\rho^{-\sigma-\varepsilon_1}$. Thus if we define $\tilde{Y}_2(\tilde{T})$ to be the union of ρ -cubes associated to $\tilde{T}$ and define $\tilde{\mathbb{T}}_2 = \tilde{\mathbb{T}}_1$ then

$$\sum_{\tilde{T} \in \tilde{\mathbb{T}}_2} |\tilde{Y}_2(\tilde{T})| \gtrsim \delta^{2\eta+2\varepsilon_1} \sum_{\tilde{T} \in \tilde{\mathbb{T}}_1} |\tilde{Y}_1(\tilde{T})| \gtrsim \delta^{2\eta+3\varepsilon_1}. \quad (14.2.51)$$

Selecting ε_1 sufficiently small, we can ensure that the LHS of (14.2.51) is at least ρ^ε .

Next, we shall define the function $\tilde{V}(p): E_{\tilde{\mathbb{T}}_2} \rightarrow S^2$ as follows. For each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}_2}$, we define $\tilde{V}(p) = V(p_Q)$ for all $p \in Q$. Clearly $\tilde{V}$ is constant on ρ -cubes, and satisfies the consistency condition with V stated in Lemma 14.2.27. It remains to show that $\tilde{V}$ is a plane map for $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$. Let $p \in E_{\tilde{\mathbb{T}}_2}$ be contained in a ρ -cube Q , and let $\tilde{T} \in \tilde{\mathbb{T}}_2(p)$. Then there exists a tube $T \in \mathbb{T}_2(T) \cap \mathbb{T}(p)$. We have

$$\begin{aligned} |\tilde{V}(p) \cdot \text{dir}(\tilde{T})| &= |V(p_Q) \cdot \text{dir}(\tilde{T})| \\ &\leq |V(p_Q) \cdot \text{dir}(T)| + |V(p_Q) \cdot (\text{dir}(\tilde{T}) - \text{dir}(T))| \\ &\leq \rho/2 + |\text{dir}(\tilde{T}) - \text{dir}(T)| \leq \rho. \end{aligned}$$

where the final inequality used (14.2.50) and the observation that $|\text{dir}(\tilde{T}) - \text{dir}(T)| \leq \rho/2$ (since $\tilde{T}$ covers T). $\square$

14.2.4.2 Lipschitz regularity of the plane map

In this section we will show that if an extremal collection of tubes has a plane map, then after restricting to a sub-collection, this plane map must be Lipschitz with controlled Lipschitz norm.

Lemma 14.2.28. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with plane map V , and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that if $p, q \in E_{\mathbb{T}'}$ are contained in a common ρ -cube, then $|V(p) - V(q)| \leq \rho$.*

Proof. Let $\varepsilon_1, \varepsilon_2 > 0$ be small constants to be chosen below. We will select η very small compared to ε_1 , ε_1 very small compared to ε_2 , and ε_2 very small compared to ε .

Let $(\mathbb{T}_1, Y_1)_\delta$ be the refinement obtained by applying Lemma 14.2.17 to $(\mathbb{T}, Y)_\delta$ with ε_1 in place of ε . Next, apply Proposition 14.2.14 to $(\mathbb{T}_1, Y_1)_\delta$ with ε_2 in place of ε , and ρ as specified in the lemma. We obtain a refinement $(\mathbb{T}_2, Y_2)_\delta$ of $(\mathbb{T}_1, Y_1)_\delta$ and a ε_2 -extremal set of ρ -tubes $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$.

After refining $(\mathbb{T}_2, Y_2)_\delta$ and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$, we may suppose that $(\mathbb{T}_2, Y_2)_\delta$ has constant multiplicity $\mu \gtrsim \delta^{-\sigma+2\eta}$, and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is still a balanced cover of

$(\mathbb{T}_2, Y_2)_\delta$. If $\eta > 0$ is selected sufficiently small (depending on ε_1 and σ), then for each $p \in E_{\mathbb{T}_2}$, there are at least $\mu^2/2$ pairs $T, T' \in \mathbb{T}_2(p)$ with $\angle(\text{dir}(T), \text{dir}(T')) \geq \delta^{\varepsilon_1}$. Thus by Item (iv) from Proposition 14.2.14, there are at least $(\mu^2/2)/((\rho/\delta)^{\sigma+\varepsilon_2})^2 \gtrsim \delta^{4\eta+2\varepsilon_2} \rho^{-2\sigma}$ pairs $\tilde{T}, \tilde{T}' \in \tilde{\mathbb{T}}$ with $p \in \bigcup_{T \in \mathbb{T}_2[\tilde{T}]} Y_2(T)$, $p \in \bigcup_{T \in \mathbb{T}_2[\tilde{T}']} Y_2(T)$, and $\angle(\text{dir}(\tilde{T}), \text{dir}(\tilde{T}')) \geq \delta^{\varepsilon_1} - 2\rho$; we will choose ε_1 sufficiently small so that $\delta^{\varepsilon_1} - 2\rho \geq \delta^{\varepsilon_1}/2$. In particular, for each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}}$, we have Let $\mathcal{A}_Q$ be the set of triples $(p, \tilde{T}, \tilde{T}')$ with $p \in Q \cap E_{\mathbb{T}_2}$, with p in both corresponding shaded unions, and with $\angle(\text{dir}(\tilde{T}), \text{dir}(\tilde{T}')) \geq \delta^{\varepsilon_1}/2$. Then

$$\begin{aligned}
 |\mathcal{A}_Q| &\gtrsim |Q \cap E_{\mathbb{T}_2}| \delta^{4\eta+2\varepsilon_2} \rho^{-2\sigma} \\
 &\gtrsim \delta^{4\eta+4\varepsilon_2} |Q \cap E_{\mathbb{T}_2}| (\#\tilde{\mathbb{T}}(Q))^2.
 \end{aligned} \tag{14.2.52}$$

where $|\cdot|$ on the LHS of (14.2.52) denotes the product of Lebesgue measure in $\mathbb{R}^3$ and counting measure, and the final inequality used Item (iii) from Proposition 14.2.14. Thus by pigeonholing, we can select two tubes from $\tilde{\mathbb{T}}$, which we will denote by $\tilde{T}_Q$ and $\tilde{T}'_Q$, with $\angle(\text{dir}(\tilde{T}_Q), \text{dir}(\tilde{T}'_Q)) \geq \delta^{\varepsilon_1}/2$, so that

$$\left| Q \cap \left(\bigcup_{T \in \mathbb{T}_2[\tilde{T}_Q]} Y_2(T) \right) \cap \left(\bigcup_{T \in \mathbb{T}_2[\tilde{T}'_Q]} Y_2(T) \right) \right| \gtrsim \delta^{4\eta+4\varepsilon_2} |Q \cap E_{\mathbb{T}_2}|. \tag{14.2.53}$$

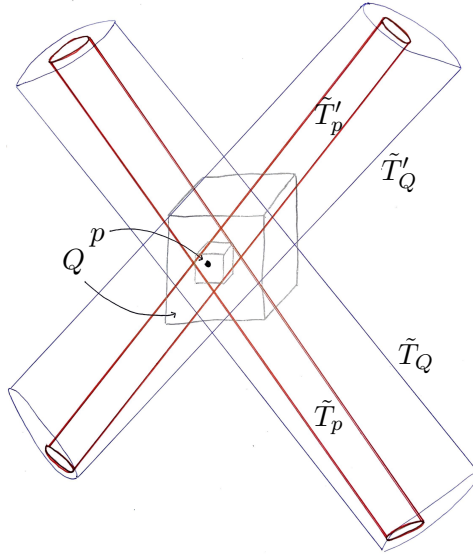


Figure 14.2.2: $V(p)$ is almost orthogonal to $\text{dir}(\tilde{T}_Q)$ and $\text{dir}(\tilde{T}'_Q)$.

Define F_Q to be the set on the LHS of (14.2.53). Let $\mathbb{T}_3 = \mathbb{T}_2$, and for each $T \in \mathbb{T}_3$ define

$$Y_3(T) = Y_2(T) \cap \bigcup_Q F_Q.$$

Then $\sum_{T \in \mathbb{T}_3} |Y_3(T)| \gtrsim \delta^{5\eta+4\varepsilon_2}$. Let Q be a ρ -cube, and let $p \in E_{\mathbb{T}_3} \cap Q$. Then there exists $T_p \in \mathbb{T}_3[\tilde{T}_Q]$ and $T'_p \in \mathbb{T}_3[\tilde{T}'_Q]$ so that $p \in Y_3(T_p) \cap Y_3(T'_p)$, and hence

$$|V(p) \cdot \text{dir}(\tilde{T}_Q)| \leq |V(p) \cdot \text{dir}(T_p)| + |V(p) \cdot (\text{dir}(\tilde{T}_Q) - \text{dir}(T_p))| \leq \delta + \rho,$$

and similarly for $|V(p) \cdot \text{dir}(\tilde{T}'_Q)|$. See Figure 14.2.2. Since $\angle(\text{dir}(\tilde{T}_Q), \text{dir}(\tilde{T}'_Q)) \gtrsim \delta^{\varepsilon_1}$, we conclude that for all $p \in Q$, we have

$$\angle\left(V(p), \frac{\text{dir}(\tilde{T}_Q) \times \text{dir}(\tilde{T}'_Q)}{|\text{dir}(\tilde{T}_Q) \times \text{dir}(\tilde{T}'_Q)|}\right) \lesssim \delta^{-\varepsilon_1} \rho. \quad (14.2.54)$$

(14.2.54) says that $(\mathbb{T}_3, Y_3)$ almost satisfies the conclusion of Lemma 14.2.28, except the RHS of (14.2.54) is $\delta^{-\varepsilon_1} \rho$ rather than ρ . To fix this, we will pigeonhole to select a set $E_4 \subset E_{\mathbb{T}_3}$ that is a union of δ -cubes so that (i): for each ρ -cube Q , $\angle(V(p), V(q)) \leq \rho$ for all $p, q \in Q$, and (ii) if we define $Y_4(T) = Y_3(T) \cap E_4$ and $\mathbb{T}_4 = \mathbb{T}_3$, then $\sum_{T \in \mathbb{T}_4} |Y_4(T)| \gtrsim \delta^{2\varepsilon_1} \sum_{T \in \mathbb{T}_3} |Y_3(T)|$. If we choose $\varepsilon_1, \varepsilon_2, \eta$ sufficiently small, then $(\mathbb{T}_4, Y_4)_\delta$ will be ε -extremal. $\square$

Applying Lemma 14.2.28 and then deleting a constant fraction of the ρ -cubes to obtain a 2ρ -separated set of cubes, we obtain the following

Corollary 14.2.29. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with plane map V , and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that if $p, q \in E_{\mathbb{T}'}$ satisfy $|p - q| \leq \rho$, then $|V(p) - V(q)| \leq 2\rho$.*

Corollary 14.2.29 says that after restricting to a sub-collection, the plane map V will be Lipschitz at a specific scale ρ . The next lemma iterates this result at many scales.

Lemma 14.2.30. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with plane map V that is constant on δ -cubes. Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$ so that V is $\delta^{-\varepsilon}$ -Lipschitz on $E_{\mathbb{T}'}$.*

Proof. After a refinement, we can suppose that if $p, q \in E_{\mathbb{T}}$ satisfy $|p - q| \leq \delta$, then p and q are contained in a common δ -cube, and hence $V(p) = V(q)$. Let $N = 2/\varepsilon$. If η, δ_0 are selected sufficiently small, then we can apply Corollary 14.2.29 iteratively to $(\mathbb{T}, Y)_\delta$ for $\rho = \rho_j = \delta^{j/N}$, $j = 1, \dots, N-1$, and the resulting sub-collection $(\mathbb{T}', Y')_\delta$ will be ε -extremal. We claim that V is $2\delta^{-1/N}$ -Lipschitz on $(\mathbb{T}', Y')_\delta$. To see this, let $p, q \in E_{\mathbb{T}'}$. If $|p - q| \leq \delta$ then $V(p) = V(q)$ and we are done. If $|p - q| > \delta^{1/N}$, then $|V(p) - V(q)| \leq 2$,

and again we are done. Otherwise, there is an index $1 \leq k \leq N - 1$ so that $\delta^{(k+1)/N} < |p - q| \leq \delta^{k/N}$, and

$$|V(p) - V(q)| \leq \delta^{k/N} \leq \delta^{-1/N} |p - q|. \quad \square$$

Combining the results of this section with Lemma 14.2.12, we have the following

Lemma 14.2.31. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and a ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ that has a $\delta^{-\varepsilon}$ -Lipschitz plane map.*

14.2.4.3 Grains

In this section, we will show that if an extremal set of tubes has a Lipschitz plane map, then after restricting to a sub-collection, the tubes arrange themselves into parallel slabs, which are called grains. In what follows, V will be a plane map for a collection of tubes. If Q is a cube with center p , we define $V(Q) = V(p_Q)$ (this is a unit vector in $\mathbb{R}^3$), and we define the projection $\pi_Q(w) = w \cdot V(Q)$.

Our goal is to show that the set of grains inside a cube arrange themselves into a AD-regular set, in the sense of Definition 14.2.20. As a first step, the next lemma controls how many grains can occur inside a cube at a given scale.

Lemma 14.2.32. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with $\delta^{-\eta}$ -Lipschitz plane map V . Let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ and let $\tau \in [\rho^{1-\varepsilon}, \rho^{1/2}]$.*

Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that for each τ -cube Q , we have

$$\mathcal{E}_\rho(V(Q) \cdot (E_{\mathbb{T}'} \cap Q)) \leq \rho^{-\varepsilon} (\tau/\rho)^{1-\sigma}. \quad (14.2.55)$$

See Figure 14.2.3.

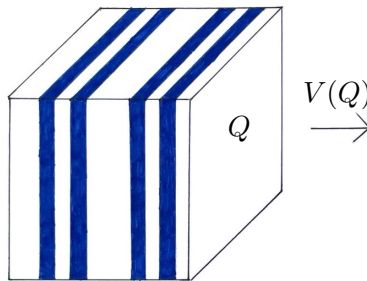


Figure 14.2.3: The set $E_{\mathbb{T}'} \cap Q$ (blue) at scale ρ , and the unit vector $V(Q)$

Proof. Let $\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0$ be small constants to be chosen below. We will select η very small compared to ε_1 , ε_1 very small compared to ε_2 , ε_2 very small compared to ε_3 , and ε_3 very small compared to ε .

Applying Proposition 14.2.14 to $(\mathbb{T}, Y)_\delta$ with $\varepsilon_1/2$ in place of ε , and then again to the output $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ with τ in place of ρ and ε_2 in place of ε , we obtain a refinement $(\mathbb{T}_1, Y_1)_\delta$ of $(\mathbb{T}, Y)_\delta$ that is covered by a ε_1 -extremal set $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$, which in turn is covered by a ε_2 -extremal set $(\tilde{\mathbb{T}}, \tilde{Y})_\tau$. Note that V is also a plane map for $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ and $(\tilde{\mathbb{T}}, \tilde{Y})_\tau$.

After refining $(\mathbb{T}_1, Y_1)_\delta$ and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$, we can ensure that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a balanced cover of $(\mathbb{T}_1, Y_1)_\delta$. In the arguments below, we will find a sub-collection of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ with certain favorable properties. Since $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a balanced cover of $(\mathbb{T}_1, Y_1)_\delta$, this sub-collection of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ will induce a sub-collection of $(\mathbb{T}_1, Y_1)_\delta$ of the same relative density.

Apply Lemma 14.2.17 to $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$, with ε_3 in place of ε . After further refinement, we can find a collection $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ with the following property (P): For each $\tilde{T} \in \tilde{\mathbb{T}}_1$ and each τ -cube Q that intersects $\tilde{Y}_1(\tilde{T})$, we have $|Q \cap \tilde{Y}_1(\tilde{T})| \geq \rho^{2+2\varepsilon_1}\tau$ (for comparison, the maximal size of an intersection of the form $\tilde{T} \cap Q$ is bounded by $100\rho^2\tau$).

Let $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$ be a constant multiplicity refinement of $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$, and let $(\tilde{\mathbb{T}}_3, \tilde{Y}_3)_\rho$ be a refinement of $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$ that again has property (P). The reason for the above refinements is as follows: If $\tilde{T} \in \tilde{\mathbb{T}}_3$ and if Q is a τ -cube that intersects $\tilde{Y}_3(\tilde{T})$, then $|Q \cap \tilde{Y}_3(\tilde{T})| \geq \rho^{2+2\varepsilon_1}\tau$. Thus we can choose a set of points $p_1, \dots, p_k \in Q \cap \tilde{Y}_3(\tilde{T})$ that are $\geq \rho^{1-\varepsilon_3}$ separated, with $k \gtrsim \rho^{2\varepsilon_1+\varepsilon_3}\rho^{-1}\tau$. For each such point p_i , we have $\#\tilde{\mathbb{T}}_2(p_i) = \mu \gtrsim \rho^{-\sigma+\varepsilon_1}$. But since we refined $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ using Lemma 14.2.17, if ε_1 is sufficiently small compared to ε_3 , then there is a tube $\tilde{T}_i \in \tilde{\mathbb{T}}_1(p_i)$ with $\angle(\text{dir}(\tilde{T}), \text{dir}(\tilde{T}_i)) \geq \rho^{\varepsilon_3}$.

We claim that the union of the sets $\{Q \cap \tilde{Y}(\tilde{T}_i)\}_{i=1}^k$ has large volume. Since the points p_i are $\rho^{1-\varepsilon_3}$ separated and the tubes $\tilde{T}_i$ make angle $\geq \rho^{\varepsilon_3}$ with $\tilde{T}$, we can label the points $p_1, \dots, p_k$ so that $\text{dist}(\tilde{T}_i \cap \tilde{T}, \tilde{T}_j \cap \tilde{T}) \gtrsim \rho^{1-\varepsilon_3}|i-j|$. Then if distinct tubes $\tilde{T}_i$ and $\tilde{T}_j$ intersect inside Q , we must have $\angle(v(\tilde{T}_i), v(\tilde{T}_j)) \gtrsim \rho^{1-\varepsilon_3}|i-j|/\tau$. But this implies that for distinct $\tilde{T}_i$ and $\tilde{T}_j$ we have

$$|Q \cap \tilde{T}_i \cap \tilde{T}_j| \lesssim \frac{\rho^3}{\angle(v(\tilde{T}_i), v(\tilde{T}_j)) + \rho} \lesssim \frac{\rho^{2+\varepsilon_3}\tau}{|i-j|}.$$

Thus we have the following Cordoba-type L^2 estimate.

$$\left| Q \cap \bigcup_{i=1}^k \tilde{Y}_1(\tilde{T}_i) \right| \geq \frac{\left(\sum_{i=1}^k |Q \cap \tilde{Y}_1(\tilde{T}_i)| \right)^2}{\left\| \sum_{i=1}^k \chi_{\tilde{T}_i} \right\|_{L^2(Q)}^2} \gtrsim \frac{k^2(\rho^{2\varepsilon_1+2}\tau)^2}{(k\rho^{2+\varepsilon_3}\tau)} \geq \rho^{6\varepsilon_1+\varepsilon_3}\tau^2\rho. \quad (14.2.56)$$

(14.2.56) says that $\bigcup_{i=1}^k \tilde{Y}_1(\tilde{T}_i)$ has large volume inside of Q . On the other hand, we claim that this set is contained inside a rectangular prism of

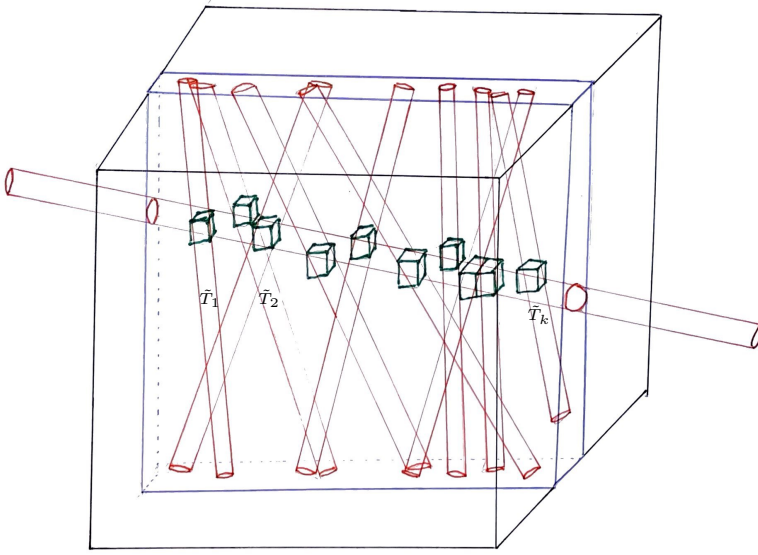


Figure 14.2.4: The union of the sets $Q \cap \tilde{Y}_1(T_i)$, $i = 1, \dots, k$ fill out most of the slab (blue) inside Q .

dimensions roughly $\delta^{-\eta} \rho \times \tau \times \tau$. Indeed, since the plane map V is $\delta^{-\eta}$ -Lipschitz and Q has diameter $\leq 2\tau$, if $q \in Q \cap \tilde{Y}_3(\tilde{T})$ then we have

$$Q \cap \bigcup_{i=1}^k \tilde{T}_i \subset Q \cap N_{4\delta^{-\eta}\rho}(q + V(Q)^\perp).$$

This is illustrated in Figure 14.2.4. Thus

$$V(q) \cdot \left(Q \cap \bigcup_{i=1}^k \tilde{T}_i \right) \subset [V(Q) \cdot q - 4\delta^{-\eta}\rho, V(Q) \cdot q + 4\delta^{-\eta}\rho]. \quad (14.2.57)$$

Note that $V(Q) \cdot (Q \cap \tilde{T})$ is also contained in this same interval.

Combining (14.2.56) and (14.2.57), we conclude that if $t \in V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}_3})$, then

$$\left| Q \cap E_{\tilde{\mathbb{T}}_1} \cap \pi_Q^{-1}([t - 4\delta^{-\eta}\rho, t + 4\delta^{-\eta}\rho]) \right| \gtrsim \rho^{6\varepsilon_1 + \varepsilon_3} \tau^2 \rho. \quad (14.2.58)$$

Informally, (14.2.58) says that each ρ -separated point $t \in V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}_3})$ corresponds to a $\rho \times \tau \times \tau$ grain (i.e. rectangular prism whose short axis is parallel to $V(Q)$) that has nearly full intersection with $Q \cap E_{\tilde{\mathbb{T}}_1}$.

If $|t - t'| > 8\delta^{-\eta}\rho$ then the sets $\pi_Q^{-1}([t - 4\delta^{-\eta}\rho, t + 4\delta^{-\eta}\rho])$ and $\pi_Q^{-1}([t' - 4\delta^{-\eta}\rho, t' + 4\delta^{-\eta}\rho])$ are disjoint, and hence (14.2.58) implies that

$$\begin{aligned} \mathcal{E}_{8\delta^{-\eta}\rho}(V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}_3})) &\lesssim \frac{|Q \cap E_{\tilde{\mathbb{T}}_1}|}{\rho^{6\varepsilon_1 + \varepsilon_3} \tau^2 \rho} \\ &\leq \frac{\rho^{\sigma - \varepsilon_1} \tau^{3 - \sigma - \varepsilon_2}}{\rho^{6\varepsilon_1 + \varepsilon_3} \tau^2 \rho} \\ &\lesssim \rho^{-7\varepsilon_1 - \varepsilon_2 - \varepsilon_3} (\tau/\rho)^{1 - \sigma}. \end{aligned}$$

and thus

$$\begin{aligned} \mathcal{E}_\rho(V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}_3})) &\leq 8\delta^{-\eta} \mathcal{E}_{8\delta^{-\eta}\rho}(V(Q) \cdot (Q \cap E_{\tilde{\mathbb{T}}_3})) \\ &\lesssim \delta^{-\eta} \rho^{-7\varepsilon_1 - \varepsilon_2 - \varepsilon_3} (\tau/\rho)^{1 - \sigma}. \end{aligned} \tag{14.2.59}$$

To conclude the proof, select $\varepsilon_1, \varepsilon_2, \varepsilon_3, \eta$ sufficiently small (depending on ε) so that the LHS of (14.2.59) is at most $\rho^{-\varepsilon} (\tau/\rho)^{1 - \sigma}$ (this is possible since $\rho \leq \delta^\varepsilon$). Let $(\mathbb{T}_3, Y_3)_\delta$ be the refinement of $(\mathbb{T}_1, Y_1)_\delta$ induced by the refinement $(\tilde{\mathbb{T}}_3, \tilde{Y}_3)_\rho$. Since $E_{\mathbb{T}_3} \subset E_{\tilde{\mathbb{T}}_3}$, (14.2.59) implies that $(\mathbb{T}_3, Y_3)_\delta$ satisfies (14.2.55). $\square$

Lemma 14.2.32 says that when restricted to a τ -cube and examined at scale ρ , the set $E_{\mathbb{T}'}$ looks like a union of about $(\tau/\rho)^{1 - \sigma}$ parallel grains, whose normal directions all coincide and are given by the plane map V . The next lemma asserts that a similar statement continues to hold if we replace a τ -cube with a (potentially much larger) $\tau \times \rho^{1/2} \times \rho^{1/2}$ rectangular prism, where the two “long” directions of the rectangular prism are orthogonal to the plane map V . To prove this lemma, we need to show that at scale ρ , the grains inside a τ -cube are pieces of the grains inside a $\rho^{1/2}$ -cube.

Lemma 14.2.33. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with $\delta^{-\eta}$ -Lipschitz plane map V . Let $\rho \in [\delta^{1 - \varepsilon}, \delta^\varepsilon]$ and let $\tau \in [\rho^{1 - \varepsilon}, \rho^{1/2}]$.*

Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that the following holds. Let Q be a $\rho^{1/2}$ -cube and let $F = V(Q) \cdot (E_{\mathbb{T}'} \cap Q)$. Then for each interval J of length τ , we have

$$\mathcal{E}_\rho(J \cap F) \leq \rho^{-\varepsilon} (\tau/\rho)^{1 - \sigma}. \tag{14.2.60}$$

Proof. For notational convenience, we will suppose that $\rho^{1/2}/\tau$ is an integer, and thus every τ -cube is contained in a single $\rho^{1/2}$ cube. The case where $\rho^{1/2}/\tau$ is not an integer can be handled similarly. Let $\varepsilon_1, \dots, \varepsilon_4 > 0$ be small constants to be chosen below. We will select η very small compared to ε_1 , and ε_i very small compared to ε_{i+1} . We will select ε_4 small compared to ε .

Apply Proposition 14.2.14 to $(\mathbb{T}, Y)_\delta$ with ε_1 in place of ε , and ρ as above. We obtain a refinement of $(\mathbb{T}, Y)_\delta$ (which we will continue to denote by $(\mathbb{T}, Y)_\delta$),

and a ε_1 -extremal collection of ρ -tubes, $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$. In what follows, we will describe various sub-collections of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. Since $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a balanced cover of $(\mathbb{T}, Y)_\delta$, each sub-collection of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ will induce an analogous sub-collection of $(\mathbb{T}, Y)_\delta$, of the same relative density.

Applying Lemma 14.2.32 to $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ with ε_2 in place of ε and τ as above, and then again to the output with ε_3 in place of ε and $\rho^{1/2}$ in place of τ , we obtain a ε_3 -extremal sub-collection $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$ of $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\rho$ with the following two properties:

- For each $\rho^{1/2}$ -cube Q , we have

$$\mathcal{E}_\rho(V(Q) \cdot (E_{\tilde{\mathbb{T}}_2} \cap Q)) \leq \rho^{-\varepsilon_3} \rho^{\frac{\sigma-1}{2}}. \quad (14.2.61)$$

- For each $\rho^{1/2}$ -cube Q , and each τ -cube $W \subset Q$, we have

$$\mathcal{E}_\rho(V(Q) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) \lesssim \delta^{-\eta} \rho^{-\varepsilon_2} (\tau/\rho)^{\sigma-1}. \quad (14.2.62)$$

Note that in (14.2.62) we have the unit vector $V(Q)$ rather than $V(W)$. This requires justification. By Lemma 14.2.32 we get the estimate $\mathcal{E}_\rho(V(W) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) \lesssim \rho^{-\varepsilon_2} (\tau/\rho)^{\sigma-1}$. To obtain (14.2.62), we bound

$$\begin{aligned} \mathcal{E}_\rho(V(Q) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) &\leq 2\delta^{-\eta} \mathcal{E}_{2\delta^{-\eta}\rho}(V(Q) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) \\ &\lesssim \delta^{-\eta} \mathcal{E}_{\delta^{-\eta}\rho}(V(W) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) \\ &\leq \delta^{-\eta} \mathcal{E}_\rho(V(W) \cdot (E_{\tilde{\mathbb{T}}_2} \cap W)) \\ &\leq \delta^{-\eta} \rho^{-\varepsilon_2} (\tau/\rho)^{\sigma-1}. \end{aligned} \quad (14.2.63)$$

The key step is the second inequality, which used the fact that $\text{diam}(W) \lesssim \rho^{1/2}$ and $\angle(V(Q), V(W)) \leq 2\delta^{-\eta} \rho^{1/2}$; the latter statement follows from the fact that $W \subset Q$, $\text{diam}(Q) \lesssim \rho^{1/2}$, and V is $\delta^{-\eta}$ -Lipschitz.

Apply Proposition 14.2.14 to $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$ with ε_4 in place of ε and $\rho^{1/2}$ in place of ρ . We obtain a refinement $(\tilde{\mathbb{T}}_3, \tilde{Y}_3)_\rho$ of $(\tilde{\mathbb{T}}_2, \tilde{Y}_2)_\rho$, which is contained in a union of $\rho^{1/2}$ -cubes of cardinality $\lesssim \rho^{-\varepsilon_4} \rho^{\frac{\sigma+3}{2}}$.

After a further refinement, we can suppose that each $\rho^{1/2}$ -cube Q that intersects $E_{\tilde{\mathbb{T}}_3}$ satisfies $|E_{\tilde{\mathbb{T}}_3} \cap Q| \gtrsim \rho^{2\varepsilon_4} \rho^{\frac{\sigma+3}{2}}$.

The two properties described above continue to hold for $(\tilde{\mathbb{T}}_3, \tilde{Y}_3)_\rho$. Therefore, if Q is a $\rho^{1/2}$ -cube that intersects $E_{\tilde{\mathbb{T}}_3}$, then by (14.2.61), $E_{\tilde{\mathbb{T}}_3} \cap Q$ is contained in a union of at most $\rho^{-\varepsilon_3} \rho^{\frac{\sigma-1}{2}}$ “grains,” which are sets of the form $Q \cap N_\rho(\pi_Q^{-1}(t))$, where $t \in \mathbb{R}$. The union of these grains has volume $\lesssim \rho^{-\varepsilon_3} \rho^{\frac{\sigma+3}{2}}$, while on the other hand $E_{\tilde{\mathbb{T}}_3} \cap Q$ has volume $\gtrsim \rho^{2\varepsilon_4} \rho^{\frac{\sigma+3}{2}}$. This means that

after further refining $(\tilde{\mathbb{T}}_3, \tilde{Y}_3)_\rho$ to obtain a pair $(\tilde{\mathbb{T}}_4, \tilde{Y}_4)_\rho$, we can suppose that for each $\rho^{1/2}$ -cube Q and each $t \in V(p_Q) \cdot (E_{\tilde{\mathbb{T}}_4} \cap Q)$, we have

$$|E_{\tilde{\mathbb{T}}_4} \cap Q \cap \pi_Q^{-1}(N_\rho(t))| \gtrsim \rho^{2\varepsilon_4 + \varepsilon_3} \rho^2. \quad (14.2.64)$$

(Note that ρ^2 is the volume of a $\rho \times \rho^{1/2} \times \rho^{1/2}$ grain).

Next, let Q be one of the $\rho^{1/2}$ cubes described above. By Fubini, there must be a point $q \in Q$ so that the line $L = q + \mathbb{R}V(p_Q)$ satisfies

$$|L \cap E_{\tilde{\mathbb{T}}_4} \cap Q| \gtrsim \rho^{2\varepsilon_4} \rho^{\frac{\sigma+3}{2}} \gtrsim \rho^{\varepsilon_2 + 2\varepsilon_4} |E_{\tilde{\mathbb{T}}_4} \cap Q|. \quad (14.2.65)$$

(The $|\cdot|$ on the LHS of (14.2.65) denotes 1-dimensional Lebesgue measure, while the $|\cdot|$ on the RHS of (14.2.65) denotes 3-dimensional Lebesgue measure).

For each cube Q , let G_Q be the union of grains that intersect $L \cap E_{\tilde{\mathbb{T}}_4} \cap Q$, i.e.

$$G_Q = Q \cap \left[\pi_Q^{-1} \left(N_\rho(\pi(L \cap E_{\tilde{\mathbb{T}}_4} \cap Q)) \right) \right].$$

Then by (14.2.64) and (14.2.65), we have

$$|E_{\tilde{\mathbb{T}}_4} \cap G_Q| \gtrsim \rho^{\varepsilon_2 + \varepsilon_3 + 4\varepsilon_4} |E_{\tilde{\mathbb{T}}_4} \cap Q|. \quad (14.2.66)$$

Let $\tilde{\mathbb{T}}_5 = \tilde{\mathbb{T}}_4$, and define the shading $\tilde{Y}_5$ by $\tilde{Y}_5(\tilde{T}) = \tilde{Y}_4(\tilde{T}) \cap \bigcup_Q G_Q$. By (14.2.66) and the bound $\varepsilon_2 + \varepsilon_3 + 4\varepsilon_4 \leq 5\varepsilon_4$, we have $\sum_{\tilde{T} \in \tilde{\mathbb{T}}_5} |\tilde{Y}_5(\tilde{T})| \gtrsim \rho^{5\varepsilon_4} \sum_{\tilde{T} \in \tilde{\mathbb{T}}_4} |\tilde{Y}_4(\tilde{T})|$.

Unwinding definitions, we see that if Q is a $\rho^{1/2}$ -cube and if $J \subset \mathbb{R}$ is an interval, then

$$\mathcal{E}_\rho \left(J \cap (V(p_Q) \cdot (E_{\tilde{\mathbb{T}}_5} \cap Q)) \right) \sim \mathcal{E}_\rho \left(V(p_Q) \cdot (L_J \cap E_{\tilde{\mathbb{T}}_3}) \right),$$

where $L_J = L \cap \pi_Q^{-1}(J)$, and L is the line from (14.2.65). In particular, if J is an interval of length τ , then L_J is contained in a union of $O(1)$ τ -cubes W , and hence

$$\mathcal{E}_\rho \left(J \cap (V(p_Q) \cdot (E_{\tilde{\mathbb{T}}_5} \cap Q)) \right) \lesssim \sum_W \mathcal{E}_\rho(V(p) \cdot (E_{\tilde{\mathbb{T}}_5} \cap W)),$$

where the sum is taken over the $O(1)$ τ -cubes W described above. Since the contribution from each of these τ -cubes is $\leq \rho^{-\varepsilon_2} (\tau/\rho)^{1-\sigma}$, we conclude that

$$\mathcal{E}_\rho \left(J \cap (V(p_Q) \cdot (E_{\tilde{\mathbb{T}}_5} \cap Q)) \right) \lesssim \rho^{-\varepsilon_2} (\tau/\rho)^{1-\sigma}. \quad (14.2.67)$$

We claim that the sub-collection $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ induced by $(\tilde{\mathbb{T}}_5, \tilde{Y}_5)_\rho$ satisfies the conclusions of Lemma 14.2.33. First, if $\varepsilon_1, \dots, \varepsilon_4$ are chosen appropriately, then $\sum_{T \in \mathbb{T}'} |Y'(T)| \geq \delta^\varepsilon$, so $(\mathbb{T}', Y')_\delta$ is ε -extremal. Next, let

Q be a $\rho^{1/2}$ cube, let $p \in Q$, and let J be an interval of length τ . Then since $E_{\mathbb{T}'} \subset E_{\tilde{\mathbb{T}}_5}$, we have

$$\mathcal{E}_\rho\left(J \cap (V(p) \cdot (E_{\mathbb{T}'} \cap Q))\right) \lesssim \delta^{-\eta} \mathcal{E}_\rho\left(J \cap (V(p_Q) \cdot (E_{\tilde{\mathbb{T}}_5} \cap Q))\right), \quad (14.2.68)$$

where we used the same argument as in (14.2.63) to replace $V(p)$ by $V(p_Q)$, at the cost of a multiplicative factor of $\delta^{-\eta}$. (14.2.60) now follows from (14.2.67). $\square$

For a fixed ρ , the next lemma records what happens when we iterate Lemma 14.2.33 for many different scales τ .

Lemma 14.2.34. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes with $\delta^{-\eta}$ -Lipschitz plane map V . Let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$.*

Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that the following holds. Let Q be a $\rho^{1/2}$ -cube. Then $V(Q) \cdot (E_{\mathbb{T}'} \cap Q)$ is a $(\rho, 1 - \sigma; \delta^{-\varepsilon})_1$ -ADset.

Proof. Let $N \geq 1/\varepsilon$ be an integer. We will select a sequence of numbers $\varepsilon/2 = \varepsilon_N > \varepsilon_{N-1} > \dots > \varepsilon_0 > 0$ so that the following holds. Define $\eta = \varepsilon_0$ and define $(\mathbb{T}_0, Y_0)_\delta = (\mathbb{T}, Y)_\delta$; this is a ε_0 -extremal collection of tubes with a $\delta^{-\varepsilon_0}$ -Lipschitz plane map. For each $j = 1, \dots, N$, apply Lemma 14.2.33 to $(\mathbb{T}_{j-1}, Y_{j-1})_\delta$ with ε_j in place of ε , ρ as above, and $\tau = \tau_j = \rho^{(j+N)/2N}$. Denote the output by $(\mathbb{T}_j, Y_j)_\delta$; this pair is ε_j -extremal and has a $\delta^{-\varepsilon_j}$ -Lipschitz plane map. We may apply Lemma 14.2.33 provided that ε_{j-1} is chosen sufficiently small depending on ε_j . Define $(\mathbb{T}', Y')_\delta = (\mathbb{T}_N, Y_N)_\delta$; this set is $\varepsilon/2$ -extremal.

We will verify that $(\mathbb{T}', Y')_\delta$ satisfies the conclusion of Lemma 14.2.34. Let Q be a $\rho^{1/2}$ -cube and let $J \subset \mathbb{R}$ be an interval of length at least ρ . Shrinking J if needed, we may suppose $J \subset V(Q) \cdot Q$, so in particular $|J| \leq 2\rho^{1/2}$. Thus there is an integer $0 \leq j < N$ so that $\rho^{(j+N+1)/2N} < |J| \leq 2\rho^{(j+N)/2N}$. Let $\tilde{J}$ be an interval of length $\tau_j = \rho^{(j+N)/2N}$ that contains J (if $|J| > \rho^{1/2}$ then a slight modification is needed; we need to cover J by a union of two intervals of length $\rho^{1/2}$). Let $F = V(Q) \cdot (E_{\mathbb{T}'} \cap Q)$. By Lemma 14.2.33, we have

$$\begin{aligned} \mathcal{E}_\rho(J \cap F) &\leq \mathcal{E}_\rho(\tilde{J} \cap F) \\ &\leq \delta^{-\varepsilon/2} (\rho^{(j+N)/2N} / \rho)^{1-\sigma} \\ &\leq \delta^{-\varepsilon/2} \rho^{-1/2N} (|J| / \rho)^{1-\sigma} \\ &\leq \delta^{-\varepsilon} (|J| / \rho)^{1-\sigma}. \quad \square \end{aligned}$$

The next lemma records what happens when we iterate Lemma 14.2.34 for many different scales ρ .

Lemma 14.2.35. *For all $\varepsilon > 0$, there exists $\delta_0, \eta > 0$ so that the following holds. Let $\delta \in (0, \delta_0]$ and let $(\mathbb{T}, Y)_\delta$ be an η -extremal collection of tubes with a $\delta^{-\eta}$ -Lipschitz plane map V .*

Then there exists a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$, so that

$$V(p) \cdot (E_{\mathbb{T}'} \cap B(p, \rho^{1/2})) \text{ is a } (\rho, 1 - \sigma; \delta^{-\varepsilon})_1\text{-ADset} \quad \text{for all } \rho \in [\delta, 1], \quad p \in \mathbb{R}^3. \quad (14.2.69)$$

Proof. Let $N \geq 2/\varepsilon$ be an integer. We will select a sequence of numbers $\varepsilon/3 = \varepsilon_N > \varepsilon_{N-1} > \dots > \varepsilon_0 > 0$ so that the following holds. Define $\eta = \varepsilon_0$ and define $(\mathbb{T}_0, Y_0)_\delta = (\mathbb{T}, Y)_\delta$; this is a ε_0 -extremal collection of tubes with a $\delta^{-\varepsilon_0}$ -Lipschitz plane map. For each $j = 1, \dots, N$, apply Lemma 14.2.34 to $(\mathbb{T}_{j-1}, Y_{j-1})_\delta$ with ε_j in place of ε and $\rho = \rho_j = \delta^{j/N}$. Denote the output by $(\mathbb{T}_j, Y_j)_\delta$; this pair is ε_j -extremal and has a $\delta^{-\varepsilon_j}$ -Lipschitz plane map. We may apply Lemma 14.2.34 provided that ε_{j-1} is chosen sufficiently small depending on ε_j . Define $(\mathbb{T}', Y')_\delta = (\mathbb{T}_N, Y_N)_\delta$; this set is $\varepsilon/3$ -extremal.

Next, let $\rho \in [\delta, 1]$, let $p \in \mathbb{R}^3$, and let J be an interval of length at least ρ . Our goal is to show that

$$\mathcal{E}_\rho \left(J \cap \left[V(p) \cdot (E_{\mathbb{T}'} \cap B(p, \rho^{1/2})) \right] \right) \leq \delta^{-\varepsilon} (|J|/\rho)^{1-\sigma}. \quad (14.2.70)$$

Case 1: $\rho \geq \delta^{1/N}$. In this case, (14.2.70) follows from the simple observation that $E_{\mathbb{T}'} \subset B(0, 1)$, and $\mathcal{E}_\rho([-1, 1]) \leq 2\rho^{-1} \leq \delta^{-\varepsilon}$.

Case 2: $\rho \in [\delta^{1-1/N}, \delta^{1/N})$. In this case, there is an integer $1 \leq j \leq N-2$ so that $\delta^{(j+1)/N} \leq \rho < \delta^{j/N}$. $B(p, \rho^{1/2})$ can be covered by a union of $O(1)$ $\rho_j^{1/2}$ -cubes. By Lemma 14.2.34, for each such cube Q we have

$$\mathcal{E}_{\rho_j} \left(J \cap (V(Q) \cdot (E_{\mathbb{T}'} \cap Q)) \right) \leq \delta^{-\varepsilon/3} (|J|/\rho_j)^{1-\sigma}. \quad (14.2.71)$$

Recall that $V(Q) = V(p_Q)$, where p_Q is the center of Q . Since Q intersects $B(p, \rho^{1/2})$, we have that $|p_Q - p| \lesssim \rho_j^{1/2}$. Since V is $\delta^{-\eta}$ -Lipschitz, this implies $\angle(V(Q), V(p)) \lesssim \delta^{-\eta} \rho_j^{1/2}$. Finally, since Q has diameter $\leq 2\rho_j^{1/2}$, we have

$$\begin{aligned} \mathcal{E}_{\delta^{-\eta}\rho_j} \left(J \cap (V(p) \cdot (E_{\mathbb{T}'} \cap Q)) \right) &\sim \mathcal{E}_{\delta^{-\eta}\rho_j} \left(J \cap (V(Q) \cdot (E_{\mathbb{T}'} \cap Q)) \right) \\ &\leq \delta^{-\varepsilon/3} (|J|/\rho_j)^{1-\sigma}, \end{aligned}$$

where the final inequality used (14.2.71).

Since (14.2.72) holds for each of the $O(1)$ cubes whose union covers $B(p, \rho^{1/2})$, we have

$$\mathcal{E}_{\delta^{-\eta}\rho_j} \left(J \cap \left[V(p) \cdot (E_{\mathbb{T}'} \cap B(p, \rho^{1/2})) \right] \right) \lesssim \delta^{-\varepsilon/3} (|J|/\rho_j)^{1-\sigma}. \quad (14.2.72)$$

This is almost what we want, except we have bounded the $\delta^{-\eta}\rho_j$ -covering number, rather than the ρ -covering number. But since each interval of length $\delta^{-\eta}\rho_j$ can be covered by $\leq \delta^{-1/N-\eta}$ intervals of length ρ , (14.2.72) implies

$$\mathcal{E}_\rho\left(J \cap \left[V(p) \cdot (E_{\mathbb{T}} \cap B(p, \rho^{1/2}))\right]\right) \lesssim \delta^{-\varepsilon/3-1/N-\eta}(|J|/\rho_j)^{1-\sigma} \leq \delta^{-\varepsilon}(|J|/\rho_j)^{1-\sigma}.$$

(14.2.70) now follows, provided δ_0 and η are chosen sufficiently small.

Case 3: $\rho \in [\delta, \delta^{1-1/N})$. This reduces to Case 2, using the observation that for every set $X \subset \mathbb{R}$, $\mathcal{E}_\rho(X) \leq \delta^{-1/N} \mathcal{E}_{\delta^{1-1/N}}(X)$. □

14.2.4.4 Proof of Proposition 14.2.22

We are now ready to prove Proposition 14.2.22. In brief, we use the results from Sections 14.2.4.1 and 14.2.4.2 to find an extremal set of δ -tubes $(\mathbb{T}, Y)_\delta$ with a plane map. Next, we use the results from Section 14.2.4.3 to show that this set of tubes arranges into grains of dimensions $\sqrt{\delta} \times \sqrt{\delta} \times \delta$. We select a tube T_0 whose $\sqrt{\delta}$ neighborhood covers about δ^{-1} tubes from $(\mathbb{T}, Y)_\delta$, and consider the unit rescaling (recall Definition 14.2.13) of this collection of tubes with respect to the $\sqrt{\delta}$ neighborhood T_0 . This is a set of $\sqrt{\delta}$ tubes, which we will denote by $(\tilde{T}, \tilde{Y})_{\sqrt{\delta}}$.

In what follows, we define $\rho = \sqrt{\delta}$. Note that the $\sqrt{\delta}$ neighborhood of T_0 can be covered by about $\delta^{-1/2}$ balls of radius $\sqrt{\delta}$, and the restriction of $E_{\mathbb{T}}$ to each of these balls consists of a union of parallel grains of dimensions $\sqrt{\delta} \times \sqrt{\delta} \times \delta$. After re-scaling, the situation is as follows: The unit ball can be covered by ρ^{-1} “slabs” (rectangular prisms) of dimensions $1 \times 1 \times \rho$, and the restriction of $E_{\tilde{\mathbb{T}}}$ to each slab consists of a union of parallel (re-scaled) grains of dimensions $1 \times \rho \times \rho$; these will be called *global grains*, and they will satisfy the properties described in Item 1 from the statement of Proposition 14.2.22.

Finally, we will apply the results from Section 14.2.4.3 to show that $(\tilde{T}, \tilde{Y})_\rho$ arranges into grains of dimensions $\sqrt{\rho} \times \sqrt{\rho} \times \rho$; these will be called *local grains*, and they will satisfy the properties described in Item 2 from the statement of Proposition 14.2.22. We now turn to the details.

Proof of Proposition 14.2.22. Let $\varepsilon_1, \dots, \varepsilon_4 > 0$ be small constants to be chosen below. We will select ε_i very small compared to ε_{i+1} , and ε_4 very small compared to ε .

Step 1: Global grains.

Define $\tau_0 = \delta_0^{\frac{2+\sigma}{\sigma}}$. By Lemma 14.2.31 followed by Lemma 14.2.35, there exists a ε_1 -extremal collection of tubes $(\mathbb{T}, Y)_\tau$ for some $\tau \in (0, \tau_0]$ that has a $\tau^{-\varepsilon_1}$ -Lipschitz plane map V , and satisfies (14.2.69) with ε_1 in place of ε .

Define $\delta = \tau^{\frac{\sigma}{2+\sigma}}$, so $\delta \in (0, \delta_0]$. Apply Proposition 14.2.14 to $(\mathbb{T}, Y)_\tau$, with τ in place of δ , δ in place of ρ , and ε_2 in place of ε . We obtain a refinement $(\mathbb{T}_1, Y_1)_\tau$ of $(\mathbb{T}, Y)_\tau$, and a ε_2 -extremal collection of δ -tubes $(\tilde{\mathbb{T}}_1, \tilde{Y}_1)_\delta$.

Fix a choice of δ -tube $\tilde{T}$; define $\mathbb{T}^\dagger = \mathbb{T}_1[\tilde{T}]$, and let $Y^\dagger$ be the restriction of Y_1 to $\mathbb{T}^\dagger$. Since the unit re-scaling of $(\mathbb{T}^\dagger, Y^\dagger)_\tau$ relative to $\tilde{T}$ is ε_2 -extremal, we can find a sub-collection $(\mathbb{T}_1^\dagger, Y_1^\dagger)_\tau$ of $(\mathbb{T}^\dagger, Y^\dagger)_\tau$, with $\sum |Y_1^\dagger(T)| \gtrsim \tau^{\varepsilon_2} \sum |Y^\dagger(T)|$, and a tube $T_0 \in \mathbb{T}_1^\dagger$ with the following property (P): for each δ -cube Q that intersects $E_{\mathbb{T}_1^\dagger}$, we have $Y_1^\dagger(T_0) \cap Q \neq \emptyset$. After enlarging $\tilde{T}$ slightly (we might have to replace δ by 2δ , but this is harmless), we may suppose that T_0 and $\tilde{T}$ have the same coaxial line.

Let $(\hat{\mathbb{T}}, \hat{Y})_{\tau/\delta}$ be the unit re-scaling of $(\mathbb{T}_1^\dagger, Y_1^\dagger)_\tau$ relative to $\tilde{T}$. This re-scaling sends T_0 to a tube $\hat{T}_0$ whose coaxial line is the z -axis. Let

$$Z = \{z \in \delta\mathbb{Z} : (0, 0, z) \in \hat{Y}(\hat{T}_0)\}.$$

Since T_0 has Property (P), for each $p = (x, y, z) \in E_{\hat{\mathbb{T}}}$ there is $z_0 \in Z$ with $|z - z_0| \lesssim \delta$.

Next, we will analyze what (14.2.69) says about $(\hat{\mathbb{T}}, \hat{Y})_{\tau/\delta}$. For each $z \in Z$, let $\phi^{-1}(z)$ be the inverse image of z under the unit-rescaling associated to $\tilde{T}$. Then $\phi^{-1}(z)$ is contained in the coaxial line of T_0 , and there is a point $p_z \in Y_1^\dagger(T_0)$ with $|\phi^{-1}(z) - p_z| \lesssim \delta$. By (14.2.69), we have

$$V(p_z) \cdot (E_{\mathbb{T}_1^\dagger} \cap B(p_z, \delta)) \text{ is a } (\delta^2, 1 - \sigma; \tau^{-\varepsilon_1})_1\text{-ADset.} \quad (14.2.73)$$

But if we consider the image of $B(p_z, \delta)$ under ϕ , (14.2.73) says that there is a unit vector $\hat{V}(z) \in S^2$ so that

$$\hat{V}(z) \cdot (E_{\hat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta))) \text{ is a } (\delta, 1 - \sigma; O(\tau^{-\varepsilon_1}))_1\text{-ADset.} \quad (14.2.74)$$

Since $|V(p_z) \cdot \text{dir}(L_0)| \leq \tau$, we have $|\hat{V}(z) \cdot (0, 0, 1)| \leq \tau/\delta \leq 1/10$ for all $z \in Z$. In particular, since V is $\tau^{-\varepsilon_1}$ -Lipschitz, we have that $\hat{V}$ is also $\tau^{-\varepsilon_1}$ -Lipschitz, and thus for each interval $I \subset [-1, 1]$ of length $\sim \tau^{\varepsilon_1}$, we must have either $|\hat{V}(z) \cdot (1, 0, 0)| \geq \frac{1}{3}$ or $|\hat{V}(z) \cdot (0, 1, 0)| \geq \frac{1}{3}$ for all $z \in I \cap Z$. Interchanging the x and y axes if necessary, we may replace Z with $Z \cap I$, and define $\hat{\mathbb{T}}_1 = \hat{\mathbb{T}}$, $\hat{Y}_1(\hat{T}) = \hat{Y}(\hat{T}) \cap (\mathbb{R}^2 \times I)$. If we choose the interval I appropriately, then $\sum_{\hat{T} \in \hat{\mathbb{T}}} |\hat{Y}_1(\hat{T})| \gtrsim \tau^{\varepsilon_1} \sum_{\hat{T} \in \hat{\mathbb{T}}} |\hat{Y}(\hat{T})|$, and $|\hat{V}(z) \cdot (1, 0, 0)| \geq \frac{1}{3}$ for all $z \in Z$. In particular, there are $3\tau^{-2\varepsilon_1}$ -Lipschitz functions $f, g: Z \rightarrow \mathbb{R}$ so that for all $z \in Z$, we have that $\hat{V}(z)$ is parallel to $(1, f(z), g(z))$ (f is given by dividing the second component of $\hat{V}$ by the first, and g is defined by dividing the third component of $\hat{V}$ by the first). Currently $f(z)$ is defined on Z , but by the Kirszbraun-Valentine Lipschitz extension theorem [30], we can

extend f to a $3\tau^{-2\varepsilon_1}$ -Lipschitz function with domain $[-1, 1]$. By (14.2.74), for each $p = (x, y, z) \in E_{\hat{\mathbb{T}}_1}$ we have

$$(1, f(z), 0) \cdot \left(E_{\hat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta]) \right) \text{ is a } (\delta, 1 - \sigma; O(\tau^{-\varepsilon_1}))_1\text{-ADset.} \quad (14.2.75)$$

Step 2: Local grains.

Apply Lemma 14.2.26 to $(\hat{\mathbb{T}}_1, \hat{Y}_1)_{\tau/\delta}$ with τ/δ in place of δ and ε_3 in place of ε ; we obtain a refinement $(\hat{\mathbb{T}}_2, \hat{Y}_2)_{\tau/\delta}$ and a map $U: E_{\hat{\mathbb{T}}_2} \rightarrow S^2$ that satisfies

$$|U(p) \cdot \text{dir}(\hat{T})| \leq (\tau/\delta)^{\sigma/2} \quad \text{for all } p \in E_{\hat{\mathbb{T}}_2}, \hat{T} \in \hat{\mathbb{T}}_2(p).$$

Next, apply Lemma 14.2.27 with τ/δ in place of δ , δ in place of ρ (note that this is a valid choice, since $\delta = (\tau/\delta)^{\sigma/2}$), and ε_3 in place of ε . We obtain a ε_3 -extremal collection of δ -tubes $(\tilde{T}, \tilde{Y})_\delta$ that covers $(\hat{\mathbb{T}}_2, \hat{Y}_2)_{\tau/\delta}$ and a plane map $W: E_{\hat{\mathbb{T}}} \rightarrow S^2$. Each δ -cube $Q \subset E_{\hat{\mathbb{T}}}$ intersects $E_{\hat{\mathbb{T}}_2}$, and thus by (14.2.75), for each $z \in Z$ we have

$$(1, f(z), 0) \cdot \left(E_{\hat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta]) \right) \text{ is a } (\delta, 1 - \sigma; O(\tau^{-\varepsilon_1}))_1\text{-ADset.} \quad (14.2.76)$$

Indeed, to verify (14.2.76) let $\tilde{X} = (E_{\hat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta]))$ and let $\hat{X} = (E_{\hat{\mathbb{T}}} \cap (\mathbb{R}^2 \times [z, z + \delta]))$. It suffices to show that each $w \in (1, f(z), 0) \cdot \tilde{X}$ is contained in the $O(\delta)$ -neighborhood of $(1, f(z), 0) \cdot \hat{X}$. For such a w , there exists a δ -cube Q that intersects $\tilde{X}$ with $w \in (1, f(z), 0) \cap Q$. But since Q is a δ -cube, it must be contained in $\tilde{X}$. Since $Q \cap E_{\hat{\mathbb{T}}} \neq \emptyset$, there must be a point $p \in Q \cap \hat{X}$. Thus $|(1, f(z), 0) \cdot w - (1, f(z), 0) \cdot p| \lesssim |w - p| \lesssim \delta$.

We will re-write (14.2.76) as

$$(1, f(z_0), 0) \cdot \left(E_{\hat{\mathbb{T}}} \cap \{z = z_0\} \right) \text{ is a } (\delta, 1 - \sigma; O(\tau^{-\varepsilon_1}))_1\text{-ADset,} \quad (14.2.77)$$

for all $z_0 \in Z$. In fact, (14.2.77) holds for all $z \in [-1, 1]$, since the set on the LHS of (14.2.77) is empty if $z_0 \in [-1, 1] \setminus Z$.

To complete the proof, apply Lemma 14.2.35 to $(\tilde{T}, \tilde{Y})_\delta$ with ε_4 in place of ε . If $\varepsilon_1, \dots, \varepsilon_4$ are chosen sufficiently small, then the resulting refinement of $(\tilde{T}, \tilde{Y})_\delta$ almost satisfies the conclusions of Proposition 14.2.22, except the function f and the plane map W are not 1-Lipschitz. This can be fixed via a mild rescaling, using Lemma 14.2.18. $\square$

14.2.5 The global grains slope function is C^2

Our goal in this section is to strengthen the conclusions of Proposition 14.2.22 by increasing the regularity of the global grains slope function. More precisely, we will prove the following.

Proposition 14.2.36. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and a ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the following properties:*

1'. Global grains with C^2 slope function.

There is a function $f: [-1, 1] \rightarrow \mathbb{R}$ with $\|f\|_{C^2} \leq 1$ so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.

2. Local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_\delta$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

We refer the reader to Section 14.2.1.1, Step 2 for an overview of the main ideas used to prove Proposition 14.2.36. Our main tool for proving Proposition 14.2.36 will be the following new result from projection theory.

Theorem 14.2.37. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\eta})$ -ADsets. Then at least one of the following must hold.*

(A) *There exist two lines ℓ and $\ell^\perp$ (where $\ell^\perp$ passes through the origin and is orthogonal to ℓ) whose δ -neighborhoods cover a significant fraction of F and G respectively. In particular,*

$$\mathcal{E}_\delta(N_\delta(\ell) \cap F) \geq \delta^{\varepsilon-1} \text{ and } \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G) \geq \delta^{\varepsilon-1}. \quad (14.2.78)$$

(B) *Let $\mathcal{H} \subset F \times G \times G$ be any set satisfying $\mathcal{E}_\delta(\mathcal{H}) \geq \delta^{\eta-3}$. Then there exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-\eta}\rho$ such that*

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2): (a, b_1, b_2) \in \mathcal{H}\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.2.79)$$

Remark 14.2.38. Note that the statement of Theorem 14.2.37 makes sense over $\mathbb{C}$ (with all exponents doubled), but the theorem is false. To construct a counter-example, let $\Omega \subset \mathbb{C}^2$ be a neighbourhood of the origin and let $F = G = \{(z, \bar{z}): z \in \mathbb{C}\} \cap \Omega$; these are $(\delta, 2, O(1))$ -sets. Let $\mathcal{H} = F \times G \times G$; then $\mathcal{E}_\delta(\mathcal{H}) \gtrsim \delta^{-6}$. Conclusion (A) fails because the δ -neighbourhood of every line $\ell = \{(a + bt, c + dt): t \in \mathbb{C}\}$ intersects only a small fraction of F ; indeed, since $(b, d) \neq (0, 0)$, we have $\mathcal{E}_\delta(N_\delta(\ell) \cap F) \leq \delta^{-1}$, which is much smaller than δ^{-2} . Conclusion (B) fails because if $\omega = (z_0, \bar{z}_0) \in F$ and $\zeta_1 = (\bar{z}_1, z_1), \zeta_2 = (\bar{z}_2, z_2) \in G$, then

$$\omega \cdot (\zeta_1 - \zeta_2) = z_0(z_1 - z_2) + \bar{z}_0(\bar{z}_1 - \bar{z}_2) = z_0(z_1 - z_2) + \overline{z_0(z_1 - z_2)} = \overline{\omega \cdot (\zeta_1 - \zeta_2)},$$

and thus

$$\{\omega \cdot (\zeta_1 - \zeta_2): (\omega, \zeta_1, \zeta_2) \in \mathcal{H}\} \subset \{z \in \mathbb{C}: \operatorname{Im}(z) = 0\}.$$

The latter set fails to satisfy the analogue of (14.2.79) (with exponent $2 - \varepsilon$ in place of $1 - \varepsilon$) for any choice of scale ρ .

The proof of Theorem 14.2.37 uses ideas and results that are different in character from the rest of the proof of Theorem 14.2.3, so we will defer the proof to Section 14.2.8. In Section 14.2.8 we will prove a mild generalization of Theorem 14.2.37 in which the non-concentration condition on F and G is weakened.

14.2.5.1 The slope function is locally linear

In this section we will use Theorem 14.2.37 to show that if $(\mathbb{T}, Y)_\delta$ is an extremal collection of tubes that satisfies the conclusions of Proposition 14.2.22, then the restriction of the global grains slope function to an interval of length $\rho^{1/2}$ looks linear at scale ρ .

To begin, we will study the interplay between the “global grains” and “local grains” structures from Proposition 14.2.22. If we select a scale $\rho \in [\delta, \delta^\varepsilon]$ and restrict $E_\mathbb{T}$ to the $\rho^{1/2}$ neighborhood of a global grain, then we obtain a set $E \subset E_\mathbb{T}$ that has the following two structures: The global grains property (Item 1 from Proposition 14.2.22) says that each “horizontal” (i.e. $z = z_0$) slice of E is a union of $1 \times \rho$ rectangles pointing in the direction $(1, f(z_0), 0)$, and the spacing of these rectangles forms a ADset in the sense of Definition 14.2.20. The local grains property (Item 2 from Proposition 14.2.22) says that the restriction of E to a ball of radius $\rho^{1/2}$ is a union of $\rho^{1/2} \times \rho^{1/2} \times \rho$ grains, whose normal direction is given by the plane map V . We will use these two structures to construct sets F , G , and $\mathcal{H} \subset F \times G \times G$ that satisfy the hypotheses of Theorem 14.2.37. We will show that Conclusion (B) from Theorem 14.2.37 would contradict the ADset spacing condition from the global grains structure, and hence the sets F and G must satisfy Conclusion (A). This in turn implies that the slope function f is linear at scale ρ over intervals of length $\rho^{1/2}$. More precisely, the slope function f agrees with a linear function at scale ρ on a substantial portion of most intervals of length $\rho^{1/2}$. The precise version is stated in Lemma 14.2.39 below. The precise hypotheses of this lemma are slightly technical; they are formulated in this fashion to allow the lemma to be applied at many scales and locations.

In Lemma 14.2.40, we will apply Lemma 14.2.39 at many scales and locations to the (re-scaled) slices of an extremalakeya set. We will conclude that for many different scales ρ , the slope function f agrees with a linear function at scale ρ on a substantial portion of most intervals of length $\rho^{1/2}$. This in turn implies that on a substantial portion of its domain, the slope function f agrees with a function that has controlled C^2 -norm.

Lemma 14.2.39. *For all $\varepsilon > 0$, there exists $\eta, \rho_0 > 0$ so that the following holds for all $\rho \in (0, \rho_0]$. Let $I \subset [-1, 1]$ be an interval of length $\rho^{1/2}$ with left endpoint z^* , and let $Z \subset I$. Let $f: I \rightarrow [0, 1]$ be 1-Lipschitz. Let $E \subset [-1, 1]^2 \times Z$ be a union of ρ -cubes contained in the $\rho^{1/2}$ -neighborhood of a line*

pointing in direction $(-f(z^*), 1, 0)$, with $|E| \geq \rho^{1+\sigma/2+\eta}$. Let $V: E \rightarrow S^2$ be 1-Lipschitz, with $|V(p) \cdot (0, 0, 1)| \leq 1/2$ for all $p \in E$. Suppose that for each $z_0 \in Z$,

$$(E \cap \{z = z_0\}) \cdot (1, f(z_0), 0) \quad \text{is a } (\rho, 1 - \sigma, \rho^{-\eta})_1\text{-ADset}, \quad (14.2.80)$$

and for each $p \in E$,

$$|(E \cap B(p, \rho^{1/2})) \cdot V(p)| \leq \rho^{\frac{1+\sigma}{2}-\eta}. \quad (14.2.81)$$

Then there exists a set $Z_{\text{lin}} \subset Z$ with $|Z_{\text{lin}}| \geq \rho^{1/2+\varepsilon}$, and a linear function $L: [z^*, z^* + \rho^{1/2}] \rightarrow [0, 1]$ so that $|f(z) - L(z)| \leq \rho$ for all $z \in Z_{\text{lin}}$.

Proof. Without loss of generality we may suppose that $I = [0, \rho^{1/2}]$ (i.e. $z^* = 0$), $f(0) = 0$ (and hence $f(Z) \subset [-\rho^{1/2}, \rho^{1/2}]$), and that E is contained in the $\rho^{1/2}$ -neighborhood of the y -axis. If not, then we can apply a (harmless) translation so that E is contained in the $\rho^{1/2}$ -neighborhood of a line passing through the origin, and then we can apply a rotation around the z -axis so that $f(0) = 0$.

Note that the image E' of E under this rotation is no longer a union of ρ -cubes (since by definition, ρ -cubes are aligned with the standard grid). However, we can replace E' by the union E'' of ρ cubes that intersect E' , and (14.2.80) and (14.2.81) will continue to hold for E'' (the quantities $\rho^{-\eta}$ and $\rho^{\frac{1+\sigma}{2}-\eta}$ in (14.2.80) and (14.2.81) are respectively weakened to $100\rho^{-\eta}$ and $100\rho^{\frac{1+\sigma}{2}-\eta}$, but this is harmless). Note that our rotation fixes each plane $\{z = z_0\}$, so after the initial shift $Z \rightarrow Z - z^*$, the set Z remains unchanged by this procedure.

Step 1: V has large x -coordinate

Divide $[0, \rho^{1/2}] \times [0, 1] \times [0, \rho^{1/2}]$ into $\rho^{-1/2}$ cubes. After pigeonholing and refining E , we may suppose that each $\rho^{1/2}$ -cube Q that intersects E satisfies $|Q \cap E| \gtrsim \rho^{3/2+\sigma/2+\eta}$, and these $\rho^{1/2}$ cubes are $\rho^{1/2}$ separated. For each such cube Q , select a point $p_Q \in Q \cap E$; since the cubes are $\rho^{1/2}$ separated, so are the points p_Q . For each such cube Q , define $\pi_Q(q) = q \cdot V(p_Q)$.

Comparing with (14.2.81), we may further refine E so that each local grain is almost full: if Q is $\rho^{1/2}$ -cube that intersects E , and $w \in V(p_Q) \cdot (Q \cap E)$, then

$$|E \cap Q \cap N_\rho(\pi_Q^{-1}(w))| \gtrsim \rho^{2+2\eta}. \quad (14.2.82)$$

Fix such a cube Q . By (14.2.82) and Fubini, there is some $z_1 \in [0, \rho^{1/2}]$ so that

$$|\{z = z_1\} \cap (E \cap Q \cap N_\rho(\pi_Q^{-1}(w)))| \gtrsim \rho^{3/2+2\eta} \quad (14.2.83)$$

(note that $|\cdot|$ in (14.2.82) refers to 3-dimensional Lebesgue measure, while $|\cdot|$ in (14.2.83) refers to 2-dimensional Lebesgue measure).

Since $|V(p_Q) \cdot (0, 0, 1)| \leq 1/2$, the set on the LHS of (14.2.83) is contained in a rectangle R (in the $\{z = z_1\}$ plane) of dimensions roughly $\rho^{1/2} \times \rho$. Thus $R \cdot (1, f(z_1))$ is an interval I , whose length depends on the angle between the vectors $V(p_Q)$ and $(1, f(z_1), 0)$. More precisely, we have

$$|I| \sim \rho^{1/2} |V(p_Q) \cdot (-f(z_1), 1, 0)|. \quad (14.2.84)$$

By (14.2.83), we have

$$\left| (1, f(z_1), 0) \cdot \left(\{z = z_1\} \cap (E \cap Q \cap N_\rho(\pi_Q^{-1}(w))) \right) \right| \gtrsim \rho^{2\eta} |I|, \quad (14.2.85)$$

but the set on the LHS of (14.2.85) is contained in $I \cap (E \cap \{z = z_1\}) \cdot (1, f(z_1), 0)$, and (14.2.80) says that

$$\left| I \cap (E \cap \{z = z_1\}) \cdot (1, f(z_1), 0) \right| \leq \rho^{1-\sigma-\eta} |I|^\sigma.$$

Re-arranging, we conclude that $|I| \leq \rho^{1-3\eta/(1-\sigma)}$, and thus by (14.2.84),

$$|V(p_Q) \cdot (-f(z_1), 1, 0)| \lesssim \rho^{1/2-3\eta/(1-\sigma)}. \quad (14.2.86)$$

Since $f(z_1) \in [0, \rho^{1/2}]$ and $V(p_Q)$ is a unit vector, if η is selected sufficiently small then (14.2.86) implies that $|V(p_Q) \cdot (0, 1, 0)| \leq 1/10$. Since $V(p_Q)$ is a unit vector and $|V(p_Q) \cdot (0, 0, 1)| \leq 1/2$, we conclude that $|V(p_Q) \cdot (1, 0, 0)| \geq 1/4$.

Step 2: Finding the function g .

Let $E_{\rho^{1/2}}$ be the union of $\rho^{1/2}$ -cubes that intersect E , and let $J \subset [0, 1]$ be the projection of $E_{\rho^{1/2}}$ to the y -axis. Each $y \in J$ is in the image of precisely one $\rho^{1/2}$ -cube that intersects E ; we will denote this cube by $Q(y)$. The set J is a union of $\rho^{1/2}$ intervals, which are $\rho^{1/2}$ separated.

Define the functions $g(y), h(y): J \rightarrow \mathbb{R}$ as follows: For each $y \in J$, $(1, h(y), g(y))$ is parallel to $V(p_Q)$, where $Q = Q(y)$. The functions g and h are constant on each $\rho^{1/2}$ interval of J . Since V is 1-Lipschitz and $|V(p_Q) \cdot (1, 0, 0)| \geq 1/4$ for each cube Q , we have that g and h are 4-Lipschitz. Abusing notation, we will replace g and h by their 4-Lipschitz extension to $[0, 1]$; the image of g and h are contained in $[-5, 5]$.

If $p = (x, y, z) \in E$ is contained in a $\rho^{1/2}$ -cube Q , then $|V(p) - V(p_Q)| \leq \sqrt{3}\rho^{1/2}$, and hence by (14.2.81) we have

$$\left| (E \cap B(p, \rho^{1/2})) \cdot (1, h(y), g(y)) \right| \lesssim \rho^{\frac{1+\sigma}{2}-\eta}.$$

In particular, for each $y_0 \in [0, 1]$ we have

$$\left| (E \cap \{y = y_0\}) \cdot (1, 0, g(y_0)) \right| \lesssim \rho^{\frac{1+\sigma}{2}-\eta}. \quad (14.2.87)$$

We remark that all of the arguments thus far also apply to the Heisenberg group example (14.2.2) (this would involve working in $\mathbb{C}^3$ rather than $\mathbb{R}^3$, but all of the arguments thus far could be translated to that setting). For the Heisenberg group example, we would have that the function $g: \mathbb{C} \rightarrow \mathbb{C}$ is given by $g(z) = \bar{z}$.

Step 3: Paths of length 3.

After refining the set E , we can select a set $Z_{\text{popular}} \subset \rho\mathbb{Z} \cap [0, \rho^{1/2}]$ so that each ρ -cube $Q \subset E$ has center with z -coordinate in Z_{popular} , and each $z \in Z_{\text{popular}}$ is the center of $\gtrsim \rho^{-3/2+\sigma/2+\eta}$ such cubes. After this refinement E is still a union of ρ -cubes, and its volume has decreased by at most a factor of $1/2$.

After pigeonholing and applying a shift of the form $(x, y, z) \mapsto (x, y - y_0, z)$ for some $y_0 \in [0, \rho^{1/2}]$, we can find a set $E_1 \subset E$ that is a union of ρ -cubes, with $|E_1| \geq \rho^{1/2+2\eta}|E|$, so that each ρ -cube $Q \subset E_1$ has center p_Q whose y -coordinate is contained in $\rho^{1/2}\mathbb{Z}$. Indeed, for each local grain, we choose a $\rho \times \rho \times \rho^{1/2}$ -segment orthogonal to the y -direction and include the intersection of this segment with E in E_1 . Note that (14.2.80) and (14.2.87) continue to hold with E_1 in place of E .

We say that two ρ -cubes Q and Q' contained in E_1 are *in the same global grain* if their centers p_Q and $p_{Q'}$ have the same z -coordinate, which we will denote by z_Q , and

$$|p_Q \cdot (1, f(z_Q), 0) - p_{Q'} \cdot (1, f(z_Q), 0)| \leq \rho.$$

We say that two ρ -cubes Q and Q' contained in E_1 are *in the same local grain* if their centers p_Q and $p_{Q'}$ have the same y -coordinate, which we will denote by y_Q , and

$$|p_Q \cdot (1, 0, g(y_Q)) - p_{Q'} \cdot (1, 0, g(y_Q))| \leq \rho.$$

Recall that E_1 is a union of at least $\rho^{-3/2+\sigma/2+3\eta}$ ρ -cubes. For each such cube Q , the y -coordinate of its center is an element of $\rho^{1/2}\mathbb{Z} \cap [0, 1]$, and thus takes one of $\rho^{-1/2}$ values. Once the y -coordinate has been fixed, by (14.2.81) there are at most $\rho^{-1/2+\sigma/2-\eta}$ choices for $p_Q \cdot (1, 0, g(y))$. Thus we can partition E_1 into $O(\rho^{-1+\sigma/2-\eta})$ sets, each of which are unions of ρ -cubes, so that any two ρ -cubes from a common set are in the same local grain. Since E_1 contains at least $\rho^{-3/2+\sigma/2+3\eta}$ ρ -cubes, by Cauchy-Schwarz there are $\gtrsim (\rho^{-3/2+\sigma/2+3\eta})^2 / \rho^{-1+\sigma/2-\eta} = \rho^{-2+\sigma/2+7\eta}$ pairs (Q, Q') , where Q and Q' are ρ -cubes in E that are contained in a common local grain.

Consider the map $s: (Q, Q') \mapsto (z_Q, z_{Q'}, r_{Q'})$, where z_Q is the z -coordinate of the center of Q (similarly for $z_{Q'}$), and $r_{Q'}$ is the number of the form $\rho\mathbb{Z}$ closest to $p_{Q'} \cdot (1, f(z_{Q'}), 0)$. There are $\leq \rho^{-1/2}$ possible values for z_Q and similarly for $z_{Q'}$. By (14.2.80), for each such value of z there are at most $\rho^{-1/2+\sigma/2-\eta}$ values for $r_{Q'}$. Thus s takes at most $\rho^{-3/2+\sigma/2-\eta}$ values.

Thus by Cauchy-Schwarz, there are $\gtrsim (\rho^{-2+\sigma/2+7\eta})^2 / \rho^{-3/2+\sigma/2-\eta} = \rho^{-5/2+\sigma/2+15\eta}$ pairs $((Q, Q'), (\tilde{Q}, \tilde{Q}'))$ where Q and Q' (resp. $\tilde{Q}$ and $\tilde{Q}'$) are contained in a common local grain, and $s(Q, Q') = s(\tilde{Q}, \tilde{Q}')$.

Let $\mathcal{Q}$ denote the set of quadruples $Q_1, \dots, Q_4$ of ρ -cubes contained in E_1 that have the following relations:

- Q_1 and Q_2 are in the same local grain.
- Q_2 and Q_3 are in the same global grain.
- Q_3 and Q_4 are in the same local grain.
- The midpoints of Q_4 and Q_1 have the same z -coordinate.

After renaming the tuple $((Q, Q'), (\tilde{Q}, \tilde{Q}'))$ as $(Q_1, \dots, Q_4)$, we conclude that

$$\#\mathcal{Q} \gtrsim \rho^{-5/2+\sigma/2+15\eta}. \quad (14.2.88)$$

Step 4: A skew transform.

In this step we will perform a minor coordinate change to reduce to the case that $f(0) = g(0) = 0$. By (14.2.88), pigeonholing, and (14.2.80), we can select a choice (z_0, w) so that there are at least $\gtrsim \rho^{-3/2+8\eta}$ quadruples $(Q_1, \dots, Q_4)$ where the center of Q_1 has z -coordinate z_0 , and

$$|p_{Q_1} \cdot (1, f(z_0), 0) - w| \leq \rho. \quad (14.2.89)$$

Denote this set of quadruples by $\mathcal{Q}'$. Define

$$A = (E \cap \{z = z_0\}) \cdot (1, f(z_0), 0) - w. \quad (14.2.90)$$

By (14.2.80), A is a $(\rho, 1 - \sigma, \rho^{-\eta})_1$ -ADset.

Define

$$\phi(x, y, z) = (x + f(z_0)y + g(0)(z - z_0), y, z - z_0).$$

Define $f^\dagger(z) = f(z) - f(z_0)$ and $g^\dagger(y) = g(y) - g(0)$. Then $f^\dagger(0) = g^\dagger(0) = 0$. Let $E^\dagger = \phi(E)$. Then for each $z_1 \in [-z_0, \rho^{1/2} - z_0]$, we have

$$(E^\dagger \cap \{z = z_1\}) \cdot (1, f^\dagger(z_1), 0) = (E \cap \{z = z_1 + z_0\}) \cdot (1, f(z_1), 0) + g(0)(z_1 - z_0), \quad (14.2.91)$$

so by (14.2.80), the LHS of (14.2.91) is a $(\rho, 1 - \sigma, \rho^{-\eta})_1$ -ADset. Similarly, for each $y_1 \in [0, 1]$,

$$(E^\dagger \cap \{y = y_1\}) \cdot (1, 0, g^\dagger(y_1)) = (E \cap \{y = y_1\}) \cdot (1, 0, g(y_1)) + f(z_0)y_1 - g(0)z_0, \quad (14.2.92)$$

so by (14.2.81), the LHS of (14.2.92) has ρ -covering number $\lesssim \rho^{\frac{-1+\sigma}{2}-\eta}$.

Step 5: Few dot products.

Let $\mathcal{W}^\dagger$ be the images of the centers of the cubes in $\mathcal{Q}'$ under ϕ . Note that for each quadruple $(p_1, \dots, p_4) \in \mathcal{W}^\dagger$, p_1 and p_4 have z -coordinate 0, and by (14.2.89), the x -value of p_1 (which we will denote by x_1) satisfies $|x_1 - w| \leq \rho$.

For each $(p_1, \dots, p_4) \in \mathcal{W}^\dagger$, we will associate a tuple (y_1, z_2, y_3) as follows: y_1 will be the y -coordinate of p_1 ; z_2 will be the z -coordinate of p_2 ; and y_3 will be the y -coordinate of p_3 . We will show that

$$z_2(g^\dagger(y_1) - g^\dagger(y_3)) + (y_1 - y_3)f^\dagger(z_2) \in N_{4\rho}(A). \quad (14.2.93)$$

To establish (14.2.93), we will argue as follows. For each index $i = 1, \dots, 4$, write $p_i = (x_i, y_i, z_i)$. First, we have $|x_1 - w| \leq \rho$ and $z_1 = 0$. Since Q_1 and Q_2 are in the same local grain, $y_2 = y_1$, and x_1, x_2, z_2 are related by $|(x_1, y_1, 0) \cdot (1, 0, g^\dagger(y_1)) - (x_2, y_2, z_2) \cdot (1, 0, g^\dagger(y_1))| \leq \rho$ i.e.

$$|x_2 - (w - z_2 g^\dagger(y_1))| \leq 2\rho.$$

Next, since Q_2 and Q_3 are in the same global grain, $z_2 = z_3$, and x_2, x_3, y_2, y_3 are related by $|(x_2, y_2, z_2) \cdot (1, f^\dagger(z_2), 0) - (x_3, y_3, z_2) \cdot (1, f^\dagger(z_2), 0)| \leq \rho$, i.e. (since $y_1 = y_2$)

$$|x_3 - (w - z_2 g^\dagger(y_1) + (y_1 - y_3)f^\dagger(z_2))| \leq 3\rho.$$

Finally, we have $z_4 = 0$. since Q_3 and Q_4 are in the same local grain, $y_4 = y_3$, and x_3, x_4, z_3 are related by $|(x_3, y_3, z_3) \cdot (1, 0, g^\dagger(y_3)) - (x_4, y_4, 0) \cdot (1, 0, g^\dagger(y_3))| \leq \rho$ i.e. (since $z_2 = z_3$)

$$|x_4 - (w - z_2(g^\dagger(y_1) - g^\dagger(y_3)) + (y_1 - y_3)f^\dagger(z_2))| \leq 4\rho. \quad (14.2.94)$$

Since $p_4 = \phi(p_Q)$ for some ρ -cube $Q \subset E$ whose center has z -coordinate z_0 , after unwinding definitions we see that $x_4 \in (E \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ and hence (14.2.93) follows from (14.2.94).

Define

$$\begin{aligned} F &= \{(\rho^{-1/2}z, \rho^{-1/2}f^\dagger(z)) : z - z_0 \in Z_{\text{popular}}\}, \\ G &= \{(y, g^\dagger(y)) : y \in [0, 1] \cap \rho^{1/2}\mathbb{Z}\}. \end{aligned}$$

Since $f^\dagger$ is 1-Lipschitz and $g^\dagger$ is 4-Lipschitz, F and G are $(\rho^{1/2}, 1, C)_2$ -ADsets for some absolute constant C . Let $\mathcal{H}$ be the set of 6-tuples

$$(y_1, g^\dagger(y_1), \rho^{-1/2}z_2, \rho^{-1/2}f^\dagger(z_2), y_3, g^\dagger(y_3)),$$

where (y_1, z_2, y_3) corresponds to a triple from $\mathcal{W}^\dagger$. By construction $\mathcal{H} \subset F \times G \times G$, and

$$\#\mathcal{H} = \#\mathcal{W}^\dagger = \#\mathcal{Q}' \gtrsim (\rho^{1/2})^{16\eta-3}.$$

Let $\tilde{A} = N_{4\rho^{1/2}}(\{\rho^{-1/2}a : a \in A\})$. Then $\tilde{A} \subset [0, 1]$ is a $(\rho^{1/2}, 1 - \sigma, 8\rho^{-\eta})_1$ -ADset, and (14.2.93) says that

$$\{p \cdot (q - q') : (p, q, q') \in \mathcal{H}\} \subset \tilde{A}.$$

Let η_1 be the output of Theorem 14.2.37 with ε as above and $\rho^{1/2}$ in place of ρ . If $\eta > 0$ is selected sufficiently small, then F, G , and $\mathcal{H}$ satisfy the hypotheses of Theorem 14.2.37. If $\eta > 0$ is selected sufficiently small depending on η_1 , then the conclusion of Item (B) cannot hold, since (14.2.79) violates the property that $\tilde{A}$ is a $(\rho^{1/2}, 1 - \sigma, 8\rho^{-\eta})_1$ -ADset.

Hence Conclusion (A) from Theorem 14.2.37 must hold. Let ℓ and $\ell^\perp$ be the orthogonal lines from Conclusion (A) of Theorem 14.2.37. Then by (14.2.78), we have

$$\mathcal{E}_\rho(\rho^{1/2}(F \cap N_{\rho^{1/2}}(\ell))) \geq (\rho^{1/2})^{\varepsilon-1}.$$

Define Z_{lin} to be the union of ρ -intervals that intersect the set

$$\{z \in Z_{\text{popular}} : (\rho^{-1/2}(z + z_0), \rho^{-1/2}f^\dagger(z + z_0)) \in F \cap N_{\rho^{1/2}}(\ell)\}.$$

We have $\rho^{-1}|Z_{\text{lin}}| \geq (\rho^{1/2})^\varepsilon \#Z_{\text{popular}}$, and thus

$$|E \cap ([-1, 1]^2 \times Z_{\text{lin}})| \gtrsim_\rho \rho^{\varepsilon/2} \rho^3 \rho^{-3/2+\sigma/2+3\eta} = \rho^{\varepsilon/2+3\eta+3/2+\sigma/2}.$$

But by (14.2.80), this implies that $|Z_{\text{lin}}| \gtrsim \rho^{1/2+\varepsilon/2+4\eta}$. We will select $\rho_0 > 0$ sufficiently small so that this quantity is at least $\rho^{1/2+\varepsilon}$.

To conclude the proof, the graph of the linear function L from the statement of Lemma 14.2.39 is the obvious re-scaling of ℓ by a factor of $\rho^{1/2}$. Then for each $z \in Z_{\text{lin}}$, we have $|f(z) - L(z)| \leq \rho$. $\square$

The next lemma records what happens when we slice an extremal Kakeya set $(\mathbb{T}, Y)_\delta$ into horizontal slabs of thickness $\rho^{1/2}$, select the $\rho^{1/2}$ neighborhood of a global grain from each of these slices, and apply Lemma 14.2.39: we conclude that the slope function f is linear at scale ρ on each interval of length $\rho^{1/2}$. In what follows, we define $\pi_z : \mathbb{R}^3 \rightarrow \mathbb{R}$ to be the projection to the third coordinate.

Lemma 14.2.40. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes, and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Suppose that there are functions $f : [-1, 1] \rightarrow \mathbb{R}$ and $V : E_{\mathbb{T}} \rightarrow S^2$ that satisfy Items 1 and 2 from the statement of Proposition 14.2.22, with η in place of ε .*

Then there is a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$ and a function $L : [-1, 1] \rightarrow \mathbb{R}$ with the following properties:

- L is 2-Lipschitz, and linear on each interval of the form $[n\rho^{1/2}, (n+1)\rho^{1/2}]$, $n \in \mathbb{Z}$.
- For each $z \in \pi_z(E_{\mathbb{T}'})$, we have $|f(z) - L(z)| \leq \rho$.

Proof. Let $\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0$ be small constants to be chosen below. We will select η very small compared to ε_1 , ε_1 very small compared to ε_2 , ε_2 very small compared to ε_3 , and ε_3 very small compared to ε .

Apply Proposition 14.2.14 to $(\mathbb{T}, Y)_\delta$ with ε_1 in place of ε , and ρ as above. We obtain a refinement of $(\mathbb{T}, Y)_\delta$, and a ε_1 -extremal collection of ρ -tubes, $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. In what follows, we will describe various sub-collections of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. Since $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a balanced cover of $(\mathbb{T}, Y)_\delta$, each of these sub-collections of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ will induce an analogous sub-collection of $(\mathbb{T}, Y)_\delta$, of the same relative density.

Apply Proposition 14.2.14 to $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ with ε_2 in place of ε_1 , and $\rho^{1/2}$ in place of ρ . We obtain a refinement of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$, and a ε_2 -extremal collection of $\rho^{1/2}$ -tubes, $(\tilde{\tilde{\mathbb{T}}}, \tilde{\tilde{Y}})_{\rho^{1/2}}$. After further refining each of these three collections, we may suppose that $(\mathbb{T}, Y)_\delta$ has constant multiplicity; $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a balanced cover of $(\mathbb{T}, Y)_\delta$, and $(\tilde{\tilde{\mathbb{T}}}, \tilde{\tilde{Y}})_{\rho^{1/2}}$ is a balanced cover of $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ (and hence also a balanced cover of $(\mathbb{T}, Y)_\delta$).

Recall that each shading $\tilde{\tilde{Y}}(\tilde{\tilde{\mathbb{T}}})$ is a union of $\rho^{1/2}$ -cubes. After further refining $(\tilde{\tilde{\mathbb{T}}}, \tilde{\tilde{Y}})_{\rho^{1/2}}$ (which again induces a refinement of $(\mathbb{T}, Y)_\delta$ and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$), we may suppose that $E_{\tilde{\tilde{\mathbb{T}}}}$ is contained in a union of $\rho^{1/2}$ -separated “slabs” of the form $\mathbb{R}^2 \times [n\rho^{1/2}, (n+1)\rho^{1/2}]$, $n \in \mathbb{Z}$, and for each such slab S that meets $E_{\tilde{\tilde{\mathbb{T}}}}$, we have $|S \cap E_{\tilde{\mathbb{T}}}| \gtrsim_\rho \rho^{1/2+\varepsilon_1} |E_{\tilde{\mathbb{T}}}|$ and hence $|S \cap E_{\mathbb{T}}| \gtrsim_\delta \rho^{1/2+\varepsilon_1} |E_{\mathbb{T}}| \gtrsim \rho^{1/2+\varepsilon_1} \delta^{\sigma+\eta}$.

We shall consider each slab S in turn. Let $Z_S \subset [n\rho^{1/2}, (n+1)\rho^{1/2}]$ be the set of values z_0 for which $|E_{\mathbb{T}} \cap \{z = z_0\}| \gtrsim_\delta \rho^{2\varepsilon_1} \delta^{\sigma+\eta}$. By Fubini, $|Z_S| \geq \rho^{1/2+\varepsilon_1} \delta^\eta$. Select a choice of $z_0 \in Z_S$. Since $(\mathbb{T}, Y)_\delta$ satisfies Item 1 from the statement of Proposition 14.2.22, we have $|(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)| \leq \delta^{\sigma-\eta}$, i.e. $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is contained in a union of at most $\delta^{\sigma-\eta-1} 1 \times \delta$ rectangles (in the $\{z = z_0\}$ plane), each of which have normal direction $(1, f(z_0), 0)$. By pigeonholing, we can select one of these rectangles that intersects $E_{\mathbb{T}} \cap \{z = z_0\}$ in a set of (two-dimensional) Lebesgue measure $\gtrsim_\delta \delta^{1+2\eta} \rho^{2\varepsilon_1}$, i.e. there exists $w \in \mathbb{R}$ so that

$$|\{p \in E_{\mathbb{T}} \cap \{z = z_0\} : |p \cdot (1, f(z_0), 0) - w| \leq \delta\}| \gtrsim_\delta \delta^{1+2\eta} \rho^{2\varepsilon_1}, \quad (14.2.95)$$

where $|\cdot|$ denotes 2-dimensional Lebesgue measure. The set on the LHS of (14.2.95) is contained in the δ -neighborhood of a unit line segment in the $\{z = z_0\}$ plane, that points in the direction $(1, f(z_0), 0)$; denote this line segment by ℓ . Then ℓ must intersect $\gtrsim \delta^{2\eta} \rho^{-1/2+2\varepsilon_1}$ $\rho^{1/2}$ -cubes from $E_{\tilde{\tilde{\mathbb{T}}}}$, and

hence

$$|N_{\rho^{1/2}}(\ell) \cap E_{\mathbb{T}}| \gtrsim (\delta^{2\eta} \rho^{-1/2+\varepsilon_1}) (\rho^{\varepsilon_1} \rho^{\frac{\sigma-3}{2}}) = \delta^{2\eta} \rho^{\sigma/2-1+2\varepsilon_1}.$$

We can now verify that if η and ε_1 are chosen sufficiently small compared to ε_3 , then the set $N_{\rho^{1/2}}(L) \cap E_{\mathbb{T}}$, along with the function f (restricted to $[n\rho^{1/2}, (n+1)\rho^{1/2}]$) and the function V (restricted to $N_{\rho^{1/2}}(\ell) \cap E_{\mathbb{T}}$) satisfy the hypotheses of Lemma 14.2.39, with ε_3 in place of ε .

Applying Lemma 14.2.39, we obtain a set $Z_{\text{lin},S} \subset [n\rho^{1/2}, (n+1)\rho)$ and a linear function L_S with domain $[n\rho^{1/2}, (n+1)\rho)$, so that $|f(z) - L_S(z)| \leq \rho$ for all $z \in Z_{\text{lin},S}$, and $|Z_{\text{lin},S}| \geq \rho^{1/2+\varepsilon_3}$. In particular, $Z_{\text{lin},S}$ contains two points z_1, z_2 that are at least $\rho^{1/2+\varepsilon_3}$ -separated. Since f is 1-Lipschitz, L_S is linear, and $|f(z_i) - L_S(z_i)| \leq \rho$ for $i = 1, 2$, we conclude that L_S has slope at most 2.

Since $Z_{\text{lin},S} \subset Z_S$ and $|Z_{\text{lin},S}| \geq \rho^{1/2+\varepsilon_3}$, we have

$$|E_{\mathbb{T}} \cap (\mathbb{R}^2 \times Z_{\text{lin},S})| \geq \rho^{1/2+\varepsilon_1+\varepsilon_3} |E_{\mathbb{T}}|. \quad (14.2.96)$$

Define $Z_{\text{lin}} = \bigcup_S Z_{\text{lin},S}$, where the union is taken over all horizontal slabs S that meet $E_{\mathbb{T}}$, a union of $\rho^{1/2}$ -cubes. Then

$$|E_{\mathbb{T}} \cap (\mathbb{R}^2 \times Z_{\text{lin}})| \gtrsim \rho^{\varepsilon_1+\varepsilon_2+\varepsilon_3} |E_{\mathbb{T}}|,$$

and hence if we define $\mathbb{T}' = \mathbb{T}$ and $Y'(T) = Y(T) \cap (\mathbb{R}^2 \times Z_{\text{lin}})$, then (since we replaced $(\mathbb{T}, Y)_{\delta}$ with a constant multiplicity refinement at an earlier step) we have $\sum_{T \in \mathbb{T}'} |Y'(T)| \gtrsim_{\delta} \delta^{\eta} \rho^{\varepsilon_1+\varepsilon_2+\varepsilon_3}$. If $\eta, \varepsilon_1, \varepsilon_2, \varepsilon_3$ are chosen sufficiently small (depending on ε), then $(\mathbb{T}', Y')$ is ε -extremal.

Finally, define L by $L(z) = L_S(z)$ if $z \in I_S$ (here we write $S = [-1, 1]^2 \times I_S$). Note that for each slab S , there are points $p = (x, y, z), p' = (x', y', z') \in E_{\mathbb{T}'}$ with $|z - z'| \geq 4\rho$. Then $|\frac{L(z)-L(z')}{z-z'}| \leq |\frac{f(z)-f(z')}{z-z'}| + \frac{2\rho}{|z-z'|} \leq 3/2$, i.e. L is $(3/2)$ -Lipschitz on I_S . Finally, we can extend L to $[-1, 1]$ by defining L to be piecewise linear; since the intervals I_S on which L was initially defined are $\rho^{1/2}$ -separated (and on each such interval, L agrees with f , which is a 1-Lipschitz function up to accuracy ρ), we have that the extension of L to $[-1, 1]$ is 2-Lipschitz. $\square$

The next lemma records what happens when we iterate Lemma 14.2.40 for many different scales. The proof below keeps the successive extremality parameters explicit and also treats the endpoint scale $\rho = \delta$.

Lemma 14.2.41. *For all $\varepsilon > 0$ and $N \geq 1$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_{\delta}$ be a η -extremal collection of tubes. Suppose that there are functions $f: [-1, 1] \rightarrow \mathbb{R}$ and $V: E_{\mathbb{T}} \rightarrow S^2$ that satisfy Items 1 and 2 from the statement of Proposition 14.2.22, with η in place of ε .*

For each $j = 1, \dots, N$, define $\rho_j = \delta^{j/N}$. Then there is a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$ and functions $L_1, \dots, L_N$ with the following properties:

- (a) Each L_j is 2-Lipschitz, and linear on each interval of the form $[n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$, $n \in \mathbb{Z}$.
- (b) For each $z \in \pi_z(E_{\mathbb{T}'})$ and each index j , we have $|f(z) - L_j(z)| \leq \rho_j$.

Proof. Fix N . We first note an endpoint extension of Lemma 14.2.40: its conclusion remains valid when $\rho = \delta$. Indeed, in that proof the first application of Proposition 14.2.14, whose only purpose is to produce the ρ -scale balanced cover, can be replaced at $\rho = \delta$ by the identity cover $(\mathbb{T}, Y)_\delta$. The second application is at scale $\rho^{1/2} = \delta^{1/2}$, which lies in the permitted range after its small parameter is chosen below $1/2$. Every subsequent estimate in the proof is unchanged. Thus the endpoint case has the same conclusion and the same quantifier order as the stated lemma.

Choose numbers

$$0 < \kappa_N \ll \kappa_{N-1} \ll \dots \ll \kappa_1 \ll \min(\varepsilon, 1/N),$$

where each κ_{j+1} is small enough to serve as the input extremality parameter required by Lemma 14.2.40 with output parameter κ_j . Finally choose η smaller than all the input parameters so obtained, and choose δ_0 smaller than the corresponding scale thresholds. Since a θ -extremal family is also θ' -extremal for $\theta' \geq \theta$, these choices allow every later application after a refinement.

Start with $(\mathbb{T}_0, Y_0) = (\mathbb{T}, Y)$. For $j = 1, \dots, N-1$, apply Lemma 14.2.40 to the current family at $\rho_j = \delta^{j/N}$. The scale condition holds because

$$\delta^{1-1/(2N)} \leq \rho_j \leq \delta^{1/(2N)}$$

after replacing $1/(2N)$ by a slightly smaller fixed number at the two endpoints. We obtain a refinement $(\mathbb{T}_j, Y_j)$ and a 2-Lipschitz function L_j , linear on every interval $[m\rho_j^{1/2}, (m+1)\rho_j^{1/2}]$, such that

$$|f(z) - L_j(z)| \leq \rho_j \quad (z \in \pi_z(E_{\mathbb{T}_j})). \quad (6.9)$$

At the last step use the endpoint extension just proved, with $\rho_N = \delta$, to construct $(\mathbb{T}_N, Y_N)$ and L_N .

Every stage is a refinement, so (6.9) for an earlier j remains true on the support of every later family. Let $(\mathbb{T}', Y') = (\mathbb{T}_N, Y_N)$. Our hierarchy of parameters ensures that the last family is κ_1 -extremal, hence ε -extremal. The functions $L_1, \dots, L_N$ have property (a) by construction, and applying (6.9) on the final support gives property (b) simultaneously for all j . $\square$

Conclusion (b) enforces some consistency between the functions L_j for different values of j —but we must be a bit careful. Let I be an interval of the form $[n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$. Note that Conclusion (b) does *not* imply that $|L_j(z) - L_{j-1}(z)| \leq \rho_j + \rho_{j-1}$ for all $z \in I$. However, if $z_1, z_2 \in I \cap \pi_z(E_{\mathbb{T}'})$, then $|L_j(z) - L_{j-1}(z)| \leq \rho_j + \rho_{j-1}$ for all $z \in [z_1, z_2]$. Luckily, after a harmless refinement of $(\mathbb{T}', Y')_\delta$ by a factor of 4^{-N} , can ensure that the output $(\mathbb{T}', Y')_\delta$ satisfies the following popularity condition: each interval $I = [n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$ that intersects $\pi_z(E_{\mathbb{T}'})$ has large intersection with I , see (14.2.97) below. We can also enforce a separation condition that will be useful later:

For each index j , let $\mathcal{I}_j$ be the set of intervals of the form $I = [n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$ that intersect $\pi_z(E_{\mathbb{T}'})$. Then in addition to Items (a) and (b) above, for each $j = 1, \dots, N$ we also have:

(c) Each interval $I \in \mathcal{I}_j$ satisfies

$$|I \cap \pi_z(E_{\mathbb{T}'})| \geq 4^{-N} \delta^\varepsilon |I|. \quad (14.2.97)$$

(d) $\mathcal{I}_j$ is $\rho_j^{1/2}$ -separated, i.e. no two adjacent intervals are contained in $\mathcal{I}_j$.

Indeed, for each $j = N, \dots, 1$, we obtain Items (c) and (d) at scale ρ_j as follows. For Item (c), we discard all intervals I for which (14.2.97) fails, and refine the shading Y' accordingly. For Item (d), we either discard all intervals $[n\rho_j^{1/2}, (n+1)\rho_j^{1/2}]$ for which n is odd, or all intervals for which n is even. This step does not destroy Items (c) and (d) for previous (larger) values of j (corresponding to smaller values of ρ_j).

Thus, for each $I \in \mathcal{I}_j$ there is an interval $I' \subset I$ of length $|I'| \geq 4^{-N} \delta^\varepsilon \rho_j^{1/2}$ so that $\pi_z(E_{\mathbb{T}'}) \cap I \subset I'$, and

$$|L_j(z) - L_{j-1}(z)| \leq \rho_j + \rho_{j-1} \quad \text{for all } z \in I'. \quad (14.2.98)$$

It will be helpful to restate Lemma 14.2.41 by replacing the piecewise-linear functions L_j with “vertical trapezoids,” which are defined as follows:

Definition 14.2.42. A *vertical trapezoid* is a set of the form

$$R = \{(z, t) \in J \times \mathbb{R} : |t - L(z)| \leq s\},$$

where $J \subset [-1, 1]$ is an interval and $L : J \rightarrow [-1, 1]$ is linear. We define $w = |J|$ to be the *length* of the trapezoid, and s to be the *height* of the trapezoid; we will refer to R as a $w \times s$ trapezoid, and we define $\text{slope}(R)$ to be the slope of L .

With this definition, (14.2.98) can be rephrased as a set-containment statement about vertical trapezoids. This allows us to restate Lemma 14.2.41 (with the additional Conclusions (c) and (d)) as follows:

Corollary 14.2.43. *For all $\varepsilon > 0$ and $N \geq 1$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a η -extremal collection of tubes. Suppose that there are functions $f: [-1, 1] \rightarrow \mathbb{R}$ and $V: E_{\mathbb{T}} \rightarrow S^2$ that satisfy Items 1 and 2 from the statement of Proposition 14.2.22, with η in place of ε .*

For each $j = 1, \dots, N$, define $\rho_j = \delta^{j/N}$. Then there is a ε -extremal sub-collection $(\mathbb{T}', Y')_\delta$ and sets $\mathcal{R}_1, \dots, \mathcal{R}_N$ of vertical trapezoids, with the following properties:

- *Each trapezoid has slope in $[-2, 2]$.*
- *For each j , each trapezoid $R \in \mathcal{R}_j$ has height ρ_j and length between $\rho_j^{1/2+\varepsilon}$ and $\rho_j^{1/2}$.*
- *For each j , the projections of the trapezoids in $\mathcal{R}_j$ to the z -coordinate are $\rho_j^{1/2}$ -separated.*
- *For each $j > 1$ and each $R \in \mathcal{R}_j$, there is a (necessarily unique) parent trapezoid $\tilde{R} \in \mathcal{R}_{j-1}$ that contains R .*
- *Let $D = \pi_z(E_{\mathbb{T}'})$. Then $\text{graph}(f|_D) \subset \bigcup_{R \in \mathcal{R}_N} R$.*

14.2.5.2 Locally linear at many scales implies C^2

In this section, we will prove that the conclusions of Corollary 14.2.43 imply that f must have small C^2 norm (or more precisely, the restriction $f|_D$ agrees to accuracy δ with a function that has small C^2 norm). The precise statement is as follows.

Proposition 14.2.44. *Let $\eta > 0$, let $N \geq 2$, and let $\delta > 0$. For each $j = 1, \dots, N$, define $\rho_j = \delta^{j/N}$ and let $\mathcal{R}_j$ be a set of vertical trapezoids with the following properties:*

- *Each trapezoid has slope in $[-2, 2]$.*
- *For each j , each trapezoid $R \in \mathcal{R}_j$ has height ρ_j and length between $\rho_j^{1/2+\eta}$ and $\rho_j^{1/2}$.*
- *For each j , the projections of the trapezoids in $\mathcal{R}_j$ to the z -coordinate are $\rho_j^{1/2}$ -separated.*
- *For each $j > 1$ and each $R \in \mathcal{R}_j$, there is a (necessarily unique) parent trapezoid $\tilde{R} \in \mathcal{R}_{j-1}$ that contains R .*

Let $D \subset [-1, 1]$ and let $f: D \rightarrow \mathbb{R}$ with $\text{graph}(f) \subset \bigcup_{R \in \mathcal{R}_N} R$.

Then there exists a function $g: [-1, 1] \rightarrow \mathbb{R}$ so that

1. $|f(x) - g(x)| \leq \delta$ for all $x \in D$.
2. $|g''(x)| \lesssim \delta^{-1/N-2\eta}$ for all $x \in [0, 1]$.
3. If $x \in \pi(R)$ for some $R \in \mathcal{R}_j$, then $|g'(x) - \text{slope}(R)| \lesssim \rho_j / \rho_{j+1}^{1/2+\eta}$.

The next lemma describes the basic building block of the function g .

Lemma 14.2.45. *Let $x_1 < x_2$ and let L be the line segment from (x_1, y_1) to (x_2, y_2) . Let $a > 0$. Then there is a function $G: \mathbb{R} \rightarrow \mathbb{R}$ with the following properties.*

- $L \subset \text{graph}(G)$.
- $G(x) = 0$ for $x \in \mathbb{R} \setminus [x_1 - a, x_2 + a]$.
- Let $h = |y_1| + |y_2|$ and let $s = \frac{y_2 - y_1}{x_2 - x_1}$. Then $|G'(x)| \lesssim h/a + |s|$ and $|G''(x)| \lesssim h/a^2 + |s|/a$ for all $x \in \mathbb{R}$.

Proof. Let $\phi: \mathbb{R} \rightarrow [0, 1]$ be a smooth (weakly) monotone increasing function with $\phi(x) = 0$ for $0 \leq x \leq 1/3$ and $\phi(x) = 1$ for $2/3 \leq x \leq 1$. We will choose ϕ so that $|\phi'(x)| \leq 10$ and $|\phi''(x)| \leq 10$ for $0 \leq x \leq 1$. Define $\phi_a(x) = \phi(x/a)$. Then $\phi_a(0) = 0$, $\phi_a(a) = 1$, $(\phi_a)'_+(0) = (\phi_a)'_-(a) = 0$, and $(\phi_a)''_+(0) = (\phi_a)''_-(a) = 0$. We also have $|\phi'_a(x)| \lesssim a^{-1}$ and $|\phi''_a(x)| \lesssim a^{-2}$ for $x \in [0, a]$.

Define $\psi_a(x) = (x - a)\phi_a(x)$. Then $\psi_a(0) = \psi_a(a) = 0$, $(\psi_a)'_+(0) = 0$, $(\psi_a)'_-(a) = 1$, and $(\psi_a)''_+(0) = (\psi_a)''_-(a) = 0$. We also have $|\psi'_a(x)| \lesssim 1$ and $|\psi''_a(x)| \lesssim a^{-1}$ for $x \in [0, a]$.

Let

$$G(x) = \begin{cases} y_1\phi_a(x - x_1 + a) + s\psi_a(x - x_1 + a), & x \leq x_1, \\ s(x - x_1) + y_1, & x_1 < x < x_2, \\ y_2\phi_a(a + x_2 - x) - s\psi_a(a + x_2 - x), & x \geq x_2. \end{cases} \quad \square$$

The function g from 14.2.44 will be a sum of the functions G from Lemma 14.2.45, with one function G for each trapezoid R . We turn to the details.

Proof of Lemma 14.2.44. For each $R \in \mathcal{R}_1$, let $\tilde{L}(R)$ be the line segment coaxial with R . Next, for $2 \leq k \leq N$ and for each $R \in \mathcal{R}_k$, let $\tilde{s}(R) = \text{slope}(R) - \text{slope}(\tilde{R})$, where $\tilde{R} \in \mathcal{R}_{k-1}$ is the parent of R . Let $(x_1, y_1), (x_2, y_2)$ be the endpoints of the line segment coaxial with R . Let $\tilde{y}_1, \tilde{y}_2$ be selected so that $(x_1, \tilde{y}_1), (x_2, \tilde{y}_2)$ lie on the line segment coaxial with $\tilde{R}$. Let $\tilde{L}(R)$ be the line segment from $(x_1, y_1 - \tilde{y}_1)$ to $(x_2, y_2 - \tilde{y}_2)$. Finally, let $G_R(x)$ be the function defined in Lemma 14.2.45 with $L = \tilde{L}(R)$ and $a = \rho_k^{1/2}/2$.

Define

$$g(x) = \sum_{k=1}^N \sum_{R \in \mathcal{R}_k} G_R(x).$$

We will verify that this function satisfies the three conclusions of Lemma 14.2.44. Conclusion 1 follows immediately from the definition of g and the definition of the functions $\{G_R\}$.

For conclusion 2, observe that for each $1 \leq k \leq N$ and each $R \in \mathcal{R}_k$, we have $x_2 - x_1 \gtrsim \rho_k^{1/2+\eta}$ and $|y_1 - \tilde{y}_1|, |y_2 - \tilde{y}_2| \leq \rho_{k-1}$. We also have $|\text{slope}(\tilde{L}(R))| \leq \rho_{k-1}/\rho_k^{1/2+\eta}$. This implies that

$$|G_R''(x)| \lesssim \frac{\rho_{k-1}}{\rho_k^{1/2+\eta}} \leq \delta^{-\frac{1}{N}-2\eta\frac{k}{N}} \quad \text{for all } x \in \mathbb{R}. \quad (14.2.99)$$

We have

$$|g''(x)| \leq \sum_{k=1}^N \left| \sum_{R \in \mathcal{R}_k} G_R''(x) \right|. \quad (14.2.100)$$

Recall that for each index $k = 1, \dots, N$, the projections of the trapezoids in $\mathcal{R}_k$ to the z -axis are $\rho_k^{1/2}$ -separated. This implies that the functions $\{G_R : R \in \mathcal{R}_k\}$ have disjoint support. Thus for each $k = 1, \dots, N$, at most one function from $\{G_R : R \in \mathcal{R}_k\}$ can contribute to the inner sum in (14.2.100). Thus by (14.2.99) we have

$$|g''(x)| \lesssim \sum_{k=1}^N \delta^{-\frac{1}{N}-2\eta\frac{k}{N}} \lesssim \delta^{-\frac{1}{N}-2\eta}.$$

For conclusion 3, observe that if $R \in \mathcal{R}_k$, then

$$|G_R'(x)| \lesssim \frac{\rho_{k-1}}{\rho_k^{1/2+\eta}} \quad \text{for all } x \in \mathbb{R}. \quad (14.2.101)$$

Next, if $R_0 \in \mathcal{R}_k$ and $x \in \pi(R_0)$, then

$$\text{slope}(R_0) = \sum_{j=1}^k \sum_{R \in \mathcal{R}_j} G_R'(x).$$

Thus

$$|g'(x) - \text{slope}(R_0)| \leq \sum_{j=k+1}^N \left| \sum_{R \in \mathcal{R}_j} G_R'(x) \right|.$$

Again, for each index $k = 1, \dots, N$ the functions $\{G_R : R \in \mathcal{R}_k\}$ have disjoint support. Thus for each $j = k+1, \dots, N$, at most one function G_R , $R \in \mathcal{R}_j$ can contribute to the above sum. Thus by (14.2.101) we have

$$|g'(x) - \text{slope}(R_0)| \leq \sum_{j=k+1}^N \frac{\rho_{j-1}}{\rho_j^{1/2+\eta}} \lesssim \frac{\rho_k}{\rho_{k+1}^{1/2+\eta}}. \quad \square$$

14.2.5.3 Proof of Proposition 14.2.36

We will prove Proposition 14.2.36 by combining Corollary 14.2.43 and Proposition 14.2.44. The details are as follows.

Proof of Proposition 14.2.36. Let $\delta_0 > 0'$ and $\varepsilon_1, \varepsilon_2 > 0$ be small constants to be determined below. We will select ε_1 very small compared to ε_2 , and ε_2 very small compared to ε . Let $\delta \in (0, \delta'_0]$ and let $(\mathbb{T}, Y)_\delta$ be a collection of δ -tubes that satisfies the conclusions of Proposition 14.2.22, with ε_1 in place of ε . Apply Corollary 14.2.43 to $(\mathbb{T}, Y)_\delta$ with ε_2 in place of ε and $N = 1/\varepsilon_2$, and let $(\mathbb{T}_1, Y_1)_\delta$ and $\{\mathcal{R}_j\}_{j=1}^N$ be the output of this lemma.

Apply Proposition 14.2.44 to the family of trapezoids $\{\mathcal{R}_j\}_{j=1}^N$. The resulting function g almost satisfies Conclusion 1' of Proposition 14.2.36, except we only have the bound $\|g\|_{C^2} \lesssim \delta^{-3\varepsilon_2}$. This can be remedied by restricting $(\mathbb{T}_1, Y_1)_\delta$ to a slab of dimensions roughly $1 \times 1 \times \delta^{3\varepsilon_2}$ and then applying a vertical re-scaling using Lemma 14.2.18. After this re-scaling, the re-scaled function g will satisfy $\|g\|_{C^2} \leq 1$, as desired, and the resulting collection of tubes $(\mathbb{T}', Y')_{\delta'}$ will continue to satisfy Conclusion 2 of Proposition 14.2.36. If δ'_0 is chosen sufficiently small (depending on δ_0 and ε) and if ε_2 is chosen sufficiently small, then we will have $\delta' \leq \delta_0$, and $(\mathbb{T}', Y')_{\delta'}$ will be ε -extremal. $\square$

14.2.6 The slope function has large slope

Our goal in this section is to strength the conclusions of Proposition 14.2.36 by showing that the global grains slope function has derivative bounded away from 0. In this step we make important use of the fact that the tubes in $\mathbb{T}$ point in different directions (though with a modification of our argument, a weaker condition specifying that the tubes in $\mathbb{T}$ do not concentrate too tightly into ρ tubes is sufficient—see [45]). More precisely, we will prove the following.

Proposition 14.2.46. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and a ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the following properties:*

1''. Global grains with nonsingular C^2 slope function.

There is a function $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$, so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.

2. Local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_\delta$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

In the remainder of the paper, only Item 1'' will be used. We keep Item 2 for completeness. We will begin with a technical lemma, which says that if

U is a segment of a tube T , and if the image of a large subset of U under a linear projection $p \mapsto p \cdot v$ is an ADset, then v must be nearly orthogonal to $\text{dir}(T)$. The precise statement is as follows.

Lemma 14.2.47. *Let $\varepsilon > 0$, $0 \leq \alpha < 1$, and $0 < \delta \leq \rho \leq 1$. Let T be a δ -tube, let $U \subset T$ be a tube segment of length $\rho^{1/2}$, and let $F \subset U$ with*

$$|F| \geq \delta^\varepsilon |U|. \quad (14.2.102)$$

Let $v \in S^2$, and suppose $v \cdot F$ is a $(\rho, \alpha, \delta^{-\varepsilon})_1$ -ADset. Then

$$|v \cdot \text{dir}(T)| \lesssim \delta^{\frac{-2\varepsilon}{1-\alpha}} \rho^{1/2}. \quad (14.2.103)$$

Proof. We will suppose that $|v \cdot \text{dir}(T)| \geq \rho^{1/2}$, since otherwise there is nothing to prove. Let $\tau = |v \cdot \text{dir}(T)|$. Divide U into sub-segments of length ρ/τ (by hypothesis $\rho/\tau \leq \rho^{1/2}$, so there is at least one such segment), and let $\mathcal{S}$ be this set of segments. For each $S \in \mathcal{S}$, $v \cdot S$ is an interval of length $\sim \rho$, and no point $x \in \mathbb{R}$ is contained in more than 10 of the sets $\{v \cdot S : S \in \mathcal{S}\}$. By (14.2.102), we have

$$\#\{S \in \mathcal{S} : S \cap F \neq \emptyset\} \geq \delta^\varepsilon \tau \rho^{-1/2},$$

and thus

$$\mathcal{E}_\rho(v \cdot F) \gtrsim \delta^\varepsilon \tau \rho^{-1/2}. \quad (14.2.104)$$

On the other hand, $v \cdot F \subset v \cdot U$ is contained in an interval $I \subset \mathbb{R}$ of length $\sim \rho^{1/2} \tau$, and hence

$$\mathcal{E}_\rho(v \cdot F) \leq \mathcal{E}_\rho((v \cdot F) \cap I) \leq \delta^{-\varepsilon} (|I|/\rho)^\alpha \lesssim \delta^{-\varepsilon} (\rho^{-1/2} \tau)^\alpha. \quad (14.2.105)$$

Comparing (14.2.104) and (14.2.105), we conclude that $\tau \lesssim \delta^{\frac{-2\varepsilon}{1-\alpha}} \rho^{1/2}$, as desired. $\square$

The next lemma says that if $(\mathbb{T}, Y)_\delta$ is an extremal set of tubes pointing in different directions with C^2 slope function f , then f cannot have small derivative on any “thick” slab that contains (at least) an average amount of the mass of $(\mathbb{T}, Y)_\delta$. This lemma contains the key geometric argument needed to prove Proposition 14.2.46.

Lemma 14.2.48. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes that satisfies the conclusions of Proposition 14.2.36, with η in place of ε . Suppose in addition that the tubes in $\mathbb{T}$ point in δ -separated directions. Let $J \subset [-1, 1]$ be an interval, with $\delta^{1/2} \leq |J| \leq \delta^\varepsilon$. Suppose that*

$$\sum_{T \in \mathbb{T}} |Y(T) \cap (\mathbb{R}^2 \times J)| \geq \delta^\eta |J|, \quad (14.2.106)$$

and that $(\mathbb{R}^2 \times J) \cap E_{\mathbb{T}}$ can be covered by a union of $\leq \delta^{-\eta}|J|^{-2+\sigma}$ cubes of side-length J .

Then the slope function $f: [-1, 1] \rightarrow \mathbb{R}$ from Proposition 14.2.36 satisfies

$$|f'(z)| \geq |J| \quad \text{for all } z \in J. \quad (14.2.107)$$

Proof. Suppose not, i.e. there exists $z_0 \in J$ with $|f'(z_0)| < |J|$. Since $\|f\|_{C^2} \leq 1$, we have $|f'(z)| \leq 2|J|$ for all $z \in J$. We will obtain a contradiction.

Define $\rho = |J|^2$, so $\delta \leq \rho \leq \delta^{2\varepsilon}$. After a translation and rotation, we may suppose that $J = [0, \rho^{1/2}]$ and $f(0) = 0$ (note that after this rotation, our shadings $Y(T)$ will be unions of cubes that are no longer axis parallel, but this is harmless for the arguments that follow).

After a refinement, we may suppose there is a number $\delta^{-\sigma+\eta} \leq \mu \leq \delta^{-\sigma-\eta}$ so that for each point $p \in (\mathbb{R}^2 \times J) \cap E_{\mathbb{T}}$, we have $\#\mathbb{T}(p) = \mu$. Define $F = (\mathbb{R}^2 \times J) \cap E_{\mathbb{T}}$. By (14.2.106), we have $\delta^{\sigma+2\eta}|J| \lesssim |F| \lesssim \delta^{\sigma-2\eta}|J|$. In the arguments that follow, we will describe several refinements of F that endow the set with additional structural properties. Each of these refinements will continue to be a union of (rotated) δ -cubes.

Refinement 1: Every global grain is popular. Recall that $(\mathbb{T}, Y)_{\delta}$ satisfies Item 1' from Proposition 14.2.36. Informally, this says that $E_{\mathbb{T}}$ is a union of global grains (rectangular prisms of dimensions $1 \times \delta \times \delta$), and that most of these grains are almost full. More precisely, we can select a subset $F_1 \subset F$ with $|F_1| \geq \delta^{2\eta}|F|$ so that the following property holds: For each $z_1 \in J$ and each $w \in (F \cap \{z = z_1\}) \cdot (1, f(z_1), 0)$, define the $1 \times \delta \times \delta$ rectangular prism

$$G_{w,z_1} = \{|z - z_1| \leq \delta\} \cap \{|(x, y) \cdot (1, f(z_1)) - w| \leq \delta\}. \quad (14.2.108)$$

Then G_{w,z_1} has large intersection with F_1 ; specifically, we have

$$|G_{w,z_1} \cap F_1| \geq \delta^{2\eta}|G_r| \gtrsim \delta^{2+2\eta}. \quad (14.2.109)$$

Refinement 2: Every global grain intersects a common $\{y = y_0\}$ plane. Each of the global grains G_{w,z_1} from (14.2.108) will intersect each plane of the form $\{y = y_0\}$ for $0 \leq y_0 \leq 1$. While it may not be true that every set $G_{w,z_1} \cap F_1$ will intersect every plane $\{y = y_0\}$, it is true that a typical set of this form is likely to intersect a typical plane. Thus we can find a choice of plane $\{y = y_0\}$ that intersects a large proportion of the sets $G_{w,z_1} \cap F_1$. We turn to the details.

Recall that F (and hence F_1) can be covered by a union of $\leq \delta^{-\eta}\rho^{-1+\sigma/2}$ cubes of side-length $\rho^{1/2}$; weakening this inequality by a constant factor, we may suppose that these cubes are aligned with the $\rho^{1/2}$ -grid. Denote the union of these cubes by K . Let $Y_{\text{bad}} \subset [-1, 1]$ be the set of y_0 for which $|\{y = y_0\} \cap K| \geq \delta^{-7\eta}\rho^{1/2+\sigma/2}$. By Fubini we have $|Y_{\text{bad}}| \leq \delta^{4\eta}$. If G_{w,z_1} is a

set of the form (14.2.108), then $|G_{w,z_1} \cap (\mathbb{R} \times Y_{\text{bad}} \times \mathbb{R})| \lesssim \delta^{4\eta+2}$, and hence if G_{w,z_1} satisfies (14.2.109), then

$$|(G_{w,z_1} \cap F_1) \setminus (\mathbb{R} \times Y_{\text{bad}} \times \mathbb{R})| \geq |G_{w,z_1} \cap F_1|/2.$$

Thus if we define $F'_1 = F_1 \setminus (\mathbb{R} \times Y_{\text{bad}} \times \mathbb{R})$, then (in light of Refinement 1) we have $|F'_1| \geq |F_1|/2$.

By Fubini, there is a choice of $y_0 \in [-1, 1] \setminus Y_{\text{bad}}$ so that

$$|\{y = y_0\} \cap F'_1| \geq \frac{1}{2}|F'_1|,$$

where on the LHS, $|\cdot|$ denotes 2-dimensional Lebesgue measure and on the RHS, $|\cdot|$ denotes 3-dimensional Lebesgue measure. Let $X = \bigcup G_{w,z_1}$, where the union is taken over all grains G_{w,z_1} with $\{y = y_0\} \cap G_{w,z_1} \cap F'_1 \neq \emptyset$, and let $F_2 = F'_1 \cap X$. By (14.2.109), we have $|F_2| \geq \delta^{4\eta}|F|$.

Step 3: Local grains at scale ρ extend to global grains. Recall that by Item 1' from Proposition 14.2.36, each slice $E_{\mathbb{T}} \cap \{z = z_0\}$ is a union of global grains. A priori, the global grains for one value of z_0 need not be related to the grains for a different value of z_0 (though of course the slopes of these grains are related by the regularity of the slope function f). In this step, we will show that the local grains coming from Item 2 of Proposition 14.2.36 ensure consistency between the global grains for different slices $\{z = z_0\}$. We turn to the details.

Cover the rectangle $[-1, 1] \times \{y = y_0\} \times J$ by interior-disjoint squares of side-length $\rho^{1/2} = |J|$. Since $y_0 \notin Y_{\text{bad}}$, at most $\delta^{-10\eta}\rho^{\frac{\sigma-1}{2}}$ of these squares intersect $F_2 \cap \{y = y_0\}$. Denote this set of squares by $\mathcal{S}$.

Let $S \in \mathcal{S}$ be a square, with center $p_S = (x_S, y_0, \rho^{1/2}/2)$. By Item 2 from Proposition 14.2.36, we have that $(S \cap F_2) \cdot V(p_S)$ is a $(\rho, 1 - \sigma, \delta^{-\eta})_1$ -ADset. Write $V(p_S) = (V_x(p_S), V_y(p_S), V_z(p_S))$. Since $S \cap F_2 \subset \{y = y_0\}$, if we define $\tilde{V}(p_S) = (V_x(p_S), 0, V_z(p_S))$, then

$$(S \cap F_2) \cdot \tilde{V}(p_S) \text{ is a } (\rho, 1 - \sigma, \delta^{-\eta})_1\text{-ADset.} \quad (14.2.110)$$

In particular, $S \cap F_2$ is contained in a union of at most $\delta^{-\eta}\rho^{\frac{\sigma-1}{2}}$ rectangles of dimensions $\rho^{1/2} \times \rho$. If $R \subset S$ is such a rectangle, and if G_{w,z_1} is a grain of the form (14.2.108) that intersects R , then since $|f(z_1)| \leq \rho$, we have $G_{w,z_1} \subset \{(x, y, z) : (x, z) \in 2R\}$, where $2R$ is the $\rho^{1/2} \times 2\rho$ rectangle that has the same major axis as R . But in light of (14.2.110), this implies that if we define the prism $P_S = \{(x, y, z) : (x, z) \in S\}$, then

$$(F_2 \cap P_S) \cdot \tilde{V}(p_S) \text{ is a } (\rho, 1 - \sigma, 2\delta^{-\eta})_1\text{-ADset.}$$

See Figure 14.2.5

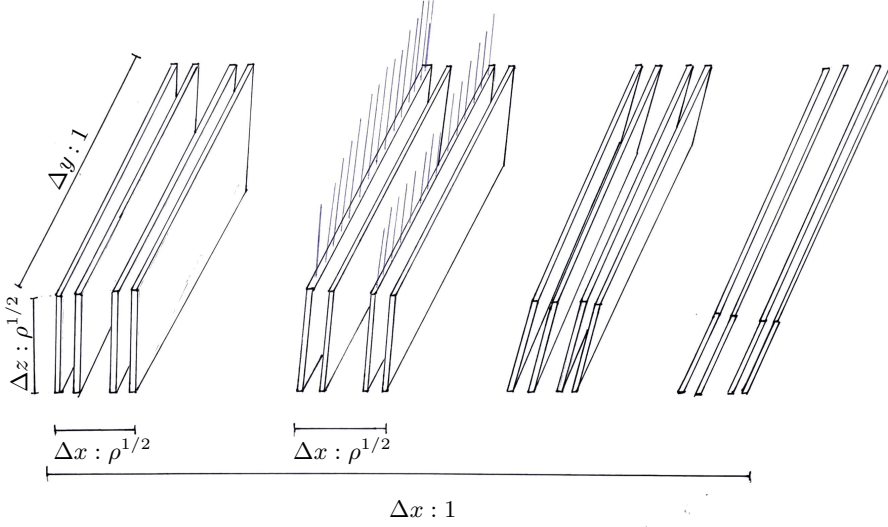


Figure 14.2.5: The rectangular prisms $\{P_S : S \in \mathcal{S}\}$, and the tubes from $\mathbb{T}$ (blue lines) that intersect them. For clarity, only a few tubes (blue lines) have been drawn.

Step 4: $\mathbb{T}$ contains parallel tubes. Many tubes from $\mathbb{T}$ must intersect each of the parallel slabs in Figure 14.2.5, and if a tube T intersects a slab, this imposes constraints on the possible directions of $v(T)$. In this step, we will show that these constraints force many tubes from $\mathbb{T}$ to point in almost the same direction. This contradicts the assumption that the tubes in $\mathbb{T}$ point in δ -separated directions. We turn to the details.

For each $T \in \mathbb{T}$, define $F_2(T) = Y(T) \cap F_2$. Then recalling (14.2.106), we have

$$\sum_{T \in \mathbb{T}} |F_2(T)| \geq \delta^{5\eta} |J| = \delta^{5\eta} \rho^{1/2}.$$

Recall that each $T \in \mathbb{T}$ intersects at most 3 prisms $P_S, S \in \mathcal{S}$, and for such a prism we have $|F_2(T) \cap P_S| \leq \rho^{1/2} \delta^2$. Thus by pigeonholing there exists a square $S \in \mathcal{S}$ and a set $\mathbb{T}_S \subset \mathbb{T}$ with

$$\#\mathbb{T}_S \gtrsim \delta^{-2} (\#\mathcal{S})^{-1} \gtrsim \delta^{11\eta-2} \rho^{\frac{1-\sigma}{2}}, \quad (14.2.111)$$

so that $|F_2(T)| \gtrsim \delta^{11\eta} \rho^{1/2} \delta^2$ for each $T \in \mathbb{T}_S$.

But for each such $T \in \mathbb{T}_S$, we have that $F_2(T) \cdot \tilde{V}(p_S) \subset (F_2 \cap P_S) \cdot \tilde{V}(p_S)$ is a $(\rho, 1 - \sigma, 2\delta^{-\eta})_1$ -ADset. Thus by Lemma 14.2.47, we have

$$|\text{dir}(T) \cdot \tilde{V}(p_S)| \leq \delta^{-C_0\eta} \rho^{1/2},$$

where the constant C_0 depends on σ . But comparing with (14.2.111) and pigeonholing, we conclude that there is a direction $v \in S^2$ so that

$$\#\{T \in \mathbb{T}_S : |\text{dir}(T) - v| \leq \delta\} \geq \delta^{C_1\eta} \rho^{-\sigma/2} \geq \delta^{C_1\eta - \sigma\varepsilon},$$

where C_1 is a constant that depends on σ . If $\eta > 0$ is selected sufficiently small (depending on σ and ε), then this contradicts the assumption that the tubes in $\mathbb{T}$ point in δ -separated directions. $\square$

We are almost ready to prove Proposition 14.2.46. First we will prove an intermediate result, which constructs an extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the property that the shadings $Y(T)$ are supported on a thick slab, and the global grains slope function f is large on this slab.

Lemma 14.2.49. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and a set of tubes $(\mathbb{T}, Y)_\delta$ that satisfy the conclusions of Proposition 14.2.36, as well as the following additional properties. First, for each $T \in \mathbb{T}$ we have $Y(T) \subset [-1, 1]^2 \times [0, \delta^\varepsilon]$. Second, there is a number $m \geq \delta^\varepsilon$ so that $m \leq |f'(z)| \leq 2m$ for all $z \in [0, \delta^\varepsilon]$.*

Proof. Let $\varepsilon_1, \varepsilon_2 > 0$ be small constants to be chosen below. We will select ε_1 very small compared to ε_2 , and ε_2 very small compared to ε .

Let $\delta \in (0, \delta_0]$ and let $(\mathbb{T}, Y)_\delta$ be a collection of tubes that satisfy the conclusions of Proposition 14.2.36, with ε_1 in place of ε . After passing to a sub-collection, we may suppose that the tubes point in δ -separated directions. If ε_1 is selected sufficiently small, then we may apply Proposition 14.2.14 with δ^{ε_2} in place of ρ , and we can select a refinement $(\mathbb{T}_1, Y_1)$ of $(\mathbb{T}, Y)$ so that $E_{\mathbb{T}_1}$ is covered by $\lesssim \delta^{-\varepsilon_2 + \sigma - 3}$ cubes of side-length δ^{ε_2} .

By pigeonholing, we can find an interval $J \subset [-1, 1]$ of length δ^{ε_2} so that $E_{\mathbb{T}_1} \cap (\mathbb{R}^2 \times J)$ is covered by $\lesssim \delta^{-2\varepsilon_2 + \sigma - 2}$ cubes of side-length δ^{ε_2} , and $\sum_{T \in \mathbb{T}_1} |Y_1(T) \cap (\mathbb{R}^2 \times J)| \gtrsim \delta^{2\varepsilon_2} |J|$.

Finally, if ε_2 is selected sufficiently small compared to ε , then we can apply Lemma 14.2.48 to conclude that $|f'(z)| \geq \delta^\varepsilon$ for all $z \in J$. Since $|f''(z)| \leq 1$, there must be a number $m \geq \delta^\varepsilon$ so that $m \leq |f'(z)| \leq 2m$ on J . Finally, replace each shading $Y_1(T)$ by $Y_1(T) \cap (\mathbb{R}^2 \times J)$, and apply a translation in the z -direction so that $J = [0, \delta^\varepsilon]$. $\square$

Proposition 14.2.46 follows by applying a re-scaling to the collection of tubes from Lemma 14.2.49. The details are as follows.

Proof of Proposition 14.2.46. Let $\varepsilon_1 > 0$ be a small constant to be chosen below. We will select ε_1 very small compared to ε . Let $\delta \in (0, \delta_0^2]$ and let $(\mathbb{T}, Y)_\delta$ satisfy the conclusions of Lemma 14.2.49 with ε_1 in place of ε (and some $m \geq \delta^{\varepsilon_1}$).

Define $\phi(x, y, z) = (x, \frac{m^2 y}{100}, \frac{100z}{m})$, and define $\tilde{f}(z) = 100m^{-2}f(mz/100)$. Then

$$1 \leq |\tilde{f}'(z)| \leq 2 \quad \text{and} \quad |\tilde{f}''| \leq 1/100, \quad z \in [0, m^{-1}\delta^{\varepsilon_1}]. \quad (14.2.112)$$

We can modify $\tilde{f}$ on $[-1, 1] \setminus [0, m^{-1}\delta^{\varepsilon}]$ so that (14.2.112) holds for all $z \in [0, 1]$. Furthermore, if $(x, y, z) \in [-1, 1]^2 \times J$, then

$$(x, y, z) \cdot (1, f(z), 0) = x + yf(z) = \phi(x, y, z) \cdot (1, \tilde{f}(\tilde{z}), 0).$$

It remains to construct the ε -extremal collection of tubes $(\tilde{\mathbb{T}}, \tilde{Y})_{\tilde{\delta}}$ that satisfies the conclusions of Proposition 14.2.46. We will do this (for some $\tilde{\delta} \leq \delta^{1-C\varepsilon_1}$; if ε_1 is selected sufficiently small, then $C\varepsilon_1 \leq 1/2$ and hence $\tilde{\delta} \leq \delta_0$, as required) by applying the scaling $\phi: [-1, -1]^2 \times [0, \delta^{\varepsilon_1}] \rightarrow [-1, 1]^3$ to $(\mathbb{T}, Y)_{\delta}$, and using Lemma 14.2.18. $\square$

14.2.6.1 Twisted projections

We will re-interpret the conclusion of Proposition 14.2.46 as a statement regarding twisted projections. First a definition.

Definition 14.2.50 (Twisted projection). For $f: [-1, 1] \rightarrow \mathbb{R}$, we define the *twisted projection* $\pi_f: [-1, 1]^3 \rightarrow \mathbb{R}^2$ by

$$\pi_f(x, y, z) = (x + f(z)y, z).$$

With this definition, the pair $(\mathbb{T}, Y)_{\delta}$ and the function f from the conclusion of Proposition 14.2.46 satisfies $|\pi_f(E_{\mathbb{T}})| \leq \delta^{\sigma-\varepsilon}$.

Applying Hölder's inequality, we have the following consequence of Proposition 14.2.46.

Corollary 14.2.51. *For all $\varepsilon > 0$, $\delta_0 > 0$, there exists $\delta \in (0, \delta_0]$; an ε -extremal collection of δ -tubes $\mathbb{T}$ (see Definition 14.2.11); and a function $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$, so that*

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(T)} \right\|_{L^{3/2}(\mathbb{R}^2)} \geq \delta^{-1-\sigma/3+\varepsilon}. \quad (14.2.113)$$

Proof. By Definition 14.2.8 (c) and Hölder's inequality,

$$\delta^{-1+\varepsilon} \leq \left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^1(\pi_f(E_{\mathbb{T}}))} \leq |\pi_f(E_f)|^{1/3} \left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^{3/2}(\mathbb{R}^2)}.$$

The conclusion follows from Proposition 14.2.46. $\square$

14.2.7 A C^2 projection theorem

In this section we will introduce a version of Wolff's circular maximal theorem [47, 31] for C^2 curves, and show how this theorem can bound expressions like the LHS of (14.2.113). The goal is to prove the following.

Proposition 14.2.52. *For all $\varepsilon > 0$, there exists $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\mathbb{T}$ be a set of δ -tubes pointing in δ -separated directions. Let $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$. Then*

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(T)} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq \delta^{-1-\varepsilon}. \quad (14.2.114)$$

Comparing Corollary 14.2.51 and Proposition 14.2.52, we conclude that $\sigma = 0$. Combined with Proposition 14.2.9, this concludes the proof of Theorem 14.2.3 and Theorem 14.2.3'. All that remains is to prove Proposition 14.2.52.

Our main tool for proving Proposition 14.2.52 is a version of Wolff's $L^{3/2}$ circular maximal function bound for C^2 curves, which is a special case of Theorem 1.7 in [37]. Before stating the result, we recall Definition 1.6 from [37].

Definition 14.2.53. Let $I \subset \mathbb{R}$ be a compact interval and let $\mathcal{F} \subset C^2(I)$. We say $\mathcal{F}$ is a family of *cinematic functions*, with cinematic constant K and doubling constant D if the following conditions hold.

1. $\mathcal{F}$ has diameter at most K .
2. $\mathcal{F}$ is a doubling metric space, with doubling constant at most D .
3. For all $f, g \in \mathcal{F}$, we have

$$\inf_t (|f(t) - g(t)| + |f'(t) - g'(t)| + |f''(t) - g''(t)|) \geq K^{-1} \|f - g\|_{C^2(I)}. \quad (14.2.115)$$

Theorem 14.2.54. *Let $I \subset \mathbb{R}$ be a compact interval and let $\mathcal{F} \subset C^2(I)$ be a family of cinematic functions. Then for all $\varepsilon > 0$, there exists $\delta_0 > 0$ depending on ε, K, D so that the following holds for all $\delta \in (0, \delta_0]$. Let $F \subset \mathcal{F}$ be a δ -separated set that satisfies the Frostman non-concentration condition*

$$\#(F \cap B) \leq \delta^{-\varepsilon} (r/\delta) \quad \text{for all balls } B \subset C^2(I) \text{ of diameter } r. \quad (14.2.116)$$

Then

$$\left\| \sum_{f \in F} \chi_{f^\delta} \right\|_{3/2} \leq \delta^{-\varepsilon},$$

where f^δ is the δ -neighborhood of $\text{graph}(f)$.

We will be interested in the following family of functions. For $f \in C^2(I)$ and $a, b, d \in \mathbb{R}$, define

$$g_{a,b,d}(t) = a + bf(t) + dtf(t). \quad (14.2.117)$$

Define $M_f = \{g_{a,b,d} : a, b, d \in [-1, 1]\}$. We will show that under certain conditions on f , this is a family of cinematic functions.

Lemma 14.2.55. *Let $f : [0, 1] \rightarrow \mathbb{R}$ with $f(0) = 0$, $1 \leq |f'(t)| \leq 2$, and $|f''(t)| \leq 1/100$ for all $t \in [0, 1]$. Then M_f is a family of cinematic functions, and the implicit constants in Definition 14.2.53 are bounded independently of the choice of f .*

Proof. Let $a_1, a_2, b_1, b_2, d_1, d_2 \in [-1, 1]$, and let $g_i = g_{a_i, b_i, d_i}$. We have $\|g_1 - g_2\|_{C^2} \sim |a_1 - a_2| + |b_1 - b_2| + |d_1 - d_2|$, so it suffices to show that

$$\begin{aligned} \inf_{t \in [0, 1]} (|g_1(t) - g_2(t)| + |g'_1(t) - g'_2(t)| \\ + |g''_1(t) - g''_2(t)|) \geq \frac{1}{200} (|a_1 - a_2| + |b_1 - b_2| + |d_1 - d_2|). \end{aligned} \quad (14.2.118)$$

Case 1: $|a_1 - a_2| \geq 10(|b_1 - b_2| + |d_1 - d_2|)$.

We have $|f(t)| \leq 2t$ and

$$\inf_{t \in [0, 1]} |g_1(t) - g_2(t)| \geq |a_1 - a_2| - 2(|b_1 - b_2| + |d_1 - d_2|) \geq \frac{1}{2}|a_1 - a_2|.$$

Thus (14.2.118) holds and we are done.

Case 2a: $|a_1 - a_2| < 10(|b_1 - b_2| + |d_1 - d_2|)$ and $|b_1 - b_2| \geq 10|d_1 - d_2|$.

We have

$$\begin{aligned} |g'_1(t) - g'_2(t)| &\geq |b_1 - b_2| |f'(t)| - |d_1 - d_2| (|f(t)| + |tf'(t)|) \\ &\geq |b_1 - b_2| - \frac{1}{10}|b_1 - b_2| (|f(t)| + |tf'(t)|) \geq \frac{1}{2}|b_1 - b_2|. \end{aligned}$$

Thus (14.2.118) holds and we are done.

Case 2b: $|a_1 - a_2| < 10(|b_1 - b_2| + |d_1 - d_2|)$ and $|b_1 - b_2| < 10|d_1 - d_2|$.

We have

$$\begin{aligned} |g''_1(t) - g''_2(t)| &\geq |d_1 - d_2| |f'(t)| - (|b_1 - b_2| + |d_1 - d_2| |t|) |f''(t)| \\ &\geq |d_1 - d_2| - 11|d_1 - d_2| |f''(t)| \\ &\geq \frac{1}{2}|d_1 - d_2|. \end{aligned}$$

Thus (14.2.118) holds and we are done. □

We are now ready to prove Proposition 14.2.52.

Proof of Proposition 14.2.52. Each $T \in \mathbb{T}$ has a coaxial line pointing in the direction $(c, d, 1)$. Without loss of generality, we may suppose that $c, d \in \delta\mathbb{Z} \cap [-1, 1]$, and the directions $(c, d, 1)$ are distinct for distinct $T \in \mathbb{T}$. By the triangle inequality, for each $c \in [-1, 1]$ it suffices to prove the estimate

$$\left\| \sum_{i \in \mathcal{I}} \chi_{\pi_f(T_i)} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq \delta^{-\varepsilon}, \quad (14.2.119)$$

where $N = \lceil \delta^{-1} \rceil$, $\mathcal{I} \subset \{-N, \dots, N\}$ (the set of indices $\mathcal{I}$ depends on c) and for each $i \in \mathcal{I}$, T_i is a δ -tube in $\mathbb{T}$ whose coaxial line L_i is of the form $(a_i, b_i, 0) + \mathbb{R}(c, i\delta, 1)$ with $a_i, b_i \in [-1, 1]$. For each tube T_i , $\pi_f(T_i)$ is contained in the 2δ -neighborhood of the plane curve $\pi_f(L_i)$, which is the graph of the function $g_i = g_{a_i, b_i, d_i}(t) + ct$ (recall that g_{a_i, b_i, d_i} is as defined in (14.2.117)). By Lemma 14.2.55, the set $\{g_{a, b, d}(t) + ct : a, b, d \in [-1, 1]\} \subset C^2([0, 1])$ is a family of cinematic functions, and the implicit constants in Definition 14.2.53 are bounded independently of c, f , and N (and hence δ).

The estimate (14.2.119) thus follows from Theorem 14.2.54, since the set of functions $\{g_i\}_{i \in \mathcal{I}}$ clearly satisfies the Frostman condition (14.2.116). $\square$

Remark 14.2.56. It is also possible to conclude that $\sigma = 0$ using restricted projections [19, 20, 21] introduced by Fässler and Orponen in [15], rather than using Theorem 14.2.54. We will briefly sketch this latter approach here. We will use the same notation as in the proof of Proposition 14.2.119. By pigeonholing, there exists $c \in [-1, 1]$ and the corresponding set $\mathcal{I} \subset \{-N, \dots, N\}$ such that $\#\mathcal{I} \geq \delta^{-\varepsilon/100}$ and for each $i \in \mathcal{I}$, T_i is a δ -tube in $\mathbb{T}$ whose coaxial line L_i is of the form $(a_i, b_i, 0) + \mathbb{R}(c, i\delta, 1)$ with $a_i, b_i \in [-1, 1]$. For this fixed c and each $z \in [-1, 1]$, define the (translated) restricted projection as

$$P_z : (a, b, d) \mapsto (a + cz + f(z)(b + dz)) = (a, b, d) \cdot (1, f(z), f(z)z) + cz. \quad (14.2.120)$$

Since f is C^2 and $1 \leq |f'(z)| \leq 2$ for each $z \in [-1, 1]$, there exists a non-degenerate curve $\gamma : [-1, 1] \rightarrow \mathbb{R}^3$ such that $\gamma'(z) = (1, f(z), f(z)z)$. Here nondegenerate means $\det(\gamma'(z), \gamma''(z), \gamma'''(z)) \neq 0$ for each $z \in [-1, 1]$. The conclusion follows from [20, Theorem 2] with $s = \sigma + \varepsilon/10$, $t = 1 - \varepsilon/10$ (note that [20, Theorem 2] still holds if we assume each δ -ball contained in H intersects $\delta^{\varepsilon/10} \#\Theta$ many slabs). See also [21, Theorem 2.1] with $m = 1, \alpha = 1$.

14.2.8 Proof of Theorem 14.2.37

In this section we prove Theorem 14.2.37. For the reader's convenience we will recall it here.

Theorem 14.2.37. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\eta})$ -ADsets. Then at least one of the following must hold.*

(A) *There exist orthogonal lines ℓ and $\ell^\perp$ whose δ -neighborhoods cover a significant fraction of F and G respectively. In particular,*

$$\mathcal{E}_\delta(N_\delta(\ell) \cap F) \geq \delta^{\varepsilon-1} \text{ and } \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G) \geq \delta^{\varepsilon-1}. \quad (14.2.78)$$

(B) *Let $\mathcal{H} \subset F \times G \times G$ be any set satisfying $\mathcal{E}_\delta(\mathcal{H}) \geq \delta^{\eta-3}$. Then there exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-\eta}\rho$ such that*

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.2.79)$$

A key ingredient in our proof is a result about radial projections from [36], which in turn builds off of ideas from [34, 41], and ultimately relies on Bourgain's discretized sum-product theorem. Our proof of Theorem 14.2.37 actually proves a slightly stronger statement that has weaker hypotheses. We first recall the following definition.

Definition 14.2.57. For $n \geq 1$, $\alpha \in (0, n]$, $C \geq 1$ and $\delta > 0$, We say a set $E \subset \mathbb{R}^n$ is a $(\delta, \alpha, C)_n$ -Frostman set if E is a union of δ -cubes, and for all $x \in \mathbb{R}^n$ and all $r \geq \delta$ we have

$$|E \cap B(x, r)| \leq Cr^\alpha |E|. \quad (14.2.121)$$

We form an analogous definition if $E \subset S^1$ is a union of δ -arcs.

In practice we will often consider sets $E \subset \mathbb{R}^2$, so we will drop the subscript 2 from our notation. Definition 14.2.57 first appeared in [6] and is closely related to Katz and Tao's notion of a $(\delta, \alpha)_n$ -set from [27] (sets obeying Katz and Tao's non-concentration condition are now sometimes referred to as $(\delta, \alpha)_n$ -Katz-Tao sets to avoid confusion with Definition 14.2.57). However, in Katz and Tao's definition from [27], the quantity $|E|$ on the RHS of (14.2.121) is replaced by $\delta^{n-\alpha}$; the former could be much larger.

We can now state the following slightly stronger variant of Theorem 14.2.37.

Theorem 14.2.37'. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\eta})$ -Frostman sets. Then at least one of the following must hold.*

(A) *There exist orthogonal lines ℓ and $\ell^\perp$ whose δ -neighborhoods cover a significant fraction of F and $G_1 \cup G_2$ respectively. In particular,*

$$\begin{aligned} \mathcal{E}_\delta(N_\delta(\ell) \cap F) &\geq \delta^{\varepsilon-1}, \\ \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G_1) &\geq \delta^{\varepsilon-1}, \quad \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G_2) \geq \delta^{\varepsilon-1}. \end{aligned} \quad (14.2.122)$$

(B) Let $\mathcal{H} \subset F \times G_1 \times G_2$ be any set satisfying $\mathcal{E}_\delta(\mathcal{H}) \geq \delta^{\eta-3}$. Then there exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-\eta}\rho$ such that

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.2.123)$$

Observe that if E is a union of δ -cubes that is also a $(\delta, \alpha, C)_n$ -ADset, and if $|E| \geq c\delta^{\eta-\alpha}$ for some $c > 0$, then E is a $(\delta, \alpha, C/c)_n$ -Frostman set, and hence Theorem 14.2.37' (with $G_1 = G_2 = G$) implies Theorem 14.2.37. Indeed, if F, G satisfy the hypotheses of Theorem 14.2.37, then $\mathcal{E}_\delta(F) \leq \delta^{-1-\eta}$ and $\mathcal{E}_\delta(G) \leq \delta^{-1-\eta}$. If at least one of these sets has δ -covering number $\leq \delta^{-1+4\eta}$, then $\mathcal{E}_\delta(F \times G \times G) \leq \delta^{2\eta-3}$, and thus Conclusion (B) of Theorem 14.2.37 is vacuously true. On the other hand, if both F and G have δ -covering number $\geq \delta^{-1+4\eta}$, then F and G are $(\delta, 1, \delta^{-5\eta})$ -Frostman sets (is harmless to suppose that F and G are unions of δ -squares, and $\mathcal{H}$ is a union of δ cubes), and thus the sets F and $G_1 = G_2 = G$ satisfy the hypotheses of Theorem 14.2.37' with 5η in place of η .

Notation conventions

For the remainder of Section 14.2.8, we adopt the convention that if X, Y are unions of δ -cubes in $\mathbb{R}^n$ (resp. δ -arcs in S^1), then we will only consider subsets $\mathcal{H} \subset X \times Y$ that are unions of sets $S_1 \times S_2$, where $S_1 \subset X$ and $S_2 \subset Y$ are δ -cubes (resp. δ -arcs). Note that the truth (or falsehood) of Theorem 14.2.37' is not affected by the adoption of this convention.

If Z is a union of δ cubes in $\mathbb{R}^n$, it will sometimes be convenient to think of Z as a set of δ cubes. For example, we will sometimes identify the (infinite) sets of points $\mathcal{H} \subset X \times Y$ with the corresponding (finite) sets of δ cubes; thus we can think of $\mathcal{H}$ as a bipartite graph on the (finite) vertex sets X and Y . In this spirit, we will sometimes use $\#Z$ to refer to the number of δ cubes in Z , i.e. $\#Z = \delta^{-n}|Z|$.

14.2.8.1 A key special case: $F \cdot (G_1 - G_2)$ is large if (G_1, G_2) have thin tubes

In this section we will introduce the results and definitions needed to state and prove a key special case of Theorem 14.2.37'; this is Proposition 14.2.68 below. Proposition 14.2.68 says that Conclusion (B) holds (with $\rho = \delta$ and $I = [-2, 2]$), provided that the graph $\mathcal{H}$ satisfies a mild uniformity condition, and (far more importantly) that the pair (G_1, G_2) satisfies a strong non-concentration condition on lines.

Hypergraphs and uniform density

The set $\mathcal{H}$ from Conclusion (B) of Theorem 14.2.37' is a 3-regular tripartite hypergraph. Some of our arguments can be stated more simply if each vertex of this hypergraph is a member of (at least) an average number of edges. The precise definition is as follows.

Definition 14.2.58. Let $A_1, \dots, A_k$ be finite sets and let $\mathcal{H} \subset A_1 \times \dots \times A_k$. For each nonempty multi-index $I \subset [k] = \{1, \dots, k\}$, define $A_I = \times_{i \in I} A_i$. For each $a_I = (a_i)_{i \in I} \in A_I$, define

$$n_{\mathcal{H}}[a_I] = \{(h_1, \dots, h_k) \in \mathcal{H} : a_i = h_i \text{ for all } i \in I\}.$$

For example if $\mathcal{H} \subset A_1 \times A_2 \times A_3$ and $(a_1, a_3) \in A_1 \times A_3$, then $n_{\mathcal{H}}[(a_1, a_3)]$ is the set of all edges of the form $(h_1, h_2, h_3) \in \mathcal{H}$ with $h_1 = a_1$ and $h_3 = a_3$.

Definition 14.2.59. If $\mathcal{H} \subset A_1 \times \dots \times A_k$ and $I \subset [k]$, we define $\pi_I : \mathcal{H} \rightarrow A_I$ to be the projection onto A_I . If $I = \{i\}$ we write π_i instead of $\pi_{\{i\}}$. Finally, if $h \in \mathcal{H}$, we write h_I to denote $\pi_I(h)$.

Remark 14.2.60. In some of the sections that follow, we will consider sets $\mathcal{H} \subset F \times G_1 \times G_2$. To avoid confusion, we will use the notation $\pi_F : \mathcal{H} \rightarrow F$ to denote the projection map to F , $\pi_{F \times G_1}$ to denote the projection to $F \times G_1$, etc.

Definition 14.2.61. Let $0 < c \leq 1$. We say a k -uniform k -partite hypergraph $\mathcal{H} \subset A_1 \times \dots \times A_k$ is *uniformly c -dense* if for each edge $h \in \mathcal{H}$ and each $I \subset [k]$ we have

$$\#n_{\mathcal{H}}[h_I] \geq c \prod_{i \notin I} \#A_i. \quad (14.2.124)$$

This definition is useful because some pigeonholing arguments are simplified when the corresponding graph is uniformly c -dense, and if $\mathcal{H}$ is uniformly c -dense then its induced subgraphs will still be dense (though possibly not uniformly dense). The next result says that every hypergraph $\mathcal{H}$ has a uniformly c -dense refinement, where c is comparable to the average density of $\mathcal{H}$.

Lemma 14.2.62 (Hypergraph refinement lemma). *Let $A_1, \dots, A_k$ be finite sets, let $\mathcal{H} \subset A_1 \times \dots \times A_k$ be a k -uniform hypergraph with density $d = \#\mathcal{H} / \prod_{i=1}^k \#A_i$, and let $0 < \varepsilon < 1$. Then there exists a uniformly $(2^{-k}\varepsilon d)$ -dense subgraph $\mathcal{H}' \subset \mathcal{H}$ with $\#\mathcal{H}' \geq (1 - \varepsilon)\#\mathcal{H}$.*

Lemma 14.2.62 was proved by Dvir and Gopi [14] in the special case $k = 2$ and $\varepsilon = 1/2$. A similar proof works in general; we give the details below.

Proof. To begin, define $A'_I = A_I$ for each multi-index I , and define $\mathcal{H}' = \mathcal{H}$. Next, if there exists a multi-index I and an element $h_I \in A'_I$ with

$$\#n_{\mathcal{H}'}[h_I] < \frac{\varepsilon}{2^k} \frac{\#\mathcal{H}}{\prod_{i \in I} \#A_i}, \quad (14.2.125)$$

then we remove h_I from A'_I , and we remove all of the edges $n_{\mathcal{H}'}[h_I]$ from $\mathcal{H}'$. We repeat this process until no $h_I \in A'_I$ fits the removal criteria (14.2.125) for any multi-index I .

Each $h_I \in A'_I$ that is removed deletes at most $\frac{\varepsilon}{2^k} \frac{\#\mathcal{H}}{\prod_{i \in I} \#A_i}$ edges from $\mathcal{H}$. Thus the number of edges deleted by this procedure is at most

$$\sum_{I \subset [k]} \frac{\varepsilon}{2^k} \frac{\#\mathcal{H}}{\prod_{i \in I} \#A_i} \#A_I = \sum_{I \subset [k]} \frac{\varepsilon}{2^k} \#\mathcal{H} = \varepsilon \#\mathcal{H}.$$

After this procedure has finished, the surviving graph $\mathcal{H}'$ is uniformly $(2^{-k}\varepsilon d)$ -dense. $\square$

Remark 14.2.63. Let $\mathcal{H} \subset A_1 \times \dots \times A_k$ be uniformly c -dense, let $1 \leq j \leq k$, and suppose that the projection map π_j is onto. Let $A'_j \subset A_j$, and define $\tilde{A}_i = A_i$ for $i \neq j$, and $\tilde{A}_j = A'_j$. Let $\mathcal{H}' = \mathcal{H} \cap (\tilde{A}_1 \times \dots \times \tilde{A}_k)$, be the corresponding induced subgraph. Then $(\#\mathcal{H}')/(\prod_{i=1}^k \#\tilde{A}_i) \geq c$.

$\mathcal{H}' \subset \tilde{A}_1 \times \dots \times \tilde{A}_k$ need not be uniformly c -dense, but by Lemma 14.2.62 it has a uniformly $2^{-k+1}c$ -dense subgraph $\mathcal{H}'' \subset \mathcal{H}'$ with $\#\mathcal{H}'' \geq \frac{1}{2}\#\mathcal{H}'$. We will use this observation frequently in the arguments below.

Kaufman's projection theorem

The next tool we will need is Kaufman's projection theorem [24]. The precise version we will need is given below; we include a proof for completeness.

Theorem 14.2.64. *Let $0 < \alpha \leq \beta \leq 1$ and let $\varepsilon > 0$. Then there exists $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $C \geq 1$, let $F \subset \mathbb{R}^2$ be a (δ, α, C) -Frostman set, and let $\Lambda \subset S^1$ be a (δ, β, C) -Frostman set. Let $\mathcal{H} \subset \Lambda \times F$, and let $d = |\mathcal{H}|/(|\Lambda| |F|)$. Then there exists $\Lambda' \subset \Lambda$ with $|\Lambda'| \gtrsim d|\Lambda|$ such that for every $\theta \in \Lambda'$, we have*

$$|\theta \cdot F_\theta| \gtrsim C^{-2} d^3 \delta^{1-\alpha+\varepsilon},$$

where $F_\theta = \{x \in F : (\theta, x) \in \mathcal{H}\}$.

Proof. Apply Lemma 14.2.62 to $\mathcal{H}$ with $\varepsilon = 1/2$, let $\mathcal{H}' \subset \mathcal{H}$ be the resulting refinement, and let $\Lambda_0 \subset \Lambda$ be the projection of $\mathcal{H}'$ to the first coordinate. Then $|\Lambda_0| \gtrsim d|\Lambda|$ (so in particular Λ_0 is a $(\delta, \beta, C/d)$ -Frostman set), and for each $\theta \in \Lambda_0$, we have $|F_\theta| \gtrsim d|F|$.

Define $\nu = \frac{\chi_{\Lambda_0}}{|\Lambda_0|}$, $\mu = \frac{\chi_F}{|F|}$, $\mu_\theta = \frac{\chi_{F_\theta}}{|F_\theta|}$. If $\theta \in \Lambda_0$ then $\frac{d\mu_\theta}{d\mu} \lesssim d^{-1}$. For $x \neq y$, define $\pi^y(x) = \frac{x-y}{|x-y|}$. Fix $0 < \gamma < \alpha$. Since μ and ν satisfy the Frostman conditions $\mu(B(x, r)) \leq Cr^\alpha$ and $\nu(B(x, r)) \leq (C/d)r^\beta$, there is a constant K depending on $\alpha - \gamma$ so that

$$\int_F \int_F |x - y|^{-\gamma} d\mu(x) d\mu(y) \leq KC,$$

and

$$\int_{\Lambda_0} |\theta \cdot e|^{-\gamma} d\nu(\theta) \leq Kd^{-1}C \quad \text{for every } e \in S^1.$$

In what follows, all implicit constants are allowed to depend on $\alpha - \gamma$. We compute

$$\begin{aligned} & \int_{\Lambda_0} \int_F \int_F |\theta \cdot x - \theta \cdot y|^{-\gamma} d\mu_\theta(x) d\mu_\theta(y) d\nu(\theta) \\ & \lesssim d^{-2} \int_F \int_F \int_{\Lambda_0} |\theta \cdot x - \theta \cdot y|^{-\gamma} d\nu(\theta) d\mu(x) d\mu(y) \\ & = d^{-2} \int_F \int_F \int_{\Lambda_0} |x - y|^{-\gamma} \cdot |\theta \cdot \pi^y(x)|^{-\gamma} d\nu(\theta) d\mu(x) d\mu(y) \\ & \lesssim Cd^{-3} \int_F \int_F |x - y|^{-\gamma} d\mu(x) d\mu(y) \\ & \lesssim C^2 d^{-3}. \end{aligned}$$

By Fubini, there exists a subset $\Lambda' \subset \Lambda_0$ with $|\Lambda'| \gtrsim |\Lambda_0|$, so that for every $\theta \in \Lambda'$ we have

$$\int_F \int_F |\theta \cdot x - \theta \cdot y|^{-\gamma} d\mu_\theta(x) d\mu_\theta(y) \lesssim C^2 d^{-3}.$$

We conclude that $|\theta \cdot F_\theta| \gtrsim C^{-2} d^3 \delta^{1-\gamma}$. The result now follows by selecting $\gamma = \alpha - \varepsilon/2$ and choosing $\delta_0 = \delta_0(\varepsilon) > 0$ sufficiently small. $\square$

Thin tubes

In [41], Shmerkin and the first author introduced a non-concentration condition on pairs of Borel measures in the plane, which prevents the supports of these measures from concentrating on a common line. We recall this definition here.

Definition 14.2.65. Let $\beta \geq 0$, $K \geq 1$, and $c \in [0, 1)$. Let ν_1, ν_2 be Borel probability measures on $\mathbb{R}^2$. We say that (ν_1, ν_2) have $(\beta, K, 1 - c)$ -thin tubes if there exists a Borel set $E \subset \text{supp}(\nu_1) \times \text{supp}(\nu_2)$ with $\nu_1 \times \nu_2(E) \geq 1 - c$ with the following property: for all $b_1 \in \text{supp}(\nu_1)$, all lines $\ell \subset \mathbb{R}^2$ containing b_1 , and all $r > 0$, we have

$$\nu_2(\{b_2 \in N_r(\ell) : (b_1, b_2) \in E\}) \leq Kr^\beta. \quad (14.2.126)$$

In our setting, we will consider unions of δ -cubes rather than probability measures. To this end, we introduce the following variant of Definition 14.2.65.

Definition 14.2.65'. Let $\beta \geq 0$, $K \geq 1$, and $c \in [0, 1)$. Let $G_1, G_2 \subset \mathbb{R}^2$ be unions of δ -squares. We say that (G_1, G_2) have δ -discretized $(\beta, K, 1 - c)$ *thin tubes* if there exists a union of δ -cubes $E \subset G_1 \times G_2$ with $|E| \geq (1 - c)|G_1||G_2|$ with the following property: for all $b_1 \in G_1$, all lines $\ell \subset \mathbb{R}^n$ containing b_1 , and all $r \geq \delta$, we have

$$|\{b_2 \in G_2 \cap N_r(\ell) : (b_1, b_2) \in E\}| \leq K \cdot r^\beta |G_2|. \quad (14.2.127)$$

We will sometimes suppress the parameters K and c , and just say that (G_1, G_2) have δ -discretized β -thin tubes.

Definitions 14.2.65 and 14.2.65' are related as follows:

Lemma 14.2.66. Let $G_1, G_2 \subset \mathbb{R}^2$ be unions of δ -squares, and define the probability measures $\nu_i = \chi_{G_i}/|G_i|$. If (ν_1, ν_2) has $(\beta, K, 1 - c)$ -thin tubes, then (G_1, G_2) has δ -discretized $(\beta, K', 1 - c)$ -thin tubes, with $K' = 4^\beta K/c$.

Proof. Let $E \subset \text{supp}(\nu_1) \times \text{supp}(\nu_2)$ be a set with measure $\nu_1 \times \nu_2(E) \geq 1 - c$ that satisfies (14.2.126) for all $b_1 \in \text{supp}(\nu_1)$, all lines $\ell \subset \mathbb{R}^2$ containing b_1 , and all $r > 0$.

Define E' to be the union of those δ cubes $S = S_1 \times S_2 \subset G_1 \times G_2$ (each S is a 4-dimensional δ -cube) satisfying $\nu_1 \times \nu_2(S \cap E) \geq c \nu_1 \times \nu_2(S) = c\delta^4/|G_1||G_2|$. We have

$$|(G_1 \times G_2) \setminus E'| = (\nu_1 \times \nu_2((G_1 \times G_2) \setminus E'))|G_1||G_2| < c|G_1||G_2|,$$

and thus

$$|E'| \geq (1 - c)|G_1||G_2|.$$

We will show that this set E' satisfies (14.2.127). Fix a choice of $b_1 \in G_1$, a line $\ell \subset \mathbb{R}^2$ containing b_1 , and $r \geq \delta$. Our goal is to establish (14.2.127) with $4^\beta K/c$ in place of K . Let $S_1 \subset G_1$ be the δ -square containing b_1 . Observe that for all $b'_1 \in S_1$, we have

$$\nu_2(\{b_2 \in N_{2r}(\ell) : (b'_1, b_2) \in E\}) \leq K(4r)^\beta. \quad (14.2.128)$$

Indeed, this is because $N_{2r}(\ell)$ is contained in the $4r$ -neighbourhood of a line passing through b'_1 ; we apply (14.2.126) to this latter line.

Let $\mathcal{S}$ be the set of δ -squares intersecting $N_r(\ell)$. We have

$$N_r(\ell) \subset \bigcup_{S \in \mathcal{S}} S \subset N_{2r}(\ell). \quad (14.2.129)$$

We compute

$$\begin{aligned}
 & |\{b_2 \in G_2 \cap N_r(\ell) : (b_1, b_2) \in E'\}| \\
 & \leq \delta^2 \#\{S_2 \in \mathcal{S} : S_1 \times S_2 \subset E'\} \\
 & = \delta^2 \#\left\{S_2 \in \mathcal{S} : (\nu_1 \times \nu_2)((S_1 \times S_2) \cap E) \geq \frac{c\delta^4}{|G_1||G_2|}\right\} \\
 & \leq c^{-1} \frac{|G_1|}{\delta^2} |G_2| \sum_{S \in \mathcal{S}} (\nu_1 \times \nu_2)((S_1 \times S_2) \cap E) \\
 & \leq c^{-1} \frac{|G_1|}{\delta^2} |G_2| \int_{S_1} \int_{N_{2r}(\ell)} \chi_E(b'_1, b_2) d\nu_2(b_2) d\nu_1(b'_1) \\
 & \leq c^{-1} \frac{|G_1|}{\delta^2} |G_2| \int_{S_1} K(4r)^\beta d\nu_1(b'_1) \\
 & \leq \frac{4^\beta K}{c} r^\beta |G_2|.
 \end{aligned} \tag{14.2.130}$$

The first inequality used the first inequality of (14.2.129). The next line follows from the definition of E' . The fourth line used the second inequality of (14.2.129), and the fifth line used (14.2.128). $\square$

$F \cdot (G_1 - G_2)$ is large if (G_1, G_2) have thin tubes

In several of the results that follow, it will be helpful to suppose that the sets F, G_1 , and G_2 are localized to small balls, and that these balls are separated from each other and from the origin. We record this assumption below

Definition 14.2.67. We say that sets $F, G_1, G_2 \subset [0, 1]^2$ satisfy the *standard separation conditions* if

$$\text{diam}(F), \text{diam}(G_1), \text{diam}(G_2) \leq \frac{1}{10}; \quad \text{dist}(G_1, G_2) \geq \frac{1}{2}; \quad \text{dist}(F, 0) \geq \frac{1}{2}. \tag{14.2.131}$$

We can now state and prove the following special case of Theorem 14.2.37'.

Lemma 14.2.68. *For all $\varepsilon > 0$, there exists $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $c \in [0, 1)$ and $K \geq 1$. Let $F, G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, K)$ -Frostman sets that satisfy the standard separation conditions, and suppose furthermore that (G_1, G_2) has δ -discretized $(1, K, 1 - c)$ -thin tubes. Let $\mathcal{H} \subset F \times G_1 \times G_2$ be uniformly $2c$ -dense. Then*

$$\mathcal{E}_\delta(\{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \gtrsim c^5 K^{-2} \delta^{\varepsilon-1}. \tag{14.2.132}$$

Proof. Since (G_1, G_2) has δ -discretized $(1, K, 1 - c)$ -thin tubes, there exists a set $E \subset G_1 \times G_2$ with $|E| \geq (1 - c)|G_1||G_2|$ that satisfies (14.2.127) with $\beta = 1$

and K as above. Since $\mathcal{H}$ is $2c$ dense, we have $|\pi_{G_1 \times G_2}(\mathcal{H})| \geq 2c|G_1||G_2|$. Thus $|E \cap \pi_{G_1 \times G_2}(\mathcal{H})| \geq (2c - c)|G_1||G_2| \geq c|G_1||G_2|$; denote this new set by E' .

By Fubini, there exists $b_1 \in G_1$ so that $|\pi_{G_2}(n_{E'}[(b_1)])| \geq c|G_2|$. Fix this choice of b_1 and define $G'_2 = \pi_{G_2}(n_{E'}[(b_1)])$. Let L be a maximal set of lines containing b_1 with δ -separated directions, each of which intersects G'_2 . By (14.2.127) with $\beta = 1$ and $r = \delta$, we have $\#L \geq \frac{|G'_2|}{K\delta|G_2|} \geq \frac{c}{K}\delta^{-1}$. Let $\Lambda \subset S^1$ be the set of (grid-aligned) δ -arcs that intersect the directions of lines in L . Then Λ is a $(\delta, 1, K/c)$ -Frostman set, and for each $\theta \in \Lambda$ there exists (at least one) $b_2 \in G_2$ so that $(b_1, b_2) \in \pi_{G_1 \times G_2}(\mathcal{H})$; $|b_1 - b_2| \geq \frac{1}{2}$; and $\frac{b_1 - b_2}{|b_1 - b_2|} = \theta$. Choose such a $b_2 \in G_2$ for each $\theta \in \Lambda$, and let $H \subset \Lambda \times F$ be the set of pairs (θ, a) so that $(a, b_1, b_2) \in \mathcal{H}$. Since $\mathcal{H}$ is $2c$ dense, for each $\theta \in \Lambda$ there are at least $2c(\#F)$ choices of (δ -separated) $a \in F$ with $(\theta, a) \in H$. Thus $|H| \geq 2c|\Lambda||F|$.

We now apply Theorem 14.2.64 to Λ , F , and H , with $\alpha = \beta = 1$ and ε as above. Let $\delta_1 > 0$ and $\Lambda' \subset \Lambda$ be the output from that lemma. Select $\theta \in \Lambda'$, let $(b_1, b_2) \in \pi_{G_1 \times G_2}(\mathcal{H})$ be the pair of points associated to θ , and let $r = |b_1 - b_2| \geq \frac{1}{2}$. We have

$$\{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\} \supset \{a \cdot \theta r : a \in (F)_\theta\}. \quad (14.2.133)$$

By Theorem 14.2.64 the latter set has δ -covering number $\gtrsim c^5 K^{-2} \delta^{\varepsilon-1}$. $\square$

Remark 14.2.69. Note that an analogue of Lemma 14.2.68 is also true over $\mathbb{C}$, i.e. this result does not distinguish between the fields $\mathbb{R}$ and $\mathbb{C}$.

14.2.8.2 Radial projection: (G_1, G_2) have thin tubes unless they are concentrated on lines

The main result of this section is the following result, which says that the conclusion of Lemma 14.2.68 continues to hold if we replace the hypothesis that (G_1, G_2) have 1-thin tubes with the hypothesis that G_1 and G_2 satisfy a weak non-concentration near lines estimate.

Proposition 14.2.70. *For all $0 < \varepsilon, \zeta < 1$, there exists $\lambda = \lambda(\varepsilon) > 0$, $\alpha = \alpha(\varepsilon, \zeta) > 0$, and $\delta_0 = \delta_0(\varepsilon, \zeta) > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\lambda})$ -Frostman sets that satisfy the standard separation conditions. Suppose that G_1 and G_2 satisfy the following non-concentration near lines estimate:*

$$|G_i \cap N_r(\ell)| \leq (\delta^{-\lambda} r)^\zeta |G_i| \quad \text{for all lines } \ell \text{ and all } r \geq \delta, \quad i = 1, 2. \quad (14.2.134)$$

Let $\mathcal{H} \subset F \times G_1 \times G_2$ be uniformly δ^α -dense. Then

$$\mathcal{E}_\delta(\{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq \delta^{\varepsilon-1}. \quad (14.2.135)$$

Proposition 14.2.70 is essentially a small variant of the radial projection theorem in [36]; we reprove it here for completeness. The proof of radial projection theorem [36] uses the “bootstrapping lemma” that first appeared in [41], which in turn uses a result from [34] that obtains an ε -improved Furstenberg set estimate. This latter result made crucial use of Bourgain’s discretized sum-product theorem [4, 5]. In particular, the analogue of Proposition 14.2.70 is not true if we replace the field $\mathbb{R}$ by $\mathbb{C}$. The “bootstrapping lemma” we state here is Lemma 2.9 from [36].

Proposition 14.2.71 ([36], Lemma 2.9). *Let $\beta, \varepsilon > 0$, $\sigma \in [\beta, 1 - \varepsilon]$, $c \in (0, 1/10)$, and $K, C \geq 1$. Then there exists $\tau = \tau(\beta, \varepsilon) > 0$ and $M(\beta, \varepsilon) \geq 1$ so that the following holds. Let ν_1, ν_2 be Borel probability measures supported in $[0, 1]^2$, with $\text{dist}(\text{supp}(\nu_1), \text{supp}(\nu_2)) \geq 1/2$, that satisfy the Frostman non-concentration estimate*

$$\nu_1(B(x, r)) \leq Cr \quad \text{and} \quad \nu_2(B(x, r)) \leq Cr,$$

for all balls $B(x, r) \subset \mathbb{R}^2$. Suppose furthermore that both (ν_1, ν_2) and (ν_2, ν_1) have $(\sigma, K, 1 - c)$ -thin tubes.

Then both (ν_1, ν_2) and (ν_2, ν_1) have $(\sigma + \tau, K', 1 - 3c)$ -thin tubes, where

$$K' = \max \left\{ K, \frac{C^2 M}{c} \right\}^M.$$

The bounded-support hypothesis displayed here is the hypothesis used in the proof and in every application below. A complete proof, including the quantitative constant reduction and the improvement from $1 - 5c$ to $1 - 3c$, is given in Chapter 14.8.

Remark 14.2.72. The statement of Lemma 2.9 in [36] differs slightly from the version given above. First, Proposition 14.2.71 considers the special case $s = 1$ from [36, Lemma 2.9], and thus $\sigma < 1$; this allows us to simplify a few expressions. Second, [36, Lemma 2.9] specifies that $\sigma \in [\beta, 1)$, and the corresponding value of τ is bounded away from 0 on compact subsets of $\{\beta \leq \sigma < 1\}$. We have fixed the compact subset $\{\beta \leq \sigma \leq 1 - \varepsilon\}$ in advance, and thus $\tau = \tau(\beta, \varepsilon) > 0$.

Most importantly, the parameter β in the statement of [36, Lemma 2.9] is not permitted to be chosen freely. Instead, [36, Lemma 2.9] specifies a particular value of β (a careful reading of [33] shows that $\beta = 1/2$, though this fact is not necessary for our arguments here). Thus the version of Proposition 14.2.71 stated above appears to be superficially stronger than what is stated in [36, Lemma 2.9]. However, examining the (short) proof of [36, Lemma 2.9], it does not actually require a specific choice of β ; any $\beta > 0$ works equally well (though the value of τ will depend on β). For our arguments, we will fix $\beta = 1/4$.

In the discretized setting (and fixing the choice $\beta = 1/4$), Proposition 14.2.71 can be restated as follows:

Proposition 14.2.71'. *Let $\varepsilon > 0$, $\sigma \in [1/4, 1 - \varepsilon]$, $c \in (0, 1/10)$, and $K, C \geq 1$. Then there exists $\tau = \tau(\varepsilon) > 0$ and $M(\varepsilon) \geq 1$ so that the following holds. Let $G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, C)$ -Frostman sets, with $\text{dist}(G_1, G_2) \geq 1/2$. Suppose that both (G_1, G_2) and (G_2, G_1) have δ -discretized $(\sigma, K, 1 - c)$ -thin tubes.*

Then both (G_1, G_2) and (G_2, G_1) have δ -discretized $(\sigma + \tau, K', 1 - 3c)$ -thin tubes, where

$$K' = \max \left\{ \frac{4K}{3c}, \frac{4C^2M}{3c^2} \right\}^M.$$

Indeed, we obtain Proposition 14.2.71' by applying Proposition 14.2.71 to the probability measures $\nu_1 = \chi_{G_1}/|G_1|$ and $\nu_2 = \chi_{G_2}/|G_2|$ and then using Lemma 14.2.66 (this final step worsens K' by a factor of $4/(3c)$).

By repeatedly applying Proposition 14.2.71' we obtain the following.

Corollary 14.2.73. *For all $\varepsilon > 0$, there exists $N = N(\varepsilon) \geq 1$ so that the following holds. Let $c \in (0, 1/10)$, $K, C \geq 1$. Let $G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, C)$ -Frostman sets, with $\text{dist}(G_1, G_2) \geq 1/2$. Suppose that both (G_1, G_2) and (G_2, G_1) have δ -discretized $(1/4, K, 1 - c)$ -thin tubes.*

Then both (G_1, G_2) and (G_2, G_1) have δ -discretized $(1 - \varepsilon, K', 1 - 3^N c)$ -thin tubes, and thus have δ -discretized $(1, \delta^{-\varepsilon} K', 1 - 3^N c)$ -thin tubes, where

$$K' = \max \left\{ \frac{4K}{3c}, \frac{4C^2N}{3c^2} \right\}^N.$$

Grabbing the bootstraps

Corollary 14.2.73 partially bridges the gap between Lemma 14.2.68 and Proposition 14.2.70. Corollary 14.2.73 says that (G_1, G_2) have δ -discretized 1-thin tubes, which is the condition needed to apply Lemma 14.2.68. Unfortunately, the hypothesis (14.2.134) of Proposition 14.2.70 only implies that (G_1, G_2) and (G_2, G_1) have δ -discretized δ^ζ -thin tubes, and if $\zeta < 1/4$ then we cannot immediately apply Corollary 14.2.73.

To fix this problem, we will first prove that the pair (G_1, G_2) from Proposition 14.2.70 has $1/4$ -thin tubes; this is Lemma 14.2.74 below. We will then be able to apply Corollary 14.2.73 and Lemma 14.2.68. The precise result we need is as follows. Results of this type first appeared in [33] (see also [32] and [40, Appendix B]), but our setup is slightly different, so we will include a proof. Note that an analogue of the arguments below would also work over $\mathbb{C}$, i.e. the argument below does not distinguish between the fields $\mathbb{R}$ and $\mathbb{C}$.

Lemma 14.2.74. *Let $\lambda, \zeta, \alpha > 0$. Then there exists $\delta_0 > 0$ so that the following is true for all $\delta \in (0, \delta_0]$. Let $G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\lambda})$ -Frostman sets with*

$\text{dist}(G_1, G_2) \geq 1/2$, and suppose that G_2 satisfies the non-concentration near lines estimate

$$|G_2 \cap N_r(\ell)| \leq (\delta^{-\lambda} r)^\zeta |G_2| \quad \text{for all lines } \ell \text{ and all } r \geq \delta. \quad (14.2.136)$$

Then (G_1, G_2) have δ -discretized $(\frac{1}{4}, K, 1 - \delta^\alpha)$ -thin tubes, where $K = \delta^{-\frac{3\alpha}{\zeta} - \lambda}$.

Proof. Fix $\beta = 1/4$; our goal in Lemma 14.2.74 is to prove that (G_1, G_2) have δ -discretized β -thin tubes. The exponent $\beta = 1/4$ is chosen for concreteness, but it is not crucial for our argument—any exponent $\beta < 1/2$ would work equally well. We will use β in place of $1/4$ to distinguish this quantity from other less important constants that occur elsewhere in the proof.

Let $\delta \leq r \leq 1$; we say a pair of δ -squares $(Q, S) \subset G_1 \times G_2$ is *bad* at scale r if there is an infinite strip T of width r that intersects Q and S , with $|T \cap G_2| \geq Kr^\beta |G_2|$. Observe that no pairs of δ -squares are bad at any scale greater than $K^{-1/\beta}$. Define

$$E_{\text{bad}} = \bigcup_{\substack{\delta \leq r \leq K^{-1/\beta} \\ r \text{ dyadic}}} \bigcup_{(Q,S) \text{ bad}} Q \times S,$$

where the first union is taken over all numbers $\delta \leq r \leq K^{-1/\beta}$ of the form $2^k \delta$, $k \in \mathbb{Z}$, and the second union is taken over all pairs (Q, S) of δ -squares that are bad at scale r . Define $E = (G_1 \times G_2) \setminus E_{\text{bad}}$. By construction, each $b_1 \in G_1$ satisfies (14.2.127) for this set E , with K as above. Clearly E is a union of δ -cubes. To prove the lemma, it remains to show that for K as above, we have $|E| \geq (1 - \delta^\alpha) |G_1| |G_2|$. To do this, it suffices to show that for each dyadic $\delta \leq r \leq K^{-1/\beta}$,

$$\left| \bigcup_{(Q,S) \text{ bad}} Q \times S \right| \leq \frac{\delta^\alpha}{|\log \delta|} |G_1| |G_2|. \quad (14.2.137)$$

Fix r . For each square $Q \subset G_1$, let $G_2^Q \subset G_2$ be the union of those squares $S \subset G_2$ so that (Q, S) is bad at scale r . Define G_1' to be the union of those squares $Q \subset G_1$ for which $|G_2^Q| \leq \frac{\delta^\alpha}{2|\log \delta|} |G_2|$, and let $G_2'' = G_2 \setminus G_2'$. The contribution to (14.2.137) from pairs (Q, S) with $Q \subset G_1'$ is at most $\frac{\delta^\alpha}{|\log \delta|} |G_1| |G_2|$, which is acceptable.

It remains to analyze the contribution from those pairs (Q, S) with $Q \subset G_1''$, and in particular it suffices to show that

$$|G_1''| \leq \frac{\delta^\alpha}{2|\log \delta|} |G_1|. \quad (14.2.138)$$

We will prove this by contradiction. Suppose that (14.2.138) fails. For each square $Q \subset G_1''$, select a set $\mathcal{T}_Q$ of width- r strips pointing in r -separated directions, each of which intersect Q and each of which satisfy

$$|T \cap G_2| \geq Kr^\beta |G_2|. \quad (14.2.139)$$

We have $|G_2 \cap \bigcup_{T \in \mathcal{T}_Q} T| \gtrsim \frac{\delta^\alpha}{|\log \delta|} |G_2|$, and since G_1 and G_2 are $1/2$ -separated, each point $b_2 \in G_2$ is contained in at most 10 of the strips from $\mathcal{T}_Q$. If $Q \subset G_1''$ and $S \subset G_2$, write $Q \sim S$ if S intersects one of the strips from $\mathcal{T}_Q$. Define

$$\mathcal{W} = \{(Q, Q', S) : Q \sim S, Q' \sim S\}.$$

If (14.2.138) fails, then by Cauchy-Schwarz we have

$$\#\mathcal{W} \gtrsim \left(\frac{\delta^\alpha}{|\log \delta|} \right)^4 (\#G_1)^2 (\#G_2) \quad (14.2.140)$$

(recall that we use $\#G_1$ to denote the number of δ -squares in G_1 , and similarly for $\#G_2$). We will show that for K as above, then (14.2.140) is impossible.

First we will bound the number of triples $(Q, Q', S) \in \mathcal{W}$ with $\text{dist}(Q, Q') \leq c_0 \delta^\lambda \delta^{-4\alpha} |\log \delta|^4$, where $c_0 > 0$ is a small absolute constant. Since G_1 is a $(\delta, 1, \delta^{-\lambda})$ -Frostman set, after Q has been chosen there are at most $c_0 \delta^{4\alpha} |\log \delta|^{-4} (\#G_2)$ choices for Q' , and thus the number of such triples is at most $c_0 \delta^{4\alpha} |\log \delta|^{-4} (\#G_1)^2 (\#G_2)$.

Next we will bound the number of triples $(Q, Q', S) \in \mathcal{T}$ where S is contained in the $s = \delta^\lambda (c_0 \frac{\delta^\alpha}{|\log \delta|})^{4/\zeta}$ neighborhood of the line $\ell_{Q, Q'}$ that connects the centers of Q and Q' . By (14.2.136), after Q and Q' have been chosen there are at $(\delta^{-\lambda} s)^\zeta \#G_2 = c_0^4 (\frac{\delta^\alpha}{|\log \delta|})^4 \#G_2$ choices for S , and thus the number of such triples is at most $c_0^4 (\frac{\delta^\alpha}{|\log \delta|})^4 (\#G_1)^2 (\#G_2)$.

Finally, we will bound the number of remaining triples $(Q, Q', S) \in \mathcal{W}$. For each such triple, there are r -tubes $T \in \mathcal{T}_Q$ and $T' \in \mathcal{T}_{Q'}$ with $\angle(T, T') \gtrsim s$ (by $\angle(T, T')$ we mean the angle between their coaxial lines), so that S intersects $T \cap T'$. Thus for each pair $Q, Q' \subset G_1''$, the corresponding set of squares $S \subset G_2$ intersect the set

$$\bigcup_{\substack{(T, T') \in \mathcal{T}_Q \times \mathcal{T}_{Q'} \\ \angle(T, T') \gtrsim s}} T \cap T'.$$

Each set $T \cap T'$ in the above union is contained in a ball of radius $\sim s^{-1}r$, and since G_2 is a $(\delta, 1, \delta^{-\lambda})$ -Frostman set, each set $T \cap T'$ can intersect at most $\delta^{-\lambda} s^{-1}r (\#G_2)$ squares from G_2 . By (14.2.139), there are at most $(\#\mathcal{T}_Q)(\#\mathcal{T}_{Q'}) \leq r^{-2\beta}$ such pairs (T, T') , and hence when δ is sufficiently small, the total number of triples of this type is at most

$$\begin{aligned} (\#G_1)^2 (\delta^{-\lambda} s^{-1}r (\#G_2)) r^{-2\beta} &\leq \delta^{-\lambda} s^{-1} K^{-2} (\#G_1)^2 (\#G_2) \\ &\leq c_0 \left(\frac{\delta^\alpha}{|\log \delta|} \right)^4 (\#G_1)^2 (\#G_2), \end{aligned}$$

where here we recall that $\beta = 1/4$ (and thus the exponent $r^{1-2\beta}$ is positive), and we used the fact that $r \leq K^{-1/\beta} = K^{-4}$ and $K = \delta^{-\frac{3\alpha}{\zeta} - \lambda}$, $s =$

$\delta^\lambda (c_0 \frac{\delta^\alpha}{|\log \delta|})^{4/\zeta}$. We conclude that if $\delta > 0$ is sufficiently small, then $\#\mathcal{W} \leq 3c_0 \left(\frac{\delta^\alpha}{|\log \delta|}\right)^4 (\#G_1)^2 (\#G_2)$. If $c_0 > 0$ is chosen sufficiently small, then this contradicts (14.2.140), and completes the proof. $\square$

Remark 14.2.75. It is tempting to avoid Lemma 14.2.74 by instead iterating Proposition 14.2.71' multiple times starting with $\sigma = \beta = \zeta$ rather than $\sigma = \beta = 1/4$. Unfortunately, this would require $O_{\zeta, \varepsilon}(1)$ iterations, and this in turn would mean that the quantity $\lambda = \lambda(\varepsilon)$ from Proposition 14.2.70 would also have to be sufficiently small depending on ζ (in particular, much smaller than ζ). This is not acceptable, because in our application below we must use a value of λ that is at least as large as ζ .

We can now combine Lemma 14.2.68, Corollary 14.2.73, and Lemma 14.2.74 to prove Proposition 14.2.70. The details are as follows.

Proof of Proposition 14.2.70. Let $\varepsilon, \zeta \in (0, 1)$. Let $\lambda = \lambda(\varepsilon)$, $\alpha = \alpha(\varepsilon, \zeta)$, and $\delta_0 = \delta_0(\varepsilon, \zeta)$ be specified below. Let $\delta \in (0, \delta_0]$ and let $F, G_1, G_2, \mathcal{H}$ satisfy the hypotheses of Proposition 14.2.70.

Let $\alpha' > 0$ be a quantity to be chosen below. Applying Lemma 14.2.74 with this choice of α' , and with ζ, λ as above, we conclude that (G_1, G_2) and (G_2, G_1) have δ -discretized $(\frac{1}{4}, \delta^{-\frac{3\alpha'}{\zeta}-\lambda}, 1 - \delta^{\alpha'})$ -thin tubes.

Applying Corollary 14.2.73 with $\varepsilon/16$ in place of ε , we conclude that there exists $N = N(\varepsilon)$ so that (G_1, G_2) and (G_2, G_1) have δ -discretized $(1, K, 1 - 3^N \delta^{\alpha'})$ -thin tubes, where

$$K = \max\{4\delta^{-\frac{4\alpha'}{\zeta}-\lambda}, 4\delta^{-2\lambda-2\alpha'} N\}^N \leq (4N)^N \delta^{-\frac{4N\alpha'}{\zeta}-2N\lambda}. \quad (14.2.141)$$

We will select

$$\lambda = \frac{\varepsilon}{100N}, \quad \alpha' = \frac{\varepsilon\zeta}{100N}.$$

Since $N = N(\varepsilon)$ depends only on ε , we have that λ depends only on ε , as required. α' depends on ε and ζ . With these choices, the RHS of (14.2.141) is at most $\delta^{-\varepsilon/8}$, provided $\delta_0 > 0$ is selected sufficiently small.

Let $\alpha = \alpha'/2$, and select δ_0 sufficiently small (depending on α' and N , which in turn depend on ε and ζ) so that $\delta_0^\alpha \geq 2 \cdot 3^N \delta_0^{\alpha'}$, and thus $\delta^\alpha \geq 2 \cdot 3^N \delta^{\alpha'}$. Applying Lemma 14.2.68 with $c = 3^N \delta^{\alpha'}$, $K = \delta^{-\varepsilon/8}$, and $\varepsilon/2$ in place of ε , we conclude (provided δ_0 is selected sufficiently small) that

$$\mathcal{E}_\delta(\{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq 3^{5N} \delta^{5\alpha'} \delta^{\varepsilon/4} \delta^{\varepsilon/2-1} \geq \delta^{\varepsilon-1},$$

where the final inequality used the fact that $5\alpha' = \frac{5\varepsilon\zeta}{100N} \leq \varepsilon/5$. $\square$

14.2.8.3 Theorem 14.2.37' for well separated sets and uniform hypergraphs

Our goal in this section is to prove the following special case of Theorem 14.2.37'.

Proposition 14.2.76. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\eta})$ -Frostman sets that satisfy the standard separation conditions. Let $\mathcal{H} \subset F \times G_1 \times G_2$ be uniformly δ^η -dense. Then at least one of the following must hold:*

(A) *There exist two lines ℓ and $\ell^\perp$ (where $\ell^\perp$ passes through the origin and is orthogonal to ℓ) that satisfy (14.2.122):*

$$\begin{aligned} \mathcal{E}_\delta(N_\delta(\ell) \cap F) &\geq \delta^{\varepsilon-1}, \\ \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G_1) &\geq \delta^{\varepsilon-1}, \quad \mathcal{E}_\delta(N_\delta(\ell^\perp) \cap G_2) \geq \delta^{\varepsilon-1}. \end{aligned} \quad (14.2.122)$$

(B) *There exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-\eta}\rho$ that satisfies (14.2.123):*

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.2.123)$$

A non-concentration condition near lines, and anisotropic rescaling

We will prove Proposition 14.2.76 by reducing it to Proposition 14.2.70. One key difference between these two results is that Proposition 14.2.70 assumes the non-concentration estimate (14.2.134). First, a two-ends reduction shows that every set $E \subset [0, 1]^2$ has a large subset that satisfies a re-scaled analogue of (14.2.134) localized inside a rectangle of some width w . A precise version and its proof follow.

Lemma 14.2.77. *Let $E \subset [0, 1]^2$ be a union of δ -squares and let $0 < \zeta \leq 1/4$. Then there exists a number $w > 0$ and a line ℓ , so that if we define $E' = E \cap N_w(\ell)$, then $|E'| \geq \frac{1}{2}w^\zeta|E|$, and for all $0 < r \leq 1$ and all lines ℓ , we have*

$$|E' \cap N_r(\ell)| \leq (r/w)^\zeta |E'|. \quad (14.2.142)$$

Proof. Parameterize a line by $\ell_{\nu,a} = \{x \in \mathbb{R}^2 : x \cdot \nu = a\}$, where $\nu \in S^1$ and $a \in [-2, 2]$; lines with offsets outside this interval do not meet $[0, 1]^2$. Consider

$$\Phi(r, \nu, a) = r^{-\zeta} |E \cap N_r(\ell_{\nu,a})|, \quad 0 < r \leq 1.$$

Since E is a finite union of squares, its intersection with a strip of width r has area $O_E(r)$ uniformly in the line. Thus Φ extends by zero at $r = 0$ because $\zeta < 1$. The resulting function is continuous after irrelevant square boundaries

of measure zero are discarded, and in any case upper semicontinuity suffices; compactness therefore gives a maximizing triple (w, ν_0, a_0) with $w > 0$. Put $\ell_0 = \ell_{\nu_0, a_0}$ and $E' = E \cap N_w(\ell_0)$.

The strip $N_1(\{y = 1/2\})$ contains $[0, 1]^2$, so maximality gives

$$w^{-\zeta}|E'| \geq |E|,$$

which is stronger than the claimed mass bound. For an arbitrary line ℓ and $0 < r \leq 1$, the same maximality gives

$$r^{-\zeta}|E' \cap N_r(\ell)| \leq r^{-\zeta}|E \cap N_r(\ell)| \leq w^{-\zeta}|E'|.$$

Multiplying by r^ζ proves (14.2.142). $\square$

If we anisotropically re-scale the output set $E' \subset N_w(\ell)$ from Lemma 14.2.77 by a factor of w^{-1} in the $l^\perp$ direction, then the non-concentration estimate (14.2.142) from Lemma 14.2.77 will resemble the non-concentration hypothesis (14.2.134) from Proposition 14.2.76. The problem is that if E was a $(\delta, 1, \delta^{-\eta})$ -Frostman set, then E' will be a $(\delta, 1, \delta^{-\eta-\zeta})$ -Frostman set, but the image of E' under this anisotropic rescaling might not be a $(\delta, 1, C)$ -Frostman set for any reasonable value of C . The next lemma says that we can fix this problem at the cost of refining E' slightly.

Lemma 14.2.78. *For all $\varepsilon > 0$, there exists $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $w \in [\delta^{1-\varepsilon}, 1]$, let $R \subset \mathbb{R}^2$ be a rectangle of dimensions $1 \times w$, and let $E \subset R$ be a $(\delta, 1, C)$ -Frostman set for some $C \geq 1$. Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an affine transform that maps R to $[0, 1]^2$. Then there exists a subset $E' \subset E$ with $|E'| \geq (\delta/w)^\varepsilon |E|$ so that the union of δ/w squares that intersect $\phi(E')$ form a $(\delta/w, 1, (w/\delta)^\varepsilon C)$ -Frostman set and each δ/w -square has about the same size of intersection with $\phi(E')$.*

Proof. Let $\mathcal{R}_0$ be the tiling of R by rectangles of dimensions $\delta \times \delta/w$ pointing in the same direction as R (we will suppose for simplicity that $1/\delta$ and w/δ are integers; if not, then we can shrink R slightly on two sides and prune the set E ; this step is harmless). By pigeonholing, we can select sets $E' \subset E$ and $\mathcal{R} \subset \mathcal{R}_0$, and a number $m > 0$, so that $|E'| \gtrsim |\log \delta|^{-1} |E|$, and $|\tilde{R} \cap E'| \in [m, 2m)$ for each $\tilde{R} \in \mathcal{R}$ with $\tilde{R} \cap E' \neq \emptyset$.

Let $F \subset [0, 1]^2$ be the union of δ/w squares that intersect $\phi(E')$. It remains to verify that F is a $(\delta/w, 1, (w/\delta)^\varepsilon C)$ -Frostman set. Let $r \geq \delta/w$, let

$B = B(x, r)$ be a ball, and let $3B$ be its 3-fold dilate. We have

$$\begin{aligned}
 \frac{|B \cap F|}{|F|} &= \frac{|\phi^{-1}(B) \cap \phi^{-1}(F)|}{|\phi^{-1}(F)|} \\
 &\leq \frac{\#\{\tilde{R} \in \mathcal{R}: \tilde{R} \cap \phi^{-1}(B) \neq \emptyset\}}{\#\mathcal{R}} \\
 &\leq \frac{|\phi^{-1}(3B) \cap E'|}{|E'|} \\
 &\lesssim \log(1/\delta) \frac{|\phi^{-1}(3B) \cap E|}{|E|} \\
 &\leq 3 \log(1/\delta) rC.
 \end{aligned} \tag{14.2.143}$$

For the final inequality, we used the fact that $\phi^{-1}(3B)$ is contained in a ball of radius $3r$, and E is a $(\delta, 1, C)$ -Frostman set. Since $(w/\delta) \geq \delta^{-\varepsilon}$, we have $3 \log(1/\delta) \leq (w/\delta)^\varepsilon$ provided δ_0 is selected sufficiently small. $\square$

We will sometimes need to move from scale δ to a coarser scale. The next lemma says that after a refinement, (δ, α, C) -Frostman sets are well-behaved under coarsening.

Lemma 14.2.79. *Let $0 < \delta \leq \rho \leq 1$ and let $E \subset \mathbb{R}^2$ be a (δ, α, C) -Frostman set for some $C \geq 1$. Then there is a set $E' \subset E$ with $|E'| \geq |\log \delta|^{-1}|E|$, so that the union of ρ squares intersecting E' is a (ρ, α, C') -Frostman set, with $C' \sim |\log \delta|C$.*

Proof. Tile $\mathbb{R}^2$ by ρ -squares. By dyadic pigeonholing, we can select $E' \subset E$ and $m > 0$ so that for each ρ -square Q in the tiling, either $|E' \cap Q| = \emptyset$ or $|E' \cap Q| = m$. Let E_ρ be the union of ρ -squares that intersect E' . Then $|E_\rho| = (|E'|/m)\rho^2 \leq |\log \delta|^{-1}(|E'|/m)\rho^2$. Let B be a ball of radius $r \geq \rho$, and let B' be its 2-fold dilate. Then $|B' \cap E'| \leq |B' \cap E| \leq C(2r)^\alpha |E|$. In particular, there are at most $C4r^\alpha |E|/m$ ρ -squares that are contained in $B' \cap E$, and hence at most $4Cr^\alpha |E|/m$ such cubes that intersect $C \cap E$. We conclude that

$$|E_\rho \cap B| \lesssim \rho^2 C r^\alpha |E|/m \lesssim |\log \delta| C r^\alpha |E_\rho|. \quad \square$$

Localization to rectangles of comparable width

We now consider the following situation. Suppose we have sets $F, G_1, G_2, \mathcal{H}$ that satisfy the hypotheses of Proposition 14.2.76, and suppose that G_1 and G_2 are contained in strips of width w_1 and w_2 , respectively. The next lemma says that either Conclusion (B) of Proposition 14.2.76 is true, or w_1 and w_2 are almost identical; G_1 and G_2 are contained in a common strip; and F is contained in an orthogonal strip of comparable thickness.

Lemma 14.2.80. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $F, G_1, G_2 \subset [0, 1]^2$ be $(\delta, 1, \delta^{-\eta})$ -Frostman sets that satisfy the standard separation conditions, and let $\mathcal{H} \subset F \times G_1 \times G_2$ be uniformly δ^η -dense. Suppose $G_1 \subset N_w(\ell)$ for some $w > 0$ and some line ℓ . Let $\ell^\perp$ be the line through the origin that is perpendicular to ℓ .*

Then at least one of the following two things must happen:

(A) $\pi_{G_2}(\mathcal{H}) \subset N_{\delta^{-\varepsilon}w}(\ell)$ and $\pi_F(\mathcal{H}) \subset N_{\delta^{-\varepsilon}w}(\ell^\perp)$.

(B) *There exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-\eta}\rho$ that satisfies (14.2.123):*

$$\mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \geq (|I|/\rho)^{1-\varepsilon}. \quad (14.2.123)$$

Proof. First, we may suppose that $w \leq \delta^\varepsilon$, or else Conclusion (A) holds and we are done.

Let $0 < \eta \ll \varepsilon_1 \ll \varepsilon$ be small quantities that will be chosen below. Without loss of generality, we can suppose that the maps $\pi_F: \mathcal{H} \rightarrow F$, $\pi_{G_1}: \mathcal{H} \rightarrow G_1$, and $\pi_{G_2}: \mathcal{H} \rightarrow G_2$ are onto (if not, replace F, G_1, G_2 by the image of $\mathcal{H}$ under the corresponding projection map; this reduces the size of each set by at most a multiplicative factor of δ^η , and hence the new sets are $(\delta, 1, \delta^{-2\eta})$ -Frostman sets). Let $t = \inf\{r: G_1 \cup G_2 \subset N_r(\ell)\}$, and let $U = N_t(\ell)$. Clearly $t \geq \delta$, since $G_1 \cup G_2$ contains at least one δ -square. Let $U^\perp = N_{\delta^{-\varepsilon_1}t}(\ell^\perp)$. Let e be a unit vector parallel to ℓ .

Step 1. Trapping F and $G_1 \cup G_2$ inside thick orthogonal strips

First, we will show that either $F \subset U^\perp$, or Conclusion (B) holds. If $t \geq \delta^{-\varepsilon_1}$ then $F \subset [0, 1]^2 \subset U^\perp$ and we are done. Thus we will suppose $t < \delta^{-\varepsilon_1}$. Suppose $F \not\subset U^\perp$, and fix an element $a \in F \setminus U^\perp$. Since F satisfies (14.2.131), we have $\text{dist}(a, 0) \geq \frac{1}{2}$, and hence $|a \cdot e| \geq \frac{1}{4}\delta^{-\varepsilon_1}t$. In particular, if $b_1, b'_1 \in G_1$ are $\geq 2t$ separated, then $|e \cdot (b_1 - b'_1)| \geq \frac{1}{4}|b_1 - b'_1|$ and hence

$$|a \cdot (b_1 - b'_1)| \geq |a \cdot e| |e \cdot (b_1 - b'_1)| - |a \cdot e^\perp| |e^\perp \cdot (b_1 - b'_1)| \geq \frac{1}{4}|a \cdot e| |b_1 - b'_1| - t. \quad (14.2.144)$$

Let $s = 8t/|a \cdot e|$. Since $a \in [0, 1]^2$, we have $|a \cdot e| \leq 2$ and hence $s \geq 4t$, and in particular $s > \delta$. We also have $s \leq 8t/(\frac{1}{4}\delta^{-\varepsilon_1}t) \leq 32\delta^{\varepsilon_1}$. Observe that if $|b_1 - b'_1| \geq s$, then by (14.2.144) we have

$$|a \cdot (b_1 - b'_1)| \geq \frac{1}{8}|a \cdot e| |b_1 - b'_1|. \quad (14.2.145)$$

Fix an element $b_2 \in G_2$ such that $n_{\mathcal{H}}[(a, b_2)]$ is nonempty, and let $G'_1 \subset G_1$ be a maximal s -separated subset of $\pi_{G_1}(n_{\mathcal{H}}[(a, b_2)])$. Since $\mathcal{H}$ is uniformly δ^η -dense, $\pi_{G_1}(n_{\mathcal{H}}[(a, b_2)])$ has measure at least $\delta^{1-\eta}$, and since G_1 is a $(\delta, 1, \delta^{-\eta})$ -Frostman set, we conclude that $\#G'_1 \geq \delta^{2\eta}s^{-1}$. Thus by (14.2.145), if we

define $\rho = 2t = |a \cdot e|s/4$, then

$$\mathcal{E}_\rho(\{a \cdot (b_1 - b_2) : b_1 \in G_1\}) \geq \mathcal{E}_\rho(\{a \cdot b_1 : b_1 \in G'_1\}) \gtrsim \#G'_1 \gtrsim \delta^{2\eta}s^{-1}.$$

Furthermore, $\{a \cdot (b_1 - b_2) : b_1 \in G_1\}$ is contained in the interval $J = a \cdot ([-1, 2]^2 \cap (U - b_2))$, which has length $|J| \sim |a \cdot e| \sim \rho/s \gtrsim \delta^{-\varepsilon_1}\rho \geq \delta^{-\eta}\rho$. We conclude that

$$\begin{aligned} \mathcal{E}_\rho(J \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) &\geq \mathcal{E}_\rho(\{a \cdot (b_1 - b_2) : b_1 \in G'_1\}) \\ &\gtrsim \delta^{2\eta}s^{-1} \gtrsim \delta^{2\eta} \frac{|J|}{\rho} \gtrsim \delta^{2\eta-\varepsilon_1} \left(\frac{|J|}{\rho}\right)^{1-\varepsilon}. \end{aligned}$$

If $\eta \leq \varepsilon_1\varepsilon/2$ and δ_0 is selected sufficiently small, then Conclusion (B) holds.

Step 2. $G_1 - G_2$ spans many directions inside U

At this point we may suppose that $F \subset U^\perp$. If $t \leq \delta^{-\varepsilon+\varepsilon_1}w$, then Conclusion (A) holds and we are done. Henceforth we will assume that $t > \delta^{-\varepsilon+\varepsilon_1}w$. We will prove that Conclusion (B) holds. Fix a point $b_2 \in G_2 \setminus N_{t/2}(\ell)$ —such a point must exist since otherwise we would have $t \leq 2w$. In particular, we have $t/2 \leq \text{dist}(b_2, N_w(\ell)) \leq t$. If $b_1, b'_1 \in G_1$ with $|b_1 - b'_1| \geq 10w/t$, then

$$|\pi^{b_2}(b_1) - \pi^{b_2}(b'_1)| \geq \frac{1}{10}t|b_1 - b'_1|,$$

where $\pi^{b_2}(x) = \frac{x-b_2}{|x-b_2|}$ denotes the radial projection from the vantage point b_2 . Indeed, this follows from the standard separation conditions, see Figure 14.2.6.

Recall that G_1 is contained in a ball B of radius $1/10$, which has distance at least $1/2$ from b_2 . Thus $G_1 \subset B \cap N_w(\ell)$, and $\pi^{b_2}(B \cap N_w(\ell))$ is contained in an arc $J \subset S^1$ of length $|J| \sim t$.

Let $u = 10w/t$, and let $G'_1 \subset G_1$ be a maximal u -separated subset of $\pi_{G_1}(n_{\mathcal{H}}[(b_2)])$. Since $\mathcal{H}$ is uniformly δ^η -dense, $\pi_{G_1}(n_{\mathcal{H}}[(b_2)])$ has measure at least $\delta^\eta|G_1|$. Since G_1 is a $(\delta, 1, \delta^{-\eta})$ -Frostman set, we conclude that $\#G'_1 \geq \delta^{2\eta}u^{-1}$. Thus we have $\pi^{b_2}(G'_1) \subset J$, and $\pi^{b_2}(G'_1)$ is dense inside J at scale w , in the sense that

$$\mathcal{E}_w(\pi^{b_2}(G'_1)) \gtrsim \#G'_1 \gtrsim \delta^{2\eta}u^{-1} \gtrsim \delta^{2\eta}tw^{-1}. \quad (14.2.146)$$

Let $X = \pi_{F \times G_1}(n_{\mathcal{H}}[(b_2)]) \cap (F \times G'_1)$; this is the set of pairs $(a, b_1) \in F \times G'_1$ so that $(a, b_1, b_2) \in \mathcal{H}$. Since $\mathcal{H}$ is uniformly δ^η -dense, we have $\#X \geq \delta^\eta(\#F)(\#G'_1)$ (recall our notation convention from Section 14.2.8). Apply Lemma 14.2.62 to X , with $\varepsilon = 1/2$, let $X' \subset X$ be the output, and let $F' = \pi_F(X')$; thus for each $a \in F'$, there are $\gtrsim \delta^\eta(\#G'_1)$ elements $b_1 \in G'_1$ with $(a, b_1) \in X$.

Let $t' = \delta^{-\varepsilon_1}t$. Let $\tilde{R}$ be a $2 \times t'$ rectangle pointing in direction e that contains $G_1 \cup G_2$, and let $\tilde{R}^\perp$ be a $2 \times t'$ rectangle pointing in direction $e^\perp$

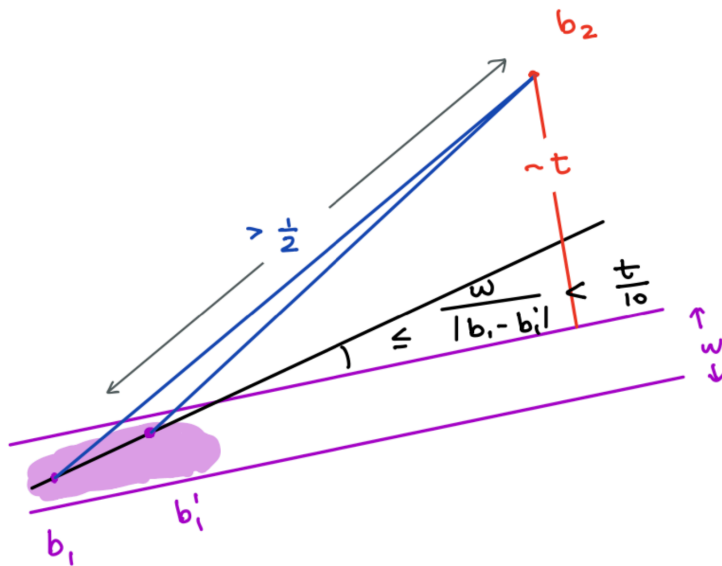


Figure 14.2.6: The purple shaded region is G_1 , which is contained in $N_w(\ell)$. The points $b_1, b_1' \in G_1$ have distance $\geq 1/2$ to $b_2 \in G_2$, which has distance $\sim t$ to $N_w(\ell)$.

that contains F . (The existence of such a rectangle $\tilde{R}$ is easy; just select a suitable subset of $N_{t'}(\ell)$. The existence of $\tilde{R}^\perp$ is guaranteed by the arguments in Step 1, which says $F \subset U^\perp = N_{t'}(\ell^\perp)$). Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an affine-linear transformation that sends $\tilde{R}$ to $[0, 1]^2$, and let ϕ^T be the transpose of ϕ ; then $\phi^T(\tilde{R}^\perp)$ is a unit square.

Apply Lemma 14.2.78 to $F \subset \tilde{R}^\perp$ with η as above, and let $F' \subset F$ be the output of that lemma. Let $\tilde{F}$ be the union of δ/t' squares that intersect $\phi^T(F')$; then $\tilde{F}$ is a $(\delta/t', 1, \delta^{-2\eta})$ -Frostman set. Applying Lemma 14.2.79 with δ/t' in place of δ and w/t' in place of ρ (this is valid since $w/t' \leq \delta^\varepsilon < 1$), we can find a subset $Y \subset \tilde{F}$, so that if $\tilde{F}'$ denotes the set of w/t' -squares that intersect Y , then $\tilde{F}'$ is a $(w/t', 1, \delta^{-3\eta})$ -Frostman set.

Since $|\pi^{b_2}(b_1) - \pi^{b_2}(b_1')| \sim t' |\pi^{\phi(b_2)}(\phi(b_1)) - \pi^{\phi(b_2)}(\phi(b_1'))|$, by (14.2.146) we have

$$\mathcal{E}_{w/t'}\left(\pi^{\phi(b_2)}(\phi(G_1'))\right) \gtrsim \mathcal{E}_w(\pi^{b_2}(G_1')) \gtrsim \delta^{2\eta} t w^{-1} = \delta^{2\eta+\varepsilon_1} t' w^{-1}.$$

Thus if δ_0 is chosen sufficiently small, then the radial projection $\Lambda = \pi^{\phi(b_2)}(\phi(G_1'))$ is a $(w/t', 1, \delta^{-3\eta-\varepsilon_1})$ -Frostman set. Each $\tilde{a} \in \tilde{F}'$ can be written as $\tilde{a} = (\phi^T)^{-1}(a)$ for some $a \in F'$. Hence there are $\gtrsim \delta^\eta \#G_1'$ unit vectors $\theta \in \Lambda$, so that $\theta = \pi^{\phi(b_2)}(\phi(b_1))$ for some $b_1 \in G_1'$ with $(a, b_1) \in X'$ (recall we have fixed b_2 at the beginning). In particular, X' has density $d \geq \delta^\eta$.

This is precisely the setting for Theorem 14.2.64. Applying Theorem 14.2.64 with $\varepsilon/2$ in place of ε , $d = \delta^\eta$ and $C = \delta^{3\eta+\varepsilon_1}$, we conclude that there exists a unit vector $\theta \in \Lambda$ of the form $\theta = \pi^{\phi(b_2)}(\phi(b_1))$, so that if $(\tilde{F}')_\theta$ denotes the set of $\tilde{a}$ (with preimage a) with $(a, b_1) \in X'$, then

$$\mathcal{E}_{w/t'}(\theta \cdot (\tilde{F}'_3)_\theta) \gtrsim \delta^{9\eta+2\varepsilon_1}(t'/w)^{1-\varepsilon/2},$$

Set

$$\begin{aligned} \mathcal{A} &= \{\phi^T(a) \cdot (\phi(b_1) - \phi(b_2)) : a \in \pi_F(n_{\mathcal{H}}[(b_1, b_2)])\}, \\ \mathcal{B} &= \{a \cdot (b_1 - b_2) : a \in \pi_F(n_{\mathcal{H}}[(b_1, b_2)])\}. \end{aligned}$$

Then

$$\mathcal{E}_{w/t'}(\mathcal{A}) \gtrsim \delta^{9\eta+2\varepsilon_1}(t'/w)^{1-\varepsilon/2}, \quad \mathcal{E}_{w/t'}(\mathcal{A}) = \mathcal{E}_w(\mathcal{B}).$$

The latter set is contained in $\tilde{R}^\perp \cdot (\tilde{R} - \tilde{R})$, which is an interval I of length about t' . Setting $\rho = w$, we conclude that I has length $t' = \delta^{-\varepsilon_1}t \geq \delta^{-\varepsilon}\rho$, and

$$\begin{aligned} \mathcal{E}_\rho(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) &\gtrsim \delta^{9\eta+2\varepsilon_1}(t'/\rho)^{1-\varepsilon/2} \\ &\gtrsim_\delta \delta^{9\eta+2\varepsilon_1}(t'/\rho)^{\varepsilon-\varepsilon/2} \left(\frac{|I|}{\rho}\right)^{1-\varepsilon} \\ &\geq \delta^{9\eta+2\varepsilon_1-\varepsilon^2/2} \left(\frac{|I|}{\rho}\right)^{1-\varepsilon}. \end{aligned}$$

If ε_1 is selected sufficiently small (depending on ε) then $9\eta + 2\varepsilon_1 - \varepsilon^2/2 < 0$. Selecting δ_0 sufficiently small, we have that Conclusion (B) holds. $\square$

The second reduction

Armed with the above lemmas, we are now ready to prove Proposition 14.2.76. In brief, we apply Lemma 14.2.77 to localize G_1 and G_2 to strips S_1 and S_2 . We then use Lemma 14.2.80 to show that these strips must coincide, and furthermore, that F is localized to an orthogonal strip of approximately the same width, up to a factor of $\delta^{-\varepsilon}$. We then re-scale these strips using Lemma 14.2.78 and apply Proposition 14.2.70 to the corresponding re-scaled sets. We now turn to the details.

Proof of Proposition 14.2.76. Let $\varepsilon > 0$. Let $\lambda = \lambda(\varepsilon/2)$ be the output from Proposition 14.2.70 with $\varepsilon/2$ in place of ε . Define

$$\varepsilon_1 = \varepsilon\lambda/30. \tag{14.2.147}$$

Let $\eta_1 = \eta_1(\varepsilon_1)$ be the output from Lemma 14.2.80 with ε_1 in place of ε . Define

$$\zeta = \min(\eta_1/2, \varepsilon\lambda/3). \tag{14.2.148}$$

Let $\alpha = \alpha(\varepsilon/2, \zeta)$ be the output from Proposition 14.2.70 with ζ as defined in (14.2.148) and $\varepsilon/2$ in place of ε . Define

$$\eta = \min(\varepsilon_1 \zeta, \varepsilon \alpha / 40), \quad (14.2.149)$$

We will be slightly less precise about our choice of δ_0 , since this plays a minor role.

Let $\delta \in (0, \delta_0]$ and let $F, G_1, G_2, \mathcal{H}$ satisfy the hypotheses of Proposition 14.2.76 for these values of η and δ . We must show that at least one of Conclusion (A) or (B) from Proposition 14.2.76 is true for ε as above.

First, apply “two-ends reduction” Lemma 14.2.77 to G_1 , with ζ as defined in (14.2.148); we obtain a width w_1 , a line ℓ_1 , and a set $G'_1 = G_1 \cap N_{w_1}(\ell_1)$. Apply Lemma 14.2.62 to the induced subgraph $\mathcal{H} \cap (F \times G'_1 \times G_2)$ with $\varepsilon = 1/2$; we obtain a uniformly c -dense hypergraph $\mathcal{H}'$ with $c \geq 2^{-4}\delta^\eta$. Replacing G_2 by its image under $\pi_{G_2}(\mathcal{H}')$, we may suppose that $\pi_{G_2}: \mathcal{H}' \rightarrow G_2$ is onto; this step might reduce the size of G_2 by a multiplicative factor of c . Next, apply Lemma 14.2.77 (with ζ as defined in (14.2.148)) to G_2 ; we obtain a width w_2 , a line ℓ_2 , and a set $G'_2 = G_2 \cap N_{w_2}(\ell_2)$. Apply Lemma 14.2.62 to the induced subgraph $\mathcal{H}' \cap (F \times G'_1 \times G'_2)$ with $\varepsilon = 1/2$; we obtain a uniformly c -dense hypergraph $\mathcal{H}''$ with $c \geq 2^{-8}\delta^\eta$. Let $G''_1 = \pi_{G_1}(\mathcal{H}'')$, and let $F'' = \pi_F(\mathcal{H}'')$.

After this step has been completed, the sets F'', G''_1, G''_2 are $(\delta, 1, \delta^{-2\zeta})$ -Frostman sets; $\mathcal{H}''$ is uniformly $\delta^{-8}\delta^\eta$ regular, and G''_1, G''_2 satisfy the following Frostman-type non-concentration on lines estimate:

$$|G''_i \cap N_r(\ell)| \leq 8\delta^{-\eta}(r/w_i)^\zeta |G''_i| \quad \text{for all lines } \ell \text{ and all } \delta \leq r \leq 1, \quad i = 1, 2. \quad (14.2.150)$$

For notational convenience, we will suppose that $w_1 \geq w_2$ (this is harmless since G''_1 and G''_2 play (anti) symmetric roles). By (14.2.148) and (14.2.149), we can apply Lemma 14.2.80 with ε_1 in place of ε (once with G''_1 and G''_2 as stated, and once with their roles reversed) to conclude that at least one of the following is true.

- Conclusion (B) of Proposition 14.2.76 holds (for ε_1 , and hence also for ε).
- $w_1 \leq \delta^{-\varepsilon_1} w_2$; $G''_1 \cup G''_2$ is contained in a strip S of width $w = \delta^{-\varepsilon_1} w_2$ (and hence contained in an appropriately chosen rectangle R of dimensions $2 \times w$); and F'' is contained in an orthogonal rectangle $R^\perp$ of dimensions $2 \times w$.

If the first item holds then we are done. Suppose instead that the second item holds. If $w \leq \delta^{1-\varepsilon/10}$, then by pigeonholing we can find orthogonal lines that satisfy (14.2.122) for the value of ε specified above, and hence Conclusion (A) of Proposition 14.2.76 holds.

Suppose instead that $w > \delta^{1-\varepsilon/10}$. Since $\eta < \varepsilon_1\zeta$, (14.2.150) can be re-written as

$$|G_i'' \cap N_r(\ell)| \lesssim \left(\frac{\delta^{-2\varepsilon_1} r}{w} \right)^\zeta |G_i''|. \quad (14.2.151)$$

Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an affine-linear transformation sending R to $[0, 1]^2$ and let ϕ^T be its transpose; then $\phi^T(R^\perp)$ is a translate of the unit square. We also have

$$a \cdot (b_1 - b_2) = w\phi^T(a) \cdot (\phi(b_1) - \phi(b_2)) \quad \text{for all } (a, b_1, b_2) \in R^\perp \times R^2. \quad (14.2.152)$$

Let $\tau = \delta/w \leq \delta^{\varepsilon/10}$. We would like to apply Proposition 14.2.70 to the sets $\phi^T(F'')$, $\phi(G_1'')$, and $\phi(G_2'')$ (and the corresponding image of $\mathcal{H}''$) at scale τ to conclude that Conclusion (B) of Proposition 14.2.76 must hold. The issue is that while F'' , G_1'' , and G_2'' are $(\delta, 1, \delta^{-2\zeta})$ -Frostman sets, their images under ϕ (resp. ϕ^T) might not be $(\tau, 1, \delta^{-2\zeta})$ -Frostman sets. We can fix this by sequentially applying Lemma 14.2.78 and Lemma 14.2.62 to each of the sets F'' , G_1'' , and G_2'' . We now turn to the details.

We apply Lemma 14.2.78 to $G_1'' \subset R$, with $\varepsilon_1\eta/10$ in place of ε ; we let G_1''' be the resulting subset. We then apply Lemma 14.2.62 to the induced subgraph $\mathcal{H}_1 = \mathcal{H}''' \cap (F'' \times G_1''' \times G_2'')$, with $\varepsilon = 1/2$; denote the output by $\mathcal{H}_1'$. Next we apply Lemma 14.2.78 to $G_2'' \subset R$, with $\varepsilon_1\eta/10$ in place of ε ; we let G_2''' be the resulting subset. We then apply Lemma 14.2.62 to the induced subgraph $\mathcal{H}_2 = \mathcal{H}_1' \cap (F'' \times G_1''' \times G_2''')$, with $\varepsilon = 1/2$; denote the output by $\mathcal{H}_2'$. Finally we apply Lemma 14.2.78 to $F'' \subset R^\perp$, with $\varepsilon_1\eta/10$ in place of ε ; we let F''' be the resulting subset. We then apply Lemma 14.2.62 to the induced subgraph $\mathcal{H}_3 = \mathcal{H}_2' \cap (F''' \times G_1''' \times G_2''')$, with $\varepsilon = 1/2$; denote the output by $\mathcal{H}_3'$.

Let $\tilde{F}$ (resp. $\tilde{G}_1, \tilde{G}_2$) be the set of τ -squares that intersect $\phi^\perp(F''')$ (resp. $\phi(G_2'''), \phi(G_1''')$), let $\tilde{\mathcal{H}}_0$ be the set of τ -cubes that intersect $(\phi \otimes \phi \otimes \phi^T)(\mathcal{H}_3')$, and let $\tilde{\mathcal{H}}_3$ be the refinement obtained by applying Lemma 14.2.62 to $\tilde{\mathcal{H}}_0$ with $\varepsilon = 1/2$. We have that $\tilde{\mathcal{H}}_3$ is uniformly $\delta^{4\eta} \geq \tau^{40\eta/\varepsilon}$ -dense, and $\tilde{G}_1, \tilde{G}_2, \tilde{F}$ are $(\tau, 1, \tau^{-3\zeta/\varepsilon})$ -Frostman sets.

After these refinements, (14.2.151) now says that for all $r \in [\delta, w]$ and all lines ℓ ,

$$|G_i''' \cap N_r(\ell)| \leq \left(\frac{\delta^{-3\varepsilon_1} r}{w} \right)^\zeta |G_i'''|, \quad i = 1, 2. \quad (14.2.153)$$

Furthermore, since each $\delta \times \delta/w$ rectangle that intersects G_1''' (resp. G_2''') has the same size of intersection with G_1''' (resp. G_2'''), (14.2.153) implies that for all $r \in [\tau, 1]$ and all lines ℓ ,

$$|\tilde{G}_i \cap N_r(\ell)| \leq (\delta^{-3\varepsilon_1} r)^\zeta |\tilde{G}_i| \leq (\tau^{-30\varepsilon_1/\varepsilon} r)^\zeta |\tilde{G}_i|, \quad i = 1, 2.$$

Let us now verify that the sets $\tilde{G}_1, \tilde{G}_2, \tilde{F}$ and $\tilde{\mathcal{H}}$ satisfy the hypotheses of Proposition 14.2.70 with $\varepsilon/2$ in place of ε , and τ in place of δ . The estimate

(14.2.134) holds with ζ as above, provided $30\varepsilon_1/\varepsilon \leq \lambda$; this is guaranteed by (14.2.147). We have that $\tilde{G}_1, \tilde{G}_2$, and $\tilde{F}$ are $(\tau, 1, \tau^{-\lambda})$ -Frostman sets, provided $3\zeta/\varepsilon \leq \lambda$; this is guaranteed by (14.2.148). Finally, $\tilde{\mathcal{H}}$ is uniformly τ^α -dense, provided $40\eta/\varepsilon \leq \alpha$; this is guaranteed by (14.2.149). We conclude that there exists τ_0 (depending on ε and ζ , which in turn depends only on ε) so that

$$\mathcal{E}_\tau(\{\tilde{a} \cdot (\tilde{b}_1 - \tilde{b}_2) : (\tilde{a}, \tilde{b}_1, \tilde{b}_2) \in \tilde{\mathcal{H}}\}) \geq \tau^{\varepsilon/2-1}, \quad (14.2.154)$$

provided $\tau \leq \tau_0$. We can ensure that $\tau \leq \tau_0$ by selecting $\delta_0 = \tau_0^{1/\varepsilon}$.

Comparing (14.2.154) and (14.2.152), we see that there is an interval I of length $w > \delta^{1-\varepsilon/10}$ so that

$$\mathcal{E}_\delta(I \cap \{a \cdot (b_1 - b_2) : (a, b_1, b_2) \in \mathcal{H}\}) \gtrsim \tau^{\varepsilon/2-1} = (|I|/\delta)^{1-\varepsilon/2}.$$

If $\delta_0 > 0$ is sufficiently small, then Conclusion (B) of Proposition 14.2.76 holds. $\square$

14.2.8.4 A final reduction

In this section we will show that Proposition 14.2.76 implies Theorem 14.2.37'. Compared with Theorem 14.2.37', the difference between these two results is that in Proposition 14.2.76, the sets F, G_1, G_2 satisfy the standard separation conditions.

Proof of Theorem 14.2.37'. Let $\varepsilon > 0$. Let η_1, δ_1 be the output from Proposition 14.2.76, with $\varepsilon/2$ in place of ε . Let $\eta = \eta_1/100$, $\delta_0 = \delta_1$. Let $F, G_1, G_2, \mathcal{H}$ satisfy the hypotheses of Theorem 14.2.37' with this value of η and some $\delta \in (0, \delta_0]$. We must show that at least one of the Conclusions (A) or (B) from Theorem 14.2.37' hold for this value of ε and η .

Let $\mathcal{B}$ be a cover of $[0, 1]^6$ by sets of the form $\mathbf{B} = B_F \times B_{G_1} \times B_{G_2}$, where $B_F, B_{G_1}, B_{G_2} \subset \mathbb{R}^2$ are balls of radius $\delta^{6\eta}$; we can construct such a cover with $\#\mathcal{B} \lesssim \delta^{-36\eta}$. This induces a decomposition $\mathcal{H} = \bigcup_{\mathbf{B}} \mathcal{H}_{\mathbf{B}}$, where $\mathcal{H}_{\mathbf{B}} = \mathcal{H} \cap (B_F \times B_{G_1} \times B_{G_2})$. Since F, G_1, G_2 are $(\delta, 1, \delta^{-\eta})$ -Frostman sets, we have

$$\#\{(a, b_1, b_2) \in \mathcal{H} : |a| \geq \delta^{5\eta}, |b_1 - b_2| \geq \delta^{5\eta}\} \geq \frac{1}{2} \#\mathcal{H}.$$

Thus by pigeonholing, there exists an element $\mathbf{B} = B_F \times B_{G_1} \times B_{G_2} \in \mathcal{B}$ with $\text{dist}(B_F, 0) \geq \delta^{5\eta} - \delta^{6\eta}$ and $\text{dist}(B_{G_1}, B_{G_2}) \geq \delta^{5\eta} - \delta^{6\eta}$ so that $\#\mathcal{H}_{\mathbf{B}} \geq \frac{1}{2} \#\mathcal{H} / \#\mathcal{B} \gtrsim \delta^{36\eta} \#\mathcal{H}$. Let $F' = F \cap B_F$, $G'_1 = G_1 \cap B_{G_1}$, and $G'_2 = G_2 \cap B_{G_2}$. After applying a common translation ϕ to the sets G'_1 and G'_2 (this translation is harmless since it preserves expressions of the form $b_1 - b_2$ with $b_1 \in G'_1$ and $b_2 \in G'_2$; we may also assume it sends δ -squares to δ -squares), we may assume that G'_1 and G'_2 are contained in $B(0, r)$, with $r = \text{dist}(B_{G_1}, B_{G_2}) \gtrsim \delta^{5\eta}$. Let

G_1'' (resp G_2'') be the image of G_1' under the map $\psi_G: (x, y) \mapsto (x/r, y/r)$. Then G_1'' and G_2'' are $(\delta, 1, \delta^{-12\eta})$ -Frostman sets that are contained in balls of radius $\leq 1/100$ that are $1/2$ -separated. Similarly, let F'' be the image of F' under the map $\psi_F: (x, y) \mapsto (x/r', y/r')$, with $r' = \text{dist}(B_{G_F}, 0) \gtrsim \delta^{5\eta}$; then F'' is a $(\delta, 1, \delta^{-12\eta})$ -Frostman set that is contained in a ball of radius $\leq 1/100$ that has distance $\geq 1/2$ from the origin.

Let $\phi_G = \psi_G \circ \phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the translation and dilation described above that was applied to G_1' and G_2' .

Define $\mathcal{H}' = \psi_F \otimes \phi_G \otimes \phi_G(\mathcal{H}_{\mathbf{B}})$. The sets F'', G_1'', G_2'' and $\mathcal{H}'$ might no longer be unions of δ -cubes, but it is harmless to replace them with the unions of δ -cubes that they intersect. The sets F'', G_1'', G_2'' now satisfy the standard separation conditions (14.2.131).

Apply Lemma 14.2.62 with $\varepsilon = \delta^{20\eta}$ to $\mathcal{H}'$, and let $\mathcal{H}''$ denote the resulting refinement; this set is uniformly $\delta^{40\eta}$ -dense. We now apply Proposition 14.2.76 to $\mathcal{H}'' \subset F'' \times G_1'' \times G_2''$ with $\varepsilon/2$ in place of ε ; we have that F'', G_1'', G_2'' satisfy the standard separation conditions (14.2.131), and since each of F'', G_1'' , and G_2'' is a $(\delta, 1, \delta^{-12\eta})$ -Frostman set, it is also a $(\delta, 1, \delta^{-100\eta})$ -Frostman set. We have that $\mathcal{H}''$ is uniformly $\delta^{100\eta}$ -dense.

Suppose Conclusion (A) of Proposition 14.2.76 is true, i.e. there are lines ℓ_0 , and $\ell_0^\perp$ (where $\ell_0^\perp$ passes through the origin and is perpendicular to ℓ_0) with

$$|N_\delta(\ell_0^\perp) \cap F''| \geq \delta^{\varepsilon/2} |F|,$$

$$|N_\delta(\ell_0) \cap G_1''| \geq \delta^{\varepsilon/2} |G_1|, \quad |N_\delta(\ell_0) \cap G_2''| \geq \delta^{\varepsilon/2} |G_2|.$$

Let $\ell = \phi_G^{-1}(\ell_0) = \phi^{-1} \circ \psi_G^{-1}$, then $\ell^\perp = \phi_F^{-1}(\ell_0^\perp)$ is perpendicular to ℓ . Then $|N_\delta(\ell) \cap F| \geq \delta^{100\eta} |N_\delta(\ell) \cap F''| \geq \delta^{1+\varepsilon/2+100\eta} \geq \delta^{1+\varepsilon}$ and similarly for $|N_\delta(\ell) \cap G_1|$ and $|N_\delta(\ell^\perp) \cap G_2|$; thus Conclusion (A) from Theorem 14.2.37' holds.

Next, suppose Conclusion (B) of Proposition 14.2.76 is true, i.e. there exists $\rho \geq \delta$ and an interval I of length at least $\delta^{-100\eta}\rho$ so that (14.2.123) holds for F'', G_1'', G_2'' and $\mathcal{H}''$ with $\varepsilon/2$ in place of ε . Unwinding definitions, we can verify that this implies Conclusion (B) from Theorem 14.2.37'. $\square$

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Chapter 14.3

The Streamlined General-to-Sticky Reduction: Full Dossier and Open Audit Defects

Provenance and reading status

This chapter reproduces, with the notation and theorem numbering absorbed into this book, the detailed proof in Larry Guth, Hong Wang, and Joshua Zahl, *A streamlined proof of the Kakeya set conjecture in $\mathbb{R}^3$* , arXiv:2601.14411v1 (2026). The authors' arXiv source is licensed CC BY 4.0. The stand-alone title page and bibliography are suppressed and the probability appendix is retained. This edition also reflows long displays, corrects evident typographical slips, supplies the slab, plank-algebra, Katz–Tao stopping-time, biased-factoring, random-thinning, rigid-motion, and scale-normalization details that were abbreviated in the source, and states explicitly the exponent separation used by the stopping argument. These additions and corrections are changes by this edition and are not attributed to the original authors.

The chapter is intended to give a self-contained deduction of the full three-dimensional Kakeya theorem from the sticky theorem proved in Chapter 14.2. It replaces the earlier survey route through high-density and four-case sketches with the authors' detailed maximal-density factoring, multiplicity, slab/plank, and bootstrap arguments. Independent-style audits found two load-bearing blockers. The printed maximal-density factorization does not preserve its parent-multiplicity and compatibility conclusions through the final child refinement; see Remark 14.3.22. Separately, the proof of Main Lemma 14.3.12 presently produces an exponent gain that may depend on the arbitrarily pre-

scribed loss ϵ , whereas the statement requires a gain depending only on β ; see Remark 14.3.55. Thus this chapter is in active human review and its deduction of Theorem 14.3.1 remains conditional until both dependencies are closed. This status was checked against the public literature on 31 August 2026. Hua–Yang, arXiv:2608.18870v3, explicitly labels its cellular repair conditional on a four-part source interface and does not supply a stronger terminal exponent. The Guth–Wang–Zahl source remains arXiv:2601.14411v1; no later arXiv version, erratum, or separate fixed-loss correction was found by that date. An arXiv DOI records a preprint and is not evidence of journal publication.

Abstract. We present a streamlined and simplified proof of the Kakeya set conjecture in $\mathbb{R}^3$.

14.3.1 Introduction to the Kakeya problem

A *Besicovitch set* is a compact set $K \subset \mathbb{R}^n$ that contains a unit line segment pointing in every direction. The Kakeya set conjecture asserts that every Besicovitch set in $\mathbb{R}^n$ has Minkowski and Hausdorff dimension n . We refer the reader to [4] and the references therein for an introduction to the Kakeya conjecture and its connections to problems in harmonic analysis.

The Kakeya set conjecture was proved in dimension two by Davies [1], and was recently proved in dimension three by the second and third authors in a sequence of papers [6, 7, 8]. The purpose of this paper is to present a new, streamlined proof of the main result from the third paper [8].

The first paper, [6], proves an important special case of this conjecture called the sticky Kakeya conjecture. The sticky case was formulated by Katz and Tao, who also outlined an approach to proving it, cf. [5]. The paper [6] proves the sticky case of the Kakeya conjecture, approximately following the outline from [5] but dealing with significant technical issues. The second paper [7] obtains a mild generalization of the sticky Kakeya theorem, and uses it to prove that every Besicovitch set in $\mathbb{R}^3$ has Assouad dimension 3; this is a weak form of the Kakeya set conjecture. The third paper, [8], reduces the general case of the Kakeya conjecture in $\mathbb{R}^3$ to the sticky case that was proved in [6, 7]. In this paper, we present a streamlined version of this reduction.

We begin with several definitions. Suppose that $\mathbb{T}$ is a set of tubes in $\mathbb{R}^3$ with radius δ and length 1. We write

$$U(\mathbb{T}) = \bigcup_{T \in \mathbb{T}} T.$$

Our goal is to understand $|U(\mathbb{T})|$, the volume of $U(\mathbb{T})$. We write $|\mathbb{T}|$ for the cardinality of $\mathbb{T}$ and we write $|T|$ for the volume of a tube $T \in \mathbb{T}$. We

will prove that if $\mathbb{T}$ satisfies suitable anti-clustering hypotheses, then $|U(\mathbb{T})|$ is almost as large as $|\mathbb{T}||T|$.

In order to describe these anti-clustering hypotheses, we need some language that describes how much convex sets cluster in other convex sets. Suppose that $\mathbb{W}$ is a finite set of convex sets in $\mathbb{R}^n$ and suppose that K is another convex set. We define

$$\mathbb{W}[K] = \{W \in \mathbb{W} : W \subset K\}, \quad (14.3.1)$$

and

$$\Delta(\mathbb{W}, K) = \frac{\sum_{W \in \mathbb{W}[K]} |W|}{|K|}. \quad (14.3.2)$$

We can think of $\Delta(\mathbb{W}, K)$ as a kind of density that measures how $\mathbb{W}$ concentrates in K . We define a maximal density

$$\Delta_{\max}(\mathbb{W}) = \max_{K \text{ convex}} \Delta(\mathbb{W}, K). \quad (14.3.3)$$

The quantity $\Delta_{\max}(\mathbb{W})$ measures the extent to which the sets from $\mathbb{W}$ cluster into larger convex sets. Quantities of this type were first introduced by Tom Wolff in [9].

If $\mathbb{W}$ is a set of convex sets, we say that Y is a shading of $\mathbb{W}$ if $Y(W) \subset W$ for each $W \in \mathbb{W}$. We define

$$U(\mathbb{W}, Y) = \bigcup_{W \in \mathbb{W}} Y(W). \quad (14.3.4)$$

If the shading Y is clear from context, we sometimes write $U(\mathbb{W})$ for $U(\mathbb{W}, Y)$. We also measure how full the shading Y is:

$$\lambda(\mathbb{W}, Y) = \frac{\sum_{W \in \mathbb{W}} |Y(W)|}{\sum_{W \in \mathbb{W}} |W|}. \quad (14.3.5)$$

With these definitions, we can now state the main result from [8].

Theorem 14.3.1 (Kakeya set conjecture in $\mathbb{R}^3$). *For every $\beta > 0$, there is some $\eta > 0$ so that the following holds for all sufficiently small $\delta > 0$.*

If $\mathbb{T}$ is a set of δ -tubes in $\mathbb{R}^3$ with $\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta}$, and Y is a shading of $\mathbb{T}$ with $\lambda(\mathbb{T}, Y) \geq \delta^\eta$, then

$$|U(\mathbb{T}, Y)| \geq \delta^\beta |\mathbb{T}||T|. \quad (14.3.6)$$

Remark 14.3.2. The original Kakeya conjecture concerns direction-separated tubes. One version says that if $\mathbb{T}$ is a set of $\delta^{-(n-1)}$ δ -tubes in $\mathbb{R}^n$ with δ -separated directions, then $|U(\mathbb{T})| \gtrsim 1$. We verify the required non-concentration estimate in dimension three. Fix a convex set K and put $M = |\mathbb{T}[K]|$. We

may replace K by $K \cap B_1$, since this does not remove any tube counted by M and can only decrease its volume. Let E be an outer John ellipsoid of $K \cap B_1$, with principal semiaxes $a \leq b \leq c$. Thus

$$K \cap B_1 \subset E, \quad |E| \lesssim |K \cap B_1|, \quad c \lesssim 1.$$

If $M > 0$, then E contains a unit δ -tube, and hence $a, b \gtrsim \delta$ and $c \gtrsim 1$. In principal-axis coordinates, the direction ω of the core segment of every tube counted by M obeys

$$|\omega_1| \lesssim a, \quad |\omega_2| \lesssim b.$$

The corresponding subset of S^2 , enlarged by a spherical δ -neighbourhood, has area $\lesssim (a + \delta)(b + \delta) \lesssim ab$. The disjoint spherical $\delta/3$ -caps about the δ -separated tube directions therefore give $M\delta^2 \lesssim ab$. Since $|K| \geq |K \cap B_1| \gtrsim |E| \sim abc \sim ab$ and $|T| \sim \delta^2$, it follows that

$$\Delta(\mathbb{T}, K) = \frac{M|T|}{|K|} \lesssim 1.$$

Taking the supremum over K proves $\Delta_{max}(\mathbb{T}) \lesssim 1$. Theorem 14.3.1 therefore implies the corresponding estimate for direction-separated tubes in $\mathbb{R}^3$. The volume bound (14.3.6) then implies that every Besicovitch set in $\mathbb{R}^3$ has Minkowski and Hausdorff dimension 3.

The goal of this paper is to give a detailed proof of Theorem 14.3.1. A key ingredient is the sticky Kakeya theorem from [6, 7]. We give a self-contained proof that Theorem 14.3.1 follows from the sticky Kakeya theorem. Our proof of Theorem 14.3.1 has the same main ideas as the one in [8], but it has been reorganized to make it more concise. A couple parts of the argument in [8], such as polynomial partitioning, are no longer needed.

The introduction of [8] gives an overview of the main ideas of the proof. The expository paper [2] gives an overview of the Kakeya problem, the sticky case, and the reduction to the sticky case. The expository paper [3] gives a detailed outline of the reduction from the general case to the sticky case. The current paper closely follows that outline.

14.3.2 Notation and Definitions

14.3.2.1 Convex sets and shadings

We will use blackboard bold letters to denote finite families of convex subsets of $\mathbb{R}^n$, and their non-bold equivalents to denote members of these sets, i.e. $\mathbb{V}$ will be a family of convex sets, and V_1, V_2 or V, V' will denote elements of $\mathbb{V}$. Similarly for $\mathbb{W}$ and W , and for $\mathbb{T}$ and T . We will often use the notation

$\sum_{\mathbb{V}} f(V)$ in place of $\sum_{V \in \mathbb{V}} f(V)$, etc. If $\mathbb{V}$ is a family of convex sets, each of which have comparable volume, then we will write $|V|$ to denote the volume of a convex set from $\mathbb{V}$.

If V is a convex set, the “dimensions” of V will be the lengths of the axes of its outer John ellipsoid, listed in increasing order. We will denote this by $v_1 \times v_2 \times \dots \times v_n$. A δ -tube is the δ neighbourhood of a unit line segment in $\mathbb{R}^n$. In $\mathbb{R}^3$, a *slab* is a convex set of dimensions $\theta \times 1 \times 1$, for some $0 < \theta \leq 1$, and a *plank* is a convex set of dimensions $a \times b \times 1$ for some $0 < a \leq b \leq 1$.

A *shading* of a set V is a measurable set $Y(V) \subset V$. We will use $(\mathbb{V}, Y)$ to denote a family of convex sets and their associated shadings $\{Y(V) : V \in \mathbb{V}\}$. For brevity we will sometimes say “let $(\mathbb{V}, Y)$ be a family of convex subsets of $\mathbb{R}^n$,” or “let $(\mathbb{T}, Y)$ be a set of δ -tubes.”

We define the multiplicity of a set of convex sets $\mathbb{W}$ with a shading Y as

$$\mu(\mathbb{W}, Y) = \frac{\sum_{W \in \mathbb{W}} |Y(W)|}{|U(\mathbb{W}, Y)|}. \quad (14.3.7)$$

When the shading Y is clear from the context, we write $\mu(\mathbb{W})$ instead.

Information about $\mu(\mathbb{W})$ is essentially equivalent to information about $|U(\mathbb{W})|$, but it can be more intuitive to think about one than the other. For instance, in Theorem 14.3.1, it would be equivalent to replace the conclusion by $\mu(\mathbb{T}) \lesssim \delta^{-\beta}$.

For $x \in \mathbb{R}^n$, we define $\mathbb{V}_Y(x) = \{V \in \mathbb{V} : x \in Y(V)\}$ and $\mu(\mathbb{V}, Y)(x) = |\mathbb{V}_Y(x)|$. We say that $(\mathbb{V}, Y)$ has constant multiplicity if $\mu(\mathbb{V}, Y)(x)$ is roughly the same for all $x \in U(\mathbb{V}, Y)$. We say that $(\mathbb{V}', Y')$ is a refinement of $(\mathbb{V}, Y)$ if $\mathbb{V}' \subset \mathbb{V}$ and $Y'(V) \subset Y(V)$ for each $V \in \mathbb{V}'$. For $c > 0$, we say that $(\mathbb{V}', Y')$ is a c -refinement of $(\mathbb{V}, Y)$ if $\sum_{V'} |Y'(V)| \geq c \sum_{V} |Y(V)|$. If Y is a shading on $\mathbb{V}$ and $\mathbb{V}'$ is a subset of $\mathbb{V}$, we will often abuse notation and continue to use Y to refer to the restriction of the shading to $\mathbb{V}'$.

14.3.2.2 Pigeonholing and $\lesssim$ notation

Many steps in our proof involve dyadic pigeonholing of the following types:

- (i) If $\mathbb{V}$ is a family of convex subsets of $\mathbb{R}^n$, each of which has a John ellipsoid with axes of lengths between δ and δ^{-1} , then we wish to pigeonhole to find numbers $v_1 \leq \dots \leq v_n$ and a subset $\mathbb{V}'$ consisting of convex sets whose John ellipsoids have axes of lengths roughly $v_1 \times \dots \times v_n$. We can always find a subset with $\sum_{V'} |V'| \gtrsim (\log 1/\delta)^{-n} \sum_{V} |V|$ with this property.
- (ii) If $\mathbb{V}$ is a family of convex subsets of $\mathbb{R}^n$ with a shading Y , then we wish to find a subset $X \subset \mathbb{R}^n$ and a number ν such that $\sum_{V} \chi_{Y(V)}(x) \sim \nu$ on X . We can always find such a set so that $\nu |X| \gtrsim (\log |\mathbb{V}|)^{-1} \sum_{V} |Y(V)|$.

- (iii) If $(\mathbb{V}, Y)$ is a family of convex subsets of $\mathbb{R}^n$ and their associated shading, then we wish to find a subset $\mathbb{V}'$ consisting of convex sets whose shadings $\{Y(V) : V \in \mathbb{V}\}$ have roughly the same measure. If $a = \min\{|Y(V)| : V \in \mathbb{V}, Y(V) \neq \emptyset\}$ and $A = \max_{\mathbb{V}} |Y(V)|$, then we can always find such a set so that $(\mathbb{V}', Y)$ is a $\log(A/a)$ -refinement of $(\mathbb{V}, Y)$.
- (iv) Sometimes we wish to perform the pigeonholing from several of the above items simultaneously. For example if $(\mathbb{V}, Y)$ is a family of convex subsets of $\mathbb{R}^n$, each of which has a John ellipsoid with axes of lengths between δ and δ^{-1} , then there exists a set $\mathbb{V}'$ as in Item ((i)), and a subset $X \subset \mathbb{R}^n$ and a number ν as in Item ((ii)), so that $\nu|X| \gtrsim (\log |\mathbb{V}|)^{-1} (\log 1/\delta)^{-n} \sum_{\mathbb{V}} |V|$.

Quantities of the form $\log(1/\delta)$ or $\log |\mathbb{V}|$ are harmless distractions, and in practice any shading $Y(V)$ with small measure (i.e. $|Y(V)| \leq \delta^n$) can be replaced by $Y(V) = \emptyset$ without affecting our arguments. To streamline our arguments, if X and Y are function of ρ , then we introduce the notation “ $X \lesssim Y$ ” to mean that for all $\varepsilon > 0$, there is a constant $C = C(\varepsilon)$ so that $X \leq C_\varepsilon \rho^{-\varepsilon} Y$. The quantity ρ will often be implicit—here is how $\lesssim$ notation is applied to the items described above.

- (i') For inequalities involving a convex set, or a collection of convex sets $\mathbb{V}$, we will implicitly define ρ to be $\min_{\mathbb{V}} \delta(V)$, where $\delta(V)$ is the ratio of the shortest to the longest axes of the John ellipsoid of V (up to harmless constants, we could also take $\delta(V)$ to be the ratio of the diameter of the largest ball contained inside V to the smallest ball that contains V). Thus for Item ((i)) above, we could write $|\mathbb{V}'| \gtrsim |\mathbb{V}|$.
- (ii') For inequalities involving the cardinality of a set $\mathbb{V}$, we will implicitly define ρ to be $|\mathbb{V}|^{-1}$. Thus for Item ((ii)) above, we could write $|\mathbb{V}'| \gtrsim |\mathbb{V}|$.
- (iii') For inequalities involving a family $(\mathbb{V}, Y)$ of convex sets and their associated shading, we will implicitly define ρ to be a/A , as described in Item ((iii)). Thus for Item ((ii)) above, $(\mathbb{V}', Y)$ is a $\gtrsim 1$ refinement of $(\mathbb{V}, Y)$.
- (iv') Items ((i)'), ((ii)'), and ((iii')) combine as described in Item ((iv)) above.

14.3.2.3 Uniform sets of tubes

It will sometimes be useful to consider sets $\mathbb{T}$ that are uniform in the following sense.

Definition 14.3.3. Let $\delta > 0$, let $M = \lceil \log \log 1/\delta \rceil$, and let $\mathbb{T}$ is a set of δ -tubes in $B_1 \subset \mathbb{R}^n$. We say that $\mathbb{T}$ is *uniform* if for each scale of the form $\rho = \delta^{k/M}$, $k = 0, \dots, M$, there is a set of ρ -tubes $\mathbb{T}_\rho$ so that

- (i) $\mathbb{T} = \bigcup_{\mathbb{T}_\rho} \mathbb{T}[T_\rho]$.
- (ii) The tubes $\mathbb{T}_\rho$ are essentially distinct. Therefore each $T \in \mathbb{T}$ lies in T_ρ for ~ 1 choice of $T_\rho \in \mathbb{T}_\rho$.
- (iii) $|\mathbb{T}[T_\rho]|$ is constant up to a factor ~ 1 as T_ρ varies in $\mathbb{T}_\rho$.

We call the numbers $N_k = |\mathbb{T}[T_{\rho_k}]|$ the *branching numbers* of $\mathbb{T}$.

Definition 14.3.4. Let $\delta > 0$, let $M = \lceil \log \log 1/\delta \rceil$, and let $(\mathbb{T}, Y)$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^n$ and their associated shading. We say that $(\mathbb{T}, Y)$ is *uniform* if the following holds:

- $\mathbb{T}$ is uniform.
- For each point $x \in U(\mathbb{T}, Y)$, the sets $\{T \in \mathbb{T} : x \in Y(T)\}$ are uniform, and have roughly the same branching numbers.

The reason for setting $M = \log \log(1/\delta)$ in our above definitions is as follows. First, since $(\log |\mathbb{T}|)^{\log \log(1/\delta)} \lesssim 1$, by pigeonholing, every set $\mathbb{T}$ of δ -tubes in $B_1 \subset \mathbb{R}^n$ has a subset $\mathbb{T}'$ of cardinality $|\mathbb{T}'| \gtrsim |\mathbb{T}|$ that is uniform, and every pair $(\mathbb{T}, Y)$ has a ≈ 1 -refinement that is uniform.

Second, if $\mathbb{T}$ is uniform, then for *every* scale $\rho \in [\delta, 1]$, there is a set of essentially distinct ρ -tubes $\mathbb{T}_\rho$ that satisfies items (i) and (ii) from Definition 14.3.3, and satisfies the following slightly weaker analogue of (iii): there is a number N_ρ so that $|\mathbb{T}[T_\rho]| \approx N_\rho$ for all $T_\rho \in \mathbb{T}_\rho$. Similarly for Definition 14.3.4; the second bullet point is weakened to: the branching numbers N_k are roughly the same up to a ≈ 1 factor.

14.3.3 Partial Keakeya estimates

In this section, we consider Keakeya-type estimates for sets of tubes obeying two important non-clustering conditions. These non-clustering conditions are called Katz-Tao conditions and Frostman conditions. They are both descended by the non-concentration conditions introduced by Wolff in [9].

Definition 14.3.5. A family of convex sets $\mathbb{V}$ is called C -convex-Katz-Tao (or C -Katz-Tao for short) if $\Delta_{\max}(\mathbb{V}) \leq C$. Informally, we say $\mathbb{V}$ is Katz-Tao if $\Delta_{\max}(\mathbb{V}) \lesssim 1$.

Definition 14.3.6. If $\mathbb{V}$ is a set of convex sets in K , we define

$$C_F(\mathbb{V}, K) = \frac{\sup_{K' \subset K} \Delta(\mathbb{V}, K')}{\Delta(\mathbb{V}, K)}, \quad (14.3.8)$$

where the supremum is taken over all convex subsets of K . We say that $\mathbb{V}$ is C -convex-Frostman (or C -Frostman for short) in K if $C_F(\mathbb{V}, K) \leq C$. Informally, we say that $\mathbb{V}$ is Frostman in K if $C_F(\mathbb{V}, K) \lesssim 1$. If K is the unit ball, then we abbreviate this as $C_F(\mathbb{V})$, and we say that $\mathbb{V}$ is C -Frostman.

Remark 14.3.7.

(A) The property of being Frostman is *inherited upwards*: Suppose $\mathbb{V} = \sqcup_{\mathbb{W}} \mathbb{V}_W$ is C -Frostman inside K , where each $W \in \mathbb{W}$ is contained in K and $V \subset W$ for each $V \in \mathbb{V}_W$. Then there is a subset $\mathbb{W}' \subset \mathbb{W}$ that is $\lesssim C$ -Frostman in K . If in addition the quantity $\sum_{V \in \mathbb{V}_W} |V|/|W|$ is approximately the same for each $W \in \mathbb{W}$, then we may take $\mathbb{W}' = \mathbb{W}$, and in fact $\mathbb{W}$ is $O(C)$ -Frostman in K .

(B) The property of being Katz-Tao is *inherited downwards*: If $\mathbb{V}$ is C -Katz-Tao, then every subset $\mathbb{V}' \subset \mathbb{V}$ is also C -Katz-Tao.

Proof. We prove (A). Recall that $\Delta(\mathbb{V}_W, W) = \frac{\sum_{V \in \mathbb{V}_W} |V|}{|W|}$. Define

$$\mathbb{W}_D = \{W \in \mathbb{W} : \Delta(\mathbb{V}_W, W) \sim D\}.$$

There are $\lesssim 1$ values of D . Choose D so that $\sum_{W \in \mathbb{W}_D} \sum_{V \in \mathbb{V}_W} |V| \approx \sum_{V \in \mathbb{V}} |V|$.

Now consider a convex set $U \subset K$. Since $\mathbb{V}$ is C -Frostman in K , we have

$$\frac{\sum_{V \in \mathbb{V}[U]} |V|}{|U|} \leq C \frac{\sum_{V \in \mathbb{V}[K]} |V|}{|K|}. \quad (14.3.9)$$

Now

$$\sum_{V \in \mathbb{V}[U]} |V| \geq \sum_{W \in \mathbb{W}_D[U]} \sum_{V \in \mathbb{V}_W} |V| \sim \sum_{W \in \mathbb{W}_D[U]} D|W|.$$

Plugging this equation into the last, we see that

$$\frac{\sum_{W \in \mathbb{W}_D[U]} D|W|}{|U|} \lesssim C \frac{\sum_{V \in \mathbb{V}[K]} |V|}{|K|}.$$

On the other hand, since $\mathbb{V} = \mathbb{V}[K]$, $\sum_{V \in \mathbb{V}} |V| \approx \sum_{W \in \mathbb{W}_D} \sum_{V \in \mathbb{V}_W} |V| \sim \sum_{W \in \mathbb{W}_D} D|W|$. Plugging this into the right-hand of the last equation and cancelling a factor of D leaves

$$\frac{\sum_{W \in \mathbb{W}_D[U]} |W|}{|U|} \lesssim C \frac{\sum_{W \in \mathbb{W}_D[K]} |W|}{|K|}.$$

In other words, $C_F(\mathbb{W}_D, K) \lesssim C = C_F(\mathbb{V}, K)$. In the special case that $\mathbb{W} = \mathbb{W}_D$, the argument above shows that $C_F(\mathbb{W}, K) \lesssim C = C_F(\mathbb{V}, K)$.

To prove (B), just note that

$$\Delta_{max}(\mathbb{V}') = \max_K \frac{\sum_{V \in \mathbb{V}'[K]} |V|}{|K|} \leq \max_K \frac{\sum_{V \in \mathbb{V}[K]} |V|}{|K|} = \Delta_{max}(\mathbb{V}). \quad \square$$

Definition 14.3.8. We write $K_{KT}(\beta)$ for the following statement. For any $\epsilon > 0$, there exists $\eta = \eta(\epsilon, \beta) > 0$ so that the following holds for any $0 < \delta \leq \delta(\epsilon, \beta)$. Let $\mathbb{T}$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^3$ with the following properties.

- $\Delta_{max}(\mathbb{T}) \leq \delta^{-\eta}$.
- Y is a shading of $\mathbb{T}$ with $\lambda(\mathbb{T}, Y) \geq \delta^\eta$.

Then

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} |\mathbb{T}|^\beta. \quad (14.3.10)$$

Note that $K_{KT}(1)$ is trivial. No preliminary improvement below exponent 1 is used in this proof: the descent begins with $K_{KT}(1)$. Our goal is to prove $K_{KT}(0)$. One important case is when $|\mathbb{T}| \sim \delta^{-2}$. In this case, $K_{KT}(\beta)$ gives $\mu(\mathbb{T}) \lesssim \delta^{-2\beta-\epsilon}$, which is equivalent to $|U(\mathbb{T})| \gtrsim \delta^{2\beta+\epsilon}$.

Definition 14.3.9. We write $K_F(\beta)$ for the following statement. For any $\epsilon > 0$, there exists $\eta = \eta(\epsilon, \beta) > 0$ so that the following holds for any $0 < \delta \leq \delta(\epsilon, \beta)$. Let $\mathbb{T}$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^3$ with the following properties.

- $C_F(\mathbb{T}, B_1) \leq \delta^{-\eta}$.
- Y is a shading of $\mathbb{T}$ with $\lambda(Y) \geq \delta^\eta$.

Then

$$\mu(\mathbb{T}) \leq \delta^{-\epsilon} \delta^{-2\beta} (|\mathbb{T}| |T|)^{1-\beta/2}. \quad (14.3.11)$$

Or equivalently,

$$|U(\mathbb{T})| \geq \delta^\epsilon \delta^{2\beta} (|\mathbb{T}| |T|)^{\beta/2}.$$

Let us parse the algebra in this bound. Since $C_F(\mathbb{T}, B_1) \leq \delta^{-\eta}$, we have $|\mathbb{T}| \geq \delta^{-2+\eta}$. One important case is when $|\mathbb{T}| \sim \delta^{-2}$. In this case, the bound $C_F(\mathbb{T}, B_1) \lesssim \delta^{-\eta}$ is equivalent to the bound $\Delta_{max}(\mathbb{T}) \lesssim \delta^{-\eta}$. So $\mathbb{T}$ is both Frostman and Katz-Tao. In this case, $K_{KT}(\beta)$ gives $|U(\mathbb{T})| \gtrsim \delta^{2\beta+\epsilon}$, and $K_F(\beta)$ gives the same bound $|U(\mathbb{T})| \gtrsim \delta^{2\beta+\epsilon}$. When $|\mathbb{T}| \gg \delta^{-2}$, then $K_F(\beta)$ gives a stronger bound, because the last factor $(|\mathbb{T}| |T|)^{\beta/2}$ is $\gg 1$. The exact exponent $\beta/2$ in this final factor is not crucial in our analysis, and the argument would work equally well with any positive exponent $c\beta$, $0 < c < 1$. There is one step (see Remark 14.3.53) where it is important that $c < 1$.

Remark 14.3.10. If $K_{KT}(\beta)$ holds, then we get for free the following slightly more general version of $K_{KT}(\beta)$: For any $\epsilon > 0$, there exists $\eta = \eta(\epsilon, \beta) > 0$ so that the following holds for any $0 < \delta \leq \delta(\epsilon, \beta)$. Suppose $\mathbb{T}$ is a set of δ -tubes in $B_1 \subset \mathbb{R}^3$, $\tau \leq \delta$, and

- $\Delta_{max}(\mathbb{T}) \leq \tau^{-\eta}$.
- Y is a shading of $\mathbb{T}$ with $\lambda(\mathbb{T}, Y) \geq \tau^\eta$.

Then

$$\mu(\mathbb{T}, Y) \leq \tau^{-\epsilon} |\mathbb{T}|^\beta.$$

Similarly, $K_F(\beta)$ implies the following version: under the hypotheses $C_F(\mathbb{T}) \leq \tau^{-\eta}$ and $\lambda(\mathbb{T}, Y) \geq \tau^\eta$,

$$\mu(\mathbb{T}, Y) \leq \tau^{-\epsilon} \delta^{-2\beta} (|\mathbb{T}| |T|)^{1-\beta/2}.$$

These slightly more general versions will be useful when we replace a set of δ -tubes $(\mathbb{T}, Y)$ with a thickening $(\mathbb{T}_\rho, Y_\rho)$, for which we only know that $\lambda(\mathbb{T}_\rho, Y_\rho) \geq \delta^\eta$.

Proof. We give both arguments. Fix $\beta, \epsilon > 0$, and let η_1 be the density exponent supplied by the relevant definition, with loss ϵ . There is a constant $Q = Q(\beta)$ such that each desired estimate is trivial with δ^{-Q} in place of the loss factor. For K_{KT} this follows from

$$\mu(\mathbb{T}, Y) \leq |\mathbb{T}| \lesssim \delta^{-4},$$

using the $O(\delta^{-4})$ bound for an essentially distinct family of δ -tubes in B_1 . Indeed, after discretization a tube is determined by two δ -spaced direction parameters and two δ -spaced transverse-position parameters. For K_F , write

$$A_F = \delta^{-2\beta} (|\mathbb{T}| |T|)^{1-\beta/2}.$$

Since $|T| \sim \delta^2$ and $|\mathbb{T}| \lesssim \delta^{-4}$,

$$\frac{|\mathbb{T}|}{A_F} \lesssim \delta^{3\beta-2} |\mathbb{T}|^{\beta/2} \lesssim \delta^{\beta-2} \leq \delta^{-2};$$

hence $\mu(\mathbb{T}, Y) \lesssim \delta^{-2} A_F$. Increasing Q absorbs all absolute constants in both cases.

Set $\eta = \epsilon \eta_1 / Q$. If $\tau \leq \delta^{Q/\epsilon}$, then $\tau^{-\epsilon} \geq \delta^{-Q}$ and the appropriate conclusion is trivial by the preceding paragraph. Otherwise,

$$\tau^{-\eta} \leq \delta^{-\eta_1}, \quad \tau^\eta \geq \delta^{\eta_1}.$$

For the Katz–Tao hypotheses, Definition 14.3.8 therefore gives

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} |\mathbb{T}|^\beta \leq \tau^{-\epsilon} |\mathbb{T}|^\beta,$$

because $\tau \leq \delta$. For the Frostman hypotheses, Definition 14.3.9 gives in exactly the same way

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} A_F \leq \tau^{-\epsilon} A_F.$$

□

The two main lemmas in the proof are

Main Lemma 14.3.11. *$K_{KT}(\beta)$ implies $K_F(\beta)$.*

Main Lemma 14.3.12. *If $\beta > 0$ and $K_{KT}(\beta)$ and $K_F(\beta)$ hold, then $K_{KT}(\beta - \nu)$ holds for $\nu = \nu(\beta) > 0$. Moreover $\nu(\beta)$ is nondecreasing in β .*

We record the exponent descent and the final conversion to volume, since both steps are needed to keep the quantifiers in Theorem 14.3.1 explicit. First, $K_{KT}(1)$ follows from the pointwise bound $\mu(\mathbb{T}, Y) \leq |\mathbb{T}|$. Assuming the two main lemmas as stated, suppose that $K_{KT}(b)$ is known. Main Lemma 14.3.11 gives $K_F(b)$ and Main Lemma 14.3.12 then gives $K_{KT}(b - \nu(b))$. If $b_* > 0$ is fixed and the descent has not yet reached b_* , monotonicity gives $\nu(b) \geq \nu(b_*) > 0$. Hence after finitely many steps it reaches an exponent $b \leq b_*$. Since $|\mathbb{T}| \geq 1$, $K_{KT}(b)$ implies $K_{KT}(b_*)$. Thus $K_{KT}(b)$, and then $K_F(b)$, hold for every $b > 0$.

Now fix the exponent $\beta_0 > 0$ in Theorem 14.3.1. Apply $K_{KT}(\beta_0/16)$ with loss $\epsilon = \beta_0/4$, and decrease its input exponent η so that $\eta \leq \beta_0/4$. An essentially distinct family of δ -tubes in $B_1 \subset \mathbb{R}^3$ has cardinality $|\mathbb{T}| \lesssim \delta^{-4}$. Therefore

$$\begin{aligned} |U(\mathbb{T}, Y)| &= \frac{\sum_{T \in \mathbb{T}} |Y(T)|}{\mu(\mathbb{T}, Y)} \\ &\gtrsim \delta^{\eta + \beta_0/4} |\mathbb{T}|^{1 - \beta_0/16} |T| \\ &\gtrsim \delta^{\eta + \beta_0/2} |\mathbb{T}| |T| \geq \delta^{\beta_0} |\mathbb{T}| |T|. \end{aligned}$$

This is exactly (14.3.6). Consequently the two main lemmas, with the parameter dependence stated in them, imply Theorem 14.3.1.

14.3.3.1 Multiplicity bounds for larger values of $\Delta_{\max}(\mathbb{T})$ or $C_F(\mathbb{T})$

The next two results say that that $K_{KT}(\beta)$ and $K_F(\beta)$ imply multiplicity bounds for sets of tubes with any value of $\Delta_{\max}(\mathbb{T})$ or $C_F(\mathbb{T})$.

Lemma 14.3.13. *Suppose that $K_{KT}(\beta)$ holds. Then for any $\epsilon > 0$, there exists $\eta = \eta(\epsilon, \beta) > 0$ such that the following holds for any $0 < \delta < \delta(\epsilon, \beta)$. Suppose that $(\mathbb{T}, Y)$ is a set of δ -tubes in B_1 with a shading with $\lambda(\mathbb{T}) \geq \delta^\eta$. Then*

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} \Delta_{\max}(\mathbb{T})^{1-\beta} |\mathbb{T}|^\beta.$$

Proof. Write $v = |T| \sim \delta^2$. We first record the sampling fact used below.

Sampling claim. Let $\mathcal{S}$ be a family of N δ -tubes in B_1 , set $D = \Delta_{\max}(\mathcal{S})$, and let

$$s = \left\lceil \frac{N}{D} \right\rceil.$$

Then there is a subset $\mathcal{S}' \subset \mathcal{S}$ of cardinality s such that $\Delta_{\max}(\mathcal{S}') \lesssim \log(2/\delta)$.

Indeed, $1 \leq D \leq N$: the lower bound follows by testing a convex set equal to one tube, and the upper bound follows because every convex set which contains a tube has volume at least v . Choose $\mathcal{S}'$ uniformly among the s -element subsets of $\mathcal{S}$. We reduce the continuum of convex tests to finitely many ellipsoid tests as follows. If a convex set K contains a tube of $\mathcal{S}'$, replace it by $K \cap B_1$ and take an outer John ellipsoid E . Thus

$$K \cap B_1 \subset E, \quad |E| \lesssim |K \cap B_1|.$$

The semiaxes of E lie between $c\delta$ and C . Discretize its centre at mesh $c_1\delta$, its orthogonal frame at mesh $c_1\delta$, and each semiaxis on a fixed-ratio geometric grid. Enlarging the selected net ellipsoid by a fixed factor if necessary gives a family $\mathcal{E}_\delta$ with

$$|\mathcal{E}_\delta| \leq \delta^{-C_0}$$

such that every such E is contained in some $\hat{E} \in \mathcal{E}_\delta$ with $|\hat{E}| \lesssim |E|$. (The centre has three parameters, the frame three, and the semiaxes three, so the displayed polynomial bound follows directly from the three discretizations.)

Fix $\hat{E} \in \mathcal{E}_\delta$, put $M = |\mathcal{S}[\hat{E}]|$, and let $X = |\mathcal{S}'[\hat{E}]|$. By the definition of D ,

$$M \leq D \frac{|\hat{E}|}{v}, \quad \mathbb{E}X = \frac{sM}{N} \leq 2 \frac{M}{D} \leq 2 \frac{|\hat{E}|}{v}.$$

Here $s \leq 2N/D$ because $N/D \geq 1$. We also give the required tail estimate explicitly. For every integer $k \geq 1$, a union bound over the k -subsets of $\mathcal{S}[\hat{E}]$ gives

$$\mathbb{P}(X \geq k) \leq \binom{M}{k} \frac{(s)_k}{(N)_k} \leq \left(\frac{eMs}{kN} \right)^k.$$

Consequently, if $L = A \log(2/\delta)$ and

$$k = \left\lceil L \max \left\{ 1, \frac{|\hat{E}|}{v} \right\} \right\rceil,$$

then, for a sufficiently large absolute A , the preceding probability is at most δ^{C_0+2} . A union bound over $\mathcal{E}_\delta$ shows that with positive probability

$$|\mathcal{S}'[\widehat{E}]| \lesssim L \max \left\{ 1, \frac{|\widehat{E}|}{v} \right\} \quad \text{for every } \widehat{E} \in \mathcal{E}_\delta.$$

If the left side is nonzero, then $|\widehat{E}| \gtrsim v$, so the maximum on the right may be replaced by $|\widehat{E}|/v$. Given any convex K with $\mathcal{S}'[K] \neq \emptyset$, choose E and $\widehat{E}$ as above. Then

$$\frac{|\mathcal{S}'[K]|v}{|K|} \leq \frac{|\mathcal{S}'[\widehat{E}]|v}{|K \cap B_1|} \lesssim L \frac{|\widehat{E}|}{|K \cap B_1|} \lesssim L.$$

This proves the sampling claim.

We apply the claim after making the shading uniform. Discard the tubes for which $|Y(T)| < \frac{1}{2}\delta^\eta v$; they contribute less than half of the total shaded mass. Divide the remaining tubes into dyadic classes according to $|Y(T)|$. There are $O(\log(2/\delta))$ relevant classes, so one class $\mathbb{T}_0$ satisfies

$$\sum_{T \in \mathbb{T}_0} |Y(T)| \gtrsim \sum_{T \in \mathbb{T}} |Y(T)|, \quad |Y(T)| \sim \lambda_0 v \quad (T \in \mathbb{T}_0), \quad \lambda_0 \geq \frac{1}{2}\delta^\eta.$$

Put $N_0 = |\mathbb{T}_0|$ and $D_0 = \Delta_{\max}(\mathbb{T}_0)$. The sampling claim gives $\mathbb{T}' \subset \mathbb{T}_0$ with

$$|\mathbb{T}'| \sim N_0/D_0, \quad \Delta_{\max}(\mathbb{T}') \lesssim \log(2/\delta), \quad \lambda(\mathbb{T}', Y) \sim \lambda_0.$$

The uniformity of the shading also gives

$$\sum_{T \in \mathbb{T}_0} |Y(T)| \lesssim D_0 \sum_{T \in \mathbb{T}'} |Y(T)|.$$

Since $U(\mathbb{T}', Y) \subset U(\mathbb{T}_0, Y) \subset U(\mathbb{T}, Y)$, the mass discarded in the two pigeonholing steps costs only a factor $O(\log(2/\delta))$, and hence

$$\mu(\mathbb{T}, Y) \lesssim D_0 \mu(\mathbb{T}', Y). \quad (14.3.12)$$

For $\beta \geq 1$ the assertion is immediate from $\mu(\mathbb{T}, Y) \leq |\mathbb{T}|$ and $1 \leq \Delta_{\max}(\mathbb{T}) \leq |\mathbb{T}|$. We may therefore suppose $0 < \beta < 1$. Apply the definition of $K_{KT}(\beta)$ with loss $\epsilon/2$, and denote its density exponent by η_0 . In the statement of the present lemma choose $0 < \eta < \eta_0$ sufficiently small. For sufficiently small δ , the preceding construction has

$$\Delta_{\max}(\mathbb{T}') \leq \delta^{-\eta_0}, \quad \lambda(\mathbb{T}', Y) \geq \delta^{\eta_0}.$$

Thus $K_{KT}(\beta)$ and (14.3.12) yield

$$\begin{aligned} \mu(\mathbb{T}, Y) &\lesssim D_0 \delta^{-\epsilon/2} |\mathbb{T}'|^\beta \\ &\lesssim \delta^{-\epsilon/2} D_0^{1-\beta} N_0^\beta \\ &\leq \delta^{-\epsilon/2} \Delta_{\max}(\mathbb{T})^{1-\beta} |\mathbb{T}|^\beta. \end{aligned}$$

Finally absorb the subpower factors into $\delta^{-\epsilon/2}$ by reducing the upper threshold for δ . This proves the stated estimate. $\square$

Lemma 14.3.14. *Suppose that $\mathbb{T}$ is a set of (essentially distinct) δ -tubes in $B_1 \subset \mathbb{R}^n$. Let $R_1, \dots, R_J$ be random rigid motions of size 1, with $J = \lceil C_F(\mathbb{T}) \rceil$, and let $\mathbb{T}' = \bigcup_{j=1}^J R_j(\mathbb{T})$. Then with high probability the tubes of $\mathbb{T}'$ are essentially distinct up to multiplicity $\lesssim 1$ and also $C_F(\mathbb{T}') \lesssim 1$. After refining $\mathbb{T}'$ to make the tubes essentially distinct, we have $|\mathbb{T}'| \approx C_F(\mathbb{T})|\mathbb{T}|$ and $C_F(\mathbb{T}') \lesssim 1$.*

We will prove Lemma 14.3.14 in Appendix A, where we collect a couple of related probability lemmas.

Lemma 14.3.15. *Suppose that $0 < \beta \leq 1$ and that $K_F(\beta)$ holds. Then for any $\epsilon > 0$, there exists $\eta = \eta(\epsilon, \beta) > 0$ such that the following holds for any $0 < \delta < \delta(\epsilon, \beta) > 0$. Suppose that $(\mathbb{T}, Y)$ is a set of δ -tubes in B_1 with a shading with $\lambda(\mathbb{T}) \geq \delta^\eta$.*

Then

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} C_F(\mathbb{T})^{1-\beta/2} \delta^{-2\beta} (\delta^2 |\mathbb{T}|)^{1-\beta/2}.$$

Proof. Put $S = \sum_{\mathbb{T}} |Y(T)|$. As in the proof of Lemma 14.3.13, discard tubes carrying less than $\frac{1}{2} \delta^\eta |T|$ shaded mass and then select one dyadic class $\mathbb{T}_0$ such that

$$S_0 := \sum_{\mathbb{T}_0} |Y(T)| \gtrsim S, \quad |Y(T)| \sim \lambda_0 |T| \quad (T \in \mathbb{T}_0), \quad \lambda_0 \geq \frac{1}{2} \delta^\eta.$$

Set $D_0 = C_F(\mathbb{T}_0)$ and $J = \lceil D_0 \rceil$. Apply Lemma 14.3.14 to J independent rigid-motion copies of $\mathbb{T}_0$, carrying the transported shadings

$$Y_j(R_j T) := R_j(Y(T)).$$

Before refinement, the total shaded mass is JS_0 , whereas the union of the transported shadings has volume at most $J|U(\mathbb{T}_0, Y)|$. Its multiplicity is therefore at least $\mu(\mathbb{T}_0, Y)$. The essentially-distinct refinement in Lemma 14.3.14 has overlap multiplicity $\lesssim 1$. In every equivalence class retain a tube whose transported shading has maximal measure. This retains a $\gtrsim 1$ fraction of the total shaded mass. Denote the resulting family and shading by $(\mathbb{T}', Y')$. We then have

$$\begin{aligned} \mu(\mathbb{T}, Y) &\lesssim \mu(\mathbb{T}_0, Y) \lesssim \mu(\mathbb{T}', Y'), & |\mathbb{T}'| &\approx D_0 |\mathbb{T}_0|, \\ C_F(\mathbb{T}') &\lesssim 1, & \lambda(\mathbb{T}', Y') &\gtrsim \lambda_0. \end{aligned}$$

The first comparison also used $S_0 \gtrsim S$ and $U(\mathbb{T}_0, Y) \subset U(\mathbb{T}, Y)$. A fixed translation and dilation place $\mathbb{T}'$ in B_1 and change all displayed quantities only by absolute constants.

Let η_0 be the density exponent in $K_F(\beta)$ with loss $\epsilon/2$, and choose the input $\eta < \eta_0$ sufficiently small. After absorbing the subpower refinement factors, $K_F(\beta)$ applies to $(\mathbb{T}', Y')$ and gives

$$\mu(\mathbb{T}, Y) \lesssim \delta^{-\epsilon/2} \delta^{-2\beta} (\delta^2 D_0 |\mathbb{T}_0|)^{1-\beta/2}.$$

Finally,

$$D_0 |\mathbb{T}_0| = \frac{\Delta_{\max}(\mathbb{T}_0)}{\Delta(\mathbb{T}_0, B_1)} |\mathbb{T}_0| \lesssim \frac{\Delta_{\max}(\mathbb{T})}{|T|} \lesssim C_F(\mathbb{T}) |\mathbb{T}|.$$

Since $1 - \beta/2 > 0$, this proves

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} C_F(\mathbb{T})^{1-\beta/2} \delta^{-2\beta} (\delta^2 |\mathbb{T}|)^{1-\beta/2},$$

after decreasing the allowed upper threshold for δ . $\square$

For completeness, we prove the scale-parameter analogue of Remark 14.3.10 used below. In the range $0 < \beta \leq 1$, if $\tau \leq \delta$ and $\lambda(\mathbb{T}, Y) \geq \tau^\eta$, then the conclusions of Lemmas 14.3.13 and 14.3.15 respectively hold with $\tau^{-\epsilon}$ in place of $\delta^{-\epsilon}$. Indeed, let η_0 be the density exponent supplied by the relevant generalized lemma with loss ϵ . Write

$$B_{KT} = \Delta_{\max}(\mathbb{T})^{1-\beta} |\mathbb{T}|^\beta$$

and

$$B_F = C_F(\mathbb{T})^{1-\beta/2} \delta^{-2\beta} (\delta^2 |\mathbb{T}|)^{1-\beta/2}.$$

Since $1 \leq \Delta_{\max}(\mathbb{T}) \leq |\mathbb{T}|$,

$$\mu(\mathbb{T}, Y) \leq |\mathbb{T}| \leq \delta^{-4} B_{KT}.$$

Also $C_F(\mathbb{T}) \geq 1$, and the calculation in the proof of Remark 14.3.10 gives $\mu(\mathbb{T}, Y) \lesssim \delta^{-2} B_F$. Choose a fixed Q large enough to absorb these two trivial losses and all absolute constants, and put $\eta = \epsilon \eta_0 / Q$. If $\tau \leq \delta^{Q/\epsilon}$, then $\tau^{-\epsilon} \geq \delta^{-Q}$, so both desired bounds are trivial. Otherwise $\tau^\eta \geq \delta^{\eta_0}$; applying the appropriate generalized lemma gives

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} B_{KT} \leq \tau^{-\epsilon} B_{KT}$$

or

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} B_F \leq \tau^{-\epsilon} B_F,$$

as asserted.

14.3.4 Organizing convex sets

If $\mathbb{V}$ is a family of convex sets with $\Delta_{\max}(\mathbb{V}) \gg 1$, it turns out to be useful to consider the convex sets W so that $\Delta(\mathbb{V}, W)$ is near maximal. These convex sets W can be used to organize $\mathbb{V}$.

Lemma 14.3.16. (*Maximal density factoring lemma*) *Let $\mathbb{V}$ be a finite family of convex subsets of $\mathbb{R}^n$. Then there is a set $\mathbb{V}' \subset \mathbb{V}$ with $\sum_{\mathbb{V}'} |V| \gtrsim \sum_{\mathbb{V}} |V|$; a family $\mathbb{W}$ of convex in $\mathbb{R}^n$, each of approximately the same dimensions; and a partition*

$$\mathbb{V}' = \bigsqcup_{W \in \mathbb{W}} \mathbb{V}_W, \quad (14.3.13)$$

where each set in $\mathbb{V}_W$ is contained in W . This partition has the following properties.

- (i) $\mathbb{W}$ is Katz-Tao, i.e. $\Delta_{\max}(\mathbb{W}) \sim 1$.
- (ii) $\mathbb{V}_W$ is Frostman in W , i.e. $|\mathbb{V}_W| \frac{|V|}{|W|} \sim \Delta(\mathbb{V}_W, W)$ for each $W \in \mathbb{W}$. Furthermore $\Delta(\mathbb{V}_W, W) \sim \Delta_{\max}(\mathbb{V}')$.

Proof. After replacing $\mathbb{V}$ by a subset for which $\sum_{\mathbb{V}} |V|$ is preserved up to a ≈ 1 factor, we may suppose that each convex set in $\mathbb{V}$ has approximately the same dimensions. We may also suppose that each set in $\mathbb{V}$ is compact (if not, then we replace each set by its closure, and then undo this step at the end of the proof). We choose sets W_j one at a time by a greedy algorithm. Choose W_0 to maximize $\Delta(\mathbb{V}, W_0)$ (a maximizer exists, since $\mathbb{V}$ is finite and the sets in $\mathbb{V}$ are compact). Then set $\mathbb{V}_1 = \mathbb{V} \setminus \mathbb{V}[W_0]$. Now choose W_1 to maximize $\Delta(\mathbb{V}_1, W_1)$. Continue in this way until $\mathbb{V}_s$ is empty. Then stop. For each j , we set $\mathbb{V}_{W_j} = \mathbb{V}_j[W_j]$; the sets $\mathbb{V}_{W_j}$ are disjoint.

Next we pigeonhole $\Delta(\mathbb{V}_j, W_j)$. For each $1 \leq \log \nu \leq \log(|\mathbb{V}|/|V|)$, we set $J(\nu) = \{j : \Delta(\mathbb{V}_j, W_j) \sim \nu\}$. We set $\mathbb{V}_\nu = \bigcup_{j \in J(\nu)} \mathbb{V}_{W_j}$. We note that $\mathbb{V} = \bigsqcup_{\nu} \mathbb{V}_\nu$. We choose ν so that $|\mathbb{V}_\nu| \gtrsim |\mathbb{V}|$. Choose a subset $\mathbb{W}$ of $\{W_j\}_{j \in J(\nu)}$ of approximately the same dimensions, so that $\mathbb{V}' = \bigsqcup_{W \in \mathbb{W}} \mathbb{V}_W$ satisfies $|\mathbb{V}'| \gtrsim |\mathbb{V}_\nu|$, and replace $J(\nu)$ with the set of indices j such that $W_j \in \mathbb{W}$. Observe that we have the desired bound on the size of $\mathbb{V}'$.

We have $\sum_{\mathbb{V}_{W_j}} |V| \sim \nu |W_j|$ for each $j \in J(\nu)$. Since $V \in \mathbb{V}$ and each $W \in \mathbb{W}$ has approximately the same dimensions, $|\mathbb{V}_{W_j}|$ is comparable for each $j \in J(\nu)$.

Note that $\Delta(\mathbb{V}_j, W_j) = \Delta_{\max}(\mathbb{V}_j)$ is non-increasing. Set j_1 to be the first number in $J(\nu)$. Since $\mathbb{V}' \subset \mathbb{V}_{j_1}$ and $\mathbb{V}_{W_j} \subset \mathbb{V}'$, we have

$$\Delta_{\max}(\mathbb{V}') \leq \Delta_{\max}(\mathbb{V}_{j_1}) = \Delta(\mathbb{V}_{W_{j_1}}, W_{j_1}) \leq \Delta_{\max}(\mathbb{V}'). \quad (14.3.14)$$

Therefore, all the inequalities above are equalities, and thus Item ((ii)) is satisfied for W_{j_1} . But since each set $\mathbb{V}_{W_j}$ has approximately the same cardinality, and each set W_j has approximately the same volume, we have that $\Delta(\mathbb{V}_W, W)$ is approximately the same for all $W \in \mathbb{W}$, and thus Item ((ii)) is satisfied for all $W \in \mathbb{W}$.

It remains to verify Item ((i)). Let $U \subset \mathbb{R}^n$ be a convex set. Then

$$|\mathbb{V}'[U]| \geq \sum_{W \in \mathbb{W}[U]} |\mathbb{V}_W| \gtrsim \sum_{W \in \mathbb{W}[U]} \Delta(\mathbb{V}_W, W) \frac{|W|}{|V|} \gtrsim |\mathbb{W}[U]| \Delta_{\max}(\mathbb{V}') \frac{|W|}{|V|}.$$

Re-arranging, we obtain

$$|\mathbb{W}[U]| \lesssim \frac{|\mathbb{V}'[U]| |V|}{\Delta_{\max}(\mathbb{V}') |W|} \leq \frac{\Delta_{\max}(\mathbb{V}') |U|}{\Delta_{\max}(\mathbb{V}') |W|} = \frac{|U|}{|W|}. \quad \square$$

Definition 14.3.17. Let $\mathbb{V}$ and $\mathbb{W}$ be families of convex subsets of $\mathbb{R}^n$, and suppose there is a partition $\mathbb{V} = \bigcup_{W \in \mathbb{W}} \mathbb{V}_W$. We say that $\mathbb{W}$ *factors* $\mathbb{V}$ if the following is true:

- $\mathbb{W}$ is Katz-Tao, i.e. $\Delta_{\max}(\mathbb{W}) \sim 1$.
- $\mathbb{V}_W$ is Frostman in W , i.e. $|\mathbb{V}_W| \frac{|V|}{|W|} \sim \Delta(\mathbb{V}_W, W)$ for each $W \in \mathbb{W}$. Furthermore $\Delta(\mathbb{V}_W, W) \sim \Delta_{\max}(\mathbb{V})$.
- Each $W \in \mathbb{W}$ has approximately the same dimensions, and each set in $\mathbb{V}_W$ is contained in W .

In the above definition, we choose the implicit constants to match those coming from Lemma 14.3.16. Thus Lemma 14.3.16 says that $\mathbb{W}$ factors $\mathbb{V}'$. We call the set $\mathbb{W}$ coming from Lemma 14.3.16 the maximal density factoring of $\mathbb{V}$, and we call Lemma 14.3.16 the maximal density factoring lemma.

Let us now discuss what we gain from the maximal density factoring lemma. We began with an arbitrary set of convex sets $\mathbb{V}$. The set $\mathbb{V}$ does not obey any “non-clustering condition” such as $\Delta_{\max}(\mathbb{V}) \lesssim 1$. Using the factoring lemma, we can often reduce the problem of understanding $\mathbb{V}$ to the problem of understanding the Frostman sets $\mathbb{V}[W]$ for $W \in \mathbb{W}$, and the problem of understanding the Katz-Tao set $\mathbb{W}$.

Our goal is to prove a bound on $\mu(\mathbb{T})$ when $\mathbb{T}$ is Katz-Tao. Our proof is by induction, and along the way we will see other sets of tubes and more generally other sets of convex sets. At first sight, these other sets are not Katz-Tao or Frostman, but by using the maximal density factoring we can reduce matters to Katz-Tao sets and Frostman sets, and we can study these by induction.

14.3.5 Multiplicities, refinements, and factoring

As discussed in the previous section, if $\mathbb{V}$ is a family of convex sets then we will bound $\mu(\mathbb{V})$ by applying the maximal density factoring lemma and then bounding $\mu(\mathbb{V}_W)$ and $\mu(\mathbb{W})$. In this section, we explain exactly how $\mu(\mathbb{V})$ is related to $\mu(\mathbb{V}_W)$ and $\mu(\mathbb{W})$.

Intuitively, it makes sense to have a bound of the form $\mu(\mathbb{V}) \lesssim \mu(\mathbb{V}_W)\mu(\mathbb{W})$. However, our set $\mathbb{V}$ has a shading Y and it is important to keep track of the shading. So we are looking for an inequality of the form $\mu(\mathbb{V}, Y) \lesssim \mu(\mathbb{V}_W, Y)\mu(\mathbb{W}, Y_{\mathbb{W}})$. We will need to define $Y_{\mathbb{W}}$. In our applications, we are usually given a lower bound for $\lambda(\mathbb{V}, Y)$, and for our product estimate to be useful, we will need to have a lower bound for $\lambda(\mathbb{W}, Y_{\mathbb{W}})$. The main work in this section is to carefully choose $Y_{\mathbb{W}}$ and prove a lower bound for $\lambda(\mathbb{W}, Y_{\mathbb{W}})$.

One main result is the following.

Proposition 14.3.18 (Factoring and multiplicity). *Let $\mathbb{V} = \sqcup_{\mathbb{W}} \mathbb{V}_W$, where $\mathbb{V}$ and $\mathbb{W}$ are families of convex sets in $\mathbb{R}^3$, and each family $\mathbb{V}_W$ is C -Frostman in W . Suppose that the sets in $\mathbb{V}$ have approximately the same dimensions. Let Y be a shading on $\mathbb{V}$.*

Then there is a refinement $(\mathbb{V}', Y')$ and a subset $\mathbb{W}' \subset \mathbb{W}$ with a shading $Y_{\mathbb{W}'}$ so that the following holds. Define $\mathbb{V}'_W = \mathbb{V}_W \cap \mathbb{V}'$ so that $\mathbb{V}' = \sqcup_{W \in \mathbb{W}'} \mathbb{V}'_W$.

1. $(\mathbb{V}', Y')$ is a $\gtrsim 1$ refinement of $(\mathbb{V}, Y)$.
2. $\lambda(\mathbb{W}', Y_{\mathbb{W}'}) \gtrsim C^{-1} \lambda(\mathbb{V}, Y)^2$.
3. $(\mathbb{W}', Y_{\mathbb{W}'})$ has constant multiplicity.
4. For each $W \in \mathbb{W}'$, $(\mathbb{V}'_W, Y')$ has constant multiplicity. This multiplicity is also the same for all $W \in \mathbb{W}'$.
5. For all $W \in \mathbb{W}'$, we have

$$\mu(\mathbb{V}, Y) \lesssim \mu(\mathbb{W}', Y_{\mathbb{W}'})\mu(\mathbb{V}_W, Y'). \quad (14.3.15)$$

6. For every $x \in U(\mathbb{V}', Y')$,

$$\mathbb{V}'_{Y'}(x) \subset \bigcup_{\substack{W \in \mathbb{W}' \\ x \in Y_{\mathbb{W}'}(W)}} (\mathbb{V}'_W)_{Y'}(x). \quad (14.3.16)$$

7. $|U(\mathbb{V}', Y') \cap B(x, w_1)|$ is roughly the same for every $x \in U(\mathbb{W}', Y_{\mathbb{W}'}),$ up to a factor $\lesssim 1$.

Remark 14.3.19. Proposition 14.3.18 is stated in $\mathbb{R}^3$. It is possible that a similar result holds in higher dimensions, but in order to obtain a lower bound on $\lambda(\mathbb{W}, Y_{\mathbb{W}})$ we use a variant of Cordoba’s maximal function estimate in the plane. To prove a version of Proposition 14.3.18 in $\mathbb{R}^n$, we would require a suitable Kakeya estimate in $\mathbb{R}^{n-1}$.

Remark 14.3.20. The hypothesis that $\mathbb{V}_W$ is C -Frostman in W can be replaced by the following weaker requirement. Let $W \in \mathbb{W}$ have dimensions $w_1 \times w_2 \times w_3$. Then instead of requiring that $\mathbb{V}_W$ is Frostman in W , we require that the w_1 -thickening of each $V \in \mathbb{V}_W$ is C -Frostman in $\mathbb{W}$.

Remark 14.3.21. The shading $Y_{\mathbb{W}'}$ on $\mathbb{W}'$ is compatible with the shading Y' on $\mathbb{V}$, in the sense that $Y_{\mathbb{W}'}(W) = W \cap N_{w_1}(U(\mathbb{V}_W, Y'))$, where w_1 is the shortest dimension of W .

Remark 14.3.22 (Compatibility audit). The proof printed below does not establish all seven conclusions of Proposition 14.3.18. There are two independent obstructions. First, the quantity $\tilde{\mu}$ selected on child points counts parents whose child support meets $B(x, w_1)$; it is not the multiplicity, at an arbitrary parent-shading point, of the clipped sets

$$W \cap N_{w_1}(U(\mathbb{V}'_W, Y')).$$

The extra intersection with the labelled parent W can make those two counts different. Thus the sentence below asserting constant multiplicity of $(\mathbb{W}', Y_{\mathbb{W}'})$ does not follow from the definition of $\Omega(\mu_{\text{inner}}, \mu_{\text{outer}}, \tilde{\mu})$.

Second, the final ballwise refinement used to obtain Item 7 is an arbitrary deletion from child shadings. Constant inner multiplicity, the induced parent shading, constant parent multiplicity, and the pointwise product bound are not monotone under such a deletion. Consequently the assertion that this final refinement “does not disturb” the earlier items also needs a new argument.

Finite labelled counterexamples to both transitions, together with a cellular replacement, are given by Z. Hua and X. Yang, *Cellular Maximal-Density Factoring and a Conditional Repair of the Streamlined Three-Dimensional Kakeya Reduction*, arXiv:2608.18870v3 (26 August 2026). That paper’s Section 9 conclusion is itself conditional on its explicit source interface (I1)–(I4), and it does not claim that the formal statement of the original very-non-sticky lemma supplies this interface. Those interface hypotheses and the cellular construction are not proved in the present chapter. Proposition 14.3.18, its downstream uses, Lemma 14.3.54, and the general Kakeya deduction must therefore be treated as conditional independently of the uniform-gain issue recorded in Remark 14.3.55.

Remark 14.3.23 (Exact interface still required for cellular closure). For review purposes, the phrase “source interface (I1)–(I4)” in the preceding audit can be

made completely explicit. A cellular replacement of the defective proposition would have to be fed one *single* labelled incidence structure with all of the following properties before any maximal-density factor is selected.

- (I1) The fine objects are a finite labelled family of essentially distinct unit δ -tubes in one fixed ball, of polynomial cardinality in δ^{-1} , with a fixed parent at every source scale and uniformly bounded ambiguity in adjacent-scale ancestry.
- (II2) There is a reference refinement $\bar{Y}$ which retains a $\delta^{o(1)}$ proportion of the original incidence mass and on which, at every source scale, both the incident-centred direction-cap counts and the corresponding local average multiplicities lie in the prescribed dyadic classes. The same $\bar{Y}$ must work at every scale; a separate refinement for each scale is not enough.
- (III3) There is a further all-source-scale refinement Y^{sc} , again with only a $\delta^{o(1)}$ loss, on which the scale- r shading comparisons used in the very-non-sticky proof hold simultaneously and the branching numbers are common across parents of the same scale.
- (IV4) All ancestry and representation maps preserve labels and have uniformly bounded fibres. Every later deletion is made through canonical windows and is saturated at the level of whole labelled incidences, so that a parent refinement lifts to its children without changing the already regularized inner and outer multiplicities.

The uniformity definition used later in this chapter supplies only part of this list: polynomial cardinality, essentially distinct tubes, and common branching after scale-by-scale pigeonholing. It does not by itself produce one reference shading satisfying the pointwise cap counts, local averages, and scale-comparison inequalities simultaneously. Nor does it force an arbitrary downstream refinement to be saturated in the sense of (I4). Iterating the ordinary pigeonhole principle does not repair this: after a later deletion, the counts on which an earlier dyadic class was defined have changed. Thus (I2)–(I4) are new hypotheses or new lemmas, rather than a renaming of the present uniformity assumptions.

The conditional cellular theorem cited above shows that (I1)–(I4), together with its tagged-cell construction, are sufficient for the corresponding Section 9 reduction. The present manuscript neither assumes these items in Lemma 14.3.54 nor proves them from its hypotheses. Consequently a citation to that conditional theorem cannot close the gap under this book’s self-contained-proof standard.

Before proving Proposition 14.3.18, we first introduce some basic facts about shadings. The following lemma says that a mild refinement cannot substantially increase the multiplicity of an arrangement.

Lemma 14.3.24. *Let $(\mathbb{W}, Y)$ be a family of convex sets, and let $(\mathbb{W}', Y')$ be a c -refinement. Then*

$$\mu(\mathbb{W}, Y) \leq c^{-1} \mu(\mathbb{W}', Y'). \quad (14.3.17)$$

In particular, if $(\mathbb{W}', Y')$ is a $\gtrsim 1$ refinement of $(\mathbb{W}, Y)$, then $\mu(\mathbb{W}', Y') \lesssim \mu(\mathbb{W}, Y)$.

Proof. We have $|U(\mathbb{W}', Y')| \leq |U(\mathbb{W}, Y)|$, and thus

$$\begin{aligned} \mu(\mathbb{W}, Y) &= \frac{\sum_{\mathbb{W}} |Y(W)|}{|U(\mathbb{W}, Y)|} \\ &\leq c^{-1} \frac{\sum_{\mathbb{W}'} |Y'(W)|}{|U(\mathbb{W}, Y)|} \\ &\leq c^{-1} \frac{\sum_{\mathbb{W}'} |Y'(W)|}{|U(\mathbb{W}', Y')|} = c^{-1} \mu(\mathbb{W}', Y'). \quad \square \end{aligned}$$

We remark that if $(\mathbb{W}', Y')$ is a $\gtrsim 1$ refinement of Y , then it can happen that $\mu(\mathbb{W}', Y') \gg \mu(\mathbb{W}, Y)$, but this is okay in our argument because the inequality goes in a favorable direction.

We say that $(\mathbb{V}, Y)$ is not empty if there is some $V \in \mathbb{V}$ with $Y(V)$ non-empty. We begin with a basic lemma about multiplicity.

Lemma 14.3.25. *Suppose that $(\mathbb{V}, Y)$ is not empty. If $\mu(\mathbb{V}, Y)(x) \geq \mu_{\text{lower}}$ for every $x \in U(\mathbb{V}, Y)$, then $\mu(\mathbb{V}, Y) \geq \mu_{\text{lower}}$. If $\mu(\mathbb{V}, Y)(x) \leq \mu_{\text{upper}}$ for every $x \in U(\mathbb{V}, Y)$, then $\mu(\mathbb{V}, Y) \leq \mu_{\text{upper}}$.*

Proof. For the lower bound, we have

$$\mu_{\text{lower}} |U(\mathbb{V}, Y)| \leq \int_{U(\mathbb{V}, Y)} \mu(\mathbb{V}, Y)(x) dx = \sum_{V \in \mathbb{V}} |Y(V)|.$$

For the upper bound, we have

$$\sum_{V \in \mathbb{V}} |Y(V)| = \int_{U(\mathbb{V}, Y)} \mu(\mathbb{V}, Y)(x) dx \leq \mu_{\text{upper}} |U(\mathbb{V}, Y)|. \quad \square$$

Proposition 14.3.18 involves a shading $Y_{\mathbb{W}'}$ on $\mathbb{W}'$. This shading will come from the shading Y on $\mathbb{V}$ according to the following procedure.

Definition 14.3.26. Let $\mathbb{V} = \bigsqcup_{\mathbb{W}} \mathbb{V}_W$, where $\mathbb{V}$ and $\mathbb{W}$ are families of convex subsets of $\mathbb{R}^n$ of dimensions roughly $v_1 \times \dots \times v_n$ and $w_1 \times \dots \times w_n$ respectively.

Suppose that each $V \in \mathbb{V}_W$ is contained in $\mathbb{W}$. Let $Y_{\mathbb{V}}$ be a shading on $\mathbb{V}$. We define the *induced shading*

$$Y_{\mathbb{W}}(W) = W \cap N_{w_1}(U(\mathbb{V}_W, Y_{\mathbb{V}})).$$

Note that the induced shading $Y_{\mathbb{W}}$ depends on the choice of partition $\sqcup_{\mathbb{W}} \mathbb{V}_W$. This choice will be stated explicitly if it is not clear from context.

We need a small variation of Lemma 14.3.25 for induced shadings.

Lemma 14.3.27. *Let $\mathbb{V} = \sqcup_{\mathbb{W}} \mathbb{V}_W$, $\mathbb{W}$, and $Y_{\mathbb{V}}$ be as in Definition 14.3.26. Suppose that $(\mathbb{V}, Y)$ is not empty. Let $Y_{\mathbb{W}}$ be the induced shading on $\mathbb{W}$. If $\mu(\mathbb{W}, Y_{\mathbb{W}})(x) \geq \mu_{\text{lower}}$ for every $x \in U(\mathbb{V}, Y_{\mathbb{V}})$, then $\mu(\mathbb{W}, Y_{\mathbb{W}}) \gtrsim \mu_{\text{lower}}$.*

Proof. For each $W \in \mathbb{W}$, define $A(W) = N_{w_1}(U(\mathbb{V}_W, Y_{\mathbb{V}}))$. We have $Y_{\mathbb{W}}(W) = A(W) \cap W$, and $|A(W)| \lesssim |Y_{\mathbb{W}}(W)|$; this is because $|W \cap B| \lesssim |B|$ for each every ball B of radius w_1 with center in W (here W has dimensions roughly to $w_1 \times \dots \times w_n$). We also have $\sum_{\mathbb{W}} \chi_{A(W)}(x) \geq \mu_{\text{lower}}$ on $N_{w_1}(U(\mathbb{V}, Y))$; denote the latter set by X . We now argue as in Lemma 14.3.25. We have

$$\begin{aligned} \mu_{\text{lower}}|U(\mathbb{W}, Y_{\mathbb{W}})| &\leq \mu_{\text{lower}}|X| \\ &\leq \int_X \sum_{W \in \mathbb{W}} \chi_{A(W)}(x) dx \\ &\lesssim \int_{\mathbb{R}^n} \mu(\mathbb{W}, Y_{\mathbb{W}})(x) dx \\ &= \sum_{W \in \mathbb{W}} |Y_{\mathbb{W}}(W)|. \end{aligned} \quad \square$$

The key estimate for the proof of Proposition 14.3.18 is the following lower bound for the density of the induced shading $(\mathbb{W}, Y_{\mathbb{W}})$. This estimate is special for $\mathbb{R}^3$, whereas all the proceeding discussion worked in any dimension.

Lemma 14.3.28. *Let $\mathbb{V} = \sqcup_{\mathbb{W}} \mathbb{V}_W$ be a collection of convex sets in $\mathbb{R}^3$, where the sets of $\mathbb{V}_W$ are contained in W . Let Y be a shading on $\mathbb{V}$, and let $Y_{\mathbb{W}}$ be the induced shading on $\mathbb{W}$, as in Definition 14.3.26. Suppose that for each $W \in \mathbb{W}$, the set $\mathbb{V}_W$ is C -Frostman in W . Then for each $W \in \mathbb{W}$, we have*

$$|Y_{\mathbb{W}}(W)| \gtrsim C^{-1} \lambda(\mathbb{V}_W, Y)^2 |W|. \quad (14.3.18)$$

Proof. We give the required L^2 argument. After dyadic pigeonholing, we may select a set $\mathbb{V}'_W \subset \mathbb{V}_W$, dimensions $v_1 \leq v_2 \leq v_3$, and a number $\lambda \gtrsim \lambda(\mathbb{V}_W)$ so that each set in $\mathbb{V}'_W$ has dimensions roughly $v_1 \times v_2 \times v_3$, $|Y'_{\mathbb{V}}(V)| \sim \lambda|V|$ for each $V \in \mathbb{V}'_W$, and $\sum_{\mathbb{V}'_W} |Y'(V)| \gtrsim \sum_{\mathbb{V}_W} |Y'(V)|$. We thus have

$$\lambda|\mathbb{V}'_W| \gtrsim |V|^{-1} \sum_{\mathbb{V}'_W} |Y'(V)| \gtrsim |V|^{-1} \sum_{\mathbb{V}_W} |Y'(V)| \gtrsim \lambda(Y_{\mathbb{V}_W})|\mathbb{V}_W|.$$

In particular, since $\mathbb{V}_W$ is C -Frostman relative to W , we have that $\mathbb{V}'_W$ is $\lesssim \frac{\lambda}{\lambda(Y_{\mathbb{V}_W})} C$ -Frostman relative to W .

Let $w_1 \times w_2 \times w_3$ be the dimensions of the set W that we fixed above. We have $v_i \leq w_i$ for $i = 1, 2, 3$. Define $p_i = \max(v_i, w_i)$, and define $\mathbb{P} = \{N_{w_1}(V) : V \in \mathbb{V}'_W\}$; each $P \in \mathbb{P}$ has dimensions comparable to $p_1 \times p_2 \times p_3$. (We note that the sets $P \in \mathbb{P}$ need not be essentially distinct. We have $|\mathbb{P}| = |\mathbb{V}'_W|$.) Define $Y_{\mathbb{P}}(P) = P \cap \bigcup_{V \in \mathbb{V}'_W} N_{w_1}(Y(V))$.

Since each $P \in \mathbb{P}$ contains a set $V \in \mathbb{V}'_W[P]$ with $|Y_{\mathbb{V}}(V)| \geq \lambda|V|$, and since $v_1 \leq w_1$, we have that the w_1 -covering number of $Y_{\mathbb{V}}(V)$ satisfies

$$\mathcal{E}_{w_1}(Y_{\mathbb{V}}(V)) \gtrsim \begin{cases} \lambda w_1^{-2} v_2 v_3, & \text{if } v_2 \geq w_1, \\ \lambda w_1^{-1} v_3, & \text{if } v_3 \geq w_1 > v_2, \\ 1, & \text{if } w_1 > v_3. \end{cases}$$

In each of the above scenarios, we have $|Y_{\mathbb{P}}(P)| \sim w_1^3 \mathcal{E}_{w_1}(Y_{\mathbb{V}}(V)) \gtrsim \lambda|P|$ for each $P \in \mathbb{P}$.

By Remark 14.3.7, there is a subset $\mathbb{P}' \subset \mathbb{P}$ that is $\lesssim \frac{\lambda}{\lambda(Y_{\mathbb{V}_W})} C$ -Frostman relative to W . Define $\theta_0 = \frac{p_2}{\min(w_2, p_3)}$. Observe that if $P, P' \in \mathbb{P}'$ with $|P \cap P'| \geq \theta|P|$ for some $\theta \in [\theta_0, 1]$, then P' must be contained in the rectangular prism of dimensions comparable to $p_1 \times \theta^{-1}p_2 \times p_3$ given by $N_{\theta^{-1}p_2}(P_1) \cap W$. This prism has volume roughly $\theta^{-1}|P|$, and thus contains $\lesssim \frac{\lambda}{\lambda(Y_{\mathbb{V}_W})} C \frac{\theta^{-1}|P|}{|W|} |\mathbb{P}'|$ sets from $\mathbb{P}'$. We now compute

$$\begin{aligned} \left\| \sum_{P \in \mathbb{P}'} \chi_P \right\|_2^2 &= \sum_{P \in \mathbb{P}'} \sum_{\substack{\theta \in [\theta_0, 1] \\ \theta \text{ dyadic}}} \sum_{\substack{P' \in \mathbb{P}' \\ |P \cap P'| \sim \theta|P|}} |P \cap P'| \\ &\lesssim \sum_{P \in \mathbb{P}'} \sum_{\substack{\theta \in [\theta_0, 1] \\ \theta \text{ dyadic}}} \left(\frac{\lambda}{\lambda(Y_{\mathbb{V}_W})} C \frac{\theta^{-1}|P|}{|W|} |\mathbb{P}'| \right) (\theta|P|) \\ &\lesssim \frac{\lambda}{\lambda(Y_{\mathbb{V}_W})} C |\mathbb{P}'|^2 |P|^2 / |W|. \end{aligned}$$

Finally, by Cauchy-Schwarz and the fact that $\lambda \gtrsim \lambda(\mathbb{V}_W)$, we have

$$\begin{aligned} |Y_{\mathbb{W}}(W)| &\geq |U(\mathbb{P}', Y)| \gtrsim \left(\lambda|P| |\mathbb{P}'| \right)^2 / \left(\left\| \sum_{P \in \mathbb{P}'} \chi_{Y(P)} \right\|_2^2 \right) \\ &\gtrsim C^{-1} \lambda \lambda(Y_{\mathbb{V}_W}) |W| \geq C^{-1} \lambda(Y_{\mathbb{V}_W})^2 |W|. \end{aligned} \quad \square$$

Remark 14.3.29. Observe that the hypothesis that $\mathbb{V}_W$ is C -Frostman in W can be replaced with the weaker hypothesis that the w_1 -thickening of each $V \in \mathbb{V}_W$ is C -Frostman in $\mathbb{W}$ (here w_1 is the smallest dimension of W). Indeed, the quantities p_1, p_2, p_3 from the above proof are not changed if we thicken each $V \in \mathbb{V}_W$ by w_1 , and the set $\mathbb{P}$ remains unchanged by this procedure.

We now have the tools to prove Proposition 14.3.18.

Proof of Proposition 14.3.18. Let us abbreviate $\lambda = \lambda(\mathbb{V}, Y)$.

By pigeonholing, we can select a set $\mathbb{V}' \subset \mathbb{V}$ and $\mathbb{W}' \subset \mathbb{W}$ so that $(\mathbb{V}', Y)$ is a $\gtrsim 1$ refinement of $(\mathbb{V}, Y)$, $|Y(V)| \gtrsim \lambda|V|$ for each $V \in \mathbb{V}'$, $\mathbb{V}' = \bigsqcup_{\mathbb{W}'} \mathbb{V}'_W$, and each set $\mathbb{V}'_W$ has approximately the same cardinality.

By Lemma 14.3.24 we have $\mu(\mathbb{V}, Y) \lesssim \mu(\mathbb{V}', Y)$. Since $\mathbb{V}' = \bigsqcup_{\mathbb{W}'} \mathbb{V}'_W$, we have

$$\mu(\mathbb{V}', Y)(x) = \sum_{\mathbb{W}'} \mu(\mathbb{V}'_W, Y)(x).$$

For each $x \in U(\mathbb{V}', Y)$ we can choose μ_{inner} , μ_{outer} , and $\tilde{\mu}$ with $\mu_{outer} \leq \tilde{\mu}$ so that

- (i) $\mu(\mathbb{V}', Y)(x) \approx \mu_{inner}\mu_{outer}$.
- (ii) There are $\sim \mu_{outer}$ sets $W \in \mathbb{W}'$ so that $\mu(\mathbb{V}'_W, Y)(x) \sim \mu_{inner}$.
- (iii) There are $\sim \tilde{\mu}$ sets $W \in \mathbb{W}'$ so that $\mu(\mathbb{V}'_W, Y)(x') \sim \mu_{inner}$ for some $x' \in B(x, w_1)$.

Next we pigeonhole the values of μ_{inner} , μ_{outer} , and $\tilde{\mu}$. Define $\Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})$ to be the set of x obeying Items (i) and (ii) and (iii) above. $U(\mathbb{V}', Y)$ is equal to the union of $\lesssim (\log |\mathbb{V}|)(\log |\mathbb{W}|)^2$ many sets $\Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})$. We choose μ_{inner} , μ_{outer} , and $\tilde{\mu}$ so that

$$\int_{\Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})} \mu(\mathbb{V}', Y)(x) dx \approx \int_{\mathbb{R}^n} \mu(\mathbb{V}', Y)(x) dx. \quad (14.3.19)$$

Next we define the shading Y' on $\mathbb{V}'$. If $V \in \mathbb{V}'_W$, we define

$$Y'(V) = Y(V) \cap \Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu}) \cap \{x : \mu(\mathbb{V}'_W, Y)(x) \sim \mu_{inner}\}.$$

We define $Y_{\mathbb{W}'}$ to be the shading on $\mathbb{W}'$ induced by the shading Y' on $\mathbb{V}' = \bigsqcup_{\mathbb{W}'} \mathbb{V}'_W$ (recall Definition 14.3.26). We have now defined all the characters and we have to check the seven items in the conclusion of Proposition 14.3.18.

Since $Y_{\mathbb{W}'}$ is induced by the shading Y' , it follows Item 6 and the claim in Remark 14.3.21 are satisfied. We have $\mu(\mathbb{V}'_W, Y')(x) \sim \mu_{inner}$ on $U(\mathbb{V}'_W, Y')$. This gives Item 4. Note that $Y'_{\mathbb{W}'}(W)$ is non-empty if and only if $(\mathbb{V}_W, Y')$ is non-empty. In this case, by Lemma 14.3.25, we get

$$\mu(\mathbb{V}_W, Y') \gtrsim \mu_{inner}. \quad (14.3.20)$$

The pair $(\mathbb{W}', Y_{\mathbb{W}'})$ has constant multiplicity $\tilde{\mu}$: in other words $\mu(\mathbb{W}', Y_{\mathbb{W}'}) \sim \tilde{\mu}$ on $U(\mathbb{W}', Y_{\mathbb{W}'})$. This gives item 3. Since $\tilde{\mu} \geq \mu_{outer}$, we have

$$\mu(\mathbb{W}', Y_{\mathbb{W}'}) \approx \tilde{\mu} \gtrsim \mu_{outer}. \quad (14.3.21)$$

We have $\mu(\mathbb{V}', Y')(x) \lesssim \mu_{inner} \mu_{outer}$ on $U(\mathbb{V}', Y')$. Thus by Lemma 14.3.25, (14.3.20), and (14.3.21), we conclude that for each $W \in \mathbb{W}'$ we get Item 5:

$$\mu(\mathbb{V}, Y) \lesssim \mu(\mathbb{V}', Y') \lesssim \mu_{inner} \mu_{outer} \lesssim \mu(\mathbb{W}', Y_{\mathbb{W}'}) \mu(\mathbb{V}'_W, Y'). \quad (14.3.22)$$

We have seen that $(\mathbb{W}', Y_{\mathbb{W}'})$ has constant multiplicity $\tilde{\mu}$. This checks Item 3.

Next we check Item 1: $(\mathbb{V}', Y')$ is a $\gtrsim 1$ refinement of $(\mathbb{V}, Y)$. Since $(\mathbb{V}', Y)$ is a $\gtrsim 1$ refinement of $(\mathbb{V}, Y)$, we have to check that $(\mathbb{V}', Y')$ is a $\gtrsim 1$ refinement of $(\mathbb{V}', Y)$, which means that

$$\sum_{V \in \mathbb{V}'} |Y'(V)| \gtrsim \sum_{V \in \mathbb{V}'} |Y(V)|. \quad (14.3.23)$$

To check the claim, the key point is that if $x \in \Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})$, then $\mu(\mathbb{V}', Y)(x) \lesssim \mu_{inner} \mu_{outer} \lesssim \mu(\mathbb{V}', Y')(x)$. Combining this with (14.3.19), we see that

$$\begin{aligned} \sum_{V \in \mathbb{V}'} |Y'(V)| &= \int \mu(\mathbb{V}', Y')(x) \geq \int_{\Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})} \mu(\mathbb{V}', Y')(x) \gtrsim \\ &\gtrsim \int_{\Omega(\mu_{inner}, \mu_{outer}, \tilde{\mu})} \mu(\mathbb{V}', Y)(x) \approx \int \mu(\mathbb{V}', Y)(x) = \sum_{V \in \mathbb{V}'} |Y(V)|. \end{aligned}$$

This proves Item 1.

In order to check Item 7, we make one more refinement. We cover $U(\mathbb{W}', Y_{\mathbb{W}'})$ by boundedly overlapping balls of radius w_1 . After dyadic pigeonholing, we may select a ≈ 1 refinement of Y' so that $|U(\mathbb{V}', Y') \cap B_{w_1}|$ is roughly the same for each of these balls. By abuse of notation, we will still call this refinement Y' . There is a corresponding refinement of $Y_{\mathbb{W}'}$, and we will still call it $Y_{\mathbb{W}'}$. These refinements do not disturb the items we have already established, and they give us Item 7. In particular, since we have made only a $\gtrsim 1$ refinement of $(\mathbb{V}', Y')$, (14.3.23) and Item 1 still hold.

Finally we check Item 2: $\lambda(\mathbb{W}', Y_{\mathbb{W}'}) \gtrsim C^{-1} \lambda^2$. For dyadic τ , we define $\mathbb{W}'_\tau$ as

$$\mathbb{W}'_\tau = \{W \in \mathbb{W}' : \sum_{V \in \mathbb{V}'_W} |Y'(V)| \sim \tau \lambda |V| |\mathbb{V}'_W|\}.$$

Combining (14.3.23) and the assumption that $|Y(V)| \gtrsim \lambda |V|$ for every $V \in \mathbb{V}'$, we see that

$$\sum_{\tau} \sum_{W \in \mathbb{W}'_\tau} \tau \lambda |V| |\mathbb{V}'_W| \sim \sum_{V \in \mathbb{V}'} |Y'(V)| \gtrsim \sum_{V \in \mathbb{V}'} |Y(V)| \gtrsim \sum_{\tau} \sum_{W \in \mathbb{W}'_\tau} \lambda |V| |\mathbb{V}'_W|.$$

Since $|\mathbb{V}'_W|$ are essentially constant in W , we can simplify this formula to

$$\sum_{\tau} \tau |\mathbb{W}'_\tau| \gtrsim |\mathbb{W}'|. \quad (14.3.24)$$

This implies that

$$\sum_{\tau \gtrsim 1} \tau |\mathbb{W}'_\tau| \gtrsim \sum_{\tau} \tau |\mathbb{W}'_\tau| \gtrsim |\mathbb{W}'|. \quad (14.3.25)$$

If $W \in \mathbb{W}_\tau$, then Lemma 14.3.28 gives the bound

$$|Y_{\mathbb{W}'}(W)| \gtrsim C^{-1} \tau^2 \lambda^2 |W|.$$

Now we are ready to estimate $\lambda(\mathbb{W}', Y_{\mathbb{W}'})$.

$$\sum_{W \in \mathbb{W}'} |Y_{\mathbb{W}'}(W)| \geq \sum_{\tau} |\mathbb{W}'_\tau| C^{-1} \tau^2 \lambda^2 |W|.$$

We can bound this using (14.3.25). If $\tau \gtrsim 1$, then $\tau^2 \gtrsim \tau$, and so

$$\sum_{\tau} |\mathbb{W}'_\tau| C^{-1} \tau^2 \lambda^2 |W| \gtrsim \sum_{\tau \gtrsim 1} |\mathbb{W}'_\tau| \tau C^{-1} \lambda^2 |W| \gtrsim |\mathbb{W}'| C^{-1} \lambda^2 |W|.$$

In total, we have

$$\sum_{W \in \mathbb{W}'} |Y_{\mathbb{W}'}(W)| \gtrsim |\mathbb{W}'| C^{-1} \lambda^2 |W|.$$

Rearranging this gives $\lambda(\mathbb{W}', Y'_{\mathbb{W}'}) \gtrsim C^{-1} \lambda^2$ as desired. $\square$

We also state an easier variant of Proposition 14.3.18 in the special case when $\mathbb{V}$ and $\mathbb{W}$ are both sets of tubes.

Lemma 14.3.30. *Suppose $\mathbb{T}$ is a uniform set of δ -tubes and Y is a shading on $\mathbb{T}$.*

Let $\rho \in [\delta, 1]$ and let $\mathbb{T} = \bigcup_{T_\rho \in \mathbb{T}_\rho} \mathbb{T}[T_\rho]$. Let Y be a shading on $\mathbb{T}$ with $|Y(T)| \sim \lambda|T|$ for each $T \in \mathbb{T}$.

Then there is a subset $\mathbb{T}'_\rho \subset \mathbb{T}_\rho$ and a shading $Y'_{\mathbb{T}'_\rho}$ on $\mathbb{T}'_\rho$ with $\lambda(\mathbb{T}'_\rho, Y'_{\mathbb{T}'_\rho}) \gtrsim \lambda$. There is shading Y' on $\mathbb{T}' = \bigsqcup_{T_\rho \in \mathbb{T}'_\rho} \mathbb{T}[T_\rho]$ so that $(\mathbb{T}', Y')$ is a ≈ 1 refinement of $(\mathbb{T}, Y)$, and for all $T_\rho \in \mathbb{T}'_\rho$ we have

$$\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{T}'_\rho, Y'_{\mathbb{T}'_\rho}) \mu(\mathbb{T}[T_\rho], Y'). \quad (14.3.26)$$

We also have the pointwise containment

$$\mathbb{T}_{Y'}(x) \subset \bigcup_{\substack{T_\rho \in \mathbb{T}'_\rho \\ x \in Y'_{\mathbb{T}'_\rho}(T_\rho)}} (\mathbb{T}[T_\rho])_{Y'}(x). \quad (14.3.27)$$

Furthermore, for each $x \in U(\mathbb{T}'_\rho, Y'_{\mathbb{T}'_\rho})$ we have

$$|U(\mathbb{T}, Y')| \gtrsim |U(\mathbb{T}'_\rho, Y'_{\mathbb{T}'_\rho})| \frac{|U(\mathbb{T}, Y') \cap B(x, \rho)|}{|B(x, \rho)|}. \quad (14.3.28)$$

The proof of (14.3.26) is the same as the proof of Proposition 14.3.18, except in place of Lemma 14.3.28 we use the fact that if $T_\rho \supset T$, then $|T_\rho \cap N_\rho(Y(T))| \gtrsim \frac{|Y(T)|}{|T|} |T_\rho|$. Thus we do not need to assume that $\mathbb{T}_{T_\rho}$ is C -Frostman in T_ρ , and we obtain the estimate $\lambda(\mathbb{T}'_\rho, Y_\rho) \gtrsim \lambda$, which is stronger than its analogue in Proposition 14.3.18.

14.3.6 Multiplicity estimates for slabs and planks

In Section 14.3.5 we reduced the problem of controlling $\mu(\mathbb{V}, Y_\mathbb{V})$ for an arbitrary collection of convex sets in $\mathbb{R}^3$ to the problem of controlling the multiplicity of two new families of convex sets, one of which is Katz-Tao and the other is Frostman. The estimates $K_{KT}(\beta)$ and $K_F(\beta)$ control the intersection multiplicity of sets of *tubes* in $\mathbb{R}^3$ that are Frostman and Katz-Tao, respectively. In this section we will show that incidence bounds for arbitrary convex sets in $\mathbb{R}^3$ can be systematically reduced to incidence bounds for tubes.

Katz-Tao multiplicity estimates. Lemma 14.3.31 below says that theakeya estimate $K_{KT}(\beta)$ implies an analogous estimate for planks of dimensions $a \times b \times 1$. Furthermore, a stronger estimate holds when $a \ll b$, provided that the planks do not concentrate into slabs (i.e. sets of dimensions $\theta \times 1 \times 1$ for $\theta \ll 1$). To make this latter statement precise, we must define what it means for planks to concentrate into slabs.

If $\mathbb{P}$ is a set of planks in $\mathbb{R}^3$ of dimensions $a \times b \times 1$, $\theta \in [a/b, 1]$, and S is a $\theta \times 1 \times 1$ slab, then we define

$$\mathbb{P}_S = \{P \in \mathbb{P} : P \subset S \text{ and } \angle(TP, TS) \leq \theta\}. \quad (14.3.29)$$

In the above, $\angle(TP, TS)$ refers to the angle between the planes TP, TS , where TP is the plane spanned by the two longest axes of P , and similarly for S (this is defined up to accuracy θ).

Lemma 14.3.31. *Suppose that $K_{KT}(\beta)$ holds. Then for every $\epsilon > 0$, there exists $\eta, b_0 > 0$ such that the following holds for any $0 < a \leq b \leq b_0$. Let $\mathbb{P}$ be a set of $a \times b \times 1$ planks in B_1 , and let $Y_\mathbb{P}$ be a shading on $\mathbb{P}$ with density $\lambda(\mathbb{P}) \geq a^\eta$. Suppose that the planks do not concentrate into slabs, in the following sense: there exists $0 \leq \gamma \leq 1$ so that for every $\theta \in [a/b, 1]$ and every $\theta \times 1 \times 1$ plank S , $|\mathbb{P}_S| \leq a^{-\eta} \theta^\gamma |\mathbb{P}|$.*

If $\mathbb{P}$ is Katz-Tao, then $\mu(\mathbb{P}) \leq a^{-\epsilon} \left(\frac{a}{b}\right)^{\gamma\beta} |\mathbb{P}|^\beta$. More generally, we have

$$\mu(\mathbb{P}) \leq a^{-\epsilon} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{a}{b}\right)^{\gamma\beta} |\mathbb{P}|^\beta. \quad (14.3.30)$$

Remark 14.3.32. We will only apply Lemma 14.3.31 in the special cases $\gamma = 0$ (in which case the non-concentration condition is vacuous, and can be ignored),

and $\gamma = 1$. When $\mathbb{P}$ is Katz-Tao and $\gamma = 0$, the RHS of (14.3.30) is $\lesssim |\mathbb{P}|^\beta$, which matches the estimate $K_{KT}(\beta)$ for tubes.

Remark 14.3.33. The inequality automatically holds in the case when $\mu(\mathbb{P}) \leq a^{-\eta}$. The right-hand side of (14.3.30) is always $\geq a^{-\eta}$, because the hypothesis implies that $|\mathbb{P}| \geq a^\eta (b/a)^\gamma$ and η is much less than ϵ .

Frostman multiplicity estimates. Lemma 14.3.34 below says that theakeya estimate $K_F(\beta)$ implies an analogous estimate for planks of dimensions $a \times b \times 1$, provided we avoid the following bad scenario: Let $\mathbb{P}$ be a set of planks of dimensions $a \times b \times 1$, let $P \in \mathbb{P}$, let $\theta \in [a/b, 1]$. We define the “thickened plank” $P_\theta = N_{\theta b}(P)$, i.e. $\tilde{P}$ has dimensions roughly $\theta b \times b \times 1$. If $\tilde{P}$ contains a very large number of planks from $\mathbb{P}$, then this can cause an unacceptably large intersection multiplicity $\mu(\mathbb{P})$. The next result says that this is the only obstruction to obtaining an analogue of $K_F(\beta)$ for planks.

Lemma 14.3.34. *Suppose that $K_F(\beta)$ holds. Then for every $\epsilon > 0$, there exists $\eta, b_0 > 0$ such that the following holds for any $0 < a \leq b \leq b_0$. Let $\mathbb{P}$ be a set of $a \times b \times 1$ planks in B_1 , and let Y be a shading on $\mathbb{P}$ with density $\lambda(Y) \geq a^\eta$. Suppose that the planks do not concentrate into thickened planks, in the following sense: there is a number $M \geq 1$ so that for each $\theta \in [a/b, 1]$ and every $P \in \mathbb{P}$, we have $|\mathbb{P}[P_\theta]| \leq M\theta$. Then*

$$|U(\mathbb{P}, Y)| \geq a^\epsilon C_F(\mathbb{P})^{\beta/2-1} b^{2\beta} (M^{-1} b^2 |\mathbb{P}|)^{\beta/2}, \quad (14.3.31)$$

and

$$\mu(\mathbb{P}) \leq a^{-\epsilon} C_F(\mathbb{P})^{1-\beta/2} M^{\beta/2} \frac{a}{b} b^{-2\beta} \left(b^2 |\mathbb{P}|\right)^{1-\beta/2}. \quad (14.3.32)$$

Remark 14.3.35. Both inequalities automatically hold in the case when $\mu(\mathbb{P}) \leq a^{-\eta}$. To see this, note that $M \geq b/a$ (by taking $\theta = a/b$) and that $C_F(\mathbb{P}) \gtrsim \frac{1}{|\mathbb{P}|ab}$.

Tubes that factor through planks. Lemmas 14.3.31 and 14.3.34 allow us to obtain improved multiplicity bounds for families of tubes that factor through planks. The precise statement is as follows.

Proposition 14.3.36. *Suppose that $K_{KT}(\beta)$ and $K_F(\beta)$ hold. Then for all $\epsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \leq \delta_0$. Let $(\mathbb{T}, Y)$ be uniform, with $\lambda(\mathbb{T}, Y) \geq \delta^\eta$.*

(A) *Suppose there is a scale ρ so that for each $T_\rho \in \mathbb{T}_\rho$, there is a family of $a \times b \times 1$ planks that factors $\mathbb{T}[T_\rho]$. Then*

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} C_F(\mathbb{T})^{1-\beta/2} \left(\frac{a}{b}\right)^{3\beta/2} \delta^{-2\beta} \left(\delta^2 |\mathbb{T}|\right)^{1-\beta/2}. \quad (14.3.33)$$

(B) Suppose there is a scale ρ and a family of $a \times b \times 1$ planks that factors $\mathbb{T}_\rho$. Then

$$\mu(\mathbb{T}, Y) \leq \delta^{-\varepsilon} \Delta_{\max}(\mathbb{T})^{1-\beta} \left(\frac{a}{b}\right)^\beta |\mathbb{T}|^\beta. \quad (14.3.34)$$

Note that the analogue of Remark 14.3.10 applies to Lemmas 14.3.31 and 14.3.34, and to Proposition 14.3.36. The remainder of this section is devoted to proving Lemmas 14.3.31 and 14.3.34, and Proposition 14.3.36.

14.3.6.1 Slabs

A standard L^2 method gives strong bounds for the incidences of slabs. A key character is the notion of typical intersection angle. Suppose that $\mathbb{S}$ is a set of $\delta \times 1 \times 1$ slabs with a shading Y . Define the set of incidence triples

$$Tri(\mathbb{S}) = \{(x, S_1, S_2) : x \in Y(S_1) \cap Y(S_2)\}.$$

There is a natural measure on $Tri(\mathbb{S})$ coming from the volume measure for x and the counting measure for S_1, S_2 . We can define the angle between two slabs S_1, S_2 , which is well defined up to additive error δ . Then we define

$$Tri_\theta(\mathbb{S}) = \{(x, S_1, S_2) \in Tri(\mathbb{S}) : \text{angle}(S_1, S_2) \sim \theta\}.$$

Definition 14.3.37. We say that θ is a typical angle of intersection for $\mathbb{S}$ if $|Tri_\theta(\mathbb{S})| \approx |Tri(\mathbb{S})|$ up to $\log(1/\delta)$ factors. If $\mu(\mathbb{S}) \geq 2$, then $\mathbb{S}$ always has a typical intersection angle $\theta \in [\delta, 1]$.

Lemma 14.3.38. Suppose $\mathbb{S}$ is a set of $\delta \times 1 \times 1$ slabs contained in $\tilde{S}$, a $\theta \times 1 \times 1$ slab, with $\delta \leq \theta \leq 1$. Suppose the slabs of $\mathbb{S}$ are shaded with $\lambda(\mathbb{S}, Y) \geq \delta^\eta$. If θ is a typical intersection angle for $\mathbb{S}$, then $|U(\mathbb{S}, Y)| \gtrsim \delta^{2\eta} |\tilde{S}|$.

Proof. We let $f = \sum_{S \in \mathbb{S}} 1_{Y(S)}$. Note that

$$\int f^2 = |Tri(\mathbb{S})| = \sum_{S_1, S_2 \in \mathbb{S}} |Y(S_1) \cap Y(S_2)|.$$

By hypothesis, we have

$$\int f^2 = |Tri(\mathbb{S})| \approx \sum_{S_1, S_2 \in \mathbb{S}, \text{angle}(S_1, S_2) \sim \theta} |Y(S_1) \cap Y(S_2)| \leq |\mathbb{S}|^2 \frac{\delta^2}{\theta}.$$

On the other hand, we know that $\int f \gtrsim |\mathbb{S}| |S| \delta^\eta = |\mathbb{S}| \delta^{1+\eta}$. By Cauchy-Schwarz, since f is supported in $U(\mathbb{S})$,

$$\int f^2 \geq \frac{(\int f)^2}{|U(\mathbb{S})|} \approx \frac{|\mathbb{S}|^2 \delta^{2+2\eta}}{|U(\mathbb{S})|}.$$

Comparing, we get

$$|U(\mathbb{S})| \gtrsim \theta \delta^{2\eta} = |\tilde{S}| \delta^{2\eta}. \quad \square$$

This L^2 bound leads to sharp estimates for the analogue of Theorem 14.3.1 where tubes are replaced by slabs that have thickness δ and length and width close to 1.

Lemma 14.3.39. *Suppose $\mathbb{S}$ is a set of $\delta \times b \times c$ slabs in B_1 shaded with $\lambda(\mathbb{S}, Y) \geq \delta^\eta$. If $b, c \geq \delta^\eta$ and $\Delta_{\max}(\mathbb{S}) \leq \delta^{-\eta}$, then $\mu(\mathbb{S}, Y) \lesssim \delta^{-8\eta}$. If $C_F(\mathbb{S}) \leq \delta^{-\eta}$, then $|U(\mathbb{S}, Y)| \geq \delta^{8\eta}$.*

Proof. Write $v = \delta bc$ for the volume of one slab and $f = \sum_{S \in \mathbb{S}} \mathbf{1}_{Y(S)}$. If the average multiplicity is less than 2, the first conclusion is immediate. In the Frostman case this low multiplicity alternative also gives the second conclusion: since $C_F(\mathbb{S}) \leq \delta^{-\eta}$ and $\Delta_{\max}(\mathbb{S}) \geq \Delta(\{S\}, S) = 1$, we have $(\#\mathbb{S})v = \Delta(\mathbb{S}, B_1) \geq \delta^\eta$. Thus

$$|U(\mathbb{S}, Y)| \geq \frac{1}{2} \sum_{S \in \mathbb{S}} |Y(S)| \geq \frac{1}{2} \delta^\eta (\#\mathbb{S})v \geq \frac{1}{2} \delta^{2\eta} \geq \delta^{8\eta}$$

after decreasing δ_0 . We may therefore suppose below that the average multiplicity is at least 2. Dyadically pigeonhole the angle in the identity

$$\int f^2 = \sum_{S, S' \in \mathbb{S}} |Y(S) \cap Y(S')|$$

and choose a typical angle $\theta \in [\delta, 1]$; the discarded logarithmic factor is absorbed by one power $\delta^{-\eta}$ below. If S and S' meet and their normal directions make angle comparable to θ , then S' is contained in a fixed constant dilate of the θ -neighbourhood of the supporting plane of S . This is a convex set of volume $O(\theta)$. Hence, under the Katz–Tao hypothesis, the number of possible S' for a fixed S is at most

$$C \Delta_{\max}(\mathbb{S}) \frac{\theta}{v} \leq C \delta^{-\eta} \frac{\theta}{\delta bc}.$$

Two δ -thick slabs meeting at angle θ have intersection volume at most $C\delta^2/\theta$: slice perpendicular to their line of intersection and use the elementary area bound for two strips. Consequently

$$\int f^2 \lesssim (\#\mathbb{S}) \delta^{-\eta} \frac{\theta}{\delta bc} \frac{\delta^2}{\theta} = (\#\mathbb{S}) \frac{\delta^{1-\eta}}{bc}.$$

On the other hand, $\int f \geq \delta^\eta (\#\mathbb{S}) \delta bc$. Cauchy–Schwarz therefore gives

$$|U(\mathbb{S}, Y)| \gtrsim (\#\mathbb{S}) \delta^{1+3\eta} b^3 c^3.$$

Dividing the total shaded volume by this lower bound, and using $b, c \geq \delta^\eta$, yields

$$\mu(\mathbb{S}, Y) \lesssim \delta^{-2\eta} b^{-2} c^{-2} \leq \delta^{-6\eta};$$

after absorbing the logarithmic pigeonholing losses this is at most $\delta^{-8\eta}$ for sufficiently small δ .

For the Frostman conclusion, use

$$\Delta_{\max}(\mathbb{S}) \leq C_F(\mathbb{S}) \Delta(\mathbb{S}, B_1) \lesssim \delta^{-\eta} (\#\mathbb{S}) v.$$

The same pair count now gives $\int f^2 \lesssim (\#\mathbb{S})^2 \delta^{2-\eta}$. Therefore

$$|U(\mathbb{S}, Y)| \gtrsim \frac{\delta^{2\eta} (\#\mathbb{S})^2 \delta^2 b^2 c^2}{(\#\mathbb{S})^2 \delta^{2-\eta}} = \delta^{3\eta} b^2 c^2 \geq \delta^{7\eta}.$$

Absorbing the single logarithmic loss gives the asserted $|U(\mathbb{S}, Y)| \geq \delta^{8\eta}$ after decreasing δ_0 . $\square$

14.3.6.2 Planks

Next we suppose that $\mathbb{P}$ is a set of $a \times b \times 1$ planks in B_1 . We will study $\mu(\mathbb{P})$ by relating it to incidence problems involving slabs and tubes. In a sense, an incidence problem for planks can be broken down into an incidence problem for slabs and an incidence problem for tubes. The incidence problem for slabs is solved completely using Lemma 14.3.38. Assuming $K_{KT}(\beta)$ or $K_F(\beta)$ we get information about the incidence problem for tubes. In this way, we are able to prove estimates for $\mu(\mathbb{P})$ based on various geometric assumptions about $\mathbb{P}$.

Let us describe this process informally first and then state a formal lemma. Given two intersecting planks, P_1, P_2 , define the angle of intersection as the angle between the planes TP_1, TP_2 , where TP is the plane spanned by the two longest axes of P . If the planks P_1 and P_2 have dimensions $a \times b \times 1$, then this angle is meaningfully defined up to accuracy a/b . Let θ denote a “typical” angle of intersection for $\mathbb{P}$.

First we note that if the angle between P_1 and P_2 is $\gg \theta$, then their intersection will not play an important role. Therefore, we can subdivide $\mathbb{P}$ into smaller collections of planks as follows. If S is a $\theta \times 1 \times 1$ slab in B_1 , we define $\mathbb{P}_S$ as in (14.3.29). We can now decompose $\mathbb{P}$ as $\mathbb{P} = \sqcup_S \mathbb{P}_S$, where S varies over some essentially distinct slabs S . If θ is a typical angle of intersection of $(\mathbb{P}, Y)$, then morally the sets $U(\mathbb{P}_S, Y)$ are disjoint and we can study them separately. Fix a single S and we study $\mathbb{P}_S$.

Next let Q denote a $\theta b \times b \times b$ box, and let $\mathbb{P}_Q$ be the set of planks $P \in \mathbb{P}_S$ so that $P \cap Q$ has dimensions $a \times b \times b$. In other words, $\mathbb{P}_Q$ are the planks that intersect Q “tangentially”, making the intersection as large as possible.

Assuming that $\mu(\mathbb{P}) \gg 1$ and assuming that $\mathbb{P}_Q$ is not empty, then morally Lemma 14.3.38 tells us that $|U(\mathbb{P}_S, Y) \cap Q| \approx |Q|$. For each plank $P \in \mathbb{P}_S$, the θb -neighborhood of P can be covered by boxes Q as above, and so we see that morally most of this neighborhood is contained in $U(\mathbb{P}_S, Y)$. This motivates defining thickened version of $\mathbb{P}_S$ as follows.

Definition 14.3.40. For each $P \in \mathbb{P}$, define P_θ to be the θb -neighborhood of P . Define $\mathbb{P}_\theta$ to be a set of essentially distinct prisms of the form P_θ , so that each $P' \in \mathbb{P}$ is contained in a set $P_\theta \in \mathbb{P}_\theta$ for which P'_θ is comparable to P_θ . Note that if P_θ and P'_θ are comparable, then they are contained in comparable slabs S and S' . Thus $\mathbb{P}_\theta$ decomposes into a boundedly overlapping union $\bigcup_S \mathbb{P}_{\theta,S}$, where the union is taken over incomparable slabs of dimensions $\theta \times 1 \times 1$.

In the previous paragraph, we saw that if θ is a typical angle of intersection for $\mathbb{P}$, then morally $U(\mathbb{P}_S)$ is almost the same as $U(\mathbb{P}_{\theta,S})$.

Next we have to study $U(\mathbb{P}_{\theta,S})$. There is a change of variables that takes S to the unit ball and takes each $P_\theta \in \mathbb{P}_{\theta,S}$ to a tube of radius b . Therefore, the problem of estimating $|U(\mathbb{P}_{\theta,S})|$ is equivalent to a problem about tubes which we can bound using $K_{KT}(\beta)$ and/or $K_F(\beta)$.

The following two lemmas gives a precise version of the moral statements above. First we carefully define the typical angle of intersection.

Lemma 14.3.41. *Let $(\mathbb{P}, Y)$ be a family of $a \times b \times 1$ tubes and their associated shading. Let $\eta > 0$ and suppose that $\mu(\mathbb{P}, Y) \geq \delta^{-\eta}$. Then there is an ≈ 1 refinement $(\mathbb{P}, Y')$ of $(\mathbb{P}, Y)$ and a number $\theta \in [a/b, 1]$ with the following properties. In what follows, let $A = e^{\sqrt{\log a^{-1}}}$*

- $(\mathbb{P}, Y')$ has constant multiplicity on $U(\mathbb{P}, Y')$ up to a factor ~ 1 .
- $\max_{P_1, P_2 \in \mathbb{P}_{Y'}(x)} \angle(TP_1, TP_2) \sim \theta$
- If $\mathbb{P}_{Y''}(x) \subset \mathbb{P}_{Y'}(x)$ with $|\mathbb{P}_{Y''}(x)| \geq \frac{1}{A} |\mathbb{P}_{Y'}(x)|$, then

$$\max_{P_1, P_2 \in \mathbb{P}_{Y''}(x)} \angle(TP_1, TP_2) \approx \theta.$$

Proof. First we replace $(\mathbb{P}, Y)$ with an ≈ 1 refinement $(\mathbb{P}, Y_1)$ that has constant multiplicity. Next, fix a point $x \in U(\mathbb{P}, Y_1)$.

Let $B = e^{(\log a^{-1})^{3/4}}$. The point here is just that $A^N \ll B$ and $B^N \ll a^{-1}$ for any real number N .

For a subset $\mathbb{P}' \subset \mathbb{P}_{Y_1}(x)$, we define the maximal angle of $\mathbb{P}'$ as

$$M(\mathbb{P}') = \max_{P_1, P_2 \in \mathbb{P}'} \angle(TP_1, TP_2).$$

We first check if there is a subset $\mathbb{P}_1 \subset \mathbb{P}_{Y_1}(x)$ so that $|\mathbb{P}_1| \geq \frac{1}{A} |\mathbb{P}_{Y_1}(x)|$ and with $M(\mathbb{P}_1) \leq \frac{1}{B}$. If not, we set $\mathbb{P}_{Y_{almost}}(x) = \mathbb{P}_{Y_1}(x)$ and $\theta(x) = 1$.

If there is such a $\mathbb{P}_1$, we select such a set $\mathbb{P}_1$, and we check if there is a subset $\mathbb{P}_2 \subset \mathbb{P}_1$ so that $|\mathbb{P}_2| \geq \frac{1}{A} |\mathbb{P}_1|$ and $M(\mathbb{P}_2) \leq (1/B)M(\mathbb{P}_1)$. If not, we output $\mathbb{P}_{Y_{almost}}(x) = \mathbb{P}_1$ and $\theta(x) = M(\mathbb{P}_1)$.

Since $|\mathbb{P}_{Y_1}(x)| \approx \mu(\mathbb{P}, Y_1) \gtrsim \mu(\mathbb{P}, Y) \gtrsim a^{-\eta}$ and since $M(\mathbb{P}_k) \geq a/b \geq a$ for every k , this process must stop eventually at some $\mathbb{P}_k$ with $k \lesssim \log_B a^{-1}$ and $|\mathbb{P}_k| \geq A^{-k} |\mathbb{P}_{Y_1}(x)| \gtrsim |\mathbb{P}_{Y_1}(x)|$. Then we output $\mathbb{P}_{Y_{almost}}(x) = \mathbb{P}_k$. Since we have $|\mathbb{P}_{Y_{almost}}(x)| \gtrsim |\mathbb{P}_{Y_1}(x)|$ for each $x \in U(\mathbb{P}, Y_1)$, we see that Y_{almost} is a $\gtrsim 1$ refinement of Y_1 .

At this moment, we have $\theta(x)$ which may be different at different points x . We let Y' be a $\gtrsim \frac{1}{\log a - 1}$ refinement of Y_{almost} so that $\theta(x)$ is roughly constant on $U(\mathbb{P}, Y')$ (up to a factor ~ 1) and so that $|\mathbb{P}_Y(x)|$ is roughly constant on $U(\mathbb{P}, Y')$ (up to a factor ~ 1). $\square$

Definition 14.3.42. We say that θ is a typical angle of intersection for $(\mathbb{P}, Y)$ if $(\mathbb{P}, Y)$ satisfies the conclusions of Lemma 14.3.41 for some $A \geq e^{(\log(a^{-1}))^{1/4}}$.

In particular, if $(\mathbb{P}, Y')$ and θ are the output of Lemma 14.3.41, then θ is a typical angle of intersection for $(\mathbb{P}, Y')$. We made the definition of “typical angle of intersection” a little bit looser than the conclusion of Lemma 14.3.41 so that some mild refinements of $(\mathbb{P}, Y')$ will still have typical angle of intersection θ . This definition of typical angle of intersection for planks is stronger than the definition that we used for typical intersection angle of slabs, Definition 14.3.37. In this definition for planks, at every point $x \in U(\mathbb{P}, Y)$, the intersecting planks have angle at most θ and there are many pairs of planks with angle $\sim \theta$. Furthermore, this remains true if the shading Y is replaced by a mild refinement.

Lemma 14.3.43. Let $\eta > 0$ and let $(\mathbb{P}, Y)$ be a set of $a \times b \times 1$ planks in the unit ball with $\lambda(\mathbb{P}, Y) \geq a^\eta$ and $\mu(\mathbb{P}, Y) \geq a^{-\eta}$.

Then there is a refinement $(\mathbb{P}', Y')$ of $(\mathbb{P}, Y)$ with $|\mathbb{P}'| \gtrsim |\mathbb{P}|$ and $\lambda(\mathbb{P}', Y') \gtrsim \lambda(\mathbb{P}, Y)$, a number $N \geq 1$, a number $\theta \in [a/b, 1]$, and a set of $\theta \times 1 \times 1$ slabs $\mathbb{S}$ so that the following holds. For each $S \in \mathbb{S}$, let $\mathbb{P}'_S$ and $\mathbb{P}'_{\theta, S}$ be as defined in (14.3.29) and Definition 14.3.40. $\mathbb{P}' = \bigcup_{S \in \mathbb{S}} \mathbb{P}'_S$. Then each $P_\theta \in \mathbb{P}'_{\theta, S}$ contains $\sim N$ planks $P \in \mathbb{P}'_S$. Furthermore, there is a shading $Y_{\theta, S}$ on $\mathbb{P}'_{\theta, S}$ so that

1. If $U(\mathbb{P}', Y') \cap B_{\theta b}$ is non-empty, then $|U(\mathbb{P}', Y') \cap B_{\theta b}| \gtrsim a^{4\eta} |B_{\theta b}|$.
2. For each $S \in \mathbb{S}$, $\lambda(\mathbb{P}'_{\theta, S}, Y_{\theta, S}) \gtrsim a^{4\eta}$ and for each $P_\theta \in \mathbb{P}'_{\theta, S}$, $Y_{\theta, S}(P_\theta) \subset U(\mathbb{P}'_S, Y')$
3. $|U(\mathbb{P}, Y)| \gtrsim \sum_{S \in \mathbb{S}} |U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})|$.

$$4. \mu(\mathbb{P}, Y) \lesssim \max_{S \in \mathbb{S}} a^{-4\eta} \frac{aN}{b\theta} \mu(\mathbb{P}'_{\theta, S}, Y_{\theta, S}).$$

Remark 14.3.44. If $(\mathbb{P}, Y)$ has a typical angle of intersection θ , then Lemma 14.3.43 holds for this value of θ .

Proof. We begin with a pair $(\mathbb{P}, Y)$ with $\mu(\mathbb{P}, Y) \geq a^{-\eta}$. Apply Lemma 14.3.41 to $(\mathbb{P}, Y)$. This yields a refinement $(\mathbb{P}, Y_1)$ and a typical angle of intersection θ obeying the conclusions of Lemma 14.3.41:

- Our set $(\mathbb{P}, Y_1)$ has roughly constant multiplicity: for each $x \in U(\mathbb{P}, Y_1)$, $|\mathbb{P}_{Y_1}(x)| \sim m$.
- For each $x \in U(\mathbb{P}, Y_1)$, $\max_{P_1, P_2 \in \mathbb{P}_{Y_1}(x)} \angle(TP_1, TP_2) \sim \theta$
- If $\mathbb{P}_{Y''}(x) \subset \mathbb{P}_{Y_1}(x)$ with $|\mathbb{P}_{Y''}(x)| \geq e^{-\sqrt{\log(a^{-1})}}m$, then

$$\max_{P_1, P_2 \in \mathbb{P}_{Y''}(x)} \angle(TP_1, TP_2) \approx \theta. \quad (14.3.35)$$

As per Remark 14.3.44, if $(\mathbb{P}, Y)$ already has a typical angle of intersection, then we can skip this step.

In the rest of the proof, we will continue to refine $\mathbb{P}$ and Y_1 . We will consider various $\tilde{\mathbb{P}} \subset \mathbb{P}$ and $\tilde{Y}$ refining Y_1 . For each refinement $(\tilde{\mathbb{P}}, \tilde{Y})$ that we consider, we will check that

$$\sum_{P \in \tilde{\mathbb{P}}} |\tilde{Y}(P)| \geq (\log 1/a)^{-O(1)} \sum_{P \in \mathbb{P}} |Y(P)| \quad (14.3.36)$$

We will call a refinement good if it obeys (14.3.36). If $\tilde{Y}$ is a good refinement, we will further refine $\tilde{Y}$ to have roughly constant multiplicity:

$$\text{For each } x \in U(\tilde{\mathbb{P}}, \tilde{Y}), |\mathbb{P}_{\tilde{Y}}(x)| \gtrsim (\log 1/a)^{-O(1)}m. \quad (14.3.37)$$

Combining (14.3.37) and (14.3.35), we see that θ is a typical angle of intersection for $(\tilde{\mathbb{P}}, \tilde{Y})$. Equation (14.3.36) also implies that each shading enjoys $\lambda(\tilde{\mathbb{P}}, \tilde{Y}) \geq (\log 1/a)^{-O(1)}\lambda(\mathbb{P}, Y_1) \geq a^\eta$ and $|\tilde{\mathbb{P}}| \geq (\log 1/a)^{-O(1)}|\mathbb{P}|$.

Informally, we find good refinements either by pigeonholing or by removing parts of our shaded set that are far below average in some respect. Now we turn to the details.

By pigeonholing, we may find a good refinement $(\mathbb{P}_2, Y_2)$ and a number $N \geq 1$ so that each set P_θ contains about N prisms $P \in \mathbb{P}_2$. Next we consider slabs $S \subset B_1$ with dimensions $\theta \times 1 \times 1$. For a given $\theta \times 1 \times 1$ slab S , define $(\mathbb{P}_2)_S$ as in (14.3.29). We note that $\mathbb{P}_2 = \sqcup_S \mathbb{P}_{2, S}$, where the union is over all essentially distinct $\theta \times 1 \times 1$ slabs $S \subset B_1$.

Because the maximal angle of intersecting planks in $(\mathbb{P}_2, Y_2)$ is $\lesssim \theta$, each point $x \in U(\mathbb{P}_2, Y_2)$ lies in $U(\mathbb{P}_{2,S}, Y_{2,S})$ for $\lesssim 1$ sets S . We know that $|\mathbb{P}_{2,Y_2}(x)| \geq (\log 1/a)^{O(1)}m$ and so there must be some S so that $|\mathbb{P}_{2,S,Y_2}(x)| \geq (\log 1/a)^{O(1)}m$. Now we can find a good refinement $(\mathbb{P}_2, Y_3)$ so that for each S and each $x \in U(\mathbb{P}_{2,S}, Y_{3,S})$, we have $|\mathbb{P}_{2,S,Y_3,S}(x)| \gtrsim (\log 1/a)^{O(1)}m$.

Let $\mathbb{S}$ be the set of S so that $\lambda(\mathbb{P}_{2,S}, Y_{3,S}) \gtrsim \lambda(\mathbb{P}_2, Y_3) \gtrsim a^\eta$. Now define $\mathbb{P}' = \sqcup_{S \in \mathbb{S}} \mathbb{P}_{2,S}$ and $\mathbb{P}'_S = \mathbb{P}_{2,S}$. Note that $(\mathbb{P}', Y_3)$ is a good refinement of $(\mathbb{P}_2, Y_3)$.

Now consider a box Q of dimensions $\theta b \times b \times b$. Define

$$\mathbb{P}'_Q = \{P \in \mathbb{P}' : |P \cap Q| \approx ab^2\}.$$

If $P \in \mathbb{P}'_Q$, then $P \cap Q$ is a slab of dimensions $a \times b \times b$. We want to use our estimates about slabs to study these slabs inside of Q . Let $\mathbb{S}_Q = \{P \cap Q : P \in \mathbb{P}'_Q\}$. (Note that the slabs in $\mathbb{S}_Q$ need not be essentially distinct.)

The slabs of $\mathbb{S}_Q$ come with a natural shading $Y_{3,Q}$, where $Y_{3,Q}(P \cap Q) = Y_3(P) \cap Q$. We will apply Lemma 14.3.38 to estimate $|U(\mathbb{S}_Q, Y_{3,Q})|$. We know that θ is a typical angle of intersection for $\mathbb{P}'_{Y_3}$ in the sense of Definition 14.3.42, and this implies that θ is also a typical angle of intersection for $(\mathbb{S}_Q, Y_{3,Q})$ in the sense of Definition 14.3.37. To apply our Lemma 14.3.38 about slabs, we need to estimate $\lambda(Y_{3,Q})$. Say Q is dense if $\lambda(Y_{3,Q}) \gtrsim \lambda(Y_{3,S}) \gtrsim a^\eta$. If Q is dense, then Lemma 14.3.38 gives the key inequality:

$$|U(\mathbb{P}'_S, Y_{3,S}) \cap Q| \geq |U(\mathbb{S}_Q, Y_{3,Q})| \gtrsim a^{3\eta}|Q|. \quad (14.3.38)$$

Notice that there is an essentially unique slab S of dimensions $\theta \times 1 \times 1$ so that $Q \subset S$ and $\angle(TQ, TS) \lesssim \theta$. For this slab S , $\mathbb{P}'_Q \subset \mathbb{P}'_S$. Moreover, we can tile S with a set $\mathbb{Q}_S$ of boxes Q so that for each $P \in \mathbb{P}'_S$,

$$P = \sqcup_{Q \in \mathbb{Q}_S} P \cap Q,$$

where $P \in \mathbb{P}'_Q$ whenever $P \cap Q$ is non-empty.

Note that a plank P lies in $\mathbb{P}_Q$ for many boxes Q , and these Q form an essentially disjoint cover of P . Let $\mathbb{Q}_P = \{Q : P \in \mathbb{P}'_Q\}$. We define

$$Y_4(P) = \bigcup_{Q \in \mathbb{Q}_P, Q \text{ dense}} Y_3(P) \cap Q.$$

Next we show that $Y_{4,S}$ is a good refinement of $Y_{3,S}$ for each $S \in \mathbb{S}$. To see this, we note that

$$\sum_{P \in \mathbb{P}'_S} |Y_{3,S}(P)| = \lambda(\mathbb{P}'_S, Y_{3,S}) \sum_{P \in \mathbb{P}'_S} |P|.$$

On the other hand,

$$\begin{aligned} \sum_{P \in \mathbb{P}'_S} |Y_{3,S}(P) \setminus Y_{4,S}(P)| &= \sum_{Q \in \mathbb{Q}_S, Q \text{ not dense}} \sum_{P \in \mathbb{P}'_Q} |Y_{3,S}(P) \cap Q| \leq \\ (1/100)\lambda(\mathbb{P}'_S, Y_{3,S}) \sum_{Q \in \mathbb{Q}_S} \sum_{P \in \mathbb{P}'_Q} |P \cap Q| &= (1/100)\lambda(\mathbb{P}'_S, Y_{3,S}) \sum_{P \in \mathbb{P}'_S} |P|. \end{aligned}$$

If $B_{\theta b} \subset Q$, we say $B_{\theta b}$ is dense if $|U(\mathbb{P}'_S, Y_{4,S}) \cap B_{\theta b}| \gtrsim a^{3\eta}$. We let $D(Q) \subset Q$ be the union of the dense balls $B_{\theta b}$. We can now further refine Y_4 so that it is contained in the union of these dense balls: if $P \in \mathbb{P}'_S$, we define

$$Y'(P) = \bigcup_{Q \in \mathbb{Q}_P, Q \text{ dense}} Y_{4,S}(P) \cap D(Q).$$

Note that if Q is dense, then $|U(\mathbb{P}', Y_4) \cap Q \setminus U(\mathbb{P}', Y') \cap Q| \ll |U(\mathbb{P}', Y') \cap Q|$. Since the multiplicity of $(\mathbb{P}'_S, Y_{4,S})$ is roughly constant, removing low density balls is a good refinement. So all together $(\mathbb{P}', Y')$ is a good refinement of $(\mathbb{P}, Y)$.

The definition of Y' in terms of dense balls gives item 1.

And now for each plank $P_\theta \in \mathbb{P}'_{\theta,S}$, define

$$Y_{\theta,S}(P_\theta) = U(\mathbb{P}'_S, Y') \cap P_\theta. \quad (14.3.39)$$

Every $P \in \mathbb{P}'_S$ is contained in its associated $P_\theta \in \mathbb{P}'_{\theta,S}$. Hence every point of $U(\mathbb{P}'_S, Y')$ belongs to at least one set $U(\mathbb{P}'_S, Y') \cap P_\theta$, while the reverse containment holds by definition. Taking the union over P_θ proves

$$U(\mathbb{P}'_{\theta,S}, Y_{\theta,S}) = U(\mathbb{P}'_S, Y').$$

If $P \subset P_{\theta,S}$, then

$$\frac{|Y_{\theta,S}(P_\theta)|}{|P_\theta|} \gtrsim a^{3\eta} \frac{\#\{Q \subset P_\theta, Q \text{ dense}\}}{\#\{Q \subset P_\theta\}} \geq a^{3\eta} \frac{|Y'(P)|}{|P|}.$$

Recall that for each $P_\theta \in \mathbb{P}'_{\theta,S}$, there are $\sim N$ planks $P \in \mathbb{P}'_S$ with $P \subset P_\theta$. Therefore, we have

$$\lambda(\mathbb{P}'_{\theta,S}, Y_{\theta,S}) \gtrsim a^{3\eta} \lambda(\mathbb{P}'_S, Y') \gtrsim a^{4\eta} \quad (14.3.40)$$

This proves item 2.

We also have

$$|U(\mathbb{P}', Y)| \geq |U(\mathbb{P}', Y')| \gtrsim \sum_{S \in \mathbb{S}} |U(\mathbb{P}'_S, Y'_S)| = \sum_{S \in \mathbb{S}} |U(\mathbb{P}'_{\theta,S}, Y_{\theta,S})|. \quad (14.3.41)$$

This proves item 3.

We can also use this setup to control $\mu(\mathbb{P}', Y)$. Notice that since each point x lies in $U(\mathbb{P}'_S, Y'_S)$ for $\lesssim 1$ slabs $S \in \mathbb{S}$, we have

$$\mu(\mathbb{P}', Y) \lesssim \max_{S \in \mathbb{S}} \mu(\mathbb{P}'_S, Y'_S).$$

Next we estimate $\mu(\mathbb{P}'_S, Y'_S)$ using that $U(\mathbb{P}'_S, Y'_S) = U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})$. We know that

$$\mu(\mathbb{P}'_S, Y'_S) = \frac{\sum_{P \in \mathbb{P}'_S} |Y'(P)|}{|U(\mathbb{P}'_S, Y'_S)|} \lesssim \frac{\sum_{P \in \mathbb{P}'_S} |P|}{|U(\mathbb{P}'_S, Y'_S)|} = \frac{|\mathbb{P}'_S| |P|}{|U(\mathbb{P}'_S, Y'_S)|} = \frac{|\mathbb{P}'_S| |P|}{|U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})|},$$

and

$$\mu(\mathbb{P}'_{\theta, S}, Y_{\theta, S}) = \frac{\sum_{P_{\theta} \in \mathbb{P}'_{\theta, S}} |Y_{\theta, S}(P_{\theta})|}{|U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})|} \gtrsim a^{4\eta} \frac{\sum_{P_{\theta} \in \mathbb{P}'_{\theta, S}} |P_{\theta}|}{|U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})|} = a^{4\eta} \frac{|\mathbb{P}'_{\theta, S}| |P_{\theta}|}{|U(\mathbb{P}'_{\theta, S}, Y_{\theta, S})|}.$$

We have

$$|\mathbb{P}'_{\theta, S}| \approx \frac{1}{N} |\mathbb{P}'_S|,$$

$$|P_{\theta}| = \frac{\theta b}{a} |P|.$$

Plugging in gives

$$\mu(\mathbb{P}, Y) \lesssim \mu(\mathbb{P}'_S, Y'_S) \lesssim a^{-4\eta} \frac{aN}{b\theta} \mu(\mathbb{P}'_{\theta, S}, Y_{\theta}).$$

This proves item 4. □

By a linear change of variables, we can convert $\mathbb{P}_{\theta, S}$ from a set of $\theta b \times b \times 1$ planks in S into a set $\mathbb{T}$ of b -tubes in B_1 . To see this more carefully, recall that $\mathbb{P}_{\theta, S}$ consists of $\theta b \times b \times 1$ planks P in the slab S so that TP is tangent to TS . The slab S has dimensions $\theta \times 1 \times 1$. Let $L : S \rightarrow B_1$ be a linear change of variables. Because of the tangency condition on P , $L(P)$ has dimensions $b \times b \times 1$.

To summarize, we have reduced the original incidence problem for $\mathbb{P}$ to an incidence problem for $\mathbb{P}_{\theta, S}$, which is equivalent to an incidence problem for tubes $\mathbb{T}$. If we assume $K_{KT}(\beta)$ or $K_F(\beta)$, then we can get bounds for $\mu(\mathbb{T})$ and hence bounds for $\mu(\mathbb{P})$. We are now ready to prove Lemmas 14.3.31 and 14.3.34.

14.3.6.3 A Katz-Tao multiplicity estimate

Proof of Lemma 14.3.31. We may suppose that $\mu(\mathbb{P}, Y) \geq a^{-\eta}$, since otherwise we are done by Remark 14.3.33. Apply Lemma 14.3.43. Abusing notation, we will continue to refer to the resulting refinement by $(\mathbb{P}, Y)$. By Lemma 14.3.43, there is a number $\theta \in [a/b, 1]$, a number N , and a slab S of dimensions $\theta \times 1 \times 1$ so that we have

$$\mu(\mathbb{P}, Y) \lesssim a^{-3\eta} \frac{aN}{b\theta} \mu(\mathbb{P}_{\theta,S}, Y_{\theta,S}). \quad (14.3.42)$$

Lemma 14.3.43 also gives a density lower bound

$$\lambda(\mathbb{P}_{\theta,S}, Y_{\theta,S}) \gtrsim a^{4\eta}.$$

By a linear change of variables, we can convert $\mathbb{P}_{\theta,S}$ to a set of b -tubes $\mathbb{T}$ in B_1 . Since convex sets are invariant with respect to linear change of variables, we have

$$\Delta_{\max}(\mathbb{T}) = \Delta_{\max}(\mathbb{P}_{\theta,S}) \leq \Delta_{\max}(\mathbb{P}_{\theta}) \leq \frac{b\theta}{aN} \Delta_{\max}(\mathbb{P}).$$

Now apply Lemma 14.3.13 (noting Remark 14.3.10) with $b, \epsilon/2$ in place of δ, ϵ . If η and b_0 are chosen sufficiently small, then

$$\begin{aligned} \mu(\mathbb{P}_{\theta,S}) &= \mu(\mathbb{T}) \leq a^{-\epsilon/2} \Delta_{\max}(\mathbb{T})^{1-\beta} |\mathbb{T}|^{\beta} \\ &\lesssim a^{-\epsilon/2} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{b\theta}{aN} \right)^{1-\beta} (|\mathbb{P}_{\theta,S}|)^{\beta} \\ &\lesssim a^{-\epsilon/2} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{b\theta}{aN} \right)^{1-\beta} (N^{-1} |\mathbb{P}_S|)^{\beta}. \end{aligned}$$

And so plugging in (14.3.42), we get

$$\begin{aligned} \mu(\mathbb{P}, Y) &\lesssim a^{-3\eta} \frac{aN}{b\theta} \mu(\mathbb{P}_{\theta,S}) \\ &\leq a^{-\epsilon/2-3\eta} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{b\theta}{aN} \right)^{-\beta} (N^{-1} |\mathbb{P}_S|)^{\beta} \\ &\leq a^{-\epsilon/2-3\eta} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{a}{b\theta} \right)^{\beta} |\mathbb{P}_S|^{\beta}. \end{aligned} \quad (14.3.43)$$

We know $a \leq b\theta$ (and thus $\frac{a}{b} \leq (\frac{a}{b})^{\gamma} \theta^{1-\gamma}$), and $|\mathbb{P}_S| \leq a^{-\eta} \theta^{\gamma} |\mathbb{P}|$. Plugging in, we get

$$\mu(\mathbb{P}) \leq a^{-\epsilon} \Delta_{\max}(\mathbb{P})^{1-\beta} \left(\frac{a}{b} \right)^{\gamma\beta} |\mathbb{P}|^{\beta}. \quad \square$$

14.3.6.4 A Frostman multiplicity estimate

Proof of Lemma 14.3.34. We may suppose that $\mu(\mathbb{P}, Y) \geq a^{-\eta}$, since otherwise we are done by Remark 14.3.35. Apply Lemma 14.3.43. Abusing notation, we will continue to refer to the resulting refinement by $(\mathbb{P}, Y)$.

By Lemma 14.3.43, there is a number $\theta \in [a/b, 1]$ and a set $\mathbb{S}$ of $\theta \times 1 \times 1$ slabs so that

$$|U(\mathbb{P}, Y)| \gtrsim |\mathbb{S}| |U(\mathbb{P}_{\theta, S}, Y_{\theta, S})|.$$

Lemma 14.3.43 also gives a density lower bound

$$\lambda(\mathbb{P}_{\theta, S}, Y_{\theta, S}) \gtrsim a^{4\eta}.$$

Define $C = C_F(\mathbb{P}, B_1)$. This does not imply that $C_F(\mathbb{P}_{\theta, S}, S) \leq C$. So we have to work out what we know about $\mathbb{P}_{\theta, S}$. Let $\mathbb{P}_\theta$ be formed by thickening the planks of $\mathbb{P}$ to dimensions $\theta b \times b \times 1$. By Lemma 14.3.43, there is a number $N \leq M\theta$ so that each plank $P_\theta \in \mathbb{P}_\theta$ contains $\sim N$ planks of $\mathbb{P}$ (recall that we have replaced $\mathbb{P}$ by a ≈ 1 refinement, but the refinement preserves the upper bound $|\mathbb{P}[P_\theta]| \leq M\theta$).

By Remark 14.3.7(A), $C_F(\mathbb{P}_\theta, B_1) \lesssim C_F(\mathbb{P}, B_1) = C$, and so

$$|\mathbb{P}_{\theta, S}| \leq |\mathbb{P}_\theta[S]| \leq C\theta |\mathbb{P}_\theta|.$$

The set $\mathbb{P}_{\theta, S}$ is not necessarily Frostman, but since $\mathbb{P}_\theta$ is Frostman, we do know that for any convex $K \subset S$,

$$\Delta(\mathbb{P}_{\theta, S}, K) \leq \Delta(\mathbb{P}_\theta, K) \leq C\Delta(\mathbb{P}_\theta, B_1) \sim C|P_\theta| |\mathbb{P}_\theta|. \quad (14.3.44)$$

Since $|\mathbb{P}_\theta| \approx |\mathbb{S}| |\mathbb{P}_{\theta, S}|$,

$$\Delta(\mathbb{P}_{\theta, S}, K) \lesssim C|\mathbb{S}| |S| \frac{|P_\theta| |\mathbb{P}_{\theta, S}|}{|S|} = C|\mathbb{S}| |S| \Delta(\mathbb{P}_{\theta, S}, S).$$

Therefore $C_F(\mathbb{P}_{\theta, S}, S) \lesssim C|\mathbb{S}| |S|$.

Now by a linear change of variables, we can convert $\mathbb{P}_{\theta, S}$ to a set of b -tubes $\mathbb{T}$ in B_1 . Let $Y_{\mathbb{T}}$ be the image of the shading $Y_{\theta, S}$ under this change of variables. Since convex sets are invariant with respect to linear change of variables, $C_F(\mathbb{T}) \lesssim C|\mathbb{S}| |S|$. Also, $\lambda(\mathbb{T}, Y_{\mathbb{T}}) = \lambda(P_{\theta, S}, Y_{\theta, S}) \gtrsim a^{4\eta}$.

If η and b_0 are chosen sufficiently small depending on ε , then by Lemma 14.3.15 (noting Remark 14.3.10) we have

$$\begin{aligned} |U(\mathbb{T})| &\gtrsim a^{\varepsilon/2} (C|\mathbb{S}| |S|)^{\beta/2-1} b^{2\beta} (b^2 |\mathbb{T}|)^{\beta/2} \\ &\approx a^{\varepsilon/2} C^{\beta/2-1} (|\mathbb{S}| |S|)^{-1} b^{2\beta} (b^2 |S| |\mathbb{P}_\theta|)^{\beta/2} \\ &= a^{\varepsilon/2} C^{\beta/2-1} (|\mathbb{S}| |S|)^{-1} b^{2\beta} (b^2 \theta |\mathbb{P}|/N)^{\beta/2} \\ &\geq a^{\varepsilon/2} C^{\beta/2-1} (|\mathbb{S}| |S|)^{-1} b^{2\beta} (M^{-1} b^2 |\mathbb{P}|)^{\beta/2}. \end{aligned}$$

Therefore if $b_0 > 0$ is sufficiently small, we have

$$|U(\mathbb{P})| \approx |\mathbb{S}| |U(\mathbb{P}_{\theta, S})| \approx |\mathbb{S}| |S| |U(\mathbb{T})| \geq a^\epsilon C^{\beta/2-1} b^{2\beta} (M^{-1} b^2 |\mathbb{P}|)^{\beta/2}.$$

This gives the desired lower bound for $|U(\mathbb{P})|$. For the multiplicity estimate, $|P| = ab$ and $Y(P) \subset P$, so the lower bound just proved gives

$$\begin{aligned} \mu(\mathbb{P}, Y) &= \frac{\sum_{P \in \mathbb{P}} |Y(P)|}{|U(\mathbb{P}, Y)|} \leq \frac{|\mathbb{P}| ab}{a^\epsilon C^{\beta/2-1} b^{2\beta} (M^{-1} b^2 |\mathbb{P}|)^{\beta/2}} \\ &= a^{-\epsilon} C^{1-\beta/2} M^{\beta/2} \frac{a}{b} b^{-2\beta} (b^2 |\mathbb{P}|)^{1-\beta/2}, \end{aligned}$$

which is the asserted multiplicity estimate. $\square$

14.3.6.5 Tubes that factor through planks

In this section we will prove Proposition 14.3.36.

Part (A). Let $C = C_F(\mathbb{T})$. After refining $\mathbb{T}_\rho$ by a $O(1)$ factor (this induces a harmless refinement on $\mathbb{T}$), we may suppose that each $T \in \mathbb{T}$ is contained in exactly one tube $T_\rho \in \mathbb{T}_\rho$. By hypothesis, for each $T_\rho \in \mathbb{T}_\rho$, there is a set $\mathbb{W}_{T_\rho}$ of $a \times b \times 1$ planks that factors $\mathbb{T}[T_\rho]$, i.e. $\mathbb{T}[T_\rho] = \bigsqcup_{\mathbb{W}_{T_\rho}} \mathbb{T}[T_\rho]_W$, where $\mathbb{T}[T_\rho]_W$ is ~ 1 Frostman in W . Define $\mathbb{W} = \bigsqcup_{\mathbb{T}_\rho} \mathbb{W}_{T_\rho}$. There is a corresponding decomposition $\mathbb{T} = \bigsqcup_{\mathbb{W}} \mathbb{T}_W$, where $\mathbb{T}_W$ is ~ 1 Frostman in W . After pigeonholing and refining $\mathbb{T}$ and $\mathbb{W}$, we may suppose that $|Y(T)|$ is approximately the same for each $T \in \mathbb{T}$, and $|\mathbb{T}_W|$ is approximately the same for each $W \in \mathbb{W}$. After this step we have that $C_F(\mathbb{T}) \lesssim C$, and $\lambda(\mathbb{T}, Y) \gtrsim \delta^\eta$.

Apply Proposition 14.3.18 to the pair $(\mathbb{T}, Y)$ and $\mathbb{W}$; we obtain a set $\mathbb{W}' \subset \mathbb{W}$, and a shading $Y_{\mathbb{W}'}$ on $\mathbb{W}'$ with $\lambda(\mathbb{W}', Y_{\mathbb{W}'}) \gtrsim \delta^{2\eta}$. We also obtain a shading $Y'(T) \subset Y(T)$ on $\mathbb{T}' = \bigcup_{\mathbb{W}'} \mathbb{T}_W$, so that $(\mathbb{T}', Y')$ is an ≈ 1 refinement of $(\mathbb{T}, Y)$, and

$$\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{W}', Y_{\mathbb{W}'}) \mu(\mathbb{T}_W, Y') \text{ for all } W \in \mathbb{W}'. \quad (14.3.45)$$

We have $C_F(\mathbb{T}_W, W) \lesssim 1$ for each $W \in \mathbb{W}'$. Since $C_F(\mathbb{T}') \lesssim C$ and $|\mathbb{T}_W|$ is approximately the same for all $W \in \mathbb{W}'$, by Remark 14.3.7(A) we have that $C_F(\mathbb{W}') \lesssim C$. Thus we will use Lemma 14.3.34 to bound each of the terms on the RHS of (14.3.45). In order to apply Lemma 14.3.34, we need to control to what extent the planks in $\mathbb{W}'$ can concentrate into thicker planks, and similarly for the (rescaled) tubes in $\mathbb{T}_W$ (this is the quantity “ M ” from Lemma 14.3.34).

We begin with $\mathbb{W}'$. For each $\theta \in [a/b, 1]$ we need to bound the maximum number of $a \times b \times 1$ planks from $\mathbb{W}'$ that are contained in a $\theta b \times b \times 1$ plank. Denote this number by $N(\theta)$. Fix a plank $W \in \mathbb{W}'$. Recall that $W \in \mathbb{W}_{T_\rho}$ for some $T_\rho \in \mathbb{T}_\rho$. Let W_θ be the corresponding $\theta b \times b \times 1$ plank. Let $W' \in \mathbb{W}'[W_\theta]$. Since W' contains at least one tube from $\mathbb{T}$, and each tube in $\mathbb{T}$ is contained

in exactly one tube from T_ρ , we have T_ρ is the unique tube containing W' , and thus $W' \in \mathbb{W}_{T_\rho}$. We conclude that $\mathbb{W}'[W_\theta] = \mathbb{W}_{T_\rho}[W_\theta]$. Recall that $\Delta_{\max}(\mathbb{W}_{T_\rho}) \lesssim 1$, and thus $|\mathbb{W}_{T_\rho}[W_\theta]| \lesssim |W_\theta|/|W| = \frac{\theta b \times b \times 1}{a \times b \times 1} = \theta(b/a)$. Thus $|\mathbb{W}_{T_\rho}[W_\theta]| \leq M\theta$ for $M = b/a$. Applying Lemma 14.3.34 (and recalling Remark 14.3.10), we have

$$\mu(\mathbb{W}', Y_{\mathbb{W}'}) \lesssim \delta^{-\epsilon/2} C_F(\mathbb{T})^{1-\beta/2} (a/b)^{1-\beta/2} b^{-2\beta} (b^2 |\mathbb{W}|)^{1-\beta/2}. \quad (14.3.46)$$

We remark that in order to apply Lemma 14.3.34, we need to verify that b is sufficiently small ($b \leq \delta^\epsilon$ will suffice). But if this is not the case, then (14.3.33) follows from Lemma 14.3.39.

Next we consider $\mathbb{T}_W$. Fix a prism $W \in \mathbb{W}'$. We first have to make a change of coordinates that maps W to B_1 and each δ tube $T \in \mathbb{T}_W$ to a $b^{-1}\delta \times a^{-1}\delta \times 1$ plank. Next we need to bound the number of $b^{-1}\delta \times a^{-1}\delta \times 1$ planks in a $\theta \times a^{-1}\delta \times a^{-1}\delta \times 1$ thick plank. Undoing the coordinate change, this is the number of tubes of $\mathbb{T}_W$ in a $\delta \times \frac{\theta b}{a} \delta \times 1$ plank. Since the tubes of $\mathbb{T}$ are essentially distinct, there are $\lesssim \left(\frac{\theta b}{a}\right)^2 \leq (b/a)^2 \theta$ such tubes. Applying Lemma 14.3.34 and remembering that the dimensions of each plank are $b^{-1}\delta \times a^{-1}\delta \times 1$ (and using our bound $M \leq (b/a)^2$), we get

$$\mu(\mathbb{T}_W, Y') \leq \delta^{-\epsilon/2} (a/b)^{1-\beta} (a^{-1}\delta)^{-2\beta} (a^{-2}\delta^2 |\mathbb{T}_W|)^{1-\beta/2}. \quad (14.3.47)$$

Combining (14.3.45), (14.3.46), and (14.3.47), we get (14.3.33).

Part (B). By hypothesis, there is a set $\mathbb{W}$ of $a \times b \times 1$ prisms that factors $\mathbb{T}_\rho$, i.e. $\mathbb{T}_\rho = \bigsqcup_{\mathbb{W}} \mathbb{T}_{\rho,W}$, and $\mathbb{T}_{\rho,W}$ is ~ 1 Frostman in W . This induces a decomposition $\mathbb{T} = \bigsqcup_{\mathbb{W}} \mathbb{T}_W$. Apply Proposition 14.3.18 to $(\mathbb{T}, Y)$ and $\mathbb{W}$; while $\mathbb{T}_W$ might not be Frostman in W , we have that $\mathbb{T}_{\rho,W}$ is Frostman in W . By Remark 14.3.20, this is enough to apply the proposition. We obtain a set $\mathbb{W}' \subset \mathbb{W}$, a shading $Y_{\mathbb{W}'}$ on $\mathbb{W}'$ with $\lambda(\mathbb{W}', Y_{\mathbb{W}'}) \gtrsim \delta^{2\eta}$, and a sub-shading $Y'(T) \subset Y(T)$ on $\mathbb{T}' = \bigsqcup_{\mathbb{W}'} \mathbb{T}_W$, so that $(\mathbb{T}', Y')$ is an ≈ 1 refinement of $(\mathbb{T}, Y)$, and

$$\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{W}', Y_{\mathbb{W}'}) \mu(\mathbb{T}_W, Y') \text{ for all } W \in \mathbb{W}'. \quad (14.3.48)$$

We have $\Delta_{\max}(\mathbb{T}_W) \leq \Delta_{\max}(\mathbb{T})$, and $\Delta_{\max}(\mathbb{W}) \lesssim 1$. We will bound the two terms on the RHS of (14.3.48) using Lemma 14.3.31 with $\gamma = 0$ and $\gamma = 1$, respectively.

We begin with $\mathbb{W}$. Applying Lemma 14.3.31 with $\gamma = 0$ (and recalling Remark 14.3.10), we obtain

$$\mu(\mathbb{W}', Y_{\mathbb{W}'}) \lesssim \delta^{-\epsilon/2} |\mathbb{W}|^\beta. \quad (14.3.49)$$

Next we consider $\mathbb{T}_W$. Select a set $W \in \mathbb{W}$ so that $\lambda(\mathbb{T}_W, Y') \gtrsim \delta^\eta$. To apply Lemma 14.3.31 with $\gamma = 1$, we make a change of coordinates that maps

W to B_1 . This change of coordinates converts $\mathbb{T}_W$, a set of δ -tubes $T \subset W$, to $\mathbb{P}$, a set of $b^{-1}\delta \times a^{-1}\delta \times 1$ planks $P \subset B_1$. Notice that the eccentricity of the planks is $\frac{a^{-1}\delta}{b^{-1}\delta} = \frac{a}{b}$. Let $Y_{\mathbb{P}}$ be the image of the shading $Y(T)$ after this transformation.

Next we need to check that $\mathbb{P}$ obeys the assumptions of Lemma 14.3.31 with $\gamma = 1$. The first assumption of Lemma 14.3.31 says that if $\theta \in [a/b, 1]$ and S is a $\theta \times 1 \times 1$ plank, then $|\mathbb{P}_S| \lesssim (b^{-1}\delta)^{-\eta}\theta|\mathbb{P}|$. If we change coordinates back from B_1 to W , then S is converted to a convex set $S' \subset W$ and our desired bound becomes $|\mathbb{T}[S']| \lesssim (b^{-1}\delta)^{-\eta} \frac{|S'|}{|W|} |\mathbb{T}_W|$. Suppose that S' has dimensions $a' \times b' \times c'$ with $a' \leq b' \leq c'$. We claim that $a' \gtrsim a \geq \rho$. We will come back to this claim below. Assuming this claim, we have

$$\frac{|\mathbb{T}[S']|}{|\mathbb{T}_W|} \approx \frac{|\mathbb{T}_{\rho}[S']|}{|\mathbb{T}_{\rho,W}|} \lesssim \frac{|S'|}{|W|} C_F(\mathbb{T}_{\rho,W}, W) \lesssim \frac{|S'|}{|W|}.$$

This gives the desired bound on $|\mathbb{T}[S']|$.

To finish we have to check that $a' \geq a$. Choose coordinates x_1, x_2, x_3 on W so that W is the rectangular box $0 \leq x_1 \leq a$, $0 \leq x_2 \leq b$, and $0 \leq x_3 \leq 1$. The change of coordinates ϕ mapping W to B_1 is $\phi(x_1, x_2, x_3) = (\frac{x_1}{a}, \frac{x_2}{b}, x_3)$.

We can assume that $\mathbb{P}_S$ is non-empty, and so there is at least one $P \in \mathbb{P}$ with $TP = TS$ (up to angle θ). Suppose $P = L(T)$ where T is a tube from $\mathbb{T}[W]$.

Recall that TS is a 2-plane. Let $TS' = \phi^*(TS)$, another 2-plane. Choose unit vectors v_1, v_2 to make a basis for TS' with v_1 in the (x_1, x_2) -plane and v_2 parallel to T . We have $|v_1 \wedge v_2| \sim 1$, since T is transverse to the (x_1, x_2) plane. Since ϕ maps TS' to $TS = TP$, we have $|\phi(v_1) \wedge \phi(v_2)| \sim a^{-1}$. Since $|\phi(v_2)| \sim 1$ we see $|\phi(v_1)| \sim a^{-1}$, and so v_1 is transverse to the x_2 -direction. Let $n(S')$ be the vector normal to TS' . Since v_2 is almost in the x_3 direction and v_1 is transverse to the x_2 -direction, we have $n(S')$ transverse to the x_1 direction. Now W can be covered by θ^{-1} parallel copies of S' and so $a'\theta^{-1} \gtrsim b$ and so $a' \geq \theta b \gtrsim a$. This confirms the first assumption in 14.3.31 with $\gamma = 1$.

We can now apply Lemma 14.3.31 with $\gamma = 1$. Recall that $\Delta_{\max}(\mathbb{P}) = \Delta_{\max}(\mathbb{T}_W) \leq \Delta_{\max}(\mathbb{T})$. We have

$$\mu(\mathbb{T}_j[W]) \leq \delta^{-\epsilon/2} \Delta_{\max}(\mathbb{T})^{1-\beta} \left(\frac{a}{b}\right)^{\beta} |\mathbb{T}_W|^{\beta}. \quad (14.3.50)$$

Combining (14.3.48), (14.3.49), and (14.3.50), we obtain (14.3.34).

14.3.7 Stickyakeya and a sticky / non-sticky dichotomy

In this section we will discuss two variants of the stickyakeya theorem. Let $\mathbb{T}$ be a uniform set of δ -tubes. Let $\rho \in [\delta, 1]$ and let $\mathbb{T}_\rho$ be the thickening of $\mathbb{T}$ at scale ρ . We begin with the following two observations

- (A) If $\mathbb{T}$ is Frostman, then by Remark 14.3.7(A), $\mathbb{T}_\rho$ is Frostman. In general, however, the tubes in $\mathbb{T}[T_\rho]$ will not be Frostman inside T_ρ .
- (B) If $\mathbb{T}$ is Katz-Tao, then by Remark 14.3.7(B), the tubes in $\mathbb{T}[T_\rho]$ will be Katz-Tao. In general, however, the tubes in $\mathbb{T}_\rho$ will not be Katz-Tao.

It will be helpful to consider the special cases where the sets $\mathbb{T}[T_\rho]$ are Frostman in T_ρ , or the set $\mathbb{T}_\rho$ is Katz-Tao.

Definition 14.3.45. Let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^n$.

- (A) We say that $\mathbb{T}$ is *Frostman at every scale with error C* if for every scale $\delta \leq \rho \leq 1$, we have

$$C_F(\mathbb{T}[T_\rho], T_\rho) \leq C \quad \text{for every } T_\rho \in \mathbb{T}_\rho.$$

- (B) We say that $\mathbb{T}$ is *Katz-Tao at every scale with error C* if for every scale $\delta \leq \rho \leq 1$, we have

$$\Delta_{\max}(\mathbb{T}_\rho) \leq C.$$

We will usually consider sets of tubes that are uniform, in the sense of Definition 14.3.3, but this is not required in the above definition.

Remark 14.3.46. Definition 14.3.45(A) is closely related to Definition 2.12 from [7]: if $\varepsilon > 0$ and $\delta > 0$ is sufficiently small depending on ε , and if $\mathbb{T}$ is a uniform set of δ -tubes that is Frostman at every scale with error $\delta^{-\varepsilon}$, then $\mathbb{T}$ “satisfies the Convex Wolff Axioms at every scale with error $\delta^{-\varepsilon}$ ” in the sense of [7], Definition 2.12.

Recall that a set $\mathbb{T}$ is sticky if it is both Frostman at every scale and Katz-Tao at every scale. In that case, the stickyakeya theorem says that $|U(\mathbb{T})| \gtrsim 1$ and $\mu(\mathbb{T}) \lesssim 1$. A mild generalization of this result says that if $\mathbb{T}$ is Frostman at every scale then $|U(\mathbb{T})| \gtrsim 1$, while if $\mathbb{T}$ is Katz-Tao at every scale, then $\mu(\mathbb{T}) \lesssim 1$.

Theorem 14.3.47. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)$ be a uniform set of δ -tubes in $B_1 \subset \mathbb{R}^3$, with $\lambda(\mathbb{T}, Y) \geq \delta^\eta$.*

(A) If $\mathbb{T}$ is Frostman at every scale with error $\delta^{-\eta}$, then

$$|U(\mathbb{T}, Y)| \geq \delta^\varepsilon.$$

(B) If $\mathbb{T}$ is Katz-Tao at every scale with error $\delta^{-\eta}$, then

$$\mu(\mathbb{T}, Y) \leq \delta^{-\varepsilon}.$$

Theorem 14.3.47(A) is Theorem 5.2 from [7] (note that [7] defines “Frostman at every scale” slightly differently, but this is not a problem—see Remark 14.3.46). To prove Theorem 14.3.47(B), we first need a simple multiscale lemma about Δ_{max} .

Lemma 14.3.48. *Let $\mathbb{T}$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^n$ and let $\delta = \rho_M \leq \rho_{M-1} \leq \dots \leq \rho_1 \leq \rho_0 = 1$ be a sequence of scales. Suppose that $\mathbb{T}$ is uniform at each scale ρ_k , in the sense that there is a set $\mathbb{T}_{\rho_k}$ of ρ_k -tubes for which Items (i), (ii) and (iii) from Definition 14.3.3 hold. Then*

$$\Delta_{max}(\mathbb{T}) \leq C(n)^M \prod_{m=1}^M \Delta_{max}(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]).$$

Proof. Suppose that $K \subset B_1$ is a convex set. We have to estimate $\Delta(\mathbb{T}, K) = \frac{|T[\mathbb{T}[K]]|}{|K|}$.

Let K_m denote the ρ_m -neighborhood of K . For a tube $T \in \mathbb{T}$, let T_{ρ_m} be the corresponding thickened tube in $\mathbb{T}_{\rho_m}$, and notice that if $T \subset K$ then $T_{\rho_m} \subset K_m$. So if $T \subset K$, then $T_{\rho_1} \subset K_1$, and $T_{\rho_2} \subset T_{\rho_1} \cap K_2$ and in general $T_{\rho_m} \subset T_{\rho_{m-1}} \cap K_m$. So

$$\begin{aligned} |\mathbb{T}[K]| &\lesssim \prod_{m=1}^M |\mathbb{T}_{\rho_m}[T_{\rho_{m-1}} \cap K_m]| \\ &\leq C(n)^M \prod_{m=1}^M \Delta_{max}(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \frac{|K_m \cap T_{\rho_{m-1}}|}{|T_{\rho_m}|}. \end{aligned} \tag{14.3.51}$$

We use the following elementary parallel-body comparison. If K is convex, $0 < r \leq R \leq 1$, $K_t = N_t(K)$, and T_R is an R -tube, then

$$\frac{|K_r \cap T_R|}{|T_R|} \lesssim_n \frac{|K_r|}{|K_R|}. \tag{14.3.52}$$

For completeness, choose a rectangular box with side lengths $a_1 \leq \dots \leq a_n \lesssim 1$ comparable, by John’s theorem, to a John ellipsoid of K (zero side lengths are allowed when K is lower dimensional). The parallel body K_t has volume

comparable to $\prod_i (a_i + t)$. Slicing the box enlarged by r along the axis of T_R and then in the $n - 1$ orthogonal directions gives

$$\frac{|K_r \cap T_R|}{|T_R|} \lesssim_n \prod_{i=1}^n \frac{a_i + r}{a_i + R} \sim_n \frac{|K_r|}{|K_R|}.$$

The slicing bound is invariant under rotating the tube axis: ordering the successive orthogonal widths only permutes the factors. This proves (14.3.52). Applying it with $r = \rho_m$ and $R = \rho_{m-1}$ yields

$$\frac{|K_m \cap T_{\rho_{m-1}}|}{|T_{\rho_m}|} \lesssim_n \frac{|K_m|}{|K_{m-1}|} \frac{|T_{\rho_{m-1}}|}{|T_{\rho_m}|}.$$

Plugging this into (14.3.51) and simplifying the telescoping product, we get

$$|\mathbb{T}[K]| \leq C(n)^M \left[\prod_{m=1}^M \Delta_{\max}(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \right] \frac{|K|}{|T|}. \quad \square$$

Lemma 14.3.49. *Let $\varepsilon, \delta > 0$ and let $\mathbb{T}$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^n$. Suppose that $\mathbb{T}$ is uniform at each of the scales $\delta = \rho_M < \dots < \rho_0 = 1$, in the sense that there is a set $\mathbb{T}_{\rho_k}$ of ρ_k -tubes for which Items (i), (ii) and (iii) from Definition 14.3.3 hold. Suppose furthermore that for each k we have*

- $\frac{\rho_k}{\rho_{k-1}} \gtrsim \delta^\varepsilon$.
- $\Delta_{\max}(\mathbb{T}_{\rho_k}[T_{\rho_{k-1}}]) \lesssim \left(\frac{\rho_{k-1}}{\rho_k} \right)^\varepsilon$.

Then provided that $\delta > 0$ is sufficiently small depending on ε, M , and n , there is a set of translations R_j of cardinality $\lesssim \max\{1, (|T||\mathbb{T}|)^{-1}\}$ such that, after the essentially-distinct refinement supplied by Lemma 14.3.50, $\bigcup_j R_j(\mathbb{T})$ is Frostman at every scale with error $\delta^{-(n+3)\varepsilon}$. If a shading has first been pigeonholed so that its members have comparable measures, this refinement retains a $\gtrsim 1$ fraction of its total shaded measure.

To prove Lemma 14.3.49, we will use random translations R_j . If $v \in \mathbb{R}^n$, we let $R_v(x) = x + v$. A random translation of size $\leq \rho$ is a translation R_v by a vector $v \in B_\rho$ selected randomly from the uniform measure on B_ρ . We will use the following probability lemma, which is a small variation on Lemma 14.3.14.

Lemma 14.3.50. *Let $\delta \leq \rho \leq 1$. Suppose that $\mathbb{T}$ is a set of δ -tubes in $T_\rho \subset \mathbb{R}^n$ with $\frac{|T_\rho|}{|\mathbb{T}||T_\delta|} \sim J \geq 1$. Let $R_1, \dots, R_J$ be a set of J random translations of size $\leq \rho$. Let $\mathbb{T}' = \bigcup_{j=1}^J R_j(\mathbb{T})$. Let $A \geq 1$. Then with probability at least $1 - (\rho/\delta)^{O(1)} e^{-A}$, the tubes of $\mathbb{T}'$ are essentially distinct up to a factor $\lesssim A$ and we have $\Delta_{\max}(\mathbb{T}') \lesssim A \Delta_{\max}(\mathbb{T})$. After refining $\mathbb{T}'$ to make the tubes essentially distinct, we have with high probability $|\mathbb{T}'| \approx J|\mathbb{T}|$ and $\Delta_{\max}(\mathbb{T}') \lesssim \Delta_{\max}(\mathbb{T})$.*

We will prove Lemma 14.3.50 in Appendix A, where we collect a couple of related probability arguments. Now we are ready to prove Lemma 14.3.49. The proof consists of using Lemma 14.3.50 at many scales.

Proof of Lemma 14.3.49. For each $k = 1, \dots, M$, let $\mathcal{R}_k$ be a set of random translations of size $\sim \rho_k$, with cardinality

$$|\mathcal{R}_k| \sim \max \left\{ 1, \frac{(\rho_k/\rho_{k-1})^{-2}}{|\mathbb{T}_{\rho_k}[T_{\rho_{k-1}}]|} \right\}. \quad (14.3.53)$$

Define the set of translations $\{R_j\}$ as $R_1 \circ R_2 \circ \dots \circ R_M$ for $R_k \in \mathcal{R}_k$, $k = 1, \dots, M$. Recall that $\mathbb{T}' = \bigcup_j R_j(\mathbb{T})$.

We will bound $\Delta_{\max}(\mathbb{T}')$ using Lemma 14.3.48. To do that, we need to estimate $\Delta_{\max}(\mathbb{T}'_{\rho_m}[T'_{\rho_{m-1}}])$. Suppose $T'_{\rho_{m-1}} \in \mathbb{T}'_{\rho_{m-1}}$. Then $T'_{\rho_{m-1}}$ is essentially equivalent to $R_1 \circ \dots \circ R_{m-1}(T_{\rho_{m-1}})$ for some $T_{\rho_{m-1}} \in \mathbb{T}_{\rho_{m-1}}$. Apply the essential-multiplicity conclusion of Lemma 14.3.50 successively at the first $m-1$ levels. Since M is fixed, with high probability the number of tuples

$$(R_1, \dots, R_{m-1}, T_{\rho_{m-1}})$$

which produce a tube essentially equivalent to the prescribed $T'_{\rho_{m-1}}$ is $\lesssim 1$. For each such tuple, the remaining ρ_m -children form a rigid image of $\bigcup_{R_m \in \mathcal{R}_m} R_m(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}])$. Maximal density is subadditive under a union of tagged families, so the $\lesssim 1$ possible tuples cost only a subpower factor. Therefore, with probability at least $3/4$, for every $m = 1, \dots, M$,

$$\Delta_{\max}(\mathbb{T}'_{\rho_m}[T'_{\rho_{m-1}}]) \lesssim \Delta_{\max} \left(\bigcup_{R_m \in \mathcal{R}_m} R_m(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \right). \quad (14.3.54)$$

Fix a tube $T_{\rho_{m-1}}$. Applying Lemma 14.3.50 with $A = \delta^{-\varepsilon/M}$, we have that with probability $1 - \delta^{-O(1)} \exp[-\delta^{-\varepsilon/M}]$,

$$\begin{aligned} \Delta_{\max} \left(\bigcup_{R_m \in \mathcal{R}_m} R_m(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \right) &\lesssim \delta^{-\varepsilon/M} \Delta_{\max}(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \\ &\lesssim \delta^{-\varepsilon/M} \left(\frac{\rho_{m-1}}{\rho_m} \right)^\varepsilon. \end{aligned} \quad (14.3.55)$$

There are $(\rho_{m-1}/\rho_m)^{O(1)} \leq \delta^{-O(1)}$ many tubes. Thus if $\delta > 0$ is sufficiently small depending on ε , M and n , we conclude that with probability at least $3/4$, (14.3.55) holds for every $T_{\rho_{m-1}} \in \mathbb{T}_{\rho_{m-1}}$.

Combining (14.3.54) and (14.3.55), we see that for each m ,

$$\Delta_{\max}(\mathbb{T}'_{\rho_m}[T'_{\rho_{m-1}}]) \lesssim \delta^{-\varepsilon/M} \left(\frac{\rho_{m-1}}{\rho_m} \right)^\varepsilon.$$

Now we can apply Lemma 14.3.48 to show that for each m ,

$$\Delta_{\max}(\mathbb{T}'[T'_{\rho_m}]) \lesssim \prod_{m'=m+1}^M \Delta_{\max}(\mathbb{T}'_{\rho_{m'}}[T_{\rho_{m'-1}}]) \lesssim \delta^{-2\varepsilon}. \quad (14.3.56)$$

On the other hand, (14.3.53) shows that

$$|\mathbb{T}'_{\rho_m}[T'_{\rho_{m-1}}]| \approx \left| \bigcup_{R_m \in \mathcal{R}_m} R_m(\mathbb{T}_{\rho_m}[T_{\rho_{m-1}}]) \right| \gtrsim \left(\frac{\rho_m}{\rho_{m-1}} \right)^{-2},$$

which implies that

$$|\mathbb{T}'[T'_{\rho_m}]| \gtrsim \left(\frac{\delta}{\rho_m} \right)^{-2}. \quad (14.3.57)$$

Combining (14.3.56) and (14.3.57), we see that for each m ,

$$C_F(\mathbb{T}'[T'_{\rho_m}], T'_{\rho_m}) \lesssim \delta^{-2\varepsilon}.$$

Finally consider an arbitrary $\rho \in [\delta, 1]$. We can choose m so that $\rho_m \leq \rho \leq \rho_{m-1}$. Since $\rho_m \geq \delta^\varepsilon \rho_{m-1}$, using the last equation for m and $m-1$ we can conclude that

$$C_F(\mathbb{T}'[T'_\rho], T'_\rho) \lesssim \delta^{-(n+3)\varepsilon}. \quad \square$$

Proof of Theorem 14.3.47. Recall that Part (A) is Theorem 5.2 from [7]. We prove Part (B), keeping separate the desired multiplicity loss and the small exponent used in Lemma 14.3.49.

Fix a desired loss $\varepsilon_0 > 0$, and apply Part (A) with $\varepsilon_A = \varepsilon_0/4$. Let $\eta_A > 0$ and $\delta_A > 0$ be its input constants. Put

$$\varepsilon_s = \eta_A/12, \quad N = \lceil \varepsilon_s^{-1} \rceil, \quad \eta = \min\{\eta_A/4, \varepsilon_s/(4N), \varepsilon_0/4\}.$$

After decreasing δ_0 , the harmless constants implicit below are absorbed by the displayed powers of δ . Pigeonhole the values $|Y(T)|$ and retain a $\gtrsim 1$ refinement, so that all surviving shaded measures are comparable. This permits every later essentially-distinct refinement to retain comparable shaded mass.

Consider $\rho_k = \delta^{k/N}$, $k = N, \dots, 0$. Then $\rho_k/\rho_{k-1} = \delta^{1/N} \gtrsim \delta^{\varepsilon_s}$. Downward heredity, followed by the affine rescaling of $T_{\rho_{k-1}}$ to B_1 , gives

$$\Delta_{\max}(\mathbb{T}_{\rho_k}[T_{\rho_{k-1}}]) \lesssim \delta^{-\eta} \leq \delta^{-\varepsilon_s/N} = \left(\frac{\rho_{k-1}}{\rho_k} \right)^{\varepsilon_s}.$$

Lemma 14.3.49 therefore supplies M translations, with

$$M \lesssim \max\{1, (|T||\mathbb{T}|)^{-1}\},$$

and an essentially distinct refinement $(\mathbb{T}', Y')$ of the tagged copied family such that $\mathbb{T}'$ is Frostman at every scale with error $\delta^{-6\varepsilon_s} = \delta^{-\eta_A/2}$ and $\lambda(\mathbb{T}', Y') \gtrsim \lambda(\mathbb{T}, Y) \geq \delta^\eta$. Decreasing η once more, and then decreasing δ_0 , absorbs the subpower losses in both the Frostman constant and the shading density. Part (A) applies and gives $|U(\mathbb{T}', Y')| \geq \delta^{\varepsilon_A}$.

Write $m = |\mathbb{T}| |\mathbb{T}|$ and $S = \sum_{T \in \mathbb{T}} |Y(T)|$. Since every copied union has the same measure as the original one,

$$|U(\mathbb{T}, Y)| = \frac{1}{M} \sum_{j=1}^M |R_j(U(\mathbb{T}, Y))| \geq \frac{1}{M} |U(\mathbb{T}', Y')| \gtrsim \delta^{\varepsilon_A} \min\{1, m\}.$$

If $m \leq 1$, the right-hand side is at least $\delta^{\varepsilon_0} S$, because $S \leq m$ and $\varepsilon_A < \varepsilon_0$. If $m > 1$, the Katz–Tao hypothesis at the bottom scale, tested on B_1 , gives $m \lesssim \delta^{-\eta}$; hence the same conclusion follows from $\varepsilon_A + \eta \leq \varepsilon_0$. Thus in both cases

$$|U(\mathbb{T}, Y)| \gtrsim \delta^{\varepsilon_0} S, \quad \text{or equivalently} \quad \mu(\mathbb{T}, Y) \lesssim \delta^{-\varepsilon_0}.$$

Absorbing the final fixed constant by decreasing δ_0 proves Part (B). $\square$

14.3.7.1 A multi-scale decomposition of $\mathbb{T}$

In order to apply Theorem 14.3.47(A) or Theorem 14.3.47(B), our collection of tubes must be Frostman at every scale or Katz–Tao at every scale, respectively. The next result says that either this is true, or else there is a long sequence of scales that are far from being Frostman (resp. Katz–Tao).

Lemma 14.3.51. *Let $N \geq 1$ be an integer. Define $\varepsilon = 1/\sqrt{N}$ and $\eta_0 = \eta$. Let*

$$\eta_0 \leq \eta_1 \leq \eta_2 \leq \dots \leq \eta_N \leq \varepsilon, \quad \eta_{j-1} \leq \varepsilon \eta_j \quad (1 \leq j \leq N).$$

Let $\delta > 0$ and let $\mathbb{T}$ be a uniform set of δ -tubes.

(A) If $C_F(\mathbb{T}) \leq \delta^{-\eta}$, then at least one of the following holds.

(i) $\mathbb{T}$ is $\lesssim \delta^{-5\varepsilon}$ Frostman at every scale.

(ii) There are scales $\delta \leq \tau \leq \theta \leq 1$ with $\tau \leq \delta^\varepsilon \theta$, and an integer $1 \leq j \leq N$ so that the following holds:

- $C_F(\mathbb{T}[T_\tau]) \lesssim (\tau/\delta)^{\eta_{j-1}}$ for all $T_\tau \in \mathbb{T}_\tau$.*
- $C_F(\mathbb{T}_\tau[T_\theta]) \lesssim (\theta/\tau)^{\eta_{j-1}}$ for all $T_\theta \in \mathbb{T}_\theta$.*
- For all $\rho \in [\tau(\theta/\tau)^\varepsilon, \theta(\tau/\theta)^\varepsilon]$, we have $C_F(\mathbb{T}_\tau[T_\rho]) \geq (\rho/\tau)^{\eta_j}$ for all $T_\rho \in \mathbb{T}_\rho$.*

(B) If $\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta}$, then at least one of the following holds.

(i) $\mathbb{T}$ is $\lesssim \delta^{-5\varepsilon}$ Katz-Tao at every scale.

(ii) There are scales $\delta \leq \tau \leq \theta \leq 1$ with $\tau \leq \delta^\varepsilon \theta$, and an integer $1 \leq j \leq N$ so that the following holds:

- $\Delta_{\max}(\mathbb{T}_\theta) \lesssim \theta^{-\eta_{j-1}}$.
- $\Delta_{\max}(\mathbb{T}_\tau[T_\theta]) \lesssim (\theta/\tau)^{\eta_{j-1}}$ for all $T_\theta \in \mathbb{T}_\theta$
- For all $\rho \in [\tau(\theta/\tau)^\varepsilon, \theta(\tau/\theta)^\varepsilon]$, we have $\Delta_{\max}(\mathbb{T}_\rho[T_\theta]) \geq (\theta/\rho)^{\eta_j}$ for all $T_\theta \in \mathbb{T}_\theta$.

Proof. We begin with part (A). To begin, we check if there is a scale $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$ so that $C_F(\mathbb{T}[T_\rho]) \leq (\rho/\delta)^{\eta_1}$.

If there is no such scale we stop. If there is such a scale, then we define $\mathbb{T}_1$ to be $\mathbb{T}[T_\rho]$ modified by a change of variables that takes T_ρ to B_1 , and we set $\mathbb{T}_2 = \mathbb{T}_\rho$. We get the following output with $J = 2$:

1. For each $j = 1, \dots, J$, $\mathbb{T}_j$ is a set of δ_j -tubes in B_1 .
2. $\prod_{j=1}^J \delta_j \approx \delta$.
3. $\prod_{j=1}^J |\mathbb{T}_j| \approx |\mathbb{T}|$.
4. $C_F(\mathbb{T}_j) \leq \delta_j^{-\eta_{J-1}}$.

General step. Suppose that for some J we have $\mathbb{T}_j$ obeying the criteria listed just above. We check whether there is a $j \in \{1, \dots, J\}$ and a scale ρ so that the following holds:

- $\delta_j < \delta^\varepsilon$ and
- $\rho \in [\delta_j^{1-\varepsilon}, \delta_j^\varepsilon]$, and
- $C_F(\mathbb{T}_j[T_\rho]) \leq (\delta_j/\rho)^{-\eta_J}$.

If there is no such j and ρ , we stop. If there is such a j , ρ , then we replace $\mathbb{T}_j$ by $\mathbb{T}_{j,\rho}$ and $\mathbb{T}_j[T_\rho]$ (rescaled to be a set of tubes in the unit ball). After this replacement, we can relabel our sets of tubes to give $\mathbb{T}_1, \dots, \mathbb{T}_{J+1}$, which obey all the criteria listed above.

Since each $\delta_j < \delta^{\varepsilon^2}$ and since $\prod_{j=1}^J \delta_j \approx \delta$, we have $J \leq N = 1/\varepsilon^2$, and therefore this process terminates after at most N steps, with some sets of tubes $\mathbb{T}_1, \dots, \mathbb{T}_J$, with $J \leq N$. Each of these sets obeys the criteria listed above (1 to 4) and also for each j , either $\delta_j \geq \delta^\varepsilon$, or else

$$\text{For each } \rho \in [\delta_j^{1-\varepsilon}, \delta_j^\varepsilon], \text{ we have } C_F(\mathbb{T}_j[T_\rho]) \geq (\delta_j/\rho)^{-\eta_J}. \quad (14.3.58)$$

Unwinding the stopping time argument, there is a sequence of $\delta = \rho_J \leq \dots \leq \rho_1 \leq \rho_0 = 1$ so that $\mathbb{T}_j$ corresponds to $\mathbb{T}_{\rho_j}[T_{\rho_{j-1}}]$. By our stopping criteria, we see that

$$C_F(\mathbb{T}_{\rho_j}[T_{\rho_{j-1}}], T_{\rho_{j-1}}) \leq (\rho_{j-1}/\rho_j)^{\eta_J} \leq (\rho_{j-1}/\rho_j)^\varepsilon.$$

By Lemma 14.3.48, we have that $C_F(\mathbb{T}[T_{\rho_j}]) \lesssim \delta^{-\varepsilon}$ for each index j .

First we consider the case where every δ_j obeys $\delta_j \geq \delta^\varepsilon$. Then each ratio ρ_j/ρ_{j-1} is at least δ^ε . We claim that $\mathbb{T}$ is $\lesssim \delta^{-5\varepsilon}$ Frostman at every scale, and hence Conclusion (i) holds. To verify this, let $\rho \in [\delta, 1]$ and choose j so that $\rho_{j+1} \leq \rho \leq \rho_j$. Then

$$\begin{aligned} C_F(\mathbb{T}[T_\rho]) &\leq C_F(\mathbb{T}[T_{\rho_j}]) \frac{|\mathbb{T}[T_{\rho_j}]|}{|\mathbb{T}[T_\rho]|} \\ &\leq C_F(\mathbb{T}[T_{\rho_j}]) \frac{|\mathbb{T}[T_{\rho_j}]|}{|\mathbb{T}[T_{\rho_{j+1}}]|} \\ &\lesssim \delta^{-\varepsilon} \left(\frac{\rho_j}{\rho_{j+1}} \right)^4 \leq \delta^{-5\varepsilon}. \end{aligned}$$

Next we consider the case where there exists at least one index j with $\delta_j < \delta^\varepsilon$ and $\mathbb{T}_j$ obeys (14.3.58). Then Conclusion (ii) holds with $\tau = \rho_j$ and $\theta = \rho_{j-1}$.

We now prove Part (B). We use the same stopping tree, but record all the Katz–Tao estimates because their direction is different from the Frostman ones. A node of the tree is a uniform family $\mathcal{A}$ of d -tubes in a unit tube, after the standard affine rescaling of that unit tube to B_1 . At generation J the leaf radii $d_1, \dots, d_J$ satisfy

$$\prod_{i=1}^J d_i \approx \delta, \quad \Delta_{\max}(\mathcal{A}_i) \lesssim d_i^{-\eta_{J-1}}. \quad (14.3.59)$$

The assertion holds at generation one by the hypothesis and $\eta_0 = \eta$.

Suppose a leaf $\mathcal{A}$ has radius $d < \delta^\varepsilon$. We ask whether there is an $s \in [d^{1-\varepsilon}, d^\varepsilon]$ for which

$$\Delta_{\max}(\mathcal{A}_s) \lesssim s^{-\eta_J}. \quad (14.3.60)$$

If so, replace the leaf by the coarse family $\mathcal{A}_s$ and the uniform fine families $\mathcal{A}[A_s]$, rescaled to radius d/s . The coarse child satisfies the required generation- $J+1$ estimate by (14.3.60). For a fine child, downward heredity of $\Delta_{\max}$ and (14.3.59) give

$$\Delta_{\max}(\mathcal{A}[A_s]) \lesssim d^{-\eta_{J-1}} \leq (d/s)^{-\eta_J}.$$

Here the last inequality follows from $s \leq d^\varepsilon$ and $\eta_{J-1} \leq \varepsilon \eta_J$ (indeed the weaker bound $\eta_{J-1} \leq (1 - \varepsilon)\eta_J$ suffices when $N > 1$). Thus (14.3.59) is preserved, while the two new radii multiply to d .

Both children of a split have radius at most $d^\varepsilon < \delta^{\varepsilon^2}$: for the fine child use $s \geq d^{1-\varepsilon}$. Consequently a branch cannot have more than $1/\varepsilon^2 = N$ leaves, because their radii have product comparable to δ . The procedure therefore stops by generation $J \leq N$. Unwinding the tree gives scales

$$1 = \rho_0 \geq \rho_1 \geq \cdots \geq \rho_J = \delta, \quad d_i = \rho_i / \rho_{i-1},$$

such that the i th factor is $\mathbb{T}_{\rho_i}[T_{\rho_{i-1}}]$ after rescaling. The submultiplicativity lemma and (14.3.59) imply, at every boundary scale,

$$\Delta_{\max}(\mathbb{T}_{\rho_i}) \lesssim \prod_{k=1}^i d_k^{-\eta_{J-1}} = \rho_i^{-\eta_{J-1}} \leq \delta^{-\varepsilon}. \quad (14.3.61)$$

If every $d_i \geq \delta^\varepsilon$, let ρ lie between two consecutive boundary scales. Coarsening a uniform tube family through a ratio at most $\delta^{-\varepsilon}$ can increase its normalized convex density by at most the four-dimensional packing factor $\delta^{-4\varepsilon}$. Applying this elementary packing bound to (14.3.61) yields

$$\Delta_{\max}(\mathbb{T}_\rho) \lesssim \delta^{-5\varepsilon} \quad (\delta \leq \rho \leq 1),$$

which is conclusion (i).

Otherwise choose a stopped leaf with $d_i < \delta^\varepsilon$ and put $\tau = \rho_i$, $\theta = \rho_{i-1}$, and $j = J$. Then $\tau \leq \delta^\varepsilon \theta$. Formula (14.3.61), stopped just before this factor, gives

$$\Delta_{\max}(\mathbb{T}_\theta) \lesssim \theta^{-\eta_{j-1}},$$

while the leaf invariant gives, for every T_θ ,

$$\Delta_{\max}(\mathbb{T}_\tau[T_\theta]) \lesssim (\theta/\tau)^{\eta_{j-1}}.$$

The leaf was not split, so (14.3.60) fails for every $s \in [d_i^{1-\varepsilon}, d_i^\varepsilon]$. With $s = \rho/\theta$, this says precisely

$$\Delta_{\max}(\mathbb{T}_\rho[T_\theta]) \gtrsim (\theta/\rho)^{\eta_j}$$

for every $\rho \in [\tau(\theta/\tau)^\varepsilon, \theta(\tau/\theta)^\varepsilon]$ and every T_θ . After absorbing the harmless dyadic-uniformity constants, these are all three bullets of conclusion (ii). $\square$

14.3.8 Main Lemma 1

In this section, we prove Main Lemma 14.3.11. We first recall the statement.

Main Lemma. $K_{KT}(\beta)$ implies $K_F(\beta)$.

Let's first digest what the lemma is saying. Suppose that $\mathbb{T}$ is a set of δ -tubes in B_1 with $C_F(\mathbb{T}) \leq \delta^{-\eta}$. We have to prove the bound

$$\mu(\mathbb{T}) \leq \delta^{-\epsilon} (\delta^{-2})^\beta \left(\delta^2 |\mathbb{T}| \right)^{1-\beta/2}. \quad (14.3.62)$$

Since $C_F(\mathbb{T}, B_1) \leq \delta^{-\eta}$, we have $|\mathbb{T}| \geq \delta^{-2+\eta}$. If $|\mathbb{T}| \approx \delta^{-2}$, then $C_F(\mathbb{T}, B_1) \leq \delta^{-\eta}$ is equivalent to $\Delta_{\max}(\mathbb{T}) \lesssim \delta^{-\eta}$, and then $K_{KT}(\beta)$ gives $\mu(\mathbb{T}) \leq \delta^{-\epsilon} \delta^{-2\beta}$, which is the desired bound. So the case $|\mathbb{T}| \approx \delta^{-2}$ is trivial. So the content of the lemma is in the case when $|\mathbb{T}| \gg \delta^{-2}$.

If $|\mathbb{T}| \gg \delta^{-2}$, then $\Delta_{\max}(\mathbb{T}) \gg 1$, so we can not immediately apply $K_{KT}(\beta)$ to $\mathbb{T}$. The sampling claim proved inside Lemma 14.3.13, applied with $D = \Delta_{\max}(\mathbb{T}) \approx \delta^2 |\mathbb{T}|$ in the present Frostman regime, produces a subset $\tilde{\mathbb{T}} \subset \mathbb{T}$ with $|\tilde{\mathbb{T}}| \approx \delta^{-2}$ and $\Delta_{\max}(\tilde{\mathbb{T}}) \lesssim 1$. We can apply $K_{KT}(\beta)$ to $\tilde{\mathbb{T}}$, and we get $\mu(\tilde{\mathbb{T}}) \lesssim \delta^{-2\beta}$ which is equivalent to $|U(\tilde{\mathbb{T}})| \gtrsim \delta^{2\beta}$. We have $|U(\mathbb{T})| \geq |U(\tilde{\mathbb{T}})|$, and this translates into

$$\mu(\mathbb{T}) \lesssim (\delta^{-2})^\beta \left(\delta^2 |\mathbb{T}| \right) \quad (14.3.63)$$

Comparing (14.3.63) with our goal (14.3.62), we see that we need to reduce the exponent on the last factor from 1 to $1 - \beta/2$. In the proof ofakeya, any exponent strictly less than 1 will work. We can also describe the lemma in terms of $|U(\mathbb{T})|$, which may be more intuitive. Since we assume $K_{KT}(\beta)$, we already know that if $C_F(\mathbb{T}) \lesssim 1$ and $|\mathbb{T}| \sim \delta^{-2}$, then $|U(\mathbb{T})| \gtrsim \delta^{2\beta}$. Now we need to show that if $C_F(\mathbb{T}) \lesssim 1$ and if $|\mathbb{T}|$ is far larger than δ^{-2} , then $|U(\mathbb{T})|$ is a little larger than $\delta^{2\beta}$. While this may sound intuitive, it is not at all trivial to prove, and the proof will depend crucially on stickyakeya.

We organize the proof as a bootstrapping or self-improving argument.

Lemma 14.3.52. *Suppose that $K_{KT}(\beta)$ holds and that $K_F(\gamma)$ holds for some $\gamma > \beta$. Then $K_F(\gamma - \nu)$ holds for $\nu = \nu(\gamma, \beta) > 0$. For β fixed, $\nu = \nu(\gamma, \beta) > 0$ is nondecreasing in γ .*

We first verify the base estimate $K_F(1)$. For an essentially distinct family of δ -tubes, the tubes through a fixed point are determined, up to $O(1)$ choices, by their δ -separated directions; hence their pointwise multiplicity is $O(\delta^{-2})$. Moreover $\Delta_{\max}(\mathbb{T}) \geq 1$, by testing the definition on one member of the family, while $\Delta(\mathbb{T}, B_1) \sim \delta^2 |\mathbb{T}|$. Thus $C_F(\mathbb{T}) \leq \delta^{-\eta}$ implies

$$\delta^2 |\mathbb{T}| \gtrsim \delta^\eta.$$

Given the loss $\epsilon > 0$ in the definition of $K_F(1)$, choose $\eta \leq \epsilon$. Then

$$\mu(\mathbb{T}, Y) \lesssim \delta^{-2} \leq \delta^{-\epsilon} \delta^{-2} (\delta^2 |\mathbb{T}|)^{1/2},$$

which is (14.3.11) with $\beta = 1$.

Proof of Lemma 14.3.52. Let ε_0 and δ_0 be the output of Theorem 14.3.47 with $\gamma/2$ in place of ε . Let $N \geq 25/\varepsilon_0^2$ be an integer to be chosen later, and define $\varepsilon = 1/\sqrt{N}$. Let $\eta \leq \eta_1 \leq \dots \leq \eta_N \leq \varepsilon/5$ be a sequence of numbers to be chosen later. We choose the constants from biggest to smallest and each η_j will be selected very small depending on η_{j+1} and β .

Apply Lemma 14.3.51(A) to $\mathbb{T}$, with N as above. If Conclusion (i) holds, then $\mathbb{T}$ is $\delta^{-\varepsilon_0}$ Frostman at every scale, and hence by Theorem 14.3.47(A) we have

$$|U(\mathbb{T}, Y)| \geq \delta^{\gamma/2} \geq \delta^{2(\gamma/2)} (|\mathbb{T}| |T|)^{(\gamma/2)/2},$$

where we used the fact that $|\mathbb{T}| \leq \delta^{-4}$. In particular, $|U(\mathbb{T}, Y)|$ satisfies the volume estimate corresponding to $K_F(\gamma/2)$. In this case, Lemma 14.3.52 holds with $\nu = \gamma/2$.

Remark 14.3.53. Recall that the statement $K_F(\beta)$ asserts the estimate $|U(\mathbb{T})| \gtrsim \delta^{2\beta} (|\mathbb{T}| |T|)^\omega$, with $\omega = \beta/2$. In the argument described above, we must have $\omega < \beta$.

Henceforth we shall assume that Conclusion (ii) of Lemma 14.3.51(A) holds. Let $\delta \leq \tau \leq \theta \leq 1$ and $1 \leq j \leq N$ be the output of that lemma. We apply Lemma 14.3.30 to $(\mathbb{T}, Y)$ and $\mathbb{T}_\tau$. We obtain a ≈ 1 refinement of $(\mathbb{T}, Y)$, a subset of $\mathbb{T}_\tau$, and a shading $Y_{\mathbb{T}_\tau}$ on $\mathbb{T}_\tau$. Abusing notation, we will continue to refer to these refinements as $(\mathbb{T}, Y)$ and $\mathbb{T}_\tau$. Apply Lemma 14.3.30 to $(\mathbb{T}_\tau, Y_{\mathbb{T}_\tau})$ and $\mathbb{T}_\theta$. We obtain a ≈ 1 refinement of $(\mathbb{T}_\tau, Y_{\mathbb{T}_\tau})$, a subset of $\mathbb{T}_\theta$, and a shading $Y_{\mathbb{T}_\theta}$ on $\mathbb{T}_\theta$. Abusing notation, we will continue to refer to these refinements as $(\mathbb{T}_\tau, Y_{\mathbb{T}_\tau})$ and $\mathbb{T}_\theta$. We may suppose that all sets of the form $(\mathbb{T}[T_\tau], Y)$, $(\mathbb{T}_\tau[T_\theta], Y_{\mathbb{T}_\tau})$ and $(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta})$ are uniform. Lemma 14.3.30 implies that for all $T_\tau \in \mathbb{T}_\tau$ and $T_\theta \in \mathbb{T}_\theta$,

$$\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{T}[T_\tau], Y) \mu(\mathbb{T}_\tau[T_\theta], Y_{\mathbb{T}_\tau}) \mu(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta}). \quad (14.3.64)$$

Our goal is to estimate the three quantities on the RHS of (14.3.64).

For each $T_\tau \in \mathbb{T}_\tau$ we have $C_F(\mathbb{T}[T_\tau], T_\tau) \leq \delta^{-\eta_{j-1}}$. Thus by applying Lemma 14.3.15 (with η_1 in place of ε) to the re-scaled sets $\mathbb{T}[T_\tau]$, we see that for each $T_\tau \in \mathbb{T}_\tau$, we have

$$\begin{aligned} \mu(\mathbb{T}[T_\tau], Y) &\leq (\delta/\tau)^{-\eta_1} \delta^{-(1-\gamma/2)\eta_{j-1}} ((\delta/\tau)^{-2})^\gamma ((\delta/\tau)^2 |\mathbb{T}[T_\tau]|)^{1-\gamma/2} \\ &\leq \delta^{-2\eta_{j-1}} ((\delta/\tau)^{-2})^\gamma ((\delta/\tau)^2 |\mathbb{T}[T_\tau]|)^{1-\gamma/2}. \end{aligned} \quad (14.3.65)$$

By Remark 14.3.7(A), $C_F(\mathbb{T}_\theta) \lesssim \delta^{-\eta}$. Thus by applying $K_F(\gamma)$ (again, with η_1 in place of ε) we have

$$\mu(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta}) \leq \delta^{-\eta_{j-1}} \left((\theta)^{-2} \right)^\gamma (\theta^2 |\mathbb{T}_\theta|)^{1-\gamma/2}. \quad (14.3.66)$$

Thus in order to prove Lemma 14.3.52 with $\nu = \eta_1$, it suffices to show that for each $T_\theta \in \mathbb{T}_\theta$, we have

$$\mu(\mathbb{T}_\tau[T_\theta], Y_{\mathbb{T}_\tau}) \leq \delta^{10\eta_{j-1}} ((\tau/\theta)^{-2})^\gamma ((\tau/\theta)^2 |\mathbb{T}_\tau[T_\theta]|)^{1-\gamma/2}. \quad (14.3.67)$$

We will show that this holds, provided each η_{j-1} is selected sufficiently small compared to η_j .

Fix a tube $T_\theta \in \mathbb{T}_\theta$, and let $(\tilde{\mathbb{T}}, \tilde{Y})$ be the image of the tubes in $\mathbb{T}_\tau[T_\theta]$ (and their shadings) under the rescaling taking $\mathbb{T}_\theta$ to a tube of thickness 1. Hence $\tilde{\mathbb{T}}$ is a family of τ/θ -tubes. Recall that $\tilde{\delta} \leq \delta^\varepsilon$, and thus to prove (14.3.67) we must show that

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \leq \tilde{\delta}^{10\eta_{j-1}/\varepsilon} \tilde{\delta}^{-2\gamma} (\tilde{\delta}^2 |\tilde{\mathbb{T}}|)^{1-\gamma/2}. \quad (14.3.68)$$

Recall that $C_F(\tilde{\mathbb{T}}) \leq \tilde{\delta}^{-\eta_{j-1}/\varepsilon}$, and

$$C_F(\tilde{\mathbb{T}}[T_\rho]) \geq (\rho/\tilde{\delta})^{\eta_j} \quad \text{for all } \rho \in [\tilde{\delta}^{1-\varepsilon}, \tilde{\delta}^\varepsilon]. \quad (14.3.69)$$

Select $\rho = \tilde{\delta}^\varepsilon$. Apply Lemma 14.3.16 to each set $\tilde{\mathbb{T}}[T_\rho]$; after pigeonholing, we may suppose that the planks in the resulting set $\mathbb{W}_{T_\rho}$ have roughly the same dimensions $a \times b \times 1$ for all $\rho \in \tilde{\mathbb{T}}_\rho$.

Applying Proposition 14.3.36(A) with η_{j-1} in place of ε and γ in place of β , we have

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \leq \tilde{\delta}^{-2\eta_{j-1}/\varepsilon} \left(\frac{a}{b}\right)^{3\gamma/2} (\tilde{\delta}^{-2})^\gamma (\tilde{\delta}^2 |\tilde{\mathbb{T}}|)^{1-\gamma/2}.$$

In particular, either (14.3.68) holds (and hence we are done), or else a is not too much smaller than b , i.e.

$$a \geq \tilde{\delta}^{\frac{10\eta_{j-1}}{\varepsilon\gamma}} b.$$

We have

$$\Delta(\tilde{\mathbb{T}}, T_b) \geq (a/b) \Delta(\tilde{\mathbb{T}}, W) \approx (a/b) \Delta_{\max}(\tilde{\mathbb{T}}).$$

Therefore,

$$C_F(\tilde{\mathbb{T}}[T_b]) \lesssim b/a \leq \tilde{\delta}^{-\frac{10\eta_{j-1}}{\varepsilon\gamma}} \leq \tilde{\delta}^{-\frac{10\eta_{j-1}}{\varepsilon\beta}}. \quad (14.3.70)$$

Since $b \leq \rho = \tilde{\delta}^\varepsilon$, if we select η_{j-1} sufficiently small depending on η_j , and γ (and depending on ε , which in turn depends on γ), then we must have $b \leq \tilde{\delta}^{1-\varepsilon}$ —if not, then (14.3.70) would violate (14.3.69) (with b in place of ρ).

Define $\eta'_{j-1} = \frac{10\eta_{j-1}}{\varepsilon\beta}$, so

$$C_F(\tilde{\mathbb{T}}[T_b]) \lesssim \tilde{\delta}^{-\eta'_{j-1}}$$

Apply Lemma 14.3.30 to $(\tilde{\mathbb{T}}, \tilde{Y})$ and $\tilde{\mathbb{T}}_b$ (abusing notation, we will continue to refer to the associated refinements of $(\tilde{\mathbb{T}}, \tilde{Y})$ and $\mathbb{T}_b$, and after a refinement we can assume that these sets are still uniform). Applying Lemma 14.3.15 to bound $\mu(\tilde{\mathbb{T}}[T_b], \tilde{Y})$, we conclude that in order to establish (14.3.68), it suffices to prove that

$$\mu(\tilde{\mathbb{T}}_b, Y_{\tilde{\mathbb{T}}_b}) \leq \tilde{\delta}^{10\eta'_{j-1}} b^{-2\gamma} (b^2 |\tilde{\mathbb{T}}_b|)^{1-\gamma/2}. \quad (14.3.71)$$

Since each tube $T_b \in \tilde{\mathbb{T}}_b[T_\rho]$ contains at least one $W \in \mathbb{W}_{T_\rho}$, we have

$$\Delta_{\max}(\tilde{\mathbb{T}}_b[T_\rho]) \lesssim \frac{|T_b|}{|W|} \Delta_{\max}(\mathbb{W}_{T_\rho}) \lesssim \frac{b}{a} \lesssim \tilde{\delta}^{-\eta'_{j-1}}.$$

Therefore

$$\Delta_{\max}(\tilde{\mathbb{T}}_b) \lesssim (\rho)^{-4} \Delta_{\max}(\tilde{\mathbb{T}}_b[T_\rho]) \lesssim \tilde{\delta}^{-4\varepsilon - \eta'_{j-1}}.$$

Applying Lemma 14.3.13, we have

$$\mu(\tilde{\mathbb{T}}_b, Y_{\tilde{\mathbb{T}}_b}) \leq \tilde{\delta}^{-4\varepsilon - \eta'_{j-1}} |\tilde{\mathbb{T}}_b|^\beta = \left(\tilde{\delta}^{-4\varepsilon - \eta'_{j-1}} (b^2 |\tilde{\mathbb{T}}_b|)^{\gamma/2 + \beta - 1} \right) b^{-2\beta} (b^2 |\tilde{\mathbb{T}}_b|)^{1-\gamma/2}.$$

We wish to bound the first bracketed term. The key point is that $(b^2 |\tilde{\mathbb{T}}_b|)$ is close to 1. To see this, we note that

$$\tilde{\delta}^{\eta'_{j-1}} \lesssim C_F(\mathbb{T}_b)^{-1} \leq b^2 |\tilde{\mathbb{T}}_b| \lesssim \Delta_{\max}(\tilde{\mathbb{T}}_b) \lesssim \tilde{\delta}^{-4\varepsilon - \eta'_{j-1}},$$

Now we do not know the sign of $\gamma/2 + \beta - 1$, but using the above upper and lower bounds for $b^2 |\tilde{\mathbb{T}}_b|$ we get

$$\mu(\tilde{\mathbb{T}}_b, Y_{\tilde{\mathbb{T}}_b}) \leq \tilde{\delta}^{-8\varepsilon - 2\eta'_{j-1} + 2(\gamma - \beta)} b^{-2\gamma} (b^2 |\tilde{\mathbb{T}}_b|)^{1-\gamma/2}.$$

Provided we select N sufficiently large (recall that $\varepsilon = 1/\sqrt{N}$) and η_N sufficiently small (recall that $\eta_{j-1} \leq \eta_N$) so that $-8\varepsilon - 2\eta'_{j-1} + 2(\gamma - \beta) \geq (\gamma - \beta)$, we have

$$\mu(\tilde{\mathbb{T}}_b, Y_{\tilde{\mathbb{T}}_b}) \lesssim \tilde{\delta}^{(\gamma - \beta)} b^{-2\gamma} (b^2 |\tilde{\mathbb{T}}_b|)^{1-\gamma/2}.$$

This implies our desired bound (14.3.71), provided we choose η_{j-1} sufficiently small so that $10\eta'_{j-1} \leq (\gamma - \beta)/2$. $\square$

Deduction of Main Lemma 14.3.11. Fix $\beta > 0$ and assume $K_{KT}(\beta)$. We have just proved $K_F(1)$. Starting at $\gamma_0 = 1$, apply Lemma 14.3.52 whenever $\gamma_m > \beta$ and set $\gamma_{m+1} = \gamma_m - \nu(\gamma_m, \beta)$. If an iterate is at most β , its estimate implies $K_F(\beta)$ after enlarging the exponent and absorbing the bounded range $1 \leq |\mathbb{T}| \lesssim \delta^{-4}$.

Otherwise the decreasing sequence has a limit $L \geq \beta$. If $L > \beta$, the stated monotonicity of the gain supplies a positive lower bound for $\nu(\gamma_m, \beta)$ once γ_m

lies in a compact subinterval of $(\beta, 1]$, contradicting convergence. Hence $L = \beta$. To pass to the endpoint, fix a desired loss $\epsilon > 0$ in $K_F(\beta)$ and choose m so that $d = \gamma_m - \beta < \epsilon/10$. Apply $K_F(\gamma_m)$ with loss $\epsilon/2$. Since $\delta^2|\mathbb{T}| \in [\delta^\eta, O(\delta^{-2})]$, decreasing its input exponent η if necessary gives

$$\delta^{-2\gamma_m}(\delta^2|\mathbb{T}|)^{1-\gamma_m/2} \leq \delta^{-\epsilon/2}\delta^{-2\beta}(\delta^2|\mathbb{T}|)^{1-\beta/2}.$$

Combining the two half-losses proves $K_F(\beta)$ for the prescribed ϵ . This is Main Lemma 14.3.11. $\square$

14.3.9 Main Lemma 2

Our last goal is to prove Main Lemma 14.3.12. We recall the statement.

Main Lemma. *If $K_{KT}(\beta)$ and $K_F(\beta)$ hold, then $K_{KT}(\beta - \nu)$, where $\nu = \nu(\beta) > 0$ is monotone.*

14.3.9.1 Reduction to the strongly non-sticky case

The stickyakeya theorem plays a crucial role in this proof. By using stickyakeya, we are able to reduce the problem to a situation which is not sticky in a strong sense.

Lemma 14.3.54. *(Very not sticky case of Main Lemma 14.3.12) For all $\beta > 0$, there is some $\epsilon_{\text{scale}} = \epsilon_{\text{scale}}(\beta) > 0$ so that for all $\zeta > 0$, there is some $\nu = \nu(\beta, \zeta) > 0$ and $\eta = \eta(\beta, \zeta) > 0$ so that the following is true:*

If $K_{KT}(\beta)$ and $K_F(\beta)$ hold and if $(\mathbb{T}, Y)$ is a uniform set of δ -tubes in B_1 obeying

- $\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta}$ and $\lambda(\mathbb{T}, Y) \geq \delta^\eta$.
- For each $\rho \in [\delta^{1-\epsilon_{\text{scale}}}, \delta^{\epsilon_{\text{scale}}}]$, $|\mathbb{T}_\rho| \geq \rho^{-2-\zeta}$.

Then

$$\mu(\mathbb{T}, Y) \leq \delta^\nu |\mathbb{T}|^\beta.$$

Our goal in this section is to show that Lemma 14.3.54 implies Main Lemma 14.3.12. We give the full deduction, organized in parallel with Main Lemma 14.3.11 but using Lemma 14.3.51(B), Proposition 14.3.36(B), and Theorem 14.3.47(B) in place of their Part (A) counterparts.

Proof of Main Lemma 14.3.12 using Lemma 14.3.54. If $\beta > 1$, take $\nu = (\beta - 1)/2$. Then $\beta - \nu = (\beta + 1)/2 \geq 1$, and the pointwise bound $\mu(\mathbb{T}, Y) \leq |\mathbb{T}| \leq |\mathbb{T}|^{\beta-\nu}$ proves the conclusion. Hence we may assume $0 < \beta \leq 1$.

Fix $\varepsilon > 0$. Let $\eta_{\text{in}} = \eta_{\text{in}}(\epsilon, \beta) > 0$ and $\delta_0 = \delta_0(\epsilon, \beta)$ be quantities to be chosen below. Let $\mathbb{T}$ be a set of δ -tubes in $B_1 \subset \mathbb{R}^3$ with $\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta_{\text{in}}}$ and $\lambda(\mathbb{T}, Y) \geq \delta^{\eta_{\text{in}}}$. Our goal is to show that if η_{in} and δ_0 are chosen appropriately, then

$$\mu(\mathbb{T}, Y) \leq \delta^{-\epsilon} |\mathbb{T}|^{\beta-\nu}. \quad (14.3.72)$$

We can assume without loss of generality that $(\mathbb{T}, Y)$ is uniform. Let ε_1 and δ_1 be the output of Theorem 14.3.47 with ε as above. We will choose δ_0 with $\delta_0 \leq \delta_1$. Let $\epsilon_{\text{scale}} = \epsilon_{\text{scale}}(\beta)$ be the quantity from Lemma 14.3.54 with β as above. Let $\varepsilon_2 = \varepsilon_2(\varepsilon_1, \epsilon_{\text{scale}}, \beta)$ be a number to be chosen later. At this stage ε_1 , and hence ε_2 , may depend on both ε and β ; the quantifier consequence of this dependence is recorded after the proof. Define $N = \lceil 25/\varepsilon_2^2 \rceil$, and choose

$$0 < \eta_0 \leq \eta_1 \leq \dots \leq \eta_N \leq \varepsilon_2, \quad \eta_{k-1} \leq \varepsilon_2 \eta_k \quad (1 \leq k \leq N),$$

with $\eta_0 \leq \beta/2$. Each η_{k-1} is selected sufficiently small depending on η_k, ε_2 , and β , as required in the argument below. Set

$$\nu = \eta_0.$$

Only after this finite hierarchy and all finitely many applications of $K_{KT}(\beta)$, $K_F(\beta)$, and Lemma 14.3.13 have been fixed do we choose $\eta_{\text{in}} > 0$ smaller than every input-density exponent they require and smaller than η_0 .

Apply Lemma 14.3.51(B), with base exponent η_0 , to $\mathbb{T}$; its initial hypothesis holds because $\eta_{\text{in}} \leq \eta_0$. If Conclusion (i) holds, then (provided we select $\varepsilon_2 \leq \varepsilon_1/5$) we have that $\mathbb{T}$ is $\delta^{-\varepsilon_1}$ Katz-Tao at every scale, and hence by Theorem 14.3.47(B) we have

$$\mu(\mathbb{T}, Y) \leq \delta^{-\varepsilon},$$

and thus (14.3.72) is satisfied.

Henceforth we shall assume that Conclusion (ii) of Lemma 14.3.51(B) holds. Let $\delta \leq \tau \leq \theta \leq 1$ and $1 \leq j \leq N$ be the output of that lemma. We apply Lemma 14.3.30 to $(\mathbb{T}, Y)$ at scale τ and then apply it again to $(\mathbb{T}_\tau, Y_{\mathbb{T}_\tau})$ at scale θ . We obtain some ≈ 1 refinements of these sets of tubes, but we will abuse notation and continue to use the same names. Lemma 14.3.30 gives that for all $T_\tau \in \mathbb{T}_\tau, T_\theta \in \mathbb{T}_\theta$, we have

$$\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{T}[T_\tau], Y) \mu(\mathbb{T}_\tau[T_\theta], Y_{\mathbb{T}_\tau}) \mu(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta}). \quad (14.3.73)$$

and

$$\lambda(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta}), \lambda(\mathbb{T}_\tau, Y_{\mathbb{T}_\tau}) \gtrsim \lambda(\mathbb{T}, Y) \geq \delta^{\eta_{\text{in}}}. \quad (14.3.74)$$

Our goal is to estimate the three quantities on the RHS of (14.3.73).

By Remark 14.3.7(B), for each $T_\tau \in \mathbb{T}_\tau$,

$$\Delta_{\max}(\mathbb{T}[T_\tau], T_\tau) \leq \delta^{-\eta_{\text{in}}}.$$

Uniformity permits us to choose the representative T_τ so that $\lambda(\mathbb{T}[T_\tau], Y) \gtrsim \delta^{\eta_{\text{in}}}$. Rescale T_τ to B_1 . The new tube radius is $\delta/\tau \geq \delta$, so the scale-parameter form of $K_{KT}(\beta)$, with auxiliary scale δ and loss η_0 , gives

$$\mu(\mathbb{T}[T_\tau], Y) \leq \delta^{-\eta_0} |\mathbb{T}[T_\tau]|^\beta. \quad (14.3.75)$$

Put $a_j = \eta_{j-1}$. Conclusion (ii) of Lemma 14.3.51(B) gives $\Delta_{\max}(\mathbb{T}_\theta) \lesssim \theta^{-a_j} \leq \delta^{-a_j}$. Apply the scale-parameter form of Lemma 14.3.13, with auxiliary scale δ and loss $\beta a_j/2$. After decreasing η_{in} to meet its input-density requirement, we obtain

$$\begin{aligned} \mu(\mathbb{T}_\theta, Y_{\mathbb{T}_\theta}) &\lesssim \delta^{-\beta a_j/2} \Delta_{\max}(\mathbb{T}_\theta)^{1-\beta} |\mathbb{T}_\theta|^\beta \\ &\lesssim \delta^{-(1-\beta/2)a_j} |\mathbb{T}_\theta|^\beta \leq \delta^{-a_j} |\mathbb{T}_\theta|^\beta. \end{aligned} \quad (14.3.76)$$

It remains to prove that for each $T_\theta \in \mathbb{T}_\theta$,

$$\mu(\mathbb{T}_\tau[T_\theta], Y_{\mathbb{T}_\tau}) \leq \delta^{10\eta_{j-1}} |\mathbb{T}_\tau[T_\theta]|^\beta. \quad (14.3.77)$$

We will show that this holds, provided each η_{j-1} is selected sufficiently small compared to η_j .

Fix a tube $T_\theta \in \mathbb{T}_\theta$, and let $(\tilde{\mathbb{T}}, \tilde{Y})$ be the image of the tubes in $\mathbb{T}_\tau[T_\theta]$ (and their shadings) under the rescaling taking T_θ to a tube of thickness 1. Hence $\tilde{\mathbb{T}}$ is a family of $\tilde{\delta} = \tau/\theta$ -tubes. Recall that $\tilde{\delta} \leq \delta^{\varepsilon_2}$, and thus to prove (14.3.77) we must show that

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \leq \tilde{\delta}^{10\eta_{j-1}/\varepsilon_2} |\tilde{\mathbb{T}}|^\beta. \quad (14.3.78)$$

Recall that

$$\Delta_{\max}(\tilde{\mathbb{T}}) \leq \tilde{\delta}^{-\eta_{j-1}/\varepsilon_2}, \quad (14.3.79)$$

and

$$\Delta_{\max}(\tilde{\mathbb{T}}_\rho) \geq \rho^{-\eta_j} \quad \text{for all } \rho \in [\tilde{\delta}^{1-\varepsilon_2}, \tilde{\delta}^{\varepsilon_2}]. \quad (14.3.80)$$

We set $\rho = \tilde{\delta}^{1-\varepsilon_2}$. Apply Lemma 14.3.16 to $\tilde{\mathbb{T}}_\rho$ and let $\mathbb{V}$ be the resulting set of planks. Lemma 14.3.16 replaces the set $\tilde{\mathbb{T}}_\rho$ with an ≈ 1 refinement. This in turn induces a ≈ 1 refinement on $(\tilde{\mathbb{T}}, \tilde{Y})$. Abusing notation, we will continue to refer to these new objects as $\tilde{\mathbb{T}}_\rho$ and $\tilde{Y}$. Suppose the planks of $\mathbb{V}$ have dimensions $a \times b \times 1$. We have two cases, depending on these dimensions.

The eccentric case.

We pick another small constant η'_{j-1} with $\eta_{j-1} \ll \eta'_{j-1} \ll \eta_j$. In fact, we pick it by $\eta'_{j-1} = \frac{12\eta_{j-1}}{\varepsilon_2\beta}$. We are in the eccentric case if

$$b \geq \tilde{\delta}^{-\eta'_{j-1}} a. \quad (14.3.81)$$

Applying Proposition 14.3.36(B) with η_{j-1} in place of ε and recalling that $a/b \leq \tilde{\delta}^{\frac{12\eta_j-1}{\varepsilon 2^\beta}}$, we have

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \leq \tilde{\delta}^{-2\eta_{j-1}/\varepsilon 2} \left(\frac{a}{b}\right)^\beta |\tilde{\mathbb{T}}|^\beta \leq \tilde{\delta}^{10\eta_{j-1}/\varepsilon 2} |\tilde{\mathbb{T}}|^\beta.$$

Thus (14.3.78) holds, and we are done.

Henceforth we shall assume that $b < \tilde{\delta}^{-\eta'_{j-1}} a$.

The non-eccentric case.

Recall that $\rho = \tilde{\delta}^{1-\varepsilon 2}$, $\mathbb{V}$ is the maximal density factoring of $\tilde{\mathbb{T}}_\rho$, and each $V \in \mathbb{V}$ has dimensions $a \times b \times 1$, with

$$b \leq \tilde{\delta}^{-\eta'_{j-1}} a.$$

We first apply Lemma 14.3.30 to $(\tilde{\mathbb{T}}, \tilde{Y})$ and $\tilde{\mathbb{T}}_b$, and, we obtain a shading $\tilde{Y}_{\tilde{\mathbb{T}}_b}$ on $\tilde{\mathbb{T}}_b$ so that

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \lesssim \mu(\tilde{\mathbb{T}}_b, \tilde{Y}_{\tilde{\mathbb{T}}_b}) \mu(\tilde{\mathbb{T}}'[T_b], \tilde{Y}').$$

Here $(\tilde{\mathbb{T}}', \tilde{Y}')$ is a $\gtrsim 1$ refinement of $(\tilde{\mathbb{T}}, \tilde{Y})$. By further pigeonholing, we can replace $\mathbb{V}$ by $\mathbb{V}' \subset \mathbb{V}$ with $|\mathbb{V}'| \gtrsim |\mathbb{V}|$ and so that for each $V \in \mathbb{V}'$,

- $|\tilde{\mathbb{T}}'[V]| \gtrsim |\tilde{\mathbb{T}}[V]|$
- $\lambda(\tilde{\mathbb{T}}'[V], \tilde{Y}') \gtrsim \lambda(\tilde{\mathbb{T}}[V], \tilde{Y})$.

In particular, this implies that for each $V \in \mathbb{V}'$,

$$C_F(\tilde{\mathbb{T}}'[V], V) \lesssim 1.$$

At this point, to ease the notation, we write $(\tilde{\mathbb{T}}, \tilde{Y})$ for $(\tilde{\mathbb{T}}', \tilde{Y}')$ and $\mathbb{V}$ for $\mathbb{V}'$.

Applying Lemma 14.3.13 with η_{j-1} in place of ε (and recalling our bound on $\Delta_{\max}(\mathbb{T}_b)$ from (14.3.83)), we have

$$\mu(\tilde{\mathbb{T}}_b, \tilde{Y}_{\tilde{\mathbb{T}}_b}) \leq \tilde{\delta}^{-2\eta'_{j-1}} |\tilde{\mathbb{T}}_b|^\beta. \quad (14.3.82)$$

The main part of the proof is to estimate $\mu(\tilde{\mathbb{T}}[T_b], \tilde{Y})$. First we estimate

$$\Delta_{\max}(\tilde{\mathbb{T}}_b) \leq \frac{|T_b|}{|V|} \Delta_{\max}(\mathbb{V}) \leq \tilde{\delta}^{-\eta'_{j-1}}. \quad (14.3.83)$$

We will select η_{j-1} small enough (depending on η_j , ε_2 , and β) so that $\eta'_{j-1} \leq \eta_j \varepsilon_2$. Comparing (14.3.80) and (14.3.83), we obtain a contradiction unless $b \geq \tilde{\delta}^{\varepsilon_2}$. We will suppose henceforth that $b \geq \tilde{\delta}^{\varepsilon_2}$.

Next we estimate

$$\Delta(\tilde{\mathbb{T}}_\rho, T_b) \geq \frac{|V|}{|T_b|} \Delta(\tilde{\mathbb{T}}_\rho, V) \gtrsim \tilde{\delta}^{\eta'_{j-1}} \Delta_{\max}(\tilde{\mathbb{T}}_\rho),$$

i.e. $C_F(\tilde{\mathbb{T}}_\rho, T_b) \lesssim \tilde{\delta}^{-\eta'_{j-1}}$, and thus by Remark 14.3.7(A), we have

$$C_F(\tilde{\mathbb{T}}_\sigma, T_b) \lesssim \tilde{\delta}^{-\eta'_{j-1}} \quad \text{for all } \sigma \in [\rho, b]. \quad (14.3.84)$$

On the other hand, using (14.3.80) and applying Lemma 14.3.48 at the scales $\sigma \in [\tilde{\delta}^{1-\varepsilon_2}, \tilde{\delta}^{\varepsilon_2}]$, we have that

$$\sigma^{-\eta_j} \leq \Delta_{\max}(\tilde{\mathbb{T}}_\sigma) \lesssim \Delta_{\max}(\tilde{\mathbb{T}}_\sigma[T_b]) \Delta_{\max}(\tilde{\mathbb{T}}_b) \leq \Delta_{\max}(\tilde{\mathbb{T}}_\sigma[T_b]) \tilde{\delta}^{-\eta'_{j-1}},$$

i.e.

$$\Delta_{\max}(\tilde{\mathbb{T}}_\sigma[T_b]) \gtrsim \tilde{\delta}^{\eta'_{j-1}} \sigma^{-\eta_j}. \quad (14.3.85)$$

Comparing (14.3.84) and (14.3.85), we conclude that

$$|\tilde{\mathbb{T}}_\sigma[T_b]| \gtrsim b^2 \tilde{\delta}^{2\eta'_{j-1}} \sigma^{-2-\eta_j} \quad \text{for all } \sigma \in [\tilde{\delta}^{1-\varepsilon_2}, \tilde{\delta}^{\varepsilon_2}]. \quad (14.3.86)$$

We bound $\mu(\tilde{\mathbb{T}}[T_b], \tilde{Y})$ using Lemma 14.3.54 with $\zeta = \eta_j/2$. Rescale T_b to B_1 , and denote the resulting family by $\mathbb{T}'$. Its tube radius is

$$\delta' = \tilde{\delta}/b \leq \tilde{\delta}^{1-\varepsilon_2}.$$

We now verify the all-scale cardinality hypothesis rather than merely citing it. Write $b = \tilde{\delta}^q$; the bound $b \geq \tilde{\delta}^{\varepsilon_2}$ says $0 \leq q \leq \varepsilon_2$. If $\sigma' \in [(\delta')^{1-\varepsilon_{\text{scale}}}, (\delta')^{\varepsilon_{\text{scale}}}]$ and $\sigma = b\sigma'$, then, provided $\varepsilon_2 \leq \varepsilon_{\text{scale}}/2$,

$$\tilde{\delta}^{1-\varepsilon_2} \leq b(\delta')^{1-\varepsilon_{\text{scale}}} \leq \sigma \leq b(\delta')^{\varepsilon_{\text{scale}}} \leq \tilde{\delta}^{\varepsilon_2}.$$

Thus (14.3.86) applies. Since thickening commutes with this affine rescaling up to fixed constants,

$$\begin{aligned} |\mathbb{T}'_{\sigma'}| &\approx |\tilde{\mathbb{T}}_{b\sigma'}[T_b]| \\ &\gtrsim \tilde{\delta}^{2\eta'_{j-1}} b^{-\eta_j} (\sigma')^{-2-\eta_j}. \end{aligned}$$

The largest possible σ' is $(\delta')^{\varepsilon_{\text{scale}}}$, and $\delta' \leq \tilde{\delta}^{1-\varepsilon_2}$. Hence $|\mathbb{T}'_{\sigma'}| \gtrsim (\sigma')^{-2-\eta_j/2}$ throughout the required interval as soon as

$$2\eta'_{j-1} \leq \frac{1}{2}(1-\varepsilon_2)\varepsilon_{\text{scale}}\eta_j.$$

Let η_{vns} and $\nu_{\text{vns}} > 0$ be the constants in Lemma 14.3.54 for $\zeta = \eta_j/2$. The affine map preserves normalized convex density. By (14.3.79) and downward heredity,

$$\Delta_{\max}(\mathbb{T}') \leq \tilde{\delta}^{-\eta_{j-1}/\varepsilon_2} \leq (\delta')^{-\eta_{\text{vns}}}$$

provided $\eta_{j-1}/\varepsilon_2 \leq (1 - \varepsilon_2)\eta_{\text{vns}}$. Similarly, $\tilde{\delta} \leq \delta^{\varepsilon_2}$ and $\delta' \leq \tilde{\delta}^{1-\varepsilon_2}$ show that the inherited shading density is at least $(\delta')^{\eta_{\text{vns}}}$ after choosing the original η_{in} so that $\eta_{\text{in}}/\varepsilon_2 \leq (1 - \varepsilon_2)\eta_{\text{vns}}$ and absorbing the subpower refinement losses. All hypotheses of Lemma 14.3.54 are now verified. It gives

$$\begin{aligned} \mu(\tilde{\mathbb{T}}[T_b], \tilde{Y}) &= \mu(\mathbb{T}', Y_{\mathbb{T}'}) \\ &\leq (\delta')^{\nu_{\text{vns}} |\mathbb{T}'|^\beta} \leq \tilde{\delta}^{(1-\varepsilon_2)\nu_{\text{vns}}} |\tilde{\mathbb{T}}[T_b]|^\beta. \end{aligned} \quad (14.3.87)$$

Put $\nu_0 = (1 - \varepsilon_2)\nu_{\text{vns}}$. Multiplying (14.3.87) and (14.3.82), and using uniformity to identify the product of the two cardinalities with $|\tilde{\mathbb{T}}|$, gives

$$\mu(\tilde{\mathbb{T}}, \tilde{Y}) \lesssim \tilde{\delta}^{\nu_0 - 2\eta'_{j-1}} |\tilde{\mathbb{T}}|^\beta.$$

Choosing η_{j-1} sufficiently small compared with the quantities already fixed makes $\nu_0 - 2\eta'_{j-1} \geq 10\eta_{j-1}/\varepsilon_2$. This establishes (14.3.78).

Returning to (14.3.73), combine (14.3.75), (14.3.77), and (14.3.76). Uniformity identifies the product of the three cardinalities with $|\mathbb{T}|$, up to a subpower factor, and gives

$$\mu(\mathbb{T}, Y) \lesssim \delta^{-\eta_0 + 10\eta_{j-1} - \eta_{j-1}} |\mathbb{T}|^\beta \leq \delta^{8\eta_0} |\mathbb{T}|^\beta,$$

because $\eta_{j-1} \geq \eta_0$. Decreasing δ_0 absorbs the remaining subpower factor into $\delta^{-\eta_0}$, so

$$\mu(\mathbb{T}, Y) \leq \delta^{7\eta_0} |\mathbb{T}|^\beta.$$

Finally, essential distinctness gives $|\mathbb{T}| \lesssim \delta^{-4}$. Since $\nu = \eta_0 \leq \beta/2$,

$$\delta^{7\eta_0} |\mathbb{T}|^\beta \lesssim \delta^{3\eta_0} |\mathbb{T}|^{\beta-\eta_0} \leq \delta^{-\epsilon} |\mathbb{T}|^{\beta-\nu}.$$

This proves (14.3.72) for the fixed target loss ϵ . □

Remark 14.3.55 (Quantifier audit for Main Lemma 14.3.12). The preceding argument is complete for each fixed target loss ϵ , but it does not, as presently parameterized, prove the stated uniform dependence $\nu = \nu(\beta)$. Indeed, ε_1 is an input exponent returned by Theorem 14.3.47 for the chosen ε ; ε_2 , N , the hierarchy $\eta_0, \dots, \eta_N$, and finally the gain $\nu = \eta_0$ may therefore depend on ε . The proof establishes a gain $\nu(\beta, \varepsilon) > 0$ at each fixed loss. To obtain Main Lemma 14.3.12 as stated, one still needs a uniform parameter-selection lemma, or an alternative exponent-descent argument that removes this ε -dependence. No such argument appears in the present source. Until that dependency is closed, the use of Main Lemma 14.3.12 in the deduction of Theorem 14.3.1 remains conditional.

Lemma 14.3.56 (Fixed-loss gains do not imply a uniform gain). *Let $P(\epsilon, \nu)$ be a two-parameter assertion. The implication*

$$(\forall \epsilon > 0 \exists \nu_\epsilon > 0 : P(\epsilon, \nu_\epsilon)) \implies (\exists \nu > 0 \forall \epsilon > 0 : P(\epsilon, \nu))$$

is false, even when $P(\epsilon, \nu)$ is monotone in the gain in the natural sense that $P(\epsilon, \nu)$ implies $P(\epsilon, \nu')$ for $0 < \nu' \leq \nu$.

Proof. Take $P(\epsilon, \nu)$ to mean $0 < \nu \leq \epsilon$. For every $\epsilon > 0$ the choice $\nu_\epsilon = \epsilon$ verifies the first statement, and the asserted monotonicity is immediate. If a fixed $\nu > 0$ verified the second statement, then choosing $\epsilon = \nu/2$ would give $\nu \leq \nu/2$, a contradiction. $\square$

Remark 14.3.57 (What would discharge the quantifier defect). Lemma 14.3.56 does not say that Main Lemma 14.3.12 is false. It says that the fixed-loss conclusion proved above cannot, by a formal rearrangement of quantifiers, establish the stated lemma. This route becomes valid if one supplies a parameter-selection result which, for each fixed $\beta > 0$, gives

$$\inf_{0 < \epsilon \leq \epsilon_*(\beta)} \eta_0(\beta, \epsilon) > 0,$$

where η_0 is the terminal gain produced by the stopping hierarchy; one may then take half this infimum and handle larger losses by monotonicity. Equivalently, a replacement all-scale theorem whose admissible input tolerance is independent of its requested output loss would decouple the stopping hierarchy from ϵ . Neither statement is among the current hypotheses or proved results.

14.3.9.2 Maximal density factoring revisited

In Lemma 14.3.16, we chose sets W to maximize $\Delta(\mathbb{V}, W)$. For some families $\mathbb{V}$ of convex sets (for example if $\mathbb{V}$ is both Katz-Tao and Frostman), there might be many equally valid candidates for W . To prove Lemma 14.3.54, it is important to choose carefully in the case of a tie or a near tie. When the maximum density is achieved by two convex sets with very different sizes, we choose W to be the convex set of smaller volume. More precisely, we adjust our maximization process as follows. We let $\eta_{\text{bias}} > 0$ be a tiny exponent to be chosen later. Then, instead of choosing W to maximize $\Delta(\mathbb{V}, W)$, we choose W to maximize

$$|W|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}, W).$$

This leads to a variation of Lemma 14.3.16 with two small changes:

Lemma 14.3.58. (*Biased maximal density factoring lemma*) Let $\mathbb{V}$ be a finite family of convex subsets of $\mathbb{R}^n$ contained in a convex set U , and suppose $\eta_{\text{bias}} > 0$. Then there is a set $\mathbb{V}' \subset \mathbb{V}$ with $\sum_{V \in \mathbb{V}'} |V| \gtrsim \sum_{V \in \mathbb{V}} |V|$; a family $\mathbb{W}$ of convex in $\mathbb{R}^n$, each of approximately the same dimensions; and a partition

$$\mathbb{V}' = \bigsqcup_{W \in \mathbb{W}} \mathbb{V}_W, \quad (14.3.88)$$

where each set in $\mathbb{V}_W$ is contained in W . This partition has the following properties:

If $K \subset W \in \mathbb{W}$,

$$\Delta(\mathbb{V}_W, K) \lesssim \left(\frac{|K|}{|W|} \right)^{\eta_{\text{bias}}} \Delta(\mathbb{V}_W, W). \quad (14.3.89)$$

In particular, $C_F(\mathbb{V}_W, W) \lesssim 1$.

For each $W \in \mathbb{W}$, $\Delta(\mathbb{V}_W, W) \gtrsim \left(\frac{|W|}{|U|} \right)^{\eta_{\text{bias}}} \Delta_{\max}(\mathbb{V}')$.

$$\Delta_{\max}(\mathbb{W}) \lesssim \left(\frac{|W|}{|U|} \right)^{-\eta_{\text{bias}}}. \quad (14.3.90)$$

Proof. After a dyadic pigeonholing in the dimensions of the members of $\mathbb{V}$, we may assume that they all have comparable volume v ; this loses only a polylogarithmic factor in the retained total volume. Put $\mathbb{V}_0 = \mathbb{V}$. Inductively, while $\mathbb{V}_m$ is nonempty, choose a convex set $W_m \subset U$ maximizing

$$\Phi_m(W) = |W|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_m, W),$$

set $\mathbb{V}_{W_m} = \mathbb{V}_m[W_m]$, and remove that block: $\mathbb{V}_{m+1} = \mathbb{V}_m \setminus \mathbb{V}_{W_m}$. As in the compactness reduction in Lemma 14.3.16, a maximizer exists after replacing the finitely many members of $\mathbb{V}$ by their closures. The blocks are disjoint and exhaust $\mathbb{V}$.

Dyadically pigeonhole the value of $\Delta(\mathbb{V}_m, W_m)$ and the dimensions of W_m , retaining an index set I for which

$$\sum_{m \in I} \sum_{V \in \mathbb{V}_{W_m}} |V| \gtrsim \sum_{V \in \mathbb{V}} |V|.$$

Let $\mathbb{W} = \{W_m : m \in I\}$ and $\mathbb{V}' = \bigsqcup_{m \in I} \mathbb{V}_{W_m}$. This gives the asserted partition and volume retention.

Fix $m \in I$ and a convex $K \subset W_m$. Every member of $\mathbb{V}_m$ contained in K belongs to $\mathbb{V}_{W_m}$, and maximality of W_m gives

$$|K|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_{W_m}, K) = |K|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_m, K) \leq |W_m|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_m, W_m).$$

This is (14.3.89); in particular its right-hand side is the density at W_m , so $C_F(\mathbb{V}_{W_m}, W_m) \lesssim 1$.

Let m_0 be the first retained index and choose a convex $K_0 \subset U$ realizing $\Delta_{\max}(\mathbb{V}')$. Since $\mathbb{V}' \subset \mathbb{V}_{m_0}$, maximality at m_0 implies

$$|W_{m_0}|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_{m_0}, W_{m_0}) \geq |K_0|^{-\eta_{\text{bias}}} \Delta(\mathbb{V}_{m_0}, K_0) \geq |K_0|^{-\eta_{\text{bias}}} \Delta_{\max}(\mathbb{V}').$$

Because $K_0 \subset U$, and because the retained W_m have comparable dimensions and densities, for every $m \in I$ this yields

$$\Delta(\mathbb{V}_{W_m}, W_m) \gtrsim (|W_m|/|U|)^{\eta_{\text{bias}}} \Delta_{\max}(\mathbb{V}'). \quad (14.3.91)$$

Finally let $K \subset U$ be convex. The disjointness of the blocks gives

$$\Delta(\mathbb{V}', K) = \frac{v \# \mathbb{V}'[K]}{|K|} \geq \frac{v}{|K|} \sum_{W \in \mathbb{W}[K]} \# \mathbb{V}_W = \frac{|W| \# \mathbb{W}[K]}{|K|} \Delta(\mathbb{V}_W, W),$$

up to the fixed dyadic comparability constants. Combining this with (14.3.91) and $\Delta(\mathbb{V}', K) \leq \Delta_{\max}(\mathbb{V}')$ gives

$$\Delta(\mathbb{W}, K) \lesssim (|W|/|U|)^{-\eta_{\text{bias}}}.$$

Taking the supremum over K proves (14.3.90), and (14.3.91) is the intervening assertion of the lemma. $\square$

We will call the decomposition $\mathbb{V} = \bigsqcup_{\mathbb{W}} \mathbb{V}_W$ a *maximal density factoring with bias η_{bias}* . Observe that we can still apply Proposition 14.3.18 to the factoring $\bigsqcup_{\mathbb{W}} \mathbb{V}_W$ —Proposition 14.3.18 requires that $\mathbb{V}_W$ be C -Frostman in W , and this remains the case with $C \sim 1$.

14.3.9.3 Setup for the proof of Lemma 14.3.54

To prove Lemma 14.3.54, we will study how $\mathbb{T}$ behaves in smaller balls at various scales. Our goal is to prove that

$$\mu(\mathbb{T}, Y) \lesssim \delta^\nu |\mathbb{T}|^\beta, \quad (14.3.92)$$

where $\nu = \nu(\beta, \zeta) > 0$. This goal can be written equivalently as

$$|U(\mathbb{T}, Y)| \gtrsim \delta^{-\nu} |T| |\mathbb{T}|^{1-\beta}, \quad (14.3.93)$$

where $\nu = \nu(\beta, \zeta) > 0$.

The argument has several cases. In some cases, we will prove (14.3.92) and in some cases we will prove (14.3.93). In some of the cases, we will prove (14.3.93) by showing that $U(\mathbb{T}, Y)$ fills a larger than expected fraction of a

typical a -ball that intersects $U(\mathbb{T}, Y)$, and we will pair this with an estimate on $U(\mathbb{T}_a)$.

We have to do some pigeonholing to prepare for our argument. First we refine $(\mathbb{T}, Y)$ so that for each $x \in U(\mathbb{T}, Y)$, the set $\mathbb{T}_{Y'}(x)$ is uniform. We identify $\mathbb{T}_Y(x)$ with a subset of S^2 corresponding to the directions of the tubes. Then for each x , we can choose a uniform subset of $U_x \subset \mathbb{T}_Y(x)$ with $|U_x| \gtrsim |\mathbb{T}_Y(x)|$. Then we define Y' so that $x \in Y'(T)$ if and only if $T \in U_x$. This is a $\gtrsim 1$ refinement. Because of the uniformity, there is a function $\mu(\rho)$ so that for any $T_0 \in \mathbb{T}$ and $x \in Y'(T_0)$,

$$\#\{T \in \mathbb{T}_{Y'}(x) : \text{angle}(T, T_0) \leq \rho\} \approx \mu(\rho) \quad (14.3.94)$$

In particular, we have $\mu(\mathbb{T}, Y) \lesssim \mu(\mathbb{T}, Y') \approx \mu(1)$. Moreover, for each $T_\rho \in \mathbb{T}_\rho$, we have $\mu(\mathbb{T}[T_\rho], Y') \approx \mu(\rho)$. Abusing notation, we will refer to Y' as Y from here on.

Next we apply Lemma 14.3.30 to $(\mathbb{T}, Y)$ and $\mathbb{T}_a$, simultaneously for the dyadic-type scales $a = \delta^{\eta j}$, $1 \leq j \leq \eta^{-1}$. We obtain a ≈ 1 refinement of $(\mathbb{T}, Y)$ (abusing notation, we will continue to refer to this as $(\mathbb{T}, Y)$) so that for each $x \in U(\mathbb{T}_a, Y_{\mathbb{T}_a})$, we have

$$|U(\mathbb{T}, Y)| \gtrsim |U(\mathbb{T}_a, Y_{\mathbb{T}_a})| \frac{|U(\mathbb{T}, Y) \cap B(x, a)|}{|B(x, a)|}. \quad (14.3.95)$$

Recall that $\lambda(\mathbb{T}_a, Y_{\mathbb{T}_a}) \gtrsim \lambda(\mathbb{T}, Y) \geq \delta^\eta$. We also have

$$\Delta_{\max}(\mathbb{T}_a) \leq \delta^{-\eta} \frac{(a/\delta)^2}{|\mathbb{T}[T_a]|}.$$

Applying Lemma 14.3.13 with $\nu/10$ in place of ε , we have (provided $\eta > 0$ is chosen sufficiently small)

$$|U(\mathbb{T}_a, Y_{\mathbb{T}_a})| \gtrsim \delta^{\nu/9} \left(|\mathbb{T}_a| |T_a| \right) / \left(\left(\frac{|T_a|/|T|}{|\mathbb{T}[T_a]|} \right)^{1-\beta} |\mathbb{T}_a|^\beta \right) = \delta^{\nu/9} \left(\frac{a}{\delta} \right)^{2\beta} |T| |\mathbb{T}|^{1-\beta}. \quad (14.3.96)$$

Chaining (14.3.95) and (14.3.96), we conclude that in order to obtain our desired estimate (14.3.93), it suffices to prove that there exists an a -ball B_a with

$$\frac{|U(\mathbb{T}) \cap B_a|}{|B_a|} \gtrsim \delta^{-2\nu} \left(\frac{\delta}{a} \right)^{2\beta}. \quad (14.3.97)$$

With these preparations, we are ready to set up the main objects in the proof. Let $r_1 = \delta^{\epsilon_{\text{scale}}}$. Divide $U(\mathbb{T}_{r_1}, Y)$ into boundedly overlapping balls B of radius r_1 .

For each ball B , we define $\mathbb{T}_B$ to be a set of essentially distinct $\delta \times \delta \times r_1$ -tubes contained in B of the form $T \cap B$ with $T \in \mathbb{T}$. For each $T_B \in \mathbb{T}_B$, we let $\mathbb{T}(T_B)$ be the set of tubes $T \in \mathbb{T}$ so that $T \cap B$ is comparable to T_B .

Next we define a shading on $\mathbb{T}_B$. For each $T_B \in \mathbb{T}_B$, we consider the multiplicity $|Y_{\mathbb{T}(T_B)}(x)|$. We choose $Y_B(T_B) \subset T_B$ by pigeonholing the multiplicity $|Y_{\mathbb{T}(T_B)}(x)|$. Finally we refine Y so that if $T \in \mathbb{T}(T_B)$, $Y(T) \cap T_B \subset Y_B(T_B)$. This is a $\gtrsim 1$ refinement. Next we refine $(\mathbb{T}_B, Y_B)$ by pigenholing $|Y_B(T_B)|$. Again we refine Y so that $Y(T) \cap T_B \subset Y_B(T_B)$ and again this is a $\gtrsim 1$ refinement.

We let $\mathbb{B}$ be the set of balls B with $\lambda(\mathbb{T}_B, Y_B) \gtrsim \lambda(\mathbb{T}, Y)$. Then we refine Y by removing the parts of $Y(T)$ that are not in the balls of $\mathbb{B}$. Again this is a $\gtrsim 1$ refinement.

In the sequel, we will use the following properties of the shading $(\mathbb{T}_B, Y_B)$:

- $\lambda(\mathbb{T}_B, Y_B) \gtrsim \delta^\eta$ for each $B \in \mathbb{B}$.
- $Y_B(T_B)$ is roughly the same for all $T_B \in \mathbb{T}_B$.
- If $Y(T) \cap B$ is non-empty, then $T \in \mathbb{T}(T_B)$ for some $T_B \in \mathbb{T}_B$ and $Y(T) \subset Y_B(T_B)$.
- The multiplicity $|\mathbb{T}(T_B)_Y(x)|$ is roughly constant for $x \in Y_B(T_B)$.

Now we apply the biased maximal density factoring, Lemma 14.3.58, to $\mathbb{T}_B$, and we let $\mathbb{W}_B$ be the resulting family of convex sets (the maximal density factoring replaces $\mathbb{T}_B$ by a large subset, but we will abuse notation and continue to refer to this set as $\mathbb{T}_B$; this is harmless because of all the uniformity in the bullet points above). By pigeonholing, we can suppose that for each B , the convex sets $W \in \mathbb{W}_B$ have dimensions $a \times b \times r_1$, with $a \leq b \leq r_1$. Thus for each ball B we obtain a decomposition $\mathbb{T}_B = \bigsqcup_{\mathbb{W}_B} \mathbb{T}_{B,W}$.

The way we proceed depends on the dimensions of W . We let $\tau > 0$ be a new small parameter, to be chosen later. If $a \geq \delta^{1-\tau}$, then we are in the *thick case*. In this case we will show that $U(\mathbb{T}, Y)$ is large by proving (14.3.97). If $a \leq \delta^{1-\tau}$, then we are in the *thin case*. This case divides into several sub-cases. In the most important sub-case, we will estimate $\mu(\mathbb{W}_B)$ by an induction on scales. We deal with these cases in the next two sections.

14.3.9.4 The thick case

In this section, we prove Lemma 14.3.54 in the thick case. Recall that we applied the maximal density factoring lemma with bias η_{bias} . Thus by (14.3.89), for any $W \in \mathbb{W}_B$ and any convex set $K \subset B$,

$$\Delta(\mathbb{T}_{B,W}, W) / \Delta(\mathbb{T}_{B,W}, K) \gtrsim \left(\frac{|W|}{|K|} \right)^{\eta_{\text{bias}}}. \quad (14.3.98)$$

In particular, taking $K = T_B$, we have

$$\Delta(\mathbb{T}_{B,W}, W) \gtrsim \left(\frac{|W|}{|T_B|} \right)^{\eta_{\text{bias}}} \Delta(\mathbb{T}_B, T_B) \approx \left(\frac{|W|}{|T_B|} \right)^{\eta_{\text{bias}}},$$

and so

$$|\mathbb{T}_{B,W}| \gtrsim \left(\frac{|W|}{|T_B|} \right)^{1+\eta_{\text{bias}}}. \quad (14.3.99)$$

In the thick case, we will prove (14.3.97) for balls of radius a :

$$\frac{|U(\mathbb{T}) \cap B_a|}{|B_a|} \gtrsim \delta^{-2\nu} \left(\frac{\delta}{a} \right)^{2\beta}.$$

This inequality says that $U(\mathbb{T}, Y)$ fills a surprisingly large fraction of a typical a -ball, and it implies our goal as explained in Subsection 14.3.9.3. For the remainder of Section 14.3.9.4 we will fix an r_1 -ball B for which $\lambda(\mathbb{T}_B, Y_B) \gtrsim \delta^\eta$. Since each $W \in \mathbb{W}_B$ has thickness a , in order to establish (14.3.97) it suffices to prove that there exists at least one $W \in \mathbb{W}_B$ for which

$$\frac{|U(\mathbb{T}_B, Y_B) \cap W|}{|W|} \gtrsim \delta^{-2\nu} \left(\frac{\delta}{a} \right)^{2\beta}. \quad (14.3.100)$$

We fix a choice of W for which $\lambda(\mathbb{T}_{B,W}, Y_B) \gtrsim \delta^\eta$.

Warmup: A simple case

We begin with a special case to illustrate the main ideas of the proof. The simplest scenario in the thick case is when W is an r_1 -ball (i.e. $W = B$), and hence $a = r_1$. The strong estimate (14.3.89) implies that $C_F(\mathbb{T}_B, B) \lesssim 1$, and so we can apply $K_F(\beta)$ to $\mathbb{T}_B$. Strictly speaking, to do the application we rescale $B = B(x, r)$ to the unit ball B_1 , which sends $\mathbb{T}_B$ to a set $\tilde{\mathbb{T}}_B$ of δ/a -tubes, and transforms the shading $Y_{\mathbb{T}_B}$ accordingly. Then we get

$$\begin{aligned} \frac{|U(\mathbb{T}_B, Y_B) \cap W|}{|W|} &= \frac{|U(\tilde{\mathbb{T}}_B, Y_{\tilde{\mathbb{T}}_B})|}{|B_1|} \sim |U(\tilde{\mathbb{T}}_B, Y_{\tilde{\mathbb{T}}_B})| \\ &\geq \delta^\epsilon (\delta/a)^{2\beta} (|\tilde{T}_B| |\tilde{\mathbb{T}}_B|)^{\beta/2}, \end{aligned}$$

where $\epsilon > 0$ is a parameter we are free to choose (provided the input density exponent is sufficiently small depending on ϵ). We choose $\epsilon = \tau \eta_{\text{bias}} \beta / 4$. By (14.3.99), we know that

$$|\mathbb{T}_B| = |\mathbb{T}_{B,W}| \gtrsim \left(\frac{|B|}{|T_B|} \right)^{1+\eta_{\text{bias}}} = \left(\frac{|B_1|}{|\tilde{T}_B|} \right)^{1+\eta_{\text{bias}}}, \quad (14.3.101)$$

and so the factor in parentheses on the right-hand side of (14.3.101) is $\gtrsim (\delta/a)^{-2\eta_{\text{bias}}}$. All together we have

$$\frac{|U(\mathbb{T}_B, Y_B)|}{|B|} \gtrsim \frac{|U(\mathbb{T}_B)|}{|B|} \gtrsim \delta^\varepsilon \left(\frac{\delta}{a}\right)^{2\beta - \eta_{\text{bias}}\beta}.$$

Since $a \geq \delta^{1-\tau}$, this gives

$$\frac{|U(\mathbb{T}_B, Y_B)|}{|B|} \gtrsim \delta^\varepsilon \delta^{-\tau\eta_{\text{bias}}\beta} \left(\frac{\delta}{a}\right)^{2\beta}.$$

This gives (14.3.100) if we choose $\nu < \tau\eta_{\text{bias}}\beta/8$.

The thick case, general version

In the general version of the thick case, W has dimensions $a \times b \times r_1$, with $\delta^{1-\tau} \leq a \leq b \leq r_1$.

We need to prove (14.3.100). To do so, we make a linear change of variables to convert W to B_1 . This converts $\mathbb{T}_{B,W}$ to a set of planks $\mathbb{P}$ in B_1 . Let $Y_{\mathbb{P}}$ denote the image of the shading Y_B under this transformation. We have $\frac{|U(\mathbb{T}_{B,W}, Y_B)|}{|W|} \sim |U(\mathbb{P}, Y_{\mathbb{P}})|$, so it suffices to prove

$$|U(\mathbb{P}, Y_{\mathbb{P}})| \gtrsim \delta^{-2\nu} \left(\frac{\delta}{a}\right)^{2\beta}.$$

If W has dimensions $a \times b \times r$, then a plank $P \in \mathbb{P}$ has dimensions $\frac{\delta}{b} \times \frac{\delta}{a} \times 1$. Note that $C_F(\mathbb{P}, B_1) \sim C_F(\mathbb{T}_{B,W}, W) \lesssim 1$ by (14.3.89), and $\lambda(\mathbb{P}, Y_{\mathbb{P}}) \sim \lambda(\mathbb{T}_{B,W}, Y_B) \gtrsim \delta^\eta$.

Now we apply Lemma 14.3.34. Before doing so, we must estimate the quantity “ M ” from that lemma. For any $\theta \in [a/b, 1]$, let $P \in \mathbb{P}$, and let P_θ be the corresponding $\theta \frac{\delta}{a} \times \frac{\delta}{a} \times 1$ thickened plank, as described in Definition 14.3.40. From the statement of Lemma 14.3.34 we can choose any M with $M \geq 1$ and $M \geq |\mathbb{P}[P_\theta]|$ for each $\theta \in [a/b, 1]$. So we need to estimate $\mathbb{P}[P_\theta]$.

Recall that $\mathbb{P}$ is related to $\mathbb{T}_B[W]$ by a linear change of variables. When we undo that change of variables, the planks $\mathbb{P}$ become $\mathbb{T}_B[W]$ and each plank P becomes a $\delta \times \delta \times r$ tube. The thickened plank P_θ becomes a rectangular solid T_θ with dimensions $\delta \times b' \times r$ with $\delta \leq b' \leq r$ (the value of b' depends on the orientation of the plank P).

It is here that we make crucial use of the modified maximal density factoring lemma. By (14.3.98), we have

$$\frac{\Delta(\mathbb{T}_B, W)}{\Delta(\mathbb{T}_B, T_\theta)} \gtrsim \left(\frac{|W|}{|T_\theta|}\right)^{\eta_{\text{bias}}}.$$

Since W has dimensions $a \times b \times r$ and T_θ has dimensions $\delta \times b' \times r$ with $b' \leq b$, we have

$$\left(\frac{|W|}{|T_\theta|}\right)^{\eta_{\text{bias}}} \geq \left(\frac{a}{\delta}\right)^{\eta_{\text{bias}}}.$$

and hence

$$\Delta(\mathbb{P}, P_\theta) = \Delta(\mathbb{T}_B, T_\theta) / \Delta(\mathbb{T}_B, W) \lesssim \left(\frac{\delta}{a}\right)^{\eta_{\text{bias}}}.$$

Now we are ready to estimate $\mathbb{P}[P_\theta]$. Note that $|\mathbb{P}[P_\theta]| = |\mathbb{T}_B[T_\theta]|$ and $|\mathbb{P}| = |\mathbb{T}_B[W]|$ and so

$$\begin{aligned} \frac{|\mathbb{P}[P_\theta]|}{|\mathbb{P}|} &= \frac{|\mathbb{T}_B[T_\theta]|}{|\mathbb{T}_B[W]|} \\ &= \frac{\Delta(\mathbb{T}_B, T_\theta)}{\Delta(\mathbb{T}_B, W)} \frac{|T_\theta|}{|W|} \\ &\lesssim \left(\frac{|T_\theta|}{|W|}\right)^{1+\eta_{\text{bias}}} = |P_\theta|^{1+\eta_{\text{bias}}} \\ &= \left(\theta \frac{\delta^2}{a^2}\right)^{1+\eta_{\text{bias}}} \leq \theta \left(\frac{\delta}{a}\right)^{2+\eta_{\text{bias}}}. \end{aligned}$$

Recalling that P_θ has dimensions $\theta \frac{\delta}{a} \times \frac{\delta}{a} \times 1$, we see that $\frac{|\mathbb{P}[P_\theta]|}{|\mathbb{P}|} \leq \theta(\delta/a)^{2+\eta_{\text{bias}}}$, and rearranging gives

$$|\mathbb{P}[P_\theta]|/\theta \lesssim (\delta/a)^{2+\eta_{\text{bias}}} |\mathbb{P}|.$$

Let us verify that $(\delta/a)^{2+\eta_{\text{bias}}} |\mathbb{P}| \geq 1$, i.e. $|\mathbb{P}| \geq (a/\delta)^{2+\eta_{\text{bias}}}$. This follows from (14.3.99), which says that

$$|\mathbb{P}| = |\mathbb{T}_{B,W}| \gtrsim \left(\frac{|W|}{|T_B|}\right)^{1+\eta_{\text{bias}}} = \left(\frac{a \times b \times r_1}{\delta \times \delta \times r_1}\right)^{1+\eta_{\text{bias}}} \geq (a/\delta)^{2+2\eta_{\text{bias}}}.$$

Thus we can apply Lemma 14.3.34 with $M = (\delta/a)^{2+\eta_{\text{bias}}} |\mathbb{P}| \geq 1$. We conclude that

$$|U(\mathbb{P}, Y_{\mathbb{P}})| \gtrsim \delta^\varepsilon \left(\frac{\delta}{a}\right)^{2\beta} \left(M^{-1} \left(\frac{\delta}{a}\right)^2 |\mathbb{P}|\right)^{\beta/2} = \delta^\varepsilon \left(\frac{\delta}{a}\right)^{2\beta - \eta_{\text{bias}}\beta/2}.$$

Since $a \geq \delta^{1-\tau}$, if we select $\varepsilon < \eta_{\text{bias}}\beta\tau/4$, then this yields (14.3.100) with $\nu = \eta_{\text{bias}}\beta\tau/8$. This finishes the proof in the thick case.

Remark 14.3.59. Notice that in the thick case, the exponent ν obeys

$$\nu \sim \tau \eta_{\text{bias}} \beta.$$

This value of ν does not directly depend on ζ . But in the thin case, we will have to choose η_{bias} and τ much smaller than ζ , so indirectly ν depends on ζ .

14.3.9.5 The thin case

Now we consider the thin case when $a \in [\delta, \delta^{1-\tau}]$. Fix a ball B . Recall that each set $W \in \mathbb{W}_B$ has dimensions $a \times b \times r_1$, with $r_1 = \delta^{\epsilon_{\text{scale}}}$. We also have that each tube segment $T_B \in \mathbb{T}_B$ has roughly the same density of shading, i.e. $|Y_B(T_B)| \sim \lambda(\mathbb{T}_B, Y_B)|T_B| \gtrsim \delta^\eta |T_B|$ for each $T_B \in \mathbb{T}_B$. After a harmless refinement we can suppose that $Y_B(T_B)$ is covered by a union of δ -balls, each of which satisfies

$$|B_\delta \cap Y_B(T_B)| \gtrsim \delta^\eta |B_\delta|. \quad (14.3.102)$$

Apply Proposition 14.3.18 to $\mathbb{T}_B = \bigsqcup_{W \in \mathbb{W}_B} \mathbb{T}_{B,W}$. We obtain a shading $Y'_B(T_B) \subset Y_B(T_B)$ on $\mathbb{T}_B$ so that $(\mathbb{T}_B, Y'_B)$ is an ≈ 1 refinement of $(\mathbb{T}_B, Y_B)$; a subset $\mathbb{W}'_B$ of $\mathbb{W}_B$; and a shading $Y_{\mathbb{W}'_B}$ on $\mathbb{W}_B$ so that

$$\lambda(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \gtrsim \delta^{2\eta}, \quad (14.3.103)$$

and

$$(\mathbb{T}_B)_{Y'_B}(x) \subset \bigcup_{\substack{W \in \mathbb{W}'_B \\ x \in Y_{\mathbb{W}'_B}(W)}} (\mathbb{T}_{B,W})_{Y'_B}(x). \quad (14.3.104)$$

Recall from Remark 14.3.21, that $Y_{\mathbb{W}'_B}(W)$ is the a -neighborhood of $U(\mathbb{T}_{B,W}, Y'_B)$. Combining this with (14.3.102) we have that

$$|U(\mathbb{T}_B, Y'_B)| \gtrsim (\delta/a)^3 \delta^\eta |U(\mathbb{W}'_B, Y_{\mathbb{W}'_B})| \gtrsim \delta^{3\tau+\eta} |U(\mathbb{W}'_B, Y_{\mathbb{W}'_B})|. \quad (14.3.105)$$

Slab case

Recall that $r_1 = \delta^{\epsilon_{\text{scale}}}$, where $\epsilon_{\text{scale}} > 0$ is the constant from Lemma 14.3.54, which is a tiny constant that we get to choose. We define the slab case to be the case $b \geq \delta^{\epsilon_{\text{scale}}} r_1 = \delta^{2\epsilon_{\text{scale}}}$. In this case, W is a giant slab of dimensions at least $a \times \delta^{2\epsilon_{\text{scale}}} \times \delta^{\epsilon_{\text{scale}}}$.

Applying Lemma 14.3.39 to $(\mathbb{W}'_B, Y_{\mathbb{W}'_B})$, we have $\mu(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \lesssim \delta^{-8\eta-8\epsilon_{\text{scale}}}$, i.e.

$$|U(\mathbb{W}'_B, Y_{\mathbb{W}'_B})| \gtrsim \delta^{8\eta+8\epsilon_{\text{scale}}} |\mathbb{W}'_B| |W|.$$

Combining this with (14.3.105), we have

$$|U(\mathbb{T}_B, Y'_B)| \gtrsim \delta^{8\eta+8\epsilon_{\text{scale}}+3\tau} \delta^\eta |\mathbb{W}'_B| |W|,$$

where the final inequality used the fact that we are in the thin case, and thus $\delta/a \geq \delta^\tau$.

Next we estimate $\Delta_{\max}(\mathbb{T}_B)$. Each tube $T_B \in \mathbb{T}_B$ is contained in a tube $T \in \mathbb{T}$. If $T_B \subset K$, then the corresponding tube T is contained in $r_1^{-1}K$, the

dilation of K by a factor r_1^{-1} . Therefore $\Delta_{\max}(\mathbb{T}_B) \leq r_1^{-2} \Delta_{\max}(\mathbb{T})$. Using this, we can see that

$$|\mathbb{W}'_B| |W| \gtrsim \Delta_{\max}(\mathbb{T}_B)^{-1} |\mathbb{T}_B| |T_B| \geq r_1^2 |\mathbb{T}_B| |T_B|.$$

Summing over the r_1 -balls B , we conclude that

$$|U(\mathbb{T}, Y)| \gtrsim \delta^{8\eta+8\epsilon_{\text{scale}}+3\tau} \delta^\eta r_1^2 |\mathbb{T}| |T| \gtrsim \delta^{9\eta+10\epsilon_{\text{scale}}+3\tau} |\mathbb{T}| |T|.$$

We choose τ, η , and $\epsilon_{\text{scale}} \ll \beta$, and so this estimate implies our goal.

This finishes the case $b \geq \delta^{\epsilon_{\text{scale}}} r_1$, and so from now on, we can assume that $b \leq \delta^{\epsilon_{\text{scale}}} r_1$.

Transverse case

Fix a ball B . We will study the typical angle of intersection of the slabs of $\mathbb{W}'_B$ by applying Lemma 14.3.41 to $(\mathbb{W}'_B, Y_{\mathbb{W}'_B})$. This gives an ≈ 1 refinement, $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ and a typical angle of intersection $\theta \in [a/b, 1]$. Let τ' be a small parameter to be determined, with $\zeta \gg \tau' > \tau$. We define the transverse case to be the case that $\theta \geq \delta^{-\tau'} a/b$.

When θ is the typical angle of intersection, Lemma 14.3.43 tells us that $U(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ morally almost fills each ball of radius θb . More precisely, Lemma 14.3.43 says that $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ has a $\gtrsim 1$ refinement $(\mathbb{W}'''_B, Y_{\mathbb{W}'''_B})$ so that for every ball $B_{\theta b}$ that intersects $U(\mathbb{W}'''_B, Y_{\mathbb{W}'''_B})$, we have

$$|U(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \cap B_{\theta b}| \geq |U(\mathbb{W}'''_B, Y_{\mathbb{W}'''_B}) \cap B_{\theta b}| \gtrsim a^{3\eta} |B_{\theta b}| \geq \delta^{3\eta} |B_{\theta b}|.$$

(Here θ is the typical angle in the paragraph above – see Remark 14.3.44.)

Recall that $Y_{\mathbb{W}'_B}$ consists of a union of a -balls that intersect $U(\mathbb{T}, Y)$. Therefore, using (14.3.102), for any ball B_r of radius $r \geq a$, we have

$$|U(\mathbb{T}, Y) \cap B_r| \gtrsim \delta^\eta (\delta/a)^3 |U(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \cap B_r|.$$

Since we are in the thin case, we have $a \leq \delta^{1-\tau}$. So for any ball B_r of radius at least a , we have

$$|U(\mathbb{T}, Y) \cap B_r| \gtrsim \delta^{3\tau+\eta} |U(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \cap B_r|.$$

In particular, for the ball $B_{\theta b}$ above, we have

$$|U(\mathbb{T}, Y) \cap B_{\theta b}| \gtrsim \delta^{3\tau+4\eta} |B_{\theta b}|. \quad (14.3.106)$$

Informally, this equation shows that $U(\mathbb{T}, Y)$ almost fills $B_{\theta b}$. Formally, we choose a radius r of the form $r = \delta^{\eta j}$ with $\theta b \leq r \leq \delta^{-\eta} \theta b$. Since $\theta b \geq \delta^{1-\tau'}$, we have

$$\frac{|U(\mathbb{T}, Y) \cap B_r|}{|B_r|} \gtrsim \delta^{3\tau+7\eta} \geq \delta^{-2\tau'\beta+3\tau+7\eta} \left(\frac{\delta}{r}\right)^{2\beta}. \quad (14.3.107)$$

Since we choose $\tau'\beta \gg \tau, \eta$, Equation (14.3.107) gives (14.3.97) with any $\nu \leq \tau'\beta - (3\tau + 7\eta)/2$. Equation (14.3.97) implies our conclusion, as explained in Subsection 14.3.9.3.

This finishes the case $\theta \geq \delta^{-\tau'}a/b$, and so from now on, we can assume that $\theta \leq \delta^{-\tau'}a/b$.

Tangential case

Recall from the start of the transverse case that we fixed a ball B and studied the typical angle of intersection of the slabs of $\mathbb{W}'_B$ by applying Lemma 14.3.41 to $(\mathbb{W}'_B, Y_{\mathbb{W}'_B})$. This gave an ≈ 1 refinement $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ with typical angle of intersection $\theta \in [a/b, 1]$. Let τ' be a small parameter to be determined, with $\zeta \gg \tau' > \tau$. We defined the transverse case to be the case that $\theta \geq \delta^{-\tau'}a/b$. Now we consider the tangential case, when $\theta < \delta^{-\tau'}a/b$.

Recall from (14.3.104) that

$$(\mathbb{T}_B)_{Y'_B}(x) \subset \bigcup_{\substack{W \in \mathbb{W}'_B \\ x \in Y_{\mathbb{W}'_B}(W)}} (\mathbb{T}_{B,W})_{Y'_B}(x),$$

Next we refine $(\mathbb{T}_B, Y'_B)$ to make it compatible with $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$. Suppose that $T_B \in \mathbb{T}_{B,W}$. If $W \in \mathbb{W}''_B$, we set $Y''_B(T_B) = Y'_B(T_B)$, and if $W \notin \mathbb{W}''_B$, we set $Y''_B(T_B)$ to be empty. With this setup, it follows that

$$(\mathbb{T}_B)_{Y''_B}(x) \subset \bigcup_{\substack{W \in \mathbb{W}''_B \\ x \in Y_{\mathbb{W}''_B}(W)}} (\mathbb{T}_{B,W})_{Y''_B}(x),$$

We claim that $(\mathbb{T}_B, Y''_B)$ is a $\gtrsim 1$ refinement of $(\mathbb{T}_B, Y'_B)$ because $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ is a $\gtrsim 1$ refinement of $(\mathbb{W}'_B, Y_{\mathbb{W}'_B})$ and $|U(\mathbb{T}_B, Y'_B) \cap B_a|$ is constant up to factors ≈ 1 and $(\mathbb{T}_{B,W}, Y'_B)$ has constant multiplicity. (The last two facts are part of Proposition 14.3.18 which was used to construct $(\mathbb{T}_B, Y'_B)$ and $\mathbb{W}'_B$.)

Next we refine $(\mathbb{T}, Y)$ to make it compatible with $(\mathbb{T}_B, Y''_B)$. Recall that $|\mathbb{T}(T_B)_Y(x)|$ is roughly constant among $x \in Y_B(T_B)$. We know that $(\mathbb{T}_B, Y''_B)$ is a $\gtrsim 1$ refinement of $(\mathbb{T}_B, Y_B)$. If $T \in \mathbb{T}(T_B)$, we define $Y'(T) = Y(T) \cap Y''_B(T_B)$. Because of the uniformity above, $(\mathbb{T}, Y')$ is a $\gtrsim 1$ refinement of $(\mathbb{T}, Y)$. From the definition of Y' we see that

$$\mathbb{T}_{Y'}(x) \subset \bigcup_{\substack{T_B \in \mathbb{T}_B \\ x \in Y''_B(T_B)}} \mathbb{T}(T_B).$$

and so

$$\mathbb{T}_{Y'}(x) \subset \bigcup_{\substack{W \in \mathbb{W}''_B \\ x \in Y_{\mathbb{W}''_B}(W)}} \bigcup_{\substack{T_B \in \mathbb{T}_{B,W} \\ x \in Y''_B(T_B)}} \mathbb{T}(T_B).$$

The angle between any two tubes in $\mathbb{T}_{B,W}$ is $\lesssim b/r_1$. Define $\rho_2 = b/r_1$. This means that for each W , there exists a unit vector $v(W)$ so that each $T \in \mathbb{T}_{B,W}$ makes angle at most ρ_2 with v_W . Thus for each $x \in U(\mathbb{T}, Y')$ we have

$$\mathbb{T}_{Y'}(x) \subset \bigcup_{\substack{W \in \mathbb{W}''_B \\ x \in Y_{\mathbb{W}''_B}(W)}} \{T \in \mathbb{T}_Y(x) : \angle(T, v(W)) \leq \rho_2\}. \quad (14.3.108)$$

Now the cardinality of the inner set is $\lesssim \mu(\rho_2) \approx \mu(\mathbb{T}[T_{\rho_2}], Y)$ for any $T_{\rho_2} \in \mathbb{T}_{\rho_2}$. Also, by Proposition 14.3.18, we know that $(\mathbb{W}', Y_{\mathbb{W}'})$ has constant multiplicity $\mu(\mathbb{W}', Y_{\mathbb{W}'})$. Since $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$ is a refinement of $(\mathbb{W}'_B, Y_{\mathbb{W}'_B})$, we get that for any $T_{\rho_2} \in \mathbb{T}_{\rho_2}$,

$$\mu(\mathbb{T}, Y) \leq \mu(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \mu(\mathbb{T}[T_{\rho_2}], Y). \quad (14.3.109)$$

Using $K_{KT}(\beta)$ with η_{bias} in place of ε , we have

$$\mu(\mathbb{T}[T_{\rho_2}], Y) \lesssim \delta^{-\eta_{\text{bias}}} |\mathbb{T}[T_{\rho_2}]|^\beta. \quad (14.3.110)$$

Next we consider $|\mathbb{T}_{\rho_2}|$. Since we are not in the slab case, $b \leq \delta^{\epsilon_{\text{scale}}} r_1$, and so $\rho_2 = b/r_1 \leq \delta^{\epsilon_{\text{scale}}}$. On the other hand, $b \geq \delta$ and so $\rho_2 = b/r_1 \geq \delta/r_1 = \delta^{1-\epsilon_{\text{scale}}}$. So $\rho_2 \in [\delta^{1-\epsilon_{\text{scale}}}, \delta^{\epsilon_{\text{scale}}}]$, and so by the hypothesis of Lemma 14.3.54, we have

$$|\mathbb{T}_{\rho_2}| \geq \rho_2^{-2-\zeta}.$$

Our next goal is to estimate $\mu(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \lesssim \mu(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$. Recall that θ is a typical angle of intersection for $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$. In particular, if $W_1, W_2 \in \mathbb{W}''_B$ and if $x \in Y_{\mathbb{W}''_B}(W_1) \cap Y_{\mathbb{W}''_B}(W_2)$, then

$$\angle(TW_1, TW_2) \lesssim \theta \leq \delta^{-\tau'} a/b. \quad (14.3.111)$$

Let S denote a slab of dimensions $(a/b)r_1 \times r_1 \times r_1$, and let $\mathbb{S}$ denote a maximal set of essentially disjoint slabs $S \subset B$. Recall from (14.3.29) that we defined

$$\mathbb{W}''_S = \{W \in \mathbb{W}''_B : W \subset S \text{ and } \angle(TW, TS) \leq \theta\},$$

and that

$$\mathbb{W}''_B = \sqcup_{S \in \mathbb{S}} \mathbb{W}''_S.$$

By (14.3.111), $\mathbb{W}''_{Y_{\mathbb{W}''_B}}(x)$ is contained in at most $\delta^{-O(\tau')}$ sets $\mathbb{W}''_S$. Therefore,

$$\mu(\mathbb{W}''_B, Y_{\mathbb{W}''_B}) \lesssim \delta^{-O(\tau')} \max_{S \in \mathbb{S}} \mu(\mathbb{W}''_S, Y_{\mathbb{W}''_S}).$$

However, we would like to only include S with $\lambda(\mathbb{W}''_S, Y_{\mathbb{W}''_S})$ not too small. Recall from (14.3.103) that $\lambda(\mathbb{W}''_B, Y_{\mathbb{W}''_B}) \gtrsim \lambda(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \gtrsim \delta^{2\eta}$. Define

$$\mathbb{S}_{dens} = \{S \in \mathbb{S} : \lambda(\mathbb{W}''_S, Y_{\mathbb{W}''_S}) \geq (1/100)\lambda(\mathbb{W}''_B, Y_{\mathbb{W}''_B})\}.$$

Then we can define $\mathbb{W}'''_B = \sqcup_{S \in \mathbb{S}_{dens}} \mathbb{W}''_S$ and $Y_{\mathbb{W}'''_B} = Y_{\mathbb{W}''_B}$, and we see that $(\mathbb{W}'''_B, Y_{\mathbb{W}'''_B})$ is a $\gtrsim 1$ refinement of $(\mathbb{W}''_B, Y_{\mathbb{W}''_B})$. Therefore

$$\mu(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \lesssim \mu(\mathbb{W}'''_B, Y_{\mathbb{W}'''_B}) \lesssim \delta^{-O(\tau')} \max_{S \in \mathbb{S}_{dens}} \mu(\mathbb{W}''_S, Y_{\mathbb{W}''_S}).$$

Select $S \in \mathbb{S}_{dens}$. We know that $\lambda(\mathbb{W}''_S, Y_{\mathbb{W}''_S}) \gtrsim \delta^{2\eta}$.

We can make a linear change of variables that converts S to B_1 and converts $\mathbb{W}''_S$ to a set $\tilde{\mathbb{T}}$ of ρ_2 -tubes in B_1 . By the maximal density factoring lemma, we know that $\Delta_{max}(\mathbb{W}''_S) \leq \Delta_{max}(\mathbb{W})$ is small. With the original lemma, we would have $\Delta_{max}(\mathbb{W}) \lesssim 1$. With our modified lemma, Lemma 14.3.58, we have $\Delta_{max}(\mathbb{W}) \lesssim \delta^{-O(\eta_{bias})}$. This implies $\Delta_{max}(\mathbb{W}''_S) \lesssim \delta^{-O(\eta_{bias})}$, and so

$$\Delta_{max}(\tilde{\mathbb{T}}) \lesssim \delta^{-O(\eta_{bias})}.$$

We also have

$$\lambda(\tilde{\mathbb{T}}, Y_{\tilde{\mathbb{T}}}) = \lambda(\mathbb{W}''_S, Y_{\mathbb{W}''_S}) \gtrsim \delta^{2\eta}.$$

Since $\tilde{\mathbb{T}}$ is a set of ρ_2 -tubes in B_1 with $\Delta_{max}(\tilde{\mathbb{T}}) \lesssim \delta^{-O(\eta_{bias})}$, we have $|\tilde{\mathbb{T}}| \lesssim \delta^{-O(\eta_{bias})} \rho_2^{-2}$. If $\eta > 0$ is sufficiently small (recalling Remark 14.3.10) we can bound $\mu(\tilde{\mathbb{T}}, Y_{\tilde{\mathbb{T}}})$ using $K_{KT}(\beta)$ with η_{bias} in place of ϵ , which gives

$$\mu(\tilde{\mathbb{T}}, Y_{\tilde{\mathbb{T}}}) \lesssim \delta^{-O(\eta_{bias})} \delta^{-\eta_{bias}(1-\beta)} |\tilde{\mathbb{T}}|^\beta \lesssim \delta^{-O(\eta_{bias})} (\rho_2^{-2})^\beta \leq \delta^{-O(\eta_{bias})} \rho_2^{\beta\zeta} |\mathbb{T}_{\rho_2}|^\beta. \quad (14.3.112)$$

Therefore, we get

$$\mu(\mathbb{W}'_B, Y_{\mathbb{W}'_B}) \lesssim \delta^{-O(\tau)} \mu(\tilde{\mathbb{T}}, Y_{\tilde{\mathbb{T}}}) \leq \delta^{-O(\eta_{bias} + \tau')} \rho_2^{\beta\zeta} |\mathbb{T}_{\rho_2}|^\beta. \quad (14.3.113)$$

Combining (14.3.109), (14.3.110), and (14.3.113), we conclude that

$$\mu(\mathbb{T}, Y) \lesssim \delta^{-O(\eta_{bias} + \tau')} \rho_2^{\beta\zeta} |\mathbb{T}|^\beta \leq \delta^{-O(\eta_{bias} + \tau') + \epsilon_{scale}\beta\zeta} |\mathbb{T}|^\beta,$$

where for the final inequality we used the fact that we are not in the slab case, and thus $\rho_2 = b/r_1 < \delta^{\epsilon_{\text{scale}}}$.

We finish by making one simultaneous parameter choice for all cases. First choose $\epsilon_{\text{scale}} = \epsilon_{\text{scale}}(\beta) > 0$ small enough that $10\epsilon_{\text{scale}} < (1 - \epsilon_{\text{scale}})2\beta/4$. Given $\zeta > 0$, choose

$$0 < \tau \ll \tau' \ll \epsilon_{\text{scale}}\beta\zeta, \quad 0 < \eta_{\text{bias}} \ll \epsilon_{\text{scale}}\beta\zeta,$$

and then choose the input exponent η smaller than all the positive gains already fixed. Finally take

$$0 < \nu < c \min\{\tau\eta_{\text{bias}}\beta, \tau'\beta, \epsilon_{\text{scale}}\beta\zeta, \beta\},$$

where the absolute constant $c > 0$ is sufficiently small. The thick, transverse, and tangential arguments above then give respectively (14.3.93), (14.3.93), and (14.3.92) with this common ν .

For completeness, the slab case also gives the same goal with this choice. Indeed the cardinality hypothesis at $\rho = \delta^{1-\epsilon_{\text{scale}}}$ implies

$$|\mathbb{T}|^\beta \geq \delta^{-(1-\epsilon_{\text{scale}})(2+\zeta)\beta}.$$

Thus its bound $|U(\mathbb{T}, Y)| \gtrsim \delta^{9\eta+10\epsilon_{\text{scale}}+3\tau} |\mathbb{T}| |T|$ dominates (14.3.93) once

$$9\eta + 10\epsilon_{\text{scale}} + 3\tau + \nu < (1 - \epsilon_{\text{scale}})(2 + \zeta)\beta,$$

which is ensured by the preceding hierarchy after decreasing ϵ_{scale} . The thick/slab/transverse/tangential alternatives exhaust all possibilities, so this proves Lemma 14.3.54. $\square$

A Appendix: Probability lemmas

In this appendix, we give probability arguments to prove Lemma 14.3.14 and Lemma 14.3.50. The proofs are based on a simple probability lemma which is a small variation on Chernoff's inequality.

Lemma 14.3.60. *Suppose that $X_1, \dots, X_N$ are i.i.d. random variables. Suppose that $0 \leq X_1 \leq M$ almost surely and that $m = \mathbb{E}(X_1)$. If $Nm \geq M$, then for every S ,*

$$\mathbb{P}[X_1 + \dots + X_N > S] \leq e^{10} e^{-\frac{S}{Nm}}.$$

If $Nm < M$, then for every S ,

$$\mathbb{P}[X_1 + \dots + X_N > S] \leq e^{10} e^{-\frac{S}{M}}.$$

Proof. Let $\lambda \in \mathbb{R}$ be a parameter to choose later. For each $\lambda \in \mathbb{R}$, using the Markov inequality and independence, we see that

$$\begin{aligned}\mathbb{P}[X_1 + \cdots + X_N > S] &\leq e^{-\lambda S} \mathbb{E}[e^{\lambda(X_1 + \cdots + X_N)}] \\ &= e^{-\lambda S} \mathbb{E}\left(\prod_{n=1}^N e^{\lambda X_n}\right) = e^{-\lambda S} \mathbb{E}(e^{\lambda X_1})^N.\end{aligned}$$

For each $\lambda \in \mathbb{R}$, the function $x \mapsto e^{\lambda x}$ is convex, and so given our distributional information about X_1 we can bound

$$\mathbb{E}(e^{\lambda X_1}) \leq \frac{m}{M} e^{\lambda M} + 1 - \frac{m}{M}.$$

Plugging in, we get

$$\mathbb{P}[X_1 + \cdots + X_N > S] \leq e^{-\lambda S} \left(\frac{m}{M} e^{\lambda M} + 1 - \frac{m}{M} \right)^N.$$

We first consider the regime $Nm \geq M$. In this regime, we choose $\lambda = \frac{1}{Nm}$. Since $Nm \geq M$, we have $0 \leq \frac{M}{Nm} \leq 1$, and so $e^{\lambda M} = e^{\frac{M}{Nm}} \leq 1 + 10\frac{M}{Nm}$. Plugging in and simplifying, we get

$$\mathbb{P}[X_1 + \cdots + X_N > S] \leq e^{-\frac{S}{Nm}} \left(1 + \frac{10}{N} \right)^N \leq e^{10} e^{-\frac{S}{Nm}}.$$

This gives the conclusion in the regime $Nm \geq M$. If $Nm < M$, choose $\lambda = M^{-1}$ directly in the moment-generating-function estimate. Since $e^{\lambda M} = e$ and $1 + u \leq e^u$, we obtain

$$\begin{aligned}\mathbb{P}[X_1 + \cdots + X_N > S] &\leq e^{-S/M} \left(1 + (e-1) \frac{m}{M} \right)^N \\ &\leq e^{-S/M} \exp \left((e-1) \frac{Nm}{M} \right) \leq e^{e-1} e^{-S/M} \leq e^{10} e^{-S/M}.\end{aligned}$$

□

Now we consider random translations and rigid motions of $\mathbb{R}^n$. If $v \in \mathbb{R}^n$, we let $R_v(x) = x + v$. A random translation of size $\leq \rho$ is a translation R_v by a vector $v \in B_\rho$ selected randomly from the uniform measure on B_ρ . A random rigid motion is a random rotation composed with a random translation of size ≤ 1 .

Now we restate and prove Lemma 14.3.14:

Lemma 14.3.14. *Suppose that $\mathbb{T}$ is a set of (essentially distinct) δ -tubes in $B_1 \subset \mathbb{R}^n$. Let $R_1, \dots, R_J$ be random rigid motions of size 1, with $J = \lceil C_F(\mathbb{T}) \rceil$, and let $\mathbb{T}' = \bigcup_{j=1}^J R_j(\mathbb{T})$. Then with high probability the tubes of $\mathbb{T}'$ are essentially distinct up to multiplicity $\lesssim 1$ and also $C_F(\mathbb{T}') \lesssim 1$. After refining $\mathbb{T}'$ to make the tubes essentially distinct, we have $|\mathbb{T}'| \approx C_F(\mathbb{T})|\mathbb{T}|$ and $C_F(\mathbb{T}') \lesssim 1$.*

Proof. Throughout the proof we regard the $J|\mathbb{T}|$ moved tubes as tagged; only at the end do we discard repetitions. Put $v = |T_\delta| \sim \delta^{n-1}$. We first record a packing bound. If a convex set $K \subset B_2$ has outer John semiaxes

$$\delta \lesssim k_1 \leq \dots \leq k_{n-1} \lesssim k_n \sim 1,$$

then

$$|\mathbb{T}[K]| \lesssim \prod_{i=1}^{n-1} (k_i/\delta)^2 \sim (|K|/v)^2. \quad (14.3.114)$$

Indeed, a tube in K is determined up to essential equivalence by its direction and by the point where its core meets a hyperplane transverse to the longest axis. In the i th short principal direction, both parameters range over an interval of length $O(k_i)$ and may be put on a $c\delta$ -grid. This gives the displayed product.

We next control repetitions. Fix a δ -tube T_0 and set

$$X_j = |R_j(\mathbb{T})[100T_0]|.$$

Essential distinctness gives $0 \leq X_j \lesssim 1$. For each fixed $T \in \mathbb{T}$,

$$\mathbb{P}[R_j(T) \subset 100T_0] \lesssim v^2. \quad (14.3.115)$$

To verify this, the rotated core direction must lie in a spherical $O(\delta)$ -cap, an event of probability $O(\delta^{n-1})$, and then the translation parameter must lie in an $O(\delta)$ -tube inside a bounded ball, again an event of probability $O(\delta^{n-1})$. Thus $m := \mathbb{E}X_j \lesssim |\mathbb{T}|v^2$. Since

$$C_F(\mathbb{T}) \sim \frac{\Delta_{\max}(\mathbb{T})}{|\mathbb{T}|v},$$

(14.3.114), applied to an extremizing John ellipsoid, gives

$$Jm \lesssim C_F(\mathbb{T})|\mathbb{T}|v^2 \lesssim \Delta_{\max}(\mathbb{T})v \lesssim 1.$$

Lemma 14.3.60 now yields

$$\mathbb{P}\left[\sum_{j=1}^J X_j > S\right] \lesssim e^{-cS} \quad (S \geq 1).$$

A δ -net for the position and direction parameters of tubes in B_2 has cardinality $\delta^{-O(n)}$; enlarging a nearest net tube by a fixed factor covers the original one. Taking $S = C \log(2/\delta)$ and using a union bound therefore controls every T_0 simultaneously. Hence every essential-equivalence class among the tagged moved tubes has cardinality $\lesssim 1$, and an essentially-distinct refinement has cardinality

$$|\mathbb{T}'| \approx J|\mathbb{T}| \approx C_F(\mathbb{T})|\mathbb{T}|.$$

It remains to control the Frostman constant. We spell out why only polynomially many tests are needed. All moved tubes lie in a fixed ball B_3 . If a convex set K contains a moved tube, replace it by $K \cap B_3$ and take an outer John ellipsoid E . Its semiaxes lie in $[c\delta, C]$. Discretizing its centre and orthogonal frame at $c_1\delta$ -mesh and its semiaxes on a fixed-ratio geometric grid produces a family $\mathcal{E}_\delta$ of at most $\delta^{-O(n^2)}$ ellipsoids such that

$$K \cap B_3 \subset E \subset \widehat{E}, \quad |\widehat{E}| \lesssim |E| \lesssim |K \cap B_3|$$

for some $\widehat{E} \in \mathcal{E}_\delta$.

Fix $\widehat{E} \in \mathcal{E}_\delta$ and put $X_j = |R_j(\mathbb{T})[\widehat{E}]|$. By the definition of $\Delta_{\max}(\mathbb{T})$,

$$0 \leq X_j \leq M := \Delta_{\max}(\mathbb{T}) \frac{|\widehat{E}|}{v}.$$

For a fixed point $x \in T$, conditioning on the rotation and using the uniform translation gives

$$\mathbb{P}[R_j(T) \subset \widehat{E}] \leq \mathbb{P}[R_j(x) \in \widehat{E}] \lesssim |\widehat{E}|.$$

Consequently $m := \mathbb{E}X_j \lesssim |\mathbb{T}| |\widehat{E}|$ and

$$Jm \lesssim C_F(\mathbb{T})|\mathbb{T}| |\widehat{E}| \sim \Delta_{\max}(\mathbb{T}) \frac{|\widehat{E}|}{v} = M.$$

Lemma 14.3.60 gives

$$\mathbb{P}\left[\sum_{j=1}^J X_j > AM\right] \lesssim e^{-cA}. \quad (14.3.116)$$

Choose $A = C \log(2/\delta)$ and take a union bound over $\mathcal{E}_\delta$. With high probability, for every convex K ,

$$|\mathbb{T}'[K]| \lesssim A \Delta_{\max}(\mathbb{T}) \frac{|K|}{v};$$

here we used the containing net ellipsoid and its volume comparison. Thus the tagged family has $\Delta_{\max}(\mathbb{T}') \lesssim A \Delta_{\max}(\mathbb{T})$. On the other hand,

$$\Delta(\mathbb{T}', B_3) \sim J|\mathbb{T}|v \sim C_F(\mathbb{T})|\mathbb{T}|v \sim \Delta_{\max}(\mathbb{T}).$$

It follows that $C_F(\mathbb{T}', B_3) \lesssim A \lesssim 1$. Refining the essential multiplicities changes this bound by at most the already obtained factor $\lesssim 1$. A fixed dilation from B_3 to B_1 preserves the conclusion up to an absolute constant. $\square$

Finally, we restate and prove Lemma 14.3.50.

Lemma 14.3.50. *Let $\delta \leq \rho \leq 1$. Suppose that $\mathbb{T}$ is a set of δ -tubes in $T_\rho \subset \mathbb{R}^n$ with $\frac{|T_\rho|}{|\mathbb{T}| |T_\delta|} \sim J \geq 1$. Let $R_1, \dots, R_J$ be a set of J random translations of size $\leq \rho$. Let $\mathbb{T}' = \bigcup_{j=1}^J R_j(\mathbb{T})$. Let $A \geq 1$. Then with probability at least $1 - (\rho/\delta)^{O(1)} e^{-A}$, the tubes of $\mathbb{T}'$ are essentially distinct up to a factor $\lesssim A$ and we have $\Delta_{\max}(\mathbb{T}') \lesssim A \Delta_{\max}(\mathbb{T})$. After refining $\mathbb{T}'$ to make the tubes essentially distinct, we have with high probability $|\mathbb{T}'| \approx J|\mathbb{T}|$ and $\Delta_{\max}(\mathbb{T}') \lesssim \Delta_{\max}(\mathbb{T})$*

We now give the complete probability and covering argument.

Proof. As above, keep the $J|\mathbb{T}|$ translated tubes tagged until the final refinement. First fix a δ -tube $T_0 \subset 2T_\rho$ and put

$$X_j = |R_j(\mathbb{T})[100T_0]|.$$

Essential distinctness gives $0 \leq X_j \lesssim 1$. Moreover,

$$\mathbb{P}[R_j(T) \subset 100T_0] \lesssim \frac{|T_\delta|}{|T_\rho|}. \quad (14.3.117)$$

Indeed, if the directions are not $O(\delta)$ -close the probability is zero; otherwise the allowed translation vectors have transverse volume $O(\delta^{n-1})$ and longitudinal length $O(\rho)$ inside the translation ball of volume $\sim \rho^n$. Hence

$$m := \mathbb{E}X_j \lesssim |\mathbb{T}| \frac{|T_\delta|}{|T_\rho|}, \quad Jm \lesssim 1.$$

Lemma 14.3.60 gives

$$\mathbb{P}\left[\sum_{j=1}^J X_j > S\right] \lesssim e^{-cS}.$$

The position and direction parameters of tubes contained in $2T_\rho$ admit a δ -net of cardinality $(\rho/\delta)^{O(n)}$. Taking the threshold to be a sufficiently large constant multiple of A and then taking a union bound controls all T_0 with failure probability at most $(\rho/\delta)^{O(1)} e^{-A}$. Thus the essential multiplicity is $\lesssim A$. In particular, taking $A = C \log(2\rho/\delta)$ gives essential multiplicity $\lesssim 1$, and an essentially-distinct refinement retains $\approx J|\mathbb{T}|$ tubes with high probability.

We now estimate $\Delta_{\max}(\mathbb{T}')$. Any convex set containing an output tube may be intersected with $2T_\rho$. Its outer John ellipsoid has short semiaxes between $c\delta$ and $C\rho$ and longest semiaxis comparable to one. Discretizing the centre and frame at mesh $c_1\delta$ and the semiaxes on a fixed-ratio geometric grid gives $(\rho/\delta)^{O(n^2)}$ net ellipsoids. Every relevant convex set is contained in one of them whose volume is comparable to the volume of the convex set.

Fix such a net ellipsoid K and set $X_j = |R_j(\mathbb{T})[K]|$. By the definition of maximal density,

$$0 \leq X_j \leq M := \Delta_{\max}(\mathbb{T}) \frac{|K|}{|T_\delta|}.$$

For a fixed $x \in T$, uniformity of the translation parameter gives

$$\mathbb{P}[R_j(T) \subset K] \leq \mathbb{P}[R_j(x) \in K] \lesssim \frac{|B_\rho(x) \cap K|}{|B_\rho|} \lesssim \frac{|K|}{|T_\rho|}. \quad (14.3.118)$$

The last inequality follows in principal-axis coordinates: the intersection has volume at most $Ck_1 \cdots k_{n-1}\rho \sim |K|\rho$, whereas $|B_\rho| \sim \rho^n$ and $|T_\rho| \sim \rho^{n-1}$. Therefore

$$m := \mathbb{E}X_j \lesssim \frac{|\mathbb{T}||K|}{|T_\rho|}, \quad Jm \lesssim \frac{|K|}{|T_\delta|} \leq M,$$

where the last step uses $\Delta_{\max}(\mathbb{T}) \geq 1$. Lemma 14.3.60, with threshold a sufficiently large constant multiple of AM , yields

$$\mathbb{P}\left[\sum_{j=1}^J X_j > CA\Delta_{\max}(\mathbb{T}) \frac{|K|}{|T_\delta|}\right] \lesssim e^{-A}. \quad (14.3.119)$$

Taking a union bound over the ellipsoid net proves, with probability at least $1 - (\rho/\delta)^{O(1)}e^{-A}$, that

$$\Delta_{\max}(\mathbb{T}') \lesssim A\Delta_{\max}(\mathbb{T}).$$

The final essentially-distinct refinement can only decrease maximal density, while the first part of the proof gives its asserted cardinality. $\square$

References for the General-to-Sticky Reduction

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Chapter 14.4

The Wang–Zahl Convex-Wolff Route: A Full Alternative Proof under Audit

Provenance, licence, and certification boundary

This chapter adapts the complete formal proof in Hong Wang and Joshua Zahl, *Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions*, arXiv:2502.17655v1, submitted 24 February 2025. The source is distributed under CC BY 4.0. Wang and Zahl are the authors of the source argument. The conversion to this book’s theorem counter, the labelled dependency adapters, all added proofs, audit remarks, and any corrections are changes by this edition and do not imply endorsement by the original authors.

The public and manuscript status stated here was checked on 31 August 2026. The arXiv record remains at v1 and the authors’ publication list describes the paper as submitted. Accordingly, this chapter is not certified merely by reproducing the source. It becomes an internal alternative proof only after every abbreviated or imported step listed in the audit at the end of the chapter has been supplied and the whole argument has passed the quantifier, refinement, exponent, and build audits.

The source article contains a long motivational vignette, an explicitly informal proof sketch, and a separate tube-doubling application. Those pages are not part of the logical proof and are omitted here. The theorem statements, definitions, complete formal argument, grains appendix, hairbrush appendix, figures, and references are retained. No survey sketch is used as a proof step.

Why this route is genuinely alternative

The streamlined chapter tries to obtain a fixed improvement in a one-parameter loss exponent and is blocked at a cellular compatibility interface and a uniform fixed-loss quantifier. The present route instead introduces two assertions $\mathcal{D}(\sigma, \omega)$ and $\mathcal{E}(\sigma, \omega)$. It proves their equivalence, proves a positive self-improvement at every point with $\omega > 0$, and reaches $(\sigma, \omega) = (0, 0)$ by relative openness and closedness. Thus it does not invoke either of the two defective streamlined lemmas. This observation establishes independence of proof architecture; it does not by itself certify the alternative route.

14.4.1 Introduction

Edition status note (checked 31 August 2026). The assertions of resolution in this adapted source are the Wang–Zahl preprint’s claims, not the present book’s certification. In Volume 14, Theorems 14.4.1 and 14.4.2 remain conditional on the four load-bearing obligations WZ–A1–A4 stated and cross-referenced at their points of use and collected in the chapter-end certification ledger. No later consequence may cite either theorem as internally proved until all four obligations are discharged.

A *Keakeya* set is a compact subset of $\mathbb{R}^n$ that contains a unit line segment pointing in every direction. The *Keakeya* set conjecture asserts that every *Keakeya* set in $\mathbb{R}^n$ has Minkowski and Hausdorff dimension n . This conjecture was proved by Davies [5] when $n = 2$, and is open in three and higher dimensions. See [17, 28] for an introduction to the *Keakeya* conjecture and a survey of historical progress on the problem. See [15, 16, 18, 20, 21, 27, 31] for current progress towards the conjecture in three and higher dimensions.

The purpose of this paper is to obtain lower bounds on the volume of unions of δ -tubes (i.e. the δ neighbourhoods of unit line segments) in $\mathbb{R}^3$ that satisfy certain non-clustering conditions. As a consequence, we resolve the *Keakeya* set conjecture in three dimensions.

Theorem 14.4.1. *Every Keakeya set in $\mathbb{R}^3$ has Minkowski and Hausdorff dimension 3.*

Theorem 14.4.1 is a corollary of the following slightly more technical result.

Theorem 14.4.2. *For all $\varepsilon > 0$, there exists $K > 1$ so that the following holds for all $\delta > 0$ sufficiently small. Let $\mathbb{T}$ be a set of δ -tubes contained in the unit ball in $\mathbb{R}^3$, and suppose that every rectangular prism of dimensions $a \times b \times 2$ contains at most $100ab\delta^{-2}$ tubes from $\mathbb{T}$ (this is true, for example, if the tubes in $\mathbb{T}$ point in δ -separated directions). For each $T \in \mathbb{T}$, let $Y(T) \subset T$*

be a measurable set with $|Y(T)| \geq \lambda|T|$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon \lambda^K \sum_{T \in \mathbb{T}} |T|. \quad (14.4.1)$$

The Kakeya maximal function conjecture asserts that for each $\varepsilon > 0$, Inequality (14.4.1) is true for $K = 3$. The Kakeya maximal function conjecture in $\mathbb{R}^2$ was proved by Cordoba [6]. While we do not resolve the Kakeya maximal function conjecture in $\mathbb{R}^3$, the weaker statement given in Theorem 14.4.2 is nonetheless sufficient to obtain Theorem 14.4.1.

The hypothesis that each $a \times b \times 2$ rectangular prism contains at most $100ab\delta^{-2}$ tubes from $\mathbb{T}$ is a type of non-clustering condition. A close variant of this hypothesis was first introduced by Wolff in [27], and sets of tubes that satisfy this hypothesis are said to satisfy the Wolff axioms.

14.4.1.1 Theorem 14.4.2 and multi-scale analysis

In [25, 26], the authors showed that Theorem 14.4.2 is true when the set $\mathbb{T}$ has a property called stickiness (see Figure 14.4.1 (left)). Roughly speaking, $\mathbb{T}$ is *sticky* if it satisfies the non-clustering condition from Theorem 14.4.2; has cardinality roughly δ^{-2} ; and for every intermediate scale $\delta \leq \rho \leq 1$, the tubes in $\mathbb{T}$ can be covered by a set of ρ tubes that satisfy the non-clustering condition from Theorem 14.4.2 with ρ in place of δ .

Unfortunately, not every set of tubes is sticky — see Figure 14.4.1 (right) for an example. The arrangement illustrated in Figure 14.4.1 (right) is challenging to analyze, because the ρ tubes intersect with large multiplicity (i.e. many ρ tubes pass through a typical point), but the arrangement of δ tubes inside each ρ tube is sparse (i.e. the union of δ tubes inside each ρ tube only fill out a small fraction of that ρ tube). To help us analyze this type of arrangement, in Section 14.4.1.2 we will introduce two variants of the non-clustering hypothesis from Theorem 14.4.2, and two variants of the volume estimate (14.4.1).

14.4.1.2 Unions of convex sets, and non-clustering

In what follows, we say a pair of sets $U, V \subset \mathbb{R}^n$ are *essentially distinct* if $|U \cap V| \leq \frac{1}{2} \max(|U|, |V|)$. $\mathbb{T}$ will denote a set of essentially distinct δ -tubes contained in the unit ball in $\mathbb{R}^3$, and $|T|$ will denote the volume of a δ -tube, i.e. $|T|$ has size about δ^2 .

Definition 14.4.3. Let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^3$.

(A) We define $C_{KT-CW}(\mathbb{T})$ to be the infimum of all $C > 0$ such that

$$\#\{T \in \mathbb{T} : T \subset W\} \leq C|W||T|^{-1} \quad \text{for all convex sets } W \subset \mathbb{R}^3.$$

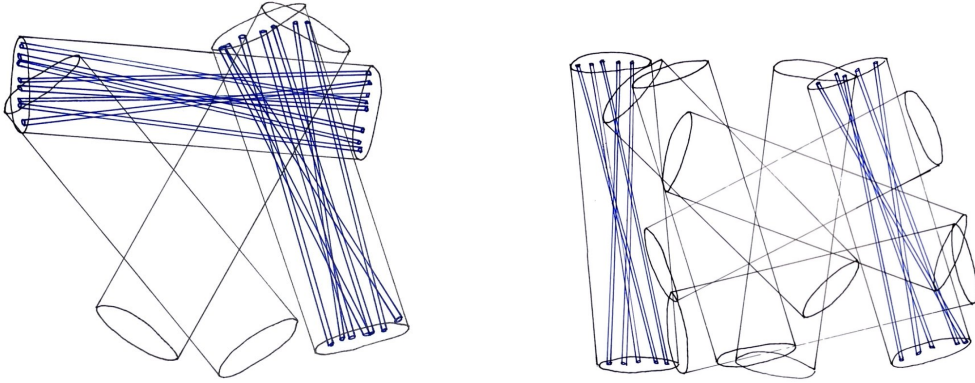


Figure 14.4.1: Left: The tubes at scale ρ (black) satisfy the non-concentration hypothesis of Theorem 14.4.2, as do the (rescaled) δ tubes (blue) inside each ρ tube. Multi-scale analysis is straightforward in this setting. This is sometimes called the “sticky” case. For clarity, not all δ tubes have been drawn.

Right: The tubes at scale ρ do not satisfy the non-concentration hypothesis of Theorem 14.4.2. The tubes at scale ρ intersect with high multiplicity, while the δ tubes inside each ρ tube are sparse.

We say that $\mathbb{T}$ obeys the Katz-Tao Convex Wolff Axioms with error $C_{KT-CW}(\mathbb{T})$.

(B) We define $C_{F-SW}(\mathbb{T})$ to be the infimum of all $C > 0$ such that

$$\#\{T \in \mathbb{T} : T \subset W\} \leq C|W|(\#\mathbb{T}) \quad \text{for all slabs } W \subset \mathbb{R}^3,$$

where a “slab” is the intersection of the unit ball with the thickened neighbourhood of a (hyper) plane. We say that $\mathbb{T}$ obeys the Frostman Slab Wolff Axioms with error $C_{F-SW}(\mathbb{T})$.

Remark 14.4.4.

(A) A note on etymology. The terms “Katz-Tao” and “Frostman” refer to different types of non-concentration conditions; they are the analogues of the well-studied non-concentration conditions $|E \cap B| \leq (r/\delta)^2$ and $|E \cap B| \leq r^2|E|$, where $E \subset \mathbb{R}^n$ is a δ -separated set and B is a ball of radius r . An arrangement of tubes arising from a Kakeya set, i.e. a set of δ -tubes with one tube pointing in each δ -separated direction, obeys both the Katz-Tao Convex Wolff Axiom and Frostman Slab Wolff Axiom with error $\lesssim 1$. The terms “convex” and “slab” refer to the class of sets for which the non-clustering condition is imposed. The term “Wolff axioms” suggests that the above definition is an analogue of the Wolff axioms from [27].

(B) The above definitions are two special cases of a non-clustering condition

(Definition 14.4.3') that will be defined in Section 14.4.3.2. In Definition 14.4.3', both tubes and convex sets (resp. slabs) are replaced by more general objects.

(C) If $\mathbb{T}$ is non empty, then by taking W to be a $\delta \times 1 \times 1$ -slab containing a tube of $\mathbb{T}$, we can see $C_{F-SW}(\mathbb{T}) \leq C$ implies $\#\mathbb{T} \geq C^{-1}\delta^{-1}$.

Next, we introduce two Kakeya-type volume estimates for unions of tubes in $\mathbb{R}^3$. These are analogues of Inequality (14.4.1) that are carefully formulated to be amenable to induction on scale. In what follows, we use the notation $(\mathbb{T}, Y)_\delta$ to denote a collection $\mathbb{T}$ of essentially distinct δ -tubes in $\mathbb{R}^3$, and a *shading* of these tubes, i.e. for each $T \in \mathbb{T}$, $Y(T)$ is a subset of T . For $\lambda > 0$, we say $(\mathbb{T}, Y)_\delta$ is λ *dense* if $\sum_{T \in \mathbb{T}} |Y(T)| \geq \lambda \sum_{T \in \mathbb{T}} |T|$.

Definition 14.4.5. Let $\sigma, \omega \geq 0$.

- We say that *Assertion $\mathcal{D}(\sigma, \omega)$ is true* if the following holds:
For all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ such that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and obey the Katz-Tao Convex Wolff Axioms and Frostman Slab Wolff Axioms, both with error at most $\delta^{-\eta}$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.2)$$

- We say that *Assertion $\mathcal{E}(\sigma, \omega)$ is true* if the following holds:
For all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ such that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} \ell (\#\mathbb{T}) |T|^{1/2})^{-\sigma}, \quad (14.4.3)$$

where $m = C_{KT-CW}(\mathbb{T})$ and $\ell = C_{F-SW}(\mathbb{T})$.

Let us examine the numerology in the estimates (14.4.2) and (14.4.3). First, in the special case $\sigma = \omega$, Assertion $\mathcal{D}(\sigma, \sigma)$ yields the estimate

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^\varepsilon (\#\mathbb{T})^{-\sigma} \sum_{T \in \mathbb{T}} |T|,$$

i.e. it says that there are $\lesssim \delta^{-\varepsilon} (\#\mathbb{T})^\sigma$ tubes passing through a typical point of the union $\bigcup Y(T)$ (for general σ and ω , this quantity is about $\delta^{\sigma-\omega-\varepsilon} (\#\mathbb{T})^\sigma$). For $\sigma > 0$ small, this means that the tubes in the union $\bigcup Y(T)$ are almost disjoint. In the arguments that follow, it will be helpful to consider situations where σ and ω are not necessarily equal.

The shape of the estimate (14.4.2) is motivated in part by the following consideration. To begin our induction on scale argument, we would like to

prove that $\mathcal{E}(\sigma, 0)$ holds for some $\sigma \in (0, 2/3]$. When $\sigma = 1/2$ and $\omega = 0$, Inequality (14.4.2) becomes the estimate

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{1/2+\varepsilon} (\delta^2 \# \mathbb{T})^{1/2}.$$

This is essentially Wolff’s hairbrush bound from [27] (here we make use of the fact that $\mathbb{T}$ obeys the Frostman Slab Wolff Axioms with small error; see Appendix 14.4.12 for details).

Assertion $\mathcal{D}(\sigma, \omega)$ is a special case of Assertion $\mathcal{E}(\sigma, \omega)$. We will explain the shape of the final bracketed term of Inequality (14.4.3). To understand the term ℓ , it is helpful to consider the following scenario. Suppose we know that Assertion $\mathcal{D}(\sigma, \omega)$ is true. Let $\delta < \rho < 1$, and let $\mathbb{T}$ be a set of δ tubes of cardinality $(\rho/\delta)^2$ that are contained inside a common ρ tube, which we will denote by T_ρ . Suppose that the tubes in $\mathbb{T}$ obey the Katz-Tao Convex Wolff Axioms with error roughly 1. This implies that $C_{F-SW}(\mathbb{T}) \sim \rho^{-1}$.

For $T \in \mathbb{T}$ (and hence $T \subset T_\rho$), we will write T^{T_ρ} to denote the image of T under the affine transformation that anisotropically dilates T_ρ by a factor of ρ^{-1} in its two “short” directions, and translates the image to the unit ball. After this rescaling and translation, the tubes in $\mathbb{T}$ become δ/ρ tubes that satisfy the Katz-Tao Convex Wolff Axioms and Frostman Slab Wolff Axioms, both with error roughly 1. Applying Assertion $\mathcal{D}(\sigma, \omega)$ to this rescaled collection of tubes, we obtain the volume bound

$$\left| \bigcup_{T \in \mathbb{T}} T^{T_\rho} \right| \geq \kappa (\delta/\rho)^\varepsilon (\# \mathbb{T}) |T^{T_\rho}| ((\# \mathbb{T}) |T^\rho|^{1/2})^{-\sigma}.$$

Undoing the anisotropic rescaling and translation (which distorted volumes by a factor of ρ^2) and noting that $|T^{T_\rho}| \sim \rho^2 |T|$, we can rewrite this as

$$\left| \bigcup_{T \in \mathbb{T}} T \right| \gtrsim \kappa \delta^\varepsilon (\# \mathbb{T}) |T| (\ell (\# \mathbb{T}) |T|^{1/2})^{-\sigma}, \quad \text{where } \ell = C_{F-SW}(\mathbb{T}) \sim \rho^{-1}.$$

As a second justification for the term ℓ , note that for every set $\mathbb{T}$ of δ tubes, we must always have $C_{F-SW}(\mathbb{T}) (\# \mathbb{T}) |T|^{1/2} \geq 1$. This is because we can always select a slab W of thickness $|T|^{1/2}$ that contains at least one tube from $\mathbb{T}$. This observation also explains the choice to write $|T|^{1/2}$ rather than δ ; any convex set $S \subset \mathbb{R}^3$ of diameter 1 can be contained in a slab of thickness $|S|^{1/2}$. Later in the proof we will consider generalizations of Assertion $\mathcal{E}(\sigma, \omega)$ in which tubes are replaced by more general families of convex sets.

To understand the terms m^{-1} and $m^{-3/2}$ in Inequality (14.4.3), it is helpful to consider the following scenario. Suppose we know that Assertion $\mathcal{D}(\sigma, \omega)$ is true. Let $\mathbb{T}$ be a set of δ tubes that obey the Frostman Slab Wolff Axioms

with error roughly 1, and the Katz-Tao Convex Wolff Axioms with error $m \gg 1$. Let $\rho = m^{1/2}\delta$, and suppose that there exists a set $\mathbb{T}_\rho$ of ρ tubes, each of which contains $m|T_\rho||T|^{-1} = m^2$ tubes from $\mathbb{T}$. Observe that this is the maximum number of essentially distinct δ tubes that can fit inside a ρ tube. In particular, the union of the δ tubes inside each ρ tube fill out essentially all of the ρ tube. We have $\#\mathbb{T}_\rho = m^{-2}(\#\mathbb{T}) = m^{-1}(|T|/|T_\rho|)(\#\mathbb{T})$, i.e. $(\#\mathbb{T}_\rho)|T_\rho| = m^{-1}(\#\mathbb{T})|T|$. It is straightforward to compute that $C_{KT-CW}(\mathbb{T}_\rho) \lesssim 1$. Applying Assertion $\mathcal{D}(\sigma, \omega)$ and using the fact that the union of δ tubes inside each ρ tube fill out most of the ρ tube, we obtain the volume bound

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} T \right| &\sim \left| \bigcup_{T_\rho \in \mathbb{T}_\rho} T_\rho \right| \geq \kappa \rho^{\omega+\varepsilon} (\#\mathbb{T}_\rho) |T_\rho| (\#\mathbb{T}_\rho) |T_\rho|^{1/2})^{-\sigma} \\ &= \kappa \rho^{\omega+\varepsilon} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} (\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \end{aligned}$$

14.4.1.3 From Assertions $\mathcal{D}$ and $\mathcal{E}$ to the Kakeya set conjecture

Clearly $\mathcal{E}(\sigma, \omega) \implies \mathcal{D}(\sigma, \omega)$. In Section 14.4.5, we will show that the reverse implication also holds:

Proposition 14.4.6. *Let $0 \leq \sigma \leq 2/3$, $\omega \geq 0$. Then $\mathcal{E}(\sigma, \omega) \iff \mathcal{D}(\sigma, \omega)$.*

As mentioned above, our proof uses induction on scale. In brief, if $\mathcal{E}(\sigma, \omega)$ is true, then we will use this fact at many locations and scales to prove that $\mathcal{D}(\sigma, \omega')$ is true for some $\omega' < \omega$ (observe that smaller values of ω are better). The precise statement is as follows.

Proposition 14.4.7. *There exists a function $g: [0, 2/3] \times (0, 1] \rightarrow (0, 1]$ so that the following is true. Let $0 \leq \sigma \leq 2/3$, $\omega > 0$. Then $\mathcal{E}(\sigma, \omega) \implies \mathcal{D}(\sigma, \omega - g(\sigma, \omega))$.*

Propositions 14.4.6 and 14.4.7 lead to a self-improving property for $\mathcal{E}(\sigma, \omega)$ (or equivalently, for $\mathcal{D}(\sigma, \omega)$). Since the collections of tubes in the definitions of $\mathcal{E}$ and $\mathcal{D}$ are essentially distinct and are contained in the unit ball, we always have $\#\mathbb{T} \lesssim \delta^{-4}$, and thus we can “trade” an improvement in ω for an improvement in σ . In particular, Proposition 14.4.7 tells us that $\mathcal{E}(\sigma, \omega) \implies \mathcal{D}(\sigma - g(\sigma, \omega)/4, \omega)$.

By applying Propositions 14.4.6 and 14.4.7, we can upgrade an initial estimate $\mathcal{D}(\sigma, \omega)$ to the improved estimate $\mathcal{D}(\sigma - g(\sigma, \omega)/4, \omega)$. We can then iterate this process. In order to begin the iteration, we must prove that $\mathcal{D}(\sigma, \omega)$ is true for some $\omega > 0$ and $0 \leq \sigma \leq 2/3$. In [27], Wolff proved that every Kakeya set in $\mathbb{R}^n$ has Hausdorff dimension at least $\frac{n+2}{2}$. In Appendix 14.4.12, we will use a similar argument to show that $\mathcal{D}(1/2, 0)$ is true:

Proposition 14.4.8. *$\mathcal{D}(1/2, 0)$ is true.*

Beginning with Proposition 14.4.8 and then iterating Propositions 14.4.6 and 14.4.7, we conclude the following.

Theorem 14.4.9. *The statements $\mathcal{D}(0, 0)$ and $\mathcal{E}(0, 0)$ are true.*

Proof. We first record all of the compactness bookkeeping suppressed in the original short argument. If a nonempty family in Definition 14.4.5 has $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$, Remark 14.4.3(C) and essential distinctness give, with absolute constants,

$$c\delta^\eta \leq q := (\#\mathbb{T})|T|^{1/2} \leq C\delta^{-3}. \quad (14.4.4)$$

The upper bound follows by taking a maximal $c\delta$ -separated set in the four-dimensional compact parameter space of unit line segments meeting the unit ball; essential distinctness makes the parameter points separated. All statements below are trivial for an empty family.

Upward stability in σ . Suppose $\mathcal{D}(\sigma, \omega)$ is true and $\sigma' \in [\sigma, 2/3]$. Given a target loss $\varepsilon > 0$, invoke $\mathcal{D}(\sigma, \omega)$ with loss $\varepsilon/2$, and decrease its output η so that $\eta(\sigma' - \sigma) \leq \varepsilon/4$. Decreasing η only strengthens the density and Wolff-axiom hypotheses. By (14.4.4),

$$q^{-\sigma'} = q^{-\sigma} q^{-(\sigma' - \sigma)} \leq C_{\sigma, \sigma'} \delta^{-\varepsilon/4} q^{-\sigma}.$$

The constant is absorbed into κ (and the bounded range of δ), so the estimate with exponent σ' and target loss ε follows.

Closedness in σ . Let $\sigma_j \rightarrow \sigma$ and suppose every $\mathcal{D}(\sigma_j, \omega)$ is true. Given $\varepsilon > 0$, choose one index j with $4|\sigma_j - \sigma| < \varepsilon/4$. Invoke that one assertion with loss $\varepsilon/2$. From (14.4.4), after decreasing its $\eta \leq 1$,

$$q^{-\sigma_j} \geq C^{-1} \delta^{-4|\sigma_j - \sigma|} q^{-\sigma}.$$

Thus its conclusion implies the conclusion for $\mathcal{D}(\sigma, \omega)$ with loss ε . Notice that the index j , and hence its constants, is chosen after ε but before δ ; no uniformity in j is being assumed. The admissible set is therefore relatively closed.

Relative openness. Fix $\omega > 0$ and suppose $\mathcal{D}(\sigma, \omega)$ is true. Proposition 14.4.6 gives $\mathcal{E}(\sigma, \omega)$, and Proposition 14.4.7 gives $\mathcal{D}(\sigma, \omega - g)$ for $g = g(\sigma, \omega) > 0$. Since $q \leq C\delta^{-3}$, that last estimate, invoked with a slightly smaller auxiliary loss, implies

$$\mathcal{D}(\max(0, \sigma - g/4), \omega).$$

Indeed the factor needed on the right is at most $q^{g/4} \leq C\delta^{-3g/4}$, while the improvement from $\delta^{\omega-g}$ supplies δ^{-g} . Upward stability then gives a relative neighbourhood of σ in the admissible set.

Now fix $\omega > 0$. Proposition 14.4.8 gives $\mathcal{D}(1/2, 0)$ and hence $\mathcal{D}(1/2, \omega)$. The nonempty admissible subset of $[0, 2/3]$ is both relatively open and relatively closed, so connectedness gives $\mathcal{D}(\sigma, \omega)$ for every $\sigma \in [0, 2/3]$; in particular, $\mathcal{D}(0, \omega)$ holds for every $\omega > 0$.

Finally, given $\varepsilon > 0$, choose $0 < \omega < \varepsilon/2$ and invoke $\mathcal{D}(0, \omega)$ with auxiliary loss $\varepsilon/2$. Since $\delta^{\omega+\varepsilon/2} \geq \delta^\varepsilon$ for $0 < \delta \leq 1$, this is exactly $\mathcal{D}(0, 0)$. Proposition 14.4.6 then yields $\mathcal{E}(0, 0)$. $\square$

The conclusion of Theorem 14.4.9 can be rephrased as follows

Corollary 14.4.10. *For all $\varepsilon > 0$, there exists K so that the following holds for all $\delta > 0$ sufficiently small. Let $(\mathbb{T}, Y)_\delta$ be λ -dense. Then*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon \lambda^K m^{-1} (\#\mathbb{T}) |T|, \quad \text{where } m = C_{KT-CW}(\mathbb{T}). \quad (14.4.5)$$

Theorem 14.4.2 is now a special case of Corollary 14.4.10 — the hypotheses of Theorem 14.4.2 ensure that $C_{KT-CW}(\mathbb{T}) \leq 1000$.

14.4.2 Notation

In the arguments that follow, $\delta > 0$ will denote a small positive quantity. From this point onward, we write $A(\delta) \lesssim_\delta B(\delta)$ if for all $\varepsilon > 0$, there exists $K_\varepsilon > 0$ so that $A(\delta) \leq K_\varepsilon \delta^{-\varepsilon} B(\delta)$. If the role of δ is apparent from context, we will often write $A \lesssim B$. For example if K is a constant independent of δ , then $\log(1/\delta)^K \lesssim 1$. Similarly, $e^{\sqrt{\log 1/\delta}} \lesssim 1$.

In some sections of the paper, it will ease notation to fix certain variables (for example the values of σ and ω from Definition 14.4.5). In such cases, we will clearly state which variables are fixed, and use bold font throughout that section to denote these fixed variables, and also to denote quantities that depend only on fixed variables. For example we might define $\beta = \sigma\omega/100$.

14.4.2.1 Convex sets and shadings

In the introduction, we defined a δ -tube to be the δ neighbourhood of a unit line segment. There are several other types of convex sets that will make frequent appearances in our arguments. A *prism* is a rectangular prism in $\mathbb{R}^n$ (usually $\mathbb{R}^3$); we will denote the dimensions by $a \times b \times c \times \dots$, with the convention that $a \leq b \leq c \leq \dots$. Informally, we say a prism in $\mathbb{R}^3$ is “flat” if it has dimensions $a \times b \times c$ with $a \ll b$, and we say it is “square” if b and c have comparable size. Finally, we will sometimes refer to the quantities a , b , and c respectively as the “thickness,” “width,” and “length” of a prism.

Rather than working with rectangular prisms, it will sometimes be convenient to work with ellipsoids, or more general convex sets. This motivates the following definition, which generalizes the definition of $(\mathbb{T}, Y)_\delta$ from the introduction.

Definition 14.4.11. For $0 < a \leq b \leq c$, we write $(\mathcal{P}, Y)_{a \times b \times c}$ to denote the following pair: $\mathcal{P}$ is a set of essentially distinct convex subsets of $\mathbb{R}^3$; for each $P \in \mathcal{P}$, the outer John ellipsoid of P has axes of lengths comparable to a, b , and c respectively. Y is a shading on $\mathcal{P}$, i.e. for each $P \in \mathcal{P}$, we have $Y(P) \subset P$.

For example, we could write $(\mathbb{T}, Y)_\delta$ as $(\mathbb{T}, Y)_{\delta \times \delta \times 1}$. Finally, we say $(\mathcal{P}, Y)_{a \times b \times c}$ is λ dense if $\sum_{P \in \mathcal{P}} |Y(P)| \geq \lambda \sum_{P \in \mathcal{P}} |P|$.

Definition 14.4.12. If $(\mathcal{P}, Y)_{a \times b \times c}$ is a set of prisms and their associated shading and $x \in \mathbb{R}^3$, we define

$$\mathcal{P}_Y(x) = \{P \in \mathcal{P} : x \in Y(P)\}.$$

Similarly, if $\mathcal{P}$ is a set of prisms (or more generally, convex sets) and no shading is present, then we define $\mathcal{P}(x) = \{P \in \mathcal{P} : x \in P\}$.

Definition 14.4.13. We say a pair $(\mathcal{P}', Y')_{a \times b \times c}$ is a t -refinement of $(\mathcal{P}, Y)_{a \times b \times c}$ if $\mathcal{P}' \subset \mathcal{P}$; $Y'(P) \subset Y(P)$ for each $P \in \mathcal{P}'$, and $\sum_{P' \in \mathcal{P}'} |Y'(P')| \geq t \sum_{P \in \mathcal{P}} |Y(P)|$. In practice, we will often have $t \approx_\delta 1$, in which case we will call it a $\approx_\delta 1$ refinement.

Note that if $(\mathcal{P}, Y)_{a \times b \times c}$ is λ dense and $(\mathcal{P}', Y')_{a \times b \times c}$ is a t -refinement, then $\#\mathcal{P}' \geq \lambda t(\#\mathcal{P})$.

Definition 14.4.14. If $W \subset \mathbb{R}^3$ is a convex set whose outer John ellipsoid E has dimensions $a \times b \times c$, we write $\text{dir}(W) \in \text{Gr}(1; \mathbb{R}^3)$ and $\Pi(W) \in \text{Gr}(2; \mathbb{R}^3)$ to denote the 1 and 2-dimensional subspaces of $\mathbb{R}^3$ spanned by the primary and secondary axes of E . We have that $\text{dir}(W)$ is meaningfully defined up to accuracy b/c , and $\Pi(W)$ is meaningfully defined up to accuracy a/b . For example, if T is a δ tube, then $\text{dir}(T)$ is meaningfully defined up to accuracy δ , while $\Pi(T)$ is only meaningfully defined up to accuracy 1 (i.e. $\Pi(T)$ is not a meaningful quantity if T is a δ tube).

We will employ the following synecdoche notation: if $\mathcal{P}$ (resp. $\mathbb{T}$, $\mathcal{W}$, etc.) is a collection of convex sets, each of the same volume, then we will use $|P|$ (resp. $|T|$, $|W|$, etc) to denote the volume of one of these convex sets. In practice, we will abuse notation slightly and continue to employ this notation if the sets in $\mathcal{P}$ have comparable (but not necessarily identical) volume.

Definition 14.4.15. Let $W \subset \mathbb{R}^n$ be a convex set. We define $\phi_W: \mathbb{R}^n \rightarrow \mathbb{R}^n$ to be an affine-linear transformation that maps the outer John ellipsoid of W to the unit ball. For concreteness, if $v_1, \dots, v_n$ are the axes of the John Ellipsoid, with lengths $\ell_1 \leq \dots \leq \ell_n$, then we select ϕ_W so that the j -th axis of the John Ellipsoid is mapped to the x_j axis in $\mathbb{R}^n$. If two or more axes have the same length, then we pick an ordering arbitrarily.

If $U \subset \mathbb{R}^n$, we define $U^W = \phi_W(U)$. In particular, if U is a convex subset of W then U^W is a convex subset of the unit ball, and $|U^W| \sim |U|/|W|$. Thus, if $T \subset T_\rho$ is a δ tube contained in a ρ tube, we write

$$T^{T_\rho} := \phi_{T_\rho}(T), \quad \mathbb{T}^{T_\rho} := \{T^{T_\rho} : T \in \mathbb{T}[T_\rho]\}.$$

These are, up to fixed comparability constants, respectively a (δ/ρ) tube and the rescaled family inside T_ρ .

Definition 14.4.16. Let $\mathcal{U}$ be a collection of convex subsets of $\mathbb{R}^n$ and let W be a convex subset of $\mathbb{R}^n$. We define

$$\mathcal{U}[W] = \{U \in \mathcal{U} : U \subset W\},$$

and

$$\mathcal{U}^W = \{U^W : U \in \mathcal{U}[W]\}.$$

If Y is a shading on $\mathcal{U}$, we will use Y^W to denote the corresponding shading on $\mathcal{U}^W$, i.e. for each $U^W \in \mathcal{U}^W$, we define $Y^W(U^W) = \phi_W(Y(U))$.

Remark 14.4.17. The expression $\mathcal{U}^W$ should not be confused with $\mathcal{U}_W$; the latter notation will be as follows: If $\mathcal{U}$ and $\mathcal{W}$ are sets of convex subsets of $\mathbb{R}^n$, then $\mathcal{U}_W$, $W \in \mathcal{W}$ will be used to denote a set of subsets of $\mathcal{U}$ that are indexed by the elements of $\mathcal{W}$.

14.4.2.2 Table of notation

To aid the reader, we will use certain notation conventions throughout this paper. For example, some symbols (such as σ and ω) will be reserved to always have the same meaning. For future reference, we record these notation conventions in the table below

Symbol	Meaning
δ, ρ, τ	These variables will denote scales. Typically $\delta \leq \rho \leq \tau$.
a, b, c	These variables will denote scales; typically the dimensions of a prism.
θ	θ will denote an angle

$\varepsilon, \eta, \zeta, \alpha$	These variables will represent (typically small) exponents, i.e. they will appear in the form δ^η , ρ^ε , etc.
κ, K	These variables will represent (positive) multiplicative constants, i.e. $ \bigcup T \geq \kappa \delta^\varepsilon$ or $C_{KT-CW}(\mathbb{T}) \leq K \delta^{-\eta}$. Typically $\kappa > 0$ is small and $K \gg 1$ is large.
σ, ω	σ and ω and their variants $\sigma', \tilde{\sigma}$, etc. will always be quantities related to the estimates $\mathcal{E}(\sigma, \omega)$ and $\mathcal{D}(\sigma, \omega)$.
σ, ω	In Sections 14.4.6 and 14.4.7, we will fix values of σ and ω that are kept constant throughout that section. We use bold symbols to denote these fixed numbers, and all subsequent quantities that depend (only) on them.
$T, P, G, S, \square$	These variables will denote convex sets. Typically T is a tube, P and G are prisms of dimensions $a \times b \times c$, S is a slab, and $\square$ is a “box” of dimensions $a \times c \times c$. We use symbols $\mathbb{T}, \mathcal{P}, \mathcal{G}, \mathcal{S}$ to denote sets of such objects.
$\mathbb{T}', \mathbb{T}_1, \tilde{\mathbb{T}}$	$\mathbb{T}'$ or $\mathbb{T}_1$ will denote a subset of $\mathbb{T}$. Similarly $\mathbb{T}_2$ will denote a subset of $\mathbb{T}_1$, etc. $\tilde{\mathbb{T}}$ will denote a new set of tubes that is related to $\mathbb{T}$, but not necessarily a subset (for example, $\tilde{\mathbb{T}}$ might consist of the 2-fold dilates of the tubes in $\mathbb{T}$).
$(\mathbb{T}', Y')_\delta$	$(\mathbb{T}', Y')_\delta$ will denote a refinement of $(\mathbb{T}, Y)_\delta$. Similarly for $(\mathbb{T}_1, Y_1)_\delta$.

14.4.3 Wolff Axioms and Factoring Convex Sets

14.4.3.1 Definitions: Wolff axioms and covers

Definition 14.4.18. Let $\mathcal{U}, \mathcal{W}$ be collections of convex sets in $\mathbb{R}^n$.

- (A) We say that $\mathcal{W}$ is a *cover* of $\mathcal{U}$ (or $\mathcal{W}$ *covers* $\mathcal{U}$) if $\bigcup_{W \in \mathcal{W}} \mathcal{U}[W] = \mathcal{U}$. We will denote this by $\mathcal{U} \prec \mathcal{W}$.
- (B) We say that $\mathcal{W}$ is a *K-almost partitioning cover* (resp. *partitioning cover*) if it is a cover, and furthermore each $U \in \mathcal{U}$ is contained in at most K sets (resp. 1 set) of the form $\mathcal{U}[W]$.
- (D) We say that $\mathcal{W}$ is a *K-balanced cover* (resp. *balanced cover*) if it is a cover, and furthermore the numbers $|W|^{-1} \sum_{U \in \mathcal{U}[W]} |U|$ are comparable for all $W \in \mathcal{W}$, up to a multiplicative factor of K (resp. 2).

The following is a mild generalization of Definition 14.4.3.

Definition 14.4.3'. Let $\mathcal{U}$ and $\mathcal{W}$ be collections of convex subsets of $\mathbb{R}^n$.

- (A) We define the *Katz-Tao Wolff constant of $\mathcal{U}$ with respect to $\mathcal{W}$* to be the

infimum of all $C > 0$ so that

$$\sum_{U \in \mathcal{U}[W]} |U| \leq C|W| \quad \text{for all } W \in \mathcal{W}. \quad (14.4.6)$$

(B) We define the *Frostman Wolff constant of $\mathcal{U}$ with respect to $\mathcal{W}$* to be the infimum of all $C > 0$ so that

$$\sum_{U \in \mathcal{U}[W]} |U| \leq C|W| \sum_{U \in \mathcal{U}} |U| \quad \text{for all } W \in \mathcal{W}. \quad (14.4.7)$$

Remark 14.4.19.

(A) To ease notation, we define $C_{KT-CW}(\mathcal{U})$ (resp. $C_{F-CW}(\mathcal{U})$) to be the Katz-Tao (resp. Frostman) Wolff constant of $\mathcal{U}$ associated to the set $\mathcal{W}$ of convex subsets of $\mathbb{R}^n$. We define $C_{F-SW}(\mathcal{U})$ to be the Frostman Wolff constant of $\mathcal{U}$ associated to the set $\mathcal{W}$ of slabs in $\mathbb{R}^n$. Note that these definitions are compatible with those from Definition 14.4.3.

(B) A set $\mathbb{T}$ of δ -tubes obeys the Wolff axioms, in the sense of [27] (see Property (*) on p655 and the preceding discussion) if the Katz-Tao Wolff constant of $\mathbb{T}$ is small with respect to the set $\mathcal{W}$ consisting of all rectangular prisms of dimensions $10\delta \times \rho \times \dots \times \rho \times 2$, with $0 < \delta \leq \rho \leq 2$.

(C) For some arguments, it will be useful to consider an analogue of the above definitions where the quantity $|W|$ on the RHS of (14.4.6) is replaced by $|W \cap B(0, 1)|/|B(0, 1)|$, and similarly for (14.4.7). This leads to a quantity that transforms naturally under affine maps such as ϕ_W from Definition 14.4.15.

(D) Note that the above definitions continue to make sense if $\mathcal{U}$ is a multiset. This will be useful in Section 14.4.9.1.

(E) If the set $\mathcal{U} \neq \emptyset$ consists of convex sets of the same size, then $C_{F-CW}(\mathcal{U}) \leq C$ implies that $\#\mathcal{U} \geq C^{-1}|U|^{-1}$. To see this, take W to be a convex set in $\mathcal{U}$. Then the LHS of (14.4.7) equals to $|U|$ while the RHS of (14.4.7) equals to $C|U|^2(\#\mathcal{U})$. Roughly speaking, if $C_{KT-CW}(\mathcal{U})$ is small, then $\mathcal{U}$ is “sparse”, while if $C_{F-CW}(\mathcal{U})$ is small, then $\mathcal{U}$ is “dense.”

Remark 14.4.20.

(A) The Frostman Wolff constant is “inherited upwards” by covers. More precisely, if $\mathcal{U}$ and $\mathcal{W}$ are collections of convex subsets of $\mathbb{R}^n$, and if $\mathcal{W}$ is a K -balanced cover of $\mathcal{U}$, then

$$C_{F-CW}(\mathcal{W}) \lesssim KC_{F-CW}(\mathcal{U}) \quad \text{and} \quad C_{F-SW}(\mathcal{W}) \lesssim KC_{F-SW}(\mathcal{U}). \quad (14.4.8)$$

(B) The Katz-Tao Wolff constant is “inherited downwards” by covers. More precisely, if $\mathcal{U}$ is a collection of convex subsets of $\mathbb{R}^n$, and if W is a convex subset of $\mathbb{R}^n$, then

$$C_{KT-CW}(\mathcal{U}^W) = C_{KT-CW}(\mathcal{U}[W]) \leq C_{KT-CW}(\mathcal{U}). \quad (14.4.9)$$

(C) The Frostman Slab Wolff Constant is “sub-multiplicative” with respect to covers. More precisely, if $\mathcal{U} \prec \mathcal{V}$ are collections of convex subsets of a convex set $W \subset \mathbb{R}^n$, then in some situations we have that $C_{F-SW}(\mathcal{U}^W)$ is controlled by $\max_{V \in \mathcal{V}} C_{F-SW}(\mathcal{U}^V) C_{F-SW}(\mathcal{V}^W)$. In certain special cases, the same is true for the Katz-Tao Convex Wolff Constant. See Section 14.4.3.4 for a precise statement.

14.4.3.2 Factoring Convex Sets

As we have observed in Remark 14.4.20, Frostman Wolff constants are inherited upwards, while Katz-Tao Wolff constants are inherited downwards. The following definition will help us exploit this observation when performing multi-scale analysis and induction on scale.

Definition 14.4.21. Let $\mathcal{U}$ and $\mathcal{W}$ be collections of convex subsets of $\mathbb{R}^n$, and let $K > 0$.

(A) We say that $\mathcal{W}$ *factors $\mathcal{U}$ from above with respect to the Katz-Tao (resp. Frostman) Convex Wolff axioms with error K* if $\mathcal{W}$ covers $\mathcal{U}$, and $\mathcal{W}$ satisfies the Katz-Tao (resp. Frostman) Convex Wolff axioms with error K .

(B) We say that $\mathcal{W}$ *factors $\mathcal{U}$ from below with respect to the Katz-Tao (resp. Frostman) Convex Wolff axioms with error K* if $\mathcal{W}$ covers $\mathcal{U}$, and for each $W \in \mathcal{W}$ the set $\mathcal{U}^W$ satisfies the Katz-Tao (resp. Frostman) Convex Wolff axioms with error K .

(C) We say that $\mathcal{W}$ *factors $\mathcal{U}$ from above (resp. below) with respect to the Katz-Tao (or Frostman) Slab Wolff axioms with error K* if the natural analogue of (A) (resp. (B)) holds, where the Convex Wolff axioms are replaced by Slab Wolff axioms.

Remark 14.4.22. Definition 14.4.21 highlights a few special cases of a more general definition: If $\mathcal{U}$, $\mathcal{W}$, and $\mathcal{V}$ are collections of convex subsets of $\mathbb{R}^n$, we can define what it means for $\mathcal{W}$ to factor $\mathcal{U}$ from above (or below) with respect to the Katz-Tao (or Frostman) Wolff axioms with respect to $\mathcal{V}$. Item (A) and (B) in Definition 14.4.21 correspond to the special case where $\mathcal{V}$ is the collection of convex sets in $\mathbb{R}^n$, while Item (C) corresponds to the case where $\mathcal{V}$ is the collection of slabs in $\mathbb{R}^n$.

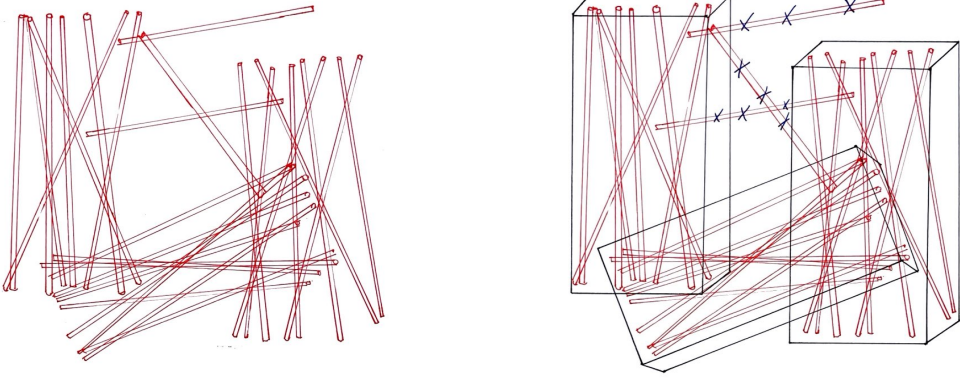


Figure 14.4.2: Left: $\mathcal{U}$ is a set of tubes (red) that cluster into rectangular prisms. Right: Proposition 14.4.23 locates these prisms (black). The tubes in $\mathcal{U} \setminus \mathcal{U}'$ have been X-ed out.

Definition 14.4.3' was carefully formulated to allow the following result, which says that for every collection $\mathcal{U}$ of convex subsets of $\mathbb{R}^n$, there exists some $\mathcal{W}$ that factors $\mathcal{U}$ from below with respect to the Frostman Convex Wolff axioms, and from above with respect to the Katz-Tao Convex Wolff axioms, both with small error. The precise statement is as follows.

Proposition 14.4.23. *Let $\mathcal{U}$ be a finite set of congruent convex subsets of the unit ball in $\mathbb{R}^n$, each of which contains a ball of radius δ . Let $K = 100^n e^{100\sqrt{\log(\delta^{-1}\#\mathcal{U})}}$ (the exact shape of K is not important; what matters is that if $\#\mathcal{U} \leq \delta^{-100}$, then $K \lesssim_\delta 1$).*

Then there exists a set $\mathcal{W}$ of congruent convex subsets of $\mathbb{R}^n$ and a set $\mathcal{U}' \subset \mathcal{U}$ with the following properties:

- i) $\#\mathcal{U}' \geq K^{-1}(\#\mathcal{U})$.
- ii) $\mathcal{W}$ is a K -balanced, K -almost partitioning cover of $\mathcal{U}'$, and

$$\#\mathcal{U}'[W] \geq K^{-1} C_{KT-CW}(\mathcal{U}') |W| |\mathcal{U}|^{-1} \quad \text{for each } W \in \mathcal{W}. \quad (14.4.10)$$

- iii) $\mathcal{W}$ factors $\mathcal{U}'$ from above respecting the Katz-Tao Convex Wolff Axioms with error K .
- iv) $\mathcal{W}$ factors $\mathcal{U}'$ from below respecting the Frostman Convex Wolff Axioms with error K .

Our proof of Proposition 14.4.23 will use the following “iterated graph pruning” lemma, which allows us to prune a bipartite graph and find an induced subgraph for which every vertex has many neighbours.

Lemma 14.4.24. *Let $G = (A \sqcup B, E)$ be a bipartite graph. Then there is a sub-graph $G' = (A' \sqcup B', E')$ so that $\#E' \geq \#E/2$; each vertex in A' has degree at least $\frac{\#E}{4\#A}$; and each vertex in B' has degree at least $\frac{\#E}{4\#B}$.*

Proof. Put $d_A = \#E/(4\#A)$ and $d_B = \#E/(4\#B)$. Starting with G , repeatedly delete any vertex of A whose current degree is smaller than d_A , or any vertex of B whose current degree is smaller than d_B , together with all edges incident to the deleted vertex. Charge an edge at the first deletion of one of its endpoints. The deletions of vertices in A remove fewer than $(\#A)d_A = \#E/4$ edges in total, and the deletions of vertices in B remove fewer than $(\#B)d_B = \#E/4$ edges in total. Hence at least $\#E/2$ edges survive. Let E' be the surviving edge set and let A' and B' be its incident vertices. The deletion process has stopped, so every vertex of A' has degree at least d_A and every vertex of B' has degree at least d_B , as required. $\square$

Proof of Proposition 14.4.23.

Step 1. Let $\mathcal{U}_0 \subset \mathcal{U}$ be a set minimizing the quantity

$$\min_{\substack{\mathcal{U}' \subset \mathcal{U} \\ \mathcal{U}' \neq \emptyset}} \exp \left[\left(\log \frac{\#\mathcal{U}}{\#\mathcal{U}'} \right)^2 \right] C_{KT-CW}(\mathcal{U}'). \quad (14.4.11)$$

Since $C_{KT-CW}(\mathcal{U}_0) \geq 1$, we have

$$\begin{aligned} \exp \left[\left(\log \frac{\#\mathcal{U}}{\#\mathcal{U}_0} \right)^2 \right] &\leq \exp \left[\left(\log \frac{\#\mathcal{U}}{\#\mathcal{U}_0} \right)^2 \right] C_{KT-CW}(\mathcal{U}_0) \\ &\leq C_{KT-CW}(\mathcal{U}) \leq \#\mathcal{U}. \end{aligned}$$

Re-arranging,

$$\#\mathcal{U}_0 \geq e^{-\sqrt{\log(\#\mathcal{U})}}(\#\mathcal{U}). \quad (14.4.12)$$

Observe that if $\mathcal{U}' \subset \mathcal{U}_0$ with $\#\mathcal{U}' \geq \frac{1}{2}(\#\mathcal{U}_0)$, then

$$\begin{aligned} \exp \left[\left(\log \frac{2(\#\mathcal{U})}{\#\mathcal{U}_0} \right)^2 \right] C_{KT-CW}(\mathcal{U}') &\geq \exp \left[\left(\log \frac{\#\mathcal{U}}{\#\mathcal{U}'} \right)^2 \right] C_{KT-CW}(\mathcal{U}') \\ &\geq \exp \left[\left(\log \frac{\#\mathcal{U}}{\#\mathcal{U}_0} \right)^2 \right] C_{KT-CW}(\mathcal{U}_0). \end{aligned}$$

Re-arranging and using (14.4.12),

$$C_{KT-CW}(\mathcal{U}') \geq \kappa_0 C_{KT-CW}(\mathcal{U}_0), \quad \kappa_0 = e^{-2 \log 2 \sqrt{\log(\#\mathcal{U})}}. \quad (14.4.13)$$

Step 2. Select closed convex sets $W_1, W_2, \dots$ in $\mathbb{R}^n$ and sets $\mathcal{U}_1 \supset \mathcal{U}_2 \supset \dots$ according to the following procedure. Beginning with $j = 1$, we select W_j to

maximize¹ the quantity $\#\mathcal{U}_{j-1}[W_j]/|W_j|$. By the definition of $C_{KT-CW}(\mathcal{U}_{j-1})$, we can select such a W_j so that

$$\#\mathcal{U}_{j-1}[W_j] = C_{KT-CW}(\mathcal{U}_{j-1}) \frac{|W_j|}{|U|}. \quad (14.4.14)$$

(Recall that $|U|$ is the volume of a set from $\mathcal{U}$; all such sets have identical volume). Define $\mathcal{U}_j = \mathcal{U}_{j-1} \setminus \mathcal{U}_{j-1}[W_j]$. Continue this process until $\#\mathcal{U}_j < \frac{1}{2}(\#\mathcal{U}_0)$.

Let $\mathcal{W}_0 = \{W_1, \dots, W_{j-1}\}$. Then

$$\#\left(\bigcup_{W \in \mathcal{W}_0} \mathcal{U}_0[W]\right) = \#(\mathcal{U}_0 \setminus \mathcal{U}_j) > \frac{1}{2}(\#\mathcal{U}_0). \quad (14.4.15)$$

Furthermore, for each $i = 1, \dots, j$, we have $\#\mathcal{U}_{i-1} \geq \frac{1}{2}(\#\mathcal{U}_0)$, and hence by (14.4.14) and (14.4.13),

$$\#\mathcal{U}_{i-1}[W_i] = C_{KT-CW}(\mathcal{U}_{i-1}) \frac{|W_i|}{|U|} \geq \kappa_0 C_{KT-CW}(\mathcal{U}_0) \frac{|W_i|}{|U|}. \quad (14.4.16)$$

Hence if $\mathcal{W}' \subset \mathcal{W}_0$, to compare $\#\left(\bigcup_{W_i \in \mathcal{W}'} \mathcal{U}_0[W_i]\right)$ and $\sum_{W_i \in \mathcal{W}'} \#\mathcal{U}_0[W_i]$,

$$\begin{aligned} \kappa_0 \frac{C_{KT-CW}(\mathcal{U}_0)}{|U|} \sum_{W_i \in \mathcal{W}'} |W_i| &\leq \sum_{W_i \in \mathcal{W}'} \#\mathcal{U}_{i-1}[W_i] \\ &= \#\left(\bigsqcup_{W_i \in \mathcal{W}'} \mathcal{U}_{i-1}[W_i]\right) \\ &\leq \#\left(\bigcup_{W_i \in \mathcal{W}'} \mathcal{U}_0[W_i]\right) \\ &\leq \sum_{W_i \in \mathcal{W}'} \#\mathcal{U}_0[W_i] \leq \frac{C_{KT-CW}(\mathcal{U}_0)}{|U|} \sum_{W_i \in \mathcal{W}'} |W_i|. \end{aligned} \quad (14.4.17)$$

The equality in (14.4.17) uses the critical fact that if $i \neq i'$, then $\mathcal{U}_{i-1}[W_i]$ and $\mathcal{U}_{i'-1}[W_{i'}]$ are disjoint.

Step 3. Each $W \in \mathcal{W}_0$ has a John ellipsoid whose axes have lengths $\ell_1, \dots, \ell_n$. Since each set $\mathcal{U}_0[W]$ is non-empty and each $U \in \mathcal{U}_0$ contains a ball of radius δ , we have that $\ell_i \geq \delta$ for each i . Since the sets in $\mathcal{U}$ are contained in the unit ball, we may suppose that $\ell_i \leq 2$ for each i . Thus by dyadic pigeonholing and (14.4.15), there exist $a_1, \dots, a_n$ and a set $\mathcal{W}_1 \subset \mathcal{W}_0$, so that the following two items hold:

¹Since $\mathcal{U}$ is a finite set of compact sets, such a maximizer exists; however the proof would work equally well if we merely approximate the maximum within a constant factor.

- (i) Each $W \in \mathcal{W}_1$ has a John ellipsoid whose axes have lengths $\ell_1 \leq \ell_2 \dots \leq \ell_n$ with $\ell_i \in [a_i/2, a_i)$.
- (ii)
$$\# \left(\bigcup_{W \in \mathcal{W}_1} \mathcal{U}_0[W] \right) \geq (100|\log \delta|)^{-n} (\#\mathcal{U}_0). \quad (14.4.18)$$

Replace each W by a congruent copy of W_0 —an ellipsoid whose axes have lengths $a_1, \dots, a_n$, and denote the corresponding set $\mathcal{W}_2$. Observe that (14.4.18) remains true with $\mathcal{W}_2$ in place of $\mathcal{W}_1$, and (14.4.17) remains true for all sets $\mathcal{W}' \subset \mathcal{W}_2$, though the first inequality has been weakened by a factor of 2^n on the RHS. Define $\mathcal{U}_2 = \bigcup_{W \in \mathcal{W}_2} \mathcal{U}_0[W]$ (recall that a sequence $\mathcal{U}_1, \mathcal{U}_2, \dots$ was defined earlier, and hence $\mathcal{U}_2$ was previously defined, but this is a harmless abuse of notation); we have that the cardinality of $\mathcal{U}_2$ is bounded below by the RHS of (14.4.18).

Since $\mathcal{U}_0[W] = \mathcal{U}_2[W]$ for all $W \in \mathcal{W}_2$, by applying (14.4.17) (beginning with the final inequality, and then using the first few inequalities) with $\mathcal{W}' = \mathcal{W}_2$ we conclude that

$$\begin{aligned} \#\{(U, W) \in \mathcal{U}_2 \times \mathcal{W}_2 : U \subset W\} &= \sum_{W \in \mathcal{W}_2} \#\mathcal{U}_0[W] \leq \frac{C_{KT-CW}(\mathcal{U}_0)}{|U|} \sum_{W \in \mathcal{W}_2} |W| \\ &\leq 2^n \kappa_0^{-1} \# \left(\bigcup_{W \in \mathcal{W}_2} \mathcal{U}_0[W] \right) = 2^n \kappa_0^{-1} (\#\mathcal{U}_2). \end{aligned} \quad (14.4.19)$$

In the above estimate, (14.4.17) was used to obtain the first and second inequalities, while the final equality follows from the definition of $\mathcal{U}_2$.

Step 4. Construct the bipartite incidence graph $(\mathcal{I}, \mathcal{U}_2 \times \mathcal{W}_2)$ whose edges consist of those pairs (U, W) with $U \subset W$. This graph has the following properties:

- (i) $\mathcal{I}$ has at most $2^n \kappa_0^{-1} (\#\mathcal{U}_2)$ edges.
- (ii) Each $U \in \mathcal{U}_2$ has at least one neighbour.
- (iii) Each $W \in \mathcal{W}_2$ has between $2^{-n} \kappa_0 C_{KT-CW}(\mathcal{U}_0) |W_0| |U|^{-1}$ and $C_{KT-CW}(\mathcal{U}_0) |W_0| |U|^{-1}$ neighbours.

Note that Items (i) and (iii) imply that

$$\#\mathcal{W}_2 \leq 2^{2n} \kappa_0^{-2} \frac{(\#\mathcal{U}_2) |U|}{C_{KT-CW}(\mathcal{U}_0) |W_0|}. \quad (14.4.20)$$

We will construct an induced subgraph of $(\mathcal{I}, \mathcal{U}_2 \times \mathcal{W}_2)$ as follows. First, remove all $U \in \mathcal{U}_2$ with more than $2^{n+1} \kappa_0^{-1}$ neighbours, and denote the resulting

induced subgraph by $(\mathcal{I}_3, \mathcal{U}_3 \times \mathcal{W}_2)$; by Items (i) and (ii), we have $\#\mathcal{U}_3 \geq \frac{1}{2}\#\mathcal{U}_2$, and $\#\mathcal{I}_3 \geq \#\mathcal{U}_3$. Next, apply Lemma 14.4.24 (iterated graph pruning) to $(\mathcal{I}_3, \mathcal{U}_3 \times \mathcal{W}_2)$. Denote the resulting induced subgraph by $(\mathcal{I}', \mathcal{U}' \times \mathcal{W})$.

Step 5. We will verify that $\mathcal{U}'$ and $\mathcal{W}$ satisfy Conclusions (i)–(iv) of Proposition 14.4.23. For Conclusion (i), we have

$$\#\mathcal{U}' \geq \frac{\kappa_0}{2^{n+1}}(\#\mathcal{I}') \geq \frac{\kappa_0}{2^{n+3}}(\#\mathcal{I}_3) \geq \frac{\kappa_0}{2^{n+3}}(\#\mathcal{U}_3) \geq \frac{\kappa_0}{2^{n+4}}(\#\mathcal{U}_2) \geq K^{-1}(\#\mathcal{U}),$$

since $\#\mathcal{U}_2$ is bounded below by the RHS of (14.4.18); $\#\mathcal{U}_0$ is bounded below by (14.4.12); and K was defined in the statement of Proposition 14.4.23.

For Conclusion (ii), Since each $U \in \mathcal{U}'$ has at most K neighbours in $(\mathcal{I}', \mathcal{U}' \times \mathcal{W})$, we have that $\mathcal{W}$ is a K -almost partitioning cover of $\mathcal{U}'$. It remains to verify (14.4.10). Since $\#\mathcal{U}'[W] \leq C_{KT-CW}(\mathcal{U}')|W||U|^{-1}$, it will then follow that $\mathcal{W}$ is a K -balanced cover of $\mathcal{U}'$. By Lemma 14.4.24 followed by (14.4.20), for each $W \in \mathcal{W}$, we have

$$\begin{aligned} \#\mathcal{U}'[W] &\geq \frac{1}{4}(\#I_3)(\#\mathcal{W}_2)^{-1} \geq \frac{1}{4}\left(\frac{1}{2}\#\mathcal{U}_2\right)\left(2^{-2n}\kappa_0^2\frac{C_{KT-CW}(\mathcal{U}_0)|W_0|}{(\#\mathcal{U}_2)|U|}\right) \\ &\geq (2^{-2n-4}\kappa_0^2)C_{KT-CW}(\mathcal{U}_0)|W_0||U|^{-1}. \end{aligned} \tag{14.4.21}$$

Since $\mathcal{U}' \subset \mathcal{U}_0$, we have $C_{KT-CW}(\mathcal{U}') \leq C_{KT-CW}(\mathcal{U}_0)$.

For Conclusion (iii), let $V \subset \mathbb{R}^n$ be a convex set. Since each $U \in \mathcal{U}_3$ has at most $2^{n+1}\kappa_0^{-1}$ neighbours and $(\mathcal{I}', \mathcal{U}' \times \mathcal{W})$ is an induced subgraph of $(\mathcal{I}_3, \mathcal{U}_3 \times \mathcal{W}_2)$, each $U \in \mathcal{U}'$ is contained in at most $2^{n+1}\kappa_0^{-1}$ sets $\mathcal{U}'[W]$, we have

$$\begin{aligned} \#\mathcal{U}'[V] &\geq 2^{-n-1}\kappa_0 \sum_{W \in \mathcal{W}[V]} \#\mathcal{U}'[W] \\ &\geq 2^{-3n-5}\kappa_0^3 C_{KT-CW}(\mathcal{U}_0)|W_0||U|^{-1}(\#\mathcal{W}[V]). \end{aligned} \tag{14.4.22}$$

where the final inequality used (14.4.21). On the other hand,

$$\#\mathcal{U}'[V] \leq C_{KT-CW}(\mathcal{U}')|V||W_0|^{-1}. \tag{14.4.23}$$

Comparing (14.4.22) and (14.4.23), we see that $\#\mathcal{W}[V] \leq K|V||W_0|^{-1}$, as desired.

Finally, for Conclusion (iv), let $W \in \mathcal{W}$ and let $V \subset W$ be a convex set. Then

$$\#(\mathcal{U}'[W])[V] \leq \#\mathcal{U}'[V] \leq C_{KT-CW}(\mathcal{U}')|V||U|^{-1}. \tag{14.4.24}$$

Comparing (14.4.24) and (14.4.10) (which we verified using (14.4.21)), we conclude that

$$\#(\mathcal{U}'[W])[V] \leq K|V||W|^{-1}(\#\mathcal{U}'[W]).$$

This is precisely the statement that $C_{F-CW}((\mathcal{U}')^W) \leq K$. $\square$

14.4.3.3 Convex Sets and the Frostman Slab Wolff Axioms

The goal of this section is to prove that after a refinement, every collection of convex sets can be partitioned into non-interacting pieces, each of which (after an appropriate rescaling) satisfies the Frostman Slab Wolff axioms. The precise statement is as follows.

Proposition 14.4.25. *For all $n \geq 2, \varepsilon > 0$, there exists $\eta > 0$ and $\kappa, K > 0$ so that the following holds for all $\delta > 0$. Let $\mathcal{U}$ be a collection of closed convex subsets of the unit ball in $\mathbb{R}^n$, each of which contains a ball of radius δ . For each $U \in \mathcal{U}$, let $Y(U) \subset U$ be a shading with $|Y(U)| \geq \delta^\eta |U|$.*

Then there exists a set $\mathcal{U}' \subset \mathcal{U}$; sets $Y'(U') \subset Y(U')$, $U' \in \mathcal{U}'$; and a set $\mathcal{W}$ of closed convex subsets of $\mathbb{R}^n$ with the following properties:

- i) $\mathcal{W}$ factors $\mathcal{U}'$ from below with respect to the Frostman Slab Wolff axioms with error $\delta^{-\varepsilon}$.*
- ii) The sets $\mathcal{U}'[W]$, $W \in \mathcal{W}$ do not interact, in the sense that the sets $\{\bigcup_{U' \in \mathcal{U}'[W]} Y'(U'), W \in \mathcal{W}\}$ are disjoint.*
- iii) The subset $\mathcal{U}'$ and the refined shading Y' preserve most of the mass of the original collection $(\mathcal{U}, Y)$, in the sense that*

$$\sum_{U' \in \mathcal{U}'} |Y'(U')| \geq \kappa \delta^\varepsilon (\log \#\mathcal{U})^{-K} \sum_{U \in \mathcal{U}} |U|. \quad (14.4.25)$$

Proposition 14.4.25 will rely on the following consequence of Brunn’s theorem:

Lemma 14.4.26. *Let $U \subset \mathbb{R}^n$ be a convex set, let $H \subset \mathbb{R}^n$ be a hyperplane, and let $s > 0$, $t \in (0, 1]$. Suppose that $|U \cap N_s(H)| = t|U|$. Then $U \subset N_{K_n s/t}(H)$, where K_n depends only on n .*

Proof. Without loss of generality we may suppose that the hyperplane H is given by $\{x_1 = 0\}$. Let $f(t) = |U \cap \{x_1 = t\}|^{\frac{1}{n-1}}$ (here $|\cdot|$ denotes $(n-1)$ -dimensional Lebesgue measure), and let $I = \text{supp}(f)$. Our task is to show that $|I| \leq K_n s/t$.

By Brunn’s theorem, f is concave on I . The result now follows by comparing the estimates $t|U| = |U \cap N_s(H)| \leq (2s)(\sup f)^{n-1}$ and $|U| \geq K_n |I| (\sup f)^{n-1}$ (the latter is a consequence of the concavity of f). $\square$

Combining Lemma 14.4.26 with a Cordoba-style L^2 argument, we obtain the following.

Lemma 14.4.27. *Let $\lambda \in (0, 1]$, let $\mathcal{U}$ be a collection of closed convex subsets of the unit ball in $\mathbb{R}^n$, each of which contains a ball of radius δ . For each $U \in \mathcal{U}$, let $Y(U) \subset U$ be a shading with $|Y(U)| \geq \lambda|U|$.*

Then there exists a set $\mathcal{U}' \subset \mathcal{U}$; sets $Y'(U) \subset Y(U)$, $U \in \mathcal{U}'$; and a set $\mathcal{S}$ of infinite slabs (i.e. the s -neighbourhood of a hyperplane in $\mathbb{R}^n$) with the following properties:

- i) $\mathcal{S}$ is a partitioning cover of $\mathcal{U}'$.*
- ii) If $S \in \mathcal{S}$ has thickness s , then the (rescaled) sets $\mathcal{U}'[S]$ have Frostman Slab Wolff constant $O(s^{-1})$. More concretely, if $\tilde{S} \subset S$ is a $s \times 2 \times \dots \times 2$ slab that contains the convex sets from $\mathcal{U}'[S]$, then $C_{F-SW}(\mathcal{U}'^{\tilde{S}}) \lesssim s^{-1}$.*
- iii) The sets $\bigcup_{U \in \mathcal{U}'[S]} Y'(U)$, $S \in \mathcal{S}$ are pairwise disjoint.*
- iv) $|Y'(U)| \geq \frac{\lambda}{2}|U|$ for each $U \in \mathcal{U}'$, and*

$$\sum_{U \in \mathcal{U}'} |U| \gtrsim \log(\lambda^{-1} \delta^{-1} \#\mathcal{U})^{-1} \lambda \sum_{U \in \mathcal{U}} |U|. \quad (14.4.26)$$

Proof.

Step 1. Define $\mathcal{U}_0 = \mathcal{U}$, and define $Y_0(U) = Y(U)$ for each $U \in \mathcal{U}_0$. For $i = 1, \dots$, let $S_i = N_{s_i}(H_i)$ be a slab maximizing the quantity

$$s_i^{-1} \sum_{U \in \mathcal{U}_{i-1}[S_i]} |U|. \quad (14.4.27)$$

Let $\mathcal{U}^{(i)} = \mathcal{U}_{i-1}[S_i]$; by the maximality of S_i , we have that $\mathcal{U}^{(i)}$ satisfies Conclusion (ii) from Lemma 14.4.27.

For each $U \in \mathcal{U}_{i-1}$, define $Y_i(U) = Y_{i-1}(U) \setminus S_i$, and define

$$\mathcal{U}_i = \{U \in \mathcal{U}_{i-1} : |Y_i(U)| \geq \frac{\lambda}{2}|U|\}.$$

In particular, $\mathcal{U}_i \cap \mathcal{U}^{(i)} = \emptyset$.

Step 2. We claim that

$$\sum_{U \in \mathcal{U}_{i-1}} |U \cap S_i| \lesssim \log(\lambda^{-1} \delta^{-1} \#\mathcal{U}) \sum_{U \in \mathcal{U}^{(i)}} |U|. \quad (14.4.28)$$

We verify (14.4.28) as follows. Since $\sum_{U \in \mathcal{U}^{(i)}} |U| \geq \frac{\lambda}{2} \delta^n$, the contribution from those $U \in \mathcal{U}_{i-1}$ with $|U \cap S_i| \leq \frac{1}{4} \lambda \delta^n (\#\mathcal{U})^{-1}$ is negligible. For each dyadic $t \in [\frac{1}{4} \lambda \delta^n (\#\mathcal{U})^{-1}, 1]$ we have

$$\begin{aligned} \sum_{\substack{U \in \mathcal{U}_{i-1} \\ |U \cap S_i| \sim t|U|}} |U \cap S_i| &\lesssim t \sum_{\substack{U \in \mathcal{U}_{i-1} \\ |U \cap S_i| \sim t|U|}} |U| \leq t \sum_{U \in \mathcal{U}_{i-1}[N_{K_n s_i/t}(H_i)]} |U| \\ &\leq K_n \sum_{U \in \mathcal{U}_{i-1}[S_i]} |U| = K_n \sum_{U \in \mathcal{U}^{(i)}} |U|. \end{aligned} \quad (14.4.29)$$

The second inequality follows from the containment

$$\{U \in \mathcal{U}_{i-1} : |U \cap S| \sim t|U|\} \subset \{U \in \mathcal{U}_{i-1} [N_{K_n s/t}(H_i)]\}$$

for an appropriately chosen constant K_n depending on n ; this is Lemma 14.4.26. The third inequality used the maximality of S_i , in the sense of (14.4.27). (14.4.28) now follows from summing (14.4.29) over dyadic values of t .

Step 3. We halt the procedure described above when $\mathcal{U}_N = \emptyset$. Define $\mathcal{U}' = \bigsqcup_{i=1}^N \mathcal{U}^{(i)}$, and for each $U \in \mathcal{U}'$, define $Y'(U) = Y_{i-1}(U)$, where i is the unique index so that $U \in \mathcal{U}^{(i)}$. Conclusions (i) and (iii) follow immediately from the above construction, as does the fact that $|Y'(U')| \geq \frac{\lambda}{2}|U'|$ for each $U' \in \mathcal{U}'$. Conclusion (ii) was already verified in Step 1.

It remains to verify (14.4.26). We claim that each $U \in \mathcal{U} = \mathcal{U}_0$ contributes at least $\frac{\lambda}{2}|U|$ to the sum

$$\sum_{i=1}^N \sum_{U \in \mathcal{U}_{i-1}} |U \cap S_i|,$$

in the sense that $\sum_{i: U \in \mathcal{U}_{i-1}} |U \cap S_i| \geq \frac{\lambda}{2}|U|$. Indeed, suppose that $j > 0$ is the smallest index with $U \notin \mathcal{U}_j$, and hence $|Y(U) \setminus \bigcup_{i=1}^j S_i| < \frac{\lambda}{2}|U|$. But since $|Y(U)| \geq \lambda|U|$, this means that

$$\sum_{i=1}^j |U \cap S_i| \geq \frac{\lambda}{2}|U|,$$

as claimed.

We conclude that

$$\sum_{i=1}^N \sum_{U \in \mathcal{U}_{i-1}} |U \cap S_i| \geq \frac{\lambda}{2} \sum_{U \in \mathcal{U}} |U|. \quad (14.4.30)$$

Comparing (14.4.28) and (14.4.30), we obtain (14.4.26). $\square$

Proposition 14.4.25 follows from repeatedly applying Lemma 14.4.27. We now turn to the details.

Proof of Proposition 14.4.25. Let $\lambda = \frac{1}{4}\delta^\eta$. After discarding those $U \in \mathcal{U}$ with $|Y(U)| < \lambda|U|$, we may suppose that $\mathcal{U}$ satisfies the hypotheses of Lemma 14.4.27.

Let $\mathcal{U}_0 = \mathcal{U}$ and for each $U \in \mathcal{U}_0$, let $Y_0(U) = Y(U)$. Let $\mathcal{P}_0 = \{B(0, 1)\}$, and let $\mathcal{W}_0 = \emptyset$. The set $\mathcal{P}_0$ corresponds to convex sets that still need to be “processed” by Lemma 14.4.27, while $\mathcal{W}_0$ will hold the convex sets that satisfy the hypotheses of Proposition 14.4.25.

We will iteratively construct sets $\mathcal{U}_i \subset \mathcal{U}_{i-1}$ and $Y_i(U) \subset Y_{i-1}(U)$; a set $\mathcal{W}_i \supset \mathcal{W}_{i-1}$ of convex subsets of $B(0, 1)$; and a set $\mathcal{P}_i$ of convex subsets of $B(0, 1)$ such that the following properties hold:

1. For each $W \in \mathcal{W}_i$, $C_{F-SW}(\mathcal{U}_i^W) \lesssim \delta^{-\varepsilon}$ (recall $\mathcal{U}_i^W = \phi_W(\mathcal{U}_i[W])$).
2. For each $P \in \mathcal{P}_i$, $|P| \leq \delta^{i\varepsilon} |B(0, 1)|$.
3. $\mathcal{W}_i \sqcup \mathcal{P}_i$ is a partitioning cover of $\mathcal{U}_i$.
4. The sets $\bigcup_{U \in \mathcal{U}_i[V]} Y_i(U)$ are pairwise disjoint, as V ranges over the convex sets in $\mathcal{P}_i \sqcup \mathcal{W}_i$.
5. $|Y_i(U)| \geq 2^{-i}\lambda|U|$ for each $U \in \mathcal{U}_i$.
6.
$$\sum_{V \in \mathcal{P}_i \cup \mathcal{W}_i} \sum_{U \in \mathcal{U}_i[V]} |Y_i(U)| \gtrsim \log(\delta^{-1} \#\mathcal{U})^i \lambda^i \sum_{U \in \mathcal{U}} |U|.$$

These six items are trivially satisfied when $i = 0$. For the i -th step, begin by setting $\mathcal{W}_i = \mathcal{W}_{i-1}$, $\mathcal{P}_i = \emptyset$, and $\mathcal{U}_i = \bigcup_{W \in \mathcal{W}_{i-1}} \mathcal{U}_{i-1}[W]$, $Y_i(U) = Y_{i-1}(U)$ for each $U \in \mathcal{U}_i$.

For each $P \in \mathcal{P}_{i-1}$, apply Lemma 14.4.27 (with $2^{-(i-1)}\lambda$ in place of λ) to each collection $\mathcal{U}_{i-1}^P = \phi_P(\mathcal{U}_{i-1}[P])$ of ellipsoids, and their associated shadings $\phi_P(Y_{i-1}(U))$. We obtain a collection of slabs $\mathcal{S}$, a set $\mathcal{U}'_{i-1}[P] \subset \mathcal{U}_{i-1}[P]$, and a shading, which we denote by $Y_i(U)$, on the ellipsoids in $\mathcal{U}'_{i-1}[P]$. Add the ellipsoids in $\mathcal{U}'_{i-1}[P]$ and their associated shading $Y_i(U)$ to $\mathcal{U}_i$. Next, we consider each slab $S \in \mathcal{S}$ in turn.

- If a slab $S \in \mathcal{S}$ has thickness $\geq \delta^\varepsilon$, then this corresponds to a convex set $W = P \cap \phi_P^{-1}(S)$ for which $C_{F-SW}((\mathcal{U}'_{i-1}[P])^W) \lesssim \delta^{-\varepsilon}$. Add this set to $\mathcal{W}_i$.
- If a slab $S \in \mathcal{S}$ has thickness $\leq \delta^\varepsilon$, then this corresponds to a convex set $P' = P \cap \phi_P^{-1}(S)$ for which

$$|P'| \leq \delta^\varepsilon |P| \leq \delta^\varepsilon (\delta^{(i-1)\varepsilon} |B(0, 1)|) = \delta^{i\varepsilon} |B(0, 1)|.$$

Add this set to $\mathcal{P}_i$.

After this procedure has been performed for each $P \in \mathcal{P}_{i-1}$, Properties 1 and 2 are immediate, while Properties 3-6 follow from their counterparts in Lemma 14.4.27.

We halt the process when the output $\mathcal{P}_N = \emptyset$. Since each $U \in \mathcal{U}$ has volume at least δ^n , the above procedure must halt after at most n/ε steps. We let $\mathcal{W} = \mathcal{W}_N$, $\mathcal{U}' = \mathcal{U}_N$, and $Y'(U) = Y_N(U)$. To obtain (14.4.25), η must be selected sufficiently small so that $\delta^{N\eta} \leq \delta^\varepsilon$, i.e. $\eta \sim \varepsilon^2/n$ and $K \sim n/\varepsilon$ will suffice. $\square$

14.4.3.4 The Frostman Slab Wolff Axioms and Covers

In this section we will state and prove a precise version of Remark 14.4.20(C). We first consider the Frostman Slab Wolff Axioms.

Lemma 14.4.28. *Let $W \subset \mathbb{R}^n$ be a convex set and let $\mathcal{U}$ and $\mathcal{V}$ be collections of convex subsets of W , with $\mathcal{U} \prec \mathcal{V}$ and $\#\mathcal{U}[V] \leq K(\#\mathcal{U})/(\#\mathcal{V})$ for each $V \in \mathcal{V}$. Suppose that each set in $\mathcal{U}$ has the same volume, and similarly for $\mathcal{V}$. Finally, suppose that each set in $\mathcal{U}^W$ has diameter $\geq 1/100$.*

Then

$$C_{F-SW}(\mathcal{U}^W) \lesssim \log(2 + |U^W|^{-1}) K \left(\sup_{V \in \mathcal{V}} C_{F-SW}(\mathcal{U}^V) \right) \left(C_{F-SW}(\mathcal{V}^W) \right). \quad (14.4.31)$$

Proof. First, to simplify notation we may suppose wlog that $W = B(0, 1)$; indeed, both the hypotheses and conclusion of Lemma 14.4.28 remain unchanged if we replace W by W^W (the latter is comparable to $B(0, 1)$); replace $\mathcal{U}$ by $\mathcal{U}^W$; and replace $\mathcal{V}$ by $\mathcal{V}^W$. In particular, each set in $\mathcal{U}$ now has diameter $\geq 1/100$.

Fix a truncated, thickened hyperplane $S = N_s(H) \cap B(0, 1)$, with $\mathcal{U}[S] \neq \emptyset$ (so in particular $s \geq |U|/K_n$, where K_n is a constant depending only on the dimension n). We may suppose that $s \leq 2$, since otherwise we can replace $N_s(H)$ by a hyperplane of the form $N_2(H')$, which has the same intersection with $B(0, 1)$.

Since $\mathcal{U} \prec \mathcal{V}$, we have

$$\mathcal{U}[S] = \bigcup_{V \in \mathcal{V}} \mathcal{U}[V \cap S] = \bigcup_{t \text{ dyadic}} \bigcup_{\substack{V \in \mathcal{V} \\ |V \cap S| \sim t|V|}} \mathcal{U}[V \cap S], \quad (14.4.32)$$

where the first union ranges over dyadic values of t between $|U||B(0, 1)|^{-1}$ and 1 (this range is sufficient, since if $|V \cap S| < |U||B(0, 1)|^{-1}|V| < |U|$, then $\mathcal{U}[V \cap S] = \emptyset$). Observe that there are $\lesssim \log(2 + |U|^{-1})$ dyadic values of t in this range.

Since $\mathcal{U}[S] \neq \emptyset$ and each element of $\mathcal{U}$ has diameter $\geq 1/100$, we have $\text{diam}(S \cap B(0, 1)) \geq 1/100$, and hence

$$|N_{s/t}(H) \cap B(0, 1)| \sim s/t \sim t^{-1}|S| \quad \text{for all } t \in [s, 1]. \quad (14.4.33)$$

Next, let $V \in \mathcal{V}$ with $|V \cap S| \sim t|V|$. This means that $|(V \cap S)^V| \sim t$. We have

$$\begin{aligned}
 \#\mathcal{U}[V \cap S] &= \#\{U \in \mathcal{U}: U \subset V \cap S\} \\
 &= \#\{U \in \mathcal{U}[V]: U \subset V \cap S\} \\
 &= \#\{U^V \in \mathcal{U}^V: U^V \subset (V \cap S)^V\} \\
 &\leq C_{F-SW}(\mathcal{U}^V)|(V \cap S)^V|(\#\mathcal{U}[V]) \\
 &\lesssim t \left(\sup_{V \in \mathcal{V}} C_{F-SW}(\mathcal{U}^V) \right) \left(K \frac{\#\mathcal{U}}{\#\mathcal{V}} \right).
 \end{aligned} \tag{14.4.34}$$

On the other hand, by Lemma 14.4.26 we have

$$\begin{aligned}
 \#\{V \in \mathcal{V}: |V \cap S| \sim t|V|\} &\leq \#\{V \in \mathcal{V}: V \subset N_{K_n s/t}(H)\} \\
 &\lesssim C_{F-SW}(\mathcal{V})|B(0, 1) \cap N_{K_n s/t}(H)|(\#\mathcal{V}) \\
 &\lesssim t^{-1} C_{F-SW}(\mathcal{V})|S|(\#\mathcal{V}),
 \end{aligned} \tag{14.4.35}$$

where the final inequality used (14.4.33).

Using (14.4.34) and (14.4.35) to control the cardinality of the union (14.4.32), we conclude that

$$\begin{aligned}
 \#\mathcal{U}[S] &\lesssim \sum_{t \text{ dyadic}} \left(t^{-1} C_{F-SW}(\mathcal{V})|S|(\#\mathcal{V}) \right) \left(t \left(\sup_{V \in \mathcal{V}} C_{F-SW}(\mathcal{U}^V) \right) \left(K \frac{\#\mathcal{U}}{\#\mathcal{V}} \right) \right) \\
 &\lesssim K_1 |S|(\#\mathcal{U}), \quad K_1 = \log(2 + |U|^{-1}) K \left(\sup_{V \in \mathcal{V}} C_{F-SW}(\mathcal{U}^V) \right) \left(C_{F-SW}(\mathcal{V}) \right).
 \end{aligned}$$

□

Next we consider Remark 14.4.20(C) for the Katz-Tao Convex Wolff Axioms. We will restrict attention to the special case where the convex sets in question are tubes.

Lemma 14.4.29. *Let $0 < \delta \leq \rho \leq 1$. Let $\mathbb{T}$ be a multiset of δ -tubes and let $\mathbb{T}_\rho$ be a cover of $\mathbb{T}$. Then*

$$C_{KT-CW}(\mathbb{T}) \lesssim \left(\sup_{T_\rho \in \mathbb{T}_\rho} C_{KT-CW}(\mathbb{T}^{T_\rho}) \right) \left(C_{KT-CW}(\mathbb{T}_\rho) \right). \tag{14.4.36}$$

Proof. Let $W \subset \mathbb{R}^3$ be a convex set with $\mathbb{T}[W] \neq \emptyset$. Replacing W by $W \cap B(0, 1)$ and then enlarging W by a constant factor, we may assume that W is a prism of dimensions $a \times b \times 2$. Since $\mathbb{T}_\rho$ covers $\mathbb{T}$, we have

$$\mathbb{T}[W] = \bigcup_{T_\rho \in \mathbb{T}_\rho} (\mathbb{T}[T_\rho])[W] = \bigcup_{T_\rho \in \mathbb{T}_\rho} \mathbb{T}[T_\rho \cap W]. \tag{14.4.37}$$

Observe that if $\mathbb{T}[T_\rho \cap W] \neq \emptyset$, then $T_\rho \cap W$ must contain a unit line segment, and thus $T_\rho \subset N_{3\rho}(W)$.

Let $\tilde{a} = \min(a, \rho)$ and $\tilde{b} = \min(b, \rho)$. Observe that

$$|N_{3\rho}(W)| \sim \left(\frac{\rho}{\tilde{a}}\right)\left(\frac{\rho}{\tilde{b}}\right)|W|,$$

and thus

$$\#\mathbb{T}_\rho[N_{3\rho}(W)] \leq C_{KT-CW}(\mathbb{T}_\rho) \frac{|N_{3\rho}W|}{|T_\rho|} \lesssim C_{KT-CW}(\mathbb{T}_\rho) \frac{|W|}{\tilde{a}\tilde{b}}. \quad (14.4.38)$$

On the other hand, if $\mathbb{T}[T_\rho \cap W]$ is non-empty, then $T_\rho \cap W$ is a convex set of dimensions bounded by $2\tilde{a} \times 2\tilde{b} \times 1$, and thus

$$\#\mathbb{T}[T_\rho \cap W] \lesssim C_{KT-CW}(\mathbb{T}[T_\rho]) \frac{|T_\rho \cap W|}{|T_\rho|} \lesssim C_{KT-CW}(\mathbb{T}^{T_\rho}) \frac{\tilde{a}\tilde{b}}{|T|}. \quad (14.4.39)$$

(14.4.36) now follows by combining (14.4.37), (14.4.38), and (14.4.39). $\square$

14.4.4 Factoring tubes into flat prisms

In this section, we will explore what happens when Proposition 14.4.23 is applied to a set $\mathbb{T}$ of δ -tubes. Recall that Proposition 14.4.23 outputs a refinement $\mathbb{T}' \subset \mathbb{T}$ and a set $\mathcal{W}$ of convex sets. If $C_{F-CW}(\mathbb{T}) = 1$, then $\mathcal{W} = \{B(0, 1)\}$. On the other hand, if $C_{KT-CW}(\mathbb{T}) = 1$, then $\mathcal{W} = \mathbb{T}$. If both $C_{KT-CW}(\mathbb{T})$ and $C_{F-CW}(\mathbb{T})$ are large, then $\mathcal{W}$ will consist of a collection of convex sets, each of which are comparable to a rectangular prism of dimensions $a \times b \times 1$, for some $\delta \leq a \leq b \leq 1$. The goal of this section is to explore the following theme: if the prisms in $\mathcal{W}$ are flat, in the sense that $a \ll b$, then the union $\bigcup T$ will have larger volume than predicted by the estimate (14.4.3) from Assertion $\mathcal{E}(\sigma, \omega)$. The precise statement is as follows.

Proposition 14.4.30. *Let $\omega > 0$, $0 < \sigma \leq 2/3$, and suppose $\mathcal{E}(\sigma, \omega)$ is true. Then for all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense. Let $\delta \leq a \leq b \leq 1$, and let $\mathcal{W}$ be a $\delta^{-\eta}$ balanced cover of $\mathbb{T}$ consisting of congruent copies of an $a \times b \times 2$ prism.*

(A) *Suppose that $\mathcal{W}$ factors $\mathbb{T}$ from below with respect to the Frostman Convex Wolff axioms and from above with respect to the Katz-Tao Convex Wolff axioms, both with error $\delta^{-\eta}$. Suppose as well that $\mathcal{W}$ is a $\delta^{-\eta}$ -balanced, $\delta^{-\eta}$ -almost partitioning cover of $\mathbb{T}$, and that $\#\mathbb{T}[W] \geq \delta^\eta C_{KT-CW}(\mathbb{T}) \frac{|W|}{|T|}$ for each $W \in \mathcal{W}$ (this condition is satisfied, for example, if $\mathcal{W}$ is the output when Proposition 14.4.23 is applied to $\mathbb{T}$). Then*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} \left(\frac{b}{a}\right)^\omega m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} \ell(\#\mathbb{T}) |T|^{1/2})^{-\sigma}, \quad (14.4.40)$$

where $m = C_{KT-CW}(\mathbb{T})$ and $\ell = C_{F-SW}(\mathbb{T})$.

(B) Suppose that $\mathcal{W}$ factors $\mathbb{T}$ from above and below with respect to the Frostman Convex Wolff Axioms, both with error $\leq \delta^{-\eta}$. Suppose as well that $\mathcal{W}$ satisfies the Katz-Tao Convex Wolff axioms at scale b in the following sense: for all $W \in \mathcal{W}$ we have $C_{KT-CW}(\mathcal{W}[N_b(W)]) \leq \delta^{-\eta}$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} \left(\frac{b}{a} \right)^\omega ((\#\mathbb{T})^{1/2} |T|)^\sigma. \quad (14.4.41)$$

Note that (14.4.41) agrees with (14.4.40) when $C_{KT-CW}(\mathbb{T}) = (\#\mathbb{T})|T|$, in which case both $C_{F-CW}(\mathbb{T})$ and $C_{F-SW}(\mathbb{T})$ have size ~ 1 .

In Section 14.4.8 we will need the following mild generalization of Proposition 14.4.30(A).

Proposition 14.4.31. *Let $\omega > 0$, $0 < \sigma \leq 2/3$, and suppose $\mathcal{E}(\sigma, \omega)$ is true. Then for all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ so that the following holds for all $0 < \delta \leq \rho \leq a \leq b \leq 1$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and let $\mathbb{T}_\rho$ be a $\delta^{-\eta}$ balanced cover of $\mathbb{T}$. Let $\mathcal{W}$ be a $\delta^{-\eta}$ balanced cover of $\mathbb{T}_\rho$ consisting of congruent copies of an $a \times b \times 2$ prism.*

Suppose that $\mathbb{T}_\rho$ factors $\mathbb{T}$ from below with respect to the Frostman Slab Wolff Axioms with error ℓ' . Suppose that $\mathcal{W}$ factors $\mathbb{T}_\rho$ from above with respect to the Katz-Tao Convex Wolff axioms and from below with respect to the Frostman Convex Wolff axioms, both with error $\delta^{-\eta}$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} \left(\frac{b}{a} \right)^\omega m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} \ell' (\#\mathbb{T}) |T|^{1/2})^{-\sigma}, \quad (14.4.42)$$

where $m = C_{KT-CW}(\mathbb{T})$ and $\ell = C_{F-SW}(\mathbb{T})$.

Proposition 14.4.30(A) is the special case of Proposition 14.4.31 where $\rho = \delta$ and $\mathbb{T}_\rho = \mathbb{T}$.

The main goal of Section 14.4.4 is to prove Propositions 14.4.30 and 14.4.31. A second goal is to introduce two cousins of the estimate $\mathcal{E}(\sigma, \omega)$, and to show that these three estimates are equivalent. The first estimate is (formally) weaker: it is the special case of the estimate $\mathcal{E}(\sigma, \omega)$ when ℓ has size about 1.

Definition 14.4.32. We say that *Assertion $\tilde{\mathcal{E}}(\sigma, \omega)$ is true* if the following holds:

For all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ such that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, and suppose $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega+\varepsilon} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} (\#\mathbb{T}) |T|^{1/2})^{-\sigma}, \quad (14.4.43)$$

where $m = C_{KT-CW}(\mathbb{T})$.

The second estimate is (formally) stronger: it is a generalization of the estimate $\mathcal{E}(\sigma, \omega)$ where δ -tubes are replaced by congruent convex sets of diameter 1.

Definition 14.4.33. We say that *Assertion $\mathcal{F}(\sigma, \omega)$ is true* if the following holds:

For all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ such that the following holds for all $0 < a \leq b \leq 1$. Let $(\mathcal{P}, Y)_{a \times b \times 1}$ be a^η dense. Then

$$\left| \bigcup_{P \in \mathcal{P}} Y(P) \right| \geq \kappa a^\varepsilon b^\omega m^{-1} (\#\mathcal{P}) |P| \times (m^{-3/2} \ell (\#\mathcal{P}) |P|^{1/2})^{-\sigma} D^{-\sigma}, \quad (14.4.44)$$

where

$$\begin{aligned} m &= C_{KT-CW}(\mathcal{P}), & \ell &= C_{F-SW}(\mathcal{P}), \\ D &= \max_{P \in \mathcal{P}} \sup_{\rho \in [a, b]} \frac{|P|}{|N_\rho(P)|} (\#\mathcal{P}[N_\rho(P)])^{1/2}. \end{aligned} \quad (14.4.45)$$

Remark 14.4.34. When $\sigma \in (0, 2/3]$, the term

$$m^{-1} (\#\mathcal{P}) |P| (m^{-3/2} \ell (\#\mathcal{P}) |P|^{1/2})^{-\sigma} D^{-\sigma}$$

is always at most 1. To see this, since $\#\mathcal{P} \leq m|P|^{-1}$ and $\sigma \in (0, 2/3]$, it remains to show that $((\#\mathcal{P})|P|)^{1/2} \ell^{-1} |P|^{1/2} \leq D$. This is true because $\#\mathcal{P} \leq \#\mathcal{P}[N_b(P)] \cdot b^{-4} \leq D^2 b^{-2} a^{-2}$, and $\ell \geq 1$.

Remark 14.4.35. If $\mathcal{P}$ is non-empty, then by selecting $\rho = a$ we see that the quantity $D = D(\mathcal{P})$ from (14.4.45) is always $\gtrsim 1$. In general, D can be as large as $(b/a)^{1/2}$: when $\rho = b$, it is possible for about $(b/a)^3$ essentially distinct $a \times b \times 1$ prisms to fit inside the b tube $N_b(P)$. If this happens, then the RHS of (14.4.45) becomes $\frac{ab}{b^2} (b/a)^{3/2} = (b/a)^{1/2}$. However, there are several important situations where we can guarantee that D has size roughly 1. We describe three of these below.

Situation 1. If $a = b$, then since the prisms in $\mathcal{P}$ are essentially distinct, we have $\#\mathcal{P}(N_a(P)) \sim 1$ and hence $D \sim 1$. In particular, this means that $\mathcal{F}(\sigma, \omega) \implies \mathcal{E}(\sigma, \omega)$.

Situation 2. Suppose that for each $P \in \mathcal{P}$, we have $C_{KT-CW}(\mathcal{P}[N_b(P)]) \leq K$, for some $K \geq 1$. This means that the Katz-Tao Convex Wolff constant of $\mathcal{P}$ might be large, but if we restrict attention to those prisms $\mathcal{P}'$ contained inside a tube of diameter b , then the Katz-Tao Convex Wolff constant of $\mathcal{P}'$ is small (this is the setup for Item (B) of Proposition 14.4.30). Then for each $\rho \in [a, b]$ and $P \in \mathcal{P}$ we have $\#\mathcal{P}[N_\rho(P)] \lesssim K \frac{\rho b-1}{a \cdot b-1} = K \frac{\rho}{a}$, and hence $D \lesssim K^{1/2}$.

Situation 3. Let $\mathbb{T}$ be a set of essentially distinct δ -tubes contained in a $s \times t \times 2$ prism W , with $0 < s \leq t \leq 2$. Let $\mathcal{P} = \phi_W(\mathbb{T})$. Then the sets in $\mathcal{P}$

are comparable to rectangular prisms of dimensions $a \times b \times 2$ with $a = \delta/t$ and $b = \delta/s$. Let us estimate the quantity $D = D(\mathcal{P})$ for this arrangement. For each $\rho \in [a, b]$ and each $P \in \mathcal{P}$, $\#\mathcal{P}[N_\rho(P)]$ counts the number of δ -tubes from $\mathbb{T}$ contained inside a rectangular box of dimensions roughly $\delta \times (\delta\rho/a) \times 2$. Since the tubes in $\mathbb{T}$ are essentially distinct, at most $O((\frac{\delta\rho/a}{\delta})^2) = O(\rho^2/a^2)$ tubes from $\mathbb{T}$ can be contained in such a box, i.e. $D \lesssim \sup_{\rho \in [a, b]} (\frac{ab}{\rho b}) (\frac{\rho}{a}) = 1$.

14.4.4.1 A few frequently used Cordoba-type L^2 arguments

In this section, we will explore several variants of the following argument: To show that a union $\bigcup_{P \in \mathcal{P}} P$ is large, it suffices to show that the quantity $\|\sum_{P \in \mathcal{P}} \chi_P\|_2^2 = \sum_{P, P' \in \mathbb{T}} |P \cap P'|$ is small, and then use Cauchy-Schwartz to conclude that

$$\left| \bigcup_{P \in \mathcal{P}} P \right| \geq \left(\sum_{P \in \mathcal{P}} |P| \right)^2 / \left\| \sum_{P \in \mathcal{P}} \chi_P \right\|_2^2.$$

This argument was used by Cordoba [6] to prove the Kakeya maximal function conjecture in $\mathbb{R}^2$, so we will call this style of argument a “Cordoba-type L^2 argument.”

A volume estimate for slabs

In this section we will use a Cordoba-type L^2 argument to estimate the volume of a union of slabs. The precise statement is as follows.

Lemma 14.4.36. *Let $\delta, \lambda > 0$. Let $\mathcal{S}$ be a collection of $\delta \times 1 \times \dots \times 1$ slabs (n.b. these slabs need not be essentially distinct), and let Y be a λ -dense shading on $\mathcal{S}$. Let $m = C_{KT-CW}(\mathcal{S})$. Then*

$$\left| \bigcup_{S \in \mathcal{S}} Y(S) \right| \gtrsim |\log \delta|^{-1} m^{-1} \lambda^2 (\#\mathcal{S}) |\mathcal{S}|. \quad (14.4.46)$$

Proof. Fix $S \in \mathcal{S}$. By Lemma 14.4.26 (applied to the outer John ellipsoid of each element of $\mathcal{S}$), we have that for each $t \in [\delta, 1]$, we have

$$\#\{S' \in \mathcal{S} : |S \cap S'| \sim t|S|\} \leq \#\{S' \in \mathcal{S} : S' \subset N_{C\delta/t}(S)\} \lesssim m \frac{\delta}{t} |S|^{-1} \sim \frac{m}{t},$$

where $C = C(n)$ depends only on n . Thus

$$\begin{aligned} \left\| \sum_{S \in \mathcal{S}} \chi_{Y(S)} \right\|_2^2 &\leq \left\| \sum_{S \in \mathcal{S}} \chi_S \right\|_2^2 \lesssim \sum_{S \in \mathcal{S}} \sum_{\substack{\delta \leq t \leq 1 \\ t \text{ dyadic}}} \sum_{\substack{S' \in \mathcal{S} \\ |S \cap S'| \sim t|S|}} t|S| \\ &\lesssim \sum_{S \in \mathcal{S}} \sum_{\substack{\delta \leq t \leq 1 \\ t \text{ dyadic}}} \left(\frac{m}{t}\right) t|S| \lesssim |\log \delta| m (\#\mathcal{S}) |\mathcal{S}|. \end{aligned} \quad (14.4.47)$$

Let $E = \bigcup_{S \in \mathcal{S}} Y(S)$. Using Cauchy-Schwartz, we have

$$\left(\lambda |S|(\#\mathcal{S})\right)^2 \leq \left(\int \chi_E \sum_{S \in \mathcal{S}} \chi_{Y(S)}\right)^2 \leq |E| \left\| \sum_{S \in \mathcal{S}} \chi_{Y(S)} \right\|_2^2. \quad (14.4.48)$$

The result now follows by comparing (14.4.47) and (14.4.48). $\square$

We will highlight a few instances where Lemma 14.4.36 will be helpful.

- $(\mathcal{S}, Y)$ is δ^η dense, and $C_{F-CW}(\mathcal{S}) \lesssim_\delta \delta^{-\eta}$. Then the RHS of (14.4.46) is $\gtrsim_\delta \delta^{3\eta}$.
- $\mathcal{R}$ is a set of $\delta \times 1$ rectangles inside a $\rho \times 2$ rectangle W : we will apply Lemma 14.4.36 to $\mathcal{R}^W$.
- $\mathbb{T}$ is a set of δ -tubes inside a $100\delta \times b \times 1$ prism W : we will apply Lemma 14.4.36 to $\mathbb{T}^W$.

Tangential vs transverse prism intersection

The next result says that if a collection of $a \times b \times 1$ prisms intersect transversely, in the sense that their tangent planes make large angle at a typical point of intersection, then the union of these prisms fills out a large fraction of a thickened neighbourhood of these prisms. The precise statement is Lemma 14.4.39 and Corollary 14.4.42 below. Before stating that result, we need a few definitions.

Definition 14.4.37. Let $W \subset \mathbb{R}^n$, let $Y(W) \subset W$ be a shading, and let $\delta > 0$. We say that the shading $Y(W)$ is *regular* at scales $\geq \delta$ if for each $x \in Y(W)$ and each $r \in [\delta, 1]$, we have

$$|Y(W) \cap B(x, r)| \geq (C_n \log(2/\delta))^{-1} |Y(W)| \left(\frac{|B(x, r) \cap W|}{|W|} \right), \quad (14.4.49)$$

where C_n is a fixed dimensional constant. Its numerical value is irrelevant below; one may take $C_n = 10^{4n}$. If the quantity δ is apparent from context, then we will omit it and say that $Y(W)$ is regular.

The next lemma says after a harmless refinement, every shading has a regular subshading. This is Lemma 2.7 from [19]; see also [26, Lemma 2.3].

Lemma 14.4.38. *Let $W \subset \mathbb{R}^n$ be a rectangular prism and let $Y(W)$ be a measurable shading. Then there is a regular shading $Y'(W) \subset Y(W)$ with $|Y'(W)| \geq \frac{1}{2} |Y(W)|$.*

Proof. Rigid motion lets us write $W = \prod_{j=1}^n [0, s_j]$. Put $L = \lceil \log_2(2/\delta) \rceil$. There are nested partitions $\mathcal{Q}_0 \prec \mathcal{Q}_1 \prec \dots \prec \mathcal{Q}_L$ of W into rectangular cells with the following property: if $Q_k(x)$ is the cell of $\mathcal{Q}_k$ containing x and $2^k \delta \leq r < 2^{k+1} \delta$, then, after shifting the index by a fixed dimensional integer,

$$Q_k(x) \subset B(x, r), \quad |Q_k(x)| \geq c_n |B(x, r) \cap W|. \quad (14.4.50)$$

Here cells on a face are taken half open. For completeness, construct the partitions one coordinate at a time: bisect $[0, s_j]$ until every interval has length between $2^{-3} n^{-1/2} \min(s_j, 2^k \delta)$ and twice this number, and at the next level merge adjacent pairs. A bounded number of end intervals may be merged with their neighbours. The product cells are nested, have diameter at most r , and have volume comparable to $\prod_j \min(s_j, r)$. Orthogonal projection of $B(x, r) \cap W$ to the coordinate axes gives the reverse upper bound $|B(x, r) \cap W| \leq C_n \prod_j \min(s_j, r)$, proving (14.4.50).

We now prune from fine scales to coarse scales. Set $A_{-1} = Y(W)$. At level k , delete the whole set $A_{k-1} \cap Q$ whenever $Q \in \mathcal{Q}_k$ and

$$|A_{k-1} \cap Q| < \frac{c_n}{4L} \frac{|Y(W)|}{|W|} |Q|;$$

call what remains A_k , and put $Y'(W) = A_L$. At one level the deleted sets are disjoint and their total measure is less than $c_n |Y(W)| / (4L)$. Enlarging the harmless dimensional constant in the definition, the sum over the $L + 1$ levels is at most $|Y(W)| / 2$.

Let $x \in Y'(W)$ and $r \in [\delta, 1]$, and choose the cell $Q_k(x)$ in (14.4.50). This cell was not deleted. Because the partitions are nested and were processed from fine to coarse, a later deletion either removes all of $Q_k(x)$ or none of it. The former is impossible since x survives. Consequently

$$\begin{aligned} |Y'(W) \cap B(x, r)| &\geq |Y'(W) \cap Q_k(x)| \\ &\geq \frac{c_n}{4L} \frac{|Y(W)|}{|W|} |Q_k(x)| \\ &\geq \frac{c_n^2}{4L} |Y'(W)| \frac{|B(x, r) \cap W|}{|W|}. \end{aligned}$$

Since $L \leq 2 \log_2(2/\delta)$, this is the asserted regularity after fixing C_n . Every use below is for a rectangular prism, so this is exactly the required form of the lemma. $\square$

The next result says that if a prism P_0 is incident to many prisms that intersect P_0 non-tangentially, then the union of these prisms fills out a thickened neighbourhood of P_0 .

Lemma 14.4.39. *Let $0 < a \leq b \leq c$ and let $\lambda > 0$. Let P_0 be a $a \times b \times c$ prism with shading $Y_0(P_0)$. Let $(\mathcal{P}, Y)_{a \times b \times c}$ be a set of prisms and their associated shading. Suppose that $|Y_0(P_0)| \geq \lambda|P_0|$; $|Y(P)| \geq \lambda|P|$ for each $P \in \mathcal{P}$; and each shading $Y(P)$ is regular, in the sense of Definition 14.4.37.*

Let $\theta \in [\frac{a}{b}, 1]$, and suppose that

$$\theta \leq \theta_{\min} := \frac{a}{b} + \inf_{x \in Y_0(P_0)} \sup_{P \in \mathcal{P}_Y(x)} \angle(\Pi(P_0), \Pi(P)).$$

Then

$$\left| N_{b\theta}(P_0) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| \gtrsim_a \lambda^4 |N_{b\theta}(P_0)|. \quad (14.4.51)$$

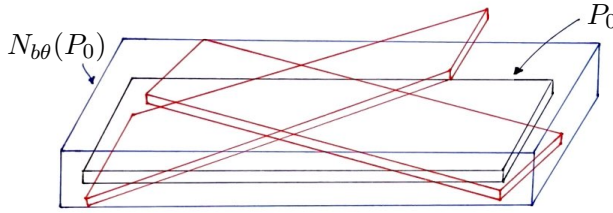


Figure 14.4.3: The geometry of Lemma 14.4.39. For each $x \in Y_0(P_0)$, there is a prism (red) from $\mathcal{P}$ that intersects P_0 at x at angle at least $\theta_{\min}$. For clarity, only two such prisms have been drawn. The union of these (red) prisms fills out a large fraction of every b -ball centered at a point of P_0 .

Remark 14.4.40. The exponent λ^4 in (14.4.54) is not important — the exponent λ^{100} would work equally well for our applications of Lemma 14.4.39.

Proof. The argument is similar in spirit to Wolff’s hairbrush argument from [27]. After dyadic pigeonholing, we can select a number $\theta_1 \in [\theta, 1]$ and a set $Y'_0(P_0) \subset Y_0(P_0)$ with $|Y'_0(P_0)| \gtrsim_a |Y_0(P_0)|$ so that

$$\sup_{P \in \mathcal{P}_Y(y)} \angle(\Pi(P_0), \Pi(P)) \sim \theta_1 \quad \text{for each } y \in Y'_0(P_0).$$

Applying Lemma 14.4.38 (and further refining $Y'_0(P_0)$ by a factor of 2), we may suppose that $Y'_0(P_0)$ is regular.

Let $b' = b\frac{\theta}{\theta_1}$; note that $a \leq b' \leq b$. Divide P_0 into sub-prisms of dimensions $a \times b' \times b'$; we have that $\gtrsim \lambda \frac{|P|}{a(b')^2}$ of these sub-prisms intersect $Y(P)$. Thus to obtain (14.4.54), it suffices to show that for each $x \in Y'(P_0)$, we have

$$\left| N_{b\theta}(P_0 \cap B(x, b')) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| \gtrsim_a \lambda^3 (b\theta)(b')^2. \quad (14.4.52)$$

Fix a choice of $x \in Y'(P_0)$. Our goal is to show that (14.4.52) holds for this choice of x . Let $E = Y'(P_0) \cap B(x, b')$. Since the shading $Y'(P_0)$ is regular, we have $|E| \gtrsim_a \lambda a(b')^2$. For each $y \in E$, let $P_y \in \mathcal{P}_Y(y)$ be a prism with $\angle(\Pi(P_0), \Pi(P_y)) \sim \theta_1$.

$B(x, b') \cap \bigcup_{y \in E} P_y$ is contained in a prism of dimensions comparable to $b\theta \times b' \times b'$ that is concentric with the $a \times b' \times b'$ prism $B(x, b') \cap P$; denote this $b\theta \times b' \times b'$ prism by Q . Then $\phi_Q(B(x, b') \cap P)$ is comparable to a prism of dimensions $\frac{a}{b\theta} \times 1 \times 1$. For notational convenience, we will select coordinates so that this prism is given by $\tilde{P} = [0, \frac{a}{b\theta}] \times [0, 1] \times [0, 1]$.

For each $y \in E$, we have that $\phi_Q(B(x, b') \cap P_y)$ is comparable to a $\frac{a}{b\theta} \times 1 \times 1$ prism, and each such prism intersects the (z_2, z_3) -plane with angle ~ 1 , i.e. the projection of the normal vector v_y of this prism to the (z_2, z_3) plane has magnitude $\gtrsim 1$. Since the shading of each $P_y \in \mathcal{P}$ is regular, we have that $\phi_Q(Y(P_y) \cap B(x, b'))$ is a subset of the prism $\phi_Q(P_y \cap B(x, b'))$ that has density $\gtrsim_a \lambda$.

After pigeonholing, we can select a set $E' \subset E$ with $|E'| \gtrsim |E|$, so that for each $y \in E'$, $\phi_Q(P_y \cap B(x, b'))$ is comparable to a $\frac{a}{b\theta} \times 1 \times 1$ prism whose normal vector v_y makes angle $\leq 1/100$ with some fixed unit vector v , and the projection of v to the (z_2, z_3) plane has magnitude $\gtrsim 1$. Let v' be the projection of v to the (z_2, z_3) plane.

By Fubini, we can find a line segment $L \subset [0, \frac{a}{b\theta}] \times [0, 1] \times [0, 1]$ pointing in direction v' , with $|L \cap \phi_Q(E')| \geq |\phi_Q(E')|$; here the left $|\cdot|$ denotes one-dimensional Lebesgue measure, while the right $|\cdot|$ denotes three-dimensional Lebesgue measure. Let $y_1, \dots, y_N$, $N \gtrsim \frac{b\theta}{a} |L \cap \phi_Q(E')| \gtrsim_a \lambda \frac{b\theta}{a}$ be a $\frac{a}{b\theta}$ -separated subset of $L \cap \phi_Q(E')$, and let

$$\mathcal{S} = \{\phi_Q(P_{y_i} \cap B(x, b')) : i = 1, \dots, N\}.$$

$\mathcal{S}$ is a set of convex subsets of $\mathbb{R}^3$, each of which is comparable to a $\frac{a}{b\theta} \times 1 \times 1$ slab. Each $S \in \mathcal{S}$ has a $\gtrsim_a \lambda$ -dense shading (the shading is the set $\phi_Q(Y(P_{y_i}) \cap B(x, b'))$), so in particular the pair $(\mathcal{S}, Y)_{\frac{a}{b\theta} \times 1 \times 1}$ is $\gtrsim_a \lambda$ dense.

We claim that $C_{KT-CW}(\mathcal{S}) \lesssim 1$. To verify this, let $W \subset \mathbb{R}^3$ be a convex set, and suppose $\#\{S \in \mathcal{S} : S \subset W\} = M$. We need to show that

$$M \lesssim |W| |S|^{-1}. \quad (14.4.53)$$

If $M = 0$ there is nothing to prove. If $M = 1$, then clearly $|W| \geq |S|$. Suppose instead that $M \geq 2$. Then $|W \cap L| \geq (M - 1) \frac{a}{b\theta} \geq \frac{M}{2b\theta} \gtrsim M |S|$, since $W \cap L$ contains at least M points on L that are $\frac{a}{b\theta}$ -separated. On the other hand, since $\mathcal{S}[W]$ is non-empty, we know that W contains a $\frac{a}{b\theta} \times 1 \times 1$ prism $S \in \mathcal{S}[W]$ whose normal direction makes angle $\leq \frac{1}{50}$ with the direction of v (whose projection to the (z_2, z_3) -plane is v' , i.e. the direction of L). Since W is convex,

W contains the convex hull of $S \cup (W \cap L)$, which has three-dimensional volume $\gtrsim |W \cap L| \gtrsim M|S|$. Thus $|W| \gtrsim M|S|$, which gives (14.4.53).

Applying Lemma 14.4.36 (a Cordoba-type L^2 argument for slabs), we conclude that

$$\left| \bigcup_{S \in \mathcal{S}} Y(S) \right| \gtrsim_a \lambda^2(\#\mathcal{S})|S| \gtrsim_a \lambda^2 \left(\lambda \frac{b\theta}{a} \right) \left(\frac{a}{b\theta} \right) = \lambda^3.$$

To conclude the proof, we verify that

$$\begin{aligned} \left| N_{b\theta}(P_0 \cap B(x, b')) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| &\geq \left| N_{b\theta}(P_0 \cap B(x, b')) \cap \bigcup_{y \in E'} Y(P_y) \right| \\ &\geq (b\theta)(b')^2 \left| \phi_Q \left(B(x, b') \cap \bigcup_{y \in E'} Y(P_y) \right) \right| \\ &\geq (b\theta)(b')^2 \left| \phi_Q \left(B(x, b') \cap \bigcup_{i=1}^N Y(P_{y_i}) \right) \right| \\ &\geq (b\theta)(b')^2 \left| \bigcup_{S \in \mathcal{S}} Y(S) \right| \gtrsim_a \lambda^3 (b\theta)(b')^2. \quad \square \end{aligned}$$

This establishes (14.4.52), as desired.

In practice, we will often use Lemma 14.4.39 in situations where each prism in $P_0 \in \mathcal{P}$ satisfies the hypotheses of the lemma. The following definitions help make that precise.

Definition 14.4.41. Let $(\mathcal{P}, Y)_{a \times b \times c}$ be a collection of prisms and their associated shading, and let $x \in \bigcup_{P \in \mathcal{P}} Y(P)$. We define

$$\theta(x) = \frac{a}{b} + \sup_{P, P' \in \mathcal{P}_Y(x)} \angle(\Pi(P), \Pi(P')),$$

where $\mathcal{P}_Y(x) = \{P \in \mathcal{P} : x \in Y(P)\}$. We define

$$\theta_{\min} = \inf \theta(x),$$

where the infimum is taken over all $x \in \bigcup_{P \in \mathcal{P}} Y(P)$. Note that $\theta_{\min}$ depends on the pair $(\mathcal{P}, Y)_{a \times b \times c}$. The choice of $\mathcal{P}$ and Y will be apparent from context.

By considering each prism $P_0 \in \mathcal{P}$ in turn, Lemma 14.4.39 now has the following corollary.

Corollary 14.4.42. *Let $0 < a \leq b \leq c$, $\lambda > 0$. Let $(\mathcal{P}, Y)_{a \times b \times c}$ be a set of prisms and their associated shading. Suppose that each shading $Y(P)$ is regular, in the sense of Definition 14.4.37, and satisfies $|Y(P)| \geq \lambda|P|$. Then for each $P_0 \in \mathcal{P}$, we have*

$$\left| N_{b\theta_{\min}}(P_0) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| \gtrsim_a \lambda^4 |N_{b\theta_{\min}}(P)|. \quad (14.4.54)$$

14.4.4.2 Assertions $\mathcal{F}$, $\mathcal{E}$, and $\tilde{\mathcal{E}}$ are equivalent

As noted in Remark 14.4.35, Situation 1, we have $\mathcal{F}(\sigma, \omega) \implies \mathcal{E}(\sigma, \omega) \implies \tilde{\mathcal{E}}(\sigma, \omega)$. In this section we will prove that the reverse implications hold. More precisely, we have the following.

Proposition 14.4.43. *Let $0 < \sigma \leq 2/3$, $\omega > 0$. Then $\mathcal{F}(\sigma, \omega) \iff \mathcal{E}(\sigma, \omega) \iff \tilde{\mathcal{E}}(\sigma, \omega)$.*

The goal of Section 14.4.4.2 is to prove Proposition 14.4.43. We begin with several lemmas that describe the structure of arrangements of rectangular prisms. Recall the quantity $\theta_{\min}$ from Definition 14.4.41. When $\theta_{\min} \sim 1$, then a typical pair of intersecting prisms intersect transversely, and by Corollary 14.4.42 this means that each prism can be replaced by a thickened neighbourhood that is comparable to an a -tube. In the next lemma, we will analyze what happens when $\theta_{\min}$ is small. I.e. we consider a collection of $a \times b \times 1$ prisms $\mathcal{P}$ that intersect tangentially, in the sense that their tangent planes make small angle at a typical point of intersection.

Informally, the argument is as follows. We will begin by describing three Moves, which we will then apply repeatedly. Note that these are *not* the grains Moves developed later in Sections 14.4.6 and 14.4.7.

Since the tangent plane of a prism is constant along the entire prism, this means that the collection of prisms can be partitioned into smaller sub-collections $\mathcal{P}_1, \dots, \mathcal{P}_N$, where for each index i , the normal vector of the tangent plane of each prism in $\mathcal{P}_i$ makes small angle with a fixed unit vector v_i . If $\mathcal{P}_i$ and $\mathcal{P}_j$ are two such sub-collections, then the corresponding sets $\bigcup_{P \in \mathcal{P}_i} P$ and $\bigcup_{P \in \mathcal{P}_j} P$ are mostly disjoint. We can cover the prisms in each set $\mathcal{P}_i$ by a set of slabs whose tangent planes have normal vector v_i . We then rescale and continue this process inside each slab. This is Move #1.

Next, we can apply Proposition 14.4.25 (factoring with respect to the Frostman Slab Wolff Axioms) to further break our collection of prisms into sub-collections, each of which satisfy the Frostman Slab Wolff Axioms with small error. This is Move #2.

Finally we apply Lemma 14.4.38 (every shading has a regular sub-shading). This is Move #3.

We iteratively apply Moves 1, 2, and 3 until our set of prisms $\mathcal{P}$ has been covered by a set $\mathcal{W}$ of convex subsets of $\mathbb{R}^3$, so that each set $\mathcal{P}^W$ satisfies the hypotheses of Corollary 14.4.42 with $\theta_{\min}$ about 1, and $\mathcal{P}^W$ satisfies the Frostman Slab Wolff Axioms with error about 1. The precise statement is as follows.

Lemma 14.4.44. *For all $\varepsilon > 0$, there exists $\eta, c > 0$ so that the following holds for all $0 < a \leq b \leq 1$. Let $(\mathcal{P}, Y)_{a \times b \times 1}$ be a^η dense. Then there exists*

a refinement $(\mathcal{P}', Y')_{a \times b \times 1}$ with $\sum_{P' \in \mathcal{P}'} |Y'(P')| \gtrsim_a a^\varepsilon \sum_{P \in \mathcal{P}} |P|$ and cover $\mathcal{W}$ of $\mathcal{P}'$ by congruent convex sets, such that the following holds.

- (A) The shading Y' is regular, in the sense of Definition 14.4.37.
- (B) $\mathcal{W}$ is a $\approx_a 1$ balanced cover of $\mathcal{P}'$, and factors $\mathcal{P}'$ from below with respect to the Frostman Slab Wolff axioms, with error $a^{-\varepsilon}$.
- (C) All of the convex sets in $\bigcup_{W \in \mathcal{W}} \mathcal{P}'^W$ have the same dimensions up to a multiplicative factor of 2, i.e. each one is comparable (after a suitable rigid transformation) to a common convex set $\tilde{P}$ (see Remark 14.4.45 below).
- (D) For each $W \in \mathcal{W}$, the sets $\mathcal{P}'^W$ and their associated shading satisfy $\theta_{\min} \geq a^\varepsilon$, in the sense of Definition 14.4.41.
- (E) The sets $\bigcup_{P' \in \mathcal{P}'[W]} Y'(P)$, $W \in \mathcal{W}$ are pairwise disjoint.

Remark 14.4.45. Note that even though the prisms in $\mathcal{P}$ (resp. $\mathcal{W}$) all have the same dimensions, it could happen that the prisms in $\mathcal{P}^W$ have differing dimensions — see Figure 14.4.4 for an example of this phenomena. Conclusion (C) (which is achieved by pigeonholing) ensures that this does not happen.



Figure 14.4.4: An example where the convex sets P_1 and P_2 are congruent, and U_1 and U_2 are congruent, but $\phi_{U_1}(P_1)$ and $\phi_{U_2}(P_2)$ are not congruent.

Proof. The main ideas of the proof were already outlined above; in brief, we apply Moves 1, 2, and 3 described above, in that order. We iteratively repeat this process until Conclusions (A), (B), (D), and (E) of Lemma 14.4.44 are satisfied. Each iteration increases the volume of each surviving convex set by at least $a^{-\varepsilon}$. On the other hand, each convex set initially had volume $ab \geq a^2$, and at each step all convex sets are contained inside the unit ball, and thus have volume at most $O(1)$. Hence the iterative process detailed above halts after at most $2/\varepsilon$ steps. If $\eta > 0$ is chosen sufficiently small (depending on ε), then the resulting refinement will satisfy $\sum_{P' \in \mathcal{P}'} |Y'(P')| \gtrsim_a a^\varepsilon \sum_{P \in \mathcal{P}} |P|$. Finally, we dyadically pigeonhole the set $\mathcal{P}'$ to select a refinement that satisfies Conclusion (C). $\square$

Audit status (checked 31 August 2026). The preceding paragraph is the complete proof text in the source, but it is not a self-contained proof of Lemma 14.4.44. In particular, Move 1 is described only informally: no lemma constructs the normal-cap partition and its slab cover, proves disjointness of the assigned shaded unions, records the mass loss, or shows that the affine child prisms have the claimed increase in normalized volume. The assertion that every iteration increases volume by $a^{-\varepsilon}$ therefore has no proved antecedent in the printed argument. Moves 2 and 3 cite proved factoring and regularization lemmas, but they do not supply Move 1. We record the missing statement as

construct a balanced, shaded-disjoint slab refinement at every node
 with mass loss $\gtrsim_a a^{O(\eta)}$ and normalized-volume gain $\geq a^{-\varepsilon}$,
(WZ–A4)

until the rescaled family has $\theta_{\min} \geq a^\varepsilon$. Lemma 14.4.44, and every later use of it, is conditional on WZ–A4 until that construction is proved.

We are now ready to prove Proposition 14.4.43. The idea is as follows. Given a set $(\mathcal{P}, Y)_{a \times b \times 1}$ of prisms, we apply Lemma 14.4.44 to cover a refinement of $\mathcal{P}$ by a collection of convex sets $\mathcal{W}$, and then apply Corollary 14.4.42 to each collection $\mathcal{P}^W$ — this gives us a collection of $\tilde{b}$ tubes associated to $\mathcal{P}^W$, for some $\tilde{b} \geq b$. The collection of $\tilde{b}$ tubes satisfies the Frostman Slab Wolff axioms with error $\lesssim 1$, and satisfies the Katz-Tao Convex Wolff axioms with some error $\tilde{m}$ that we will analyze later. We now apply the estimate $\tilde{\mathcal{E}}(\sigma, \omega)$ to this collection of $\tilde{b}$ tubes, and then undo the transformation ϕ_W . Summing the contributions from each $W \in \mathcal{W}$, we obtain an estimate for $\left| \bigcup Y(P) \right|$ that becomes better as $\tilde{m}$ becomes larger. Thus we are faced with the task of estimating the size of $\tilde{m}$ — this quantity is closely related to the quantity D from (14.4.45). We now turn to the details.

Proof of Proposition 14.4.43. It suffices to prove that $\tilde{\mathcal{E}}(\sigma, \omega) \implies \mathcal{F}(\sigma, \omega)$. Fix $0 < \sigma \leq 2/3$, $\omega > 0$, and suppose $\tilde{\mathcal{E}}(\sigma, \omega)$ is true. Fix $\varepsilon > 0$. Then there exists $\eta_0 > 0$ so that the volume estimate (14.4.43) holds for all pairs $(\mathbb{T}, Y)_\delta$ that are δ^{η_0} dense and satisfy $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta_0}$.

Let $\kappa, \eta > 0$ be quantities to be determined below. Let $0 < a \leq b \leq 1$ and let $(\mathcal{P}, Y)_{a \times b \times 1}$ be a^η dense. Our goal is to prove that

$$\left| \bigcup_{P \in \mathcal{P}} Y(P) \right| \geq \kappa a^\varepsilon b^\omega m^{-1} (\#\mathcal{P}) |P| (m^{-3/2} \ell(\#\mathcal{P}) |P|^{1/2})^{-\sigma} D^{-\sigma}, \quad (14.4.55)$$

with m, ℓ , and D as defined in (14.4.45).

Step 1. Let η_1 be a small quantity to be determined later (we will select η_1 very small compared to η_0 , and η very small compared to η_1). We will choose $\eta > 0$ sufficiently small so that we can apply Lemma 14.4.44 with $\eta_1/4$ in place

of ε to $(\mathcal{P}, Y)_{a \times b \times 1}$. Denote the output of that lemma by $(\mathcal{P}_1, Y_1)_{a \times b \times 1}$ and $\mathcal{W}$. By Conclusion (B) of Lemma 14.4.44 we have $C_{F-SW}(\mathcal{P}_1^W) \leq a^{-\eta_1/4}$ for each $W \in \mathcal{W}_1$. By Conclusion (E) of Lemma 14.4.44, we have

$$\left| \bigcup_{P \in \mathcal{P}} Y(P) \right| \geq \sum_{W \in \mathcal{W}} \left| \bigcup_{P \in \mathcal{P}_1[W]} Y_1(P) \right|. \quad (14.4.56)$$

By Conclusion (C) of Lemma 14.4.44, there are numbers $\tilde{a} \leq \tilde{b} \leq 1$ with $\tilde{b} \geq b$ so that for each $W \in \mathcal{W}$ and each $P^W \in \mathcal{P}_1^W$, P^W is comparable to a $\tilde{a} \times \tilde{b} \times 1$ prism. Conclusions (A) and (D) of Lemma 14.4.44 say that for each $W \in \mathcal{W}$, the pair $(\mathcal{P}_1^W, Y_1^W)_{\tilde{a} \times \tilde{b} \times 1}$ satisfies the hypotheses of Corollary 14.4.42, and hence we can apply Corollary 14.4.42 to conclude that for each $P^W \in \mathcal{P}_1^W$ we have

$$\left| N_{\tilde{b}}(P^W) \cap \bigcup_{P'^W \in \mathcal{P}_1^W} Y_1^W(P'^W) \right| \gtrsim a^{\eta_1} |N_{\tilde{b}}(P^W)|. \quad (14.4.57)$$

In words, (14.4.57) says the following: for each $P^W \in \mathcal{P}_1^W$, the set $N_{\tilde{b}}(P^W)$ is comparable to a $\tilde{b}$ -tube. This $\tilde{b}$ -tube has almost full intersection with the union of shadings $\bigcup_{P'^W \in \mathcal{P}_1^W} Y_1^W(P'^W)$. See Figure 14.4.3 for an illustration of this situation.

Step 2. As noted above, each set $N_{\tilde{b}}(P^W)$ is comparable to a $\tilde{b}$ tube. Let $\mathbb{T}_W$ be a maximal, essentially distinct subset of $\{N_{\tilde{b}}(P^W) : P^W \in \mathcal{P}_1^W\}$. For each $T \in \mathbb{T}_W$, define the shading

$$Y(T) = T \cap \bigcup_{P^W \in \mathcal{P}_1^W} Y_1^W(P^W).$$

Then (14.4.57) says that $(\mathbb{T}_W, Y)_{\tilde{b}}$ is $\gtrsim a^{\eta_1}$ dense. After pigeonholing and refining $\mathcal{W}$ and $\mathbb{T}_W$ (which induces a refinement on $\mathcal{P}_1^W$), we can ensure that $\#\mathcal{P}_1^W[T]$ is roughly the same (up to a factor of 2) for each $T \in \mathbb{T}_W$, $W \in \mathcal{W}$. Abusing notation, we continue to refer to these refined sets as $\mathcal{W}, \mathbb{T}_W$ and $\mathcal{P}_1^W$.

At this point, $\mathbb{T}_W$ is a balanced cover of $\mathcal{P}_1^W$, and we still have $C_{F-SW}(\mathcal{P}_1^W) \lesssim_a a^{-2\eta_1}$. Since $\mathbb{T}_W$ is a balanced cover of $\mathcal{P}_1^W$, by Remark 14.4.20(A) (i.e. Frostman Wolff constants are inherited upwards), we have that $\tilde{\ell} := \max_{W \in \mathcal{W}} C_{F-SW}(\mathbb{T}_W) \lesssim_a a^{-2\eta_1}$. Finally, define $\tilde{m} := \max_{W \in \mathcal{W}} C_{KT-CW}(\mathbb{T}_W)$.

Step 3. At this point, we have covered our set of prisms $\mathcal{P}_1$ by a collection of convex sets $\mathcal{W}$. For each rescaled set $\mathcal{P}_1^W$, we have located a collection of tubes $\mathbb{T}_W$, whose shadings are almost full. The relation between the volumes of these objects is as follows:

$$\left| \bigcup_{P \in \mathcal{P}_1[W]} Y_1(P) \right| \sim |W| \left| \bigcup_{P^W \in \mathcal{P}_1^W} Y_1^W(P^W) \right| \geq |W| \left| \bigcup_{T \in \mathbb{T}_W} Y(T) \right|. \quad (14.4.58)$$

Our next task is to estimate the volume of $\bigcup_{T \in \mathbb{T}_W} Y(T)$. First we consider the case where $\tilde{b} \leq a^{\varepsilon/5}$. In this case, $(\mathbb{T}_W, Y)_{\tilde{b}}$ is $\gtrsim \tilde{b}^{5\eta_1/\varepsilon}$ dense, and $C_{F-SW}(\mathbb{T}_W) \lesssim_a \tilde{b}^{-10\eta_1/\varepsilon}$. If η_1 is selected sufficiently small depending on η_0 and ε (for example, $\eta_1 = \varepsilon\eta_0/20$ will suffice), then we can apply the estimate $\tilde{\mathcal{E}}(\sigma, \omega)$ with $\varepsilon/2$ in place of ε to conclude that

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}_W} Y(T) \right| &\gtrsim \tilde{b}^{\omega + \varepsilon/2} \tilde{m}^{-1}(\#\mathbb{T}_W) |T| \left(\tilde{m}^{-3/2} \tilde{\ell}(\#\mathbb{T}_W) |T|^{1/2} \right)^{-\sigma} \\ &\gtrsim_a a^{\varepsilon/2 + 2\eta_1\sigma} b^\omega \tilde{m}^{-1}(\#\mathbb{T}_W) |T| \left(\tilde{m}^{-3/2} (\#\mathbb{T}_W) |T|^{1/2} \right)^{-\sigma}. \end{aligned} \quad (14.4.59)$$

On the other hand, if $\tilde{b} \geq a^{\varepsilon/5}$, then the estimate (14.4.59) follows from the fact that for every $T \in \mathbb{T}_W$ we have

$$\left| \bigcup_{T \in \mathbb{T}_W} Y(T) \right| \geq |Y(T)| \gtrsim a^{\eta_1} \tilde{b}^2 \geq a^{\varepsilon/2}, \quad (14.4.60)$$

which is stronger than (14.4.59).

Step 4. We have estimated the volume of $\bigcup_{T \in \mathbb{T}_W} Y(T)$ for each $W \in \mathcal{W}$. Our next task is to combine these estimates in order to estimate the volume of $\bigcup_{P \in \mathcal{P}} Y(P)$.

We know that each prism $W \in \mathcal{W}$ has the same dimensions up to a factor of 2; call these dimensions $s \times t \times 1$. Then for each $W \in \mathcal{W}$ and each $T \in \mathbb{T}_W$, we have that $\phi_W^{-1}(T)$ is comparable to a rectangular prism of dimensions $\tilde{s} \times \tilde{t} \times 1$.

Let $\hat{\mathcal{P}}_W = \{\phi_W^{-1}(T) : T \in \mathbb{T}_W\}$, i.e. $\hat{\mathcal{P}}_W$ is a set of $\tilde{s} \times \tilde{t} \times 1$ prisms contained in W ; $|T| \sim |\hat{P}|/|W|$; and $\#\hat{\mathcal{P}}_W = \#\mathbb{T}_W$. For each $W \in \mathcal{W}$ and each $\hat{P} = \phi_W^{-1}(T) \in \hat{\mathcal{P}}_W$, define the natural shading $\hat{Y}(\hat{P}) = \phi_W^{-1}(Y(T))$.

(14.4.56) allows us to combine the (rescaled) volume estimates (14.4.58) and (14.4.59) from each $W \in \mathcal{W}$. Defining $\hat{\mathcal{P}} = \bigsqcup \hat{\mathcal{P}}_W$, we have

$$\begin{aligned} \left| \bigcup_{P \in \mathcal{P}} Y(P) \right| &\gtrsim_a |W| \sum_{W \in \mathcal{W}} a^{\varepsilon/2 + \eta_1\sigma} b^\omega \tilde{m}^{-1}(\#\hat{\mathcal{P}}_W) \frac{|\hat{P}|}{|W|} \\ &\quad \times \left(\tilde{m}^{-3/2} (\#\hat{\mathcal{P}}_W) \left(\frac{|\hat{P}|}{|W|} \right)^{1/2} \right)^{-\sigma} \\ &\approx_a a^{\varepsilon/2 + \eta_1\sigma} b^\omega \tilde{m}^{-1}(\#\hat{\mathcal{P}}) |\hat{\mathcal{P}}| \\ &\quad \times \left(\tilde{m}^{-3/2} \frac{\#\hat{\mathcal{P}}}{\#\mathcal{W}} \left(\frac{|\hat{P}|}{|W|} \right)^{1/2} \right)^{-\sigma}. \end{aligned} \quad (14.4.61)$$

Step 5. To understand the RHS of (14.4.61) we must estimate $\tilde{m}$. Recall that in Step 2, we have pigeonholed to ensure that $\#\mathcal{P}_1^W[T]$ is roughly the same

for each $T \in \mathbb{T}_W$, $W \in \mathcal{W}$. Thus each $\hat{P} \in \hat{\mathcal{P}}$ satisfies $\#\mathcal{P}_1[\hat{P}] \approx_a \frac{\#\mathcal{P}_1}{\#\hat{\mathcal{P}}}$. Thus we have

$$\tilde{m} \lesssim_a m \frac{|\hat{P}|}{|P|} \frac{\#\hat{\mathcal{P}}}{\#\mathcal{P}_1}. \quad (14.4.62)$$

Thus we can estimate the RHS of (14.4.61) as follows.

$$\begin{aligned} \left| \bigcup_{P \in \mathcal{P}} Y(P) \right| &\gtrsim_a a^{\varepsilon/2 + \eta_1 \sigma} b^\omega m^{-1} (\#\mathcal{P}_1) |P| \\ &\quad \times \left[m^{-3/2} \left(\frac{|P|}{|\hat{P}|} \frac{\#\mathcal{P}_1}{\#\hat{\mathcal{P}}} \right)^{3/2} \frac{\#\hat{\mathcal{P}}}{\#\mathcal{W}} \left(\frac{|\hat{P}|}{|W|} \right)^{1/2} \right]^{-\sigma} \\ &\gtrsim_a a^{\varepsilon/2 + 3\eta_1} b^\omega m^{-1} (\#\mathcal{P}) |P| \left(m^{-3/2} (\#\mathcal{P}) |P|^{1/2} \right)^{-\sigma} \\ &\quad \times \left[(\#\mathcal{W}) |W|^{1/2} \right]^\sigma \left[\frac{|P|}{|\hat{P}|} \left(\frac{\#\mathcal{P}}{\#\hat{\mathcal{P}}} \right)^{1/2} \right]^{-\sigma}. \end{aligned} \quad (14.4.63)$$

In the above computation, we used the fact that $\#\mathcal{P}_1 \gtrsim_a \delta^{-\eta_1} (\#\mathcal{P})$.

Step 6. Compare the RHS of (14.4.63) with (14.4.55). It remains to analyze the final two terms on the RHS of (14.4.63). We begin with the penultimate term. Since the sets in $\mathcal{W}$ are convex and have diameter ~ 1 , each $W \in \mathcal{W}$ is contained in a slab of volume $O(|W|^{1/2})$. Thus

$$\#\mathcal{P}_1[W] \lesssim C_{F-SW}(\mathcal{P}) |W|^{1/2} (\#\mathcal{P}).$$

Since $\#\mathcal{P}_1[W] \approx_a \frac{\#\mathcal{P}_1}{\#\mathcal{W}} \gtrsim_a a^{\eta_1} \frac{\#\mathcal{P}}{\#\mathcal{W}}$, we conclude that $(\#\mathcal{W}) |W|^{1/2} \gtrsim_a a^{\eta_1} C_{F-SW}(\mathcal{P})^{-1} = a^{\eta_1} \ell^{-1}$.

Step 7. We now turn to the final term on the RHS of (14.4.63). Recall that $\mathcal{P}$ is a collection of prisms of dimensions $a \times b \times 1$, while $\hat{\mathcal{P}}$ is a collection of prisms that all have the same, but unknown, dimensions $\tilde{s}b \times \tilde{t}b \times 1$ — call these dimensions $a' \times b' \times 1$, with $a \leq a' \leq b'$ and $b' \geq b$. Our desired estimate (14.4.55) will follow from (14.4.63) and the estimate

$$\frac{|P|}{|\hat{P}|} \left(\frac{\#\mathcal{P}}{\#\hat{\mathcal{P}}} \right)^{1/2} \lesssim D = \max_{P \in \mathcal{P}} \sup_{\rho \in [a, b]} \frac{|P|}{|N_\rho(P)|} (\#\mathcal{P}[N_\rho(P)])^{1/2}. \quad (14.4.64)$$

Fix $\hat{P} \in \hat{\mathcal{P}}$. Let $\mathcal{P}^\dagger$ be a maximal set of essentially distinct $\min(a', b) \times b \times 1$ prisms contained in $\hat{P}$, so that each $P \in \mathcal{P}[\hat{P}]$ is contained in at least one $P^\dagger \in \mathcal{P}^\dagger$. We claim that

$$\#\mathcal{P}^\dagger \sim (|\hat{P}|/|P^\dagger|)^2. \quad (14.4.65)$$

Indeed, when $a' \leq b$, the RHS of (14.4.65) is $(b'/b)^2$ and the numerology comes from the fact that a $b' \times 1$ rectangle can be filled with about $(b'/b)^2$ essentially

distinct $b \times 1$ rectangles. When $a' \geq b$ the RHS of (14.4.65) is $(a'b'/b^2)^2$, and the numerology comes from the fact that a $a' \times b' \times 1$ prism can be filled with about $(b'/a')^2$ essentially distinct $a' \times a' \times 1$ tubes, and each of these tubes can be filled with about $(a'/b)^4$ essentially distinct $b \times b \times 1$ tubes.

By the definition of D , we have $D \geq \frac{|P|}{|P^\dagger|} (\#\mathcal{P}[P^\dagger])^{1/2}$, i.e. $\#\mathcal{P}[P^\dagger] \leq (D \frac{|P^\dagger|}{|P|})^2$. Thus by (14.4.65),

$$\#\mathcal{P}[\hat{P}] \lesssim \left(\frac{|\hat{P}|}{|P^\dagger|} \right)^2 \#\mathcal{P}[P^\dagger] \lesssim \left(\frac{|\hat{P}|}{|P^\dagger|} \right)^2 \left(D \frac{|P^\dagger|}{|P|} \right)^2,$$

which is (14.4.64). $\square$

14.4.4.3 Proof of Proposition 14.4.30: Tubes that factor through flat boxes

With Proposition 14.4.43 in hand, we are now ready to prove Proposition 14.4.30. Fix $\omega > 0$, $0 < \sigma \leq 2/3$, and suppose $\mathcal{E}(\sigma, \omega)$ is true (and thus by Proposition 14.4.43, $\mathcal{F}(\sigma, \omega)$ is true). Let $\kappa, \eta > 0$ be small quantities to be specified below. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, let $\delta \leq a \leq b \leq 1$, and let $\mathcal{W}$ be a set of congruent copies of an $a \times b \times 2$ prism W_0 , as described in the statement of Proposition 14.4.30.

Step 1. After dyadic pigeonholing, we can find a refinement $(\mathbb{T}_1, Y_1)_\delta$ of $(\mathbb{T}, Y)_\delta$ with $\sum_{T \in \mathbb{T}_1} |Y_1(T)| \gtrsim_\delta \sum_{T \in \mathbb{T}} |Y(T)|$, and a subset $\mathcal{W}_1 \subset \mathcal{W}$ so that for each $W \in \mathcal{W}_1$ and each $x \in \bigcup_{T \in \mathbb{T}_1[W]} Y_1(T)$, we have

$$\left| B(x, a) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \sim \lambda |B(x, a)|, \quad \text{with} \quad \lambda \geq |W|^{-1} \left| \bigcup_{T \in \mathbb{T}_1[W]} Y_1(T) \right|, \quad (14.4.66)$$

where the “density” $\lambda = \lambda(W)$ is the same (up to a factor of 2) for all $W \in \mathcal{W}$. In words, (14.4.66) says that if we blur the shading $\bigcup_{T \in \mathbb{T}_1[W]} Y_1(T)$ at scale a (for example by convolving with the characteristic function of $B(0, a)$), then each point in the shading has density $\sim \lambda$.

After further pigeonholing, we can ensure that each set $(\mathbb{T}_1[W], Y_1)_\delta$, $W \in \mathcal{W}_1$ is $\approx_\delta \delta^{O(\eta)}$ dense; $\mathcal{W}_1$ is a $\approx_\delta \delta^{-O(\eta)}$ balanced cover of $\mathbb{T}_1$; and each set $\mathbb{T}_1^W$ obeys the Frostman Convex Wolff axioms with error $\lesssim_\delta \delta^{-O(\eta)}$. Abusing notation, we will continue to refer to the output of this pigeonholing by $(\mathbb{T}_1, Y_1)_\delta$ and $\mathcal{W}_1$.

Step 2. Fix $W \in \mathcal{W}_1$. Then $(\mathbb{T}_1^W, Y_1^W)_{\delta/b \times \delta/a \times 1}$ is a collection of convex sets, each comparable to a $\delta/b \times \delta/a \times 1$ prism, with a $\delta^{O(\eta)}$ -dense shading. We first consider the case where $b \geq \delta^{1-\varepsilon/100}$, so $\delta/b \leq \delta^{\varepsilon/100}$. In this case, the shading on $(\mathbb{T}_1^W, Y_1^W)_{\delta/b \times \delta/a \times 1}$ is $(\delta/b)^{O(\eta)/\varepsilon}$ dense. If η is selected sufficiently

small, then we can apply the estimate $\mathcal{F}(\sigma, \omega)$ with $\varepsilon/2$ in place of ε (recall that $\mathcal{F}(\sigma, \omega)$ is true in light of Proposition 14.4.43) to conclude that

$$\begin{aligned} \left| \bigcup_{T^W \in \mathbb{T}_1^W} Y_1^W(T^W) \right| &\geq \kappa_\varepsilon \left(\frac{\delta}{b} \right)^{\varepsilon/2} \left(\frac{\delta}{a} \right)^\omega m_W^{-1} (\#\mathbb{T}^W) |T^W| \\ &\quad \times \left(m_W^{-3/2} \ell_W (\#\mathbb{T}^W) |T^W|^{1/2} \right)^{-\sigma}, \end{aligned} \quad (14.4.67)$$

where $m_W = C_{KT-CW}(\mathbb{T}^W)$ and $\ell_W = C_{F-SW}(\mathbb{T}^W) \leq C_{F-CW}(\mathbb{T}^W) \lesssim_\delta \delta^{-O(\eta)}$. Recall as well that $|T^W| \sim |T|/|W|$. Note that the estimate $\mathcal{F}(\sigma, \omega)$ involves the additional term “ D ” defined in (14.4.45), but as discussed in Remark 14.4.35, Situation 3, we have $D \lesssim 1$.

Since $C_{F-CW}(\mathbb{T}^W) \lesssim_\delta \delta^{-O(\eta)}$, we have that $m_W \lesssim_\delta \delta^{-O(\eta)} |T^W| (\#\mathbb{T}^W)$, and thus (14.4.67) allows us to estimate the density λ (recall that λ was defined in (14.4.66)):

$$\begin{aligned} \lambda &\geq \left| \bigcup_{T^W \in \mathbb{T}^W} Y^W(T^W) \right| \\ &\gtrsim_\delta \kappa_\varepsilon \left(\frac{\delta}{b} \right)^{\varepsilon/2} \delta^{O(\eta)} \left(\frac{\delta}{a} \right)^\omega \left((\#\mathbb{T}^W)^{1/2} |T^W| \right)^\sigma. \end{aligned} \quad (14.4.68)$$

Note that Inequality (14.4.68) is currently only valid when $b \geq \delta^{1-\varepsilon/100}$. Next we consider the case where $b < \delta^{1-\varepsilon/100}$, so each $T^W \in \mathbb{T}_1^W$ has dimensions comparable to $d_1 \times d_2 \times 1$ for some $1 \geq d_2 \geq d_1 \geq \delta^{\varepsilon/100}$. This is true because

$$\lambda \geq |Y^W(T^W)| \geq \delta^{O(\eta)} |T^W| \geq \delta^{\varepsilon/50+O(\eta)} \geq \delta^{\varepsilon/10}.$$

Step 3. For each $W \in \mathcal{W}_1$, define the shading

$$\tilde{Y}_1(W) = W \cap N_a \left(\bigcup_{T \in \mathbb{T}_1[W]} Y_1(T) \right).$$

By (14.4.66), we have

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \lambda \left| \bigcup_{W \in \mathcal{W}_1} \tilde{Y}_1(W) \right|. \quad (14.4.69)$$

Our next task is to show that $(\mathcal{W}_1, \tilde{Y}_1)_{a \times b \times 1}$ is $\delta^{O(\eta)}$ dense. The argument is shown in Figure 14.4.5. Fix $W \in \mathcal{W}$. Since $(\mathbb{T}_1^W, Y_1^W)$ is $\gtrsim \delta^{O(\eta)}$ dense, we can select a refinement so that each $T \in \mathbb{T}_1[W]$ satisfies $|Y_1(T)| \gtrsim \delta^{O(\eta)} |T|$. After pigeonholing, we can select a set $\mathbb{T}_a$ of essentially distinct a -tubes that form a $\approx_\delta 1$ balanced cover of $\mathbb{T}_1[W]$. Since $(\mathbb{T}_1^W, Y_1^W)$ satisfies the Frostman Convex Wolff axioms with error $\lesssim_\delta \delta^{-O(\eta)}$, we have that $\mathbb{T}_a$ satisfies the Frostman Convex Wolff axioms with error $\lesssim_\delta \delta^{-O(\eta)}$. The shading

$Y_a(T_a) = T_a \cap N_a(\bigcup_{T \in \mathbb{T}[W]} Y_1(T))$ is $\approx_\delta \delta^{O(\eta)}$ dense. Applying Lemma 14.4.36 to $(\mathbb{T}_a^W, Y_a^W)_{a/b \times 1 \times 1}$ and then undoing the scaling ϕ_W , we conclude that

$$\left| W \cap N_a \left(\bigcup_{T \in \mathbb{T}_1[W]} Y_1(T) \right) \right| \geq \left| \bigcup_{T_a \in \mathbb{T}_a} Y_a(T_a) \right| \gtrsim_\delta \delta^{O(\eta)} |W|.$$

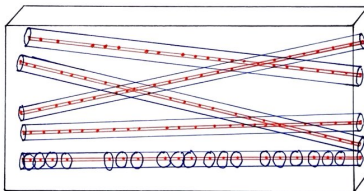


Figure 14.4.5: Each a -tube in $\mathbb{T}_a$ (blue tubes) contains at least one δ -tube from $\mathbb{T}_1[W]$ (red tube); this δ -tube has a dense shading (red dots), which in turn gives us a dense shading on the a -tube that contains it (blue balls; for clarity we have only drawn these for one of the tubes in $\mathbb{T}_a$). Finally, since W (black box) has thickness a , and the a -tubes inside W satisfy rescaled Frostman Convex Wolff axioms, an L^2 argument says that the union of a -tubes has almost full volume.

Step 4. It remains to estimate the RHS of (14.4.69). We first consider the case where $a \leq \delta^{\varepsilon/100}$. In this case, $(\mathcal{W}_1, \tilde{Y}_1)_{a \times b \times 1}$ is $a^{O(\eta/\varepsilon)}$ dense. If η is selected sufficiently small depending on ε , then we can apply the estimate $\mathcal{F}(\sigma, \omega)$ to conclude that

$$\left| \bigcup_{W \in \mathcal{W}_1} \tilde{Y}_1(W) \right| \geq \kappa_\varepsilon a^{\varepsilon/2} b^\omega \tilde{m}^{-1} (\#\mathcal{W}_1) |W| \left(\tilde{m}^{-3/2} \tilde{\ell} (\#\mathcal{W}_1) |W|^{1/2} \right)^{-\sigma}, \quad (14.4.70)$$

where $\tilde{m} = C_{KT-CW}(\mathcal{W}_1)$ and $\tilde{\ell} = C_{F-SW}(\mathcal{W}_1)$. Note that the estimate $\mathcal{F}(\sigma, \omega)$ involves the additional term “ D ” defined in (14.4.45). However, we claim that in both Parts (A) and (B) of Proposition 14.4.30, we have

$$D \lesssim \delta^{-\eta}. \quad (14.4.71)$$

We verify this claim as follows. In Part (A) of Proposition 14.4.30, we have the hypothesis that $C_{KT-CW}(\mathcal{W}) \leq \delta^{-\eta}$. As discussed in Remark 14.4.35, Situation 1, this ensures that $D \lesssim \delta^{-\eta}$. In Part (B) of Proposition 14.4.30, we have the hypothesis that

$$C_{KT-CW}(\mathcal{W}[N_b(W)]) \leq \delta^{-\eta} \quad \text{for all } W \in \mathcal{W}.$$

As discussed in Remark 14.4.35, Situation 2, this ensures that $D \lesssim \delta^{-\eta}$. This establishes (14.4.71).

In summary, we have established (14.4.70) when $a \leq \delta^{\varepsilon/100}$. If instead $a > \delta^{\varepsilon/100}$, then (14.4.70) follows from the estimate

$$\left| \bigcup_{W \in \mathcal{W}_1} \tilde{Y}_1(W) \right| \gtrsim \delta^{\varepsilon/20} \gtrsim \delta^{\varepsilon/20} \tilde{m}^{-1}(\#\mathcal{W}_1) |W|. \quad (14.4.72)$$

Combining (14.4.68), (14.4.69), and (14.4.70) (when $a \leq \delta^{\varepsilon/100}$) or (14.4.72) (when $a \geq \delta^{\varepsilon/100}$), we conclude that

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\gtrsim_{\delta} \delta^{\omega+\varepsilon+O(\eta)-\varepsilon^2/200} \left(\frac{b}{a}\right)^{\omega} \tilde{m}^{-1}(\#\mathcal{W}_1) |W| \\ &\quad \times \left(\tilde{m}^{-3/2} \tilde{\ell}(\#\mathcal{W}_1) |W|^{1/2} \right)^{-\sigma} \left(\left(\frac{\#\mathbb{T}}{\#\mathcal{W}_1} \right)^{1/2} \frac{|T|}{|W|} \right)^{\sigma} \\ &= \delta^{\omega+\varepsilon-\varepsilon^2/300} \left(\frac{b}{a}\right)^{\omega} \tilde{m}^{-1}(\#\mathcal{W}_1) |W| \\ &\quad \times \left(\tilde{m}^{-3/2} \tilde{\ell}(\#\mathcal{W}_1)^{3/2} |W|^{3/2} (\#\mathbb{T})^{-1/2} |T|^{-1} \right)^{-\sigma}. \end{aligned} \quad (14.4.73)$$

Step 5. It remains to analyze the RHS of (14.4.73). Our analysis will differ for Parts (A) and (B) of Proposition 14.4.30. We begin with Part (A). We have $\tilde{m} \lesssim \delta^{-\eta}$, and since $\mathcal{W}_1$ is a $\approx_{\delta} \delta^{-O(\eta)}$ balanced cover of $\mathbb{T}_1$, by Remark 14.4.20(A) (i.e. Frostman Wolff constants are inherited upwards) we have $\delta^{\eta} \tilde{\ell} \lesssim_{\delta} C_{F-SW}(\mathbb{T}) = \ell$. Since $\mathcal{W}$ is a $\delta^{-\eta}$ -balanced, $\delta^{-\eta}$ -almost partitioning cover of $\mathbb{T}$, $\#\mathbb{T}[W] \geq \delta^{\eta} C_{KT-CW}(\mathbb{T}) \frac{|W|}{|T|}$ for each $W \in \mathcal{W}$, and $\mathcal{W}_1$ is a refinement of $\mathcal{W}$, we have $(\#\mathcal{W}_1) |W| \gtrsim_{\delta} \delta^{\eta} m^{-1}(\#\mathbb{T}) |T|$. The RHS of (14.4.73) becomes

$$\delta^{\omega+\varepsilon} \left(\frac{b}{a}\right)^{\omega} m^{-1}(\#\mathbb{T}) |T| \left(m^{-3/2} \ell(\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma},$$

as claimed.

Next we consider part (B). We have $\tilde{m} \lesssim_{\delta} \delta^{-\eta} |W|(\#\mathcal{W}) \lesssim_{\delta} \delta^{-\eta} |W|(\#\mathcal{W}_1)$, and $\tilde{\ell} \lesssim_{\delta} \delta^{-\eta}$. Thus the RHS of (14.4.73) becomes

$$\delta^{\omega+\varepsilon} \left(\frac{b}{a}\right)^{\omega} \left((\#\mathbb{T})^{1/2} |T| \right)^{\sigma}.$$

This concludes the proof of Proposition 14.4.30.

14.4.4.4 Proof of Proposition 14.4.31: Factoring at two scales

Proof of Proposition 14.4.31. Fix $\varepsilon > 0$. All refinements below lose a power $\delta^{C\eta}$ and a fixed power of $\log(2/\delta)$; at the end choose η so small that their product is at least $\delta^{\varepsilon/20}$.

Step 1: simultaneous balancing and the blurred density. Compose the balanced cover of $\mathbb{T}$ by $\mathbb{T}_\rho$ with the balanced cover of $\mathbb{T}_\rho$ by $\mathcal{W}$. Dyadically pigeonhole the two fibre cardinalities and the densities of the shadings in those fibres. This gives a refinement $(\mathbb{T}_1, Y_1)$ and a subcollection $\mathcal{W}_1$ such that

$$\#\mathbb{T}_1[W] \sim \frac{\#\mathbb{T}_1}{\#\mathcal{W}_1}, \quad W \in \mathcal{W}_1,$$

up to $\delta^{-C\eta}$, and each rescaled pair $(\mathbb{T}_1^W, Y_1^W)$ is $\delta^{C\eta}$ -dense. A second dyadic pigeonhole, applied to the convolution of the union shading with $|B(0, a)|^{-1}\mathbf{1}_{B(0, a)}$, gives one number λ for which

$$|B(x, a) \cap \bigcup_{T \in \mathbb{T}} Y(T)| \sim \lambda |B(x, a)| \quad (14.4.74)$$

at every surviving shaded point, and

$$\lambda \geq |W|^{-1} \left| \bigcup_{T \in \mathbb{T}_1[W]} Y_1(T) \right|.$$

The sum of the retained shaded masses is at least $\delta^{C\eta}$ times the original one. These assertions follow by sorting positive quantities into dyadic intervals; the number of intervals is $O(\log(2/\delta))$ because all relevant volumes and cardinalities lie between fixed powers of δ .

Step 2: the fine estimate inside one W . For a convex set $V \subset B(0, 1)$,

$$\#(\mathbb{T}^W[V]) = \#\mathbb{T}[\phi_W^{-1}(V)] \leq m \frac{|\phi_W^{-1}(V)|}{|T|} \sim m \frac{|V|}{|T^W|},$$

so $C_{KT-CW}(\mathbb{T}^W) \lesssim m$. The function $u \mapsto u^{-1+3\sigma/2}$ is non-increasing for $0 < \sigma \leq 2/3$; hence replacing the local Katz–Tao constant by m only weakens the desired lower bound. Moreover, transitivity of the slab Frostman constant gives

$$C_{F-SW}(\mathbb{T}^W) \lesssim \max_{T_\rho \subset W} C_{F-SW}(\mathbb{T}^{T_\rho}) C_{F-SW}(\mathbb{T}_\rho^W) \leq \delta^{-C\eta} \ell'. \quad (14.4.75)$$

Here the first factor is at most ℓ' by the lower factorisation of $\mathbb{T}$ through $\mathbb{T}_\rho$, and the second is at most $\delta^{-\eta}$ because Frostman-convex factorisation controls slabs in particular.

Apply $\mathcal{F}(\sigma, \omega)$ to the $\delta/b \times \delta/a \times 1$ prisms in $\mathbb{T}_1^W$. The same large-scale alternative used in Step 2 of Proposition 14.4.30 covers the case in which δ/b is not small enough to invoke the assertion. Using (14.4.75) and $|T^W| \sim |T|/|W|$ yields

$$\begin{aligned} \left| \bigcup_{T^W \in \mathbb{T}_1^W} Y_1^W(T^W) \right| &\gtrsim \delta^{C\eta} (\delta/b)^{\varepsilon/2} (\delta/a)^\omega m^{-1} (\#\mathbb{T}_1[W]) |T^W| \\ &\cdot (m^{-3/2} \ell' (\#\mathbb{T}_1[W]) |T^W|^{1/2})^{-\sigma}. \end{aligned} \quad (14.4.76)$$

After undoing ϕ_W and using the definition of λ , this becomes

$$\lambda \gtrsim \delta^{C\eta} (\delta/b)^{\varepsilon/2} (\delta/a)^\omega m^{-1} (\#\mathbb{T}_1[W]) |T| (m^{-3/2} \ell' (\#\mathbb{T}_1[W]) (|T|/|W|)^{1/2})^{-\sigma}. \quad (14.4.77)$$

Step 3: a dense shading on the coarse prisms. Define

$$\tilde{Y}_1(W) = W \cap N_a\left(\bigcup_{T \in \mathbb{T}_1[W]} Y_1(T)\right).$$

The lower Frostman-convex factorisation of $\mathbb{T}_\rho$ through $\mathcal{W}$ implies, after rescaling W , that any convex subregion which contains a proportion q of the volume contains at most $\delta^{-C\eta} q$ of the ρ -tube mass. Cover the rescaled W by a/b -tubes and apply the proved Córdoba L^2 lemma 14.4.36. Since every surviving fine tube has $\delta^{C\eta}$ -dense shading, the resulting a -neighbourhood occupies at least $\delta^{C\eta} |W|$. Thus $(\mathcal{W}_1, \tilde{Y}_1)$ is $\delta^{C\eta}$ -dense. Equation (14.4.74) also gives, by integrating the convolution and using bounded overlap of radius- a balls,

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \lambda \left| \bigcup_{W \in \mathcal{W}_1} \tilde{Y}_1(W) \right|. \quad (14.4.78)$$

Step 4: the coarse estimate. Apply $\mathcal{F}(\sigma, \omega)$ to the $a \times b \times 1$ prisms $\mathcal{W}_1$, again using the one-prism bound when a is larger than the small-scale threshold. We obtain

$$\left| \bigcup_{W \in \mathcal{W}_1} \tilde{Y}_1(W) \right| \gtrsim a^{\varepsilon/2} b^\omega \tilde{m}^{-1} (\#\mathcal{W}_1) |W| (\tilde{m}^{-3/2} \tilde{\ell} (\#\mathcal{W}_1) |W|^{1/2})^{-\sigma}. \quad (14.4.79)$$

The upper Katz–Tao factorisation gives $1 \leq \tilde{m} = C_{KT-CW}(\mathcal{W}_1) \leq \delta^{-C\eta}$. Because the composed cover of $\mathbb{T}_1$ by $\mathcal{W}_1$ is balanced, Frostman constants are inherited upwards, and therefore

$$\tilde{\ell} = C_{F-SW}(\mathcal{W}_1) \leq \delta^{-C\eta} \ell. \quad (14.4.80)$$

Step 5: multiply and cancel the fibre sizes. Put $N = \#\mathbb{T}_1$ and $M = \#\mathcal{W}_1$. Balance gives $\#\mathbb{T}_1[W] \sim N/M$ up to $\delta^{-C\eta}$. Multiply (14.4.77) and (14.4.79), insert the result into (14.4.78), and use

$$(\delta/b)^{\varepsilon/2} a^{\varepsilon/2} \geq \delta^\varepsilon, \quad (\delta/a)^\omega b^\omega = \delta^\omega (b/a)^\omega.$$

The outside factors $(N/M)|T|/|W|$ and $M|W|$ multiply to $N|T|$. The two bracketed factors multiply to

$$m^{-3/2} \ell' N |T|^{1/2} \tilde{m}^{-3/2} \tilde{\ell}.$$

Since $\tilde{m} \geq 1$ and (14.4.80) holds, this is at most $\delta^{-C\eta} m^{-3/2} \ell' N |T|^{1/2}$. Also $\tilde{m}^{-1} \geq \delta^{C\eta}$. Hence

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \delta^{\omega + \varepsilon + C\eta} (b/a)^\omega m^{-1} N |T| (m^{-3/2} \ell' N |T|^{1/2})^{-\sigma}.$$

The refinement has $N \geq \delta^{C\eta} \# \mathbb{T}$; because the total power of N is $1 - \sigma > 0$, restoring $\# \mathbb{T}$ costs another $\delta^{C\eta}$. Choose η sufficiently small relative to the spare ε used at the start and rename 2ε as ε . This is the stated estimate. $\square$

14.4.4.5 Tubes organized into slabs

We conclude this section by using the tools developed thus far to prove the following. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes and their associated shading. Suppose that for each $T \in \mathbb{T}$, there is a $\delta \times b \times 1$ slab $S \supset T$ that has large intersection with $\bigcup_{\mathbb{T}} Y(T)$. Then provided $C_{KT-CW}(\mathbb{T})$ is small, $\bigcup_{\mathbb{T}} Y(T)$ has larger volume than one would expect from the estimate $\mathcal{E}(\sigma, \omega)$; the estimate becomes better as $C_{KT-CW}(\mathbb{T})$ becomes smaller and b becomes larger. The precise statement is as follows.

Lemma 14.4.46. *Let $\omega > 0, \sigma \in [0, 2/3]$ and suppose $\mathcal{E}(\sigma, \omega)$ is true. For all $\varepsilon > 0$, there exist $\kappa, \eta > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes and their associated shading. Let $b \geq \delta$ and suppose that for each $T \in \mathbb{T}$ there exists a $\delta \times b \times 1$ prism $S \supset T$ with $|S \cap \bigcup_{T \in \mathbb{T}} Y(T)| \geq \delta^\eta |S|$. Then*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^\varepsilon b^\omega m^{-1} (\# \mathbb{T}) |T| (m^{-1} \ell (\# \mathbb{T}) |T|^{1/2})^{-\sigma} \left(\frac{|S|}{|T|} \right)^{\sigma/2}, \quad (14.4.81)$$

where $m = C_{KT-CW}(\mathbb{T})$ and $\ell = C_{F-SW}(\mathbb{T})$.

Remark 14.4.47. Note that the exponent of m in (14.4.81) is only $m^{\sigma-1}$, rather than the usual estimate $m^{(3/2)\sigma-1}$ from $\mathcal{E}(\sigma, \omega)$. While this is likely not optimal, in practice we will only apply Lemma 14.4.46 with m of size about 1, so the distinction will not be important.

Proof. After pigeonholing, we can select a $\approx_\delta 1$ refinement $(\mathbb{T}_1, Y_1)_\delta$ of $(\mathbb{T}, Y)_\delta$ and a set $\mathcal{S}$ of essentially distinct $\delta \times b \times 1$ slabs with the following properties:

- For each $T \in \mathbb{T}_1$ with corresponding slab $S(T)$, there is a slab $S \in \mathcal{S}$ comparable to $S(T)$. We denote this by $T \sim S$
- There is an integer N so that each slab in $\mathcal{S}$, there are between N and $2N$ tubes $T \in \mathbb{T}_1$ with $T \sim S$.

Abusing notation, we will replace each slab in $\mathcal{S}$ with its 10-fold dilate. Then

- (i) $\mathcal{S}$ covers $\mathbb{T}_1$,
- (ii) $\#\mathbb{T}[S] \geq N$ for each $S \in \mathcal{S}$.
- (iii) $\#\mathcal{S} \sim N^{-1}(\#\mathbb{T}_1)$.
- (iv) The shading $Y(S) = S \cap \bigcup_{T \in \mathbb{T}} Y(T)$ is $\gtrsim \delta^\eta$ dense.

From Items (ii) and (iii) we conclude that

$$C_{F-SW}(\mathcal{S}) \lesssim C_{F-SW}(\mathbb{T}_1) \lesssim_\delta \delta^{-\eta} \ell,$$

and

$$C_{KT-CW}(\mathcal{S}) \lesssim C_{KT-CW}(\mathbb{T}_1) \frac{\#\mathcal{S}}{\#\mathbb{T}_1} \frac{|S|}{|T|} \lesssim_\delta \delta^{-\eta} m \frac{\#\mathcal{S}}{\#\mathbb{T}} \frac{|S|}{|T|}. \quad (14.4.82)$$

We would like to use Proposition 14.4.43 and apply the estimate $\mathcal{F}(\sigma, \omega)$ to obtain a lower bound for the volume of $\bigcup Y(S)$. Before doing so, we should estimate the quantity D from (14.4.45). By Remark 14.4.35, Situation 2, and (14.4.82), we have

$$D \lesssim \left(\sup_{S \in \mathcal{S}} C_{KT-CW}(\mathcal{S}[N_b(S)]) \right)^{1/2} \lesssim_\delta \left(\delta^{-\eta} m \frac{\#\mathcal{S}}{\#\mathbb{T}} \frac{|S|}{|T|} \right)^{1/2}.$$

If $\eta, \kappa > 0$ are chosen sufficiently small depending on ω, σ , and ε , then we can apply the estimate $\mathcal{F}(\sigma, \omega)$ to conclude that

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\geq \left| \bigcup_{S \in \mathcal{S}} Y(S) \right| \\ &\geq \kappa \delta^{\varepsilon/2} b^\omega C_{KT-CW}(\mathcal{S})^{-1} (\#\mathcal{S}) |S| D^{-\sigma} \\ &\quad \times \left(C_{KT-CW}(\mathcal{S})^{-3/2} C_{F-SW}(\mathcal{S}) (\#\mathcal{S}) |S|^{1/2} \right)^{-\sigma} \\ &\gtrsim_\delta \kappa \delta^{\varepsilon/2+2\eta} b^\omega m^{-1} (\#\mathbb{T}) |T| \left(m^{-1} \ell (\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma} \\ &\quad \times \left(m^{1/2} \frac{|S|}{|T|} \frac{(\#\mathcal{S})^{1/2}}{(\#\mathbb{T})^{1/2}} D^{-1} \right)^\sigma \\ &\gtrsim_\delta \kappa \delta^{\varepsilon/2+2\eta} b^\omega m^{-1} (\#\mathbb{T}) |T| \left(m^{-1} \ell (\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma} \left(\frac{|S|}{|T|} \right)^{\sigma/2}. \quad \square \end{aligned}$$

Often, we will use the following weaker version of Lemma 14.4.46.

Corollary 14.4.48. *Let $\omega > 0, \sigma \in (0, 2/3]$ and suppose $\mathcal{E}(\sigma, \omega)$ is true. For all $\varepsilon > 0$, there exist $\eta, c > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes and their associated shading. Let $\rho \geq \delta$ and suppose*

that for each $T \in \mathbb{T}$ there exists a ρ tube $T_\rho \supset T$ with $|T_\rho \cap \bigcup_{T \in \mathbb{T}} Y(T)| \geq \delta^\eta |T_\rho|$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^\varepsilon \rho^\omega m^{-1} (\#\mathbb{T}) |T| (m^{-1} \ell (\#\mathbb{T}) |T|^{1/2})^{-\sigma} \left(\frac{\rho}{\delta} \right)^{\sigma/2}, \quad (14.4.83)$$

where $m = C_{KT-CW}(\mathbb{T})$ and $\ell = C_{F-SW}(\mathbb{T})$.

14.4.5 Assertions $\mathcal{D}$ and $\mathcal{E}$ are equivalent

Our goal in this section is to prove Proposition 14.4.6. To do so, we will need a result from [26], which (informally) says that if a set of δ -tubes satisfies the Frostman Convex Wolff Axioms at many different scales, then the union of these tubes must have large volume. To state the result precisely, we recall Definition 2.12 from [26].

Definition 14.4.49. Let $K \geq 1, \delta > 0$. We say a set $\mathbb{T}$ of δ -tubes in $\mathbb{R}^3$ satisfies the *Frostman Convex Wolff Axioms at every scale with error K* if the tubes in $\mathbb{T}$ are essentially distinct, and for every $\rho_0 \in [\delta, 1]$, there exists $\rho \in [\rho_0, K\rho_0]$ and a set of ρ -tubes $\mathbb{T}_\rho$ that satisfies the following properties.

- (i) $\mathbb{T}_\rho$ is a K -balanced partitioning cover of $\mathbb{T}$.
- (ii) For each $T_\rho \in \mathbb{T}_\rho$, $\mathbb{T}^{T_\rho}$ satisfies the Frostman Convex Wolff Axioms with error K .

Next we recall Theorem 5.2 from [26].

Theorem 14.4.50. For all $\varepsilon > 0$, there exists $\eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $\mathbb{T}$ be a set of δ -tubes that satisfy the Frostman Convex Wolff Axioms at every scale with error $\delta^{-\eta}$, and let $Y(T)$ be a δ^η dense shading. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^\varepsilon. \quad (14.4.84)$$

Proof. This is not left as an external citation in this edition. Apply the fully proved all-scale convex–Wolff theorem Theorem 14.5.38, printed later in this volume. Its “Convex Wolff Axioms” are the Frostman normalization

$$\#\{T \in \mathbb{T} : T \subset W\} \leq C|W| \#\mathbb{T},$$

which is exactly the normalization in Definition 14.4.49. Its uniform partitioning cover is the balanced partitioning cover used here. The two unit-rescaling conventions differ only by a fixed ellipsoid, so changing between them multiplies the error by an absolute constant; after decreasing η , that constant is absorbed by $\delta^{-\eta}$.

Given the present ε , invoke Theorem 14.5.38 with $\varepsilon/2$. It supplies $\eta_0, \delta_0 > 0$ and gives $|\bigcup_T Y(T)| \geq \delta^{\varepsilon/2} \geq \delta^\varepsilon$ whenever $0 < \delta \leq \delta_0$. Take the present η smaller than $\eta_0/2$ to absorb the fixed convention constants. If $\delta \in [\delta_0, 1]$, the average-density hypothesis gives a tube T with $|Y(T)| \geq \delta^\eta |T|$. Since $|T| \geq c\delta_0^2$, we have $|\bigcup_T Y(T)| \geq c\delta_0^{2+\eta}$. Thus the desired conclusion holds for all $\delta > 0$ after taking $\kappa = \min(1, c\delta_0^{2+\eta})$. $\square$

When $\varepsilon < \omega$, the conclusion of Theorem 14.4.50 gives a volume estimate that is superior to the volume estimate (14.4.3) from Assertion $\mathcal{E}(\sigma, \omega)$. However, the hypotheses of Assertion $\mathcal{E}(\sigma, \omega)$ are weaker. The next result says that for every collection $\mathbb{T}$ of tubes that satisfies the Frostman Convex Wolff axioms, at least one of the following must occur:

- (A) $\mathbb{T}$ satisfies the hypothesis of Theorem 14.4.50. This is good, provided we select $\varepsilon < \omega$.
- (B) At a suitable scale, $\mathbb{T}$ satisfies the Katz-Tao Convex Wolff axioms with error roughly 1. This is good, if we know that $\mathcal{D}(\sigma, \omega_1)$ is true for some $\omega_1 < \omega$.
- (C) At a suitable scale, $\mathbb{T}$ is factored by flat rectangular prisms, and thus satisfies the hypotheses of Proposition 14.4.30. This is good, since it gives a stronger volume estimate than $\mathcal{E}(\sigma, \omega)$.

The precise statement is as follows.

Proposition 14.4.51. *Let $\zeta_1 \geq \zeta_2 \geq \zeta_3 > 0$. Then there exists $\eta > 0$ such that the following holds for all $\delta > 0$. Let $\mathbb{T}$ be a set of δ -tubes satisfying the Frostman Convex Wolff Axioms with error $\leq \delta^{-\eta}$. Then after replacing $\mathbb{T}$ by a $\approx_\delta 1$ refinement, at least one of the following is true.*

- (A) *$\mathbb{T}$ satisfies the Frostman Convex Wolff Axioms at every scale with error $\delta^{-\zeta_1}$, in the sense of Definition 14.4.49.*
- (B) *There exists $\delta \leq \tau < \rho \leq 1$, with $\tau \leq \delta^{\zeta_1/5} \rho$; a balanced partitioning cover $\mathbb{T}_\tau$ of $\mathbb{T}$; and a balanced partitioning cover $\mathbb{T}_\rho$ of $\mathbb{T}_\tau$ such that the following is true:*
 - (i) $C_{F-CW}(\mathbb{T}_{\tau}) \lesssim \delta^{-\zeta_2}$ for each $T_\tau \in \mathbb{T}_\tau$.
 - (ii) $C_{KT-CW}(\mathbb{T}_\tau^{T_\rho}) \lesssim \delta^{-\zeta_2}$ and $\#\mathbb{T}_\tau^{T_\rho} \geq \delta^{\zeta_2} (\rho/\tau)^2$ for each $T_\rho \in \mathbb{T}_\rho$.
 - (iii) $C_{F-CW}(\mathbb{T}_\rho) \lesssim_\delta \delta^{-\eta}$.
- (C) *There exists $\delta \leq a < b \leq 1$ with $a \leq \delta^{\zeta_2/100} b$, and a set $\mathcal{W}$ of $a \times b \times 1$ prisms that satisfies the hypotheses of Proposition 14.4.30(B): $\mathcal{W}$ factors $\mathbb{T}$ above and below with respect to the Frostman Convex*

Wolff Axioms, with error $O(\delta^{-\zeta_3})$. And for each $W \in \mathcal{W}$, we have $C_{KT-CW}(\mathcal{W}[N_b(W)]) \lesssim \delta^{-\zeta_3}$.

We will defer the proof of Proposition 14.4.51 to Section 14.4.51.

Using Proposition 14.4.51, we will prove the following weaker form of Proposition 14.4.6; recall that $\tilde{\mathcal{E}}$ is defined in Definition 14.4.32.

Lemma 14.4.52. *Let $0 < \sigma \leq 2/3$. For all $\omega, t > 0$, there exists $\alpha > 0$ so that the following holds for all $\omega' \geq \omega + t$. Suppose $\mathcal{D}(\sigma, \omega)$ and $\mathcal{E}(\sigma, \omega')$ are true. Then $\tilde{\mathcal{E}}(\sigma, \omega' - \alpha)$ is true.*

Proof. Let $\alpha = \alpha(\sigma, \omega, t) > 0$ be a small number to be specified below. Let $\eta, \kappa > 0$ be small numbers that depend on ω, ω' , and σ . Our goal is to prove that if $(\mathbb{T}, Y)_\delta$ is δ^η dense with $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$, then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega' - \alpha} m^{-1} (\#\mathbb{T}) |T| \left(m^{-3/2} (\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma}, \quad (14.4.85)$$

with $m = C_{KT-CW}(\mathbb{T})$. Note that the estimate (14.4.85) is slightly stronger than the desired estimate $\tilde{\mathcal{E}}(\sigma, \omega' - \alpha)$, since there is no additional δ^ε loss; this stronger estimate is possible since $\omega' - \omega > 0$, which gives us a bit of “wiggle room.”

Step 1. Without loss of generality we may suppose that $|Y(T)| \geq \delta^\eta |T|$ for each $T \in \mathbb{T}$. Let $\zeta_i = \zeta_i(\sigma, \omega, t)$, $i = 1, \dots, 5$ be small quantities to be chosen below; we will have ζ_{i+1} very small compared to ζ_i , and η very small compared to ζ_5 .

We first consider the case where there exists a subset $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \delta^\eta (\#\mathbb{T})$, and $C_{F-CW}(\mathbb{T}') \leq \delta^{-\zeta_4}$ (this assumption will remain until Step 4, where we will consider the case where no such subset exists). Abusing notation slightly, we will continue to use $\mathbb{T}$ to refer to this subset. In particular, we have that

$$\#\mathbb{T} \gtrsim_\delta \delta^{\eta + \zeta_4} m |T|^{-1}, \quad \text{and} \quad C_{F-CW}(\mathbb{T}) \leq \delta^{-\zeta_4}. \quad (14.4.86)$$

If ζ_4 and η are chosen sufficiently small compared to $\zeta_1, \zeta_2, \zeta_3$, then we can apply Proposition 14.4.51 to $(\mathbb{T}, Y)_\delta$, with $\zeta_1, \zeta_2, \zeta_3$ as specified above. We will select $\zeta_1 = \zeta_1(\omega)$ and η sufficiently small so that if Conclusion (A) of Proposition 14.4.51 holds, then we can use Theorem 14.4.50 to conclude that

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \delta^\omega,$$

and hence (14.4.85) holds provided we choose $\alpha(\omega, t) \leq t$. Henceforth we shall assume that Conclusion (A) of Proposition 14.4.51 does *not* hold.

Step 2. Suppose that Conclusion (B) of Proposition 14.4.51 holds. We will define shadings Y_τ and Y_ρ on the sets of tubes $\mathbb{T}_\tau$ and $\mathbb{T}_\rho$ as follows. For each $T_\tau \in \mathbb{T}_\tau$, we refine the shading on $\mathbb{T}[T_\tau]$ to have average density on balls of radius τ (see (14.4.66) and the surrounding discussion). We define the shading $Y_\tau(T_\tau)$ to be the union of those τ -balls that intersect $\bigcup_{\mathbb{T}[T_\tau]} Y(T)$. We perform the analogous procedure to define Y_ρ (this induces a refinement on the shadings Y_τ and Y).

After these steps have been performed, $(\mathbb{T}_\rho, Y_\rho)_\rho$ is $\delta^{O(\eta)}$ dense; each pair $(\mathbb{T}_\tau^{T_\rho}, Y_\tau^{T_\rho})_{\tau/\rho}$ is $\delta^{O(\eta)}$ dense; and each pair $(\mathbb{T}^{T_\tau}, Y^{T_\tau})_{\delta/\tau}$ is $\delta^{O(\eta)}$ dense. Furthermore, we have Put

$$\begin{aligned} V_\rho &:= \left| \bigcup_{T_\rho \in \mathbb{T}_\rho} Y_\rho(T_\rho) \right|, \\ V_{\tau|\rho} &:= \inf_{T_\rho \in \mathbb{T}_\rho} \left| \bigcup_{T_\tau^{T_\rho} \in \mathbb{T}_\tau^{T_\rho}} Y_\tau^{T_\rho}(T_\tau^{T_\rho}) \right|, \\ V_{\delta|\tau} &:= \inf_{T_\tau \in \mathbb{T}_\tau} \left| \bigcup_{T^{T_\tau} \in \mathbb{T}^{T_\tau}} Y^{T_\tau}(T^{T_\tau}) \right|. \end{aligned}$$

Then the density construction and Fubini give

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq V_\rho V_{\tau|\rho} V_{\delta|\tau}. \quad (14.4.87)$$

Our next task is to estimate the three terms on the RHS of (14.4.87) as follows.

$$\left| \bigcup_{T_\rho \in \mathbb{T}_\rho} Y_\rho(T_\rho) \right| \geq \delta^{\varepsilon_1} \rho^{\omega'} ((\#\mathbb{T}_\rho)^{1/2} |T_\rho|)^\sigma, \quad (14.4.88)$$

$$\inf_{T_\rho \in \mathbb{T}_\rho} \left| \bigcup_{T_\tau^{T_\rho} \in \mathbb{T}_\tau^{T_\rho}} Y_\tau^{T_\rho}(T_\tau^{T_\rho}) \right| \geq \delta^{\varepsilon_1} \left(\frac{\tau}{\rho} \right)^\omega \left(\left(\frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_\rho} \right)^{1/2} \frac{|T_\tau|}{|T_\rho|} \right)^\sigma, \quad (14.4.89)$$

$$\inf_{T_\tau \in \mathbb{T}_\tau} \left| \bigcup_{T^{T_\tau} \in \mathbb{T}^{T_\tau}} Y^{T_\tau}(T^{T_\tau}) \right| \geq \delta^{\varepsilon_1} \left(\frac{\delta}{\tau} \right)^{\omega'} \left(\left(\frac{\#\mathbb{T}}{\#\mathbb{T}_\tau} \right)^{1/2} \frac{|T|}{|T_\tau|} \right)^\sigma, \quad (14.4.90)$$

where $\varepsilon_1 = \frac{\zeta_1}{24}(\omega - \omega')$.

First, observe that if we choose η sufficiently small, then (14.4.88) (resp. (14.4.89) or (14.4.90)) immediately holds if $\rho > \delta^{\varepsilon_1/2}$ (resp. $\delta/\tau > \delta^{\varepsilon_1/2}$). This is because the volume of the union (14.4.88) is bounded by the volume of a single tube. In particular, we have the following

- Either (14.4.88) automatically holds, or $(\mathbb{T}_\rho, Y_\rho)_\rho$ is $\rho^{O(\eta/\varepsilon_1)}$ dense and satisfies $C_{F-CW}(\mathbb{T}_\rho) \lesssim \rho^{-2\zeta_2/\varepsilon_1}$.

- Since $\varepsilon_1 \leq \zeta_1/24$ and $\tau \leq \delta^{\zeta_1/5}\rho$, each pair $(\mathbb{T}_\tau^{T_\rho}, Y_\rho^{T_\rho})_{\tau/\rho}$ is $(\tau/\rho)^{O(\eta/\varepsilon_1)}$ dense and satisfies $C_{KT-CW}(\mathbb{T}_\tau^{T_\rho}) \lesssim (\tau/\rho)^{-2\zeta_2/\varepsilon_1}$ and $\#\mathbb{T}_\tau^{T_\rho} \geq (\tau/\rho)^{2\zeta_2/\varepsilon_1}(\rho/\tau)^2$.
- Either (14.4.90) automatically holds, or each pair $(\mathbb{T}^{T_\tau}, Y^{T_\tau})_{\delta/\tau}$ is $(\delta/\tau)^{O(\eta/\varepsilon_1)}$ dense and satisfies $C_{KT-CW}(\mathbb{T}^{T_\tau}) \lesssim (\delta/\tau)^{-2\zeta_2/\varepsilon_1}$.

If we select η and ζ_2 sufficiently small, depending on ω and t (recall that $t \leq \omega - \omega'$) (η and ζ_2 also depend on ε_1 , but ε_1 only depends on $\zeta_1 = \zeta_1(\omega)$, ω and t), then in light of the three bullet points above, the estimates (14.4.88), (14.4.89), and (14.4.90) follow from $\mathcal{E}(\sigma, \omega')$, $\mathcal{D}(\sigma, \omega)$, and $\mathcal{E}(\sigma, \omega')$, respectively.

Combining (14.4.87), (14.4.88), (14.4.89), and (14.4.90), we conclude that

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^{3\varepsilon_1} \left(\frac{\rho}{\tau} \right)^{\omega - \omega'} \delta^{\omega'} \left((\#\mathbb{T})^{1/2} |T| \right)^\sigma \geq \delta^{\frac{\zeta_1(\omega - \omega')}{8}} \delta^{\omega'} \left((\#\mathbb{T})^{1/2} |T| \right)^\sigma. \quad (14.4.91)$$

Combining (14.4.86) and (14.4.91), we verify that (14.4.85) holds, provided $\alpha(\omega, t) \leq \frac{\zeta_1 t}{16}$; $\zeta_4 < \frac{\zeta_1 t}{17}$; and $\eta > 0$ is chosen sufficiently small. Henceforth we shall assume that Conclusion (B) of Proposition 14.4.51 does *not* hold.

Step 3. Suppose that Conclusion (C) of Proposition 14.4.51 holds, i.e. there is a set $\mathcal{W}$ of $a \times b \times 1$ prisms, with $a \leq \delta^{\zeta_2/100}b$, so that $\mathcal{W}$ satisfies the hypotheses of Proposition 14.4.30(B): $\mathcal{W}$ factors $\mathbb{T}$ above and below with respect to the Frostman Convex Wolff Axioms with error $O(\delta^{-\zeta_3})$. And for each $W \in \mathcal{W}$ we have $C_{KT-CW}(\mathcal{W}[N_b(W)]) \lesssim \delta^{-\zeta_3}$.

Let $\varepsilon_2 = \zeta_2\omega/200$. If ζ_3 is selected sufficiently small depending on ζ_2 and ε_2 (both of these numbers in turn ultimately only depend on ω and t) and if $\eta > 0$ is selected sufficiently small, then by Proposition 14.4.30(B) (with ε_2 in place of ε) we have

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\gtrsim \delta^{\omega' + \varepsilon_2} \left(\frac{b}{a} \right)^{\omega'} \left((\#\mathbb{T})^{1/2} |T| \right)^\sigma \\ &\geq \delta^{\omega' + \varepsilon_2} \delta^{-\frac{\zeta_2\omega}{100}} \left((\#\mathbb{T})^{1/2} |T| \right)^\sigma \\ &\geq \delta^{\omega' - \frac{\zeta_2\omega}{200}} \left((\#\mathbb{T})^{1/2} |T| \right)^\sigma. \end{aligned} \quad (14.4.92)$$

Combining (14.4.86) and (14.4.92), we verify that (14.4.85) holds, provided $\alpha(\omega, t) \leq \frac{\zeta_2\omega}{400}$; $\zeta_4 < \frac{\zeta_2\omega}{500}$; and $\eta > 0$ is chosen sufficiently small.

Step 4. It remains to consider the case where every subset $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \delta^\eta(\#\mathbb{T})$ satisfies $C_{F-CW}(\mathbb{T}') > \delta^{-\zeta_4}$. Apply Proposition 14.4.23 (factoring a collection of convex sets) to $\mathbb{T}$, and denote the output by $\mathbb{T}'$ and $\mathcal{W}$. Then Item ii) of Proposition 14.4.23 implies $\#\mathbb{T}[W] \approx_\delta C_{KT-CW}(\mathbb{T}') \frac{|W|}{|\mathbb{T}|}$. Since $\#\mathbb{T}' \geq \delta^\eta(\#\mathbb{T})$, we have $C_{F-CW}(\mathbb{T}') > \delta^{-\zeta_4}$. We also have $C_{F-CW}(\mathbb{T}'^W) \lesssim_\delta 1$

for each $W \in \mathcal{W}$, from which it follows that $|W| \lesssim_\delta \delta^{\zeta_4}$ (recall that the sets $W \in \mathcal{W}$ are congruent, and thus they all have identical volume).

If the prisms in $\mathcal{W}$ are flat, in the sense that they are comparable to $a \times b \times 1$ prisms with $a \leq \delta^{\zeta_5} b$, then we can apply Proposition 14.4.30(A) with $\varepsilon_3 = \zeta_5 \omega' / 2$ to conclude that

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\gtrsim \delta^{\omega' + \varepsilon_3} \left(\frac{b}{a} \right)^{\omega'} (m')^{-1} (\#\mathbb{T}') |T| ((m')^{-3/2} (\ell') (\#\mathbb{T}) |T|^{1/2})^{-\sigma} \\ &\gtrsim \delta^{\omega' - \frac{\zeta_5 \omega'}{2} + 2\eta} m^{-1} (\#\mathbb{T}) |T| (m^{-3/2} (\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \end{aligned} \quad (14.4.93)$$

In the second inequality, we used the fact that $\#\mathbb{T}' \geq \delta^\eta \#\mathbb{T}$; $\ell' := C_{F-SW}(\mathbb{T}') \lesssim \delta^{-\eta}$; $m' := C_{KT-CW}(\mathbb{T}') \leq m$; and $\sigma \leq 2/3$. We conclude that (14.4.85) holds, provided $\alpha(\omega, t) \leq \frac{\zeta_5 \omega}{4} \leq \frac{\zeta_5 \omega'}{4}$ and $\eta > 0$ is chosen sufficiently small.

Finally, we consider the case where the prisms in $\mathcal{W}$ are not flat, in the sense that $a \geq \delta^{\zeta_5} b$. In this case we can replace each prism $W \in \mathcal{W}$ by its b -neighbourhood, and then refine the corresponding set of b -tubes $\mathbb{T}_b$ by a factor of $(b/a)^3 \leq \delta^{-3\zeta_5}$ (this is the number of essentially distinct $a \times b \times 1$ prisms that can fit inside a b tube) so that the tubes in $\mathbb{T}_b$ are essentially distinct. Since $|W| \lesssim_\delta \delta^{\zeta_4}$ and $a \geq \delta^{\zeta_5} b$, we have

$$b \lesssim_\delta \delta^{\zeta_4/2 - \zeta_5} \leq \delta^{\zeta_4/3}. \quad (14.4.94)$$

To recap, the set $\mathbb{T}_b$ has the following properties.

- $C_{KT-CW}(\mathbb{T}_b) \lesssim \frac{b}{a} C_{KT-CW}(\mathcal{W}) \lesssim_\delta \delta^{-\zeta_5} \leq b^{-3\zeta_5/\zeta_4}$.
- $C_{F-SW}(\mathbb{T}_b) \lesssim \delta^{-3\zeta_5} C_{F-SW}(\mathcal{W}) \lesssim_\delta \delta^{-3\zeta_5} C_{F-SW}(\mathbb{T}) \lesssim_\delta \delta^{-3\zeta_5 - \eta}$.
- For each $T_b \in \mathbb{T}_b$, we have $C_{F-CW}(\mathbb{T}^{T_b}) \lesssim \frac{b}{a} C_{F-CW}(\mathbb{T}^W) \lesssim_\delta \delta^{-\zeta_5}$, where $\mathcal{W} \ni W \subset T_b$ is the prism containing all of the tubes in $\mathbb{T}[T_b]$.
- For each $T_b \in \mathbb{T}_b$, we have $\#\mathbb{T}^{T_b} \lesssim_\delta C_{KT-CW}(\mathcal{W}) \frac{b}{a} \#\mathbb{T}^W \lesssim_\delta \frac{b}{a} m \frac{|W|}{|T|} \lesssim_\delta \delta^{-2\zeta_5} m \frac{|T_b|}{|T|}$.

(For the second item above, we make crucial use of the fact that $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$, which is why the conclusion of Lemma 14.4.52 only says that $\tilde{\mathcal{E}}(\sigma, \omega' - \alpha)$ is true, rather than the superficially stronger statement $\mathcal{E}(\sigma, \omega' - \alpha)$).

Mirroring the argument in Step 2, refine the shadings $Y(T)$ on each set of tubes $\mathbb{T}[T_b]$ to have average density on balls of radius b . We define the shading $Y_b(T_b)$ to be the union of those b -balls that intersect $\bigcup_{T \in \mathbb{T}[T_b]} Y(T)$. Then $(\mathbb{T}_b, Y_b)_b$ is $\delta^\eta \geq b^{3\eta/\zeta_4}$ dense. We thus have the following analogue of (14.4.87):

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \left(\left| \bigcup_{T_b \in \mathbb{T}_b} Y_b(T_b) \right| \right) \left(\inf_{T_b \in \mathbb{T}_b} \left| \bigcup_{T^{T_b} \in \mathbb{T}^{T_b}} Y^{T_b}(T^{T_b}) \right| \right). \quad (14.4.95)$$

Let $\varepsilon_4 = \frac{\zeta_4(\omega - \omega')}{12}$. If ζ_5 and η are selected sufficiently small depending on ζ_4 and ε_4 (which in turn depend on ω and t), then we can use the estimate (14.4.2) from Assertion $\mathcal{D}(\sigma, \omega)$ to conclude that

$$\left| \bigcup_{T_b \in \mathbb{T}_b} Y_\rho(T_b) \right| \gtrsim b^{\omega + \varepsilon_4} (\#\mathbb{T}_b) |T_b| ((\#\mathbb{T}_b) |T_b|^{1/2})^{-\sigma}. \quad (14.4.96)$$

Finally, we would like to obtain the estimate

$$\begin{aligned} \left| \bigcup_{T^{T_b} \in \mathbb{T}^{T_b}} Y^{T_b}(T^{T_b}) \right| &\gtrsim \delta^{\varepsilon_4} \left(\frac{\delta}{b} \right)^{\omega'} \\ &\times \left((\#\mathbb{T}^{T_b})^{1/2} \frac{|T|}{|T_b|} \right)^\sigma. \end{aligned} \quad (14.4.97)$$

When $\delta/b > \delta^{\varepsilon_4/2}$, we have that (14.4.97) follows from the elementary fact that the shading on each $Y(T)$ is $\gtrsim \delta^\eta$ dense, and the union on the LHS of (14.4.97) is bounded by the volume of a single tube. On the other hand, when $\delta/b \leq \delta^{\varepsilon_4/2}$, we have that (14.4.97) follows from the estimate (14.4.3) from Assertion $\mathcal{E}(\sigma, \omega')$, provided ζ_5 is chosen sufficiently small depending on ζ_4 and ε_4 , since $C_{F-CW}(\mathbb{T}^{T_b}) \lesssim \delta^{-\zeta_5}$.

Since $\#\mathbb{T}^{T_b} \lesssim m \delta^{-2\zeta_5} \frac{|T|}{|T_b|}$ for each $T_b \in \mathbb{T}_b$ (where $m = C_{KT-CW}(\mathbb{T})$), (14.4.97) becomes

$$\begin{aligned} \left| \bigcup_{T^{T_b} \in \mathbb{T}^{T_b}} Y^{T_b}(T^{T_b}) \right| &\gtrsim \delta^{\varepsilon_4 + 2\zeta_5} \left(\frac{\delta}{b} \right)^{\omega'} m^{-1} (\#\mathbb{T}^{T_b}) \frac{|T|}{|T_b|} \\ &\times \left(m^{-3/2} (\#\mathbb{T}^{T_b}) \left(\frac{|T|}{|T_b|} \right)^{1/2} \right)^{-\sigma}. \end{aligned} \quad (14.4.98)$$

Combining (14.4.95), (14.4.96), (14.4.98), and (14.4.94), we conclude that if we select $\zeta_5 \leq \zeta_4(\omega - \omega')/12$, then

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\geq \delta^{\omega' + \varepsilon_4 + 2\zeta_5} b^{\omega - \omega' + \varepsilon_4} m^{-1} (\#\mathbb{T}) |T| \left(m^{-3/2} (\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma} \\ &\geq \delta^{\omega' - \frac{\zeta_4(\omega - \omega')}{12}} m^{-1} (\#\mathbb{T}) |T| \left(m^{-3/2} (\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma}. \end{aligned} \quad (14.4.99)$$

We conclude that (14.4.85) holds, provided $\alpha(\omega, t) \leq \frac{\zeta_4 t}{12}$, and $\eta > 0$ is chosen sufficiently small. $\square$

We now use Lemma 14.4.52 to prove Proposition 14.4.6.

Proof of Proposition 14.4.6. Let $0 \leq \sigma \leq 2/3$, $\omega \geq 0$, and suppose that Assertion $\mathcal{D}(\sigma, \omega)$ is true. Fix $t > 0$ and let $\alpha = \alpha(\sigma, \omega, t) > 0$ be the output of Lemma 14.4.52. Since a δ -tube has volume $\sim \delta^2$, we always have that $\mathcal{E}(\sigma, 2)$ is true. Now suppose that $\mathcal{E}(\sigma, \omega')$ is true for some $\omega' \in [\omega + t, 2]$. Applying Lemma 14.4.52 followed by Proposition 14.4.43, we have

$$\mathcal{E}(\sigma, \omega') \implies \tilde{\mathcal{E}}(\sigma, \omega' - \alpha) \implies \mathcal{E}(\sigma, \omega' - \alpha).$$

Iterating the above argument, we conclude that $\mathcal{E}(\sigma, \omega'')$ is true for some $\omega'' \leq \omega + t$, so in particular $\mathcal{E}(\sigma, \omega + t)$ is true.

However, $t > 0$ was arbitrary, and by the definition of $\mathcal{E}$, it is clear that the set

$$\{\omega' \in [\omega, 2] : \mathcal{E}(\sigma, \omega') \text{ is true}\}$$

is a closed interval. We conclude that $\mathcal{E}(\sigma, \omega)$ is true. $\square$

This concludes the proof of Proposition 14.4.6, except that we must still prove Proposition 14.4.51. We do this below.

14.4.5.1 Proof of Proposition 14.4.51: A factoring trichotomy

Step 1. We begin by regularizing the set $\mathbb{T}$. Let $\eta > 0$ be a small quantity to be determined below, with $N = 1/\eta$ an integer. Define $\delta_i = \delta^{i/N}$, $i = 1, \dots, N$. By iterated pigeonholing and replacing $\mathbb{T}$ by a $|\log \delta|^{-N}$ -refinement, we may suppose that for each $i = 1, \dots, N$ there exists a set $\mathbb{T}_{\delta_i}$ of δ_i -tubes that is a balanced partitioning cover of $\mathbb{T}_{\delta_{i+1}}$. We will call numbers of the form δ_i “admissible scales.” In particular, for each admissible scale δ_i , we have

$$\#\mathbb{T}[T_{\delta_i}] \approx_{\delta} \frac{\#\mathbb{T}}{\#\mathbb{T}_{\delta_i}} \quad \text{for every } T_{\delta_i} \in \mathbb{T}_{\delta_i}. \quad (14.4.100)$$

Next we apply Proposition 14.4.23 to each set $\mathbb{T}^{T_{\delta_i}}$; the output of Proposition 14.4.23 is a ≈ 1 refinement of $\mathbb{T}^{T_{\delta_i}}$ (this induces a ≈ 1 refinement of $\mathbb{T}$ and all sets $\mathbb{T}_{\delta_j}$ for $j > i$; observe that (14.4.100) remains true after this refinement; to simplify notation, we still use $\mathbb{T}_{\delta_i}$ and $\mathbb{T}$ to denote these refinement) and a set $\mathcal{W}$ of convex subsets of $\mathbb{R}^3$. If $|W| \geq \delta^{\zeta_1/2}$, then $C_{F-CW}(\mathbb{T}^{T_{\delta_i}}) \leq \delta^{-\zeta_1/2}$. In this case, we say $\mathbb{T}^{T_{\delta_i}}$ is of *Type 1*. If $|W| < \delta^{\zeta_1/2}$, then we say we say $\mathbb{T}^{T_{\delta_i}}$ is of *Type 2*.

We now proceed as follows. For each $i = N - 1, \dots, 1$, if at least half the sets $\mathbb{T}^{T_{\delta_i}}$, $T_{\delta_i} \in \mathbb{T}_{\delta_i}$ are of Type 1, then we refine $\mathbb{T}_{\delta_i}$ to only consist of those T_{δ_i} for which $\mathbb{T}^{T_{\delta_i}}$ are of Type 1; observe that (14.4.100) remains true after this refinement. We say that $\mathbb{T}$ has passed stage i . On the other hand, if at least half the sets $\mathbb{T}^{T_{\delta_i}}$, $T_{\delta_i} \in \mathbb{T}_{\delta_i}$ are of Type 2, then we say that $\mathbb{T}$ has failed stage i .

Suppose that $\mathbb{T}$ passes every stage $i = N - 1, \dots, 1$. Then by (14.4.100) we have that for each $i = 1, \dots, N$, the set $\mathbb{T}_{\delta_i}$ (this consists of those δ_i -tubes that survived the refinements described above) is a $\approx_\delta 1$ -balanced partitioning cover of $\mathbb{T}$. Furthermore, since $\mathbb{T}$ passed stage i , we have that for each $T_{\delta_i} \in \mathbb{T}_{\delta_i}$ we have that $C_{F-CW}(\mathbb{T}^{T_{\delta_i}}) \lesssim_\delta \delta^{-\zeta_1/2}$. We conclude that $\mathbb{T}$ satisfies the Frostman Convex Wolff Axioms at every scale with error $O(\delta^{-\zeta_1})$, and hence Conclusion (A) of Proposition 14.4.51 holds.

Step 2. Suppose that $\mathbb{T}$ fails some stage $i \geq 1$. After pigeonholing and replacing $\mathbb{T}_{\delta_i}$ and $\mathbb{T}$ by a $\approx_\delta 1$ refinement, we may suppose that there exists $\delta \leq a \leq b \leq 1$ so that for each $T_{\delta_i} \in \mathbb{T}_{\delta_i}$, the output of Proposition 14.4.23 applied to $\mathbb{T}^{T_{\delta_i}}$ consists of a set $\mathcal{W}_{T_{\delta_i}}$ of $\frac{a}{\delta_i} \times \frac{b}{\delta_i} \times 1$ prisms that forms a $\approx_\delta 1$ balanced cover of $\mathbb{T}^{T_{\delta_i}}$, and factors $\mathbb{T}^{T_{\delta_i}}$ from above (resp. below) with respect to the Katz-Tao Convex Wolff Axioms (resp. Frostman Convex Wolff Axioms) with error $\lesssim_\delta 1$.

Since the tubes in $\mathbb{T}_{\delta_i}$ are essentially distinct, we can further refine $\mathbb{T}_{\delta_i}$ by a ~ 1 factor so that every pair of distinct tubes $T_{\delta_i}, T'_{\delta_i} \in \mathbb{T}_{\delta_i}$ that intersect must satisfy $\angle(\text{dir}(T_{\delta_i}), \text{dir}(T'_{\delta_i})) \geq 100\delta_i$, so in particular we have

$$\text{diam}(2T_{\delta_i} \cap 2T'_{\delta_i}) \leq \frac{1}{2}. \quad (14.4.101)$$

Define

$$\mathcal{W} = \bigsqcup_{T_{\delta_i} \in \mathbb{T}_{\delta_i}} \phi_{T_{\delta_i}}^{-1}(\mathcal{W}_{T_{\delta_i}}).$$

Then $\mathcal{W}$ is a collection of convex sets, each of which is comparable to a $a \times b \times 1$ prism. Recall that since $\mathbb{T}$ failed stage i , we have that the prisms in $\mathcal{W}$ are substantially smaller than the tubes in $\mathbb{T}_{\delta_i}$; specifically, we have

$$|W| \lesssim \delta^{\zeta_1/2} |T_{\delta_i}|. \quad (14.4.102)$$

We claim that the convex sets in $\mathcal{W}$ are essentially distinct. To verify this claim, we argue as follows. Every pair of convex sets W, W' from the same set $\mathcal{W}_{T_{\delta_i}}$ are essentially distinct. On the other hand, if $W \in \mathcal{W}_{T_{\delta_i}}$ and $W' \in \mathcal{W}_{T'_{\delta_i}}$ for distinct tubes T_{δ_i} and T'_{δ_i} , then $\text{diam}(W \cap W') \leq \text{diam}(T_{\delta_i} \cap T'_{\delta_i}) \leq \frac{1}{2}$ by (14.4.101), from which it follows that W and W' are essentially distinct.

Since $\mathcal{W}$ is a $\approx_\delta 1$ balanced cover of $\mathbb{T}$, and $C_{F-CW}(\mathbb{T}) \lesssim_\delta \delta^{-\eta}$, by Remark 14.4.20(A) (Frostman Wolff Axioms are inherited upwards), we have $C_{F-CW}(\mathcal{W}) \lesssim_\delta \delta^{-\eta}$; we will select $\eta > 0$ sufficiently small so that $C_{F-CW}(\mathcal{W}) \lesssim \delta^{-\zeta_3}$.

Step 3. Suppose that the prisms in $\mathcal{W}$ are flat, in the sense that $a \leq \delta^{\zeta_2/100}b$. Our task is to show that $\mathcal{W}$ and our refined set $\mathbb{T}$ satisfies the conditions of Conclusion (C) from Proposition 14.4.51.

Each $W \in \mathcal{W}$ came from some set $\mathcal{W}_{T_{\delta_i}}$. We claim that if $W' \in \mathcal{W}$ satisfies $W' \subset N_b(W)$, then we must have that W' came from the same set $\mathcal{W}_{T_{\delta_i}}$, i.e.

$$\mathcal{W}[N_b(W)] = (\mathcal{W}[T_{\delta_i}])[N_b(W)] = \phi_{T_{\delta_i}}^{-1}(\mathcal{W}_{T_{\delta_i}})[N_b(W)]. \quad (14.4.103)$$

To verify this claim, we argue by contradiction: suppose instead that W' came from a distinct tube T'_{δ_i} , then we would have $W' \subset N_b(W) \cap T'_{\delta_i} \subset 2T_{\delta_i} \cap T'_{\delta_i}$, and in particular the latter set would have diameter ≥ 1 . But this is forbidden by (14.4.101).

(14.4.103) implies that

$$C_{KT-CW}(\mathcal{W}[N_b(W)]) = C_{KT-CW}((\mathcal{W}[T_{\delta_i}])[N_b(W)]) \leq C_{KT-CW}(\mathcal{W}_{T_{\delta_i}}) \lesssim_\delta 1.$$

Thus $\mathcal{W}$ and our refined set $\mathbb{T}$ satisfies the conditions of Conclusion (C) from Proposition 14.4.51.

Step 4. Now we consider the case where the prisms in $\mathcal{W}$ are not flat, in the sense that $a > \delta^{\zeta_2/100}b$. Define τ to be the smallest admissible scale greater than or equal to b . Since at most $O((b/a)^3)$ essentially distinct $a \times b \times 1$ prisms can fit inside a b -tube, and at most $O(\delta^{-4/N}) = O(\delta^{-4\eta})$ essentially distinct b -tubes can fit inside a τ -tube, we have that $\mathcal{W}[T_\tau] \lesssim \delta^{-4\eta}(b/a)^3 \lesssim \delta^{-4\zeta_2/100}$ for every τ -tube T_τ (for the last inequality we select $\eta \leq \zeta_2/400$). After pigeonholing $\mathbb{T}$, $\mathcal{W}$, and $\mathbb{T}_\tau$, we may suppose that $\mathbb{T}_\tau$ is a balanced partitioning cover of $\mathcal{W}$. We have

$$1 \leq \#\mathcal{W}[T_\tau] \lesssim \delta^{-4\zeta_2/100} \quad \text{for each } T_\tau \in \mathbb{T}_\tau. \quad (14.4.104)$$

Since $C_{F-CW}(\mathbb{T}[W]) \lesssim_\delta 1$ for each $W \in \mathcal{W}$, we conclude from (14.4.104) that $C_{F-CW}(\mathbb{T}[T_\tau]) \lesssim_\delta \delta^{-4\zeta_2/100}$ for each $T_\tau \in \mathbb{T}_\tau$; this gives us Conclusion (B.i).

At this point, we have correctly identified the scale τ from Conclusion (B) of Proposition 14.4.51. What about the scale ρ ? One candidate is δ_i ; by (14.4.102) we have $\tau \lesssim \delta^{\zeta_1/4-2\zeta_2/100}\delta_i \leq \delta^{\zeta_1/5}\delta_i$, as specified in Conclusion (B).

The scale δ_i satisfies some of the required properties of Conclusion (B). Recall that for each $T_{\delta_i} \in \mathbb{T}_{\delta_i}$, we have $C_{KT-CW}(\mathcal{W}[T_{\delta_i}]) \lesssim_\delta 1$. Since $|T_\tau| = (b/a)|W| \leq \delta^{-\zeta_2/100}|W|$, by (14.4.104) we have

$$C_{KT-CW}(\mathbb{T}_\tau[T_{\delta_i}]) \leq \delta^{-5\zeta_2/100} \quad \text{for each } T_{\delta_i} \in \mathbb{T}_{\delta_i}. \quad (14.4.105)$$

This is half of Conclusion (B.ii). If $\frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_{\delta_i}} \geq \delta^{100\eta}(\delta_i/\tau)^2$ (in fact a weaker estimate $\frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_{\delta_i}} \geq \delta^{\zeta_2}(\delta_i/\tau)^2$ suffices), then after a refinement, $\mathbb{T}_{\delta_i}$ satisfies Conclusion (B.ii). Conclusion (B.iii) then follows from Remark 14.4.20(A) (Frostman Wolff Axioms are inherited upwards), and we are done.

Suppose instead that $\frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_{\delta_i}} < \delta^{100\eta}(\delta_i/\tau)^2$. Let ρ be the minimum of all admissible scales in $[\delta_i, 1]$ for which

$$\frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_\rho} \geq \delta^{100\eta}(\rho/\tau)^2. \quad (14.4.106)$$

Such a choice of $\rho \in [\rho_i, 1]$ must exist, since $C_{F-CW}(\mathbb{T}_\tau) \lesssim_\delta \delta^{-\eta}$, and hence $\#\mathbb{T}_\tau \gtrsim_\delta \delta^\eta \tau^{-2}$, from which it follows that $\rho = 1$ satisfies (14.4.106) and hence $\rho = 1$ is a valid candidate.

In particular, for this choice of ρ we have

$$\delta^{100\eta}(\rho/\tau)^2 \leq \frac{\#\mathbb{T}_\tau}{\#\mathbb{T}_\rho} \leq \delta^{50\eta}(\rho/\tau)^2, \quad (14.4.107)$$

since if the RHS of (14.4.107) failed, then (14.4.106) would hold for a smaller admissible scale, which would contradict the minimality of ρ .

Suppose for the moment that there exists a $\gtrsim_\delta$ 1-refinement of $\mathbb{T}_\tau$ (abusing notation, we will continue to call this set $\mathbb{T}_\tau$) such that

$$C_{F-CW}(\mathbb{T}_\tau^{T_\rho}) \leq \delta^{-\zeta_2/2} \quad \text{for at least half of the tubes } T_\rho \in \mathbb{T}_\rho. \quad (14.4.108)$$

Then (14.4.107) plus (14.4.108) implies that Conclusion (B.ii) holds, and Conclusion (B.iii) follows from Remark 14.4.20(A) (Frostman Wolff axioms are inherited upwards). Thus if (14.4.108) is true, then Conclusion (B) of Proposition 14.4.51 holds, and we are done.

Step 5. We claim that either (14.4.108) holds (and we are done, as discussed in the previous step), or else Conclusion (C) of Proposition 14.4.51 holds.

We will verify this claim as follows. Suppose that (14.4.108) fails. Applying Proposition 14.4.23 to each set $\mathbb{T}_\tau^{T_\rho}$ and then undoing the scaling ϕ_{T_ρ} , we obtain a refinement of $\mathbb{T}[T_\rho]$ (abusing notation, we will continue to call this set $\mathbb{T}[T_\rho]$), and a set $\mathcal{U}_{T_\rho}$ of $s \times t \times 1$ prisms contained in T_ρ (so in particular $t \leq \rho$) that factors $\mathbb{T}[T_\rho]$ from below with respect to the Frostman Convex Wolff axioms with error $\lesssim_\delta 1$ and each $T \in \mathbb{T}[T_\rho]$ is contained in $\lesssim_\delta 1$ sets $U \in \mathcal{U}_{T_\rho}$.

Pigeonhole and refine $\mathbb{T}_\rho$ to consist of those tubes T_ρ for which $C_{F-CW}(\mathbb{T}_\tau^{T_\rho}) > \delta^{-\zeta_2/2}$ (such a refinement exists because (14.4.108) fails) and the corresponding sets $\mathcal{U}_{T_\rho}$ are comparable to $s \times t \times 1$ prisms for a common pair of numbers (s, t) . In particular, this implies

$$|U| \lesssim_\delta \delta^{\zeta_2/2} |T_\rho|. \quad (14.4.109)$$

(Cf. (14.4.102) when $s \geq \tau$. When $s \leq \tau$, this is true because $\tau \lesssim \delta^{\zeta_1/5} \delta_i \leq \delta^{\zeta_1/5} \rho$.)

Suppose for the moment that $s \geq \tau$. Then $C_{F-CW}(\mathbb{T}_\tau^U) \lesssim_\delta 1$ by Remark 14.4.20(A) (Frostman Wolff axioms are inherited upwards). By (14.4.107) and our hypothesis that $C_{F-CW}(\mathbb{T}_\tau^{T_\rho}) > \delta^{-\zeta_2/2}$, we have

$$C_{KT-CW}(\mathbb{T}_\tau[T_\rho]) \geq \delta^{-\zeta_2/2+100\eta}. \quad (14.4.110)$$

We conclude that by Item ii) of Proposition 14.4.23,

$$\#(\mathbb{T}[T_\rho])[U] \gtrsim_\delta C_{KT-CW}(\mathbb{T}[T_\rho]) \left(\frac{|U|}{|T|} \right) \quad \text{for each } U \in \mathcal{U}_{T_\rho}.$$

Since $s \geq \tau$ and τ is an admissible scale, we have

$$C_{KT-CW}(\mathbb{T}[T_\rho]) \gtrsim_\delta \frac{\#\mathbb{T}}{\#\mathbb{T}_\tau} \frac{|T|}{|T_\tau|} C_{KT-CW}(\mathbb{T}_\tau[T_\rho]),$$

and so

$$\begin{aligned} \#(\mathbb{T}_\tau[T_\rho])[U] &\gtrsim_\delta C_{KT-CW}(\mathbb{T}_\tau[T_\rho]) \frac{|U|}{|T_\tau|} \\ &\geq \delta^{-\zeta_2/2+100\eta} \frac{|U|}{|T_\tau|}, \quad U \in \mathcal{U}_{T_\rho}. \end{aligned} \quad (14.4.111)$$

We will show that

$$s < \delta^{\zeta_2/100} t. \quad (14.4.112)$$

To verify (14.4.112), suppose to the contrary that $s \geq \delta^{\zeta_2/100} t$; we will obtain a contradiction. If $s \geq \delta^{\zeta_2/100} t$ then $t \leq \delta^{\zeta_2/4} \rho$. By (14.4.111) and the fact that each tube in $\mathbb{T}_\tau[T_\rho]$ is contained in $\lesssim_\delta 1$ sets $U \in \mathcal{U}_{T_\rho}$, we have

$$\begin{aligned} \#\mathcal{U}_{T_\rho} &\lesssim_\delta \frac{\#\mathbb{T}_\tau[T_\rho]}{\inf_{U \in \mathcal{U}_{T_\rho}} \#(\mathbb{T}_\tau[T_\rho])[U]} \\ &\lesssim_\delta \delta^{\zeta_2/2-100\eta} (\#\mathbb{T}_\tau[T_\rho]) \frac{|T_\tau|}{|U|} \\ &\leq \delta^{48\zeta_2/100} (\#\mathbb{T}_\tau[T_\rho]) \left(\frac{\tau}{t} \right)^2. \end{aligned}$$

and thus if t' is the smallest admissible scale greater than or equal to t , then

$$\begin{aligned} \#\mathbb{T}_{t'} &\leq \# \left(\bigcup_{T_\rho \in \mathbb{T}_\rho} \mathcal{U}_{T_\rho} \right) \\ &\lesssim_\delta \delta^{48\zeta_2/100} (\#\mathbb{T}_\tau) \left(\frac{\tau}{t} \right)^2 \\ &\leq \delta^{47\zeta_2/100} (\#\mathbb{T}_\tau) \left(\frac{\tau}{t'} \right)^2. \end{aligned} \quad (14.4.113)$$

From (14.4.113) we see that $t' \geq \delta_i$ — if not, then there would exist a t' -tube $T_{t'}$ with $\#\mathbb{T}_\tau[T_{t'}] \gtrsim_\delta \delta^{-\frac{47}{100}\zeta_2} \frac{|T_{t'}|}{|T_\tau|}$, and this t' -tube would be contained in some δ_i -tube; but this violates (14.4.105).

Comparing (14.4.113) and (14.4.106), we see that $t' \in [t, \delta^{\zeta_2/5}\rho] \subset [\delta_i, \delta^{\zeta_2/5}\rho]$ is an admissible scale that satisfies (14.4.107). But this contradicts the assumption that ρ was the minimal such scale. We conclude that (14.4.112) must hold.

When (14.4.112) holds, we verify Conclusion (C) directly. Put $\mathcal{U} = \bigcup_{T_\rho \in \mathbb{T}_\rho} \mathcal{U}_{T_\rho}$. After splitting the essentially distinct ρ -tubes into a bounded number of classes, distinct intersecting parents in a class make angle at least 100ρ ; consequently

$$\text{diam}(2T_\rho \cap 2T'_\rho) \leq \frac{1}{2} \quad (T_\rho \neq T'_\rho). \quad (14.4.114)$$

The child families $\mathcal{U}_{T_\rho}$ are essentially distinct by the factorization proposition. If children from different parents were comparable, their intersection would have diameter comparable to one, in contradiction with (14.4.114); hence $\mathcal{U}$ is essentially distinct.

Each $\mathcal{U}_{T_\rho}$ is a balanced, almost-partitioning cover of $\mathbb{T}[T_\rho]$, with both its upper Katz–Tao and lower Frostman constants $\lesssim_\delta 1$. The parent cover $\mathbb{T}_\rho$ is balanced, so after the standard dyadic degree refinement their union is a balanced cover of $\mathbb{T}$ and loses only an $\approx_\delta 1$ fraction of the incidences. Frostman inheritance upward gives

$$C_{F-CW}(\mathcal{U}) \lesssim_\delta C_{F-CW}(\mathbb{T}) \leq \delta^{-\eta}, \quad \sup_{U \in \mathcal{U}} C_{F-CW}(\mathbb{T}^U) \lesssim_\delta 1.$$

Taking η smaller than $\zeta_3/4$ makes both bounds at most $\delta^{-\zeta_3}$.

It remains to check the local Katz–Tao condition in Conclusion (C). Fix $U \in \mathcal{U}_{T_\rho}$. If $U' \in \mathcal{U}$ and $U' \subset N_t(U)$, then $N_t(U)$ has longitudinal diameter comparable to one. Thus the parent of U' meets $2T_\rho$ in a set of diameter comparable to one; by (14.4.114), that parent must be T_ρ . Therefore

$$\begin{aligned} \mathcal{U}[N_t(U)] &= (\mathcal{U}_{T_\rho})[N_t(U)], \\ C_{KT-CW}(\mathcal{U}[N_t(U)]) &\leq C_{KT-CW}(\mathcal{U}_{T_\rho}) \lesssim_\delta 1 \leq \delta^{-\zeta_3}. \end{aligned}$$

Finally (14.4.112) says $s < \delta^{\zeta_2/100}t$. Thus, with $(a, b) = (s, t)$, every clause of Conclusion (C) is verified.

Now suppose (14.4.112) does not hold, so $\delta^{\zeta_2/100}t \leq s < \tau$. If $s \geq \tau\delta^{\zeta_2/20}$, then together with $s \geq \delta^{\zeta_2/100}t$,

$$C_{KT-CW}(\mathbb{T}_\tau[T_\rho]) \leq \delta^{-\zeta_2/4} C_{KT-CW}(\mathcal{U}_{T_\rho}) \lesssim_\delta \delta^{-\zeta_2/4},$$

which is a contradiction to (14.4.110). Hence the only remaining branch has

$$\delta^{\zeta_2/100}t \leq s \leq \delta^{\zeta_2/20}\tau, \quad \frac{s}{\rho} \leq \delta^{\zeta_2/20}\frac{\tau}{\delta_i}. \quad (14.4.115)$$

We now make the iteration explicit. At generation k , record the outer parent scale r_k , the inner covering-tube scale q_k , and the child prism dimensions $a_k \times b_k \times 1$. For the first generation these are $r_0 = \delta_i$, $q_0 = \tau$, and $(a_0, b_0) = (a, b)$. In the remaining branch above set

$$r_{k+1} = \rho, \quad q_{k+1} = s, \quad (a_{k+1}, b_{k+1}) = (s, t), \quad \mathcal{W}_{k+1} = \mathcal{U}.$$

All hypotheses used at the beginning of Step 4 are inherited: the parent and child covers are balanced after one dyadic degree refinement, the lower Frostman constant is inherited from the factorization at the new parent, and the upper Katz–Tao constant is supplied by that same factorization. The first alternative in the repeated Step 4 gives Conclusion (B); the flat alternative gives Conclusion (C) by the calculation immediately above. If neither occurs, (14.4.115) gives the strict invariant

$$\frac{q_{k+1}}{r_{k+1}} \leq \delta^{\zeta_2/20} \frac{q_k}{r_k}. \quad (14.4.116)$$

Every scale belongs to $[\delta, 1]$, so $q_k/r_k \geq \delta$. Since $q_0/r_0 \leq 1$, iterating (14.4.116) more than $\lceil 20/\zeta_2 \rceil$ times would give $q_k/r_k < \delta$, a contradiction. Thus one of Conclusions (B) or (C) occurs after at most that many generations. Each generation costs an $\approx_\delta 1$ refinement; the number of generations depends only on ζ_2 , so their product is still an $\approx_\delta 1$ refinement. This completes the proof of Proposition 14.4.51.

14.4.6 A two-scale grains decomposition for tubes in $\mathbb{R}^3$

In [16], Katz, Łaba and Tao proved that every union of δ tubes in $\mathbb{R}^3$ coming from the discretization of a (hypothetical) Kakeya set with upper Minkowski dimension $5/2$ can be written as a union of “grains,” (i.e. rectangular prisms) of dimensions roughly $\delta \times \delta^{1/2} \times \delta^{1/2}$. Guth [9] generalized this result and proved that every union of δ tubes in $\mathbb{R}^3$ satisfying a certain broadness hypothesis can be written as a union of grains of dimensions roughly $\delta \times t \times t$, where the diameter t is related to the number of tubes in the arrangement and the volume of their union.

The purpose of this section is to prove a structural statement for unions of δ tubes in $\mathbb{R}^3$, in the spirit of the Katz–Łaba–Tao and Guth results described above. This is Proposition 14.4.57 below. As discussed in the introduction,

Proposition 14.4.57 is a key step in the proof of Proposition 14.4.7—Proposition 14.4.57 helps us find the correct scales and arrangements of convex sets to which we can apply Assertion $\mathcal{E}(\sigma, \omega)$.

In brief, Proposition 14.4.57 explores what happens when we cover an arrangement of δ -tubes by ρ -tubes, apply (a variant of) Guth’s grains theorem inside each re-scaled ρ -tube, and then analyze how the resulting grains coming from the δ tubes inside different ρ -tubes interact. The specific hypotheses and conclusions of Proposition 14.4.57 are somewhat technical; they were adapted to match the needs of the arguments in Section 14.4.8. In order to state Proposition 14.4.57 we will require a few definitions.

Definition 14.4.53. Let $\lambda > 0$, $0 < \delta \leq \rho \leq 1$ and $\delta \leq a \leq b \leq c$ with $\rho = b/c$. Let $(\mathbb{T}, Y)_\delta$ be a set of δ tubes and their associated shading, and let $\mathbb{T}_\rho$ be a balanced partitioning cover of $\mathbb{T}$. We say $(\mathcal{P}, Y)_{a \times b \times c}$ is a *robustly λ -dense two-scale grains decomposition of $(\mathbb{T}, Y)_\delta$ with regard to (wrt) $\mathbb{T}_\rho$* if the following is true:

- (i) For each $P \in \mathcal{P}$, there is a unique $T_\rho \in \mathbb{T}_\rho$ satisfying $P \subset T_\rho$ and $\angle(\text{dir}(P), \text{dir}(T_\rho)) \leq 2\rho$. This induces a partition $\mathcal{P} = \bigsqcup_{T_\rho} \mathcal{P}_{T_\rho}$.
- (ii) For each $T_\rho \in \mathbb{T}_\rho$, the sets $\{Y(P) : P \in \mathcal{P}_{T_\rho}\}$ are disjoint, and we have

$$\bigcup_{T \in \mathbb{T}[T_\rho]} Y(T) = \bigsqcup_{P \in \mathcal{P}_{T_\rho}} Y(P). \quad (14.4.117)$$

- (iii) The pair $(\mathcal{P}, Y)_{a \times b \times c}$ is λ -dense, and furthermore there exists a number μ so that for each $T_\rho \in \mathbb{T}_\rho$ and each $x \in (14.4.117)$, we have $\#\mathbb{T}[T_\rho]_Y(x) \sim \mu$.
- (iv) For each $T_\rho \in \mathbb{T}_\rho$ and each pair $T \in \mathbb{T}[T_\rho]$ and $P \in \mathcal{P}_{T_\rho}$ with $Y(T) \cap Y(P) \neq \emptyset$, we have that T exits P through the “long end” (See Figure 14.4.6), and $Y(T) \cap P \subset Y(P)$.

Remark 14.4.54. Conclusion (iv) implies that $a \geq 2\delta$. Usually we will be interested in the case where $a \sim \delta$, though it will sometimes be useful to consider larger values of a .

Definition 14.4.55. Let P be a $a \times b \times c$ prism. Define $\square(P)$ to be the $\frac{ac}{b} \times c \times c$ prism containing P with the same center and normal direction as P (the latter condition means that $\Pi(P) = \Pi(\square(P))$). Observe that both $\Pi(P)$ and $\Pi(\square(P))$ are defined up to accuracy a/b .

Let $\mathcal{P}$ be a set of $a \times b \times c$ prisms, and let W be a convex set. Define

$$\mathcal{P}\langle W \rangle = \{P \in \mathcal{P} : \square(P) \subset W\}.$$

Note that since $P \subset \square(P)$, we have $\mathcal{P}\langle W \rangle \subset \mathcal{P}[W]$.

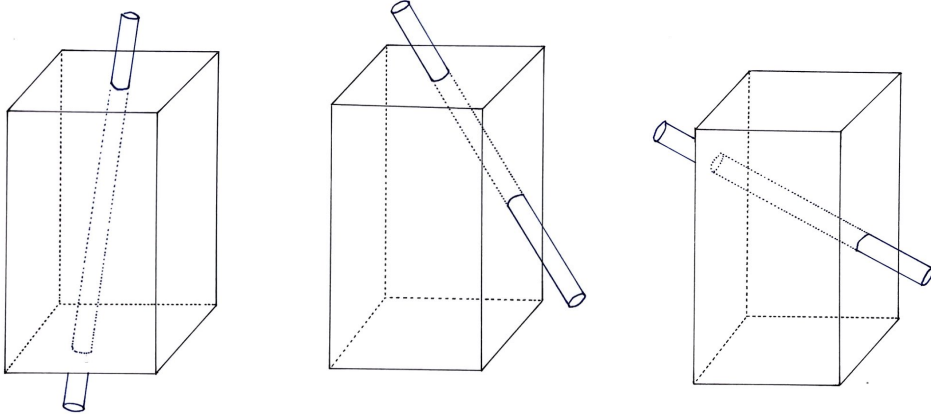


Figure 14.4.6: A tube exiting a grain through the long ends (left) vs failing to do so (middle and right).

Observe that if P, P' are intersecting $a \times b \times c$ prisms and $\angle(\Pi(P), \Pi(P')) \leq K \frac{a}{b}$ for some $K \geq 1$, then the K -fold thickenings (i.e. the prisms of dimensions $\frac{ac}{b} \times c \times c$ with the same center, direction, and tangent plane) of $\square(P)$ and $\square(P')$ are comparable up to factors of K . In particular, if $(\mathcal{P}, Y)_{a \times b \times c}$ is a pair of prisms and their associated shadings, with $\angle(\Pi(P), \Pi(P')) \leq K \frac{a}{b}$ for all pairs $P, P' \in \mathcal{P}$ for which $Y(P) \cap Y(P') \neq \emptyset$, then we can find a set $\mathcal{W}$ of prisms of dimensions comparable to $K \frac{ac}{b} \times c \times c$ so that each $P \in \mathcal{P}$ is contained in at least 1, and at most $O(1)$ sets of the form $\mathcal{P}\langle W \rangle$, $W \in \mathcal{W}$. Furthermore, the sets $\{\bigcup_{P \in \mathcal{P}\langle W \rangle} Y(P), W \in \mathcal{W}\}$ are $O(1)$ overlapping.

In our arguments below, we will exploit the above observation when $K \leq \delta^{-\varepsilon}$ for a small $\varepsilon > 0$. Since we will analyze each set $\mathcal{P}\langle W \rangle$ individually, we will be interested in the quantity $C_{KT-CW}(\mathcal{P}\langle W \rangle)$, rather than the (potentially much larger) quantity $C_{KT-CW}(\mathcal{P})$. Each set W contains at most $100K^{100}$ essentially distinct prisms of the form $\square(P)$, and thus if K is not too large then $C_{KT-CW}(\mathcal{P}\langle W \rangle)$ is controlled by $\sup_P C_{KT-CW}(\mathcal{P}\langle 2\square(P) \rangle)$, where $2\square(P)$ denotes the 2-fold dilate of $\square(P)$. This motivates the following definition.

Definition 14.4.56. Let $\mathcal{P}$ be a set of $a \times b \times c$ prisms. Define

$$C_{KT-CW}^{\text{loc}}(\mathcal{P}) = \max_{P \in \mathcal{P}} C_{KT-CW}(\mathcal{P}\langle 2\square(P) \rangle).$$

With these definitions, we can now state the main result of Section 14.4.6.

Proposition 14.4.57. *Let $\omega > 0$, $\sigma \in (0, 2/3]$, and $\zeta \in (0, \omega/1000)$. Suppose that $\mathcal{E}(\sigma, \omega)$ is true. Then there exists $\alpha, \eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}.$$

(B) *There exist the following:*

- Numbers ρ and $\delta \leq a \leq b \leq c \leq 1$, with $\rho = b/c$.
- A δ^ζ refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$.
- A set $\mathbb{T}_\rho$ of ρ tubes.
- A pair $(\mathcal{G}, Y)_{a \times b \times c}$.

These objects have the following properties:

- (i) $(\mathbb{T}', Y')_\delta$ is δ^ζ dense, $C_{KT-CW}(\mathbb{T}') \leq \delta^{-\zeta}$, and $C_{F-SW}(\mathbb{T}') \leq \delta^{-\zeta}$.
- (ii) $\mathbb{T}_\rho$ is a balanced partitioning cover of $\mathbb{T}'$ that factors $\mathbb{T}'$ above and below with respect to the Frostman Slab Wolff Axioms with error $\delta^{-\zeta}$.
- (iii) $(\mathcal{G}, Y)_{a \times b \times c}$ is a robustly δ^ζ -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}_\rho$.
- (iv)
$$c \geq \delta^\zeta \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}_\delta}. \quad (14.4.118)$$
- (v) $\delta^{1-\omega/100} \leq \rho \leq \delta^{\omega/100}$.
- (vi) $C_{KT-CW}^{\text{loc}}(\mathcal{G}) \leq \delta^{-\zeta}$.

We now give the proof of Proposition 14.4.57; its single-scale input and the three iterative Moves are stated and proved in the subsections below.

Fixing ω and σ . In our proof of Proposition 14.4.57, the values of $\omega > 0$, $\sigma \in (0, 2/3]$ and $\zeta > 0$ will never change. Thus to simplify our exposition below, we will fix values of ω , σ , and ζ , which will remain unchanged throughout Sections 14.4.6 and 14.4.7. In particular, some of our definitions (such as Definition 14.4.58, which defines broadness) will depend on these quantities. By fixing them in advance, we can suppress this dependence.

We will make crucial use of Guth's methods to find a grains decomposition inside each re-scaled set $\mathbb{T}^{T_\rho}$. In order to apply Guth's techniques, the tubes need to satisfy a broadness condition. We describe and prove the required form below.

14.4.6.1 Broadness

For the definition that follows, we will fix $\beta = \omega\zeta/100$.

Definition 14.4.58. Let $\delta > 0$ and $K \geq 1$. We say a multi-set $\mathcal{V} \subset S^{n-1}$ of unit vectors in $\mathbb{R}^n$ is *broad with error K at scales $\geq \delta$* if for all unit vectors $v_0 \in \mathbb{R}^n$ and all $r \in [\delta, 1]$ we have

$$\#\{v \in \mathcal{V} : \angle(v, v_0) \leq r\} \leq Kr^\beta(\#\mathcal{V}).$$

More generally, let $\mathcal{V}$ be contained in a disk $D \subset S^{n-1}$ of radius ρ . We say $\mathcal{V}$ is broad with error K at scales $\geq \delta$ inside D if for all unit vectors $v_0 \in \mathbb{R}^n$ and all $r \in [\delta, \rho]$ we have

$$\#\{v \in \mathcal{V} : \angle(v, v_0) \leq r\} \leq K(r/\rho)^\beta(\#\mathcal{V}).$$

Definition 14.4.59. Let $(\mathbb{T}, Y)_\delta$ be a collection of tubes and their associated shading, and let $\mathbb{T}_\rho$ be a cover of $\mathbb{T}$.

- (A) We say that $(\mathbb{T}, Y)_\delta$ is *broad with error K* if for each $x \in \bigcup_{T \in \mathbb{T}} Y(T)$, the set of unit vectors $\{\text{dir}(T) : x \in Y(T)\}$ is broad with error K at scales $\geq \delta$.
- (B) We say that $(\mathbb{T}, Y)_\delta$ is *broad with error K relative to the cover $\mathbb{T}_\rho$* if for each $T_\rho \in \mathbb{T}_\rho$, the set $(\mathbb{T}^{T_\rho}, Y^{T_\rho})_{\delta/\rho}$ is broad with error K .

The next result is a variant of the “two ends” reduction. In general, a set $\mathcal{V}$ of unit vectors need not be broad (with small error). However, the next result says that every set of unit vectors is broad when localized inside ρ disks, for some value of ρ . The precise statement is as follows.

Lemma 14.4.60. *Let $\delta > 0$ and let $\mathcal{V}$ be a set of vectors in $\mathbb{R}^n$ pointing in δ -separated directions. Then there exists a scale $\rho \in [\delta, 1]$; a set $\mathcal{B}$ of disjoint balls $B \subset S^{n-1}$ of radius ρ ; and sets $\mathcal{V}_B \subset \mathcal{V} \cap B$ so that the following holds.*

- (i) *Each set $\mathcal{V}_B$ has cardinality $\gtrsim \rho^\beta(\#\mathcal{V})$.*
- (ii) *$\mathcal{V}_B$ is broad with error 100 at scales $\geq \delta$ inside B .*
- (iii) *$\bigcup_B \mathcal{V}_B \gtrsim (\log 1/\delta)^{-1}(\#\mathcal{V})$.*

Proof. We will greedily construct a sequence of sets $\mathcal{V} = \mathcal{V}_0 \supset \mathcal{V}_1 \supset \dots$ as follows. For each index $i \geq 1$, let v_i be a unit vector and $r_i \in [\delta, 1]$ a radius that maximizes the quantity

$$r_i^{-\beta}(\#\mathcal{V}_{i-1} \cap B(v_i, r_i)). \quad (14.4.119)$$

Let $\mathcal{W}_i = \mathcal{V}_{i-1} \cap B(v_i, r_i)$ and let $\mathcal{V}_i = \mathcal{V}_{i-1} \setminus B(v_i, 100r_i)$. Note that $\#(\mathcal{V}_{i-1} \cap B(v_i, 3r_i)) \leq 100^\beta(\#\mathcal{W}_i)$. Continue this process until $\#\mathcal{V}_i \leq \frac{1}{2}\#\mathcal{V}$.

We claim that for each index i , each unit vector v , and each $r \geq \delta$, we have

$$\#(\mathcal{W}_i \cap B(v, r)) \leq 100(r/r_i)^\beta (\#\mathcal{W}_i). \quad (14.4.120)$$

This will give Conclusion (ii). To verify (14.4.120), suppose to the contrary that (14.4.120) failed for some pair (v, r) , then this would contradict the maximality of (v_i, r_i) in (14.4.119).

Note as well that since $r_i = 1$ is a valid choice of r , we have $\#\mathcal{W}_i \geq r_i^\beta (\#\mathcal{V}_{i-1}) \geq \frac{1}{2}r_i^\beta (\#\mathcal{V})$. Finally, note that if $r_i \leq r_j \leq 2r_i$, then the balls $2B(v_i, r_i)$ and $2B(v_j, r_j)$ must be disjoint. Indeed, we may suppose that both $\mathcal{W}_i$ and $\mathcal{W}_j$ are non-empty. If $i < j$ then we must have $B(v_j, r_j) \cap B(v_i, 100r_i)^c \neq \emptyset$, while if $j < i$ then we must have $B(v_i, r_i) \cap B(v_j, 100r_j)^c \neq \emptyset$. In either case, since $r_i \leq r_j \leq 2r_i$, this means that $2B(v_i, r_i) \cap 2B(v_j, r_j) = \emptyset$.

To conclude the proof, use dyadic pigeonholing to select a scale $\rho \in [\delta, 1]$ so that $\#\bigcup_{i: \rho/2 \leq r_i \leq \rho} \mathcal{W}_i \gtrsim (\log 1/\delta)^{-1} (\#\mathcal{V})$. This gives us the collection $\mathcal{B}$ of disjoint balls (it is harmless for us to replace each ball $B(v_i, r_i)$ with $B(v_i, \rho)$; as noted above, the balls remain disjoint). $\square$

Remark 14.4.61. Lemma 14.4.60 is similar to the standard “two-ends” broadness reduction. However, the standard two-ends broadness reduction typically replaces $\mathcal{V}$ by $\mathcal{V} \cap B$ for a *single* ρ ball B . The resulting set has cardinality $\gtrsim \rho^\beta (\#\mathcal{V})$. For our applications, this would have introduced an unacceptably large reduction in the cardinality of $\mathcal{V}$.

Corollary 14.4.62. *Let $(\mathbb{T}, Y)_\delta$ be a set of δ tubes and their corresponding shading. Then there exists a $\gtrsim_\delta 1$ refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$; a scale $\rho \in [\delta, 1]$; and a balanced partitioning cover $\mathbb{T}_\rho$ of $(\mathbb{T}', Y')$, so that the following holds:*

(i) *Each point $x \in \mathbb{R}^3$ is contained $\lesssim_\delta \rho^{-\beta}$ sets of the form $\bigcup_{T \in \mathbb{T}_\rho} Y'(T)$, $T_\rho \in \mathbb{T}_\rho$.*

(ii) *$\mathbb{T}'$ is broad with error $\approx_\delta 1$ relative to the cover $\mathbb{T}_\rho$.*

Proof. Using dyadic pigeonholing, we can select a number $\mu \geq 1$ and a $(\log 1/\delta)^{-1}$ refinement $(\mathbb{T}, Y_1)_\delta$ of $(\mathbb{T}, Y)$ with the property that $\#\{T \in \mathbb{T}: x \in Y_1(T)\} \sim \mu$ for each $x \in \bigcup_{T \in \mathbb{T}} Y_1(T)$. Apply Lemma 14.4.60 to each point $x \in \bigcup_{T \in \mathbb{T}} Y_1(T)$. We obtain a scale $\rho_x \in [\delta, 1]$, and a set of disjoint ρ_x balls $\mathcal{B}_x$ (each ball in $\mathcal{B}_x$ is a subset of S^2). After further dyadic pigeonholing we can select the following:

- A common scale ρ ;
- Multiplicities $\nu \geq 1$ and $N \lesssim \rho^{-\beta}$;

- A set $\mathcal{B}$ of ρ balls (each ball in $\mathcal{B}$ is a subset of S^2), whose 100ρ neighbourhoods are disjoint;
- $A \gtrsim (\log 1/\delta)^{-1}$ refinement $(\mathbb{T}, Y_2)_\delta$ of $(\mathbb{T}, Y_1)_\delta$.

We can select the above numbers and sets so that the following property holds: for each $x \in \bigcup_{\mathbb{T}} Y_2(T)$, the set of unit vectors $\{\text{dir}(T) : x \in Y_2(T)\} \subset S^2$ can be covered by a union of N balls from $\mathcal{B}$, and for each such ball, we have that the set $B \cap \{\text{dir}(T) : x \in Y_2(T)\}$ has cardinality ν and is broad with error $O(1)$ inside B .

After refining $\mathbb{T}$ by a factor of $\approx_\delta 1$, we obtain a new pair $(\mathbb{T}_2, Y_2)_\delta$ that is a $\approx_\delta 1$ refinement of $(\mathbb{T}, Y_2)_\delta$, and a set $\mathbb{T}_\rho$ of ρ tubes with the property that $\mathbb{T}_\rho$ is a balanced partitioning cover of $\mathbb{T}_2$, and furthermore the family of convex sets $\{3T_\rho : T_\rho \in \mathbb{T}_\rho\}$ is a partitioning cover of $\mathbb{T}_2$. This means that for each $T_\rho \in \mathbb{T}_\rho$, $\mathbb{T}_2[T_\rho] = \mathbb{T}_2[3T_\rho]$.

For each $T \in \mathbb{T}_2$ with $T \subset T_\rho \in \mathbb{T}_\rho$, define

$$Y_3(T) = \{x \in Y_2(T) : \#\{T' \in \mathbb{T}_2[T_\rho] : x \in Y_2(T')\} \geq \kappa_0 \nu\}.$$

Since $(\mathbb{T}_2, Y_2)_\delta$ is a $\approx_\delta 1$ refinement of $(\mathbb{T}, Y_2)_\delta$, if $\kappa_0 > 0$ is chosen sufficiently small (depending on the implicit constant mentioned previously), then $(\mathbb{T}_2, Y_3)_\delta$ is a $\approx_\delta 1$ refinement of $(\mathbb{T}_2, Y_2)_\delta$. Furthermore, for each point $x \in \bigcup_{T \in \mathbb{T}[3T_\rho]} Y_3(T) = \bigcup_{T \in \mathbb{T}[T_\rho]} Y_3(T)$, we have that the set of unit vectors $\{\text{dir}(T) : T \in \mathbb{T}_2[3T_\rho] = \mathbb{T}_2[T_\rho], x \in Y_3(T)\}$ is broad inside $B(\text{dir}(T_\rho), 2\rho)$ with error $O(1)$, in the sense of Definition 14.4.58. We conclude that the pair $(\mathbb{T}_2, Y_3)_\delta$ and $\{3T_\rho, T_\rho \in \mathbb{T}_\rho\}$ satisfy the conclusions of Corollary 14.4.62 (with 3ρ in place of ρ). $\square$

The next result says that every pair $(\mathbb{T}, Y)_\delta$ of δ tubes and their associated shading is either broad at *some* scale $\delta \ll \rho \ll 1$, or else the tubes in $(\mathbb{T}, Y)_\delta$ are almost disjoint.

Lemma 14.4.63. *Let $\delta > 0$ and let $(\mathbb{T}, Y)_\delta$ be a pair of δ tubes and their associated shading. Then at least one of the following must occur:*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_\delta \delta^{\omega/2} \sum_{T \in \mathbb{T}} |Y(T)|. \quad (14.4.121)$$

(B) *There is a scale $\rho \in [\delta^{1-\omega/100}, \delta^{\omega/100}]$, a $\approx_\delta 1$ refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)$, and a balanced partitioning cover $\mathbb{T}_\rho$ of $\mathbb{T}$, so that $(\mathbb{T}', Y')_\delta$ is broad with error $O(1)$ relative to the cover $\mathbb{T}_\rho$.*

Proof. Let $r = \delta^{\omega/100}$. After replacing $(\mathbb{T}, Y)_\delta$ by a ~ 1 refinement, we can find a partitioning cover $\mathbb{T}_r$ of $\mathbb{T}$. Apply Corollary 14.4.62 to each set $(\mathbb{T}^{T_r}, Y^{T_r})_{\delta/r}$. After dyadic pigeonholing $\mathbb{T}_r$ and $\mathbb{T}$, we can suppose that the resulting scale,

which we will call $\tilde{\rho}$, is the same for each $T_r \in \mathbb{T}_r$. Define $\rho = \tilde{\rho}r$, and let $(\mathbb{T}', Y')$ be the $\approx_\delta 1$ refinement of $(\mathbb{T}, Y)_\delta$ consisting of the tubes and their associated shadings coming from the conclusion of Corollary 14.4.62 for each $T_r \in \mathbb{T}_r$.

Recall that the tubes $\mathbb{T}_r$ form a partitioning cover of $\mathbb{T}'$, and for each $T_r \in \mathbb{T}_r$, the (re-scaled) ρ tubes coming from Corollary 14.4.62 form a partitioning cover of $\mathbb{T}'[T_r]$. We conclude that if we define $\mathbb{T}_\rho$ to be the union of these ρ tubes, then $\mathbb{T}_\rho$ is a partitioning cover of $\mathbb{T}'$, and furthermore $(\mathbb{T}', Y')_\delta$ is broad with error $O(1)$ relative to the cover $\mathbb{T}_\rho$.

At this point, we have constructed the pair $(\mathbb{T}', Y')_\delta$ and $\mathbb{T}_\rho$, which satisfies all of the requirements for Conclusion (B) of Lemma 14.4.63 with one exception — we know that $\rho \in [\delta, \delta^{\omega/100}]$, but we do not know that $\rho \in [\delta^{1-\omega/100}, \delta^{\omega/100}]$. Our next task is to show that if $\rho < \delta^{1-\omega/100}$, then Conclusion (A) holds.

Since the tubes in $\mathbb{T}_r$ are essentially distinct, $O(r^{-2})$ tubes can pass through a common point. By Conclusion (i) from Corollary 14.4.62, we have that for each tube T_r and each $x \in T_r$ we have

$$\#\{T_\rho \in \mathbb{T}_\rho[T_r] : x \in \bigcup_{T \in \mathbb{T}'[T_\rho]} Y'(T)\} \lesssim_\delta \rho^{-\beta} \leq \delta^{-\omega/100}.$$

Recall $\beta = \omega\zeta/100$ was fixed at the beginning of Section 7.1.

Finally, for each $T_\rho \in \mathbb{T}_\rho$, at most $(\rho/\delta)^2$ tubes from $\mathbb{T}'[T_\rho]$ can pass through a common point and at most r^{-2} distinct $T_r \in \mathbb{T}_r$ can pass through a common point. We conclude that

$$\#\mathbb{T}'_{Y'}(x) \lesssim \delta^{-\omega/100} \left(\frac{\rho}{\delta r}\right)^2 \quad \text{for each } x \in \mathbb{R}^3.$$

In summary, if $\rho < \delta^{1-\omega/100}$, then Conclusion (A) holds. Otherwise, Conclusion (B) holds. $\square$

We conclude this section with two results on the union of broad sets of vectors.

Lemma 14.4.64. *Let $\delta > 0$, let $B \subset S^{n-1}$ be a disk, and let $\mathcal{V}_i$, $i = 1, \dots, N$ be sets of unit vectors in B . Suppose that each set $\mathcal{V}_i$ is broad with error K at scales $\geq \delta$ inside B . Then the multi-set $\bigsqcup_{i=1}^N \mathcal{V}_i$ is broad with error K at scales $\geq \delta$ inside B .*

Proof. Fix a unit vector v_0 and $r \in [\delta, \text{rad}(B)]$. For each i , broadness gives

$$\#\{v \in \mathcal{V}_i : \angle(v, v_0) \leq r\} \leq K \left(\frac{r}{\text{rad}(B)}\right)^\beta \#\mathcal{V}_i.$$

Sum these inequalities over i . Both cardinalities are counted with multiplicity, so their left sides sum to the corresponding cardinality in $\bigsqcup_i \mathcal{V}_i$, and their

right sides sum to $K(r/\text{rad}(B))^\beta \#(\bigsqcup_i \mathcal{V}_i)$. This is precisely the definition of broadness inside B . $\square$

The second result described how broadness combines across scales.

Lemma 14.4.65. *Let $0 < \delta < \rho \leq 1$. Let $\mathcal{U}$ be a set of unit vectors in $\mathbb{R}^n$. For each $u \in \mathcal{U}$, let $\mathcal{V}_u \subset B(u, \rho) \subset S^{n-1}$ be contained in the disk of radius ρ centered at u . Suppose that $\mathcal{U}$ is broad with error K_1 at scales $\geq \rho$, and that each set $\mathcal{V}_u$ is broad with error K_2 at scales $\geq \delta$ inside $B(u, \rho)$. Suppose furthermore that each set $\mathcal{V}_u$ has comparable cardinality (up to a factor of 2). Then the multi-set $\bigsqcup_{u \in \mathcal{U}} \mathcal{V}_u$ is broad with error $O(K_1 K_2)$ at scales $\geq \delta$.*

Proof. Let $\mathcal{V} = \bigsqcup_{u \in \mathcal{U}} \mathcal{V}_u$. By hypothesis, there is a number M_1 so that $M_1 \leq \#\mathcal{V}_u \leq 2M_1$ for each set $\mathcal{V}_u$. Let $X = \sup_{u_0} \#\{u \in \mathcal{U} : \angle(u, u_0) \leq \rho\}$, where the supremum is taken over all unit vectors $u_0 \in S^{n-1}$. Since $\mathcal{U}$ is broad with error K_1 at scales $\geq \rho$, if we choose a unit vector u_0 achieving the above supremum, then

$$X \leq \#\{u \in \mathcal{U} : \angle(u, u_0) \leq \rho\} \leq K_1 \rho^\beta (\#\mathcal{U}),$$

and thus $\#\mathcal{U} \geq K_1^{-1} \rho^{-\beta} X$.

First we consider the case where $r \in [\delta, \rho]$. For each unit vector v_0 , there are at most $O(X)$ vectors $u \in \mathcal{U}$ for which $\{v \in \mathcal{V}_u : \angle(v, v_0) \leq r\}$ is non-empty. For each such u , we have

$$\#\{v \in \mathcal{V}_u : \angle(v, v_0) \leq r\} \lesssim K_2 (r/\rho)^\beta (2M_1) \lesssim K_1 K_2 r^\beta (\#\mathcal{U}) X^{-1} M_1.$$

Thus the total contribution from all such u is $\lesssim K_1 K_2 r^\beta (\#\mathcal{V})$, as desired.

Next we consider the case where $r \in [\rho, 1]$. Then

$$\begin{aligned} \#\{v \in \mathcal{V} : \angle(v, v_0) \leq r\} &\lesssim 2M_1 \#\{u \in \mathcal{U} : \angle(u, v_0) \leq 2r\} \\ &\leq M_1 K_1 r^\beta (\#\mathcal{U}) = K_1 r^\beta (\#\mathcal{V}). \quad \square \end{aligned}$$

14.4.6.2 Broadness and the Frostman Slab Wolff axioms

Given a pair $(\mathbb{T}, Y)_\delta$, Lemma 14.4.63 allows us to find a set $\mathbb{T}_\rho$ for which the pairs $(\mathbb{T}^{T_\rho}, Y^{T_\rho})_{\delta/\rho}$ are broad. The next result says that under suitable hypotheses, $\mathbb{T}_\rho$ will factor $\mathbb{T}$ with respect to the Frostman Slab Wolff axioms. The precise statement is as follows.

Lemma 14.4.66. *Suppose that $\mathcal{E}(\sigma, \omega)$ is true, and let $\varepsilon > 0$. Then there exists $\alpha, \eta, \kappa > 0$ so that the following holds for all $0 < \delta \leq \rho \leq 1$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Let $\mathbb{T}_\rho$ be a balanced cover of $\mathbb{T}$, and suppose $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$ relative to the cover $\mathbb{T}_\rho$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.122)$$

(B) *There exists a δ^ε refinement $(\mathbb{T}', Y')_\delta$ and a set $\mathbb{T}'_\rho \subset \mathbb{T}_\rho$, so that $\mathbb{T}'_\rho$ factors $\mathbb{T}'$ above and below with respect to the Frostman Slab Wolff Axioms with error $\delta^{-\varepsilon}$.*

In brief, the proof is as follows. We apply Proposition 14.4.25 inside each set $\mathbb{T}^{T_\rho}$ to cover the tubes in $\mathbb{T}[T_\rho]$ by prisms $\mathcal{W}$, with the property that each set $\mathbb{T}^W$ satisfies the Frostman Slab Wolff axioms with small error, and the tubes in $\mathbb{T}[T_\rho]$ from distinct prisms do not interact (i.e. their shadings are almost disjoint). Each prism W is contained inside its corresponding ρ tube T_ρ . If W is almost as large as T_ρ , then $C_{F-SW}(\mathbb{T}^{T_\rho})$ must be almost as small as $C_{F-SW}(\mathbb{T}^W)$, which in turn has size about 1. If this happens, then Conclusion (B) holds.

Suppose instead that the prisms are much smaller than T_ρ ; we will refer to the dimensions of these prisms as $s \times t \times 1$, with $\delta \leq s \leq t$. Since each prism is contained inside a ρ tube, we also have $t \leq \rho$. Thus if the prisms are much smaller than the ρ tubes, then in particular we must have $s \ll \rho$.

Next, we will make use of the assumption that the tubes are broad relative to the cover $\mathbb{T}_\rho$ in order to show that $t \sim \rho$. Since $s \ll \rho$, this means that the prisms are flat. Specifically, broadness ensures that a typical pair of tubes passing through a common point $x \in \bigcup_{T \in \mathbb{T}[T_\rho]} Y(T)$ make angle roughly ρ . Such a pair of tubes is contained in a common prism W , from which it follows that W must have “width” ρ , i.e. each prism W has dimensions roughly $s \times \rho \times 1$.

To summarize, at this point in the argument each δ tube is contained inside a flat prism W of dimensions roughly $s \times \rho \times 1$ with $s \ll \rho$. Furthermore, at a typical point $x \in Y(T)$, there is at least one other tube from $\mathbb{T}$ contained inside the same flat prism W , and this second tube intersects T at angle roughly ρ . Thus the hairbrush of T (i.e. the union of set of tubes intersecting T), when restricted to the $\frac{\rho}{s}\delta$ neighbourhood of T , fills out a rectangular slab of dimensions roughly $\delta \times \frac{\rho}{s}\delta \times 1$. But this is precisely the setting where Lemma 14.4.46 asserts that $\bigcup_{\mathbb{T}} Y(T)$ is larger than we would expect from the estimate $\mathcal{E}(\sigma, \omega)$, and thus Conclusion (A) holds. We now turn to the details.

Proof of Lemma 14.4.66.

Step 1. We may suppose that $\rho \leq \delta^{\varepsilon/10}$, or else Conclusion (B) follows by selecting a single ρ -tube that contains at least $\rho^4(\#\mathbb{T})$ tubes from $\mathbb{T}$. We may also suppose that $\rho \geq \delta^{\varepsilon/10}$, or else Conclusion (B) holds trivially by taking $\mathbb{T}' = \mathbb{T}$ and $\mathbb{T}'_\rho = \mathbb{T}_\rho$.

By pigeonholing and replacing $\mathbb{T}_\rho$ by a subset $\mathbb{T}_{\rho,1}$, we can find a $\approx_\delta 1$ refinement $(\mathbb{T}_1, Y_1)_\delta$ of $(\mathbb{T}, Y)_\delta$ so that $\mathbb{T}_{\rho,1}$ is a balanced cover of $\mathbb{T}_1$, and furthermore each set $(\mathbb{T}_1^{T_\rho}, Y^\rho)_{\delta/\rho}$ is $\gtrsim_\delta \delta^\eta \gtrsim_\rho \rho^{10\eta/\varepsilon}$ dense.

Let $\varepsilon_1 = \varepsilon\beta/1600$. Apply Proposition 14.4.25 with ε_1 in place of ε to each set $\mathbb{T}_1^{T_\rho}$; this gives us a collection of convex sets $\mathcal{W}_{T_\rho}$ that factors a $(\delta/\rho)^{\varepsilon_1}$ -fraction of $\mathbb{T}_1^{T_\rho}$ (abusing notation, we continue to use $\mathbb{T}_1^{T_\rho}$ to denote this $(\delta/\rho)^{\varepsilon_1}$ -fraction) from below with respect to the Frostman Slab Wolff axioms with error $(\delta/\rho)^{-\varepsilon_1} \leq \delta^{-\varepsilon_1}$. We may do this, provided $\eta > 0$ is selected sufficiently small depending on ε and ε_1 .

After further pigeonholing (which induces a further $\approx_\delta 1$ refinement $(\mathbb{T}_2, Y_2)_\delta$ of $(\mathbb{T}_1, Y_1)$ and replaces $\mathbb{T}_{\rho,1}$ by a subset $\mathbb{T}_{\rho,2}$), we may suppose that the convex sets in $\phi_{T_\rho}^{-1}(\mathcal{W}_{T_\rho})$ all have common dimensions (up to a factor of 2) for each $T_\rho \in \mathbb{T}_{\rho,2}$; call these dimensions $s \times t \times 1$. We may also suppose that the value of $C_{F-SW}(\mathbb{T}_2^{T_\rho})$ is the same (up to a factor of 2) for all $T_\rho \in \mathbb{T}_{\rho,2}$, and that the mass $\sum_{T \in \mathbb{T}[T_\rho]} |Y_2(T)|$ is the same (up to a factor of 2) for each $T_\rho \in \mathbb{T}_{\rho,2}$.

Let $\mathcal{W} = \bigcup_{T_\rho \in \mathbb{T}_\rho} \phi_{T_\rho}^{-1}(\mathcal{W}_{T_\rho})$. To summarize, the situation is as follows:

- (i) We have $\sum_{T \in \mathbb{T}_2[T_\rho]} |Y_2(T)| \gtrsim_\delta \delta^{\varepsilon_1} \sum_{T \in \mathbb{T}[T_\rho]} |Y(T)|$ for each $T_\rho \in \mathbb{T}_{\rho,2}$.
- (ii) We have a collection $\mathcal{W}$ of convex sets of dimensions $s \times t \times 1$, with $\mathbb{T}_2 \prec \mathcal{W} \prec \mathbb{T}_\rho$.
- (iii) For each $T_\rho \in \mathbb{T}_{\rho,2}$, the sets $\bigcup_{W \in \mathcal{W}[T_\rho]} Y_2(T)$, $W \in \mathcal{W}[T_\rho]$ are disjoint.
- (iv) For each $W \in \mathcal{W}$, we have $C_{F-SW}(\mathbb{T}_2^W) \leq \delta^{-\varepsilon_1}$.

By Items (ii) and (iv), for each $T_\rho \in \mathbb{T}_\rho$ we have

$$C_{F-SW}(\mathbb{T}_2^{T_\rho}) \leq \left(\sup_{W \in \mathcal{W}[T_\rho]} C_{F-SW}(\mathbb{T}_2^W) \right) \left(\frac{|T_\rho|}{|W|} \right)^{100} \leq \delta^{-\varepsilon/2} \left(\frac{|T_\rho|}{|W|} \right)^{100}. \quad (14.4.123)$$

The last term in the above inequality is an (intentionally) crude estimate for the number of essentially distinct $s \times t \times 1$ prisms that can fit inside a ρ -tube.

If $\left(\frac{|T_\rho|}{|W|} \right)^{100} \leq \delta^{-\varepsilon/2}$, then Conclusion (B) holds with $(\mathbb{T}', Y')_\delta = (\mathbb{T}_2, Y_2)$ and $\mathbb{T}_\rho = \mathbb{T}_{\rho,2}$ and we are done.

Step 2. We shall suppose henceforth that

$$|W| \leq \delta^{\varepsilon/200} |T_\rho|. \quad (14.4.124)$$

Our goal is to prove the (14.4.122) holds for an appropriate choice of α .

Each prism in $\mathcal{W}$ has dimensions $s \times t \times 1$, with $t \leq \rho$. We claim the reverse inequality is almost true. Specifically, we have

$$t \gtrsim_{\delta} \delta^{\frac{2\varepsilon_1}{\beta}} \rho. \quad (14.4.125)$$

Recall from Definition 14.4.59 that for each $T_{\rho} \in \mathbb{T}_{\rho,2}$ and each $x \in \bigcup_{T \in \mathbb{T}[T_{\rho}]} Y(T)$, we have, for $v \in S^2$ and $r \geq \delta$,

$$\begin{aligned} \#\{T \in \mathbb{T}[T_{\rho}] : x \in Y(T), \angle(v, \text{dir}(T)) \leq r\} &\leq \delta^{-\eta} \left(\frac{r}{\rho}\right)^{\beta} \\ &\times \#\{T \in \mathbb{T}[T_{\rho}] : x \in Y(T)\}, \end{aligned} \quad (14.4.126)$$

By Item (i) above, there exists at least one point $x \in \mathbb{R}^3$ for which

$$\#\{T \in \mathbb{T}_2[T_{\rho}] : x \in Y_2(T)\} \gtrsim_{\delta} \delta^{\varepsilon_1} \#\{T \in \mathbb{T}[T_{\rho}] : x \in Y(T)\} > 0, \quad (14.4.127)$$

and hence for this choice of x we have, for $v \in S^2$ and $r \geq \delta$,

$$\begin{aligned} \#\{T \in \mathbb{T}_2[T_{\rho}] : x \in Y_2(T), \angle(v, \text{dir}(T)) \leq r\} &\leq \delta^{-\eta-\varepsilon_1} \left(\frac{r}{\rho}\right)^{\beta} \\ &\times \#\{T \in \mathbb{T}_2[T_{\rho}] : x \in Y_2(T)\}, \end{aligned} \quad (14.4.128)$$

On the other hand, by Item (iii) above, the tubes $\{T \in \mathbb{T}_2[T_{\rho}] : x \in Y_2(T)\}$ are all contained in a common prism $W \in \mathcal{W}$, and thus must all make angle $\leq 2t$ with the direction v of this prism. Selecting $r = 2t$ in (14.4.128) and comparing with (iii), we obtain (14.4.125) (provided we select $\eta \leq \varepsilon_1$).

Comparing (14.4.124) and (14.4.125), we see that the prisms in $\mathcal{W}$ must be flat, i.e.

$$s \sim t^{-1}|W| \lesssim t^{-1} \delta^{\varepsilon/200} \rho^2 \lesssim_{\delta} \delta^{\frac{\varepsilon}{200} - \frac{4\varepsilon_1}{\beta}} t \leq \delta^{\frac{\varepsilon}{400}} t. \quad (14.4.129)$$

Step 3. Apply Lemma 14.4.38 to replace each shading $Y_2(T)$, $T \in \mathbb{T}_2$ with a regular sub-shading $Y_3(T) \subset Y_2(T)$. Define $\mathbb{T}_3 = \mathbb{T}_2$. We say a point $x \in \bigcup_{T \in \mathbb{T}} Y(T)$ has *survived* if

$$\#\{T \in \mathbb{T}_3 : x \in Y_3(T)\} \geq \kappa_0 \delta^{2\varepsilon_1} \#\{T \in \mathbb{T} : x \in Y(T)\}.$$

Let $Y_4(T) \subset Y_3(T)$ consist of surviving points and let $\mathbb{T}_4 = \mathbb{T}_3$; we will choose the constant κ_0 sufficiently small so that $(\mathbb{T}_4, Y_4)_{\delta}$ is a $\gtrsim_{\delta} \delta^{\varepsilon_1}$ refinement of $(\mathbb{T}_3, Y_3)$.

Observe that if the point x has survived, then there is a unique prism $W \in \mathcal{W}$ with $x \in \bigcup_{T \in \mathbb{T}_3[W]} Y(T)$, and at least two tubes $T, T' \in \mathbb{T}_3[W]$ with $x \in Y_3(T)$, $x \in Y_3(T')$ with $\angle(\text{dir}(T), \text{dir}(T')) \gtrsim_{\delta} \delta^{\frac{2\varepsilon_1}{\beta}} \rho$.

For each $T \in \mathbb{T}_4$, let $S(T) \supset T$ be the $\delta \times \frac{t}{s}\delta \times 1$ prism with coaxial line T , and plane parallel to $\Pi(W)$, where $W \in \mathcal{W}$ is the unique prism covering T . Recall that by (14.4.129), we have $\frac{t}{s}\delta \geq \delta^{1-\frac{\varepsilon}{400}}$. A Cordoba style L^2 argument (see i.e. Lemma 14.4.36 and its proof for an example of a similar argument) shows that for each $T \in \mathbb{T}_4$,

$$\left| S(T) \cap \bigcup_{T \in \mathbb{T}_3} Y_3(T) \right| \gtrsim_{\delta} \delta^{4\varepsilon_1} \frac{|Y_4(T)|}{|T|} |S|.$$

Let $b = \frac{t}{s}\delta \geq \delta^{1-\varepsilon/400}$. Applying Lemma 14.4.46, we obtain Conclusion (A) for $\alpha = \varepsilon\omega/500$, provided we select $\eta > 0$ sufficiently small. $\square$

14.4.6.3 The iteration base case: Guth’s grains decomposition

We next state precisely the single-scale grains decomposition that is used in the iteration.

Proposition 14.4.67. *Let $\varepsilon > 0$. Then there exists $\eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_{\delta}$ be δ^{η} dense and be broad with error $\delta^{-\eta}$. Suppose that the tubes in $\mathbb{T}$ are contained in a common 1 tube T_1 .*

Then there is a δ^{ε} refinement $(\mathbb{T}', Y')$ that is δ^{ε} dense and is broad with error $\leq \kappa^{-1}\delta^{-\varepsilon}$, and a number $\mu \geq 1$ so that $\mu \sim \#\mathbb{T}'_{Y'}(x)$ for each $x \in \bigcup_{\mathbb{T}'} Y'(T)$. In addition, there is a number $c \geq \kappa\mu\delta^{\varepsilon}(\delta\#\mathbb{T})^{-1}$; and a pair $(\mathcal{G}, Y)_{\delta \times c \times c}$, so that $(\mathcal{G}, Y)_{\delta \times c \times c}$ is a robustly δ^{ε} -dense two-scale grains decomposition of $(\mathbb{T}', Y')_{\delta}$ wrt $\{T_1\}$ (the latter is a set consisting of a single 1 tube).

Remark 14.4.68. Proposition 14.4.67 says that $(\mathcal{G}, Y)_{\delta \times c \times c}$ is a two-scale grains decomposition of $(\mathbb{T}', Y')_{\delta}$ wrt $\{T_1\}$. A two-scale grains decomposition is defined in Definition 14.4.53, and Item (iv) from that definition specifies that if $Y'(T) \cap Y(G) \neq \emptyset$, then T exits G through its long ends, in the sense of Figure 14.4.6. Since the grains $(\mathcal{G}, Y)_{\delta \times c \times c}$ from Proposition 14.4.67 are square, the definition is somewhat ambiguous in this setting. However, Proposition 14.4.67 is only used to prove Corollary 14.4.69. Thus the 1 tube T_1 should be thought of as the anisotropic rescaling of a ρ tube T_{ρ} . What is needed is the following: the images of the tubes in $\mathbb{T}'$ under the anisotropic scaling sending T_1 to T_{ρ} must exit the images of the grains in $\mathcal{G}$ through their long ends.

Proposition 14.4.67 is a variant of Guth’s grains decomposition from [9]. Since this precise statement does not appear in [9] (the hypotheses in [9] are stated slightly differently), we will provide a proof in Appendix 14.4.11.

If $(\mathbb{T}, Y)_{\delta}$ is broad relative to a set of ρ tubes $\mathbb{T}_{\rho}$, then we can apply Proposition 14.4.67 to each re-scaled set $(\mathbb{T}^{T_{\rho}}, Y^{T_{\rho}})_{\delta}$.

Corollary 14.4.69. *Let $\varepsilon > 0$. Then there exists $\kappa, \eta > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_{\delta}$ be δ^{η} dense and let $\mathbb{T}_{\rho}$ be a balanced partitioning*

cover of $\mathbb{T}$. Suppose that $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$ relative to the cover $\mathbb{T}_\rho$, and that $|\bigcup_{T \in \mathbb{T}} Y(T)| \leq \delta^\varepsilon (\#\mathbb{T})|T|$.

Then there is a δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ that is δ^ε dense; a subset $\mathbb{T}'_\rho \subset \mathbb{T}_\rho$; a number $c \gtrsim \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}}$; and a pair $(\mathcal{P}, Y)_{\delta \times b \times c}$ with $\rho = b/c$ that is a robustly δ^ε -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}'_\rho$. Finally, $(\mathbb{T}', Y')_\delta$ is broad with error $\leq \kappa^{-1} \delta^{-\varepsilon}$ relative to the cover $\mathbb{T}'_\rho$.

Note that the estimate on the size of c in Corollary 14.4.69 omits the term μ (though this term is used to compensate for the δ^ε loss in Proposition 14.4.67); this is because the weaker estimate $c \gtrsim \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}}$ will be sufficient in the arguments that follow.

Corollary 14.4.69 has two important consequences. First, when combined with Lemma 14.4.63, it says that if $(\mathbb{T}, Y)_\delta$ is an arrangement for which $\mathcal{E}(\sigma, \omega)$ is tight, then $(\mathbb{T}, Y)_\delta$ admits a two-scale grains decomposition. We call this the Guth grains decomposition of $\mathbb{T}$. The precise statement is as follows.

Lemma 14.4.70. *Suppose that $\mathcal{E}(\sigma, \omega)$ is true and let $\varepsilon > 0$. Then there exists $\alpha, \eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.130)$$

(C) *There exist the following:*

- A scale $\rho \in [\delta^{1-\omega/100}, \delta^{\omega/100}]$.
- A δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$.
- A balanced partitioning cover $\mathbb{T}_\rho$ of $\mathbb{T}'$.
- Numbers $\delta \leq b \leq c$ with $b/c = \rho$ and $c \gtrsim \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}}$.
- A pair $(\mathcal{P}, Y)_{\delta \times b \times c}$.

So that the following holds:

- (i) $(\mathbb{T}', Y')_\delta$ is broad with error $\leq \delta^{-\varepsilon}$ relative to the cover $\mathbb{T}_\rho$.
- (ii) $(\mathcal{P}, Y)_{\delta \times b \times c}$ is a robustly δ^ε -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}_\rho$.

Remark 14.4.71. Note that the Conclusions of Lemma 14.4.70 are labelled (A) and (C), rather than (A) and (B). We chose this convention in order to have parallelism with Conclusions (A), (B), and (C) of Moves #1, #2, and #3 below.

Lemma 14.4.70 is the starting point for the iterative process. The following subsections state and prove its three Moves.

14.4.6.4 Moves #1, #2, #3: Parallel structure

In the following sections, we will describe three Moves, which we will iteratively apply to the two-scale grains decomposition that we obtained from Lemma 14.4.70. Each of these Moves are expressed as a lemma, and these three lemmas have similar structure. In particular, the three lemmas have the same hypotheses, and have similar conclusions.

Common setup for Moves #1, #2, #3: Hypotheses

Suppose that $\mathcal{E}(\sigma, \omega)$ is true and let $\varepsilon \in (0, \zeta/2]$. Then there exists $\alpha, \eta, \kappa > 0$ so that the following holds for all $0 < \delta \leq 1$, $\rho \geq \delta^{1-\omega/100}$, and all $\delta \leq a \leq b \leq c$ with $b/c = \rho$.

Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Let $\mathbb{T}_\rho$ be a balanced partitioning cover of $\mathbb{T}$, and suppose that $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$ relative to $\mathbb{T}_\rho$. Let $(\mathcal{P}, Y)_{a \times b \times c}$ be a robustly δ^η -dense two-scale grains decomposition of $(\mathbb{T}, Y)_\delta$ wrt $\mathbb{T}_\rho$.

Each of Moves #1, #2, and #3 will have three possible conclusions, which we label (A), (B), and (C). Conclusion (A) is the same for all three moves.

Common setup for Moves #1, #2, #3: Conclusion (A).

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.131)$$

In Inequality (14.4.131), $\kappa, \alpha > 0$ are the quantities from the **Common setup for Moves #1, #2, #3: Hypotheses** described above.

Conclusion (B) is not identical for the three Moves, but shares many common elements. We describe these below

Common setup for Moves #1, #2, #3: Conclusion (B).

There are δ^ε refinements $(\mathbb{T}', Y')_\delta$ and $(\mathcal{P}', Y')_{a \times b \times c}$ of $(\mathbb{T}, Y)_\delta$ and $(\mathcal{P}, Y)_{a \times b \times c}$, respectively, and a set $\mathbb{T}'_\rho \subset \mathbb{T}_\rho$, so that the following holds.

- (i) $(\mathbb{T}', Y')_\delta$ is δ^ε dense, $C_{KT-CW}(\mathbb{T}') \leq \delta^{-\varepsilon}$, and $C_{F-SW}(\mathbb{T}') \leq \delta^{-\varepsilon}$.
- (ii) $\mathbb{T}'_\rho$ is a balanced partitioning cover of $\mathbb{T}'$, and $(\mathbb{T}', Y')_\delta$ is broad with error $\delta^{-\varepsilon}$ relative to $\mathbb{T}'_\rho$.
- (iii) $(\mathcal{P}', Y')_{a \times b \times c}$ is a robustly δ^ε -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}'_\rho$.
- (iv) Moves #1, #2, #3 will have additional conclusions specific to that Move.

Conclusion (C) is not identical for the three Moves, but shares many common elements. We describe these below

Common setup for Moves #1, #2, #3: Conclusion (C).

There exist the following:

- Numbers $\tilde{\rho}$ and $\tilde{a} \leq \tilde{b} \leq \tilde{c} \leq 1$, with $\tilde{\rho} = \tilde{b}/\tilde{c}$.
- A δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$.
- A set $\mathbb{T}_{\tilde{\rho}}$ of $\tilde{\rho}$ tubes.
- A pair $(\tilde{\mathcal{P}}, \tilde{Y})_{\tilde{a} \times \tilde{b} \times \tilde{c}}$.

These objects have the following properties:

- (i) $(\mathbb{T}', Y')_\delta$ is δ^ε dense, $C_{KT-CW}(\mathbb{T}') \leq \delta^{-\varepsilon}$, and $C_{F-SW}(\mathbb{T}') \leq \delta^{-\varepsilon}$.
- (ii) $\mathbb{T}_{\tilde{\rho}}$ is a balanced partitioning cover of $\mathbb{T}'$, and $(\mathbb{T}', Y')_\delta$ is broad with error $\delta^{-\varepsilon}$ relative to $\mathbb{T}_{\tilde{\rho}}$.
- (iii) $(\tilde{\mathcal{P}}, \tilde{Y})_{\tilde{a} \times \tilde{b} \times \tilde{c}}$ is a robustly δ^ε -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}_{\tilde{\rho}}$.
- (iv) $\delta^{1-\omega/100} \leq \tilde{\rho} \leq 1$.
- (v) Moves #1, #2, #3 will have additional conclusions specific to that Move.

Remark 14.4.72. Observe that Items (i), (ii), and (iii) of Conclusion (B) (resp. Items (i) – (iv) of Conclusion (C)) say that the output $(\mathbb{T}', Y')_\delta$; $\mathbb{T}'_\rho$; and $(\mathcal{P}', Y')_{\delta \times b \times c}$ (resp. $\mathbb{T}_{\tilde{\rho}}$ and $(\tilde{\mathcal{P}}, \tilde{Y})_{\tilde{a} \times \tilde{b} \times \tilde{c}}$) of Moves #1, #2, and #3 match the “input” hypotheses (i.e. the **Common setup for Moves #1, #2, #3: Hypotheses**), except that exponent η has been weakened to ε . This will allow us to iteratively apply these Moves many times.

14.4.6.5 Using Moves #1, #2, #3 to prove Proposition 14.4.57

We will state and prove Moves #1, #2, and #3 in Section 14.4.7. However, by using the parallel structure described above we can already state the hypotheses and conclusions of these three moves. We do so in the table below.

Move	Lemma	Conclusion (B), Item (iv)	Conclusion (C), Item (v)
#1	14.4.73	$c \geq \delta^\zeta(\rho/\delta)$ ($\#\mathbb{T}_\rho/\#\mathbb{T}_\delta$)	$\tilde{c} \geq \delta^{-\zeta}c$, $\tilde{a} = \delta$, and $\tilde{\rho} = \rho$
#2	14.4.74	$\rho \leq \delta^{\omega/100}$	$\tilde{c} \geq \delta^{-\omega/100}c$

#3	14.4.75	$C_{KT-CW}^{\text{loc}}(\mathcal{P}') \leq \delta^{-\zeta}$	$\tilde{\rho} \geq \delta^{-\zeta/1000} \rho$ and $\tilde{c} \geq c$
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We will now match the Conclusions of Proposition 14.4.57 to their counterparts from Moves #1, #2, #3.

- Conclusion (A) for Moves #1, #2, #3 matches Conclusion (A) of Proposition 14.4.57.
- Conclusion (B), Items (i) and (iii) for Moves #1, #2, #3 matches Conclusion (B), Items (i) and (iii), respectively, of Proposition 14.4.57.
- Conclusion (B), Item (ii) for Moves #1, #2, #3, plus Lemma 14.4.66 either yields Conclusion (A), or Conclusion (B), Item (ii) of Proposition 14.4.57.
- Conclusion (B), Item (iv) for Move #1 matches Conclusion (B), Item (iv) of Proposition 14.4.57.
- Conclusion (B), Item (iv) for Move #2, plus the hypothesis $\rho \geq \delta^{1-\omega/100}$, matches Conclusion (B), Item (v) of Proposition 14.4.57.
- Conclusion (B), Item (v) for Move #3 matches Conclusion (B), Item (vi) of Proposition 14.4.57.

Proof of Proposition 14.4.57. We proceed as follows. Let $\eta_0 < \eta_1 < \dots < \eta_N$ be a sequence of numbers that we will determine below. Let $(\mathbb{T}, Y)_\delta$ be δ^{η_0} dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta_0}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta_0}$. If η_0 is sufficiently small depending on η_1 , then we can apply the iteration base case, Lemma 14.4.70, with η_1 in place of ε . If Conclusion (A) of Lemma 14.4.70 holds, then Conclusion (A) of Proposition 14.4.57 holds, and we are done.

Suppose instead that Conclusion (B) of Lemma 14.4.70 holds. The output of Conclusion (B) is precisely the set of objects required for the **Common setup for Moves #1, #2, #3: Hypotheses**, with η_1 in place of η .

We now repeatedly apply Moves #1, #2, #3, with η_{j+1} in place of ε at stage j . We may do so, provided η_j is sufficiently small compared to η_{j+1} .

- If Conclusion (A) occurs at any point, then Conclusion (A) of Proposition 14.4.57 holds, and we halt.
- If Conclusion (B) occurs for Move # i , for some $i = 1, 2, 3$, then we switch to a different move.

- If Conclusion (B) occurs for all three moves in succession, then Items (i), (iii), and (iv) of Conclusion (B) of Proposition 14.4.57 hold. We halt and apply Lemma 14.4.66 to show either Conclusion (B), Item (ii) holds, or else Conclusion (A) holds. We conclude that at least one of Conclusion (A) or Conclusion (B) of Proposition 14.4.57 holds.
- If Conclusion (C) occurs for Move # i, then at least one of the following must occur:
 - c does not decrease, and ρ becomes larger by $\delta^{-\zeta/1000}$; this can occur at most $1000/\zeta$ times in a row.
 - c becomes larger by $\min\{\delta^{-\zeta}, \delta^{-\omega/100}\}$. This can occur at most $1/\zeta + 100/\omega$ times total.

In particular, the iterative process described above must halt after at most $N = \frac{1000}{\zeta}(\frac{2}{\zeta} + \frac{100}{\omega})$ steps. We will choose η_{N+1} below, and then select each of $\eta_N, \eta_{N-1}, \dots, \eta_0$ in turn. Finally, we define η (the quantity from Proposition 14.4.57) by $\eta = \eta_0$.

The refinement $(\mathbb{T}', Y')_\delta$, the set $\mathbb{T}_\rho$, and the pair $(\mathcal{G}, Y)_{a \times b \times c}$ satisfy all of the conclusions of Proposition 14.4.57, Conclusion (B), except that for Item (ii), we have not yet shown that $\mathbb{T}_\rho$ factors $\mathbb{T}'$ from below with respect to the Frostman Slab Wolff Axioms with error $\delta^{-\zeta}$. However, since $(\mathbb{T}', Y')_\delta$ is broad with respect to $\mathbb{T}_\rho$, we can apply Lemma 14.4.66 and conclude that either this is indeed the case (after replacing $(\mathbb{T}', Y')_\delta$, $\mathbb{T}_\rho$, and $(\mathcal{G}, Y)_{a \times b \times c}$ by a suitable refinement), or else Conclusion (A) of Proposition 14.4.57 holds.

This completes the proof of Proposition 14.4.57, except that we have not proved Moves #1, #2, #3. We shall do so in the next section. $\square$

14.4.7 Moves #1, #2, and #3

14.4.7.1 Move #1: Replacing grains with longer grains to ensure $c \geq \delta^\zeta \frac{\rho}{\delta} (\#\mathbb{T}_\rho) / (\#\mathbb{T})$

Our goal in this section is to state and prove Move #1, whose role in the parallel-structure alternative was described in Section 14.4.6.4.

Lemma 14.4.73. *We assume the **Common setup for Moves #1, #2, #3: Hypotheses** from Section 14.4.6.4. Then at least one of the following must hold.*

- (A) Conclusion (A) of the common setup for Moves #1, #2, #3.
- (B) Conclusion (B) of the common setup for Moves #1, #2, #3. In addition,

$$(iv) \quad c \geq \delta^\zeta \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}_\delta}. \quad (14.4.132)$$

(C) Conclusion (C) of the common setup for Moves #1, #2, #3. In addition,

$$(v) \quad \tilde{a} = \delta, \tilde{\rho} = \rho, \text{ and } \tilde{c} \geq \delta^{-\zeta} c.$$

Proof. If Conclusion (B) fails, then we discard the cover $(\mathcal{P}, Y)_{a \times b \times c}$ and replace it with the cover coming from Corollary 14.4.69 applied to $(\mathbb{T}, Y)_\delta$ and $\mathbb{T}_\rho$. Conclusion (C) of Lemma 14.4.73 was chosen to match the output of Corollary 14.4.69. Note that the prisms coming from Corollary 14.4.69 have length $\tilde{c} \geq \frac{\rho}{\delta} \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}}$. If Conclusion (B) fails, then this quantity is $\geq \delta^{-\zeta} c$, as claimed. $\square$

14.4.7.2 Move #2: Replacing square grains with longer grains

In this section we will use geometric arguments in the spirit of Cordoba’s proof of the Kakeya maximal function conjecture in $\mathbb{R}^2$ to show the following: if $(\mathbb{T}, Y)_\delta$ has a two-scale grains decomposition consisting of square grains, i.e. grains of dimensions $a \times b \times c$ with $\rho = b/c > \delta^{\omega/100}$, then either $\bigcup Y(T)$ is large, or else we can construct a new two-scale grains decomposition of $(\mathbb{T}, Y)$ with significantly longer grains. The precise statement is as follows.

Lemma 14.4.74. *We assume the **Common setup for Moves #1, #2, #3: Hypotheses** from Section 14.4.6.4. Then at least one of the following must hold.*

(A) Conclusion (A) of the common setup for Moves #1, #2, #3.

(B) Conclusion (B) of the common setup for Moves #1, #2, #3. In addition,

$$(iv) \quad \rho \leq \delta^{\omega/100}.$$

(C) Conclusion (C) of the common setup for Moves #1, #2, #3. In addition,

$$(v) \quad \tilde{c} \geq \delta^{-\omega/100} c.$$

Proof.

Step 1. Let $0 < \varepsilon_1 < \varepsilon_2$ be small quantities to be chosen below. We will choose ε_1 very small compared to ε_2 ; we will choose ε_2 very small compared to ε ; we will choose α, η very small compared to ε_1 .

First, we claim that either

$$a \leq \delta^{1-\varepsilon_1}, \quad (14.4.133)$$

or else Conclusion (A) immediately holds, provided we choose α and η sufficiently small depending on ε_1 .

We verify this claim as follows. Suppose that (14.4.133) fails. Then after replacing $(\mathbb{T}, Y)_\delta$ by an $\approx_\delta 1$ refinement, we have that for each $x \in \bigcup_{T \in \mathbb{T}} Y(T)$,

$$\left| B(x, \delta^{1-\varepsilon_1}) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \delta^\eta |B(x, \delta^{1-\varepsilon_1})|.$$

But from this it follows (see Corollary 14.4.48) that Conclusion (A) holds, provided we select $\alpha \leq \omega \varepsilon_1 / 2$ and $\eta > 0$ sufficiently small. Henceforth we shall suppose that (14.4.133) holds.

Step 2. We will regularize the set $\mathbb{T}$. By dyadic pigeonholing and replacing $(\mathbb{T}, Y)_\delta$ by a $\gtrsim (\log 1/\delta)^{-1/\varepsilon_1}$ refinement $(\mathbb{T}_1, Y_1)_\delta$, we can suppose that

- (i) For each scale of the form $\tau_i = \delta^{\varepsilon_1 i}$, $i = 1, \dots, \varepsilon_1^{-1}$, there exists a balanced partitioning cover $\mathbb{T}_{\tau_i}$ of $\mathbb{T}$, and a number μ_i so that $\#(\mathbb{T}_1[T_{\tau_i}])_{Y_1}(x) \sim \mu_i$ for each $T_{\tau_i} \in \mathbb{T}_{\tau_i}$ and each $x \in \bigcup_{T \in \mathbb{T}_1[T_{\tau_i}]} Y_1(T)$.
- (ii) There exists a number μ_{fine} so that for each $T_\rho \in \mathbb{T}_\rho$ and each $x \in \bigcup_{T \in \mathbb{T}_1[T_\rho]} Y_1(T)$, we have $\#(\mathbb{T}_1[T_\rho])_{Y_1}(x) \sim \mu_{\text{fine}}$.
- (iii) For each $T_\rho \in \mathbb{T}_\rho$ and each $x \in \bigcup_{T \in \mathbb{T}_1[T_\rho]} Y_1(T)$, we have

$$(\#\mathbb{T}_1[T_\rho])_{Y_1}(x) \gtrsim (\log 1/\delta)^{-1/\varepsilon_1} (\#\mathbb{T}[T_\rho]_Y(x)).$$

Item (iii) implies that $(\mathbb{T}_1, Y_1)_\delta$ is broad with error $\lesssim_\delta \delta^{-\eta}$ relative to $\mathbb{T}_\rho$.

Since the tubes in $\mathbb{T}_\rho$ are essentially distinct, at most $O(\rho^{-2})$ tubes from $\mathbb{T}_\rho$ can pass through a common point. We conclude that either Conclusion (B) holds, or

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \mu_{\text{fine}}^{-1} \rho^2 \sum_{T \in \mathbb{T}} |Y_1(T)| \geq \delta^{\frac{\omega}{50} + \eta} \mu_{\text{fine}}^{-1} (\#\mathbb{T}) |T|.$$

If μ_{fine} is small, then Conclusion (A) holds. More precisely, if Conclusion (A) fails then

$$\mu_{\text{fine}} \geq \delta^{-\omega + \frac{\omega}{50} + 2\eta + \alpha} \left((\#\mathbb{T}) |T|^{1/2} \right)^\sigma \geq \delta^{-\frac{24}{25}\omega}, \quad (14.4.134)$$

where for the final inequality we used the fact that $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$ to conclude that $(\#\mathbb{T}) |T|^{1/2} \geq \delta^\eta$. Henceforth we shall suppose that (14.4.134) is true.

Step 3. Let $\tilde{c} = \delta^{-\omega/100} c$; a bit later in the argument we will abuse notation and replace $\tilde{c}$ by a number of the form $K \delta^{-\omega/100} c$ for $1 \leq K \lesssim 1$. We will describe a procedure that finds a prism $\tilde{P}$ of dimensions $a \times \tilde{\rho} \tilde{c} \times \tilde{c}$, for some $\tilde{\rho} = \tilde{\rho}(\tilde{P}) \in [\delta^{1-\omega/100}, \delta^{\omega/100}]$, that satisfies some of the properties from Conclusion (C). This procedure is illustrated in Figure 14.4.7. In Step 4, we will iterate this procedure multiple times.

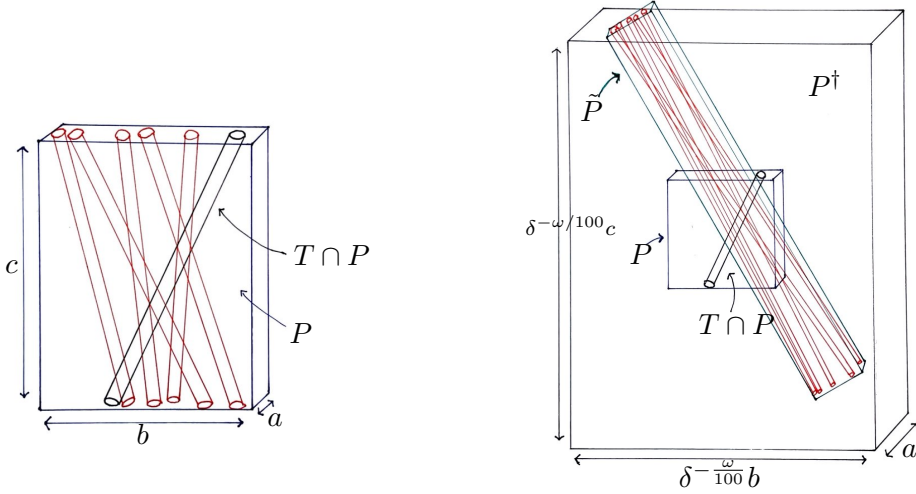


Figure 14.4.7: Finding a prism $\tilde{P}$

Recall that $(\mathbb{T}_1, Y_1)_\delta$ is $\gtrsim_\delta \delta^\eta$ dense. This means that for a typical tube and a typical point $x \in Y_1(T)$, we expect $|B(x, \tilde{c}) \cap Y_1(T)|$ to have size $\gtrsim_\delta \delta^\eta \tilde{c} \delta^2$. This should also hold for any (reasonably dense) sub-shading $Y'(T) \subset Y_1(T)$. The next definition makes this heuristic precise: Given a shading $Y'(T) \subset T$, we define $Y'_{\text{reg}}(T) \subset Y'(T)$ to consist of those points $x \in Y(T)$ for which

$$|B(x, \tilde{c}) \cap Y'(T)| \geq \frac{1}{100} \delta^{2\eta} \tilde{c} \delta^2, \quad (14.4.135)$$

i.e. Y'_{reg} consists of those points where $Y'(T)$ has at density at least $\delta^{2\eta}/100$ at scale $\tilde{c}$. Observe that if $(\mathbb{T}_1, Y')_\delta$ is $\delta^{2\eta}$ dense, then $(\mathbb{T}, Y'_{\text{reg}})$ is a ~ 1 refinement of $(\mathbb{T}_1, Y')$.

With the above definition, we proceed as follows. Let $(\mathbb{T}_1, Y')_\delta$ be any refinement of $(\mathbb{T}_1, Y_1)_\delta$ that satisfies $\sum |Y'(T)| \geq \frac{1}{2} \sum |Y_1(T)|$ (so in particular, $(\mathbb{T}_1, Y')_\delta$ is $\gtrsim_\delta \delta^\eta$ dense).

We claim that we can select a ρ tube T_ρ ; a δ tube $T_{\text{stem}} \in \mathbb{T}_1[T_\rho]$; a prism $P \in \mathcal{P}_{T_\rho}$ (recall Definition 14.4.53, Item (i)) and a number $\kappa \sim 1$ so that the set

$$E = \{x \in Y'(T_{\text{stem}}) \cap Y(P) : \#\mathbb{T}_1[T_\rho]_{Y'_{\text{reg}}}(x) \geq \kappa \mu_{\text{fine}}\} \quad (14.4.136)$$

satisfies

$$|E| \gtrsim \delta^{2\eta} c \delta^2. \quad (14.4.137)$$

To verify this claim, let us temporarily define the refinement $(\mathbb{T}_1, Y'')_\delta$ given by

$$Y''(T) = Y'(T) \cap \{x : \#\mathbb{T}_1[T_\rho]_{Y'_{\text{reg}}}(x) \geq \kappa \mu_{\text{fine}}\},$$

where T_ρ is the unique ρ tube from $\mathbb{T}_\rho$ containing T_{stem} . If $\kappa \sim 1$ is chosen sufficiently small, then $(\mathbb{T}_1, Y'')_\delta$ is a $1/2$ refinement of $(\mathbb{T}_1, Y')_\delta$. By pigeonholing we can select a tube T_{stem} and a point $x \in T_{\text{stem}}$ so that the two adjacent tube segments $T_{\text{seg}}^{(1)}, T_{\text{seg}}^{(2)}$ of length $c/10$ whose intersection contains x (see Figure 14.4.8, Left) satisfy $|Y''(T_{\text{stem}}) \cap T_{\text{seg}}^{(i)}| \gtrsim \delta^{2\eta} c \delta^2$, $i = 1, 2$. Next, select a prism $P \in \mathcal{P}_{T_\rho}$ with $x \in Y(P)$ (by Definition 14.4.53, Item (ii), such P is unique); by Definition 14.4.53, Item (iv), at least one of the segments $T_{\text{seg}}^{(i)}$ must be almost contained in P , in the sense that $T_{\text{seg}}^{(i)} \setminus N_\delta(x) \subset P$ (see Figure 14.4.8, Right), and thus the set E from (14.4.136) contains at least one of the sets $(Y''(T_{\text{stem}}) \cap T_{\text{seg}}^{(i)}) \setminus N_\delta(x)$. This yields the volume bound (14.4.137).

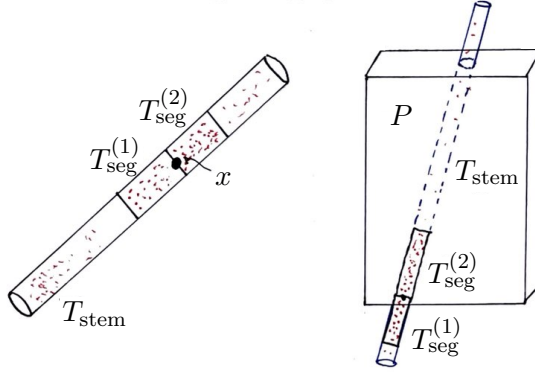


Figure 14.4.8: Left: The two tube segments on either side of the point x . Both tube segments have rich shadings.
Right: Since $x \in P$ and T_{stem} exits P through its “long ends,” at least one of the tube segments must be almost contained in P , in the sense that $T_{\text{seg}}^{(i)} \setminus N_\delta(x) \subset P$.

Since $(\mathbb{T}_1, Y_1)_\delta$ is broad with error $\lesssim_\delta \delta^{-\eta}$ relative to $\mathbb{T}_\rho$, we have that for each $x \in E$, there are $\gtrsim_\delta \mu_{\text{fine}}$ tubes $T \in \mathbb{T}_1[T_\rho]_{Y'_{\text{reg}}}(x)$ with $\angle(T, T_{\text{stem}}) \gtrsim_\delta \delta^{\eta/\beta} \rho$. Since each such tube intersects T_{stem} in a set of dimensions at most $\delta \times \delta \times \frac{\delta}{\delta^{\eta/\beta} \rho}$, we conclude that there is a set $\mathcal{H} = \mathcal{H}(T_{\text{stem}}) \subset \mathbb{T}_1[T_\rho]$ with the following properties:

- $\#\mathcal{H} \gtrsim_\delta (\mu_{\text{fine}} \delta^{-1} c) (\delta^{\eta/\beta} \rho) \delta^{2\eta}$.
- Each $T \in \mathcal{H}$ intersects P , and exits P through the “long ends.”
- Each $T \in \mathcal{H}$ satisfies $|B \cap Y'(T)| \gtrsim \delta^{2\eta} |B \cap T| \sim \delta^{2\eta} \tilde{c} \delta^2$, where B is the ball of radius $\tilde{c}$ with the same center as P .

The second item follows from Definition 14.4.53 Item (iv), plus the fact that $T \in \mathcal{H}$ implies that T and P are both associated to the same ρ tube, and

$Y(T) \cap Y(P) \neq \emptyset$. The third item follows from the fact that the set E from (14.4.136) was defined with respect to the shading Y'_{reg} (recall (14.4.135) for the definition of Y'_{reg}).

As a consequence, using $a \in [\delta, \delta^{1-\varepsilon_1}]$, $\tilde{c} = c\delta^{-\omega/100}$, and (14.4.134), we have

$$\begin{aligned} \sum_{T \in \mathcal{H}} |B \cap Y'(T)| &\gtrsim \left(\mu_{\text{fine}} \delta^{-1+2\eta+\eta/\beta} c\rho \right) \left(\delta^{2\eta} \tilde{c} \delta^2 \right) \\ &\gtrsim \mu_{\text{fine}} \delta^{\eta/\beta+4\eta+\omega/100+\varepsilon_1} (a\rho \tilde{c}^2) \\ &\gtrsim \delta^{-\omega/2} (a\rho \tilde{c}^2). \end{aligned}$$

where B is the ball of radius $\tilde{c}$ with the same center as P . To ensure that the final inequality holds, we select $\eta \leq \beta\omega/100$ and $\varepsilon_1 \leq \omega/100$.

We claim that by pigeonholing, we can select a prism $P^\dagger \supset P$ (the larger prism on the right side of Figure 14.4.7) of dimensions $2a \times 2\rho\tilde{c} \times \tilde{c}$ so that

$$\begin{aligned} \sum_{\substack{T \in \mathcal{H} \\ T \text{ long end } P^\dagger}} |P^\dagger \cap Y'(T)| &\gtrsim \delta^{2\eta} \sum_{\substack{T \in \mathcal{H} \\ T \text{ long end } P^\dagger}} |P^\dagger \cap T| \\ &\gtrsim \delta^{-\omega/4} |P^\dagger|, \end{aligned} \tag{14.4.138}$$

where “ T long end $P^\dagger$ ” means that T exits $P^\dagger$ through its long ends.

To verify this claim, note that for each tube $T \in \mathcal{H}$, there exists at least one $a \times \rho\tilde{c} \times \tilde{c}$ prism $P^\dagger$ so that T exits $P^\dagger$ through its long end. On the other hand, there are only $\lesssim (\tilde{c}/c)^2 = \delta^{-\frac{1}{50}\omega}$ essentially distinct prisms of dimensions $a \times \rho\tilde{c} \times \tilde{c}$ that contain P . The result now follows from pigeonholing. (Note that if T exits a prism $P_1^\dagger$ through its long ends, and if $P_2^\dagger$ is comparable to $P_1^\dagger$, then T exits the 2-fold thickening of $P_2^\dagger$ through its long ends, where the 2-fold thickening is the prism obtained by increasing the two smaller dimensions of $P_2^\dagger$ by a factor of 2. This is why the dimension of our prisms have increased to $2a \times 2\rho\tilde{c} \times \tilde{c}$ at this step).

Apply Corollary 14.4.62 (finding a broad scale) to the set $\mathcal{H}$ with the shading $P^\dagger \cap Y'(T)$, $T \in \mathcal{H}$. We obtain a scale $\tilde{\rho} \in [\delta, 1]$; a set $\mathcal{H}' \subset \mathcal{H}$ (each $T \in \mathcal{H}'$ exits $P^\dagger$ through its long ends); a sub-shading of the shadings $\{P^\dagger \cap Y'(T), T \in \mathcal{H}'\}$, which we will denote by $Y_{P^\dagger}(T)$; and a balanced partitioning cover $\mathbb{T}_{\tilde{\rho}}^{\mathcal{H}'}$ of $\mathcal{H}'$.

Note that $Y_{P^\dagger}(T) \subset P^\dagger \cap Y'(T)$, and the latter set is contained in a tube segment of dimensions comparable to $\delta \times \delta \times \tilde{c}$. In particular, $Y_{P^\dagger}(T)$ is *not* a $\delta^{O(\eta)}$ dense shading of T . However, Corollary 14.4.62 guarantees that the shadings are “relatively” dense inside $T \cap P^\dagger$, thus

$$\sum_{T \in \mathcal{H}'} |Y_{P^\dagger}(T)| \gtrsim_{\delta} \delta^{2\eta} \tilde{c} \delta^2 (\#\mathcal{H}'). \tag{14.4.139}$$

Note that (14.4.138) remains true if the shading $P^\dagger \cap Y'(T)$ on the LHS of (14.4.138) is replaced by $Y_{P^\dagger}(T)$, provided the RHS is weakened by an additional $\approx_\delta 1$ factor, i.e. we have

$$\sum_{T \in \mathcal{H}'} |Y_{P^\dagger}(T)| \gtrsim_\delta \delta^{-\frac{1}{4}\omega} |P^\dagger|. \quad (14.4.140)$$

We claim that

$$\tilde{\rho} \geq \delta^{1-\omega/100}. \quad (14.4.141)$$

We verify this claim as follows. Each point $x \in P^\dagger$ is contained in at most $(\tilde{\rho}/\delta)^2(\tilde{\rho})^{-2\beta}$ of the shadings $\{Y_{P^\dagger}(T), T \in \mathcal{H}\}$. Thus if (14.4.141) failed, then by (14.4.140) we would have

$$|P^\dagger| \geq \left| \bigcup_{T \in \mathcal{H}} P^\dagger \cap Y_{P^\dagger}(T) \right| \gtrsim_\delta \left(\left(\frac{\tilde{\rho}}{\delta} \right)^2 (\tilde{\rho})^{-2\beta} \right)^{-1} \left(\delta^{-\frac{1}{4}\omega} |P^\dagger| \right) \gtrsim \delta^{-\frac{1}{8}\omega} |P^\dagger|,$$

which is impossible. For the final inequality, we used the assumption that $\beta \leq \omega/100$.

By (14.4.139) and pigeonholing, we can select $T_1 \in \mathcal{H}'$ with $|Y_{P^\dagger}(T_1)| \gtrsim_\delta \delta^{2\eta} \tilde{c} \delta^2$. Let $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}^{\mathcal{H}}$ be the (unique) $\tilde{\rho}$ tube with $T_1 \in \mathcal{H}'[T_{\tilde{\rho}}]$. For each $x \in Y_{P^\dagger}(T_1)$, the directions of the tubes in $\mathcal{H}'[T_{\tilde{\rho}}]_{Y_{P^\dagger}}(x)$ are broad with error $\lesssim_\delta 1$ inside the $\tilde{\rho}$ cap centered at $\text{dir}(T_{\tilde{\rho}})$. In particular, the intersection of each of these tubes with $P^\dagger$ is contained in $P^\dagger \cap N_{\tilde{\rho}\tilde{c}}(T_1)$; the latter set is contained in a $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism; call this prism $\tilde{P}$ —this is the green prism in Figure 14.4.7. Note that since each $T \in \mathcal{H}'$ exits $P^\dagger$ through its long ends, each of the tubes $T \in \mathcal{H}'[T_{\tilde{\rho}}]_{Y_{P^\dagger}}(x)$ described above exit $\tilde{P}$ through its long ends.

Applying a standard Cordoba-type L^2 argument², we conclude that if we define

$$\mathbb{T}_{\tilde{P}} = \{T \in \mathcal{H}'[T_{\tilde{\rho}}] : T \cap \tilde{P} \neq \emptyset, T \text{ exits } \tilde{P} \text{ through its long ends}\},$$

and define the shading $Y_{\tilde{P}}(T) = Y_{P^\dagger}(T) \cap \tilde{P}$ (note if $T \in \mathbb{T}_{\tilde{P}}$, T exits $\tilde{P}$ through its long ends, so $Y_{\tilde{P}}(T) = Y_{P^\dagger}(T)$, we rename it to be $Y_{\tilde{P}}(T)$ just for notational convenience), then the set

$$Y(\tilde{P}) = \bigcup_{T \in \mathbb{T}_{\tilde{P}}} Y_{\tilde{P}}(T)$$

satisfies $|Y(\tilde{P})| \gtrsim \delta^{4\eta+4\varepsilon_1} |\tilde{P}|$ (here the $\delta^{4\varepsilon_1}$ loss comes from the fact that the prism $\tilde{P}$ is not a $\delta \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism, but rather a $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism with

²In brief, we select a set of tubes $T' \in \mathcal{H}'$ that make angle $\sim \tilde{\rho}$ with T_1 and intersect T_1 at $\delta/\tilde{\rho}$ separated points; this latter collection of tubes, restricted to the $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism described above, satisfies the Katz-Tao Convex Wolff Axioms with error $\lesssim 1$. Each of these tubes has a shading $Y'(T) \cap \tilde{P}$ that satisfies $|Y'(T) \cap \tilde{P}| \gtrsim \delta^{2\eta} \tilde{c} \delta^2$, and hence the union of these shadings is almost disjoint.

$a \in [\delta, \delta^{1-\varepsilon_1}]$). Furthermore, for each $x \in Y(\tilde{P})$, the set of unit vectors $\{\text{dir}(T) : T \in (\mathbb{T}_{\tilde{P}})_{Y_{\tilde{P}}}(x)\}$ is broad with error $\lesssim_\delta 1$ inside the $\tilde{\rho}$ -cap $B(\text{dir}(\tilde{P}), \tilde{\rho})$. **Audit status (checked 31 August 2026).** The footnote identifies the intended selection, but does not prove the displayed density. A complete proof must (i) extract from the three-dimensional shaded set on T_1 a $\delta/\tilde{\rho}$ -separated set of stem parameters of the required cardinality; (ii) control repetitions when one transverse tube meets several selected parameters; (iii) prove the Katz–Tao bound for the anisotropically truncated $\delta \times \delta \times \tilde{c}$ tube segments; and (iv) run the shaded L^2 calculation with the affine Jacobian of the $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ parent. None of the preceding lemmas supplies that four-part implication. We denote it

the local square-grain shaded Cordoba estimate above. (WZ–A3)

The rest of Lemma 14.4.74 is conditional on WZ–A3.

Step 4. We summarize the conclusion from Step 3. Given a refinement $(\mathbb{T}_1, Y')_\delta$ of $(\mathbb{T}_1, Y_1)_\delta$ that satisfies $\sum |Y'(T)| \geq \frac{1}{2} \sum |Y_1(T)|$, we have located the following objects:

- A scale $\tilde{\rho}$.
- A $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism $\tilde{P}$ and a shading $Y(\tilde{P})$ on this prism.
- A set of tubes $\mathbb{T}_{\tilde{P}}$ and a shading $Y_{\tilde{P}}(T) \subset Y'(T) \cap \tilde{P}$ on these tubes.

These objects have the following properties:

- Each $T \in \mathbb{T}_{\tilde{P}}$ exits $\tilde{P}$ through its long ends, in the sense of Figure 14.4.6, Left.
- $Y(\tilde{P}) = \bigcup_{T \in \mathbb{T}_{\tilde{P}}} Y_{\tilde{P}}(T)$, and $|Y(\tilde{P})| \gtrsim_\delta \delta^{4\eta+4\varepsilon_1} |\tilde{P}|$.
- For each $x \in Y(P)$, the tubes in $(\mathbb{T}_{\tilde{P}})_{Y_{\tilde{P}}}(x)$ point in directions that are broad with error $\lesssim_\delta 1$ inside the $\tilde{\rho}$ cap $B(\text{dir}(\tilde{P}), \tilde{\rho})$.

We will now iteratively apply the argument from Step 3. We begin by setting $(\mathbb{T}, Y') = (\mathbb{T}, Y_1)$ and $\tilde{\mathcal{P}}_0 = \emptyset$. As long as $\sum |Y'(T)| \geq \frac{1}{2} \sum |Y_1(T)|$, we proceed as follows:

- Apply the argument from Step 3.
- Place the prism $\tilde{P}$ located in Step 3 into the multiset $\tilde{\mathcal{P}}_0$ (i.e. if the prism is already present, then we add another copy).
- For each $T \in \mathbb{T}_{\tilde{P}}$, replace the shading $Y'(T)$ with $Y'(T) \setminus Y_{\tilde{P}}(T)$.

We repeat the above steps until $\sum |Y'(T)| < \frac{1}{2} \sum |Y_1(T)|$, at which point we halt.

Let us examine the output from the above procedure. First, we have

$$\sum_{\tilde{P} \in \tilde{\mathcal{P}}_0} \sum_{T \in \mathbb{T}_{\tilde{P}}} |Y_{\tilde{P}}(T)| \geq \frac{1}{2} \sum_{T \in \mathbb{T}} |Y_1(T)| \gtrsim_{\delta} \sum_{T \in \mathbb{T}} |Y(T)|. \quad (14.4.142)$$

After dyadic pigeonholing, we can select a multiset $\tilde{\mathcal{P}}_1 \subset \tilde{\mathcal{P}}_0$ so that each $\tilde{P} \in \tilde{\mathcal{P}}_1$ has a common value of $\tilde{\rho}$ (up to a factor of 2). Abusing notation slightly, we will denote this value by $\tilde{\rho}$. We will choose $\tilde{\mathcal{P}}_1$ so that the bound (14.4.142) (the first and final terms) remains true with $\tilde{\mathcal{P}}_1$ in place of $\tilde{\mathcal{P}}_0$.

For each $T \in \mathbb{T}_1$, define $Y_2(T) = \bigcup_{\tilde{P}} Y_{\tilde{P}}(T)$, where the union is taken over those $\tilde{P} \in \tilde{\mathcal{P}}_1$ with $T \in \mathbb{T}_{\tilde{P}}$. For notational consistency, define $\mathbb{T}_2 = \mathbb{T}_1$. Then $(\mathbb{T}_2, Y_2)_{\delta}$ is an $\approx_{\delta} 1$ refinement of $(\mathbb{T}_1, Y_1)_{\delta}$.

Step 5. We will summarize the conclusion from Step 4. We have located the following objects:

- A $\approx_{\delta} 1$ refinement $(\mathbb{T}_2, Y_2)_{\delta}$ of $(\mathbb{T}_1, Y_1)_{\delta}$, which in turn is an $\approx_{\delta} 1$ refinement of $(\mathbb{T}, Y)_{\delta}$.
- A scale $\tilde{\rho} \geq \delta^{1-\omega/100}$.
- A multiset $\tilde{\mathcal{P}}_1$ of $2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prisms, and a shading $Y_1(\tilde{P})$ on these prisms (In Step 4 this shading was called $Y(\tilde{P})$). Note that the prisms in $\tilde{\mathcal{P}}_1$ might not be essentially distinct.
- For each prism $\tilde{P} \in \tilde{\mathcal{P}}_1$, a set of tubes $\mathbb{T}_{\tilde{P}} \subset \mathbb{T}_2$.

These objects have the following properties:

- For each $\tilde{P} \in \tilde{\mathcal{P}}_1$, each $T \in \mathbb{T}_{\tilde{P}}$ exits $\tilde{P}$ through the long ends.
- For each $\tilde{P} \in \tilde{\mathcal{P}}_1$, we have $Y_1(\tilde{P}) = \tilde{P} \cap \bigcup_{T \in \mathbb{T}_{\tilde{P}}} Y_2(T)$, and $|Y_1(\tilde{P})| \gtrsim_{\delta} \delta^{4\eta+4\varepsilon_1} |\tilde{P}|$.
- For each $\tilde{P} \in \tilde{\mathcal{P}}_1$ and for each $x \in Y_1(\tilde{P})$, the tubes in $(\mathbb{T}_{\tilde{P}})_{Y_2}(x)$ point in directions that are broad with error $\lesssim_{\delta} 1$ inside the $\tilde{\rho}$ cap $B(\text{dir}(\tilde{P}), \tilde{\rho})$.

Let us compare the above items to Conclusion (C) of Lemma 14.4.74. We have that Items (i) and (iv) of Conclusion (C) are currently satisfied. We will work towards satisfying the other Items.

First, the prisms in $\tilde{\mathcal{P}}_1$ might not be essentially distinct. This is not a minor failure fixable by a ~ 1 refinement, but instead a dramatic failure — it is possible that a very large number of prisms from $\tilde{\mathcal{P}}_1$ are all pairwise comparable, or even identical. We can fix this problem by merging comparable

$2a \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prisms in $\tilde{\mathcal{P}}_1$ into a single $4a \times \tilde{\rho}(2\tilde{c}) \times (2\tilde{c})$ prism. We will refer to this new, post-merger set of prisms as $\tilde{\mathcal{P}}_2$. Our shading $Y_2(\tilde{\mathcal{P}})$ on our newly constructed prisms is given by the union of the shadings of the corresponding prisms from $\tilde{\mathcal{P}}_1$, and the set $\mathbb{T}_{\tilde{\mathcal{P}}} \subset \mathbb{T}_2$ is the union of the sets $\{\mathbb{T}_{\tilde{\mathcal{P}}_1}\}$ from the corresponding prisms from $\tilde{\mathcal{P}}_1$.

Item (a) from the start of Step 5 might no longer hold for our newly constructed prisms $\tilde{\mathcal{P}}_2$, but this is a minor failure — we can restore it by replacing each $4a \times 2\tilde{\rho}(2\tilde{c}) \times (2\tilde{c})$ prism by the prism of dimensions $100a \times 100\tilde{\rho}\tilde{c} \times 2\tilde{c}$ with the same center and axes (see Figure 14.4.9). Annoyingly, this might destroy the property that the prisms are essentially distinct, but this time, this is only a minor failure — essential distinctness can be restored by a ~ 1 refinement of the prisms (this refinement induces a ~ 1 refinement of the shading Y_2 on $\mathbb{T}_2$). Denote the new set of prisms created through this process by $\tilde{\mathcal{P}}_3$. Abusing notation slightly, we will redefine the quantities $\tilde{c}$, and $\tilde{\rho}$ and let $\tilde{a} = 100a$ (increasing each by a ~ 1 multiplicative factor) so that the prisms in $\tilde{\mathcal{P}}_3$ still have dimensions $\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}$.

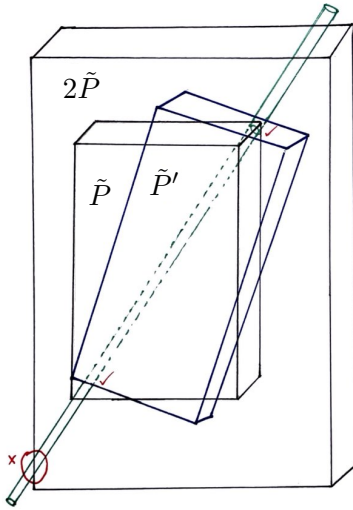


Figure 14.4.9: The prisms $\tilde{\mathcal{P}}$ and $\tilde{\mathcal{P}}'$ are comparable, and thus both are replaced by a common $4a \times \tilde{\rho}(2\tilde{c}) \times (2\tilde{c})$ prism (which happens to be $2\tilde{\mathcal{P}}$, i.e. the 2-fold dilate of $\tilde{\mathcal{P}}$). This creates a problem (circled in red): A tube (green) that exits the prism $\tilde{\mathcal{P}}'$ through its long ends might fail to exit $2\tilde{\mathcal{P}}$ through its long ends. However, this problem can be fixed by replacing the $4a \times \tilde{\rho}(2\tilde{c}) \times (2\tilde{c})$ prism by a slightly wider and thicker (but not taller) prism.

Note that Item (b) above continues to hold for our newly constructed prisms and their associated shading. Crucially, Item (c) also continues to hold; this follows from Lemma 14.4.64 (a union of sets of broad vectors is broad). More specifically, the sets of broad vectors through each point are disjoint

because in Step 4, for each $T \in \mathbb{T}_{\tilde{P}}$, we have replaced the shading $Y'(T)$ with $Y'(T) \setminus Y_{\tilde{P}}(T)$, so through each point x , there is at most one $\tilde{P}$ such that $x \in Y_{\tilde{P}}(T)$. Since the sets are disjoint, their union is a set instead of multi-set, so Lemma 14.4.64 implies that the union (as a set) of sets of broad vectors is broad.

Step 6. In Step 5 we constructed a set $\tilde{\mathcal{P}}_3$ of essentially distinct $\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prisms, and a shading Y_3 on these prisms. We shall refer to this pair as $(\tilde{\mathcal{P}}_3, Y_3)_{\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}}$. For each $\tilde{P} \in \tilde{\mathcal{P}}_3$, we have a set $\mathbb{T}_{\tilde{P}} \subset \mathbb{T}_2$; each of these tubes exits $\tilde{P}$ through its long ends.

Our current task is to further refine the pair $(\tilde{\mathcal{P}}_3, Y_3)_{\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}}$ and $(\mathbb{T}_2, Y_2)_\delta$ to more closely match Conclusion (C) of Lemma 14.4.74. Item (ii) of Conclusion (C) refers to a partitioning cover $\mathbb{T}_{\tilde{\rho}}$ of $\mathbb{T}'$. We will construct this as follows. To begin, let $\{T_{\tilde{\rho}}\}$ be a set of $\tilde{\rho}$ tubes with the following properties:

- Every δ tube is contained in at least one tube from $\{T_{\tilde{\rho}}\}$.
- For every $\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}$ prism $\tilde{P}$, there is at least one $\tilde{\rho}$ tube $T_{\tilde{\rho}} \in \{T_{\tilde{\rho}}\}$ with $\tilde{P} \subset T_{\tilde{\rho}}$ and $\angle(\text{dir}(\tilde{P}), \text{dir}(T_{\tilde{\rho}})) \leq 2\tilde{\rho}$.
- The tubes in $\{T_{\tilde{\rho}}\}$ are *weakly* essentially distinct, in the following sense: for each fixed $T_{\tilde{\rho}} \in \{T_{\tilde{\rho}}\}$, there are $O(1)$ other tubes from $\{T_{\tilde{\rho}}\}$ that are comparable to $T_{\tilde{\rho}}$.

Next, by pigeonholing the set $\{T_{\tilde{\rho}}\}$ by a $O(1)$ factor, we can select a set $\mathbb{T}_{\tilde{\rho}} \subset \{T_{\tilde{\rho}}\}$ that is *strongly* essentially distinct, in the following sense: for each pair of distinct tubes $T_{\tilde{\rho}}, T'_{\tilde{\rho}}$ from $\mathbb{T}_{\tilde{\rho}}$, we have that $N_{100\tilde{\rho}}(T_{\tilde{\rho}}) \cap N_{100\tilde{\rho}}(T'_{\tilde{\rho}})$ has diameter at most $1/2$, and in particular no δ tube can be contained in both $N_{100\tilde{\rho}}(T_{\tilde{\rho}})$ and $N_{100\tilde{\rho}}(T'_{\tilde{\rho}})$. We will select the set $\mathbb{T}_{\tilde{\rho}}$ so that the following properties hold:

- If we define $\mathbb{T}_4$ to be the set of tubes $T \in \mathbb{T}_2$ contained in some $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$, i.e. $\mathbb{T}_4 = \bigcup_{T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}} \mathbb{T}_2[T_{\tilde{\rho}}]$, and define Y_4 to be the restriction of Y_2 to $\mathbb{T}_4$, then $(\mathbb{T}_4, Y_4)_\delta$ is a ~ 1 refinement of $(\mathbb{T}_2, Y_2)_\delta$.
- Similarly to the previous item, if we define $\tilde{\mathcal{P}}_4$ to be the set of prisms $\tilde{P} \in \tilde{\mathcal{P}}_3$ with the property that there exists $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$ with $\tilde{P} \subset T_{\tilde{\rho}}$ and $\angle(\text{dir}(\tilde{P}), \text{dir}(T_{\tilde{\rho}})) \leq 2\tilde{\rho}$, and define

$$Y_4(\tilde{P}) = Y_3(\tilde{P}) \cap \bigcup_{T \in \mathbb{T}_{\tilde{P}} \cap \mathbb{T}_4} Y_4(T), \quad (14.4.143)$$

then $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}}$ is a $\gtrsim 1$ refinement of $(\tilde{\mathcal{P}}_3, Y_3)_{\tilde{a} \times \tilde{\rho}\tilde{c} \times \tilde{c}}$.

- (a) A consequence of Item (i) is that $\mathbb{T}_{\tilde{\rho}}$ is a partitioning cover of $\mathbb{T}_4$, and in fact more is true: the sets $\{N_{100\tilde{\rho}}(T_{\tilde{\rho}}) : T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}\}$ form a partitioning cover of $\mathbb{T}_4$.
- (b) A consequence of Item (ii) is that for each $\tilde{P} \in \tilde{\mathcal{P}}_4$, there is a unique $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$ that satisfies the two properties $\tilde{P} \subset T_{\tilde{\rho}}$ and $\angle(\text{dir}(\tilde{P}), \text{dir}(T_{\tilde{\rho}})) \leq 2\tilde{\rho}$. This induces a partition

$$\tilde{\mathcal{P}}_4 = \bigsqcup_{\mathbb{T}_{\tilde{\rho}}} (\tilde{\mathcal{P}}_4)_{T_{\tilde{\rho}}}. \quad (14.4.144)$$

(Compare Definition 14.4.53, Item (i).)

- (c) A consequence of Items (i) and (ii) is that if $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$, $\tilde{P} \in (\tilde{\mathcal{P}}_4)_{T_{\tilde{\rho}}}$, and $T \in \tilde{\mathcal{P}} \cap \mathbb{T}_4$, then $T \in \mathbb{T}_4[T_{\tilde{\rho}}]$. This is because T exits $\tilde{P}$ through its long ends, and hence $\angle(\text{dir}(T), \text{dir}(\tilde{P})) \leq 10\tilde{\rho}$.

Our next task is to estimate the quantity $\sum_{\tilde{P} \in \tilde{\mathcal{P}}_4} |\tilde{P}|$. Let τ_i be the scale from Step 2 satisfying $\delta^{\varepsilon_1} \rho \leq \tau_i < \rho$. Since $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ is $\gtrsim_{\delta} \delta^{4\eta+4\varepsilon_1}$ dense, we have

$$\begin{aligned} \sum_{\tilde{P} \in \tilde{\mathcal{P}}_4} |\tilde{P}| &\lesssim_{\delta} \delta^{-4\eta-4\varepsilon_1} \sum_{\tilde{P} \in \tilde{\mathcal{P}}_4} |Y_4(\tilde{P})| \\ &= \delta^{-4\eta-4\varepsilon_1} \sum_{T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}} \left| \bigsqcup_{\tilde{P} \in (\tilde{\mathcal{P}}_4)_{T_{\tilde{\rho}}}} Y_4(\tilde{P}) \right| \\ &\leq \delta^{-4\eta-4\varepsilon_1} \sum_{T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}} \left| \bigcup_{T \in \mathbb{T}_4[T_{\tilde{\rho}}]} Y_4(T) \right| \\ &\lesssim \delta^{-4\eta-4\varepsilon_1} \sum_{T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}} \left(\mu_i^{-1} \sum_{T \in \mathbb{T}_1[T_{\tilde{\rho}}]} |Y_1(T)| \right) \\ &= \delta^{-4\eta-4\varepsilon_1} \mu_i^{-1} \sum_{T \in \mathbb{T}_1} |Y_1(T)| \\ &\lesssim \delta^{-4\eta-4\varepsilon_1} \mu_i^{-1} \sum_{T \in \mathbb{T}_2} |Y_2(T)|. \end{aligned} \quad (14.4.145)$$

For the third line we used (14.4.143). For the fourth line, we used the fact that each $\tilde{\rho}$ tube contains some τ_i -tube and Item (i) in Step 2. For the last line, we used (14.4.142) and the definition of $Y_2(T) = \bigcup_{\tilde{P}} Y_{\tilde{P}}(T)$.

Step 7.

We would like to show that after a suitable refinement, $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ is a robustly δ^{ε} -dense two-scale grains decomposition of $(\mathbb{T}_4, Y_4)_{\delta}$ wrt $\mathbb{T}_{\tilde{\rho}}$, in the sense of Definition 14.4.53. Currently, the biggest obstacle is Item (ii)

from Definition 14.4.53. In particular, it need not be the case that the sets $\{Y_4(\tilde{P}): \tilde{P} \in (\tilde{\mathcal{P}}_4)_{T_{\tilde{\rho}}}\}$ are pairwise disjoint.

We will fix this problem as follows. We claim that either Conclusion (A) of Lemma 14.4.74 is true (and thus we are done), or there exists a refinement $(\mathbb{T}_5, Y_5)_\delta$ of $(\mathbb{T}_4, Y_4)_\delta$, and a refinement $(\tilde{\mathcal{P}}_5, Y_5)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ of $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ with the following properties:

- For each $T_\rho \in \mathbb{T}_\rho$, the sets $\{Y_5(\tilde{P}): \tilde{P} \in (\tilde{\mathcal{P}}_5)_{T_{\tilde{\rho}}}\}$ are pairwise disjoint (here $(\tilde{\mathcal{P}}_5)_{T_{\tilde{\rho}}} = \tilde{\mathcal{P}}_5 \cap (\tilde{\mathcal{P}}_4)_{T_{\tilde{\rho}}}$; recall (14.4.144)).
- $(\tilde{\mathcal{P}}_5, Y_5)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ is a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement of $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$.
- The pair $(\mathbb{T}_5, Y_5)_\delta$ is a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement of $(\mathbb{T}_4, Y_4)_\delta$, where $\mathbb{T}_5 = \mathbb{T}_4$, and the shading Y_5 is given by

$$Y_5(T) = Y_4(T) \cap \bigcup_{\substack{\tilde{P} \in (\tilde{\mathcal{P}}_5)_{T_{\tilde{\rho}}} \\ T \in \mathbb{T}_{\tilde{P}} \cap \mathbb{T}_4}} Y_5(\tilde{P}), \quad (14.4.146)$$

where $T_{\tilde{\rho}}$ is the unique $\tilde{\rho}$ tube containing T .

- For each $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$, we have

$$\bigcup_{T \in \mathbb{T}_5[T_{\tilde{\rho}}]} Y_5(T) = \bigsqcup_{\tilde{P} \in (\tilde{\mathcal{P}}_5)_{T_{\tilde{\rho}}}} Y_5(\tilde{P}). \quad (14.4.147)$$

We will prove this claim in Step 8 below. Let us accept this claim for the moment.

The pair $(\tilde{\mathcal{P}}_5, Y_5)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ and $(\mathbb{T}_5, Y_5)_\delta$ now satisfy Items (i), (ii), and (iv) from Definition 14.4.53. Items (i) and (ii) are immediate. We can verify Item (iv) as follows: If T and $\tilde{P}$ are associated to a common ρ tube, and if $Y_5(T) \cap Y_5(\tilde{P}) \neq \emptyset$, then we must have $T \in \mathbb{T}_4 \cap \mathbb{T}_{\tilde{P}}$, and hence we have that T exits $\tilde{P}$ through its long end, and also $Y_5(T) \cap \tilde{P} \subset Y_5(\tilde{P})$ (this follows from the definition of the shading Y_5 from (14.4.146)), as desired.

It remains to obtain Item (iii) from Definition 14.4.53. By dyadic pigeonholing, there is a number μ ; a set $\mathbb{T}'_\rho$; and an $\approx_\delta 1$ refinement $(\mathbb{T}_6, Y_6)_\delta$ of $(\mathbb{T}_5, Y_5)_\delta$ so that the following holds:

- $\mathbb{T}'_\rho$ is a balanced partitioning cover of $\mathbb{T}_6$.
- For each $T_{\tilde{\rho}} \in \mathbb{T}'_\rho$ and each $x \in \bigcup_{T \in \mathbb{T}_6[T_{\tilde{\rho}}]} Y_6(T)$, we have $\#((\mathbb{T}_6[T_{\tilde{\rho}}])_{Y_6}(x)) \sim \mu$.

Let $\tilde{\mathcal{P}}_6 = \bigcup_{T_{\tilde{\rho}} \in \mathbb{T}'_{\tilde{\rho}}} (\tilde{\mathcal{P}}_5)_{T_{\tilde{\rho}}}$ and let $Y_6(\tilde{P}) \subset Y_5(\tilde{P})$ be the shading so that (14.4.147) continues to hold with $(\mathbb{T}_6, Y_6)_{\delta}$ in place of $(\mathbb{T}_5, Y_5)_{\delta}$, and $(\tilde{\mathcal{P}}_6, Y_6)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ in place of $(\tilde{\mathcal{P}}_5, Y_5)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$.

The triple $(\mathbb{T}_6, Y_6)_{\delta}$, $(\tilde{\mathcal{P}}_6, Y_6)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$, and $\mathbb{T}_{\tilde{\rho}}$ continue to satisfy Items (i), (ii), and (iv) from Definition 14.4.53. To verify Item (iii), we need to estimate the density of the shading on $(\tilde{\mathcal{P}}_6, Y_6)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$. From (14.4.145), we have

$$\sum_{\tilde{P} \in \tilde{\mathcal{P}}_6} |\tilde{P}| \lesssim_{\delta} \delta^{-4\eta-4\varepsilon_1} \mu_i^{-1} \sum_{T \in \mathbb{T}_2} |Y_2(T)|. \quad (14.4.148)$$

Note that $\mu \lesssim \mu_i \delta^{-2\varepsilon_1}$ (recall that μ_i is the multiplicity associated to scale τ_i , which was chosen in Step 6). Thus we can compute

$$\begin{aligned} \sum_{\tilde{P} \in \tilde{\mathcal{P}}_6} |Y_6(\tilde{P})| &= \sum_{T_{\tilde{\rho}} \in \mathbb{T}'_{\tilde{\rho}}} \left| \bigsqcup_{\tilde{P} \in (\tilde{\mathcal{P}}_6)_{T_{\tilde{\rho}}}} Y_6(\tilde{P}) \right| \\ &= \sum_{T_{\tilde{\rho}} \in \mathbb{T}'_{\tilde{\rho}}} \left| \bigcup_{T \in \mathbb{T}_6[T_{\tilde{\rho}}]} Y_6(T) \right| \\ &\gtrsim \sum_{T_{\tilde{\rho}} \in \mathbb{T}'_{\tilde{\rho}}} \left(\mu^{-1} \sum_{T \in \mathbb{T}_6[T_{\tilde{\rho}}]} |Y_6(T)| \right) \\ &\gtrsim \mu_i^{-1} \delta^{2\varepsilon_1} \sum_{T \in \mathbb{T}_6} |Y_6(T)| \\ &\gtrsim_{\delta} \mu_i^{-1} \delta^{\varepsilon_2+2\varepsilon_1} \sum_{T \in \mathbb{T}_6} |Y_2(T)|. \end{aligned} \quad (14.4.149)$$

Comparing (14.4.148) and (14.4.149), we conclude that $(\tilde{\mathcal{P}}_6, Y_6)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ is $\gtrsim_{\delta} \delta^{4\eta+6\varepsilon_1+\varepsilon_2} \geq \delta^{2\varepsilon_2}$ dense. We now select ε_2 sufficiently small (depending on ε) so that this quantity is $\geq \delta^{\varepsilon}$. We conclude that Conclusion (C) of Lemma 14.4.74 holds.

This concludes the proof of Lemma 14.4.74, except that we must still prove the Claim stated at the beginning of Step 7. We will do this below.

Step 8. Our final task is to prove the Claim from Step 7. For notational convenience, we will abuse notation and rename the set $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ as $(\tilde{\mathcal{P}}, Y)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$. Informally, the idea is as follows: If the shadings $\{Y(\tilde{P}) : \tilde{P} \in \tilde{\mathcal{P}}_{T_{\tilde{\rho}}}\}$ have small overlap, then we can refine these shadings to be disjoint. On the other hand, if the shadings have large overlap, then since the prisms in $\tilde{\mathcal{P}}_{T_{\tilde{\rho}}}$ are essentially distinct and all satisfy $\angle(\text{dir}(\tilde{P}), \text{dir}(T_{\tilde{\rho}})) \lesssim \tilde{\rho}$, we have that the prisms in $\tilde{\mathcal{P}}_Y(x)$ (i.e. the prisms passing through a typical point) must have differing tangent planes (i.e. there must exist prisms $\tilde{P}, \tilde{P}' \in \tilde{\mathcal{P}}_Y(x)$ for which $\angle(\Pi(\tilde{P}), \Pi(\tilde{P}'))$ is large). We then apply Lemma 14.4.39 to show that the thickened neighbourhood of a typical prism in $\tilde{\mathcal{P}}$ has large intersection with $\bigcup_{\tilde{P} \in \tilde{\mathcal{P}}} Y(\tilde{P})$, and this in turn means that the thickened neighbourhood of

a typical tube in $\mathbb{T}_4$ has large intersection with $\bigcup_{\mathbb{T}} Y(T)$. By Corollary 14.4.48, this yields Conclusion (A) of Lemma 14.4.74. We now turn to the details.

Using Lemma 14.4.38 (every shading has a regular sub-shading), we may select a $\gtrsim 1$ refinement $(\tilde{\mathcal{P}}', Y')_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ with the property that each shading $Y'(\tilde{P})$, $\tilde{P} \in \tilde{\mathcal{P}}'$ is regular (recall Definition 14.4.37) and satisfies $|Y'(\tilde{P})| \gtrsim_{\delta} \delta^{4\eta+4\varepsilon_1} |\tilde{P}|$.

After dyadic pigeonholing, we may suppose there exists a number ν and a $\gtrsim_{\delta}$ refinement $(\tilde{\mathcal{P}}'', Y'')_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ of $(\tilde{\mathcal{P}}', Y')_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ so that for each $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$ and each point $x \in \bigcup_{\tilde{P} \in (\tilde{\mathcal{P}}'')_{T_{\tilde{\rho}}}} Y''(\tilde{P})$, we have $\#((\tilde{\mathcal{P}}''_{T_{\tilde{\rho}}})_{Y''}(x)) \sim \nu$.

First, we will consider the case where

$$\nu \geq \delta^{-\varepsilon_2}. \quad (14.4.150)$$

We will show that Conclusion (A) of Lemma 14.4.74 is true for a suitably chosen value of α . Observe that the prisms in $((\tilde{\mathcal{P}}')_{T_{\tilde{\rho}}})_{Y'}(x)$ are essentially distinct, and they all satisfy $\angle(\text{dir}(T_{\tilde{\rho}}), \text{dir}(\tilde{P})) \leq 2\tilde{\rho}$. Furthermore, they all (by definition) pass through the common point x . Thus for each point $x \in \bigcup_{\tilde{P} \in (\tilde{\mathcal{P}}'')_{T_{\tilde{\rho}}}} Y''(\tilde{P})$, there must exist a pair of prisms $\tilde{P}, \tilde{P}'$ from this set with

$$\angle(\Pi(\tilde{P}^{T_{\tilde{\rho}}}), \Pi((\tilde{P}')^{T_{\tilde{\rho}}})) \gtrsim \nu^{1/2}(\tilde{a}/(\tilde{\rho}\tilde{c})).$$

(For comparison, $\Pi(\tilde{P}^{T_{\tilde{\rho}}})$ and $\Pi((\tilde{P}')^{T_{\tilde{\rho}}})$ are defined up to uncertainty $\tilde{a}/(\tilde{\rho}\tilde{c})$).

From the above discussion, we see that for each $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$, each $\tilde{P}_0 \in \tilde{\mathcal{P}}''_{T_{\tilde{\rho}}}$ and each $x \in Y''(\tilde{P}_0)$, we have

$$\sup_{\tilde{P} \in ((\tilde{\mathcal{P}}'')_{T_{\tilde{\rho}}})_{Y''}(x)} \angle(\Pi(\tilde{P}_0^{T_{\tilde{\rho}}}), \Pi(\tilde{P}^{T_{\tilde{\rho}}})) \gtrsim \nu^{1/2} \tilde{a}/(\tilde{\rho}\tilde{c}),$$

and thus

$$\inf_{x \in Y''(\tilde{P}_0)} \sup_{\tilde{P} \in ((\tilde{\mathcal{P}}'')_{T_{\tilde{\rho}}})_{Y''}(x)} \angle(\Pi(\tilde{P}_0^{T_{\tilde{\rho}}}), \Pi(\tilde{P}^{T_{\tilde{\rho}}})) \gtrsim \nu^{1/2} \tilde{a}/(\tilde{\rho}\tilde{c}).$$

But this is precisely the condition we need to apply Lemma 14.4.39 with $\lambda \approx_{\delta} \delta^{4\eta+4\varepsilon_1}$. Let $\tilde{\mathcal{P}}'''$ be the set of those prisms $\tilde{P}_0 \in \tilde{\mathcal{P}}''$ satisfying $|Y''(\tilde{P}_0)| \gtrsim_{\delta} \delta^{4\eta+4\varepsilon_1}$. Undoing the scaling, we conclude that for each $\tilde{P}_0 \in \tilde{\mathcal{P}}'''$ we have

$$\left| N_{\nu^{1/2}\tilde{a}}(\tilde{P}_0) \cap \bigcup_{\tilde{P} \in \tilde{\mathcal{P}}} Y(\tilde{P}) \right| \gtrsim_{\delta} \delta^{16\eta+16\varepsilon_1} |N_{\nu^{1/2}\tilde{a}}(\tilde{P}_0)|. \quad (14.4.151)$$

But this means that after refining $(\mathbb{T}_4, Y_4)_{\delta}$ by an $\approx_{\delta} 1$ factor, there is a pair $(\mathbb{T}'_4, Y'_4)_{\delta}$ so that for each $x \in \bigcup_{T \in \mathbb{T}'_4} Y'_4(T)$, we have

$$\left| B(x, \nu^{1/2}\tilde{a}) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_{\delta} \delta^{O(\eta+\varepsilon_1)} |B(x, \nu^{1/2}\tilde{a})|. \quad (14.4.152)$$

By Corollary 14.4.48 (and using (14.4.150)), we conclude that Conclusion (A) holds, provided $\alpha \leq \omega \varepsilon_2 / 2$, and provided ε_1 and η are selected sufficiently small (depending on ω, ε_2 , and the implicit constant on the RHS of (14.4.152)).

Finally, we will consider the case where (14.4.150) fails, i.e.

$$\nu \leq \delta^{-\varepsilon_2}. \quad (14.4.153)$$

This means that for each $T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}$, the sets $\{Y''(\tilde{P}) : \tilde{P} \in (\tilde{\mathcal{P}}'')_{T_{\tilde{\rho}}}\}$ are $\leq \delta^{-\varepsilon_2}$ overlapping. By pigeonholing, we can select a refinement $(\tilde{\mathcal{P}}_5, Y_5)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ of $(\tilde{\mathcal{P}}_4, Y_4)_{\tilde{a} \times \tilde{\rho} \tilde{c} \times \tilde{c}}$ satisfying the four Items listed in Step 7. $\square$

14.4.7.3 Move #3: Replacing grains with wider grains with small C_{KT-CW}^{loc}

Lemma 14.4.75. *We assume the Common setup for Moves #1, #2, #3: Hypotheses from Section 14.4.6.4. Then at least one of the following must hold.*

Suppose that $\mathcal{E}(\sigma, \omega)$ is true and let $\varepsilon > 0$. Then there exists $\alpha, \eta, c > 0$ so that the following holds for all $0 < \delta \leq \rho \leq 1$, and all $\delta \leq a \leq b \leq c$ with $b/c = \rho$.

Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Let $\mathbb{T}_\rho$ be a balanced partitioning cover of $\mathbb{T}$, and suppose that $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$ relative to $\mathbb{T}_\rho$. Let $(\mathcal{P}, Y)_{a \times b \times c}$ be a robustly δ^η -dense two-scale grains decomposition of $(\mathbb{T}, Y)_\delta$ wrt $\mathbb{T}_\rho$.

Then at least one of the following must hold.

- (A) *Conclusion (A) of the common setup for Moves #1, #2, #3.*
- (B) *Conclusion (B) of the common setup for Moves #1, #2, #3. In addition,*
 - (iv) $C_{KT-CW}^{\text{loc}}(\mathcal{P}') \leq \delta^{-\zeta}$.
- (C) *Conclusion (C) of the common setup for Moves #1, #2, #3. In addition,*
 - (v) $\tilde{c} \geq c, \delta^{-\zeta/400} \rho \leq \tilde{\rho} \leq 1$.

Proof. Step 1. Choose small quantities

$$0 < \varepsilon_1 < \varepsilon_2 < \varepsilon_3 < \varepsilon_4 < \varepsilon.$$

Quantitatively, choose ε_i sufficiently small in terms of ε_{i+1} for $1 \leq i \leq 3$, then choose ε_4 in terms of ε , and finally choose α, η in terms of ε_1 .

First, we may suppose that

$$a \leq \delta^{1-\varepsilon_1}, \quad (14.4.154)$$

or else Conclusion (A) immediately holds, provided we choose α and η sufficiently small depending on ε_1 .

Here are the details. If $a > \delta^{1-\varepsilon_1}$, use the robust δ^η density of the grains and then discard dyadic classes of grain and tube multiplicity. The defining disjointness and incidence properties of a two-scale grains decomposition imply that the resulting pair $(\mathbb{T}_0, Y_0)_\delta$ is an $\approx_\delta 1$ refinement and that, for every $x \in \bigcup_{T \in \mathbb{T}_0} Y_0(T)$,

$$\left| B(x, a) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_\delta \delta^{2\eta} |B(x, a)|.$$

Indeed, choose the unique shaded grain P associated to x inside the relevant ρ -parent. Its thickness is a , its shading has density at least δ^η , and regularizing that shading by Lemma 14.4.38 gives the displayed estimate in the a -ball; summing over parents is legitimate because their assigned shadings are disjoint. For each surviving δ -tube choose an a -tube containing it and centred on one of these a -balls. Corollary 14.4.48, with $\rho = a$ and with its auxiliary loss exponent denoted by $\gamma > 0$, gives

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_\delta \delta^{\gamma+O(\eta)} a^\omega (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma} \left(\frac{a}{\delta} \right)^{\sigma/2},$$

after using $C_{KT-CW}(\mathbb{T}), C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Since $a > \delta^{1-\varepsilon_1}$,

$$a^\omega (a/\delta)^{\sigma/2} \geq \delta^{\omega-\varepsilon_1(\omega+\sigma/2)}.$$

Choose $\gamma < \varepsilon_1(\omega + \sigma/2)/4$, and then choose η so that the $O(\eta)$ loss is smaller than the same quantity. The resulting bound is Conclusion (A), for example with $\alpha = \varepsilon_1(\omega + \sigma/2)/2$.

Next we will regularize the set $\mathbb{T}$. By dyadic pigeonholing and replacing $(\mathbb{T}, Y)_\delta$ by a $\gtrsim (\log 1/\delta)^{-1/\varepsilon_1}$ refinement $(\mathbb{T}_1, Y_1)_\delta$, we can suppose that

- (a) For each scale of the form $\tau_i = \delta^{\varepsilon_1 i}$, $i = 1, \dots, \varepsilon_1^{-1}$, there exists a “density” λ_i so that

$$\left| B(x, \tau_i) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \sim \lambda_i |B(x, \tau_i)| \quad \text{for every } x \in \bigcup_{T \in \mathbb{T}_1} Y_1(T). \quad (14.4.155)$$

- (b) For each $i = 1, \dots, \varepsilon_1^{-1}$, there is a pair $(\mathbb{T}_{\tau_i}, Y_{\tau_i})_{\tau_i}$ that is $\gtrsim_\delta \delta^\eta$ dense. Furthermore, $\mathbb{T}_{\tau_i}$ is a balanced partitioning cover of $\mathbb{T}_1$; and for each T_{τ_i} we have

$$Y_{\tau_i}(T_{\tau_i}) \subset T_{\tau_i} \cap \bigcup_{T \in \mathbb{T}_1} N_{\tau_i}(Y_1(T)).$$

From the above items, we have that $C_{F-SW}(\mathbb{T}_{\tau_i}) \lesssim C_{F-SW}(\mathbb{T}_1) \lesssim (\log 1/\delta)^{-1/\varepsilon_1} \delta^{-\eta}$, and

$$C_{KT-CW}(\mathbb{T}_{\tau_i}) \lesssim C_{KT-CW}(\mathbb{T}_1) \frac{\#\mathbb{T}_{\tau_i} |T_{\tau_i}|}{\#\mathbb{T}_1 |T|}.$$

If $\eta > 0$ is selected sufficiently small depending on ε_1 , then we can apply the estimate $\mathcal{E}(\boldsymbol{\sigma}, \boldsymbol{\omega})$ (with ε_1 in place of ε) to conclude that

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\gtrsim \delta^{2\varepsilon_1} \lambda_i \tau_i^\omega C_{KT-CW}(\mathbb{T}_{\tau_i})^{-1} (\#\mathbb{T}_{\tau_i}) |T_{\tau_i}| \\ &\quad \times \left(C_{KT-CW}(\mathbb{T}_{\tau_i})^{-3/2} C_{F-SW}(\mathbb{T}_{\tau_i}) (\#\mathbb{T}_{\tau_i}) |T_{\tau_i}|^{1/2} \right)^{-\sigma} \\ &\gtrsim_\delta \delta^{2\varepsilon_1 + O(\eta)} \left[\lambda_i \left(\frac{\tau_i}{\delta} \right)^\omega \left(\frac{|T_{\tau_i}| (\#\mathbb{T}_{\tau_i})^{1/2}}{|T| (\#\mathbb{T})^{1/2}} \right)^\sigma \right] \\ &\quad \times \delta^\omega (\#\mathbb{T}) |T| \left((\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma} \\ &\gtrsim \delta^{3\varepsilon_1} \left[\lambda_i \left(\frac{\tau_i}{\delta} \right)^\omega \frac{|T_{\tau_i}|^{\sigma/2}}{|T|^{\sigma/2}} \right] \\ &\quad \times \delta^\omega (\#\mathbb{T}) |T| \left((\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma}. \end{aligned} \tag{14.4.156}$$

For the last inequality, we used the fact that $C_{KT-CW}(\mathbb{T}_1) \lesssim_\delta \delta^{-\eta}$, and so $(\#\mathbb{T})/(\#\mathbb{T}_{\tau_i}) \lesssim_\delta \delta^{-\eta} |T_{\tau_i}|/|T|$. In particular, if there is an index i for which $\lambda_i \left(\frac{\tau_i}{\delta} \right)^\omega \frac{|T_{\tau_i}|^{\sigma/2}}{|T|^{\sigma/2}}$ is substantially larger than 1, then we will obtain Conclusion (A) of Lemma 14.4.75.

Step 2. Let $\mathcal{P}_1 = \mathcal{P}$. For each $P \in \mathcal{P}_1$, define $Y_1(P) = Y(P) \cap \bigcup_{T \in \mathbb{T}_1[T_\rho]} Y_1(T)$, where $T_\rho \in \mathbb{T}_\rho$ is the unique ρ tube with $P \in T_\rho$ and $\angle(\text{dir}(P), \text{dir}(T_\rho)) \leq 2\rho$. Since $(\mathbb{T}_1, Y_1)_\delta$ is an $\approx_\delta 1$ refinement of $(\mathbb{T}, Y)_\delta$, by Definition 14.4.53 Items (ii) and (iii) we have that $(\mathcal{P}_1, Y_1)_{a \times b \times c}$ is a $\approx_\delta 1$ refinement of $(\mathcal{P}, Y)_{a \times b \times c}$.

Apply Lemma 14.4.38 (every shading has a regular sub-shading) to each shading $Y_1(P)$, $P \in \mathcal{P}_1$. This gives us a regular sub-shading $Y_2(P) \subset Y_1(P)$. Let $\mathcal{P}_2 \subset \mathcal{P}_1$ be those prisms for which $|Y_2(P)| \geq \delta^{2\eta} |P|$; we have that $(\mathcal{P}_2, Y_2)_{a \times b \times c}$ is a $\gtrsim 1$ refinement of $(\mathcal{P}_1, Y_1)_{a \times b \times c}$.

Let $\mathcal{P}_3 = \mathcal{P}_2$. By dyadic pigeonholing, we can select a number $\theta_0 \in [\frac{a}{b}, 1]$ and a $(\log 1/\delta)^{-1}$ refinement $(\mathcal{P}_3, Y_3)_{a \times b \times c}$ of $(\mathcal{P}_2, Y_2)_{a \times b \times c}$ so that for each $x \in \bigcup_{P \in \mathcal{P}_3} Y_3(P)$, we have $\theta(x) \sim \theta_0$, where $\theta(x)$ is as defined in Definition 14.4.41.

We first consider the case where $\theta_0 \geq \delta^{-\varepsilon_1}(a/b)$. Our goal is to show that Conclusion (A) holds. Let $\mathcal{P}'_3 \subset \mathcal{P}_3$ be the set of those prisms for which $|Y_3(P_0)| \geq \frac{1}{100} \delta^{2\eta} |P_0|$. Then for each $x \in Y_3(P_0)$, we have

$$\frac{a}{b} + \sup_{P \in \mathcal{P}_2} \angle(\Pi(P_0), \Pi(P)) \gtrsim \theta_0.$$

We have $|Y_3(P_0)| \geq \frac{1}{100} \delta^{2\eta} |P_0|$; each $P \in \mathcal{P}_2$ satisfies $|Y_2(P)| \geq \frac{1}{100} \delta^{2\eta} |P_0|$; and $Y_2(P)$ is regular. Thus we can apply Lemma 14.4.39 (with $Y_3(P_0)$ in place of $Y_0(P_0)$ and $(\mathcal{P}_2, Y_2)_{a \times b \times c}$ in place of $(\mathcal{P}, Y)_{a \times b \times c}$) to conclude that

$$\left| N_{b\theta_0}(P_0) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| \gtrsim_{\delta} \delta^{8\eta} |N_{b\theta_0}(P_0)|. \quad (14.4.157)$$

Recall that (14.4.157) holds for each $P_0 \in \mathcal{P}'_3$, and $(\mathcal{P}'_3, Y_3)_{a \times b \times c}$ is a $\gtrsim_{\delta} 1$ refinement of $(\mathcal{P}, Y)_{a \times b \times c}$. After replacing $(\mathcal{P}'_3, Y_3)_{a \times b \times c}$ by a further ~ 1 refinement $(\mathcal{P}'_3, Y'_3)_{a \times b \times c}$, we can suppose that for each $x \in \bigcup_{P \in \mathcal{P}'_3} Y'_3$, we have

$$\left| N_{b\theta_0}(x) \cap \bigcup_{P \in \mathcal{P}} Y(P) \right| \gtrsim_{\delta} \delta^{8\eta} |N_{b\theta_0}(x)|.$$

Finally, if $(\mathbb{T}_1, Y'_1)_{\delta}$ is the refinement of $(\mathbb{T}_1, Y_1)_{\delta}$ induced by $(\mathcal{P}'_3, Y'_3)_{a \times b \times c}$, then by Definition 14.4.53, Item (ii), $(\mathbb{T}_1, Y'_1)_{\delta}$ is a $\approx_{\delta}$ 1-refinement of $(\mathbb{T}_1, Y_1)_{\delta}$, and for each $x \in \bigcup_{T \in \mathbb{T}'_1} Y'_1(T)$ we have

$$\left| N_{b\theta_0}(x) \cap \bigcup_{T \in \mathbb{T}'_1} Y(T) \right| \gtrsim_{\delta} \delta^{8\eta} |N_{b\theta_0}(x)|. \quad (14.4.158)$$

Since $b\theta_0 \geq \delta^{-\varepsilon_1} a$, by Corollary 14.4.48 we see that Conclusion (A) holds, provided we select $\eta > 0$ sufficiently small depending on ε_1 , and select $\alpha \leq \varepsilon_1 \omega/2$.

Henceforth we shall suppose that $\theta_0 \leq \delta^{-\varepsilon_1}(a/b)$, i.e.

$$\sup_x \sup_{P, P' \in (\mathcal{P}_3)_{Y'_3}(x)} \angle(\Pi(P), \Pi(P')) \leq \delta^{-\varepsilon_1}(a/b). \quad (14.4.159)$$

Step 3. Recall the discussion following Definition 14.4.55 (the quantity “ K ” in that discussion is $\delta^{-\varepsilon_1}$ in this context); after replacing $(\mathcal{P}_3, Y_3)_{a \times b \times c}$ by a ~ 1 refinement, which we will denote by $(\mathcal{P}_4, Y_4)_{a \times b \times c}$ (which in turn induces a ~ 1 refinement $(\mathbb{T}_4, Y_4)_{\delta}$ of $(\mathbb{T}_1, Y_1)_{\delta}$), we can find a set $\mathcal{U}$ of pairwise distinct prisms of dimensions $\delta^{-\varepsilon_1} \frac{ac}{b} \times c \times c$ so that the following holds.

- (a) The sets $\{\mathcal{P}_4 \langle U \rangle : U \in \mathcal{U}\}$ are a partition of $\mathcal{P}_4$.
- (b) The sets $\{\bigcup_{P \in \mathcal{P}_4 \langle U \rangle} Y_4(P) : U \in \mathcal{U}\}$ are disjoint.

Each $U \in \mathcal{U}$ is a prism of dimensions $\delta^{-\varepsilon_1} \frac{ac}{b} \times c \times c = \delta^{-\varepsilon_1} \frac{a}{\rho} \times c \times c$. Thus for each $U \in \mathcal{U}$, there is a set $\{Z\}$ of $\lesssim \delta^{-3\varepsilon_1}$ prisms of dimensions comparable

to $\frac{a}{\rho} \times c \times c$ with the property that for each $P \in \mathcal{P}_4\langle U \rangle$, there is a prism Z from this collection with $\square(P) \subset Z$ (recall that $\rho = b/c$, and thus $\square(P)$ is a prism of dimensions comparable to $\frac{a}{\rho} \times c \times c$).

Let Z_U be a prism of dimensions comparable to $\frac{a}{\rho} \times c \times c$ that maximizes

$$\#\{P \in \mathcal{P}_4: \square(P) \subset Z\},$$

so in particular $\#\mathcal{P}_4\langle Z_U \rangle \gtrsim \delta^{3\varepsilon_1}(\#\mathcal{P}_4\langle U \rangle)$. Let $\mathcal{Z} = \{Z_U: U \in \mathcal{U}\}$; let $\mathcal{P}_5 = \bigcup_{U \in \mathcal{U}} \mathcal{P}_4\langle Z_U \rangle$; and let Y_5 be the restriction of Y_4 to $\mathcal{P}_5$. Then $(\mathcal{P}_5, Y_5)_{a \times b \times c}$ is a $\gtrsim \delta^{3\varepsilon_1}$ refinement of $(\mathcal{P}_4, Y_4)_{a \times b \times c}$, and we have the following analogue of Items (a) and (b).

(a') The sets $\{\mathcal{P}_4\langle Z \rangle: Z \in \mathcal{Z}\}$ become a partition of $\mathcal{P}_5$.

(b') The sets $\{\bigcup_{P \in \mathcal{P}_5\langle Z \rangle} Y_5(P): Z \in \mathcal{Z}\}$ are disjoint.

For each $Z \in \mathcal{Z}$, the sets in $(\mathcal{P}_5\langle Z \rangle)^Z$ are convex sets of dimensions comparable to $\rho \times \rho \times 1$, i.e. the sets are comparable to ρ tubes. To record this useful fact, we will define $(\tilde{\mathbb{T}}_Z, \tilde{Y}_5)_\rho = ((\mathcal{P}_5\langle Z \rangle)^Z, Y_5^Z)_{\rho \times \rho \times 1}$.

After replacing $(\mathcal{P}_5, Y_5)_{a \times b \times c}$ and $\mathcal{Z}$ with $\approx_\delta 1$ refinements, we may suppose that each set $\tilde{\mathbb{T}}_Z$ has approximately the same size (up to a factor of 2) for each $Z \in \mathcal{Z}$, and similarly each set $|\tilde{Y}_5(\tilde{T})|$ has approximately the same size for each $\tilde{T} \in \tilde{\mathbb{T}}_Z$. Furthermore, we can suppose that each pair $(\tilde{\mathbb{T}}_Z, \tilde{Y}_5)_\rho$ is $\gtrsim_\delta \delta^\eta$ dense (indeed, recall that $(\mathcal{P}, Y)_{a \times b \times c}$ is δ^η dense; $(\mathcal{P}_4, Y_4)_{a \times b \times c}$ is a $\gtrsim_\delta 1$ -refinement of $(\mathcal{P}, Y)_{a \times b \times c}$; and $(\mathcal{P}_5\langle Z \rangle, Y_5)_{a \times b \times c}$ is a $\approx_\delta 1$ refinement of $(\mathcal{P}_4\langle Z \rangle, Y_4)_{a \times b \times c}$).

Step 4. For notational convenience, we will fix a prism $Z \in \mathcal{Z}$. In what follows, we will find certain quantities (for example certain scales, multiplicities, etc.), and navigate between different cases depending on the specifics of the arrangement $\tilde{\mathbb{T}}_Z$. However, by pigeonholing the set $\mathcal{Z}$, we may suppose that all quantities described below are the same (up to a factor of 2) for each $Z \in \mathcal{Z}$, and thus the same cases occur for each $Z \in \mathcal{Z}$.

Apply Proposition 14.4.23 (factoring convex sets) to $\tilde{\mathbb{T}}_Z$. We obtain a number $m \geq 1$; a $\approx_\delta 1$ refinement $\tilde{\mathbb{T}}'_Z$ of $\tilde{\mathbb{T}}_Z$, and a partitioning cover $\mathcal{W}_Z$ of $\tilde{\mathbb{T}}'_Z$ consisting of congruent prisms; we shall denote the dimensions of these prisms by $s \times t \times 1$ (since each prism contains at least one tube, we know that the longest dimension is ~ 1). We have that $\mathcal{W}_Z$ factors $\tilde{\mathbb{T}}'_Z$ from below with respect to the Frostman Convex Wolff Axioms, and from above with respect to the Katz–Tao Convex Wolff Axioms, both with error $\lesssim_\delta 1$. Finally,

$$C_{KT-CW}(\tilde{\mathbb{T}}'_Z) \leq m, \quad \text{and} \quad \#\tilde{\mathbb{T}}'_Z[W] \approx_\delta m \frac{|W|}{|\tilde{T}|} \quad \text{for each } W \in \mathcal{W}_Z. \quad (14.4.160)$$

We first consider the case where

$$m \leq \delta^{-\zeta}. \quad (14.4.161)$$

Our goal is to show that Conclusion (B) of Lemma 14.4.75 holds.

As described at the beginning of Step 4, we can suppose that (14.4.161) is true for at least half the prisms $Z \in \mathcal{Z}$. Let $\mathcal{P}_6 \subset \mathcal{P}_5$ be given by $\mathcal{P}_6 = \bigcup_Z \phi_Z^{-1}(\tilde{\mathbb{T}}'_Z)$, where the union is taken over those prisms in $\mathcal{Z}$ for which (14.4.161) holds (note that $\mathcal{P}_6 \subset \mathcal{P}_5$, since each tube in $\tilde{\mathbb{T}}'_Z \subset \tilde{\mathbb{T}}_Z$ is the image of a prism from $\mathcal{P}_5$ under the map ϕ_Z). Let Y_6 be the restriction of Y_5 to $\mathcal{P}_6$.

We have that $(\mathcal{P}_6, Y_6)_{a \times b \times c}$ is a $\gtrsim_\delta 1$ refinement of $(\mathcal{P}_5, Y_5)$. Undoing the scaling ϕ_Z , we have that for each $P \in \mathcal{P}_6$,

$$C_{KT-CW}(\mathcal{P}_6(2\Box(P))) \lesssim \sup_{Z \in \mathcal{Z}} C_{KT-CW}(\mathcal{P}_6(Z)) \leq m.$$

We conclude that

$$C_{KT-CW}^{\text{loc}}(\mathcal{P}_6) \lesssim m.$$

Applying a final dyadic pigeonholing, we can select a $\approx_\delta 1$ refinement $(\mathcal{P}', Y')_{a \times b \times c}$ of $(\mathcal{P}_6, Y_6)_{a \times b \times c}$ (this in turn induces a $\gtrsim_\delta \delta^{3\varepsilon_1}$ refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}_4, Y_4)_\delta$) and a set $\mathbb{T}'_\rho \subset \mathbb{T}_\rho$ so that the following holds: $\mathbb{T}'_\rho$ is a balanced partitioning cover of $(\mathbb{T}', Y')_\delta$, and $(\mathcal{P}', Y')_{a \times b \times c}$ is a robustly $\approx_\delta \delta^{3\varepsilon_1}$ -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\mathbb{T}'_\rho$.

Since $C_{KT-CW}^{\text{loc}}(\mathcal{P}') \lesssim m$, Conclusion (B), Item (iv) of Lemma 14.4.75 is satisfied. $(\mathcal{P}', Y')_{a \times b \times c}$ satisfies Conclusion (B), Item (iii) by construction. $(\mathbb{T}', Y')_\delta$ satisfies Conclusion (B), Item (i), provided ε_1 is chosen sufficiently small depending on ε . Finally, since

$$\begin{aligned} \#(\mathbb{T}'[T_\rho]_{Y'}(x)) &\gtrsim_\delta \delta^{3\varepsilon_1} \#(\mathbb{T}[T_\rho]_Y(x), \\ T_\rho \in \mathbb{T}'_\rho, \quad x &\in \bigcup_{T \in \mathbb{T}'[T_\rho]} Y'(T). \end{aligned}$$

and since (by hypothesis) $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$ relative to $\mathbb{T}_\rho$, we conclude that $(\mathbb{T}', Y')_\delta$ is broad with error $\lesssim_\delta \delta^{-\eta-3\varepsilon_1}$ relative to $\mathbb{T}'_\rho$. We will select η and ε_1 sufficiently small so that this quantity is $\leq \delta^{-\varepsilon}$. This verifies Conclusion (B), Item (ii) ³.

In summary, if (14.4.161) holds, then Conclusion (B) of Lemma 14.4.75 is satisfied. Henceforth we will suppose that (14.4.161) fails, i.e.

$$m > \delta^{-\zeta}. \quad (14.4.162)$$

³To be precise, we can only ensure that the error is $\leq \delta^{-\varepsilon}$ provided $\delta > 0$ is sufficiently small, depending on the implicit constant in the above $\gtrsim$ notation. However if this fails then Conclusion (A) holds, provided we select $\kappa > 0$ sufficiently small.

Step 5. In Step 4, we fixed a choice of prism $Z \in \mathcal{Z}$. We will continue to fix this choice of Z , and in addition we will fix a choice of $W \in \mathcal{W}_Z$. As in Step 4, we can assume (by dyadic pigeonholing) that all relevant scales, multiplicities, etc. are approximately the same (up to a factor of 2) for each $Z \in \mathcal{Z}$ and each $W \in \mathcal{W}_Z$.

In the arguments that follow, we will analyze the pairs $(\tilde{\mathbb{T}}'_Z[W], \tilde{Y}_5)_\rho$ constructed in Step 4. For notational convenience, we will refer to such a pair as $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. Recall that this pair is $\gtrsim_\delta \delta^\eta$ dense; the cardinality of $\tilde{\mathbb{T}}$ is given by (14.4.160); and m satisfies (14.4.162). In particular, we have

$$C_{F-CW}(\tilde{\mathbb{T}}^W) \lesssim_\delta 1. \quad (14.4.163)$$

Apply Corollary 14.4.62 (finding a broad scale) to the set $\tilde{\mathbb{T}}$. Denote the “output” scale of this Corollary by τ (in Corollary 14.4.62, this output scale is called ρ , but that variable is already in use). Abusing notation, we will continue to use $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ to refer to the output of Corollary 14.4.62. Thus there is a set $\mathbb{T}_\tau$ that forms a balanced partitioning cover of $\tilde{\mathbb{T}}$, and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is broad with error $\lesssim_\delta 1$ relative to $\mathbb{T}_\tau$. Furthermore,

$$\text{the sets } \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}[T_\tau]} \tilde{Y}(\tilde{T}), \quad T_\tau \in \mathbb{T}_\tau \quad \text{are } \lesssim \tau^{-\beta} \text{ overlapping.} \quad (14.4.164)$$

We claim that

$$\tau \geq \rho^{1-\omega/8}, \quad (14.4.165)$$

or else Conclusion (A) of Lemma 14.4.75 holds, and we are done. To verify this claim, note that if (14.4.165) failed, then the sets $\{\tilde{Y}(\tilde{T}) : \tilde{T} \in \tilde{\mathbb{T}}\}$ are $\lesssim \rho^{-\omega/4} \tau^{-\beta} \leq \delta^{-\omega/2}$ overlapping; but this fact, combined with Item (b') from Step 3, gives Conclusion (A) of Lemma 14.4.75.

Step 6. We would like to apply the estimate $\mathcal{E}(\sigma, \omega)$ to each set $(\tilde{\mathbb{T}}^{T_\tau}, \tilde{Y}^{T_\tau})_{\rho/\tau}$. However, we do not currently have a good estimate for $C_{F-SW}(\tilde{\mathbb{T}}^{T_\tau})$. To fix this problem, apply Proposition 14.4.25 (factoring convex sets with respect to the Frostman Slab Wolff Axioms) to each set $(\tilde{\mathbb{T}}[T_\tau], \tilde{Y})_\rho$, with ε_2 in place of ε . We can do so, provided ε_1 and η is selected sufficiently small compared to ε_2 . This gives us a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement $(\tilde{\mathbb{T}}'[T_\tau], \tilde{Y}')_\rho$ of $(\tilde{\mathbb{T}}[T_\tau], \tilde{Y})_\rho$, and a family of convex subsets of T_τ , which we denote by $\mathcal{V}_{T_\tau}$, that factors $\tilde{\mathbb{T}}'[T_\tau]$ from below with respect to the Frostman Slab Wolff Axioms with error $\delta^{-\varepsilon_2}$. In addition,

$$\text{the sets } \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}'[V]} \tilde{Y}'(\tilde{T}), \quad V \in \mathcal{V}_{T_\tau} \quad \text{are disjoint.} \quad (14.4.166)$$

After pigeonholing, we may suppose that the sets in $\mathcal{V}_{T_\tau}$ have the same dimensions, and furthermore these dimensions are common across all $T_\tau \in \mathbb{T}_\tau$.

Denote these dimensions $\theta \times \tau' \times 1$. Since $V \subset T_\tau$, we must have $\tau' \leq \tau$. We claim that this inequality is almost tight, in the sense that

$$\delta^{\varepsilon_2/\beta} \tau \lesssim_\delta \tau' \leq \tau. \quad (14.4.167)$$

To verify this claim, note that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is broad with error $\lesssim_\delta 1$ relative to the cover $\mathbb{T}_\tau$. Since $(\tilde{\mathbb{T}}'[T_\tau], \tilde{Y}')_\rho$ is a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement of $(\tilde{\mathbb{T}}[T_\tau], \tilde{Y})_\rho$, by pigeonholing there is at least one point x for which the tubes in $\tilde{\mathbb{T}}'_{Y'}(x)$ point in directions that are broad with error $\lesssim_\delta \delta^{-\varepsilon_2}$ inside a cap of radius τ . This means that there are at least two tubes from this set that make an angle $\gtrsim_\delta \delta^{\varepsilon_2/\beta} \tau$. On the other hand, by (14.4.166) we have that the pair of tubes described above must be contained in a common $\theta \times \tau' \times 1$ prism. This establishes (14.4.167).

Let $\tilde{\mathbb{T}}'$ be the union of the sets $\tilde{\mathbb{T}}'[T_\tau]$, as T_τ ranges over the elements of $\mathbb{T}_\tau$. Let $\tilde{Y}'$ be the associated shading on $\tilde{\mathbb{T}}'$ coming from the pairs $(\tilde{\mathbb{T}}'[T_\tau], \tilde{Y}')_\rho$. Abusing notation, we will rename this pair $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$. At this point in the argument, this pair is $\gtrsim_\delta \delta^{\eta+\varepsilon_2} \geq \delta^{2\varepsilon_2}$ dense.

Step 7. We first consider the case where the prisms $\mathcal{V}_{T_\tau}$ from Step 6 are almost tubes, in the sense that

$$\theta \geq \delta^{\zeta/10} \tau'. \quad (14.4.168)$$

If (14.4.168) is true, then each set $\tilde{\mathbb{T}}^V$ consists of prisms of dimensions $\frac{\rho}{\tau'} \times \frac{\rho}{\theta} \times 1$. The pair $(\tilde{\mathbb{T}}^V, \tilde{Y}^V)_{\rho/\tau' \times \rho/\theta \times 1}$ is $\gtrsim_\delta \delta^{2\varepsilon_2}$ dense. Let us suppose for the moment that

$$\rho \leq \delta^{\sqrt{\varepsilon_2}}. \quad (14.4.169)$$

If we select ε_2 sufficiently small so that $\varepsilon_2 \leq \frac{1}{2} \sqrt{\varepsilon_2} \beta \omega$, and hence $\frac{8\varepsilon_2 \beta}{\sqrt{\varepsilon_2} \beta \omega - 8\varepsilon_2} \leq \frac{16\sqrt{\varepsilon_2}}{\omega}$, then by (14.4.165) and (14.4.167), we have $\delta^{2\varepsilon_2} \geq \left(\frac{\rho}{\tau'}\right)^{\frac{32\sqrt{\varepsilon_2}}{\omega}}$.

Thus if ε_2 is chosen sufficiently small compared to ε_3 and ω , and if (14.4.169) is true, then we can apply the estimate $\mathcal{F}(\sigma, \omega)$ (recall Definition 14.4.33 and Remark 14.4.35, Situation 3) to estimate the volume of each (rescaled) set $(\tilde{\mathbb{T}}^V, \tilde{Y}^V)_{\rho/\tau' \times \rho/\theta \times 1}$, with ε_3 in place of ε and $32\sqrt{\varepsilon_2}/\omega$ in place of η . This gives the estimate

$$\left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}[V]} \tilde{Y}^V(\tilde{T}^V) \right| \gtrsim \delta^{\varepsilon_3} \left(\frac{\rho}{\theta}\right)^\omega \tilde{m}^{-1} (\#\tilde{\mathbb{T}}[V]) |\tilde{T}^V| \left(\tilde{m}^{-3/2} \tilde{\ell} (\#\tilde{\mathbb{T}}[V]) |\tilde{T}^V|^{1/2} \right)^{-\sigma}, \quad (14.4.170)$$

where we define

$$\tilde{m} = C_{KT-CW}(\tilde{\mathbb{T}}[V]) \leq C_{KT-CW}(\tilde{\mathbb{T}}) \leq m \quad \text{and} \quad \tilde{\ell} = C_{F-SW}(\tilde{\mathbb{T}}[V]) \lesssim \delta^{-\varepsilon_2}. \quad (14.4.171)$$

(for the first inequality, we used (14.4.160)). On the other hand, if (14.4.169) is false, then (14.4.170) follows from the fact that the LHS of (14.4.170) is bounded below by the volume of a single shading $|\tilde{Y}^V(T^V)|$, and we can select a tube with volume

$$|\tilde{Y}^V(T^V)| \gtrsim \delta^{2\varepsilon_2} |T^V| = \delta^{2\varepsilon_2} \frac{\rho^2}{\tau'\theta} \geq \delta^{2\varepsilon_2} \rho^2 \geq \delta^{3\varepsilon_2} \geq \delta^{\varepsilon_3}. \quad (14.4.172)$$

Since $\sigma \in (0, 2/3]$, by Remark 14.4.34 and Remark 14.4.35, Situation 3,

$$\tilde{m}^{-1}(\#\tilde{\mathbb{T}}[V])|\tilde{T}^V| \left(\tilde{m}^{-3/2} \tilde{\ell}(\#\tilde{\mathbb{T}}[V])|\tilde{T}^V|^{1/2} \right)^{-\sigma} \lesssim 1. \quad (14.4.173)$$

Combining (14.4.172) and (14.4.173) establishes (14.4.170) in the case where (14.4.169) is false. We conclude that (14.4.170) holds, independently of whether (14.4.169) is true or false.

Undoing the scaling ϕ_V and substituting the values of $\tilde{m}$ and $\tilde{\ell}$ from (14.4.171), we have

$$\left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}[V]} \tilde{Y}(\tilde{T}) \right| \gtrsim \delta^{2\varepsilon_3} \left(\frac{\rho}{\theta} \right)^\omega m^{-1}(\#\tilde{\mathbb{T}}[V])|\tilde{T}| \left(m^{-3/2}(\#\tilde{\mathbb{T}}[V]) \frac{|\tilde{T}|^{1/2}}{|V|^{1/2}} \right)^{-\sigma}. \quad (14.4.174)$$

By (14.4.164), (14.4.166) (recall that we have renamed $\tilde{Y}'$ as $\tilde{Y}$), and (14.4.174), we have

$$\begin{aligned} \left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right| &\gtrsim \tau^\beta \sum_{T_\tau \in \mathbb{T}_\tau} \sum_{V \in \mathcal{V}_{T_\tau}} \left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}[V]} \tilde{Y}(\tilde{T}) \right| \\ &\gtrsim \delta^{2\varepsilon_3+\beta} \left(\frac{\rho}{\theta} \right)^\omega m^{-1}(\#\tilde{\mathbb{T}})|\tilde{T}| \left(m^{-3/2}(\#\tilde{\mathbb{T}}[V]) \frac{|\tilde{T}|^{1/2}}{|V|^{1/2}} \right)^{-\sigma}. \end{aligned} \quad (14.4.175)$$

Next, by (14.4.163) and using the fact that $(\tilde{\mathbb{T}}'[T_\tau], Y')_\rho$ is a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement of $(\mathbb{T}[T_\tau], \tilde{Y})_\rho$, we have that $C_{F-CW}(\tilde{\mathbb{T}}^W) \lesssim_\delta \delta^{-2\varepsilon_2}$. By (14.4.160), $\#\tilde{\mathbb{T}} \gtrsim_\delta \delta^{2\varepsilon_2} \left(m \frac{|W|}{|\tilde{T}|} \right)$. Since

$$\#\tilde{\mathbb{T}}[V] \leq C_{F-CW}(\tilde{\mathbb{T}}^W) \frac{|V|}{|W|} (\#\tilde{\mathbb{T}}) \lesssim_\delta \delta^{-2\varepsilon_2} \frac{|V|}{|W|} \left(m \frac{|W|}{|\tilde{T}|} \right),$$

using (14.4.162), (14.4.167), (14.4.168), and (14.4.175) we conclude that

$$\begin{aligned} \left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right| &\gtrsim_\delta \delta^{2\varepsilon_3+\beta-\zeta\sigma/2} \left(\frac{\rho}{\theta} \right)^\omega m^{-1} \left(m \frac{|W|}{|\tilde{T}|} \right) |\tilde{T}| \left(m^{-1} \left(m \frac{|V|}{|\tilde{T}|} \right) \frac{|\tilde{T}|^{1/2}}{|V|^{1/2}} \right)^{-\sigma} \\ &\gtrsim_\delta \delta^{2\varepsilon_3+\beta-\zeta\sigma/4} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |W|. \end{aligned} \quad (14.4.176)$$

Step 8. Let us analyze (14.4.176). First, the set on the LHS of (14.4.176) is contained in W , which is a prism of dimensions $s \times t \times 1$. Second, the quantities ε_3 and β are chosen after ζ and σ , so we can select the former quantities to ensure that $\delta^{-2\varepsilon_3+\beta-\zeta\sigma/4} \geq \delta^{-\zeta\sigma/5}$.

Thus by pigeonholing, we can select a ball B of radius θ (recall that $\rho \leq \theta \leq s$) with the property that

$$\left| B \cap \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(T) \right| \gtrsim \delta^{-\zeta\sigma/5} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |B|. \quad (14.4.177)$$

Recall that at the beginning of Step 4 we fixed a prism $Z \in \mathcal{Z}$. The set of ρ tubes $\tilde{\mathbb{T}}$ are the images of prisms from $\mathcal{P}_3\langle Z \rangle$ under the linear map ϕ_Z . Let $B^\dagger = \phi_Z^{-1}(B)$, where B is the ball described above. Then $B^\dagger$ is an ellipsoid of dimensions $\theta \frac{a}{\rho} \times \theta c \times \theta c$. By (14.4.177), we have

$$\left| B^\dagger \cap \bigcup_{P \in \mathcal{P}_1} Y_1(P) \right| \gtrsim \delta^{-\zeta\sigma/5} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |B^\dagger|. \quad (14.4.178)$$

Let $1 \leq i \leq \varepsilon_1^{-1}$ be the index so that $\delta^{i\varepsilon_1} \leq \theta \frac{a}{\rho} < \delta^{(i-1)\varepsilon_1}$. Such an index exists since $\theta \in [\rho, 1]$ and $\delta \leq \frac{a}{\rho} = \frac{ac}{b} \leq c \leq 1$. Recall that we defined $\tau_i = \delta^{i\varepsilon_1}$, so $\theta a/\rho \leq \tau_i \delta^{-\varepsilon_1}$ implies $\rho/\theta \geq \delta^{\varepsilon_1} a/\tau_i \geq \delta^{1+\varepsilon_1}/\tau_i$. Then there exists a ball B_{τ_i} of radius τ_i , so that

$$\lambda_i \geq \left| B_{\tau_i} \cap \bigcup_{T \in \mathbb{T}_1} Y_1(T) \right| |B_{\tau_i}|^{-1} \geq \delta^{3\varepsilon_1-\zeta\sigma/5} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} \geq \delta^{4\varepsilon_1-\zeta\sigma/5} \left(\frac{\delta}{\tau_i} \right)^\omega \frac{|T|^{\sigma/2}}{|T_{\tau_i}|^{\sigma/2}}. \quad (14.4.179)$$

Combining (14.4.156) and (14.4.179), we conclude that Conclusion (A) of Lemma 14.4.75 holds, provided we select $\varepsilon_1 \leq \frac{1}{100}\zeta\sigma$ and select α sufficiently small.

This concludes our analysis of the case where (14.4.168) holds (the analysis of this case began at the start of Step 7). Henceforth we shall suppose that (14.4.168) fails.

Step 9. We shall now return to the start of Step 7, except, instead of assuming (14.4.168), we will instead suppose that

$$\theta < \delta^{\zeta/10}\tau'. \quad (14.4.180)$$

Informally, (14.4.180) says that the prisms $V \in \mathcal{V}_{T_\tau}$ are flat.

Recall that in Steps 4 and 5, we fixed a prism $Z \in \mathcal{Z}$ and a prism $W \in \mathcal{W}_Z$. In this step, we will fix a τ tube $T_\tau \in \mathbb{T}_\tau$ and a $\theta \times \tau' \times 1$ prism $V \in \mathcal{V}_{T_\tau}$. Define $\mathbb{T}^\dagger = \tilde{\mathbb{T}}[V]$ and let $Y^\dagger$ be the restriction of $\tilde{Y}$ to $\mathbb{T}^\dagger$. Thus $(\mathbb{T}^\dagger, Y^\dagger)_\rho$ is a set of ρ tubes contained in V , and

$$C_{F-SW}((\mathbb{T}^\dagger)^V) \lesssim_\delta \delta^{-\varepsilon_2}. \quad (14.4.181)$$

After pigeonholing, we may suppose that $(\mathbb{T}^\dagger, Y^\dagger)_\rho$ is $\gtrsim_\delta \delta^{2\varepsilon_2}$ dense. For each “stem” $T_0^\dagger \in \mathbb{T}^\dagger$, define the “hairbrush”

$$\mathcal{H}(T_0^\dagger) = \{T^\dagger \in \mathbb{T}^\dagger : Y^\dagger(T_0^\dagger) \cap Y^\dagger(T^\dagger) \neq \emptyset\}.$$

We claim that for each tube $T_0^\dagger \in \mathbb{T}^\dagger$, the set

$$N_{(\tau'/\theta)\rho}(T_0^\dagger) \cap \bigcup_{T^\dagger \in \mathcal{H}(T_0^\dagger)} Y^\dagger(T^\dagger) \quad (14.4.182)$$

is contained in a rectangular prism of dimensions comparable to $\rho \times (\tau'/\theta)\rho \times 1$; we will call this rectangular prism $X = X(T_0^\dagger)$. The following coordinate calculation verifies the claim. Define $Y(X)$ to be the set (14.4.182), so $Y(X) \subset X$.

We record the geometry explicitly. Put $R = (\tau'/\theta)\rho$ and choose orthonormal coordinates in which the long axis of V is e_1 and its thin normal is e_3 . The core direction v of every ρ -tube contained in V satisfies $|v \cdot e_3| \lesssim \theta/\tau'$. If $T^\dagger$ belongs to the hairbrush, the core lines of $T^\dagger$ and $T_0^\dagger$ come within $O(\rho)$ at a point of their shaded intersection. Hence, at arclength at most R from the stem, the displacement normal to the plane generated by e_1 and the in-plane transverse direction is at most

$$O(\rho) + O(R\theta/\tau') = O(\rho).$$

The in-plane transverse displacement is at most $O(R)$, while all core segments have length $O(1)$. Thus the set in (14.4.182) lies in a coaxial $C\rho \times CR \times C$ prism. Enlarging a fixed dimensional constant gives the asserted X .

Next, we claim that if ε_2 is chosen sufficiently small compared to ε_3 , then for a $\gtrsim_\delta 1$ fraction of the tubes $T_0^\dagger \in \mathbb{T}^\dagger$ we have

$$|Y(X)| \gtrsim_\delta \delta^{\varepsilon_3} |X|, \quad \text{where } X = X(T_0^\dagger). \quad (14.4.183)$$

The estimate (14.4.183) says that the shading $Y(X)$ is $\gtrsim_\delta \delta^{\varepsilon_3}$ dense. We now prove it. Let $Y^\dagger(T^\dagger) \subset Y^\dagger(T^\dagger)$, $T^\dagger \in \mathbb{T}^\dagger$ be a regular shading, in the sense of Definition 14.4.37, with $|Y^\dagger(T^\dagger)| \geq \frac{1}{2}|Y^\dagger(T^\dagger)|$. By pigeonholing, a $\gtrsim_\delta 1$ fraction of the tubes $T_0^\dagger \in \mathbb{T}^\dagger$ satisfy

$$|\{x \in Y^\dagger(T_0^\dagger) : \#\mathbb{T}_{Y^\dagger}^\dagger(x) \geq \frac{1}{4}\#\mathbb{T}_{Y^\dagger}^\dagger(x)\}| \gtrsim_\delta \delta^{2\varepsilon_1} |T_0^\dagger|. \quad (14.4.184)$$

For each point x in the set on the left of (14.4.184), broadness at the stopping scale τ supplies a tube $T_x^\dagger$ with $x \in Y^\dagger(T_x^\dagger)$ and

$$c_\delta \tau \leq \angle(\text{dir}(T_0^\dagger), \text{dir}(T_x^\dagger)) \leq C\tau.$$

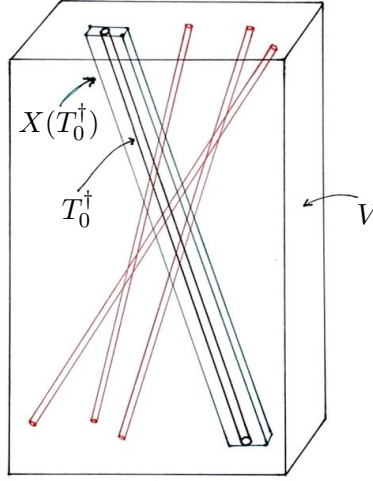


Figure 14.4.10: All tubes (red) in this figure intersect the stem tube $T_0^\dagger$ (black). Since each tube is contained in the $\theta \times \tau' \times 1$ prism V , each tube makes angle $\lesssim \theta/\tau'$ with the plane $\Pi(V)$. Since each red tube intersect $T_0^\dagger$, the union of these tubes, intersected with the $(\tau'/\theta)\rho$ neighbourhood of $T_0^\dagger$, are contained in the prism $X = X(T_0^\dagger)$ (green) of dimensions $\frac{\theta}{\tau'} \cdot (\frac{\tau'}{\theta})\rho \times (\frac{\tau'}{\theta})\rho \times 1 = \rho \times (\frac{\tau'}{\theta})\rho \times 1$.

The upper bound holds because all these tubes lie in the same τ -parent. A dyadic angular pigeonhole and a bounded covering of the circle let us retain an $\approx_\delta 1$ part on which the projected normal directions occupy one fixed small cap.

Set $R = (\tau'/\theta)\rho$ and $q = \rho/R = \theta/\tau'$. Let ϕ_X be the affine normalization of X . The preceding coordinate calculation shows that every $T_x^\dagger \cap X$ is sent to a set comparable to a $q \times 1 \times 1$ slab; its thin normal has a fixed non-zero projection onto the two-dimensional cross-section of the normalized stem. Fubini applied to the normalized image of the set in (14.4.184) gives a line segment L in that cross-section and a subset of one-dimensional measure $\gtrsim_\delta \delta^{2\varepsilon_1}$. Choose a maximal q -separated sequence $x_1, \dots, x_N$ on it. Then

$$N \gtrsim_\delta \delta^{2\varepsilon_1} q^{-1}. \quad (14.4.185)$$

The chosen tubes may be made distinct: a ρ -tube making angle $\gtrsim_\delta \tau$ with the stem meets it in an interval of length $O(\rho/\tau)$, and the preliminary angular and spatial pigeonholing retains one representative whenever a bounded cluster occurs. The loss is an $\approx_\delta 1$ factor already allowed in (14.4.185).

Write $S_j = \phi_X(T_{x_j}^\dagger \cap X)$. We claim

$$C_{KT-CW}(\{S_j : 1 \leq j \leq N\}) \lesssim_\delta 1. \quad (14.4.186)$$

To verify it, let a convex set W contain M of the S_j . If $M = 0$ there is nothing to prove, and if $M = 1$, then $|W| \gtrsim |S_j| \sim q$. For $M \geq 2$, the corresponding points of $L \cap W$ are q -separated, so $|L \cap W| \gtrsim Mq$. Since W also contains one of the slabs and its normal has a fixed non-zero projection transverse to L , the convex hull of that slab with $L \cap W$ has volume $\gtrsim_\delta Mq$. Hence $M \lesssim_\delta |W|/|S_j|$, which is exactly (14.4.186).

Regularity of $Y^\dagger(T_{x_j}^\dagger)$ at radius R gives

$$|Y^\dagger(T_{x_j}^\dagger) \cap X| \gtrsim_\delta \delta^{2\varepsilon_2} |T_{x_j}^\dagger \cap X|.$$

Thus the normalized slabs have $\gtrsim_\delta \delta^{2\varepsilon_2}$ -dense shadings. Lemma 14.4.36, (14.4.185), and (14.4.186) yield

$$\left| \bigcup_{j=1}^N \phi_X(Y^\dagger(T_{x_j}^\dagger) \cap X) \right| \gtrsim_\delta \delta^{4\varepsilon_2} Nq \gtrsim_\delta \delta^{2\varepsilon_1+4\varepsilon_2}.$$

Undoing ϕ_X , the union on the left is contained in $Y(X)$. Choosing $2\varepsilon_1 + 4\varepsilon_2 < \varepsilon_3$ (and absorbing logarithms by a further strict inequality) proves (14.4.183). Notice that the separation scale in normalized coordinates is $q = \theta/\tau'$; no unexplained δ/τ scale is used.

Abusing notation, we will refine $\mathbb{T}^\dagger$ so that each $T^\dagger \in \mathbb{T}^\dagger$ satisfies (14.4.183). By dyadic pigeonholing and replacing $\mathbb{T}^\dagger$ by a $\approx_\delta 1$ refinement (abusing notation, we will continue to refer to this set as $\mathbb{T}^\dagger$), we can select a number M and a set $\mathcal{X}$ of essentially distinct $\rho \times \frac{\tau'\rho}{\theta} \times 1$ prisms of cardinality $\#\mathcal{X} = M^{-1}(\#\mathbb{T}^\dagger)$, so that

- (a) For each $X \in \mathcal{X}$, there are $\sim M$ tubes $T^\dagger \in \mathbb{T}^\dagger$ with $T^\dagger \subset X$ and $X(T^\dagger)$ comparable to X . Denote this latter set by $\mathbb{T}_X^\dagger$.
- (b) Each set $\mathbb{T}^\dagger[X]$, $X \in \mathcal{X}$ has the same cardinality (up to a factor of 2).

Note that $\mathbb{T}_X^\dagger \subset \mathbb{T}^\dagger[X]$, but the two sets need not be equal; it could be the case that $\#\mathbb{T}^\dagger[X]$ is much larger than $\#\mathcal{X}$. In particular, $\mathbb{T}^\dagger = \bigsqcup_{X \in \mathcal{X}} \mathbb{T}_X^\dagger$, but $\mathcal{X}$ might not be a partitioning cover of $\mathbb{T}^\dagger$. For example, there could exist a tube in $\mathbb{T}^\dagger[X]$ that is contained in a different prism V' ; such a tube will be also be contained in the set $\mathbb{T}_{X'}^\dagger$, where X' a prism with orientation compatible with V' (recall Figure 14.4.10). In particular, it is possible that X and X' intersect transversely.

Step 10. In Step 9 we fixed a choice of τ tube $T_\tau \in \mathbb{T}_\tau$ and a $\theta \times \tau' \times 1$ prism $V \in \mathcal{V}_{T_\tau}$. We then constructed a pair $(\mathcal{X}, Y)_{\rho \times \frac{\tau'\rho}{\theta} \times 1}$. The set $\mathcal{X}$ depended on the choice of prism $V \in \mathcal{V}_{T_\tau}$; we will highlight this dependence by writing $\mathcal{X}_V$. In this step we will analyze the interaction between different collections $\mathcal{X}_V$.

Define $\mathcal{V} = \bigcup_{T_\tau \in \mathbb{T}_\tau} \mathcal{V}_{T_\tau}$. Note that $\mathcal{V}$ is a set of $\theta \times \tau' \times 1$ prisms contained in W (the set W was fixed at Step 5). The quantity M from Step 9 depends on the choice of V , but after pigeonholing and refining $\mathcal{V}$ we may suppose that this number is the same (up to a factor of 2) for every $V \in \mathcal{V}$.

We will first consider the case where

$$\#\mathbb{T}_X^\dagger \sim M \leq \delta^{-\zeta/100} \frac{|X|}{|T^\dagger|}. \quad (14.4.187)$$

We will show that Conclusion (A) of Lemma 14.4.75 holds.

Recall from Figure 14.4.10 that the prisms $X \in \mathcal{X}_V$ and V have compatible orientations, in the sense that X^V is a prism of dimensions comparable to $\frac{\rho}{\theta} \times \frac{\rho}{\theta} \times 1$. Thus we will refer to the set $(\mathcal{X}_V^V, Y^V)_{\frac{\rho}{\theta} \times \frac{\rho}{\theta} \times 1}$ as $(\mathbb{T}_{\theta, V}^{\frac{\rho}{\theta}}, Y_V)_{\frac{\rho}{\theta}}$.

We claim that

$$\text{the sets } \left\{ \bigcup_{X \in \mathcal{X}_V} Y(X), V \in \mathcal{V} \right\} \text{ are } \leq \delta^{-\beta} \text{ overlapping.} \quad (14.4.188)$$

This claim follows from combining (14.4.164) and (14.4.166) (and noting that $\tau^{-\beta} \leq \delta^{-\beta}$).

Observe that for each $V \in \mathcal{V}$, we have

$$C_{F-SW}(\mathbb{T}_{\theta, V}^{\frac{\rho}{\theta}}) \lesssim_\delta \delta^{-2\varepsilon_2}. \quad (14.4.189)$$

This is because $C_{F-SW}(\tilde{\mathbb{T}}^V) \lesssim_\delta \delta^{-2\varepsilon_2}$ (recall that $\mathcal{V}$ factors $\tilde{\mathbb{T}}$ from below with small error with respect to the Frostman Slab Wolff Axioms), and by Item (b) from Step 9, each $X \in \mathcal{X}_V$ contains the same number (up to a factor of 2) of tubes from $\tilde{\mathbb{T}}[V]$. This means that $C_{F-SW}(\mathcal{X}_V^V) \lesssim_\delta \delta^{-2\varepsilon_2}$, which is precisely (14.4.189).

Define

$$m_{\frac{\rho}{\theta}} = mM^{-1} \frac{|X|}{|\tilde{T}|}, \quad (14.4.190)$$

where m is as defined in (14.4.160), and $|X| = \rho \times \frac{\tau'}{\theta} \rho \times 1 = \frac{\tau'}{\theta} \rho^2$ is the volume of a prism from $\mathcal{X}_V$. We have

$$C_{KT-CW}(\mathbb{T}_{\theta, V}^{\frac{\rho}{\theta}}) \lesssim_\delta mM^{-1} \frac{|X|}{|\tilde{T}|} \leq m_{\frac{\rho}{\theta}}. \quad (14.4.191)$$

Finally, we compute

$$\sum_{V \in \mathcal{V}} (\#\mathbb{T}_{\theta, V}^{\frac{\rho}{\theta}}) \gtrsim_\delta M^{-1} \#\tilde{\mathbb{T}}[W] \gtrsim mM^{-1} \frac{|W|}{|\tilde{T}|}, \quad (14.4.192)$$

where W is the prism fixed at Step 5, and the final inequality used (14.4.160).

Fix a choice of $V \in \mathcal{V}$. To keep the calculation readable, put

$$\begin{aligned} q &= \rho/\theta, & m_q &= m_{\rho/\theta}, & \mathbb{T}_{q,V} &:= \mathbb{T}_{\rho/\theta,V}, & T_q &:= T_{\rho/\theta}, \\ n_V &= \#\mathbb{T}_{\rho/\theta,V}, & n_* &= \sup_{V \in \mathcal{V}} n_V. \end{aligned}$$

Applying the estimate $\mathcal{E}(\boldsymbol{\sigma}, \boldsymbol{\omega})$ to $(\mathbb{T}_{q,V}, Y_V)_q$ with ε_4 in place of ε and using (14.4.189), we have

$$\begin{aligned} \left| \bigcup_{T_q \in \mathbb{T}_{q,V}} Y_V(T_q) \right| &\gtrsim_{\delta} \delta^{\varepsilon_4 + 2\varepsilon_2} q^{\omega} m_q^{-1} n_V |T_q| \\ &\times \left(m_q^{-3/2} n_V |T_q|^{1/2} \right)^{-\boldsymbol{\sigma}}. \end{aligned} \quad (14.4.193)$$

In the above inequality, we used (14.4.191) plus the fact that $\boldsymbol{\sigma} \leq 2/3$ (the latter inequality allows us to replace $C_{KT-CW}(\mathbb{T}_{\theta,V})$ with the potentially larger quantity m_{θ}). Undoing the scaling ϕ_V and using (14.4.188), we conclude that

$$\begin{aligned} \left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} Y(\tilde{T}) \right| &\geq \left| \bigcup_{V \in \mathcal{V}} \bigcup_{X \in \mathcal{X}_V} Y(X) \right| \\ &\geq \delta^{\beta} \sum_{V \in \mathcal{V}} \left| \bigcup_{X \in \mathcal{X}_V} Y(X) \right| \\ &\geq \delta^{\beta} \frac{|X|}{|T_q|} \sum_{V \in \mathcal{V}} \left| \bigcup_{T_q \in \mathbb{T}_{q,V}} Y_V(T_q) \right| \\ &\gtrsim_{\delta} \delta^{\beta + \varepsilon_4 + 2\varepsilon_2} \frac{|X|}{|T_q|} q^{\omega} m_q^{-1} \left(\sum_{V \in \mathcal{V}} n_V \right) |T_q| \\ &\quad \times \left(m_q^{-3/2} n_* |T_q|^{1/2} \right)^{-\boldsymbol{\sigma}} \\ &\gtrsim_{\delta} \delta^{\beta + 2\varepsilon_4} \frac{|X|}{|T_q|} q^{\omega} m_q^{-1} \left(mM^{-1} \frac{|W|}{|\tilde{T}|} \right) |T_q| \\ &\quad \times \left(m_q^{-3/2} n_* |T_q|^{1/2} \right)^{-\boldsymbol{\sigma}}. \end{aligned} \quad (14.4.194)$$

where the final inequality used (14.4.192). Substituting (14.4.190) and simpli-

fying, we obtain

$$\begin{aligned}
 \text{LHS of (14.4.194)} &\gtrsim_{\delta} \delta^{\beta+2\varepsilon_4} \frac{|X|}{|T_q|} q^{\omega} \left(mM^{-1} \frac{|X|}{|\tilde{T}|} \right)^{-1} \left(mM^{-1} \frac{|W|}{|\tilde{T}|} \right) |T_q| \\
 &\quad \times \left[\left(mM^{-1} \frac{|X|}{|\tilde{T}|} \right)^{-3/2} n_* |T_q|^{1/2} \right]^{-\sigma} \\
 &\gtrsim_{\delta} \delta^{\beta+2\varepsilon_4} q^{\omega} |W| \left(mM^{-1} \frac{|X|}{|\tilde{T}|} \right)^{\sigma/2} \\
 &\quad \times \left[\left(mM^{-1} \frac{|X|}{|\tilde{T}|} \right)^{-1} M^{-1} \sup_{V \in \mathcal{V}} \# \tilde{\mathbb{T}}[V] |T_q|^{1/2} \right]^{-\sigma} \\
 &\gtrsim_{\delta} \delta^{-\sigma\zeta/4} q^{\omega} |W| \left[m^{-1} \frac{|\tilde{T}|}{|X|} (\# \tilde{\mathbb{T}}[V]) |T_q|^{1/2} \right]^{-\sigma}.
 \end{aligned} \tag{14.4.195}$$

where the second inequality used the fact that $\# \mathbb{T}_{\theta, V} \sim M^{-1} (\# \tilde{\mathbb{T}}[V])$ for each $V \in \mathcal{V}$, and the third inequality used (14.4.162), (14.4.187), $|T^{\dagger}| = |\tilde{T}|$, and the fact that β and ε_4 are small compared to $\sigma\zeta$.

Observe that the set on the LHS of (14.4.195) is contained in $|W|$, while the RHS involves the term $|W|$. Thus (14.4.195) gives a lower bound for the density of $\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T})$ inside W . This bound contains the term $\delta^{-\sigma\zeta/4}$ — this quantity is much larger than 1, and this will eventually allow us to conclude that Conclusion (A) of Lemma 14.4.75 holds.

Let us analyze the final term in brackets $(\dots)^{-\sigma}$. We have

$$\begin{aligned}
 m^{-1} \frac{|\tilde{T}|}{|X|} (\# \tilde{\mathbb{T}}[V]) |T_{\theta}|^{1/2} &\leq m^{-1} \frac{|\tilde{T}|}{|X|} \left(C_{F-CW}(\tilde{\mathbb{T}}^W) \frac{|V|}{|W|} (\# \tilde{\mathbb{T}}[W]) \right) |T_{\theta}|^{1/2} \\
 &\lesssim_{\delta} m^{-1} \frac{|\tilde{T}|}{|X|} \left(\frac{|V|}{|W|} \left(m \frac{|W|}{|\tilde{T}|} \right) \right) |T_{\theta}|^{1/2} \\
 &= \frac{|V|}{|X|} |T_{\theta}|^{1/2} \\
 &= \frac{\theta}{\rho},
 \end{aligned} \tag{14.4.196}$$

where the second inequality used (14.4.163) and (14.4.160).

Combining (14.4.195) and (14.4.196), we conclude that

$$\left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} Y(\tilde{T}) \right| \gtrsim_{\delta} \delta^{-\sigma\zeta/4} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |W|. \tag{14.4.197}$$

Step 11. We now spell out the density-to-gain calculation. The set on the left of (14.4.197) is contained in W , a prism of dimensions $s \times t \times 1$, with

$\delta \leq \rho \leq \theta \leq s \leq 1$. Cover W by radius- θ balls with bounded overlap and with the sum of their volumes comparable to $|W|$. Summing the intersections with the set on the left of (14.4.197) and averaging selects a ball B_θ for which

$$\left| B_\theta \cap \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} Y(\tilde{T}) \right| \gtrsim_\delta \delta^{-\sigma\zeta/4} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |B_\theta|.$$

Recall that at the beginning of Step 4 we fixed a prism $Z \in \mathcal{Z}$. The set of ρ tubes $\tilde{\mathbb{T}}$ are the images of prisms from $\mathcal{P}_3\langle Z \rangle$ under the linear map ϕ_Z . Let $B^\dagger = \phi_Z^{-1}(B_\theta)$; $B^\dagger$ is an ellipsoid of dimensions $\theta \frac{a}{\rho} \times \theta c \times \theta c$, which satisfies

$$\left| B^\dagger \cap \bigcup_{T \in \mathbb{T}_1} Y_1(T) \right| \gtrsim_\delta \delta^{-\sigma\zeta/4} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma} |B^\dagger|.$$

This is the analogue of (14.4.178). We now prove the resulting implication, using $a \in [\delta, 1]$, with all scales displayed. Put

$$u = \theta a / \rho.$$

Because $\theta \geq \rho$ and $a \geq \delta$, while $a/\rho = ac/b \leq c \leq 1$, we have $u \in [\delta, 1]$. Choose $1 \leq i \leq \varepsilon_1^{-1}$ so that

$$\tau_i = \delta^{i\varepsilon_1} \leq u < \delta^{-\varepsilon_1} \tau_i.$$

The ellipsoid $B^\dagger$ has dimensions $u \times \theta c \times \theta c$. A bounded-overlap cover by τ_i -balls, followed by averaging, loses at most the harmless factor $\delta^{3\varepsilon_1}$ and therefore gives a ball occurring in the definition of λ_i with

$$\lambda_i \gtrsim_\delta \delta^{3\varepsilon_1 - \sigma\zeta/4} \left(\frac{\rho}{\theta} \right)^{\omega+\sigma}. \quad (14.4.198)$$

Moreover, $u < \delta^{-\varepsilon_1} \tau_i$ and $a \geq \delta$ imply

$$\frac{\rho}{\theta} = \frac{a}{u} \geq \delta^{\varepsilon_1} \frac{a}{\tau_i} \geq \frac{\delta^{1+\varepsilon_1}}{\tau_i}.$$

Since $|T| \sim \delta^2$ and $|T_{\tau_i}| \sim \tau_i^2$, (14.4.198) consequently yields

$$\lambda_i \gtrsim_\delta \delta^{4\varepsilon_1 - \sigma\zeta/4} \left(\frac{\delta}{\tau_i} \right)^\omega \frac{|T|^{\sigma/2}}{|T_{\tau_i}|^{\sigma/2}}, \quad (14.4.199)$$

after making the displayed $4\varepsilon_1$ loss larger, if necessary, to absorb the bounded exponents $\omega + \sigma$. Substitute (14.4.199) into (14.4.156). The bracket in that estimate contributes $\delta^{4\varepsilon_1 - \sigma\zeta/4}$, in addition to its printed $\delta^{3\varepsilon_1}$ factor. Choosing $7\varepsilon_1 < \sigma\zeta/20$ leaves a fixed gain at least $\delta^{-\sigma\zeta/5}$, and any $\alpha < \sigma\zeta/10$ is then admissible after absorbing the sub-power constants. This proves Conclusion (A) in the case (14.4.187).

Step 12. We shall now return to the start of Step 10, except, instead of assuming (14.4.187), we will instead suppose that

$$M > \delta^{-\zeta/100} \frac{|X|}{|\tilde{T}|}. \quad (14.4.200)$$

We summarize the situation thus far:

- We have fixed a choice of $Z \in \mathcal{Z}$ (these are prisms of dimensions $\frac{a}{\rho} \times c \times c$) and $W \in \mathcal{W}_Z$.
- We have a set $\tilde{\mathbb{T}}$ of ρ tubes contained in W .
- We have a set $\mathcal{V}$ of $\theta \times \tau' \times 1$ prisms contained in W .
- For each $V \in \mathcal{V}$, we have a set $\mathcal{X}_V$ of $\rho \times \frac{\tau'}{\theta} \rho \times 1$ prisms, and a partition $\tilde{\mathbb{T}}[V] = \bigsqcup_{X \in \mathcal{X}_V} (\tilde{\mathbb{T}}[V])_X$.
- Each set $(\tilde{\mathbb{T}}[V])_X$ has cardinality roughly M , where M satisfies (14.4.200).
- We have a shading $\tilde{Y}$ on $\tilde{\mathbb{T}}$ so that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is $\gtrsim_\delta \delta^{2\varepsilon_2}$ dense.

Let $\mathcal{X} = \bigcup_{V \in \mathcal{V}} \mathcal{X}_V$ (this is a slight abuse of notation, since in Step 9 we defined $\mathcal{X}$ to be a set of the form $\mathcal{X}_V$, where the prism V was fixed in advance). To simplify notation, define $\tilde{\mathbb{T}}_X = (\tilde{\mathbb{T}}[V])_X$, where V is the (unique) prism for which $X \in \mathcal{X}_V$. Since $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is $\gtrsim_\delta \delta^{2\varepsilon_2}$ dense, after a harmless refinement of $\mathcal{X}$ we may suppose that each pair $(\tilde{\mathbb{T}}_X, \tilde{Y})_\rho$ is $\gtrsim_\delta \delta^{2\varepsilon_2}$ dense, where $\tilde{Y}$ is the restriction of the shading on $\tilde{\mathbb{T}}$ to the set $\tilde{\mathbb{T}}_X$.

Apply Corollary 14.4.62 (finding a broad scale) to each pair $(\tilde{\mathbb{T}}_X, \tilde{Y})_\rho$, for each $X \in \mathcal{X}$. This yields a scale $\tilde{\rho}$ and a $\gtrsim_\delta 1$ refinement $(\tilde{\mathbb{T}}'_X, \tilde{Y}')_\rho$ that is broad with error $\lesssim_\delta 1$ with regard to a balanced partitioning cover of $\tilde{\rho}$ tubes.

After dyadic pigeonholing and refining the set $\mathcal{X}$, we can suppose that the scale $\tilde{\rho}$ from Corollary 14.4.62 is the same for each prism X . We claim that

$$\tilde{\rho} \gtrsim_\delta \delta^{-\zeta/200} \rho. \quad (14.4.201)$$

To verify this claim, observe that for each $X \in \mathcal{X}$, the sets $\{\tilde{Y}'(\tilde{T}) : \tilde{T} \in \tilde{\mathbb{T}}'_X\}$ are $\lesssim_\delta (\tau'/\tilde{\rho})^\beta (\tilde{\rho}/\rho)$ overlapping. This is because the tubes from $\tilde{\mathbb{T}}'_X$ whose shadings pass through a common point x must point in directions confined to a $\rho \times \tau'$ region in the unit sphere $S^2 \subset \mathbb{R}^3$ (we identify S^2 with the set of directions of tubes in $\mathbb{R}^3$). Since these tubes are essentially distinct and pass through a common point, they must point in ρ separated directions. Thus at most $\rho/\tilde{\rho}$ tubes can point in directions confined to a $\rho \times \tilde{\rho}$ region in S^2 (this corresponds to those ρ tubes contained in a single $\tilde{\rho}$ tube), and at most $(\tau'/\tilde{\rho})^\beta$ distinct $\tilde{\rho}$ tubes can contribute to the count. This implies that

$$\left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}'_X} \tilde{Y}'(\tilde{T}) \right| \gtrsim_\delta \left(\frac{\tilde{\rho}}{\tau'} \right)^\beta \left(\frac{\rho}{\tilde{\rho}} \right) \sum_{\tilde{T} \in \tilde{\mathbb{T}}'_X} |\tilde{Y}'(\tilde{T})| \geq \delta^{-\zeta/200} \left(\frac{\rho}{\tilde{\rho}} \right) |X|,$$

where the second inequality used (14.4.200) and $\beta < \zeta/200$. Since the set on the LHS is contained in X , we obtain (14.4.201), as claimed.

Next, for each prism $X \in \mathcal{X}$ there exists a set $\{U\}$ of essentially distinct $\rho \times \tilde{\rho} \times 1$ prisms, each of which are contained in X , so that these sets form a partitioning cover of $\tilde{\mathbb{T}}'_X$, and for each such U , the pair $(\tilde{\mathbb{T}}'_X[U], \tilde{Y}')_\rho$ is broad with error $\lesssim_\delta 1$ relative to the cap of diameter $\tilde{\rho}$ centered at the point $\text{dir}(U)$. See Figure 14.4.11.

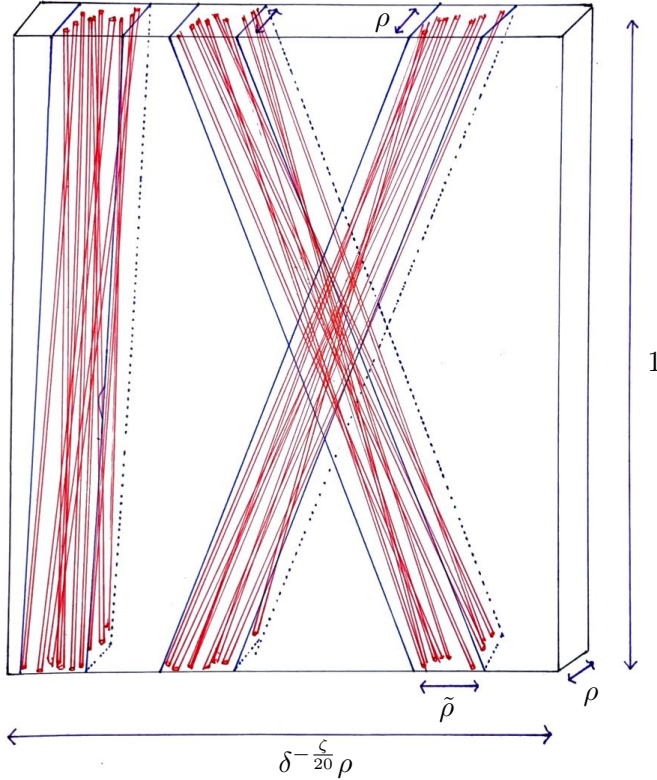


Figure 14.4.11: We find a set of $\rho \times \tilde{\rho} \times 1$ prisms (blue) inside X (black prism), which forms a partitioning cover of $\tilde{\mathbb{T}}'_X$ (red lines). A typical pair of tubes from $\tilde{\mathbb{T}}'_X$ inside a common (blue) $\rho \times \tilde{\rho} \times 1$ prism intersect at angle roughly $\tilde{\rho}$.

For each such prism U , define

$$Y(U) = \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}'_X[U]} \tilde{Y}'(\tilde{T}).$$

The source now invokes a “Cordoba-style L^2 argument” to assert

$$|Y(U)| \gtrsim_\delta \delta^{8\epsilon_2} |U| \quad \text{for every retained } U. \quad (14.4.202)$$

Here is the exact reduction that an internal proof of that assertion must close. Put $q = \rho/\tilde{\rho}$ and normalize U affinely to a unit box. A ρ -tube in U becomes a set comparable to a $q \times 1 \times 1$ slab. Write

$$N_U = \#\tilde{\mathbb{T}}'_X[U], \quad m_U = C_{KT-CW}(\{\phi_U(\tilde{T} \cap U) : \tilde{T} \in \tilde{\mathbb{T}}'_X[U]\}).$$

After regularizing the tube shadings, their normalized density is still $\gtrsim_\delta \delta^{2\varepsilon_2}$. Lemma 14.4.36 gives the fully justified estimate

$$\frac{|Y(U)|}{|U|} \gtrsim_\delta \delta^{4\varepsilon_2} m_U^{-1} q N_U. \quad (14.4.203)$$

Thus (14.4.202) follows if one proves

$$m_U^{-1} q N_U \gtrsim_\delta \delta^{4\varepsilon_2} \quad \text{for every retained } U. \quad (\text{WZ-A1})$$

Audit status (checked 31 August 2026). Condition (WZ-A1) has not been derived from the preceding printed hypotheses. Essential distinctness only separates the two-parameter offset-slope data of the normalized slabs; it does not by itself imply $m_U \lesssim 1$. Nor does the fact that the shadings on the individual tubes are $\delta^{2\varepsilon_2}$ dense give a lower bound for qN_U in each parent U . Pointwise broadness gives only $N_U \gtrsim_\delta q^{-\beta}$ when $Y(U)$ is nonempty, which yields $qN_U \gtrsim_\delta q^{1-\beta}$ and is insufficient uniformly for the asserted $\delta^{4\varepsilon_2}$ bound. A dyadic refinement can make m_U and N_U comparable across retained parents, but cannot manufacture the missing lower bound without an additional total-incidence estimate controlling the number of parents.

Accordingly, the remainder of Steps 13–14 is a complete derivation *conditional on* (WZ-A1). This is a mathematical audit boundary, not a notational or bibliographic omission. Let $\mathcal{U}$ denote the parents for which (14.4.202) holds, conditionally on (WZ-A1), as X ranges over $\mathcal{X}$. The prisms in $\mathcal{U}$ need not yet be essentially distinct.

Step 13. Our task in this step is to unwind the various transformations and rescalings from the previous step, and to understand what the prisms U (and their associated shadings $Y(U)$) correspond to in the original space in which the tubes $\mathbb{T}$ and prisms $\mathcal{P}$ reside.

In Step 12, we fixed a choice of $Z \in \mathcal{Z}$ and $W \in \mathcal{W}_Z$. Recall that $\tilde{\mathbb{T}}$ comes from $(\mathcal{P}_5\langle Z \rangle)^Z$, and is a set of $\rho \times \rho \times 1$ -tubes. We obtained a set $\mathcal{U} = \mathcal{U}_{W,Z}$ of $\rho \times \tilde{\rho} \times 1$ prisms (this $\mathcal{U}$ is different from the set $\mathcal{U}$ in Step 3, the latter $\mathcal{U}$ was defined to introduce $\mathcal{Z}$ and only appeared within Step 3), and a shading $Y(U)$ on these prisms. These prisms are contained inside W . Undoing the linear transformation ϕ_Z , we have a set $\phi_Z^{-1}(\mathcal{U}_{W,Z})$ of prisms. After pigeonholing, we may assume that these prisms are of dimensions $\tilde{a} \times \tilde{b} \times c$, for some $\tilde{a} \geq a$ and $\tilde{b} \geq b$. Since the linear transformation ϕ_Z distorts volume by a factor of

$(a/\rho)c^2$, we have that $(\tilde{a}\tilde{b}) = ac\tilde{\rho}$. The dimensions $\tilde{a}, \tilde{b}$ depend on the choice of W and Z , but after pigeonholing, we may suppose that these values are the same (up to a factor of 2) for all $Z \in \mathcal{Z}$ and all $W \in \mathcal{W}_Z$. We claim (provided ε_2 is chosen sufficiently small compared to ε_3) that either

$$\tilde{a} \leq \delta^{-\varepsilon_3} a, \quad (14.4.204)$$

or else Conclusion (A) of Lemma 14.4.75 holds (provided α is chosen sufficiently small, depending on ε_3).

We give the quantitative implication. If (14.4.204) fails, then, since $a \geq \delta$,

$$\tilde{a} > \delta^{-\varepsilon_3} a \geq \delta^{1-\varepsilon_3}.$$

Replace every $Y(Q)$ by the regular subshading supplied by Lemma 14.4.38. The preceding construction and the Jacobian of ϕ_Z^{-1} give $|Y(Q)| \gtrsim_\delta \delta^{8\varepsilon_2} |Q|$. Hence, after the usual dyadic incidence refinement, every surviving $x \in \bigcup_{T \in \mathbb{T}'} Y'(T)$ lies in a ball of radius comparable to $\tilde{a}$ satisfying

$$\left| B(x, C\tilde{a}) \cap \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_\delta \delta^{9\varepsilon_2} |B(x, C\tilde{a})|.$$

Choose a $C\tilde{a}$ -tube around every surviving δ -tube and apply Corollary 14.4.48. If its auxiliary loss is $\gamma > 0$, the same calculation as in Step 1 gives the factor

$$\delta^{\gamma + O(\varepsilon_2 + \eta)} \tilde{a}^\omega (\tilde{a}/\delta)^{\sigma/2} \geq \delta^{\omega - \varepsilon_3(\omega + \sigma/2) + \gamma + O(\varepsilon_2 + \eta)}.$$

First choose γ , then ε_2 and η , smaller than a fixed multiple of $\varepsilon_3(\omega + \sigma/2)$. This is Conclusion (A), with for example $\alpha = \varepsilon_3(\omega + \sigma/2)/2$.

Henceforth we shall assume that (14.4.204) holds. Abusing notation, we will re-define $\tilde{\rho} = \tilde{b}/c$; this re-definition might decrease the value of $\tilde{\rho}$ by as much as δ^{ε_3} , and (14.4.201) might be weakened to $\tilde{\rho} \geq \delta^{-\zeta/400} \rho$. With this re-definition of $\tilde{\rho}$, we clearly have $\tilde{b} = \tilde{\rho}c$.

Let

$$\mathcal{Q} = \bigcup_{Z \in \mathcal{Z}} \bigcup_{W \in \mathcal{W}_Z} \phi_Z^{-1}(\mathcal{U}_{Z,W}).$$

For each $Q \in \mathcal{Q}$ of the form $Q = \phi_Z^{-1}(U)$, define the shading $Y(Q) = \phi_Z^{-1}(Y(U))$ and define the set $\mathcal{P}_Q \subset \mathcal{P}[Q]$ as $\phi_Z^{-1}(\tilde{\mathbb{T}}'_X[U])$, where X is the $\rho \times \frac{\tau'}{\theta} \rho \times 1$ -prism associated to U (see Step 12). Summarizing our conclusions thus far, we have the following.

- Each $Q \in \mathcal{Q}$ is a prism of dimensions $\tilde{a} \times \tilde{b} \times c$, with $\tilde{b} = \tilde{\rho}c$.
- For each $Q \in \mathcal{Q}$, there is a set $\mathcal{P}_Q \subset \mathcal{P}[Q]$. Define $\mathcal{P}' = \bigcup_{Q \in \mathcal{Q}} \mathcal{P}_Q$. There is a shading $Y'(P) \subset Y(P)$, so that $(\mathcal{P}', Y')_{a \times b \times c}$ is a $\gtrsim_\delta \delta^{2\varepsilon_2}$ refinement of $(\mathcal{P}, Y)_{a \times b \times c}$.

- Each $Q \in \mathcal{Q}$ has a shading $Y(Q)$ given by

$$Y(Q) = \bigcup_{P \in \mathcal{P}_Q} Y'(P).$$

We have $|Y(Q)| \gtrsim_{\delta} \delta^{8\varepsilon_2} |Q|$ for each $Q \in \mathcal{Q}$.

- For each $Q \in \mathcal{Q}$ and each $x \in Y(Q)$, the prisms $P \in \mathcal{P}_Q$ with $x \in Y'(P)$ point in directions that are broad with error $\lesssim_{\delta} 1$ inside a cap of radius $\tilde{b}/c$ centered at $\text{dir}(Q)$ (these correspond to the tubes $\tilde{\mathbb{T}}'_X[U]$ and their associated shading, which are broad with error $\lesssim_{\delta} 1$ inside the $\rho \times \tilde{\rho} \times 1$ prism U).
- The refinement of $(\mathbb{T}, Y)_{\delta}$ induced by the refinement $(\mathcal{P}', Y')_{a \times b \times c}$ of $(\mathcal{P}, Y)_{a \times b \times c}$ is $\gtrsim_{\delta} \delta^{2\varepsilon_2}$ dense. If we denote this refinement by $(\mathbb{T}', Y')_{\delta}$, then (by the definition of being an induced refinement) we have

$$\bigcup_{T \in \mathbb{T}'} Y'(T) = \bigcup_{Q \in \mathcal{Q}} Y(Q).$$

The pair $(\mathcal{Q}, Y)_{\tilde{a} \times \tilde{b} \times c}$ has some of the desired properties from Conclusion (C) of Lemma 14.4.75. Observe that for each $Q \in \mathcal{Q}$, $\text{dir}(Q)$ is defined up to uncertainty $\tilde{\rho}$. Thus after a refinement of $\mathcal{Q}$ and $\mathbb{T}'$, we can find a set of $\tilde{\rho}$ tubes $\mathbb{T}_{\tilde{\rho}}$ and a partition $\mathcal{Q} = \bigsqcup_{T_{\tilde{\rho}} \in \mathbb{T}_{\tilde{\rho}}} \mathcal{Q}_{T_{\tilde{\rho}}}$, so that each $Q \in \mathcal{Q}_{T_{\tilde{\rho}}}$ is contained in $T_{\tilde{\rho}}$ and satisfies $\angle(\text{dir}(Q), \text{dir}(T_{\tilde{\rho}})) \leq \tilde{\rho}$.

Fix a prism $Q \in \mathcal{Q}_{T_{\tilde{\rho}}}$ and a point $x \in Y(Q)$. The set of prisms $P \in \mathcal{P}_Q$ with $x \in Y'(P)$ point in directions that satisfy $\angle(\text{dir}(P), \text{dir}(Q)) \leq \tilde{\rho}$, and this set of directions is ρ -separated and broad with error $\lesssim_{\delta} 1$ at scales $\geq \rho$ inside a cap of diameter $\tilde{\rho}$ centered at $\text{dir}(Q)$. For each such P (contained in a ρ tube T_{ρ}), the set of tubes $T \in \mathbb{T}[T_{\rho}]$ with $x \in Y'(T)$ are broad with error $\lesssim_{\delta} \delta^{-2\varepsilon_2}$ at scales $\geq \delta$, inside a cap of diameter ρ centered at $\text{dir}(P)$. Thus by Lemma 14.4.65 (broadness combines across scales) and Items (ii) and (iii) of Definition 14.4.53, we have that the set of tubes $T \in \mathbb{T}'$ associated to the point $x \in Y(Q)$ point in directions that are broad with error $\lesssim_{\delta} \delta^{-2\varepsilon_2}$ at scales $\geq \delta$ inside a cap of diameter $\tilde{\rho}$ centered at $\text{dir}(Q)$.

Note that even though the conclusion of Lemma 14.4.65 is a statement about multi-sets, here it is also true in the sense of sets. Indeed, Item (ii) of Definition 14.4.53 says that for each point $x \in \bigcup_{Q \in \mathcal{Q}_{T_{\tilde{\rho}}}} Y(Q)$, we have that the sets $\{\text{dir}(T) : T \in \mathbb{T}[T_{\rho}] \text{ and } x \in Y'(T) \cap Y'(P)\}$ are disjoint, as P ranges over the elements of $\bigcup_{Q \in \mathcal{Q}_{T_{\tilde{\rho}}}} \mathcal{P}_Q$ that are contained in T_{ρ} . In particular, we can construct a set $\mathbb{T}_{\tilde{\rho}}$ of $\tilde{\rho}$ tubes that covers $\mathbb{T}'$, so that $(\mathbb{T}', Y')_{\delta}$ is broad with error $\lesssim_{\delta} \delta^{-2\varepsilon_2}$ relative to $\mathbb{T}_{\tilde{\rho}}$.

Step 14.

We would like to show that $(\mathbb{T}', Y')_\delta$, $\mathcal{Q}$ and its associated shading Y , and $\tilde{\mathbb{T}}$ satisfy Conclusion (C) from Lemma 14.4.75. We supply all of the merging and disjointization bookkeeping.

Choose greedily a maximal subcollection of pairwise non-comparable prisms from $\mathcal{Q}$. Assign every discarded Q to the first selected prism comparable to it, and replace the selected prism by a fixed 100-fold transverse enlargement containing every assigned Q . A bounded-colouring of the resulting comparability graph, followed by retention of its heaviest colour, makes these enlarged prisms essentially distinct and loses only an absolute factor. Denote the set by $\tilde{\mathcal{P}}$. Its members have dimensions comparable to $\tilde{a} \times \tilde{b} \times c$. Define the shading on a merged prism to be the union of the assigned shadings and put

$$\mathcal{P}_{\tilde{P}} = \bigcup_{\substack{Q \in \mathcal{Q} \\ Q \subset \tilde{P}}} \mathcal{P}_Q.$$

If $P \in \mathcal{P}_{\tilde{P}}$, then $\angle(\text{dir}(P), \text{dir}(\tilde{P})) \lesssim \tilde{b}/c = \tilde{\rho}$. The fixed enlargement can therefore be chosen so that every tube meeting P and exiting P through its long ends also meets $\tilde{P}$ and exits $\tilde{P}$ through its long ends. This is seen in coordinates: during a length- c traversal the transverse displacement is $O(c\tilde{\rho}) = O(\tilde{b})$, and the displacement in the thin normal is $O(\tilde{a})$.

We next construct the parent cover. Take a maximal separated net in the four-dimensional position–direction parameter space of $\tilde{\rho}$ -tubes. Fixed dilates of the net tubes cover every δ -tube and every $\tilde{P}$, while the $100\tilde{\rho}$ neighbourhoods of two retained tubes have intersection of longitudinal diameter at most $1/2$. Colour the bounded-overlap net and keep the heaviest colour. After the induced refinement, each surviving tube and grain has one parent, the fibres have cardinalities within a factor two, and the parent set, denoted again by $\mathbb{T}_{\tilde{\rho}}$, is a balanced partitioning cover. The broadness estimate obtained at the end of Step 13 loses only the dyadic factor, hence has error $\lesssim_\delta \delta^{-3\varepsilon_2}$.

It remains to make the grain shadings disjoint. Regularize every merged grain shading, losing a factor two, and pigeonhole its pointwise multiplicity within each $\tilde{\rho}$ -parent. Thus one number ν satisfies

$$\#\{\tilde{P} : x \in \tilde{Y}(\tilde{P}), \tilde{P} \subset T_{\tilde{\rho}}\} \sim \nu$$

at every retained point. Fix a small $\varepsilon_5 > 0$. If $\nu \geq \delta^{-\varepsilon_5}$, essential distinctness and the common parent imply that at every such point two grain tangent planes make angle

$$\gtrsim \nu^{1/2} \frac{\tilde{a}}{\tilde{b}}.$$

Indeed, tangent planes lying in an angular interval of length s have only $O(1 + s\tilde{b}/\tilde{a})$ possible separated normal parameters; the two-dimensional packing

bound for ν parameters gives the displayed square-root separation. Lemma 14.4.39, with $\theta = \nu^{1/2}\tilde{a}/\tilde{b}$, then says that the $\nu^{1/2}\tilde{a}$ neighbourhood of every retained grain has $\delta^{C\varepsilon_2}$ density in the original union. Corollary 14.4.48 supplies the gain $\nu^{\omega/2} \geq \delta^{-\varepsilon_5\omega/2}$, which dominates the losses when ε_2, η , and its auxiliary exponent are chosen sufficiently small. This is Conclusion (A).

We may therefore assume $\nu < \delta^{-\varepsilon_5}$. Order the grains in every parent and assign each retained point to the first grain containing it. The assigned subsets are disjoint and retain at least a ν^{-1} fraction of the total grain-shading mass. Pigeonhole once more so every surviving grain retains that fraction, and define the tube shading to be its old shading intersected with the union of the assigned grains incident to it. Then

$$\bigcup_{T \in \mathbb{T}'[T_{\tilde{\rho}}]} Y'(T) = \bigsqcup_{\tilde{P} \subset T_{\tilde{\rho}}} \tilde{Y}(\tilde{P}),$$

and the grain density is at least $\gtrsim_{\delta} \delta^{8\varepsilon_2 + \varepsilon_5}$. A final dyadic pigeonhole makes $\#\mathbb{T}'[T_{\tilde{\rho}}]_{Y'}(x)$ comparable to one number μ at all retained points. Thus Items (i)–(iii) of Definition 14.4.53 hold; the coordinate exit calculation above gives Item (iv). Choosing $8\varepsilon_2 + \varepsilon_5 < \varepsilon$ absorbs every logarithmic loss and also gives the density, C_{KT-CW} , C_{F-SW} , and broadness bounds in Conclusion (C), Items (i)–(iii). Finally $\tilde{\rho} \geq \delta^{-\zeta/400}\rho$ is precisely the additional scale assertion, and $\tilde{c} = c$. Hence Conclusion (C) of Lemma 14.4.75 holds. $\square$

14.4.8 A refined induction-on-scales argument

The goal in this section is to show that if $(\mathbb{T}, Y)_{\delta}$ is a set of δ -tubes, then either $\bigcup_{\mathbb{T}} Y(T)$ has larger volume than one would expect from the estimate (14.4.3) from Assertion $\mathcal{E}(\sigma, \omega)$, or else there exists a scale $\delta \ll \rho \ll 1$ and a set of ρ tubes that factors $\mathbb{T}$ above and below with respect to the Katz-Tao Convex Wolff Axioms and Frostman Slab Wolff Axioms. The precise statement is as follows.

Proposition 14.4.76. *Let $\omega, \zeta > 0$ and $\sigma \in (0, 2/3]$, and suppose that $\mathcal{E}(\sigma, \omega)$ is true. Then there exists $\alpha, \eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_{\delta}$ be δ^{η} dense, and suppose that $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega - \alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \quad (14.4.205)$$

(B) *There exists a refinement $(\mathbb{T}', Y')_{\delta}$ of $(\mathbb{T}, Y)_{\delta}$ that is δ^{ζ} dense, a number $\rho \in [\delta^{1-\omega/100}, \delta^{\omega/100}]$, and a set $\mathbb{T}_{\rho}$ that factors $\mathbb{T}'$ above and below with respect to both the Katz-Tao Convex Wolff Axioms and the Frostman Slab Wolff Axioms, both with error $\leq \delta^{-\zeta}$.*

Proof. Step 1. Let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be small numbers to be chosen below. We will select ε_1 very small compared to ε_2 and ε_2 very small compared to ε_3 . These numbers depend on ω, σ , and ζ . We will select η and α very small compared to ε_1 .

Apply Proposition 14.4.57 (two scale grains decomposition) to $(\mathbb{T}, Y)$ with ε_1 in place of ζ , and let $\alpha_1 = \alpha_1(\omega, \sigma, \varepsilon_1)$ be the output of that proposition. If Conclusion (A) of Proposition 14.4.57 holds, then (14.4.205) is true (provided we select $\alpha \leq \alpha_1$), and we are done.

Next, suppose Conclusion (B) of Proposition 14.4.57 holds. Let $\rho, c, (\mathbb{T}_1, Y_1)_\delta, \mathbb{T}_\rho$, and $(\mathcal{G}, Y)_{a \times \rho c \times c}$ be the output from Proposition 14.4.57, Conclusion (B).

Apply Proposition 14.4.23 (factoring convex sets) to $\mathbb{T}_\rho$. We obtain a $\approx_\rho 1$ refinement of $\mathbb{T}_\rho$, which in turn induces a $\approx_\rho 1$ refinement of $(\mathbb{T}_1, Y_1)_\delta$ (abusing notation, we will continue to refer to these objects as $\mathbb{T}_\rho$ and $(\mathbb{T}_1, Y_1)_\delta$) and a collection $\mathcal{Z}$ of congruent convex sets that factors $\mathbb{T}_\rho$ from above with respect to the Katz-Tao Convex Wolff Axioms and from below with respect to the Frostman Convex Wolff Axioms, both with error $\lesssim_\rho 1$. Furthermore,

$$\#\mathbb{T}_\rho[Z] \gtrsim_\rho C_{KT-CW}(\mathbb{T}_\rho)|Z||T_\rho|^{-1} \quad \text{for each } Z \in \mathcal{Z}. \quad (14.4.206)$$

If $C_{KT-CW}(\mathbb{T}_\rho) \leq \delta^{-\zeta}$, then provided we select $\varepsilon_1 \leq \zeta/2$, we have that $(\mathbb{T}_1, Y_1)$ and $\mathbb{T}_\rho$ satisfy Conclusion (B) of Proposition 14.4.76, and we are done. Indeed; by Remark 14.4.20(A) we have $C_{F-SW}(\mathbb{T}_\rho) \lesssim C_{F-SW}(\mathbb{T}_1) \lesssim_\delta \delta^{-\eta_1 - \varepsilon_1} \leq \delta^{-\zeta}$, while by Remark 14.4.20(B) we have $C_{KT-CW}(\mathbb{T}_1^{T_\rho}) \lesssim C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ for each $T_\rho \in \mathbb{T}_\rho$.

Step 2. We now consider the case where $C_{KT-CW}(\mathbb{T}_\rho) > \delta^{-\zeta}$, and hence

$$\#\mathbb{T}_\rho[Z] \gtrsim \delta^{-\zeta} \frac{|Z|}{|T_\rho|} \quad \text{for each } Z \in \mathcal{Z}. \quad (14.4.207)$$

Our goal is to show that Conclusion (A) of Proposition 14.4.76 holds, provided $\alpha > 0$ is chosen appropriately.

First, we claim that either Conclusion (A) of Proposition 14.4.76 holds, or else the prisms in $\mathcal{Z}$ are almost tubes. Indeed, let $t \times \theta \times 2$ be the dimensions of the prisms in $\mathcal{Z}$. If ε_1 is chosen sufficiently small depending on ε_2, ω , and σ , then by applying Proposition 14.4.31 with ε_2 in place of ε , we have

$$\left| \bigcup_{T \in \mathbb{T}_1} Y_1(T) \right| \gtrsim \delta^{\omega + \varepsilon_2} \left(\frac{\theta}{t} \right)^\omega (\#\mathbb{T})|T|((\#\mathbb{T})|T|^{1/2})^{-\sigma}, \quad (14.4.208)$$

where the implicit constant depends on ε_2 . In particular, we may suppose that

$$t \geq \delta^{\varepsilon_3} \theta, \quad (14.4.209)$$

or else Conclusion (A) of Proposition 14.4.76 holds, provided $\varepsilon_2 \leq \varepsilon_3\omega/2$ and $\alpha \leq \varepsilon_3\omega/2$. Replace each $t \times \theta \times 2$ prism $Z \in \mathcal{Z}$ with its coaxial θ -tube. After dyadic pigeonholing and replacing $(\mathbb{T}_1, Y_1)_\delta$ and $\mathbb{T}_\rho$ with a $\approx_\delta 1$ refinement, we can find a balanced cover $\mathbb{T}_\theta$ of $\mathbb{T}_\rho$ that factors $\mathbb{T}_\theta$ from below with respect to the Frostman Convex Wolff Axioms and from above with respect to the Katz-Tao Wolff Axioms, both with error $\lesssim_\delta \delta^{-\varepsilon_3}$.

(14.4.207) implies that for each $T_\theta \in \mathbb{T}_\theta$ we have $\#\mathbb{T}_\rho[T_\theta] \gtrsim_\delta \delta^{\varepsilon_3-\zeta}(\theta/\rho)^2$, and hence

$$\#\mathbb{T}_\rho \gtrsim_\delta \delta^{\varepsilon_3-\zeta}(\theta/\rho)^2(\#\mathbb{T}_\theta). \quad (14.4.210)$$

Step 3. For each $T_\theta \in \mathbb{T}_\theta$, define

$$\mathcal{G}_{T_\theta} = \bigcup_{T_\rho \in \mathbb{T}_\rho[T_\theta]} \mathcal{G}_{T_\rho}.$$

We claim that either Conclusion (A) holds (for a suitably chosen value of α), or else there is a $\approx_\delta 1$ refinement of $(\mathcal{G}, Y)_{a \times \rho c \times c}$ so that the following holds: $a \in [\delta, \delta^{1-\varepsilon_1}]$ and for each $T_\theta \in \mathbb{T}_\theta$ and each $G \in \mathcal{G}_{T_\theta}$, the set of grains $G' \in \mathcal{G}_{T_\theta}$ with $Y(G) \cap Y(G') \neq \emptyset$ is contained in a prism of dimensions comparable to $\frac{\theta\delta^{1-\varepsilon_2}}{\rho} \times \theta c \times c$ (compare this with the dimensions of G , which are $a \times \rho c \times c$).

We prove the claim. If $a > \delta^{1-\varepsilon_1}$, regularize the grain shadings and retain the incident tube points. Every surviving point then has a $\delta^{C\varepsilon_1}$ -dense a -ball in the original union. Applying Corollary 14.4.48 at scale a , exactly as in the calculation following (14.4.154), yields Conclusion (A) after choosing its auxiliary exponent and η smaller than $\varepsilon_1(\omega + \sigma/2)$. We may therefore assume $a \leq \delta^{1-\varepsilon_1}$.

Fix T_θ and apply its affine normalization. A grain $G \in \mathcal{G}_{T_\theta}$ becomes a prism whose three dimensions are comparable to

$$a/\theta, \quad \rho c/\theta, \quad c.$$

Regularize its shading. For each retained point x and incident grain G_0 , consider the largest angle between the tangent plane of G_0 and that of another incident grain. Pigeonhole this angle. If it is at least

$$\delta^{-\varepsilon_1} \frac{a/\theta}{\rho c/\theta} = \delta^{-\varepsilon_1} \frac{a}{\rho c},$$

Lemma 14.4.39 fills a $\delta^{-\varepsilon_1}a$ -neighbourhood of the normalized grain to $\delta^{C\varepsilon_1}$ density. Undoing the normalization and applying Corollary 14.4.48 gives Conclusion (A); the gain is a fixed power of $\delta^{-\varepsilon_1}$ and all refinement losses are chosen smaller.

In the remaining case, every pair of grains with intersecting retained shadings has tangent-plane angle at most $C\delta^{-\varepsilon_1}a/(\rho c)$. The geometric observation

following Definition 14.4.55 now applies with $K = C\delta^{-\varepsilon_1}$. It places all grains meeting a fixed grain, in the normalized T_θ coordinates, in a prism of dimensions

$$C\delta^{-\varepsilon_1} \frac{a}{\rho} \times Cc \times Cc.$$

The inverse affine map multiplies the first two transverse dimensions by θ and leaves the longitudinal one unchanged. Thus the container in the original coordinates has dimensions

$$C\theta\delta^{-\varepsilon_1} \frac{a}{\rho} \times C\theta c \times Cc.$$

Since $a \leq \delta^{1-\varepsilon_1}$ and $2\varepsilon_1 < \varepsilon_2$, $\delta^{-\varepsilon_1}a \leq \delta^{1-2\varepsilon_1} \leq \delta^{1-\varepsilon_2}$. Enlarging the first dimension gives exactly the container asserted in the claim. The choices were made uniformly in T_θ , so one global dyadic refinement completes the proof.

We shall suppose henceforth that for each $T_\theta \in \mathbb{T}_\theta$ and each $G \in \mathcal{G}_{T_\theta}$, the set of grains $G' \in \mathcal{G}_{T_\theta}$ with $Y(G) \cap Y(G') \neq \emptyset$ is contained in a prism of dimensions comparable to $\frac{\theta\delta^{1-\varepsilon_2}}{\rho} \times \theta c \times c$.

Step 4. By dyadic pigeonholing we can find a number μ_{fine} and a $\approx_\delta 1$ refinement of $(\mathbb{T}_1, Y_1)_\delta$ and $\mathbb{T}_\rho$ so that for each $T_\rho \in \mathbb{T}_\rho$ and each $x \in \bigcup_{T \in \mathbb{T}_1[T_\rho]} Y_1(T)$, we have

$$\#((\mathbb{T}_1[T_\rho])_{Y_1}(x)) \sim \mu_{\text{fine}}.$$

We can choose these refinements so it continues to be the case that $(\mathbb{T}_1^{T_\rho}, Y_1^{T_\rho})_{\delta/\rho}$ is $\gtrsim_\delta \delta^{\varepsilon_1}$ dense and $C_{F-SW}(\mathbb{T}_1^{T_\rho}) \lesssim_\delta \delta^{-\varepsilon_1}$ for each $T_\rho \in \mathbb{T}_\rho$ (recall that we still have $C_{KT-CW}(\mathbb{T}_1^{T_\rho}) \leq \delta^{-\eta}$).

If ε_1 is chosen sufficiently small depending on ε_2 , then we can apply the estimate $\mathcal{E}(\sigma, \omega)$ to conclude that for each $T_\rho \in \mathbb{T}_\rho$, we have

$$\left| \bigcup_{T \in \mathbb{T}_1[T_\rho]} Y_1(T) \right| \gtrsim \left(\frac{\delta}{\rho} \right)^{\omega + \varepsilon_2} (\#\mathbb{T}_1[T_\rho]) |T| \left((\#\mathbb{T}_1[T_\rho]) \left(\frac{|T|}{|T_\rho|} \right)^{1/2} \right)^{-\sigma},$$

where the implicit constant depends on ε_2 , and hence by (14.4.210),

$$\mu_{\text{fine}} \lesssim \left(\frac{\delta}{\rho} \right)^{-\omega - \varepsilon_2} \left(\frac{\#\mathbb{T}_1}{\#\mathbb{T}_\rho} \frac{\delta}{\rho} \right)^\sigma \lesssim_\delta \delta^{-2\varepsilon_3 + \sigma\zeta} \left(\frac{\delta}{\rho} \right)^{-\omega} \left(\frac{\#\mathbb{T}_1}{\#\mathbb{T}_\theta} \frac{\delta\rho}{\theta^2} \right)^\sigma. \quad (14.4.211)$$

Step 5. In previous applications of induction on scale, the estimate (14.4.211) would be paired with a multiplicity estimate on the tubes in $\mathbb{T}_\rho$. Our innovation, however, is to pair the estimate (14.4.211) with a multiplicity estimate on $\mathcal{G}$.

After refining the pair $(\mathcal{G}, Y)_{a \times \rho c \times c}$ by a $\approx_\delta 1$ factor (this in turn refines $(\mathbb{T}_1, Y_1)_\delta$ by a similar quantity), we can find a number μ_{medium} so that for each $T_\theta \in \mathbb{T}_\theta$ and each $x \in \bigcup_{G \in \mathcal{G}_{T_\theta}} Y(G)$, we have

$$\#\{G \in \mathcal{G}_{T_\theta} : x \in Y(G)\} \sim \mu_{\text{medium}}.$$

Our task is to estimate μ_{medium} . Recalling the conclusion of Step 3, we can cover T_θ by rectangular prisms P of dimensions comparable to $\frac{\theta\delta^{1-\varepsilon_2}}{\rho} \times \theta c \times c$, so that every pair of grains $G, G' \in \mathcal{G}_{T_\theta}$ with $Y(G) \cap Y(G') \neq \emptyset$ are contained in a common prism. Let $\mathcal{P}$ denote this set of prisms; then for each $P \in \mathcal{P}$, $\mathcal{G}_{T_\theta}^P$ is a set of prisms, each of which has dimensions roughly $\frac{\rho}{\theta} \times \frac{\rho}{\theta} \times 1$ (more precisely, each prism in $\mathcal{G}_{T_\theta}^P$ has dimensions comparable to $s \times t \times 1$, where $s, t \in [\delta^{\varepsilon_2} \frac{\rho}{\theta}, \frac{\rho}{\theta}]$; this additional δ^{ε_2} factor will be harmless). After pigeonholing, we may assume that the lengths s and t are the same for every $T_\theta \in \mathbb{T}_\theta$ and every $P \in \mathcal{P}$.

We have $C_{KT-CW}(\mathcal{G}_{T_\theta}^P) \lesssim \delta^{-\varepsilon_2} C_{KT-CW}^{\text{loc}}(\mathcal{G}) \leq \delta^{-\varepsilon_1-\varepsilon_2}$, while

$$\begin{aligned} C_{F-SW}(\mathcal{G}_{T_\theta}^P) &\leq C_{F-CW}(\mathcal{G}_{T_\theta}^P) \\ &\leq C_{KT-CW}(\mathcal{G}_{T_\theta}^P)(\rho/\theta)^{-2}(\#\mathcal{G}_{T_\theta}^P)^{-1} \\ &\leq \delta^{-\varepsilon_1-2\varepsilon_2}(\rho/\theta)^{-2}(\#\mathcal{G}_{T_\theta}^P)^{-1}. \end{aligned}$$

If ε_2 is selected sufficiently small depending on ε_3 , ω , and σ , then we can apply Assertion $\mathcal{E}(\sigma, \omega)$ to conclude that

$$\begin{aligned} \left| \bigcup_{G^P \in \mathcal{G}_{T_\theta}^P} Y^P(G^P) \right| &\geq \left(\frac{\rho}{\theta} \right)^{\omega+\varepsilon_3} \delta^{5\varepsilon_2} (\#\mathcal{G}_{T_\theta}^P) |G^P| \\ &\quad \times \left[\left(\frac{\rho}{\theta} \right)^{-2} (\#\mathcal{G}_{T_\theta}^P)^{-1} (\#\mathcal{G}_{T_\theta}^P) |G^P|^{1/2} \right]^{-\sigma} \\ &\gtrsim_\delta \left(\frac{\rho}{\theta} \right)^{\omega+\varepsilon_3} \delta^{5\varepsilon_2} (\#\mathcal{G}_{T_\theta}^P) |G^P| \left(\frac{\rho}{\theta} \right)^\sigma. \end{aligned}$$

and thus

$$\mu_{\text{medium}} \lesssim_\delta \delta^{-2\varepsilon_3} \left(\frac{\rho}{\theta} \right)^{-\omega-\sigma}. \quad (14.4.212)$$

Step 6. At this point, we have estimated the quantities μ_{fine} and μ_{medium} . The former allows us to control the number of δ tubes that contribute to (a specific point in) a grain, while the latter allows us to control the number of grains that contribute to (a specific point in) a θ tube.

It remains to place a dense shading on $\mathbb{T}_\theta$ and obtain a corresponding multiplicity estimate for the number of θ tubes that contribute to (a specific point in) $\mathbb{R}^3$. After dyadic pigeonholing, we can refine $(\mathbb{T}_1, Y_1)_\delta$ and $\mathbb{T}_\rho$ so that for each $T_\theta \in \mathbb{T}_\theta$ and each $x \in \bigcup_{T \in \mathbb{T}_1[T_\theta]} Y_1(T)$, we have that $|B(x, \theta) \cap \bigcup_{T \in \mathbb{T}_1[T_\theta]} Y(T)|$ has roughly the same volume. Let $Y(T_\theta) = T_\theta \cap N_\theta(\bigcup_{T \in \mathbb{T}_1[T_\theta]} Y_1(T))$; then $(\mathbb{T}_\theta, Y)_\theta$ is $\gtrsim_\delta \delta^{\varepsilon_1}$ dense. After further pigeonholing we can find a number μ_{coarse} so that

$$\#(\mathbb{T}_\theta)_Y(x) \sim \mu_{\text{coarse}} \quad \text{for each } x \in \bigcup_{T_\theta \in \mathbb{T}_\theta} Y(T_\theta).$$

By Remark 14.4.20(A), we have $C_{F-SW}(\mathbb{T}_\theta) \lesssim C_{F-SW}(\mathbb{T}) \lesssim_\delta \delta^{-\varepsilon_1}$, and $C_{KT-CW}(\mathbb{T}_\theta) \lesssim \delta^{-\varepsilon_3} C_{KT-CW}(\mathbb{Z}) \lesssim_\delta \delta^{-\varepsilon_3}$. Thus if ε_1 is chosen sufficiently small compared to ε_2 , then we can apply $\mathcal{E}(\sigma, \omega)$ to conclude that

$$\left| \bigcup_{T_\theta \in \mathbb{T}_\theta} Y(T_\theta) \right| \gtrsim \theta^{\omega + \varepsilon_2 + \varepsilon_3} (\#\mathbb{T}_\theta) |T_\theta| \left((\#\mathbb{T}_\theta) |T_\theta|^{1/2} \right)^{-\sigma},$$

and hence

$$\mu_{\text{coarse}} \lesssim \theta^{-\omega - \varepsilon_2 - \varepsilon_3} \left((\#\mathbb{T}_\theta) \theta \right)^\sigma. \quad (14.4.213)$$

Combining (14.4.211), (14.4.212), and (14.4.213), we conclude that for each $x \in \mathbb{R}^3$ we have

$$\begin{aligned} \#\{T \in \mathbb{T}_1 : x \in Y_1(T)\} &\leq \mu_{\text{fine}} \mu_{\text{medium}} \mu_{\text{coarse}} \\ &\lesssim \delta^{-4\varepsilon_3 + \sigma\zeta} \delta^{-\omega} \left((\#\mathbb{T}) \delta \right)^\sigma. \end{aligned} \quad (14.4.214)$$

We conclude that

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim_\delta \delta^{\omega + 4\varepsilon_3 - \sigma\zeta} (\#\mathbb{T}_1) |T| \left((\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma}.$$

Since $\#\mathbb{T}_1 \gtrsim_\delta \delta^{\varepsilon_1} (\#\mathbb{T})$, we have that Conclusion (A) holds, provided we select $\varepsilon_3 < \sigma\zeta/10$ and $\alpha < \sigma\zeta/10$. $\square$

14.4.9 Sticky Kakeya for tubes satisfying the Katz-Tao Convex Wolff Axioms at every Scale

In Section 14.4.5, we recalled a version of the Sticky Kakeya Theorem that was proved in [26]; this is Theorem 14.4.50. Theorem 14.4.50 applies to families of tubes that satisfy the Frostman Convex Wolff Axioms at every scale, in the sense of Definition 14.4.49. In this section, we will prove an analogue of Theorem 14.4.50 for sets of tubes that satisfy the *Katz-Tao* Convex Wolff Axioms at every scale.

Definition 14.4.77. Let $K \geq 1, \delta > 0$. We say a set $\mathbb{T}$ of essentially distinct δ -tubes satisfies the *Katz-Tao Wolff Axioms at every scale with error K* if for every $\rho_0 \in [\delta, 1]$, there exists $\rho \in [\rho_0, K\rho_0]$ and a set of ρ -tubes $\mathbb{T}_\rho$ that satisfies the following properties.

- (i) $\mathbb{T}_\rho$ is a K -balanced partitioning cover of $\mathbb{T}$.
- (ii) $C_{KT-CW}(\mathbb{T}_\rho) \leq K$.

Theorem 14.4.78. *For all $\varepsilon > 0$, there exists $\eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $\mathbb{T}$ be a set of δ -tubes that satisfy the Katz-Tao Convex Wolff Axioms at every scale with error $\delta^{-\eta}$, and let $Y(T)$ be a δ^η dense shading. Then*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^\varepsilon (\#\mathbb{T}) |T|. \quad (14.4.215)$$

In the next section, we will combine Theorem 14.4.78 with Proposition 14.4.76 to prove Proposition 14.4.7. Theorem 14.4.78 is proved by combining Theorem 14.4.50 with the following Nikishin-Stein-Pisier Factorization type result.

Proposition 14.4.79. *Let $\varepsilon > 0$. Then there exists $K_\varepsilon, \eta > 0$ so that the following holds for all $\delta > 0$. Let $\mathbb{T}$ be a non-empty set of δ tubes inside the unit ball in $\mathbb{R}^3$ that satisfy the Katz-Tao Convex Wolff axioms at every scale with error $\delta^{-\eta}$. Then there exist rigid transformations $A_1, \dots, A_N$, $N \leq K_\varepsilon (\#\mathbb{T})^{-1} |T|^{-1}$ so that each set $A_i(\mathbb{T})$ is contained inside $B(0, 2)$, and $\bigcup_{i=1}^N A_i(\mathbb{T})$ contains a subset of essentially distinct tubes that satisfies the Frostman Convex Wolff Axioms at every scale with error $K_\varepsilon \delta^{-\varepsilon}$.*

Proof of Theorem 14.4.78 using Proposition 14.4.79. Fix $\varepsilon > 0$ and let $\eta = \eta(\varepsilon) > 0$ be a small quantity to be determined below. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, and suppose that $\mathbb{T}$ satisfies the Katz-Tao Wolff Axioms at every scale with error $\delta^{-\eta}$. After a harmless refinement we may suppose $|Y(T)| \geq \delta^{2\eta} |T|$ for each $T \in \mathbb{T}$.

Apply Proposition 14.4.79 with a small value ε_1 in place of ε . We may do this, provided η is selected sufficiently small depending on ε_1 . Let $\tilde{\mathbb{T}} \subset \bigcup_{i=1}^N A_i(\mathbb{T})$ be the output from Proposition 14.4.79. Note that each $\tilde{T} \in \tilde{\mathbb{T}}$ is of the form $\tilde{T} = A_i(T)$ for some index i and some $T \in \mathbb{T}$, and hence we can define the shading $\tilde{Y}(\tilde{T}) = A_i(Y(T))$; we have $|\tilde{Y}(\tilde{T})| = |Y(T)| \geq \delta^{2\eta} |T|$, and hence $(\tilde{\mathbb{T}}, \tilde{Y})_\delta$ is $\delta^{2\eta}$ dense.

If ε_1 and η are chosen sufficiently small depending on ε , then we can apply Theorem 14.4.50 to conclude that

$$\begin{aligned} N \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &= \sum_{i=1}^N \left| \bigcup_{T \in \mathbb{T}} A_i(Y(T)) \right| \\ &\geq \left| \bigcup_{i=1}^N \bigcup_{T \in \mathbb{T}} A_i(Y(T)) \right| \\ &\geq \left| \bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right| \geq \kappa_\varepsilon \delta^\varepsilon. \end{aligned}$$

Re-arranging and noting that $N \leq K_\varepsilon (\#\mathbb{T})^{-1} |T|^{-1}$, we obtain (14.4.215), with $\kappa = \kappa_\varepsilon K_\varepsilon^{-1}$. $\square$

It remains to prove Proposition 14.4.79. We will do so below.

14.4.9.1 Nikishin-Stein-Pisier Factorization and the Convex Wolff Axioms

We begin with a single-scale version of Proposition 14.4.79. We first need the following definition.

Definition 14.4.80. We say that a set $\mathbb{T}$ of δ tubes is regular with granularity $\tau \in [\delta, 1]$ if for every scale $\rho \in [\delta, 1]$ of the form $\rho = \delta\tau^{-\ell}$, $\ell \in \mathbb{N}$, we have that $\mathbb{T}$ has a balanced partitioning cover by ρ tubes.

Definition 14.4.81. For $\rho > 0$, we define $\mathfrak{A}_\rho$ to be the set of rigid transformations $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that satisfy $|Ax - x| \leq \rho$ for all $x \in B(0, 1)$.

Lemma 14.4.82. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $K_\varepsilon \geq 1$ so that the following holds for all $0 < \delta \leq \rho \leq 1$. Let $K, M \geq 1$ and let $\mathbb{T}_1, \dots, \mathbb{T}_K$ be sets of δ tubes in $B(0, 1) \subset \mathbb{R}^3$, each of cardinality at most M . Suppose that the tubes in each set $\mathbb{T}_j$ are regular with granularity δ^η , and furthermore each set $\mathbb{T}_j$ is contained in a ρ tube.*

Then there exists a set of rigid transformations $\mathcal{A} \subset \mathfrak{A}_\rho$ with $\#\mathcal{A} = \lceil \frac{\rho^2}{M\delta^2} \rceil$, so that

$$C_{KT-CW} \left(\bigsqcup_{A \in \mathcal{A}} A(\mathbb{T}_j) \right) \leq K_\varepsilon \delta^{-\varepsilon} (\log(2 + K)) C_{KT-CW}(\mathbb{T}_j), \quad j = 1, \dots, K. \quad (14.4.216)$$

Remark 14.4.83. Note that for distinct $A, A' \in \mathcal{A}$, the sets $A(\mathbb{T}_j)$ and $A'(\mathbb{T}_j)$ might contain common tubes, and thus the disjoint union on the LHS of (14.4.216) should be interpreted as a multiset. By Remark 14.4.19(D), the LHS of (14.4.216) is well-defined.

Proof.

Step 1. Define $\tilde{\delta} = \delta/\rho$. First, we may suppose that $M < \tilde{\delta}^{-2}$, or else we can define $\mathcal{A} = \{I\}$ (here $I: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the identity map) and we are done. Similarly, we may suppose $\rho \geq \delta^{1-\varepsilon/2}$, or else we can define $\mathcal{A}$ to be $\lceil \frac{\rho^2}{M\delta^2} \rceil$ infinitesimally perturbed copies of I , and (14.4.216) follows from the fact that

$$C_{KT-CW} \left(\bigsqcup_{A \in \mathcal{A}} A(\mathbb{T}_j) \right) \leq (\#\mathcal{A}) C_{KT-CW}(\mathbb{T}_j).$$

Fix an index $j \in [1, \dots, K]$ and let $\mathbb{T} = \mathbb{T}_j$. By hypothesis, all of the tubes in $\mathbb{T}$ are contained in a common ρ tube, which we will denote by T_ρ . Fix numbers $\delta \leq a \leq b \leq 2\rho$, with both a and b of the form $\delta^{\ell\eta}$. Let $\nu \geq 1$ be a power of 2. By hypothesis, $\mathbb{T}$ has a balanced partitioning cover $\mathbb{T}_a$.

Let $\mathcal{W}_\nu$ be a maximal set of essentially distinct $a \times b \times 2$ prisms, each of which satisfy $\#\mathbb{T}[W] \in [\nu \frac{\#\mathbb{T}}{\#\mathbb{T}_a}, 2\nu \frac{\#\mathbb{T}}{\#\mathbb{T}_a})$. Note that each $W \in \mathcal{W}_\nu$ is contained in $N_{2\rho}(T_\rho)$. Observe that if W is an $a \times b \times 2$ prism and $T \in \mathbb{T}[T_a]$ with $T \subset W$, then $T_a \subset 2W$. In particular, since $\mathbb{T}_a$ is a balanced partitioning cover of $\mathbb{T}$, we have

$$\#\mathbb{T}_a[2W] \geq \nu/2 \quad \text{for each } W \in \mathcal{W}_\nu.$$

Each tube $T_a \in \mathbb{T}_a$ is contained in $\lesssim \frac{b}{a}$ essentially distinct $2a \times 2b \times 4$ prisms. Thus by double-counting we have

$$\#\mathcal{W}_\nu \lesssim (\#\mathbb{T}_a) \frac{b}{a} \nu^{-1}. \quad (14.4.217)$$

The above estimate is useful when ν is not too large. When $\nu \frac{\#\mathbb{T}}{\#\mathbb{T}_a} \geq C_{KT-CW}(\mathbb{T})(ab)|T|^{-1}$, then $\mathcal{W}_\nu = \emptyset$.

Step 2. We say two rigid motions $A, A' \in \mathfrak{A}_\rho$ are δ -separated if there exists a point $x \in B(0, 1)$ with $|A(x) - A'(x)| \geq \delta$. Let $\mathfrak{A}_\rho^\delta$ be a maximal δ -separated subset of $\mathfrak{A}_\rho$; since $\tilde{\delta} = \delta/\rho$, the six translation-rotation parameters give

$$\#\mathfrak{A}_\rho^\delta \sim \tilde{\delta}^{-6} = \left(\frac{\rho}{\delta}\right)^6.$$

Let

$$N = 2 \left\lceil \frac{\rho^2}{M\delta^2} \right\rceil, \quad (14.4.218)$$

and let $A_1, \dots, A_N$ be chosen uniformly and independently at random from $\mathfrak{A}_\rho^\delta$. We have

$$\mathbb{P}(\#\{A_1, \dots, A_N\} \geq N/2) \geq 3/4, \quad (14.4.219)$$

where $\#\{A_1, \dots, A_N\}$ denotes the number of distinct rigid motions in the set $\{A_1, \dots, A_N\}$.

Fix a $a \times b \times 2$ prism W_0 . We would like to estimate the probability that

$$\#\left(\bigcup_{i=1}^N A_i(\mathbb{T})\right)[W_0] \geq K_\varepsilon \delta^{-\varepsilon/2} \log(2 + K) C_{KT-CW}(\mathbb{T}) |W_0| |T|^{-1}, \quad (14.4.220)$$

i.e. we would like to estimate the probability that

$$\sum_{i=1}^N \#\mathbb{T}[A_i^{-1}(W_0)] \geq K_\varepsilon \delta^{-\varepsilon/2} \log(2 + K) C_{KT-CW}(\mathbb{T}) |W_0| |T|^{-1}. \quad (14.4.221)$$

Note that if (14.4.221) occurs, then by pigeonholing, there must be some dyadic ν so that

$$\#\{(W, i) \in \mathcal{W}_\nu \times \{1, \dots, N\} : A_i(W) \subset 10W_0\} \geq Z_\nu, \quad (14.4.222)$$

where

$$Z_\nu = (4 \log(1/\delta))^{-1} K_\varepsilon \delta^{-\varepsilon/2} \log(2 + K) \times C_{KT-CW}(\mathbb{T}) |W_0| |T|^{-1} \left(\sup_{W \in \mathcal{W}_\nu} \# \mathbb{T}[W] \right)^{-1}. \quad (14.4.223)$$

We may suppose that

$$\sup_{W \in \mathcal{W}_\nu} \# \mathbb{T}[W] \sim \nu \frac{\# \mathbb{T}}{\# \mathbb{T}_a} \leq C_{KT-CW}(\mathbb{T})(ab) |T|^{-1}, \quad (14.4.224)$$

since otherwise $\mathcal{W}_\nu = \emptyset$, and thus it is impossible for either of Inequality (14.4.222) or (14.4.221) to be true. Since $|W_0| = 2ab$, we can use (14.4.224) to bound Z_ν . We conclude that

$$Z_\nu \geq (2 \log(1/\delta))^{-1} K_\varepsilon \delta^{-\varepsilon/2} (\log(2 + K)).$$

We will choose the constant K_ε sufficiently large so that $Z_\nu \geq 2$, and in particular

$$Z_\nu - 1 \sim Z_\nu. \quad (14.4.225)$$

This will be relevant in Step 3 when we apply Chernoff's inequality.

Step 3. We will estimate the probability that (14.4.222) occurs for a fixed choice of ν and W_0 . For two $a \times b \times 2$ prisms W, W_0 , both of which are contained inside $N_{2\rho}(T_\rho)$ we have that the number of $A \in \mathfrak{A}_\rho^\delta$ for which $A(W)$ is comparable to W_0 (or equivalently, $A^{-1}(W_0)$ is comparable to W) is $\lesssim \delta^{-6} a^2 b^2 \rho \min\{a/b, \rho\}$. We will write this as $\tilde{\delta}^{-6} \tilde{a}^2 \tilde{b}^2 \min\{\tilde{a}/(\tilde{b}\rho), 1\} \leq \tilde{\delta}^{-6} \tilde{a}^2 \tilde{b}^2$, where we define $\tilde{a} = a/\rho$ and $\tilde{b} = b/\rho$.

The reason for this numerology is as follows. Without loss of generality, assume $W_0 = [0, a] \times [0, b] \times [0, 2]$. Then a rigid motion A is determined by $A(v_i)$ with $v_0 = (0, 0, 0)$, $v_1 = (0, 1, 0)$, $v_2 = (0, 0, 1)$. Since $A \in \mathfrak{A}_\rho^\delta$, the number of δ -separated choice for $A(v_0)$ is $\leq \frac{|W \cap B(0, \rho)|}{\delta^3} \sim \frac{ab\rho}{\delta^3}$. Once $A(v_0)$ is fixed, the number of δ -separated choices for $A(v_2)$ is $\leq \frac{ab}{\delta^2}$. Once $A(v_0), A(v_2)$ are fixed, the number of δ -separated choices for $A(v_1)$ is $\leq \min\{\frac{a}{b}, \rho\} \delta^{-1}$.

Thus if we define X_i to be the event that there exists $W \in \mathcal{W}_\nu$ such that $A_i(W)$ is comparable to W_0 , then

$$\mathbb{P}(X_i) \lesssim \tilde{\delta}^{-6} \tilde{a}^2 \tilde{b}^2 \cdot \# \mathfrak{A}_\rho^\delta \cdot \# \mathcal{W}_\nu \lesssim \tilde{a}^2 \tilde{b}^2 (\# \mathcal{W}_\nu). \quad (14.4.226)$$

Since the prisms in $\mathcal{W}_\nu$ are essentially distinct, there can exist at most $O(1)$ $W \in \mathcal{W}_\nu$ such that $A_i(W)$ is comparable to W_0 . Thus by linearity of expectation,

we have

$$\begin{aligned}
 & \mathbb{E}\left(\#\{(W, i) \in (\mathcal{W}_\nu \times \{1, \dots, N\}) : A_i(W) \text{ is comparable to } W_0\}\right) \\
 &= \mathbb{E}(X_1 + \dots + X_N) \\
 &\lesssim N \tilde{a}^2 \tilde{b}^2 (\#\mathcal{W}_\nu) \\
 &\lesssim 2 \left\lceil \frac{\rho^2}{M\delta^2} \right\rceil \tilde{a}^2 \tilde{b}^2 (\#\mathcal{W}_\nu)
 \end{aligned}$$

On the third line we used (14.4.226); on the fourth line we used (14.4.218) and (14.4.217).

Define $X = X_1 + \dots + X_N$. Recalling (14.4.223) and (14.4.225), we have

$$\begin{aligned}
 \gamma := \frac{Z_\nu}{\mathbb{E}X} &\gtrsim \frac{K_\varepsilon \delta^{-\varepsilon/2} \log(2+K)}{\log(1/\delta)} C_{KT-CW}(\mathbb{T}) |W_0| |T|^{-1} \\
 &\quad \times \left[2 \left\lceil \frac{\rho^2}{M\delta^2} \right\rceil \tilde{a}^2 \tilde{b}^2 \left(\sup_{W \in \mathcal{W}_\nu} \#\mathbb{T}[W] \right) (\#\mathcal{W}_\nu) \right]^{-1} \\
 &\gtrsim \frac{K_\varepsilon \delta^{-\varepsilon/2} \log(2+K)}{\log(1/\delta)} C_{KT-CW}(\mathbb{T}) |W_0| |T|^{-1} \\
 &\quad \times \left[2 \left\lceil \frac{\rho^2}{M\delta^2} \right\rceil \tilde{a}^2 \tilde{b}^2 \frac{b}{a} (\#\mathbb{T}) \right]^{-1} \\
 &\gtrsim \log(1/\delta)^{-1} K_\varepsilon \delta^{-\varepsilon/2} \log(2+K).
 \end{aligned}$$

On the second line we used (14.4.217) and the inequality $\#\mathbb{T}[W] \sim \nu(\#\mathbb{T})/(\#\mathbb{T}_a)$; on the third line we used $|T| \sim \delta^2$ and $\tilde{a} = a/\rho$; and on the final line we used $C_{KT-CW}(\mathbb{T}) \geq 1$, $\tilde{b}^2 \leq 4$, $|W_0| = 2ab$, and $\#\mathbb{T} \leq M$.

Hence we can apply the multiplicative Chernoff's inequality to conclude that

$$\begin{aligned}
 \mathbb{P}(X \geq Z_\nu) &\leq \left(\frac{e^{\gamma-1}}{\gamma} \right)^{\mathbb{E}X} \lesssim e^{-\gamma \mathbb{E}X} = e^{-Z_\nu} \\
 &\lesssim (2+K)^{-(\log(1/\delta))^{-1} K_\varepsilon \delta^{-\varepsilon/2}} \\
 &\lesssim K^{-1} \exp[-K_\varepsilon \delta^{-\varepsilon/3}].
 \end{aligned}$$

For the second inequality, we used the fact that K_ε is sufficiently large, $\frac{e^{\gamma-1}}{\gamma} \leq e^{-\gamma}$ for all $\delta > 0$ and $K \geq 1$.

Step 4. There are $\log(1/\delta)$ choices of ν ; at most $a^{-3}b^{-1} \leq \delta^{-4}$ essentially distinct prisms W_0 ; and η^{-2} choices of numbers (a, b) . Thus the probability that there exists some prism $W \subset N_{2\rho}(T_\rho)$ of dimensions $a \times b \times 2$ for some pair $a \leq b$ of the form δ^{η^ℓ} for which (14.4.220) holds is $\lesssim \delta^{-6} K^{-1} \exp[-K_\varepsilon \delta^{-\varepsilon/3}]$.

We will select K_ε sufficiently large so that this quantity is at most $(2K)^{-1}$. If no such prism exists, then provided $\eta \leq \varepsilon/4$ we have

$$C_{KT-CW} \left(\bigsqcup_{i=1}^N A_i(\mathbb{T}) \right) \leq K_\varepsilon \delta^{-\varepsilon} (\log(2+K)) C_{KT-CW}(\mathbb{T}). \quad (14.4.227)$$

Indeed, for every prism $W \subset \mathbb{R}^3$ of dimensions $a \times b \times 2$, we can select a prism $W' \subset W$ of dimensions $a' \times b' \times 2$ with a', b' of the form $\delta^{\ell\eta}$ and $|W'| \leq \delta^{-2\eta}|W| \leq \delta^{-\varepsilon/2}|W|$.

In particular, we have that (14.4.227) holds with probability at least $1 - (2K)^{-1}$. Since there are K sets of tubes $\mathbb{T}_1, \dots, \mathbb{T}_K$, we conclude that with probability at least $1/2$, we have that (14.4.227) is true for every set $\mathbb{T}_1, \dots, \mathbb{T}_K$. Finally, by (14.4.219), we have that the probability that $\#\mathcal{A} \geq N/2$ and (14.4.227) is true for every set $\mathbb{T}_1, \dots, \mathbb{T}_K$ is at least $1/4$. We conclude that there exists a choice of $A_1, \dots, A_N$ and a set $\mathcal{A} \subset \{A_1, \dots, A_N\}$ of cardinality $\left\lceil \frac{\rho^2}{M\delta^2} \right\rceil$ so that (14.4.216) holds. $\square$

Lemma 14.4.82 has the following consequence.

Corollary 14.4.84. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $K_\varepsilon \geq 1$ so that the following holds for all $0 < \delta \leq \rho \leq 1$. Let $\mathbb{T}$ be a set of δ -tubes, and let $\mathbb{T}_\rho$ be a balanced partitioning cover of $\mathbb{T}$. Suppose that for each $T_\rho \in \mathbb{T}_\rho$, we have that $\mathbb{T}^{T_\rho}$ is regular with granularity $(\delta/\rho)^\eta$.*

Then there exists a set of rigid transformations $\mathcal{A}$ with $\#\mathcal{A} = 2^{\lceil \frac{\#\mathbb{T}_\rho}{\#\mathbb{T}} \frac{|T_\rho|}{|T|} \rceil}$ and

$$A(T_\rho) \subset N_{3\rho}(T_\rho) \quad \text{for each } A \in \mathcal{A}, T_\rho \in \mathbb{T}_\rho, \quad (14.4.228)$$

so that

$$C_{KT-CW} \left(\left(\bigsqcup_{A \in \mathcal{A}} A(\mathbb{T}) \right) [N_{3\rho}(T_\rho)] \right) \leq K_\varepsilon \delta^{-\varepsilon} C_{KT-CW}(\mathbb{T}) \quad \text{for each } T_\rho \in \mathbb{T}_\rho. \quad (14.4.229)$$

Remark 14.4.85. An immediate application of Lemma 14.4.82 yields the slightly weaker statement

$$C_{KT-CW} \left(\left(\bigsqcup_{A \in \mathcal{A}} A(\mathbb{T}[T_\rho]) \right) \right) \leq K_\varepsilon \delta^{-\varepsilon} C_{KT-CW}(\mathbb{T}) \quad \text{for each } T_\rho \in \mathbb{T}_\rho,$$

but (14.4.229) follows from the hypothesis that the tubes in $\mathbb{T}_\rho$ are essentially distinct, and hence there are $\lesssim 1$ tubes $T'_\rho \in \mathbb{T}_\rho$ for which $(\bigsqcup_{A \in \mathcal{A}} A(\mathbb{T}[T'_\rho]))[T_\rho]$ is non-empty.

We are now ready to prove Proposition 14.4.79.

Proof of Proposition 14.4.79. Step 1. Fix $\varepsilon > 0$. Let $\varepsilon_1 > 0$ be a small quantity to be chosen below. We will choose ε_1 small compared to ε , and η small compared to ε_1 . Let $\mathbb{T}$ be a set of δ tubes that satisfy the Katz-Tao Convex Wolff axioms at every scale with error $\delta^{-\eta}$.

Provided we select $\eta \leq \varepsilon$, there exists a number $J \sim \varepsilon^{-1}$ and scales $1 = \rho_0 > \rho_1 > \dots > \rho_J = \delta$ with $\delta^\varepsilon \leq \rho_{j+1}/\rho_j < 1/8$ for each index j , so that the following holds: for each index $j = 0, \dots, J-1$, there is a $\delta^{-\eta}$ -balanced partitioning cover $\mathbb{T}_{\rho_j}$ of $\mathbb{T}$, with $C_{KT-CW}(\mathbb{T}_{\rho_j}) \leq \delta^{-\eta}$. Define $\mathbb{T}_{\rho_J} = \mathbb{T}$.

Currently, $\mathbb{T}_{\rho_j}$ covers $\mathbb{T}$, but unfortunately, it might not be the case that $\mathbb{T}_{\rho_j}$ covers $\mathbb{T}_{\rho_{j+1}}$. We can fix this as follows. We know that for each index $j > 0$, we have $\mathbb{T}[T_{\rho_{j+1}}] \neq \emptyset$ for each $T_{\rho_{j+1}} \in \mathbb{T}_{\rho_{j+1}}$. Since $\mathbb{T}_{\rho_j}$ covers $\mathbb{T}$, we have that for each $T_{\rho_{j+1}} \in \mathbb{T}_{\rho_{j+1}}$, there exists $T_{\rho_j} \in \mathbb{T}_{\rho_j}$ so that $\mathbb{T}[T_{\rho_{j+1}}] \cap \mathbb{T}[T_{\rho_j}] \neq \emptyset$, and hence $T_{\rho_{j+1}} \cap T_{\rho_j}$ contains a unit line segment. But this implies that $T_{\rho_{j+1}} \subset N_{2\rho_{j+1}}(T_{\rho_j})$. Thus, if we replace each set $\mathbb{T}_{\rho_j}$ with $\{N_{2\rho_{j+1}}(T_{\rho_j}) : T_{\rho_j} \in \mathbb{T}_{\rho_j}\}$ for $j = 0, \dots, J-1$ (leaving $\mathbb{T}_{\rho_J}$ unchanged), then for each index j we have that $\mathbb{T}_{\rho_j}$ covers $\mathbb{T}_{\rho_{j+1}}$. It is still the case that $\mathbb{T}_{\rho_j}$ is a $\lesssim \delta^{-\eta}$ -balanced cover of $\mathbb{T}$, and hence $\mathbb{T}_{\rho_j}$ is a $\lesssim \delta^{-2\eta}$ -balanced cover of $\mathbb{T}_{\rho_{j+1}}$. Abusing notation, we will continue to refer to these sets as $\mathbb{T}_{\rho_j}$ (even though this set is technically a collection of $(\rho_j + 2\rho_{j+1})$ tubes; since $\rho_{j+1} \leq \rho_j/4$, this distinction will not be important). Note that it might no longer be the case that the tubes in $\mathbb{T}_{\rho_j}$ are essentially distinct, nor that $\mathbb{T}_{\rho_j}$ is a partitioning cover of $\mathbb{T}$.

Step 2.

After dyadic pigeonholing, we can replace each set $\mathbb{T}_{\rho_j}$ by a subset (abusing notation, we will continue to refer to this subset as $\mathbb{T}_{\rho_j}$) so that $\#\mathbb{T}_{\rho_j} \gtrsim \#\mathbb{T}$, and the following holds for each $j = 0, \dots, J$. (To simplify notation, we define $\tilde{T}_{\rho_j} = N_{3\rho_j}(T_{\rho_j})$ and $\tilde{\mathbb{T}}_{\rho_j} = \{\tilde{T}_{\rho_j} : T_{\rho_j} \in \mathbb{T}_{\rho_j}\}$).

- (i) The tubes in $\tilde{\mathbb{T}}_{\rho_j}$ are essentially distinct.
- (ii) For each $T_{\rho_j} \in \mathbb{T}_{\rho_j}$, we have $\tilde{\mathbb{T}}_{\rho_{j+1}}[\tilde{T}_{\rho_j}] = \tilde{\mathbb{T}}_{\rho_{j+1}}[T_{\rho_j}]$.
- (iii) $\mathbb{T}_{\rho_j}$ is a balanced partitioning cover of $\tilde{\mathbb{T}}_{\rho_{j+1}}$.
- (iv) for each $T_{\rho_j} \in \mathbb{T}_{\rho_j}$, we have that $(\tilde{\mathbb{T}}_{\rho_{j+1}})^{T_{\rho_j}}$ is regular with granularity $(3\rho_j/\rho_{j+1})^{\eta_1}$.

For each index $j = 1, \dots, J$, apply Corollary 14.4.84 with ε_1 in place of ε , $\tilde{\mathbb{T}}_{\rho_j}$ in place of $\mathbb{T}$, $\mathbb{T}_{\rho_{j-1}}$ in place of $\mathbb{T}_\rho$, $3\rho_j$ in place of δ , and $3\rho_j/\rho_{j-1}$ in place of ρ . Let $\mathcal{A}_j$ be the output of Corollary 14.4.84.

By (14.4.228) plus Item (ii) above, for each $T_{\rho_{j-1}} \in \mathbb{T}_{\rho_{j-1}}$, we have

$$\bigsqcup_{A \in \mathcal{A}_j} A(\tilde{\mathbb{T}}_{\rho_j}[\tilde{T}_{\rho_{j-1}}]) = \bigsqcup_{A \in \mathcal{A}_j} A(\tilde{\mathbb{T}}_{\rho_j}[T_{\rho_{j-1}}]) \subset \tilde{T}_{\rho_{j-1}}, \quad (14.4.230)$$

and hence by Item (iii),

$$\tilde{\mathbb{T}}_{\rho_{j-1}} \text{ covers } \bigsqcup_{A \in \mathcal{A}_j} A(\tilde{\mathbb{T}}_{\rho_j}). \quad (14.4.231)$$

Here the disjoint union is in the sense of multi-sets.

For every $T_{\rho_{j-1}} \in \mathbb{T}_{\rho_{j-1}}$ we have

$$\begin{aligned} \#\left(\left(\bigsqcup_{A \in \mathcal{A}_j} A(\tilde{\mathbb{T}}_{\rho_j})\right)[\tilde{T}_{\rho_{j-1}}]\right) &\geq (\#\mathcal{A}_j) \frac{\#\tilde{\mathbb{T}}_{\rho_j}}{2\#\mathbb{T}_{\rho_{j-1}}} \\ &\gtrsim \frac{|T_{\rho_{j-1}}|}{|T_{\rho_j}|}. \end{aligned} \quad (14.4.232)$$

By (14.4.229), there exists $C_{\varepsilon_1} \geq 1$ so that, for every $\tilde{T}_{\rho_{j-1}} \in \tilde{\mathbb{T}}_{\rho_{j-1}}$,

$$C_{KT-CW}\left(\left(\bigsqcup_{A \in \mathcal{A}_j} A(\tilde{\mathbb{T}}_{\rho_j})\right)[\tilde{T}_{\rho_{j-1}}]\right) \leq C_{\varepsilon_1} \delta^{-\varepsilon_1-2\eta}. \quad (14.4.233)$$

Step 3. Define $\mathcal{A}^{(0)} = \{I\}$ and for $j = 1, \dots, J$, define

$$\mathcal{A}^{(j)} = \{A_1 \circ A_2 \circ \dots \circ A_j : A_i \in \mathcal{A}_i \text{ for each } i = 1, \dots, j\}.$$

Define $\mathcal{A} = \mathcal{A}^{(J)}$; this set of transformations will be the output of Proposition 14.4.79.

For each index $j = 0, \dots, J$, define

$$\hat{\mathbb{T}}_{\rho_j} = \bigsqcup_{A \in \mathcal{A}^{(j)}} A(\tilde{\mathbb{T}}_{\rho_j}).$$

By (14.4.231), $\hat{\mathbb{T}}_{\rho_{j-1}}$ covers $\hat{\mathbb{T}}_{\rho_j}$ for each $j = 1, \dots, J$.

By (14.4.232), for each $1 \leq j \leq J$ and each $T_{\rho_{j-1}} \in \mathbb{T}_{\rho_{j-1}}$ we have

$$\#\hat{\mathbb{T}}_{\rho_j}[\tilde{T}_{\rho_{j-1}}] \gtrsim \frac{|T_{\rho_{j-1}}|}{|T_{\rho_j}|}.$$

Thus after dyadic pigeonholing, we can replace each set $\hat{\mathbb{T}}_{\rho_j}$ by a refinement $\hat{\mathbb{T}}'_{\rho_j}$ (in the sense of multi-sets), so that $\hat{\mathbb{T}}'_{\rho_{j-1}}$ is a balanced partitioning cover of $\hat{\mathbb{T}}'_{\rho_j}$ for each $j = 1, \dots, J$, and for each $j = 1, \dots, J$ and each $T_{\rho_{j-1}} \in \mathbb{T}_{\rho_{j-1}}$ we have

$$\#\hat{\mathbb{T}}'_{\rho_j}[\tilde{T}_{\rho_{j-1}}] \gtrsim \delta^\eta \frac{|T_{\rho_{j-1}}|}{|T_{\rho_j}|},$$

and hence

$$\#\hat{\mathbb{T}}'_{\rho_j}[\tilde{T}_{\rho_j}] \gtrsim \delta^{\eta/\varepsilon} \frac{|T_{\rho_j}|}{|T|} \quad \text{for each } \tilde{T}_{\rho_j} \in \hat{\mathbb{T}}'_{\rho_j}. \quad (14.4.234)$$

Step 4. Using Lemma 14.4.29 and (14.4.233), we have

$$C_{KT-CW}(\hat{\mathbb{T}}'_{\rho_J}) \leq C_{KT-CW}(\hat{\mathbb{T}}_{\rho_J}) \lesssim \prod_{j=1}^J (C_{\varepsilon_1} \delta^{-\varepsilon_1-2\eta}) \lesssim \delta^{-2\varepsilon_1 J}.$$

In particular, for each $T \in \hat{\mathbb{T}}'_{\rho_J}$, there are $\lesssim \delta^{-2\varepsilon_1 J}$ tubes $T' \in \hat{\mathbb{T}}'_{\rho_J}$ comparable to T . Recall that $J \sim \varepsilon^{-1}$. We will choose ε_1 sufficiently small so that $2\varepsilon_1 J < \varepsilon/4$. Thus we can select a set $\mathbb{T}^\dagger \subset \hat{\mathbb{T}}'_{\rho_J} \subset \bigcup_{A \in \mathcal{A}} A(\hat{\mathbb{T}}_{\rho_J})$ with $\#\mathbb{T}^\dagger \gtrsim \delta^{\varepsilon/4} (\#\hat{\mathbb{T}}'_{\rho_J}) \geq \delta^\varepsilon |T|^{-1}$ that consists of essentially distinct tubes. The set $\mathbb{T}^\dagger$ will be the output of Proposition 14.4.79.

It remains to verify that $\mathbb{T}^\dagger$ satisfies the Frostman Convex Wolff Axioms at every scale, with error $\delta^{-\varepsilon}$. To do so, it suffices to note that for each index $j = 0, \dots, J-1$, we have that $\hat{\mathbb{T}}'_{\rho_j}$ is a $\delta^{-\varepsilon/4} 2^{J-j}$ -balanced partitioning cover of $\mathbb{T}^\dagger$, and thus if we choose $\eta < \varepsilon^2/2$, then for each $\tilde{T}_{\rho_j} \in \hat{\mathbb{T}}'_{\rho_j}$ we can use (14.4.234) to conclude that

$$\begin{aligned} C_{F-CW}((\mathbb{T}^\dagger)^{\tilde{T}_{\rho_j}}) &\leq \frac{C_{KT-CW}(\mathbb{T}^\dagger)}{\#\mathbb{T}^\dagger[\tilde{T}_{\rho_j}]} \frac{|\tilde{T}_{\rho_j}|}{|T|} \\ &\lesssim \delta^{-\eta/\varepsilon} C_{KT-CW}(\mathbb{T}^\dagger) \\ &\leq \delta^{-\eta/\varepsilon - \varepsilon/4} C_{KT-CW}(\hat{\mathbb{T}}'_{\rho_J}) \leq \delta^{-\varepsilon}. \quad \square \end{aligned}$$

14.4.10 Multi-scale analysis and the proof of Proposition 14.4.7

Our goal in this section is to prove Proposition 14.4.7 by combining Proposition 14.4.76 and Theorem 14.4.78.

Lemma 14.4.86. *Let $\sigma \in (0, 2/3]$; $\omega, \varepsilon > 0$ and $N \geq 1$. Suppose that $\mathcal{E}(\sigma, \omega)$ is true. Then there exists $\eta, \alpha, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}.$$

(B) *There is a δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$; a number $J \leq 2^N$; scales $\delta = \rho_J < \rho_{J-1} < \dots < \rho_0 = 1$; and sets $\mathbb{T}_{\rho_j}$, $j = 0, \dots, J$, so that the following holds*

$$(i) \quad \rho_{i+1}/\rho_i \geq \delta^{(1-\omega/100)^N} \text{ for each } i = 0, \dots, J-1.$$

- (ii) $\mathbb{T}' = \mathbb{T}_{\rho_J}$, and $\mathbb{T}_{\rho_0}$ consists of a single 1 tube.
- (iii) For each $i = 1, \dots, J-1$ and each $T_{\rho_{i-1}} \in \mathbb{T}_{\rho_{i-1}}$, we have that $\mathbb{T}_{\rho_i}^{T_{\rho_{i-1}}}$ factors $\mathbb{T}_{\rho_{i+1}}^{T_{\rho_{i-1}}}$ from above and below with regard to the Katz-Tao Convex Wolff Axioms and Frostman Slab Wolff Axioms, both with error $\delta^{-\varepsilon}$.

Proof.

Step 1. We will prove the result by induction on N . When $N = 0$ there is almost nothing to prove; since the tubes in $\mathbb{T}$ are contained in the unit ball, we can select a ~ 1 refinement $\mathbb{T}' \subset \mathbb{T}$ that is contained in a single 1 tube. We have $J = 1$, and Conclusion (B) always holds (note that Item (iii) of Conclusion B is vacuously true when $J = 1$).

Suppose now that the result has been proved for $N - 1$. Let $\theta \in (0, 2/3]$ and let $\varepsilon, \omega > 0$. Let $\varepsilon_1, \varepsilon_2, \eta, \alpha, c > 0$ be small quantities to be determined below. We will choose ε_1 depending on $\varepsilon, N, \theta, \omega$; ε_2 small compared to ε_1 ; and η, α, c small compared to ε_2 .

Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and satisfy $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. If η is chosen sufficiently small depending on ω, σ , and ε_2 , then we can apply the induction hypothesis with ε_2 in place of ε ; let $\alpha_1 = \alpha_1(\omega, \sigma, \varepsilon_2, N - 1)$ be the corresponding value of α .

If Conclusion (A) holds when we apply the induction hypothesis, then Conclusion (A) holds and we are done, provided we select $\alpha \leq \alpha_1$. Suppose instead that Conclusion (B) holds, i.e. there is a δ^{ε_2} refinement $(\mathbb{T}', Y')_\delta$; scales $\delta = \rho_J < \rho_{J-1} < \dots < \rho_0 = 1$ with $J \leq 2^{N-1}$ and $\rho_{i+1}/\rho_i \geq \delta^{(1-\omega/100)^{N-1}}$; and sets $\mathbb{T}_{\rho_j}$ satisfying Conclusion (B) with ε_2 in place of ε .

Replacing $(\mathbb{T}', Y')_\delta$ and each set $\mathbb{T}_{\rho_i}$ with a $(\log 1/\delta)^{-J} \geq (\log 1/\delta)^{-2^N}$ refinement, we can construct shadings Y_{ρ_i} on $\mathbb{T}_{\rho_i}$ so that the following holds.

- (a) $|Y_{\rho_0}(T_{\rho_0})| \gtrsim (\log 1/\delta)^{-2^N} \delta^\eta$, where T_{ρ_0} is the unique tube in $\mathbb{T}_{\rho_0}$.
- (b) For each $T_{\rho_i} \in \mathbb{T}_{\rho_i}$, we have that $(\mathbb{T}_{\rho_{i+1}}^{T_{\rho_i}}, Y_{\rho_{i+1}}^{T_{\rho_i}})_{\rho_{i+1}/\rho_i}$ is $\gtrsim (\log 1/\delta)^{-2^N} \delta^\eta$ dense, and $(Y_{\rho_J}, \mathbb{T}_{\rho_J})_\delta$ is a $\gtrsim (\log 1/\delta)^{-2^N}$ -refinement of $(\mathbb{T}', Y')_\delta$.
- (c) Item (iii) of Conclusion (B) remains true, with the error $\delta^{-\varepsilon_2}$ weakened to $\lesssim (\log 1/\delta)^{2^N} \delta^{-\varepsilon_2}$.
- (d) For each $i = 1, \dots, J - 1$ and each $T_{\rho_{i-1}} \in \mathbb{T}_{\rho_{i-1}}$, we have that $\mathbb{T}_{\rho_i}^{T_{\rho_{i-1}}}$ is a balanced partitioning cover of $\mathbb{T}_{\rho_{i+1}}^{T_{\rho_{i-1}}}$.
- (e) For each $i = 0, \dots, J - 1$, the quantity

$$\left| \bigcup_{T \in \mathbb{T}_{\rho_{i+1}}[T_{\rho_i}]} Y_{\rho_{i+1}}(T_{\rho_{i+1}}) \right|$$

is approximately the same (up to a multiplicative factor of 2) for each $T_{\rho_i} \in \mathbb{T}_{\rho_i}$.

(f) We have

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \prod_{i=0}^{J-1} \left(\sup_{T_{\rho_i} \in \mathbb{T}_{\rho_i}} \frac{1}{|T_{\rho_i}|} \left| \bigcup_{T_{\rho_{i+1}} \in \mathbb{T}_{\rho_{i+1}}[T_{\rho_i}]} Y_{\rho_{i+1}}(T_{\rho_{i+1}}) \right| \right). \quad (14.4.235)$$

We now construct the shadings and verify (a)–(f). This bookkeeping is included because the product estimate is load-bearing.

Put $L = 20 \log(2/\delta)$. First turn every cover into a partitioning cover. If a child tube lies in several parents, assign it to the first parent in a fixed ordering. Since the original cover has bounded overlap after the inductive refinement, one of the dyadic classes of assignment multiplicity retains at least an L^{-1} proportion of the tube-shading mass. Next sort the non-empty fibres by their cardinality and keep one dyadic class. Thus, at the cost of L^{-2} at one level, every surviving child has one parent and all non-empty fibres have cardinalities within a factor of two. A restriction to a q proportion of a fibre can enlarge either its upper or lower factorisation constant by at most q^{-1} ; this follows immediately by inserting the restricted cardinality into the defining inequalities. After at most $J \leq 2^N$ levels, Item (iii) therefore persists with error $L^{2J} \delta^{-\varepsilon_2}$, proving (c) and (d).

We construct the spatial shadings from the bottom upwards. Start with the given shading on $\mathbb{T}_{\rho_J} = \mathbb{T}'$ and discard one dyadic density class so that it remains an L^{-1} refinement. Suppose $Y_{\rho_{i+1}}$ has been defined. For a parent $P \in \mathbb{T}_{\rho_i}$ put

$$U_i(P) = \bigcup_{T \in \mathbb{T}_{\rho_{i+1}}[P]} Y_{\rho_{i+1}}(T).$$

Partition P into half-open cubes of side between $\rho_i/20$ and $\rho_i/10$, using a fixed translated grid and intersecting boundary cubes with P . These cells have bounded overlap after a fixed enlargement. Sort the non-zero ratios $|U_i(P) \cap Q|/|Q|$ into dyadic intervals, first inside each parent and then across the parents. Retain one common interval, discard parents for which it carries less than an L^{-2} fraction of $|U_i(P)|$, and define

$$Y_{\rho_i}(P) = \bigcup_{Q \text{ retained}} (U_i(P) \cap Q).$$

The averaging identity

$$\sum_P \sum_Q |U_i(P) \cap Q| = \sum_P |U_i(P)|$$

shows that the retained parent-and-cell class carries at least L^{-2} of the mass at this level. It also shows that

$$\frac{|Y_{\rho_i}(P) \cap Q|}{|Q|} \sim q_i \quad (14.4.236)$$

for one number q_i , uniformly in all surviving P, Q , with a factor two. Finally sort the quantities $|U_i(P)|$ into dyadic intervals and retain one interval. This costs one more factor L^{-1} and proves (e).

Repeat upwards to $i = 0$. The total loss is at most L^{-3J} , which is at least $(\log(2/\delta))^{-2^{N+2}}$ after increasing the displayed harmless logarithmic power. Since the original bottom shading is δ^η dense, averaging over the unique root tube gives (a). The same argument inside each parent, followed by the affine normalization by that parent, gives (b); the density is at least $L^{-2^{N+2}}\delta^\eta$. Relabelling the fixed logarithmic exponent as 2^N only changes the implicit constants used later.

It remains to prove (f). For each level choose one point in every retained grid cell and compare the cell with a concentric ball of radius comparable to ρ_i ; bounded overlap changes measure by an absolute factor. Equation (14.4.236) says that, conditionally on a point surviving to level i , the proportion which survives to level $i + 1$ is between $q_i/2$ and $2q_i$. Induction and Fubini therefore give

$$|Y_{\rho_J}| \geq 2^{-J} |Y_{\rho_0}| \prod_{i=0}^{J-1} q_i.$$

The root estimate (a), the definition of q_i , and $|T_{\rho_0}| \sim 1$ turn this into (14.4.235), with the fixed factor 2^{-J} absorbed in the implicit constant. By (e), replacing the supremum there by the infimum costs at most another 2^J . This completes the verification of all six items.

If ε_2 is chosen sufficiently small depending on ε_1 , then for each index i and each $T_{\rho_i} \in \mathbb{T}_{\rho_i}$ we have

$$\begin{aligned} \frac{1}{|T_{\rho_i}|} \left| \bigcup_{T_{\rho_{i+1}} \in \mathbb{T}_{\rho_{i+1}}[T_{\rho_i}]} Y_{\rho_{i+1}}(T_{\rho_{i+1}}) \right| &\gtrsim \delta^{\varepsilon_1} \left(\frac{\rho_{i+1}}{\rho_i} \right)^\omega \frac{\#\mathbb{T}_{\rho_{i+1}}}{\#\mathbb{T}_{\rho_i}} |T_{\rho_{i+1}}| \\ &\times \left(\frac{\#\mathbb{T}_{\rho_{i+1}}}{\#\mathbb{T}_{\rho_i}} \left(\frac{|T_{\rho_{i+1}}|}{|T_{\rho_i}|} \right)^{1/2} \right)^{-\sigma}. \end{aligned} \quad (14.4.237)$$

Indeed, (14.4.237) follows from the estimate $\mathcal{E}(\sigma, \omega)$; this estimate can be applied, for example, when $\frac{\rho_{i+1}}{\rho_i} < \delta^{\varepsilon_1}$ and ε_2 is selected sufficiently small depending on ω, σ , and ε_1 . If $\frac{\rho_{i+1}}{\rho_i} \geq \delta^{\varepsilon_1}$, then (14.4.237) follows from the elementary fact that the volume of the LHS of (14.4.237) is bounded below by the volume of $Y_{\rho_{i+1}}(T_{\rho_{i+1}})$ for every $T_{\rho_{i+1}} \in \mathbb{T}_{\rho_{i+1}}[T_{\rho_i}]$.

Step 2.

We say an index i is of Type C if $\rho_{i+1}/\rho_i \geq 2^{-N}\delta^{(1-\omega/100)^N}$; in this case no further splitting is required. If instead $\rho_{i+1}/\rho_i < 2^{-N}\delta^{(1-\omega/100)^N}$, then apply Proposition 14.4.76 (refined induction on scales) with ω and σ as above, and $\zeta = \varepsilon$ to each arrangement $(\mathbb{T}_{\rho_{i+1}}^{T_{\rho_i}}, Y_{\rho_{i+1}}^{T_{\rho_i}})_{\rho_{i+1}/\rho_i}$; we can do this provided ε_2 is chosen sufficiently small depending on ω, σ, N , and ε . If Conclusion (A) holds for at least one $T_{\rho_i} \in \mathbb{T}_{\rho_i}$, then we say the index i is of Type A. Otherwise we say that the index i is of Type B.

Suppose there exists at least one index of Type A; call this index i_0 . Let $\alpha_2 = \alpha_2(\omega, \sigma, \varepsilon)$ be the output of Proposition 14.4.76. To make the ensuing telescoping transparent, set

$$A_i := \delta^{\varepsilon_1} \left(\frac{\rho_{i+1}}{\rho_i} \right)^{\omega} \frac{\#\mathbb{T}_{\rho_{i+1}}}{\#\mathbb{T}_{\rho_i}} |T_{\rho_{i+1}}| \left(\frac{\#\mathbb{T}_{\rho_{i+1}}}{\#\mathbb{T}_{\rho_i}} \left(\frac{|T_{\rho_{i+1}}|}{|T_{\rho_i}|} \right)^{1/2} \right)^{-\sigma}.$$

Applying (14.4.237) to each index $i \neq i_0$, using Conclusion (A) of Proposition 14.4.76 at i_0 , and then using (14.4.235), we have

$$\begin{aligned} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| &\gtrsim \left(\prod_{\substack{1 \leq i \leq J-1 \\ i \neq i_0}} A_i \right) A_{i_0} \left(\frac{\rho_{i_0+1}}{\rho_{i_0}} \right)^{-\alpha_2} \\ &\geq \delta^{\varepsilon_1 J} \left(\frac{\rho_{i_0+1}}{\rho_{i_0}} \right)^{-\alpha_2} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma} \\ &\gtrsim \delta^{\varepsilon_1 2^N - \alpha_2 (1-\omega/100)^N} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}. \end{aligned} \quad (14.4.238)$$

Thus Conclusion (A) of Lemma 14.4.86 holds, provided we select $\alpha \leq \frac{1}{2}\alpha_2(1 - \omega/100)^N$ and $\varepsilon_1 < 2^{-N-1}\alpha_2(1 - \omega/100)^N$.

Step 3. Suppose instead that every index is of Type B or Type C. We will show that Conclusion (B) of Lemma 14.4.86 holds. For each index i of Type B, starting with the lowest, we will perform the following procedure. After dyadic pigeonholing, there is a number $s \in [(\frac{\rho_{i+1}}{\rho_i})^{1-\frac{\omega}{100}}, (\frac{\rho_{i+1}}{\rho_i})^{\frac{\omega}{100}}]$ and a set $\mathbb{T}'_{\rho_i} \subset \mathbb{T}_{\rho_i}$ with $\#\mathbb{T}'_{\rho_i} \gtrsim_{\delta} \#\mathbb{T}_{\rho_i}$ such that the output “ ρ ” from Proposition 14.4.76 is between s and $2s$ for each $T_{\rho_i} \in \mathbb{T}'_{\rho_i}$. Define $\rho_{i+1/2} = s\rho_i$. Then

$$\frac{\rho_{i+1/2}}{\rho_i} \geq s \geq \left(\frac{\rho_{i+1}}{\rho_i} \right)^{1-\frac{\omega}{100}} \geq (2^{-N+1}\delta^{(1-\omega/100)^{N-1}})^{1-\frac{\omega}{100}} \geq 2^{-N}\delta^{(1-\omega/100)^N}. \quad (14.4.239)$$

Similarly, we have

$$\frac{\rho_{i+1}}{\rho_{i+1/2}} = \frac{\rho_{i+1}}{\rho_i} \frac{\rho_i}{\rho_{i+1/2}} \geq \frac{\rho_{i+1}}{\rho_i} \frac{1}{2s} \geq \frac{1}{2} \left(\frac{\rho_{i+1}}{\rho_i} \right)^{1-\frac{\omega}{100}} \geq 2^{-N}\delta^{(1-\omega/100)^N}. \quad (14.4.240)$$

This verifies Item (i) from Conclusion (B) of Lemma 14.4.86.

Let $\mathbb{T}_{\rho_{i+1/2}}$ be the union of the corresponding sets from Proposition 14.4.76, and let

$$\mathbb{T}'_{\rho_{i+1}} = \bigcup_{T_{\rho_{i+1/2}} \in \mathbb{T}_{\rho_{i+1/2}}} \mathbb{T}_{\rho_{i+1}}[T_{\rho_{i+1/2}}].$$

Then for each $T_{\rho_i} \in \mathbb{T}'_{\rho_i}$, we have that $\mathbb{T}_{\rho_{i+1/2}}^{T_{\rho_i}}$ factors $(\mathbb{T}'_{\rho_{i+1}})^{T_{\rho_i}}$ above and below with respect to the Katz-Tao Convex Wolff Axioms and Frostman Slab Wolff Axioms, both with error $\delta^{-\varepsilon}$. This verifies Item (iii) from Conclusion (B) of Lemma 14.4.86.

To conclude the proof, we re-index our sets $\mathbb{T}'_{\rho_i}$ and $\mathbb{T}_{\rho_{i+1/2}}$ using consecutive integers $1, 2, 3, \dots$, and let $\mathbb{T}' = \mathbb{T}_{\rho_{J'}}$, where J' is the final index after re-indexing. Note that $J' \leq 2J - 1 \leq 2^N$. After re-indexing, we still have $\rho_0 = 1$ and $\rho_{J'} = \delta$. This concludes the induction step. $\square$

Lemma 14.4.87. *Let $\sigma \in (0, 2/3]$ and $\omega, \varepsilon > 0$. Suppose that $\mathcal{E}(\sigma, \omega)$ is true. Then there exists $\eta, \alpha, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then at least one of the following must hold.*

$$(A) \quad \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{\omega-\alpha} (\#\mathbb{T}) |T| ((\#\mathbb{T}) |T|^{1/2})^{-\sigma}.$$

(B) *There is a δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ that satisfies the Katz-Tao Convex Wolff Axioms at every scale with error $\delta^{-\varepsilon}$ (recall Definition 14.4.77).*

Proof. Let $\varepsilon_1 > 0$ be a small quantity to be chosen below. Select N sufficiently large so that $(1 - \omega/100)^N < \varepsilon$. Apply Lemma 14.4.86 to $(\mathbb{T}, Y)_\delta$ with this choice of N , with ε_1 in place of ε , and with ω, σ as above. If Conclusion (A) of Lemma 14.4.86 holds, then Conclusion (A) of Lemma 14.4.87 holds and we are done.

Suppose instead that Conclusion (B) of Lemma 14.4.86 holds. After a harmless refinement we can suppose that each set $\mathbb{T}_{\rho_i}$ is a balanced partitioning cover of $\mathbb{T}_{\rho_{i+1}}$, and hence $\mathbb{T}_{\rho_i}$ is a $2^J \leq \delta^{-\varepsilon}$ balanced partitioning cover of $\mathbb{T}'$.

We claim that if ε_1 is chosen sufficiently small (depending on N and ε), then the output $(\mathbb{T}', Y')_\delta$ of Lemma 14.4.86 satisfies the Katz-Tao Convex Wolff Axioms at every scale with error $\delta^{-\varepsilon}$. We verify this as follows. Since $(1 - \omega/100)^N < \varepsilon$, for each $\rho \in [\delta, 1]$ the interval $[\rho, \delta^{-\varepsilon}\rho]$ contains at least one ρ_i . We have already noted that $\mathbb{T}_{\rho_i}$ satisfies Item (i) from Definition 14.4.77. It remains to verify Item (ii). Applying Lemma 14.4.29, we have that for each index i ,

$$C_{KT-CW}(\mathbb{T}_i) \lesssim \prod_{j=i+1}^J \delta^{-2\varepsilon_1} \lesssim \delta^{-2^{N+1}\varepsilon_1}.$$

We will select ε_1 sufficiently small so that $C_{KT-CW}(\mathbb{T}_i) \leq \delta^{-\varepsilon}$ for all sufficiently small $\delta > 0$. If $\delta > 0$ is not sufficiently small, then Conclusion (A) always holds, provided we select $\kappa > 0$ sufficiently small. $\square$

Proof of Proposition 14.4.7. Fix $\omega, \sigma > 0$ and suppose that $\mathcal{E}(\sigma, \omega)$ is true. Let η_1 be the output of Theorem 14.4.78 with $\omega/2$ in place of ε . Let α_2, η_2, c_2 be the output of Lemma 14.4.87 with η_1 in place of ε . Then for every pair $(\mathbb{T}, Y)_\delta$ that is δ^{η_2} dense and satisfies $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta_2}$, $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta_2}$, we have

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \gtrsim \delta^{\omega - \min(\alpha_2, \omega/2)} (\#\mathbb{T}) |T| \left((\#\mathbb{T}) |T|^{1/2} \right)^{-\sigma}. \quad (14.4.241)$$

In particular, we have that $\mathcal{D}(\sigma, \omega - g(\sigma, \omega))$ is true, where $g(\sigma, \omega) = \min(\alpha_2, \omega/2)$. $\square$

14.4.11 A grains decomposition for tubes in $\mathbb{R}^3$

Our goal in this section is to prove Proposition 14.4.67. For the reader's convenience, we reproduce it here. In what follows, recall that broadness is defined in Definition 14.4.59. This definition assumes a small parameter $\beta > 0$, which was defined in Section 14.4.6.1. In Section 14.4.6.1, an explicit value of β was chosen, which depends on ω and ζ . In the results below, however, we will allow β to be any positive number (though in $\mathbb{R}^n$, we will always have $\beta \in (0, n-1]$), and subsequent quantities (such as η and K_1) may depend on β . In particular, the results proved in this section are applicable in Section 14.4.6.

14.4.11.1 A self-contained partitioning tool for the grains argument

The grains decomposition below uses polynomial partitioning for finite measures. We include the exact form required, together with its topological input. This prevents the word “partitioning” from hiding a second proof dependency.

Lemma 14.4.88 (Tucker's antipodal labelling lemma). *Let K be a triangulation of the n -dimensional cross-polytope*

$$B_1^n = \{x \in \mathbb{R}^n : |x_1| + \cdots + |x_n| \leq 1\}$$

whose boundary triangulation is antipodally symmetric. If the vertices are labelled by $\{\pm 1, \dots, \pm n\}$, with opposite boundary vertices receiving opposite labels, then some edge has labels i and $-i$.

Proof. Assume that there is no complementary edge. Let L be the boundary complex of the cross-polytope with vertices $\{\pm 1, \dots, \pm n\}$. The labels on each simplex then contain no opposite pair and hence span a simplex of L ; the labelling is therefore a simplicial map $f : K \rightarrow L$.

We use finite simplicial chains with coefficients in $\mathbb{F}_2$. For an oriented notation in which signs are immaterial, put

$$\partial[v_0, \text{ldots}, v_k] = \sum_{j=0}^k [v_0, \text{ldots}, \widehat{v_j}, \text{ldots}, v_k].$$

Every codimension-two face occurs twice, so $\partial^2 = 0$. A simplicial map induces a map $f_{\#}$ on chains (a degenerate image is declared zero), and a direct check on one simplex gives $\partial f_{\#} = f_{\#} \partial$. Finally, the boundary of the sum of all top simplices of a triangulated manifold is the sum of its boundary facets, because every interior facet occurs twice.

Let A_n be the sum of the n -simplices of K and A_{n-1} the sum of its boundary facets. Since L has dimension $n - 1$, $f_{\#} A_n = 0$, and hence

$$f_{\#} A_{n-1} = f_{\#} \partial A_n = \partial f_{\#} A_n = 0. \quad (14.4.242)$$

On the other hand the boundary map is antipodal. We show directly that the top chain of an antipodal simplicial map of an $(n - 1)$ -sphere to L is not zero. Split the domain sphere into two antipodal hemispheres H_k^+ and H_k^- successively along the coordinate equators, for $k = n - 1, n - 2, \dots, 0$, and let $C_k^{\pm}$ be the images of their fundamental k -chains. Their common equator gives

$$\partial C_k^+ = \partial C_k^- = C_{k-1}^+ + C_{k-1}^-.$$

If $C_k^+ + C_k^- = 0$, then $C_k^+ = C_k^-$. Taking one representative from each antipodal pair of k -simplices and applying ∂ shows that $C_{k-1}^+ + C_{k-1}^- = 0$. Descending to $k = 0$ would say that the sum of one antipodal pair of vertices is zero, which is false in the free $\mathbb{F}_2$ -vector space on the vertices. Thus $f_{\#} A_{n-1} \neq 0$, contradicting (14.4.242). $\square$

Lemma 14.4.89 (Borsuk–Ulam). *Every continuous map $g : S^N \rightarrow \mathbb{R}^N$ has a point x with $g(x) = g(-x)$.*

Proof. Set $h(x) = g(x) - g(-x)$. If h has no zero, fix a triangulation of a closed hemisphere which is antipodally symmetric on the equator and whose mesh is smaller than a prescribed number $a > 0$. Such triangulations are obtained by repeated barycentric subdivision. Label a vertex v by the signed index of the first coordinate of $h(v)$ having maximal absolute value. The labelling is

antipodal on the boundary. By Theorem 14.4.88, a complementary edge vv' exists. Uniform continuity gives

$$\min\{\|h(v)\|_\infty, \|h(v')\|_\infty\} \leq \|h(v) - h(v')\|_\infty = o_{a \rightarrow 0}(1).$$

Choosing meshes tending to zero and then taking a convergent subsequence of the corresponding vertices produces x with $h(x) = 0$. $\square$

Lemma 14.4.90 (Ham sandwich for finite measures). *For finite Borel measures $\nu_1, \dots, \nu_N$ on $\mathbb{R}^N$, there is an affine hyperplane H such that neither open half-space bounded by H carries more than $\nu_i(\mathbb{R}^N)/2$ for any i .*

Proof. First suppose that all measures give zero mass to hyperplanes. Represent an oriented affine hyperplane by $u = (u_0, u_1, \dots, u_N) \in S^N$ and set

$$\Phi_i(u) = \nu_i\{x : u' \cdot x > u_0\} - \nu_i\{x : u' \cdot x < u_0\}.$$

Dominated convergence shows that $\Phi : S^N \rightarrow \mathbb{R}^N$ is continuous, and it is odd. By Theorem 14.4.89, $\Phi(u) = 0$ for some u .

For arbitrary measures, convolve them with a smooth approximate identity. The smoothed measures have bisecting hyperplanes H_k . Compactness of the unit parameter sphere gives a convergent subsequence. It cannot converge to an empty hyperplane unless every measure is zero. If H is the limiting hyperplane, every closed set a positive distance inside one of its open half-spaces eventually lies in the corresponding half-space of H_k . Weak convergence and then monotone exhaustion of the open half-space give the two required inequalities. $\square$

Lemma 14.4.91 (Polynomial ham sandwich). *For finite Borel measures $\mu_1, \dots, \mu_M$ on $\mathbb{R}^n$, there is a non-zero polynomial P of degree at most $C_n M^{1/n}$ such that neither $\{P > 0\}$ nor $\{P < 0\}$ carries more than half of any μ_j .*

Proof. Choose D least so that $N = \binom{D+n}{n} - 1 \geq M$. The Veronese map consisting of all non-constant monomials of degree at most D sends $\mathbb{R}^n$ into $\mathbb{R}^N$. Apply Theorem 14.4.90 to the pushforwards of the M measures, appending zero measures. Pulling the affine bisecting hyperplane back gives a non-zero polynomial of degree at most D . Since $\binom{D+n}{n} \geq D^n/n!$ and D was minimal, $D \leq C_n M^{1/n}$. $\square$

Lemma 14.4.92 (Controlled polynomial partitioning). *Let μ be a finite Borel measure on $\mathbb{R}^n$ and $D \geq 1$. There is a non-zero polynomial P of degree at most D and at most $C_n D^n$ pairwise disjoint open sign-history regions O covering $\mathbb{R}^n \setminus Z(P)$ such that*

$$\mu(O) \leq C_n D^{-n} \mu(\mathbb{R}^n).$$

Every line not contained in $Z(P)$ meets at most $D + 1$ of these regions.

Proof. At round $i+1$, apply Theorem 14.4.91 simultaneously to the restrictions of μ to the at most 2^i sign-history regions already constructed. This gives a factor P_{i+1} of degree $O_n(2^{i/n})$ and bisects the mass in every current region. After j rounds the product $P = P_1 \cdots P_j$ has degree $O_n(2^{j/n})$, while each of its at most 2^j sign-history regions has mass at most $2^{-j}\mu(\mathbb{R}^n)$. Choose j maximal so that the degree is at most D ; changing constants handles bounded D and gives $2^j \asymp_n D^n$.

If a line is not contained in $Z(P)$, the univariate restriction of P has at most D zeros. The complement consists of at most $D+1$ intervals, and the complete sign history is constant on each interval. Hence the line meets at most $D+1$ regions. $\square$

Proposition 14.4.67. Let $\varepsilon > 0$. Then there exists $\eta, \kappa > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and be broad with error $\delta^{-\eta}$. Suppose that the tubes in $\mathbb{T}$ are contained in a common 1 tube T_1 .

Then there is a δ^ε refinement $(\mathbb{T}', Y')$ that is δ^ε dense and is broad with error $\leq \kappa^{-1}\delta^{-\varepsilon}$, and a number $\mu \geq 1$ so that $\mu \sim \#\mathbb{T}'_{Y'}(x)$ for each $x \in \bigcup_{\mathbb{T}} Y'(T)$. In addition, there is a number $c \geq \kappa\mu\delta^\varepsilon(\delta\#\mathbb{T})^{-1}$ and a pair $(\mathcal{G}, Y)_{\delta \times c \times c}$, so that $(\mathcal{G}, Y)_{\delta \times c \times c}$ is a robustly δ^ε -dense two-scale grains decomposition of $(\mathbb{T}', Y')_\delta$ wrt $\{T_1\}$ (the latter is a set consisting of a single 1 tube).

To prove Proposition 14.4.67, we shall repeatedly apply the Guth-Katz polynomial partitioning theorem [13] to decompose a (large portion of) $\bigcup_{\mathbb{T}} Y(T)$ into a union of thin neighbourhoods of semi-algebraic sets. This idea was first used by Guth [11, 12] in the context of the Fourier restriction problem, and later by Guth and the second author [14] in the context of the Kakeya problem.

Before stating this result precisely, we will need several definitions. First, recall that a *semi-algebraic set* is a set $S \subset \mathbb{R}^n$ that is defined by a Boolean combination of polynomial equalities and inequalities.

Definition 14.4.93. Let $\rho > 0$. A *semi-algebraic grain* of thickness ρ is a set of the form

$$G = N_\rho(S),$$

where $S \subset \mathbb{R}^3$ is a semi-algebraic set with $\dim(S) \leq 2$. If there exists a non-zero polynomial $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$ with $S \subset Z(Q)$, then we say that G is defined by a polynomial of degree at most $\deg(Q)$.

Next we prove the tubular-neighbourhood estimate that will be used below. The argument is Wongkew’s grid induction [30]; the component count needed inside that induction is included so that the result is not being used as an unprinted algebraic-geometry black box.

Lemma 14.4.94 (The algebraic count used in the grid argument). *Let P be a non-zero real polynomial of degree D in n variables. The number of connected components of $Z(P)$, and the number of connected components of its complement in a ball, are at most $C_n(1 + D)^n$.*

Proof. We first record the finite form of Bézout that is needed in the proof. If $F_1, \dots, F_n \in \mathbb{C}[z_1, \dots, z_n]$ have degrees $d_1, \dots, d_n$ and a finite common zero set, then

$$\#Z(F_1, \dots, F_n) \leq d_1 \cdots d_n. \quad (14.4.243)$$

Here distinct zeros, rather than zeros with multiplicity, are counted. For completeness, homogenize the F_i with a new variable z_0 and perturb their coefficients generically. The perturbed homogeneous equations have no common zero on $z_0 = 0$ and meet transversely. Successively adjoining an equation of degree d_i gives the exact sequence

$$0 \longrightarrow A_{i-1}(-d_i) \xrightarrow{F_i} A_{i-1} \longrightarrow A_i \longrightarrow 0, \quad A_i = \mathbb{C}[z_0, \dots, z_n]/(F_1, \dots, F_i).$$

Indeed, genericity makes F_i a non-zero-divisor modulo the preceding equations: a zero-divisor would vanish on one of their finitely many irreducible components, and that condition is a proper linear condition on the coefficients of F_i . The displayed exact sequences multiply the Hilbert series by $(1 - t^{d_i})$. After all n equations, the eventual value of the Hilbert function—equivalently the number of projective points counted with their local quotient dimensions—is

$$\lim_{t \rightarrow 1} \prod_{i=1}^n \frac{1 - t^{d_i}}{1 - t} = d_1 \cdots d_n.$$

Every transverse point has local quotient dimension one. For the original system, take disjoint small polydiscs about its isolated zeros. The iterated one-variable argument principle on their distinguished boundaries is stable under a sufficiently small coefficient perturbation, so each original zero, with its positive local quotient dimension, accounts for at least one nearby perturbed zero. This proves (14.4.243). Notice that this last passage only gives an upper bound, which is all that is required.

We now count components. Replace P by its square-free part; this does not change its zero set or increase its degree. Choose a generic linear function L . If t is a regular value of P , every compact component of $Z(P - t)$ has a maximum of L . At such a maximum the tangent space is annihilated by dL . In coordinates with $L = x_1$, the point therefore solves

$$P = t, \quad \partial_2 P = \cdots = \partial_n P = 0.$$

For generic L and t this system is finite; otherwise an arbitrarily small generic change of L makes it finite. By (14.4.243), it has at most $D(D - 1)^{n-1}$ points.

The same estimate, with $2D$ in place of D , applies to a possibly singular zero set. To see this without assuming smoothness, choose any finite list of bounded components of $Z(P)$. They have pairwise disjoint bounded neighbourhoods whose boundaries do not meet $Z(P)$. For all sufficiently small regular $s > 0$, the level $P^2 = s$ has a compact component in each of these neighbourhoods. The preceding critical-point count bounds the length of the list by $2D(2D - 1)^{n-1}$. Since the list was arbitrary, this bounds all bounded components. For the unbounded ones, choose R outside the finite exceptional set of critical values of $|x|$ on a Whitney stratification of $Z(P)$. Every unbounded component then meets S_R , and distinct components meet disjoint connected components of $Z(P) \cap S_R$. Cover S_R by two stereographic charts. After substituting

$$x = R \frac{(2u, 1 - |u|^2)}{1 + |u|^2}$$

and multiplying by $(1 + |u|^2)^D$, the equation $P(x) = 0$ becomes a polynomial equation in $u \in \mathbb{R}^{n-1}$ of degree at most $2D$. Assign components meeting the equator to the first chart and the remaining components to the second. The induction hypothesis in dimension $n-1$ bounds their total by $2C_{n-1}(1+2D)^{n-1}$, which is absorbed by $C_n(1+D)^n$. This proves the claimed bound for the unbounded components as well.

Finally, for a component U of the complement inside a ball, the continuous function P has constant sign on U . After replacing P by $-P$ when necessary and making an arbitrarily small regular perturbation, an interior maximum of P in each component again solves $\partial_1 P = \dots = \partial_n P = 0$; components whose maximum lies on the sphere are counted by the already proved $(n-1)$ -dimensional boundary case. Bézout and induction give $C_n(1+D)^n$. Passing back to the unperturbed polynomial cannot create more components than the upper limit of the nearby regular fibres: disjoint compact subsets of distinct components have disjoint neighbourhoods and persist under a sufficiently small perturbation. This completes the component count. $\square$

Theorem 14.4.95 (Wongkew tubular-neighbourhood estimate). *Let $Z = Z(Q) \subset \mathbb{R}^n$ be an algebraic variety of dimension d , and let $B \subset \mathbb{R}^n$ be a ball of radius r . There is a constant K_n such that, for $0 < \rho \leq r$,*

$$|N_\rho(Z \cap B)| \leq K_n \sum_{j=n-d}^n (1 + \deg Q)^j \rho^j r^{n-j} \leq K'_n (1 + \deg Q)^n \rho^{n-d} r^d.$$

Proof. We prove the first inequality by induction on the ambient dimension n ; the second follows term by term from $\rho \leq r$ and $1 + \deg Q \geq 1$. Translation and dilation reduce the bookkeeping to a ball centred at the origin, but we retain r and ρ to display the powers.

For $n = 1$, a non-zero polynomial has at most $D = \deg Q$ real zeros, and the ρ -neighbourhood of their intersection with an interval has length at most $2D\rho$. Assume the assertion in dimensions below n . Choose n orthonormal vectors in generic position relative to all strata of Z . In each of these directions place parallel hyperplanes at spacing between $\rho/(3\sqrt{n})$ and $\rho/(2\sqrt{n})$, translating the grids so that every slice has dimension at most $d - 1$ and degree at most D . Only $O_n(r/\rho)$ hyperplanes of each family meet $B(0, r + 2\rho)$.

Let $\mathcal{H}$ be the union of all grid hyperplanes. Every point of $N_\rho(Z \cap B)$ either lies in

$$N_{(1+1/n)\rho}(Z \cap H \cap B(0, r + 2\rho)) \quad \text{for some } H \in \mathcal{H},$$

or its nearest point of Z belongs to a component of Z contained in the interior of one grid box. This dichotomy follows by joining the nearest point to the boundary of its grid box: if its component exits the box, the first exit lies in $Z \cap H$; if it does not, the component is contained in that box. Enlarging the radius from ρ to $(1 + 1/n)\rho$ covers the joining segment and removes the harmless choice of a nearest point.

Apply the induction hypothesis in each $H \cong \mathbb{R}^{n-1}$ and then thicken in the normal direction by $O(\rho)$. Summing over the $O_n(r/\rho)$ slices in one family gives

$$C_n \sum_{j=n-d}^{n-1} (1 + D)^j \rho^j r^{n-j}.$$

The genericity of the grids is used only to lower dimension on a slice; a non-generic translate is obtained by a limit, since the neighbourhood volume is upper semicontinuous under translation of the grid.

It remains to bound components trapped in grid boxes. By Theorem 14.4.94, at most $C_n(1 + D)^n$ boxes contain such a component. Every grid box has diameter less than ρ , so its ρ -neighbourhood lies in a ball of radius 3ρ and has volume at most $C_n\rho^n$. Their total contribution is therefore at most $C_n(1 + D)^n\rho^n$, the $j = n$ term. Adding the slice and trapped-component contributions closes the induction. $\square$

Corollary 14.4.96. *Let $G \subset \mathbb{R}^3$ be a semi-algebraic grain of thickness ρ that is defined by a polynomial of degree at most D . Then*

$$|G| \lesssim D^3 \rho \operatorname{diam}(G)^2.$$

Next, we will need the following grains decomposition result, which is similar to Proposition 3.2 from [14].

Proposition 14.4.97. *Let $\varepsilon > 0$. Then there exists $K_\varepsilon > 0$ so that the following holds. Let $E \subset B(0, 1) \subset \mathbb{R}^3$ be open. Then there exists*

- A set $\mathcal{G}$ of semi-algebraic grains of thickness 100δ , each of which has diameter $\leq K_\varepsilon \delta^{-\varepsilon} (\#\mathcal{G})^{-1/3}$ and is defined by a polynomial of degree at most K_ε .
- For each $G \in \mathcal{G}$, a set $E_G \subset E \cap G$. The sets $\{E_G : G \in \mathcal{G}\}$ are disjoint.

Furthermore, we have

$$\sum_{G \in \mathcal{G}} |E_G| \geq K_\varepsilon^{-1} \delta^\varepsilon |E|,$$

and for each $G \in \mathcal{G}$ we have

$$K_\varepsilon^{-1} \delta^\varepsilon \frac{|E|}{\#\mathcal{G}} \leq |E_G| \leq K_\varepsilon \delta^{-\varepsilon} \frac{|E|}{\#\mathcal{G}}.$$

Finally, for every δ tube T , we have

$$\#\{G \in \mathcal{G} : T \cap E_G \neq \emptyset\} \leq K_\varepsilon \delta^{-\varepsilon} (\#\mathcal{G})^{1/3}. \quad (14.4.244)$$

Proof. We give the recursive construction, including the diameter and tube-crossing bookkeeping. It is convenient to spend $\varepsilon/20$ at each of the finitely many pigeonholing stages; at the end these losses are relabelled as ε .

Fix a large integer $D = D(\varepsilon)$. A node of the recursion consists of a rectangular box B_ν , a measurable set $E_\nu \subset E \cap B_\nu$, and its depth s . At that node apply Theorem 14.4.92 to $\mathbf{1}_{E_\nu} dx$, with degree D . Multiply the resulting polynomial by the equations of $O(D)$ coordinate planes which divide each side of B_ν into D equal intervals. The product, denoted P_ν , has degree at most CD . Its complementary components can be assigned to the coordinate subboxes; hence they have diameter at most $CD^{-1} \text{diam}(B_\nu)$. The partitioning part of the product ensures that each component carries at most $CD^{-3} |E_\nu|$ of the measure. By Theorem 14.4.94, the number of components is at most CD^3 .

Remove the 20δ -neighbourhood of $Z(P_\nu)$. For every complementary component O , create a child with

$$B_{\nu O} = B_\nu \cap B_O, \quad E_{\nu O} = E_\nu \cap O \setminus N_{20\delta}(Z(P_\nu)),$$

where B_O is its coordinate subbox. The removed part is assigned, without overlap, to the grain

$$G_\nu = N_{100\delta}(S_\nu), \quad S_\nu = Z(P_\nu) \cap B_\nu; \quad E_{G_\nu} = E_\nu \cap N_{20\delta}(Z(P_\nu)).$$

The set S_ν is semi-algebraic, has dimension at most two, and is contained in the zero set of a polynomial of degree at most CD . Empty assigned sets are discarded. When a box has diameter at most 50δ , stop and assign its

remaining set to the 100δ -neighbourhood of a plane through the centre of the box. Consequently the recursion stops after

$$S \leq C \frac{\log(1/\delta)}{\log D}$$

levels, and the assigned sets form a measurable partition of E up to a null set.

At depth s , every surviving node satisfies

$$\begin{aligned} |E_\nu| &\leq (CD^{-3})^s |E|, & \text{diam}(B_\nu) &\leq (CD^{-1})^s, \\ \#\{\nu : |\nu| = s\} &\leq (CD^3)^s. \end{aligned} \tag{14.4.245}$$

The harmless powers C^s are sub-power losses. Choose D so large that $C^S \leq \delta^{-\varepsilon/100}$ for all $0 < \delta < 1/2$; this follows from the displayed bound for S . A pigeonhole in the depth and then in the dyadic size of $|E_{G_\nu}|$ selects a collection $\mathcal{G}$ whose assigned sets have comparable measure and whose total measure is at least $K_\varepsilon^{-1} \delta^{\varepsilon/4} |E|$. After weakening $\varepsilon/4$ to ε , this is the asserted mass conclusion and gives both displayed upper and lower bounds for each $|E_G|$.

Write $M = \#\mathcal{G}$ and let s be its selected depth. From the first and third estimates in (14.4.245), together with the retained mass, we obtain

$$K_\varepsilon^{-1} \delta^{\varepsilon/2} D^{3s} \leq M \leq K_\varepsilon \delta^{-\varepsilon/2} D^{3s}.$$

The diameter estimate in (14.4.245) therefore yields

$$\text{diam}(G) \leq K_\varepsilon \delta^{-\varepsilon} M^{-1/3}.$$

This is the explicit sub-power form used in the rest of the proof. It avoids silently treating the iterated constants in the often-stated lossless diameter variant as a constant independent of δ .

It remains to count grains met by a tube. Erode every child region by 20δ as above. If a δ -tube T meets E_{G_ν} at depth s , then its axis meets the corresponding chain of un-eroded parent regions. At one node, either the axis is contained in $Z(P_\nu)$, in which case it meets no eroded complementary region, or the one-variable polynomial obtained by restricting P_ν to the axis has at most CD zeros and the axis meets at most $CD + 1$ regions. Induction down the tree gives at most $(CD + 1)^s$ relevant depth- s nodes. Using the lower bound for M above and increasing $D(\varepsilon)$ once more,

$$(CD + 1)^s \leq K_\varepsilon \delta^{-\varepsilon} M^{1/3}.$$

This is (14.4.244) and completes the proof. $\square$

Remark 14.4.98 (Why the displayed diameter has an explicit loss). The cited formulation in [14] supplies the mass and tube-crossing parts; the source

manuscript adds coordinate planes and states a lossless $K_\varepsilon M^{-1/3}$ diameter bound. Iterating the component-count constants actually gives the sub-power factor displayed above. Every later use already has free δ^ε margins, so replacing ε by a sufficiently small multiple proves the same downstream statements. We retain the loss rather than hiding it in a symbol declared independent of δ .

Using Proposition 14.4.97, we have the following structure theorem about arrangements of tubes.

Lemma 14.4.99. *Let $\varepsilon, \beta > 0$. Then there exists $K, \eta > 0$ so that the following holds. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and broad with error $\delta^{-\eta}$ (and exponent β). Suppose there is a number μ so that $\#\mathbb{T}_Y(x) \in [\mu, 2\mu)$ for all $x \in \bigcup_{T \in \mathbb{T}} Y(T)$. Then there exists the following:*

- A δ^ε refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$.
- A partition $E = E_1 \sqcup \dots \sqcup E_N$ of $\bigcup_{\mathbb{T}} Y'(T)$.

These objects have the following properties:

$$(a) \quad (i) \quad 1 \leq N \leq K\delta^{-\varepsilon}(\delta\mu^{-1} \#\mathbb{T})^3. \quad (14.4.246)$$

$$(ii) \quad r \geq K^{-1}\delta^\varepsilon N^{-1/3}.$$

(b) Each set E_i has diameter at most $2r$, and volume

$$K^{-1}N^{-1}\delta^\varepsilon \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq |E_i| \leq K\delta(\text{diam}(E_i))^2. \quad (14.4.247)$$

(c) $(\mathbb{T}', Y')_\delta$ is broad with error $K\delta^{-\varepsilon}$ (and exponent β), and $\#\mathbb{T}'_{Y'}(x) \geq K^{-1}\delta^\varepsilon\mu$ for each $x \in \bigcup_{T \in \mathbb{T}'} Y'(T)$.

(d) For each $T \in \mathbb{T}'$, there exists a set $\{T^{(j)}\}$ of pairwise $100r$ -separated tube segments $T^{(j)} \subset T$, each of length r . This set has the following properties:

- (i) Each segment $T^{(j)}$ intersects $Y'(T)$, and $Y'(T) \subset \bigcup T^{(j)}$.
- (ii) If $T^{(j)}$ is a tube-segment, then $T^{(j)} \cap Y'(T)$ intersects exactly one set E_i .

Proof.

Step 1. Let ε_1 be a small quantity to be chosen below. Let $\mathbb{T}_1 = \{T \in \mathbb{T} : |Y(T)| \geq \frac{1}{2}\delta^\eta|T|\}$, and let Y_1 be the restriction of Y to $\mathbb{T}_1$. Then $(\mathbb{T}_1, Y_1)_\delta$ is a $(1/2)$ -refinement of $(\mathbb{T}, Y)_\delta$. Apply Proposition 14.4.97 to $E = \bigcup_{T \in \mathbb{T}_1} Y_1(T)$

with ε_1 in place of ε , and let $\mathcal{G}$ and $\{E_G: G \in \mathcal{G}\}$ be the output of that Proposition.

For each $T \in \mathbb{T}_1$, there are at most $K_{\varepsilon_1} \delta^{-\varepsilon_1} (\#\mathcal{G})^{1/3}$ grains $G \in \mathcal{G}$ for which $T \cap E_G \neq \emptyset$. Define the set of “significant” grains

$$\mathcal{G}_{T,\text{sig}} = \{G \in \mathcal{G}: |T \cap E_G| \geq (\frac{1}{2}|Y_1(T)|)(K_{\varepsilon_1} \delta^{-\varepsilon_1} (\#\mathcal{G})^{1/3})^{-1}\}.$$

Then

$$\left| Y_1(T) \cap \bigsqcup_{G \in \mathcal{G}_{T,\text{sig}}} E_G \right| \geq \frac{1}{2} |Y_1(T)|.$$

Each set $Y_1(T) \cap E_G$ is contained in a sub-tube of length $\text{diam}(T \cap E_G)$, and we have

$$\delta^{\eta+\varepsilon_1} K_{\varepsilon_1}^{-1} (\#\mathcal{G})^{-1/3} \lesssim \text{diam}(T \cap E_G) \leq K_{\varepsilon_1} \delta^{-\varepsilon_1} (\#\mathcal{G})^{-1/3}.$$

The first inequality follows from the volume bound $|T \cap E_G| \gtrsim \delta^{\eta+\varepsilon_1} |T| K_{\varepsilon_1}^{-1} (\#\mathcal{G})^{-1/3}$, while the second follows from the diameter bound on G coming from Proposition 14.4.97.

Define

$$r = \kappa \delta^{\eta+\varepsilon_1} K_{\varepsilon_1}^{-1} (\#\mathcal{G})^{-1/3}, \quad (14.4.248)$$

where $\kappa \sim 1$ is chosen so that each set $T \cap E_G$ has diameter at least r . Then for each grain $G \in \mathcal{G}_{T,\text{sig}}$, we can choose a sub-tube $T_G \subset T$ of length r , so that

$$|T_G \cap E_G| \gtrsim \delta^{\eta+2\varepsilon_1} K_{\varepsilon_1}^{-2} |T \cap E_G| \gtrsim \delta^{2\eta+3\varepsilon_1} K_{\varepsilon_1}^{-3} |T| (\#\mathcal{G})^{-1/3} \gtrsim \delta^{\eta+2\varepsilon_1} K_{\varepsilon_1}^{-2} r |T|. \quad (14.4.249)$$

Furthermore, we can choose these sub-tubes so that every pair of sub-tubes in the set $\{T_G: G \in \mathcal{G}_{T,\text{sig}}\}$ either coincide or are interior-disjoint.

Since the sets $\{E_G\}$ are disjoint, the volume bound (14.4.249) says that for each tube T , at most $\delta^{-\eta-2\varepsilon_1} K_{\varepsilon_1}^2$ tubes from the set $\{T_G: G \in \mathcal{G}_{T,\text{sig}}\}$ can pairwise coincide (the sets T_G are sub-tubes of T , as described previously). Thus for each $T \in \mathbb{T}_1$, we can select a set $Y_2(T) \subset Y_1(T)$ with the following properties:

•

$$|Y_2(T)| \gtrsim \delta^{2\eta+4\varepsilon_1} K_{\varepsilon_1}^{-4} |Y_1(T)|.$$

(This follows from the first inequality in (14.4.249), plus the fact that at most $\delta^{-\eta-2\varepsilon_1} K_{\varepsilon_1}^2$ can coincide).

- There is a set $\{T^{(j)}\}$ of sub-tubes of T , each of length r . These tubes are $100r$ separated. Each of these sub-tubes $T^{(j)}$ intersects $Y_2(T)$, and for each sub-tube $T^{(j)}$ there is exactly one $G \in \mathcal{G}$ for which $Y_2(T) \cap T^{(j)} \cap E_G \neq \emptyset$.

Step 2. In Step 1 we processed the tubes. In this step, we will process the grains in $\mathcal{G}$. Fix a grain G . Cover G by boundedly overlapping balls $\{B_i\}$ of radius $2r$ (recall that r was defined in Step 1). We will select a $100r$ -separated subset of $\{B_i\}$ (after re-indexing, we will denote this set by $\{B_i\}_{i=1}^M$ for some $M = M(G) \geq 1$) so that

$$\sum_{i=1}^M \sum_{T \in \mathbb{T}_1} |Y_2(T) \cap E_G \cap B_i| \gtrsim \sum_{T \in \mathbb{T}_1} |Y_2(T) \cap E_G|. \quad (14.4.250)$$

Since the balls are $100r$ -separated, each sub-tube $T^{(j)}$ of length r intersects at most one ball. Thus for each $T \in \mathbb{T}_1$ and each sub-tube $T^{(j)}$, there is at most one ball B_i from the above collection for which $Y_2(T) \cap T^{(j)} \cap E_G \cap B_i \neq \emptyset$.

We perform the above operation for each $G \in \mathcal{G}$. Let $\mathcal{G}'$ be the union of sets $\{B_i \cap G : i = 1, \dots, M(G)\}$, as G ranges over the grains in $\mathcal{G}$. For each $G' \in \mathcal{G}'$, define $E_{G'} = G' \cap E_G$, where $G \in \mathcal{G}$ is the (unique) grain that gave rise to G' . We have that the sets $\{E_{G'} : G' \in \mathcal{G}'\}$ are disjoint.

Define $Y_3(T) = Y_2(T) \cap \bigcup_{G' \in \mathcal{G}'} E_{G'}$. By (14.4.250) we have that $(\mathbb{T}_1, Y_3)_\delta$ is a $\gtrsim 1$ refinement of $(\mathbb{T}_1, Y_2)_\delta$, which in turn is a $\gtrsim \delta^{2\eta+4\varepsilon_1} K_{\varepsilon_1}^{-4}$ refinement of $(\mathbb{T}, Y)_\delta$. Furthermore, for each $T \in \mathbb{T}_1$ it is still the case that there is a set of $100r$ -separated sub-tubes $\{T^{(j)}\}$, each of which has length r and intersects $Y_3(T)$ (indeed, any sub-tubes $T^{(j)}$ from the previous collection that fail to intersect $Y_3(T)$ can be discarded), so that for each sub-tube $T^{(j)}$, there is exactly one $G' \in \mathcal{G}'$ with $T^{(j)} \cap E_{G'} \neq \emptyset$.

Step 3. Let $\mu' \leq \mu$ and let $(\mathbb{T}_1, Y_4)_\delta$ be a $\gtrsim_\delta 1$ refinement (we use $\gtrsim_\delta$ to absorb the $K_{\varepsilon_1}^{-4}$ factor) of $(\mathbb{T}_1, Y_3)_\delta$ with the property that $\#(\mathbb{T}_1)_{Y_4}(x) \sim \mu'$ for each $x \in \bigcup_{T \in \mathbb{T}_1} Y_4(T)$. Since $(\mathbb{T}_1, Y_4)_\delta$ is a $\approx_\delta \delta^{2\eta+4\varepsilon_1}$ refinement of $(\mathbb{T}, Y)_\delta$, we have

$$\mu' \gtrsim_\delta \delta^{2\eta+4\varepsilon_1} \mu, \quad (14.4.251)$$

and hence $(\mathbb{T}_1, Y_4)_\delta$ is broad with error $\lesssim_\delta \delta^{-3\eta-4\varepsilon_1}$ and exponent β . We will select η and ε_1 sufficiently small, and K sufficiently large so that $(\mathbb{T}_1, Y_4)_\delta$ is broad with error $\leq K\delta^{-\varepsilon}$, and $\mu' \geq K^{-1}\delta^\varepsilon\mu$ (c.f. Conclusion (c) from Proposition 14.4.99).

For each $G' \in \mathcal{G}'$, define $E'_{G'} = E_{G'} \cap \bigcup_{T \in \mathbb{T}_1} Y_4(T)$. After dyadic pigeonholing, we can select a set $\mathcal{G}'' \subset \mathcal{G}'$ so that $|E'_{G'}|$ is the same (up to a factor of 2) for each $G' \in \mathcal{G}''$. For each $T \in \mathbb{T}_1$, define $Y_5(T) = Y_4(T) \cap \bigcup_{G'' \in \mathcal{G}''} E'_{G''}$.

Define $N = \#\mathcal{G}''$. Abusing notation, we will enumerate the elements of $\{E'_{G''} : G'' \in \mathcal{G}''\}$ as $E_1, \dots, E_N$. Since $\#\mathbb{T}_Y(x) \sim \mu$ and $K_{\varepsilon_1}^{-1}\delta^{\varepsilon_1} \frac{|E|}{\#\mathcal{G}} \leq |E_G| \leq K_{\varepsilon_1}\delta^{-\varepsilon_1} \frac{|E|}{\#\mathcal{G}}$ for each $G \in \mathcal{G}$ (the latter is true since $\mathcal{G}$ was the output of Proposition 14.4.97), we have

$$\#\mathcal{G} \geq N = \#\mathcal{G}'' \gtrsim_\delta \delta^{2\varepsilon_1} \#\mathcal{G}. \quad (14.4.252)$$

Define $\mathbb{T}' = \mathbb{T}_1$ and $Y'(T) = Y_5(T)$. This choice of $(\mathbb{T}', Y')_\delta$ satisfies Conclusion (c) from Proposition 14.4.99.

We have

$$\bigcup_{T \in \mathbb{T}'} Y'(T) = \bigsqcup_{i=1}^N E_i,$$

and if the constant $K = K(\varepsilon_1)$ (recall that ε_1 depends on ε) is chosen sufficiently large, then for each index $i = 1, \dots, N$ we have

$$N^{-1} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \lesssim_\delta N^{-1} \left| \bigcup_{T \in \mathbb{T}'} Y'(T) \right| \lesssim |E_i| \leq K\delta(\text{diam}(E_i))^2,$$

where the final inequality is Corollary 14.4.96. Thus we have established Conclusion (b) from Proposition 14.4.99. Recall that for each $T \in \mathbb{T}'$, we have a set $\{T^{(j)}\}$ of tube-segments, each of which has length r . These segments are $100r$ -separated. Discarding segments if necessary, we may suppose that each tube segment $T^{(j)}$ intersects $Y'(T)$. It is still the case that for each surviving tube segments, there is exactly one index i so that $Y'(T) \cap T^{(j)} \cap E_i \neq \emptyset$. This establishes Conclusions (d.i) and (d.ii).

Step 4. It remains to establish Conclusion (a) from Proposition 14.4.99. We begin with Conclusion (a.i), i.e. we must show that N satisfies Inequality (14.4.246). In Step 3 we established Conclusion (d.ii). Combining this with (14.4.248) and (14.4.252), we conclude that for each $T \in \mathbb{T}$, there are $\lesssim_\delta \delta^{-2\varepsilon_1} N^{1/3}$ indices i for which $Y'(T) \cap E_i \neq \emptyset$. Thus by pigeonholing, there is an index i so that

$$\#\{T \in \mathbb{T} : Y'(T) \cap E_i \neq \emptyset\} \lesssim_\delta \delta^{-2\varepsilon_1} N^{-2/3} (\#\mathbb{T}).$$

Fix this index i , and denote the above set by $\mathbb{T}_i$. We have

$$\begin{aligned} \nu\{(x, T, T') \in E_i \times \mathbb{T}_i^2 : x \in Y'(T) \cap Y'(T'), \\ \angle(\text{dir}(T), \text{dir}(T')) \gtrsim_\delta \delta^{(3\eta+2\varepsilon_1)/\beta}\} & \quad (14.4.253) \\ \lesssim_\delta \delta^{3-(3\eta+2\varepsilon_1)/\beta} (\#\mathbb{T}_i)^2, \end{aligned}$$

where ν denotes the product of Lebesgue measure on E_i with counting measure on $\mathbb{T}_i^2$. The above inequality is justified by the observation that for each pair $T, T' \in \mathbb{T}_i$ with $\angle(\text{dir}(T), \text{dir}(T')) \geq \delta^{(3\eta+2\varepsilon_1)/\beta}$, we have $|T \cap T'| \lesssim \delta^{3-(3\eta+2\varepsilon_1)/\beta}$.

On the other hand, we have $|E_i| \gtrsim_\delta \delta^{\varepsilon_1} N^{-1} (\mu^{-1} \delta^2 (\#\mathbb{T}))$, and since $(\mathbb{T}', Y')_\delta$ is broad with error $\lesssim_\delta \delta^{-(3\eta+2\varepsilon_1)}$, we have that for each $x \in E_i$ there are $\gtrsim (\mu')^2$ pairs $T, T' \in \mathbb{T}_i$ so that (x, T, T') is in the above set. We

conclude that by (14.4.251),

$$\begin{aligned}
 \mu \delta^{2+6\eta+4\varepsilon_1} N^{-1} (\#\mathbb{T}) &\lesssim_\delta (\mu')^2 \left(\delta^{4\eta+2\varepsilon_1} N^{-1} \mu'^{-1} \delta^2 (\#\mathbb{T}) \right) \\
 &\lesssim \text{LHS (14.4.253)} \\
 &\lesssim \text{RHS (14.4.253)} \\
 &= \delta^{3-(3\eta+2\varepsilon_1)/\beta} (\#\mathbb{T}_i)^2 \\
 &\lesssim K_{\varepsilon_1}^2 \delta^{3-(3\eta+2\varepsilon_1)/\beta-4\varepsilon_1} N^{-4/3} (\#\mathbb{T})^2.
 \end{aligned}$$

Re-arranging, we have

$$N^{1/3} \lesssim_\delta K_{\varepsilon_1}^2 \mu^{-1} \delta^{-(3\eta+2\varepsilon_1)/\beta-6\eta-8\varepsilon_1+1} (\#\mathbb{T}).$$

The claimed bound on N now follows by selecting $3\eta + 2\varepsilon_1 < \varepsilon\beta/20$, and choosing K appropriately.

Finally, Conclusion (a.ii) follows from the definition of r and (14.4.252). $\square$

Lemma 14.4.99 says nothing about the shape of the sets $\{E_i\}$. The next result will help us show that a substantial portion of each set E_i must be contained in the δ neighbourhood of a plane. The idea is that since the volume of E_i is small, each point $x \in E_i$ is broad, and $E_i \cap T$ is dense for each tube $T \in \mathbb{T}_i$, we can find many “triangles” formed by three tubes from $\mathbb{T}_i$ that pairwise intersect at distinct points. Since every triple of lines that pairwise intersect at distinct points must be coplanar, this in turn forces a significant subset of E_i to be planar.

Lemma 14.4.100. *Let $\varepsilon, \beta > 0$. Then there exists $\kappa, \eta > 0$ so that the following holds. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense and broad with error $\leq \delta^{-\eta}$ (and exponent β). Suppose that*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{1-\eta}.$$

Then there is a plane Π so that if we define $\mathbb{T}' = \{T \in \mathbb{T} : T \subset N_{2\delta}(\Pi)\}$, then there is a shading $\{Y'(T) : T \in \mathbb{T}'\}$ and a number μ so that

$$(a) \quad \#\mathbb{T}'_{Y'}(x) \sim \mu \text{ for each } x \in \bigcup_{T \in \mathbb{T}'} Y'(T).$$

$$(b) \quad \sum_{T \in \mathbb{T}'} |Y'(T)| \geq \kappa \delta^\varepsilon \sum_{T \in \mathbb{T}} |Y(T)|. \quad (14.4.254)$$

$$(c) \quad \left| \bigcup_{T \in \mathbb{T}'} Y'(T) \right| \geq \kappa \delta^{1+\varepsilon}, \quad (14.4.255)$$

Proof. Replacing β by $\min(\beta, 1/2)$ only weakens the broadness hypothesis, because $r^\beta \leq r^{\min(\beta, 1/2)}$ for $0 < r \leq 1$. We may therefore assume throughout this proof that $\beta \leq 1/2$. First, after dyadic pigeonholing and replacing $(\mathbb{T}, Y)_\delta$ by a refinement, we may suppose that there is a number μ so that $\#\mathbb{T}_Y(x) \sim \mu$ for all $x \in \bigcup_{T \in \mathbb{T}} Y(T)$; we still have that $(\mathbb{T}, Y)_\delta$ is $\gtrsim_\delta \delta^\eta$ dense, and $(\mathbb{T}, Y)_\delta$ is broad with error $\lesssim_\delta \delta^{-\eta}$. Observe that

$$\delta^{2+\eta}(\#\mathbb{T}) \lesssim_\delta \sum_{T \in \mathbb{T}} |Y(T)| \lesssim \mu \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{1-\eta} \mu,$$

and thus

$$\#\mathbb{T} \lesssim_\delta \delta^{-1-2\eta} \mu. \quad (14.4.256)$$

Define

$$\begin{aligned} \mathbb{T}' &= \{T \in \mathbb{T} : |Y(T)| \geq \delta^{2\eta}|T|\}, \\ E &= \{x \in \mathbb{R}^3 : \#\mathbb{T}'_Y(x) \geq \delta^{2\eta}\mu\}, \\ \mathbb{T}'' &= \{T \in \mathbb{T} : |Y(T) \cap E| \geq \delta^{4\eta}|T|\}. \end{aligned}$$

By double-counting we have $\#\mathbb{T}'' \geq \delta^{5\eta}(\#\mathbb{T})$.

Fix a sufficiently small constant $c_\beta > 0$. For each $T_0 \in \mathbb{T}''$ and each $x \in Y(T_0) \cap E$, there are at least $\frac{1}{2}\delta^{2\eta}\mu$ tubes $T \in \mathbb{T}'$ with $x \in Y(T)$ and

$$\angle(\text{dir}(T), \text{dir}(T_0)) \geq c_\beta \delta^{3\eta/\beta}.$$

Indeed, broadness bounds the number in the complementary angular cap by

$$\delta^{-\eta}(c_\beta \delta^{3\eta/\beta})^\beta \mu = c_\beta^\beta \delta^{2\eta} \mu,$$

which is at most half the total after fixing c_β . Integrating this incidence count over $Y(T_0) \cap E$, using $|Y(T_0) \cap E| \geq \delta^{4\eta}|T_0|$, and observing that one transverse tube meets T_0 in volume $O_\beta(\delta^3 \delta^{-3\eta/\beta})$, shows that there are at least

$$c_\beta \delta^{6\eta} \mu (\delta^{3\eta/\beta} \delta^{-1})$$

distinct such tubes. Since each $T \in \mathbb{T}'$ satisfies $|Y(T)| \geq \delta^{2\eta}|T|$ and $|T \cap N_{\delta^{10\eta/\beta}}(T_0)| \leq \frac{1}{2}\delta^{2\eta}|T|$, we conclude that

$$\begin{aligned} \sum_{\substack{T \in \mathbb{T}' \\ Y(T) \cap Y(T_0) \neq \emptyset \\ \angle(\text{dir}(T), \text{dir}(T_0)) \geq c_\beta \delta^{3\eta/\beta}}} |Y(T) \setminus N_{\delta^{10\eta/\beta}}(T_0)| &\gtrsim_\beta (\delta^{6\eta+3\eta/\beta} \mu \delta^{-1}) (\delta^{2\eta}|T|) \\ &\gtrsim \delta^{10\eta/\beta} \delta \mu. \end{aligned}$$

For the final inequality we use $\beta \leq 7/8$, which is harmless here because the application fixed β to be a much smaller positive number. On the other hand,

$$\left| \bigcup_{\substack{T \in \mathbb{T}' \\ Y(T) \cap Y(T_0) \neq \emptyset \\ \angle(\text{dir}(T), \text{dir}(T_0)) \geq \delta^{3\eta/\beta}}} Y(T) \setminus N_{\delta^{10\eta/\beta}}(T_0) \right| \leq \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{1-\eta},$$

Thus if we define F_{T_0} to be the set of points $x \in \mathbb{R}^3 \setminus N_{\delta^{10\eta/\beta}}(T_0)$ satisfying

$$\sum_{\substack{T \in \mathbb{T}' \\ Y(T) \cap Y(T_0) \neq \emptyset \\ \angle(\text{dir}(T), \text{dir}(T_0)) \geq \delta^{3\eta/\beta}}} \chi_{Y(T)}(x) \geq \delta^{11\eta/\beta} \mu,$$

then $|F_{T_0}| \geq \delta^{11\eta/\beta+1}$ because $\#\mathbb{T}_Y(x) \lesssim \mu$. Since $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$, for each point $x \in F_{T_0}$, there are $\gtrsim (\delta^{11\eta/\beta} \mu)^2$ pairs $T, T' \in \mathbb{T}'$ satisfying

$$\begin{aligned} Y(T) \cap Y(T_0) &\neq \emptyset, & Y(T') \cap Y(T_0) &\neq \emptyset, \\ Y(T) \cap Y(T') &\neq \emptyset, \\ \angle(\text{dir}(T_0), \text{dir}(T)) &\geq \delta^{3\eta/\beta}, & \angle(\text{dir}(T_0), \text{dir}(T')) &\geq \delta^{3\eta/\beta}, \\ \angle(\text{dir}(T), \text{dir}(T')) &\geq \delta^{12\eta/\beta^2}, \\ (T \cap T') \setminus N_{\delta^{10\eta/\beta}}(T_0) &\neq \emptyset. \end{aligned} \tag{14.4.257}$$

For each such pair, we have $|Y(T) \cap Y(T')| \leq |T \cap T'| \lesssim \delta^{3-12\eta/\beta^2}$. We conclude that for T_0 fixed, there are $\gtrsim (\delta^{11\eta/\beta} \mu)^2 (|F_{T_0}| / \delta^{3-12\eta/\beta^2}) \gtrsim \delta^{23\eta/\beta^2} \mu^2 \delta^{-2}$ distinct pairs (T, T') satisfying (14.4.257). Thus by (14.4.256) there are $\gtrsim \delta^{23\eta/\beta^2} \mu^2 \delta^{-2} (\#\mathbb{T}'') \gtrsim_\delta \delta^{24\eta/\beta^2} (\#\mathbb{T}')^2 (\#\mathbb{T})$ triples $(T_0, T_1, T_2) \in \mathbb{T}'' \times \mathbb{T}' \times \mathbb{T}'$ that satisfy (14.4.257).

By pigeonholing, we can select a pair (T, T') which is a member of $\gtrsim_\delta \delta^{24\eta/\beta^2} (\#\mathbb{T})$ such triples. Fix this pair T, T' , and let $\mathbb{T}_0$ denote the set of tubes T_0 so that (T_0, T, T') satisfies (14.4.257). We have the following:

- T and T' intersect, and make angle $\gtrsim \delta^{12\eta/\beta^2}$.
- For each $T_0 \in \mathbb{T}_0$, we have that $Y(T_0)$ intersects both T and T' , and makes angle $\geq \delta^{3\eta/\beta}$ with T and T' .
- $(T \cap T') \setminus N_{\delta^{10\eta/\beta}}(T_0) \neq \emptyset$.

The above items imply that each $T_0 \in \mathbb{T}_0$ is contained in the $\delta^{1-100\eta/\beta^2}$ neighbourhood of a plane — this is the plane spanned by the tubes T and T' (the latter plane is technically not well defined, but is instead defined up to uncertainty $\frac{\delta}{\angle(\text{dir}(T), \text{dir}(T'))} \leq \delta^{1-11\eta/\beta^2}$). In particular, there is a set

of $\lesssim \delta^{-200\eta/\beta^2}$ planes $\{\Pi_i\}$ so that each $T_0 \in \mathbb{T}_0$ is contained in the 2δ neighbourhood of some plane from this collection. By pigeonholing, we can select a plane Π so that

$$\sum_{\substack{T \in \mathbb{T}_0 \\ T \subset N_{2\delta}(\Pi)}} |Y(T)| \gtrsim_{\delta} (\delta^{200\eta/\beta^2})(\delta^{2+2\eta})(\delta^{24\eta/\beta^2}(\#\mathbb{T})) \gtrsim_{\delta} \delta^{300\eta/\beta^2} \sum_{T \in \mathbb{T}} |Y(T)|.$$

Define $\mathbb{T}' = \{T \in \mathbb{T} : T \subset N_{2\delta}(\Pi)\}$. By pigeonholing, we can select a number μ' so that if we define the shading $Y'(T) = \{x \in Y(T) : \#\mathbb{T}'_{Y'}(x) \sim \mu'\}$, then $(\mathbb{T}', Y')_{\delta}$ is an $\approx_{\delta} 1$ refinement of $(\mathbb{T}', Y)_{\delta}$. This gives Conclusions (a) with μ' in place of μ and (b).

The source invokes a planar Cordoba argument for Conclusion (c). The exact missing input is the following shaded version, with constants uniform in the displayed parameters:

$$\begin{aligned} \mathbb{T}' \subset N_{2\delta}(\Pi), \quad (\mathbb{T}', Y')_{\delta} \text{ is } K\text{-broad with exponent } \beta, \quad |Y'(T)| \geq \lambda|T|, \\ \#\mathbb{T}'_{Y'}(x) \sim \mu' \quad \implies \quad \left| \bigcup_{T \in \mathbb{T}'} Y'(T) \right| \geq c_{\beta} K^{-C_{\beta}} \lambda^{C_{\beta}} \delta. \end{aligned} \tag{WZ-A2}$$

With $K = \delta^{-400\eta/\beta^2}$ and the retained density supplied by Conclusion (b), choosing η sufficiently small compared with ε would give (14.4.255). However, neither Lemma 14.4.36 (which assumes a global slab Katz–Tao constant) nor the preceding incidence count proves (WZ–A2). Consequently Conclusion (c), this lemma, and its rescaled corollary remain conditional on WZ–A2. $\square$

Next, we record the following re-scaled variant of Lemma 14.4.100

Corollary 14.4.101. *Let $\varepsilon, \beta > 0$. Then there exists $\kappa, \eta > 0$ so that the following holds. Let $0 < \delta < r \leq 1$ and let $B \subset \mathbb{R}^3$ be a ball of radius r . Let $(\mathbb{T}, Y)_{\delta}$ be broad with error $\delta^{-\eta}$ (and exponent β). Suppose that $Y(T) \subset B$ for each $T \in \mathbb{T}$, and $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^{\eta} \delta^2 r (\#\mathbb{T})$. Suppose furthermore that the tubes in $\mathbb{T}$ are contained in a 1 tube T_1 , and that*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq r^2 \delta^{1-\eta}.$$

Then there is a set $\mathbb{T}' \subset \mathbb{T}$; a number μ ; a shading $Y'(T) \subset Y(T)$, $T \in \mathbb{T}'$; a prism P of dimensions comparable to $\delta \times r \times r$ so that each $T \in \mathbb{T}'$ satisfies $T \cap P \neq \emptyset$, and T exits P through its long sides, in the sense of Remark 14.4.68. Furthermore, we have

$$(a) \quad \#\mathbb{T}'_{Y'}(x) \sim \mu \text{ for each } x \in \bigcup_{T \in \mathbb{T}'} Y'(T).$$

(b)

$$\sum_{T \in \mathbb{T}'} |Y'(T)| \geq \kappa \delta^\varepsilon \sum_{T \in \mathbb{T}} |Y(T)|. \quad (14.4.258)$$

(c)

$$\left| \bigcup_{T \in \mathbb{T}'} Y'(T) \right| \geq \kappa \delta^{1+\varepsilon} r^2. \quad (14.4.259)$$

We are now ready to prove Proposition 14.4.67. In brief, we use Lemma 14.4.99 to cover a substantial portion of $(\mathbb{T}, Y)_\delta$ by a union of disjoint sets $\{E_i\}$, and then we use Corollary 14.4.101 to trap a substantial portion of each set E_i in a prism P_i of dimensions comparable to $\delta \times r \times r$, for an appropriately chosen diameter r .

Proof of Proposition 14.4.67.

Step 1. Let $\delta_0, \varepsilon_1, \varepsilon_2$ be small numbers to be chosen below. We will select δ_0 very small compared to ε_1 , ε_1 very small compared to ε_2 , and η, κ very small compared to ε_1 . These numbers depend on ε and β . We may suppose that $\delta \leq \delta_0$, or else the result is immediate provided we choose κ sufficiently small so that $\kappa \delta^\varepsilon (\delta \# \mathbb{T})^{-1} \mu \leq \delta$. In this case the set $\mathcal{G}$ consists of $\delta \times \delta \times \delta$ balls, and there is nothing to prove. Henceforth we shall assume that $\delta \leq \delta_0$.

After dyadic pigeonholing and replacing $(\mathbb{T}, Y)_\delta$ by a $\sim (\log 1/\delta)^{-1}$ refinement, we may suppose there exists a number $\mu \geq 1$ so that $\# \mathbb{T}_Y(x) \in [\mu, 2\mu]$ for every $x \in \bigcup_{T \in \mathbb{T}} Y(T)$; we still have that $(\mathbb{T}, Y)_\delta$ is broad with error $\delta^{-\eta}$.

Apply Lemma 14.4.99 to $(\mathbb{T}, Y)_\delta$ with ε_1 in place of ε and β as above; we can do this, provided we select $\eta > 0$ sufficiently small depending on ε_1 and β . Let $K_1, N, (\mathbb{T}_1, Y_1)_\delta$, and $E_1, \dots, E_N$ be the output of that lemma. In particular, $(\mathbb{T}_1, Y_1)$ is broad with error $K_1 \delta^{-\varepsilon_1}$. Note that by Item (a), we have

$$r \geq K_1^{-2} \delta^{2\varepsilon_1} (\delta \# \mathbb{T})^{-1} \mu. \quad (14.4.260)$$

Let I_1 denote the set of indices in $\{1, \dots, N\}$ for which

$$\#\{T \in \mathbb{T}_1 : Y_1(T) \cap E_i \neq \emptyset\} \leq 2N^{-1}r^{-1}(\# \mathbb{T}),$$

and let I_2 denote the remaining indices. For each $T \in \mathbb{T}_1$, there are at most $(100r)^{-1}$ sets E_i for which $Y_1(T) \cap E_i \neq \emptyset$. Thus

$$\#I_2 \leq \left(2N^{-1}r^{-1}(\# \mathbb{T})\right)^{-1} \sum_{T \in \mathbb{T}} \#\{i : Y_1(T) \cap E_i \neq \emptyset\} \leq N/2.$$

We conclude that $\#I_1 \geq N/2$. For each $i \in I_1$, define

$$\mathbb{T}^{(i)} = \{T \in \mathbb{T} : Y_1(T) \cap E_i \neq \emptyset\}.$$

We have

$$\begin{aligned}
 \sum_{T \in \mathbb{T}^{(i)}} |Y_1(T) \cap E_i| &\geq K_1^{-1} \delta^{\varepsilon_1} \mu |E_i| \\
 &\geq \mu K_1^{-2} \delta^{2\varepsilon_1} N^{-1} \left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \\
 &\geq K_1^{-1} N^{-1} \delta^{2\varepsilon_1 + \eta} (|T| \# \mathbb{T}) \\
 &\geq K_1^{-2} \delta^{2\varepsilon_1 + \eta} N^{-1} (Nr/2) (|T| \# \mathbb{T}^{(i)}) \\
 &\geq K_1^{-3} \delta^{3\varepsilon_1 + \eta} (\delta^2 r) (\# \mathbb{T}^{(i)}).
 \end{aligned}$$

In the above, the first inequality used Conclusion (c) of Lemma 14.4.99; the second inequality used (14.4.247); the third inequality used the fact that $(\mathbb{T}, Y)_\delta$ is δ^η dense, and $\# \mathbb{T}_Y(x) \sim \mu$ for each $x \in \bigcup Y(T)$; and the fourth inequality used the fact that $i \in I_1$.

Step 2. By (14.4.247) we have $|E_i| \leq K\delta(\text{diam}(E_i))^2$, and by Conclusion (b) of Lemma 14.4.99, $\text{diam}(E_i) \leq 2r$. Let B_i be a ball of radius r that contains E_i . For each $T \in \mathbb{T}^{(i)}$ define $Y^{(i)}(T) = Y_1(T) \cap E_i \subset B_i$. If δ_0 and ε_1 are chosen sufficiently small depending on ε_2 and β , then by Step 1, $(\mathbb{T}^{(i)}, Y^{(i)})_\delta$ satisfies the hypotheses of Corollary 14.4.101. Applying this Corollary, we obtain a set G_i comparable to a $\delta \times r \times r$ prism; a number μ_i ; a set $\mathbb{T}^{(i)'} \subset \mathbb{T}^{(i)}$; and a sub-shading $Y^{(i)'}(T) \subset Y^{(i)}(T)$, $T \in \mathbb{T}^{(i)'}$, so that the following holds:

- Each $T \in \mathbb{T}^{(i)'}$ intersects G_i and exits G_i through its long ends, in the sense of Remark 14.4.68.

•

$$\sum_{T \in \mathbb{T}^{(i)'}} |Y^{(i)'}(T)| \geq \kappa_2 \delta^{\varepsilon_2} \sum_{T \in \mathbb{T}^{(i)}} |Y^{(i)}(T)|.$$

•

$$\left| \bigcup_{T \in \mathbb{T}^{(i)'}} Y^{(i)'}(T) \right| \geq \kappa_2 \delta^{1+\varepsilon_2} r^2 \gtrsim \kappa_2 \delta^{\varepsilon_2} |G_i|.$$

- For each $x \in \bigcup_{T \in \mathbb{T}^{(i)'}} Y^{(i)'}(T)$ we have $\# \mathbb{T}_{Y^{(i)'}}^{(i)'}(x) \sim \mu_i$.

After dyadic pigeonholing, we can select a number μ' and a set of indices $I'_1 \subset I_1$ so that $\mu_i \sim \mu'$ for each $i \in I'_1$. We will choose μ' in such a way that if we define $\mathcal{G} = \{G_i : i \in I'_1\}$, define the shading

$$Y(G_i) = \bigcup_{T \in \mathbb{T}^{(i)'}} Y^{(i)'}(T),$$

and define the shading

$$Y_2(T) = Y_1(T) \cap \bigcup_G Y(G),$$

where the union is taken over those $G \in \mathcal{G}$ for which $T \in \mathbb{T}^{(i)'}$, then $(\mathbb{T}_1, Y_2)_\delta$ is a $\gtrsim_\delta \delta^{\varepsilon_2}$ refinement of $(\mathbb{T}_1, Y_1)_\delta$, and thus a $\gtrsim_\delta \delta^{\varepsilon_1 + \varepsilon_2}$ refinement of $(\mathbb{T}, Y)_\delta$.

Note that for each $G_i \in \mathcal{G}$, we have $Y(G_i) \subset G_i \cap E_i$, and thus the sets $\{Y(G) : G \in \mathcal{G}\}$ are disjoint. Furthermore, $|Y_1(G)| \geq \kappa_2 \delta^{\varepsilon_2} |G_i|$, and thus $(\mathcal{G}, Y)_{\delta \times r \times r}$ is $\kappa_2 \delta^{\varepsilon_2}$ dense.

We have the following:

- (i) $\bigcup_{\mathbb{T}} Y_2(T) = \bigsqcup_{G \in \mathcal{G}} Y(G)$.
- (ii) If $Y_2(T) \cap Y(G) \neq \emptyset$, then T exits G through its long ends, in the sense of Remark 14.4.68.
- (iii) If $Y_2(T) \cap Y(G) \neq \emptyset$, then $Y_2(T) \cap G \subset Y(G)$.

Item (iii) follows from the fact that if $Y_2(T) \cap Y(G_i) \neq \emptyset$ for some $G_i \in \mathcal{G}$ (recall that G_i is a prism associated to a set E_i), then there is a length- r sub-tube $T^{(j)} \subset T$ so that $Y_2(T) \cap T^{(j)} \subset E_i \subset Y(G_i)$. Since the sub-tubes $\{T^{(j)}\}$ associated to T are $100r$ separated, and $\text{diam}(G) \leq 10r$, we conclude that $T^{(j)}$ is the only sub-tube from the above collection that intersects G .

We conclude that $(\mathcal{G}, Y)_{\delta \times r \times r}$ is a robustly $\kappa_2 \delta^{\varepsilon_2}$ -dense two-scale grains decomposition of $(\mathbb{T}_1, Y_2)_\delta$ wrt the single 1-tube $\{T_1\}$. Finally, the desired bound on r is given by (14.4.260). $\square$

14.4.12 Wolff’s hairbrush argument and the proof of Proposition 14.4.8

The goal of this section is to prove Proposition 14.4.8. We will restate it here in an expanded form.

Proposition 14.4.8, expanded. For all $\varepsilon > 0$, there exists $\kappa, \eta > 0$ so that the following holds for all $\delta > 0$. Let $(\mathbb{T}, Y)_\delta$ be δ^η dense, with $C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$ and $C_{F-SW}(\mathbb{T}) \leq \delta^{-\eta}$. Then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \kappa \delta^{3/2 + \varepsilon} (\#\mathbb{T})^{1/2}. \quad (14.4.261)$$

Proof. Fix $\varepsilon > 0$. We shall prove the result with a loss $\delta^{C\eta}$ and then choose $\eta < \varepsilon/(2C)$. We discretize every shading by half-open δ -cubes, changing all volumes by absolute factors.

Step 1: angular stopping time. At a shaded cube Q , begin with a unit cap containing all directions of $\mathbb{T}_Y(Q)$. If a dyadic subcap of relative radius $s \leq 1/2$ contains more than s^η of the current directions, retain that subcap and repeat. If the successive relative radii are $s_1, \dots, s_k$, the retained fraction

is greater than $(s_1 \cdots s_k)^\eta$, while the final radius is $\theta_Q = s_1 \cdots s_k \geq \delta$. Hence at least a δ^η fraction remains. At termination, for every $r \in [\delta, \theta_Q]$ and every direction w ,

$$\#\{T \in \mathbb{T}_Y(Q) : \angle(\text{dir}T, w) \leq r\} \leq 2(r/\theta_Q)^\eta \#\mathbb{T}_Y(Q). \quad (14.4.262)$$

The factor two accounts for replacing an arbitrary cap by a dyadic cap. Pigeonhole the dyadic value of θ_Q and the multiplicity at Q ; retain the corresponding tube–cube incidences. The retained incidence mass is at least $\delta^{2\eta}$ of the original after absorbing logarithms.

Choose a maximal essentially distinct family $\mathbb{T}_\theta$ of $C\theta$ -tubes. The tubes contributing at a retained cube Q lie in a $C\theta$ -tube with direction equal to the terminal cap centre. Assign the cube, and all incidences retained there, to the first comparable member of $\mathbb{T}_\theta$. Assign each fine tube to one such parent and pigeonhole the parent fibre sizes and shaded masses. We obtain a balanced partitioning cover and disjoint coarse shadings, so that

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^{C\eta} \sum_{S \in \mathbb{T}_\theta} \left| \bigcup_{T \in \mathbb{T}[S]} Y_S(T) \right|. \quad (14.4.263)$$

After the affine map taking S to a unit tube, the members of $\mathbb{T}^S$ are h -tubes, $h = \delta/\theta$; their shading has density $\lambda \geq \delta^{C\eta}$ and satisfies

$$\#\{T \in \mathbb{T}_{Y^S}^S(Q) : \angle(\text{dir}T, w) \leq r\} \leq 2r^\eta \#\mathbb{T}_{Y^S}^S(Q), \quad h \leq r \leq 1. \quad (14.4.264)$$

The affine invariance calculation used in (14.4.75) also gives $C_{KT-CW}(\mathbb{T}^S) \lesssim C_{KT-CW}(\mathbb{T}) \leq \delta^{-\eta}$.

Step 2: the two-broad shaded hairbrush lemma. We prove the precise estimate used for each rescaled fibre. Let $(\mathcal{A}, Z)_h$ be a family of essentially distinct h -tubes satisfying (14.4.264), let $K = C_{KT-CW}(\mathcal{A})$, let $N = \#\mathcal{A}$, and suppose its shading is λ -dense. Then

$$\left| \bigcup_{T \in \mathcal{A}} Z(T) \right| \geq c_\eta \lambda^{5/2} K^{-1/2} (\log(2/h))^{-1/2} h^{3/2} N^{1/2}. \quad (14.4.265)$$

Refine so that every tube contains between M and $2M$ shaded h -cubes, where $M \geq c\lambda h^{-1}$, and every shaded cube belongs to between μ and 2μ shadings. Dyadic pigeonholing loses only powers of $\log(2/h)$, which may be inserted into the right side of (14.4.265). Since an interval of length $c\lambda$ contains at most $C\lambda h^{-1}$ h -cubes, choosing c small shows that each tube supplies at least cM^2 ordered pairs (Q, Q') of its shaded cubes with $\text{dist}(Q, Q') \geq c\lambda$.

Choose $r_\eta > 0$ so small that $2r_\eta^\eta \leq 1/2$. At Q' , condition (14.4.264) leaves at least $\mu/2$ tubes T_0 with $\angle(T, T_0) \geq r_\eta$. Thus the number of quadruples

$$(T, Q, Q', T_0), \quad Q, Q' \subset Z(T), \quad Q' \subset Z(T_0), \\ \text{dist}(Q, Q') \geq c\lambda, \quad \angle(T, T_0) \geq r_\eta,$$

is at least $cNM^2\mu$. Averaging in T_0 selects a stem T_0 for which there are at least $cM^2\mu$ triples. A transverse tube T meets T_0 in only $O_\eta(1)$ h -cubes. Once such an intersection cube Q' exists, add to the brush every shaded cube of T at distance at least $c\lambda/2$ from Q' . Each selected brush tube then contributes at least cM cubes, and the number H of brush tubes satisfies

$$H \geq c_\eta M\mu \geq c_\eta \lambda h^{-1} \mu. \quad (14.4.266)$$

Every selected brush cube lies at distance at least $c_\eta \lambda$ from the axis of T_0 .

Sort the brush into leaves. A leaf is the Ch -neighbourhood of a plane through the axis of T_0 ; choose the planes with h -separated angular parameters and assign each brush tube to a nearest plane. At distance t from the stem, the angular interval of planes whose Ch -neighbourhood contains a point has length $O(h/t)$. Since the plane parameters are h -separated, the leaf overlap is $O(1+t^{-1})$. On the selected cubes $t \geq c_\eta \lambda$, so

$$\left| \bigcup_{T \in \mathcal{H}} Z(T) \right| \geq c_\eta \lambda \sum_j \left| \bigcup_{T \in \mathcal{H}_j} Z_j(T) \right|. \quad (14.4.267)$$

We next prove the planar L^2 estimate in one leaf. Fix T in that leaf. The intersection of two h -tubes making angle $\alpha \geq h$ has volume at most Ch^3/α . Tubes in the same leaf which meet T and make angle at most α with it are contained in one convex prism of dimensions $Ch \times C\alpha \times C$; the Katz–Tao definition bounds their number by $CK\alpha/h$. Summing the dyadic angular shells and adding the diagonal gives

$$\int \left(\sum_{T \in \mathcal{H}_j} \mathbf{1}_T \right)^2 \leq CK \log(2/h), h^2 \# \mathcal{H}_j.$$

Every brush tube has at least cM selected cubes. Cauchy–Schwarz therefore gives

$$\left| \bigcup_{T \in \mathcal{H}_j} Z_j(T) \right| \geq c \frac{(Mh^3 \# \mathcal{H}_j)^2}{K \log(2/h) h^2 \# \mathcal{H}_j} \geq c \frac{\lambda^2 h^2}{K \log(2/h)} \# \mathcal{H}_j.$$

Sum this estimate in (14.4.267) and use (14.4.266):

$$|U| := \left| \bigcup_{T \in \mathcal{A}} Z(T) \right| \geq c_\eta \frac{\lambda^4 h \mu}{K \log(2/h)}. \quad (14.4.268)$$

On the other hand, the multiplicity refinement gives

$$\mu |U| \geq c \sum_{T \in \mathcal{A}} |Z(T)| \geq c \lambda N h^2. \quad (14.4.269)$$

Multiplying (14.4.268) and (14.4.269) cancels μ and proves (14.4.265).

Step 3: return from the fibres and sum. For a fixed $S \in \mathbb{T}_\theta$, apply (14.4.265) with $h = \delta/\theta$, $\lambda \geq \delta^{C\eta}$, and $K \leq \delta^{-\eta}$. Absorb the logarithm into another δ^η . Undoing the affine map multiplies volume by θ^2 , and hence

$$\left| \bigcup_{T \in \mathbb{T}[S]} Y_S(T) \right| \geq c_\eta \delta^{C\eta} \theta^2 (\delta/\theta)^{3/2} (\#\mathbb{T}^S)^{1/2} = c_\eta \delta^{C\eta} \delta^{3/2} \theta^{1/2} (\#\mathbb{T}^S)^{1/2}.$$

The cover is balanced, so if $M_\theta = \#\mathbb{T}_\theta$ and $N = \#\mathbb{T}$, then $\sum_S (\#\mathbb{T}^S)^{1/2} \gtrsim (M_\theta N)^{1/2}$. Each S is contained in a slab of volume $C\theta$, and therefore

$$\#\mathbb{T}[S] \leq C\theta C_{F-SW}(\mathbb{T})N \leq C\delta^{-\eta}\theta N.$$

Balance implies $M_\theta \geq c\delta^\eta\theta^{-1}$. Using these facts in (14.4.263) cancels the factor $\theta^{1/2}$ and gives

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq c_\eta \delta^{3/2+C\eta} N^{1/2}.$$

Choose η so that $C\eta < \varepsilon$; decreasing the constant handles δ bounded away from zero. This is (14.4.261). $\square$

References for the Wang–Zahl Alternative Proof

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14.4.13 Certification ledger for the alternative route

The complete source argument above is now present, but source completeness and strict internal proof closure are different claims. The status in this ledger was checked on 31 August 2026. The following adapters are load-bearing and are audited individually:

1. the all-scale Frostman convex-Wolff theorem, matched to the complete proof in Chapter 14.5;
2. the elementary graph-pruning, regular-shading, and broadness-union lemmas, whose full proofs must appear at their points of use;
3. the grains base, including the tubular-volume and polynomial-partitioning inputs;
4. the two-broad Wolff hairbrush base, including the broadness refinement and the passage from the hairbrush estimate to the shaded volume bound;
5. the terminal relative-closedness argument for the sets of admissible (σ, ω) , with all constant and exponent dependences made uniform in the required variables.

The first, second, and fifth lines now have printed proofs or explicit adapters. The grains and hairbrush lines have been expanded substantially, but the following load-bearing obligations remain open:

WZ–A1. In Step 12 of Lemma 14.4.75, prove the parentwise normalized-slab estimate

$$m_U^{-1}(\rho/\tilde{\rho})\#\tilde{\mathbb{T}}'_X[U] \gtrsim_{\delta} \delta^{4\varepsilon_2}.$$

The source’s density and broadness statements do not presently imply this estimate; the exact reduction is (14.4.203)– (WZ–A1).

- WZ–A2. Replace the final planar Cordoba invocation in Lemma 14.4.100 by a proved shaded, pointwise-broad planar L^2 lemma with the stated parameter dependence.
- WZ–A3. Complete the local square-grain Cordoba calculation in Step 3 of Lemma 14.4.74, including the affine distortion and the claimed $\delta^{4\eta+4\varepsilon_1}$ density.
- WZ–A4. Replace the one-paragraph proof of Lemma 14.4.44 by the missing tangential-isolation lemma: construct the pointwise normal-cap/slab refinement, prove shaded disjointness and the mass bound, and establish the asserted $a^{-\varepsilon}$ normalized-volume gain at every iteration. The exact missing interface is (WZ–A4).

Until WZ–A1–A4 and the downstream exponent/refinement audit are discharged, this chapter is labelled *alternative proof under audit*. In particular, it does not promote the generalakeya theorem in the Volume 14 certificate.

Chapter 14.5

The Complete Convex–Wolff and Assouad Bridge

Provenance and role in the proof chain

This chapter gives the detailed multi-scale theorem that converts the convex Wolff axioms into a sticky, self-similar configuration at suitable separated scales. It is the exact input used in the general-to-sticky reduction in the preceding chapter, and it also proves the full Assouad-dimension consequence.

The proof below is adapted from Hong Wang and Joshua Zahl, *The Assouad dimension ofakeya sets in $\mathbb{R}^3$* , arXiv:2401.12337v2, licensed CC BY 4.0. The notation, labels, and bibliography have been isolated for use inside the book. Where the source refers back to the earlier sticky paper, this edition inserts an explicit deduction from the complete sticky proof printed in the preceding chapter; it also supplies the grain, C^2 -slope, large-slope, and twisted projection transfers and corrects the sign in the terminal projection-volume bound. These are changes by this edition and are not attributed to the original authors. The corresponding cinematic proof also appears here, so the dependency can be followed without leaving the series.

Abstract. This paper studies the structure ofakeya sets in $\mathbb{R}^3$. We show that for everyakeya set $K \subset \mathbb{R}^3$, there exist well-separated scales $0 < \delta < \rho \leq 1$ so that the δ neighborhood of K is almost as large as the ρ neighborhood of K . As a consequence, everyakeya set in $\mathbb{R}^3$ has Assouad dimension 3 and every Ahlfors-David regularakeya set in $\mathbb{R}^3$ has Hausdorff dimension 3. We also show that everyakeya set in $\mathbb{R}^3$ that has “stably equal” Hausdorff and packing dimension (this is a new notion, which is introduced to avoid certain obvious obstructions) must have Hausdorff dimension 3.

The above results follow from certain multi-scale structure theorems for arrangements of tubes and rectangular prisms in three dimensions, and

a generalization of the sticky Kakeya theorem previously proved by the authors.

14.5.1 Introduction

A Kakeya set is a compact subset of $\mathbb{R}^n$ that contains a unit line segment pointing in every direction. The Kakeya set conjecture asserts that every Kakeya set in $\mathbb{R}^n$ has Minkowski and Hausdorff dimension n . This conjecture is proved in the plane [1, 2], and is open in three and higher dimensions. See [9, 18] for a survey of progress on the Kakeya conjecture.

In this paper, we study the structure of Kakeya sets in $\mathbb{R}^3$. We show that for every Kakeya set $K \subset \mathbb{R}^3$, there exist well-separated scales $0 < \delta < \rho \leq 1$ so that the δ neighborhood of K is almost as large as the ρ neighborhood of K . The precise statement is given in Theorem 14.5.5 below. As a consequence, we prove several weaker variants of the Kakeya set conjecture: we prove that every Kakeya set in $\mathbb{R}^3$ has Assouad dimension 3, and we prove that every Kakeya set in $\mathbb{R}^3$ with “stably equal” Hausdorff and packing dimension (see Definition 14.5.3 below) must have Hausdorff and packing dimension 3. In order to explain these statements precisely, we require the following definitions.

Definition 14.5.1. Let $E \subset \mathbb{R}^n$ be non-empty. The *Assouad dimension* of E , denoted $\dim_A(E)$, is the infimum of all $\beta \geq 0$ for which there exist positive constants C and r_0 so that for all $0 < \rho < r \leq r_0$, we have

$$\sup_{x \in \mathbb{R}^n} \mathcal{E}_\rho(E \cap B(x, r)) \leq C(r/\rho)^\beta.$$

In the above, $\mathcal{E}_\rho(X)$ denotes the ρ -covering number of X .

The Assouad dimension is always at least as large as the Hausdorff and Minkowski dimensions, i.e. if $E \subset \mathbb{R}^n$ is bounded then we have $0 \leq \dim_H(E) \leq \underline{\dim}_M(E) \leq \overline{\dim}_M(E) \leq \dim_A(E) \leq n$. If E is bounded and Ahlfors-David regular, then all of these dimensions are equal. See [5] for further details on Assouad dimension and its properties.

Our first result is the following weak version of the Kakeya set conjecture in $\mathbb{R}^3$.

Theorem 14.5.2. *Every Kakeya set in $\mathbb{R}^3$ has Assouad dimension 3.*

Remarks

1. Theorem 14.5.2 holds for a slightly more general class of sets, where the lines satisfy a mild strengthening of the Wolff axioms. See the remarks in Section 14.5.8 for details.

2. Our proof of Theorem 14.5.2 establishes a slightly stronger (though more technical) result, which asserts that the analogue of Theorem 14.5.2 continues to hold if we restrict to pairs of scales $\rho < r$ in Definition 14.5.1 that are “exponentially separated,” in the sense that $\rho < r^{1+\varepsilon}$. This shows that every Kakeya set in $\mathbb{R}^3$ has quasi-Assouad dimension 3. (The Assouad dimension of a set is always bounded below by its quasi-Assouad dimension; see [5] for further details).

3. Fraser, Olson and Robinson [4] proved that every half-extended Kakeya set in $\mathbb{R}^n$ (i.e. a subset of $\mathbb{R}^n$ containing a half-infinite line segment in every direction) has Assouad dimension n . However this is a somewhat different question. In brief, Fraser, Olson and Robinson showed that the Assouad dimension does not increase under a “zooming out” rescaling, which transforms an extended Kakeya set into one where all lines pass through the origin (the latter type of set has full Assouad dimension).

Next, we consider Kakeya sets with equal Hausdorff and packing dimension. Recall that for $E \subset \mathbb{R}^n$, the packing dimension is given by

$$\dim_P(E) = \inf \{ \sup \overline{\dim}_M E_i \},$$

where $\overline{\dim}_M$ denotes the upper Minkowski dimension, and the infimum is taken over all decompositions $E = \bigcup_i E_i$ into countably many sets. We have $\dim_P(E) \geq \dim_H(E)$ for every set $E \subset \mathbb{R}^n$, and if $\dim_P(E) = \dim_H(E) = \alpha$, then for every $\varepsilon > 0$ there exists a constant C and a measure μ supported on E so that for μ -a.e. $x \in E$ we have the Frostman-type estimate

$$C^{-1}r^{\alpha+\varepsilon} \leq \mu(B(x, r)) \leq Cr^{\alpha-\varepsilon} \quad \text{for all } 0 < r \leq 1.$$

We would like to say that every Kakeya set in $\mathbb{R}^3$ with equal Hausdorff and packing dimension must have dimension 3. Unfortunately, we cannot prove this statement, and indeed this would imply that every Kakeya set in $\mathbb{R}^3$ has packing dimension 3: let $K \subset \mathbb{R}^3$ be a Kakeya set, and let $X \subset \mathbb{R}^3$ be a compact set with equal Hausdorff and packing dimension $\dim_H(X) = \dim_P(X) = \dim_P(K)$. Then $K' = K \cup X$ is a Kakeya set with equal Hausdorff and packing dimension, and this common value is $\dim_P(K)$. In the above example, the Kakeya set K' has equal Hausdorff and packing dimension for the somewhat trivial reason that it is a union of a Kakeya set with potentially unequal Hausdorff and packing dimension, and a set of equal Hausdorff and packing dimension¹. In particular, if $\dim_H K < \dim_H K' = \alpha$, then an α -dimensional Frostman measure supported on K' would be null on K . To exclude this type of situation, we introduce the following definition.

¹We could even select X to be a union of unit line segments pointing in different directions (specifically a $\dim X - 1$ dimensional set of directions), and modify K so that K' is a union of unit line segments, with one line segment pointing in each direction.

Definition 14.5.3. We say a set compact set $K \subset \mathbb{R}^n$ is a *Keakeya set with stably equal Hausdorff and packing dimension* if for all $\varepsilon > 0$, there is a constant C and a measure μ supported on K with the following properties:

- For all $x \in \text{supp}(\mu)$ and all $0 < r \leq 1$, we have

$$C^{-1}r^{\alpha+\varepsilon} \leq \mu(B(x, r)) \leq Cr^{\alpha-\varepsilon},$$

where $\alpha = \dim_H(K)$.

- There is a positive (two-dimensional Lebesgue) measure set of directions $\Omega \subset S^2$, so that for each direction $e \in \Omega$, there is a line ℓ_e pointing in direction e for which $\ell_e \cap \text{supp}(\mu)$ has positive one-dimensional Lebesgue measure.

In particular, if K is a Keakeya set with unequal Hausdorff and packing dimension, then the set $K' = K \cup X$ described above will not have stably equal Hausdorff and packing dimension².

Theorem 14.5.4. *Every Keakeya set in $\mathbb{R}^3$ with stably equal Hausdorff and packing dimension has Hausdorff and packing dimension 3.*

Theorems 14.5.2 and 14.5.4 will follow from a common discretized Keakeya estimate, which is the main technical contribution of this paper. In the statement below, $N_\rho(X)$ denotes the ρ -neighborhood of the set X ; a δ -tube is the δ -neighborhood of a unit line segment; and we say two δ -tubes are essentially distinct if neither tube is contained in the 2-fold dilate of the other. See Section 14.5.2 for precise definitions.

Theorem 14.5.5. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\mathbb{T}$ be a set of essentially distinct δ -tubes in $B(0, 1) \subset \mathbb{R}^3$, and suppose that each convex set of volume V contains at most $\delta^{-\eta}V(\#\mathbb{T})$ tubes from $\mathbb{T}$. For each $T \in \mathbb{T}$, let $Y(T) \subset T$ be a measurable set, and suppose that $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^\eta \sum_{T \in \mathbb{T}} |T|$.*

Then there exists $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\eta r$ and a ball B of radius r so that

$$\left| B \cap N_\rho \left(\bigcup_{T \in \mathbb{T}} Y(T) \right) \right| \geq (\rho/r)^\varepsilon |B|. \quad (14.5.1)$$

Remark 14.5.6. In the above theorem, we have required that the tubes be essentially distinct and that every convex set $U \subset \mathbb{R}^3$ contain at most $\delta^{-\eta}|U|(\#\mathbb{T})$ tubes from $\mathbb{T}$. This condition is satisfied, for example, if $\#\mathbb{T} = \delta^{-2}$ and the tubes in $\mathbb{T}$ point in δ -separated directions. Indeed; for the latter condition we need only consider convex sets $U \subset B(0, 1) \subset \mathbb{R}^3$ of diameter roughly

²Unless the set X is itself a Keakeya set with stably equal Hausdorff and packing dimension.

1, and every set of this form is comparable to an ellipsoid whose axes have lengths $s \leq t \leq w$ with $w \sim 1$. If T is a δ -tube contained in U , then the direction of T lies in a rectangular sector of S^2 of dimensions roughly $s \times t$; if the tubes in $\mathbb{T}$ point in δ -separated directions, we conclude that $\#\{T \in \mathbb{T} : T \subset U\} \lesssim st\delta^{-2} \lesssim |U|(\#\mathbb{T})$. The anti-clustering condition imposed in Theorem 14.5.5 will be discussed further in Section 14.5.2.2.

Theorem 14.5.5 can be amplified to obtain the following variant, which says that if $\mathbb{T}$ is a collection of δ -tubes satisfying the anti-clustering condition described above, then there are two well-separated scales $\rho < r$ so that the ρ -neighborhood of $\bigcup_{\mathbb{T}} T$ is almost as large as its r -neighborhood.

Corollary 14.5.7. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\mathbb{T}$ be a set of essentially distinct δ -tubes in $B(0, 1) \subset \mathbb{R}^3$, and suppose that each convex set of volume V contains at most $\delta^{-\eta} V(\#\mathbb{T})$ tubes from $\mathbb{T}$. For each $T \in \mathbb{T}$, let $Y(T) \subset T$ be a measurable set, and suppose that $\sum_{\mathbb{T}} |Y(T)| \geq \delta^\eta \sum_{\mathbb{T}} |T|$.*

Then there exist sets $Y'(T) \subset Y(T)$, $T \in \mathbb{T}$, with $\sum_{\mathbb{T}} |Y'(T)| \gtrsim |\log \delta|^{-2} \sum_{\mathbb{T}} |Y(T)|$, and there exist scales $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\eta r$ so that

$$\left| N_\rho \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right| \geq (\rho/r)^\varepsilon \left| N_r \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right|. \quad (14.5.2)$$

14.5.1.1 Main ideas, and a sketch of the proof

In [16], the authors proved that every sticky Kakeya set in $\mathbb{R}^3$ has Hausdorff and Minkowski dimension 3. In brief, a *Sticky Kakeya Set* in $\mathbb{R}^n$ is a Kakeya set that, when discretized at scale δ , gives rise to a collection of δ -tubes with the following two properties:

- (a) The tubes point in δ -separated directions.
- (b) For each $\delta \leq \rho \leq 1$, the tubes can be covered by a set of ρ -tubes, with few (i.e. about 1) ρ -tubes pointing in each direction.

Sticky Kakeya sets have two useful properties. First, for $\delta \leq \rho \leq 1$, the ρ -neighborhood of a sticky Kakeya set at scale δ is a sticky Kakeya set at scale ρ . In other words, if we thicken a sticky Kakeya set, we obtain a sticky Kakeya set at a coarser scale. Second, for $\delta \leq \rho \leq 1$, the ρ -neighborhood of a typical tube in our sticky Kakeya set will contain about $(\rho/\delta)^{1-n}$ δ -tubes; if we rescale this ρ -tube to be comparable to the unit ball, then we obtain a sticky Kakeya set at scale δ/ρ . In other words, if we zoom-in and rescale a sticky Kakeya set, we obtain a sticky Kakeya set at a coarser scale.

The Sticky Kakeya Theorem (or more precisely, a generalization that we will prove in Section 14.5.6) will play an important role in the proof of

Theorem 14.5.5. In Section 14.5.4 we show that every set of δ -tubes satisfying the hypotheses of Theorem 14.5.5 must either satisfy a suitable analogue of stickiness (and hence we can apply our generalization of the Stickyakeya Theorem and obtain the conclusion of Theorem 14.5.5), or else at least one of the following three things must happen:

- (i) After thickening the tubes at a suitable scale, we obtain a richer collection of tubes, i.e. a collection of tubes that has similar structural properties, but larger cardinality (relative to the new scale).
- (ii) After zooming-in and rescaling at a suitably chosen location and scale, we obtain a richer collection of tubes.
- (iii) After possibly coarsening and zooming-in, the tubes arrange themselves into “planks” (like planks of wood, these sets are long (i.e. length 1) in one direction, much shorter in a second direction, and shorter still in the final direction).

In Section 14.5.4 we show that if Item (iii) occurs, then the corresponding collection of tubes satisfies the conclusion of Theorem 14.5.5. This observation is the main new geometric input in the proof of Theorem 14.5.5. In brief, the argument is as follows. Suppose each plank has dimensions roughly $s \times t \times 1$, with $\delta \leq s \ll t \leq 1$. This means that each plank has an associated plane, which is the span of the two vectors pointing in the longest and second-longest directions. First we consider the case in which a typical pair of intersecting planks have corresponding planes that make angle $\theta \gg s/t$. If this occurs, then the union of the s -neighborhood of these planks will contain a ball of radius $r = \theta t$. This is much larger than s , and hence (14.5.1) holds. Second, we consider the case in which a typical pair of intersecting planks have corresponding planes that make angle $\leq s/t$. This forces the planks to cluster into bigger planks dimensions $s' \times t' \times 1$, where $s' \ll t' \leq 1$ are substantially larger than s and t , respectively. We can then repeat this argument at a larger scale. After finitely many iterations of this argument, we must find ourselves in the First Case. This concludes the argument.

Summarizing the arguments above, if $\mathbb{T}$ is a counter-example to Theorem 14.5.5, then there exists either a thickening or a zooming-in and rescaling that produces a richer collection of tubes. However, Theorem 14.5.5 is agnostic to location and scale, and this means that if $\mathbb{T}$ is a counter-example to Theorem 14.5.5, then any thickening or zooming-in and rescaling of $\mathbb{T}$ remains a counter-example. Thus to prove Theorem 14.5.5, we show that if a counter-example to Theorem 14.5.5 exists, then there must exist a (nearly) maximally rich counter-example. But this is impossible, since some thickening or zooming-in and rescaling of this (supposedly) maximally rich counter-example is an even richer counter-example.

In Section 14.5.2.2 we precisely define the relevant generalization of stickiness, and in Section 14.5.3 we define what it means for $\mathbb{T}$ to be a maximally rich counter-example to Theorem 14.5.5. In Section 14.5.4 we carry out the bulk of the geometric arguments described above. In Section 14.5.5 we show that a (hypothetical) counterexample to Theorem 14.5.5 must satisfy the hypotheses of (a generalization of) the stickyakey Theorem. In Sections 14.5.6 and 14.5.7 we prove the necessary generalization of the Stickyakey Theorem (this is Theorem 14.5.38), thereby proving Theorem 14.5.5. Finally, in Section 14.5.8 we show how Theorem 14.5.5 implies Theorems 14.5.2 and 14.5.4.

14.5.1.2 Thanks

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14.5.2 Notation and definitions

We write $A \lesssim B$ or $A = O(B)$ to mean there is a constant C (which might depend on the ambient dimension n) so that $A \leq CB$. We write $A \sim B$ if $A \lesssim B$ and $B \lesssim A$. If the constant C is allowed to depend on an additional parameter such as ε , then we denote this by $A \lesssim_\varepsilon B$. Many of our results will involve a small positive parameter, which we will call δ . We write $A \lesssim_\delta B$ to mean that there is a constant C so that $A \leq C(\log 1/\delta)^C B$. As above, if the constant C is allowed to depend on an additional parameter such as ε , then we denote this by $A \lesssim_{\varepsilon,\delta} B$. Finally, during proof sketches and other informal remarks, we will sometimes write $A \ll B$ to suggest to the reader that the real number A is much smaller than the real number B .

14.5.2.1 Tubes and their shadings

Recall that a δ -tube in $\mathbb{R}^n$ is the δ -neighborhood of a unit line segment. We will use T to denote a δ -tube, and $\ell(T)$ to denote its coaxial line. If T is a δ -tube (or more generally, the translation of a centrally symmetric convex set) and $R > 0$, we define RT to be the R -fold dilate of T around its center. We say two tubes T, T' are *essentially distinct* if $T \not\subset 2T'$ and $T' \not\subset 2T$.

We will sometimes be concerned with collections of tubes, plus shadings (i.e. distinguished subsets) of these tubes. To this end, we introduce the

following definitions.

Definition 14.5.8. Let $\mathcal{S}$ be a collection of subsets of $\mathbb{R}^n$, and let $\tau \in (0, 1]$. A collection of sets $\{Y(S) \subset S : S \in \mathcal{S}\}$ is called a τ -dense shading of $\mathcal{S}$ if $\sum_{S \in \mathcal{S}} |Y(S)| \geq \tau \sum_{S \in \mathcal{S}} |S|$. $\{Y(S), S \in \mathcal{S}\}$ is called a *uniformly* τ -dense shading of $\mathcal{S}$ if $|Y(S)| \geq \tau |S|$ for each $S \in \mathcal{S}$.

In practice, $\mathcal{S}$ will either be a collection of δ -tubes, or a collection of $s \times t \times 1$ rectangular prisms. We will use the notation $(\mathbb{T}, Y)_\delta$ to refer to a collection of δ -tubes and an associated shading $\{Y(T) : T \in \mathbb{T}\}$.

Definition 14.5.9. Let $S \subset \mathbb{R}^n$, let $Y(S) \subset S$, and let $\delta > 0$. We say the shading $Y(S)$ is *regular* at scales $\geq \delta$ if for each $x \in Y(S)$ and each $r \in [\delta, 1]$, we have

$$|Y(S) \cap B(x, r)| \geq (100^n \log(1/\delta))^{-1} |Y(S)| \left(\frac{|B(x, r) \cap S|}{|S|} \right). \quad (14.5.3)$$

If the quantity δ is apparent from context, then we will omit it and say that $Y(S)$ is regular.

The following is Lemma 2.7 from [10].

Lemma 14.5.10. *Let $\delta > 0$, let $S \subset B(0, 1) \subset \mathbb{R}^n$, and let $Y(S)$ be a shading. Then there is a shading $Y'(S) \subset Y(S)$ that is regular at scales $\geq \delta$ and satisfies $|Y'(S)| \geq \frac{1}{2} |Y(S)|$.*

The proof in [10] considers the special case where S is a unit line segment (and hence $|B(x, r) \cap S|/|S| \sim r$), but the proof is identical. In brief, we consider the quantity $|Y(S) \cap B(x, r)|$ for each dyadic scale $r = 2^k \delta$ and each ball $B(x, r)$ aligned to the dyadic grid. We delete those balls for which $|Y(S) \cap B(x, r)|$ is too small (i.e. for which (14.5.3) fails), and we denote the surviving subset of $Y(S)$ by $Y'(S)$. At each dyadic scale we have only deleted a small fraction of $Y(S)$, so at least half of $Y(S)$ survives this process.

14.5.2.2 Wolff axioms and their generalizations

In this section we will define several anti-clustering conditions that we can impose on our collection of tubes. In [17] (see also Definition 13.1 from [7]), Wolff introduced what are now called the Wolff axioms; this is a non-concentration condition that forbids a collection of δ -tubes from clustering inside a fatter ρ -tube, and also forbids a collection of δ -tubes from clustering near (affine) 2-planes. The precise definition is as follows.

Definition 14.5.11. Let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^n$. We say $\mathbb{T}$ satisfies the *Wolff axioms* if:

- Every ρ -tube in $\mathbb{R}^n$ can contain at most $C(\rho/\delta)^{n-1}$ tubes from $\mathbb{T}$.
- Every rectangular prism in $\mathbb{R}^n$ of dimensions $2\delta \times 2\delta \times \cdots \times 2\delta \times \rho \times 2$ can contain at most $C(\rho/\delta)$ tubes from $\mathbb{T}$.

Remarks

1. In the above definition, the constant C is generally chosen to be a large number depending only on the ambient dimension n .
2. Every set of δ -tubes pointing in δ -separated directions will satisfy the Wolff axioms with constant $C = 100^n$.
3. Every set of δ -tubes contained in $B(0, 1) \subset \mathbb{R}^n$ that satisfies the Wolff axioms has cardinality at most $100^n C \delta^{1-n}$.

In the arguments that follow, we will need several variants of the Wolff axioms. We define these below.

Definition 14.5.12. Let $C \geq 1$. We say a multi-set $\mathcal{S}$ of sets in $\mathbb{R}^n$ satisfies the *Convex Wolff Axioms with error C* if for all convex sets $W \subset \mathbb{R}^n$, we have

$$\#\{S \in \mathcal{S} : S \subset W\} \leq C|W|(\#\mathcal{S}). \quad (14.5.4)$$

Informally, when C has size close to 1, we will sometimes suppress the role of C and say that $\mathcal{S}$ satisfies the Convex Wolff Axioms. In practice we will have $n = 3$, and the sets in $\mathcal{S}$ will be either δ -tubes, or $s \times t \times 1$ rectangular prisms (not necessarily axis-parallel) for some $\delta \leq s \leq t \leq 1$.

Remark 14.5.13. If the elements of $\mathcal{S}$ are convex, then $\#\mathcal{S} \geq C^{-1}(\inf_{S \in \mathcal{S}} |S|)^{-1}$. In particular, a set of δ -tubes in $\mathbb{R}^3$ satisfying the Convex Wolff Axioms must have cardinality $\gtrsim \delta^{-2}$.

Remark 14.5.14. Let $\mathcal{S}$ be a multi-set of subsets of $\mathbb{R}^n$. For each $S \in \mathcal{S}$, let $T(S) \supset S$, and define the multi-set $\mathcal{S}' = \{T(S) : S \in \mathcal{S}\}$, i.e. the multi-sets $\mathcal{S}$ and $\mathcal{S}'$ have the same cardinality. If $\mathcal{S}$ satisfies the Convex Wolff Axioms with error C , then $\mathcal{S}'$ also satisfies the Convex Wolff Axioms with error C .

In our arguments below, we will often consider a *Takeya set* (or more accurately a collection of δ -tubes) at many different scales. Typically, for a collection of δ -tubes $\mathbb{T}$, we will want to examine the coarsening of $\mathbb{T}$ at a larger scale ρ , and also examine the collection of δ -tubes from $\mathbb{T}$ that are contained inside each of these coarser ρ -tubes. As observed in Remark 14.5.14 above, if $\mathbb{T}$ satisfies the Convex Wolff Axioms, then the coarsening of $\mathbb{T}$ at a larger scale will also satisfy these axioms³. However, the (rescaled) collection of δ -tubes from $\mathbb{T}$ contained inside each coarser ρ -tube might not satisfy the Convex Wolff Axioms. Our next task is to introduce a slightly stricter variant of the Convex Wolff Axioms that addresses this issue.

³In contrast, the more traditional condition that the tubes in $\mathbb{T}$ point in δ -separated directions is *not* preserved after coarsening.

Definition 14.5.15. Let U be a convex set and let $S \subset U$. Let $U^* \supset U$ be the John ellipsoid that circumscribes U , and let ϕ_U be an affine transformation that sends U^* to the unit ball. The *unit rescaling of S relative to U* is the set $\phi_U(S)$. If $\mathcal{S}$ be a multi-set of subsets of $\mathbb{R}^n$ that are contained in U , then the *unit rescaling of $\mathcal{S}$ relative to U* is the multi-set $\{\phi_U(S) : S \in \mathcal{S}\}$.

Remark 14.5.16. In [16], the authors considered a similar definition in the special case where $\mathcal{S}$ is a set of δ -tubes and U is a ρ -tube. In order for this definition to be compatible with the definition from [16] of a tube’s shading, the map ϕ_U was defined slightly differently, so that image of U^* was the ellipsoid $\{Cx^2 + Cy^2 + z^2 \leq 1\}$ (for some fixed constant C), rather than the unit ball.

Definition 14.5.17. Let $\mathcal{A}$ be a set and $\mathcal{B}$ be a multi-set of convex subsets of $\mathbb{R}^n$. We say that $\mathcal{A}$ is a *cover of $\mathcal{B}$* if each set $B \in \mathcal{B}$ is contained in at least one set from $\mathcal{A}$. For each $A \in \mathcal{A}$, we write $\mathcal{B}[A]$ to denote the multi-set $\{B \in \mathcal{B} : B \subset A\}$. We say that $\mathcal{A}$ is a *K -uniform cover of $\mathcal{B}$* if $\#\mathcal{B}[A] \leq K(\#\mathcal{B}[A'])$ for each pair of sets $A, A' \in \mathcal{A}$. When $K = 1$, we will abbreviate this to a *uniform cover*. Finally, we say that $\mathcal{A}$ is a *partitioning cover of $\mathcal{B}$* if $\mathcal{A}$ is a cover of $\mathcal{B}$, and for every pair of distinct $A, A' \in \mathcal{A}$, the sets $\mathcal{B}[2A]$ and $\mathcal{B}[2A']$ are disjoint, where $2A$ denotes the 2-fold dilate of A . In practice, $\mathcal{A}$ and $\mathcal{B}$ will be collections of ρ -tubes and δ -tubes respectively, for some $\rho > \delta$.

Remark 14.5.18. Note that if $\mathcal{B}$ satisfies the Convex Wolff Axioms with error C , and if $\mathcal{A}$ is a K -uniform partitioning cover of $\mathcal{B}$, then $\mathcal{A}$ satisfies the Convex Wolff Axioms with error KC .

Definition 14.5.19. Let $C \geq 1, \delta > 0$. We say that a set $\mathbb{T}$ of δ -tubes in $\mathbb{R}^n$ satisfies the *Convex Wolff Axioms at every scale with error C* if the tubes in $\mathbb{T}$ are essentially distinct, and for every $\rho_0 \in [\delta, 1]$, there exists $\rho \in [\rho_0, C\rho_0)$ and a set of ρ -tubes $\mathbb{T}_\rho$ that satisfies the following properties:

- (i) $\mathbb{T}_\rho$ is a C -uniform partitioning cover of $\mathbb{T}$.
- (ii) For each $T_\rho \in \mathbb{T}_\rho$, the unit rescaling of $\mathbb{T}[T_\rho]$ relative to T_ρ satisfies the Convex Wolff Axioms with error C .

As the name suggests, Definition 14.5.19 is a multi-scale property. In particular, if $\mathbb{T}$ satisfies the Convex Wolff Axioms at every scale, then we can cover $\mathbb{T}$ by fatter ρ -tubes; both the ρ -tubes and the (rescaled) tubes from $\mathbb{T}$ inside each ρ -tube will again satisfy the Convex Wolff Axioms at every scale. The precise statement is as follows.

Lemma 14.5.20. *Let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^n$ that satisfies the Convex Wolff Axioms at every scale with error C . Let $\rho_0 \in [\delta, 1]$. Then there exists*

$\rho \in [\rho_0, C\rho_0]$ and a set of ρ -tubes $\mathbb{T}_\rho$, so that $\mathbb{T}_\rho$ satisfies the *Convex Wolff Axioms* at every scale with error $O(C)$, and for each $T_\rho \in \mathbb{T}_\rho$, the unit rescaling of $\mathbb{T}[T_\rho]$ relative to T_ρ satisfies the *Convex Wolff Axioms* at every scale with error $O(C)$.

Proof. Let N be the largest integer with $C^{N+1} \leq \delta^{-1}$. For $i = 1, \dots, N$, let $\mathbb{T}_{\tau_i}$ be a set of τ_i -tubes, with $\tau_i \in [\delta C^i, \delta C^{i+1})$ that satisfies Items (i) and (ii) from Definition 14.5.19. We claim that for $i < j$, if $T_{\tau_i} \in \mathbb{T}_{\tau_i}$, $T_{\tau_j} \in \mathbb{T}_{\tau_j}$, and $\mathbb{T}[T_{\tau_i}] \cap \mathbb{T}[T_{\tau_j}] \neq \emptyset$, then $\mathbb{T}[T_{\tau_i}] \subset \mathbb{T}[T_{\tau_j}]$. To see this, observe that if the intersection contains at least one tube $T \in \mathbb{T}$, then $2T_{\tau_j} \supset N_{\tau_j}(T) \supset T_{\tau_i}$. Since $\mathbb{T}_{\tau_j}$ covers $\mathbb{T}$, and the sets $\{\mathbb{T}[2T'_{\tau_j}]: T'_{\tau_j} \in \mathbb{T}_{\tau_j}\}$ are disjoint, we conclude that $\mathbb{T}[T_{\tau_i}] \subset \mathbb{T}[T_{\tau_j}]$.

The consequence of the above observation is the following: if we partially order the sets $\{\mathbb{T}[T_{\tau_i}]: T_{\tau_i} \in \mathbb{T}_{\tau_i}, i = 1, \dots, N\}$ under inclusion, then this forms a tree with N levels—the vertices at level i are precisely the sets $\mathbb{T}[T_{\tau_i}]$. By Item (i) of Definition 14.5.19, for each $i = 1, \dots, N$ the sets $\{\mathbb{T}[T_{\tau_i}]: T_{\tau_i} \in \mathbb{T}_{\tau_i}\}$ have comparable cardinality, up to a multiplicative factor of C . This tree is precisely what is needed to verify Lemma 14.5.20: given $\rho_0 \in [\delta, 1]$, we select an index i so that $\tau_i \in [\rho_0, C\rho_0)$, and we choose $\rho = \tau_i$, $\mathbb{T}_\rho = \mathbb{T}_{\tau_i}$; note that since $\mathbb{T}_\rho$ forms a partitioning cover of $\mathbb{T}$, the ρ -tubes in $\mathbb{T}_\rho$ are essentially distinct. $\square$

Finally, we will introduce two additional anti-concentration conditions that will play a technical role in the arguments to follow.

Definition 14.5.21. Let $C \geq 1$ and let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^n$. We say that $\mathbb{T}$ satisfies a *Frostman condition at dimension σ with error C* if the tubes in $\mathbb{T}$ are essentially distinct, and for every $\rho \geq \delta$ and every ρ -tube T_ρ , we have $\#\mathbb{T}[T_\rho] \leq C\rho^\sigma(\#\mathbb{T})$.

Definition 14.5.22. Let $C \geq 1$ and let $\mathbb{T}$ be a set of δ -tubes in $\mathbb{R}^n$. We say that $\mathbb{T}$ satisfies the *self-similar Convex Wolff Axioms with error C* if the tubes in $\mathbb{T}$ are essentially distinct, and for every $\rho_0 \in [\delta, 1]$, there exists $\rho \in [\rho_0, C\rho_0)$ and a set of ρ -tubes $\mathbb{T}_\rho$ that satisfies the following properties:

- (i) $\mathbb{T}_\rho$ is a C -uniform partitioning cover of $\mathbb{T}$.
- (ii) For each $T_\rho \in \mathbb{T}_\rho$, the unit rescaling of $\mathbb{T}[T_\rho]$ relative to T_ρ satisfies the *Convex Wolff Axioms* with error C .
- (iii) For each $T_\rho \in \mathbb{T}_\rho$, we have

$$C^{-1}(\rho/\delta)^\sigma \leq \#\mathbb{T}[T_\rho] \leq C(\rho/\delta)^\sigma, \quad (14.5.5)$$

where $\sigma > 0$ is the unique number satisfying $\#\mathbb{T} = \delta^{-\sigma}$.

Note that Items (i) and (ii) above are identical to their counterparts in Definition 14.5.19. In particular, we have

$$\begin{aligned} \text{Self-similar CWA with error } C &\implies \text{CWA at every scale with error } C \\ &\implies \text{CWA with error } C. \end{aligned} \tag{14.5.6}$$

14.5.2.3 Discretized Assouad dimension

The conclusion of Theorem 14.5.5 involves a discretized analogue of Assouad dimension. We formalize this as follows.

Definition 14.5.23. Let $E \subset \mathbb{R}^n$, let $\beta \in [0, n]$, let $\delta > 0$, and let $A \geq 1$. We say that E has *discretized Assouad dimension at least β , at scale δ and scale separation A* if there exist scales $\delta \leq \rho \leq r \leq 1$ with $r \geq A\rho$, and a ball B of radius r , so that

$$|B \cap N_\rho(E)| \geq (\rho/r)^{n-\beta} |B|.$$

If the scale δ is apparent from context, then for brevity we may say “ E has discretized Assouad dimension at least β at scale separation A .”

In this language, the conclusion of Theorem 14.5.5 says that the set $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale separation $\delta^{-\eta}$.

14.5.3 Assouad-extremalakey sets

In this section we will suppose that Theorem 14.5.5 is false, and we seek to construct a “worst possible” counter-example to Theorem 14.5.5. In this and later sections, we will restrict attention to $\mathbb{R}^3$. Our example will be worst possible in two respects. First, Inequality (14.5.1) will fail as dramatically as possible, i.e. a reverse inequality will hold where the term $(\rho/r)^\varepsilon$ on the RHS of (14.5.1) will be replaced by $(\rho/r)^\omega$ for the largest possible value of $\omega > 0$. Second, of all collections of tubes for which such a reverse inequality holds for this value of ω , we will choose a set $\mathbb{T}$ of largest possible cardinality. The hypotheses of Theorem 14.5.5 ensure that $\mathbb{T}$ has size at least δ^{-2} ; we will choose a set of tubes of size $\delta^{-\alpha}$, for the largest possible value of α .

We now turn to the task of carefully defining the quantities ω and α ; this will be done in Sections 14.5.3.1 and 14.5.3.2, respectively. This task is complicated by the fact that Theorem 14.5.5 involves a sequence of quantifiers, and we must unwind these quantifiers in the correct order.

14.5.3.1 Defining ω

Let $(\mathbb{T}, Y)_\delta$ be a set of tubes and their associated shading. Let $\delta \leq \rho < r \leq 1$, and let B be a ball of radius r whose intersection with $\bigcup_{T \in \mathbb{T}} Y(T)$ has positive measure. Define $\zeta = \zeta(\mathbb{T}, Y, B, \rho)$ to be the unique number satisfying

$$\left| B \cap N_\rho \left(\bigcup_{T \in \mathbb{T}} Y(T) \right) \right| = \left(\frac{\rho}{r} \right)^\zeta |B|. \quad (14.5.7)$$

Next, for $\eta > 0$, define

$$\zeta(\mathbb{T}, Y; \eta) = \inf_{B, \rho} \zeta(\mathbb{T}, Y, B, \rho), \quad (14.5.8)$$

where the infimum is taken over all pairs (B, ρ) , where $\rho \in [\delta, \delta^\eta]$ and B is a ball of radius $r \in [\delta^{-\eta}\rho, 1]$ whose intersection with $\bigcup_{T \in \mathbb{T}} Y(T)$ has positive measure. Observe that as $\eta \searrow 0$, the set of admissible pairs (B, ρ) becomes larger, and hence $\zeta(\mathbb{T}, Y; \eta)$ weakly decreases, i.e. $\zeta(\mathbb{T}, Y; \eta)$ is a weakly increasing function of η .

For $s, \eta, \delta > 0$ and $u \geq 2$, define

$$\omega(\varepsilon, \eta, \delta; u) = \sup_{(\mathbb{T}, Y)_\delta} \zeta(\mathbb{T}, Y; \eta), \quad (14.5.9)$$

where the supremum is taken over all choices of $(\mathbb{T}, Y)_\delta$ with the following properties:

- $\mathbb{T}$ is a set of essentially distinct δ -tubes contained in $B(0, 1)$ that satisfies the Convex Wolff Axioms with error $\delta^{-\varepsilon}$, and Y is a δ^ε -dense shading.
- $\#\mathbb{T} \geq \delta^{-u+\varepsilon}$.

Again, $\omega(\varepsilon, \eta, \delta; u)$ is a weakly increasing function of η . It is also a weakly increasing function of ε and a weakly decreasing function of u , since as ε becomes larger or u becomes smaller, the set of admissible pairs $(\mathbb{T}, Y)_\delta$ becomes larger.

Define

$$\omega(u) = \lim_{\varepsilon \rightarrow 0^+} \lim_{\eta \rightarrow 0^+} \limsup_{\delta \rightarrow 0^+} \omega(\varepsilon, \eta, \delta; u) = \inf_{\varepsilon, \eta > 0} \limsup_{\delta \rightarrow 0^+} \omega(\varepsilon, \eta, \delta; u).$$

Definition 14.5.24. Define $\omega = \omega(2)$.

Unwrapping the above definitions, we have the following.

Lemma 14.5.25. *Let $\omega \geq 0$ be as in Definition 14.5.24. For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of essentially distinct tubes that satisfy the Convex Wolff Axioms with error $\delta^{-\eta}$, with Y a δ^η -dense shading.*

Then $\bigcup_{T \in \mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \omega - \varepsilon$ at scale separation $\delta^{-\eta}$.

Proof. Select $\eta > 0$ sufficiently small so that $\limsup_{\delta \rightarrow 0^+} \omega(\eta, \eta, \delta; 2) \leq \omega + \varepsilon/2$. Then there exists $\delta_0 > 0$ so that $\omega(\eta, \eta, \delta; 2) \leq \omega + \varepsilon$ for all $\delta \in (0, \delta_0]$. The conclusion of Lemma 14.5.25 now holds for this choice of η and δ_0 . $\square$

In light of Lemma 14.5.25, in order to prove Theorem 14.5.5 it suffices to prove that $\omega = 0$.

14.5.3.2 Defining α

Note that $\omega(u)$ is weakly monotone decreasing. We also have that $\omega(4) = 0$, since if $\mathbb{T}$ is a set of δ^{-4} essentially distinct δ -tubes in $B(0, 1)$, then $\bigcup_{\mathbb{T}} T$ must have volume $\gtrsim 1$. Define

$$\alpha = \sup\{u \in [2, 4] : \omega(u) = \omega(2)\}. \quad (14.5.10)$$

The quantity α has the following interpretation. We have defined $3 - \omega$ to be the smallest possible discretized Assouad dimension of a collection of essentially distinct tubes that satisfy the Convex Wolff Axioms. Every such collection of tubes must have cardinality at least δ^{-2} , and in general we might expect that larger collections of tubes should have larger discretized Assouad dimension. Any collection of essentially distinct tubes of cardinality substantially larger than $\delta^{-\alpha}$ that satisfies the Convex Wolff Axioms must have discretized Assouad dimension larger than $3 - \omega$.

14.5.3.3 The existence of Assouad-extremalakeya sets

Now that we have defined ω and α , we are ready to construct a (hypothetical) “worst possible” counter-example to Theorem 14.5.5, in the sense described at the beginning of Section 14.5.3.

Definition 14.5.26. For $\varepsilon > 0$, we say a pair $(\mathbb{T}, Y)_\delta$ is ε Assouad-extremal if it satisfies the following properties:

- (i) $\#\mathbb{T} \geq \delta^{-\alpha+\varepsilon}$.
- (ii) The tubes are contained in $B(0, 1)$, essentially distinct, and satisfy the Convex Wolff Axioms with error $\delta^{-\varepsilon}$.
- (iii) The shading Y is δ^ε -dense.
- (iv) For all $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\varepsilon r$ and all balls B of radius r , we have

$$\left| B \cap N_\rho \left(\bigcup_{T \in \mathbb{T}} Y(T) \right) \right| \leq (\rho/r)^{\omega-\varepsilon} |B|. \quad (14.5.11)$$

Note that if $\mathbb{T}$ is ε Assouad-extremal, then it is also ε' Assouad-extremal for all $\varepsilon' \geq \varepsilon$.

Lemma 14.5.27. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε Assouad-extremal set of tubes $(\mathbb{T}, Y)_\delta$.*

Proof. Fix $u \geq \max(2, \alpha - \varepsilon/2)$ with $\omega(u) = \omega(2) = \omega$. Since $\omega(s, \eta, \delta; u)$ is a weakly increasing function of s and η , we have $\limsup_{\delta \rightarrow 0^+} \omega(\varepsilon/2, \varepsilon, \delta; u) \geq \omega(u)$. Thus there exists $\delta \in (0, \delta_0]$ with $\omega(\varepsilon/2, \varepsilon, \delta; u) \geq \omega - \varepsilon/2$.

In light of (14.5.9), there exists a set of tubes and their associated shading $(\mathbb{T}, Y)_\delta$, so that the following holds:

- (a) $\#\mathbb{T} \geq \delta^{-u+\varepsilon/2} \geq \delta^{-\alpha+\varepsilon}$ (cf. Definition 14.5.26, Item (i)).
- (b) The tubes are contained in $B(0, 1)$, essentially distinct, and satisfy the Convex Wolff Axioms with error $\delta^{-\varepsilon}$ (cf. Definition 14.5.26, Item (ii)).
- (c) Y is a δ^ε -dense shading (cf. Definition 14.5.26, Item (iii)).
- (d) We have $\zeta(\mathbb{T}, Y; \eta) \geq \omega - \varepsilon$.

Thus by (14.5.8), for every pair of scales $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\varepsilon r$ and every ball B of radius r , we have

$$\zeta(\mathbb{T}, Y, B, \rho) \geq \zeta((\mathbb{T}, Y); \eta) \geq \omega - \varepsilon.$$

But this is precisely (14.5.11), and thus $(\mathbb{T}, Y)_\delta$ satisfies Definition 14.5.26, Item (iv). $\square$

14.5.4 Convex Wolff Axioms and multi-scale structure

In this section we will carefully execute the arguments described in Section 14.5.1.1. The main result is Proposition 14.5.28, which says that under suitable hypotheses, for every collection of δ -tubes $\mathbb{T}$, at least one of the following must occur:

- (A) $\mathbb{T}$ has discretized Assouad dimension close to 3 (as we will see later in this section, this is forced to happen if the tubes in $\mathbb{T}$ arrange themselves into planks).
- (B) $\mathbb{T}$ satisfies an analogue of stickiness.
- (C) A suitable thickening of $\mathbb{T}$ yields a richer collection of tubes.

- (D) A suitable zooming-in and rescaling of $\mathbb{T}$ yields a richer collection of tubes.

The precise statement is as follows.

Proposition 14.5.28. *For all $\varepsilon > 0$, there exists $\tau > 0$, so that for all $\eta_1 > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\beta \geq 2$ and let $(\mathbb{T}, Y)_\delta$ be a set of $\delta^{-\beta}$ essentially distinct δ -tubes that satisfy the Convex Wolff Axioms with error δ^η , with Y a δ^η -uniformly dense shading. Then at least one of the following must hold:*

- (A) $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ , with scale separation $\delta^{-\eta}$.
- (B) There exists $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \delta^{\eta_1}(\#\mathbb{T})$, and $\mathbb{T}'$ satisfies the self-similar Convex Wolff Axioms with error $\delta^{-\varepsilon}$.
- (C) There exists $\rho \in [\delta, \delta^\varepsilon]$ and a set $\tilde{\mathbb{T}}$ of ρ -tubes, with the following properties:
 - (C.i) The tubes in $\tilde{\mathbb{T}}$ are essentially distinct, and $\#\tilde{\mathbb{T}} \geq \rho^{-\beta-\tau}$.
 - (C.ii) $\tilde{\mathbb{T}}$ satisfies the Convex Wolff Axioms with error $\leq \rho^{-\eta_1}$.
 - (C.iii) For each $T_\rho \in \tilde{\mathbb{T}}$, there exists $T \in \mathbb{T}$ with $T \subset T_\rho$.
- (D) There exists $\rho \in [\delta^{1-\varepsilon}, 1]$, a ρ -tube T_ρ , and a set $\tilde{\mathbb{T}} \subset \mathbb{T}[T_\rho]$, so that the following holds:
 - (D.i) $\#\tilde{\mathbb{T}} \geq (\delta/\rho)^{-\beta-\tau}$.
 - (D.ii) The unit rescaling of $\tilde{\mathbb{T}}$ relative to T_ρ satisfies the Convex Wolff Axioms with error $(\delta/\rho)^{-\eta_1}$.

To prove Proposition 14.5.28, we will consider arrangements $\mathbb{T}$ for which conclusions (B) and (C) fail. This will force the existence of a ρ -tube T_ρ that satisfies Item (D.i). However, it is not clear that this ρ -tube will also satisfy Item (D.ii); the main concern is that there might exist many convex sets $W \subset T_\rho$ of dimensions $a \times b \times 1$ with $a \ll b$ that contain an illegally large number of tubes from $\mathbb{T}[T_\rho]$ (convex sets of dimensions $a \times b \times 1$ with $a \sim b$ can be handled by replacing ρ with a smaller scale). Our goal is to show that if this happens, then Conclusion (A) holds. To do this, we will show that if this happens, then the tubes in $\mathbb{T}[T_\rho]$ must cluster into planks; this step is straightforward and will be deferred to later. The second step is to show that under suitable hypotheses, every arrangement of planks must have discretized Assouad dimension close to 3; we will do this in the next sub-section.

14.5.4.1 Arrangements of rectangular prisms

We begin the process described at the end of the previous section. Let $\mathcal{R}$ be a set of planks, i.e. rectangular prisms of dimensions roughly $s \times t \times 1$ for some $\delta \leq s \leq t \leq 1$. If a typical pair of planks from $\mathcal{R}$ intersects tangentially, then this forces the planks to cluster into wider (and somewhat thicker) planks; crucially, the aspect ratio (i.e. thickness vs. width) remains the same. If instead a typical pair of planks does not intersect tangentially, then the hairbrush of a typical plank $R \in \mathcal{R}$ (i.e. the union of the other planks that intersect R) fills out a thickened neighbourhood of R , and this in turn means that the union of planks has discretized Assouad dimension close to 3. The precise statement is as follows.

Lemma 14.5.29. *For all $\varepsilon, \tau \in (0, 1]$, there exists $\delta_0, \eta > 0$ so that the following is true for all $\delta \in (0, \delta_0]$. Let $\delta \leq s \leq t \leq 1$ with $s \leq \delta^\varepsilon t$. Let $\mathcal{R}$ be a multi-set of $s \times t \times 1$ rectangular prisms contained in $B(0, 1)$, and suppose that $\mathcal{R}$ satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$. Let $\{Y(R), R \in \mathcal{R}\}$ be a δ^η -dense shading.*

Then at least one of the following two things must happen:

- (A) $\bigcup_{R \in \mathcal{R}} Y(R)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale s , with scale separation $\delta^{-\eta}$. This means that there exists $r \in [\delta^{-\eta}s, 1]$ and a r -ball B so that

$$\left| B \cap \bigcup_{R \in \mathcal{R}} Y(R) \right| \geq (s/r)^\varepsilon |B|. \quad (14.5.12)$$

- (B) There are scales $s' \in [\delta^{-\varepsilon/4}s, \delta^{\varepsilon/2}]$ and $t' = \min(1, t \frac{s'}{s})$, and a multi-set $\mathcal{R}'$ of $s' \times t' \times 1$ prisms that satisfies the Convex Wolff Axioms with error $\delta^{-\tau}$, so that the shading

$$Y'(R') = R' \cap N_{s'} \left(\bigcup_{\substack{R \in \mathcal{R} \\ R \subset 2R'}} Y(R) \right) \quad (14.5.13)$$

is δ^τ -dense.

Remark 14.5.30. Note that as $\varepsilon \searrow 0$, Conclusion (A) becomes stronger, while conclusion (B) becomes weaker (indeed, if $\varepsilon = 0$ then Conclusion (B) vacuously holds with $s' = s$ and $t' = t$). As $\tau \searrow 0$, Conclusion (A) is unchanged, while Conclusion (B) becomes stronger.

Proof.

Step 1: Robust broadness. After replacing $\mathcal{R}$ by a subset of cardinality $\gtrsim \delta^\eta(\#\mathcal{R})$ if necessary, we may suppose that $|Y(R)| \geq \frac{1}{2}\delta^\eta|R|$ for each $R \in \mathcal{R}$.

Let $Y_1(R) \subset Y(R)$ be a regular (in the sense of Definition 14.5.9) refinement of Y , with $|Y_1(R)| \geq \frac{1}{2}|Y(R)|$ for each $R \in \mathcal{R}$.

Next, we perform a “two-ends” style robust broadness reduction. Let $\varepsilon_1 > 0$ be a small quantity to be chosen below. For each $x \in \bigcup_{R \in \mathcal{R}} Y_1(R)$, let $u_x \in [t, 1]$ and let T_{u_x} be a u_x -tube that comes within a multiplicative factor of 2 of maximizing the quantity

$$u_x^{-\varepsilon_1} \# \{R \in \mathcal{R} : x \in Y_1(R), R \subset T_{u_x}\}.$$

Since there exists a 1-tube that contains a $\gtrsim 1$ fraction of $\mathcal{R}$, and since $u_x \geq \delta$, we have

$$\# \{R \in \mathcal{R} : x \in Y_1(R), R \subset T_{u_x}\} \gtrsim \delta^{\varepsilon_1} \# \{R \in \mathcal{R} : x \in Y_1(R)\}.$$

Define the shading

$$Y_2(R) = \{x \in Y_1(R) : R \subset T_{u_x}\}.$$

Then $\sum_{R \in \mathcal{R}} |Y_2(R)| \gtrsim \delta^{\varepsilon_1} \sum_{R \in \mathcal{R}} |Y_1(R)|$. For each $x \in \bigcup_{R \in \mathcal{R}} Y_2(R)$, each $t \leq r \leq u_x$, and each r -tube T_r , we have

$$\# \{R \in \mathcal{R} : x \in Y_2(R), R \subset T_r\} \leq 2(r/u_x)^{\varepsilon_1} \# \{R \in \mathcal{R} : x \in Y_2(R)\}. \quad (14.5.14)$$

By dyadic pigeonholing, we can select a multiplicity μ and a number $t \leq u \leq 1$ so that if we define

$$Y_3(R) = Y_2(R) \cap \left\{x \in \mathbb{R}^3 : \sum_{R \in \mathcal{R}} \chi_{Y_2(R)}(x) \in [\mu, 2\mu), u_x \in [u/2, u)\right\},$$

then provided we choose δ_0 sufficiently small and $\eta < \varepsilon_1/2$, we have $\sum_{R \in \mathcal{R}} |Y_3(R)| \geq \delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}} |R|$.

We first consider the case where $\mu \leq \delta^{-\varepsilon^2+2\varepsilon_1}(st\#\mathcal{R})$. Then

$$\left| \bigcup_{R \in \mathcal{R}} Y(R) \right| \geq \mu^{-1} \sum_{R \in \mathcal{R}} |Y_3(R)| \geq \delta^{\varepsilon^2-2\varepsilon_1} \delta^{2\varepsilon_1} \frac{\sum_{R \in \mathcal{R}} |R|}{st(\#\mathcal{R})} \geq \delta^{\varepsilon^2} \geq s^\varepsilon,$$

where the final inequality used the hypothesis that $s \leq \delta^\varepsilon t$ and hence $s \leq \delta^\varepsilon$. Thus Conclusion (A) holds with $r = 1$ and B the unit ball, and we are done.

Our choice of ε_1 will satisfy $\varepsilon_1 \leq \varepsilon^3/6$, and thus for the remainder of the proof we may suppose that

$$\mu > \delta^{-(2/3)\varepsilon^2}(st\#\mathcal{R}). \quad (14.5.15)$$

Step 2: Localization. Let $\mathbb{T}_u$ be a set of essentially distinct u tubes with the property that for every u -tube T_u , there is at least one, and at most $O(1)$,

tubes from $\mathbb{T}_u$ whose 2-fold dilate contains T_u . For each $x \in \bigcup_{R \in \mathcal{R}} Y_3(R)$, select a tube $T_u(x) \in \mathbb{T}_u$ with the property that $T_{u_x} \subset 2T_u$. Define the shading $Y(T_u) = \{x \in \mathbb{R}^3 : T_u(x) = T_u\}$. We have $\bigcup_{R \in \mathcal{R}} Y_3(R) = \bigsqcup_{T_u \in \mathbb{T}_u} Y(T_u)$. Note that technically, $Y(T_u)$ is a subset of $2T_u$, rather than a subset of T_u . Observe that if $R \in \mathcal{R}$ with $x \in Y_3(R)$, then $R \subset T_{u_x} \subset 2T_u(x)$.

Since the tubes in $\mathbb{T}_u$ are essentially distinct, for each $T_u \in \mathbb{T}_u$ there are $O(1)$ tubes $T'_u \in \mathbb{T}_u$ with the property that $2T_u \cap 2T'_u$ contains a unit line segment — indeed, if $2T_u \cap 2T'_u$ contains a unit line segment, then T_u and T'_u must make angle $\lesssim u$. Thus by pigeonholing, we can replace $\mathbb{T}_u$ by a $O(1)$ refinement $\mathbb{T}'_u$ with the following properties:

- For each pair of distinct tubes $T_u, T'_u \in \mathbb{T}'_u$, $2T_u \cap 2T'_u$ does not contain a unit line segment. In particular, each $R \in \mathcal{R}$ is contained in at most one dilated tube $2T_u$.

•

$$\left| \bigsqcup_{T_u \in \mathbb{T}'_u} Y(T_u) \right| \gtrsim \left| \bigsqcup_{T_u \in \mathbb{T}_u} Y(T_u) \right| = \left| \bigcup_{R \in \mathcal{R}} Y_3(R) \right|.$$

For each $R \in \mathcal{R}$, define $Y_4(R) = Y_3(R) \cap Y(T_u)$, where T_u is the unique tube from $\mathbb{T}'_u$ for which $R \subset 2T_u$. If no such tube exists, then define $Y_4(R) = \emptyset$. We have

$$\begin{aligned} \sum_{R \in \mathcal{R}} |Y_4(R)| &\geq \mu \sum_{T_u \in \mathbb{T}'_u} |Y(T_u)| \gtrsim \mu \sum_{T_u \in \mathbb{T}_u} |Y(T_u)| \\ &\geq \frac{1}{2} \sum_{R \in \mathcal{R}} |Y_3(R)| \geq \delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}} |R|. \end{aligned}$$

Finally, let $\mathbb{T}''_u \subset \mathbb{T}'_u$ consist of those $T_u \in \mathbb{T}'_u$ for which

$$\sum_{R \in \mathcal{R}[2T_u]} |Y_4(R)| \geq c\delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}[2T_u]} |R|. \quad (14.5.16)$$

We will choose the constant $c \sim 1$ sufficiently small so that

$$\sum_{T_u \in \mathbb{T}''_u} \sum_{R \in \mathcal{R}[2T_u]} |Y_4(R)| \gtrsim \sum_{T_u \in \mathbb{T}'_u} \sum_{R \in \mathcal{R}[2T_u]} |Y_4(R)|.$$

In summary, the situation is as follows: For each $T_u \in \mathbb{T}''_u$, we have

$$\sum_{R \in \mathcal{R}[2T_u]} |Y_4(R)| \gtrsim \delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}[2T_u]} |R|, \quad (14.5.17)$$

and for each $x \in \bigcup_{R \in \mathcal{R}[2T_u]} Y_4(R)$, we have

$$\{R \in \mathcal{R}[2T_u] : x \in Y_4(R)\} = \{R \in \mathcal{R} : x \in Y_3(R)\} = \{R \in \mathcal{R} : x \in Y_2(R)\}. \quad (14.5.18)$$

The cardinality of the above set is between μ and 2μ .

For each $T_u \in \mathbb{T}_u''$ and each $R \in \mathcal{R}[2T_u]$, the unit rescaling of R relative to $2T_u$ is comparable to a $s/u \times t/u \times 1$ prism. Define $\tilde{s} = s/u$ and $\tilde{t} = t/u$. Abusing notation slightly, we will call this set a $\tilde{s} \times \tilde{t} \times 1$ prism, and we will denote the (multi) set of these prisms by $\tilde{\mathcal{R}}^{T_u}$. Let $\tilde{Y}_1$ (resp. $\tilde{Y}_4$) be the images of the shadings Y_1 (resp. Y_4) under the unit rescaling described above. Thus for each $T_u \in \mathbb{T}_u$, we have that $(\tilde{\mathcal{R}}^{T_u}, \tilde{Y}_1)$ is a set of $\tilde{s} \times \tilde{t} \times 1$ prisms and a shading on these prisms, and similarly for $(\tilde{\mathcal{R}}^{T_u}, \tilde{Y}_4)$. By (14.5.18) and our definition of Y_3 , for each $T_u \in \mathbb{T}_u$ and each $x \in \bigcup_{\tilde{R} \in \tilde{\mathcal{R}}} \tilde{Y}_4(\tilde{R})$, we have

$$\mu \leq \#\{\tilde{R} \in \tilde{\mathcal{R}}^{T_u} : x \in \tilde{Y}_4(\tilde{R})\} < 2\mu. \quad (14.5.19)$$

For each $\tilde{R} \in \tilde{\mathcal{R}}^{T_u}$, let $v(\tilde{R})$ be the direction of a tube of minimal diameter that contains $\tilde{R}$. Thus $v(\tilde{R})$ is the “direction” of the tube $\tilde{R}$. We have that $v(\tilde{R})$ is meaningfully defined up to accuracy $\tilde{t}$, in the sense that if $\tilde{R}$ and $\tilde{R}'$ are comparable, then $\angle(v(\tilde{R}), v(\tilde{R}')) = O(\tilde{t})$.

As a consequence of (14.5.14), we have that for each $x \in \bigcup_{\tilde{R} \in \tilde{\mathcal{R}}^{T_u}} \tilde{Y}_4(\tilde{R})$, each $\tilde{t} \leq r \leq 1$, and each unit vector v , we have

$$\#\{\tilde{R} \in \tilde{\mathcal{R}}^{T_u} : x \in \tilde{Y}_4(\tilde{R}), \angle(v(\tilde{R}), v) \leq r\} \lesssim r^{\varepsilon_1} \mu. \quad (14.5.20)$$

Step 3: Dealing with transverse intersection. Fix a u -tube $T_u \in \mathbb{T}_u''$, and define $\tilde{\mathcal{R}} = \tilde{\mathcal{R}}^{T_u}$. Each $\tilde{R} \in \tilde{\mathcal{R}}$ is comparable to a rectangular prism of dimensions $\tilde{s} \times \tilde{t} \times 1$, i.e. $\tilde{R}$ is comparable to the intersection of the following three sets: the $\tilde{s}$ neighbourhood of a plane $\Pi(\tilde{R})$; the $\tilde{t}$ neighbourhood of a tube $T(\tilde{R})$; and the unit ball. Let $n(\tilde{R})$ be the unit normal vector of $\Pi(\tilde{R})$; this is meaningfully defined up to accuracy $\tilde{s}/\tilde{t} = s/t$, in the sense that if $\tilde{R}$ and $\tilde{R}'$ are comparable, then $\angle(n(\tilde{R}), n(\tilde{R}')) = O(\tilde{s}/\tilde{t})$ (here $\angle(\cdot, \cdot)$ refers to unsigned angle, since we do not distinguish between the vectors $n(\tilde{R})$ and $-n(\tilde{R})$).

For each $x \in \bigcup_{\tilde{R} \in \tilde{\mathcal{R}}} \tilde{Y}_4(\tilde{R})$, define the “plane spread”

$$\text{P-spread}(x) = \max \left\{ \angle(n(\tilde{R}), n(\tilde{R}')) \right\},$$

where the maximum is taken over all pairs $\tilde{R}, \tilde{R}'$ for which $x \in \tilde{Y}_4(\tilde{R})$ and $x \in \tilde{Y}_4(\tilde{R}')$. Since $n(\tilde{R})$ and $n(\tilde{R}')$ are defined up to uncertainty $O(\tilde{s}/\tilde{t}) = O(s/t)$, $\text{P-spread}(x)$ is meaningfully defined if it is substantially larger than s/t . If $x \notin \bigcup_{\tilde{R} \in \tilde{\mathcal{R}}} \tilde{Y}_4(\tilde{R})$, then define $\text{P-spread}(x) = 0$.

Suppose there exists a prism $\tilde{R}_0 \in \tilde{\mathcal{R}}$ and a number $\theta \geq \delta^{-7\varepsilon_1/\varepsilon}(\tilde{s}/\tilde{t})$ so that

$$|\{x \in \tilde{R}_0 : \theta \leq \text{P-spread}(x) < 2\theta\}| \geq \frac{c}{100} |\log \delta|^{-2} \delta^{2\varepsilon_1} |\tilde{R}_0|, \quad (14.5.21)$$

where $c \sim 1$ is the constant from (14.5.16).

Since $\tilde{R}_0$ is a $\tilde{s} \times \tilde{t} \times 1$ prism, for each $x_0 \in \tilde{R}_0$ we have $|\tilde{R}_0 \cap B(x_0, \tilde{t})| \sim \tilde{s}\tilde{t}^2$. By pigeonholing, we can select a point $x_0 \in \tilde{R}_0$ so that

$$|\{x \in \tilde{R}_0 \cap B(x_0, \tilde{t}) : \theta \leq \text{P-spread}(x) < 2\theta\}| \gtrsim \delta^{2\varepsilon_1} \tilde{s}\tilde{t}^2. \quad (14.5.22)$$

After a harmless translation and rotation, we may suppose that $\Pi(\tilde{R}_0)$ is the yz -plane, i.e. $\text{span}\{e_2, e_3\}$, and that $\tilde{R}_0 \cap B(x_0, \tilde{t})$ is contained in the “square” $S = [-\tilde{s}, \tilde{s}] \times [-\tilde{t}, \tilde{t}] \times [-\tilde{t}, \tilde{t}]$ (note as well that $|S| \sim |\tilde{R}_0 \cap B(x_0, \tilde{t})|$). Let $S^\dagger$ denote the thickened square $[-10\tilde{t}\theta, 10\tilde{t}\theta] \times [-10\tilde{t}, 10\tilde{t}] \times [-10\tilde{t}, 10\tilde{t}]$, i.e. $S^\dagger$ is comparable to the $\tilde{t}\theta$ neighborhood of S .

Denote the set on the LHS of (14.5.22) by E ; then for each $x \in E$ there is a prism $\tilde{R}_x \in \mathcal{R}$ with $x \in \tilde{Y}_4(\tilde{R})$ and $\angle(n(\tilde{R}_x), e_1) \in [\theta, 2\theta]$. This implies that $\tilde{R}_x \cap B(x, \tilde{t}) \subset \tilde{R}_x \cap B(x_0, 10\tilde{t}) \subset S^\dagger$. Recall that the shading $\tilde{Y}_1(\tilde{R}_x)$ is regular, and thus we have

$$|\tilde{Y}_1(\tilde{R}_x) \cap B(x_0, 2\tilde{t})| \gtrsim |\tilde{Y}_1(\tilde{R}_x)| \frac{|\tilde{R}_x \cap B(x_0, 2\tilde{t})|}{|\tilde{R}_x|} \gtrsim \delta^{2\varepsilon_1} \tilde{s}\tilde{t}^2. \quad (14.5.23)$$

Define the linear map $\tilde{A}(x, y, z) = \left(\frac{x}{10\tilde{t}\theta}, \frac{y}{10\tilde{t}}, \frac{z}{10\tilde{t}}\right)$, so $A(S^\dagger) = [-1, 1]^3$. Define

$$\hat{S} = A(S) = \left[-\frac{\tilde{s}}{10\tilde{t}\theta}, \frac{\tilde{s}}{10\tilde{t}\theta}\right] \times [-1/10, 1/10] \times [-1/10, 1/10],$$

and let $\hat{E} = A(E) \subset \hat{S}$. For each $x \in E$, let $\hat{R}_x = A(\tilde{R}_x \cap S^\dagger)$; this is comparable to a rectangular prism of dimensions roughly $\frac{\tilde{s}}{\tilde{t}\theta} \times 1 \times 1$, and this prism makes angle ~ 1 with the yz -plane, i.e. there is a constant $c_0 \sim 1$ (independent of the choice of $x \in E$) so that the prism has a normal direction $n(\hat{R}_x)$ (defined up to uncertainty $O(\frac{\tilde{s}}{\tilde{t}\theta})$) with $\angle(n(\hat{R}_x), e_1) \geq c_0$. By (14.5.23) we have

$$|\hat{Y}_2(\hat{R}_x)| \gtrsim \delta^{2\varepsilon_1} \frac{\tilde{s}}{\tilde{t}\theta}. \quad (14.5.24)$$

By pigeonholing, we can find a unit vector $e \in \mathbb{R}^3$ and a set $\hat{E}' \subset \hat{E}$ with $|\hat{E}'| \sim |\hat{E}|$, so that $\angle(e, n(\hat{R}_x)) \leq c_0/2$ for each $x \in \hat{E}'$. In particular this means that $\angle(e, e_1) \geq c_0/2 \gtrsim 1$, and hence the projection of e to the yz -plane has magnitude ~ 1 ; denote the image of e under this projection by e' .

(14.5.22) says that $|E| \gtrsim \delta^{2\varepsilon_1} \tilde{s}\tilde{t}^2$ and hence $|\hat{E}'| \gtrsim |\hat{E}| \gtrsim \delta^\eta \frac{\tilde{s}}{\tilde{t}\theta}$. On the other hand, $\hat{E}' \subset \hat{S}$, and the latter set has volume $\sim \frac{\tilde{s}}{\tilde{t}\theta}$. Since $\hat{S} = \left[-\frac{\tilde{s}}{10\tilde{t}\theta}, \frac{\tilde{s}}{10\tilde{t}\theta}\right] \times [-1/10, 1/10] \times [-1/10, 1/10]$ and e' is contained in the yz plane, by Fubini we can select a line ℓ pointing in direction e' that satisfies $|\ell \cap \hat{E}'| \gtrsim \delta^{2\varepsilon_1}$.

Select a $\frac{\tilde{s}}{\tilde{t}\theta}$ -separated set of points $x_1, \dots, x_N$ (here $N \gtrsim \delta^{2\varepsilon_1} \frac{\tilde{t}\theta}{\tilde{s}}$) from $|\ell \cap \hat{E}'|$. We will index these points so that $|x_i - x_j| \geq \frac{\tilde{s}}{\tilde{t}\theta} |i - j|$. Note that $\hat{R}_{x_i} \cap \ell$ is a line segment of length between $\frac{\tilde{s}}{\tilde{t}\theta}$ and $10\frac{\tilde{s}}{\tilde{t}\theta}$, and hence if i, j is a

pair of indices with $|x_i - x_j| \geq 100 \frac{\tilde{s}}{\tilde{t}\theta}$, then $\text{dist}(\hat{R}_{x_i} \cap \ell, \hat{R}_{x_j} \cap \ell) \gtrsim \frac{\tilde{s}}{\tilde{t}\theta} |i - j|$. Since the line ℓ points in direction e' , and $|e' \cdot n(\hat{R}_{x_j})| \sim 1$, we conclude that $\text{dist}(x_i, \hat{R}_{x_j}) \gtrsim \frac{\tilde{s}}{\tilde{t}\theta} |i - j|$, so in particular $\hat{R}_{x_i}$ contains a point that has distance $\gtrsim \frac{\tilde{s}}{\tilde{t}\theta} |i - j|$ from $\hat{R}_{x_j}$. Thus either $\hat{R}_{x_i}$ and $\hat{R}_{x_j}$ do not intersect, or if they do intersect then their normal vectors satisfy $\angle(n(\hat{R}_{x_i}), n(\hat{R}_{x_j})) \gtrsim \frac{\tilde{s}}{\tilde{t}\theta} |i - j|$. In either case, we have

$$|\hat{R}_{x_i} \cap \hat{R}_{x_j}| \lesssim \frac{(\frac{\tilde{s}}{\tilde{t}\theta})^2}{\frac{\tilde{s}}{\tilde{t}\theta}(1 + |i - j|)} = \frac{\tilde{s}}{\tilde{t}\theta(1 + |i - j|)}. \quad (14.5.25)$$

The above inequality vacuously holds if i, j are a pair of indices with $|x_i - x_j| < 100 \frac{\tilde{s}}{\tilde{t}\theta}$, since in this case $|i - j| \sim 1$ and we can use the trivial bound $|\hat{R}_{x_i} \cap \hat{R}_{x_j}| \leq |\hat{R}_{x_i}|$.

We conclude that

$$\left\| \sum_{i=1}^N \chi_{\hat{Y}_2(\hat{R}_i)} \right\|_2^2 \leq \sum_{i,j=1}^N |\hat{R}_i \cap \hat{R}_j| \lesssim N \frac{\tilde{s}}{\tilde{t}\theta} |\log(\frac{\tilde{s}}{\tilde{t}\theta})|, \quad (14.5.26)$$

and hence

$$\left| \bigcup_{i=1}^N \hat{Y}_2(\hat{R}_i) \right| \geq \frac{\left(\sum_i |\hat{Y}_2(\hat{R}_i)| \right)^2}{\left\| \sum_i \chi_{\hat{Y}_2(\hat{R}_i)} \right\|_2^2} \gtrsim \frac{\left(N \delta^{2\varepsilon_1} \frac{\tilde{s}}{\tilde{t}\theta} \right)^2}{N \frac{\tilde{s}}{\tilde{t}\theta} |\log(\frac{\tilde{s}}{\tilde{t}\theta})|} \gtrsim \delta^{4\varepsilon_1} N \frac{\tilde{s}}{\tilde{t}\theta} \gtrsim \delta^{6\varepsilon_1},$$

where the second inequality used (14.5.24) and (14.5.26). Undoing the scaling transformation A , we conclude that

$$\left| S^\dagger \cap \bigcup_{\tilde{R} \in \tilde{\mathcal{R}}} \tilde{Y}_1(\tilde{R}) \right| \gtrsim \delta^{6\varepsilon_1} |S^\dagger|.$$

Recall that $S^\dagger$ is a rectangular prism of dimensions roughly $\tilde{t}\theta \times \tilde{t} \times \tilde{t}$.

Next, undo the unit rescaling with respect to $2T_u$; recall that this unit rescaling anisotropically dilated two directions by a multiplicative factor of roughly u^{-1} , and left the third direction unchanged. Thus the preimage of $S^\dagger$ under this rescaling is comparable to a rectangular prisms whose smallest dimension is at least $\gtrsim \tilde{t}\theta u = t\theta \geq \delta^{-7\varepsilon_1/\varepsilon} s$.

By pigeonholing, we can find a ball B of radius $r \gtrsim \delta^{-7\varepsilon_1/\varepsilon} s$ contained in the preimage of $S^\dagger$, so that

$$\left| B \cap \bigcup_{R \in \mathcal{R}} Y(R) \right| \gtrsim \delta^{6\varepsilon_1} |B| \geq \delta^{-\varepsilon_1} (s/r)^\varepsilon |B|. \quad (14.5.27)$$

If δ_0 is selected sufficiently small, then the LHS of (14.5.27) is at least $(s/r)^\varepsilon |B|$, and thus Conclusion (A) holds.

Step 4: Ensuring that u is large. At this point we may assume that for every $T_u \in \mathbb{T}_u''$, every prism $\tilde{R}_0 \in \tilde{\mathcal{R}}^{T_u}$, and every $\theta \geq \delta^{-7\varepsilon_1/\varepsilon}(\tilde{s}/\tilde{t})$, we have that (14.5.21) fails.

For each $T_u \in \mathbb{T}_u''$ and each $\tilde{R} \in \tilde{\mathcal{R}}^{T_u}$, define

$$\tilde{Y}_5(\tilde{R}) = \{x \in \tilde{Y}_4(R) : \text{P-spread}(x) \leq \delta^{-7\varepsilon_1/\varepsilon}(\tilde{s}/\tilde{t})\}.$$

For each $R \in \bigcup_{T_u \in \mathbb{T}_u''} \mathcal{R}[2T_u]$, define $Y_5(R) \subset Y_4(R)$ to be the preimage of $\tilde{Y}_5(\tilde{R})$ under the unit rescaling associated to $2T_u$. Define $Y_5(R) = \emptyset$ if $R \notin \bigcup_{T_u \in \mathbb{T}_u''} \mathcal{R}[2T_u]$. We have that

- For each $T_u \in \mathbb{T}_u$,

$$\sum_{R \in \mathcal{R}[2T_u]} |Y_5(R)| \gtrsim \sum_{R \in \mathcal{R}[2T_u]} |Y_4(R)| \gtrsim \delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}[2T_u]} |R|. \quad (14.5.28)$$

- $\sum_{R \in \mathcal{R}} |Y_5(R)| \gtrsim \sum_{R \in \mathcal{R}} |Y_4(R)| \gtrsim \delta^{2\varepsilon_1} \sum_{R \in \mathcal{R}} |R|.$
- For each $T_u \in \mathbb{T}_u''$ and each $x \in \bigcup_{R \in \mathcal{R}[2T_u]} Y_5(R)$, we have

$$\mu \leq \#\{R \in \mathcal{R}[2T_u] : x \in Y_5(R)\} < 2\mu. \quad (14.5.29)$$

We claim that

$$u \geq \delta^{-\varepsilon^2/4} t. \quad (14.5.30)$$

To verify this claim, fix a choice of $T_u \in \mathbb{T}_u''$ and a choice of $x \in \bigcup_{R \in \mathcal{R}[2T_u]} Y_5(R)$. The prisms in the set (14.5.29) are contained in a prism of volume $\lesssim \delta^{-7\varepsilon_1/\varepsilon} s u^2 / t$; this is because the unit rescaling of these prisms are contained in a rectangular prism of dimensions comparable to $\delta^{-7\varepsilon_1/\varepsilon}(\tilde{s}/\tilde{t}) \times 1 \times 1$. This latter set has volume roughly $\delta^{-7\varepsilon_1/\varepsilon} s / t$, and undoing the unit rescaling distorts volume by a factor of u^2 .

On the other hand by (14.5.30) and (14.5.15), at least $\mu \geq \delta^{-(2/3)\varepsilon^2} (st \# \mathcal{R})$ prisms from $\mathcal{R}$ must be contained in this set. Since $\mathcal{R}$ satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$, we conclude that

$$\delta^{-(2/3)\varepsilon^2} (st \# \mathcal{R}) \leq \delta^{-\eta} (\delta^{-7\varepsilon_1/\varepsilon} s u^2 / t) (\# \mathcal{R}).$$

re-arranging, we obtain (14.5.30) provided we select η and ε_1 sufficiently small.

Step 5: Constructing larger prisms. Define $s_1 = \delta^{-7\varepsilon_1/\varepsilon} s$. Since $s \leq \delta^\varepsilon t$, if we select ε_1 sufficiently small (depending on ε), then $s_1 \leq t$. For each prism $R \in \mathcal{R}_0$, let $\hat{R}$ be the prism of dimensions $\frac{s_1 u}{t} \times u \times 1$ obtained by

anisotropically dilating R . We claim that if $R_0, R \in \mathcal{R}$ are prisms and if $x \in Y_5(R_0) \cap Y_4(R)$, then $R \subset 2\hat{R}_0$. We verify this claim as follows: we have that R_0 and R are contained in the 2-fold dilate of a common u -tube T_u , and furthermore $\angle(n(\tilde{R}_0), n(\tilde{R})) \leq s_1/t$ (here $\tilde{R}_0$ and $\tilde{R}$ are the unit rescalings of R_0 and R with respect to $2T_u$). This means that $\tilde{R}$ is contained in the s_1/t neighbourhood of the plane $\Pi(\tilde{R}_0)$; but after undoing the unit rescaling with respect to $2T_u$, the preimage of $N_{s_1/t}(B(0, 1) \cap \Pi(\tilde{R}_0))$ is comparable to $\hat{R}$.

In fact a superficially stronger statement is true — if we define $w = \max(\frac{s_1 u}{t}, t)$, then $N_{s_1 u/t}(R)$ is comparable to a prism of dimensions $\frac{s_1 u}{t} \times w \times 1$, and this prism is contained in $2\hat{R}_0$. Furthermore, $N_{s_1 u/t}(R) \cap R_0$ is comparable to a rectangular prism of dimensions $s \times t \times w/u$.

Let $\mathcal{R}_0 \subset \mathcal{R}$ consist of those rectangular prisms for which $|Y_5(R)| \geq c_1 \delta^{2\varepsilon_1} |R|$. We will choose $c_1 \sim 1$ sufficiently small so that $\sum_{R \in \mathcal{R}_0} |Y_5(R)| \gtrsim \sum_{R \in \mathcal{R}} |Y_5(R)|$. Let $R_0 \in \mathcal{R}_0$. Recall that R_0 is a $s \times t \times 1$ rectangular prism; for notational simplicity we will suppose $R_0 = [-s/2, s/2] \times [-t/2, t/2] \times [-1/2, 1/2]$, and that the intersection of $Y_5(R_0)$ with the z -axis has measure $\gtrsim \delta^\eta$. Then $2\hat{R}_0 = [-\frac{s_1 u}{t}, \frac{s_1 u}{t}] \times [-u, u] \times [-1, 1]$.

Select $\frac{w}{u}$ -separated points $z_1, \dots, z_N \in Y_5(R_0)$ on the z -axis, with $N \gtrsim \delta^{2\varepsilon_1} \frac{u}{w}$. We will label the points so that $|z_i - z_j| \geq \frac{w}{u} |i - j|$. For each index i , select a prism $R_i \in \mathcal{R}$ with $z_i \in Y_4(R_i)$ and $\angle((0, 0, 1), \ell(R)) \in [c_2 u, u]$. If we select $c_2 \sim 1$ sufficiently small, then such a prism R_i always exists by (14.5.19) and (14.5.20).

For each index i , let $R_i^\dagger$ be the $\frac{s_1 u}{t}$ neighborhood of R_i , and let $Y_1^\dagger(R_i^\dagger) = N_{\frac{s_1 u}{t}}(Y_1(R_i))$. Since $|Y_1(R_i)| \geq \frac{1}{2} \delta^\eta |R_i|$, we have

$$|Y_1^\dagger(R_i^\dagger)| \gtrsim \delta^\eta |R_i^\dagger|. \quad (14.5.31)$$

$R_i^\dagger$ is comparable to a prism of dimensions $\frac{s_1 u}{t} \times w \times 1$ whose coaxial line makes angle $\sim u$ with the z axis. We have

$$|R_i^\dagger \cap R_j^\dagger| \lesssim \left(\frac{s_1 u}{t}\right) \frac{w}{1 + |i - j|},$$

and hence

$$\left\| \sum_{i=1}^N \chi_{Y_1^\dagger(R_i^\dagger)} \right\|_2^2 \leq \sum_{i,j=1}^N |R_i^\dagger \cap R_j^\dagger| \lesssim N \frac{s_1 u w}{t} |\log(u/w)|.$$

We conclude that

$$\left| \bigcup_{i=1}^N Y_1^\dagger(R_i^\dagger) \right| \geq \frac{\left(\sum |Y_1^\dagger(R_i^\dagger)| \right)^2}{\left\| \sum \chi_{Y_1^\dagger(R_i^\dagger)} \right\|_2^2} \gtrsim \frac{\left(N \delta^{2\varepsilon_1} \frac{s_1 u}{t} w \right)^2}{N \frac{s_1 u w}{t}} = N \delta^{4\varepsilon_1} \frac{s_1 u}{t} w \gtrsim \delta^{6\varepsilon_1} \frac{s_1}{t} u^2. \quad (14.5.32)$$

On the other hand, each set R'_i is contained in $\hat{R}_0$ (a set of volume roughly $s_1 u^2/t$), and thus

$$\left| 2\hat{R}_0 \cap N_{\frac{s_1 u}{t}} \left(\bigcup_{\substack{R \in \mathcal{R} \\ R \subset \hat{R}_0}} Y(R) \right) \right| \gtrsim \delta^{6\varepsilon_1} |2\hat{R}_0|. \quad (14.5.33)$$

Recall that $2\hat{R}_0$ is a prism of dimensions $2\frac{s_1 u}{t} \times 2u \times 2$, and $s_1 = \delta^{-7\varepsilon_1/\varepsilon} s$. Define $s' = \frac{s_1 u}{t}$ and $t' = \min(1, t\frac{s'}{s}) = \min(1, \delta^{-7\varepsilon_1/\varepsilon} u)$. By (14.5.30), we have $s' \in [\delta^{-\varepsilon^2/4} s, \frac{s_1}{t}]$. Since $s \leq \delta^\varepsilon t$, we can select η sufficiently small so that $\frac{s_1}{t} \leq \delta^{\varepsilon/2}$, and hence $s' \in [\delta^{-\varepsilon^2/4} s, \delta^{\varepsilon/2}]$.

By (14.5.33) we conclude that for each $R_0 \in \mathcal{R}_0$, if $\hat{R}'_0$ is the corresponding $s' \times t' \times 1$ prism and ε_1 is selected sufficiently small compared to ε and τ , then the shading $\hat{Y}'(\hat{R}'_0)$ given by (14.5.13) satisfies

$$|\hat{Y}'(\hat{R}'_0)| \gtrsim \delta^{6\varepsilon_1} |\hat{R}_0| \geq \delta^{6\varepsilon_1 + 7\varepsilon_1/\varepsilon} |\hat{R}'_0| \geq \delta^\tau |\hat{R}'_0|. \quad (14.5.34)$$

To finish the proof, define

$$\mathcal{R}' = \{\hat{R}'_0 : \hat{R}_0 \in \mathcal{R}_0\}.$$

(14.5.34) ensures that $|Y'(R')| \geq \delta^\tau |R'|$ for each $R' \in \mathcal{R}'$. Since $\mathcal{R}$ satisfies the Convex Wolff axioms with error $\delta^{-\eta}$ and $\#\mathcal{R}_0 \gtrsim \delta^{2\varepsilon_1} \#\mathcal{R}$, we have that $\mathcal{R}_0$ satisfies the Convex Wolff axioms with error $\lesssim \delta^{-2\varepsilon_1 - \eta}$, and hence by Remark 14.5.14 we have that $\mathcal{R}'$ satisfies the Convex Wolff Axioms with error $\lesssim \delta^{-2\varepsilon_1 - \eta}$. If ε_1, η , and δ_0 and η is selected sufficiently small, then this is $\leq \delta^{-\tau}$. $\square$

Soon, we will iterate Lemma 14.5.29 multiple times. If Conclusion (B) happens repeatedly, then eventually our planks will have width 1, i.e. our planks will become rectangular slabs of dimensions $s \times 1 \times 1$ for some $s \ll 1$. The next lemma says that the union of such a collection of slabs has volume close to 1.

Lemma 14.5.31. *Let $0 < s \leq 1/2$ and $K \geq 1$. Let $\mathcal{R}$ be a multi-set of $s \times 1 \times 1$ rectangular prisms, and suppose that $\mathcal{R}$ satisfies the Convex Wolff Axioms with error K . Let $\{Y(R), R \in \mathcal{R}\}$ be a K^{-1} -dense shading. Then*

$$\left| \bigcup_{R \in \mathcal{R}} Y(R) \right| \gtrsim K^{-3} (\log 1/s)^{-1}. \quad (14.5.35)$$

Proof. Let $E = \bigcup_{R \in \mathcal{R}} Y(R)$. Since the shading Y is K^{-1} dense, by Cauchy-Schwartz we have

$$K^{-1} s (\#\mathcal{R}) \leq \int \chi_E \cdot \sum_{R \in \mathcal{R}} \chi_{Y(R)} \leq \int \chi_E \cdot \sum_{R \in \mathcal{R}} \chi_R \leq |E|^{1/2} \left\| \sum_{R \in \mathcal{R}} \chi_R \right\|_2. \quad (14.5.36)$$

In the argument that follows, we will obtain an upper bound for $\|\sum_{\mathcal{R}} \chi_R\|_2$.

For each prism $R \in \mathcal{R}$, there is an associated normal vector $v(R) \in \mathbb{S}^2$, which is well-defined up to uncertainty $O(s)$. For $R, R' \in \mathcal{R}$, define $\angle(R, R')$ to be the (unsigned) angle between $v(R)$ and $v(R')$.

Fix $R \in \mathcal{R}$. Observe that if $R \cap R'$ is non-empty and $\angle(R, R') = \theta \sim 2^j s$, then R' is contained in the $O(\theta) \times 1 \times 1$ rectangular prism concentric with R . Since $\mathcal{R}$ obeys the Convex Wolff Axioms, at most $K(2^j s)(\#\mathcal{R})$ slabs $R' \in \mathcal{R}$ can have this property.

Let $j_1 \sim \log(1/s)$ be the smallest integer with $2^{j_1} s \geq 1$. Using the above observation, we compute

$$\begin{aligned} \left\| \sum_{R \in \mathcal{R}} \chi_R \right\|_2^2 &= \sum_{R' \in \mathcal{R}} |R \cap R'| = \sum_{j=0}^{j_1} \sum_{\substack{R' : R \cap R' \neq \emptyset, \\ \angle(R, R') \sim 2^j s}} |R \cap R'| \\ &\lesssim \sum_{j=0}^{j_1} \left(K(2^j s \#\mathcal{R}) \right) \left(\frac{s^2}{2^j s} \right) \lesssim K s^2 (\#\mathcal{R}) \log(1/s). \end{aligned} \tag{14.5.37}$$

Chaining (14.5.36) and (14.5.37), we obtain (14.5.35). $\square$

The culmination of Section 14.5.4.1 is the following result, which says that every arrangement of planks satisfying the Convex Wolff Axioms must have discretized Assouad dimension close to 3.

Proposition 14.5.32. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $s, t \in [\delta, 1]$ with $s \leq \delta^\varepsilon t$. Let $\mathcal{R}$ be a multi-set of $s \times t \times 1$ prisms contained in $B(0, 1)$, and suppose that $\mathcal{R}$ satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$. Let $\{Y(R), R \in \mathcal{R}\}$ be a δ^η -dense shading. Then $\bigcup_{\mathcal{R}} Y(R)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ and scale separation $\delta^{-\eta}$.*

In brief, we prove Proposition 14.5.32 by iterating Lemma 14.5.29. If Conclusion (A) holds at any step in the iteration, then our union of planks has discretized Assouad dimension close to 3, and we are done. If Conclusion (B) holds, then we repeat the process with the resulting wider collection of planks. Eventually, our planks will have width 1, at which point we can apply Lemma 14.5.31 to again conclude that our union of planks has discretized Assouad dimension close to 3. We now turn to the details.

Proof. Fix $\varepsilon > 0$ and let η, δ_0 be small quantities to be determined below. Let $\delta \in (0, \delta_0]$, $s, t \in [\delta, 1]$ with $s \leq \delta^\varepsilon t$; let $\mathcal{R}$ be a set of $s \times t \times 1$ prisms that satisfy the Convex Wolff Axioms with error $\delta^{-\eta}$; and let $Y(R)$ be a δ^η -dense shading.

Define $\mathcal{R}_1 = \mathcal{R}$ and define the shading $Y_1 = Y$. Let $s_1 = s$ and $t_1 = t$. Let $\eta = \tau_1 < \tau_2 < \tau_3 < \dots < \tau_N$, $N = \lfloor 4/\varepsilon^2 \rfloor$ be small quantities to be specified below. We will perform the following iterative process for each $i = 1, 2, \dots, N$. At the beginning of step i , the following properties will hold:

- (a) We have a set $\mathcal{R}_i$ of $s_i \times t_i \times 1$ prisms, with $s_i \leq \delta^\varepsilon t_i$.
- (b) $\mathcal{R}_i$ satisfies the Convex Wolff axioms with error $\delta^{-\tau_i}$.
- (c) We have a shading $\{Y_i(R) : R \in \mathcal{R}_i\}$ that is δ^{τ_i} dense.
- (d) For each $R_i \in \mathcal{R}_i$, we have

$$Y_i(R_i) \subset N_{s_i} \left(\bigcup_{R \in \mathcal{R}} Y(R) \right). \quad (14.5.38)$$

When $i = 1$, the hypotheses of Proposition 14.5.32 guarantee that the above properties hold.

Suppose that the above properties hold at step i . Apply Lemma 14.5.29 to $(\mathcal{R}_i, Y_i)$ with ε as above and τ_{i+1} in place of τ . If $\tau_i > 0$ is sufficiently small depending on ε and τ_{i+1} , then the Lemma can be applied (indeed, τ_i must be less than or equal to the quantity “ η ” from the conclusion of Lemma 14.5.29). The quantity δ_0 (and hence δ) will be chosen sufficiently small so that Lemma 14.5.29 can be applied at each step.

Case (A). Suppose that Conclusion (A) from Lemma 14.5.29 holds, i.e. there exists $r \in [\delta^{-\tau_i} s_i, 1]$ and an r -ball B so that

$$\left| B \cap \bigcup_{R \in \mathcal{R}_i} Y_i(R_i) \right| \geq (s_i/r)^\varepsilon |B|.$$

Using (14.5.38), we conclude that

$$\left| B \cap N_{s_i} \left(\bigcup_{R \in \mathcal{R}} Y(R) \right) \right| \geq (s_i/r)^\varepsilon |B|,$$

i.e. $\bigcup_{R \in \mathcal{R}} Y(R)$ has discretized Assouad dimension $\geq 3 - \varepsilon$ at scale δ and scale separation $\delta^{-\tau_i}$. If this occurs, then we halt the iteration and conclude the proof.

Case (B). Next, suppose that Conclusion (B) from Lemma 14.5.29 holds. Define $s_{i+1} = s'_i$ and $t_{i+1} = t'_i$ (here s'_i and t'_i are the output of Lemma 14.5.29, Conclusion (B)). Let $\mathcal{R}_{i+1}$ be the set of $s_{i+1} \times t_{i+1} \times 1$ prisms, with shading $Y_{i+1} = Y'_i$, as described by Lemma 14.5.29, Conclusion (B). We have that $\mathcal{R}_{i+1}$ satisfies the Convex Wolff Axioms with error $\delta^{-\tau_{i+1}}$, and Y_{i+1} is $\delta^{\tau_{i+1}}$ dense. Furthermore, the shading Y_{i+1} satisfies (14.5.38) with $i + 1$ in place of i . In

particular, $\mathcal{R}_{i+1}$ and Y_{i+1} satisfy our induction hypotheses (b), (c), and (d) above.

Case (B.i). Suppose that Conclusion (B) holds and that $t_{i+1} = 1$. Recall that $s_{i+1} \leq \delta^{\varepsilon/2}$. Thus we can apply Lemma 14.5.31 with s_{i+1} in place of δ . We have

$$\left| N_{s_{i+1}} \left(\bigcup_{R \in \mathcal{R}} Y(R) \right) \right| \geq \left| \bigcup_{R \in \mathcal{R}_{i+1}} Y_{i+1}(R) \right| \gtrsim \delta^{3\tau_{i+1}} |\log s_{i+1}|^{-1} \gtrsim s_{i+1}^{7\tau_{i+1}/\varepsilon}. \quad (14.5.39)$$

We can select τ_{i+1} and δ_0 sufficiently small so that the LHS of (14.5.39) has size at least $s_{i+1}^{\varepsilon} |B(0, 1)|$. Since the set on the LHS of (14.5.39) is contained in $B(0, 1)$, we conclude that $\bigcup_{\mathcal{R}} Y(R)$ has discretized Assouad dimension $\geq 3 - \varepsilon$ at scale δ and scale separation $\delta^{-\varepsilon/2} \geq \delta^{-\eta}$. If this occurs, then we halt the iteration and conclude the proof.

Case (B.ii). Suppose $t_{i+1} < 1$; then $s_{i+1}/t_{i+1} = s_i/t_i$, so in particular $s_{i+1} \leq \delta^{\varepsilon} t_{i+1}$. This means that the induction hypothesis (a) also holds, and thus we have verified all of the properties required to continue the iteration, with $i + 1$ in place of i . Note that $s_{i+1} \geq \delta^{-\varepsilon^2/4} s_i$; $t_{i+1} \geq \min(1, \delta^{\varepsilon} s_{i+1})$, and the iterative process halts if $t_{i+1} = 1$. We conclude that this iterative process will halt after at most $N = \lfloor 4/\varepsilon^2 \rfloor$ steps. $\square$

14.5.4.2 Arrangements of tubes that cluster into rectangular prisms

In Section 14.5.4.1, we showed that unions of planks have discretized Assouad dimension close to 3. In this section we will use these results to show that if an arrangement of δ -tubes clusters into planks, then the union of these tubes must have discretized Assouad dimension close to 3. This is made precise in Lemmas 14.5.34 and 14.5.36 below.

As a first step, the following lemma says that if $\mathbb{T}$ is a set of δ tubes of cardinality $\gg \rho/\delta$, all of which are contained in a common $\delta \times \rho \times 1$ prism, then these tubes must cluster into $\delta \times \omega \times 1$ planks (for some $\delta \ll \omega \leq \rho$), and the union of these tubes must fill out a substantial fraction of each $\delta \times \omega \times 1$ plank. Later, we will show that the existence of such (mostly full) $\delta \times \omega \times 1$ planks will force $\bigcup_{\mathbb{T}} T$ to have discretized Assouad dimension close to 3.

Lemma 14.5.33. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\rho \in [20\delta, 1]$ and let $K \geq 1$. Let $(\mathbb{T}, Y)_{\delta}$ be a set of essentially distinct δ -tubes, with $\#\mathbb{T} \geq K\rho/\delta$, and Y a uniformly δ^{η} -dense shading. Let R_0 be a $10\delta \times \rho \times 1$ rectangle, and suppose each $T \in \mathbb{T}$ is contained in R_0 .*

Then there exists a number $K\delta \lesssim \omega \leq \rho$ and a set $\mathcal{R}$ of essentially distinct $10\delta \times \omega \times 1$ prisms, so that

$$\#\left(\bigcup_{R \in \mathcal{R}} \{T \in \mathbb{T} : T \subset R\}\right) \gtrsim \#\mathbb{T}, \quad (14.5.40)$$

and for each prism $R \in \mathcal{R}$ we have

$$\left| \bigcup_{\substack{T \in \mathbb{T} \\ T \subset R}} Y(T) \right| \geq \delta^\varepsilon |R|. \quad (14.5.41)$$

Proof. For each $x \in \mathbb{R}^3$, define the “line spread”

$$\text{L-spread}(x) = \max \{ \angle(T, T') \},$$

where the maximum is taken over all pairs of tubes T, T' intersecting x . We define $\text{L-spread}(x) = 0$ if at most one tube intersects x . Let C be a large number to be determined below. We claim that there exists a choice of $\theta \in [\frac{K\delta}{C}, \rho]$ so that the shading

$$Z(T) = \{x \in T : \text{L-spread}(x) \in [\theta, 2\theta]\}$$

satisfies

$$\sum_{T \in \mathbb{T}} |Z(T)| \geq (C |\log \delta|)^{-1} \delta^2 (\#\mathbb{T}). \quad (14.5.42)$$

(Note that the shading $Z(T)$ is not necessarily a subset of $Y(T)$). We justify this claim as follows. Since there are at most $10 \log(C/\delta)$ dyadic values of θ in the interval $[\frac{K\delta}{C}, \rho]$, if we could not select a $\theta \in [\frac{K\delta}{C}, \rho]$ so that (14.5.42) holds, then (provided C is selected sufficiently large) the shading

$$Z'(T) = \left\{ x \in T : \text{L-spread}(x) \leq \frac{K\delta}{C} \right\}$$

satisfies $\sum_{T \in \mathbb{T}} |Z'(T)| \geq \frac{1}{100} \delta^2 (\#\mathbb{T})$. But for each $x \in \mathbb{R}^3$, the set of tubes $T \in \mathbb{T}$ with $x \in Z'(T)$ are contained in the 10δ -neighborhood of a plane (since they are contained in R_0), and also make angle $\leq \frac{K\delta}{C}$ with a common unit vector; since the tubes in $\mathbb{T}$ are essentially distinct, we conclude that $O(K/C)$ tubes $T \in \mathbb{T}$ can satisfy $x \in Z'(T)$, and hence

$$\left| \bigcup_{T \in \mathbb{T}} T \right| \geq \left| \bigcup_{T \in \mathbb{T}} Z'(T) \right| \gtrsim (C/K) \sum_{T \in \mathbb{T}} |Z'(T)| \gtrsim C\rho\delta. \quad (14.5.43)$$

If C is selected sufficiently large then this is impossible, since the set on the LHS of (14.5.43) is contained in R_0 , which has volume $10\rho\delta$.

At this point we have established the existence of an angle $\theta \in [\frac{K\delta}{C}, \rho]$ so that the associated shading $Z(T)$ satisfies (14.5.42). In the arguments that follow, the quantity C has been fixed, and all implicit constants are allowed to depend on C . Let

$$\mathbb{T}_1 = \{T \in \mathbb{T} : |Z(T)| \geq (C|\log \delta|)^{-1}|T|\}. \quad (14.5.44)$$

Then (14.5.42) guarantees that $\#\mathbb{T}_1 \gtrsim \#\mathbb{T}$. For each $T_1 \in \mathbb{T}_1$, let $x_2, \dots, x_N$, $N \gtrsim \theta/\delta$ be a set of δ/θ -separated points from $Y_1(T_1)$. For each x_i , select $T_i \in \mathbb{T}$ with $x_i \in T_i$ and $\angle(T_1, T_i) \in [\theta, 2\theta)$. An L^2 argument (see e.g. (14.5.26)) plus the fact that $|Y(T_i)| \geq \delta^\eta |T|$ shows that

$$\left| \bigcup_{i=2}^N Y(T_i) \right| \gtrsim \delta^{2\eta} \sum_{i=2}^N |T_i| \gtrsim \delta^{2\eta} \delta \theta. \quad (14.5.45)$$

On the other hand, the set on the LHS of (14.5.45) is contained in $R_0 \cap N_{2\theta}(T_1)$, which in turn is contained in a rectangular prism of dimensions $10\delta \times 4\theta \times 1$; denote this prism by $R(T_1)$. We have just shown that

$$\left| \bigcup_{\substack{T \in \mathbb{T} \\ T \subset R(T_1)}} Y(T) \right| \gtrsim \delta^{2\eta} |R(T_1)|. \quad (14.5.46)$$

Finally, let $\omega = 4\theta$ and let $\mathcal{R}_1 = \{R(T_1) : T_1 \in \mathbb{T}_1\}$. We claim that (14.5.40) holds for $\mathcal{R}_1$. Indeed, this follows from the fact that $T_1 \subset R(T_1)$ for each $T_1 \in \mathbb{T}_1$, and $\#\mathbb{T}_1 \gtrsim \#\mathbb{T}$. By (14.5.46) we see that (14.5.41) holds for each $R \in \mathcal{R}_1$, provided we select η and δ_0 sufficiently small. To conclude the proof, greedily select a set $\mathcal{R} \subset \mathcal{R}_1$ of essentially distinct rectangular prisms; we can perform this selection so that (14.5.40) continues to hold. $\square$

The following lemmas says that if $\mathbb{T}$ clusters into planks, then it must have discretized Assouad dimension close to 3.

Lemma 14.5.34. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\mathbb{T}$ be a set of essentially distinct δ -tubes that satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$. Let Y be a uniformly δ^η -dense shading. Then at least one of the following must occur:*

- (A) $\bigcup_{T \in \mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ and scale separation $\delta^{-\eta}$.
- (B) There is a set $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \frac{1}{2}(\#\mathbb{T})$ so that for every $\delta \leq \rho \leq 1$ and every $\delta \times \rho \times 1$ rectangle R , at most $\delta^{-\varepsilon}(\rho/\delta)$ tubes from $\mathbb{T}'$ are contained in $100R$.

Remark 14.5.35. Since $\mathbb{T}$ satisfies the Convex Wolff Axioms, we have that at most $100^3 \delta^{-\eta} (\rho \delta) (\#\mathbb{T})$ tubes are contained in each $100\delta \times 100\rho \times 100$ rectangle. If $\#\mathbb{T}$ is substantially larger than δ^{-2} , then the estimate from Conclusion (B) is stronger.

Proof. We say a $100\delta \times 100\rho \times 100$ rectangle is “heavy” if it contains more than $\delta^{-\varepsilon} (\rho/\delta)$ tubes from $\mathbb{T}$. If fewer than $\frac{1}{2}(\#\mathbb{T})$ of the tubes are contained in heavy rectangles, let $\mathbb{T}'$ be the set of tubes that are not contained in heavy rectangles. Then Conclusion (B) holds, and we are done.

Suppose instead that at least $\frac{1}{2}(\#\mathbb{T})$ tubes are contained in heavy rectangles. Since the tubes in $\mathbb{T}$ are essentially distinct, each $100\delta \times 100\rho \times 100$ rectangle can contain at most $O((\rho/\delta)^2)$ tubes from $\mathbb{T}$. Thus if a $100\delta \times 100\rho \times 100$ rectangle is heavy, then we must have $\rho \gtrsim \delta^{1-\varepsilon}$. Let $\mathcal{R}_0$ denote this set of heavy rectangles (note that the rectangles in $\mathcal{R}_0$ can have different dimensions). Let ε_1 be a small quantity to be chosen below. Apply Lemma 14.5.33 to each rectangle $R_0 \in \mathcal{R}_0$, with ε_1 in place of ε , and $K = \delta^{-\varepsilon}$. We obtain a number $\omega(R_0) \gtrsim \delta^{1-\varepsilon}$ and a set $\mathcal{R}(R_0)$ of $\delta \times \omega(R_0) \times O(1)$ rectangles. We have

$$\# \left(\bigcup_{R_0 \in \mathcal{R}_0} \bigcup_{R \in \mathcal{R}(R_0)} \{T \in \mathbb{T} : T \subset R\} \right) \gtrsim \#\mathbb{T}. \quad (14.5.47)$$

After dyadic pigeonholing, we can select $\omega_0 \in [\delta^{1-\varepsilon}, 1]$ so that if we define $\mathcal{R}_1 = \{R_0 \in \mathcal{R}_0 : \omega(R_0) \in [\omega_0, 2\omega_0]\}$, then (14.5.47) continues to hold with $\mathcal{R}_1$ in place of $\mathcal{R}_0$.

Define the multi-set

$$\mathcal{R}_2 = \bigsqcup_{R_1 \in \mathcal{R}_1} \mathcal{R}(R_1).$$

(In the above, $\bigsqcup$ is used to suggest that if the same rectangle is present in multiple sets $\mathcal{R}(R_1)$, then this rectangle should occur multiple times in $\mathcal{R}_2$; i.e. the union might create a multi-set). Let

$$\mathbb{T}_2 = \bigcup_{R \in \mathcal{R}_2} \{T \in \mathbb{T} : T \subset R\}.$$

Since $\mathbb{T}$ satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$, by (14.5.47) we have that the set (i.e. not multi-set) $\mathbb{T}_2$ satisfies the Convex Wolff Axioms with error $\lesssim \delta^{-\eta}$. Finally, for each $T \in \mathbb{T}_2$, let $R(T)$ be some rectangle in $\mathcal{R}_2$ with $T \subset R$. Define the multi-set $\mathcal{R} = \{R(T) : T \in \mathbb{T}_2\}$, i.e. $\mathcal{R}$ is a multi-set with the same cardinality as $\mathbb{T}_2$. By Remark 14.5.14, we conclude that $\mathcal{R}$ satisfies the Convex Wolff Axioms with error $\lesssim \delta^{-\eta}$.

For each $R \in \mathcal{R}$, define the shading

$$Y(R) = R \cap \bigcup_{T \in \mathbb{T}} Y(T).$$

By Lemma 14.5.33, we have $|Y(R)| \geq \delta^{\varepsilon_1}|R|$ for each $R \in \mathcal{R}$. To conclude the proof, apply Proposition 14.5.32 with ε as above, $s = \delta$, and $t = \omega_0$; to ensure that Proposition 14.5.32 can be applied, we must select ε_1 sufficiently small depending on ε (our choice of ε_1 depends on the quantity “ η ” from Proposition 14.5.32), and then η sufficiently small depending on ε_1 . Since $\bigcup Y(T) \subset \bigcup Y(R)$, we conclude that Conclusion (A) holds. $\square$

Lemma 14.5.34 says that either a union of δ -tubes $\mathbb{T}$ has large discretized Assouad dimension, or the tubes in $\mathbb{T}$ cannot concentrate into $\delta \times \rho \times 1$ rectangular prisms. The next lemma is similar, except the latter conclusion is strengthened: either a union of δ -tubes has large discretized Assouad dimension, or these tubes cannot concentrate into $s \times t \times 1$ rectangular prisms, for any $\delta \leq s \leq t \leq 1$. To prove this result, we apply Lemma 14.5.34 to the s -thickening of the tubes in our arrangement, for many different values of s . The precise statement is as follows.

Lemma 14.5.36. *For all $\varepsilon = 1/N > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of essentially distinct δ -tubes that satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$, and let Y be a uniformly δ^η -dense shading. Then at least one of the following must occur:*

- (A) $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ and scale separation $\delta^{-\eta}$.
- (B) *There is a set $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \delta^\varepsilon(\#\mathbb{T})$ so that the following holds. For every number $\delta \leq s \leq 1$ of the form $\delta^{i\varepsilon}$, there is a uniform partitioning cover $\mathbb{T}_s$ of $\mathbb{T}'$ (see Definition 14.5.17), where $\mathbb{T}_s$ is a set of essentially distinct s -tubes. Furthermore, for every $t \in [s, 1]$ and every $s \times t \times 1$ rectangular prism R , at most $\delta^{-\varepsilon} \frac{t}{s} \frac{(\#\mathbb{T}')}{(\#\mathbb{T}_s)}$ tubes from $\mathbb{T}'$ are contained in $100R$.*

Proof. Step 1.

Our first step is to regularize the set $\mathbb{T}$ at scales of the form $s_i = \delta^{i/N}$, with $i = 1, \dots, N$ (we do not need to consider $s_0 = 1$, since in that case there is nothing to prove). After repeated pigeonholing, we can select a set $\mathbb{T}^\dagger \subset \mathbb{T}$ with $\#\mathbb{T}^\dagger \geq (100 \log(1/\delta))^{-N}(\#\mathbb{T})$ and sets of s_i -tubes $\mathbb{T}_{s_i}$, $i = 0, \dots, N$, so that $\mathbb{T}_{s_N} = \mathbb{T}^\dagger$, and for each $i = 1, \dots, N - 1$, the set $\mathbb{T}_{s_i}$ is a uniform partitioning cover of $\mathbb{T}_{s_{i+1}}$ (and hence also a uniform partitioning cover of $\mathbb{T}_{s_j}$ for all $j > i$). See e.g. Lemma 3.4 from [11] for an explicit description of this iterated pigeonholing process. We will choose δ_0 sufficiently small so that $\#\mathbb{T}^\dagger \geq \delta^{\eta/2}(\#\mathbb{T})$. In particular, by Remark 14.5.18 this means that $\mathbb{T}^\dagger$ and each set $\mathbb{T}_{s_i}$, $i = 1, \dots, N$ satisfies the Convex Wolff Axioms with error $\delta^{-2\eta} \leq s_i^{-2\eta N}$. For each index i and each $T_{s_i} \in \mathbb{T}_{s_i}$, define the shading

$Y_{s_i}(T_{s_i}) = T_{s_i} \cap N_{s_i}(Y(T))$, where T is a tube from $\mathbb{T}[T_{s_i}]$. In particular, $|Y_{s_i}(T_{s_i})| \gtrsim \delta^\eta |T_{s_i}| \gtrsim s_i^{\eta N} |T_{s_i}|$, i.e. Y_{s_i} is a uniformly $s_i^{\eta N}$ -dense shading of $\mathbb{T}_{s_i}$.

Step 2.

For each $i = 1, \dots, N$, apply Lemma 14.5.34 to $(\mathbb{T}_{s_i}, Y_{s_i})_{s_i}$ with ε as above; we can do this provided $\eta > 0$ and δ_0 are selected sufficiently small depending on ε (recall that $N = 1/\varepsilon$). If Conclusion (A) of Lemma 14.5.34 holds, then $\bigcup_{\mathbb{T}_{s_i}} Y_{s_i}(T_{s_i})$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale s_i and scale separation $s_i^{-\eta'}$, for some η' depending only on ε . This in turn implies that $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ (this is true since $\delta \leq s_i$) and scale separation $\delta^{-\eta'/N}$. Thus Conclusion (A) of Lemma 14.5.36 holds, provided we select $\eta \leq \eta'/N = \eta'\varepsilon$. If this happens then we are done.

Suppose instead that Conclusion (B) of Lemma 14.5.34 holds. This gives us a set $\mathbb{T}'_{s_i} \subset \mathbb{T}_{s_i}$ with $\#\mathbb{T}'_{s_i} \geq \frac{1}{2}(\#\mathbb{T}_{s_i})$ so that for every $s_i \leq t \leq 1$ and every $s_i \times t \times 1$ rectangular prism R , at most $s_i^{-\varepsilon}(t/s_i) \leq \delta^{-\varepsilon}(t/s_i)$ tubes from $\mathbb{T}'_{s_i}$ are contained in $100R$. Abusing notation slightly, we will replace each set $\mathbb{T}_{s_j}$, $j > i$ with $\bigcup_{\mathbb{T}'_{s_i}} \mathbb{T}_{s_j}[T_{s_i}]$, i.e. we only keep those tubes from $\mathbb{T}_{s_j}$ that are contained in some tube from $\mathbb{T}'_{s_i}$. This process discards at most half the tubes from $\mathbb{T}_{s_j}$, so at each stage in this process, $\mathbb{T}_{s_j}$ still satisfies the convex Wolff axioms with error at most $2^N s_j^{-2\eta N}$ and hence we can continue to iterate Step 2.

Step 3.

Suppose that Conclusion (B) of Lemma 14.5.34 holds for each $i = 1, \dots, N$ in Step 2. While each set $\mathbb{T}_{s_i}$ is a uniform partitioning cover of $\mathbb{T}^\dagger$, this property might no longer hold for the sets $\mathbb{T}'_{s_i}$. We will fix this with another round of pigeonholing. This gives us sets $\mathbb{T}'_{s_i} \subset \mathbb{T}_{s_i}$, $i = 1, \dots, N$ and a set $\mathbb{T}' \subset \mathbb{T}^\dagger$, so that $\#\mathbb{T}' \geq (100 \log(1/\delta))^{-N} (\#\mathbb{T}^\dagger)$; each set $\mathbb{T}'_{s_i}$ is a uniform partitioning cover of $\mathbb{T}'$; and for each index i and each $s_i \times t \times 1$ prism R we have

$$\#\mathbb{T}'[100R] \leq \#\mathbb{T}'_{s_i}[100R] \frac{\#\mathbb{T}'}{\#\mathbb{T}'_{s_i}} \leq \delta^{-\varepsilon} \frac{t}{s_i} \frac{(\#\mathbb{T}')}{(\#\mathbb{T}'_{s_i})}.$$

Thus $\mathbb{T}'$ and $\mathbb{T}'_{s_i}$, $i = 1, \dots, N$ satisfy Conclusion (B) of Lemma 14.5.36. $\square$

14.5.4.3 Proof of Proposition 14.5.28

We are now ready to prove Proposition 14.5.28. For the reader's convenience, we reproduce it here

Proposition 14.5.28. For all $\varepsilon > 0$, there exists $\tau > 0$, so that for all $\eta_1 > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\beta \geq 2$ and let $(\mathbb{T}, Y)_\delta$ be a set of $\delta^{-\beta}$ essentially distinct δ -tubes that satisfy the

Convex Wolff Axioms with error δ^η , with Y a δ^η -uniformly dense shading. Then at least one of the following must be true:

- (A) $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \varepsilon$ at scale δ , with scale separation $\delta^{-\eta}$.
- (B) There exists $\mathbb{T}' \subset \mathbb{T}$ with $\#\mathbb{T}' \geq \delta^{\eta_1}(\#\mathbb{T})$, and $\mathbb{T}'$ satisfies the self-similar Convex Wolff Axioms with error $\delta^{-\varepsilon}$.
- (C) There exists $\rho \in [\delta, \delta^\varepsilon]$ and a set $\tilde{\mathbb{T}}$ of ρ -tubes, with the following properties:
 - (C.i) The tubes in $\tilde{\mathbb{T}}$ are essentially distinct, and $\#\tilde{\mathbb{T}} \geq \rho^{-\beta-\tau}$.
 - (C.ii) $\tilde{\mathbb{T}}$ satisfies the Convex Wolff Axioms with error $\rho^{-\eta_1}$.
 - (C.iii) For each $T_\rho \in \tilde{\mathbb{T}}$, there exists $T \in \mathbb{T}$ with $T \subset T_\rho$.
- (D) There exists $\rho \in [\delta^{1-\varepsilon}, 1]$, a ρ -tube T_ρ , and a set $\tilde{\mathbb{T}} \subset \mathbb{T}[T_\rho]$, so that the following holds:
 - (D.i) $\#\tilde{\mathbb{T}} \geq (\delta/\rho)^{-\beta-\tau}$.
 - (D.ii) The unit rescaling of $\tilde{\mathbb{T}}$ relative to T_ρ satisfies the Convex Wolff Axioms with error $(\delta/\rho)^{-\eta_1}$.

Proof. Step 1.

Decreasing η_1 if necessary, we may suppose that $\eta_1 \leq 1$. Fix a number $\eta_2 \leq \eta_1 \varepsilon / 40$ of the form $\eta_2 = 1/N > 0$. Apply Lemma 14.5.36 with η_2 in place of ε . If Conclusion (A) of Lemma 14.5.36 holds, then Conclusion (A) of Proposition 14.5.28 holds (since $\eta_2 \leq \varepsilon$) and we are done. If instead Conclusion (B) of Lemma 14.5.36 holds, then there exists:

- A set $\mathbb{T}' \subset \mathbb{T}$,
- For each number ρ of the form $\delta^{i/N}$, a uniform partitioning cover $\mathbb{T}_\rho$ of $\mathbb{T}'$.

These objects have the properties described in Conclusion (B) of Lemma 14.5.36.

Suppose there exists a scale $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$ so that $\#\mathbb{T}_\rho \geq \delta^{-\beta-\tau}$. Then by Remark 14.5.18, $\mathbb{T}_\rho$ satisfies the Convex Wolff Axioms with error $\delta^{-2\eta} \leq \rho^{-2\eta/\varepsilon} \leq \rho^{-\eta_1}$. Thus Conclusion (C) holds, with $\tilde{\mathbb{T}} = \mathbb{T}_\rho$.

Henceforth we shall suppose that for each scale $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$, we have

$$\#\mathbb{T}_\rho \leq \delta^{-\beta-\tau}. \quad (14.5.48)$$

Step 2. Our goal in this step is to show that either Conclusion (D) of Proposition 14.5.28 holds, or for every $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$, we have

$$\#\mathbb{T}_\rho \geq \rho^{-\beta+\varepsilon/3}, \quad \rho \in [\delta, \delta^\varepsilon]. \quad (14.5.49)$$

First, we consider the case where $\beta \leq 2 + \varepsilon/4$; in this case (14.5.49) will always hold. Indeed, let $\rho \in [\delta, \delta^\varepsilon]$ be of the form $\rho = \delta^{i/N}$. Then for each $T_\rho \in \mathbb{T}_\rho$, we have

$$\frac{\#\mathbb{T}'}{\#\mathbb{T}_\rho} = \#\mathbb{T}'[T_\rho] \leq \delta^{-2\eta} \rho^2 \#\mathbb{T}',$$

where the second inequality follows from the fact that $\mathbb{T}'$ satisfies the Convex Wolff Axioms with error $\delta^{-2\eta}$. We conclude that $\#\mathbb{T}_\rho \geq \delta^{2\eta} \rho^{-2}$. In particular, if $\beta \leq 2 + \varepsilon/4$, and if we select η sufficiently small, then

$$\#\mathbb{T}_\rho \geq \delta^{2\eta} \rho^{-2} \geq \rho^{2\eta/\varepsilon+\varepsilon/4} \rho^{-\beta} \geq \rho^{-\beta+\varepsilon/3}, \quad (14.5.50)$$

and hence (14.5.49) holds.

Next we consider the case where $\beta > 2 + \varepsilon/4$. Suppose there exists some $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$ for which $\#\mathbb{T}_\rho < \rho^{-\beta+\varepsilon/4}$. Let $\rho \in [\delta, \delta^\varepsilon]$ be the smallest number of the form $\delta^{i/N}$ for which this is true. Note that $\rho \geq \delta^{1-\varepsilon/20}$; this is because the tubes in $\mathbb{T}'$ are essentially distinct, and thus $\#\mathbb{T}_\rho \geq (\delta/\rho)^4 (\#\mathbb{T}') \geq (\delta/\rho)^4 \delta^{-\beta+2\eta}$.

Let $T_\rho \in \mathbb{T}_\rho$. By Conclusion (B) of Lemma 14.5.36, for each $s \in [\delta, \rho]$ of the form δ^{jN} , each $t \in [s, \rho]$, and each $s \times t \times 1$ rectangular prism $R \subset T_\rho$, we have

$$\begin{aligned} \#\{T \in \mathbb{T}'[T_\rho] : T \subset R\} &\leq \delta^{-\eta_2} \frac{t}{s} \frac{\#\mathbb{T}'}{\#\mathbb{T}_s} \\ &\leq \delta^{-\eta_2} \frac{t}{s} \frac{\delta^{-\beta}}{\rho^{-\beta+\varepsilon/4}} \left(\frac{s}{\rho}\right)^{\beta-\varepsilon/4} \\ &\leq \delta^{-\eta_2} \frac{t}{s} \frac{\delta^{-\beta}}{\rho^{-\beta+\varepsilon/4}} \left(\frac{s}{\rho}\right)^2 \\ &\leq \delta^{-2\eta_2} \frac{t}{s} \frac{\#\mathbb{T}'}{\#\mathbb{T}_\rho} \left(\frac{s}{\rho}\right)^2 \\ &\leq \delta^{-2\eta_2} \frac{st}{\rho^2} \#\mathbb{T}'[T_\rho], \end{aligned}$$

where the second inequality used the fact that $\#\mathbb{T}_s \geq s^{-\beta+\varepsilon/4}$ (this follows from the minimality of ρ); the third inequality used the assumption $\beta - \varepsilon/4 \geq 2$; and the fourth inequality used the assumptions $\#\mathbb{T}' \geq \delta^{-\beta+\eta_1}$ and $\#\mathbb{T}_\rho \leq \rho^{-\beta+\varepsilon/4}$.

Define $\tilde{\mathbb{T}} = \mathbb{T}'[T_\rho]$, and let $W \subset T_\rho$ be a convex set that contains at least one tube from $\tilde{\mathbb{T}}$. Then W is comparable to a $s \times t \times 1$ prism, for some

$\delta \leq s \leq t \leq \rho$; increasing s (and possibly t) by a factor of at most $\delta^{-1/N}$, we may suppose that s is of the form δ^{jN} . Thus we have

$$\begin{aligned} \#\{T \in \tilde{\mathbb{T}} : T \subset W\} &\lesssim \delta^{-4\eta_2} \frac{|W|}{|T_\rho|} \#\tilde{\mathbb{T}} \\ &\leq (\delta/\rho)^{-40\eta_2/\varepsilon} \frac{|W|}{\rho^2} \#\tilde{\mathbb{T}} \\ &\leq (\delta/\rho)^{-\eta_1} \frac{|W|}{\rho^2} \#\tilde{\mathbb{T}}. \end{aligned}$$

Thus our set $\tilde{\mathbb{T}}$ satisfies Conclusion (D) from Proposition 14.5.28, and we are done. Henceforth we shall assume that (14.5.49) holds.

Step 3. At this point, we have reduced to the case where (14.5.48) and (14.5.49) hold for all $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$. We will show that Conclusion (B) holds. To begin, note that Items (i) and (iii) from Definition 14.5.22 hold for $\mathbb{T}'$: for each $\rho_0 \in [\delta, \delta^\varepsilon]$, we can select $\rho \in [\rho_0, \delta^{-\varepsilon}\rho_0]$ of the form $\rho = \delta^{i/N}$; $\mathbb{T}_\rho$ is a partitioning cover of $\mathbb{T}'$, so Item (i) holds, and Item (iii) holds by (14.5.49) (if $\rho_0 > \delta^\varepsilon$ then we can select $\rho = 1$ and there is nothing to prove).

It remains to show that Item (ii) holds for $\mathbb{T}'$. It suffices to consider ρ -tubes T_ρ with $\rho \in [\delta, \delta^\varepsilon]$ of the form $\rho = \delta^{i/N}$. Initially, we will consider convex sets R that are $s \times t \times 1$ prisms, with $s \in [\delta, \rho]$ of the form $s = \delta^{j/N}$ and $t \in [s, \rho]$. We would like to estimate the number of tubes from $\mathbb{T}'[T_\rho]$ contained in R .

By Conclusion (B) of Lemma 14.5.36 and (14.5.49), we have

$$\begin{aligned} \#\{T \in \mathbb{T}'[T_\rho] : T \subset R\} &\leq \delta^{-\eta_2} \frac{t}{s} \frac{\#\mathbb{T}'}{\#\mathbb{T}_s} \\ &\lesssim \delta^{-\eta_2 - \varepsilon/4} (ts) (s^{\beta-2} \delta^{-\beta}) \\ &\lesssim \delta^{-\eta_2 - \varepsilon/4} (ts) (\rho^{\beta-2} \delta^{-\beta}) \\ &= \delta^{-\eta_2 - \varepsilon/4} \frac{ts}{\rho^2} \left(\frac{\rho}{\delta}\right)^\beta \\ &\leq \delta^{-2\eta_2 - \varepsilon/4} \frac{ts}{\rho^2} \#\mathbb{T}'[T_\rho] \\ &\leq \delta^{-\varepsilon/2} \frac{|R|}{|T_\rho|} \#\mathbb{T}'[T_\rho]. \end{aligned}$$

Finally, every convex set $W \subset T_\rho$ can be contained in a bounded number of $s \times t \times 1$ rectangular prisms R , where s is of the form $\delta^{j/N}$, $t \in [s, \rho]$, and $|R| \leq \delta^{-\eta_2} |W|$. Thus for every convex set $W \subset T_\rho$, we have

$$\#\{T \in \mathbb{T}'[T_\rho] : T \subset W\} \leq \delta^{-\varepsilon} \frac{|W|}{|T_\rho|} (\#\mathbb{T}'[T_\rho]).$$

We conclude that $\mathbb{T}$ satisfies Condition (ii) from Definition 14.5.22, and hence Conclusion (B) of Proposition 14.5.28 holds. $\square$

14.5.5 Stickyakeya and the self-similar Convex Wolff Axioms

Our goal in this section is to prove Theorem 14.5.5. Our first task is to analyze the structure of extremal collections of tubes. If $\omega > 0$, then such collections must satisfy the self-similar Convex Wolff Axioms.

14.5.5.1 Extremalakeya sets satisfy the self-similar Convex Wolff Axioms

Proposition 14.5.37. *Suppose ω from Definition 14.5.24 is positive. Then for all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be η Assouad-extremal. Then there exists $\mathbb{T}' \subset \mathbb{T}$ so that $(\mathbb{T}', Y)_\delta$ is ε Assouad-extremal, and $\mathbb{T}'$ satisfies the self-similar Convex Wolff axioms with error $\delta^{-\varepsilon}$.*

Proof. Step 1. By decreasing ε if necessary, we can suppose that $\varepsilon < \omega/2$. Let $\tau > 0$ be the value from Proposition 14.5.28, with this choice of ε . Let η_1 be a small quantity to be chosen below (η_1 will depend on ω and ε), and let η', δ'_0 be the values from Proposition 14.5.28 for this choice of ε and η_1 .

Let η and δ_0 be small positive quantities to be chosen below. η and δ_0 will depend on the following quantities: $\omega, \varepsilon, \tau, \eta_1, \eta'$. In addition, δ_0 will depend on δ'_0 .

Let $\delta \in (0, \delta_0]$ and let $(\mathbb{T}, Y)_\delta$ be η Assouad-extremal. By pigeonholing, we can select a subset $\mathbb{T}_1 \subset \mathbb{T}$ with $\#\mathbb{T}_1 \gtrsim \delta^\eta(\#\mathbb{T})$ so that $|Y(T)| \geq \frac{1}{2}\delta^\eta|T|$ for each $T \in \mathbb{T}_1$. Then $\#\mathbb{T}_1 = \delta^{-\beta}$ for some $\beta \geq \alpha - 2\eta$ (recall that α is defined in (14.5.10)), and $\mathbb{T}_1$ satisfies the Convex Wolff Axioms with error $\delta^{-2\eta}$. Furthermore, for all ρ_1, r_1 with $\rho_1 \leq \delta^\eta r_1$, and all balls B of radius r_1 , we have

$$\left| B \cap N_{\rho_1} \left(\bigcup_{T \in \mathbb{T}_1} Y(T) \right) \right| \leq (\rho_1/r_1)^{\omega-\eta} |B|. \quad (14.5.51)$$

Step 2. Apply Proposition 14.5.28 to $(\mathbb{T}_1, Y)_\delta$. We will show that Conclusion (A) from Proposition 14.5.28 cannot hold. Suppose to the contrary that Conclusion (A) holds, i.e. there exists $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^{\eta'} r$ and a r -ball B such that

$$\left| B \cap N_\rho \left(\bigcup_{T \in \mathbb{T}_1} Y(T) \right) \right| \geq (\rho/r)^\varepsilon |B|. \quad (14.5.52)$$

If we select $\eta \leq \min(\eta', \omega_2)$ (recall that $\varepsilon \leq \omega/2$), then (14.5.52) contradicts (14.5.51). Hence Conclusion (A) cannot hold.

Step 3. We will show that Conclusion (C) from Proposition 14.5.28 cannot hold. Suppose to the contrary that Conclusion (C) holds, i.e. there exists $\rho \in [\delta, \delta^\varepsilon]$ and a set of ρ -tubes $\tilde{\mathbb{T}}$ satisfying Items (C.i), (C.ii) and (C.iii) from Conclusion (C). By Item (C.iii), we have that the shading

$$\tilde{Y}(\tilde{T}) = \tilde{T} \cap N_\rho\left(\bigcup_{T \in \mathbb{T}_1} Y(T)\right), \quad \tilde{T} \in \tilde{\mathbb{T}}$$

satisfies $|\tilde{Y}(\tilde{T})| \gtrsim \delta^\eta |\tilde{T}| \gtrsim \rho^{\eta/\varepsilon} |\tilde{T}|$ for each $\tilde{T} \in \tilde{\mathbb{T}}$; we will select $\eta < \eta'\varepsilon$ and δ_0 sufficiently small that $|\tilde{Y}(\tilde{T})| \leq \rho^{\eta'} |\tilde{T}|$ for each $\tilde{T} \in \tilde{\mathbb{T}}$.

Define

$$\gamma = \omega - \omega(\beta + \tau). \quad (14.5.53)$$

(In the above equation, $\omega(\beta + \tau)$ refers to the function ω evaluated at the (positive) number $\beta + \tau$). Provided η_1 is selected sufficiently small (depending on τ), we have $\beta + \tau \geq \alpha - 2\eta_1 + \tau > \alpha$, and hence by the definition of α from (14.5.10), we have $\gamma > 0$. Hence if η_1 is sufficiently small, then

$$\limsup_{x \searrow 0} \omega(\eta_1, \eta_1, x, \beta + \tau) \leq \omega - \gamma/2,$$

i.e. there exists $x_0 > 0$ so that

$$\omega(\eta_1, \eta_1, x, \beta + \tau) \leq \omega - \gamma/4 \quad \text{for all } x \in (0, x_0].$$

Select δ_0 sufficiently small so that $\delta_0^\varepsilon \leq x_0$, and thus $\rho \in [\delta, \delta^\varepsilon]$ implies $\rho \leq x_0$. In particular, $\omega(\eta_1, \eta_1, \rho, \beta + \tau) \leq \omega - \gamma/4$, i.e. there exists $\rho_1, r_1 \in [\rho, 1]$ with $\rho_1 \leq \rho^{m_1} r_1$, and a r_1 -ball B such that

$$\left| B \cap N_{\rho_1}\left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T})\right) \right| \geq (\rho_1/r_1)^{\omega - \gamma/8} |B|. \quad (14.5.54)$$

But if we select $\eta \leq \min(\eta_1 \varepsilon, \gamma/9)$, then $\rho_1 \leq \delta^\eta r_1$, and (14.5.54) contradicts (14.5.51). Hence Conclusion (C) cannot hold.

Step 4. We will show that Conclusion (D) from Proposition 14.5.28 cannot hold. Suppose to the contrary that Conclusion (D) holds, i.e. there exists $\rho \in [\delta^{1-\varepsilon}, 1]$, a ρ -tube T_ρ , and a set $\tilde{\mathbb{T}} \subset \mathbb{T}_1[T_\rho]$ satisfying Items (D.i) and (D.ii) from Conclusion (D). For each $T \in \tilde{\mathbb{T}}$, let $\tilde{Y}(\tilde{T}) \subset Y(\tilde{T})$ be the sub-shading produced by Lemma 14.5.10; this shading is regular at scales $\geq \delta$ and satisfies $|\tilde{Y}(\tilde{T})| \geq \frac{1}{2} |Y(\tilde{T})|$.

Let $\tilde{\delta} = \delta/\rho \in [\delta, \delta^\varepsilon]$ and let $\tilde{\mathbb{T}}_{\tilde{\delta}}$ be the unit rescaling of $\tilde{\mathbb{T}}$ relative to T_ρ , i.e. the tubes in $\tilde{\mathbb{T}}_{\tilde{\delta}}$ are the images of the tubes in $\tilde{\mathbb{T}}$ under the affine map

ϕ_{T_ρ} . Let $\tilde{Y}_\delta(\tilde{T}_\delta)$ be the image of $\tilde{Y}(\tilde{T})$ under this same rescaling. We have $|\tilde{Y}_\delta(\tilde{T})| \gtrsim \delta^\eta |\tilde{T}_\delta| \geq \tilde{\delta}^{\eta/\varepsilon} |\tilde{T}_\delta|$, and we will select η and δ_0 sufficiently small so that $|\tilde{Y}_\delta(\tilde{T}_\delta)| \geq \tilde{\delta}^\eta |\tilde{T}_\delta|$ for all $\tilde{T}_\delta \in \tilde{\mathbb{T}}_\delta$. The tubes in $\tilde{\mathbb{T}}_\delta$ are essentially distinct; by Item (D.i) we have $\#\tilde{\mathbb{T}}_\delta \geq \tilde{\delta}^{-\beta-\tau}$; and by Item (D.ii) we have that $\tilde{\mathbb{T}}_\delta$ satisfies the Convex Wolff Axioms with error $\lesssim \tilde{\delta}^{-\eta_1}$.

Arguing as in Step 3, we conclude that there exists $\rho_2, r_2 \in [\tilde{\delta}, 1]$ with $\rho_2 \leq \tilde{\delta}^{\eta_1} r_2$, and a r_2 -ball B_2 such that

$$\left| B_2 \cap N_{\rho_2} \left(\bigcup_{\tilde{T}_\delta \in \tilde{\mathbb{T}}_\delta} \tilde{Y}_\delta(\tilde{T}_\delta) \right) \right| \geq (\rho_2/r_2)^{\omega-\gamma/8} |B_2|. \quad (14.5.55)$$

Let $E_2 = \phi_{T_\rho}^{-1}(B_2)$; this is an ellipsoid of dimensions $\rho r_2 \times \rho r_2 \times \rho$. We claim that

$$\left| E_2 \cap N_{\rho \rho_2} \left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right) \right| \gtrsim \delta^{-\eta} (\rho_2/r_2)^{\omega-\gamma/8} |E_2|. \quad (14.5.56)$$

To verify this claim, we argue as follows. Let $\tilde{T}_\delta \in \tilde{\mathbb{T}}_\delta$, let $\tilde{x} \in \tilde{Y}_\delta(\tilde{T}_\delta)$, and let $\tilde{B} = B(\tilde{x}, \rho_2)$. Let $E = \phi_{T_\rho}^{-1}(\tilde{B})$; this is an ellipsoid of dimensions $\rho_2 \rho \times \rho_2 \rho \times \rho_2$, and let $x = \phi_{T_\rho}^{-1}(\tilde{x})$ be the center of E . Similarly, let $\tilde{T}_\delta$ and $\tilde{Y}(\tilde{T})$ be the δ tube and its shading corresponding to $\tilde{T}_\delta$ and the shading $\tilde{Y}_\delta(\tilde{T}_\delta)$. Then $E \cap \tilde{T}$ contains a tube-segment of length $\geq \rho_2/2$ centered at x , i.e. $E \cap \tilde{T} \supset B(x, \rho_2/2)$. Since $\tilde{Y}(\tilde{T})$ is regular at scales $\geq \delta$ (and $\rho_2 \geq \tilde{\delta} = \delta/\rho$), we have that $|E \cap \tilde{Y}(\tilde{T})| \gtrsim \delta^\eta \rho_2 \delta^2$, i.e. $E \cap \tilde{Y}(\tilde{T})$ contains $\gtrsim \delta^\eta (\rho_2)/(\rho \rho_2) = \delta^\eta \rho^{-1}$ points that are $10\rho \rho_2$ -separated. For each such point y , we have that $B(y, \rho \rho_2) \subset 2E$, and these balls are disjoint. Since each of these balls has volume roughly $(\rho \rho_2)^3$, we conclude that

$$|2E \cap N_{\rho \rho_2} \left(\bigcup_{\tilde{T}' \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}') \right)| \geq |2E \cap N_{\rho \rho_2}(\tilde{Y}(\tilde{T}))| \gtrsim \delta^\eta \rho^2 \rho_2^3. \quad (14.5.57)$$

Next, let $\{\tilde{x}_i\}_{i=1}^N$ be a $2\rho_2$ -separated subset of $B_2 \cap \bigcup_{\tilde{T}_\delta \in \tilde{\mathbb{T}}_\delta} \tilde{Y}_\delta(\tilde{T}_\delta)$ of cardinality

$$N \sim \rho_2^{-3} \left| B_2 \cap N_{\rho_2} \left(\bigcup_{\tilde{T}_\delta \in \tilde{\mathbb{T}}_\delta} \tilde{Y}_\delta(\tilde{T}_\delta) \right) \right| \geq \rho_2^{-3} (\rho_2/r_2)^{\omega-\gamma/8} |B_2|,$$

with the property that $B(\tilde{x}_i, 2\rho_2) \subset B_2$ for each index i .

For each index i , let $E_i = \phi_{T_\rho}^{-1}(B(\tilde{x}_i, \rho_2))$; this is an ellipsoid of dimensions $\rho_2 \rho \times \rho_2 \rho \times \rho_2$ (this introduces a slight clash of notation, since we previously introduced the ellipsoid E_2 of dimensions $\rho r_2 \times \rho r_2 \times \rho$). The ellipsoids $\{2E_i\}_{i=1}^N$ are disjoint and contained in E_2 . Applying (14.5.57) to each such ellipsoid, we

have

$$\begin{aligned}
 \left| E_2 \cap N_{\rho\rho_2} \left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right) \right| &\geq \sum_{i=1}^N \left| 2E_i \cap N_{\rho\rho_2} \left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right) \right| \\
 &\gtrsim N \delta^\eta \rho^2 \rho_2^3 \\
 &\gtrsim \delta^\eta \rho^2 (\rho_2/r_2)^{\omega-\gamma/8} |B_2| \\
 &= \delta^\eta (\rho_2/r_2)^{\omega-\gamma/8} |E_2|.
 \end{aligned} \tag{14.5.58}$$

This establishes (14.5.56), as claimed.

Next, by (14.5.56) and pigeonholing we can select a ball $B \subset E_2$ of radius $r_2\rho$ so that

$$\left| B \cap N_{\rho\rho_2} \left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right) \right| \gtrsim \delta^\eta (\rho_2/r_2)^{\omega-\gamma/8} |B| \geq (\rho_2/r_2)^{\omega-\gamma/8+\eta/(\varepsilon\eta_1)} |B|.$$

If we select δ_0 and η sufficiently small depending on γ, ε , and η_1 , then

$$\left| B \cap N_{\rho\rho_2} \left(\bigcup_{\tilde{T} \in \tilde{\mathbb{T}}} \tilde{Y}(\tilde{T}) \right) \right| \geq 2(\rho_2/r_2)^{\omega-\eta} |B|. \tag{14.5.59}$$

Finally, observe that the scale separation $\frac{\rho\rho_2}{\rho r_2}$ satisfies $\frac{\rho\rho_2}{\rho r_2} = \frac{\rho_2}{r_2} \leq \rho^{\eta_1} \leq \delta^{\eta_1\varepsilon} \leq \delta^\eta$, provided we select η sufficiently small. But then (14.5.59) contradicts (14.5.51). Hence Conclusion (D) cannot hold.

Step 5. At this point, we shown that Conclusions (A), (C), and (D) from Proposition 14.5.28 cannot hold. Thus Conclusion (B) must hold. This completes the proof of Proposition 14.5.37. $\square$

14.5.5.2 Proof of Theorem 14.5.5

Our proof of Theorem 14.5.5 follows by combining two main ingredients. The first is Proposition 14.5.37. The second is a variant of the Stickyakeya Theorem, which was proved by the authors in [16]. The precise statement we need is as follows.

Theorem 14.5.38. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $\mathbb{T}$ be a set of δ -tubes that satisfy the Convex Wolff Axioms at every scale with error $\delta^{-\eta}$, and let $Y(T)$ be a δ^η dense shading. Then*

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon. \tag{14.5.60}$$

As noted above, Theorem 14.5.38 is similar to the Stickyakeya Theorem proved by the authors in [16]. In Section 14.5.6 we will discuss the proof of

Theorem 14.5.38, and how it differs from the arguments in [16]. Assuming this result for now, we can combine these two ingredients to prove Theorem 14.5.5 as follows

Proof of Theorem 14.5.5. It suffices to show that $\omega = 0$. Suppose instead that $\omega > 0$. Let η_0, δ_0 be the output from Theorem 14.5.38, with $\varepsilon = \omega/3$. By Proposition 14.5.37, there exists $\delta \in (0, \delta_0]$ and a set $(\mathbb{T}', Y)_\delta$ with discretized Assouad dimension $\leq 3 - \omega/3$ that satisfies the self-similar Convex Wolff Axioms (and hence the Convex Wolff Axioms at every scale) with error $\delta^{-\eta_0}$, and for which $\{Y(T)\}$ is a δ^{η_0} dense shading. But then (14.5.60) says that $\bigcup_{\mathbb{T}} Y(T)$ has discretized Assouad dimension at least $3 - \omega/3$, which is a contradiction. $\square$

Thus we have completed the proof of Theorem 14.5.5, except that it remains to prove Theorem 14.5.38; we will do so in the next section.

14.5.6 Keakeya estimates for tubes satisfying the Convex Wolff Axioms at every scale

Our goal in this section is to prove Theorem 14.5.38. Theorem 14.5.38 is closely related to the discretized analogue of [16, Theorem 1.1], and specifically the estimate $\sigma_3 = 0$ from that paper; the latter result says that if $\mathbb{T}$ is a collection of δ -tubes in $\mathbb{R}^3$ of cardinality roughly δ^{-2} that point in δ -separated directions, and if $\mathbb{T}$ satisfies a property called “stickiness,” then the union of the tubes must have volume $\gtrsim 1$. Stickiness is a technical property, which roughly speaking says that for each $\rho \in [\delta, 1]$, the tubes in $\mathbb{T}$ can be covered by a collection of ρ -tubes that point in ρ -separated directions (more accurately, the tubes point in ρ -separated directions up to multiplicity $\lesssim 1$).

Stickiness is a multi-scale property, in the following sense: if $\mathbb{T}$ is sticky, then for each $\rho \geq \delta$ we can cover $\mathbb{T}$ by a sticky collection of ρ -tubes. This multi-scale property was exploited throughout the arguments in [16]. Theorem 14.5.38 differs from [16, Theorem 1.1], because the stickiness hypothesis has been replaced by the hypothesis that the tubes satisfy the Convex Wolff axioms at every scale. As we will see below, however, this latter hypothesis is sufficient in order to repeat the arguments from [16]. Indeed, with two exceptions that will be explained below, the arguments from [16] can be repeated mutatis mutandis to obtain Theorem 14.5.38.

The proof of [16, Theorem 1.1] is divided into several sections. The paper begins by supposing that the result is false, and studying the structure of a (hypothetical) counter-example. In Section 2, the authors define what it means for a pair $(\mathbb{T}, Y)$ of tubes and their associated shading to be a worst possible counter-example to [16, Theorem 1.1]; this is called an ε -extremal pair, and we will adopt this terminology in our discussion below. In [16],

Section 3, it is shown that an ε -extremal pair $(\mathbb{T}, Y)$ must look coarsely self similar at every intermediate scale $\rho \in [\delta, 1]$. More precisely, for every such ρ it is possible to cover the tubes in $\mathbb{T}$ by a collection of ρ -tubes, so that the corresponding collection of ρ -tubes is a worst possible counter-example to [16, Theorem 1.1] at scale ρ , and the δ -tubes inside each ρ -tube also form a (rescaled) worst-possible counter-example at scale δ/ρ . The complete proof needed here is obtained below from Proposition 14.2.14, with its covering input replaced by Lemma 14.5.20.

In Sections 3, 4, and 5 of [16], the authors use the multi-scale self-similarity established in Section 2 of [16] to show that an extremal pair $(\mathbb{T}, Y)$ must have certain structural properties called planiness and graininess, and in particular there must exist local grains that are described by a Lipschitz plane map, and global grains that are described by a C^2 slope function f ; this will be described in greater detail below. The arguments from [16] can be repeated without modification to establish the same conclusions in the present setting. In particular, the hypothesis from [16] that the tubes point in δ -separated directions has not been used up until this step.

In Section 6 of [16], the authors show that the derivative f' of the slope function has magnitude roughly 1 (and in particular, this quantity is bounded away from 0). It is during this step that the authors use the hypothesis that the tubes point in δ -separated directions. With some small modifications, we can instead use the hypothesis that the tubes satisfy the Convex Wolff axioms at every scale; we will elaborate on this step in Section 14.5.6.4.

Finally, in Section 7 of [16], the authors recall an analogue of Wolff’s circular maximal function estimate [19, 14], and they show how this estimate contradicts the assumption that there exists a collection of sticky tubes whose union has volume much smaller than 1, thereby completing the proof of the theorem. This section of [16] uses the assumption that the tubes point in different directions in order to obtain a rich planar arrangement of curves to which Wolff’s circular maximal theorem can be applied. With some modifications, the direction-separated assumption can be replaced with a weaker ball condition. We will explain this in detail in Sections 14.5.6.5 and 14.5.7.

14.5.6.1 Relevant definitions from [16]

Before describing the arguments in [16], it will be helpful to recall a few key definitions. First, a minor technical annoyance: tubes and cubes in [16] are defined slightly differently than in the present paper. In [16], a δ -tube in $\mathbb{R}^3$ is defined to be a set of the form $N_{6\delta}(\ell) \cap [-1, 1]^3$, where ℓ is a line in $\mathbb{R}^3$. This definition is adopted for technical reasons, and for consistency with the results in [16] we will use this definition throughout Section 14.5.6; the distinction between this definition and our previous definition in Section 14.5.1 is not

important in the arguments that follow. Second, in [16], a shading $Y(T) \subset T$ is defined to be a union of axis-aligned δ -cubes contained in T . Again, the distinction between this definition and our previous one from Section 14.5.2.1 is harmless: replace each tube T (in the sense of Section 14.5.1) and a each shading $Y(T)$ (in the sense of Section 14.5.2.1) with the corresponding tube in the sense of [16] and the shadings consisting of the union of axis-aligned δ -cubes Q that satisfy $|Q \cap Y(T)| \geq \frac{1}{100}|Y(T)|/|T|$. This gives us a collection of tubes and shadings in the sense of [16], and if the conclusion of Theorem 14.5.5 holds for this latter collection of tubes and shadings, then it also holds for the original collection of tubes and shadings, except Inequality (14.5.1) has been weakened by an additional (harmless) factor of $1/1000$.

Tubes, covers, rescaling. We will often refer to a collection of δ -tubes $\mathbb{T}$, and a shading $Y(T)$ of these tubes. We will refer to this pair as $(\mathbb{T}, Y)_\delta$. We use $E_\mathbb{T}$ to denote the set $\bigcup_{T \in \mathbb{T}} Y(T)$; the shading Y will always be apparent from context. For $p \in \mathbb{R}^3$, we define

$$\mathbb{T}(p) = \{T \in \mathbb{T} : p \in Y(T)\}. \quad (14.5.61)$$

We say a pair $(\mathbb{T}', Y')_\delta$ is a *refinement* of $(\mathbb{T}, Y)_\delta$ if $\mathbb{T}' \subset \mathbb{T}$; $Y'(T') \subset Y(T')$ for each $T \in \mathbb{T}'$; and $\sum_{T'} |Y'(T')| \geq C^{-1} |\log \delta|^{-C} \sum_{\mathbb{T}} |Y(T)|$ for some absolute constant C (in practice, $C = 100$ will always suffice).

Let $0 < \delta \leq \rho \leq 1$. In [16], the authors say that a ρ -tube $\tilde{T}$ *covers* a δ -tube T if their respective coaxial lines satisfy $d(\ell, \tilde{\ell}) \leq \rho/2$, where $d(\cdot, \cdot)$ is an appropriately chosen metric on the affine Grassmannian of lines in $\mathbb{R}^3$; see Section 2 of [16] for details. Note that if $\tilde{T}$ covers T , then $T \subset \tilde{T}$, and hence this definition is consistent with ours from Definition 14.5.17 (the definition in [16] is slightly stronger than Definition 14.5.17, but the distinction is harmless).

If $(\mathbb{T}, Y)_\delta$ and $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ are pairs of δ (resp. ρ) tubes and their associated shadings, then we say $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ *covers* $(\mathbb{T}, Y)_\delta$ if each $T \in \mathbb{T}$ is covered by at least one $\tilde{T} \in \tilde{\mathbb{T}}$, and for each $\tilde{T} \in \tilde{\mathbb{T}}$ covering T , we have $Y(T) \subset \tilde{Y}(\tilde{T})$. If this is the case, we write $\mathbb{T}[\tilde{T}]$ to denote the set of tubes in $\mathbb{T}$ covered by $\tilde{T}$. We say that $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is a *balanced cover* of $(\mathbb{T}, Y)_\delta$ if it is a cover, and in addition $|E_\mathbb{T} \cap Q|$ is the same for each ρ -cube $Q \subset E_{\tilde{\mathbb{T}}}$.

Let $(\mathbb{T}, Y)_\delta$ be a set of δ -tubes and their associated shading. Let $\tilde{T}$ be a ρ -tube, and suppose each tube in $\mathbb{T}$ is covered by $\tilde{T}$. We define a new pair $(\hat{\mathbb{T}}, \hat{Y})_{\delta/\rho}$, which we call the *unit rescaling of $(\mathbb{T}, Y)_\delta$ relative to $\tilde{T}$* . Informally, our definition is as follows: consider the linear transformation that sends the coaxial line of $\tilde{T}$ to the z axis, and dilates the x and y directions by ρ^{-1} ; the image of $\tilde{T}$ under this transformation is comparable to the unit ball. The tubes in $\hat{\mathbb{T}}$ are those (δ/ρ) -tubes whose coaxial lines are the images of the coaxial lines of the tubes in $\mathbb{T}$ under this transformation. The shadings $\hat{Y}(\hat{T})$ consist

of the union of axis-aligned δ/ρ -cubes that intersect the image of $Y(T)$ under the above transformation. See Section 3 of [16] for details. As above, the definition of unit rescaling in [16] is slightly different than Definition 14.5.15, but the distinction is harmless.

The plane map. In the arguments in [16], the authors show that certain collections of tubes possess an important structural property called planiness. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes and their associated shading. We say a function $V: E_{\mathbb{T}} \rightarrow S^2$ is a *plane map* for $(\mathbb{T}, Y)_\delta$ if

$$|\operatorname{dir}(T) \cdot V(p)| \leq \delta \quad \text{for all } (T, p) \in \mathbb{T} \times E_{\mathbb{T}} \text{ with } p \in Y(T),$$

where $\operatorname{dir}(T) \in S^2$ is a unit vector parallel to the line coaxial with T .

Discretized Ahlfors-David regular sets. We say a set $E \subset \mathbb{R}$ is a (δ, α, C) -ADset if for all $\rho \geq \delta$, all $r \geq \rho$, and all intervals I of length r , we have $\mathcal{E}_\rho(E \cap I) \leq C(r/\rho)^\alpha$, where $\mathcal{E}_\rho(E \cap I)$ means the ρ -covering number of $E \cap I$. In practice we will have $\alpha \in (0, 1]$, and C will be a small power of $1/\delta$. See Section 4 of [16] for further discussion and motivation for this definition.

14.5.6.2 Extremal collection of tubes, and multi-scale structure

We are ready to begin the proof of Theorem 14.5.38. Suppose to the contrary that the result was false. Then there exists $\sigma' > 0$ so that for all $\eta, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$; a set $\mathbb{T}$ of δ -tubes that satisfy the Convex Wolff Axioms at every scale with error $\delta^{-\eta}$; and a δ^η -dense shading $Y(T)$, so that

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{\sigma'}.$$

Let $\sigma > 0$ be the supremum of all numbers σ' with the above property; this supremum exists, since by hypothesis the set of admissible σ' is non-empty, and it is bounded above by 3. Our goal is to show that we must in fact have $\sigma = 0$, and thereby obtain a contradiction.

The following definition is the analogue of [16, Definition 3.1].

Definition 14.5.39. Let $\varepsilon, \delta > 0$. We say a pair $(\mathbb{T}, Y)_\delta$ is ε -*extremal* if the following holds:

1. $\mathbb{T}$ satisfies the Convex Wolff Axioms at all scales with error $\delta^{-\varepsilon}$.
2. Y is δ^ε -dense.
3. $\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \leq \delta^{\sigma - \varepsilon}$.

It is immediate from the above definitions that for every $\varepsilon, \delta_0 > 0$, there exists an ε -extremal collection of tubes $(\mathbb{T}, Y)_\delta$ for some $\delta \in (0, \delta_0]$; cf. [16, Lemma 3.1].

Our next task is to prove an analogue of [16, Proposition 3.2]. The precise statement is as follows.

Proposition 14.5.40. *For all $\varepsilon > 0$, there exists $\eta > 0$ and $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be an η -extremal collection of tubes, and let $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$. Then there is a refinement $(\mathbb{T}', Y')_\delta$ of $(\mathbb{T}, Y)_\delta$ and a balanced cover $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ of $(\mathbb{T}', Y')_\delta$ with the following properties:*

- (i) $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is ε -extremal.
- (ii) For each $\tilde{T} \in \tilde{\mathbb{T}}$, the unit rescaling of $(\mathbb{T}'[\tilde{T}], Y')_\delta$ relative to $\tilde{T}$ is ε -extremal.
- (iii) For each $p \in \mathbb{R}^3$, $\#\tilde{\mathbb{T}}(p) \leq \rho^{2-\sigma-\varepsilon}(\#\tilde{\mathbb{T}})$.
- (iv) For each $p \in \mathbb{R}^3$ and each $\tilde{T} \in \tilde{\mathbb{T}}$, $\#\mathbb{T}'[\tilde{T}](p) \leq (\delta/\rho)^{2-\sigma-\varepsilon}(\#\mathbb{T}'[\tilde{T}])$.

Proof. We apply the complete proof of Proposition 14.2.14, printed earlier in this volume, with one input replaced. That proof uses the direction-separated sticky hypothesis only in its first covering step. Here Lemma 14.5.20 supplies exactly that step: after a refinement it gives a cover whose coarse family and every rescaled fine fibre satisfy the Convex Wolff axioms at all scales, with a loss $\delta^{-O(\eta)}$. The rest of the proof of Proposition 14.2.14 consists solely of balanced shading selection, constant-multiplicity refinements, and the definition of the extremal exponent. Those operations do not use direction separation and apply verbatim to the present convex-Wolff extremal class.

We spell out the resulting parameter bookkeeping. We begin by reducing to the case where each $T \in \mathbb{T}$ satisfies $|Y(T)| \geq \frac{1}{2}\delta^\eta|T|$. This can be done by discarding those tubes with $|Y(T)| < \frac{1}{2}\delta^\eta|T|$ and then repeatedly pigeonholing to re-establish a collection of tubes that satisfies the Convex Wolff Axioms at every scale (with a slightly worse constant, say $\delta^{-2\eta}$); after this process, the cardinality of our collection of tubes has been reduced by at most a factor of δ^η .

Next, we apply Lemma 14.5.20 with $\rho_0 = \delta^{2\eta}\rho$. We get a number $\rho' \in [\delta^{2\eta}\rho, \rho]$ and a collection $\mathbb{T}_{\rho'}$ of ρ' -tubes, as described by that Lemma. We may replace each ρ' tube by its corresponding ρ -tube, and denote the resulting collection by $\tilde{\mathbb{T}}$. After pigeonholing to re-establish the property that the tubes are essentially distinct, it is still the case that $\tilde{\mathbb{T}}$ satisfies the Convex Wolff Axioms at every scale with error $\delta^{-O(\eta)}$, and each of the rescaled collections $\mathbb{T}[\tilde{T}]$ satisfies the Convex Wolff Axioms at every scale with error $\delta^{-O(\eta)}$.

After refining the shadings $Y(T)$, we may suppose that for each collection $\mathbb{T}[\tilde{T}]$, the same number of tubes pass through each δ -cube in $E_{\mathbb{T}[\tilde{T}]}$ in the sense

of (14.5.61); after further pigeonholing we may suppose that this number is the same (up to a multiplicative factor of 2) for each set $\mathbb{T}[\tilde{T}]$; call this number μ_{fine} . Since $\mathbb{T}[\tilde{T}]$ is a (rescaled) collection of tubes satisfying the Convex Wolff Axioms at every scale, if η and δ_0 are chosen sufficiently small then we must have $\mu_{\text{fine}} \leq (\delta/\rho)^{2-\sigma-\varepsilon/10}(\#\mathbb{T}'[\tilde{T}])$; this is Conclusion (iv) above.

Our next task is to define a shading $\tilde{Y}$ on the tubes of $\tilde{\mathbb{T}}$. To begin, for each $\tilde{T}$ we consider the union of all ρ -cubes $\tilde{Q} \subset \tilde{T}$ that intersect $E_{\mathbb{T}[\tilde{T}]}$. After pigeonholing, we can refine this shading so that each ρ -cube intersects $E_{\mathbb{T}[\tilde{T}]}$ in a set of roughly the same size, and the same number of tubes from $\tilde{\mathbb{T}}$ pass through each cube of $E_{\tilde{\mathbb{T}}}$; call this number μ_{coarse} . We refine the corresponding shadings $Y(T)$, $T \in \mathbb{T}$ so that $Y(T) \subset \tilde{Y}(\tilde{T})$ whenever $T \in \mathbb{T}[\tilde{T}]$; in particular $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is now a balanced cover of $(\mathbb{T}, Y)_\delta$. Note that Conclusion (iv) remains true after this procedure. Since $\tilde{\mathbb{T}}$ satisfies the Convex Wolff Axioms with error $\delta^{-O(\eta)}$, if η and δ_0 are chosen sufficiently small then we must have $\mu_{\text{coarse}} \leq \rho^{2-\sigma-\varepsilon/10}(\#\tilde{\mathbb{T}})$; this is Conclusion (iii) above.

Finally, observe that $\mu_{\text{fine}}\mu_{\text{coarse}} \leq \delta^{2-\sigma-\varepsilon/5}(\#\mathbb{T})$, but on the other hand since $\mathbb{T}$ was η -extremal, we must have $\mu_{\text{fine}}\mu_{\text{coarse}} \gtrsim \delta^{2-\sigma+\eta}(\#\mathbb{T})$. We conclude that

$$\begin{aligned} (\delta/\rho)^{2-\sigma+\varepsilon/10}(\#\mathbb{T}'[\tilde{T}]) &\leq \mu_{\text{fine}} \leq (\delta/\rho)^{2-\sigma-\varepsilon/10}(\#\mathbb{T}'[\tilde{T}]), \quad \tilde{T} \in \tilde{\mathbb{T}}, \\ \rho^{2-\sigma+\varepsilon/10}(\#\tilde{\mathbb{T}}) &\leq \mu_{\text{coarse}} \leq \rho^{2-\sigma-\varepsilon/10}(\#\tilde{\mathbb{T}}), \end{aligned}$$

and in particular Conclusions (i) and (ii) are satisfied. To match the numbering with Proposition 14.2.14, note that the Convex Wolff axioms give $\#\tilde{\mathbb{T}} \lesssim \rho^{-2}$ and $\#\mathbb{T}'[\tilde{T}] \lesssim (\rho/\delta)^2$. Thus its absolute bounds $\#\tilde{\mathbb{T}}(p) \leq \rho^{-\sigma-O(\varepsilon)}$ and $\#\mathbb{T}'[\tilde{T}](p) \leq (\rho/\delta)^{\sigma+O(\varepsilon)}$ are exactly (iii) and (iv) after division by these cardinalities. Replacing the input epsilon there by $\varepsilon/20$ absorbs every displayed $O(\varepsilon)$ loss. This verifies all hypotheses and all outputs of the earlier complete proof, and completes the present proposition. $\square$

14.5.6.3 Local and global grains

Using Proposition 14.5.40 in place of [16, Proposition 3.2], we can follow an identical argument as in [16] to obtain the analogue of [16, Proposition 4.1], which we recall here.

Proposition 14.5.41. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the following two properties:*

1. *$E_{\mathbb{T}}$ is a union of global grains with Lipschitz slope function.*

There is a 1-Lipschitz function $f : [-1, 1] \rightarrow \mathbb{R}$ so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.

2. $E_{\mathbb{T}}$ is a union of local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_{\delta}$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

Proof. The complete proof of Proposition 14.2.22, printed earlier in this volume, applies with Proposition 14.5.40 in place of Proposition 14.2.14. We verify the transfer rather than appealing to an external source. After the respective notation is identified, both propositions provide: a balanced coarse cover at every $\rho \in [\delta^{1-\epsilon}, \delta^{\epsilon}]$; extremality of the coarse family; extremality of every unit-rescaled fine fibre; and the same coarse and fine pointwise multiplicity bounds. The apparent difference between $\rho^{-\sigma}$ and $\rho^{2-\sigma} \# \tilde{\mathbb{T}}$ disappears because the Convex Wolff axioms give $\# \tilde{\mathbb{T}} \lesssim \rho^{-2}$, and similarly at the fine scale.

Inspecting the proof of Proposition 14.2.22, every subsequent step uses only those four outputs, the multilinear Takeya theorem 14.2.25, dyadic refinements, and affine rescaling. In particular, its plane-map construction, Lipschitz selection, local grain estimate, and global slice argument do not use direction separation. Run those displayed arguments with $E_{\mathbb{T}}$ and the present extremal exponent σ . They produce respectively the plane map and the global slope function asserted in items 2 and 1 above. Choosing the input error there to be ϵ/C for the finite bookkeeping constant C , and then taking η sufficiently small in Proposition 14.5.40, converts all $\delta^{-O(\eta)}$ losses into $\delta^{-\epsilon}$. These are exactly the two conclusions stated here. $\square$

We next upgrade the global grains slope function f in Proposition 14.5.41 from Lipschitz to C^2 . The proof is deduced below from the complete radial-projection argument already printed in this volume. The precise statement is as follows.

Proposition 14.5.42. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε -extremal set of tubes $(\mathbb{T}, Y)_{\delta}$ with the following properties:*

1' Global grains with C^2 -slope function.

There is a function $f : [-1, 1] \rightarrow \mathbb{R}$ with $\|f\|_{C^2} \leq 1$ so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.

2 Local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_{\delta}$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

Proof. Apply the complete proof of Proposition 14.2.36, with Proposition 14.5.41 in place of Proposition 14.2.22. The hypotheses match term for term: the same Lipschitz global slope function, the same one-dimensional AD bounds in each horizontal slice, and the same Lipschitz local plane map are

available. That proof applies the internally proved nested-rectangle corollary and the C^2 interpolation proposition to obtain a function g with $\|g\|_{C^2} \lesssim \delta^{-3\epsilon_2}$. It then restricts to a slab of height $\delta^{3\epsilon_2}$ and rescales vertically. Linear images and preimages of convex sets are convex, so the defining Convex Wolff estimates are preserved under this bounded-power affine rescaling, with a $\delta^{-O(\epsilon_2)}$ loss. The local plane-map conclusion is preserved by the same change of variables. Choosing $\epsilon_2 \ll \epsilon$ and the input error in Proposition 14.5.41 still smaller gives $\|g\|_{C^2} \leq 1$, an ϵ -extremal output, and precisely items 1' and 2. $\square$

14.5.6.4 The slope function has large slope

Our next task is to show that the global grains slope function f in Proposition 14.5.41 has derivative f' with magnitude bounded away from 0. As discussed above, the proof of the analogous statement [16, Proposition 6.1] used the property that $\mathbb{T}$ contains tubes pointing in many different directions. Here, we will use the property that $\mathbb{T}$ satisfies the Convex Wolff Axioms at every scale. The precise statement we need is as follows.

Proposition 14.5.43. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε -extremal set of tubes $(\mathbb{T}, Y)_\delta$ with the following properties:*

1'' Global grains with nonsingular C^2 -slope function. *There is a function $f : [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$, so that $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is a $(\delta, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset for each $z_0 \in [-1, 1]$.*

2 Local grains with Lipschitz plane map.

$(\mathbb{T}, Y)_\delta$ has a 1-Lipschitz plane map V . For all $\rho \in [\delta, 1]$ and all $p \in \mathbb{R}^3$, $V(p) \cdot (B(p, \rho^{1/2}) \cap E_{\mathbb{T}})$ is a $(\rho, 1 - \sigma, \delta^{-\varepsilon})_1$ -ADset.

Before proving 14.5.43, we will establish the following technical result, which is the analogue of [16, Lemma 6.3].

Lemma 14.5.44. *For all $\varepsilon > 0$, there exists $\eta > 0, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of tubes that satisfies the conclusions of Proposition 14.5.42, with η in place of ε . Let $J \subset [-1, 1]$ be an interval, with $\delta^{1/2} \leq |J| \leq \delta^\varepsilon$. Suppose that*

$$\sum_{T \in \mathbb{T}} |Y(T) \cap (\mathbb{R}^2 \times J)| \geq \delta^\eta |J|, \quad (14.5.62)$$

and that $(\mathbb{R}^2 \times J) \cap \bigcup_{\mathbb{T}} Y(T)$ can be covered by a union of $\leq \delta^{-\eta} |J|^{-2+\sigma}$ cubes of side-length J .

Then the slope function $f : [-1, 1] \rightarrow \mathbb{R}$ from Proposition 14.5.42 satisfies

$$|f'(z)| \geq |J| \text{ for all } z \in J.$$

Proof. We use Steps 1–3 of the complete proof of Lemma 14.2.48, printed earlier in this volume. Those steps use only the two grain conclusions of Proposition 14.5.42, the density and covering hypotheses displayed above, Fubini, and dyadic refinement; direction separation enters only in Step 4 of that earlier proof. For clarity, we record every output of those steps in the present notation. After a refinement, we may suppose that for each point $p \in F := (\mathbb{R}^2 \times J) \cap \bigcup_{\mathbb{T}} Y(T)$,

$$\delta^{2-\sigma+2\eta}(\#\mathbb{T}) \leq \#\mathbb{T}(p) \leq \delta^{2-\sigma-2\eta}(\#\mathbb{T}). \quad (14.5.63)$$

After a translation followed by a rotation that fixes the z axis (i.e. a rotation of the form $(x, y, z) \mapsto (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z)$), we may suppose that $J = [0, \rho^{1/2}]$ and $f(0) = 0$.

Suppose for contradiction that $|f'(z)| \leq |J|$ for some $z \in J$. Since $|f''(z)| \lesssim 1$, we have $|f'(z)| \lesssim |J|$ for all $z \in J$, and thus $|f(z)| \lesssim |J|^2$. We find a subset $F_2 \subset F$ with $|F_2| \geq \delta^{4\eta}|F|$ so that the follows holds:

1. For each $z_1 \in J$ and each $r \in (F \cap \{z = z_1\}) \cdot (1, f(z_1), 0)$, the $1 \times \delta \times \delta$ -rectangular prism

$$G_{r,z_1} = \{|z - z_1| \leq \delta\} \cap \{|(x, y) \cdot (1, f(z_1)) - r| \leq \delta\} \quad (14.5.64)$$

has a large intersection with F_2 , i.e.

$$|G_{r,z_1} \cap F_2| \geq \delta^{4\eta}|G_{r,z_1}| \geq \delta^{2+4\eta}. \quad (14.5.65)$$

2. Let $\Pi_{x,z} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection to the first and third coordinates, then

$$|N_{\rho^{1/2}}\Pi_{x,z}(F_2)| \leq \delta^{-7\eta}\rho^{1/2+\sigma/2}.$$

3. There is a $y_0 \in [-1, 1]$ such that

$$|\{y = y_0\} \cap F_2| \geq |F_2|/2,$$

where the $|\cdot|$ on the LHS denote 2-dimensional measure and $|\cdot|$ on the RHS denote 3-dimensional measure, and each G_{r,z_1} defined in (14.5.64) satisfies

$$G_{r,z_1} \cap \{y = y_0\} \cap F_2 \neq \emptyset.$$

Cover $[-1, 1] \times \{y = y_0\} \times J$ by a set $\mathcal{S}$ of interior-disjoint squares of side-length $\rho^{1/2} = |J|$. Let $S \in \mathcal{S}$ be a square with center $p_S = (x_S, y_0, \rho^{1/2}/2)$. By Item 2 from Proposition 14.5.42, $S \cap F_2 \cdot V(p_S)$ is a $(\rho, 1 - \sigma, \delta^{-\eta})_1$ -ADset. Write $V(p_S) = (V_x(p_S), V_y(p_S), V_z(p_S))$ and define $\tilde{V}(p_S) = (V_x(p_S), 0, V_z(p_S))$. Since $S \cap F_2 \subset \{y = y_0\}$,

$$(S \cap F_2) \cdot \tilde{V}(p_S) \text{ is a } (\rho, 1 - \sigma, \delta^{-\eta})_1\text{-ADset.} \quad (14.5.66)$$

Define $P_S = \{(x, y, z) : (x, z) \in S\}$, then

$$(F_2 \cap P_S) \cdot \tilde{V}(p_S) \text{ is a } (\rho, 1 - \sigma, 2\delta^{-\eta})_1\text{-ADset.}$$

This completes Steps 1–3 and introduces all geometric objects used below.

In Step 4, we will show that the constraints from the above steps will force many tubes from $\mathbb{T}$ to be contained in a $\delta^{-2C_0\eta}\rho^{1/2} \times 1 \times 1$ rectangular prism (henceforth called a *slab*, to match the terminology in [16]) for some constant $C_0 = C_0(\sigma)$, which contradicts the assumption that $\mathbb{T}$ satisfies the Convex Wolff Axioms (see Definition 14.5.39 and Definition 14.5.12).

For each $T \in \mathbb{T}$, define $F_2(T) = Y(T) \cap F_2$. Recalling (14.5.62) and (14.5.63), we have

$$\sum_{T \in \mathbb{T}} |F_2(T)| \geq \delta^{10\eta} |J| = \delta^{10\eta} \rho^{1/2}. \quad (14.5.67)$$

Recall that each $T \in \mathbb{T}$ intersects at most 3 prisms P_S , $S \in \mathcal{S}$ and for each such prism we have $|F_2(T) \cap P_S| \leq \rho^{1/2}\delta^2$. By pigeonholing, there exists a square $S \in \mathcal{S}$ and a set $\mathbb{T}_S = \{T \in \mathbb{T} : T \cap P_S \neq \emptyset\}$ with

$$\#\mathbb{T}_S \gtrsim (\#\mathbb{T})(\#\mathcal{S})^{-1} \gtrsim \delta^\eta \rho^{-\sigma/2+1/2} (\#\mathbb{T}). \quad (14.5.68)$$

But for each $T \in \mathbb{T}_S$, we have

$$F_2(T) \cdot \tilde{V}(p_S) \subset (F_2 \cap P_S) \cdot \tilde{V}(p_S) \text{ is a } (\rho, 1 - \sigma, 2\delta^{-\eta})_1\text{-ADset.} \quad (14.5.69)$$

Recall that for each $T \in \mathbb{T}_S$, the (shaded) tube segment $F_2(T)$ is almost full. If $\tilde{V}(p_S)$ was parallel to $\text{dir}(T)$, then $F_2(T) \cdot \tilde{V}(p_S)$ would fill out most of an interval of length $\rho^{1/2}$. However, (14.5.69) says that $F_2(T) \cdot \tilde{V}(p_S)$ is an ADset with dimension $1 - \sigma$. Apply the internally proved Lemma 14.2.47, with $\alpha = 1 - \sigma$ and with the density loss from the refinement absorbed into $C_0\eta$. It gives

$$|\text{dir}(T) \cdot \tilde{V}(p_S)| \leq \delta^{-C_0\eta} \rho^{1/2}. \quad (14.5.70)$$

It follows that $\mathbb{T}_S$ is contained in the slab

$$\Sigma_S = \{(x, y, z) \in [0, 1]^3 : |(x - x_S, 0, z) \cdot \tilde{V}(p_S)| \leq \delta^{-2C_0\eta} \rho^{1/2}\}.$$

By the Convex Wolff Axioms,

$$\#\mathbb{T}_S \leq |\Sigma_S| \delta^{-\eta} (\#\mathbb{T}) \leq \delta^{-(2C+1)\eta} \rho^{1/2} (\#\mathbb{T}),$$

which yields a contradiction to (14.5.68). □

Proof of Proposition 14.5.43. We first repeat the stopping argument in the proof of the internally proved Lemma 14.2.49. Start with the output of Proposition 14.5.42 with error ϵ_1 . Apply Proposition 14.5.40 at $\rho = \delta^{\epsilon_2}$, pigeonhole a horizontal interval J of that length carrying at least a $\delta^{O(\epsilon_2)}$ fraction of the total shaded mass, and retain the corresponding shading. The balanced-cover conclusion supplies a cover of the part over J by at most $\delta^{-O(\epsilon_1)}|J|^{-2+\sigma}$ cubes of side $|J|$. Therefore Lemma 14.5.44 applies, provided $\epsilon_1 \ll \epsilon_2$, and yields $|f'| \geq |J|$ throughout J . Since $\|f\|_{C^2} \leq 1$, dyadic pigeonholing gives a number $m \geq \delta^{O(\epsilon_2)}$ and a subinterval, still denoted J , on which $m \leq |f'| \leq 2m$ and which retains a power-sized portion of the shading.

Translate J to start at zero and apply exactly the affine normalization in the proof of Proposition 14.2.46:

$$\Phi(x, y, z) = (x, m^2 y/100, 100z/m), \quad \tilde{f}(z) = 100m^{-2}f(mz/100).$$

The chain rule gives $1 \leq |\tilde{f}'| \leq 2$ and $|\tilde{f}''| \leq 1/100$; extend $\tilde{f}$ outside the rescaled interval with the same bounds as in that earlier proof. The identity

$$x + yf(z) = \Phi(x, y, z) \cdot (1, \tilde{f}(100z/m), 0)$$

transfers the global-grain statement. Affine images of convex sets are convex, and the singular values of Φ are bounded by fixed powers of $\delta^{-\epsilon_2}$, so the Convex Wolff axioms, density, extremality, and the local plane-map statement survive with a $\delta^{-O(\epsilon_2)}$ loss. Taking $\epsilon_2 \ll \epsilon$ and then $\epsilon_1 \ll \epsilon_2$ absorbs this loss and proves both conclusions. $\square$

14.5.6.5 Twisted projections, and the completion of the proof

Recalling Definition 6.1 from [16], for $f: [-1, 1] \rightarrow \mathbb{R}$ we define the twisted projection $\pi_f(x, y, z) = (x + f(z)y, z)$. The consequence of Proposition 14.5.43 is the following analogue of [16, Corollary 6.5]:

Lemma 14.5.45. *For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$; a pair $(\mathbb{T}, Y)$ of δ -tubes satisfying the Convex Wolff Axioms with error $\delta^{-\varepsilon}$ with a δ^ε -dense shading; and a function $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$, so that*

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \leq \delta^{\sigma-\varepsilon}. \quad (14.5.71)$$

Proof. Take the pair and slope function supplied by Proposition 14.5.43 with an auxiliary error $\eta \ll \varepsilon$. For fixed z , its global-grain conclusion says that

$$A_z = (E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$$

is a $(\delta, 1 - \sigma, \delta^{-\eta})_1$ -AD set. With $r \sim 1$ and $\rho = \delta$ in the AD estimate, $\mathcal{E}_\delta(A_z) \lesssim \delta^{-1+\sigma-\eta}$. Because $E_{\mathbb{T}}$ is a union of δ -cubes, A_z is contained in the

$O(\delta)$ -neighbourhood of that many points, and hence $|A_z| \lesssim \delta^{\sigma-O(\eta)}$. Fubini in z gives

$$|\pi_f(E_{\mathbb{T}})| \lesssim \delta^{\sigma-O(\eta)} \text{leq} \delta^{\sigma-\epsilon}$$

after choosing η small and then δ_0 . Since $\bigcup_T \pi_f(Y(T)) = \pi_f(E_{\mathbb{T}})$, this is the desired estimate. The output with auxiliary error η also satisfies the weaker $\delta^{-\epsilon}$ Convex Wolff and δ^ϵ density requirements. $\square$

To complete the proof of Theorem 14.5.38, it suffices to establish the following estimate, which is the analogue of [16, Proposition 7.1]:

Proposition 14.5.46. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of δ -tubes satisfying the Convex Wolff Axioms with error $\delta^{-\eta}$, and let Y be a δ^η -dense shading. Let $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$. Then*

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \geq \delta^\varepsilon.$$

If $\sigma > 0$, choose $\epsilon < \sigma/3$. Lemma 14.5.45 gives an upper bound $\delta^{\sigma-\epsilon}$, while Proposition 14.5.46 gives the lower bound δ^ϵ ; these are incompatible for sufficiently small δ . Hence $\sigma = 0$, which completes the proof of Theorem 14.5.38. It remains to prove Proposition 14.5.46, which is done in the next section.

14.5.7 Twisted projections of tubes satisfying the Convex Wolff Axioms

Our goal in this section is to prove Proposition 14.5.46. In fact, we will prove a slightly stronger statement. We say a set of tubes $\mathbb{T}$ satisfies the *Tube Wolff Axioms* with error C , if for every ρ -tube T_ρ , at most $C|T_\rho|(\#\mathbb{T})$ tubes from $\mathbb{T}$ are contained in T_ρ ; since ρ -tubes are convex, this is a special case of the Convex Wolff Axioms. We will prove the following mild generalization of Proposition 14.5.46:

Proposition 14.5.47. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of δ -tubes satisfying the Tube Wolff Axioms with error $\delta^{-\eta}$, and let Y be a δ^η -dense shading. Let $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$. Then*

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \geq \delta^\varepsilon.$$

Before proving Proposition 14.5.47, we will need to introduce two closely related non-concentration conditions for subsets of $\mathbb{R}^n$, which arise from discretizing fractal sets. The first was proposed by Katz and Tao [8], while the second was explored by Orponen and Shmerkin [12].

Definition 14.5.48. Let $\delta > 0$, $s \in (0, n]$, and $C > 0$. A non-empty bounded set $A \subset \mathbb{R}^n$ is called a (δ, s, C) -Katz-Tao set if

$$\mathcal{E}_\delta(A \cap B(x, r)) \leq C(\delta/r)^s, \quad \forall x \in \mathbb{R}^n, r \in [\delta, 1].$$

Definition 14.5.49. Let $\delta > 0$, $s \in (0, n]$, and $C > 0$. A non-empty bounded set $A \subset \mathbb{R}^n$ is called a (δ, s, C) -Frostman set if

$$\mathcal{E}_\delta(A \cap B(x, r)) \leq Cr^s \mathcal{E}_\delta(A), \quad \forall x \in \mathbb{R}^n, r \in [\delta, 1].$$

Remark 14.5.50. If $\mathbb{T}$ is a set of δ -tubes contained in a bounded set such as $B(0, 1)$, then we can identify the coaxial line of each tube with a point in $\mathbb{R}^4$. If $\mathbb{T}$ satisfies the Tube Wolff Axioms, then the corresponding set of points in $\mathbb{R}^4$ forms a $(\delta, 2, O(1))$ -Frostman set.

Remark 14.5.51. If $A \subset \mathbb{R}^n$ is a (δ, s, C) -Katz-Tao set (resp. (δ, s, C) -Frostman set), then $N_\delta(A)$ is a (δ, s, C') -Katz-Tao set (resp. (δ, s, C') -Frostman set), where $C' = O_n(C)$, and conversely.

Lemma 14.5.52. Let $A \subset \mathbb{R}^n$ be a (δ, s, C) -Frostman set. Then there exists a $(\delta, s, 100)$ -Katz-Tao set $A' \subset A$, with $\mathcal{E}_\delta(A') \gtrsim_{n,s} \delta^{-s} |\log \delta|^{-1} C^{-1}$.

Proof. Recall that the s -dimensional Hausdorff content is defined as

$$\mathcal{H}_\infty^s(A) := \inf \left\{ \sum_i r(B_i)^s : A \subset \cup_i B_i \right\},$$

where each B_i in the covering is a cube and $r(B_i)$ is the side-length of B_i . Next we recall Lemma 3.13 from [3]

Lemma 14.5.53. Let $\delta, s > 0$ and suppose that $\mathcal{H}_\infty^s(A) = \kappa > 0$. Then there exists a $(\delta, s, 100)$ -Katz-Tao set $A' \subset A$ with $\mathcal{E}_\delta(A') \gtrsim \kappa \delta^{-s}$.

(In [3], the authors do not explicitly state the constant $C = 100$, but this follows from their proof; any fixed constant would work equally well for our argument).

In light of Lemma 14.5.53, in order to prove Lemma 14.5.52, it suffices to obtain the bound

$$\mathcal{H}_\infty^s(A) \gtrsim_{n,s} |\log \delta|^{-1} C^{-1}. \quad (14.5.72)$$

The remainder of the proof is devoted to establishing (14.5.72). By Remark 14.5.51, we can assume A is a union of δ -balls (this introduces a harmless

$O_n(1)$ loss). Let $\{B_i\}$ be a cover of A by balls. Sort the balls according to their side-length; it suffices to consider balls of side-length $\geq \delta$ since A is a union of δ -balls. By dyadic pigeonholing, there exists a dyadic number $\lambda \in [\delta, 1]$ such that

$$\mathcal{E}_\delta\left(\bigcup_{r(B_i) \sim \lambda} B_i \cap A\right) \gtrsim |\log \delta|^{-1} \mathcal{E}_\delta(A).$$

Since A is a (δ, s, C) -Frostman set, $\mathcal{E}_\delta(A \cap B_i) \leq C\lambda^s \mathcal{E}_\delta(A)$. The number of balls B_i in the cover with $r(B_i) \sim \lambda$ is $\gtrsim_{n,s} \lambda^{-s} |\log \delta|^{-1} C^{-1}$. This establishes (14.5.72) and concludes the proof. $\square$

The following definition describes when a discretized set is homogeneous at many different scales.

Definition 14.5.54 (η -uniform set down to scale δ). Let $\eta > 0$ be a small parameter. A set $A \subset [0, 1]^n$ is called an η -uniform set up to scale δ if for any $\rho \in 2^{-\eta^{-1}\mathbb{N}} \cap [\delta, 1]$, if two ρ -cubes $Q_1, Q_2 \subset N_\rho(A)$, then the δ -covering numbers are comparable, in the sense that

$$\frac{1}{100} \mathcal{E}_\delta(Q_1 \cap A) \leq \mathcal{E}_\delta(Q_2 \cap A) \leq 100 \mathcal{E}_\delta(Q_1 \cap A).$$

Let $\eta > 0$ be a small parameter and $\delta \in 2^{-\eta^{-1}\mathbb{N}}$. Given an arbitrary set $A \subset [0, 1]^n$, [13, Lemma 2.15] says there exists an η uniform set $A' \subset A$ down to scale δ and $\mathcal{E}_\delta(A') \gtrsim |\log \delta|^{-\eta^{-1}} \mathcal{E}_\delta(A)$. We call A' an η -uniform refinement (usually δ is clear from context and we omit it from notation).

Even though [13, Lemma 2.15] is for dyadic cubes, the same argument works for any set A and the relevant ρ -balls $Q \subset N_\rho(A)$, which we restate in our language below.

Lemma 14.5.55 (Lemma 2.15 from [13] in our notation). *Let $A \subset [0, 1]^n$ and let $\eta > 0, T \in \mathbb{N}$, $\delta = 2^{-\eta^{-1}T}$. Then there exists an η -uniform refinement $A' \subset A$ such that*

$$\mathcal{E}_\delta(A') \geq (2T)^{-\eta^{-1}} \mathcal{E}_\delta(A).$$

In particular, if $\varepsilon > 0$ and $T^{-1} \log(2T) \leq \varepsilon$, then $\mathcal{E}_\delta(A') \geq \delta^\varepsilon \mathcal{E}_\delta(A)$.

Given an η -uniform set A with δ -covering number δ^{-s} , the next lemma finds a Frostman-set if one zooms in at sufficiently small scale.

Lemma 14.5.56. *For any $\varepsilon > 0$, there exists $\eta > 0$ such that for any $\delta \in 2^{-\eta^{-1}\mathbb{N}}$, the following holds. Let $A \subset [0, 1]^n$ be an η -uniform set with $\mathcal{E}_\delta(A) \geq \delta^{-s}$. Then there exists $\rho \in (\delta^{1-\varepsilon}, 1]$ such for any ρ -cube $Q \in N_\rho(A)$,*

$$A_Q := S_Q(A \cap Q) \text{ is a } (\delta/\rho, s, (\delta/\rho)^{-4d\varepsilon})\text{-Frostman set,}$$

where $S_Q : Q \rightarrow [0, 1]^n$ is an affine transformation that maps Q to the unit cube.

Proof. Let $\eta \ll \varepsilon$ to be determined later. We shall apply the multi-scale decomposition techniques developed by Keleti and Shmerkin [11]. By [15, Lemma 5.21], there exists a decomposition of scales such that

- $a_1 = \eta^{1/2}$, $a_{J+1} = 1$, $a_{j+1} - a_j \geq \tau$ where $\tau = \tau(\varepsilon) > 0$ depends only on ε .
- For each δ^{a_j} -cube $Q \subset N_{\delta^{a_j}}(A)$, $A_Q := S_Q(A \cap Q)$ is a $(\delta^{a_{j+1}-a_j}, t_j, \delta^{-\eta})$ -set, where $S_Q : Q \rightarrow [0, 1]^3$ is an affine transformation that maps Q to the unit cube.
- $\sum t_j(a_{j+1} - a_j) \geq s - \varepsilon^2$.

In principle there is no control on the values of t_j . However, we can fix this as follows. The “Merging Lemma,” [15, Lemma 5.20] says that if $t_j \geq t_{j+1}$, then for each δ^{a_j} -cube $Q \subset N_{\delta^{a_j}}(A)$, $A_Q := S_Q(A \cap Q)$ is a $(\delta^{a_{j+2}-a_j}, t'_j, \delta^{-\eta})$ -set, where $t'_j = \frac{(a_{j+1}-a_j)t_j + (a_{j+2}-a_{j+1})t_{j+1}}{a_{j+2}-a_j}$.

Start with the output of [15, Lemma 5.21] applied to A . If there exists $t_j \geq t_{j+1}$, apply [15, Lemma 5.20] to merge the two intervals $[a_j, a_{j+1}]$ and $[a_{j+1}, a_{j+2}]$ to $[a_j, a_{j+2}]$ and the two numbers t_j, t_{j+1} becomes t'_j . Rename the set of intervals and t_j s and replace J by $J-1$. Iterate the process until $t_j < t_{j+1}$ for all j . Now we have obtained a decomposition of scales $\{[a_j, a_{j+1}]\}_{j=1}^J$ such that

- $a_1 = \eta^{1/2}$, $a_{J+1} = 1$, $a_{j+1} - a_j \geq \tau$ where $\tau = \tau(\varepsilon) > 0$ depends only on ε .
- For each δ^{a_j} -cube $Q \subset N_{\delta^{a_j}}(A)$, $A_Q := S_Q(A \cap Q)$ is a $(\delta^{a_{j+1}-a_j}, t_j, \delta^{-\eta})$ -set, where $S_Q : Q \rightarrow [0, 1]^3$ is an affine transformation that maps Q to the unit cube.
- $t_j < t_{j+1}$ and $\sum t_j(a_{j+1} - a_j) \geq s - \varepsilon^2$.

Let J_0 be the smallest j such that $t_j \geq s - \varepsilon$. It follows from

$$s - \varepsilon^2 \leq \sum_j t_j(a_{j+1} - a_j) \leq (s - \varepsilon)a_{J_0} + d(1 - a_{J_0})$$

that $a_{J_0} \leq 1 - \frac{\varepsilon}{2d}$.

Define $\rho = \delta^{a_{J_0}}$. For any ρ -cube $Q \subset N_\rho(A)$ and $A_Q = S_Q(A \cap Q)$. Since $s - \varepsilon \leq t_{J_0} < t_{J_0+1} < \dots < t_J$, A_Q is a union of δ/ρ -balls that is a $(\delta/\rho, s, (\delta/\rho)^{-2\varepsilon})$ -set if $\eta \ll \varepsilon^2$. The conclusion holds with ε replaced by $2d\varepsilon$. $\square$

Lemma 14.5.57. *Let $s \in (0, n]$. Suppose $Q_0 \subset [0, 1]^n$ is a cube of side-length ρ and $A_0 \subset Q_0$ is a (δ, s, C) -Katz-Tao set with $\mathcal{E}_\delta(A_0) \gtrsim (\rho/\delta)^s$. Then there exists a $(\delta, s, O(C))$ -Katz-Tao set A with $\mathcal{E}_\delta(A) \gtrsim \delta^{-s}$ so that for each axis-aligned ρ -cube Q that intersects A , we have that $Q \cap A$ is a translation of A_0 .*

Proof. Divide the unit cube $[0, 1]^n$ into a union of cubes $\tilde{Q}$ of side length $\rho^{s/d} \leq \rho$. For each $\tilde{Q}$, let $A_{\tilde{Q}}$ denote the translation that maps the center of Q_0 to the center of $\tilde{Q}$. Define $A = \bigsqcup A_{\tilde{Q}}(A_0)$. Since the side-length of $\tilde{Q}$ is greater than the side-length of Q_0 , $A_{\tilde{Q}}(A_0)$ are disjoint, and so $\mathcal{E}_\delta(A) \gtrsim \delta^{-s}$.

It remains to verify that A is a (δ, s, C) -Katz-Tao set. By the construction of A , it suffices to check that for each $r \geq \rho$,

$$\mathcal{E}_\delta(A \cap B_r) \lesssim C(r/\delta)^s.$$

To see this, when $r \in (\rho, \rho^{s/n})$, we have $\mathcal{E}_\delta(A \cap B_r) \sim \mathcal{E}_\delta(A_0) \lesssim C(\rho/\delta)^s$. On the other hand, when $r \geq \rho^{s/n}$, we have

$$\mathcal{E}_\delta(A \cap B_r) \lesssim C\left(\frac{r}{\rho^{s/n}}\right)^n \cdot \left(\frac{\rho}{\delta}\right)^s \lesssim C\left(\frac{r}{\delta}\right)^s. \quad \square$$

Next we will prove a weaker variant of Proposition 14.5.47, which relates the area of the twisted projection $\bigcup_{\mathbb{T}} \pi_f(Y'(T))$ to the area of this projection at a coarser resolution.

Lemma 14.5.58. *For all $\varepsilon > 0$, there exists $\eta, \delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $(\mathbb{T}, Y)_\delta$ be a set of δ -tubes satisfying the Tube Wolff Axioms with error $\delta^{-\eta}$, and let Y be a δ^η -dense shading. Let $f: [-1, 1] \rightarrow \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1, 1]$. Then there exists $\rho \in (\delta^{1-\varepsilon^2}, 1)$ and $\delta^{4\eta}$ -shading $Y' \subset Y$ such that*

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y'(T)) \right| \geq (\delta/\rho)^\varepsilon \left| N_\rho\left(\bigcup_{T \in \mathbb{T}} \pi_f(Y'(T)) \right) \right|. \quad (14.5.73)$$

The quantity on the RHS of (14.5.73) will be closely related to the graph of cinematic functions (see [16, Definition 7.1]). To estimate it, we recall [16, Theorem 7.2]:

Theorem 14.5.59. *Let $I \subset \mathbb{R}$ be a compact interval and let $M \subset C^2(I)$ be a family of cinematic functions (see [16, Definition 1.6]). Then for all $\varepsilon > 0$, there exists $\delta_0 > 0$ so that the following holds for all $\delta \in (0, \delta_0]$. Let $G \subset M$ be a δ -separated set that satisfies the following Frostman non-concentration condition:*

$$\#(G \cap B) \leq \delta^{-\varepsilon}(r/\delta) \text{ for all balls } B \subset C^2(I) \text{ of diameter } r. \quad (14.5.74)$$

Then

$$\left\| \sum_{g \in G} \chi_{g^\delta} \right\|_{3/2} \leq \delta^{-\varepsilon},$$

where g^δ is the δ -neighborhood of $\text{graph}(g)$.

Proof of Lemma 14.5.58. Let $X_0 = \bigcup_{\mathbb{T}} \pi_f(Y(T))$, and let $0 < \eta \ll \varepsilon$ be a small parameter to be chosen later. For each dyadic number $1 < \lambda < \delta^{-4}$, let X_λ be the set of $x \in X_0$ such that $\sum_{p \in \pi_f^{-1}(x)} \#\mathbb{T}(p) \sim \lambda$, where the sum is over a δ -separated set of $p \in \pi_f^{-1}(x)$ and $\#\mathbb{T}(p)$ was defined in (14.5.61). By pigeonholing, there exists λ such that $Y_\lambda(T) := Y(T) \cap \pi_f^{-1}(X_\lambda)$ is a $|\log \delta|^{-2} \delta^\eta$ -dense shading.

Apply Lemma 14.5.55 to find an η -uniform subset $X \subset X_\lambda$ such that $|X| \geq \delta^{\eta/2} |X_\lambda|$. By double counting, this implies that for a $\geq \delta^{2\eta}$ -fraction of $\mathbb{T}$, $|Y(T) \cap \pi_f^{-1}(X)| \geq \delta^{2\eta} |T|$. Define $Y'(T) = Y_\lambda(T) \cap \pi_f^{-1}(X)$.

Observe that the image of the line $l_{a,b,c,d} := \{(a + ct, b + dt, t) : t \in \mathbb{R}\}$ under the twisted projection π_f is the curve

$$\pi_f(l_{a,b,c,d}) = \{(a + ct + f(t)(b + dt), t)\}.$$

After possibly refinement and rotation, each tube $T \in \mathbb{T}$ is δ -neighborhood of a line of the form $l_{a,b,c,d}$ intersecting the unit cube. Let $A(\mathbb{T})$ denote the set of tuples (a, b, c, d) corresponding to these lines, i.e.

$$A(\mathbb{T}) := \{(a, b, c, d) : T_{a,b,c,d} := N_\delta(l_{a,b,c,d}) \cap [0, 1]^3 \in \mathbb{T}\}.$$

Since $\mathbb{T}$ satisfies the Tube Wolff axioms, by Remark 14.5.50 we have that $A(\mathbb{T}) \subset \mathbb{R}^4$ is a $(\delta, 2, \delta^{-\eta})$ -Frostman set.

Let $A_c = \{(a, b, d) : (a, b, c, d) \in A(\mathbb{T})\}$. By Fubini, there exists a $c \in [0, 1]$ such that $\mathcal{E}_\delta(A_c) := \delta^{-s} \geq \delta^{-1+\eta} \geq \delta^{-1+\varepsilon^2}$. Apply Lemma 14.5.55 to replace A_c with an η -uniform subset, which we will continue to denote by A_c .

At this point we have a bound on the δ -covering number of A_c , but we do not know if this set satisfies our desired non-concentration condition. To apply Theorem 14.5.59, we need to know that A_c satisfies (14.5.74). By Lemma 14.5.56, there exists $\rho' \in (\delta^{1-\varepsilon^2}, 1]$ such that for any ρ -cube $Q \subset N_{\rho'}(A_c)$, $A_Q := S_Q(A_c \cap Q)$ is a $(\rho, 1 - \varepsilon^2, \rho^{-12\varepsilon^2})$ -set with $\rho = \delta/\rho'$.

By Lemma 14.5.52, since A_Q is a $(\rho, 1 - \varepsilon^2, \rho^{-12\varepsilon^2})$ -set, there exists a $(\rho, 1 - \varepsilon^2, 100)$ -Katz-Tao set $A'_Q \subset A_Q$ with $\mathcal{E}_\rho(A'_Q) \gtrsim \rho^{14\varepsilon^2-1}$. Then $S_Q^{-1}(A'_Q)$ is a $(\delta, 1 - \varepsilon^2, 100)$ -Katz-Tao set contained in Q , a cube of side-length $\rho' = \delta/\rho$. By Lemma 14.5.57, there exist $\sim \rho'^{-1}$ translations A_i such that the set $A'_c := \bigsqcup_i A_i(S_Q^{-1}(A'_Q))$ is a $(\rho, 1 - \varepsilon^2, 100)$ -Katz-Tao set with $\mathcal{E}_\delta(A'_c) \gtrsim \rho^{14\varepsilon^2} \delta^{-1}$. For each A_i and $(a, b, d) \in S_Q^{-1}(A'_Q)$, define the shading on the tube $T' = T_{a',b',c,d'}$, where $(a', b', d') = A_i(a, b, d)$, as

$$(a' + ct, b' + d't, t) \in Y'(T') \text{ if and only if } (a + ct, b + dt, t) \in Y'(T).$$

For each t , define $\pi_{f,t} = (x + f(t)y)$. Then for each fixed pair (A_i, A_j) ,

$$\left| \bigcup_{A_i} \pi_{f,t}(Y'(T)) \right| \sim \left| \bigcup_{A_j} \pi_{f,t}(Y'(T)) \right|, \quad (14.5.75)$$

where the union $\bigcup_{A_i}$ is taken over all $(a, b, d) \in A_i(S_Q^{-1}(A'_Q))$ and the corresponding $T = T_{a,b,c,d}$. To see that (14.5.75) holds, let $A_j \circ A_i^{-1}(x) = x + (a_0, b_0, d_0)$. Then $a + ct + f(t)(b + dt) \in \Omega_{i,t} := \bigcup_{A_i} \pi_{f,t}(Y'(T))$ if and only if

$$a + a_0 + ct + f(t)(b + b_0 + (d + d_0)t) \in \Omega_{j,t} := \bigcup_{A_j} \pi_{f,t}(Y'(T)).$$

Therefore,

$$\Omega_{i,t} = \Omega_{j,t} + a_0 + b_0 f(t) + d_0 f(t)t,$$

which establishes (14.5.75). Finally, integrating (14.5.75) in t yields

$$\left| \bigcup_{A_i} \pi_f(Y'(T)) \right| \sim \left| \bigcup_{A_j} \pi_f(Y'(T)) \right|. \quad (14.5.76)$$

By [16, Lemma 7.3], the set of functions

$$M_f := \{g_{a,b,d}(t) = a + bf(t) + dtf(t) : a, b, d \in [-1, 1]\}$$

is a family of cinematic functions (see [16, Definition 7.1]). Apply Theorem 14.5.59 with $I = [-1, 1]$, $M = M_f$ and $G = \{g_{a,b,d} : (a, b, d) \in A'_c\}$. We conclude that

$$\left\| \sum_{g \in G} \chi_{g^{2\delta}} \right\|_{3/2} \lesssim \delta^{-\varepsilon^2}.$$

If $g = g_{a,b,d}$ then $\pi_f(T) \subset g^{2\delta}$ with $T = T_{a,b,c,d}$. Let $\mathbb{T}_0 = \{T_{a,b,c,d} : (a, b, d) \in A'_c\}$. Therefore,

$$\begin{aligned} \rho^{15\varepsilon^2} &\leq \sum_{T \in \mathbb{T}_0} |Y'(T)| \leq \left| \bigcup_{T \in \mathbb{T}_0} \pi_f(Y'(T)) \right|^{1/3} \left\| \sum_{T \in \mathbb{T}_0} \chi_{\pi_f(Y'(T))} \right\|_{3/2} \\ &\leq \left| \bigcup_{T \in \mathbb{T}_0} \pi_f(Y'(T)) \right|^{1/3} \left\| \sum_{T \in \mathbb{T}_0} \chi_{\pi_f(T)} \right\|_{3/2} \\ &\leq \left| \bigcup_{T \in \mathbb{T}_0} \pi_f(Y'(T)) \right|^{1/3} \left\| \sum_{g \in G} \chi_{g^{2\delta}} \right\|_{3/2}. \end{aligned}$$

Therefore,

$$\left| \bigcup_{T \in \mathbb{T}_0} \pi_f(Y'(T)) \right| \gtrsim \rho^{45\varepsilon^2}.$$

Since A'_c consists of $\lesssim \rho/\delta$ translated copies of $S_Q^{-1}(A'_Q)$, and the latter is a subset of $A_c \cap Q$, (14.5.76) implies

$$\left| \bigcup_{(a,b,c,d) \in A_c \cap Q} \pi_f(Y'(T_{a,b,c,d})) \right| \gtrsim \rho^{45\varepsilon^2} \delta / \rho. \quad (14.5.77)$$

Note that the LHS of (14.5.77) is contained in the $10\delta/\rho$ -neighborhood of a curve $\pi_f(l_{a,b,c,d})$, $(a,b,d) \in Q$, whose volume is comparable to the RHS of (14.5.77), up to a factor of $\rho^{-45\varepsilon^2}$. We conclude that the η -uniform set $X = \bigcup_{\mathbb{T}} \pi_f(Y'(T))$ satisfies

$$|N_{\delta/\rho} X| \leq \rho^{-50\varepsilon^2} |X|. \quad \square$$

We prove Proposition 14.5.47 by iterating Lemma 14.5.58. The details are as follows.

Proof of Proposition 14.5.47. Let $0 < \eta \ll \varepsilon$ be a small number to be determined below. Let $\mathbb{T}' \subset \mathbb{T}$ be the set of tubes with $|Y(T)| \geq \delta^{2\eta}|T|$; then $\#\mathbb{T}' \geq \delta^\eta(\#\mathbb{T})$. Apply Lemma 14.5.55 to find an η -uniform subset $\mathbb{T}'' \subset \mathbb{T}'$ with $\#\mathbb{T}'' \geq \delta^\eta(\#\mathbb{T}')$. Then $(\mathbb{T}'', Y)_\delta$ becomes a set of δ -tubes satisfying the Tube Wolff Axioms with error $\delta^{-3\eta}$, and Y is a $\delta^{2\eta}$ -dense shading. To ease notation, we will rename $\mathbb{T}''$ to be $\mathbb{T}$.

By Lemma 14.5.58, we can find a scale $\rho_1 \in (\delta^{1-\varepsilon^2}, 1)$ and a $\delta^{4\eta}$ -shading $Y_1 \subset Y$ such that

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \geq \left| \bigcup_{T \in \mathbb{T}} \pi_f(Y_1(T)) \right| \geq (\delta/\rho_1)^\varepsilon \left| N_{\rho_1} \left(\bigcup_{T \in \mathbb{T}} \pi_f(Y_1(T)) \right) \right|. \quad (14.5.78)$$

Consider $\mathbb{T}_{\rho_1}$ as the set of distinct ρ_1 -tubes containing at least one tube in $\mathbb{T}$. Since $\mathbb{T}$ is η -uniform, $\mathbb{T}_{\rho_1}$ satisfies the Tube Wolff Axioms with error $\delta^{-3\eta}$. Define Y_{ρ_1} to be the shading on $\tilde{T} \in \mathbb{T}_{\rho_1}$ given by $Y_{\rho_1} = N_{\rho_1}(\bigcup_{T \in [\tilde{T}]} Y_1(T))$. Therefore, Y_{ρ_1} is a $\delta^{2\eta}$ -dense shading. Apply Lemma 14.5.58 to $(\mathbb{T}_{\rho_1}, Y_{\rho_1})_{\rho_1}$ and find $\rho_2 \in (\rho_1^{1-\varepsilon^2}, 1)$ and corresponding $Y_2 \subset Y_{\rho_1}$ such that

$$\left| \bigcup_{\tilde{T} \in \mathbb{T}_{\rho_1}} \pi_f(Y_2(\tilde{T})) \right| \geq (\rho_1/\rho_2)^\varepsilon \left| N_{\rho_2} \left(\bigcup_{\tilde{T} \in \mathbb{T}_{\rho_1}} \pi_f(Y_2(\tilde{T})) \right) \right|.$$

Since for any $\tilde{T} \in \mathbb{T}_{\rho_1}$, we have

$$Y_2(\tilde{T}) \subset Y_{\rho_1}(\tilde{T}) = N_{\rho_1} \left(\bigcup_{T \in [\tilde{T}]} Y_1(T) \right),$$

we have

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \geq (\delta/\rho_2)^\varepsilon \left| N_{\rho_2} \left(\bigcup_{\tilde{T} \in \mathbb{T}_{\rho_1}} \pi_f(Y_2(\tilde{T})) \right) \right|.$$

Iterate the above process and obtain correspondingly scales $\rho_3, \dots, \rho_N$ until $\rho_N \geq \delta^{\varepsilon^2}$. Since $\rho_{k+1} \geq \rho_k^{1-\varepsilon^2}$, $N \leq \varepsilon^{-2}$. When $\eta > 0$ is sufficiently small depending on ε ,

$$\left| \bigcup_{T \in \mathbb{T}} \pi_f(Y(T)) \right| \geq (\delta/\rho_2)^\varepsilon \left| N_{\rho_N} \left(\bigcup_{\tilde{T} \in \mathbb{T}_{\rho_{N-1}}} \pi_f(Y_N(\tilde{T})) \right) \right| \geq \delta^\varepsilon \rho_N^2 \geq \delta^{2\varepsilon}.$$

We conclude with ε replaced by $\varepsilon/2$. □

14.5.8 The consequences of Theorem 14.5.5

In this section we will show how Theorem 14.5.5 implies Theorem 14.5.2, Theorem 14.5.4, and Corollary 14.5.7.

Proof of Theorem 14.5.2. Suppose to the contrary that there exists a Kakeya set $K \subset \mathbb{R}^3$ with $\dim_{\mathcal{M}}(K) = 3 - \beta$ for some $\beta > 0$, i.e. there exists $C, r_0 > 0$ so that

$$\sup_{x \in \mathbb{R}^3} \mathcal{E}_\rho(K \cap B(x, r)) \leq C(r/\rho)^{3-\beta} \quad \text{for all } x \in \mathbb{R}^3, \ 0 < \rho < r \leq r_0. \quad (14.5.79)$$

Replacing C by $(1 + 100r_0^{-3})C$, we may suppose that (14.5.79) holds with $r_0 = 1$.

Let η, δ_0 be the values from Theorem 14.5.5, with $\varepsilon = \beta/2$. Let $\delta \in (0, \delta_0]$ be chosen sufficiently small (we will describe the choice of δ in further detail below), and let $\mathbb{T}$ be a set of δ^{-2} δ -tubes contained in $N_\delta(K)$ that point in δ separated directions; in particular the tubes in $\mathbb{T}$ are essentially distinct and satisfy the Convex Wolff Axioms with error 100; we will select $\delta \in (0, \delta_0]$ sufficiently small so that $100 \leq \delta^{-\eta}$. For each $T \in \mathbb{T}$, let $Y(T) = T$; this is a 1-dense shading.

Apply Theorem 14.5.5 with $\varepsilon = \beta/2$. We conclude that there exists $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\eta r$, and a ball B of radius r so that (14.5.1) holds, and hence

$$\mathcal{E}_\rho(K \cap B) \gtrsim \rho^{-3} \left| B \cap N_\rho \left(\bigcup_{T \in \mathbb{T}} Y(T) \right) \right| \gtrsim (\rho/r)^\varepsilon (r/\rho)^3 \geq \delta^{-\beta\eta/2} (r/\rho)^{3-\beta}. \quad (14.5.80)$$

Comparing (14.5.79) and (14.5.80), we obtain a contradiction, provided $\delta > 0$ is selected sufficiently small depending on C, β, η , and the implicit constant in (14.5.80). We conclude that every Kakeya set in $\mathbb{R}^3$ has Assouad dimension 3. □

Remark 14.5.60. As noted in the introduction, the conclusion of Theorem 14.5.2 continues to hold if we remove the hypotheses that the lines point in

different directions, and replace this with the requirement that the lines satisfy a variant of the Wolff axioms. This is most easily stated in the discretized setting: For all $\varepsilon > 0$, there exists $\eta > 0$ so that the following holds. Let $\mathbb{T}$ be a set of δ -tubes that satisfy the Convex Wolff Axioms with error $\delta^{-\eta}$. Then $\bigcup T$ has discretized Assouad dimension $\geq 3 - \varepsilon$ at scale δ and scale separation $\delta^{-\eta}$.

Proof of Theorem 14.5.4. Suppose to the contrary that there exists a Kakeya set K with stably equal Hausdorff and packing dimension $\alpha < 3$. Without loss of generality, we can assume $K \subset B(0, 1)$. Let $\varepsilon = (3 - \alpha)/4$; in what follows, all implicit constant may depend on ε . Let $\eta > 0$ and $\delta_0 > 0$ be the quantities from Theorem 14.5.5 for this choice of ε . Let $\eta_1 = (3 - \varepsilon)\eta/6$. Since K has stably equal Hausdorff and packing dimension, we can find a measure μ supported on K so that

$$r^{\alpha+\eta_1} \lesssim \mu(B(x, r)) \lesssim r^{\alpha-\eta_1}, \quad x \in \text{supp}(\mu), \quad 0 \leq r \leq 1, \quad (14.5.81)$$

and there exists $\tau > 0$ and a set $\Omega \subset S^2$ of positive measure, so that for each $e \in \Omega$ there is a line ℓ_e pointing in direction e , with $|\ell_e \cap \text{supp}(\mu)| \geq \tau$.

Let Ω_δ be a δ -separated subset of Ω of cardinality $\#\Omega_\delta \gtrsim \delta^{-2}|\Omega|$. For each $e \in \Omega_\delta$, let T_e be a δ -tube with coaxial line ℓ_e , which satisfies $|T_e \cap \ell_e \cap \text{supp}(\mu)| \geq \tau/2$. Define the shading

$$Y(T_e) = \bigcup_{x \in T_e \cap \ell_e \cap \text{supp}(\mu)} B(x, \delta).$$

We have $|Y(T_e)| \gtrsim \tau|T|$ for each tube T_e ; we will select $\delta \in (0, \delta_0]$ sufficiently small so that $|Y(T_e)| \geq \delta^\eta|T|$. Let $\mathbb{T} = \{T_e : e \in \Omega_\delta\}$. Since the tubes in $\mathbb{T}$ point in δ -separated directions, the tubes are essentially distinct and satisfy the Convex Wolff Axioms with error $O(\delta^{-2}/(\#\mathbb{T})) = O(|\Omega|^{-1})$; we will select $\delta \in (0, \delta_0]$ sufficiently small so that $\mathbb{T}$ satisfies the convex Wolff Axioms with error at most $\delta^{-\eta}$. Thus we can apply Theorem 14.5.5 to conclude that there exist scales $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\eta r$, and a ball B of radius r so that (14.5.1) holds.

Let $E \subset B$ be a ρ -separated subset of $B \cap N_\rho\left(\bigcup_{T \in \mathbb{T}} Y(T)\right)$ of cardinality $\gtrsim (r/\rho)^{3-\varepsilon}$. Then by (14.5.81) we have

$$r^{\alpha-\eta_1} \gtrsim \mu(B) \gtrsim \sum_{x \in E} \mu(B(x, \rho)) \gtrsim (r/\rho)^{3-\varepsilon} \rho^{\alpha+\eta_1}.$$

Re-arranging and using the fact that $\rho/r \leq \delta^\eta$, as well as the definition of ε and η_1 , we conclude that

$$\delta^{-\eta(3-\varepsilon)/2} = \delta^{-\eta(3-\varepsilon-\alpha)} \leq (r/\rho)^{-\eta(3-\varepsilon-\alpha)} \lesssim (r\rho)^{-\eta_1} \lesssim \delta^{-2\eta_1} = \delta^{-\eta(3-\varepsilon)/3}.$$

If $\delta \in (0, \delta_0]$ is selected sufficiently small, then this is impossible. We conclude that every Kakeya set in $\mathbb{R}^3$ with stably equal Hausdorff and packing dimension must have dimension 3. $\square$

Proof of Corollary 14.5.7. Let $\varepsilon > 0$. Let $\eta_0 > 0$ and $\delta_0 > 0$ be the quantities from Theorem 14.5.5 with $\varepsilon/2$ in place of ε . Let $\eta = \eta_0/2$, let $\mathbb{T}$ be a set of essentially distinct δ -tubes in $B(0, 1)$ that satisfies the Convex Wolff Axioms with error $\delta^{-\eta}$, and let $Y(T) \subset T$ be a shading with $\sum_{\mathbb{T}} |Y(T)| \geq \delta^\eta \sum_{\mathbb{T}} |T|$.

Let $Y_0(T) = Y(T)$. For each $i = 1, \dots$, apply Theorem 14.5.5 to $\mathbb{T}$ with the shading Y_{i-1} ; let r_i, ρ_i and B_i be the output from this theorem. Define $Y_i(T) = Y_{i-1}(T) \setminus B_i$. We repeat this process until $\sum_{\mathbb{T}} |Y_i(T)| < \frac{1}{2} \delta^\eta \sum_{\mathbb{T}} |T|$, at which point we have

$$\sum_{T \in \mathbb{T}} \left| Y(T) \cap \bigcup_i B_i \right| \geq \frac{1}{2} \sum_{T \in \mathbb{T}} |Y(T)|.$$

After dyadic pigeonholing, we can select $\rho, r \in [\delta, 1]$ with $\rho \leq \delta^\eta r$, and a set of indices I with the following properties:

- For each index $i \in I$, we have $\rho_i \in [\rho, 2\rho)$ and $r_i \in [r, 2r)$.
- $\sum_{T \in \mathbb{T}} \left| Y(T) \cap \bigcup_{i \in I} B_i \right| \gtrsim |\log \delta|^{-2} \sum_{\mathbb{T}} |Y(T)|$.

Next, we can greedily select a set of indices $I' \subset I$ so that the above properties continue to hold (with the implicit constant in the second item worsened slightly), and the sets $\{B_i : i \in I'\}$ are pairwise disjoint.

We claim that the values of ρ and r described above, plus the shading $Y'(T) = Y(T) \cap \bigcup_{i \in I'} B_i$ satisfy the conclusions of Corollary 14.5.7. Indeed, we can compute

$$\begin{aligned} \left| N_\rho \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right| &= \left| N_\rho \left(\bigcup_{i \in I'} B_i \cap \bigcup_{T \in \mathbb{T}} Y'(T) \right) \right| \\ &\gtrsim \sum_{i \in I'} \left| B_i \cap N_\rho \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right| \\ &\gtrsim \sum_{i \in I'} (r/\rho)^{\varepsilon/2} |B_i| \\ &\gtrsim (r/\rho)^{\varepsilon/2} \left| N_r \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right| \\ &\geq \delta^{-\eta_0 \varepsilon/2} (r/\rho)^\varepsilon \left| N_r \left(\bigcup_{T \in \mathbb{T}} Y'(T) \right) \right|. \end{aligned}$$

After decreasing δ_0 (and hence δ) if necessary, the quantity $\delta^{-\eta_0 \varepsilon/2}$ dominates the implicit constants in the above inequality, and we obtain (14.5.2). $\square$

References for the Convex–Wolff and Assouad Bridge

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Chapter 14.6

The Complete Cinematic Maximal Theorem

Provenance and role in the proof chain

This chapter proves the restricted $L^{3/2}$ maximal estimate for cinematic families that is used by both the sticky Kakeya proof and the convex–Wolff bridge. It includes the reductions, tangency geometry, pseudo-circle lens encoding, perturbation, two-ends analysis, and the restricted projection application rather than only the overview appearing in the earlier survey chapters.

The proof is adapted from Malabika Pramanik, Tongou Yang, and Joshua Zahl, *A Furstenberg-type problem for circles, and a Kaufman-type restricted projection theorem in $\mathbb{R}^3$* , arXiv:2207.02259v3, licensed CC BY 4.0. Internal labels and bibliography keys have been namespaced for the composite book; long displays have been reflowed and the coarse-tangency reduction has been expanded from its abbreviated source form. These are changes by this edition and are not attributed to the original authors. The one combinatorial lens-counting theorem used in the source is proved separately in the next proof-dossier chapter, together with the sampling argument that turns the balanced estimate into the stated incidence bound.

[Restricted projection to lines in $\mathbb{R}^3$]A Furstenberg-type problem for circles, and a Kaufman-type restricted projection theorem in $\mathbb{R}^3$ Malabika Pramanik, Tongou Yang and Joshua Zahl

Abstract. We resolve a conjecture of Fässler and Orponen on the dimension of exceptional projections to one-dimensional subspaces indexed by a space curve in $\mathbb{R}^3$. We do this by obtaining sharp L^p bounds for a variant of the Wolff circular maximal function over fractal sets for a class of C^2 curves related to Sogge’s cinematic curvature condition. A key new tool is the use of lens cutting techniques from discrete geometry.

14.6.1 Introduction

In 1954, Marstrand [20] proved that if $Z \subset \mathbb{R}^2$ is Borel, then $\dim(Z \cdot v) = \min(\dim Z, 1)$ for a.e. $v \in S^1$. Here and throughout, “dim” will mean Hausdorff dimension. This was sharpened by Kaufman [16] in 1968, who bounded the size of the set of exceptional directions for which the above inequality fails.

Theorem 14.6.1 (Kaufman). *Let $Z \subset \mathbb{R}^2$ be Borel, and let $0 \leq s < \min\{\dim Z, 1\}$. Then*

$$\dim\{\theta \in S^1: \dim(Z \cdot \theta) < s\} \leq s. \quad (14.6.1)$$

Marstrand and Kaufman’s results have been generalized to higher dimensions and other settings, and they are the foundation of projection theory, which is an active area in geometric measure theory. See [22] for an introduction to the topic, or [5] for a recent survey of the area.

In this paper, we will be interested in analogues of the above results for Borel sets $Z \subset \mathbb{R}^3$. As a starting point, one analogue of Marstrand’s theorem says that $\dim(Z \cdot v) = \min(\dim Z, 1)$ for a.e. $v \in S^2$. There are also analogues of Kaufman’s theorem; for example, if $0 \leq s < \min\{\dim Z, 1\}$ then

$$\dim\{v \in S^2: \dim(Z \cdot v) < s\} \leq 1 + s. \quad (14.6.2)$$

See [21] for details. In particular, (14.6.2) says that the set of $v \in S^2$ for which $\dim(Z \cdot v) = 0$ has dimension at most 1, and this can be sharp—for example, if Z is a subset of the z -axis then $Z \cdot v = \{0\}$ for each v in the great circle $S^2 \cap \{z = 0\}$. It is possible that stronger bounds hold if we restrict the set of projection directions to a subset of S^2 that does not concentrate near great circles. In this direction, Fässler and Orponen [6] conjectured the following restricted Marstrand-type estimate.

Conjecture 14.6.2. Let $I \subset \mathbb{R}$ be a compact interval and let $\gamma: I \rightarrow S^2$ be a C^2 curve that satisfies the “escaping great circle” condition

$$\text{span}\{\gamma(\theta), \dot{\gamma}(\theta), \ddot{\gamma}(\theta)\} = \mathbb{R}^3 \quad \text{for all } \theta \in I. \quad (14.6.3)$$

Let $Z \subset \mathbb{R}^3$ be analytic. Then for almost every $\theta \in I$, $\dim(\gamma(\theta) \cdot Z) = \min(\dim Z, 1)$.

In [6], Fässler and Orponen proved Conjecture 14.6.2 in the special case $\dim(Z) \leq 1/2$. In [12], He considered the related problem of estimating the size of the set where $\dim(\gamma(\theta) \cdot Z)$ is close to $(\dim Z)/3$. In [13], Käenmäki, Orponen, and Venieri used ideas from incidence geometry, and specifically a circle tangency bound due to Wolff [29] to prove Conjecture 14.6.2 (for all Z) in the special case $\gamma(\theta) = \frac{1}{\sqrt{2}}(\cos \theta, \sin \theta, 1)$. Our contribution is the following Kaufman-type sharpening of Conjecture 14.6.2.

Theorem 14.6.3. *Let $I \subset \mathbb{R}$ be a compact interval and let $\gamma : I \rightarrow S^2$ be a C^2 curve that satisfies the “escaping great circle” condition (14.6.3). Let $Z \subset \mathbb{R}^3$ be analytic and let $0 \leq s < \min\{\dim Z, 1\}$. Then*

$$\dim\{\theta \in I : \dim(Z \cdot \gamma(\theta)) < s\} \leq s. \quad (14.6.4)$$

In particular, Conjecture 14.6.2 is true.

Added November 14, 2022: There have been several exiting recent developments related to Theorem 14.6.3. First, Gan, Guth and Maldague [9] have also recently and independently resolved Conjecture 14.6.2, using a different method related to decoupling. Second, Gan, Guo, Guth, Harris, Maldague and Wang [8] solved a closely related conjecture of Fässler and Orponen, where projections onto one-dimensional subspaces of $\mathbb{R}^3$ are replaced by projections onto two-dimensional subspaces. Finally, Harris [11] proved that if $\dim Z > 1$, then $\gamma(\theta) \cdot Z$ has positive Lebesgue measure for almost every θ .

14.6.1.1 A Furstenberg-type problem for circles

Theorem 14.6.3 follows from an L^p bound for a variant of Wolff’s circular maximal function, where circles have been replaced by a class of C^2 curves that are the graphs of we call “cinematic functions,” and the domain of integration is restricted to a special type of fractal set. To state this result, we will introduce several definitions. First, we recall the definition of a $(\delta, \alpha; C)_1$ -set from [14].

Definition 14.6.4. Let $0 < \alpha \leq 1$, $\delta > 0$, $C \geq 1$. A set $E \subset [0, 1]$ is called a $(\delta, \alpha; C)_1$ -set if

- (a) E is a union of intervals of length δ .
- (b) For every interval $I \subset [0, 1]$, we have the Frostman-type non-concentration condition

$$|E \cap I| \leq C\delta^{1-\alpha}|I|^\alpha. \quad (14.6.5)$$

Since (14.6.5) always holds for intervals of length $\leq \delta$, every $(\delta, \alpha; C)_1$ -set is also a $(\delta, \beta; C)_1$ -set if $\beta \geq \alpha$. We will be concerned with certain subsets of the plane, which are fibered products of $(\delta, \alpha; C)_1$ -sets.

Definition 14.6.5. Let $0 < \alpha, \beta \leq 1$, $\delta > 0$, and $C \geq 1$. A set $E \subset [0, 1]^2$ is called a $(\delta, \alpha; C)_1 \times (\delta, \beta; C)_1$ quasi-product if E is Borel and can be expressed in the form

$$E = \bigcup_{a \in A} \{a\} \times B_a,$$

where A is a $(\delta, \alpha; C)_1$ -set and B_a is a $(\delta, \beta; C)_1$ -set for each $a \in A$.

Our main result is a maximal function estimate, where the domain of integration is a quasi-product. Our next task is to introduce the class of objects that can be analyzed using our maximal function estimate. The following definition was inspired by Sogge's notion of cinematic curvature [24], which was considered by Kolasa and Wolff [17] in the context of the Wolff circular maximal function.

Definition 14.6.6. Let $I \subset \mathbb{R}$ be a compact interval and let $\mathcal{F} \subset C^2(I)$. We say $\mathcal{F}$ is a *family of cinematic functions*, with *cinematic constant* K and *doubling constant* D , if the following conditions hold.

1. $\mathcal{F}$ has diameter at most K (here and in what follows, we use the usual metric on $C^2(I)$).
2. $\mathcal{F}$ is a doubling metric space, with doubling constant at most D .
3. For all $f, g \in \mathcal{F}$, we have

$$\inf_t (|f(t) - g(t)| + |f'(t) - g'(t)| + |f''(t) - g''(t)|) \geq K^{-1} \|f - g\|_{C^2(I)}. \quad (14.6.6)$$

With these definitions, we can now state the following variant of Wolff's circular maximal function theorem.

Theorem 14.6.7. Let $\varepsilon > 0$, $0 < \alpha \leq \zeta \leq 1$, and $D, K \geq 1$. Then the following is true for all $\delta > 0$ sufficiently small (depending on D, K, ε only). Let $\mathcal{F}$ be a family of cinematic functions, with cinematic constant K and doubling constant D . Let E be a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ quasi-product.

Let $F \subset \mathcal{F}$ be a set of functions that satisfies the Frostman-type non-concentration condition

$$\#(F \cap B) \leq \delta^{-\varepsilon} (r/\delta)^\zeta \quad \text{for all balls } B \subset C^2(I) \text{ of radius } r \geq \delta. \quad (14.6.7)$$

Then

$$\int_E \left(\sum_{f \in F} \mathbf{1}_{f^\delta} \right)^{3/2} \leq \delta^{2-\alpha/2-\zeta/2-C\varepsilon} (\#F), \quad (14.6.8)$$

where $C = C(D)$ depends only on D , and f^δ is the δ -neighborhood of the graph of f .

Remarks.

1. Since $[0, 1]^2$ is a $(\delta, 1; 1)_1 \times (\delta, 1; 1)_1$ quasi-product and $I, \mathcal{F}$ are bounded, as a special case of Theorem 14.6.7 we obtain the estimate

$$\left\| \sum_{f \in F} \mathbf{1}_{f^\delta} \right\|_{3/2} \leq \delta^{-C\varepsilon} (\delta \#F)^{2/3}, \quad (14.6.9)$$

where F satisfies (14.6.7) for $\zeta = 1$, and the order of parameters are the same as in the statement of Theorem 14.6.7.

2. As a special case of (14.6.9), Theorem 14.6.7 generalizes Wolff's $L^{3/2}$ circular maximal function bound from [27]. The (upper) halves of circles with centers in a small neighborhood of the origin and radii between 1 and 2 form a family of cinematic functions $\mathcal{C}$ over the interval $[-1/2, 1/2]$. If we parameterize a circle by its center-radius pair (x, y, r) , then a set $C \subset \mathcal{C}$ obeys the non-concentration condition (14.6.7) if the corresponding collection of centre-radius pairs $P_C \subset \mathbb{R}^3$ obeys a corresponding Frostman-type non-concentration condition; namely,

$$\#(P_C \cap B) \leq \delta^{-\varepsilon}(r/\delta) \quad \text{for all balls } B \subset \mathbb{R}^3 \text{ of radius } r \geq \delta. \quad (14.6.10)$$

Thus if C is a set of circles with centers in $[0, 1]^2$ and radii between 1 and 2, whose center-radius pairs obey (14.6.10), then Theorem 14.6.7 says that

$$\left\| \sum_{c \in C} \mathbf{1}_{c^\delta} \right\|_{3/2} \leq \delta^{-C\varepsilon} (\delta \#C)^{2/3}. \quad (14.6.11)$$

In [27], Wolff proved (14.6.7) in the special case where the circles in C had δ -separated radii.

3. Theorem 14.6.7 generalizes the third author's variable coefficient Wolff circular maximal function bound [31] from curves given by a C^∞ defining function to curves given by a C^3 defining function (the latter is the minimal amount of regularity required to state Sogge's cinematic curvature condition for the defining function). A precise statement is slightly technical, and is given in Appendix A.
4. Theorem 14.6.7 has proved useful in other contexts beyond the proof of Theorem 14.6.3. For example, Wang and the third author [26] used Theorem 14.6.7 to prove a special case of the Kakeya conjecture for sticky Kakeya sets in $\mathbb{R}^3$, and Katz, Wu, and the third author [15] used Theorem 14.6.7 to prove a special case of the Kakeya conjecture for SL_2 Kakeya sets. Chang, Dosidis, and Kim [3] used a corollary of Theorem 14.6.7 to analyze a Nikodym-type maximal function associated to spheres in $\mathbb{R}^n$. Fässler and Orponen [7] used Theorem 14.6.7 to study vertical projections in the first Heisenberg group. We remark that in [26], it is crucial that Theorem 14.6.3 holds for C^2 curves; previous maximal function bounds are insufficient.

Theorem 14.6.7 will be discussed further in Section 14.6.1.3, and the complete proof will be given in Section 14.6.5.

14.6.1.2 From maximal functions to restricted projections

We will briefly sketch the proof of Theorem 14.6.3 and describe the connection between Theorems 14.6.3 and 14.6.7. For the purpose of this proof sketch, we will ignore many technical details. A complete proof is given in Section 14.6.2. Let $Z \subset \mathbb{R}^3$ be Borel with $\dim Z = \zeta \leq 1$, and let $0 \leq s < \dim Z$. Let $\Theta = \{\theta \in I : \dim(Z \cdot \gamma(\theta)) < s\}$, and suppose for contradiction that $\dim \Theta = \alpha > s$.

In [13], Käenmäki, Orponen, and Venieri observed that this problem could be transformed to a question about curves in the plane. For each $z \in \mathbb{R}^3$, define the plane curve

$$\Gamma_z = \{(\theta, z \cdot \gamma(\theta)) : \theta \in I\}. \quad (14.6.12)$$

Then for each $\theta \in \Theta$, the vertical line $L_\theta = \mathbb{R} \times \{\theta\}$ satisfies $\dim(L_\theta \cap \bigcup \Gamma_z) < s$.

Next, we discretize the arrangement at a small scale $\delta > 0$. The idea (ignoring some technical details) is that we replace Z by a δ -separated subset Z_δ of cardinality roughly $\delta^{-\zeta}$ that satisfies a Frostman-type non-concentration condition, and we replace Θ by a $(\delta, \alpha; \delta^{-\varepsilon})_1$ -set, which we denote by Θ_δ . For each $z \in Z_\delta$, let Γ_z^δ be the δ -neighborhood of the curve Γ_z . Then for most $\theta \in \Theta_\delta$, $L_\theta \cap \bigcup_{z \in Z_\delta} \Gamma_z^\delta$ will have one-dimensional Lebesgue measure at most $\delta^{1-\eta-\varepsilon}$, and it will be contained in a $(\delta, s, \delta^{-\varepsilon})_1$ -set. Thus by Hölder's inequality we have that

$$\int_{\Theta_\delta \times \mathbb{R}} \left(\sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta} \right)^{3/2} \geq \delta^{2+\frac{1}{2}s-\frac{3}{2}\zeta-\alpha+\varepsilon}. \quad (14.6.13)$$

Most of the integrand in (14.6.13) is supported on a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ quasi-product. We can use the “escaping great circle” condition (14.6.3) to show that the set of curves $\{\Gamma_z^\delta : z \in Z_\delta\}$ are the graphs of functions from a family of cinematic functions, and the Frostman-type non-concentration condition on Z_δ implies the analogous bound (14.6.7). Thus we can apply Theorem 14.6.7 to upper-bound (14.6.13). Comparing (14.6.8) and (14.6.13) (and recalling that Z_δ has cardinality roughly $\delta^{-\zeta}$), we conclude that $\alpha \leq s$. This gives us our desired contradiction and completes the proof.

Remarks.

1. The main challenge when executing the proof strategy described above is that we must discretize the set Z , the set Θ , and the sets $L_\theta \cap \bigcup \Gamma_z$ at a common scale δ . We refer the reader to Section 14.6.2 for details.
2. Our bound (14.6.8) in Theorem 14.6.7 becomes stronger as the dimension α of the quasi-product E decreases. With this sensitivity to E , Theorem

14.6.7 is related to the Furstenberg set problem for curves; see [18] for a related result on this topic. If we replace the domain of integration in (14.6.8) by $\mathbb{R}^2$ or $[0, 1]^2$, then it is possible for the LHS of (14.6.8) to be as large as $\delta \# F$. On the other hand, it is easy to verify that if the RHS of (14.6.8) were replaced by the weaker estimate $\delta \# F$, then this would not be sufficient to prove the Kaufman-type estimate from Theorem 14.6.3 (though it would still be strong enough to prove Conjecture 14.6.2).

14.6.1.3 Proof sketch of Theorem 14.6.7

We conclude the introduction with a discussion of the main ideas in Theorem 14.6.7 and a brief sketch the proof. In [28], Wolff proved that for each $\varepsilon > 0$ and all $\delta > 0$ sufficiently small, if C is a set of circles with δ -separated radii $1 \leq r \leq 2$, then

$$\left\| \sum_{c \in C} \mathbf{1}_{c^\delta} \right\|_p \leq \delta^{-\varepsilon} (\delta \# C)^{1/p}, \quad p = 3/2, \quad (14.6.14)$$

where c^δ denotes the δ -neighborhood of the circle c . Wolff's proof was quite technical, but it has been simplified in several follow-up works [29, 23]. The main ingredient in these later proofs was a bound on the number of “tangency rectangles” determined by the arrangement C . Informally, a tangency rectangle is a curvilinear rectangle R of dimensions $\delta^{1/2} \times \delta$ (other dimensions are possible, but we will ignore this for now) that is contained in at least two annuli c_1^δ, c_2^δ , with $c_1, c_2 \in C$. Heuristically, we may suppose that every pair of circles $c_1, c_2 \in C$ either intersect transversely, in which case $|c_1^\delta \cap c_2^\delta| \sim \delta^2$, or they intersect tangentially, in which case $|c_1^\delta \cap c_2^\delta| \sim \delta^{3/2}$ and the intersection is localized to a tangency rectangle.

The contribution to the LHS of (14.6.14) from the first type of intersection is negligible. Indeed, if every pair of circles intersected transversely, then (14.6.14) would hold for $p = 2$, which is a stronger estimate. Thus the goal is to control the number of tangential intersections, or more precisely, the number of tangency rectangles where these intersections occur. Let $\mathcal{R}$ be a set of tangency rectangles that cover all the tangential intersections determined by C . For each $R \in \mathcal{R}$, let $m(R)$ be the number circles $c \in C$ with $R \subset c^\delta$. If the rectangles in $\mathcal{R}$ were disjoint and if the LHS of (14.6.14) was concentrated on these rectangles, then we would have a $\ell^{3/2}(\mathcal{R}) \rightarrow L^{3/2}(\mathbb{R}^2)$ bound of the form

$$\left\| \sum_{c \in C} \mathbf{1}_{c^\delta} \right\|_{3/2} \leq \delta \left(\sum_{R \in \mathcal{R}} m(R)^{3/2} \right)^{2/3}. \quad (14.6.15)$$

We caution that reader that in general, (14.6.15) is not true as stated. However, the heuristic (14.6.15) was made precise by Schlag [23], who used induction

on scales and a bilinear $\implies$ linear argument to prove that (14.6.14) would follow from a bilinear version of the weak $\ell^{3/2}$ bound

$$\#\{R \in \mathcal{R}: m(R) \geq k\} \leq (\#C/k)^{3/2}. \quad (14.6.16)$$

Previously in [29], Wolff had proved a suitable bilinear version of (14.6.16) using the “cuttings” method developed by researchers in computational geometry in the late 1980s (see [4, 1]), and these two results combined to give a new proof of (14.6.14).

The cuttings method makes crucial use of the property that circles are algebraic curves. In [30], the third author extended Wolff’s arguments and the cuttings method to obtain an analogue of (14.6.14) where circles are replaced by algebraic curves of bounded degree that satisfy Sogge’s cinematic curvature condition (see also [17] for earlier work in this direction). In [31], the third author used the newly developed polynomial partitioning technique of Guth and Katz [10] to extend (14.6.14) to smooth curves with cinematic curvature. All of these techniques, however, require that the curves are algebraic (or in the case of smooth curves, that they can be accurately approximated by moderately low degree algebraic curves). These methods do not work, and new ideas are needed, for curves that are merely C^k .

The main difficulty in proving Theorem 14.6.7 is establishing an analogue of (14.6.16) for C^2 curves. We do this by adapting a result due to Marcus and Tardos [19] from topological graph theory, which concerns objects called lenses. In brief, if $\mathcal{C}$ is a set of Jordan curves in the plane, then a *lens* is a pair of curve-segments $\lambda_1 \subset \gamma_1$; $\lambda_2 \subset \gamma_2$; $\gamma_1, \gamma_2 \in \mathcal{C}$, so that λ_1 and λ_2 intersect at their endpoints, and $\lambda_1 \cup \lambda_2$ is homeomorphic to S^1 . For example, if two circles c_1, c_2 with well-separated centers share a common tangency rectangle, then after a small perturbation we expect to find arcs of c_1 and c_2 of length roughly $\delta^{1/2}$ that create a lens.

Lemma 10 from [19] says that an arrangement of n closed Jordan curves that satisfy a certain “pseudo-circle” property determines $O(n^{3/2} \log n)$ pairwise non-overlapping lenses. We use this result to obtain an analogue of (14.6.16) by carefully perturbing our arrangement of curves, so that the number of tangency rectangles in $\mathcal{R}$ is comparable to the number of lenses. See Proposition 14.6.26 for a precise statement, and Section 14.6.4 for the details of this argument.

With Proposition 14.6.26 in hand, it remains to prove Theorem 14.6.7. We encounter several difficulties. First, our bilinear analogue of (14.6.16) is more general than the corresponding estimates from [29, 30, 31] because it applies to C^2 curves rather than algebraic (or smooth) curves. However, our bilinear analogue of (14.6.16) requires slightly stronger assumptions, which do not work well with induction on scales. Thus we need a new bilinear $\implies$ linear argument that does not use induction on scales. Closely related to this difficulty is the annoyance that intersections between curves need not be completely

tangential or transverse—there are many intermediate amounts of tangency or transversality. We handle these difficulties by combining a L^2 argument at fine scales with a suitable bilinear version of (14.6.16) at coarse scales. In brief, we define a parameter Δ that measures the typical amount of tangency between curves; if $\Delta = \delta$ then the main contribution to the LHS of (14.6.14) comes from pairs of curves that intersect completely tangentially, while if $\Delta = 1$ then the main contribution comes from pairs of curves that intersect completely transversely. We then divide $\mathbb{R}^2$ into rectangles of dimensions $\Delta \times \Delta^{1/2}$. If two curves intersect, then typically this intersection is localized to one such rectangle, and the curves are tangent at scale Δ . Thus we can use a bilinear analogue of (14.6.16) to control the number of tangency rectangles at scale Δ . Within each rectangle, most pairs of curves intersect with a precisely prescribed amount of transversality. Thus we can use a L^2 argument to control the contribution from each rectangle. This is the most technical part of the paper, and the details are in Section 14.6.5.

14.6.1.4 Notation

To reduce clutter, we will write $A \lesssim B$ or $A = O(B)$ if there is a constant $C > 0$ so that $A \leq CB$. The constant C will be allowed to depend on several quantities, which are context-dependent. Based on context, C may depend on any of the quantities $|I|$, γ , and $\dim Z$ from the statement of Theorem 14.6.3, and also the quantities K , D , and ε from the statement of Theorem 14.6.7.

In the arguments that follow, we will frequently introduce a small parameter $\delta > 0$, and we will employ dyadic pigeonholing arguments that replace a set X with a subset of size roughly $|\log \delta|^{-1}(\#X)$. We will write $A \lesssim B$ or $B \gtrsim A$ if $A \lesssim |\log \delta|^{C_0} B$ for some absolute constant C_0 (in practice, we will always have $C_0 \leq 100$). When we write $A \lesssim B$, the relevant quantity δ will always be apparent from context.

14.6.1.5 Thanks

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14.6.2 From Theorem 14.6.7 to Theorem 14.6.3

In this section, we will expand upon the sketch from Section 14.6.1.2. The following version of Theorem 14.6.7 is a main ingredient of the proof.

Proposition 14.6.8. *Let us fix $\varepsilon > 0$ and dimensional parameters $0 < \alpha \leq \zeta \leq 1$. Let I be a compact interval and let $\gamma: I \rightarrow S^2$ be a curve satisfying (14.6.3). Then there exists a positive constant δ_0 depending only on these quantities, such that the following is true for all $0 < \delta \leq \delta_0$.*

Let E be a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ quasi-product. Let $Z_\delta \subset B(0, 1) \subset \mathbb{R}^3$ be a δ -separated set that satisfies the Frostman-type non-concentration condition

$$\#(Z_\delta \cap B) \leq \delta^{-\varepsilon} (r/\delta)^\zeta \quad \text{for all balls } B \subset \mathbb{R}^3 \text{ of radius } r \geq \delta. \quad (14.6.17)$$

Then

$$\int_E \left(\sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta} \right)^{3/2} \leq \delta^{2-\alpha/2-\zeta/2-C\varepsilon} (\#Z_\delta), \quad (14.6.18)$$

where $C > 0$ is an absolute constant, and Γ_z^δ is the δ -neighborhood of the graph Γ_z defined in (14.6.12).

Proof. Proposition 14.6.8 follows from Theorem 14.6.7, applying the latter to the family of functions

$$F = \{f_z : z \in Z_\delta\} \subset \mathcal{F} = \{f_z : z \in B(0; 1)\},$$

where $f_z(\theta) = \gamma(\theta) \cdot z$ and $\gamma: I \rightarrow S^2$ is a curve satisfying (14.6.3). In order for Theorem 14.6.7 to be applicable, we need to verify that $\mathcal{F}$ is cinematic (with cinematic constant that depends only on γ) and that F satisfies a Frostman-type non-concentration bound (14.6.7). The verification proceeds as follows. First, we have

$$\begin{aligned} \sup_{f \in \mathcal{F}} \|f\|_{C^2(I)} &= \sup_{z \in B(0; 1)} \|f_z\|_{C^2(I)} \\ &\leq \sup_{\theta \in I} (|\gamma(\theta)| + |\gamma'(\theta)| + |\gamma''(\theta)|) |z| \\ &\leq \|\gamma\|_{C^2(I)} < \infty, \end{aligned}$$

and depends only on γ . Next, if $z_1, z_2 \in B(0, 1)$ then $\|f_{z_1} - f_{z_2}\|_{C^2(I)} \leq \|\gamma\|_{C^2(I)} |z_1 - z_2|$. Furthermore, the non-degeneracy condition (14.6.3) implies that

$$\begin{aligned} \inf_{\theta \in I} (|f_{z_1}(\theta) - f_{z_2}(\theta)| + |f'_{z_1}(\theta) - f'_{z_2}(\theta)| \\ + |f''_{z_1}(\theta) - f''_{z_2}(\theta)|) \gtrsim \|f_{z_1} - f_{z_2}\|_{C^2(I)}, \end{aligned}$$

where the implicit constant depends only on γ . As a consequence,

$$\|f_{z_1} - f_{z_2}\|_{C^2(I)} \sim |z_1 - z_2| \text{ for } z_1, z_2 \in B(0; 1). \quad (14.6.19)$$

Thus $\mathcal{F}$ is a doubling metric space (with the same doubling constant as $\mathbb{R}^3$). Finally, identifying F with Z_δ , we observe from (14.6.19) that the Frostman-type non-concentration condition (14.6.7) for F follows from the analogous condition (14.6.17) for the Euclidean set Z_δ . $\square$

We are now ready to prove Theorem 14.6.3. Let $\gamma: I \rightarrow S^2$ satisfy (14.6.3) and let $Z \subset \mathbb{R}^3$ be Borel. After a harmless re-scaling, we can suppose $Z \subset B(0, 1)$ and after possibly replacing Z by a subset, we may suppose that $\dim Z \leq 1$. Let $0 \leq s < \dim Z$. Aiming for a contradiction, suppose if possible that (14.6.4) fails, i.e. $\dim \Theta_I > s$, where

$$\Theta_I = \{\theta \in I : \dim(Z \cdot \gamma(\theta)) < s\}.$$

Then Θ_I admits a subset Θ with the property that $s < \dim \Theta \leq \dim Z$. Set $\alpha = \dim \Theta$. We will show that the existence of such a set $\Theta \subset \Theta_I$ contradicts the validity of Proposition 14.6.8.

Define

$$\varepsilon = \frac{\alpha - s}{2C + 11} > 0, \quad (14.6.20)$$

where C is the constant from (14.6.18).

14.6.2.1 Initial discretization

Bourgain [2] proved a discretized projection theorem, which bounded the number of exceptional directions where the projection of a δ -discretized set can be small. Bourgain then showed how this discretized theorem implies an analogous Hausdorff dimension bound. The next set of arguments are similar to the reduction from Theorem 4 to Theorem 3 in [2], but with a twist. The location of the twist will be noted below.

Let $\zeta = \dim Z$ and let $\mathbb{P}$ be a probability measure supported on Z that satisfies the Frostman condition

$$\mathbb{P}(B) \lesssim r^{\zeta - \varepsilon} \quad \text{for all balls } B \subset \mathbb{R}^3 \text{ of radius } r > 0. \quad (14.6.21)$$

Let ν be a probability measure supported on Θ with

$$\nu(J) \lesssim |J|^{\alpha - \varepsilon} \quad \text{for all intervals } J \subset \mathbb{R}. \quad (14.6.22)$$

Let $k_0 \in \mathbb{Z}$ be chosen sufficiently large, depending on γ , ζ , s and α . The choice of k_0 will be determined as follows: our arguments below will select a number $0 < \delta \leq 2^{-\lfloor (1+\varepsilon)k_0 \rfloor}$ (as in [14, Section 7]; we do not get to choose δ , but our choice of k_0 gives an upper bound for δ). Eventually, we will arrive at the conclusion that a certain inequality involving δ is impossible, provided $\delta > 0$ is sufficiently small (depending on γ , ζ , s and α). In order to arrive at a contradiction, we will select $k_0 \in \mathbb{Z}$ sufficiently large so that any positive $\delta \leq 2^{-\lfloor (1+\varepsilon)k_0 \rfloor}$ obeys Proposition 14.6.8.

For each $\theta \in I$, define $\rho_\theta: \mathbb{R}^3 \rightarrow \mathbb{R}$ by $\rho_\theta(z) = z \cdot \gamma(\theta)$. For each $\theta \in \text{supp}(\nu) \subset \Theta$, cover $\rho_\theta(Z)$ by a union of sets $\{G_{\theta,k}\}_{k=k_0}^\infty$, where $G_{\theta,k}$ is a $(2^{-\lfloor (1+\varepsilon)k \rfloor}, s; 2^{\lfloor (1+\varepsilon)k \rfloor})_1$ -set (in Definition 14.6.4 a $(\delta, \alpha; C)_1$ -set must be

contained in $[0, 1]$, but it is harmless to apply a translation); see i.e. [14, Lemma 7.5] for details on how to do this. This is the “twist” alluded to above; in [2], Bourgain simply covers $\rho_\theta(A)$ by unions of 2^{-k} intervals; he does not ask that these unions satisfy the non-concentration condition (14.6.5).

For each $\theta \in \text{supp}(\nu)$ we have

$$1 = \mathbb{P}(Z) = \mathbb{P}(Z \cap \rho_\theta^{-1}(\rho_\theta(Z))) = \mathbb{P}\left(Z \cap \bigcup_{k \geq k_0} \rho_\theta^{-1}(G_{\theta,k})\right).$$

Thus by integrating over I with respect to ν we conclude that

$$\int_I \sum_{k \geq k_0} \mathbb{P}(Z \cap \rho_\theta^{-1}(G_{\theta,k})) d\nu(\theta) \geq \nu(\Theta) = 1.$$

In particular, there exists $k \geq k_0$ so that

$$\int_I \mathbb{P}(Z \cap \rho_\theta^{-1}(G_{\theta,k})) d\nu(\theta) \gtrsim \frac{1}{k^2}. \quad (14.6.23)$$

We will fix this choice of k , and for the remainder of this proof, define $\delta = 2^{-\lfloor (1+\varepsilon)^k \rfloor}$ and $G_\theta = G_{\theta,k}$; thus the right hand side of (14.6.23) becomes essentially $\left(\frac{\log(1+\varepsilon)}{\log \log(1/\delta)}\right)^2$, which is $\geq |\log \delta|^{-2}$ if δ is sufficiently small relative to ε .

Recalling the notation introduced in Section 14.6.1.4, we will write $A \gtrsim B$ to mean $A \gtrsim |\log \delta|^{-C} B$, where C is an absolute constant (in practice, we will always have $C \leq 100$).

14.6.2.2 Replacing $\mathbb{P}$ by a sum over curves

Given the choice of δ above, let us consider a covering of Z using cubes of sidelength $\delta/2$. After dyadic pigeonholing, we can find a set $\mathcal{Q}$ of such cubes and a “weight” w , so that $w \leq \mathbb{P}(Q) < 2w$ for each $Q \in \mathcal{Q}$, and

$$\int_I \mathbb{P}\left(\bigcup_{Q \in \mathcal{Q}} Q \cap \rho_\theta^{-1}(G_\theta)\right) d\nu(\theta) \gtrsim 1. \quad (14.6.24)$$

Let $G'_\theta = N_\delta(G_\theta)$; here and henceforth, $N_\delta(X)$ denotes the δ -neighbourhood of the set X . Then G'_θ is a $(3\delta, s; 3\delta^{-\varepsilon})_1$ -set, and if $Q \in \mathcal{Q}$ is a cube that intersects $\rho_\theta^{-1}(G_\theta)$, then $Q \subset \rho_\theta^{-1}(G'_\theta)$. Thus (14.6.24) implies that

$$\int_I w \#\{Q \in \mathcal{Q} : Q \subset \rho_\theta^{-1}(G'_\theta)\} d\nu(\theta) \gtrsim 1,$$

or

$$\int_I \#\{Q \in \mathcal{Q} : \rho_\theta(Q) \subset G'_\theta\} d\nu(\theta) \gtrsim w^{-1}. \quad (14.6.25)$$

Since Q is a cube of side-length $\delta/2$ and $\gamma(\theta) \in S^2$, we have that $\rho_\theta(Q)$ is an interval in $\mathbb{R}$ of length $\sim \delta$. Thus $\int_{G'_\theta} \mathbf{1}_{\rho_\theta(Q)}(y) dy \sim \delta$ for each $Q \in \mathcal{Q}$ with $\rho_\theta(Q) \subset G'_\theta$. Hence (14.6.25) implies that

$$\int_I \int_{G'_\theta} \sum_{Q \in \mathcal{Q}} \mathbf{1}_{\rho_\theta(Q)}(y) dy d\nu(\theta) \gtrsim \delta w^{-1}. \quad (14.6.26)$$

Note that $\delta^3 \lesssim w \lesssim \delta^{\zeta-\varepsilon}$, and $1 \lesssim w \# \mathcal{Q} \leq 1$. Let $A = \delta^{\zeta-\varepsilon} w^{-1} \gtrsim 1$, so that $\# \mathcal{Q} \gtrsim A \delta^{-\zeta+\varepsilon}$ and $w = \delta^{\zeta-\varepsilon} A^{-1}$. Thus the right hand side of (14.6.26) becomes $A \delta^{1-\zeta+\varepsilon}$.

Let Z'_δ denotes the set of centers of the cubes in $\mathcal{Q}$; then the Frostman condition (14.6.21) implies that

$$\#(Z'_\delta \cap B) \lesssim w^{-1} \mathbb{P}(B) \lesssim w^{-1} r^{\zeta-\varepsilon} = A(r/\delta)^{\zeta-\varepsilon} \quad (B \text{ a ball of radius } r \geq \delta). \quad (14.6.27)$$

Note that if $Q \in \mathcal{Q}$ has center $z \in Z'_\delta$, then $Q \subset N_\delta(z)$ (this is why we required that the cubes in $\mathcal{Q}$ have side-length $\delta/2$, rather than δ). Thus for each $\theta \in I$, we have the implication

$$y \in \rho_\theta(Q) \implies |\gamma(\theta) \cdot z - y| \leq \delta.$$

This leads to the inequality

$$\sum_{Q \in \mathcal{Q}} \mathbf{1}_{\rho_\theta(Q)}(y) \leq \sum_{z \in Z'_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y). \quad (14.6.28)$$

Combining (14.6.26) and (14.6.28), we obtain

$$\int_I \int_{G'_\theta} \sum_{z \in Z'_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy d\nu(\theta) \gtrsim A \delta^{1-\zeta+\varepsilon}. \quad (14.6.29)$$

Let Z_δ be a subset of Z'_δ obtained by randomly and independently selecting each point of Z'_δ with probability $p = A^{-1}$; we will select the set Z_δ so that the following properties hold.

$$\#(Z_\delta \cap B) \leq |\log \delta| A^{-1} \#(Z'_\delta \cap B) \leq \delta^{-\varepsilon} (r/\delta)^\zeta \quad (B \text{ a ball of radius } r \geq \delta), \quad (14.6.30)$$

and

$$\int_I \int_{G'_\theta} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy d\nu(\theta) \gtrsim \delta^{1-\zeta+\varepsilon}. \quad (14.6.31)$$

Let us briefly discuss the selection of Z_δ . Apriori, (14.6.30) appears to be an assertion about an uncountable set of balls $\{B \subset \mathbb{R}^3 : \text{radius}(B) \geq \delta\}$. However, it suffices to establish (14.6.30) (with the RHS replaced by $\frac{1}{2} \delta^{-\varepsilon} (r/\delta)^\zeta$) for a

set of roughly δ^{-3} balls: For each $j = 0, \dots, |\log \delta|$, let $\mathcal{B}_j$ be a set of about $(2^j \delta)^{-3}$ balls of radius $2^{j+1} \delta$, so that every ball of radius $2^j \delta$ is contained in at most $O(1)$ balls from $\mathcal{B}_j$. Then it suffices to prove (14.6.30) for the balls in $\mathcal{B} = \bigcup_{j=0}^{|\log \delta|} \mathcal{B}_j$; this set has cardinality $\lesssim \delta^{-3}$. If δ is small enough, then using Chernoff's inequality and (14.6.27) we can ensure that the probability that (14.6.30) fails for any particular ball in $\mathcal{B}$ is smaller than δ^4 . Applying the union bound over $B \in \mathcal{B}$, we deduce that with probability at least $1 - O(\delta)$, the estimate (14.6.30) holds for all balls $B \in \mathcal{B}$ and hence in general for all balls of radius $r \geq \delta$. Choosing such an outcome with the additional requirement (also derivable from Chernoff's inequality) that

$$\#Z_\delta > \frac{1}{100A} \#Z'_\delta,$$

the property (14.6.31) follows from (14.6.29) and an expected value calculation (expectation is linear) and Markov's inequality.

14.6.2.3 Replacing ν by an integral over a quasi-product

Let $\mathcal{J}_0$ be the set of intervals in I (recall I is the domain of $\gamma(\theta)$) of the form $[n\delta, (n+1)\delta)$, for $n \in \mathbb{Z}$. For each $J \in \mathcal{J}_0$, let $\theta(J) \in J$ be a value of θ that gets within a factor of 2 of maximizing the quantity

$$\int_{G'_\theta} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy.$$

For each $J \in \mathcal{J}_0$, we have

$$\begin{aligned} \nu(J) \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta(J), y) dy &= \int_J \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta(J), y) dy d\nu(\theta) \\ &\geq \frac{1}{2} \int_J \int_{G'_\theta} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy d\nu(\theta). \end{aligned} \quad (14.6.32)$$

The first equality is just the statement that $\nu(J) = \int_J d\nu(\theta)$, while the second inequality follows from the definition of $\theta(J)$. To simplify notion, define

$$K(J) = \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta(J), y) dy,$$

so the left hand side of (14.6.32) is $\nu(J)K(J)$. Summing (14.6.32) over $J \in \mathcal{J}_0$, we obtain

$$\begin{aligned} \sum_{J \in \mathcal{J}_0} \nu(J)K(J) &\geq \frac{1}{2} \sum_{J \in \mathcal{J}_0} \int_J \int_{G'_\theta} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy d\nu(\theta) \\ &= \frac{1}{2} \int_I \int_{G'_\theta} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta, y) dy d\nu(\theta) \\ &\gtrsim \delta^{1-\zeta+\varepsilon}. \end{aligned}$$

Observe that $K(J) \lesssim \delta \# Z_\delta \lesssim \delta^{1-\zeta+\varepsilon}$ for each $J \in \mathcal{J}_0$, and $\sum_{J \in \mathcal{J}_0} \nu(J) = 1$. Thus by dyadic pigeonholing, there is a set $\mathcal{J}_1 \subset \mathcal{J}_0$ and a weight $\delta \lesssim w' \leq \delta^{\alpha-\varepsilon}$ so that $w' \leq \nu(J) < 2w'$ for each $J \in \mathcal{J}_1$, and $1 \lesssim w' \# \mathcal{J}_1 \leq 1$. We can write $A' = \delta^{\alpha-\varepsilon} w'^{-1} \gtrsim 1$, and thus $\# \mathcal{J}_1 \approx A' \delta^{-\alpha+\varepsilon}$, and we have

$$\sum_{J \in \mathcal{J}_1} \delta^{\alpha-\varepsilon} (A')^{-1} K(J) \gtrsim \delta^{1-\zeta+\varepsilon},$$

i.e.

$$\sum_{J \in \mathcal{J}_1} K(J) \gtrsim A' \delta^{1-\zeta-\alpha+2\varepsilon}.$$

We will now perform a random sampling argument analogous to how we selected $Z_\delta \subset Z'_\delta$ at the end of Section 14.6.2.2. Let $p = (A')^{-1}$, and select $\mathcal{J} \subset \mathcal{J}_1$ by randomly and independently choosing each $J \in \mathcal{J}_1$ with probability p . By (14.6.22), with positive probability, we may choose $\mathcal{J}$ so that the following two properties hold.

$$|\Lambda \cap \bigcup_{J \in \mathcal{J}} J| \leq \delta |\log \delta| (|\Lambda|/\delta)^{\alpha-\varepsilon} \leq \delta^{1-\varepsilon} (|\Lambda|/\delta)^{\alpha-\varepsilon} \quad \text{for all intervals } \Lambda, \quad (14.6.33)$$

(i.e. $\bigcup_{J \in \mathcal{J}} J$ is a $(\delta, \alpha - \varepsilon; \delta^{-\varepsilon})_1$ -set), and

$$\sum_{J \in \mathcal{J}} K(J) \gtrsim \delta^{1-\zeta-\alpha+2\varepsilon}. \quad (14.6.34)$$

(14.6.33) and (14.6.34) are the analogues of (14.6.30) and (14.6.31), respectively, and the argument is the same.

Next, we claim that if $(\theta, y) \in \Gamma_z^\delta$, and if $|\theta - \theta'| \leq \delta$, then $\theta' \in \Gamma_z^{L\delta}$, where

$$L = \sup_{z \in Z_\delta} \sup_{\theta \in I} \left| \frac{d}{d\theta} \rho_\theta(z) \right| \lesssim 1.$$

Thus for each $z \in Z_\delta$, each $y \in \mathbb{R}$, and each $J \in \mathcal{J}$, we have

$$\int_J \mathbf{1}_{\Gamma_z^\delta}(\theta(J), y) d\theta \leq \int_J \mathbf{1}_{\Gamma_z^{L\delta}}(\theta, y) d\theta. \quad (14.6.35)$$

Integrating (14.6.35) over $G'_{\theta(J)}$ and summing over $z \in Z_\delta$ (and interchanging the order of integration and summation), we obtain

$$\begin{aligned} \delta K(J) &= \int_J \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^\delta}(\theta(J), y) dy d\theta \\ &\leq \int_J \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^{L\delta}}(\theta, y) dy d\theta. \end{aligned}$$

Combining this with (14.6.34), we conclude that

$$\begin{aligned} \delta^{2-\zeta-\alpha+2\varepsilon} &\lesssim \sum_{J \in \mathcal{J}} \delta K(J) \\ &\leq \sum_{J \in \mathcal{J}''} \int_J \int_{G'_{\theta(J)}} \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^{L_\delta}}(\theta, y) dy d\theta. \end{aligned} \quad (14.6.36)$$

Finally, define

$$E = \bigcup_{J \in \mathcal{J}} J \times G'_{\theta(J)}.$$

Then E is a $(\delta, \alpha - \varepsilon; \delta^{-\varepsilon})_1 \times (3\delta, s; 3\delta^{-\varepsilon})_1$ quasi-product, and (14.6.36) can be re-written as

$$\int_E \sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^{L_\delta}} \gtrsim \delta^{2-\zeta-\alpha+2\varepsilon}. \quad (14.6.37)$$

Since

$$|E| \lesssim (\delta^{1-\alpha+\varepsilon} \delta^{-\varepsilon}) \times ((3\delta)^{1-s} 3\delta^{-\varepsilon}) \sim \delta^{2-\alpha-s-2\varepsilon},$$

the relation (14.6.37) together with Hölder's inequality implies that

$$\int_E \left(\sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^{L_\delta}} \right)^{3/2} \gtrsim \delta^{2+\frac{1}{2}s-\alpha-\frac{3}{2}\zeta+4\varepsilon}. \quad (14.6.38)$$

14.6.2.4 Applying Proposition 14.6.8

Since E is a $(\delta, \alpha - \varepsilon; \delta^{-\varepsilon})_1 \times (3\delta, s; 3\delta^{-\varepsilon})_1$ quasi-product and $s < \alpha$, E can be expressed as a union of $O(1)$ many $(\delta, \alpha, \delta^{-\varepsilon})_1 \times (\delta, \alpha, \delta^{-\varepsilon})_1$ quasi-products. Thus we can apply Proposition 14.6.8 with $\#Z_\delta \lesssim \delta^{-\zeta-\varepsilon}$ and use the triangle inequality to conclude that

$$\int_E \left(\sum_{z \in Z_\delta} \mathbf{1}_{\Gamma_z^{L_\delta}} \right)^{3/2} \lesssim \delta^{2-\frac{1}{2}\alpha-\frac{3}{2}\zeta-C\varepsilon-\varepsilon}. \quad (14.6.39)$$

Comparing (14.6.38) and (14.6.39), we conclude that $\delta^{s-\alpha} \lesssim \delta^{-(2C+10)\varepsilon}$. By selecting $\delta > 0$ sufficiently small (depending on ε), we conclude that $\alpha - s \leq (2C + 10)\varepsilon$. But comparing with (14.6.20), we obtain a contradiction. This finishes the proof of Theorem 14.6.3.

14.6.3 The geometry of cinematic functions

In this section, we will explore some geometric properties of families of cinematic functions. Many of the results in this section are inspired by analogous results in [13, 17, 29]. For the definitions that follow, we will fix a family of cinematic

functions $\mathcal{F}$ over a compact interval J , with cinematic constant K and doubling constant D .

For $f \in \mathcal{F}$, we write f^λ to denote the closed λ -vertical neighborhood of the graph of f , i.e. the set $\{(\theta, y) \in J \times \mathbb{R} : |f(\theta) - y| \leq \lambda\}$. This definition differs slightly from the definition of f^δ given in the statement of Theorem 14.6.7. This distinction does not matter, however, since f^λ is comparable to the λ -neighborhood of the graph of f .

Definition 14.6.9 (Curvilinear rectangles). Let $0 < \delta \leq 1$, $\delta \leq t \leq 1$, and $f \in \mathcal{F}$. A (δ, t) -rectangle is defined to be a set of the form

$$f^\delta(I) = \{(\theta, y) \in I \times \mathbb{R} : |y - f(\theta)| \leq \delta\},$$

where $I \subset J$ is an interval of length $\sqrt{\delta/t}$.

Definition 14.6.10 (Comparability). We say two (δ, t) -rectangles R, R' are λ -comparable, if there is another $(\lambda\delta, t)$ -rectangle R'' that contains $R \cup R'$. Otherwise we say R, R' are λ -incomparable.

Definition 14.6.11 (Tangency). We say a (δ, t) -rectangle R is λ -tangent to the function $f \in \mathcal{F}$ (or f is λ -tangent to R) if $R \subset f^{\lambda\delta}$. In most cases we take $\lambda = 5$, and we will simply refer to this as R is tangent to f .

Definition 14.6.12 (Separation). Let $\mathcal{W}, \mathcal{B} \subset \mathcal{F}$. For $t > 0$, we say $(\mathcal{W}, \mathcal{B})$ is t -separated if $\|w - b\| \geq t$ for all $w \in \mathcal{W}$, $b \in \mathcal{B}$.

14.6.3.1 Preliminary reductions

We start with the following lemma, which is similar to [13, Lemma 3.1].

Lemma 14.6.13. *If $I \subset J$ is an interval of length $|I| \leq (6K)^{-1}$, $f, g \in \mathcal{F}$, and k is either of the functions $f - g$ or $f' - g'$, then either of the following alternatives hold (depending on the choice of I and k):*

$$(S) \quad |k(\theta)| < (3K)^{-1} \|f - g\| \text{ for all } \theta \in I.$$

$$(L) \quad |k(\theta)| \geq (6K)^{-1} \|f - g\| \text{ for all } \theta \in I.$$

Moreover, if for I the alternative (S) holds for both $k = f - g$ and $k = f' - g'$, then the alternative (L) must hold for $k = f'' - g''$.

Proof. Let $\overline{B}$ be the closed unit ball of $C^2(J)$. Then if $|\theta - \theta'| < (6K)^{-1}$ and $h \in \overline{B}$, we have $|h(\theta) - h(\theta')| < (6K)^{-1}$ and $|h'(\theta) - h'(\theta')| < (6K)^{-1}$.

Next, fix $I \subset J$ with $|I| < (6K)^{-1}$, and suppose that the alternative (L) fails for $k = f - g$. So there is $\theta_0 \in I$ such that $|k(\theta_0)| < (6K)^{-1} \|f - g\|$.

Then, if $\theta \in I$ is arbitrary, we have $|\theta - \theta_0| \leq (6K)^{-1}$, and so if we define $l = k/\|f - g\| \in \overline{B}(0, 1)$, then

$$\frac{|k(\theta)|}{\|f - g\|} \leq \frac{|k(\theta_0)|}{\|f - g\|} + \left| \frac{k(\theta)}{\|f - g\|} - \frac{k(\theta_0)}{\|f - g\|} \right| < \frac{1}{6K} + |l(\theta) - l(\theta_0)| < \frac{1}{3K}.$$

Thus the alternative (S) holds for k . The proof of the case $k = f' - g'$ is the same. The last assertion follows from the cinematic curvature assumption (14.6.6). $\square$

To prove Theorem 14.6.7, it suffices to prove (14.6.8) for each individual subinterval I of length $\leq (6K)^{-1}$, since the restriction of the functions in $\mathcal{F}$ to each such sub-interval I is still family of cinematic functions with cinematic constant K and doubling constant D . By an abuse of notation, we continue to write $I = J$, and assume without loss of generality that $|J| \leq (6K)^{-1}$.

The following lemma is an analogue of [13, Lemma 3.2].

Lemma 14.6.14. *For distinct $f, g \in \mathcal{F}$, the map $\theta \mapsto f(\theta) - g(\theta)$ has at most two zeros in J . Moreover, if $\theta \mapsto f'(\theta) - g'(\theta)$ has two zeros in J , then the alternative (L) holds for $\theta \mapsto f(\theta) - g(\theta)$.*

Proof. We start with the second claim. Assume $\theta \mapsto f'(\theta) - g'(\theta)$ has two zeros in J . This implies, by Rolle's theorem, that $\theta \mapsto f''(\theta) - g''(\theta)$ has a zero in J , so in particular the alternative (L) cannot hold for $\theta \mapsto f''(\theta) - g''(\theta)$. Now Lemma 14.6.13 implies that the alternative (S) holds for J and $\theta \mapsto f'(\theta) - g'(\theta)$. Thus by the last sentence of Lemma 14.6.13, we must have that (L) holds for $\theta \mapsto f(\theta) - g(\theta)$.

The first claim follows from the second one: if $\theta \mapsto f(\theta) - g(\theta)$ had three zeros in J , then $\theta \mapsto f'(\theta) - g'(\theta)$ would have two zeros in J again by the Rolle's theorem. But then, by the second claim, $\theta \mapsto f(\theta) - g(\theta)$ satisfies the alternative (L) in J , and hence cannot have zeros in J . $\square$

14.6.3.2 The tangency parameter Δ

Next, we shall define a quantity that measures how close two functions $f, g \in \mathcal{F}$ are to being tangent. Our definition is similar to the quantity (3.1) from [13], which in turn was inspired by [17, 29].

Definition 14.6.15. For $f, g \in \mathcal{F}$, we define

$$\Delta(f, g) = \min_{\theta \in J/2} |f(\theta) - g(\theta)| + |f'(\theta) - g'(\theta)|, \quad (14.6.40)$$

where $J/2$ is the interval of length $|J|/2$ with the same midpoint as J .

In particular, we have

$$\Delta(f, g) \leq \|f - g\|. \quad (14.6.41)$$

$\Delta(f, g)$ measures the tangency between the graphs of functions f, g over the interval $J/2$. If $\Delta(f, g) = 0$, then f and g are tangent at some point $\theta \in J/2$. Note that $\Delta(f, g)$ is a pseudo-metric, since it is symmetric and satisfies the triangle inequality. The restriction to $J/2$ but not J is needed for technical reasons, as is also the case of [17] and [13].

The next proposition is a close cousin of [17, Lemma 3.3]. It describes the key geometric information about cinematic functions.

Lemma 14.6.16. *Let $f, g \in \mathcal{F}$, and define $h = f - g$, $t = \|f - g\|$, and $\Delta = \Delta(f, g)$.*

1. *There exists $c_1 = c_1(K) > 0$ so that if $\Delta \leq c_1 t$, then the following are true.*
 - (a) *There is a unique $\theta_0 \in \frac{3}{5}J$ such that $h'(\theta_0) = 0$. We also have $|h(\theta_0)| \lesssim \Delta$.*
 - (b) *We have $|h(\theta)| \gtrsim t > 0$ for every $\theta \in J \setminus (\frac{4}{5}J)$. If in addition h changes sign in J , then h has exactly 2 zeros $\theta_1 < \theta_2$, both within $\frac{4}{5}J$, such that $|\theta_i - \theta_0| \lesssim \sqrt{\Delta/t}$, $i = 1, 2$.*
2. *For each $0 < \delta \leq t/(6K)$, the set*

$$E_\delta = \{\theta \in J/4 : |h(\theta)| \leq \delta\} \quad (14.6.42)$$

satisfies the following:

- (a) *E_δ is contained in an interval of length $O(\sqrt{(\Delta + \delta)/t})$.*
- (b) *E_δ is either a closed interval of length $O(\delta/\sqrt{(\Delta + \delta)t})$ or the union of two closed intervals of length $O(\delta/\sqrt{(\Delta + \delta)t})$.*
- (c) *If in addition there is some $\tilde{\theta} \in J/4$ such that $|h(\tilde{\theta})| \leq \delta/2$, and I is the interval constituting E_δ that contains $\tilde{\theta}$, then $|I| \gtrsim \delta/\sqrt{(\Delta + \delta)t}$.*

Proof. Let $\theta_\Delta \in J/2$ be such that

$$\Delta = |h(\theta_\Delta)| + |h'(\theta_\Delta)|.$$

Thus $|h(\theta_\Delta)| \leq \Delta$ and $|h'(\theta_\Delta)| \leq \Delta$.

We first prove Part (1). If we choose $c_1 < (3K)^{-1}$, then using the definition of $\Delta(f, g)$, we have the alternative (S) holds for both h, h' . Thus the alternative (L) must hold for h'' . Without loss of generality we may assume $h''(\theta) \gtrsim t$.

For Part (1a), using $h''(\theta) \gtrsim t$, if c_1 is chosen to be small enough, then there is a unique $\theta_0 \in (\frac{1}{2} + O(\Delta/t))J \subset \frac{3}{5}J$ with $h'(\theta_0) = 0$. Also, $|\theta_0 - \theta_\Delta| \lesssim \Delta/t$, and thus

$$|h(\theta_0) - h(\theta_\Delta)| \lesssim t|\theta_0 - \theta_\Delta| \lesssim \Delta.$$

Since $|h(\theta_\Delta)| \leq \Delta$, we also have $|h(\theta_0)| \lesssim \Delta$.

For Part (1b), write $J = [a, b]$. Using $|h(\theta_0)| \lesssim \Delta$, for any $\theta \in [\theta_0, b]$ we have

$$\begin{aligned} |h(\theta)| &\geq \left| \int_{\theta_0}^{\theta} h'(s) ds \right| - C\Delta = \int_{\theta_0}^{\theta} |h'(s)| ds - C\Delta \\ &= \int_{\theta_0}^{\theta} \int_{\theta_0}^s h''(u) du ds - C\Delta \geq ct(\theta - \theta_0)^2 - C\Delta. \end{aligned} \quad (14.6.43)$$

Here C, c are absolute constants. The same is true for $\theta \in [a, \theta_0]$. If c_1 is small enough, we then have $|h(\theta)| \gtrsim t$ for every $\theta \in J \setminus (\frac{4}{5}J)$, since $\theta_0 \in \frac{3}{5}J$ and $|J| \sim 1$. The statements about the zeros follow from the intermediate value theorem and a similar application of (14.6.43).

Now we come to Part (2). First, for Part (2a), if $\Delta > c_1 t$ then the result is trivial. If $\Delta \leq c_1 t$, then the argument of Part (1) applies. Let $\alpha = \inf E_\delta$ and $\beta = \sup E_\delta$, and hence $E_\delta \subset [\alpha, \beta]$. Using (14.6.43) we have $\beta - \theta_0 \leq C\sqrt{(\Delta + \delta)/t}$ and $\theta_0 - \alpha \leq C\sqrt{(\Delta + \delta)/t}$. Thus the result follows.

For Part (2b), we first note that the alternative (L) cannot hold for h , otherwise it will contradict the assumption $\delta < t/(6K)$. Thus (L) holds for h' or h'' . In either case, we see E_δ is a closed interval or the union of two closed intervals.

For the length estimate, we first consider the easier case $\Delta > c_1 t$. In this case, if $\delta > c_1 t/2$, then the result is trivial since then $\delta \sim \Delta \sim t$. If $\delta \leq c_1 t/2$, then for $\theta \in E_\delta$ we have

$$c_1 t < \Delta \leq \min_{\theta \in E_\delta} (|h(\theta)| + |h'(\theta)|) \leq \delta + \min_{\theta \in E_\delta} |h'(\theta)|,$$

and thus $|h'(\theta)| \gtrsim t$ on E_δ , and thus each interval constituting E_δ has length $\lesssim \delta/t = \delta/\sqrt{(\Delta + \delta)t}$.

For the harder case $\Delta \leq c_1 t$, the argument of Part (1) is applicable. We also have two subcases. First, if $\delta \geq c_1 \Delta$, then the result follows from Part (2a), since then $\sqrt{(\Delta + \delta)/t} \sim \delta/\sqrt{(\Delta + \delta)t}$.

If $\delta < c_1 \Delta$, then we may further assume $\theta_0 \in J/2$. Otherwise, using (14.6.43) we have $h(\theta) \gtrsim t$ for every $\theta \in J/4$ if c_1 is small enough. In particular, $E_\delta = \emptyset$ and we have nothing to prove. Thus we have $\theta_0 \in J/2$.

But by the definition of Δ and θ_0 , we have $|h(\theta_0)| = |h(\theta_0)| + |h'(\theta_0)| \geq \Delta$. Thus for every $\theta \in E_\delta \cap [\theta_0, b]$, we have

$$\begin{aligned} \delta &\geq |h(\theta)| \geq \Delta - \int_{\theta_0}^{\theta} |h'(s)| ds \\ &\geq \Delta - \int_{\theta_0}^{\theta} \int_{\theta_0}^s |h''(u)| du ds \geq \Delta - C't(\theta - \theta_0)^2. \end{aligned}$$

Here C' is an absolute constant. For $\theta \in E_\delta \cap [a, \theta_0]$ the argument is the same. This implies that

$$|\theta - \theta_0| \gtrsim \sqrt{\Delta/t}, \quad \forall \theta \in E_\delta. \quad (14.6.44)$$

Since $h''(\theta) \gtrsim t$ in J , this further implies that for $\theta \in E_\delta$

$$|h'(\theta)| = |h'(\theta) - h'(\theta_0)| \gtrsim t|\theta - \theta_0| \gtrsim \sqrt{t\Delta}. \quad (14.6.45)$$

Thus each interval constituting E_δ has length $O(\delta/\sqrt{t\Delta}) = O(\delta/\sqrt{(\Delta + \delta)t})$.

Now we come to Part (2c). Let I be the interval constituting E_δ that contains $\tilde{\theta}$. If $I = J/4$ then the result is trivial, so we assume $I \neq J/4$. In this case, there exists an endpoint θ_I of I so that $|h(\theta_I)| = \delta$.

If $\Delta > c_1 t$, then we have

$$\delta/2 \leq |h(\theta_I)| - |h(\tilde{\theta})| \lesssim |\theta_I - \tilde{\theta}|t,$$

and so $|\theta_I - \tilde{\theta}| \gtrsim \delta/t \sim \delta/\sqrt{(\Delta + \delta)t}$.

If $\Delta \leq c_1 t$, then the argument of Part (1) is applicable. For every $\theta \in E_\delta$, the argument of Part (2a) implies that $|\theta - \theta_0| \lesssim \sqrt{(\Delta + \delta)t}$. Thus

$$|h'(\theta)| = |h'(\theta) - h'(\theta_0)| \lesssim t|\theta - \theta_0| \lesssim \sqrt{(\Delta + \delta)t}, \quad \forall \theta \in E_\delta.$$

Hence

$$\delta/2 \leq |h(\theta_I)| - |h(\tilde{\theta})| \lesssim |\theta_I - \tilde{\theta}|\sqrt{(\Delta + \delta)t},$$

which implies that $|\theta_I - \tilde{\theta}| \gtrsim \delta/\sqrt{(\Delta + \delta)t}$. In particular, $|I| \gtrsim \delta/\sqrt{(\Delta + \delta)t}$. $\square$

14.6.3.3 Further geometric estimates

We conclude this section with some simple arguments that relate the quantities $\|f - g\|$, $\Delta(f, g)$, the volume of $f^\delta \cap g^\delta$, and the number of tangency rectangles associated to the intersection.

Lemma 14.6.17. *Let $f, g \in \mathcal{F}$ and let $t > 0$. If there is a (δ, t) -rectangle R over the interval $J/4$ (i.e. the interval I from Definition 14.6.9 is contained in $J/4$) that is tangent to both f and g , then $(\Delta(f, g) + \delta)\|f - g\| \lesssim \delta t$.*

Proof. Let R be a (δ, t) -rectangle over an interval $I \subset J/4$. By Definition 14.6.11 and the triangle inequality, for each $\theta \in I$ we have $|f(\theta) - g(\theta)| \leq 10\lambda\delta$. Since $I \subset J/4$, by Part (2b) of Lemma 14.6.16, we have

$$\sqrt{\delta/t} = |I| \lesssim \frac{\delta}{\sqrt{(\Delta(f, g) + \delta)\|f - g\|}}. \quad \square$$

Lemma 14.6.18. *Let $I \subset J/4$ be an interval, let $\lambda \geq 1$, let $f, g \in \mathcal{F}$, and let $0 < \delta \leq \|f - g\|/(6K)$. If $|f(\theta) - g(\theta)| \leq \delta$ on I , then $|f(\theta) - g(\theta)| \lesssim \lambda^2\delta$ on $\lambda I \cap J$.*

Proof. Let $k = f - g$, let $t = \|f - g\|$, and let $\Delta = \Delta(f, g)$. Since $I \subset J/4$, by Part (2b) of Lemma 14.6.16 we have $|I| \lesssim \delta/\sqrt{(\Delta + \delta)t} \leq \sqrt{\delta/t}$.

We will first show that $|k'(\theta)| \leq C\lambda\delta|I|^{-1}$ on $\lambda I \cap J$, for some $C \lesssim 1$. Suppose not. Then there is some $\theta_1 \in \lambda I \cap J$ with $|k'(\theta_1)| > C\lambda\delta|I|^{-1}$. We have $|k''(\theta)| \leq t$ in J . Since $|I| \lesssim \sqrt{\delta/t}$, for any $\theta \in \lambda I \cap J$ we have

$$|k'(\theta)| \geq C\lambda\delta|I|^{-1} - \lambda t|I| \geq C\lambda\delta|I|^{-1}/2,$$

provided $C \sim 1$ is chosen appropriately. But since $k \in C^2(I)$ and $\lambda > 1$, if $C > 4$ then it is impossible for $|k'(\theta)| \geq C\lambda\delta|I|^{-1}/2$ and $|k(\theta)| \leq \delta$ to simultaneously hold for all $\theta \in I$. Hence $|k'(\theta)| \lesssim \lambda\delta|I|^{-1}$ on $\lambda I \cap J$. By the triangle inequality, we thus have $|k(\theta)| \lesssim \lambda^2\delta$ on $\lambda I \cap J$. $\square$

Definition 14.6.19. For $R = f^\delta(I)$ where $I \subset J$, we define the λ -dilation of R to be

$$\lambda R = f^{\lambda\delta}(\lambda I \cap J). \quad (14.6.46)$$

We will record several technical lemmas related to comparability.

Lemma 14.6.20. *Let $f, g \in \mathcal{F}$ and $\lambda \geq 1$. Let $R = f^\delta(I)$ and $R' = g^\delta(I')$ be two λ -comparable (δ, t) -rectangles over $J/4$. Let I'' be the convex hull of $I \cup I'$. Then $|I''| \leq \sqrt{\lambda\delta/t}$, and*

$$|f(\theta) - g(\theta)| \lesssim \lambda^3\delta \quad \text{for all } \theta \in I''. \quad (14.6.47)$$

In particular, $R' \subset C\lambda R$ for some $C \sim 1$.

Proof. By definition of comparability, there is some $(\lambda\delta, t)$ -rectangle $R'' = h^{\lambda\delta}(I_0)$ such that $R \cup R' \subset R''$ and $|I_0| = \sqrt{\lambda\delta/t}$. Since $I'' \subset I_0$, we have $|I''| \leq \sqrt{\lambda\delta/t}$.

It remains to prove (14.6.47). If $\|f - g\| < 12K\lambda\delta$ then (14.6.47) is immediate and we are done.

Next, suppose $\|f - g\| \geq 12K\lambda\delta$. We have

$$|f(\theta) - h(\theta)| \leq \lambda\delta \quad \text{for all } \theta \in I.$$

Using Lemma 14.6.18 (with $2\lambda\delta$ in place of δ ; note that $2\lambda\delta \leq \|f - g\|/(6K)$), we have

$$|f(\theta) - h(\theta)| \lesssim \lambda^3 \delta \quad \text{for all } \theta \in I''.$$

The same is true with f replaced by g . Then the result follows from the triangle inequality. $\square$

Corollary 14.6.21. *Let R_1, R_2, R_3 be (δ, t) rectangles over $J/4$. Suppose that R_1 is λ_1 -comparable to R_2 and R_2 is λ_2 -comparable to R_3 . Then R_1 is λ_3 -comparable to R_3 , for some λ_3 depending on K, λ_1 , and λ_2 .*

Lemma 14.6.22. *Let $0 < c < 1$. If C is large enough depending on c and K , then the following holds. Let $f, g \in \mathcal{F}$ and $R = f^\delta(I)$ and $R' = g^\delta(I')$ be two C -incomparable (δ, t) -rectangles over subintervals of $J/4$. For $0 < c < 1$, let $\tilde{R} = f^\delta(\tilde{I})$ and $\tilde{R}' = g^\delta(\tilde{I}')$ where $\tilde{I} = cI$, $\tilde{I}' = cI'$. Then $\tilde{R}$ and $\tilde{R}'$ are 100-incomparable.*

Proof. Suppose towards contradiction that $\tilde{R}$ and $\tilde{R}'$ are 100-comparable. Then

$$\text{dist}(cI, cI') \leq \sqrt{100c\delta/t},$$

and thus $\text{diam}(I \cup I') \lesssim \sqrt{\delta/t}$. Also, we have $|f(\theta) - g(\theta)| \leq 100\delta$ for $\theta \in \tilde{I}$, and thus by Lemma 14.6.20, we have $|f(\theta) - g(\theta)| \lesssim \delta$ for $\theta \in I \cup I'$. Thus R, R' are C -comparable for a sufficiently large C , from which a contradiction arises. $\square$

Lemma 14.6.23. *Let $\mathcal{R}$ be a family of pairwise 100-incomparable (δ, t) -rectangles $R = f^\delta(I)$ such that all $f \in \mathcal{F}$ are contained in a ball $B \subset C^2(I)$ of radius $3t$. Suppose on the other hand that all $R \in \mathcal{R}$ are contained in a $(\lambda\delta, t)$ -rectangle $\tilde{R}$ where $\lambda \geq 100$. Then $\#\mathcal{R} \lesssim \lambda^{5/2}$.*

Proof. We will show that every point in $\mathbb{R}^2$ can be contained in $\lesssim \lambda$ many rectangles $R/2$ where $R \in \mathcal{R}$. (Here $R/2 := f^\delta(I/2)$.) Then the lemma follows from the simple inequality

$$(\#\mathcal{R})\delta^{3/2}t^{-1/2} \lesssim \int_{\tilde{R}} \sum_{R \in \mathcal{R}} \mathbf{1}_{R/2} \lesssim \lambda |\tilde{R}| \lesssim \lambda^{5/2} \delta^{3/2} t^{-1/2}.$$

Suppose there is some $(\theta_0, y_0) \in \mathbb{R}^2$ that is contained in N pairwise 100-incomparable (δ, t) -rectangles of the form $R_i/2$, $i = 1, \dots, N$. We want to show $N \lesssim \lambda$.

It suffices to show that for every pair $1 \leq i \neq j \leq N$,

$$|f'_i(\theta_0) - f'_j(\theta_0)| \leq 10\lambda\sqrt{\delta t}, \quad (14.6.48)$$

$$|f'_i(\theta_0) - f'_j(\theta_0)| > \sqrt{\delta t}. \quad (14.6.49)$$

Suppose $(\theta_0, y_0) \in (R_i/2) \cap (R_j/2)$. In particular, $\theta_0 \in (I_i/2) \cap (I_j/2)$.

Denote $h = h_{i,j} = f_i - f_j$ and $I = I_i \cap I_j$. Then $\theta_0 \in I$, $\sqrt{\delta/t} \leq |I| \leq 2\sqrt{\delta/t}$ and $|h(\theta_0)| \leq \delta$.

For (14.6.48), suppose $|h'(\theta_0)| > 10\lambda\sqrt{\delta t}$. Then since $\|f_i - f_j\| \leq 6t$, we have in particular $\|h''\|_\infty \leq 6t$, and thus for every $\theta \in I$, we have

$$|h'(\theta)| > 10\lambda\sqrt{\delta t} - 12t\sqrt{\delta/t} \geq 5\lambda\sqrt{\delta t},$$

since $\lambda \geq 100$. Since $|h(\theta_0)| \leq \delta$, this implies that for some $\theta \in I$ we have

$$|h(\theta)| \geq 5\lambda\sqrt{\delta t}\sqrt{\delta/t}/2 - \delta > \lambda\delta,$$

contradicting the assumption that all rectangles $R_i \subset \tilde{R}$. This establishes (14.6.48).

For (14.6.49), we argue similarly. Suppose that $|h'(\theta_0)| \leq \sqrt{\delta t}$. Then using $\|h''\|_\infty \leq 6t$, we have for every $\theta' \in I$

$$|h'(\theta)| \leq \sqrt{\delta t} + 12t\sqrt{\delta/t} \leq 13\sqrt{\delta t}.$$

Thus for every $\theta \in I$ we have

$$|h(\theta)| \leq |h(\theta_0)| + 26\sqrt{\delta t}\sqrt{\delta/t} \leq 100\delta,$$

contradicting the assumption that R_i, R_j are 100-incomparable. This establishes (14.6.49). $\square$

Lemma 14.6.24. *Let $\mathcal{R}$ be a finite family of pairwise 100-incomparable (δ, t) -rectangles $R = f^\delta(I)$ such that all $f \in \mathcal{F}$ are contained in a ball of radius $3t$. If $C \geq 100$, then there is a C -incomparable subcollection $\mathcal{R}' \subset \mathcal{R}$ such that $\#\mathcal{R}' \sim \#\mathcal{R}$.*

Proof. Start with $R_1 \in \mathcal{R}$. Choose a maximally C -incomparable subcollection $\mathcal{R}'$ containing R_1 . For each $R \in \mathcal{R}'$, let $S(R)$ be the set of rectangles in $\mathcal{R}$ which are C -comparable to R . Then $\mathcal{R} = \cup_{R \in \mathcal{R}'} S(R)$ by maximality. To prove that $\#\mathcal{R} \sim \#\mathcal{R}'$, it suffices to show that $\#S(R) \lesssim 1$ for every $R \in \mathcal{R}'$. This follows from Lemma 14.6.20 with $\lambda = C$ and Lemma 14.6.23 with $\lambda = C^3$. $\square$

We end this section with a coarse bound that will be used in some dyadic pigeonholing argument later.

Lemma 14.6.25. *Let $\lambda \geq 100$. If $\mathcal{R}$ is a collection of pairwise λ -incomparable (δ, t) -rectangles of the form $R = f^\delta(I)$ where $f \in \mathcal{F}$, then we have a coarse bound $\#\mathcal{R} \leq \delta^{-C}$ for some $C = C(D)$ for δ sufficiently small (depending on K).*

Proof. Let $R = f^\delta(I)$ and $R' = g^\delta(I')$ where $f, g \in \mathcal{F}$. It is easy to see that if the centres of I, I' are no more than $\sqrt{\delta/t}$ -separated and $\|f - g\| \leq \delta$, then R, R' must be λ -comparable. The result now follows from the fact that $\mathcal{F}$ has diameter K and doubling constant D , and choosing $\delta \leq K^{-1}$ sufficiently small. $\square$

14.6.4 Lens counting and a bipartite curve-rectangle tangency bound

In this section, we will use Marcus and Tardos' lens-counting theorem from [19] to prove a bipartite version of the weak $\ell^{3/2}$ bound (14.6.16) from Section 14.6.1.3. The precise statement is as follows. We remark that all propositions in this section make no assumption on the doubling property in the definition of cinematic functions.

Proposition 14.6.26. *Let $\mathcal{F}$ be a family of cinematic functions with cinematic constant K . Then there is a constant $C = C(K) > 0$ so that the following holds.*

Let $A \geq 1$. Let $\delta, t > 0$ with $0 < \delta \leq At$. Let $\mathcal{W}, \mathcal{B} \subset \mathcal{F}$ be finite sets. Suppose $\mathcal{W} \cup \mathcal{B}$ has diameter at most $6t$, and the pair $(\mathcal{W}, \mathcal{B})$ is t/A -separated.

Let $\mathcal{R}$ be a collection of pairwise 100-incomparable (δ, t) -rectangles of the form $f^\delta(I)$ where $f \in \mathcal{F}$ and $I \subset J/4$. Suppose each rectangle in $\mathcal{R}$ is tangent to at least μ elements of $\mathcal{W}$ and at least ν elements of $\mathcal{B}$. Then

$$\#\mathcal{R} \leq C A^C \left(\frac{\#\mathcal{W}}{\mu} + \frac{\#\mathcal{B}}{\nu} \right)^{3/2} \log \left(\frac{\#\mathcal{W}}{\mu} + \frac{\#\mathcal{B}}{\nu} \right). \quad (14.6.50)$$

14.6.4.1 Preliminary reductions

Lemma 14.6.27. *Let $c > 0$. Suppose there exists a constant $C_1 = C_1(K) > 0$ so that Proposition 14.6.26 holds whenever $\delta \leq ct$. Then there is a constant $C = C(C_1, K, c)$ so that Proposition 14.6.26 holds for all $\delta \leq At$.*

Proof. If $\delta \leq ct$ then we are done. Suppose instead that $\delta > ct$, and let $\mathcal{W}, \mathcal{B}, \mathcal{R}, \mu$, and ν be as in the statement of the Proposition. Let $C_1 \sim 1$ be a constant to be specified below. Apply Lemma 14.6.24 to find a subcollection $\mathcal{R}' \subset \mathcal{R}$ that is pairwise C_1 -incomparable, with $\#\mathcal{R}' \sim \#\mathcal{R}$. (By Lemma 14.6.25 we may suppose that $\mathcal{R}$ is finite.) Next, shrink the length of each $R \in \mathcal{R}'$ to be $\sqrt{\delta/t'}$ where $t' = c^{-1}\delta > t$, and so $\delta = ct'$. Denote by $\mathcal{R}''$ the resulting collection of (δ, t') -rectangles. If C_1 is chosen large enough, then using Lemma 14.6.22, we have that $\mathcal{R}''$ is 100-incomparable. Applying Proposition 14.6.26 with t' in place of t , we are done. $\square$

Similar arguments yield the following.

Lemma 14.6.28. *Suppose there exists a constant $C_2 = C_2(K) > 0$ so that Proposition 14.6.26 holds whenever $(\mathcal{W}, \mathcal{B})$ is $2t$ -separated. Then there is a constant $C = C(C_2, K)$ so that Proposition 14.6.26 holds when $(\mathcal{W}, \mathcal{B})$ is t/A -separated.*

Lemma 14.6.29. *Suppose there exists a constant $C_3 = C_3(K)$ so that Proposition 14.6.26 holds whenever $\mathcal{R}$ is $2C_c$ -incomparable, where C_c is as in Lemma 14.6.34 below. Then there is a constant $C = C(C_3, K)$ so that Proposition 14.6.26 holds when $\mathcal{R}$ is 100 -incomparable.*

From now on we will assume $0 < \delta \leq c_2 t$ where $c_2 = c_2(K)$ will be determined in Lemma 14.6.34 below. We also assume $(\mathcal{W}, \mathcal{B})$ is $2t$ -separated, and that $\mathcal{R}$ is $2C_c$ -incomparable.

Remark. In the arguments below, we will choose a few constants depending on K . The order of dependence is as follows: c_1 as in Lemma 14.6.16; λ as in Lemma 14.6.37; c_2 and C_c as in Lemma 14.6.34.

14.6.4.2 Pseudo-circles and lenses

Now we introduce some terminology from Marcus-Tardos [19] that will be used in the proof of Proposition 14.6.26.

Definition 14.6.30. Let $\mathcal{C}$ be a family of closed Jordan plane curves (i.e. the homeomorphic image of S^1). We say $\mathcal{C}$ forms a family of pseudo-circles if for every pair of distinct $c, c' \in \mathcal{C}$, the following conditions are satisfied:

1. $\#(c \cap c') \leq 2$.
2. If $p \in c \cap c'$, then c and c' intersect properly at p . That is, for all sufficiently small circles $C(p)$ centered at p , the points in $C(p) \cap c$ and $C(p) \cap c'$ appear alternatively cyclicly counterclockwise. (See Figure 14.6.1 below).

The first condition is key. The second condition was not explicitly stated in [19], so we included it here for completeness. It is a technical assumption included to make our settings fit rigorously to the discrete geometry scenario in [19]. Of course, in the Kakeya type maximal inequality, if two functions f, g are exactly tangent at some point, then the intersection may be improper. To avoid this issue, we will slightly perturb the cinematic functions f by $O(\delta)$, such that no exact tangency could occur. This will do no harm, since if f and g satisfy (14.6.6) and also $\|f - g\| \geq C\delta$ for a sufficiently large constant C , then $f + O(\delta)$ and g will continue to satisfy (14.6.6) (with K replaced by $2K$). See Section 14.6.4.5 for the details of this perturbation process.

Now we recall the definition of lenses from [19].

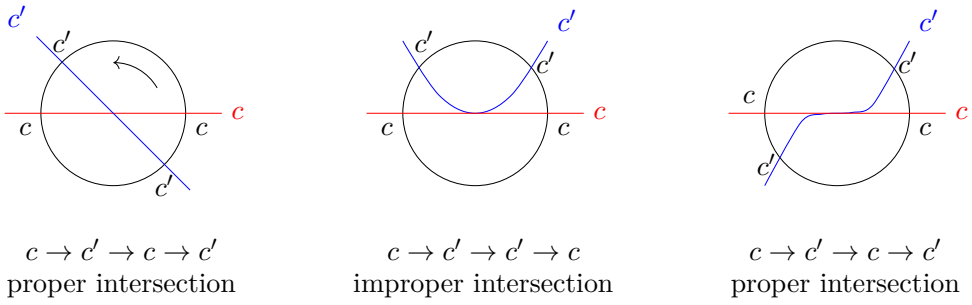


Figure 14.6.1: Proper and improper intersection.

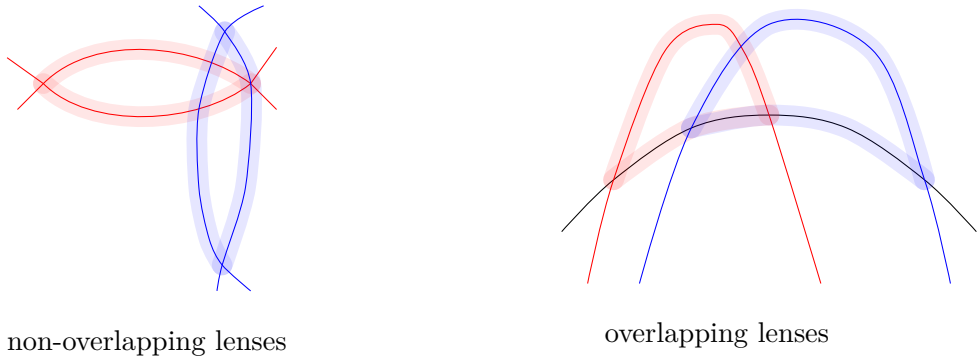


Figure 14.6.2: Overlapping and non-overlapping lenses.

Definition 14.6.31. Let $\mathcal{C}$ be a set of pseudo-circles. Let $c, c' \in \mathcal{C}$ be two curves that intersect exactly twice, at the points p, p' . This divides c, c' into four closed segments, that intersect only at their endpoints. We say $\ell \subset c \cup c'$ is a *lens* if it is a simple closed curve that is the union of two of these segments.

Note that there are exactly $\binom{4}{2} = 6$ lenses formed by c, c' ; two of these are the curves c and c' themselves. See [19, Figure 1]. As a technical aside, we only consider “first-generation lenses”, that is, we ignore lenses formed by lenses. Thus the total number of lenses determined by an arrangement $\mathcal{C}$ is at most $4\binom{\#\mathcal{C}}{2} + \#\mathcal{C}$.

Definition 14.6.32 (Overlapping lenses). Let $\mathcal{C}$ be a family of pseudo-circles, and let ℓ, ℓ' be two lenses formed by $\mathcal{C}$. We say ℓ, ℓ' *overlap* if they share a segment of positive length (see Figure 14.6.2).

Note that if two lenses intersect (properly) at only finitely many points, then they do not overlap.

We can now state Lemma 10 from [19].

Theorem 14.6.33. *Let $\mathcal{C}$ be a family of n pseudo-circles. Then every set of non-overlapping lenses has cardinality $O(n^{3/2} \log n)$.*

A complete proof, including the intersection-reverse cyclic-order estimate and the planar order reduction, is given in Chapter 14.7, Theorem 14.7.6.

14.6.4.3 Lenses and pseudo-parabolas

We return to our case of a finite family $F \subset \mathcal{F}$ of cinematic functions. By Lemma 14.6.14, their graphs intersect at most twice, but they are not yet pseudo-circles because they are not closed curves. However, we can extend them to be pseudo-circles in the following way.

We need a mild technical assumption first. Denote $J = [\alpha, \beta]$. We require that both sets

$$\{f(\alpha) : f \in F\}, \quad \{f(\beta) : f \in F\}$$

consist of distinct numbers, respectively. We also need that a pair of different cinematic functions f, g never intersect exactly tangentially at any point in J . As mentioned before, this can be guaranteed by a careful perturbation; see Section 14.6.4.5 for details. Also, we note that all curves are confined in $J \times [-M, M]$ for a finite M . We index the curves by $f_j, 1 \leq j \leq \#F$, and rearrange so that $f_j(\alpha)$ is increasing. We first extend each $f_j(\theta)$ to be equal to $f_j(\alpha)$ for $\theta \in [\alpha - j, \alpha]$ and equal to $f_j(\beta)$ for $\theta \in [\beta, \beta + j]$. We then extend by adding the line segments $\{\alpha - j\} \times [-M - j, f_j(\alpha)]$ and $\{\beta + j\} \times [-M - j, f_j(\beta)]$. Finally, close the loop by adding the line segment $[\alpha - j, \beta + j] \times \{-M - j\}$. In this way we have extended the graphs of the functions in F to a family $\mathcal{C}$ of pseudo-circles, with all intersections proper. See Figure 14.6.3 below.

We observe that if two functions f, g intersect exactly twice in the strip $J \times \mathbb{R}$, then they give rise to exactly one lens ℓ that lies entirely in the strip $J \times \mathbb{R}$. See Figure 14.6.3. Denote the collection of these lenses by $\mathcal{L}$. They are the only lenses we are interested in.

Now we explain the connection between overlapping lenses and curve tangencies.

Lemma 14.6.34. *If $\delta/t \leq c_2 \leq c_1$ (where c_1 is as in Part (1) of Lemma 14.6.16) for some small enough constant $c_2 = c_2(K)$, then the following holds. Let $w, b_1, b_2 \in \mathcal{F}$. For $i = 1, 2$, let $t_i = \|w - b_i\|$. Suppose $t_i \geq t$, and that w and b_i intersect exactly twice in J , thus forming a lens ℓ_i that lies over some subinterval of J . Let R_i be a (δ, t) -rectangle that is 2λ -tangent to both w and b_i , where $\lambda = \lambda(K)$ is to be determined in Lemma 14.6.37.*

Suppose that ℓ_1 and ℓ_2 overlap. Then R_1 and R_2 are C_c -comparable, where C_c (c stands for “comparable”) depends on λ, K .

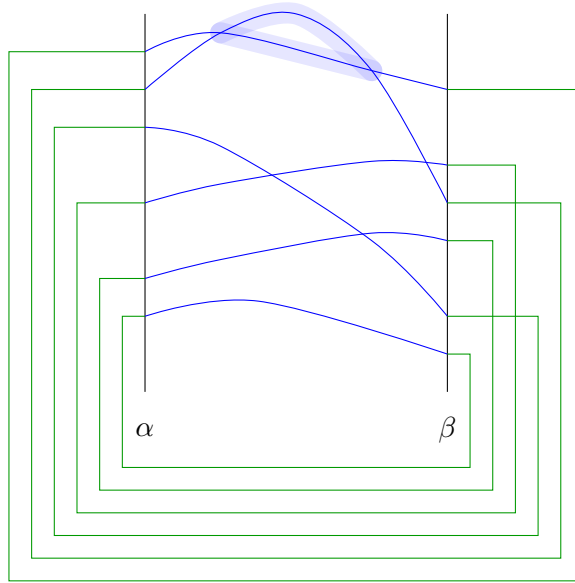


Figure 14.6.3: Extending the graph of $f \in F$ to a closed curve.

Proof. For $i = 1, 2$, define the interval $I_i = [\alpha_i, \beta_i] \subset J$ so that R_i is a curvilinear rectangle over I_i . Since R_1, R_2 are both tangent to w , it suffices to show that $\text{diam}(I_1 \cup I_2) = O(\sqrt{\delta/t})$.

By Lemma 14.6.17, we have $\Delta(w, b_i) \lesssim \delta$. If δ/t is small enough, then $\Delta(w, b_i) \leq c_1 t$. Thus there is a unique $\theta_0^{(i)} \in \frac{3}{5}J$ such that $|\alpha_i - \theta_0^{(i)}| = O(\sqrt{\delta/t_i}) = O(\sqrt{\delta/t})$ and $|\beta_i - \theta_0^{(i)}| = O(\sqrt{\delta/t})$. Since $(\alpha_1, \beta_1) \cap (\alpha_2, \beta_2) \neq \emptyset$, we have $\theta_0^{(2)} - \theta_0^{(1)} = O(\sqrt{\delta/t})$.

Now we study I_i . Using the triangle inequality, for $\theta \in I_i$ we have $|w(\theta) - b_i(\theta)| \lesssim \delta$. By Part (2a) of Lemma 14.6.16, I_i is contained in an interval of length $O(\sqrt{\delta/t})$ centered at $\theta_0^{(i)}$. But since $\theta_0^{(2)} - \theta_0^{(1)} = O(\sqrt{\delta/t})$, we see that $\text{diam}(I_1 \cup I_2) = O(\sqrt{\delta/t})$, as required. $\square$

14.6.4.4 Rectangle tangency bounds for properly intersecting curves

We first prove a special case of Proposition 14.6.26, where the curves intersect properly.

Lemma 14.6.35. *Let $\mathcal{F}$ be a family of cinematic functions, with cinematic constant K . Then there is a constant $C = C(K) > 0$ so that the following holds.*

Let $\delta, t > 0$ with $0 < \delta \leq c_2 t$ (here c_2 is the quantity from Lemma 14.6.34). Let $\mathcal{W}, \mathcal{B} \subset \mathcal{F}$ be a pair of t -separated finite sets.

Let $\mathcal{R}$ be a collection of pairwise C_c -incomparable (δ, t) -rectangles of the form $f^\delta(I)$, where $f \in \mathcal{F}$, $I \subset J/4$ and C_c is as in Lemma 14.6.34. Suppose each rectangle $R \in \mathcal{R}$ is tangent to at least one $w_R \in \mathcal{W}$ and at least one $b_R \in \mathcal{B}$. Suppose furthermore that the following holds.

A1. For each $R \in \mathcal{R}$, we have that w_R, b_R intersect exactly twice in J .

A2. If $J = [\alpha, \beta]$, then the sets

$$\{f(\alpha) : f \in \mathcal{W} \cup \mathcal{B}\}, \quad \{f(\beta) : f \in \mathcal{W} \cup \mathcal{B}\}$$

respectively consist of distinct elements.

A3. For all $f \neq g \in \mathcal{W} \cup \mathcal{B}$, f, g never intersect tangentially in J , that is, the two quantities $f(\theta) - g(\theta)$ and $f'(\theta) - g'(\theta)$ are never simultaneously 0 for any $\theta \in J$.

Then

$$\#\mathcal{R} \leq C(\#\mathcal{W} + \#\mathcal{B})^{3/2} \log(\#\mathcal{W} + \#\mathcal{B}). \quad (14.6.51)$$

Proof. With the additional assumptions A1–A3, each $R \in \mathcal{R}$ is tangent to a pair (w_R, b_R) intersecting exactly twice in J . After extending them to pseudo-circles in the way described in Section 14.6.4.3, we see the pair (w_R, b_R) gives rise to a lens ℓ_R in the collection $\mathcal{L}$.

By Lemma 14.6.34, the lenses in $\mathcal{L} = \{\ell_R : R \in \mathcal{R}\}$ are non-overlapping. Since the mapping $R \mapsto \ell_R$ is clearly a bijection, using Theorem 14.6.33, we conclude that

$$\#\mathcal{R} = \#\mathcal{L} \lesssim (m + n)^{3/2} \log(m + n). \quad \square$$

14.6.4.5 Perturbing the curves

Our next step is to remove the additional assumptions A1–A3 in Lemma 14.6.35 by perturbing the functions in $\mathcal{W}$ and $\mathcal{B}$. Namely, we would like to prove the following version of Proposition 14.6.26.

Lemma 14.6.36. *Let $\mathcal{F}$ be a family of cinematic functions, with cinematic constant K . Then there is a constant $C = C(K) > 0$ so that the following holds.*

Let $\delta, t > 0$ with $0 < \delta \leq c_2 t/2$ (here c_2 is the quantity from Lemma 14.6.34). Let $\mathcal{W}, \mathcal{B} \subset \mathcal{F}$ be a pair of $2t$ -separated finite sets.

Let $\mathcal{R}$ be a collection of pairwise $2C_c$ -incomparable (δ, t) -rectangles of the form $f^\delta(I)$ where $f \in \mathcal{F}$ and $I \subset J/4$. Suppose each rectangle in $\mathcal{R}$ is tangent to at least one element of $\mathcal{W}$ and at least one element of $\mathcal{B}$.

Then

$$\#\mathcal{R} \leq C(\#\mathcal{W} + \#\mathcal{B})^{3/2} \log(\#\mathcal{W} + \#\mathcal{B}). \quad (14.6.52)$$

The reduction from Lemma 14.6.35 to Lemma 14.6.36 requires two perturbations. The first perturbation (which has size roughly δ) ensures that property A1 holds for a considerable fraction of the tangency rectangles in $\mathcal{R}$. The second perturbation (which is infinitesimally small) ensures that properties A2 and A3 hold.

Lemma 14.6.37. *Let $0 < \delta < c_1 t$ and let $f, g \in \mathcal{F}$ with $\|f - g\| \geq 2t$. Suppose f and g are tangent to a common (δ, t) -rectangle R over $J/4$. If $\lambda = \lambda(K) \geq 1$ is chosen to be large enough, then there exists $\eta \in \{-1, 0, 1\}$ so that the function $h = f + \eta\lambda\delta$ is 2λ -tangent to R , and the functions g and h intersect exactly twice in J .*

Proof. By Part (1b) of Lemma 14.6.16, it suffices to show that for a suitable $\eta \in \{-1, 0, 1\}$, we have $k = h - g$ changes sign within J . We have three cases. If $f - g$ changes sign in J , then we just take $\eta = 0$. Otherwise, we must have $f - g \geq 0$ on J or $f - g \leq 0$ on J . In the former case, we take $\eta = -1$, and so $k = f - \lambda\delta - g$. Also, by Lemma 14.6.17, we have $\Delta(f, g) \lesssim \delta$, and thus $\min_{\theta \in J} f(\theta_0) - g(\theta_0) \leq C'\delta$ for some $C' = C'(K)$, using Part (1a) of Lemma 14.6.16. If we choose $\lambda > C'$, then we have $\min_{\theta \in J} k(\theta_0) < 0$, and hence k changes sign in J .

In the case $f - g \leq 0$ on J , we take $\eta = 1$ and the argument is similar. $\square$

Corollary 14.6.38. *Let $\delta, t > 0$ with $0 < \delta \leq c_2 t/2$. Let $\mathcal{W}, \mathcal{B} \subset \mathcal{F}$ be a pair of $2t$ -separated finite sets.*

Let $\mathcal{R}$ be a collection of pairwise $2C_c$ -incomparable (δ, t) -rectangles of the form $f^\delta(I)$ where $f \in \mathcal{F}$ and $I \subset J/4$. Suppose each rectangle in $\mathcal{R}$ is tangent to at least one $w_R \in \mathcal{W}$ and at least one $b_R \in \mathcal{B}$.

Then there exists $\eta \in \{-1, 0, 1\}$ and a set $\mathcal{R}' \subset \mathcal{R}$ with the following properties.

- $\#\mathcal{R}' \geq \frac{1}{3}\#\mathcal{R}$.
- If we define $\mathcal{W}' = \{w + \eta\lambda\delta : w \in \mathcal{W}\}$, then each rectangle in $\mathcal{R}'$ is 2λ -tangent to at least one $w_R \in \mathcal{W}'$ and at least one $b_R \in \mathcal{B}$.
- For each $R \in \mathcal{R}'$, we have that w_R, b_R intersect exactly twice in J .

We are now ready to prove Lemma 14.6.36. Applying Corollary 14.6.38 to $\mathcal{W}$ and $\mathcal{B}$, we obtain a perturbed set $\mathcal{W}'$ and a subset $\mathcal{R}' \subset \mathcal{R}$. We claim that $\mathcal{W}' \cup \mathcal{B}$ is a finite subset of a family $\mathcal{F}'$ that has cinematic constant at most $2K$. Indeed, we can simply define $\mathcal{F}' = (\mathcal{W} - \eta\lambda\delta) \cup \mathcal{B}$ (recall that $\mathcal{W} \subset C^2(J)$, so $\mathcal{W} - \eta\lambda\delta = \{f - \eta\lambda\delta : f \in \mathcal{W}\}$). It is easy to check that since $\mathcal{W}$ and $\mathcal{B}$ are $2t$ -separated, (14.6.6) continues to hold with $2K$ in place of K (the only interesting case to check is when $f \in \mathcal{W} - \eta\lambda\delta$ and $g \in \mathcal{B}$, or vice-versa). Also, $\mathcal{W}' \cup \mathcal{B}$ is at least t separated, if c_2 is chosen to be small enough.

Our collection $\mathcal{W}'$, $\mathcal{B}$, and $\mathcal{R}'$ satisfies property A1 from the statement of Lemma 14.6.35. After applying a second harmless infinitesimal perturbation, we can ensure that $\mathcal{W}'$, $\mathcal{B}$, and $\mathcal{R}'$ satisfy Properties A2 and A3, while maintaining property A1. Lemma 14.6.36 now follows from Lemma 14.6.35.

Finally, in light of the preliminary reductions given by Lemmas 14.6.27 and 14.6.28, Proposition 14.6.26 follows from Lemma 14.6.36 via a standard random sampling argument. See i.e. the proof of Lemma 1.4 from [29].

14.6.5 Proof of Theorem 14.6.7

We have now assembled the necessary ingredients to execute the proof sketch described in Section 14.6.1.3. We will begin with a few preliminary reductions. First, if $\mathcal{F} \subset C^2(J)$ is a family of cinematic functions, the results from Sections 14.6.3 and 14.6.4 allow us to control the functions $f \in \mathcal{F}$ on a slightly shorter interval $J/4$. This is inconvenient, since the domain of integration in (14.6.8) might involve the entire interval J . Our first task is to show that Theorem 14.6.7 will follow from the following (superficially) weaker version, in which the domain of integration has been restricted to a slightly smaller set.

Lemma 14.6.39. *Let $\varepsilon > 0$, $0 < \alpha \leq \zeta \leq 1$, and $D, K \geq 1$. Then the following is true for all $\delta > 0$ sufficiently small (depending on D, K , and ε). Let $\mathcal{F} \subset C^2(J)$ be a family of cinematic functions, with cinematic constant K and doubling constant D . Let E be a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ quasi-product.*

Let $F \subset \mathcal{F}$ be a set of functions that satisfies the Frostman-type non-concentration condition

$$\#(F \cap B) \leq \delta^{-\varepsilon}(r/\delta)^\zeta \quad \text{for all balls } B \subset C^2(I) \text{ of radius } r \geq \delta.$$

Let $J' = \frac{1}{16}J$ and let $E' = E \cap (J' \times \mathbb{R})$. Then

$$\int_{E'} \left(\sum_{f \in F} \mathbf{1}_{f^\delta} \right)^{3/2} \leq \delta^{2-\alpha/2-\zeta/2-C_0\varepsilon}(\#F), \quad (14.6.53)$$

where $C_0 = C_0(D) > 0$ depends on D .

Using Lemma 14.6.39 we can prove Theorem 14.6.7 as follows. Let $J_1, \dots, J_N$, $N \sim |\log \delta|$ be a sequence of sub-intervals of J , so that $J \setminus \bigcup_{i=1}^N \frac{1}{16}J_i$ consists of two intervals of length δ (these intervals will be at the two endpoints of J). Note that if J_0 is an interval of length δ and if $E_0 = E \cap (J_0 \times \mathbb{R})$, then

$$\int_{E_0} \left(\sum_{f \in F} \mathbf{1}_{f^\delta} \right)^{3/2} \lesssim \delta^2(\#F)^{3/2} \leq \delta^{2-\alpha/2-\zeta/2-C_0\varepsilon}(\#F). \quad (14.6.54)$$

For each index i , apply Lemma 14.6.39 to the restriction $\mathcal{F}|_{J_i} \subset C^2(J_i)$; this is still a family of cinematic functions, with cinematic constant K and doubling constant D . Using the triangle inequality, we conclude that

$$\int_E \left(\sum_{f \in F} \mathbf{1}_{f^\delta} \right)^{3/2} \lesssim |\log \delta| \delta^{2-\alpha/2-\zeta/2-C_0\varepsilon} (\#F).$$

Theorem 14.6.7 now follows, provided we select $C > C_0$ and restrict $\delta > 0$ sufficiently small.

Next, we remark that when proving Lemma 14.6.39, we may suppose that F is δ -separated. This follows from the triangle inequality and observation that by (14.6.7) each δ -ball contains at most $\delta^{-\varepsilon}$ elements of F .

Finally, we claim that Lemma 14.6.39 follows from the following restricted weak-type estimate. For each $x \in \mathbb{R}^2$, let $F(x) = \{f \in F : x \in f^\delta\}$. For each $x \in \mathbb{R}^2$, $\#F(x) \leq \#F \lesssim \delta^{-\varepsilon-\zeta}$. Thus by dyadic pigeonholing, there is a set $E_0 \subset E$ and an integer μ so that $\mu \leq \#F(x) < 2\mu$ for all $x \in E_0$, and $\mu^{3/2}|E_0| \approx \text{LHS of (14.6.53)}$. Thus in order to prove Lemma 14.6.39 (and hence Theorem 14.6.7), it suffices to establish the estimate

$$|E_0| \lesssim \delta^{2-\frac{\alpha}{2}-\frac{\zeta}{2}-C_1\varepsilon} \#F \mu^{-3/2}, \quad (14.6.55)$$

for some $C_1 = C_1(D)$.

14.6.5.1 Two ends reductions

In this section, we will find quantities called t and Δ , so that for a typical point $x \in E_0$, the functions in $F(x)$ are localized to a ball in $C^2(J)$ of radius t , and for a typical pair of functions $f, g \in F(x)$, we have $\Delta(f, g) \sim \Delta$. Our main tool will be the “two-ends” reduction. See [25] for an introduction to the topic.

Finding the diameter of $F(x)$

Since $F \subset C^2(J)$ is bounded, so is the set $F(x)$ for each $x \in E_0$. However, it is possible that $F(x)$ (or a large fraction thereof) has diameter much smaller than $\text{diam}(F)$ for a typical $x \in E_0$. In this section, we will find the “true” diameter of $F(x)$ for a typical $x \in E_0$.

For $t > 0$ and $x \in \mathbb{R}^2$, define

$$n_1(t, x) = \sup_{g \in C^2(J)} \#F(x) \cap B(g, t), \quad (14.6.56)$$

where $B(g, t) \subset C^2(J)$ is the ball centered at $g \in C^2(J)$ of radius t . With this, we further define

$$n_2(x) = \sup_{t \in [\delta, K]} n_1(t, x) t^{-\varepsilon}. \quad (14.6.57)$$

Hence, given x , there is some $t(x) \in [\delta, K]$ such that $n_2(x) \leq 2n_1(t(x), x)t(x)^{-\varepsilon}$. There is also some g_x such that

$$\#F(x) \cap B(g_x, t(x)) \geq n_2(t(x), x)/2, \quad (14.6.58)$$

so we have

$$n_2(x) \leq 4t(x)^{-\varepsilon} \#F(x) \cap B(g_x, t(x)). \quad (14.6.59)$$

For simplicity we denote $B_x = B(g_x, t(x))$.

Since $t(x) \in [\delta, K]$, by a dyadic pigeonholing, there is a Borel set $E_1 \subset E_0$ and a number $t \in [\delta, 2K]$ such that $t(x) \in (t/2, t]$ for every $x \in E_1$, and $|E_1| \approx |E_0|$. It then suffices to show

$$|E_1| \lesssim \delta^{2-\frac{\alpha}{2}-\frac{\zeta}{2}-C_1\varepsilon} \#F \mu^{-3/2}. \quad (14.6.60)$$

Since $\mathcal{F}$ is doubling, we can cover it by finitely overlapping balls B of radius t , and denote

$$F_B = F \cap (3B),$$

so $\{F_B\}$ covers F and is finitely overlapping. Also, let E_B be the set of $x \in E_1$ such that B_x intersects B , and thus $\{E_B\}$ is also finitely overlapping. Thus it suffices to show for each B

$$|E_B| \lesssim \delta^{2-\frac{\alpha}{2}-\frac{\zeta}{2}-C_1\varepsilon} \#F_B \mu^{-3/2}. \quad (14.6.61)$$

Now fix a ball B of radius t and let $x \in E_B$. By (14.6.59), (14.6.57) and (14.6.56), we have

$$\#F(x) \cap B_x \gtrsim t^\varepsilon n_2(x) \gtrsim t^\varepsilon n_1(K, x) = t^\varepsilon \#F(x) \sim t^\varepsilon \mu.$$

By another dyadic pigeonholing, we may find a subset $E'_B \subset E_B$ and a scale μ_1 with

$$t^\varepsilon \mu \lesssim \mu_1 \leq \mu, \quad (14.6.62)$$

such that $|E'_B| \approx |E_B|$ and for every $x \in E'_B$ we have $\#F(x) \cap B_x \sim \mu_1$. It suffices to show (14.6.61) with E_B replaced by E'_B .

Now let $x \in E'_B$. If B intersects B_x then we have $B_x \subset 3B$. Hence

$$\#F_B(x) = \#F(x) \cap (3B) \cap B_x = \#F(x) \cap B_x \sim \mu_1. \quad (14.6.63)$$

From now on we fix such a ball B and aim to prove (14.6.61). We end with a useful lemma.

Lemma 14.6.40. *Fix $x \in E'_B$ and $g \in C^2(J)$. If $\delta t^{-1} < \lambda < 1$, then for at most $4 \cdot (2\lambda)^\varepsilon$ fraction of $f \in F_B(x)$ we have $\|f - g\| \leq \lambda t$.*

Proof. Let $g \in C^2(J)$. By (14.6.59) and (14.6.63), we have

$$n_2(x) \leq 4t(x)^{-\varepsilon} \#F(x) \cap B_x \leq 4 \cdot 2^\varepsilon t^{-\varepsilon} \#F_B(x).$$

But by the definition of n_2 in (14.6.57) and $\lambda t \in [\delta, K]$, we also have

$$n_2(x) \geq n_1(\lambda t, x)(\lambda t)^{-\varepsilon} \geq \#F(x) \cap B(g, \lambda t) \lambda^{-\varepsilon} t^{-\varepsilon}$$

Combining the lower and upper bounds of $n_2(x)$ gives

$$\#F(x) \cap B(g, \lambda t) \leq 4 \cdot (2\lambda)^\varepsilon \#F_B(x),$$

from which the result follows. $\square$

Finding the typical tangency of curve-curve intersection

As discussed in Section 14.6.1.3, a typical pair of intersecting curves in F might intersect transversely, tangentially, or something in-between. In this section, we will find the typical degree of tangency when two curves intersect. We will do this by using a two-ends reduction argument similar to that in Section 14.6.5.1. This two-ends reduction will use a much smaller quantity in place of ε , so that the resulting refinement retains the two-ends property from Section 14.6.5.1.

For $x \in E_B$ and $\Delta > 0$, define

$$n_3(\Delta, x) = \max_{k \in F_B(x)} \#\{f \in F_B(x) : \Delta(f, k) \leq \Delta\}. \quad (14.6.64)$$

Since F_B is contained in $3B$ which has diameter $6t$, invoking (14.6.63) and the relation $\Delta(f, k) \leq \|f - g\|$, we have

$$n_3(6t, x) = \#F_B(x). \quad (14.6.65)$$

We further define

$$n_4(x) = \sup_{\Delta \in [\delta, 6t]} n_3(\Delta, x) \Delta^{-\varepsilon^2}. \quad (14.6.66)$$

Given x , there is some $\Delta_x \in [\delta, 6t]$ such that

$$n_4(x) \leq 2n_3(\Delta_x, x) \Delta_x^{-\varepsilon^2}. \quad (14.6.67)$$

There is also some $k_x \in F_B(x)$ attaining the maximum in (14.6.64), that is,

$$\#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta_x\} = n_3(\Delta_x, x), \quad (14.6.68)$$

and hence

$$n_4(x) \leq 2\Delta_x^{-\varepsilon^2} \#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta_x\}. \quad (14.6.69)$$

Since $\Delta_x \in [\delta, 6t]$, by a dyadic pigeonholing, there is a Borel set $E_B'' \subset E_B'$ and a number $\Delta \in [\delta, 12t]$ such that $\Delta_x \in (\Delta/2, \Delta]$ for every $x \in E_B''$, and $|E_B''| \approx |E_B'|$. It then suffices to show

$$|E_B''| \lesssim \delta^{2-\frac{\alpha}{2}-\frac{\zeta}{2}-C_1\varepsilon} \#F_B \mu^{-3/2}. \quad (14.6.70)$$

Shadings

At this point, the set E_0 from (14.6.55) has been divided into (mostly) disjoint pieces E_B . For each $x \in E_B$, most of the functions $f \in F(x)$ are contained in the ball B , and only a small fraction of these functions are contained in any ball substantially smaller than B . For a typical pair of functions $f, g \in F(x)$, $\Delta(f, g)$ has size roughly Δ . For technical reasons, it will be convenient to restrict the sets f^δ to a slightly smaller “shading” $Y_1(f)$, so that the above properties hold whenever $x \in Y_1(f)$. More precisely, we define

$$Y_1(f) = \{x \in E_B'' \cap f^\delta : \Delta(f, k_x) \leq \Delta\}. \quad (14.6.71)$$

In view of (14.6.69), (14.6.65) and $\Delta_x \leq \Delta$, for every $x \in E_B''$ we have

$$\#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta\} \gtrsim (\Delta/t)^{\varepsilon^2} \#F_B(x) \sim (\Delta/t)^{\varepsilon^2} \mu_1. \quad (14.6.72)$$

On the other hand, we have a trivial bound

$$\#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta\} \leq \#F_B(x) \sim \mu_1.$$

Now we apply another dyadic pigeonholing to find a subset $E_2 \subset E_B''$ with $|E_2| \approx |E_B''|$ and a scale μ_2 with

$$(\Delta/t)^{\varepsilon^2} \mu_1 \lesssim \mu_2 \leq \mu_1, \quad (14.6.73)$$

such that for every $x \in E_2$ we have

$$\#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta\} \sim \mu_2. \quad (14.6.74)$$

Equivalently, we have

$$\#\{f \in F_B : x \in Y_1(f)\} = \sum_{f \in F_B} \mathbf{1}_{Y_1(f)}(x) \sim \mu_2, \quad \forall x \in E_2. \quad (14.6.75)$$

It then suffices to prove

$$|E_2| \lesssim \delta^{2-\frac{\alpha}{2}-\frac{\zeta}{2}-C_1\varepsilon} (\#F_B) \mu^{-3/2}. \quad (14.6.76)$$

We record a coarse bound that is used in some dyadic pigeonholing argument later. Since $\#F_B(x) \gtrsim t^\varepsilon \mu$ and $\mu \leq \#F \lesssim \delta^{-1-\varepsilon}$, the right hand side of (14.6.76) is trivially bounded below by δ^3 , for δ small enough. If $|E_2| \leq \delta^3$ then we are done, and so in the following we may assume

$$|E_2| > \delta^3. \quad (14.6.77)$$

14.6.5.2 Fine-scale rectangles

In this section, we will show that E_2 can be covered by curvilinear rectangles of dimensions roughly $\delta \times \delta/\sqrt{t}\Delta$, so that these rectangles are (mostly) disjoint, and for most pairs $f, g \in F_B(x)$, we have that $Y_1(f) \cap Y_1(g)$ is localized to roughly one of these rectangles.

Lemma 14.6.41. *There is a constant $C_R = C_R(K)$ (R stands for “rectangle”) such that the following is true. Let*

$$t' = C_R t \Delta / \delta. \quad (14.6.78)$$

For each $x \in E_2$, there is a (δ, t') -rectangle R_x over $J/8$, such that $x \in R_x$, and every $f \in F_B$ with $x \in Y_1(f)$ is tangent to R_x .

Proof. Denote $x = (\theta_0, y_0)$ where $x \in J/16$. Let I be the interval of length $\sqrt{\delta/t'}$ centred at θ_0 . Since $t \gtrsim \Delta \geq \delta$, we have $I \subset J/8$ if C_R is large enough. With this, define the curvilinear rectangle

$$R_x = k_x^\delta(I), \quad (14.6.79)$$

where k_x is as in (14.6.68). We now show that R_x is as required.

First we have $x \in R_x$ since $k_x \in F_B(x)$ by (14.6.64). Also, let f be such that $x \in Y_1(f)$, then in particular, $x \in f^\delta$. By the triangle inequality, $|f(\theta_0) - k_x(\theta_0)| \leq 2\delta$. By Part (2c) of Lemma 14.6.16 at the scale 4δ , we have

$$E_{4\delta} = \{\theta \in J/4 : |f(\theta) - k_x(\theta)| \leq 4\delta\}$$

contains an interval I' obeying $\theta_0 \in I'$ and

$$|I'| \gtrsim \delta d(f, k_x)^{-1/2} (\delta + \Delta(f, k_x))^{-1/2}.$$

Since $x \in Y_1(f)$ we have $\Delta(f, k_x) \leq \Delta$. Also $d(f, k(x)) \leq 6t$. If C_R is chosen to be large enough, then the interval I containing θ_0 will have length $\sqrt{\delta/t'} \leq |I'|$, and thus $I \subset I'$ and R_x is tangent to f . $\square$

Shadings of fine-scale rectangles

For each (δ, t') -rectangle R , we define its shading as

$$Y_2(R) = \{x \in C_s R \cap E_2 : R_x \text{ is } C_s\text{-comparable to } R\}, \quad (14.6.80)$$

where $C_s = C_s(K) \geq 100$ (s standing for “shadings”) is a suitable constant to be determined. Intuitively, $Y_2(R)$ is an important subset of R since it is where the δ -neighbourhoods of the curves in F_B overlap greatly.

For technical purposes, we also define a slightly smaller shading of R as

$$Y'_2(R) = \{x \in R \cap E_2 : R_x \text{ is 100-comparable to } R\}.$$

By Lemma 14.6.41, we have

$$E_2 = \bigcup_{x \in E_2} Y'_2(R_x). \quad (14.6.81)$$

We also record an elementary lemma on the shadings of curvilinear rectangles.

Lemma 14.6.42. *If C_s is suitably chosen, then for any pair (R, R') of 100-comparable (δ, t) -rectangles, we have $Y'_2(R') \subset Y_2(R)$.*

Proof. If R, R' are 100-comparable, then by Lemma 14.6.20, we have $R' \subset C_s R$ if C_s is large enough. Thus if $x \in Y'_2(R')$, then $x \in C_s R \cap E_2$. Also, R_x is 100-comparable to R' , and since R' is 100-comparable to R , Corollary 14.6.21 implies that R_x is C_s -comparable to R if C_s is large enough. Thus $Y'_2(R') \subset Y_2(R)$. $\square$

Finding an incomparable set of fine-scale rectangles

Lemma 14.6.43. *There is a pairwise 100-incomparable subcollection $\mathcal{R}$ of $\{R_x : x \in E_2\}$ such that the following holds:*

1. *There is some $\Lambda \in (0, 1)$ such that for each $R_x \in \mathcal{R}$ we have $\Lambda \leq |Y_2(R_x)| < 2\Lambda$.*
2. *$Y_2(R) \cap Y_2(R') = \emptyset$ for $R \neq R' \in \mathcal{R}$, and*

$$|E_2| \lesssim \sum_{R \in \mathcal{R}} |Y_2(R)| \sim \Lambda \#\mathcal{R}. \quad (14.6.82)$$

3. *We have the coarse bounds*

$$\#\mathcal{R} \leq \delta^{-C}, \quad \Lambda \geq \delta^{C'}, \quad (14.6.83)$$

for δ sufficiently small (depending on K, D), where C, C' depend on D only.

Proof. Before we construct $\mathcal{R}$, we first construct a larger collection $\mathcal{R}^*$ in the following greedy way.

Denote $\Lambda_1 = \sup_{x \in E_2} |Y'_2(R_x)|$. Since we have assumed $|E_2| > \delta^3$ we have $\Lambda_1 > 0$ by (14.6.81). So we pick $R_1 = R_{x_1}$ such that $|Y'_2(R_1)| > \Lambda_1/2$.

To choose R_2 , we consider all (δ, t) -rectangles R_x that are 100-incomparable to R_1 ; denote this collection by $\mathcal{R}^{(1)}$. Denote $\Lambda_2 = \sup_{R_x \in \mathcal{R}^{(1)}} |Y'_2(R_x)|$. If

$\Lambda_2 = 0$, then we just let $\mathcal{R}^* = \{R_1\}$. With this, (14.6.81) and Lemma 14.6.42, we have (here $\sim_{100}$ means 100-comparable)

$$\begin{aligned} |E_2| &\leq \left| \bigcup_{x \in E_2, R_x \sim_{100} R_1} Y'_2(R_x) \right| + \left| \bigcup_{x \in E_2, R_x \not\sim_{100} R_1} Y'_2(R_x) \right| \\ &\leq |Y_2(R_1)| + \sum_{R_x \in \mathcal{R}^{(1)}} |Y'_2(R_x)| \\ &= |Y_2(R_1)|. \end{aligned}$$

If $\Lambda_2 > 0$, then we pick $R_2 \in \mathcal{R}^{(1)}$ such that $|Y'_2(R_2)| > \Lambda_2/2$. To choose R_3 , we consider all (δ, t) -rectangles that are 100-incomparable to both R_1 and R_2 ; denote this collection by $\mathcal{R}^{(2)}$. Denote $\Lambda_3 = \sup_{R_x \in \mathcal{R}^{(2)}} |Y'_2(R_x)|$. If $\Lambda_3 = 0$, we just let $\mathcal{R}^* = \{R_1, R_2\}$. With this, we similarly have

$$|E_2| \leq \left| \bigcup_{x \in E_2, R_x \sim_{100} R_1} Y'_2(R_x) \right| + \left| \bigcup_{x \in E_2, R_x \sim_{100} R_2} Y'_2(R_x) \right| \leq |Y_2(R_1)| + |Y_2(R_2)|.$$

We can continue arguing in this fashion. Note that Lemma 14.6.25 implies that the process must stop at some finite time $N < \delta^{-C}$ for some constant $C = C(D)$, proving the first inequality in (14.6.83). Thus $\Lambda_{N+1} = 0$, $\Lambda_N > 0$, $\mathcal{R}^* = \{R_1, \dots, R_N\}$, and

$$|E_2| \leq \sum_{n=1}^N |Y_2(R_n)|.$$

We may assume without loss of generality that $|Y_2(R_n)| \geq \delta^{C+3}$ for all n . Indeed, we may throw away all n such that $|Y_2(R_n)| < \delta^{C+3}$, since the sum of all such $|Y_2(R_n)|$ is less than $\delta^{C+3}N < \delta^3 < |E_2|/2$, as $|E_2| > \delta^3$ by our assumption.

Now for $j \geq 0$, we denote

$$\mathcal{R}^{(j)} = \{R_n \in \mathcal{R}^* : 2^{j-1}\Lambda_N \leq |Y_2(R_n)| < 2^j\Lambda_N\}.$$

There are at most $O(|\log \delta|)$ many such j 's, since $\delta^{C+3} \leq |Y_2(R_n)| \leq 1$. By pigeonholing, there is some j such that

$$\sum_{n=1}^N |Y_2(R_n)| \lesssim |\log \delta| \sum_{n: R_n \in \mathcal{R}^{(j)}} |Y_2(R_n)|.$$

We now define

$$\mathcal{R} = \mathcal{R}^{(j)}, \quad \Lambda = 2^{j-1}\Lambda_N > 0.$$

Thus

$$|E_2| \leq \sum_{n=1}^N |Y_2(R_n)| \lesssim \Lambda \# \mathcal{R}.$$

Also, we have $\Lambda \geq \delta^{C+4}$, since $|E_2| > \delta^3$ and $\# \mathcal{R} \leq N < \delta^{-C}$. □

We now observe the following. Since $Y_2(R) \subset C_s R \cap E_2$, R is over an interval of length $\sim \sqrt{\delta/t'}$, and E_2 is further contained in a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ quasi-product, we have

$$\Lambda \sim |Y_2(R)| \lesssim \delta^{2-\varepsilon} (\sqrt{\delta/t'}/\delta)^\alpha \sim \delta^{2-\varepsilon} \Delta^{-\alpha/2} t^{-\alpha/2}.$$

With this, we write

$$\Lambda = \tau \delta^{2-\varepsilon} \Delta^{-\alpha/2} t^{-\alpha/2}, \quad (14.6.84)$$

where

$$0 < \tau \lesssim 1. \quad (14.6.85)$$

To establish (14.6.76), it thus suffices to show

$$\# \mathcal{R} \lesssim \tau^{-1} \delta^{-\frac{\alpha}{2} - \frac{\zeta}{2} - C_2 \varepsilon} \Delta^{\alpha/2} t^{\alpha/2} \# F_B \mu^{-3/2}, \quad (14.6.86)$$

for some $C_2 = C_2(D)$.

14.6.5.3 Coarse-scale rectangles

In this section, we will show that the fine-scale rectangles from Section 14.6.5.2 are contained in coarse-scale rectangles of dimensions roughly $\Delta \times \sqrt{\Delta/t}$. If a fine-scale rectangle R is contained in a coarse-scale rectangle $\tilde{R}$, then $\tilde{R}$ is (roughly) a dilate of R .

For each $R \in \mathcal{R}$, we define $\tilde{R}(R)$ as follows. If $R = f^\delta(I)$, then $\tilde{R}(R) = f^\Delta(\tilde{I})$, where $\tilde{I}$ is the Δ/δ dilation of I , and hence it obeys $|\tilde{I}| = \sqrt{\frac{\Delta}{C_R t}}$ where C_R is as in Lemma 14.6.41. Thus $\tilde{R}(R)$ is a $(\Delta, C_R t)$ -rectangle, and trivially we have $R \subset \tilde{R}(R)$. Also, since $I \subset J/8$, for C_R large enough, we have $\tilde{I} \subset J/4$.

Lemma 14.6.44. *Let $R = R_x$ and $f \in F_B$ with $x \in Y_1(f)$, such that f is tangent to R by Lemma 14.6.41. If C_R was chosen to be sufficiently large, then f is also tangent to $\tilde{R}$.*

Proof. Write $x = (\theta_0, y_0)$ and recall that $R_x = k_x^\delta(I)$, while $\tilde{R} = k_x^\Delta(\tilde{I})$ with $|\tilde{I}| = \sqrt{\Delta/(C_R t)}$ and centre θ_0 . If $x \in Y_1(f)$, then $x \in f^\delta \cap k_x^\delta$, and hence

$$|f(\theta_0) - k_x(\theta_0)| \leq 2\delta \leq 2\Delta. \quad (14.6.87)$$

Moreover, the construction of Y_1 gives $\Delta(f, k_x) \leq \Delta$, and the fact that $f, k_x \in F_B \subset 3B$ gives $d(f, k_x) \leq 6t$.

If $d(f, k_x) < 24K\Delta$, then the definition of the C^2 metric gives $|f - k_x| \leq 24K\Delta$ on all of J , so f is tangent to $\tilde{R}$ after enlarging the fixed tangency constant. Otherwise Lemma 14.6.16(2c), applied to $f - k_x$ at scale 4Δ , is applicable. In view of (14.6.87), the component I' of

$$\{\theta \in J/4 : |f(\theta) - k_x(\theta)| \leq 4\Delta\}$$

containing θ_0 has length

$$|I'| \gtrsim \frac{\Delta}{\sqrt{(\Delta + \Delta(f, k_x))d(f, k_x)}} \gtrsim \sqrt{\frac{\Delta}{t}}.$$

The proof of Lemma 14.6.16(2c) also shows that θ_0 lies a comparable distance from each relevant endpoint; equivalently, its conclusion applies to the centred subinterval of I' . Choosing C_R larger than the reciprocal square of the implicit constant therefore ensures $\tilde{I} \subset I'$. Thus $|f - k_x| \leq 4\Delta$ throughout $\tilde{I}$, which is precisely tangency of f to a fixed dilate of $\tilde{R}$. $\square$

Next, we pick a maximally 100-incomparable subcollection $\tilde{\mathcal{R}}$ of $\{\tilde{R}(R) : R \in \mathcal{R}\}$, and we will denote each member of $\tilde{\mathcal{R}}$ as $\tilde{R}$. In this way we have partitioned $\mathcal{R}$ into subcollections of the form $\mathcal{R}(\tilde{R})$ consisting some of those $R \in \mathcal{R}$ such that the enlarged $\tilde{R}(R)$ of R is 100-comparable to $\tilde{R}$.

For each $R \in \mathcal{R}(\tilde{R})$, we have $R \subset \tilde{R}(R)$, and since $\tilde{R}(R)$ is 100-comparable to $\tilde{R}$, Lemma 14.6.20 shows that $\tilde{R}(R) \subset C\tilde{R}$ for some suitable absolute constant $C \geq 1$. Thus $R \subset C\tilde{R}$, and so $R \cap E_2 \subset C\tilde{R} \cap E_2$.

Recall we want to prove (14.6.86). Using (14.6.83) and dyadic pigeonholing, we may find a subcollection $\tilde{\mathcal{R}}_1 \subset \tilde{\mathcal{R}}$ such that for some M we have

$$\#\mathcal{R}(\tilde{R}) \sim M \tag{14.6.88}$$

for each $\tilde{R} \in \tilde{\mathcal{R}}_1$; moreover, this M is chosen such that if we define $\mathcal{R}_1 = \cup_{\tilde{R} \in \tilde{\mathcal{R}}_1} \mathcal{R}(\tilde{R})$, then $\#\mathcal{R}_1 \approx \#\mathcal{R}$. As a result, we have $\#\mathcal{R} \approx \#\tilde{\mathcal{R}}_1 M$. Note that by Lemma 14.6.43, we also have $|E_2| \lesssim \Lambda \#\mathcal{R}_1$.

For this reason, in the subsequent argument we will slightly abuse notation and denote $\mathcal{R} = \mathcal{R}_1$ and $\tilde{\mathcal{R}}_1 = \tilde{\mathcal{R}}$. Thus we have

$$\#\mathcal{R} \approx \#\tilde{\mathcal{R}} M, \tag{14.6.89}$$

and so it suffices to bound $\#\tilde{\mathcal{R}}$ and M .

We will now give an easy upper bound of M . Since E_2 is contained in a $(\delta, \alpha; \delta^{-\varepsilon})_1 \times (\delta, \alpha; \delta^{-\varepsilon})_1$ -quasi product and $C\tilde{R}$ is a the $O(\Delta)$ -neighbourhood of a curve of length $\sim \sqrt{\Delta/t}$, we have

$$|C\tilde{R} \cap E_2| \lesssim \delta^{1-\varepsilon} (\Delta/\delta)^\alpha \delta^{1-\varepsilon} (\sqrt{\Delta/t}/\delta)^\alpha = \delta^{2-2\alpha-2\varepsilon} \Delta^{3\alpha/2} t^{-\alpha/2}.$$

Recall $Y_2(R)$ defined as in (14.6.80). For $R \in \mathcal{R}(\tilde{R})$, we have $Y_2(R) \subset C_s R \cap E_2 \subset C_s \tilde{R} \cap E_2$. But by Lemma 14.6.43, $Y_2(R)$'s are disjoint, and thus we have a measure bound

$$M \lesssim \frac{|C_s \tilde{R} \cap E_2|}{|Y_2(R)|} \lesssim \Lambda^{-1} \delta^{2-2\alpha-2\varepsilon} \Delta^{3\alpha/2} t^{-\alpha/2} = \tau^{-1} \delta^{-2\alpha-2\varepsilon} \Delta^{2\alpha}, \quad (14.6.90)$$

where in the last equality we have used (14.6.84). Later we will give a second bound on M using Proposition 14.6.26.

14.6.5.4 A bilinear L^2 bound

In this section, we will analyze the interaction between fine and coarse-scale rectangles. We will show that for each fine-scale rectangle R , there are about μ^2 pairs of functions f, g that are tangent to R , so that $f^\delta \cap g^\delta$ is localized to R . Thus if N functions $f \in F$ are tangent to a coarse-scale rectangle $\tilde{R}$, then $\tilde{R}$ contains about N^2/μ^2 fine-scale rectangles. This is the “ L^2 argument” alluded to in the proof sketch from Section 14.6.1.3.

We turn to the details. By construction, each $R \in \mathcal{R}$ is indexed by some $x \in E_2$, namely, $R = R_x$. We will write $x(R)$ to denote this value of x . Let $f \in F_B$ and $R \in \mathcal{R}$. We say the pair (f, R) is *good* if $x(R) \in Y_1(f)$. Note that if (f, R) is good, then f is tangent to R by Lemma 14.6.41.

Denote by $\mathcal{G}$ the collection of all good pairs. For each $f \in F_B$, we define

$$\mathcal{G}(f) = \{R \in \mathcal{R} : (f, R) \in \mathcal{G}\}. \quad (14.6.91)$$

For each $R \in \mathcal{R}$, we define

$$\mathcal{G}(R) = \{f \in F_B : (f, R) \in \mathcal{G}\}. \quad (14.6.92)$$

For each $x \in E_2$, we have $x \in Y_1(f)$ if and only if $R_x \in \mathcal{G}(f)$. Thus by (14.6.75) and unravelling the definitions above, for each $R \in \mathcal{R}$ we have

$$\#\mathcal{G}(R) \sim \mu_2. \quad (14.6.93)$$

Pigeonholing the good pairs

For each $\tilde{R} \in \tilde{\mathcal{R}}$, we consider the quantity

$$\#\mathcal{G}(f) \cap \mathcal{R}(\tilde{R}) = \#\{R \in \mathcal{R}(\tilde{R}) : (f, R) \in \mathcal{G}\}.$$

By the disjointness of $\mathcal{R}(\tilde{R})$, we have

$$\#\mathcal{G} = \sum_{f \in F_B} \sum_{\tilde{R} \in \tilde{\mathcal{R}}} \#\mathcal{G}(f) \cap \mathcal{R}(\tilde{R}).$$

In view of (14.6.83), we may apply a dyadic pigeonholing to find an integer q and a set $\mathcal{G}' \subset \mathcal{G}$ with $\#\mathcal{G}' \gtrsim |\log \delta|^{-1} \#\mathcal{G}$, so that for each $\tilde{R} \in \tilde{\mathcal{R}}$ and each $f \in F_B$, the quantity $\#\mathcal{G}'(f) \cap \mathcal{R}(\tilde{R})$ is either 0 or $\sim q$. (Here we define $\mathcal{G}'(f)$ and $\mathcal{G}'(R)$ similarly as (14.6.91) and (14.6.92).)

For each $\tilde{R} \in \tilde{\mathcal{R}}$, define

$$F(\tilde{R}) = \{f \in F_B : \mathcal{G}'(f) \cap \mathcal{R}(\tilde{R}) \neq \emptyset\}. \quad (14.6.94)$$

For each $f \in F(\tilde{R})$, we have in particular $\mathcal{G}(f) \cap \mathcal{R}(\tilde{R}) \neq \emptyset$, and thus there is some $R \in \mathcal{R}(\tilde{R})$ such that $(f, R) \in \mathcal{G}$. Hence f is tangent to R , and using Lemma 14.6.44, we also have f is tangent to $\tilde{R}$.

By construction, $\#\mathcal{G}'(f) \cap \mathcal{R}(\tilde{R}) \sim q$ for each $f \in F(\tilde{R})$, and hence

$$\#F(\tilde{R}) \sim \frac{1}{q} \#\{(f, R) \in \mathcal{G}' : R \in \mathcal{R}(\tilde{R})\} = \frac{1}{q} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R). \quad (14.6.95)$$

More generally, for every ball $B(g, r) \subset C^2(J)$ we have

$$\#F(\tilde{R}) \cap B(g, r) \sim \frac{1}{q} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \cap B(g, r). \quad (14.6.96)$$

By (14.6.93) and (14.6.88), for each $\tilde{R} \in \tilde{\mathcal{R}}'$ we have

$$\sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \leq \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}(R) \sim \mu_2 M,$$

and hence

$$\#\mathcal{G}' = \sum_{\tilde{R} \in \tilde{\mathcal{R}}} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \lesssim \mu_2 M \#\tilde{\mathcal{R}}.$$

On the other hand, we have

$$\#\mathcal{G}' \gtrsim |\log \delta|^{-1} \#\mathcal{G} = |\log \delta|^{-1} \sum_{\tilde{R} \in \tilde{\mathcal{R}}} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}(R) \sim |\log \delta|^{-1} \mu_2 M \#\tilde{\mathcal{R}}, \quad (14.6.97)$$

where the final quasi-inequality used (14.6.93). Combining the upper and lower bounds, we see that for a $\gtrsim |\log \delta|^{-1}$ fraction of $\tilde{R} \in \tilde{\mathcal{R}}$ we have

$$\sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \gtrsim |\log \delta|^{-1} \mu_2 M. \quad (14.6.98)$$

Denote this subcollection as $\tilde{\mathcal{R}}'$. Similarly, for each $\tilde{R} \in \tilde{\mathcal{R}}'$, for a $\gtrsim |\log \delta|^{-1}$ fraction of $R \in \mathcal{R}(\tilde{R})$ we have

$$\#\mathcal{G}'(R) \gtrsim |\log \delta|^{-1} \mu_2. \quad (14.6.99)$$

Denote this subcollection as $\mathcal{R}'(\tilde{R})$, which has cardinality $\gtrsim |\log \delta|^{-1}M$. Denote

$$\mathcal{R}' = \bigsqcup_{\tilde{R} \in \tilde{\mathcal{R}}'} \mathcal{R}'(\tilde{R}). \quad (14.6.100)$$

Thus $\#\mathcal{R}' \gtrsim |\log \delta|^{-2} \#\mathcal{R}$. Since each $\#\mathcal{R}'(\tilde{R}) \approx M$, it suffices to bound $\#\mathcal{R}'$.

Each fine-scale rectangle has many good pairs

Lemma 14.6.45. *If $\delta > 0$ is sufficiently small, then for each $R \in \mathcal{R}'$, there are at least $(\#\mathcal{G}'(R))^2/3$ pairs $(f, g) \in (\mathcal{G}'(R))^2$ such that $\delta^{2\varepsilon}t \leq \|f - g\| \leq 6t$ and $\delta^\varepsilon \Delta \leq \Delta(f, g) + \delta \leq 2\Delta$.*

Proof. Let $R = R_x \in \mathcal{R}'$ where $x \in E_2$. Fix $g \in \mathcal{G}'(R)$. We deal with $\|f - g\|$ first. If $t \leq \delta^{1-\varepsilon}$, then we immediately have $\|f - g\| \geq \delta \geq \delta^{2\varepsilon}t$ for every $f \in \mathcal{G}'(R) \setminus \{g\}$, since the elements of F are δ -separated.

Next, suppose $t > \delta^{1-\varepsilon}$, and let $\lambda = \delta^{2\varepsilon}$; by hypothesis we have $\delta t^{-1} < \lambda < 1$. Thus we can apply Lemma 14.6.40 to conclude that

$$\begin{aligned} & \#\{f \in \mathcal{G}'(R) : \|f - g\| \leq \lambda t\} \\ & \leq \#\{f \in F_B(x) : \|f - g\| \leq \lambda t\} \\ & \lesssim \lambda^\varepsilon \mu_1 \\ & \lesssim \lambda^\varepsilon \delta^{-\varepsilon^2} \mu_2 \\ & = \delta^{\varepsilon^2} \mu_2 \\ & \lesssim \delta^{\varepsilon^2} |\log \delta| \#\mathcal{G}'(R), \end{aligned}$$

where the third inequality used (14.6.73) and in the last inequality we have used (14.6.99). If $\delta > 0$ is sufficiently small (depending on ε), then

$$\#\{f \in \mathcal{G}'(R) : \|f - g\| \leq \lambda t\} \leq \#\mathcal{G}'(R)/3. \quad (14.6.101)$$

Next we deal with $\Delta(f, g)$. If $\Delta \leq \delta^{1-\varepsilon}$ then we immediately have $\Delta(f, g) + \delta \geq \delta^\varepsilon \Delta$ for every $f \in \mathcal{G}'(R)$. If $\Delta > \delta^{1-\varepsilon}$, then we choose

$$\sigma = |\log \delta|^{-2},$$

and thus $\sigma^{\varepsilon-2} \Delta \geq \delta^\varepsilon \Delta > \delta$ for δ small enough. Using (14.6.66) and (14.6.67), we have

$$2n_3(\Delta_x, x) \Delta_x^{-\varepsilon^2} \geq n_4(x) \geq n_3(\sigma^{\varepsilon-2} \Delta, x) (\sigma^{\varepsilon-2} \Delta)^{-\varepsilon^2},$$

and thus $n_3(\sigma^{\varepsilon-2} \Delta, x) \lesssim \sigma n_3(\Delta_x, x)$ since $\Delta_x \sim \Delta$. Now recalling the definition of n_3 in (14.6.64), taking $k = g \in F_B(x)$ shows that

$$n_3(\sigma^{\varepsilon-2} \Delta, x) \geq \#\{f \in F_B(x) : \Delta(f, g) \leq \sigma^{\varepsilon-2} \Delta\}.$$

Combining the lower and upper bounds for $n_3(\sigma^{\varepsilon-2}\Delta, x)$ we have

$$\begin{aligned}
& \#\{f \in \mathcal{G}'(R) : \Delta(f, g) \leq \sigma^{\varepsilon-2}\Delta\} \\
& \leq \#\{f \in F_B(x) : \Delta(f, g) \leq \sigma^{\varepsilon-2}\Delta\} \\
& \lesssim \sigma n_3(\Delta_x, x) \\
& = \sigma \#\{f \in F_B(x) : \Delta(f, k_x) \leq \Delta_x\} \\
& \leq \sigma \mu_2 \\
& \lesssim \sigma |\log \delta| \#\mathcal{G}'(R) \\
& \leq \#\mathcal{G}'(R)/3,
\end{aligned} \tag{14.6.102}$$

for δ small enough, where in the fourth line we have used (14.6.68) and in the sixth line we have used $\Delta_x \leq \Delta$, (14.6.74) and (14.6.99).

By (14.6.101) and (14.6.102), we conclude that at least a $1/3$ fraction of the functions $f \in \mathcal{G}'(R)$ satisfy $\|f - g\| \geq \delta^{2\varepsilon}t$ and $\Delta(f, g) + \delta \geq \delta^\varepsilon\Delta$. The corresponding upper bounds $\|f - g\| \leq 6t$ and $\Delta(f, g) + \delta \leq 2\Delta$ hold for every $f \in \mathcal{G}'(R)$. $\square$

Lemma 14.6.46. *For each pair $f, g \in F_B \times F_B$ such that $\|f - g\| \geq \delta^{2\varepsilon}t$ and $\Delta(f, g) + \delta \geq \delta^\varepsilon\Delta$, there are $\lesssim \delta^{-4\varepsilon}$ rectangles $R \in \mathcal{R}'$ such that both $f, g \in \mathcal{G}'(R)$.*

Proof. If $f \in \mathcal{G}'(R)$, then $f \in \mathcal{G}(R)$ and thus f is tangent to R . It then suffices to show that there are at most $\lesssim \delta^{-3\varepsilon}$ many $R \in \mathcal{R}'$ that are tangent to both f, g .

Let f, g with $\|f - g\| \geq \delta^{2\varepsilon}t$ and $\Delta(f, g) + \delta \geq \delta^\varepsilon\Delta$, and suppose $R = k^\delta(I)$ is tangent to both f and g . Then over I we have $|f(\theta) - g(\theta)| \leq 10\delta$. By Part (2b) of Lemma 14.6.16 at the scale 10δ , the union of all R that are tangent to both f and g projects to the θ -axis into a set of Lebesgue measure

$$\lesssim \delta / \sqrt{\|f - g\|(\Delta(f, g) + \delta)} \lesssim \delta^{1-(3/2)\varepsilon} (t\Delta)^{-1/2} \sim \delta^{-(3/2)\varepsilon} \sqrt{\delta/t'}.$$

Using Lemma 14.6.23 with $\lambda \lesssim \delta^{-(3/2)\varepsilon}$, we see there are $\lesssim \delta^{-(15/4)\varepsilon} \leq \delta^{-4\varepsilon}$ $R \in \mathcal{R}'$ that are tangent to both f, g . $\square$

A tangency bound on $\#\mathcal{R}(\tilde{R})$

Now we give another bound on M in addition to (14.6.90), using the conclusions in Section 14.6.5.4. For $\tilde{R} \in \tilde{\mathcal{R}}'$, define

$$\mathcal{P}(\tilde{R}) = \bigcup_{R \in \mathcal{R}'(\tilde{R})} \{(f, g) \in \mathcal{G}'(R)^2 : \|f - g\| \geq \delta^{2\varepsilon}t, \Delta(f, g) + \delta \geq \delta^\varepsilon\Delta\}.$$

By Lemma 14.6.45, Lemma 14.6.46 and (14.6.99) we have

$$\#\mathcal{P}(\tilde{R}) \gtrsim \delta^{4\varepsilon} \mu_2^2 M. \quad (14.6.103)$$

On the other hand, we have a trivial bound (recall (14.6.94))

$$\#\mathcal{P}(\tilde{R}) \leq (\#F(\tilde{R}))^2. \quad (14.6.104)$$

Combining (14.6.103) and (14.6.104) and recalling (14.6.73) and (14.6.62), we conclude that

$$M \lesssim \delta^{-4\varepsilon} \mu_2^{-2} (\#F(\tilde{R}))^2 \lesssim \delta^{-6\varepsilon} \mu^{-2} (\#F(\tilde{R}))^2. \quad (14.6.105)$$

This is our second bound on M . Taking a suitable geometric average of these bounds, we have

$$M \lesssim (14.6.90)^{1/4} (14.6.105)^{3/4} \lesssim \tau^{-1/4} \delta^{-\alpha/2-6\varepsilon} \Delta^{\alpha/2} \mu^{-3/2} (\#F(\tilde{R}))^{3/2}. \quad (14.6.106)$$

Note that (14.6.106) holds for every $\tilde{R} \in \tilde{\mathcal{R}}'$, and the LHS is independent of the choice of $\tilde{R}$. Thus if we define

$$l = \min_{\tilde{R} \in \tilde{\mathcal{R}}'} \#F(\tilde{R}),$$

Then using (14.6.85) and (14.6.106), (14.6.86) would follow from the estimate

$$\#\tilde{\mathcal{R}}' \leq \delta^{-C_3\varepsilon} \delta^{-\zeta/2} t^{\alpha/2} l^{-3/2} \#F_B, \quad (14.6.107)$$

for some $C_3 = C_3(D)$.

In the next section we will establish this estimate.

14.6.5.5 Proving Inequality (14.6.107)

Our goal in this section is to bound the size of $\tilde{\mathcal{R}}'$. Our main tool will be Proposition 14.6.26. However, before the proposition can be applied we need to construct suitable sets $\mathcal{W}$ and $\mathcal{B}$.

Non-concentration of functions tangent to $\tilde{R}$

In this section, we will show that for each coarse-scale rectangle $\tilde{R}$, the functions from F tangent to $\tilde{R}$ cannot be concentrated inside a ball of diameter much smaller than t .

Lemma 14.6.47. *Let $\tilde{R} \in \tilde{\mathcal{R}}'$. For every $r > 0$ and every $g \in C^2(J)$, we have*

$$\#F(\tilde{R}) \cap B(g, r) \lesssim |\log \delta| (r/t)^\varepsilon (t/\Delta)^{\varepsilon^2} \#F(\tilde{R}). \quad (14.6.108)$$

Proof. Recall (14.6.96) that

$$\#F(\tilde{R}) \cap B(g, r) \sim q^{-1} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \cap B(g, r).$$

For each $R \in \mathcal{R}(\tilde{R})$, write $R = R_x$. using Lemma 14.6.40 with $\lambda = r/t$ and (14.6.63) we have

$$\#\mathcal{G}'(R) \cap B(g, r) \leq \#F_B(x) \cap B(g, r) \lesssim (r/t)^\varepsilon \#F_B(x) \sim (r/t)^\varepsilon \mu_1.$$

Thus

$$\#F(\tilde{R}) \cap B(g, r) \lesssim q^{-1} \#\mathcal{R}(\tilde{R}) (r/t)^\varepsilon \mu_1 \sim q^{-1} M (r/t)^\varepsilon \mu_1.$$

On the other hand, by (14.6.95) and (14.6.98), we have

$$\begin{aligned} \#F(\tilde{R}) &\sim q^{-1} \sum_{R \in \mathcal{R}(\tilde{R})} \#\mathcal{G}'(R) \\ &\gtrsim q^{-1} M |\log \delta|^{-1} \mu_2 \gtrsim q^{-1} M |\log \delta|^{-1} (\Delta/t)^{\varepsilon^2} \mu_1. \end{aligned}$$

Thus the result follows. $\square$

By Lemma 14.6.47, we now take

$$r = tc^{1/\varepsilon} |\log \delta|^{-1/\varepsilon} (\Delta/t)^\varepsilon, \quad (14.6.109)$$

where $c = c(D)$ is a sufficiently small constant to be determined, such that for every ball $B(g, r) \subset C^2(J)$ and every $\tilde{R} \in \tilde{\mathcal{R}}'$ we have

$$\#F(\tilde{R}) \cap B(g, r) \leq c \#F(\tilde{R}). \quad (14.6.110)$$

Note that $t \leq \delta^{-2\varepsilon} r/3$ for δ small enough.

Applying Proposition 14.6.26

Since $\mathcal{F}$ has doubling constant D , there exists $C_D \geq 1$ so that we can cover $3B$ by a set of $(3t/r)^{C_D} \leq \delta^{-2C_D\varepsilon}$ balls of radius r , so that these balls are boundedly overlapping (with the bound depending on D). Denote this set of balls by $\mathcal{O}$.

If $c = c(D)$ is chosen to be small enough, then for any $\tilde{R} \in \tilde{\mathcal{R}}'$, by pigeonholing and (14.6.110), there are two balls $O_1(\tilde{R}), O_2(\tilde{R}) \in \mathcal{O}$ with their centres $\geq 10r$ separated, such that for $i = 1, 2$, we have

$$\#F(\tilde{R}) \cap O_i(\tilde{R}) \gtrsim \delta^{2C_D\varepsilon} \#F(\tilde{R}) \geq \delta^{2C_D\varepsilon} l. \quad (14.6.111)$$

Since $(\#\mathcal{O})^2 \lesssim \delta^{-4C_D\varepsilon}$, by pigeonholing, there is a pair $(O_1, O_2) \in \mathcal{O}^2$ so that (14.6.111) holds for $\gtrsim \delta^{-4C_D\varepsilon} \#\tilde{\mathcal{R}}'$ rectangles in $\tilde{\mathcal{R}}'$. Fix this choice of O_1 and O_2 , and let $\tilde{\mathcal{R}}'' \subset \#\tilde{\mathcal{R}}'$ be the set of rectangles for which (14.6.111) holds.

To establish (14.6.107), it suffices to prove that

$$\#\tilde{\mathcal{R}}'' \lesssim \delta^{-\zeta/2-C_3\varepsilon} t^{\alpha/2} l^{-3/2} \#F_B, \quad (14.6.112)$$

for some $C_3 = C_3(D)$.

Let $\mathcal{W} = F_B \cap O_1(\tilde{R})$ and $\mathcal{B} = F_B \cap O_2(\tilde{R})$. Then $(\mathcal{W}, \mathcal{B})$ is r -separated. Also, by the discussion right after (14.6.94), each $(\Delta, C_R t)$ -rectangle $\tilde{R} \in \tilde{\mathcal{R}}''$ is in particular tangent to $\geq l$ many $f \in \mathcal{W}$ and $\geq l$ many $f \in \mathcal{B}$.

With this, we can now apply Proposition 14.6.26 to the family $\#\tilde{\mathcal{R}}''$ with $\mu = \nu = l$, $C_R t$ in place of t , and $A \leq \delta^{-2\varepsilon}$ to get

$$\#\tilde{\mathcal{R}}'' \lesssim \delta^{-C_4\varepsilon} (\#F_B/l)^{3/2}, \quad (14.6.113)$$

for some $C_4 = C_4(D)$.

By (14.6.7) with $3t$ in place of r , we have

$$\#F_B \leq \delta^{-\varepsilon} (t/\delta)^\zeta. \quad (14.6.114)$$

Combining (14.6.113) and (14.6.114) and using the condition $\alpha \leq \zeta$, we obtain (14.6.112) for some suitable C_3 . This concludes the proof of Theorem 14.6.7.

Remark 14.6.48. For some applications, it may be helpful to replace the Frostman condition (14.6.7) with the weaker inequality

$$\#(F \cap B) \leq A(r/\delta)^\zeta \quad \text{for all balls } B \subset C^2(I) \text{ of radius } r \geq \delta, \quad (14.6.115)$$

for some $A \geq 1$. By randomly selecting each $f \in F$ with probability $1/A$ and then applying Theorem 14.6.7, we obtain the following analogue of (14.6.8).

$$\int_E \left(\sum_{f \in F} \mathbf{1}_{f^\delta} \right)^{3/2} \leq A^{1/2} \delta^{2-\alpha/2-\zeta/2-C\varepsilon} (\#F). \quad (14.6.116)$$

A Sogge's cinematic curvature, and cinematic functions

In this section, we will explain the connection between cinematic functions (see Definition 14.6.6) and Sogge's cinematic curvature condition, as used by Kolasa and Wolff in [17]. First, we will show that if $\Phi: \mathbb{R}^4 \rightarrow \mathbb{R}$ satisfies the cinematic curvature condition (for simplicity, we will state this condition in a form that is convenient to compute with), then there are sets $Y \times R \subset \mathbb{R}^3$ and $I \times J \subset \mathbb{R}^2$, so that for each $(y, r) \in Y \times R$, the curve $\{x \in I \times J: \Phi(x, y) = r\}$ is the graph of a function $f_{(y,r)}: I \rightarrow J$, and $\{f_{(y,r)}\}$ is a family of cinematic functions. The proof of this statement requires careful computation using the inverse function theorem. These computations are described below.

A.1 Cinematic curvature for normalized defining functions

Let $U = I \times J \times Y \subset \mathbb{R}^4$, where $I, J \subset \mathbb{R}$ are open intervals containing 0, and $Y \subset \mathbb{R}^2$ is a neighborhood of 0. Let $\Phi(x, y): U \rightarrow \mathbb{R}$, with

$$\|\Phi\|_{C^3(U)} = C_0. \quad (14.6.117)$$

We will require

$$\Phi(x, 0) = x_2 + O(x^3), \quad (14.6.118)$$

$$d_y \left(\frac{\partial_{x_1} \Phi(x, y)}{\partial_{x_1}^2 \Phi(x, y) / |\nabla_x \Phi(x, y)|} \right) \Big|_{(x, y) = (0, 0)} \text{ is invertible.} \quad (14.6.119)$$

Conditions (14.6.118) and (14.6.119) are a formulation of Sogge's cinematic curvature condition for the defining function Φ . See, for example Section 3 or the beginning of Section 5 from [17].

Shrinking I and Y if necessary, by the inverse function theorem we can find an interval R containing 0 so that for each $(y, r) \in Y \times R$, the curve $\{x \in I \times J: \Phi(x, y) = r\}$ is the graph $x_2 = f_{y,r}(x_1)$, where $f_{y,r} \in C^2(I)$. We will show that after possibly further shrinking I and Y , the set

$$\mathcal{F} = \{f_{y,r}: (y, r) \in Y \times R\} \subset C^2(I) \quad (14.6.120)$$

is a cinematic family of functions. Specifically, we will show there exists $\varepsilon > 0$ so that for all $(y, r), (y', r') \in Y \times R$ and all $x_1 \in I$, we have

$$\begin{aligned} & |f_{y,r}(x_1) - f_{y',r'}(x_1)| + |\dot{f}_{y,r}(x_1) - \dot{f}_{y',r'}(x_1)| \\ & + |\ddot{f}_{y,r}(x_1) - \ddot{f}_{y',r'}(x_1)| \geq \varepsilon |(y, r) - (y', r')|. \end{aligned} \quad (14.6.121)$$

Clearly the metric space $\mathcal{F}$ has bounded diameter, and doubling.

Step 1

After shrinking Y , I , and J if necessary (and re-defining $U = I \times J \times Y$), we can suppose that

$$1/2 \leq \partial_{x_2} \Phi(x, y) \leq 2 \quad \text{for all } (x, y) \in U, \quad (14.6.122)$$

and there exists $c > 0$ so that

$$\left| \det \left(d_y \left(\frac{\partial_{x_1} \Phi(x, y)}{\partial_{x_1}^2 \Phi(x, y) / |\nabla_x \Phi(x, y)|} \right) \right) \right| \geq c \quad \text{for all } (x, y) \in U. \quad (14.6.123)$$

(14.6.122) is a consequence of (14.6.118), while (14.6.123) is a consequence of (14.6.119).

Let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be small positive numbers. We will choose ε_3 sufficiently small depending on c and C_0 . We will choose ε_2 sufficiently small depending on c , C_0 and ε_3 . We will choose ε_1 sufficiently small depending on c , C_0 and $\varepsilon_2, \varepsilon_3$.

After further shrinking Y , I , and J if necessary, we can suppose that

$$|\partial_{x_1}\Phi(x, y)| \leq \varepsilon_1, \quad |\nabla_x^2\Phi(x, y)| \leq \varepsilon_1 \quad \text{for all } (x, y) \in U. \quad (14.6.124)$$

Step 2 Let $(y, r), (y', r') \in Y \times R$. Let $x_1 \in I$, and let $x_2, x'_2 \in J$ so that $\Phi(x_1, x_2, y) = r$, $\Phi(x_1, x'_2, y') = r'$. We will define $x = (x_1, x_2)$ and $x' = (x_1, x'_2)$, so $|x - x'| = |x_2 - x'_2|$. Finally, we will adopt the notation $\Phi_1(x, y) = \partial_{x_1}\Phi(x, y)$; $\Phi_2(x, y) = \partial_{x_2}\Phi(x, y)$; $\Phi_{11}(x, y) = \partial_{x_1}^2\Phi(x, y)$, etc.

If $|x_2 - x'_2| \geq \frac{1}{4}|r - r'|$, then (14.6.121) holds with $\varepsilon = 1/4$ and we are done. Thus we can suppose that

$$|x_2 - x'_2| < \frac{1}{4}|r - r'|. \quad (14.6.125)$$

We have

$$\begin{aligned} |r - r'| &= |\Phi(x, y) - \Phi(x', y')| \leq |\Phi(x, y) - \Phi(x', y)| \\ &\quad + |\Phi(x', y) - \Phi(x', y')| \\ &\leq 2|x - x'| + C|y - y'|. \end{aligned}$$

Using (14.6.125) and re-arranging, we conclude that $|r - r'| \leq 2C|y - y'|$, so in particular

$$|(y, r) - (y', r')| \leq 3C|y - y'|. \quad (14.6.126)$$

Step 2

Let ε_2 be a small constant to be determined later. If $|x_2 - x'_2| \geq \varepsilon_2|y - y'|$, then by (14.6.126), (14.6.121) holds with $\varepsilon = \frac{\varepsilon_2}{3C}$ and we are done. Thus we can suppose that

$$|x_2 - x'_2| < \varepsilon_2|y - y'|. \quad (14.6.127)$$

Next, let ε_3 be a small constant to be determined later. Suppose that

$$|\Phi_1(x', y) - \Phi_1(x, y')| \geq \varepsilon_3|y - y'|. \quad (14.6.128)$$

We will show that if ε_1 and ε_2 were selected sufficiently small (depending on c, C_0 and ε_3), then

$$|\dot{f}_{y,r}(x_1) - \dot{f}_{y',r'}(x_1)| \geq \frac{\varepsilon_3}{10}|y - y'|, \quad (14.6.129)$$

and thus by (14.6.126), (14.6.121) holds with $\varepsilon = \frac{\varepsilon_3}{30C}$.

To establish (14.6.129), we will make use of the following inequality

$$\left| \frac{A}{B} - \frac{A'}{B'} \right| \geq |A - A'| \frac{|B|}{|BB'|} - |A| \frac{|B - B'|}{|BB'|} \quad (14.6.130)$$

In practice, when we apply Inequality (14.6.130), B and B' will always have $\frac{1}{2} \leq |B| \leq 2$ and similarly for B' ; $|A - A'| \gtrsim |y - y'|$; $|B - B'| \lesssim |y - y'|$ (with implicit constants that might depend on c and C_0), and $|A| \leq \varepsilon_1$. If A, A', B, B' satisfy these conditions, then we will have $|A/B - A'/B'| \geq \frac{1}{8}|A - A'|$, provided ε_1 is chosen sufficiently small.

We have

$$\begin{aligned} |\dot{f}_{y,r}(x_1) - \dot{f}_{y',r'}(x_1)| &= \left| \frac{\Phi_1(x, y)}{\Phi_2(x, y)} - \frac{\Phi_1(x', y')}{\Phi_2(x', y')} \right| \\ &\geq \left| \frac{\Phi_1(x', y)}{\Phi_2(x', y)} - \frac{\Phi_1(x', y')}{\Phi_2(x', y')} \right| - \left| \frac{\Phi_1(x, y)}{\Phi_2(x, y)} - \frac{\Phi_1(x', y)}{\Phi_2(x', y)} \right|. \end{aligned} \quad (14.6.131)$$

By (14.6.117), (14.6.122), and (14.6.127), we have

$$\left| \frac{\Phi_1(x, y)}{\Phi_2(x, y)} - \frac{\Phi_1(x', y)}{\Phi_2(x', y)} \right| \leq 4C|x - x'| \leq 4C_0\varepsilon_2|y - y'|.$$

Applying Inequality (14.6.130) and using assumption (14.6.128), we conclude that

$$\left| \frac{\Phi_1(x', y)}{\Phi_2(x', y)} - \frac{\Phi_1(x', y')}{\Phi_2(x', y')} \right| \geq \frac{1}{8}|\Phi_1(x', y) - \Phi_1(x', y')| \geq \frac{\varepsilon_3}{8}|y - y'|.$$

Thus if ε_2 is chosen sufficiently small depending on C_0 and ε_3 , then (14.6.129) follows from (14.6.131).

Step 3

Suppose now that (14.6.128) is false, i.e.

$$|\Phi_1(x', y) - \Phi_1(x, y')| < \varepsilon_3|y - y'|. \quad (14.6.132)$$

Then by (14.6.123), we have

$$\left| \frac{\Phi_{11}(x', y)}{|\nabla \Phi(x, y)|} - \frac{\Phi_{11}(x', y')}{|\nabla \Phi(x', y')|} \right| \geq \frac{c}{4}|y - y'|. \quad (14.6.133)$$

We will show that if $\varepsilon_1, \varepsilon_2, \varepsilon_3$ were chosen sufficiently small, then

$$|\ddot{f}_{y,r}(x_1) - \ddot{f}_{y',r'}(x_1)| \geq \frac{c}{10}|y - y'|, \quad (14.6.134)$$

and thus by (14.6.126), (14.6.121) holds with $\varepsilon = \frac{c}{30C}$. Thus (14.6.134) implies that set M defined in (14.6.120) is a cinematic family of functions. It remains to establish (14.6.134). The proof of this inequality uses the same ideas as the proof of (14.6.129), though the steps are slightly more complicated.

Define

$$F(a, b) = \frac{\Phi_{11}(a, b)\Phi_2(a, b)^2 - 2\Phi_{12}(a, b)\Phi_1(a, b)\Phi_2(a, b) + \Phi_{22}(a, b)\Phi_1(a, b)^2}{\Phi_2(a, b)^3}$$

Then

$$\begin{aligned} |\ddot{f}_{y,r}(x_1) - \ddot{f}_{y',r'}(x_1)| &= |F(x, y) - F(x', y')| \\ &\geq |F(x', y) - F(x, y')| - |F(x, y) - F(x', y)|. \end{aligned}$$

By (14.6.117), (14.6.122), and (14.6.127), we have

$$|F(x, y) - F(x', y)| \lesssim C|x - x'| \lesssim C\varepsilon_2|y - y'|. \quad (14.6.135)$$

Thus our goal is to establish a lower bound for $|F(x', y) - F(x, y')|$. We will estimate

$$\begin{aligned} |F(x', y) - F(x, y')| &\geq \left| \frac{\Phi_{11}(x', y)}{\Phi_2(x', y)} - \frac{\Phi_{11}(x', y')}{\Phi_2(x', y')} \right| \\ &\quad - 2 \left| \frac{\Phi_{12}(x', y)\Phi_1(x', y)}{\Phi_2(x', y)^2} - \frac{\Phi_{12}(x', y')\Phi_1(x', y')}{\Phi_2(x', y')^2} \right| \\ &\quad - \left| \frac{\Phi_{22}(x', y)\Phi_1(x', y)^2}{\Phi_2(x', y)^3} - \frac{\Phi_{22}(x', y')\Phi_1(x', y')^2}{\Phi_2(x', y')^3} \right|. \end{aligned} \quad (14.6.136)$$

For the first term, we will use the inequality $AB - A'B' = (A - A')B' + (B - B')A$ to write

$$A_y = \frac{\Phi_{11}(x', y)}{|\nabla\Phi(x', y)|}, \quad B_y = \left(\frac{\Phi_1^2(x', y)}{\Phi_2^2(x', y)} + 1 \right)^{1/2},$$

and define $A_{y'}$ and $B_{y'}$ by replacing y with y' . Then

$$\begin{aligned} \frac{\Phi_{11}(x', y)}{\Phi_2(x', y)} - \frac{\Phi_{11}(x', y')}{\Phi_2(x', y')} &= A_y B_y - A_{y'} B_{y'} \\ &= (A_y - A_{y'})B_{y'} + (B_y - B_{y'})A_{y'}. \end{aligned}$$

Using (14.6.124) and (14.6.122), and then (14.6.133), We have

$$|(A_y - A_{y'})B_{y'}| \geq \frac{1}{2}|A_y - A_{y'}| \geq \frac{c}{4}|y - y'|,$$

and by (14.6.124),

$$|(B_y - B_{y'})A_{y'}| \leq 2\varepsilon_1|B_y - B_{y'}| \leq (2C)|y - y'| (2\varepsilon_1).$$

Thus if ε_1 is selected sufficiently small, we have

$$\left| \frac{\Phi_{11}(x', y)}{\Phi_2(x', y)} - \frac{\Phi_{11}(x', y')}{\Phi_2(x', y')} \right| \geq \frac{c}{5} |y - y'|.$$

Bounding the second and third term on the RHS of (14.6.136) is straightforward, using the estimates (14.6.122), (14.6.124), and (14.6.132). We conclude that

$$|F(x', y) - F(x, y')| \geq \frac{c}{6} |y - y'|.$$

Combining this with (14.6.135), we obtain (14.6.134).

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Chapter 14.7

Lens Counting and the Missing Sampling Reductions

This chapter closes the two combinatorial gaps left visible in the preceding cinematic-maximal proof. First we prove the non-overlapping-lens theorem invoked as Theorem 14.6.33. We include the complete intersection-reverse-sequence argument, not merely the geometric reduction. Second we give the sampling and scale-reduction details suppressed after Lemma 14.6.36. Thus Proposition 14.6.26, and hence the cinematic maximal theorem, no longer has an external combinatorial proof dependency.

The sequence argument is an expanded adaptation of Adam Marcus and Gábor Tardos, *Intersection reverse sequences and geometric applications*, whose Lemma 10 is the result used in the preceding chapter. We spell out the topological order lemma as well as every averaging inequality because those are precisely the steps hidden by the short citation in the source proof.

14.7.1 Intersection-reverse cyclic orders

Definition 14.7.1 (Reverse cyclic orders). Let A and B be cyclic orders on subsets of a finite alphabet X . They are *intersection reverse* if, after writing their common symbols as $a_1, \dots, a_q$ in the cyclic order induced by A , the cyclic order induced by B is $a_q, \dots, a_1$, up to a cyclic shift. This convention makes the condition automatic when $q \leq 2$.

If C is a linearly ordered block and $a \neq b$, put

$$f(C, a, b) = \begin{cases} 0, & \text{if } a \notin C \text{ or } b \notin C, \\ 1, & \text{if } a \text{ precedes } b \text{ in } C, \\ -1, & \text{if } b \text{ precedes } a \text{ in } C, \end{cases}$$

and set $f(C, D, a, b) = f(C, a, b)f(D, a, b)$. Thus $f(C, D, a, b)$ is 1 when C, D agree on the order of a, b , is -1 when they disagree, and is zero unless both symbols occur in both blocks.

Lemma 14.7.2 (The block lemma). *Suppose intersection-reverse cyclic orders A, A' are cut into linearly ordered consecutive blocks $B_1, \dots, B_r$ and $B'_1, \dots, B'_r$. At most one pair (B_i, B'_j) fails to be intersection reverse as a pair of linear orders. If $q = |B_i \cap B'_j|$, then for that exceptional pair*

$$\sum_{a \neq b} f(B_i, B'_j, a, b) \leq q.$$

For every nonexceptional pair the same sum equals $q - q^2$.

Proof. Write the common symbols in their order in B_i as $a_1, \dots, a_q$. Reversal of the ambient cyclic orders says that their order in B'_j must be

$$a_x, a_{x-1}, \dots, a_1, a_q, a_{q-1}, \dots, a_{x+1}$$

for some $x \in \{1, \dots, q\}$. The pair of blocks is linearly reverse exactly when $x = q$. A cut in each of the two cyclic orders can split the reversed common cycle in this way for at most one pair of blocks, proving the first assertion.

For an exceptional pair, the ordered pairs with one member in $\{a_1, \dots, a_x\}$ and the other in $\{a_{x+1}, \dots, a_q\}$ contribute $2x(q-x)$, while ordered pairs within the two groups contribute $-x(x-1) - (q-x)(q-x-1)$. Their sum is

$$q - (q - 2x)^2 \leq q.$$

For a reverse pair every ordered pair of distinct common symbols contributes -1 , so the sum is $-q(q-1) = q - q^2$. $\square$

Theorem 14.7.3 (Uniform intersection-reverse bound). *Let $A^1, \dots, A^m$ be pairwise intersection-reverse cyclic orders, each containing exactly d distinct symbols from an alphabet of size n . Then*

$$d \leq C \left(\sqrt{n} \log(2n) + \frac{n}{\sqrt{m}} \right)$$

for an absolute constant C .

Proof. If $dm \leq 2n$, then $d \leq 2n/m \leq 2n/\sqrt{m}$ and there is nothing to prove. Assume $dm > 2n$. Put

$$p_{ij} = |A^i \cap A^j|, \quad p = \sum_{i \neq j} p_{ij}.$$

If d_a is the number of orders containing the symbol a , then

$$p = \sum_a (d_a^2 - d_a) \geq \frac{1}{n} \left(\sum_a d_a \right)^2 - \sum_a d_a = \frac{d^2 m^2}{n} - dm \geq \frac{d^2 m^2}{2n}. \quad (6.1)$$

Also Cauchy–Schwarz gives

$$\sum_{i \neq j} p_{ij}^2 \geq \frac{p^2}{m^2 - m} > \frac{p^2}{m^2}. \quad (6.2)$$

Let $k = \lceil \log_2 d \rceil$. Bisect every A^i into two consecutive blocks whose sizes differ by at most one; bisect the descendants in the same way, and continue for k generations. For a binary word s of length ℓ , denote the corresponding block by A_s^i . Then $|A_s^i| < d2^{-\ell} + 1$, and the terminal blocks have size at most one.

For positive weights $w_1, \dots, w_k$, define

$$S_{ij} = \sum_{\ell=1}^k w_\ell \sum_{\substack{|s|=|t|=\ell \\ a \neq b}} f(A_s^i, A_t^j, a, b).$$

We first obtain a lower bound after summing over $i \neq j$. For fixed a, b, ℓ , summing over all ordered pairs i, j produces the square

$$\left(\sum_i \sum_{|s|=\ell} f(A_s^i, a, b) \right)^2 \geq 0.$$

The diagonal term is at most the number of ordered pairs in each block. Consequently

$$\begin{aligned} \sum_{i \neq j} S_{ij} &\geq - \sum_i \sum_{\ell=1}^k 2w_\ell \sum_{|s|=\ell} \binom{|A_s^i|}{2} \\ &\geq -md^2 \sum_{\ell=1}^k \frac{w_\ell}{2^\ell}. \end{aligned} \quad (6.3)$$

Fix $i \neq j$. Write $r_{st} = |A_s^i \cap A_t^j|$ and $Q_{st} = \sum_{a \neq b} f(A_s^i, A_t^j, a, b)$. At a fixed generation, the blocks partition the two cyclic orders, so

$$\sum_{|s|=|t|=\ell} r_{st} = p_{ij}. \quad (6.4)$$

Call (s, t) a *leader* if the two blocks are linearly reverse and either $|s| = 1$, or their parent pair is the exceptional pair in Lemma 14.7.2. At each generation

there are at most four leaders: an exceptional parent has only four pairs of children. Every common symbol belongs to exactly one leader pair, namely the first descendant pair after the last exceptional ancestor. Hence, if L is the set of leaders,

$$\sum_{(s,t) \in L} r_{st} = p_{ij}, \quad \sum_{(s,t) \in L} \frac{1}{w_{|s|}} \leq 4 \sum_{\ell=1}^k \frac{1}{w_{\ell}}. \quad (6.5)$$

Lemma 14.7.2 gives $Q_{st} \leq r_{st}$ for every pair and $Q_{st} = r_{st} - r_{st}^2$ for leaders. Using (6.4), and then weighted Cauchy–Schwarz with (6.5), we obtain

$$\begin{aligned} S_{ij} &\leq p_{ij} \sum_{\ell=1}^k w_{\ell} - \sum_{(s,t) \in L} w_{|s|} r_{st}^2 \\ &\leq p_{ij} \sum_{\ell=1}^k w_{\ell} - \frac{p_{ij}^2}{4 \sum_{\ell=1}^k 1/w_{\ell}}. \end{aligned} \quad (6.6)$$

Put $W = \sum w_{\ell}$, $V = \sum 1/w_{\ell}$, and $U = \sum w_{\ell}/2^{\ell}$. Combining (6.2), (6.3), and (6.6) gives

$$-md^2U \leq pW - \frac{p^2}{4m^2V}.$$

Thus either $p^2/(4m^2V) \leq 2pW$, or $p^2/(4m^2V) \leq 2md^2U$. In the first case (6.1) yields

$$d \leq 4\sqrt{nWV};$$

in the second it yields

$$d \leq 6 \frac{n}{\sqrt{m}} \sqrt{UV}.$$

Choose

$$w_{\ell} = \frac{1}{1 + k2^{-\ell/2}}.$$

Then $W \leq k$, $V \leq 4k$, and $U \leq 3/k$: the first estimate is immediate, the second follows by summing $1 + k2^{-\ell/2}$, and the third follows from $w_{\ell} \leq 2^{\ell/2}/k$. Therefore either $d \leq 8k\sqrt{n}$ or $d \leq 21n/\sqrt{m}$. Since $k \leq 1 + \log_2 n$, the asserted bound follows. $\square$

Corollary 14.7.4 (Unequal lengths). *If $A^1, \dots, A^m$ are pairwise intersection-reverse cyclic orders on subsets of an n -symbol alphabet, then*

$$\sum_{i=1}^m |A^i| \text{leq} C(m\sqrt{n} \log(2n) + n\sqrt{m}).$$

In particular, for $m = n$ the sum is $O(n^{3/2} \log(2n))$.

Proof. For $u > 0$, let $M(u)$ be the number of orders of length at least u . Truncating those orders to u consecutive symbols preserves the intersection-reverse property. Theorem 14.7.3 therefore implies

$$u \leq C_0 \left(\sqrt{n} \log(2n) + \frac{n}{\sqrt{M(u)}} \right) \quad \text{whenever } M(u) > 0. \quad (6.7)$$

Set

$$u_0 = 2C_0 \left(\sqrt{n} \log(2n) + \frac{n}{\sqrt{m}} \right).$$

For $u \geq u_0$, (6.7) implies $M(u) \leq 4C_0^2 n^2 / u^2$. Because every order has length at most n , dyadic summation gives

$$\begin{aligned} \sum_i |A^i| &\leq mu_0 + \sum_{q \geq 0} 2^{q+1} u_0 M(2^q u_0) \\ &\leq mu_0 + C \frac{n^2}{u_0} \leq C(m\sqrt{n} \log(2n) + n\sqrt{m}), \end{aligned}$$

where empty dyadic ranges contribute zero. □

14.7.2 From lenses to cyclic orders

We now supply the planar-topology step that is often conveyed only by a picture. Orient every Jordan curve counterclockwise. If a lens side lies on c , call that side positive when the other side lies locally on the interior side of c , and negative otherwise. A lens therefore has a sign pair $(+, +)$, $(-, -)$, $(+, -)$, or $(-, +)$ after its two curves have been ordered.

Lemma 14.7.5 (Annular order lemma). *Let a, b be two members of a pseudo-circle family and let $x_1, \dots, x_q$ be distinct further members. Suppose that, for each r , there is a lens between a and x_r and a lens between b and x_r ; all these lenses are pairwise non-overlapping. Suppose also that the two lenses associated with every base curve have one fixed sign type. In the mixed-sign case, record a lens at the curve carrying its positive side. Then the cyclic order of $x_1, \dots, x_q$ around a is the reverse of their cyclic order around b .*

Proof. We give the cut-open argument in detail. Replace each lens side on a and b by a very short closed attachment interval in the same side. This can be done independently because non-overlap says that the interiors of all such intervals are disjoint. Delete the interiors of thin annular neighborhoods of a and b . Each x_r now meets the resulting surface in two disjoint connector arcs issuing from its attachment interval at a and two issuing from its attachment interval at b . The signs specify on which boundary component each connector

lies. For $(+, +)$ and $(-, -)$ the two connectors use matching boundary sides; for a mixed lens, recording it only at the positive side makes the same statement after reversing one boundary orientation.

Cut every component of the resulting surface along one connector belonging to x_1 . Each relevant component becomes a disk whose boundary contains two linearly ordered sets of attachment intervals, one inherited from a and one from b . In a disk, two disjoint crosscuts joining the two boundary sets have opposite endpoint orders: if their endpoints occurred in the same order, the first crosscut together with either boundary subarc would be a Jordan curve separating the two endpoints of the second crosscut. The second crosscut would then have to meet the first.

Apply this observation to each of the two connector arcs of x_r and x_s . If x_r, x_s occur in the same order on the two boundary sets, one connector of x_r must meet the corresponding connector of x_s , and the other connector pair must meet as well. These are two proper crossings. If three symbols x_r, x_s, x_t appeared in the same cyclic order around a and b , choose a cut just before x_r . The middle symbol x_s is separated from x_r by one connector of x_t in each of the two cut-open components. The disk argument then forces two further proper crossings between the same pair of pseudo-circles among x_r, x_s, x_t . That pair would have at least four intersection points, contradicting the pseudo-circle axiom that two members meet at most twice.

Therefore no three common symbols can have the same cyclic orientation. For two cyclic orders, this is equivalent to reversal: after fixing one common symbol as the cut point, every pair among the remaining symbols has the opposite linear order. The conclusion follows. The sign discussion in the first paragraph also shows why mixed lenses must be entered at their positive, not their negative, side. $\square$

Theorem 14.7.6 (Non-overlapping-lens bound). *Let $\mathcal{C}$ be a family of n pseudo-circles. Every family $\mathcal{L}$ of non-overlapping first-generation lenses determined by $\mathcal{C}$ has cardinality*

$$|\mathcal{L}| \leq Cn^{3/2} \log(2n).$$

This proves Theorem 14.6.33.

Proof. There are only n degenerate lenses, so discard them. Partition the remaining lenses by the sign types $(+, +)$, $(-, -)$, and mixed. For mixed lenses retain the orientation that points from the curve carrying the positive side to the curve carrying the negative side. For a fixed ordered pair of pseudo-circles, each sign type determines at most one lens: after the two intersection points are removed, each curve has two complementary arcs, one on either side of the other curve, and the two selected signs determine the two arcs uniquely.

Fix one type. For every $c \in \mathcal{C}$, form a cyclic list S_c of the curves c' that make a lens of this type with c , ordered by the disjoint lens sides encountered around c . In the mixed case make the entry only when c carries the positive side. The lists contain no repeated symbols by the preceding uniqueness observation, and Lemma 14.7.5 says that they are pairwise intersection reverse. Corollary 14.7.4, with $m = n$, yields

$$\sum_{c \in \mathcal{C}} |S_c| = O(n^{3/2} \log(2n)).$$

Every lens of the fixed type is counted at least once in this sum. Summing over the four ordered sign types and restoring the n degenerate lenses proves the theorem. $\square$

14.7.3 The balanced tangency estimate and random sampling

The preceding chapter proves, after perturbation, the following balanced case: if $\mathcal{W}, \mathcal{B}$ are $2t$ -separated and every rectangle in a pairwise $2C_c$ -incomparable family is tangent to at least one member of each colour, then

$$\#\mathcal{R} \leq C(\#\mathcal{W} + \#\mathcal{B})^{3/2} \log(2 + \#\mathcal{W} + \#\mathcal{B}). \quad (6.8)$$

Here and below changing a fixed tangency constant changes only C .

Lemma 14.7.7 (Sampling with multiplicities). *Assume (6.8). If every $R \in \mathcal{R}$ is tangent to at least μ members of $\mathcal{W}$ and at least ν members of $\mathcal{B}$, then*

$$\#\mathcal{R} \leq C \left(\frac{\#\mathcal{W}}{\mu} + \frac{\#\mathcal{B}}{\nu} \right)^{3/2} \log \left(2 + \frac{\#\mathcal{W}}{\mu} + \frac{\#\mathcal{B}}{\nu} \right).$$

Proof. Select every $w \in \mathcal{W}$ independently with probability $p = \min(1, 2/\mu)$, and every $b \in \mathcal{B}$ independently with probability $q = \min(1, 2/\nu)$. Call the selected sets $\mathcal{W}_0, \mathcal{B}_0$ and put $Z = \#\mathcal{W}_0 + \#\mathcal{B}_0$. A fixed rectangle is hit by the selected white family with probability at least $1 - e^{-2}$ (or probability one when $p = 1$), and independently has the same lower bound for the blue family. Hence it is hit by both with probability at least $\alpha = (1 - e^{-2})^2$.

Moreover

$$\mathbb{E}Z = p\#\mathcal{W} + q\#\mathcal{B} \leq 2 \left(\frac{\#\mathcal{W}}{\mu} + \frac{\#\mathcal{B}}{\nu} \right) =: 2M.$$

By Markov's inequality, $\mathbb{P}(Z > 8\mathbb{E}Z) \leq 1/8$. If $\mathcal{R}_0$ denotes the rectangles hit in both colours, then

$$\mathbb{E}[\#\mathcal{R}_0; Z \leq 8\mathbb{E}Z] \geq (\alpha - 1/8)\#\mathcal{R}.$$

Thus one realization has $Z \leq 16M$ and $\#\mathcal{R}_0 \geq (\alpha - 1/8)\#\mathcal{R}$. Apply (6.8) to this realization and absorb the absolute constants. $\square$

14.7.4 Completion of the preliminary reductions

We finally justify the two short reductions that were stated without proofs before the pseudo-circle construction.

Lemma 14.7.8 (Changing the separation scale). *Suppose the tangency estimate is known when the two colour classes are $2t$ -separated. Then, for $A \geq 1$, it holds for a t/A -separated pair, with an additional factor $A^{C(K,D)}$.*

Proof. Put $r = t/(10A)$. The ball of radius $6t$ containing $\mathcal{W} \cup \mathcal{B}$ can, by the doubling hypothesis, be covered by at most CA^D balls of radius r . Assign every curve to the first such ball that contains it. If R has μ white and ν blue tangencies, two covering balls contain at least $\mu/(CA^D)$ of those white curves and at least $\nu/(CA^D)$ of those blue curves. Because the original pair is $t/A = 10r$ separated, the two selected balls are at distance at least $8r$; their curve sets are therefore $2r$ -separated. There are at most CA^{2D} ordered pairs of covering balls, so assigning each rectangle to its first maximizing pair loses only another factor CA^{2D} .

It remains to compare the rectangle scales. Divide the interval of a (δ, t) -rectangle into descendants at ratio r/t , and divide its vertical $O(\delta)$ -neighborhood into translates of thickness $\delta r/t$. There are at most CA^2 descendants. If a curve is tangent to the parent, its graph lies in one of the enlarged descendants; pigeonholing the tangencies chooses a descendant tangent to at least a CA^{-2} fraction of each selected colour. Lemma 14.6.18 shows that the enlargement changes the tangency constant by at most a power of A , and Lemma 14.6.22 shows that, after retaining one of at most $A^{C(K)}$ classes, the descendants remain 100-incomparable. These two claims follow directly from their proofs: horizontal dilation by s changes the vertical error by $O(s^2)$, while a failure of descendant incomparability makes the parents comparable after the same dilation.

Apply the estimate at scale r to every retained colour-ball pair and every descendant class. The number of classes and every loss in μ, ν is a fixed power of A ; summing gives the asserted factor $A^{C(K,D)}$. $\square$

Lemma 14.7.9 (Changing the incomparability constant). *If the estimate is known for pairwise $2C_c$ -incomparable rectangles, it is valid for pairwise 100-incomparable rectangles, with a constant depending only on K .*

Proof. Apply Lemma 14.6.24 with $C = 2C_c$. Its greedy proof partitions a 100-incomparable family into $O_K(1)$ subfamilies, each pairwise $2C_c$ -incomparable: a maximal such subfamily covers only $O_K(1)$ rectangles per selected member, by Lemma 14.6.23. Apply the assumed estimate to the largest subfamily and absorb this bounded factor. $\square$

Completion of Proposition 14.6.26. Lemma 14.6.27 reduces to $\delta \leq c_2 t$. Lemma 14.7.8 reduces the pair to the $2t$ -separated case, at the permitted cost A^C , and Lemma 14.7.9 supplies the required $2C_c$ -incomparability. The perturbation argument ending in (6.8) applies. Finally Lemma 14.7.7 inserts the tangency multiplicities μ, ν . This is exactly inequality 14.6.50, including its logarithmic factor. $\square$

Chapter 14.8

The Complete Radial-Projection Bootstrap

This chapter proves Proposition 14.2.71. The earlier radial projection chapter records the shape of the argument, but the proof below contains the missing quantifiers, the common tube discretisation, the single-scale selection, the discrete Frostman extraction, and both sides of the two-ends dichotomy. In particular, no “pigeonhole a good scale” or “the other case is similar” step is being suppressed.

There is one genuinely deeper input: the epsilon-improved Furstenberg incidence theorem stated in Theorem 14.8.1. Its proof is given in Chapter 14.10; here we prove every deduction from that theorem to the radial bootstrap.

Throughout the chapter, an r -tube is the closed r -neighbourhood of an affine line. If $E \subset X \times Y$, write

$$E_x = \{y : (x, y) \in E\}, \quad E^y = \{x : (x, y) \in E\}.$$

The definition of $(\sigma, K, 1 - c)$ -thin tubes is Definition 14.2.65.

14.8.1 The single incidence input

We use the following normalised version of the Orponen–Shmerkin improvement. For a bounded subset P of a metric space, write $\mathcal{E}_r(P)$ for its r -covering number. We call P an (r, t, D) -set if

$$\mathcal{E}_r(P \cap B(z, \rho)) \leq D\rho^t \mathcal{E}_r(P), \quad \rho \geq r. \quad (14.8.1)$$

The same terminology is used for tubes after identifying a tube with its axis in the usual two-dimensional metric on affine lines.

Theorem 14.8.1 (Epsilon-improved Furstenberg incidence theorem). *For $0 < a < 1$ there are numbers $e(a) > 0$ and $r_F(a) > 0$ with the following property. Let $0 < r \leq r_F(a)$. Suppose $P \subset B(0, 3)$ is an $(r, 1, r^{-e(a)})$ -set. For every $x \in P$, let $\mathcal{T}_x$ be an $(r, a, r^{-e(a)})$ -set of r -tubes containing x . Then*

$$\mathcal{E}_r \left(\bigcup_{x \in P} \mathcal{T}_x \right) \geq r^{-2a-e(a)}. \quad (14.8.2)$$

The functions $e(a)$ and $r_F(a)$ may be chosen uniformly when a ranges over a compact subinterval of $(0, 1)$.

The apparent restriction to $B(0, 3)$ is harmless below because both measures are supported in $[0, 1]^2$. The small changes between round and dyadic tubes, and between tubes meeting $B(0, 3)$ and their axes, are proved as part of the full incidence theorem in Chapter 14.10.

14.8.2 Three elementary discretisation lemmas

Lemma 14.8.2 (A common net of tubes). *There are absolute constants $A \geq 10$ and $a_0 > 0$ such that, for every $0 < r < 1/100$, there is a finite family $\mathcal{U}_r$ of Ar -tubes meeting $B(0, 3)$ with the following properties.*

1. *The axes are $a_0 r$ -separated in line space and $\#\mathcal{U}_r \leq Ar^{-2}$.*
2. *If T is an r -tube, then $T \cap B(0, 3)$ is contained in some $U \in \mathcal{U}_r$.*
3. *If $u, v \in B(0, 3)$ and $|u - v| \geq r$, then*

$$\#\{U \in \mathcal{U}_r : u, v \in U\} \leq A|u - v|^{-1}. \quad (14.8.3)$$

4. *If $x \in [0, 1]^2$ and $Z \subset [0, 1]^2$ has $\text{dist}(x, Z) \geq 1/2$, then every subfamily of tubes in $\mathcal{U}_r$ containing x has overlap at most $2A$ on Z .*
5. *Let $y \in [0, 1]^2$, $r \leq \xi \leq 1$, and let $\mathcal{V} \subset \mathcal{U}_r$ consist of tubes containing y whose directions are $A(r/\xi)$ -separated. Then $\mathcal{V}$ has overlap at most A on the annulus $\{z : \xi \leq |z - y| < 2\xi\}$.*

Proof. Parameterise an unoriented line meeting $B(0, 3)$ by (θ, b) , where $0 \leq \theta < \pi$ is its normal angle and $|b| \leq 3$. Choose an r -net in this rectangle and take a sufficiently large fixed multiple of the r -neighbourhood of each corresponding line. This gives the first two properties: changing either parameter by $O(r)$ moves the line by $O(r)$ inside $B(0, 3)$.

Let $d = |u - v| \geq r$. An Ar -tube containing both points has direction in an interval of length $O(r/d)$ about the direction of $v - u$. Once the direction

has been chosen, its offset belongs to an interval of length $O(r)$. The r -net therefore supplies $O((r/d)/r) = O(d^{-1})$ choices, proving (14.8.3). If $d \geq 1/2$, the same estimate is an absolute bound; applying it pointwise on Z proves the fourth assertion.

Finally, at distance comparable to ξ from y , two axes whose directions differ by at least a fixed multiple of r/ξ have disjoint Ar -thickenings, apart from a bounded number of immediate neighbours. This proves the last assertion. Enlarging A once accommodates the finitely many endpoint and unoriented-angle conventions. $\square$

The next lemma is the finite Frostman extraction that is often hidden behind the phrase “apply the discrete Frostman lemma.” We include its proof because the exponent of its loss is used in the incidence theorem.

Lemma 14.8.3 (Discrete Frostman extraction). *Let $0 < t \leq 2$, let λ be a finite Borel measure on $B(0, 3)$ satisfying*

$$\lambda(B(z, \rho)) \leq C\rho^t \quad (z \in \mathbb{R}^2, \rho > 0),$$

and suppose $\lambda(A) \geq m > 0$. If $0 < r < r_0(C, m, t)$, then A contains a finite (r, t, D) -set P with

$$D \leq C_t \frac{C}{m} \log(2/r). \quad (14.8.4)$$

In particular, if C is fixed, $m \geq r^\eta$, and r is sufficiently small in terms of C, t, η , then one may take $D \leq r^{-2\eta}$.

Proof. First, the mass-distribution estimate gives

$$\mathcal{H}_\infty^t(A) \geq C_t^{-1} m/C : \quad (14.8.5)$$

indeed, for every ball cover $A \subset \bigcup_j B_j$, countable subadditivity and the Frostman bound give $m \leq \sum_j \lambda(B_j) \leq C \sum_j (\text{diam } B_j)^t$.

We recall a finite form of Frostman’s lemma and prove the part needed here. Partition $B(0, 3)$ into dyadic r -squares and retain those meeting A . On the resulting finite dyadic tree, maximise $\sum_Q w_Q$ subject to $w_Q \geq 0$ and

$$\sum_{Q \subset R} w_Q \leq \ell(R)^t \quad (14.8.6)$$

for every dyadic ancestor R . Linear-programming duality says that the maximum equals the minimum cost of a fractional cover by dyadic ancestors. Because these ancestors form a laminar family, a bottom-up pruning replaces every fractional cover by an integral one without increasing its cost: at a vertex, either retain the total choices made below it, or replace them by the single vertex, whichever is cheaper. Thus the dual minimum is the dyadic t -content

of the retained squares. Balls and dyadic squares compare after using one of three shifted dyadic grids, so (14.8.5) implies that weights can be chosen with

$$W := \sum_Q w_Q \geq c_t m / C. \quad (14.8.7)$$

Discard weights below $W/(2N)$, where $N = O(r^{-2})$ is the number of retained r -squares. At least $W/2$ remains. Since (14.8.6) gives $w_Q \leq r^t$, the remaining weights belong to $O_t(\log(2/r) + \log(C/m))$ dyadic size classes. Select one class carrying maximal total weight and choose one point of A from each square in that class; call the resulting set P . All selected weights are comparable. Consequently (14.8.6), first for dyadic squares and then for balls, gives

$$\frac{\mathcal{E}_r(P \cap B(z, \rho))}{\mathcal{E}_r(P)} \leq C_t \frac{C}{m} (\log(2/r) + \log(C/m)) \rho^t.$$

For the range of r in the statement, the second logarithm is absorbed by the first, proving (14.8.4). If $m \geq r^\eta$, then $C_t C m^{-1} \log(2/r) \leq r^{-2\eta}$ once r is small enough. $\square$

Lemma 14.8.4 (Two separated pieces). *Let U be an r -tube, $R \geq 10r$, and let ω be a finite measure on U of mass m . If no ball of radius $4R$ has ω -mass at least $m/3$, then there are Borel sets $Y_1, Y_2 \subset U$ with*

$$\text{dist}(Y_1, Y_2) \geq R, \quad \omega(Y_1), \omega(Y_2) \geq m/3.$$

Proof. Project orthogonally onto the axis of U and choose lower and upper one-third quantiles $q_1 \leq q_2$ of the pushforward measure. If $q_2 - q_1 < 2R$, the part whose projected coordinate lies between the two quantiles has mass at least $m/3$ and lies in a ball of radius $4R$, contrary to the hypothesis. Hence $q_2 - q_1 \geq 2R$. The inverse images of $(-\infty, q_1]$ and $[q_2, \infty)$ each have mass at least $m/3$, and their distance is at least $2R - 2r \geq R$. $\square$

14.8.3 Uniform preparation of the parameters

Fix $\beta, \varepsilon > 0$ as in Proposition 14.2.71. The compact interval

$$I = [\beta, 1 - \varepsilon]$$

lies in $(0, 1)$. By the uniform part of Theorem 14.8.1, there are $e_* > 0$ and $r_F > 0$ which work for every $a \in I$.

Choose $0 < \eta < 1$ so small that

$$\eta < \frac{\varepsilon}{100}, \quad 8\eta < \sqrt{\eta} < e_*, \quad 6\eta + \frac{14\eta}{\varepsilon} < \sqrt{\eta}. \quad (14.8.8)$$

We shall prove the proposition with $\tau = \eta$. For $\sigma \in I$, put

$$\kappa = \frac{14\eta}{1 - \sigma}. \quad (14.8.9)$$

Then $0 < \kappa < 1/2$, after decreasing η once more if necessary, and

$$6\eta + \kappa < \sqrt{\eta}. \quad (14.8.10)$$

We also record how the compact formula for K' absorbs every small-scale requirement. There is an integer $M = M(\beta, \varepsilon)$, enlarged whenever necessary below, such that if

$$R_0 = \max \left\{ K, \frac{C^2 M}{c} \right\}, \quad K' = R_0^M, \quad (14.8.11)$$

and $L = K'/A_1$ for a sufficiently large absolute constant A_1 , then

$$r_* := L^{-1/(\sigma+\eta)} \leq r_F, \quad K + C + A_1 \leq r_*^{-\eta}, \quad 4 \sum_{\substack{r \leq r_* \\ r \text{ dyadic}}} r^\eta < c, \quad (14.8.12)$$

and all fixed constants appearing below are at most $r_*^{-\eta}$.

For completeness, this is only elementary exponent bookkeeping. Since $\sigma + \eta < 2$,

$$r_*^\eta = L^{-\eta/(\sigma+\eta)} \leq A_1 R_0^{-M\eta/2}.$$

The number R_0 is at least M/c , at least K , and at least a fixed positive power of C . Choosing M larger than a constant multiple of η^{-2} makes the right side smaller than every fixed power of $c/K/C/M$ needed in (14.8.12). The geometric sum is at most $C_\eta r_*^\eta$. This proves all three assertions uniformly for $\sigma \in I$. Notice that M depends only on β, ε , as required.

14.8.4 Failure of improvement produces one rich scale

Proof of Proposition 14.2.71. Write $X = \text{spt } \nu_1$, $Y = \text{spt } \nu_2$, and set $\mu = \nu_1$, $\nu = \nu_2$. Let E_{12} witness the $(\sigma, K, 1 - c)$ -thin-tube property of (μ, ν) , and let E_{21} witness the corresponding property of (ν, μ) . The symmetric good relation

$$G = E_{12} \cap \{(x, y) : (y, x) \in E_{21}\} \quad (14.8.13)$$

has $(\mu \times \nu)(G) \geq 1 - 2c$ and satisfies, for every tube T through the distinguished point,

$$\nu(T \cap G_x) \leq K\rho^\sigma \quad (x \in X, T \text{ a } \rho\text{-tube through } x), \quad (14.8.14)$$

$$\mu(T \cap G^y) \leq K\rho^\sigma \quad (y \in Y, T \text{ a } \rho\text{-tube through } y). \quad (14.8.15)$$

We prove that (μ, ν) has $(\sigma + \eta, K', 1 - 3c)$ -thin tubes. Suppose otherwise. For every dyadic r , use the family $\mathcal{U}_r$ from Lemma 14.8.2. For $x \in X$, let

$$\mathcal{U}'_{x,r} = \{U \in \mathcal{U}_r : x \in U, \nu(U \cap G_x) \geq Lr^{\sigma+\eta}\},$$

and define the Borel relation

$$H_r = \{(x, y) \in G : y \in U \text{ for some } U \in \mathcal{U}'_{x,r}\}, \quad H = \bigcup_{r \text{ dyadic}} H_r. \quad (14.8.16)$$

The functions $x \mapsto \nu(U \cap G_x)$ are Borel, so no measurability is hidden here.

If $(\mu \times \nu)(G \setminus H) \geq 1 - 3c$, then $G \setminus H$ witnesses $(\sigma + \eta, K', 1 - 3c)$ -thin tubes. Indeed, enlarge an arbitrary ρ -tube through x to a member of $\mathcal{U}_r$ at the first dyadic scale $r \geq \rho$, changing widths and masses by only the absolute factor absorbed in A_1 . Since no point of $G \setminus H$ lies in a heavy member, its fibre mass in the original tube is at most $Lr^{\sigma+\eta} \leq K'\rho^{\sigma+\eta}$. Our contrary assumption therefore implies

$$(\mu \times \nu)(H) > c. \quad (14.8.17)$$

This direct definition of H_r inside G is the bookkeeping that yields $1 - 3c$ rather than losing two further copies of c .

The tubes through a fixed x have bounded overlap on Y , by the separation of the supports and Lemma 14.8.2. Hence

$$\#\mathcal{U}'_{x,r} \leq AL^{-1}r^{-\sigma-\eta}. \quad (14.8.18)$$

Moreover, a heavy tube can exist only when $Lr^{\sigma+\eta} \leq 1$, hence only when $r \leq r_*$. It follows from (14.8.17) and the last inequality in (14.8.12) that for some dyadic $r \leq r_*$,

$$(\mu \times \nu)(H_r) \geq 2r^\eta. \quad (14.8.19)$$

Fix this r for the rest of the proof. In particular,

$$K + C + A_1 \leq r^{-\eta}. \quad (14.8.20)$$

Set

$$\mathbf{X} = \{x \in X : \nu((H_r)_x) \geq r^\eta\}. \quad (14.8.21)$$

Since every fibre has mass at most one, (14.8.19) gives $\mu(\mathbf{X}) \geq r^\eta$. For $x \in \mathbf{X}$, retain the tubes

$$\mathcal{T}_x = \{U \in \mathcal{U}'_{x,r} : \nu(U \cap (H_r)_x) \geq r^{\sigma+2\eta}\}, \quad Y_x = (H_r)_x \cap \bigcup_{U \in \mathcal{T}_x} U. \quad (14.8.22)$$

By (14.8.18), the discarded tubes carry total mass at most $AL^{-1}r^\eta$. Our choice of A_1 and M therefore gives

$$\nu(Y_x) \geq \frac{1}{2}r^\eta. \quad (14.8.23)$$

The forward thin-tube bound and (14.8.20) imply

$$r^{\sigma+2\eta} \leq \nu(U \cap Y_x) \leq AKr^\sigma \leq r^{\sigma-2\eta}, \quad U \in \mathcal{T}_x, \quad (14.8.24)$$

where, as throughout, M was chosen large enough to absorb the fixed geometric constants by an extra power of r^η . Consequently

$$r^{-\sigma+3\eta} \lesssim \#\mathcal{T}_x \leq r^{-\sigma-3\eta}. \quad (14.8.25)$$

Here the lower bound follows from covering Y_x and the upper mass in (14.8.24); the upper bound follows from bounded overlap and the lower mass in that display. We have weakened the exponents by one η to absorb absolute constants.

If $\mathcal{S} \subset \mathcal{T}_x$ has its axes in a line-space ball of radius $\rho \geq r$, then the members of $\mathcal{S}$, inside $[0, 1]^2$, lie in an $A\rho$ -tube through x . Bounded overlap, the lower bound in (14.8.24), and (14.8.14) give

$$\#\mathcal{S} r^{\sigma+2\eta} \leq A\nu(N_{A\rho}(\ell) \cap G_x) \leq AK(A\rho)^\sigma \leq r^{-2\eta} \rho^\sigma.$$

Together with the lower bound in (14.8.25), this shows that

$$\mathcal{T}_x \text{ is an } (r, \sigma, r^{-8\eta})\text{-set of tubes.} \quad (14.8.26)$$

The two-ends dichotomy

Use κ from (14.8.9). Call $U \in \mathcal{T}_x$ *concentrated* if some ball B_U of radius $4r^\kappa$ satisfies

$$\nu(U \cap B_U \cap Y_x) \geq \frac{1}{3}\nu(U \cap Y_x). \quad (14.8.27)$$

For each x , either at least half the tubes are non-concentrated or at least half are concentrated. One of these alternatives holds on a subset $\mathbf{X}' \subset \mathbf{X}$ with

$$\mu(\mathbf{X}') \geq \frac{1}{2}\mu(\mathbf{X}) \geq \frac{1}{2}r^\eta. \quad (14.8.28)$$

Case 1: non-concentrated tubes

Assume first that at least half of $\mathcal{T}_x$ is non-concentrated for every $x \in \mathbf{X}'$. Lemma 14.8.3, applied to $\mu|_{\mathbf{X}'}$ with $t = 1$, gives an $(r, 1, r^{-2\eta})$ -set $P \subset \mathbf{X}'$; the required smallness of r is included in (14.8.12). Let $\mathcal{T}'_x$ be the non-concentrated half and set $\mathcal{T}' = \bigcup_{x \in P} \mathcal{T}'_x$. Passing to a half-sized subfamily worsens the constant in (14.8.26) by at most two. Since $8\eta < \sqrt{\eta} < e_*$, the hypotheses of Theorem 14.8.1 hold, and its conclusion gives

$$\#\mathcal{T}' \gtrsim \mathcal{E}_r(\mathcal{T}') \geq r^{-2\sigma-\sqrt{\eta}}. \quad (14.8.29)$$

The first comparison uses the $a_0 r$ -separation of the common tube net.

For $U \in \mathcal{T}'$, Lemma 14.8.4 and (14.8.24) provide sets $Y_{U,1}, Y_{U,2} \subset U \cap Y_x$ separated by r^κ and satisfying

$$\nu(Y_{U,j}) \geq \frac{1}{3}r^{\sigma+2\eta} \geq r^{\sigma+3\eta}, \quad j = 1, 2. \quad (14.8.30)$$

The last inequality follows by making r_* smaller by a fixed amount. Count ordered pairs of points in the two pieces. By (14.8.3),

$$\begin{aligned} r^{2\sigma+6\eta} \#\mathcal{T}' &\leq \sum_{U \in \mathcal{T}'} (\nu \times \nu)(Y_{U,1} \times Y_{U,2}) \\ &\leq \iint \#\{U \in \mathcal{U}_r : y_1, y_2 \in U\} d\nu(y_1) d\nu(y_2) \\ &\leq Ar^{-\kappa}. \end{aligned} \quad (14.8.31)$$

Thus

$$\#\mathcal{T}' \leq Ar^{-2\sigma-6\eta-\kappa},$$

which contradicts (14.8.29), (14.8.10), and sufficiently small r .

Case 2: concentrated tubes

We are left with the case in which at least half of $\mathcal{T}_x$ is concentrated for every $x \in \mathbf{X}'$. Denote this half by $\mathcal{T}_x^c$ and choose a ball B_U as in (14.8.27) for each member. Define

$$H' = \{(x, y) : x \in \mathbf{X}', y \in U \cap B_U \cap Y_x \text{ for some } U \in \mathcal{T}_x^c\}. \quad (14.8.32)$$

Since the tubes through x have bounded overlap on Y , the lower bounds (14.8.25), (14.8.24), and (14.8.28) yield

$$(\mu \times \nu)(H') \gtrsim r^\eta r^{-\sigma+3\eta} r^{\sigma+3\eta} = r^{7\eta}. \quad (14.8.33)$$

Notice that $H' \subset H_r \subset G$.

For every $(x, y) \in H'$, choose one witnessing $U = U(x, y)$. Because $y \in B_U$ and B_U has radius $4r^\kappa$,

$$\nu(U \cap B(y, 8r^\kappa) \cap Y_x) \geq r^{\sigma+3\eta}. \quad (14.8.34)$$

On the other hand, the Frostman estimate and $4\eta < 1 - \sigma$ give

$$\nu(B(y, Ar)) \leq ACr \leq \frac{1}{2}r^{\sigma+3\eta} \quad (14.8.35)$$

once r_* is small enough. Decompose $B(y, 8r^\kappa) \setminus B(y, Ar)$ into $O(\log(2/r))$ dyadic annuli. From (14.8.34)–(14.8.35), for each $(x, y) \in H'$ there is a dyadic $\xi(x, y) \in [Ar, 16r^\kappa]$ such that

$$\nu(U(x, y) \cap \{z : \xi(x, y) \leq |z - y| < 2\xi(x, y)\} \cap Y_x) \geq r^{\sigma+4\eta}. \quad (14.8.36)$$

Indeed, $1/\log(2/r) \geq r^\eta$ at sufficiently small scales.

There are only $O(\log(2/r))$ possible values of ξ . A second pigeonhole and (14.8.33) therefore give a common ξ and a Borel subrelation $H'' \subset H'$ with

$$(\mu \times \nu)(H'') \geq r^{8\eta}. \quad (14.8.37)$$

Choose $y \in Y$ with

$$\mu((H'')^y) \geq r^{8\eta}. \quad (14.8.38)$$

Let $\mathcal{T}_y$ be the set of witnessing tubes $U(x, y)$ as x ranges over $(H'')^y$. Each member contains y and satisfies the annular lower bound (14.8.36).

Partition the directions of $\mathcal{T}_y$ into intervals of length Ar/ξ . All points x whose witnessing tube belongs to one interval lie in an $A(r/\xi)$ -tube through y : here we use $|x - y| \geq 1/2$. Since $H'' \subset G$, the reverse thin-tube estimate (14.8.15) and (14.8.20) show that one direction interval accounts for at most

$$AK(r/\xi)^\sigma \leq r^{-\eta}(r/\xi)^\sigma \quad (14.8.39)$$

of the μ -mass of $(H'')^y$. Hence a greedy selection gives a subfamily $\mathcal{T}'_y \subset \mathcal{T}_y$ with $A(r/\xi)$ -separated directions and

$$\#\mathcal{T}'_y \gtrsim r^{9\eta}(\xi/r)^\sigma. \quad (14.8.40)$$

By the last part of Lemma 14.8.2, these annular tube pieces have bounded overlap. Summing (14.8.36) over $\mathcal{T}'_y$, then using the Frostman bound for ν , gives

$$r^{\sigma+4\eta}\#\mathcal{T}'_y \lesssim \nu(B(y, 2\xi)) \leq 2C\xi. \quad (14.8.41)$$

Combining (14.8.40) and (14.8.41) gives

$$r^{13\eta} \lesssim C\xi^{1-\sigma} \leq C16^{1-\sigma}r^{14\eta}, \quad (14.8.42)$$

where the last equality in the exponent is $\kappa(1 - \sigma) = 14\eta$. Since $r^{13\eta}/r^{14\eta} = r^{-\eta}$, this is impossible once r_* is small enough in terms of C and η . This contradiction closes the concentrated case.

Both alternatives have contradicted the assumption that (μ, ν) fails the improved thin-tube property. Therefore (μ, ν) has $(\sigma + \eta, K', 1 - 3c)$ -thin tubes. The proof is symmetric in μ, ν ; interchanging them gives the same conclusion for (ν, μ) . With $\tau = \eta$, K' as in (14.8.11), and $M = M(\beta, \varepsilon)$, this is exactly Proposition 14.2.71. $\square$

14.8.5 What has and has not been used

The proof above uses only the following ingredients: the two original σ -thin-tube relations, the one-dimensional Frostman bounds, separated supports in a fixed bounded region, elementary tube geometry, finite Frostman extraction, and Theorem 14.8.1. In particular, no radial projection theorem is being invoked to prove the bootstrap: the logical direction is the reverse. The next proof-dossier chapter supplies the full proof of the Furstenberg incidence input, including its Bourgain projection ingredient.

Chapter 14.9

A Complete Proof of Bourgain’s Discretised Projection Theorem

The epsilon-improved Furstenberg estimate ultimately needs a projection theorem with two features that the classical almost-everywhere theorem does not provide: a polynomial gain at a specified small scale, and a conclusion which remains valid for every sufficiently dense subset of the original set. This chapter proves exactly that version.

The proof follows the streamlined real-variable route of William O’Regan, Pablo Shmerkin, and Hong Wang, but is rewritten in the notation of this book. We include the positive-measure expansion argument and all analytic reductions. Three finite additive-combinatorial facts are used in their already-proved forms: Ruzsa’s triangle inequality (Volume 10, Theorem 10.3.4), the Plünnecke–Ruzsa inequality (Volume 10, Theorem 10.3.5), and the Balog–Szemerédi–Gowers theorem (Volume 10, Theorem 10.4.2). Their covering-number versions follow by replacing a bounded set by the δ -grid cells which meet it; a bounded change of scale only changes absolute constants.

14.9.1 Conventions and the theorem

If $A \subset \mathbb{R}^d$ is a union of closed δ -balls, then

$$|A| \asymp_d \delta^d \mathcal{E}_\delta(A).$$

All sumsets below are harmlessly thickened by $O(\delta)$ before their Lebesgue measure is taken. Thus an estimate in Lebesgue measure is equivalent, up to absolute factors, to the corresponding estimate for δ -covering numbers.

We shall also use the following elementary scaling comparison. If $A \subset \mathbb{R}$ is bounded and $0 < t \leq 1$, then

$$\mathcal{E}_\delta(A) \leq \mathcal{E}_\delta(t^{-1}A) \leq 3t^{-1}\mathcal{E}_\delta(A). \quad (14.9.1)$$

Indeed, $\mathcal{E}_\delta(t^{-1}A) = \mathcal{E}_{t\delta}(A)$. The first inequality follows from monotonicity in the covering scale, while every interval of length 2δ is covered by at most $3t^{-1}$ intervals of length $2t\delta$. This proves the second inequality. The same comparison, with a different absolute constant, holds for the thickened covering numbers used below.

Definition 14.9.1 (Discretised and continuous Frostman objects). Let $0 < q \leq d$.

1. A union A of δ -balls in $\mathbb{R}^d$ is a (δ, q, D) -set if

$$|A \cap B(z, r)| \leq Dr^q |A|, \quad z \in \mathbb{R}^d, \quad r \geq \delta.$$

2. A Borel probability measure λ is a (q, D) -measure if

$$\lambda(B(z, r)) \leq Dr^q, \quad z \in \mathbb{R}^d, \quad r > 0.$$

For a unit vector θ , $\pi_\theta(z) = z \cdot \theta$. We prove the following form of Bourgain's theorem.

Theorem 14.9.2 (Robust discretised projection theorem). *Let $0 < \alpha < 2$ and let $\beta, \kappa > 0$. There are*

$$\eta = \eta(\alpha, \beta, \kappa) > 0, \quad \varepsilon = \varepsilon(\alpha, \beta, \kappa) > 0, \quad \delta_0 > 0$$

such that the following holds for $0 < \delta < \delta_0$. Let λ be a $(\kappa, \delta^{-\varepsilon})$ probability measure on S^1 , and let $E \subset B(0, \delta^{-\varepsilon})$ be a $(\delta, \beta, \delta^{-\varepsilon})$ -set satisfying

$$|E| \leq \delta^{2-\alpha}.$$

Then there is $\Theta \subset \text{spt } \lambda$ with $\lambda(\Theta) > 1 - \delta^\varepsilon$ such that, for every $\theta \in \Theta$ and every union of δ -balls $G \subset E$ with $|G| > \delta^\varepsilon |E|$,

$$\mathcal{E}_\delta(\pi_\theta G) > \delta^{-\eta} \mathcal{E}_\delta(E)^{1/2}. \quad (14.9.2)$$

The theorem remains true if κ or β is replaced by a smaller positive number. We shall use this monotonicity repeatedly to keep all energy exponents strictly below their Frostman exponents.

14.9.2 Energy and projection lemmas

For a probability measure μ on $\mathbb{R}^d$, let

$$I_q(\mu) = \iint |x - y|^{-q} d\mu(x) d\mu(y).$$

Lemma 14.9.3 (Energy of a discretised Frostman set). *If A is a (δ, q, D) -set and $\mu_A = |A|^{-1} \mathbf{1}_A dx$, then*

$$I_u(\mu_A) \leq C_{d,q,u} D, \quad 0 < u < q.$$

Proof. For fixed x , split $\mathbb{R}^d$ into the annuli $2^{-j-1} < |x - y| \leq 2^{-j}$ down to scale δ . The contribution of the j th annulus is at most

$$2^{(j+1)u} \mu_A(B(x, 2^{-j})) \leq 2^u D 2^{-j(q-u)}.$$

The resulting geometric series converges. Below scale δ , the bounded density of μ_A gives the same or a smaller contribution. $\square$

Lemma 14.9.4 (Removing high-density cubes). *Let $I_q(\mu) \leq K$, let $L \geq 1$, and let $\delta = 2^{-m}$. There is a Borel set $A \subset \text{spt } \mu$ with $\mu(A) \geq 1 - C_q/L$ such that the conditioned measure μ_A satisfies*

$$\mu_A(B(z, r)) \leq C_{d,q} K L r^{q/2}, \quad r \geq \delta.$$

Consequently the union of the δ -cubes meeting A is a $(\delta, q/2, C_{d,q} K L)$ -set.

Proof. For each dyadic scale $2^{-j} \geq \delta$, let $\mathcal{Q}_j$ be the dyadic cubes of that scale. Since two points in a common cube are $O(2^{-j})$ apart,

$$2^{jq} \sum_{Q \in \mathcal{Q}_j} \mu(Q)^2 \lesssim_d I_q(\mu) \leq K.$$

Let $\mathcal{H}_j$ consist of cubes satisfying $\mu(Q) > K L 2^{-jq/2}$. Then

$$\mu\left(\bigcup \mathcal{H}_j\right) \leq \frac{2^{jq/2}}{K L} \sum_Q \mu(Q)^2 \lesssim_d L^{-1} 2^{-jq/2}.$$

The sum over j is $O_q(L^{-1})$. Delete the union of all these heavy cubes. Every surviving dyadic cube of side $r \geq \delta$ has mass at most $K L r^{q/2}$, and every ball is covered by $O_d(1)$ such cubes. Conditioning on the surviving set changes the constant by at most a factor two once L is a sufficiently large constant depending on q . $\square$

Lemma 14.9.5 (Averaged projection energies). *Let $0 < q < a \leq 1$. If ν is an (a, D) -measure on S^1 , then for every probability measure μ on $\mathbb{R}^2$,*

$$\int_{S^1} I_q(\pi_\theta \mu) d\nu(\theta) \leq C_{a,q} D I_q(\mu).$$

Proof. Fix $v \neq 0$. The set of θ for which $|\theta \cdot v| \leq r|v|$ is contained in two arcs of length $O(r)$. The Frostman bound and dyadic layer-cake decomposition therefore give

$$\int |\theta \cdot v|^{-q} d\nu(\theta) \lesssim_{a,q} D |v|^{-q} \sum_{j \geq 0} 2^{jq} \nu\{|\theta \cdot v| \leq 2^{-j}|v|\} \lesssim_{a,q} D |v|^{-q}.$$

Integrating this inequality with $v = x - y$ against $\mu \times \mu$ and applying Tonelli proves the claim. $\square$

We also need a higher-dimensional, one-dimensional projection fact.

Lemma 14.9.6 (A positive-measure linear image). *Let μ be a compactly supported probability measure on $\mathbb{R}^n$ with $I_1(\mu) < \infty$. If $\Omega \subset S^{n-1}$ has positive surface measure, then some $v \in \Omega$ satisfies*

$$|\pi_v(\text{spt } \mu)| \geq c_{n,\Omega} I_1(\mu)^{-1}.$$

Proof. For almost every v , $\pi_v \mu$ has a density f_v . Plancherel, polar coordinates, and the Fourier representation of the Riesz energy give

$$\int_{S^{n-1}} \|f_v\|_2^2 d\sigma(v) = c_n \int_{\mathbb{R}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{1-n} d\xi = c'_n I_1(\mu).$$

The same upper bound holds after restricting the integral to Ω , so some $v \in \Omega$ has $\|f_v\|_2^2 \leq C_{n,\Omega} I_1(\mu)$. Since $\int f_v = 1$, Cauchy-Schwarz gives $1 \leq |\text{spt } f_v| \|f_v\|_2^2$, which is the stated lower bound. $\square$

14.9.3 Expansion to positive measure

Lemma 14.9.7 (Blichfeldt's principle). *If $V \subset \mathbb{R}^n$ is measurable and L is invertible, some translate of $L\mathbb{Z}^n$ contains at least $|\det L|^{-1}|V|$ points of V .*

Proof. For $y \in [0, 1]^n$, set $N(y) = \#(L^{-1}V \cap (\mathbb{Z}^n + y))$. Tonelli and the tiling of $\mathbb{R}^n$ by $[0, 1]^n + k$ give

$$\int_{[0,1]^n} N(y) dy = \sum_{k \in \mathbb{Z}^n} \int_{[0,1]^n} \mathbf{1}_{L^{-1}V}(k + y) dy = |L^{-1}V|.$$

Some y is at least the average; translating back by L proves the lemma. $\square$

Lemma 14.9.8 (Eliminating one coefficient). *Let $A \subset \mathbb{R}$ be compact with diameter $D > 0$, let $v = (v_1, \dots, v_n) \in [1/2, 1]^n$, $n \geq 2$, and suppose*

$$\lambda := |v_1 A + \dots + v_n A| > 0.$$

There is an integer $N = N(n, D, \lambda)$ and a reordering of the coordinates such that, with $A_1 = NA^{(2)} - NA^{(2)}$,

$$|v_1 A_1 + \dots + v_{n-1} A_1| \geq D\lambda.$$

The integer is bounded when $n, D, D^{-1}, \lambda^{-1}$ are bounded.

Proof. Let $\pi(z) = v \cdot z$. Choose an orthogonal map U with $U(\mathbb{R}^{n-1} \times \{0\}) = \ker \pi$, and for $R \geq 1$ set

$$V_R = U(B^{n-1}(0, R) \times [-1, 1]).$$

Then $|V_R| \asymp_n R^{n-1}$ and $\pi(V_R)$ lies in a fixed interval of length $O_n(1)$. Apply Lemma 14.9.7 to the lattice $2D\mathbb{Z}^n$. Once R is large enough in terms of n, D, λ , a translate of this lattice supplies points $q_1, \dots, q_M \in V_R$ with

$$M\lambda > 2(nD + O_n(1)).$$

The sets $A^n + q_j$ are pairwise disjoint: a non-zero coordinate difference between two lattice points is at least $2D$, whereas each coordinate range of A^n has length D . Their π -images all have measure λ and all lie in a common interval of length $nD + O_n(1)$. Hence two of those images overlap.

Thus there are $x, y \in A^n$ and a non-zero $\ell \in \mathbb{Z}^n$ with

$$v \cdot (x - y - 2D\ell) = 0, \quad 2D\ell \in V_R - V_R.$$

Reorder coordinates so that $\ell_n \neq 0$, and put $d_i = x_i - y_i - 2D\ell_i$. Then $|d_n| \geq D$ and

$$v_n d_n = - \sum_{i < n} v_i d_i.$$

Multiplying the set $v_1 A + \dots + v_n A$ by d_n and substituting this identity yields

$$d_n(v_1 A + \dots + v_n A) \subset \sum_{i < n} v_i (d_n A - d_i A).$$

Because $D = \max A - \min A \in A - A$, and $|\ell_i| \leq C_n R/D$, every set $d_n A - d_i A$ is contained in $NA^{(2)} - NA^{(2)}$ for an integer $N = O_n(1 + R/D)$. Taking measures in the last inclusion gives the result. $\square$

Theorem 14.9.9 (Algebraic expansion of a Frostman support). *Let $0 < q < 1$ and $D \geq 1$. There is an integer $N = N(q, D)$ such that every (q, D) probability measure μ supported on $[-D, D]$, with $K = \text{spt } \mu$, satisfies*

$$|NK^{(N)} - NK^{(N)}| \geq c(q, D) > 0.$$

Proof. Let n be the least integer with $nq > 1$. The product measure μ^n obeys

$$\mu^n(B(z, r)) \leq D^n(2r)^{nq},$$

so $I_1(\mu^n) \lesssim_{q,D} 1$. Apply Lemma 14.9.6 to the positive-measure patch of directions whose coordinates, after normalisation, lie in $[1/2, 1]$. We obtain $v \in [1/2, 1]^n$ such that

$$|v_1 K + \cdots + v_n K| \gtrsim_{q,D} 1.$$

Also $\text{diam } K \gtrsim_{q,D} 1$, since $1 = \mu(K) \leq D(\text{diam } K)^q$. Lemma 14.9.8 removes one coefficient while replacing K by a bounded signed sum of products of two copies of K . The output again has diameter bounded above and below and has a positive-measure linear sum, with constants depending only on q, D . Repeat $n - 1$ times. Expanding all brackets, the final one-dimensional set is contained in $NK^{(N)} - NK^{(N)}$ for a single $N = N(q, D)$, and it has measure bounded below by a positive constant. $\square$

14.9.4 The sum–product hybrid

We now convert positive-measure algebraic expansion into growth of $A + xA$. The argument is quantitative only through fixed powers of δ ; exact numerical values of the powers are immaterial, but every choice is made in the proof.

Proposition 14.9.10 (Hybrid expansion for a strong direction measure). *Let $0 < q \leq \sigma < 1$. There are $c, \varepsilon_0, \delta_0 > 0$, depending only on q, σ , such that the following holds. Let $A \subset [-2, 2]$ be a $(\delta, q, \delta^{-\varepsilon_0})$ -set with $|A| \leq \delta^{1-\sigma}$, and let μ be a $(q, 2)$ probability measure on $[0, 1]$. If $\delta^c < t \leq 1$, then some $x \in \text{spt } \mu$ satisfies*

$$|t^{-1}A + xA| \geq \delta^{-c}|t^{-1}A|. \quad (14.9.3)$$

Proof. All implicit constants in this proof depend only on q, σ . The Frostman condition lets us discard $[0, a_q]$ for a fixed $a_q > 0$, retaining at least half the mass. After conditioning, $\text{spt } \mu \subset [a_q, 1]$ and the Frostman constant remains bounded.

We may also arrange that $1 \in \text{spt } \mu$. Indeed, if $b = \max \text{spt } \mu \asymp_q 1$, push μ forward by $x \mapsto x/b$. Apply the proposition proved below with bt in place of t and then multiply the resulting sumset by b ; this returns $t^{-1}A + xA$ with x in the original support. Thus write $K = \text{spt } \mu \subset [a_q, 1]$ with $1 \in K$.

By Theorem 14.9.9, for an integer $N = O_q(1)$,

$$X = NK^{(N)} - NK^{(N)} \quad \text{satisfies} \quad |X| \gtrsim_q 1.$$

Let m be the least integer with $mq > 1$. The uniform probability measure on A^m is an $(mq, \delta^{-m\varepsilon_0})$ measure above scale δ , and Lemma 14.9.3 gives

$I_1(\mu_{A^m}) \lesssim \delta^{-m\varepsilon_0}$. Since X^m has positive Lebesgue measure in a fixed ball, its radial image contains a positive-measure set of directions. Apply Lemma 14.9.6 on that set of directions. We find $x_1, \dots, x_m \in X$ such that

$$|x_1 A + \dots + x_m A| \gtrsim \delta^{m\varepsilon_0}. \quad (14.9.4)$$

Expand each x_i as a signed sum of N products from K . With $k = mN$, (14.9.4) supplies $y_1, \dots, y_{2k} \in K^{(N)}$ for which

$$|y_1 A + \dots + y_k A - y_{k+1} A - \dots - y_{2k} A| \geq \delta^{-q_0} |t^{-1} A| \quad (14.9.5)$$

for some $q_0 = q_0(\sigma) > 0$, provided first $c < (1 - \sigma)/10$ and then $m\varepsilon_0 < (1 - \sigma)/10$. Indeed, the right side is at most $\delta^{-q_0 - c} |A|$, while the left side of (14.9.4) is at least $\delta^{m\varepsilon_0}$.

We spell out the reduction in additive complexity. Put $Z = t^{-1} A$. If all the sets $Z \pm y_i A$ had covering number at most $\delta^{-q_0/(4k)} \mathcal{E}_\delta(Z)$, the Plünnecke–Ruzsa inequality, applied with base set Z , would bound the signed sum in (14.9.5) by $\delta^{-q_0/2} \mathcal{E}_\delta(Z)$, a contradiction for small δ . Thus, for some $y \in K^{(N)}$ and one sign,

$$\mathcal{E}_\delta(Z \pm y A) \geq \delta^{-q_0/(4k)} \mathcal{E}_\delta(Z).$$

The covering-number form of $|X - Y| \lesssim |X + Y|^3 / |Y|^2$, obtained from Ruzsa and Plünnecke–Ruzsa, changes the plus sign to a minus sign at the cost of dividing the exponent by three. Hence

$$\mathcal{E}_\delta(Z - y A) \geq \delta^{-q_1} \mathcal{E}_\delta(Z), \quad q_1 = \frac{q_0}{12k}. \quad (14.9.6)$$

Write $y = w_1 \cdots w_N$, $w_j \in K$, and $z_j = w_1 \cdots w_j$, with $z_0 = 1$. Ruzsa’s triangle inequality gives

$$\mathcal{E}_\delta(Z - z_j A) \lesssim \frac{\mathcal{E}_\delta(Z - z_{j-1} A) \mathcal{E}_\delta(A - w_j A)}{\mathcal{E}_\delta(A)}$$

and, again,

$$\mathcal{E}_\delta(A - w_j A) \lesssim \frac{\mathcal{E}_\delta(Z - A) \mathcal{E}_\delta(Z - w_j A)}{\mathcal{E}_\delta(Z)}.$$

Let

$$L = \max_{w \in \{1, w_1, \dots, w_N\}} \frac{\mathcal{E}_\delta(Z - w A)}{\mathcal{E}_\delta(Z)}.$$

Multiplying the preceding inequalities for $j = 1, \dots, N$ and using the lower bound $\mathcal{E}_\delta(A) \geq (t/3) \mathcal{E}_\delta(Z)$ from (14.9.1) yields

$$\mathcal{E}_\delta(Z - y A) \lesssim t^{-N} L^{2N+1} \mathcal{E}_\delta(Z).$$

Together with (14.9.6) and $t \geq \delta^c$, this gives

$$L \gtrsim \delta^{(cN - q_1)/(2N+1)}.$$

Choose $c < q_1/(2N)$. Then $L \geq \delta^{-c_1}$ for some $c_1 = c_1(q, \sigma) > 0$. Since $1, w_1, \dots, w_N$ all belong to K , one of them is the required x . Replacing a difference by a sum with the sum–difference inequality loses only a factor three in the exponent. Renaming the final positive exponent as c proves (14.9.3). $\square$

Theorem 14.9.11 (Sum–product hybrid). *Let $0 < q \leq \sigma < 1$. There are $c, \varepsilon, \delta_0 > 0$, depending only on q, σ , such that, for $0 < \delta < \delta_0$, the following holds. If $A \subset [-2, 2]$ is a $(\delta, q, \delta^{-\varepsilon})$ -set with $|A| \leq \delta^{1-\sigma}$ and μ is a $(q, \delta^{-\varepsilon})$ probability measure on $[0, 1]$, then some $x \in \text{spt } \mu$ satisfies*

$$|A + xA| \geq \delta^{-c}|A|.$$

Proof. For a closed interval I of length $r \in [\delta, 1]$, set

$$\Phi(I) = \frac{\mu(I)}{r^{q/2}}.$$

Choose a maximising interval $I_0 = [x_0, x_0 + r_0]$, choosing one of minimal length among maximisers. Since $\Phi(I_0) \geq \Phi([0, 1]) = 1$ and μ is $(q, \delta^{-\varepsilon})$,

$$r_0^{q/2} \leq \mu(I_0) \leq \delta^{-\varepsilon} r_0^q, \quad \text{so} \quad r_0 \geq \delta^{2\varepsilon/q}.$$

Maximality gives, for every interval $J \subset I_0$ of length $r \leq r_0$,

$$\frac{\mu(J)}{\mu(I_0)} \leq (r/r_0)^{q/2}.$$

After mapping I_0 affinely onto $[0, 1]$ and conditioning, we therefore obtain a $(q/2, 2)$ probability measure. Minimality ensures that an endpoint of I_0 belongs to $\text{spt } \mu$.

Choose ε so small that $r_0 \geq \delta^{c_0}$, where c_0 is the exponent in Proposition 14.9.10. Apply that proposition with $t = r_0$. For some $x \in \text{spt } \mu \cap I_0$, writing $y = (x - x_0)/r_0$, it gives

$$|r_0^{-1}A + yA| \geq \delta^{-c_0}|r_0^{-1}A|.$$

Multiplication by r_0 yields

$$|A + xA - x_0A| \geq \delta^{-c_0}|A|. \tag{14.9.7}$$

If each of $A + A$, $A + xA$, and $A - x_0A$ had size $< \delta^{-c_0/4}|A|$, Plünnecke–Ruzsa would contradict (14.9.7). Thus one is large. The second is the desired

conclusion. The sum–difference inequality converts the third to the desired conclusion with x_0 and a smaller exponent. In the first case choose $x_2 = \max \text{spt } \mu$. Since $1 \leq \delta^{-\varepsilon} x_2^q$, one has $x_2 \geq \delta^{\varepsilon/q}$; the Ruzsa corollary

$$\mathcal{E}_\delta(A + A) \lesssim \frac{\mathcal{E}_\delta(A + x_2 A)^2}{\mathcal{E}_\delta(x_2 A)}$$

then gives growth of $A + x_2 A$, with the loss $\delta^{\varepsilon/(2q)}$. Taking ε sufficiently small relative to $c_0 q$ completes the proof. $\square$

14.9.5 Dense subsets of approximate products

Lemma 14.9.12 (Bourgain’s product covering inequality). *Let $A, B \subset \mathbb{R}$ be bounded unions of δ -intervals, let $H \subset A \times B$ be a union of δ -squares, and put $\pi_x(a, b) = a + xb$. Then*

$$\mathcal{E}_\delta(\pi_x H) \gtrsim \frac{\mathcal{E}_\delta(H) \mathcal{E}_\delta(A + xA)}{\mathcal{E}_\delta(A - A) \mathcal{E}_\delta(B - A)}. \quad (14.9.8)$$

Proof. It is enough to prove the finite-grid version. Let A_δ, B_δ be the grid indices meeting A, B , and let H_δ be the corresponding bipartite relation. For every $(a, a') \in A_\delta^2$ and every $(b_1, b_2) \in H_\delta$, put

$$u = b_1 - a \in A_\delta - A_\delta, \quad v = b_2 - a' \in B_\delta - A_\delta.$$

Then

$$a + xa' = (b_1 + xb_2) - (u + xv).$$

Consequently

$$\mathbf{1}_{A_\delta + xA_\delta} \leq \frac{1}{\#H_\delta} \sum_{\substack{u \in A_\delta - A_\delta \\ v \in B_\delta - A_\delta}} \mathbf{1}_{\pi_x(H_\delta) - (u + xv)}$$

after changing all sets by $O(1)$ grid cells. Summing the indicator inequality gives

$$\#(A_\delta + xA_\delta) \lesssim \frac{\#(A_\delta - A_\delta) \#(B_\delta - A_\delta)}{\#H_\delta} \# \pi_x(H_\delta),$$

which is (14.9.8). $\square$

Lemma 14.9.13 (Normalising three directions). *Let $\theta_1, \theta_2, \theta_3 \in S^1$ be pairwise separated by at least $\rho > 0$. There is an invertible linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose induced map on the projective line sends the three directions to the horizontal, vertical, and diagonal directions. Moreover*

$$\|L\| + \|L^{-1}\| \leq C\rho^{-C}.$$

The induced maps on points, directions, and affine projection coordinates are $C\rho^{-C}$ -bi-Lipschitz on bounded sets.

Proof. Choose representatives v_1, v_2 for the first two directions and let L_0 send them to the coordinate vectors. The determinant of (v_1, v_2) has magnitude $\gtrsim \rho$, so $\|L_0\| + \|L_0^{-1}\| \lesssim \rho^{-1}$. Write $L_0 v_3 = (a, b)$. Separation of the third direction from the first two gives $|a|, |b| \gtrsim \rho^C$ after normalisation. Composing L_0 with the diagonal map $(u, v) \mapsto (u/a, v/b)$ sends the third direction to the diagonal and has condition number ρ^{-C} . The final assertion follows by differentiating the induced fractional-linear map on slopes, away from its single pole, and covering the projective line by two affine charts. $\square$

Proposition 14.9.14 (A large core with robust projections). *Under the hypotheses of Theorem 14.9.2, there are absolute $C_0 \geq 1$ and parameters $\eta > 10\varepsilon > 0$ such that one can find*

$$F \subset E, \quad |F| \geq \delta^{C_0 \eta} |E|, \quad \Theta \subset \text{spt } \lambda, \quad \lambda(\Theta) > 1 - \delta^\varepsilon,$$

with the following property: for every $\theta \in \Theta$ and every union of δ -balls $G \subset F$ satisfying $|G| > \delta^\varepsilon |F|$,

$$\mathcal{E}_\delta(\pi_\theta G) > \delta^{-\eta} \mathcal{E}_\delta(E)^{1/2}.$$

Proof. We prove the proposition by contradiction. Throughout, constants concealed by $\lesssim$ may depend on α, β, κ , while a notation $X \lesssim_\eta Y$ means $X \leq \delta^{-C_\eta} Y$. First reduce κ so that

$$0 < \kappa < \min\{\beta/4, (2 - \alpha)/8, 1/4\}. \quad (14.9.9)$$

This is legitimate by the monotonicity noted after the theorem. By Lemma 14.9.3,

$$I_{(\kappa+\beta)/2}(\mu_E) \lesssim \delta^{-\varepsilon}, \quad \mu_E = |E|^{-1} \mathbf{1}_E dx.$$

Lemma 14.9.5, followed by Markov, shows that

$$\lambda(\Theta_{\text{en}}) \lesssim \delta^{2\varepsilon}, \quad \Theta_{\text{en}} = \{\theta : I_\kappa(\pi_\theta \mu_E) > \delta^{-3\varepsilon}\}. \quad (14.9.10)$$

Selecting three separated bad directions

Suppose the proposition fails for the proposed η , a sufficiently large C_0 , and a much smaller ε . We claim that there are nested sets

$$E = E_0 \supset E_1 \supset E_2 \supset E_3$$

and directions $\theta_1, \theta_2, \theta_3$ such that

$$|E_i| \geq \delta^\varepsilon |E_{i-1}|, \quad i = 1, 2, 3, \quad (14.9.11)$$

$$|\theta_i - \theta_j| \geq \delta^{10\varepsilon/\kappa}, \quad i \neq j, \quad (14.9.12)$$

$$\mathcal{E}_\delta(\pi_{\theta_i} E_i) \leq \delta^{-\eta} \mathcal{E}_\delta(E)^{1/2}, \quad i = 1, 2, 3, \quad (14.9.13)$$

$$I_\kappa(\pi_{\theta_i} \mu_E) \leq \delta^{-3\varepsilon}, \quad i = 1, 2, 3. \quad (14.9.14)$$

Indeed, suppose $E' \subset E$ has $|E'| \geq \delta^{3\varepsilon}|E|$ and at most two directions have already been selected. Since $C_0\eta > 3\varepsilon$, this E' is an admissible candidate for F . The negation of the proposition says that the set of directions θ for which some $E'' \subset E'$, $|E''| \geq \delta^\varepsilon|E'|$, has a small projection has λ -measure at least δ^ε . Each ball of radius $\delta^{10\varepsilon/\kappa}$ has λ -measure at most $\delta^{9\varepsilon}$, and (14.9.10) has measure $O(\delta^{2\varepsilon})$. Deleting those sets leaves a permissible new direction. Iterating three times proves the claim.

By Lemma 14.9.13, after a linear change of variables whose condition number is at most $\delta^{-C\varepsilon/\kappa}$, we may assume

$$\pi_{\theta_1}(u, v) = u, \quad \pi_{\theta_2}(u, v) = v, \quad \pi_{\theta_3}(u, v) = u + v.$$

The transformation changes every Frostman constant, density threshold, and scale by at most $\delta^{-C\varepsilon/\kappa}$. Choosing $\varepsilon \ll_\kappa \eta$ absorbs these changes. We work in these coordinates until the end of the proof.

Extracting an approximate Cartesian product

Condition μ_E on E_3 . By (14.9.11) and (14.9.14),

$$I_\kappa(\pi_{\theta_i}\mu_{E_3}) \leq \delta^{-6\varepsilon}I_\kappa(\pi_{\theta_i}\mu_E) \leq \delta^{-9\varepsilon}, \quad i = 1, 2.$$

Lemma 14.9.4, with a fixed large L , gives $(\delta, \kappa/2, \delta^{-C\eta})$ -sets $A_i \subset \pi_{\theta_i}(E_3)$ such that

$$\mu_{E_3}(\pi_{\theta_i}^{-1}A_i) \geq 3/4, \quad i = 1, 2.$$

Hence

$$E' = E_3 \cap (A_1 \times A_2) \quad \text{satisfies} \quad |E'| \geq \frac{1}{2}|E_3| \geq \frac{1}{2}\delta^{3\varepsilon}|E|.$$

The small-projection bounds give

$$|A_i| \lesssim \delta^{-\eta}|E|^{1/2}, \quad i = 1, 2,$$

where we use the equivalence between measure and covering number for δ -sets. Since $E' \subset A_1 \times A_2$,

$$|E'| \geq \delta^{C\eta}|A_1||A_2|, \tag{14.9.15}$$

$$|A_i| \geq \delta^{C\eta}|E|^{1/2}, \quad i = 1, 2. \tag{14.9.16}$$

All constants C in exponents are absolute and may increase from line to line.

The diagonal projection of E' is a partial sumset:

$$A_1 +_{E'} A_2 = \{a_1 + a_2 : (a_1, a_2) \in E'\}.$$

Using (14.9.13), (14.9.15), and (14.9.16), the covering-number Balog–Szemerédi–Gowers theorem produces subsets $B_i \subset A_i$ such that

$$|B_i| \geq \delta^{C\eta} |A_i| \geq \delta^{C\eta} |E|^{1/2}, \quad i = 1, 2, \quad (14.9.17)$$

$$\mathcal{E}_\delta(B_1 + B_2) \leq \delta^{-C\eta} \mathcal{E}_\delta(B_1)^{1/2} \mathcal{E}_\delta(B_2)^{1/2}, \quad (14.9.18)$$

$$|E' \cap (B_1 \times B_2)| \geq \delta^{C\eta} |E|. \quad (14.9.19)$$

Put

$$F = E' \cap (B_1 \times B_2),$$

enlarging by $O(\delta)$ so that it is a union of δ -balls. Equation (14.9.19) gives the required lower bound for $|F|$. Since A_i are discretised Frostman sets and B_i retain a $\delta^{C\eta}$ fraction, each B_i is a $(\delta, \kappa/2, \delta^{-C\eta})$ -set.

Applying the hybrid theorem

The assumed failure for this particular F gives a set Θ_{bad} with

$$\lambda(\Theta_{\text{bad}}) \geq \delta^\varepsilon$$

such that for every $\theta \in \Theta_{\text{bad}}$ there is $G_\theta \subset F$ satisfying

$$|G_\theta| \geq \delta^\varepsilon |F| \geq \delta^{C\eta} |E|, \quad (14.9.20)$$

$$\mathcal{E}_\delta(\pi_\theta G_\theta) \leq \delta^{-\eta} \mathcal{E}_\delta(E)^{1/2}. \quad (14.9.21)$$

After retaining one of two affine charts of the projective line, represent these projections as

$$\pi_x(u, v) = u + xv, \quad x \in [-2, 2].$$

Conditioning λ on Θ_{bad} and pushing it to the slope coordinate gives a $(\kappa/2, \delta^{-C\varepsilon})$ probability measure $\tilde{\lambda}$.

Choose η small enough that $\alpha/2 + C\eta < 1$, and then fix

$$\max\{\kappa/2, \alpha/2 + C\eta\} < \sigma < 1. \quad (14.9.22)$$

It follows that

$$|B_1| \leq \delta^{1-\sigma}.$$

Indeed, the first small-coordinate projection and $|E| \leq \delta^{2-\alpha}$ give $|B_1| \leq \delta^{-C\eta} |E|^{1/2} \leq \delta^{1-\alpha/2-C\eta} \leq \delta^{1-\sigma}$. Theorem 14.9.11, applied to B_1 and $\tilde{\lambda}$, supplies a bad slope x such that

$$\mathcal{E}_\delta(B_1 + xB_1) \geq \delta^{-c} \mathcal{E}_\delta(B_1) \geq \delta^{-c+C\eta} \mathcal{E}_\delta(E)^{1/2}, \quad (14.9.23)$$

where $c = c(\alpha, \beta, \kappa) > 0$.

Ruzsa's triangle inequality and (14.9.18) give

$$\max_{i=1,2} \mathcal{E}_\delta(B_1 - B_i) \leq \delta^{-C\eta} \mathcal{E}_\delta(E)^{1/2}. \quad (14.9.24)$$

Apply Lemma 14.9.12 with $A = B_1$, $B = B_2$, and $H = G_x \subset F \subset B_1 \times B_2$. Equations (14.9.20), (14.9.23), and (14.9.24) yield

$$\mathcal{E}_\delta(\pi_x G_x) \geq \delta^{-c+C\eta} \mathcal{E}_\delta(E)^{1/2}.$$

Choose $\eta < c/(2C)$ and then $\varepsilon \ll_{\alpha,\beta,\kappa} \eta$. The last display contradicts (14.9.21). This proves the proposition in the normalised coordinates. Lemma 14.9.13 and the choice $\varepsilon \ll \eta\kappa$ transfer the result back to the original coordinates. $\square$

14.9.6 Exhaustion: proof of the robust theorem

Proof of Theorem 14.9.2. First fix the gain η in Proposition 14.9.14. Its proof permits the auxiliary exponent to be decreased after η is fixed. Choose $e > 0$ so small that the proposition is valid with auxiliary exponent $8e$. The hypotheses with Frostman constant δ^{-e} imply the same hypotheses with the weaker constant δ^{-8e} . Apply the proposition to obtain $F_1 \subset E$ and Θ_1 . If $|E \setminus F_1| \leq \delta^{4e}|E|$, stop. Otherwise apply the proposition to $E \setminus F_1$, and continue.

This iteration is legitimate. As long as the remainder R has $|R| > \delta^{4e}|E|$,

$$|R \cap B(z, r)| \leq \delta^{-e} r^\beta |E| \leq \delta^{-5e} r^\beta |R| \leq \delta^{-8e} r^\beta |R|,$$

so R has the required discretised Frostman property. The sets F_j are disjoint, and each has at least a fixed δ -power times the current remainder, so the finite process stops with

$$\left| E \setminus \bigcup_{j=1}^N F_j \right| \leq \delta^{4e} |E|.$$

For every j ,

$$\lambda(\Theta_j) > 1 - \delta^{8e},$$

and if $\theta \in \Theta_j$, every $G \subset F_j$ with $|G| > \delta^{8e}|F_j|$ has the projection lower bound.

For a direction θ , put

$$B(\theta) = \sum_{j: \theta \notin \Theta_j} |F_j|.$$

Fubini gives

$$\int B(\theta) d\lambda(\theta) \leq \delta^{8e} \sum_j |F_j| \leq \delta^{8e} |E|.$$

Therefore Markov's inequality shows that the set

$$\Theta = \left\{ \theta : B(\theta) \leq \delta^{2e} |E| \right\}$$

has $\lambda(\Theta) \geq 1 - \delta^{6e} > 1 - \delta^e$.

Fix $\theta \in \Theta$ and $G \subset E$ with $|G| > \delta^e |E|$. If $|G \cap F_j| \leq \delta^{8e} |F_j|$ for every j with $\theta \in \Theta_j$, then

$$|G| \leq \delta^{8e} \sum_j |F_j| + B(\theta) + \left| E \setminus \bigcup_j F_j \right| \leq (\delta^{8e} + \delta^{2e} + \delta^{4e}) |E| < \delta^e |E|$$

for all sufficiently small δ , a contradiction. Thus some j has $|G \cap F_j| > \delta^{8e} |F_j|$. Since $\pi_\theta(G)$ contains $\pi_\theta(G \cap F_j)$, the conclusion of Proposition 14.9.14 gives (14.9.2). Relabel e as ε to match the theorem statement. $\square$

14.9.7 The form used by the Furstenberg argument

Corollary 14.9.15 (Projection obstruction for a family of dense fibres). *Fix $0 < s < 1$ and $\tau > 0$. There is $\eta = \eta(s, \tau) > 0$ such that the following holds for all sufficiently small δ . Let $Y \subset [0, 1]$ be a $(\delta, \tau, \delta^{-\eta})$ -set, and for each $y \in Y$ let $X_y \subset [0, 1]$ be a $(\delta, s, \delta^{-\eta})$ -set. Put*

$$Z = \bigcup_{y \in Y} X_y \times \{y\}.$$

For every $z \in Z$, suppose $\mathcal{T}(z)$ is a $(\delta, s, \delta^{-\eta})$ -set of dyadic δ -tubes containing z . Then

$$\# \bigcup_{z \in Z} \mathcal{T}(z) \geq \delta^{-2s-\eta}.$$

The exponent and small-scale threshold may be chosen uniformly for (s, τ) in a compact subset of $(0, 1) \times (0, \infty)$.

Proof. Assume the union $\mathcal{T}$ has fewer than $\delta^{-2s-\eta}$ members. We first make the slope coordinate legitimate. Cover the projective line by finitely many angular charts on which a tube axis has the form $x = ay + b$ or $y = ax + b$, with the slope coordinate uniformly bi-Lipschitz. At each $z \in Z$, one chart contains a fixed fraction of $\mathcal{T}(z)$. For each y , one chart is selected by a fixed fraction of the points $x \in X_y$, and one final pigeonhole retains a fixed fraction of the levels $y \in Y$ using the same chart. A subset of a discretised Frostman set of fixed relative size is again such a set, with only an absolute change in its constant. After interchanging the two coordinates if necessary, we may therefore identify every retained dyadic tube with the (a, b) -parameter of its axis $x = ay + b$. We relabel all retained objects by their original symbols.

For fixed y , the subfamily

$$\mathcal{T}_y = \bigcup_{x \in X_y} \mathcal{T}(x, y)$$

has cardinality $\gtrsim \delta^{-2s+C\eta}$: families attached to distinct x have bounded overlap, and both the set of x and every fibre family have size $\gtrsim \delta^{-s+C\eta}$. Moreover $\mathcal{T}_y$ is a $(\delta, 2s, \delta^{-C\eta})$ -set. Indeed, in an r -square of parameter space, the map

$$\pi_y(a, b) = ay + b$$

takes values in an $O(r)$ -interval. Only $\delta^{-s+C\eta}r^s$ values of x can contribute, and for each x only $\delta^{-s+C\eta}r^s$ tubes can contribute. Multiplying proves the $2s$ -bound.

We give the refinement argument, since it is what permits the robust projection theorem to be applied to one common set. Consider the incidence relation

$$\mathcal{I} = \{(T, y) : y \in Y, T \in \mathcal{T}_y\}.$$

The preceding estimates give $\#\mathcal{I} \geq \delta^{-2s+C\eta}\#Y$. Pigeonhole the tubes according to their degree

$$d(T) = \#\{y : T \in \mathcal{T}_y\}$$

and retain a class $\overline{\mathcal{T}}$ which carries at least a $C \log(2/\delta)^{-1}$ fraction of all incidences and on which $d(T) \asymp d$. Since $d \leq \#Y$,

$$\#\overline{\mathcal{T}} \geq \delta^{-2s+C\eta}. \quad (14.9.25)$$

For every parameter-space square R of side $r \geq \delta$,

$$\begin{aligned} d \#(\overline{\mathcal{T}} \cap R) &\lesssim \sum_{y \in Y} \#(\mathcal{T}_y \cap R) \\ &\leq \delta^{-C\eta} r^{2s} \delta^{-2s} \#Y. \end{aligned}$$

On the other hand $d \#\overline{\mathcal{T}} \geq \delta^{C\eta} \delta^{-2s} \#Y$. Dividing the two displays proves that $\overline{\mathcal{T}}$ is a $(\delta, 2s, \delta^{-C\eta})$ -set.

The identity

$$\sum_{y \in Y} \#(\overline{\mathcal{T}} \cap \mathcal{T}_y) = d \#\overline{\mathcal{T}}$$

and one more dyadic pigeonhole provide $\overline{Y} \subset Y$ with $\#\overline{Y} \geq \delta^{C\eta} \#Y$ and

$$\#(\overline{\mathcal{T}} \cap \mathcal{T}_y) \geq \delta^{C\eta} \#\overline{\mathcal{T}}, \quad y \in \overline{Y}.$$

Thus $\overline{Y}$ is a $(\delta, \tau, \delta^{-C\eta})$ -set. Put $G_y = \overline{\mathcal{T}} \cap \mathcal{T}_y$.

For every $y \in \bar{Y}$, incidence of (x, y) with the tube $x = ay + b$ gives

$$\pi_y(G_y) \subset (X_y)_{O(\delta)},$$

so

$$\mathcal{E}_\delta(\pi_y G_y) \lesssim \delta^{-s-C\eta}.$$

Let $E_{\mathcal{T}}$ be the union of fixed-radius δ -balls centred at the parameters in $\bar{\mathcal{T}}$, and let the same symbol G_y denote the corresponding union of balls centred at its discrete members. The normalised Lebesgue measure on $E_{\mathcal{T}}$ is a $(\delta, \beta, \delta^{-C\eta})$ -set for any fixed $0 < \beta < \min\{2s, 1\}$, has measure at most $\delta^{2-2s-C\eta}$, and each G_y occupies a $\delta^{C\eta}$ fraction of it. Thicken $\bar{Y}$ by disjoint intervals of length comparable to δ and transport the normalised Lebesgue measure on their union to the corresponding unit directions. This is a $(\tau/2, \delta^{-C\eta})$ probability measure. If a slope y' lies in the interval associated with y , then $\pi_{y'}(G_y)$ is contained in an $O(\delta)$ -neighbourhood of X_y , so the preceding projection upper bound remains valid.

Now fix

$$\alpha_* = 1 + s < 2, \quad 0 < \beta < \min\{2s, 1\}, \quad 0 < \kappa_* < \min\{\tau/2, 1/2\}.$$

Let $\gamma > 0$ and $\varepsilon_* > 0$ be the gain and admissible Frostman exponent supplied by Theorem 14.9.2 for these three fixed parameters. Choose the present structural exponent η so small that

$$C\eta < \min\{(1-s)/4, \gamma/4, \varepsilon_*/4\}. \quad (14.9.26)$$

Then $|E_{\mathcal{T}}| \leq \delta^{2-\alpha_*}$, all Frostman constants are at most $\delta^{-\varepsilon_*}$, and every G_y has relative measure larger than δ^{ε_*} . The robust theorem therefore supplies a slope y' in the thickened support, associated with some $y \in \bar{Y}$, for which

$$\mathcal{E}_\delta(\pi_{y'} G_y) \geq \delta^{-\gamma} \mathcal{E}_\delta(E_{\mathcal{T}})^{1/2} \geq \delta^{-s-\gamma+C\eta}.$$

This contradicts $\mathcal{E}_\delta(\pi_{y'} G_y) \lesssim \delta^{-s-C\eta}$ by (14.9.26). Hence the assumed small union cannot exist. For the uniform clause, keep s in a sufficiently small neighbourhood of a fixed value and spend half of the displayed gain to absorb the resulting changes in the Frostman exponents and cardinality bounds. If $\tau \geq \tau_0 > 0$, a τ -Frostman estimate implies the τ_0 -Frostman estimate. The same choices therefore work in a neighbourhood with a fixed positive lower bound for τ . A finite cover of a compact parameter set and the minimum of the finitely many gains prove the assertion. $\square$

Corollary 14.9.15 is the exact product-configuration obstruction used in the regular part of the epsilon-improved Furstenberg incidence theorem.

Chapter 14.10

A Complete Epsilon-Improved Furstenberg Incidence Proof

This chapter proves Theorem 14.8.1. We prove the point-dimension-one form needed by the radial bootstrap, rather than carrying an unused second dimension parameter through the argument. Every component of the proof is included: the elementary incidence estimate, the thick-tube selection, the incidence factorisation across two scales, the regular-scale product obstruction, the iteration over many scales, and the decomposition of an arbitrary uniform set into regular, normal, and negligible scales.

The only non-elementary ingredient is the robust discretised projection theorem proved in Chapter 14.9. We use it only through Corollary 14.9.15. Thus the present chapter, together with the preceding one, contains the full proof chain behind the radial incidence input.

14.10.1 Dyadic conventions and the main statement

For $r = 2^{-k}$, let $\mathcal{D}_r$ denote the half-open dyadic squares of side r . We use the duality

$$(a, b) \longleftrightarrow \ell_{a,b} := \{(x, y) : x = ay + b\}.$$

If $R = I \times J$ is a dyadic square of side r in the (a, b) -plane, the associated dyadic r -tube is

$$\mathbf{D}(R) = \bigcup_{(a,b) \in R} \ell_{a,b}.$$

Only tubes meeting a fixed bounded square will occur. Consequently their parameters lie in a fixed bounded rectangle, and all constants caused by the choice of that rectangle are absolute.

If A is a bounded set, write $|A|_r = \mathcal{E}_r(A)$. A nonempty union of r -squares is an (r, u, D) -set if

$$|A \cap B(z, \rho)|_r \leq D\rho^u |A|_r, \quad r \leq \rho \leq 1. \quad (14.10.1)$$

Families of dyadic tubes are called (r, u, D) -sets when their parameter squares have this property. Replacing balls by dyadic squares, or conversely, changes D by an absolute factor.

We shall prove the following dyadic theorem first.

Theorem 14.10.1 (Dyadic point-dimension-one incidence improvement). *Let $0 < s < 1$. There are $\epsilon_0 = \epsilon_0(s) > 0$ and $r_0 = r_0(s) > 0$ such that the following holds for dyadic $0 < r < r_0$. Let $\mathcal{P} \subset \mathcal{D}_r([0, 1]^2)$ be an $(r, 1, r^{-\epsilon_0})$ -set. For every $p \in \mathcal{P}$, suppose that $\mathcal{T}(p)$ is an $(r, s, r^{-\epsilon_0})$ -set of dyadic r -tubes, all meeting p . Then*

$$\# \bigcup_{p \in \mathcal{P}} \mathcal{T}(p) \geq r^{-2s - \epsilon_0}. \quad (14.10.2)$$

The same constants work uniformly for s in a compact subinterval of $(0, 1)$.

The proof is quantitative only through fixed powers of r . We use

$$A \lesssim_{\log r} B \iff A \leq C[\log(2/r)]^C B$$

with an absolute C ; subscripts on the symbol allow dependence on the listed fixed parameters. Every logarithmic loss will eventually be absorbed into a smaller positive power of r .

Lemma 14.10.2 (Tubes through one square and their slopes). *Let $p \in \mathcal{D}_r([0, 1]^2)$ and let $\mathcal{U}$ be dyadic r -tubes meeting p . The slope map $T \mapsto a(T)$ is at most ten-to-one on $\mathcal{U}$. Moreover, $\mathcal{U}$ is an (r, u, D) -set if and only if its slope set is an (r, u, CD) -set, for an absolute C .*

Proof. Write $T_j = \mathbf{D}([a, a+r) \times [b_j, b_j+r))$ and choose $(x_j, y_j) \in p \cap T_j$. There are $a_j \in [a, a+r)$ and $b'_j \in [b_j, b_j+r)$ with $x_j = a_j y_j + b'_j$. Since $p \subset [0, 1]^2$,

$$|b_1 - b_2| \leq |x_1 - x_2| + |a_1||y_1 - y_2| + |a_1 - a_2||y_2| + 2r \leq 5r.$$

Thus at most ten dyadic intercept intervals occur for a fixed slope interval.

If the slopes lie in an interval of length ρ , the preceding calculation, with the left endpoint of p fixed, places all corresponding intercepts in an interval of length $C\rho$. Hence the relevant parameter squares lie in C dyadic squares of side comparable to ρ . This proves both directions of the Frostman comparison, as well as the ten-to-one assertion. $\square$

14.10.2 Trimming finite Frostman families

Lemma 14.10.3 (Trimming). *If A is an (r, u, D) -set in a fixed bounded region, then it contains an r -separated $(r, u, CD \log(2/r))$ -subset A' with*

$$(CD)^{-1}r^{-u} \leq \#A' \leq 2r^{-u}.$$

If $A_0 \subset A$ has $|A_0|_r \geq L^{-1}|A|_r$, then A_0 is an (r, u, DL) -set.

Proof. The second assertion follows immediately from (14.10.1). We prove the first assertion without invoking a continuous Frostman lemma. Put $N = |A|_r$ and choose one point from each occupied r -cell. Applying (14.10.1) to a ball of radius Cr around any one occupied cell gives

$$N \geq cD^{-1}r^{-u}. \quad (14.10.3)$$

If $N \leq 2r^{-u}$, keep all the chosen points and we are done.

Suppose $N > 2r^{-u}$ and put $k = \lceil r^{-u} \rceil$. Select uniformly at random a k -element subset A' of the chosen points. For a dyadic cube Q of side $\rho \geq r$, the random variable $X_Q = \#(A' \cap Q)$ is hypergeometric and

$$\mathbb{E}X_Q = k \frac{\#(A \cap Q)}{N} \leq CD\rho^u k.$$

The standard sampling-without-replacement tail estimate

$$\Pr\{X_Q \geq e\mathbb{E}X_Q + a\} \leq e^{-a} \quad (a \geq 0) \quad (14.10.4)$$

is elementary here. If n_Q of the N population points lie in Q , Maclaurin's inequality for elementary symmetric means, applied to the N numbers consisting of n_Q copies of e and $N - n_Q$ copies of 1, gives

$$\mathbb{E}e^{X_Q} \leq \left(1 + (e-1)\frac{n_Q}{N}\right)^k \leq e^{(e-1)\mathbb{E}X_Q}.$$

Markov's inequality, with $e\mathbb{E}X_Q + a$ in place of the threshold, proves (14.10.4). Since $\rho^u k \geq 1$, (14.10.4), with $a = C_0 \log(2/r)D\rho^u k$, shows that

$$X_Q \leq C_1 D \log(2/r) \rho^u k$$

simultaneously for every relevant dyadic Q with positive probability: there are at most $Cr^{-d} \log(2/r)$ such cubes, and C_0 is chosen larger than $d + 3$. Hence a deterministic choice with these bounds exists. Every ball is covered by $O_d(1)$ dyadic cubes of comparable side, so A' is an $(r, u, CD \log(2/r))$ -set. Its cardinality is $k \in [r^{-u}, 2r^{-u}]$; together with (14.10.3), this proves the claim.

For the one-dimensional statement, take the dyadic intervals I of lengths between ρ and 1. If $n_I = \#(A \cap I)$, the interval Frostman bound is $n_I/N \leq D\rho^u$ for $|I| < r$ and $n_I/N \leq D(|I|/r)^u \rho^u$ for $|I| \geq r$. Select the same uniformly random k -point subset. The preceding hypergeometric computation, now with X_I in place of X_Q , gives

$$\mathbb{P}(X_I > e\mathbb{E}X_I + a) \leq e^{-a}.$$

There are at most $2r^{-1} \log_2(2/r)$ relevant intervals. Taking $a = C_0 D \rho^u k \log(2/r)$ with $C_0 > 4$ and using $\rho^u k \geq 1$, the union bound is strictly smaller than one. Hence one deterministic choice satisfies all interval estimates simultaneously. Every real interval is covered by at most two dyadic intervals of comparable length, which yields the asserted one-dimensional Frostman constant and completes that case. $\square$

14.10.3 The elementary incidence estimate

Proposition 14.10.4 (Elementary incidence bound). *Let $0 \leq s \leq t \leq 1$. Suppose $\mathcal{P}$ is an (r, t, D_P) -set of r -squares and every $p \in \mathcal{P}$ carries an (r, s, D_T) -set $\mathcal{T}(p)$ of M tubes meeting p . Put $\mathcal{T} = \bigcup_p \mathcal{T}(p)$. Then*

$$\#\mathcal{T} \gtrsim_{\log r} (D_P D_T)^{-1} M r^{-s} (M r^s)^{(t-s)/(1-s)}. \quad (14.10.5)$$

For $s = t = 1$, the last exponent is interpreted as one.

Proof. Let $I = \{(p, T) : T \in \mathcal{T}(p)\}$, so $\#I = M\#\mathcal{P}$. Cauchy–Schwarz in T gives

$$(\#I)^2 \leq \#\mathcal{T} \left(\#I + \sum_{p \neq q} \#(\mathcal{T}(p) \cap \mathcal{T}(q)) \right). \quad (14.10.6)$$

Let $d = d(p, q) \geq r$. A tube meeting both squares has slope in an interval of length Cr/d . Lemma 14.10.2 and the Frostman property give the two bounds

$$\#(\mathcal{T}(p) \cap \mathcal{T}(q)) \lesssim \min\{D_T M (r/d)^s, d^{-1}\}.$$

Put $\theta = (1 - t)/(1 - s)$, with $\theta = 0$ if $s = t = 1$. Since $t = s\theta + 1 - \theta$ and $\min(A, B) \leq A^\theta B^{1-\theta}$,

$$\#(\mathcal{T}(p) \cap \mathcal{T}(q)) \lesssim (D_T M r^s)^\theta d^{-t}.$$

For fixed p , dyadic annuli and the (r, t, D_P) condition yield

$$\sum_{q \neq p} d(p, q)^{-t} \lesssim D_P \#\mathcal{P} \log(2/r).$$

Insert this in (14.10.6). If the diagonal term dominates, $\#\mathcal{T} \gtrsim M\#\mathcal{P} \gtrsim D_P^{-1}Mr^{-t}$. Otherwise,

$$\#\mathcal{T} \gtrsim_{\log r} D_P^{-1}D_T^{-\theta}M^{2-\theta}r^{-s\theta}.$$

Because $2 - \theta = 1 + (t - s)/(1 - s)$ and $-s\theta = -s + s(t - s)/(1 - s)$, this is (14.10.5). The diagonal lower bound is no smaller, since $M \lesssim r^{-1}$ and $t \geq s$. $\square$

14.10.4 Selecting a Frostman family of thick tubes

Proposition 14.10.5 (Thick-tube selection). *Let $0 < r \leq R \leq 1$ be dyadic. Let $\mathcal{P}$ be finite and suppose each $p \in \mathcal{P}$ carries an (r, s, D) -set $\mathcal{T}(p)$ of between $M/2$ and M dyadic r -tubes, every one meeting p (and all lying in the fixed bounded slope chart). There are $\overline{\mathcal{P}} \subset \mathcal{P}$ and a family $\overline{\mathcal{T}}_R$ of dyadic R -tubes such that*

1. $\#\overline{\mathcal{P}} \gtrsim_{\log R} \#\mathcal{P}$;
2. $\overline{\mathcal{T}}_R$ is an $(R, s, CD[\log(2/R)]^C)$ -set;
3. every $\mathbf{T} \in \overline{\mathcal{T}}_R$ contains at least

$$H \gtrsim_{\log R} \frac{M\#\mathcal{P}}{\#\overline{\mathcal{T}}_R} \quad (14.10.7)$$

pairs (p, T) with $p \in \overline{\mathcal{P}}$ and $T \in \mathcal{T}(p)$.

Proof. Let $\mathcal{A}_R$ be the minimal dyadic R -tube cover of $\bigcup_p \mathcal{T}(p)$. It has $O(R^{-2})$ members. For fixed p , pigeonhole the ancestors $\mathbf{T} \in \mathcal{A}_R$ according to the dyadic number m_1 of descendants from $\mathcal{T}(p)$ they contain. Classes with $m_1 < cMR^2$ carry fewer than $M/4$ descendants in total. Among the remaining $O(\log(2/R))$ classes, one class carries $\gtrsim M/\log(2/R)$ descendants. It consists of $m_2 \asymp_{\log R} M/m_1$ ancestors.

There are only logarithmically many pairs (m_1, m_2) . Retain a logarithmic fraction $\mathcal{P}_1$ of the points for which the pair is fixed, and retain only the selected descendants. Count the incidences between $\mathcal{P}_1$ and their R -ancestors. A second dyadic pigeonhole, now on the number h of points using an ancestor, gives a class $\overline{\mathcal{T}}_R$ such that

$$h\#\overline{\mathcal{T}}_R \asymp_{\log R} m_2\#\mathcal{P}_1. \quad (14.10.8)$$

Each retained ancestor contains at least hm_1 pairs, and (14.10.8) gives $hm_1 \asymp_{\log R} M\#\mathcal{P}/\#\overline{\mathcal{T}}_R$. This proves (14.10.7).

It remains to prove nonconcentration. Let $\mathbf{U}$ be a dyadic ρ -tube, $\rho \geq R$, containing N members of $\overline{\mathcal{T}}_R$. Double-count their point incidences and use (14.10.8); some $p_0 \in \mathcal{P}_1$ uses at least

$$c_{\log R} N m_2 / \# \overline{\mathcal{T}}_R$$

of those ancestors. They contain at least m_1 distinct members of $\mathcal{T}(p_0)$ each. The (r, s, D) condition therefore gives

$$\frac{N m_1 m_2}{\# \overline{\mathcal{T}}_R} \lesssim_{\log R} D M \rho^s.$$

Since $m_1 m_2 \asymp_{\log R} M$, this is the required $N \lesssim_{\log R} D \rho^s \# \overline{\mathcal{T}}_R$. $\square$

14.10.5 Incidence factorisation across two scales

Definition 14.10.6 (Nice configuration). A pair $(\mathcal{P}_0, \mathcal{T}_0)$ is (r, s, D, M) -nice if, for every $p \in \mathcal{P}_0$, there is an (r, s, D) -set $\mathcal{T}_0(p) \subset \mathcal{T}_0$ of exactly M tubes meeting p .

For $Q \in \mathcal{D}_R$, let S_Q be the homothety taking Q to $[0, 1]^2$.

Proposition 14.10.7 (Two-scale factorisation). *Let $r \leq R$ be dyadic and let $(\mathcal{P}_0, \mathcal{T}_0)$ be (r, s, D, M) -nice. After passing to subsets, there are*

- a set $\mathcal{P} \subset \mathcal{P}_0$ whose occupied R -squares and whose populations inside those squares are each reduced by at most a factor $[\log(2/r)]^C$;
- an (R, s, D_R, M_R) -nice configuration $(\mathcal{D}_R(\mathcal{P}), \mathcal{T}_R)$;
- for every $Q \in \mathcal{D}_R(\mathcal{P})$, an $(r/R, s, D_Q, M_Q)$ -nice configuration $(S_Q(\mathcal{P} \cap Q), \mathcal{T}_Q)$,

where $D_R, D_Q \leq CD[\log(2/r)]^C$, such that

$$\frac{\# \mathcal{T}_0}{M} \gtrsim_{\log r} \frac{\# \mathcal{T}_R}{M_R} \max_Q \frac{\# \mathcal{T}_Q}{M_Q}. \quad (14.10.9)$$

The retained family at every point has at least $M/[\log(2/r)]^C$ members.

Proof. We divide the construction into five finite pigeonholing steps.

Step 1: coarse families. For each occupied R -square Q , apply Proposition 14.10.5 to $\mathcal{P}_0 \cap Q$ and the families $\mathcal{T}_0(p)$. This gives a large subset $\mathcal{P}_Q \subset \mathcal{P}_0 \cap Q$ and an $(R, s, D_R(Q))$ -family $\mathcal{A}_R(Q)$ of ancestors. Pigeonholing

in Q makes $\#\mathcal{A}_R(Q) = M_R$ and $D_R(Q) \leq CD[\log(2/r)]^C$ independent of Q , after retaining a logarithmic fraction of the occupied squares. Proposition 14.10.5 also ensures that each $\mathbf{T} \in \mathcal{A}_R(Q)$ supports at least

$$H_Q \gtrsim_{\log r} \frac{M\#\mathcal{P}_Q}{M_R} \quad (14.10.10)$$

pairs (p, T) with $p \in \mathcal{P}_Q$ and $T \in \mathcal{T}_0(p)$ contained in $\mathbf{T}$.

Put $\mathcal{T}_R = \bigcup_Q \mathcal{A}_R(Q)$. Because all ancestors in $\mathcal{A}_R(Q)$ meet Q , the pair $(\mathcal{D}_R(\mathcal{P}), \mathcal{T}_R)$ is (R, s, D_R, M_R) -nice once $\mathcal{P} = \bigcup_Q \mathcal{P}_Q$ has received the later refinements.

Step 2: the fine multiplicity of a coarse tube. Pigeonhole the members of $\mathcal{T}_R$ according to $\#(\mathcal{T}_0 \cap \mathbf{T})$. Retain a dyadic class which carries a logarithmic fraction of the incidences in (14.10.10), and then retain those Q and p for which a logarithmic fraction survives. With no change of notation,

$$\#(\mathcal{T}_0 \cap \mathbf{T}) \asymp N_R, \quad \mathbf{T} \in \mathcal{T}_R. \quad (14.10.11)$$

The thick-tube Frostman property and all population estimates persist, with one more logarithmic factor. Since distinct dyadic R -ancestors contain disjoint parameter cells,

$$\#\mathcal{T}_0 \geq N_R \#\mathcal{T}_R. \quad (14.10.12)$$

Step 3: keep many fine tubes at each point. For $p \in \mathcal{P}_Q$, keep only tubes whose R -ancestor lies in $\mathcal{A}_R(Q)$. Summing (14.10.10) over the ancestors shows that the average retained cardinality is $\gtrsim_{\log r} M$. Since it never exceeds M , a logarithmic fraction of the points retain $\gtrsim_{\log r} M$ tubes. Pass to those points and call their families $\mathcal{T}(p)$. This proves the asserted pointwise retention and preserves the population statement in the proposition.

Step 4: packets and rescaling inside Q . Fix Q and put $\bar{r} = r/R$. Partition $\mathcal{T}(p)$ into packets by their dyadic $\bar{r}$ -ancestor in parameter space. Dyadically pigeonhole the nonempty packet sizes. After retaining a logarithmic fraction, every $p \in \mathcal{P}_Q$ has M_Q packets, each containing

$$m_Q \asymp_{\log r} M/M_Q \quad (14.10.13)$$

fine tubes. The original slope Frostman bound implies that the set of packet slopes is an $(\bar{r}, s, CD[\log(2/r)]^C)$ -set: if an interval I has length $\rho \geq \bar{r}$, the packets over I contain $\asymp m_Q$ original slopes each, whence

$$\#\{\text{packet slopes in } I\} \lesssim \frac{DM\rho^s}{m_Q} \asymp_{\log r} DM_Q\rho^s.$$

Choose one representative r -tube from each packet which meets p and has the left endpoint of the packet's slope interval as its slope. Such a tube exists

after changing the intercept by at most Cr . Under S_Q , its intersection with Q is covered by at most four dyadic $\bar{r}$ -tubes of that same packet slope. Choose one which meets $S_Q(p)$. The resulting family $\mathcal{T}_Q(S_Q p)$ has M_Q members and, by Lemma 14.10.2, is an $(\bar{r}, s, D_Q)$ -set. This constructs the asserted nice configuration.

Step 5: counting distinct packets. Let $\mathcal{T}(Q) = \bigcup_{p \in \mathcal{P}_Q} \mathcal{T}(p)$. It lies in the M_R selected R -ancestors, so

$$\#\mathcal{T}(Q) \leq M_R N_R. \quad (14.10.14)$$

On the other hand, every member of $\mathcal{T}_Q$ represents a packet of m_Q fine tubes. We verify that a constant fraction of the representatives may be chosen with disjoint represented packets. At a fixed packet slope, write a representative as $\mathbf{D}([a, a+r) \times [b, b+r))$. If the Cr -thickenings of two such tubes meet $[0, 1)^2$, then their intercepts differ by at most $C'r$. Selecting every $(2C' + 1)$ st intercept therefore gives a constant-size separated subfamily. Representatives of distinct packet slopes correspond to disjoint slope intervals automatically. If two separated representatives had a common fine tube, that fine tube would meet the two underlying r -squares and would lie within Cr of both representatives inside Q , contradicting separation. Thus the represented packets are disjoint.

Consequently, using (14.10.13),

$$\#\mathcal{T}(Q) \gtrsim_{\log r} \#\mathcal{T}_Q m_Q \asymp_{\log r} \frac{\#\mathcal{T}_Q}{M_Q} M. \quad (14.10.15)$$

Combining (14.10.14) and (14.10.15) gives $N_R/M \gtrsim_{\log r} (\#\mathcal{T}_Q)/(M_R M_Q)$. Multiply by (14.10.12) and maximise over Q to obtain (14.10.9). $\square$

14.10.6 The regular-scale obstruction

Definition 14.10.8 (One-intermediate-scale regularity). For dyadic r with $r^{1/2}$ dyadic, an (r, u, D) -set $\mathcal{P}$ is (r, u, D, K) -regular if

$$\#\mathcal{D}_{r^{1/2}}(\mathcal{P}) \leq K r^{-u/2}.$$

The next proposition is the only place where Bourgain's projection theorem enters the Furstenberg argument.

Proposition 14.10.9 (Regular alternative). *Fix $0 < s < 1$ and $s < u \leq 2$. There is $c = c(s, u) > 0$ such that, for every sufficiently small dyadic r whose square root is dyadic, the following holds. Let $\mathcal{P}$ be an (r, u, r^{-c}, r^{-c}) -regular set, and let every $p \in \mathcal{P}$ carry an (r, s, r^{-c}) -set $\mathcal{T}(p)$ of tubes meeting p . With $\mathcal{T} = \bigcup_p \mathcal{T}(p)$, either*

$$\#\mathcal{T} \geq r^{-2s-c} \quad \text{or} \quad \#\mathcal{D}_{r^{1/2}}(\mathcal{T}) \geq r^{-s-c}. \quad (14.10.16)$$

The exponent and threshold may be chosen uniformly on compact subsets of $\{(s, u) : 0 < s < 1, s < u \leq 2\}$.

Proof. Put $\Delta = r^{1/2}$. We argue by contradiction and assume both inequalities in (14.10.16) fail. In this proof, write $A \lesssim_c B$ when $A \leq C_s r^{-C_c} B$, and similarly for $\asymp_c$. Logarithmic factors are included in this notation.

Initial uniformisation. By Lemma 14.10.3 and dyadic pigeonholing, we may arrange

$$\#\mathcal{P} \asymp_c r^{-u}, \quad \#\mathcal{T}(p) = M \asymp_c r^{-s} \quad (p \in \mathcal{P}). \quad (14.10.17)$$

Let $\mathcal{Q} = \mathcal{D}_\Delta(\mathcal{P})$. Regularity gives $\#\mathcal{Q} \lesssim_c \Delta^{-u}$, while the Frostman condition and (14.10.17) give the reverse inequality. Discarding light members of $\mathcal{Q}$ and pigeonholing their populations, we may assume

$$\#\mathcal{Q} \asymp_c \Delta^{-u}, \quad \#(\mathcal{P} \cap Q) \asymp_c \Delta^{-u} \quad (Q \in \mathcal{Q}). \quad (14.10.18)$$

The family $\mathcal{Q}$ is itself a (Δ, u, r^{-C_c}) -set: for a dyadic ρ -square R , divide the bound for $\#(\mathcal{P} \cap R)$ by the lower population in (14.10.18).

Let $\mathcal{A}_\Delta$ be the minimal dyadic Δ -tube cover of $\mathcal{T}$. The counter-assumption says

$$\#\mathcal{A}_\Delta < r^{-s-c} = \Delta^{-2s-2c}. \quad (14.10.19)$$

For every Q , Proposition 14.10.5 supplies a large subset $\mathcal{P}_Q \subset \mathcal{P} \cap Q$ and an (Δ, s, r^{-C_c}) -set $\mathcal{A}_\Delta(Q) \subset \mathcal{A}_\Delta$ of tubes meeting Q . Pigeonholing in Q , write $\#\mathcal{A}_\Delta(Q) \asymp M_\Delta$. Put $u_b = \min\{u, 1\}$. The set $\mathcal{Q}$ is also an (Δ, u_b, r^{-C_c}) -set. Proposition 14.10.4, applied at scale Δ with this exponent, gives

$$\#\mathcal{A}_\Delta \gtrsim_c M_\Delta \Delta^{-s} (M_\Delta \Delta^s)^{(u_b-s)/(1-s)}. \quad (14.10.20)$$

Since an (Δ, s, r^{-C_c}) -set has at least $\Delta^{-s} r^{C_c}$ members, (14.10.19) and (14.10.20) imply

$$M_\Delta \asymp_c \Delta^{-s}, \quad \#\mathcal{A}_\Delta \asymp_c \Delta^{-2s}. \quad (14.10.21)$$

Pigeonhole $\mathbf{T} \in \mathcal{A}_\Delta$ according to the number of fine tubes it contains, and retain a class carrying a logarithmic fraction of the incidences $(Q, \mathbf{T})$ with $\mathbf{T} \in \mathcal{A}_\Delta(Q)$. After a further refinement of $\mathcal{Q}$ and its families, we have

$$\#(\mathcal{T} \cap \mathbf{T}) \asymp N \lesssim_c r^{-s} = \Delta^{-2s} \quad (\mathbf{T} \in \mathcal{A}_\Delta). \quad (14.10.22)$$

Indeed, the classes are disjoint in parameter space, so $N\#\mathcal{A}_\Delta \leq \#\mathcal{T} < r^{-2s-c}$, and (14.10.21) proves the bound. All estimates (14.10.18)–(14.10.21) persist.

A projection lemma stable under dense restriction. We claim that, for each Q , at least half of the slopes in $a(\mathcal{A}_\Delta(Q))$ have this property: whenever $\mathcal{P}'_Q \subset \mathcal{P}_Q$ and $\#\mathcal{P}'_Q \geq r^{B_c} \#\mathcal{P}_Q$, the projection

$$\pi_a(x, y) = x - ay \quad (14.10.23)$$

of $S_Q(\bigcup \mathcal{P}'_Q)$ contains a $(\Delta, s, r^{-C(B+1)c})$ -set.

Here is the complete energy proof. Let μ_Q be normalised Lebesgue measure on the rescaled Δ -squares $S_Q(p)$, $p \in \mathcal{P}_Q$, and let ν_Q be normalised Lebesgue measure on the Δ -intervals about the retained slopes. The heavy-square and tube Frostman bounds give

$$\mu_Q(B(z, \rho)) \lesssim_c \rho^u, \quad \nu_Q(I) \lesssim_c |I|^s.$$

For $z \neq w$, the set of a satisfying $|\pi_a(z) - \pi_a(w)| \leq \lambda$ is an interval of length $O(\lambda/|z - w|)$, after splitting into the two coordinate charts. A layer-cake integration, split at the smoothing scale Δ , therefore gives

$$\int I_s(\pi_a \mu_Q) d\nu_Q(a) \lesssim_c \log(2/\Delta) I_s(\mu_Q) \lesssim_c 1,$$

where the last inequality follows by summing the u -Frostman estimate over dyadic annuli and using $u > s$; the logarithm is included in $\lesssim_c$. More explicitly, the inner layer-cake integral is at most $Cr^{-C_c}|z - w|^{-s} \log(2/\Delta)$. Markov's inequality leaves a set of ν_Q -measure at least one half on which $I_s(\pi_a \mu_Q) \lesssim_c 1$. Since the slope intervals defining ν_Q are disjoint and have equal ν_Q -mass, at least half of them meet this good set.

Restricting to a relative r^{B_c} portion and renormalising multiplies this energy by at most r^{-2B_c} . If a probability measure ρ has $I_s(\rho) \leq K$, then

$$\mathcal{H}_\infty^s(\text{spt } \rho) \geq K^{-1}. \quad (14.10.24)$$

To see this, refine a cover to disjoint intervals I_j and use

$$1 = \left(\sum_j \rho(I_j) \right)^2 \leq \left(\sum_j |I_j|^s \right) \left(\sum_j \rho(I_j)^2 |I_j|^{-s} \right) \leq I_s(\rho) \sum_j |I_j|^s.$$

The finite Frostman extraction in Lemma 14.10.3, applied to the projected support at scale Δ , proves the claim. Replacing a slope by the left endpoint of its Δ -interval changes every projection by $O(\Delta)$ and hence changes only absolute constants.

Keep only these good directions in every $\mathcal{A}_\Delta(Q)$. They still form (Δ, s, r^{-C_c}) -sets of cardinality $\asymp_c \Delta^{-s}$.

Finding a thick tube with regularly distributed squares. For distinct $Q, Q' \in \mathcal{Q}$, a common tube in $\mathcal{A}_\Delta(Q) \cap \mathcal{A}_\Delta(Q')$ must have slope in an interval of length $C\Delta/d(Q, Q')$. Hence

$$\#(\mathcal{A}_\Delta(Q) \cap \mathcal{A}_\Delta(Q')) \lesssim_c d(Q, Q')^{-s}. \quad (14.10.25)$$

Multiplying by $d(Q, Q')^{-(u-s)}$, summing, and using the u -Frostman condition for $\mathcal{Q}$ gives

$$\sum_{\mathbf{T} \in \mathcal{A}_\Delta} \sum_{\substack{Q \neq Q' \\ \mathbf{T} \in \mathcal{A}_\Delta(Q) \cap \mathcal{A}_\Delta(Q')}} d(Q, Q')^{-(u-s)} \lesssim_c \Delta^{-2u}. \quad (14.10.26)$$

Put $\tau = u - s$. For a large absolute K , call a coarse tube exceptional if the left side below exceeds $r^{-Kc} \Delta^{-2\tau}$. By (14.10.26), the number of exceptional tubes is at most

$$r^{Kc} \Delta^{2\tau} r^{-C_1 c} \Delta^{-2u} = r^{(K-C_1)c} \Delta^{-2s}.$$

Choose K larger than the implicit C . If at least half of the tubes in $\mathcal{A}_\Delta(Q)$ were exceptional for at least half of the squares Q , Proposition 14.10.4, applied to those squares and those half-families, would produce at least $r^{C_1 c} \Delta^{-2s}$ exceptional tubes, for a fixed C_1 . Taking $K > C + C_1$ contradicts the preceding upper bound when r is small. We may therefore retain half of the squares and, in each retained family, its nonexceptional half. Trimming these dense subfamilies changes only the hidden power of r^{-c} . Every remaining $\mathbf{T}$ satisfies

$$\sum_{\substack{Q \neq Q' \\ \mathbf{T} \in \mathcal{A}_\Delta(Q) \cap \mathcal{A}_\Delta(Q')}} d(Q, Q')^{-(u-s)} \leq r^{-Kc} \Delta^{-2(u-s)}. \quad (14.10.27)$$

The retained incidence count and (14.10.21) give a tube $\mathbf{T}_0$ incident to

$$\#\mathcal{Q}(\mathbf{T}_0) := \#\{Q : \mathbf{T}_0 \in \mathcal{A}_\Delta(Q)\} \gtrsim_c \Delta^{-(u-s)}. \quad (14.10.28)$$

Equations (14.10.27) and (14.10.28), followed by Markov on the pointwise potential, produce a subset, still denoted $\mathcal{Q}(\mathbf{T}_0)$, which is a $(\Delta, u - s, r^{-C_1 c})$ -set. Explicitly, after increasing C , retain those Q for which

$$\sum_{Q' \neq Q} d(Q, Q')^{-(u-s)} \leq r^{-C_1 c} \Delta^{-(u-s)}.$$

At least half survive; for a ball of radius ρ , each contained Q' adds at least $\rho^{-(u-s)}$, giving the required local count.

The product configuration. The pair lower bound in Proposition 14.10.5 and (14.10.21) show that, for every $Q \in \mathcal{Q}(\mathbf{T}_0)$, there is a subset $\mathcal{P}_Q^0 \subset \mathcal{P}_Q$ with

$$\#\mathcal{P}_Q^0 \gtrsim_c \Delta^{-u}, \quad \#(\mathcal{T}(p) \cap \mathbf{T}_0) \gtrsim_c \Delta^{-s} \quad (p \in \mathcal{P}_Q^0). \quad (14.10.29)$$

To justify the first inequality, the total number of pairs (p, T) in $\mathbf{T}_0$ is $\gtrsim_c M \Delta^{-u} / \Delta^{-s}$, while Lemma 14.10.2 bounds the number contributed by one p by $C M \Delta^s \asymp_c \Delta^{-s}$. Pigeonhole the nonzero point degrees dyadically. A class

carrying a logarithmic fraction of the pair count has at least $\gtrsim_c \Delta^{-u}$ points, each of degree $\gtrsim_c \Delta^{-s}$; otherwise its total pair count would be too small. This is (14.10.29).

Apply the dense-restriction projection claim to $\mathcal{P}_Q^0$ in the slope $a(\mathbf{T}_0)$. It yields a (Δ, s, r^{-C_c}) -set of projected coordinates in every such Q . To make the ensuing coordinates precise, write

$$\mathbf{T}_0 = \mathbf{D}([a_0, a_0 + \Delta) \times [b_0, b_0 + \Delta)).$$

The affine map $F(x, y) = (x - a_0 y - b_0, y)$ sends this tube to $\mathbf{D}([0, \Delta)^2)$ and translates every tube-parameter cell by $(-a_0, -b_0)$. Its condition number is absolute. Images of spatial dyadic squares are parallelograms of diameter $O(r)$ or $O(\Delta)$; cover each by $O(1)$ dyadic squares of the same scale and retain a most populous covering cell. The covering has bounded overlap in both directions, so all cardinality, density, and Frostman estimates above survive with absolute changes. We henceforth relabel the images and assume $\mathbf{T}_0 = \mathbf{D}([0, \Delta)^2)$.

Write $Q = [x_Q, x_Q + \Delta) \times [y_Q, y_Q + \Delta)$. Since all Q meet the normalised vertical tube, at most an absolute number have a given y_Q . After thinning,

$$Y = \{y_Q : Q \in \mathcal{Q}(\mathbf{T}_0)\}$$

is a $(\Delta, u - s, r^{-C_c})$ -set. The projection claim gives a set $\Pi^Q \subset [0, 1]$ of left endpoints of Δ -intervals in the first coordinate of $S_Q(\mathcal{P}_Q^0)$. For every $\xi \in \Pi^Q$, choose one $p \in \mathcal{P}_Q^0$ whose projected cell has that left endpoint, and put

$$x_p = x_Q + \Delta\xi, \quad \Pi_Q = \{x_p : \xi \in \Pi^Q\}, \quad X_{y_Q} = \Delta^{-1}\Pi_Q = \Pi^Q + x_Q/\Delta.$$

Thus every X_y is a (Δ, s, r^{-C_c}) -set. Since Q meets the normalised vertical tube, x_Q/Δ ranges in a fixed bounded interval, so the same is true of all X_y . Put

$$Z = \bigcup_{y \in Y} X_y \times \{y\}.$$

For $z = (\Delta^{-1}x_p, y_Q) \in Z$, use the fine tubes in $\mathcal{T}(p) \cap \mathbf{T}_0$ from (14.10.29). If $T = \mathbf{D}([a, a + r) \times [b, b + r))$ meets $p \subset Q$, then for suitable $(x, y) \in p \cap T$ and $(a', b') \in [a, a + r) \times [b, b + r)$,

$$x_p = a'y_Q + b' + O(r + \Delta^2) = a'y_Q + b' + O(r),$$

because $0 \leq a', b' < \Delta$ inside $\mathbf{T}_0$ and $\Delta^2 = r$. After the horizontal dilation $(x, y) \mapsto (\Delta^{-1}x, y)$, this becomes a point-tube incidence at width $O(\Delta)$ with parameters $(\Delta^{-1}a, \Delta^{-1}b)$. Cover by at most four dyadic Δ -tubes and choose one containing z . These choices have bounded multiplicity. The resulting slope family has size $\gtrsim_c \Delta^{-s}$. If I is an interval of length $\rho \geq \Delta$ in the rescaled

slope variable, its preimage has length $\Delta\rho$; hence the original slope Frostman estimate gives at most

$$r^{-c}M(\Delta\rho)^s \lesssim_c \Delta^{-s}\rho^s \lesssim_c \#\mathcal{T}(z)\rho^s$$

members. Lemma 14.10.2 now shows that every z carries a (Δ, s, r^{-C_c}) -family of incident tubes.

All resulting tubes arise, with multiplicity at most four, from the universal family $\mathcal{T} \cap \mathbf{T}_0$. By (14.10.22), their union has cardinality

$$\lesssim_c \Delta^{-2s}. \quad (14.10.30)$$

Corollary 14.9.15, at scale Δ and with $\tau = u - s > 0$, gives instead at least $\Delta^{-2s-\eta(s,u-s)}$ tubes. To meet the harmless $[0, 1]$ normalisation in that corollary, partition the fixed bounded interval containing the X_y into $O(1)$ unit intervals, retain a fixed fraction of each X_y in one interval, and pigeonhole a fixed fraction of the levels y using the same interval; a translation then puts all retained fibres in $[0, 1]$. Choose $c > 0$ so small that every $r^{-C_c} = \Delta^{-2C_c}$ loss above is bounded by $\Delta^{-\eta/4}$. Then (14.10.30) is at most $\Delta^{-2s-\eta/2}$ for small r , contradicting the corollary's lower bound. This proves the regular alternative. On a compact parameter set, $u - s$ and $1 - s$ stay uniformly away from zero, and the compact-uniform form of Corollary 14.9.15 supplies a common η . Taking the finite minimum of the remaining smallness conditions proves the uniform clause. $\square$

14.10.7 The full regular-scale estimate

Theorem 14.10.10 (Incidence improvement for regular point sets). *Fix $0 < s < 1$ and $s < u \leq 2$. There is $c_1 = c_1(s, u) > 0$ such that, for every sufficiently small dyadic r whose square root is dyadic, if $\mathcal{P}$ is $(r, u, r^{-c_1}, r^{-c_1})$ -regular and every $p \in \mathcal{P}$ carries an (r, s, r^{-c_1}) -set $\mathcal{T}(p)$ of incident tubes, then*

$$\#\bigcup_p \mathcal{T}(p) \geq r^{-2s-c_1}.$$

Moreover, c_1 and the small-scale threshold may be chosen uniformly when (s, u) ranges over a compact subset of $\{(s, u) : 0 < s < 1, s < u \leq 2\}$.

Proof. Let $c > 0$ be the exponent in Proposition 14.10.9, and choose an auxiliary $0 < \lambda \ll c$. Trimming and pigeonholing reduce to

$$\#\mathcal{P} \asymp r^{-u+O(\lambda)}, \quad \#\mathcal{T}(p) = M \asymp r^{-s+O(\lambda)}.$$

Put $R = r^{1/2}$. Remove light R -squares so that every occupied Q has $\#(\mathcal{P} \cap Q) \asymp R^{-u+O(\lambda)}$. Apply Proposition 14.10.7. Its coarse point family is an

$(R, u, r^{-O(\lambda)})$ -set, and every rescaled local point family is also an $(R, u, r^{-O(\lambda)})$ -set. Put $u_b = \min\{u, 1\}$. Both point families are also u_b -Frostman sets. Therefore Proposition 14.10.4 gives, for a constant $C = C(s, u)$,

$$\frac{\#\mathcal{T}_R}{M_R} \gtrsim r^{C\lambda} R^{-s} (M_R R^s)^q, \quad q = \frac{u_b - s}{1 - s} > 0, \quad (14.10.31)$$

$$\frac{\#\mathcal{T}_Q}{M_Q} \gtrsim r^{C\lambda} R^{-s} (M_Q R^s)^q \geq r^{C\lambda} R^{-s}. \quad (14.10.32)$$

In the second line we used the lower cardinality bound $M_Q R^s \geq r^{C'\lambda}$ supplied by the tube Frostman condition and then enlarged C to absorb the additional factor $(M_Q R^s)^q$.

Apply Proposition 14.10.9 to the retained fine-scale configuration. It remains regular because the factorisation loses only a logarithmic proportion in every occupied R -square. If the first alternative holds, there is nothing to prove. Otherwise

$$\#\mathcal{T}_R \geq r^{-s-c/2} = R^{-2s-c}, \quad (14.10.33)$$

after decreasing c to absorb dyadic-cover constants.

Set $x = M_R R^s > 0$. The first line above and (14.10.33) give, for an absolute C ,

$$\frac{\#\mathcal{T}_R}{M_R} \gtrsim R^{-s} \max\{R^{2C\lambda} x^q, R^{-c}/x\}.$$

The minimum of the maximum over $x > 0$ occurs where its two entries agree. Consequently

$$\max\{R^{2C\lambda} x^q, R^{-c}/x\} \geq R^{(2C\lambda - cq)/(q+1)}.$$

Choose $\lambda \leq cq/(4C)$ and put $c_2 = cq/[2(q+1)]$. After making r small enough to absorb the absolute implicit constant, we obtain

$$\frac{\#\mathcal{T}_R}{M_R} \geq R^{-s-c_2}, \quad (14.10.34)$$

which is the required coarse-scale gain.

Now combine (14.10.32), (14.10.34), and (14.10.9):

$$\#\mathcal{T} \gtrsim r^{C\lambda} M R^{-2s-c_2} \geq r^{-2s-c_2/3}$$

once $\lambda \ll c_2$. Taking $c_1 < c_2/3$ proves the theorem. On a compact parameter set, the alternative has a common exponent, while $q = (\min\{u, 1\} - s)/(1 - s)$ has a positive lower bound. The displayed choices of λ, c_2, c_1 can therefore be made uniformly, proving the last assertion. $\square$

14.10.8 Uniform sets and preservation under refinement

Let

$$r = R_n < R_{n-1} < \cdots < R_1 < R_0 = 1$$

be dyadic scales. A set P is $(R_j)_{j=1}^n$ -uniform if there are integers N_j such that every occupied R_{j-1} -square contains exactly N_j occupied R_j -squares.

Lemma 14.10.11 (Bottom-to-top uniformisation). *Every finite union P of r -squares contains an $(R_j)_{j=1}^n$ -uniform subset P' satisfying*

$$|P'|_r \geq [4n^{-1} \log(1/r)]^{-n} |P|_r. \quad (14.10.35)$$

Proof. At the bottom scale, pigeonhole the occupied R_{n-1} -squares according to the dyadic range containing their number of r -children. One range carries at least $[2 \log(R_{n-1}/r)]^{-1}$ of all r -cells. In every parent in that range, retain exactly the lower endpoint N_n of the range; this loses at most an additional factor two. Proceed upward. At stage j , retain whole R_j -subtrees, so the uniformities already established below R_j are not disturbed. The total retention is at least

$$\prod_{j=1}^n [4 \log(R_{j-1}/R_j)]^{-1} \geq [4n^{-1} \log(1/r)]^{-n},$$

where the last inequality is the arithmetic–geometric mean inequality. $\square$

We say that a uniform set P is an (u, D) -set between R_j and R_{j-1} if $S_Q(P \cap Q)$ is an $(R_j/R_{j-1}, u, D)$ -set for every occupied R_{j-1} -square Q . It is (u, D, K) -regular between those scales if those rescaled sets are regular in the sense of Definition 14.10.8.

Lemma 14.10.12 (Refinement preserves scale information). *Suppose P is $(R_j)_{j=1}^n$ -uniform and $P_0 \subset P$ has $|P_0|_r \geq L^{-1} |P|_r$. There is a uniform $P_1 \subset P_0$ with*

$$|P_1|_r \geq M^{-1} |P|_r, \quad M = L[4 \log(1/r)]^n,$$

such that every between-scale Frostman constant of P increases by at most M , while every between-scale regular-cover upper bound is unchanged.

Proof. Apply Lemma 14.10.11 to P_0 . Let N_j and N'_j be the branching numbers of P and the resulting P_1 . Since $N'_j \leq N_j$ and

$$\prod_j N'_j = |P_1|_r \geq M^{-1} |P|_r = M^{-1} \prod_j N_j,$$

each $N'_j \geq M^{-1} N_j$. Thus, inside every parent, P_1 retains at least an M^{-1} fraction of the children of P at the next selected scale. Dividing the old local

Frostman inequality by this lower density multiplies its constant by at most M . Since $P_1 \subset P$, any upper bound on the number of half-scale cells remains valid. $\square$

14.10.9 A self-contained decomposition of the branching function

Lemma 14.10.13 (Long affine pieces). *For every $L \geq 1$ and $0 < \varepsilon < 1$ there is $\tau = \tau(L, \varepsilon) > 0$ such that every L -Lipschitz function $f : [0, m] \rightarrow \mathbb{R}$ admits disjoint intervals I_j with*

1. $|I_j| \geq \tau m$;

2. on $I_j = [a_j, b_j]$,

$$|f(x) - f(a_j) - s_j(x - a_j)| \leq \varepsilon |I_j|, \quad s_j = \frac{f(b_j) - f(a_j)}{|I_j|};$$

3. the intervals cover all but at most εm of $[0, m]$.

Proof. By rescaling, take $m = 1$. For a dyadic interval I , let s_I be its endpoint slope and define the slope energy at generation k by

$$E_k = \sum_{I \in \mathcal{D}_{2^{-k}}([0,1])} |I| s_I^2.$$

The identity

$$\frac{1}{2}(s_{I_-}^2 + s_{I_+}^2) - s_I^2 = \frac{1}{4}(s_{I_-} - s_{I_+})^2$$

shows that E_k is nondecreasing; it is bounded by L^2 .

Choose an integer h with $4L2^{-h} < \varepsilon/2$. If f is not ε -linear on a dyadic interval I , choose a point where its distance from the endpoint chord exceeds $\varepsilon|I|$. A closest generation- h grid point in I still has deviation at least $\varepsilon|I|/2$. Let d_J be the endpoint slopes of the generation- h descendants $J \subset I$. At a grid point the deviation is a partial sum of $|J|(d_J - s_I)$. Cauchy–Schwarz therefore gives

$$\sum_{\substack{J \subset I \\ |J|=2^{-h}|I|}} |J|(d_J - s_I)^2 \geq \frac{\varepsilon^2}{4}|I|.$$

The left side is exactly the increase of slope energy inside I from its top to its generation- h descendants.

Inspect generations $0, h, 2h, \dots$. If at each of the first N such generations the non- ε -linear intervals had total length at least ε , the energy would increase

by at least $N\varepsilon^3/4$. Taking $N > 4L^2\varepsilon^{-3}$ is impossible. Hence at one of these generations the good intervals cover length greater than $1 - \varepsilon$. They all have length at least $\tau = 2^{-Nh}$ and satisfy the required estimate. $\square$

For an interval $[a, b]$, write $s_f(a, b) = (f(b) - f(a))/(b - a)$. We say f is ε -superlinear there if

$$f(x) \geq f(a) + s_f(a, b)(x - a) - \varepsilon(b - a) \quad (a \leq x \leq b).$$

Lemma 14.10.14 (Merging low-slope pieces). *Fix $0 < s < t \leq 2$. For every sufficiently small $\varepsilon > 0$ there is $\tau = \tau(s, t, \varepsilon) > 0$ such that the following holds. If f is 2-Lipschitz on $[0, m]$, $f(0) = 0$, and*

$$f(x) \geq tx - \varepsilon m \quad (0 \leq x \leq m), \quad (14.10.36)$$

then there are disjoint intervals $[a_j, b_j]$, each of length at least τm , covering all but $C_{s,t}\varepsilon m$, such that either

1. *f is $C\varepsilon$ -linear on $[a_j, b_j]$ and $s_f(a_j, b_j) \geq s$; or*
2. *f is $C\varepsilon$ -superlinear on $[a_j, b_j]$ and $s_f(a_j, b_j) = s$.*

Proof. Apply Lemma 14.10.13 with error ε^2 and list the resulting intervals $[c_j, d_j]$ from left to right. Intervals whose chord slope is at least s already satisfy the first alternative. If there is a low-slope interval, let $[c_k, d_k]$ be the last one.

If $d_k \leq 2\varepsilon m/(t - s)$, discard everything to its left; this costs $O_{s,t}(\varepsilon m)$. Otherwise

$$s_f(0, d_k) \geq t - \varepsilon m/d_k > s.$$

By continuity there is a largest $c' < c_k$ with $s_f(c', d_k) = s$. For $c' < x < c_k$, maximality of c' implies $s_f(x, d_k) < s$; since $s_f(c', d_k)$ is the length-weighted average of $s_f(c', x)$ and $s_f(x, d_k)$, this gives $f(x) > f(c') + s(x - c')$. For $c_k \leq x \leq d_k$, the ε^2 -linearity on $[c_k, d_k]$ gives

$$f(x) \geq f(d_k) - s_f(c_k, d_k)(d_k - x) - \varepsilon^2(d_k - c_k) > f(c') + s(x - c') - \varepsilon^2(d_k - c_k).$$

Since $d_k - c_k \leq d_k - c'$, the new interval is ε^2 -superlinear and has chord slope exactly s . On a later pass the low-slope working interval may be a truncated one with 3ε -linearity rather than an original one with ε^2 -linearity. The identical calculation then gives 3ε -superlinearity on the new compound interval, which is still the asserted bound.

Replace all original intervals met by $(c', d_k]$ with this new interval. If c' lies in a gap, continue with the intervals to its left. If it lies in an interval $[c_\ell, d_\ell]$, then either $c' - c_\ell \leq \varepsilon(d_\ell - c_\ell)$, in which case discard that short initial

piece, or retain $[c_\ell, c']$. In the latter case, restricting an ε^2 -linear function to a subinterval of relative length at least ε makes it 3ε -linear. Return this truncated interval to the working list: if its new chord slope is below s , the next pass merges it as well; otherwise it is a final interval of the first type. Continue to the left.

Every pass consumes at least one of the original long intervals, so there are at most τ_0^{-1} passes, where τ_0 is supplied by Lemma 14.10.13. The discarded portions total $O_{s,t}(\varepsilon m)$, and every retained interval contains at least an ε fraction of an original interval. Hence its length is at least $\varepsilon\tau_0 m$. Each compound interval obtains its superlinearity estimate from its final low-slope constituent, so errors do not accumulate across passes. An original constituent has error ε^2 , while a previously truncated constituent has error $O(\varepsilon)$; in either case the resulting compound interval has error $O(\varepsilon)$ relative to its own length. This proves the stated form. $\square$

Let $0 < \Delta < 1$ be dyadic, let $r = \Delta^m$, and suppose P is uniform for the scales $1, \Delta, \dots, \Delta^m$. If N_i is the common number of Δ^i -children in each occupied Δ^{i-1} -square, define its branching function by

$$f(0) = 0, \quad f(j) = \sum_{i=1}^j \frac{\log N_i}{\log(1/\Delta)}, \quad (14.10.37)$$

and interpolate linearly. Since $1 \leq N_i \leq \Delta^{-2}$, f is 2-Lipschitz.

Lemma 14.10.15 (Branching-function dictionary). *Suppose f is ε -superlinear on an integer interval $[a, b]$ and put $v = s_f(a, b)$. Then P is a $(v, C_\Delta(\Delta^a/\Delta^b)^\varepsilon)$ -set between Δ^b and Δ^a . If f is ε -linear there, then P is also regular between those scales with the same form of constant.*

Proof. Rescale an occupied Δ^a -square to the unit square and put $m' = b - a$. The descendant branching function is $g(j) = f(a + j) - f(a)$. For an occupied Δ^j -square inside it, uniformity and superlinearity give

$$\#\{\Delta^{m'}\text{-descendants in the square}\} = \Delta^{-g(m')+g(j)} \leq \Delta^{-\varepsilon m'} \Delta^{jv} \Delta^{-g(m')}.$$

This is the Frostman inequality at the grid scales; an arbitrary intermediate dyadic scale lies within a factor Δ^{-1} of one of them.

If f is linear to error $\varepsilon m'$, evaluate at $j = \lceil m'/2 \rceil$. The number of occupied half-scale cells is at most

$$\Delta^{-g(j)} \leq C_\Delta \Delta^{-jv} \Delta^{-\varepsilon m'} \leq C_\Delta (\Delta^{m'})^{-v/2-\varepsilon},$$

which is the regularity condition. $\square$

Proposition 14.10.16 (Multiscale decomposition at point dimension one). *Fix $0 < s < 1$ and $\varepsilon > 0$. There is $\tau = \tau(s, \varepsilon) > 0$ such that the following holds when m is sufficiently large. Let $r = \Delta^m$ and let P be uniform at the powers of Δ and be an $(r, 1, r^{-\varepsilon})$ -set. Then there are scales*

$$r = R_n < \cdots < R_1 < R_0 = 1,$$

each a power of Δ , exponents $v_j \in [s, 2]$, and a partition into structured and bad indices $\mathcal{S} \sqcup \mathcal{B}$ such that

1. $R_{j-1}/R_j \geq r^{-\tau}$ for $j \in \mathcal{S}$, while

$$\prod_{j \in \mathcal{B}} R_{j-1}/R_j \leq r^{-C_s \varepsilon}; \quad (14.10.38)$$

2. for $j \in \mathcal{S}$, P is a $(v_j, (R_{j-1}/R_j)^{C_s \varepsilon})$ -set between the two scales, and it is regular there whenever $v_j > s$;

- 3.

$$\prod_{j \in \mathcal{S}} (R_{j-1}/R_j)^{v_j} \geq r^{-1+C_s \varepsilon}; \quad (14.10.39)$$

4. no two bad indices are consecutive, and $n \leq 2\tau^{-1} + 1$.

Proof. Let f be the branching function. If Q is an occupied Δ^i -square, the Frostman condition gives

$$N_{i+1} \cdots N_m \leq r^{-\varepsilon} \Delta^i |P|_r.$$

After cancelling the full product $N_1 \cdots N_m$, this becomes $f(i) \geq i - \varepsilon m$; the same holds between integers by linearity. Apply Lemma 14.10.14 with $t = 1$. Perturb the endpoints to integers, changing every error by $O(1/m)$, and take m sufficiently large.

The selected intervals $[a_i, b_i]$ produce structured scale intervals $[\Delta^{b_i}, \Delta^{a_i}]$; the gaps between them, and the two endpoint gaps, produce the bad intervals. Their total logarithmic length is at most $C_s \varepsilon m$, which is exactly (14.10.38). Every structured interval has length at least τm , so its scale ratio is at least $r^{-\tau}$. The dictionary lemma gives the Frostman and regularity claims, with its constant absorbed by starting with ε/C_s .

Since f is 2-Lipschitz and the gaps have total length $O_s(\varepsilon m)$,

$$\sum_i (b_i - a_i) v_i = \sum_i [f(b_i) - f(a_i)] \geq f(m) - C_s \varepsilon m \geq (1 - C_s \varepsilon) m.$$

Exponentiating proves (14.10.39). A gap occurs only between two selected intervals or at an endpoint, so bad intervals cannot be consecutive. There are at most τ^{-1} selected intervals and at most one more gap than selected intervals, proving the bound for n . $\square$

14.10.10 Combining information from many scales

Proposition 14.10.17 (Scale-combination estimate). *Fix $0 < s < u_0 \leq 1$, $0 < \tau < 1$, and an integer n . There are $\eta = \eta(s, u_0) > 0$ and constants $C, C' > 0$ with the following property. Let $(\mathcal{P}, \mathcal{T})$ be an $(r, s, r^{-\lambda}, M)$ -nice configuration, let*

$$r = R_n < R_{n-1} < \cdots < R_1 < R_0 = 1,$$

and suppose $\mathcal{P}$ is uniform for these scales. Partition the indices into normal, good, and bad sets $\mathcal{N}, \mathcal{G}, \mathcal{B}$. Assume $R_j/R_{j-1} \leq r^\tau$ for $j \in \mathcal{N} \cup \mathcal{G}$, and:

1. *if $j \in \mathcal{N}$, then $\mathcal{P}$ is an $(s, L(R_{j-1}/R_j)^{\epsilon_N})$ -set between the two scales;*
2. *if $j \in \mathcal{G}$, then for some $u_j \in [u_0, 2]$, $\mathcal{P}$ is $(u_j, L(R_{j-1}/R_j)^{\epsilon_G}, L(R_{j-1}/R_j)^{\epsilon_G})$ -regular there;*

where $L = [\log(2/r)]^{C_P}$, $0 < \epsilon_N \leq \epsilon_G$, and ϵ_G, η are sufficiently small in terms of s, u_0 . If λ is sufficiently small in terms of the displayed fixed parameters, then

$$\#\mathcal{T} \geq [\log(2/r)]^{-C} M r^{C'\lambda} r^{-s+\epsilon_N} \prod_{j \in \mathcal{G}} (R_{j-1}/R_j)^\eta \prod_{j \in \mathcal{B}} (R_j/R_{j-1}). \quad (14.10.40)$$

Proof. We first establish the three one-scale estimates. Write the single scale as ρ . At a bad scale, the trivial inequality $\#\mathcal{T} \geq M$ is stronger than $M\rho^{-s+\epsilon_N}\rho$, since $s < 1$ and ϵ_N is small. At a normal scale, Proposition 14.10.4 with point exponent s gives

$$\#\mathcal{T} \geq [\log(2/\rho)]^{-C} M \rho^{C\lambda} \rho^{-s+\epsilon_N}. \quad (14.10.41)$$

At a good scale the point family is, in particular, an u_0 -Frostman set, so the elementary estimate may be used with $q = (u_0 - s)/(1 - s) > 0$. For the small-fibre case, apply Theorem 14.10.10 at the actual regularity exponent u_j . Its compact-uniform clause supplies a common exponent $c_0 > 0$ for all $u_j \in [u_0, 2]$. Choose $\gamma = c_0/3$ and then

$$0 < \eta < \min\{q\gamma/4, c_0/4\}. \quad (14.10.42)$$

If $M \geq \rho^{-s-\gamma}$, the elementary estimate at exponent u_0 gives

$$\#\mathcal{T} \geq M \rho^{-s-q\gamma+O(\epsilon_G+\lambda)} \geq M \rho^{-s-\eta}.$$

If $M < \rho^{-s-\gamma}$, the regular theorem at exponent u_j gives

$$\#\mathcal{T} \geq \rho^{-2s-c_0} \geq M \rho^{-s-\eta}.$$

Thus

$$\#\mathcal{T} \geq M\rho^{-s-\eta} \quad (14.10.43)$$

at every good scale, after taking ϵ_G and λ sufficiently small. This proves Proposition 14.10.17 when $n = 1$.

Assume the proposition proved for fewer than n scales. Put $R = R_1$ and $\bar{r} = r/R$. Apply Proposition 14.10.7. It gives a coarse configuration $(\mathcal{P}_R, \mathcal{T}_R)$ at scale R and, for each coarse square Q , a configuration $(\mathcal{P}^Q, \mathcal{T}_Q)$ at scale $\bar{r}$, with

$$\frac{\#\mathcal{T}}{M} \gtrsim_{\log r} \frac{\#\mathcal{T}_R}{M_R} \frac{\#\mathcal{T}_Q}{M_Q}. \quad (14.10.44)$$

By Lemma 14.10.12, we may make $\mathcal{P}^Q$ uniform for the normalised scales $R_2/R, \dots, R_n/R$ while multiplying every Frostman constant by only a power of $\log(2/r)$.

If all indices $2, \dots, n$ are bad, use $\#\mathcal{T}_Q/M_Q \geq 1$. The product of their bad factors is r/R , and

$$1 \geq \bar{r}^{-s+\epsilon_N} \bar{r}$$

because $s < 1$. The desired estimate then follows from the appropriate one-scale estimate at R and (14.10.44).

Otherwise some later index is normal or good. Hence $\bar{r} \leq r^\tau$. Consequently

$$r^{-C\lambda} [\log(2/r)]^C \leq \bar{r}^{-C_\tau\lambda}$$

for small r . Thus the local tube Frostman constant delivered by Proposition 14.10.7 has the form required by the induction hypothesis, with λ replaced by $C_\tau\lambda$. The hypotheses between the remaining scales are preserved by Lemma 14.10.12. Applying the induction hypothesis gives

$$\frac{\#\mathcal{T}_Q}{M_Q} \geq [\log(2/r)]^{-C} \bar{r}^{-C'\lambda} \bar{r}^{-s+\epsilon_N} \prod_{\substack{j \in \mathcal{G} \\ j \geq 2}} (R_{j-1}/R_j)^\eta \prod_{\substack{j \in \mathcal{B} \\ j \geq 2}} (R_j/R_{j-1}). \quad (14.10.45)$$

For the first scale, the three one-scale estimates say respectively

$$\begin{aligned} \frac{\#\mathcal{T}_R}{M_R} &\geq 1, & 1 \in \mathcal{B}, \\ \frac{\#\mathcal{T}_R}{M_R} &\geq [\log(2/r)]^{-C} r^{C\lambda} R^{-s+\epsilon_N}, & 1 \in \mathcal{N}, \\ \frac{\#\mathcal{T}_R}{M_R} &\geq R^{-s-\eta}, & 1 \in \mathcal{G}. \end{aligned}$$

In the bad case, the missing factor is legitimate because $1 \geq R^{-s+\epsilon_N} R$. Multiply the applicable line by (14.10.45) and use $R^{-s+\epsilon_N} \bar{r}^{-s+\epsilon_N} = r^{-s+\epsilon_N}$. Then (14.10.44) gives (14.10.40), after increasing C, C' by constants depending on n and τ . This completes the induction. $\square$

14.10.11 Proof of the dyadic theorem

Proof of Theorem 14.10.1. Fix

$$u_0 = \frac{1+s}{2}, \quad \kappa_s = \frac{1-u_0}{2-u_0} > 0,$$

and let $\eta > 0$ be the gain in Proposition 14.10.17 for (s, u_0) . We now specify the order of all small parameters. Choose a number $\varepsilon_G > 0$ small enough for that proposition. Next choose

$$0 < \varepsilon \ll_s \varepsilon_G \eta. \quad (14.10.46)$$

Let $\tau = \tau(s, \varepsilon)$ and $n_0 = 2\tau^{-1} + 1$ be supplied by Proposition 14.10.16, and only then choose

$$0 < \lambda \ll_{s, \tau, n_0} \varepsilon_G \eta.$$

Choose the base dyadic $\Delta = 2^{-2K}$ sufficiently small in terms of ε ; then every ratio of two powers of Δ has dyadic square root, as required by the regular-scale theorem. Finally choose

$$0 < \epsilon_0 < \min\{\varepsilon/4, \lambda/4, \kappa_s \eta/4\}.$$

All subsequent small-scale thresholds are imposed by decreasing r_0 .

First suppose $r = \Delta^m$. Apply Lemma 14.10.11 to the powers of Δ . Since

$$[4m^{-1} \log(1/r)]^{-m} = [4 \log(1/\Delta)]^{-m} = r^{\zeta(\Delta)}, \quad \zeta(\Delta) = \frac{\log(4 \log(1/\Delta))}{\log(1/\Delta)},$$

and $\zeta(\Delta) \rightarrow 0$ as $\Delta \rightarrow 0$, the retained set is still an $(r, 1, r^{-\varepsilon})$ -set after choosing $\zeta(\Delta) < \varepsilon/2$ and recalling $\epsilon_0 < \varepsilon/4$. Pigeonholing the fibre cardinalities and trimming them makes $(\mathcal{P}, \mathcal{T})$ an $(r, s, r^{-\lambda}, M)$ -nice configuration with

$$M \geq r^{-s+\lambda}. \quad (14.10.47)$$

Apply Proposition 14.10.16. Split its structured indices into

$$\mathcal{G} = \{j : v_j \geq u_0\}, \quad \mathcal{N} = \mathcal{S} \setminus \mathcal{G}.$$

At a normal scale, the v_j -Frostman estimate implies the s -Frostman estimate. At a good scale, retain the actual exponent $v_j \in [u_0, 2]$ in the regularity conclusion; this is exactly the range allowed in Proposition 14.10.17. Hence all assumptions of Proposition 14.10.17 hold, with $\epsilon_N = C_s \varepsilon$ and ϵ_G as chosen in (14.10.46).

There must be a quantitatively large product of good ratios. Put $G = \prod_{j \in \mathcal{G}} R_{j-1}/R_j$ and $N = \prod_{j \in \mathcal{N}} R_{j-1}/R_j$. Since $v_j \leq 2$ on good scales and $v_j < u_0$ on normal scales, (14.10.39) gives

$$r^{-1+C_s \varepsilon} \leq G^2 N^{u_0}.$$

Also $GN \leq r^{-1}$, since the omitted bad ratios are at least one. Thus

$$G \geq r^{-\kappa_s + C_s \varepsilon}. \quad (14.10.48)$$

Insert (14.10.47), (14.10.38), and (14.10.48) into (14.10.40). We obtain

$$\#\mathcal{T} \geq [\log(2/r)]^{-C} r^{-2s} r^{(C'+1)\lambda + C_s \varepsilon + \epsilon_N} r^{-\kappa_s \eta + C_s \varepsilon \eta}.$$

The order (14.10.46) makes the total positive loss in the exponent less than $\kappa_s \eta/2$. Absorbing the logarithm gives

$$\#\mathcal{T} \geq r^{-2s - \kappa_s \eta/3}.$$

Our prior choice of ϵ_0 is smaller than this gain and supports every preceding application. This proves the theorem for scales $r = \Delta^m$.

For an arbitrary small dyadic r , choose m with $\Delta^{m+1} < r \leq \Delta^m =: \tilde{r}$. Replace each r -square and r -tube by one of the $O_\Delta(1)$ dyadic $\tilde{r}$ objects covering it, and pigeonhole the choices. Frostman constants and cardinalities change by $O_\Delta(1)$, while $\tilde{r} \asymp_\Delta r$. Decreasing ϵ_0 once absorbs those constants and proves the stated dyadic theorem.

For s in a compact subinterval of $(0, 1)$, the gaps $u_0 - s = (1 - s)/2$ and κ_s have positive lower bounds. The compact- uniform clauses in the regular theorem and the Bourgain product obstruction, followed by a finite cover of the parameter interval, allow the same finite hierarchy of minima. This proves uniformity. $\square$

14.10.12 From dyadic tubes to the radial incidence input

Proof of Theorem 14.8.1. Fix $0 < a < 1$ and let $\epsilon_0(a)$ be supplied by Theorem 14.10.1. Choose

$$e(a) < \epsilon_0(a)/(100C_0), \quad (14.10.49)$$

where C_0 is a fixed absolute constant larger than every loss in the round-to-dyadic comparisons below. Let $P \subset B(0, 3)$ and the round r -tube families $\mathcal{T}_x$ satisfy the hypotheses of Theorem 14.8.1 with $s = a$. Replace r by a comparable dyadic scale r_d . Cover the bounded region by finitely many unit squares and retain one containing a fixed fraction of P ; translate and dilate

it to $[0, 1]^2$. Lemma 14.10.3 and a bounded-overlap assignment of points to dyadic r_d -squares produce an $(r_d, 1, r_d^{-2e(a)})$ family $\mathcal{P}$.

First apply Lemma 14.10.3 in line space to make every fibre r_d -separated. Cover each retained round tube, inside the bounded reference square, by at most an absolute number of dyadic Cr_d -tubes in one of two slope charts. At each point, retain a common chart and a common covering choice carrying a fixed fraction of the family. Because the axes were r_d -separated and parameter coordinates are uniformly bi-Lipschitz in the selected chart, any one dyadic parameter cell receives only $O(1)$ round tubes. Lemma 14.10.2, followed by trimming, now produces an $(Cr_d, a, r_d^{-Ce(a)})$ family of dyadic tubes meeting the assigned square. The bounded multiplicities are harmless because of (14.10.49).

The dyadic theorem now yields at least

$$r_d^{-2a-\epsilon_0(a)/2}$$

distinct dyadic tubes. Each selected dyadic tube has an assigned original round tube, and the separation just noted makes this assignment bounded-to-one after passing to r -covering cells in line space. Therefore the round union has r -covering number at least a constant times the number of distinct dyadic tubes. Since $r_d \asymp r$ and (14.10.49) leaves a fixed exponent margin, decreasing the small-scale threshold gives

$$\mathcal{E}_r \left(\bigcup_{x \in P} \mathcal{T}_x \right) \geq r^{-2a-e(a)}.$$

The compact-uniform assertion follows from the compact-uniform conclusion of Theorem 14.10.1 and the absolute nature of the dyadic reductions. $\square$

14.10.13 Dependency closure

The proof just completed uses no survey-level or external black box. Its sole research input is Corollary 14.9.15, proved in the preceding chapter from the robust discretised projection theorem. All remaining ingredients are finite counting, energy, dyadic pigeonholing, and the explicitly proved branching-function decomposition. Therefore the forward reference in the radial-bootstrap chapter is now discharged.

Chapter 14.11

Why the Proof Is Three-Dimensional

The preceding dossiers prove the Kakeya set conjecture in $\mathbb{R}^3$. They do not claim a higher-dimensional theorem. This chapter is a dependency audit: it identifies the exact places where the proof uses dimension three and separates those facts from non-load-bearing historical context.

14.11.1 Context above dimension three

The Kakeya set conjecture remains open in dimensions at least four. Two frequently cited benchmarks are the Katz–Zahl planebrush lower bound in $\mathbb{R}^4$ and the Katz–Tao arithmetic lower bounds in high dimension. They are mentioned only to locate the theorem proved here; neither statement is used anywhere in this volume. Their original papers, not this contextual paragraph, are the sources for their precise hypotheses and numerical forms.

This distinction matters. A contextual record may change as the literature develops, whereas the proof chain in Chapters 14.2 and 14.3 is a fixed mathematical implication. The theorem ledger therefore classifies the two higher-dimensional benchmarks as context, not as internally proved results.

14.11.2 The dimensional audit

The proof can be divided into the following layers.

Layer	Role in the proof	Dimensional status
Discretisation and density selection	turn a small Kakeya set into a finite tube configuration	formally available in every dimension
Convex–Wolff and Assouad bridge	extract separated scales and self-similar structure	the version proved here is tailored to the three-dimensional reduction
Multilinear incidence control	rules out sufficiently transverse concentration	higher-dimensional analogues exist, but produce different narrow objects
Planiness and grains	place the narrow configuration in slab-like cells convert grain	the useful geometry is specifically plane geometry in $\mathbb{R}^3$
Projection and slope regularity	structure into a controlled planar family	uses two-dimensional projection theory
Cinematic maximal and lens bounds	count the final projected curve configuration	proved here for planar curves, not higher-dimensional surfaces

The first genuine bottleneck is not the existence of a hyperplane containing many directions. Multilinear Kakeya can still force a narrow alternative in higher dimensions. The problem is what happens next. In three dimensions a codimension-one narrow object is a plane, two separated grains project to curves in a plane, and the curvature and two-ends hypotheses can be expressed by scalar determinants. The complete cinematic proof in Chapter 14.6 and the complete lens count in Chapter 14.7 use precisely this structure.

In dimension $n \geq 4$, the projected objects are no longer curves and the pairwise lens geometry no longer captures all possible tangencies. A direct extension would require both of the following new ingredients:

1. a projection theorem for the relevant subsets of $\mathbb{R}^{n-1}$ with a robust exceptional-set estimate at every scale used by the grain decomposition;
2. a maximal or incidence theorem for curved families of positive-dimensional submanifolds, with a quantitative replacement for the planar cinematic-curvature and lens hypotheses.

Neither ingredient follows formally from its planar counterpart. In particular, replacing every occurrence of “plane” by “hyperplane” leaves the final counting problem underdetermined.

14.11.3 What does transfer

Several mechanisms in the proof are reusable even though the theorem is not. Finite-scale uniformisation, density pigeonholing, convex-cover estimates, and the abstract principle of extracting a self-similar block do not depend on planar curve geometry. The robust projection chapter also isolates a general lesson: a dense-subset conclusion is essential when an induction on scales repeatedly refines the configuration.

These transferable mechanisms should not be confused with a partial proof in higher dimension. The terminal contradiction in the sticky theorem still passes through the two-dimensional cinematic and lens estimates, so the logical output of this volume remains exactly the three-dimensional theorem.

14.11.4 A concrete research checklist

A higher-dimensional analogue of the present strategy would have to verify, in order:

1. a stable narrow-object description after every rescaling;
2. a grain decomposition whose descendants retain quantitative transversality;
3. a higher-dimensional robust projection theorem compatible with dense restrictions;
4. a replacement for the planar cinematic-curvature condition;
5. a lens or multiplicity estimate strong enough to close the final exponent inequality.

The first two items extend existing structural ideas. The last three are new theorems, not bookkeeping modifications. This is the precise sense in which the proof is three-dimensional.

Drills

Drill 14.11.1. Trace the complete sticky proof and identify the first point at which the projected objects are required to be curves rather than higher-dimensional submanifolds.

Drill 14.11.2. Explain why a higher-dimensional multilinearakeya estimate does not by itself supply a higher-dimensional cinematic maximal theorem.

Drill 14.11.3. Separate the dimension-free steps of finite uniformisation from the dimension-specific terminal incidence estimate.

Chapter 14.12

Distance Sets: the Certified Planar Record

The purpose of this short chapter is orientation, not substitution. The next chapter proves the Guth–Iosevich–Ou–Wang theorem in full: it constructs the good–bad wave-packet decomposition, proves that the bad part is small, proves the refined decoupling estimate for the good part, derives Liu’s identity, and performs the final energy calculation. Here we state the result in the language of the earlier distance-set volumes and give a map of that proof.

14.12.1 Pinned distance sets

Definition 14.12.1. For a nonempty set $E \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, set

$$\Delta_x(E) = \{|x - y| : y \in E\}, \quad \Delta(E) = \{|x - y| : x, y \in E\}.$$

The first set is the *distance set pinned at x* ; the second is the unpinned distance set.

Lemma 14.12.2 (Pinned implies unpinned). *If $x \in E$ and $m(\Delta_x(E)) > 0$, then $m(\Delta(E)) > 0$.*

Proof. For every $y \in E$, the pair (x, y) belongs to $E \times E$. Hence $\Delta_x(E) \subseteq \Delta(E)$, and monotonicity of Lebesgue measure gives the claim. $\square$

Theorem 14.12.3 (The planar 5/4 theorem). *Let $E \subseteq \mathbb{R}^2$ be compact. If $\dim_{\text{H}} E > \frac{5}{4}$, then there is $x \in E$ such that $m(\Delta_x(E)) > 0$. Consequently $m(\Delta(E)) > 0$.*

Proof. The pinned conclusion is Theorem 14.13.7, proved in the next chapter. The unpinned conclusion follows from Theorem 14.12.2. $\square$

The theorem is stronger than an unpinned statement: a single point x must see a positive-measure set of distances. Pinning also exposes the useful one-dimensional object. If μ is a measure on E , then

$$d_*^x \mu(A) = \mu\{y : |x - y| \in A\}$$

is a measure on the radius line. If $d_*^x \mu$ has a nonzero L^2 density, Cauchy–Schwarz forces its support, and hence $\Delta_x(E)$, to have positive Lebesgue measure.

14.12.2 The proof certificate

The proof of Theorem 14.12.3 has two quantitative outputs. Start with an exponent α satisfying

$$\frac{5}{4} < \alpha < \dim_{\mathrm{H}} E$$

and use Theorem 14.13.2 to obtain separated probability measures μ_1, μ_2 with a common α -Frostman bound. The microlocal decomposition of Theorem 14.13.3 writes μ_1 as a good part plus a bad part, up to a rapidly decreasing error. Then:

1. Theorem 14.13.5 finds a set of pins of nearly full μ_2 -measure for which the bad part changes the pinned measure by less than 10^{-3} in L^1 . Its proof uses the stationary-phase estimate Theorem 14.13.8, the single-packet estimate Theorem 14.13.9, and the internally proved radial thin-tube bootstrap through Theorem 14.13.11.
2. Theorem 14.13.6 proves that the good pinned measure has finite L^2 norm for μ_2 -almost every pin. Its analytic core is the refined decoupling theorem Theorem 14.13.12, followed by the multiplicity estimate Theorem 14.13.17, Liu’s exact identity Theorem 14.13.13, and the Fourier-energy calculation in Theorem 14.13.16.

The two conclusions hold simultaneously for at least one x . For that pin, the good part has almost unit L^1 mass and finite L^2 norm. The final Cauchy–Schwarz argument is written out in the proof of Theorem 14.13.7.

Where we are. This is a dependency map, not a replacement for any proof. Every named Volume 14 input in the preceding two paragraphs is proved in the proof-dossier chapters of this volume. The ordinary decoupling theorem and the Fourier energy identities used as foundations were proved in earlier volumes and are identified explicitly at their points of use.

14.12.3 Why the exponent is $5/4$

The last summation makes the threshold transparent. At frequency r , the good-part estimate supplies the decay

$$r^{-(\alpha+1)/3+\varepsilon}.$$

After Liu’s identity and the polar-energy bookkeeping, convergence is exactly the inequality

$$2 - \frac{\alpha + 1}{3} < \alpha.$$

Multiplying by three and rearranging gives $5 < 4\alpha$, or $\alpha > 5/4$. This calculation is proved, with all weights and cutoffs, in Theorem 14.13.6; Worked Example 14.13.1 compares it with the train-track obstruction to the undecomposed Mattila integral.

WORKED EXAMPLE 14.12.1— TIER A **Certified distance-set scoreboard**

Hypothesis on $\dim_{\mathbb{H}} E$	Conclusion and method	Proof location in this series
$> (n + 1)/2$	positive measure; Fourier decay	Volume 8
$> n/2 + 1/3$	positive measure; spherical averages	Volume 12
$> 4/3$ in $\mathbb{R}^2$	positive measure; circular means	Volume 13
$> 5/4$ in $\mathbb{R}^2$	positive pinned measure; good–bad packets and refined decoupling	Chapter 14.13
$> n/2$	Falconer conjecture	open

The rows are not obtained by changing a number in one estimate. The first three are organized around spherical averages and the Mattila integral. The $5/4$ row instead removes a geometrically exceptional microlocal part and controls the remainder pin by pin. That change of framework is the substantive advance.

14.12.4 What remains open

In the plane the conjectural threshold is 1. Thus the present gap is $5/4 - 1 = 1/4$. Neither the train-track example nor the proof of Theorem 14.12.3 shows that $5/4$ is sharp: it is the point at which the current refined-decoupling exponent ceases to close the energy integral. A future improvement must strengthen

that estimate, exploit additional structure of the good packets, or introduce a different decomposition. The problem is recorded without a speculative theorem claim in Theorem 14.15.4.

Reviewer checks

1. Verify the set inclusion in Theorem 14.12.2; no regularity of E is used there.
2. In Theorem 14.13.7, check that the L^1 error and finite L^2 conclusions are obtained for the same pin.
3. In Theorem 14.13.6, track the radial measure $r \, dr$ and the additional factor in Liu's identity; this is where a normalization error would change the threshold.
4. Check the arithmetic $2 - (\alpha + 1)/3 < \alpha \iff \alpha > 5/4$.

Notes

The theorem proved in Chapter 14.13 is due to Guth, Iosevich, Ou, and Wang. The chapter follows the Euclidean planar formulation needed here. Historical claims in the scoreboard are orientation only; no later proof depends on their priority or on the assertion that a listed threshold is permanently current.

Chapter 14.13

The Guth–Iosevich–Ou–Wang Theorem

This chapter proves the planar pinned-distance theorem at the threshold $\frac{5}{4}$. The argument has two halves, and its organising idea is a *decomposition of the measure* into a part that behaves well and a part supported on the configurations that obstruct every earlier method. The only non-elementary geometric input in the bad-part estimate is the radial thin-tube bootstrap, proved in Chapter 14.8; the analytic half is proved below through refined decoupling and Liu’s identity.

14.13.1 The obstacle

Proposition 14.13.1 (Train track examples). *For every $\alpha < \frac{4}{3}$ and every B there is a probability measure μ on the unit ball of $\mathbb{R}^2$ with $\mu(B(x, r)) \lesssim r^\alpha$ such that, with $d(x, y) = |x - y|$ and $d^x(y) = |x - y|$,*

$$\|d_*(\mu \times \mu)\|_{L^2} > B \quad \text{and} \quad \|d_*^x(\mu)\|_{L^2} > B \quad \text{for every } x \in \text{spt } \mu.$$

Proof. It is enough to treat $1 < \alpha < 4/3$; if the requested exponent is at most one, choose any α' strictly between 1 and $4/3$ and use the α' -Frostman example, which is automatically α -Frostman after enlarging the constant. Take R to be a sufficiently large perfect power so that the quantities below are integers. Let $A_R \subset [0, 1]^2$ consist of the points satisfying

$$0 \leq x_1 \leq R^{-1/2}, \quad kR^{-\alpha/2} \leq x_2 \leq kR^{-\alpha/2} + R^{-1}$$

for some integer $0 \leq k < cR^{\alpha/2}$. Thus A_R is one train track, made of $\asymp R^{\alpha/2}$ horizontal slats of dimensions $R^{-1/2} \times R^{-1}$. Put

$$A_{R,l} = A_R + (lR^{-(\alpha-1)/2}, 0), \quad 0 \leq l < cR^{(\alpha-1)/2},$$

where the fixed $c > 0$ keeps all tracks in the unit square, and let E_R be their union. Let μ_R be normalized area measure on E_R . Up to fixed endpoint constants, E_R is the union of

$$R^{(\alpha-1)/2} R^{\alpha/2} R^{1/2} = R^\alpha$$

pairwise disjoint squares of side R^{-1} , each of μ_R -mass $\asymp R^{-\alpha}$.

Step 1. *The Frostman condition at every radius.*

For $r = AR^{-1}$ with $1 \leq A \leq R^{1/2}$, a ball meets only one track. It meets at most $1 + AR^{-1+\alpha/2}$ slats and at most CA small squares in each slat. Hence the number $N(B)$ of small squares met by the ball satisfies

$$N(B) \lesssim A + A^2 R^{-1+\alpha/2} \leq 2A^\alpha.$$

The first term is at most A^α , and the second is at most A^α because $A^{2-\alpha} \leq R^{(2-\alpha)/2}$.

If $R^{1/2} \leq A \leq R$, the ball meets at most $1 + AR^{(\alpha-3)/2}$ tracks. In each such track it meets at most $CAR^{(\alpha-2)/2}$ slats, each containing $R^{1/2}$ small squares. Therefore

$$N(B) \lesssim AR^{(\alpha-1)/2} + A^2 R^{\alpha-2} l e 2A^\alpha.$$

Here the two inequalities are respectively $A^{1-\alpha} R^{(\alpha-1)/2} \leq 1$ and $(A/R)^{2-\alpha} \leq 1$. Multiplying by the mass $R^{-\alpha}$ of a small square proves $\mu_R(B(x, r)) \lesssim r^\alpha$ for $R^{-1} \leq r \leq 1$. For $r < R^{-1}$ the density is $\asymp R^{2-\alpha}$, so $\mu_R(B(x, r)) \lesssim R^{2-\alpha} r^2 = r^\alpha (Rr)^{2-\alpha} \leq r^\alpha$; for $r \geq 1$ the assertion is trivial. The Frostman constant is independent of R .

Step 2. *The distance measure concentrates.*

Fix x in a slat of one track. On at least one side of that slat there are $\asymp R^{\alpha/2}$ slats whose vertical indices differ from its index by $M \asymp R^{\alpha/2}$. If y belongs to such a slat, write the vertical separation as $u = MR^{-\alpha/2} + O(R^{-1})$ and the horizontal separation as h , where $|h| \leq R^{-1/2}$. Since $u \asymp 1$,

$$|x - y| = \sqrt{u^2 + h^2} = u + O(h^2) \in I_M := [MR^{-\alpha/2} - CR^{-1}, MR^{-\alpha/2} + CR^{-1}].$$

The intervals I_M are disjoint after increasing their index spacing by a fixed factor, because $R^{-\alpha/2} \gg R^{-1}$. A slat has μ_R -mass $\asymp R^{-\alpha+1/2}$. Thus, uniformly for every $x \in \text{spt } \mu_R$ and for $\asymp R^{\alpha/2}$ indices M ,

$$d_*^x(\mu_R)(I_M) \gtrsim R^{-\alpha+1/2}.$$

Integrating this inequality in x gives the same lower bound for $d_*(\mu_R \times \mu_R)(I_M)$. By Cauchy–Schwarz on each disjoint interval,

$$\int_{I_M} |d_*(\mu_R \times \mu_R)|^2 \gtrsim \frac{R^{-2\alpha+1}}{|I_M|} \approx R^{-2\alpha+2},$$

and the same estimate holds with $d_*^x(\mu_R)$ in place of the unpinned measure. Summing over the $\asymp R^{\alpha/2}$ intervals gives, uniformly in x ,

$$\int |d_*(\mu_R \times \mu_R)|^2 + \inf_{x \in \text{spt } \mu_R} \int |d_*^x(\mu_R)|^2 \gtrsim R^{-\frac{3}{2}\alpha+2}. \quad (14.13.1)$$

Step 3. Conclude.

The exponent $-\frac{3}{2}\alpha + 2$ is positive exactly when $\alpha < \frac{4}{3}$, so the right-hand side tends to infinity. Choose R so large that the square root of the lower bound exceeds B , and take $\mu = \mu_R$. This gives both assertions of the proposition with one measure and with a Frostman constant independent of the chosen R . $\square$

Where we are. Theorem 14.13.1 is why $\frac{4}{3}$ resisted for twenty years. *Both* frameworks — Mattila’s, using $d_*(\mu \times \mu)$, and Liu’s, using $d_*^x\mu$ — have an infinite L^2 integral below $\frac{4}{3}$, so neither can be pushed further as it stands. Numerically, at $\alpha = \frac{5}{4}$ the exponent in (14.13.1) is $\frac{1}{8} > 0$; at $\alpha = \frac{4}{3}$ it is 0; at $\frac{3}{2}$ it is $-\frac{1}{4}$. Wolff’s theorem sits exactly at the boundary. *But the distance set is still large.* If T is a train track far from x , then $d_*^x(\mu|_T)$ is close to the pushforward of the *uniform* measure on T and is evenly spread. Only the track *through* x produces the spikes on $\bigcup I_M$. So the support of $d_*^x\mu$ has positive measure — the example obstructs the *method*, not the theorem. That observation is what the proof exploits.

14.13.2 The good part of the measure

We first make the reduction from Hausdorff dimension to two separated probability measures explicit.

Lemma 14.13.2 (Separated Frostman pieces). *If $E \subset \mathbb{R}^2$ is compact and $0 < \alpha < \dim_H E$, then, after a similarity of the plane, there are disjoint compact sets $E_1, E_2 \subset E \cap B(0, 1)$, with $\text{dist}(E_1, E_2) \geq 1/2$, and probability measures μ_i supported on E_i such that*

$$\mu_i(B(z, r)) \leq C_\alpha r^\alpha \quad (z \in \mathbb{R}^2, r > 0).$$

Proof. Choose s with $\alpha < s < \dim_H E$. Frostman’s lemma gives a probability measure μ on E with $\mu(B(z, r)) \leq C_s r^s$. This measure has no atoms. Its support therefore contains two distinct points a, b . Choose disjoint closed balls B_a, B_b , centred at these points and of a common sufficiently small radius, with $\mu(B_a), \mu(B_b) > 0$. Restrict μ to each ball and renormalize. The resulting probabilities obey an s -Frostman bound with a larger constant, hence an α -Frostman bound for radii at most one; the bound for larger radii is automatic.

A final translation and dilation put the supports in $B(0, 1)$ and their separation at least $1/2$, changing only the constant. $\square$

For the remainder of the proof fix

$$\frac{5}{4} < \alpha < \dim_{\mathbb{H}} E, \quad 0 < \delta < \min\{10^{-4}, (\alpha - 1)/1000\},$$

and take E_i, μ_i from Lemma 14.13.2.

Definition 14.13.3 (Good and bad tubes). Fix R_0 large and $R_j = 2^j R_0$. Choose a smooth finite-overlap partition of frequency space

$$1 = \psi_0 + \sum_{j \geq 1} \sum_{\tau} \psi_{j,\tau},$$

where ψ_0 is supported in $\{|\omega| \leq 2R_0\}$ and, on the annulus $R_{j-1} \leq |\omega| \leq R_j$, the function $\psi_{j,\tau}$ is supported in a radial block of dimensions $R_j^{1/2+\delta} \times R_j$. The block centres have angular separation comparable to $R_j^{-1/2+\delta}$, and all derivatives have the corresponding symbol bounds. Using the slightly larger tangential scale makes the physical and angular discretisations agree; its only analytic cost is $R_j^{O(\delta)}$, which is retained explicitly below.

For each (j, τ) cover $B(0, 2)$ with bounded overlap by tubes T of dimensions $R_j^{-1/2+\delta} \times 4$ parallel to the radial axis of τ . Choose smooth functions η_T , supported in T , whose sum is one on $B(0, 2)$ and which satisfy $|\partial_{\perp}^a \partial_{\parallel}^b \eta_T| \lesssim R_j^{a(1/2-\delta)}$. Set

$$M_T f = \eta_T (\psi_{j,\tau} \hat{f})^{\vee}.$$

Also put $M_0 f = (\psi_0 \hat{f})^{\vee}$. Call $T \in \mathbb{T}_{j,\tau}$ *bad* if $\mu_2(4T) \geq R_j^{-1/2+100\delta}$, and *good* otherwise. This fixed dilation convention ensures that every packet tail used below remains inside the set tested for goodness. Define

$$\mu_{1,\text{good}} = M_0 \mu_1 + \sum_{T \in \mathbb{T}, T \text{ good}} M_T \mu_1. \quad (14.13.2)$$

Lemma 14.13.4 (The microlocal partition reconstructs f). *If $f \in L^1(\mathbb{R}^2)$ is supported in $B(0, 1)$, then*

$$\left\| f - M_0 f - \sum_{j \geq 1} \sum_{\tau} \sum_{T \in \mathbb{T}_{j,\tau}} M_T f \right\|_{L^1} \leq C_N R_0^{-N} \|f\|_{L^1}$$

for every N . The series converges in tempered distributions and locally in L^1 after the displayed rapidly decreasing error is grouped by j .

Proof. Write $M_{j,\tau}f = (\psi_{j,\tau}\widehat{f})^\vee$. The frequency partition gives $f = M_0f + \sum_{j,\tau} M_{j,\tau}f$ in tempered distributions. For a fixed (j, τ) ,

$$M_{j,\tau}f - \sum_T M_Tf = \left(1 - \sum_T \eta_T\right) (\psi_{j,\tau}^\vee * f).$$

The multiplier symbol bounds imply that $\psi_{j,\tau}^\vee$ is rapidly decreasing away from the dual rectangle of dimensions $R_j^{-1/2-\delta} \times R_j^{-1}$. Since f is supported in $B(0, 1)$ and $\sum_T \eta_T = 1$ on $B(0, 2)$, the last display has L^1 norm at most $C_N R_j^{-N} \|f\|_1$. There are only $O(R_j^{1/2-\delta})$ blocks at level j ; increase N and sum the resulting geometric series. This proves the claim and the convergence statement. $\square$

Where we are. The definition isolates exactly the train track feature. A track T has two properties: $\mu(T)$ is large, and — because the slats are perpendicular to T — the Fourier transform of $\mu|_T$ concentrates on frequencies *in the direction of* T . So M_T selects the part of μ_1 with physical support in T and frequency support in the dual direction, and (14.13.2) removes it.

There are $\approx R_j^{1/2-\delta}$ tubes in $\mathbb{T}_{j,\tau}$, so an even distribution of μ_2 would give $\mu_2(T) \approx R_j^{-1/2+\delta}$. A tube is bad when it carries significantly more.

Two examples. For the train track measure of Theorem 14.13.1, $\mu_{1,\text{good}}$ is approximately area measure on the union of the large rectangles — the spikes are gone. But if the slats are tilted to 45° while the rectangles stay vertical, then $\mu_{1,\text{good}} = \mu_1$: the frequency support no longer aligns with the tube direction, and nothing is removed. *The definition sees the alignment, not merely the concentration.*

Proposition 14.13.5 (The bad part is small in L^1 ; proved via Theorem 14.13.10 and Theorem 14.13.11). *If $\alpha > 1$, $0 < \delta < (\alpha - 1)/1000$, and R_0 is sufficiently large, there is $E'_2 \subseteq E_2$ with $\mu_2(E'_2) \geq 1 - \frac{1}{1000}$ such that for each $x \in E'_2$,*

$$\|d_*^x(\mu_1) - d_*^x(\mu_{1,\text{good}})\|_{L^1} < \frac{1}{1000}.$$

Proposition 14.13.6 (The good part is finite in L^2). *If $\alpha > \frac{5}{4}$ then*

$$\int_{E_2} \|d_*^x(\mu_{1,\text{good}})\|_{L^2}^2 d\mu_2(x) < \infty.$$

Theorem 14.13.7 (Guth–Iosevich–Ou–Wang). *If $E \subseteq \mathbb{R}^2$ is compact with $\dim_{\mathbb{H}} E > \frac{5}{4}$, there is $x \in E$ with $m(\Delta_x(E)) > 0$.*

Proof. By Theorem 14.13.5 and Theorem 14.13.6 there is $x \in E_2$ with both conclusions. Since $d_*^x \mu_1$ is a probability measure, the first gives $\int |d_*^x \mu_{1,\text{good}}| \geq$

$1 - \frac{1}{1000}$, and since $\text{spt } d_*^x \mu_1 \subseteq \Delta_x(E)$,

$$\begin{aligned} \int_{\Delta_x(E)} |d_*^x \mu_{1,\text{good}}| & \geq 1 - \frac{1}{1000} - \int |d_*^x \mu_1 - d_*^x \mu_{1,\text{good}}| \\ & \geq 1 - \frac{2}{1000}. \end{aligned}$$

By Cauchy–Schwarz, $\int_{\Delta_x(E)} |d_*^x \mu_{1,\text{good}}| \leq |\Delta_x(E)|^{1/2} (\int |d_*^x \mu_{1,\text{good}}|^2)^{1/2}$, and the second factor is finite. Hence $|\Delta_x(E)| > 0$. $\square$

14.13.3 Why the bad part is small

Lemma 14.13.8 (Distant bad tubes are negligible). *If $T \in \mathbb{T}_{j,\tau}$, $x \in E_2$ and $x \notin 2T$, then $\|d_*^x(M_T \mu_1)\|_{L^1} \lesssim \text{RapDec}(R_j)$.*

Proof. For a finite measure g , the pinned pushforward is defined distributionally by $\langle d_*^x g, \varphi \rangle = \langle g, \varphi(|x - \cdot|) \rangle$. When g is smooth its density is, up to the harmless normalisation of arc length,

$$d_*^x g(t) = \int_{S^1(x,t)} g(y) \, d\mathcal{H}^1(y).$$

The kernel $\psi_{j,\tau}^\vee$ is rapidly decreasing outside the dual rectangle, while $\text{dist}(E_1, E_2) \geq 1/2$; hence the pushforward below is rapidly decreasing unless $t \asymp 1$. Expanding $M_T \mu_1$ and using $|\hat{\mu}_1| \leq 1$, it is enough, after choosing the decay order large enough to absorb the volume of τ , to prove uniformly for $t \asymp 1$ and $\omega \in \tau$ that

$$\int_{S^1(x,t)} \eta_T(y) e^{2\pi i \omega y} \, dl(y) \leq \text{RapDec}(R_j). \quad (14.13.3)$$

Rotate so $\omega = (0, \omega_2)$ with $|\omega_2| \asymp R_j$; then T is vertical up to an angle $O(R_j^{-1/2+\delta})$. First suppose that T meets $S^1(x, t/2)$. Each of the two arcs $T \cap S^1(x, t)$ is a graph $y_2 = h(y_1)$ over an interval of length $O(R_j^{-1/2+\delta})$, and h and all of its derivatives are bounded there. Because $x \notin 2T$, the arc stays at horizontal distance $\gtrsim R_j^{1/2+\delta}$ from the top and bottom points of the circle. The identity $h'(y_1) = -(y_1 - x_1)/(h(y_1) - x_2)$ therefore gives

$$|h'(y_1)| \gtrsim R_j^{-1/2+\delta} \quad \text{on } I(T). \quad (14.13.4)$$

The graph Jacobian has bounded derivatives. With $\phi(y_1) = 2\pi\omega_2 h(y_1)$, one has $|\phi'| \gtrsim R_j^{1/2+\delta}$ and $|\partial^k(\eta_T(y_1, h(y_1)))| \lesssim_k R_j^{k(1/2-\delta)}$. Every integration by parts

with $L = (i\phi')^{-1} \frac{d}{dy_1}$ therefore gains at least a fixed power $R_j^{-2\delta}$; derivatives of $1/\phi'$ are no worse, by the explicit circle formula and the lower bound above. Iterating arbitrarily often proves (14.13.3) in this case.

If T does not meet $S^1(x, t/2)$, each relevant arc is instead a graph $y_1 = k(y_2)$ with uniformly bounded derivatives. Parametrising by y_2 makes the phase $2\pi\omega_2 y_2$, whose derivative has size $\asymp R_j$; the amplitude derivatives cost only $R_j^{1/2-\delta}$. Repeated integration by parts is then even stronger. This proves the pointwise rapid decay. The relevant t -interval has bounded length and the tails are rapidly decreasing, so integration in t gives the asserted L^1 estimate. $\square$

Lemma 14.13.9 (An L^1 bound for one microlocal piece). *For $T \in \mathbb{T}_{j,\tau}$ and $f \in L^1$ supported in $B(0, 1)$,*

$$\|M_T f\|_{L^1} \lesssim \|f\|_{L^1(2T)} + \text{RapDec}(R_j) \|f\|_{L^1}.$$

Consequently $\|d_^x(M_T f)\|_{L^1} \leq \|M_T f\|_{L^1}$ for every x .*

Proof. The second assertion is the contraction of total variation under a pushforward. For the first, write $M_T f = \eta_T(\psi_{j,\tau}^\vee * f)$. The kernel is rapidly decreasing outside a rectangle of transverse width $O(R_j^{-1/2-\delta})$, strictly smaller than the transverse slack between T and $2T$. Its L^1 norm is uniformly bounded. Integrating first in the output variable and then in the input variable therefore gives the contribution $C\|f\|_{L^1(2T)}$ from inputs in $2T$; repeated integration by parts in the formula for the kernel gives $C_N R_j^{-N} \|f\|_1$ from the complement. $\square$

Lemma 14.13.10 (Reduction to bad tubes through x). *For $x \in E_2$, writing $\text{Bad}_j(x) = \bigcup \{4T : T \in \mathbb{T}_j, x \in 2T, T \text{ bad}\}$,*

$$\|d_*^x(\mu_{1,\text{good}}) - d_*^x(\mu_1)\|_{L^1} \lesssim \sum_{j \geq 1} \mu_1(\text{Bad}_j(x)) + \text{RapDec}(R_0).$$

Proof. Lemma 14.13.4 gives

$$\mu_{1,\text{good}} - \mu_1 = - \sum_{j \geq 1} \sum_{T \in \mathbb{T}_j, T \text{ mbad}} M_T \mu_1 + \mathcal{E}, \quad \|\mathcal{E}\|_{L^1} \leq \text{RapDec}(R_0).$$

For $x \in 2T$ apply Lemma 14.13.9; for $x \notin 2T$ apply Lemma 14.13.8. At a fixed j , the axes represented by the blocks τ have angular separation comparable to the tube width $R_j^{-1/2+\delta}$. Since $|x - y| \geq 1/2$ for $x \in E_2, y \in E_1$, the direction of an axis of a doubled tube containing both x and y is confined to an interval of this same size. Hence only $O(1)$ blocks, and then only $O(1)$ translates in each block, can occur. Fubini consequently gives

$$\sum_{\substack{T \in \mathbb{T}_j \text{ mbad} \\ x \in 2T}} \mu_1(2T) \lesssim \mu_1(\text{Bad}_j(x)).$$

Summing over j , including the rapidly decreasing errors, proves the lemma. $\square$

Lemma 14.13.11 (Power decay for the heavy-tube relation). *Let $\alpha > 1$, and let $0 < \delta < (\alpha - 1)/1000$. There is $c_{\alpha,\delta} > 0$ such that, for*

$$\mathcal{B}_j = \{(y, x) \in E_1 \times E_2 : y, x \in 4T \text{ for some bad } T \in \mathbb{T}_j\},$$

one has

$$(\mu_1 \times \mu_2)(\mathcal{B}_j) \lesssim_{\alpha,\delta} R_j^{-c_{\alpha,\delta}\delta}.$$

Proof. Put

$$w = R_j^{-1/2+\delta}, \quad m = R_j^{-1/2+100\delta}, \quad s = 1 - \delta.$$

We first derive, at this one scale, an $(s, K_j, 1 - e_j)$ -thin-tube relation for (μ_1, μ_2) . Covering a unit-length r -tube by $O(r^{-1})$ balls of radius r gives

$$\mu_i(T) \lesssim r^{\alpha-1} \quad (0 < r \leq 1). \quad (14.13.5)$$

Thus both ordered pairs have $(\sigma_0, K_0, 1 - c)$ -thin tubes, for every $c > 0$, where $\sigma_0 = (\alpha - 1)/2$ and K_0 depends only on the Frostman constants.

Apply Proposition 14.2.71 repeatedly, with $\beta = \sigma_0$ and compact upper gap parameter $\varepsilon = \delta$, until the exponent first reaches s . If the last application overshoots s , weaken its conclusion to exponent s . The number $N = N(\alpha, \delta)$ of applications is finite and independent of j . Starting with $c_0 = R_j^{-q}/3^N$, the exceptional masses after the successive applications are $c_n = 3^n c_0$. From the displayed formula for K' in Proposition 14.2.71, a finite induction gives constants $A, Q < \infty$, depending only on α, δ and the Frostman constants, such that the final relation $G_j \subset E_1 \times E_2$ satisfies

$$(\mu_1 \times \mu_2)(G_j) \geq 1 - R_j^{-q}, \quad \mu_2(T \cap (G_j)_y) \leq AR_j^{qQ} \rho^s \quad (14.13.6)$$

for every $y \in E_1$ and every ρ -tube T through y . Indeed, if $K_n \leq A_n R_j^{qQ_n}$, then $K_{n+1} \leq A_{n+1} R_j^{qM(Q_n+1)}$; this proves the asserted finite power bound by induction.

Fix $y \in E_1$. To every bad tube T with $y \in 4T$ associate the arc $A_y(T) = P_y(4T \cap E_2)$, enlarged to length Cw . The separation of E_1, E_2 makes this enlargement uniform. Moreover, since T is bad,

$$P_y \mu_2(A_y(T)) \geq \mu_2(4T) \geq m. \quad (14.13.7)$$

Choose by the one-dimensional Vitali lemma a disjoint subcollection $\{A_y(T_k)\}_{k=1}^L$ whose fivefold dilates cover all the arcs. Equation (14.13.7) and disjointness give $L \leq m^{-1}$. The inverse image under P_y of a fivefold arc is contained, on E_2 , in a $C'w$ -tube through y . Hence (14.13.6) yields

$$\begin{aligned} \mu_2(\{x : (y, x) \in \mathcal{B}_j\} \cap (G_j)_y) &\leq CLAR_j^{qQ} w^s \\ &\leq CAR_j^{qQ} \frac{w^s}{m}. \end{aligned} \quad (14.13.8)$$

Integrating in y and adding the exceptional relation gives

$$(\mu_1 \times \mu_2)(\mathcal{B}_j) \leq R_j^{-q} + C A R_j^{qQ-D}, \quad D = s(\tfrac{1}{2} - \delta) - \tfrac{1}{2} + 100\delta. \quad (14.13.9)$$

Since $s = 1 - \delta$, $D = \frac{197}{2}\delta + \delta^2 > 98\delta$. Choose $q = D/(2(Q+1))$. Both exponents on the right of (14.13.9) are then negative by a fixed positive multiple of δ . Decreasing that multiple if necessary proves the claim, including the finitely many small values of j by enlarging the implicit constant. $\square$

Proof of Proposition 14.13.5. Integrate Lemma 14.13.10 against $\mu_2(x)$. Fubini and the definition of $\mathcal{B}_j$ give

$$\int_{E_2} \mu_1(\text{Bad}_j(x)) \, d\mu_2(x) \leq (\mu_1 \times \mu_2)(\mathcal{B}_j) \lesssim R_j^{-c_{\alpha,\delta}}.$$

Consequently

$$\int_{E_2} \|d_*^x(\mu_1) - d_*^x(\mu_{1,\text{good}})\|_{L^1} \, d\mu_2(x) \lesssim \sum_{j \geq 1} R_j^{-c_{\alpha,\delta}} + \text{RapDec}(R_0).$$

The geometric series tends to zero as $R_0 \rightarrow \infty$. Choose R_0 so large that the last display is below 10^{-6} . Markov's inequality then shows that the set on which the norm is at least 10^{-3} has μ_2 -measure at most 10^{-3} . Its complement is the required E'_2 . $\square$

Where we are. Theorem 14.13.11 is where the argument's structural difficulty lies, and Theorem 14.13.1 explains why. Analysing one train track is easy. The problem is that *a single point may lie in many tracks pointing in different directions*, and there is no bound on that from the Frostman condition alone.

The completed radial bootstrap supplies the missing uniformity. Iterated to an exponent just below one, it says that outside a power-small relation, no pin can see too much mass in too many narrow directions. The Vitali selection above converts that statement directly into the power decay needed here. This proves exactly the consequence used by GIOW, without importing a separate radial-projection L^p theorem.

14.13.4 Refined decoupling

Theorem 14.13.12 (Refined decoupling). *Let $S \subseteq \mathbb{R}^d$ be strictly convex and C^2 with Gaussian curvature ≈ 1 , and let $2 \leq p \leq \frac{2(d+1)}{d-1}$. Decompose the R^{-1} -neighbourhood of S into finitely overlapping caps θ of dimensions $R^{-1/2} \times \dots \times R^{-1/2} \times R^{-1}$. For each θ , let $\mathbb{T}_\theta$ be a bounded-overlap family of*

tubes in B_R of length R and radius $R^{1/2+\delta}$, directed along the normal to S at θ .

Let $\mathbb{W} \subset \bigcup_\theta \mathbb{T}_\theta$, put $W = \#\mathbb{W}$, and suppose $f = \sum_{T \in \mathbb{W}} f_T$, where $\widehat{f}_T$ is rapidly decreasing off $\theta(T)$, f_T is rapidly decreasing off T , and the non-zero $\|f_T\|_{L^p}$ are within a factor two of one another. If Y is a union of $R^{1/2}$ -cubes in B_R , each meeting at most M tubes of $\mathbb{W}$, then for every $\varepsilon > 0$

$$\|f\|_{L^p(Y)} \lesssim R^\varepsilon \left(\frac{M}{W} \right)^{\frac{1}{2} - \frac{1}{p}} \left(\sum_{T \in \mathbb{W}} \|f_T\|_{L^p}^2 \right)^{1/2}. \quad (14.13.10)$$

Proof. Rapid tails may be truncated at a loss $R^{-N} \sum_T \|f_T\|_p$, absorbed by taking N large; incidences may thus be read using fixed dilates. Dyadically pigeonhole the values $\|f\|_{L^p(Q)}$ over the $R^{1/2}$ -cubes $Q \subset Y$. One class carries a logarithmic fraction of $\|f\|_{L^p(Y)}^p$, and its cube norms differ by at most two. Replace Y by this class. Losses denoted $\lesssim$ are powers of $\log R$ and will be absorbed by $R^{\varepsilon/4}$.

We induct on dyadic R . For bounded R the assertion follows from the triangle inequality. Assume the result at every smaller dyadic radius with exponent $\varepsilon/2$.

Step 1. *Two-scale decomposition.*

Cover the R^{-1} -neighbourhood of S by coarser blocks τ of tangential diameter $R^{-1/4}$ and normal thickness $R^{-1/2}$. For each τ , cover B_R with bounded overlap by cylinders l of radius $R^{3/4+O(\delta)}$ and length R , directed along the normal at τ . After a bounded-colour split, every fine packet belongs to one such l for its parent τ . Put

$$\mathbb{W}_l = \{T \in \mathbb{W} : \theta(T) \subset \tau(l), T \subset Cl\}, \quad f_l = \sum_{T \in \mathbb{W}_l} f_T.$$

On an $R^{1/2}$ -cube Q , ordinary ℓ^2 decoupling at radius $R^{1/2}$ (Volume 13, Theorem 13.5.2) applied to the coarser caps gives

$$\|f\|_{L^p(Q)} \lesssim \left(\sum_l \|f_l\|_{L^p(Q)}^2 \right)^{1/2}. \quad (14.13.11)$$

Step 2. *Induction on scales.*

Parabolic rescaling sends l to a ball of radius $R^{1/2}$, the fine caps to caps at that radius, and the fine tubes to its wave packets. An $R^{1/2} \times R^{3/4+O(\delta)}$ subcylinder becomes an $R^{1/4}$ -cube up to bounded distortion. Let Y_{l,M_1} be the union of those subcylinders meeting $\sim M_1$ members of $\mathbb{W}_l$. The induction hypothesis gives

$$\|f_l\|_{L^p(Y_{l,M_1})} \lesssim R^{\varepsilon/2} \left(\frac{M_1}{\#\mathbb{W}_l} \right)^{\frac{1}{2} - \frac{1}{p}} \left(\sum_{T \in \mathbb{W}_l} \|f_T\|_{L^p}^2 \right)^{1/2}. \quad (14.13.12)$$

Step 3. *Three pigeonholings.*

There are $O(\log R)$ nonempty dyadic values of M_1 . Pigeonholing first in M_1 and then in the contribution to (14.13.11) supplies a common M_1 and a logarithmic fraction of the cubes for which the sum may be restricted to l with $Q \subset Y_{l,M_1}$. Next pigeonhole $\#\mathbb{W}_l \sim W_1$, put $\mathbb{B} = \{l : \#\mathbb{W}_l \sim W_1\}$, and retain another logarithmic fraction. Finally pigeonhole the retained cubes by the number of $l \in \mathbb{B}$ for which they lie in Y_{l,M_1} . We obtain $Y' \subset Y$ and M_2 such that

$$\|f\|_{L^p(Y')} \gtrsim \|f\|_{L^p(Y)}, \quad \#\{l \in \mathbb{B} : Q \subset Y_{l,M_1}\} \sim M_2$$

for every $Q \subset Y'$. Two counting facts result:

$$M_1 M_2 \leq M, \quad W_1 \#\mathbb{B} \leq W. \quad (14.13.13)$$

The first holds because the M_1 tubes contributed by each of the M_2 cylinders are disjoint members of $\mathbb{W}$ and all meet Q ; their total is at most M . The second holds because the $\mathbb{W}_l$ are disjoint subsets of $\mathbb{W}$.

Step 4. *Assemble.*

By (14.13.11) and Hölder over the $\approx M_2$ terms, $\|f\|_{L^p(Q)} \lesssim M_2^{\frac{1}{2}-\frac{1}{p}} (\sum_l \|f_l\|_{L^p(Q)}^p)^{1/p}$. Raising to the p -th power, summing over $Q \subseteq Y'$, and inserting (14.13.12),

$$\begin{aligned} \|f\|_{L^p(Y)}^p &\lesssim R^{p\varepsilon/2} \left(\frac{M_1 M_2}{W_1} \right)^{\frac{p}{2}-1} \sum_{l \in \mathbb{B}} \left(\sum_{T \in \mathbb{W}_l} \|f_T\|_{L^p}^2 \right)^{p/2} \\ &\lesssim R^{p\varepsilon/2} \left(\frac{M_1 M_2}{W} \right)^{\frac{p}{2}-1} \frac{\#\mathbb{B} W_1}{W} \left(\sum_T \|f_T\|_{L^p}^2 \right)^{p/2}, \end{aligned}$$

The factor-two comparability of packet norms gives

$$\sum_{T \in \mathbb{W}_l} \|f_T\|_p^2 \asymp \frac{W_1}{W} \sum_{T \in \mathbb{W}} \|f_T\|_p^2.$$

Applying (14.13.13), taking p -th roots, and absorbing all polylogarithmic losses into $R^{\varepsilon/2}$ yields (14.13.10). This closes the induction. $\square$

Where we are. Theorem 14.13.12 strengthens Volume 13, Theorem 13.5.2 exactly when the wave packets are *spatially* separated as well as frequency-separated.

The comparison. If f_θ is non-zero for N caps, each a sum of P wave packets, ordinary decoupling reads $\|f\|_{L^p(B_R)} \lesssim P^{-(\frac{1}{2}-\frac{1}{p})} (\sum_T \|f_T\|_{L^p}^2)^{1/2}$. An $R^{1/2}$ -cube lies in $O(1)$ tubes per direction, so $M \leq N$ and $W = NP$, giving $\frac{M}{W} \leq \frac{1}{P}$. Hence (14.13.10) is *at least as strong*, and strictly stronger when

$M \ll N$ — that is, when each cube sees only a small fraction of the caps. *The two extremes.* $M = W$ recovers ordinary decoupling; $M = 1$ gives $\|f\|_{L^p} \lesssim (\sum_T \|f_T\|_{L^p}^p)^{1/p}$, the trivial bound for disjointly supported pieces. The theorem interpolates between them, and the ratio $\frac{M}{W}$ measures how disjoint the packets are.

14.13.5 Why the good part is finite

Theorem 14.13.13 (Liu’s identity). *For every Schwartz function $f : \mathbb{R}^2 \rightarrow \mathbb{C}$ and $x \in \mathbb{R}^2$,*

$$\int_0^\infty |f * \sigma_t(x)|^2 t \, dt = \int_0^\infty |f * \widehat{\sigma}_r(x)|^2 r \, dr,$$

where σ_t is normalised arc length on the circle of radius t . The identity extends to either side whenever one, hence both, is finite.

Proof. Fix x and set

$$A_x(\rho) = \int_{S^1} \widehat{f}(\rho\omega) e^{2\pi i x \cdot (\rho\omega)} \, d\sigma(\omega),$$

where σ is normalized arc length on the unit circle. Since $\widehat{\sigma}_t(\xi) = J_0(2\pi t|\xi|)$, polar coordinates give

$$f * \sigma_t(x) = 2\pi \int_0^\infty A_x(\rho) J_0(2\pi t\rho) \rho \, d\rho. \quad (14.13.14)$$

On the other hand, Fourier inversion and the evenness of σ_r give

$$f * \widehat{\sigma}_r(x) = \int \widehat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\sigma_r(\xi) = A_x(r). \quad (14.13.15)$$

The map

$$A(\rho) \mapsto 2\pi \int_0^\infty A(\rho) J_0(2\pi t\rho) \rho \, d\rho$$

is the order-zero Hankel transform with the present Fourier normalization. It is unitary on $L^2((0, \infty), \rho \, d\rho)$: this is precisely Plancherel’s theorem applied to the radial function on $\mathbb{R}^2$ whose radial profile is A . Apply this isometry to (14.13.14) and use (14.13.15). Approximation by Schwartz functions proves the extension assertion. $\square$

Where we are. The identity is a *self-duality*: in $\mathbb{R}^2$ the families $\{\sigma_t\}$ and $\{\widehat{\sigma}_r\}$ share the radial profile J_0 , so the two sides are Hankel transforms of one another and the measures $t \, dt$ and $r \, dr$ match. Numerically, evaluating both sides at $x = 0$ for a Gaussian over 600 radii in $[0.02, 12]$ gives 0.497270 on each side.

Why it matters here. The left side is what $\|d_*^x \mu\|_{L^2}$ produces; the right side is a Fourier integral, to which decoupling applies. It plays for pinned distance sets the rôle Theorem 12.8.2 of Volume 12 plays for the unpinned problem.

Lemma 14.13.14 (Spatial tail of the good distribution). *Let $A = \{z : \text{dist}(z, E_1) > R_0^{-1/2+2\delta}\}$. For every N ,*

$$\int_A |\mu_{1,\text{good}}(z)| \, dz + \sup_{z \in A} |\mu_{1,\text{good}}(z)| \leq C_N R_0^{-N}.$$

In particular, for $x \in E_2$ the contribution of $\mu_{1,\text{good}}$ to circles with radius outside a fixed compact subinterval of $(0, \infty)$ is rapidly decreasing in R_0 .

Proof. For the low-frequency term, $M_0 \mu_1 = \psi_0^\vee * \mu_1$, and the usual rescaled Schwartz bounds for $\psi_0^\vee$ give the assertion outside the $R_0^{-1/2+2\delta}$ -neighbourhood of E_1 .

For a fixed (j, τ) , bounded overlap and $0 \leq \eta_T \leq 1$ give

$$\sum_{T \in \mathbb{T}_{j,\tau}} |\eta_T(\psi_{j,\tau}^\vee * \mu_1)| \lesssim |\psi_{j,\tau}^\vee * \mu_1|.$$

The kernel on the right is Schwartz-adapted to the dual rectangle of dimensions $R_j^{-1/2-\delta} \times R_j^{-1}$. If $\text{dist}(z, E_1) > R_0^{-1/2+2\delta}$, the ratio of this distance to the longer dual scale is at least a positive power of R_0 , uniformly in j . The pointwise and integrated tails are therefore at most $C_N R_j^{-N}$ after increasing N . There are $O(R_j^{1/2-\delta})$ blocks τ at level j ; summing first over τ and then over the geometric sequence R_j proves both estimates. The final statement follows from $\text{dist}(E_1, E_2) \geq 1/2$. $\square$

Lemma 14.13.15 (Pinned density and Liu reduction). *For $x \in E_2$,*

$$\|d_*^x(\mu_{1,\text{good}})\|_2^2 \lesssim \int_0^\infty |\mu_{1,\text{good}} * \widehat{\sigma_r}(x)|^2 r \, dr + \text{RapDec}(R_0).$$

Proof. For a smooth density g , the coarea formula and normalized circle measure give

$$d_*^x(g \, dy)(t) = 2\pi t (g * \sigma_t)(x).$$

Regularization proves the same identity distributionally. Hence the square of the pinned L^2 norm is a fixed multiple of $\int |g * \sigma_t(x)|^2 t^2 \, dt$. By Lemma 14.13.14, apart from a term whose L^1 and L^∞ norms are both rapidly decreasing, the relevant radii belong to a fixed interval $[c, C] \subset (0, \infty)$. On this interval $t^2 \lesssim t$; the discarded term is rapidly decreasing in L^2 by $\|h\|_2^2 \leq \|h\|_1 \|h\|_\infty$. Liu's identity, Theorem 14.13.13, now gives the asserted right-hand side. $\square$

Proposition 14.13.16 (The key estimate). *For every $\varepsilon > 0$, after choosing $\delta > 0$ sufficiently small in terms of ε , and for every $r \geq 1$,*

$$\int_{E_2} |\mu_{1,\text{good}} * \widehat{\sigma}_r(x)|^2 d\mu_2(x) \leq C_\varepsilon(R_0) r^{-\frac{\alpha+1}{3}+\varepsilon} r^{-1} \int |\widehat{\mu}_1|^2 \psi_r d\xi, \quad (14.13.16)$$

with $\psi_r(\xi) = (1 + |r - |\xi||)^{-100}$.

Proof of Theorem 14.13.6 from Theorem 14.13.16. Choose $\varepsilon > 0$ so small that $\beta := 2 - (\alpha + 1)/3 + \varepsilon < \alpha$. Integrate Lemma 14.13.15 in x and apply Tonelli. The part $0 < r < 1$ is finite because the multiplier is smooth and all measures are finite. On $r \geq 1$, insert (14.13.16). Since

$$\int_1^\infty r^{-(\alpha+1)/3+\varepsilon} \psi_r(\xi) dr \lesssim (1 + |\xi|)^{-(\alpha+1)/3+\varepsilon},$$

we obtain

$$\int_{E_2} \|d_*^x \mu_{1,\text{good}}\|_2^2 d\mu_2(x) \lesssim_{R_0, \varepsilon} 1 + \int_{|\xi| \geq 1} |\xi|^{\beta-2} |\widehat{\mu}_1(\xi)|^2 d\xi.$$

The Fourier representation of Riesz energy identifies the last integral, up to a constant and the harmless low-frequency part, with $I_\beta(\mu_1)$. The Frostman bound makes this finite for $\beta < \alpha$. Such an ε exists precisely when $2 - (\alpha + 1)/3 < \alpha$, equivalently $\alpha > 5/4$. $\square$

Lemma 14.13.17 (The multiplicity of good tubes). *Let $\mathbb{W}$ be any subfamily of good tubes at scales $R_j \asymp r$. With $\mathcal{Q}_{\gamma, M}$ the $r^{-1/2}$ -squares q with $\mu_2(q) \asymp \gamma$ meeting $\asymp M$ tubes of $\mathbb{W}$, and $Y_{\gamma, M} = \bigcup \mathcal{Q}_{\gamma, M}$,*

$$\mu_2(Y_{\gamma, M}) \lesssim \frac{\#\mathbb{W} r^{-1/2+100\delta}}{M}. \quad (14.13.17)$$

Proof. Double count $\mathcal{I} = \{(q, T) : q \in \mathcal{Q}_{\gamma, M}, T \in \mathbb{W}, q \cap T \neq \emptyset\}$. Each T is good, so $\mu_2(2T) \lesssim r^{-1/2+100\delta}$ and T meets $\lesssim \gamma^{-1} r^{-1/2+100\delta}$ squares; hence $\mathcal{I} \lesssim \gamma^{-1} r^{-1/2+100\delta} \#\mathbb{W}$. Each q meets $\gtrsim M$ tubes, so $\mathcal{I} \gtrsim \#\mathcal{Q}_{\gamma, M} M$. Comparing and multiplying by γ gives (14.13.17). $\square$

Proof of Theorem 14.13.16. We first treat $r > 10R_0$.

Step 1. *Select the frequency levels and identify the packets.*

Multiplication by $\widehat{\sigma}_r$ in frequency restricts the expansion of $\mu_{1,\text{good}}$ to the $O(1)$ indices with $R_j \asymp r$; all other indices contribute $\text{RapDec}(r)$ by the symbol tails. Let η_1 be one on $B(0, 1)$ and supported in $B(0, 2)$, and define

$$f_T = \eta_1(M_T \mu_1 * \widehat{\sigma}_r) \quad (R_j \asymp r, T \text{ good}).$$

Then on E_2 ,

$$\mu_{1,\text{good}} * \widehat{\sigma}_r = \sum_{R_j \asymp r} \sum_{\tau} \sum_{T \text{ mgood}} f_T + \text{RapDec}(r). \quad (14.13.18)$$

Let $\theta(T)$ be the unit neighbourhood of $3\tau \cap S_r^1$. Since $\widehat{M_T \mu_1}$ is rapidly decreasing off 2τ , convolution with $\widehat{\eta}_1$ shows that $\widehat{f_T}$ is rapidly decreasing off $\theta(T)$, a cap of tangential length $r^{1/2+O(\delta)}$ and normal thickness one. To locate f_T in space, insert a smooth multiplier $\widehat{\psi}_\tau$ equal to one on 2τ :

$$M_T \mu_1 * \widehat{\sigma}_r = M_T \mu_1 * (\widehat{\psi}_\tau \sigma_r)^\wedge + \text{RapDec}(r).$$

In coordinates parallel and perpendicular to the radial axis of τ , the phase on $S_r^1 \cap 3\tau$ has non-vanishing transverse derivative away from the dual tube of width $r^{-1/2+O(\delta)}$. Repeated integration by parts, exactly as in Lemma 14.13.8, shows that the last kernel is rapidly decreasing outside that tube through the origin. Since $M_T \mu_1$ is supported in T , f_T is rapidly decreasing off a fixed dilate of T . After dilating physical space by r , these are precisely the packets in Theorem 14.13.12, with tubes of length r and radius $r^{1/2+O(\delta)}$.

Step 2. *Finite dyadic sorting.*

The number of relevant tubes is polynomial in r , and each f_T has a polynomial L^6 bound. Discard the classes $\|f_T\|_6 < r^{-A}$ with A so large that their total contribution is $\text{RapDec}(r)$, and group the rest into

$$\mathbb{W}_\lambda = \{T : \|f_T\|_6 \asymp \lambda\}, \quad f_\lambda = \sum_{T \in \mathbb{W}_\lambda} f_T.$$

There are $O_A(\log r)$ classes. Likewise partition the unit ball into $r^{-1/2}$ -squares. Squares of mass below r^{-A} have total mass at most $O(r^{1-A})$ and are negligible after increasing A . Group the remaining squares according to

$$\mu_2(q) \asymp \gamma, \text{quad} \# \{T \in \mathbb{W}_\lambda : q \cap CT \neq \emptyset\} \asymp M,$$

and write their union as $Y_{\gamma,M}$. There are only $O_A((\log r)^2)$ pairs (γ, M) . Thus it is enough to prove the desired bound on each f_λ and $Y_{\gamma,M}$, paying $r^{O(\delta)}$ for all logarithmic and fixed-dilation losses.

Step 3. *Replace the Frostman measure by a local density.*

If a function has Fourier support in a fixed enlargement of the unit neighbourhood of S_r^1 , the reproducing-kernel formula and Cauchy–Schwarz give

$$|f(x)|^2 \lesssim (|f|^2 * \Phi_{1/r})(x),$$

where $\Phi_{1/r}$ is a non-negative Schwartz bump of integral one at scale r^{-1} . Fubini therefore bounds the integral of $|f_\lambda|^2$ against μ_2 by the integral against $\mu_2 * \Phi_{1/r}$

on a fixed enlargement of $Y_{\gamma,M}$; because its component squares have side $r^{-1/2}$, this enlargement changes neither dyadic class nor multiplicity beyond a fixed factor. Hölder gives

$$\int_{Y_{\gamma,M}} |f_\lambda|^2 d\mu_2 \lesssim \left(\int_{Y_{\gamma,M}} |f_\lambda|^6 \right)^{1/3} \left(\int_{Y_{\gamma,M}} |\mu_2 * \Phi_{1/r}|^{3/2} \right)^{2/3}. \quad (14.13.19)$$

After the dilation $x \mapsto rx$, Theorem 14.13.12 with $d = 2, p = 6$ gives

$$\|f_\lambda\|_{L^6(Y_{\gamma,M})} \lesssim r^{O(\delta)} \left(\frac{M}{\#\mathbb{W}_\lambda} \right)^{1/3} \left(\sum_{T \in \mathbb{W}_\lambda} \|f_T\|_6^2 \right)^{1/2}.$$

Lemma 14.13.17 rearranges to

$$\frac{M}{\#\mathbb{W}_\lambda} \lesssim \frac{r^{-1/2+100\delta}}{\mu_2(Y_{\gamma,M})}.$$

The Frostman bound gives

$$\|\mu_2 * \Phi_{1/r}\|_\infty \lesssim r^{2-\alpha}. \quad (14.13.20)$$

Indeed, sum the kernel over dyadic annuli around its centre and use $\mu_2(B(z, 2^k/r)) \lesssim (2^k/r)^\alpha$; the Schwartz decay makes the sum converge. Moreover $\int_{Y_{\gamma,M}} \mu_2 * \Phi_{1/r} \lesssim \mu_2(Y_{\gamma,M})$, with the same fixed enlargement convention. Therefore

$$\int_{Y_{\gamma,M}} |\mu_2 * \Phi_{1/r}|^{3/2} \lesssim r^{1-\alpha/2} \mu_2(Y_{\gamma,M}). \quad (14.13.21)$$

Substitution into (14.13.19) cancels the powers of $\mu_2(Y_{\gamma,M})$ and yields

$$\int_{Y_{\gamma,M}} |f_\lambda|^2 d\mu_2 \lesssim r^{O(\delta)} r^{(1-\alpha)/3} \sum_{T \in \mathbb{W}_\lambda} \|f_T\|_6^2. \quad (14.13.22)$$

Step 4. *The norm of one packet.*

In the unit ball, a packet occupies area $r^{-1/2+O(\delta)}$, hence

$$\|f_T\|_6 \lesssim r^{-1/12+O(\delta)} \|f_T\|_\infty.$$

Fourier inversion on the normalized circle gives

$$f_T(x) = \eta_1(x) \int_{S_r^1} \widehat{M_T \mu_1}(\xi) e^{2\pi i x \cdot \xi} d\sigma_r(\xi).$$

The integrand is negligible off $\theta(T)$, and $\sigma_r(\theta(T)) \lesssim r^{-1/2+O(\delta)}$. Cauchy–Schwarz therefore gives

$$\|f_T\|_6 \lesssim r^{-1/3+O(\delta)} \|\widehat{M_T \mu_1}\|_{L^2(d\sigma_r)}. \quad (14.13.23)$$

Step 5. *Orthogonality with all weights displayed.*

We claim

$$\sum_{R_j \asymp r} \sum_{\tau} \sum_{T \in \mathbb{T}_{j,\tau}} \int |\widehat{M_T \mu_1}|^2 d\sigma_r \lesssim r^{-1} \int |\widehat{\mu_1}(\xi)|^2 \psi_r(\xi) d\xi. \quad (14.13.24)$$

Choose a weight $\psi_{j,\tau,r}$ equal to one on the unit-thickening of $\tau \cap S_r^1$ and rapidly decreasing away from it. Because $M_T \mu_1$ is supported in a fixed ball, its Fourier transform satisfies the unit-scale submean inequality. Covering S_r^1 by unit arcs and using that normalized circle measure is $\asymp r^{-1}$ times arc length gives

$$\int |\widehat{M_T \mu_1}|^2 d\sigma_r \lesssim r^{-1} \int |\widehat{M_T \mu_1}(\xi)|^2 \psi_{j,\tau,r}(\xi) d\xi.$$

Since $\widehat{M_T \mu_1} = \widehat{\eta_T} * (\psi_{j,\tau} \widehat{\mu_1})$, enlarge the weight slightly to $\widetilde{\psi}_{j,\tau,r}$ so the convolution may be restricted to it, with a rapidly decreasing error. Plancherel then gives

$$\begin{aligned} \sum_{T \in \mathbb{T}_{j,\tau}} \int |\widehat{M_T \mu_1}|^2 d\sigma_r &\lesssim r^{-1} \sum_T \int |\eta_T(x)|^2 |(\widetilde{\psi}_{j,\tau,r} \widehat{\mu_1})^\vee(x)|^2 dx \\ &\lesssim r^{-1} \int |\widetilde{\psi}_{j,\tau,r}(\xi) \widehat{\mu_1}(\xi)|^2 d\xi. \end{aligned}$$

The second inequality is exactly the bounded overlap of the physical tube partition. Finally, the enlarged frequency weights have bounded overlap as $R_j \asymp r$ and τ vary, and their squared sum is bounded by a constant multiple of ψ_r . Summation proves (14.13.24).

Combining (14.13.22), (14.13.23), and (14.13.24), and then summing the finitely many dyadic classes, gives

$$\int_{E_2} |\mu_{1,\text{good}} * \widehat{\sigma_r}|^2 d\mu_2 \lesssim r^{O(\delta)} r^{-(\alpha+1)/3} r^{-1} \int |\widehat{\mu_1}|^2 \psi_r.$$

Choose δ so small that the implicit multiple of δ is below ε .

Step 6. *The bounded-frequency range.*

For $1 \leq r \leq 10R_0$, Cauchy–Schwarz on the normalized circle gives

$$\int_{E_2} |\mu_{1,\text{good}} * \widehat{\sigma_r}|^2 d\mu_2 \leq \|\widehat{\mu_{1,\text{good}}}\|_{L^2(d\sigma_r)}^2.$$

The proof of (14.13.24), now including the low-frequency piece $M_0\mu_1$ and only finitely many $R_j \lesssim R_0$, bounds this by $C(R_0)r^{-1} \int |\widehat{\mu}_1|^2 \psi_r$. Since the remaining power of r is bounded above and below in this range, it is absorbed into $C_\varepsilon(R_0)$. This completes the proof. $\square$

Remark 14.13.18 (Dependency closure for the GIOW theorem). *What is proved here.* Theorem 14.13.1 in full, including the Frostman verification and the divergence exponent; the good/bad decomposition; Theorem 14.13.8 by stationary phase; Theorem 14.13.10; Theorem 14.13.11 from the internally proved radial thin-tube bootstrap; Theorem 14.13.12 in full, with its three pigeonholings and the counting facts (14.13.13); Theorem 14.13.17; the derivation of Theorem 14.13.6 from Theorem 14.13.16 including the arithmetic giving $\frac{5}{4}$; and Theorem 14.13.7.

Transitive status. The geometric dependency is Proposition 14.2.71, whose proof is in Chapter 14.8, resting in turn on the fully proved Furstenberg incidence theorem. The analytic dependencies—refined decoupling, Liu’s identity, the wave-packet reduction, and the energy calculation—are proved in this chapter. Thus no quoted projection theorem remains in the proof chain.

On the generality of Liu’s identity. The proof above is the Euclidean case. Liu states it for a general norm on $\mathbb{R}^d$, where the circles are replaced by the boundary of the unit ball and J_0 by Herz’s asymptotic for $\widehat{\sigma_K}$; the argument is the same once the radial profile is identified.

WORKED EXAMPLE 14.13.1— TIER A **Where $\frac{5}{4}$ comes from**

Two computations meet.

The obstruction. Theorem 14.13.1 gives $\int d_*(\mu \times \mu)^2 \gtrsim R^{-\frac{3}{2}\alpha+2}$, divergent for $\alpha < \frac{4}{3}$:

α	exponent	
$\frac{5}{4}$	$+\frac{1}{8}$	diverges
$\frac{4}{3}$	0	borderline — Wolff’s threshold
$\frac{3}{2}$	$-\frac{1}{4}$	converges

The gain. Theorem 14.13.16 gives decay $r^{-(\alpha+1)/3}$, and the energy integral converges when $2 - \frac{\alpha+1}{3} < \alpha$, that is $\alpha > \frac{5}{4}$. Symbolically the two sides are equal exactly at $\frac{5}{4}$.

Reading the two together. Below $\frac{4}{3}$ the naive integral diverges; removing the bad part restores convergence down to $\frac{5}{4}$. *The gap between $\frac{4}{3}$ and $\frac{5}{4}$ is precisely what the decomposition buys*, and $\frac{5}{4}$ is where the good part’s own estimate runs out.

What would improve it. A better exponent in Theorem 14.13.16 than $\frac{\alpha+1}{3}$. That exponent comes from L^6 refined decoupling; the conjectured $\frac{n}{2} = 1$

would need $2 - \beta(\alpha) < \alpha$ at $\alpha = 1$, so $\beta(1) > 1$ — roughly three times what is available.

WORKED EXAMPLE 14.13.2— TIER B **Refined decoupling as a general tool**

Theorem 14.13.12 was discovered independently by Du and Zhang and is of interest beyond this theorem.

(a) *What it adds.* Ordinary decoupling is sharp when $|f_\theta|$ is roughly constant on B_R ; it is wasteful when the f_θ have disjoint supports, where the trivial ℓ^p bound is better. Refined decoupling interpolates via the parameter M/W .

(b) *Where it came from.* Refined Strichartz estimates, in the work of Du–Guth–Li on the Schrödinger maximal function. Volume 13, Theorem 13.9.1 proved the unrefined periodic version.

(c) *Why it suits this problem.* Theorem 14.13.17 bounds M/W using the *goodness* of the tubes — each carries little μ_2 -mass. So the decomposition of Theorem 14.13.3 is designed precisely to make the refined estimate applicable.

(d) *The pattern.* Volume 13, Worked Example 13.9.4 observed that problems falling to decoupling *alone* are the L^2 ones, and that progress elsewhere came from combining it with geometry. This chapter is a case study: decoupling supplies the analysis, the radial thin-tube bootstrap the geometry, and neither suffices alone.

WORKED EXAMPLE 14.13.3— TIER C: RESEARCH TASTE **Decomposing the measure: a recurring device**

Theorem 14.13.3 splits μ_1 into a part that behaves and a part carrying the obstruction. That device appears repeatedly in this series and is worth abstracting.

(a) *The pattern.* Identify the configuration that defeats the method; show it occupies a small part of the object; estimate the remainder by the method and the small part by hand.

(b) *Instances.* Volume 3’s Calderón–Zygmund decomposition — bounded part plus mean-zero pieces; Volume 12’s broad–narrow split; the multiscale regular/irregular alternatives in Chapter 14.10; and the present good/bad wave-packet split.

(c) *What makes it work here.* Two things. The obstruction is *identifiable* — Theorem 14.13.1 exhibits it explicitly, so one knows what to remove. And it is *rare* — Theorem 14.13.11 shows the bad tubes carry little mass, using the completed radial bootstrap.

(d) *What makes it fail elsewhere.* For the restriction conjecture the extremal configurations are Knapp caps, which are identifiable but *not* rare — every cap is one after rescaling. Volume 12, Worked Example 12.5.3(c) records this, and it is why no analogous decomposition is known there.

(e) *The diagnostic.* Before attempting a decomposition, ask whether the bad set is rare in a quantifiable sense. If the obstruction is scale-invariant, it is not, and a different idea is needed.

(f) *A further subtlety here.* Theorem 14.13.3 removes not the *tubes* with large mass but the part of μ_1 with physical support in them *and* frequency support in their direction. The tilted-slat example shows the difference is real: concentration alone is not the obstruction, only concentration aligned with the frequency support.

Drills

Drill 14.13.1. Verify Step 1 of Theorem 14.13.1 in the case $A \leq R^{1/2}$.

Drill 14.13.2. Confirm that $-\frac{3}{2}\alpha + 2 > 0$ iff $\alpha < \frac{4}{3}$, and evaluate at $\alpha = \frac{5}{4}$.

Drill 14.13.3. In Theorem 14.13.8, verify (14.13.4) and the worst case of the integration by parts.

Drill 14.13.4. Check the counting facts (14.13.13) and the two extremes $M = W$, $M = 1$ of Theorem 14.13.12.

Notes

This chapter is Guth, Iosevich, Ou and Wang (2020), Sections 2–6. Theorem 14.13.12 was found independently by Du and Zhang. The train track example descends from Katz and Tao. Theorem 14.13.18 records the two remaining inputs.

Chapter 14.14

Proof Certification and Reviewer Map

This chapter is the mathematical handoff for human review. It records what the active Volume 14 actually proves, where each long argument lives, which earlier results it uses, and which checks remain editorial rather than mathematical. It supersedes the proof-status tables attached to the retired survey route.

Status date: 31 August 2026.

All uses of *conditional*, *internally proved*, and *human-review pending* in this chapter refer to the manuscript and public source status checked on that date. A later preprint or publication does not silently change this certificate; it must be incorporated and audited first.

The operative standard is the following. A theorem is called *internally proved* when the active manuscript contains a complete proof and every non-elementary input is either proved in this volume or identified as a theorem already proved earlier in the series. A source citation records provenance; it does not replace a missing proof. Historical results that are mentioned only for comparison are not vertices of the proof graph.

14.14.1 Volume 14 theorem certificate

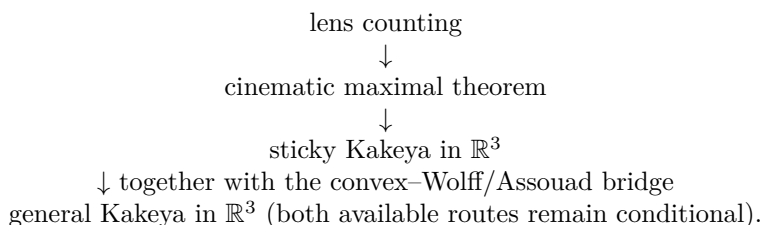
Output review	under	Proof/review site	Principal inputs or blocker
Sticky Kakeya in $\mathbb{R}^3$	in	Chapter 14.2	cinematic maximal estimate, projection/incidence theory

Output review	under	Proof/review site	Principal inputs or blocker
General Takeya in $\mathbb{R}^3$ (conditional as of 31 August 2026)		Chapters 14.3 and 14.4	streamlined factoring-interface and uniform-gain defects at Remarks 14.3.22 and 14.3.55; alternative-route obligations WZ–A1–A4 in Section 14.4.13
Convex–Wolff and Assouad bridge		Chapter 14.5	sticky theorem and cinematic maximal estimate
Cinematic maximal theorem		Chapter 14.6	pseudo-circle geometry and the complete lens count
Lens incidence theorem		Chapter 14.7	block decomposition, order comparison, random sampling
Radial thin-tube bootstrap		Chapter 14.8	Furstenberg incidence input and dyadic extraction
Bourgain projection theorem		Chapter 14.9	energy extraction, Blichfeldt counting, additive expansion
Planar Furstenberg improvement		Chapter 14.10	regular/irregular alternatives and multiscale combination
Pinned Falconer theorem above 5/4		Chapter 14.13	radial bootstrap, refined decoupling, Liu identity, energy summation

Except for the row explicitly marked conditional, the words “complete proof site” are literal. The active include list does not use the former chapters whose titles advertised schemes, architecture, assertions, or omitted bookkeeping. Those files remain in the source archive for provenance, but they are not part of either the Volume 14 book or the fourteen-volume composite.

14.14.2 The two main dependency chains

The Takeya chain is



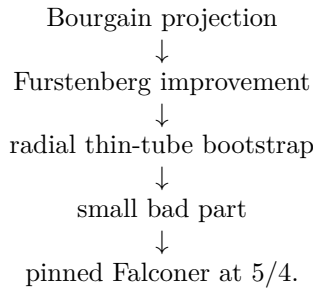
The arrows mean logical use, not merely historical influence. The first line is supplied by Chapters 14.2, 14.6 and 14.7; the second by Chapters 14.3 and 14.5. The streamlined route’s two blockers are Remarks 14.3.22 and 14.3.55. The independent route’s current blockers are WZ–A1–A4 in Section 14.4.13.

The fourth closure attempt sharpened these blockers rather than concealing them. Remark 14.3.23 lists the four labelled, simultaneous source-interface

assertions needed by the available cellular repair and explains which parts do not follow from the present uniformity definition. Lemma 14.3.56 gives a self-contained logical counterexample to the attempted passage from a gain for each fixed loss to one gain uniform in the loss, while Remark 14.3.57 records an exact sufficient replacement. These additions turn the boundary into checkable proof obligations; they do not discharge those obligations.

The long Wang–Zahl paper arXiv:2502.17655v1 uses a different two-parameter self-improvement and is therefore a plausible independent route around the two streamlined defects. Its formal argument is reproduced in Chapter 14.4, together with added proofs for several previously imported or abbreviated inputs. The audit nevertheless found three unresolved Cordoba interfaces, WZ–A1–A3, and an unstated tangential-isolation move, WZ–A4. In particular, WZ–A1 is the exact missing parentwise occupancy estimate (WZ–A1); it is not implied by the density and broadness statements currently printed. The alternative therefore remains conditional rather than serving as an unconditional input.

The projection–distance chain is



The analytic branch meeting the last arrow is

ordinary decoupling $\longrightarrow$ refined decoupling $\longrightarrow$ good-part L^2 estimate.

These branches are joined in Theorem 14.13.7. In particular, the phrase “projection input” in a provenance note should not be read as an unproved black box: the relevant projection, Furstenberg, and radial arguments are the full proofs in Chapters 14.9, 14.10, and 14.8.

14.14.3 Foundational inputs from earlier volumes

Self-containment for a series does not mean re-proving Hölder’s inequality in every later volume. The following earlier-volume results are intentionally reused. They are foundations, not Volume 14 omissions.

Input	Earlier proof	Use in Volume 14
Frostman measures and energy identities	Volumes 7–8	projection, radial, Furstenberg, and distance arguments
Loomis–Whitney and multilinear Kakeya	Volume 11	multilinear tube control inside the Kakeya dossiers incidence
Polynomial partitioning foundations	Volume 11	decompositions and geometric induction microlocal
Wave packets and restriction estimates	Volumes 6, 11–12	decompositions and packet orthogonality the ordinary
Parabola decoupling	Volume 13	decoupling input to Theorem 14.13.12
Fourier–Bessel and Riesz-energy calculus	Volumes 2, 7–8	Liu’s identity and the final Falconer summation

At each use the hypotheses must still be checked. In particular, the GIOW chapter performs the separated-support reduction before stationary phase, matches angular cap width to physical tube width, and verifies the precise Frostman exponent needed by the energy integral.

14.14.4 Source provenance and adaptation boundary

Several long proofs are adapted from author-released sources. Their provenance notes occur at the beginning of the relevant chapters. The active book makes four kinds of changes:

1. stand-alone title pages, global counters, appendices, and bibliographies are isolated so that the sources can be included in one book;
2. theorem labels and cross-references are namespaced or locally contained to avoid collisions;
3. dependencies that the source papers cite to other papers are supplied by adjacent proof dossiers when they are part of the Volume 14 chain.

4. abbreviated or omitted local steps are expanded, and evident statement, sign, delimiter, and normalization slips are corrected, with each mathematical intervention identified in the chapter provenance.

No mathematical step may be deleted merely to shorten the exposition. When a source proof contains a routine reduction whose omission would prevent a reader from reconstructing the constants or quantifiers, the book expands that reduction. The lens sampling argument, radial exceptional-set iteration, and the GIOW microlocal reconstruction are examples.

For the four adapted source chapters, the material interventions in this edition are explicit. The sticky chapter adds the omitted two-ends-on-lines proof. The general-to-sticky chapter adds the slab estimate, final plank algebra, Katz–Tao stopping tree, and biased maximal-density factoring proof, makes the stopping exponents explicit, and supplies the random-thinning, rigid-motion, all-scale, and fixed-loss bookkeeping. Its human-review audit records the unresolved parent–child factoring interface and uniform-gain dependency in Remarks 14.3.22 and 14.3.55. The Assouad chapter replaces external “same proof” instructions by exact deductions from complete sticky arguments printed earlier in this volume and corrects the terminal projection-volume sign. The cinematic chapter expands the coarse-tangency lemma. Layout-only reflow is not a mathematical intervention; all four chapter notices nevertheless record it as an adaptation under CC BY 4.0.

Where we are. Licensing and mathematical certification are separate questions. The provenance notes establish the adaptation basis and identify the source. The certificate above concerns proof closure in the manuscript. A reviewer should check both, but success of one does not imply success of the other.

14.14.5 Human-review routes

The manuscript is long enough that a single linear pass is inefficient. Four routes expose different failure modes.

Route A: theorem and quantifier audit

Read each principal theorem, then jump to every invoked lemma. Record the order of parameter choices. The most sensitive choices are the hierarchy of Kakeya scales, the common interval and density parameters in projection arguments, the radial-bootstrap exceptional mass, and the order

$$\alpha > \frac{5}{4}, \quad 0 < \delta \ll \alpha - 1, \quad \text{quad} R_0 \gg_{\alpha, \delta} 1$$

in Chapter 14.13. A constant may depend only on parameters chosen to its left.

Route B: normalization and scaling audit

Check every rescaling map together with its determinant, tube dimensions, frequency-cap dimensions, and Frostman constant. In the distance proof, verify the Hankel Plancherel measure in Theorem 14.13.13; in the refined decoupling proof, verify the parabolic rescaling of both caps and physical tubes; in the projection chapters, verify that discretized masses agree with covering numbers after trimming.

Route C: exceptional-set and refinement audit

At each refinement, write down the retained mass and the number of possible dyadic classes. Check that the cumulative loss is subpower or explicitly absorbed. This is the natural route through the general-to-sticky, Furstenberg, and radial chapters. It also catches a common logical error: having a good parameter for each scale does not automatically provide one parameter good at all scales.

Route D: analytic convergence audit

Replace every rapidly decreasing term by $C_N R^{-N}$ and choose N only after counting the summands. Verify all uses of Tonelli or Fubini first for nonnegative or absolutely integrable quantities. In Chapter 14.13, audit the low-frequency term, the sum over packet classes, the spatial tail of the good measure, and the terminal Riesz-energy integral separately.

14.14.6 What the automated audit can and cannot certify

The build audit can detect undefined references, duplicate destinations, missing files, malformed index entries, fatal typesetting errors, and many layout defects. A textual audit can detect forbidden status phrases in the active include graph and compare declarations with the theorem manifest. Neither audit proves a theorem. Human review remains essential for hidden changes of quantifier, circular induction, invalid disposal of exceptional sets, and exponent arithmetic.

The release labels therefore have two layers:

Mathematical proof closure. The complete proof is present and its declared dependencies are closed in the series.

Human-review readiness. The source builds cleanly, references resolve, the PDF is visually inspected, provenance is exposed, and the reviewer routes above can be followed without consulting retired survey chapters.

This volume does not currently claim the first for the entire active Volume 14 theorem graph: the general Kekeya output is blocked by Remarks 14.3.22 and 14.3.55. The closure criteria are made explicit in Remarks 14.3.23 and 14.3.57. Human-review readiness is a property of the release artifact and must be rechecked whenever the source changes.

Reviewer sign-off sheet

1. Confirm that the active include list contains the nine proof dossiers, the higher-dimensional dependency audit, the distance certificate, and these closing chapters, but none of the retired survey chapters.
2. Confirm that every unconditional principal result in the first table has a proof body, not only a statement, source citation, or architecture paragraph, and resolve the defect attached to the conditional row.
3. Follow both dependency chains and verify that each arrow lands on a proved theorem with matching hypotheses.
4. Run the reference and duplicate-label audit on both the Volume 14 book and the fourteen-volume composite.
5. Build the index, build twice more, render representative pages from every proof dossier, and inspect equations, figures, tables, headers, and bibliography boundaries.
6. Record any mathematical concern by theorem label and exact hypothesis; record any editorial concern by file and page.

Notes

The machine-readable companion to this chapter is `THEOREM-MANIFEST.txt`. The manifest is a navigation and governance artifact; when prose and proof disagree, the proof must be repaired and the manifest updated rather than treating the manifest as mathematical evidence.

Chapter 14.15

Research Frontier and Open Proof Obligations

The proof-audit revision changes the function of the final chapter. Most formerly missing Volume 14 arguments now appear in active proof dossiers, but the streamlined generalakeya route retains the two explicitly dated obligations recorded in the Volume 14 preface and Chapter 14.14. This chapter records those manuscript obligations separately from genuinely open research directions and from historical results mentioned only for context. The status was checked on 31 August 2026.

An open problem is not a theorem obligation. Its discussion may identify a barrier or a plausible interface, but neither is used in any proof in this volume. Numerical records move, so the durable content below is the precise question and the obstruction revealed by the completed proofs.

14.15.1akeya beyond three dimensions

Open Problem 14.15.1 (akeya in dimension at least four). If $E \subseteq \mathbb{R}^n$ contains a unit line segment in every direction and $n \geq 4$, prove

$$\dim_{\mathrm{H}} E = n.$$

The three-dimensional proof in Chapters 14.2 and 14.3 uses structures that are genuinely three-dimensional: planes control clusters of tubes, grains are plate-like, and the relevant projection and cinematic estimates have two-dimensional parameter geometry. A higher-dimensional proof needs an analogue that is stable under the same multiscale refinements; merely replacing “plane” by “hyperplane” does not supply the required incidence estimate.

Open Problem 14.15.2 (A higher-dimensional sticky theory). Formulate and prove a stickyakeya theorem in $\mathbb{R}^n$, $n \geq 4$, whose hypotheses are strong

enough to force full dimension but weak enough to arise from a generalakeya set after a complete general-to-sticky reduction.

The qualification is essential. A structural theorem that produces a sticky object with uncontrolled ancestry, incompatible refinements, or scale-dependent exceptional directions will not compose with the final argument. The proof in Chapter 14.3 gives a checklist of the compatibility properties that a higher-dimensional replacement must preserve.

Open Problem 14.15.3 (Quantitativeakeya maximal estimates in $\mathbb{R}^3$). Upgrade the dimension conclusion supplied by the completed three-dimensional proof to sharp quantitative bounds for theakeya maximal operator, with explicit dependence on the density and scale parameters.

The dimension theorem tolerates subpower losses and non-effective scale thresholds. A maximal estimate does not. The relevant audit is therefore not only exponent counting: one must track the number of refinements, the retained shading density, and the dependence of every induction constant.

14.15.2 Distance sets and radial projections

Open Problem 14.15.4 (The planar Falconer conjecture). Let $E \subseteq \mathbb{R}^2$ be compact with $\dim_{\text{H}} E > 1$. Prove $m(\Delta(E)) > 0$; ideally prove that $m(\Delta_x(E)) > 0$ for some $x \in E$.

The completed argument in Chapter 14.13 reaches $\alpha > 5/4$. Its terminal condition is

$$2 - \frac{\alpha + 1}{3} < \alpha.$$

Thus the present proof cannot reach $\alpha = 1$ without improving the good-packet exponent, finding additional cancellation, or changing the decomposition. This is a limitation of the method, not a counterexample at $5/4$.

Open Problem 14.15.5 (Falconer in higher dimensions). For $n \geq 3$, prove that every compact $E \subseteq \mathbb{R}^n$ with $\dim_{\text{H}} E > n/2$ has a positive-measure distance set.

The earlier spherical-average method and the planar pinned method use different geometry. A higher-dimensional pinned proof would require a radial projection theorem, a wave-packet decomposition for spheres, and a multiplicity estimate that survives the larger family of tangential directions. Each interface must be quantitative enough for the final energy integral.

Open Problem 14.15.6 (Higher-dimensional radial thin tubes). Find a sharp analogue of the radial thin-tube bootstrap in Chapter 14.8 for measures in $\mathbb{R}^n$, including an exceptional-set estimate uniform across scales.

The planar proof ultimately converts radial concentration into a planar Furstenberg incidence problem. In higher dimension, a tube through a pin

has a larger direction space and the corresponding incidence problem is not a formal iteration of the planar one.

14.15.3 Projection and Furstenberg questions

Open Problem 14.15.7 (Furstenberg sets in higher dimension). Determine the sharp Hausdorff-dimension lower bound for a set in $\mathbb{R}^n$ that meets a quantitatively large family of affine lines in subsets of prescribed dimension.

The planar proof in Chapter 14.10 depends on a line-parameter duality, dyadic regularization, and a Bourgain-type projection gain. Higher-dimensional line space has more parameters, so both the correct statement and the regular/irregular dichotomy must change.

Open Problem 14.15.8 (Endpoint and effective projection gains). Quantify the dependence of the projection improvement in Chapter 14.9 on the Frostman constants, separation parameters, and distance from the endpoint. Determine which subpower losses can be replaced by logarithmic or explicit power bounds.

Such estimates would sharpen the radial bootstrap and clarify whether the losses there are structural or only artifacts of repeated dyadic pigeonholing.

14.15.4 Analytic interfaces

Open Problem 14.15.9 (Refined decoupling with stronger incidence gain). Improve the dependence on packet multiplicity in Theorem 14.13.12, or identify additional hypotheses on the packet family that give a stronger estimate.

For the distance problem, goodness is converted into a bound for the number of active packets through a typical cube. Any stronger estimate must therefore state exactly which geometric statistic replaces or supplements that multiplicity. A gain that assumes uniform packet size or support without proving it will not feed the Frostman application.

Open Problem 14.15.10 (Endpoint decoupling constants). Replace R^ε losses in the decoupling inputs used by this series by effective polylogarithmic bounds wherever possible, and propagate the constants through refined decoupling and the geometric applications.

The qualitative induction closes because a fixed power gain absorbs every $R^{O(\delta)}$ loss. An effective theorem requires a different ledger: the number of scales, the loss at each pigeonholing, and the rate in every induction-on-scales step must all be recorded.

Open Problem 14.15.11 (Restriction and Bochner–Riesz in higher dimensions). Prove the sharp restriction and Bochner–Riesz conjectures in dimensions where they remain open.

The claimed Takeya theorem in $\mathbb{R}^3$ is conditional in this manuscript: the streamlined route has two recorded audit defects, and the independent Wang–Zahl reconstruction has the four obligations WZ–A1–A4 recorded in Section 14.4.13. A proof closing either complete route would remove one geometric obstruction but does not automatically control arbitrary bounded Fourier data. Decoupling is an ℓ^2 theory over caps, whereas the sharp restriction problem must exploit information compatible with L^∞ data. A valid transfer from Takeya must bridge that norm mismatch, not only cite the implication heuristic.

14.15.5 Robustness and verification problems

Open Problem 14.15.12 (Stability of near-extremal configurations). Classify configurations that nearly saturate the lens, projection, Furstenberg, or refined-decoupling estimates proved in this volume. Prove a quantitative statement saying that near equality forces closeness to one of the model configurations.

Such a theorem would turn several contradiction arguments into robust estimates and could reveal whether the same obstruction is being rediscovered in different coordinate systems.

Open Problem 14.15.13 (Machine-checkable dependency certificates). Develop a proof-aware manifest that records theorem hypotheses, parameter dependencies, and exact source locations, then mechanically checks the composition of those interfaces.

The current manifest can verify that a label has a proof and that references resolve. It cannot decide that a Frostman constant is uniform after rescaling or that an exceptional set is independent of scale. Even a partial formal language for these two patterns would materially strengthen future audits.

14.15.6 Context-only material

Historical sharp bounds, alternative proofs, and unused stronger theorems may be mentioned in notes or bibliographies. They are deliberately not promoted to numbered theorem environments unless their proofs are included or they are explicitly accepted as foundational inputs under the series policy. In particular:

- a newer numerical record does not invalidate a proved weaker theorem;
- an alternate route does not become a dependency merely because it is historically related;
- a conjectural extension is not evidence for a step in the proved dimension;

- a citation attached to an open problem is a reading pointer, not a proof certificate.

This distinction keeps the active proof graph stable while the research frontier moves.

Closing perspective

Volume 14 presently closes the planar pinned-distance chain above $5/4$ at the internal-proof level. It does not yet certify the general Kakeya chain in three dimensions. As of 31 August 2026, the streamlined route remains open at the cellular compatibility and uniform fixed-loss gates, while the independent Wang–Zahl route remains open at WZ–A1–A4, most concretely the parentwise normalized-slab occupancy estimate in (WZ–A1). The research problems above begin beyond the claimed endpoints. Their value for a reviewer is diagnostic: each one identifies a specific new interface without concealing the separate proof obligations already recorded inside this manuscript.

Notation

Referee C asked for a single list; the per-volume appendices remain, and this is the global one. Conventions for the asymptotic relations $\lesssim, \gtrsim, \approx, \lesssim$ are in the preface.

Sets, measures, dimensions

$m(E), E $	Lebesgue measure of E
$\mathcal{H}^s$	s -dimensional Hausdorff measure, Vol 7
$\dim_H, \dim_M, \underline{\dim}_M, \dim_P$	Hausdorff, upper and lower Minkowski, packing dimension
$\mathcal{E}_\rho(E)$	least number of ρ -balls covering E
$I_\alpha(\mu)$	α -energy $\iint x - y ^{-\alpha} d\mu d\mu$
$\mathcal{N}_t(X)$	the t -neighbourhood of X
$\mathbb{S}^{n-1}, d\sigma$	unit sphere and its surface measure
$\mathbb{T}^n$	the n -torus $\mathbb{R}^n/\mathbb{Z}^n$
$\Delta(E), \Delta_x(E)$	distance set and pinned distance set, Vols 8, 14
$\sigma(\mu, r)$	spherical average $\int_{\mathbb{S}^{n-1}} \widehat{\mu}(r\omega) ^2 d\sigma$

Analysis

$L^p, \ \cdot\ _p$	Lebesgue space and norm
$\widehat{f}, f^\vee$	Fourier transform and its inverse
$\mathcal{E}, \mathcal{R}$	extension and restriction operators, Vol 6
$\mathcal{M}^\#$	sharp maximal function, Vol 4
$\text{RapDec}(R)$	a quantity $\lesssim_N R^{-N}$ for every N
$D_p(\delta)$	decoupling constant, Vol 13
$\mathcal{N}_\delta$	δ -neighbourhood of the surface under discussion
$\mathbb{P}, \mathbb{C}, \Gamma_k$	paraboloid, cone, moment curve

Tubes and incidence geometry

$\mathbb{T}, \mathfrak{L}$	families of tubes, of lines
$Y(T), E_{\mathbb{T}}$	shading of a tube, and $\bigcup_T Y(T)$, Vol 14
$\Delta_{\max}(\mathbb{T})$	maximal convex density, Vol 14
$C_{\text{KT}}, C_{\text{F-S}}, C_{\text{F-C}}$	Katz–Tao convex, Frostman slab, Frostman convex Wolff constants
$\text{dir}(T)$	direction of a tube
$P_r(\mathfrak{L})$	points on at least r lines, Vol 11
$\mathcal{I}(P, \mathbb{T})$	incidences between points and tubes

Dependency Summary

What each principal theorem rests on, and which of those are quoted. The complete omission list is Chapter 14.15.

The main chain

Result	Where	Rests on
Hölder	Vol 1	—
Loomis–Whitney	Vol 11	Hölder, induction
Multilinear Kakeya	Vol 11	Loomis– Whitney Tucker’s
Polynomial ham sandwich	Vol 11	lemma, Borsuk–Ulam partitioning,
Line incidences in $\mathbb{R}^3$	Vol 11	Bézout, Szemerédi– Trotter discrete
Multilinear restriction	Vol 12	Loomis– Whitney
Tao’s bilinear theorem	Vol 12	wave packets, two-ends
Erdős’s threshold	Vol 12	Tao’s theorem, Mattila’s criterion
Decoupling, parabola	Vol 13	ball inflation, multilinear Kakeya
Decoupling, moment curve	Vol 13	Vandermonde, lower-degree induction
Vinogradov	Vol 13	moment curve decoupling
Sticky Kakeya in $\mathbb{R}^3$	Vol 14	grain structure, sum-product sticky case; streamlined
Kakeya in $\mathbb{R}^3$ (conditional as of 31 August 2026)	Vol 14	reduction has open cellular and fixed-loss audit gates

What is quoted, and what carries it

Quoted	Carries
Freiman’s theorem	the arithmetic Kakeya bound only
Orponen–Shmerkin ε -improvement	the radial projection bootstrap
Shmerkin’s starting point	the same
Fu–Ren incidence bound	the Falconer form, Furstenberg branch (iii)
Ren–Wang branch (ii)	nothing
Bourgain’s sum-product form	one expository remark
The circle method	Waring, zero-free regions
Wolff’s dyadic sum	the bootstrap base case, via the Frostman Slab axiom
Katz–Zahl, planebrush, Katz–Tao	nothing
Wooley’s congruencing	nothing

Status checked 31 August 2026. This historical table is not a proof certificate. Volume 13’s lower-degree moment-curve lemma, cone decoupling and microlocal local-smoothing implication now have internal proofs, as do the strict Volume 11–12 inputs they use. Its Wolff chain remains transitively pending only at the shared Volume 14 fractal-radii cinematic link. For the current transitive status use `THEOREM-MANIFEST.txt`; in particular, the general Kakeya row above remains conditional until an audited route closes its stated obligations.